Singularities in Hermitian pluripotential theory

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Introduction

A singular potential can be studied through its growth near its poles, through the holomorphic functions that remain integrable there, or through its Monge–Ampère measure. These descriptions lead to ordinary singularity types, multiplier ideals and curvature measures. Global pluripotential theory relates them and asks which singularities can be approximated by analytic ones. This book develops that theory for Hermitian backgrounds, starting from *Singularities in global pluripotential theory* [Xia26].

In a fixed Kähler class, all bounded potentials have the same Monge–Ampère mass. The non-pluripolar mass of a singular potential can therefore be compared with a fixed total mass, and this comparison connects analytic approximation, envelopes and energy. A nonclosed Hermitian background changes this relation at its first step: the mass can vary even among smooth potentials. Ordinary singularity type, which identifies potentials with bounded difference, no longer determines a single mass. Envelopes and multiplier ideals remain available, but their connection to mass requires different arguments.

Closedness of the background is a separate issue from the geometry of the manifold. A non-Kähler manifold can carry closed positive currents, and a Kähler manifold can be equipped with a nonclosed Hermitian form. Let X be a connected compact complex manifold of dimension $n \geq 1$, and let θ be a smooth real $(1,1)$-form on X. A function u is θ-plurisubharmonic if it is locally the sum of a plurisubharmonic function and a smooth function and the current $\theta_u = \theta + \mathrm{dd}^{\mathrm{c}} u$ is positive; the normalization of dd^{c} is fixed in the Conventions that follow this Introduction. If θ is closed, this current is locally a complex Hessian. For a general background, a smooth local correction still makes u plurisubharmonic, but leaves a smooth remainder in the current. Compactness, regularization and multiplier ideals thus survive locally; the new difficulties arise when these constructions are related globally.

Closedness, mass and energy

Let ω be a Hermitian form on X, that is, a smooth strictly positive $(1,1)$-form. If ω is Kähler, every smooth real potential u with $\omega + \mathrm{dd^c} u > 0$ satisfies

$$\int_X (\omega + \mathrm{dd^c} u)^n = \int_X \omega^n.$$

For a general Hermitian form, integration by parts gives instead

$$\int_X (\omega + \mathrm{dd^c} u)^n - \int_X \omega^n = \sum_{k=1}^{n-1} \binom{n}{k} \int_X u (\mathrm{dd^c} u)^{k-1} \wedge \mathrm{dd^c}(\omega^{n-k})$$

for every smooth u. The derivatives of the background already affect the mass before singularities appear. Chapter 2 proves that mass invariance is equivalent to $\mathrm{dd^c}(\omega^j) = 0$ for $1 \le j \le n-1$. The usual global comparison inequality would force this invariance, so a comparison argument in an arbitrary Hermitian background must account for these derivatives.

Once mass varies, its upper and lower bounds become separate questions. On the scalar Hopf manifolds $(\mathbb{C}^n \setminus \{0\})/\langle z \mapsto \mathrm{e}^{-1} z \rangle$, $n \ge 3$, the masses of smooth potentials with positive perturbed form are unbounded, while for the Hopf metric on the standard Hopf threefold they have a positive infimum. A different compact threefold carries a balanced metric with bounded masses whose infimum is zero (Chapter 9 and Section 27.18). Neither bound follows from the other. The envelope and measure constructions below accommodate this variation; Part IX develops volume comparison and equation theory under explicit mass bounds.

Mass conservation does not by itself restore the Kähler energy. At a smooth potential u with $\omega + \mathrm{dd^c} u > 0$, consider the one-form

$$\alpha_u(v) = \int_X v (\omega + \mathrm{dd^c} u)^n$$

on the space of such potentials, where v is a smooth real tangent direction. For a Kähler form, it is the differential of the Monge–Ampère energy. In dimension at least two, a primitive even near the zero potential forces $\mathrm{d}\omega = 0$ (Theorem 3.1.1). On a non-Kähler surface, a Gauduchon metric conserves mass but has no such primitive. Perron interpolation and Legendre duality can still be constructed by envelopes; an interpretation through a metric energy needs a separate justification.

A Gauduchon metric supplies a conserved linear trace in place of the varying nonlinear mass. Its form γ satisfies $\mathrm{dd^c}(\gamma^{n-1}) = 0$, and hence, for every θ-plurisubharmonic potential,

$$\int_X (\theta + \mathrm{dd^c} u) \wedge \gamma^{n-1} = \int_X \theta \wedge \gamma^{n-1}.$$

This trace measure determines u up to an additive constant. In a fixed background, for a smooth positive volume form $\mathrm{d}V$ and the normalization $\int_X u \, \mathrm{d}V = 0$, weak

convergence of trace measures is equivalent to L^1 convergence of potentials (Chapter 6). The trace stays finite even when the nonlinear Monge–Ampère mass is infinite. Extended non-pluripolar products and finite damped measures make it possible to study such potentials, while passage from damped to undamped measures requires control near the poles.

Singularity types and envelopes

Multiplier ideals record the holomorphic functions locally square integrable against e^{-ku}, for every $k > 0$. This gives a second equivalence relation on potentials, coarser than ordinary singularity type. To compare the two, fix $u \in \mathrm{PSH}(X, \theta)$ and take envelopes in the same background. The ordinary envelope $P_\theta[u]$ is the upper semicontinuous regularization of the supremum of the θ-plurisubharmonic $h \leq 0$ with $h \leq u + C$ for some constant C. The multiplier-ideal envelope $P_\theta[u]_{\mathcal{I}}$ uses instead the $h \leq 0$ with the same multiplier ideals as u at every positive weight.

In a closed big class on a compact Kähler manifold, for potentials of positive mass, the equality of these two envelopes characterizes $\mathcal{I}$-good singularities and is equivalent to a mass identity and approximation by analytic singularities. In the Hermitian setting, analytic approximation still relates the envelopes even when a mass identity is unavailable. Correcting the curvature errors keeps the dyadic approximants in a positive Hermitian background, and their envelopes decrease to the multiplier-ideal envelope. In such a positive background, the two envelopes, together with the ideal and Lelong data controlled by these approximants, define a uniform topology on ordinary classes in which equality of the envelopes characterizes the closure of the analytic classes, including classes with infinite-mass representatives (Theorem 17.6.1). This topology can identify distinct ordinary classes.

The obstacle in an envelope is part of this construction. A smooth change $\theta \mapsto \theta + \mathrm{dd}^{\mathrm{c}}\chi$ transforms potentials by $u \mapsto u - \chi$ and transforms their obstacles as well. Keeping only the obstacle zero need not give an invariant topology; allowing all smooth obstacles does. Whether equality of the envelopes at zero implies equality at every smooth obstacle is therefore a substantive question. It holds in a smooth big background with bounded mass when the multiplier-ideal envelope has positive lower volume, as well as on curves and on certain product loci without those volume conditions (Proposition 31.8.1 and Corollaries 31.14.1 and 32.3.3). The general case at zero volume remains open.

Envelopes and mass give complementary ways to compare singularities. An ordinary model is a fixed point $P_\theta[u] = u$, and the infimum and supremum of mass over an ordinary type define its lower and upper singularity volumes. These volumes need not agree. In a smooth big background with bounded mass, volume comparison gives a completely metrizable topology on the models of positive lower volume. At zero volume, the comparison can fail to distinguish models, so an extension must retain information that mass has lost.

On curves, the polar curvature—the part of $\theta + \mathrm{dd^c} u$ carried by the pole set—determines a model. Total variation of this measure gives a complete metric that includes zero-volume models, with the ideal models as the closure of the analytic models. The construction extends to certain higher-dimensional product loci without an ambient mass bound (Proposition 32.3.7). A general higher-dimensional model contains more information: residual top-degree mass may lie on a diffuse pole set although every polar curvature current on every modification vanishes. Distinct models can even have the same residual measure and both volumes, while their comparison distance is positive (Proposition 32.5.9). Thus a classification by measures already has limitations in closed Kähler backgrounds.

Restriction raises a related question. A potential may be identically $-\infty$ on a submanifold even though its generic Lelong number there is zero. In this case, the restrictions of its ambient multiplier ideals define finite divisorial data on every modification. Trace envelopes turn this data into constraints on potentials of the submanifold. Exact realization is possible on curves and in sufficiently large backgrounds on Kähler targets, and projective pullback data also admit exact realizations on Hopf manifolds of unbounded mass (Theorem 20.8.1). Thus failure of a mass bound does not itself prevent realization. Exact divisorial data nevertheless need not determine the Monge–Ampère mass, or a single ordinary type.

Equations with prescribed data

The varying mass also changes the normalization of an equation. Given a smooth real function F on X, one seeks a smooth pair (u, c) satisfying

$$(\omega + \mathrm{dd^c} u)^n = c\, \mathrm{e}^F \, \mathrm{d}V, \qquad \omega + \mathrm{dd^c} u > 0, \qquad \sup_X u = 0.$$

In a Kähler background, integration determines $c > 0$ before solving for u. In a Hermitian background the left-hand mass depends on the unknown potential, so the constant must be found together with it. Such a smooth pair exists and is unique. The linearization gives an adjoint Gauduchon probability measure describing the variation of $\log c$ with the density, and leads to a variational volume defined by Gauduchon test forms. This volume remains finite even on manifolds with unbounded upper Monge–Ampère mass.

Positive variational volume, however, need not imply that a background is big, meaning that it admits a potential ρ with $\theta + \mathrm{dd^c}\rho \geq \delta\omega$ for some $\delta > 0$. A nef nonclosed background on a projective manifold has positive variational volume and fails this condition (Theorem 27.13.1). Up to a smooth change of representative, the same example occurs at a finite maximal Chern–Ricci time (Theorem 27.20.3). An argument for geometric convergence that needs bigness therefore cannot substitute positive variational volume for it.

For a nonnegative, nonzero L^p density, $p > 1$, continuous solutions still exist and the positive constant is still unique, but uniqueness of the potential is a separate matter.

A smooth density with zeros may have distinct normalized smooth solutions. For a continuous density whose positive set is connected and dense, bounded solutions are unique up to constants (Theorem 12.4.1). Selection by approximation must also be distinguished from uniqueness. Moreover, when singularities are prescribed, distinct divisorial types can have the same normalization functional and supporting Gauduchon probabilities (Chapter 16); the scalar response does not recover the full type.

Replacing the determinant by an intermediate elementary symmetric polynomial leads to the Hessian measure $H_m(u) = (\omega + \mathrm{dd^c}u)^m \wedge \omega^{n-m}$, for $2 \le m < n$. Its admissible cone permits negative eigenvalues of the perturbed form, so bounded products require capacity estimates beyond the plurisubharmonic calculus. Derivatives of the background again obstruct the usual comparison inequality. Rescaling one potential and working on a sufficiently small sublevel set, however, allows the torsion error to be absorbed. The resulting modified comparison gives strict domination on an arbitrary Hermitian background. For example, for every $a > 0$ and smooth real F, the equation $H_m(u) = \mathrm{e}^{au+F}\omega^n$ has exactly one bounded admissible solution, and it is smooth (Corollary 13.12.2). Thus the increasing dependence on the potential can recover uniqueness and regularity even when mass is not conserved.

To prescribe singularities as well as the measure, one fixes a reference potential and uses relative envelopes to control the solution. For general finite non-pluripolar measures, Part IX proves existence in two regimes: semipositive big backgrounds of positive lower mass, with a model reference admitting a competitor in a smaller background, and arbitrary smooth big backgrounds of bounded mass, with a model reference of positive lower volume. These solutions belong to a relative envelope class. Membership in the prescribed ordinary type requires the stronger conclusion that their difference from the reference is bounded; additional estimates give this conclusion for suitable data.

Geometric constructions

Logarithmic poles give concrete tests of the theory. On a resolution, removing a prescribed divisorial pole can leave a strictly positive residual form, where a bounded equation determines the remaining potential. A divisor forced by the background behaves differently: it belongs to the type of minimal singularities, and removing it can force the current to descend. The constructions in Sections 15.21 and 15.22 distinguish a freely prescribed pole from such a compulsory one.

The regularity of a potential also affects the geometry of its current. When that current is smooth and strictly positive off an analytic set, a Hölder potential places the analytic set at finite distance, making the regular metric incomplete if the set is nonempty (Theorem 1.18.1). Boundedness alone is insufficient: radial examples in closed Kähler classes have continuous minimal residuals and complete regular ends. The integral controlling the potential differs from the one controlling path length. This distinction comes from regularity, even before closedness is lost.

Holomorphic sections link singularities to geometry through a closed line-bundle curvature, so counting sections cannot measure an arbitrary Hermitian Monge–Ampère mass. Gauduchon degrees control section growth and flag valuations, while fibrations can concentrate equilibrium measures on a lower-dimensional base. Partial Bergman kernels connect these section spaces to envelopes of singular weights. Closed Bott–Chern classes and locally conformally Kähler structures give further settings where the differential constraints can be used more precisely.

Reading the book

We assume familiarity with complex manifolds, differential forms and currents. Chapter 1 develops the plurisubharmonic tools used throughout. On a first reading, the maximum principle, normalized compactness and exponential integrability are the starting points; the longer approximation and descent arguments can be consulted when needed. The mass and energy calculations explain the obstructions above, and Gauduchon traces, non-pluripolar products and the Hopf construction complete the foundations. Chapter 5 applies Gauduchon degrees to the curvature of vector bundles; the later chapters do not depend on it.

The equation chapters develop smooth and weak estimates and comparison principles; the Hessian chapter examines how these arguments change when positivity is weakened. Envelopes and analytic approximation then develop the comparison of ordinary and ideal types; interpolation, restriction and holomorphic sections connect it to geometry. Part IX gives the volume and equation theory under mass hypotheses, together with the zero-volume models on products and fibrations of Chapter 32, some of which need no mass bound. The first appendix explains descent to compact unibranch spaces and solves equations on cones, normalized curves and cuspidal products; the second computes the cotangent torsors and section algebras of a split ruled surface over an elliptic curve. Open questions are collected in Chapter 35, and the sources and precise dependencies in Chapter 36; monograph references also appear where they are used.

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Conventions

Unless otherwise stated, X denotes a connected compact complex manifold of dimension $n \geq 1$. A Hermitian form is a smooth real strictly positive $(1,1)$-form. The letter ω is reserved for such a form, while θ denotes an arbitrary smooth real $(1,1)$-form. Whenever θ or ω is assumed to be closed, this is stated explicitly.

We follow the normalizations of [Xia26]:

$$\mathrm{d}^{\mathrm{c}} = \frac{\mathrm{i}}{4\pi}(\bar{\partial} - \partial), \qquad \mathrm{dd}^{\mathrm{c}} = \frac{\mathrm{i}}{2\pi}\partial\bar{\partial}.$$

Thus, on a disk in $\mathbb{C}$ with coordinate z, $\mathrm{dd}^{\mathrm{c}} \log |z|^2$ is the unit Dirac measure at the origin. Multiplier ideals of a quasi-plurisubharmonic function u are defined with the weight $\exp(-u)$, and Monge–Ampère masses carry no factor $1/n!$.

Frequently used notation

The following symbols are used throughout the book; the last column gives the place where each is defined. Notation that is used in only one chapter is introduced in that chapter.

Symbol	Meaning	Defined in
$\mathrm{PSH}(X,\theta)$, θ_u	θ-plurisubharmonic functions; $\theta_u = \theta + \mathrm{dd}^{\mathrm{c}}u$	Definition 1.1.1 and Section 1.7
$u \preceq v$, $u \sim v$	comparison and equality of ordinary singularity types	Definition 1.12.1
$\mathcal{I}(u)$	multiplier ideal sheaf of u	Definition 1.12.2
$u \preceq_{\mathcal{I}} v$, $u \sim_{\mathcal{I}} v$	comparison and equality of multiplier ideals at all weights	Section 1.12
$\nu(u,x)$	Lelong number of u at x	Section 1.13
$P_\theta(f)$, V_θ	envelope below an obstacle f; $V_\theta = P_\theta(0)$	Section 1.19
$P_\theta[u]$	envelope of the ordinary singularity type; its fixed points are the ordinary models	Definition 1.19.1
$P_\theta[u]_{\mathcal{I}}$	multiplier-ideal envelope	Definition 17.2.1
$\mathrm{MA}_\theta(u)$	Monge–Ampère measure of a bounded potential; where a chapter says so, also the extended measure	Section 1.7
$\mathrm{MA}^{\mathrm{ext}}_\theta(u)$	extended non-pluripolar measure, also written $\langle(\theta+\mathrm{dd}^{\mathrm{c}}u)^n\rangle$	Theorem 7.1.1
Cap_ω	Hermitian capacity	Section 1.8
$\nu_-(\omega)$, $\nu_+(\omega)$	lower and upper mass of a Hermitian form; bounded mass property	Definition 7.5.1
Gauduchon, balanced and pluriclosed metrics	Hermitian forms γ with $\mathrm{dd}^{\mathrm{c}}(\gamma^{n-1})=0$, $\mathrm{d}(\gamma^{n-1})=0$ and $\mathrm{dd}^{\mathrm{c}}\gamma=0$, respectively	Definition 4.1.1
$\mathcal{T}_\gamma$, $d_{\gamma,d}$	Gauduchon trace map and trace metric	Sections 6.1 and 6.2
$\underline{\mathrm{vol}}_\theta$, $\overline{\mathrm{vol}}_\theta$	lower and upper volumes of a singularity type	Definition 14.3.1
$\mathcal{E}_{\mathrm{env}}(X,\theta,\varphi)$	relative envelope class of a reference φ	Definition 14.4.1
d_θ	volume comparison of ordinary models	Definition 31.6.1
$b_h(f)$	subsolution threshold of a density f	Section 15.27
$\mathcal{D}_h$, ρ_h	space of density roots and root functional	Proposition 15.31.2 and Section 15.28
$\mathcal{V}_\mu(\theta)$, $\lambda_\mu(\theta)$	Gauduchon variational volume and root volume	Definition 10.8.1

Part I
The Hermitian foundations

Chapter 1
Plurisubharmonic functions and singularities

A function whose complex Hessian is bounded below by a smooth form becomes plurisubharmonic after a local smooth correction. This brings the local theory of plurisubharmonic functions to a compact complex manifold, provided the smooth remainder left by the correction is kept when potentials are glued, when their Hessians are multiplied and when one integrates by parts. Normalization by the supremum then controls the resulting families globally, giving compactness, uniform exponential integrability and capacity estimates on the whole manifold. Products of Hessians require more care than the potentials themselves: without a common bound, convergence of potentials in L^1, or even in capacity, does not prevent their Monge–Ampère measures from concentrating.

To keep track of singularities under these operations, we compare potentials in two ways: by domination up to an additive constant, which leads to ordinary singularity types and their envelopes, and by the local integrability of holomorphic functions against exponential weights, which leads to multiplier ideals. Lelong numbers, analytic approximation and modifications are the main tools for computing the latter.

1.1 Plurisubharmonic potentials

Throughout the chapter, X is a connected compact complex manifold of dimension $n \geq 1$, θ is a smooth real $(1,1)$-form and ω is a Hermitian form on X, unless a statement specifies another setting.

On an open set U in a complex manifold, a *plurisubharmonic function* is a locally integrable upper semicontinuous function satisfying the submean inequality on every complex line in local coordinates. We write $\mathrm{PSH}(U)$ for the set of these functions. A function is *quasi-plurisubharmonic* if it is locally the sum of a plurisubharmonic function and a smooth real function; the set of such functions is denoted by $\mathrm{QPSH}(U)$. The following relative positivity condition makes sense on any such open set as well.

Definition 1.1.1 A quasi-plurisubharmonic function u on X is *θ-plurisubharmonic* if

$$\theta + \mathrm{dd^c} u \geq 0$$

as a current. We denote the set of these functions by $\mathrm{PSH}(X, \theta)$.

Since $\omega > 0$, every constant function belongs to $\mathrm{PSH}(X, \omega)$.

Proposition 1.1.1 *Every real distribution u on X with $\theta + \mathrm{dd^c} u \geq 0$ has a unique θ-plurisubharmonic representative. On each relatively compact coordinate ball with coordinates $z = (z_1, \ldots, z_n)$, there is a constant $A > 0$ such that $u + A|z|^2$ is plurisubharmonic. Conversely, every quasi-plurisubharmonic function u on X belongs to $\mathrm{PSH}(X, C\omega)$ for some constant $C > 0$.*

Proof On a coordinate ball with compact closure in a larger chart, smoothness of θ gives

$$\theta \leq A\mathrm{dd^c}|z|^2$$

for a sufficiently large A. Consequently,

$$\mathrm{dd^c}(u + A|z|^2) \geq A\mathrm{dd^c}|z|^2 - \theta \geq 0.$$

The local distributional characterization of plurisubharmonic functions, [Dem12, Chapter I, Theorem 5.8], gives a unique plurisubharmonic representative of $u + A|z|^2$. Subtracting the smooth correction gives a quasi-plurisubharmonic representative of u on the ball. These representatives agree on overlaps: after a common local smooth correction, two of them become plurisubharmonic functions that are equal almost everywhere, hence everywhere. They therefore glue to a unique θ-plurisubharmonic representative.

For the converse, cover X by finitely many open sets $V_j \Subset U_j$, where each U_j is a chart on which $u = p_j + f_j$, with p_j plurisubharmonic and f_j smooth. On the compact set $\overline{V}_j$, the smooth form $\mathrm{dd^c} f_j$ is bounded below by $-C_j\omega$ for some constant C_j, so $\mathrm{dd^c} u \geq \mathrm{dd^c} f_j \geq -C_j\omega$ on V_j. Any constant $C > 0$ larger than all the C_j gives $\mathrm{dd^c} u \geq -C\omega$ on X. □

For $u \in \mathrm{PSH}(X, \theta)$, choose coordinates z and a constant $A > 0$ as in Proposition 1.1.1, and put $p = u + A|z|^2$. Then

$$\theta + \mathrm{dd^c} u = \mathrm{dd^c} p + (\theta - A\mathrm{dd^c}|z|^2).$$

Thus local plurisubharmonic theory applies to p, while the smooth remainder $\theta - A\mathrm{dd^c}|z|^2$ records the original background; this remainder must be kept when forming products or integrating by parts.

Proposition 1.1.2 *If $\theta = \mathrm{dd^c}\rho$ on an open set for a smooth function ρ, then $\mathrm{d}\theta = 0$ on that set. In particular, at a point where $\mathrm{d}\theta \neq 0$, no neighborhood admits such a representation.*

Proof Apply d and use $\mathrm{ddd^c} = 0$. □

Conversely, a closed real $(1,1)$-form has smooth local dd^{c}-potentials on sufficiently small coordinate balls, by the local $\partial\bar{\partial}$-lemma; this local fact does not require X to admit a Kähler metric.

The exterior derivative also constrains the regularity of a potential whose current is pointwise proportional to a Hermitian form. In dimension at least two, differentiating such an identity expresses the derivative of the proportionality factor through smooth coefficients.

Lemma 1.1.1 *Let X be a complex manifold of dimension $n \geq 2$, let θ be a smooth real $(1,1)$-form and let γ be a Hermitian form on X. Suppose that real-valued $u, a \in L^1_{\mathrm{loc}}(X)$ satisfy*

$$\theta + \mathrm{dd}^{\mathrm{c}} u = a\gamma$$

as currents. Then a and u are smooth.

Proof Taking the exterior derivative and multiplying by γ^{n-2} gives

$$\mathrm{d}a \wedge \gamma^{n-1} = \mathrm{d}\theta \wedge \gamma^{n-2} - a\,\mathrm{d}\gamma \wedge \gamma^{n-2}.$$

At each point, $\xi \mapsto \xi \wedge \gamma^{n-1}$ is an isomorphism from real one-forms to real $(2n-1)$-forms, as one sees in a unitary frame. Its inverse depends smoothly on the point. Let B and C be the images of $\mathrm{d}\theta \wedge \gamma^{n-2}$ and $\mathrm{d}\gamma \wedge \gamma^{n-2}$ under this inverse. They are smooth real one-forms, and

$$\mathrm{d}a = B - aC$$

in the sense of distributions. Since $a \in L^1_{\mathrm{loc}}$, the right-hand side is locally integrable, so $a \in W^{1,1}_{\mathrm{loc}}$. Differentiating the equation and using the smoothness of its coefficients, we obtain inductively $a \in W^{k,1}_{\mathrm{loc}}$ for every integer k, and the local Sobolev embedding shows that a is smooth. Consequently $\mathrm{dd}^{\mathrm{c}} u = a\gamma - \theta$ is smooth; taking its trace in local coordinates and applying elliptic regularity shows that u is smooth as well. □

1.2 Changes of background

A positive current can be written as a smooth form plus a complex Hessian in more than one way. Changing the decomposition by a smooth dd^{c}-exact form changes the potential by a bounded function, and positive linear combinations allow us to compare potentials belonging to different backgrounds. In this section, η denotes a second smooth real $(1,1)$-form on X.

Proposition 1.2.1 *Let f be a smooth real function on X, and set $\theta' = \theta + \mathrm{dd}^{\mathrm{c}} f$. Then*

$$\mathrm{PSH}(X,\theta) \longrightarrow \mathrm{PSH}(X,\theta'), \qquad u \longmapsto u - f$$

is a bijection. It preserves the positive current and changes each potential by a bounded function. Moreover, $\mathrm{d}\theta' = \mathrm{d}\theta$.

Proof For every $u \in \mathrm{PSH}(X,\theta)$,

$$\theta' + \mathrm{dd}^{\mathrm{c}}(u - f) = \theta + \mathrm{dd}^{\mathrm{c}} u.$$

The inverse map is $v \mapsto v + f$, and f is bounded because X is compact. The last assertion follows from $\mathrm{ddd}^{\mathrm{c}} f = 0$. □

Thus the natural object is the class of θ modulo smooth dd^{c}-exact forms. This quotient of all smooth real $(1,1)$-forms contains nonclosed classes; the usual Bott–Chern cohomology group $H^{1,1}_{\mathrm{BC}}(X,\mathbb{R})$ is the corresponding quotient of closed forms (Lemma 1.17.4).

Proposition 1.2.2 *If $u \in \mathrm{PSH}(X,\theta)$ and $v \in \mathrm{PSH}(X,\eta)$, then, for $a, b \geq 0$,*

$$au + bv \in \mathrm{PSH}(X, a\theta + b\eta).$$

In particular, $\mathrm{PSH}(X,\theta)$ *is convex. If $\eta \geq \theta$, then* $\mathrm{PSH}(X,\theta) \subseteq \mathrm{PSH}(X,\eta)$.

Proof Omitting a term whose coefficient is zero, we see that $au + bv$ is quasi-plurisubharmonic. Moreover,

$$a\theta + b\eta + \mathrm{dd}^{\mathrm{c}}(au + bv) = a(\theta + \mathrm{dd}^{\mathrm{c}} u) + b(\eta + \mathrm{dd}^{\mathrm{c}} v) \geq 0.$$

For the inclusion, add the semipositive smooth form $\eta - \theta$ to $\theta + \mathrm{dd}^{\mathrm{c}} u$. □

By Proposition 1.1.1, any two quasi-plurisubharmonic functions on X belong to $\mathrm{PSH}(X, C\omega)$ for a common sufficiently large constant C. This allows us to compare their singularities within one positive background.

1.3 Maxima and the maximum principle

Maxima allow us to truncate potentials and to modify them on an open set without losing the lower curvature bound. For a smooth background, the local maximum and gluing principles for plurisubharmonic functions keep this bound: one freezes the coefficients of the background on small balls and then lets the resulting curvature error tend to zero.

Lemma 1.3.1 *If $u, v \in \mathrm{PSH}(X,\theta)$, then* $\max(u,v) \in \mathrm{PSH}(X,\theta)$. *Consequently, if $\theta \geq 0$ and $u \in \mathrm{PSH}(X,\theta)$, then* $\max(u,-t)$ *is θ-plurisubharmonic for every real t.*

Proof Adding a common local smooth correction to u and v shows that $\max(u, v)$ is quasi-plurisubharmonic. To obtain the sharp lower bound for its Hessian, fix a positive constant $(1, 1)$-form κ on a coordinate ball. For each $\varepsilon > 0$, every point of the ball has a neighborhood on which θ is close to a constant form θ', in the sense that $-\varepsilon\kappa \le \theta - \theta' \le \varepsilon\kappa$. Write $\theta' + \varepsilon\kappa = \mathrm{dd}^{\mathrm{c}} r$ with a quadratic polynomial r. Then $\mathrm{dd}^{\mathrm{c}}(u + r) \ge \theta + \mathrm{dd}^{\mathrm{c}} u \ge 0$, so $u + r$ and, likewise, $v + r$ are plurisubharmonic; hence so is their maximum. It follows that

$$\theta + 2\varepsilon\kappa + \mathrm{dd}^{\mathrm{c}} \max(u, v) \ge 0$$

on each such neighborhood, and letting $\varepsilon \to 0$ gives $\theta + \mathrm{dd}^{\mathrm{c}} \max(u, v) \ge 0$. The last assertion follows because constants belong to $\mathrm{PSH}(X, \theta)$ when $\theta \ge 0$. □

The proof is local. Hence the first assertion of Lemma 1.3.1 holds for θ-plurisubharmonic functions on any open subset of a complex manifold, with θ a smooth real $(1, 1)$-form there.

Lemma 1.3.2 *Let Y be a complex manifold, let β be a smooth real $(1, 1)$-form on Y, and let $D \subset Y$ be open. Suppose that u is β-plurisubharmonic on Y, that v is β-plurisubharmonic on D, and that*

$$\limsup_{D \ni y \to x} v(y) \le u(x) \qquad (x \in \partial D).$$

Then the function equal to $\max(u, v)$ on D and to u on $Y \setminus D$ is β-plurisubharmonic.

Proof Denote the glued function by w; it is upper semicontinuous by the boundary inequality. Freeze β on a small coordinate ball as in the proof of Lemma 1.3.1, and choose a quadratic correction r such that $u + r$ and $v + r$ are plurisubharmonic on their respective domains. Off ∂D, the function $w + r$ is plurisubharmonic, being locally either $u + r$ or the maximum of two plurisubharmonic functions. At a point of ∂D, it equals $u + r$ and dominates $u + r$ on the ball, so it satisfies the submean inequality on every complex line through that point. Thus $w + r$ is plurisubharmonic on the whole ball. Letting the error of the frozen coefficients tend to zero gives $\beta + \mathrm{dd}^{\mathrm{c}} w \ge 0$. □

For quasi-plurisubharmonic functions, an inequality that holds almost everywhere holds everywhere. This allows comparisons proved for Sobolev or Lebesgue classes to be applied to the potentials themselves.

Lemma 1.3.3 *Let a, b be quasi-plurisubharmonic functions on a complex manifold. If $a \le b$ almost everywhere, then $a \le b$ everywhere. In particular, equality almost everywhere implies equality everywhere.*

Proof On a coordinate ball, add the same smooth function q to both functions so that $a + q$ and $b + q$ are plurisubharmonic. By the submean inequality and upper semicontinuity, a plurisubharmonic function is, at each point, the limit of its averages over balls centered there as the radius tends to zero. The almost-everywhere inequality orders these averages, so passing to the limit gives $a + q \le b + q$. Subtracting q proves the claim. □

Lemma 1.3.4 *Let a, b be quasi-plurisubharmonic functions on a complex manifold. If $a(x) < b(x)$ at some point x, then $\{a < b\}$ has positive volume with respect to any smooth positive volume form.*

Proof Choose a number c with $a(x) < c < b(x)$. By upper semicontinuity, $a < c$ on a neighborhood V of x. If $\{b > c\} \cap V$ had zero volume, then $b \le c$ almost everywhere on V, and Lemma 1.3.3 would give $b(x) \le c$, a contradiction. Thus $\{b > c\} \cap V$, which is contained in $\{a < b\}$, has positive volume. □

Lemma 1.3.5 *A plurisubharmonic function on $\mathbb{C}^m$, $m \ge 1$, that is bounded above is constant.*

Proof This is the Liouville theorem of [GZ17, Lemma 2.28]. □

For a function f, we write $f^*(x) = \limsup_{y \to x} f(y)$ for its upper semicontinuous regularization. A subset of a complex manifold is *pluripolar* if each of its points has a neighborhood U such that the part of the subset in U is contained in $\{p = -\infty\}$ for some $p \in \mathrm{PSH}(U)$. The local upper-envelope theorem recalled in [Xia26] states that if a nonempty family of plurisubharmonic functions is locally uniformly bounded above, then the regularization of its supremum is plurisubharmonic and differs from the supremum only on a pluripolar set.

Periodicity also forces rigidity when an entire subharmonic function has only logarithmic upper growth.

Lemma 1.3.6 *Let h be subharmonic on $\mathbb{C}$, with $h(\zeta + 1) = h(\zeta)$. If there are real constants $A \ge 0$ and B such that*

$$h(\zeta) \le A \log(1 + |\zeta|) + B \qquad (\zeta \in \mathbb{C}),$$

then h is constant.

Proof By the growth bound, the real translates $h(\cdot + s)$, $s \in [0, 1]$, form a family of subharmonic functions that is locally uniformly bounded above. The local upper-envelope theorem therefore shows that

$$H(\zeta) = \Big(\sup_{s \in [0,1]} h(\zeta + s) \Big)^*$$

is subharmonic. By periodicity, H is invariant under every real translation, so it depends only on $\operatorname{Im} \zeta$, and its subharmonicity means that it is a convex function of $\operatorname{Im} \zeta$. On the imaginary axis it satisfies $H(\mathrm{i}y) \le A \log(2 + |y|) + B$. Since a nonconstant convex function on $\mathbb{R}$ grows at least linearly at one end, this bound shows that H is constant. Thus $h \le H$ is bounded above, and Lemma 1.3.5 shows that h is constant. □

Lemma 1.3.7 *Let U be a complex manifold, let θ be a smooth real $(1,1)$-form on U, and let $Z \subset U$ be closed and locally pluripolar. Let u be θ-plurisubharmonic on $U \setminus Z$ and locally bounded above near Z. Then its upper-limit extension, defined at $x \in Z$ by $u(x) = \limsup_{U \setminus Z \ni y \to x} u(y)$, is θ-plurisubharmonic on U. In particular, this applies when Z is a nowhere dense analytic subset of U.*

Proof First suppose that u is plurisubharmonic. Near a point of Z, choose a plurisubharmonic function $\psi \le 0$ with $\psi = -\infty$ on Z. For $t > 0$, extend $u + t\psi$ by $-\infty$ on Z. The local upper bound for u makes this extension upper semicontinuous. It is plurisubharmonic off Z, and its submean inequality on complex lines is automatic at points of Z, where its value is $-\infty$. It is therefore plurisubharmonic on the whole neighborhood. As $t \downarrow 0$, these functions increase, with a common local upper bound. By the local upper-envelope theorem, the regularization of their limit is a plurisubharmonic function that agrees with u almost everywhere off Z, hence everywhere off Z by Lemma 1.3.3. By upper semicontinuity and the submean inequality, its value at a point of Z is the upper limit of u off Z, so it is the upper-limit extension of u.

For a smooth background, fix a positive constant form κ on a coordinate chart and let $\epsilon > 0$. On a sufficiently small ball, freeze θ to a constant form θ' with $-\epsilon\kappa \le \theta - \theta' \le \epsilon\kappa$, and choose a quadratic potential q for $\theta' + \epsilon\kappa$. Applying the preceding argument to the plurisubharmonic function $u + q$ shows that the upper-limit extension satisfies $\theta + \mathrm{dd}^{\mathrm{c}} u \ge -2\epsilon\kappa$ on the ball. Since the extension is defined by upper limits, it does not depend on the ball, and letting $\epsilon \downarrow 0$ gives $\theta + \mathrm{dd}^{\mathrm{c}} u \ge 0$. Finally, a nowhere dense analytic subset is locally pluripolar: locally, it is contained in the $-\infty$ locus of the logarithm of the sum of the squared moduli of local defining functions, not all identically zero. □

Lemma 1.3.8 *Let X be a complex manifold of dimension $n \ge 2$, let θ be a smooth real $(1,1)$-form on X, and let $p \in X$. Every θ-plurisubharmonic function on $X \setminus \{p\}$ is locally bounded above near p and extends uniquely to a member of* $\mathrm{PSH}(X, \theta)$. *If X is compact, it is bounded above on $X \setminus \{p\}$.*

Proof Choose coordinates centered at p and write $\Delta_R = \{z \in \mathbb{C} : |z| < R\}$. On the coordinate polydisk Δ_R^n, add a smooth quadratic function to the given function to obtain a plurisubharmonic function h on $\Delta_R^n \setminus \{0\}$. Write the coordinates as (z_1, z_2, z'), with z' absent when $n = 2$, and choose $0 < r < R$. The compact set

$$K = \{|z_1| \le r,\ |z_2| = r,\ |z_j'| \le r\}$$

avoids the puncture, so $h \le M$ on K for some finite M. Consider a point of the punctured closed polydisk of radius r. If $z_1 \ne 0$, the disk in the z_2-direction through the point avoids the puncture and has its boundary in K, so the maximum principle gives $h \le M$ at the point. If $z_1 = 0$ and $(z_2, z') \ne 0$, the disk in the z_1-direction through the point avoids the puncture, and its boundary consists of points with $z_1 \ne 0$, where the bound has already been proved; the maximum principle again gives $h \le M$. Subtracting the smooth correction proves the local upper bound.

By Lemma 1.3.7, applied with $Z = \{p\}$, the upper-limit extension belongs to $\mathrm{PSH}(X, \theta)$. It is unique by Lemma 1.3.3. If X is compact, the extension is an upper semicontinuous function on a compact space and is therefore bounded above. □

Lemma 1.3.9 *Let u be θ-plurisubharmonic near a point x. If a C^2 function a touches u from above at x, that is, $a \ge u$ near x and $a(x) = u(x)$, then $\theta(x) + \mathrm{dd}^{\mathrm{c}} a(x) \ge 0$.*

Proof In coordinates centered at x, let q be a quadratic polynomial with $\mathrm{dd^c} q = \theta(x)$, and let $\varepsilon > 0$. On a sufficiently small ball about x, we have $\theta \le \theta(x) + \varepsilon \mathrm{dd^c}|z|^2$, so $u + q + \varepsilon|z|^2$ is plurisubharmonic, and $a + q + \varepsilon|z|^2$ touches it from above at x. By the submean inequality on complex lines through x, a C^2 function touching a plurisubharmonic function from above at x has nonnegative complex Hessian at x. Hence $\theta(x) + \mathrm{dd^c} a(x) + \varepsilon \mathrm{dd^c}|z|^2 \ge 0$, and letting $\varepsilon \to 0$ proves the lemma. □

1.4 Normalized compactness

Fix a smooth positive probability volume form $\mathrm{d}V$ on X, used for L^1 norms. Consider a sequence of plurisubharmonic functions on a connected domain that is locally uniformly bounded above. By the local compactness theorem for plurisubharmonic functions [GZ17, Theorem 1.46], it has a subsequence that either converges in L^1_{loc} or converges locally uniformly to $-\infty$. For normalized θ-plurisubharmonic functions on X, a point at which the maximum is attained excludes the second alternative, and connectedness of X propagates the first one across X.

The submean inequality also controls upper limits at moving points.

Lemma 1.4.1 *Let u_j, u be quasi-plurisubharmonic functions on an open set U in a complex manifold. Suppose that $u_j \to u$ in $L^1_{\mathrm{loc}}(U)$ and that $\mathrm{dd^c} u_j \ge -\beta$ for a fixed smooth real $(1,1)$-form β on U. If $x_j \to x$ in U, then*

$$\limsup_{j\to\infty} u_j(x_j) \le u(x).$$

Moreover, the tail envelopes $h_j = (\sup_{k\ge j} u_k)^*$ decrease to u.

Proof On a coordinate ball about x, add a common smooth correction q so that $u_j + q$ and $u + q$ are plurisubharmonic. For a fixed small radius, the submean inequality bounds $(u_j + q)(x_j)$ by the average of $u_j + q$ over the ball of that radius centered at x_j. As $j \to \infty$, these averages converge to the average of $u + q$ over the ball of the same radius centered at x: the contribution of $u_j - u$ tends to zero in L^1, and the integral of the fixed locally integrable function $u + q$ over a ball depends continuously on its center. Letting the radius decrease to zero proves the inequality.

The same estimate gives local uniform upper bounds for the u_j with j large. By coefficient freezing and the local upper-envelope theorem, the tail envelopes are therefore quasi-plurisubharmonic, with the same lower Hessian bound. They decrease with j. If their limit exceeded $u(x)$ at some point x, the definition of the regularization would give indices $k_j \ge j$ and points $x_j \to x$ contradicting the first assertion for the subsequence (u_{k_j}). On the other hand, an almost everywhere convergent subsequence of (u_j) shows that $h_j \ge u$ almost everywhere, hence everywhere by Lemma 1.3.3. Thus $h_j \downarrow u$. □

Theorem 1.4.1 *For every smooth real $(1,1)$-form θ on X, the family*

$$\mathcal{K}_\theta = \{u \in \mathrm{PSH}(X,\theta) : \sup_X u = 0\}$$

is compact in $L^1(X, \mathrm{d}V)$. *In particular, there is a constant* C_θ *such that*

$$\int_X |u|\,\mathrm{d}V \le C_\theta \quad (u \in \mathcal{K}_\theta).$$

Proof Let (u_j) be a sequence in $\mathcal{K}_\theta$. By upper semicontinuity, there are points x_j with $u_j(x_j) = 0$; passing to a subsequence, we may assume that $x_j \to x$. Cover X by finitely many connected coordinate balls, and on a slightly larger ball around each of them choose a fixed smooth function r_i such that every $u_j + r_i$ is plurisubharmonic; such corrections exist because θ is smooth. Since $u_j \le 0$, these local functions are uniformly bounded above.

On a ball containing x in its interior, locally uniform convergence to $-\infty$ is impossible, because the values $(u_j + r_i)(x_j) = r_i(x_j)$ stay bounded below. Local compactness therefore gives a subsequence converging in L^1 on this ball. If a second ball meets the first, local compactness on the second ball cannot yield convergence to $-\infty$ on compact subsets of the intersection, since the first limit is finite almost everywhere there. Thus a further subsequence converges in L^1 on the second ball. Since X is connected, the intersection graph of the finite cover by connected balls is connected. Proceeding along this graph and passing to subsequences finitely many times, we obtain local L^1 convergence on every ball, and a partition of unity then gives convergence in $L^1(X)$ to a function u.

The inequalities $\theta + \mathrm{dd}^{\mathrm{c}} u_j \ge 0$ pass to the limit in the sense of distributions. By Proposition 1.1.1, u has a θ-plurisubharmonic representative, still denoted by u. Since $u_j \le 0$, we have $u \le 0$ almost everywhere, hence everywhere by Lemma 1.3.3.

By Lemma 1.4.1, $0 = \lim_j u_j(x_j) \le u(x)$. Thus $u(x) = 0$, and the limit belongs to $\mathcal{K}_\theta$. This proves sequential compactness, which is compactness in the metric space $L^1(X)$. The norm bound follows, since a compact subset of $L^1(X)$ is bounded. □

Corollary 1.4.1 *Let* (u_j) *be a decreasing sequence in* $\mathrm{PSH}(X,\theta)$ *with uniformly bounded* L^1 *norms. Then its pointwise limit* u *belongs to* $\mathrm{PSH}(X,\theta)$, *and* $u_j \to u$ *in* L^1.

Proof On each connected coordinate ball, after a common smooth correction, the limit of a decreasing sequence of plurisubharmonic functions is either plurisubharmonic or identically $-\infty$. Fatou's lemma, applied to the negative parts, excludes the second alternative because of the L^1 bound. Hence u is quasi-plurisubharmonic, and the inequality $\theta + \mathrm{dd}^{\mathrm{c}} u \ge 0$ passes to the decreasing limit. Finally, the nonnegative functions $u_1 - u_j$ increase to $u_1 - u$, so monotone convergence gives $u_j \to u$ in L^1.□

1.5 Uniform exponential integrability

For a single quasi-plurisubharmonic function v on X, Skoda's local integrability theorem [Dem11, Lemma 5.6] applies after local smooth corrections. Taking the smallest exponent over a finite cover gives $\mathrm{e}^{-a(v)v} \in L^1(X)$ for some $a(v) > 0$. Normalized compactness makes the exponent and the bound for the integral uniform: otherwise, a convergent weighted sum of counterexamples would violate this qualitative statement.

Let K_ω be the constant of Theorem 1.4.1 for the background ω, so that every $v \in \mathrm{PSH}(X, \omega)$ with $\sup_X v = 0$ satisfies

$$\int_X |v| \, \mathrm{d}V \le K_\omega. \tag{1.1}$$

Lemma 1.5.1 *There are constants $\alpha > 0$ and $C < +\infty$, depending only on $(X, \omega, \mathrm{d}V)$, such that*

$$\int_X \mathrm{e}^{-\alpha v} \, \mathrm{d}V \le C \quad \textit{for all } v \in \mathrm{PSH}(X, \omega) \textit{ with } \sup_X v = 0.$$

Proof Suppose that no such pair exists. Set $a_j = 2^{-j}$ and choose normalized potentials $v_j \in \mathrm{PSH}(X, \omega)$ with

$$\int_X \exp\left(-\frac{a_j}{j} v_j\right) \mathrm{d}V \ge 2^j.$$

The partial sums $V_N = \sum_{j \le N} a_j v_j$ decrease, since every $v_j \le 0$. They belong to $\mathrm{PSH}(X, \omega)$ by Proposition 1.2.2, because $\sum_j a_j \le 1$, and their L^1 norms are uniformly bounded by (1.1). By Corollary 1.4.1, their limit V is therefore ω-plurisubharmonic. Since all summands are nonpositive, $V \le a_j v_j$ for every j.

Fix $p > 0$. For every $j \ge 1/p$,

$$\int_X \mathrm{e}^{-pV} \, \mathrm{d}V \ge \int_X \mathrm{e}^{-p a_j v_j} \, \mathrm{d}V \ge \int_X \mathrm{e}^{-(a_j/j) v_j} \, \mathrm{d}V \ge 2^j.$$

Thus e^{-pV} is not integrable for any $p > 0$, which contradicts the integrability of $\mathrm{e}^{-a(V)V}$ obtained above from Skoda's theorem. □

1.6 Local products and capacity

Bedford–Taylor products of bounded plurisubharmonic functions, and the capacity defined through them, measure the exceptional sets of plurisubharmonic functions and control bounded approximation. We recall these classical facts of local pluripotential theory from [GZ17, Chapters 3 and 4] and [Xia26, Sections 1.3, 2.1 and 2.3]; in

particular, Theorem 1.6.1 collects [Xia26, Proposition 2.1.1, Theorem 2.1.1 and Proposition 2.2.1]. The *plurifine topology* is the coarsest topology that makes all local plurisubharmonic functions continuous. It is finer than the ordinary topology, and quasi-plurisubharmonic functions are plurifine continuous; in particular, their strict superlevel sets are plurifine open.

Theorem 1.6.1 (Bedford–Taylor) *Let $1 \le k \le n$. If $p_1, \dots, p_k$ are locally bounded plurisubharmonic functions on a domain in $\mathbb{C}^n$, their mixed product $\mathrm{dd^c} p_1 \wedge \cdots \wedge \mathrm{dd^c} p_k$ is a positive closed current of order zero. It is symmetric, multilinear in its current factors, and continuous under simultaneous locally bounded decreasing approximation. It does not charge pluripolar sets and is local in the plurifine topology.*

Here plurifine locality means that if two families of such functions agree on a plurifine open Borel set, then their products have the same restriction to that set.

The decreasing continuity of Theorem 1.6.1 persists after the products are multiplied by a bounded potential.

Lemma 1.6.1 *Let $a_j \downarrow a$ and $b_{i,j} \downarrow b_i$, $1 \le i \le r \le n$, be locally uniformly bounded plurisubharmonic functions on a domain in $\mathbb{C}^n$. Then*

$$a_j \, \mathrm{dd^c} b_{1,j} \wedge \cdots \wedge \mathrm{dd^c} b_{r,j} \longrightarrow a \, \mathrm{dd^c} b_1 \wedge \cdots \wedge \mathrm{dd^c} b_r$$

weakly as currents. The same holds for quasi-plurisubharmonic functions that become plurisubharmonic after fixed smooth corrections, and after wedging with smooth forms that converge locally in C^∞.

Proof For plurisubharmonic functions this is the decreasing convergence theorem [GZ17, Theorem 3.18] with $T = 1$; there $\mathrm{dd^c}$ is twice ours, which multiplies both sides by 2^r. For quasi-plurisubharmonic functions, add the fixed smooth corrections to the decreasing sequences, apply the plurisubharmonic case to the corrected functions, and expand; the terms involving the correction to a_j converge by the decreasing continuity in Theorem 1.6.1. Finally, the local Chern–Levine–Nirenberg bounds show that these finitely many products have uniformly bounded variation on compact sets, so smooth forms converging locally in C^∞ may replace fixed forms in the tests. □

For concentric coordinate balls $B' \Subset B$ and a Borel set $E \subset B'$, the *relative Bedford–Taylor capacity* is

$$\operatorname{cap}(E, B) = \sup\Big\{ \int_E (\mathrm{dd^c} v)^n : v \in \mathrm{PSH}(B),\ 0 \le v \le 1 \Big\}.$$

If v is bounded, nonconstant and plurisubharmonic on B, applying the definition to $(v - \inf_B v)/(\sup_B v - \inf_B v)$ gives

$$(\mathrm{dd^c} v)^n(E) \le (\sup_B v - \inf_B v)^n \operatorname{cap}(E, B);$$

for constant v both sides vanish. Write $\mathrm{Vol}(E) = \int_E \mathrm{d}V$, where $\mathrm{d}V$ is a fixed smooth positive volume form. We use two standard local estimates on these balls, with constants $C, a > 0$ depending only on the nested balls and on $\mathrm{d}V$:

(1) If $w \le 0$ is plurisubharmonic on B and v is bounded and plurisubharmonic on B, then
$$\int_{B'} |w| (\mathrm{dd^c} v)^n \le C \|w\|_{L^1(B)} \|v\|^n_{L^\infty(B)}.$$
Since adding a constant to v does not change $(\mathrm{dd^c} v)^n$, one may first normalize v by such a constant. This is the weighted Chern–Levine–Nirenberg inequality.

(2) For every Borel set $E \subset B'$,
$$\mathrm{Vol}(E) \le C \exp\big(-a \operatorname{cap}(E, B)^{-1/n}\big),$$
where the right-hand side is zero when the capacity is zero. This is the classical local volume–capacity estimate.

The first is [GZ17, Theorem 3.14]. The second is the local form of the volume–capacity estimate on compact Kähler manifolds given by [GZ17, Lemma 12.2 and equation (12.1.3) in the proof of Proposition 12.6]. Applied on $\mathbb{P}^n \supset B$ with the Fubini–Study form ω_{FS}, that estimate gives the local one, because a bounded local potential of ω_{FS} turns each competitor $0 \le u \le 1$ for the Monge–Ampère capacity of $(\mathbb{P}^n, \omega_{\mathrm{FS}})$ into a plurisubharmonic function on B of bounded range; inner regularity of the volume passes the bound from compact to Borel sets. In [GZ17, Section 2.1.3], $\mathrm{dd^c}$ is normalized as $\frac{\mathrm{i}}{\pi}\partial\bar\partial$, twice ours, which changes only the constants. The Bedford–Taylor products of [Xia26, Section 2.1] use our normalization, and the capacity defined there through global potentials has the same null sets, namely the pluripolar sets [Xia26, Theorem 2.3.2].

The dependence on the range of the potential can be read off from the usual cutoff argument. Let $M > 0$ and let p be plurisubharmonic on B with $0 \le p \le M$. Let $\kappa = \mathrm{dd^c}|z|^2$ be the Euclidean form, and choose a cutoff function χ supported in B, with $0 \le \chi \le 1$, equal to one on a smaller ball and satisfying $-C\kappa \le \mathrm{dd^c}\chi \le C\kappa$. Integration by parts gives, for $1 \le j \le n$,

$$\int \chi(\mathrm{dd^c} p)^j \wedge \kappa^{n-j} = \int p\, \mathrm{dd^c}\chi \wedge (\mathrm{dd^c} p)^{j-1} \wedge \kappa^{n-j} \leq CM \int_{\operatorname{supp}\chi} (\mathrm{dd^c} p)^{j-1} \wedge \kappa^{n-j+1}.$$

Iterating this estimate with nested cutoffs between B' and B, and stopping at degree one or at degree zero, we obtain respectively

$$\begin{aligned} \int_{B'} (\mathrm{dd^c} p)^j \wedge \kappa^{n-j} &\leq C_j M^{j-1} \int_B \mathrm{dd^c} p \wedge \kappa^{n-1}, \\ \int_{B'} (\mathrm{dd^c} p)^j \wedge \kappa^{n-j} &\leq C_j M^j. \end{aligned} \tag{1.2}$$

The constants depend only on the nested balls and on j. For bounded p, the integrations by parts are justified by local decreasing regularization.

Definition 1.6.1 For a compact set E in an open Euclidean ball $B \subset \mathbb{C}^n$, the *relative extremal function* is

$$h_{E,B} = \sup\{u \in \mathrm{PSH}(B) : -1 \leq u \leq 0,\ u|_E \leq -1\}.$$

The constant -1 is a competitor, and the family is bounded above by zero. The local upper-envelope theorem therefore gives $h^*_{E,B} \in \mathrm{PSH}(B)$ with $-1 \leq h^*_{E,B} \leq 0$. Allowing all competitors $u \leq 0$ with $u|_E \leq -1$ gives the same supremum, because the truncation $\max(u, -1)$ of such a competitor is a competitor in the original family.

A sequence of functions f_j converges to f *in local capacity* if $\mathrm{cap}(K \cap \{|f_j - f| > \epsilon\}, B) \to 0$ for every $\epsilon > 0$, every coordinate ball B and every compact set $K \subset B$. An assertion holds *quasi-everywhere*, abbreviated q.e., if it holds outside a pluripolar set; in particular, differences of functions that may both be $-\infty$ are considered only off such a set. A positive measure is *non-pluripolar* if it gives zero mass to every pluripolar set.

A function is *quasi-continuous* if, for every $\epsilon > 0$, it is continuous outside an open set of capacity less than ϵ. Plurisubharmonic functions have this property for the local Bedford–Taylor capacity, and a Borel set has zero capacity if and only if it is pluripolar. A locally uniformly bounded monotone sequence of plurisubharmonic functions converges in capacity: to its limit if it decreases, and to the upper semicontinuous regularization of its limit if it increases. If uniformly bounded plurisubharmonic functions converge in capacity, their local products converge weakly, and this remains true after the products are multiplied by uniformly bounded quasi-continuous coefficients converging in capacity. These local results, recalled in [Xia26, Section 2.3], are [GZ17, Theorems 4.20 and 4.40, Proposition 4.25 and Theorem 4.26]; they also apply to quasi-plurisubharmonic functions after a common smooth correction.

Theorem 1.6.2 (Mixed Monge–Ampère inequality) *Let $U \subset \mathbb{C}^n$ be a domain, and let $\beta = \mathrm{dd^c}|z|^2$ be the Euclidean form. Let a, b be locally bounded plurisubharmonic functions on U, and let f, g be nonnegative measurable functions on U with*

$$(\mathrm{dd^c} a)^n \geq f\beta^n, \qquad (\mathrm{dd^c} b)^n \geq g\beta^n.$$

Then, for $0 \leq k \leq n$,

$$(\mathrm{dd^c} a)^k \wedge (\mathrm{dd^c} b)^{n-k} \geq f^{k/n} g^{(n-k)/n} \beta^n.$$

The comparison principle of Bedford and Taylor is stated for $(\mathrm{dd^c}\cdot)^n$ in [GZ17, Theorem 3.29]; its proof, which rests on the maximum principle [GZ17, Theorem 3.27], also gives mixed forms. Let a, b be locally bounded plurisubharmonic functions on U such that $E = \{a < b\} \Subset U$, and let S be a positive closed $(n-1, n-1)$-current that is a product of Hessians of bounded plurisubharmonic functions and smooth closed positive forms. Then

$$\int_E \mathrm{dd^c} b \wedge S \leq \int_E \mathrm{dd^c} a \wedge S. \tag{1.3}$$

Indeed, apply the Bedford–Taylor integration by parts formula to $b_\epsilon = \max\{a, b - \epsilon\}$, which equals a outside E, compare the restrictions to $\{a < b - \epsilon\}$ and to $\{a > b - \epsilon\}$ by plurifine locality, and let $\epsilon \downarrow 0$; the sets $\{a < b - \epsilon\}$ increase to E. Replacing one factor at a time, we obtain, with $P = \mathrm{dd^c} a$ and $Q = \mathrm{dd^c} b$,

$$\int_E P^{n-j} \wedge Q^j \leq \int_E P^n, \qquad 0 \leq j \leq n. \tag{1.4}$$

The local envelope theorem for bounded obstacles shows where an envelope can carry Monge–Ampère mass. For continuous obstacles it is [GZ17, Exercise 5.10]; for quasi-continuous ones, the proof of the compact Kähler version [Xia26, Theorem 2.4.4], by balayage and approximation in capacity, applies on a ball. Let B be a ball with a smooth strictly plurisubharmonic defining function $h < 0$, and let f be a bounded quasi-continuous obstacle with $Ah \leq f \leq 0$ for some $A > 0$. Then the envelope

$$q = \big(\sup\{v \in \mathrm{PSH}(B) : v \leq f \text{ q.e.}\}\big)^*$$

satisfies $Ah \leq q \leq 0$, tends to zero at the boundary, and

$$\mathbf{1}_{\{q<f\}} (\mathrm{dd^c} q)^n = 0. \tag{1.5}$$

For a locally bounded quasi-plurisubharmonic function v and a smooth convex increasing function F on a neighborhood of the range of v, the bounded Bedford–Taylor chain rule reads

$$\mathrm{dd^c} F(v) = F'(v)\mathrm{dd^c} v + F''(v)\, \mathrm{d}v \wedge \mathrm{d^c} v. \tag{1.6}$$

The gradient term, and its products with positive Bedford–Taylor factors of bounded potentials, are nonnegative. For bounded plurisubharmonic v, the gradient term and its positivity are [GZ17, Definition 3.2 and Proposition 3.3], and (1.6) is [GZ17, Exercise 3.4]; the quasi-plurisubharmonic case follows by decreasing regularization, and bounded potentials have the local Sobolev regularity needed to define the gradient term. In particular, inequalities derived from this formula survive local smoothing with a lower curvature error tending to zero.

Lemma 1.6.2 *Let B be a coordinate ball and let τ be a finite non-pluripolar measure with compact support in B. Then there is a bounded plurisubharmonic function v on B such that $\tau \ll (\mathrm{dd^c} v)^n$.*

Proof For $n \geq 2$, we use Cegrell's decomposition theorem. On a bounded hyperconvex domain in $\mathbb{C}^n$, $n \geq 2$, let $\mathcal{E}_0$ consist of the bounded negative plurisubharmonic functions with zero boundary values and finite total Monge–Ampère mass. The theorem states that every positive compactly supported measure τ has a decomposition

$$\tau = g(\mathrm{dd^c} v)^n + \tau_s, \qquad v \in \mathcal{E}_0, \qquad 0 \leq g \in L^1((\mathrm{dd^c} v)^n),$$

where τ_s is a positive measure carried by a pluripolar set. If τ is non-pluripolar, then $\tau_s = 0$, so $\tau \ll (\mathrm{dd^c} v)^n$.

In dimension one, the assertion follows directly from the Green potential. Let $a \leq 0$ be the Green potential of τ on the disk, normalized by $\mathrm{dd^c} a = \tau$. Then $v = \mathrm{e}^a$ is bounded and subharmonic, and

$$\mathrm{dd^c} v \geq \mathrm{e}^a \tau.$$

To justify this inequality, set $a_M = \max(a, -M)$. The chain rule (1.6) in dimension one gives $\mathrm{dd^c} \mathrm{e}^{a_M} \geq \mathrm{e}^{a_M} \mathrm{dd^c} a_M$, and the right-hand side is at least $\mathrm{e}^a \tau$ on $\{a > -M\}$. Let $M \to \infty$, using the decreasing convergence of the bounded subharmonic functions e^{a_M} on the left and monotone convergence on the right. Since $\{a = -\infty\}$ is polar and τ charges no polar set, $\mathrm{e}^a > 0$ almost everywhere for τ. Thus $\tau \ll \mathrm{dd^c} v$. □

For a signed measure ν, write $|\nu|$ for its total variation measure and $\|\nu\|_{\mathrm{TV}(B)} = |\nu|(B)$. On a curve, the Green representation controls capacity by the total variation of the Laplacian, together with an L^1 term for the harmonic remainder.

Lemma 1.6.3 *Let $B' \Subset B$ be concentric disks in $\mathbb{C}$. There is a constant C, depending only on these disks, such that for all quasi-subharmonic functions u, v on a neighborhood of $\overline{B}$ and all $t > 0$,*

$$\mathrm{cap}(B' \cap \{|u - v| > t\}, B) \leq \frac{C}{t} \left(\|u - v\|_{L^1(B)} + \|\mathrm{dd^c}(u - v)\|_{\mathrm{TV}(B)} \right). \tag{1.7}$$

Here the difference $u - v$ is formed off the pole sets of u and v.

Proof Put $f = u - v$ and write the Jordan decomposition on B as $\mathrm{dd^c} f = \alpha^+ - \alpha^-$. Let $g^\pm \leq 0$ be the Green potentials of $\alpha^\pm$ on B with zero boundary values. The $L^1(B)$ norm of the Green kernel of the disk is bounded uniformly in its pole, so

$$\|g^+\|_{L^1(B)} + \|g^-\|_{L^1(B)} \le C\|\mathrm{dd^c} f\|_{\mathrm{TV}(B)}.$$

Indeed, on the unit disk the kernel is $\log|(z-\zeta)/(1-\overline{\zeta}z)|^2$; its logarithmic singularity is uniformly integrable, and $|1-\overline{\zeta}z| \le 2$. Scaling gives the bound on B.

The distribution $h = f - g^+ + g^-$ is harmonic, and the interior mean-value estimate gives

$$\|h\|_{L^\infty(B')} \le C\left(\|f\|_{L^1(B)} + \|\mathrm{dd^c} f\|_{\mathrm{TV}(B)}\right).$$

The identity $u + g^- = v + g^+ + h$ holds almost everywhere, hence everywhere by Lemma 1.3.3. Thus $|u - v| \le -g^+ - g^- + |h|$ off the polar sets of u and v. Let $0 \le a \le 1$ be a subharmonic test function on B. Since the measure $\mathrm{dd^c} a$ charges no polar set, this pointwise bound, the weighted Chern–Levine–Nirenberg estimate and (1.2) yield

$$\int_{B'} |u - v|\,\mathrm{dd^c} a \le C\left(\|f\|_{L^1(B)} + \|\mathrm{dd^c} f\|_{\mathrm{TV}(B)}\right).$$

Restricting the integral to $B' \cap \{|u - v| > t\}$, dividing by t and taking the supremum over a proves (1.7). □

1.6.1 Polar parts on curves

Every locally finite signed measure ν on a Riemann surface splits uniquely as the sum of its *polar part* ν^{pol}, which is carried by a polar Borel set, and a measure that charges no polar set. To construct the splitting locally, choose countably many polar Borel sets whose $|\nu|$-masses approach the supremum over all polar Borel sets, and restrict ν to their union, which is again polar. Uniqueness allows these local decompositions to be patched together and shows that $\nu \mapsto \nu^{\mathrm{pol}}$ is linear.

For a locally quasi-subharmonic function u, the *polar curvature* of u is the positive measure $\mu_u = (\mathrm{dd^c} u)^{\mathrm{pol}} = \mathbf{1}_{\{u=-\infty\}}\mathrm{dd^c} u$. The second equality holds because smooth corrections do not change polar parts, while bounded truncations and plurifine locality show that $\mathbf{1}_{\{u>-j\}}\mathrm{dd^c} u$ charges no polar set. The following inverse maximum principle shows that an upper bound for the difference of two potentials orders their polar curvatures.

Lemma 1.6.4 *Let $B \subset \mathbb{C}$ be a disk and let $f \in L^1_{\mathrm{loc}}(B)$ have a distributional Laplacian that is a signed Radon measure. If $f \le 0$ almost everywhere, then $(\Delta f)^{\mathrm{pol}} \ge 0$.*

Proof It suffices to prove $\Delta f(K) \ge 0$ for every compact polar set $K \subset B$; restriction to smaller disks and inner regularity then give the assertion. Choose open neighborhoods O_j of K with smooth boundaries such that $\overline{O}_{j+1} \subset O_j \Subset B$ and $\bigcap_j \overline{O}_j = K$. Let h_j be the function that equals one on $\overline{O}_j$, is harmonic on $B \setminus \overline{O}_j$, and vanishes on ∂B. The Dirichlet problem on this smooth domain shows that h_j is continuous with $0 \le h_j \le 1$. Its outward normal derivative on ∂O_j is nonpositive, so the jump of the derivative across ∂O_j gives $\Delta h_j \le 0$ on B in the sense of distributions. The maximum principle gives $h_{j+1} \le h_j$.

Since the functions h_j decrease and satisfy $0 \le h_j \le 1$, they converge in $L^1_{\mathrm{loc}}(B)$. Let h be the superharmonic representative of this limit; it may differ from the pointwise limit on K. It is harmonic off K, so its Laplacian is supported on the polar set K. On the other hand, $-h$ is bounded and subharmonic, so its Laplacian charges no polar set, by Theorem 1.6.1. Hence h is harmonic on B. Since $0 \le h \le h_1$ and h_1 tends to zero at ∂B, the maximum principle gives $h = 0$. It follows that $h_j \to 0$ smoothly on compact subsets of $B \setminus K$, while $h_j = 1$ on K.

Choose $0 \le \chi \in C_c^\infty(B)$ equal to one near K. For large j, regularize h_j by convolution on a neighborhood of $\operatorname{supp}\chi$, obtaining smooth functions $h_{j,\epsilon}$ with $\Delta h_{j,\epsilon} \le 0$. Since $f \le 0$,

$$\begin{aligned}\int_B \chi h_{j,\epsilon}\, \mathrm{d}(\Delta f) &= \int_B f\, \Delta(\chi h_{j,\epsilon})\, \mathrm{d}x\, \mathrm{d}y \\ &\ge \int_B f \left(2\nabla\chi \cdot \nabla h_{j,\epsilon} + h_{j,\epsilon}\Delta\chi\right) \mathrm{d}x\, \mathrm{d}y.\end{aligned}$$

For fixed j, the convolutions converge uniformly to h_j on $\operatorname{supp}\chi$, because h_j is continuous. On the support of the derivatives of χ, the function h_j is harmonic for all large j. We may therefore first let $\epsilon \downarrow 0$ and then let $j \to \infty$. The smooth convergence away from K makes the right-hand side tend to zero, and dominated convergence for the signed measure Δf makes the left-hand side tend to $\Delta f(K)$. This proves $\Delta f(K) \ge 0$. □

Corollary 1.6.1 *Let u, v be locally quasi-subharmonic functions on a Riemann surface. If $u \le v + C$ on an open set for some constant C, then $\mu_u \ge \mu_v$ there. Moreover,*

$$\mu_{\max(u,v)} = \mu_u \wedge \mu_v, \tag{1.8}$$

where the right-hand side is the greatest measure dominated by both μ_u and μ_v.

Proof Apply Lemma 1.6.4 to $u - v - C$ in local coordinates. Since smooth functions have zero polar part, linearity of the polar part gives the first assertion. Applying it to $u - \max(u, v)$ and to $v - \max(u, v)$ gives the inequality $\le$ in (1.8).

For the reverse inequality, add a common smooth correction to make u and v subharmonic on a coordinate disk. On a smaller disk, let g be the Green potential of the restriction of $\lambda = \mu_u \wedge \mu_v$ to that disk, so that $\mathrm{dd}^{\mathrm{c}} g = \lambda$. The distributions $u - g$ and $v - g$ have positive Hessians and hence subharmonic representatives a and b. Since $\max(u, v) = g + \max(a, b)$ almost everywhere, the Hessian of $\max(u, v)$ dominates λ. Restricting this inequality to a polar Borel set carrying λ proves the reverse inequality on the smaller disk, and these local statements give it on the whole surface. □

Polar parts need not be atomic: polar sets can carry diffuse measures.

Lemma 1.6.5 *Every nonempty open subset of a Riemann surface contains a compact polar set carrying a probability measure without atoms.*

Proof Choose a holomorphic coordinate z on part of the open set whose image contains the segment $[0,1]$. Put $r_k = \exp(-4^{k+1})$, so that $2r_{k+1} < r_k$ and $r_0 < 1$. Choose 2^k pairwise disjoint closed intervals of length r_k in $[0,1]$, for $k \geq 0$, such that each interval of level k contains two of level $k+1$, and let K be the intersection over k of the unions of the intervals of level k. Coding a point of K by its nested intervals identifies K with $\{0,1\}^{\mathbb{N}}$; let σ be the image of the product of the uniform probability measures on $\{0,1\}$. It gives mass 2^{-k} to each interval of level k, so it has no atoms. The logarithmic potential

$$u(z) = \int_K \log|z-y|^2 \, \mathrm{d}\sigma(y)$$

is subharmonic on $\mathbb{C}$ and finite off K. For $x \in K$, the interval of level k containing x has mass 2^{-k} and its points satisfy $|x-y| \leq r_k$, while $|x-y| \leq 1$ on the rest of K; hence $u(x) \leq 2^{1-k} \log r_k = -2^{k+3}$ for every k. Thus $K \subset \{u = -\infty\}$ is polar. □

1.6.2 Products at small pole sets

A top-degree product can carry mass on a pole set, and the non-pluripolar product discards this mass. When the unbounded locus is sufficiently small, the full local current product is well defined. Singularity order then compares the restrictions of these products to the pole set, which we call *residual measures*, and bounded perturbations leave these measures unchanged.

Theorem 1.6.3 *Let $\Omega \subset \mathbb{C}^n$ be a domain and $S \subset \Omega$ a closed set of zero one-dimensional Hausdorff measure. Suppose that $u_1, \dots, u_q$, $1 \leq q \leq n$, are plurisubharmonic on Ω and locally bounded on $\Omega \setminus S$. Then their inductively defined full current product*

$$\left(\mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_q\right)_{\mathrm{full}}$$

is a closed positive current of order zero. It is symmetric, multilinear, local in the ordinary topology, and continuous under simultaneous decreasing approximation by plurisubharmonic functions. The weighted current $u_1\left(\mathrm{dd^c} u_2 \wedge \cdots \wedge \mathrm{dd^c} u_q\right)_{\mathrm{full}}$ has coefficients of locally finite variation. These full products may charge S.

Proof Apply the classical theorem on products with unbounded potentials, together with its monotone continuity statement, in [Dem12, Chapter III, Theorem 4.5 and Proposition 4.9, pp. 152–156], with the underlying current $T = 1$. For every choice of m of the functions, that theorem requires the intersection of their unbounded loci to have zero Hausdorff measure of dimension $2n - 2m + 1$. This intersection lies in S, and $2n - 2m + 1 \geq 1$, so its Hausdorff measure of that dimension is zero. The same hypothesis gives local integrability of the weighted current. Symmetry, multilinearity

and locality on open sets follow from the local construction of the currents and decreasing regularization. □

The subscript distinguishes this product from the non-pluripolar product used later; the two coincide for locally bounded functions.

Lemma 1.6.6 *Let $\Omega \subset \mathbb{C}^n$ be a domain and $S \subset \Omega$ a closed set of zero one-dimensional Hausdorff measure. Let a, b be plurisubharmonic on Ω and locally bounded off S, with $\{a = -\infty\} = S$. If $a \le b + C$ near S for some finite constant C, then, as positive Radon measures,*

$$\mathbf{1}_S (\mathrm{dd^c} b)^n_{\mathrm{full}} \le \mathbf{1}_S (\mathrm{dd^c} a)^n_{\mathrm{full}}.$$

If $a - b$ is bounded near S, equality holds.

Proof Fix a compact set $E \subset S$ and $\varepsilon > 0$, and put $w_c = \max\{b - c, (1+\varepsilon)a\}$ for $c > 0$. Since $a = -\infty$ on the compact set E, upper semicontinuity gives a neighborhood of E on which $a < -(C + c)/\varepsilon$ and $a \le b + C$. Off S, on this neighborhood, $b - c - (1+\varepsilon)a \ge -C - c - \varepsilon a > 0$, while on S the second branch of the maximum equals $-\infty$. Thus $w_c = b - c$ near E, and locality on open sets gives $(\mathrm{dd^c} w_c)^n_{\mathrm{full}}(E) = (\mathrm{dd^c} b)^n_{\mathrm{full}}(E)$.

As $c \to \infty$, the functions w_c decrease to $(1+\varepsilon)a$. Theorem 1.6.3 applies to every w_c with the same set S, so their full measures converge weakly to $(1+\varepsilon)^n (\mathrm{dd^c} a)^n_{\mathrm{full}}$. Since the mass of a compact set is upper semicontinuous under weak convergence of positive measures, we obtain

$$(\mathrm{dd^c} b)^n_{\mathrm{full}}(E) \le (1+\varepsilon)^n (\mathrm{dd^c} a)^n_{\mathrm{full}}(E).$$

Here we apply the semicontinuity after multiplying the measures by a fixed nonnegative cutoff function with compact support, equal to one near E; both masses on E are finite by the theorem. Letting $\varepsilon \downarrow 0$ and applying inner regularity to all compact sets $E \subset S$ gives the asserted inequality of measures. If $a - b$ is bounded near S, then also $\{b = -\infty\} = S$, so the roles of a and b can be interchanged, and equality follows. □

1.6.3 Spherical Monge–Ampère measures

A bounded radial maximum gives a measure concentrated on a real hypersurface. This supplies singular measures that still give zero mass to pluripolar sets.

Lemma 1.6.7 *For $r > 0$, let σ_r be the normalized Euclidean surface measure on $\{|z| = r\} \subset \mathbb{C}^n$. Then*

$$\left(\mathrm{dd^c} \log \max\{|z|^2, r^2\}\right)^n = \sigma_r.$$

In particular, σ_r does not charge pluripolar sets.

Proof Write $t = \log|z|^2$ and approximate $\max\{t, \log r^2\}$ uniformly by smooth convex increasing functions $F_\epsilon(t)$ whose derivatives equal zero below $\log r^2 - \epsilon$ and one above $\log r^2 + \epsilon$. Their compositions with t extend smoothly across zero. Since $\mathrm{dd^c} t$ has rank $n-1$ off zero, $(\mathrm{dd^c} t)^n = 0$ there, and the chain rule gives

$$(\mathrm{dd^c} F_\epsilon(t))^n = n(F_\epsilon')^{n-1} F_\epsilon'' \, \mathrm{d}t \wedge \mathrm{d^c} t \wedge (\mathrm{dd^c} t)^{n-1}.$$

The integral of $\mathrm{d^c} t \wedge (\mathrm{dd^c} t)^{n-1}$ over each sphere is one. Indeed, $\mathrm{d^c} t$ integrates to one on each circle of the Hopf fibration, and $\mathrm{dd^c} t$ is the pullback of the form on $\mathbb{P}^{n-1}$ given on an affine chart by $\mathrm{dd^c}\log(1+|w|^2)$; the top power of that form has density $(n-1)!\pi^{-(n-1)}(1+|w|^2)^{-n}$ on the affine chart, whose integral is one. Thus each of the radial measures $(\mathrm{dd^c} F_\epsilon(t))^n$ has total mass $\int ((F_\epsilon')^n)' \, \mathrm{d}t = 1$. Their supports shrink to the sphere, and their invariance under unitary rotations identifies every weak limit as σ_r. On the other hand, the bounded potentials $F_\epsilon(t)$ converge uniformly, hence in capacity, to $\log\max\{|z|^2, r^2\}$, so their Monge–Ampère measures converge weakly to the product in the statement. Non-pluripolarity follows from Theorem 1.6.1. □

1.6.4 Measure concentration under integrable convergence

Decreasing approximation controls Bedford–Taylor products, but convergence of the potentials in L^1 alone does not. If plurisubharmonic functions tend to zero in L^1_{loc} and are locally uniformly bounded, the range estimate (1.2) shows that their Monge–Ampère measures tend to zero. Without a common bound, smooth bounded potentials that tend to zero, even in capacity, can have Monge–Ampère measures converging to a point mass.

Lemma 1.6.8 *Let $\Omega \subset \mathbb{C}^n$ be a domain. Suppose that $u_j \in \mathrm{PSH}(\Omega)$ tend to zero in $L^1_{\mathrm{loc}}(\Omega)$ and satisfy $\sup_j \|u_j\|_{L^\infty(K)} < \infty$ for every compact set $K \subset \Omega$. Then $(\mathrm{dd^c} u_j)^n(K) \to 0$ for every such K.*

Proof Fix concentric balls $B' \Subset B \Subset B_1 \Subset \Omega$, and put $\beta = \mathrm{dd^c}|z|^2$. If ψ is a nonnegative smooth cutoff function supported in B_1 and equal to one near $\overline{B}$, integration by parts gives

$$0 \le \int_B \mathrm{dd^c} u_j \wedge \beta^{n-1} \le \int_{B_1} u_j \mathrm{dd^c}\psi \wedge \beta^{n-1} \le C\|u_j\|_{L^1(B_1)} \longrightarrow 0.$$

Choose $M > 0$ with $|u_j| \le M$ on B for all j, and apply the first estimate in (1.2) to $u_j + M$, whose values lie in $[0, 2M]$. It bounds the top-degree mass of B' by $C(2M)^{n-1}$ times the trace integral over B, which tends to zero; for $n = 1$ the trace estimate itself suffices. Covering K by finitely many such balls proves the lemma. □

The construction uses a regularized maximum of a small multiple of a logarithm in one variable and a potential in the remaining variables. Its Monge–Ampère measure

can be computed exactly by integration over circles in the first variable, and this will keep its top-degree mass fixed while the logarithmic coefficient tends to zero.

Lemma 1.6.9 *Let $n \geq 2$, let $U \subset \mathbb{C}^{n-1}$ be a domain, and let v be a smooth plurisubharmonic function on U. Fix $\varepsilon, R, \eta > 0$ and $A \in \mathbb{R}$. Let χ_η be the convolution of $t \mapsto \max(t, 0)$ with a smooth even probability density supported in $[-\eta, \eta]$, and put*

$$q(\zeta) = \varepsilon \log(|\zeta|^2/R^2), \qquad T(\zeta, z') = v(z') - A - q(\zeta),$$
$$h(\zeta, z') = q(\zeta) + \chi_\eta(T(\zeta, z'))$$

for $\zeta \neq 0$. Then h extends to a smooth plurisubharmonic function on $\mathbb{C} \times U$. Set

$$w_\eta(t) = n\chi_\eta'(t)^{n-1}\chi_\eta''(t), \qquad r_t(z') = R\exp\big((v(z') - A - t)/(2\varepsilon)\big).$$

The function w_η is nonnegative, supported in $[-\eta, \eta]$, and has integral one. For every $f \in C_c(\mathbb{C} \times U)$,

$$\int f(\mathrm{dd^c} h)^n = \varepsilon \int_U \left[\int_{-\eta}^{\eta} w_\eta(t) \int_0^{2\pi} f(r_t(z')\mathrm{e}^{\mathrm{i}\vartheta}, z') \frac{\mathrm{d}\vartheta}{2\pi}\,\mathrm{d}t\right](\mathrm{dd^c} v)^{n-1}. \quad (1.9)$$

Proof By convexity, $0 \leq \chi_\eta' \leq 1$ and $\chi_\eta'' \geq 0$. The function χ_η vanishes on $(-\infty, -\eta]$ and equals t on $[\eta, \infty)$. Since $T \to +\infty$ as $\zeta \to 0$, we have $h = v - A$ near each point with $\zeta = 0$, which gives the smooth extension. Off the hyperplane $\{\zeta = 0\}$, the function q is pluriharmonic and

$$\mathrm{dd^c} h = \chi_\eta'(T)\mathrm{dd^c} v + \chi_\eta''(T)\,\mathrm{d}T \wedge \mathrm{d^c} T \geq 0.$$

The second summand has rank at most one, and the first has rank at most $n - 1$. Hence

$$(\mathrm{dd^c} h)^n = n\chi_\eta'(T)^{n-1}\chi_\eta''(T)\,\mathrm{d}T \wedge \mathrm{d^c} T \wedge (\mathrm{dd^c} v)^{n-1}.$$

The last factor has top degree in the variables z', so the derivatives of v in $\mathrm{d}T \wedge \mathrm{d^c} T$ contribute nothing. Write $s = \log(|\zeta|^2/R^2)$. In polar coordinates, $\mathrm{d}q = \varepsilon\,\mathrm{d}s$ and $\mathrm{d^c} q = \varepsilon\,\mathrm{d}\vartheta/(2\pi)$. At fixed z', the change of variables from s to $t = v - A - \varepsilon s$ gives (1.9). There is no contribution near $\zeta = 0$, where $h = v - A$ and the form $\mathrm{dd^c} h$ has rank at most $n - 1$. Finally, w_η is the derivative of $(\chi_\eta')^n$, whose values at $-\eta$ and η are zero and one. □

Proposition 1.6.1 *Let $\Omega \subset \mathbb{C}^n$, $n \geq 2$, be a bounded domain and $a \in \Omega$. For every $c > 0$, there are smooth bounded plurisubharmonic functions $u_j \leq 0$ on Ω such that*

$$\|u_j\|_{L^p(\Omega)} \longrightarrow 0 \quad (1 \leq p < \infty), \qquad (\mathrm{dd^c} u_j)^n \longrightarrow c\delta_a \quad \textit{vaguely}.$$

The same sequence tends to zero in local capacity. Here the norms are taken with respect to Lebesgue measure, and vague convergence means convergence against every compactly supported continuous test function.

Proof This is the case $b = 0$, $\nu = c\delta_a$ of Theorem 1.6.4 below, whose proof does not use the proposition. □

Each approximating measure has a smooth density and charges no pluripolar set, whereas the limit is a point mass. Since the masses of a compact neighborhood of a do not tend to zero, Lemma 1.6.8 shows that the potentials have no common lower bound on some compact set. Thus convergence in capacity is compatible with concentration of the Monge–Ampère measures once common local bounds are lost.

1.6.5 Adding a prescribed local measure

We now construct the potentials of Proposition 1.6.1 on top of a given locally bounded potential b, and repeat the construction so as to add an arbitrary positive measure to $(\mathrm{dd^c} b)^n$. Adding a perturbation h to b creates the mixed terms $(\mathrm{dd^c} b)^{n-r} \wedge (\mathrm{dd^c} h)^r$. A mixed Chern–Levine–Nirenberg estimate controls them, provided that the lower-degree currents of h are kept small.

Lemma 1.6.10 *Fix the data $n, U, v, \varepsilon, R, \eta, \chi_\eta$ of Lemma 1.6.9, and write h_A for the smooth plurisubharmonic function h of that lemma, regarded as a function of A. Put $D = \{0\} \times U$, and write $[D]$ for the current of integration over this hypersurface. Then, for every $1 \le r \le n$,*

$$(\mathrm{dd^c} h_A)^r \longrightarrow \varepsilon [D] \wedge (\mathrm{dd^c} v)^{r-1} \quad \text{as currents on } \mathbb{C} \times U,\ A \to \infty.$$

The product on the right is integration of the restriction of the smooth form $(\mathrm{dd^c} v)^{r-1}$ over D.

Proof Write $S = \mathrm{dd^c} v$, $q = \varepsilon \log(|\zeta|^2/R^2)$ and $T_A = v - A - q$. The rank-one calculation in the proof of Lemma 1.6.9 gives

$$(\mathrm{dd^c} h_A)^r = \chi'_\eta(T_A)^r S^r + r\chi'_\eta(T_A)^{r-1} \chi''_\eta(T_A)\, \mathrm{d}T_A \wedge \mathrm{d^c} T_A \wedge S^{r-1}$$

off D; since h_A is smooth and D has measure zero, this formula determines the current. Fix a smooth compactly supported test form. On the projection of its support to U, the function v and its derivatives are bounded. Hence the support of $\chi'_\eta(T_A)$ lies in $\{|\zeta| \le s_A\}$, where $s_A = C \exp(-A/(2\varepsilon))$ and C depends only on this compact set and the fixed data. The coefficients involving v and χ_η are bounded at the fixed parameters, so the first summand has pairing $O(s_A^2)$. In the second summand, the term with two derivatives of v also has pairing $O(s_A^2)$, and the two cross terms have pairing $O(s_A)$: the derivative of q in the ζ direction has size $O(|\zeta|^{-1})$, and its area integral over the disk $\{|\zeta| \le s_A\}$ is $O(s_A)$.

It remains to integrate the term containing $\mathrm{d}q \wedge \mathrm{d^c} q$. The logarithmic change of the radial variable used for (1.9) leaves the factor ε and the probability weight

$$r\chi'_\eta(t)^{r-1} \chi''_\eta(t)\, \mathrm{d}t, \qquad -\eta \le t \le \eta.$$

Only the component of the test form in the z' directions enters this term, and on the shrinking circles its coefficients tend uniformly to their values at $\zeta = 0$. Integration in the angle therefore gives, in the limit, exactly the pairing with $\varepsilon[D] \wedge S^{r-1}$. This proves the convergence in every degree. □

Lemma 1.6.11 *Let $\Omega \subset \mathbb{C}^n$ be a domain, let K and L be compact subsets of Ω with K contained in the interior L° of L, and put $\beta = \mathrm{dd^c}|z|^2$. Let b be a locally bounded plurisubharmonic function on Ω, and let T be a smooth positive closed (r,r)-form, $1 \le r < n$. Then*

$$\int_K (\mathrm{dd^c} b)^{n-r} \wedge T \le C_{K,L,n}(1 + \|b\|_{L^\infty(L)})^{n-r} \int_L T \wedge \beta^{n-r}.$$

Proof Put $M = 1 + \|b\|_{L^\infty(L)}$. Cover K by finitely many balls whose larger concentric balls have closures in L°. Replacing b by $b + M$, which has the same Hessian and takes values in $[0, 2M]$ on L, we may assume $0 \le b \le 2M$ there. For nested cutoff functions ψ, integration by parts against closed forms gives

$$\int \psi \mathrm{dd^c} b \wedge W = \int b \mathrm{dd^c}\psi \wedge W \le CM \int_{\operatorname{supp}\psi} \beta \wedge W,$$

where W is T multiplied by the remaining positive closed factors. Apply this successively to the $n - r$ Hessians of b. For bounded b, the integrations by parts are justified by local decreasing regularization and Theorem 1.6.1, with one common bound on the range. Summing over the finite cover gives the asserted constant. □

Proposition 1.6.2 *Let $\Omega \subset \mathbb{C}^n$, $n \ge 2$, be a bounded domain, let b be a locally bounded plurisubharmonic function on Ω, and let $a \in \Omega$ and $c > 0$. Given finitely many $f_i \in C_c(\Omega)$ and numbers $\tau, t > 0$, there is a smooth bounded plurisubharmonic function $h \le 0$ on Ω such that*

$$\|h\|_{L^p(\Omega)} \le C_{\Omega,p} t \quad (1 \le p < \infty),$$
$$\left| \int_\Omega f_i \big((\mathrm{dd^c}(b+h))^n - (\mathrm{dd^c} b)^n \big) - c f_i(a) \right| < \tau \quad \text{for every } i.$$

For concentric balls $B' \Subset B \Subset \Omega$ and every $w \in \mathrm{PSH}(B)$ with $0 \le w \le 1$, it also satisfies

$$\int_{B'} |h| (\mathrm{dd^c} w)^n \le C_{B',B,\Omega} t.$$

These constants are independent of b, a, c, the test functions and the tolerances.

Proof We take for h a function from Lemma 1.6.9 whose top-degree measure approximates $c\delta_a$ and whose lower-degree currents are small; Lemma 1.6.11 then controls the mixed terms.

Write $a = (a_1, a')$ and $m = n - 1$, and fix $R = 2 \sup_\Omega |z_1| + 2$. For $\delta > 0$, put $H_\delta(z') = \log(|z' - a'|^2 + \delta^2)$. With $s = |z' - a'|^2$, the complex Hessian of H_δ has the

eigenvalue $(s+\delta^2)^{-1}$ with multiplicity $m-1$ in the directions orthogonal to $z'-a'$ and the eigenvalue $\delta^2(s+\delta^2)^{-2}$ in the radial direction. Taking the determinant and integrating in polar coordinates gives

$$(\mathrm{dd^c} H_\delta)^m = \frac{m!}{\pi^m} \frac{\delta^2}{(|z'-a'|^2+\delta^2)^{m+1}}\, \mathrm{d}\lambda_{2m},$$
$$\int_{|z'-a'|<\rho} (\mathrm{dd^c} H_\delta)^m = \left(\frac{\rho^2}{\rho^2+\delta^2}\right)^m,$$

where λ_{2m} is Lebesgue measure. These probability measures tend to $\delta_{a'}$ as $\delta \downarrow 0$, since their mass outside each ball about a' tends to zero. Extend every f_i by zero outside Ω, and choose δ such that

$$\left| c \int_{\mathbb{C}^m} f_i(a_1, z')(\mathrm{dd^c} H_\delta)^m - c f_i(a) \right| < \tau/3 \quad \text{for all } i.$$

Choose compact sets $K, L \subset \Omega$ with $K \subset L^\circ$ and K containing the supports of all f_i, and a nonnegative smooth cutoff function ψ with compact support in Ω, equal to one near L. Put $\beta = \mathrm{dd^c}|z|^2$. Expanding the top power of $\mathrm{dd^c}(b+h)$, summing the bounds of Lemma 1.6.11 with their binomial coefficients and the norms $\|f_i\|_\infty$, we find $\gamma > 0$ such that the mixed terms contribute less than $\tau/3$ to every test integral whenever h is smooth and plurisubharmonic and

$$\int_L (\mathrm{dd^c} h)^r \wedge \beta^{n-r} < \gamma, \qquad 1 \le r < n.$$

Fix $0 < \eta \le t/2$. Apply Lemma 1.6.9 on $\mathbb{C} \times \mathbb{C}^m$ with transverse variable $\zeta = z_1 - a_1$ and $v = c^{1/m}\varepsilon^{-1/m} H_\delta$. For $r < n$, Lemma 1.6.10 gives

$$(\mathrm{dd^c} h_A)^r \longrightarrow \varepsilon^{(n-r)/(n-1)} c^{(r-1)/(n-1)} [\{z_1 = a_1\}] \wedge (\mathrm{dd^c} H_\delta)^{r-1}$$

as $A \to \infty$. For the fixed δ, the integrals of these limits against $\psi\beta^{n-r}$ are finite, and their exponents of ε are positive. Choose $0 < \varepsilon \le t/2$ so that each of them is less than $\gamma/2$. With these parameters fixed, choose A so large that the integrals $\int_L (\mathrm{dd^c} h_A)^r \wedge \beta^{n-r}$ are less than γ, that $v - A \le 0$ on the projection of Ω to the z' variables, and that

$$\left| \int_\Omega f_i (\mathrm{dd^c} h_A)^n - c \int_{\mathbb{C}^m} f_i(a_1, z')(\mathrm{dd^c} H_\delta)^m \right| < \tau/3 \quad \text{for all } i.$$

The last choice follows from (1.9) and the uniform continuity of the f_i: as $A \to \infty$, the radii of the circles there tend to zero uniformly for z' in the projection of Ω, while $\varepsilon(\mathrm{dd^c} v)^m = c(\mathrm{dd^c} H_\delta)^m$ has mass c.

Set $h = h_A - \eta$, a smooth plurisubharmonic function. In the notation of Lemma 1.6.9, now $q = \varepsilon \log(|z_1 - a_1|^2/R^2)$ and $T = v - A - q$. The convolution satisfies $0 \le \chi_\eta - \max(\cdot, 0) \le \eta$, and $q + \max(T, 0) = \max(q, v - A)$. On Ω, both branches of this maximum are nonpositive, by the choices of R and A, and the second is bounded

below. Hence h is bounded on Ω, and

$$q - \eta \le h \le 0, \qquad \text{so} \qquad |h| \le \varepsilon\bigl|\log(|z_1 - a_1|^2/R^2)\bigr| + \eta.$$

The logarithm has $L^p(\Omega)$ norm bounded uniformly for a_1 in the projection of Ω to the first coordinate: after translation, its one-variable integral is bounded by polar integration on one fixed disk. This gives the L^p bound. The weighted Chern–Levine–Nirenberg estimate, applied to this nonpositive plurisubharmonic logarithm, and the range estimate (1.2) for the constant term give the bound for the integrals over B'. Since $\varepsilon + \eta \le t$, both estimates hold with constants depending only on Ω, p, B' and B.

Finally, multilinearity of bounded products gives

$$(\mathrm{dd^c}(b+h))^n = (\mathrm{dd^c}b)^n + (\mathrm{dd^c}h)^n + \sum_{r=1}^{n-1} \binom{n}{r} (\mathrm{dd^c}b)^{n-r} \wedge (\mathrm{dd^c}h)^r .$$

The choice of δ, the choice of A for the top-degree term and the estimate of the mixed terms each contribute less than $\tau/3$ to every test integral. This proves the required inequalities, with one choice of h for all finite p. □

For a domain $\Omega \subset \mathbb{C}^n$, write $\mathcal{M}_+(\Omega)$ for the set of positive Radon measures on Ω, including zero and measures of infinite total mass, with the vague topology defined by integration against $C_c(\Omega)$. This topology is metrizable.

Lemma 1.6.12 *There are real functions $f_i \in C_c^\infty(\Omega)$ such that*

$$d_v(\alpha, \beta) = \sum_{i \ge 1} 2^{-i} \frac{\left|\int_\Omega f_i(\alpha - \beta)\right|}{1 + \left|\int_\Omega f_i(\alpha - \beta)\right|}$$

is a metric inducing the vague topology on $\mathcal{M}_+(\Omega)$. The finite atomic measures with nonnegative rational weights at points of a fixed countable dense subset of Ω are dense in this metric space; in particular, it is separable.

Proof Choose relatively compact open sets Ω_k exhausting Ω, each with closure contained in the next one. On each Ω_k, take a countable family of functions in $C_c^\infty(\Omega_k)$ that is uniformly dense in the space of continuous functions on $\overline{\Omega}_k$ vanishing on $\partial\Omega_k$, and add a function $\chi_k \in C_c^\infty(\Omega)$ with $0 \le \chi_k \le 1$ and $\chi_k = 1$ on $\overline{\Omega}_k$. Enumerate the union of these families as (f_i), and call the maps $\alpha \mapsto \int_\Omega f_i \, \mathrm{d}\alpha$ the coordinates.

Fix $\alpha \in \mathcal{M}_+(\Omega)$, $g \in C_c(\Omega)$ and $0 < \epsilon \le 1$. Choose k with $\operatorname{supp} g \subset \Omega_k$ and a member f_i of the family of Ω_k with $|g - f_i| < \epsilon$. For every $\beta \in \mathcal{M}_+(\Omega)$,

$$\left|\int_\Omega g \, \mathrm{d}(\beta - \alpha)\right| \le \epsilon \int_\Omega \chi_k \, \mathrm{d}(\alpha + \beta) + \left|\int_\Omega f_i \, \mathrm{d}(\beta - \alpha)\right| .$$

On the set where the coordinates of χ_k and f_i differ from those of α by less than ϵ, the right-hand side is at most $\epsilon(2\int \chi_k \, \mathrm{d}\alpha + 2)$. Thus integration against g is continuous

for the topology generated by the coordinates. Since the coordinates are vaguely continuous, they generate exactly the vague topology, and they separate measures.

The function $s \mapsto s/(1+s)$ is increasing and subadditive on $[0,\infty)$. Applying it to the triangle inequalities for the coordinates gives the triangle inequality for d_v, and d_v is positive on distinct measures because the coordinates separate them. Each coordinate is continuous for this metric, and the tails of the series are uniformly small, so the metric induces exactly the topology generated by the coordinates.

For density, fix α and j, and let $K \subset \Omega$ be a compact set containing the supports of $f_1, \dots, f_j$. Partition K into finitely many Borel sets E_s of diameter less than η, and choose points a_s of the dense subset within distance η of E_s and rational weights $c_s \geq 0$ with $\sum_s |c_s - \alpha(E_s)| < \eta$. If η is small, the measure $\sum_s c_s \delta_{a_s}$ has coordinates $i \leq j$ arbitrarily close to those of α, by uniform continuity of $f_1, \dots, f_j$ and finiteness of $\alpha(K)$. The tail of the series beyond j is at most 2^{-j}, which proves density. □

The metric controls limits along corresponding subsequences: if $d_v(\alpha_j, \beta_j) \to 0$ and one of the sequences converges vaguely to a fixed measure along a subsequence, then, by the triangle inequality, so does the other. Without common local mass bounds, however, $d_v(\alpha_j, \beta_j) \to 0$ need not imply that $\int g(\alpha_j - \beta_j) \to 0$ for every $g \in C_c(\Omega)$. For example, fix $a \in \Omega$, a unit vector e, and numbers $r_j \downarrow 0$ with $a + r_j e \in \Omega$, and put

$$\alpha_j = r_j^{-1/2} \delta_{a + r_j e}, \qquad \beta_j = r_j^{-1/2} \delta_a .$$

For every smooth compactly supported test, the difference of the two integrals is $O(r_j^{1/2})$, so $d_v(\alpha_j, \beta_j) \to 0$ because the tails of the series are small. On the other hand, if $\chi \in C_c^\infty(\Omega)$ equals one near a, the continuous test function $g(z) = \chi(z)|z-a|^{1/2}$ gives $\int g(\alpha_j - \beta_j) = 1$ for all large j.

Theorem 1.6.4 *Let $\Omega \subset \mathbb{C}^n$, $n \geq 2$, be a bounded domain, let b be a locally bounded plurisubharmonic function on Ω, and let ν be a positive Radon measure on Ω. There are plurisubharmonic functions $u_j \leq b$ on Ω, with $u_j - b$ bounded, such that*

$$\|u_j - b\|_{L^p(\Omega)} \longrightarrow 0 \quad (1 \leq p < \infty),$$
$$(\mathrm{dd}^c u_j)^n \longrightarrow (\mathrm{dd}^c b)^n + \nu \quad \textit{vaguely}.$$

The same sequence converges to b in local capacity. If b is smooth, every u_j can be chosen smooth.

Proof Let (f_i) and d_v be as in Lemma 1.6.12. For each j, the density of finite atomic measures in that lemma gives a finite positive atomic measure

$$\nu_j = \sum_{s=1}^{N_j} c_{j,s} \delta_{a_{j,s}}, \qquad \left| \int_\Omega f_i (\nu_j - \nu) \right| < j^{-1} \quad (i \leq j).$$

Atoms of weight zero are omitted. If $N_j = 0$, set $u_j = b$. Otherwise start with $b_{j,0} = b$ and apply Proposition 1.6.2 successively to $b_{j,s-1}$, $a_{j,s}$ and $c_{j,s}$, with the tests $f_1, \dots, f_j$ and $\tau = t = (jN_j)^{-1}$. This gives perturbations $h_{j,s}$, and we define

$$b_{j,s} = b_{j,s-1} + h_{j,s}, \qquad u_j = b_{j,N_j}.$$

Every intermediate potential is locally bounded and plurisubharmonic, so the proposition applies at each step, and each perturbation serves all finite p at once. The difference $u_j - b = \sum_s h_{j,s}$ is bounded and nonpositive, and summing the norm estimates yields $\|u_j - b\|_{L^p(\Omega)} \le C_{\Omega,p}/j$. Summing the measure errors gives

$$\left| \int_\Omega f_i \big((\mathrm{dd^c} u_j)^n - (\mathrm{dd^c} b)^n - \nu \big) \right| < 2/j \quad (i \le j).$$

Hence $d_v\big((\mathrm{dd^c} u_j)^n, (\mathrm{dd^c} b)^n + \nu\big) \le 2/j + 2^{-j}$, and Lemma 1.6.12 proves the asserted vague limit, also when $\nu(\Omega) = \infty$.

For fixed concentric balls $B' \Subset B \Subset \Omega$, the perturbations are nonpositive, so $|u_j - b| = \sum_s |h_{j,s}|$. The last estimate of Proposition 1.6.2 bounds the integral of this function against $(\mathrm{dd^c} w)^n$ over B' by $C_{B',B,\Omega}/j$, for every $w \in \mathrm{PSH}(B)$ with $0 \le w \le 1$. Restricting to the set where $|u_j - b|$ exceeds a fixed threshold, dividing by the threshold and taking the supremum over w proves convergence in local capacity. If b is smooth, the finite sums defining u_j are smooth as well. □

The construction takes place in a bounded Euclidean domain, without prescribed boundary values and without a common bound on the approximating potentials. These freedoms allow concentration even under convergence in local capacity; a fixed global Hermitian background imposes further constraints.

1.6.6 Full cluster sets

Theorem 1.6.4 prescribes one limit of the Monge–Ampère measures, but a single sequence of potentials can have many subsequential limits. On a bounded domain in dimension at least two, the sets of all subsequential limits for potentials that converge from below in local capacity to a locally bounded potential b are exactly the vaguely closed sets of measures dominating $(\mathrm{dd^c} b)^n$ (Theorem 1.6.5); for $b = 0$, every vaguely closed set occurs.

Let $\Omega \subset \mathbb{C}^n$ be a domain, $n \ge 1$, and let b be a locally bounded plurisubharmonic function on Ω. If potentials approach b from below in local capacity, every limit of their Monge–Ampère measures must dominate $(\mathrm{dd^c} b)^n$. Truncation at a fixed distance below b provides a common bound, and plurifine locality keeps the measure outside the small set where the truncation changes the potential.

Lemma 1.6.13 *Let b, u_j be locally bounded plurisubharmonic functions on Ω with $u_j \le b$ and $u_j \to b$ in local capacity. Then, for every nonnegative $f \in C_c(\Omega)$,*

$$\liminf_{j\to\infty} \int_\Omega f (\mathrm{dd^c} u_j)^n \ge \int_\Omega f (\mathrm{dd^c} b)^n.$$

In particular, every vague subsequential limit of $(\mathrm{dd^c} u_j)^n$ dominates $(\mathrm{dd^c} b)^n$.

Proof First suppose that f is supported in B', where $B' \Subset B \Subset \Omega$ are concentric balls. Put $v_j = \max(u_j, b-1)$ and $E_j = \{u_j \le b-1\}$. The functions v_j lie between $b-1$ and b and tend to b in local capacity. Because of their common bounds, the continuity of bounded products recalled in Section 1.6 gives $(\mathrm{dd^c} v_j)^n \to (\mathrm{dd^c} b)^n$ vaguely on B. With $R = \sup_B b - \inf_B b + 1$, the capacity bound for bounded potentials gives

$$(\mathrm{dd^c} v_j)^n(E_j \cap \operatorname{supp} f) \le R^n \operatorname{cap}(E_j \cap \operatorname{supp} f, B) \longrightarrow 0,$$

where the limit holds because $E_j \subset \{|u_j - b| > 1/2\}$. On the plurifine open Borel set $\{u_j > b-1\}$ we have $v_j = u_j$, so Theorem 1.6.1 shows that the two measures agree there. By positivity,

$$\int f(\mathrm{dd^c} u_j)^n \ge \int f(\mathrm{dd^c} v_j)^n - \|f\|_\infty (\mathrm{dd^c} v_j)^n(E_j \cap \operatorname{supp} f).$$

Taking the lower limit proves the assertion for such f. For a general f, use a finite nonnegative partition of unity subordinate to a cover of its support by such balls. Applying the inequality along a vaguely convergent subsequence proves the final assertion. □

Theorem 1.6.5 *Assume that Ω is bounded and $n \ge 2$. For a set $F \subset \mathcal{M}_+(\Omega)$, the following are equivalent:*

(1) *F is vaguely closed and every $\mu \in F$ satisfies $\mu \ge (\mathrm{dd^c} b)^n$;*
(2) *there are locally bounded plurisubharmonic functions $u_j \le b$ on Ω, converging to b in $L^p(\Omega)$ for every $1 \le p < \infty$ and in local capacity, whose full vague cluster set is F:*

$$F = \left\{\mu \in \mathcal{M}_+(\Omega) : \ (\mathrm{dd^c} u_{j_k})^n \longrightarrow \mu \text{ vaguely for some } j_1 < j_2 < \cdots\right\}.$$

The empty set is allowed. In the second condition, the u_j can be chosen with $u_j - b$ bounded, and smooth if b is smooth.

Proof In a metric space, the full cluster set of a sequence is the intersection of the closures of its tails: a point of this intersection can be approximated within $1/k$ at an index larger than the preceding one, successively for $k = 1, 2, \ldots$. Hence, if the second condition holds, F is closed by Lemma 1.6.12, and Lemma 1.6.13 gives the measure inequality.

Conversely, assume the first condition. If F is nonempty, choose a dense sequence $\mu_1, \mu_2, \ldots$ in F, repetitions allowed; this is possible because a subset of a separable metric space is separable. Concatenating the finite lists (μ_1), (μ_1, μ_2), (μ_1, μ_2, μ_3), and so on, we obtain a sequence (ν_j) in F in which every μ_k appears arbitrarily late. Hence every point of F is the limit of a subsequence, and since F is closed, no other subsequential limit occurs. If F is empty, take $\nu_j = (\mathrm{dd^c} b)^n + j\delta_a$ for a fixed $a \in \Omega$. A nonnegative compactly supported test function equal to one at a has integral at least j against ν_j, so this sequence has no vaguely convergent subsequence. In either case, $\nu_j - (\mathrm{dd^c} b)^n$ is a positive Radon measure.

Let (f_i) and d_v be as in Lemma 1.6.12. For each j, perform the j-th step of the proof of Theorem 1.6.4 with the excess $\nu_j - (\mathrm{dd^c} b)^n$ in place of ν: approximate it against $f_1, \dots, f_j$ by a finite positive atomic measure with errors less than $1/j$, and add its atoms successively to b by Proposition 1.6.2. As in that proof, $u_j \le b$, $u_j - b$ is bounded, u_j is smooth when b is smooth, and

$$\|u_j - b\|_{L^p(\Omega)} \le C_{\Omega,p}/j, \qquad \left| \int_\Omega f_i \big((\mathrm{dd^c} u_j)^n - \nu_j \big) \right| < 2/j \quad (i \le j).$$

The capacity estimates in that proof do not depend on the measure, so $u_j \to b$ in local capacity. Neither convergence of the ν_j nor a common local mass bound for them is needed.

The error in the first j coordinates and the tail of the series yield

$$d_v\big((\mathrm{dd^c} u_j)^n, \nu_j \big) \le 2/j + 2^{-j} \longrightarrow 0.$$

By the remark following Lemma 1.6.12, the two sequences of measures have the same limits along corresponding subsequences. Hence their full vague cluster sets coincide. □

For $b = 0$, the first condition says only that F is vaguely closed: every vaguely closed subset of $\mathcal{M}_+(\Omega)$, including the empty set, is the full vague cluster set of smooth bounded plurisubharmonic functions $u_j \le 0$ tending to zero in every $L^p(\Omega)$, $1 \le p < \infty$, and in local capacity. On the unit ball, the function $b = |z|^2 - 1$ has nonzero Monge–Ampère measure. For every point a of the ball, the theorem allows the cluster sets $\{(\mathrm{dd^c} b)^n\}$, $\{(\mathrm{dd^c} b)^n + \delta_a\}$ and the empty set, but it excludes $\{0\}$; this exclusion relies on the convergence in local capacity. In dimension one, by contrast, L^1_{loc} convergence alone forces $\mathrm{dd^c} u_j \to \mathrm{dd^c} b$ in the sense of distributions, hence vaguely by positivity, so the full cluster set is always $\{\mathrm{dd^c} b\}$; the freedom in the theorem begins in dimension two.

1.7 Products with smooth backgrounds

The Bedford–Taylor construction also defines products of currents of the form $\theta + \mathrm{dd^c} u$ with u locally bounded: one expands them into Hessians of plurisubharmonic functions and smooth forms. Since the result does not depend on the local correction, these products are defined globally. Their positivity is not visible term by term; it follows from the lower curvature bound by freezing the coefficients of the background. For a smooth real $(1,1)$-form θ and a function v, we write $\theta_v = \theta + \mathrm{dd^c} v$, and similarly $\omega_v = \omega + \mathrm{dd^c} v$ for other backgrounds. Fix $1 \le k \le n$.

Let u be locally bounded and θ-plurisubharmonic. On a relatively compact coordinate ball with coordinates z, choose $A > 0$ such that $p = u + q$ is plurisubharmonic, where $q = A|z|^2$. Set $\beta = \theta - \mathrm{dd^c} q$ and define

$$(\theta + \mathrm{dd^c} u)^k := \sum_{j=0}^{k} \binom{k}{j} (\mathrm{dd^c} p)^j \wedge \beta^{k-j}. \tag{1.10}$$

Each summand is a product from Theorem 1.6.1 multiplied by a smooth form.

Proposition 1.7.1 *The construction* (1.10) *and its mixed version are independent of the local correction q and of the decomposition of each current into a smooth form and the Hessian of a potential. They define global products of currents with locally bounded quasi-plurisubharmonic potentials. These products are symmetric, multilinear and plurifinely local, and they do not charge pluripolar sets. If every factor is positive, their mixed product is positive.*

Proof On the intersection of two coordinate balls, the two corrected potentials differ by a smooth function. The rule $\mathrm{dd^c}(p + f) = \mathrm{dd^c} p + \mathrm{dd^c} f$, for bounded p and smooth f in bounded local products, follows by smooth approximation; together with the binomial formula, it shows that the two expansions agree. The same argument handles a smooth change in the decomposition of a current. Moreover, any two decompositions of the same $(1,1)$-current differ by a smooth function: the difference of the potentials has a smooth Hessian, and taking its local Euclidean trace and applying elliptic regularity shows that it is smooth. Symmetry, multilinearity and locality then follow term by term. Bounded Bedford–Taylor products wedged with smooth forms have locally finite variation and do not charge pluripolar sets, so these properties pass to the finite expansion as well.

It remains to prove positivity. Write the factors as $T_i = \theta_i + \mathrm{dd^c} u_i$, $1 \le i \le k$, where each θ_i is a smooth real $(1,1)$-form and each u_i is locally bounded and quasi-plurisubharmonic, and assume $T_i \ge 0$. Fix $\varepsilon > 0$, a point of the coordinate ball and a positive constant form κ. On a sufficiently small neighborhood of the point, each θ_i is close to a constant, hence closed, form θ_i':

$$-\varepsilon\kappa \le \theta_i - \theta_i' \le \varepsilon\kappa.$$

Then $S_i = \theta_i' + \varepsilon\kappa + \mathrm{dd^c} u_i \ge T_i$ is a positive closed current there, with a bounded local plurisubharmonic potential, and

$$T_i + 2\varepsilon\kappa = S_i + (\theta_i - \theta_i' + \varepsilon\kappa),$$

where the last term is a smooth semipositive form. Expanding the product of the currents $T_i + 2\varepsilon\kappa$ therefore gives a sum of positive currents, so this product is positive near the chosen point. Such neighborhoods, although they depend on ε, cover the coordinate ball, so the product of the $T_i + 2\varepsilon\kappa$ is positive on the whole ball. By multilinearity, it converges weakly to the product of the T_i as $\varepsilon \to 0$, which proves positivity. □

The local expansion also gives continuity along decreasing sequences of uniformly locally bounded potentials: apply Theorem 1.6.1 to the corrected potentials and sum the finitely many terms. Since X is compact, every bounded top-degree product has finite total mass. For bounded $u \in \mathrm{PSH}(X,\theta)$ we write $\mathrm{MA}_\theta(u) = (\theta + \mathrm{dd}^{\mathrm{c}} u)^n$.

Decreasing maxima compare the products of two ordered potentials on the set where they are equal. The comparison also holds in the presence of fixed positive factors, which will be useful in integration by parts.

Lemma 1.7.1 *Let $a \le b$ be locally bounded quasi-plurisubharmonic functions on an n-dimensional complex manifold, and let θ be a smooth real $(1,1)$-form such that $\theta_a, \theta_b \ge 0$. For $1 \le k \le n$, let W be a product of $n-k$ positive currents with smooth backgrounds and locally bounded potentials, with $W = 1$ when $k = n$. Then*

$$\mathbf{1}_{\{a=b\}} \theta_a^k \wedge W \le \mathbf{1}_{\{a=b\}} \theta_b^k \wedge W.$$

Proof The functions $m_\delta = \max(a+\delta, b)$ satisfy $b \le m_\delta \le b+\delta$ and decrease to b as $\delta \downarrow 0$. By Lemma 1.3.1 and Proposition 1.7.1, the products $\theta_{m_\delta}^k \wedge W$ are positive, and by bounded decreasing continuity they converge weakly to $\theta_b^k \wedge W$. On the plurifine open set $\{a + \delta > b\}$, which contains $\{a = b\}$, we have $m_\delta = a + \delta$, so plurifine locality identifies the restriction of $\theta_{m_\delta}^k \wedge W$ to this set with that of $\theta_a^k \wedge W$. Hence, for every compact set $K \subset \{a = b\}$ in a coordinate chart, the upper semicontinuity of the mass of compact sets under weak convergence gives

$$\int_K \theta_a^k \wedge W = \lim_{\delta \downarrow 0} \int_K \theta_{m_\delta}^k \wedge W \le \int_K \theta_b^k \wedge W.$$

Since the measures are locally finite, inner regularity extends this inequality to all Borel subsets of the contact set, and a countable cover by charts proves the assertion.□

The products commute with pullback by holomorphic submersions. Here the pullback of a current is defined in local product coordinates by pulling back its distributional coefficients.

Lemma 1.7.2 *Let $\pi\colon Z \to Y$ be a holomorphic submersion of complex manifolds. For $1 \le r \le \dim Z$, let $u_1, \dots, u_r$ be locally bounded quasi-plurisubharmonic functions on Y, and let $\theta_1, \dots, \theta_r$ be smooth real $(1,1)$-forms on Y. Then*

$$\bigwedge_{i=1}^{r} \bigl(\pi^*\theta_i + \mathrm{dd^c}(u_i \circ \pi)\bigr) = \pi^*\left(\bigwedge_{i=1}^{r} (\theta_i + \mathrm{dd^c} u_i)\right).$$

For $r > \dim Y$, the product on the left vanishes, and the right-hand side is interpreted as zero.

Proof Work in product coordinates in which π is a coordinate projection. On a smaller chart of Y, choose smooth quadratic functions q_i such that $p_i = u_i + q_i$ is plurisubharmonic, and approximate each p_i by a decreasing sequence of smooth plurisubharmonic functions. Their pullbacks have the same properties on the product chart, and for smooth forms pullback commutes with wedge products. Applying bounded decreasing continuity to the local expansions (1.10) on both charts gives the asserted identity. Here we use that pullback of currents by the projection is weakly continuous: pairing the pullback with a compactly supported test form amounts to pairing the original current with the integral of the test form over the fiber variables. If $r > \dim Y$, every smooth approximating product vanishes for degree reasons, and so does its limit. □

For $C^{1,1}$ potentials, these products of currents agree with the products of their measurable Hessian coefficients.

Proposition 1.7.2 *Let $u_1, \dots, u_k$ be locally $C^{1,1}$ quasi-plurisubharmonic functions, and let $\theta_1, \dots, \theta_k$ be smooth real $(1,1)$-forms. Then the bounded product*

$$(\theta_1 + \mathrm{dd^c} u_1) \wedge \cdots \wedge (\theta_k + \mathrm{dd^c} u_k)$$

is the form with locally bounded measurable coefficients obtained by taking the wedge product of the factors pointwise almost everywhere.

Proof On a smaller coordinate ball, add a common smooth quadratic function q such that each $u_i + q$ is plurisubharmonic, and convolve with a smooth radial probability kernel. The Hessians of the convolutions are uniformly locally bounded and converge to the weak Hessians in every L^s space with $s < \infty$, so their finite products converge in L^1 on smaller balls. Since the potentials converge uniformly, the continuity of bounded Bedford–Taylor products identifies this limit with the pluripotential product. Expanding with respect to the smooth correction, as in (1.10), proves the assertion for the original factors. □

When the Hessian coefficients are bounded, a positive lower bound for the top-degree product gives a lower bound for the current itself.

Proposition 1.7.3 *Let h be a Hermitian form on X, and let $u \in \mathrm{PSH}(X,\theta) \cap C^{1,1}(X)$. If*

$$(\theta + \mathrm{dd^c} u)^n \geq a h^n$$

for a constant $a > 0$, then $\theta + \mathrm{dd^c} u \geq \delta h$ as a current for some $\delta > 0$.

Proof Put $T = \theta + \mathrm{dd^c} u$. Its measurable coefficients are bounded on the compact manifold X, so there is $K > 0$ such that, almost everywhere, the eigenvalues of T relative to h lie in $[0, K]$. By Proposition 1.7.2, their product is the density of T^n with respect to h^n, and hence it is at least a. Each eigenvalue is therefore at least a/K^{n-1} almost everywhere. This inequality of matrices almost everywhere gives the asserted inequality of currents with $\delta = a/K^{n-1}$. □

The mixed Monge–Ampère inequality of Theorem 1.6.2 also holds after a smooth background is added. In the following three lemmas, $U \subset \mathbb{C}^n$ is an open set and $\beta = \mathrm{dd^c}|z|^2$.

Lemma 1.7.3 *Let $T_i = \vartheta_i + \mathrm{dd^c} u_i \geq 0$, $i = 1, 2$, where the ϑ_i are smooth real $(1,1)$-forms on U and the u_i are locally bounded quasi-plurisubharmonic functions on U. If*

$$T_i^n \geq F_i \beta^n, \qquad F_i \geq 0 \text{ measurable},$$

then, for every integer $0 \leq k \leq n$,

$$T_1^k \wedge T_2^{n-k} \geq F_1^{k/n} F_2^{(n-k)/n} \beta^n. \tag{1.11}$$

Consequently, for $a, b \geq 0$,

$$(aT_1 + bT_2)^n \geq \left(a F_1^{1/n} + b F_2^{1/n}\right)^n \beta^n. \tag{1.12}$$

Proof Fix $W \Subset U$. Since the forms ϑ_i are smooth, there is a constant C such that, for every $x \in W$ and every small ball $B_r(x)$, the constant forms

$$A_i = \vartheta_i(x) + Cr\beta$$

satisfy $0 \leq S_i \leq 2Cr\beta$ on $B_r(x)$, where $S_i = A_i - \vartheta_i$. If q_i is a quadratic potential for A_i, then $u_i + q_i$ is plurisubharmonic on $B_r(x)$ and

$$\left(\mathrm{dd^c}(u_i + q_i)\right)^n = (T_i + S_i)^n \geq F_i \beta^n.$$

Apply Theorem 1.6.2 to these two closed Hessians. After expansion, every term other than $T_1^k \wedge T_2^{n-k}$ contains some S_i and is therefore bounded by $C' r \mu$, where

$$\mu = \sum_{\ell + m \leq n-1} T_1^\ell \wedge T_2^m \wedge \beta^{n-\ell-m}$$

is a locally finite measure.

A finite cover and a partition of unity give, on W,

$$T_1^k \wedge T_2^{n-k} \geq F_1^{k/n} F_2^{(n-k)/n} \beta^n - C' r \mu.$$

Letting $r \downarrow 0$ proves (1.11), and the binomial expansion then gives (1.12). □

Lemma 1.7.4 *Let ϑ be a smooth real $(1,1)$-form on U. Suppose that u is continuous, $T = \vartheta + \mathrm{dd}^{\mathrm{c}} u \geq 0$, and $T^n \geq g^n \beta^n$ with $g \geq 0$ continuous. For a nonnegative convolution kernel χ_δ of mass one,*

$$(T * \chi_\delta)^n \geq (g * \chi_\delta)^n \beta^n \tag{1.13}$$

on every smaller domain on which the convolution is defined.

Proof Approximate the convolution by finite convex combinations of translates, and apply (1.12), extended by induction to finitely many terms. Along these approximations, the background forms converge smoothly and the potentials converge uniformly on compact sets. To pass to the limit, add a common quadratic function and write the difference of two n-th powers as a telescoping sum. After pairing with a test form, integration by parts bounds each term by the uniform distance between the potentials times the mass of a mixed current, and successive integration against nested cutoff functions bounds these masses uniformly. Thus the products of the bounded potentials converge, and (1.13) follows. □

For a positive $(1,1)$-current T on U, write T_{ac} for the measurable $(1,1)$-form whose coefficients are the absolutely continuous parts of the coefficients of T, and define $\det_\beta T_{\mathrm{ac}}$ by $(T_{\mathrm{ac}})^n = (\det_\beta T_{\mathrm{ac}})\beta^n$. When the top-degree product has a continuous density, the convolution estimate identifies this determinant.

Lemma 1.7.5 *Let u be continuous and $T = \vartheta + \mathrm{dd}^{\mathrm{c}} u \geq 0$, with ϑ a smooth real $(1,1)$-form on U. If $T^n = F\beta^n$ for a continuous function $F \geq 0$, then*

$$\det_\beta T_{\mathrm{ac}} = F \qquad \textit{almost everywhere.}$$

Proof Take a nonnegative smooth radial mollifier χ_δ supported in the ball of radius δ. Then $T * \chi_\delta \to T_{\mathrm{ac}}$ almost everywhere. Indeed, the convolutions of the absolutely continuous parts converge at Lebesgue points of their densities, while the convolution of the singular part λ_s of a coefficient measure is bounded by $C\delta^{-2n}|\lambda_s|(B_\delta)$, which tends to zero almost everywhere by the differentiation theorem for measures. Applying Lemma 1.7.4 with $g = F^{1/n}$ and letting $\delta \downarrow 0$ gives $\det_\beta T_{\mathrm{ac}} \geq F$.

For the reverse inequality, write the convolved current as the Hessian of a decreasing regularization of a locally bounded plurisubharmonic function plus a convergent smooth correction. The continuity of bounded products gives $(T * \chi_\delta)^n \rightharpoonup T^n$. Fatou's lemma, applied against a nonnegative compactly supported test function, then gives $\det_\beta T_{\mathrm{ac}}\, \beta^n \leq T^n = F\beta^n$. □

Lemma 1.7.6 *Let $h \le a$ be locally bounded θ-plurisubharmonic functions, and let χ be a smooth function on a neighborhood of $(-\infty, 0]$ with $0 \le \chi' \le 1$ and $\chi'' \ge 0$. Then $w = a + \chi(h-a)$ is θ-plurisubharmonic and*

$$\theta_w = (1-\chi'(h-a))\theta_a + \chi'(h-a)\theta_h + \chi''(h-a)\,\mathrm{d}(h-a) \wedge \mathrm{d}^{\mathrm{c}}(h-a).$$

In particular, $\theta_w^n \ge \chi'(h-a)^n \theta_h^n$. The positivity of θ_w and this inequality also hold for convex functions χ of class $C^{1,1}$ on compact intervals with $0 \le \chi' \le 1$.

Proof The function $G(x,y) = x + \chi(y-x)$, for $y \le x$, is convex and nondecreasing in each variable: its first derivatives are $1-\chi'$ and χ', and its Hessian is

$$\chi'' \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \ge 0.$$

For smooth potentials, the chain rule gives the identity. For bounded potentials, add a common quadratic correction and convolve. Convolution preserves the order of the potentials, and since $G(x+r, y+r) = r + G(x,y)$, adding the same correction to both arguments adds it to G. The bounded chain rule (1.6) and its products with the gradient term give the identity in the limit. Freezing the smooth background as in the proof of Proposition 1.7.1 preserves the exact lower bound. Every term on the right is positive, and the positivity of bounded mixed products gives the last assertion.

For a convex weight of class $C^{1,1}$, extend it convexly across zero with derivative in $[0,1]$, and smooth it in one variable. The smooth convex weights and their first derivatives converge uniformly on compact intervals, and the derivatives stay in $[0,1]$. On each smaller coordinate ball, the corresponding potentials converge uniformly to w. The continuity of bounded products gives convergence of their top powers, while the weights on the right-hand side converge uniformly against the fixed measure θ_h^n. Passing to the limit proves the positivity and the product inequality with the original smooth background. □

1.8 Hermitian capacity

After a fixed rescaling, a local capacity test can be glued to a constant function and so becomes a global potential on X. This compares the local Bedford–Taylor capacity with a capacity defined by global potentials, and it transfers to X both the volume–capacity estimate and an estimate for the sublevel sets of normalized potentials.

For a Borel set $E \subset X$, the *Hermitian capacity* of E is

$$\mathrm{Cap}_\omega(E) = \sup_{v \in \mathrm{PSH}(X,\omega),\ 0 \le v \le 1} \int_E (\omega + \mathrm{dd}^{\mathrm{c}} v)^n,$$

where the products are those of Proposition 1.7.1. Their local expansion and (1.2) show that $\mathrm{Cap}_\omega(X) < +\infty$. On relatively compact coordinate balls, a fixed smooth correction bounds the measures of global tests by a constant times the corresponding local capacity. Conversely, the gluing construction in the proof of Lemma 1.8.1 below bounds local capacity by global capacity. Consequently, the quasi-continuity of quasi-plurisubharmonic functions and the characterization of sets of capacity zero recalled in Section 1.6 hold for Cap_ω. A sequence u_j converges to u *in capacity* if

$$\mathrm{Cap}_\omega(\{|u_j - u| > \delta\}) \longrightarrow 0 \quad \text{for every } \delta > 0,$$

where pole sets may be omitted. To compare with another Hermitian form ω', choose $C \geq 1$ with $\omega' \leq C\omega$. Every test $0 \leq v \leq 1$ for $\mathrm{Cap}_{\omega'}$ gives a test v/C for Cap_ω, and the positivity of mixed products gives

$$(\omega' + \mathrm{dd^c} v)^n \leq C^n (\omega + \mathrm{dd^c}(v/C))^n .$$

Thus $\mathrm{Cap}_{\omega'} \leq C^n \mathrm{Cap}_\omega$. Interchanging the two forms shows that the capacities are equivalent up to fixed multiplicative constants, so convergence in capacity does not depend on the choice of ω. Finally, the continuity statements for bounded products in Section 1.6 pass to the products with smooth backgrounds, by their local expansion and a finite cover.

1.8.1 Capacity estimates

In this subsection we abbreviate $\mathrm{Cap} = \mathrm{Cap}_\omega$.

Lemma 1.8.1 *There are constants $a, C > 0$ such that*

$$\mathrm{Vol}(E) \leq C \exp\bigl(-a\,\mathrm{Cap}(E)^{-1/n}\bigr) \tag{1.14}$$

for every Borel set $E \subset X$. Moreover, for every $u \in \mathrm{PSH}(X, \omega)$ with $\sup_X u = 0$ and every $t > 0$,

$$\mathrm{Cap}(\{u < -t\}) \leq C/t. \tag{1.15}$$

Proof Cover X by finitely many coordinate balls B'_j, with larger concentric balls B_j, $B'_j \Subset B_j$. A local capacity test $0 \leq v \leq 1$ on B_j can be turned into a global test as follows. Choose a smooth function χ_j with $-2 \leq \chi_j \leq 0$, equal to zero near $\overline{B'_j}$ and to -2 near ∂B_j. Choose $\delta > 0$ so small that

$$\omega + \delta \mathrm{dd^c} \chi_j \geq 0, \qquad 3\delta/2 \leq 1.$$

The function

$$V = \max\{\delta(v + \chi_j), -\delta/2\} + \delta/2 \quad \text{on } B_j, \qquad V = 0 \quad \text{outside } B_j,$$

is ω-plurisubharmonic on X, because near ∂B_j the constant branch is the larger one, and it takes values between zero and one. On B_j' we have $\omega_V \geq \delta \mathrm{dd}^{\mathrm{c}} v$. By the positivity of bounded products,

$$\operatorname{cap}(E \cap B_j', B_j) \leq \delta^{-n} \operatorname{Cap}(E).$$

The local volume–capacity estimate, summed over the finite cover, proves (1.14).

For the second estimate, choose on each B_j a smooth negative quadratic function r_j with $\mathrm{dd}^{\mathrm{c}} r_j \geq \omega$. For every global test $0 \leq v \leq 1$, the functions $u + r_j \leq 0$ and $v + r_j$ are plurisubharmonic on B_j. Thus

$$\begin{aligned}
\int_{\{u<-t\}} \omega_v^n &\leq t^{-1} \int_X (-u)\omega_v^n \\
&\leq t^{-1} \sum_j \int_{B_j'} (-u - r_j)\bigl(\mathrm{dd}^{\mathrm{c}}(v + r_j)\bigr)^n \\
&\leq \frac{C}{t}\left(1 + \int_X |u| \, \mathrm{d}V\right).
\end{aligned}$$

Here the second inequality uses $\omega_v \leq \mathrm{dd}^{\mathrm{c}}(v + r_j)$, and the third is the weighted Chern–Levine–Nirenberg inequality, applied to local test functions with a common bound. The L^1 bound for normalized potentials in Theorem 1.4.1 completes the proof after taking the supremum over v. □

Corollary 1.8.1 *Fix $p > 1$. There is a constant C_p such that every nonnegative $F \in L^p(X, \mathrm{d}V)$ satisfies*

$$\int_E F \, \mathrm{d}V \leq C_p \|F\|_p \operatorname{Cap}(E)^2$$

for every Borel set $E \subset X$.

Proof By Hölder's inequality and (1.14), the integral is at most

$$C\|F\|_p \exp(-a \operatorname{Cap}(E)^{-1/n}/p'), \qquad p' = p/(p-1).$$

The function $x \mapsto x^{-2}\exp(-ax^{-1/n}/p')$ is bounded on $0 < x \leq \operatorname{Cap}(X) < \infty$, and sets of capacity zero have zero volume. □

The same proof bounds the integral of an L^p density by any fixed positive power of the capacity. The exponent two will be convenient for the iteration over sublevel sets in Lemma 11.3.2.

1.8.2 Integration against a Hölder reference

A Hölder modulus for a bounded potential gives a modulus for integration against its Monge–Ampère measure. The proof uses a smooth approximation of the reference, with integration by parts controlling the change in its measure.

Lemma 1.8.2 *Let $B' \Subset B$ be concentric balls in $\mathbb{C}^n$, and let $0 < \alpha \le 1$. Suppose $p \in \mathrm{PSH}(B) \cap C^{0,\alpha}(B)$ has $C^{0,\alpha}$ norm at most N. For every $M \ge 1$ there is a constant C_M, depending only on M, N, α and the balls, such that all $a, b \in \mathrm{PSH}(B)$ with $|a|, |b| \le M$ satisfy*

$$\int_{B'} |a-b|(\mathrm{dd^c} p)^n \le C_M \|a-b\|_{L^1(B)}^{\alpha/(\alpha+2n)}.$$

Proof Choose intermediate concentric balls and a smooth cutoff $0 \le \chi \le 1$ supported in the innermost of them and equal to one on B'. All integrations below take place away from ∂B. Let p_ε be the convolution of p with a nonnegative radial mollifier of radius ε, defined on a larger intermediate ball. It is plurisubharmonic, and, for small $\varepsilon > 0$,

$$\|p - p_\varepsilon\|_\infty \le C\varepsilon^\alpha, \qquad 0 \le \mathrm{dd^c} p_\varepsilon \le C\varepsilon^{-2}\mathrm{dd^c}|z|^2.$$

Both p and p_ε take values in $[-N, N]$. Set $f = |a-b| = 2\max(a,b) - a - b$, and put

$$T = \sum_{j=0}^{n-1} (\mathrm{dd^c} p)^j \wedge (\mathrm{dd^c} p_\varepsilon)^{n-1-j}.$$

This is a closed positive current. Factorization and integration by parts give

$$\int \chi f \big((\mathrm{dd^c} p)^n - (\mathrm{dd^c} p_\varepsilon)^n\big) = \int (p - p_\varepsilon)\mathrm{dd^c}(\chi f) \wedge T.$$

We check that the signed measure $\mathrm{dd^c}(\chi f) \wedge T$ has total variation at most $C(1+M)^2$, uniformly in ε, where C depends only on N, α and the balls. Expand

$$\mathrm{dd^c}(\chi f) = \chi \mathrm{dd^c} f + f\mathrm{dd^c}\chi + \mathrm{d}\chi \wedge \mathrm{d^c} f + \mathrm{d}f \wedge \mathrm{d^c}\chi.$$

The term $\chi \mathrm{dd^c} f \wedge T$ is a linear combination of the positive measures formed with $\max(a,b), a, b$, whose masses are at most CM by local Chern–Levine–Nirenberg estimates. The term $f\mathrm{dd^c}\chi \wedge T$ has variation at most CM.

For the two terms involving $\mathrm{d}f$, use a cutoff η equal to one near $\operatorname{supp}\chi$. If q is any of $a, b, \max(a,b)$, translate it so that $0 \le q \le 2M$. The bounded chain rule and closedness of T give

$$\int \eta\, \mathrm{d}q \wedge \mathrm{d^c} q \wedge T = \frac{1}{2}\int q^2 \mathrm{dd^c}\eta \wedge T - \int \eta q \mathrm{dd^c} q \wedge T \le CM^2.$$

The last term is nonnegative before subtraction, and the first is bounded using the local mass of T. Since $f = 2\max(a,b) - a - b$, its gradient energy is also bounded

by CM^2. Cauchy–Schwarz against the positive current T now bounds the variation of $\mathrm{d}\chi \wedge \mathrm{d}^{\mathrm{c}} f \wedge T$ and $\mathrm{d}f \wedge \mathrm{d}^{\mathrm{c}}\chi \wedge T$ by CM. This accounts for all four terms of the expansion.

The bounded formulas follow first for local smooth regularizations. To pass to bounded potentials, regularize a, b, p decreasingly on the intermediate balls, write f as the displayed linear combination, and apply Lemma 1.6.1 to the expanded identities. The energy bounds are uniform; thus the integration by parts and its estimate survive this limit.

Writing $L = \|a - b\|_{L^1(B)}$ and using $(\mathrm{dd}^{\mathrm{c}} p_\varepsilon)^n \le C\varepsilon^{-2n}(\mathrm{dd}^{\mathrm{c}}|z|^2)^n$ for the second term, we have proved

$$\int \chi f (\mathrm{dd}^{\mathrm{c}} p)^n \le C(1+M)^2 \varepsilon^{\alpha} + C\varepsilon^{-2n} L. \tag{1.16}$$

For small positive L, take $\varepsilon = L^{1/(\alpha+2n)}$. A fixed admissible radius and the bounded local mass handle the remaining L. For $L = 0$, let $\varepsilon \downarrow 0$ in the same inequality. □

Corollary 1.8.2 *Let $u \in \mathrm{PSH}(X,\omega) \cap C^{0,\alpha}(X)$, $0 < \alpha \le 1$. If uniformly bounded quasi-plurisubharmonic functions a, b have a fixed common smooth lower curvature bound, then*

$$\int_X |a-b|\, \omega_u^n \le C \|a-b\|_{L^1(\mathrm{d}V)}^{\alpha/(\alpha+2n)}.$$

The constant depends on the bounds for a, b, their lower curvature bound and the $C^{0,\alpha}$ norm of u.

Proof On each of finitely many coordinate balls, choose a fixed smooth strictly plurisubharmonic quadratic Q such that $a+Q, b+Q, u+Q$ are plurisubharmonic and $\mathrm{dd}^{\mathrm{c}} Q \ge \omega$. There $0 \le \omega_u \le \mathrm{dd}^{\mathrm{c}}(u+Q)$, so positivity of bounded products gives $\omega_u^n \le (\mathrm{dd}^{\mathrm{c}}(u+Q))^n$. Apply Lemma 1.8.2 on smaller balls covering X. The quadratic cancels in the difference, and the local volumes are equivalent to $\mathrm{d}V$. Summing proves the assertion. □

The estimate before optimization also controls the tails of unbounded potentials. We need the following local form with a compact ambient space, so that the exponential bound for smooth volume is uniform.

Lemma 1.8.3 *Let (Y,Ω) be a connected compact Hermitian manifold, let $B' \Subset B \subset Y$ be coordinate balls, and let $p \in \mathrm{PSH}(B) \cap C^{0,\alpha}(B)$ have norm at most N, with $0 < \alpha \le 1$. There are $c, C > 0$, uniform for these references, such that*

$$(\mathrm{dd}^{\mathrm{c}} p)^n \big(B' \cap \{v < -t\}\big) \le C\mathrm{e}^{-ct} \quad (t \ge 0)$$

for every $v \in \mathrm{PSH}(Y,\Omega)$ with $\sup_Y v = 0$.

Proof Choose a smooth quadratic Q with $\mathrm{dd}^{\mathrm{c}} Q \ge \Omega$ on B and a fixed constant D such that $w = v + Q - D \le 0$. For integers $m \ge 1$, set

$$a_m = \max(w, -m), \qquad b_m = \max(w, -m-1), \qquad f_m = a_m - b_m.$$

These are bounded plurisubharmonic truncations, and $0 \le f_m \le 1$; moreover, f_m vanishes on $\{w \ge -m\}$ and equals one on $\{w < -m-1\}$. By Lemma 1.5.1 on (Y, Ω) and the bounded correction $Q - D$,

$$\|f_m\|_{L^1(B)} \le Ce^{-am}$$

for a fixed $a > 0$. Use the proof of Lemma 1.8.2, with $M = m+1$, keeping its estimate (1.16) before optimization. Taking $\varepsilon = e^{-am/(\alpha+2n)}$ gives

$$\int \chi f_m (dd^c p)^n \le C(1+m)^2 \exp\left(-\frac{a\alpha m}{\alpha+2n}\right) \le Ce^{-cm}$$

for some $c > 0$. For large m the convolution radius ε is admissible, and the common local mass handles the other integers. Finally, $w - v = Q - D$ is bounded on B. Hence, for large t, the set $B' \cap \{v < -t\}$ lies in $\{w < -m-1\}$, where $\chi f_m = 1$, for an integer $m \ge t - C$. This proves the estimate for large real t, and the common local mass covers the remaining $t \ge 0$. □

1.8.3 Capacity bounds for Hölder references

To pass from the exponential tail estimate to a capacity estimate, we use a closed background on an auxiliary projective space. The local capacity comparison then returns the result to the original Hermitian manifold.

For the Fubini–Study form ω_{FS} on $\mathbb{P}^n$, normalized by $\int \omega_{FS}^n = 1$, define for a compact set K

$$T(K) = \inf_{v \in PSH(\mathbb{P}^n, \omega_{FS}),\ \sup v = 0} \exp(\sup_K v).$$

We recall the classical global extremal-function theorem in this closed setting [GZ17, Theorem 9.17]. Its dichotomy is stated there in terms of pole sets of ω_{FS}-plurisubharmonic functions; by Josefson's theorem [GZ17, Theorem 12.5], the subsets of such pole sets are exactly the pluripolar sets. If K is non-pluripolar, the upper semicontinuous regularization of

$$V_K = \sup\{v \in PSH(\mathbb{P}^n, \omega_{FS}) : v|_K \le 0\}$$

is bounded, nonnegative and ω_{FS}-plurisubharmonic. Its Monge–Ampère probability is supported on K. If K is pluripolar, $\sup V_K = +\infty$ and $T(K) = 0$.

Lemma 1.8.4 *For every compact $K \subset \mathbb{P}^n$,*

$$T(K) \le e \exp\big(-\mathrm{Cap}_{\omega_{FS}}(K)^{-1/n}\big).$$

The right-hand side is zero for capacity zero.

Proof If K is pluripolar, both sides vanish, by the recalled theorem and the characterization of sets of capacity zero. Otherwise let $M = \sup V_K^* < \infty$. Subtracting

$\sup_K v$ from a normalized competitor, and conversely normalizing competitors for V_K, shows directly that $T(K) = \mathrm{e}^{-M}$.

If $M \geq 1$, the function $q = V_K^*/M$ satisfies $0 \leq q \leq 1$ and is an admissible capacity test. Since the background is closed,

$$1 = \int_K \omega_{V_K^*}^n \leq M^n \int_K \omega_q^n \leq M^n \operatorname{Cap}_{\omega_{\mathrm{FS}}}(K).$$

Here $\omega_{V_K^*} \leq M\omega_q$ follows from $M \geq 1$. Thus $M \geq \operatorname{Cap}_{\omega_{\mathrm{FS}}}(K)^{-1/n}$. If $M < 1$, V_K^* itself is a capacity test, so the same supported probability gives $\operatorname{Cap}_{\omega_{\mathrm{FS}}}(K) = 1$. Since $T(K) \leq 1$, the stated factor e covers this case. □

Proposition 1.8.1 *Fix $0 < \alpha \leq 1$ and $N > 0$. There are $a, C > 0$ such that every $u \in \mathrm{PSH}(X, \omega) \cap C^{0,\alpha}(X)$ with $\|u\|_{C^{0,\alpha}} \leq N$ satisfies*

$$\omega_u^n(E) \leq C \exp\bigl(-a \operatorname{Cap}_\omega(E)^{-1/n}\bigr) \quad (E \subset X \text{ Borel}).$$

Consequently, for each $s > 0$ there is a uniform C_s with $\omega_u^n(E) \leq C_s \operatorname{Cap}_\omega(E)^s$.

Proof Fix nested coordinate balls $B' \Subset B$ and a smooth quadratic Q with $\mathrm{dd^c} Q \geq \omega$ on B. The function $p = u + Q$ is plurisubharmonic with a common Hölder bound, and $\omega_u^n \leq (\mathrm{dd^c} p)^n$ there. Regard B as a ball in the affine chart $\mathbb{C}^n \subset \mathbb{P}^n$ and restrict this local measure to B', extending it by zero. By Lemma 1.8.3 on $(\mathbb{P}^n, \omega_{\mathrm{FS}})$, there are $c, C > 0$, independent of u, such that this measure gives mass at most $C\mathrm{e}^{-ct}$ to $\{v < -t\}$, $t \geq 0$, for every $v \in \mathrm{PSH}(\mathbb{P}^n, \omega_{\mathrm{FS}})$ with $\sup v = 0$. Its total mass is uniformly bounded by local Chern–Levine–Nirenberg estimates.

For a non-pluripolar compact $K \subset B'$, choose a normalized potential v with $\sup_K v \leq \log T(K) + 1$. Then $K \subset \{v < -t\}$ for every $t < -\log T(K) - 1$. For small $T(K)$ the tail estimate therefore gives

$$(\mathrm{dd^c} p)^n(K) \leq CT(K)^c \leq C \exp\bigl(-c \operatorname{Cap}_{\omega_{\mathrm{FS}}}(K)^{-1/n}\bigr).$$

The common mass bound handles the remaining $T(K)$, and pluripolar sets have zero local product mass. A bounded local potential for ω_{FS} turns every global test $0 \leq v \leq 1$ into a plurisubharmonic function of a fixed bounded range on B. Hence

$$\operatorname{Cap}_{\omega_{\mathrm{FS}}}(K) \leq C_1 \operatorname{cap}(K, B) \leq C_2 \operatorname{Cap}_\omega(K),$$

where the last inequality is the gluing construction of Lemma 1.8.1 on the original manifold. These inequalities have the required direction after taking negative inverse powers. Inner regularity of the finite local product extends the bound from compact to Borel subsets of B'. Summing over a fixed finite cover proves the first assertion. The second follows by bounding $x^{-s} \exp(-ax^{-1/n})$ on $0 < x \leq \operatorname{Cap}_\omega(X)$. □

1.8.4 Convergence of potentials and measures

For quasi-continuous obstacles, a decreasing lower semicontinuous approximation keeps the original bounds and confines the error to sets of vanishing capacity.

Lemma 1.8.5 *Every bounded quasi-continuous function f on X admits bounded lower semicontinuous functions f_j with common bounds such that $f_j \downarrow f$ quasi-everywhere and $f_j \to f$ in capacity.*

Proof Choose open sets O_j of capacity at most 2^{-j} such that the restriction of f to $X \setminus O_j$ is continuous, and set $G_j = \bigcup_{k \geq j} O_k$. Then the sets G_j decrease, their capacities tend to zero, and f is continuous on the closed complement of each G_j. Choose $M \geq \sup_X f$ and define

$$f_j = f \text{ on } X \setminus G_j, \qquad f_j = M \text{ on } G_j.$$

At a boundary point of G_j, the values of f_j on the closed complement tend to the value of f there, while the values inside G_j are at least that value. Thus f_j is lower semicontinuous. The functions f_j decrease to f outside $\bigcap_j G_j$, a set of capacity zero. Since $f_j - f$ is supported in G_j, they also converge to f in capacity. □

Uniform bounds on potentials give a common capacity bound for their mixed measures.

Lemma 1.8.6 *Let $A \geq 0$ and let $0 \leq k \leq n$ be an integer. There is a constant C_A such that all the positive mixed measures*

$$\omega_{v_1} \wedge \cdots \wedge \omega_{v_k} \wedge \omega^{n-k}, \qquad v_i \in \mathrm{PSH}(X, \omega), \quad \|v_i\|_\infty \leq A,$$

are bounded above on Borel sets by $C_A \operatorname{Cap}_\omega$. In particular, their total masses are uniformly bounded. If uniformly bounded functions tend to zero in capacity, then their integrals against these measures tend to zero uniformly in the v_i.

Proof Translate the v_i so that $0 \leq v_i \leq 2A$, and put $q = (v_1 + \cdots + v_k)/B$, where $B \geq \max\{2k, 2kA, 1\}$. Then $0 \leq q \leq 1$ and

$$\omega_q = B^{-1} \sum_{i=1}^{k} \omega_{v_i} + (1 - k/B)\omega,$$

where all coefficients are positive. Expanding ω_q^n shows that the given mixed product is bounded by a constant multiple of ω_q^n, and the definition of capacity proves the first claim. The case $k = 0$ follows by testing with $v = 0$. For a bounded function z, split its integral into the parts over $\{|z| > t\}$ and over its complement; first use convergence in capacity for fixed $t > 0$, and then let $t \downarrow 0$. □

The same control applies to the total variation of a product formed with a different smooth background.

Lemma 1.8.7 *Let Ω be a Hermitian form on X. Suppose that Ω-plurisubharmonic functions v_j have common finite upper and lower bounds and decrease to v. Then the signed measures*

$$\nu_j = (\theta + \mathrm{dd}^{\mathrm{c}} v_j)^n$$

converge to $(\theta + \mathrm{dd}^{\mathrm{c}} v)^n$ against every fixed bounded quasi-continuous function. Their total variations are bounded, uniformly in j, by a constant multiple of the local capacity on the balls of a fixed finite cover.

Proof On a relatively compact coordinate ball, choose a fixed quadratic function q such that the functions $v_j + q$ are plurisubharmonic; they still have common bounds. Expand

$$(\theta + \mathrm{dd}^{\mathrm{c}} v_j)^n = \sum_{\ell=0}^{n} \binom{n}{\ell} \big(\mathrm{dd}^{\mathrm{c}}(v_j + q)\big)^{\ell} \wedge (\theta - \mathrm{dd}^{\mathrm{c}} q)^{n-\ell}.$$

The absolute value of the ℓ-th summand is bounded by a fixed constant times

$$\big(\mathrm{dd}^{\mathrm{c}}(v_j + q)\big)^{\ell} \wedge \kappa_0^{n-\ell},$$

where κ_0 is a fixed positive Euclidean form. This positive mixed measure is bounded, uniformly in j, by a constant times the local capacity: add a fixed strictly plurisubharmonic quadratic function to $v_j + q$, expand the top power of its Hessian, and rescale its uniformly bounded range into $[0, 1]$ to apply the definition of capacity. This also covers $\ell = 0$. Summing over a finite cover gives the asserted control of the total variations. The ordinary decreasing continuity of Bedford–Taylor products gives weak convergence term by term. For a bounded quasi-continuous coefficient, remove an open set of small capacity and extend the restriction of the coefficient to the compact complement to a continuous function on X. The variation bound controls the resulting error uniformly for the sequence and its limit, and letting the capacity of the removed set tend to zero proves the assertion. □

A finite non-pluripolar measure is uniformly absolutely continuous with respect to capacity, although it need not be bounded by a power of the capacity.

Lemma 1.8.8 *Let μ be a finite non-pluripolar Borel measure on X. Then*

$$\lim_{\delta \downarrow 0} \sup\{\mu(E) : E \text{ Borel}, \ \mathrm{Cap}_\omega(E) \le \delta\} = 0.$$

Consequently, convergence in capacity implies convergence in μ-measure.

Proof If the first assertion failed, there would be $\epsilon > 0$ and Borel sets E_j of capacity at most 2^{-j} with $\mu(E_j) \ge \epsilon$. By countable subadditivity of capacity,

$$\mathrm{Cap}_\omega\left(\limsup_j E_j\right) = 0.$$

Hence $\limsup_j E_j$ is pluripolar and has μ-measure zero. On the other hand, since μ is finite, continuity from above applied to the sets $\bigcup_{k \ge j} E_k$ shows that $\limsup_j E_j$ has measure at least ϵ, a contradiction. If $u_j \to u$ in capacity, applying the first assertion to the sets $\{|u_j - u| > \delta\}$ gives the consequence. □

Lemma 1.8.9 *Let $u_j \in \mathrm{PSH}(X, \theta)$. Suppose either that u_j decreases to a function $u \not\equiv -\infty$, or that u_j increases, the sequence is uniformly bounded above, and $u = (\lim_j u_j)^*$. Then $u_j \to u$ in capacity.*

Proof In the decreasing case u is a common lower bound for the sequence and its limit; in the increasing case u_1 is one. Choose a Hermitian form $\Omega \ge \theta$ and truncate all functions at $-T$. The truncations are bounded, monotone and Ω-plurisubharmonic, so the convergence in capacity of locally bounded monotone sequences, recalled in Section 1.6, applies to them on the balls of a finite cover. Let w be the common lower bound. Then $w - \sup_X w$ is Ω-plurisubharmonic with supremum zero, so (1.15), applied with Ω in place of ω, and the equivalence of the capacities Cap_Ω and Cap_ω show that the capacity of the sublevel set $\{w < -T\}$ tends to zero as $T \to \infty$. Since the functions agree with their truncations outside $\{w < -T\}$, letting $T \to \infty$ proves the lemma. □

A fixed bounded reference potential also controls a mixed trace in which one normalized potential varies.

Lemma 1.8.10 *Fix a bounded $\psi \in \mathrm{PSH}(X, \theta)$. There is a constant C_ψ, depending only on an upper bound for $\|\psi\|_\infty$, such that*

$$\int_X \theta_v \wedge \theta_\psi^{n-1} \le C_\psi$$

for every bounded $v \in \mathrm{PSH}(X, \theta)$ with $\sup_X v = 0$.

Proof For $n = 1$, Stokes' theorem gives $\int_X \theta_v = \int_X \theta$, so assume $n \ge 2$. Choose a fixed Hermitian form $\Omega \ge \theta$. All currents on the left-hand side are positive, and $\Omega_\psi \ge \theta_\psi$, so $\theta_v \wedge \theta_\psi^{n-1} \le \theta_v \wedge \Omega_\psi^{n-1}$. Integrating and applying Stokes' theorem for bounded potentials to $\mathrm{dd}^c v \wedge \Omega_\psi^{n-1}$, we obtain

$$\int_X \theta_v \wedge \theta_\psi^{n-1} \le \int_X \theta \wedge \Omega_\psi^{n-1} + \int_X v \, \mathrm{dd^c}(\Omega_\psi^{n-1}).$$

The first term on the right does not depend on v; since $\theta \le \Omega$, it is at most the mass of $\Omega \wedge \Omega_\psi^{n-1}$. To estimate the second, differentiate:

$$\mathrm{dd^c}(\Omega_\psi^{n-1}) = (n-1)\mathrm{dd^c}\Omega \wedge \Omega_\psi^{n-2} + (n-1)(n-2)\mathrm{d}\Omega \wedge \mathrm{d^c}\Omega \wedge \Omega_\psi^{n-3},$$

where the second term is absent for $n = 2$. Since $\mathrm{dd^c}\Omega$ and $\mathrm{d}\Omega \wedge \mathrm{d^c}\Omega$ are smooth forms, the absolute values of these terms are bounded by fixed multiples of the positive measures $\Omega^2 \wedge \Omega_\psi^{n-2}$ and $\Omega^3 \wedge \Omega_\psi^{n-3}$. The weighted local Chern–Levine–Nirenberg inequality of Section 1.6 bounds the integrals of $|v|$ against these measures. Indeed, on each larger coordinate ball, add fixed smooth corrections to v and ψ, shift the corrected v so that it becomes negative, expand in the corrected positive Hessian of ψ, and apply that inequality on a smaller ball. The resulting bound is

$$C(1 + \|\psi\|_\infty)^n (1 + \|v\|_{L^1(X)}).$$

The same expansion and the range estimate (1.2) bound the mass of $\Omega \wedge \Omega_\psi^{n-1}$ by $C(1 + \|\psi\|_\infty)^{n-1}$. By Theorem 1.4.1, normalized θ-plurisubharmonic functions have uniformly bounded L^1 norms, and summing the local bounds proves the assertion.□

1.8.5 Vanishing area on curves

On a compact curve, the integral of the background is the total mass of every curvature current. When this integral vanishes, positivity forces the current to vanish. Normalized compactness and the local estimate of capacity by total variation make this uniqueness stable as the area tends to zero.

Throughout this subsection, Y is a compact connected Riemann surface and η is a smooth positive $(1,1)$-form on Y with $\int_Y \eta = 1$, which we use for L^1 norms and for the capacity Cap_η on Y.

Lemma 1.8.11 *For a smooth real $(1,1)$-form σ on Y, put $A = \int_Y \sigma$. There is a smooth real function h with $\sigma = A\eta + \mathrm{dd^c} h$. The class $\mathrm{PSH}(Y, \sigma)$ is nonempty if and only if $A \ge 0$. If $A = 0$, it consists of the functions $-h + c$, $c \in \mathbb{R}$, and its unique member with supremum zero is $u_\sigma = -h + \min_Y h$.*

Proof The smooth form $\sigma - A\eta$ has integral zero, so the Poisson equation on the compact connected surface Y has a smooth solution h, unique up to an additive constant. If $u \in \mathrm{PSH}(Y, \sigma)$, positivity and Stokes' theorem give

$$0 \le (\sigma + \mathrm{dd^c} u)(Y) = A.$$

Conversely, $\sigma + \mathrm{dd^c}(-h) = A\eta$, so $-h \in \mathrm{PSH}(Y, \sigma)$ when $A \ge 0$. If $A = 0$, every positive curvature measure has total mass zero and therefore vanishes. Thus

$\mathrm{dd}^{\mathrm{c}}(u+h)=0$ for every $u\in \mathrm{PSH}(Y,\sigma)$, and elliptic regularity and the uniqueness of harmonic functions on Y show that $u+h$ is constant. Normalizing the supremum gives the stated formula. □

Theorem 1.8.1 *Let σ_j be smooth real $(1,1)$-forms on Y that converge uniformly to a smooth form σ_0, and suppose that*

$$A_j=\int_Y \sigma_j \geq 0, \qquad \int_Y \sigma_0 = 0.$$

Let u_0 be the normalized σ_0-plurisubharmonic potential of Lemma 1.8.11. Then, for every $t>0$,

$$\sup_{u\in \mathrm{PSH}(Y,\sigma_j),\ \sup_Y u=0} \|u-u_0\|_{L^1(Y)} \longrightarrow 0,$$

$$\sup_{u\in \mathrm{PSH}(Y,\sigma_j),\ \sup_Y u=0} \mathrm{Cap}_\eta(\{|u-u_0|>t\}) \longrightarrow 0.$$

Moreover, the curvature measures $\sigma_j+\mathrm{dd}^{\mathrm{c}}u$ of these potentials tend to zero in total variation, uniformly in u.

Proof Choose $C>0$ with $\sigma_j\leq C\eta$ for every j. Every sequence of normalized σ_j-plurisubharmonic functions then lies in the compact family of normalized $C\eta$-plurisubharmonic functions of Theorem 1.4.1. Every L^1 limit u of a subsequence is normalized and, by passing to the limit in the sense of distributions, satisfies $\sigma_0+\mathrm{dd}^{\mathrm{c}}u\geq 0$. Lemma 1.8.11 gives $u=u_0$. If the first uniform limit failed, a sequence violating it would have a subsequence converging to this same limit, a contradiction.

For a potential u in the j-th normalized family, put $T=\sigma_j+\mathrm{dd}^{\mathrm{c}}u$. Positivity and Stokes' theorem give $\|T\|_{\mathrm{TV}}=A_j$, and $A_j\to\int_Y\sigma_0=0$. Since $\sigma_0+\mathrm{dd}^{\mathrm{c}}u_0=0$,

$$\|\mathrm{dd}^{\mathrm{c}}(u-u_0)\|_{\mathrm{TV}}\leq A_j+\|\sigma_j-\sigma_0\|_{\mathrm{TV}}\longrightarrow 0$$

uniformly over that family. Apply Lemma 1.6.3 on a fixed finite cover of Y by concentric coordinate disks. The first limit and the bound above make each local capacity tend to zero uniformly, and the comparison between local and global capacity at the beginning of this section proves the second limit. The identity $\|T\|_{\mathrm{TV}}=A_j$ proves the final assertion. □

1.9 Comparing local regularizations

To glue local convolutions, we need them to agree on overlaps up to an error tending uniformly to zero. A bounded plurisubharmonic function need not be continuous, so this comparison cannot come from uniform convergence to the original function. Instead, the logarithmic convexity of its suprema over balls controls both the change of scale and the change of coordinates.

Let $U \subset \mathbb{C}^n$ be a domain and u a bounded plurisubharmonic function on U. For z in a fixed relatively compact subset of U and small $\delta > 0$, put

$$S_\delta u(z) = \sup_{|\zeta - z| \le \delta} u(\zeta), \qquad A_\delta u(z) = \fint_{|\zeta - z| = \delta} u(\zeta)\, d\sigma(\zeta), \qquad u_\delta = u * \rho_\delta,$$

where $d\sigma$ is the Euclidean surface measure and ρ is a smooth nonnegative radial mollifier of integral one, supported in the unit ball, with $\rho_\delta(\zeta) = \delta^{-2n}\rho(\zeta/\delta)$. All balls below lie in a fixed larger relatively compact subset of U, and B denotes a bound for the oscillation of u on that subset.

Lemma 1.9.1 *For each fixed $A > 1$,*

$$S_{A\delta}u - S_\delta u \longrightarrow 0, \qquad S_\delta u - u_\delta \longrightarrow 0$$

uniformly on compact subsets as $\delta \downarrow 0$.

Proof The maximum of a plurisubharmonic function over concentric balls is a convex function of the logarithm of the radius. This follows by restricting to each complex line through the center and comparing, on an annulus, with the harmonic function that is affine in $\log|\zeta|$. Fix a radius $r > A\delta$ for which all the balls of radius r about points of the given compact set lie in the larger subset. Comparing secant slopes of this convex function between $\log\delta$, $\log(A\delta)$ and $\log r$ gives

$$0 \le S_{A\delta}u - S_\delta u \le \frac{B \log A}{\log(r/\delta)}, \tag{1.17}$$

which proves the first assertion.

To compare the maximum with the spherical mean, we use the Poisson kernel in real dimension $2n$. For a point at distance at most $\delta/2$ from the center, the normalized Poisson kernel of the ball of radius δ is bounded below by a positive dimensional constant c_n. Since u is bounded above by $S_\delta u$ and by its harmonic extension from the sphere of radius δ, we obtain

$$S_\delta u - S_{\delta/2}u \ge c_n(S_\delta u - A_\delta u).$$

These inequalities hold for bounded plurisubharmonic functions by local smoothing followed by a decreasing limit. By (1.17), $S_\delta u - A_\delta u$ therefore tends uniformly to zero.

For $0 < t \le 1$, the spherical mean is increasing in the radius, and

$$0 \le A_\delta u - A_{t\delta} u \le (S_\delta u - S_{t\delta} u) + (S_{t\delta} u - A_{t\delta} u).$$

By logarithmic convexity, the first term is at most $B \log(1/t)/\log(r/(t\delta))$, and by the preceding estimate the second is at most $C_n B/\log(r/(t\delta))$. Write the radial convolution as an average of the spherical means $A_{t\delta} u$ with respect to the probability density $\widehat{\rho}(t) = |\mathbb{S}^{2n-1}| t^{2n-1} \rho(t)$ on $[0, 1]$, where $\rho(t)$ denotes the value of ρ at radius t. Since $\int_0^1 (1 + \log(1/t)) \widehat{\rho}(t)\, \mathrm{d}t < \infty$, integration gives

$$0 \le A_\delta u - u_\delta \le \frac{C_{n,\rho} B}{\log(r/\delta)}.$$

Combining the two comparisons proves the second assertion. □

Corollary 1.9.1 *Let $U, V \subset \mathbb{C}^n$ be domains, let $F\colon U \to V$ be biholomorphic, and let u be bounded and plurisubharmonic on U. Then*

$$u_\delta - \big((u \circ F^{-1}) * \rho_\delta\big) \circ F$$

tends locally uniformly to zero as $\delta \downarrow 0$.

Proof On a compact subset, F and its inverse have uniformly bounded first derivatives. Thus, for a fixed $A > 1$, the inverse image under F of the ball of radius δ about $F(z)$ lies between the balls about z of radii δ/A and $A\delta$. Consequently, $S_\delta(u \circ F^{-1})(F(z))$ and $S_\delta u(z)$ both lie between $S_{\delta/A} u(z)$ and $S_{A\delta} u(z)$, and the first assertion of Lemma 1.9.1 shows that their difference tends to zero uniformly. The second assertion of the lemma, applied to u and to $u \circ F^{-1}$, compares each of the two convolutions with the corresponding supremum regularization. □

1.10 Gluing with a small curvature error

The overlap estimate allows local regularizations to be glued by a regularized maximum. Small corrections near the boundary of each chart make the corresponding local branch irrelevant there, while convexity of the regularized maximum preserves a common lower curvature bound. The curvature error is measured by the Hermitian form ω.

For an integer $N \ge 1$, choose independent real random variables $Z_1, \dots, Z_N$ with a common smooth even probability density supported in $[-1, 1]$, and define

$$M_\tau(t_1, \dots, t_N) = \mathbb{E} \max_i (t_i + \tau Z_i) \quad (\tau > 0).$$

As a convolution, M_τ is smooth. It is convex and increasing in every argument, and, with $\mathbf{1} = (1, \dots, 1)$, it satisfies

$$\max_i t_i \le M_\tau(t) \le \max_i t_i + \tau, \qquad M_\tau(t + c\mathbf{1}) = M_\tau(t) + c$$

for $t \in \mathbb{R}^N$ and $c \in \mathbb{R}$. If $t_i < \max_{j\neq i} t_j - 2\tau$, then the i-th argument can be removed without changing the value. Moreover, $M_\tau(t)$ is nondecreasing in τ, because it is a convex even function of τ, the distribution of each Z_i being symmetric.

The second identity expresses invariance under translations. If smooth real functions f_i, $1 \le i \le N$, satisfy $\mathrm{dd^c} f_i \ge -\gamma$ for a common smooth real $(1,1)$-form γ, then

$$\mathrm{dd^c} M_\tau(f_1, \dots, f_N) \ge -\gamma. \tag{1.18}$$

Indeed, in the chain rule the first partial derivatives of M_τ are nonnegative and, by the translation identity, sum to one, while the real Hessian of the convex function M_τ contributes a semipositive $(1,1)$-form.

Proposition 1.10.1 *Let $u \in \mathrm{PSH}(X,\theta) \cap L^\infty(X)$. There are smooth functions v_j and positive numbers $\varepsilon_j \downarrow 0$ such that*

$$v_j \downarrow u, \qquad \theta + \varepsilon_j \omega + \mathrm{dd^c} v_j \ge 0.$$

Proof For fixed $\epsilon > 0$, we first glue the local convolutions of u by a regularized maximum, obtaining smooth functions that decrease to u and have curvature error $2\epsilon\omega$; a diagonal sequence then proves the proposition.

Step 1: local regularizations. Fix $\epsilon > 0$. By continuity of θ, we can choose finitely many nested coordinate neighborhoods $V_i \Subset W_i \Subset U_i$, with the V_i covering X, and smooth quadratic functions r_i such that

$$0 \le \mathrm{dd^c} r_i - \theta \le \epsilon\omega \quad \text{on } U_i.$$

To do so, freeze the coefficients of θ at the center of a chart, add a small positive constant form, and shrink the coordinate neighborhood. Then $p_i = u + r_i$ is a bounded plurisubharmonic function on U_i; let $p_{i,\delta}$ be its convolution in the coordinates of U_i.

On an overlap, express both p_i and p_j in the coordinates of U_j. Their difference is the smooth function $r_i - r_j$, so the difference of their convolutions in these coordinates tends uniformly to $r_i - r_j$. By Corollary 1.9.1, changing the coordinates in which p_i is convolved introduces another uniformly vanishing error. Thus, on the intersections of the closures of the W_i,

$$(p_{i,\delta} - r_i) - (p_{j,\delta} - r_j) \longrightarrow 0 \quad \text{uniformly.} \tag{1.19}$$

Step 2: gluing. Choose functions $\eta_i \in C^\infty(U_i, [-1,0])$ that vanish on V_i and equal -1 near the boundary of W_i, and a constant $C \ge 1$ with $\mathrm{dd^c}\eta_i \ge -C\omega$. The cover, the functions r_i, η_i and the constant C may depend on ϵ; they are fixed while δ tends to zero. Set

$$f_{i,\delta} = p_{i,\delta} - r_i + (\epsilon/C)\eta_i \quad \text{on } W_i.$$

Each of these functions satisfies $\mathrm{dd^c} f_{i,\delta} \ge -\theta - 2\epsilon\omega$. Every boundary point of W_i lies in some V_j, and there $\eta_j = 0$ while $\eta_i = -1$. By (1.19), for all small δ,

$$f_{i,\delta} < f_{j,\delta} - \epsilon/(2C)$$

on a neighborhood of that boundary point. Since the cover is finite, this separation holds uniformly.

Let $\tau(\delta) \downarrow 0$ as $\delta \downarrow 0$, with $\tau(\delta) < \epsilon/(8C)$, and define the value at each point as the regularized maximum $M_{\tau(\delta)}$ of the functions $f_{i,\delta}$ for those i with the point in W_i. A branch disappears only near the boundary of its W_i, where it lies below another branch by more than 2τ. By the removal property of M_τ, these local formulas therefore define a global smooth function $\psi_{\epsilon,\delta}$, and (1.18) gives

$$\theta + 2\epsilon\omega + \mathrm{dd}^{\mathrm{c}}\psi_{\epsilon,\delta} \geq 0.$$

At every point there is a branch with $\eta_i = 0$, and $p_{i,\delta} \geq p_i$; hence $\psi_{\epsilon,\delta} \geq u$. For fixed ϵ, the local convolutions decrease with δ, and the regularized maximum decreases when both its arguments and τ decrease. Its pointwise limit is

$$\max_{i:x\in W_i} \big(u(x) + (\epsilon/C)\eta_i(x)\big) = u(x).$$

Thus $\psi_{\epsilon,\delta} \downarrow u$ as $\delta \downarrow 0$, and also in L^1.

Step 3: a diagonal sequence. Take $\epsilon_j \downarrow 0$ and put $v_j = \psi_{\epsilon_j,\delta_j} + 2^{-j}$, where δ_j is chosen so small that

$$\|\psi_{\epsilon_j,\delta_j} - u\|_{L^1} \leq 2^{-j}, \qquad v_j < v_{j-1} \quad (j \geq 2).$$

The second requirement can be met because $v_{j-1} \geq u + 2^{-j+1}$. Indeed, as $\delta \downarrow 0$, the continuous functions $\psi_{\epsilon_j,\delta} + 2^{-j} - v_{j-1}$ decrease pointwise to a function bounded above by -2^{-j}. Their sets of nonnegative values are decreasing compact sets with empty intersection, so one of them is empty. The resulting sequence (v_j) decreases, converges to u in L^1, and stays above u. Its pointwise limit is $(\theta + 2\epsilon_k\omega)$-plurisubharmonic for every k, hence θ-plurisubharmonic, and it equals u almost everywhere; by Lemma 1.3.3, it equals u everywhere. Taking $\varepsilon_j = 2\epsilon_j$ proves the proposition. □

For continuous potentials, the approximants of Proposition 1.10.1 converge uniformly.

Lemma 1.10.1 *Let α be a smooth real $(1,1)$-form on X, and let $u \in \mathrm{PSH}(X,\alpha) \cap C^0(X)$. There are smooth real functions u_j and numbers $\varepsilon_j \downarrow 0$ such that $u_j \to u$ uniformly and*

$$\alpha + \mathrm{dd}^{\mathrm{c}} u_j \geq -\varepsilon_j\omega.$$

Proof The continuous function u is bounded, so Proposition 1.10.1, applied with $\theta = \alpha$, gives smooth functions $u_j \downarrow u$ and numbers $\varepsilon_j \downarrow 0$ with $\alpha + \varepsilon_j\omega + \mathrm{dd}^{\mathrm{c}} u_j \geq 0$. The continuous functions u_j decrease to the continuous function u on the compact manifold X, so the convergence is uniform by Dini's theorem. □

1.11 Regularization in a fixed Hermitian background

For $\theta = \omega$, rescaling the approximants absorbs the curvature error into the positive background. Subtracting a constant first makes the approximants negative, so that this rescaling preserves their decreasing order.

Theorem 1.11.1 *For every $u \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$, there are smooth functions $u_j \downarrow u$ such that*

$$\omega + \mathrm{dd}^{\mathrm{c}} u_j > 0.$$

The functions u_j are uniformly bounded.

Proof Subtracting a constant, we may assume $u \le -2$. Apply Proposition 1.10.1 with $\theta = \omega$, obtaining smooth functions $v_j \downarrow u$ and positive numbers $\varepsilon_j \downarrow 0$ with $(1+\varepsilon_j)\omega + \mathrm{dd}^{\mathrm{c}} v_j \ge 0$. The compact sets $\{v_j \ge -1\}$ decrease to the empty set, since $u \le -2$. After discarding finitely many terms, we may therefore assume $v_j < 0$ everywhere. Put

$$u_j = \frac{v_j}{1+2\varepsilon_j}.$$

Then

$$\omega + \mathrm{dd}^{\mathrm{c}} u_j \ge \frac{\varepsilon_j}{1+2\varepsilon_j}\omega > 0.$$

Since v_j is negative and decreases, while the positive denominator decreases to one, u_j also decreases, and it converges to u. Moreover, $u \le v_j \le u_j \le u_1$, which gives uniform boundedness. Adding back the subtracted constant proves the theorem. □

This regularization provides the approximation used for products and mass in Chapter 7. For an arbitrary smooth form θ, Proposition 1.10.1 only gives approximants whose curvature error tends to zero.

1.12 Singularity type and multiplier ideals

Domination up to an additive constant compares the singularities of quasi-plurisubharmonic functions on X; multiplier ideals record only which holomorphic functions are locally square integrable against the weights e^{-ku}, $k > 0$.

Definition 1.12.1 Let u, v be quasi-plurisubharmonic functions on X. We write $u \preceq v$ if there is a constant $C \in \mathbb{R}$ such that $u \le v + C$ on X, and $u \sim v$ if both $u \preceq v$ and $v \preceq u$; in the latter case u and v have the same *singularity type*. When this has to be distinguished from the multiplier-ideal type introduced below, we speak of the *ordinary singularity type*, or simply the *ordinary type*.

Thus $u \preceq v$ means that u is at least as singular as v. For instance, a quasi-plurisubharmonic function with a logarithmic pole lies below every bounded potential in this order, but not conversely.

Proposition 1.12.1 *The relation $\preceq$ is a preorder, and $\sim$ is an equivalence relation. If u is quasi-plurisubharmonic and f is a smooth function on X, then $u + f \sim u$. In particular, all smooth functions on X have the same singularity type.*

Proof Reflexivity follows with $C = 0$. If $u \le v+C$ and $v \le w+D$, then $u \le w+C+D$, which proves transitivity. As for every preorder, comparison in both directions defines an equivalence relation. Since X is compact, a smooth function f satisfies $|f| \le A$ for some constant A, and then $u - A \le u + f \le u + A$. Taking $u = 0$ shows that every smooth function has the singularity type of the constants. □

Some later results require the singularities of a potential to be concentrated on a small closed set. A closed subset $A \subseteq X$ is *complete pluripolar* if every point of X has a neighborhood U with a plurisubharmonic function p on U such that $A \cap U = \{p = -\infty\}$. A quasi-plurisubharmonic function has *small unbounded locus* if it is locally bounded outside a closed complete pluripolar set.

Definition 1.12.2 Let $\mathcal{O}_X$ denote the sheaf of holomorphic functions on X. For a quasi-plurisubharmonic function u on X, the *multiplier ideal* $\mathcal{I}(u) \subseteq \mathcal{O}_X$ is the ideal sheaf whose sections over an open set U are the holomorphic functions g on U for which $|g|^2 \mathrm{e}^{-u}$ is locally integrable on U with respect to a smooth positive volume form. Its stalk at x is denoted by $\mathcal{I}(u)_x$.

Changing the volume form does not change local integrability: on relatively compact coordinate neighborhoods, the ratio of two smooth positive volume forms is bounded above and below by positive constants. In particular, the definition does not involve a choice of Hermitian metric, and it applies verbatim to quasi-plurisubharmonic functions on any complex manifold.

Proposition 1.12.2 *If $u \preceq v$, then $\mathcal{I}(ku) \subseteq \mathcal{I}(kv)$ for every real $k > 0$. If $u \sim v$, then these ideals agree for every $k > 0$.*

Proof Let g be a local holomorphic function on X. If $u \le v+C$, then $\mathrm{e}^{-kv} \le \mathrm{e}^{kC}\mathrm{e}^{-ku}$. Multiplying by $|g|^2$ shows that local integrability of $|g|^2\mathrm{e}^{-ku}$ implies local integrability of $|g|^2\mathrm{e}^{-kv}$. If $u \sim v$, the same argument applies in both directions. □

Example 1.12.1 On a disk in $\mathbb{C}$ centered at the origin, with coordinate z, let $\lambda \ge 0$ and put $u(z) = \lambda \log |z|^2$ for $\lambda > 0$, while $u = 0$ for $\lambda = 0$. Then

$$\mathcal{I}(u)_0 = (z^{\lfloor \lambda \rfloor}).$$

Indeed, if a nonzero holomorphic germ has order m at the origin, its squared modulus is comparable to $|z|^{2m}$ near the origin. In polar coordinates, its weighted integral over a disk of sufficiently small radius $\epsilon > 0$ is finite precisely when

$$\int_0^{\epsilon} r^{2m-2\lambda+1}\,\mathrm{d}r < \infty, \qquad \text{equivalently} \qquad m > \lambda - 1.$$

The smallest nonnegative integer satisfying this strict inequality is $\lfloor \lambda \rfloor$.

The example also shows that a single multiplier ideal does not determine the singularity: different coefficients λ can have the same integer part. For a real weight $k > 0$, the same calculation applied to ku gives $\mathcal{I}(ku)_0 = (z^{\lfloor k\lambda \rfloor})$. The ideals at all weights $k > 0$ recover λ: if $0 \le \lambda < \mu$, the integer parts of $k\lambda$ and $k\mu$ differ as soon as $k(\mu - \lambda) > 1$.

We therefore compare multiplier ideals at all positive weights. For quasi-plurisubharmonic functions u, v, we write $u \preceq_{\mathcal{I}} v$ if $\mathcal{I}(ku) \subseteq \mathcal{I}(kv)$ for every real $k > 0$, and $u \sim_{\mathcal{I}} v$ if $\mathcal{I}(ku) = \mathcal{I}(kv)$ for every real $k > 0$. We call $\preceq_{\mathcal{I}}$ the *ideal order*; if $u \sim_{\mathcal{I}} v$, then u and v have the same *multiplier-ideal type*, or *ideal type*. By Proposition 1.12.2, $u \preceq v$ implies $u \preceq_{\mathcal{I}} v$, and $u \sim v$ implies $u \sim_{\mathcal{I}} v$. These relations make sense on any complex manifold. Their relation to envelopes and to global masses requires further arguments; see Chapter 17.

1.13 Analytic types and restriction

The simplest singularities of quasi-plurisubharmonic functions are logarithmic poles along the common zeros of finitely many holomorphic functions. The logarithmic growth of a general singularity at a point is measured by its Lelong number. Let u be a quasi-plurisubharmonic function on a complex manifold and let x be a point. Choose holomorphic coordinates centered at x and a smooth function q such that $p = u + q$ is plurisubharmonic near x, and write $B(x, r)$ for the Euclidean coordinate ball of radius r. The *Lelong number* of u at x is

$$\nu(u, x) = \lim_{r \downarrow 0} \frac{\sup_{B(x,r)} p}{\log r^2}.$$

By the local growth theorem recalled in [Xia26], the limit exists and does not depend on the coordinates or on the correction q. With this normalization, $\log |z|^2$ has Lelong number one at the origin. For an irreducible analytic subset Z, the *generic Lelong number* of u along Z is

$$\nu(u, Z) = \inf_{x \in Z} \nu(u, x).$$

By Siu's theorem, recalled in Section 1.13.1, $\nu(u, x) = \nu(u, Z)$ for all $x \in Z$ outside a countable union of proper analytic subsets of Z.

For analytic singularities we follow the convention of [Xia26]: we say that a quasi-plurisubharmonic function u has *analytic singularities* if near every point there

are holomorphic functions $f_1, \dots, f_N$, not all identically zero, and a rational number $c > 0$ such that

$$u = c \log\bigl(|f_1|^2 + \cdots + |f_N|^2\bigr) + O(1), \qquad c \in \mathbb{Q}_{>0}.$$

The remainder is only required to be bounded; if it can be chosen smooth, we say that u has analytic singularities *with smooth remainder*. Given a coherent ideal sheaf $\mathfrak{a}$ and a common coefficient c, we say that u has *analytic type* $(c, \mathfrak{a})$ if this holds with local generators $f_1, \dots, f_N$ of $\mathfrak{a}$. The choice of generators does not matter: two finite systems of generators of the same ideal change the logarithm only by a bounded amount on smaller relatively compact open sets.

Proposition 1.13.1 *Analytic singularities are preserved by addition, finite maxima, and multiplication by positive rational numbers. Let u have analytic type $(c, \mathfrak{a})$, and let $Y \subset X$ be a connected complex submanifold on which $\mathfrak{a}$ does not vanish identically. Then the restriction $u|_Y$ is a quasi-plurisubharmonic function on Y of analytic type*

$$(c, \mathfrak{a}\mathcal{O}_Y).$$

Here $\mathfrak{a}\mathcal{O}_Y$ is the image ideal, generated by the restrictions to Y of local sections of $\mathfrak{a}$.

Proof For multiplication by a positive rational number, multiply the coefficient. Consider two presentations with coefficients c_1, c_2 and ideals $\mathfrak{a}_1, \mathfrak{a}_2$, and choose a positive rational number d with $c_1 = kd$ and $c_2 = \ell d$, where k, ℓ are positive integers. Raising a sum of squared moduli of generators of $\mathfrak{a}_1$ to the power k gives, up to bounded factors, the corresponding sum for $\mathfrak{a}_1^k$; similarly for $\mathfrak{a}_2^\ell$. Products of generators of $\mathfrak{a}_1^k$ and $\mathfrak{a}_2^\ell$ therefore present the sum with coefficient d, since a product of sums of squares is a sum of squares of pairwise products.

For the maximum, apply the elementary inequalities

$$\max\{A, B\} \le A + B \le 2 \max\{A, B\}, \qquad A, B \ge 0,$$

to the two sums of squared moduli of generators of $\mathfrak{a}_1^k$ and $\mathfrak{a}_2^\ell$. Taking logarithms shows that the maximum differs from d times the logarithm of their total by a bounded amount. These are local assertions; on the compact manifold, finitely many charts give the corresponding comparisons of ordinary singularity types.

For the restriction, restrict to Y the two-sided bounds that define the remainder. Since Y is connected and is not contained in the zero set of $\mathfrak{a}$, the common zero set of the restricted generators is a proper analytic subset of Y, so $u|_Y$ is finite on a dense open subset of Y. After a smooth local correction, the restriction theorem for plurisubharmonic functions shows that $u|_Y$ is quasi-plurisubharmonic, and the restricted bounds give the asserted presentation. □

For an arbitrary quasi-plurisubharmonic function, a vanishing generic Lelong number along a submanifold does not prevent the restriction from being identically $-\infty$. For example, on a small polydisk in $\mathbb{C}^n$ with coordinates $z = (z_1, \dots, z_n)$ and $|z_1| < \mathrm{e}^{-1}$, the function

$$u(z) = -\sqrt{-\log |z_1|^2}$$

is plurisubharmonic, because $t \mapsto -\sqrt{-t}$ is increasing and convex for $t < 0$. Its Lelong number vanishes at every point of $\{z_1 = 0\}$, but its restriction to this hyperplane is identically $-\infty$. For the analytic approximants constructed below, the analytic hypothesis in Proposition 1.13.1 excludes this behavior.

1.13.1 Lelong level sets

Siu's analyticity theorem states that, for every quasi-plurisubharmonic function u and every $c > 0$, the upper level set $\{x : \nu(u, x) \geq c\}$ is an analytic subset. After a smooth local correction, this is the theorem for closed positive currents, which we use in the form recalled in [Xia26]; see Section 36.1 for the original source.

For a closed positive current, the Lelong monotonicity formula also bounds the trace mass of small balls. The resulting Hausdorff estimate shows that a closed positive (j, j)-current puts no mass on sets of zero Hausdorff $2(n - j)$-measure.

Corollary 1.13.1 *Let T be a closed positive (j, j)-current on a complex manifold of dimension n, with $1 \leq j < n$. Its coefficient measures vanish on every Borel set of zero Hausdorff $2(n - j)$-measure.*

Proof In a coordinate chart put $\kappa = \mathrm{dd}^{\mathrm{c}}|z|^2$ and $\mu = T \wedge \kappa^{n-j}$. By the Lelong monotonicity formula in bidimension $(n - j, n - j)$, the function $r \mapsto r^{-2(n-j)}\mu(B(x, r))$ is nondecreasing; see [Dem12, Chapter III, formulas (5.5)–(5.8), pp. 158–159]. Fix a relatively compact inner chart and a radius $r_0 > 0$ such that all balls of radius r_0 centered in it lie in a fixed compact subset of the chart. Their masses are bounded by the mass of that compact set. Hence

$$\mu(B(x, r)) \leq C r^{2(n-j)}, \qquad 0 < r \leq r_0,$$

uniformly for centers in the inner chart. A Borel set in the inner chart with zero Hausdorff $2(n - j)$-measure can be covered by balls $B(x_i, r_i)$ with $\sum_i r_i^{2(n-j)}$ arbitrarily small, so it has zero μ-mass. Since T is positive, the total variation of each of its coefficient measures is bounded by a constant multiple of μ. A countable cover by charts proves the assertion. □

The non-strict inequality in Siu's theorem cannot be replaced by a strict one, even on a compact Kähler manifold.

Example 1.13.1 Let $X = \mathbb{P}^2$, with homogeneous coordinates $[x : y : z]$, and let σ be the Fubini–Study form normalized by $\int_X \sigma^2 = 1$. Put $\|Z\|^2 = |x|^2 + |y|^2 + |z|^2$, $a_j = 2^{-j}$ and $\lambda_j = 2^{-j-1}$ for $j \geq 1$. The global function

$$u([Z]) = \log \frac{|y|^2}{\|Z\|^2} + \sum_{j \geq 1} \lambda_j \log \frac{|x - a_j z|^2}{(1 + a_j^2)\|Z\|^2}$$

belongs to $\mathrm{PSH}(X, \frac{3}{2}\sigma)$. For the line $D = \{y = 0\}$ and the points $p_j = [a_j : 0 : 1]$, it satisfies

$$\{\nu(u, \cdot) \geq 1\} = D, \qquad \{\nu(u, \cdot) > 1\} = \{p_j : j \geq 1\}.$$

The second set is not analytic.

Proof For a linear form ℓ with coefficient vector of norm one, the function $v_\ell = \log(|\ell(Z)|^2/\|Z\|^2)$ is nonpositive and satisfies $\mathrm{dd^c} v_\ell = [\ell = 0] - \sigma$. Its $L^1(X, \sigma^2)$ norm is finite and independent of ℓ, because the unitary group $U(3)$ acts on homogeneous coordinates, preserves σ, and acts transitively on such forms. Since $\sum_j \lambda_j = 1/2$, the partial sums of the series defining u decrease to an L^1 function. They have the common curvature lower bound $-\frac{3}{2}\sigma$, so their limit is quasi-plurisubharmonic and

$$T := \tfrac{3}{2}\sigma + \mathrm{dd^c} u = [D] + \sum_{j \geq 1} \lambda_j [L_j], \qquad L_j = \{x - a_j z = 0\}.$$

The series of positive closed currents converges, since every line has σ-trace mass one and the coefficients are summable.

We compute the Lelong numbers from the local mass formula. In affine coordinates $w = (w_1, w_2)$ centered at a point q, put $\kappa = \mathrm{dd^c}|w|^2$. An affine complex line H at distance d from q satisfies

$$\int_{H \cap B(q,r)} \kappa = (r^2 - d^2)_+.$$

Consequently $r^{-2} \int_{H \cap B(q,r)} \kappa$ is at most one and tends to $\mathbf{1}_H(q)$ as $r \downarrow 0$; a line that does not meet the chart contributes zero. Tonelli's theorem and dominated convergence for the summable coefficients therefore give

$$\nu(u, q) = \nu(T, q) = \mathbf{1}_D(q) + \sum_{j \geq 1} \lambda_j \mathbf{1}_{L_j}(q).$$

Here we used that a smooth local correction does not change the Lelong number. The computation also covers points at which infinitely many lines accumulate.

The lines L_j meet D at the distinct points p_j and meet one another only at $p_\infty = [0 : 1 : 0] \notin D$. Thus $\nu(u, p_j) = 1 + \lambda_j$, whereas $\nu(u, q) = 1$ for $q \in D \setminus \{p_j : j \geq 1\}$. Outside D, the Lelong number is at most $\sum_j \lambda_j = 1/2$. This proves the two level-set identities. Finally, $p_j \to p_* = [0 : 0 : 1]$ and $\nu(u, p_*) = 1$, so the strict level set is not closed and hence not analytic. □

1.13.2 Sobolev regularity at divisorial poles

For a potential that is a convex function of $\log|f|$, with f holomorphic, the growth of the convex function at $-\infty$ determines which derivatives are integrable near the

divisor of f. The gradient and the classical Laplacian have different thresholds. Moreover, the distributional Laplacian keeps a measure on the divisor unless the slope of the convex function tends to zero.

Let $U \subset \mathbb{C}^n$ be a sufficiently small neighborhood of 0, and let $f \not\equiv 0$ be a holomorphic function on U with $f(0) = 0$. Write

$$f = g \prod_{j=1}^{N} h_j^{m_j}, \qquad m_0 = \min_j m_j,$$

where g is a holomorphic unit and the h_j are distinct irreducible germs represented on U. Fix a convex nondecreasing function $\chi \in C^2((-\infty, A])$ and shrink U so that $\log|f| < A - 1$. Set $v = \chi(\log|f|)$ off $Z = \{f = 0\}$; this function is plurisubharmonic and bounded above, and we extend it across Z by Lemma 1.3.7. In this subsection, gradients, Laplacians, Sobolev spaces and the volume form $\mathrm{d}V$ are the real Euclidean ones. Write $|\partial f|^2 = \sum_j |\partial f/\partial z_j|^2$, so that $|\nabla |f|| = |\partial f|$ off Z.

Lemma 1.13.1 *After shrinking U, there is a constant $C > 0$ such that*

$$|\partial f| \le C|f|^{1-1/m_0} \quad \text{on } U \setminus Z.$$

For each compact set $K \subset U$, there are $C_K, r_K > 0$ with

$$\int_{K \cap \{|f| = r\}} |\partial f| \, \mathrm{d}S \le C_K r \qquad (0 < r < r_K),$$

where $\mathrm{d}S$ is the Euclidean surface measure on the real hypersurface $\{|f| = r\}$. In particular, for every nonnegative measurable function Ψ,

$$\int_{K \cap \{0 < |f| < r_K\}} \Psi(|f|) |\partial f|^2 \, \mathrm{d}V \le C_K \int_0^{r_K} \Psi(r) r \, \mathrm{d}r.$$

Proof Shrink U so that $|h_j| \le 1$ and the unit g, its inverse and the first derivatives of g and of all h_j are bounded. For each j, $|f|^{1/m_0} \le C|h_j|^{m_j/m_0} \le C|h_j|$. Logarithmic differentiation of the factorization therefore gives

$$\frac{|\partial f|}{|f|} \le C\left(1 + \sum_j |h_j|^{-1}\right) \le C|f|^{-1/m_0},$$

which proves the first bound.

For small $\zeta \in \mathbb{C}$, let μ_ζ be the Euclidean volume measure of the fiber divisor $\{f = \zeta\}$, counted with multiplicity. As $\zeta \to 0$, we have $\log|f - \zeta| \to \log|f|$ in L^1_{loc}: by the local compactness theorem recalled at the beginning of Section 1.4, it suffices to identify subsequential limits, and the convergence is locally uniform off Z, which identifies every such limit and excludes convergence to $-\infty$. The Poincaré–Lelong formula therefore gives weak convergence of the fiber currents, and hence of their

Euclidean trace measures. Pairing with a fixed nonnegative cutoff function equal to one near K shows that $\sup_{|\zeta|<r_K} \mu_\zeta(K) < \infty$ for some $r_K > 0$.

The critical locus of f has finitely many local components, and f is constant on each of them. After shrinking U, every remaining critical component passes through 0, where f vanishes. Thus the small nonzero fibers are smooth. On $\{|f| = r\}$, the coarea formula for the map f to the circle of radius r gives

$$\int_{K\cap\{|f|=r\}} |\partial f| \, \mathrm{d}S = \int_{|\zeta|=r} \mu_\zeta(K) \, \mathrm{d}S(\zeta) \le C_K r.$$

The coarea formula for the function $|f|$, together with $|\nabla|f|| = |\partial f|$, proves the last inequality. The same formulas hold in dimension one, where the fiber volume is the counting measure. □

Proposition 1.13.2 *For each* $2 \le p < \infty$,

$$v \in W^{1,p}_{\mathrm{loc}}(U) \quad \Longleftrightarrow \quad \int_{-\infty}^{A} |\chi'(s)|^p \mathrm{e}^{(2-p)s/m_0} \, \mathrm{d}s < \infty.$$

In this case, the weak gradient of v *is the zero extension of* $\chi'(\log|f|)\nabla \log|f|$ *from* $U \setminus Z$.

Proof At a smooth point of a component of multiplicity m_0, outside the other components, multiplying the defining function of the component by a holomorphic m_0-th root of the remaining unit gives a normal coordinate w_1 with $f = w_1^{m_0}$. Coordinate changes with bounded derivatives and Jacobians preserve local first-order Sobolev integrability. With $r = |w_1|$ and $s = m_0 \log r$, polar integration of the classical gradient gives, up to fixed positive factors,

$$\int_0^{r_0} |\chi'(m_0 \log r)|^p r^{1-p} \, \mathrm{d}r = \frac{1}{m_0} \int_{-\infty}^{m_0 \log r_0} |\chi'(s)|^p \mathrm{e}^{(2-p)s/m_0} \, \mathrm{d}s.$$

Such smooth points occur in every neighborhood of 0. This proves necessity, since changing the finite upper endpoint does not affect convergence.

For sufficiency, put

$$s_\varepsilon = \tfrac{1}{2}\log(|f|^2 + \varepsilon^2), \qquad v_\varepsilon = \chi(s_\varepsilon).$$

For small $\varepsilon > 0$, these are C^2 plurisubharmonic functions decreasing to v. By convexity, $|\chi(s)| \le C(1 + |s|)$ for $s \le A$, so the local integrability of $|f|^{-c}$ for small $c > 0$, a case of Skoda's lemma [Dem11, Lemma 5.6], gives convergence in every local L^p space with $p < \infty$. By the chain rule,

$$|\nabla v_\varepsilon| = \chi'(s_\varepsilon) \frac{|f|}{|f|^2 + \varepsilon^2} |\partial f|.$$

Apply Lemma 1.13.1 with the coarea formula, keeping two powers of $|\partial f|$ for the level estimate and bounding the other $p-2$ powers by the first inequality of the lemma. The local integral of $|\nabla v_\varepsilon|^p$ over $K \cap \{0 < |f| < r_K\}$ is then bounded by

$$C \int_0^{r_K} |\chi'(\tfrac{1}{2}\log(r^2+\varepsilon^2))|^p \, \frac{r^{p+(p-2)(1-1/m_0)+1}}{(r^2+\varepsilon^2)^p} \, \mathrm{d}r.$$

With $s = \frac{1}{2}\log(r^2+\varepsilon^2)$, we have $\mathrm{d}s = r\,\mathrm{d}r/(r^2+\varepsilon^2)$ and $r \le (r^2+\varepsilon^2)^{1/2}$, so this is at most

$$C \int_{-\infty}^{A} |\chi'(s)|^p \mathrm{e}^{(2-p)s/m_0} \, \mathrm{d}s.$$

The integral over the part where $|f| \ge r_K$ is uniformly bounded as well. Weak compactness in Sobolev spaces and the local L^p convergence therefore show that $v \in W^{1,p}_{\mathrm{loc}}$. Its weak gradient agrees with the classical one off Z, and Z has Lebesgue measure zero, which proves the final assertion. □

Proposition 1.13.3 *Let G be the zero extension of $\chi''(\log|f|)|\nabla \log|f||^2$ from $U \setminus Z$, and put $\ell = \lim_{s\to-\infty} \chi'(s)$. For each $1 \le p < \infty$,*

$$G \in L^p_{\mathrm{loc}}(U) \quad \Longleftrightarrow \quad \int_{-\infty}^{A} |\chi''(s)|^p \mathrm{e}^{2(1-p)s/m_0} \, \mathrm{d}s < \infty.$$

In the sense of distributions,

$$\Delta v = G + \ell \, \Delta \log|f|,$$

where $\Delta\log|f|$ is a positive measure carried by the divisor of f. In particular, Δv is represented by a locally L^p function exactly when $\ell = 0$ and the integral above is finite.

Proof The limit $\ell \ge 0$ exists because χ' is nonnegative and nondecreasing. Near a smooth point of a component of multiplicity m_0, outside the other components, the coordinate reduction to $f = w_1^{m_0}$ makes the positive complex Hessian of v off Z a single factor in the normal direction. Its Euclidean traces in the two coordinate systems are uniformly comparable, so polar integration gives necessity. In general, $G = \chi''(\log|f|)|\partial f|^2/|f|^2$ off Z. The coarea formula and Lemma 1.13.1, now bounding the remaining $2p-2$ powers of $|\partial f|$ by the first inequality of the lemma, give

$$\int_{K\cap\{0<|f|<r_K\}} G^p \, \mathrm{d}V \le C \int_0^{r_K} |\chi''(\log r)|^p r^{-1-2(p-1)/m_0} \, \mathrm{d}r.$$

The substitution $s = \log r$ proves sufficiency. For $p = 1$ the integral is always finite, since $\int_{-\infty}^{A} \chi'' = \chi'(A) - \ell$.

For the distributional identity, use s_ε and v_ε from the preceding proof. Direct differentiation gives

$$\Delta v_\varepsilon = \chi''(s_\varepsilon)\frac{|f|^2|\partial f|^2}{(|f|^2+\varepsilon^2)^2} + \chi'(s_\varepsilon)\Delta s_\varepsilon,$$
$$\Delta s_\varepsilon = \frac{2\varepsilon^2|\partial f|^2}{(|f|^2+\varepsilon^2)^2}.$$

Off Z, the first term tends locally uniformly to G. By the level estimate of Lemma 1.13.1 and the same substitution, its integral over $K \cap \{|f| < r_0\}$ is at most

$$C\int_{-\infty}^{\frac{1}{2}\log(r_0^2+\varepsilon^2)} \chi''(s)\,\mathrm{d}s.$$

This bound tends to zero if we first let $\varepsilon \to 0$ and then $r_0 \to 0$, so the first term converges to G in L^1_{loc}.

The positive measures Δs_ε converge weakly to $\Delta \log|f|$, with locally bounded masses, and the Poincaré–Lelong formula shows that this limit is a measure supported on Z. On $\{|f| < r_0\}$,

$$0 \le \chi'(s_\varepsilon) - \ell \le \chi'(\tfrac{1}{2}\log(r_0^2+\varepsilon^2)) - \ell,$$

while $\Delta s_\varepsilon \to 0$ uniformly on compact subsets of $\{|f| \ge r_0\}$. Taking the same successive limits shows that $(\chi'(s_\varepsilon) - \ell)\Delta s_\varepsilon \to 0$ weakly. Since $v_\varepsilon \to v$ in L^1_{loc}, passing to the limit in the formula for Δv_ε gives the identity. If $\ell > 0$, the measure $\ell\,\Delta\log|f|$ is nonzero and singular with respect to Lebesgue measure; uniqueness of the Lebesgue decomposition therefore proves the final criterion. □

Example 1.13.2 Shrink U so that $\log|f| < -2$, and let $0 < a < 1$. Then the plurisubharmonic function

$$v_a = -(-\log|f|)^a$$

has $\Delta v_a \in L^1_{\mathrm{loc}}$, while

$$v_a \in W^{1,2}_{\mathrm{loc}} \quad\Longleftrightarrow\quad a < \tfrac{1}{2}.$$

It belongs to no $W^{1,p}_{\mathrm{loc}}$ with $p > 2$ and to no $W^{2,1}_{\mathrm{loc}}$.

Proof Here $\chi(s) = -(-s)^a$ is of class C^2 on $(-\infty,-1]$, so Propositions 1.13.2 and 1.13.3 apply with $A = -1$. Moreover, $\chi'(s) = a(-s)^{a-1}$ and $\chi''(s) = a(1-a)(-s)^{a-2}$, so $\ell = 0$. The two propositions give the assertion about the Laplacian and the first-order thresholds. At $a = 1/2$, the first-order integral diverges logarithmically; for $p > 2$, its exponential factor forces divergence.

For second derivatives, take a smooth point of a component of multiplicity m, outside the other components, with a normal coordinate w_1 such that $f = w_1^m$. Write $F(r) = \chi(m\log r)$ with $r = |w_1|$. The real second derivative of v_a in the unit direction tangent to the circles $|w_1| = r$ equals $F'(r)/r = m\chi'(m\log r)/r^2$. Thus the integral

of its absolute value over the normal disk is bounded below, up to a fixed positive factor, by

$$\int_0^{r_0} \chi'(m\log r)\frac{\mathrm{d}r}{r} = \frac{1}{m}\int_{-\infty}^{m\log r_0} \chi'(s)\,\mathrm{d}s = +\infty.$$

Hence the classical Hessian of v_a off the divisor is not locally integrable, whereas local integrability is necessary for $W^{2,1}_{\mathrm{loc}}$. Since smooth coordinate changes preserve this Sobolev class, the argument is complete. □

1.14 The local Bergman estimates

Demailly's approximation of plurisubharmonic functions by functions with analytic singularities is built from Bergman kernels of weighted spaces of holomorphic functions. In this section we recall the local construction and its estimates. Fix a bounded pseudoconvex domain $\Omega \subset \mathbb{C}^n$ and a plurisubharmonic function ψ on Ω. Write $\mathrm{d}\lambda$ for Euclidean volume and $B(x,r)$ for the Euclidean ball of radius $r>0$ about x. For a positive integer m, define the weighted Hilbert space of holomorphic functions

$$\mathcal{H}_m = \left\{ f \in \mathcal{O}(\Omega) : \|f\|_m^2 := \int_\Omega |f|^2 \mathrm{e}^{-m\psi}\,\mathrm{d}\lambda < \infty \right\}.$$

Choose an orthonormal basis $(g_{m,j})_j$ of $\mathcal{H}_m$ and define $\beta_m = m^{-1}\log\sum_j |g_{m,j}|^2$. The sum is the squared norm of point evaluation,

$$\sum_j |g_{m,j}(x)|^2 = \sup_{\|f\|_m \le 1} |f(x)|^2,$$

and is therefore independent of the basis. By the local finite generation part of Bergman regularization, β_m has analytic type $(1/m, \mathcal{I}(m\psi))$ on relatively compact open subsets of Ω. This follows from the generation of the multiplier ideal by weighted L^2 holomorphic functions and from local L^2 division; see [Dem11, Proposition 5.7 and Section 14.A]. We use this local statement, together with the point-extension estimate in the proof below, as a standard input.

Proposition 1.14.1 *There are constants C_1, C_2 independent of m such that, for $B(x,r) \Subset \Omega$,*

$$\psi(x) - \frac{C_1}{m} \le \beta_m(x) \le \sup_{B(x,r)} \psi + \frac{1}{m}\log\frac{C_2}{r^{2n}}. \tag{1.20}$$

Consequently $\beta_m \to \psi$ pointwise and in L^1_{loc}, and

$$\max\{\nu(\psi,x) - n/m, 0\} \le \nu(\beta_m, x) \le \nu(\psi,x). \tag{1.21}$$

Proof For $\|f\|_m \le 1$, the submean inequality gives

$$|f(x)|^2 \le C_2 r^{-2n} \int_{B(x,r)} |f|^2 \, d\lambda \le C_2 r^{-2n} e^{m \sup_{B(x,r)} \psi}.$$

Taking the supremum over the unit ball of $\mathcal{H}_m$ and then the logarithm proves the upper bound.

For the lower bound, suppose that $\psi(x) > -\infty$. The local Ohsawa–Takegoshi point-extension theorem gives a holomorphic f with $f(x) = 1$ and

$$\int_\Omega |f|^2 e^{-m\psi} \, d\lambda \le C e^{-m\psi(x)},$$

where the constant is independent of the plurisubharmonic weight and of m. After normalizing f in $\mathcal{H}_m$, we see that the squared norm of point evaluation is at least $C^{-1} e^{m\psi(x)}$, which proves the lower bound. At a pole of ψ the lower bound holds trivially.

Choosing $r = 1/m$ in the upper bound and using upper semicontinuity of ψ proves pointwise convergence. On a fixed relatively compact set, the same estimates bound β_m above by a constant and below by $\psi - C_1/m$. Since ψ is locally integrable, dominated convergence also gives convergence in L^1_{loc}.

For the Lelong estimate, take suprema over $B(x,r)$ in the two inequalities, bounding the right-hand side of the upper one by a supremum over $B(x,2r)$. Divide by $\log r^2 < 0$ and let $r \downarrow 0$. After the division, the term $-(2n/m)\log r$ contributes $-n/m$, while the constant terms disappear. This proves the two inequalities in (1.21); since Lelong numbers are nonnegative, the lower bound may be replaced by its maximum with zero. □

In [Dem11], the local weight is $e^{-2m\phi}$ and Lelong numbers are normalized by $\log|z|$. With $\psi = 2\phi$, these conventions give exactly the formulas of Proposition 1.14.1; in particular, the loss is n/m for the weight $e^{-m\psi}$ and the normalization by $\log|z|^2$.

The same construction and estimates hold for every real $m > 0$. The coefficient $1/m$ of the resulting logarithmic presentation is rational exactly when m is rational, and we speak of an analytic type only in that case. The global sequence constructed below is indexed by integers.

For a quasi-plurisubharmonic function u, the generators of $\mathcal{I}(mu)$ also control the multiplier ideals of u at nearby weights; a small loss in the weight compensates for the finite Bergman index.

Lemma 1.14.1 *Let u be quasi-plurisubharmonic, let $m > 0$, and let a_m be a function which locally satisfies $a_m = m^{-1} \log \sum_k |g_k|^2 + O(1)$, where the g_k generate $\mathcal{I}(mu)$. For $t > 0$ and $s > 1$,*

$$\mathcal{I}(s t a_m) \subseteq \mathcal{I}(tu) \qquad \text{whenever} \qquad m \ge \frac{st}{s-1}.$$

Moreover, $u \preceq a_m$ locally, and hence $\mathcal{I}(tu) \subseteq \mathcal{I}(t a_m)$ for every $t > 0$.

Proof On a sufficiently small coordinate neighborhood, choose generators $g_1, \dots, g_N$ of $\mathcal{I}(mu)$ and put $B = (\sum_k |g_k|^2) e^{-mu}$. After shrinking the neighborhood, B is

integrable and $a_m = m^{-1}\log\sum_k |g_k|^2 + O(1)$ with a bounded remainder. If $f \in \mathcal{I}(sta_m)$, shrink once more so that f is bounded and $A = |f|^2 e^{-sta_m}$ is integrable. Then, almost everywhere,

$$|f|^2 e^{-tu} \le C|f|^{2-2/s} A^{1/s} B^{t/m}.$$

Hölder's inequality with exponents s and $s/(s-1)$ applies because $st/(m(s-1)) \le 1$, and it shows that the left side is locally integrable, also in the case of equality in the bound for m. Finally, the local Bergman lower estimate of Proposition 1.14.1 gives $u \preceq a_m$ locally; bounded smooth corrections affect neither this comparison nor any of the ideals. □

1.15 The global curvature bound

We return to the compact manifold X with its Hermitian form ω. Demailly's global regularization theorem glues the local Bergman constructions while controlling the curvature error. We use it as a classical input in the following form.

Theorem 1.15.1 (Demailly regularization) *Let u be quasi-plurisubharmonic on X and let γ be a continuous real $(1,1)$-form with $\mathrm{dd^c}u \ge \gamma$. There are quasi-plurisubharmonic functions u_m, defined for all sufficiently large positive integers m, and numbers $\epsilon_m \downarrow 0$ such that*

(1) *u_m has analytic type $(1/m, \mathcal{I}(mu))$;*
(2) *$u_m \ge u$ and $u_m \to u$ pointwise and in $L^1(X)$;*
(3) *$\mathrm{dd^c}u_m \ge \gamma - \epsilon_m\omega$;*
(4) *$\max\{\nu(u,x) - n/m, 0\} \le \nu(u_m,x) \le \nu(u,x)$ for all x.*

The sequence (u_m) need not be monotone.

This is the form of [Dem11, Theorem 14.12] used in this book. The local generators in the construction are the weighted Bergman generators of $\mathcal{I}(mu)$; smooth local corrections do not change this ideal. On overlaps the local Bergman sums are comparable, so the glued logarithm keeps the analytic type $(1/m, \mathcal{I}(mu))$. The construction freezes the continuous lower bound γ on sufficiently small coordinate balls, and its curvature error combines the modulus of continuity of γ with an overlap term that tends to zero as m increases. It is this global curvature estimate, and not merely the local convergence of Proposition 1.14.1, that we take from the theorem.

Corollary 1.15.1 *Let $u \in \mathrm{PSH}(X,\theta)$. The approximants u_m of Theorem 1.15.1 for $\gamma = -\theta$ have analytic type $(1/m, \mathcal{I}(mu))$ and satisfy*

$$\theta + \mathrm{dd^c}u_m \ge -\epsilon_m\omega.$$

Proof Apply Theorem 1.15.1 to u with the lower bound $\gamma = -\theta$; the theorem does not require γ to be closed. Adding θ to the curvature inequality gives the assertion.

In the normalization of [Dem11], the theorem is applied to the potential $u/2$, since $\frac{\mathrm{i}}{\pi}\partial\bar\partial(u/2) = \mathrm{dd}^{\mathrm{c}}u$. □

The multiplier ideals of these approximants determine the limit of their ordinary envelopes; see Theorem 17.5.1.

1.15.1 Regularization of closed currents

We also use Demailly's regularization of closed currents, in a smooth and in an analytic form; the sources are given in Section 36.1. Let $T = \eta + \mathrm{dd}^{\mathrm{c}}v$ be a closed real $(1,1)$-current, where η is a smooth closed form and v is quasi-plurisubharmonic, and set $\nu(T,x) = \nu(v,x)$. Suppose that $T \geq \gamma$ for a smooth real $(1,1)$-form γ. Smooth regularization gives smooth representatives $T_j = \eta + \mathrm{dd}^{\mathrm{c}}v_j$ with

$$T_j \geq \gamma - C\lambda_j\omega,$$

where $C > 0$ is a fixed constant and the continuous functions λ_j decrease pointwise to $\nu(T,\cdot)$; the additional loss $\varepsilon_j\omega$ with constants $\varepsilon_j \downarrow 0$ in Demailly's statement has been absorbed into λ_j. If all Lelong numbers of T vanish, Dini's theorem shows that $\lambda_j \to 0$ uniformly, so the loss of positivity becomes uniformly small. Analytic regularization [Dem11, Corollary 14.13] instead gives representatives S_j with analytic singularities and

$$S_j \geq \gamma - \varepsilon_j\omega, \qquad \varepsilon_j \downarrow 0, \qquad \nu(S_j,x) \leq \nu(T,x).$$

Thus analytic regularization preserves strict positivity up to a small loss and creates no positive Lelong number at a point where the Lelong number of T vanishes.

1.15.2 Smooth remainders after resolution

For arguments on a modification, the bounded remainder in an analytic presentation is sometimes insufficient. The following version of regularization gives a smooth remainder after resolution.

Theorem 1.15.2 (Demailly regularization) *Let α be a smooth real $(1,1)$-form and let $u \in \mathrm{PSH}(X,\alpha)$. There are numbers $\delta_j \downarrow 0$ and functions $\widetilde{u}_j \in \mathrm{PSH}(X, \alpha + \delta_j\omega)$ decreasing to u with the following properties. Each $\widetilde{u}_j$ is smooth outside an analytic pole set, and there is a modification $\pi_j : Y_j \to X$, obtained by a finite sequence of blowups with smooth centers, such that locally on Y_j*

$$\pi_j^*\widetilde{u}_j = \sum_\ell a_\ell \log|s_\ell|^2 + b, \qquad a_\ell > 0, \quad b \in C^\infty.$$

Here the s_ℓ are local holomorphic defining functions of divisors, which may be taken to have simple normal crossings.

This classical theorem is applied to the closed current $\mathrm{dd^c}u$ with the smooth lower bound $-\alpha$; see Section 36.1 for its source. The lower bound need not be closed, and X need not be Kähler. The sequence $(\widetilde{u}_j)$ need not be the multiplier-ideal sequence of Theorem 1.15.1. If $\theta + \mathrm{dd^c}u \geq \varepsilon\omega$ for some $\varepsilon > 0$, applying the theorem with $\alpha = \theta - \varepsilon\omega$ and discarding the terms with $\delta_j \geq \varepsilon$ gives approximants with $\theta + \mathrm{dd^c}\widetilde{u}_j \geq (\varepsilon - \delta_j)\omega > 0$; thus strict positivity is preserved.

On such a modification, a Hermitian form pulled back from X becomes positive after a small exceptional correction. For an effective real divisor $D = \sum_j a_j D_j$ on a compact complex manifold, with prime divisors D_j and $a_j > 0$, choose a smooth Hermitian metric on each line bundle $\mathcal{O}(D_j)$, with canonical section s_j and Chern curvature form η_j, and put

$$\sigma_D = \sum_j a_j \log|s_j|^2, \qquad \eta_D = \sum_j a_j\eta_j.$$

The form η_D is closed, and the Poincaré–Lelong formula reads $\mathrm{dd^c}\sigma_D = [D] - \eta_D$.

Lemma 1.15.1 *Let $p : Z \to X$ be a finite composition of blowups along smooth centers of codimension at least two, and let h_X be a Hermitian form on X. There are a constant $C_0 > 0$, an effective real divisor D on Z whose components are exceptional for p, and a curvature form η_D as above such that*

$$h_Z = C_0 p^* h_X - \eta_D > 0. \tag{1.22}$$

Every component of D is mapped by p into a set of complex codimension at least two in X.

Proof First let $q : M' \to M$ be one blowup along a smooth center, with exceptional divisor E, and let h_M be a Hermitian form on M. The exceptional divisor E is the projectivized normal bundle of the center, and $\mathcal{O}(-E)|_E$ is the dual of its tautological line subbundle; on each projective fiber of E it restricts to $\mathcal{O}(1)$. A Hermitian metric on the normal bundle of the center therefore induces a metric on $\mathcal{O}(-E)|_E$ whose curvature restricts to a Fubini–Study form on each fiber, and so is positive on the tangent spaces of the fibers. Extend this metric to $\mathcal{O}(-E)$ on M' by extending the smooth logarithm of its ratio to a fixed reference metric from the closed submanifold

E, and let β be its curvature. The semipositive form q^*h_M is positive off E, and at points of E its kernel is exactly the tangent space of the fiber, on which β is positive. Hence $Cq^*h_M + \beta > 0$ for every sufficiently large C: otherwise, unit tangent vectors violating the inequality for $C \to \infty$ would converge, by compactness, to a vector in the kernel of q^*h_M on which β is nonpositive. With $\eta_E = -\beta$, the curvature of the induced metric on $\mathcal{O}(E)$, this reads $Cq^*h_M - \eta_E > 0$.

For a composition of blowups, apply this step successively, with h_M the positive form already obtained on the previous blowup. Each step multiplies the previous divisor by a positive constant, pulls it back and adds the new exceptional divisor, so the divisor stays effective and exceptional. This proves (1.22). Each exceptional divisor maps into the image of its center, which has dimension at most $n - 2$. □

1.16 Multiplier ideals and logarithmic poles

For a logarithmic pole along a submanifold, weighted integrability of a holomorphic germ is determined by its vanishing order in the normal directions. For general singularities, the valuative characterization of multiplier ideals and the strong openness theorem, both recalled in [Xia26], make the comparison of multiplier ideals compatible with maxima, sums and increasing limits.

In local coordinates $(z, w) \in \mathbb{C}^d \times \mathbb{C}^k$, with $d \geq 0$ and $k \geq 1$, let $Y = \{w = 0\}$ and denote its ideal by $\mathcal{I}_Y$.

Lemma 1.16.1 *If a quasi-plurisubharmonic function has the form $u = c \log |w|^2 + O(1)$ with $c > 0$ and locally bounded remainder, then for every $t > 0$,*

$$\mathcal{I}(tu) = \mathcal{I}_Y^{\max\{\lfloor tc \rfloor - k + 1, 0\}}.$$

The zeroth power is the full structure sheaf.

Proof The bounded remainder changes the weight only by bounded positive factors. Expand a nonzero holomorphic germ $g(z, w)$ in homogeneous polynomials in w, and let a be the smallest degree whose term does not vanish identically. On smaller product neighborhoods, $|g(z, w)| \leq C|w|^a$. Hence $a + k > tc$ suffices for weighted integrability, since the corresponding radial integral in $\rho = |w|$ is

$$\int_0^{\varepsilon} \rho^{2a - 2tc + 2k - 1} \, \mathrm{d}\rho < +\infty.$$

Conversely, let z be a point near the origin at which the term of degree a does not vanish identically in w. Integration of $|g(z, w)|^2$ over the spheres $|w| = \rho$ gives a positive leading term of order ρ^{2a}, so the radial integral in w diverges when $a + k \leq tc$, logarithmically in the case of equality. Such points z form a nonempty open set in every sufficiently small neighborhood, and Fubini's theorem shows that g is not weighted-integrable near the origin. For $d = 0$ this is the radial calculation at a point.

Thus g is weighted-integrable exactly when $a \geq \lfloor tc \rfloor - k + 1$, that is, when g lies in the power of $\mathcal{I}_Y$ given in the statement. □

Cutoff functions allow us to insert small isolated logarithmic poles into any Hermitian background.

Lemma 1.16.2 *Let ω be a Hermitian form on X, and fix distinct points $p_1, \ldots, p_N$ of X. There are constants $C_j > 0$ such that, whenever $a_j > 0$ and $\sum_j C_j a_j < 1$, there is $u \in \mathrm{PSH}(X, \omega)$, smooth off these points, with*

$$u = a_j \log |z|^2 + O(1) \quad \text{near } p_j, \qquad \omega + \mathrm{dd}^{\mathrm{c}} u \geq \left(1 - \sum_j C_j a_j\right) \omega.$$

Here z denotes local coordinates centered at p_j. The function can be chosen to vanish outside arbitrarily small disjoint coordinate neighborhoods, with the constants depending on those neighborhoods.

Proof Choose disjoint coordinate balls centered at the points p_j and smooth cutoffs χ_j supported in them and equal to one near their centers. The function $\chi_j \log |z|^2$, extended by zero, is plurisubharmonic near its pole and smooth elsewhere, so its complex Hessian is bounded below by $-C_j \omega$. Set $u = \sum_j a_j \chi_j \log |z|^2$. These lower bounds give the asserted current inequality, and the remaining properties follow from the disjointness of the supports. □

Theorem 1.16.1 *Let u, v be plurisubharmonic functions near a point of $\mathbb{C}^n$. By a* local modification *we mean a modification π, with smooth source, of a neighborhood of this point.*

(1) *The inclusions $\mathcal{I}(ku) \subseteq \mathcal{I}(kv)$ for every real $k > 0$ are equivalent to the inequalities $\nu(u \circ \pi, y) \geq \nu(v \circ \pi, y)$ at every point y of the source of every local modification π. Likewise, equality of the multiplier ideals at every positive weight is equivalent to equality of all these Lelong numbers.*

(2) *If p_j is an increasing sequence of plurisubharmonic functions converging almost everywhere to a plurisubharmonic function p, then, for each point x and each $\lambda > 0$,*

$$\mathcal{I}(\lambda p_j)_x = \mathcal{I}(\lambda p)_x \quad \text{for all sufficiently large } j.$$

In the first item, the neighborhoods may be shrunk as necessary; we never need to extend a local modification to a modification of a compact manifold. The second item is the increasing-limit form of the strong openness theorem used in [Xia26]. Since adding a smooth function changes neither multiplier ideals nor Lelong numbers, both items apply to quasi-plurisubharmonic functions after the local correction of Proposition 1.1.1.

Lemma 1.16.3 *Let u, u', v, v' be quasi-plurisubharmonic functions on X. For any quasi-plurisubharmonic h, one has*

$$u + h \sim_{\mathcal{I}} v + h \quad \Longleftrightarrow \quad u \sim_{\mathcal{I}} v.$$

If $u \sim_{\mathcal{I}} v$ and $u' \sim_{\mathcal{I}} v'$, then

$$\max(u,u') \sim_{\mathcal{I}} \max(v,v'), \qquad u+u' \sim_{\mathcal{I}} v+v', \qquad au \sim_{\mathcal{I}} av \quad (a>0).$$

Proof Fix a coordinate neighborhood $U \subset X$ and a modification $\pi\colon Y \to U$ with smooth source. At every $y \in Y$, the elementary rules for the Lelong numbers of maxima, sums and positive multiples give

$$\begin{aligned} \nu(\max(u,u')\circ\pi, y) &= \min\{\nu(u\circ\pi,y), \nu(u'\circ\pi,y)\},\\ \nu((u+u')\circ\pi, y) &= \nu(u\circ\pi,y) + \nu(u'\circ\pi,y),\\ \nu(au\circ\pi, y) &= a\nu(u\circ\pi,y). \end{aligned}$$

These rules are unaffected by smooth local corrections. Since $u \sim_{\mathcal{I}} v$ and $u' \sim_{\mathcal{I}} v'$, the first item of Theorem 1.16.1 shows that the right-hand sides do not change when u, u' are replaced by v, v'. The converse direction of the same item, applied on each coordinate neighborhood, gives the three asserted equivalences.

For the cancellation rule, suppose that $u + h \sim_{\mathcal{I}} v + h$. By the first item of Theorem 1.16.1, the pullbacks $(u+h)\circ\pi$ and $(v+h)\circ\pi$ have the same Lelong numbers. The common summand $h\circ\pi$ is quasi-plurisubharmonic, so its Lelong numbers are finite, and subtracting them shows that $u\circ\pi$ and $v\circ\pi$ have the same Lelong numbers. Hence $u \sim_{\mathcal{I}} v$ by the same item. The reverse implication is the addition rule proved above. □

A quasi-plurisubharmonic function with zero Lelong numbers can still be unbounded, but adding it to another potential changes no multiplier ideal.

Lemma 1.16.4 *Let q be quasi-plurisubharmonic on a complex manifold and suppose that $\nu(q,x) = 0$ at every point. Then e^{-tq} is locally integrable for every $t > 0$. For every quasi-plurisubharmonic v and every $t > 0$,*

$$\mathcal{I}\big(t(v+q)\big) = \mathcal{I}(tv).$$

Proof At each point x, Skoda's theorem [Dem11, Lemma 5.6], applied after a local smooth correction, makes e^{-tq} integrable near x, since $\nu(tq,x) = 0 < 1$. Thus $q \sim_{\mathcal{I}} 0$; the addition rule in Lemma 1.16.3, whose proof is local, gives $v + q \sim_{\mathcal{I}} v$, which is precisely the asserted equality at every weight. □

Let v, w be quasi-plurisubharmonic functions on an open set, and let Y be a complex submanifold on which $v|_Y$ is not identically $-\infty$. The restriction and subadditivity theorems give the two inclusions recalled in [Xia26] and [Dem11, Section 15]:

$$\mathcal{I}(v|_Y) \subseteq \mathcal{I}(v)\mathcal{O}_Y, \qquad \mathcal{I}(v+w) \subseteq \mathcal{I}(v)\mathcal{I}(w). \tag{1.23}$$

The first follows from the local Ohsawa–Takegoshi extension theorem. For the second, the weighted Hilbert space of holomorphic functions on a product is the Hilbert tensor product of the two weighted spaces, so the multiplier ideal of $v(x)+w(y)$ on the product is generated by products of generators of $\mathcal{I}(v)$ and $\mathcal{I}(w)$. The first inclusion, applied to the diagonal, then gives the second. These local results for plurisubharmonic functions extend to quasi-plurisubharmonic functions, in the weight convention of this book, by smooth corrections.

Multiplier ideals commute with pullback by holomorphic submersions. Here the extension $\mathcal{I}\mathcal{O}_Z$ of an ideal sheaf $\mathcal{I}$ on Y is the ideal sheaf on Z generated by the pullbacks of its local sections.

Lemma 1.16.5 *Let $p\colon Z \to Y$ be a holomorphic submersion of complex manifolds and let $u \in \mathrm{QPSH}(Y)$. For every $t > 0$,*

$$\mathcal{I}(tu \circ p) = \mathcal{I}(tu)\mathcal{O}_Z.$$

Proof Work in product coordinates (z,ζ) with $p(z,\zeta) = z$. Write α for a multi-index in the fiber variables and $\mathrm{d}\lambda_\zeta$ for their Euclidean volume. A germ in the ideal on the right is a finite sum of bounded holomorphic multiples of pullbacks of germs that are square integrable against $\mathrm{e}^{-tu(z)}$. Fubini's theorem shows that it is weighted-integrable on a sufficiently small product neighborhood.

Conversely, let $F(z,\zeta)$ be a weighted-integrable holomorphic germ and expand it in the fiber variables,

$$F(z,\zeta) = \sum_{\alpha} f_\alpha(z)\zeta^\alpha.$$

The coefficient estimate for a holomorphic function on a polydisk gives, on smaller product neighborhoods,

$$|f_\alpha(z)|^2 \le C_\alpha \int |F(z,\zeta)|^2 \,\mathrm{d}\lambda_\zeta.$$

Multiplying by $\mathrm{e}^{-tu(z)}$ and integrating in z shows that each coefficient germ f_α belongs to $\mathcal{I}(tu)$. Let J be the extension of this ideal to the local ring R of holomorphic germs on the product. Every finite Taylor sum of F belongs to J, and the remainder of total fiber degree at least N belongs to $(\zeta)^N$. Thus the class of F in R/J lies in $(\zeta)^N(R/J)$ for every N. By the Krull intersection theorem for the finitely generated module R/J over the Noetherian local ring R, the intersection of these submodules is zero. Hence $F \in J$. Smooth corrections and smooth positive volume densities do not change the integrability conditions. □

Example 1.16.1 The identity of Lemma 1.16.5 can fail at a critical point of the map. On a polydisk in $\mathbb{C}^n$, $n \geq 1$, let $f(z_0, \dots, z_{n-1}) = z_0^2$ and put $v(t) = \frac{1}{2}\log|t|^2$. At the origin,

$$\mathcal{I}(v) = \mathcal{O}, \qquad \mathcal{I}(v \circ f) = (z_0) \neq \mathcal{I}(v)\mathcal{O}.$$

Proof On the curve, the coefficient of v at the origin is $1/2$, so Lemma 1.16.1, with $k = 1$, gives the zeroth power of the maximal ideal. Upstairs, $v \circ f = \log|z_0|^2$ has coefficient one along the smooth divisor $\{z_0 = 0\}$, and the same lemma gives the first power of its ideal (z_0). □

This calculation concerns a fixed weight. For maps with connected fibers from a connected compact manifold onto a compact Riemann surface, pullback still preserves and reflects all-weight ideal types (Corollary 1.22.1).

1.16.1 Weighted distance from a sum of multiplier ideals

Membership in a multiplier ideal is a local integrability condition. On a pseudoconvex domain one can ask, more quantitatively, how small the weighted norm of a holomorphic function can be made by subtracting global sections of such ideals. A sharp L^2 estimate answers this question for one weight. Applied successively, with the remaining weights kept in the auxiliary weight at each step, it gives a product bound for several weights.

Throughout this subsection, $\Omega \subset \mathbb{C}^n$ is a pseudoconvex domain, $n \geq 1$, and dV denotes Euclidean volume. For a plurisubharmonic weight ρ on Ω, put

$$\|f\|_\rho^2 = \int_\Omega |f|^2 e^{-\rho}\, dV.$$

For a plurisubharmonic function v on Ω, we write $\Gamma(\Omega, \mathcal{I}(v))$ for the space of holomorphic functions on Ω whose germs belong to $\mathcal{I}(v)$ at every point. Membership requires only local weighted integrability, not a finite global norm.

We first prove a division estimate with two summands. Its sharp constant comes from keeping the full sum of the twisting functions in the $\bar\partial$-estimate.

Lemma 1.16.6 *Let ψ, ϕ_1, ϕ_2 be plurisubharmonic functions on Ω, and suppose that $S = e^{\phi_1} + e^{\phi_2} < 1$. If f is holomorphic and*

$$J = \int_\Omega \frac{|f|^2 e^{-\psi}}{S^2}\, dV < +\infty,$$

then there are holomorphic F_1, F_2 with $F_1 + F_2 = f$ and

$$\int_\Omega \left(|F_1|^2 e^{-\phi_1} + |F_2|^2 e^{-\phi_2}\right) e^{-\psi}\, dV \leq J. \tag{1.24}$$

Proof *Step 1: a twisted $\bar\partial$-estimate.* Suppose first that Ω is bounded, that the weights are smooth on a neighborhood of $\overline{\Omega}$ with $S<1$ there, and that $\Phi=\log S$ is strictly plurisubharmonic. Set $\chi=\phi_1+\phi_2+\psi$ and $A=(\Phi_{j\bar k})$. For a $(0,1)$-form b, write $|b|_A^2$ for its squared norm with respect to the dual of this positive Hermitian matrix. We claim that every $\bar\partial$-closed $(0,1)$-form b for which the right-hand side below is finite admits a solution u of

$$\bar\partial u=b,\qquad \int |u|^2 \mathrm{e}^{\Phi-\chi}\,\mathrm{d}V \le \int (1-\mathrm{e}^{\Phi})|b|_A^2\mathrm{e}^{-\chi}\,\mathrm{d}V. \tag{1.25}$$

To prove this, define for $x<0$

$$T(x)=-\log(1-\mathrm{e}^{x}),\qquad \eta(x)=\frac{-x}{1-\mathrm{e}^{x}}-1,\qquad \lambda(x)=-\frac{\eta'(x)}{T'(x)}.$$

The inequality $-x>1-\mathrm{e}^{x}$ gives $\eta>0$. Also $\eta'=-[1-(1-x)\mathrm{e}^{x}]/(1-\mathrm{e}^{x})^2<0$, whereas $T'>0$, so $\lambda>0$. Direct differentiation gives

$$\eta T'-\eta'=1,\qquad \eta T''-\eta''-\frac{(\eta')^2}{\lambda}=0,\qquad \frac{\mathrm{e}^{-T}}{\eta+\lambda}=\mathrm{e}^{x}.$$

Use the weight $h=\mathrm{e}^{-\chi-T(\Phi)}$ on the trivial line bundle and evaluate the twisting functions at Φ. The twisted curvature is

$$\begin{aligned} B_0 &= \eta\,\mathrm{i}\partial\bar\partial(\chi+T(\Phi))-\mathrm{i}\partial\bar\partial\eta-\mathrm{i}\lambda^{-1}\partial\eta\wedge\bar\partial\eta\\ &= \eta\,\mathrm{i}\partial\bar\partial\chi+\mathrm{i}\partial\bar\partial\Phi\ge \mathrm{i}\partial\bar\partial\Phi. \end{aligned}$$

Let Λ denote contraction by the Euclidean Kähler form and $\bar\partial_h^{*}$ the weighted adjoint. For a compactly supported $(n,1)$-form v, the twisted Bochner identity and the weighted Cauchy–Schwarz inequality give

$$\langle [B_0,\Lambda]v,v\rangle_h \le \|\sqrt{\eta+\lambda}\,\bar\partial_h^{*}v\|_h^2+\|\sqrt{\eta}\,\bar\partial v\|_h^2. \tag{1.26}$$

More explicitly, the cross term is bounded by $\lambda|\bar\partial_h^{*}v|^2+\lambda^{-1}|(\bar\partial\eta)^{*}v|^2$; the latter term is exactly the commutator contribution of $\mathrm{i}\lambda^{-1}\partial\eta\wedge\bar\partial\eta$. The terms of type $(n+1,1)$ vanish, and the remaining connection norm has a favorable sign.

The Hilbert space existence argument applied to (1.26) therefore gives

$$\int \frac{|u|^2 h}{\eta+\lambda}\,\mathrm{d}V\le\int |b|_A^2 h\,\mathrm{d}V.$$

To see the coefficient, project a test form to $\ker\bar\partial$. On this kernel the second term of (1.26) vanishes, and the Cauchy–Schwarz inequality in the curvature norm bounds the functional $v\mapsto\langle b,v\rangle_h$ by $(\int|b|_A^2h)^{1/2}\|\sqrt{\eta+\lambda}\,\bar\partial_h^{*}v\|_h$. The Riesz representation theorem gives a solution with the asserted norm. The twisting functions are bounded on Ω, since Φ is smooth and bounded away from both $-\infty$ and 0. The argument first uses a complete Kähler metric; approximating the Euclidean metric by the Euclidean

metric plus small positive multiples of a complete metric gives the Euclidean statement. Under the identification with canonical-bundle-valued forms, the $(n,0)$ norm integral and the curvature-dual $(n,1)$ integral have the same coordinate expressions as above. Since $h = \mathrm{e}^{-\chi}(1-\mathrm{e}^{\Phi})$, the identities for T, η, λ now give (1.25).

Step 2: division for smooth weights. Put $v_i = \mathrm{e}^{\phi_i}/S$ and $\ell = (v_1 f, v_2 f)$. Equip $\mathbb{C}^2$ with the diagonal metric $(\mathrm{e}^{-\phi_1}, \mathrm{e}^{-\phi_2})$. Then ℓ is orthogonal to the kernel line of the sum map, generated by $(1,-1)$, and $|\ell|^2 = |f|^2/S$. Write $\delta = \phi_1 - \phi_2$. The Hessian formula gives

$$A = v_1(\phi_{1,j\bar{k}}) + v_2(\phi_{2,j\bar{k}}) + v_1 v_2(\delta_j \overline{\delta_k}),$$
$$\bar{\partial}(v_1 f) = f v_1 v_2 \bar{\partial}\delta.$$

Thus $v_1 v_2 |\bar{\partial}\delta|_A^2 \le 1$ and, for $b = \bar{\partial}(v_1 f)$,

$$|b|_A^2 \mathrm{e}^{-\chi} \le \frac{|f|^2 \mathrm{e}^{-\psi}}{S^2}.$$

Solve (1.25) for this b and set $F = \ell - u(1,-1)$. Since $v_1 + v_2 = 1$, the vector $F = (F_1, F_2)$ is holomorphic and its components sum to f. The squared norm of $(1,-1)$ is $\mathrm{e}^{-\phi_1} + \mathrm{e}^{-\phi_2} = S\mathrm{e}^{-\phi_1-\phi_2}$. Orthogonality and (1.25) consequently give

$$\begin{aligned}\int |F|^2 \mathrm{e}^{-\psi} &= \int \frac{|f|^2 \mathrm{e}^{-\psi}}{S} + \int |u|^2 \mathrm{e}^{\Phi-\chi} \\ &\le \int \frac{|f|^2 \mathrm{e}^{-\psi}}{S^2}.\end{aligned}$$

This proves (1.24) for smooth weights.

Step 3: general weights. Exhaust Ω by bounded pseudoconvex domains with compact closures in Ω. On each of them, convolve the weights on a slightly larger neighborhood and add a sufficiently small positive multiple of $|z|^2$. These smooth plurisubharmonic weights dominate the original weights, converge to them almost everywhere, preserve $S < 1$ on the chosen closure, and make Φ strictly plurisubharmonic. The corresponding integrals J are bounded by the original one. On every compact subset the weights have common upper bounds, so (1.24) gives uniform unweighted local L^2 bounds for both components. The holomorphic mean-value inequality and a diagonal extraction give holomorphic limits. Almost-everywhere convergence of the weights and Fatou's lemma yield (1.24) on Ω, and $F_1 + F_2 = f$ passes to the limit. □

Proposition 1.16.1 *Let ψ and ϕ be plurisubharmonic functions on Ω with $\phi < 0$, and let $p > 1$. For holomorphic f with $I = \|f\|^2_{\psi+\phi} < +\infty$, define*

$$c = \inf\{\|g\|^2_{\psi} : g \in \mathcal{O}(\Omega), \quad f - g \in \Gamma(\Omega, \mathcal{I}(p\phi + \psi))\}.$$

Then $c \le (1 - 1/p)I$.

Proof For $t > 0$, apply Lemma 1.16.6 with

$$\phi_1 = \log \frac{t}{1+t} + p\phi, \qquad \phi_2 = -\log(1+t).$$

Their exponential sum is $S = (1 + te^{p\phi})/(1+t) < 1$. The integral J of the lemma is finite, since $\|f\|^2_{\psi} \le I$ and $S \ge 1/(1+t)$. The first component F_1 belongs to $\Gamma(\Omega, \mathcal{I}(p\phi + \psi))$, so the second component $F_2 = f - F_1$ is admissible for c, and the norm estimate gives

$$\frac{c}{1+t} \le \int_\Omega \frac{|f|^2 e^{-\psi}}{(1+te^{p\phi})^2}\, dV.$$

Set $t = x^p$ and integrate over $x > 0$. With $\kappa_k = \int_0^\infty (1+x^p)^{-k}\, dx$, Tonelli's theorem and the change of variable $x \mapsto xe^{\phi}$ give $c\kappa_1 \le I\kappa_2$; the pole set of ϕ is Lebesgue null. Since $p > 1$, both integrals are finite, and integrating the derivative of $x/(1+x^p)$ over $(0, \infty)$ gives $\kappa_2 = (1 - 1/p)\kappa_1$. The argument does not require the infimum to be attained. □

Theorem 1.16.2 *Let ψ and $\phi_1, \dots, \phi_r$ be plurisubharmonic functions on Ω, with $r \ge 1$ and $\phi_i < 0$, and choose $p_i > 1$. For holomorphic f with*

$$I = \|f\|^2_{\psi + \sum_i \phi_i} < +\infty,$$

put

$$c = \inf\left\{ \|g\|^2_{\psi} : g \in \mathcal{O}(\Omega), \quad f - g \in \sum_{i=1}^{r} \Gamma(\Omega, \mathcal{I}(p_i\phi_i + \psi)) \right\}.$$

Then

$$c \le \prod_{i=1}^{r} \left(1 - \frac{1}{p_i}\right) I. \tag{1.27}$$

For $r \le n$, this constant is optimal for the stated class of data.

Proof Set $\psi_i = \psi + \sum_{j>i} \phi_j$, $a_i = 1 - 1/p_i$, and $f_0 = f$. For arbitrary $\varepsilon_i > 0$, Proposition 1.16.1 successively gives holomorphic f_i such that

$$\|f_i\|^2_{\psi_i} \le a_i \|f_{i-1}\|^2_{\psi_i + \phi_i} + \varepsilon_i,$$
$$f_{i-1} - f_i \in \Gamma(\Omega, \mathcal{I}(p_i\phi_i + \psi_i)).$$

Since $\psi_i \le \psi$, these ideals are contained in $\mathcal{I}(p_i\phi_i + \psi)$. As $f - f_r$ is the sum of the differences $f_{i-1} - f_i$, the function f_r is admissible for c. Moreover, $\psi_i + \phi_i = \psi_{i-1}$ for $i \ge 1$ and $\psi_r = \psi$, so iteration gives

$$c \le \left(\prod_i a_i\right) I + \sum_{i=1}^{r} \varepsilon_i \prod_{j>i} a_j .$$

Letting every ε_i tend to zero proves (1.27).

For sharpness when $r \le n$, take the unit polydisk $\Omega = \Delta^n$, $f = 1$, $\psi = 0$, and $\phi_i = p_i^{-1} \log |z_i|^2$ for $i \le r$. By Lemma 1.16.1, $\mathcal{I}(p_i\phi_i) = (z_i)$. Every admissible g equals one on $Y = \{z_1 = \cdots = z_r = 0\}$, hence $g(0) = 1$. Orthogonality of the monomials on the polydisk gives $\|g\|_0^2 \ge \pi^n$, with equality for $g = 1$. Direct radial integration gives $I = \pi^n \prod_i p_i/(p_i - 1)$, so equality holds in (1.27). □

The congruence condition in Theorem 1.16.2 involves a sum of global sections, and the successive remainders in its proof are global; a condition on a single germ would not produce them.

1.16.2 The optimal constant in dimension one

In this subsection, $\Omega \subset \mathbb{C}$ is a pseudoconvex domain with Euclidean area dV, and we use the weighted norms and spaces of global sections of Section 1.16.1. In one variable, every multiplier ideal is determined by vanishing orders at points. This allows us to combine the weights into a single one before applying the one-weight estimate, and leads to a better constant than the product bound.

Lemma 1.16.7 *For a subharmonic function u on Ω and $x \in \Omega$,*

$$\mathcal{I}(u)_x = (z - x)^{\lfloor \nu(u,x) \rfloor} \mathcal{O}_{\Omega,x}. \tag{1.28}$$

These ideals form a locally principal ideal sheaf.

Proof With the normalization of dd^c fixed in the conventions, $\mu = dd^c u$ is a positive Radon measure and $a = \mu(\{x\}) = \nu(u, x)$. After removal of this atom, the distribution $u - a \log |z-x|^2$ has positive complex Hessian, so it has a subharmonic representative v, and $dd^c v$ has no atom at x. Since subharmonic functions that agree almost everywhere agree everywhere,

$$u = a \log |z - x|^2 + v.$$

Since $dd^c v$ has no atom at x, the same identification of Lelong numbers with atoms gives $\nu(v, x) = 0$. Hence, for each $t > 0$, Skoda's theorem [Dem11, Lemma 5.6] makes e^{-tv} integrable on a neighborhood of x.

A nonzero holomorphic germ of order k at x is a unit times $(z - x)^k$. Since v is locally bounded above, weighted integrability requires $k + 1 > a$; equality is excluded by the logarithmic divergence of the radial integral. Conversely, if $k + 1 > a$,

choose $b > 1$ such that $b(a-k) < 1$. Then $|z-x|^{2k-2a}$ belongs locally to L^b, and e^{-v} belongs locally to $L^{b/(b-1)}$ by Skoda's theorem, so Hölder's inequality proves integrability. Thus the germ belongs to $\mathcal{I}(u)_x$ exactly when $k \geq \lfloor a \rfloor$.

The points y with $\nu(u,y) \geq 1$ form a locally finite set, since each of them carries an atom of mass at least one of the locally finite measure $\mathrm{dd^c}u$. Near any point, therefore, the ideal sheaf is generated by a single power of a coordinate, which proves the last assertion. □

The distance condition below involves a sum of spaces of global sections. In one variable, this sum coincides with the space of global sections of the sum of the ideal sheaves.

Lemma 1.16.8 *If $u_1, \ldots, u_r$ are subharmonic functions on Ω, then*

$$\Gamma\left(\Omega, \sum_{i=1}^{r} \mathcal{I}(u_i)\right) = \sum_{i=1}^{r} \Gamma(\Omega, \mathcal{I}(u_i)). \tag{1.29}$$

Proof For $r = 1$ there is nothing to prove, so let $r \geq 2$. Put $J_i = \mathcal{I}(u_i)$ and $J = \sum_i J_i$. By Lemma 1.16.7, these sheaves are locally free of rank one. At each point, the order of J is the smallest of the orders of the J_i, and a local generator of an ideal J_i attaining this order also generates J. Consequently the sum map $\bigoplus_i J_i \to J$ is locally split, and its kernel E is a holomorphic vector bundle of rank $r-1$.

Fix $f \in \Gamma(\Omega, J)$. Gluing local holomorphic lifts of f with a smooth partition of unity gives a smooth section s of $\bigoplus_i J_i$ whose image is f. Since f is holomorphic, $\beta = \bar\partial s$ is an E-valued $(0,1)$-form. It suffices to solve $\bar\partial v = \beta$ for a section v of E, for then $s - v$ is a holomorphic lift of f.

We solve this equation using a bundle metric of positive curvature. Choose a smooth strictly subharmonic exhaustion function $\rho \geq 1$ of Ω. The Kähler metric

$$\omega_* = \mathrm{i}\,\mathrm{d}z \wedge \mathrm{d}\bar z + \mathrm{i}\partial\bar\partial\rho^2$$

is complete: it dominates $2\mathrm{i}\partial\rho \wedge \bar\partial\rho$, so ρ has bounded gradient with respect to it, and therefore every curve leaving all compact subsets has infinite length.

Equip E with a smooth Hermitian metric h. Choose a smooth increasing convex function χ on $[1, +\infty)$ so that the metric $h_\chi = h\mathrm{e}^{-\chi(\rho)}$ has

$$\mathrm{i}\Theta_{E,h_\chi} \geq \omega_* \,\mathrm{Id}_E, \qquad \int_\Omega |\beta|^2_{h_\chi,\omega_*}\, \mathrm{d}V_{\omega_*} < +\infty.$$

These two requirements can be imposed successively on the compact sublevel sets of ρ. We increase χ' so that $\chi'(\rho)$ times the positive Hessian of ρ dominates ω_* plus the negative part of the curvature of h, and we increase χ so that the integral over each shell $\{m \leq \rho < m+1\}$ is at most 2^{-m}. Increasing the slopes preserves the inequalities already achieved.

Use the global frame $\mathrm{d}z$ of the canonical bundle K_Ω to identify E-valued $(0,q)$-forms with $(E \otimes K_\Omega^{-1})$-valued $(1,q)$-forms, and give K_Ω^{-1} its flat coordinate metric. Its curvature is zero, and ω_* dominates the Euclidean metric, so the identified form β

remains L^2. The Bochner–Kodaira inequality for compactly supported forms w of type $(1,1)$ gives

$$\|\bar\partial^* w\|^2 \geq \langle [\mathrm{i}\Theta_{E,h_\chi}, \Lambda_{\omega_*}] w, w\rangle \geq \|w\|^2.$$

Here we used that there are no $(1,2)$-forms. By completeness, cutoffs and smoothing extend this inequality to the domain of the adjoint. Since β is L^2, the Cauchy–Schwarz inequality bounds the functional $w \mapsto \langle w, \beta\rangle$ by $\|\beta\|\|\bar\partial^* w\|$, and the Riesz representation theorem gives a distributional solution v of $\bar\partial v = \beta$. Thus $s - v$ is distributionally holomorphic, hence holomorphic in every bundle chart, and it is the required global lift. No global weighted norm condition is imposed on its components. □

Theorem 1.16.3 *Let ψ and $\phi_1, \dots, \phi_r$ be subharmonic functions on Ω, with $r \geq 1$ and $\phi_i < 0$, and choose $p_i > 1$. For holomorphic f with $I = \|f\|^2_{\psi+\sum_i \phi_i} < +\infty$, define*

$$c = \inf\left\{ \|g\|^2_\psi : g \in \mathcal{O}(\Omega), \quad f - g \in \sum_{i=1}^r \Gamma(\Omega, \mathcal{I}(p_i\phi_i + \psi)) \right\}, \qquad s = \sum_{i=1}^r \frac{1}{p_i}.$$

Then

$$c \leq \max\{1 - s, 0\}\, I. \tag{1.30}$$

For every fixed choice of the exponents, this constant is optimal over the stated class of data. If $s \geq 1$, the function $g = 0$ is admissible.

Proof Put $J_i = \mathcal{I}(p_i\phi_i + \psi)$ and $J = \sum_i J_i$. At a point x, write $A = \nu(\psi, x)$ and $B_i = \nu(\phi_i, x)$. By Lemma 1.16.7, the order of J_x is $\min_i \lfloor A + p_i B_i \rfloor$.

Suppose first that $s < 1$, and put $P = 1/s > 1$. The numbers P/p_i sum to one, so

$$\min_i p_i B_i \leq \sum_i \frac{P}{p_i} p_i B_i = P \sum_i B_i.$$

Taking integer parts and using Lemma 1.16.7 gives $\mathcal{I}(P\sum_i \phi_i + \psi) \subseteq J$. By Lemma 1.16.8, every global section of this smaller ideal sheaf lies in $\sum_i \Gamma(\Omega, J_i)$, so every remainder admissible for the one-weight problem is admissible for c. Proposition 1.16.1 with $\phi = \sum_i \phi_i$ and $p = P$ therefore gives $c \leq (1 - 1/P)I = (1-s)I$.

Now let $s \geq 1$. Since $I < +\infty$, we have $f \in \Gamma(\Omega, \mathcal{I}(\psi + \sum_i \phi_i))$. If $f \equiv 0$ there is nothing to prove; otherwise let k be its order at x. Then $k + 1 > A + \sum_i B_i$. If its germ did not belong to J_x, we would have $k + 1 \leq A + p_i B_i$ for every i. With $L = k + 1 - A > 0$, this would give $\sum_i B_i \geq sL \geq L$, a contradiction. Hence $f \in \Gamma(\Omega, J)$, and Lemma 1.16.8 shows that $g = 0$ is admissible.

For sharpness when $s < 1$, take the unit disk, $\psi = 0$, $f = 1$, and $\phi_i = p_i^{-1} \log|z|^2$. Then $J_i = (z)$ for all i, so an admissible g has $g(0) = 1$. Orthogonality of the monomials on the disk gives $c = \pi$, attained by $g = 1$, whereas $I = \pi/(1 - s)$. If $s \geq 1$, keep these weights and take $f = z^q$ with an integer $q \geq 0$ satisfying $q + 1 > s$.

Then $I = \pi/(q+1-s) > 0$ and $c = 0$, so the constant 0 cannot be replaced by a negative number. □

Consequently, for $r \geq 2$ the product constant of Theorem 1.16.2 is not optimal on plane domains. The argument above relies on all local ideals being principal; it does not determine the optimal constant when $n \geq 2$ and $r > n$.

1.16.3 Distance for several hyperplane weights

In this subsection, $n \geq 2$, $B \subset \mathbb{C}^n$ is the unit ball, and dV is Euclidean volume. If the weights have logarithmic poles along several hyperplanes through the origin, every admissible remainder must take a prescribed value at their common zero. On the other hand, the input norm stays finite as long as the combined radial exponent is below the dimension. Together these observations give a positive distance even in part of the range where the planar distance of Theorem 1.16.3 vanishes.

Example 1.16.2 Fix $r > n$ and $p > r/n$. There are negative plurisubharmonic functions $\phi_1, \dots, \phi_r$ on B for which

$$I = \int_B e^{-\sum_i \phi_i}\, dV < +\infty$$

and

$$\inf\left\{ \int_B |g|^2\, dV : g \in \mathcal{O}(B), \quad 1 - g \in \sum_{i=1}^{r} \Gamma(B, \mathcal{I}(p\phi_i)) \right\} = \frac{\pi^n}{n!} > 0. \tag{1.31}$$

In particular, for $r/n < p \leq r$, the distance is positive although $\sum_i 1/p = r/p \geq 1$.

Proof Choose distinct real numbers $t_1, \dots, t_r$ and unit linear forms

$$\ell_i(z) = \frac{\sum_{k=0}^{n-1} t_i^k z_{k+1}}{(\sum_{k=0}^{n-1} t_i^{2k})^{1/2}}.$$

Any n of these forms are linearly independent, since the determinant of their unnormalized coefficients is a nonzero Vandermonde determinant. Put $\phi_i = p^{-1} \log |\ell_i|^2$. Since the coefficient vectors of the ℓ_i have norm one, $|\ell_i(z)| < 1$ on B, so the ϕ_i are negative plurisubharmonic functions. By Lemma 1.16.1, $\mathcal{I}(p\phi_i) = (\ell_i)$.

At a nonzero point, at most $n-1$ of the forms vanish. These forms are linearly independent and can be used as part of a holomorphic coordinate system, while the other forms are bounded away from zero on a small neighborhood of the point. Hence

$$W(z) = \prod_{i=1}^{r} |\ell_i(z)|^{-2/p}$$

is locally integrable off the origin, because each factor $|\ell_i|^{-2/p}$ has exponent $2/p < 2$ (note that $p > r/n > 1$). By compactness, W is integrable on every closed annulus. Set $s = r/p$ and let $\mathrm{d}\sigma$ denote surface measure on the unit sphere. Homogeneity and polar integration over such an annulus show that

$$A = \int_{|\zeta|=1} \prod_i |\ell_i(\zeta)|^{-2/p} \, \mathrm{d}\sigma(\zeta) < +\infty.$$

Tonelli then gives

$$I = A \int_0^1 \rho^{2n-1-2s} \, \mathrm{d}\rho = \frac{A}{2(n-s)} < +\infty,$$

since $s < n$. This takes care of the common zero at the origin as well as of the individual hyperplanes.

Every global section of (ℓ_i) vanishes at the origin. Thus every admissible g in (1.31) satisfies $g(0) = 1$. For each $z \in B$, the holomorphic function $\lambda \mapsto g(\lambda z)$ is defined on a disk of radius greater than one, and the submean inequality for the subharmonic function $|g(\lambda z)|^2$ gives

$$\frac{1}{2\pi} \int_0^{2\pi} |g(\mathrm{e}^{\mathrm{i}\tau} z)|^2 \, \mathrm{d}\tau \geq |g(0)|^2 = 1.$$

Integrating in z and using the rotation invariance of B proves $\int_B |g|^2 \, \mathrm{d}V \geq \mathrm{Vol}(B) = \pi^n/n!$, also when the integral is infinite. The constant $g = 1$ is admissible and attains this bound. □

Hence, for these equal exponents, the optimal universal constant in an estimate of the distance by a multiple of I is at least $\pi^n/(n!I) > 0$. In particular, the vanishing of the optimal constant for $s \geq 1$ in Theorem 1.16.3 does not extend to higher dimensions. The example does not determine the optimal constant for $r > n$.

1.17 Lelong numbers and divisorial estimates

Lelong numbers measure the logarithmic concentration of a positive current. Along a divisor, a lower bound for the generic Lelong number allows us to subtract the corresponding multiple of the divisor without losing positivity. At points, the local mass formula controls Lelong numbers under convergence and uniformly in a fixed background. For a prime divisor E on a complex manifold, $[E]$ denotes its current of integration, and $\nu(w, E)$ is the generic Lelong number along E of a quasi-plurisubharmonic function w, as defined in Section 1.13.

We use the following local consequence of Siu's decomposition theorem, recalled in [Xia26] and [Dem11]. If T is a closed positive $(1,1)$-current and E is a prime divisor, the restriction of (the coefficient measures of) T to E equals the generic

Lelong number of T along E times $[E]$. In particular, removing finitely many of these divisorial components from T leaves a positive current. Distinct prime divisors do not contribute overlapping mass, since their intersections have codimension at least two. All these are local statements about closed currents.

Lemma 1.17.1 *Let M be a complex manifold, let β be a smooth real $(1,1)$-form on M, let $w \in \mathrm{PSH}(M,\beta)$, and let $E_1,\dots,E_N$ be distinct prime divisors on M. Then*

$$\beta + \mathrm{dd^c} w - \sum_{j=1}^{N} \nu(w,E_j)[E_j] \geq 0.$$

Proof Work on a coordinate ball and choose a smooth function ρ with $\mathrm{dd^c}\rho \geq \beta$. The current

$$T = \mathrm{dd^c}(w+\rho) = (\beta + \mathrm{dd^c} w) + (\mathrm{dd^c}\rho - \beta)$$

is closed and positive. A smooth form, viewed through its coefficient measures, assigns zero mass to every proper analytic set. Thus T and $\beta + \mathrm{dd^c} w$ have the same restriction to $\bigcup_j E_j$, and the consequence of Siu's theorem stated above identifies this restriction with $\sum_j \nu(w,E_j)[E_j]$. Subtracting it from the positive current $\beta + \mathrm{dd^c} w$ leaves the restriction of $\beta + \mathrm{dd^c} w$ to the complement of $\bigcup_j E_j$, which is positive. This proves the inequality locally, hence globally. □

Order makes the support theorem a rigidity statement for potentials: two ordered quasi-plurisubharmonic functions whose Hessians agree outside an analytic set already differ by a constant.

Lemma 1.17.2 *Let $u \leq v$ be quasi-plurisubharmonic on a connected compact complex manifold X, and let $Z \subsetneq X$ be analytic. If $\mathrm{dd^c} u = \mathrm{dd^c} v$ on $X \setminus Z$, then $v = u + C$ for a real constant C.*

Proof The current $S = \mathrm{dd^c}(v-u)$ is real, closed, of order zero and supported on Z. By the support theorem in [Dem12, Chapter III, Corollaries 2.11 and 2.14, pp. 141–143],

$$S = \sum_{j=1}^{m} c_j [Z_j], \qquad c_j = \nu(v,Z_j) - \nu(u,Z_j) \leq 0, \tag{1.32}$$

where Z_j are the prime divisorial components of Z. Indeed, the support theorem first gives real constants c_j such that $S - \sum_j c_j[Z_j]$ is supported in codimension at least two; this closed current of order zero then vanishes by the same theorem. Near a generic smooth point of Z_j, with local equation $z_1 = 0$, the function $v - u - c_j \log|z_1|^2$ has zero complex Hessian and is therefore smooth and pluriharmonic. This identifies c_j with the difference of the generic Lelong numbers, and $c_j \leq 0$ because $u \leq v$.

Thus $\mathrm{dd^c}(u-v) = -S \geq 0$. Hence the locally integrable distribution $u - v$ has a global plurisubharmonic representative, which is constant by the maximum principle on the compact connected manifold X. Since quasi-plurisubharmonic functions that agree almost everywhere agree everywhere, $v = u + C$ for a constant C. □

Every closed real $(1,1)$-current is the sum of a smooth closed form and the complex Hessian of a global distribution, and positivity of the current makes this potential quasi-plurisubharmonic. This decomposition computes Bott–Chern cohomology with currents, and it allows us to attach an ordinary singularity type to a divisorial current.

Lemma 1.17.3 *Let Z be a complex manifold and T a closed real $(1,1)$-current on Z. There are a smooth closed real $(1,1)$-form β and a real distribution v such that*

$$T = \beta + \mathrm{dd}^{\mathrm{c}} v.$$

If $T \geq 0$, then v is quasi-plurisubharmonic. Any two distributional potentials in such decompositions differ by a smooth function.

Proof The local dd^{c}-Poincaré lemma gives real distributional potentials v_i on the sets of a cover of Z by coordinate neighborhoods, with $T = \mathrm{dd}^{\mathrm{c}} v_i$; for positive T these potentials are plurisubharmonic. See [Dem12, Chapter III, Section 1.18 and Proposition 1.19]. On overlaps $\mathrm{dd}^{\mathrm{c}}(v_i - v_j) = 0$. Taking the Euclidean trace shows that $v_i - v_j$ is a harmonic distribution, so it is smooth by elliptic regularity.

Choose a smooth locally finite partition of unity (χ_i) subordinate to the cover and put $v = \sum_i \chi_i v_i$, each term being extended by zero as a distribution. On the jth neighborhood,

$$v - v_j = \sum_i \chi_i (v_i - v_j)$$

is smooth. Thus $\beta = T - \mathrm{dd}^{\mathrm{c}} v = -\mathrm{dd}^{\mathrm{c}}(v - v_j)$ is smooth on each neighborhood, hence on Z, and it is real and closed. For positive T, the same identity expresses v locally as a plurisubharmonic function plus a smooth function. Finally, if $T = \beta + \mathrm{dd}^{\mathrm{c}} v = \beta' + \mathrm{dd}^{\mathrm{c}} v'$ are two such decompositions, then $\mathrm{dd}^{\mathrm{c}}(v - v') = \beta' - \beta$ is smooth; taking a smooth Hermitian trace and applying elliptic regularity shows that $v - v'$ is smooth. □

For a complex manifold Z, let $\Lambda^{1,1}_{\mathbb{R}}$ denote the bundle of real $(1,1)$-forms and define

$$H^{1,1}_{\mathrm{BC}}(Z, \mathbb{R}) = \frac{\{\beta \in C^\infty(Z, \Lambda^{1,1}_{\mathbb{R}}) : \mathrm{d}\beta = 0\}}{\{\mathrm{dd}^{\mathrm{c}} f : f \in C^\infty(Z, \mathbb{R})\}}.$$

Lemma 1.17.4 *Replacing smooth forms and smooth functions in this quotient by real currents and real distributions gives the same space. In particular, every closed positive $(1,1)$-current T on Z can be written*

$$T = \beta + \mathrm{dd}^{\mathrm{c}} w$$

with β smooth and closed and w a global quasi-plurisubharmonic function.

Proof The decomposition of Lemma 1.17.3 shows that the natural map from the smooth quotient to the current quotient is surjective, and it gives a quasi-plurisubharmonic potential for positive currents. The same lemma shows that two distributional potentials in such decompositions differ by a smooth function; equivalently, a distribution whose dd^{c} is smooth is itself smooth. Hence two smooth forms

that are identified in the current quotient are already identified in the smooth quotient, and the map is injective. □

For a potential whose complex Hessian is a divisorial current minus a smooth form, ordinary domination, ideal domination and the corresponding inequalities for generic Lelong numbers along the divisors are all equivalent. Countably many divisors are allowed, provided the divisorial current has finite mass.

Lemma 1.17.5 *Let Y be a connected compact complex manifold. Let (D_j) be a finite or countable family of distinct prime divisors on Y and $c_j > 0$ real coefficients such that the current $S = \sum_j c_j [D_j]$ has finite mass, and let β be a smooth real $(1,1)$-form on Y. Let $v \in \mathrm{QPSH}(Y)$ satisfy*

$$\mathrm{dd}^{\mathrm{c}} v = S - \beta.$$

Then, for $a \in \mathrm{QPSH}(Y)$, the following conditions are equivalent:

$$a \preceq v, \qquad a \preceq_{\mathcal{I}} v, \qquad \nu(a, D_j) \geq c_j \quad \text{for every } j.$$

Proof We first note that $\nu(v, D_j) = c_j$. Indeed, the restriction of S to D_j is $c_j [D_j]$: distinct divisors intersect in codimension at least two, and such sets carry no mass of any divisor current, also after countable summation. Smooth forms give every divisor zero mass. The local form of Siu's decomposition, applied after a smooth correction, therefore gives $\nu(v, D_j) = c_j$.

Ordinary domination implies ideal domination by Proposition 1.12.2, and ideal domination implies the asserted inequalities for generic Lelong numbers by Theorem 1.16.1. It remains to prove that these inequalities imply ordinary domination.

Choose a Hermitian form Λ with $a \in \mathrm{PSH}(Y, \Lambda)$. By Lemma 1.17.1 and the inequalities $\nu(a, D_j) \geq c_j$, each finite partial sum of S can be subtracted from $\Lambda + \mathrm{dd}^{\mathrm{c}} a$ without losing positivity. Letting the partial sums increase to S gives

$$\Lambda + \mathrm{dd}^{\mathrm{c}} a - S \geq 0.$$

Consequently $\mathrm{dd}^{\mathrm{c}}(a - v) \geq \beta - \Lambda$. Hence the integrable function $a - v$ has a quasi-plurisubharmonic representative, which is bounded above on the compact manifold Y. This yields $a \leq v + C$ almost everywhere, and Lemma 1.3.3 extends this inequality everywhere. Thus $a \preceq v$. □

Lemma 1.17.3 attaches a canonical ordinary singularity type to a countable divisorial current of finite mass, and Lemma 1.17.5 describes, through generic Lelong numbers along the divisors, the potentials dominated by it.

Corollary 1.17.1 *Let Z be a connected compact complex manifold, let (D_j) be a finite or countable family of distinct prime divisors on Z, and let $c_j > 0$. Suppose that the current $S = \sum_j c_j [D_j]$ has finite mass. Then there are quasi-plurisubharmonic functions v on Z satisfying*

$$\mathrm{dd}^{\mathrm{c}} v = S - \beta$$

for a smooth closed real $(1,1)$-form β, and any two such functions differ by a smooth function. In particular, they determine a canonical ordinary singularity type, which depends only on S.

Proof As a convergent sum of closed positive currents, S is closed and positive. Lemma 1.17.3, applied to $T = S$, gives the decomposition, with v quasi-plurisubharmonic since $S \geq 0$, and shows that any two such potentials differ by a smooth function. By Proposition 1.12.1, they have the same ordinary singularity type. □

Lemma 1.17.6 *Let u_j, u be quasi-plurisubharmonic functions on a complex manifold with a common smooth lower curvature bound, that is, $\mathrm{dd}^{\mathrm{c}} u_j \geq -\beta$ and $\mathrm{dd}^{\mathrm{c}} u \geq -\beta$ for one smooth real $(1,1)$-form β. If $u_j \to u$ in L^1_{loc}, then at every point x,*

$$\nu(u, x) \geq \limsup_j \nu(u_j, x).$$

Proof On a coordinate ball about x, add one smooth function to all the potentials to make them plurisubharmonic, and write p_j, p for the resulting functions. Let n be the dimension and κ the Euclidean form of the chart. Distributional convergence gives weak convergence of the positive trace measures $\mathrm{dd}^{\mathrm{c}} p_j \wedge \kappa^{n-1}$ to $\mathrm{dd}^{\mathrm{c}} p \wedge \kappa^{n-1}$. For every sufficiently small radius r such that the sphere of radius r about x has zero mass for the limit measure, the masses of the balls converge. The Lelong monotonicity formula bounds each $\nu(p_j, x)$ by the normalized trace mass of p_j on such a ball. Taking the upper limit in j, and then letting these radii decrease to zero, gives $\limsup_j \nu(p_j, x) \leq \nu(p, x)$. Since smooth additions do not change Lelong numbers, this proves the lemma. □

Lemma 1.17.7 *Let Z be a connected compact complex manifold and let β be a smooth real $(1,1)$-form. There is a finite constant C_β such that*

$$\nu(v, z) \leq C_\beta \qquad \text{for every } v \in \mathrm{PSH}(Z, \beta) \text{ and every } z \in Z.$$

In particular, the generic Lelong numbers of all $v \in \mathrm{PSH}(Z, \beta)$ along all prime divisors are bounded by C_β.

Proof Since adding a constant does not change Lelong numbers, we may assume that $\sup_Z v = 0$. By Theorem 1.4.1, these normalized potentials have a common L^1 bound. Cover Z by finitely many triples of concentric coordinate balls, a smaller, an

intermediate and a larger one, such that the smaller balls cover Z. On each larger ball choose a fixed smooth quadratic function r with $\mathrm{dd^c} r \geq \beta$, so that $p = v + r$ is plurisubharmonic there. The functions p obtained in this way have uniformly bounded local L^1 norms.

Write $d = \dim Z$ and let κ be the Euclidean form in the chart. Choose a nonnegative smooth cutoff χ supported in the larger ball and equal to one on an intermediate ball. Positivity and distributional integration by parts give

$$\int_{\text{intermediate ball}} \mathrm{dd^c} p \wedge \kappa^{d-1} \leq \int \chi \, \mathrm{dd^c} p \wedge \kappa^{d-1} = \int p \, \mathrm{dd^c} \chi \wedge \kappa^{d-1} \leq C \|p\|_{L^1}.$$

Every point of the smaller ball is the center of a Euclidean ball of a fixed radius contained in the intermediate ball. The monotonicity of the trace mass ratio [Dem12, Chapter III, (5.7)–(5.8)] bounds the Lelong number at this center by a fixed multiple of the trace mass of the intermediate ball. The finite cover gives one bound on Z. Since smooth additions and additive constants do not change Lelong numbers, this proves the assertion for the original potentials. □

The Bergman estimates also control Lelong numbers on a fixed local modification. A *local divisor over* X is a prime divisor E on the source of a local modification $\pi \colon Z \to U$ of a coordinate neighborhood $U \subset X$, that is, a modification with smooth source Z. For a quasi-plurisubharmonic function u on X, let $\nu(u \circ \pi, E)$ be the generic Lelong number of $u \circ \pi$ along E. For a holomorphic function or an ideal sheaf, ord_E denotes the generic vanishing order along E (for an ideal, the minimum of the orders of its local generators). Let $\mathrm{Jac}\, \pi$ be the holomorphic Jacobian determinant of π in local coordinates. Write

$$\nu_E(u) = \nu(u \circ \pi, E), \qquad A_U(E) = 1 + \mathrm{ord}_E(\mathrm{Jac}\, \pi).$$

The second number is the *log discrepancy* of E. Local modifications are not required to extend over the compact manifold.

Lemma 1.17.8 *For a quasi-plurisubharmonic function u on X, every local divisor E over X, and every real $m > 0$, one has*

$$\max\left\{\nu_E(u) - \frac{A_U(E)}{m}, 0\right\} \leq \frac{1}{m} \mathrm{ord}_E \mathcal{I}(mu) \leq \nu_E(u).$$

Here $\mathrm{ord}_E \mathcal{I}(mu)$ is the order along E of the pulled-back ideal. Consequently, if locally $a_m = m^{-1} \log \sum_k |f_{m,k}|^2 + O(1)$, where the $f_{m,k}$ generate $\mathcal{I}(mu)$, then $\nu_E(a_m) \to \nu_E(u)$ as $m \to \infty$.

Proof The upper estimate follows from the local Bergman lower bound in Proposition 1.14.1: the logarithm of the sum of squared moduli of local generators of $\mathcal{I}(mu)$, divided by m, is at least u up to a locally bounded term. Pulling back and taking generic Lelong numbers along E reverses this inequality.

For the lower estimate, take a local generator f of the ideal and set $k = \mathrm{ord}_E(f \circ \pi)$. Near a generic point of E, choose a local coordinate z_1 defining E. Divisorial removal

(Lemma 1.17.1), after a smooth correction, gives

$$u \circ \pi \le \nu_E(u) \log |z_1|^2 + C.$$

Shrink around a point where the factors of $f \circ \pi$ and of the Jacobian complementary to the powers of z_1 do not vanish. The change-of-variables formula and local integrability of $|f|^2 e^{-mu}$ then require integrability in z_1 of

$$|z_1|^{2(k+A_U(E)-1-m\nu_E(u))}.$$

This implies $k+A_U(E) > m\nu_E(u)$, and in particular the weaker non-strict lower bound in the statement. Taking the minimum over a finite set of local generators and using that orders are nonnegative proves the lower estimate. Since $\nu_E(a_m) = m^{-1} \operatorname{ord}_E \mathcal{I}(mu)$, the last assertion follows by letting $m \to \infty$. □

For rational $m > 0$, the function a_m has analytic singularities in the sense of Section 1.13; the estimate for real m does not require this. Properness of π is not used either: the upper estimate is a pullback, and the change of variables in the lower one takes place on relatively compact patches of Z, whose images are relatively compact in U. Hence the lemma, with $A_U(E)$ given by the same formula, holds for every holomorphic map $\pi\colon Z \to U$ from a complex manifold of dimension $\dim U$ that is injective off a nowhere dense analytic subset of Z.

Blowing up a point converts the divisorial estimate into a pointwise order bound on any local modification.

Lemma 1.17.9 *Let U be a complex manifold of dimension n, let u be quasi-plurisubharmonic on U, and let $\pi\colon Y \to U$ be a modification with smooth source. Fix $y \in Y$ and a real number $m > 0$. Write J_π for the holomorphic Jacobian in local coordinates near y. If a nonzero holomorphic germ g at $\pi(y)$ belongs to $\mathcal{I}(mu)$, then*

$$\operatorname{ord}_y(g \circ \pi) \ge m\nu(u \circ \pi, y) - n - \operatorname{ord}_y J_\pi.$$

Proof If $n \ge 2$, let $\rho\colon \widetilde{Y} \to Y$ be the blowup of y and let E be its exceptional divisor. For $n = 1$, take ρ to be the identity and $E = \{y\}$. Set $F = \pi \circ \rho$ and write J_F for its local holomorphic Jacobian determinant. Generic orders along E give

$$\nu(u \circ F, E) = \nu(u \circ \pi, y), \qquad \operatorname{ord}_E(g \circ F) = \operatorname{ord}_y(g \circ \pi).$$

On a standard blowup chart the Jacobian of ρ has order $n-1$ along E. The chain rule therefore gives the log discrepancy over U as

$$A_U(E) = 1 + \operatorname{ord}_E J_F = n + \operatorname{ord}_y J_\pi.$$

Since $g \in \mathcal{I}(mu)$, the order of $g \circ F$ along E is at least the order of this ideal. The lower estimate in Lemma 1.17.8, applied to F, therefore proves the inequality. □

Lemma 1.17.10 *Let v be quasi-plurisubharmonic on X, and let a have analytic singularities on X. If $v \preceq_{\mathcal{I}} a$, then $v \preceq a$. More precisely, it suffices to assume the*

finitely many inequalities

$$\nu(v \circ \pi, E) \geq \nu(a \circ \pi, E)$$

for the prime components E of the divisor obtained by principalizing, on one log resolution π, the ideal of an analytic presentation of a.

Proof Principalize the coherent ideal of an analytic presentation of a by a log resolution $\pi\colon Y \to X$ with smooth Y. In suitable local coordinates $z_1, \dots, z_n$ on Y,

$$a \circ \pi = \sum_{i=1}^{r} b_i \log |z_i|^2 + B, \qquad b_i \geq 0, \qquad B \in L^\infty_{\mathrm{loc}},$$

where $r \leq n$. The finitely many inequalities in the statement give $\nu(v \circ \pi, \{z_i = 0\}) \geq b_i$; if $v \preceq_{\mathcal{I}} a$, these inequalities follow from Theorem 1.16.1. Add a smooth local function to $v \circ \pi$ to make it plurisubharmonic. Divisorial removal for this closed positive complex Hessian shows that

$$v \circ \pi - \sum_i b_i \log |z_i|^2$$

has a quasi-plurisubharmonic representative; here the difference is first interpreted off the divisor, or as a locally integrable distribution. Such a representative is locally bounded above. Since B is locally bounded, $v \circ \pi \leq a \circ \pi + C$ on each relatively compact chart of Y, away from the divisor. Since π is proper, finitely many such charts cover the inverse image of a smaller coordinate neighborhood in X, which makes C uniform there. Push the inequality down off the analytic exceptional set; as quasi-plurisubharmonic functions are determined by their values almost everywhere, it then holds everywhere. Finitely many neighborhoods cover the compact manifold, so $v \leq a + C$ globally. □

Lemma 1.17.11 *Let u, v be quasi-plurisubharmonic on X, and let s have analytic singularities on X. Suppose that $u + v \leq s + O(1)$, and let U_m, V_m be quasi-plurisubharmonic functions of analytic types $(1/m, \mathcal{I}(mu))$ and $(1/m, \mathcal{I}(mv))$. Then there are numbers $\delta_m \downarrow 0$ such that*

$$U_m + V_m \leq (1 - \delta_m)s + O(1) \quad \textit{for all sufficiently large integers } m.$$

The bounded error may depend on m.

Proof If s is bounded, upper bounds for U_m and V_m give the assertion with $\delta_m = 1/m$ for $m > 1$. Otherwise principalize the ideal of s by a log resolution $\pi\colon \widetilde{X} \to X$ with smooth $\widetilde{X}$. Let E_i be the components of the resulting divisor, with positive coefficients c_i in $s \circ \pi$; in local coordinates z_i defining these components, the singular part of $s \circ \pi$ is $\sum_i c_i \log |z_i|^2$. Set $A_i = 1 + \mathrm{ord}_{E_i}(\mathrm{Jac}\,\pi)$. The hypothesis and Lemma 1.17.8 give

$$\nu(U_m\circ\pi,E_i)+\nu(V_m\circ\pi,E_i)\geq\nu(u\circ\pi,E_i)+\nu(v\circ\pi,E_i)-\frac{2A_i}{m}\geq c_i-\frac{2A_i}{m}.$$

Choose a positive integer $N\geq\max_i 2A_i/c_i$ and put $\delta_m=N/m$, which is less than one for large m. Then the right-hand side above is at least $(1-\delta_m)c_i$. After a smooth local correction, divisorial removal from the positive Hessian of $(U_m+V_m)\circ\pi$ shows that

$$(U_m+V_m)\circ\pi-(1-\delta_m)\sum_i c_i\log|z_i|^2$$

has a quasi-plurisubharmonic representative, which is locally bounded above. A finite cover of $\widetilde{X}$ by charts and descent along the proper map π, as in the proof of Lemma 1.17.10, therefore give $U_m+V_m\leq(1-\delta_m)s+O(1)$. □

1.18 Potential regularity and metric distance

A Hölder modulus of continuity of the potential controls the trace mass of a positive current on small balls. Averaging the lengths of segments against this trace measure then bounds the distance of the metric defined by the current on its regular locus. This turns regularity of the potential into a metric estimate without any pointwise bound on the Hessian.

Let $U\subset\mathbb{C}^n$ be a coordinate domain with Euclidean form $\omega_{\mathrm{E}}=\mathrm{dd^c}|z|^2$ and Euclidean volume $\mathrm{d}\lambda$. Let $0<\alpha\leq 1$, let θ be a smooth real $(1,1)$-form on U, and let $v\in C^{0,\alpha}(U)$ satisfy $T=\theta+\mathrm{dd^c}v\geq 0$. Write $\mu=T\wedge\omega_{\mathrm{E}}^{n-1}$ for the trace measure of T.

Theorem 1.18.1 *For every open set $V\Subset U$ there are constants $C,r_0>0$ such that*

$$\mu(B(x,r))\leq Cr^{2n-2+\alpha},\qquad x\in V,\quad 0<r<r_0. \tag{1.33}$$

The total variation of every coefficient measure of T is bounded by a constant multiple of μ. Hence the coefficient measures of T vanish on sets of zero Hausdorff $(2n-2+\alpha)$-measure, and all Lelong numbers of v vanish.

Suppose in addition that T is a smooth Hermitian form η on $U\setminus A$ for a proper analytic subset $A\subset U$. For $x,y\in U\setminus A$, let $d_\eta^U(x,y)$ be the infimum of the η-lengths of piecewise smooth paths in $U\setminus A$ joining x and y. Then every point of U has a neighborhood $V\Subset U$ such that, for a constant C,

$$d_\eta^U(x,y)\leq C|x-y|^{\alpha/2},\qquad x,y\in V\setminus A. \tag{1.34}$$

In particular, all sequences in $V\setminus A$ converging to a given point of $A\cap V$ are Cauchy for d_η^U and have a common limit in the metric completion. If these local hypotheses hold in charts covering X, with a proper analytic subset $A\subset X$ and a Hermitian form η on $X\setminus A$, then the metric completion of $X\setminus A$ for its intrinsic η-distance is compact,

and the identity of $X \setminus A$ extends to a continuous surjection of X onto it. The metric η is incomplete when $A \neq \varnothing$. Both exponents in (1.33) *and* (1.34) *are sharp.*

Proof Choose a nonnegative cutoff function χ equal to one on $B(x,r)$ and supported in $B(x,2r)$, with second derivatives bounded by Cr^{-2}. Positivity and integration by parts, after subtracting the constant $v(x)$ from v, give

$$\mu(B(x,r)) \le C_1 r^{2n} + C_2 r^{2n-2} \operatorname{osc}_{B(x,2r)} v \le C r^{2n-2+\alpha}.$$

Here we used that the constant $v(x)$ integrates to zero against $\mathrm{dd^c}\chi \wedge \omega_{\mathrm{E}}^{n-1}$. The matrix (a_{ij}) of Radon–Nikodym derivatives of the coefficient measures with respect to μ is positive semidefinite almost everywhere, so $|a_{ij}| \le (a_{ii} + a_{jj})/2$. A ball cover proves the assertion about Hausdorff null sets. After a smooth local correction making v plurisubharmonic, dividing (1.33) by r^{2n-2} shows that all Lelong numbers vanish.

For the distance estimate, let $m = 2n$ and let τ be the Euclidean trace of η. Then $\tau \, \mathrm{d}\lambda \le C\mu$ on the regular locus and $|\xi|_\eta \le C\sqrt{\tau}|\xi|_{\mathrm{E}}$ for tangent vectors ξ. Polar integration bounds the average η-length of the segments from x to the points of $B(x,R)$ by a constant times $\int_{B(x,R)} \sqrt{\tau}(\zeta)|\zeta - x|^{1-m} \, \mathrm{d}\lambda(\zeta)$. On a dyadic annulus of radius s, the Cauchy–Schwarz inequality and (1.33) bound the contribution to this integral by $Cs^{1-m} s^{m/2} s^{(m-2+\alpha)/2} = Cs^{\alpha/2}$. Summing over the annuli gives the bound $CR^{\alpha/2}$.

Now let $x, y \in V \setminus A$, and average the lengths of the broken segments $[x,z] \cup [z,y]$ over z in the ball of radius $|x-y|/2$ about the midpoint of x and y. By the preceding bound, some such path has length at most $C|x-y|^{\alpha/2}$. For almost every z, both segments avoid A: proper analytic sets are locally countable unions of smooth strata of real codimension at least two, and the cones from a fixed regular point over these strata have zero volume. If V is small enough, all these paths stay inside U.

By this estimate, all regular sequences converging to a point of A have a common limit in the completion, and the identity of the regular locus extends continuously across A. By compactness, the image of this extension is the entire completion; different points of A may be identified. A sequence converging to a point of A cannot converge in the metric to a regular point, since η is bounded below by a positive multiple of the Euclidean metric on a small closed neighborhood of that point. Hence the metric η is incomplete when A is nonempty. Finally, the local model $v = |z|^\alpha + |w|^2$, in coordinates $(z,w) \in \mathbb{C} \times \mathbb{C}^{n-1}$, has trace mass of exact order $r^{2n-2+\alpha}$ on balls centered on $\{z = 0\}$, and the distance to $\{z = 0\}$ is of exact order $|z|^{\alpha/2}$; this proves both sharpness claims. □

1.18.1 Dini moduli and limits of distances

The annular argument of Theorem 1.18.1 also applies to a continuous potential whose modulus of continuity is not a power; what matters is a Dini condition on the square

root of the modulus. This gives uniform distance estimates for families whose Hölder constants are not controlled.

Keep the coordinate domain $U \subset \mathbb{C}^n$ with its Euclidean form ω_{E} and volume $\mathrm{d}\lambda$, but now let $v \in C^0(U)$ and let θ be a smooth real $(1,1)$-form with $T = \theta + \mathrm{dd}^{\mathrm{c}} v \geq 0$. On the relatively compact charts under consideration, choose a nondecreasing function $m\colon (0, r_0) \to [0, +\infty)$ with $m(r) \to 0$ as $r \to 0$ such that

$$|v(x) - v(y)| \leq m(|x - y|), \qquad |x - y| < r_0.$$

Put

$$D(r) = r + \int_0^r \frac{\sqrt{m(t)}}{t}\, \mathrm{d}t. \tag{1.35}$$

Theorem 1.18.2 *Assume that $D(r_0/2) < +\infty$ and that T is a smooth Hermitian form η on $U \setminus A$ for a proper analytic subset $A \subset U$. Then every point of U has a neighborhood $V \Subset U$ with constants $C, r_1 > 0$ such that, for $x, y \in V$,*

$$d_\eta^U(x, y) \leq C D(8|x - y|), \qquad x, y \notin A, \quad |x - y| < r_1.$$

The constants depend only on the charts and an upper bound for $|\theta|$; the dependence on v enters only through D. All sequences in $V \setminus A$ converging to a given point of $A \cap V$ have a common limit in the metric completion. If these hypotheses hold in charts covering X, the metric completion of the regular locus is compact, and the identity of the regular locus extends to a continuous surjection of X onto it. If A is nonempty, the metric η on the regular locus is incomplete.

Proof Write $d = 2n$ and $\mu = T \wedge \omega_{\mathrm{E}}^{n-1}$. The same cutoff calculation as in Theorem 1.18.1, with a constant subtracted from v, gives

$$\mu(B(x, s)) \leq C\big(s^d + s^{d-2} m(2s)\big).$$

If τ is the Euclidean trace of η, then $\tau\, \mathrm{d}\lambda \leq C\mu$ on the regular locus. Polar integration bounds the average length of the segments from a regular point x to the points of $B(x, R)$ by

$$C \int_{B(x,R)} \sqrt{\tau(\zeta)}\, |\zeta - x|^{1-d}\, \mathrm{d}\lambda(\zeta).$$

On an annulus of outer radius s, the Cauchy–Schwarz inequality and the trace bound make the contribution to this integral at most

$$C s^{1-d} s^{d/2} \big(s^d + s^{d-2} m(2s)\big)^{1/2} \leq C\big(s + \sqrt{m(2s)}\big).$$

For $s_j = 2^{-j} R$, monotonicity gives

$$\sum_{j \geq 0} \sqrt{m(2s_j)} \leq \frac{1}{\log 2} \int_0^{4R} \frac{\sqrt{m(t)}}{t}\, \mathrm{d}t.$$

Indeed, $\sqrt{m(2s_j)}\log 2$ is at most the integral of $\sqrt{m(t)}/t$ over $[2s_j, 4s_j]$, and these intervals have disjoint interiors; no doubling condition on m is needed. The average length is therefore at most $CD(4R)$.

For $x, y \in V \setminus A$, average the lengths of the broken segments $[x, z] \cup [z, y]$ over z in the ball of radius $|x - y|/2$ about the midpoint of x and y, applying the preceding bound with $R = 2|x - y|$ at each endpoint. Since the volume of this ball is comparable to R^d, some broken path has length at most $CD(8|x - y|)$. As in the proof of Theorem 1.18.1, almost every such path avoids A, and for small V it stays inside U.

Since $D(r_0/2) < +\infty$, we have $D(r) \to 0$ as $r \to 0$. Hence any two regular sequences converging to the same point are Cauchy and have the same limit in the completion. The argument for the completion in the proof of Theorem 1.18.1 applies with this modulus. A positive lower bound for η near a regular point prevents a sequence converging to a point of A from converging in the metric to the regular point. This also proves incompleteness when A is nonempty. □

For a power modulus $m(r) \le Cr^{\alpha}$ with $0 < \alpha \le 1$, one has $D(r) \le C' r^{\alpha/2}$ for small r, and the theorem gives the bound (1.34) for $|x - y| < r_1$.

Corollary 1.18.1 *In the setting of Theorem 1.18.2, let V be as in the theorem. If*

$$m(r) \le C(-\log r)^{-a}, \qquad a > 2,$$

then for sufficiently close $x, y \in V \setminus A$,

$$d_{\eta}^{U}(x, y) \le C'(-\log|x - y|)^{1-a/2}.$$

Proof The substitution $s = -\log t$ gives

$$\int_0^r \frac{(-\log t)^{-a/2}}{t}\,\mathrm{d}t = \frac{(-\log r)^{1-a/2}}{a/2 - 1}.$$

The term r in $D(r)$ is bounded by the asserted rate for small r. The theorem now gives the estimate; the fixed dilation of the argument of D changes only the constants. □

Corollary 1.18.2 *Let ω_0 be a Hermitian form on X and d_{ω_0} the distance of the associated Riemannian metric. For each j, let $T_j = \theta_j + \mathrm{dd}^{\mathrm{c}} v_j \geq 0$, where θ_j is a smooth real $(1,1)$-form, v_j is continuous, and T_j is a smooth Hermitian form off a proper analytic subset A_j. Assume*

$$|\theta_j|_{\omega_0} \leq \Lambda, \qquad |v_j(x) - v_j(y)| \leq m(d_{\omega_0}(x,y)),$$

with one fixed Λ and one nondecreasing modulus m tending to zero such that the function in (1.35) satisfies $D(r_0/2) < +\infty$. By Theorem 1.18.2, the identity of $X \setminus A_j$ extends to a continuous surjection of X onto the metric completion of $X \setminus A_j$; let d_j be the pseudometric on X obtained by pulling back the completed distance. Then the diameters of the d_j are uniformly bounded, and a subsequence converges uniformly on $X \times X$ to a continuous pseudometric. If $T_j \geq c\omega_0$ on the regular loci for one $c > 0$, the limit is a metric and dominates $\sqrt{c}\, d_{\omega_0}$.

Proof A finite fixed chart cover and the theorem give common constants $C, r_1 > 0$ such that

$$d_j(x,y) \leq CD(Cd_{\omega_0}(x,y)), \qquad d_{\omega_0}(x,y) < r_1.$$

For a pair meeting A_j, approximate it by regular pairs in the manifold topology. The completed distance and D are continuous, so the same estimate holds for all such pairs in $X \times X$. No uniform bound on the geometry of the analytic sets enters the constants. Dividing a minimizing geodesic of d_{ω_0} into a uniformly bounded number of short pieces and using the triangle inequality gives a uniform diameter bound. The triangle inequality also gives

$$|d_j(x,y) - d_j(x',y')| \leq d_j(x,x') + d_j(y,y'),$$

so the distance functions are equicontinuous. The Arzelà–Ascoli theorem gives a subsequence d_{j_k} converging uniformly on $X \times X$ to a function d_∞. Nonnegativity, symmetry, vanishing on the diagonal and the triangle inequality pass to the limit. The lower bound $T_j \geq c\omega_0$ gives $d_j \geq \sqrt{c}\, d_{\omega_0}$, first for regular pairs and then for all pairs, which proves the last assertion.

Without this lower bound the limit may degenerate. For $\varepsilon_j \downarrow 0$, take $\theta_j = \varepsilon_j \omega_0$ and $v_j = 0$, with $A_j = \varnothing$. These data satisfy the hypotheses with $m(r) = r^2$, and $d_j = \sqrt{\varepsilon_j}\, d_{\omega_0}$ converges uniformly to zero. □

1.19 Obstacle envelopes

Given an obstacle function, one looks for the largest θ-plurisubharmonic function lying below it. The local upper-envelope theorem constructs this envelope as soon as the competitors have a common upper bound, and normalized compactness provides such a bound for an obstacle that is finite almost everywhere or prescribed on a non-pluripolar compact set. Letting the obstacles $\min(u + C, 0)$ increase with C produces the envelope $P_\theta[u]$ of the ordinary singularity type of u.

A smooth real $(1,1)$-form θ is called *big* if there are $\rho \in \mathrm{PSH}(X,\theta)$ and $\delta > 0$ such that

$$\theta + \mathrm{dd}^{\mathrm{c}}\rho \geq \delta\omega.$$

Here ω is any Hermitian form; changing it changes δ but not the condition. The definition does not require θ to be closed.

Let θ be a smooth real $(1,1)$-form. For an extended-real function f on X, defined at least outside a pluripolar set (the *obstacle*), write

$$P_\theta(f) = \left(\sup\{v \in \mathrm{PSH}(X,\theta) : v \leq f \text{ q.e.}\}\right)^* .$$

As in Section 1.6, q.e. (quasi-everywhere) means outside a pluripolar set. The functions v in the supremum are called *competitors*; the empty supremum is $-\infty$. For obstacles $f_1, \ldots, f_N$, we write $P_\theta(f_1, \ldots, f_N) = P_\theta(\min_i f_i)$; when the f_i belong to $\mathrm{PSH}(X,\theta)$, this is their *rooftop envelope*. For an upper semicontinuous obstacle, the pointwise constraint $v \leq f$ gives the same competitors: by the local submean inequality and upper semicontinuity of f, a quasi-plurisubharmonic function lying below f almost everywhere lies below it everywhere.

If the family of competitors is nonempty and uniformly bounded above, then $P_\theta(f) \in \mathrm{PSH}(X,\theta)$. Indeed, freeze the coefficients of θ on small coordinate balls as in the proof of Lemma 1.3.1, so that one quadratic function makes all competitors plurisubharmonic there, and apply the local upper-envelope theorem; letting the freezing error tend to zero gives the exact lower bound $-\theta$. The same local theorem shows that the regularization changes the supremum only on a pluripolar set.

The following observation gives the upper bound for an unbounded obstacle. Suppose that $f \leq A$, for a real constant A, on a Borel set E with $\mathrm{Vol}(E) = \int_E \mathrm{d}V > 0$, where $\mathrm{d}V$ is the volume form fixed in Theorem 1.4.1. If h is a competitor, put $\widetilde{h} = h - \sup_X h$. By Theorem 1.4.1,

$$(\sup_X h - A)\,\mathrm{Vol}(E) \leq \int_E |\widetilde{h}|\,\mathrm{d}V \leq C_\theta \quad \text{whenever } \sup_X h > A.$$

Thus $\sup_X h \leq A + C_\theta/\mathrm{Vol}(E)$ for every competitor. In particular, the competitors for a measurable obstacle that is finite almost everywhere have a common upper bound: apply this to a sublevel set $\{f \leq A\}$ of positive volume.

An obstacle bound on a non-pluripolar compact set also bounds every competitor globally, even when that set has zero smooth volume.

Lemma 1.19.1 *Let θ be a smooth real $(1,1)$-form on X, let $K \subseteq X$ be compact and non-pluripolar, and let $w\colon K \to \mathbb{R}$ be bounded above. There is a constant $C = C(\theta, K, w)$ such that every $a \in \mathrm{PSH}(X,\theta)$ satisfying $a \leq w$ quasi-everywhere on K has $\sup_X a \leq C$.*

Proof Otherwise there are such functions a_j with

$$M_j := \sup_X a_j \geq 2^j + \sup_K w.$$

The normalized functions $v_j = a_j - M_j \le 0$ have uniformly bounded L^1 norms by Theorem 1.4.1. Choose $A > 0$ with $\theta \le A\omega$. The partial sums

$$s_N = \sum_{j=1}^{N} 2^{-j} v_j$$

decrease, have uniformly bounded L^1 norms, and satisfy $\mathrm{dd}^c s_N \ge -A\omega$. By Corollary 1.4.1, their limit s is a quasi-plurisubharmonic function; in particular, s is not identically $-\infty$.

Let $N_j \subseteq K$ be a pluripolar set outside which $a_j \le w$ on K. On $K \setminus \bigcup_j N_j$ one has $v_j \le -2^j$ for every j, hence $s = -\infty$. The pole set $\{s = -\infty\}$ of the quasi-plurisubharmonic function s is pluripolar, and so is the countable union of the N_j. Their union would contain all of K, contradicting the hypothesis on K. □

Suppose that $\mathrm{PSH}(X, \theta)$ is nonempty. Then the envelope of the zero obstacle,

$$V_\theta = P_\theta(0),$$

belongs to $\mathrm{PSH}(X, \theta)$: the competitors are bounded above by zero, and they include $u - \sup_X u$ for every $u \in \mathrm{PSH}(X, \theta)$. In particular,

$$u - \sup_X u \le V_\theta \qquad (u \in \mathrm{PSH}(X, \theta)).$$

A potential $v \in \mathrm{PSH}(X, \theta)$ has *minimal singularities* if $u \preceq v$ for every $u \in \mathrm{PSH}(X, \theta)$. By the last inequality, V_θ has minimal singularities, and every potential v with minimal singularities satisfies $v \sim V_\theta$. If $\theta \ge 0$, then 0 is a competitor, and hence $V_\theta = 0$.

Lemma 1.19.2 *Let η be a smooth semipositive real $(1,1)$-form on X. If an obstacle f is finite almost everywhere and bounded below quasi-everywhere by a constant, then $P_\eta(f)$ is a bounded η-plurisubharmonic function.*

Proof Since $\eta \ge 0$, constants are η-plurisubharmonic, so the constant lower bound of f is a competitor. The argument for obstacles that are finite almost everywhere, given before Lemma 1.19.1, gives a common upper bound for all competitors. Hence $P_\eta(f)$ is an η-plurisubharmonic function lying between these two constants. □

Proposition 1.19.1 *If $f \in C^0(X)$, then $P_\omega(f)$ is continuous. Moreover,*

$$\|P_\omega(f) - P_\omega(g)\|_\infty \le \|f - g\|_\infty \quad (f, g \in C^0(X)).$$

Proof The constant $\min_X f$ is a competitor, so $p = P_\omega(f)$ is bounded. The local envelope theorem gives $p \le f$ quasi-everywhere, and continuity of f makes this inequality hold everywhere. By Theorem 1.11.1, there are smooth ω-plurisubharmonic functions $v_j \downarrow p$. Fix $\epsilon > 0$. The continuous functions $v_j - f$ decrease to a function that is at most zero, so $v_j \le f + \epsilon$ everywhere for all large j; otherwise the decreasing compact sets $\{v_j - f \ge \epsilon\}$ would have nonempty intersection. Hence $v_j - \epsilon$ is a competitor, and

$$p \le v_j \le p + \epsilon.$$

Thus $v_j \to p$ uniformly, which proves continuity. The last inequality follows from the monotonicity of P_ω in the obstacle and the identity $P_\omega(g + c) = P_\omega(g) + c$ for constants c, applied with $c = \|f - g\|_\infty$. □

Definition 1.19.1 For $u \in \mathrm{PSH}(X, \theta)$, set

$$P_\theta[u] = \left(\lim_{C \to +\infty} P_\theta(u + C, 0) \right)^* = \left(\sup\{h \in \mathrm{PSH}(X, \theta) : h \preceq u,\ h \le 0\} \right)^*.$$

We call u an *envelope fixed point*, or an *ordinary model*, if $P_\theta[u] = u$; in the Kähler literature such potentials are called *model potentials*. For $F \in C^\infty(X, \mathbb{R})$, replacing the constraint $h \le 0$ by $h \le F$ defines $P_{\theta,F}[u]$.

The two expressions agree, because the family in the second one is the increasing union over C of the families defining $P_\theta(u + C, 0)$. They do not change if zero is replaced by V_θ, since every θ-plurisubharmonic function below zero lies below V_θ. The competitors include $u - \sup_X u$ and are bounded above by zero, so $P_\theta[u] \in \mathrm{PSH}(X, \theta)$ and $u - \sup_X u \le P_\theta[u] \le 0$. In particular, $u \preceq P_\theta[u]$; this does not assert that $P_\theta[u] - u$ is bounded.

Directly from the definition, $P_\theta[u]$ depends only on the ordinary singularity type of u, and

$$u \preceq v \quad \Longrightarrow \quad P_\theta[u] \le P_\theta[v].$$

Replacing the condition $h \preceq u$ by $h \sim_{\mathcal{I}} u$ gives the multiplier-ideal envelope $P_\theta[u]_{\mathcal{I}}$ of Definition 17.2.1. This is a different operator; its relation to $P_\theta[u]$ is studied in Chapter 17.

For an obstacle bounded below in a big background, exceptional sets in the competitor inequality can be removed by a small singular perturbation.

Lemma 1.19.3 *Let θ be big, let f be a real function on X that is bounded below, and let $u \in \mathrm{PSH}(X, \theta)$ satisfy $u \le f$ quasi-everywhere. Then there are $u_t \in \mathrm{PSH}(X, \theta)$, $t > 0$, such that $u_t \le f$ everywhere and $u_t \to u$ almost everywhere as $t \downarrow 0$.*

Proof Choose $\rho \in \mathrm{PSH}(X,\theta)$ with $\theta + \mathrm{dd^c}\rho \geq \delta\omega$ for some $\delta > 0$. The exceptional set $E = \{u > f\}$ is pluripolar. The global form of Josefson's theorem on compact complex manifolds gives a quasi-plurisubharmonic function χ with $\chi = -\infty$ on E; see [Xia26, Theorem 1.1.5]. After rescaling and adding a constant, we may assume that $\chi \leq 0$ and $\mathrm{dd^c}\chi \geq -\omega$. Subtract a constant from ρ so that $\rho \leq \inf_X f$. With $s = t/\delta$, put

$$u_t = \frac{u + t\chi + s\rho}{1+s}.$$

All three functions in the numerator are finite almost everywhere, so u_t is not identically $-\infty$. Moreover,

$$\theta + \mathrm{dd^c}u_t = \frac{\theta + \mathrm{dd^c}u + t\mathrm{dd^c}\chi + s(\theta + \mathrm{dd^c}\rho)}{1+s} \geq 0.$$

Outside E, both u and ρ are at most f, while $\chi \leq 0$, so $u_t \leq f$. On E the numerator is $-\infty$. Finally, $u_t \to u$ wherever all three functions are finite, which is a set of full measure. □

For an obstacle prescribed only on a compact set, the same perturbation shows that pluripolar exceptions in the constraint do not change the regularized envelope.

Proposition 1.19.2 *Let θ be big, let $K \subseteq X$ be compact and non-pluripolar, and let ψ be a bounded real function on K. Set*

$$R = \sup\{u \in \mathrm{PSH}(X,\theta) : u \leq \psi \text{ on } K\},$$
$$Q = \sup\{u \in \mathrm{PSH}(X,\theta) : u \leq \psi \text{ q.e. on } K\}.$$

Then $R^* = Q^*$ belongs to $\mathrm{PSH}(X,\theta)$ and has minimal singularities; it is bounded if $\theta \geq 0$. Moreover, this regularized envelope does not change if the pointwise constraint is imposed only on $K \setminus E$, for any pluripolar subset $E \subseteq K$.

Proof Put $m = \inf_K \psi$ and $M = \sup_K \psi$. Since θ is big, $\mathrm{PSH}(X,\theta)$ is nonempty, and $V_\theta \leq 0$; hence $V_\theta + m$ belongs to the family defining R. Applying Lemma 1.19.1 with the constant obstacle M gives a common upper bound C for the larger family defining Q. The upper-envelope argument at the beginning of Section 1.19 therefore shows that R^* and Q^* belong to $\mathrm{PSH}(X,\theta)$.

Now fix a competitor u for Q and define a bounded function on X by

$$f(x) = \begin{cases} \psi(x), & x \in K, \\ \max\{\sup_X u, M\}, & x \notin K. \end{cases}$$

Then $u \leq f$ quasi-everywhere. By Lemma 1.19.3, there are $u_t \in \mathrm{PSH}(X,\theta)$ satisfying $u_t \leq f$ everywhere and $u_t \to u$ almost everywhere. Each u_t is a competitor for R, so $R^* \geq u$ almost everywhere, and Lemma 1.3.3 gives this inequality everywhere. Taking the supremum over u and regularizing yields $R^* \geq Q^*$; the inclusion of the competitor families gives the reverse inequality.

Write $F = R^* = Q^*$. Since $F \leq C$, the potential $F - C$ is a competitor for V_θ. Thus

$$V_\theta + m \le F \le V_\theta + C,$$

which proves minimal singularities and, when $\theta \ge 0$, boundedness.

Finally, if $E \subseteq K$ is pluripolar, the family constrained on $K \setminus E$ lies between the families defining R and Q, so its regularized supremum also equals F. □

A potential with minimal singularities stays away from the poles of a strictly positive potential with analytic singularities, in the following sense.

Lemma 1.19.4 *Let $\rho \in \mathrm{PSH}(X,\theta)$ have analytic singularities with smooth remainder and satisfy $\theta + \mathrm{dd}^c\rho \ge \varepsilon\omega$ for some $\varepsilon > 0$. Set $\Omega = \{\rho > -\infty\}$. If $\varphi \in \mathrm{PSH}(X,\theta)$ has minimal singularities and is continuous on Ω, then $\varphi - \rho$ is continuous, bounded below, and proper on Ω: all its sublevel sets are compact in Ω.*

Proof Adding constants to ρ and φ does not affect the assertions, so we may assume that $\rho \le 0$ and $\sup_X \varphi = 0$. Since φ has minimal singularities, there is a constant C with $\varphi \ge V_\theta - C$. Choose $M \ge 0$ with $\theta \ge -M\omega$. If $a \in (0,1)$ is sufficiently close to one, then

$$\theta + \mathrm{dd}^c(a\rho) = a(\theta + \mathrm{dd}^c\rho) + (1-a)\theta \ge \big(a\varepsilon - (1-a)M\big)\omega > 0.$$

Thus $a\rho$ is a competitor for V_θ, and therefore

$$0 \ge \varphi \ge V_\theta - C \ge a\rho - C,$$

and hence

$$\varphi - \rho \ge (1-a)(-\rho) - C. \tag{1.36}$$

By hypothesis φ is continuous on Ω, and ρ is smooth there because it has analytic singularities with smooth remainder. Hence $\varphi - \rho$ is continuous on Ω, and by (1.36) it is bounded below. Since $\rho \to -\infty$ at $X \setminus \Omega$, the sets $\{\rho \ge -c\}$ are compact subsets of Ω, and (1.36) places every sublevel set of $\varphi - \rho$ in one of them. Being closed in Ω, each sublevel set is therefore compact. □

1.19.1 A hyperplane Green envelope on the ball

For a logarithmic pole along a hyperplane section of the unit ball, the envelope can be computed explicitly by restricting to the complex lines normal to the hyperplane. The example also shows how envelopes of bounded obstacles recover the finite part of a logarithmic pole, and that continuity inside the domain does not imply continuity up to the boundary.

In this subsection let $n \ge 2$ and $m > 0$, and write

$$B = \{(\zeta, w) \in \mathbb{C} \times \mathbb{C}^{n-1} : |\zeta|^2 + |w|^2 < 1\}, \qquad D = B \cap \{\zeta = 0\}.$$

Define G_m as the pointwise supremum of the functions $u \in \mathrm{PSH}(B)$ such that $u \le 0$ and, near every point of D, $u \le m \log|\zeta|^2 + C$ for some local constant C. In particular, $u = -\infty$ on D. The constants need not stay bounded as the point of D approaches ∂B.

Proposition 1.19.3 *The envelope G_m is given by*

$$G_m(\zeta, w) = m \log \frac{|\zeta|^2}{1 - |w|^2}. \tag{1.37}$$

It is plurisubharmonic and continuous on B with values in $[-\infty, 0)$. The finite part $G_m - m \log|\zeta|^2$, first defined off D, extends smoothly across D as the plurisubharmonic function

$$H_m(w) = -m \log(1 - |w|^2).$$

Proof The function $-\log(1 - |w|^2)$ is smooth and plurisubharmonic on $|w| < 1$: its complex Hessian is

$$\left(\frac{\delta_{jk}}{1 - |w|^2} + \frac{\overline{w}_j w_k}{(1 - |w|^2)^2} \right)_{j,k}.$$

Thus the right-hand side of (1.37) is plurisubharmonic; it is negative on B and satisfies the pole condition, so it is a competitor. For the reverse inequality, fix a competitor u and a point w with $|w| < 1$. The restriction $u(\cdot, w)$ is either identically $-\infty$ or subharmonic on the disk of radius $R = (1 - |w|^2)^{1/2}$. In the latter case, $u(\zeta, w) - m \log|\zeta|^2$ is subharmonic on the punctured disk and, by the pole condition, bounded above near zero, so it extends as a subharmonic function across zero.

Since $u \le 0$, this extension is at most $-m \log r^2$ on every circle $|\zeta| = r < R$. The maximum principle and then $r \uparrow R$ give

$$u(\zeta, w) \le m \log \frac{|\zeta|^2}{R^2}.$$

On D both sides are $-\infty$. Taking the supremum over u proves the formula. Near a point of D, the function H_m is bounded and the logarithmic term tends to $-\infty$, which proves continuity there as an extended-real function. □

For a finite continuous obstacle Φ on B, put

$$P_B\Phi = \left(\sup\{v \in \mathrm{PSH}(B) : v \le \Phi\}\right)^*.$$

The local upper-envelope theorem used in Section 1.19 applies whenever this family is nonempty and locally uniformly bounded above. Continuity of Φ ensures that the regularized envelope still satisfies $P_B\Phi \le \Phi$.

Proposition 1.19.4 *For $t > 0$, define*

$$\Phi_t(\zeta, w) = \min\{-m \log|\zeta|^2, t\}, \qquad V_t = P_B \Phi_t,$$

with $\Phi_t = t$ on D. Then $0 \le V_t \le t$, and

$$-m \log\bigl(1 - (1 - \mathrm{e}^{-t/m})|w|^2\bigr) \le V_t(\zeta, w) \le H_m(w). \tag{1.38}$$

In particular, $V_t \uparrow H_m$ pointwise on all of B, including D. The convergence is uniform on every compact subset, with

$$0 \le H_m - V_t \le \frac{m\mathrm{e}^{-t/m} R^2}{1 - R^2} \qquad \textit{where } |w| \le R < 1. \tag{1.39}$$

Proof The obstacle is continuous, nonnegative and at most t. Zero is a competitor, so the upper-envelope theorem gives $V_t \in \mathrm{PSH}(B)$ with $0 \le V_t \le \Phi_t \le t$. Consequently $m \log|\zeta|^2 + V_t$ is a competitor for G_m, and Proposition 1.19.3 gives $V_t \le H_m$ off D. At a point $(0, w)$, the submean inequality on any sufficiently small circle in the normal line gives the same bound, since H_m does not depend on ζ.

For the lower bound, set $a_t = 1 - \mathrm{e}^{-t/m}$ and $F_t(w) = -m \log(1 - a_t|w|^2)$. The Hessian calculation in the proof of Proposition 1.19.3, with $|w|^2$ replaced by $a_t|w|^2$, shows that F_t is plurisubharmonic. Moreover, $0 \le F_t \le t$, and $F_t \le H_m \le -m \log|\zeta|^2$ on $B \setminus D$ because $|\zeta|^2 < 1 - |w|^2$. Thus $F_t \le \Phi_t$ everywhere, and hence $F_t \le V_t$. Finally,

$$H_m - F_t = m \log\left(1 + \frac{\mathrm{e}^{-t/m}|w|^2}{1 - |w|^2}\right) \le \frac{m\mathrm{e}^{-t/m}|w|^2}{1 - |w|^2}.$$

This proves (1.39) and the convergence; the monotonicity of V_t in t follows from that of Φ_t. □

Example 1.19.1 At every point $p = (0, w_0) \in \partial B$ with $|w_0| = 1$, the cluster set of G_m at p, that is, the set of limits of G_m along sequences in B converging to p, is all of $[-\infty, 0]$. Indeed, set $w_s = (1 - s^2)^{1/2} w_0$ for $0 < s < 1$. For every fixed $0 < a < 1$, the points (as, w_s) belong to B, tend to p as $s \to 0$, and satisfy $G_m = m \log a^2$. Choosing $a = a_s < 1$ with $a_s \to 1$ gives the limit zero, whereas the points (s^2, w_s) give $G_m = m \log s^2 \to -\infty$. These realize every value in $[-\infty, 0]$, and negativity excludes all others. Thus G_m, although continuous on B, has no continuous extension to any point of ∂B in the closure of D.

1.20 The minimum principle

An infimum of plurisubharmonic functions need not be plurisubharmonic. Kiselman's minimum principle describes an important exception: the infimum over a real parameter t of a family u_t whose complexification $u_{\operatorname{Re} z}$ is plurisubharmonic in the joint variables. Freezing the coefficients of the background form, we show that the principle preserves the exact lower curvature bound $-\theta$.

Lemma 1.20.1 *Let $I \subset \mathbb{R}$ be a nonempty open interval, set $S_I = \{z \in \mathbb{C} : \operatorname{Re} z \in I\}$, and let $\pi \colon X \times S_I \to X$ be the projection. Let $(u_t)_{t\in I}$ be a family of functions on X such that $U(x,z) = u_{\operatorname{Re} z}(x)$ is quasi-plurisubharmonic on $X \times S_I$ and $\pi^*\theta + \mathrm{dd^c} U \geq 0$. For each x, the function $t \mapsto u_t(x)$ is finite and convex or identically $-\infty$. The finite alternative holds for almost every x.*

Proof A local smooth correction making U plurisubharmonic can be chosen to depend only on x. Hence the restriction of U to $\{x\} \times S_I$ is subharmonic or identically $-\infty$. In the first case its distributional Laplacian is the second derivative of $t \mapsto u_t(x)$, which is therefore nonnegative. The resulting convex representative is finite on I, and by the submean inequality it agrees everywhere with the subharmonic function. Finally, U is locally integrable on the product, and Fubini's theorem, applied on products of coordinate balls with a fixed rectangle in S_I, excludes the identically $-\infty$ alternative for almost every x. □

Corollary 1.20.1 *Let $I \subset \mathbb{R}$ be a nonempty open interval. A radial function $z \mapsto q(\log|z|^2)$ is subharmonic on the annulus $\{z : \log|z|^2 \in I\}$ if and only if q is finite and convex on I.*

Proof The holomorphic covering map $z = \mathrm{e}^{w/2}$ pulls the function back to $q(\operatorname{Re} w)$, and the argument of Lemma 1.20.1 proves necessity. Conversely, a finite convex function q has nonnegative distributional second derivative. Thus $q(\operatorname{Re} w)$ is subharmonic, and subharmonicity descends through the local biholomorphisms of the covering. □

Proposition 1.20.1 *Under the hypotheses of Lemma 1.20.1, the function*

$$q(x) = \inf_{t\in I} u_t(x)$$

is either θ-plurisubharmonic or identically $-\infty$. If I is bounded, then $q \in \mathrm{PSH}(X,\theta)$.

Proof The classical local Kiselman principle states that if a plurisubharmonic function on the product of a domain with a strip does not depend on the imaginary part of the strip variable, then its infimum over the real part is plurisubharmonic or identically $-\infty$. The interval of the real part may be unbounded. We use this local result from [Xia26].

Fix $x_0 \in X$, choose local holomorphic coordinates centered at x_0, and let κ be their Euclidean form. Fix $\eta > 0$. On a sufficiently small coordinate ball the constant form $\beta = \theta(x_0)$ satisfies

$$\beta - \eta\kappa \le \theta \le \beta + \eta\kappa.$$

Choose a quadratic function r with $\mathrm{dd}^{\mathrm{c}} r = \beta + \eta\kappa$. Then $U + r \circ \pi$ is plurisubharmonic on the product of the ball with S_I. Its infimum over the parameter is $q + r$, so the local principle shows that $q + r$ is plurisubharmonic or identically $-\infty$ on the ball; in the first case

$$\mathrm{dd}^{\mathrm{c}} q \ge -\beta - \eta\kappa \ge -\theta - 2\eta\kappa.$$

In a fixed atlas, the Euclidean forms are bounded by a fixed multiple of ω, so letting $\eta \downarrow 0$ proves the exact lower bound $\mathrm{dd}^{\mathrm{c}} q \ge -\theta$. Since a plurisubharmonic function cannot be identically $-\infty$ on a nonempty open set, the local alternative propagates across overlapping balls on the connected manifold X. Thus either $q \in \mathrm{PSH}(X, \theta)$ or $q \equiv -\infty$. No regularization is needed, because an infimum of upper semicontinuous functions is upper semicontinuous.

Now suppose that I is bounded. By Lemma 1.20.1, the function $t \mapsto u_t(x)$ is finite and convex for almost every x, and a supporting affine function at any point of I bounds it from below on the bounded interval I. Hence $q(x) > -\infty$ almost everywhere, which excludes the alternative $q \equiv -\infty$. □

For example, the proposition applies to $u_t - t\tau$ for a fixed real number τ. On an unbounded interval the infimum may be identically $-\infty$; this alternative will encode the cutoff of a test curve (Definition 19.1.1).

1.20.1 Pointwise Legendre transforms

The minimum principle can also show that a supremum is already upper semicontinuous. For a concave family of potentials, the regularized Legendre transform is a potential on a product. The minimum principle identifies its inverse transforms with the original family everywhere, and applying the one-variable duality once more identifies the regularization with the pointwise supremum. We first state this duality precisely, including its behavior at the endpoints; the general Legendre–Fenchel theorem is recalled in [Xia26, Appendix A].

Lemma 1.20.2 *Let $c \in \mathbb{R}$. Suppose $g\colon \mathbb{R} \to [-\infty, 0]$ is upper semicontinuous, concave and decreasing, finite somewhere, and equal to $-\infty$ on (c, ∞). Then*

$$f(t) = \sup_{\tau < c} \bigl(g(\tau) + t\tau\bigr), \qquad t > 0,$$

is finite and convex, satisfies $f(t) \le ct$, and obeys

$$g(\tau) = \inf_{t>0} \bigl(f(t) - t\tau\bigr), \qquad \lim_{t \downarrow 0} f(t) = \sup_{\tau < c} g(\tau).$$

Conversely, every finite convex f on $(0, \infty)$ with $f(t) \le ct$ is recovered by the first formula from $g(\tau) = \inf_{t>0}(f(t) - t\tau)$, and satisfies the same endpoint identity. The formulas also hold for $g \equiv -\infty$, with $f \equiv -\infty$.

Proof Since g is decreasing and finite somewhere, it is finite on a half-line $(-\infty, \tau_1)$. Hence the defining supremum is finite for $t > 0$: it is bounded above by ct and below by any one of its finite terms. Concavity and upper semicontinuity give $g(c) = \lim_{\tau\uparrow c} g(\tau)$, so including $\tau = c$ does not change the supremum. The Legendre transform $t \mapsto \sup_\tau(g(\tau) + t\tau)$ of $-g$ is infinite for $t < 0$. At $t = 0$ it equals $m = \sup_{\tau<c} g(\tau)$, and the bounds

$$g(\tau) + t\tau \le f(t) \le m + ct$$

show that $f(t) \to m$ as $t \downarrow 0$. The biconjugacy theorem for the closed convex function $-g$ therefore gives the inverse formula; the value at $t = 0$ may be omitted without changing the infimum.

Conversely, start with f. At each $t > 0$, choose the slope p of a supporting line of f. The upper bound $f(s) \le cs$ as $s \to \infty$ forces $p \le c$. If $p < c$, the supporting line gives $g(p) = f(t) - tp$, so the first formula recovers $f(t)$. If $p = c$, then $f(s) - cs$ is constant for $s \ge t$, because this function is convex and bounded above, hence decreasing, while its supporting line at t is horizontal. Consequently, for every $\tau < c$,

$$f(t) - ct \le g(\tau) \le f(t) - t\tau,$$

and letting $\tau \uparrow c$ again recovers $f(t)$. In particular, g is finite somewhere. As an infimum of affine functions with slopes $-s < 0$, it is upper semicontinuous, concave and decreasing. Letting $s \downarrow 0$ in the bound $f(s) - s\tau \le s(c - \tau)$ gives $g \le 0$, and letting $s \to \infty$ gives $g = -\infty$ for $\tau > c$. The endpoint identity now follows from the first part. The case of functions identically equal to $-\infty$ follows directly from the formulas. □

Fix $c \in \mathbb{R}$, and let $\Gamma_\tau \in \mathrm{PSH}(X, \theta)$ for $\tau < c$ be functions with $\Gamma_\tau \le 0$ such that $\tau \mapsto \Gamma_\tau(x)$ is concave and decreasing for every x. Extend the family to $\tau = c$ by the decreasing limit $\Gamma_c = \lim_{\tau\uparrow c} \Gamma_\tau$, and by $\Gamma_\tau \equiv -\infty$ for $\tau > c$. Define the forward transform of the family by

$$U_t(x) = \sup_{\tau<c}\big(\Gamma_\tau(x) + t\tau\big), \qquad t > 0. \tag{1.40}$$

For a family $(u_t)_{t>0}$, the inverse transform is

$$\widehat{u}_\tau(x) = \inf_{t>0}\big(u_t(x) - t\tau\big). \tag{1.41}$$

Lemma 1.20.3 *At each x, the extended function $\tau \mapsto \Gamma_\tau(x)$ is upper semicontinuous and concave, or identically $-\infty$. Its forward transform satisfies*

$$\Gamma_\tau(x) = \inf_{t>0}\big(U_t(x) - t\tau\big) \qquad (\tau \in \mathbb{R}).$$

Proof Fix $\tau < c$ and let $\tau_j \uparrow \tau$. The decreasing limit of the functions Γ_{τ_j} dominates Γ_τ, so it belongs to $\mathrm{PSH}(X, \theta)$. Choose $r \in (\tau, c)$. Outside the pole set of Γ_r, the concave function $s \mapsto \Gamma_s(x)$ is finite near τ and hence continuous there. Thus the

decreasing limit equals Γ_τ almost everywhere, and hence everywhere by Lemma 1.3.3. This proves left continuity at every $\tau < c$; at c it holds by definition. Since the function is decreasing, left continuity gives upper semicontinuity, also where the value is $-\infty$, and concavity survives the extension. The inverse formula is Lemma 1.20.2 at each x. □

Theorem 1.20.1 *For the preceding family $(\Gamma_\tau)_{\tau<c}$, the pointwise supremum in (1.40) is already upper semicontinuous on $X\times(0,\infty)$. Its complexification $U(x,z) = U_{\operatorname{Re} z}(x)$ is quasi-plurisubharmonic on $X \times \{\operatorname{Re} z > 0\}$ and satisfies $\pi^*\theta + \mathrm{dd}^{\mathrm{c}}U \geq 0$, where $\pi\colon X \times \{\operatorname{Re} z > 0\} \to X$ is the projection. Each slice belongs to $\mathrm{PSH}(X,\theta)$.*

Proof Each term $(x,z) \mapsto \Gamma_\tau(x) + \tau \operatorname{Re} z$ of the supremum satisfies the current inequality on the product, since $\operatorname{Re} z$ is pluriharmonic. The bound $\Gamma_\tau \leq 0$ gives $U_t \leq ct$, while any fixed $\tau_0 < c$ gives the minorant $\Gamma_{\tau_0} + t\tau_0$. By the local upper-envelope theorem, as in Section 1.19, the upper semicontinuous regularization V of U on the product is quasi-plurisubharmonic, satisfies $\pi^*\theta + \mathrm{dd}^{\mathrm{c}}V \geq 0$, and agrees with U almost everywhere. It does not depend on $\operatorname{Im} z$ and satisfies $V_t \leq ct$. The pointwise supremum U is measurable, since by the left continuity in Lemma 1.20.3 the parameters may be restricted to rational numbers.

By Fubini's theorem, there is a set $Z \subset X$ of zero smooth volume, containing the pole set of Γ_{τ_0}, such that $U_t(x) = V_t(x)$ for almost every $t > 0$ whenever $x \notin Z$. For such x, both sides are finite convex functions of t, by Lemmas 1.20.1 and 1.20.2, so by continuity the equality holds for every $t > 0$.

For each $\tau < c$, Proposition 1.20.1 gives

$$\chi_\tau = \inf_{t>0}(V_t - t\tau) \in \mathrm{PSH}(X,\theta) \cup \{-\infty\}.$$

Off Z it equals Γ_τ by Lemma 1.20.3. In particular, χ_τ is not identically $-\infty$, and Lemma 1.3.3 gives $\chi_\tau = \Gamma_\tau$ everywhere. At each x, the convex function $t \mapsto V_t(x)$ can be recovered from these inverse transforms by Lemma 1.20.2, also when it is identically $-\infty$. We obtain

$$V_t(x) = \sup_{\tau<c}\bigl(\chi_\tau(x) + t\tau\bigr) = U_t(x).$$

Finally, the minorant $\Gamma_{\tau_0} + t\tau_0$ shows that no slice U_t is identically $-\infty$, so every slice belongs to $\mathrm{PSH}(X,\theta)$. □

1.21 Chern-connection regularization

Convolution in local coordinates does not directly produce global approximants with a controlled lower curvature bound. Demailly's regularization instead averages over the fibers of the tangent bundle, using the Chern connection of ω, and a Kiselman-type minimum of these averages controls the loss of positivity. The classical curvature

estimate for this minimum holds in a Hermitian background, while the L^1 and gradient bounds for the averages are elementary.

Fix $C_0 > 0$ and $u \in \mathrm{PSH}(X, \omega)$ with $-C_0 \le u \le 0$. Distances, gradients and norms of tensors are taken with respect to the Riemannian metric associated with ω.

For small $t > 0$, let $R_t u$ be Demailly's regularization of u formed with the Chern connection of ω. It averages u over the image of the ball of radius t in each tangent space, with a fixed smooth nonnegative radial kernel of integral one. The map used for this purpose is a smooth realization of the holomorphic part of the formal fiberwise expansion of the exponential map; it need not be holomorphic in the fiber. The following classical estimate controls the Kiselman-type transform of these averages, including its mixed base–fiber terms; see Section 36.1 for the sources.

Lemma 1.21.1 *There are constants $A, K > 0$ and $t_* > 0$, depending only on (X, ω) and C_0, such that $R_t u$ is smooth for $0 < t < t_*$,*

$$-C_0 \le R_t u \le 0, \qquad R_t u + K t^2 \downarrow u \quad \text{as } t \downarrow 0.$$

For $0 < \delta < t_*$ and $c > 0$, the bounded function

$$\Phi_{c,\delta} = \inf_{0<t\le\delta} \left\{ R_t u + K(t^2 - \delta^2) + K(t - \delta) - c \log(t/\delta) \right\}$$

is quasi-plurisubharmonic and satisfies

$$\omega + \mathrm{dd^c} \Phi_{c,\delta} \ge -(Ac + 2K\delta)\omega.$$

At every point, the infimum is attained at some $t > 0$.

For the background ω, this lemma is the only input we take from the classical theory. The linear term $K(t - \delta)$ is needed in the Hermitian case, where it absorbs the mixed base–fiber terms in the complex Hessian of the regularization. Since $-c \log(t/\delta)$ tends to $+\infty$ as $t \downarrow 0$ while u is bounded, the infimum is attained away from zero. We need neither smoothness of the minimizing radius nor smoothness of the transform across the region where this radius equals δ.

Next we prove two elementary estimates for the averaging operators.

Lemma 1.21.2 *For all sufficiently small $t > 0$,*

$$\|R_t u - u\|_{L^1(\mathrm{d}V)} \le C t^2, \qquad \|\nabla R_t u\|_\infty \le C/t.$$

The constants are uniform for $-C_0 \le u \le 0$.

Proof Let z be local coordinates on X and $\xi \in \mathbb{C}^n$ the fiber variable with respect to a smooth orthonormal frame of the tangent bundle. Write $\rho(|\xi|^2)$ for the fixed radial averaging kernel, normalized with respect to Lebesgue measure $\mathrm{d}\xi$ and supported in $|\xi| \le 1$. In these coordinates the averaging map F gives

$$R_t u(z) = \int u(F(z, t\xi)) \rho(|\xi|^2)\, \mathrm{d}\xi, \qquad F(z, 0) = z.$$

The map F is smooth in these variables, and its fiber differential at zero is an isomorphism. Let R_t^* be the formal adjoint of R_t with respect to $\mathrm{d}V$, so that $\int_X (R_t a) b \, \mathrm{d}V = \int_X a \, R_t^* b \, \mathrm{d}V$. We first claim that

$$R_t^* 1 = 1 + O(t^2)$$

uniformly. To see this, insert a smooth partition of unity in the variable z, each term being compactly supported in a chart on which the frame is defined. For fixed ξ, change variables $x = F(z, t\xi)$ on a neighborhood of the support of a term. The inverse map and its Jacobian have uniform Taylor expansions in t. The term of order zero in the transformed density is the chosen partition function at x, and the term of order one is linear in ξ, so it integrates to zero against the even radial kernel. The remaining density is bounded by $Ct^2 \mathrm{d}V(x)$, since the second derivatives of all maps, partition functions and volume coefficients are bounded on these finitely many compact charts. Summing over the partition functions proves the claim.

Consequently,

$$\left| \int_X (R_t u - u) \, \mathrm{d}V \right| = \left| \int_X u (R_t^* 1 - 1) \, \mathrm{d}V \right| \leq C t^2.$$

By Lemma 1.21.1, $R_t u - u \geq -K t^2$. A real function $a \geq -K t^2$ satisfies $|a| \leq a + 2K t^2$; applying this to $a = R_t u - u$ and integrating gives the first estimate.

For the derivative bound, change variables in the fiber to write $R_t u(z) = \int_X u(x) k_t(z, x) \, \mathrm{d}V(x)$. In local coordinates the smooth kernel k_t is supported where $|x - z| \leq Ct$ and satisfies

$$|\nabla_z k_t(z, x)| \leq C t^{-2n-1}.$$

Indeed, the inverse fiber map and its Jacobian have uniformly bounded derivatives, and differentiating the rescaled radial argument introduces at most one factor t^{-1} in addition to t^{-2n}. Since the kernel is smooth with compact support, there are no boundary terms. Integration over a ball of volume at most Ct^{2n} gives $\int |\nabla_z k_t| \, \mathrm{d}V \leq C/t$, and multiplication by the uniform bound for u proves the assertion. □

1.21.1 The general curvature modulus

Let θ now be any smooth real $(1,1)$-form and let $u \in \mathrm{PSH}(X, \theta) \cap L^\infty(X)$. We use the same averaging operators and the same metric.

We first state the classical input for this background. The classical Chern-connection regularization theorem applies to the closed current $\mathrm{dd}^{\mathrm{c}}u$ with the smooth, possibly nonclosed, lower bound $-\theta$; the curvature of the Chern connection on the tangent bundle is bounded by a fixed multiple of ω. The theorem gives a constant $K_0 > 0$ such that

$$R_t u + K_0 t^2 \downarrow u.$$

To state its curvature estimate, use local coordinates z on X and a complex radius variable w, and put $t = |w|$. Write $\zeta \in T_z^{1,0}X$ and $\eta \in \mathbb{C}$ for the base and radius components of a tangent vector, and set $\lambda(z,t) = t\partial_t(R_t u(z) + K_0 t^2)$. Then λ is nonnegative and increasing in t. Moreover, there are constants $A, C > 0$ and a function $\epsilon_0(t) \geq 0$ with $\epsilon_0(t) \downarrow 0$ as $t \downarrow 0$ with the following property: the complex Hessian of $(z,w) \mapsto R_{|w|}u(z)$ is bounded below by the base form $-\theta - A\lambda(z,t)\omega - \epsilon_0(t)\omega$, evaluated on ζ, minus mixed and vertical error terms bounded by $C(|\zeta||\eta| + |\eta|^2)$. This is the classical regularization estimate for a smooth lower bound.

Lemma 1.21.3 *For K sufficiently large, there is an increasing function $\epsilon(\delta) \geq 0$ tending to zero at zero such that, for every small $\delta > 0$ and every $c > 0$, the function*

$$\Phi_{c,\delta} = \inf_{0<t\leq\delta}\{R_t u + K(t^2 - \delta^2) + K(t - \delta) - c\log(t/\delta)\}$$

is bounded and quasi-plurisubharmonic and satisfies

$$\theta + \mathrm{dd}^{\mathrm{c}}\Phi_{c,\delta} \geq -(Ac + \epsilon(\delta))\omega.$$

The infimum is attained away from zero. Moreover, $R_t u + Kt^2 \downarrow u$ as $t \downarrow 0$.

Proof Take $K \geq K_0$ and put $H(z,t) = R_t u(z) + Kt^2 + Kt$. As a function of $\log t$, H is increasing and strictly convex, because $R_t u + K_0 t^2$ has nonnegative increasing logarithmic derivative and the added terms have positive second logarithmic derivative. Its logarithmic derivative $\widetilde{\lambda} = t\partial_t H$ is at least λ.

The complex Hessian of $K|w|$ has a positive vertical term proportional to $K/|w|$. For sufficiently small $|w| = t$, this term absorbs the vertical error and, by the inequality

$$C|\zeta||\eta| \leq C't|\zeta|^2 + \frac{K}{8t}|\eta|^2,$$

also the mixed error. After increasing the constants and replacing λ by the larger $\widetilde{\lambda}$, we find that the complex Hessian of $(z,w) \mapsto H(z,|w|)$ is bounded below by the base form

$$-\theta - (A\widetilde{\lambda} + \epsilon(t))\omega,$$

where ϵ is an increasing majorant of $\epsilon_0(t) + C't$.

Since u is bounded, the term $-c\log(t/\delta)$ tends uniformly to $+\infty$ as $t \downarrow 0$, so the minimum occurs on a compact interval of radii away from zero. Strict convexity in $\log t$ makes the minimizing radius unique. At an interior minimum $\widetilde{\lambda} = c$, while at the upper endpoint $t = \delta$ one has $\widetilde{\lambda} \leq c$. No differentiability of the minimizing radius is needed.

In more detail, localize near a point of X and fix $a > 0$. By continuity of the minimizers and of $\widetilde{\lambda}$, there is an interval of radii containing the minimizers at all nearby points on which $\widetilde{\lambda} \le c + a$. If the minimum is attained at the upper endpoint, keep the upper endpoint δ; an open interval approaching it has the same infimum. Because of the uniform logarithmic term, the lower endpoint can be chosen below all minimizers, and since $\widetilde{\lambda}$ is increasing, the bound $\widetilde{\lambda} \le c + a$ holds throughout the chosen interval. The local Kiselman minimum principle for functions depending only on $|w|$, applied after a local quadratic correction, now gives the lower curvature bound with error $A(c + a) + \epsilon(\delta)$, up to an arbitrarily small error from freezing the smooth base forms. Letting these errors and a tend to zero proves the asserted current inequality and the quasi-plurisubharmonicity. Finally, $K \ge K_0$ preserves the monotone convergence of $R_t u + Kt^2$ to u. □

The classical theorem only gives some modulus ϵ; we do not claim that $\epsilon(\delta)$ is linear in δ. The cancellation and kernel arguments in the proof of Lemma 1.21.2 still give

$$\|R_t u - u\|_{L^1(\mathrm{d}V)} \le Ct^2, \qquad \|\nabla R_t u\|_\infty \le C/t,$$

because the averaging map is unchanged, u is bounded, and these arguments use only the monotone lower bound $R_t u - u \ge -Kt^2$.

1.21.2 Hölder reference approximation and weighted smoothing

The Chern averages of a Hölder potential have a common Hölder bound. Their small lower-curvature loss can be absorbed by rescaling, without changing the limiting potential.

Lemma 1.21.4 *If $u \in \mathrm{PSH}(X, \omega) \cap C^{0,\alpha}(X)$ with $0 < \alpha \le 1$, there are smooth $u_j \in \mathrm{PSH}(X, \omega)$ tending uniformly to u, with a common $C^{0,\alpha}$ bound. If $\sup_X u = 0$, the approximants can also be normalized by $\sup_X u_j = 0$.*

Proof Fix a sufficiently small sequence $t_j \downarrow 0$. The classical estimate stated before Lemma 1.21.3, with ω in place of θ and vanishing radius component, gives constants $K_0, A_0 > 0$ and a function $\epsilon_0(t) \to 0$ with

$$\omega + \mathrm{dd^c} R_t u \ge -(A_0 \lambda(\cdot, t) + \epsilon_0(t))\omega, \qquad \lambda(x, t) = t\partial_t (R_t u(x) + K_0 t^2).$$

The function λ is nonnegative, continuous for $t > 0$, and increasing in t. For fixed x, the convex function $s \mapsto R_{\mathrm{e}^s} u(x) + K_0 \mathrm{e}^{2s}$ has the finite limit $u(x)$ as $s \to -\infty$. Its nonnegative derivative must therefore tend to zero, since a positive limit of the derivative would make the function tend to $-\infty$. Thus $\lambda(x, t) \to 0$ as $t \to 0$, and Dini's theorem gives $\|\lambda(\cdot, t_j)\|_\infty \to 0$. The averaging kernels have mass one and only sample points at distance at most Ct from the base point, so uniform continuity of u gives $\|R_t u - u\|_\infty \to 0$. With a decreasing positive majorant $\varepsilon_j \to 0$ of $A_0 \|\lambda(\cdot, t_j)\|_\infty + \epsilon_0(t_j)$, we obtain

$$\omega + \mathrm{dd}^{\mathrm{c}} R_{t_j} u \geq -\varepsilon_j \omega, \qquad R_{t_j} u \longrightarrow u \quad \text{uniformly}.$$

We next prove a uniform Hölder bound for the averages. In a coordinate chart with a smooth orthonormal tangent frame, the averaging formula is

$$R_t u(x) = \int u(F(x, t\xi))\rho(|\xi|^2)\, \mathrm{d}\xi.$$

On a smaller chart, smoothness of F bounds $\operatorname{dist}(F(x,t\xi), F(y,t\xi))$ by $C \operatorname{dist}(x,y)$, uniformly for small t and $|\xi| \leq 1$. Integrating the Hölder inequality for u gives the same bound for $R_t u$. A finite cover and the common supremum bound handle pairs outside a common smaller chart.

Set $u_j = R_{t_j} u/(1+\varepsilon_j)$. Then

$$\omega + \mathrm{dd}^{\mathrm{c}} u_j = \frac{\varepsilon_j \omega + (\omega + \mathrm{dd}^{\mathrm{c}} R_{t_j} u)}{1+\varepsilon_j} \geq 0.$$

These functions converge uniformly with a common Hölder bound. Subtracting their suprema gives the asserted normalization when $\sup_X u = 0$. □

Lemma 1.21.5 *Let $u \in \mathrm{PSH}(X,\omega) \cap C^{0,\alpha}(X)$, $0 < \alpha \leq 1$, and let μ be a positive measure with $\mu \leq A\omega_u^n$. For bounded $\varphi \in \mathrm{PSH}(X,\omega)$ with $-M \leq \varphi \leq 0$, the Chern averages satisfy*

$$\|R_t\varphi - \varphi\|_{L^1(\mu)} \leq C t^{2\alpha/(\alpha+2n)}$$

for small $t > 0$. The constant is uniform under fixed bounds for A, M and $\|u\|_{C^{0,\alpha}}$.

Proof First we check a common lower curvature bound for the averages. The base part of the classical Chern curvature estimate stated before Lemma 1.21.3 gives

$$\omega + \mathrm{dd}^{\mathrm{c}} R_t\varphi \geq -(A_0\lambda(\cdot,t) + \epsilon_0(t))\omega, \qquad \lambda(x,t) = t\partial_t(R_t\varphi(x) + K_0 t^2).$$

Its constants are fixed for potentials with values in $[-M, 0]$. To bound λ, write the averages, as in the proof of Lemma 1.21.2, with a local kernel of the form

$$k_t(x,y) = t^{-2n} a\big(x, (y-x)/t, t\big),$$

where a has uniformly bounded first derivatives and compact support in its second variable. Differentiating with respect to t at fixed x, y and changing variables shows that

$$\int |t\partial_t k_t(x,y)|\, \mathrm{d}y \leq C.$$

Thus $|t\partial_t R_t\varphi| \leq CM$, so λ is uniformly bounded and the displayed curvature estimate supplies one fixed lower bound $\mathrm{dd}^{\mathrm{c}} R_t\varphi \geq -C\omega$. Both $R_t\varphi$ and φ also lie in $[-M, 0]$.

Apply Corollary 1.8.2 to the pair $R_t\varphi, \varphi$, whose common lower curvature bound has just been established. By Lemma 1.21.2, $\|R_t\varphi - \varphi\|_{L^1(\mathrm{d}V)} \leq Ct^2$. Domination by $A\omega_u^n$ therefore gives

$$\int |R_t\varphi - \varphi|\,\mathrm{d}\mu \le C\|R_t\varphi - \varphi\|_{L^1(\mathrm{d}V)}^{\alpha/(\alpha+2n)} \le Ct^{2\alpha/(\alpha+2n)}.$$

1.21.3 A linear error for a smooth background

Throughout this subsection, θ is a smooth real $(1,1)$-form that is big. By Theorem 1.15.2 and the remark following it, there is a strict potential $\rho \in \mathrm{PSH}(X,\theta)$, that is, $\theta + \mathrm{dd^c}\rho \ge \varepsilon\omega$ for some $\varepsilon > 0$, which is smooth outside an analytic pole set and has a smooth remainder after resolution. Put $\Omega = X \setminus \{\rho = -\infty\}$. For normalized potentials that are locally bounded on Ω, the next lemma improves the curvature error of Lemma 1.21.3 to a linear one.

Lemma 1.21.6 *Let $a \in \mathrm{PSH}(X,\theta)$ satisfy $\sup_X a = 0$, and suppose that a is locally bounded on Ω. There are constants $A, K, C > 0$ such that the averaging operators R_t of Lemma 1.21.1 satisfy*

$$R_t a + Kt^2 \downarrow a, \qquad \|R_t a - a\|_{L^1(\mathrm{d}V)} \le Ct^2.$$

For small $\delta > 0$ and $c > 0$, the transform on Ω

$$\Phi_{c,\delta} = \inf_{0<t\le\delta}\{R_t a + K(t^2 - \delta^2) + K(t-\delta) - c\log(t/\delta)\}$$

extends canonically to X and satisfies

$$a - K(\delta^2 + \delta) \le \Phi_{c,\delta} \le R_\delta a \le 0, \qquad \theta + \mathrm{dd^c}\Phi_{c,\delta} \ge -(Ac + C\delta)\omega. \tag{1.42}$$

At each point of Ω the infimum is attained at a positive radius. On every compact set $K_1 \Subset \Omega$, for sufficiently small t, $\|\nabla R_t a\|_{L^\infty(K_1)} \le C_{K_1}/t$.

Proof *Step 1: the classical expansion.* Apply the classical Chern-connection regularization expansion to the closed current $\mathrm{dd^c}a$ and the smooth lower bound $-\theta$. It gives monotonicity and logarithmic convexity: for a constant $K_0 > 0$, the function $\lambda(x,t) = t\partial_t(R_t a(x) + K_0 t^2)$ is nonnegative and increasing in t. Instead of inferring a rate from the unspecified modulus in that theorem, we estimate the error terms of the expansion directly, using the smoothness of θ.

Choose the auxiliary curvature form to be $A\omega$, with A large enough that the Chern curvature plus $A\omega \otimes \mathrm{Id}$ is nonnegative on every decomposable tensor. Equivalently, after absorbing the normalization constants, the Chern curvature coefficients $R_{j\bar{k}\ell\bar{m}}$ in a unitary frame satisfy

$$\sum R_{j\bar{k}\ell\bar{m}}\xi_j\bar{\xi}_k\zeta_\ell\bar{\zeta}_m \ge -A|\xi|^2|\zeta|^2$$

for all tangent vectors ξ, ζ. Such an A exists by compactness. Pairing this inequality with the nonnegative eigenvalues of a positive complex Hessian bounds its curvature contraction from below by $-A$ times its trace, with no further error term.

Step 2: a frozen quadratic correction. Fix $x \in X$ and a complex radius variable $w \neq 0$, and put $t = |w|$. Write $\xi \in T_x^{1,0}X$ and $\tau \in \mathbb{C}$ for the base and radius components of a tangent vector. In a coordinate chart of uniform size about x, write $\beta = \mathrm{dd}^{\mathrm{c}}|z|^2$ and choose a quadratic function $q_{x,t}$ with

$$\mathrm{dd}^{\mathrm{c}} q_{x,t} = \theta(x) + Bt\beta,$$

where the constant B is chosen so large that $a + q_{x,t}$ is plurisubharmonic on a ball of radius of order t containing the support of the averages at x and at nearby points. This is possible by the bound $\|\theta(y) - \theta(x)\| \leq C|y - x|$ on such a ball. For the Hessian calculation at (x, w), the quadratic function is held fixed; its coefficients and all needed derivatives are uniformly bounded.

Step 3: the error terms. By normalized compactness, $\|a\|_{L^1}$ is bounded. On each fixed larger chart, add one fixed strictly plurisubharmonic quadratic function Q to a. For the trace measure $\mu = \mathrm{dd}^{\mathrm{c}}(a + Q) \wedge \beta^{n-1}$, the local Lelong mass inequality and a bound for the mass of a fixed larger ball give $\mu(B(x,t)) \leq Ct^{2n-2}$. The coefficients of $\partial\bar{\partial}(a + q_{x,t})$ are controlled in total variation by those of $\mathrm{dd}^{\mathrm{c}}(a + Q)$ plus a uniformly bounded smooth form. Write $|\partial\bar{\partial}(a + q_{x,t})|$ for the sum of the total variations of its coefficient measures. The rescaled averaging kernels are supported in $B(x, Ct)$ and bounded by a constant times t^{-2n}, so their integrals against this measure are bounded by

$$Ct^{-2n}|\partial\bar{\partial}(a + q_{x,t})|(B(x, Ct)) \leq C't^{-2}.$$

Thus the terms of the expansion with a coefficient $O(t^3)$ contribute $O(t)$, and the remainder of the expansion of order two is $O(t)$ as well. The mixed and vertical contributions are bounded by $C(|\xi||\tau| + |\tau|^2)$. By the choice of A, the curvature term is bounded below by $-A\lambda(x,t)|\xi|^2$; here the radial term of the same expansion identifies the relevant mass with λ up to $O(t^2)$. Subtracting the regularization of $q_{x,t}$ restores $-\theta(x)$: its base Hessian differs from $\mathrm{dd}^{\mathrm{c}} q_{x,t}$ by $O(t)$, its mixed and vertical Hessians are bounded, and its radial logarithmic derivative is $O(t^2)$. The extra term $Bt\beta$ also costs only $O(t)$. For a real $(1,1)$-form α, write $\alpha[v]^2$ for its associated Hermitian quadratic form evaluated on v. Consequently, for a fixed constant $C_1 > 0$, the complex Hessian of $(x, w) \mapsto R_{|w|}a(x)$ satisfies

$$\mathrm{dd}^{\mathrm{c}}_{x,w} R_{|w|}a[\xi, \tau]^2 \geq -\theta_x[\xi]^2 - (A\lambda(x,t) + Ct)|\xi|^2 - C_1(|\xi||\tau| + |\tau|^2).$$

All constants are independent of x, t, c, δ and uniform for normalized a. In particular, setting $\xi = 0$ in the same expansion gives a uniform lower bound for the vertical Hessian. Adding a fixed multiple of $|w|^2$ makes the function subharmonic in w, which gives the logarithmic convexity and monotonicity of Step 1 with a uniform constant K_0.

Step 4: the minimum. Take $K \geq K_0$ and add $K(t^2 + t)$ to the regularization. The positive vertical Hessian of $K|w|$ absorbs the mixed term by the inequality

$$C_1|\xi||\tau| \leq C't|\xi|^2 + \frac{K}{8t}|\tau|^2,$$

and it absorbs the remaining bounded vertical error for small t. The logarithmic derivative of the corrected function is at least λ. The Kiselman minimum argument with an upper endpoint, as in the proof of Lemma 1.21.3, therefore gives the last inequality in (1.42), now with error $C\delta$. On compact subsets of Ω, local boundedness of a makes the term $-c\log(t/\delta)$ dominate uniformly as $t \downarrow 0$, so that the minimum is attained away from zero.

Step 5: the remaining bounds. The other bounds in (1.42) follow from the monotonicity of $R_t a + Kt^2$ and from evaluation at $t = \delta$. In particular, the transform is bounded above near the analytic set $X \setminus \Omega$. The removable singularity theorem therefore extends it across this set, and local freezing and passage to the limit preserve the exact smooth lower bound for its Hessian. Its lower bound $a - K(\delta^2 + \delta)$ also extends, since quasi-plurisubharmonic functions are determined by their values off an analytic set.

The adjoint computation in the proof of Lemma 1.21.2 gives $R_t^* 1 = 1 + O(t^2)$. It only needs $a \in L^1$, since

$$\left| \int_X (R_t a - a)\, \mathrm{d}V \right| \le Ct^2 \|a\|_1.$$

Together with $R_t a - a \ge -Kt^2$, this proves the stated L^1 estimate. Finally, near K_1 the smoothing kernel only samples a fixed larger compact subset of Ω, on which a is bounded. The kernel derivative estimate in the proof of Lemma 1.21.2 then gives the bound C_{K_1}/t. □

1.22 Pullback and descent

Pullback by a holomorphic map preserves the inequality $\theta + \mathrm{dd}^{\mathrm{c}} u \ge 0$. Conversely, on a submersion with compact connected fibers, a potential that is positive with respect to a pulled-back background, which has no vertical component, must be constant on the fibers and therefore descends. For maps to a curve, descent also holds across critical fibers, and pullback preserves and reflects all-weight ideal types. If the fibers are disconnected, potentials descend to the curve of fiber components rather than the original base. For a finite map between curves, the maximum over each fiber pushes potentials down to the base, and measures pull back with the local degrees as weights. Potentials also descend under modifications, and there a Jacobian factor accounts for the change in multiplier-ideal integrability.

1.22.1 Holomorphic disks

A smooth background form need not have local dd^{c}-potentials, but its pullback to a disk does, since a real $(1,1)$-form on a disk is closed. Testing on holomorphic disks therefore characterizes the exact lower curvature bound in a way that is preserved by holomorphic maps.

Lemma 1.22.1 *Let θ be a smooth real $(1,1)$-form on a complex manifold M, and let u be quasi-plurisubharmonic on M. Then $\theta + \mathrm{dd}^{\mathrm{c}} u \geq 0$ if and only if the following condition holds: for every holomorphic map $f\colon \Delta \to M$ from the unit disk such that $u \circ f$ is not identically $-\infty$, and for every smooth function a on an open subset of Δ with $\mathrm{dd}^{\mathrm{c}} a = f^{*}\theta$, the function $u \circ f + a$ is subharmonic there.*

Proof Suppose first that $\theta + \mathrm{dd}^{\mathrm{c}} u \geq 0$. On a coordinate ball, convolve u and θ with a smooth nonnegative radial kernel. The resulting smooth functions u_ϵ satisfy

$$\mathrm{dd}^{\mathrm{c}} u_\epsilon \geq -\theta_\epsilon, \qquad \theta_\epsilon \longrightarrow \theta \quad \text{smoothly}.$$

Write $u = p - r$ locally, with p plurisubharmonic and r smooth. The convolutions p_ϵ decrease to p. On a smaller disk on which $u \circ f$ is not identically $-\infty$, $p_\epsilon \circ f$ therefore converges in L^1_{loc} to $p \circ f$, while $r_\epsilon \circ f$ converges smoothly. Pulling back the smooth inequality and passing to the limit in the sense of distributions gives $\mathrm{dd}^{\mathrm{c}}(u \circ f) + f^{*}\theta \geq 0$. Since $u \circ f = p \circ f - r \circ f$ is the sum of a subharmonic and a smooth function, this distributional inequality makes $u \circ f + a$ subharmonic.

Conversely, test on affine complex lines in a coordinate ball. For a fixed nonzero constant direction ξ, local integrability of u implies that the restriction of u to almost every line parallel to ξ is not identically $-\infty$. Fubini's theorem, applied to nonnegative smooth test functions, then gives positivity of the scalar distribution obtained by evaluating the Hermitian coefficient matrix of $\theta + \mathrm{dd}^{\mathrm{c}} u$ on ξ. Positivity for a countable dense set of constant directions proves positivity of the Hermitian matrix of coefficient measures, which is the current inequality. □

Consequently, holomorphic pullback preserves the inequality $\theta + \mathrm{dd}^{\mathrm{c}} u \geq 0$ wherever the pulled-back function is not identically $-\infty$: apply the disk criterion to the compositions with the given map. We also use descent of subharmonicity under the finite map $\rho(t) = t^k$, $k \geq 1$, of the disk. If $h \circ \rho$ is subharmonic, local inverse branches of ρ show that h is subharmonic off zero. Moreover, h is bounded above near zero, and its value at zero is its upper limit there, since

$$h(0) = (h \circ \rho)(0) = \limsup_{t \to 0,\ t \neq 0} h(t^k) = \limsup_{z \to 0,\ z \neq 0} h(z).$$

By the removable-singularity theorem, h is subharmonic at zero as well.

1.22.2 Compact fibers

The connected components of the fibers of a submersion determine a finite covering of the base, and potentials first descend to this covering.

Lemma 1.22.2 *Let $p\colon M \to Y$ be a surjective holomorphic submersion between connected compact complex manifolds. There is a factorization*

$$M \xrightarrow{\pi} Y' \xrightarrow{r} Y, \qquad p = r \circ \pi,$$

where Y' is a connected compact complex manifold, r is a finite unramified holomorphic covering and π is a surjective holomorphic submersion with connected fibers. The points of $r^{-1}(y)$ correspond to the connected components of $p^{-1}(y)$, which are the fibers of π over those points.

Proof Fix a Hermitian metric on M. The orthogonal complement of $\ker \mathrm{d}p$ gives a smooth horizontal lift of tangent vectors on Y. Around $y_0 \in Y$ choose a coordinate ball B centered at y_0, with compact closure inside a larger coordinate ball. For $y \in B$, lift the straight segment from y_0 to y. The ordinary differential equation for horizontal transport has a solution along the whole segment, since its projection remains in the compact closure of B, whose inverse image is compact. Smooth dependence on the starting point and on y, together with transport along the reversed segment, gives a smooth trivialization

$$p^{-1}(B) \simeq B \times p^{-1}(y_0)$$

over B.

The compact manifold $p^{-1}(y_0)$ has finitely many connected components. The trivialization partitions $p^{-1}(B)$ into finitely many open and closed pieces, each of which meets every fiber in one connected component. On overlaps, two such local partitions identify the same fiber components, and their labels differ by locally constant permutations. Glue one copy of each base ball for every component, using these identifications. Equivalently, take the quotient of M that identifies points in the same connected component of a fiber. The local partitions make this quotient a Hausdorff covering space: distinct points over one base point lie on different sheets, and points over different base points are separated by their images in Y.

Call the quotient Y' and its projection r. Coordinates of Y on each sheet give holomorphic charts with holomorphic transition maps, so r is a local biholomorphism. It is a finite covering, and Y' is compact because a finite covering of a compact space is compact. Locally, the quotient map π is p restricted to one of the open pieces, so it is a holomorphic submersion with the asserted connected fibers. Surjectivity of π and connectedness of M make Y' connected. □

Proposition 1.22.1 *Let $\pi\colon Y \to X$ be a surjective holomorphic submersion between connected compact complex manifolds, with connected fibers. Let θ be a smooth real $(1,1)$-form on X. Pullback gives a bijection*

$$\mathrm{PSH}(X,\theta) \longrightarrow \mathrm{PSH}(Y,\pi^*\theta).$$

Every member on the right is constant on each fiber, with the value $-\infty$ allowed on an entire fiber.

Proof On a coordinate ball $V \subset X$, choose a smooth function ρ with $\mathrm{dd^c}\rho \geq \theta$. If $u \in \mathrm{PSH}(Y,\pi^*\theta)$, then $u + \rho\circ\pi$ is plurisubharmonic on $\pi^{-1}(V)$. Its restriction to a fiber is plurisubharmonic or identically $-\infty$, and since the fiber is compact and connected, the maximum principle shows that it is constant in either case. As $\rho\circ\pi$ is constant on fibers, $u = v\circ\pi$ for a uniquely defined function v on X.

A holomorphic submersion admits local holomorphic sections. Composing $u+\rho\circ\pi$ with such a section shows that $v+\rho$ is plurisubharmonic or identically $-\infty$ on a smaller ball. The latter alternative would make $u = -\infty$ on a nonempty open set, which is impossible for a quasi-plurisubharmonic function. Hence v is quasi-plurisubharmonic. Distributional differentiation in local product coordinates gives

$$\pi^*(\theta + \mathrm{dd^c}v) = \pi^*\theta + \mathrm{dd^c}u \geq 0.$$

Positivity of a pulled-back current of this form implies positivity on the base: test in product coordinates against products with a nonnegative compactly supported volume form of integral one in the fiber variables. It follows that $v \in \mathrm{PSH}(X,\theta)$.

Conversely, by Lemma 1.22.1, the pullback of a θ-plurisubharmonic function by π is $\pi^*\theta$-plurisubharmonic; in local product coordinates it is clearly not identically $-\infty$. The pullback map is injective because π is surjective. □

1.22.3 Maps to a curve

Let $p\colon X \to C$ be a surjective holomorphic map with connected fibers, where X is a connected compact complex manifold and C is a connected compact Riemann surface. Critical fibers are allowed. Fix any smooth real $(1,1)$-form β on C.

Proposition 1.22.2 *Pullback gives a bijection*

$$\mathrm{PSH}(C,\beta) \longrightarrow \mathrm{PSH}(X, p^*\beta).$$

It preserves the supremum, and it preserves and reflects the ordinary singularity order.

Proof The critical set is a proper compact analytic subset of X. It has finitely many irreducible components, and p is constant on each: its differential vanishes on the smooth locus of that component, which is connected and dense in the component, and continuity extends the constant to the component. Thus the set $D \subset C$ of critical values is finite.

Let $U \in \mathrm{PSH}(X, p^*\beta)$. Over $C \setminus D$, the map p is a proper submersion with connected fibers, and the local argument in the proof of Proposition 1.22.1 gives a function v on $C \setminus D$ with $U = v \circ p$ there; if a is a smooth function on a coordinate disk with $\mathrm{dd}^{\mathrm{c}} a = \beta$, then $a + v$ is subharmonic on the disk minus D.

The function v is bounded above by $\sup_X U$. The removable singularity theorem extends $a + v$ subharmonically across the finitely many punctures. The extensions glue, giving $v \in \mathrm{PSH}(C,\beta)$. Its pullback is $p^*\beta$-plurisubharmonic by Lemma 1.22.1. It agrees with U off the proper analytic set $p^{-1}(D)$, hence almost everywhere and everywhere by Lemma 1.3.3.

Conversely, pullback preserves the curvature inequality by the disk criterion. The pullback is not identically $-\infty$, because p is surjective. Surjectivity of p also shows that pullback is injective and preserves the supremum, and that an inequality $v \le w + c$ on C is equivalent to $v \circ p \le w \circ p + c$ on X. □

Lemma 1.22.3 *Let $f\colon (\mathbb{C}^m, 0) \to (\mathbb{C}, 0)$ be a nonconstant holomorphic germ, and let $e = \mathrm{ord}_0 f$. For every quasi-subharmonic germ v at zero,*

$$\nu(v \circ f, 0) = e\nu(v, 0).$$

Proof A smooth correction of v pulls back to a smooth function, so we may assume that v is subharmonic. The Taylor expansion of f gives $|f(z)| \le M|z|^e$ on a small ball. Choose a unit vector ξ on which the leading homogeneous term is nonzero. Then

$$f(\zeta\xi) = a\zeta^e + O(\zeta^{e+1}), \qquad a \neq 0.$$

For all small r, the remainder has modulus at most $|a|r^e/4$ on $|\zeta| = r$. If $|t| < |a|r^e/4$, Rouché's theorem shows that $f(\zeta\xi) - t$ has e zeros in $|\zeta| < r$. Consequently, with $c = |a|/4$,

$$\{|t| < cr^e\} \subset f(B(0,r)) \subset \{|t| < Mr^e\}.$$

Hence the supremum of $v \circ f$ on $B(0,r)$ lies between the suprema of v on the two disks. Divide by $\log r^2 < 0$. Since $\log(cr^e)^2/\log r^2 \to e$, and similarly for M, both disk quotients tend to $e\nu(v,0)$ by the definition in Section 1.13. The middle quotient therefore has the same limit, proving the formula. □

Corollary 1.22.1 *For $v, w \in \mathrm{QPSH}(C)$, the inclusions $\mathcal{I}(tv) \subseteq \mathcal{I}(tw)$ for every $t > 0$ hold if and only if*

$$\mathcal{I}(t(v \circ p)) \subseteq \mathcal{I}(t(w \circ p)) \quad \text{for every } t > 0.$$

In particular, pullback preserves and reflects all-weight multiplier-ideal equivalence.

Proof On a smooth curve, local modifications are isomorphisms. By Theorem 1.16.1, the base inclusions are therefore equivalent to $\nu(v,s) \geq \nu(w,s)$ for every $s \in C$. Let π be any local modification of a neighborhood in X with smooth source, and let y be a point of its source. In a coordinate at $s = p(\pi(y))$, the germ $p \circ \pi - s$ is nonconstant. Indeed, a modification is an isomorphism on a dense open set, and the surjective map p is not constant on any nonempty open subset of the connected manifold X. Its order e_y is therefore positive and finite. Lemma 1.22.3 gives

$$\nu(v \circ p \circ \pi, y) = e_y \nu(v,s), \qquad \nu(w \circ p \circ \pi, y) = e_y \nu(w,s).$$

The base inequalities thus imply all inequalities upstairs, and Theorem 1.16.1, applied on X, gives the ideal inclusions. Conversely, the upstairs inclusions give the inequalities for the identity modification at any $x \in p^{-1}(s)$. Dividing by $\mathrm{ord}_x(p - s) > 0$ recovers the base inequalities. Apply this argument in both directions for equality. □

Lemma 1.22.4 *If $v_j, v \in \mathrm{PSH}(C, \beta)$ and $v_j \to v$ in $L^1(C)$, then $v_j \circ p \to v \circ p$ in $L^1(X)$ for every smooth positive volume form on X.*

Proof Put $M_j = \sup_C v_j$ and $M = \sup_C v$. Hartogs' lemma, Lemma 1.4.1, applied at maximum points of the v_j, gives $\limsup M_j \leq M$. If the suprema of a subsequence converged to a limit below M, possibly $-\infty$, then v would lie below that limit almost everywhere, hence everywhere by Lemma 1.3.3, a contradiction. Thus $M_j \to M$.

By surjectivity, $(v_j - M_j) \circ p$ has supremum zero. Normalized compactness, Theorem 1.4.1, makes these pullbacks precompact in $L^1(X)$. On every relatively compact product chart over $C \setminus D$, where D is the finite set of critical values, Fubini's theorem and bounded smooth volume densities give local L^1 convergence to $(v - M) \circ p$. Every subsequential limit therefore equals this pullback almost everywhere off $p^{-1}(D)$, and hence almost everywhere, because that analytic set has smooth volume zero. All subsequential limits agree, so the entire normalized sequence converges. Adding $M_j \to M$ proves the assertion. □

Lemma 1.22.5 *A function $v \in \mathrm{QPSH}(C)$ has analytic singularities if and only if $v \circ p$ has analytic singularities.*

Proof Composing a local analytic presentation with p proves the forward implication. The pulled-back generators are not all identically zero, since a nonconstant map to a curve is open.

Conversely, fix $s \in C$ and $x \in p^{-1}(s)$. In coordinates, choose a complex line through x on which $p - s$ has its finite order e and is nonconstant, as in the proof of Lemma 1.22.3. On that line write

$$(p - s)(\zeta) = h(\zeta)^e,$$

where h is a local biholomorphism fixing zero; take an analytic eth root of the nonvanishing factor of the germ. The restriction of $v \circ p$ to the line is not identically $-\infty$, since v is finite almost everywhere and the restriction of p to the line is open. By Proposition 1.13.1, it has analytic singularities. In one variable its presentation is $r \log |\zeta|^2 + O(1)$ with $r \in \mathbb{Q}_{\geq 0}$, where r is the coefficient times the least vanishing order of the finitely many generators. Changing to $w = h(\zeta)$ gives

$$v(s + w^e) = r \log |w|^2 + O(1).$$

Every sufficiently small t has an eth root w, so $v(s + t) = (r/e) \log |t|^2 + O(1)$. This is an analytic presentation on the base, including a bounded one when $r = 0$. □

1.22.4 The curve of fiber components

Dropping the connected-fiber hypothesis requires replacing the base by the curve of fiber components. The following construction is the curve case of Stein factorization. It also identifies the multiplicities needed when pulling back curvature measures.

Proposition 1.22.3 *Let $p\colon X \to C$ be a surjective holomorphic map, where X is a connected compact complex manifold and C is a connected compact Riemann surface. There is a factorization*

$$X \xrightarrow{q} C' \xrightarrow{r} C, \qquad p = r \circ q,$$

where C' is a connected compact Riemann surface, q is surjective and holomorphic with connected fibers, and r is a finite holomorphic map. The points of $r^{-1}(s)$ correspond to the connected components of $p^{-1}(s)$, which are the fibers of q over those points. The factorization is unique up to a unique isomorphism commuting with q and r.

If $t \in r^{-1}(s)$ has ramification index e_t, then, as effective divisors with multiplicities,

$$[p^{-1}(s)] = \sum_{t \in r^{-1}(s)} e_t [q^{-1}(t)].$$

The degree of r is the number of components of a regular fiber of p.

Proof We first construct the covering over the regular values, complete it across the critical values, and then extend its component quotient to X.

The critical values form a finite set $D \subset C$, by the critical-set argument in the proof of Proposition 1.22.2; that argument does not use connectedness of the fibers. Over $C^\circ = C \setminus D$, the map p is a proper submersion. The local construction in Lemma 1.22.2 applies there: horizontal transport is defined over every disk with compact closure in C°, because the inverse image of this closure is compact. Compact fibers have finitely many components, and the resulting local labels are glued by locally constant permutations. We obtain

$$X^\circ \xrightarrow{q^\circ} C'^\circ \xrightarrow{r^\circ} C^\circ, \qquad X^\circ = p^{-1}(C^\circ),$$

with q° a holomorphic submersion with connected fibers and r° a finite unramified covering of degree d. The set X° is connected, as the complement of the hypersurface $p^{-1}(D)$ in the connected manifold X, and hence so is its image C'°. Indeed, in a ball, the complement of the zero set of a nonzero holomorphic function is path connected: two points outside the zero set lie on the disk cut out by the complex line through them; the restriction of the function to this disk is nonzero, because it does not vanish at these points, so its isolated zeros can be avoided. Applying this local argument along a chain of balls proves the global connectedness.

For each $s \in D$, take a small coordinate disk centered at s and disjoint from the other critical values. A connected component of the inverse image under r° of the punctured disk is a finite connected covering of the punctured disk, of some degree e. It has a coordinate w in which $r^\circ(w) = w^e$: the universal cover of the punctured disk is a half-plane with projection $z = \exp\zeta$, and the index-e subgroup of its deck group is generated by $\zeta \mapsto \zeta + 2\pi i e$. The coordinate $w = \exp(\zeta/e)$ gives the asserted form. Attach one point $w = 0$ to each such component. There are finitely many, with their degrees summing to d. Gluing these disks to C'° gives a connected Riemann

surface C' and a finite holomorphic map $r\colon C' \to C$. The new points have disjoint disk neighborhoods, so this gluing is Hausdorff. Moreover, C' is compact. Indeed, the inverse image under r of a smaller closed disk centered at a point of D is a finite union of closed disks, while over the complement of the open disks, the finite covering r° is proper.

We extend q°. Fix $x \in p^{-1}(s)$, $s \in D$. On a small coordinate ball B about x, the image of p lies in the chosen disk. The complement $B \setminus p^{-1}(s)$ is connected, by the local argument above. Its image under q° therefore lies in a single connected component of the inverse image under r° of the punctured base disk, that is, in one of the punctured disks completed above. In its coordinate w, the function $w \circ q^\circ$ is holomorphic and bounded, and

$$(w \circ q^\circ)^e = p - s.$$

The bounded holomorphic removable-singularity theorem extends it across $p^{-1}(s) \cap B$. To see this without assuming that the divisor is smooth, first extend at its smooth points by the one-variable theorem in transverse coordinates. The singular locus of the reduced divisor has complex codimension at least two in the ambient manifold, so Hartogs extension across that locus completes the extension. In dimension one there are only isolated points. Its eth power remains $p - s$ by the identity theorem, and hence its value is zero on that fiber. This defines a holomorphic map from B to the completed disk. Two local extensions agree with q° on the dense part of their common domain that lies in X°, and therefore glue to a holomorphic map $q\colon X \to C'$ with $p = r \circ q$. Its image is closed, by compactness, and contains the dense set C'°, so q is surjective.

Every fiber of q is connected. Otherwise write one fiber as a union of two disjoint nonempty compact subsets and put them in disjoint open neighborhoods U_1, U_2. Properness gives a disk V about their base point t such that

$$q^{-1}(V) \subset U_1 \cup U_2.$$

The map q is open near each point x of $q^{-1}(t)$: the germ of $q - t$ at x is nonconstant, since q is surjective and X is connected, and on a complex line through x on which it has its first nonzero homogeneous term, its restriction is a nonconstant, hence open, function of one variable. Consequently, each $q(U_i)$ contains a neighborhood of t. A nearby point of C'° then has a fiber meeting both U_1 and U_2, which contradicts the connectedness of the fibers of q°.

Thus each fiber $p^{-1}(s)$ is the disjoint union of the connected compact q-fibers over the finitely many points of $r^{-1}(s)$, and these are its components. Locally at t, the formula for a finite map is $r(w) - s = w^{e_t}$ after changing the coordinate by a holomorphic root of a nonvanishing factor. Pulling back this equality by q gives $p - s = (w \circ q)^{e_t}$ near $q^{-1}(t)$, hence the divisor formula. Over C°, r is unramified and each sheet is one component, proving the degree assertion.

Finally, let $X \xrightarrow{q_1} C_1 \xrightarrow{r_1} C$ be another factorization with the stated properties. Since $p = r_1 \circ q_1$ has no critical points over C° and q_1 is surjective, neither r_1 nor

q_1 has critical points over C°; in particular, r_1 is unramified there. Matching points that label the same fiber component of p gives a bijection $\phi\colon r_1^{-1}(C^\circ) \to C'^\circ$ with $\phi \circ q_1 = q^\circ$ on X°. Composing q° with local holomorphic sections of q_1 shows that ϕ is holomorphic. On every punctured disk, ϕ identifies two completions of the same finite covering, and the coordinates w^e above extend it biholomorphically across the added point. The extension is an isomorphism $\phi\colon C_1 \to C'$. By density, it satisfies $\phi \circ q_1 = q$ and $r \circ \phi = r_1$, and surjectivity of q_1 makes it unique. □

1.22.5 Finite maps between curves

Let $r\colon C' \to C$ be a finite surjective holomorphic map of degree d between connected compact Riemann surfaces. Write e_x for its local degree at $x \in C'$, so that $\sum_{r(x)=s} e_x = d$ for every $s \in C$. Fix a smooth real $(1,1)$-form β on C and put $\sigma = r^*\beta$. Potentials in $\mathrm{PSH}(C', \sigma)$ need not descend to C. Taking the maximum on each fiber nevertheless gives a base potential that dominates the original potential after pullback. Its Lelong number at a point of C is the smallest Lelong number on the fiber over that point, after division by the local degree.

Lemma 1.22.6 *For $h \in \mathrm{PSH}(C', \sigma)$, the function*

$$m_h(s) = \max_{r(x)=s} h(x)$$

belongs to $\mathrm{PSH}(C, \beta)$ and satisfies $h \le m_h \circ r$. For every $s \in C$,

$$\nu(m_h, s) = \min_{r(x)=s} \frac{\nu(h, x)}{e_x}.$$

If $v \in \mathrm{PSH}(C, \beta)$, then $h \preceq v \circ r$ implies $m_h \preceq v$. The analogous implication holds for the all-weight ideal order $\preceq_I$.

Proof On a base disk choose a smooth function a with $\mathrm{dd}^{\mathrm{c}} a = \beta$. Off the branch values, $m_h + a$ is the maximum of finitely many subharmonic functions obtained from $h + a \circ r$ by inverse branches. It is subharmonic there and locally bounded above. The removable-singularity theorem extends it across the branch values, and the extension at a branch value s is $m_h(s) + a(s)$. Indeed, upper semicontinuity of $h + a \circ r$ and properness of r bound the upper limit of $m_h + a$ at s by this value. Conversely, at each point x of the fiber, the subharmonic function $h + a \circ r$ equals its upper limit over punctured neighborhoods of x, which r maps into punctured neighborhoods of s; this gives the reverse bound. Hence $m_h + a$ is subharmonic on the whole disk. The function m_h is not identically $-\infty$, since h is finite at some point, so $m_h \in \mathrm{PSH}(C, \beta)$. The inequality $h \le m_h \circ r$ holds by definition.

Choose a coordinate t centered at s, and coordinates z on the disjoint disks over s, each centered at a point x of the fiber, such that r is given by $t = z^{e_x}$ on the disk centered at x. Then

$$\sup_{|t|<\delta} m_h(t) = \max_{r(x)=s} \sup_{|z|<\delta^{1/e_x}} h(z).$$

Divide by $\log \delta^2 < 0$ and use the supremum definition of the Lelong number in Section 1.13. Each quotient on the right tends to $\nu(h,x)/e_x$, and the maximum becomes the minimum after division. Smooth local corrections do not affect these limits.

An ordinary bound $h \le v \circ r + c$ gives $m_h \le v + c$, by taking the maximum over each fiber. On a curve, local modifications are isomorphisms, so the all-weight ideal order is equivalent to the pointwise Lelong inequalities, by Theorem 1.16.1. The pullback formula of Lemma 1.22.3, applied to r in local coordinates, makes $h \preceq_I v \circ r$ equivalent to $\nu(h,x) \ge e_x \nu(v, r(x))$ at every x. The minimum formula then gives $\nu(m_h, s) \ge \nu(v,s)$ for every $s \in C$, that is, $m_h \preceq_I v$. □

The pullback of measures by r counts each preimage with its local degree, so that ramification is visible at atoms. For a continuous real function f on C', put

$$(\mathrm{Tr}_r\, f)(s) = \sum_{r(x)=s} e_x f(x).$$

This function is continuous: in the local description $t = z^e$, the e values of f over $t \neq 0$ tend to its value at the branch point. For a finite positive measure μ on C, the functional $f \mapsto \int_C \mathrm{Tr}_r\, f \,\mathrm{d}\mu$ is positive, and it is bounded because $|\mathrm{Tr}_r\, f| \le d\|f\|_\infty$. By the Riesz representation theorem, it defines a finite positive measure $r^*\mu$ on C' with

$$\int_{C'} f \,\mathrm{d}(r^*\mu) = \int_C \mathrm{Tr}_r\, f \,\mathrm{d}\mu.$$

We extend the definition to finite signed measures by linearity.

Lemma 1.22.7 *For every finite signed measure μ and Borel set $B \subset C$,*

$$r_* r^* \mu = d\mu, \qquad \|r^*\mu\|_{\mathrm{TV}} = d\|\mu\|_{\mathrm{TV}},$$
$$r^*(\mathbf{1}_B \mu) = \mathbf{1}_{r^{-1}(B)} r^*\mu, \qquad r^*\delta_s = \sum_{r(x)=s} e_x \delta_x.$$

For $v \in \mathrm{PSH}(C,\beta)$, write $T_v = \beta + \mathrm{dd}^{\mathrm{c}} v$, and let $\mu_v = \mathbf{1}_{\{v=-\infty\}} T_v$ be the polar curvature of v, as in Section 1.6.1; for potentials in $\mathrm{PSH}(C',\sigma)$ the background σ replaces β. Then

$$T_{v\circ r} = r^* T_v, \qquad \mu_{v\circ r} = r^* \mu_v.$$

Images and inverse images of polar sets under r are polar. Pullback and pushforward of positive measures commute with taking the atomic part.

Proof The trace of $g \circ r$ is dg, giving the pushforward identity. The same weighted fiber formula for Borel test functions follows from the monotone-class theorem and gives the restriction and atom formulas. The positive and negative Jordan parts of a signed measure have disjoint Borel carriers; their pullbacks are positive with

disjoint inverse-image carriers. Each has d times its original mass, which proves the total-variation identity.

Choose a local smooth a with $\mathrm{dd}^{\mathrm{c}} a = \beta$; then $v + a$ is a subharmonic potential of T_v. On a smaller disk Δ, its Riesz representation is the logarithmic potential $\int_\Delta \log|t-\tau|^2 \, \mathrm{d}T_v(\tau)$ plus a harmonic function. In local coordinates in which r is $t = z^e$, the identity

$$\mathrm{dd}^{\mathrm{c}}_z \log|z^e - \tau|^2 = \sum_{z^e=\tau} \mathrm{ord}_z(z^e - \tau)\delta_z,$$

valid for each fixed τ, shows that pulling back this potential gives the weighted measure r^*T_v. Fubini's theorem applies locally: factorization of $z^e - \tau$ into its e linear factors gives a uniform local L^1 bound for these logarithmic kernels as τ varies on a smaller disk. The harmonic term has zero Hessian after pullback. Since $\sigma + \mathrm{dd}^{\mathrm{c}}(v \circ r) = \mathrm{dd}^{\mathrm{c}}\big((v+a)\circ r\big)$ locally, this proves $T_{v\circ r} = r^*T_v$. The pole set of $v \circ r$ is exactly $r^{-1}\{v = -\infty\}$, so the restriction formula proves the polar identity.

Inverse images of polar sets are polar by composition of local subharmonic witnesses. For images, take local witnesses on the finitely many inverse-image disks of a small base disk. Off the branch values, the sum of their values over the inverse branches is subharmonic and is $-\infty$ wherever any witness has a pole. It is locally bounded above and is not identically $-\infty$. Its extension across the branch values is subharmonic. Adding the finitely many branch values to the resulting polar set still gives a polar set. This proves the image assertion locally, hence globally.

The pullback of a nonatomic measure has no atoms, by the restriction and atom formulas. The pushforward of a nonatomic measure has no atoms because every fiber is finite. Atomic measures pull back and push forward to atomic measures. Splitting a positive measure into these two parts proves the last assertion. □

1.22.6 Modifications

Proposition 1.22.4 *Let $\pi : Y \to X$ be a modification of connected complex manifolds, and let θ be a smooth real $(1,1)$-form on X. Pullback gives a bijection*

$$\mathrm{PSH}(X,\theta) \longrightarrow \mathrm{PSH}(Y,\pi^*\theta).$$

Proof Let $S \subset X$ be a proper analytic subset over whose complement π is an isomorphism, and put $D = \pi^{-1}(S)$. By Lemma 1.22.1, pullback preserves the curvature inequality, and the pullback of a θ-plurisubharmonic function is not identically $-\infty$, since π is an isomorphism off D. The pullback map is injective because π is surjective.

To prove surjectivity, let $b \in \mathrm{PSH}(Y,\pi^*\theta)$, and define $u = b \circ \pi^{-1}$ off S. On a coordinate neighborhood in X, choose a smooth function ρ with $\mathrm{dd}^{\mathrm{c}}\rho \geq \theta$. Then $b + \rho\circ\pi$ is plurisubharmonic on the inverse image of this neighborhood. By properness,

its local upper bounds give a local upper bound for $u + \rho$ near S, so the removable-singularity theorem extends $u + \rho$ plurisubharmonically across S. The resulting quasi-plurisubharmonic extensions of u agree almost everywhere on overlaps, hence everywhere, and glue to a function on X.

The current $\theta + \mathrm{dd}^{\mathrm{c}} u$ is positive off S. Locally it is the sum of the positive current $\mathrm{dd}^{\mathrm{c}}(u + \rho)$ and the smooth form $\theta - \mathrm{dd}^{\mathrm{c}}\rho$. The smooth form gives S zero mass, so the restriction of $\theta + \mathrm{dd}^{\mathrm{c}} u$ to S is positive as well. Thus $u \in \mathrm{PSH}(X, \theta)$. The plurisubharmonic functions $b + \rho \circ \pi$ and $(u + \rho) \circ \pi$ agree off D, hence almost everywhere, hence everywhere. Therefore $b = u \circ \pi$. □

Pullback by a modification preserves L^1 convergence within a family with a common lower curvature bound. Compactness controls the pullbacks near the exceptional locus, where a direct change of variables would involve a vanishing Jacobian.

Lemma 1.22.8 *Let $\pi\colon Y \to X$ be a modification between connected complex manifolds, and let θ be a smooth real $(1,1)$-form on X. If $v_j \to v$ in $L^1_{\mathrm{loc}}(X)$ and all v_j, v belong to $\mathrm{PSH}(X, \theta)$, then*

$$v_j \circ \pi \longrightarrow v \circ \pi \quad \text{in } L^1_{\mathrm{loc}}(Y).$$

In particular, on compact manifolds this is convergence in L^1.

Proof By Lemma 1.4.1, the functions v_j are uniformly bounded above on relatively compact subsets of X. The pullbacks are therefore locally uniformly bounded above and have the common smooth lower curvature bound $-\pi^*\theta$. After a smooth local correction, the compactness theorem for plurisubharmonic functions shows that every subsequence has a further subsequence that either converges in L^1_{loc} or tends locally uniformly to $-\infty$ on each connected coordinate region in Y.

Every such region meets the complement of the exceptional locus. There π is a biholomorphism, and change of variables gives L^1_{loc} convergence to $v \circ \pi$, which excludes convergence to $-\infty$. Each subsequential limit agrees with $v \circ \pi$ off the exceptional locus, hence almost everywhere, since this analytic set has smooth volume zero. Thus the limit is unique, and applying this argument on a countable cover by coordinate regions proves convergence of the whole sequence in $L^1_{\mathrm{loc}}(Y)$. □

Convergence in capacity is also preserved by modifications. The comparison takes place away from small neighborhoods of the exceptional sets, so no uniform comparison of capacities on the whole manifolds is needed.

Lemma 1.22.9 *Let $\pi\colon Y \to X$ be a modification between connected compact complex manifolds, equipped with Hermitian forms ω_Y, ω_X. Write $\mathrm{Cap}_Y = \mathrm{Cap}_{\omega_Y}$ and $\mathrm{Cap}_X = \mathrm{Cap}_{\omega_X}$ for the Hermitian capacities of Section 1.8. For quasi-plurisubharmonic functions v_j, v on X,*

$$v_j \longrightarrow v \text{ in capacity on } X \quad\Longleftrightarrow\quad v_j \circ \pi \longrightarrow v \circ \pi \text{ in capacity on } Y.$$

Proof Choose a proper analytic subset $S \subset X$ such that π is an isomorphism outside $D = \pi^{-1}(S)$. Both S and D have capacity zero. For $\epsilon > 0$, choose an open neighborhood V of D of capacity less than ϵ. By properness, there is a neighborhood U_0 of S with $\pi^{-1}(U_0) \subset V$; shrink it to an open set U with $\mathrm{Cap}_X(U) < \epsilon$. The inverse of π is biholomorphic near the compact set $X \setminus U$. A finite cover by nested coordinate charts and the comparison of local and global capacities in Section 1.8 give a constant C_ϵ such that, for every Borel set $E \subset X$,

$$\begin{aligned} \mathrm{Cap}_Y(\pi^{-1}E) &\le \epsilon + C_\epsilon\, \mathrm{Cap}_X(E), \\ \mathrm{Cap}_X(E) &\le \epsilon + C_\epsilon\, \mathrm{Cap}_Y(\pi^{-1}E). \end{aligned}$$

Apply these inequalities to $E = \{|v_j - v| > \delta\}$ for each fixed $\delta > 0$, omitting the pluripolar pole sets, and let first $j \to \infty$ and then $\epsilon \downarrow 0$. □

1.22.7 Multiplier ideals under modifications

Throughout this subsection, $\pi\colon Y \to X$ is a modification of complex manifolds of dimension n, that is, a proper holomorphic bimeromorphic map; here X need not be compact. There is an analytic subset $S \subset X$ of codimension at least two such that π is an isomorphism over $X \setminus S$. Let K_X and K_Y be the canonical line bundles. Set $D = \pi^{-1}(S)$ and

$$R_\pi = K_Y \otimes \pi^* K_X^{-1}.$$

The determinant of the differential of π is a canonical holomorphic section J_π of R_π; in local coordinates its coefficient is the holomorphic Jacobian determinant $\mathrm{Jac}\,\pi$, and its zeros are contained in D. These standard facts about modifications of complex manifolds include the case $n = 1$, in which every modification is an isomorphism.

Lemma 1.22.10 *Let a be quasi-plurisubharmonic on X. Pullback followed by multiplication with J_π induces an isomorphism*

$$\pi_*\big(R_\pi \otimes \mathcal{I}(a \circ \pi)\big) = \mathcal{I}(a).$$

More precisely, for every holomorphic line bundle M on X, the map

$$s \longmapsto J_\pi\, \pi^* s$$

is an isomorphism

$$H^0\big(X, M \otimes \mathcal{I}(a)\big) \simeq H^0\big(Y, \pi^* M \otimes R_\pi \otimes \mathcal{I}(a \circ \pi)\big).$$

Proof The assertion is local on X. In coordinates and local frames, the map sends a holomorphic coefficient f to $(f \circ \pi)\operatorname{Jac}\pi$. By the change-of-variables formula, local integrability of

$$|f|^2 \mathrm{e}^{-a} \quad \text{and} \quad |f \circ \pi|^2 |\operatorname{Jac}\pi|^2 \mathrm{e}^{-a\circ\pi}$$

are equivalent. For the implication from integrability on Y to integrability near a point of X, take a smaller relatively compact neighborhood of the point; by properness, its inverse image has compact closure, so finitely many charts on which the integrability holds suffice. Smooth positive coordinate densities are comparable on these charts, and analytic exceptional sets have smooth volume zero, so they do not change the integrals.

It remains to show that every section on Y is of the form $J_\pi \pi^* s$. Divide it by J_π off D and use the inverse of π there; this gives a holomorphic section of M off S. Since S has codimension at least two, Hartogs' extension theorem gives a holomorphic section s on X. Its image $J_\pi \pi^* s$ agrees with the given section off D, hence everywhere by the identity theorem. The integrability equivalence then places s in the required multiplier ideal. The same argument on each open set proves the sheaf identity as well as the global identification. □

Note that the lemma does not identify $\pi^* \mathcal{I}(a)$ with $\mathcal{I}(a \circ \pi)$: the Jacobian is part of the holomorphic coefficient on Y and cannot be removed from the integrability test.

We also need the invariance of the ideal order $\preceq_{\mathcal{I}}$ of Section 1.12, which involves all positive weights. The proof is local, so that we never have to extend a modification of a small open subset of Y to a global one.

Lemma 1.22.11 *For quasi-plurisubharmonic a, b on X,*

$$a \preceq_{\mathcal{I}} b \quad \Longleftrightarrow \quad a \circ \pi \preceq_{\mathcal{I}} b \circ \pi.$$

Proof The reverse implication follows at every positive weight from the sheaf identity in Lemma 1.22.10: tensor the inclusion of ideals on Y with R_π and apply the left-exact direct image functor.

For the forward implication, fix a coordinate neighborhood $W \subset Y$, a local modification $\sigma\colon Z \to W$ with smooth source, and a prime divisor E on Z. After shrinking W around the point under consideration, its image is contained in a coordinate neighborhood $U \subset X$. Put $F = \pi \circ \sigma$ and

$$A_F(E) = 1 + \operatorname{ord}_E \operatorname{Jac} F.$$

The map F is holomorphic and injective off a nowhere dense analytic subset of Z, but it need not be proper over U.

Let m be a positive integer, and let ν_E and ord_E be the generic Lelong number and the generic vanishing order along E, as in Section 1.17. By the remark following Lemma 1.17.8, that lemma applies to F and gives

$$\max\left\{\nu_E(a \circ F) - \frac{A_F(E)}{m}, 0\right\} \le \frac{1}{m} \operatorname{ord}_E\big(\mathcal{I}(ma)\mathcal{O}_Z\big) \le \nu_E(a \circ F).$$

The same estimates hold for b.

The hypothesis gives $\mathcal{I}(ma) \subseteq \mathcal{I}(mb)$ for every $m > 0$, so the orders of the pulled-back ideals satisfy the reverse inequality. Letting $m \to \infty$ in the two estimates proves

$$\nu_E(a \circ F) \ge \nu_E(b \circ F).$$

This holds for every local divisor over Y. Blowing up a point on the source of any local modification recovers the Lelong number at that point as a generic Lelong number along the exceptional divisor; in dimension one the point already is a divisor. The local criterion in Theorem 1.16.1 now gives $a \circ \pi \preceq_{\mathcal{I}} b \circ \pi$. □

Lemma 1.22.12 *For quasi-plurisubharmonic a, b on X,*

$$a \sim_{\mathcal{I}} b \quad \Longleftrightarrow \quad a \circ \pi \sim_{\mathcal{I}} b \circ \pi.$$

Proof Apply Lemma 1.22.11 to both (a, b) and (b, a). Together, the two order relations are equivalent to equality of the multiplier ideals at every positive weight.□

1.22.8 Restriction of analytic approximants

A quasi-plurisubharmonic function with zero generic Lelong number along a submanifold may still restrict to the function identically equal to $-\infty$ (Section 1.13). Analytic regularization removes this obstruction: the restrictions of the analytic approximants are quasi-plurisubharmonic, and a fixed lower curvature bound controls their generic Lelong numbers along divisors on every fixed modification.

Corollary 1.22.2 *Let θ be a smooth real $(1,1)$-form and $u \in \mathrm{PSH}(X,\theta)$, and let u_m, with curvature errors ϵ_m, be the approximants of Corollary 1.15.1. Let $Y \subset X$ be a connected compact complex submanifold along which the generic Lelong number of u vanishes:*

$$\nu(u,Y) = \inf_{y\in Y} \nu(u,y) = 0.$$

Then each restriction $u_m|_Y$ is quasi-plurisubharmonic on Y; in particular, it is not identically $-\infty$. If $\pi\colon Z \to Y$ is a modification with smooth source and $E \subset Z$ is a prime divisor, then

$$\sup_m \nu\big((u_m|_Y)\circ\pi, E\big) < \infty.$$

Proof For each m, choose $y \in Y$ with $\nu(u,y) < 1/m$. Skoda's local integrability criterion gives $\mathcal{I}(mu)_y = \mathcal{O}_{X,y}$, so the image ideal $\mathcal{I}(mu)\mathcal{O}_Y$ does not vanish identically. Since u_m has analytic type $(1/m, \mathcal{I}(mu))$, Proposition 1.13.1 shows that $u_m|_Y$ is quasi-plurisubharmonic on Y. Its pullback to Z is quasi-plurisubharmonic as well, since π is biholomorphic off a proper analytic subset.

Restricting and pulling back the curvature inequality of Corollary 1.15.1, which is allowed by Lemma 1.22.1, gives

$$\pi^*(\theta|_Y) + \mathrm{dd^c}\big((u_m|_Y)\circ\pi\big) \geq -\epsilon_m \pi^*(\omega|_Y).$$

Since the errors ϵ_m are bounded, a sufficiently large multiple of a single Hermitian form on Z dominates $\pi^*(\theta|_Y) + \epsilon_m\pi^*(\omega|_Y)$ for all m. Lemma 1.17.7, applied on Z in this fixed background, therefore bounds the generic Lelong numbers along E uniformly in m. □

1.23 Integration against non-pluripolar measures

For uniformly bounded potentials, L^1 convergence with respect to a smooth volume form can be transferred to any fixed finite measure that does not charge pluripolar sets. Passing to bounded exponentials extends this to families that are only bounded above. The same mechanism applies to moving mixed Monge–Ampère measures of bounded potentials and to smoothings of a fixed measure.

1.23.1 A fixed measure

Cegrell's fixed-measure convergence lemma states that a uniformly bounded sequence of plurisubharmonic functions converging in L^1_{loc} converges in $L^1(\nu)$ on compact subsets, for every finite non-pluripolar measure ν. The usual formulation, which allows passing to a subsequence, gives the same conclusion: apply it to every subsequence after extracting a further subsequence converging almost everywhere for the smooth volume, and use the subsequence criterion for convergence. On compact Kähler manifolds, [GZ17, Lemma 11.5] gives $\int \psi_j \,\mathrm{d}\nu \to \int \psi \,\mathrm{d}\nu$ if ψ_j, ψ are uniformly bounded and ω-plurisubharmonic for a Kähler form ω, with $\psi_j \to \psi$ in L^1; applied also to $\max(\psi_j, \psi)$, it gives convergence in $L^1(\nu)$, because $|\psi_j - \psi| = 2\max(\psi_j, \psi) - \psi_j - \psi$. Its proof is local, and we use this local lemma as a standard input.

Lemma 1.23.1 *Let ν be a finite non-pluripolar measure on X. If uniformly bounded quasi-plurisubharmonic functions a_j, a have a common smooth lower curvature bound and $a_j \to a$ in $L^1(X)$, then*

$$\int_X |a_j - a| \,\mathrm{d}\nu \longrightarrow 0.$$

If instead $u_j, u \in \mathrm{PSH}(X, \theta)$ have a common upper bound and $u_j \to u$ in $L^1(X)$, then, for every $b > 0$,

$$\int_X |\mathrm{e}^{bu_j} - \mathrm{e}^{bu}| \,\mathrm{d}\nu \longrightarrow 0.$$

Proof For the first statement, add a fixed smooth strictly plurisubharmonic function on each of finitely many coordinate balls, so that the resulting functions are uniformly bounded and plurisubharmonic there. Apply the local lemma on smaller compact balls covering X and sum the integrals; the added functions cancel in the differences.

For the second statement, choose $A > 0$ with $A\omega \geq \theta$. If $u_j \leq C$, the convex chain rule gives $\mathrm{dd^c e}^{bu_j} \geq -bA\mathrm{e}^{bC}\omega$, also for unbounded u_j by bounded truncation; here the exponential is given the value zero at a pole. Thus the functions e^{bu_j} are uniformly bounded and quasi-plurisubharmonic, with a common lower curvature bound. They converge in $L^1(\mathrm{d}V)$, because the exponential is Lipschitz with constant $b\mathrm{e}^{bC}$ on $(-\infty, C]$. The first statement now applies. □

1.23.2 Vanishing exponents

Proposition 1.23.1 *Let $\varphi \in \mathrm{PSH}(X,\theta)$ be unbounded below, and for each $\lambda > 0$ let u_λ be a quasi-plurisubharmonic function with $u_\lambda \preceq \varphi$. Then, for each $\lambda > 0$, the function λu_λ has infinite distance in $L^\infty(\mathrm{d}V)$ from every constant. In particular, λu_λ cannot converge uniformly on X to a finite constant as $\lambda \downarrow 0$.*

Proof Since $u_\lambda \le \varphi + C$ for a constant C, the function u_λ is unbounded below. Every nonempty strict sublevel set $\{u_\lambda < t\}$ of the upper semicontinuous function u_λ is open, hence of positive smooth volume. Thus u_λ, and with it λu_λ, is essentially unbounded below, and both conclusions follow. □

A vanishing exponent gives convergence of the exponentials to one, even when the normalized potentials vary.

Lemma 1.23.2 *Let $v_j \in \mathrm{PSH}(X,\theta)$ satisfy $\sup_X v_j = 0$, and let $\lambda_j > 0$ tend to zero. For every finite non-pluripolar measure μ,*

$$\int_X \left|\mathrm{e}^{\lambda_j v_j} - 1\right| \mathrm{d}\mu \longrightarrow 0.$$

If $f \in L^p(\mathrm{d}V)$ for some $p > 1$, then more precisely

$$\int_X \left|\mathrm{e}^{\lambda_j v_j} - 1\right| |f| \,\mathrm{d}V \le C_p \lambda_j \|f\|_{L^p(\mathrm{d}V)}.$$

The constant depends on the fixed background, p and $\mathrm{d}V$.

Proof Choose a Hermitian form $\Omega \ge \theta$, so that $v_j \in \mathrm{PSH}(X,\Omega)$. By Lemma 1.8.1, for every $T > 1$,

$$\mathrm{Cap}_\Omega(\{v_j < -T\}) \le C/T$$

uniformly in j. Since $v_j \le 0$ and $1 - \mathrm{e}^{-x} \le x$ for $x \ge 0$, splitting the integral at this sublevel set gives

$$\int_X (1 - \mathrm{e}^{\lambda_j v_j}) \,\mathrm{d}\mu \le \lambda_j T \mu(X) + \mu(\{v_j < -T\}).$$

By Lemma 1.8.8 and the equivalence of Cap_Ω and Cap_ω (Section 1.8), the second term is uniformly small when T is large, and for this fixed T the first term tends to zero as $j \to \infty$. This proves the general assertion; here the exponential is given the value zero at a pole, and pole sets have zero μ-measure.

For the quantitative assertion, Lemma 1.5.1, applied in the background Ω, gives a uniform L^q bound for $-v_j$, where $q = p/(p-1)$. Hölder's inequality and the bound

$$0 \le 1 - \mathrm{e}^{\lambda_j v_j} \le \lambda_j |v_j|$$

give the claimed estimate for every j. □

1.23.3 Measures with bounded potentials

The fixed measure can be replaced by moving mixed Monge–Ampère measures of uniformly bounded potentials, provided these potentials converge in capacity. Since ω need not be closed, the proof has to control the torsion terms that appear in integration by parts.

Lemma 1.23.3 *Let $a_j, a, w_j, w \in \mathrm{PSH}(X,\omega)$ be uniformly bounded, and write $\omega_v = \omega + \mathrm{dd^c} v$. If $a_j \to a$ in $L^1(X)$ and $w_j \to w$ in capacity, then, for $0 \le k \le n$,*

$$\int_X |a_j - a|\, \omega_{w_j}^k \wedge \omega^{n-k} \longrightarrow 0.$$

The same conclusion holds if, instead of $a_j, a \in \mathrm{PSH}(X,\omega)$, the uniformly bounded functions a_j, a have any fixed smooth lower curvature bound.

Proof By Lemma 1.4.1, the tail envelopes $A_j = (\sup_{\ell \ge j} a_\ell)^*$ decrease to a. Since $a \le \max(a_j, a) \le A_j$, monotone convergence in capacity of bounded potentials (Lemma 1.8.9) implies

$$\alpha_j := \max(a_j, a) \longrightarrow a \quad \text{in capacity.}$$

We split $|a_j - a| = (\alpha_j - a) + (\alpha_j - a_j)$. The integral of the first part against the moving products tends to zero by Lemma 1.8.6.

For the second part, put $h_j = \alpha_j - a_j \ge 0$. We compare its integrals against the measures with indices j and ℓ. The case $k = 0$ follows from the assumed L^1 convergence. For $k \ge 1$, put $z = w_j - w_\ell$ and

$$T = \sum_{s=0}^{k-1} \omega_{w_j}^s \wedge \omega_{w_\ell}^{k-1-s} \wedge \omega^{n-k}.$$

Factorization and Stokes' theorem on the compact manifold X give

$$\int_X h_j \bigl(\omega_{w_j}^k - \omega_{w_\ell}^k\bigr) \wedge \omega^{n-k} = \int_X h_j \mathrm{dd^c} z \wedge T = \int_X z\, \mathrm{dd^c}(h_j T).$$

Since ω is not assumed to be closed, T need not be closed, and the product rule reads

$$\mathrm{dd^c}(h_j T) = \mathrm{dd^c} h_j \wedge T + \mathrm{d}h_j \wedge \mathrm{d^c} T - \mathrm{d^c} h_j \wedge \mathrm{d}T + h_j \mathrm{dd^c} T. \tag{1.43}$$

We estimate the four terms. Since $\mathrm{dd^c} h_j = \omega_{\alpha_j} - \omega_{a_j}$, the absolute contribution of the first is at most

$$\int_X |z| (\omega_{\alpha_j} + \omega_{a_j}) \wedge T.$$

It tends to zero as $j, \ell \to \infty$ by Lemma 1.8.6, since $w_j - w_\ell \to 0$ in capacity. Differentiating each factor shows that $\mathrm{d}T = \mathrm{d}\omega \wedge R$ and $\mathrm{d}^{\mathrm{c}}T = \mathrm{d}^{\mathrm{c}}\omega \wedge R$, where R is a positive linear combination of mixed products of bidegree $(n-2, n-2)$. Moreover, $\mathrm{dd}^{\mathrm{c}}T$ is a sum of terms consisting of $\mathrm{dd}^{\mathrm{c}}\omega$ times mixed products of degree $n-2$, or of $\mathrm{d}\omega \wedge \mathrm{d}^{\mathrm{c}}\omega$ times mixed products of degree $n-3$; terms of negative degree are absent. These smooth torsion factors are bounded in absolute value by fixed multiples of ω^2 and ω^3, respectively. As h_j is uniformly bounded, the contribution of $h_j \mathrm{dd}^{\mathrm{c}}T$ also tends to zero, by the same capacity estimate.

For the two gradient terms, the pointwise Cauchy–Schwarz inequality for a positive mixed product gives, for example,

$$\left| \int_X z \, \mathrm{d}h_j \wedge \mathrm{d}^{\mathrm{c}}\omega \wedge R \right|^2 \leq C \left(\int_X |z| \, \mathrm{d}h_j \wedge \mathrm{d}^{\mathrm{c}}h_j \wedge \omega \wedge R \right) \left(\int_X |z| \, \omega^2 \wedge R \right).$$

The second factor tends to zero, and the first is uniformly bounded. Indeed, if p is any of the uniformly bounded potentials and S any positive mixed product of degree $n-1$ occurring here, then

$$\int_X \mathrm{d}p \wedge \mathrm{d}^{\mathrm{c}}p \wedge S = \frac{1}{2} \int_X p^2 \mathrm{dd}^{\mathrm{c}}S - \int_X p \, \mathrm{dd}^{\mathrm{c}}p \wedge S.$$

Expand $\mathrm{dd}^{\mathrm{c}}S$ with all its torsion terms as above, and write $\mathrm{dd}^{\mathrm{c}}p = \omega_p - \omega$. Boundedness of p and Lemma 1.8.6 bound both integrals uniformly. Finally, $\mathrm{d}h_j \wedge \mathrm{d}^{\mathrm{c}}h_j \leq 2\mathrm{d}\alpha_j \wedge \mathrm{d}^{\mathrm{c}}\alpha_j + 2\mathrm{d}a_j \wedge \mathrm{d}^{\mathrm{c}}a_j$ as positive quadratic forms, which proves the asserted bound. The other gradient term is estimated in the same way. In dimension one, $T = 1$ and all torsion terms in (1.43) vanish.

These integration-by-parts and energy identities hold for smooth potentials. They extend to bounded potentials through smooth decreasing approximations in the Hermitian background, using the local Bedford–Taylor chain rule, its gradient products and quasi-continuity; equivalently, they follow from the bounded current calculus of Section 1.7. Only the common bound on the potentials enters the estimates.

We have shown that the difference between the integrals of h_j against the measures with indices j and ℓ tends to zero as $j, \ell \to \infty$. For a fixed index ℓ, the measure $\omega_{w_\ell}^k \wedge \omega^{n-k}$ is a fixed finite non-pluripolar measure, and Lemma 1.23.1, applied separately to $\alpha_j \to a$ and $a_j \to a$ in $L^1(X)$, gives

$$\int_X h_j \, \omega_{w_\ell}^k \wedge \omega^{n-k} \longrightarrow 0 \qquad \text{as } j \to \infty.$$

Choosing first ℓ large and then letting $j \to \infty$, we conclude that the integrals of h_j against $\omega_{w_j}^k \wedge \omega^{n-k}$ tend to zero. Together with the estimate for $\alpha_j - a$, this proves the result. If the a_j only have some other fixed lower curvature bound, choose a Hermitian form $\Omega \geq \omega$ such that they are Ω-plurisubharmonic. By Section 1.8, convergence in capacity is unchanged by this replacement. Apply the result just proved with Ω in place of ω; by positivity, the original mixed measures are bounded by their counterparts for Ω. □

1.23.4 Smoothing the measure

Convolution in coordinate charts gives smooth approximations of a finite measure. For a varying quasi-plurisubharmonic integrand, the submean inequality transfers the error to the original measure; this gives the additional control needed beyond weak convergence of the measures themselves.

For a smooth real $(1,1)$-form θ with $\mathrm{PSH}(X,\theta) \neq \varnothing$ and a probability measure ν on X, put

$$A_\theta(\nu) = \sup\left\{ \int_X (-w)\,\mathrm{d}\nu : w \in \mathrm{PSH}(X,\theta),\ \sup_X w = 0 \right\}.$$

This number may be infinite. It is finite for Monge–Ampère measures of bounded potentials.

Lemma 1.23.4 *Let $\psi \in \mathrm{PSH}(X,\omega)$ be bounded, and put $\nu = (\omega + \mathrm{dd^c}\psi)^n$. There is a finite constant C, depending only on ω, θ, $\mathrm{d}V$ and $\|\psi\|_\infty$, such that*

$$\int_X (-w)\,\mathrm{d}\nu \le C \qquad (w \in \mathrm{PSH}(X,\theta),\ \sup_X w = 0).$$

Proof Choose finitely many pairs of concentric coordinate balls $B'_a \Subset B_a$ such that the B'_a cover X, and fixed smooth functions q_a, r_a on B_a with $\mathrm{dd^c} q_a \ge \omega$ and $\mathrm{dd^c} r_a \ge \theta$, with $r_a \le 0$. Then $\psi + q_a$ is bounded and plurisubharmonic, $w + r_a$ is negative and plurisubharmonic, and positivity of bounded products gives

$$\int_{B'_a} (-w)\,\mathrm{d}\nu \le \int_{B'_a} (-w - r_a)\big(\mathrm{dd^c}(\psi + q_a)\big)^n \le C_a\big(1 + \|w\|_{L^1(B_a)}\big).$$

The last step is the weighted Chern–Levine–Nirenberg estimate of Section 1.6. Summing over the cover and using the normalized L^1 bound of Theorem 1.4.1 proves the lemma. □

To construct the smoothings, choose finitely many coordinate balls B_a and smooth functions $\eta_a \ge 0$ with compact support in B_a and $\sum_a \eta_a = 1$. Let $\chi \ge 0$ be a smooth radial kernel of integral one supported in the unit ball of $\mathbb{C}^n$, and fix $\epsilon_j \downarrow 0$ so small that all convolutions below stay in the corresponding coordinate ball. For a function F, let $S_{a,j}F$ be its convolution, in the chart of B_a, with the kernel $\epsilon_j^{-2n}\chi(\cdot/\epsilon_j)$. For a probability measure ν on X, define a smooth probability measure ν_j by convolving each coordinate measure $\eta_a\nu$ with the same kernel and adding the extensions by zero. It is characterized by the identity

$$\int_X F\,\mathrm{d}\nu_j = \sum_a \int_X \eta_a S_{a,j} F\,\mathrm{d}\nu, \tag{1.44}$$

for Borel functions F; since the kernel is even, the same convolution occurs on the right. This identity first holds for nonnegative Borel functions and then for integrable signed functions. In particular, $\nu_j \rightharpoonup \nu$ weakly.

Lemma 1.23.5 *Let ν be a probability measure with $A_\theta(\nu) < \infty$, and let ν_j be its smoothings defined by* (1.44)*. Then*

$$\sup_j A_\theta(\nu_j) < \infty.$$

Moreover, let z_j, z be quasi-plurisubharmonic functions with a common upper bound and a common smooth lower curvature bound, and suppose that $z_j \to z$ both in $L^1(X)$ and in $L^1(\nu)$. Then

$$\int_X |z_j - z| \, \mathrm{d}\nu_j \longrightarrow 0. \tag{1.45}$$

Consequently, for every $\lambda > 0$,

$$\mathrm{e}^{\lambda z_j}\nu_j \rightharpoonup \mathrm{e}^{\lambda z}\nu.$$

Proof On each coordinate ball B_a choose a fixed smooth function q_a with $\mathrm{dd}^{\mathrm{c}} q_a \geq \theta$. If $w \in \mathrm{PSH}(X, \theta)$ and $\sup_X w = 0$, then $w + q_a$ is plurisubharmonic, and the submean inequality gives

$$-S_{a,j} w \leq -w + S_{a,j} q_a - q_a \leq -w + C,$$

with a constant independent of w and j. Integrating and using (1.44) proves $A_\theta(\nu_j) \leq A_\theta(\nu) + C$.

For the second assertion, the tail envelopes

$$H_j = \left(\sup_{k \geq j} z_k \right)^*$$

decrease to z by Lemma 1.4.1. Enlarge the functions q_a so that $H_j + q_a$, $z_j + q_a$ and $z + q_a$ are all plurisubharmonic. Since $H_j \geq z_j$ and $H_j \geq z$, we have $|z_j - z| \leq 2H_j - z_j - z$, and the identity (1.44) and the submean inequality imply

$$\begin{aligned} \int_X |z_j - z| \, \mathrm{d}\nu_j &\leq \sum_a \int_X \eta_a S_{a,j}(2H_j - z_j - z) \, \mathrm{d}\nu \\ &\leq 2 \sum_a \int_X \eta_a \big(S_{a,j}(H_j + q_a) - (z + q_a) \big) \, \mathrm{d}\nu + \int_X (z - z_j) \, \mathrm{d}\nu. \end{aligned}$$

All these integrals are well defined: each function on the larger side of a submean inequality is bounded above, and its negative part is controlled by that of the integrable function on the smaller side plus a constant.

Radial averages of a plurisubharmonic function decrease when the radius decreases. Therefore $S_{a,j}(H_j + q_a)$ decreases to $z + q_a$ on the support of η_a: to identify the

limit, fix an index k, use $H_j \le H_k$ for $j \ge k$, let $j \to \infty$ in the radius, and then let $k \to \infty$; the reverse inequality is the submean inequality. The nonnegative integrand $\eta_a\big(S_{a,j}(H_j + q_a) - (z + q_a)\big)$ in the second line of the last display is dominated on this compact support by a constant minus z, which is ν-integrable, so its integral tends to zero by dominated convergence. The final integral tends to zero by convergence in $L^1(\nu)$. This proves (1.45).

Since the exponential is Lipschitz on the common range of the functions, (1.45) also gives $\int_X |\mathrm{e}^{\lambda z_j} - \mathrm{e}^{\lambda z}|\,\mathrm{d}\nu_j \to 0$. If $\zeta > 0$ is smooth, then $\zeta \mathrm{e}^{\lambda z} = \mathrm{e}^{\lambda z + \log \zeta}$ is bounded and quasi-plurisubharmonic. For any fixed bounded quasi-plurisubharmonic function F, the same local submean argument and dominated convergence in (1.44) give

$$\int_X F\,\mathrm{d}\nu_j \longrightarrow \int_X F\,\mathrm{d}\nu.$$

Applying this to $F = \zeta \mathrm{e}^{\lambda z}$ gives $\int_X \zeta \mathrm{e}^{\lambda z_j}\,\mathrm{d}\nu_j \to \int_X \zeta \mathrm{e}^{\lambda z}\,\mathrm{d}\nu$ for every smooth $\zeta > 0$. Since every smooth function is a difference of two positive smooth functions, the convergence holds for all smooth test functions, and uniformly bounded masses together with uniform approximation of continuous functions by smooth ones prove weak convergence. □

1.24 Curvature of line bundles

Let M be a complex manifold and let θ be a smooth real $(1,1)$-form on M. To relate potentials to holomorphic sections, we ask when a current $\theta + \mathrm{dd}^{\mathrm{c}} u$, with u locally integrable, can be the curvature current of a metric on a line bundle. The transition law for the local weights of such a metric forces a closedness condition on the background.

Proposition 1.24.1 *The curvature current of a singular Hermitian metric with locally integrable weights on a holomorphic line bundle is closed. In particular, a nonclosed form θ cannot itself be the curvature current of such a metric, and $\theta + \mathrm{dd}^{\mathrm{c}} u$ is not closed on any neighborhood of a point where $\mathrm{d}\theta \neq 0$.*

Proof Let h be a singular Hermitian metric on a holomorphic line bundle $L \to M$. Choose a local holomorphic frame e of L and write its squared norm as $|e|_h^2 = \mathrm{e}^{-\varphi}$; the curvature current is then locally $\mathrm{dd}^{\mathrm{c}}\varphi$. If $e' = ge$ is another frame, with g holomorphic and nowhere zero, the new weight is $\varphi - \log|g|^2$. Since $\log|g|^2$ is pluriharmonic, these local currents agree on overlaps and glue to a global current, which is closed because $\mathrm{ddd}^{\mathrm{c}} = 0$. The final assertion follows from the identity

$$\mathrm{d}(\theta + \mathrm{dd}^{\mathrm{c}} u) = \mathrm{d}\theta$$

of currents. □

Accordingly, when we study sections subject to multiplier-ideal conditions, we specify both the line bundle, whose curvature is closed, and any auxiliary Hermitian form on M. An algebraic interpretation then requires relating the invariants of the line bundle to this analytic background.

1.25 Polar curvature

For $u \in \mathrm{PSH}(X, \theta)$, the restriction of the current $\theta + \mathrm{dd}^{\mathrm{c}} u$ to the pole set $\{u = -\infty\}$ is called the *polar curvature* of u; on curves it is the polar curvature of Section 1.6.1, since smooth forms do not charge polar sets. A bounded change of the potential preserves more than its Lelong numbers: it preserves the whole polar curvature. Indeed, a local energy estimate puts the bounded difference of two potentials in a Sobolev space, while pole sets admit smooth test functions of arbitrarily small energy, and these test functions show that no coefficient measure of the Hessian of the difference charges the pole set. Polar curvature is closed even when the pole set is not closed, so a local elliptic equation recovers it from its scalar Hermitian trace.

Lemma 1.25.1 *Let $U \subset \mathbb{R}^d$ be open and let $w \in L^\infty_{\mathrm{loc}}(U)$. If Δw is a locally finite signed Radon measure, then $w \in W^{1,2}_{\mathrm{loc}}(U)$.*

Proof Fix a smooth function $0 \le \psi \le 1$ with compact support in U, and a relatively compact open neighborhood V of its support. Mollify w inside V with a standard mollifier ρ_ε, writing $w_\varepsilon = w * \rho_\varepsilon$ and $\nu_\varepsilon = (\Delta w) * \rho_\varepsilon$. For small ε, integration by parts and Young's inequality give

$$\begin{aligned}\int \psi^2 |\nabla w_\varepsilon|^2 &= -\int \psi^2 w_\varepsilon \nu_\varepsilon - 2\int \psi w_\varepsilon \nabla\psi \cdot \nabla w_\varepsilon \\ &\le \|w\|_{L^\infty(V)} |\Delta w|(V) + \tfrac{1}{2}\int \psi^2 |\nabla w_\varepsilon|^2 + 2\int w_\varepsilon^2 |\nabla\psi|^2.\end{aligned}$$

Absorbing the middle term on the right, we obtain a bound for $\int \psi^2 |\nabla w_\varepsilon|^2$ that is independent of ε. Taking $\psi = 1$ on a prescribed smaller compact set, weak compactness in $W^{1,2}$ and convergence of the mollifications in L^2 identify the weak limit with w. This proves the assertion. □

Lemma 1.25.2 *Let v be quasi-plurisubharmonic on a complex manifold, let U be a coordinate ball with compact closure in a larger chart, and let $K \subset U \cap \{v = -\infty\}$ be compact. For every open $O \subset U$ containing K, there are real $\varphi_j \in C^\infty_c(O)$ with $0 \le \varphi_j \le 1$, equal to one near K, such that $\|\varphi_j\|_{W^{1,2}(U)} \to 0$. The norm uses the Euclidean coordinates.*

Proof Adding a smooth quadratic function and subtracting a constant, we obtain a negative plurisubharmonic function h on U with the same pole set as v. Put $h_M = \max(h, -M)$ for $M \ge 1$, and choose $\psi \in C^\infty_c(O)$ with $0 \le \psi \le 1$ and $\psi = 1$ near K. On a fixed compact neighborhood of $\operatorname{supp}\psi$, the positive measures Δh_M have

uniformly bounded mass: choose a nonnegative smooth function χ with compact support in U, equal to one on this neighborhood, and use

$$\int \chi \, \Delta h_M = \int h_M \Delta \chi, \qquad |h_M| \le |h| \in L^1_{\mathrm{loc}}(U).$$

Apply the integration-by-parts estimate of the preceding proof to mollifications of h_M. Their absolute values are at most M, and their squares are at most M times their absolute values. The uniform bounds for the mass of the Laplacians and for the local L^1 norms therefore give, after passing to the weak limit in $W^{1,2}$,

$$\int \psi^2 |\nabla h_M|^2 \le CM, \qquad \left\| \frac{-\psi h_M}{M} \right\|^2_{W^{1,2}(U)} \le \frac{C}{M}.$$

Here C depends on the fixed cutoffs and the local L^1 norm of h, but not on M. The second inequality also uses $h_M^2 \le M|h|$ in the terms containing ψ and $\nabla\psi$.

The Sobolev function $-\psi h_M / M$ takes values between zero and one. It equals one on a neighborhood of K, since h is upper semicontinuous and therefore $\{h < -M\}$ is an open neighborhood of K. Extend it by zero and mollify with a sufficiently small radius: the support stays in O, the function remains equal to one on a smaller neighborhood of K, and its Sobolev norm does not increase. Letting $M \to \infty$ gives the required smooth functions. □

For a current T of order zero and a Borel set B, we write $\mathbf{1}_B T$ for the restriction of T to B, taken coefficientwise on the Radon coefficient measures of T. These restrictions commute with smooth changes of coordinates.

Proposition 1.25.1 *Let u, v be quasi-plurisubharmonic functions on a complex manifold such that every point has a neighborhood on which $u - C \le v \le u + C$ for some constant C. Then u and v have the same pole set $S = \{u = -\infty\} = \{v = -\infty\}$, and*

$$\mathbf{1}_S \mathrm{dd^c} u = \mathbf{1}_S \mathrm{dd^c} v.$$

Consequently, for any smooth real $(1,1)$-forms θ, β,

$$\mathbf{1}_S (\theta + \mathrm{dd^c} u) = \mathbf{1}_S (\beta + \mathrm{dd^c} v).$$

Proof The two local inequalities show that the pole sets agree. Off S put $w = u - v$, and give w arbitrary bounded values on S. Since S has Lebesgue measure zero, this defines the locally bounded distribution $u - v$. The coefficient measures of $\mathrm{dd^c} w$ are locally finite, since after a smooth correction the complex Hessians of u and v are positive currents of order zero. Taking the Euclidean trace shows that Δw is a signed Radon measure, so Lemma 1.25.1 gives $w \in W^{1,2}_{\mathrm{loc}}$.

Consequently, the real and imaginary parts of the coefficient measures of $\mathrm{dd^c} w$ are continuous linear functionals on $W^{1,2}_0$ of a smaller ball, since a second distributional derivative of w is a first derivative of its L^2 gradient. Let ν be any one of these real signed measures, with Jordan decomposition $\nu = \nu^+ - \nu^-$. Choose a compact set

K contained in the intersection of S with a positive set of a Hahn decomposition, so that $\nu^-(K) = 0$. For $\epsilon > 0$, take an open neighborhood O of K in the ball with $\nu^-(O) < \epsilon$. The test functions φ_j of Lemma 1.25.2, applied to u, satisfy

$$\nu^+(K) \le \int \varphi_j \, \mathrm{d}\nu^+ = \langle \nu, \varphi_j \rangle + \int \varphi_j \, \mathrm{d}\nu^- \le |\langle \nu, \varphi_j \rangle| + \epsilon.$$

By continuity on the Sobolev space, the first term tends to zero. Letting $\epsilon \downarrow 0$ and using inner regularity gives $\nu^+(S) = 0$ locally, and the same argument applied to $-\nu$ gives $\nu^-(S) = 0$. Thus all coefficient measures of $\mathrm{dd^c} w$ vanish on S, which proves the first identity. Since smooth forms also give S zero mass, the second identity follows.□

For closed positive currents, local potentials turn equality of the scalar traces into an elliptic equation. If both currents are carried by a Lebesgue null set, the resulting smooth difference vanishes.

Lemma 1.25.3 *Let T, S be closed positive $(1,1)$-currents on a complex manifold of dimension $n \ge 1$, both carried coefficientwise by a Borel set of Lebesgue measure zero. For any Hermitian form η,*

$$T \wedge \eta^{n-1} = S \wedge \eta^{n-1} \quad \Longrightarrow \quad T = S.$$

Proof On a small coordinate ball, take plurisubharmonic local potentials a, b with $T = \mathrm{dd^c} a$ and $S = \mathrm{dd^c} b$, by [Dem12, Chapter III, Proposition 1.19, p. 135]. Put $h = a - b \in L^1_{\mathrm{loc}}$. The trace identity gives $\mathrm{dd^c} h \wedge \eta^{n-1} = 0$ in the sense of distributions. Dividing by the smooth positive volume form η^n gives a scalar second-order operator with smooth coefficients whose principal matrix is a positive multiple of the inverse of the Hermitian matrix of η; it is uniformly elliptic on every smaller ball. Local elliptic regularity therefore makes h smooth. Thus $T - S = \mathrm{dd^c} h$ is smooth and carried by a Lebesgue null set, so its coefficient measures vanish. This proves equality locally, and hence globally. □

Pushforward by a proper map commutes with restriction to the inverse image of a Borel set. Hence polar curvature is compatible with modifications, even when the pole set is not closed.

Proposition 1.25.2 *Let $\pi : Y \to X$ be a modification of connected complex manifolds, let θ be a smooth real $(1,1)$-form on X, and let $u \in \mathrm{PSH}(X, \theta)$. Put $S = \{u = -\infty\}$ and*

$$T = \theta + \mathrm{dd^c} u, \qquad T_Y = \pi^* \theta + \mathrm{dd^c}(u \circ \pi).$$

Then

$$\pi_* \big(\mathbf{1}_{\pi^{-1}(S)} T_Y \big) = \mathbf{1}_S T. \tag{1.46}$$

Proof By Proposition 1.22.4, $u \circ \pi \in \mathrm{PSH}(Y, \pi^* \theta)$. Change of variables on the locus where π is an isomorphism, whose complement has Lebesgue measure zero in both manifolds, gives $\pi_*(u \circ \pi) = u$ as distributions and $\pi_* \pi^* \theta = \theta$ as currents. Since pushforward by a proper map commutes with $\mathrm{dd^c}$, we get $\pi_* T_Y = T$.

For any Borel set $B \subset X$, the coefficient measures of $\pi_* T_Y$ are pushforwards of the coefficient measures of T_Y, multiplied by the smooth coefficients that arise from pulling back a test form. Restriction therefore gives $\pi_*(\mathbf{1}_{\pi^{-1}(B)} T_Y) = \mathbf{1}_B \pi_* T_Y$. Applying this to the Borel set S proves (1.46). □

The restriction of a closed positive current to a closed complete pluripolar set is closed; the classical proof uses bounded cutoff functions. Since these cutoff functions converge pointwise to the indicator function of the complement of the pole set, the argument also applies to pole sets that are not closed.

Proposition 1.25.3 *Let X be a complex manifold of dimension $n \geq 1$, let u be quasi-plurisubharmonic on X, and let θ be a smooth real $(1,1)$-form. Then*

$$\mathrm{Pol}_\theta(u) := \mathbf{1}_{\{u=-\infty\}}(\theta + \mathrm{dd^c} u)$$

is a closed positive $(1,1)$-current.

Proof Locally, adding a smooth quadratic function to u and subtracting a constant gives a negative plurisubharmonic function a with pole set $S = \{u = -\infty\}$. Put $\Theta = \mathrm{dd^c} a$. Since smooth forms give S zero mass, the polar current equals $\mathbf{1}_S \Theta$ and is positive. In dimension one it is closed for degree reasons.

Now suppose that $n \geq 2$. Choose a smooth convex nondecreasing function χ on $\mathbb{R}$ with $\chi(t) = 0$ for $t \leq 1/2$ and $\chi(1) = 1$, and a smooth function $0 \leq \vartheta \leq 1$ that vanishes near zero and equals one near one. Set

$$w_k = \chi(\mathrm{e}^{a/k}), \qquad k \geq 1.$$

These functions are bounded and plurisubharmonic, with $0 \leq w_k \leq 1$. They vanish on S and tend to one at every point outside S. Hence $\vartheta(w_k)\Theta$ converges locally, coefficientwise in total variation, to $\mathbf{1}_{X\setminus S}\Theta$. Moreover, $\vartheta'(w_k)$ vanishes on S and tends to zero everywhere.

Fix a constant strongly positive $(n-2, n-2)$-form β and put $\Xi = \Theta \wedge \beta$. The bounded Bedford–Taylor chain rule gives

$$\mathrm{d}\big(\vartheta(w_k)\Xi\big) = \vartheta'(w_k)\, \mathrm{d}w_k \wedge \Xi.$$

Indeed, approximate w_k on a smaller ball by a decreasing sequence of smooth plurisubharmonic functions. Dominated convergence gives convergence of the corresponding ϑ-weighted currents, while the convergence of bounded Bedford–Taylor energies and the Cauchy–Schwarz inequality give convergence of the differentiated expressions; this proves the chain rule for w_k. For a smooth compactly supported test one-form α, choose a nonnegative smooth cutoff τ equal to one near its support. Since $w_k \geq 0$ and Ξ is closed and positive, bounded integration by parts gives

$$2\int \tau\, \mathrm{d}w_k \wedge \mathrm{d^c} w_k \wedge \Xi \leq \int \tau\, \mathrm{dd^c}(w_k^2) \wedge \Xi = \int w_k^2 \mathrm{dd^c}\tau \wedge \Xi \leq C.$$

Here C depends only on the local mass of Ξ and the fixed cutoff. The Cauchy–Schwarz inequality for a positive current now yields

$$\left|\langle \vartheta'(w_k)\,\mathrm{d}w_k \wedge \Xi, \alpha\rangle\right|^2 \le C_\alpha \int_{\operatorname{supp}\alpha} |\vartheta'(w_k)|^2\, \Xi \wedge \kappa \longrightarrow 0,$$

where κ is the Euclidean form; the limit follows by dominated convergence with respect to the fixed locally finite trace measure. Hence $\mathrm{d}(\vartheta(w_k)\Xi) \to 0$, and $\mathbf{1}_{X\setminus S}\Theta \wedge \beta$ is closed. Since constant strongly positive forms span all constant forms of bidegree $(n-2,n-2)$, the current $\mathbf{1}_{X\setminus S}\Theta$ is closed. Subtracting it from Θ shows that $\mathbf{1}_S\Theta$ is closed. The local currents agree on overlaps and give the asserted global current. $\square$

Chapter 2
Torsion and the variation of mass

Throughout this chapter, ω is a Hermitian form on X. If ω were closed, Stokes' theorem would make the total Monge–Ampère mass $\int_X (\omega + \mathrm{dd^c} u)^n$ independent of the potential u. For a nonclosed Hermitian form this invariance can fail. Integration by parts expresses the mass of a smooth potential as a polynomial whose coefficients pair the potential with the complex Hessians of the powers of ω, and polarization shows that the vanishing of these Hessians is also necessary for invariance. When they do not vanish, even smooth potentials of the same singularity type can have different masses, and the usual global comparison principle fails. The masses may nevertheless remain bounded, and whether they do depends only on the manifold, not on the chosen Hermitian form.

2.1 Integration by parts with a nonclosed form

Integration by parts moves the complex Hessian of a function onto the form against which it is integrated. When that form is a power of a nonclosed $(1,1)$-form, its complex Hessian contains, besides the complex Hessian of the form itself, a quadratic term in its exterior derivative.

Lemma 2.1.1 *For a smooth real function f and a smooth form T of bidegree $(n-1, n-1)$,*

$$\int_X \mathrm{dd^c} f \wedge T = \int_X f \, \mathrm{dd^c} T.$$

Proof Since $\mathrm{d^c} f$ is a one-form, $\mathrm{d}(\mathrm{d^c} f \wedge T) = \mathrm{dd^c} f \wedge T - \mathrm{d^c} f \wedge \mathrm{d}T$, and Stokes' theorem gives the first equality in

$$\int_X \mathrm{dd^c} f \wedge T = \int_X \mathrm{d^c} f \wedge \mathrm{d}T = \int_X f \, \mathrm{dd^c} T.$$

For the second, recall that the integral of $\mathrm{d}^{\mathrm{c}}\sigma$ vanishes for every form σ of total degree $2n-1$, by Stokes' theorem for ∂ and $\bar{\partial}$. We apply this to $\sigma = f\,\mathrm{d}T$ and use $\mathrm{d}^{\mathrm{c}}\mathrm{d} = -\mathrm{d}\mathrm{d}^{\mathrm{c}}$. □

Lemma 2.1.2 *For a smooth real $(1,1)$-form θ and an integer $q \geq 1$,*

$$\mathrm{d}\mathrm{d}^{\mathrm{c}}(\theta^q) = q\,\theta^{q-1} \wedge \mathrm{d}\mathrm{d}^{\mathrm{c}}\theta + q(q-1)\,\theta^{q-2} \wedge \mathrm{d}\theta \wedge \mathrm{d}^{\mathrm{c}}\theta,$$

where the second term is omitted when $q = 1$.

Proof For $q = 1$ there is nothing to prove. For $q \geq 2$, since θ has even degree, $\mathrm{d}^{\mathrm{c}}(\theta^q) = q\,\theta^{q-1} \wedge \mathrm{d}^{\mathrm{c}}\theta$. Applying d and using again that θ^{q-1} has even degree, we obtain

$$\mathrm{d}\mathrm{d}^{\mathrm{c}}(\theta^q) = q(q-1)\,\theta^{q-2} \wedge \mathrm{d}\theta \wedge \mathrm{d}^{\mathrm{c}}\theta + q\,\theta^{q-1} \wedge \mathrm{d}\mathrm{d}^{\mathrm{c}}\theta.$$

Thus the quadratic torsion term $\mathrm{d}\omega \wedge \mathrm{d}^{\mathrm{c}}\omega$ can make the higher powers of ω fail to be $\mathrm{d}\mathrm{d}^{\mathrm{c}}$-closed even when $\mathrm{d}\mathrm{d}^{\mathrm{c}}\omega = 0$. The mass expansion below involves the complex Hessians of all these powers separately.

2.2 The mass polynomial

For a smooth real function u on X, recall the notation $\omega_u = \omega + \mathrm{d}\mathrm{d}^{\mathrm{c}}u$ of Section 1.7, and set

$$\mathcal{H}_\omega = \{u \in C^\infty(X,\mathbb{R}) : \omega + \mathrm{d}\mathrm{d}^{\mathrm{c}}u > 0\}, \qquad \mathrm{M}_\omega(u) = \int_X (\omega + \mathrm{d}\mathrm{d}^{\mathrm{c}}u)^n.$$

The set $\mathcal{H}_\omega$ of smooth strictly ω-plurisubharmonic functions is convex and open in the C^2 topology, while the mass $\mathrm{M}_\omega(u)$ is defined for every smooth real u. Expanding it along a smooth direction gives a polynomial whose nonconstant coefficients detect the complex Hessians of the powers of the background.

Theorem 2.2.1 *For every smooth real function u,*

$$\mathrm{M}_\omega(u) = \int_X \omega^n + \sum_{k=1}^{n-1} \binom{n}{k} \int_X u(\mathrm{d}\mathrm{d}^{\mathrm{c}}u)^{k-1} \wedge \mathrm{d}\mathrm{d}^{\mathrm{c}}(\omega^{n-k}). \tag{2.1}$$

Proof Expand $(\omega + \mathrm{d}\mathrm{d}^{\mathrm{c}}u)^n$ by the binomial theorem. The term with $k = n$ integrates to zero, since it is exact:

$$(\mathrm{d}\mathrm{d}^{\mathrm{c}}u)^n = \mathrm{d}\big(\mathrm{d}^{\mathrm{c}}u \wedge (\mathrm{d}\mathrm{d}^{\mathrm{c}}u)^{n-1}\big).$$

For $1 \leq k \leq n-1$, apply Lemma 2.1.1 with $f = u$ and $T = (\mathrm{d}\mathrm{d}^{\mathrm{c}}u)^{k-1} \wedge \omega^{n-k}$. Since both d and d^{c} annihilate $\mathrm{d}\mathrm{d}^{\mathrm{c}}u$,

$$\mathrm{d}\mathrm{d}^{\mathrm{c}}T = (\mathrm{d}\mathrm{d}^{\mathrm{c}}u)^{k-1} \wedge \mathrm{d}\mathrm{d}^{\mathrm{c}}(\omega^{n-k}),$$

which gives the kth term of (2.1). When $n = 1$, the sum is empty and the mass is constant. □

If every $dd^c(\omega^j)$ with $1 \le j \le n-1$ vanishes, (2.1) shows that the mass is constant. Conversely, varying the potential in several independent directions recovers each of these forms by polarization.

Theorem 2.2.2 (Chiose) *The following conditions are equivalent:*

(1) $\int_X (\omega + dd^c u)^n$ *is independent of* $u \in \mathcal{H}_\omega$*;*
(2) $dd^c(\omega^j) = 0$ *for* $1 \le j \le n-1$*;*
(3) $dd^c\omega = 0$ *and* $d\omega \wedge d^c\omega = 0$*.*

They are also equivalent to the condition that $\int_X (\omega + dd^c u)^n$ *be the same for all bounded* ω*-plurisubharmonic functions* u*, where* $(\omega + dd^c u)^n$ *is the Bedford–Taylor product of Section 1.7. Under these conditions the common mass is* $\int_X \omega^n$*.*

Proof *Step 1: the first condition implies the second.* Let u be a smooth real function. For $|t|$ small, tu belongs to $\mathcal{H}_\omega$ for both signs of t, so the polynomial $t \mapsto \int_X (\omega + t dd^c u)^n$ is constant on an interval and its nonconstant coefficients vanish. For $1 \le j \le n-1$ this gives

$$\int_X \omega^j \wedge (dd^c u)^{n-j} = 0.$$

Replace u by a linear combination of $n-j$ smooth real functions $u_1, \dots, u_{n-j}$ and compare the coefficients of the monomial in which every parameter appears once. We obtain

$$\int_X \omega^j \wedge dd^c u_1 \wedge \cdots \wedge dd^c u_{n-j} = 0.$$

Integrating by parts in the first factor and using that u_1 is arbitrary, we obtain the pointwise identity

$$dd^c(\omega^j) \wedge dd^c u_2 \wedge \cdots \wedge dd^c u_{n-j} = 0$$

for every choice of the remaining functions. At a given point, the complex Hessians of these functions can be prescribed arbitrarily: multiply a quadratic polynomial in a coordinate chart by a cutoff function equal to one near that point. Since the identity is multilinear, we may complexify it and use the quadratic monomials $z_a \bar{z}_b$. Wedge products of their complex Hessians give all coordinate forms of the complementary bidegree, and pairing with these forms shows that every coefficient of $dd^c(\omega^j)$ vanishes. When $j = n-1$ there are no remaining factors, and the same conclusion follows directly from the arbitrariness of u_1. This proves the second condition.

Step 2: the remaining equivalences for smooth potentials. The second condition implies the first by Theorem 2.2.1. For $n \ge 3$, the identities for $j = 1, 2$ and Lemma 2.1.2 give $dd^c\omega = 0$ and

$$0 = dd^c(\omega^2) = 2\omega \wedge dd^c\omega + 2d\omega \wedge d^c\omega,$$

so the second condition implies the third. For $n = 2$, the second condition is $\mathrm{dd^c}\omega = 0$, while $\mathrm{d}\omega \wedge \mathrm{d^c}\omega$ vanishes for degree reasons; for $n = 1$, both forms in the third condition vanish for degree reasons. In both cases the implication is immediate. Conversely, Lemma 2.1.2 derives all the identities of the second condition from the third.

Step 3: bounded potentials. Assume the first condition and let u be a bounded ω-plurisubharmonic function. By Theorem 1.11.1, there are uniformly bounded functions $u_j \in \mathcal{H}_\omega$ decreasing to u. Their masses all equal $\int_X \omega^n$, and the continuity of the bounded products of Section 1.7 along such sequences passes this common value to the limit. Conversely, every element of $\mathcal{H}_\omega$ is a bounded ω-plurisubharmonic function, so constancy for bounded potentials implies the first condition. The zero potential shows that the common mass is $\int_X \omega^n$. □

In particular, a closed background has invariant mass. Under the equivalent conditions above, (2.1) even gives $\mathrm{M}_\omega(u) = \int_X \omega^n$ for every smooth real u, not only for $u \in \mathcal{H}_\omega$.

Mass invariance can also hold in a nonclosed background. In Theorem 27.17.3, a pluriclosed metric with nonzero exterior derivative satisfies the second condition of Theorem 2.2.2, so all bounded potentials have the same mass in this background, although the perturbed forms of a family of smooth potentials have least eigenvalue, relative to a flat metric, tending to zero.

The first variation of the mass involves only the complex Hessian of the power ω_u^{n-1} of the perturbed form.

Proposition 2.2.1 *For smooth real functions u, v on X,*

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathrm{M}_\omega(u + tv) = n \int_X v\,\mathrm{dd^c}(\omega_u^{n-1}). \tag{2.2}$$

If M_ω is constant on $\mathcal{H}_\omega$, then $\mathrm{dd^c}(\omega_u^{n-1}) = 0$ for every $u \in \mathcal{H}_\omega$.

Proof Since $\mathrm{M}_\omega(u + tv)$ is a polynomial in t, its derivative at $t = 0$ is

$$n \int_X \mathrm{dd^c} v \wedge \omega_u^{n-1},$$

and Lemma 2.1.1 turns this into the right-hand side of (2.2). For the second assertion, fix $u \in \mathcal{H}_\omega$. For every smooth real direction v, the segment $u + tv$ lies in $\mathcal{H}_\omega$ when $|t|$ is sufficiently small, so the right-hand side of (2.2) vanishes for all smooth real v. Hence the smooth top-degree form $\mathrm{dd^c}(\omega_u^{n-1})$ vanishes. □

Corollary 2.2.1 *If $n = 2$, then for every smooth real function u on X,*

$$\mathrm{M}_\omega(u) = \int_X \omega^2 + 2 \int_X u\,\mathrm{dd^c}\omega.$$

Consequently, M_ω is constant on $\mathcal{H}_\omega$ if and only if $\mathrm{dd^c}\omega = 0$.

Proof The formula is the case $n = 2$ of (2.1), and the equivalence is the case $n = 2$ of Theorem 2.2.2. □

The identity $\mathrm{dd^c}(\omega^{n-1}) = 0$ makes the first variation vanish at $u = 0$; Hermitian forms satisfying it are the Gauduchon metrics of Chapter 4. On a surface this identity already gives mass invariance. In dimension three, however, (2.1) reads

$$\mathrm{M}_\omega(u) = \int_X \omega^3 + 3\int_X u\,\mathrm{dd^c}(\omega^2) + 3\int_X u\,\mathrm{dd^c}u \wedge \mathrm{dd^c}\omega.$$

Vanishing of $\mathrm{dd^c}(\omega^2)$ removes the linear term, while $\mathrm{dd^c}\omega$ still governs the quadratic term. Thus stationarity of the mass at one potential must be distinguished from its invariance on $\mathcal{H}_\omega$; an explicit example is given in Example 4.8.1.

2.3 Mass variation on a complex torus

By Corollary 2.2.1, the mass on a surface varies precisely through the pairing $2\int_X u\,\mathrm{dd^c}\omega$ of the potential with the complex Hessian of the background. A product of elliptic curves gives an explicit background and a family of smooth potentials for which this pairing is nonzero.

Example 2.3.1 Let $X = (\mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z}))^2$. For $j = 1, 2$, let $z_j = x_j + \mathrm{i}y_j$ be the standard coordinates on the universal cover $\mathbb{C}^2$, with x_j, y_j real, and put

$$\beta_j = \frac{\mathrm{i}}{2\pi}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j.$$

The forms β_j descend to X. For a real parameter t, define

$$b(x_1) = 2 + \cos(2\pi x_1), \qquad \omega = \beta_1 + b(x_1)\beta_2, \qquad u_t = t\cos(2\pi x_1).$$

Since $1 \le b \le 3$, the form ω is Hermitian. It is not closed, because $\mathrm{d}\omega = \mathrm{d}b \wedge \beta_2$ is not identically zero.

For a function f of x_1 alone, one has $\mathrm{dd^c}f = (f''/4)\beta_1$, where the prime denotes differentiation in x_1. Hence

$$\omega_{u_t} = \big(1 - \pi^2 t\cos(2\pi x_1)\big)\beta_1 + b\beta_2 > 0 \quad \text{if} \quad |t| < \pi^{-2}.$$

Writing $B = \int_X \beta_1 \wedge \beta_2 > 0$, we have

$$\mathrm{M}_\omega(u_t) = (4 - \pi^2 t)B. \tag{2.3}$$

Indeed,

$$\omega_{u_t}^2 = 2\big(1 - \pi^2 t\cos(2\pi x_1)\big)\big(2 + \cos(2\pi x_1)\big)\beta_1 \wedge \beta_2,$$

and over a period the average of $\cos(2\pi x_1)$ is zero while that of its square is $1/2$.

Although the complex torus is Kähler, the background ω chosen above is not closed, and this already allows the mass to vary. On non-Kähler manifolds even an upper bound for the mass can fail; the Hopf manifolds of Chapter 9 give examples.

Corollary 2.3.1 *In general, the total Monge–Ampère mass is not determined by the ordinary singularity type or by the multiplier ideals, even among smooth strictly ω-plurisubharmonic potentials.*

Proof For $|t| < \pi^{-2}$, the potentials u_t of Example 2.3.1 belong to $\mathcal{H}_\omega$. They are smooth on the compact manifold X, so any two of them differ by a bounded function and have the same singularity type in the sense of Section 1.12. For the same reason, their multiplier ideals are the structure sheaf for every positive multiple of the weight. By (2.3), their masses differ. □

2.4 Comparison and mass invariance

The global comparison principle of Kähler geometry compares the Monge–Ampère measures of two potentials on the set where one lies below the other. Adding a constant to a potential leaves its Monge–Ampère measure unchanged, and after a large constant shift this set becomes the whole manifold. Such an inequality would therefore compare total masses, and the varying mass of Example 2.3.1 rules it out.

Proposition 2.4.1 *Suppose that every pair $u, v \in \mathcal{H}_\omega$ satisfies*

$$\int_{\{u<v\}} \omega_v^n \le \int_{\{u<v\}} \omega_u^n. \tag{2.4}$$

Then M_ω is constant on $\mathcal{H}_\omega$.

Proof Fix $u, v \in \mathcal{H}_\omega$. By compactness there is a constant C such that $u < v + C$ everywhere. Since $\omega_{v+C} = \omega_v$, applying (2.4) to u and $v + C$ gives $\mathrm{M}_\omega(v) \le \mathrm{M}_\omega(u)$. Exchanging the roles of u and v gives the opposite inequality. □

Corollary 2.4.1 *There is a Hermitian form ω on a compact complex manifold for which (2.4) fails for some pair $u, v \in \mathcal{H}_\omega$.*

Proof For the form ω of Example 2.3.1, M_ω is not constant on $\mathcal{H}_\omega$ by (2.3); apply Proposition 2.4.1. □

The modified comparison principle of Theorem 11.2.1 compensates for the derivatives of the background. Conversely, under the equivalent conditions of Theorem 2.2.2, the ordinary comparison principle holds for bounded potentials, by Theorem 12.5.2.

2.5 Uniform mass bounds

Although the mass varies, it may remain bounded. Set

$$\overline{\mathrm{v}}_{\mathrm{sm}}(\omega) = \sup_{u \in \mathcal{H}_\omega} \mathrm{M}_\omega(u) \in (0, +\infty].$$

Comparing two Hermitian forms and rescaling the potentials shows that the finiteness of this supremum depends only on X.

Proposition 2.5.1 *If η is another Hermitian form and $\omega \leq C\eta$ for a constant $C > 0$, then*

$$\overline{\mathrm{v}}_{\mathrm{sm}}(\omega) \leq C^n \overline{\mathrm{v}}_{\mathrm{sm}}(\eta).$$

In particular, finiteness of $\overline{\mathrm{v}}_{\mathrm{sm}}$ is independent of the Hermitian form.

Proof Let $u \in \mathcal{H}_\omega$. Since $C\eta + \mathrm{dd^c} u \geq \omega + \mathrm{dd^c} u > 0$, the function u/C belongs to $\mathcal{H}_\eta$. The top exterior power is monotone on positive $(1,1)$-forms: for positive Hermitian matrices $A \leq B$, all eigenvalues of $A^{-1/2} B A^{-1/2}$ are at least one, so $\det A \leq \det B$. Applying this pointwise gives

$$\int_X (\omega + \mathrm{dd^c} u)^n \leq C^n \int_X \big(\eta + \mathrm{dd^c}(u/C)\big)^n \leq C^n \overline{\mathrm{v}}_{\mathrm{sm}}(\eta).$$

Taking the supremum over u proves the inequality. Since any two Hermitian forms on the compact manifold X are comparable in both directions, the last assertion follows. □

The smooth supremum considered here is also the upper mass over all bounded ω-plurisubharmonic potentials: Proposition 7.5.1 proves this using approximation by smooth strict potentials and continuity of bounded products. Chapter 7 then passes to truncations of unbounded potentials. By Corollary 7.6.1, finiteness of this common supremum is equivalent to finiteness of the extended non-pluripolar mass of every ω-plurisubharmonic potential.

Chapter 3
Monge–Ampère energy and torsion

Throughout this chapter, ω is a Hermitian form on X; the notation $\omega_u = \omega + \mathrm{dd^c} u$ and the space $\mathcal{H}_\omega$ of smooth strictly ω-plurisubharmonic functions are as in Chapter 2.

For a Kähler form, the Monge–Ampère operator is the gradient of an energy: there is a functional on $\mathcal{H}_\omega$ whose derivative at u in every smooth real direction v is $\int_X v\,\omega_u^n$. For a Hermitian form, such a primitive exists exactly when ω is closed. Symmetry of its mixed second derivatives first forces $\mathrm{d}(\omega_u^{n-1}) = 0$, and varying the potential then forces $\mathrm{d}\omega = 0$; this is stronger than the mass invariance of Chapter 2. For a nonclosed background, integrating the Monge–Ampère one-form along radial segments still defines a functional, and its first variation shows which torsion terms enter the corresponding Euler equation. A different energy is available for Hermitian symplectic forms on fourfolds: along potential variations, this completion energy changes by minus the change of the volume. It is therefore constant along potential variations exactly when the mass is invariant, although even then the Monge–Ampère one-form has no primitive unless ω is closed.

3.1 Primitives of the Monge–Ampère one-form

On $\mathcal{H}_\omega$ consider the *Monge–Ampère one-form*

$$\alpha_u(v) = \int_X v\,\omega_u^n, \qquad v \in C^\infty(X, \mathbb{R}).$$

Let $U \subseteq \mathcal{H}_\omega$ be open in the C^2 topology on smooth functions. A *primitive* of α on U is a functional E on U such that, for every $u \in U$ and every smooth real direction v, the directional derivative

$$DE_u(v) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} E(u + tv)$$

exists and equals $\alpha_u(v)$. Restrict E to the intersection of U with a finite-dimensional affine plane. Its first derivatives are values of α, which depend smoothly on the parameters, so its mixed second derivatives agree. The following integration-by-parts criterion identifies the differential condition imposed by this symmetry.

Lemma 3.1.1 *For a smooth real $(n-1,n-1)$-form Ω, the pairing $B(v,w) = \int_X w\,\mathrm{dd^c} v \wedge \Omega$ is symmetric on smooth real functions if and only if $\mathrm{d}\Omega = 0$.*

Proof By Lemma 2.1.1, $B(v,w) = \int_X v\,\mathrm{dd^c}(w\Omega)$. Since v is arbitrary, symmetry is therefore equivalent to the pointwise identity

$$\mathrm{dd^c}(w\Omega) - \mathrm{dd^c} w \wedge \Omega = w\,\mathrm{dd^c}\Omega + \mathrm{d}w \wedge \mathrm{d^c}\Omega - \mathrm{d^c} w \wedge \mathrm{d}\Omega = 0$$

for every smooth real w. Taking $w = 1$ gives $\mathrm{dd^c}\Omega = 0$. The remaining expression is linear in the first jet of w, so we may complexify the test functions. At a chosen point, take a holomorphic linear function multiplied by a cutoff function, so that $\partial w = \lambda$ is an arbitrary $(1,0)$-covector and $\bar\partial w = 0$ at that point. There the first-order expression equals

$$\frac{\mathrm{i}}{2\pi}\big(\partial w \wedge \bar\partial\Omega - \bar\partial w \wedge \partial\Omega\big).$$

It follows that $\lambda \wedge \bar\partial\Omega = 0$ for every $(1,0)$-covector λ. Since the wedge pairing between $(1,0)$-covectors and forms of bidegree $(n-1,n)$ is nondegenerate, $\bar\partial\Omega = 0$, and $\partial\Omega = 0$ because Ω is real. Conversely, if $\mathrm{d}\Omega = 0$, then $\partial\Omega = \bar\partial\Omega = 0$ by type, so $\mathrm{d^c}\Omega = 0$, and the same product formula proves symmetry. □

Lemma 3.1.2 *Let T be a form of type $(2,1)+(1,2)$ on a complex vector space of dimension $n \geq 2$. If $T \wedge A^{n-2} = 0$ for every real $(1,1)$-form A in a nonempty open set, then $T = 0$.*

Proof For $n = 2$ the assertion is immediate. Otherwise the expression $T \wedge A^{n-2}$ is polynomial in the real coefficients of A, so it vanishes for every real $(1,1)$-form A. Polarization and then complexification give $T \wedge A_1 \wedge \cdots \wedge A_{n-2} = 0$ for arbitrary complex $(1,1)$-forms A_j, and such products span all forms of bidegree $(n-2,n-2)$.

Since these wedges preserve type, the identity holds separately for the two type components of T. Wedging the identity for $T^{2,1}$ with arbitrary $(0,1)$-covectors, we obtain test forms spanning bidegree $(n-2,n-1)$, which pairs perfectly with bidegree $(2,1)$ into top degree. Thus $T^{2,1} = 0$. The component $T^{1,2}$ is treated in the same way with $(1,0)$-covectors. □

Theorem 3.1.1 *The one-form α has a primitive on $\mathcal{H}_\omega$ if and only if $\mathrm{d}\omega = 0$. In fact, a primitive on a C^2-open neighborhood of 0 already forces $\mathrm{d}\omega = 0$.*

Proof If $n = 1$, then $\mathrm{d}\omega = 0$ for degree reasons. Let $n \geq 2$, and suppose that a primitive E exists on such a neighborhood U. For $u \in U$ and smooth real directions v, w, the mixed second derivatives of $(s,t) \mapsto E(u + sv + tw)$ at the origin agree. After division by n, this equality reads

$$\int_X w\,\mathrm{dd^c} v \wedge \omega_u^{n-1} = \int_X v\,\mathrm{dd^c} w \wedge \omega_u^{n-1}. \tag{3.1}$$

By Lemma 3.1.1,

$$0 = \mathrm{d}(\omega_u^{n-1}) = (n-1)\mathrm{d}\omega \wedge (\omega + \mathrm{dd^c} u)^{n-2}.$$

Fix a point x. A real quadratic polynomial in local coordinates, multiplied by a cutoff function equal to one near x, realizes any prescribed real $(1,1)$-form H as $\mathrm{dd^c} u(x)$. These functions can be chosen to depend linearly on H, with global C^2 norms bounded by a fixed constant times the coefficient norm of H. Thus every sufficiently small H is realized by a potential in U, and the preceding identity holds at x with $\omega(x) + H$ ranging over an open neighborhood of the positive form $\omega(x)$. Applying Lemma 3.1.2 with $T = \mathrm{d}\omega(x)$, we conclude that $\mathrm{d}\omega(x) = 0$.

Since x was arbitrary, $\mathrm{d}\omega = 0$. Conversely, Proposition 3.2.1 below constructs a primitive for a closed background. □

For $n \geq 2$, the symmetry condition at a single potential u is $\mathrm{d}(\omega_u^{n-1}) = 0$; the theorem uses it on an open set of potentials. Mass invariance alone does not suffice. Indeed, let X be a compact complex surface admitting no Kähler metric, and let ω be a Gauduchon form on X, that is, $\mathrm{dd^c}\omega = 0$ (Definition 4.1.1); such forms exist by Theorem 4.1.1. The mass of ω is constant by Corollary 2.2.1, but ω is not closed, so its Monge–Ampère one-form has no primitive.

3.2 Radial energy and closed backgrounds

Integrating the Monge–Ampère one-form along the segment from 0 to u defines a functional for an arbitrary background. Let θ be a smooth real $(1,1)$-form on X. For $u \in C^\infty(X, \mathbb{R})$, with $\theta_u = \theta + \mathrm{dd^c} u$ as in Section 1.7, define the *radial energy*

$$\begin{aligned} E_\theta(u) &= \int_0^1 \int_X u(\theta + t\,\mathrm{dd^c} u)^n \,\mathrm{d}t \\ &= \frac{1}{n+1} \sum_{j=0}^{n} \int_X u\,\theta_u^j \wedge \theta^{n-j}. \end{aligned} \tag{3.2}$$

To obtain the second expression, write $\theta + t\,\mathrm{dd^c} u = (1-t)\theta + t\theta_u$, expand, and use

$$\binom{n}{j} \int_0^1 t^j (1-t)^{n-j} \,\mathrm{d}t = \frac{1}{n+1}.$$

We write $DE_{\theta,u}(v)$ for the derivative of E_θ at u in the direction v. When θ is closed, integration by parts makes the derivatives of the mixed terms telescope.

Proposition 3.2.1 *If* $\mathrm{d}\theta = 0$*, then* $DE_{\theta,u}(v) = \int_X v\theta_u^n$ *for every pair of smooth real functions* u, v.

Proof Differentiate the finite sum in (3.2). The term indexed by j contributes

$$\int_X v\theta_u^j \wedge \theta^{n-j} + j \int_X u\mathrm{dd}^{\mathrm{c}} v \wedge \theta_u^{j-1} \wedge \theta^{n-j},$$

the second summand being absent for $j = 0$. For $j \geq 1$, the form $\theta_u^{j-1} \wedge \theta^{n-j}$ is closed, so by Lemma 3.1.1 we may interchange u and v in the second summand. Using $\mathrm{dd}^{\mathrm{c}} u = \theta_u - \theta$, we can then write the contribution of the term with $j \geq 1$ as

$$(j+1) \int_X v\theta_u^j \wedge \theta^{n-j} - j \int_X v\theta_u^{j-1} \wedge \theta^{n-j+1}.$$

Together with the contribution $\int_X v\theta^n$ of the term $j = 0$, these expressions telescope to $(n+1) \int_X v\theta_u^n$. Dividing by $n+1$ completes the proof. □

Thus, when $\mathrm{d}\omega = 0$, the restriction of E_ω to $\mathcal{H}_\omega$ is the primitive required in Theorem 3.1.1.

3.3 First variation in a nonclosed background

For the Hermitian background $\theta = \omega$, the functional (3.2) integrates α along the segment from 0 to $u \in \mathcal{H}_\omega$. When ω is not closed, the derivative of E_ω differs from α by the terms produced when integration by parts differentiates the background powers. Computing this difference gives the Euler equation of E_ω even when no primitive exists.

Proposition 3.3.1 *For $u, v \in C^\infty(X, \mathbb{R})$, put $\Omega_t = (\omega + t\mathrm{dd}^{\mathrm{c}} u)^{n-1}$ for $0 \leq t \leq 1$. Then*

$$DE_{\omega,u}(v) - \int_X v\omega_u^n = n \int_0^1 t \int_X v\big(u\mathrm{dd}^{\mathrm{c}}\Omega_t + \mathrm{d}u \wedge \mathrm{d}^{\mathrm{c}}\Omega_t - \mathrm{d}^{\mathrm{c}} u \wedge \mathrm{d}\Omega_t\big)\,\mathrm{d}t. \tag{3.3}$$

Proof Differentiating the radial integral in (3.2) in the direction v gives

$$DE_{\omega,u}(v) = \int_0^1 \left(\int_X v(\omega + t\mathrm{dd}^{\mathrm{c}} u)^n + nt \int_X u\mathrm{dd}^{\mathrm{c}} v \wedge \Omega_t \right) \mathrm{d}t.$$

On the other hand,

$$\begin{aligned} \int_X v\omega_u^n &= \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t} \left(t \int_X v(\omega + t\mathrm{dd}^{\mathrm{c}} u)^n \right) \mathrm{d}t \\ &= \int_0^1 \left(\int_X v(\omega + t\mathrm{dd}^{\mathrm{c}} u)^n + nt \int_X v\mathrm{dd}^{\mathrm{c}} u \wedge \Omega_t \right) \mathrm{d}t. \end{aligned}$$

The terms without a factor nt cancel in the difference, leaving

$$\begin{aligned} DE_{\omega,u}(v) - \int_X v\omega_u^n &= n\int_0^1 t\int_X (u\mathrm{dd}^{\mathrm{c}}v - v\mathrm{dd}^{\mathrm{c}}u)\wedge\Omega_t\,\mathrm{d}t \\ &= n\int_0^1 t\int_X v\big(\mathrm{dd}^{\mathrm{c}}(u\Omega_t) - \mathrm{dd}^{\mathrm{c}}u\wedge\Omega_t\big)\,\mathrm{d}t. \end{aligned}$$

The second equality is Lemma 2.1.1. Expanding the last parenthesis by the product rule in the proof of Lemma 3.1.1 gives (3.3). When $n = 1$, one has $\Omega_t = 1$ and the right-hand side vanishes. □

Proposition 3.3.2 *If $n = 2$, then every $u \in C^\infty(X,\mathbb{R})$ satisfies*

$$E_\omega(u) = \int_X u\omega^2 + \int_X u\omega\wedge\mathrm{dd}^{\mathrm{c}}u + \frac{1}{3}\int_X u(\mathrm{dd}^{\mathrm{c}}u)^2.$$

For every smooth real direction v, its differential satisfies

$$DE_{\omega,u}(v) - \int_X v\omega_u^2 = \int_X v\big(u\mathrm{dd}^{\mathrm{c}}\omega + \mathrm{d}u\wedge\mathrm{d}^{\mathrm{c}}\omega - \mathrm{d}^{\mathrm{c}}u\wedge\mathrm{d}\omega\big). \tag{3.4}$$

Proof Expanding the radial integral gives the first identity. For $n = 2$, one has $\Omega_t = \omega + t\mathrm{dd}^{\mathrm{c}}u$, and since d and d^{c} annihilate $\mathrm{dd}^{\mathrm{c}}u$, the derivatives $\mathrm{d}\Omega_t$, $\mathrm{d}^{\mathrm{c}}\Omega_t$ and $\mathrm{dd}^{\mathrm{c}}\Omega_t$ are those of ω. Formula (3.3) and $2\int_0^1 t\,\mathrm{d}t = 1$ give the second identity. □

Consequently, if $n = 2$ and μ is a fixed smooth real $(2,2)$-form on X, a critical point $u \in \mathcal{H}_\omega$ of the functional $E_\omega(u) - \int_X u\,\mu$ satisfies

$$\omega_u^2 + u\mathrm{dd}^{\mathrm{c}}\omega + \mathrm{d}u\wedge\mathrm{d}^{\mathrm{c}}\omega - \mathrm{d}^{\mathrm{c}}u\wedge\mathrm{d}\omega = \mu$$

rather than the Monge–Ampère equation $\omega_u^2 = \mu$. On a Gauduchon surface the term $u\mathrm{dd}^{\mathrm{c}}\omega$ vanishes, but the first-order torsion terms can persist. The radial Euler equation can therefore differ from the Monge–Ampère equation even when the total mass is invariant.

3.4 Symplectic completions and potential variations

The radial energy is built from the Monge–Ampère one-form, and its first variation exhibits the torsion of the background. When a Hermitian form is the $(1,1)$-part of a closed real two-form, there is another natural energy: the squared L^2 norm of the smallest $(2,0)$-part needed to close it. In complex dimension four, adding a complex Hessian to the form changes this energy by minus the change of the volume. Its potential variations are therefore governed by the volume, and not by a primitive of the Monge–Ampère one-form.

Throughout this section we assume $n = 4$. A Hermitian form ω is *Hermitian symplectic* if there is a smooth $(2,0)$-form ρ, called a *completion* of ω, such that

$$\partial\rho = 0, \qquad \bar\partial\rho = -\partial\omega.$$

Equivalently, the real two-form $\omega+\rho+\bar\rho$ is closed. Applying $\bar\partial$ to the second equation shows that ω is pluriclosed, meaning $\mathrm{dd^c}\omega = 0$. Two completions differ by a closed holomorphic $(2,0)$-form, so the completions of ω form an affine space over the finite-dimensional space of closed holomorphic $(2,0)$-forms. The quadratic norm

$$\|\rho\|^2_{L^2(\omega)} = \frac{1}{2}\int_X \rho\wedge\bar\rho\wedge\omega^2$$

is positive definite on $(2,0)$-forms, and minimizing it over this affine space gives a unique *canonical completion* ρ_ω. It is characterized by being orthogonal to every closed holomorphic $(2,0)$-form. Define the *completion energy* and the volume of ω by

$$F(\omega) = \|\rho_\omega\|^2_{L^2(\omega)}, \qquad V(\omega) = \frac{1}{4!}\int_X \omega^4.$$

The completion energy is nonnegative and vanishes exactly when ω is Kähler: a zero canonical completion forces $\partial\omega = 0$, while for a closed ω the zero form is a completion of minimal norm.

Since ω is pluriclosed, it defines an Aeppli cohomology class, whose real $(1,1)$-representatives are the forms $\omega+\partial\bar\tau+\bar\partial\tau$, with τ a smooth $(1,0)$-form. We call a real form $\psi = \partial\bar\tau+\bar\partial\tau$ an *Aeppli variation*. If ρ completes ω and $\omega+t\psi$ is positive for a real t, then $\omega+t\psi$ has the completion $\rho+t\partial\tau$. In particular, the positive representatives $\omega+\partial\bar\tau+\bar\partial\tau > 0$ of the Aeppli class of ω are again Hermitian symplectic. Every positive potential variation $\omega+\mathrm{dd^c}\varphi$ belongs to this class and has the same exterior derivative as ω. We write $DF_\omega(\psi)$ for the derivative of $t\mapsto F(\omega+t\psi)$ at $t=0$.

Lemma 3.4.1 *Let ω be Hermitian symplectic, with canonical completion $\rho=\rho_\omega$. Every Aeppli variation $\psi=\partial\bar\tau+\bar\partial\tau$ satisfies*

$$DF_\omega(\psi) = \int_X \Big(\mathrm{Re}(\partial\tau\wedge\bar\rho\wedge\omega^2) + \rho\wedge\bar\rho\wedge\psi\wedge\omega\Big). \tag{3.5}$$

Proof The canonical completion of $\omega+t\psi$ is $\rho+t\partial\tau-h(t)$, where $h(t)$ is a closed holomorphic $(2,0)$-form and $h(0)=0$. In a fixed basis of these holomorphic forms, the orthogonal projection defining $h(t)$ solves a positive definite Gram system whose coefficients are smooth in t. Since ρ is orthogonal to closed holomorphic $(2,0)$-forms, the contribution of $h'(0)$ to the first derivative of the squared norm vanishes. We may therefore differentiate using the completion $\rho+t\partial\tau$. Differentiating its two factors and the factor $(\omega+t\psi)^2/2$ gives the formula. □

Proposition 3.4.1 *Let ω be Hermitian symplectic. For every smooth real function φ with $\omega_\varphi=\omega+\mathrm{dd^c}\varphi>0$, one has $\rho_{\omega_\varphi}=\rho_\omega$ and*

$$F(\omega_\varphi)+V(\omega_\varphi) = F(\omega)+V(\omega). \tag{3.6}$$

Moreover, $DF_\omega(\mathrm{dd^c}\varphi) = 0$ *for every smooth real* φ *if and only if* $\mathrm{dd^c}(\omega^3) = 0$*. Thus a critical point of* F *in its Aeppli class is Gauduchon.*

Proof Fix any completion ζ of ω and put $\Theta_\zeta = \omega + \zeta + \bar\zeta$. Expansion by type gives

$$\frac{1}{4!}\int_X \Theta_\zeta^4 = V(\omega) + \frac{1}{2}\int_X \zeta\wedge\bar\zeta\wedge\omega^2 + \frac{1}{4}\int_X \zeta^2\wedge\bar\zeta^2.$$

Since $\partial\omega_\varphi = \partial\omega$, the forms ω_φ and ω have the same completions. With ζ fixed, replacing ω by ω_φ changes the closed form Θ_ζ by an exact form, so the integral on the left is unchanged; so is the last integral on the right, which does not involve ω. Hence

$$\|\zeta\|^2_{L^2(\omega_\varphi)} - \|\zeta\|^2_{L^2(\omega)} = V(\omega) - V(\omega_\varphi).$$

This difference is independent of ζ, so the two norms have the same minimizer, which gives $\rho_{\omega_\varphi} = \rho_\omega$ and (3.6). Differentiating at $t = 0$ along $\omega + t\mathrm{dd^c}\varphi$ and using Lemma 2.1.1, we obtain

$$DF_\omega(\mathrm{dd^c}\varphi) = -\frac{1}{6}\int_X \varphi\mathrm{dd^c}(\omega^3).$$

Since φ is arbitrary, this proves the remaining assertions. □

The *positive index* of a real quadratic form on a real vector space is the supremum of the dimensions of the finite-dimensional subspaces on which it is positive definite; the *negative index* is defined in the same way.

Theorem 3.4.1 *Let* ω *be Hermitian symplectic, and let* H_ω *be the Hessian of* F *at* ω *along the affine space of real representatives of the Aeppli class of* ω*, so that* $H_\omega(\psi,\psi)$ *is the second derivative of* $t \mapsto F(\omega + t\psi)$ *at* $t = 0$*. If* $\mathrm{dd^c}(\omega^2) \neq 0$*, the restriction of* H_ω *to potential directions* $\psi = \mathrm{dd^c}\varphi$ *has infinite positive and negative index. Consequently, every local minimum of* F *in an Aeppli class satisfies* $\mathrm{dd^c}(\omega^2) = 0$*.*

Proof *Step 1: the Hessian formula.* Pluriclosedness and Stokes' theorem give $\int_X \omega\wedge(\mathrm{dd^c}\varphi)^3 = \int_X(\mathrm{dd^c}\varphi)^4 = 0$. Thus the volume and, by (3.6), the energy along a potential line $t \mapsto \omega + t\mathrm{dd^c}\varphi$ are polynomials of degree at most two in t. Differentiating twice and integrating by parts, we obtain

$$\begin{aligned} H_\omega(\mathrm{dd^c}\varphi, \mathrm{dd^c}\varphi) &= -\frac{1}{2}\int_X \omega^2\wedge(\mathrm{dd^c}\varphi)^2 \\ &= \frac{1}{2}\int_X \mathrm{dd^c}(\omega^2)\wedge\mathrm{d}\varphi\wedge\mathrm{d^c}\varphi. \end{aligned} \tag{3.7}$$

For the second equality, apply Lemma 2.1.1 once and then use the identity $\mathrm{dd^c}(\varphi^2/2) = \varphi\mathrm{dd^c}\varphi + \mathrm{d}\varphi\wedge\mathrm{d^c}\varphi$ against the closed form $\mathrm{dd^c}(\omega^2)$.

Step 2: both indices are infinite if $\mathrm{dd^c}(\omega^2) \neq 0$. Put $\Psi = \mathrm{dd^c}(\omega^2)$. At each point, $\xi \mapsto \Psi\wedge \mathrm{i}\xi\wedge\bar\xi$, divided by a positive volume form, is a Hermitian quadratic form

in the $(1,0)$-covector ξ, and these quadratic forms determine Ψ. Their traces with respect to ω are positive multiples of the density $\Psi \wedge \omega$, whose integral vanishes by Stokes' theorem and $\mathrm{dd^c}\omega = 0$. If the quadratic forms took no strictly positive value, they would be negative semidefinite everywhere, with nonpositive traces of integral zero; the traces, and hence Ψ, would vanish identically. The same argument applies with the signs reversed. Therefore, if $\Psi \neq 0$, the quadratic forms take a strictly positive value somewhere and a strictly negative value somewhere.

Fix one of the two signs. Choose holomorphic coordinates on a ball such that, with $x = \operatorname{Re} z_1$, the density $\Psi \wedge \mathrm{d}x \wedge \mathrm{d^c}x$ has that sign on the ball; this is possible because $\mathrm{d}x \wedge \mathrm{d^c}x$ is a positive multiple of $\mathrm{i}\,\mathrm{d}z_1 \wedge \mathrm{d}\bar{z}_1$. For a nonzero real cutoff function a supported in the ball, set $\varphi_N = a\cos(Nx)$. Formula (3.7) gives

$$H_\omega(\mathrm{dd^c}\varphi_N, \mathrm{dd^c}\varphi_N) = \frac{N^2}{2}\int_X a^2 \sin^2(Nx)\Psi \wedge \mathrm{d}x \wedge \mathrm{d^c}x + O(N),$$

where the constant in $O(N)$ depends on Ψ, the chart and the cutoff, but not on N. By integration by parts in x, the integral tends to half the same integral without the factor $\sin^2(Nx)$. Hence the Hessian has the chosen sign for large N. For a positive integer k, apply this construction on k disjoint smaller balls inside this ball. Since the integrand in (3.7) is local, the resulting nonzero directions, which have disjoint supports, are linearly independent and pairwise orthogonal for the Hessian, so their span is a k-dimensional subspace on which H_ω is definite of the chosen sign. Repeating the argument with the opposite sign proves that both indices are infinite. On each of these finite-dimensional spans, $\omega + \mathrm{dd^c}\varphi$ remains positive for all sufficiently small φ, so these directions are realized by positive forms in the Aeppli class of ω.

Finally, a local minimum cannot have a potential direction along which the Hessian is negative, so it satisfies $\mathrm{dd^c}(\omega^2) = 0$ by Step 2. □

Corollary 3.4.1 *For a Hermitian symplectic form ω on X, the following conditions are equivalent:*

(1) *$F(\omega + \mathrm{dd^c}\varphi)$ is independent of the smooth real function φ with $\omega + \mathrm{dd^c}\varphi > 0$;*
(2) *$\mathrm{dd^c}(\omega^2) = 0$;*
(3) *$\int_X (\omega + \mathrm{dd^c}u)^4$ is independent of the bounded ω-plurisubharmonic function u.*

The common mass in the third condition is $24V(\omega)$.

Proof By (3.6), the energy is constant along smooth positive potential variations if and only if their volume is. Since $\mathrm{dd^c}\omega = 0$, Lemma 2.1.2 and Theorem 2.2.2 show that this smooth mass invariance is equivalent to $\mathrm{dd^c}(\omega^2) = 0$ and to mass invariance for bounded potentials. The zero potential gives the common mass. □

3.4.1 Completion energy on products of complex tori

Products of two complex two-tori give explicit Hermitian symplectic forms with $\mathrm{dd^c}(\omega^2) \neq 0$ and with $\mathrm{dd^c}(\omega^2) = 0$, the regimes of Theorem 3.4.1 and Corollary 3.4.1.

Each factor carries a one-parameter family of nonclosed forms with explicit completions. If both factors carry a suitable member of this family, the product is critical for F under every Aeppli variation although its potential Hessian is indefinite. If only one factor carries a nonclosed form, the energy is constant along potential variations but varies in the Aeppli class.

Consider the square complex two-torus $S = \mathbb{C}^2/(\mathbb{Z} + \mathrm{i}\mathbb{Z})^2$. Let $z_j = x_j + \mathrm{i}y_j$ be the coordinates on its cover and put $\eta = \mathrm{i}\sum_{j=1}^{2} \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j$. For $0 < s < 1$, the periodic forms

$$v_s = \frac{s}{\pi}\mathrm{e}^{2\pi\mathrm{i}x_1}\mathrm{d}z_2, \qquad \alpha_s = \eta + \partial\bar{v}_s + \bar{\partial}v_s$$

are well defined on S. The Hermitian matrix of α_s with respect to the forms $\mathrm{i}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$ is

$$\begin{pmatrix} 1 & -s\mathrm{e}^{-2\pi\mathrm{i}x_1} \\ -s\mathrm{e}^{2\pi\mathrm{i}x_1} & 1 \end{pmatrix},$$

with eigenvalues $1 - s$ and $1 + s$, so α_s is a Hermitian form. It has the completion $\rho_s = \partial v_s = \mathrm{i}s\mathrm{e}^{2\pi\mathrm{i}x_1}\mathrm{d}z_1 \wedge \mathrm{d}z_2$, and with $\mu_0 = \eta^2/2$,

$$\frac{\alpha_s^2}{2} = (1 - s^2)\mu_0, \qquad \rho_s \wedge \bar{\rho}_s = s^2\mu_0.$$

Example 3.4.1 On $X = S \times S$, let ω be the sum of the pullbacks of $\alpha_{1/\sqrt{2}}$ from the two factors. This form is Hermitian symplectic but not Kähler, and it is critical for F under every Aeppli variation, including those with mixed factor components. If $V_0 = \int_S \mu_0$, then $F(\omega) = V_0^2/2 > 0$. The potential Hessian of F at ω has infinite positive and negative index, even though potential variations preserve $\mathrm{d}\omega$.

Proof Denote the two factor forms by α_i, ρ_i, and put $\mu_i = \alpha_i^2/2$ and $\nu_i = \rho_i \wedge \bar{\rho}_i$; we suppress pullbacks from the notation. Every holomorphic form on a complex torus has constant coefficients. On each factor, the L^2 pairing of $\rho_i = \partial v_i$ with a holomorphic two-form is a multiple of their wedge pairing, which vanishes by Stokes' theorem. On the product, holomorphic two-forms pulled back from the first factor, those pulled back from the second factor and mixed ones are mutually pointwise orthogonal. The pairing of ρ_i with a form pulled back from the same factor is the vanishing pairing on that factor times the volume of the other factor. Thus $\rho = \rho_1 + \rho_2$ is the canonical completion of ω. Since $\mu_i = \nu_i$ is half the pullback of μ_0, expansion gives $F(\omega) = \int_X (\nu_1 \wedge \mu_2 + \mu_1 \wedge \nu_2) = V_0^2/2$.

Put $\Theta_i = \alpha_i + \rho_i + \bar{\rho}_i$ and let $\Theta = \Theta_1 + \Theta_2$, which is closed. For an arbitrary real $(1,1)$-form ψ and $(2,0)$-form σ on X, expansion on the two factors gives

$$\mathrm{Re}(\sigma \wedge \bar{\rho} \wedge \omega^2) + \rho \wedge \bar{\rho} \wedge \psi \wedge \omega = \frac{1}{12}(\psi + \sigma + \bar{\sigma}) \wedge \Theta^3.$$

Indeed, the mixed components of ψ and σ have odd degree on both factors and contribute zero. The pure components on the left give $(\psi+\sigma+\bar{\sigma}) \wedge (\Theta_1 \wedge \mu_2 + \Theta_2 \wedge \mu_1)$, and since $\Theta_i^2 = 4\mu_i$, the cube on the right is twelve times the same complementary form. Apply Lemma 3.4.1 with $\psi = \partial\bar{\tau} + \bar{\partial}\tau$ and $\sigma = \partial\tau$. The derivative is

$\frac{1}{12}\int_X \mathrm{d}(\tau+\bar\tau)\wedge\Theta^3 = 0$, since Θ is closed. This proves criticality under every Aeppli variation.

Each $\partial\alpha_i = -\bar\partial\rho_i$ is nonzero, so ω is not Kähler. In $\partial\omega\wedge\bar\partial\omega$, the terms coming from a single factor vanish for dimension reasons, while the two cross terms are nonzero and have different bidegrees on the factors, so they cannot cancel. Hence $\mathrm{d}\omega\wedge\mathrm{d}^{\mathrm{c}}\omega \neq 0$, and since $\mathrm{dd}^{\mathrm{c}}\omega = 0$, Lemma 2.1.2 gives $\mathrm{dd}^{\mathrm{c}}(\omega^2) \neq 0$. It remains to apply Theorem 3.4.1. □

If only one factor carries a nonclosed form, the energy becomes constant along potential variations, while the same explicit family shows that it still varies in the Aeppli class.

Example 3.4.2 For $0 < s < 1$, let $\widehat{\omega}_s$ on $X = S\times S$ be the sum of the pullbacks of α_s from the first factor and of η from the second, and set $V_0 = \int_S \mu_0$. Then $\mathrm{d}\widehat{\omega}_s \neq 0$ and

$$F(\widehat{\omega}_s) = s^2V_0^2, \qquad V(\widehat{\omega}_s) = (1-s^2)V_0^2.$$

The energy is constant on all positive smooth potential variations of $\widehat{\omega}_s$, and every bounded $\widehat{\omega}_s$-plurisubharmonic function has mass $24(1-s^2)V_0^2$. The form $\widehat{\omega}_s$ is not a critical point of F in its Aeppli class, and its Monge–Ampère one-form has no primitive on any C^2-open neighborhood of zero.

Proof The pullback of ρ_s from the first factor is a completion of $\widehat{\omega}_s$. It is canonical: for the product metric it is pointwise orthogonal to the holomorphic two-forms pulled back from the second factor and to the mixed ones, and its pairing with those pulled back from the first factor vanishes by the Stokes argument in the proof of Example 3.4.1. The energy is therefore $\int_X \rho_s\wedge\bar\rho_s\wedge\eta^2/2 = s^2V_0^2$. The volume is the product of the two surface volumes, which gives the second formula.

The exterior derivatives of $\widehat{\omega}_s$ come entirely from the first factor. Their wedge product $\mathrm{d}\widehat{\omega}_s\wedge\mathrm{d}^{\mathrm{c}}\widehat{\omega}_s$ vanishes for degree reasons, while $\mathrm{dd}^{\mathrm{c}}\widehat{\omega}_s = 0$. Thus $\mathrm{dd}^{\mathrm{c}}(\widehat{\omega}_s^2) = 0$ by Lemma 2.1.2, and Corollary 3.4.1 gives the constancy of the energy and the common bounded mass.

The forms $\widehat{\omega}_{(1+t)s}$ are positive for small $|t|$ and lie in the Aeppli class of $\widehat{\omega}_s$, since $v_{(1+t)s} = (1+t)v_s$. The derivative of their energy at $t=0$ is $2s^2V_0^2 > 0$, so $\widehat{\omega}_s$ is not a critical point in its Aeppli class. Finally, $\partial\alpha_s = -\bar\partial\rho_s \neq 0$. Hence $\mathrm{d}\widehat{\omega}_s \neq 0$, and Theorem 3.1.1 shows that no primitive exists. □

Chapter 4
Gauduchon metrics and linear mass

Throughout this chapter, ω is a fixed Hermitian form on X, and $n \geq 2$ unless a section specifies another setting.

The mass variation in Chapter 2 leaves a linear conservation law. If γ is a Gauduchon metric, so that $\mathrm{dd}^{\mathrm{c}}(\gamma^{n-1}) = 0$, integration against γ^{n-1} annihilates every complex Hessian. Thus the trace mass $\int_X (\theta + \mathrm{dd}^{\mathrm{c}} u) \wedge \gamma^{n-1}$ of a positive current depends only on its smooth background θ, even when u is unbounded. Gauduchon's theorem produces such a metric in every conformal class, providing conserved pairings when the background is not closed.

These pairings determine positivity as well as mass. A smooth form θ admits a positive current $\theta + \mathrm{dd}^{\mathrm{c}} u$ exactly when all its Gauduchon degrees are nonnegative. Normalizing the degrees gives the largest lower curvature bound obtainable by adding a complex Hessian. This bound is always attained by a potential, and a minimizing test exists in the weak closure of the normalized smooth Gauduchon forms, although a smooth minimizing metric generally does not exist. This closure can be smaller than the set of all normalized positive dd^{c}-closed $(n-1, n-1)$-currents: on the blowup of $\mathbb{P}^2$ at a point, the normalized integration current along the exceptional curve lies outside it. The smooth tests can have uniformly bounded pairings with forms concentrating transversely to a local hypersurface only if a global hypersurface contains it. The weak limits of these tests are positive pluriharmonic currents; positivity and the vanishing of their dd^{c} control their concentration on a smooth hypersurface. A manifold carrying a balanced metric, a pluriclosed metric and a connection with parallel torsion that preserves a Hermitian metric and the complex structure is Kähler, and pluriclosed metrics persist under small deformations of the complex structure with constant Aeppli number $h^{1,1}_A$. Applied to ω_u, the conformal construction also gives the adjoint probability of the linearized Monge–Ampère operator. It identifies the compatibility condition for that operator and explains the extra term in entropy dissipation for a nonclosed background. Chapter 5 applies Gauduchon degrees and this compatibility condition to the curvature of vector bundles.

4.1 Conformal Gauduchon metrics

By Proposition 2.2.1, the first variation of the mass at $u = 0$ vanishes exactly when $\mathrm{dd}^{\mathrm{c}}(\omega^{n-1}) = 0$. This condition can be achieved in every conformal class: finding a conformal factor amounts to finding a positive generator of the kernel of the adjoint of a scalar elliptic operator.

Definition 4.1.1 A Hermitian form γ is *Gauduchon*, or a *Gauduchon metric*, if

$$\mathrm{dd}^{\mathrm{c}}(\gamma^{n-1}) = 0.$$

It is *balanced* if $\mathrm{d}(\gamma^{n-1}) = 0$, and *pluriclosed* if $\mathrm{dd}^{\mathrm{c}}\gamma = 0$.

Theorem 4.1.1 (Gauduchon's theorem) *Every Hermitian form η has a conformal Gauduchon representative. More precisely, there is a smooth strictly positive function ρ, unique up to multiplication by a positive constant, such that*

$$\mathrm{dd}^{\mathrm{c}}(\rho\eta^{n-1}) = 0.$$

The associated Gauduchon form is $\gamma = \rho^{1/(n-1)}\eta$.

Proof For a smooth real function v on X, define

$$Lv = n\frac{\mathrm{dd}^{\mathrm{c}}v \wedge \eta^{n-1}}{\eta^n}.$$

This real second-order operator has positive definite principal symbol and no zeroth-order term. By the maximum principle, its kernel consists of the constants. Its index is zero, since the linear path joining it to a Laplace operator consists of elliptic operators. Fredholm theory and elliptic regularity therefore show that the formal adjoint L^* of L with respect to η^n has a one-dimensional kernel, generated by a nonzero smooth real function ρ.

We first prove that $\int_X \rho\eta^n \neq 0$. Otherwise the constant function 1 would be orthogonal to the kernel of L^*, and the Fredholm alternative would solve $Lv = 1$, which is impossible at a maximum point of v. We may therefore rescale so that $\int_X \rho\eta^n = 1$.

Next we show that $\rho \geq 0$. Suppose instead that ρ is negative somewhere. Choose a nonzero smooth function $h \geq 0$ supported where $\rho < 0$, and put $c = \int_X h\rho\eta^n < 0$. The function $h - c$ is strictly positive and orthogonal to the kernel of L^*, so the Fredholm alternative would solve $Lv = h - c > 0$, which is again impossible at a maximum point of v.

In real local coordinates $(x_1, \ldots, x_{2n})$, write $L^* = A^{ij}\partial_i\partial_j + B^i\partial_i + C$, where $\partial_i = \partial/\partial x_i$, the coefficients are smooth, and repeated indices are summed from 1 to $2n$. The leading symbol (A^{ij}) is positive definite. Since $\rho \geq 0$,

$$(L^* - \max(C, 0))\rho = -\max(C, 0)\rho \leq 0,$$

and the operator $L^* - \max(C, 0)$ has nonpositive zeroth-order term. By the strong minimum principle, a zero of ρ would force ρ to vanish identically on the connected manifold X. Hence $\rho > 0$.

Finally, integration by parts gives

$$\int_X \rho L v \, \eta^n = n \int_X v \, \mathrm{dd^c}(\rho \eta^{n-1}).$$

Thus the equation $L^*\rho = 0$ is precisely $\mathrm{dd^c}(\rho\eta^{n-1}) = 0$, and the one-dimensional adjoint kernel gives uniqueness up to positive scaling. Since $\gamma^{n-1} = \rho\eta^{n-1}$, the form γ is Gauduchon. □

A Kähler form, that is, a closed Hermitian form, is both balanced and pluriclosed, and every balanced form is Gauduchon. On surfaces, balanced forms are Kähler, and the Gauduchon and pluriclosed conditions coincide. The first statement has a converse.

Lemma 4.1.1 *If a Hermitian form ζ satisfies $\mathrm{d}\zeta^{n-1} = 0$ and $\mathrm{dd^c}\zeta = 0$, then $\mathrm{d}\zeta = 0$.*

Proof For $n = 2$, the balanced condition is the assertion $\mathrm{d}\zeta = 0$. Suppose $n \geq 3$ and set $\alpha = \partial\zeta$. Since ζ is balanced, $\alpha \wedge \zeta^{n-2} = 0$, so α is a primitive $(2,1)$-form. Applying $\partial\bar\partial$ to the balanced power gives

$$0 = \partial\bar\partial\zeta^{n-1} = (n-1)\partial\bar\partial\zeta \wedge \zeta^{n-2} + (n-1)(n-2)\alpha \wedge \bar\alpha \wedge \zeta^{n-3}.$$

The first term vanishes because ζ is pluriclosed. For a primitive $(2,1)$-form, pointwise Hermitian linear algebra gives

$$-\mathrm{i}\,\alpha \wedge \bar\alpha \wedge \frac{\zeta^{n-3}}{(n-3)!} = |\alpha|^2_\zeta \frac{\zeta^n}{n!}.$$

Thus $\alpha = 0$, and since ζ is real, also $\bar\partial\zeta = 0$, so $\mathrm{d}\zeta = 0$. □

4.2 Conservation of the trace mass

Proposition 4.2.1 *Let θ be a smooth real $(1,1)$-form, let u be a θ-plurisubharmonic function, and let γ be a Gauduchon metric. Then*

$$\int_X (\theta + \mathrm{dd^c}u) \wedge \gamma^{n-1} = \int_X \theta \wedge \gamma^{n-1}. \tag{4.1}$$

Proof Since $\mathrm{dd^c}(\gamma^{n-1}) = 0$, distributional integration by parts gives

$$\langle \mathrm{dd^c}u, \gamma^{n-1} \rangle = \langle u, \mathrm{dd^c}(\gamma^{n-1}) \rangle = 0.$$

The product on the left of (4.1) is a positive measure, whose total mass is therefore the finite number on the right. □

We call $\int_X \theta \wedge \gamma^{n-1}$ the *Gauduchon degree* of θ with respect to γ. Here $\theta + \mathrm{dd^c} u$ is the full distributional current, including any mass carried by the pole set of u. The non-pluripolar Monge–Ampère measure of Chapter 7 discards this mass; Example 8.5.1 illustrates the difference on a curve.

Corollary 4.2.1 *There are constants $0 < a \le b < +\infty$ such that every $u \in \mathrm{PSH}(X, \omega)$ satisfies*

$$a \le \int_X (\omega + \mathrm{dd^c} u) \wedge \omega^{n-1} \le b.$$

Proof Fix a Gauduchon form γ on X and choose positive constants c, C with $c\gamma^{n-1} \le \omega^{n-1} \le C\gamma^{n-1}$. Wedging these inequalities with the positive current $\omega + \mathrm{dd^c} u$ and integrating, we obtain, by Proposition 4.2.1, the bounds cA and CA, where $A = \int_X \omega \wedge \gamma^{n-1} > 0$. □

The trace bound controls the mass of positive currents and hence their weak compactness. By itself it gives no bound for the Monge–Ampère mass, since a bound on the integral of a trace does not bound the integral of its nth power.

The pointwise arithmetic–geometric mean inequality relates the conserved trace to the density in a smooth Monge–Ampère equation. Fix a smooth positive volume form $\mathrm{d}V$ on X.

Proposition 4.2.2 *For any two Hermitian forms η, γ,*

$$\eta \wedge \gamma^{n-1} \ge \left(\frac{\eta^n}{\gamma^n}\right)^{1/n} \gamma^n \tag{4.2}$$

pointwise. Consequently, if θ is a smooth real $(1,1)$-form and $u \in C^\infty(X, \mathbb{R})$ satisfies $\theta + \mathrm{dd^c} u > 0$ and $(\theta + \mathrm{dd^c} u)^n = c\, f\, \mathrm{d}V$ with a constant $c > 0$ and a function f, then

$$c^{1/n} \int_X \left(\frac{f\, \mathrm{d}V}{\gamma^n}\right)^{1/n} \gamma^n \le \int_X \theta \wedge \gamma^{n-1} \tag{4.3}$$

for every Gauduchon metric γ.

Proof In a frame where γ is the identity, let $\lambda_1, \dots, \lambda_n > 0$ be the eigenvalues of η. The ratios of the two sides of (4.2) to γ^n are $n^{-1} \sum_j \lambda_j$ and $(\prod_j \lambda_j)^{1/n}$, respectively, and their comparison is the arithmetic–geometric mean inequality.

For the second assertion, apply (4.2) with $\eta = \theta + \mathrm{dd^c} u$, integrate, and use (4.1). □

The corresponding density inequality for unbounded potentials in a Hermitian background is proved in Lemma 7.4.2.

4.3 The Gauduchon criterion for positivity

The trace identity gives a necessary condition for $\mathrm{PSH}(X,\theta)$ to be nonempty: every Gauduchon degree of θ must be nonnegative. The condition is also sufficient. A Hahn–Banach separation produces a positive current with the prescribed pairings, and a scalar elliptic equation shows that its difference from θ is dd^{c}-exact.

Theorem 4.3.1 (The generalized Lamari criterion) *For a smooth real $(1,1)$-form θ, the following conditions are equivalent:*

(1) *There exists $u \in \mathrm{PSH}(X,\theta)$.*
(2) $\int_X \theta \wedge \gamma^{n-1} \geq 0$ *for every Gauduchon metric γ.*

To apply separation, we express the test in terms of smooth strictly positive dd^{c}-closed $(n-1,n-1)$-forms Γ. Every strictly positive $(n-1,n-1)$-form Γ has a unique strictly positive $(1,1)$-root γ, that is, $\Gamma = \gamma^{n-1}$. Indeed, in a frame in which Γ is diagonal, and with the same wedge normalization on both sides, the coefficients g_i of Γ are the products $\prod_{j\neq i}\lambda_j$ of the eigenvalues λ_j of γ. Hence

$$\lambda_i = \frac{(\prod_j g_j)^{1/(n-1)}}{g_i}.$$

This spectral formula uses only a positive determinant and a matrix inverse, so the root depends smoothly on the coefficients of Γ, also at repeated eigenvalues, and pointwise uniqueness makes the local roots agree on overlaps. Consequently, $\mathrm{dd}^{\mathrm{c}}\Gamma = 0$ says exactly that the root γ is Gauduchon.

Proof (of Theorem 4.3.1) Necessity is Proposition 4.2.1. For sufficiency, let F be the real vector space of smooth real $(n-1,n-1)$-forms, and let

$$E = \{\Gamma \in F : \mathrm{dd}^{\mathrm{c}}\Gamma = 0\}, \qquad \ell(\Gamma) = \int_X \theta \wedge \Gamma \quad (\Gamma \in E).$$

Fix a strictly positive $\Gamma_0 \in E$, which exists by Theorem 4.1.1. By the root calculation above, the hypothesis says that ℓ is nonnegative on the strictly positive elements of E. Adding $\epsilon\Gamma_0$ and letting $\epsilon \downarrow 0$, we see that ℓ is also nonnegative on the semipositive elements of E.

Step 1: a positive extension of ℓ. For $\Phi \in F$ put

$$p(\Phi) = \inf\{\ell(\Gamma) : \Gamma \in E,\ \Gamma \geq \Phi\}.$$

This is a finite sublinear function. To see finiteness, choose C with $-C\Gamma_0 \leq \Phi \leq C\Gamma_0$. The admissible element $C\Gamma_0$ gives an upper bound, while every admissible Γ satisfies $\Gamma + C\Gamma_0 \geq 0$ and hence $\ell(\Gamma) \geq -C\ell(\Gamma_0)$. Sublinearity follows by adding and rescaling competitors. For $\Phi \in E$, the nonnegativity of $\ell(\Gamma - \Phi)$ for admissible Γ shows that $p(\Phi) = \ell(\Phi)$.

The Hahn–Banach theorem gives a real linear extension T of ℓ to F with $T(\Phi) \leq p(\Phi)$ for all $\Phi \in F$. If $\Phi \geq 0$, the zero form is admissible for $p(-\Phi)$, so $T(-\Phi) \leq 0$,

that is, $T(\Phi) \geq 0$. Moreover, the bounds by $C\Gamma_0$ show that T is continuous for the uniform norm. It is therefore a positive $(1,1)$-current, and the current $S = T - \theta$ annihilates E.

Step 2: S is dd^{c}-exact. To recover a potential from S, consider the scalar operator

$$Dv = \frac{\mathrm{dd}^{\mathrm{c}}(v\Gamma_0)}{\mathrm{d}V},$$

where $\mathrm{d}V$ is the smooth positive volume form fixed above. Since $\mathrm{dd}^{\mathrm{c}}\Gamma_0 = 0$, this is an elliptic scalar operator with no zeroth-order term. Its kernel consists of the constants and its index is zero, while Stokes' theorem gives $\int_X Dv\,\mathrm{d}V = 0$. Consequently its range is exactly the space of smooth functions of integral zero, and by elliptic estimates its inverse on functions normalized by $\int_X v\,\mathrm{d}V = 0$ is continuous in the smooth topology.

For a smooth top-degree form ζ with $\int_X \zeta = 0$, solve $Dv = \zeta/\mathrm{d}V$ with this normalization and set $R\zeta = v\Gamma_0$. Then $\mathrm{dd}^{\mathrm{c}}R\zeta = \zeta$, and R is continuous. Define a distribution u on such top-degree forms by

$$\langle u, \zeta\rangle = S(R\zeta),$$

and extend it to all top-degree forms by assigning an arbitrary value on one fixed volume form of integral one. For every $\Phi \in F$, the form $\Phi - R(\mathrm{dd}^{\mathrm{c}}\Phi)$ belongs to E, and S annihilates it. It follows that

$$\langle \mathrm{dd}^{\mathrm{c}}u, \Phi\rangle = S(R(\mathrm{dd}^{\mathrm{c}}\Phi)) = S(\Phi).$$

Thus $\theta + \mathrm{dd}^{\mathrm{c}}u = T \geq 0$, and Proposition 1.1.1 gives a representative of u in $\mathrm{PSH}(X,\theta)$. □

Applying the criterion to $\theta - a\omega$, for $a \in \mathbb{R}$, determines the largest lower curvature bound attainable in the background θ.

Corollary 4.3.1 *For a smooth real $(1,1)$-form θ, set*

$$\kappa_\omega(\theta) = \inf_{\gamma \text{ Gauduchon}} \frac{\int_X \theta \wedge \gamma^{n-1}}{\int_X \omega \wedge \gamma^{n-1}}.$$

Then $\kappa_\omega(\theta)$ is the largest real number a for which there is a quasi-plurisubharmonic function u satisfying

$$\theta + \mathrm{dd}^{\mathrm{c}}u \geq a\omega.$$

In particular, θ is big, in the sense of Section 1.19, precisely when $\kappa_\omega(\theta) > 0$.

Proof Since θ is smooth, $-C\omega \leq \theta \leq C\omega$ for some C, so the infimum lies in $[-C, C]$. If $\theta + \mathrm{dd}^{\mathrm{c}}u \geq a\omega$, the trace identity applied to the background $\theta - a\omega$ shows that a is at most each of the ratios in the definition, hence $a \leq \kappa_\omega(\theta)$. Conversely, for $a = \kappa_\omega(\theta)$, all Gauduchon degrees of $\theta - a\omega$ are nonnegative, and Theorem 4.3.1 applied to this smooth form gives the required u. The last assertion follows, since

bigness means the existence of a θ-plurisubharmonic u with $\theta + \mathrm{dd}^{\mathrm{c}} u \geq \delta\omega$ for some $\delta > 0$. □

Whether the infimum defining $\kappa_\omega(\theta)$ is attained by a smooth Gauduchon metric is decided by a rigidity property of the trace: a positive current $\theta + \mathrm{dd}^{\mathrm{c}} u$ with vanishing Gauduchon degree is zero.

Lemma 4.3.1 *Let θ be a smooth real $(1,1)$-form with $\mathrm{PSH}(X,\theta) \neq \varnothing$. If*

$$\int_X \theta \wedge \gamma^{n-1} = 0$$

for one Gauduchon metric γ, then $\theta = \mathrm{dd}^{\mathrm{c}} f$ for a smooth real function f. Conversely, a form $\mathrm{dd}^{\mathrm{c}} f$ with f smooth has zero degree with respect to every Gauduchon metric. The same conclusions hold for $n = 1$, with $\gamma^{n-1} = 1$.

Proof Choose $u \in \mathrm{PSH}(X,\theta)$ and put $S = \theta + \mathrm{dd}^{\mathrm{c}} u \geq 0$. By Proposition 4.2.1, $\int_X S \wedge \gamma^{n-1} = 0$. Since the trace measure $S \wedge \gamma^{n-1}$ of the positive current S controls all its coefficient measures, $S = 0$. Taking the trace of the equation $\mathrm{dd}^{\mathrm{c}} u = -\theta$ with respect to ω gives an elliptic equation with smooth right-hand side, so u is smooth by elliptic regularity, and $f = -u$ has the required property. The converse is Stokes' theorem. For $n = 1$, the trace measure is S itself and the same argument applies. □

For a smooth real $(1,1)$-form θ, put $a = \kappa_\omega(\theta)$. By Corollary 4.3.1, the form $\theta - a\omega$ admits a positive current $\theta - a\omega + \mathrm{dd}^{\mathrm{c}} u$. A Gauduchon metric γ attains the infimum defining $\kappa_\omega(\theta)$ precisely when the Gauduchon degree of $\theta - a\omega$ with respect to γ vanishes. Hence Lemma 4.3.1 shows that the infimum is attained by a smooth Gauduchon metric if and only if

$$\theta - a\omega = \mathrm{dd}^{\mathrm{c}} f \quad \text{for some } f \in C^\infty(X,\mathbb{R}),$$

and in that case every Gauduchon metric attains it. Thus a positive current at the threshold always exists, whereas a smooth minimizing Gauduchon metric exists only in this exceptional situation.

4.4 Compactness and duality of Gauduchon currents

When no smooth Gauduchon metric attains the threshold, a minimizer can still be found among weak limits of normalized Gauduchon forms. The normalization bounds the coefficient measures and hence gives compactness. Let $\mathcal{G}_\omega$ be the set of positive $(n-1,n-1)$-currents T on X satisfying

$$\mathrm{dd}^{\mathrm{c}} T = 0, \qquad \int_X \omega \wedge T = 1.$$

We call them *normalized Gauduchon currents*. Let $\mathscr{C}_\omega$ be the weak closure of the set of smooth strictly positive members of $\mathscr{G}_\omega$. By the root calculation preceding the proof of Theorem 4.3.1, these smooth members are exactly the normalized powers $\gamma^{n-1}/\int_X \omega \wedge \gamma^{n-1}$ of Gauduchon metrics γ.

Proposition 4.4.1 *The sets $\mathscr{G}_\omega$ and $\mathscr{C}_\omega$ are nonempty, compact and convex. For every smooth real $(1,1)$-form θ,*

$$\kappa_\omega(\theta) = \min_{T \in \mathscr{C}_\omega} \int_X \theta \wedge T. \tag{4.4}$$

Moreover, a current $T \in \mathscr{G}_\omega$ belongs to $\mathscr{C}_\omega$ if and only if

$$\int_X \theta \wedge T \geq 0 \quad \text{for every smooth real } \theta \text{ with } \mathrm{PSH}(X,\theta) \neq \varnothing. \tag{4.5}$$

Proof Gauduchon's theorem gives a smooth strictly positive member of $\mathscr{G}_\omega$ after normalization. The fixed trace mass bounds the total variation of all coefficient measures in finitely many local frames, so weak compactness of finite measures gives compactness of the family of positive currents with this trace. Positivity, the trace normalization and the equation $\mathrm{dd^c} T = 0$ pass to weak limits. On this bounded set, testing against smooth forms or against forms with continuous coefficients gives the same topology, by uniform approximation, and this topology is metrizable since the space of continuous test forms is separable. Thus the compactness can also be used sequentially. The defining conditions of $\mathscr{G}_\omega$ are convex. Its smooth strictly positive members form a convex set, so their closure $\mathscr{C}_\omega$ is compact and convex as well.

Pairing with a fixed smooth θ is weakly continuous. Its minimum on the closure of the smooth members is therefore their infimum, which proves (4.4). By Proposition 4.2.1, (4.5) holds for every smooth strictly positive member, and hence for every member of their weak closure.

Conversely, suppose that $T \in \mathscr{G}_\omega$ does not belong to $\mathscr{C}_\omega$, and separate this point from the compact convex set $\mathscr{C}_\omega$ in the space of currents with the weak topology. A continuous linear functional for that topology is evaluation against a smooth form. After changing its sign if necessary, we obtain a smooth real $(1,1)$-form β such that

$$\int_X \beta \wedge T < \min_{R \in \mathscr{C}_\omega} \int_X \beta \wedge R = \kappa_\omega(\beta).$$

By Corollary 4.3.1, the smooth form $\beta - \kappa_\omega(\beta)\omega$ admits a positive current. Its pairing with T is negative, because $\int_X \omega \wedge T = 1$. This contradicts (4.5) and proves the converse. □

Corollary 4.4.1 *The function κ_ω is concave, positively homogeneous and invariant under the addition of $\mathrm{dd^c} f$ for $f \in C^\infty(X,\mathbb{R})$. Let θ and η be smooth real $(1,1)$-forms and let $\varepsilon \geq 0$. If $-\varepsilon\omega \leq \theta - \eta \leq \varepsilon\omega$, then*

$$|\kappa_\omega(\theta) - \kappa_\omega(\eta)| \leq \varepsilon.$$

If smooth real $(1,1)$-forms θ_j converge uniformly to a smooth form θ and $T_j \in \mathscr{C}_\omega$ minimize their pairings, every weak cluster point of (T_j) minimizes the pairing with θ.

Proof By (4.4), κ_ω is an infimum of linear functionals, which proves concavity and positive homogeneity. Invariance under the addition of $\mathrm{dd}^{\mathrm{c}} f$ follows from $\mathrm{dd}^{\mathrm{c}} T = 0$. The normalization of the trace gives

$$\left| \int_X (\theta - \eta) \wedge T \right| \leq \varepsilon \quad (T \in \mathscr{C}_\omega),$$

and taking infima proves the estimate. For the last assertion, pass to a subsequence $T_j \rightharpoonup T \in \mathscr{C}_\omega$. The estimate and the uniform convergence of the backgrounds give

$$\int_X \theta \wedge T = \lim_j \int_X \theta_j \wedge T_j = \lim_j \kappa_\omega(\theta_j) = \kappa_\omega(\theta).$$

Thus T is a minimizer. □

In general, the closure $\mathscr{C}_\omega$ cannot be replaced by all of $\mathscr{G}_\omega$. Testing against all of $\mathscr{G}_\omega$ instead detects smooth positivity and nefness, as shown in Theorem 27.20.1. The difference between the two sets comes from positive currents that pair negatively with a class which is nonnegative on all smooth tests. Integration currents along negative curves give such currents, and the following example separates the two sets by a smooth closed form.

Example 4.4.1 The weak closure $\mathscr{C}_\omega$ of the smooth strictly positive normalized Gauduchon forms can be smaller than $\mathscr{G}_\omega$, even on a Kähler surface. Let X be the blowup of $\mathbb{P}^2$ at one point and C its exceptional curve. Fix a Kähler form ω and a smooth closed representative β_C of the class of $[C]$. Then

$$T_C = \frac{[C]}{\int_C \omega} \in \mathscr{G}_\omega, \qquad \int_X \beta_C \wedge T_C = \frac{-1}{\int_C \omega} < 0,$$

whereas every smooth strictly positive $T \in \mathscr{G}_\omega$ satisfies $\int_X \beta_C \wedge T = \int_C T > 0$. In particular $T_C \notin \mathscr{C}_\omega$.

Proof The integration current $[C]$ is positive and closed, so its normalization belongs to $\mathscr{G}_\omega$. Its pairing with the smooth form β_C is the self-intersection number $C^2 = -1$ of the exceptional curve. For smooth T, the Poincaré–Lelong formula and $\mathrm{dd}^{\mathrm{c}} T = 0$ give the second identity. Since the pairing with the fixed smooth form β_C is continuous under weak convergence of currents, it is nonnegative on the weak closure of the smooth tests, and this closure does not contain T_C. □

4.5 Transverse concentration of Gauduchon tests

Let $\mathcal{T}_\omega$ consist of the smooth strictly positive members Θ of the set $\mathcal{G}_\omega$ of normalized Gauduchon currents of Section 4.4, so that $\mathrm{dd^c}\Theta = 0$ and $\int_X \omega \wedge \Theta = 1$. These tests form a weakly relatively compact family, but their pairings with a form concentrating transversely to a hypersurface need not stay bounded. A uniform bound would, by the Lamari criterion, produce an ω-plurisubharmonic function whose current charges the local hypersurface. Analyticity of Lelong level sets then forces a global hypersurface.

Choose a product coordinate chart $(z, \zeta) \in U \times \Delta_r$, with $z \in \mathbb{C}^{n-1}$, and a smooth compactly supported function $0 \le \chi \le 1$ on U, equal to one on a nonempty connected smaller polydisk U'. Put $\sigma_\zeta = (\mathrm{i}/(2\pi))\, \mathrm{d}\zeta \wedge \mathrm{d}\bar{\zeta}$. Fix $p \in C_c^\infty(\mathbb{C})$, nonnegative and supported in the closed unit disk, with $\int p\, \sigma_\zeta = 1$. For small $\delta > 0$ extend

$$B_\delta = \chi(z)\delta^{-2} p(\zeta/\delta)\sigma_\zeta$$

by zero to X, and set

$$C_\delta = \sup_{\Theta \in \mathcal{T}_\omega} \int_X B_\delta \wedge \Theta.$$

The form B_δ is smooth and semipositive. For fixed δ it is bounded above by a finite multiple of ω, so $C_\delta < \infty$.

Theorem 4.5.1 *If C_{δ_j} stays bounded along a sequence $\delta_j \downarrow 0$, then some global analytic hypersurface contains $U' \times \{0\}$. In particular, on a manifold without analytic hypersurfaces, $C_\delta \to \infty$ as $\delta \downarrow 0$.*

Proof Suppose $C_{\delta_j} \le M$ and put $\kappa = (M+1)^{-1}$. For every Gauduchon metric γ, the normalized form $\Theta = \gamma^{n-1} / \int_X \omega \wedge \gamma^{n-1}$ belongs to $\mathcal{T}_\omega$, so

$$\int_X (\omega - \kappa B_{\delta_j}) \wedge \Theta \ge 1 - \kappa M > 0.$$

The generalized Lamari criterion, Theorem 4.3.1, applied to the smooth form $\omega - \kappa B_{\delta_j}$, gives v_j with

$$\omega + \mathrm{dd^c} v_j \ge \kappa B_{\delta_j}, \qquad \sup_X v_j = 0.$$

Since $B_{\delta_j} \ge 0$, the potentials belong to the family $\mathcal{K}_\omega$ of normalized ω-plurisubharmonic functions. By Theorem 1.4.1, a subsequence converges in L^1 to $v \in \mathrm{PSH}(X, \omega)$. As currents, the forms B_{δ_j} converge weakly to $\chi[\zeta = 0]$, where $[\zeta = 0]$ is the current of integration on the local hypersurface $\{\zeta = 0\}$. Passing to the limit in distributions therefore gives

$$\omega + \mathrm{dd^c} v \ge \kappa \chi [\zeta = 0].$$

Near each point of $U' \times \{0\}$, choose a smooth local function ρ with $\mathrm{dd^c}\rho \ge \omega$. Since $\chi = 1$ on U', the closed positive current $\mathrm{dd^c}(v + \rho)$ dominates $\kappa[\zeta = 0]$ there. Hence the Lelong number of v, which does not change under the smooth correction ρ, is

at least κ at every point of $U' \times \{0\}$. By Siu's theorem in Section 1.13.1, the set $Z = \{\nu(v, \cdot) \geq \kappa\}$ is analytic. It lies in the pole set of v, which has measure zero, so Z has dimension at most $n-1$. The compact analytic set Z has finitely many global irreducible components. Their intersections with the connected hypersurface $U' \times \{0\}$ are analytic subsets covering it. At least one intersection is the whole hypersurface, since finitely many proper analytic subsets cannot cover an open polydisk. That component has dimension $n-1$ and is the required global hypersurface. If no such hypersurface exists, there is no bounded sequence C_{δ_j} with $\delta_j \downarrow 0$, which proves the asserted divergence. □

Proposition 4.5.1 *Suppose an effective Cartier divisor D agrees, with multiplicity one, with $\{\zeta = 0\}$ throughout the chosen coordinate chart. Then C_δ is uniformly bounded for all sufficiently small δ.*

Proof Let s be the canonical section of $O_X(D)$. In the chart write $s = \zeta e$ with e a nonvanishing holomorphic frame. Choose a smooth Hermitian metric h with $|e|_h = 1$ on a neighborhood of $\operatorname{supp}\chi \times \{|\zeta| \leq \delta_0\}$ for a fixed small $\delta_0 > 0$. The smooth global functions $\psi_\delta = \log(|s|_h^2 + \delta^2)$ satisfy $\mathrm{dd}^c\psi_\delta \geq -C\omega$ with a constant $C \geq 1$ independent of δ. Off the divisor their complex Hessians are a bounded scalar multiple of the negative curvature of h, plus a semipositive gradient term; continuity gives the same lower bound at the divisor. Fix $0 < \epsilon \leq (2C)^{-1}$, so $\omega + \epsilon\mathrm{dd}^c\psi_\delta \geq \omega/2$. Where $|e|_h = 1$, we have $|s|_h^2 = |\zeta|^2$, and therefore

$$\mathrm{dd}^c\psi_\delta = \frac{\delta^2}{(|\zeta|^2 + \delta^2)^2}\sigma_\zeta \geq \frac{1}{4\delta^2}\sigma_\zeta \quad (|\zeta| \leq \delta).$$

If $P = \|p\|_\infty$, then $B_\delta \leq P\delta^{-2}\sigma_\zeta$ on its support. For $\delta \leq \delta_0$, this support lies in the region of the preceding display, where $\omega + \epsilon\mathrm{dd}^c\psi_\delta \geq \epsilon\mathrm{dd}^c\psi_\delta$; off the support, $\omega + \epsilon\mathrm{dd}^c\psi_\delta \geq \omega/2 \geq 0$. Therefore

$$\omega + \epsilon\mathrm{dd}^c\psi_\delta \geq \frac{\epsilon}{4P}B_\delta$$

globally for $\delta \leq \delta_0$. Since $\mathrm{dd}^c\Theta = 0$, Stokes' theorem gives, for every $\Theta \in \mathcal{T}_\omega$,

$$1 = \int_X (\omega + \epsilon\mathrm{dd}^c\psi_\delta) \wedge \Theta \geq \frac{\epsilon}{4P}\int_X B_\delta \wedge \Theta.$$

Taking the supremum proves $C_\delta \leq 4P/\epsilon$. □

The proposition requires the divisor to agree with $\{\zeta = 0\}$ on the entire chart. The global hypersurface given by Theorem 4.5.1 contains $U' \times \{0\}$ but can have further branches in the chart, so the proposition gives a converse to the theorem only in this restricted situation. Divergence of the transverse pairings also does not, by itself, bound from below the volume $\int_X \gamma^n$ of the root γ of a test $\Theta = \gamma^{n-1}$.

4.6 Pluriharmonic currents

A current is *pluriharmonic* if its dd^{c} vanishes. The normalized Gauduchon currents of Section 4.4 are positive pluriharmonic currents of bidegree $(n-1, n-1)$. This section proves four basic properties of positive pluriharmonic currents; the first three hold in arbitrary bidegree. Positivity and the second-order equation control their concentration on a smooth hypersurface. On compact Kähler manifolds, a closed test form bounds the mass of the pullback of a smooth positive pluriharmonic form, and the heat equation smooths positive pluriharmonic forms with continuous coefficients. Finally, a positive pluriharmonic $(1,1)$-current with vanishing vertical trace descends along a fiber bundle whose fibers are entire lines or compact curves; this is used for elliptic fibrations in Chapter 29.

Lemma 4.6.1 *Let Z be a complex manifold of dimension m, and let $j\colon D \hookrightarrow Z$ be a smooth closed hypersurface. If a positive (q,q)-current R, $1 \le q \le m$, is supported on D and $\mathrm{dd}^{\mathrm{c}} R$ has order zero, then there is a unique positive $(q-1, q-1)$-current S on D such that $R = j_* S$. If $\mathrm{dd}^{\mathrm{c}} R = 0$, then $\mathrm{dd}^{\mathrm{c}} S = 0$. In particular, if $q = 1$, D is compact and connected, and $\mathrm{dd}^{\mathrm{c}} R = 0$, then $R = c[D]$ for some $c \ge 0$.*

Proof Use local coordinates $(z, w_1, \dots, w_{m-1})$ with $D = \{z = 0\}$. Both R and $\mathrm{dd}^{\mathrm{c}} R$ have measure coefficients supported on D. Suppose first that $q < m$. For every compactly supported smooth form η of bidegree $(m-q-1, m-q-1)$,

$$0 = \langle \mathrm{dd}^{\mathrm{c}} R, |z|^2 \eta \rangle = \langle R, \mathrm{dd}^{\mathrm{c}}(|z|^2 \eta) \rangle = \langle R, \mathrm{dd}^{\mathrm{c}} |z|^2 \wedge \eta \rangle,$$

where the first equality holds because $\mathrm{dd}^{\mathrm{c}} R$ is a measure supported on D, on which $|z|^2$ vanishes, and the last one because the terms containing $|z|^2$ or its first derivatives vanish on D. Thus $R \wedge \mathrm{i}\, \mathrm{d}z \wedge \mathrm{d}\bar{z} = 0$, which removes the coefficients of R containing neither normal differential.

Fix a constant positive Hermitian form τ in the w variables. The current $M_\tau = R \wedge \tau^{m-q-1}$ is positive of bidegree $(m-1, m-1)$, and $M_\tau \wedge \mathrm{i}\, \mathrm{d}z \wedge \mathrm{d}\bar{z} = 0$. Its Hermitian matrix of measure densities therefore has zero diagonal entry in the normal direction, and positivity removes the corresponding row and column, so

$$M_\tau \wedge \mathrm{d}z = 0, \qquad M_\tau \wedge \mathrm{d}\bar{z} = 0.$$

Put $s = m - q - 1$. The powers τ^s, as τ varies over the positive Hermitian forms, span the real horizontal (s,s)-forms with constant coefficients. Indeed, a linear functional vanishing on these powers vanishes on a polynomial over an open cone; polarization makes it vanish on products of arbitrary real $(1,1)$-forms, and these products span the forms of that bidegree. Multiplying by smooth functions, we obtain $R \wedge \Phi \wedge \mathrm{d}z = R \wedge \Phi \wedge \mathrm{d}\bar{z} = 0$ for every horizontal (s,s)-form Φ. A coefficient of R containing exactly one normal differential is detected by test forms $\Phi \wedge \beta \wedge \mathrm{d}z$ or $\Phi \wedge \beta \wedge \mathrm{d}\bar{z}$, where β is a horizontal one-form, so all these coefficients vanish. For $q = m$, every top-degree coefficient already contains both normal differentials, and this reduction is unnecessary.

The remaining coefficients are measures on $\{z = 0\}$ multiplying both normal differentials. They therefore define a unique current S on D with $R = j_*S$. Testing against extensions of strongly positive forms on D proves positivity of S, and uniqueness makes these local currents agree on overlaps. Pushforward commutes with dd^{c} and is injective for a closed embedding, so $\mathrm{dd}^{\mathrm{c}}R = 0$ implies $\mathrm{dd}^{\mathrm{c}}S = 0$.

When $q = 1$, the current S is a nonnegative distribution on D. The equation $\mathrm{dd}^{\mathrm{c}}S = 0$ and Weyl's lemma make it a smooth nonnegative pluriharmonic function. If D is compact and connected, the maximum principle makes it a constant $c \geq 0$, proving the last assertion. □

Lemma 4.6.2 *Let $F\colon Y \to X$ be a holomorphic map between compact Kähler manifolds of dimensions $N \geq n$, with Kähler forms Ω and ω. For $1 \leq p \leq n$, there is a constant C such that every smooth positive (p,p)-form S on X with $\mathrm{dd}^{\mathrm{c}}S = 0$ satisfies*

$$\int_Y F^*S \wedge \Omega^{N-p} \leq C \int_X S \wedge \omega^{n-p}.$$

Proof The positive closed current $\Gamma = F_*(\Omega^{N-p})$ has bidegree $(n-p, n-p)$ on X. The classical dd^{c}-lemma for currents on compact Kähler manifolds gives a smooth real closed form γ of this bidegree and a current U with $\Gamma = \gamma + \mathrm{dd}^{\mathrm{c}}U$. For $p = n$, a closed current of degree zero is already a smooth locally constant function, and one may take $U = 0$. Since S is smooth, pairing it with U is legitimate, and

$$\int_Y F^*S \wedge \Omega^{N-p} = \langle \Gamma, S\rangle = \int_X \gamma \wedge S + \langle U, \mathrm{dd}^{\mathrm{c}}S\rangle = \int_X \gamma \wedge S.$$

The pairing with the fixed smooth form γ is bounded in absolute value by a constant times the trace mass $\int_X S \wedge \omega^{n-p}$ of the positive form S. This proves the estimate.□

Lemma 4.6.3 *Let (X, ω) be a compact Kähler manifold of dimension n, and let $1 \leq p \leq n$. Every continuous positive real (p,p)-form R satisfying $\mathrm{dd}^{\mathrm{c}}R = 0$ in the sense of currents is a uniform limit of smooth positive real (p,p)-forms R_j with $\mathrm{dd}^{\mathrm{c}}R_j = 0$. If R is closed, the approximants can also be chosen closed.*

Proof Let Δ be the real Hodge Laplacian of the Kähler metric. Its heat operators $H_t = \mathrm{e}^{-t\Delta}$ are smoothing for $t > 0$. The Kähler identities give $\Delta = 2\Delta_\partial = 2\Delta_{\bar\partial}$, so H_t preserves bidegree, reality and the equation $\mathrm{dd}^{\mathrm{c}}R = 0$.

We recall why $H_tR \to R$ uniformly for a continuous form R. The Weitzenböck formula writes $\Delta = \nabla^*\nabla + A$, with a bounded curvature term A of order zero. For smooth initial data β, the heat equation and the maximum principle give

$$\|H_t\beta\|_{C^0} \leq \mathrm{e}^{Ct}\|\beta\|_{C^0} \qquad (0 < t \leq 1),$$

where C depends only on the fixed metric and the degree. Indeed, the scalar equation for $|H_t\beta|^2$ has a nonpositive gradient term and a curvature term bounded above by $2C|H_t\beta|^2$. The classical smooth heat solution converges uniformly to its smooth initial data. Approximating a continuous form by smooth forms and using this bound proves

uniform convergence for continuous initial data. The resulting heat solution is smooth for $t > 0$, and since the commutation with dd^{c} extends to currents, $\mathrm{dd}^{\mathrm{c}} H_t R = 0$.

Set $\epsilon_t = \|H_t R - R\|_{C^0}$, which tends to zero. A fixed constant $c > 0$ satisfies

$$E + c\|E\|_{C^0}\omega^p \geq 0$$

for every continuous real (p,p)-form E; to see this, evaluate on decomposable unit p-vectors, on which ω^p has a uniformly positive value. Consequently

$$R_t = H_t R + c\epsilon_t \omega^p$$

is smooth and positive. Since ω is Kähler, $\mathrm{dd}^{\mathrm{c}}\omega^p = 0$, and hence $\mathrm{dd}^{\mathrm{c}} R_t = 0$. Moreover, $R_t \to R$ uniformly. The Hodge heat flow also commutes with d, and ω^p is closed, so $\mathrm{d}R = 0$ implies $\mathrm{d}R_t = 0$. Choosing any sequence $t_j \downarrow 0$ proves the assertion. □

We next consider a positive $(1,1)$-current with vanishing vertical trace on the total space of a fiber bundle. Positivity removes its vertical and mixed coefficients. For a closed current the remaining coefficients are constant along the fibers. For a pluriharmonic current they are only harmonic along the fibers, but positivity gives the same conclusion when the fiber is the whole complex line or a compact connected curve.

Lemma 4.6.4 *Let Y be a complex manifold of dimension $\ell \geq 1$, let η be a Hermitian form on Y, and let $\pi\colon Z \to Y$ be a holomorphic fiber bundle whose fiber is either $\mathbb{C}$ or a compact connected Riemann surface. If S is a positive real $(1,1)$-current on Z with $\mathrm{dd}^{\mathrm{c}} S = 0$ and $S \wedge \pi^*\eta^\ell = 0$, then there is a unique positive dd^{c}-closed $(1,1)$-current R on Y such that $S = \pi^* R$. If S is closed, then R is closed.*

Proof Fix a local trivialization $\pi^{-1}(U) = U \times F$ of the bundle, with base coordinates $w = (w_1, \ldots, w_\ell)$ and a local fiber coordinate z. The coefficient measures of S have a positive Hermitian matrix of densities relative to their trace measure. The identity $S \wedge \pi^*\eta^\ell = 0$ makes the vertical diagonal entry zero, and the inequality $|s_{\alpha\bar{z}}|^2 \leq s_{\alpha\bar{\alpha}} s_{z\bar{z}}$ then removes all mixed entries. Thus

$$S = \frac{\mathrm{i}}{2\pi} \sum_{\alpha,\beta=1}^{\ell} s_{\alpha\bar{\beta}}(w,z)\, \mathrm{d}w_\alpha \wedge \mathrm{d}\bar{w}_\beta,$$

with distribution coefficients. In the equation $\partial\bar{\partial} S = 0$, the coefficient containing both vertical differentials and the horizontal pair (α,β) gives $\partial_z \partial_{\bar{z}} s_{\alpha\bar{\beta}} = 0$.

Fix $v \in \mathbb{C}^\ell$ and a nonnegative smooth function ξ with compact support in U. Pairing the positive coefficient distribution $\sum_{\alpha,\beta} s_{\alpha\bar{\beta}} v_\alpha \bar{v}_\beta$, in the base variables, with ξ times the Euclidean volume form defines a nonnegative harmonic distribution on the whole fiber F. Its definition does not depend on the fiber coordinate, since the remaining differentials are horizontal. Weyl's lemma makes it a nonnegative harmonic function. Such a function is constant on $\mathbb{C}$ by the Liouville theorem for harmonic functions, and on a compact connected curve by the maximum principle.

Polarization in v and linearity in the base test function show that every coefficient $s_{\alpha\bar\beta}$ is the pullback of a distribution $r_{\alpha\bar\beta}$ on U. To identify this distribution, pair against any nonnegative smooth fiber density of total integral one; this also proves positivity of the resulting current R_U on U and injectivity of the pullback. Consequently $\mathrm{dd^c} S = \pi^*(\mathrm{dd^c} R_U) = 0$ implies $\mathrm{dd^c} R_U = 0$. On overlaps, injectivity makes the currents R_U agree, so they glue to a current R, and injectivity also proves uniqueness. If $\mathrm{d}S = 0$, the same argument gives $\mathrm{d}R = 0$. □

The assumption that the fibers are entire lines or compact curves is essential. On the bidisk $\Delta_z \times \Delta_w$, the current

$$S = (2 + \operatorname{Re} z)\frac{\mathrm{i}}{2\pi}\,\mathrm{d}w \wedge \mathrm{d}\bar w$$

is positive and $\mathrm{dd^c}$-closed, and its vertical trace vanishes. Its horizontal coefficient depends on z, so it does not descend under the projection to Δ_w.

4.7 Balanced and pluriclosed metrics

By Lemma 4.1.1, a Hermitian form that is both balanced and pluriclosed is closed. For a balanced metric and a pluriclosed metric that may be different, the first subsection obtains a Kähler metric when a connection with parallel torsion preserves a Hermitian metric and the complex structure; the proof averages over the holonomy and maximizes the volume on a bounded slice of fixed trace. The second subsection shows that pluriclosed metrics persist under small deformations of the complex structure with constant Aeppli number $h^{1,1}_A$.

4.7.1 Parallel torsion

For a balanced metric β, Stokes' theorem shows that the closed form β^{n-1} pairs to zero with every exact real two-form. Parallel torsion gives a setting in which this integral constraint becomes pointwise. Write J for the complex structure of X, let g be a Hermitian metric on X, and let D be a real connection on TX that preserves g and J and has parallel torsion T^D:

$$Dg = 0, \qquad DJ = 0, \qquad DT^D = 0, \qquad T^D(U,V) = D_U V - D_V U - [U,V].$$

The balanced and pluriclosed metrics considered below need not equal g. Write

$$E^k_D = \{\alpha \in \Omega^k(X,\mathbb{R}) : D\alpha = 0\}$$

for the space of D-parallel real k-forms. Since X is connected, evaluation at one point is injective on this space, so it is finite-dimensional. Its complexification inherits the decomposition by bidegree because $DJ = 0$.

Lemma 4.7.1 *There is a linear projection $S\colon \Omega^\bullet(X,\mathbb{R}) \to E_D^\bullet$ with $S\mathrm{d} = \mathrm{d}S$. Its complexification preserves bidegree and commutes with ∂ and $\bar\partial$. It preserves strict positivity in bidegrees $(1,1)$ and $(n-1,n-1)$.*

Proof *Step 1: the projection.* For a parallel form α, the formula for exterior differentiation in terms of a connection expresses $\mathrm{d}\alpha$ as an algebraic contraction of T^D with α. Since $DT^D = 0$, it follows that $\mathrm{d}E_D^k \subseteq E_D^{k+1}$. The Hodge star of g is also parallel and maps E_D^k onto E_D^{2n-k}. Consequently the pairing

$$E_D^k \times E_D^{2n-k} \longrightarrow \mathbb{R}, \qquad (\alpha,\beta) \longmapsto \int_X \alpha\wedge\beta$$

is nondegenerate: pairing a nonzero α with $*_g\alpha$ gives its squared L^2 norm, which is positive. Hence every smooth real k-form α determines a unique $S_k\alpha \in E_D^k$ such that

$$\int_X S_k\alpha\wedge\beta = \int_X \alpha\wedge\beta \qquad (\beta \in E_D^{2n-k}). \tag{4.6}$$

This defines a projection onto E_D^k. For $\gamma \in E_D^{2n-k-1}$ we have $\mathrm{d}\gamma \in E_D^{2n-k}$, so Stokes' theorem gives

$$\int_X \mathrm{d}S_k\alpha\wedge\gamma = (-1)^{k+1}\int_X \alpha\wedge\mathrm{d}\gamma = \int_X S_{k+1}(\mathrm{d}\alpha)\wedge\gamma.$$

Both forms being compared are parallel, so nondegeneracy proves $\mathrm{d}S_k\alpha = S_{k+1}\mathrm{d}\alpha$.

Step 2: positivity. Fix $p \in X$. Let H be the closure of the full D-holonomy group at p, a compact subgroup of the unitary group, with probability Haar measure m_H. For a path from p to x, write $P_x\colon T_pX \to T_xX$ for parallel transport along it. The average

$$R_\alpha(x) = \int_H h^*(P_x^*\alpha_x)\,\mathrm{d}m_H(h)$$

is independent of the path, since a change of path acts by an element of H. It is an H-invariant tensor at p depending smoothly on x, because paths can be chosen smoothly on each coordinate neighborhood. Average it over X:

$$a_p = \frac{1}{\mathrm{Vol}_g(X)}\int_X R_\alpha(x)\,\mathrm{d}V_g(x).$$

Being H-invariant, the tensor a_p extends uniquely to a global parallel form $\widetilde{S}_k\alpha$. For parallel β, write $\alpha\wedge\beta = f\,\mathrm{d}V_g$. Since parallel transport is unitary and β is invariant under holonomy, averaging gives

$$R_\alpha(x)\wedge\beta_p = f(x)\,\mathrm{d}V_{g,p}.$$

Integrating this equality shows that $\widetilde{S}_k\alpha$ satisfies (4.6); hence $\widetilde{S}_k = S_k$.

Every parallel transport and every holonomy element preserves reality, bidegree and the positive cones. A smooth strictly positive (q,q)-form, for $q = 1$ or $n-1$, is bounded below by a positive multiple of the qth power of the fundamental form of g, uniformly on the compact manifold, and both probability averages preserve that lower bound. Therefore S preserves strict positivity and bidegree. Comparing bidegrees in $S\mathrm{d} = \mathrm{d}S$ proves that S commutes with ∂ and $\bar\partial$. □

Theorem 4.7.1 *Let D be a connection on TX as above, preserving g and J and with parallel torsion. If X admits a balanced metric and a pluriclosed metric, then it admits a D-parallel Kähler metric.*

Proof Let β be a balanced metric and η a pluriclosed metric. By Lemma 4.7.1, $S(\beta^{n-1})$ is strictly positive, parallel and closed. It has a unique smooth positive $(n-1)$-st root B. Indeed, after the fixed normalization of wedge powers, the corresponding map on matrices is $G \mapsto C = (\det G)G^{-1}$, whose inverse is

$$G = (\det C)^{1/(n-1)}C^{-1}.$$

By uniqueness the root is invariant under parallel transport. Thus $DB = 0$ and $\mathrm{d}B^{n-1} = 0$. Similarly, $P = S\eta$ is a parallel positive pluriclosed form.

Consider the finite-dimensional space of real parallel directions of exact type

$$\mathcal{A}_D = \{(\mathrm{d}\xi)^{1,1} : \xi \in E^1_D\}.$$

Every such direction is $\partial\bar\partial$-closed, since

$$(\mathrm{d}\xi)^{1,1} = \partial\xi^{0,1} + \bar\partial\xi^{1,0}.$$

For a parallel positive form Q, the volume functional has derivative

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \frac{1}{n!}\int_X (Q + t(\mathrm{d}\xi)^{1,1})^n = \frac{1}{(n-1)!}\int_X \xi \wedge \mathrm{d}Q^{n-1}, \tag{4.7}$$

because the $(2,0)$ and $(0,2)$ parts of $\mathrm{d}\xi$ pair to zero with Q^{n-1}. Since $\mathrm{d}Q^{n-1}$ is parallel, nondegeneracy of the parallel pairing shows that all these derivatives vanish exactly when Q is balanced.

Apply this to the balanced form B. For $a \in \mathcal{A}_D$, the function $\mathrm{tr}_B\, a$ is parallel and hence constant. Its integral against B^n vanishes by (4.7), so it vanishes identically. Therefore every form in the positive affine slice

$$\mathcal{P} = \{Q \in P + \mathcal{A}_D : Q > 0\}$$

has the same pointwise trace $\mathrm{tr}_B\, Q = \mathrm{tr}_B\, P =: c_0 > 0$.

Evaluate the parallel forms at one point. The positive eigenvalues of $Q \in \mathcal{P}$ relative to B sum to c_0, so they are bounded above by c_0. Thus $\mathcal{P}$ is bounded in its finite-dimensional affine space. Its closure is

$$\overline{\mathcal{P}} = \{Q \in P + \mathcal{A}_D : Q \geq 0\},$$

because $(1-t)Q+tP$ is positive for $0 < t \leq 1$ whenever Q is semipositive; in particular, this closure is compact. Since both forms are parallel, the relative eigenvalues and the relative determinant are constant on X. Hence

$$\frac{1}{n!}\int_X Q^n = \det(B^{-1}Q)\frac{1}{n!}\int_X B^n.$$

At a relative boundary point of $\overline{\mathcal{P}}$ the semipositive matrix is singular, so this volume is zero, whereas it is positive at P. A maximum of the volume on the compact set $\overline{\mathcal{P}}$ therefore lies in $\mathcal{P}$.

At this interior maximum all the derivatives in (4.7) vanish, so the maximizing form is balanced; it is pluriclosed because it lies in $P + \mathcal{A}_D$. By Lemma 4.1.1 it is closed. Since it is positive and D-parallel, this proves the theorem. □

4.7.2 Pluriclosed metrics under deformation

Pluriclosed metrics persist under small deformations of the complex structure when the Aeppli number $h_A^{1,1}$ stays constant. Let M be a compact smooth manifold of real dimension $2n$, $n \geq 1$, and let J_t, $t \in U \subset \mathbb{R}^\ell$, be a smooth family of integrable complex structures, where U is an open neighborhood of 0. Write $\Omega_t^{p,q}$ for the smooth complex forms of type (p,q) for J_t. A pluriclosed form belongs to the kernel of $\partial_t\bar\partial_t$, and the corresponding Aeppli group is

$$H_A^{1,1}(M,J_t) = \frac{\ker(\partial_t\bar\partial_t : \Omega_t^{1,1} \to \Omega_t^{2,2})}{\partial_t\Omega_t^{0,1} + \bar\partial_t\Omega_t^{1,0}}, \qquad h_A^{1,1}(M,J_t) = \dim_{\mathbb{C}} H_A^{1,1}(M,J_t).$$

Constancy of this dimension gives a smooth family of projections onto the forms satisfying the closedness constraint, and positivity then persists near a given strictly positive form.

Lemma 4.7.2 *The number $h_A^{1,1}(M,J_t)$ is finite. If it is constant near 0, then, after shrinking U, the orthogonal projections*

$$\mathcal{P}_t : \Omega_t^{1,1} \longrightarrow \ker(\partial_t\bar\partial_t)$$

for any smooth family of auxiliary Hermitian metrics vary smoothly with t on smooth forms and preserve reality.

Proof *Step 1: an elliptic operator.* Fix t and suppress it from the notation; the auxiliary metric defines the adjoints. Put $C = \partial\bar\partial$ and $E = \bar\partial\partial^*$. The nonnegative self-adjoint operator

$$\mathcal{L} = C^*C + CC^* + E^*E + EE^* + \partial\partial^* + \bar\partial\bar\partial^* \tag{4.8}$$

is elliptic of order four. Indeed, at a nonzero real covector ξ, the principal symbols of ∂ and $\bar\partial$ are, up to the common factor i, exterior multiplication by the $(1,0)$ and $(0,1)$ parts of ξ. These parts are orthogonal and have the same norm $r > 0$. Split the exterior algebra according to which of these two directions are present. The four terms of order four act on the four resulting summands respectively, and the sum of their symbols is r^4 Id. The terms of lower order do not affect ellipticity.

The quadratic form of $\mathcal{L}$ is a sum of squared norms, which gives the inclusion $\subseteq$ in

$$\ker \mathcal{L} = \ker C \cap \ker \partial^* \cap \ker \bar\partial^*.$$

For the reverse inclusion, note that $C^* = \bar\partial^*\partial^*$, $E = \bar\partial\partial^*$ and $E^* = \partial\bar\partial^*$ all vanish under these three conditions. Elliptic theory on the compact manifold gives a finite-dimensional space of smooth forms $\ker \mathcal{L}$, the orthogonal projection $\mathcal{H}$ onto it and the Green operator $\mathcal{G}$, with $\mathcal{L}\mathcal{G} = \mathrm{Id} - \mathcal{H}$.

Step 2: the projection. Set $R = \mathcal{L} - C^*C$. Each term of R has image in $\operatorname{im}\partial + \operatorname{im}\bar\partial$. Thus every smooth $(1,1)$-form α has the decomposition

$$\alpha = \mathcal{H}\alpha + R\mathcal{G}\alpha + C^*C\mathcal{G}\alpha. \tag{4.9}$$

The first two terms lie in $\ker C$, since $C\partial = C\bar\partial = 0$, and the last is orthogonal to $\ker C$. Consequently

$$\mathcal{P} = \mathrm{Id} - C^*C\mathcal{G} \tag{4.10}$$

is the orthogonal projection onto $\ker C$. If $C\alpha = 0$, the last term in (4.9) vanishes, so the Aeppli class of α is represented by $\mathcal{H}\alpha$. A harmonic form in $\operatorname{im}\partial + \operatorname{im}\bar\partial$ is zero, by orthogonality. This identifies $H_A^{1,1}$ with $\ker \mathcal{L}$ and proves finiteness.

Step 3: dependence on t. Now identify the varying type bundles and their L^2 spaces smoothly with those at $t = 0$, using a unitary identification for the auxiliary metrics. The coefficients of $\mathcal{L}_t$ then vary smoothly. Choose a circle $|z| = \varepsilon$ enclosing no nonzero eigenvalue of $\mathcal{L}_0$. The resolvents of $\mathcal{L}_0$ on this circle are bounded from H^s to H^{s+4}. The Neumann series for their perturbations gives smooth resolvents on the circle for small t, and the contour projections have constant rank. Constancy of $h_A^{1,1}$ makes that rank equal to $\dim\ker \mathcal{L}_t$. Hence the contour projections are exactly $\mathcal{H}_t$, and $\mathcal{L}_t$ has no additional small eigenvalues.

The operators

$$\mathcal{G}_t = (\mathcal{L}_t + \mathcal{H}_t)^{-1} - \mathcal{H}_t$$

vary smoothly on every Sobolev scale, by the same inverse argument and elliptic regularity. Formula (4.10) therefore gives the asserted smooth family $\mathcal{P}_t$. Complex conjugation sends C_t to $-C_t$, exchanges E_t with E_t^*, and exchanges the last two terms of (4.8). Thus $\mathcal{L}_t$, $\mathcal{H}_t$, $\mathcal{G}_t$ and $\mathcal{P}_t$ commute with conjugation. □

Theorem 4.7.2 *Suppose that (M, J_0) carries a pluriclosed Hermitian form ω_0 and that $h_A^{1,1}(M, J_t)$ is constant near 0. Then for small t there is a smooth family of pluriclosed Hermitian forms ω_t on (M, J_t) with the prescribed value ω_0 at $t = 0$.*

Proof Let $\pi_t^{1,1}$ denote the projection onto forms of type $(1,1)$ for J_t, and use the operators of Lemma 4.7.2. Define

$$\omega_t = \mathcal{P}_t\big(\pi_t^{1,1}\omega_0\big).$$

These are smooth real $(1,1)$-forms satisfying $\partial_t\bar\partial_t\omega_t = 0$. At $t = 0$ the projection fixes ω_0, since ω_0 already satisfies this equation. The tensors $\omega_t(\cdot, J_t\cdot)$ converge uniformly to the positive definite tensor $\omega_0(\cdot, J_0\cdot)$, which has a uniform positive lower bound by compactness. Hence they remain positive definite after shrinking U. □

The construction projects onto $\partial_t\bar\partial_t$-closed forms and obtains positivity by continuity near ω_0; closedness and conditions on the curvature would impose further equations.

4.8 The adjoint measure of the linearization

The derivative of the logarithmic Monge–Ampère operator $u \mapsto \log(\omega_u^n/\mathrm{d}V)$ annihilates the constants, so its linearized equation can be solved only under a compatibility condition. The Gauduchon multiplier of Theorem 4.1.1 identifies this condition. Fix $u \in \mathcal{H}_\omega$ and write $\eta = \omega_u$. For a smooth real direction v, logarithmic differentiation gives the linearized operator

$$L_u v = n\,\frac{\mathrm{d}\mathrm{d}^{\mathrm{c}} v \wedge \eta^{n-1}}{\eta^n}. \tag{4.11}$$

Choose the Gauduchon multiplier ρ_u of η given by Theorem 4.1.1, normalized by

$$\int_X \rho_u \eta^n = 1, \qquad \text{and put} \qquad \mathrm{d}\mu_u := \rho_u \eta^n.$$

The *adjoint measure* μ_u is a smooth positive probability measure.

Proposition 4.8.1 *For every smooth real v,*

$$\int_X L_u v \,\mathrm{d}\mu_u = 0. \tag{4.12}$$

Moreover, for smooth real h, the equation $L_u v = h$ has a smooth solution if and only if $\int_X h\,\mathrm{d}\mu_u = 0$. When it exists, the solution is unique modulo constants.

Proof The integral in (4.12) is

$$n\int_X \mathrm{d}\mathrm{d}^{\mathrm{c}} v \wedge \rho_u \eta^{n-1} = n\int_X v\,\mathrm{d}\mathrm{d}^{\mathrm{c}}(\rho_u\eta^{n-1}) = 0.$$

This proves that the compatibility condition is necessary.

By the Fredholm argument in the proof of Theorem 4.1.1, the range of $L_u\colon C^{k+2,\alpha}(X) \to C^{k,\alpha}(X)$ has codimension one for every integer $k \geq 0$ and $0 < \alpha < 1$. The nonzero functional $h \mapsto \int_X h\,\mathrm{d}\mu_u$ annihilates that range, so its kernel

is the range. Elliptic regularity gives a smooth solution, and since the kernel of L_u consists of the constants, the solution is unique modulo constants. □

By its defining equation and the uniqueness in Theorem 4.1.1, the multiplier ρ_u is constant precisely when η is Gauduchon. Thus

$$\mu_u = \frac{\eta^n}{\int_X \eta^n} \quad \text{if } \mathrm{dd^c}(\eta^{n-1}) = 0.$$

On a surface, $\mathrm{dd^c}\omega_u = \mathrm{dd^c}\omega$, so every positive potential perturbation of a Gauduchon form is still Gauduchon. In higher dimension this persistence requires more than the Gauduchon condition at $u = 0$. Write $M = \int_X \omega^n$.

Proposition 4.8.2 *Every form ω_u with $u \in \mathcal{H}_\omega$ is Gauduchon if and only if*

$$\int_X \omega_u^n = M \quad \textit{for every } u \in \mathcal{H}_\omega.$$

In that case the normalized multiplier is $\rho_u = M^{-1}$ and $\mu_u = M^{-1}\omega_u^n$ for every $u \in \mathcal{H}_\omega$.

Proof If all the forms ω_u are Gauduchon, the mass derivative (2.2) vanishes at every point of $\mathcal{H}_\omega$ and in every direction. Since $\mathcal{H}_\omega$ is convex, integration along the segment from 0 to u gives the mass identity. Conversely, Proposition 2.2.1 shows that constant mass forces $\mathrm{dd^c}(\omega_u^{n-1}) = 0$ for every $u \in \mathcal{H}_\omega$. In that case the normalized multiplier is the constant M^{-1}, by Theorem 4.1.1. □

By Theorem 2.2.2, this persistence is equivalent to $\mathrm{dd^c}(\omega^j) = 0$ for all integers $1 \le j \le n-1$. The following example distinguishes it from the single Gauduchon identity in dimension three.

Example 4.8.1 Let $E = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$ and $X = E^3$. For $1 \le j \le 3$, write $z_j = x_j + \mathrm{i}y_j$ for the coordinates on the covering space, and let η_j be the form on X induced by $(\mathrm{i}/2)\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j$. The periodic functions

$$b = \exp(\cos(2\pi x_1)), \qquad c = b^{-1}$$

define a Hermitian form

$$\omega = \eta_1 + b\eta_2 + c\eta_3.$$

Since $bc = 1$,

$$\omega^2 = 2b\eta_1 \wedge \eta_2 + 2c\eta_1 \wedge \eta_3 + 2\eta_2 \wedge \eta_3.$$

The functions b and c depend only on x_1, so $\mathrm{dd^c}b$ and $\mathrm{dd^c}c$ are multiples of η_1, and $\mathrm{dd^c}(\omega^2) = 0$. Thus ω is Gauduchon.

Put $v = \cos(2\pi x_2)$ and $h = -\pi\cos(2\pi x_2)$, so that $\mathrm{dd^c}v = h\eta_2$. For a real s with $0 < |s| < 1/(\pi\mathrm{e})$, the form

$$\omega_{sv} = \eta_1 + (b + sh)\eta_2 + c\eta_3$$

is positive, since $b \geq \mathrm{e}^{-1}$ and $|h| \leq \pi$. However,

$$\mathrm{dd^c}(\omega_{sv}^2) = 2s\,\mathrm{dd^c}\omega \wedge \mathrm{dd^c}v = 2sh\,\mathrm{dd^c}c \wedge \eta_2 \wedge \eta_3 \neq 0.$$

Indeed, $\mathrm{dd^c}c = (c''/(4\pi))\eta_1$, where the prime denotes differentiation in x_1, and $c''(0) = 4\pi^2\mathrm{e}^{-1}$, while $h = -\pi$ at $x_2 = 0$. Thus a Gauduchon form can have arbitrarily small positive potential perturbations that are not Gauduchon.

When persistence holds, the one-form $v \mapsto \int_X v\,\mathrm{d}\mu_u$ on $\mathcal{H}_\omega$ is the Monge–Ampère one-form of Chapter 3 divided by M. By Theorem 3.1.1, it therefore has a primitive exactly when ω is closed.

4.8.1 Convex entropies along a Monge–Ampère flow

The logarithmic Monge–Ampère operator generates a parabolic flow whose linearization is the operator L_u of (4.11). Along this flow, the ratio of the Monge–Ampère measure to a fixed target measure satisfies a heat-type equation. Integrating a convex function of this ratio by parts produces, besides the usual nonpositive gradient term, a term involving $\mathrm{dd^c}(\omega_{u_t}^{n-1})$. If the forms along the flow are Gauduchon, this additional term vanishes and all convex entropies are nonincreasing. Conversely, requiring a nonpositive initial derivative of an L^p entropy for every smooth positive target characterizes the Gauduchon condition on the initial form.

Recall that $M = \int_X \omega^n$, and fix a smooth strictly positive probability volume form ν on X. Consider a smooth solution on an interval $[0, T)$ of the *logarithmic Monge–Ampère flow*

$$\dot{u}_t = \log \rho_t, \qquad \rho_t = \frac{\omega_{u_t}^n}{M\nu}, \qquad u_0 = 0, \qquad \omega_{u_t} > 0. \tag{4.13}$$

We keep the initial normalizing constant M fixed; the evolving mass $\int_X \omega_{u_t}^n$ need not equal M. Write $\Delta_t = L_{u_t}$ for the linearized operator (4.11) at u_t. Logarithmic differentiation gives $\partial_t \log \rho_t = \Delta_t \dot{u}_t = \Delta_t \log \rho_t$. Integration by parts against $\omega_{u_t}^{n-1}$ then produces the same derivative of the background power that appears in the mass variation of Proposition 2.2.1.

Proposition 4.8.3 *Let $\Phi \in C^2((0,+\infty), \mathbb{R})$, and choose a function G with $G'(r) = \Phi'(r)/r$. Along (4.13),*

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} \int_X \Phi(\rho_t)\nu = \frac{n}{M} \int_X G(\rho_t)\mathrm{dd^c}(\omega_{u_t}^{n-1}) \\ - \frac{n}{M} \int_X \frac{\Phi''(\rho_t)}{\rho_t} \mathrm{d}\rho_t \wedge \mathrm{d^c}\rho_t \wedge \omega_{u_t}^{n-1}. \end{aligned} \tag{4.14}$$

In particular, for $p > 1$,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\int_X \rho_t^p \nu &= \frac{np}{(p-1)M}\int_X \rho_t^{p-1}\mathrm{dd^c}(\omega_{u_t}^{n-1}) \\ &\quad - \frac{np(p-1)}{M}\int_X \rho_t^{p-3}\mathrm{d}\rho_t\wedge\mathrm{d^c}\rho_t\wedge\omega_{u_t}^{n-1}.\end{aligned} \tag{4.15}$$

Proof Using $\partial_t\rho_t = \rho_t\Delta_t\log\rho_t$ and $\rho_t\nu = M^{-1}\omega_{u_t}^n$, we find that the derivative of the entropy is

$$\frac{n}{M}\int_X \Phi'(\rho_t)\mathrm{dd^c}\log\rho_t\wedge\omega_{u_t}^{n-1}.$$

The chain rule and the definition of G give

$$\Phi'(r)\mathrm{dd^c}\log r = \mathrm{dd^c}G(r) - \frac{\Phi''(r)}{r}\mathrm{d}r\wedge\mathrm{d^c}r,$$

and Lemma 2.1.1 applies to the first term. Adding a constant to G does not change the result, since $\int_X \mathrm{dd^c}(\omega_{u_t}^{n-1}) = 0$. Taking $\Phi(r) = r^p$ and $G(r) = pr^{p-1}/(p-1)$ proves the second formula. □

Corollary 4.8.1 *Suppose that every form ω_{u_t} in (4.13) is Gauduchon. Then $\int_X \rho_t\nu = 1$, and $t\mapsto\int_X\Phi(\rho_t)\nu$ is nonincreasing for every convex function $\Phi\in C^2((0,+\infty),\mathbb{R})$. In particular,*

$$\|\rho_t\|_{L^p(\nu)} \le \|\rho_s\|_{L^p(\nu)} \quad (0\le s\le t<T,\ p>1).$$

If $\Phi''>0$, the derivative of the entropy vanishes at a time t precisely when $\rho_t = 1$, in which case $\dot u_t = 0$.

The hypothesis holds if $\mathrm{dd^c}(\omega^j) = 0$ for $1\le j\le n-1$, in particular on a Gauduchon surface.

Proof The derivative of the mass is

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_X \rho_t\nu = \frac{n}{M}\int_X \dot u_t\mathrm{dd^c}(\omega_{u_t}^{n-1}) = 0,$$

and its initial value is one. In (4.14) the first term vanishes, while the second is nonpositive, because $\mathrm{d}\rho_t\wedge\mathrm{d^c}\rho_t$ is a semipositive $(1,1)$-form and $\Phi''\ge 0$. If $\Phi''>0$, the vanishing of this term forces $\mathrm{d}\rho_t = 0$, and connectedness together with the mass normalization gives $\rho_t = 1$. The converse and the assertion about $\dot u_t$ follow from the equation.

Under the identities $\mathrm{dd^c}(\omega^j) = 0$ for $1\le j\le n-1$, Theorem 2.2.2 and Proposition 4.8.2 show that ω_u is Gauduchon for every $u\in\mathcal{H}_\omega$, in particular for $u = u_t$. On a surface these identities reduce to the Gauduchon condition $\mathrm{dd^c}\omega = 0$. □

For $\Phi(r) = r\log r$, the same formula gives the dissipation of the relative entropy. In higher dimension, the Gauduchon condition on the initial form alone need not persist along potential perturbations, as Example 4.8.1 shows. The next criterion determines the role of this condition at the initial time, without assuming that the flow exists.

Proposition 4.8.4 *Fix $p > 1$. For each smooth strictly positive probability volume form ν, put $\rho_\nu = \omega^n/(M\nu)$ and define*

$$\mathcal{D}_p(\omega, \nu) := \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \int_X \left(\frac{(\omega + t\mathrm{dd}^{\mathrm{c}} \log \rho_\nu)^n}{M\nu} \right)^p \nu.$$

Then $\mathcal{D}_p(\omega, \nu) \le 0$ for every such ν if and only if ω is Gauduchon.

Proof The form in the definition is positive for sufficiently small $|t|$. The path $u_t = t \log \rho_\nu$ starts at $u_0 = 0$ with the velocity $\log \rho_\nu$ prescribed by (4.13) at time zero. The proof of Proposition 4.8.3 at a given time uses only the potential and its velocity at that time, so (4.15) holds for the derivative at $t = 0$. If ω is Gauduchon, its first term vanishes and the derivative is nonpositive.

Conversely, suppose that $\lambda := \mathrm{dd}^{\mathrm{c}}(\omega^{n-1}) \neq 0$. Write $\lambda = f\omega^n/M$, where f is smooth and real. Stokes' theorem gives $\int_X f\omega^n = 0$, whereas

$$\int_X f\lambda = \frac{1}{M} \int_X f^2 \omega^n > 0.$$

For small $\varepsilon > 0$, set

$$c_\varepsilon = \frac{1}{M} \int_X \mathrm{e}^{-\varepsilon f} \omega^n, \qquad r_\varepsilon = c_\varepsilon \mathrm{e}^{\varepsilon f}, \qquad \nu_\varepsilon = \frac{\omega^n}{M r_\varepsilon}.$$

Then ν_ε is a smooth strictly positive probability volume form, and $\rho_{\nu_\varepsilon} = r_\varepsilon$. Since $c_\varepsilon = 1 + O(\varepsilon^2)$, $\int_X \lambda = 0$ and $\mathrm{d}r_\varepsilon = \varepsilon r_\varepsilon \mathrm{d}f$, the two terms in (4.15) give

$$\mathcal{D}_p(\omega, \nu_\varepsilon) = \frac{np}{M} \varepsilon \int_X f\lambda + O(\varepsilon^2) > 0$$

for sufficiently small $\varepsilon > 0$. This proves necessity. □

The criterion involves only the initial velocity: every smooth solution of (4.13) with target ν has the same initial entropy derivative, independently of its existence for later times or its convergence. Quantification over all targets is essential. For the target $\nu = \omega^n/M$, the flow $u_t = 0$ is stationary for every Hermitian ω, and all entropy derivatives vanish.

4.8.2 Total variation between two evolving probabilities

The entropies of Section 4.8.1 compare one evolving probability with the fixed target. Two solutions of the flow with different initial potentials can nevertheless move apart. On a flat torus, even with a closed background, their total-variation distance can increase at the initial time, while no convex entropy relative to the target increases. We show this by computing the initial derivatives explicitly; they depend only on the initial velocities, so no existence theorem for the flow is needed. By a *logarithmic*

Monge–Ampère flow with target ν we mean a smooth solution of the equation in (4.13), with an arbitrary smooth initial potential.

For this example, let $E = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$ and $X = E^2$, with covering coordinates $z_j = x_j + \mathrm{i}y_j$. Write

$$\eta_j = \frac{\mathrm{i}}{2}\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j, \qquad \kappa = \eta_1 + \eta_2, \qquad \mu = \eta_1 \wedge \eta_2 = \frac{\kappa^2}{2}.$$

Thus μ is a probability volume form and $\int_X \kappa^2 = 2$. We use κ as the background form ω in (4.13), so that $M = 2$. Put

$$c_j = \cos(2\pi x_j), \qquad S = c_1 + c_2, \qquad P = c_1 c_2.$$

Lemma 4.8.1 *The averages of $|S|$ and $P|S|$ are*

$$\int_X |S|\mu = \frac{8}{\pi^2}, \qquad \int_X P|S|\mu = \frac{8}{3\pi^2}.$$

Proof The integrands do not depend on y_1, y_2. The degree-two map of real tori $(a, b) \mapsto (a+b, a-b)$ preserves normalized Haar averages. With the angles $A = 2\pi a$ and $B = 2\pi b$, it gives

$$S = 2\cos A \cos B, \qquad P = \frac{\cos(2A) + \cos(2B)}{2}.$$

The one-variable averages of $|\cos A|$ and $|\cos A|^3$ are $2/\pi$ and $4/(3\pi)$, respectively, so the average of $\cos(2A)|\cos A|$ is $2/(3\pi)$. Multiplying the averages over the two independent angles gives both formulas. □

For a fixed real number $K > 3$, define the smooth target probability ν_K by

$$Z_K = \int_X \mathrm{e}^{KP}\mu, \qquad f_K = Z_K^{-1}\mathrm{e}^{KP}, \qquad \nu_K = f_K\mu.$$

For $0 < \varepsilon < 1/\pi$, the initial potentials εS and zero give positive forms. Set

$$A_\varepsilon = 1 - \pi\varepsilon c_1, \qquad B_\varepsilon = 1 - \pi\varepsilon c_2, \qquad r_\varepsilon = A_\varepsilon B_\varepsilon,$$

so that $\kappa + \mathrm{dd}^{\mathrm{c}}(\varepsilon S) = A_\varepsilon\eta_1 + B_\varepsilon\eta_2$ and its normalized Monge–Ampère probability is $r_\varepsilon\mu$.

Consider the short smooth paths

$$U_\varepsilon(t) = \varepsilon S + t\log\frac{r_\varepsilon}{f_K}, \qquad V(t) = -t\log f_K. \tag{4.16}$$

They remain positive for sufficiently small $|t|$, and at $t = 0$ their velocities are exactly those prescribed by the logarithmic Monge–Ampère flow with target ν_K. Their normalized probabilities are

$$\mu_\varepsilon(t) = \frac{(\kappa + \mathrm{dd^c} U_\varepsilon(t))^2}{2}, \qquad \mu_0(t) = \frac{(\kappa + \mathrm{dd^c} V(t))^2}{2},$$

which have mass one because the background κ is closed. Define their total-variation distance, with the convention for probabilities, by

$$D_\varepsilon(t) = \frac{1}{2}\int_X |\mu_\varepsilon(t) - \mu_0(t)|.$$

Proposition 4.8.5 *For the paths* (4.16)*,*

$$D'_\varepsilon(0) = 4\varepsilon\left(\frac{K}{3} - 1\right) + o(\varepsilon) \quad \text{as } \varepsilon \downarrow 0.$$

In particular, this derivative is positive for sufficiently small $\varepsilon > 0$. For any two logarithmic Monge–Ampère flows with target ν_K starting from these two initial potentials, the initial derivative of their total-variation distance is $D'_\varepsilon(0)$.

Proof For a real function h depending only on x_1, x_2, the coefficient matrix of $\mathrm{dd^c} h$ with respect to the forms $(\mathrm{i}/2)\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$ is $(4\pi)^{-1}(h_{x_j x_k})$. Put $R_\varepsilon(t) = \mathrm{d}\mu_\varepsilon(t)/\mathrm{d}\mu$ and $R_0(t) = \mathrm{d}\mu_0(t)/\mathrm{d}\mu$. Differentiating the determinant at the diagonal initial matrices gives

$$\begin{aligned}\partial_t(R_\varepsilon - R_0)|_{t=0} = \frac{1}{4\pi}\big(&B_\varepsilon \partial_{x_1}^2 \log A_\varepsilon + A_\varepsilon \partial_{x_2}^2 \log B_\varepsilon \\ &- K(B_\varepsilon - 1)\partial_{x_1}^2 P - K(A_\varepsilon - 1)\partial_{x_2}^2 P\big).\end{aligned}$$

Since $\partial_{x_j}^2 P = -4\pi^2 P$, Taylor expansion in ε yields the uniform estimate

$$\partial_t(R_\varepsilon - R_0)|_{t=0} = \pi^2 \varepsilon S(1 - KP) + O(\varepsilon^2).$$

At $t = 0$, the difference of the densities is

$$q_\varepsilon = r_\varepsilon - 1 = -\pi\varepsilon S + \pi^2 \varepsilon^2 P.$$

For each fixed x_2, its coefficient of c_1 is $-\pi\varepsilon(1 - \pi\varepsilon c_2) \neq 0$, so as a function of x_1 it has at most two zeros in a period, and its zero set has measure zero. Differentiation under the integral sign, justified by the uniform bound on the derivatives of the smooth densities, gives

$$D'_\varepsilon(0) = \frac{1}{2}\int_X \operatorname{sgn}(q_\varepsilon)\, \partial_t(R_\varepsilon - R_0)|_{t=0}\, \mu.$$

The set $\{S = 0\}$ also has measure zero, and $\operatorname{sgn}(q_\varepsilon) \to -\operatorname{sgn}(S)$ off this set. Dominated convergence and Lemma 4.8.1 give

$$\lim_{\varepsilon \downarrow 0} \frac{D'_\varepsilon(0)}{\varepsilon} = \frac{\pi^2}{2}\int_X |S|(KP - 1)\mu = 4\left(\frac{K}{3} - 1\right).$$

Every flow with the stated initial data has the velocities (4.16) at $t = 0$, so its initial density derivatives, and hence the derivative of the distance, coincide with this computation. □

Corollary 4.8.2 *For these paths, the initial derivative of every C^2 convex entropy of each probability relative to ν_K is nonpositive, while the total-variation distance between the two probabilities has positive initial derivative for sufficiently small $\varepsilon > 0$.*

Proof The initial velocities are those of the logarithmic flow, and both initial forms are closed. The chain-rule calculation in the proof of Proposition 4.8.3 therefore applies at $t = 0$: its background term vanishes and its gradient term is nonpositive. The preceding proposition gives the assertion about the pair. □

4.9 Limits of adjoint measures

When every positive potential perturbation of ω is Gauduchon, the adjoint probability is the normalized Monge–Ampère measure. Mass invariance and the continuity of bounded products then extend this description to bounded potentials.

Proposition 4.9.1 *Assume that ω_v is Gauduchon for every $v \in \mathcal{H}_\omega$. For every $u \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$, the measure*

$$\mu_u := M^{-1}(\omega + \mathrm{dd^c} u)^n$$

is a probability measure, which agrees with the adjoint measure of Proposition 4.8.1 when $u \in \mathcal{H}_\omega$. If $u_j, u \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$ are uniformly bounded and $u_j \to u$ in capacity, then $\mu_{u_j} \rightharpoonup \mu_u$.

Proof By Proposition 4.8.2 and Theorem 2.2.2, the positive bounded product $(\omega + \mathrm{dd^c} u)^n$ has mass M, and for $u \in \mathcal{H}_\omega$ its normalization is the smooth adjoint measure by Proposition 4.8.2. The continuity of bounded products under convergence in capacity, in the sense of Section 1.8, gives weak convergence of the products, and division by M preserves it. □

The proposition applies in particular to Gauduchon surfaces. For a general Hermitian background, passing from smooth $u_j \in \mathcal{H}_\omega$ to a limit $u \in \mathrm{PSH}(X,\omega)$ requires control of the multipliers ρ_{u_j} as well as of the potentials. The supporting probabilities of Chapter 16 recover μ_u in the smooth case. If the solution of the normalized equation with a continuous logarithmic density is bounded and has a continuous strictly positive complex Hessian, Theorem 16.2.4 gives a unique supporting probability, which is stationary for the linearized operator, and hence approximation independence under uniform convergence of the logarithmic densities. We do not know whether, in general, the limit is independent of the approximating sequence, or whether a weak linearization satisfies an adjoint identity.

Chapter 5
Gauduchon degrees and positivity of vector bundles

Throughout this chapter, ω is a fixed Hermitian form on X, and $n \geq 2$ unless a section specifies another setting. The inputs from Chapter 4 are Gauduchon's theorem, the trace identity, the Gauduchon roots of strictly positive dd^{c}-closed $(n-1,n-1)$-forms found before the proof of Theorem 4.3.1, and the compatibility condition of Proposition 4.8.1.

For a holomorphic line bundle, the conserved Gauduchon pairing of a curvature form is the degree of the bundle, and a positive degree excludes a metric with nonpositive curvature. Fiber averaging extends this obstruction to plurisubharmonic norms near a direct sum of line bundles of positive degree. Polar duality and homogeneous regularization then turn it into an obstruction to semipositive Finsler curvature. The stable ample bundles constructed on Hirzebruch surfaces show that ampleness alone does not remove the obstruction; their projective complements give a geometric way to measure it.

The compatibility condition for the scalar trace equation also improves curvature when the mean curvature is nonnegative: a conformal change spreads directional positivity from a nonempty open set to the whole manifold. Ricci-flat Kähler surfaces illustrate the distinction between this directional condition and positivity on every tangent and fiber pair. Holomorphic sectional curvature evaluates the same tangent vector in both places. An integral identity against a Kähler metric with nonnegative Ricci curvature excludes quasi-negative holomorphic sectional curvature for every Hermitian metric, closed or not, and on a complex torus it forces a metric with nonpositive holomorphic sectional curvature to be translation invariant. On a projective manifold, positivity on all tangent and fiber pairs admits another duality criterion, expressed by positive measures over the dual bundle; on a branched double cover of the projective line, its untwisted test fails for a line bundle with a positively curved metric. Finally, for a uniformly negative bundle over a Kähler base with a plurisubharmonic exhaustion, the squared norm gives logarithmic weights along the zero section and a complete Kähler metric on the punctured total space.

5.1 Convex norms and Gauduchon degrees

For holomorphic line bundles, the trace identity of Proposition 4.2.1 becomes a statement about curvature. Let F be a holomorphic line bundle on X and k a smooth Hermitian metric on F. In a local holomorphic frame e of F we identify k with the positive function $|e|_k^2$, and the curvature form of k is $R_k = -\mathrm{dd^c}\log k$. Let Θ be a smooth strictly positive real $(n-1,n-1)$-form with $\mathrm{dd^c}\Theta = 0$; by the root calculation preceding the proof of Theorem 4.3.1, $\Theta = \gamma^{n-1}$ for a unique Gauduchon metric γ. The *degree* of F with respect to Θ is

$$\deg_\Theta F = \int_X R_k \wedge \Theta.$$

Two metrics on F differ by a factor e^{-u} with u smooth, which adds $\mathrm{dd^c}u$ to the curvature, so the degree is independent of k by Lemma 2.1.1. It is the Gauduchon degree of the curvature form with respect to γ; the form Θ need not be d-closed.

The squared norm $v \mapsto |v|_k^2$ is plurisubharmonic on the total space of F off the zero section exactly when $\log k$ is plurisubharmonic, that is, when $R_k \le 0$; this follows from the Levi form computation in the proof of Lemma 5.1.6 below. A line bundle of positive degree therefore admits no metric with plurisubharmonic squared norm. This section extends the obstruction to direct sums of line bundles of positive degree, to all holomorphic structures sufficiently close to such a direct sum, and to norms that need not be Hermitian. The mechanism is again integration against Θ: averaging the horizontal complex Hessians of a norm near the line summands produces a negative multiple of the degrees, and this negativity is stable under small changes of the holomorphic structure.

For a smooth complex vector bundle F, a *smooth complex norm* is a continuous function $p\colon F \to [0,+\infty)$, positive away from the zero section and smooth there, such that

$$p(\lambda v) = |\lambda| p(v), \qquad p(v+w) \le p(v) + p(w)$$

for $\lambda \in \mathbb{C}$ and vectors v, w in the same fiber. Its square $G = p^2$ satisfies $G(\lambda v) = |\lambda|^2 G(v)$. Near each line summand of a direct sum, the triangle inequality controls the fiber derivatives of G, and these estimates control the averaged horizontal complex Hessians.

Lemma 5.1.1 *Fix a smooth positive volume form μ on X. Every holomorphic line bundle F has a Hermitian metric k satisfying*

$$R_k \wedge \Theta = \lambda\mu, \qquad \lambda = \frac{\deg_\Theta F}{\int_X \mu}.$$

Proof Start with any metric k^0, and put $\mathcal{L}u = (\mathrm{dd^c}u \wedge \Theta)/\mu$. This scalar operator is uniformly elliptic, has no zeroth-order term, and its kernel consists of the constants by the maximum principle. On Hölder spaces, the homotopy to a Laplace operator used in the proof of Theorem 4.1.1 shows that its index is zero, so its cokernel

is one-dimensional. Since $\mathrm{dd}^{\mathrm{c}}\Theta = 0$, integration by parts gives $\int_X \mathcal{L}u\,\mu = 0$, and therefore its image consists exactly of the smooth functions of zero μ-mean. Hence we can solve

$$\mathrm{dd}^{\mathrm{c}}u \wedge \Theta = \lambda\mu - R_{k^0} \wedge \Theta,$$

whose right-hand side has integral zero, and elliptic regularity makes u smooth. Then $k = k^0\mathrm{e}^{-u}$ has the required curvature. □

Let $F_0 = \bigoplus_{s=1}^{r} F_s$ be a direct sum of holomorphic line bundles. Choose Hermitian metrics k_s on F_s and write $k = \bigoplus_s k_s$. Let P_s be the orthogonal projection onto F_s, and let

$$\Sigma = \{v \in F_0 : |v|_k = 1\}$$

be the unit sphere bundle, with fibers Σ_x. On each fiber sphere Σ_x we use the rotation-invariant probability measure $\mathrm{d}\sigma_x$. Fix a nonnegative smooth function χ supported in $(-1, 1)$ and positive near zero. For $0 < \varepsilon < 1/2$, define weights ρ_s on Σ by

$$\rho_s(v) = c_\varepsilon \chi\left(\frac{|(I - P_s)v|_k^2}{\varepsilon^2}\right), \qquad \int_{\Sigma_x} \rho_s \,\mathrm{d}\sigma_x = 1.$$

In unitary frames respecting the splitting, these weights and the normalizing constant c_ε are independent of the base point and of the index s. Independent phase rotations of the summands preserve them, so they are globally defined. On the support of ρ_s we have $|(I - P_s)v|_k < \varepsilon$, so this support lies within distance 2ε of the unit circle of F_s.

Lemma 5.1.2 *For a smooth complex norm on F_0 with square G, put*

$$a_s(x) = G(e_s), \qquad N(x) = \sum_s a_s(x), \qquad D_jG(v) = \mathrm{d}_v G(P_j v),$$

where e_s is any k_s-unit vector in the fiber of F_s at x; complex homogeneity makes a_s well defined. Then

$$G(v) \le N|v|_k^2, \qquad |\mathrm{d}_v G(w)| \le 2N|v|_k|w|_k.$$

On the support of ρ_s,

$$|D_jG| \le 2\varepsilon N \quad (j \ne s), \qquad |D_sG - 2a_s| \le 10\varepsilon N.$$

Proof The triangle inequality and the Cauchy–Schwarz inequality give $p(v) \le \sum_s |v_s|_k\sqrt{a_s} \le \sqrt{N}|v|_k$. Differentiating $|p(v + tw) - p(v)| \le |t|p(w)$ in the real variable t gives $|\mathrm{d}_v p(w)| \le p(w)$, and both global bounds follow.

For v in the support of ρ_s, set $w = (I - P_s)v$ and $e_s = P_s v/|P_s v|_k$. Then $|w|_k < \varepsilon$ and $|v - e_s|_k \le |w|_k + 1 - |P_s v|_k < 2\varepsilon$. The segment from e_s to v stays in the unit ball and avoids zero, so integrating the derivative bound along it gives $|G(v) - a_s| \le 4\varepsilon N$. For $j \ne s$, we have $|P_j v|_k < \varepsilon$, which gives the first axis bound. Euler's identity $\mathrm{d}_v G(v) = 2G(v)$ yields

$$D_s G - 2a_s = 2(G(v) - a_s) - \sum_{j \neq s} D_j G.$$

The sum on the right is $\mathrm{d}_v G(w)$, whose absolute value is at most $2\varepsilon N$. This proves the second axis bound. □

Write $\pi\colon F_0 \to X$ for the bundle projection. For $v \in F_0$ with $\pi(v) = x$, let $H_{0,v}\colon T_x X \to T_v F_0$ be the horizontal lift for the Chern connection of k. It is complex linear, and at unit vectors it is tangent to Σ, since the Chern connection preserves k. For a smooth real function G on $F_0 \setminus \{0\}$, define a $(1,1)$-form $\beta_s(G)$ and functions U_s, V_{sj} on X by

$$\beta_s(G) = \int_{\Sigma_x} \rho_s H_{0,v}^*(\mathrm{dd^c} G)\, \mathrm{d}\sigma_x,$$
$$U_s = \int_{\Sigma_x} G \rho_s\, \mathrm{d}\sigma_x,$$
$$V_{sj} = \int_{\Sigma_x} D_j G \rho_s\, \mathrm{d}\sigma_x.$$

Lemma 5.1.3 *With R_j the curvature of k_j,*

$$\beta_s(G) = \mathrm{dd^c} U_s - \frac{1}{2} \sum_j R_j V_{sj}.$$

Proof At a fixed base point, choose holomorphic frames of the line bundles with $k_j = \mathrm{e}^{-\varphi_j}$ and $\mathrm{d}\varphi_j = 0$ at that point. There the horizontal lift has zero vertical component. Write the smooth unitary fiber coordinates as $\zeta_j = \mathrm{e}^{-\varphi_j/2} v_j$ and $G(z, v) = g(z, \zeta)$. At the base point, the first base derivatives of ζ_j vanish, and its mixed second base derivatives are $-\frac{1}{2}(\varphi_j)_{\alpha\bar\beta}\zeta_j$ and their conjugates. The chain rule therefore gives

$$G_{\alpha\bar\beta} = g_{\alpha\bar\beta} - \frac{1}{2} \sum_j (\varphi_j)_{\alpha\bar\beta} (\zeta_j g_{\zeta_j} + \bar\zeta_j g_{\bar\zeta_j}).$$

The expression in parentheses is $D_j G$. Both the spherical measure and ρ_s are independent of z in unitary coordinates, so averaging the identity gives the result, since $R_j = \mathrm{dd^c}\varphi_j$. □

Theorem 5.1.1 *Suppose that for each s there is a smooth strictly positive real $(n-1, n-1)$-form Θ_s satisfying*

$$\mathrm{dd^c}\Theta_s = 0, \qquad \deg_{\Theta_s} F_s > 0.$$

Then there is a C^4 neighborhood of the split Dolbeault operator on the underlying smooth bundle of F_0 such that no integrable Dolbeault operator in this neighborhood admits a smooth complex norm whose square is plurisubharmonic off the zero section.

Proof *Step 1: the split structure.* Fix μ and use Lemma 5.1.1 to choose k_s with $R_s \wedge \Theta_s = \lambda_s \mu$, where $\lambda_s > 0$. Put

$$\lambda_* = \min_s \lambda_s, \qquad C = 5\sum_s \lambda_s + \sum_s \sum_{j\neq s} \sup_X \left| \frac{R_j \wedge \Theta_s}{\mu} \right|.$$

For a squared norm G, define $I_0(G) = \sum_s \int_X \beta_s(G) \wedge \Theta_s$. By Lemma 5.1.3, the term $\mathrm{dd}^{\mathrm{c}} U_s$ integrates to zero against Θ_s, and the axis estimates of Lemma 5.1.2 then give

$$I_0(G) = -\frac{1}{2} \sum_{s,j} \int_X V_{sj} R_j \wedge \Theta_s \leq -(\lambda_* - \varepsilon C) \int_X N\mu.$$

Choose ε so that $\varepsilon C < \lambda_*/2$, and keep all weights fixed.

Step 2: nearby holomorphic structures. Write a nearby operator as $\bar\partial_A = \bar\partial_0 + A$, with $\bar\partial_A^2 = 0$. Its Chern connection for k is $D_A = D_0 + A - A^{*k}$, where the adjoint changes the type from $(0,1)$ to $(1,0)$. Let J_A be the corresponding complex structure on the total space and $H_{A,v}$ the horizontal lift of D_A. These depend smoothly on A, and $H_{A,v}$ is complex linear and tangent to the fixed sphere bundle Σ.

For $g \in C^\infty(\Sigma, \mathbb{R})$, define its homogeneous extension by

$$(\mathcal{E}g)(v) = |v|_k^2 g(v/|v|_k),$$

and define a real differential operator on Σ by

$$(\mathcal{P}_A g)(v) = \sum_s \rho_s(v) \frac{H_{A,v}^* (\mathrm{dd}^{\mathrm{c}}_{J_A} \mathcal{E}g)_v \wedge \Theta_s(x)}{\mu(x)}, \qquad x = \pi(v).$$

It has order at most two. By the identity $\mathrm{d}^{\mathrm{c}}_J f = -(4\pi)^{-1} \mathrm{d} f \circ J$, its coefficients depend on A and its first derivatives, as well as on the fixed smooth data. Consequently they converge in C^2 as $A \to 0$ in C^4.

Step 3: the adjoint operator. Take the formal adjoint $\mathcal{P}_A^*$ for the fixed density $\mathrm{d}\nu = \mathrm{d}\sigma_x\, \mu(x)$ on the compact sphere bundle, and put $b_A = \mathcal{P}_A^* 1$. Locally, if $\mathrm{d}\nu = m(y)\,\mathrm{d}y$ and $\mathcal{P}_A = \sum_{|\alpha|\leq 2} c_{\alpha,A} \partial^\alpha$, then

$$b_A = m^{-1} \sum_{|\alpha|\leq 2} (-1)^{|\alpha|} \partial^\alpha (m c_{\alpha,A}).$$

Thus $\|b_A - b_0\|_{C^0} \to 0$ as $A \to 0$ in C^4. For $g = G|_\Sigma$, homogeneity gives $\mathcal{E}g = G$ and

$$I_A(G) := \int_\Sigma \mathcal{P}_A g \,\mathrm{d}\nu = \int_\Sigma g b_A \,\mathrm{d}\nu.$$

Since $0 < g \leq N \circ \pi$ and the spherical measures have mass one,

$$|I_A(G) - I_0(G)| \leq \|b_A - b_0\|_{C^0} \int_X N\mu.$$

Choose the neighborhood so that $\|b_A - b_0\|_{C^0} < \lambda_*/4$ in it. Then $I_A(G) \leq -\frac{1}{4}\lambda_* \int_X N\mu < 0$ for every smooth squared complex norm G. If G were pluri-

subharmonic for J_A, all horizontal pullbacks of $\mathrm{dd}^{\mathrm{c}}_{J_A} G$ would be semipositive, and their pairings with Θ_s, weighted by $\rho_s \geq 0$, would give $I_A(G) \geq 0$, a contradiction.□

The neighborhood depends on the bundle and on the test forms Θ_s, and different line summands may require different tests. Its size is independent of all derivatives of the candidate norms: integration over the fixed sphere bundle transfers those derivatives to the coefficients of the adjoint operator.

To apply Theorem 5.1.1 to norms obtained by duality, we first need to regularize continuous norms while preserving a strict convexity margin (Sections 5.1.1 and 5.1.2). Polar duality then turns the obstruction into a failure of semipositive Kobayashi curvature on the dual bundle (Section 5.1.3). Stable ample bundles on Hirzebruch surfaces provide concrete examples (Sections 5.1.4 to 5.1.6), and a fixed Stein hypersurface detects the failure for every Finsler metric on such a bundle (Section 5.1.7).

Dual support norms also connect this obstruction with the geometry of neighborhoods of the zero section, which leads to bounds for the convexity of the complement of the zero section in a projective completion (Section 5.1.8). Curvature gives bounds in the opposite direction, through explicit exhaustions (Sections 5.1.9 and 5.1.10). Finally, homogeneous plurisubharmonic functions on the punctured bundle and restrictions to closed submanifolds give lower bounds for the convexity index of the projective complement; on Hopf manifolds, together with the curvature bounds, they determine it for line bundles whose de Rham first Chern class vanishes (Section 5.1.11).

5.1.1 Homogeneous convex regularization

Theorem 5.1.1 concerns smooth norms, whereas the norms produced by duality later in this section are in general only continuous. We show that a continuous homogeneous function with strict positivity margins can be approximated by smooth ones that keep strict margins of the same kind, so the obstruction applies to it as well. The local step is Wu's convolution in projective coordinates, written as an average of holomorphic linear shears; the local approximations are glued by the argument of Greene and Wu. We give both steps in order to keep track of strict plurisubharmonicity and of strong convexity in the fibers; see Section 36.4 for attribution.

Let $\pi\colon F \to X$ be a holomorphic vector bundle, and write $F^{\times}$ for the complement of its zero section. A continuous real function Q on F is *complex homogeneous of degree two* if

$$Q(\lambda v) = |\lambda|^2 Q(v) \qquad (\lambda \in \mathbb{C}).$$

We call Q *locally strongly convex in the fibers* if, in a local bundle frame and near every point of $F^{\times}$, there is a constant $c > 0$ such that $Q(z, v) - c|v|^2$ is convex on sufficiently small real fiber balls. For continuous functions, strict plurisubharmonicity means that subtracting a sufficiently small positive multiple of the Euclidean squared norm in local coordinates of the total space leaves a plurisubharmonic function. Both

conditions are local, and on compact subsets of $F^\times$ they hold with uniform positive margins.

Local strong convexity in the fibers requires one constant on a neighborhood in the total space. Strong convexity of each fixed fiber is weaker, even for a plurisubharmonic squared complex norm.

Example 5.1.1 On the trivial bundle $D \times \mathbb{C}^2$, where $D = \{z \in \mathbb{C} : |z| < 1/2\}$, put

$$Q(z, v) = \max\left\{|v_1|^2 + |v_2|^2, \frac{|v_1|^2}{1 - |z|^2} + |z|^2|v_2|^2\right\}.$$

This is a continuous plurisubharmonic function, positive off the zero section and complex homogeneous of degree two. Its square root is a complex norm on every fiber, and for each fixed z, the real function $Q(z, \cdot)$ is strongly convex. Nevertheless, Q is not locally strongly convex in the fibers near $(z, v) = (0, (1, 0))$ in the sense defined above.

Proof Write $t = |z|^2$. Both branches are continuous, positive semidefinite Hermitian quadratic forms in v, and the first is positive definite. Their square roots satisfy the triangle inequality and complex homogeneity; their maximum therefore is a complex norm. In particular Q is positive off zero and continuous, also on the zero section.

The first branch is plurisubharmonic on the total space. For the second, the locally uniformly convergent expansion

$$\frac{|v_1|^2}{1 - |z|^2} + |z|^2|v_2|^2 = \sum_{k=0}^{\infty} |z^k v_1|^2 + |z v_2|^2$$

expresses it as a limit of sums of squared moduli of holomorphic functions. Thus it too is plurisubharmonic, and so is their maximum.

For $0 < t < 1/4$, subtracting $t|v|^2$ gives

$$Q(z, v) - t|v|^2 = \max\left\{(1 - t)|v|^2, \left(\frac{1}{1 - t} - t\right)|v_1|^2\right\},$$

which is convex. At $z = 0$, one has $Q(0, v) = |v|^2$. Hence each fixed fiber has a positive strong convexity margin.

To test a margin on a neighborhood of $(0, (1, 0))$, fix any $c > 0$ and choose a nonzero z arbitrarily close to zero with $t < c$. At $v = (1, 0)$ the second branch is strictly larger than the first, so it remains the maximum on a small fiber ball. Along the real line $v = (1, s)$ in this ball,

$$Q(z, (1, s)) - c|(1, s)|^2 = \frac{1}{1 - t} - c + (t - c)s^2.$$

Its second derivative in s is $2(t - c) < 0$. Thus no positive c makes $Q - c|v|^2$ convex on the fiber balls in any such total-space neighborhood. A fiber-linear change of smooth frame and its inverse are uniformly bounded on a smaller base neighborhood,

so a positive quadratic margin in another frame would give one in this frame. This proves the failure of the local margin. □

Lemma 5.1.4 *Let M be a compact smooth manifold and let $\mathcal{S}$ be a sheaf of germs of continuous real functions on M, whose elements we call sections. Suppose that $\mathcal{S}$ is closed under maxima and has the following two properties.*

Stability: *if f is a section near a compact set K, then $f + u$ is a section near K whenever u is smooth and sufficiently small in the C^∞ topology on a neighborhood of K.*

Local approximation: *every point has a neighborhood U such that, for every compact $K \Subset U$, a section near K can be approximated uniformly on K by smooth sections near K, with arbitrarily small errors in any prescribed finite collection of derivative norms on compact subsets where the section is already smooth.*

Then every global section can be approximated uniformly on M by smooth global sections.

Proof We first describe a gluing step. Let f be a global section that is smooth near a compact set L_1, and let L_2 be a compact subset of one of the neighborhoods U in the statement. Choose relatively compact open sets W_3, W_4 with $L_1 \subset W_3$, f smooth near $\overline{W}_3$, and $L_2 \subset W_4 \Subset U$. Choose a smooth function $0 \le \rho \le 1$ equal to one near $\overline{W}_3$, with compact support in the region where f is smooth. Put $V = W_3 \cup W_4$, and choose a smooth function $-1 \le \psi \le 1$ equal to one near $L_1 \cup L_2$ and equal to minus one near $M \setminus V$.

By stability, for sufficiently small $\xi > 0$ the function $f_\xi = f + \xi\psi$ is still a global section. Approximate f_ξ on $\overline{W}_4$ by a smooth section g, with uniform error less than $\delta < \xi/2$, and with arbitrarily small derivative errors near the compact set $\overline{W}_4 \cap \operatorname{supp}\rho$, where f_ξ is smooth. On V define

$$h = f_\xi \quad \text{on } W_3, \qquad h = \rho f_\xi + (1-\rho)g \quad \text{on } W_4.$$

These definitions agree on the overlap, and h is smooth. Where ρ vanishes, h is the section g. Where ρ can be nonzero, $h - f_\xi = (1-\rho)(g - f_\xi)$, and stability applied to the smooth section f_ξ makes h a section if the derivative errors of $g - f_\xi$ are sufficiently small. Moreover, $|h - f_\xi| < \delta$.

Define $f_1 = \max\{f, h\}$ on V and $f_1 = f$ outside V. Near $L_1 \cup L_2$ we have $h > f$, and near the boundary of V we have $h < f$. Thus f_1 is continuous, it is a section because $\mathcal{S}$ is closed under maxima, and it is smooth near $L_1 \cup L_2$. Also $\|f_1 - f\|_{C^0} < \xi + \delta$. Near L_1 it equals $f + \xi$, so the previous smoothness is preserved.

Now cover M by finitely many compact sets L_2 of this kind and repeat the step, starting with L_1 empty and then taking for L_1 the union of the previously treated compact sets. At every step choose ξ and δ so small that the sum of the finitely many uniform errors is below the prescribed tolerance. The final section is smooth everywhere and has the required uniform error. □

Lemma 5.1.5 *In a holomorphic bundle chart, write $v = (v', v_r)$ and use the affine projective coordinate $w = v'/v_r$ on the cone $v_r \neq 0$. Suppose that a positive continuous complex homogeneous function Q of degree two is strictly plurisubharmonic and locally strongly convex in the fibers on this cone, and put $q(z, w) = Q(z, (w, 1))$. For a nonnegative smooth approximate identity $\eta_\delta(s, t)$ of integral one, define*

$$q_\delta(z, w) = \int q(z - s, w - t)\eta_\delta(s, t)\,\mathrm{d}s\,\mathrm{d}t, \qquad Q_\delta(z, v) = |v_r|^2 q_\delta(z, v'/v_r).$$

On every relatively compact projective subchart, for sufficiently small δ, Q_δ is a smooth positive complex homogeneous function of degree two that is strictly plurisubharmonic and locally strongly convex in the fibers. As $\delta \to 0$, it converges to Q uniformly on compact subsets of the cone consisting of vectors of norm one, and in every finite derivative norm wherever Q is already smooth.

Proof Complex homogeneity gives the identity

$$Q_\delta(z, v) = \int Q(z - s, T_t v)\eta_\delta(s, t)\,\mathrm{d}s\,\mathrm{d}t, \qquad T_t(v', v_r) = (v' - v_r t, v_r).$$

For fixed (s, t), the map $(z, v) \mapsto (z - s, T_t v)$ is holomorphic and invertible, and its restriction to each fiber is linear. On a compact subset of the cone consisting of vectors of norm one, all these maps with (s, t) sufficiently small stay in a fixed compact subset of the original cone, and their differentials and inverses are uniformly bounded there. A positive local lower bound for the Levi form of Q, interpreted through quadratic barriers, therefore gives a positive lower bound for every pullback and for their average. The same argument with real quadratic barriers in the fibers preserves local strong convexity. Only fiber segments contained in the local cone neighborhood are used here.

Ordinary convolution makes q_δ smooth, and the formula for Q_δ makes it smooth on the cone and complex homogeneous. Uniform convergence, derivative convergence on compact regions where Q is smooth, and positivity for small δ follow from the corresponding properties of convolution and from the positive minimum of Q on compact sets of vectors of norm one. □

Theorem 5.1.2 *Let $Q\colon F \to [0, +\infty)$ be continuous, positive on $F^\times$, complex homogeneous of degree two, strictly plurisubharmonic on $F^\times$, and locally strongly convex in the fibers. Then there are continuous complex homogeneous functions Q_j of degree two, smooth on $F^\times$, strictly plurisubharmonic there and locally strongly convex in the fibers, which converge to Q uniformly on the unit sphere bundle of any fixed smooth Hermitian metric. Each $p_j = \sqrt{Q_j}$ is a smooth complex norm.*

Proof Fix a smooth Hermitian squared norm G on F, and let $M = \mathbb{P}(F)$ be the bundle of complex lines in F. The quotient $m = Q/G$ is a positive continuous function on the compact manifold M. Define a sheaf $\mathcal{S}$ on M by requiring that the corresponding function Gm on the cone be positive, strictly plurisubharmonic, and locally strongly convex in the fibers.

This sheaf is closed under maxima. Indeed, multiplication by $G > 0$ commutes with maxima. Near a point of the cone, subtract a common positive quadratic barrier from both functions; their maximum remains plurisubharmonic, and on sufficiently small fiber balls it remains convex. Positivity is also preserved.

For stability on a compact subset K of M, restrict to vectors of norm one lying over K, and to a fixed compact annulus around them. Finitely many bundle charts provide positive common lower bounds for both barriers. If u is a smooth function on M that is sufficiently small in C^2 near K, the Levi form and the real fiber Hessian of Gu are smaller than half these bounds on that annulus. The perturbed function is therefore still positive and has both strict margins. Homogeneity transports the local conditions to the rest of the cone over K. This proves the stability required in Lemma 5.1.4.

On a sufficiently small projective coordinate chart, Lemma 5.1.5 gives local approximation, including derivative control on compact subsets where m is smooth; dividing its regularizations by G gives local smooth sections of $\mathcal{S}$. By Lemma 5.1.4, there are smooth positive global sections $m_j \to m$ uniformly. Put $Q_j = Gm_j$ off the zero section and $Q_j = 0$ on the zero section. Homogeneity gives continuity at zero, uniformly over the compact base, as well as the asserted convergence on a fixed unit sphere bundle.

It remains to see that p_j is a norm. Local convexity in the fibers away from zero implies convexity on the whole fiber. Indeed, the restriction of Q_j to a real line avoiding zero is locally convex and hence convex, while on a real line through zero, homogeneity gives $Q_j(tv) = t^2 Q_j(v)$ for real t, which is convex including at zero. Thus the set $B_x = \{v : Q_j(v) \le 1\}$ is convex, balanced and absorbing, and its gauge is $\sqrt{Q_j}$. If v, w are nonzero, convexity applied to $v/p_j(v)$ and $w/p_j(w)$, with weights proportional to $p_j(v)$ and $p_j(w)$, gives the triangle inequality; the cases with a zero vector follow directly. Positivity, smoothness off zero and complex homogeneity give the remaining properties of a smooth complex norm. □

Proposition 5.1.1 *Suppose that F admits a smooth positive complex homogeneous function H of degree two that is strictly plurisubharmonic on $F^\times$. If a positive continuous complex homogeneous function Q of degree two is plurisubharmonic on $F^\times$ and locally strongly convex in the fibers, then F admits a smooth complex norm whose square is strictly plurisubharmonic on $F^\times$ and locally strongly convex in the fibers.*

Proof For $\tau > 0$, set $Q_\tau = Q + \tau H$; it is positive, complex homogeneous and strictly plurisubharmonic. On a compact annulus of vectors of norm close to one, local strong convexity of Q in the fibers gives a positive uniform margin in finitely many frames, while the real fiber Hessian of H is uniformly bounded there. Choose τ so small that the possible negative part of τ times this Hessian consumes less than half the margin. Homogeneity then gives local strong convexity of Q_τ in the fibers everywhere off zero. Apply Theorem 5.1.2 to Q_τ and take one of its smooth regularizations Q_j; then $\sqrt{Q_j}$ is the required smooth complex norm. □

Corollary 5.1.1 *For every holomorphic structure in the neighborhood of Theorem 5.1.1, there is no positive continuous complex homogeneous function of degree two that is both strictly plurisubharmonic off zero and locally strongly convex in the fibers. If that structure admits a function H as in Proposition 5.1.1, there is also no positive continuous complex homogeneous function of degree two that is plurisubharmonic off zero and locally strongly convex in the fibers.*

Proof For the first assertion, Theorem 5.1.2 would produce a smooth complex norm with plurisubharmonic square, contrary to Theorem 5.1.1. For the second, use Proposition 5.1.1 and the same obstruction. □

The second assertion requires the strictly plurisubharmonic homogeneous function H, which need not exist for an arbitrary holomorphic bundle; Section 5.1.2 shows that it exists when the dual bundle is ample. Without a strong convexity margin in the fibers, the regularization gives no obstruction for continuous fiber norms.

5.1.2 Strict positivity from the tautological metric

The second part of Corollary 5.1.1 requires a smooth strictly plurisubharmonic function on $F^\times$ that is complex homogeneous of degree two. Such a function is the squared norm of a vector, viewed as a point of the tautological line it spans, for a suitable metric on the tautological line bundle. Its Levi form is strictly positive exactly when the dual tautological metric is positively curved, so the function exists when F^* is ample.

Let $F \to X$ be a holomorphic vector bundle of positive rank. In this subsection, $\mathbb{P}(F)$ parametrizes complex lines in F, $\mathcal{O}_{\mathbb{P}(F)}(-1)$ is the tautological line bundle whose fiber over a line is that line, and

$$L = \mathcal{O}_{\mathbb{P}(F)}(1) = \mathcal{O}_{\mathbb{P}(F)}(-1)^*.$$

The map $v \mapsto ([v], v)$ identifies $F^\times$ holomorphically with the complement of the zero section in $\mathcal{O}_{\mathbb{P}(F)}(-1)$. We use the curvature convention $R_h = -\mathrm{dd}^{\mathrm{c}} \log h$ of Section 5.1.

Lemma 5.1.6 *A smooth Hermitian metric h on L determines a positive smooth complex homogeneous function H of degree two on $F^\times$: the squared norm of v in the dual metric on the tautological line $\mathcal{O}_{\mathbb{P}(F)}(-1)_{[v]}$. The function H is strictly plurisubharmonic if and only if R_h is strictly positive on $\mathbb{P}(F)$.*

Proof Take local holomorphic coordinates y on $\mathbb{P}(F)$ and a holomorphic frame e of the tautological line bundle, and write the metric of the dual frame as $h = \mathrm{e}^{-\varphi}$. The dual metric gives

$$H(y, \lambda e) = |\lambda|^2 \mathrm{e}^{\varphi(y)}, \qquad R_h = \mathrm{dd}^{\mathrm{c}} \varphi.$$

Let $\mathcal{L}$ denote the complex Hessian, without the factor $\mathrm{i}/(2\pi)$ of the curvature normalization. Differentiating at a point with $\lambda \neq 0$ gives

$$\mathcal{L}H(u,t) = H\left(\mathcal{L}\varphi(u) + \left|\frac{t}{\lambda} + \partial\varphi(u)\right|^2\right), \qquad (u,t) \in T^{1,0}_y\mathbb{P}(F) \oplus \mathbb{C}.$$

If $\mathcal{L}\varphi$ is positive definite, the right-hand side is positive for every nonzero (u,t): the first term detects $u \neq 0$, and the second detects $t \neq 0$ when $u = 0$. Conversely, for $u \neq 0$, set $t = -\lambda\partial\varphi(u)$; strict positivity of $\mathcal{L}H$ then forces $\mathcal{L}\varphi(u) > 0$. This is exactly positivity of R_h. The squared tautological norm is independent of the chosen frame and is complex homogeneous, so the local functions define the asserted global function H. □

A line bundle is very ample when its complete linear system defines a holomorphic embedding, and ample when some positive tensor power is very ample. With the convention above, a vector bundle F^* is called ample when L is ample.

Proposition 5.1.2 *If F^* is ample, there is a positive complex homogeneous function H of degree two on $F^\times$ that is smooth and strictly plurisubharmonic there and, after extension by zero, continuous on F.*

Proof Choose $m \geq 1$ such that L^m is very ample. Its complete linear system gives an embedding $\iota\colon \mathbb{P}(F) \to \mathbb{P}^N$ with $L^m = \iota^* O_{\mathbb{P}^N}(1)$. Pull back the Fubini–Study metric to a metric k on L^m, and take its positive mth root h on L: in a local frame $h = k^{1/m}$, and the transformation law for L^m makes these local roots a metric on L. Its curvature is

$$R_h = \frac{1}{m}\iota^*\omega_{\mathrm{FS}} > 0,$$

where ω_{FS} is the curvature form of the Fubini–Study metric on $O_{\mathbb{P}^N}(1)$; the pullback is strictly positive because the differential of the embedding is injective. Apply Lemma 5.1.6. Since the base is compact, the resulting H is bounded on the unit sphere bundle of a fixed Hermitian metric. Homogeneity then gives $0 < H(v) \leq C|v|^2$ off zero, which proves continuity of the extension by zero. □

Corollary 5.1.2 *Let F carry a holomorphic structure in the neighborhood of Theorem 5.1.1. If F^* is ample, there is no positive continuous complex homogeneous function of degree two on F that is plurisubharmonic off zero and locally strongly convex in the fibers.*

Proof By Proposition 5.1.2, ampleness of F^* gives the function H required in Proposition 5.1.1, so the second assertion of Corollary 5.1.1 applies. □

The function H is not the square of a Hermitian norm in general, since the metric on the tautological line may vary with the line. It provides the strict plurisubharmonicity on the total space needed for regularization, whereas local strong convexity in the fibers remains a separate hypothesis in the continuous obstruction.

5.1.3 Polar norms of semipositive Finsler metrics

For a holomorphic vector bundle that is not assumed to carry a positively curved Hermitian metric, positivity can be measured by Finsler metrics and their Kobayashi curvature. Polar duality converts a Finsler metric on E with semipositive Kobayashi curvature into a continuous norm on E^* whose square is plurisubharmonic and uniformly convex in the fibers. With the regularization of Section 5.1.1, Theorem 5.1.1 therefore excludes such Finsler metrics on an ample bundle whose dual is close to a split bundle with positive Gauduchon degrees.

Let $E \to X$ be a holomorphic vector bundle of positive rank; in this subsection $n \geq 1$. Write $E^\times$ for the complement of its zero section. A *squared Finsler metric* on E is a positive smooth function G on $E^\times$ with

$$G(x, tv) = |t|^2 G(x, v), \qquad t \in \mathbb{C} \setminus \{0\},$$

extended continuously by zero to the zero section. It is *strongly pseudoconvex* if its vertical complex Hessian $(G_{i\bar{j}})$ is positive definite; this does not require convexity in the real fiber variables.

In local holomorphic base and fiber coordinates (z, v), with Greek indices for base coordinates, Latin indices for fiber coordinates and summation over repeated indices, put

$$S_{\alpha\bar{\beta}} = G_{\alpha\bar{\beta}} - G_{\alpha\bar{j}} G^{\bar{j}i} G_{i\bar{\beta}},$$
$$\mathcal{K}_G = -\frac{\mathrm{i}}{2\pi G} S_{\alpha\bar{\beta}}\, \mathrm{d}z^\alpha \wedge \mathrm{d}\bar{z}^\beta,$$

where $(G^{\bar{j}i})$ is the inverse of the vertical Hessian. We use this convention for the *Kobayashi curvature* $\mathcal{K}_G$, a horizontal $(1,1)$-form at each point of $E^\times$. Thus $\mathcal{K}_G \geq 0$ means $S \leq 0$, and $\mathcal{K}_G > 0$ means that S is negative definite.

Lemma 5.1.7 *Let G be a strongly pseudoconvex squared Finsler metric. If $\mathcal{K}_G > 0$, then for every $v_0 \in E^\times_{x_0}$ there is a local holomorphic section σ with $\sigma(x_0) = v_0$ and*

$$G(z, \sigma(z)) \leq G(x_0, v_0)$$

near x_0. Moreover, without any curvature assumption, on a coordinate ball, for $\varphi = |z|^2$ and $\varepsilon > 0$, the metric $G_\varepsilon = \mathrm{e}^{-\varepsilon\varphi} G$ is strongly pseudoconvex and satisfies

$$\mathcal{K}_{G_\varepsilon} = \mathcal{K}_G + \varepsilon \mathrm{d}\mathrm{d}^{\mathrm{c}} \varphi.$$

Proof Since the vertical Hessian is positive, the Levi-orthogonal complement $H_{G,v}$ of the vertical tangent space at v projects isomorphically onto the base tangent space. The Levi form of G on $H_{G,v}$ is the Schur complement S, which is negative definite when $\mathcal{K}_G > 0$. For the vertical radial vector $R = v_i \partial_{v_i}$, complex homogeneity gives

$$\partial G(R) = G, \qquad \mathcal{L}_G(a, \bar{R}) = \partial G(a)$$

for every tangent vector a, where $\mathcal{L}_G$ is the Levi form. Since R is vertical, the second identity implies $\partial G|_{H_{G,v}} = 0$.

Choose a holomorphic section σ_0 through v_0 whose tangent space at x_0 is H_{G,v_0}. In coordinates centered at x_0, the function $G \circ \sigma_0$ has zero first derivative and Taylor expansion

$$G(x_0, v_0) + P(z) + \overline{P(z)} + S_{\alpha\bar{\beta}} z^\alpha \bar{z}^\beta + O(|z|^3),$$

where P is a homogeneous holomorphic quadratic polynomial. The identity $\partial_v G(v_0)(v_0) = G(v_0) > 0$ gives a vertical vector b with $\partial_v G(v_0)(b) = 1$. Set $\sigma = \sigma_0 - bP$ in the local frame. This changes only the pure quadratic terms and cancels $P + \bar{P}$. The remaining quadratic form is negative definite, and therefore dominates the cubic remainder on a sufficiently small ball. The section stays nonzero there and satisfies the asserted inequality.

For the last assertion, differentiate $\mathrm{e}^{-\varepsilon\varphi} G$ in the Schur complement. Euler's identities give $G_{\bar{j}} = v_i G_{i\bar{j}}$ and $G_\alpha = G_{\alpha\bar{j}} \bar{v}_j$. Hence the terms containing $\partial\varphi$, including its square, cancel between the base Hessian and the mixed block product, leaving

$$S(G_\varepsilon) = \mathrm{e}^{-\varepsilon\varphi}\big(S(G) - \varepsilon G(\varphi_{\alpha\bar{\beta}})\big).$$

Dividing by G_ε gives the curvature formula. The vertical Hessian of G_ε is the positive multiple $\mathrm{e}^{-\varepsilon\varphi}(G_{i\bar{j}})$, so G_ε is again strongly pseudoconvex. □

Proposition 5.1.3 *Let G be a strongly pseudoconvex squared Finsler metric with $\mathcal{K}_G \geq 0$. Its* polar square

$$Q(x, \xi) = \max_{[v] \in \mathbb{P}(E_x)} \frac{|\xi(v)|^2}{G(x, v)}, \qquad \xi \in E_x^*,$$

is continuous and plurisubharmonic on the total space of E^*, positive off the zero section and complex homogeneous of degree two. The function $\log Q$ *is plurisubharmonic off the zero section, and $\sqrt{Q}$ is a complex norm on each fiber.*

Proof The ratio is invariant under complex rescaling of v, so the maximum can be taken over the compact unit sphere in a local frame, which proves continuity. Positive upper and lower bounds for G on that sphere compare Q with a Hermitian squared dual norm; this gives positivity off zero and continuity at zero, and complex homogeneity is clear. Moreover,

$$\sqrt{Q(x, \xi)} = \sup_{v \neq 0} \frac{|\xi(v)|}{\sqrt{G(x, v)}}.$$

Each term satisfies the triangle inequality in ξ, and so does their supremum.

On a coordinate ball, set $G_\varepsilon = \mathrm{e}^{-\varepsilon|z|^2} G$. By Lemma 5.1.7, its Kobayashi curvature is strictly positive, and its polar square is $Q_\varepsilon = \mathrm{e}^{\varepsilon|z|^2} Q$. At a point (x_0, ξ_0) with $\xi_0 \neq 0$, choose a maximizing line and a vector v_0 on it. The lemma, applied to G_ε, gives a holomorphic section σ such that

$$\log Q_\varepsilon(z,\xi) \geq \log|\xi(\sigma(z))|^2 - \log G_\varepsilon(x_0,v_0),$$

with equality at (x_0,ξ_0). The pairing $\xi(\sigma(z))$ is nonzero near that point, so the right-hand side is pluriharmonic there. Restriction to sufficiently small holomorphic disks centered at the point therefore gives the submean inequality for $\log Q_\varepsilon$. This argument applies at every point off the zero section, and together with continuity it shows that $\log Q_\varepsilon$ is plurisubharmonic there.

As $\varepsilon \downarrow 0$, the functions $\log Q_\varepsilon = \log Q + \varepsilon|z|^2$ converge locally uniformly off the zero section. Passing to the limit in the submean inequalities on disks proves that $\log Q$ is plurisubharmonic off the zero section. Since the exponential is increasing and convex, Jensen's inequality shows that $Q = \exp(\log Q)$ is plurisubharmonic there. At a point of the zero section, $Q = 0$ and $Q \geq 0$, so every centered disk satisfies the submean inequality there. Together with continuity and the submean inequalities at nonzero points, this proves plurisubharmonicity on the whole total space of E^*. □

Lemma 5.1.8 *Let k be a smooth Hermitian metric on E, and let G be a positive smooth complex homogeneous function of degree two on $E^\times$, extended by zero. Then its polar square Q, defined as in Proposition 5.1.3, has a constant $c > 0$ such that*

$$Q(x,\xi) - c|\xi|^2_{k^*}$$

is real convex on every dual fiber. Neither strong pseudoconvexity nor real convexity of G is required here.

Proof The real fiber Hessian of G is bounded above by $2L$ times k on the compact unit sphere bundle of k, for some $L > 0$, and homogeneity extends this bound to every nonzero vector. Also, G is C^1 at zero with derivative zero, since its value is $O(|v|^2)$ and its first fiber derivative is $O(|v|)$, uniformly over the base. Integrating the Hessian estimate along a segment gives

$$G(v+w) \leq G(v) + \mathrm{d}_v G(w) + L|w|^2_k; \tag{5.1}$$

a segment passing through zero is handled by splitting it there and using the continuity of the first derivative.

Complex homogeneity and completing the square on each complex line give the identity

$$Q(x,\xi) = \max_{v\in E_x}\{2\operatorname{Re}\xi(v) - G(x,v)\}. \tag{5.2}$$

The maximum exists because $G(v) \geq a|v|^2_k$ for a uniform constant $a > 0$. Fix a maximizer v_0 for ξ_0. Differentiation, including the case $\xi_0 = v_0 = 0$, gives $\mathrm{d}G(v_0) = 2\operatorname{Re}\xi_0$. Evaluate the expression in (5.2) for $\xi_0 + \zeta$ at $v_0 + w$ and apply (5.1). The terms involving $\xi_0(w)$ cancel, and we obtain

$$Q(\xi_0+\zeta) \geq Q(\xi_0) + 2\operatorname{Re}\zeta(v_0) + 2\operatorname{Re}\zeta(w) - L|w|^2_k.$$

Taking the supremum over w yields

$$Q(\xi_0 + \zeta) \geq Q(\xi_0) + 2\,\mathrm{Re}\,\zeta(v_0) + L^{-1}|\zeta|^2_{k^*}.$$

Set $c = L^{-1}$ and $\Phi = Q - c|\cdot|^2_{k^*}$. After subtracting the square, this inequality gives a real affine function on the whole dual fiber that lies below Φ and equals Φ at ξ_0. Hence Φ is the supremum of such affine functions, and therefore convex. Neither uniqueness of the maximizer nor real convexity of G has been used. □

Corollary 5.1.3 *If E is ample and has a strongly pseudoconvex squared Finsler metric with semipositive Kobayashi curvature, then E^* admits a smooth complex norm whose square is strictly plurisubharmonic off zero and locally strongly convex in the real fibers.*

Proof By Proposition 5.1.3, the polar square is a positive continuous plurisubharmonic complex homogeneous function of degree two, and by Lemma 5.1.8 it is locally strongly convex in the fibers, since in a local frame the quadratic form $|\cdot|^2_{k^*}$ dominates a positive multiple of the Euclidean one over compact sets. Applying Proposition 5.1.2 to $F = E^*$, whose dual $F^* = E$ is ample, gives the function H on E^* required in Proposition 5.1.1, and that proposition gives the asserted smooth norm.□

Corollary 5.1.4 *Let F be the underlying smooth bundle of F_0 in Theorem 5.1.1, with a holomorphic structure in the neighborhood given there. If F^* is ample, then F^* admits no strongly pseudoconvex squared Finsler metric with semipositive Kobayashi curvature.*

Proof Apply Corollary 5.1.3 to $E = F^*$. Such a metric would produce a smooth complex norm on F with plurisubharmonic square, contrary to Theorem 5.1.1. □

The corollary applies only near a split bundle whose line summands have positive Gauduchon degrees, and it is not vacuous only if such a neighborhood contains holomorphic structures with ample dual. On Hirzebruch surfaces, Theorem 5.1.3 gives stable ample bundles whose duals lie in this neighborhood. Since the polar square need not be smooth, the regularization of Section 5.1.1 is an essential step of the argument.

5.1.4 Ample comparison quotients on Hirzebruch surfaces

To apply Corollary 5.1.4, we need ample vector bundles whose duals are close to a split bundle $F_1 \oplus F_2$ with $\deg_{\Theta_s} F_s > 0$ for suitable test forms Θ_s. The split bundle $F_1^* \oplus F_2^*$ is not ample itself: otherwise its quotient line bundles F_s^* would be ample, hence carry positively curved metrics, and each F_s would have negative degree for every test form. The ample bundles must therefore be obtained by deformation. On a Hirzebruch surface we choose a split bundle $E_0 = L_1 \oplus L_2$ whose dual summands have positive degree for suitable Kähler forms, and we construct an ample bundle E_{amp} with the same Chern classes as a quotient of a trivial bundle. Both bundles

belong to the family of bundles with these Chern classes that are prioritary for the ruling. This family is irreducible, which will allow us to find ample bundles in every neighborhood of E_0.

Let $e \geq 0$ be an integer, and let

$$X = \mathbb{F}_e = \mathbb{P}_{\mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}(e)), \qquad \pi\colon X \longrightarrow \mathbb{P}^1,$$

be the Hirzebruch surface, where the projective bundle parametrizes lines. Write S for the section defined by the subbundle $\mathcal{O}(e)$ and f for a fiber of π, called a ruling fiber; we identify line bundles with divisor classes and write $\mathcal{O}_X(aS + bf)$. The normal bundle of S is $\mathcal{O}(-e)$, so

$$S^2 = -e, \qquad S \cdot f = 1, \qquad f^2 = 0.$$

The dual of the tautological line bundle is $\mathcal{O}_X(S)$. A line bundle of degree zero on the ruling fibers is pulled back from $\mathbb{P}^1$: its direct image has rank one, and the evaluation map from the pullback of the direct image is an isomorphism on each fiber. Consequently $\mathrm{Pic}(X) = \mathbb{Z}S \oplus \mathbb{Z}f$.

Lemma 5.1.9 *For integers $a \geq 0$ and b,*

$$\pi_*\mathcal{O}_X(aS + bf) = \bigoplus_{i=0}^{a} \mathcal{O}_{\mathbb{P}^1}(b - ie), \qquad R^1\pi_*\mathcal{O}_X(aS + bf) = 0.$$

A class $aS + bf$ is effective exactly when $a, b \geq 0$. If $a \geq 1$ and $b > ae$, its line bundle is very ample. In particular,

$$H = S + (3e + 1)f, \qquad H^2 = 5e + 2, \qquad h^0(X, \mathcal{O}_X(H)) = 5e + 4.$$

The canonical class is $K_X = -2S - (e + 2)f$.

Proof Over a trivializing chart of the base, sections of $\mathcal{O}_X(aS)$ are homogeneous polynomials of degree a in the fiber coordinates. Their transition law gives

$$\pi_*\mathcal{O}_X(aS) = \mathrm{Sym}^a(\mathcal{O} \oplus \mathcal{O}(-e)),$$

and twisting by the base gives the formula. The first cohomology of $\mathcal{O}_{\mathbb{P}^1}(a)$ on each fiber vanishes, which gives the assertion on the higher direct image. If $a < 0$, restriction to a general ruling fiber excludes a nonzero section. For $a \geq 0$, the direct-image formula gives a nonzero section exactly when $b \geq 0$.

Suppose that $a \geq 1$ and $b > ae$. All line bundles in the direct image then have degree at least one. Fiber polynomials of degree a separate points and tangent directions within a fiber. To separate points over different base points, choose a fiber monomial that is nonzero at the first point and a coefficient section vanishing at the second base point but not at the first. At any point, choose such a nonzero monomial and a coefficient section with a simple zero at its base point; the resulting global section vanishes there, has nonzero derivative in a prescribed base direction and has

zero vertical derivative. The complete linear system therefore separates points and first jets, so it defines an embedding. The formulas for H^2 and $h^0(H)$ follow from the intersection numbers and from the direct image $\mathcal{O}(3e+1) \oplus \mathcal{O}(2e+1)$. Finally, the relative Euler sequence gives $K_{X/\mathbb{P}^1} = -2S - ef$, and adding the pullback of the canonical class of $\mathbb{P}^1$ proves the last identity. □

Put

$$L_1 = \mathcal{O}_X(5S - f), \qquad L_2 = \mathcal{O}_X(-S + (12e+5)f), \qquad E_0 = L_1 \oplus L_2.$$

Their Chern classes satisfy

$$c_1(E_0) = 4H, \qquad c_2(E_0) = 65e + 26 = 13H^2.$$

An ample bundle with these Chern classes can be constructed as a quotient of a trivial bundle. The main point of the proof is to make the fibers of the associated map to projective space finite, which requires excluding common curves of the defining sections uniformly over the projective parameter.

Proposition 5.1.4 *There exist linearly independent sections $a_0, \dots, a_3$ of $\mathcal{O}_X(H)$ with no common zero. For any such quadruple and a general quadruple $b_0, \dots, b_3$ of sections of $\mathcal{O}_X(3H)$, the sequence*

$$0 \longrightarrow \mathcal{O}_X(-H) \oplus \mathcal{O}_X(-3H) \xrightarrow{(a,b)} \mathcal{O}_X^{\oplus 4} \longrightarrow E_{\mathrm{amp}} \longrightarrow 0$$

has a locally free ample cokernel E_{amp} of rank two, with $c_1(E_{\mathrm{amp}}) = 4H$ and $c_2(E_{\mathrm{amp}}) = 13H^2$. Here general *means outside a proper Zariski-closed subset of the space of quadruples.*

Proof *Step 1: the quadruple a.* Since $\mathcal{O}_X(H)$ is globally generated, a common zero at a fixed point imposes four independent linear conditions on a quadruple a. The incidence variety of pairs consisting of a point of the surface and a quadruple vanishing there therefore has dimension two less than that of the space of quadruples, and its image in that space is closed by properness. Hence quadruples with a common zero form a proper closed subset. Linear independence is also a nonempty open condition, because $h^0(H) \geq 4$, and the two open conditions intersect. Fix such an a and write

$$V = H^0(X, \mathcal{O}_X(3H))^{\oplus 4}, \qquad N = \dim V.$$

Step 2: local freeness. At a fixed point x, the map (a, b) drops rank exactly when $b(x)$ is proportional to the nonzero vector $a(x)$. Since evaluation on V is surjective, this is a condition of codimension three. Its incidence over X has dimension at most $N - 1$, and its image in V is closed. Outside that image the cokernel is locally free of rank two.

Step 3: the map to projective space. Dualizing the sequence identifies E_{amp}^* with a subbundle of $(\mathbb{C}^4)^* \otimes \mathcal{O}_X$. This inclusion defines a proper morphism

$$q_b \colon \mathbb{P}(E_{\mathrm{amp}}^*) \longrightarrow \mathbb{P}((\mathbb{C}^4)^*), \qquad \mathcal{O}_{\mathbb{P}(E_{\mathrm{amp}}^*)}(1) = q_b^* \mathcal{O}_{\mathbb{P}^3}(1).$$

For $w = [w_0 : \cdots : w_3]$, its fiber is the common zero scheme of

$$s_w = \sum_i w_i a_i \in H^0(X, O_X(H)), \qquad t_w = \sum_i w_i b_i \in H^0(X, O_X(3H)).$$

Linear independence of a gives $s_w \neq 0$ for every w.

Step 4: no common curves. A nonzero effective subdivisor D of $\operatorname{div}(s_w)$ has class $uS + vf$, where

$$u \in \{0, 1\}, \qquad 0 \leq v \leq 3e + 1, \qquad (u, v) \neq (0, 0).$$

Write $|D| = \mathbb{P}H^0(X, O_X(D))$ for the complete linear system. For each of these finitely many classes D_0, let T_{D_0} parametrize the triples $(w, [\sigma], [\tau])$ with

$$[\sigma] \in |D_0|, \qquad [\tau] \in |H - D_0|, \qquad [s_w] = [\sigma\tau].$$

Since products of nonzero sections are nonzero, this is a closed projective parameter scheme. Its fiber over w consists of the finitely many effective subdivisors of the fixed divisor $\operatorname{div}(s_w)$, counted with their possible multiplicities. Thus $\dim T_{D_0} \leq 3$.

Vanishing on a nonzero effective divisor D imposes at least four independent conditions on $H^0(X, O_X(3H))$. Indeed, choose an integral curve C in its support. Very ampleness gives sections r_0, r_1 of $O_X(H)$ with nonconstant ratio on C. The four cubic monomials in their restrictions are linearly independent, since a linear relation would make the nonconstant ratio algebraic over $\mathbb{C}$, hence constant.

For fixed $(w, [\sigma], [\tau])$, the map $V \to H^0(X, O_X(3H))$, $b \mapsto t_w$, is surjective. The condition that t_w vanish on $D = \operatorname{div}(\sigma)$ therefore has codimension at least four in V. This condition is closed in families: the universal multiplication map

$$H^0(X, O_X(3H - D_0)) \otimes O_{|D_0|}(-1) \longrightarrow H^0(X, O_X(3H)) \otimes O_{|D_0|}$$

is injective on fibers, so its image is a subbundle, and the condition is the vanishing of the image of t_w in the quotient bundle. Its incidence over T_{D_0} has dimension at most $N - 1$, and its image in V is closed.

Step 5: conclusion. Choose b outside these finitely many proper closed images and outside the image of Step 2. Then s_w and t_w have no common curve for any w, so every fiber of q_b is finite. A proper morphism with finite fibers is finite, and the pullback of $O_{\mathbb{P}^3}(1)$ by a finite morphism is ample; hence E_{amp} is ample. Finally, the Whitney formula gives

$$c(E_{\mathrm{amp}}) = \frac{1}{(1 - H)(1 - 3H)} = 1 + 4H + 13H^2.$$

A vector bundle E on X is called *prioritary for the ruling* if $\operatorname{Ext}^2_X(E, E(-f)) = 0$. Since all ruling fibers are linearly equivalent, this condition does not depend on the chosen fiber, and by Serre duality it is equivalent to

$$\operatorname{Hom}_X(E, E(K_X + f)) = 0.$$

Prioritary bundles of fixed rank $r \geq 2$ and fixed Chern classes form an irreducible family, by the theorem of Walter recalled in Section 5.1.6.

Lemma 5.1.10 *Both E_0 and E_{amp} are prioritary for the ruling.*

Proof Write $D = K_X + f = -2S - (e+1)f$. The summands of $\mathrm{End}(E_0)(D)$ have classes

$$D, \quad D, \quad 4S - (13e+7)f, \quad -8S + (11e+5)f.$$

Each has a negative coefficient in the basis S, f, so by Lemma 5.1.9 none of them has a nonzero section. This proves the required vanishing of Hom for E_0.

Precomposition with the surjection $\mathcal{O}_X^4 \to E_{\mathrm{amp}}$ gives an injection

$$\mathrm{Hom}(E_{\mathrm{amp}}, E_{\mathrm{amp}}(D)) \hookrightarrow H^0(X, E_{\mathrm{amp}}(D))^4.$$

Twist the defining sequence by D. Its middle term has no sections. For $j = 1, 3$, Serre duality and the formula of Lemma 5.1.9 give

$$H^1(X, \mathcal{O}_X(D - jH)) = H^1(X, \mathcal{O}_X(jH - f))^*,$$
$$\pi_* \mathcal{O}_X(jH - f) = \bigoplus_{i=0}^{j} \mathcal{O}_{\mathbb{P}^1}(j(3e+1) - 1 - ie).$$

The higher direct image vanishes, and every degree in this sum is at least $j(2e+1) - 1 \geq 0$. Hence all these first cohomology groups vanish, and the long exact sequence of the twisted sequence gives $H^0(X, E_{\mathrm{amp}}(D)) = 0$. This proves the vanishing of Hom for E_{amp}. □

Proposition 5.1.5 *For $F_s = L_s^*$, $s = 1, 2$, there are Kähler forms η_s with $\deg_{\eta_s} F_s = 1$. Consequently, the split bundle $E_0^* = F_1 \oplus F_2$ satisfies the hypotheses of Theorem 5.1.1.*

Proof The integral classes

$$A_1 = 6S + (6e+1)f, \qquad A_2 = S + (13e+6)f$$

are very ample by Lemma 5.1.9. Pullbacks of normalized Fubini–Study forms under the corresponding embeddings give Kähler forms η_s representing $c_1(\mathcal{O}_X(A_s))$. A direct computation of intersection numbers gives

$$c_1(L_1) \cdot A_1 = -1, \qquad c_1(L_2) \cdot A_2 = -1,$$

so the degrees of the duals are equal to one. Since $n = 2$, the closed positive $(1,1)$-forms η_s are admissible test forms Θ_s in Theorem 5.1.1. □

5.1.5 Stable extensions approaching the split comparison bundle

The bundles near E_0 that we construct will also be slope-stable for a fixed polarization. The split bundle E_0 is unstable, and the ample bundle E_{amp} need not be stable. Nonsplit extensions of L_2 by L_1 have the same Chern classes as E_0, are stable, and converge to E_0 as the extension class tends to zero; they are not ample, however. The next subsection combines them with E_{amp}.

Keep the Hirzebruch surface $X = \mathbb{F}_e$, its ruling and the line bundles L_1, L_2 and $E_0 = L_1 \oplus L_2$ of Section 5.1.4. We use the very ample polarization $P = 4H - f = 4S + (12e + 3)f$. For a torsion-free coherent sheaf E of positive rank on X, put

$$\deg_P E = c_1(E) \cdot P, \qquad \mu_P(E) = \frac{\deg_P E}{\operatorname{rank} E}.$$

A vector bundle V is *μ_P-stable* if $\mu_P(M) < \mu_P(V)$ for every coherent subsheaf $M \subset V$ with $0 < \operatorname{rank} M < \operatorname{rank} V$. Since $\mu_P(L_2) > \mu_P(E_0)$ by the computation below, the split bundle E_0 is not μ_P-stable.

Proposition 5.1.6 *The extension space has dimension*

$$\dim \operatorname{Ext}^1_X(L_2, L_1) = 105e + 35.$$

Every nonsplit extension

$$0 \longrightarrow L_1 \longrightarrow V \xrightarrow{q} L_2 \longrightarrow 0$$

is a locally free, μ_P-stable bundle with $c_1(V) = 4H$ and $c_2(V) = 13H^2$.

Proof Extensions of line bundles are locally free: the quotient is free on a small trivializing neighborhood, so the sequence splits there. Their classes form

$$\operatorname{Ext}^1_X(L_2, L_1) = H^1(X, \mathcal{O}_X(6S - (12e + 6)f)).$$

By Lemma 5.1.9, the higher direct image of this line bundle vanishes and its direct image is the sum of the seven line bundles $\mathcal{O}_{\mathbb{P}^1}(-12e - 6 - ie)$, $0 \le i \le 6$. Their first cohomology groups have total dimension

$$\sum_{i=0}^{6} (12e + 5 + ie) = 105e + 35 > 0.$$

The Chern classes follow from the extension sequence. Direct computations of intersection numbers give

$$\mu_P(L_1) = 40e + 11, \qquad \mu_P(V) = 40e + 14, \qquad \mu_P(L_2) = 40e + 17.$$

It suffices to test line-bundle subsheaves $M \hookrightarrow V$. Indeed, a rank-one torsion-free subsheaf on a smooth surface has a line bundle as its double dual, and its inclusion

extends across the finitely many missing points by Hartogs extension in local bundle frames. The extended map remains injective and has the same first Chern class. This argument does not require the quotient to be locally free.

If the composite $M \to L_2$ is zero, then M maps nontrivially into L_1, so $\deg_P M \leq \mu_P(L_1) < \mu_P(V)$. Otherwise its zero divisor D is effective and $M = L_2(-D)$. If $D = 0$, the composite is an isomorphism and its inverse splits q, contrary to the hypothesis. Hence $D \neq 0$, and

$$\deg_P M = \mu_P(V) + 3 - D \cdot P.$$

Every effective class is $aS + bf$ with $a, b \geq 0$, and

$$S \cdot P = 8e + 3, \qquad f \cdot P = 4.$$

For $e \geq 1$, every nonzero effective divisor has P-degree at least four, which proves the strict slope inequality.

For $e = 0$, the only nonzero effective class of P-degree at most three is S. A lift of multiplication by a nonzero section $s \in H^0(X, \mathcal{O}_X(S))$ from $L_2(-S)$ to V would make the pullback of the nonzero extension class vanish. That pullback is the image of the class under the map

$$H^1(X, \mathcal{O}_X(6S - 6f)) \xrightarrow{\cdot s} H^1(X, \mathcal{O}_X(7S - 6f)).$$

On $X = \mathbb{P}^1 \times \mathbb{P}^1$, order the factors so that $\mathcal{O}_X(S) = \mathcal{O}(1, 0)$ and $\mathcal{O}_X(f) = \mathcal{O}(0, 1)$. The Künneth formula identifies this map with

$$\left[H^0(\mathbb{P}^1, \mathcal{O}(6)) \xrightarrow{\cdot s} H^0(\mathbb{P}^1, \mathcal{O}(7))\right] \otimes \mathrm{Id}_{H^1(\mathbb{P}^1, \mathcal{O}(-6))}.$$

Multiplication by a nonzero section is injective on global sections, and tensoring with a vector space preserves injectivity. Thus the nonzero extension class cannot pull back to zero. This excludes the possible equality of slopes and proves stability also for $e = 0$. □

Proposition 5.1.7 *Every extension of L_2 by L_1, including the split extension, is prioritary for the ruling and is not ample. Its restriction to every ruling fiber is $\mathcal{O}_{\mathbb{P}^1}(5) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$.*

Proof Put $D = K_X + f$. Since E_0 is prioritary by Lemma 5.1.10, the group $\mathrm{Hom}_X(E_0, E_0(D)) = \bigoplus_{i,j} H^0(X, L_i^* \otimes L_j(D))$ vanishes. Applying $\mathrm{Hom}_X(L_i, \cdot)$ to the extension twisted by D gives $\mathrm{Hom}_X(L_i, V(D)) = 0$ for $i = 1, 2$. Applying $\mathrm{Hom}_X(\cdot, V(D))$ to the original extension then gives $\mathrm{Hom}_X(V, V(D)) = 0$. By Serre duality this is the prioritary condition $\mathrm{Ext}^2_X(V, V(-f)) = 0$.

Restriction to a ruling fiber is exact, because the quotient is locally free. The restricted line bundles have degrees five and minus one, and the restricted extension splits, since $\mathrm{Ext}^1_{\mathbb{P}^1}(\mathcal{O}(-1), \mathcal{O}(5)) = H^1(\mathbb{P}^1, \mathcal{O}(6)) = 0$. The quotient of negative degree on this curve excludes ampleness. □

Proposition 5.1.8 *On the smooth bundle underlying E_0 there is a holomorphic one-parameter family of Dolbeault operators $\bar\partial_t = \bar\partial_0 + tA$, $t \in \mathbb{C}$, such that for $t \neq 0$ the holomorphic bundle defined by $\bar\partial_t$ is μ_P-stable and prioritary, but not ample. As $t \to 0$, these operators converge to $\bar\partial_0$ in the C^m norm for every integer $m \geq 0$.*

Proof Choose a nonzero extension class and a smooth Dolbeault representative $\eta \in \Omega^{0,1}(X, L_2^* \otimes L_1)$ of it, so that $\bar\partial\eta = 0$. On the smooth direct sum, set

$$A = \begin{pmatrix} 0 & \eta \\ 0 & 0 \end{pmatrix}.$$

The covariant Dolbeault derivative of A is zero and $A \wedge A = 0$, so $\bar\partial_t^2 = 0$. Adding the Dolbeault operator of the base $\mathbb{C}$ of the parameter t also gives an integrable operator on $X \times \mathbb{C}$, because the family depends holomorphically on t. This defines a holomorphic family of bundles whose extension class is $t[\eta]$. It is nonsplit for $t \neq 0$, so the two preceding propositions apply. For a fixed smooth background connection and fixed metrics,

$$\|\bar\partial_t - \bar\partial_0\|_{C^m} = |t|\,\|A\|_{C^m} \longrightarrow 0,$$

and the norm $\|A\|_{C^m}$ is finite for every m because X is compact. □

Thus stability and ampleness have been realized separately in the prioritary family of E_0: the nonsplit extensions are stable but not ample, while E_{amp} is ample. The deformation argument of Theorem 5.1.3 joins the stable locus to the ample one and provides stable ample bundles arbitrarily close to E_0.

5.1.6 Deforming to stable ample bundles

We keep the Hirzebruch surface $X = \mathbb{F}_e$ and the data H, P, L_1, L_2, E_0 of Sections 5.1.4 and 5.1.5. The ample quotient E_{amp} and a nonsplit extension are two prioritary bundles with the same Chern classes as E_0; the former is ample and the latter is stable. Prioritary bundles with these invariants form an irreducible family in which ampleness and stability are open conditions, so bundles with both properties are dense in the family, and in particular they accumulate at E_0. We first prove the required openness assertions for algebraic families.

Throughout this subsection, T denotes a smooth complex algebraic variety. A *flat family* is a coherent sheaf $\mathcal{E}$ on $X \times T$ that is flat over T, and its fiber $\mathcal{E}_t$ at $t \in T$ is its restriction to $X \times \{t\}$.

Lemma 5.1.11 *If a flat family $\mathcal{E}$ has a locally free fiber at $t_0 \in T$, then $\mathcal{E}$ is a vector bundle on $X \times T'$ for some Zariski-open neighborhood T' of t_0.*

Proof At a point of $X \times \{t_0\}$, lift a local basis of the fiber to $\mathcal{E}$. By Nakayama's lemma, the resulting map from a free module of finite rank is surjective near that point. Tensor its kernel sequence with the residue field of t_0. Flatness over T makes

the first Tor term zero, so the kernel restricts to the kernel of the chosen isomorphism on the fiber, namely zero, and Nakayama's lemma shows that the kernel vanishes near that point. Thus $\mathcal{E}$ is locally free near every point of the central fiber. The locus where $\mathcal{E}$ is not locally free is closed. Since the projection $X \times T \to T$ is proper, the image of that locus is closed and misses t_0, and we remove this image. □

Lemma 5.1.12 *For a vector bundle $\mathcal{E}$ on $X \times T$, the parameters t for which $\mathcal{E}_t$ is ample form a Zariski-open subset of T.*

Proof Put $Y = \mathbb{P}_{X\times T}(\mathcal{E}^*)$, with the line convention, and let $p\colon Y \to T$ be the projection, a projective flat morphism. Write $\mathcal{L} = \mathcal{O}_Y(1)$. If $\mathcal{E}_{t_0}$ is ample, then $\mathcal{L}_{t_0}$ is ample. Choose k so that its kth power is very ample and has vanishing higher cohomology. By Serre vanishing, semicontinuity, and cohomology and base change, after shrinking T around t_0 the sheaf $\mathcal{V} = p_*\mathcal{L}^k$ is locally free and commutes with restriction to fibers. After a further shrinking, the evaluation map $p^*\mathcal{V} \to \mathcal{L}^k$ is surjective: its cokernel vanishes on the central fiber, and the image of its support under the proper map p is closed and misses t_0.

Evaluation gives a proper morphism

$$g\colon Y \longrightarrow \mathbb{P}_T(\mathcal{V}^*), \qquad g^*\mathcal{O}(1) = \mathcal{L}^k,$$

whose restriction to the central fiber is a closed immersion. The locus where g is quasi-finite is open; its complement has closed image in T by properness and misses t_0, so after a further shrinking g is proper and quasi-finite, hence finite. For a finite morphism, pushforward commutes with base change. The cokernel of

$$\mathcal{O}_{\mathbb{P}_T(\mathcal{V}^*)} \longrightarrow g_*\mathcal{O}_Y$$

therefore vanishes on the central fiber, and its support projects to a closed subset of T missing t_0. After removing that subset, the map is surjective, so g is a closed immersion. Consequently $\mathcal{L}^k$ is very ample on every fiber, and $\mathcal{E}_t$ is ample there. □

Lemma 5.1.13 *For a family $\mathcal{E}$ of rank-two vector bundles on $X \times T$ whose fibers have a fixed first Chern class, the parameters t for which $\mathcal{E}_t$ is μ_P-stable form a Zariski-open subset of T.*

Proof Fix a stable fiber at t_0 and put $\nu = c_1(\mathcal{E}_t)\cdot P/2$, which is independent of t. Choose $k \geq 1$ such that $\mathcal{E}_{t_0}^*(kP)$ is globally generated and has vanishing higher cohomology. The argument with Serre vanishing, base change and evaluation used in the previous proof gives, over a neighborhood of t_0, maps

$$\mathcal{E}_t \longrightarrow \mathcal{O}_X(kP)^{\oplus N}$$

that are injective on every fiber. By the double-dual argument for rank-one subsheaves in the proof of Proposition 5.1.6, a destabilizing subsheaf of $\mathcal{E}_t$ yields a nonzero map from a line bundle $M = \mathcal{O}_X(aS + bf)$ with $\deg_P M \geq \nu$. At least one component of

its composite with the injection above is nonzero, and since effective classes have nonnegative coefficients,

$$a \le 4k, \qquad b \le (12e+3)k.$$

Since $\deg_P M = (8e+3)a + 4b$, the inequality $\deg_P M \ge \nu$ also gives

$$a \ge \frac{\nu - 4(12e+3)k}{8e+3}, \qquad b \ge \frac{\nu - 4k(8e+3)}{4}.$$

Only finitely many integral pairs (a, b) satisfy these four bounds. For each such pair with $\deg_P M \ge \nu$, semicontinuity makes the set

$$\{t : H^0(X, \mathcal{E}_t \otimes M^*) \ne 0\}$$

closed, and it does not contain t_0. Conversely, a nonzero map from such an M is injective as a map of sheaves and violates stability. Thus the set of parameters with unstable fibers is, on this neighborhood, exactly the finite union of these closed subsets, and removing the union proves the claim. □

Lemma 5.1.14 *Let T be irreducible, let $U \subset T$ be a nonempty Zariski-open subset, and let $\mathcal{E}$ be a vector bundle on $X \times T$. For every $t_0 \in T$, there are points $t_j \in U$ tending to t_0 in the analytic topology, together with smooth complex-linear identifications of the bundles underlying $\mathcal{E}_{t_j}$ and $\mathcal{E}_{t_0}$, under which the Dolbeault operators of the fibers converge in the C^m norm for every integer $m \ge 0$. The dual operators converge in the same sense.*

Proof The proper algebraic closed subset $T \setminus U$ has empty interior in the analytic topology. In an analytic coordinate ball B around t_0, choose $t_j \in U \cap B$ with $t_j \to t_0$. Choose a smooth complex connection on $\mathcal{E}|_{X \times B}$. Parallel transport along radial paths in B, keeping $x \in X$ fixed, gives identifications

$$\Phi_t : \mathcal{E}_{t_0}^{\mathrm{sm}} \longrightarrow \mathcal{E}_t^{\mathrm{sm}}, \qquad \Phi_{t_0} = \mathrm{id},$$

depending smoothly on (x, t), where the superscript sm denotes the underlying smooth complex bundle. These identifications need not be holomorphic. Transport the Dolbeault operators by these maps and write

$$\bar\partial_t = \Phi_t^{-1} \bar\partial_{\mathcal{E}_t} \Phi_t, \qquad A_t = \bar\partial_t - \bar\partial_{\mathcal{E}_{t_0}}.$$

Since both operators satisfy the same Leibniz rule, A_t is an endomorphism-valued $(0,1)$-form. Its coefficients are smooth in (x, t), and $A_{t_0} = 0$. Fix background metrics and a smooth connection on the central bundle. Compactness of X and continuity of every finite collection of derivatives give

$$\|A_{t_j}\|_{C^m(X)} \longrightarrow 0 \qquad \text{for every integer } m \ge 0.$$

The dual identifications are $(\Phi_t^{-1})^*$. In transported dual frames the perturbation is $-A_t^{\mathsf{T}}$, so the same convergence holds for the dual Dolbeault operators. □

We use Walter's theorem on prioritary stacks: on a birationally ruled surface, a nonempty stack of prioritary torsion-free sheaves with fixed rank $r \geq 2$ and fixed numerical Chern classes is smooth and irreducible. Here prioritary means that $\operatorname{Ext}^2(E, E(-f)) = 0$ for every ruling fiber f; on X all ruling fibers are linearly equivalent. The theorem concerns the full prioritary stack and has no semistability hypothesis.

Theorem 5.1.3 *For every $e \geq 0$, there are integrable Dolbeault operators on the smooth bundle underlying E_0 that converge to its split operator in every C^m norm, $m \geq 0$, and define ample μ_P-stable bundles of rank two with $c_1 = 4H$ and $c_2 = 13H^2$. The induced dual operators converge to the split dual operator in every C^m norm.*

Proof Let $\mathcal{P}$ be the prioritary stack with invariants $(2, 4H, 13H^2)$. It contains E_0 by Proposition 5.1.7, and Walter's theorem makes it smooth and irreducible. Let $\mathcal{A}$ and $\mathcal{S}$ be the loci of ample and of μ_P-stable vector bundles in $\mathcal{P}$. They are open by Lemmas 5.1.12 and 5.1.13, which apply on smooth local charts of $\mathcal{P}$; Lemma 5.1.11 allows restriction to locally free families near a point given by a vector bundle. The ample bundle of Proposition 5.1.4 is prioritary by Lemma 5.1.10, so $\mathcal{A}$ is nonempty. A nonsplit extension is stable and prioritary by Propositions 5.1.6 and 5.1.7, so $\mathcal{S}$ is nonempty. By irreducibility, $\mathcal{U} = \mathcal{A} \cap \mathcal{S}$ is nonempty and dense.

Choose a smooth local chart $q\colon T \to \mathcal{P}$ of finite type over $\mathbb{C}$, with a point t_0 representing E_0, and let $\mathcal{E}$ be the corresponding flat family on $X \times T$. Fix an identification $\mathcal{E}_{t_0} \simeq E_0$. The variety T is smooth because the stack and the chart are smooth. Shrink T to an irreducible neighborhood of t_0 and use Lemma 5.1.11 to make $\mathcal{E}$ a vector bundle on the whole product. The inverse image $q^{-1}(\mathcal{U})$ is dense: every nonempty open $T' \subset T$ has nonempty open image $q(T')$ under the smooth map q, and this image meets $\mathcal{U}$. Apply Lemma 5.1.14 to this dense open set. Each transported operator defines a bundle isomorphic to the corresponding ample stable fiber. The fixed invariants of the stack give its Chern classes, and the lemma also gives the convergence of the dual operators. □

Corollary 5.1.5 *For every $e \geq 0$, there is an ample μ_P-stable bundle E of rank two on X with $c_1(E) = 4H$ and $c_2(E) = 13H^2$ that admits no smooth strongly pseudoconvex squared Finsler metric with semipositive Kobayashi curvature.*

Proof By Proposition 5.1.5, the two line summands of E_0^* have positive degrees with respect to suitable test forms. The preceding theorem gives an ample stable bundle E whose dual Dolbeault operator is as close to the split dual operator in C^4 as required. Apply Corollary 5.1.4 to $F = E^*$. □

5.1.7 A fixed Stein hypersurface for all Finsler metrics

Corollary 5.1.5 shows that no strongly pseudoconvex Finsler metric on E has semipositive Kobayashi curvature, but it does not locate the failure. For a single metric, a point where the curvature is not semipositive gives a local hypersurface through that point on which the metric has a nondegenerate critical point with many positive Levi directions. Holomorphic approximation on a Stein manifold allows one hypersurface to serve for all metrics at once.

In this subsection, X is a smooth projective variety of dimension $n \geq 1$, and $E \to X$ is an algebraic vector bundle of rank $r \geq 2$, viewed as a holomorphic bundle. We use the squared Finsler metrics and the convention for Kobayashi curvature of Section 5.1.3. A compact subset K of a Stein manifold Y is *holomorphically convex* if it coincides with its holomorphic hull $\{y \in Y : |f(y)| \leq \sup_K |f| \text{ for all } f \in \mathcal{O}(Y)\}$. We use the classical Oka–Weil theorem in the following form: a function holomorphic near a holomorphically convex compact subset of a Stein manifold can be approximated uniformly there by global holomorphic functions. We also use polynomial Runge approximation in one variable on a union of two disjoint closed disks.

Lemma 5.1.15 *Let K, L be holomorphically convex compact subsets of a Stein manifold Y. If a global holomorphic function maps them into disjoint closed disks in $\mathbb{C}$, then $K \cup L$ is holomorphically convex.*

Proof Write h for the function and Δ_K, Δ_L for the disks. Their union is polynomially convex, so testing with polynomials in h shows that h maps the holomorphic hull of $K \cup L$ into $\Delta_K \cup \Delta_L$. Suppose that $p \notin K \cup L$ and $h(p) \in \Delta_K$. Holomorphic convexity of K gives $a \in \mathcal{O}(Y)$ with $|a(p)| > \sup_K |a|$. Choose polynomials tending uniformly to one on Δ_K and to zero on Δ_L. Their compositions with h, multiplied by a, converge to a at p and on K, and to zero on L. A sufficiently good approximation therefore separates p from $K \cup L$. The case $h(p) \in \Delta_L$ is identical, with K and L interchanged. Thus no point outside $K \cup L$ belongs to its hull. □

Lemma 5.1.16 *Let U be a smooth affine variety and let $q \geq 0$ be an integer. Put $B = U \times \mathbb{C}^* \times \mathbb{C}^q$, with coordinates (x, t, s), and for $\lambda \in \mathbb{C}^*$ let $d_\lambda(x, t, s) = (x, \lambda t, \lambda s)$. For each $j \geq 1$, let $C_j \subset B$ be a holomorphically convex compact set, let g_j be holomorphic near C_j, and let $\delta_j > 0$. Suppose that $t(C_j)$ lies in the disk $|t - t_j| \leq |t_j|/4$ for some $t_j \neq 0$. Then there are $\lambda_j \in \mathbb{C}^*$ and $f \in \mathcal{O}(B)$ such that*

$$\sup_{d_{\lambda_j}(C_j)} |f - \lambda_j g_j \circ d_{\lambda_j}^{-1}| < |\lambda_j| \delta_j \qquad (j \geq 1).$$

Proof Choose an exhaustion $(L_j)_{j \geq 0}$ of the affine, hence Stein, manifold B by holomorphically convex compact sets with $L_j \subset \operatorname{int} L_{j+1}$. We construct $f_j \in \mathcal{O}(B)$ and compact sets K_j inductively, starting with $f_0 = 0$ and $K_0 = L_0$.

At stage $j \geq 1$, choose $M > 0$ with $t(K_{j-1}) \subset \{|t| \leq M\}$, and choose $R > 4M/3$. Dilation by $\lambda_j = R/t_j$ moves the t-projection of C_j into the disk $|t - R| \leq R/4$, which is disjoint from $|t| \leq M$. For the rest of the induction we replace C_j, g_j and

δ_j by the dilated compact set $d_{\lambda_j}(C_j)$, the function $\lambda_j g_j \circ d_{\lambda_j}^{-1}$ and the tolerance $|\lambda_j|\delta_j$, keeping the same notation. Lemma 5.1.15, applied with the function t, makes $K_{j-1} \cup C_j$ holomorphically convex.

On disjoint neighborhoods of K_{j-1} and C_j, prescribe f_{j-1} and g_j, respectively. Oka–Weil approximation gives $f_j \in \mathcal{O}(B)$ with

$$\sup_{K_{j-1}} |f_j - f_{j-1}| < \epsilon_j, \qquad \sup_{C_j} |f_j - g_j| < \epsilon_j, \qquad \epsilon_j = 2^{-j-2} \min(1, \delta_1, \dots, \delta_j).$$

Let K_j be a member of the fixed exhaustion containing $K_{j-1} \cup C_j \cup L_j$ in its interior. The interiors of the sets K_j exhaust B, and the summable errors make f_j converge locally uniformly to some $f \in \mathcal{O}(B)$. For a fixed j, every later approximation controls its error on C_j, since $C_j \subset K_{m-1}$ for $m > j$. Consequently

$$\sup_{C_j} |f - g_j| \le \sum_{m \ge j} \epsilon_m \le 2^{-j-1} \delta_j < \delta_j.$$

Returning to the original notation gives the stated tolerances. □

Theorem 5.1.4 *Suppose that E admits no smooth strongly pseudoconvex squared Finsler metric with semipositive Kobayashi curvature. Then there is a connected smooth Stein hypersurface $\mathcal{S} \subset E^{\times}$, locally closed in the analytic topology, with the following property: for every smooth strongly pseudoconvex squared Finsler metric G on E, the restriction $G|_{\mathcal{S}}$ has a nondegenerate critical point with at least r positive Levi eigenvalues, and hence with at most $n-1$ negative ones.*

Proof *Step 1: coordinates.* An algebraic frame trivializes E over a nonempty affine open subset $U \subset X$. Since X is smooth and connected, it is irreducible, so U is dense and connected. Write the fiber coordinates as $(w, t, s) \in \mathbb{C} \times \mathbb{C} \times \mathbb{C}^{r-2}$, and put $B = U \times \mathbb{C}^* \times \mathbb{C}^{r-2}$. We seek $\mathcal{S}$ as a graph $w = f(x, t, s)$ over B; such a graph avoids the zero section because $t \neq 0$.

Step 2: a local test for one metric. Fix a candidate metric G. Since its Kobayashi curvature is not semipositive, its horizontal Schur complement has a positive direction somewhere. This condition is open, so it holds at a point $v_0 = (x_0, w_0, t_0, s_0)$ with $x_0 \in U$, $t_0 \neq 0$ and $G_w(v_0) \neq 0$. Indeed, the first two conditions define a dense open set, and the zero set of G_w has empty interior, since vanishing of G_w on an open set would give $G_{w\bar{w}} = 0$ there, contradicting the positivity of the vertical Hessian.

Let H be the Levi-orthogonal complement of the vertical tangent space V at v_0. The Euler identities in the proof of Lemma 5.1.7 give

$$K := \ker \partial G(v_0) = H \oplus (V \cap K), \qquad \dim_{\mathbb{C}}(V \cap K) = r - 1.$$

Indeed, ∂G vanishes on H, and its value on the vertical radial vector is $G(v_0) > 0$. The two summands are Levi orthogonal, the vertical summand is positive definite, and the horizontal summand has a positive direction. Thus the Levi form of G restricted to K has at least r positive eigenvalues.

Since $G_w(v_0) \neq 0$, the space K is the graph of a linear map defined on $T^{1,0}_{b_0}B$, where $b_0 = (x_0, t_0, s_0)$. Choose a local holomorphic function g near b_0 with $g(b_0) = w_0$ whose graph has tangent space K at v_0. Then $F = G \circ \mathrm{Graph}(g)$ has a critical point at b_0, and its Levi form there is the restriction of the Levi form of G to K. Keep this first jet of g fixed and vary its quadratic jet. The mixed second derivatives of F do not change, while in local coordinates z on B the pure second derivatives satisfy

$$F_{\alpha\beta}(b_0) = A_{\alpha\beta} + G_w(v_0) g_{\alpha\beta}(b_0),$$

where A is fixed. Since $G_w(v_0) \neq 0$, we can prescribe the pure quadratic part of F to be $\tau \,\mathrm{Re} \sum_\alpha z_\alpha^2$, whose real Hessian has eigenvalues $\pm 2\tau$. If τ is larger than half the operator norm of the real Hessian of the fixed mixed part, the full real Hessian is invertible by the Neumann-series criterion. The critical point is then nondegenerate, with the same Levi form.

Step 3: stability of the test. Choose compact neighborhoods $C' \subset \mathrm{int}\, C$ of b_0 on which g is defined, with C holomorphically convex and $t(C) \subset \{|t - t_0| \leq |t_0|/4\}$. For a closed affine embedding of B, the intersection of B with a small closed Euclidean ball centered at b_0 is such a set C, since a closed Euclidean ball is convex and hence polynomially convex. Fix a smooth Hermitian background metric on E and its compact unit sphere bundle Σ. By homogeneity, the C^2 distance between metrics on Σ controls their C^2 distance on each fixed compact subset of $E^\times$. Persistence of nondegenerate critical points and openness of the conditions on the Levi eigenvalues therefore give a neighborhood $\mathcal{V}_G$ of $G|_\Sigma$ for the C^2 topology and a number $\delta_G > 0$ with the following property. For every metric whose restriction to Σ lies in $\mathcal{V}_G$ and every holomorphic function within δ_G of g uniformly on C, the restriction of the metric to the graph of the function still has a nondegenerate critical point over C' with at least r positive Levi eigenvalues. Here the Cauchy estimates on C' turn uniform closeness on C into C^2 closeness on C'.

Step 4: one hypersurface for all metrics. The restrictions of the candidate metrics to Σ form a subspace of the separable metrizable space $C^2(\Sigma)$. This subspace is second countable, hence Lindelöf, so the cover by the neighborhoods $\mathcal{V}_G$ has a countable subcover $(\mathcal{V}_j)_{j \geq 1}$, with associated data C_j, g_j, δ_j. For $D_\lambda(x, w, t, s) = (x, \lambda w, \lambda t, \lambda s)$,

$$G \circ D_\lambda = |\lambda|^2 G.$$

Dilation therefore preserves critical points of restrictions to graphs, their real nondegeneracy and their Levi inertia, for every candidate metric. It maps the graph of a function g over C to the graph of $\lambda g \circ d_\lambda^{-1}$ over $d_\lambda(C)$, with d_λ as in Lemma 5.1.16. A function within $|\lambda|\delta_j$ of $\lambda g_j \circ d_\lambda^{-1}$ on $d_\lambda(C_j)$ pulls back under the dilation to a function within δ_j of g_j on C_j. In particular, the metric neighborhood $\mathcal{V}_j$ needs no change.

Applying Lemma 5.1.16, we obtain one function $f \in \mathcal{O}(B)$ that satisfies all the dilated tests. Its graph $\mathcal{S}$ is biholomorphic to the connected affine Stein manifold B, and it is smooth and closed in the open subset $\{x \in U,\ t \neq 0\}$ of $E^\times$. Every candidate metric belongs to some $\mathcal{V}_j$, so its restriction to $\mathcal{S}$ has the required nondegenerate

critical point. Since $\dim_{\mathbb{C}} \mathcal{S} = n + r - 1$, at least r positive Levi eigenvalues leave at most $n - 1$ negative ones. □

Corollary 5.1.6 *For each $e \geq 0$, the bundle E of Corollary 5.1.5 can be chosen so that $E^\times$ contains a connected smooth Stein hypersurface on which the restriction of every smooth strongly pseudoconvex squared Finsler metric on E has a nondegenerate critical point with at least two positive Levi eigenvalues and at most one negative Levi eigenvalue.*

Proof The bundle constructed in the proof of Theorem 5.1.3 is a fiber of an algebraic family on a chart of the prioritary stack, so it is algebraic. Apply Theorem 5.1.4 with $n = r = 2$, using the curvature obstruction of Corollary 5.1.5. □

5.1.8 Concave neighborhoods and dual norms

A relatively compact neighborhood Ω of the zero section of a vector bundle E determines a norm on E^*, the support function $\xi \mapsto \max_{v \in \overline{\Omega} \cap E_x} |\xi(v)|$ of its fibers. If the boundary of Ω is sufficiently concave in the horizontal directions, holomorphic sections touching the boundary from inside show that this support norm has plurisubharmonic square, and supporting balls give it uniform convexity in the fibers. Consequently, when the dual norms are obstructed as in Theorem 5.1.1, such neighborhoods cannot exist. In terms of exhaustion functions, this bounds the convexity of the complement of the zero section in the projective completion of E.

In this subsection, $E \to X$ is a holomorphic vector bundle of rank $r \geq 1$, and $n \geq 1$. Fix a smooth Hermitian metric k on E, with real inner product $k_{\mathbb{R}} = \operatorname{Re} k$ and dual norm $|\cdot|_{k^*}$. For a smooth real function f, write $\mathcal{L}_f$ for its complex Hessian. Let Ω be an open set with smooth boundary, defined near $\partial\Omega$ by $f < 0$ with $\mathrm{d}f \neq 0$ on $\partial\Omega$. We say that $\partial\Omega$ has *at least n negative Levi directions* if, at every boundary point, $\mathcal{L}_f$ has at least n negative eigenvalues on $\ker \partial f$.

Lemma 5.1.17 *Fix $\delta, B > 0$. On a finite-dimensional Hermitian vector space, put*

$$s_c(\xi) = |\xi(c)| + \delta|\xi|$$

for covectors ξ and vectors c with $|c| \leq B$. There is $\gamma > 0$, depending only on δ and B, such that $s_c^2 - \gamma|\xi|^2$ is real convex for every such c.

Proof For $0 < \tau \leq 1$, replace s_c by the norm

$$s_{c,\tau}(\xi) = \sqrt{|\xi(c)|^2 + \tau|\xi|^2} + \delta|\xi|,$$

which is smooth off zero. Its Lipschitz constant is at most $L = \sqrt{B^2 + 1} + \delta$, uniformly in c and τ. All derivatives below are real derivatives on the dual space. At $\xi \neq 0$, set $u = \xi/|\xi|$ and decompose a vector as $w = \alpha u + b$ with b orthogonal to u for the

real inner product. Write $s = s_{c,\tau}$ and $A = \mathrm{d}s_\xi(w)$. Since the first summand of s is convex and $s \geq \delta|\xi|$,

$$D^2 s(w,w) \geq \delta \frac{|b|^2}{|\xi|}, \qquad D^2(s^2)(w,w) \geq 2A^2 + 2\delta^2|b|^2.$$

Homogeneity gives $\mathrm{d}s_\xi(u) = s(\xi)/|\xi| \geq \delta$. Since $|\mathrm{d}s_\xi(b)| \leq L|b|$, we have

$$|w|^2 \leq \frac{2A^2}{\delta^2} + \left(1 + \frac{2L^2}{\delta^2}\right)|b|^2.$$

Take $\gamma = \delta^4/(\delta^2 + 2L^2)$. Then $\gamma \leq \delta^2/2$ and $\gamma(1 + 2L^2/\delta^2) = \delta^2$, so $D^2(s^2)(w,w) \geq 2\gamma|w|^2$.

The homogeneous function $s^2 - \gamma|\xi|^2$ of degree two extends as a C^1 function at zero with differential zero. Its restriction to every real line is convex, including at zero, because its second derivative is nonnegative away from zero and its first derivatives match there. Thus it is convex on the whole space. Finally,

$$0 \leq s_{c,\tau}(\xi) - s_c(\xi) \leq \sqrt{\tau}\,|\xi|,$$

and the locally uniform limit as $\tau \downarrow 0$ proves the assertion with the same γ. □

Theorem 5.1.5 *Suppose that a relatively compact neighborhood $\Omega \Subset E$ of the zero section has smooth boundary with at least n negative Levi directions. Then*

$$p(x,\xi) = \max_{v \in \overline{\Omega} \cap E_x} |\xi(v)|$$

is a continuous complex norm on E^* whose square $Q = p^2$ is plurisubharmonic on the total space of E^*. For some $\gamma > 0$, each function $Q(x,\cdot) - \gamma|\cdot|^2_{k^*}$ is real convex. If E is ample, E^* also admits a smooth complex norm with strictly plurisubharmonic square off the zero section.

Proof *Step 1: basic bounds.* Put $A_x = \overline{\Omega} \cap E_x$. By compactness of X there are $0 < a_0 < b_0$ such that, in every fiber, the closed k-ball of radius a_0 is contained in Ω and $\overline{\Omega}$ lies in the open ball of radius b_0. The maximum defining p is attained, and

$$a_0|\xi|_{k^*} \leq p(x,\xi) \leq b_0|\xi|_{k^*}.$$

As a supremum of absolute values of linear functionals, p is complex homogeneous and satisfies the triangle inequality.

Step 2: support points. Fix $\xi_0 \neq 0$ and a maximizing point $v_0 \in A_{x_0}$. It lies on $\partial\Omega$, since at an interior point a small vertical displacement would increase $|\xi_0(v)|$. Multiply ξ_0 by a complex number of modulus one to obtain η with $\eta(v_0) = p(x_0,\xi_0) = a > 0$. Then $\ell = \operatorname{Re}\eta$ satisfies $\ell \leq a$ on A_{x_0}. Let f be a defining function as in the definition of negative Levi directions, so that $\Omega = \{f < 0\}$ near $\partial\Omega$ and $\mathrm{d}f \neq 0$ on $\partial\Omega$.

Let $V = T^{1,0}_{v_0} E_{x_0}$ be the vertical tangent space and $T = \ker \partial f(v_0)$. Choose an n-dimensional complex subspace $W \subset T$ on which $\mathcal{L}_f$ is negative definite. We first

show that the vertical differential $\mathrm{d}_v f(v_0)$ is nonzero. Otherwise $V \subset T$, and counting dimensions gives $W \cap V \neq \{0\}$. A vector in this intersection has negative vertical Levi form, and the two corresponding real directions show that the real vertical Hessian of f has a negative direction. Its negative cone is open, hence contains a direction w with $\ell(w) > 0$. Since the vertical first derivative vanishes, Taylor expansion gives $f(v_0 + tw) < 0$ for small $t > 0$, contradicting $\ell \leq a$ on A_{x_0}.

The boundary of A_{x_0} is therefore smooth at v_0. Since the supporting functional ℓ attains its maximum on A_{x_0} there,

$$\mathrm{d}_v f(v_0) = c\,\mathrm{d}\ell, \qquad c > 0;$$

the sign follows because an inward direction along which ℓ increases would contradict the maximality. On the affine hyperplane $\{\ell = a\}$ of the fiber, the restriction of f has a local minimum zero at v_0: a point there with $f < 0$ would allow an additional displacement increasing ℓ within Ω. The real vertical Hessian of f is thus nonnegative on $\ker \mathrm{d}\ell$. Consequently $\mathcal{L}_f$ is nonnegative on the complex space $V \cap T$, of dimension $r - 1$. Hence $W \cap V = \{0\}$, and the projection $W \to T^{1,0}_{x_0} X$ is an isomorphism.

Step 3: plurisubharmonicity. In local coordinates z centered at x_0, choose a holomorphic section σ_0 through v_0 with tangent space W. The function $f \circ \sigma_0$ has zero first derivative and negative definite mixed Hessian. Its quadratic expansion is

$$f(\sigma_0(z)) = \sum H_{\alpha\bar{\beta}} z_\alpha \bar{z}_\beta + 2\,\mathrm{Re}\,P(z) + O(|z|^3),$$

where P is a holomorphic quadratic polynomial. Choose a vertical vector b with $\partial_v f(b) = 1$, and replace σ_0 by $\sigma = \sigma_0 - bP$. This preserves the first jet and cancels the pure quadratic part. Thus, after shrinking the chart,

$$f(\sigma(z)) \leq -\epsilon |z|^2, \qquad \sigma(0) = v_0,$$

for some $\epsilon > 0$. In particular $\sigma(z) \in A_z$ throughout the chart.

Compactness of $\overline{\Omega}$ makes p upper semicontinuous: for convergent covectors, the corresponding maximizing vectors have a subsequence whose limit lies in the limiting fiber of $\overline{\Omega}$. The supporting section gives the opposite inequality $p(z, \xi) \geq |\xi(\sigma(z))|$, with equality at (x_0, ξ_0). Therefore p is continuous off the zero section, and its upper bound gives continuity at zero. The holomorphic pairing $\xi(\sigma(z))$ is nonzero near the contact point, and its logarithmic absolute value is a pluriharmonic function lying below $\log p$ and equal to it at (x_0, ξ_0). The mean-value property on small holomorphic disks therefore gives the submean inequality for $\log p$ at every point off the zero section. Thus $\log p$, and then $Q = \exp(2 \log p)$ by Jensen's inequality, are plurisubharmonic there. At the zero section the continuous nonnegative function Q vanishes, so its submean inequality on disks also holds there.

Step 4: uniform convexity in the fibers. Consider all maximizing triples (x, ξ, v) with $|\xi|_{k^*} = 1$. They form a compact set, since p is continuous, and by Step 2 the vertical differential $\mathrm{d}_v f$ does not vanish at these support points. Thus there are uniform constants $m > 0$, $\rho > 0$ and $C \geq 1$ such that, for vertical vectors w,

$$|\mathrm{d}_v f| \geq m, \qquad f(v+w) \leq \mathrm{d}_v f(w) + C|w|_k^2 \quad (|w|_k \leq \rho).$$

Choose $0 < \delta < \min\{a_0/2, \rho/2, m/(2C)\}$. At a support point, write N for the outward real vertical unit normal and set $c = v - \delta N$. If $|w + \delta N|_k \leq \delta$, then

$$k_{\mathbb{R}}(w, N) \leq -\frac{|w|_k^2}{2\delta}, \qquad f(v+w) \leq -\left(\frac{m}{2\delta} - C\right)|w|_k^2 \leq 0.$$

All these points lie in the neighborhood where f is defined, so the closed ball $\overline{B}_k(c, \delta)$ is contained in A_x.

The support function of this ball is $s_c(\xi) = |\xi(c)| + \delta|\xi|_{k^*} \leq p(x, \xi)$. Let ξ be a covector of unit norm for which v is a maximizing point, and let η be ξ with its phase normalized as in Step 2. Then $\operatorname{Re}\eta$ has real dual norm one, and its metric dual vector is N. Hence $\operatorname{Re}\eta(N) = 1$, and since $|\eta(N)| \leq 1$, also $\eta(N) = 1$. We obtain $\eta(c) = a - \delta > 0$, and therefore $s_c(\xi) = a$. Homogeneity now gives

$$Q(x, \xi) = \sup_c s_c(\xi)^2,$$

where c ranges over the centers of the supporting balls in the fiber. Every such center satisfies $|c|_k \leq b_0 + \delta$, so by Lemma 5.1.17 all the functions $s_c^2 - \gamma|\xi|_{k^*}^2$ are convex for the same $\gamma > 0$. Their supremum, which is $Q(x, \cdot) - \gamma|\cdot|_{k^*}^2$, is finite and therefore convex, as asserted.

Step 5: smoothing. Finally, if E is ample, Proposition 5.1.2 gives the smooth strictly plurisubharmonic homogeneous function on E^* required in Proposition 5.1.1, and applying that proposition to Q gives the smooth norm. □

For a complex manifold of dimension N and an integer $1 \leq q \leq N$, we say that the manifold is *q-convex* if it has a smooth real exhaustion function with at least $N - q + 1$ positive Levi eigenvalues outside a compact set; here an exhaustion function is one whose sublevel sets are compact. A q-convex manifold is also q'-convex for $q \leq q' \leq N$. We call such integers q the *smooth convexity indices* of the manifold, and the smallest of them, when one exists, its *least smooth convexity index*.

Lemma 5.1.18 *Let M be a complex manifold of dimension N, and let $S \subset M$ be a closed complex submanifold of dimension $D \geq 1$. If S has least smooth convexity index κ, then every smooth convexity index of M is at least κ.*

Proof Suppose that M is p-convex for an integer $1 \leq p < \kappa$. Then a smooth exhaustion function ψ has at least $N - p + 1$ positive Levi eigenvalues outside a compact set $K \subset M$. Its restriction to the closed submanifold S is an exhaustion function, since its sublevel sets are closed subsets of the compact sublevel sets of ψ, and $K \cap S$ is compact.

At a point of $S \setminus K$, choose a complex subspace $P \subset T^{1,0}M$ of dimension $N - p + 1$ on which the Levi form of ψ is positive definite. Then

$$\dim_{\mathbb{C}}(P \cap T^{1,0}S) \geq (N - p + 1) + D - N = D - p + 1.$$

The Levi form of $\psi|_S$ is the restriction of the Levi form of ψ, since the inclusion is holomorphic, so it has at least $D - p + 1$ positive eigenvalues. Since $p < \kappa \le D$, this shows that S is p-convex, which contradicts the definition of κ. □

In the following results, projectivizations parametrize complex lines.

Corollary 5.1.7 *Suppose that E is ample and that E^* admits no smooth complex norm with plurisubharmonic square off the zero section. Then the complement*

$$Z \setminus Y, \qquad Z = \mathbb{P}(\mathcal{O}_X \oplus E), \quad Y = \mathbb{P}(\mathcal{O}_X),$$

is not r-convex.

Proof The chart of Z where the $\mathcal{O}_X$-coordinate is nonzero is identified with the total space of E, and Y is its zero section. Suppose that $Z \setminus Y$ is r-convex. Since $\dim Z = n + r$, a smooth exhaustion function ψ then has at least $n + 1$ positive Levi eigenvalues outside a compact set K. Choose a relatively compact disk-bundle neighborhood $D \Subset E$ of Y. The set $Z \setminus D$ is compact and disjoint from Y. Choose a regular value c of ψ greater than its maximum on $Z \setminus D$ and on K.

Since $\{\psi \le c\}$ is compact in $Z \setminus Y$, the set $\Omega = \{\psi > c\} \cup Y$ is an open neighborhood of Y. Its closure is a compact subset of D, and its boundary is the regular level set $\{\psi = c\}$. The defining function $f = c - \psi$ has at least $n + 1$ negative Levi eigenvalues there. Restriction to the complex tangent hyperplane removes at most one negative direction, leaving at least n. Since E is ample, Theorem 5.1.5 produces a smooth complex norm on E^* with plurisubharmonic square, a contradiction. □

Corollary 5.1.8 *For every $e \ge 0$, there is an ample rank-two bundle E on the Hirzebruch surface $X = \mathbb{F}_e$ such that $\mathbb{P}(\mathcal{O}_X \oplus E) \setminus \mathbb{P}(\mathcal{O}_X)$ is not 2-convex. The bundle may be chosen with the slope-stability and Chern classes of Theorem 5.1.3.*

Proof Use Theorem 5.1.3 to choose E whose dual Dolbeault operator lies in the C^4 neighborhood of Theorem 5.1.1; the split dual bundle satisfies the hypotheses of that theorem by Proposition 5.1.5. Thus E^* admits no smooth complex norm with plurisubharmonic square, and Corollary 5.1.7 with $r = 2$ gives the assertion. □

5.1.9 Curvature and projective complement exhaustions

Corollary 5.1.7 excludes r-convexity of the complement of the zero section in the projective completion, but it does not decide which larger indices occur. Upper bounds come from curvature. The reciprocal of the squared norm of a Hermitian metric is an exhaustion near the removed zero section, and every positive direction of the curvature, evaluated at a fiber vector, contributes a positive Levi eigenvalue. For a stable bundle of positive Gauduchon degree, the Hermitian–Einstein metric provides such a direction at every fiber vector.

In this subsection, $E \to X$ is a holomorphic vector bundle of rank $r \geq 1$, $n \geq 1$, and h is a smooth Hermitian metric on E. In a holomorphic frame, write the matrix of h as M, using column vectors and an inner product linear in its first argument, so that $h(v, v) = v^* M v$. Put

$$F_h = \bar\partial(M^{-1}\partial M) = \sum_{\alpha,\beta} F_{\alpha\bar\beta}\, \mathrm{d}z_\alpha \wedge \mathrm{d}\bar z_\beta, \qquad R_h(u,\bar u) = \sum_{\alpha,\beta} F_{\alpha\bar\beta} u_\alpha \bar u_\beta.$$

For $v \in E_x$, the real $(1,1)$-form $\alpha_{h,v} = \mathrm{i}\, h(F_h v, v)$ has the Hermitian coefficient form $u \mapsto h(R_h(u,\bar u)v, v)$. For a Hermitian form ω on X, the *mean curvature* of h is the h-Hermitian endomorphism $K_{\omega,h} = \Lambda_\omega(\mathrm{i}F_h)$. Here Λ_ω denotes contraction with ω, normalized so that $\Lambda_\omega(\beta)\,\omega^n = n\beta \wedge \omega^{n-1}$ for $(1,1)$-forms β; we also call $\mathrm{tr}_\omega\,\beta = \Lambda_\omega(\beta)$ the *trace* of β with respect to ω. The curvature F_h is unscaled: for a line bundle, the curvature $-\mathrm{dd^c}\log h$ of Section 5.1 is $\mathrm{i}F_h/(2\pi)$.

Use the line convention for projectivization and put

$$U_E = \mathbb{P}(\mathcal{O}_X \oplus E) \setminus \mathbb{P}(\mathcal{O}_X).$$

The chart $[1 : v]$ identifies the total space of E with an open subset of $\mathbb{P}(\mathcal{O}_X \oplus E)$, in which $\mathbb{P}(\mathcal{O}_X)$ is the zero section; thus U_E contains $E^\times$. Its complex dimension is $n+r$. Convexity is understood in the sense of the smooth exhaustions of Section 5.1.8.

Lemma 5.1.19 *Put $G(v) = h(v,v)$ and $\rho = G^{-1}$ on $E^\times$. In a holomorphic frame that is unitary and* Chern-normal *at x, meaning that $M(x)$ is the identity and the first derivatives of M vanish at x, the complex Hessian of ρ at (x,v) is*

$$\mathcal{L}_\rho(u,w) = \frac{h(R_h(u,\bar u)v, v)}{G^2} - \frac{|w|_h^2}{G^2} + \frac{2|h(w,v)|^2}{G^3}, \qquad u \in T^{1,0}_x X, \quad w \in E_x,$$

where (u,w) are the horizontal and vertical components of a tangent vector. There are no mixed horizontal–vertical terms, and the vertical block has exactly one positive and $r-1$ negative eigenvalues.

Proof Such a frame is obtained by a constant change of frame making it unitary at x, followed by a holomorphic change of frame, linear in the coordinates, that cancels ∂M. Then $F_{\alpha\bar\beta} = -\partial_\alpha\partial_{\bar\beta} M$ at x. The horizontal first derivatives of G and its mixed horizontal–vertical second derivatives vanish. The horizontal complex Hessian of G is $-h(R_h(u,\bar u)v, v)$, its vertical complex Hessian is $|w|_h^2$, and its vertical first derivative in the direction w is $h(w,v)$. Apply

$$\partial\bar\partial(G^{-1}) = 2G^{-3}\partial G \wedge \bar\partial G - G^{-2}\partial\bar\partial G.$$

The vertical block equals $G^{-1} > 0$ at $w = v$ and equals $-|w|_h^2/G^2$ when $w \perp_h v$. These two subspaces are orthogonal for the vertical block, which proves the statement about its eigenvalues. □

Theorem 5.1.6 *Let s be an integer with $0 \leq s \leq n$. Suppose that for every $x \in X$ and every $0 \neq v \in E_x$, the coefficient form of $\alpha_{h,v}$ has at least s positive eigenvalues. Then U_E is $(n+r-s)$-convex.*

Proof By Lemma 5.1.19, the horizontal block of $\mathcal{L}_\rho$ has at least s positive eigenvalues and the vertical block has one. Their direct sum thus has at least $s+1$ positive eigenvalues at every point of $E^\times$.

Choose $\epsilon > 0$ and a smooth function $\chi\colon [0,+\infty) \to [0,1]$ equal to one on $[0,\epsilon]$ and to zero on $[2\epsilon,+\infty)$. Define

$$\psi([1:v]) = \frac{\chi(G(v))}{G(v)} \quad (v \neq 0),$$

and extend ψ by zero on the divisor at infinity $\mathbb{P}(E)$. This extension is smooth. Indeed, near a point $[0:v]$ with $v \neq 0$, use a local representative $[u:v]$ in a projective chart. On its intersection with the affine chart, $G(v/u) = h(v,v)/|u|^2$ tends to infinity as $u \to 0$, so $\chi(G)$ vanishes in a neighborhood of the point.

The function ψ tends to infinity near the removed zero section. Since the ambient projective bundle is compact, every sublevel set of ψ is compact in U_E: it is closed there and cannot accumulate on the removed section. Put $D = \{G < \epsilon\} \subset E$, including the zero section, and $K = \mathbb{P}(\mathcal{O}_X \oplus E) \setminus D$. Then K is a compact subset of U_E, and $\psi = \rho$ on $U_E \setminus K$. Thus the exhaustion ψ has at least $s+1$ positive Levi eigenvalues outside a compact set. In dimension $n+r$, this is $(n+r-s)$-convexity.□

Taking $s = 0$ gives $(n+r)$-convexity for every bundle. For $s = 1$, the curvature hypothesis is precisely RC-positivity of the metric, as in Definition 5.2.1. A positive definite mean curvature provides such a direction by taking a trace.

To apply this to stable bundles, we extend the degree of line bundles of Section 5.1 to vector bundles through their determinants.

Definition 5.1.1 For a Gauduchon form ω on X, define

$$\deg_\omega E = \deg_{\omega^{n-1}} \det E = \int_X \frac{\mathrm{i}\operatorname{tr} F_h}{2\pi} \wedge \omega^{n-1}, \qquad \mu_\omega(E) = \frac{\deg_\omega E}{r}.$$

For a nonzero torsion-free coherent sheaf $\mathcal{F}$, use the determinant line bundle $\det \mathcal{F} = (\bigwedge^{\operatorname{rank}\mathcal{F}} \mathcal{F})^{**}$ to define its degree and slope. The bundle E is *ω-slope-stable* if $\mu_\omega(\mathcal{F}) < \mu_\omega(E)$ for every coherent subsheaf $\mathcal{F} \subset E$ with $0 < \operatorname{rank}\mathcal{F} < r$ and torsion-free quotient.

Here $\operatorname{tr} F_h = \bar\partial\partial \log\det M$, so $\mathrm{i}\operatorname{tr} F_h/(2\pi)$ is the curvature form $-\mathrm{dd}^{\mathrm{c}} \log\det M$ of the metric $\det h$ on $\det E$. If $\det h' = \mathrm{e}^f \det h$, the two curvature representatives differ by $-\mathrm{dd}^{\mathrm{c}} f$, whose integral against ω^{n-1} vanishes by Lemma 2.1.1. Thus the degree depends only on the Bott–Chern class $c_1^{\mathrm{BC}}(E)$ of the closed form $\mathrm{i}\operatorname{tr} F_h/(2\pi)$, and it is the pairing of this class with ω^{n-1}. The Hopf example in Example 5.1.2 shows that the de Rham first Chern class of E does not determine the degree.

Corollary 5.1.9 *If $K_{\omega,h}$ is positive definite for a Hermitian form ω on X, then U_E is $(n+r-1)$-convex. This holds, in particular, when ω is Gauduchon, E is ω-slope-stable, and $\deg_\omega E > 0$.*

Proof For $v \neq 0$, the trace of $\alpha_{h,v}$ with respect to ω is $h(K_{\omega,h}v, v) > 0$, and a Hermitian form with positive trace has a positive eigenvalue. Apply Theorem 5.1.6 with $s = 1$.

Under the Gauduchon and stability hypotheses, the classical existence theorem for Hermitian–Einstein metrics on stable bundles gives a metric h with $K_{\omega,h} = \lambda \operatorname{Id}_E$, where λ is a constant. Taking the trace and integrating the contraction identity gives

$$r\lambda \int_X \omega^n = n \int_X \operatorname{tr}(\mathrm{i}F_h) \wedge \omega^{n-1} = 2\pi n \deg_\omega E > 0.$$

Thus $\lambda > 0$, and the first assertion applies. □

Corollary 5.1.10 *For every $e \geq 0$, there is a stable ample rank-two bundle E on $X = \mathbb{F}_e$ as in Corollary 5.1.8 for which U_E is 3-convex but not 2-convex. Its least smooth convexity index is three.*

Proof Recall the ruling section S, the fiber f and the class $H = S + (3e + 1)f$ of Lemma 5.1.9, and the polarization $P = 4H - f$ of Section 5.1.5. Choose E as in Corollary 5.1.8, μ_P-stable and with $c_1(E) = 4H$. The intersection numbers of Lemma 5.1.9 give

$$c_1(E) \cdot P = 4H \cdot (4H - f) = 80e + 28 > 0.$$

By the same lemma, $P = 4S + (12e + 3)f$ is very ample. Take for ω the pulled-back Fubini–Study Kähler form representing $c_1(\mathcal{O}_X(P))$. Degrees with respect to ω are intersection numbers with P, so E is ω-slope-stable with $\deg_\omega E > 0$, and Corollary 5.1.9 gives 3-convexity, since $n = r = 2$. By the choice of E, the complement is not 2-convex. Finally, 1-convexity would imply 2-convexity, so the least index is three. □

5.1.10 External tensor products and complement convexity

On a product of k manifolds, the external tensor product of bundles with positive mean curvature has, at every nonzero fiber vector, a positive curvature direction tangent to each factor. Indeed, the curvature of the tensor product metric has no components mixing different factors, and the mean curvature of each factor, extended by the identity on the other factors, is positive definite on the whole tensor product, not only on decomposable tensors. Theorem 5.1.6 then improves the convexity bound by k.

Let $k \geq 1$. For $1 \leq i \leq k$, let X_i be a connected compact complex manifold of dimension $n_i \geq 1$, let $E_i \to X_i$ be a holomorphic vector bundle of rank $r_i \geq 1$, and

choose a smooth Hermitian metric h_i on E_i and a Hermitian form ω_i on X_i. With $p_i \colon X = \prod_i X_i \to X_i$ the projections, put

$$E = \bigotimes_{i=1}^{k} p_i^* E_i, \qquad h = \bigotimes_{i=1}^{k} p_i^* h_i, \qquad n = \sum_i n_i, \quad r = \prod_i r_i.$$

For an endomorphism A_i of E_i, write $\widehat{A_i}$ for its action on the ith tensor factor, with the identity on every other factor. We use the Chern curvature, the forms $\alpha_{h,v}$ and the mean curvatures of Section 5.1.9.

Lemma 5.1.20 *The curvature of h is*

$$F_h = \sum_i \widehat{p_i^* F_{h_i}}.$$

For every $v \in E_x$, the coefficient form of $\alpha_{h,v}$ is block diagonal with respect to the decomposition $T^{1,0}_x X = \bigoplus_i T^{1,0}_{x_i} X_i$, and

$$\operatorname{tr}_{\omega_i}\big(\alpha_{h,v}|_{T_{x_i}X_i}\big) = h\big(\widehat{K_i} v, v\big), \qquad K_i = K_{\omega_i, h_i}.$$

Proof In product holomorphic frames, the matrix of the metric is $M = \bigotimes_i M_i$, and its connection matrix is

$$M^{-1}\partial M = \sum_i \widehat{M_i^{-1}\partial M_i}.$$

Each summand depends only on the coordinates of X_i. Applying $\bar\partial$ gives the curvature formula and shows that all components mixing different factors vanish. Pairing with v gives the block diagonal form. Contracting the ith block with ω_i commutes with the identity actions on the other tensor factors, which gives the trace identity. □

Theorem 5.1.7 *If every K_i is positive definite, then*

$$\mathbb{P}(\mathcal{O}_X \oplus E) \setminus \mathbb{P}(\mathcal{O}_X)$$

is $(n + r - k)$-convex.

Proof At each point, choose an h_i-orthonormal eigenbasis of K_i. The resulting product basis is h-orthonormal, and $\widehat{K_i}$ has the positive eigenvalues of K_i, repeated according to the dimensions of the other tensor factors. Hence $h(\widehat{K_i} v, v) > 0$ for every nonzero v, whether or not v is decomposable.

By Lemma 5.1.20, each diagonal block of $\alpha_{h,v}$ has positive trace and therefore a positive eigenvalue, so the block diagonal form has at least k positive eigenvalues. Since $k \le n$, Theorem 5.1.6 with $s = k$ gives the assertion. □

Corollary 5.1.11 *Suppose that each ω_i is Gauduchon, that E_i is ω_i-slope-stable as in Definition 5.1.1, and that $\deg_{\omega_i} E_i > 0$. Then the complement $\mathbb{P}(\mathcal{O}_X \oplus E) \setminus \mathbb{P}(\mathcal{O}_X)$ is $(n+r-k)$-convex. For k stable ample Hirzebruch examples as in Corollary 5.1.10, with their polarizations, this gives $(2^k + k)$-convexity.*

Proof As in the proof of Corollary 5.1.9, choose a Hermitian–Einstein metric on each E_i. Its mean curvature is $\lambda_i \operatorname{Id}_{E_i}$, with $\lambda_i > 0$ by the positivity of the degree and the integrated trace identity. Apply Theorem 5.1.7. For the Hirzebruch factors, $n_i = r_i = 2$, so $n + r - k = 2k + 2^k - k = 2^k + k$. □

5.1.11 Homogeneous plurisubharmonic functions and Hopf complements

Curvature gives upper bounds for the least smooth convexity index of a projective complement. Lower bounds come from plurisubharmonic functions: a smooth positive plurisubharmonic function on the punctured bundle that is homogeneous of degree two excludes every improvement of the trivial bound, and closed submanifolds transfer such lower bounds by Lemma 5.1.18. On Hopf manifolds these tools determine the least index for line bundles whose de Rham first Chern class vanishes. The answer depends on the sign of their Gauduchon degree, which therefore detects information invisible to de Rham cohomology.

Let $F \to B$ be a holomorphic vector bundle of rank $r \geq 1$ over a connected compact complex manifold B of dimension $n \geq 1$. Write $F^\times$ for the complement of its zero section, and put $N = n + r$ and

$$U_F = \mathbb{P}(\mathcal{O}_B \oplus F) \setminus \mathbb{P}(\mathcal{O}_B).$$

The reciprocal-norm exhaustion in Theorem 5.1.6 with $s = 0$ shows that U_F is N-convex.

Proposition 5.1.9 *Suppose that $F^\times$ admits a smooth positive plurisubharmonic function G satisfying $G(cv) = |c|^2 G(v)$ for every $c \in \mathbb{C}^*$ and $v \in F^\times$. Then the least smooth convexity index of U_F is N.*

Proof Since every smaller convexity index would imply it, it suffices to exclude $(N-1)$-convexity. Suppose that a smooth exhaustion function ψ of U_F has at least two positive Levi eigenvalues outside a compact set $K \subset U_F$.

Fix a reference Hermitian norm on F. On its compact unit sphere bundle, G has a positive minimum and a finite maximum, so by homogeneity G is comparable to the squared reference norm uniformly on $F^\times$. Thus the level sets

$$\Sigma_t = \{G = t^2\}, \qquad t > 0,$$

are compact, and multiplication by t maps Σ_1 onto Σ_t. They are smooth hypersurfaces by the Euler identity: if $R_v = v$ is the real radial vector at v, then $\mathrm{d}G(R_v) = 2G(v) > 0$.

Since K stays away from the removed zero section, there is $\epsilon > 0$ such that Σ_t is disjoint from K for $0 < t < \epsilon$.

Put

$$M(t) = \max_{v \in \Sigma_1} \psi([1 : tv]), \qquad 0 < t < \epsilon.$$

On any closed interval in $(0, \epsilon)$, the radial derivatives of the smooth function under the maximum are uniformly bounded over the compact level set, so M is locally Lipschitz. Since ψ is an exhaustion, $M(t) \to +\infty$ as $t \to 0$: otherwise maximizing points along a sequence $t \to 0$ would lie in one compact sublevel set while accumulating on the removed section. Choose $0 < t_0 < t_1 < \epsilon$ with $M(t_0) > M(t_1)$. Since a Lipschitz function is absolutely continuous and is the integral of its derivative, there is a point $t \in (t_0, t_1)$ at which M is differentiable with $M'(t) < 0$.

Choose $v \in \Sigma_1$ attaining $M(t)$, and set $p = tv$. For both signs of h, the inequality

$$M(t + h) \geq \psi([1 : (t + h)v])$$

is an equality at $h = 0$, so differentiability of M gives $\frac{\mathrm{d}}{\mathrm{d}t}\psi([1 : tv]) = M'(t)$. Since ψ restricted to Σ_t attains its maximum at p and $\mathrm{d}G \neq 0$, there is a real number λ with $\mathrm{d}\psi = \lambda\, \mathrm{d}G$ at p. Evaluating on the radial direction gives

$$\lambda = \frac{M'(t)}{2t} < 0.$$

The function $\psi - \lambda G$ has zero differential at p and nonpositive real Hessian on $T_p\Sigma_t$. Indeed, its restriction to Σ_t is ψ minus a constant, which has a maximum at p, and since its differential vanishes at p, its second derivative along a curve in Σ_t is the Hessian evaluated on the tangent vector. On the complex tangent hyperplane $H = T_p\Sigma_t \cap \mathrm{i}T_p\Sigma_t$, this implies

$$\mathcal{L}_\psi|_H \leq \lambda \mathcal{L}_G|_H \leq 0.$$

The first inequality follows by adding the real Hessian inequalities for a tangent vector and its complex rotation; the second uses plurisubharmonicity of G and $\lambda < 0$. A two-dimensional complex subspace on which $\mathcal{L}_\psi$ is positive definite would meet the hyperplane H nontrivially, so ψ has at most one positive Levi eigenvalue at $p \notin K$. This contradiction excludes $(N - 1)$-convexity. Together with Theorem 5.1.6 for $s = 0$, it proves the proposition. □

Corollary 5.1.12 *If F admits a smooth Hermitian metric h with*

$$h(R_h(u,\bar u)v,v)\le 0 \quad \text{for every } x\in B,\ u\in T^{1,0}_xB,\ v\in F_x,$$

then the least smooth convexity index of U_F is $n+r$.

Proof Put $G(v)=h(v,v)$. At $x\in B$, take a holomorphic frame that is unitary and Chern-normal at x, and write a tangent vector to F at $v\in F_x$ as (u,w), with $u\in T^{1,0}_xB$ and $w\in F_x$. The calculation in the proof of Lemma 5.1.19 gives

$$\mathcal{L}_G(u,w)=-h(R_h(u,\bar u)v,v)+|w|_h^2\ge 0,$$

with zero mixed block. Thus G is smooth, plurisubharmonic and positive off the zero section, and it satisfies $G(cv)=|c|^2G(v)$. Apply Proposition 5.1.9. □

Corollary 5.1.13 *If F admits a Hermitian metric with flat Chern connection, then the least smooth convexity index of U_F is $n+r$.*

Proof The curvature vanishes, so this is the case $R_h=0$ of Corollary 5.1.12. □

Fix $n\ge 2$ and a real number $0<q<1$. On the scalar Hopf manifold

$$X_q=(\mathbb{C}^n\setminus\{0\})/\langle z\mapsto qz\rangle,$$

a complex number $a\neq 0$ defines the holomorphic line bundle E_a, the quotient of $(\mathbb{C}^n\setminus\{0\})\times\mathbb{C}$ by $(z,v)\sim(qz,av)$. For an integer $r\ge 1$ and complex numbers $a_1,\dots,a_r\neq 0$, put

$$F=\bigoplus_{j=1}^{r}E_{a_j},\qquad b_j=-\frac{\log|a_j|}{\log q},\qquad G(z,v)=\sum_{j=1}^{r}|z|^{2b_j}|v_j|^2.$$

Since $q^{2b_j}|a_j|^2=1$, the function G descends to the squared norm of the direct sum h of Hermitian metrics on the line bundles E_{a_j}.

Let α be the form on X_q induced by $\mathrm{dd^c}\log|z|^2$; it descends because the contraction changes $\log|z|^2$ by a constant. For $u\in T^{1,0}_z(\mathbb{C}^n\setminus\{0\})$, its coefficient form is, up to the positive factor $1/(2\pi)$,

$$\frac{|u|^2}{|z|^2}-\frac{|\langle u,z\rangle|^2}{|z|^4},$$

where $\langle\cdot,\cdot\rangle$ is the Euclidean Hermitian inner product. By the Cauchy–Schwarz inequality, it is positive semidefinite with kernel $\mathbb{C}z$, hence of rank $n-1$. The curvature of h is diagonal, the line metrics have the curvature forms $-\mathrm{dd^c}\log|z|^{2b_j}=-b_j\alpha$, and therefore

$$\frac{\mathrm{i}}{2\pi}h(F_hv,v)=\left(\sum_{j=1}^{r}(-b_j)|z|^{2b_j}|v_j|^2\right)\alpha.$$

The map $X_q \to \mathbb{P}^{n-1}$, $[z] \mapsto [z]$, has compact embedded elliptic curves $C \simeq \mathbb{C}^*/q^{\mathbb{Z}}$ as fibers. On a lifted fiber $z = \zeta z_0$, the logarithm of each metric coefficient is $b_j \log|\zeta|^2 + b_j \log|z_0|^2$, which is pluriharmonic in ζ. Thus $F|_C$ has flat Chern connection, and Corollary 5.1.13 gives least index $r+1$ for $U_{F|_C}$. This is a closed complex submanifold of U_F, being its inverse image over C, so Lemma 5.1.18 shows that every smooth convexity index of U_F is at least $r+1$, for all parameters a_j.

Proposition 5.1.10 *The least smooth convexity index of U_F is $r+1$ if every $|a_j| < 1$, and $n+r$ if every $|a_j| \geq 1$.*

Proof If every $|a_j| < 1$, then every $b_j < 0$. The coefficient of α above is then strictly positive for $v \neq 0$, so Theorem 5.1.6 with $s = n-1$ shows that U_F is $(r+1)$-convex, and the lower bound $r+1$ was proved above. If every $|a_j| \geq 1$, then every $b_j \geq 0$, so $h(R_h(u,\bar{u})v, v) \leq 0$ for all u and v, and Corollary 5.1.12 applies. □

Example 5.1.2 The least smooth convexity index of U_{E_a} is two when $0 < |a| < 1$, and $n+1$ when $|a| \geq 1$. For every Gauduchon form γ on X_q, the degree of E_a with respect to γ^{n-1} is positive, zero or negative according as $|a| < 1$, $|a| = 1$ or $|a| > 1$. The de Rham first Chern class of E_a vanishes in all cases.

Proof The first assertion is the case $r = 1$, $a_1 = a$ of Proposition 5.1.10. With $b = b_1$, the curvature of the metric on E_a is $-b\alpha$. The form $\alpha \wedge \gamma^{n-1}$ is a smooth positive volume form, because α is semipositive and nonzero at every point, so the degree has the sign of $-b$, which gives the three stated cases. The map $z \mapsto (z/|z|, \log|z|)$ identifies the covering space with $S^{2n-1} \times \mathbb{R}$, and the contraction translates the second coordinate by $\log q$. Hence X_q is diffeomorphic to $S^{2n-1} \times S^1$, and $H^2_{\mathrm{dR}}(X_q, \mathbb{R}) = 0$ since $n \geq 2$. □

For a mixed sum, let $t = \#\{j : |a_j| \geq 1\}$ and assume $0 < t < r$. If $\kappa(F)$ denotes the least smooth convexity index of U_F, then

$$\max\{r+1, n+t\} \leq \kappa(F) \leq n+r.$$

Indeed, the lower bound $r+1$ was proved above. The direct sum of the t summands with $|a_j| \geq 1$ is a holomorphic subbundle $F_0 \subset F$, and U_{F_0} is a closed complex submanifold of U_F. Its least index is $n+t$ by Proposition 5.1.10, and Lemma 5.1.18 applies again. The upper bound is Theorem 5.1.6 with $s = 0$. In the mixed case these bounds leave the least index undetermined.

5.2 Conformal changes and directional curvature positivity

The compatibility condition of Proposition 4.8.1 decides which right-hand sides a scalar trace equation can have, and it can be used to improve the curvature of a Hermitian metric on a vector bundle. A conformal change $h \mapsto \mathrm{e}^{-tf}h$ adds the same form $t\,\mathrm{i}\partial\bar\partial f$ to the curvature in every fiber direction. Solving a Poisson equation for a Gauduchon form, we can make the trace of this form positive outside any prescribed ball. If the original curvature already has a positive direction for every fiber vector over that ball, a small conformal change produces such a direction everywhere.

Throughout this section, $E \to X$ is a holomorphic vector bundle of positive rank. For a smooth Hermitian metric h on E, we use the Chern curvature F_h, the Hermitian endomorphisms $R_h(u,\bar u)$, the directional forms $\alpha_{h,v}$ and the mean curvature $K_{\omega,h}$ of Section 5.1.9; recall that h is linear in its first argument.

Definition 5.2.1 The metric h is *RC-positive* at x if, for every $0 \neq v \in E_x$, there is a tangent vector $u \in T^{1,0}_x X$ such that

$$h(R_h(u,\bar u)v, v) > 0.$$

It is *uniformly RC-positive* at x if one tangent vector u works for every nonzero $v \in E_x$. A metric has either property on a set if it has that property at every point of the set.

Lemma 5.2.1 *Suppose that $K_{\omega,h}(x) \geq 0$. Then h is RC-positive at x if and only if $\alpha_{h,v} \neq 0$ for every $0 \neq v \in E_x$.*

Proof The trace of $\alpha_{h,v}$ with respect to ω is $h(K_{\omega,h}v, v) \geq 0$. A nonzero Hermitian form with nonnegative trace has a positive eigenvalue, since if all its eigenvalues were nonpositive, their nonnegative sum would force all of them to vanish. This gives the required tangent direction. Conversely, a tangent vector with positive evaluation shows that $\alpha_{h,v} \neq 0$. □

Theorem 5.2.1 *Suppose that $K_{\omega,h} \geq 0$ on X and that h is RC-positive on a nonempty open set U. Then there is a smooth real function f such that $h_t = \mathrm{e}^{-tf}h$ is RC-positive on all of X for every sufficiently small $t > 0$.*

Proof By Theorem 4.1.1, choose a conformal Gauduchon form $\gamma = \mathrm{e}^a\omega$. Then $K_{\gamma,h} = \mathrm{e}^{-a}K_{\omega,h} \geq 0$. Choose a closed coordinate ball $Q \subset U$ and a smooth function $\chi \geq 0$ supported in its interior, with $\int_X \chi\gamma^n > 0$, and set

$$\rho = 1 - \frac{\int_X \gamma^n}{\int_X \chi\gamma^n}\chi.$$

Thus $\rho = 1$ on $X \setminus Q$ and $\int_X \rho\gamma^n = 0$. Since γ is Gauduchon, Proposition 4.8.1, applied with γ in place of ω and $u = 0$, gives a smooth real solution f of

$$P_\gamma f := \mathrm{tr}_\gamma(\mathrm{i}\partial\bar\partial f) = \rho. \tag{5.3}$$

Indeed, with $L_\gamma f = n\mathrm{dd}^c f \wedge \gamma^{n-1}/\gamma^n$ as in (4.11), we have $P_\gamma = 2\pi L_\gamma$, and the adjoint probability of L_γ is $\gamma^n / \int_X \gamma^n$.

The curvature transforms by

$$F_{h_t} = F_h + t\partial\bar\partial f \ \mathrm{Id}_E, \qquad \alpha_{h_t,v} = \mathrm{e}^{-tf}\big(\alpha_{h,v} + t|v|_h^2 \mathrm{i}\partial\bar\partial f\big). \tag{5.4}$$

On $X \setminus Q$, the γ-trace of the expression in parentheses is at least $t|v|_h^2\rho = t|v|_h^2 > 0$ for $v \neq 0$, so it has a positive eigenvalue there.

Over Q, the unit sphere bundle $\{|v|_h = 1\}$ is compact. The largest γ-eigenvalue of $\alpha_{h,v}$ is continuous and strictly positive on it, so it has a positive minimum δ. Let C be the maximum on X of the operator norm of $\mathrm{i}\partial\bar\partial f$ relative to γ. For every unit vector over Q, the largest eigenvalue of the expression in parentheses in (5.4) is at least $\delta - tC$, which is positive for small $t > 0$. Homogeneity in v and the positive factor e^{-tf} prove the assertion on Q as well. □

In this conclusion, the positive direction may depend on the fiber vector v. The distinction from uniform RC-positivity already occurs for a local metric with zero mean curvature.

Example 5.2.1 On a sufficiently small ball in $\mathbb{C}^2$, equip the trivial bundle of rank two with the metric

$$H(z) = \mathrm{Id} - zz^* + \tfrac{1}{2}|z|^2 \,\mathrm{Id}\,.$$

At the origin this metric has zero mean curvature for the Euclidean Hermitian form and is RC-positive, but it is not uniformly RC-positive.

Proof The two eigenvalues of $H(z)$ are $1 - \frac{1}{2}|z|^2$ and $1 + \frac{1}{2}|z|^2$, so the metric is positive on a small ball. Its first derivatives vanish at the origin, where

$$F_{i\bar\jmath} = e_i e_j^* - \tfrac{1}{2}\delta_{ij}\,\mathrm{Id}, \qquad h(R_h(u,\bar u)v, v) = |v^* u|^2 - \tfrac{1}{2}|u|^2|v|^2.$$

The sum $F_{1\bar 1} + F_{2\bar 2}$ is zero. For each nonzero v, taking $u = v$ gives a positive evaluation. For each nonzero u, taking $v \perp u$ gives a negative evaluation, so no single u works for every v. □

Proposition 5.2.1 *If* $\det E$ *admits a flat Hermitian metric, then no smooth Hermitian metric on* E *is uniformly RC-positive on* X *or has positive definite mean curvature relative to a Hermitian form. This applies in particular when* $\det E$ *is holomorphically trivial.*

Proof If h were uniformly RC-positive, then at each point some u would make the h-Hermitian endomorphism $R_h(u,\bar u)$ positive definite, and its trace would be strictly positive. Since the curvature of the induced metric on $\det E$ is $\mathrm{tr}\, F_h$, the determinant metric would be RC-positive as a metric on a line bundle. If instead $K_{\omega,h} > 0$, the ω-trace of the determinant curvature would be $\mathrm{tr}\, K_{\omega,h} > 0$, which again gives a positive tangent evaluation.

Fix a flat Hermitian metric h_0 on $\det E$ and write $\det h = \mathrm{e}^b h_0$ for a smooth real function b. The determinant curvature is then $-\partial\bar\partial b$. At a global minimum of b, this

curvature has no positive tangent evaluation, contradicting the RC-positivity of the determinant metric. Finally, a holomorphic trivialization of $\det E$ gives a flat metric, by assigning constant norm to the trivializing frame. □

Consequently, if $\det E$ admits a flat Hermitian metric and the hypotheses of Theorem 5.2.1 hold, the conformally changed metric is RC-positive, whereas E admits neither a uniformly RC-positive metric nor a metric with positive definite mean curvature. The next section shows that the tangent bundle of a nonflat Ricci-flat Kähler surface has these properties, and the local example above shows the pointwise distinction.

5.3 Ricci-flat Kähler surfaces

The tangent bundle of a Ricci-flat Kähler surface gives an example for the conformal construction of Section 5.2. Ricci-flatness makes the mean curvature of the tangent metric zero and its determinant metric flat, but it does not make every directional curvature zero. In real dimension four, the remaining curvature acts on the three-dimensional space of anti-self-dual two-forms, and its invertibility gives exactly the positive directions needed for the conformal construction.

In this section, X is a compact complex surface, so $n = 2$, and ω is a Ricci-flat Kähler form on X. Write h for the Hermitian metric on $T^{1,0}X$ defined by ω, and g for the associated Riemannian metric, so that $\omega(S,T) = g(JS,T)$.

Use g to identify bivectors and two-forms. For the complex orientation, the Hodge star gives

$$\Lambda^2 = \Lambda^+ \oplus \Lambda^-, \qquad \Lambda^\pm = \{\zeta : *\zeta = \pm\zeta\}.$$

Each summand has real rank three. The first is spanned by ω and the real parts of $(2,0)$-forms, and the second consists of the real primitive $(1,1)$-forms; both descriptions follow by writing the star in a unitary real frame.

Let R^∇ be the curvature of the Levi–Civita connection, with $R^\nabla(S,T) = \nabla_S\nabla_T - \nabla_T\nabla_S - \nabla_{[S,T]}$, and define the self-adjoint curvature operator $\mathscr{R}$ on Λ^2 by

$$\langle \mathscr{R}(S \wedge T), U \wedge V\rangle = g(R^\nabla(S,T)V,U).$$

Since the curvature of a Kähler metric has type $(1,1)$, the operator $\mathscr{R}$ annihilates the real parts of $(2,0)$-forms. Its value on ω is the Ricci form, which vanishes. Thus

$$\mathscr{R} = 0 \oplus A, \qquad A\colon \Lambda^- \longrightarrow \Lambda^-.$$

The endomorphism A is symmetric and trace free, since the trace of $\mathscr{R}$ is half the scalar curvature; in this Ricci-flat setting it is the anti-self-dual Weyl operator. Norms of curvature tensors below are Hilbert–Schmidt norms of these endomorphisms.

Lemma 5.3.1 *At a point x, the metric h is RC-positive if and only if A_x is invertible.*

Proof For a real tangent vector V, set $v = \frac{1}{2}(V - \mathrm{i}JV)$. The Chern and Levi–Civita connections agree on a Kähler manifold. Their curvature pairings give

$$\alpha_{h,v} = \tfrac{1}{2}\big(\mathscr{R}(V \wedge JV)\big)^\flat.$$

Indeed, $h(v,w) = \frac{1}{2}(g(V,W) - \mathrm{i}g(JV,W))$ for $w = \frac{1}{2}(W - \mathrm{i}JW)$, and R^∇ commutes with J. Since R^∇ is skew-adjoint, the real part of $h(F_h v, v)$ vanishes, which gives the formula.

The anti-self-dual projection of $V \wedge JV$ is

$$\Pi(V) = V \wedge JV - \tfrac{1}{2}|V|^2\omega.$$

It is nonzero for $V \neq 0$, and every nonzero element of Λ^- is of this form. To see this, diagonalize a nonzero real primitive $(1,1)$-form in a unitary real frame. It has the form $c(e_1 \wedge Je_1 - e_2 \wedge Je_2)$, with $c > 0$ after interchanging the two complex lines. Taking $V = \sqrt{2c}\, e_1$ gives exactly this form as $\Pi(V)$. Also $|\Pi(V)|^2 = |V|^4/2$.

Consequently, since $\mathscr{R}$ vanishes on Λ^+, we have $\alpha_{h,v} \neq 0$ for every nonzero v if and only if $\ker A_x = 0$. Since $K_{\omega,h} = 0$, Lemma 5.2.1 gives the assertion. □

Lemma 5.3.2 *If g is not flat, the open set $\{x : \det A_x \neq 0\}$ is nonempty.*

Proof Suppose that $\det A = 0$ everywhere. On the open set where $A \neq 0$, its three eigenvalues are $\lambda_0, -\lambda_0, 0$ with $\lambda_0 > 0$; they are distinct, so A has smooth local eigenframes there.

Step 1: a Bochner identity. On an open set where A has distinct eigenvalues, we compute $\Delta|A|^2$, keeping track of the normalizations.

Choose an oriented local eigenframe ξ_1, ξ_2, ξ_3 of Λ^- with $|\xi_i|^2 = 2$, and denote the corresponding eigenvalues by λ, μ, ν. As skew-adjoint endomorphisms, these forms satisfy the multiplication rules of the quaternions. Write

$$\nabla\xi_1 = c \otimes \xi_2 - b \otimes \xi_3, \quad \nabla\xi_2 = -c \otimes \xi_1 + a \otimes \xi_3, \quad \nabla\xi_3 = b \otimes \xi_1 - a \otimes \xi_2.$$

Here a, b, c are real one-forms. The Ricci identity and the contracted Bianchi identity, using $\mathrm{Ric}(g) = 0$, give

$$\mathrm{d}a + b \wedge c = \lambda\xi_1, \qquad \mathrm{d}\lambda = (\lambda - \mu)\xi_3 c + (\lambda - \nu)\xi_2 b,$$

and their simultaneous cyclic permutations. In the second formula the two-forms act as skew-adjoint endomorphisms on one-forms. Put $D = \mathrm{d}^*\mathrm{d} = -\operatorname{div}_g \nabla$ on functions. Applying d^* to the second formula yields

$$D\lambda = 2\lambda^2 + 4\mu\nu + 2(\nu - \lambda)|b|^2 + 2(\mu - \lambda)|c|^2.$$

For example, the calculation uses

$$\mathrm{d}^*(\xi_3 c) = -\langle b, \xi_1 c\rangle + \langle a, \xi_2 c\rangle + \langle a, \xi_3 b\rangle - 2\nu,$$
$$\mathrm{d}^*(\xi_2 b) = -\langle b, \xi_1 c\rangle - \langle a, \xi_2 c\rangle - \langle a, \xi_3 b\rangle - 2\mu,$$

which follow from the connection and curvature formulas above; substitution and the cyclic formulas for $\mathrm{d}\mu, \mathrm{d}\nu$ cancel the mixed pairings.

Now $\lambda + \mu + \nu = 0$ and $|A|^2 = \lambda^2 + \mu^2 + \nu^2$. Multiplying the three scalar equations by their eigenvalues and adding gives

$$\lambda D\lambda + \mu D\mu + \nu D\nu = 18\lambda\mu\nu - 2\big((\mu-\nu)^2|a|^2 + (\nu-\lambda)^2|b|^2 + (\lambda-\mu)^2|c|^2\big).$$

The connection expression also gives

$$|\nabla A|^2 = |\mathrm{d}\lambda|^2 + |\mathrm{d}\mu|^2 + |\mathrm{d}\nu|^2 + 2\big((\mu-\nu)^2|a|^2 + (\nu-\lambda)^2|b|^2 + (\lambda-\mu)^2|c|^2\big).$$

The product rule therefore gives, with $\Delta = \mathrm{div}_g \nabla = -D$,

$$\Delta|A|^2 = 2|\nabla A|^2 - 36\det A = 2|\nabla A|^2. \tag{5.5}$$

This identity also holds on the interior of $\{A = 0\}$. Both sides are smooth functions, and the two open sets have dense union in X, so the identity holds everywhere. Integration over X gives $\nabla A = 0$.

Step 2: a parallel operator A forces flatness. If g is not flat, then $A \neq 0$ somewhere, and since A is parallel and $\det A = 0$, it is nonzero everywhere and has three distinct constant eigenvalues. Its eigenlines are parallel; local unit sections of them are parallel as well. The induced connection on Λ^- therefore has zero curvature. Under $\mathfrak{so}(4) = \Lambda^+ \oplus \Lambda^-$, the two summands commute, while the action of Λ^- on itself is faithful. These facts follow directly from the bases $e_1 \wedge e_2 \pm e_3 \wedge e_4$, $e_1 \wedge e_3 \mp e_2 \wedge e_4$, $e_1 \wedge e_4 \pm e_2 \wedge e_3$. Thus zero induced curvature forces $\mathscr{R}(S \wedge T)$ to lie in Λ^+ for every S, T. It already lies in Λ^-, so $\mathscr{R} = 0$, a contradiction. □

Theorem 5.3.1 *If X has a nonflat Ricci-flat Kähler metric, then $T^{1,0}X$ admits an RC-positive Hermitian metric whose associated $(1,1)$-form is nonclosed. It admits neither a uniformly RC-positive Hermitian metric nor a metric with positive definite mean curvature relative to any Hermitian form.*

Proof By Lemmas 5.3.1 and 5.3.2, the original tangent metric h is RC-positive on a nonempty open set. Its mean curvature is zero, so Theorem 5.2.1 gives a metric $h_t = \mathrm{e}^{-tf}h$ that is RC-positive everywhere.

In that construction we may take the Gauduchon form to be ω itself, and the closed coordinate ball to be a proper subset of X. Since $P_\omega f = 1$ off that ball, f is not constant. The associated form is $\omega_t = \mathrm{e}^{-tf}\omega$, with

$$\mathrm{d}\omega_t = -t\mathrm{e}^{-tf}\mathrm{d}f \wedge \omega.$$

On a real four-dimensional symplectic vector space, $\zeta \mapsto \zeta \wedge \omega$ is injective on one-forms, as a unitary real frame shows. Thus $\mathrm{d}\omega_t \neq 0$ somewhere.

Finally, the induced determinant metric of h is flat because its Chern curvature is the Ricci form of ω, which vanishes. The two obstructions follow from Proposition 5.2.1.□

Nonflat Ricci-flat Kähler metrics exist, for example, on K3 surfaces, by Yau's solution of the Calabi conjecture.

Combined with Theorem 5.1.6, the RC-positive tangent metric of Theorem 5.3.1 bounds the convexity of the complement of the zero section in the projective completion of the tangent bundle. This bundle admits no metric with positive definite mean curvature, so Corollary 5.1.9 does not apply to it.

Corollary 5.3.1 *If X has a nonflat Ricci-flat Kähler metric, then the complement*

$$\mathbb{P}(\mathcal{O}_X \oplus T^{1,0}X) \setminus \mathbb{P}(\mathcal{O}_X)$$

is 3*-convex.*

Proof Apply Theorem 5.1.6, with $n = r = 2$ and $s = 1$, to the RC-positive tangent metric of Theorem 5.3.1. □

5.4 Holomorphic sectional curvature

The curvature conditions above test a tangent direction and a fiber vector separately. For a metric on the tangent bundle, holomorphic sectional curvature tests the same vector in both places. Averaging these diagonal evaluations controls a symmetrized curvature trace. A second trace occurs when one differentiates the trace of the metric itself; their difference contains torsion and an exact form. With a Kähler metric for the contraction, integration removes the exact form and the torsion terms cancel. This gives an obstruction even when the metric whose curvature is tested is not closed.

In this section, $n \geq 1$, g is a Kähler metric on X and h is any smooth Hermitian metric on $T^{1,0}X$. In local holomorphic coordinates, their forms are $\omega_g = \mathrm{i}\sum_{k,l} g_{k\bar{l}}\,\mathrm{d}z^k \wedge \mathrm{d}\bar{z}^l$ and $\omega_h = \mathrm{i}\sum_{k,l} h_{k\bar{l}}\,\mathrm{d}z^k \wedge \mathrm{d}\bar{z}^l$. The unscaled Chern curvature convention of Section 5.1.9 gives the coefficients

$$R^h_{i\bar{j}k\bar{l}} = -\partial_i\partial_{\bar{j}} h_{k\bar{l}} + h^{p\bar{q}}(\partial_i h_{k\bar{q}})(\partial_{\bar{j}} h_{p\bar{l}}).$$

The *Chern holomorphic sectional curvature* is

$$H_h(x,[v]) = \frac{R^h(v,\bar{v},v,\bar{v})}{|v|_h^4}, \qquad 0 \neq v \in T^{1,0}_x X. \tag{5.6}$$

It is *quasi-negative* if it is nonpositive everywhere and strictly negative in every direction at some point. Put

$$u = \operatorname{tr}_g h, \qquad q = \operatorname{tr}\big((g^{-1}h)^2\big).$$

In a g-unitary frame, define the real contractions

$$Q_g(h) = \sum_{i,k} R^h_{i\bar{i}k\bar{k}}, \qquad B_g(h) = \sum_{i,k} R^h_{i\bar{k}k\bar{i}}, \qquad \mathcal{R}_g(h) = \tfrac{1}{2}\big(Q_g(h) + B_g(h)\big).$$

These expressions are invariant under a change of unitary frame. For a general h, the two traces $Q_g(h)$ and $B_g(h)$ can differ.

Lemma 5.4.1 *At a point where $H_h \le -\kappa$ for some $\kappa \ge 0$, one has*

$$\mathcal{R}_g(h) \le -\frac{\kappa}{2}(u^2 + q).$$

In particular, nonpositive holomorphic sectional curvature gives $\mathcal{R}_g(h) \le 0$.

Proof In a g-unitary frame, let $Z_1, \dots, Z_n$ be independent standard complex Gaussian variables, with density $\pi^{-1}\mathrm{e}^{-|z|^2}$ on $\mathbb{C}$. Their second and fourth moments are $\mathbb{E}|Z_i|^2 = 1$ and $\mathbb{E}|Z_i|^4 = 2$. Independence and invariance under coordinate phases give

$$\mathbb{E}[Z_i\overline{Z_j}Z_k\overline{Z_l}] = \delta_{ij}\delta_{kl} + \delta_{il}\delta_{kj}.$$

Thus $\mathbb{E}R^h(Z, \bar{Z}, Z, \bar{Z}) = Q_g(h) + B_g(h)$ and $\mathbb{E}|Z|^4_h = u^2 + q$. Taking expectations in the homogeneous inequality $R^h(Z, \bar{Z}, Z, \bar{Z}) \le -\kappa|Z|^4_h$ proves the result. □

Let $A = \nabla^h - \nabla^g$ be the difference of the Chern connections, viewed as a tensor with two holomorphic covariant indices and a tangent-bundle value. Write

$$S(v, w) = \tfrac{1}{2}\big(A(v, w) + A(w, v)\big), \qquad T(v, w) = A(v, w) - A(w, v).$$

Since g is Kähler, T is the Chern torsion of h. Norms such as $|S|^2_{g,h} = \sum_{i,k} |S(e_i, e_k)|^2_h$, where (e_i) is a g-unitary frame, use g in the covariant indices and h in the value. Let $r_{i\bar{j}}$ be the coefficients of the Ricci form $\mathrm{Ric}(g) = -\mathrm{i}\partial\bar{\partial}\log\det(g_{i\bar{j}})$, and in a g-unitary frame set

$$I_g(h) = \sum_{i,k} r_{i\bar{k}}h_{k\bar{i}}, \qquad \mathrm{d}V_g = \frac{\omega_g^n}{n!}, \qquad P_g f = \mathrm{tr}_{\omega_g}(\mathrm{i}\partial\bar{\partial}f).$$

Proposition 5.4.1 *The two metrics satisfy*

$$\int_X \big(|S|^2_{g,h} + I_g(h) - \mathcal{R}_g(h)\big)\,\mathrm{d}V_g = 0. \tag{5.7}$$

Proof We first compute the exact term that distinguishes the two curvature traces. Define

$$\begin{aligned} D_{i\bar{j}k\bar{l}} = {} & \partial_i\partial_{\bar{j}}h_{k\bar{l}} + \partial_k\partial_{\bar{l}}h_{i\bar{j}} \\ & - \partial_i\partial_{\bar{l}}h_{k\bar{j}} - \partial_k\partial_{\bar{j}}h_{i\bar{l}}. \end{aligned}$$

This is a tensor: putting holomorphic differentials first gives

$$\mathrm{i}\partial\bar\partial\omega_h = \sum_{i<k,\, j<l} D_{i\bar j k\bar l}\, \mathrm{d}z^i \wedge \mathrm{d}z^k \wedge \mathrm{d}\bar z^j \wedge \mathrm{d}\bar z^l.$$

Put $D_g = \sum_{i,k} D_{i\bar i k\bar k}$ in a g-unitary frame. For $n = 1$, the only coefficient $D_{1\bar 1 1\bar 1}$ vanishes, so $D_g = 0$. For $n \geq 2$, expansion of the wedge product gives

$$D_g\, \mathrm{d}V_g = 2\mathrm{i}\partial\bar\partial\omega_h \wedge \frac{\omega_g^{n-2}}{(n-2)!}.$$

Indeed, each pair $i < k$ contributes $D_{i\bar i k\bar k}\mathrm{d}V_g$ to the wedge product and occurs twice in the contraction, while the terms with $i = k$ vanish. Closedness of ω_g and Stokes' theorem therefore give $\int_X D_g\, \mathrm{d}V_g = 0$ in all dimensions.

The four-term antisymmetrization of the curvature formula is

$$\begin{aligned} R^h_{i\bar j k\bar l} + R^h_{k\bar l i\bar j} - R^h_{i\bar l k\bar j} - R^h_{k\bar j i\bar l} \\ = h_{p\bar q} T^p_{ik} \overline{T^q_{jl}} - D_{i\bar j k\bar l}. \end{aligned}$$

The second derivatives give $-D_{i\bar j k\bar l}$. The quadratic term of $R^h_{i\bar j k\bar l}$ is $h_{p\bar q}\Gamma^p_{ik}\overline{\Gamma^q_{jl}}$, where $\Gamma^p_{ik} = h^{p\bar q}\partial_i h_{k\bar q}$ are the Chern connection coefficients of h. Since g is Kähler, $\Gamma^p_{ik} - \Gamma^p_{ki} = T^p_{ik}$, so the four quadratic terms combine to $h_{p\bar q}T^p_{ik}\overline{T^q_{jl}}$. Contracting gives $2(Q_g(h) - B_g(h)) = |T|^2_{g,h} - D_g$, hence

$$Q_g(h) = \mathcal{R}_g(h) + \tfrac{1}{4}|T|^2_{g,h} - \tfrac{1}{4}D_g.$$

Now choose holomorphic coordinates that are normal for g at a given point, so that g is the identity and its first derivatives vanish there; then A has the coefficients Γ^p_{ik} at this point. When $u = g^{k\bar l}h_{k\bar l}$ is differentiated twice, the second derivatives of the inverse metric contribute $I_g(h)$, by the Kähler curvature symmetries of g, while the second derivatives of h contribute $-Q_g(h) + |A|^2_{g,h}$. Thus

$$\begin{aligned} P_g u &= |A|^2_{g,h} + I_g(h) - Q_g(h) \\ &= |S|^2_{g,h} + I_g(h) - \mathcal{R}_g(h) + \tfrac{1}{4}D_g, \end{aligned}$$

where symmetric and skew tensors are orthogonal and $|A|^2_{g,h} = |S|^2_{g,h} + |T|^2_{g,h}/4$. Integrating removes both $P_g u$ and D_g, proving (5.7). □

Corollary 5.4.1 *Suppose that* $\mathrm{Ric}(g) \geq 0$. *Every Hermitian tangent metric* h *with* $H_h \leq 0$ *satisfies* $\mathrm{Ric}(g) = 0$ *and* $H_h = 0$ *in every direction. In particular,* X *admits no Hermitian tangent metric with quasi-negative Chern holomorphic sectional curvature.*

Proof The Ricci assumption gives $I_g(h) \geq 0$, and Lemma 5.4.1 gives $-\mathcal{R}_g(h) \geq 0$. All three continuous terms in (5.7) are nonnegative, so each vanishes everywhere. A positive definite h pairs strictly positively with every nonzero semipositive Ricci matrix; hence $I_g(h) = 0$ forces $\mathrm{Ric}(g) = 0$.

At each point, the Gaussian expectation in the averaging proof is now zero. Its integrand $R^h(Z, \bar{Z}, Z, \bar{Z})$ is nonpositive. If it were negative in one direction, continuity and the strictly positive Gaussian density would make its expectation negative. It is therefore zero in every direction, which gives $H_h = 0$ and excludes quasi-negativity.□

For the Ricci-flat surfaces of Section 5.3, this obstruction concerns all Hermitian tangent metrics, including nonclosed ones. It coexists with the RC-positive metric of Theorem 5.3.1: that condition allows the positive tangent direction to depend on the fiber vector, while holomorphic sectional curvature evaluates the same direction twice.

5.4.1 Rigidity on complex tori

For a flat reference metric, the vanishing of the symmetric part S of the connection difference has a stronger consequence. Periodic integration by parts forces the second metric to be translation invariant, and hence to have zero full Chern curvature.

Proposition 5.4.2 *Let* $Y = \mathbb{C}^m/\Lambda$ *be a complex torus, with* $m \geq 1$. *A smooth Hermitian tangent metric* h *on* Y *has nonpositive Chern holomorphic sectional curvature if and only if it is translation invariant. Such a metric is Kähler and has zero Chern curvature.*

Proof Suppose that $H_h \leq 0$, and take the flat Kähler metric g induced by a constant Hermitian inner product on $\mathbb{C}^m$. Choose linear coordinates in which this inner product is the identity. The coordinate vector fields descend to Y, even though the coordinates themselves need not. Since $I_g(h) = 0$ for the flat metric g and $\mathcal{R}_g(h) \leq 0$ by Lemma 5.4.1, (5.7) gives $S = 0$. Since g has zero connection coefficients, multiplying $S = 0$ by h gives

$$\partial_i h_{k\bar{l}} + \partial_k h_{i\bar{l}} = 0 \qquad \text{for all } i, k, l.$$

Fix l and put $a_k = h_{k\bar{l}}$. With respect to the flat volume, integration by parts and commutation of constant coordinate derivatives give

$$\sum_{i,k} \int_Y \partial_i a_k \, \overline{\partial_k a_i} \, \mathrm{d}V_g = \int_Y \left| \sum_k \partial_{\bar{k}} a_k \right|^2 \mathrm{d}V_g.$$

Consequently,

$$0 = \sum_{i,k} \int_Y |\partial_i a_k + \partial_k a_i|^2 \, \mathrm{d}V_g$$
$$= 2 \sum_{i,k} \int_Y |\partial_i a_k|^2 \, \mathrm{d}V_g + 2 \int_Y \left| \sum_k \partial_{\bar{k}} a_k \right|^2 \mathrm{d}V_g .$$

Thus every $\partial_i h_{k\bar{l}}$ vanishes. Hermitian symmetry gives the vanishing of the conjugate derivatives as well, so all coefficients of h are constant.

Conversely, a constant positive Hermitian matrix defines a closed form and has zero connection and curvature coefficients. This proves all the assertions. □

Corollary 5.4.2 *Let $\pi : Z \to Y$ be an unramified holomorphic covering of a complex manifold Y by a complex torus Z. Every smooth Hermitian tangent metric on Y with nonpositive Chern holomorphic sectional curvature is Kähler and Chern flat.*

Proof The covering is locally biholomorphic, so the pullback metric again has nonpositive holomorphic sectional curvature. By Proposition 5.4.2, its associated form is closed and its Chern curvature is zero. Both conclusions descend through the covering. □

5.5 Projective first jets and positive bundle metrics

RC-positivity asks for one positive tangent direction for each fiber vector, whereas Griffiths positivity asks for positivity on every pair of nonzero tangent and fiber vectors. The Lamari criterion of Theorem 4.3.1 characterizes, by a duality with Gauduchon tests, the smooth real $(1,1)$-forms θ that admit a positive current $\theta + \mathrm{dd}^{\mathrm{c}} u$, and Griffiths positivity on a projective manifold admits an analogous duality. A strictly Griffiths-positive metric on E amounts to a strictly plurisubharmonic squared norm on the dual bundle, and the Levi form of this squared norm depends linearly on the dual Hermitian form. Twisting by powers of a very ample line bundle gives finite-dimensional spaces of candidate forms. At each level, either some candidate is strictly plurisubharmonic or a finite atomic positive measure annihilates the Levi forms of all candidates, and if every level fails, a weak limit of these measures annihilates the Levi forms of all smooth Hermitian forms. This measure lives over the dual bundle, whereas the Gauduchon currents in Theorem 4.3.1 live on the base.

Throughout this section, X is a projective manifold of any dimension $n \geq 1$, $E \to X$ is a holomorphic vector bundle of positive rank, and $F = E^*$. Fix a very ample line bundle A, a basis $t_1, \ldots, t_d$ of $H^0(X, A)$, and the Fubini–Study metric a on A determined by this basis. Fix also an integer $m_0 \geq 0$ for which $E \otimes A^{m_0}$ is globally generated; this integer is part of the data. In a local frame of A, put

$$\rho = \sum_{\alpha=1}^{d} |t_\alpha|^2, \qquad a = \mathrm{e}^{-\phi}, \qquad \phi = \log \rho .$$

We use the curvature conventions of Section 5.1.9. A Hermitian metric g on a holomorphic vector bundle over X is *strictly Griffiths-positive* if $g(R_g(u,\bar{u})v,v) > 0$ for every $x \in X$, every nonzero $u \in T^{1,0}_x X$ and every nonzero vector v in the fiber over x; it is *strictly Griffiths-negative* if the reverse strict inequality holds.

For a smooth Hermitian form h on F, not initially assumed positive, let $G_h(v) = h(v,v)$ on its total space. Write $\mathcal{L}G_h(\xi)$ for the complex Hessian evaluated on $\xi \in T^{1,0}F$; in local coordinates it is $\sum (G_h)_{j\bar{k}}\xi_j\bar{\xi}_k$, the coefficient form of $\mathrm{i}\partial\bar{\partial}G_h$, without the factor $1/(2\pi)$.

Lemma 5.5.1 *The function G_h is strictly plurisubharmonic off the zero section if and only if h is a smooth positive Hermitian metric with strictly Griffiths-negative curvature. Equivalently, its dual metric on E is strictly Griffiths-positive.*

Proof On vertical tangent vectors the Levi form is h itself, so strict positivity forces $h > 0$. For $h > 0$, write a tangent vector to F at v by its horizontal and vertical components (u,w) in a holomorphic frame that is unitary and Chern-normal at the base point. The calculation in the proof of Lemma 5.1.19 gives

$$\mathcal{L}G_h(u,w) = |w|^2_h - h(R_h(u,\bar{u})v,v),$$

without mixed terms. For $v \neq 0$, this is positive for every nonzero (u,w) precisely when the curvature evaluation is negative for every nonzero u. The curvature of the dual metric is the negative transpose of the original curvature, with respect to the dual Hermitian pairing, which gives the last assertion. □

The Levi forms will be tested on a compact set. Choose any smooth Hermitian metric k on F, let $S = \{v : k(v,v) = 1\}$ be its unit sphere bundle, and choose a smooth positive Hermitian form b on the complex vector bundle $W = T^{1,0}F|_S$ over S. The set

$$K = \{(v,\xi) : v \in S,\ \xi \in W_v,\ b(\xi,\xi) = 1\}$$

is compact. Define the real-linear operator D by $Dh = \mathcal{L}G_h|_S$, viewed as a Hermitian form on W. It is continuous from the C^2 norm of Hermitian forms on F to the uniform norm on W, since its coefficients involve at most two base derivatives of h and the fiber vectors range over a compact set. Fiber dilation $v \mapsto cv$ gives

$$G_h(cv) = |c|^2 G_h(v), \qquad \mathcal{L}G_h|_{cv}(\mathrm{d}(v \mapsto cv)\xi) = |c|^2 \mathcal{L}G_h|_v(\xi).$$

Thus strict positivity of $\mathcal{L}G_h$ on K is equivalent to strict positivity off the zero section.

5.5.1 The finite-dimensional alternative

For $m \geq m_0$, choose a basis of $V_m = H^0(X, E \otimes A^m)$ and write $N_m = \dim V_m$. Pairing these sections with $v \in F_x$ and using a local frame of A^m, we obtain a column $p_m(v)$

of local holomorphic functions on the total space of F. For any Hermitian $N_m \times N_m$ matrix B, set

$$G_{m,B}(v) = \mathrm{e}^{-m\phi(x)}\, p_m(v)^* B p_m(v).$$

This function is globally defined and quadratic on each fiber, so $G_{m,B} = G_{h_{m,B}}$ for a smooth Hermitian form $h_{m,B}$ on F. The matrix B is allowed to have negative eigenvalues.

For $(v,\xi) \in K$, let u be the projection of ξ to $T^{1,0}_x X$, write $\phi_u = \partial\phi(u)$, and put

$$r_m = \mathrm{d}p_m(\xi) - m\phi_u p_m(v), \qquad \kappa = \phi_{u\bar{u}},$$
$$\Psi_m(v,\xi) = \mathrm{e}^{-m\phi}\bigl(r_m r_m^* - m\kappa p_m p_m^*\bigr).$$

Differentiating the weighted quadratic function gives

$$(Dh_{m,B})_v(\xi,\xi) = \operatorname{tr}(B\Psi_m(v,\xi)). \tag{5.8}$$

All entries involve only the first jets of the sections: if $t = (t_\alpha)$ is the column of the sections of A, then

$$\phi_u = \frac{t^*\mathrm{d}t(u)}{\rho}, \qquad \kappa = \frac{\rho|\mathrm{d}t(u)|^2 - |t^*\mathrm{d}t(u)|^2}{\rho^2}.$$

Changing the frame of A by a nowhere-zero holomorphic factor λ changes p_m to $\lambda^{-m}p_m$ and ϕ to $\phi - \log|\lambda|^2$. It changes r_m by the same factor and leaves κ unchanged, so Ψ_m is frame independent. A change of the basis of V_m replaces Ψ_m by a congruent matrix, so whether the convex hull of the values of Ψ_m contains zero does not depend on that choice.

Proposition 5.5.1 *At each fixed m, exactly one of the following holds:*

(1) *Some Hermitian matrix B satisfies $\operatorname{tr}(B\Psi_m) > 0$ everywhere on K.*
(2) *There is a probability measure μ on K, supported on at most $N_m^2 + 1$ points, such that $\int_K \Psi_m \,\mathrm{d}\mu = 0$.*

The first alternative holds precisely when $0 \notin \operatorname{conv}(\Psi_m(K))$.

Proof The real vector space of Hermitian $N_m \times N_m$ matrices has dimension N_m^2. Carathéodory's theorem expresses every point of the convex hull as a convex combination of at most $N_m^2 + 1$ points of the compact image $\Psi_m(K)$, so the hull is compact. If it contains zero, the coefficients of such an expression define the required atomic probability measure μ, and then $\int_K \operatorname{tr}(B\Psi_m)\,\mathrm{d}\mu = 0$ excludes the first alternative.

If the hull misses zero, let Z_0 be its point of least Hilbert–Schmidt norm. Differentiating the squared norm along the segment from Z_0 to any Z in the hull gives

$$\operatorname{tr}(Z_0 Z) \geq \operatorname{tr}(Z_0^2) > 0.$$

Taking $B = Z_0$ proves the first alternative. □

5.5.2 Approximation with two derivatives

Let $\mathcal{H}_m = \{h_{m,B} : B = B^*\}$. To pass from these finite-dimensional tests to all smooth metrics, we need approximation in C^2, since D involves two derivatives.

Lemma 5.5.2 *The matrix-valued map $Y = tt^*/\rho$ embeds X as a compact smooth submanifold of the real vector space of Hermitian $d \times d$ matrices. Restrictions of real polynomials in its real coordinates are dense in $C^2(X, \mathbb{R})$ for the C^2 norm.*

Proof The normalized rank-one projector Y determines the point $[t]$ of projective space, and its differential vanishes only in directions in which $\mathrm{d}t$ is proportional to t. Since A is very ample, Y is therefore injective with injective differential, and compactness makes it an embedding. A C^2 function on this image extends to a compactly supported C^2 function f on the ambient space $\mathbb{R}^{d^2}$: use local coordinates in which the image is a graph, extend constantly in the normal coordinates, and combine these extensions with a finite smooth partition of unity and a cutoff.

The convolutions of f with the Gaussian kernels $G_\varepsilon(y) = (\pi\varepsilon)^{-d^2/2}\mathrm{e}^{-|y|^2/\varepsilon}$ converge to f in C^2 as $\varepsilon \to 0$, since $\partial^\alpha(f * G_\varepsilon) = (\partial^\alpha f) * G_\varepsilon$ for $|\alpha| \le 2$ and these derivatives of f are uniformly continuous. Each $f * G_\varepsilon$ is the restriction of an entire function on $\mathbb{C}^{d^2}$, because f has compact support, so its Taylor polynomials at the origin converge to it in C^2 on every compact set. Thus real polynomials approximate f in C^2 on a neighborhood of the image. Restrict to the embedded image and use the chain rule in a finite collection of charts to obtain the asserted C^2 approximation. □

Lemma 5.5.3 *The spaces $\mathcal{H}_m$ are nested, and their union is C^2 dense in the space of C^2 Hermitian forms on F.*

Proof For nesting, multiply the sections in V_m by each t_α and express the products in the basis of V_{m+1}. The identity $\sum_\alpha \mathrm{e}^{-\phi}|t_\alpha|^2 = 1$ writes $G_{m,B}$ as the sum of the corresponding weighted quadratic functions at level $m+1$, which combine into one Hermitian coefficient matrix at that level. Surjectivity of the multiplication map is not needed.

For density, write $p_{m_0}(v) = P_0 v$ in a local frame of F, and put $L = \mathrm{e}^{-m_0\phi/2}P_0$. Global generation of $E \otimes A^{m_0}$ means that L is injective, so $M = L^*L$ is positive definite. If a C^2 Hermitian form has matrix H, define

$$C(x) = LM^{-1}HM^{-1}L^*.$$

Then $L^*CL = H$. A change of frame of F cancels in this expression; a change of frame of A multiplies L by a scalar of modulus one, which also cancels. Thus C is a globally defined C^2 Hermitian matrix function on X.

Applying Lemma 5.5.2 to its real coordinates in the vector space of Hermitian matrices, we obtain Hermitian matrix polynomials in the entries of Y that converge to C in C^2; their degrees may increase along the sequence. Let $C_q(Y)$ be such a polynomial, of degree at most q. Every monomial of degree k in Y has the form $t^I\bar{t}^J/\rho^k$, with $|I| = |J| = k$. For a polynomial of degree at most q, multiply each numerator by ρ^{q-k} to use the common denominator ρ^q. Consequently

$$e^{-m_0\phi} p_{m_0}^* C_q(Y) p_{m_0}$$

is a quadratic expression in the product sections $\sigma_i t^I$ of $E \otimes A^{m_0+q}$, with weight $e^{-(m_0+q)\phi}$. Express these products in the basis of V_{m_0+q} and replace the coefficient matrix by its Hermitian part, which does not change the real function. The result is an element of $\mathcal{H}_{m_0+q}$. Since $L^*CL = H$ and multiplication by the fixed smooth matrix L is continuous in C^2, these elements converge to H. □

5.5.3 The metric and measure alternatives

A *positive matrix-valued measure* on $W \to S$ is a linear functional T on the space of continuous Hermitian forms Φ on W such that $\langle T, \Phi\rangle \geq 0$ whenever $\Phi \geq 0$. Its trace mass relative to b is $\langle T, b\rangle$. We write $D^*T = 0$ if $\langle T, Dh\rangle = 0$ for every smooth Hermitian form h on F.

Theorem 5.5.1 *With the fixed projective data above, the following are equivalent:*

(1) *E admits a smooth strictly Griffiths-positive Hermitian metric.*
(2) *For some $m \geq m_0$, a Hermitian matrix B makes $G_{m,B}$ strictly plurisubharmonic off the zero section.*
(3) *For some $m \geq m_0$, $0 \notin \operatorname{conv}(\Psi_m(K))$.*
(4) *There is no nonzero positive matrix-valued measure T on $W \to S$ satisfying $D^*T = 0$.*

Proof Given a metric as in the first assertion, its dual metric h on F satisfies $Dh \geq \delta b$ on S for some $\delta > 0$, by Lemma 5.5.1 and compactness. The C^2 density in Lemma 5.5.3 and the continuity of D give a form $h_{m,B}$ with $Dh_{m,B} \geq \delta b/2$, which proves the second assertion. Conversely, if $G_{m,B}$ is strictly plurisubharmonic, its vertical Levi form shows that $h_{m,B}$ is positive, and Lemma 5.5.1 shows that its dual metric is strictly Griffiths-positive. The second and third assertions are equivalent by Proposition 5.5.1.

If the first assertion holds, every nonzero positive matrix-valued measure T has positive trace mass, and

$$\langle T, Dh\rangle \geq \delta\langle T, b\rangle > 0,$$

so such a measure cannot satisfy $D^*T = 0$. Conversely, suppose that the third assertion fails at every level. By Proposition 5.5.1, choose at each level m an atomic probability measure μ_m on K with $\int_K \Psi_m \, d\mu_m = 0$, so that μ_m annihilates Dh for every $h \in \mathcal{H}_m$. A subsequence converges weakly to a probability measure μ on the compact set K. Since the spaces $\mathcal{H}_m$ are nested, the limit annihilates Dh for every h in each fixed $\mathcal{H}_j$, and the C^2 density and the continuity of D extend this to every smooth Hermitian form.

Define

$$\langle T, \Phi\rangle = \int_K \Phi_v(\xi, \xi) \, d\mu(v, \xi).$$

This is a positive matrix-valued measure on $W \to S$ with $\langle T, b\rangle = 1$ and $D^*T = 0$. Its entries in local frames are represented by scalar measures: positivity bounds the functional on Hermitian forms of uniform norm at most one by the trace mass, so the Riesz representation theorem applies to the coordinate functionals, and a finite partition of unity combines these local representations. This contradicts the fourth assertion and completes the proof. □

The equation $D^*T = 0$ concerns a bundle over the sphere bundle $S \subset F = E^*$; it should not be confused with the equation $\mathrm{dd}^{\mathrm{c}}T = 0$ for currents on X. The theorem gives no bound for the first successful level, and no finite procedure for deciding that all levels fail.

Corollary 5.5.1 *If E is globally generated and $m_0 = 0$, the level-zero test succeeds exactly when the complete evaluation map $\Phi\colon F \to H^0(X, E)^*$ is an immersion off the zero section.*

Proof At $m = 0$, one has $\Psi_0 = \mathrm{d}\Phi(\xi)\mathrm{d}\Phi(\xi)^*$, where Φ is written in the coordinates given by the chosen basis. If $\mathrm{d}\Phi$ is injective off the zero section, $B = \mathrm{Id}$ gives a strictly positive Levi form. If $\mathrm{d}\Phi$ has a nonzero kernel vector there, fiber dilation and normalization of the tangent vector place such a vector in K, where $\Psi_0 = 0$, and no Hermitian matrix can separate the image from zero. □

When this untwisted test fails, the higher twists remain available. The finite alternatives use all Hermitian coefficient matrices and the complete spaces of sections; testing only positive coefficient matrices or a smaller collection of sections gives a different criterion.

5.6 Ramified section tests and a positive root metric

The untwisted test of Corollary 5.5.1 can fail even for a line bundle with a positively curved metric, because the evaluation map of a globally generated bundle may fail to be an immersion where a holomorphic map ramifies. We give an explicit example on a branched double cover of the projective line, where the complete spaces of sections can be computed. The same example gives a family of weighted tests, built from a root of the Fubini–Study weight of a very ample power instead of the twists of the preceding section, whose first successful level is explicit and can be made arbitrarily large.

Fix an integer $q \geq 2$. Choose a section $\beta \in H^0(\mathbb{P}^1, O(2q))$ with $2q$ distinct zeros. Let $\pi\colon C \to \mathbb{P}^1$ be the double cover defined in the total space of $O(q)$ by the equation $w^2 = \beta$, where w is the tautological coordinate. In this section $X = C$, so $n = 1$, and

$$L = \pi^*O(1), \qquad A = L^q, \qquad F = L^{-1}.$$

Lemma 5.6.1 *The curve C is smooth, connected and projective. The line bundle L is globally generated, A is very ample, and for every integer $k \geq 0$,*

$$H^0(C, L^k) = H^0(\mathbb{P}^1, \mathcal{O}(k)) \oplus H^0(\mathbb{P}^1, \mathcal{O}(k-q))\, w.$$

In particular,

$$h^0(C, L^k) = k + 1 + \max(k - q + 1, 0).$$

For a basis $\sigma_1, \ldots, \sigma_N$ of $H^0(C, A)$, the Fubini–Study weight and the corresponding root weight, in a local frame of L and the induced frame of $A = L^q$, are

$$\phi_A = \log \sum_{i=1}^{N} |\sigma_i|^2, \qquad \phi = \phi_A / q.$$

The metric $\mathrm{e}^{-\phi}$ on L has strictly positive curvature. Thus it is strictly Griffiths-positive, and $G(v) = \mathrm{e}^{\phi}|v|^2$ is strictly plurisubharmonic on F off its zero section.

Proof The direct image of the structure sheaf of C is $\mathcal{O} \oplus \mathcal{O}(-q)\, w$, with multiplication of the second summand given by $w^2 = \beta$. Tensoring by $\mathcal{O}(k)$, the local basis $1, w$ gives the stated decomposition of sections, and the dimensions follow since line bundles of negative degree on $\mathbb{P}^1$ have no nonzero sections. Simple zeros of β make the cover smooth: near a branch point the equation gives a coordinate w with the base coordinate a nonzero multiple of w^2 to leading order. The cover is connected because the rational function representing β has a simple zero and therefore is not a square. Finiteness over compact $\mathbb{P}^1$ makes C compact. The degree-one base sections generate L.

The complete sections of $A = L^q$ are the degree-q base monomials together with w. The map to projective space defined by these sections separates the base points by the Veronese coordinates, separates the two sheets by w, and has injective differential. At the ramification points the coordinate w gives the missing derivative, and elsewhere the base coordinates do. Compactness makes this map an embedding. This shows that C is projective and that A is very ample.

Its complete linear system therefore defines an immersion, so the Fubini–Study metric on A has strictly positive curvature. Dividing its weight by q defines a metric on L, since the transition functions of $A = L^q$ are the qth powers of those of L, and its curvature is $1/q$ times that positive form. The final assertion follows from Lemma 5.5.1. □

5.6.1 The weighted family and its first successful level

Keep the root metric $\mathrm{e}^{-\phi}$ on L, and choose the sphere bundle S and the compact set K for F as in Section 5.5. For each integer $j \geq 0$, use the complete space

$$\widetilde{V}_j = H^0(C, L^{j+1})$$

and pair its basis with $v \in F_x$ to get a local column $p_j(v)$, written in a local frame of L^j. For every Hermitian coefficient matrix B, define

$$\widetilde{G}_{j,B}(v) = \mathrm{e}^{-j\phi(x)} p_j(v)^* B p_j(v).$$

These are globally defined functions, quadratic on each fiber. Their Levi matrices are

$$\widetilde{\Psi}_j(v,\xi) = \mathrm{e}^{-j\phi}(r_j r_j^* - j\kappa p_j p_j^*),$$
$$r_j = \mathrm{d}p_j(\xi) - j\phi_u p_j, \qquad \kappa = \phi_{u\bar{u}},$$

where u is the projection of ξ to the base. As in (5.8), direct differentiation gives $\mathcal{L}\widetilde{G}_{j,B}(\xi) = \mathrm{tr}(B\widetilde{\Psi}_j)$, and $\widetilde{\Psi}_j$ transforms in the same way under changes of frames.

Proposition 5.6.1 *The first integer $j \geq 0$ for which some $\widetilde{G}_{j,B}$, with B an arbitrary Hermitian matrix, is strictly plurisubharmonic off the zero section is $j = q-1$. At every lower level, zero belongs to* $\mathrm{conv}(\widetilde{\Psi}_j(K))$*; one atom witnesses this at $j = 0$, and two atoms suffice at each $1 \leq j < q-1$.*

Proof At $j = q-1$, choose the basis σ_i that defines ϕ_A and take $B = \mathrm{Id}$. Since $p_j(v) = v(\sigma_i(x))_i$ in the chosen frames,

$$\widetilde{G}_{q-1,\mathrm{Id}}(v) = \mathrm{e}^{-(q-1)\phi}|v|^2 \mathrm{e}^{q\phi} = \mathrm{e}^{\phi}|v|^2.$$

This function is strictly plurisubharmonic by Lemma 5.6.1.

Now suppose that $0 \leq j < q-1$. Then $j+1 < q$, so by Lemma 5.6.1 every section in $\widetilde{V}_j$ comes from the first summand of the decomposition, which does not involve w. At a ramification point of C, choose a local coordinate t on C, and use frames of the bundles pulled back from $\mathbb{P}^1$. The column has the form $p_j(v) = vU_j(\pi(t))$. It is nonzero for $v \neq 0$, since L^{j+1} is globally generated, and since π ramifies at the point, its derivative in the t direction vanishes there when v is held fixed.

Choose such a fiber vector v on the sphere bundle S. Write tangent vectors to the total space of F by their components in the t direction and in the fiber direction. Let $\xi_v = (0, v)$ be the vertical radial vector, let $u = \partial/\partial t$, and let $\xi_h = (u, j\phi_u v)$. At this point $\mathrm{d}p_j(\xi_v) = p_j$ and $r_j(\xi_h) = 0$. Thus

$$\widetilde{\Psi}_j(v,\xi_v) = \mathrm{e}^{-j\phi} p_j p_j^*,$$
$$\widetilde{\Psi}_j(v,\xi_h) = -j\kappa \mathrm{e}^{-j\phi} p_j p_j^*, \qquad \kappa > 0.$$

Let a_v and a_h be the squared norms of ξ_v and ξ_h for the form b, and divide each tangent vector by its norm to obtain points of K. At $j = 0$ the second matrix is zero, which gives the witness with one atom. For $j > 0$, the matrix at the second normalized point is $-c$ times the matrix at the first, with $c = j\kappa a_v/a_h > 0$. The probability measure giving these two points the weights $c/(1+c)$ and $1/(1+c)$ has zero matrix integral, which excludes strict positivity for every Hermitian B. This proves failure at all lower levels, so the first successful level is $j = q-1$. □

5.6.2 Relation to the fixed very ample criterion

Corollary 5.6.1 *For the data $E = L$ and $A = L^q$, with the Fubini–Study metric of a basis of $H^0(C, A)$, the criterion of Theorem 5.5.1 applies with $m_0 = 0$. Its untwisted level fails, although E carries a strictly Griffiths-positive metric, and a higher level succeeds. The atom witnessing the failure at level zero annihilates only the Levi forms of the level-zero test space, and does not give a measure with $D^*T = 0$.*

Proof By Lemma 5.6.1, L is globally generated and A is very ample, as the criterion requires. At a ramification point, the complete untwisted evaluation map has a nonzero tangent vector in its kernel, in the direction of the cover, as in the preceding proof; hence Corollary 5.5.1 gives failure at level zero. The positive root metric and Theorem 5.5.1 give success at some higher level. Moreover, the Levi form of the dual of this positive metric pairs strictly positively with every nonzero positive matrix-valued measure on $W \to S$, so the atom at level zero cannot annihilate the Levi forms of all smooth Hermitian forms. □

The level m of the criterion in Theorem 5.5.1 corresponds to $j = qm$ in the root-weight family, since $E \otimes A^m = L^{qm+1}$ and $m\phi_A = qm\phi$. The first success at $j = q - 1$ therefore gives no bound on the first successful level m of that criterion. The root-weight family is nested in steps of q, because multiplication by the sections of A uses $\sum_i |\sigma_i|^2 = \mathrm{e}^{q\phi}$; we do not claim that it is nested in steps of one. As q grows, the root-weight family has arbitrarily many failed initial levels, although the bundle carries a strictly positive metric.

5.7 Negative bundle metrics and logarithmic weights

For positive curvature, the reciprocal norm in Lemma 5.1.19 is useful near a removed zero section. With negative curvature, the squared norm itself is plurisubharmonic. Adding its logarithm gives a weight whose Levi form controls both the base and the fiber directions.

In this section the base X has dimension $n \geq 1$ and need not be compact. Suppose that it has a Kähler form ω and a smooth nonnegative plurisubharmonic exhaustion ρ, in the sense of Section 5.1.8; on a compact base a constant function qualifies. Let $\pi\colon E \to X$ be a holomorphic vector bundle of rank $r \geq 1$, with a smooth Hermitian metric h, and let Z be its zero section. Use the unscaled curvature and coefficient forms $\alpha_{h,v}$ of Section 5.1.9, and assume

$$\alpha_{h,v} \leq -|v|_h^2 \omega \qquad (x \in X,\ v \in E_x).$$

Proposition 5.7.1 *Put $U(v) = |v|_h^2$, $q = U\mathrm{e}^U$ and $\Psi = \rho \circ \pi + U$. Then Ψ is a smooth plurisubharmonic exhaustion of E and*

$$\Omega = \pi^*\omega + 2\pi \mathrm{dd}^{\mathrm{c}} U$$

is a Kähler form on E. The function $\log q$, extended by $-\infty$ on Z, is plurisubharmonic and satisfies

$$2\pi \mathrm{dd}^{\mathrm{c}} \log q \geq \Omega \quad \text{on } E \setminus Z.$$

Both U and q are strictly plurisubharmonic off Z. For the real radial vector field $\mathcal{R}$, whose flow is $v \mapsto \mathrm{e}^{s} v$, one has $\mathcal{R} q = 2(1+U)q$. In particular q has no critical points off Z.

Proof At x, choose a Chern-normal unitary holomorphic frame as in Lemma 5.1.19. Write a tangent vector at (x, v) as (a, b), with a horizontal and b vertical. The first base derivatives and mixed base–fiber derivatives of U vanish there. Let $\mathcal{L}$ denote the unscaled complex Hessian, corresponding to $2\pi \mathrm{dd}^{\mathrm{c}} = \mathrm{i}\partial\bar{\partial}$. The normal-frame calculation of that lemma gives

$$\mathcal{L}_U(a,b) = -h(R_h(a,\bar{a})v, v) + |b|_h^2 \geq U\omega(a,\bar{a}) + |b|_h^2,$$
$$\mathcal{L}_{\log U}(a,b) = -\frac{h(R_h(a,\bar{a})v, v)}{U} + \frac{|b|_h^2 U - |h(b,v)|^2}{U^2} \geq \omega(a,\bar{a}) \quad (v \neq 0).$$

The last vertical term is nonnegative by Cauchy–Schwarz. These formulas show that U is plurisubharmonic, that Ω is positive definite even on Z, and that $2\pi \mathrm{dd}^{\mathrm{c}}(\log U + U) \geq \Omega$ off Z. The form Ω is closed because ω is closed. The first formula gives strict positivity for U off Z; exponentiating $\log q$ gives it for q. Locally, U is comparable to the squared Euclidean fiber norm, so $\log q$ tends to $-\infty$ at Z and is locally bounded above. By Lemma 1.3.7, the upper-limit extension is plurisubharmonic and has value $-\infty$ on Z.

The function Ψ is plurisubharmonic. Its sublevel set of height c lies over the compact set $\{\rho \leq c\}$ and in the disk bundle $\{U \leq c\}$ over that set, which is compact. The sublevel set is closed, so Ψ is an exhaustion. Finally, $\mathcal{R}U = 2U$ gives the radial identity and excludes critical points where $q > 0$. □

Corollary 5.7.1 *For every $m > 0$,*

$$\mathcal{I}(m \log q) = \mathcal{I}_Z^{\max\{\lfloor m \rfloor - r + 1, 0\}}.$$

In particular, if $k \geq 1$ is an integer and $m \geq r + k - 1$, a holomorphic function that is locally square integrable against q^{-m} times a smooth positive volume form vanishes to order at least k along Z. If $m < r + k - 1$, this conclusion fails for some local holomorphic functions.

Proof In a local bundle frame, $\log q = \log |v|^2 + O(1)$ near Z. Apply Lemma 1.16.1 in codimension r. Its radial integrability condition for a homogeneous fiber term of degree a is $a + r > m$. For $m < r + k - 1$, a monomial of degree $k - 1$ satisfies this condition and has insufficient vanishing order. Smooth positive volume forms do not change local integrability. □

Corollary 5.7.2 *The manifold $E \setminus Z$ admits a complete Kähler metric.*

Proof Choose a smooth convex increasing function η on $\mathbb{R}$ equal to $-\log(-t)$ for $t \le -1$. On $E \setminus Z$ put

$$\widehat{\Omega} = \Omega + 2\pi \mathrm{dd}^{\mathrm{c}}(\Psi^2) + 2\pi \mathrm{dd}^{\mathrm{c}}\eta(\log q).$$

The added forms are nonnegative, so this is Kähler. Since $\Psi \ge 0$ is plurisubharmonic, $2\pi \mathrm{dd}^{\mathrm{c}}(\Psi^2) \ge 2\mathrm{i}\partial\Psi \wedge \bar{\partial}\Psi$. Thus Ψ has bounded gradient for $\widehat{\Omega}$, and a curve of finite length stays in a compact sublevel of Ψ in E. The only possible escape from that compact set within $E \setminus Z$ is approach to Z. Where $\log q \le -1$, set $\tau = \log(-\log q)$. Convexity and the chain rule give

$$2\pi \mathrm{dd}^{\mathrm{c}}\eta(\log q) \ge \mathrm{i}\partial\tau \wedge \bar{\partial}\tau.$$

Hence τ also has bounded gradient there, while $\tau \to +\infty$ on approach to Z. A finite-length curve cannot escape in this way either. The criterion that every escaping curve has infinite length proves completeness. □

For example, let $B \to X$ have a smooth Hermitian metric with positive unscaled curvature form ω. The direct sum $E = (B^*)^{\oplus r}$ with its induced metric satisfies $\alpha_{h,v} = -|v|_h^2\omega$. Since ω is closed, the construction applies whenever X also has a plurisubharmonic exhaustion as above, in particular when X is compact.

If the base is connected, compact and has positive dimension, one may take $\rho = 0$. Then U is an exhaustion that is strictly plurisubharmonic off Z, but the total space E is not Stein: every holomorphic function on E restricts to a constant on the compact connected zero section and therefore fails to separate its distinct points. Strict positivity away from a compact analytic set does not remove this obstruction.

Chapter 6
Gauduchon traces and the topology of potentials

Throughout this chapter, θ is a smooth real $(1,1)$-form on X, $\mathrm{d}V$ is a smooth positive probability volume form used for L^1 norms, and γ is a Gauduchon metric (Definition 4.1.1); in dimension one every Hermitian form is Gauduchon, since $\gamma^{n-1} = 1$.

The current $\theta + \mathrm{dd}^{\mathrm{c}} u$ has a conserved linear trace even when its non-pluripolar Monge–Ampère measure has infinite mass, as we shall see in Chapter 7. By Proposition 4.2.1, the positive measure $(\theta + \mathrm{dd}^{\mathrm{c}} u) \wedge \gamma^{n-1}$ has the same total mass for every $u \in \mathrm{PSH}(X, \theta)$. The ellipticity of $f \mapsto \mathrm{dd}^{\mathrm{c}} f \wedge \gamma^{n-1}$ makes the trace more informative than this mass alone: the kernel of this operator consists of the constants, so the trace measure determines the potential up to an additive constant.

On mean-normalized potentials, this correspondence also identifies the topologies: weak convergence of trace measures is equivalent to L^1 convergence of potentials. Lipschitz tests give a complete metric for this topology, and the inverse elliptic operator compares it with the L^1 norm by sharp power bounds. Extending the conserved pairing to a positive pluriharmonic current requires more than this elliptic inversion. For continuous potentials, a gradient-energy estimate defines the derivative terms in the product and gives continuity under uniform approximation.

6.1 The trace map

Adding a constant to a potential does not change its current, so we normalize potentials by their mean and consider

$$\mathcal{K}^0_{\theta,\mathrm{d}V} = \left\{ u \in \mathrm{PSH}(X,\theta) : \int_X u \, \mathrm{d}V = 0 \right\}.$$

For $u \in \mathcal{K}^0_{\theta,\mathrm{d}V}$ we call the measure

$$\mathcal{T}_\gamma(u) = (\theta + \mathrm{dd}^{\mathrm{c}} u) \wedge \gamma^{n-1}$$

the *Gauduchon trace* of u, and $\mathcal{T}_\gamma$ the *trace map*. By Proposition 4.2.1 (by Stokes' theorem when $n = 1$), $\mathcal{T}_\gamma(u)$ is a positive measure of total mass

$$M_{\theta,\gamma} = \int_X \theta \wedge \gamma^{n-1}.$$

The trace map is governed by the linear operator

$$Af = \frac{\mathrm{dd^c} f \wedge \gamma^{n-1}}{\mathrm{d}V}, \qquad f \in C^\infty(X, \mathbb{R}).$$

It is a real second-order elliptic operator without zeroth-order term, and $\mathcal{T}_\gamma(u) - \mathcal{T}_\gamma(v) = A(u - v)\,\mathrm{d}V$ in the sense of distributions.

Theorem 6.1.1 *The trace map $\mathcal{T}_\gamma$ is an affine homeomorphism from the compact convex set $\mathcal{K}^0_{\theta,\mathrm{d}V}$, with its L^1 topology, onto a compact convex set of positive measures of mass $M_{\theta,\gamma}$, with the weak topology. In other words, for $u_j, u \in \mathcal{K}^0_{\theta,\mathrm{d}V}$,*

$$u_j \longrightarrow u \text{ in } L^1 \quad \Longleftrightarrow \quad \mathcal{T}_\gamma(u_j) \rightharpoonup \mathcal{T}_\gamma(u).$$

If $M_{\theta,\gamma} = 0$ and $\mathrm{PSH}(X, \theta) \neq \varnothing$, both sets consist of a single point.

Proof The set $\mathcal{K}^0_{\theta,\mathrm{d}V}$ is convex. It is the image of the compact set $\mathcal{K}_\theta$ of Theorem 1.4.1 under the continuous map $v \mapsto v - \int_X v\,\mathrm{d}V$, and is therefore compact.

If $u_j \to u$ in L^1, then $\mathrm{dd^c} u_j \to \mathrm{dd^c} u$ as currents, so the trace measures converge in the sense of distributions. Since they are positive measures of the same finite mass, and every continuous function is a uniform limit of smooth functions, they also converge weakly as measures. Thus $\mathcal{T}_\gamma$ is continuous.

To prove injectivity, suppose that $\mathcal{T}_\gamma(u) = \mathcal{T}_\gamma(v)$. Then $A(u-v) = 0$ in the sense of distributions. Elliptic regularity for A shows that $u - v$ agrees almost everywhere with a smooth function, and the maximum principle shows that this function is constant. Its mean is zero, so $u = v$ almost everywhere, and hence everywhere by Lemma 1.3.3.

A continuous injection from a compact space into a Hausdorff space is a homeomorphism onto its image; here the target is the space of positive measures of mass $M_{\theta,\gamma}$ with the weak topology. Since $\mathcal{T}_\gamma$ is affine, its image is convex. If $M_{\theta,\gamma} = 0$, every trace measure is zero, so injectivity shows that $\mathcal{K}^0_{\theta,\mathrm{d}V}$ has at most one point; it has exactly one when $\mathrm{PSH}(X, \theta)$ is nonempty, since the mean of any θ-plurisubharmonic function can be subtracted. □

The inverse of the trace map can be described by a Green operator for A. As for the operator in the proof of Theorem 4.1.1, the maximum principle gives $\ker A = \mathbb{R}$, and A has Fredholm index zero. Let A^* be the formal adjoint of A with respect to $\mathrm{d}V$. Stokes' theorem gives $\int_X Af\,\mathrm{d}V = 0$ for all smooth f, that is, $A^*1 = 0$; hence $\ker A^* = \mathbb{R}$ as well. By the Fredholm alternative, A maps smooth functions onto the smooth functions g with $\int_X g\,\mathrm{d}V = 0$, and the same holds for A^*. In particular, A^* has a continuous inverse on smooth functions of mean zero, normalized to take values of mean zero. By duality with this inverse, every top-degree distribution σ of total

mass zero, and in particular every finite signed measure with $\sigma(X) = 0$, determines a unique distribution $G_\gamma \sigma$ with mean zero against $\mathrm{d}V$ such that

$$\mathrm{dd^c}(G_\gamma \sigma) \wedge \gamma^{n-1} = \sigma. \tag{6.1}$$

Inverting the trace map at a positive measure therefore amounts to solving (6.1) and then checking positivity of the full current.

Proposition 6.1.1 *A positive measure ν of mass $M_{\theta,\gamma}$ belongs to the image of $\mathcal{T}_\gamma$ if and only if the distribution*

$$w = G_\gamma\big(\nu - \theta \wedge \gamma^{n-1}\big)$$

satisfies $\theta + \mathrm{dd^c} w \geq 0$. In that case the θ-plurisubharmonic representative of w is the unique $u \in \mathcal{K}^0_{\theta,\mathrm{d}V}$ with $\mathcal{T}_\gamma(u) = \nu$.

Proof If $\nu = \mathcal{T}_\gamma(u)$ with $u \in \mathcal{K}^0_{\theta,\mathrm{d}V}$, then u is a mean-zero solution of (6.1) with $\sigma = \nu - \theta \wedge \gamma^{n-1}$, so uniqueness gives $w = u$ as distributions. Conversely, if $\theta + \mathrm{dd^c} w \geq 0$, then Proposition 1.1.1 gives a θ-plurisubharmonic representative u of w. It has mean zero by construction, and (6.1) gives $\mathcal{T}_\gamma(u) = \nu$. □

In dimension one the positivity condition is automatic, because $\theta + \mathrm{dd^c} w = \nu$; thus every positive measure of mass $M_{\theta,\gamma}$ is the trace of a normalized potential. In higher dimensions, positivity of the trace measure ν does not imply positivity of the $(1,1)$-current $\theta + \mathrm{dd^c} w$, and the condition in Proposition 6.1.1 must be checked separately.

6.2 A metric on normalized potentials

By Theorem 6.1.1, every metric for the weak topology on positive measures of mass $M_{\theta,\gamma}$ pulls back to a metric for the L^1 topology on $\mathcal{K}^0_{\theta,\mathrm{d}V}$. We use the bounded Lipschitz metric on measures, which can be compared quantitatively with the L^1 norm through the inverse of A. Fix a distance d on X induced by a smooth Riemannian metric. For a real Lipschitz function f on X, write $\mathrm{Lip}_d(f) = \sup_{x \neq y} |f(x) - f(y)|/d(x,y)$, and for $u, v \in \mathcal{K}^0_{\theta,\mathrm{d}V}$ put

$$d_{\gamma,d}(u,v) = \sup_{\substack{\|f\|_\infty \leq 1 \\ \mathrm{Lip}_d(f) \leq 1}} \left| \int_X f \, \mathrm{d}\big(\mathcal{T}_\gamma(u) - \mathcal{T}_\gamma(v)\big) \right|.$$

We call $d_{\gamma,d}$ the *Gauduchon trace metric*.

Corollary 6.2.1 *The function $d_{\gamma,d}$ is a bounded complete metric on $\mathcal{K}^0_{\theta,\mathrm{d}V}$ inducing its L^1 topology. Changing γ or d gives the same topology and a uniformly equivalent metric on this compact space.*

Proof Symmetry and the triangle inequality follow directly from the definition, and $d_{\gamma,d} \le 2M_{\theta,\gamma}$ because both trace measures have mass $M_{\theta,\gamma}$. Let $\mathcal{F}$ denote the family of test functions in the supremum. After rescaling, $\mathcal{F}$ contains a positive multiple of every Lipschitz function, and Lipschitz functions are uniformly dense in $C(X)$. Hence $\mathcal{F}$ separates finite measures, and together with the injectivity of $\mathcal{T}_\gamma$ this shows that $d_{\gamma,d}$ is a metric. The same density, together with the common mass bound, shows that convergence in $d_{\gamma,d}$ implies weak convergence of the trace measures. Conversely, $\mathcal{F}$ is uniformly bounded and equicontinuous, so the Arzelà–Ascoli theorem gives a finite uniform ϵ-net in $\mathcal{F}$ for every $\epsilon > 0$. Weak convergence on the finitely many functions of the net, and the bound $2M_{\theta,\gamma}\epsilon$ for the remaining error, give convergence in $d_{\gamma,d}$.

Thus $d_{\gamma,d}$ is the pullback by the homeomorphism of Theorem 6.1.1 of a metric for the weak topology. The resulting metric space is compact, hence complete. For two choices of the auxiliary data, both metrics induce the same compact L^1 topology, so the identity map is uniformly continuous in both directions. □

Proposition 6.2.1 *There is a constant $C \ge 1$, depending only on $X, \theta, \gamma, \mathrm{d}V$ and d, such that*

$$C^{-1}\|u - v\|_{L^1} \le d_{\gamma,d}(u, v) \le C\|u - v\|_{L^1}^{1/2} \qquad (u, v \in \mathcal{K}^0_{\theta,\mathrm{d}V}).$$

Proof The lower bound follows by duality from the inverse of A^*, and the upper bound by smoothing the test functions. Put $w = u - v$, $\sigma = \mathcal{T}_\gamma(u) - \mathcal{T}_\gamma(v)$ and $M = M_{\theta,\gamma}$, and write $\|\sigma\|_{\mathrm{TV}}$ for the total variation of σ. Then $\int_X w\,\mathrm{d}V = 0$, $\sigma = \mathrm{d}\mathrm{d}^{\mathrm{c}} w \wedge \gamma^{n-1}$ and $\|\sigma\|_{\mathrm{TV}} \le 2M$. If $M = 0$, Theorem 6.1.1 gives $u = v$, and both estimates hold trivially.

For the lower bound, let g be a smooth real function. Since $\ker A = \ker A^* = \mathbb{R}$, the equation

$$A^*\phi = g - \int_X g\,\mathrm{d}V, \qquad \int_X \phi\,\mathrm{d}V = 0,$$

has a unique smooth solution. For $p > 2n$, elliptic $W^{2,p}$ estimates on functions of mean zero and the Sobolev embedding theorem give $\|\phi\|_\infty + \mathrm{Lip}_d(\phi) \le C\|g\|_\infty$.

Pairing the distributional equation $Aw\,\mathrm{d}V = \sigma$ with ϕ, and using $\int_X w\,\mathrm{d}V = 0$, we obtain

$$\left|\int_X wg\,\mathrm{d}V\right| = \left|\int_X \phi\,\mathrm{d}\sigma\right| \le C\|g\|_\infty d_{\gamma,d}(u, v).$$

Taking the supremum over smooth real g with $\|g\|_\infty \le 1$ gives $\|w\|_{L^1} \le C d_{\gamma,d}(u, v)$.

For the upper bound, let f be any test function in the definition of $d_{\gamma,d}$. Using a fixed finite coordinate cover, a smooth partition of unity and convolution at scale $0 < \epsilon < \epsilon_0$, we obtain smooth functions f_ϵ satisfying

$$\|f - f_\epsilon\|_\infty \le C\epsilon, \qquad \|A^* f_\epsilon\|_\infty \le C\epsilon^{-1},$$

with constants uniform over this family of test functions. To see the second bound, localize f by the fixed partition of unity. The localized functions have uniformly

bounded values and first weak derivatives. In each second derivative of their convolutions, move one derivative onto the mollifier, whose derivative has L^1 norm $O(\epsilon^{-1})$. The lower-order terms of A^* have fixed smooth coefficients and satisfy the same bound. Since $\int_X f_\epsilon \, \mathrm{d}\sigma = \int_X w A^* f_\epsilon \, \mathrm{d}V$, it follows that

$$\left| \int_X f \, \mathrm{d}\sigma \right| \leq C\bigl(M\epsilon + \epsilon^{-1} \|w\|_{L^1}\bigr).$$

If $0 < \|w\|_{L^1} < \epsilon_0^2$, choose $\epsilon = \|w\|_{L^1}^{1/2}$. For larger norms use $d_{\gamma,d}(u, v) \leq 2M$, and for $w = 0$ use injectivity. Taking the supremum over f proves the upper bound. □

Both exponents in Proposition 6.2.1 are optimal, already on an elliptic curve.

Example 6.2.1 Let $X = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$, with coordinate $z = x + \mathrm{i}y$. Take $\theta = \gamma = \eta$, where $\eta = (\mathrm{i}/2)\mathrm{d}z \wedge \mathrm{d}\bar{z}$, together with $\mathrm{d}V = \eta$ and the flat distance d. For integers $k \geq 1$, set

$$u_k = \frac{\cos(2\pi k x)}{2\pi k^2}.$$

These smooth potentials have mean zero and satisfy

$$\theta + \mathrm{dd}^{\mathrm{c}} u_k = \left(1 - \frac{1}{2}\cos(2\pi k x)\right)\eta > 0, \qquad \|u_k\|_{L^1} = \frac{1}{\pi^2 k^2}.$$

The function $f_k = \cos(2\pi k x)/(2\pi k)$ has supremum norm at most one and Lipschitz constant at most one. It follows that

$$d_{\gamma,d}(u_k, 0) \geq \left| \int_X f_k \, \mathrm{dd}^{\mathrm{c}} u_k \right| = \frac{1}{8\pi k}.$$

Consequently, the exponent $1/2$ in Proposition 6.2.1 cannot be replaced by any larger exponent, even for smooth potentials with strictly positive currents on a fixed curve. Nor can the power one in the lower bound be replaced by a smaller positive power: for $0 < t \leq 1$, both $\|t u_1\|_{L^1}$ and $d_{\gamma,d}(t u_1, 0)$ equal t times their positive values at $t = 1$.

6.3 Continuous potentials and pluriharmonic traces

The Gauduchon trace pairs $\theta + \mathrm{dd}^{\mathrm{c}} u$ with the smooth form γ^{n-1}, which is positive and pluriharmonic. In this section, ω is a Hermitian form on X, and G is a positive pluriharmonic $(n-1, n-1)$-current, that is, a positive current with $\mathrm{dd}^{\mathrm{c}} G = 0$ (see Section 4.6). Such currents include γ^{n-1}, but in general their coefficients are only measures, and the product $(\theta + \mathrm{dd}^{\mathrm{c}} u) \wedge G$ is not defined a priori. Its integration-by-parts formula contains gradient terms, since G need not be d-closed. The equation $\mathrm{dd}^{\mathrm{c}} G = 0$ converts uniform control of a potential difference into a bound for its gradient energy against G. This estimate defines the gradient currents for a continuous potential and makes the product stable under uniform approximation.

Lemma 6.3.1 *Put $\mu = \omega \wedge G$ and $M = \mu(X)$. If a, b are smooth real functions and $C \geq 0$ is a constant with $\mathrm{dd^c} a, \mathrm{dd^c} b \geq -C\omega$, then*

$$0 \leq \int_X \mathrm{d}(a-b) \wedge \mathrm{d^c}(a-b) \wedge G \leq 2CM\|a-b\|_\infty.$$

For every smooth real function v, the total variation norms of the order-zero currents $\mathrm{d}v \wedge G$ and $\mathrm{d^c} v \wedge G$ satisfy

$$\|\mathrm{d}v \wedge G\|_{\mathrm{TV}} + \|\mathrm{d^c} v \wedge G\|_{\mathrm{TV}} \leq C_\omega M^{1/2} \left(\int_X \mathrm{d}v \wedge \mathrm{d^c} v \wedge G \right)^{1/2}.$$

Here the total variation norms are computed with respect to ω, and C_ω depends only on (X, ω).

Proof Set $v = a - b$. The identity $\mathrm{dd^c}(v^2) = 2\mathrm{d}v \wedge \mathrm{d^c} v + 2v\mathrm{dd^c} v$, integrated against G, and the equation $\mathrm{dd^c} G = 0$ give

$$\int_X \mathrm{d}v \wedge \mathrm{d^c} v \wedge G = -\int_X v \mathrm{dd^c} v \wedge G.$$

The measures $(\mathrm{dd^c} a + C\omega) \wedge G$ and $(\mathrm{dd^c} b + C\omega) \wedge G$ are positive, and each has mass CM, because the terms $\mathrm{dd^c} a \wedge G$ and $\mathrm{dd^c} b \wedge G$ integrate to zero. Their difference $\mathrm{dd^c} v \wedge G$ therefore has total variation at most $2CM$, which proves the upper bound. The lower bound holds because $\mathrm{d}v \wedge \mathrm{d^c} v = (\mathrm{i}/2\pi)\partial v \wedge \bar{\partial} v$ is a positive $(1,1)$-form.

For the second estimate, work in a local unitary frame for ω. Positivity of G means that its coefficients form a positive semidefinite matrix of measures. These entries are absolutely continuous with respect to the trace measure μ: the diagonal entries are dominated by the trace, and the off-diagonal entries by the Cauchy–Schwarz inequality for positive semidefinite matrices. The resulting matrix of densities has trace bounded by a constant depending only on the normalization. Pairing $\mathrm{d}v \wedge G$ or $\mathrm{d^c} v \wedge G$ with a test one-form η, and applying the matrix Cauchy–Schwarz inequality and then the integral Cauchy–Schwarz inequality, we obtain

$$|\langle \mathrm{d}v \wedge G, \eta \rangle| + |\langle \mathrm{d^c} v \wedge G, \eta \rangle| \leq C_\omega \|\eta\|_\infty M^{1/2} \left(\int_X \mathrm{d}v \wedge \mathrm{d^c} v \wedge G \right)^{1/2}.$$

A finite cover by such frames and a partition of unity give the global estimate. Taking the supremum over smooth test forms of norm at most one yields the total variation bound. When $M = 0$, positivity gives $G = 0$, and both estimates hold trivially. □

Theorem 6.3.1 *Let α be a smooth real $(1,1)$-form, and let $u \in \mathrm{PSH}(X,\alpha) \cap C^0(X)$. The product*

$$(\alpha + \mathrm{dd^c} u) \wedge G$$

has a canonical definition as a positive measure of mass $\int_X \alpha \wedge G$, which agrees with the usual product if u or G is smooth. If $u_j, u \in \mathrm{PSH}(X,\alpha) \cap C^0(X)$ and $u_j \to u$ uniformly, and if positive pluriharmonic currents G_j converge weakly to G, then

$$(\alpha + \mathrm{dd^c} u_j) \wedge G_j \rightharpoonup (\alpha + \mathrm{dd^c} u) \wedge G.$$

Proof We first define the gradient currents $\mathrm{d}u \wedge G$ and $\mathrm{d^c} u \wedge G$ as limits, then define the product by an integration-by-parts formula, and finally prove joint continuity.

Step 1: gradient currents. Choose smooth approximants $a_k \to u$ as in Lemma 1.10.1. There is a common constant $C > 0$ such that $\mathrm{dd^c} a_k \geq -C\omega$. Write $M = \int_X \omega \wedge G$. Since a_k converges uniformly, Lemma 6.3.1 shows that $\mathrm{d}a_k \wedge G$ and $\mathrm{d^c} a_k \wedge G$ are Cauchy sequences in total variation. Denote their limits, which are currents of order zero, by $\mathrm{d}u \wedge G$ and $\mathrm{d^c} u \wedge G$. Applying the same lemma to two uniformly convergent smooth sequences with a common lower Hessian bound shows that the limits do not depend on the sequence. Uniform convergence also gives $a_k G \to uG$ in total variation.

Step 2: the product. Define

$$(\alpha + \mathrm{dd^c} u) \wedge G = \alpha \wedge G + \mathrm{d}(\mathrm{d^c} u \wedge G) - \mathrm{d^c}(\mathrm{d}u \wedge G) - \mathrm{dd^c}(uG) \tag{6.2}$$

in the sense of distributions. For smooth u, the Leibniz rule and $\mathrm{dd^c} G = 0$ show that this agrees with the usual product. Consequently $(\alpha + \mathrm{dd^c} a_k) \wedge G$ converges in distributions to the right-hand side of (6.2). With $\varepsilon_k \to 0$ as in Lemma 1.10.1, the measures $(\alpha + \mathrm{dd^c} a_k + \varepsilon_k \omega) \wedge G$ are positive, with masses $\int_X \alpha \wedge G + \varepsilon_k M$. Their distributional limit is therefore a positive measure, and testing against the constant function one gives its mass. The uniform mass bound upgrades distributional convergence to weak convergence of measures. If G is smooth, the uniform convergence of a_k also identifies the limit with the product of the current $\alpha + \mathrm{dd^c} u$ and the smooth form G.

Step 3: independence of the decomposition. Adding a smooth function h to u adds $\mathrm{d}h \wedge G$ and $\mathrm{d^c} h \wedge G$ to the gradient currents, so replacing (α, u) by $(\alpha - \mathrm{dd^c} h, u + h)$ leaves the product unchanged. If two continuous decompositions define the same $(1,1)$-current, the difference of their potentials has smooth $\mathrm{dd^c}$; taking its trace in a coordinate chart and applying elliptic regularity shows that this difference is smooth. Thus the product depends only on the current $\alpha + \mathrm{dd^c} u$ and on G.

Step 4: joint continuity. For u and for each u_j, keep only those smooth approximants from Lemma 1.10.1 whose curvature errors are at most one. Choosing C with $\alpha \leq (C-1)\omega$, all these approximants satisfy the same lower Hessian bound $-C\omega$. The numbers $M_j = \int_X \omega \wedge G_j$ are bounded, by weak convergence. Fix a smooth approximant a_k of u, and put $\delta_{j,k} = \|u_j - a_k\|_\infty$. Applying Lemma 6.3.1 to a_k and to smooth approximants of u_j, and passing to the gradient limits, we find that the

total variations of the differences $(\mathrm{d}u_j - \mathrm{d}a_k) \wedge G_j$ and $(\mathrm{d}^{\mathrm{c}}u_j - \mathrm{d}^{\mathrm{c}}a_k) \wedge G_j$ are at most $C' M_j \sqrt{\delta_{j,k}}$, where C' is independent of j and k.

For a smooth real test function ϕ, formula (6.2) now gives

$$\left|\left\langle (\alpha + \mathrm{dd}^{\mathrm{c}}u_j) \wedge G_j - (\alpha + \mathrm{dd}^{\mathrm{c}}a_k) \wedge G_j, \phi \right\rangle\right| \leq C_\phi M_j (\sqrt{\delta_{j,k}} + \delta_{j,k}),$$

where C_ϕ depends on the first two derivatives of ϕ, on ω and on C. The same estimate holds with u_j, G_j replaced by u, G. For fixed k, the products $(\alpha + \mathrm{dd}^{\mathrm{c}}a_k) \wedge G_j$ of a fixed smooth form with G_j converge weakly to $(\alpha + \mathrm{dd}^{\mathrm{c}}a_k) \wedge G$. Let first $j \to \infty$ and then $k \to \infty$; since $u_j \to u$ uniformly, the error tends to zero. This proves convergence in distributions, and positivity together with the bounded masses $\int_X \alpha \wedge G_j$ upgrades it to weak convergence of measures. □

For $G = \gamma^{n-1}$ and $\alpha = \theta$, the product in Theorem 6.3.1 is the Gauduchon trace of a continuous potential. Its conserved mass controls a linear pairing of $\theta + \mathrm{dd}^{\mathrm{c}}u$; the nonlinear powers of this current, constructed for unbounded potentials in Chapter 7, require a separate mass analysis.

Chapter 7
Non-pluripolar Monge–Ampère measures

Throughout this chapter, ω is a Hermitian form on X and $\mathrm{d}V$ is a smooth positive probability volume form used for L^1 norms and integrals. As in Section 1.7, we write $\theta_v = \theta + \mathrm{dd}^{\mathrm{c}} v$, and $\mathrm{MA}_\theta(v) = (\theta + \mathrm{dd}^{\mathrm{c}} v)^n$ when v is bounded; convergence in capacity refers to the capacity Cap_ω of Section 1.8.

For an unbounded θ-plurisubharmonic function u, the Bedford–Taylor products of Proposition 1.7.1 are not defined. On the set $\{u > -t\}$, however, u coincides with its bounded truncation $\max(u, -t)$, and since bounded products are plurifinely local, the restricted measures are compatible as t increases. Their limit is a measure on the complement of the pole set, the non-pluripolar Monge–Ampère measure of u. For a general smooth background θ, the truncation need not be θ-plurisubharmonic, so positivity of the restricted products requires a local argument. The total mass of the limit may also be infinite. For a Hermitian background, finiteness of this mass for every potential is equivalent to a uniform bound on the Monge–Ampère masses of bounded potentials, the bounded mass property; it holds whenever X has a modification from a compact Kähler manifold, in particular when X is Moishezon. When the property fails, we construct a potential of infinite mass whose multiplier ideals are all trivial; its complex Hessian even has integrable coefficients, although its determinant is not integrable.

7.1 Truncation and non-pluripolar measures

Let θ be a smooth real $(1,1)$-form and $u \in \mathrm{PSH}(X, \theta)$. For $t > 0$, put

$$u_t = \max(u, -t), \qquad D_t = \{u > -t\}.$$

Choose a Hermitian form $\Omega \geq \theta$. Then u_t is bounded and Ω-plurisubharmonic, so Proposition 1.7.1 defines the product $(\theta + \mathrm{dd}^{\mathrm{c}} u_t)^n$ as a signed measure. It need not be positive on X, but its restriction to the plurifinely open set D_t is positive. The proof

replaces u_t near each point by a local truncation that preserves positivity up to an arbitrarily small smooth error.

Lemma 7.1.1 *With the notation above,*

$$0 \le \mathbf{1}_{D_t}(\theta + \mathrm{dd^c} u_t)^n \le \mathbf{1}_{D_t}(\Omega + \mathrm{dd^c} u_t)^n.$$

More generally, let $\theta_1, \dots, \theta_n$ be smooth real $(1,1)$-forms, let $u^i \in \mathrm{PSH}(X, \theta_i)$, and let D be the set where $u^i > -t$ for every i. Then the mixed product of the currents $\theta_i + \mathrm{dd^c} \max(u^i, -t)$ is positive on D.

Proof Work in a coordinate ball with a fixed positive constant form κ, and fix $\epsilon > 0$. Every point has a smaller neighborhood carrying a constant real form θ' such that

$$-\epsilon\kappa \le \theta - \theta' \le \epsilon\kappa.$$

Take a smooth local potential r for $\theta' + \epsilon\kappa$. Then $u + r$ is plurisubharmonic. On a relatively compact smaller neighborhood choose $k > t + \sup|r|$ and set

$$h_k = \max(u + r, -k) - r.$$

This function is locally bounded and equals $u_t = u$ on D_t. Moreover,

$$\theta + \mathrm{dd^c} h_k + 2\epsilon\kappa = \mathrm{dd^c} \max(u + r, -k) + (\theta - \theta' + \epsilon\kappa)$$

is the sum of a positive closed current with bounded potential and a smooth positive form, so its bounded products are positive. By plurifine locality, the restriction of $(\theta + \mathrm{dd^c} u_t + 2\epsilon\kappa)^n$ to D_t is positive on this neighborhood.

These neighborhoods cover the coordinate ball. Now let $\epsilon \downarrow 0$. The product with the fixed bounded potential u_t is a polynomial in ϵ whose coefficients have locally finite variation, so it converges in local total variation, also after restriction to D_t. This proves the first inequality. The same construction, applied to each factor, gives the positivity of mixed products.

For the second inequality, expand the difference of the two powers by the telescoping identity, using the smooth positive form $\Omega - \theta$. Every summand is a mixed product of factors that are positive on D_t, as just proved. The difference is therefore nonnegative there. □

Theorem 7.1.1 (Extended non-pluripolar measure) *Let θ be a smooth real $(1,1)$-form and $u \in \mathrm{PSH}(X,\theta)$. The measures $\mathbf{1}_{D_t}(\theta + \mathrm{dd^c} u_t)^n$ increase with t. Their limit*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) := \lim_{t\to+\infty} \mathbf{1}_{\{u>-t\}}(\theta + \mathrm{dd^c}\max(u,-t))^n \tag{7.1}$$

is a positive Borel measure that puts no mass on pluripolar sets; it is finite on each D_t and σ-finite on X. It depends only on the current $\theta + \mathrm{dd^c} u$; in particular, it is unchanged when a constant is added to u. If Ω is a Hermitian form with $\theta \le \Omega$, then

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) \le \mathrm{MA}_\Omega^{\mathrm{ext}}(u).$$

The total mass of $\mathrm{MA}_\theta^{\mathrm{ext}}(u)$ may be infinite.

Proof For $s > t$, plurifine locality on $D_t \subset D_s$ gives

$$\mathbf{1}_{D_t}(\theta + \mathrm{dd^c} u_s)^n = \mathbf{1}_{D_t}(\theta + \mathrm{dd^c} u_t)^n.$$

Together with Lemma 7.1.1, this proves that the measures increase. Their increasing limit is countably additive by monotone convergence, and it vanishes on pluripolar sets because each bounded product does. The comparison with Ω passes to the limit. By the compatibility just proved,

$$\mathbf{1}_{D_t}\mathrm{MA}_\theta^{\mathrm{ext}}(u) = \mathbf{1}_{D_t}(\theta + \mathrm{dd^c} u_t)^n,$$

and the mass of this measure is bounded by that of the bounded product $(\Omega + \mathrm{dd^c} u_t)^n$. The sets D_t exhaust the complement of $\{u = -\infty\}$, which has zero mass, so the limit is σ-finite.

For a smooth real function f, consider $\theta' = \theta + \mathrm{dd^c} f$ and $u' = u - f$. On a common superlevel set of u and u', sufficiently deep truncations of u and u' define the same current with bounded potentials, so their restricted products agree. These sets exhaust the complement of the common pole set, which proves invariance under this change of decomposition. By the elliptic-regularity argument in the proof of Proposition 1.7.1, any two smooth decompositions of the same current differ in this way, and adding a constant is a special case. □

We call $\mathrm{MA}_\theta^{\mathrm{ext}}(u)$ the *extended non-pluripolar Monge–Ampère measure* of u; the word *extended* indicates that its total mass may be infinite. On compact X it defines a current of order zero exactly when its total mass is finite. In that case it is the non-pluripolar Monge–Ampère measure in the usual sense, namely the extension by zero across $\{u = -\infty\}$ of the plurifinely local bounded products. If u is bounded, then $D_t = X$ for large t, and $\mathrm{MA}_\theta^{\mathrm{ext}}(u) = \mathrm{MA}_\theta(u)$. In accordance with the usual notation for non-pluripolar products, we also write $\langle(\theta + \mathrm{dd^c} u)^n\rangle$ for $\mathrm{MA}_\theta^{\mathrm{ext}}(u)$; lower powers $\langle(\theta + \mathrm{dd^c} u)^k\rangle$ are defined by the same truncation where they are used.

7.2 Contact sets

The bounded contact inequality of Lemma 1.7.1 extends to unbounded potentials on common finite levels. As in the previous section, local positive corrections are needed, because the global truncations need not remain θ-plurisubharmonic.

Proposition 7.2.1 *Let θ be a smooth real $(1,1)$-form and let $u, v \in \mathrm{PSH}(X,\theta)$ satisfy $u \le v$. Then*

$$\mathbf{1}_{\{u=v\}}\mathrm{MA}_\theta^{\mathrm{ext}}(u) \le \mathbf{1}_{\{u=v\}}\mathrm{MA}_\theta^{\mathrm{ext}}(v).$$

Proof Fix $t > 0$, put $D_t = \{u > -t\}$, and choose a Hermitian form $\Omega \ge \theta$. The truncations $u_t = \max(u, -t)$ and $v_t = \max(v, -t)$ are bounded and Ω-plurisubharmonic, so their signed products with background θ are defined. Fix $\epsilon > 0$ and a point of a coordinate ball, and let κ, r and $k > t + \sup|r|$ be as in the proof of Lemma 7.1.1. On a neighborhood of the point, the bounded functions

$$a_k = \max(u + r, -k) - r, \qquad b_k = \max(v + r, -k) - r$$

satisfy $a_k \le b_k$, and as shown there, $\theta + 2\epsilon\kappa + \mathrm{dd^c} a_k \ge 0$ and $\theta + 2\epsilon\kappa + \mathrm{dd^c} b_k \ge 0$. On D_t we have $v \ge u > -t$, so $a_k = u_t$ and $b_k = v_t$ there. The top-degree case of Lemma 1.7.1, with background $\theta + 2\epsilon\kappa$, and plurifine locality give

$$\mathbf{1}_{\{u=v\}\cap D_t}(\theta + 2\epsilon\kappa + \mathrm{dd^c} u_t)^n \le \mathbf{1}_{\{u=v\}\cap D_t}(\theta + 2\epsilon\kappa + \mathrm{dd^c} v_t)^n$$

on this neighborhood, hence on the whole ball. As in the proof of Lemma 7.1.1, both sides converge in local total variation as $\epsilon \downarrow 0$, which gives the same inequality for the background θ. Since $D_t \subset \{v > -t\}$, Theorem 7.1.1 identifies its two sides with the restrictions of $\mathrm{MA}_\theta^{\mathrm{ext}}(u)$ and $\mathrm{MA}_\theta^{\mathrm{ext}}(v)$ to $\{u = v\} \cap D_t$. Finally let $t \to +\infty$: the sets $\{u = v\} \cap D_t$ exhaust the contact set off the pluripolar set $\{u = -\infty\}$, which has zero extended mass. □

7.3 Weighted lower semicontinuity

Products of uniformly bounded potentials are continuous under convergence in capacity. Multiplying the extended measure by a cutoff in the potential that vanishes below a fixed level reduces it to such bounded products, and letting the level tend to $-\infty$ gives lower semicontinuity of the extended measures, even when their masses are infinite. The weights are allowed to vary with the potentials.

Lemma 7.3.1 *Let θ be a smooth real $(1,1)$-form. Suppose $u_j, u \in \mathrm{PSH}(X,\theta)$ have a common upper bound and $u_j \to u$ in capacity. If b_j, b are nonnegative uniformly bounded quasi-continuous functions and $b_j \to b$ in capacity, then*

$$\int_X b\,\mathrm{MA}_\theta^{\mathrm{ext}}(u) \le \liminf_j \int_X b_j\,\mathrm{MA}_\theta^{\mathrm{ext}}(u_j).$$

Both sides may be infinite.

Proof Subtracting a common constant, we may assume that all the potentials are nonpositive. Choose a Hermitian form $\Omega \ge \theta$. For $T > 0$, let $g_T\colon [-\infty, 0] \to [0,1]$ be a continuous nondecreasing cutoff, equal to zero on $[-\infty, -2T]$ and to one on $[-T, 0]$, chosen so that g_T increases pointwise to one on $(-\infty, 0]$ as $T \to \infty$. Put $v_j = \max(u_j, -2T)$ and $v = \max(u, -2T)$. These are uniformly bounded Ω-plurisubharmonic functions, and they converge in capacity.

Fix T. The weights $b_j g_T(v_j)$ are uniformly bounded and quasi-continuous, and they converge in capacity to $b g_T(v)$. The weighted capacity continuity of bounded products (Sections 1.6 and 1.8), applied term by term to the local expansion of the signed products $(\theta + \mathrm{dd^c} v_j)^n$, gives the convergence below. Since the cutoff vanishes at the truncation level $-2T$, locality gives the identity

$$\begin{aligned}\int_X b_j g_T(u_j)\,\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) &= \int_X b_j g_T(v_j)(\theta + \mathrm{dd^c} v_j)^n \\ &\longrightarrow \int_X b g_T(u)\,\mathrm{MA}_\theta^{\mathrm{ext}}(u).\end{aligned}$$

The first integral is nonnegative and at most the corresponding integral without the cutoff, although the bounded product $(\theta + \mathrm{dd^c} v_j)^n$ may be signed. Hence the limit is bounded by the lower limit in the assertion. Letting $T \to \infty$ and using monotone convergence completes the proof. □

7.4 Density and trace inequalities

The mixed density inequality of Lemma 1.7.3 passes to unbounded potentials on common finite levels. It gives a lower bound for the extended measure of a convex combination of potentials and, by comparison with a smooth metric, a lower bound for the full trace measure.

Lemma 7.4.1 *Let θ be a smooth semipositive $(1,1)$-form and let $f, g \geq 0$ be measurable functions on X. Suppose $u, v \in \mathrm{PSH}(X,\theta)$ satisfy*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) \geq f\,\mathrm{d}V, \qquad \mathrm{MA}_\theta^{\mathrm{ext}}(v) \geq g\,\mathrm{d}V.$$

Then, for $0 \leq t \leq 1$,

$$\mathrm{MA}_\theta^{\mathrm{ext}}(tu + (1-t)v) \geq \bigl(t f^{1/n} + (1-t) g^{1/n}\bigr)^n\,\mathrm{d}V. \tag{7.2}$$

Proof The cases $t = 0$ and $t = 1$ are the hypotheses. Let $0 < t < 1$, translate u, v to be nonpositive, and put $u_k = \max(u, -k)$ and $v_k = \max(v, -k)$. Since θ is semipositive, these are bounded θ-plurisubharmonic functions, and by locality

$$\mathrm{MA}_\theta(u_k) \geq \mathbf{1}_{\{u > -k\}} f\,\mathrm{d}V, \qquad \mathrm{MA}_\theta(v_k) \geq \mathbf{1}_{\{v > -k\}} g\,\mathrm{d}V.$$

On $E_k = \{u > -k\} \cap \{v > -k\}$, the inequality (1.12) of Lemma 1.7.3, with $a = t$ and $b = 1 - t$, bounds $\mathrm{MA}_\theta(tu_k + (1-t)v_k)$ below by the measure on the right of (7.2). On the same set, plurifine locality identifies the left-hand side with the extended measure of $tu + (1-t)v$. As $k \to \infty$, the sets E_k exhaust the complement of the two pole sets, which have zero smooth volume. Monotone convergence on Borel sets proves the inequality. □

The same mixed inequality gives a lower bound for the full trace measure of an unbounded potential. Unlike the extended measure, the full current $\omega + \mathrm{d}\mathrm{d}^{\mathrm{c}} u$ keeps its mass on the pole set; the bound below concerns only its absolutely continuous part.

Lemma 7.4.2 *Let γ be a Hermitian form. If $u \in \mathrm{PSH}(X,\omega)$ and a measurable function $g \geq 0$ satisfy $\mathrm{MA}_\omega^{\mathrm{ext}}(u) \geq g\,\mathrm{d}V$, then*

$$(\omega + \mathrm{d}\mathrm{d}^{\mathrm{c}} u) \wedge \gamma^{n-1} \geq (g\,\mathrm{d}V/\gamma^n)^{1/n}\gamma^n.$$

Proof For bounded u, Lemma 1.7.3, applied in local coordinates with $T_1 = \omega + \mathrm{d}\mathrm{d}^{\mathrm{c}} u$, $T_2 = \gamma$ and $k = 1$, gives the asserted inequality.

In general, subtract a constant so that $u \leq 0$, and put $u_k = \max(u, -k)$. Positivity and locality give

$$\mathrm{MA}_\omega(u_k) \geq \mathbf{1}_{\{u > -k\}} g\,\mathrm{d}V,$$

so the bounded case applies to u_k with the density $\mathbf{1}_{\{u > -k\}} g$. Since $u_k \to u$ in L^1, the positive measures $(\omega + \mathrm{d}\mathrm{d}^{\mathrm{c}} u_k) \wedge \gamma^{n-1}$ converge weakly to $(\omega + \mathrm{d}\mathrm{d}^{\mathrm{c}} u) \wedge \gamma^{n-1}$. The densities on the right increase to $(g\,\mathrm{d}V/\gamma^n)^{1/n}$ off the pole set, which has zero smooth volume. Testing against nonnegative continuous functions and using monotone convergence on the right proves the assertion. □

7.5 The bounded mass property

When ω is a Kähler form, all bounded ω-plurisubharmonic functions have the same Monge–Ampère mass $\int_X \omega^n$. For a general Hermitian form, the mass can depend on the potential, as in Example 2.3.1, and the following two numbers measure its range.

Definition 7.5.1 Let ω be a Hermitian form on X. The *lower mass* and the *upper mass* of ω are

$$v_-(\omega) = \inf_{v \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)} \int_X (\omega + \mathrm{dd}^{\mathrm{c}} v)^n,$$
$$v_+(\omega) = \sup_{v \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)} \int_X (\omega + \mathrm{dd}^{\mathrm{c}} v)^n.$$

The manifold X has the *bounded mass property*, abbreviated BMP, if $v_+(\omega) < +\infty$. The form ω has *positive lower mass* if $v_-(\omega) > 0$.

Testing with $v = 0$ gives $v_-(\omega) \le \int_X \omega^n \le v_+(\omega)$. By the next proposition, the bounded mass property does not depend on the choice of ω. Since the extended measure (7.1) is built from products of bounded potentials, the upper mass also controls the extended measures of unbounded potentials (Proposition 7.5.2).

Proposition 7.5.1 *The bounded mass property is independent of the Hermitian form. Moreover, $v_+(\omega) = \overline{\mathrm{v}}_{\mathrm{sm}}(\omega)$, the supremum of the masses of smooth strictly ω-plurisubharmonic potentials in Proposition 2.5.1; in particular, the supremum in Definition 7.5.1 is unchanged if only smooth potentials are used.*

Proof Smooth strictly ω-plurisubharmonic potentials are bounded, so $\overline{\mathrm{v}}_{\mathrm{sm}}(\omega) \le v_+(\omega)$. Conversely, approximate a bounded potential by Theorem 1.11.1. The smooth approximants are strictly ω-plurisubharmonic, decrease and remain uniformly bounded, so the continuity of bounded products along decreasing sequences gives convergence of their total masses. This proves $v_+(\omega) = \overline{\mathrm{v}}_{\mathrm{sm}}(\omega)$, and metric independence follows from Proposition 2.5.1. □

A compact complex manifold of dimension n is *Moishezon* if it has n algebraically independent meromorphic functions. By Moishezon's theorem, a compact irreducible complex space of dimension n with n algebraically independent meromorphic functions admits a modification from a smooth projective manifold; the source of this form of the theorem is given in Section 36.5. Such a modification gives X the bounded mass property: the pulled-back Hermitian form is dominated by a fixed Kähler form, whose Monge–Ampère mass does not change when a smooth potential is added.

Lemma 7.5.1 *If a compact complex manifold X admits a modification $\pi\colon Z \to X$ from a compact Kähler manifold, in particular if X is Moishezon, then X has the bounded mass property.*

Proof For a Moishezon manifold, Moishezon's theorem gives such a modification from a smooth projective, hence Kähler, manifold. Fix a Hermitian form ω on X and a Kähler form κ on Z, and choose $C > 0$ with $\pi^*\omega \le C\kappa$. By Proposition 7.5.1, it suffices to bound the masses of smooth functions $u \in \mathrm{PSH}(X, \omega)$. Their pullbacks satisfy

$$0 \le \pi^*\omega + \mathrm{dd^c}(u \circ \pi) \le C\kappa + \mathrm{dd^c}(u \circ \pi).$$

Taking top powers and integrating gives

$$\begin{aligned}\int_X (\omega + \mathrm{dd^c} u)^n &= \int_Z (\pi^*\omega + \mathrm{dd^c}(u \circ \pi))^n \\ &\le \int_Z (C\kappa + \mathrm{dd^c}(u \circ \pi))^n = C^n \int_Z \kappa^n.\end{aligned}$$

The first equality follows by change of variables on the set where π is biholomorphic, since smooth top-degree forms give the exceptional analytic sets zero mass. The inequality follows by expanding in the smooth semipositive difference $C\kappa - \pi^*\omega$, and the last equality follows from Stokes' theorem because κ is closed. The bound does not depend on u, which proves the bounded mass property. □

Proposition 7.5.2 *If θ is a smooth real $(1,1)$-form with $\theta \le C\omega$ for some $C > 0$, and $u \in \mathrm{PSH}(X, \theta)$, then*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u)(X) \le C^n v_+(\omega).$$

Proof The comparison in Theorem 7.1.1, applied with the Hermitian form $C\omega$, gives

$$\mathbf{1}_{D_t} \mathrm{MA}_\theta^{\mathrm{ext}}(u) \le \mathbf{1}_{D_t} (C\omega + \mathrm{dd^c} u_t)^n.$$

Since u_t / C is bounded and ω-plurisubharmonic, the right-hand side has mass at most $C^n v_+(\omega)$. Let $t \to \infty$. □

7.6 Infinite mass with trivial multiplier ideals

If the bounded mass property fails, there are smooth ω-plurisubharmonic functions u_j whose masses M_j tend to infinity. In a convex combination $U = \sum_j a_j u_j$, positivity of mixed products shows that each summand contributes at least $a_j^n M_j$ to the extended measure of U. Choosing the masses M_j to outgrow a_j^{-n} therefore forces infinite mass. On the other hand, each tail $\sum_{j>N} a_j u_j$ is a small multiple of a convex combination of normalized potentials, and the uniform exponential integrability of Lemma 1.5.1 applies to it at every exponent.

Theorem 7.6.1 *If $v_+(\omega) = +\infty$, there is $U \in \mathrm{PSH}(X,\omega)$, $U \le 0$, such that*

$$\mathrm{MA}_\omega^{\mathrm{ext}}(U)(X) = +\infty, \qquad \int_X \mathrm{e}^{-pU}\,\mathrm{d}V < +\infty \quad \textit{for every } p > 0.$$

In particular, $\mathcal{I}(pU) = \mathcal{O}_X$ for every $p > 0$.

Proof Set $a_j = 2^{-j}$ for $j \ge 1$. By Proposition 7.5.1, we can choose smooth ω-plurisubharmonic functions u_j with $\sup_X u_j = 0$ and

$$M_j := \int_X \omega_{u_j}^n \ge 2^{j(n+1)}.$$

The partial sums $U_N = \sum_{j\le N} a_j u_j$ decrease and satisfy

$$\omega + \mathrm{dd^c} U_N = \sum_{j\le N} a_j \omega_{u_j} + \left(1 - \sum_{j\le N} a_j\right)\omega \ge 0.$$

Their L^1 norms are bounded by (1.1), so their limit U is ω-plurisubharmonic and nonpositive.

We first prove that the mass is infinite. Fix N, and put

$$S_N = \sum_{j\le N} a_j \omega_{u_j}, \qquad \delta_N = \sum_{j>N} a_j = 2^{-N}, \qquad V_N = U - U_N.$$

Passing to the limit in the remaining partial sums gives $\delta_N\omega + \mathrm{dd^c} V_N \ge 0$. For $t > 0$, the function $W_{N,t} = U_N + \max(V_N, -t)$ is bounded, and its current is the sum of two positive currents with bounded potentials:

$$\omega + \mathrm{dd^c} W_{N,t} = S_N + \big(\delta_N\omega + \mathrm{dd^c}\max(V_N,-t)\big).$$

On $D_t = \{U > -t\}$, the inequality $U_N \le 0$ gives $V_N \ge U > -t$, and hence $W_{N,t} = U = \max(U,-t)$ there. Plurifine locality and positivity of bounded mixed products yield

$$\mathbf{1}_{D_t}\omega^n_{\max(U,-t)} \ge \mathbf{1}_{D_t} S_N^n.$$

The sets D_t exhaust $X \setminus \{U = -\infty\}$. This pole set is pluripolar, hence has zero smooth volume, so the smooth measure S_N^n gives it no mass. Letting $t \to +\infty$, we obtain

$$\mathrm{MA}_{\omega}^{\mathrm{ext}}(U) \geq S_N^n. \tag{7.3}$$

For $j \leq N$, positivity gives $S_N^n \geq a_j^n \omega_{u_j}^n$, so the mass of U is at least $a_j^n M_j \geq 2^j$. Since j is arbitrary, the mass is infinite.

For the integrability, fix $p > 0$, let α be the exponent of Lemma 1.5.1, and take N so large that $p\delta \leq \alpha$, where $\delta = \sum_{j>N} a_j = 2^{-N}$. Write $U = A_N + V_N$ with $A_N = \sum_{j \leq N} a_j u_j$ smooth. For a finite tail ending at $m > N$, Jensen's inequality for the convex exponential function, with the unused weight assigned to the value zero, gives

$$\exp\left(-p \sum_{N<j\leq m} a_j u_j\right) \leq \sum_{N<j\leq m} \frac{a_j}{\delta} \mathrm{e}^{-p\delta u_j} + 1 - \sum_{N<j\leq m} \frac{a_j}{\delta}.$$

Because $u_j \leq 0$ and $p\delta \leq \alpha$, the integrals of the exponentials on the right are uniformly bounded. Integrating and letting $m \to +\infty$ by monotone convergence, we obtain $\int_X \mathrm{e}^{-pV_N} \mathrm{d}V < +\infty$. Multiplication by the bounded smooth function e^{-pA_N} proves $\int_X \mathrm{e}^{-pU} \mathrm{d}V < +\infty$.

Consequently, the constant germ 1 is locally integrable against each weight e^{-pU}, and every multiplier ideal $\mathcal{I}(pU)$ is the whole structure sheaf. □

Corollary 7.6.1 *The bounded mass property holds if and only if* $\mathrm{MA}_{\omega}^{\mathrm{ext}}(u)(X) < +\infty$ *for every* $u \in \mathrm{PSH}(X, \omega)$. *If the bounded mass property fails, there are potentials with infinite extended mass and trivial multiplier ideals* $\mathcal{I}(pu) = \mathcal{O}_X$, $p > 0$, *arbitrarily close to zero in* L^1.

Proof The first assertion follows from Proposition 7.5.2 and Theorem 7.6.1. For the second, take εU, $0 < \varepsilon \leq 1$, with U as in Theorem 7.6.1. It is ω-plurisubharmonic and has the required exponential integrability. On $D_t = \{U > -t\}$, its truncation at level εt is $\varepsilon \max(U, -t)$, and

$$\omega + \mathrm{dd}^{\mathrm{c}}\big(\varepsilon \max(U, -t)\big) = \varepsilon \omega_{\max(U,-t)} + (1 - \varepsilon)\omega.$$

Positivity of mixed products bounds the restricted measure of εU below by ε^n times that of U. Its limiting mass is therefore infinite. Finally, $\|\varepsilon U\|_{L^1} \to 0$ as $\varepsilon \to 0$. □

7.7 Integrable Hessians and infinite mass

In the construction of Theorem 7.6.1, the Gauduchon identity of Proposition 4.2.1 fixes the trace integral of every positive summand $\omega + \mathrm{dd}^{\mathrm{c}} u_j$. The series defining $\omega + \mathrm{dd}^{\mathrm{c}} U$ therefore converges in the coefficient L^1 norm, although the determinant of its sum is not integrable. Thus the trace measures of the partial sums converge in total variation, yet the Monge–Ampère mass of U is infinite. In Chapter 8, weights that decay on deep sublevel sets will control these masses.

Fix a Gauduchon form γ and set $M = \int_X \omega \wedge \gamma^{n-1}$. For a real $(1,1)$-form T with measurable coefficients, let A_T be its Hermitian matrix relative to γ, and let $|A_T| = (A_T^2)^{1/2}$ be its positive matrix absolute value. We use the norm

$$\|T\|_{L^1(\gamma)} = \frac{1}{n} \int_X \operatorname{tr} |A_T| \, \gamma^n.$$

For $T \geq 0$ this equals $\int_X T \wedge \gamma^{n-1}$. On compact X, any other smooth Hermitian form gives an equivalent norm and hence the same space of integrable coefficients. When the coefficients of T are integrable, we write $\det_\gamma T = \det A_T$; this function is defined almost everywhere, and $T^n = (\det_\gamma T)\gamma^n$ when T is smooth.

Proposition 7.7.1 *Suppose $v_+(\omega) = +\infty$. For the potential U and its smooth decreasing approximants U_N constructed in the proof of Theorem 7.6.1, the current $T = \omega + \mathrm{dd^c} U$ has integrable coefficients and*

$$\|\omega + \mathrm{dd^c} U_N - T\|_{L^1(\gamma)} \leq 2^{1-N} M.$$

Its pointwise determinant satisfies

$$\mathrm{MA}_\omega^{\mathrm{ext}}(U) \geq (\det_\gamma T)\gamma^n, \qquad \int_X (\det_\gamma T)\gamma^n = +\infty.$$

Moreover,

$$\int_X (\omega + \mathrm{dd^c} U_N)^n \longrightarrow +\infty.$$

Proof Write $T_j = \omega + \mathrm{dd^c} u_j$ for the smooth positive forms in that construction, and, as in its proof, put

$$S_N = \sum_{j \leq N} a_j T_j, \qquad \delta_N = \sum_{j > N} a_j = 2^{-N}.$$

The smooth positive forms S_N increase pointwise, and the Gauduchon identity gives

$$\int_X S_N \wedge \gamma^{n-1} = (1 - \delta_N) M.$$

By monotone convergence of their traces, the series of matrices has a finite positive limit S almost everywhere. The coefficients of S are integrable, because in a γ-unitary frame the absolute value of every coefficient of a positive matrix is bounded by its trace. Positivity and the same trace identity give

$$\|S - S_N\|_{L^1(\gamma)} = \delta_N M.$$

In particular, $S_N \to S$ in the sense of distributions. Since

$$\omega + \mathrm{dd^c} U_N = S_N + \delta_N \omega \longrightarrow \omega + \mathrm{dd^c} U$$

in the sense of distributions, by L^1 convergence of the potentials, we obtain $T = S$. The triangle inequality now gives

$$\|\omega + \mathrm{dd^c} U_N - T\|_{L^1(\gamma)} \le \|S_N - S\|_{L^1(\gamma)} + \delta_N \|\omega\|_{L^1(\gamma)} = 2\delta_N M.$$

To compare $\mathrm{MA}_\omega^{\mathrm{ext}}(U)$ with the determinant, recall (7.3): $\mathrm{MA}_\omega^{\mathrm{ext}}(U) \ge S_N^n$ for every N. The positive matrices S_N increase to T almost everywhere, so their determinants increase to $\det_\gamma T$. Monotone convergence on every Borel set gives the asserted inequality between $\mathrm{MA}_\omega^{\mathrm{ext}}(U)$ and the determinant measure.

Finally, for every $N \ge j$, positivity gives $T \ge S_N \ge a_j T_j$ and $\omega + \mathrm{dd^c} U_N \ge S_N$. Hence both the measurable limit and the smooth approximating forms satisfy

$$\int_X (\det_\gamma T)\gamma^n \ge a_j^n \int_X T_j^n \ge 2^j, \qquad \int_X (\omega + \mathrm{dd^c} U_N)^n \ge a_j^n \int_X T_j^n \ge 2^j.$$

The first bound holds for every j, and the second for every $N \ge j$; together they prove the two assertions about infinite mass. □

Corollary 7.7.1 *If the bounded mass property fails, there are nonpositive potentials $w_\varepsilon \in \mathrm{PSH}(X, \omega)$, $0 < \varepsilon < 1$, such that*

$$\omega + \mathrm{dd^c} w_\varepsilon \ge (1 - \varepsilon)\omega, \qquad \mathcal{I}(p w_\varepsilon) = \mathcal{O}_X \quad (p > 0),$$

and such that both the integral of the pointwise determinant $\det_\gamma(\omega + \mathrm{dd^c} w_\varepsilon)$ against γ^n and the extended Monge–Ampère mass of w_ε are infinite. As $\varepsilon \downarrow 0$, the potentials converge to zero in L^1, their currents converge to ω in the coefficient L^1 norm, and their full trace measures $(\omega + \mathrm{dd^c} w_\varepsilon) \wedge \gamma^{n-1}$ converge to $\omega \wedge \gamma^{n-1}$ in total variation.

Proof Take $w_\varepsilon = \varepsilon U$, with U and $T = \omega + \mathrm{dd^c} U$ as in Proposition 7.7.1. Its current is

$$T_\varepsilon = (1 - \varepsilon)\omega + \varepsilon T.$$

The lower bound and the two L^1 convergence assertions follow immediately. The full trace measures have integrable densities, and their distance from $\omega \wedge \gamma^{n-1}$ in total variation is at most

$$\varepsilon \|T - \omega\|_{L^1(\gamma)}.$$

Every multiplier ideal is trivial by the exponential integrability of U proved in Theorem 7.6.1. Since $T_\varepsilon \ge \varepsilon T$ almost everywhere, the determinant integral of T_ε is at least ε^n times that of T, hence infinite. The extended mass of w_ε is infinite by the proof of Corollary 7.6.1. □

For smooth potentials, the total masses can be prescribed in advance while both the currents and the exponential weights converge.

Corollary 7.7.2 *Suppose $v_+(\omega) = +\infty$. Let $(\varepsilon_j)_{j\geq 1}$ and $(b_j)_{j\geq 1}$ be real sequences with $0 < \varepsilon_j < 1$, $\varepsilon_j \to 0$, and $b_j > \int_X \omega^n$. There are smooth potentials $w_j \in \mathrm{PSH}(X,\omega)$ with $\sup_X w_j = 0$ such that*

$$\omega + \mathrm{dd^c} w_j \geq (1-\varepsilon_j)\omega, \qquad \int_X (\omega + \mathrm{dd^c} w_j)^n = b_j,$$
$$\|\mathrm{dd^c} w_j\|_{L^1(\gamma)} \leq 2M\varepsilon_j.$$

Moreover, $w_j \to 0$ in $L^1(\mathrm{d}V)$ and $\mathrm{e}^{-pw_j} \to 1$ in $L^1(\mathrm{d}V)$ for every fixed $p > 0$. The full Gauduchon trace measures converge to $\omega \wedge \gamma^{n-1}$ in total variation.

Proof By Proposition 7.5.1, we can choose a smooth $u_j \in \mathrm{PSH}(X,\omega)$ with $\sup_X u_j = 0$ and

$$\int_X (\omega + \mathrm{dd^c} u_j)^n > \varepsilon_j^{-n} b_j.$$

Put $T_j = \omega + \mathrm{dd^c} u_j$. The polynomial

$$F_j(s) = \int_X \big((1-s)\omega + sT_j\big)^n, \qquad 0 \leq s \leq \varepsilon_j,$$

satisfies $F_j(0) < b_j$, and also $F_j(\varepsilon_j) \geq \varepsilon_j^n \int_X T_j^n > b_j$ by positivity of the mixed terms. The intermediate value theorem gives $s_j \in (0,\varepsilon_j)$ with $F_j(s_j) = b_j$. Set $w_j = s_j u_j$. Then $\sup_X w_j = 0$, $\omega + \mathrm{dd^c} w_j = (1-s_j)\omega + s_j T_j \geq (1-\varepsilon_j)\omega$, and w_j has the prescribed mass b_j.

The Gauduchon identity gives $\|T_j\|_{L^1(\gamma)} = \|\omega\|_{L^1(\gamma)} = M$. Hence

$$\|\mathrm{dd^c} w_j\|_{L^1(\gamma)} = s_j \|T_j - \omega\|_{L^1(\gamma)} \leq 2M\varepsilon_j.$$

Since the absolute value of the trace of a Hermitian matrix is bounded by the trace of its absolute value, the same bound holds for the total variation of $\mathrm{dd^c} w_j \wedge \gamma^{n-1}$. With K_ω as in (1.1), we also have $\|w_j\|_{L^1(\mathrm{d}V)} \leq K_\omega \varepsilon_j$.

Finally, let α, C be the constants of Lemma 1.5.1 for $(X,\omega,\mathrm{d}V)$, and fix $p > 0$. For all sufficiently large j, one has $ps_j \leq \alpha$. Convexity of the exponential function, applied with weight ps_j/α, gives

$$0 \leq \mathrm{e}^{-pw_j} - 1 \leq \frac{ps_j}{\alpha}\big(\mathrm{e}^{-\alpha u_j} - 1\big).$$

Integrating, we bound the $L^1(\mathrm{d}V)$ norm of $\mathrm{e}^{-pw_j} - 1$ by $pC\varepsilon_j/\alpha$, which proves the last convergence. □

Chapter 8
Weighted Monge–Ampère measures

Throughout this chapter, ω is a Hermitian form on X. Capacity refers to the capacity Cap_ω of Section 1.8. For nonpositive $u \in \mathrm{PSH}(X,\omega)$ we write

$$\mu_u = \mathrm{MA}_\omega^{\mathrm{ext}}(u)$$

for the extended non-pluripolar Monge–Ampère measure of Theorem 7.1.1, and we put

$$r_n = \max(n-2,0).$$

By Chapter 7, the measure μ_u can have infinite mass when the bounded mass property fails. Its mass above depth t, on the superlevel set $\{u \geq -t\}$, is nevertheless bounded by a polynomial of degree r_n in t, because the masses of bounded potentials grow at most polynomially in their range. A weight that decreases sufficiently fast in the depth $-u$ compensates for this growth. If $w\colon [0,+\infty) \to [0,+\infty)$ is a nonincreasing function with $w(t) \to 0$ as $t \to +\infty$, we call

$$w(-u)\,\mu_u$$

the *damped Monge–Ampère measure* of u with weight w; the weight is set to zero on the pole set $\{u = -\infty\}$, which μ_u does not charge. The main examples are the exponential weights e^{-pt} and the polynomial weights $(1+t)^{-q}$. For suitable weights the damped measures are finite, uniformly absolutely continuous with respect to capacity, and continuous under convergence in capacity, a property that the extended measures themselves lack. Removing the weight requires uniform control of the mass on deep sublevel sets; we characterize this control and show by examples that such mass can persist although the potentials converge in capacity.

8.1 Range and restricted mass

We first bound the Monge–Ampère mass of a bounded potential polynomially in its range. The first mixed mass $\int_X (\omega + \mathrm{dd^c} v) \wedge \omega^{n-1}$ is bounded independently of v by the Gauduchon trace. Local Chern–Levine–Nirenberg estimates turn this into a bound of degree $j-1$ in the range for the jth mixed mass, and Stokes' theorem expresses the top mass in terms of the lower ones.

Proposition 8.1.1 *There is a constant C_ω such that, for every $A \geq 0$,*

$$\int_X (\omega + \mathrm{dd^c} v)^n \leq C_\omega (1+A)^{r_n} \quad \text{if } v \in \mathrm{PSH}(X,\omega),\ 0 \leq v \leq A.$$

Proof Choose finitely many coordinate balls whose concentric smaller balls, with compact closure in the larger ones, cover X. On each larger ball choose a fixed quadratic function q with $\mathrm{dd^c} q \geq \omega$, shifted so that $p = v + q$ is plurisubharmonic with $0 \leq p \leq A + C$. Let κ be the Euclidean form. In the identity

$$\mathrm{dd^c} p = (\omega + \mathrm{dd^c} v) + (\mathrm{dd^c} q - \omega),$$

the integral of the first term against κ^{n-1} is bounded by a fixed multiple of the global trace mass, which is bounded by Corollary 4.2.1, and the second term is smooth and fixed. Thus $\int \mathrm{dd^c} p \wedge \kappa^{n-1}$ is bounded uniformly in v and A; in dimension one, Stokes' theorem replaces the trace bound. By (1.2), on each smaller ball,

$$\int (\mathrm{dd^c} p)^j \wedge \kappa^{n-j} \leq C_j (1+A)^{j-1} \quad (1 \leq j \leq n).$$

Expand $(\omega + \mathrm{dd^c} v)^j = (\mathrm{dd^c} p + \omega - \mathrm{dd^c} q)^j$ and sum over the finite cover. Since the smooth factors have bounded coefficients, the positive mixed masses satisfy

$$0 \leq M_j(v) := \int_X (\omega + \mathrm{dd^c} v)^j \wedge \omega^{n-j} \leq C_j (1+A)^{\max(j-1,0)} \quad (0 \leq j \leq n).$$

For smooth v, the exact form $(\mathrm{dd^c} v)^n$ integrates to zero, as in the proof of Theorem 2.2.1; the regularization of Theorem 1.11.1 and the continuity of bounded products give the same identity for bounded v. Thus

$$0 = \int_X (\mathrm{dd^c} v)^n = \sum_{j=0}^{n} (-1)^{n-j} \binom{n}{j} M_j(v).$$

Since the mixed masses are nonnegative, it follows that

$$M_n(v) \leq \sum_{j=0}^{n-1} \binom{n}{j} M_j(v) \leq C_\omega (1+A)^{r_n}.$$

The largest power is $n-2$ when $n \geq 2$; for $n = 1$, only the constant term remains. □

For $n \leq 2$ the bound is uniform, so every compact complex curve or surface has the bounded mass property. In higher dimensions the bound grows with the range, and truncation turns it into a bound for the mass of an unbounded potential above a finite depth.

Lemma 8.1.1 *Let $u \in \mathrm{PSH}(X,\omega)$ be nonpositive. For $t \geq 0$ and every Borel set $E \subset X$,*

$$\mu_u(\{u \geq -t\}) \leq C_\omega (1+t)^{r_n}, \qquad \mu_u(E \cap \{u \geq -t\}) \leq (1+t)^n \operatorname{Cap}_\omega(E). \quad (8.1)$$

Proof Put $v_t = \max(u, -t-1) + t + 1$. It is ω-plurisubharmonic and takes values in $[0, t+1]$, and by plurifine locality its Monge–Ampère measure equals μ_u on $\{u \geq -t\} \subset \{u > -t-1\}$. The first estimate therefore follows from Proposition 8.1.1, after enlarging the constant. For the second, $v_t/(1+t)$ is a test function for the capacity, and positivity gives

$$(\omega + \mathrm{dd^c} v_t)^n \leq (1+t)^n \big(\omega + \mathrm{dd^c}(v_t/(1+t))\big)^n.$$

Restricting this inequality to $E \cap \{u \geq -t\}$ proves the capacity bound. □

8.2 Exponential damping

The exponential weight can be handled directly. A convex increasing function with bounded range transforms u into a bounded ω-plurisubharmonic function, and the chain rule shows that the transformed current dominates an exponential multiple of $\omega + \mathrm{dd^c} u$. The Monge–Ampère measure of the bounded transform therefore controls the damped measure.

For $a > 0$ put $F_a(t) = a^{-1} \mathrm{e}^{at}$ for $t \leq 0$, with $F_a(-\infty) = 0$. Then $0 \leq F_a \leq a^{-1}$, while $0 < F_a' \leq 1$ and $F_a'' \geq 0$ at finite arguments.

Proposition 8.2.1 *If $u \in \mathrm{PSH}(X,\omega)$ and $u \leq 0$, then $F_a(u)$ is a bounded ω-plurisubharmonic function, and*

$$\mathrm{e}^{nau} \mu_u \leq \mathrm{MA}_\omega(F_a(u)).$$

Proof First suppose that u is bounded. The chain rule (1.6) gives

$$\omega + \mathrm{dd^c} F_a(u) = \mathrm{e}^{au}(\omega + \mathrm{dd^c} u) + (1 - \mathrm{e}^{au})\omega + a\mathrm{e}^{au}\, \mathrm{d}u \wedge \mathrm{d^c} u.$$

All three summands are positive, and positivity of their bounded mixed products yields the claimed inequality.

For general u, use the truncations $u_t = \max(u, -t)$. The bounded functions $F_a(u_t)$ decrease and take values between zero and a^{-1}, so their limit $F_a(u)$ is a bounded

ω-plurisubharmonic function. On the plurifinely open set $D_t = \{u > -t\}$, it agrees with $F_a(u_t)$. Locality and the bounded case therefore give

$$\mathrm{e}^{nau}\mathbf{1}_{D_t}\mu_u \le \mathbf{1}_{D_t}\mathrm{MA}_\omega(F_a(u)).$$

Let $t \to +\infty$. The pole set is not charged by μ_u, and it has zero mass for the bounded product on the right. □

Theorem 8.2.1 *For $p > 0$, the damped measures $\nu_{p,u} = \mathrm{e}^{pu}\mu_u$ satisfy*

$$\nu_{p,u}(X) \le C_\omega(1 + p^{-1})^{r_n}, \qquad \nu_{p,u}(E) \le \max(1, n/p)^n \operatorname{Cap}_\omega(E)$$

for every Borel set $E \subset X$, uniformly over all nonpositive $u \in \mathrm{PSH}(X, \omega)$.

Proof Take $a = p/n$ in Proposition 8.2.1. Since $F_{p/n}(u)$ takes values in $[0, n/p]$, Proposition 8.1.1 with $A = n/p$ gives the total mass bound, after enlarging C_ω. For the capacity bound put $B = \max(1, n/p)$ and $v = F_{p/n}(u)$. Then v/B is a test function for the capacity, and

$$B\big(\omega + \mathrm{dd^c}(v/B)\big) = (\omega + \mathrm{dd^c}v) + (B-1)\omega \ge \omega + \mathrm{dd^c}v.$$

Consequently $\nu_{p,u} \le \mathrm{MA}_\omega(v) \le B^n\mathrm{MA}_\omega(v/B)$, which gives the second bound. □

Since μ_u does not charge the pole set, multiplication by e^{-pu} recovers μ_u from $\nu_{p,u}$, including when its mass is infinite. An additive shift $u \mapsto u + c$ multiplies the damped measure by e^{pc}.

8.3 Decreasing weights and capacity continuity

For a general decreasing weight w, writing $w(s) = \int_s^\infty (-w'(t))\,\mathrm{d}t$ and integrating the finite-depth estimate (8.1) in t bounds both the total mass and the mass on deep sublevel sets. Above any fixed depth, the damped measure is a weighted product of bounded potentials, which is continuous under convergence in capacity; the tail bound then removes the restriction to a fixed depth.

Theorem 8.3.1 *Let $w\colon [0,+\infty) \to [0,+\infty)$ be continuously differentiable and nonincreasing, with $w(t) \to 0$ as $t \to +\infty$. For $T \geq 0$ define*

$$B_w(T) = \int_T^\infty (1+t)^{r_n}(-w'(t))\,\mathrm{d}t.$$

Assume $B_w(0) < +\infty$. For nonpositive $u \in \mathrm{PSH}(X,\omega)$, the damped measures $\nu_{w,u} = w(-u)\mu_u$ satisfy

$$\nu_{w,u}(X) \leq C_\omega B_w(0), \qquad \nu_{w,u}(\{u < -T\}) \leq C_\omega B_w(T).$$

For every Borel set E and $T \geq 0$,

$$\nu_{w,u}(E) \leq w(0)(1+T)^n \operatorname{Cap}_\omega(E) + C_\omega B_w(T).$$

In particular, these measures are uniformly absolutely continuous with respect to capacity. If nonpositive $u_j, u \in \mathrm{PSH}(X,\omega)$ satisfy $u_j \to u$ in capacity, then $\nu_{w,u_j} \rightharpoonup \nu_{w,u}$.

Proof Write $s = -u$. Since $w(s) = \int_s^\infty (-w'(t))\,\mathrm{d}t$ for finite s, and since μ_u does not charge the pole set $\{s = +\infty\}$, Tonelli's theorem and Lemma 8.1.1 give

$$\int_X w(s)\,\mathrm{d}\mu_u = \int_0^\infty (-w'(t))\mu_u(\{s \leq t\})\,\mathrm{d}t \leq C_\omega B_w(0).$$

Restricting the integral to $\{s > T\}$ gives the same bound with $B_w(T)$ in place of $B_w(0)$, since the inner set is then $\{T < s \leq t\}$, which is empty unless $t > T$.

For the capacity bound, the second estimate in (8.1) gives

$$\int_{E\cap\{s\leq T\}} w(s)\,\mathrm{d}\mu_u \leq w(0)(1+T)^n \operatorname{Cap}_\omega(E).$$

Adding the tail estimate for $\{s > T\}$ proves the asserted bound. Since $B_w(T) \to 0$, choosing first T large and then the capacity small gives uniform absolute continuity.

To prove continuity, fix a continuous nonincreasing $\eta\colon [0,+\infty) \to [0,1]$, equal to one on $[0,1]$ and zero on $[2,+\infty)$, and put $\eta_T(s) = \eta(s/T)$. For fixed $T > 0$, set $v_{j,T} = \max(u_j, -2T)$ and $g_T(t) = \eta_T(-t)w(-t)$ on $[-2T, 0]$. Since $g_T(-2T) = 0$, locality gives

$$\eta_T(-u_j)w(-u_j)\mu_{u_j} = g_T(v_{j,T})\,\mathrm{MA}_\omega(v_{j,T}). \tag{8.2}$$

Truncation preserves convergence in capacity. The functions $g_T(v_{j,T})$ are uniformly bounded and quasi-continuous, and they converge in capacity to the corresponding function for u. The weighted continuity of bounded products in Sections 1.6 and 1.8 therefore gives weak convergence of the measures in (8.2) for each fixed T. The measure $(1-\eta_T(-u_j))\nu_{w,u_j}$ is carried by $\{u_j < -T\}$, so by the tail bound its integral against a continuous test function f is at most $C_\omega B_w(T)\|f\|_\infty$, uniformly in j; the same holds for u. Letting $T \to \infty$ completes the proof. □

For $w(t) = \mathrm{e}^{-pt}$, $p > 0$, the integral $B_w(0)$ is finite because the exponential dominates the polynomial factor. Thus the exponentially damped measures of Theorem 8.2.1 are continuous under convergence in capacity. By that theorem, their tails also satisfy

$$\nu_{p,u}(\{u < -T\}) \le \mathrm{e}^{-pT/2}\nu_{p/2,u}(X) \le C_p \mathrm{e}^{-pT/2}.$$

For polynomial weights, the general estimate takes the following explicit form.

Theorem 8.3.2 *For every $q > r_n$, the damped measures*

$$\sigma_{q,u} := (1-u)^{-q}\mu_u$$

have uniformly bounded mass for $u \in \mathrm{PSH}(X,\omega)$, $u \le 0$. More precisely,

$$\sigma_{q,u}(\{u < -T\}) \le C\frac{q}{q-r_n}(1+T)^{r_n-q} \quad (T \ge 0),$$

where C depends only on ω. There is also a constant C_q such that, for every Borel set $E \subset X$,

$$\sigma_{q,u}(E) \le C_q \,\mathrm{Cap}_\omega(E)^{(q-r_n)/(q+n-r_n)}.$$

Proof For $w(t) = (1+t)^{-q}$, direct integration gives

$$B_w(T) = q\int_T^\infty (1+t)^{r_n-q-1}\,\mathrm{d}t = \frac{q}{q-r_n}(1+T)^{r_n-q}.$$

Thus Theorem 8.3.1 gives the mass and tail estimates. For the capacity bound put $\delta = \mathrm{Cap}_\omega(E)$. If $0 < \delta \le 1$, choose $T = \delta^{-1/(q+n-r_n)} \ge 1$ in that theorem. Both $T^n\delta$ and T^{r_n-q} equal $\delta^{(q-r_n)/(q+n-r_n)}$, so its two terms have the required bound. If $\delta = 0$, let $T \to \infty$; if $\delta > 1$, use the total mass bound. □

For $n \ge 2$, the capacity exponent is $1 - n/(q+2)$. The threshold $q > n-2$ comes from the mixed-mass estimate in Proposition 8.1.1.

Logarithmic factors give weights at the threshold. For $n \ge 3$ and $a > 1$, put $r = n-2 > 0$ and

$$w(t) = (1+t)^{-r}[\log(\mathrm{e}+t)]^{-a}.$$

Differentiation gives

$$(1+t)^r(-w'(t)) = \frac{r}{(1+t)[\log(\mathrm{e}+t)]^a} + \frac{a}{(\mathrm{e}+t)[\log(\mathrm{e}+t)]^{a+1}}.$$

The ratios $(\mathrm{e}+t)/(1+t)$ are bounded. The substitution $y = \log(\mathrm{e}+t)$ therefore bounds the integrals of the two terms over $[T,+\infty)$ by fixed multiples of $[\log(\mathrm{e}+T)]^{1-a}$ and $[\log(\mathrm{e}+T)]^{-a}$, respectively. Since $a > 1$ and $\log(\mathrm{e}+T) \ge 1$, we obtain $B_w(T) \le C_{n,a}[\log(\mathrm{e}+T)]^{1-a}$. Hence, by Theorem 8.3.1, the damped measures $(1-u)^{-(n-2)}[\log(\mathrm{e}-u)]^{-a}\mu_u$ have uniformly bounded mass, their mass on $\{u < -T\}$ is at most a constant times $[\log(\mathrm{e}+T)]^{1-a}$, and they are continuous under convergence

in capacity. In dimensions one and two, $r_n = 0$ and $B_w(T) = w(T)$, so the finiteness condition $B_w(0) = w(0) < +\infty$ is automatic and every weight satisfying the remaining hypotheses of Theorem 8.3.1 can be used.

Damping can restore convergence when every undamped measure has infinite mass. When the bounded mass property fails, take the potential U constructed in the proof of Theorem 7.6.1, whose current has integrable coefficients by Proposition 7.7.1, and normalize it by $\sup_X U = 0$. For $0 < \varepsilon < 1$, the potentials εU converge to zero in capacity as $\varepsilon \downarrow 0$: by (1.15), the capacity of $\{\varepsilon U < -\delta\}$ is at most $C\varepsilon/\delta$ for each fixed $\delta > 0$ and small ε. Consequently, for $p > 0$ and $q > r_n$,

$$e^{p\varepsilon U}\mu_{\varepsilon U} \rightharpoonup \omega^n, \qquad (1 - \varepsilon U)^{-q}\mu_{\varepsilon U} \rightharpoonup \omega^n,$$

although $\mu_{\varepsilon U}(X) = +\infty$ for every $0 < \varepsilon < 1$ by Corollary 7.7.1. The full currents also converge to ω in the coefficient L^1 norm, and their Gauduchon traces converge in total variation.

8.4 Uniform tails and undamped convergence

On deep sublevel sets the damping weight tends to zero, so convergence of the weighted measures can leave their undamped mass uncontrolled. Uniformly vanishing deep tails supply the additional control: the following condition gives a finite limit measure and weak convergence of the undamped measures μ_{u_j}. It can equivalently be expressed by an increasing integrability weight.

Theorem 8.4.1 *Let $u_j, u \in \mathrm{PSH}(X, \omega)$ be nonpositive and suppose $u_j \to u$ in capacity. Assume*

$$\lim_{T\to\infty} \sup_j \mu_{u_j}(\{u_j < -T\}) = 0.$$

Then μ_u is finite and $\mu_{u_j} \rightharpoonup \mu_u$.

Proof Choose T_0 such that every tail $\mu_{u_j}(\{u_j < -T\})$ with $T \geq T_0$ is at most one. By Lemma 8.1.1, $\mu_{u_j}(X) \leq C(1 + T_0)^{r_n} + 1$, so the tail condition already gives a uniform bound for the total masses.

With the cutoffs η_T of (8.2), the fixed-level argument in the proof of Theorem 8.3.1, applied with $w \equiv 1$, gives

$$\eta_T(-u_j)\mu_{u_j} \rightharpoonup \eta_T(-u)\mu_u \quad \text{for every fixed } T > 0.$$

The masses of the measures on the left are bounded by the uniform bound for the total masses. Since $\eta_T(-u)$ increases to one off the pole set as $T \to \infty$, monotone convergence proves $\mu_u(X) < +\infty$.

For a continuous test function $f\colon X \to \mathbb{R}$, the error made by inserting $\eta_T(-u_j)$ is at most $\|f\|_\infty \mu_{u_j}(\{u_j < -T\})$, which is small for large T, uniformly in j. For the limit potential, the corresponding error tends to zero because μ_u is finite and does not charge the pole set. Let first $j \to \infty$ with T fixed, and then $T \to \infty$. □

Let $\chi\colon [0,+\infty) \to [1,+\infty)$ be continuous, nondecreasing and unbounded. For a nonpositive potential $u \in \mathrm{PSH}(X,\omega)$ define

$$J_{\chi,\omega}(u) = \int_X \chi(-u)\,\mathrm{d}\mu_u,$$

with the value $+\infty$ allowed. Since $\chi(T) \to +\infty$, a uniform bound for this functional forces the sublevel tails to vanish uniformly.

Corollary 8.4.1 *Let $u_j, u \in \mathrm{PSH}(X,\omega)$ be nonpositive. If $u_j \to u$ in capacity and $\sup_j J_{\chi,\omega}(u_j) \le C$, then*

$$J_{\chi,\omega}(u) \le C, \qquad \mu_{u_j} \rightharpoonup \mu_u.$$

Proof Since $\chi \ge 1$, the total masses are at most C, and

$$\mu_{u_j}(\{u_j < -T\}) \le C/\chi(T).$$

Thus Theorem 8.4.1 applies. For the bound on $J_{\chi,\omega}(u)$, use (8.2) with the compactly supported continuous function $s \mapsto \chi(s)\eta_T(s)$ in place of $w\eta_T$. The corresponding weighted integrals converge for each fixed T, and they are at most C. Let $T \to \infty$ and use monotone convergence. □

Proposition 8.4.1 *For a family of nonpositive potentials in $\mathrm{PSH}(X,\omega)$, the uniform tail condition of Theorem 8.4.1 is equivalent to a uniform bound on $J_{\chi,\omega}$ for some continuous nondecreasing unbounded function $\chi \ge 1$.*

Proof One implication was proved in Corollary 8.4.1. Conversely, write the family as $(u_i)_{i\in I}$. As in the proof of Theorem 8.4.1, the tail condition and Lemma 8.1.1 give a uniform bound for the total masses. Choose numbers $T_k \to +\infty$, with $T_{k+1} > T_k + 1$, such that $\mu_{u_i}(\{-u_i > T_k\}) \le 2^{-k}$ for every $i \in I$. Let ρ_k be a continuous nondecreasing function, equal to zero on $[0,T_k]$ and to one on $[T_k+1,+\infty)$, and define

$$\chi(s) = 1 + \sum_{k\ge 1} \rho_k(s).$$

The sum is locally finite, so χ is continuous, nondecreasing and unbounded. Tonelli's theorem gives

$$J_{\chi,\omega}(u_i) \le \mu_{u_i}(X) + \sum_{k\ge 1} \mu_{u_i}(\{-u_i > T_k\}) \le \sup_{i\in I} \mu_{u_i}(X) + 1,$$

so this weight gives a uniform integrability bound for the whole family. □

8.5 Mass concentration

The tail condition excludes mass carried by sublevel sets that disappear in the capacity limit. On a curve this mass can concentrate at a pole of the limit. When the bounded mass property fails, it can also persist along smooth potentials converging to zero.

Example 8.5.1 On $X = \mathbb{P}^1$, let z be the standard affine coordinate and let δ_0 denote the unit Dirac measure at its origin. Take the Fubini–Study form

$$\omega = \mathrm{dd^c} \log(1 + |z|^2), \qquad \int_X \omega = 1,$$

and the global nonpositive potential

$$u(z) = \log \frac{|z|^2}{1 + |z|^2}, \qquad u_j = \max(u, -j).$$

At infinity, in the coordinate $\zeta = 1/z$, the function u equals $-\log(1 + |\zeta|^2)$, so it is smooth there. The Poincaré–Lelong formula gives

$$\omega + \mathrm{dd^c} u = \delta_0, \qquad \mathrm{MA}_\omega^{\mathrm{ext}}(u) = 0.$$

The second identity holds because the extended measure discards the pole set and the current vanishes on its complement.

Each u_j is bounded and ω-plurisubharmonic. Its measure $\mu_j = \mu_{u_j} = \omega + \mathrm{dd^c} u_j$ has mass one by Stokes' theorem, and $\mu_j = 0$ on the open set $\{u > -j\}$ by locality. Hence

$$\operatorname{supp} \mu_j \subset \{u \le -j\} = \left\{ |z|^2 \le \frac{\mathrm{e}^{-j}}{1 - \mathrm{e}^{-j}} \right\}.$$

These disks shrink to the origin, so $\mu_j \rightharpoonup \delta_0$.

On their supports, $u_j = -j$. Consequently, for $j > T$,

$$\mu_j(\{u_j < -T\}) = 1, \qquad \mathrm{e}^{p u_j} \mu_j = \mathrm{e}^{-p j} \mu_j \rightharpoonup 0 \quad (p > 0).$$

Nevertheless, $u_j \to u$ in capacity. Indeed, outside any neighborhood of the origin the functions eventually agree with u, and the origin has arbitrarily small neighborhoods of arbitrarily small capacity. Thus bounded mass and capacity convergence do not prevent the undamped measures from acquiring mass at a pole of the limit. The damped limit is the extended measure of u, as predicted by Theorem 8.3.1.

We return to a general compact Hermitian manifold (X, ω).

Corollary 8.5.1 *Assume $v_+(\omega) = +\infty$ and put $V_\omega = \int_X \omega^n$. Fix a real number $b > V_\omega$ and a sequence $0 < \varepsilon_j < 1$ tending to zero. The smooth potentials w_j of Corollary 7.7.2, chosen with mass b, converge to zero in capacity. For every $T > 0$,*

$$\bigl\|\mathbf{1}_{\{w_j \geq -T\}}(\omega + \mathrm{dd^c} w_j)^n - \omega^n\bigr\|_{\mathrm{TV}} \longrightarrow 0, \qquad \int_{\{w_j < -T\}} (\omega + \mathrm{dd^c} w_j)^n \longrightarrow b - V_\omega.$$

After passage to a subsequence, there is a positive measure σ of mass $b - V_\omega$ such that

$$(\omega + \mathrm{dd^c} w_j)^n \rightharpoonup \omega^n + \sigma, \qquad \mathbf{1}_{\{w_j < -T\}}(\omega + \mathrm{dd^c} w_j)^n \rightharpoonup \sigma \quad \textit{for every } T > 0.$$

Proof The construction in Corollary 7.7.2 gives $w_j = s_j u_j$, where u_j is smooth and ω-plurisubharmonic, $\sup_X u_j = 0$ and $0 < s_j < \varepsilon_j$. The sublevel estimate (1.15) gives, for every fixed $\delta > 0$ and all sufficiently large j,

$$\mathrm{Cap}_\omega(\{|w_j| > \delta\}) = \mathrm{Cap}_\omega(\{u_j < -\delta/s_j\}) \leq C s_j/\delta \longrightarrow 0.$$

Put $\mu_j = (\omega + \mathrm{dd^c} w_j)^n$. Fix $T > 0$ and write $E_{j,T} = \{w_j \geq -T\}$. The lower bound in Corollary 7.7.2 gives

$$\mathbf{1}_{E_{j,T}} \mu_j \geq \lambda_{j,T} := (1 - \varepsilon_j)^n \mathbf{1}_{E_{j,T}} \omega^n.$$

Since $w_j \to 0$ in L^1, the smooth volume of $X \setminus E_{j,T}$ tends to zero, and hence $\lambda_{j,T} \to \omega^n$ in total variation. For every $p > 0$, Theorem 8.3.1 gives $\int_X \mathrm{e}^{p w_j}\, \mathrm{d}\mu_j \to V_\omega$, and therefore

$$\limsup_j \mu_j(E_{j,T}) \leq \mathrm{e}^{pT} \lim_j \int_X \mathrm{e}^{p w_j}\, \mathrm{d}\mu_j = \mathrm{e}^{pT} V_\omega.$$

Letting $p \downarrow 0$ and combining with the lower bound, we obtain $\mu_j(E_{j,T}) \to V_\omega$. Positivity now yields

$$\begin{aligned} \|\mathbf{1}_{E_{j,T}} \mu_j - \omega^n\|_{\mathrm{TV}} &\leq \mu_j(E_{j,T}) - \lambda_{j,T}(X) + \|\lambda_{j,T} - \omega^n\|_{\mathrm{TV}} \\ &\longrightarrow 0. \end{aligned}$$

Subtracting this shallow mass from $\mu_j(X) = b$ proves the assertion about the mass on $\{w_j < -T\}$.

The positive measures μ_j have fixed mass on the compact manifold X, so a subsequence converges weakly to a measure ν of mass b. The inequality $\mu_j \geq (1 - \varepsilon_j)^n \omega^n$ gives $\nu \geq \omega^n$. Set $\sigma = \nu - \omega^n$. For each $T > 0$, subtracting the total-variation limit of the shallow restrictions from $\mu_j \rightharpoonup \nu$ gives the last assertion along the same subsequence, for every T. □

Thus the defect σ in Corollary 8.5.1 carries the whole excess mass $b - V_\omega$, although the currents $\omega + \mathrm{dd^c} w_j$ converge to ω in the coefficient L^1 norm and their

Gauduchon traces converge in total variation. The location of σ is constrained by the limits of the intermediate powers of $\mathrm{dd^c} w_j$, as we now show.

8.5.1 Intermediate currents of a shrinking sequence

We formulate the result for a Hermitian form h on X, since it will be applied with different metrics in Chapter 9. Consider smooth functions

$$v_j = s_j u_j, \qquad u_j \in \mathrm{PSH}(X,h), \qquad 0 < s_j \longrightarrow 0,$$

and suppose that

$$(h + \mathrm{dd^c} v_j)^n \rightharpoonup h^n + \sigma.$$

The lower bound $h + \mathrm{dd^c} v_j \geq (1 - s_j)h$ shows that σ is positive. Although $\mathrm{dd^c} v_j \to 0$ in coefficient total variation, its intermediate powers can have nonzero limits, and these limits determine σ.

Proposition 8.5.1 *After passage to a common subsequence, the positive forms*

$$H_j = \mathrm{dd^c} v_j + s_j h = s_j (h + \mathrm{dd^c} u_j)$$

satisfy $H_j^k \rightharpoonup K_k$ *for* $1 \leq k \leq n$. *The currents* K_k *are positive and closed, and* $K_1 = K_n = 0$. *For every smooth real* $(n-k, n-k)$*-form* η *with* $\mathrm{d}\eta = 0$ *or* $\mathrm{dd^c}\eta = 0$,

$$\int_X K_k \wedge \eta = 0.$$

Moreover,

$$\sigma = \sum_{k=2}^{n-1} \binom{n}{k} h^{n-k} \wedge K_k. \tag{8.3}$$

If ϑ *is a smooth closed semipositive* $(1,1)$*-form on* X, *then* σ *is supported on* $\{\vartheta^n = 0\}$. *In dimension three the same conclusion holds for every smooth semipositive* $(1,1)$*-form* ϑ *with* $\mathrm{dd^c}\vartheta = 0$.

Proof The masses of the weakly convergent measures $(h + \mathrm{dd^c} v_j)^n$ are bounded. For $s_j < 1/2$, positivity in the expansion

$$(h + \mathrm{dd^c} v_j)^n = \sum_{k=0}^{n} \binom{n}{k} (1 - s_j)^{n-k} h^{n-k} \wedge H_j^k$$

gives a uniform bound for each $\int_X h^{n-k} \wedge H_j^k$. In local unitary coordinates, the coefficient matrix of the smooth positive form H_j^k on exterior powers is positive semidefinite, so its entries are bounded in absolute value by its trace, up to a dimensional constant. The trace bounds therefore bound every coefficient in total

variation, and weak compactness of measures gives a common subsequence and positive limits K_k.

For $n \geq 2$, choose a Gauduchon form γ. By Proposition 4.2.1,

$$\int_X H_j \wedge \gamma^{n-1} = s_j \int_X h \wedge \gamma^{n-1}.$$

Since γ and h are equivalent smooth metrics, it follows that $H_j \to 0$ in coefficient total variation. For $n = 1$ the same conclusion follows from $\int_X H_j = s_j \int_X h$. Thus $K_1 = 0$ in all dimensions. Since $\mathrm{d}H_j = s_j \mathrm{d}h$,

$$\mathrm{d}(H_j^k) = k s_j \mathrm{d}h \wedge H_j^{k-1} \longrightarrow 0$$

as currents, because the preceding trace bounds control multiplication by the fixed smooth form $\mathrm{d}h$. Hence every K_k is closed.

The smooth form $(\mathrm{dd^c} v_j)^k$ is both d-exact and $\mathrm{dd^c}$-exact:

$$(\mathrm{dd^c} v_j)^k = \mathrm{d}\big(\mathrm{d^c} v_j \wedge (\mathrm{dd^c} v_j)^{k-1}\big) = \mathrm{dd^c}\big(v_j (\mathrm{dd^c} v_j)^{k-1}\big).$$

Its integral against either kind of test form η in the statement is therefore zero by Stokes' formula. Expanding $\mathrm{dd^c} v_j = H_j - s_j h$ gives

$$\int_X \eta \wedge H_j^k = -\sum_{r=0}^{k-1} \binom{k}{r} (-s_j)^{k-r} \int_X \eta \wedge h^{k-r} \wedge H_j^r = O(s_j).$$

Each integral on the right is bounded by the trace bound for H_j^r. Taking $k = n$ and $\eta = 1$ shows that the positive measures H_j^n tend to zero in total variation, which proves $K_n = 0$; the other vanishing integrals follow by weak convergence. Finally, letting $j \to \infty$ in the positive expansion above, the terms with $k = 0$ tend to h^n and those with $k = 1$ and $k = n$ vanish in the limit, which proves (8.3).

For closed semipositive ϑ, apply the vanishing integrals to $\eta = \vartheta^{n-k}$. The measure $K_k \wedge \vartheta^{n-k}$ is positive and has zero mass. On every relatively compact open set where $\vartheta > 0$, it dominates a positive multiple of the trace measure of K_k, so K_k vanishes there. The decomposition gives the asserted support. In dimension three only K_2 occurs, and the same argument applies to the $\mathrm{dd^c}$-closed test form ϑ itself. □

In dimensions at most two the decomposition forces $\sigma = 0$. A threefold carrying a pluriclosed Hermitian metric, that is, a Hermitian form ϑ with $\mathrm{dd^c}\vartheta = 0$, has no nonzero defect σ of this form either. In higher dimensions, the vanishing integrals impose necessary conditions on each intermediate current; they do not by themselves produce a smooth sequence with prescribed intermediate limits.

8.6 Damping for a smooth background

For a general smooth background θ, truncation at a constant level may fail to preserve θ-plurisubharmonicity. A larger Hermitian form makes the truncations plurisubharmonic and dominates the extended measure, and the local expansion of the θ-product gives continuity at each fixed truncation level.

Theorem 8.6.1 *Let θ be a smooth real $(1,1)$-form, and for nonpositive $u \in \mathrm{PSH}(X,\theta)$ put $\mu_{\theta,u} = \mathrm{MA}_\theta^{\mathrm{ext}}(u)$. The mass bounds, capacity estimates and capacity-continuity statements for damped measures in this chapter, including Theorem 8.3.1, hold for the measures $\mu_{\theta,u}$ in place of μ_u, with constants depending on θ and ω. In particular, for $p > 0$ and $q > r_n$,*

$$\int_X \mathrm{e}^{pu}\,\mathrm{d}\mu_{\theta,u} \le C(1+p^{-1})^{r_n}, \qquad \int_X (1-u)^{-q}\,\mathrm{d}\mu_{\theta,u} \le C_q.$$

The undamped tail criterion of Theorem 8.4.1 and the integrability criterion of Corollary 8.4.1 also hold with $\mu_{\theta,u}$ in place of μ_u.

Proof Choose $C_0 \ge 1$ such that $\theta \le \Omega := C_0\omega$. Then $u \in \mathrm{PSH}(X,\Omega)$, and Theorem 7.1.1 gives

$$0 \le \mu_{\theta,u} \le \mathrm{MA}_\Omega^{\mathrm{ext}}(u).$$

All the mass, tail and capacity bounds therefore follow from the estimates already proved for Ω. The metric comparison in Section 1.8 gives, for every Borel set $E \subset X$,

$$\mathrm{Cap}_\omega(E) \le \mathrm{Cap}_\Omega(E) \le C_0^n\,\mathrm{Cap}_\omega(E).$$

For continuity, let nonpositive $u_j, u \in \mathrm{PSH}(X,\theta)$ converge in capacity, and use (8.2) with the θ-product. The truncations are bounded and Ω-plurisubharmonic. The weighted capacity continuity of Sections 1.6 and 1.8 applies term by term to the local expansion of their signed θ-products, with the same bounded quasi-continuous coefficients. Since the cutoff vanishes at the truncation level, the result is exactly the restricted positive measure. The uniform tails obtained from the domination by $\mathrm{MA}_\Omega^{\mathrm{ext}}$ remove the cutoff. Domination also gives the restricted mass estimate (8.1), so the preceding proofs of the undamped tail and integrability criteria apply without change. □

Chapter 9
Unbounded mass on Hopf manifolds

By Proposition 8.1.1, every compact complex curve or surface has the bounded mass property. The property can fail in dimension three even when the metric is Gauduchon. We prove this on a Hopf threefold, then extend the construction to the scalar Hopf manifolds $(\mathbb{C}^n \setminus \{0\})/\langle z \mapsto \mathrm{e}^{-1} z\rangle$ of all dimensions $n \geq 3$.

The mass polynomial of Theorem 2.2.1 provides the starting point. On a threefold, apart from the constant $\int_X \omega^3$, it has only a linear and a quadratic term. The linear term vanishes for a Gauduchon metric, and for the standard Hopf metric the quadratic term is a positive multiple of the Dirichlet energy along the elliptic fibers over $\mathbb{P}^2$. We therefore seek smooth potentials with large vertical energy whose perturbed forms stay positive. A fiber heat kernel supplies the concentration. Varying its time across the base controls the mixed derivatives; a function of the base then restores positivity without changing the vertical energy. The higher-dimensional extension follows by comparing the angular densities.

We next use the fiber structure to study minimum mass, damping and weak limits. Fiber averaging identifies the bounded potentials of minimum mass. For normalized potentials invariant under unitary rotations of the last two coordinates, a positive fiber flux yields the sharp square-root damping threshold. At the endpoint of this threshold, concentration on a divisor separates individual finiteness from uniform tails and capacity continuity. Finally, we classify the excess measures produced by shrinking smooth potentials. They are exactly the lifts of base measures by the normalized area measures on the elliptic fibers, for the standard metric and for every other Hermitian metric on this threefold.

Theorem 9.0.1 (Xia–Zhang) *The compact threefold*

$$X = (\mathbb{C}^3 \setminus \{0\})/\langle z \mapsto \mathrm{e}^{-1} z\rangle$$

admits a Gauduchon form ω and smooth real functions V_j such that

$$\omega + \mathrm{dd^c} V_j > 0, \qquad \int_X (\omega + \mathrm{dd^c} V_j)^3 \longrightarrow +\infty.$$

In particular, X does not have the bounded mass property.

9.1 The Hopf metric and its mass

The projection to $\mathbb{P}^2$ separates the horizontal and vertical parts of the metric. We normalize the fiber measure to have mass one; this fixes the constants in the mass formula.

On the covering space $\mathbb{C}^3 \setminus \{0\}$, with coordinates $z = (z_0, z_1, z_2)$, put

$$s = \log |z|^2, \qquad \alpha = \mathrm{dd^c} s, \qquad \beta = \mathrm{d}s \wedge \mathrm{d^c} s = \frac{\mathrm{i}}{2\pi} \partial s \wedge \bar\partial s, \qquad \omega = \alpha + \beta.$$

The contraction $z \mapsto \mathrm{e}^{-1} z$ changes s by the constant -2, so these forms descend to X. A closed spherical shell between two successive radii is a compact fundamental domain; this also proves that the quotient is compact. Let ω_{FS} be the Fubini–Study form on $\mathbb{P}^2$, given by $\mathrm{dd^c} \log(1 + |\xi|^2)$ in affine coordinates $\xi = (\xi_1, \xi_2)$. Projectivization gives a holomorphic map

$$\pi \colon X \longrightarrow \mathbb{P}^2, \qquad \alpha = \pi^* \omega_{\mathrm{FS}}, \qquad \int_{\mathbb{P}^2} \omega_{\mathrm{FS}}^2 = 1.$$

The form α has rank two and is positive transversally to the fibers, while the rank-one form β is positive on the vertical line. Their kernels intersect only in zero, and therefore $\omega > 0$.

We shall use the identities

$$\alpha^3 = 0, \quad \beta^2 = 0, \quad \mathrm{d}\alpha = 0, \quad \mathrm{dd^c}\beta = -\alpha^2, \quad \mathrm{dd^c}(\omega^2) = 0. \tag{9.1}$$

The first three follow from the definitions and the dimension of the base. For the fourth, differentiating $\beta = \mathrm{i}\partial s \wedge \bar\partial s/(2\pi)$ gives $\mathrm{dd^c}\beta = (\partial\bar\partial s)^2/(4\pi^2) = -\alpha^2$. Finally, $\omega^2 = \alpha^2 + 2\alpha \wedge \beta$, so $\mathrm{dd^c}(\omega^2) = -2\alpha^3 = 0$. Thus ω is Gauduchon, although $\mathrm{dd^c}\omega = -\alpha^2 \neq 0$.

On the affine base chart $U_0 = \{[z_0 : z_1 : z_2] \in \mathbb{P}^2 : z_0 \neq 0\} \simeq \mathbb{C}^2$ write

$$\xi_j = z_j / z_0 \quad (j = 1, 2), \qquad w = \log z_0.$$

The choice of logarithm and the contraction change w by elements of $\Lambda = \mathbb{Z} + 2\pi\mathrm{i}\mathbb{Z}$. Hence

$$\pi^{-1}(U_0) \simeq U_0 \times E, \qquad E = \mathbb{C}/\Lambda, \qquad \mathrm{d}A_E = \frac{\mathrm{i}}{4\pi} \mathrm{d}w \wedge \mathrm{d}\bar w, \quad \int_E \mathrm{d}A_E = 1.$$

On another base chart the fiber coordinate changes by a holomorphic function of the base. Consequently, the vertical derivative ∂_w and the fiber integral below are globally defined. For a smooth function V on X, write $V_w = \partial V/\partial w$.

Proposition 9.1.1 *For every $V \in C^\infty(X, \mathbb{R})$,*

$$\int_X (\omega + \mathrm{dd^c} V)^3 = 6 + 6 \int_{\mathbb{P}^2} \left(\int_{\pi^{-1}(y)} |V_w|^2 \, \mathrm{d}A_E \right) \omega_{\mathrm{FS}}^2(y). \tag{9.2}$$

Proof Since β restricts to $2\mathrm{d}A_E$ on each fiber,

$$\int_X \omega^3 = 3 \int_X \alpha^2 \wedge \beta = 6.$$

By (2.1) and (9.1), followed by an integration by parts against the closed form α^2,

$$\begin{aligned} \int_X (\omega + \mathrm{dd^c} V)^3 - \int_X \omega^3 &= 3 \int_X V \mathrm{dd^c}\omega \wedge \mathrm{dd^c} V \\ &= -3 \int_X V \alpha^2 \wedge \mathrm{dd^c} V = 3 \int_X \mathrm{d}V \wedge \mathrm{d^c} V \wedge \alpha^2. \end{aligned}$$

Wedging with α^2 removes every component of $\mathrm{d}V \wedge \mathrm{d^c}V$ except the vertical one, which is $\mathrm{i}|V_w|^2 \mathrm{d}w \wedge \mathrm{d}\bar{w}/(2\pi) = 2|V_w|^2 \mathrm{d}A_E$. Integration over the fibers proves the formula. □

9.2 The radial–vertical block

We look for potentials that depend on the base point only through its distance from the origin of the affine chart. In a suitable frame, the matrix of $\omega + \mathrm{dd^c}V$ then consists of one tangential horizontal coefficient and a two-by-two radial–vertical block, whose Schur complement isolates the mixed derivative that must be controlled. Subscripts denote partial derivatives in the indicated variables.

For $\xi \neq 0$ introduce

$$x = \log |\xi|^2, \qquad \mu = \frac{\mathrm{e}^x}{1 + \mathrm{e}^x}, \qquad \mu_x = \mu(1 - \mu).$$

We use smooth real functions $V \colon \mathbb{R} \times E \to \mathbb{R}$, written $V = V(x, w)$, that are independent of x for x near $-\infty$ and constant in both variables for x near $+\infty$. These conditions allow V to be extended smoothly across $\xi = 0$, and over the complement of the affine chart by the same constant. The region of the base where V is nonconstant may depend on V.

Lemma 9.2.1 *For such a function, at $\xi \neq 0$ the inequality $\omega + \mathrm{dd^c} V > 0$ is equivalent to*

$$1 + V_{w\bar{w}} > 0, \qquad \mu + V_x > 0, \qquad Q[V] := V_{xx} + \mu - \frac{|\mu + V_{x\bar{w}}|^2}{1 + V_{w\bar{w}}} > 0. \tag{9.3}$$

At $\xi = 0$, the inequality $\omega + \mathrm{dd^c} V > 0$ is equivalent to $1 + V_{w\bar{w}} > 0$.

Proof In these coordinates,

$$s = w + \bar{w} + \log(1 + \mathrm{e}^x), \qquad \partial s = \mathrm{d}w + \mu \partial x,$$

and

$$2\pi\alpha = \mathrm{i}\mu\partial\bar{\partial}x + \mathrm{i}\mu(1-\mu)\partial x \wedge \bar{\partial}x.$$

Choose horizontal vectors of unit length for $2\pi\alpha$, one orthogonal to the radial direction and the other

$$e_r = \frac{\xi_1 \partial_{\xi_1} + \xi_2 \partial_{\xi_2}}{\sqrt{\mu(1-\mu)}}.$$

Adjoin the vertical vector ∂_w. In this frame the matrix of $2\pi(\omega + \mathrm{dd^c} V)$ is

$$\begin{pmatrix} (\mu + V_x)/\mu & 0 & 0 \\ 0 & (\mu + V_{xx})/[\mu(1-\mu)] & (\mu + V_{x\bar{w}})/\sqrt{\mu(1-\mu)} \\ 0 & (\mu + V_{wx})/\sqrt{\mu(1-\mu)} & 1 + V_{w\bar{w}} \end{pmatrix}.$$

Taking the Schur complement of the last diagonal entry gives (9.3). At $\xi = 0$, the derivatives of V in the base directions vanish and $\partial s = \mathrm{d}w$, which gives the last assertion. □

A rapid cutoff of a concentrated fiber potential can make the mixed derivative $V_{x\bar{w}}$ too large for this criterion. We therefore spread the concentration by the heat equation before cutting off, so that the fiber derivatives are already exponentially small in the cutoff region.

9.3 Heat potentials and the mixed derivative

The fiber potential will be built from the heat kernel of E. It has large vertical energy at small times and decays at large times, where it can be cut off, and a lower bound for its logarithmic Hessian controls vertical positivity. To control the radial–vertical block when the time varies, we also need an estimate relating the kernel to its logarithmic gradient.

For $t > 0$ define

$$k_t(w) = \frac{2}{t} \sum_{\lambda \in \Lambda} \exp\left(-\frac{|w - \lambda|^2}{t}\right). \tag{9.4}$$

This is a smooth positive Λ-periodic function of integral one: unfolding the integral reduces it to the Gaussian integral on $\mathbb{C}$, since $\mathrm{d}A_E = \mathrm{d}a\,\mathrm{d}b/(2\pi)$ for $w = a + \mathrm{i}b$. For $(p, q) \in \mathbb{Z}^2$ set

$$\nu_{p,q} = p + \frac{\mathrm{i}q}{2\pi}, \qquad \chi_{p,q}(w) = \mathrm{e}^{2\pi \mathrm{i} p \operatorname{Re} w + \mathrm{i} q \operatorname{Im} w}.$$

These characters form an orthonormal basis of $L^2(E, \mathrm{d}A_E)$. Computing the Fourier coefficients by the same unfolding gives

$$k_t = \sum_{p,q} \mathrm{e}^{-\pi^2 |\nu_{p,q}|^2 t} \chi_{p,q}, \qquad \partial_t k_t = \partial_w \partial_{\bar{w}} k_t. \tag{9.5}$$

All differentiated series converge locally uniformly for $t > 0$.

Lemma 9.3.1 *Let $\eta > 0$ and define*

$$F_\eta(t, w) = -\eta \int_t^\infty (k_r(w) - 1)\,\mathrm{d}r.$$

Then F_η is smooth and real, and

$$(F_\eta)_t = (F_\eta)_{w\bar{w}} = \eta(k_t - 1), \tag{9.6}$$

$$(\log k_t)_{w\bar{w}} \geq -1/t, \tag{9.7}$$

$$\int_E |(F_\eta)_w(t, w)|^2 \mathrm{d}A_E = 2\eta^2 \log(1/t) + O(\eta^2) \quad (0 < t \leq 1). \tag{9.8}$$

The error term in (9.8) is uniform in t and η. For fixed nonnegative integers a, m there is also a uniform estimate

$$\|\partial_t^a F_\eta(t, \cdot)\|_{C^m(E)} \leq C_{a,m} \eta \mathrm{e}^{-t/4} \quad (t \geq 1). \tag{9.9}$$

Proof Integrating the nonconstant Fourier modes gives

$$F_\eta(t, w) = -\frac{\eta}{\pi^2} \sum_{(p,q) \neq (0,0)} \frac{\mathrm{e}^{-\pi^2 |\nu_{p,q}|^2 t}}{|\nu_{p,q}|^2} \chi_{p,q}(w).$$

This proves smoothness, reality and (9.6). The least nonzero value of $\pi^2 |\nu_{p,q}|^2$ is $1/4$, and any fixed number of differentiations introduces coefficients of polynomial growth in $|\nu_{p,q}|$. Factoring out $\mathrm{e}^{-t/4}$ for $t \geq 1$ therefore leaves a convergent Gaussian series, which proves (9.9).

To prove (9.7), fix a lift of w and give Λ the probability weights

$$P_\lambda = \frac{\mathrm{e}^{-|w-\lambda|^2/t}}{\sum_{\lambda'} \mathrm{e}^{-|w-\lambda'|^2/t}}.$$

Logarithmic differentiation of the Gaussian sum yields

$$t\partial_{\bar w}\log k_t = -\sum_{\lambda}(w-\lambda)P_\lambda,$$

and one more derivative gives

$$(\log k_t)_{w\bar w} = -\frac{1}{t} + \frac{1}{t^2}\left(\sum_{\lambda}|w-\lambda|^2 P_\lambda - \left|\sum_{\lambda}(w-\lambda)P_\lambda\right|^2\right).$$

The expression in parentheses is a variance and is nonnegative.

Finally, Parseval's identity gives, with $\mathcal{E}_\eta(t)$ denoting the energy in (9.8),

$$\mathcal{E}_\eta(t) = \frac{\eta^2}{\pi^2}\sum_{(p,q)\neq(0,0)}\frac{\mathrm{e}^{-2\pi^2|\nu_{p,q}|^2 t}}{|\nu_{p,q}|^2}, \qquad \mathcal{E}_\eta'(t) = -2\eta^2(k_{2t}(0)-1).$$

The Gaussian formula at the origin implies

$$k_{2t}(0) = \frac{1}{t} + O\left(t^{-1}\mathrm{e}^{-1/(4t)}\right) \quad (0<t\le 1).$$

To see this, isolate the zero lattice vector and use $\mathrm{e}^{-|\lambda|^2/(2t)} \le \mathrm{e}^{-1/(4t)}\mathrm{e}^{-|\lambda|^2/4}$ for $\lambda\neq 0$. The error term is integrable near $t=0$, and $\mathcal{E}_\eta(1) = O(\eta^2)$. Integrating the derivative between t and 1 proves (9.8). □

The Schur complement subtracts a square involving the mixed derivative. When the time parameter is allowed to vary, the heat equation produces a compensating positive square, and the following bound relating the heat kernel to its logarithmic gradient controls the region where that square is small.

Lemma 9.3.2 *For a constant $C_0 \ge 1$ depending only on E,*

$$t k_t(w) \le C_0 \exp\left(-\frac{|t\partial_{\bar w}\log k_t(w)|^2}{C_0 t}\right) \quad (0<t\le 1). \tag{9.10}$$

Proof Choose a nearest lattice point λ_0, and put $d = |w-\lambda_0|$ and $s_\lambda = |w-\lambda|^2 - d^2 \ge 0$. The distance d is bounded by the diameter of a fundamental rectangle. By the triangle inequality, $s_\lambda \ge |\lambda-\lambda_0|^2/4 - C$, and outside a fixed finite set of λ we have $s_\lambda \ge |\lambda-\lambda_0|^2/8$. Consequently,

$$\sum_{\lambda}(1+|w-\lambda|)\mathrm{e}^{-s_\lambda/t} \le C, \qquad t k_t(w) \le C\mathrm{e}^{-d^2/t}.$$

The formula for $t\partial_{\bar w}\log k_t$ in the proof of Lemma 9.3.1 gives, with $G = |t\partial_{\bar w}\log k_t|$,

$$G \le d + \sum_{\lambda\neq\lambda_0}|w-\lambda|\mathrm{e}^{-s_\lambda/t}.$$

If $d \geq 1/4$, this is at most Cd. If $d < 1/4$, the nonzero lattice distances are at least one and

$$s_\lambda \geq |\lambda - \lambda_0|(|\lambda - \lambda_0| - 2d) \geq \tfrac{1}{2}|\lambda - \lambda_0|^2.$$

Summing the resulting Gaussian bound gives $G \leq C(d + \mathrm{e}^{-c/t})$ in all cases. When $G > 2C\mathrm{e}^{-c/t}$, we have $d \geq G/(2C)$, and the bound $tk_t \leq C\mathrm{e}^{-d^2/t}$ gives (9.10). In the remaining case G^2/t is bounded, and (9.10) follows from $tk_t \leq C$ after increasing C_0. □

Lemma 9.3.3 *There is a constant C depending only on E with the following property. For $0 < \kappa < 1/(200C_0)$, $0 < \eta < 1/10$, $0 < t \leq 1$ and $1/2 \leq \mu \leq 1$, define $\sigma > 0$ and the functions h, ζ of $w \in E$ by*

$$\sigma^2 = \frac{2\kappa}{\eta} \log \frac{1}{\eta t}, \qquad h = \eta k_t, \qquad \zeta = \frac{\sigma}{\mu}\, t\partial_{\bar{w}} \log k_t.$$

Then

$$\kappa t k_t - \frac{\mu^2(1-\eta)h}{1-\eta+h}\left|\zeta - \frac{1}{1-\eta}\right|^2 \leq C\kappa \exp\left(-\frac{1}{C\sigma^2 t}\right). \tag{9.11}$$

Proof If $\sigma^2 t \geq 1/100$, the bound $tk_t \leq C_0$ gives (9.11) after enlarging C. Assume therefore that $\sigma^2 t < 1/100$. If $|\zeta - (1-\eta)^{-1}| \geq 1/4$, the subtracted expression is at least $h/[100(1+h)]$. Since $\log(1/(\eta t)) > 2$, the definition of σ gives

$$\kappa t(1+h) \leq \frac{\eta}{400} + C_0\kappa\eta < \frac{\eta}{100}.$$

Multiplication by $k_t/(1+h)$ shows that the left side of (9.11) is nonpositive.

Otherwise $|\zeta| \geq 3/4$, so

$$|t\partial_{\bar{w}} \log k_t| = \frac{\mu|\zeta|}{\sigma} \geq \frac{3}{8\sigma}.$$

Discarding the nonnegative square and applying Lemma 9.3.2 gives (9.11) in this case as well. □

9.4 Variation of the heat time

The heat time will remain small on a fixed base ball and then increase with the radial variable until the fiber potential can be cut off. We choose its rate of increase so that the negative part of the Schur complement has small integral in the radial variable. Fix κ as in Lemma 9.3.3.

All estimates below concern the limit $\eta \downarrow 0$, with κ fixed. For each η choose the heat time ε on the central ball with $0 < \varepsilon \leq \eta^{1/\eta^2}$; the proof of Theorem 9.0.1 takes $\varepsilon = \eta^{1/\eta^2}$, and Section 9.8 uses smaller values. Write

$$L = \log(1/\eta), \quad q_0 = \sqrt{\log \frac{1}{\eta\varepsilon}}, \quad a = \sqrt{\frac{\kappa}{2\eta}}.$$

Fix a smooth nondecreasing $\chi\colon \mathbb{R} \to [0,1]$, independent of η, equal to zero on $(-\infty, 0]$ and one on $[1, +\infty)$, with $\int_0^1 \chi = 1/2$. Define

$$q(x) = q_0 - \int_0^{ax} \chi(s)\,\mathrm{d}s, \qquad t(x) = \eta^{-1}\mathrm{e}^{-q(x)^2}. \tag{9.12}$$

Thus $t = \varepsilon$ on $x \le 0$, and $q = q_0 + 1/2 - ax$ on $x \ge a^{-1}$. Put

$$b = \frac{q_0 - 1/2}{a}, \qquad x_* = \frac{q_0 + 1/2}{a}.$$

For small η, we have $0 < a^{-1} < b < x_*$, with

$$q(b) = 1, \qquad q(x_*) = 0. \tag{9.13}$$

Fix also a smooth cutoff $\vartheta\colon \mathbb{R} \to [0,1]$, independent of η, zero on $(-\infty, 0]$ and one on $[1, +\infty)$, and define

$$W(x, w) = \vartheta(q(x)) F_\eta(t(x), w).$$

This function equals $F_\eta(\varepsilon, w)$ for $x \le 0$ and vanishes for $x \ge x_*$. Moreover, by (9.6),

$$1 + W_{w\bar{w}} = 1 + \vartheta(q)\eta(k_t - 1) > 1 - \eta. \tag{9.14}$$

Lemma 9.4.1 *For the functions just constructed, let*

$$d_\eta(x) = \max_{w \in E} \max\{-Q[W](x, w), 0\}.$$

There are $C, \eta_0 > 0$, depending only on κ, χ and ϑ, such that the following holds whenever $0 < \eta < \eta_0$ and $\eta x_* \le 1/(10\mathrm{e})$. The function d_η is continuous and supported in $[0, b]$, $Q[W] > 0$ outside $[0, b] \times E$, $Q[W] \ge 2/27$ on $[-2, -1] \times E$, and

$$\int_{\mathbb{R}} d_\eta(x)\,\mathrm{d}x \le C\big(\eta b + \sqrt{\eta L}\big), \qquad \sup_{\mathbb{R}\times E} \max\{-W_x, 0\} \le C\sqrt{\eta L}. \tag{9.15}$$

Proof We first derive an exact formula for $Q[W]$ on $[0, b]$, then bound its negative part separately on $[0, a^{-1}]$ and on $[a^{-1}, b]$, integrate these bounds, and finally examine the ends of the radial line.

Step 1: the Schur complement on $[0, b]$. Write $F = F_\eta$ for the heat potential. On $[0, b]$ we have $q \ge 1$, hence $\vartheta(q) = 1$ and $W = F(t(x), w)$. Set

$$h = \eta k_t, \qquad \sigma = (\log t)_x, \qquad \zeta = \frac{\sigma t}{\mu}\partial_{\bar{w}} \log k_t.$$

The heat equation gives $F_t = h - \eta$, and the chain rule gives

$$W_x = \sigma t F_t, \quad W_{xx} = (\sigma_x + \sigma^2) t F_t + \sigma^2 t^2 F_{tt}, \quad W_{x\bar{w}} = \sigma t F_{t\bar{w}}.$$

Substituting these in $Q[W]$, using $h_t = h((\log k_t)_{w\bar{w}} + |\partial_{\bar{w}} \log k_t|^2)$, and completing the square, we obtain the exact identity

$$\begin{aligned} Q[W] = \sigma^2 t\{h[1 + t(\log k_t)_{w\bar{w}}] - \eta\} + \sigma_x t F_t + \mu(1-\mu) - \frac{\mu^2 \eta}{1-\eta} \\ + \frac{\mu^2 (1-\eta) h}{1 - \eta + h} \left| \zeta - \frac{1}{1-\eta} \right|^2 . \end{aligned} \tag{9.16}$$

By (9.7), the part $\sigma^2 t h[1 + t(\log k_t)_{w\bar{w}}]$ is nonnegative.

Step 2: the initial interval $[0, a^{-1}]$. Here

$$\sigma = 2aq\chi(ax), \qquad \sigma_x = -\frac{\kappa}{\eta}\chi(ax)^2 + \frac{\kappa}{\eta} q\chi'(ax).$$

Moreover $q_0 - 1/2 \le q \le q_0$ and $t \le \varepsilon e^{q_0}$. Since $\log(1/\varepsilon) \ge L/\eta^2$ and $q_0^2 \le 2\log(1/\varepsilon)$, for small η both t and $q_0^2 t$ are at most $\varepsilon^{1/4} \le \eta^{1/(4\eta^2)}$, which tends to zero faster than any power of η. In particular,

$$\kappa q \chi'(ax) t \le \eta, \qquad \sigma^2 t \le \eta^2.$$

At each point where $\chi(ax) > 0$, apply Lemma 9.3.3 with $\kappa\chi(ax)^2$ in place of κ; its constant is uniform in x. The positive square in (9.16) then absorbs the negative term $-\kappa\chi^2 t k_t$ coming from $\sigma_x t F_t$. The remaining part of $\sigma_x t F_t$, which involves χ', is at least $-\eta$, because $F_t \ge -\eta$. The other negative terms in (9.16) are $O(\eta)$, and so is the exponential error of Lemma 9.3.3, since $\sigma^2 t \le \eta^2$. Thus $d_\eta \le C\eta$ on this interval. At points where $\chi(ax) = 0$, the smooth nonnegative function χ has a minimum, so $\chi'(ax) = 0$ and directly

$$Q[W] = \mu - \frac{\mu^2}{1 - \eta + h} > 0,$$

since $\mu \le \mu(a^{-1}) < 1 - \eta$ for small η.

Step 3: the interval $[a^{-1}, b]$. Here the formulas simplify to

$$\sigma = 2aq, \qquad \sigma_x = -\kappa/\eta, \qquad \mathrm{d}x = -\frac{\mathrm{d}q}{a} = \sqrt{\frac{\eta}{2\kappa}} \frac{\mathrm{d}t}{tq}. \tag{9.17}$$

For $t \le 1$, Lemma 9.3.3 and (9.16) give

$$d_\eta(x) \le 2\eta + 2\kappa q^2 t + C\kappa \exp\left(-\frac{\eta}{C\kappa q^2 t}\right). \tag{9.18}$$

Indeed, $-\eta\sigma^2 t = -2\kappa q^2 t$ and $\sigma_x t F_t = -\kappa t k_t + \kappa t$; these are the two terms for which the scale was chosen. When $t \geq 1$, we insert the exponential derivative estimates (9.9) directly in the definition of Q. The denominator stays above $1-\eta$, and the chain rule gives

$$d_\eta(x) \leq C\kappa q^2 t^2 \mathrm{e}^{-t/4}. \tag{9.19}$$

The perturbing derivatives are bounded by constants times $\eta\mathrm{e}^{-t/4}$, $\eta\sigma t\mathrm{e}^{-t/4}$ and $\eta[(|\sigma_x|+\sigma^2)t+\sigma^2 t^2]\mathrm{e}^{-t/4}$. Since $q, t \geq 1$ here, each is bounded by a constant times the right side of (9.19). Since $1 + W_{w\bar{w}} \geq 1-\eta$, subtracting the two quotients gives

$$|Q[W] - Q[0]| \leq |W_{xx}| + C\big(|W_{x\bar{w}}| + |W_{x\bar{w}}|^2 + |W_{w\bar{w}}|\big).$$

The square of the mixed derivative is at most $C\kappa\eta q^2 t^2 \mathrm{e}^{-t/2}$, so it satisfies the same bound. Since $Q[0] > 0$, this proves (9.19).

Step 4: integration of the negative part. The constant 2η and the initial interval contribute $O(\eta b)$. In the small-time region, $q \geq \sqrt{L}$ and $t = \eta^{-1}\mathrm{e}^{-q^2}$. Therefore

$$\int 2\kappa q^2 t \,\mathrm{d}x \leq \frac{2\kappa}{a\eta} \int_{\sqrt{L}}^{\infty} q^2 \mathrm{e}^{-q^2} \,\mathrm{d}q \leq C\sqrt{\eta L}.$$

For the exponential term the integral is at most

$$\frac{C\kappa}{a} \int_{\sqrt{L}}^{q_0} \exp\left(-\frac{\eta^2 \mathrm{e}^{q^2}}{C\kappa q^2}\right) \mathrm{d}q.$$

The interval $[\sqrt{L}, 2\sqrt{L}]$ has length $O(\sqrt{L})$. For $q \geq 2\sqrt{L}$ and sufficiently small η, $\eta^2 \mathrm{e}^{q^2}/q^2 \geq \mathrm{e}^{q^2/4}$. The integrand on this remaining interval is bounded by $\exp(-\mathrm{e}^{q^2/4}/(C\kappa))$, which is integrable on $[1, +\infty)$. The prefactor is $O(\sqrt{\eta})$, and the initial interval contributes $O(\sqrt{\eta L})$, so the full expression is $O(\sqrt{\eta L})$. In the large-time region, (9.17) gives

$$\int d_\eta(x) \,\mathrm{d}x \leq C\sqrt{\eta} \int_1^{1/(\mathrm{e}\eta)} q t \mathrm{e}^{-t/4} \,\mathrm{d}t \leq C\sqrt{\eta L} \int_1^{\infty} t \mathrm{e}^{-t/4} \,\mathrm{d}t.$$

This proves the integral bound on $[0, b]$.

Step 5: the radial slope. The negative part of W_x is uniformly small as well. For $t \leq 1$, we have $F_t \geq -\eta$ and $0 \leq \sigma \leq 2aq$, so

$$\max\{-W_x, 0\} \leq \sqrt{2\kappa\eta}\, t \sqrt{\log \frac{1}{\eta t}} \leq C\sqrt{\eta L}.$$

The last inequality follows because $t\sqrt{L + \log(1/t)}$ is increasing on $(0, 1]$ when $L > 1$. For $t \geq 1$ and $q \geq 1$, use $|W_x| \leq C\sqrt{\eta}\, q t \mathrm{e}^{-t/4}$ and $q \leq \sqrt{L}$.

Step 6: the ends. It remains to consider the two ends, including the cutoff region. If $b \le x \le x_*$, then $0 \le q \le 1$ and $1/(\mathrm{e}\eta) \le t \le 1/\eta$. The derivatives of the cutoffs and scale have at most polynomial growth in η^{-1}. Thus (9.9) implies

$$\sup_w \bigl(|W_x| + |Q[W] - Q[0]|\bigr) \le C\mathrm{e}^{-1/(5\mathrm{e}\eta)}.$$

But $Q[0] = \mu(1-\mu) \ge \mathrm{e}^{-x_*}/4 \ge \mathrm{e}^{-1/(10\mathrm{e}\eta)}/4$ on this interval, because $\eta x_* \le 1/(10\mathrm{e})$. Hence $Q[W] > 0$ there for small η. For $x \ge x_*$, $W = 0$ and positivity is immediate. For $x \le 0$, $W_x = 0$ and

$$Q[W] \ge \mu\left(1 - \frac{\mu}{1-\eta}\right) > 0,$$

since $\mu \le 1/2$. On $[-2,-1]$, the bounds $1/9 \le \mu \le 3/10$ and $1-\eta \ge 9/10$ give $Q[W] \ge 2/27$. Finally, $Q[W]$ is continuous, with positive denominator, so the maximum of its negative part over the compact fiber E is continuous in x. This proves the assertions about support, positivity and continuity, and completes the proof. □

9.5 A base correction and unbounded mass

Adding a function of x alone changes $Q[W]$ by its second derivative and leaves the vertical energy unchanged. The integral bound for d_η lets us choose this second derivative larger than d_η and compensate its integral on an interval where $Q[W]$ is uniformly positive. The correction has small first derivative, so the remaining horizontal coefficient stays positive as well.

Lemma 9.5.1 *Fix a smooth $\psi \ge 0$ supported in $(-2,-1)$ with $\int_{\mathbb{R}} \psi = 1$, and put $m_0 = \min\{\mu(-2)/2,\ 1/(27\|\psi\|_\infty)\}$. Let η and W be as in Lemma 9.4.1, with $\max\{-W_x, 0\} \le 1/4$. Let $r \ge 0$ be a smooth function supported in $[-1/2, b+1/2]$ with $r > d_\eta$ on $[0,b]$ and $m := \int_{\mathbb{R}} r\,\mathrm{d}x < m_0$, and put*

$$g(x) = \int_{-\infty}^{x} (x-s)(r(s) - m\psi(s))\,\mathrm{d}s, \qquad V = W + g.$$

Then $-m \le g' \le 0$, $g = 0$ for $x \le -2$, g is constant for $x \ge b + 1/2$, and $-m(x+2) \le g(x) \le 0$ for $x \ge -2$. The function V extends to a smooth function on X with $\omega + \mathrm{dd}^{\mathrm{c}} V > 0$.

Proof Since $g'' = r - m\psi$, where ψ is supported to the left of the support of r and both terms have integral m, we have $g' = -m\int_{-\infty}^{x} \psi$ for $x \le -1/2$ and $g' = -m + \int_{-\infty}^{x} r$ for $x \ge -1/2$. This gives $-m \le g' \le 0$, with $g' = 0$ for $x \le -2$ and for $x \ge b + 1/2$, and hence the assertions about g. In particular, V is independent of x near $-\infty$ and constant near $+\infty$, so it extends smoothly to X.

Since g is independent of w, the Schur complement changes by exactly its second derivative:

$$Q[V] = Q[W] + r - m\psi.$$

This is positive on $[0, b]$ because $r > d_\eta$ there. On $(-2, -1)$ it is at least $2/27 - m\|\psi\|_\infty > 0$. Elsewhere $Q[W] > 0, \psi = 0$ and $r \geq 0$. The vertical coefficient remains above $1 - \eta$ by (9.14). For the remaining coefficient, if $x \leq -2$, both W_x and g' vanish; if $-2 \leq x \leq 0$, then $W_x = 0$ and $\mu \geq \mu(-2) > m \geq |g'|$; and if $x \geq 0$, then $\mu \geq 1/2$, $W_x \geq -1/4$ and $g' \geq -m > -1/8$. In all cases $\mu + V_x > 0$. Therefore Lemma 9.2.1 gives $\omega + \mathrm{dd^c} V > 0$ over U_0. Near the fibers over $\mathbb{P}^2 \setminus U_0$, the function V is constant, so $\omega + \mathrm{dd^c} V = \omega > 0$ there as well. □

Such functions r exist with $\int_{\mathbb{R}} r \, \mathrm{d}x \leq \int_{\mathbb{R}} d_\eta \, \mathrm{d}x + \delta$ for every $\delta > 0$: add a small positive constant to a uniform smooth approximation of the compactly supported continuous function d_η, and multiply by a cutoff equal to one on $[0, b]$ and supported in $[-1/2, b + 1/2]$.

Proof (of Theorem 9.0.1) Take $\varepsilon = \eta^{1/\eta^2}$. Then $q_0^2 = (1 + \eta^{-2})L$, so $b, x_* = O(\sqrt{L/\eta})$ and $\eta x_* \to 0$. For small η, Lemma 9.4.1 therefore gives

$$\int_{\mathbb{R}} d_\eta \, \mathrm{d}x = o(1), \qquad \sup_{\mathbb{R}\times E} \max\{-W_x, 0\} = o(1).$$

Choose r_η as in Lemma 9.5.1 with $m_\eta = \int_{\mathbb{R}} r_\eta \, \mathrm{d}x \leq \int_{\mathbb{R}} d_\eta \, \mathrm{d}x + \eta = o(1)$, and let g_η and $V_\eta = W + g_\eta$ be the corresponding functions. For small η, $m_\eta < m_0$ and $\max\{-W_x, 0\} \leq 1/4$, and that lemma shows that V_η is smooth on X with $\omega + \mathrm{dd^c} V_\eta > 0$.

On the fixed ball $B = \{|\xi|^2 < \mathrm{e}^{-3}\} \Subset U_0$, one has $V_\eta = F_\eta(\varepsilon, w)$, including at $\xi = 0$. Thus

$$\int_{\pi^{-1}(y)} |(V_\eta)_w|^2 \mathrm{d}A_E = 2\eta^2 \log(1/\varepsilon) + O(\eta^2) = 2\log(1/\eta) + O(\eta^2) \longrightarrow +\infty$$

uniformly for $y \in B$. Since $\int_B \omega_{\mathrm{FS}}^2 > 0$, the mass identity (9.2) shows that $\int_X (\omega + \mathrm{dd^c} V_\eta)^3 \to +\infty$, and the theorem follows by taking $V_j = V_{\eta_j}$ along any sequence $\eta_j \downarrow 0$. □

9.6 Higher-dimensional Hopf manifolds

The threefold mass formula locates the source of unbounded mass, but it does not by itself give a construction in higher dimension. The same heat potentials do work there: the radial–vertical block is unchanged, and each extra angular direction contributes a factor whose lower bound is uniform where the threefold mass diverges.

For $n \geq 2$, let

$$H_n = (\mathbb{C}^n \setminus \{0\})/\langle z \mapsto \mathrm{e}^{-1} z\rangle, \qquad p_n \colon H_n \longrightarrow \mathbb{P}^{n-1},$$

with coordinates $z = (z_0, \dots, z_{n-1})$ on $\mathbb{C}^n$. Put $s = \log|z|^2$, $\alpha_n = \mathrm{dd^c} s = p_n^* \omega_{\mathrm{FS}}$, $\beta_n = \mathrm{d}s \wedge \mathrm{d^c} s$ and $\omega_n = \alpha_n + \beta_n$, where ω_{FS} is the Fubini–Study form on $\mathbb{P}^{n-1}$, so that $\int_{\mathbb{P}^{n-1}} \omega_{\mathrm{FS}}^{n-1} = 1$. The forms descend and ω_n is positive for the same reasons as in dimension three. The fiber is the same elliptic curve E, with $\int_E \beta_n = 2$. The form α_n is closed of rank $n-1$, $\beta_n^2 = 0$, and $\mathrm{dd^c}\beta_n = -\alpha_n^2$ by the computation following (9.1). Hence $\omega_n^n = n\alpha_n^{n-1} \wedge \beta_n$ and

$$\int_{H_n} \omega_n^n = 2n, \qquad \mathrm{dd^c}(\omega_n^{n-1}) = -(n-1)\alpha_n^n = 0.$$

Thus ω_n is Gauduchon.

Potentials pulled back from the projective base reduce to the closed manifold $\mathbb{P}^{n-1}$. For the invariant backgrounds $a\alpha_n + b\beta_n$, positivity of such a potential is decided on the base, also in families over a parameter manifold. Since $\beta_n^2 = 0$, the vertical form contributes exactly one factor of the Monge–Ampère measure. Later chapters use this reduction for subgeodesics over strips, trace envelopes, descended Monge–Ampère equations and envelope systems.

Lemma 9.6.1 *Let $n \geq 2$, let $a, b \geq 0$ be real numbers, and let S be a complex manifold, possibly a point. Put $P = p_n \times \mathrm{id}_S \colon H_n \times S \to \mathbb{P}^{n-1} \times S$, and write* pr *for the projections of $H_n \times S$ to H_n and of $\mathbb{P}^{n-1} \times S$ to $\mathbb{P}^{n-1}$.*

(1) *For every quasi-plurisubharmonic function V on $\mathbb{P}^{n-1} \times S$,*

$$a\,\mathrm{pr}^*\,\alpha_n + b\,\mathrm{pr}^*\,\beta_n + \mathrm{dd^c} P^* V \geq 0 \quad \Longleftrightarrow \quad a\,\mathrm{pr}^*\,\omega_{\mathrm{FS}} + \mathrm{dd^c} V \geq 0.$$

(2) *A quasi-plurisubharmonic function on $H_n \times S$ that is constant on the fibers of P, equivalently invariant under scalar multiplication on the first factor, is P^*V for a unique quasi-plurisubharmonic function V on $\mathbb{P}^{n-1} \times S$.*

(3) *For a positive Borel measure ν on $\mathbb{P}^{n-1}$, let $\beta_n \wedge p_n^* \nu$ denote the measure $F \mapsto \int_{\mathbb{P}^{n-1}} \big(\int_{p_n^{-1}(y)} F\beta_n\big)\mathrm{d}\nu(y)$ on H_n. For every $v \in \mathrm{PSH}(\mathbb{P}^{n-1}, a\omega_{\mathrm{FS}})$,*

$$\mathrm{MA}^{\mathrm{ext}}_{a\alpha_n + b\beta_n}(p_n^* v) = nb\,\beta_n \wedge p_n^* \mathrm{MA}^{\mathrm{ext}}_{a\omega_{\mathrm{FS}}}(v).$$

Consequently, for every nonnegative Borel function F on $\mathbb{P}^{n-1}$,

$$\int_{H_n} p_n^* F\, \mathrm{MA}^{\mathrm{ext}}_{a\alpha_n + b\beta_n}(p_n^* v) = 2nb \int_{\mathbb{P}^{n-1}} F\, \mathrm{MA}^{\mathrm{ext}}_{a\omega_{\mathrm{FS}}}(v).$$

The total masses of $\mathrm{MA}^{\mathrm{ext}}_{a\omega_{\mathrm{FS}}}(v)$ and $\mathrm{MA}^{\mathrm{ext}}_{a\alpha_n + b\beta_n}(p_n^* v)$ are at most a^{n-1} and $2nba^{n-1}$, with equality for bounded v. In particular, $\mathrm{MA}^{\mathrm{ext}}_{a\alpha_n}(p_n^* v) = 0$.

Proof (1) In local product coordinates for the submersion P, the current $\mathrm{dd^c} P^* V$ is the pullback of $\mathrm{dd^c} V$. Thus the current on the left is $T = P^*\Theta + b\,\mathrm{pr}^*\,\beta_n$, where $\Theta = a\,\mathrm{pr}^*\,\omega_{\mathrm{FS}} + \mathrm{dd^c} V$. If $\Theta \geq 0$, then $P^*\Theta \geq 0$, because holomorphic pullback

preserves positivity (Lemma 1.22.1), and $T \geq 0$ because $\beta_n \geq 0$. Conversely, suppose that $T \geq 0$. Its wedge product with the smooth semipositive form $\mathrm{pr}^* \beta_n$ is a positive current, and $\beta_n^2 = 0$ gives $T \wedge \mathrm{pr}^* \beta_n = P^*\Theta \wedge \mathrm{pr}^* \beta_n$. Pairing a pulled-back current with a test form amounts to pairing the original current with the fiber integral of the form, as in the proof of Lemma 1.7.2. The fibers of P are those of p_n, on which β_n has integral two, so

$$P_*\big(T \wedge \mathrm{pr}^* \beta_n\big) = 2\Theta.$$

The map P is proper, so this push-forward is positive, and $\Theta \geq 0$.

(2) Each fiber of P is an orbit of scalar multiplication on the first factor, so the two conditions agree. Since P is surjective, a function U satisfying them is P^*V for a unique function V. Near each point of $\mathbb{P}^{n-1} \times S$, write $V = U \circ \gamma$ for a local holomorphic section γ of P; as in the second paragraph of the proof of Proposition 1.22.1, V is quasi-plurisubharmonic, since otherwise U would equal $-\infty$ on a nonempty open set.

(3) By (1), $p_n^* v$ belongs to $\mathrm{PSH}(H_n, a\alpha_n + b\beta_n)$, so its extended measure is defined. Adding a constant to v changes neither side, so we may assume $v \leq 0$. The functions $v_j = \max(v, -j)$ are bounded and $a\omega_{\mathrm{FS}}$-plurisubharmonic, and $p_n^* v_j = \max(p_n^* v, -j)$. Put $\nu_j = (a\omega_{\mathrm{FS}} + \mathrm{dd^c} v_j)^{n-1}$. By Lemma 1.7.2, applied with $n-1$ and with n factors,

$$(a\alpha_n + \mathrm{dd^c} p_n^* v_j)^{n-1} = p_n^* \nu_j, \qquad (a\alpha_n + \mathrm{dd^c} p_n^* v_j)^n = 0.$$

Expanding the nth power of $(a\alpha_n + \mathrm{dd^c} p_n^* v_j) + b\beta_n$ by multilinearity of bounded products (Proposition 1.7.1), and using $\beta_n^2 = 0$, we obtain

$$(a\alpha_n + b\beta_n + \mathrm{dd^c} p_n^* v_j)^n = nb\, \beta_n \wedge p_n^* \nu_j.$$

On the right, the wedge product is taken in local product coordinates. Wedging with the pullback of a top-degree measure on the base keeps only the restriction of β_n to the fibers, so this is the measure defined in the statement. Restrict both sides to the set $\{p_n^* v > -j\} = p_n^{-1}\{v > -j\}$. As $j \to \infty$, the measures $\mathbf{1}_{\{v > -j\}} \nu_j$ increase to $\mathrm{MA}^{\mathrm{ext}}_{a\omega_{\mathrm{FS}}}(v)$. Monotone convergence in the outer integral defining $\beta_n \wedge p_n^* \nu$ and the definition (7.1) give the measure identity. Since β_n has integral two on each fiber, the integral identity follows. Finally, ν_j is a positive measure of mass $\int_{\mathbb{P}^{n-1}} (a\omega_{\mathrm{FS}})^{n-1} = a^{n-1}$, by Stokes' theorem for the closed form ω_{FS}, and $\mathbf{1}_{\{v > -j\}} \nu_j \leq \nu_j$. For bounded v and large j, nothing is removed. □

For the rest of the section, let $n \geq 3$. On the affine chart $\{z_0 \neq 0\}$ write $\xi = (z_1/z_0, \ldots, z_{n-1}/z_0) \in \mathbb{C}^{n-1}$, $x = \log|\xi|^2$, $w = \log z_0 \pmod{\Lambda}$ and $\mu = \mathrm{e}^x/(1+\mathrm{e}^x)$ as before. For a smooth real function $V(x, w)$ with the endpoint properties required before Lemma 9.2.1, let $J_n(V; x, w)$ be the density obtained by integrating $(\omega_n + \mathrm{dd^c} V)^n$ over the angular variables $\xi/|\xi|$ against their rotation-invariant probability measure. Thus

$$\int_{H_n} (\omega_n + \mathrm{dd^c} V)^n = \int_{\mathbb{R} \times E} J_n(V; x, w)\, \mathrm{d}x\, \mathrm{d}A_E.$$

The endpoints and the complement of the affine chart have zero smooth volume.

Lemma 9.6.2 *The positivity criterion* (9.3) *holds in every dimension* $n \geq 3$*, and*

$$J_n(V; x, w) = \frac{n(n-1)}{6}(\mu + V_x)^{n-3} J_3(V; x, w). \tag{9.20}$$

Here J_3 *is evaluated for the same function* V *on the threefold.*

Proof As in the proof of Lemma 9.2.1, choose horizontal vectors forming an orthonormal frame for $2\pi\alpha_n$: $n-2$ angular vectors orthogonal to the radial direction, and the radial vector e_r. Adjoin the vertical vector ∂_w. In this frame, each angular diagonal entry of the matrix of $2\pi(\omega_n + \mathrm{dd^c} V)$ is $A = (\mu + V_x)/\mu$, the off-diagonal entries in the angular rows and columns vanish, and the radial–vertical block is

$$K_V = \begin{pmatrix} (\mu + V_{xx})/[\mu(1-\mu)] & (\mu + V_{x\bar{w}})/\sqrt{\mu(1-\mu)} \\ (\mu + V_{wx})/\sqrt{\mu(1-\mu)} & 1 + V_{w\bar{w}} \end{pmatrix}.$$

This follows from the formula for $2\pi\alpha$ in the proof of Lemma 9.2.1: the form $\mathrm{i}\partial\bar{\partial}x$ vanishes in the radial direction and equals $1/\mu$ on each angular unit vector, where ∂x vanishes, so that $\mathrm{i}\partial\bar{\partial}V$ contributes V_x/μ there. The Schur complement of the last entry gives the stated positivity criterion. At $\xi = 0$ the same endpoint argument as in dimension three applies.

Set $d = n-1$. After averaging over the angular variables, ω_{FS}^d becomes the probability measure $d\,\mu^{d-1}\,\mathrm{d}\mu = d\,\mu^d(1-\mu)\,\mathrm{d}x$; this follows from the polar-coordinate determinant of $\mathrm{dd^c}\log(1+|\xi|^2)$. The background block K_0 has determinant one. Since $\omega_n^n = n\alpha_n^{n-1} \wedge \beta_n$ and the fiber mass of β_n is two, therefore,

$$J_n(V) = 2n\,d\,\mu^d(1-\mu)A^{n-2}\det K_V.$$

For $n = 3$ this is $12\mu^2(1-\mu)A\det K_V$. Comparing the two expressions proves (9.20). □

Theorem 9.6.1 *For every* $n \geq 3$*, the Gauduchon metric* ω_n *admits smooth potentials* V_η *with* $\omega_n + \mathrm{dd^c} V_\eta > 0$ *whose total Monge–Ampère masses tend to infinity as* $\eta \downarrow 0$*.*

Proof Use the functions $V_\eta = W + g_\eta$ of the proof of Theorem 9.0.1, with $\varepsilon = \eta^{1/\eta^2}$. Their endpoint properties give smooth potentials on every H_n, and Lemma 9.6.2 gives strict positivity. Since $\mu \geq 1/2$ on $x \geq 0$, the slope bound $\max\{-W_x, 0\} = o(1)$, the inequalities $-m_\eta \leq g_\eta' \leq 0$ of Lemma 9.5.1 and $m_\eta = o(1)$ imply

$$\mu + (V_\eta)_x \geq \tfrac{1}{4} \qquad (x \geq 0)$$

for all sufficiently small η.

We next show that the threefold mass of V_η on $\{x < 0\}$ stays bounded, so that it diverges on $\{x \geq 0\}$. On $x < 0$, the heat potential is $W = F_\eta(\varepsilon, w)$ and has no x derivatives. The determinant formula in the preceding proof and $\int_E (F_\eta)_{w\bar{w}}(\varepsilon, \cdot)\,\mathrm{d}A_E = 0$ give

$$\int_E J_3(V_\eta; x, w)\,\mathrm{d}A_E = 12(\mu + g'_\eta)\big(g''_\eta + \mu(1-\mu)\big).$$

Here $0 < \mu + g'_\eta \le \mu$, and $g''_\eta = r_\eta - m_\eta\psi$ with $\int r_\eta = m_\eta$ and $\int \psi = 1$. Hence

$$\int_{x<0} J_3(V_\eta)\,\mathrm{d}x\,\mathrm{d}A_E \le 12\int_{-\infty}^{0} \mu^2(1-\mu)\,\mathrm{d}x + 12\int_{\mathbb{R}} |g''_\eta|\,\mathrm{d}x \le \tfrac{3}{2} + 24m_\eta.$$

No bound on the amplitude of the fiber Hessian is required; its mean has been integrated exactly. Since the total threefold mass tends to infinity, its mass on $x \ge 0$ tends to infinity. Since $J_n \ge 0$ by positivity, the dimensional comparison on $x \ge 0$ now gives

$$\int_{H_n} (\omega_n + \mathrm{dd^c} V_\eta)^n \ge \frac{n(n-1)}{6}\, 4^{-(n-3)} \int_{x\ge 0} J_3(V_\eta)\,\mathrm{d}x\,\mathrm{d}A_E \longrightarrow +\infty.$$

Corollary 9.6.1 *For $n \ge 3$, every Hermitian form h on H_n satisfies $\nu_+(h) = +\infty$. In particular, H_n does not have the bounded mass property.*

Proof For a Hermitian form h, compactness gives $h \ge c\omega_n$ for some $c > 0$. The smooth potentials cV_η satisfy $h + \mathrm{dd^c}(cV_\eta) \ge c(\omega_n + \mathrm{dd^c}V_\eta) > 0$, and positivity of determinants bounds their masses below by c^n times the divergent masses in Theorem 9.6.1. Hence $\nu_+(h) = +\infty$ (Definition 7.5.1). □

9.7 Minimum mass and fiber averaging

We return to the threefold $X = H_3$ and its form ω.

Averaging along the elliptic fibers removes the variation measured by (9.2). The Poincaré inequality on the fibers bounds the distance from a potential to its average by the excess mass. Bounded regularization preserves this inequality, which therefore identifies all bounded potentials of minimal mass and shows that potentials of almost minimal mass are close to them.

Let E act on X by scalar multiplication: $\tau_a[z] = [\mathrm{e}^a z]$ for $a \in E$. The action preserves α, β and ω, and $\mathrm{d}A_E$ is the Haar probability measure of E. For a bounded Borel function $u\colon X \to \mathbb{R}$, set

$$\mathrm{d}\nu = \frac{\omega^3}{6} = \frac{1}{2}\alpha^2 \wedge \beta, \qquad Au(p) = \int_E u(\tau_a(p))\,\mathrm{d}A_E(a) \quad (p \in X).$$

Thus Au is constant on each fiber, and ν is the probability measure whose disintegration along π has fiber measures $\mathrm{d}A_E$ and base measure ω_{FS}^2.

Proposition 9.7.1 *For every $u \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$, the average Au is the pullback of a bounded ω_{FS}-plurisubharmonic function on $\mathbb{P}^2$, and*

$$\|u - Au\|^2_{L^2(X,\nu)} \le \frac{2}{3}\left(\int_X (\omega + \mathrm{dd^c} u)^3 - 6\right). \tag{9.21}$$

The coefficient $2/3$ *is optimal, even for smooth* u *with* $\omega + \mathrm{dd^c} u > 0$*. Moreover,*

$$\int_X (\omega + \mathrm{dd^c} u)^3 = 6 \quad \Longleftrightarrow \quad u = \pi^* v \quad \textit{for some } v \in \mathrm{PSH}(\mathbb{P}^2, \omega_{\mathrm{FS}}) \cap L^\infty(\mathbb{P}^2).$$

Proof First suppose that u is smooth and $\omega + \mathrm{dd^c} u > 0$. Averaging the positive forms $\tau_a^*(\omega + \mathrm{dd^c} u)$ gives $\omega + \mathrm{dd^c} Au > 0$. Since Au is smooth and constant on the fibers, $Au = \pi^* v$ for a smooth function v on $\mathbb{P}^2$. At each point, the kernel of ∂s maps isomorphically onto the tangent space of the base, and β vanishes on it. Restricting $\omega + \mathrm{dd^c} Au = \pi^*(\omega_{\mathrm{FS}} + \mathrm{dd^c} v) + \beta$ to this kernel therefore shows that $\omega_{\mathrm{FS}} + \mathrm{dd^c} v > 0$.

For a smooth function $f\colon E \to \mathbb{R}$, the nonconstant Fourier characters in (9.5), indexed by $(p, q) \in \mathbb{Z}^2 \setminus \{(0,0)\}$, have eigenvalues $\pi^2 p^2 + q^2/4 \ge 1/4$ for $-\partial_w \partial_{\bar w}$. Parseval's identity therefore gives

$$\int_E \left| f - \int_E f \, \mathrm{d}A_E \right|^2 \mathrm{d}A_E \le 4 \int_E |f_w|^2 \, \mathrm{d}A_E.$$

Applying this inequality on every fiber, integrating against ω_{FS}^2 and using the mass identity (9.2), we obtain (9.21).

For bounded u, choose uniformly bounded smooth functions $u_j \downarrow u$ with $\omega + \mathrm{dd^c} u_j > 0$ by Theorem 1.11.1. Their averages decrease pointwise to Au by dominated convergence. The corresponding smooth base potentials v_j decrease to a bounded ω_{FS}-plurisubharmonic function v, so $Au = \pi^* v$. Dominated convergence gives $u_j \to u$ and $Au_j \to Au$ in $L^2(X, \nu)$, while bounded Bedford–Taylor continuity gives convergence of the total masses. Thus (9.21) passes to u.

If the mass is 6, the inequality gives $u = Au$ almost everywhere. Since both functions are bounded and quasi-plurisubharmonic, Lemma 1.3.3 gives equality everywhere. Conversely, if $u = \pi^* v$ with bounded $v \in \mathrm{PSH}(\mathbb{P}^2, \omega_{\mathrm{FS}})$, then Lemma 9.6.1(3), applied on $X = H_3$ with $\omega = \omega_3$ and $a = b = 1$, gives mass 6.

It remains to prove that the constant $2/3$ is optimal. Choose a nonzero real $\rho \in C_c^\infty(U_0)$. For $\delta \in \mathbb{R}$, the function

$$u(\xi, w) = \delta \rho(\xi) \cos(\mathrm{Im}\, w)$$

extends by zero to a smooth function on X. For sufficiently small nonzero real δ, it satisfies $\omega + \mathrm{dd^c} u > 0$. Its fiber average is zero, and its only fiber frequencies are $(p, q) = (0, \pm 1)$, with eigenvalue $1/4$. Hence equality holds in the fiber inequality, and therefore in (9.21), with nonzero left-hand side. No smaller coefficient can hold.□

Corollary 9.7.1 *The Hopf metric satisfies*

$$v_-(\omega) = 6, \qquad v_+(\omega) = +\infty.$$

Proof The lower bound $v_-(\omega) \geq 6$ follows from (9.21), and it is attained by constants. Theorem 9.0.1 gives $v_+(\omega) = +\infty$. □

9.8 Sharp damping under unitary rotations

The general damping estimate of Theorem 8.3.1 controls mass by the depth of a potential. On the standard Hopf threefold, unitary symmetry gives a stronger estimate: after integration along the elliptic fibers, a positive radial flux has total mass one. A semiconvexity estimate bounds the remaining radial coefficient by the square root of the depth. The resulting damping threshold is sharp, and at its endpoint finite measures can still concentrate on a pole divisor.

We keep the standard threefold X, its form ω, the projection $\pi\colon X \to \mathbb{P}^2$, the fiber E and the probability area $\mathrm{d}A_E$ of this chapter. Let $\mathcal{R}$ be the class of $u \in \mathrm{PSH}(X,\omega)$ with $\sup_X u = 0$ invariant under

$$(z_0, z') \longmapsto (z_0, Az'), \qquad A \in U(2), \quad z' = (z_1, z_2).$$

This holomorphic action preserves ω. For a continuous nonincreasing function $W\colon [0,+\infty) \to (0,+\infty)$, write

$$\mu_u = \mathrm{MA}_\omega^{\mathrm{ext}}(u), \qquad \nu_{W,u} = W(-u)\mu_u,$$
$$\Theta_T(W) = \sup_{t \geq T} (1+\sqrt{2t})W(t), \qquad \Theta(W) = \Theta_0(W).$$

The weighted measure gives zero mass to the pole set, as does the extended nonpluripolar measure. Capacity always means Cap_ω of Section 1.8.

9.8.1 Positive fiber flux

Let V be a smooth invariant ω-plurisubharmonic function. In the affine chart it has the form $V(x,w)$. The matrix calculation in the proof of Lemma 9.2.1 applies at every finite x, without assuming the special endpoint behavior of the heat family. Put

$$A_V = \mu + V_x, \qquad B_V = 1 + V_{w\bar{w}}, \qquad R_V = \mu + V_{xx},$$
$$C_V = \mu + V_{x\bar{w}}, \qquad \Delta_V = B_V R_V - |C_V|^2.$$

Let $\mathrm{d}\Omega$ be the rotation-invariant probability measure in the angular variable $\xi/|\xi|$ of $\mathbb{C}^2$. In these coordinates $\omega_{\mathrm{FS}}^2 = 2\mu^2(1-\mu)\,\mathrm{d}x\,\mathrm{d}\Omega$, and $\omega^3 = 6\omega_{\mathrm{FS}}^2 \mathrm{d}A_E$. Semipositivity gives $A_V, B_V, R_V, \Delta_V \geq 0$ and, without dividing by B_V,

$$\mathrm{MA}_\omega(V) = 12 A_V \Delta_V \,\mathrm{d}x\,\mathrm{d}\Omega\,\mathrm{d}A_E. \tag{9.22}$$

The set not covered by these coordinates, where $\xi = 0$ or $z_0 = 0$, has zero mass for every smooth top form.

Lemma 9.8.1 *For every smooth invariant $V \in \mathrm{PSH}(X, \omega)$,*

$$\int_{\mathbb{R}\times E} \Delta_V \, \mathrm{d}x \, \mathrm{d}A_E = 1.$$

If in addition $V \le 0$, then

$$|V_x|^2 \le -2V, \qquad 0 \le A_V \le 1 + \sqrt{-2V}.$$

Proof Set $m(x) = \int_E V \, \mathrm{d}A_E$ and $e(x) = \int_E |V_w|^2 \, \mathrm{d}A_E$. Periodic integration by parts gives

$$\int_E V_{w\bar{w}} V_{xx} \, \mathrm{d}A_E = -\operatorname{Re} \int_E V_{\bar{w}} V_{xxw} \, \mathrm{d}A_E,$$
$$e'' = 2 \int_E |V_{xw}|^2 \, \mathrm{d}A_E + 2 \operatorname{Re} \int_E V_{\bar{w}} V_{xxw} \, \mathrm{d}A_E.$$

The mean of every fiber derivative vanishes. Expanding Δ_V yields

$$\int_E \Delta_V \, \mathrm{d}A_E = m'' + \mu(1-\mu) - \tfrac{1}{2} e''.$$

Thus $f(x) = \log(1 + \mathrm{e}^x) + m(x) - e(x)/2$ is convex.

Both m and e are smooth global functions on the projective base: local fiber coordinates change by holomorphic translations, preserving fiber Haar area, the vertical derivative and the Dirichlet integral. They are $U(2)$-invariant. Smoothness and radial invariance at $\xi = 0$ give $m', e' \to 0$ at the negative end. At the positive end use a normal coordinate z_0/z_1 to the hyperplane $\{z_0 = 0\}$ on the base. The invariant base functions are constant along this hyperplane and smooth across it; their x derivatives tend to zero. This does not require that the pointwise potential be constant on its elliptic fibers at that end. Consequently $f'(-\infty) = 0$ and $f'(+\infty) = 1$. Integrating the nonnegative f'' proves the flux identity.

For each fixed w, $R_V \ge 0$ implies $V_{xx} \ge -1$ on all of $\mathbb{R}$. Hence for every real h,

$$V(x + h, w) \ge V(x, w) + hV_x(x, w) - h^2/2.$$

Taking $h = V_x(x, w)$ and using $V \le 0$ proves the derivative bound. Finally $\mu \le 1$ proves the bound for A_V. □

If $V \le 0$, then $W(-V)A_V \le (1 + \sqrt{-2V})W(-V)$ by Lemma 9.8.1. Since V is invariant, integrating (9.22) and using the flux identity therefore proves

$$\int_X W(-V)\mathrm{MA}_\omega(V) \le 12\Theta(W), \qquad \int_{\{-V>T\}} W(-V)\mathrm{MA}_\omega(V) \le 12\Theta_T(W). \tag{9.23}$$

The positive flux supplies the reference integral even though the unweighted Monge–Ampère masses are unbounded.

Lemma 9.8.2 *Every bounded invariant ω-plurisubharmonic function u, with $-L \le u \le 0$, admits smooth invariant ω-plurisubharmonic approximants with the same bounds converging to u in capacity.*

Proof The compact group $G = U(3) \times (\mathbb{R}/\mathbb{Z})$ acts transitively and holomorphically on X by unitary transformations and positive scalar multiplication. It preserves ω. Choose a nonnegative smooth central approximate identity ρ_ϵ on G, supported near its identity, and put

$$u_\epsilon(p) = \int_G u(gp)\rho_\epsilon(g)\,\mathrm{d}g.$$

Convolution on a compact homogeneous space is smooth. Averaging preserves the current inequality and the bounds, and centrality makes convolution commute with the $U(2)$ action. Transporting the bounded tests defining capacity shows that Cap_ω is G-invariant.

Suppose $L > 0$. By quasi-continuity, choose an open set O of arbitrarily small capacity such that u is continuous on $X \setminus O$. When g is sufficiently close to the identity, $u(gp)$ and $u(p)$ are uniformly close whenever both points lie outside O. Thus, outside O and for small kernel support,

$$\{|u_\epsilon - u| > \delta\} \subset \{P_\epsilon \mathbf{1}_O > \delta/(4L)\}, \qquad P_\epsilon \mathbf{1}_O(p) = \int_G \mathbf{1}_O(gp)\rho_\epsilon(g)\,\mathrm{d}g.$$

Markov's inequality, followed by capacity invariance, gives

$$\begin{aligned}\mathrm{Cap}_\omega\{P_\epsilon \mathbf{1}_O > \delta/(4L)\} &\le \frac{4L}{\delta} \sup_{0\le v\le 1} \int_X P_\epsilon \mathbf{1}_O\,(\omega + \mathrm{dd}^{\mathrm{c}} v)^3 \\ &\le \frac{4L}{\delta}\,\mathrm{Cap}_\omega(O),\end{aligned}$$

where the supremum runs over $v \in \mathrm{PSH}(X, \omega)$. Letting the capacity of O tend to zero proves the claim. If $L = 0$, $u = 0$ and no approximation is needed. □

Proposition 9.8.1 *For every $u \in \mathcal{R}$ and $T \geq 0$,*

$$\nu_{W,u}(X) \leq 12\Theta(W), \qquad \nu_{W,u}(\{-u > T\}) \leq 12\Theta_T(W). \tag{9.24}$$

If $\sqrt{t}\,W(t) \to 0$, the family $\{\nu_{W,u} : u \in \mathcal{R}\}$ is uniformly absolutely continuous with respect to capacity, and $u \mapsto \nu_{W,u}$ is continuous from convergence in capacity to weak convergence of finite measures.

Proof For bounded $u \leq 0$, apply Lemma 9.8.2. The approximants are uniformly bounded, so the continuity under convergence in capacity of bounded products with bounded quasi-continuous coefficients (Sections 1.6 and 1.8) passes the total estimate (9.23) to u. For the tail at $T > 0$, use continuous depth cutoffs majorizing $\mathbf{1}_{\{-u>T\}}$ and supported in $\{-u > T - \delta\}$, where $0 < \delta < T$. The same product continuity gives a bound by $12\Theta_{T-\delta}(W)$. Continuity of $(1 + \sqrt{2t})W(t)$ gives $\Theta_{T-\delta}(W) \to \Theta_T(W)$. For $T = 0$, use the total estimate.

For unbounded u, the canonical truncations $u^K = \max(u, -K)$ remain bounded and invariant. On $\{u > -K\}$ their measures agree with the finite-level restrictions defining μ_u. These restrictions increase to μ_u, so monotone convergence proves (9.24).

Splitting at depth T and rescaling u^T into a capacity test gives, for every Borel set B,

$$\nu_{W,u}(B) \leq W(0)(1+T)^3\,\mathrm{Cap}_\omega(B) + 12\Theta_T(W). \tag{9.25}$$

Indeed $\omega + \mathrm{dd^c}u^T \leq (1+T)(\omega + \mathrm{dd^c}(u^T/(1+T)))$, and the latter potential has range of length at most one. If $\sqrt{t}\,W(t) \to 0$, then $\Theta_T(W) \to 0$. Taking first T large and then the capacity small proves uniform absolute continuity.

Suppose $u_j \to u$ in capacity in $\mathcal{R}$. For fixed K, the truncations u_j^K converge in capacity to u^K, and the same continuity gives $\nu_{W,u_j^K} \rightharpoonup \nu_{W,u^K}$. Outside $\{u \leq -K\}$, locality identifies ν_{W,u^K} with $\nu_{W,u}$. Both measures give this exceptional set mass at most $12\Theta_{K-1}(W)$ when $K > 1$, by their tail estimates; this also includes any truncation mass on $\{u = -K\}$. Hence

$$\|\nu_{W,u^K} - \nu_{W,u}\|_{\mathrm{TV}} \leq 24\Theta_{K-1}(W)$$

uniformly over $u \in \mathcal{R}$. Letting $K \to +\infty$ removes the truncations and proves the claimed continuity. □

9.8.2 Heat concentration at large depth

The flux estimate suggests the depth scale at which a counterexample must place its mass. A smaller initial heat time than in the unbounded-mass construction gives mass of order η^{-1} at depth of order η^{-2}. Its integrated radial defect remains small after a fixed parameter is chosen, but no longer tends to zero. We therefore need a correction with a small fixed total slope.

We use the heat kernel k_t of (9.4) and its potential F_η of Lemma 9.3.1. The following value estimates complement the derivative bounds already proved.

Lemma 9.8.3 *Uniformly in η and w, $F_\eta(t,w) \le C\eta$ for $t > 0$, and $F_\eta(t,w) \ge -2\eta\log(1/t) - C\eta$ for $0 < t \le 1$. Uniformly on $\sqrt{t} \le |w| \le 2\sqrt{t}$ near the origin of E, as $t \to 0$,*

$$\begin{aligned} F_\eta(t,w) &= 2\eta\log t + O(\eta),\\ tk_t(w) &= 2\mathrm{e}^{-|w|^2/t} + O(\mathrm{e}^{-1/(Ct)}),\\ t\partial_{\bar w}\log k_t(w) &= -w + O(\mathrm{e}^{-1/(Ct)}). \end{aligned}$$

Proof For the upper bound use $k_r \ge 0$ on $(0,1]$ and (9.9). The nonnegative Fourier coefficients imply $k_r(w) \le k_r(0)$, and Gaussian summation gives $k_r(0) \le 2/r + Cr^{-1}\mathrm{e}^{-1/(2r)}$ for $r \le 1$. Its integrable error proves the lower bound. On the indicated annulus the nonzero-lattice terms and their derivatives are exponentially small relative to the zero-lattice term, proving the last two identities. For the first, split the defining integral at a fixed small r_0. On $[t, r_0]$ the nonzero-lattice contributions have bounded integral, and the difference between $(2/r)\mathrm{e}^{-|w|^2/r}$ and $2/r$ has integral at most $2|w|^2/t \le 8$. The remaining integral is uniformly bounded. □

Theorem 9.8.1 *There are $A, B, D, \eta_0 > 0$ such that for every $0 < \eta < \eta_0$ there exist $u_\eta \in C^\infty(X,\mathbb{R}) \cap \mathcal{R}$ and a measurable $K_\eta \subset X$ satisfying*

$$\begin{aligned} &\omega + \mathrm{dd^c} u_\eta > 0, \qquad -u_\eta \le B\eta^{-2} \quad \text{on } X,\\ &-u_\eta \ge A\eta^{-2} \quad \text{on } K_\eta, \qquad \int_{K_\eta} (\omega + \mathrm{dd^c} u_\eta)^3 \ge D\eta^{-1}. \end{aligned}$$

Proof Let $c > 0$ be a small constant, fixed below. Apply the construction preceding Lemma 9.4.1 with the heat time $\varepsilon = \mathrm{e}^{-c\eta^{-3}}$, which is at most η^{1/η^2} for small η, and write Z_η for the resulting function W, since W now denotes a weight. Thus $q_0 = \sqrt{c\eta^{-3} + L}$ with $L = \log(1/\eta)$, $Z_\eta = F_\eta(\varepsilon,\cdot)$ for $x \le 0$, $Z_\eta = 0$ for $x \ge x_*$, and

$$\eta b, \eta x_* \longrightarrow \ell := \sqrt{2c/\kappa}.$$

Let C' be the constant of Lemma 9.4.1 and m_0 that of Lemma 9.5.1; neither depends on c. Choose c so small that $\ell < 1/(10\mathrm{e})$ and $2C'\ell < m_0$. Then $\eta x_* < 1/(10\mathrm{e})$ for small η, and Lemma 9.4.1 gives

$$\int_{\mathbb{R}} d_\eta\,\mathrm{d}x \le C'\bigl(\eta b + \sqrt{\eta L}\bigr) < 2C'\ell, \qquad \max\{-(Z_\eta)_x, 0\} \le C'\sqrt{\eta L} \le \tfrac14. \quad (9.26)$$

Choose r_η as in Lemma 9.5.1 with $m_\eta = \int r_\eta \le \int d_\eta + \eta < m_0$, and put $V_\eta = Z_\eta + g_\eta$. By that lemma, V_η is smooth on X, $\omega + \mathrm{dd^c} V_\eta > 0$, $g_\eta \le 0$ and $|g_\eta| \le m_\eta(b+3) = O_c(\eta^{-1})$.

On $x < -3$, $V_\eta = F_\eta(\varepsilon,\cdot)$ has fiber mean zero, so $0 \le \sup_X V_\eta \le C\eta$ by Lemma 9.8.3 and $g_\eta \le 0$. Set $u_\eta = V_\eta - \sup_X V_\eta$. Where the cutoff $\vartheta(q)$ is nonzero,

$0 < q \le q_0$, so $t \ge \varepsilon$, and Lemma 9.8.3 and (9.9) give $F_\eta(t,w) \ge -2c\eta^{-2} - C\eta$. Multiplication by $\vartheta(q) \in [0,1]$ preserves this lower bound for Z_η, including where F_η is positive, and $Z_\eta = 0$ where $\vartheta(q) = 0$. Together with $\sup_X V_\eta \le C\eta$ and the bound for $|g_\eta|$, this gives

$$0 \le -u_\eta \le 2c\eta^{-2} + O_c(\eta^{-1}) + O(\eta). \tag{9.27}$$

Mass and depth on the concentration region. Let K_η be the set of points of the affine chart, with arbitrary angular part $\xi/|\xi|$, such that

$$q_0/2 \le q(x) \le 3q_0/4, \qquad \sqrt{t(x)} \le |w| \le 2\sqrt{t(x)},$$

where w is lifted to a small disk about 0. Let σ, h and ζ be as in Step 1 of the proof of Lemma 9.4.1. This set lies in the interval $[a^{-1}, b]$ where q is affine, with $\vartheta = 1$, $x \asymp_c \eta^{-1} \to \infty$, $\sigma = 2aq \asymp_c \eta^{-2}$, and superexponentially small t. Lemma 9.8.3 gives, uniformly,

$$0 < c_1 \le tk_t \le 3, \quad h \to \infty, \quad \zeta \to 0, \quad \eta\sigma^2 t \to 0, \quad \mu \to 1.$$

In (9.16) the first h contribution is nonnegative, $\sigma_x t F_t = -\kappa t(k_t - 1) \ge -3\kappa$, and the completed square tends to one. Its other negative terms tend to zero. Thus $Q[Z_\eta] \ge 1/4$ for small η. On the core $\psi = 0$ and $g_\eta'' = r_\eta \ge 0$, so $Q[V_\eta] \ge 1/4$. Also

$$\mu + (V_\eta)_x \ge c_2\eta\sigma, \qquad 1 + (V_\eta)_{w\bar w} \ge c_1\eta/t.$$

Indeed $(Z_\eta)_x = \eta\sigma(tk_t - t)$ and $\mu + g_\eta' > 0$ here; the vertical inequality follows from $1 - \eta + \eta k_t$.

In (9.22), $\Delta_{V_\eta} = (1 + (V_\eta)_{w\bar w})Q[V_\eta]$. The fiber annulus has $\mathrm{d}A_E$-area $3t/2$. Integrating the preceding lower bounds and using $\sigma\,\mathrm{d}x = -2q\,\mathrm{d}q$ yields

$$\int_{K_\eta} (\omega + \mathrm{dd}^{\mathrm{c}} u_\eta)^3 \ge c_3\eta^2 \int_{q_0/2}^{3q_0/4} 2q\,\mathrm{d}q = c_4\eta^2 q_0^2 \ge c_5\eta^{-1}.$$

The value estimate of Lemma 9.8.3 on the annulus also gives

$$-u_\eta = 2\eta q^2 + O_c(\eta^{-1}) + O(\eta\log(1/\eta)).$$

As $\eta q_0^2 = c\eta^{-2} + \eta\log(1/\eta)$, this is bounded above and below by fixed positive multiples of η^{-2} on the core. Together with (9.27), this proves the asserted mass and depth bounds. All these potentials are bounded: truncations beyond their individual depth equal them everywhere, so the defining extended product equals (9.22). □

Corollary 9.8.1 *Let $W\colon [0,\infty) \to (0,\infty)$ be nonincreasing. A uniform bound for the weighted masses*

$$\int_X W(-u)\mathrm{MA}_\omega^{\mathrm{ext}}(u)$$

over all $u \in \mathrm{PSH}(X,\omega)$ with $\sup_X u = 0$ requires

$$\sup_{T\gg 1} \sqrt{T}\, W(T) < \infty.$$

Uniform deep tails over that class, that is, $\sup_u \nu_{W,u}(\{-u > T\}) \to 0$ as $T \to +\infty$, require $\sqrt{T}\, W(T) \to 0$.

Proof On K_η, monotonicity of W gives

$$\int_X W(-u_\eta)\mathrm{MA}_\omega^{\mathrm{ext}}(u_\eta) \geq D\eta^{-1} W(B\eta^{-2}).$$

Every sufficiently large T equals $B\eta^{-2}$ for some sufficiently small η, proving the necessary bound. Also $K_\eta \subset \{-u_\eta > A\eta^{-2}/2\}$, so uniform deep tails force this lower bound to tend to zero. □

Theorem 9.8.2 *Suppose $W\colon [0,+\infty) \to (0,+\infty)$ is nonincreasing and*

$$\sup_{T\gg 1} \sqrt{T} W(T) = \infty.$$

Then there is a single $U \in \mathcal{R}$ such that

$$\int_X W(-U)\mathrm{MA}_\omega^{\mathrm{ext}}(U) = \infty.$$

Proof First select the correction with additional local control. Put

$$x_{\mathrm{pre}} = (q_0 + 1/2 - 2\sqrt{L})/a.$$

Then $t(x_{\mathrm{pre}}) = \eta^3$, $x_{\mathrm{pre}} \to \infty$, and $b - x_{\mathrm{pre}} = O(\sqrt{\eta L}) \to 0$. For a fixed large K, the pointwise bounds in the proof of Lemma 9.4.1 give $d_\eta \leq (K/2)\eta$ on $[0, x_{\mathrm{pre}}]$: on $[0, a^{-1}]$ Step 2 gives $d_\eta \leq C\eta$, and on $[a^{-1}, x_{\mathrm{pre}}]$ we have $q^2 t \leq 4L\eta^3$, so that the exponential error in (9.18) is $O(\exp(-1/(C\kappa\eta^2 L))) = o(\eta)$. By Step 4 of that proof and the length of $[x_{\mathrm{pre}}, b]$, the integral of d_η over this interval tends to zero.

Choose a fixed left cutoff, zero for $x \leq -1/2$ and one for $x \geq -1/4$, and a translated right cutoff, one for $x \leq x_{\mathrm{pre}}$ and zero for $x \geq x_{\mathrm{pre}} + 1/4$. Let r_η^0 be $K\eta$ times their product. It strictly exceeds d_η on $[0, x_{\mathrm{pre}}]$. The continuous function $e_\eta = \max(d_\eta - r_\eta^0, 0)$ is supported in $[x_{\mathrm{pre}}, b]$. As after Lemma 9.5.1, add a smooth nonnegative majorant $s_\eta > e_\eta$ on that interval, supported in $(x_{\mathrm{pre}} - 1/4, b + 1/2)$, with $\int s_\eta \leq \int e_\eta + \eta$. Set $r_\eta = r_\eta^0 + s_\eta$. Then

$$r_\eta > d_\eta \quad \text{on } [0,b], \qquad m_\eta = \int r_\eta \leq K\eta b + O(\sqrt{\eta L}) + O(\eta).$$

After shrinking the fixed c so that $K\ell < m_0$, we have $m_\eta < m_0$ for small η. Thus r_η satisfies the hypotheses of Lemma 9.5.1, and the conclusions of Theorem 9.8.1 hold for this choice, with fixed A, B, D. On every fixed compact interval $I \Subset (0,\infty)$, for sufficiently small η we have $r_\eta = K\eta$ and exactly

$$g_\eta(x) = -m_\eta(x-\mu_\psi) + K\eta(x^2/2 + b_0 x + b_1), \qquad \mu_\psi = \int s\psi(s)\,\mathrm{d}s. \tag{9.28}$$

Here b_0, b_1 depend only on the fixed left cutoff. Globally $0 \le -g_\eta(x) \le m_0(x+2)$ for $x \ge 0$.

Let $G(w) = -\int_0^\infty (k_r(w) - 1)\,\mathrm{d}r$ for $w \ne 0$ on E. Gaussian integration gives the uniform bound

$$-F_\eta(t,w) \le C\eta\big(1 + \log^+(1/\mathrm{dist}_E(w,0)^2)\big), \quad t > 0. \tag{9.29}$$

Indeed the nearest Gaussian has integral bounded by $C + 2\log^+(1/\mathrm{dist}_E(w,0)^2)$ on $(0,1]$; the other lattice terms have bounded integral and large times decay. The same bound applies after the cutoff. On a fixed $I \Subset (0,\infty)$, $\vartheta = 1$ for small η and $t_\eta(x) \le \exp(-c/(2\eta^3))$. Every fixed x derivative of t_η has this bound times a polynomial in η^{-1}. On compact fiber sets away from zero,

$$F_\eta(t_\eta(x), w)/\eta \longrightarrow G(w) \quad \text{in } C^\ell \text{ for every fixed } \ell.$$

To verify derivatives, put $d = \mathrm{dist}_E(w,0)$ and note that

$$F_\eta(t,\cdot)/\eta - G = \int_0^t (k_r - 1)\,\mathrm{d}r = -t + O\big(\exp(-d^2/(Ct))\big)$$

when d stays positive. Differentiated Gaussian errors have only polynomial factors in t^{-1}, and repeated x differentiation preserves their decay. Together with (9.28) and $0 \le \sup V_\eta \le C\eta$, this gives uniform C^ℓ bounds for u_η on each such fixed compact product set, for all sufficiently small η. The threshold may depend on the set and derivative order.

Fix $a_j = 2^{-j-2}$, $j \ge 1$, so $\sum a_j = 1/4$. Choose scales $\eta_j \downarrow 0$ recursively and write $u_j = u_{\eta_j}$, $K_j = K_{\eta_j}$. Require the minimum x on K_j to exceed every earlier $x_{*,i}$ by two. Choose a relatively compact product neighborhood O_j of K_j, with positive x interval and fiber closure away from zero, on which the earlier u_i are constant. The K_j are pairwise disjoint. With $S_j = \sum_{i<j} a_i \|u_i\|_\infty$, impose

$$S_j \le a_j \eta_j^{-2}, \qquad \eta_j \le a_j,$$
$$a_j \eta_j \le 2^{-j} a_i \eta_i \quad (i < j).$$

For each earlier O_i, require the preceding uniform derivative bounds to hold for u_j through order j. All are finite constraints at stage j, achievable by making η_j smaller. They ensure that the future series $\sum_{k>i} a_k u_k$ converges in C^∞ on compact subsets of

O_i: for every fixed derivative order the sufficiently late terms have a uniform bound there, and $\sum a_k < \infty$.

These constraints also give $\sum_{k>j} a_k\eta_k \le a_j\eta_j$. On K_j the coordinate and distance estimates are

$$x \le C_c\eta_j^{-1}, \qquad \log^+(1/\mathrm{dist}_E(w,0)^2) \le C_c\eta_j^{-3}.$$

Using (9.29), the global slope bound on g_k, and the supremum constants, the future depth $-\sum_{k>j} a_k u_k$ is at most

$$C_c\eta_j^{-3}\sum_{k>j} a_k\eta_k + C_c\eta_j^{-1}\sum_{k>j} a_k \le C_c a_j\eta_j^{-2} + C_c\eta_j^{-1}/4.$$

The last term is at most a fixed multiple of $a_j\eta_j^{-2}$ since $\eta_j \le a_j$. Including the earlier and current terms, there is a fixed $C_* > 0$, independent of the scales, such that

$$-\sum_i a_i u_i \le C_* a_j\eta_j^{-2} \quad \text{on } K_j. \tag{9.30}$$

Fix this constant before the scale choice. Since $\sqrt{T}W(T)$ is unbounded, at each stage choose arbitrarily large T_j such that

$$\frac{D}{\sqrt{C_*}} a_j^{5/2}\sqrt{T_j}W(T_j) \ge 1, \qquad \eta_j = \sqrt{C_* a_j/T_j},$$

and so small a scale that all finite preceding constraints hold. Such values occur arbitrarily far out, since $\sqrt{T}W(T) \le \sqrt{T_0}W(0)$ for $T \le T_0$. Thus the recursive choice is possible.

The smooth nonpositive partial sums $v_N = \sum_{j\le N} a_j u_j$ decrease, and

$$\omega + \mathrm{dd^c} v_N = (1 - \sum_{j\le N} a_j)\omega + \sum_{j\le N} a_j(\omega + \mathrm{dd^c} u_j) \ge \tfrac{3}{4}\omega.$$

By Theorem 1.4.1, $\int_X (-v_N)\,\mathrm{d}V \le C\sum_j a_j < +\infty$. Thus Corollary 1.4.1 gives an ω-plurisubharmonic limit v with $\omega + \mathrm{dd^c} v \ge 3\omega/4$. Its supremum is finite and nonpositive.

On O_j the earlier terms are constant and the future series converges smoothly. Thus v is smooth there and $\omega + \mathrm{dd^c} v \ge a_j(\omega + \mathrm{dd^c} u_j)$ pointwise. The determinant inequality for positive Hermitian matrices gives on K_j

$$\mathrm{MA}_\omega^{\mathrm{ext}}(v) \ge a_j^3(\omega + \mathrm{dd^c} u_j)^3.$$

On these sets the extended product is the smooth product: on a compact neighborhood of K_j all sufficiently deep canonical truncations equal v. Set $U = v - \sup_X v$. Since $\sup v \le 0$, $-U \le -v$, and its local currents are unchanged. Monotonicity of W and (9.30) give

$$\int_{K_j} W(-U)\mathrm{MA}_\omega^{\mathrm{ext}}(U) \geq D a_j^3 \eta_j^{-1} W(C_* a_j \eta_j^{-2}) = \frac{D}{\sqrt{C_*}} a_j^{5/2} \sqrt{T_j} W(T_j) \geq 1.$$

The disjoint K_j prove infinite total mass for this single potential. They lie in its smooth locus, so no mass is assigned to poles. □

9.8.3 Concentration at the critical weight

Lemma 9.8.4 *Suppose $\sqrt{t}\,W(t)$ is bounded but does not tend to zero. There are smooth $u_j \in \mathcal{R}$ converging in capacity to $g \in \mathcal{R}$ such that a weak cluster measure of ν_{W,u_j} gives positive mass to the pole divisor*

$$H = \pi^{-1}\{[z_0 : z_1 : z_2] : z_0 = 0\},$$

whereas $\nu_{W,g}(H) = 0$.

Proof Use the correction chosen in the proof of Theorem 9.8.2. Its plateau and its late supplement satisfy

$$\int r_\eta^0 = K\eta x_{\mathrm{pre}} + O(\eta), \qquad \int s_\eta = O(\sqrt{\eta L}) + O(\eta), \qquad \eta x_{\mathrm{pre}} \to \ell.$$

Therefore $m_\eta \to m := K\ell$, and $0 < m < m_0$. On each fixed finite x interval, r_η and all its derivatives tend to zero; the late supplement escapes the interval. Thus g_η converges smoothly there to

$$g(x) = -m \int_{-\infty}^{x} (x - s)\psi(s)\,\mathrm{d}s.$$

It is zero for $x \leq -2$ and equals $-m(x - \mu_\psi)$ for $x \geq -1$. Since $m < m_0$, also $\mu + g' > 0$ and $g'' + \mu(1 - \mu) \geq 0$ at finite x. Across H, its extension has logarithmic singularity of coefficient m: a local defining function ζ gives $g = m \log|\zeta|^2 + O(1)$. Its upper semicontinuous extension is ω-plurisubharmonic, has supremum zero and belongs to $\mathcal{R}$.

The bound (9.29) and $F_\eta \leq C\eta$ imply that, over compact base sets away from H, the set $\{|Z_\eta| > \delta\}$, for fixed $\delta > 0$ and small η, lies in a tube about the local holomorphic zero section $w = 0$ whose radius shrinks exponentially as $\eta \downarrow 0$. These tubes have capacity tending to zero. Indeed, on a relatively compact coordinate chart, $\log|w|^2$ is locally integrable and plurisubharmonic; the weighted Chern–Levine–Nirenberg estimate of Section 1.6 bounds the relative capacity of $\{\log|w|^2 < -T\}$ by C/T. A finite cover and the local capacity comparison transfer this estimate to Cap_ω.

The corrections converge uniformly on compact base sets away from H, including $\xi = 0$, and the supremum normalization is $O(\eta)$. Since g has its logarithmic pole on H, the estimate (1.15) gives neighborhoods of H with arbitrarily small capacity. Removing one such neighborhood proves $u_\eta \to g$ in capacity.

Now use the hypothesis on W. Since $\sqrt{t}\,W(t)$ does not tend to zero, there are $\eta_j \downarrow 0$ for which the lower bound $D\eta_j^{-1}W(B\eta_j^{-2})$ for the weighted mass of the core K_{η_j} (see the proof of Corollary 9.8.1) stays above some $c_0 > 0$; put $u_j = u_{\eta_j}$. Since $\sqrt{t}\,W(t)$ is bounded, Proposition 9.8.1 makes the full weighted measures ν_{W,u_j} uniformly finite, so a subsequence has a weak cluster measure ν. For each fixed closed neighborhood N of H, all sufficiently late cores lie in N. Portmanteau gives $\nu(N) \geq c_0$; decreasing continuity of the finite measure along closed neighborhoods shrinking to H gives $\nu(H) \geq c_0$. But $\nu_{W,g}(H) = 0$ by its defining non-pluripolar truncation. □

Theorem 9.8.3 *Within $\mathcal{R}$ the following assertions hold.*

(1) *Every individual $\nu_{W,u}$ is finite if and only if $\sup_{t\gg 1}\sqrt{t}\,W(t) < \infty$. This is also equivalent to a uniform total-mass bound. In this case*

$$\sup_{u\in\mathcal{R}} \nu_{W,u}(X) \leq 12\Theta(W).$$

If the condition fails there is one $u \in \mathcal{R}$ with infinite measure.

(2) *Uniform deep tails hold if and only if $\sqrt{t}\,W(t) \to 0$. For all $u \in \mathcal{R}$ and $T \geq 0$,*

$$\nu_{W,u}(\{-u > T\}) \leq 12\Theta_T(W).$$

The inequality is understood in the extended sense.

(3) *Uniform absolute continuity relative to Cap_ω holds if and only if $\sqrt{t}\,W(t) \to 0$.*

(4) *The assignment $u \mapsto \nu_{W,u}$ is a map into finite measures and is continuous from capacity convergence in $\mathcal{R}$ to weak measure convergence if and only if $\sqrt{t}\,W(t) \to 0$. When $\sqrt{t}\,W(t)$ is bounded but does not tend to zero, there are smooth $u_j \in \mathcal{R}$ converging in capacity to $g \in \mathcal{R}$, whose weighted measures have a weak cluster measure charging the pole divisor of g.*

Proof The sufficient conditions are Proposition 9.8.1. The necessary uniform mass and tail conditions follow from Corollary 9.8.1, whose smooth potentials are invariant. If $\sqrt{t}\,W(t)$ is unbounded, Theorem 9.8.2 disproves universal individual finiteness and prevents a map into finite measures.

For uniform absolute continuity, the concentration regions K_η constructed in the proof of Theorem 9.8.1 lie in $\{x \geq c'/\eta\}$ for a fixed $c' > 0$, hence in shrinking neighborhoods of H. The globally defined $h_H = \log(|z_0|^2/|z|^2)$ has supremum zero and $\omega + \mathrm{dd}^{\mathrm{c}} h_H = \beta + [H] \geq 0$. Its sublevel estimate (1.15) shows that these neighborhoods have capacity tending to zero. If $\limsup_{t\to+\infty}\sqrt{t}\,W(t) > 0$, choose the scales so that $D\eta^{-1}W(B\eta^{-2})$ stays bounded below. The weighted masses on these regions then violate uniform absolute continuity. Finally, Lemma 9.8.4 disproves capacity-to-weak continuity whenever $\sqrt{t}\,W(t)$ is bounded but does not tend to zero.□

Corollary 9.8.2 *For a weight W satisfying the assumptions of this section and agreeing for large t with*

$$W_{p,a}(t) = (1+t)^{-p}[\log(\mathrm{e}+t)]^{-a},$$

universal individual finiteness and uniform total-mass bounds in $\mathcal{R}$ hold exactly when $p > 1/2$, or $p = 1/2$ and $a \geq 0$. Uniform deep tails, uniform capacity absolute continuity and capacity-to-weak continuity hold exactly when $p > 1/2$, or $p = 1/2$ and $a > 0$.

Proof Apply Theorem 9.8.3 to $\sqrt{t}\, W_{p,a}(t) \sim t^{1/2-p}(\log t)^{-a}$. The conditions of that theorem concern only large t. □

The necessary conditions and the single infinite-measure example also apply to the class of all normalized potentials on X. The sufficient conditions proved here use the $U(2)$ symmetry. For the unrestricted class and C^1 weights tending to zero, Theorem 8.3.1 gives the sufficient condition $\int_0^{+\infty}(1+t)(-W'(t))\,\mathrm{d}t < +\infty$; the optimal damping rate remains open.

9.9 Limits of shrinking smooth potentials

When the bounded mass property fails, Corollary 8.5.1 produces smooth potentials converging to zero whose Monge–Ampère measures keep an excess mass in the limit. On the Hopf threefold we can determine exactly which excess measures occur: they are the lifts of arbitrary base measures by the fiber probability measures. The proof combines the intermediate currents of Proposition 8.5.1 with averaging over the automorphism group.

On a general Hermitian manifold, the restrictions on the intermediate currents in Proposition 8.5.1 are necessary; we do not know whether they are sufficient, or what the support and pluripolar behavior of such excess measures can be in general.

We keep the Hopf threefold X, its projection $\pi\colon X \to \mathbb{P}^2$, and the forms $\omega = \alpha + \beta$ of the previous sections. For $p \in \mathbb{P}^2$, let h_p be the Haar probability measure on the elliptic fiber $\pi^{-1}(p)$, that is, the measure $\mathrm{d}A_E$ transported by the action of E. For a finite positive measure μ on the base, define its *fiberwise lift* $\mathcal{L}\mu$ by

$$\int_X \varphi\,\mathrm{d}\mathcal{L}\mu = \int_{\mathbb{P}^2}\left(\int_{\pi^{-1}(p)} \varphi\,\mathrm{d}h_p\right)\mathrm{d}\mu(p), \qquad \varphi \in C(X).$$

Since fiber integrals of continuous functions are continuous in p, the map $\mathcal{L}$ is weakly continuous. The normalization $\int_{\pi^{-1}(p)}\beta = 2$ gives

$$\mathcal{L}\mu = \tfrac{1}{2}\beta \wedge \pi^*\mu, \qquad \mathcal{L}(\omega_{\mathrm{FS}}^2) = \tfrac{1}{6}\omega^3. \tag{9.31}$$

Here $\pi^*\mu$ denotes the pullback of a top-degree current by the submersion π; locally it is the base measure with constant coefficients in the fiber directions.

For a Hermitian form Ω on X, put

$$M_\Omega(f) = \int_X (\Omega + \mathrm{dd^c} f)^3, \qquad V_\Omega = \int_X \Omega^3$$

for smooth real f, and let b be a real number. A positive measure σ on X is called a *shrinking defect* for (Ω, b) if, for every sequence $0 < \varepsilon_j < 1$ tending to zero, there are smooth $U_j \in \mathrm{PSH}(X, \Omega)$ and numbers $0 < r_j < \varepsilon_j$ such that

$$\sup_X U_j = 0, \qquad w_j = r_j U_j, \qquad M_\Omega(w_j) = b, \qquad (\Omega + \mathrm{dd^c} w_j)^3 \rightharpoonup \Omega^3 + \sigma. \tag{9.32}$$

Here weak convergence means convergence against every continuous function on X. We first classify the shrinking defects for the metric ω, for which $V_\omega = 6$; we then write $m = b - 6 > 0$. Afterwards we pass to an arbitrary Hermitian form.

9.9.1 Positive quadratic measures

The mass formula of Proposition 9.1.1 gives

$$M_\omega(\lambda f) - 6 = \lambda^2 \big(M_\omega(f) - 6\big)$$

for every smooth real f and real λ. To use this scalar identity in a weak limit, we also need a positive measure that represents the quadratic term.

Lemma 9.9.1 *Let f be smooth, $0 < t < 1$, and $\mathrm{dd^c} f \geq -t\omega$. Set $H = \mathrm{dd^c} f + t\omega$, $d = M_\omega(f) - 6$, and $\nu = 3\omega \wedge H^2$. Then $d \geq 0$ and*

$$\int_X \omega^2 \wedge H = 6t, \qquad \int_X \omega \wedge H^2 = d/3 + 6t^2, \qquad \int_X H^3 = td + 6t^3,$$
$$\big\| (\omega + \mathrm{dd^c} f)^3 - \omega^3 - \nu \big\|_{\mathrm{TV}} \leq 36t + 2td + 24t^3.$$

Proof The Gauduchon identity $\mathrm{dd^c}\omega^2 = 0$ gives $\int_X \omega^2 \wedge \mathrm{dd^c} f = 0$, and Stokes' formula gives $\int_X (\mathrm{dd^c} f)^3 = 0$. Expanding the three integrals with these identities proves the asserted values, and Proposition 9.1.1 gives $d \geq 0$. The identity of measures

$$(\omega + \mathrm{dd^c} f)^3 = (1-t)^3\omega^3 + 3(1-t)^2\omega^2 \wedge H + (1-t)\nu + H^3$$

has four positive terms. After subtracting $\omega^3 + \nu$, the total masses of the remaining terms bound the total variation by $18t + 18t + t(d + 18t^2) + td + 6t^3$, as asserted. □

9.9.2 Orbital mixtures

Averaging a shrinking defect over automorphisms of X produces new shrinking defects. The compact group $G = U(3) \times (\mathbb{R}/\mathbb{Z})$ of the proof of Lemma 9.8.2 acts transitively on X by $g[z] = [\mathrm{e}^t A z]$, $g = (A, t)$, preserving ω. Let h_G be its Haar probability measure. For a probability measure ρ on G and a finite measure σ on X, write

$$\mathcal{A}_\rho\sigma = \int_G g^*\sigma \, \mathrm{d}\rho(g).$$

Here the pullback of a measure is defined by $\int_X \varphi \, \mathrm{d}(g^*\sigma) = \int_X \varphi \circ g^{-1} \, \mathrm{d}\sigma$. In particular, transitivity gives

$$\int_G g^*\zeta \, \mathrm{d}h_G(g) = \frac{\zeta(X)}{6}\omega^3$$

for every finite measure ζ on X.

Proposition 9.9.1 *If σ is a shrinking defect for $(\omega, 6+m)$, then so is $\mathcal{A}_\rho\sigma$ for every probability measure ρ on G.*

Proof Fix a sequence ε_j as in the definition. Take a sequence $f_k = t_k u_k$ realizing σ as in (9.32), so that f_k has mass $6+m$, and put $H_k = \mathrm{dd^c} f_k + t_k\omega$. By Lemma 9.9.1, the positive measures $\nu_k = 3\omega \wedge H_k^2$ have masses $m + 18t_k^2$ and converge weakly to σ.

The idea is to replace f_k by a normalized sum of N random translates. First suppose that ρ has density at most L relative to h_G. Choose N independent random group elements $g_1, \dots, g_N$ with law ρ, and set

$$W = N^{-1/2}\sum_{a=1}^{N} f_k \circ g_a, \qquad \tau = t_k\sqrt{N}, \qquad K = \mathrm{dd^c}W + \tau\omega = N^{-1/2}\sum_{a=1}^{N} g_a^* H_k \geq 0.$$

The quadratic measure of W is

$$3\omega \wedge K^2 = D + C, \qquad D = N^{-1}\sum_{a=1}^{N} g_a^*\nu_k, \qquad C = \frac{6}{N}\sum_{a<c} \omega \wedge g_a^*H_k \wedge g_c^*H_k \geq 0.$$

Averaging $u_k \circ g$ against h_G gives a constant, hence $\int_G g^*H_k \, \mathrm{d}h_G = t_k\omega$. Consequently $B = \int_G g^*H_k \, \mathrm{d}\rho \leq Lt_k\omega$, and independence gives

$$\mathbb{E}[C(X)] = 3(N-1)\int_X \omega \wedge B^2 \leq 18(N-1)L^2t_k^2.$$

For any real $\varphi \in C(X)$ with $\|\varphi\|_\infty \leq 1$, the variance of $D(\varphi)$ is at most $(m+18t_k^2)^2/N$.

For general ρ, convolve it with the normalized restrictions of h_G to shrinking neighborhoods of the identity, which have positive Haar measure. This gives probability measures $\rho_j \rightharpoonup \rho$ with bounded densities L_j. Fix a countable dense set (φ_l) in the real unit ball of $C(X)$ and numbers $0 < \delta_j < \min(1, m/4)$ tending to zero. Choose N_j large enough that

$$\frac{j(m+18)^2}{N_j\delta_j^2} < \frac{1}{4}.$$

Next choose k_j large enough that $t_{k_j} < 1$, $\tau_j = t_{k_j}\sqrt{N_j} < \min(\varepsilon_j/4, 1/j, 1/4)$, $18N_jt_{k_j}^2 < m/4$, and the expected cross-term mass is less than $\delta_j/4$. Chebyshev's

inequality for the first j test functions and Markov's inequality for $C(X)$ show that some choice of N_j group elements satisfies

$$|D_j(\varphi_l) - (\mathcal{A}_{\rho_j}\nu_{k_j})(\varphi_l)| < \delta_j \quad (l \le j), \qquad C_j(X) < \delta_j.$$

We may also arrange $k_j \to \infty$. For each fixed test function φ, the family $\{\varphi \circ g^{-1} : g \in G\}$ is compact in $C(X)$, so a finite uniform net and the bounded masses of ν_k show that $\int \varphi \circ g^{-1}\,\mathrm{d}\nu_k \to \int \varphi \circ g^{-1}\,\mathrm{d}\sigma$ uniformly in g. It follows that $\mathcal{A}_{\rho_j}\nu_{k_j} \rightharpoonup \mathcal{A}_\rho\sigma$. The dense test functions and the bounded masses now imply $D_j \rightharpoonup \mathcal{A}_\rho\sigma$, while $C_j \to 0$ in total variation.

By the integral identity of Lemma 9.9.1,

$$d_j = M_\omega(W_j) - 6 = m + C_j(X) - 18(N_j - 1)t_{k_j}^2 \longrightarrow m, \qquad d_j > 3m/4.$$

Put $\lambda_j = \sqrt{m/d_j}$, so $\lambda_j \to 1$ and $\lambda_j < 2$. The average $a_j = N_j^{-1}\sum_a u_{k_j} \circ g_a$ is smooth and ω-plurisubharmonic. With

$$U_j = a_j - \sup_X a_j, \qquad r_j = \lambda_j \tau_j < \varepsilon_j,$$

the potential $r_j U_j$ differs from $\lambda_j W_j$ by a constant and has exact mass $6+m$. Its positive quadratic measure is $\lambda_j^2(D_j + C_j)$. The total-variation estimate of Lemma 9.9.1, with parameter r_j, proves the required weak limit. □

Since $v_+(\omega) = +\infty$, Corollary 8.5.1 gives a sequence with some defect of mass m. Passing to a further subsequence meets any prescribed bounds ε_j without changing the defect, so a shrinking defect of mass m exists. Applying Proposition 9.9.1 with $\rho = h_G$ shows that $(m/6)\omega^3$ is a shrinking defect. In particular, the excess mass can persist as a smooth limiting density.

9.9.3 Fiber measures

The intermediate currents force every shrinking defect to be a lift of a base measure, and the automorphisms of X realize all such lifts.

Lemma 9.9.2 *Let $f_j = t_j u_j$, where $u_j \in \mathrm{PSH}(X,\omega)$ are smooth, $\sup_X u_j = 0$ and $0 < t_j \to 0$. Suppose that $M_\omega(f_j) = 6 + m$ and $(\omega + \mathrm{dd^c} f_j)^3 \rightharpoonup \omega^3 + \sigma$. Then there is a unique positive measure μ on $\mathbb{P}^2$, of mass m, such that*

$$\sigma = \mathcal{L}\mu, \qquad (\mathrm{dd^c} f_j + t_j\omega)^2 \rightharpoonup \tfrac{1}{6}\pi^*\mu.$$

Proof Put $H_j = \mathrm{dd^c} f_j + t_j\omega$. By Proposition 8.5.1, applied with $h = \omega$ and $n = 3$, every convergent subsequence of H_j^2 has a positive closed limit K with $\sigma = 3\omega \wedge K$; moreover $\alpha \wedge K = 0$, because α is closed and semipositive, so that $\alpha \wedge K$ is a positive measure with $\int_X \alpha \wedge K = 0$. In holomorphic coordinates (ξ_1, ξ_2, w) on a

local trivialization of the elliptic bundle, the coefficient matrix of K on the three exterior pairs is positive semidefinite as a matrix of measures. To see this directly, note that positivity of the matrices of H_j^2 passes to the limit when they are tested against constant vectors, and take Radon–Nikodym derivatives with respect to the diagonal trace. Since α is strictly positive in the horizontal directions, the identity $\alpha \wedge K = 0$ makes both diagonal entries containing $\mathrm{d}w$ vanish. Positivity then makes their rows and columns vanish, so only the horizontal top-degree coefficient remains.

Since K is closed, both real fiber derivatives of that coefficient vanish in the sense of distributions. Testing against a compactly supported base function times a nonconstant Fourier mode on $E = \mathbb{C}/(\mathbb{Z} + 2\pi\mathrm{i}\mathbb{Z})$ shows that every such mode vanishes. By density of trigonometric polynomials in $C(E)$, the coefficient is a base measure times the constant fiber area. Changes of bundle coordinates translate w by a holomorphic function of the base and preserve that area. Thus $K = \pi^*\tau$ for a positive measure τ on $\mathbb{P}^2$.

Since $\alpha \wedge K = 0$, (9.31) gives $\sigma = 3\beta \wedge \pi^*\tau = \mathcal{L}(6\tau)$. Therefore $\mu = 6\tau = \pi_*\sigma$ is uniquely determined by σ, and it has mass m. In particular, every subsequential limit of H_j^2 is the same, which proves convergence of the entire sequence. □

Lemma 9.9.3 *For every $p \in \mathbb{P}^2$, the measure mh_p is a shrinking defect for $(\omega, 6+m)$.*

Proof Start with smooth $f_j = t_j u_j$ realizing the shrinking defect $(m/6)\omega^3 = \mathcal{L}(m\,\omega_{\mathrm{FS}}^2)$, and put $H_j = \mathrm{dd}^{\mathrm{c}} f_j + t_j\omega$. By Lemma 9.9.2, $H_j^2 \rightharpoonup (m/6)\alpha^2$, while H_j has trace mass $\int_X \omega^2 \wedge H_j = 6t_j$ by Lemma 9.9.1. Fix $A \in GL(3,\mathbb{C})$, regarded as an automorphism of X, and let a be the induced projective transformation of $\mathbb{P}^2$. Choose $C_A \geq 1$ with $A^*\omega \leq C_A\omega$, and put

$$R_A = C_A\omega - A^*\omega \geq 0, \qquad H_j^A = \mathrm{dd}^{\mathrm{c}}(f_j \circ A) + C_A t_j\omega = A^*H_j + t_j R_A.$$

For this fixed A, the trace mass of A^*H_j is $O_A(t_j)$, and the positive difference $(H_j^A)^2 - A^*(H_j^2)$ has trace mass $O_A(t_j^2)$. Since $\alpha = \pi^*\omega_{\mathrm{FS}}$, it follows that

$$(H_j^A)^2 \rightharpoonup \frac{m}{6}\pi^*(a^*\omega_{\mathrm{FS}}^2), \qquad 3\omega \wedge (H_j^A)^2 \rightharpoonup \mathcal{L}(ma^*\omega_{\mathrm{FS}}^2).$$

The limiting measure has mass m, so the integral identity of Lemma 9.9.1, applied with the parameter $C_A t_j$, gives $d_j^A = M_\omega(f_j \circ A) - 6 \to m$. For large j, multiply $f_j \circ A$ by $\sqrt{m/d_j^A} \to 1$. The mass is then exactly $6 + m$, and the weak limit is unchanged. The underlying normalized potential is $u_j \circ A/C_A \in \mathrm{PSH}(X,\omega)$, with parameter $C_A t_j\sqrt{m/d_j^A} \to 0$. The total-variation estimate of Lemma 9.9.1 therefore shows that $\mathcal{L}(m\,a^*\omega_{\mathrm{FS}}^2)$ is a shrinking defect; any prescribed bounds ε_j are met by passing to a sufficiently late subsequence.

Now take $A_l = \mathrm{diag}(R_l^{-1}, 1, 1)$, $R_l \to +\infty$. Since

$$a_l^*\omega_{\mathrm{FS}}^2 = (a_l^{-1})_*\omega_{\mathrm{FS}}^2,$$

and $a_l^{-1}[q_0 : q_1 : q_2] = [R_l q_0 : q_1 : q_2] \to [1 : 0 : 0]$ outside the hyperplane $\{q_0 = 0\}$, which has zero volume, dominated convergence gives $a_l^* \omega_{\mathrm{FS}}^2 \rightharpoonup \delta_{[1:0:0]}$. By weak continuity of $\mathcal{L}$, the shrinking defects $\mathcal{L}(m\, a_l^* \omega_{\mathrm{FS}}^2)$ converge to $m h_{[1:0:0]}$. For each l, choose one corrected potential for A_l, sufficiently late in its sequence that its parameter is less than ε_l and its measure differs from the corresponding limit by less than $1/l$ on the first l functions of a fixed dense sequence of test functions. This diagonal sequence has the exact mass $6 + m$ and the limit $\omega^3 + m h_{[1:0:0]}$. Unitary automorphisms move this fiber to any prescribed fiber. □

Theorem 9.9.1 *For $b > 6$, the shrinking defects for (ω, b) are exactly $\mathcal{L}\mu$, where μ ranges over all positive measures on $\mathbb{P}^2$ of mass $b - 6$. Each such defect has support $\pi^{-1}(\operatorname{supp}\mu)$ and has no point atoms. It charges no pluripolar sets if and only if μ charges no pluripolar sets on $\mathbb{P}^2$.*

Proof Put $m = b - 6$. Necessity follows from Lemma 9.9.2. Finite convex combinations of the measures $m h_p$ are shrinking defects by Lemma 9.9.3 and Proposition 9.9.1, the latter applied to an atomic measure ρ supported on unitary maps that move a fixed fiber to the desired fibers. Approximate an arbitrary base measure of mass m by finite atomic measures of the same mass, using finite partitions of diameter tending to zero. Weak continuity of $\mathcal{L}$ and the diagonal choice on dense test functions used in the preceding proof give a realizing sequence for every prescribed sequence ε_j.

Every nonempty open subset of a fiber has positive Haar measure. Local trivializations therefore show that $\mathcal{L}\mu$ has support $\pi^{-1}(\operatorname{supp}\mu)$, and the fiber measures have no point atoms. If B is a pluripolar subset of the base, local plurisubharmonic functions whose pole sets contain B pull back by the submersion to such functions on X. Hence $\pi^{-1}(B)$ is pluripolar, and $\mathcal{L}\mu(\pi^{-1}(B)) = \mu(B)$; thus $\mathcal{L}\mu$ charges a pluripolar set whenever μ does.

Conversely, suppose that μ charges no pluripolar subset of the base. Cover a pluripolar set in X by countably many local product charts, each carrying a plurisubharmonic function $q(\xi, w) \not\equiv -\infty$ whose pole set contains the part of the set in that chart. For a relatively compact disk D in the fiber direction, the function

$$F(\xi) = \frac{1}{|D|} \int_D q(\xi, w)\, \mathrm{d}A(w)$$

is plurisubharmonic and not identically $-\infty$ on a smaller base chart. Indeed, local integrability of q and Fubini's theorem show that F is locally integrable, and in particular finite almost everywhere; a common upper bound and the reverse Fatou lemma give upper semicontinuity; and integrating the submean inequalities in the base directions gives the submean property. The restriction of q to a fiber is either subharmonic with a pole set of zero area or identically $-\infty$. The fibers of the second kind lie over the pluripolar set $\{F = -\infty\}$, which has zero μ-measure. Integrating the Haar measures of the fibers against μ therefore shows that each local pole set has zero $\mathcal{L}\mu$-measure, and the countable cover proves the assertion for this pluripolar set. For arbitrary pluripolar subsets, the same argument applies to Borel pole sets containing them, with outer measure. Similarly, for the first implication, a Borel $\mathcal{L}\mu$-null set

containing $\pi^{-1}(B)$ has a measurable fiber-mass function equal to one on B, which gives the corresponding outer-measure statement. □

For every nonempty closed set $F \subseteq \mathbb{P}^2$, a measure $m \sum_{l\geq 1} 2^{-l}\delta_{p_l}$ carried by a countable dense subset $\{p_l\}$ of F has support F. Thus every closed set of the form $\pi^{-1}(F)$ is the support of a shrinking defect. The fibers themselves are pluripolar in a global sense. For the fiber over $[1:0:0]$, the function

$$\psi = \log \frac{|z_1|^2 + |z_2|^2}{|z|^2}$$

satisfies $\mathrm{dd}^c\psi \geq -\alpha$, hence $\omega + \mathrm{dd}^c\psi \geq \beta \geq 0$. Thus ψ is ω-plurisubharmonic, and its pole set is exactly that fiber; unitary maps give the other fibers. By Theorem 9.9.1, for $0 \leq \lambda \leq 1$ and $p \in \mathbb{P}^2$, the measure

$$m\lambda h_p + \frac{m(1-\lambda)}{6}\omega^3$$

is a shrinking defect. It gives mass $m\lambda$ to the pluripolar fiber $\pi^{-1}(p)$, while every pluripolar set has mass at most $m\lambda$. For $0 < \lambda < 1$ its support is all of X, despite this positive mass on a pluripolar set.

9.9.4 Changing the Hermitian metric

For a general Hermitian form, the fiber Haar measures are replaced by the normalized area measures of the form. Let Ω be any Hermitian form on the Hopf threefold X. Its restriction to each elliptic fiber has positive area

$$A_\Omega(p) = \int_{\pi^{-1}(p)} \Omega.$$

These areas depend smoothly on p and are bounded above and below by positive constants. Define the probability measures $h_p^\Omega = \Omega|_{\pi^{-1}(p)}/A_\Omega(p)$ on the fibers, and define the lift $\mathcal{L}_\Omega\mu$ of a finite positive measure μ on $\mathbb{P}^2$ by the formula defining $\mathcal{L}\mu$, with h_p^Ω in place of h_p.

Theorem 9.9.2 *For every Hermitian form Ω on X and every $b > V_\Omega$, the shrinking defects for (Ω, b) are exactly*

$$\mathcal{L}_\Omega\mu, \qquad \mu \geq 0, \qquad \mu(\mathbb{P}^2) = b - V_\Omega.$$

Their supports are $\pi^{-1}(\operatorname{supp}\mu)$, they have no point atoms, and they charge no pluripolar sets exactly when μ charges no base pluripolar sets.

Proof Put $m = b - V_\Omega > 0$. For necessity, let σ be a shrinking defect for (Ω, b). Proposition 8.5.1, applied with $h = \Omega$, gives a positive closed limit K of the squares

of the intermediate forms with $\sigma = 3\Omega \wedge K$, and $\alpha \wedge K = 0$ as in the proof of Lemma 9.9.2. The local coefficient and Fourier argument of that proof gives $K = \pi^*\tau$ for a positive measure τ on $\mathbb{P}^2$. Consequently

$$\sigma = \mathcal{L}_\Omega \mu, \qquad \mu = 3A_\Omega \tau = \pi_* \sigma,$$

and μ has mass m.

Conversely, fix such a μ and set $\tau = \mu/(3A_\Omega)$. By Theorem 9.9.1, choose a sequence $f_j = t_j u_j$ realizing the shrinking defect $\mathcal{L}(6\tau)$ for ω. The positive forms $H_j = \mathrm{dd}^c f_j + t_j \omega$ have trace mass $O(t_j)$, and $H_j^2 \rightharpoonup \pi^*\tau$ by Lemma 9.9.2. Choose $C \geq 1$ with $\omega \leq C\Omega$, and put

$$R = C\Omega - \omega \geq 0, \qquad G_j = \mathrm{dd}^c f_j + Ct_j\Omega = H_j + t_j R \geq 0.$$

The trace mass of G_j is $O(t_j)$, and $G_j^2 - H_j^2$ has trace mass $O(t_j^2)$. Thus $G_j^2 \rightharpoonup \pi^*\tau$ and $3\Omega \wedge G_j^2 \rightharpoonup \mathcal{L}_\Omega \mu$.

Unlike ω, the form Ω need not be Gauduchon, so the mass polynomial may have a linear term. Define

$$\ell_j = 3\int_X \Omega^2 \wedge \mathrm{dd}^c f_j = O(t_j), \qquad q_j = 3\int_X \Omega \wedge (\mathrm{dd}^c f_j)^2 \longrightarrow m.$$

These estimates follow by expanding $\mathrm{dd}^c f_j = G_j - Ct_j\Omega$ and using the preceding trace bounds. Stokes' formula removes the cubic integral, so

$$M_\Omega(\lambda f_j) = V_\Omega + \lambda \ell_j + \lambda^2 q_j.$$

For large j, $q_j > 0$. The positive root

$$\lambda_j = \frac{-\ell_j + \sqrt{\ell_j^2 + 4mq_j}}{2q_j} \longrightarrow 1 \tag{9.33}$$

gives $M_\Omega(\lambda_j f_j) = b$. The normalized function $U_j = u_j/C$ is Ω-plurisubharmonic, and $w_j = \lambda_j f_j = r_j U_j$ with $r_j = \lambda_j C t_j \to 0$.

To identify the full weak limit, first note that expanding $G_j = \mathrm{dd}^c f_j + Ct_j\Omega$ gives

$$\int_X G_j^3 = Ct_j q_j + (Ct_j)^2 \ell_j + (Ct_j)^3 V_\Omega = O(t_j).$$

Hence $G_j^3 \to 0$ in total variation. The positive expansion

$$(\Omega + \mathrm{dd}^c w_j)^3 = (1-r_j)^3\Omega^3 + 3(1-r_j)^2\lambda_j\Omega^2 \wedge G_j + 3(1-r_j)\lambda_j^2 \Omega \wedge G_j^2 + \lambda_j^3 G_j^3$$

converges weakly to $\Omega^3 + \mathcal{L}_\Omega\mu$, since the second and fourth terms tend to zero in total variation. Given prescribed bounds ε_j, pass to a sufficiently late subsequence

with $r_j < \varepsilon_j$ at every index. This preserves the normalization, the exact mass and the weak limit.

The fiber probabilities h_p^{Ω} have smooth densities relative to h_p that are bounded above and below by positive constants, uniformly in p. Thus $\mathcal{L}_{\Omega}\mu$ and $\mathcal{L}\mu$ are bounded by fixed positive multiples of each other, and the assertions about support, point atoms and pluripolar sets follow from Theorem 9.9.1. □

Part II
Monge–Ampère and Hessian equations

Chapter 10
Smooth Monge–Ampère equations

Throughout this chapter, ω is a Hermitian form on X and $\mathrm{d}V$ is a smooth positive volume form. Given $F \in C^\infty(X, \mathbb{R})$, we seek a smooth real potential u and a scalar $b \in \mathbb{R}$ satisfying

$$\omega_u^n = \mathrm{e}^{F+b}\mathrm{d}V, \qquad \omega_u > 0. \tag{10.1}$$

Here $\omega_u = \omega + \mathrm{dd}^{\mathrm{c}}u$, and adding a constant to u does not change the equation. For a closed background, integration determines b from the prescribed density. For a Hermitian background, the total mass can depend on u, so the potential and the scalar must be found together.

The existence proof combines weighted integration, which bounds the normalized potential, with a trace estimate for the complex Hessian, which gives uniform ellipticity. The adjoint probability measure of Proposition 4.8.1 supplies the compatibility condition for the linearized equation. Including b as an unknown removes this obstruction, and the estimates give the compactness needed for the continuity method. When the density contains the exponential of a positive multiple of the potential, the equation is strictly monotone, and existence holds without an additional scalar.

After proving existence, we differentiate the solution with respect to the prescribed logarithmic density. The same adjoint probability measure gives the first and second variations of b, proving concavity and a Lipschitz bound. When $\mathrm{e}^F\mathrm{d}V$ is a probability measure, $\mathrm{e}^{b/n}$ is an infimum of quotients over Gauduchon metrics. This infimum defines the Gauduchon variational volume of every pseudoeffective background, and its extrema over the probability measures are the lower and upper Monge–Ampère masses. On the Hopf threefold of Chapter 9, the energy of the solution along the elliptic fibers measures exactly how b differs from the closed-background value (10.2).

The last part of the chapter prescribes the volume form of a Gauduchon metric, which requires a different variation of the metric. For $n \geq 2$, we deform the $(n-1)$st power of the metric through a scalar Aeppli ansatz. If the initial metric is Gauduchon, this deformation preserves the Gauduchon condition and the Aeppli class of the $(n-1)$st power. The first-order torsion terms have a specific matrix structure, which makes the second-derivative estimate possible; rescaling then reduces the gradient

bound to a Liouville theorem on $\mathbb{C}^n$. The resulting uniqueness concerns the scalar ansatz. We finish by studying the variation of the normalization constant of this equation.

10.1 Scalar bounds and uniqueness

Evaluating the equation at the extrema of the potential bounds the normalization scalar in terms of the prescribed data. Applied to the difference of two solutions, the same argument shows that their scalars agree, and ellipticity then determines the potential up to a constant. The scalar bounds will also give uniform control along the continuity path.

Theorem 10.1.1 (Tosatti–Weinkove) *For every smooth real F, (10.1) has a unique pair $(u, b) \in C^\infty(X, \mathbb{R}) \times \mathbb{R}$ with $\sup_X u = 0$.*

In dimension one, integration determines $\mathrm{e}^b = \int_X \omega / \int_X \mathrm{e}^F \mathrm{d}V$, and the remaining equation $\mathrm{dd^c} u = \mathrm{e}^{F+b}\mathrm{d}V - \omega$ is a Poisson equation whose right-hand side has integral zero, so it has a smooth solution, unique up to a constant; positivity of ω_u is automatic. Thus the theorem holds in that dimension, and we prove it for $n \geq 2$ at the end of Section 10.5.

Proposition 10.1.1 *Every solution of (10.1) satisfies*

$$\inf_X \left(\log \frac{\omega^n}{\mathrm{d}V} - F \right) \leq b \leq \sup_X \left(\log \frac{\omega^n}{\mathrm{d}V} - F \right).$$

Proof At a maximum point x_+ of u, the real $(1,1)$-form $\mathrm{dd^c} u$ is nonpositive, so $0 < \omega_u \leq \omega$ there. Hence $\omega_u^n(x_+) \leq \omega^n(x_+)$, and $b \leq \log(\omega^n/\mathrm{d}V)(x_+) - F(x_+)$. At a minimum point the reverse argument gives the lower bound. □

When $\mathrm{d}\omega = 0$, integration gives the familiar explicit formula

$$b = \log \frac{\int_X \omega^n}{\int_X \mathrm{e}^F \mathrm{d}V}. \tag{10.2}$$

The same formula holds whenever the mass-invariance criterion of Theorem 2.2.2 is satisfied. In general, integrating the equation gives only

$$\mathrm{e}^b = \frac{\mathrm{M}_\omega(u)}{\int_X \mathrm{e}^F \mathrm{d}V},$$

where $\mathrm{M}_\omega(u) = \int_X \omega_u^n$ is the mass of Theorem 2.2.1; the numerator depends on the unknown potential.

Proposition 10.1.2 *Suppose (u, b) and (v, c) are smooth solutions of (10.1) with the same F and $\mathrm{d}V$. Then $b = c$ and $u - v$ is constant.*

Proof At a maximum of $u - v$, one has $\omega_u \le \omega_v$, so the equations give $\mathrm{e}^b \le \mathrm{e}^c$. At a minimum one obtains $\mathrm{e}^b \ge \mathrm{e}^c$. Therefore $b = c$.

Put $h = u - v$ and $\eta_t = \omega_v + t\mathrm{dd^c} h = (1-t)\omega_v + t\omega_u$ for $0 \le t \le 1$. These forms are strictly positive. Subtracting the logarithms of the two equations and writing the difference as the integral of its derivative along the segment, we obtain

$$0 = \int_0^1 n \, \frac{\mathrm{dd^c} h \wedge \eta_t^{n-1}}{\eta_t^n} \, \mathrm{d}t.$$

The right-hand side is a linear second-order operator applied to h. Its coefficients are smooth, its principal part is uniformly elliptic on the compact manifold, and it has no zeroth-order term. The strong maximum principle therefore implies that h is constant. □

10.2 The potential bound from weighted integration

We now show that an upper bound for the density controls the normalized potential, following the method of Cherrier. Integrating the equation against e^{-pu} produces a gradient term and mixed weighted masses; a finite recurrence absorbs the latter together with the torsion error. Moser iteration then gives a set of uniformly positive volume near the minimum, and normalized L^1 compactness bounds that minimum. Norms below are computed with respect to the fixed metric ω.

Lemma 10.2.1 *There is a constant C depending only on ω such that, for every positive $(1,1)$-form η, every smooth real f, every $\delta > 0$, and $0 \le k \le n-2$,*

$$|\mathrm{d^c} f \wedge \mathrm{d}\omega \wedge \eta^k \wedge \omega^{n-k-2}| \le \frac{C}{\delta} \, \mathrm{d}f \wedge \mathrm{d^c} f \wedge \eta^k \wedge \omega^{n-k-1} + C\delta \, \eta^k \wedge \omega^{n-k}.$$

Here the absolute value of a top-degree form is the absolute value of its coefficient relative to the positive volume form ω^n.

Proof At a given point, choose an ω-unitary frame in which η is diagonal, with positive eigenvalues $\lambda_1, \dots, \lambda_n$. Expand η^k as a sum over k-element subsets I of $\{1, \dots, n\}$, with coefficients that are positive dimensional multiples of $\lambda_I = \prod_{i \in I} \lambda_i$. After wedging with $\mathrm{d^c} f \wedge \mathrm{d}\omega \wedge \omega^{n-k-2}$, each surviving term is a bounded coefficient of $\mathrm{d}\omega$ times λ_I times one first derivative f_a with $a \notin I$. Young's inequality bounds its absolute value by

$$C\lambda_I |f_a| \le (C/\delta)\lambda_I |f_a|^2 + C\delta\lambda_I.$$

The sum of the first terms is bounded by a fixed multiple of the first form on the right of the assertion, whose diagonal expansion consists of the terms $\lambda_I |f_a|^2$ with $a \notin I$. The second form on the right contains all the λ_I. Since there are finitely many indices and $\mathrm{d}\omega$ is bounded, the constant C is uniform. □

Lemma 10.2.2 *Suppose u is smooth, $\eta = \omega + \mathrm{dd^c}u > 0$, and $\eta^n \le \Lambda\omega^n$ for a fixed $\Lambda > 0$. There are constants $p_0 \ge 1$ and C depending only on ω and Λ such that*

$$\int_X \mathrm{d}(\mathrm{e}^{-pu/2}) \wedge \mathrm{d^c}(\mathrm{e}^{-pu/2}) \wedge \omega^{n-1} \le Cp \int_X \mathrm{e}^{-pu}\omega^n \quad (p \ge p_0).$$

Proof For a fixed real $p \ge 1$, define M_k for $0 \le k \le n$ and J_k for $0 \le k \le n-1$ by

$$M_k = \int_X \mathrm{e}^{-pu}\eta^k \wedge \omega^{n-k}, \qquad J_k = \int_X \mathrm{e}^{-pu}\mathrm{d}u \wedge \mathrm{d^c}u \wedge \eta^k \wedge \omega^{n-k-1}, \quad J = \sum_{k=0}^{n-1} J_k.$$

All these numbers are nonnegative. The proof first bounds pJ in terms of the masses M_k, and then bounds these masses by a recurrence. Let $T = \sum_{k=0}^{n-1} \eta^k \wedge \omega^{n-k-1}$, so that $\eta^n - \omega^n = \mathrm{dd^c}u \wedge T$. Since $\mathrm{d}\eta = \mathrm{d}\omega$, differentiation gives

$$\mathrm{d}T = n \sum_{k=0}^{n-2} \eta^k \wedge \omega^{n-k-2} \wedge \mathrm{d}\omega.$$

The sum is empty when $n = 1$. Stokes' theorem gives the exact identity

$$\begin{aligned} \int_X \mathrm{e}^{-pu}(\eta^n - \omega^n) &= \int_X \mathrm{e}^{-pu}\mathrm{dd^c}u \wedge T \\ &= pJ + \int_X \mathrm{e}^{-pu}\mathrm{d^c}u \wedge \mathrm{d}T. \end{aligned}$$

The density assumption bounds the left-hand side above by $(\Lambda + 1)M_0$. Apply Lemma 10.2.1 to the last integral, with $\delta = K/p$ for a sufficiently large fixed K. The contribution proportional to J/δ is then at most $pJ/2$, and after absorbing it we obtain

$$pJ \le CM_0 + \frac{C}{p} \sum_{k=0}^{n-2} M_k. \tag{10.3}$$

This estimate alone does not suffice, because the mixed weighted masses on the right have not yet been bounded.

For $1 \le k \le n-2$, a second integration by parts gives

$$M_k - M_{k-1} = pJ_{k-1} + R_k, \qquad R_k = \int_X \mathrm{e}^{-pu}\mathrm{d^c}u \wedge \mathrm{d}(\eta^{k-1} \wedge \omega^{n-k}). \tag{10.4}$$

The derivative in R_k has two terms, with coefficients $k-1$ and $n-k$. Applying the pointwise estimate with $\delta = 1/p$ to each of them, and omitting terms with negative indices, we obtain

$$|R_k| \le Cp(J_{k-2} + J_{k-1}) + \frac{C}{p}(M_{k-2} + M_{k-1}), \qquad M_{-1} = J_{-1} = 0.$$

Consequently, for $p \ge 1$,

$$M_k \le C(M_{k-1} + M_{k-2} + pJ).$$

Induction over the finitely many values of k proves

$$\sum_{k=0}^{n-2} M_k \le C(M_0 + pJ).$$

For $n = 2$ this is immediate, and for $n = 1$ the sum is empty. Inserting this bound in (10.3) gives $pJ \le CM_0 + C'J$. For $p \ge 2C'$ the last term can be absorbed, and $pJ \le CM_0$.

Finally,

$$\int_X \mathrm{d}(\mathrm{e}^{-pu/2}) \wedge \mathrm{d}^{\mathrm{c}}(\mathrm{e}^{-pu/2}) \wedge \omega^{n-1} = \frac{p^2}{4} J_0 \le \frac{p^2}{4} J \le CpM_0.$$

This proves the claimed estimate for all $p \ge p_0$. □

Theorem 10.2.1 *For each $\Lambda > 0$ there is C_Λ such that every smooth u with*

$$\omega + \mathrm{dd}^{\mathrm{c}} u > 0, \qquad \sup_X u = 0, \qquad (\omega + \mathrm{dd}^{\mathrm{c}} u)^n \le \Lambda \omega^n$$

satisfies $\|u\|_\infty \le C_\Lambda$.

Proof Let $\mathrm{d}\mu = \omega^n / \int_X \omega^n$. The Sobolev inequality for the Riemannian metric of ω gives, for some $\beta > 1$ and every smooth real function w,

$$\left(\int_X |w|^{2\beta} \, \mathrm{d}\mu \right)^{1/\beta} \le C \left(\int_X \mathrm{d}w \wedge \mathrm{d}^{\mathrm{c}} w \wedge \omega^{n-1} + \int_X |w|^2 \, \mathrm{d}\mu \right).$$

We may take $\beta = n/(n-1)$ if $n \ge 2$, and for example $\beta = 2$ if $n = 1$. Choose p_0 as in Lemma 10.2.2. Applying the Sobolev inequality to $w = \mathrm{e}^{-pu/2}$ and using that lemma, we obtain, with $Z = \mathrm{e}^{-u}$,

$$\|Z\|_{L^{p\beta}(\mu)} \le (Cp)^{1/p} \|Z\|_{L^p(\mu)} \quad (p \ge p_0).$$

Iterating with $p = p_0 \beta^j$, $j = 0, 1, 2, \ldots$, we obtain a bound for $\sup_X Z$; the product of the constants is finite because $\sum_j (1 + \log(p_0 \beta^j))/(p_0 \beta^j) < \infty$. Hence, for a uniform $C_1 \ge 1$ and $m = \inf_X u$,

$$\mathrm{e}^{-p_0 m} \le C_1 \int_X \mathrm{e}^{-p_0 u} \, \mathrm{d}\mu. \tag{10.5}$$

The right-hand side still involves the unknown potential. Divide by $\mathrm{e}^{-p_0 m}$ and set $A = p_0^{-1} \log(2C_1)$. The integrand $\mathrm{e}^{-p_0(u-m)}$ is at most one everywhere and at most $(2C_1)^{-1}$ outside $S = \{u \le m + A\}$. Therefore (10.5) implies

$$\mu(S) \ge \frac{1}{2C_1} =: \delta > 0.$$

Since $\sup_X u = 0$, Theorem 1.4.1 gives $\int_X (-u)\,\mathrm{d}\mu \le C_0$. On S we have $-u \ge -m - A$, so

$$C_0 \ge \int_X (-u)\,\mathrm{d}\mu \ge \delta(-m - A)$$

when $-m \ge A$; otherwise the desired bound is immediate. Thus $-m \le A + C_0/\delta$, proving the theorem. □

10.3 The linearized equation and smooth dependence

Recall from (4.11) that the linearization of $u \mapsto \log(\omega_u^n/\mathrm{d}V)$ at u is $L_u v = n\,\mathrm{d}\mathrm{d}^{\mathrm{c}} v \wedge \omega_u^{n-1}/\omega_u^n$. This operator annihilates constants, and by Proposition 4.8.1 its range consists of the functions with zero mean under the adjoint probability measure μ_u. Including the scalar b as an unknown removes this obstruction. In dimension one, $\mu_u = \omega_u / \int_X \omega_u$, and the compatibility condition follows from Stokes' theorem.

The normalization $\sup_X u = 0$ is convenient for compactness, but it need not depend differentiably on the data. To differentiate solutions, we use instead the linear normalization

$$\int_X u\,\mathrm{d}V = 0.$$

Passing from one normalization to the other only adds a constant to u and leaves b unchanged.

Proposition 10.3.1 *With this linear normalization, the solution (u, b) of (10.1) depends smoothly on F in Hölder spaces, and hence smoothly on the parameters of every smooth finite-dimensional family of smooth data.*

Proof Fix an integer $k \ge 0$ and a real exponent $0 < \alpha < 1$. Consider the map

$$(u, b, F) \longmapsto \log \frac{(\omega + \mathrm{d}\mathrm{d}^{\mathrm{c}} u)^n}{\mathrm{d}V} - F - b$$

on the open set of potentials in $C^{k+2,\alpha}$ with $\omega_u > 0$, restricted to $\int_X u\,\mathrm{d}V = 0$. Its derivative in (u, b) is

$$(v, a) \longmapsto L_u v - a.$$

Given $h \in C^{k,\alpha}$, integrating $L_u v - a = h$ against μ_u forces $a = -\int_X h\,\mathrm{d}\mu_u$. For this value, the equation for v satisfies the compatibility condition of Proposition 4.8.1, and it has a unique solution with zero $\mathrm{d}V$-mean. Schauder estimates show that this inverse is bounded. The implicit function theorem in Banach spaces therefore gives a smooth local solution map around any given solution; this step does not use the global existence theorem. Uniqueness identifies the local maps wherever they overlap, and elliptic regularity allows arbitrary k for smooth data. □

10.4 Second derivatives and ellipticity

The equation controls the product of the eigenvalues of ω_u relative to ω, so an upper bound for their sum gives uniform ellipticity. We obtain this trace bound by completing a square in the differentiated equation. The remaining torsion term is of first order in the trace, and it can be absorbed at the maximum of a suitable function of the trace and the potential.

In holomorphic coordinates $(z^1, \dots, z^n)$, subscripts denote coordinate derivatives, inverse matrices carry upper indices, and repeated indices run from 1 to n. Write

$$\omega = \frac{\mathrm{i}}{2\pi} g_{i\bar{j}}\, \mathrm{d}z^i \wedge \mathrm{d}\bar{z}^j, \qquad \omega_u = \frac{\mathrm{i}}{2\pi} h_{i\bar{j}}\, \mathrm{d}z^i \wedge \mathrm{d}\bar{z}^j, \qquad h_{i\bar{j}} = g_{i\bar{j}} + u_{i\bar{j}}.$$

For a smooth real function v, set $\Delta_g v = g^{i\bar{j}} v_{i\bar{j}}$ and $\Delta_h v = h^{i\bar{j}} v_{i\bar{j}}$. Thus $\Delta_h = L_u$ and $\Delta_h u = n - \mathrm{tr}_h\, g$. Write also $|\partial v|_h^2 = h^{i\bar{j}} v_i v_{\bar{j}}$. Let ∇ be the Chern connection of the fixed metric g, with

$$\Gamma^p_{ik} = g^{p\bar{q}} \partial_i g_{k\bar{q}}, \qquad T^p_{ik} = \Gamma^p_{ik} - \Gamma^p_{ki}, \qquad \tau_k = \sum_i T^i_{ik}.$$

Here T is the torsion of ∇ and τ its trace; all bounds for this connection, its torsion and its curvature depend only on g.

Lemma 10.4.1 *Suppose $h = g + (u_{i\bar{j}}) > 0$, and put*

$$S = \mathrm{tr}_g\, h, \qquad B = \mathrm{tr}_h\, g, \qquad G = \log \frac{\det h}{\det g}.$$

There is a constant C_g, depending only on g, such that, at every point where $S \geq 1$,

$$\Delta_h \log S \geq \frac{2}{S^2} \mathrm{Re}\big(h^{k\bar{q}} \tau_k S_{\bar{q}}\big) + \frac{\Delta_g G}{S} - C_g B. \tag{10.6}$$

Proof The identity $\mathrm{d}\omega_u = \mathrm{d}\omega$ is equivalent to

$$\nabla_i h_{k\bar{j}} - \nabla_k h_{i\bar{j}} = T^p_{ik}(g_{p\bar{j}} - h_{p\bar{j}}), \tag{10.7}$$

together with its complex conjugate; this can also be checked directly, since the third derivatives of u cancel.

For tensors with two holomorphic and one antiholomorphic lower index, we use the norm

$$\|A\|^2 = g^{i\bar{\ell}} h^{k\bar{q}} h^{p\bar{j}} A_{ik\bar{j}} \overline{A_{\ell q\bar{p}}}.$$

Set $A_{ik\bar{j}} = \nabla_i h_{k\bar{j}}$ and $D_{ik\bar{j}} = T^p_{ik} h_{p\bar{j}}$. We first claim

$$\Delta_h S = \|A + D\|^2 + \Delta_g G + E, \qquad |E| \leq C_g(1 + S)B. \tag{10.8}$$

To prove this, start with $\Delta_h S = h^{i\bar{j}} g^{k\bar{\ell}} \nabla_i \nabla_{\bar{j}} h_{k\bar{\ell}}$. Switch $\bar{j}, \bar{\ell}$ by the conjugate of (10.7), commute $\nabla_i, \nabla_{\bar{\ell}}$, and then switch i, k by (10.7). The terms with a derivative of h are

$$h^{i\bar{j}} g^{k\bar{\ell}} \nabla_{\bar{\ell}} \nabla_k h_{i\bar{j}} - 2\,\mathrm{Re}\big(h^{i\bar{j}} g^{k\bar{\ell}} T^p_{ik} \nabla_{\bar{\ell}} h_{p\bar{j}}\big).$$

The first switch also produces the term $-h^{i\bar{j}} g^{k\bar{\ell}} \overline{T^q_{j\ell}} \nabla_i h_{k\bar{q}}$. Applying (10.7) to it gives one of the two conjugate derivative terms displayed above, together with

$$h^{i\bar{j}} g^{k\bar{\ell}} T^p_{ik} \overline{T^q_{j\ell}} (h_{p\bar{q}} - g_{p\bar{q}}).$$

The part of this expression containing $h_{p\bar{q}}$ is $\|D\|^2$. Differentiating the logarithmic determinant twice gives

$$\Delta_g G = h^{i\bar{j}} g^{k\bar{\ell}} \nabla_{\bar{\ell}} \nabla_k h_{i\bar{j}} - \|A\|^2.$$

By the antisymmetry of T, the two remaining derivative terms combine to $2\,\mathrm{Re}\langle A, D\rangle$, where $\langle \cdot, \cdot \rangle$ is the inner product of the norm above. These identities give (10.8).

To bound the remainder E explicitly, let

$$C_{i\bar{\ell}k\bar{j}} = (\nabla_i \nabla_{\bar{\ell}} - \nabla_{\bar{\ell}} \nabla_i) h_{k\bar{j}}.$$

The remainder is the contraction by $h^{i\bar{j}} g^{k\bar{\ell}}$ of

$$\begin{aligned}(\nabla_i \overline{T^q_{j\ell}})(g_{k\bar{q}} - h_{k\bar{q}}) + (\nabla_{\bar{\ell}} T^p_{ik})(g_{p\bar{j}} - h_{p\bar{j}}) \\ + C_{i\bar{\ell}k\bar{j}} - T^p_{ik} \overline{T^q_{j\ell}} g_{p\bar{q}}.\end{aligned}$$

The commutator C is the curvature of ∇ acting linearly on h, as one sees by inserting $\nabla = \partial + \Gamma$ and canceling the derivatives of h. In a g-unitary frame diagonalizing h, every coefficient in this remainder is bounded by $C_g(1 + S)$, so the contraction is bounded by $C_g(1 + S) \sum_i h^{i\bar{i}} = C_g(1 + S)B$, as asserted. In particular, no derivative of h remains in E.

It remains to control the gradient term introduced by the logarithm. Define

$$H_{ik\bar{j}} = h_{i\bar{j}} \frac{S_k}{S}, \qquad K = \|A + D - H\|^2 \geq 0.$$

By (10.7),

$$g^{i\bar{\ell}} (A + D)_{ik\bar{\ell}} = S_k + \tau_k.$$

Moreover $\|H\|^2 = |\partial S|^2_h / S$. Expanding the square therefore gives

$$K = \|A + D\|^2 - \frac{|\partial S|^2_h}{S} - \frac{2}{S}\,\mathrm{Re}(h^{k\bar{q}} \tau_k S_{\bar{q}}).$$

Combining this with (10.8) and with $\Delta_h \log S = \Delta_h S / S - |\partial S|^2_h / S^2$, we obtain the exact identity

$$\Delta_h \log S = \frac{K}{S} + \frac{2}{S^2}\operatorname{Re}(h^{k\bar{q}}\tau_k S_{\bar{q}}) + \frac{\Delta_g G}{S} + \frac{E}{S}.$$

Since $K \geq 0$ and $S \geq 1$, and since $E/S \geq -2C_g B$, enlarging the constant proves the lemma. □

Theorem 10.4.1 *Fix $D, K > 0$. Suppose that u and G are smooth real functions on X such that $\omega_u > 0$ and*

$$\operatorname{osc}_X u \leq D, \qquad \omega_u^n = \mathrm{e}^G \omega^n, \qquad |G| \leq K, \qquad \Delta_g G \geq -K.$$

There is a constant $C = C(\omega, D, K)$ such that

$$C^{-1}\omega \leq \omega_u \leq C\omega.$$

Proof Use $h = g + (u_{i\bar{j}})$, $S = \operatorname{tr}_g h$, and $B = \operatorname{tr}_h g$ as in Lemma 10.4.1. Write $s = u - \inf_X u + 1$, so that $1 \leq s \leq D + 1$, and consider the maximum of

$$Q = \log S - Au + s^{-1}$$

on X, where $A \geq 1$ will be fixed below. Let x be a maximum point. If $S(x) < 1$, an upper bound for S on X already follows from the oscillation bound and the definition of Q, as at the end of the proof. We may therefore assume $S(x) \geq 1$ and use Lemma 10.4.1.

Since the eigenvalues of h relative to g are positive, $SB \geq n^2$, and hence $\Delta_g G/S \geq -KB/n^2$. Moreover, $\partial Q = 0$ at x gives

$$\frac{S_k}{S} = (A + s^{-2})u_k.$$

Because $|\tau|_h^2 \leq C_g B$, the absolute value of the torsion term is bounded by

$$\begin{aligned}\left|\frac{2}{S^2}\operatorname{Re}(h^{k\bar{q}}\tau_k S_{\bar{q}})\right| &\leq \frac{C_g(A+1)}{S}\sqrt{B}\,|\partial u|_h \\ &\leq \frac{|\partial u|_h^2}{s^3} + \frac{C_g(A+1)^2 s^3}{S^2}B.\end{aligned} \tag{10.9}$$

If $S(x)^2 \leq (A+1)^2(D+1)^3$, then $S(x)$ is bounded once A is fixed, and again so is S on X. In the remaining case the coefficient of B in the last term is at most C_g.

On the other hand, direct computation gives

$$\Delta_h Q = \Delta_h \log S + (A + s^{-2})(B - n) + 2s^{-3}|\partial u|_h^2.$$

The trace inequality and (10.9) imply at x

$$0 \geq \Delta_h Q \geq (A - C_g - K/n^2)B - n(A+1),$$

after a fixed enlargement of C_g. Choosing $A > C_g + K/n^2 + 1$, we conclude that $B(x)$ is bounded. If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of h relative to g, then $\prod_i \lambda_i = \mathrm{e}^G$, and for $n \geq 2$

$$S = \sum_i \lambda_i = \mathrm{e}^G \sum_i \prod_{j \neq i} \lambda_j^{-1} \leq n \mathrm{e}^K B^{n-1}.$$

For $n = 1$, directly $S = \mathrm{e}^G$. Thus $S(x)$ is bounded in all cases. For every $y \in X$, the maximality of Q at x gives

$$\log S(y) \leq \log S(x) + A(u(y) - u(x)) + s(x)^{-1} - s(y)^{-1} \leq \log S(x) + AD + 1.$$

This proves $S \leq C$ on X. The lower bound for the determinant then gives $\lambda_i \geq \mathrm{e}^{-K} C^{-(n-1)}$, which proves the uniform equivalence. □

For the Monge–Ampère equation, the trace estimate of Theorem 10.4.1 does not involve the gradient. For other equations the maximum principle may bound the trace only in terms of the gradient. Since traces such as $\mathrm{tr}_g\, h = n + \Delta_g u$ are positive, an upper bound for the trace bounds $|\Delta_g u|$; if it depends at most linearly on the gradient, the following lemma closes the estimate: elliptic regularity for the fixed operator Δ_g bounds a Hölder norm of the gradient linearly in its supremum, and interpolation with the oscillation bounds the supremum. A quadratic dependence is invariant under the rescalings $u \mapsto u(x + z/R)$ in local coordinates, and the argument does not apply to it. The second-order estimates for the Gauduchon ansatz and for complex Hessian equations, Theorem 10.14.1 and Proposition 13.3.2, are quadratic in the gradient; there the gradient is bounded instead by rescaling at points of maximal gradient and a Liouville theorem for the limit (Propositions 10.16.1 and 13.2.1).

Lemma 10.4.2 *Let $A, M \geq 0$. There is $C = C(g, A, M)$ such that every smooth real function v on X satisfying*

$$\operatorname{osc}_X v \leq M, \qquad \|\Delta_g v\|_{L^\infty} \leq A\big(1 + \|\mathrm{d}v\|_{L^\infty(g)}\big)$$

also satisfies

$$\|\mathrm{d}v\|_{L^\infty(g)} + \|\Delta_g v\|_{L^\infty} \leq C.$$

Here all volume and function-space norms use the fixed smooth metric g.

Proof Subtract $\inf_X v$, so that $0 \leq v \leq M$, and put $K = \|\mathrm{d}v\|_{L^\infty(g)}$. Fix $p > 2n$ and $0 < \alpha < 1 - 2n/p$. The fixed operator Δ_g is uniformly elliptic on the underlying real manifold. The local $W^{2,p}$ estimates, patched by a finite atlas, and Morrey's inequality give

$$\|v\|_{C^{1,\alpha}} \leq C_g \|v\|_{W^{2,p}} \leq C_g\big(\|v\|_{L^p} + \|\Delta_g v\|_{L^p}\big) \leq C_g\big(M + A(1 + K)\big).$$

These are the standard linear elliptic L^p estimates of [GT01, Chapter 9] and the Sobolev embedding of [GT01, Chapter 7]; the constants depend on the fixed metric and on the chosen p, α.

For $0 < t \leq t_0$, with t_0 fixed by the atlas, integration along a coordinate segment of length t gives

$$K \le C_g t^{-1} M + C_g t^{\alpha} \|v\|_{C^{1,\alpha}}.$$

Indeed, the difference quotient is bounded by M/t, since $0 \le v \le M$, and its difference from the derivative at the initial point is bounded by $C_g t^{\alpha} \|v\|_{C^{1,\alpha}}$. A finite atlas with smaller charts controls every point and coordinate direction; equivalence with the g-norm gives the displayed estimate. Consequently,

$$K \le C_g t^{-1} M + C_g^2 t^{\alpha} (M + A) + C_g^2 A t^{\alpha} K.$$

Choose $t \le t_0$ so small that $C_g^2 A t^{\alpha} \le 1/2$. This choice depends only on g, A, including when $A = 0$. Absorbing the last term bounds K in terms of g, A, M, and the assumed trace estimate then bounds $\|\Delta_g v\|_{L^\infty}$. □

Corollary 10.4.1 *Fix ω. For smooth solutions of*

$$(\omega + \mathrm{dd^c} u)^n = \mathrm{e}^G \omega^n, \qquad \omega + \mathrm{dd^c} u > 0,$$

a uniform C^0 bound for u and uniform bounds for the smooth real logarithmic density G in every C^k norm imply uniform bounds for u in every C^k norm.

Proof Theorem 10.4.1 gives uniform ellipticity and a bound for $\Delta_g u = S - n$. The linear elliptic $W^{2,p}$ estimates for the fixed operator Δ_g first bound u in $C^{1,\alpha}$ for every $0 < \alpha < 1$. The local Evans–Krylov estimate for the concave logarithmic determinant then gives a uniform $C^{2,\beta}$ bound for some $0 < \beta < 1$. To apply it in its real form, project real symmetric Hessians to their complex Hermitian part and extend the logarithmic determinant concavely from the compact set of positive matrices given by the preceding theorem. The extension is uniformly elliptic on real symmetric matrices because the projection preserves positivity and the trace, so the usual Evans–Krylov theorem applies. Finally, differentiating the equation gives linear uniformly elliptic equations with Hölder coefficients, and Schauder estimates and induction give all higher bounds. No regularity results beyond these standard local elliptic estimates are needed. □

The same conclusion holds for equations $\omega_u^n = \mathrm{e}^{au+F} \omega^n$ with a in a fixed bounded interval of nonnegative numbers, u uniformly bounded, and F uniformly bounded in every C^k norm. Indeed, with $G = au + F$,

$$\Delta_g (au + F) = a(S - n) + \Delta_g F \ge -an + \Delta_g F,$$

so Theorem 10.4.1 applies without any prior bound for derivatives of u. The estimate for the fixed operator Δ_g then gives a $C^{1,\alpha}$ bound, after which the same local regularity and bootstrapping apply.

10.5 The continuity method

We join the background density to the prescribed one by a path of equations. The local inverse of Proposition 10.3.1 shows that the set of solvable parameters is open. The extremum bound for the scalar, the potential estimate and the derivative estimates give compactness along the path, and uniform ellipticity preserves positivity at a limiting parameter.

Proof (of Theorem 10.1.1) The case $n = 1$ was treated after the statement of the theorem. For $n \geq 2$ put $G = F + \log(\mathrm{d}V/\omega^n)$ and consider

$$(\omega + \mathrm{dd}^{\mathrm{c}} u_t)^n = \mathrm{e}^{tG+b_t}\omega^n, \qquad \sup_X u_t = 0, \qquad 0 \leq t \leq 1.$$

At $t = 0$, take $u_0 = 0$ and $b_0 = 0$. The local implicit-function argument in Proposition 10.3.1, with the additive constant temporarily fixed by a linear mean, shows that the set of solvable parameters is open.

At every solvable parameter, Proposition 10.1.1 gives

$$-\sup_X(tG) \leq b_t \leq -\inf_X(tG), \qquad \mathrm{e}^{tG+b_t} \leq \mathrm{e}^{\operatorname{osc}_X G}.$$

The constants b_t are therefore uniformly bounded, and Theorem 10.2.1 bounds u_t in C^0. By Corollary 10.4.1, the solutions are bounded in every C^k norm. Along any sequence of solvable parameters converging in $[0, 1]$, a subsequence of (u_t, b_t) therefore converges smoothly to a solution; uniform equivalence of the metrics preserves strict positivity, and the supremum normalization passes to the limit. Thus the solvable set is closed. Since it is nonempty, open and closed, connectedness of $[0, 1]$ gives a solution at $t = 1$, and Proposition 10.1.2 gives uniqueness. □

10.6 The smooth exponential equation

When the logarithmic density contains a positive multiple of the potential, the extrema of the potential are bounded directly, and the linearization acquires a negative zeroth-order term. The continuity method then needs neither an additive normalization nor an extra scalar unknown.

Proposition 10.6.1 *Let $\lambda > 0$, let F be smooth, and let v, w be smooth strictly ω-plurisubharmonic functions. If*

$$\omega_v^n \geq \mathrm{e}^{\lambda v+F}\omega^n, \qquad \omega_w^n \leq \mathrm{e}^{\lambda w+F}\omega^n,$$

then $v \leq w$. In particular, two smooth solutions coincide.

Proof Suppose that $v - w$ has a positive maximum at a point x. Then $\omega_v(x) \leq \omega_w(x)$, and taking determinants and applying the two assumptions gives $\mathrm{e}^{\lambda v(x)+F(x)} \leq$

$e^{\lambda w(x)+F(x)}$, which contradicts $v(x) > w(x)$. For two solutions, apply the comparison in both orders. □

Theorem 10.6.1 *For every $\lambda > 0$ and every smooth real F, there is a unique smooth u with $\omega_u > 0$ such that*

$$\omega_u^n = e^{\lambda u + F}\omega^n.$$

The solution satisfies

$$-\lambda^{-1}\sup_X F \le u \le -\lambda^{-1}\inf_X F.$$

Proof At a minimum of u, the equation gives $\lambda u + F \ge 0$; at a maximum it gives $\lambda u + F \le 0$. These inequalities prove the stated bounds.

Consider the equations

$$\omega_{u_t}^n = e^{\lambda u_t + tF}\omega^n, \qquad 0 \le t \le 1.$$

At $t = 0$, $u_0 = 0$ is a solution. The linearization in u is $L_u - \lambda$, whose kernel is zero by the maximum principle. Its Fredholm index is zero: deform its principal coefficients to those of a fixed Laplacian while keeping the negative zeroth-order term; the latter operator is invertible, and the index is constant along this deformation of elliptic operators. Thus $L_u - \lambda$ is an isomorphism between the usual Hölder spaces, with bounded inverse by Schauder theory, and the implicit function theorem proves that the set of solvable parameters is open.

The extremum bounds give $\|u_t\|_\infty \le \lambda^{-1}\|F\|_\infty$. The semilinear version of Theorem 10.4.1 and Corollary 10.4.1, stated in the paragraph following that corollary, then gives uniform ellipticity and bounds in every C^k norm. Along any convergent sequence of solvable parameters, a subsequence of the solutions therefore converges smoothly to a strictly positive solution at the limit parameter. This proves closedness, and hence existence at $t = 1$. Uniqueness is Proposition 10.6.1. □

For a fixed smooth real obstacle f on X and a parameter $\beta > 0$, the choice $\lambda = \beta$ and $F = -\beta f$ gives

$$\omega_{u_\beta}^n = e^{\beta(u_\beta - f)}\omega^n.$$

These equations penalize values above the obstacle. In Proposition 14.1.1 we use them to construct the envelope of f and to show that its Monge–Ampère measure is supported on the contact set.

10.7 Variation of the normalization scalar

Integration against the adjoint probability measure eliminates the derivatives of the potential from the first variation of b. In the second variation, the second derivative of the logarithmic determinant contributes a nonnegative square with a minus sign. This gives concavity of b and identifies the equality case.

Write $b(F)$ for the scalar in (10.1), and use the linear normalization $\int_X u\,dV = 0$. For a smooth family F_t, with t in an open interval $I \subset \mathbb{R}$, write (u_t, b_t) for the

corresponding solutions and put $\eta_t = \omega_{u_t}$. By Proposition 10.3.1 these depend smoothly on t; a dot denotes differentiation in t.

Theorem 10.7.1 *Let $A_t = \eta_t^{-1}\mathrm{dd^c}\dot{u}_t$ be the endomorphism of $T^{1,0}X$ that represents the real $(1,1)$-form $\mathrm{dd^c}\dot{u}_t$ with respect to η_t; it is Hermitian with respect to η_t. The normalization constant satisfies*

$$\dot{b}_t = -\int_X \dot{F}_t \,\mathrm{d}\mu_{u_t}, \tag{10.10}$$

$$\ddot{b}_t = -\int_X \ddot{F}_t \,\mathrm{d}\mu_{u_t} - \int_X \mathrm{tr}(A_t^2)\,\mathrm{d}\mu_{u_t}. \tag{10.11}$$

In particular, for smooth real functions F_0, G on X and the affine family $F_t = F_0 + tG$, the function b_t is concave. Its second derivative vanishes at a parameter value if and only if G is constant on X.

Proof Differentiating the logarithm of (10.1) gives

$$L_{u_t}\dot{u}_t = \dot{F}_t + \dot{b}_t.$$

Integration against μ_{u_t} proves (10.10).

For a smooth family of positive Hermitian matrices H_t, differentiation of the inverse matrix gives

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\log\det H_t = \mathrm{tr}(H_t^{-1}\ddot{H}_t) - \mathrm{tr}\big((H_t^{-1}\dot{H}_t)^2\big).$$

Apply this pointwise to the matrices of η_t, using $\dot{\eta}_t = \mathrm{dd^c}\dot{u}_t$ and $\ddot{\eta}_t = \mathrm{dd^c}\ddot{u}_t$. The second derivative of the logarithmic equation is

$$L_{u_t}\ddot{u}_t - \mathrm{tr}(A_t^2) = \ddot{F}_t + \ddot{b}_t.$$

Integrating against μ_{u_t} at this fixed parameter proves (10.11); there is no need to differentiate the measure.

The eigenvalues of the Hermitian endomorphism A_t are real, so $\mathrm{tr}(A_t^2) \geq 0$. For affine F_t we have $\ddot{F}_t = 0$, so the second derivative of b_t is nonpositive. If it vanishes, strict positivity of the smooth density of μ_{u_t} implies $A_t = 0$ everywhere. Thus $\mathrm{dd^c}\dot{u}_t = 0$, and the maximum principle shows that $\dot{u}_t$ is constant. The first differentiated equation then gives $G = -\dot{b}_t$, a constant. Conversely, if G is constant, the solution metric is independent of t and $b_t = b_0 - tG$. □

Corollary 10.7.1 *For two smooth functions F, G, the normalization constants satisfy*

$$-\sup_X(G-F) \le b(G)-b(F) \le -\inf_X(G-F).$$

In particular, $|b(G)-b(F)| \le \|G-F\|_\infty$. If $G-F$ is nonconstant, both inequalities in the first display are strict.

Proof Apply (10.10) along $F_t = F + t(G-F)$ and integrate over $0 \le t \le 1$. Each μ_{u_t} is a probability measure with strictly positive smooth density, so the integral of a nonconstant continuous function against it lies strictly between the minimum and the maximum of that function. □

Thus b is a decreasing, concave, Lipschitz functional of the logarithmic density, with $b(F+a) = b(F) - a$ for every real constant a. By uniform approximation, it extends uniquely to a Lipschitz functional of continuous F. This extension concerns the scalar functional only; the existence and regularity of corresponding potentials require further estimates.

10.8 Gauduchon quotients and the normalization constant

When the background has a strictly positive representative, the normalization constant can be computed from Gauduchon metrics alone. Integrating the arithmetic–geometric mean inequality against the $(n-1)$st power of a Gauduchon metric γ bounds the nth root of the constant from above by a quotient whose numerator $\int_X \theta \wedge \gamma^{n-1}$ is linear in the background and unchanged by smooth dd^{c} changes of it, and the Gauduchon metric conformal to the solution form gives equality. The infimum of these quotients makes sense for every pseudoeffective background and defines the Gauduchon variational volume, which Chapter 27 studies as a function of the background. Varying the probability measure realizes the mass of every smooth strictly positive potential, so the extrema of the volume over all measures are the lower and upper Monge–Ampère masses.

In this section θ is a smooth real $(1,1)$-form on X, which we call the *background*, and μ is a smooth strictly positive probability form on X. The background θ is called *pseudoeffective* if $\mathrm{PSH}(X,\theta) \neq \varnothing$, that is, if $\theta + \mathrm{dd}^{\mathrm{c}} u \ge 0$ for some quasi-plurisubharmonic u; by Theorem 4.3.1 this holds exactly when $\int_X \theta \wedge \gamma^{n-1} \ge 0$ for every Gauduchon metric γ. Bigness is defined in Section 1.19. Neither notion requires θ to be closed, and both are unchanged by a smooth dd^{c} change of θ.

For a Gauduchon metric γ, Stokes' theorem shows that the pairing $\int_X \theta \wedge \gamma^{n-1}$ is unchanged by a smooth dd^{c} change of θ; this pairing will be the numerator of our quotient. For the denominator, put

$$D_\mu(\gamma) = \int_X \left(\frac{\mu}{\gamma^n}\right)^{1/n} \gamma^n.$$

This number is positive. Like the numerator, it is multiplied by r^{n-1} when γ is multiplied by a positive constant r, so the quotient below depends only on the ray of γ.

Definition 10.8.1 For a pseudoeffective smooth real $(1,1)$-form θ on X, define

$$\lambda_\mu(\theta) = \inf_{\gamma \text{ Gauduchon}} \frac{\displaystyle\int_X \theta \wedge \gamma^{n-1}}{D_\mu(\gamma)}, \qquad \mathcal{V}_\mu(\theta) = \lambda_\mu(\theta)^n.$$

We call $\mathcal{V}_\mu(\theta)$ the *Gauduchon variational volume* of θ with respect to μ, and $\lambda_\mu(\theta)$ its *root volume*. In dimension one every Hermitian metric is Gauduchon, and every quotient equals $\int_X \theta$.

Proposition 10.8.1 *For a pseudoeffective θ, the number $\lambda_\mu(\theta)$ is finite and nonnegative. It is unchanged by a smooth dd^{c} change of θ, positively homogeneous of degree one, and concave on the cone of pseudoeffective forms. If β is a smooth real $(1,1)$-form and $\beta - \theta$ is pseudoeffective, then*

$$\lambda_\mu(\beta) \geq \lambda_\mu(\theta).$$

Finally, λ_μ is upper semicontinuous under smooth convergence within this cone.

Proof Choose $u \in \mathrm{PSH}(X,\theta)$. For every Gauduchon metric γ, Stokes' theorem for currents and the Gauduchon condition give

$$\int_X \theta \wedge \gamma^{n-1} = \int_X (\theta + \mathrm{dd}^{\mathrm{c}} u) \wedge \gamma^{n-1} \geq 0.$$

Thus every quotient is nonnegative, and testing one fixed Gauduchon metric gives a finite upper bound. The same identity shows that each quotient is unchanged by a smooth dd^{c} change of the background, and, applied to $\beta - \theta$, it proves the monotonicity.

Each quotient is a linear function of the background, so their infimum is concave and positively homogeneous. Each individual quotient is also continuous in the smooth topology. Given a smoothly converging sequence, we bound the infimum by any one quotient, take the upper limit, and finally take the infimum over the test metric. This proves upper semicontinuity. □

The smooth equation identifies both the value of the quotient and every minimizing metric. Let $\mathscr{P}$ be the set of backgrounds θ admitting a smooth real function u with $\theta + \mathrm{dd}^{\mathrm{c}} u > 0$. Choosing such a Hermitian representative of θ and writing μ as a smooth positive density relative to the volume form of this representative, we may apply Theorem 10.1.1. It gives, for each $\theta \in \mathscr{P}$, a unique positive form Ω_θ, which we call the *solution form*, and a unique constant $c_\theta(\mu) > 0$ such that

$$\Omega_\theta = \theta + \mathrm{dd}^{\mathrm{c}} u_\theta, \qquad \Omega_\theta^n = c_\theta(\mu)\mu.$$

The potential u_θ is unique after any fixed additive normalization. In the notation of (10.1), with the chosen representative of θ as ω and $\mathrm{e}^F \mathrm{d}V = \mu$, the constant $c_\theta(\mu)$ is e^b.

Theorem 10.8.1 *For every $\theta \in \mathscr{P}$,*

$$\lambda_\mu(\theta) = c_\theta(\mu)^{1/n}, \qquad \mathcal{V}_\mu(\theta) = c_\theta(\mu).$$

The infimum in Definition 10.8.1 is attained. When $n \geq 2$, its minimizing Gauduchon metrics are precisely the constant multiples of the Gauduchon metric conformal to Ω_θ.

Proof Suppose first that $n \geq 2$. By Proposition 4.2.2, applied with $\mathrm{d}V = \mu$ and $f = 1$, every test quotient is at least $c_\theta(\mu)^{1/n}$; this bound integrates the pointwise inequality (4.2) for $\eta = \Omega_\theta$, whose right side is $c_\theta(\mu)^{1/n}(\mu/\gamma^n)^{1/n}\gamma^n$. Take the Gauduchon metric $\gamma = \mathrm{e}^g\Omega_\theta$ conformal to Ω_θ, given by Theorem 4.1.1, where g is a smooth real function. Since all eigenvalues of Ω_θ relative to γ are equal at every point, equality holds pointwise and hence in the quotient. This proves the formula and attainment. Conversely, if a Gauduchon metric attains the infimum, the nonnegative smooth difference of the two sides of the pointwise inequality has zero integral. It therefore vanishes everywhere, and the equality case of the arithmetic–geometric mean inequality makes γ conformal to Ω_θ. Since the Gauduchon metric in a conformal class is unique up to a constant factor, this proves the last assertion.

For $n = 1$, we integrate the equation and use $\int_X \mathrm{d}\mathrm{d}^{\mathrm{c}} u_\theta = 0$. This gives $c_\theta(\mu) = \int_X \theta$, which equals every quotient in the definition. □

For a Hermitian background ω, a smooth positive probability volume $\mathrm{d}V$ and a nonzero nonnegative density $f \in L^1(\mathrm{d}V)$, the infimum of the analogous quotient, with μ replaced by $f\,\mathrm{d}V$, is the nth root of the subsolution threshold $b_\omega(f)$ of Section 15.27; see Theorem 15.28.1. Attainment in this extremal problem, and the regularity of the extremal tests, are studied from Section 27.5 on.

Corollary 10.8.1 *If θ is big, then $\lambda_\mu(\theta) > 0$.*

Proof By the definition of bigness, there are a Hermitian form h and $\delta > 0$ such that $\theta - \delta h$ is pseudoeffective. Monotonicity, homogeneity and the theorem, applied to $h \in \mathscr{P}$, give $\lambda_\mu(\theta) \geq \delta\lambda_\mu(h) > 0$. □

The converse fails: Theorem 27.13.1 gives, on a projective manifold, a smooth semipositive nonclosed background of positive volume that is not big.

10.8.1 Varying the probability measure

Changing the smooth positive probability measure μ multiplies each denominator $D_\mu(\gamma)$ by a factor bounded above and below independently of γ. The value of the volume may therefore change, but its vanishing and positivity do not. Moreover, every smooth potential whose form is strictly positive solves the normalized equation for a suitable measure. Consequently, on $\mathscr{P}$ the volume is independent of the measure exactly when the total Monge–Ampère mass is independent of the potential.

Proposition 10.8.2 *For smooth positive probability measures μ, ν, choose constants $m, M > 0$ such that $m\mu \le \nu \le M\mu$. Then for every pseudoeffective θ,*

$$M^{-1/n}\lambda_\mu(\theta) \le \lambda_\nu(\theta) \le m^{-1/n}\lambda_\mu(\theta).$$

In particular, whether $\lambda_\mu(\theta)$ vanishes does not depend on the smooth positive probability measure μ.

Proof Since $D_\nu(\gamma) = \int_X (\nu/\mu)^{1/n}(\mu/\gamma^n)^{1/n}\gamma^n$, the inequalities $m\mu \le \nu \le M\mu$ give

$$m^{1/n}D_\mu(\gamma) \le D_\nu(\gamma) \le M^{1/n}D_\mu(\gamma)$$

for every admissible metric γ. The numerators of the quotients are nonnegative and do not involve the measure. Dividing and taking infima proves the assertion. □

Proposition 10.8.3 *Let $\theta \in \mathscr{P}$. The value $\mathcal{V}_\mu(\theta)$ is the same for all smooth positive probability measures μ if and only if*

$$\int_X (\theta + \mathrm{dd^c}v)^n$$

is independent of the smooth potential v with $\theta + \mathrm{dd^c}v > 0$.

Proof For a smooth potential v with $\theta + \mathrm{dd^c}v > 0$, put

$$M_v = \int_X (\theta + \mathrm{dd^c}v)^n, \qquad \mu_v = \frac{(\theta + \mathrm{dd^c}v)^n}{M_v}.$$

Then μ_v is a smooth positive probability measure, and v solves the normalized equation for μ_v with scalar M_v. Uniqueness of the scalar and Theorem 10.8.1 give $\mathcal{V}_{\mu_v}(\theta) = M_v$. Hence, if the volume does not depend on the measure, neither does the mass. Conversely, suppose that all these masses equal one constant M. For an arbitrary smooth positive probability measure μ, integrating the normalized equation gives $c_\theta(\mu) = M$, so $\mathcal{V}_\mu(\theta) = M$ for every μ. □

For instance, the nonclosed torus metric of Example 2.3.1 has volumes that depend on the measure. If $n = 2$ and ω is a Gauduchon metric, Stokes' theorem shows that the smooth masses are all equal to $\int_X \omega^2$, so $\mathcal{V}_\mu(\omega) = \int_X \omega^2$ for every μ. For the standard metric ω on the Hopf threefold of Chapter 9, each $\mathcal{V}_\mu(\omega)$ is finite and positive, although the upper mass $v_+(\omega)$ is infinite.

10.8.2 Upper and lower masses

As the probability measure varies, the smooth equation realizes the mass of every smooth strictly positive potential, by the proof of Proposition 10.8.3. The extrema of the volume over all measures are therefore the extrema of these masses, and regularization of bounded potentials identifies them with the lower and upper masses $v_-(\omega)$ and $v_+(\omega)$ of Definition 7.5.1. The infimum simplifies further: at each Gauduchon metric an explicit measure maximizes the denominator, and this removes the density from the formula entirely.

Throughout this subsection ω is a Hermitian form on X, and $\mathscr{M}$ denotes the set of smooth strictly positive probability measures on X.

Theorem 10.8.2 *One has*

$$\inf_{\mu\in\mathscr{M}} \mathcal{V}_\mu(\omega) = v_-(\omega), \qquad \sup_{\mu\in\mathscr{M}} \mathcal{V}_\mu(\omega) = v_+(\omega),$$

where the supremum may be infinite. Moreover,

$$v_-(\omega)^{1/n} = \inf_{\gamma \text{ Gauduchon}} \frac{\displaystyle\int_X \omega\wedge\gamma^{n-1}}{\displaystyle\left(\int_X \gamma^n\right)^{(n-1)/n}}. \tag{10.12}$$

Proof For each $\mu \in \mathscr{M}$, the smooth normalized equation realizes $\mathcal{V}_\mu(\omega)$ as the mass $\int_X(\omega + \mathrm{dd^c}v)^n$ of a smooth potential v with $\omega + \mathrm{dd^c}v > 0$. Conversely, every such potential realizes its mass as $\mathcal{V}_{\mu_v}(\omega)$ for the measure μ_v in the proof of Proposition 10.8.3. Thus the set of volumes $\mathcal{V}_\mu(\omega)$, $\mu \in \mathscr{M}$, coincides with the set of these smooth masses.

It remains to show that the smooth masses have the same extrema as the masses of bounded potentials, which define $v_\pm(\omega)$. If $u \in \mathrm{PSH}(X,\omega)\cap L^\infty$, Theorem 1.11.1 gives uniformly bounded smooth strictly ω-plurisubharmonic functions u_j decreasing to u. Their Monge–Ampère measures converge by the monotone continuity of bounded products, and so do their total masses, since X is compact. Every bounded mass is therefore a limit of smooth strictly positive masses. Conversely, smooth potentials are bounded. This proves the first two identities.

For the last formula, take nth roots and interchange the infima over μ and over γ in Theorem 10.8.1. For a fixed Gauduchon metric γ, Hölder's inequality and $\int_X \mu = 1$ give

$$D_\mu(\gamma) \le \left(\int_X \gamma^n\right)^{(n-1)/n},$$

with equality for the smooth probability measure $\mu = \gamma^n/\int_X\gamma^n$. The numerator is positive and does not depend on μ, so the infimum of the quotient over μ is obtained by maximizing the denominator. This gives (10.12). For $n = 1$ both denominators equal one, and the same argument applies. □

Thus $v_-(\omega) > 0$ if and only if $\int_X \omega \wedge \gamma^{n-1}$ has a positive lower bound over all Gauduchon metrics γ with $\int_X \gamma^n = 1$. This lower bound fails on the threefold constructed in Theorem 27.18.1. Corollary 33.1.1 in Chapter 33 shows that positivity of $v_-(\omega)$ is also equivalent to solvability, in the background ω, of the normalized equation for every nonzero finite non-pluripolar measure. In the same way, each individual volume $\mathcal{V}_\mu(\omega)$ is finite, while their supremum $v_+(\omega)$ can be infinite.

10.9 Normalization on the Hopf threefold

On the Hopf threefold of Chapter 9, the deviation of b from the closed-background formula (10.2) can be computed: it is the energy of the solution along the elliptic fibers. In this section, X is that threefold and ω its Gauduchon metric; we use the projection $\pi\colon X \to \mathbb{P}^2$, the probability measure $\nu = \omega^3/6$ and the averaging operator A of Proposition 9.7.1, and we write $\beta = \omega - \pi^*\omega_{\mathrm{FS}}$ for the rank-one form of that construction. As in Chapter 9, the fiber derivative u_w does not depend on the local trivialization of the elliptic bundle.

For $F \in C^\infty(X,\mathbb{R})$, let $(u_F, b(F))$ be the solution of

$$(\omega + \mathrm{dd^c} u_F)^3 = \mathrm{e}^{F+b(F)}\omega^3, \qquad \omega + \mathrm{dd^c} u_F > 0, \qquad \int_X u_F \,\mathrm{d}\nu = 0.$$

Define

$$D(F) = b(F) + \log \int_X \mathrm{e}^F \,\mathrm{d}\nu.$$

For a closed background, the corresponding quantity vanishes by (10.2).

Proposition 10.9.1 *For every smooth real F on this threefold,*

$$\mathrm{e}^{D(F)} = 1 + \int_X |(u_F)_w|^2 \,\mathrm{d}\nu, \tag{10.13}$$

$$\|u_F - Au_F\|^2_{L^2(X,\nu)} \le 4\big(\mathrm{e}^{D(F)} - 1\big). \tag{10.14}$$

Thus $D(F) \ge 0$, with equality if and only if $F = \pi^*H$ for some $H \in C^\infty(\mathbb{P}^2,\mathbb{R})$.

Proof Integration of the equation and (9.2) give

$$\mathrm{e}^{D(F)} = \frac{1}{6}\int_X (\omega + \mathrm{dd^c} u_F)^3 = 1 + \int_X |(u_F)_w|^2 \,\mathrm{d}\nu.$$

Substituting the first equality into (9.21) proves (10.14).

If $D(F) = 0$, then $(u_F)_w = 0$ everywhere, because u_F is smooth. Since u_F is real, it is constant on each connected fiber, so it descends to a smooth real function q on $\mathbb{P}^2$, and $\omega_{\mathrm{FS}} + \mathrm{dd^c} q > 0$ as in the proof of Proposition 9.7.1. Since $\beta^2 = 0$ and $\omega = \pi^*\omega_{\mathrm{FS}} + \beta$, we have

$$(\omega + \mathrm{dd}^{\mathrm{c}}\pi^{*}q)^{3} = 3\pi^{*}(\omega_{\mathrm{FS}} + \mathrm{dd}^{\mathrm{c}}q)^{2} \wedge \beta.$$

Dividing by $\omega^{3} = 3\pi^{*}\omega_{\mathrm{FS}}^{2} \wedge \beta$ and taking logarithms proves that F is a smooth base pullback.

Conversely, suppose $F = \pi^{*}H$. Apply Theorem 10.1.1 on $(\mathbb{P}^{2}, \omega_{\mathrm{FS}})$ to obtain a smooth real q and a scalar c with

$$\omega_{\mathrm{FS}} + \mathrm{dd}^{\mathrm{c}}q > 0, \qquad (\omega_{\mathrm{FS}} + \mathrm{dd}^{\mathrm{c}}q)^{2} = \mathrm{e}^{H+c}\omega_{\mathrm{FS}}^{2}.$$

Since the base form is closed, Stokes' theorem gives $\mathrm{e}^{c}\int_{\mathbb{P}^{2}} \mathrm{e}^{H}\omega_{\mathrm{FS}}^{2} = 1$. The preceding cube identity shows that $\pi^{*}q$, after subtraction of its ν-mean, solves the equation on X with scalar c, and uniqueness gives $b(F) = c$. Since the base measure of ν is ω_{FS}^{2}, it follows that $D(F) = 0$. □

In particular, if a smooth positive density f satisfies $\int_{X} f\,\mathrm{d}\nu = 1$, the equation $(\omega + \mathrm{dd}^{\mathrm{c}}u)^{3} = \mathrm{e}^{b} f\omega^{3}$ has $b \geq 0$, with equality if and only if f is the pullback of a function on the base. Fixing the total integral of the density therefore does not determine the scalar on this nonclosed background.

Corollary 10.9.1 *Let $G \in C^{\infty}(X, \mathbb{R})$, let $L_{0}v = 3\,\mathrm{dd}^{\mathrm{c}}v \wedge \omega^{2}/\omega^{3}$ be the linearized operator at $u = 0$, and let v be the unique smooth real function satisfying*

$$L_{0}v = G - \int_{X} G\,\mathrm{d}\nu, \qquad \int_{X} v\,\mathrm{d}\nu = 0.$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}t} D(tG)\Big|_{t=0} = 0, \qquad \frac{\mathrm{d}^{2}}{\mathrm{d}t^{2}} D(tG)\Big|_{t=0} = 2\int_{X} |v_{w}|^{2}\,\mathrm{d}\nu.$$

The second derivative vanishes if and only if G is a smooth base pullback.

Proof For real t near zero put $u_{t} = u_{tG}$ and $b_{t} = b(tG)$. By Proposition 10.3.1, this is a smooth family, with $u_{0} = b_{0} = 0$. Differentiating the logarithmic equation at $t = 0$ gives $L_{0}\dot{u}_{0} = G + \dot{b}_{0}$. Since ω is Gauduchon, ν is the adjoint probability measure of L_{0}, and integration gives $\dot{b}_{0} = -\int_{X} G\,\mathrm{d}\nu$. Hence $\dot{u}_{0} = v$; this also proves the existence and uniqueness of v.

Smooth dependence gives $(u_{t})_{w} = tv_{w} + O(t^{2})$ uniformly on X, so (10.13) gives

$$\mathrm{e}^{D(tG)} = 1 + t^{2}\int_{X} |v_{w}|^{2}\,\mathrm{d}\nu + O(t^{3}).$$

Taking logarithms proves the derivative formulas.

If the second derivative vanishes, then $v_{ww} = 0$, so $v = \pi^* q$ for a smooth real function q on the base. Expanding $\omega^2 = \pi^*\omega_{\mathrm{FS}}^2 + 2\pi^*\omega_{\mathrm{FS}} \wedge \beta$ gives

$$L_0(\pi^* q) = \pi^* \left(\frac{2\mathrm{dd}^{\mathrm{c}} q \wedge \omega_{\mathrm{FS}}}{\omega_{\mathrm{FS}}^2} \right).$$

Hence $G = L_0 v + \int_X G \, \mathrm{d}\nu$ is a base pullback. Conversely, when G is a base pullback, Proposition 10.9.1 gives $D(tG) = 0$ for every real t. □

This is a second-variation formula at the background density only. It does not determine whether D is convex globally, because the second derivatives of b and of the logarithmic integral have opposite signs.

10.10 Prescribing a Gauduchon volume form

From now on, X is again arbitrary, of dimension $n \geq 2$. To stay in a fixed Aeppli class of Gauduchon metrics, we vary the $(n-1)$st power of the metric by a term of the form $\partial\xi + \bar\partial\bar\xi$. The resulting scalar ansatz contains first derivatives of the potential as well as its complex Hessian.

Theorem 10.10.1 (Székelyhidi–Tosatti–Weinkove) *Fix Hermitian forms γ_0, γ and a smooth real function F. There is a unique pair $(u, b) \in C^\infty(X, \mathbb{R}) \times \mathbb{R}$ with $\sup_X u = 0$ such that the $(n-1, n-1)$-form*

$$\Gamma_u = \gamma_0^{n-1} + \mathrm{dd}^{\mathrm{c}} u \wedge \gamma^{n-2} + \mathrm{Re}\left(\frac{\mathrm{i}}{2\pi} \partial u \wedge \bar\partial(\gamma^{n-2}) \right)$$

is strictly positive, and the Hermitian form η_u with $\eta_u^{n-1} = \Gamma_u$ satisfies

$$\eta_u^n = \mathrm{e}^{F+b} \gamma^n.$$

Here we use the linear-algebra fact that a real $(n-1, n-1)$-form is strictly positive if and only if it is the $(n-1)$st power of a Hermitian form, and that this Hermitian form is unique; the matrix form of this correspondence is described in Section 10.13. Keep γ_0 and γ fixed for the remainder of the chapter, and write $\mathcal{D}_\gamma u = \Gamma_u - \gamma_0^{n-1}$ for the part of the ansatz that is linear in u.

To see which geometry is preserved, set $\xi_u = \mathrm{i}\,\bar\partial u \wedge \gamma^{n-2}/(4\pi)$. The Leibniz rule gives $\Gamma_u - \gamma_0^{n-1} = \partial\xi_u + \bar\partial\overline{\xi_u}$. Since $\mathrm{dd}^{\mathrm{c}}(\partial\xi_u + \bar\partial\overline{\xi_u}) = 0$, if γ_0 is Gauduchon then so is η_u, and the $(n-1)$st powers of γ_0 and η_u have the same Aeppli class. The uniqueness in Theorem 10.10.1 concerns this scalar ansatz; it does not assert uniqueness among all Gauduchon metrics in the Aeppli class.

10.11 Bounds for the Gauduchon potential

Taking the trace of the positive form Γ_u gives a lower bound for a fixed linear elliptic operator applied to the potential, and its Green kernel then gives an L^1 bound. Near a minimum, supporting affine functions select a contact set on which the gradient is small and the real Hessian is bounded below; the determinant equation forces this set to have uniformly positive volume. Combining the two estimates bounds the minimum. This is an Alexandrov–Bakelman–Pucci type argument.

Proposition 10.11.1 *Every smooth solution (u, b) of the equation in Theorem 10.10.1 with $\sup_X u = 0$ satisfies*

$$|b| + \|u\|_{L^\infty(X)} \le C.$$

The constant depends only on X, γ_0, γ and an upper bound for $\|F\|_\infty$.

Proof We first bound the scalar. At a maximum of u, the first derivatives of u vanish and $\mathrm{dd^c} u \le 0$, so $\Gamma_u \le \gamma_0^{n-1}$; at a minimum the reverse inequality holds. The determinant is monotone with respect to the order of positive matrices, and $(\eta_u^n/\gamma^n)^{n-1}$ is the determinant of the coefficient matrix of Γ_u relative to γ (see (10.19) below). Evaluating the equation at these two points gives

$$\inf_X \left(\log \frac{\gamma_0^n}{\gamma^n} - F \right) \le b \le \sup_X \left(\log \frac{\gamma_0^n}{\gamma^n} - F \right).$$

This proves the scalar bound.

Consider the fixed real linear elliptic operator

$$Lu = \frac{\gamma \wedge \mathcal{D}_\gamma u}{\gamma^n}.$$

Its principal part is $\mathrm{dd^c} u \wedge \gamma^{n-1}/\gamma^n$, its remaining term is of first order, and it has no zeroth-order term. Since Γ_u is positive, $\gamma \wedge \Gamma_u \ge 0$, and therefore $Lu \ge -C_0$ for a fixed constant C_0.

By the Fredholm and maximum-principle argument in the proof of Theorem 4.1.1, the operator L has an adjoint probability measure ν with smooth strictly positive density, that is, $\int_X Lv \, \mathrm{d}\nu = 0$ for every smooth v. As for the Laplacian, $Lv \le 0$ at a maximum of v, and the normalized Green kernel of L gives the representation

$$v(x) - \int_X v \, \mathrm{d}\nu = -\int_X K(x,y) L v(y) \, \mathrm{d}\nu(y),$$
$$K \ge 0, \qquad \sup_x \int_X K(x,y) \, \mathrm{d}\nu(y) < +\infty.$$

Here the kernel is smooth off the diagonal and locally integrable. To obtain this representation, construct the inverse of L on functions of zero ν-mean by the Fredholm alternative. With the minus sign above, the local fundamental solution has a positive leading singularity, so compactness gives a lower bound for the kernel.

Adding a constant to K makes it nonnegative and does not change the formula, since $\int_X Lv \, d\nu = 0$. The local estimates for the Green kernel give the uniform integral bound.

Apply the formula to u and use $Lu \geq -C_0$. At a maximum point, where $u = 0$, we obtain

$$-\int_X u \, d\nu \leq C_1.$$

Because $u \leq 0$ and ν has a fixed smooth positive density, this is a uniform L^1 bound for u with respect to any fixed smooth volume form.

Now let $M = \min_X u$, and choose holomorphic coordinates on a ball centered at a minimum point. By compactness, the coordinate radius can be fixed, with uniform bounds for the metric coefficients and their derivatives; rescale the ball to the unit ball $B \subset \mathbb{R}^{2n}$. Write x for the real coordinates, D and D^2 for the Euclidean gradient and Hessian, dx for Euclidean volume, and I for an identity matrix of the appropriate size. Fix a small number $\varepsilon > 0$, depending only on those bounds, and set

$$v(x) = u(x) + \varepsilon |x|^2.$$

For every real vector p with $|p| < \varepsilon/2$, the function $v(x) - p \cdot x$ attains its minimum at an interior point, because its boundary values are larger than $M + \varepsilon/2$, whereas its value at the center is M. At such a point,

$$Dv = p, \qquad D^2 v \geq 0, \qquad v \leq M + \varepsilon/2.$$

Let $A \subset B$ be the set of all points satisfying $|Dv| \leq \varepsilon/2$, $D^2 v \geq 0$, and $v \leq M + \varepsilon/2$. Since $u \geq M$, this set lies in $\{|x| \leq 1/\sqrt{2}\}$. It is closed there, hence compact in B, and by the preceding paragraph its image under the gradient map Dv contains the ball $B_{\varepsilon/2}$ of radius $\varepsilon/2$. The area formula for the smooth gradient map therefore gives

$$|B_{\varepsilon/2}| \leq \int_A \det D^2 v \, dx. \tag{10.15}$$

The Hessian is nonnegative on A, so no absolute value is needed in the determinant.

We now bound this determinant pointwise on A. There,

$$|Du| \leq \frac{5}{2}\varepsilon, \qquad D^2 u \geq -2\varepsilon I.$$

The complex Hessian of u has the corresponding lower bound. Wedging with the fixed positive form γ^{n-2} and bounding the first-order term of Γ_u by the gradient estimate, we find that $\Gamma_u \geq \gamma_0^{n-1} - C\varepsilon\gamma^{n-1}$. Let Q_u be the coefficient matrix of Γ_u in a fixed local frame of $(n-1,0)$-forms. Since γ_0^{n-1} is strictly positive, choosing ε small gives

$$Q_u \geq cI$$

with a uniform $c > 0$. The equation and the bounds for $F + b$ already proved bound $\det Q_u$ from above, because the determinant of this matrix is, up to a fixed positive

factor depending on the frame, the $(n-1)$st power of the density of η_u^n. All eigenvalues of Q_u are therefore bounded above as well: their product is bounded, and each of them is at least c.

In particular, $\gamma \wedge \Gamma_u / \gamma^n$ is uniformly bounded on A. Subtracting its fixed background term and its bounded first-order term, we obtain an upper bound for

$$\frac{\mathrm{d}\mathrm{d}^{\mathrm{c}} u \wedge \gamma^{n-1}}{\gamma^n}.$$

This is a fixed positive multiple of the γ-trace of the complex Hessian of u, which differs from that of v by a bounded term. The complex Hessian of v is nonnegative on A, since the real Hessian is nonnegative there, so comparison of metrics bounds its Euclidean trace from above. The real trace of $D^2 v$ is four times this complex trace. It follows that

$$0 \le \det D^2 v \le \left(\frac{\mathrm{tr}_{\mathbb{R}}\, D^2 v}{2n} \right)^{2n} \le C_2 \quad \text{on } A.$$

Inequality (10.15) now gives $|A| \ge c_2 > 0$, where c_2 is independent of u. On A we also have $u \le v \le M + \varepsilon/2$. Consequently the uniform L^1 bound gives

$$C_3 \ge \int_A (-u)\, \mathrm{d}x \ge c_2(-M - \varepsilon/2).$$

This bounds $-M$ and completes the proof. □

10.12 The adjoint equation for Gauduchon volume

As for the Monge–Ampère equation, solving the linearized Gauduchon volume equation requires a compatibility condition, given by an adjoint probability measure. Taking the $(n-1)$st root multiplies the trace linearization by the factor $n/(n-1)$, and the first-order term of the ansatz introduces a drift term. We compute the linearized operator at any potential with $\Gamma_u > 0$, and then express its adjoint condition as an equation for a positive $(1,1)$-form. Throughout this section we put $K = \gamma^{n-2}$.

Proposition 10.12.1 *On the open set of smooth potentials u with $\Gamma_u > 0$, the derivative of*

$$\mathcal{G}(u) = \log \frac{\eta_u^n}{\gamma^n}$$

is the real elliptic operator

$$\mathcal{L}_u v = \frac{n}{n-1} \frac{\eta_u \wedge \mathcal{D}_\gamma v}{\eta_u^n}. \tag{10.16}$$

It has no zeroth-order term, and its kernel consists of the constants. There is a unique smooth positive function ρ_u, normalized by $\int_X \rho_u \eta_u^n = 1$, such that the probability measure

$$\mathrm{d}\mu_u^\gamma = \rho_u \eta_u^n$$

satisfies $\int_X \mathcal{L}_u v \, \mathrm{d}\mu_u^\gamma = 0$ for every smooth real v.

Proof For a smooth path of potentials $u(t)$, write $\eta = \eta_{u(t)}$ and $\Gamma = \Gamma_{u(t)}$. Differentiating $\eta^{n-1} = \Gamma$ gives $(n-1)\dot\eta \wedge \eta^{n-2} = \dot\Gamma$, where $\dot\Gamma = \mathcal{D}_\gamma \dot u$. Wedging with η and dividing by η^n, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t} \log \eta^n = \frac{n}{n-1} \frac{\eta \wedge \dot\Gamma}{\eta^n},$$

which proves (10.16).

Only the second-order term $\mathrm{dd^c} v \wedge \gamma^{n-2}$ of $\mathcal{D}_\gamma v$ contributes to the principal symbol. For a nonzero complex covector ζ of type $(1,0)$, the form

$$\eta_u \wedge \mathrm{i}\zeta \wedge \overline{\zeta} \wedge \gamma^{n-2}$$

is a positive volume form. Hence the principal symbol is positive definite, and by compactness the operator is uniformly elliptic for each fixed u. The remaining term is of first order, and constants are annihilated, so the strong maximum principle gives $\ker \mathcal{L}_u = \mathbb{R}$.

The Fredholm and positivity arguments in the proof of Theorem 4.1.1 apply to this real elliptic operator with first-order terms and no zeroth-order term. Its index is zero by deformation to a Laplacian, so its formal adjoint with respect to η_u^n has a one-dimensional kernel spanned by a smooth strictly positive function. Normalizing its integral to one gives ρ_u and proves uniqueness. □

The adjoint condition can be expressed as an equation for the positive $(1,1)$-form $\rho_u \eta_u$. For a smooth real $(1,1)$-form A, set

$$\mathcal{A}_\gamma(A) = \mathrm{dd^c} A \wedge \gamma^{n-2} + \mathrm{Re}\left(\frac{\mathrm{i}}{2\pi} \partial A \wedge \bar\partial(\gamma^{n-2}) \right).$$

This is a real form of top degree, and $\mathcal{A}_\gamma$ is the formal adjoint of $\mathcal{D}_\gamma$ under the pairing of $(1,1)$-forms with $(n-1,n-1)$-forms, as the proof below shows.

Proposition 10.12.2 *The function ρ_u of Proposition 10.12.1 is characterized by*

$$\mathcal{A}_\gamma(\rho_u\eta_u) = 0, \qquad \int_X \rho_u \eta_u^n = 1. \tag{10.17}$$

When $n = 2$, its first equation is $\mathrm{dd^c}(\rho_u\eta_u) = 0$. When γ is Gauduchon and $\eta_u = \gamma$, the function ρ_u is constant and $\mathrm{d}\mu_u^\gamma = \gamma^n / \int_X \gamma^n$.

Proof We first prove that, for every smooth real $(1,1)$-form A and every smooth real function v on X,

$$\int_X A \wedge \mathcal{D}_\gamma v = \int_X v\, \mathcal{A}_\gamma(A). \tag{10.18}$$

Write $c = \mathrm{i}/(2\pi)$. Since A, K, v are real, expanding the real part gives

$$\mathcal{D}_\gamma v = c\, \partial\bar\partial v \wedge K + \frac{c}{2}\big(\partial v \wedge \bar\partial K - \bar\partial v \wedge \partial K\big).$$

Integrating the first term by parts twice gives $\int_X v\, c\partial\bar\partial(A \wedge K)$, and integrating each first-order term by parts once gives the correction

$$\int_X v\, \frac{c}{2}\big(-\partial A \wedge \bar\partial K + \bar\partial A \wedge \partial K - 2A \wedge \partial\bar\partial K\big).$$

Finally,

$$\partial\bar\partial(A \wedge K) = \partial\bar\partial A \wedge K - \bar\partial A \wedge \partial K + \partial A \wedge \bar\partial K + A \wedge \partial\bar\partial K.$$

Adding these contributions cancels the terms containing $A \wedge \partial\bar\partial K$ and leaves $\mathcal{A}_\gamma(A)$. This proves (10.18). Applied to $A = \rho_u\eta_u$, it shows by (10.16) that $\int_X \mathcal{L}_u v\, \mathrm{d}\mu_u^\gamma$ is a fixed positive multiple of $\int_X v\, \mathcal{A}_\gamma(\rho_u\eta_u)$, so the adjoint condition is equivalent to (10.17).

For $n = 2$, we have $K = 1$, which gives the first special case. For the second, the product rule and the Gauduchon condition give

$$\mathcal{A}_\gamma(\gamma) = \frac{1}{n-1}\mathrm{dd^c}(\gamma^{n-1}) = 0.$$

By uniqueness, ρ_u is consequently the constant $(\int_X \gamma^n)^{-1}$. □

For $n > 2$ the equation for $\rho_u\eta_u$ involves the first derivatives of γ^{n-2}, so the fixed auxiliary metric γ remains part of the adjoint equation; it is not simply the Gauduchon condition for η_u.

10.13 The matrix structure of the Gauduchon equation

The Hodge star converts the positive $(n-1,n-1)$-form Γ_u into a Hermitian matrix whose logarithmic determinant is a concave function of the complex Hessian. Two features will control the largest eigenvalue in the second-order estimate: the positive background gives coercivity, and the torsion coefficients omit exactly those gradient indices that would otherwise carry the largest inverse weight. We derive both properties before differentiating the equation.

For the matrix and derivative calculations, set $v = u/(2\pi)$, so that $\mathrm{dd}^{\mathrm{c}}u = \mathrm{i}\partial\bar\partial v$. All covariant derivatives are taken with respect to the Chern connection ∇ of γ. In holomorphic coordinates $z^1,\dots,z^n$, write $v_i = \partial_i v$ and $v_{\bar\jmath} = \partial_{\bar\jmath} v$; indices range from 1 to n, and δ_{ij} is the Kronecker symbol. Write $\gamma = \mathrm{i}\gamma_{a\bar b}\,\mathrm{d}z^a \wedge \mathrm{d}\bar z^b$, with repeated indices summed. We denote by $\star_\gamma$ the Hodge star of γ, and by $\det_\gamma$ the determinant of the Hermitian matrix of a real $(1,1)$-form relative to γ. Write the torsion of ∇ as T^c_{ab} and lower the last index with γ. Thus

$$T_{ab\bar c} = \partial_a \gamma_{b\bar c} - \partial_b \gamma_{a\bar c}, \qquad T_{ab\bar c} = -T_{ba\bar c}.$$

Let $Q_0 = \star_\gamma \gamma_0^{n-1}/(n-1)!$ and

$$Z = \frac{1}{(n-1)!}\,\star_\gamma \operatorname{Re}\bigl(\mathrm{i}\partial v \wedge \bar\partial \gamma^{n-2}\bigr).$$

Here the real $(1,1)$-forms on the right are identified with their Hermitian coefficient matrices relative to γ. Set

$$\chi = (\operatorname{tr}_\gamma Q_0)\gamma - (n-1)Q_0, \quad W = (\operatorname{tr}_\gamma Z)\gamma - (n-1)Z, \quad g = \chi + \mathrm{i}\partial\bar\partial v + W.$$

The linear map $P_\gamma(A) = ((\operatorname{tr}_\gamma A)\gamma - A)/(n-1)$ on real $(1,1)$-forms, or Hermitian matrices, is invertible, and the equation of Theorem 10.10.1 becomes

$$\begin{aligned} Q_u &:= P_\gamma(g) = Q_0 + P_\gamma(\mathrm{i}\partial\bar\partial v) + Z > 0, \\ \log\det_\gamma Q_u &= h = (n-1)(F+b). \end{aligned} \tag{10.19}$$

Here $Q_u = \star_\gamma \Gamma_u/(n-1)!$ is the coefficient matrix of Γ_u. The tensor g need not be positive. To check the Hessian term, diagonalize $\mathrm{i}\partial\bar\partial v$ in a unitary frame: the coefficient of $\star_\gamma(\mathrm{i}\partial\bar\partial v \wedge \gamma^{n-2})/(n-1)!$ in each diagonal position is the sum of the other diagonal entries divided by $n-1$. If η_u has eigenvalues a_i relative to γ, then Q_u has eigenvalues $\prod_{j\neq i} a_j$, whose product is $(\prod_i a_i)^{n-1}$. This proves the determinant identity.

Lemma 10.13.1 *At a point in a unitary frame,*

$$2(n-1)Z_{i\bar{j}} = \delta_{ij}\sum_{p,k} v_p\overline{T_{pk\bar{k}}} - v_i\sum_k \overline{T_{jk\bar{k}}} - \sum_k v_k\overline{T_{kj\bar{i}}} + \delta_{ij}\sum_{p,k} v_{\bar{p}}T_{pk\bar{k}} - v_{\bar{j}}\sum_k T_{ik\bar{k}} - \sum_k v_{\bar{k}}T_{ki\bar{j}}. \tag{10.20}$$

In particular $Z_{i\bar{j}}$ contains no v_j or $v_{\bar{i}}$. Moreover

$$2(n-1)Z_{i\bar{i}} = \sum_{p\neq i,\ k\neq i}\left(v_p\overline{T_{pk\bar{k}}} + v_{\bar{p}}T_{pk\bar{k}}\right). \tag{10.21}$$

Covariant differentiation preserves these coefficient identities. For example $\nabla_1 Z_{1\bar{1}}$ contains no pure Hessian v_{11} or its conjugate, and $\nabla_{\bar{1}}\nabla_1 Z_{1\bar{1}}$ contains no $v_{11\bar{1}}$ or its conjugate, nor a pure v_{11} or its conjugate.

Proof For $n = 2$, the defining expression of Z is zero, and so are the formulas below, because the generalized Kronecker symbol with three indices vanishes in two dimensions. Suppose $n \geq 3$ and use coordinates that are unitary at the point. Direct antisymmetrization gives

$$\bar{\partial}\gamma = -\frac{\mathrm{i}}{2}\sum_{a,b,c}\overline{T_{bc\bar{a}}}\,\mathrm{d}z^a\wedge\mathrm{d}\bar{z}^b\wedge\mathrm{d}\bar{z}^c.$$

We use $\bar{\partial}\gamma^{n-2} = (n-2)\bar{\partial}\gamma\wedge\gamma^{n-3}$. To write the contraction explicitly, put

$$\delta^{paj}_{bci} = \det\begin{pmatrix}\delta_{pb} & \delta_{pc} & \delta_{pi}\\ \delta_{ab} & \delta_{ac} & \delta_{ai}\\ \delta_{jb} & \delta_{jc} & \delta_{ji}\end{pmatrix}.$$

Let $A = \star_\gamma(\mathrm{i}\partial v\wedge\bar{\partial}\gamma^{n-2})/(n-1)!$. Expanding the $n-3$ remaining copies of γ leaves the three complementary indices and gives

$$A_{i\bar{j}} = \frac{1}{2(n-1)}\sum_{p,a,b,c} v_p\overline{T_{bc\bar{a}}}\,\delta^{paj}_{bci}.$$

Indeed, the factor is $(n-2)(n-3)!/[2(n-1)!]$, and the determinant records the signs of the remaining holomorphic and antiholomorphic indices. Expanding its six terms and using $T_{bc\bar{a}} = -T_{cb\bar{a}}$, we obtain

$$A_{i\bar{j}} = \frac{1}{n-1}\left(\delta_{ij}\sum_{p,k} v_p\overline{T_{pk\bar{k}}} - v_i\sum_k\overline{T_{jk\bar{k}}} - \sum_k v_k\overline{T_{kj\bar{i}}}\right).$$

Since $Z = (A + A^*)/2$, where A^* denotes the Hermitian adjoint matrix, this gives (10.20).

For $i \neq j$, the coefficient of v_j in this expression vanishes because $T_{jj\bar{i}} = 0$, and the coefficient of $v_{\bar{i}}$ vanishes in the conjugate expression. For $i = j$, canceling the terms with $p = i$ and $k = i$ gives (10.21). These are tensor identities valid in every unitary frame. Since the Chern connection preserves the metric, they can be differentiated in a unitary frame that is parallel at the point. For instance, the first derivative of a summand in (10.21) is

$$(\nabla_1 v_p)\overline{T_{pk\bar{k}}} + v_p \nabla_1 \overline{T_{pk\bar{k}}}, \qquad p, k \neq 1,$$

together with its conjugate counterpart. A further derivative in the direction $\bar{1}$ still has $p \neq 1$. This gives the stated exclusions, including the terms in which a derivative falls on a coefficient. □

The second-order estimate uses these exclusions; an arbitrary first-order perturbation could not be substituted for Z.

Lemma 10.13.2 *Diagonalize g at a point and order its eigenvalues as $\lambda_1 \geq \cdots \geq \lambda_n$. Put*

$$\mu_i = \frac{\sum_{j \neq i} \lambda_j}{n-1} > 0, \quad t_i = \mu_i^{-1}, \quad f_i = \frac{1}{n-1} \sum_{j \neq i} t_j, \quad \mathcal{F} = \sum_i f_i = \sum_i t_i.$$

Then $\mu_1 \leq \cdots \leq \mu_n$, $f_1 \leq \cdots \leq f_n$, and

$$\sum_i f_i \lambda_i = n, \qquad \frac{\mathcal{F}}{n(n-1)} \leq f_k \leq \frac{\mathcal{F}}{n-1} \quad (k > 1), \qquad t_i \leq (n-1) f_1 \quad (i > 1), \tag{10.22}$$

$$f_n \geq \mathcal{F}/n.$$

Fix $\tau, D > 0$ and assume $Q_0 \geq \tau\gamma$ and $\det P_\gamma(g) \leq D$. For λ_1 larger than a constant depending only on n, τ, D,

$$\sum_i f_i(\chi_{i\bar{i}} - \lambda_i) \geq \frac{\tau}{2}\mathcal{F}. \tag{10.23}$$

There is also a uniform positive lower bound for $\mathcal{F}$ depending only on n, D.

Proof The ordering follows from the formula for μ_i, and then from that for f_i. The identity $\sum_i f_i \lambda_i = n$, which reflects the relation $\log\det P_\gamma(sg) = \log\det P_\gamma(g) + n \log s$, can be checked directly:

$$\sum_i f_i \lambda_i = \sum_j \frac{t_j}{n-1} \sum_{i \neq j} \lambda_i = \sum_j t_j \mu_j = n.$$

For $k > 1$, the sum defining f_k contains the largest weight $t_1 \geq \mathcal{F}/n$. The upper bound for f_k and the estimate for t_i are immediate, and the largest weight f_n is at least the average $\mathcal{F}/n$.

The map P_γ is self-adjoint for the matrix trace pairing. Consequently

$$\sum_i f_i \chi_{i\bar{\imath}} = \sum_i t_i (Q_0)_{i\bar{\imath}} \geq \tau \mathcal{F}.$$

Moreover $\sum_i \lambda_i = \sum_i \mu_i > \lambda_1 > 0$, where the strict inequality follows from $\mu_1 > 0$. Hence $\mu_n > \lambda_1/n$. Since the other $n-1$ numbers μ_i are at least μ_1,

$$\mu_1^{n-1} \mu_n \leq \prod_i \mu_i \leq D, \qquad \mathcal{F} \geq \mu_1^{-1} \geq \left(\frac{\lambda_1}{nD} \right)^{1/(n-1)}.$$

For λ_1 sufficiently large this is at least $2n/\tau$, so $n \leq \tau \mathcal{F}/2$. Subtracting $\sum_i f_i \lambda_i = n$ therefore proves (10.23). Finally, the arithmetic–geometric mean inequality applied to the t_i gives $\mathcal{F} \geq n D^{-1/n}$. □

The concavity of the logarithmic determinant will supply a positive term with an explicit formula. For the matrix function $\Phi(g) = \log\det P_\gamma(g)$, the negative second differential in a complex matrix direction B and its adjoint is

$$-D^2 \Phi_g(B, B^*) = \operatorname{tr}\bigl(P_\gamma(g)^{-1} P_\gamma(B) P_\gamma(g)^{-1} P_\gamma(B)^* \bigr) \geq 0. \tag{10.24}$$

This follows by differentiating the inverse matrix. In a unitary frame diagonalizing g, the off-diagonal terms of this expression include

$$\frac{|B_{k\bar{1}}|^2}{(n-1)^2 \mu_k \mu_1}.$$

When $\lambda_k < \lambda_1$, that coefficient is exactly

$$\frac{f_k - f_1}{\lambda_1 - \lambda_k} = \frac{1}{(n-1)^2 \mu_k \mu_1}.$$

The expression on the right also makes sense when the two eigenvalues coincide. This positive square and the weights f_i enter the largest-eigenvalue estimate in the next section.

10.14 The largest-eigenvalue estimate

The matrix structure now gives a bound for the complex Hessian in terms of the squared gradient. We apply the maximum principle to the largest eigenvalue of g, corrected by functions of the gradient and of the potential. The missing torsion indices allow the concavity square to absorb the third derivatives. If the smallest eigenvalue is not too negative, the positive background closes the estimate; otherwise that eigenvalue produces a large positive Hessian square.

We keep the notation of Section 10.13. Plain subscripts on scalar functions and subscripts after a comma on matrix coefficients denote coordinate derivatives, and the coefficients of the Chern connection of γ are written Γ^q_{ik}. Choose $\tau > 0$ with $Q_0 \geq \tau\gamma$.

Theorem 10.14.1 *For a smooth real function h on X, suppose that v is a smooth solution of*

$$Q = Q_0 + P_\gamma(\mathrm{i}\partial\bar\partial v) + Z(\partial v) > 0, \qquad \log\det_\gamma Q = h,$$

with the tensors of Section 10.13. If $\|v\|_{C^0}$ and $\|h\|_{C^2}$ are bounded, then

$$\sup_X |\mathrm{i}\partial\bar\partial v|_\gamma \leq C\bigl(1 + \sup_X |\partial v|^2_\gamma\bigr),$$

where C depends only on these bounds and the fixed smooth data.

Proof Adding a constant, we may assume $\sup_X v = 0$, and we write $-M \leq v \leq 0$. Set

$$K = 1 + \sup_X |\partial v|^2_\gamma, \qquad \phi(s) = -\tfrac{1}{2}\log(1 - s/(2K)), \qquad \psi(t) = D_1 \mathrm{e}^{-D_2 t}.$$

The positive constants D_1, D_2 will be chosen below using only the fixed data. On $0 \leq s \leq K$,

$$\frac{1}{4K} \leq \phi' \leq \frac{1}{2K}, \qquad \phi'' = 2(\phi')^2, \qquad 0 \leq \phi \leq \tfrac{1}{2}\log 2.$$

Let λ_1 be the largest eigenvalue of $g = \chi + \mathrm{i}\partial\bar\partial v + W$ relative to γ. It is positive, because $\mathrm{tr}_\gamma\, g = \mathrm{tr}_\gamma\, Q > 0$. Consider a maximum point of

$$H = \log\lambda_1 + \phi(|\partial v|^2_\gamma) + \psi(v).$$

It suffices to bound λ_1/K at this point, because the other two terms of H are uniformly bounded once D_1, D_2 are fixed. We may assume that this ratio exceeds any fixed constant specified in the course of the argument.

Choose holomorphic coordinates that are unitary at the maximum point, with g diagonal there and eigenvalues ordered as $\lambda_1 \geq \cdots \geq \lambda_n$. All estimates below are at this point, and repeated indices are summed only when a sum is written. Put $\lambda = \lambda_1$ and use the weights t_i, f_k and their sum $\mathcal{F}$ from Lemma 10.13.2. Since $\sum_i \lambda_i > 0$ and the eigenvalues are ordered,

$$|\lambda_i| \leq (n-1)\lambda, \qquad |v_{i\bar j}| \leq C\lambda \quad \text{if } \lambda \geq K.$$

The constant C may change from line to line; it is independent of K and, until D_1, D_2 are fixed below, also of D_1, D_2. The pure second derivatives v_{ij} are not covered by this bound.

Differentiating the largest eigenvalue.

Since λ_1 need not be smooth where it is a multiple eigenvalue, we first perturb g. Choose a smooth local unitary frame that is parallel for the Chern connection at the point, with first vector an eigenvector for λ. Let B be the tensor that is diagonal in this frame, with eigenvalues $0, \epsilon_0, \ldots, \epsilon_0$, where $\epsilon_0 > 0$ is fixed and small. Then $B \geq 0$, $\nabla B = 0$ at the point, and its second derivatives are bounded by fixed constants. The largest eigenvalue $\widetilde{\lambda}$ of $g - B$ is simple near the point, is at most $\lambda_1(g)$, and equals λ at the point. Hence the smooth function $\widetilde{H}$, defined like H with $\widetilde{\lambda}$ in place of λ_1, also has a local maximum there. Only the largest eigenvalue needs to be simple.

For a family of self-adjoint matrices with a simple largest eigenvalue, differentiate the equation for the normalized eigenvector. Its component in the pth eigenspace is the corresponding derivative of the matrix divided by the eigenvalue gap. Differentiating the eigenvalue once more therefore gives

$$\begin{aligned} a_k &:= \widetilde{\lambda}_k = \nabla_k g_{1\bar{1}}, \\ \widetilde{\lambda}_{k\bar{k}} &= \nabla_{\bar{k}} \nabla_k g_{1\bar{1}} + O(1) + \sum_{p>1} \frac{|\nabla_k g_{1\bar{p}}|^2 + |\nabla_k g_{p\bar{1}}|^2}{\lambda - \lambda_p + \epsilon_0}. \end{aligned}$$

This formula remains valid if some of the lower eigenvalues coincide, with the sum taken over any orthonormal basis of their eigenspaces. The gaps are positive and at most $n\lambda + \epsilon_0$. In holomorphic coordinates, $a_k = g_{1\bar{1},k} + O(\lambda)$, while converting the second covariant derivative to partial derivatives introduces an error of at most $C \sum_p |g_{1\bar{p},k}| + C\lambda$. We also convert the off-diagonal covariant derivatives in the positive squares, with errors $O(\lambda)$, and absorb the linear off-diagonal terms by Young's inequality. Thus, after decreasing a fixed positive coefficient,

$$\widetilde{\lambda}_{k\bar{k}} \geq g_{1\bar{1},k\bar{k}} + \frac{1}{8n\lambda} \sum_{p>1} (|g_{1\bar{p},k}|^2 + |g_{p\bar{1},k}|^2) - C(|g_{1\bar{1},k}| + \lambda). \tag{10.25}$$

The precise positive constant will not be used.

Write $V_k = \partial_k(|\partial v|^2_\gamma)$. The first derivative of $\widetilde{H}$ gives

$$\frac{a_k}{\lambda} = -\phi' V_k - \psi' v_k, \qquad |V_k| \leq C\sqrt{K} \sum_p (|v_{pk}| + |v_{p\bar{k}}|) + CK. \tag{10.26}$$

In the second derivative, the derivatives of the fixed metric in $|\partial v|^2_\gamma$ cost only a small fraction of the positive Hessian squares and a constant. Therefore

$$\begin{aligned} \widetilde{H}_{k\bar{k}} \geq \frac{\widetilde{\lambda}_{k\bar{k}}}{\lambda} - \frac{|a_k|^2}{\lambda^2} + \phi' \sum_p (v_p v_{\bar{p}k\bar{k}} + v_{\bar{p}} v_{pk\bar{k}}) + \frac{1}{5K} \sum_p (|v_{pk}|^2 + |v_{p\bar{k}}|^2) \\ + \phi'' |V_k|^2 + \psi'' |v_k|^2 + \psi' v_{k\bar{k}} - C. \end{aligned} \tag{10.27}$$

The third and fourth terms are obtained by differentiating $\sum \gamma^{p\bar{q}} v_p v_{\bar{q}}$ twice and applying $ab \le \varepsilon a^2 + C_\varepsilon b^2$ to each term containing a derivative of the metric.

The third derivatives and the torsion term.

We now estimate the third derivatives in this inequality. Define

$$E = \sum_{k,p} f_k (|v_{pk}|^2 + |v_{p\bar{k}}|^2), \qquad S = \sum_{k,p} f_k (|v_{pk}| + |v_{p\bar{k}}|),$$
$$C = -D^2 \Phi_g (\nabla_1 g, (\nabla_1 g)^*) \ge 0.$$

The first differentiated equation is

$$\sum_k f_k \nabla_p g_{k\bar{k}} = h_p. \tag{10.28}$$

Since $P_\gamma(W) = Z$ and the connection preserves γ, one has the exact contractions

$$\sum_k f_k \nabla_p W_{k\bar{k}} = \sum_i t_i \nabla_p Z_{i\bar{i}}, \qquad \sum_k f_k \nabla_{\bar{1}} \nabla_1 W_{k\bar{k}} = \sum_i t_i \nabla_{\bar{1}} \nabla_1 Z_{i\bar{i}}. \tag{10.29}$$

The weights in these identities are evaluated at the point; they are not differentiated.

The large weights are controlled as follows. By (10.22), the only weight t_i not bounded by $(n-1) f_1$ is t_1. In $Z_{1\bar{1}}$, every gradient index differs from 1, by (10.21). Thus a pure second derivative in $\nabla_p Z_{1\bar{1}}$ has an index $q > 1$, and it can be weighted by f_q, which is comparable to $\mathcal{F}$; here we use that pure derivatives commute in these coordinates. A mixed second derivative can similarly be placed in column q by conjugate symmetry. All other t_i are bounded by a constant times every f_k, and derivatives of coefficients cost $C\sqrt{K}\mathcal{F}$. Applying these observations to the first identity in (10.29), and then to (10.28), we obtain, for each p,

$$\left| \sum_k f_k v_{k\bar{k}p} \right| + \left| \sum_k f_k v_{k\bar{k}\bar{p}} \right| \le C(S + \sqrt{K}\mathcal{F}). \tag{10.30}$$

The connection terms in the differentiated equation contribute weighted mixed second derivatives and terms bounded by $C\sqrt{K}\mathcal{F}$, which are already on the right. The bounded derivatives of h are absorbed using the positive lower bound for $\mathcal{F}$ in Lemma 10.13.2.

Next we insert the second derivative of the equation into the eigenvalue estimate. Since partial derivatives of v commute,

$$g_{1\bar{1},k\bar{k}} = g_{k\bar{k},1\bar{1}} + W_{1\bar{1},k\bar{k}} - W_{k\bar{k},1\bar{1}} + O(1).$$

Differentiating the equation twice covariantly gives

$$\sum_k f_k \nabla_{\bar 1}\nabla_1 g_{k\bar k} = h_{1\bar 1} + C.$$

The conversion to partial derivatives uses

$$\nabla_{\bar 1}\nabla_1 g_{k\bar k} = g_{k\bar k,1\bar 1} - 2\,\mathrm{Re}\sum_q \overline{\Gamma^q_{1k}} g_{k\bar q,1} + O(\lambda).$$

We rewrite the derivative of g in this term using $g_{k\bar q,1} = g_{1\bar q,k} + W_{k\bar q,1} - W_{1\bar q,k} + O(1)$ and its conjugate. The first part is absorbed by the off-diagonal squares in (10.25), at the cost of $C\lambda\mathcal{F} + C\sum_k f_k|g_{1\bar 1,k}|$. The part involving $W_{1\bar q,k}$ is bounded by $C\lambda\mathcal{F} + C\sum_{k,p} f_k|v_{pk}|$. For the remaining part we use

$$2\,\mathrm{Re}\sum_q \overline{\Gamma^q_{1k}} W_{k\bar q,1} = W_{k\bar k,1\bar 1} - \nabla_{\bar 1}\nabla_1 W_{k\bar k} + O(\sqrt{K}).$$

This cancels the partial second derivative $-W_{k\bar k,1\bar 1}$ above, and we obtain

$$\begin{aligned}\sum_k f_k \widetilde{\lambda}_{k\bar k} \ge C &+ \sum_k f_k (W_{1\bar 1,k\bar k} - \nabla_{\bar 1}\nabla_1 W_{k\bar k}) \\ &- C\left(\sum_k f_k|g_{1\bar 1,k}| + \sum_{k,p} f_k|v_{pk}| + \lambda\mathcal{F}\right). \end{aligned} \tag{10.31}$$

The second contraction in (10.29) bounds the negative torsion term here. In the part with weight t_1, the possible third derivatives are $v_{q1\bar 1}$ and their conjugates with $q > 1$; all other parts have weights bounded by Cf_1. We commute partial derivatives and use

$$v_{1\bar 1,k} = g_{1\bar 1,k} - \chi_{1\bar 1,k} - W_{1\bar 1,k}.$$

The exclusions in Lemma 10.13.1 ensure that any remaining pure derivative v_{11} carries only the small weight Cf_1, and derivatives of the connection contribute second derivatives with the same weighted bounds. It follows that

$$\left|\sum_k f_k \nabla_{\bar 1}\nabla_1 W_{k\bar k}\right| \le C\left(\sum_k f_k|g_{1\bar 1,k}| + \sum_{k,p} f_k|v_{pk}| + \lambda\mathcal{F}\right).$$

For the positive torsion term, write $W_{1\bar 1} = \sum_p (c^p v_p + \overline{c^p} v_{\bar p})$ with fixed smooth coefficients c^p. Two derivatives give $c^p \sum_k f_k v_{pk\bar k}$ and its conjugate, plus terms bounded by $C(S + \sqrt{K}\mathcal{F})$. The former are bounded by (10.30), and in S the mixed second derivatives can be bounded by $C\lambda$, giving

$$\left|\sum_k f_k W_{1\bar 1,k\bar k}\right| \le C\left(\sum_{k,p} f_k|v_{pk}| + \lambda\mathcal{F}\right).$$

Substitute these estimates in (10.31), multiply (10.27) by f_k and sum over k. The third derivatives in the latter are bounded by (10.30) and $\phi'|\partial v| \le C/\sqrt{K}$, and the resulting linear terms in the second derivatives are absorbed by a small part of E/K, at the cost of $C\mathcal{F}$. Since $\widetilde{H}$ has a local maximum, $\sum_k f_k \widetilde{H}_{k\bar{k}} \le 0$, and thus

$$0 \ge \frac{C}{\lambda} - \sum_k f_k \frac{|a_k|^2}{\lambda^2} + \frac{E}{6K} + \phi'' \sum_k f_k |V_k|^2 + \psi'' \sum_k f_k |v_k|^2 + \psi' \sum_k f_k v_{k\bar{k}} - C\left(\sum_k f_k \frac{|g_{1\bar{1},k}|}{\lambda} + \mathcal{F} \right). \tag{10.32}$$

Cancellation when the smallest eigenvalue is controlled.

Fix $0 < \delta < 1/2$, to be chosen after D_1, D_2. First suppose that $\lambda_n \ge -\delta\lambda$. Put $I = \{k : f_k > \delta^{-1} f_1\}$, so that $1 \notin I$. Equation (10.26) and $\phi'' = 2(\phi')^2$ give

$$-\sum_{k\notin I} f_k |a_k|^2/\lambda^2 \ge -\phi'' \sum_{k\notin I} f_k |V_k|^2 - 2(\psi')^2 \delta^{-1} f_1 K,$$
$$-2\delta \sum_{k\in I} f_k |a_k|^2/\lambda^2 \ge -2\delta\phi'' \sum_{k\in I} f_k |V_k|^2 - 4\delta(\psi')^2 \sum_{k\in I} f_k |v_k|^2.$$

Choose δ small enough that $4\delta(\psi')^2 \le \psi''/2$ throughout $[-M, 0]$. By (10.24),

$$\frac{C}{\lambda} \ge \sum_{k\in I} \frac{f_k - f_1}{\lambda(\lambda - \lambda_k)} |\nabla_1 g_{k\bar{1}}|^2 \ge (1 - 2\delta) \sum_{k\in I} \frac{f_k |\nabla_1 g_{k\bar{1}}|^2}{\lambda^2}.$$

Here we used $(1-\delta)/(1+\delta) \ge 1 - 2\delta$.

We need the following comparison of the two derivatives $\nabla_1 g_{k\bar{1}}$ and $a_k = \nabla_k g_{1\bar{1}}$. For every $\varepsilon > 0$, provided λ/K is sufficiently large depending on the fixed function ψ,

$$\sum_{k\in I} f_k \frac{|\nabla_1 g_{k\bar{1}}|^2 - |a_k|^2}{\lambda^2} \ge -\frac{E}{12K} - C\mathcal{F} - \varepsilon C|\psi'|\mathcal{F} - C_\varepsilon |\psi'| \sum_k f_k |v_k|^2. \tag{10.33}$$

To prove this, let $U = \sum_{p>1,q} |v_{pq}|$. Commuting partial derivatives in g gives, for $k > 1$,

$$\nabla_1 g_{k\bar{1}} - a_k = W_{k\bar{1},1} - W_{1\bar{1},k} + O(\lambda), \qquad |W_{k\bar{1},1}| + |W_{1\bar{1},k}| \le C(\lambda + U).$$

The second inequality uses the absence of v_1 in $Z_{k\bar{1}}$, so that differentiation in the direction 1 introduces no pure derivative v_{11}. All mixed second derivatives are bounded by $C\lambda$, and every other pure second derivative occurs in U.

Use $|a+e|^2-|a|^2\ge -2|a||e|$, and then (10.26). For $k>1$, the pure second derivatives in V_k all occur in U. With $H_k=\sum_p |v_{p\bar k}|$, the resulting negative terms, divided by λ^2, are bounded by a constant times

$$K^{-1/2}H_k+K^{-1/2}U+\frac{U^2}{\lambda\sqrt{K}}+1+|\psi'||v_k|+\frac{|\psi'|}{\lambda}|v_k|U.$$

Here we used $H_k\le C\lambda$ and $\lambda\ge K$ to simplify the cross terms. The weight comparison (10.22), the symmetry of pure derivatives and the Cauchy–Schwarz inequality give $\mathcal{F}U^2\le CE$. Consequently, after multiplication by f_k and summation, the first two terms are bounded by any fixed small fraction of E/K plus $C\mathcal{F}$. The third is at most $CE/(\lambda\sqrt{K})$, which is absorbed by another such fraction when λ/K is large. For the last two terms we use

$$|v_k|\le \varepsilon+C_\varepsilon|v_k|^2,\qquad |v_k|U\le |v_k|^2+U^2.$$

The contribution of U^2 is at most $C|\psi'|E/\lambda$, which can again be absorbed when λ/K is large. Choosing the fractions with sum $1/12$ proves (10.33).

Insert this cancellation and the preceding inequalities into (10.32). The remaining derivative error satisfies, by (10.26),

$$C\sum_k f_k|g_{1\bar 1,k}|/\lambda\le \frac{E}{60K}+C\mathcal{F}+\varepsilon C|\psi'|\mathcal{F}+C_\varepsilon|\psi'|\sum_k f_k|v_k|^2,$$

after Young's inequality and an adjustment of constants. The torsion contraction also satisfies

$$\left|\sum_k f_k W_{k\bar k}\right|=\left|\sum_i t_i Z_{i\bar i}\right|\le C\sum_k f_k|v_k|\le \varepsilon C\mathcal{F}+C_\varepsilon\sum_k f_k|v_k|^2.$$

The first inequality uses again the missing gradient index in $Z_{1\bar 1}$. Since $\psi'<0$ and $v_{k\bar k}=\lambda_k-\chi_{k\bar k}-W_{k\bar k}$, we conclude, after decreasing the remaining coefficient of E if necessary, that

$$\begin{aligned}0\ge f_1\left(\frac{\lambda^2}{40K}-2(\psi')^2\delta^{-1}K\right)+\left(\frac{\psi''}{2}-C_\varepsilon|\psi'|\right)\sum_k f_k|v_k|^2\\ -C_0\mathcal{F}-\varepsilon C_0|\psi'|\mathcal{F}+|\psi'|\sum_k f_k(\chi_{k\bar k}-\lambda_k).\end{aligned}\tag{10.34}$$

For the first term we used $|v_{1\bar 1}|^2\ge \lambda^2/2-CK$, which follows from $\chi+W=O(\sqrt{K})$, and $E\ge f_1|v_{1\bar 1}|^2$.

For large λ, Lemma 10.13.2 bounds the last sum below by $\tau\mathcal{F}/2$. Choose first ε with $\varepsilon C_0<\tau/4$. Next choose D_2 so that $\psi''/2>C_\varepsilon|\psi'|$, and then choose D_1 so that $|\psi'|\tau/4>C_0$ on $[-M,0]$. Finally choose δ as above. All these choices depend

only on the fixed data. Inequality (10.34) then forces $\lambda^2 \le 80\,\delta^{-1} \sup_{[-M,0]} |\psi'|^2 K^2$, which proves the desired bound in the first case.

The remaining eigenvalue configuration.

Suppose now that $\lambda_n < -\delta\lambda$, keeping the same constants. Since $f_n \ge \mathcal{F}/n$ and $|\chi + W| \le C\sqrt{K}$,

$$\frac{E}{6K} \ge \frac{f_n}{6K}|v_{n\bar{n}}|^2 \ge \frac{\delta^2\lambda^2}{12nK}\mathcal{F} - C\mathcal{F}.$$

In (10.32), discard C and the term with ψ''. Since ψ is now fixed, $|\psi' \sum_k f_k v_{k\bar{k}}| \le C\lambda\mathcal{F}$. Using $g_{1\bar{1},k} = a_k + O(\lambda)$ and Young's inequality, we bound the last error term by $\frac{1}{2}\sum_k f_k|a_k|^2/\lambda^2 + C\mathcal{F}$. This leaves

$$0 \ge -\tfrac{3}{2}\sum_k f_k|a_k|^2/\lambda^2 + \frac{\delta^2\lambda^2}{12nK}\mathcal{F} + \phi''\sum_k f_k|V_k|^2 - C\lambda\mathcal{F}.$$

By (10.26) and the scalar inequality $\frac{3}{2}|a+b|^2 \le 2|a|^2 + 6|b|^2$,

$$\tfrac{3}{2}\sum_k f_k|a_k|^2/\lambda^2 \le \phi''\sum_k f_k|V_k|^2 + CK\mathcal{F}.$$

Since $\lambda \ge K$, combining the last two inequalities and dividing by $\lambda\mathcal{F} > 0$ gives $\lambda/K \le C$. This finishes both cases. Since H attains its maximum at the chosen point, the bound for λ_1/K there gives a bound at all points, and $|\lambda_i| \le (n-1)\lambda_1$ then bounds every entry of the complex Hessian, as asserted. □

10.15 A Liouville theorem for the limiting cone

Rescaling at a point of large gradient removes the background and first-order terms of the Gauduchon equation and leaves a degenerate equation on $\mathbb{C}^n$, whose solutions are maximal for a cone of matrices. We show that its bounded Lipschitz solutions are constant. The proof argues by induction on the dimension: if the energy of a solution in some complex direction becomes small on large balls, a limit is independent of one variable; if all directional energies stay positive, they give a strictly admissible comparison function.

On $\mathbb{C}^n$ we use Euclidean coordinates $z = (z_1, \dots, z_n)$, the identity matrix I, the Euclidean balls $B(x, r)$ and Euclidean volume $\mathrm{d}y$. In this section ∇ and Δ denote the Euclidean real gradient and Laplacian, and ∂_ξ denotes differentiation in a complex direction ξ.

For $n \geq 2$ and a Hermitian matrix H, set

$$P_n(H) = \frac{(\operatorname{tr} H)I - H}{n-1}.$$

This is the map P_γ of Section 10.13 for the Euclidean metric. It preserves nonnegativity and satisfies $P_n(I) = I$ and $\operatorname{tr} P_n(H) = \operatorname{tr} H$. For a real function v write $H(v) = (v_{i\bar{j}})$ in Euclidean coordinates. A continuous function v on an open set of $\mathbb{C}^n$ is *admissible* if $P_n(H(v)) \geq 0$ in the sense of matrix-valued distributions. It is *maximal for this cone* if it is admissible and every smooth function h touching v from below at a point satisfies

$$P_n(H(h)) \not> 0$$

at that point. Here touching means that $v - h$ has a local minimum with value zero, which need not be a strict minimum. Maximality is thus the viscosity supersolution condition for the degenerate equation $\det P_n(H(v)) = 0$ on the boundary of the cone, together with distributional admissibility.

Taking the trace shows that admissible functions are subharmonic. Since P_n is linear, convolution preserves admissibility, and so do nonnegative linear combinations and integrals of admissible smooth functions. Translations and unitary changes of coordinates preserve both conditions, as do multiplication by a positive constant and addition of a constant.

Lemma 10.15.1 *Local uniform limits of admissible maximal functions are admissible and maximal. If v is maximal on a bounded open set $D \subset \mathbb{C}^n$, h is smooth and admissible on D, both functions are continuous on $\overline{D}$, and $h \leq v$ on ∂D, then $h \leq v$ on D.*

Proof Let v_j be admissible maximal functions converging locally uniformly to v. Distributional positivity is preserved under this convergence. For maximality, suppose that a smooth function h touches v from below at z_0 and $P_n(H(h)(z_0)) > 0$. Choose a small ball on which $v - h \geq 0$. Replacing h by $h - \delta|z - z_0|^2$, with $\delta > 0$ sufficiently small, keeps its matrix strictly positive near z_0 and makes the minimum strict. Uniform convergence then produces interior minimum points of $v_j - h + \delta|z - z_0|^2$ tending to z_0. Adding the minimum value to the test function gives a function touching v_j from below with strictly positive matrix, a contradiction.

For comparison, choose R with $\overline{D} \subset \{|z| \leq R\}$. If $h > v$ somewhere, then for small $\delta > 0$ the smooth function

$$h_\delta = h + \delta(|z|^2 - R^2)$$

still exceeds v at some interior point and is at most v on the boundary. Hence the negative minimum of $v - h_\delta$ is attained in the interior. But $P_n(H(h_\delta)) = P_n(H(h)) + \delta I > 0$ there, which contradicts maximality after the minimum value is added to h_δ. No boundary regularity of D is needed for this argument. □

Theorem 10.15.1 *A bounded globally Lipschitz function on $\mathbb{C}^n$, $n \geq 2$, that is maximal for this cone is constant.*

Proof Let v be a bounded globally Lipschitz maximal function on $\mathbb{C}^n$. We argue by induction on n. For $n = 2$, $P_2(H) \geq 0$ is equivalent to $H \geq 0$, so v is plurisubharmonic, and Lemma 1.3.5 shows that it is constant.

We first explain the reduction in dimension for $n \geq 3$. If a function is independent of z_n, its complex Hessian has the block form $H \oplus 0$. The first block of $P_n(H \oplus 0)$ is

$$\frac{n-2}{n-1} P_{n-1}(H),$$

and the last entry is $\operatorname{tr} H/(n-1)$. Nonnegativity of $P_{n-1}(H)$ implies $\operatorname{tr} H \geq 0$, and positive definiteness implies $\operatorname{tr} H > 0$. Thus $P_n(H \oplus 0)$ is nonnegative, respectively positive definite, exactly when $P_{n-1}(H)$ is. The distributional version follows from the same linear identities. A test function in $n-1$ variables extends to $\mathbb{C}^n$ independently of z_n, so maximality also passes to $n-1$ variables. Consequently the induction hypothesis applies to bounded Lipschitz maximal functions that are independent of one complex variable.

Suppose now that v is nonconstant. Adding a constant and multiplying by a positive constant, we may assume that $\inf v = 0$ and $\sup v = 1$. Let L be the Lipschitz constant of v. Choose a smooth radial nonnegative probability kernel η, supported in the closed unit ball and strictly positive in its interior. For a bounded measurable function f and $r > 0$, write

$$[f]_r(x) = \int_{\mathbb{C}^n} f(x+y)\eta_r(y)\,\mathrm{d}y, \qquad \eta_r(y) = r^{-2n}\eta(y/r), \qquad v_\epsilon = [v]_\epsilon.$$

Both v and v^2 are subharmonic. For v^2, use $v_\epsilon \geq 0$ and $\Delta(v_\epsilon^2) = 2v_\epsilon \Delta v_\epsilon + 2|\nabla v_\epsilon|^2 \geq 0$, and pass to the uniform limit as $\epsilon \to 0$.

For $f = v$ and for $f = v^2$, the averages $[f]_r(x)$ increase with r, by the monotonicity of spherical means of subharmonic functions. Their limit as $r \to \infty$ is independent of x, since

$$|[f]_r(x) - [f]_r(y)| \leq \|f\|_\infty \|\eta_r(\cdot - x) - \eta_r(\cdot - y)\|_{L^1} \leq C\frac{|x-y|}{r}.$$

By the submean inequality this limit is at least $f(x)$ for every x, while it is at most $\sup f = 1$. Therefore

$$[v]_r(x) \longrightarrow 1, \qquad [v^2]_r(x) \longrightarrow 1. \tag{10.35}$$

We distinguish two cases. First suppose that there exist $\rho > 0$, sequences $\epsilon_k \downarrow 0$ and $r_k \to \infty$, points x_k and unit complex directions ξ_k such that

$$[v^2]_{r_k}(x_k) + [v]_\rho(x_k) - 2v(x_k) \geq \frac{4}{3}, \qquad \int_{B(x_k, r_k)} |\partial_{\xi_k} v_{\epsilon_k}|^2\,\mathrm{d}y \longrightarrow 0. \tag{10.36}$$

Translate x_k to the origin and make ξ_k the last complex direction by a unitary change of coordinates. The resulting functions have the same bounds and Lipschitz constant, so by the Arzelà–Ascoli theorem and Lemma 10.15.1 a subsequence converges locally

uniformly to a maximal function a. The corresponding mollifications at scale ϵ_k have the same local uniform limit, and by the integral condition their derivatives in the last complex direction tend to zero in L^2 on every fixed ball. Hence $\partial a/\partial z_n = 0$ in the sense of distributions. Since a is real and continuous, it is independent of z_n.

The dimension reduction and the induction hypothesis show that a is a constant $c \in [0,1]$. In the first inequality of (10.36), the first term is at most one, while the average at the fixed radius ρ and the value at the origin both tend to c. It would follow that $4/3 \le 1 - c \le 1$, a contradiction.

In the remaining case, for every fixed $\rho > 0$ there is $C_\rho \ge 1$ such that, whenever $0 < \epsilon < C_\rho^{-1}$, $r > C_\rho$, and

$$[v^2]_r(x) + [v]_\rho(x) - 2v(x) \ge \frac{4}{3},$$

one has

$$\int_{B(x,r)} |\partial_\xi v_\epsilon|^2 \, dy \ge C_\rho^{-1} \quad \text{for every unit complex direction } \xi. \tag{10.37}$$

Indeed, if this failed for $C_\rho = k$ for every integer k, a diagonal choice would give sequences as in (10.36).

After a translation we may assume that $v(0) < 1/12$. By (10.35), choose ρ with $[v]_\rho(0) > 3/4$, and then $r > C_\rho$ with $[v^2]_r(0) > 3/4$. The open set

$$D = \{x : 2v(x) < [v^2]_r(x) + [v]_\rho(x) - 4/3\}$$

contains the origin. We claim that, for some $c > 0$,

$$[v^2]_r - c|z|^2$$

is smooth and admissible on D.

To prove the claim, differentiate the square of the smooth mollification:

$$H(v_\epsilon^2) = 2v_\epsilon H(v_\epsilon) + 2(\partial_i v_\epsilon \, \overline{\partial_j v_\epsilon})_{ij}.$$

Denote by A_ϵ and B_ϵ the convolutions at the fixed radius r of these two terms. The first satisfies $P_n(A_\epsilon) \ge 0$, because $v_\epsilon \ge 0$, v_ϵ is admissible, and P_n is linear.

The second satisfies $B_\epsilon \ge c_0 I$ on D for small ϵ, with a constant $c_0 > 0$. Here the passage from the unweighted integral in (10.37) to the convolution needs care. The bound $|\nabla v_\epsilon| \le L$ makes the energy of a sufficiently thin boundary annulus of $B(x,r)$ at most $C_\rho^{-1}/2$, uniformly in x, ϵ, ξ. On the remaining smaller closed ball, the fixed kernel η_r has a positive minimum. For points $x \in D$, (10.37) then gives a uniform positive lower bound for the weighted integral in every direction, which is the asserted matrix inequality for B_ϵ.

Choose $0 < c \le c_0$. Since P_n preserves nonnegative matrices,

$$P_n\big(H([v_\epsilon^2]_r) - cI\big) \ge 0 \quad \text{on } D.$$

As $\epsilon \downarrow 0$, $v_\epsilon^2 \to v^2$ uniformly, and convolution at the fixed positive radius r gives convergence of all derivatives. This proves the claim.

The smooth function

$$h = \tfrac{1}{2}\bigl([v^2]_r - c|z|^2 + [v]_\rho - 4/3\bigr)$$

is therefore admissible on D. The set $D' = \{v < h\}$ contains the origin and lies in D. It is bounded, because $v \geq 0$ while $h \to -\infty$ at infinity, the averages being bounded. Moreover $v = h$ on $\partial D'$. Lemma 10.15.1, applied on D', gives $h \leq v$ there, which contradicts the definition of D'. This excludes the second case and proves the theorem. □

The Lipschitz bound is used both for the compactness of translates and for the annulus estimate. For $n > 1$, the conclusion would not follow from boundedness and subharmonicity alone, since $\mathbb{C}^n = \mathbb{R}^{2n}$ carries nonconstant bounded subharmonic functions.

10.16 The gradient bound by rescaling

If the gradients of a family of solutions were unbounded, rescaling at points of maximal gradient would produce functions on growing Euclidean balls with gradient one at the origin. The Hessian estimate makes these functions compact in C^1 on every fixed ball, so a limit is a bounded Lipschitz function on $\mathbb{C}^n$ with nonzero gradient at the origin. The rescaled determinant tends to zero, so the limit is maximal for the cone of Section 10.15, which contradicts Theorem 10.15.1.

Consider a family of smooth solutions (u, b) of the equation in Theorem 10.10.1, normalized by $\sup_X u = 0$, with $\|F\|_{C^0} \leq M$ for a fixed $M > 0$. By Proposition 10.11.1, $\|u\|_{C^0}$ and $|b|$ are uniformly bounded. The following proposition shows that a Hessian estimate in terms of the squared gradient implies bounds for the gradient and the complex Hessian.

Proposition 10.16.1 *Suppose there is a constant K such that every solution in this family satisfies*

$$\sup_X |\mathrm{dd^c} u|_\gamma \leq K\bigl(1 + \sup_X |\partial u|_\gamma^2\bigr). \tag{10.38}$$

Then $\sup_X |\partial u|_\gamma$ and $\sup_X |\mathrm{dd^c} u|_\gamma$ are bounded by a constant depending only on X, γ_0, γ, M and K.

Proof We use the matrices Q_u of (10.19). Their background term Q_0 is fixed, and Z is linear in the first derivatives of $u/(2\pi)$.

Suppose that the gradient bound fails. Then there are solutions (u_j, b_j) with prescribed functions F_j such that

$$A_j = \sup_X |\partial u_j|_\gamma = |\partial u_j(x_j)|_\gamma \longrightarrow \infty.$$

After passing to a subsequence, the points x_j lie in one fixed coordinate neighborhood. Translate the coordinates so that x_j becomes the origin, and apply a uniformly controlled complex linear change of coordinates so that the matrix of γ at x_j is the identity; a coordinate ball of fixed radius remains available around every x_j. On balls whose radii tend to infinity define

$$v_j(z) = u_j(x_j + z/A_j),$$

in these coordinates. The coefficient matrices $g_j(z)$ of γ at the points $x_j + z/A_j$ converge smoothly to the identity on every fixed ball. In this notation

$$|v_j| \le C, \qquad |\partial v_j|_{g_j} \le 1, \qquad |\partial v_j(0)| = 1.$$

The assumed estimate (10.38) gives a uniform bound for all complex Hessian entries of v_j on every fixed ball, since two derivatives contribute the factor A_j^{-2}. Hence the Euclidean Laplacians of the v_j are uniformly bounded, and the interior estimates for the Laplacian give local $W^{2,p}$ bounds for every finite p. Choosing $p > 2n$ and using Sobolev embedding and compactness, we extract a diagonal subsequence converging in C^1 on compact subsets to a function v on $\mathbb{C}^n$. This function is bounded and globally Lipschitz, and

$$|\partial v(0)| = 1.$$

Now divide the matrix in (10.19) by A_j^2 and evaluate it at the scaled points. The background term becomes $O(A_j^{-2})$, and the first-order term becomes $O(A_j^{-1})$, since one derivative of u_j is A_j times a derivative of v_j. Since the complex Hessians of v_j are uniformly bounded and $g_j \to I$, we obtain on any fixed ball

$$A_j^{-2} Q_{u_j} = P_n\left(\frac{H(v_j)}{2\pi}\right) + o(1), \qquad \det(A_j^{-2} Q_{u_j}) = A_j^{-2n} \mathrm{e}^{(n-1)(F_j + b_j)} \longrightarrow 0. \tag{10.39}$$

To be precise, we compute in the unitary frames obtained from the positive square roots $S_j = g_j^{-1/2}$. In these frames the principal matrix is $P_n(S_j H(v_j) S_j/(2\pi))$, which differs from $P_n(H(v_j)/(2\pi))$ by an error tending to zero uniformly; no derivatives of the frames are taken, and the determinant identity is exact in unitary frames. Since the matrices on the left of (10.39) are positive and the positive factor $1/(2\pi)$ does not change the cone, passing to the limit in the sense of distributions gives $P_n(H(v)) \ge 0$.

It remains to check maximality. Suppose that a smooth function h touches v from below at z_0 and $P_n(H(h)(z_0)) > 0$. Subtract $\delta|z - z_0|^2$ with $\delta > 0$ small; on a small ball this makes the contact strict and preserves strict positivity of the matrix. Local uniform convergence gives points $z_j \to z_0$ at which v_j minus this perturbed test function has an interior local minimum. At such a minimum,

$$H(v_j)(z_j) \ge H(h)(z_j) - \delta I.$$

Because P_n preserves nonnegative matrices, (10.39) now implies $A_j^{-2} Q_{u_j}(z_j) \ge cI$ for a fixed $c > 0$ and all sufficiently large j. The determinant of this matrix is then at

least c^n, which contradicts the limit zero in (10.39). Thus v is maximal for the cone of Section 10.15.

By Theorem 10.15.1, v is constant, which contradicts $|\partial v(0)| = 1$. This proves the uniform gradient bound, and substitution in (10.38) gives the uniform complex Hessian bound. □

For prescribed functions F bounded in C^2, Proposition 10.11.1 bounds u and b, and Theorem 10.14.1, applied to $v = u/(2\pi)$ and $h = (n-1)(F+b)$, proves (10.38). Thus both the gradient and the complex Hessian are uniformly bounded.

10.17 Existence for the Gauduchon ansatz

The potential, gradient and Hessian bounds put the matrices Q_u in a compact set of positive matrices. This yields uniform ellipticity after the real Hessian is projected to its complex part and transformed by P_n, so local elliptic regularity gives the compactness needed for closedness in the continuity method; the adjoint probability measure gives openness. The maximum principle then proves uniqueness within the scalar ansatz.

Lemma 10.17.1 *If the prescribed functions F range over a set of smooth functions bounded in every C^k norm, then the smooth solutions (u, b) of the equation in Theorem 10.10.1 with $\sup_X u = 0$ are bounded in every C^k norm.*

Proof Work with $v = u/(2\pi)$ and write $Q = Q_u$ for the matrix in (10.19). By Propositions 10.11.1 and 10.16.1 and Theorem 10.14.1, the function v, its gradient and its complex Hessian are uniformly bounded, and so are the constants b. Hence the eigenvalues of Q are uniformly bounded above, and by the equation its determinant is uniformly bounded below. Every eigenvalue is therefore also bounded below by a positive constant. Choose fixed numbers $0 < a < b_0$ such that $aI \le Q \le b_0 I$ in unitary frames for every solution.

The function $\Delta_\gamma v = \operatorname{tr}_\gamma(\mathrm{i}\partial\bar\partial v)$ is uniformly bounded, and Δ_γ is a fixed uniformly elliptic operator. Its interior $W^{2,p}$ estimates, applied on a finite cover, bound v in $W^{2,p}$ for every finite p. Choose $p > 2n$ and $0 < \alpha < 1 - 2n/p$; Sobolev embedding gives a uniform $C^{1,\alpha}$ bound. Thus the matrix $R(x) = Q_0(x) + Z_x(\partial v(x))$ is uniformly bounded in C^α.

On a coordinate ball, let $\mathcal{H}_x(A)$ denote the Hermitian matrix of $\mathrm{i}\partial\bar\partial v$ at x in a fixed smooth unitary frame, expressed as a function of the real Hessian $A = D^2_{\mathbb{R}} v(x)$. It is a real linear function of A depending smoothly on x. If a real symmetric matrix N is nonnegative, then $\mathcal{H}_x(N) \ge 0$, and its trace is comparable to $\operatorname{tr}_{\mathbb{R}} N$ with uniform positive constants on the ball. This follows by evaluating on a vector and its rotation by i, or from the coordinate formula for $v_{i\bar\jmath}$.

Freeze the function $\partial v(x)$ already obtained in $R(x)$, and define for all real symmetric matrices A

$$\widetilde{\Phi}(A, x) = \inf_{aI \le B \le b_0 I} \left\{ \log\det B + \operatorname{tr}\left[B^{-1}\left(R(x) + P_n(\mathcal{H}_x(A)) - B \right) \right] \right\}.$$

The infimum is over a fixed compact set of positive Hermitian matrices, and each expression in braces is an affine function of A, so $\widetilde{\Phi}$ is concave in A. If $R(x) + P_n(\mathcal{H}_x(A))$ lies in that compact set, then $\widetilde{\Phi}(A, x)$ equals its logarithmic determinant: each expression in braces is a tangent affine function of the concave logarithmic determinant, and the tangent at that matrix itself attains equality.

For $N \geq 0$, each of these affine functions increases by

$$\operatorname{tr}\bigl(B^{-1} P_n(\mathcal{H}_x(N))\bigr)$$

when A is replaced by $A + N$. Since P_n preserves positivity and trace, this increment lies between fixed positive multiples of $\operatorname{tr}_{\mathbb{R}} N$, uniformly in B and x. Taking infima preserves both inequalities, so $\widetilde{\Phi}$ is uniformly elliptic on all real symmetric matrices.

Smoothness of $\mathcal{H}_x$ and the uniform C^α bound for R also give

$$|\widetilde{\Phi}(A, x) - \widetilde{\Phi}(A, y)| \leq C|x - y|^\alpha (1 + \|A\|).$$

These are the hypotheses of the local Evans–Krylov estimate for concave uniformly elliptic equations with Hölder dependence on x, in the formulation cited in Section 36.6. The value $\widetilde{\Phi}(0, x)$ is uniformly bounded, by the compactness of the set of matrices B and the bound for R. By the choice of a and b_0, the given smooth function solves

$$\widetilde{\Phi}(D^2_{\mathbb{R}} v, x) = (n - 1)(F + b)$$

on the ball. The Evans–Krylov estimate, applied on smaller balls of a finite cover, yields a uniform $C^{2,\beta}$ bound for some $\beta > 0$.

Differentiating the original matrix equation gives linear uniformly elliptic equations for the first derivatives of v, whose coefficients, including the first-order coefficients coming from Z, are now Hölder continuous. Local Schauder estimates give $C^{3,\beta}$ bounds, and differentiation and induction give all higher bounds. The finite coordinate cover makes the bounds global. The only regularity results used here are the standard $W^{2,p}$ estimates for the Laplacian, the Evans–Krylov estimate for concave equations, and the Schauder estimates. □

Proof (of Theorem 10.10.1) Let

$$F_0 = \log \frac{\gamma_0^n}{\gamma^n}, \qquad F_t = (1 - t)F_0 + tF, \quad 0 \leq t \leq 1.$$

We use the continuity method along the path F_t. At $t = 0$, the pair $u = 0$, $b = 0$ solves the equation, since $\eta_0 = \gamma_0$. During the continuity argument we normalize potentials by $\int_X u\, \gamma^n = 0$; adding a constant does not affect the equation, and at the end we impose $\sup_X u = 0$ instead.

For openness, the derivative of the equation with respect to the mean-zero potential and the scalar is $(w, a) \mapsto \mathcal{L}_u w - a$. As in the proof of Proposition 10.3.1, with the adjoint probability measure of Proposition 10.12.1 in place of μ_u, it is invertible between Hölder spaces with bounded inverse. Since positivity of Γ_u is an open condition, the implicit function theorem gives solutions for all nearby parameters.

For closedness, note that the functions F_t are uniformly bounded in every C^k norm. After supremum normalization, Proposition 10.11.1 bounds u and b, the second-order and rescaling estimates bound the gradient and the complex Hessian, and Lemma 10.17.1 gives all higher bounds. Subtracting the mean preserves all derivative bounds and changes the potentials by uniformly bounded constants. A subsequence of mean-zero solutions along a convergent sequence of parameters therefore converges smoothly to a solution at the limiting parameter, and the uniform lower bound for the eigenvalues of Q_u obtained in the proof of Lemma 10.17.1 preserves strict positivity in the limit. Thus the set of solvable parameters is nonempty, open and closed, hence equal to $[0, 1]$.

Finally, consider two solutions (u, b) and (w, c) for the same F. At a maximum of $u - w$ their gradients agree and their complex Hessians are ordered, so the first-order terms are equal there and, since P_n preserves the order, $Q_u \le Q_w$. The equation then gives $b \le c$, and at a minimum the reverse inequality gives $b \ge c$. Hence $b = c$. Along the segment between u and w, the matrices Q remain positive because the ansatz is affine in the potential. Integrating the derivative of the logarithmic determinant along that segment produces a uniformly elliptic scalar operator with first-order terms and no zeroth-order term that annihilates $u - w$. The maximum principle shows that $u - w$ is constant, and the supremum normalization makes the constant zero. □

10.18 Normalization for the Gauduchon equation

By Theorem 10.10.1, every smooth real function F determines a scalar $b_\gamma(F)$, the constant b of that theorem. As for the Monge–Ampère equation, its variation is governed by the adjoint probability measure μ_u^γ of Proposition 10.12.1 and by the concavity of the logarithmic determinant. To differentiate solutions, we fix a smooth positive volume form $\mathrm{d}V_0$ and use the normalization $\int_X u \, \mathrm{d}V_0 = 0$ in place of $\sup_X u = 0$.

Lemma 10.18.1 *The solution $(u, b_\gamma(F))$, normalized by $\int_X u \, \mathrm{d}V_0 = 0$, depends smoothly on the parameters of every smooth finite-dimensional family of functions F.*

Proof The map $\Gamma \mapsto \eta$ with $\eta^{n-1} = \Gamma$ is smooth on positive forms, since in a frame it is the inverse of the smooth map $\eta \mapsto \eta^{n-1}$, whose differential is invertible. The openness argument in the proof of Theorem 10.10.1, with the $\mathrm{d}V_0$-mean in place of the γ^n-mean, gives a bounded inverse for the derivative with respect to the potential and the scalar between the relevant Hölder spaces. The implicit function theorem gives local smooth solution maps, and uniqueness shows that they agree on overlaps. Applying this for every order of Hölder regularity gives the assertion for smooth data. □

For a smooth family of prescribed functions $F_t \in C^\infty(X, \mathbb{R})$, with t in an open interval of $\mathbb{R}$, let u_t be the corresponding solutions with $\int_X u_t \, dV_0 = 0$, and set $Q_t = Q_{u_t}$ as in (10.19). The expression

$$\operatorname{tr}\big((Q_t^{-1}\dot{Q}_t)^2\big)$$

is independent of the chosen frame and nonnegative, because $Q_t^{-1}\dot{Q}_t$ is similar to the Hermitian matrix $Q_t^{-1/2}\dot{Q}_t Q_t^{-1/2}$.

Theorem 10.18.1 *For a smooth family F_t and its solutions,*

$$\dot{b}_\gamma(F_t) = -\int_X \dot{F}_t \, d\mu_{u_t}^\gamma,$$
$$\ddot{b}_\gamma(F_t) = -\int_X \ddot{F}_t \, d\mu_{u_t}^\gamma - \frac{1}{n-1}\int_X \operatorname{tr}\big((Q_t^{-1}\dot{Q}_t)^2\big) \, d\mu_{u_t}^\gamma.$$

In particular, b_γ is concave, and along $F_t = F_0 + tG$ its second derivative is strictly negative unless G is constant. Moreover, for smooth real functions F, G on X and every real constant a,

$$b_\gamma(F + a) = b_\gamma(F) - a, \qquad |b_\gamma(G) - b_\gamma(F)| \le \|G - F\|_\infty.$$

Proof By (10.19),

$$\mathcal{G}(u) = \frac{1}{n-1}\log\det_\gamma Q_u.$$

Differentiating $\mathcal{G}(u_t) = F_t + b_\gamma(F_t)$ once gives $\mathcal{L}_{u_t}\dot{u}_t = \dot{F}_t + \dot{b}_\gamma(F_t)$, and integration against the adjoint probability measure gives the first formula.

Because Γ_u is affine in u, its second derivative along the family is $\mathcal{D}_\gamma \ddot{u}_t$. The formula for the second derivative of $\log\det$ used in the proof of Theorem 10.7.1 therefore gives

$$\mathcal{L}_{u_t}\ddot{u}_t - \frac{1}{n-1}\operatorname{tr}\big((Q_t^{-1}\dot{Q}_t)^2\big) = \ddot{F}_t + \ddot{b}_\gamma(F_t).$$

Integrating against $\mu_{u_t}^\gamma$ at this fixed parameter gives the second formula; no derivative of the measure is needed.

For affine F_t, we have $\ddot{F}_t = 0$, and positivity of the density of $\mu_{u_t}^\gamma$ proves concavity. If the second derivative vanishes, then $\dot{Q}_t = 0$, so $\dot{\Gamma}_{u_t} = 0$ and $\mathcal{L}_{u_t}\dot{u}_t = 0$. By Proposition 10.12.1, $\dot{u}_t$ is constant, and the first differentiated equation forces G to be constant. Conversely, adding a constant to F changes only b, by the opposite constant. Finally, integrate the first derivative along the segment from F to G: since $\mu_{u_t}^\gamma$ is a probability measure, the derivative lies between $-\sup_X(G - F)$ and $-\inf_X(G - F)$, which proves the Lipschitz bound. □

The Lipschitz bound extends b_γ uniquely to continuous functions F. Constructing corresponding weak solutions would require, in addition, a volume measure for the $(n-1)$st roots of singular forms in the Aeppli ansatz, compactness of their potentials, and stability of the adjoint equation.

On a surface with Gauduchon background form γ_0, conservation of the total mass makes the scalar and the adjoint probability measure explicit.

Example 10.18.1 If $n = 2$ and γ_0 is Gauduchon, then $\Gamma_u = \gamma_0 + \mathrm{dd}^{\mathrm{c}} u = \eta_u$, and η_u is also Gauduchon. The mass identity of Theorem 2.2.2 gives $\int_X \eta_u^2 = \int_X \gamma_0^2$. Therefore

$$b_\gamma(F) = \log \int_X \gamma_0^2 - \log \int_X \mathrm{e}^F \gamma^2, \qquad \mathrm{d}\mu_u^\gamma = \frac{\mathrm{e}^F \gamma^2}{\int_X \mathrm{e}^F \gamma^2}.$$

The adjoint equation of Proposition 10.12.2 gives the same measure, because $\mathrm{dd}^{\mathrm{c}} \eta_u = 0$ and hence ρ_u is constant. Along $F + tG$, the second derivative of b_γ is minus the variance of G with respect to this probability measure.

In Chapter 11 we return to the Monge–Ampère equation (10.1) for $\omega + \mathrm{dd}^{\mathrm{c}} u$. Smooth approximation will extend its solvability to nonnegative, nonzero L^p densities, $p > 1$. Control of the normalization constant and uniform convergence of the potentials will replace the derivative estimates used here.

Chapter 11
Continuous solutions and normalization

Throughout this chapter, X is a connected compact complex manifold of dimension $n \geq 1$, ω is a Hermitian form and $\mathrm{d}V$ is a smooth positive volume form on X. As in Chapter 10, we write $\omega_u = \omega + \mathrm{dd}^{\mathrm{c}} u$; for a bounded ω-plurisubharmonic function u, the power ω_u^n is the Bedford–Taylor product of Proposition 1.7.1. We fix $p > 1$ and a nonnegative $f \in L^p(X, \mathrm{d}V)$ with $\int_X f \, \mathrm{d}V > 0$. Norms and almost-everywhere statements refer to $\mathrm{d}V$ unless another measure is specified.

We seek continuous ω-plurisubharmonic solutions of $\omega_u^n = c f \, \mathrm{d}V$, with an unknown constant $c > 0$, and of $\omega_u^n = \mathrm{e}^{\lambda u} f \, \mathrm{d}V$, with $\lambda > 0$. Smooth positive approximations of f give solutions by Chapter 10. To pass to a limit, we need uniform control of both their constants and their potentials. Integration against a Gauduchon metric bounds the constants from above, and hence bounds the L^p norms of the resulting densities. Modified comparison then turns normalized L^1 compactness into uniform compactness, which identifies the limiting Monge–Ampère measure.

For the exponential equation, a weighted Gauduchon mean also controls the additive constants. Letting λ decrease to zero determines the normalization constant c from f, even when the normalized potential is not unique. When the density of a bounded solution is a smooth positive function of the point and the potential, we replace the potential in the density by smooth approximations. The resulting exponential equations satisfy a uniform trace estimate, and their solutions converge uniformly to the given one; elliptic regularity then shows that it is smooth.

On the Hopf threefold, the constant has a sharp lower bound, with equality exactly for densities pulled back from the projective plane. For densities of unit integral with a common L^p bound, $1 < p < \infty$, near equality forces normalized potentials to approach their fiber averages uniformly, and the differences between the densities and their fiber averages to converge weakly to zero in L^p. Smooth oscillating examples show why strong L^1 convergence of the densities need not follow.

11.1 Continuous solvability

There are two natural ways to normalize the prescribed density: by an unknown positive constant, or by an exponential of the potential.

Theorem 11.1.1 (Kołodziej–Nguyen) *There are $c > 0$ and $u \in \mathrm{PSH}(X, \omega) \cap C(X)$ such that*

$$(\omega + \mathrm{dd^c} u)^n = c f \, \mathrm{d}V.$$

The potential can be normalized by $\sup_X u = 0$.

The constant c is uniquely determined by f, by Proposition 11.6.1. The normalized potential, however, need not be unique: Theorem 12.6.1 gives a Hermitian form on $\mathbb{P}^3$ and a smooth density, vanishing on a nonempty open set, with two distinct normalized solutions.

Theorem 11.1.2 (Nguyen) *For every $\lambda > 0$, there is a unique bounded ω-plurisubharmonic function u_λ satisfying*

$$(\omega + \mathrm{dd^c} u_\lambda)^n = \mathrm{e}^{\lambda u_\lambda} f \, \mathrm{d}V, \tag{11.1}$$

and u_λ is continuous.

11.2 The modified comparison principle

For a Hermitian background, the total Monge–Ampère mass varies with the potential, and the ordinary comparison principle fails in general (see Theorem 12.5.2). A weaker form survives. Replacing the comparison potential v by $(1 - \varepsilon)v$ leaves the strictly positive form $\varepsilon\omega$ in reserve, and this reserve absorbs the torsion terms. On a sublevel set of height s, integration by parts makes the error proportional to s, and a finite induction controls the intermediate mixed products. The resulting multiplicative correction tends to one as s tends to zero with ε fixed.

Since X is compact and ω is positive, we can choose $B_0 \geq 1$ such that

$$-B_0\omega^2 \leq \mathrm{dd^c}\omega \leq B_0\omega^2, \qquad -B_0\omega^3 \leq \mathrm{d}\omega \wedge \mathrm{d^c}\omega \leq B_0\omega^3.$$

These inequalities are understood after wedging with positive forms of complementary degree; forms of degree larger than $2n$ vanish, and the corresponding terms below are omitted.

Theorem 11.2.1 *Let $u, v \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$ and $0 < \varepsilon < 1$. For $s > 0$, put*

$$m_\varepsilon = \inf_X \big(u - (1-\varepsilon)v\big), \qquad U_{\varepsilon,s} = \{u < (1-\varepsilon)v + m_\varepsilon + s\}.$$

If $0 < s < \varepsilon^3/(32B_0n^2)$, then, with $a = B_0n^2s\varepsilon^{-3}$,

$$(1-4a)^n \int_{U_{\varepsilon,s}} \omega^n_{(1-\varepsilon)v} \le \int_{U_{\varepsilon,s}} \omega_u^n. \tag{11.2}$$

Proof Write $v_\varepsilon = (1-\varepsilon)v$, $U = U_{\varepsilon,s}$ and

$$\phi = \max(u, v_\varepsilon + m_\varepsilon + s), \qquad T_k = \omega_u^k \wedge \omega_\phi^{n-k}, \qquad I_k = \int_U T_k \quad (0 \le k \le n).$$

Set $I_{-1} = I_{-2} = 0$. Then $0 \le \phi - u \le s$, this difference vanishes outside U, and on U plurifine locality gives $\omega_\phi = \omega_{v_\varepsilon} \ge \varepsilon\omega$ in all bounded products.

By Lemma 1.7.1, for every fixed positive mixed product W of $n-1$ bounded-potential factors,

$$\mathbf{1}_{\{u=\phi\}}\omega_u \wedge W \le \mathbf{1}_{\{u=\phi\}}\omega_\phi \wedge W.$$

For $0 \le k \le n-1$, apply this with $W = \omega_u^k \wedge \omega_\phi^{n-k-1}$. Since $u = \phi$ on $X \setminus U$, the difference $T_k - T_{k+1} = (\omega_\phi - \omega_u) \wedge W$ is nonnegative there. Consequently

$$I_k - I_{k+1} \le \int_X \mathrm{dd^c}(\phi - u) \wedge W = \int_X (\phi - u)\mathrm{dd^c}W. \tag{11.3}$$

This integration by parts is valid for bounded potentials. To see this, approximate u and ϕ by decreasing sequences of smooth, uniformly bounded ω-plurisubharmonic functions, using Theorem 1.11.1. For smooth functions the identity is ordinary integration by parts. The form $\mathrm{dd^c}W$ expands as below into bounded Hessian products and smooth forms, and bounded Bedford–Taylor convergence, also against bounded quasi-continuous coefficients converging in capacity, passes each term to the limit.

Set $r = n-k-1$. Since $\mathrm{d}\omega_u = \mathrm{d}\omega_\phi = \mathrm{d}\omega$, expansion gives

$$\begin{aligned}\mathrm{dd^c}W = {}& k\,\mathrm{dd^c}\omega \wedge \omega_u^{k-1} \wedge \omega_\phi^r + r\,\mathrm{dd^c}\omega \wedge \omega_u^k \wedge \omega_\phi^{r-1} \\ & + k(k-1)\mathrm{d}\omega \wedge \mathrm{d^c}\omega \wedge \omega_u^{k-2} \wedge \omega_\phi^r \\ & + 2kr\,\mathrm{d}\omega \wedge \mathrm{d^c}\omega \wedge \omega_u^{k-1} \wedge \omega_\phi^{r-1} \\ & + r(r-1)\mathrm{d}\omega \wedge \mathrm{d^c}\omega \wedge \omega_u^k \wedge \omega_\phi^{r-2}.\end{aligned}$$

Terms containing a negative power have coefficient zero and are omitted. On U we may replace each ω^j in an upper bound by $\varepsilon^{-j}\omega_\phi^j$, since $\omega_\phi \ge \varepsilon\omega$ there. Using $\phi - u \le s$ and $\varepsilon^{-2} \le \varepsilon^{-3}$ in (11.3), we obtain

$$I_k - I_{k+1} \le B_0 s\varepsilon^{-3}\big(k(k-1)I_{k-2} + k(1+2r)I_{k-1} + r^2 I_k\big).$$

Each coefficient is at most n^2, and $a = B_0n^2s\varepsilon^{-3}$. Hence

$$I_k - I_{k+1} \le a(I_{k-2} + I_{k-1} + I_k). \tag{11.4}$$

Put $q = 1 - 4a$; the assumption on s gives $a < 1/32$, so $q > 7/8$. Suppose inductively that $qI_j \le I_{j+1}$ for $j < k$. Then $I_{k-1} \le q^{-1} I_k$ and $I_{k-2} \le q^{-2} I_k$ whenever the indices exist. Since

$$1 + q^{-1} + q^{-2} \le 1 + 8/7 + 64/49 < 4,$$

(11.4) gives $qI_k \le I_{k+1}$. The case $k = 0$ follows from the same formula with the two absent terms set to zero. By induction, $q^n I_0 \le I_n$. Since $T_0 = \omega_{v_\varepsilon}^n$ on U and $T_n = \omega_u^n$, this is the theorem. In dimension one all torsion terms vanish and the same argument applies. □

The reverse factor $(1 - 4a)^{-n}$ in (11.2) tends to one as $s \downarrow 0$ for fixed ε. A strict multiplicative inequality between the Monge–Ampère measures on $\{u < v\}$ therefore absorbs the error in modified comparison. One first chooses the positivity reserve ε, and then a sublevel set thin enough to preserve the strict inequality.

Theorem 11.2.2 *Let $u, v \in \mathrm{PSH}(X, \omega) \cap L^\infty(X)$, and let $0 \le c < 1$. If*

$$\mathbf{1}_{\{u<v\}} \omega_u^n \le c\, \mathbf{1}_{\{u<v\}} \omega_v^n,$$

then $u \ge v$. In particular, $\omega_u^n(\{u < v\}) = 0$ implies $u \ge v$.

Proof Suppose $m = \inf_X(u - v) < 0$. Choose $\varepsilon > 0$ so small that

$$(1 - \varepsilon)^n > c, \qquad 2\varepsilon \|v\|_\infty < -m/4.$$

Let m_ε and $U_{\varepsilon,s}$ be defined as in Theorem 11.2.1 for this pair u, v. Since $|m_\varepsilon - m| \le \varepsilon\|v\|_\infty$, for all sufficiently small $s > 0$ the set $U = U_{\varepsilon,s}$ is contained in $\{u < v\}$: there

$$u - v < m_\varepsilon + s - \varepsilon v \le m + 2\varepsilon\|v\|_\infty + s < 0.$$

The set U is nonempty by the definition of the infimum and has positive volume by Lemma 1.3.4.

Decrease s further so that Theorem 11.2.1 applies and $(1 - 4B_0 n^2 s\varepsilon^{-3})^n (1-\varepsilon)^n > c$. Write the first factor as q^n. Positivity of every mixed term gives

$$\omega_{(1-\varepsilon)v}^n \ge (1 - \varepsilon)^n \omega_v^n + \varepsilon^n \omega^n.$$

Modified comparison and the assumption would therefore imply

$$q^n (1 - \varepsilon)^n \int_U \omega_v^n + q^n \varepsilon^n \int_U \omega^n \le \int_U \omega_u^n \le c \int_U \omega_v^n.$$

All integrals are finite, so the choice $q^n(1 - \varepsilon)^n > c$ and the positive volume of U make this impossible. Hence $m \ge 0$. □

Corollary 11.2.1 *Every bounded ω-plurisubharmonic function has strictly positive Monge–Ampère mass.*

Proof Let u be a bounded ω-plurisubharmonic function. If $\omega_u^n = 0$, then Theorem 11.2.2, applied with $c = 0$ and the constant potential $v = \sup_X u + 1$, would give $u \geq v$, a contradiction. □

The positive lower bound in Corollary 11.2.1 depends on the potential. A bound that is uniform over all bounded potentials is the condition $v_-(\omega) > 0$ of Definition 7.5.1.

Corollary 11.2.2 *If $u, v \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$ and $\lambda > 0$, then*

$$\mathrm{e}^{-\lambda v}\omega_v^n \geq \mathrm{e}^{-\lambda u}\omega_u^n \quad \Longrightarrow \quad v \leq u.$$

Proof For every $t > 0$, the assumption gives $\omega_u^n \leq \mathrm{e}^{-\lambda t}\omega_v^n$ on $\{u < v - t\}$. Since $\omega_{v-t} = \omega_v$, Theorem 11.2.2, applied to u and $v - t$, gives $u \geq v - t$. Letting $t \downarrow 0$ proves $v \leq u$. □

11.3 Capacity estimates and uniform compactness

Let $\mathrm{Cap} = \mathrm{Cap}_\omega$ denote the capacity of Section 1.8. Modified comparison bounds the capacity of a thin sublevel set by the Monge–Ampère mass of a slightly larger one. If the Monge–Ampère measure is bounded by a power $1 + \alpha$ of the capacity, with $\alpha > 0$, the resulting iteration converges. For an L^p density, Corollary 1.8.1 supplies the case $\alpha = 1$. Iterating the two inequalities gives a lower bound for the capacity of every sublevel set of fixed height above the minimum. Since the capacity of deep sublevel sets of normalized potentials is small by the tail estimate (1.15), this bounds the depth of a normalized solution. Applied to the difference of two solutions, the same estimates give a one-sided stability bound, and hence uniform compactness.

Lemma 11.3.1 *Let $u, v \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$, with $-H \leq v \leq 0$ and $H \geq 1$. Fix $0 < \varepsilon < 1$, and put*

$$m = \inf_X (u - (1-\varepsilon)v), \qquad U_s = \{u < (1-\varepsilon)v + m + s\}.$$

There are $\delta_\varepsilon > 0$, depending only on (X, ω, ε), and $C_H > 0$, depending only on (n, H), such that

$$t^n \,\mathrm{Cap}(U_s) \leq C_H \int_{U_{s+t}} \omega_u^n \qquad (0 < s, t,\ s + t < \delta_\varepsilon). \tag{11.5}$$

Every U_s, $s > 0$, has positive capacity.

Proof Let B_0 be the constant of Theorem 11.2.1, and set

$$\delta_\varepsilon = \min\{1 - \varepsilon, \varepsilon^3/(128 B_0 n^2)\}.$$

For a capacity test $0 \le \rho \le 1$, put

$$\tau = \frac{t}{2(1-\varepsilon)(H+1)}, \quad w = (1-\tau)v + \tau\rho, \quad m' = \inf_X (u - (1-\varepsilon)w).$$

Then $0 < \tau < 1$ and $0 \le (1-\varepsilon)(w-v) \le t/2$, so $m - t/2 \le m' \le m$. Consequently

$$U_s \subset V := \{u < (1-\varepsilon)w + m' + s + t/2\} \subset U_{s+t}.$$

We apply modified comparison to u, w at height $s + t/2 < \delta_\varepsilon$; by the choice of δ_ε, its inverse factor is at most $(8/7)^n$. Moreover,

$$\omega_{(1-\varepsilon)w} \ge (1-\varepsilon)\tau\omega_\rho = \frac{t}{2(H+1)}\omega_\rho.$$

Integrating over $U_s \subset V$, applying modified comparison on V, and taking the supremum over ρ, we obtain (11.5) with $C_H = [2(H+1)]^n (8/7)^n$. Finally, each U_s is nonempty by the definition of the infimum, so it has positive smooth volume by Lemma 1.3.4, and hence positive capacity, as the constant test function shows. □

Lemma 11.3.2 *In the setting of Lemma 11.3.1, suppose that $A, \alpha > 0$ and*

$$\omega_u^n(E) \le A \operatorname{Cap}_\omega(E)^{1+\alpha} \qquad (E \subset X \text{ Borel}).$$

For every $0 < D < \delta_\varepsilon$,

$$D \le C_\alpha (C_H A)^{1/n} \operatorname{Cap}_\omega(U_D)^{\alpha/n}, \qquad C_\alpha = \frac{2}{1-2^{-\alpha}}. \tag{11.6}$$

In particular, when $\alpha = 1$,

$$\operatorname{Cap}(U_D) \ge \frac{D^n}{4^n C_H A}. \tag{11.7}$$

Proof Put $a(s) = \operatorname{Cap}_\omega(U_s)^{1/n} > 0$ and $K = (C_H A)^{1/n}$. The thin-sublevel estimate (11.5) and the hypothesis, applied to $E = U_{s+t}$, give

$$t\, a(s) \le K a(s+t)^{1+\alpha} \qquad (0 < s, t,\ s + t < \delta_\varepsilon).$$

Suppose $C_\alpha K a(D)^\alpha < D$. Starting with $s_0 = D$, set

$$t_j = 2K a(s_j)^\alpha, \qquad s_{j+1} = s_j - t_j.$$

If $a(s_i) \le 2^{-i} a(D)$ for $0 \le i \le j$, then

$$\sum_{i=0}^{j} t_i \le 2K a(D)^\alpha \sum_{i=0}^{j} 2^{-i\alpha} < C_\alpha K a(D)^\alpha < D.$$

Thus $s_{j+1} > 0$, and the sublevel estimate, applied with $s = s_{j+1}$ and $t = t_j$, gives $a(s_{j+1}) \le a(s_j)/2$. Induction justifies every step. The decreasing sequence s_j has a limit

$$s_\infty \ge D - C_\alpha K a(D)^\alpha > 0.$$

Monotonicity now gives $0 < a(s_\infty) \le 2^{-j} a(D)$ for every j, a contradiction. This proves (11.6), including the case in which $U_D = X$. □

Theorem 11.3.1 *There are constants $C_p, c_p > 0$, depending only on $(X, \omega, \mathrm{d}V, p)$, such that, whenever $u \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$, $\sup_X u = 0$, and*

$$\omega_u^n = F\,\mathrm{d}V, \qquad F \in L^p(X, \mathrm{d}V), \quad F \ge 0,$$

we have

$$\|u\|_\infty \le C_p(1 + \|F\|_p), \qquad \|F\|_p \ge c_p. \tag{11.8}$$

Proof By Corollary 11.2.1, F is not almost everywhere zero. Apply Lemma 11.3.2 with $v = 0$, $H = 1$, $\varepsilon = 1/2$, and $A = C\|F\|_p > 0$, where C is the constant of Corollary 1.8.1. Fix $D = \delta_{1/2}/2$, and put $M = -\inf_X u$. Since $U_D = \{u < -M + D\}$, the resulting inequality reads

$$\mathrm{Cap}(\{u < -M + D\}) \ge c/\|F\|_p.$$

If $M > D + 1$, (1.15) bounds this capacity above by $C/(M - D)$, and the first assertion follows; the case $M \le D + 1$ is immediate. Bounding the same capacity above by $\mathrm{Cap}(X)$ gives the second assertion. □

11.3.1 Stability under a power capacity bound

The capacity lower bound of Lemma 11.3.2 turns an integral error, measured against the Monge–Ampère measure of the solution, into a uniform error. The argument uses only the power capacity bound on this measure and therefore applies to singular measures as well.

Proposition 11.3.1 *Fix $A, \alpha > 0$ and $H \ge 1$. Let $u, v \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$ satisfy $\sup_X u = 0$ and $-H \le v \le 0$, and suppose*

$$\mu(E) \le A\,\mathrm{Cap}_\omega(E)^{1+\alpha}, \qquad \mu = \omega_u^n, \qquad E \subset X \text{ Borel}.$$

There is a constant C, depending only on $(X,\omega), A, \alpha, H$, such that

$$\sup_X (v - u)_+ \le C\left(\int_X (v-u)_+\,\mathrm{d}\mu\right)^{\sigma_\alpha}, \qquad \sigma_\alpha = \frac{\alpha}{3n(1+\alpha)+\alpha}. \tag{11.9}$$

For the square-capacity bound, $\alpha = 1$, the exponent is $1/(6n+1)$.

Proof Write $L = \int_X (v-u)_+ \, d\mu$ and $S = \sup_X (v-u)_+$. Apply Lemma 11.3.2 with 0 in place of v, with $H = 1$, $\varepsilon = 1/2$ and one fixed $0 < D < \delta_{1/2}$. Then $U_D = \{u < \inf_X u + D\}$, and (11.6) bounds its capacity below by a positive constant depending only on $(X, \omega), A, \alpha$. The tail estimate (1.15) therefore bounds $-\inf_X u$ in terms of $(X, \omega), A, \alpha$. Since $v \le 0$, we have $S \le -\inf_X u$, so S has a fixed upper bound.

For $0 < \varepsilon \le 1/2$, choose a fixed small $c > 0$ and put $D = c\varepsilon^3$. The formula for δ_ε in Lemma 11.3.1 allows c to be chosen so that $2D < \delta_\varepsilon$ and $4D \le \varepsilon$. With

$$m = \inf_X (u - (1-\varepsilon)v), \qquad U_s = \{u < (1-\varepsilon)v + m + s\},$$

we have $m \le \inf_X (u - v) = -S$ when $S > 0$, since $u - (1-\varepsilon)v \le u - v$. On U_{2D}, therefore,

$$v - u > S - \varepsilon H - 2D.$$

If $S > 2\varepsilon H + 4D$, this is greater than $S/2$. In this case (11.6), (11.5) and Chebyshev's inequality give

$$c_0 D^{n(1+1/\alpha)} \le D^n \operatorname{Cap}_\omega(U_D) \le C_H \mu(U_{2D}) \le \frac{2C_H L}{S},$$

where $c_0 > 0$ depends only on $(X, \omega), A, \alpha, H$. Hence $S \le CL\varepsilon^{-3n(1+1/\alpha)}$ in this case. In the other case, $S \le (2H+1)\varepsilon$.

For sufficiently small $L > 0$, choose $\varepsilon = L^{\sigma_\alpha}$. The two alternatives give (11.9). The remaining positive values of L are covered by the fixed upper bound for S. If $L = 0$, the first alternative is impossible for every small ε, and the second gives $S = 0$ on letting $\varepsilon \downarrow 0$. □

For L^p densities, this stability estimate upgrades L^1 compactness to uniform compactness. Put $p' = p/(p-1)$.

Theorem 11.3.2 *Suppose $u_j \in \mathrm{PSH}(X, \omega) \cap C(X)$, $\sup_X u_j = 0$, and*

$$\omega_{u_j}^n = F_j \, dV, \qquad F_j \ge 0, \qquad B := \sup_j \|F_j\|_p < \infty$$

for the fixed $p > 1$. If (u_j) is Cauchy in $L^1(X, dV)$, then it is Cauchy uniformly. In particular, every such family has a subsequence converging uniformly to a continuous ω-plurisubharmonic function.

Proof By Theorem 11.3.1, there is a common $H \ge 1$ with $-H \le u_j \le 0$. By Corollary 1.8.1, every measure $\omega_{u_j}^n$ satisfies the hypothesis of Proposition 11.3.1 with $\alpha = 1$ and $A = CB$, where C is the constant of Corollary 1.8.1. Since $|u_j - u_k| \le 2H$, Hölder's inequality gives, for $l \in \{j, k\}$,

$$\int_X |u_j - u_k| F_l \, dV \le B \|u_j - u_k\|_{p'} \le (2H)^{1/p} B \|u_j - u_k\|_1^{1/p'}.$$

That proposition, applied to the pairs $(u, v) = (u_k, u_j)$ and (u_j, u_k), therefore gives

$$\|u_j - u_k\|_\infty \leq C' \|u_j - u_k\|_1^{1/((6n+1)p')}$$

with C' independent of j, k. This proves the first assertion. For the last assertion, the normalized compactness theorem, Theorem 1.4.1, gives an L^1-Cauchy subsequence of any such family. Its uniform limit is continuous, and distributional convergence preserves ω-plurisubharmonicity. □

11.4 Construction of continuous solutions

Smooth approximation gives solutions with unknown normalization constants. A Gauduchon trace bounds these constants above, and the lower bound for the density norm in Theorem 11.3.1 bounds them away from zero. The resulting uniform L^p bound puts the potentials in the compact family of Theorem 11.3.2, so their limit solves the prescribed equation.

Proof (of Theorem 11.1.1) Choose smooth strictly positive functions $f_j \to f$ in $L^p(X, \mathrm{d}V)$. By Theorem 10.1.1, there are smooth u_j with $\omega_{u_j} > 0$ and constants c_j such that

$$\omega_{u_j}^n = c_j f_j \, \mathrm{d}V, \qquad \sup_X u_j = 0, \qquad c_j > 0.$$

We first bound the constants. For $n \geq 2$, fix a Gauduchon metric γ. By Proposition 4.2.2,

$$c_j^{1/n} A_j \leq \int_X \omega \wedge \gamma^{n-1}, \qquad A_j = \int_X (f_j \mathrm{d}V/\gamma^n)^{1/n} \gamma^n.$$

Since $|x^{1/n} - y^{1/n}| \leq |x - y|^{1/n}$ for $x, y \geq 0$, Hölder's inequality gives $A_j \to A > 0$, where $A = \int_X (f \mathrm{d}V/\gamma^n)^{1/n} \gamma^n$. Thus c_j is bounded above. In dimension one, Stokes' theorem gives the same estimate, with $A_j = \int_X f_j \, \mathrm{d}V$.

The lower bound for the density norm in Theorem 11.3.1 gives $c_j \|f_j\|_p \geq c_p$, so the constants are bounded away from zero. The densities $c_j f_j$ are therefore uniformly bounded in L^p, and the same theorem gives a common bound for the u_j. By Theorem 1.4.1, we may pass to a subsequence with $c_j \to c > 0$ and u_j Cauchy in L^1. By Theorem 11.3.2, $u_j \to u$ uniformly, with u continuous and $\sup_X u = 0$.

Bounded Bedford–Taylor continuity gives $\omega_{u_j}^n \rightharpoonup \omega_u^n$, while $c_j f_j \to cf$ in L^p. Hence $\omega_u^n = cf \, \mathrm{d}V$, as required. □

For the exponential equation, the role of the normalization constant is played by $\mathrm{e}^{\lambda \sup_X u}$. Its upper bound comes from Jensen's inequality: weighting the Gauduchon volume by the nth root of the density relates $\sup_X u$ to the Gauduchon trace of ω_u, which does not depend on u. The lower bound for the density norm again gives the reverse bound.

Lemma 11.4.1 *Fix $\lambda > 0$. Suppose that smooth functions u_j with $\omega_{u_j} > 0$ solve*

$$\omega_{u_j}^n = \mathrm{e}^{\lambda u_j} f_j \, \mathrm{d}V,$$

where the smooth strictly positive functions f_j converge in L^p to a nonnegative nonzero f. Then $M_j = \sup_X u_j$ is bounded above and below, uniformly in j.

Proof Absorbing the fixed smooth positive ratio $\mathrm{d}V/\omega^n$ into f_j and f preserves their L^p convergence, so we may assume $\mathrm{d}V = \omega^n$. Choose $\rho > 0$ smooth with $\mathrm{dd^c}(\rho\omega^{n-1}) = 0$, using Theorem 4.1.1; for $n = 1$, take $\rho = 1$. Put

$$\alpha = \int_X \rho\omega^n, \quad A_j = \int_X f_j^{1/n} \rho\omega^n, \quad \psi_j = u_j - M_j \le 0.$$

Set $A = \int_X f^{1/n}\rho\omega^n$. The inequality for nth roots used in the preceding existence proof gives $A_j \to A > 0$. Since $\mathrm{dd^c}(\rho\omega^{n-1}) = 0$, the trace $\int_X \omega_{u_j} \wedge \rho\omega^{n-1}$ equals α; the arithmetic–geometric mean inequality therefore implies

$$\alpha = \int_X \omega_{u_j} \wedge \rho\omega^{n-1} \ge \mathrm{e}^{\lambda M_j/n} \int_X \mathrm{e}^{\lambda\psi_j/n} f_j^{1/n} \rho\omega^n.$$

Applying Jensen's inequality to the probability measure $A_j^{-1} f_j^{1/n}\rho\omega^n$ and taking logarithms, we obtain

$$\lambda M_j \le n \log(\alpha/A_j) + \frac{\lambda}{A_j} \int_X (-\psi_j) f_j^{1/n} \rho\omega^n. \tag{11.10}$$

By Lemma 1.5.1, the normalized potentials ψ_j have uniform L^q bounds for every finite q. With $q = np/(np-1)$, Hölder's inequality gives a uniform bound for the last integral, since $\|f_j^{1/n}\|_{np} = \|f_j\|_p^{1/n}$. This proves the upper bound for M_j.

Finally, the density $F_j = \mathrm{e}^{\lambda u_j} f_j$ satisfies $\|F_j\|_p \le \mathrm{e}^{\lambda M_j}\|f_j\|_p$. The lower-density bound of Theorem 11.3.1, applied to ψ_j, gives $c_p \le \mathrm{e}^{\lambda M_j}\|f_j\|_p$, which bounds M_j below. □

Proof (of Theorem 11.1.2) Choose smooth strictly positive functions $f_j \to f$ in L^p. By Theorem 10.6.1, there are smooth u_j with $\omega_{u_j} > 0$ and $\omega_{u_j}^n = \mathrm{e}^{\lambda u_j} f_j \, \mathrm{d}V$. The preceding lemma bounds $M_j = \sup_X u_j$, and consequently bounds $F_j = \mathrm{e}^{\lambda u_j} f_j$ in L^p. The normalized potentials $\psi_j = u_j - M_j$ are uniformly bounded by Theorem 11.3.1. By Theorem 1.4.1, we may pass to a subsequence with $M_j \to M$ and ψ_j Cauchy in L^1. Theorem 11.3.2 gives uniform convergence $u_j \to u \in \mathrm{PSH}(X,\omega) \cap C(X)$.

Uniform convergence and $f_j \to f$ in L^p give $\mathrm{e}^{\lambda u_j} f_j \to \mathrm{e}^{\lambda u} f$ in L^p, and continuity of bounded products therefore gives $\omega_u^n = \mathrm{e}^{\lambda u} f \, \mathrm{d}V$. Uniqueness among bounded solutions follows from Corollary 11.2.2. □

For the exponential equation, a larger density gives a smaller solution, since the factor $\mathrm{e}^{\lambda u}$ has to compensate for the increase. Bounded exponential comparison gives this monotonicity even where the densities vanish.

Lemma 11.4.2 *Let $\lambda > 0$, and let $f_a, f_b \geq 0$ be measurable. Suppose that $a, b \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$ solve*

$$\omega_a^n = \mathrm{e}^{\lambda a} f_a \, \mathrm{d}V, \qquad \omega_b^n = \mathrm{e}^{\lambda b} f_b \, \mathrm{d}V.$$

If $f_a \leq f_b$ almost everywhere, then $b \leq a$ on X.

Proof Since $f_a \leq f_b$, the two equations give $\mathrm{e}^{-\lambda b}\omega_b^n \geq \mathrm{e}^{-\lambda a}\omega_a^n$. Hence $b \leq a$ by Corollary 11.2.2, applied with $u = a$ and $v = b$. □

Corollary 11.4.1 *Every bounded ω-plurisubharmonic solution of $\omega_u^n = f \, \mathrm{d}V$ is continuous.*

Proof Since u is bounded, the density $\mathrm{e}^{-u} f$ is nonnegative, nonzero and belongs to L^p, and u is a bounded solution of $\omega_u^n = \mathrm{e}^u(\mathrm{e}^{-u} f)\, \mathrm{d}V$. By Theorem 11.1.2, applied with $\lambda = 1$ to this density, u is continuous. □

11.5 Smoothness of bounded solutions

When the density is merely integrable, the solutions constructed above are only known to be continuous. If the density is a smooth positive function of the point and the potential, however, every bounded solution is smooth. Differentiating that density directly would require derivatives of the solution. Instead, we replace the potential in the density by a smooth approximation v, the prescribed potential, and solve an exponential equation as in Theorem 10.6.1. Adding a multiple of the square of v, shifted to be positive, to the test function of the maximum principle gives a trace estimate that does not depend on the derivatives of v.

We first prove this estimate, using the matrix notation g, h, the traces S, B and the torsion trace τ of Lemma 10.4.1.

Lemma 11.5.1 *Fix $M \geq 0$ and a smooth real function G on $X \times [-M, M]$. There is C, depending only on ω, M and $\|G\|_{C^2}$, with the following property. If v, u are smooth, $\omega_v \geq 0$, $\omega_u > 0$, $|v| \leq M$, and*

$$\omega_u^n = \mathrm{e}^{u + G(x,v)} \omega^n,$$

then

$$|u| \leq \|G\|_{C^0}, \qquad C^{-1}\omega \leq \omega_u \leq C\omega.$$

Proof The extrema of u give the first bound, as in Theorem 10.6.1. Put $k = g + (v_{i\bar{j}})$, $S_v = \mathrm{tr}_g\, k$, $D = 2\|G\|_{C^0}$ and $H = u + G(x, v)$. Then $\log(\det h / \det g) = H$, so H plays the role of the function G of Lemma 10.4.1. The chain rule and the Cauchy–Schwarz inequality give

$$\Delta_g H \geq -C_0\big(1 + S_v + |\partial v|_g^2\big).$$

Indeed $\Delta_g u = S - n \geq -n$, and $\Delta_g [G(x, v)]$ is the sum of $G_t(x, v)\Delta_g v = G_t(x, v)(S_v - n)$, where G_t is the derivative of G in its second variable, and of terms containing at

most two first derivatives of v, which are bounded below by $-C_0(1+|\partial v|_g^2)$. Here and below, constants depend only on ω, M and $\|G\|_{C^2}$.

Set $r = v + M + 1$, $R = 2M + 1$ and $s = u - \inf_X u + 1$, so $1 \le r \le R$ and $1 \le s \le D + 1$. For constants $A, L > 0$ to be chosen, maximize

$$Q = \log S + Ar^2 - Lu + s^{-1}$$

on X. If $S < 1$ at the maximum, the bounded ranges of r, u, s already bound S everywhere. Otherwise Lemma 10.4.1 applies there. Positivity gives

$$SB \ge n^2, \qquad S_v \le S \operatorname{tr}_h k, \qquad |\partial v|_g^2 \le S|\partial v|_h^2.$$

The middle inequality follows by diagonalizing h relative to g: every eigenvalue of h is at most S, and the diagonal entries of k are nonnegative. Consequently,

$$\frac{\Delta_g H}{S} \ge -C_1 B - C_0 \operatorname{tr}_h k - C_0|\partial v|_h^2.$$

At the maximum,

$$\frac{\partial S}{S} = (L + s^{-2})\partial u - 2Ar\partial v.$$

Since $|\tau|_h^2 \le C_g B$, Young's inequality bounds the torsion term by

$$\left|\frac{2}{S^2}\operatorname{Re}(h^{i\bar{j}}\tau_i S_{\bar{j}})\right| \le s^{-3}|\partial u|_h^2 + A|\partial v|_h^2 + \frac{C_g\big((L+1)^2(D+1)^3 + AR^2\big)}{S^2}B.$$

Also

$$\begin{aligned}\Delta_h(Ar^2) &\ge 2A \operatorname{tr}_h k - 2ARB + 2A|\partial v|_h^2,\\ \Delta_h(-Lu + s^{-1}) &= (L + s^{-2})(B - n) + 2s^{-3}|\partial u|_h^2.\end{aligned}$$

Combining these inequalities with (10.6) yields

$$\begin{aligned}0 \ge \Delta_h Q \ge &\left(L - 2AR - C_2 - \frac{C_g((L+1)^2(D+1)^3 + AR^2)}{S^2}\right)B - n(L+1)\\ &+ (2A - C_0)\operatorname{tr}_h k + (A - C_0)|\partial v|_h^2 + s^{-3}|\partial u|_h^2.\end{aligned}$$

Choose $A > C_0 + 1$ and then $L > 2AR + C_2 + 2$. If the fraction in the coefficient of B is at least one, S^2 at the maximum is bounded by its numerator. Otherwise the displayed inequality gives $B \le n(L+1)$ there. For $n \ge 2$, the bounded determinant gives

$$S \le n\mathrm{e}^{2\|G\|_{C^0}} B^{n-1}.$$

For $n = 1$, $S = \mathrm{e}^{u+G(x,v)}$ directly. Thus S is bounded at the maximum in every case. Maximality of Q and the bounded ranges of r, u, s give the same bound globally.

The positive lower determinant bound then bounds every eigenvalue of h away from zero. □

Theorem 11.5.1 *Let $F \in C^\infty(X \times \mathbb{R}, \mathbb{R})$. Every bounded $u \in \mathrm{PSH}(X, \omega)$ satisfying*

$$(\omega + \mathrm{dd^c} u)^n = \mathrm{e}^{F(x,u)}\, \mathrm{d}V$$

is smooth on X, and $\omega + \mathrm{dd^c} u > 0$.

Proof The density $\mathrm{e}^{F(x,u)}$ is bounded and strictly positive, so Corollary 11.4.1 makes u continuous. Choose smooth strictly ω-plurisubharmonic $v_j \downarrow u$ by Theorem 1.11.1. They have a common bound M and converge uniformly by Dini's theorem. Define

$$G(x,t) = F(x,t) - t + \log \frac{\mathrm{d}V}{\omega^n}.$$

By Theorem 10.6.1, there is a smooth strictly ω-plurisubharmonic a_j with

$$\omega_{a_j}^n = \mathrm{e}^{a_j + G(x, v_j)} \omega^n.$$

Put $\delta_j = \|G(x, v_j) - G(x, u)\|_{C^0} \to 0$. The given u solves the same equation with $G(x, v_j)$ replaced by $G(x, u)$. Since $|G(x, v_j) - G(x, u)| \le \delta_j$, the functions $a_\pm = a_j \pm \delta_j$ satisfy $\mathrm{e}^{-a_+} \omega_{a_+}^n \le \mathrm{e}^{-u} \omega_u^n \le \mathrm{e}^{-a_-} \omega_{a_-}^n$, so Corollary 11.2.2, applied twice, gives

$$a_j - \delta_j \le u \le a_j + \delta_j.$$

Hence $a_j \to u$ uniformly. By Lemma 11.5.1, $C^{-1}\omega \le \omega_{a_j} \le C\omega$ uniformly. In particular $\Delta_g a_j$ is uniformly bounded. Fixed-operator elliptic estimates give uniform $W^{2,q}$ bounds for every finite q, so the a_j are bounded in $C^{1,\alpha}(X)$ uniformly in j, for every $0 < \alpha < 1$, and $u \in C^{1,\alpha}(X)$.

The right-hand sides $a_j + G(x, v_j)$ need not be uniformly Hölder, because the v_j are not. We therefore solve once more, with the a_j as prescribed potentials. By Theorem 10.6.1, there is a smooth strictly ω-plurisubharmonic b_j with

$$\omega_{b_j}^n = \mathrm{e}^{b_j + G(x, a_j)} \omega^n.$$

The a_j are smooth, strictly ω-plurisubharmonic and uniformly bounded, since $|a_j| \le \sup_X |u| + \delta_j$. Hence Lemma 11.5.1, applied with a_j in place of v, gives $C^{-1}\omega \le \omega_{b_j} \le C\omega$ uniformly, and the elliptic estimates above bound the b_j in $C^{1,\alpha}(X)$ uniformly in j. Moreover $|G(x, a_j) - G(x, u)| \le C|a_j - u| \to 0$ uniformly, so the comparison argument above gives $b_j \to u$ uniformly. The right-hand sides $b_j + G(x, a_j)$ are now bounded in $C^\alpha(X)$ uniformly in j.

We pass to second derivatives using the concave extension from Corollary 10.4.1. On a relatively compact coordinate chart, let P send a real Hessian to its complex Hermitian part, with the coordinate normalization of $\mathrm{dd^c}$. By the uniform equivalence just proved, all matrices $g(x) + P(D^2 b_j)$ lie in a fixed compact set $\mathcal{K} = \{A : \lambda I \le A \le \Lambda I\}$ of positive Hermitian matrices. Define

$$\mathcal{F}(x,N) = \inf_{A\in\mathcal{K}} \left\{ \log\det A + \operatorname{tr}\left[A^{-1}(g(x)+P(N)-A)\right]\right\} - \log\det g(x).$$

The tangent inequality for the concave logarithmic determinant shows that this is the original operator whenever $g(x)+P(N)\in\mathcal{K}$. It is concave and uniformly elliptic on real symmetric matrices: for $N'\geq 0$, $P(N')\geq 0$ and its trace is a fixed positive multiple of $\operatorname{tr} N'$, while $A^{\pm 1}$ have uniform bounds. Its dependence on x is uniformly Lipschitz on this chart. Thus the b_j are classical solutions of

$$\mathcal{F}(x, D^2 b_j) = b_j + G(x, a_j),$$

with the same operator for every j and right-hand sides bounded in C^α uniformly in j. The local Evans–Krylov estimate bounds the b_j in $C^{2,\beta}$ on smaller charts, uniformly in j, for some $\beta>0$. Since $b_j\to u$ uniformly, the Arzelà–Ascoli theorem gives $u\in C^{2,\beta}$. The metric bounds pass to the limit, so the original equation is now uniformly elliptic with $\omega+\mathrm{dd^c}u>0$. Differentiating it and applying Schauder estimates repeatedly proves $u\in C^\infty(X)$. □

11.6 The vanishing exponent and scalar uniqueness

To show that the constant in Theorem 11.1.1 is unique, we compare the solution u_λ of (11.1) with constant translates of a solution u of the normalized equation. Translating u by its maximum and by its minimum gives two barriers for the exponential equation, whose separation is the oscillation of u. After multiplication by λ, both barriers converge to the logarithm of the constant as $\lambda\downarrow 0$. Since u_λ depends only on f and λ, the constant is determined by f.

Proposition 11.6.1 *Suppose $u\in \mathrm{PSH}(X,\omega)\cap L^\infty(X)$ and $c>0$ satisfy $\omega_u^n = c f\,\mathrm{d}V$. Set $m=\inf_X u$ and $M=\sup_X u$. Then, for every $\lambda>0$, the solution of (11.1) satisfies*

$$u + \frac{\log c}{\lambda} - M \leq u_\lambda \leq u + \frac{\log c}{\lambda} - m. \tag{11.11}$$

In particular,

$$\|\lambda u_\lambda - \log c\|_\infty \leq \lambda(M-m), \qquad \mathrm{e}^{\lambda u_\lambda}\longrightarrow c \quad \text{uniformly on } X \text{ as } \lambda\downarrow 0.$$

Consequently, c is the only positive constant c' for which the equation $\omega_v^n = c' f\,\mathrm{d}V$ has a solution $v\in\mathrm{PSH}(X,\omega)\cap L^\infty(X)$.

Proof Set $a = u+\lambda^{-1}\log c - M$ and $b = u+\lambda^{-1}\log c - m$. These translates of u have the same Monge–Ampère measure as u, and since $m\leq u\leq M$,

$$\mathrm{e}^{\lambda a} f\,\mathrm{d}V \leq c f\,\mathrm{d}V = \omega_a^n = \omega_b^n \leq \mathrm{e}^{\lambda b} f\,\mathrm{d}V.$$

Thus $\mathrm{e}^{-\lambda a}\omega_a^n \geq f\,\mathrm{d}V = \mathrm{e}^{-\lambda u_\lambda}\omega_{u_\lambda}^n \geq \mathrm{e}^{-\lambda b}\omega_b^n$. Applying Corollary 11.2.2 first to the pair (u_λ, a) and then to the pair (b, u_λ) gives $a\leq u_\lambda\leq b$, which is (11.11).

Multiplying these inequalities by λ and using $m \le u \le M$ gives the norm bound, and the uniform convergence follows.

If (v, c') is another such pair, the same argument applied to v gives $\lambda u_\lambda \to \log c'$, because u_λ depends only on f and λ. Hence $c' = c$. □

The barriers also give $\operatorname{osc}_X u_\lambda \le 2(M - m)$, and the densities $e^{\lambda u_\lambda} f$ converge to cf in L^p. By Theorem 11.3.2, the normalized potentials $u_\lambda - \sup_X u_\lambda$ are therefore precompact in $C^0(X)$ as $\lambda \downarrow 0$, and every cluster point solves the equation with constant c. Uniqueness of the constant does not, however, single out one cluster point; Proposition 15.19.1 describes the symmetries that these limits inherit.

11.7 Normalization on the Hopf threefold

In this section, X is the Hopf threefold of Chapter 9, with its Hermitian form ω and projection $\pi\colon X \to \mathbb{P}^2$. Recall that $\alpha = \pi^*\omega_{\mathrm{FS}}$ and $\beta = \omega - \alpha$, and that $\nu = \omega^3/6$ is a probability measure. We write E for the elliptic fiber, τ_a for the translation by $a \in E$, $\mathrm{d}A_E$ for the probability Haar measure of E, and A for the fiber-averaging operator of Proposition 9.7.1.

The minimum-mass characterization in that proposition gives a lower bound for the normalization constant, and equality forces every solution to descend to the projective base. The uniform L^p control of this chapter makes the equality case stable under limits of the data, although the densities themselves need only converge weakly.

Here we take $\mathrm{d}V = \omega^3$; the density f is still a nonnegative nonzero function in L^p. By Theorem 11.1.1 and Proposition 11.6.1, there is a unique positive constant c_f for which the equation

$$(\omega + \mathrm{dd}^{\mathrm{c}} u)^3 = c_f f\, \omega^3$$

has a continuous ω-plurisubharmonic solution. Put $I_f = \int_X f\, \mathrm{d}\nu$.

Proposition 11.7.1 *One has $c_f I_f \ge 1$, and every continuous solution u satisfies*

$$\|u - Au\|^2_{L^2(X,\nu)} \le 4(c_f I_f - 1). \tag{11.12}$$

Moreover, $c_f I_f = 1$ if and only if $f = \pi^* g$ almost everywhere with respect to ν, for some nonnegative $g \in L^p(\mathbb{P}^2, \omega^2_{\mathrm{FS}})$. In this case every continuous solution is the pullback of a continuous ω_{FS}-plurisubharmonic function.

Proof Integrating the equation gives $\int_X \omega_u^3 = 6 c_f I_f$. Both the inequality $c_f I_f \ge 1$ and (11.12) therefore follow from (9.21), whose left-hand side is nonnegative.

Suppose $c_f I_f = 1$. By Proposition 9.7.1, $u = \pi^* q$ for a bounded ω_{FS}-plurisubharmonic function q; local holomorphic sections of π show that q is continuous. Thus $u \circ \tau_a = u$ for every $a \in E$. Since bounded products are natural under biholomorphisms and ω is invariant under the translations, it follows that

$$\tau_a^*(f\nu) = f\nu \qquad (a \in E).$$

As f is defined only almost everywhere, we use this invariance as an identity of measures.

By Fubini's theorem, the fiber average Af is defined almost everywhere. It equals $\pi^* g$ for a measurable function g on the base, because local trivializations differ by translations of the Haar variable. Jensen's inequality gives

$$\int_{\mathbb{P}^2} |g|^p \, \omega_{\mathrm{FS}}^2 \leq \int_X |f|^p \, \mathrm{d}\nu.$$

For every continuous real test function χ on X, the invariance of the measure gives $\int_X \chi f \, \mathrm{d}\nu = \int_X (\chi \circ \tau_a) f \, \mathrm{d}\nu$. Integrating in a, and using Fubini's theorem together with the invariance of Haar measure under $a \mapsto -a$, we obtain $\int_X \chi f \, \mathrm{d}\nu = \int_X \chi A f \, \mathrm{d}\nu$. Hence $f = Af = \pi^* g$ almost everywhere, with $g \geq 0$.

Conversely, suppose $f = \pi^* g$ almost everywhere. Disintegration gives $I_f = \int_{\mathbb{P}^2} g \omega_{\mathrm{FS}}^2$. Applying Theorem 11.1.1 on the base, we find a continuous ω_{FS}-plurisubharmonic function q and a constant $k > 0$ with $(\omega_{\mathrm{FS}} + \mathrm{dd^c} q)^2 = k g \, \omega_{\mathrm{FS}}^2$. Put $u = \pi^* q$. By Lemma 9.6.1(3), applied on $X = H_3$ with $a = b = 1$, the measure $(\omega_{\mathrm{FS}} + \mathrm{dd^c} q)^2$ has mass one, so $k = I_f^{-1}$, and

$$(\omega + \mathrm{dd^c} u)^3 = 3\beta \wedge \pi^* \big(I_f^{-1} g \, \omega_{\mathrm{FS}}^2 \big).$$

Since β restricts to $2\mathrm{d}A_E$ on each fiber and ν disintegrates into $\mathrm{d}A_E$ on the fibers and ω_{FS}^2 on the base, the right-hand side is $6 I_f^{-1} f \, \mathrm{d}\nu = I_f^{-1} f \, \omega^3$. Uniqueness of the constant implies $c_f = I_f^{-1}$. Every continuous solution then has mass 6, so the mass characterization in Proposition 9.7.1 makes it a pullback from the base, and the descended potential is continuous by local sections, as above. □

When $c_f I_f$ is close to one, compactness gives uniform convergence of subsequences of potentials and weak convergence of their densities, and the equality case identifies the limits.

Proposition 11.7.2 *Fix $1 < p < \infty$. Suppose $f_j \geq 0$ satisfy*

$$\int_X f_j \, d\nu = 1, \qquad \sup_j \|f_j\|_{L^p(X,\nu)} < \infty, \qquad c_{f_j} \longrightarrow 1.$$

For each j, let u_j be any continuous solution of $\omega_{u_j}^3 = c_{f_j} f_j \omega^3$ normalized by $\sup_X u_j = 0$. Then

$$\|u_j - Au_j\|_{C^0(X)} \longrightarrow 0, \qquad f_j - Af_j \rightharpoonup 0 \quad \textit{weakly in } L^p(X,\nu).$$

Every subsequence has a further subsequence with $u_j \to \pi^* q$ uniformly and $f_j \rightharpoonup \pi^* g$ weakly in L^p, where q is a continuous ω_{FS}-plurisubharmonic function, $g \geq 0$, $\int_{\mathbb{P}^2} g\omega_{\mathrm{FS}}^2 = 1$, and

$$(\omega_{\mathrm{FS}} + \mathrm{dd^c} q)^2 = g\omega_{\mathrm{FS}}^2.$$

Proof Since the constants c_{f_j} converge, the densities $c_{f_j} f_j$ have a common L^p bound. By Theorem 11.3.2, every subsequence has a further subsequence on which $u_j \to u$ uniformly. Since L^p is reflexive, a further extraction gives $f_j \rightharpoonup f$ weakly in L^p. Positivity and the unit integral pass to the limit, and continuity of bounded products under uniform convergence gives

$$(\omega + \mathrm{dd^c} u)^3 = f\omega^3.$$

Thus $c_f = 1$. Since also $I_f = 1$, Proposition 11.7.1 gives $u = \pi^* q$, with q continuous, and $f = \pi^* g$ with $g \geq 0$; in particular, both limits are fixed by A. For continuous $F \geq 0$ on $\mathbb{P}^2$, the integral identity in Lemma 9.6.1(3) gives

$$6\int_{\mathbb{P}^2} F(\omega_{\mathrm{FS}} + \mathrm{dd^c} q)^2 = \int_X \pi^* F \, f\omega^3 = 6\int_{\mathbb{P}^2} Fg\,\omega_{\mathrm{FS}}^2,$$

which is the asserted base equation, and $\int_{\mathbb{P}^2} g\,\omega_{\mathrm{FS}}^2 = I_f = 1$.

Haar averaging is a contraction on $C^0(X)$ and on $L^p(X,\nu)$. It is also self-adjoint with respect to ν: Fubini's theorem and the invariance of ν under the fiber translations give $\int \chi Af \, d\nu = \int A\chi f \, d\nu$ for $\chi \in L^{p'}$. Along the extracted subsequence, therefore, $u_j - Au_j \to 0$ uniformly and $f_j - Af_j \rightharpoonup 0$ in L^p. If either convergence failed for the original sequence, some subsequence would stay away from the limit (for the weak convergence, when tested against one fixed function in $L^{p'}$), and a further subsequence of the kind just constructed would give a contradiction. This proves both conclusions for the whole sequence. □

Example 11.7.1 Choose $0 < a < 1$, and use the function $s = \log|z|^2$ on the covering space $\mathbb{C}^3 \setminus \{0\}$ from Chapter 9. Since s changes by -2 under the defining contraction, every 2-periodic function of s descends to X. For an integer $m \geq 1$, define the smooth normalized potential

$$u_m = \frac{a}{\pi^2 m^2}\big(\cos(\pi m s) - 1\big), \qquad \|u_m - Au_m\|_{C^0} = \frac{a}{\pi^2 m^2} \longrightarrow 0.$$

Primes below denote derivatives with respect to s. Since $\mathrm{dd^c} s = \alpha$ and $\mathrm{d}s \wedge \mathrm{d^c} s = \beta$, the chain rule gives

$$\omega + \mathrm{dd^c} u_m = (1 + u_m')\alpha + (1 + u_m'')\beta, \quad u_m' = -\frac{a}{\pi m}\sin(\pi m s), \quad u_m'' = -a\cos(\pi m s).$$

Both coefficients are bounded below by positive constants independent of m. The rank identities $\alpha^3 = \beta^2 = 0$ imply

$$(\omega + \mathrm{dd^c} u_m)^3 = H_m \omega^3, \qquad H_m = \Big(1 - \frac{a}{\pi m}\sin(\pi m s)\Big)^2 \big(1 - a\cos(\pi m s)\big).$$

On each fiber, s is twice the real Haar coordinate plus a constant depending on the base point, so its distribution modulo 2 is uniform. When the periodic expression $H_m = (1+u_m')^2(1+u_m'')$ is averaged, the terms u_m', u_m'', $u_m' u_m''$ and $(u_m')^2 u_m''$ contribute zero, since they are derivatives of periodic functions. Hence

$$I_m := \int_X H_m \, \mathrm{d}\nu = AH_m = 1 + \frac{a^2}{2\pi^2 m^2}.$$

Set $f_m = H_m / I_m$. These smooth densities have unit integral and common strictly positive lower and finite upper bounds. Uniqueness of the constant gives $c_{f_m} = I_m \to 1$, and $Af_m = 1$. Nevertheless,

$$\|f_m - 1 + a\cos(\pi m s)\|_{C^0(X)} = O(m^{-1}), \qquad \lim_{m\to\infty} \|f_m - Af_m\|_{L^1(X,\nu)} = \frac{2a}{\pi} > 0.$$

Here the constant in $O(m^{-1})$ depends only on a, and the last identity uses the fact that the average of $|\cos|$ over a period is $2/\pi$. Thus near equality does not imply strong L^1 convergence to the fiber average, even for uniformly positive bounded smooth densities. For any pullback g of a function on the base, the contraction property gives $\|f_m - Af_m\|_1 \leq 2\|f_m - g\|_1$, so approximation by such pullbacks also fails in L^1.

In Chapter 12 we return to the Monge–Ampère equation for $\omega + \mathrm{dd^c} u$ and ask when two potentials with the same density agree up to a constant. The uniqueness of the normalization scalar does not determine the potential: where the density vanishes, the polarized linear equation for the difference can be degenerate.

Chapter 12
Uniqueness for degenerate equations

Let X be a connected compact complex manifold of dimension $n \geq 1$, let ω be a Hermitian form, and let $\mathrm{d}V$ be a smooth positive volume form. For a nonzero nonnegative density $f \in L^p(X, \mathrm{d}V)$, $p > 1$, the preceding chapter shows that f determines the constant c in $(\omega + \mathrm{dd}^{\mathrm{c}} u)^n = c f \, \mathrm{d}V$. We now ask whether the potential is determined up to a constant.

For real potentials u, v, with the regularity stated in each result, set

$$A = \omega + \mathrm{dd}^{\mathrm{c}} u, \qquad B = \omega + \mathrm{dd}^{\mathrm{c}} v, \qquad w = u - v, \qquad T = \sum_{j=0}^{n-1} A^j \wedge B^{n-1-j}.$$

Then

$$\mathrm{dd}^{\mathrm{c}} w \wedge T = A^n - B^n. \tag{12.1}$$

If $A, B \geq 0$ and $A^n = B^n$, then w solves the linear equation $\mathrm{dd}^{\mathrm{c}} w \wedge T = 0$. Its polarized cofactor T is semipositive, but it may be degenerate and need not be closed. Coefficient products for Sobolev potentials are interpreted almost everywhere. For $C^{1,1}$ solutions with a common continuous density, perturbing w locates its extrema in regions where this equation is uniformly elliptic, and the maximum principle propagates the extremal values over the corresponding components.

For C^2 potentials, a continuous weight $h > 0$ with $\mathrm{dd}^{\mathrm{c}}(hT) = 0$ in distributions gives a global energy identity forcing $\mathrm{d}w$ to vanish in the positive directions of the cofactor.

For bounded solutions, an energy measure replaces the pointwise gradient calculation. A continuous density whose positive set is connected and dense then gives uniqueness. When the background has invariant mass, ordinary comparison proves uniqueness for nonzero nonnegative L^p densities, $p > 1$, even if the density vanishes on an open set.

The examples on $\mathbb{P}^3$ have the bounded mass property and a smooth nonclosed Hermitian background, yet distinct smooth normalized solutions share one density. The zero set of this density separates the elliptic regions: it can be an open band or

two hypersurfaces with quadratic vanishing, even for a real analytic density. Finite upper mass therefore does not ensure uniqueness.

12.1 Extrema and density sublevels

To apply the strong maximum principle to $w = u - v$, we first locate a maximum point and a minimum point of w in the region where the linear equation $\mathrm{dd^c} w \wedge T = 0$ is uniformly elliptic. Maximizing a small perturbation of w by a function χ with $\omega + \mathrm{dd^c}\chi > 0$ does this, even when the Hessians are only measurable. The strong maximum principle then makes w constant on the components of this region that contain these points, and w is constant if the closures of the two components meet.

In this section, $u, v \in C^{1,1}(X)$ satisfy $A, B \geq 0$ almost everywhere and

$$A^n = B^n = f\,\mathrm{d}V, \qquad f \in C^0(X), \qquad f \geq 0. \tag{12.2}$$

Fix a smooth positive $(1,1)$-form β, put $H = A + B$, and let E_{unif} be the set of points near which H is uniformly positive:

$$E_{\mathrm{unif}} = \left\{ x : \begin{array}{l} \text{there are a neighborhood } V \ni x \text{ and } c > 0 \\ \text{with } H \geq c\beta \text{ almost everywhere on } V \end{array} \right\}. \tag{12.3}$$

This open set does not depend on β or on the choice of representatives of the measurable coefficients.

Theorem 12.1.1 *Assume* (12.2). *If $\chi \in C^2(X, \mathbb{R})$ satisfies $C := \omega + \mathrm{dd^c}\chi > 0$, put*

$$P_\chi = \{x \in X : f(x)\mathrm{d}V_x \geq C_x^n\}, \qquad U = \{f > 0\}.$$

Then P_χ is a nonempty compact subset of U, and $U \subset E_{\mathrm{unif}}$. There are components G_+, G_- of E_{unif}, each meeting P_χ, such that $w = u - v$ satisfies

$$w \equiv \max_X w \quad \text{on } \overline{G_+}, \qquad w \equiv \min_X w \quad \text{on } \overline{G_-}. \tag{12.4}$$

If w is nonconstant, these closures are disjoint. Consequently w is constant if any one of the following holds:

(1) *every two components of E_{unif} meeting P_χ have intersecting closures;*
(2) *E_{unif} is connected;*
(3) *E_{unif} is dense and has at most two components.*

The assertions about components, including the three criteria, also hold with U in place of E_{unif}. In particular, w is constant when $\{f = 0\}$ is a nonempty connected embedded C^1 real hypersurface; this hypersurface need not be orientable or coorientable.

Since the Hessians of $C^{1,1}$ functions exist only almost everywhere, the contact argument must use points where they exist. The following lemma finds such points

near a maximum; it also provides the strict comparison step in the measurable strong maximum principle.

Lemma 12.1.1 *In Euclidean space $\mathbb{R}^m$, let $B(p,r)$ be the ball with center p and radius $r > 0$, and let I be the identity matrix. Let $g \in C^{1,1}$ near $\overline{B(p,r)} \subset \mathbb{R}^m$, suppose $g \le g(p)$ on the ball, and let E be any subset of full measure in the ball at whose points g has a second-order Taylor expansion. For every $\delta > 0$ and $0 < \rho < r$, there is $x \in E \cap B(p,\rho)$ such that*

$$D^2 g(x) \le 2\delta I.$$

Proof Put $g_\delta(x) = g(x) - \delta|x-p|^2$. The gradient map $F = \nabla g_\delta$ is Lipschitz on the ball, so it maps the null set $N = \overline{B(p,r)} \setminus E$ to a null set. Choose $s \in B(0,\delta\rho) \setminus F(N)$ and maximize $g_\delta(x) - s \cdot (x-p)$ on the closed ball. At any point with $|x-p| \ge \rho$, its value is strictly smaller than its value at p, because

$$g(x) - g(p) - \delta|x-p|^2 - s \cdot (x-p) \le -\delta|x-p|^2 + |s|\,|x-p| < 0.$$

Thus a maximizer x lies in $B(p,\rho)$. The first-order condition at x gives $F(x) = s$, so $x \notin N$. The second-order Taylor expansion at this upper contact point gives $D^2 g(x) \le 2\delta I$. Finally, sets E as in the statement exist: by Rademacher's theorem, applied to the Lipschitz gradient of g, the function g has a second-order Taylor expansion almost everywhere. □

Lemma 12.1.2 *Let $a = (a^{ij})$ be a measurable symmetric matrix field on a connected domain $\Omega \subset \mathbb{R}^m$. For each compact $K \Subset \Omega$, suppose there are constants $0 < \lambda_K \le \Lambda_K < \infty$ such that*

$$\lambda_K I \le a(x) \le \Lambda_K I$$

almost everywhere on K, where I is the identity matrix. Repeated indices $i, j = 1, \dots, m$ are summed, and $z_{ij} = \partial_i \partial_j z$. If $z \in W^{2,\infty}_{\mathrm{loc}}(\Omega)$, $a^{ij} z_{ij} \ge 0$ almost everywhere, and z attains its supremum at a point of Ω, then z is constant.

Proof We first prove strict weak comparison on a relatively compact domain: if $b \in C^{1,1}$, $Lb := a^{ij} b_{ij} \ge \kappa > 0$ almost everywhere, and $b \le M$ on the boundary, then $b \le M$ inside. Otherwise an interior maximum point lies in a closed ball on which $a \le \Lambda I$. Apply Lemma 12.1.1 to the set of full measure where both the differential inequality and the Taylor expansion hold. Choosing $2\delta m\Lambda < \kappa$ gives the contradiction $\kappa \le Lb \le 2\delta \operatorname{tr} a \le 2\delta m\Lambda$.

Take the $C^{1,1}$ representative of z, write $M = \sup_\Omega z$, and suppose that $K = \{z = M\}$ is a nonempty proper subset of Ω. Near a relative boundary point of K, choose $y \notin K$ so that the ball of radius $r = \operatorname{dist}(y, K) > 0$ is compactly contained in Ω and touches K at x_0. On the closed annulus $r/2 \le |x-y| \le r$, use fixed bounds $\lambda I \le a \le \Lambda I$ and set

$$b(x) = \mathrm{e}^{-\alpha|x-y|^2} - \mathrm{e}^{-\alpha r^2}.$$

For sufficiently large α,

$$Lb = \mathrm{e}^{-\alpha|x-y|^2}\bigl(4\alpha^2 a^{ij}(x_i - y_i)(x_j - y_j) - 2\alpha \operatorname{tr} a\bigr) \geq \kappa > 0$$

almost everywhere on the annulus. On its inner sphere $z < M$ uniformly; on its outer sphere $b = 0$. Choose $\eta > 0$ small enough that $z + \eta b \leq M$ on both spheres. Strict weak comparison, applied to $z + \eta b$, gives this inequality throughout the annulus. At x_0, however, $z = M$, $b = 0$, $\nabla z = 0$, and b increases to first order in the inward radial direction, which contradicts that inequality. Thus K has no relative boundary point in Ω, and connectedness proves the assertion. The argument uses no derivatives of the measurable coefficients. □

At a point where A, B are semipositive, define the cofactor quadratic form on real cotangent vectors

$$Q_x(\xi)\mathrm{d}V_x = \mathrm{i}\,\xi^{1,0} \wedge \overline{\xi^{1,0}} \wedge T_x. \tag{12.5}$$

Here $\xi^{1,0}$ is the $(1,0)$-part of ξ, and forms of bidegree $(n-1, n-1)$ are ordered by testing against $\mathrm{i}\xi^{1,0} \wedge \overline{\xi^{1,0}}$ in this way. Mixed products of semipositive $(1,1)$-forms are semipositive in this order. Expanding H^{n-1}, where $H = A + B$, gives

$$2^{-(n-1)} H^{n-1} \leq T \leq H^{n-1}. \tag{12.6}$$

At a point, choose a β-unitary coframe $\mathrm{d}z_1, \ldots, \mathrm{d}z_n$ that diagonalizes H, and write $\lambda_1, \ldots, \lambda_n$ for the eigenvalues of H. For a $(1,0)$-covector $\alpha = \sum_{k=1}^n b_k\,\mathrm{d}z_k$ with complex coefficients b_k, the coefficient of $\mathrm{i}\alpha \wedge \bar\alpha \wedge H^{n-1}$ in this coframe is a positive constant depending only on n, times

$$\sum_{k=1}^{n} |b_k|^2 \prod_{\ell \neq k} \lambda_\ell. \tag{12.7}$$

For $n \geq 2$, (12.6) and (12.7) show that Q is locally uniformly elliptic exactly on E_{unif}, that is, where H has a local essential positive lower bound. The implication from ellipticity of Q to a lower bound for H uses the local upper bound for H: testing each eigendirection bounds every product $\prod_{\ell \neq k} \lambda_\ell$ below, and choosing k different from an index of a smallest eigenvalue then bounds that eigenvalue below. For $n = 1$ one has $T = 1$, so the linearized equation is elliptic everywhere.

Proof (of Theorem 12.1.1) We first locate the extrema. Fix $s > 0$ and consider a maximum point of

$$w - s(v - \chi). \tag{12.8}$$

For smooth data, the complex Hessian of this function is nonpositive at a maximum point, that is,

$$(1 + s)B \geq A + sC. \tag{12.9}$$

Since $sC > 0$, this gives $B > 0$, and since $A^n = B^n$, also $A > 0$. In a B-unitary frame put $J = B^{-1/2} A B^{-1/2}$, $K = B^{-1/2} C B^{-1/2}$. Since $\det J = 1$, arithmetic–geometric mean and (12.9) give

$$n + s \operatorname{tr} K \leq \operatorname{tr} J + s \operatorname{tr} K \leq n(1 + s).$$

Consequently $\operatorname{tr} K \le n$, $B \ge C/n$, and $\det K \le (\operatorname{tr} K/n)^n \le 1$. In particular $B^n \ge C^n$.

For $C^{1,1}$ potentials, fix a maximum point p of (12.8) and apply Lemma 12.1.1 on a coordinate ball, with radii tending to zero and errors $\delta_j \downarrow 0$. For the set of full measure in that lemma, take the points where both Hessians are defined and semipositive and have the common determinant given by (12.2). With $\beta_0 = \mathrm{dd}^{\mathrm{c}}|z|^2$, the resulting points $x_j \to p$ satisfy

$$A(x_j) + sC_j(x_j) \le (1+s)B(x_j), \qquad C_j = C - (\delta_j/s)\beta_0 > 0$$

for large j. The matrix argument just given yields $f(x_j)\mathrm{d}V_{x_j} \ge C_j(x_j)^n$. Since f, $\mathrm{d}V$ and C are continuous, letting $j \to \infty$ gives $p \in P_\chi$; no continuity of the Hessians is needed. Thus every maximum point of the perturbed function lies in P_χ. The set P_χ is closed, by continuity of f, $\mathrm{d}V$ and C, hence compact, and it is contained in U because $C^n > 0$. Letting $s \downarrow 0$, a limit point of these maximum points is a maximum point of w lying in P_χ. Applying the same argument to $-w$ gives a minimum point of w in P_χ as well. In particular, P_χ is nonempty.

On a relatively compact ball in U, the determinants of A and B have a positive lower bound, while their coefficients have an essential upper bound. Every eigenvalue is therefore bounded below by the determinant lower bound divided by a fixed power of the upper bound. Thus $U \subset E_{\mathrm{unif}}$.

By (12.1), w solves a linear equation in nondivergence form with measurable coefficients, whose principal quadratic form is a fixed positive multiple of Q. On each relatively compact coordinate ball in E_{unif}, these coefficients are uniformly elliptic. Lemma 12.1.2, applied to w and to $-w$, makes w constant on the component of E_{unif} containing the maximum point found above, and constant on the component containing the minimum point found above. Continuity extends the two constant values to the closures, proving (12.4). Distinct extrema force disjoint closures.

The same argument applies on U itself, since U contains the extremal points found in P_χ and the cofactor is locally uniformly elliptic on $U \subset E_{\mathrm{unif}}$. This proves the assertions with U in place of E_{unif}.

The component criteria are immediate except for the dense two-component case. There we may assume $G_+ \ne G_-$, so that G_+ and G_- are the two components. Their union is dense, so their closures are nonempty closed sets whose union is X, and connectedness of X forces these closures to meet. Finally, when $Z = \{f = 0\}$ is a connected embedded real hypersurface, flattening charts assign to each point of Z the unordered pair of components containing its two local sides. This pair is locally constant along Z, so connectedness makes it independent of the point; the two labels may coincide or be exchanged along a loop. Every component of $X \setminus Z$ has a nonempty boundary in Z, since otherwise it would be both open and closed in connected X. Near such a boundary point it contains a local side. Thus the pair accounts for every component, and U has at most two components. It is dense because a hypersurface has empty interior, so the third criterion applies with U in place of E_{unif}. This proves the last assertion without choosing a global side. □

In this argument, the $C^{1,1}$ bound controls the measurable coefficients, and continuity of the density places the limiting contact points in $\{f > 0\}$. The next criterion gives a quantitative condition for $\{f > 0\}$ to be connected.

12.1.1 A square-root threshold

A nonnegative $C^{1,1}$ density grows at most quadratically away from each of its zeros. If its zero set separates a small ball, a family of parallel segments crossing the zero set therefore contributes at least a constant times $\sqrt{t}$ to the volume of $\{f \le t\}$ for small $t > 0$. Excluding this lower bound makes $\{f > 0\}$ connected, and Theorem 12.1.1 then gives uniqueness.

Write $\mathrm{Vol}(E) = \int_E \mathrm{d}V$ for Borel sets E.

Theorem 12.1.2 *Let $u, v \in C^{1,1}(X)$ satisfy $A, B \ge 0$ almost everywhere and $A^n = B^n = f\,\mathrm{d}V$, where $f \ge 0$ is $C^{1,1}$. If*

$$\liminf_{t \downarrow 0} \frac{\mathrm{Vol}\{f \le t\}}{\sqrt{t}} = 0, \tag{12.10}$$

then $\{f > 0\}$ is connected and $u - v$ is constant. In particular this holds when $f^{-1/2} \in L^1(\mathrm{d}V)$, with $f^{-1/2} = +\infty$ on $\{f = 0\}$. The exponent $1/2$ is sharp in every dimension $n \ge 3$.

Proof Write $Z = \{f = 0\}$. Condition (12.10) first implies $\mathrm{Vol}(Z) = 0$. We show that the complement of Z in every sufficiently small coordinate ball is connected. Fix such a ball with convex Euclidean image. If the complement of Z in it were disconnected, choose points a, b in different components and small closed disks centered at a and b, perpendicular to the segment $[a, b]$ and contained in those respective components. By making the disks small, all parallel segments between corresponding points stay in the ball. Each segment meets Z, since otherwise it would connect the two components.

On a fixed coordinate region containing the closed ball, the gradient of f is Lipschitz with some constant $K > 0$. Since $f \ge 0$, we have $\nabla f(z) = 0$ at every zero z, hence

$$f(x) \le \frac{K}{2}|x - z|^2 \tag{12.11}$$

whenever the segment from z to x stays in that region. The endpoint disks are compact and disjoint from Z, so the parts of the segments near them contain no zeros. Thus every parallel segment contains a zero at a fixed positive distance from its endpoints. For small $t > 0$, (12.11) gives on that segment an interval of length $2\sqrt{2t/K}$ contained in $\{f \le t\}$. Integrating over the transverse disk, which has positive measure, and using that $\mathrm{d}V$ is comparable to coordinate volume, we obtain

$$\mathrm{Vol}\{f \le t\} \ge c\sqrt{t}$$

for all sufficiently small t, contradicting (12.10). This proves the local connectedness assertion.

For each such ball, the complement of Z in it is nonempty, since Z has zero volume, and it lies in one component of $U = X \setminus Z$. Two overlapping balls give the same component, because their open overlap contains a point outside Z. Since X is connected, all these balls give the same component. Hence U is connected, and Theorem 12.1.1 shows that $u - v$ is constant.

If $f^{-1/2} \in L^1$, then, outside the null zero set,

$$\frac{\operatorname{Vol}\{f \le t\}}{\sqrt{t}} \le \int_{\{f \le t\}} f^{-1/2}\, \mathrm{d}V \longrightarrow 0$$

by dominated convergence. This proves the sufficient integrability condition. The sharpness assertion follows from the examples in Section 12.9, as quantified at the end of that section. □

12.2 Polarized cofactors and propagation

Uniform ellipticity is sufficient for the propagation of an extremum, but it is not necessary: the directions in which the cofactor is positive may already connect enough points to force constancy. For C^2 potentials, the quadratic form Q in (12.5) is continuous, and its polarization defines $Q^\sharp \colon T^*X \to TX$ by $\eta(Q^\sharp \xi) = Q(\xi, \eta)$. The *reachable set* of a point consists of the endpoints of absolutely continuous curves starting there whose velocity lies in $\operatorname{im} Q^\sharp$ almost everywhere. The *control orbit* of a point consists of the endpoints of finite concatenations of flow lines, starting there, of locally Lipschitz vector fields with values in $\operatorname{im} Q^\sharp$.

A positive weight h satisfying $\mathrm{dd}^{\mathrm{c}}(hT) = 0$ gives an energy identity that controls $\mathrm{d}w$ everywhere. Without such a weight, a degenerate maximum principle gives propagation only from the extremal sets. This distinction accounts for the different density assumptions on reachable sets and on orbits in the following theorem.

Theorem 12.2.1 *Let $u, v \in C^2(X)$ satisfy $A, B \ge 0$ and $A^n = B^n$, and put $w = u - v$.*

(1) *If there is a continuous $h > 0$ such that $\mathrm{dd}^{\mathrm{c}}(hT) = 0$ in distributions, then*

$$\mathrm{i}\,\partial w \wedge \bar\partial w \wedge T \equiv 0. \tag{12.12}$$

It follows that w is constant if the set where T is strictly positive is dense, or if the reachable set of one point is dense.

(2) *Without the weight hypothesis, the sets where w attains its maximum and its minimum are invariant under the flow of every locally Lipschitz vector field with values in $\operatorname{im} Q^\sharp$. If every control orbit is dense, then w is constant. For smooth data it is enough that smooth vector fields with values in $\operatorname{im} Q^\sharp$, for example the columns of the symbol a in (12.18) below, together with their iterated Lie brackets, span TX at every point.*

12.2.1 Weighted energy

Testing the weighted cofactor against w^2 produces a nonnegative gradient integral. Only distributional $\mathrm{dd^c}$-closedness of hT is needed: for a continuous $h > 0$, the condition is

$$\int_X \mathrm{dd^c}\varphi \wedge hT = 0 \qquad (\varphi \in C^\infty(X, \mathbb{R})). \tag{12.13}$$

Proof (of Theorem 12.2.1(1)) The product rule gives

$$\mathrm{dd^c}(w^2) = 2w\mathrm{dd^c}w + 2\mathrm{d}w \wedge \mathrm{d^c}w = 2w\mathrm{dd^c}w + \frac{\mathrm{i}}{\pi}\partial w \wedge \bar\partial w. \tag{12.14}$$

Approximate w^2 in C^2 by smooth functions φ in (12.13). Since hT is continuous and $\mathrm{dd^c}w \wedge T = 0$ by (12.1), this gives

$$0 = \frac{1}{\pi}\int_X h\,\mathrm{i}\partial w \wedge \bar\partial w \wedge T. \tag{12.15}$$

The continuous integrand is nonnegative, so it vanishes identically, which is (12.12).

By (12.5), (12.12) says that $Q(\mathrm{d}w) = 0$. For the nonnegative quadratic form Q, this implies $Q(\mathrm{d}w, \eta) = 0$ for every η. Therefore

$$\mathrm{d}w_x\big(Q_x^\sharp \eta\big) = 0 \qquad (x \in X,\ \eta \in T_x^*X). \tag{12.16}$$

By the chain rule and (12.16), w is constant along every absolutely continuous curve whose velocity lies in $\operatorname{im} Q^\sharp$ almost everywhere. Thus w is constant on each reachable set, and by continuity one dense reachable set suffices. If instead Q is positive definite on a dense set, then $Q(\mathrm{d}w) = 0$ gives $\mathrm{d}w = 0$ there, and hence everywhere by continuity. □

The argument uses only the test function w^2. For general test functions, integration by parts introduces derivatives of hT; the condition $\mathrm{dd^c}(hT) = 0$ alone does not make $L\varphi = (\mathrm{dd^c}\varphi \wedge T)/\mathrm{d}V$ a symmetric operator in divergence form.

12.2.2 The real symbol and exterior-ball propagation

To propagate an extremum without a weight, we write the linearized operator in real coordinates. Its symbol is continuous and semipositive. At a point where a ball outside the maximum set touches it, the normal direction must lie in the kernel of the symbol; this forces the maximum set to be invariant under the flows of vector fields with values in the range of the symbol.

In a holomorphic chart $z_j = x_j + \mathrm{i}y_j$, let C_{cf} be the Hermitian matrix characterized by

$$\frac{\big(\mathrm{i}\sum H_{j\bar k}\mathrm{d}z_j \wedge \mathrm{d}\bar z_k\big) \wedge T}{\mathrm{d}V} = \operatorname{Tr}(C_{\mathrm{cf}}H) \tag{12.17}$$

for every Hermitian matrix $H = (H_{j\bar{k}})$. Positivity of mixed products makes C_{cf} positive semidefinite. Write $C_{\mathrm{cf}} = P_{\mathrm{cf}} + \mathrm{i}Q_{\mathrm{cf}}$, where P_{cf} is real symmetric and Q_{cf} is real skew-symmetric.

In the real coordinate order $(x_1, \dots, x_n, y_1, \dots, y_n)$,

$$L\varphi := \frac{\mathrm{dd^c}\varphi \wedge T}{\mathrm{d}V} = \mathrm{Tr}(aD^2\varphi), \qquad a = \frac{1}{8\pi}\begin{pmatrix} P_{\mathrm{cf}} & Q_{\mathrm{cf}} \\ -Q_{\mathrm{cf}} & P_{\mathrm{cf}} \end{pmatrix}. \tag{12.18}$$

Indeed $\varphi_{j\bar{k}} = (\varphi_{x_jx_k} + \varphi_{y_jy_k} + \mathrm{i}\varphi_{x_jy_k} - \mathrm{i}\varphi_{y_jx_k})/4$; inserting this expression in (12.17) proves (12.18). For real coordinate vectors $\xi, \eta \in \mathbb{R}^n$, moreover,

$$(\xi, \eta)^T a(\xi, \eta) = \frac{1}{8\pi}(\xi - \mathrm{i}\eta)^* C_{\mathrm{cf}}(\xi - \mathrm{i}\eta) \geq 0.$$

Thus a is a continuous positive semidefinite principal symbol, the operator L has no first-order term in holomorphic coordinates, and the range of a is that of $Q^\sharp$.

An exterior-ball argument propagates the maximum for this operator. Let $W \in C^2$ satisfy $LW \geq 0$ in a chart, suppose $W \leq M$, and put $K = \{W = M\}$. If $z \in K$ is a point of K nearest to a point $y \notin K$ of the chart, then

$$(y - z)^T a(z)(y - z) = 0. \tag{12.19}$$

To prove this, suppose the left side is positive, set $r_0 = |y - z|$, and shrink the tangent ball toward z: $y' = z + \theta(y - z)$, $R = \theta r_0$, with $0 < \theta < 1$ small. The closed ball $\overline{B(y', R)}$ lies in the chart and meets K only at z. This follows from the triangle inequality; equality in $|k - y| \leq |k - y'| + |y' - y| \leq r_0$, for $k \in K$, forces $k = z$. For large $\alpha > 0$, the function

$$\psi(x) = \mathrm{e}^{-\alpha R^2} - \mathrm{e}^{-\alpha|x - y'|^2}$$

satisfies

$$L\psi = 2\alpha \mathrm{e}^{-\alpha|x-y'|^2}\big(\mathrm{tr}\, a - 2\alpha(x - y')^T a(x)(x - y')\big) < 0$$

in a neighborhood of z, because $z - y'$ is a nonzero multiple of $y - z$, so that $(z - y')^T a(z)(z - y') > 0$ by assumption. Choose a sufficiently small $r < R$, and consider the lens $D = B(z, r) \cap B(y', R)$. On the part of its boundary lying on the sphere $\partial B(y', R)$, we have $\psi = 0$ and $W - M \leq 0$. On the remaining compact boundary portion, $W - M < 0$, since the tangent ball meets K only at z. Hence, for sufficiently small $\varepsilon > 0$, $W - M \leq \varepsilon\psi$ on ∂D. At a positive interior maximum of $W - M - \varepsilon\psi$, the Hessian would be nonpositive, whereas $L(W - M - \varepsilon\psi) > 0$; this is impossible since $a \geq 0$. Hence the inequality $W - M \leq \varepsilon\psi$ holds in D. Outside the ball $B(y', R)$ we have $\psi \geq 0$, so this inequality holds in a full neighborhood of z. Since equality holds at z, the function $W - M - \varepsilon\psi$ has a local maximum at z, and therefore

$$0 \leq LW(z) \leq \varepsilon L\psi(z) < 0,$$

proving (12.19).

Since $a \geq 0$, its range is the orthogonal complement of its kernel, and (12.19) implies that $y - z$ lies in this kernel. Consequently $(y - z) \cdot Z(z) = 0$ for every vector field Z with values in the range of a. If Z is locally Lipschitz and γ is one of its flow curves starting in K, let $d(t)$ denote the squared distance from $\gamma(t)$ to K in a sufficiently small compact chart neighborhood. Shortening the time interval ensures that nearest points z_t lie in the chart interior. Write $D^+d(t) = \limsup_{h \downarrow 0}(d(t+h) - d(t))/h$ for the upper right derivative. Holding such a nearest point fixed when differentiating the distance from above gives

$$\begin{aligned} D^+d(t) &\leq 2(\gamma(t) - z_t) \cdot Z(\gamma(t)) \\ &= 2(\gamma(t) - z_t) \cdot \big(Z(\gamma(t)) - Z(z_t)\big) \leq 2\,\mathrm{Lip}(Z) d(t). \end{aligned}$$

As $d(0) = 0$, Gronwall's inequality gives $d(t) = 0$. The same argument for $-Z$, followed by continuation, proves invariance under both signs of time and under finite concatenations of such flows.

Proof (of Theorem 12.2.1(2)) Apply the preceding argument to $W = w$ and to $W = -w$ in each chart; both satisfy $LW = 0$ by (12.1). The maximum set of w is nonempty, closed, and contains every control orbit starting in it. If every orbit is dense, this maximum set is all of X, proving constancy. The argument for the minimum set is identical.

It remains to prove the bracket criterion. Suppose that a family of smooth vector fields with values in $\operatorname{im} Q^\sharp$ is bracket generating. At p, let $\mathcal{D}_p$ be the span of the values of all original fields pushed forward by finite compositions of their local flows and inverse flows. This distribution contains the original fields and is invariant under their flows. Differentiation of pushed-forward fields along a flow shows that it contains their nested Lie brackets; finite-dimensional subspaces are closed, so the derivative remains in $\mathcal{D}_p$. Induction and the Jacobi identity give every iterated bracket. The assumed rank condition therefore gives $\mathcal{D}_p = T_pX$.

Choose $2n$ such pushed-forward fields independent at p. The flow of each is a conjugate of an original flow and hence preserves its orbit. The map obtained by successively flowing for $2n$ small times has invertible derivative at zero. The inverse function theorem puts a neighborhood of p in its orbit. All orbits are open, and their partition of connected X has only one member. This proves the Hörmander-type sufficient condition; it requires neither a constant-rank hypothesis nor a smooth matrix square root of a. In particular, the columns of a smooth symbol a may be used as the original vector fields. □

For $C^{2,1}$ potentials, the columns of a are locally Lipschitz, so the orbit criterion still applies to them. The bracket criterion requires smooth vector fields.

12.2.3 The unweighted energy defect

Integration against w^2 also measures the obstruction to using the unweighted cofactor, that is, the case $h = 1$. For smooth potentials, the error is exactly half the pairing of w^2 with $\mathrm{dd^c}T$. On a surface, the Gauduchon condition makes this error vanish for every pair of potentials; in dimension three, $\mathrm{dd^c}T$ contains an additional term depending on the potentials.

Proposition 12.2.1 *Let u, v be smooth ω-plurisubharmonic functions with $A^n = B^n = f\,\mathrm{d}V$. Then*

$$\int_X \mathrm{d}w \wedge \mathrm{d^c}w \wedge T = \frac{1}{2}\int_X w^2 \mathrm{dd^c}T. \tag{12.20}$$

If $\mathrm{dd^c}T = 0$ and $\{f > 0\}$ is dense in X, then $u - v$ is constant.

Proof By (12.1), $\mathrm{dd^c}w \wedge T = 0$. Integrating (12.14) against T, and applying Stokes' theorem twice to the left-hand side, yields

$$\int_X \mathrm{dd^c}(w^2) \wedge T = \int_X w^2 \mathrm{dd^c}T.$$

Substitution proves (12.20).

For the uniqueness assertion, apply Theorem 12.2.1(1) with $h = 1$. At every point where $f > 0$, the form B is positive definite, so the term B^{n-1} makes the cofactor T strictly positive. The set where T is strictly positive is therefore dense, as required.□

For $n = 2$, one has $T = A + B$ and $\mathrm{dd^c}T = 2\mathrm{dd^c}\omega$. On a Gauduchon surface the proposition therefore makes $u - v$ constant whenever $\{f > 0\}$ is dense; Corollary 12.3.1 below gives the same conclusion on every Hermitian surface.

In dimension three, direct expansion gives

$$\mathrm{dd^c}T = 3\mathrm{dd^c}(\omega^2) + 3\mathrm{dd^c}\omega \wedge \mathrm{dd^c}(u + v).$$

The Gauduchon condition cancels the first term but need not cancel the second. For instance, take both potentials equal to the perturbation sv in Example 4.8.1. The common form $A = B$ is positive and satisfies $\mathrm{dd^c}(A^2) \neq 0$, so $\mathrm{dd^c}T = 3\mathrm{dd^c}(A^2) \neq 0$. Thus even for two equal positive solution forms, the cofactor need not be $\mathrm{dd^c}$-closed.

12.3 Integral rigidity in dimensions two and three

In dimensions two and three, if two positive forms A, B have equal determinants, an algebraic expression quadratic in $A - B$ has a strict sign unless the two forms agree. On a surface its integral vanishes because it is exact. On a threefold the integral is a term involving $\mathrm{dd^c}\omega$, so a sign condition on $\mathrm{dd^c}\omega$ leads to the same conclusion. Approximation in the Sobolev norm justifies the integral identities for $W^{2,n}$ potentials.

In this section $n \in \{2,3\}$. Norms of coefficients of forms are taken with respect to a fixed smooth metric and $\mathrm{d}V$.

Theorem 12.3.1 *Let $u, v \in W^{2,n}(X,\mathbb{R})$, and assume that $A, B \geq 0$ almost everywhere and*

$$A^n = B^n = f\,\mathrm{d}V, \qquad f > 0 \quad \text{almost everywhere}.$$

If $n = 3$, assume in addition that

$$\mathrm{dd^c}\omega \wedge \mathrm{i}\eta \wedge \bar{\eta} \leq 0 \tag{12.21}$$

for every $(1,0)$-covector η. Then $\mathrm{dd^c}(u-v) = 0$ in the sense of distributions, and $u - v$ is almost everywhere constant. For the ω-plurisubharmonic representatives of u and v, the difference is constant everywhere.

Proof Put

$$s = (u+v)/2, \qquad D = \mathrm{dd^c}w = A - B,$$
$$C = (A+B)/2 = \omega + \mathrm{dd^c}s.$$

The coefficients of C and D lie in L^n, so their products of top degree are integrable by Hölder's inequality. At almost every point the common determinant is positive, so A and B are positive definite there, and we may diagonalize them simultaneously by congruence: for a suitable coframe of $(1,0)$-covectors $\theta_1, \dots, \theta_n$, write $A = \sum e_j$, $B = \sum \lambda_j e_j$, where $e_j = \mathrm{i}\theta_j \wedge \bar{\theta}_j$, $\lambda_j > 0$, and $\prod \lambda_j = 1$.

In dimension two,

$$D^2 = A^2 + B^2 - 2A \wedge B = 2(2 - \lambda_1 - \lambda_2)e_1 \wedge e_2 \leq 0, \tag{12.22}$$

with equality if and only if $A = B$. In dimension three,

$$2C \wedge D^2 = A^3 + B^3 - A^2 \wedge B - A \wedge B^2 \leq 0. \tag{12.23}$$

Indeed, with $E = e_1 \wedge e_2 \wedge e_3$, the four terms on the right are

$$6E, \qquad 6E, \qquad 2\Big(\sum_j \lambda_j\Big)E, \qquad 2\Big(\sum_{i<j} \lambda_i\lambda_j\Big)E.$$

By the arithmetic–geometric mean inequality and $\prod_j \lambda_j = 1$, each sum is at least three, and equality forces all $\lambda_j = 1$, that is, $A = B$.

Choose smooth real functions $w_k \to w$ in $W^{2,n}$ and set $D_k = \mathrm{dd^c}w_k$; such approximations are obtained by a partition of unity and convolution, and they need not preserve semipositivity. For $n = 2$, Stokes' theorem gives $\int_X D_k^2 = 0$, since $D_k^2 = \mathrm{d}(\mathrm{d^c}w_k \wedge D_k)$. For a fixed norm-comparison constant $C_0 > 0$,

$$\|D_k^2 - D^2\|_{L^1} \leq C_0\|D_k - D\|_{L^2}(\|D_k\|_{L^2} + \|D\|_{L^2}) \longrightarrow 0.$$

Thus $\int_X D^2 = 0$, and the nonpositive integrand in (12.22) vanishes almost everywhere. By the equality case, $D = 0$ almost everywhere.

For $n = 3$, also choose smooth functions $s_k \longrightarrow s$ in $W^{2,3}$, and put $C_k = \omega + \mathrm{dd^c} s_k$. We need the identity

$$\int_X C_k \wedge D_k^2 = \int_X \omega \wedge D_k^2 = -\int_X \mathrm{d}w_k \wedge \mathrm{d^c} w_k \wedge \mathrm{dd^c}\omega. \tag{12.24}$$

The first equality holds because $\mathrm{dd^c} s_k \wedge D_k^2 = \mathrm{d}(\mathrm{d^c} s_k \wedge D_k^2)$. For the second equality, including its sign, put $J_k = \mathrm{d}w_k \wedge \mathrm{d^c} w_k$. Using $\mathrm{d^c d} = -\mathrm{dd^c}$ and $(\mathrm{d^c})^2 = 0$,

$$\mathrm{d^c} J_k = -D_k \wedge \mathrm{d^c} w_k, \qquad \mathrm{dd^c} J_k = -D_k^2.$$

Stokes' theorem, applied first to $\mathrm{d}(\omega \wedge \mathrm{d^c} J_k)$ and then to $\mathrm{d^c}(\mathrm{d}\omega \wedge J_k)$, gives

$$\int_X \omega \wedge \mathrm{dd^c} J_k = -\int_X \mathrm{d}\omega \wedge \mathrm{d^c} J_k,$$
$$0 = -\int_X \mathrm{dd^c}\omega \wedge J_k - \int_X \mathrm{d}\omega \wedge \mathrm{d^c} J_k.$$

Combining them yields (12.24).

With a fixed norm-comparison constant $C_0 > 0$, strong L^3 convergence implies convergence in L^1 of all products of three factors. For example,

$$\|(C_k - C) \wedge D_k^2\|_1 \le C_0 \|C_k - C\|_3 \|D_k\|_3^2 \longrightarrow 0,$$

and

$$\|C \wedge (D_k^2 - D^2)\|_1 \le C_0 \|C\|_3 \|D_k - D\|_3 (\|D_k\|_3 + \|D\|_3) \longrightarrow 0.$$

Also $\mathrm{d}w_k \wedge \mathrm{d^c} w_k \longrightarrow \mathrm{d}w \wedge \mathrm{d^c} w$ in $L^{3/2}$, so the right side of (12.24) converges. Consequently

$$\int_X C \wedge D^2 = -\int_X \mathrm{d}w \wedge \mathrm{d^c} w \wedge \mathrm{dd^c}\omega \ge 0. \tag{12.25}$$

The sign follows from $\mathrm{d}w \wedge \mathrm{d^c} w = (\mathrm{i}/(2\pi))\partial w \wedge \bar\partial w$ and (12.21). The integrand on the left is nonpositive by (12.23), so it vanishes almost everywhere, and the equality case, which applies where the common determinant is positive, gives $D = 0$ almost everywhere.

In both dimensions, therefore, $\mathrm{dd^c} w = 0$ in the sense of distributions. By Proposition 1.1.1 with background zero, w has a plurisubharmonic representative on X. It attains its maximum on the compact manifold X, so the maximum principle and connectedness make it constant.

For the ω-plurisubharmonic representatives, Lemma 1.3.3 upgrades the almost-everywhere equality $u = v$+constant to equality everywhere. In particular, normalizing the means gives equality in the Sobolev class, and normalizing the suprema gives equality of the plurisubharmonic representatives. □

Corollary 12.3.1 *Let $n \in \{2, 3\}$, and let $u, v \in C^2(X)$ satisfy $A, B \ge 0$ and $A^n = B^n = f\,\mathrm{d}V$, where f is positive on a dense set. If $n = 3$, assume (12.21); for instance, $\mathrm{dd^c}\omega = 0$ suffices. Then $u - v$ is constant.*

Proof Put $D = A - B$ and $C = (A + B)/2$. At a point where the common top power vanishes, the algebraic expressions in the proof of Theorem 12.3.1 are still nonpositive: in dimension two $D^2 = -2A \wedge B$, and in dimension three $2C \wedge D^2 = -A^2 \wedge B - A \wedge B^2$. The same integral identities force these continuous expressions to vanish everywhere. The equality cases give $D = 0$ on $\{f > 0\}$, and density and continuity then give $D = 0$ on X. As in the proof of the theorem, $u - v$ is then constant. □

For example, the Gauduchon metric ω of the Hopf threefold of Chapter 9 satisfies $\mathrm{dd}^{\mathrm{c}}\omega = -\alpha^2$ by (9.1), where $\alpha = \pi^*\omega_{\mathrm{FS}}$ is pulled back by the projection π to $\mathbb{P}^2$ and is semipositive; hence (12.21) holds. Theorem 12.3.1 therefore applies to this metric, even though its upper mass $v_+(\omega)$ of Definition 7.5.1 is infinite by Theorem 9.0.1. The following sections return to a general Hermitian background.

12.4 Bounded uniqueness and connectedness

For a continuous density whose positive set is connected and dense, bounded solutions are unique up to a constant. An energy measure of $w = u - v$ replaces the pointwise gradient calculation available for smooth solutions. Its pushforward under w has a Lipschitz density vanishing at the extreme values. If w were nonconstant, these properties would force both extremal sets to have zero mass for the common Monge–Ampère measure. A smooth test potential with a larger density near one of these sets then contradicts modified comparison.

Theorem 12.4.1 *Let $F \in C^0(X)$ satisfy $F \geq 0$, with $U = \{F > 0\}$ connected and dense. If bounded ω-plurisubharmonic functions u, v satisfy*

$$\omega_u^n = \omega_v^n = F\omega^n,$$

then $u - v$ is constant. In particular, $\sup_X u = \sup_X v$ *implies $u = v$.*

We use modified comparison in the following form. If b is bounded and $\omega_b \geq \varepsilon\omega$ for some $0 < \varepsilon < 1$, then $b/(1 - \varepsilon)$ is ω-plurisubharmonic, so Theorem 11.2.1 applies with $(1 - \varepsilon)v = b$.

Proposition 12.4.1 *Let u, v be continuous ω-plurisubharmonic functions with $\omega_u^n = \omega_v^n = F\omega^n$, where $F \geq 0$ is measurable. Let q be a smooth strictly ω-plurisubharmonic function, and write $\omega_q^n = G\omega^n$. Then for every $0 < \kappa < 1$ and $\eta > 0$,*

$$F\omega^n\big(\{u - v < \min_X(u - v) + \eta\} \cap \{F > \kappa G\}\big) > 0. \tag{12.26}$$

The analogous assertion holds for the set $\{u - v > \max_X(u - v) - \eta\}$.

Proof Choose $\delta \in (0, 1]$ with $\omega_q \geq \delta\omega$. For $0 < t < 1$, put $b_t = (1 - t)v + tq$, $m_t = \min_X(u - b_t)$, and $E_s = \{u < b_t + m_t + s\}$. Then $\omega_{b_t} \geq t\delta\omega$. Write $\mu = F\omega^n$, $d = \kappa^{-1/n} - 1$, and $\gamma = nd$. By (1.12),

$$\omega_{b_t}^n \geq \big((1-t)F^{1/n} + tG^{1/n}\big)^n \omega^n.$$

On $L = \{F \leq \kappa G\}$ this is at least $(1+\gamma t)\mu$, since $G^{1/n} \geq (1+d)F^{1/n}$ there, and on $R = \{F > \kappa G\}$ it is at least $(1-nt)\mu$. Choose $s > 0$ so small that the reverse comparison factor in (11.2), with $\varepsilon = t\delta$, is at most $1 + \gamma t/2$. Modified comparison on E_s, with $x = \mu(E_s \cap L)$ and $y = \mu(E_s \cap R)$, gives

$$(1+\gamma t)x + (1-nt)y \leq (1+\gamma t/2)(x+y), \qquad \text{and hence} \qquad y \geq \frac{1-\kappa^{1/n}}{2}\,\mu(E_s).$$

The open set E_s is nonempty by the definition of m_t. As $\omega_{b_t}^n \geq (t\delta)^n\omega^n$, (11.2) forces $\mu(E_s) > 0$, hence $y > 0$.

Let $a = \min_X(u-v)$ and $\Lambda = \|v-q\|_\infty$. Since $u - b_t = u - v + t(v-q)$, we have $|m_t - a| \leq t\Lambda$ and $E_s \subset \{u - v < a + 2t\Lambda + s\}$. Given η, choose t and then s with $2t\Lambda + s < \eta$. Since $y = \mu(E_s \cap R) > 0$, this proves (12.26). Interchanging u and v gives the assertion for the near-maximum set. □

12.4.1 The energy of the difference

In this subsection, u, v are continuous ω-plurisubharmonic functions with $A^n = B^n = F\omega^n$ for a measurable function $F \geq 0$, and $w = u - v$. Although the cofactor T is now only a current, its product with $\mathrm{d}w \wedge \mathrm{d}^c w$ still defines a positive measure. The identity $\mathrm{d}\mathrm{d}^c w \wedge T = 0$ determines how this energy is distributed over the level sets of w.

Proposition 12.4.2 *There is a finite positive measure $E = \mathrm{d}w \wedge \mathrm{d}^{\mathrm{c}}w \wedge T$, defined below as a limit of regularizations, with the following properties.*

For every smooth real function χ on $\mathbb{R}$,

$$\int_X \chi''(w)\,\mathrm{d}E = \int_X \chi(w)\,\mathrm{d}\mathrm{d}^{\mathrm{c}}T. \tag{12.27}$$

The signed current $\mathrm{d}\mathrm{d}^{\mathrm{c}}T$ is a finite measure. If $a = \min_X w$, $b = \max_X w$, and

$$k(t) = \int_X (w - t)_+\,\mathrm{d}\mathrm{d}^{\mathrm{c}}T,$$

then k is nonnegative, globally Lipschitz, supported on $[a, b]$, and

$$w_* E = k(t)\,\mathrm{d}t. \tag{12.28}$$

In particular, E charges no level set of w.

Let V be a relatively compact coordinate domain such that F has a positive essential lower bound on a neighborhood of $\overline{V}$. Then there is $C_V < \infty$ such that every piecewise C^1 Lipschitz function p on $[a, b]$ satisfies

$$\int_V |\mathrm{d}(p(w))|\,\mathrm{d}\lambda \le C_V \left(\int_V |p'(w)|^2\,\mathrm{d}E \right)^{1/2}, \tag{12.29}$$

where $\mathrm{d}\lambda$ is Euclidean coordinate volume.

Proof Locally, expand T into powers of the smooth form ω multiplied by closed Bedford–Taylor products of $\mathrm{d}\mathrm{d}^{\mathrm{c}}u$ and $\mathrm{d}\mathrm{d}^{\mathrm{c}}v$. The operator $\mathrm{d}\mathrm{d}^{\mathrm{c}}$ then acts only on the powers of ω, and the local mass bounds in Section 1.6 show that $\mathrm{d}\mathrm{d}^{\mathrm{c}}T$ is a finite signed measure.

We first construct E and prove (12.27). On a coordinate ball, convolve u, v and the coefficients of ω with a common nonnegative radial mollifier. Write the resulting smooth positive forms as $A_\rho = \omega_\rho + \mathrm{d}\mathrm{d}^{\mathrm{c}}u_\rho$ and $B_\rho = \omega_\rho + \mathrm{d}\mathrm{d}^{\mathrm{c}}v_\rho$, and let T_ρ be their mixed sum. On a slightly larger ball, add the same fixed multiple of $|z|^2$ to each potential to obtain plurisubharmonic functions. Their radial regularizations decrease to the originals, so Bedford–Taylor continuity applies to all closed Hessian products. Since u, v are continuous, u_ρ, v_ρ, w_ρ converge uniformly on compact subsets.

For a closed smooth product S_ρ appearing in T_ρ, the identity

$$\mathrm{d}w_\rho \wedge \mathrm{d}^{\mathrm{c}}w_\rho \wedge S_\rho = \tfrac{1}{2}\mathrm{d}\mathrm{d}^{\mathrm{c}}(w_\rho^2 S_\rho) - w_\rho \mathrm{d}\mathrm{d}^{\mathrm{c}}w_\rho \wedge S_\rho \tag{12.30}$$

has a distributional limit: both Hessian products on the right converge weakly with uniformly bounded local variation, and their weights converge uniformly. The same holds after multiplication by the convergent smooth factors coming from ω_ρ. Thus the positive measures $E_\rho = \mathrm{d}w_\rho \wedge \mathrm{d}^{\mathrm{c}}w_\rho \wedge T_\rho$ converge to a finite positive measure E. By (12.30), this limit does not depend on the regularization, so the local limits agree on overlapping charts.

For smooth regularizations, the chain rule gives

$$\mathrm{dd^c}\chi(w_\rho)\wedge T_\rho = \chi''(w_\rho)E_\rho + \chi'(w_\rho)\mathrm{dd^c}w_\rho\wedge T_\rho.$$

The last current tends to zero, because $\mathrm{dd^c}w_\rho\wedge T_\rho = A_\rho^n - B_\rho^n$ tends weakly to $A^n - B^n = 0$ with uniformly bounded variation, and $\chi'(w_\rho)$ converges uniformly. Passing to the limit and integrating by parts on the compact manifold X gives (12.27). A partition of unity justifies this passage without a global regularization; the terms containing derivatives of the cutoff functions cancel after summation.

Let $\sigma = w_*(\mathrm{dd^c}T)$ and $\nu = w_*E$. Equation (12.27) says that $\nu'' = \sigma$ as distributions on $\mathbb{R}$. Stokes' theorem and $\mathrm{dd^c}w\wedge T = 0$ give

$$\sigma(\mathbb{R}) = \int_X \mathrm{dd^c}T = 0, \qquad \int_{\mathbb{R}} s\,\mathrm{d}\sigma(s) = \int_X w\,\mathrm{dd^c}T = \int_X \mathrm{dd^c}w\wedge T = 0.$$

Consequently $k(t) = \int (s-t)_+\,\mathrm{d}\sigma(s)$ is globally Lipschitz and vanishes outside $[a,b]$. It satisfies $k'' = \sigma$, so the compactly supported distribution $\nu - k(t)\,\mathrm{d}t$ has second derivative zero and therefore vanishes. Positivity of ν and continuity of k imply $k \geq 0$. This proves (12.28), and the assertion about level sets follows because ν is absolutely continuous.

We now prove (12.29). On a coordinate neighborhood of $\overline{V}$, set $\beta = \mathrm{dd^c}|z|^2$ and choose $c_V > 0$ with $F\omega^n \geq c_V\beta^n$. Let A_{ac} denote the absolutely continuous part of the current A, whose coefficients are locally integrable functions. Applying Lemma 1.7.4 with the constant function $g = c_V^{1/n}$ and differentiating the coefficient measures, we obtain $\det_\beta A_{\mathrm{ac}} \geq c_V$ almost everywhere. Bounded plurisubharmonic functions lie in $W^{1,2}_{\mathrm{loc}}$ (Section 1.6), hence so does w after the quadratic correction above, and its mollifications converge to it strongly in $W^{1,2}_{\mathrm{loc}}$, because the gradient of a mollification is the mollification of the gradient. Thus $w_\rho \to w$ strongly in $W^{1,2}_{\mathrm{loc}}$, and a subsequence of the gradients converges almost everywhere. Moreover, $A_\rho \to A_{\mathrm{ac}}$ almost everywhere, by differentiation of the coefficient measures.

Since T_ρ contains the term A_ρ^{n-1} and all its terms are positive, Fatou's lemma, applied to E_ρ, yields the measure inequality

$$E \geq \mathrm{d}w\wedge\mathrm{d^c}w\wedge A_{\mathrm{ac}}^{n-1}. \tag{12.31}$$

The cofactor matrix of A_{ac} has smallest eigenvalue at least $\det A_{\mathrm{ac}}/\operatorname{tr} A_{\mathrm{ac}} \geq c_V/\operatorname{tr} A_{\mathrm{ac}}$, where traces and determinants are taken with respect to β. The trace is locally integrable because A has locally finite mass. The chain rule for Sobolev functions, (12.31) and the Cauchy–Schwarz inequality give

$$\int_V |\mathrm{d}(p(w))|\,\mathrm{d}\lambda \leq C\left(\int_V |p'(w)|^2\,\mathrm{d}E\right)^{1/2}\left(\int_V \frac{\operatorname{tr}A_{\mathrm{ac}}}{c_V}\,\mathrm{d}\lambda\right)^{1/2}.$$

The last factor is finite and is absorbed into C_V. The values of p' at the finitely many corners of p are immaterial, because E charges no level set of w. □

Lemma 12.4.1 *In the setting of Proposition 12.4.2, assume in addition that F is continuous and that $U = \{F > 0\}$ is connected and dense. If $w = u - v$ is nonconstant, then*

$$F\omega^n(\{w = \min_X w\}) = F\omega^n(\{w = \max_X w\}) = 0.$$

Proof Put $a = \min_X w < b = \max_X w$, and let k be the function of Proposition 12.4.2. It is Lipschitz and nonnegative, and $k(a) = 0$. Choose $\epsilon_j \downarrow 0$ with $a + \epsilon_j < b$. For $\delta > 0$, set

$$J_j(\delta) = \int_a^{a+\epsilon_j} \frac{\mathrm{d}t}{k(t) + \delta}.$$

If L is a Lipschitz constant for k, then $k(t) \le L(t - a)$ on this interval, so $J_j(\delta) \to \infty$ as $\delta \downarrow 0$. Choose $\delta_j > 0$ with $J_j = J_j(\delta_j) \ge j$. Let $p_j = 0$ on $(-\infty, a]$, $p_j = 1$ on $[a + \epsilon_j, \infty)$, and

$$p_j'(t) = \frac{1}{J_j(k(t) + \delta_j)} \qquad (a < t < a + \epsilon_j).$$

The pushforward identity (12.28) gives

$$\int_X |p_j'(w)|^2 \, \mathrm{d}E = \frac{1}{J_j^2} \int_a^{a+\epsilon_j} \frac{k(t)}{(k(t) + \delta_j)^2} \, \mathrm{d}t \le \frac{1}{J_j} \longrightarrow 0.$$

Every point of U has a coordinate neighborhood V as in Proposition 12.4.2, since F is continuous. By (12.29), therefore, $\mathrm{d}(p_j(w)) \to 0$ in $L^1_{\mathrm{loc}}(U)$. The bounded functions $p_j(w)$ converge pointwise and in L^1_{loc} to $\mathbf{1}_{\{w>a\}}$, whose distributional gradient therefore vanishes on U. Since U is connected, this indicator function is almost everywhere constant on U. As w is nonconstant, $\{w > a\}$ is a nonempty open set, and its intersection with the dense open set U is nonempty and open; hence the constant is one. Thus $\{w = a\} \cap U$ has zero volume, and since $F = 0$ off U, the first assertion follows. Applying the argument to $-w$ gives the assertion for the maximum set. □

12.4.2 Concentrating smooth test potentials

The next proposition produces smooth test potentials whose Monge–Ampère density dominates a fixed multiple of a prescribed function g, with a constant independent of $\sup_X g$ and of the derivatives of g. This uniformity allows the test densities to concentrate near a set of measure zero.

Proposition 12.4.3 *For every $B_0 > 0$ there is $\varepsilon = \varepsilon(X, \omega, B_0) > 0$ such that every smooth function $g \geq 1$ with $\|g\|_{L^2(\omega^n)} \leq B_0$ admits a smooth strictly ω-plurisubharmonic function q satisfying*

$$\omega_q^n \geq \varepsilon g \omega^n. \tag{12.32}$$

Proof Choose finitely many coordinate balls $B_j = \{|z_j| < R_j\}$, with closures in their charts, such that the concentric smaller balls $\{|z_j| < r_j\}$ cover X. On B_j, let ψ_j solve the Dirichlet problem

$$(\mathrm{dd^c}\psi_j)^n = g\omega^n, \qquad \psi_j|_{\partial B_j} = 0. \tag{12.33}$$

The density of $g\omega^n$ with respect to coordinate volume is smooth and strictly positive. The classical smooth Dirichlet theorem therefore applies, since $C_j(|z_j|^2 - R_j^2)$ is a smooth strict subsolution with zero boundary values for sufficiently large $C_j > 0$. Thus ψ_j is smooth and strictly plurisubharmonic on $\overline{B}_j$, and the maximum principle gives $\psi_j \leq 0$.

We need a lower bound for ψ_j independent of $\sup_X g$. Write the coordinate equation as $\det(\partial_a \bar\partial_b \psi_j) = D_j$, and put $M_j = -\min_{B_j} \psi_j$. Let x_j be a minimum point. For each real slope p with $|p| < M_j/(2R_j)$, the function $\psi_j - p \cdot (x - x_j)$ attains its minimum at an interior point, because its value at x_j is $-M_j$, while its boundary values are larger than $-M_j$. At such a contact point the real Hessian S of ψ_j is nonnegative, and the gradient of ψ_j maps the contact set onto the ball of these slopes. The area formula and the matrix inequality

$$\det_{\mathbb{R}} S \leq 4^n \big(\det_{\mathbb{C}}(\partial_a \bar\partial_b \psi_j)\big)^2 = 4^n D_j^2$$

therefore give $M_j \leq C(n, R_j)\|D_j\|_{L^2(B_j)}^{1/n}$. To prove the matrix inequality, let J be the standard complex structure on the real coordinate space and average S with $J^{\mathsf{T}} S J$. Concavity of log det, and the fact that the eigenvalues of this J-invariant average occur in pairs, give the displayed factor; singular matrices are handled by approximation. Since the ratio of ω^n to coordinate volume is fixed and bounded, $\|D_j\|_2$ is controlled by B_0. Consequently

$$-M_0 \leq \psi_j \leq 0 \quad \text{on every } B_j, \tag{12.34}$$

where M_0 depends only on X, ω, B_0 and the chosen finite cover.

Next we glue the local solutions. Put $\beta_j = \mathrm{dd^c}|z_j|^2$. Choose $a_j > 0$ such that $a_j \beta_j \leq \omega/2$ on B_j, and let $t_j = (R_j^2 + r_j^2)/2$. Fix $\tau > 0$ so small that

$\tau M_0 < a_j(R_j^2 - r_j^2)/4$ for every j. The smooth function

$$s_j = \tau\psi_j + a_j(t_j - |z_j|^2)$$

is bounded below by a fixed positive constant on the smaller ball, negative on a collar of ∂B_j, and satisfies $\omega_{s_j} \geq \omega/2 + \tau dd^c\psi_j$. Choose a smooth convex function $\phi_j : \mathbb{R} \to \mathbb{R}$ approximating $s \mapsto \max(s, 0)$, equal to zero below a small negative threshold and to s above the corresponding positive threshold, with $0 \leq \phi_j' \leq 1$ and $\phi_j'' \geq 0$; the thresholds are chosen smaller in absolute value than both margins of s_j. Set $Q_j = \phi_j(s_j)$ on B_j and extend it by zero. This extension is smooth, and

$$\omega_{Q_j} = \phi_j'(s_j)\omega_{s_j} + \big(1 - \phi_j'(s_j)\big)\omega + \phi_j''(s_j)\, ds_j \wedge d^c s_j \geq \omega/2.$$

On the smaller ball, $Q_j = s_j$ and $\omega_{Q_j} \geq \tau dd^c\psi_j$. If N is the number of balls, put $q = N^{-1}\sum_j Q_j$. Then $\omega_q = N^{-1}\sum_j \omega_{Q_j} \geq \omega/2$ everywhere. Each point belongs to at least one smaller ball $\{|z_j| < r_j\}$, so positivity of the other summands and (12.33) give

$$\omega_q^n \geq (\tau/N)^n (dd^c\psi_j)^n = (\tau/N)^n g\omega^n.$$

Thus (12.32) holds with $\varepsilon = (\tau/N)^n$, independent of $\sup g$ and of all derivatives of g. □

Proof (of Theorem 12.4.1) By Corollary 11.4.1, u, v are continuous. In dimension one, $dd^c w = 0$, so the maximum principle makes w constant. Assume $n \geq 2$.

Suppose that $w = u - v$ is nonconstant. Let $a = \min_X w$, $K = \{w = a\}$, and $\mu = F\omega^n$. By Lemma 12.4.1, $\mu(K) = 0$. Choose $B_0 > \|1\|_{L^2(\omega^n)}$ and let $\varepsilon > 0$ be the constant from Proposition 12.4.3. Since $\mu(K) = 0$, the compact set $K_0 = K \cap \{F \geq \varepsilon/2\}$ has zero ω^n-volume. Fix $M > 1$ with $\varepsilon M > \max_X F$. By outer regularity of the smooth volume measure, choose an open neighborhood O of K_0 so small that

$$\int_X \omega^n + (M^2 - 1)\int_O \omega^n < B_0^2.$$

Take a smooth $0 \leq \chi \leq 1$ equal to one near K_0 and supported in O, and set $g = 1 + (M - 1)\chi$. If $K_0 = \varnothing$, take $\chi = 0$. Then $g \geq 1$, $g = M$ on K_0, and $\|g\|_{L^2(\omega^n)} < B_0$, since $g^2 \leq 1 + (M^2 - 1)\chi$.

By Proposition 12.4.3, there is a smooth strictly ω-plurisubharmonic q with $\omega_q^n = G\omega^n \geq \varepsilon g\omega^n$. On K_0 we have $G \geq \varepsilon M > F$; on $K \setminus K_0$ we have $F < \varepsilon/2 < \varepsilon \leq G$. Thus $G > F$ throughout the compact set K. By compactness and continuity, we can choose $0 < \kappa < 1$ such that $F < \kappa G$ on K; put $W = \{F < \kappa G\}$. The complement of this open neighborhood of K is compact, and $w - a$ has a strictly positive minimum there if it is nonempty. Hence some $\eta > 0$ satisfies $\{w < a + \eta\} \subset W$.

Proposition 12.4.1 asserts that $\mu(\{w < a + \eta\} \cap \{F > \kappa G\}) > 0$. By the preceding inclusion, however, this set is empty. This contradiction shows that w is constant. If $\sup_X u = \sup_X v$, the constant is zero. □

For a bounded measurable density, a positive lower bound gives the same conclusion. In this case coercivity holds on all of X, and the concentrating test potential can dominate the essential upper bound of the density on a whole neighborhood of the extremal set.

Corollary 12.4.1 *If $0 < c_0 \le F \le C_0 < \infty$ almost everywhere and bounded ω-plurisubharmonic functions u, v satisfy $\omega_u^n = \omega_v^n = F\omega^n$, then $u - v$ is constant.*

Proof The functions u, v are continuous by Corollary 11.4.1, and in dimension one the assertion follows from $\mathrm{dd}^{\mathrm{c}}(u - v) = 0$ and the maximum principle. Let $n \ge 2$, suppose that $w = u - v$ is nonconstant, and put $K = \{w = \min_X w\}$. The proof of Lemma 12.4.1 applies with $U = X$, since Proposition 12.4.2 gives the pushforward identity and local coercivity on every chart. Thus K has zero volume.

Fix $B_0 > \|1\|_{L^2(\omega^n)}$ and the corresponding ε in Proposition 12.4.3. Choose $M > 1$ with $\varepsilon M > C_0$, and a smooth cutoff function χ, equal to one on a neighborhood W of K and supported in an open set of volume so small that $g = 1 + (M - 1)\chi$ satisfies $\|g\|_2 < B_0$. The resulting test potential q has density $G \ge \varepsilon g$. Choose $\kappa < 1$ with $C_0 < \kappa\varepsilon M$. Then $F < \kappa G$ almost everywhere on W. By continuity of w, a sufficiently thin set $\{w < \min_X w + \eta\}$ lies inside W, which contradicts Proposition 12.4.1. Hence w is constant. □

12.5 Comparison and invariant mass

The preceding argument controls the extremal level sets of $u - v$ through connectedness of the set where the density is positive. When the background satisfies the Guan–Li condition below, invariance of the total mass gives the ordinary comparison principle instead. An auxiliary solution whose density is larger on the set where $u > v$ yields small perturbations q of u for which $\{v < q\}$ has zero mass for the common Monge–Ampère measure, and the domination principle then proves uniqueness.

Set $V = \int_X \omega^n$. The *Guan–Li condition*

$$\mathrm{dd}^{\mathrm{c}}\omega = 0, \qquad \mathrm{dd}^{\mathrm{c}}(\omega^2) = 0 \tag{12.35}$$

is equivalent to invariance of the total mass $\int_X \mathrm{MA}_\omega(u)$ by Lemma 2.1.2 and Theorem 2.2.2. In dimension one it holds automatically, for degree reasons.

Theorem 12.5.1 *Assume* (12.35)*, let* $p > 1$*, and let* $f \geq 0$ *belong to* $L^p(X, \omega^n)$ *with positive integral. Then there is exactly one pair* (c, u)*, with* $c > 0$*,* $u \in \mathrm{PSH}(X, \omega) \cap L^\infty(X)$ *and* $\sup_X u = 0$*, satisfying*

$$(\omega + \mathrm{dd}^c u)^n = c f \, \omega^n.$$

Moreover

$$c = \frac{\int_X \omega^n}{\int_X f \, \omega^n},$$

and u *is continuous. Thus two bounded solutions with the same* L^p *density differ by a constant, without any positive lower bound on* f*.*

We first show that the ordinary comparison principle is equivalent to invariance of the mass.

Theorem 12.5.2 *Condition* (12.35) *holds if and only if every pair of bounded* ω*-plurisubharmonic functions* a, b *satisfies*

$$\int_{\{a<b\}} \mathrm{MA}_\omega(b) \leq \int_{\{a<b\}} \mathrm{MA}_\omega(a). \tag{12.36}$$

For the converse implication, it suffices to require this inequality for $a, b \in \mathcal{H}_\omega$*.*

Proof Under (12.35), Theorem 2.2.2 gives $\int_X \mathrm{MA}_\omega(a) = V$ for every bounded potential a; for $n = 1$ this is Stokes' theorem.

Set $q = \max(a, b)$. On $\{a < b\}$, plurifine locality gives $\mathrm{MA}_\omega(q) = \mathrm{MA}_\omega(b)$. On $\{a \geq b\}$, one has $a = q$, and $a \leq q$ everywhere, so bounded ordered contact (Lemma 1.7.1) gives

$$\mathbf{1}_{\{a \geq b\}} \mathrm{MA}_\omega(a) \leq \mathbf{1}_{\{a \geq b\}} \mathrm{MA}_\omega(q).$$

The total masses of a, b and q all equal V. Hence

$$\int_{\{a<b\}} \mathrm{MA}_\omega(b) = V - \int_{\{a \geq b\}} \mathrm{MA}_\omega(q) \leq V - \int_{\{a \geq b\}} \mathrm{MA}_\omega(a) = \int_{\{a<b\}} \mathrm{MA}_\omega(a),$$

which is (12.36).

Conversely, if (12.36) holds for all $a, b \in \mathcal{H}_\omega$, then Proposition 2.4.1 makes the mass constant on $\mathcal{H}_\omega$. By Theorem 2.2.2 and Lemma 2.1.2, this implies (12.35) when $n \geq 2$; for $n = 1$ that condition holds for degree reasons. □

For the uniqueness argument we need, in addition to comparison, auxiliary solutions and the mixed inequality (1.12). The auxiliary solutions come from continuous existence: if $G \geq 0$ lies in $L^p(X, \omega^n)$ and $\int_X G\omega^n = V$, Theorem 11.1.1 gives a continuous solution of $\mathrm{MA}_\omega(\rho) = cG\omega^n$ with a constant $c > 0$, and since all bounded potentials have mass V, this constant equals one. Thus

$$\mathrm{MA}_\omega(\rho) = G\omega^n \tag{12.37}$$

has a continuous solution ρ for every such G, in particular for the density with an indicator factor used below.

Proof (of Theorem 12.5.1) We use the domination principle in the form of the case $c = 0$ of Theorem 11.2.2: if $\mathrm{MA}_\omega(a)(\{a < b\}) = 0$ for bounded ω-plurisubharmonic functions a, b, then $a \geq b$.

We first prove that two bounded solutions u, v of $\mathrm{MA}_\omega(u) = \mathrm{MA}_\omega(v) = \mu = F\omega^n$, with $F \in L^p$, $p > 1$, differ by a constant. The mass identity gives $\mu(X) = V$. If $u - v$ is nonconstant, add a constant to u so that $u - v$ has supremum $s > 0$ and infimum $-s$. Since $\{v < u\}$ and $\{u < v\}$ are disjoint, after interchanging u and v if necessary, the set $S = \{v < u\}$ satisfies $\mu(S) \leq V/2$. Define

$$b_0 = \frac{V - 2\mu(S)}{V} \geq 0, \qquad G = 2F\mathbf{1}_S + b_0.$$

This nonnegative L^p function has total integral V, so we can choose a continuous solution ρ of (12.37). Take C_0 with $\rho - C_0 \leq u$, and set

$$q_\delta = (1 - \delta)u + \delta(\rho - C_0), \qquad E_\delta = \{v < q_\delta\} \subset S, \qquad 0 < \delta < 1.$$

Since $G \geq 2F$ on S, the mixed inequality (1.12) gives, on S,

$$\mathrm{MA}_\omega(q_\delta) \geq \left((1 - \delta) + \delta 2^{1/n}\right)^n \mu =: A_\delta \mu, \qquad A_\delta > 1. \tag{12.38}$$

Ordinary comparison (12.36), applied to the pair v, q_δ, yields

$$A_\delta \mu(E_\delta) \leq \int_{E_\delta} \mathrm{MA}_\omega(q_\delta) \leq \int_{E_\delta} \mathrm{MA}_\omega(v) = \mu(E_\delta).$$

Thus $\mu(E_\delta) = 0$, that is, $\mathrm{MA}_\omega(v)(\{v < q_\delta\}) = 0$. Bounded domination gives $v \geq q_\delta$ everywhere, and letting $\delta \downarrow 0$ gives $v \geq u$, which contradicts $\sup_X(u - v) = s > 0$. Therefore $u - v$ is constant. The argument never divides by F, so F may vanish on an open set.

Finally, for the density f in the theorem, put $I = \int_X f\omega^n > 0$ and $c = V/I$. By (12.37) with $G = cf$, there is a continuous solution, which we normalize by subtracting its supremum. Any other bounded solution, with constant c', satisfies $V = c'I$ by mass invariance, hence $c' = c$; the uniqueness just proved and the common normalization then identify the potentials. In particular, every bounded solution with this L^p density is continuous, although continuity was not assumed. □

Remark 12.5.1 Uniqueness does not require the Guan–Li condition: the other criteria in this chapter apply without it. The condition must hold for ω itself: on the torus of Example 2.3.1, the flat metric satisfies (12.35), but (12.36) fails for MA_ω by Corollary 2.4.1.

12.6 Smooth nonuniqueness for a degenerate density

The first example places the freedom in the potential inside an open region where the density vanishes. Three independent semipositive forms of rank one reduce equality of the Monge–Ampère measures to equality of products of three scalar coefficients. We construct two semipositive forms with the same cube, and then add an exact correction to their average to obtain a strictly positive background.

In this section and the following ones, $X = \mathbb{P}(\mathbb{C}^2 \oplus \mathbb{C}^2) = \mathbb{P}^3$, with block homogeneous coordinates $[A : B]$, where $A, B \in \mathbb{C}^2$, and ω_X is its Fubini–Study form. Here A and B denote coordinates, not the forms of (12.1); the two solution forms will be written A_0 and B_0.

Theorem 12.6.1 *On $X = \mathbb{P}^3$ there exist a smooth Hermitian form ω with $\mathrm{d}\omega \neq 0$, a smooth nonnegative nonzero function f, and distinct smooth functions $u, v \in \mathrm{PSH}(X, \omega)$ such that*

$$\sup_X u = \sup_X v = 0, \qquad (\omega + \mathrm{dd^c} u)^3 = f\omega_X^3 = (\omega + \mathrm{dd^c} v)^3.$$

The density vanishes on a nonempty open set, and both potentials have minimal singularities.

12.6.1 Three rank-one directions

On $U = \{A \neq 0, B \neq 0\}$ put

$$r = \log \frac{|A|^2}{|B|^2}, \qquad P = \mathrm{dd^c} \log |A|^2, \quad Q = \mathrm{dd^c} \log |B|^2, \quad R = \mathrm{d}r \wedge \mathrm{d^c} r.$$

The logarithms are computed in a local homogeneous lift, but their Hessians do not depend on the lift. The forms P, Q are pullbacks of Fubini–Study forms from the two projective lines $\mathbb{P}(\mathbb{C}^2)$. All three forms are semipositive of rank one, and

$$P^2 = Q^2 = R^2 = 0, \qquad P \wedge Q \wedge R > 0. \tag{12.39}$$

Indeed, in local coordinates $A = (1, a)$ and $B = \tau(1, b)$, P and Q are positive multiples of $\mathrm{i}\,\mathrm{d}a \wedge \mathrm{d}\bar{a}$ and $\mathrm{i}\,\mathrm{d}b \wedge \mathrm{d}\bar{b}$, while

$$\partial r = \frac{\bar{a}\,\mathrm{d}a}{1 + |a|^2} - \frac{\mathrm{d}\tau}{\tau} - \frac{\bar{b}\,\mathrm{d}b}{1 + |b|^2}.$$

Its $\mathrm{d}\tau$ component is nonzero, so the three rank-one forms correspond to independent directions. In particular, for nonnegative x, y, z,

$$(xP + yQ + zR)^3 = 6xyz\, P \wedge Q \wedge R, \tag{12.40}$$

and the form is positive exactly when all three coefficients are positive.

Since $\mathrm{dd^c} r = P - Q$, the chain rule gives, for a smooth function $G\colon \mathbb{R} \to \mathbb{R}$,

$$\mathrm{dd^c} G(r) = G'(r)(P - Q) + G''(r)R. \tag{12.41}$$

Let $\sigma(r) = \mathrm{e}^r/(1 + \mathrm{e}^r)$. Since $\log(|A|^2 + |B|^2) = \log|B|^2 + \log(1 + \mathrm{e}^r)$, we have

$$\omega_X = \sigma(r)P + \sigma(-r)Q + \sigma'(r)R. \tag{12.42}$$

A function of r that is constant near each end of $\mathbb{R}$ extends smoothly across $X \setminus U$. A form $xP + yQ + zR$ whose coefficients equal those of (12.42) for $|r| \geq 2$ extends smoothly by ω_X across $X \setminus U$.

12.6.2 The scalar coefficients

On the central interval $-1 \leq r \leq 1$, the coefficients will make a different rank-one factor vanish in each of the two solution forms. Outside a larger interval they coincide with the Fubini–Study coefficients, so the resulting forms extend across the omitted projective lines. Primes denote derivatives with respect to r.

Choose an even smooth function β supported in $[-1, 1]$, strictly positive on $(-1, 1)$, with $\beta' > 0$ on $(-1, 0)$, and normalized by $\int_{-1}^{1} \beta = 2$; for instance, a positive multiple of $\exp(-1/(1 - r^2))$ on $(-1, 1)$, extended by zero. Define

$$F(r) = -1 + \int_{-1}^{r} \beta(t)\,\mathrm{d}t \quad (-1 < r < 1), \qquad F = -1 \ (r \leq -1), \quad F = 1 \ (r \geq 1),$$

and put $p = \beta/2$, $q = \beta'/2$. Then F is smooth and odd, p is even, and q is odd, positive on $(-1, 0)$ and negative on $(0, 1)$. Both p and q vanish to infinite order at the endpoints.

Take smooth $a \geq 0$ with $a = 1$ on $(-\infty, -2/3]$ and $a = 0$ on $[-1/3, \infty)$, and put $b(r) = a(-r)$. Choose an even nonnegative $\chi \in C_c^\infty((-1/3, 1/3))$ with $\chi(0) > 0$. On $[-1, 1]$ set

$$z_0 = (q^2 + \chi^2)^{1/2}.$$

It is smooth: the expression under the square root is positive in the interior, and near each endpoint $z_0 = |q|$, where q has a fixed sign. It is positive in the interior and vanishes to infinite order at the endpoints. Also,

$$z_0 \geq |q|, \qquad a(z_0 - q) = 0, \qquad b(z_0 + q) = 0. \tag{12.43}$$

The first cancellation holds because $\chi = 0$ on the support of a and $q \geq 0$ there; the second follows by reflection.

On $(-\infty, -1]$ choose smooth functions ℓ, s such that $0 \leq \ell \leq 1$, $\ell = 0$ on $(-\infty, -2]$, $\ell = 1$ on $[-3/2, -1]$, while $s > 0$ for $r < -1$, $s = 1$ on $(-\infty, -3/2]$, and s vanishes to infinite order at -1. Such functions are easily obtained from smooth cutoff functions and a function like $\exp(-1/(r+1)^2)$. Define

$$\begin{aligned} &x = (1-\ell)\sigma + \ell, \quad y = (1-\ell)\sigma(-r) + \ell s, \quad z = (1-\ell)\sigma' + \ell s \quad (r \leq -1), \\ &x = p + a, \qquad\qquad y = p + b, \qquad\qquad z = z_0 \qquad\qquad (-1 \leq r \leq 1), \end{aligned}$$

and for $r \geq 1$ use $x(r) = y(-r)$, $y(r) = x(-r)$, $z(r) = z(-r)$.

The definitions join smoothly: at -1 both definitions give the values $(1, 0, 0)$, and their differences vanish to infinite order there, since $a = 1$ and $b = 0$ near -1. The central definitions are symmetric under reflection, so the same holds at 1. We have

$$x, y, z > 0 \quad \text{for } r \neq \pm 1, \qquad (x, y, z)(-1) = (1, 0, 0), \quad (x, y, z)(1) = (0, 1, 0).$$

For $|r| \geq 2$ their values are $(\sigma(r), \sigma(-r), \sigma'(r))$.

12.6.3 Two forms with the same cube

The cancellation identities now turn the scalar coefficients into two semipositive forms with the same cube. Their average is strictly positive except at the two ends of the central interval.

By the extensions described above, the form and the function

$$\omega_0 = x(r)P + y(r)Q + z(r)R, \qquad w = F(r)$$

are smooth on X. The form is semipositive, and positive except on the hypersurfaces $M_- = \{r = -1\}$ and $M_+ = \{r = 1\}$. Put

$$H = \tfrac{1}{2} \mathrm{dd^c} w = p(P - Q) + qR, \qquad A_0 = \omega_0 - H, \quad B_0 = \omega_0 + H.$$

For $|r| \geq 1$ we have $H = 0$. On $|r| < 1$,

$$A_0 = aP + (2p + b)Q + (z_0 - q)R, \qquad B_0 = (2p + a)P + bQ + (z_0 + q)R.$$

All coefficients are nonnegative, and by (12.40) and (12.43), $A_0^3 = B_0^3 = 0$ on this band. Outside it both cubes equal ω_0^3. Thus A_0, B_0 are smooth semipositive forms on X with equal cubes. This common top-degree form vanishes on the closed band $|r| \leq 1$ and is positive on its complement, including near $X \setminus U$.

12.6.4 Restoring strict positivity of the background

The forms A_0, B_0 already have the required common measure. We obtain a Hermitian background by adding an exact correction $\mathrm{dd^c}\rho$ to their average ω_0 and subtracting ρ from both potentials. The correction must restore positivity at $M_\pm$ while preserving it in the transition regions.

Take an even $\kappa \in C_c^\infty((-2,2))$, equal to one on $[-3/2, 3/2]$. For a parameter $\varepsilon > 0$ to be chosen, set

$$\rho(r) = \varepsilon(r^2 - 1)\kappa(r), \qquad \omega = \omega_0 + \mathrm{dd^c}\rho.$$

By (12.41), its coefficients on U are $x + \rho'$, $y - \rho'$ and $z + \rho''$. On $[-3/2, 3/2]$ one has $\rho' = 2\varepsilon r$ and $\rho'' = 2\varepsilon$, so the last coefficient is positive there. The first coefficient $x + 2\varepsilon r$ is positive for $r \geq 0$, since $x \geq 0$ and $x(0) > 0$, and on $[-3/2, 0]$ it is at least $m - 3\varepsilon$, where $m > 0$ is the minimum of x there. Reflection gives the corresponding estimate for $y - 2\varepsilon r$. On the compact region $3/2 \leq |r| \leq 2$, the form ω_0 is positive and the perturbation is $O(\varepsilon)$. For $|r| \geq 2$, including near $X \setminus U$, one has $\omega = \omega_0$. Hence ω is strictly positive on X for sufficiently small $\varepsilon > 0$. This argument uses only that $x, y, z \geq 0$, $y(r) = x(-r)$, $x > 0$ on $[-3/2, 0]$, and ω_0 is positive on $\{|r| \geq 3/2\}$ and near $X \setminus U$.

The form ω is not closed. Indeed, since $\mathrm{d}R = -\mathrm{d}r \wedge (P - Q)$,

$$\mathrm{d}\omega = \mathrm{d}\omega_0 = \mathrm{d}r \wedge \big((x' - z)P + (y' + z)Q\big). \tag{12.44}$$

At $r = 0$, $x' = y' = 0$ and $z = \chi(0) > 0$. Hence $\mathrm{d}\omega = z\,\mathrm{d}r \wedge (Q - P) \neq 0$ there. The nonzero $\mathrm{d}\tau$ component used in (12.39) also shows that $\mathrm{d}r \wedge P$ and $\mathrm{d}r \wedge Q$ are linearly independent.

Proof (of Theorem 12.6.1) Set $\widetilde{u} = -w/2 - \rho$ and $\widetilde{v} = w/2 - \rho$. Since $\mathrm{dd^c}w = 2H$, the construction gives

$$\omega + \mathrm{dd^c}\widetilde{u} = A_0 \geq 0, \qquad \omega + \mathrm{dd^c}\widetilde{v} = B_0 \geq 0.$$

Subtracting their respective suprema gives u, v. Their Hessians are unchanged, and $v - u = w + \text{constant}$ is nonconstant. The function $f = A_0^3/\omega_X^3 = B_0^3/\omega_X^3$ is smooth, nonnegative and nonzero, and it vanishes on the nonempty open band $|r| < 1$. Both potentials are smooth, hence bounded, so they have minimal singularities: for a Hermitian background, the envelope V_ω of Section 1.19 is zero. □

Every Hermitian background on $\mathbb{P}^3$ has the bounded mass property of Definition 7.5.1: since the Fubini–Study form is closed, Stokes' theorem for bounded potentials shows that all its bounded potentials have the same mass, and Proposition 7.5.1 shows that the property does not depend on the metric. This applies to all the backgrounds on $\mathbb{P}^3$ constructed below.

The linearized equation also explains why the maximum principle argument fails. The form

$$T = A_0^2 + A_0 \wedge B_0 + B_0^2$$

satisfies $\mathrm{dd}^c w \wedge T = B_0^3 - A_0^3 = 0$, but $T = 0$ on $M_\pm$, where $A_0 = B_0 = \omega_0$ has rank one. Thus the linear equation for w is not elliptic everywhere, even though both potentials and their common density are smooth.

12.6.5 A continuum of radial solutions

The open band where the density vanishes allows more solutions than the two smooth ones constructed above. Writing a further potential as $G(r) - \rho$, with G a function of one real variable, reduces the equation on the band to choosing the slope G' so that one of the three coefficients vanishes. Switching the vanishing coefficient at two suitable levels produces a continuum of $C^{1,1}$ solutions for the same background ω and density f.

Recall that $q = p'$ and $z_0 \geq |q|$. The strict inequality $z_0 > |p'|$ near zero, where $\chi > 0$, gives

$$\int_0^1 z_0(r)\,\mathrm{d}r > \int_0^1 |p'(r)|\,\mathrm{d}r = p(0). \tag{12.45}$$

Proposition 12.6.1 *For the smooth background ω and the smooth density f of Theorem 12.6.1, the equation*

$$\mathrm{MA}_\omega(u) = f\omega_X^3$$

has uncountably many distinct $C^{1,1}$ solutions normalized by $\sup_X u = 0$. One of them, constructed explicitly, is invariant under the block exchange $\iota[A : B] = [B : A]$.

Proof Consider pairs $-1 < I < 0 < J < 1$ satisfying

$$\int_I^J z_0(r)\,\mathrm{d}r = p(I) + p(J). \tag{12.46}$$

We first show that such pairs exist, with I ranging over an interval. Define

$$K(I, J) = \int_I^J z_0(r)\,\mathrm{d}r - p(I) - p(J).$$

We have $K(0, 0) = -2p(0) < 0$ and $K(0, 1) > 0$ by (12.45). By continuity, there is $\epsilon > 0$ such that $K(I, 0) < 0 < K(I, 1)$ for every $I \in [-\epsilon, 0]$. For $0 < J < 1$,

$$\partial_J K(I, J) = z_0(J) - p'(J) > 0.$$

The intermediate value theorem therefore gives a unique $J = J(I) \in (0, 1)$ satisfying (12.46) for every $I \in [-\epsilon, 0)$; we may assume $\epsilon < 1$.

For any pair satisfying (12.46), define the slope on $[-1, 1]$ by

$$h_{I,J}(r) = \begin{cases} p(r), & -1 \le r \le I, \\ p(I) - \displaystyle\int_I^r z_0(s)\,\mathrm{d}s, & I \le r \le J, \\ -p(r), & J \le r \le 1, \end{cases}$$

and extend it by zero outside that interval. The balance identity makes it continuous at J, and continuity at I is immediate. Each piece has bounded derivative, so $h_{I,J}$ is Lipschitz on $\mathbb{R}$; it vanishes to infinite order at ± 1. Let $G_{I,J}$ be a primitive of $h_{I,J}$. It is $C^{1,1}$ and constant near each end of the real line. Consequently $G_{I,J}(r)$ extends to a $C^{1,1}$ function on X that is smooth near $X \setminus U$.

On the middle interval $[I, J]$ the slope lies between $-p$ and p. Indeed,

$$(h_{I,J} - p)' = -z_0 - p' \le 0, \qquad (h_{I,J} + p)' = -z_0 + p' \le 0.$$

The first difference vanishes at I, and the second vanishes at J. Since both are nonincreasing, $h_{I,J} \le p$ and $h_{I,J} \ge -p$ on $[I, J]$.

Put $\widetilde{u}_{I,J} = G_{I,J}(r) - \rho$. By (12.41), almost everywhere on U,

$$\omega + \mathrm{dd}^{\mathrm{c}}\widetilde{u}_{I,J} = \omega_0 + \mathrm{dd}^{\mathrm{c}} G_{I,J}(r) = (x + h_{I,J})P + (y - h_{I,J})Q + (z + h'_{I,J})R. \quad (12.47)$$

On $I < r < J$, the first two coefficients are nonnegative because $x = p + a$, $y = p + b$ and $-p \le h_{I,J} \le p$, and the third is zero. On $-1 < r < I$, we have $h_{I,J} = p$ and $b = 0$, since $I < 0$; so the second coefficient is zero, and the others are nonnegative because $z_0 + p' \ge 0$. On $J < r < 1$, we have $h_{I,J} = -p$ and $a = 0$; so the first coefficient is zero, and the others are nonnegative because $z_0 - p' \ge 0$. Outside the band, $h_{I,J} = h'_{I,J} = 0$, so the form equals ω_0.

Formula (12.47) has no additional term supported on the hypersurfaces $r = I$ and $r = J$: the function $G_{I,J}$ is $C^{1,1}$, so its first derivative is continuous at both transition points, and its distributional Hessian is its bounded almost-everywhere Hessian. The preceding coefficient inequalities therefore prove $\omega + \mathrm{dd}^{\mathrm{c}}\widetilde{u}_{I,J} \ge 0$ as a current.

By Proposition 1.7.2, its Monge–Ampère measure is the pointwise cube of (12.47), computed almost everywhere. By the rank-one identities (12.39) and (12.40), this cube is zero almost everywhere on the band, since one coefficient vanishes there, and it equals ω_0^3 off the band. These are exactly the values of $f\omega_X^3$ in the preceding construction. Thus $\widetilde{u}_{I,J}$ solves the equation, and subtracting its supremum gives a normalized solution $u_{I,J}$.

Different values of $I \in [-\epsilon, 0)$ give different normalized solutions. If $I_1 < I_2 < 0$ and $I_1 < r < I_2$, then the slope belonging to I_2 is $p(r)$, whereas

$$p(r) - h_{I_1,J(I_1)}(r) = \int_{I_1}^r (p'(s) + z_0(s))\,\mathrm{d}s > 0,$$

because $p' > 0$ on $(-1, 0)$. The two primitives therefore do not differ by a constant. Since the interval $[-\epsilon, 0)$ is uncountable, this proves the first assertion. None of these slopes equals p or $-p$ on all of $(-1, 1)$, so the constructed solutions are also distinct from the two smooth solutions of Theorem 12.6.1.

Finally, the function

$$L(t) = \int_0^t z_0(r)\,\mathrm{d}r - p(t), \qquad 0 \le t \le 1,$$

is negative at zero, positive at one, and strictly increasing on $(0, 1)$. Its unique zero t_* gives a symmetric pair $I = -t_*$, $J = t_*$ satisfying (12.46). Since z_0 and p are even, $h_{I,J}$ is odd: on the middle interval it equals $-\int_0^r z_0(s)\,\mathrm{d}s$. Choose its primitive even. Since ι changes r into $-r$ and ρ is even, the potential $G_{I,J}(r) - \rho$ and its normalization are fixed by ι. This proves the last assertion. □

For these symmetric data, the normalized solutions of the equations with a positive exponent converge, as the exponent tends to zero, to the symmetric solution just constructed; see Theorem 15.19.1.

12.7 Nonuniqueness with a zero set of empty interior

We now keep the Hermitian background of Theorem 12.6.1, but change the density so that it is positive off $M_- \cup M_+$, where $M_\pm = \{r = \pm 1\}$. For potentials of the form $G(r) - \rho$, the equation on the middle band becomes a first-order ordinary differential equation for the slope G'. Barriers keep its solutions between the admissible slopes, and infinite-order vanishing at the endpoints then makes their extensions by zero smooth.

Theorem 12.7.1 *There are a smooth nonclosed Hermitian form ω on $\mathbb{P}^3$ and a smooth nonnegative density g, positive off two disjoint smooth real hypersurfaces, such that the equation*

$$\mathrm{MA}_\omega(u) = g\omega_X^3$$

has uncountably many distinct smooth solutions normalized by $\sup_X u = 0$. The upper mass $v_+(\omega)$ is finite, that is, $\mathbb{P}^3$ has the bounded mass property.

Thus a zero set with empty interior, even one of measure zero, does not ensure uniqueness of smooth normalized solutions.

We use the coefficients of Section 12.6.2, with the explicit choice

$$p(r) = C\exp\!\left(-\frac{1}{1-r^2}\right), \qquad -1 < r < 1, \qquad \int_{-1}^{1} p(r)\,\mathrm{d}r = 1,$$

extended by zero, where the integral condition determines $C > 0$; then $q = p'$. Fix $0 < s < 1$, and set

$$k_s(r) = (x(r) + sp(r))(y(r) - sp(r))(z(r) + sq(r)).$$

For $|r| > 1$ this is xyz. If $G' = sp$, then the potential $G(r) - \rho$ has $\omega + \mathrm{dd}^{\mathrm{c}}(G(r) - \rho) = \omega_0 + sH$. This form is strictly positive for $-1 < r < 1$, since

$$x + sp = a + (1+s)p > 0, \quad y - sp = b + (1-s)p > 0, \quad z_0 + sq \geq (1-s)z_0 > 0.$$

It is also strictly positive outside the closed band, and at $r = \pm 1$ it equals the rank-one form ω_0. Consequently

$$g\omega_X^3 = (\omega_0 + sH)^3 \tag{12.48}$$

defines a smooth density g that is positive exactly off $M_- \cup M_+$. These hypersurfaces have measure zero and empty interior. On the band $|r| < 1$, a potential $G(r) - \rho$ with slope $h = G'$ solves the equation exactly when $(x + h)(y - h)(z + h') = k_s$, which is the differential equation (12.49) below.

Lemma 12.7.1 *There exists $\eta_0 > 0$ such that for every $0 < \eta < \eta_0$ the initial-value problem*

$$h' = \frac{k_s(r)}{(x(r) + h)(y(r) - h)} - z(r), \qquad h(0) = sp(0) - \eta, \tag{12.49}$$

has a smooth solution on $(-1, 1)$ with $-x < h < y$ and $h < sp$. Moreover, on some fixed endpoint neighborhoods there are constants $0 < c_L < s < t < 1$, independent of η, such that

$$c_L p < h < sp \quad \textit{near } -1, \qquad -tp < h < sp \quad \textit{near } 1.$$

Proof The function $h_s = sp$ solves (12.49) and lies strictly between $-x$ and y throughout $(-1, 1)$. Write $F_s(r, h)$ for the right-hand side of the equation, which is smooth on the open domain $-x(r) < h < y(r)$.

Near -1 the coefficients are exactly

$$x = 1 + p, \qquad y = p, \qquad z = q > 0.$$

For a constant $c \in (-1, 1)$, the product of coefficients for the trial slope cp is

$$k_c = (1 - c^2)\big(1 + (1 + c)p\big)pq.$$

Fix $c_L \in (0, s)$. At $p = 0$, the factor preceding pq for $c = c_L$ exceeds that for $c = s$ by $s^2 - c_L^2 > 0$. Thus, on a sufficiently small fixed neighborhood of the left endpoint,

$k_{c_L} > k_s$ and

$$F_s(r, c_L p) - (c_L p)' = \frac{k_s - k_{c_L}}{(x + c_L p)(y - c_L p)} < 0. \tag{12.50}$$

Thus $c_L p$ is a lower barrier when the equation is solved backward, that is, toward -1.

Near 1 the exact coefficients are

$$x = p, \qquad y = 1 + p, \qquad z = -q > 0,$$

and the trial slope cp has product of coefficients

$$k_c = (1 - c^2)\big(1 + (1 - c)p\big)p(-q).$$

Fix $s < t < 1$. At $p = 0$, the factor for $c = s$ exceeds that for $c = -t$ by $t^2 - s^2 > 0$. On a sufficiently small fixed neighborhood of the right endpoint, therefore, $k_s > k_{-t}$, so

$$F_s(r, -tp) - (-tp)' = \frac{k_s - k_{-t}}{(x - tp)(y + tp)} > 0. \tag{12.51}$$

Thus $-tp$ is a lower barrier when the equation is solved forward, toward 1.

Choose $r_L < 0 < r_R$ in these respective neighborhoods. The graph of h_s over $[r_L, r_R]$ is a compact subset of the domain of F_s. Existence and continuous dependence on initial values for smooth ordinary differential equations therefore give $\eta_0 > 0$ such that, for every $0 < \eta < \eta_0$, the solution with initial value $sp(0) - \eta$ exists on $[r_L, r_R]$, stays in the domain, and satisfies

$$h(r_L) > c_L p(r_L), \qquad h(r_R) > -tp(r_R).$$

By local uniqueness, two solutions of the equation cannot cross. Since $h(0) < h_s(0)$, we have $h < h_s$ wherever h exists.

Follow the solution backwards from r_L; it cannot cross $c_L p$. Indeed, at a first crossing point the difference $h - c_L p$ would vanish and be positive immediately to its right, so its derivative there would be nonnegative, whereas (12.50) shows that this derivative is strictly negative. Thus $c_L p < h < sp$ on the part of the interval of existence to the left of r_L. Likewise, (12.51) prevents a first downward crossing of $-tp$ to the right of r_R, giving $-tp < h < sp$ there.

These bounds also show that the maximal interval of existence is all of $(-1, 1)$. Indeed, near a possible endpoint inside $(-1, 1)$, the factors $x + h$ and $y - h$ stay bounded away from zero, so the graph of h stays in a compact subset of the domain of F_s, and the continuation theorem for ordinary differential equations extends the solution beyond that endpoint. Hence the solution exists on $(-1, 1)$ and satisfies the claimed bounds. □

Lemma 12.7.2 *Every solution furnished by Lemma 12.7.1, extended by zero outside* $(-1,1)$*, is smooth on* $\mathbb{R}$ *and vanishes to infinite order at* ± 1.

Proof Let h be one of the solutions in Lemma 12.7.1. On either endpoint neighborhood put $\xi = h/p$; since $p > 0$ on $(-1,1)$, this quotient is smooth there, and the trapping bounds place ξ in a fixed compact subinterval of $(-1,1)$. Define the rational functions

$$R_L(p,\xi) = \frac{(1-s^2)(1+(1+s)p)}{(1+(1+\xi)p)(1-\xi)} - 1,$$

$$R_R(p,\xi) = 1 - \frac{(1-s^2)(1+(1-s)p)}{(1+\xi)(1+(1-\xi)p)}.$$

The explicit formulas for the coefficients near the endpoints give

$$h' = q\,R_L(p,\xi) \quad \text{on the left,} \qquad h' = q\,R_R(p,\xi) \quad \text{on the right.}$$

Both rational functions are smooth, with all derivatives bounded, on a neighborhood of the relevant set of pairs (p,ξ) with $p \geq 0$ small and ξ in the compact interval above. Writing $R_{L/R}$ for the function corresponding to the endpoint under consideration, we obtain from $h' = p'\xi + p\xi'$ and $q = p'$

$$\xi' = \frac{p'}{p}\big(R_{L/R}(p,\xi) - \xi\big). \tag{12.52}$$

For the chosen bump,

$$\frac{p'}{p} = -\frac{2r}{(1-r^2)^2}.$$

This rational function and each of its derivatives grow at most like a power of the inverse distance to the endpoint. Moreover, each derivative of p is p times a rational function with the same property.

An induction using (12.52) shows that every derivative $\xi^{(j)}$ grows at most polynomially in the inverse distance to the endpoint. The case $j = 0$ is the trapping bound on ξ. For the induction step, differentiate the right-hand side of (12.52), including its term $-\xi$, j times. Every resulting term is a product of a derivative of p'/p, a bounded partial derivative of $R_{L/R}$, and derivatives of p and of ξ of order at most j. All of these factors have polynomial bounds, by the induction hypothesis for the derivatives of ξ and since every derivative of p is bounded. This gives the bound for $\xi^{(j+1)}$.

Leibniz' rule for $h = p\xi$ now gives, for each j, constants C_j, N_j such that

$$|h^{(j)}(r)| \leq C_j p(r)(1-|r|)^{-N_j}$$

near the endpoints. The right-hand side tends to zero at the endpoints, because p decays exponentially in the inverse distance to the endpoint. Thus all derivatives of h tend to zero at ± 1. Extend each derivative by zero outside $(-1,1)$. By the mean value theorem, the difference quotient of $h^{(j)}$ at an endpoint tends to the limit of $h^{(j+1)}$,

which is zero. Hence these extensions are the successive derivatives of the extension of h, which is therefore smooth and vanishes to infinite order at ± 1. □

Proof (of Theorem 12.7.1) Let η_0 be as in Lemma 12.7.1, and for $0 < \eta < \eta_0$ denote by h_η the solution with initial value $sp(0) - \eta$. Extend h_η by zero, which gives a smooth function by the preceding lemma, and take a primitive G_η. It is constant near each end of the real line, so $G_\eta(r)$ extends smoothly to all of $\mathbb{P}^3$, including the two omitted projective lines. Set

$$\widetilde{u}_\eta = G_\eta(r) - \rho.$$

On the open band,

$$\omega + \mathrm{dd}^{\mathrm{c}}\widetilde{u}_\eta = (x + h_\eta)P + (y - h_\eta)Q + (z + h'_\eta)R.$$

The coefficients of P and Q are strictly positive by Lemma 12.7.1, while the equation gives

$$z + h'_\eta = \frac{k_s}{(x + h_\eta)(y - h_\eta)} > 0.$$

Outside the band the form is ω_0, and on the two boundary hypersurfaces it is also the rank-one form ω_0, since h_η vanishes to infinite order there. It is therefore semipositive on X, and by (12.40) its cube equals $6k_s P \wedge Q \wedge R = (\omega_0 + sH)^3$ on the band and $\omega_0^3 = (\omega_0 + sH)^3$ elsewhere. Thus its cube is everywhere the measure $g\omega_X^3$ of (12.48). Finally, normalize by subtracting $\sup_X \widetilde{u}_\eta$.

Distinct values of η give different slopes at $r = 0$, so the corresponding potentials do not differ by a constant, and their normalized versions are distinct. This gives an uncountable family. The background is that of Theorem 12.6.1, and $v_+(\omega)$ is finite because every Hermitian background on $\mathbb{P}^3$ has the bounded mass property. □

The density vanishes to infinite order on the two hypersurfaces. Their complement has three components, and on each of the two outer components, two different solutions differ by a constant, with different constants on the two components. The differential equation on the middle component joins these constants smoothly across the degenerate hypersurfaces. This is why ellipticity at every point of the positive-density set does not give a maximum principle argument across its boundary.

12.8 Quadratic density zeros

Infinite-order vanishing of the density is not needed for smooth nonuniqueness. We change the coefficients of ω_0 so that its coefficient of R stays positive and one of its coefficients of P and Q vanishes quadratically at each of the two hypersurfaces $M_\pm$. The resulting singular differential equation forces the slopes of differences of solutions to decay like $\exp(-A/d)$, where d is the distance to the hypersurface and $A > 0$. The slopes therefore extend smoothly by zero, even though the density vanishes only to finite order.

Theorem 12.8.1 *On $\mathbb{P}^3$ there exist a smooth nonclosed Hermitian form ω and a smooth nonnegative density g such that:*

(1) *The zero set of g consists of two disjoint smooth real hypersurfaces. Near either hypersurface, g is the square of a smooth defining function times a smooth strictly positive function.*
(2) *The equation $(\omega + \mathrm{dd^c} u)^3 = g\omega_X^3$ has uncountably many distinct smooth ω-plurisubharmonic solutions normalized by $\sup_X u = 0$.*

The upper mass $v_+(\omega)$ is finite, that is, $\mathbb{P}^3$ has the bounded mass property.

We keep the rank-one forms P, Q, R and the function r of Section 12.6.1, and use new scalar coefficients.

Let a and $b(r) = a(-r)$ be the cutoff functions of Section 12.6.2. Choose a smooth even cutoff χ valued in $[0,1]$, zero for $|r| \le 5/4$ and one for $|r| \ge 3/2$. Set $p(r) = (1-r^2)^2$ and

$$\begin{aligned} x &= (1-\chi)(p+a) + \chi\sigma, \\ y &= (1-\chi)(p+b) + \chi(1-\sigma), \\ z &= 1 - \chi + \chi\sigma', \qquad \omega_0 = xP + yQ + zR. \end{aligned} \tag{12.53}$$

Since these coefficients coincide with those of (12.42) for $|r| \ge 3/2$, the form ω_0 extends by ω_X across the two omitted projective lines. The coefficient z is strictly positive, and x, y are strictly positive except that $y(-1) = 0$ and $x(1) = 0$. Thus ω_0 is positive off $M_\pm = \{r = \pm 1\}$ and has rank two there.

With the correction $\rho = \varepsilon(r^2-1)\kappa(r)$ of Section 12.6.4, put $\omega = \omega_0 + \mathrm{dd^c}\rho$. The coefficients (12.53) are nonnegative, $y(r) = x(-r)$, $x > 0$ on $[-3/2, 0]$, and $\omega_0 = \omega_X$ for $|r| \ge 3/2$. The argument of Section 12.6.4 therefore shows that ω is strictly positive for sufficiently small $\varepsilon > 0$. The form ω is not closed: at $r = 0$, $x' = y' = 0$ and $z = 1$, so by (12.44)

$$\mathrm{d}\omega = \mathrm{d}r \wedge \big((x'-z)P + (y'+z)Q\big) \neq 0.$$

Define the density g by

$$g\omega_X^3 = \omega_0^3.$$

It is smooth and positive exactly off $M_- \cup M_+$. By (12.42) and (12.40), $g = xyz/(\sigma(r)(1-\sigma(r))\sigma'(r))$. Near either hypersurface, $xyz = (1+p)p$, and hence

$$g(r) = \frac{(1+p(r))(1-r^2)^2}{\sigma(r)(1-\sigma(r))\sigma'(r)}.$$

The denominator is smooth and strictly positive at ± 1, and $1 - r^2$ is a defining function for $M_\pm$ near these hypersurfaces. This proves the asserted quadratic vanishing. The potential $-\rho$ is one smooth solution.

Lemma 12.8.1 *For every sufficiently small $\eta > 0$, the equation*

$$h' = \frac{xyz}{(x+h)(y-h)} - z, \qquad h(0) = -\eta \tag{12.54}$$

has a smooth solution on $(-1, 1)$ with $-x < h < y$ and $h < 0$. There are a constant $0 < c < 1$ and fixed neighborhoods of both endpoints, independent of η, on which

$$-cp < h < 0.$$

Every such solution, extended by zero, is a smooth function on $\mathbb{R}$ that vanishes to infinite order at ± 1.

Proof The right-hand side $F(r, h)$ is smooth on $-x < h < y$, and $h = 0$ is a solution. Near -1, $(x, y, z) = (1 + p, p, 1)$; near 1, $(x, y, z) = (p, 1 + p, 1)$. Fix $0 < c < 1$. For the trial function $-cp$, near the left endpoint

$$F(r, -cp) + cp' = \frac{1+p}{(1+(1-c)p)(1+c)} - 1 + cp' \longrightarrow -\frac{c}{1+c} < 0,$$

whereas near the right endpoint

$$F(r, -cp) + cp' = \frac{1+p}{(1-c)(1+(1+c)p)} - 1 + cp' \longrightarrow \frac{c}{1-c} > 0$$

as $r \to \pm 1$, since $p, p' \to 0$ at both endpoints. The strict inequalities therefore hold on fixed neighborhoods of the endpoints, and $-cp$ is a lower barrier for the equation solved backwards near -1 and forwards near 1.

Choose a compact interval $[r_L, r_R] \ni 0$ with endpoints in those neighborhoods. Continuous dependence on initial values, around the solution $h = 0$, gives $\eta_0 > 0$ such that for every $0 < \eta < \eta_0$ the solution exists on $[r_L, r_R]$, stays in the domain of F, and lies above $-cp$ at r_L and r_R. By uniqueness of solutions, it cannot meet the solution zero, so $h < 0$. The barrier argument used in the proof of Lemma 12.7.1 then gives $-cp < h < 0$ near both ends. On each compact subinterval of $(-1, 1)$ the trapped graph stays at a positive distance from the boundary of the domain of F; hence, by the continuation theorem, the solution exists on all of $(-1, 1)$.

The equation factors as

$$h' = h\, L(r, h), \qquad L(r, h) = \frac{z(x - y + h)}{(x+h)(y-h)}. \tag{12.55}$$

Since $h < 0$, we have $(\log(-h))' = L(r, h)$. Near the left endpoint, the trapping bounds and the explicit coefficients imply

$$L(r, h) = \frac{1+h}{(1+p+h)(p-h)} \geq \frac{C_1}{p}$$

after shrinking that neighborhood, with $C_1 > 0$. Near the right endpoint they give

$$L(r,h) = \frac{-1+h}{(p+h)(1+p-h)} \le -\frac{C_2}{p}, \qquad C_2 > 0.$$

Indeed, on the left, $1+h \ge 1-cp \ge 1/2$ and the denominator is at most $(1+p)(1+c)p$; on the right, the numerator is at most -1, and the denominator is at most $p(1+(1+c)p)$. Since p is comparable to the square of the distance $d = 1-|r|$ to the endpoint, integrating these inequalities backwards on the left and forwards on the right gives, for some $A > 0$,

$$|h(r)| \le C\mathrm{e}^{-A/d} \tag{12.56}$$

near either endpoint. The constants may depend on the solution; uniformity in η is not needed for the smooth extension.

It remains to control all derivatives. On the trapped graph, all partial derivatives in (r,h) of the rational function L grow at most polynomially in d^{-1}. Indeed, near an endpoint x, y are polynomials in r, $z = 1$, h is bounded, and of the two factors in the denominator, one is bounded below by a positive constant and the other by $(1-c)p$. Repeated differentiation of the quotient therefore only introduces finite powers of p^{-1}.

Inductively, for every $j \ge 0$ there are constants C_j, N_j with

$$|h^{(j)}(r)| \le C_j \mathrm{e}^{-A/d} d^{-N_j}.$$

The case $j = 0$ is (12.56). Differentiate $h' = hL(r,h(r))$ j times. Every resulting term is the product of a derivative of the factor h, of order at most j, and a derivative of the composite function $L(r,h(r))$. The latter involves only partial derivatives of L and derivatives of h of order at most j, which are polynomially bounded by the induction hypothesis, while the first factor has the exponential bound. This proves the case $j+1$. All derivatives of h therefore tend to zero at ± 1. Extending them by zero and applying the mean value theorem to the difference quotients at the endpoints, as in Lemma 12.7.2, shows that the extension of h by zero is smooth and vanishes to infinite order at ± 1. □

Proof (of Theorem 12.8.1) Let $\eta_0 > 0$ be such that Lemma 12.8.1 applies for $0 < \eta < \eta_0$, and let h_η denote the solution with $h_\eta(0) = -\eta$. Take a primitive G_η of its extension by zero. Since G_η is constant near each end of the real line, $G_\eta(r)$ defines a smooth function on X. The potential $u_\eta = G_\eta(r) - \rho$ satisfies

$$\omega + \mathrm{dd}^{\mathrm{c}} u_\eta = (x+h_\eta)P + (y-h_\eta)Q + (z+h_\eta')R.$$

Inside the band, the first two coefficients are positive by the lemma, and the equation gives

$$z + h_\eta' = \frac{xyz}{(x+h_\eta)(y-h_\eta)} > 0.$$

Outside the band the form is ω_0, and on the two boundary hypersurfaces it is also the rank-two form ω_0, since h_η vanishes to infinite order there. Thus it is semipositive on X, and by (12.40) its cube is $6xyzP \wedge Q \wedge R = \omega_0^3 = g\omega_X^3$. Different initial values give potentials that do not differ by constants, so after normalization by the supremum

we obtain an uncountable family. Bounded mass follows from Proposition 7.5.1, as for Theorem 12.6.1. □

At each of the hypersurfaces $M_{\pm}$, the difference of two of these solutions agrees to infinite order with a constant, because its derivative is a difference of slopes that vanish to infinite order there. The constants at the two hypersurfaces need not agree. Thus the singular differential equation produces differences that are flat at $M_{\pm}$, even though the density has only quadratic zeros.

12.9 A real analytic density

A globally real analytic density can occur in an equation with a smooth nonclosed Hermitian background and distinct smooth normalized solutions. We prescribe it as a polynomial in a real analytic function on projective space. Two symmetric radial perturbations give the required determinant on the middle band; patching the coefficients while preserving their product extends the construction across the projective ends. The two hypersurfaces where the density vanishes quadratically separate regions with different extremal values of the difference of the solutions.

Theorem 12.9.1 *On $X = \mathbb{P}^3$, with block homogeneous coordinates $[A : B]$ and Fubini–Study form ω_X, put*

$$t = \frac{|A|^2}{|A|^2 + |B|^2}, \qquad a = \frac{1}{1+\mathrm{e}}, \qquad b = \frac{\mathrm{e}}{1+\mathrm{e}}, \qquad f_*(t) = (t-a)^2(t-b)^2.$$

There exist a smooth nonclosed Hermitian form ω and distinct smooth ω-plurisubharmonic functions u, v, normalized by $\sup_X u = \sup_X v = 0$, with

$$(\omega + \mathrm{dd}^{\mathrm{c}} u)^3 = (\omega + \mathrm{dd}^{\mathrm{c}} v)^3 = f_*(t)\omega_X^3. \tag{12.57}$$

Both solution forms are positive away from the two connected real analytic hypersurfaces $\{t = a\}$, $\{t = b\}$, and have complex rank two on them. The density is real analytic on X, and it vanishes exactly to second order in the direction transverse to these hypersurfaces. For every $n \geq 4$ the same assertions hold on $\mathbb{P}^3 \times \mathbb{P}^{n-3}$, with the pulled-back density and the product Fubini–Study volume, and with rank $n-1$ on the two product hypersurfaces.

Proof We use the forms P, Q, R and the function r of Section 12.6.1. By (12.40), the construction reduces to finding scalar coefficients with a prescribed product. By (12.42),

$$\omega_X^3 = 6(\sigma')^2 P \wedge Q \wedge R, \qquad \sigma' = \sigma(r)\sigma(-r) > 0. \tag{12.58}$$

12.9.1 Prescribing the determinant

On U, one has $t = \sigma(r)$. We first construct two triples of coefficients, positive on the middle band, with product $f_*(\sigma)(\sigma')^2$. At each endpoint of the band, exactly one of the coefficients of P and Q will vanish.

Put $d_*(r) = f_*(\sigma(r))(\sigma'(r))^2$. Since $a = \sigma(-1)$, $b = \sigma(1)$, define the positive smooth divided differences

$$k_-(r) = \int_0^1 \sigma'(-1 + s(r+1))\,\mathrm{d}s, \qquad k_+(r) = \int_0^1 \sigma'(1 + s(r-1))\,\mathrm{d}s,$$

and $H_*(r) = k_-(r)k_+(r)\sigma'(r) > 0$. The fundamental theorem of calculus gives

$$\sigma(r) - a = (r+1)k_-(r), \quad \sigma(r) - b = (r-1)k_+(r), \quad d_* = (1-r^2)^2 H_*^2. \tag{12.59}$$

These formulas show that H_* is smooth and positive also at the endpoints, without any division by a vanishing function. The symmetry $\sigma(-r) = 1 - \sigma(r)$ makes d_*, and hence H_*, even.

Choose the function

$$p(r) = \begin{cases} \mathrm{e}^{-1/(1-r^2)}, & |r| < 1, \\ 0, & |r| \geq 1, \end{cases} \qquad q = p'.$$

All its derivatives vanish at $r = \pm 1$: each derivative is a finite sum of polynomials in r times powers of $(1-r^2)^{-1}$, multiplied by the exponential, and the exponential factor dominates every such power. On $|r| < 1$, $q = -2rp/(1-r^2)^2$. Define

$$z_b = \sqrt{q^2 + 2H_*^2}, \qquad S_* = \sqrt{r^2 + 4p^2 + 2z_b(1-r^2)^2}, \tag{12.60}$$

$$x_b = (S_* - r)/2, \qquad y_b = (S_* + r)/2. \tag{12.61}$$

Both radicands are strictly positive everywhere, so these formulas define smooth functions on $\mathbb{R}$, including at the endpoints. They satisfy

$$z_b > |q|, \qquad x_b - y_b = -r, \qquad x_b y_b = p^2 + \frac{z_b}{2}(1-r^2)^2. \tag{12.62}$$

The coefficients x_b, y_b are nonnegative, positive away from the endpoints, with values $(1,0)$ at -1 and $(0,1)$ at 1. On the open band,

$$S_*^2 - (2p + |r|)^2 = 2(1-r^2)^2(z_b - |q|) > 0,$$

so $x_b > p$, $y_b > p$. Consequently both triples

$$(x_b - p, y_b + p, z_b - q), \qquad (x_b + p, y_b - p, z_b + q) \tag{12.63}$$

are strictly positive there.

We now verify that the two triples have the same product. Set $D_* = x_b y_b - p^2$, $E_* = p(x_b - y_b) = -rp$. Equations (12.62) and the formula for q give

$$D_* = (z_b/2)(1-r^2)^2, \qquad qD_* = z_b E_*.$$

The two products in (12.63) therefore equal

$$(D_* + E_*)(z_b - q) = (D_* - E_*)(z_b + q) = \frac{D_*}{z_b}(z_b^2 - q^2) = d_*. \tag{12.64}$$

For $|r| \geq 1$, $p = q = 0$ and the same identity reads $x_b y_b z_b = d_*$. Thus both triples have exactly the prescribed product, and at the endpoints exactly one of their entries vanishes.

12.9.2 Extension over the projective ends

The coefficients just constructed have the right product but need not extend across the projective ends. Away from the zero levels of $f_*(t)$, we interpolate the coefficients of P and Q to those of $f_*(t)^{1/3}\omega_X$, and determine the coefficient of R by the prescribed product.

Choose a smooth even cutoff $0 \leq \eta \leq 1$, equal to one on $|r| \leq 3/2$ and zero on $|r| \geq 2$. On $|r| < 3/2$ set $(x, y, z) = (x_b, y_b, z_b)$. On $|r| > 5/4$, where $f_*(\sigma(r)) > 0$, put

$$\begin{aligned} \kappa &= f_*(\sigma(r))^{1/3}, \\ x &= \eta x_b + (1-\eta)\kappa\sigma(r), \\ y &= \eta y_b + (1-\eta)\kappa\sigma(-r), \qquad z = d_*/(xy). \end{aligned}$$

These coefficients are smooth and positive on the outer region. On the overlap $5/4 < |r| < 3/2$, we have $\eta = 1$, $p = q = 0$ and $x_b y_b z_b = d_*$, so the two definitions agree. For $|r| \geq 2$ they equal $\kappa(\sigma(r), \sigma(-r), \sigma'(r))$. Thus

$$\omega_0 = xP + yQ + zR$$

extends smoothly across both omitted projective lines, being equal near them to $f_*(t)^{1/3}\omega_X$, which is smooth and positive there since $f_*(0) = f_*(1) = a^2 b^2 > 0$. On X, the form ω_0 is semipositive, positive away from $r = \pm 1$, and of rank two on these levels.

The function

$$F(r) = 2\int_0^r p(s)\,\mathrm{d}s$$

is constant on each of the intervals $(-\infty, -1]$ and $[1, \infty)$, and so $F(r)$ extends to a smooth nonconstant function on X. By (12.41), half of its complex Hessian is $H_0 = \mathrm{dd}^{\mathrm{c}} F/2 = p(P - Q) + qR$. By (12.64) and (12.58), the forms $A_0 = \omega_0 - H_0$,

$B_0 = \omega_0 + H_0$ satisfy

$$A_0^3 = B_0^3 = f_*(t)\omega_X^3$$

on U, and hence on X. Their ranks and positivity were determined above, after (12.63). The construction is symmetric: $x(-r) = y(r)$, and z is even.

12.9.3 A positive equation background

As in the first example, an exact correction makes the average form strictly positive without changing the two solution forms. Here the coefficient of R is already positive, and the correction restores the missing coefficient of P or Q at each endpoint.

For $\varepsilon > 0$, set $\rho = \varepsilon(r^2 - 1)\eta(r)$ and $\omega = \omega_0 + \mathrm{dd}^{\mathrm{c}}\rho$. The coefficients of ω_0 are nonnegative, $y(r) = x(-r)$, $x = x_b > 0$ on $[-3/2, 0]$, and ω_0 is positive on $\{|r| \geq 3/2\}$ and near $X \setminus U$. The argument of Section 12.6.4, with its cutoff replaced by η, therefore shows that ω is strictly positive for sufficiently small $\varepsilon > 0$.

The correction is exact, so $\mathrm{d}\omega = \mathrm{d}\omega_0$. At $r = 0$, where $x = x_b$ and $y = y_b$, evenness of S_* gives $x'(0) = -1/2$ and $y'(0) = 1/2$. Hence (12.44) gives

$$\mathrm{d}\omega_0 = (z_b(0) + 1/2)\,\mathrm{d}r \wedge (Q - P) \neq 0,$$

since $\mathrm{d}r \wedge P$ and $\mathrm{d}r \wedge Q$ are linearly independent, as noted after (12.44). Thus the Hermitian background ω is not closed.

Now put $\widetilde{u} = -F/2 - \rho$ and $\widetilde{v} = F/2 - \rho$. Then $\omega + \mathrm{dd}^{\mathrm{c}}\widetilde{u} = A_0$, $\omega + \mathrm{dd}^{\mathrm{c}}\widetilde{v} = B_0$. Subtracting their respective suprema gives u, v with $\sup_X u = \sup_X v = 0$, and (12.57) holds. Their difference is nonconstant because F is nonconstant.

12.9.4 Zeros and higher dimensions

It remains to examine the regularity of the density and the geometry of its zero levels. Taking a product with projective space then preserves the equality of the Monge–Ampère measures and gives examples in every higher dimension.

The function t is real analytic in every affine chart of X, being a quotient of real polynomials with positive denominator. Hence $f_*(t)$ is real analytic on X. Its two zero levels are regular hypersurfaces because $t = \sigma(r)$, $\sigma' > 0$, and $\mathrm{d}r \neq 0$. Each level is connected, since it is the quotient of the product of two three-spheres of fixed nonzero radii by the diagonal action of the circle. Each level is compact and has zero volume. The divided differences in (12.59) show near the two levels that

$$f_*(\sigma(r)) = (r + 1)^2 g_-(r), \qquad f_*(\sigma(r)) = (r - 1)^2 g_+(r), \tag{12.65}$$

respectively, with smooth strictly positive functions $g_\pm$.

For $n \geq 4$, let θ be the Fubini–Study form on $Y = \mathbb{P}^{n-3}$, let $\pi_1 \colon X \times Y \to X$ and $\pi_2 \colon X \times Y \to Y$ be the projections, and set, on $X \times Y$,

$$\omega_n = \pi_1^* \omega + \pi_2^* \theta, \qquad \Omega_n = \pi_1^* \omega_X + \pi_2^* \theta, \qquad u_n = \pi_1^* u, \quad v_n = \pi_1^* v.$$

In the expansion of an n-fold product, only the terms with exactly three factors pulled back from X survive. Consequently

$$(\omega_n + \mathrm{dd^c} u_n)^n = \binom{n}{3} \pi_1^*(A_0^3) \wedge \pi_2^*(\theta^{n-3}) = f_*(\pi_1^* t) \Omega_n^n,$$

and the same holds for v_n. On the two product levels, the solution forms have rank $2 + (n-3) = n - 1$. Restriction to a slice $X \times \{y\}$ recovers $\mathrm{d}\omega \neq 0$, so ω_n is not closed. Normalization, distinctness and real analyticity are preserved by pullback. □

Write Vol for the volume with respect to ω_X^3, or Ω_n^n in dimension n. The quadratic zeros complete the proof of the sharpness assertion in Theorem 12.1.2. In finitely many tubular charts around the zero levels, the density is bounded above and below by positive multiples of the square of a real transverse coordinate, and the volume form has a bounded positive density with respect to coordinate volume. Away from these charts, the density is bounded below by a positive constant. Thus, for $s > 0$ and $\alpha > 0$,

$$\mathrm{Vol}\{f_* \leq s\} \asymp \sqrt{s} \quad (s \downarrow 0), \qquad f_*^{-\alpha} \in L^1 \quad \Longleftrightarrow \quad \alpha < \tfrac{1}{2}. \tag{12.66}$$

Thus, for every $0 < \alpha < 1/2$ and every dimension $n \geq 3$, there are smooth examples of nonuniqueness whose density f satisfies $f^{-\alpha} \in L^1$; this condition is weaker than the hypothesis $f^{-1/2} \in L^1$ of Theorem 12.1.2.

Remark 12.9.1 For this example, the positive-density set $\{f_*(t) > 0\}$ has three components: the middle band and the two outer regions. The two solution forms have the same missing direction on each zero level, so $E_{\mathrm{unif}} = \{f_*(t) > 0\}$. The maximum and the minimum of the difference of the two potentials are attained on the two outer components, whose closures are disjoint, in accordance with Theorem 12.1.1. Although the closure of the middle component meets both outer closures, the extremal values do not propagate through the middle component.

Chapter 13
Complex Hessian equations

Let X be a connected compact complex manifold of dimension n, let ω be a Hermitian form, and fix $2 \le m < n$. The complex Hessian operator $(\omega + \mathrm{dd}^{\mathrm{c}} u)^m \wedge \omega^{n-m}$ replaces the determinant by an intermediate symmetric polynomial. Its admissible cone allows negative eigenvalues, so the product calculus for plurisubharmonic potentials from Chapter 1 no longer applies directly. The cone, the normalized symmetric functions σ_j and the admissible potentials are fixed in Section 13.1.

For a continuous admissible weak solution whose smooth positive density depends on the potential, the difficulty is to obtain derivative estimates before smoothness is known. Proper exponential equations associate a smooth solution with each smooth admissible approximation of the weak solution. A Liouville theorem reduces their gradient bound to a trace bound (Section 13.2), and the trace estimate of Section 13.3 is independent of derivatives of the prescribed approximation. The resulting complex-Hessian bound therefore survives passage to the limit (Section 13.4). Uniform convergence defines the continuous mixed products, while obstacle envelopes give the domination principle that identifies the limit with the original solution (Section 13.6). A concave uniformly elliptic extension then supplies classical interior regularity (Section 13.7). The determinant case is treated in Theorem 11.5.1; the argument here concerns the intermediate Hessian degrees. Uniform stability also constructs solutions for strictly increasing exponents, by contraction between constant barriers.

Bounded admissible potentials require decreasing approximation (Section 13.5.1). The recursion of Section 13.8 first gives currents with measure coefficients below degree m. Sobolev capacity controls first Hessians but not products of two (Section 13.9); local Hessian capacities control multiplication by the discontinuous potentials in all degrees (Section 13.10). Quasicontinuity and weighted convergence make the products symmetric and allow all approximation indices to tend to their limits together. At degree m, the positive product after wedging with ω^{n-m} is a Radon measure.

These weighted limits pass locality and the maximum inequality to bounded potentials (Section 13.11). In global comparison (Section 13.12), the maximum inequality leaves a signed total-mass difference. The complex Hessians of the background powers determine when this difference vanishes for every bounded

admissible list. On small sublevel sets, a modified comparison estimate absorbs the torsion error on arbitrary Hermitian backgrounds. It yields strict domination, hence nonvanishing of the Hessian measure of every bounded potential, bounded stability for proper equations and smoothness for their solutions with smooth data between constant barriers.

The last two sections do not use the fixed degree m, and there $n \geq 1$ is arbitrary. For a smooth real $(1,1)$-form χ, they treat equations defined by a concave symmetric function of the eigenvalues of $\omega^{-1}(\chi + \mathrm{dd^c} u)$, such as $\sigma_m^{1/m}$ on Γ_m. An admissible reference with a strict asymptotic gap controls an auxiliary Monge–Ampère comparison, giving stability as positive right-hand sides tend to zero (Section 13.13). Equicontinuous right-hand sides then give uniform compactness, and nearby smooth admissible functions yield a gradient bound if the first derivatives of the right-hand sides are uniformly bounded (Section 13.14).

13.1 The admissible cone

For $\lambda \in \mathbb{R}^n$, let $S_j(\lambda)$ be the elementary symmetric polynomial of degree j, put $\sigma_j = S_j / \binom{n}{j}$ and $\sigma_0 = 1$, and set

$$\Gamma_m = \{\lambda : S_j(\lambda) > 0,\ 1 \leq j \leq m\}.$$

A real $(1,1)$-form belongs to $\Gamma_m(\omega)$ if its eigenvalues relative to ω belong to Γ_m. Write $C_{n,m}$ for the corresponding closed cone of Hermitian matrices. The algebraic assertions below also apply when $m = n$.

We use the classical Gårding facts: the cone $C_{n,m}$ is convex, contains the nonnegative matrices, and is preserved by adding them; adding a positive matrix puts a matrix of $C_{n,m}$ in its interior. The normalized root $F = \sigma_m^{1/m}$ is concave and increasing on Γ_m. Polarized products of k forms in $\Gamma_k(\omega)$ with ω^{n-k} are positive, and on Γ_k, hence by continuity on $\overline{\Gamma_k}$, Maclaurin's inequalities give $\sigma_j^{1/j} \geq \sigma_k^{1/k}$ for $1 \leq j \leq k$. If α has eigenvalues λ relative to ω, then $\alpha^j \wedge \omega^{n-j} = \sigma_j(\lambda)\, \omega^n$; this is the normalization of all Hessian measures in the chapter.

A continuous real function v on X is *admissible* if

$$(\omega + \mathrm{dd^c} v) \wedge \gamma_1 \wedge \cdots \wedge \gamma_{m-1} \wedge \omega^{n-m} \geq 0 \tag{13.1}$$

in distributions for every smooth collection $\gamma_1, \ldots, \gamma_{m-1} \in \Gamma_m(\omega)$. For smooth v this is equivalent, by Gårding duality, to $\omega + \mathrm{dd^c} v \in \overline{\Gamma_m(\omega)}$ at every point; a smooth v is *strictly admissible* if $\omega + \mathrm{dd^c} v \in \Gamma_m(\omega)$ everywhere. Maxima and convex combinations preserve continuous admissibility. To check the maximum property, fix $\gamma_1, \ldots, \gamma_{m-1}$ in (13.1). By polarized positivity, the left-hand side is $(Lv + g)\, \omega^n$ for a linear elliptic operator L with smooth coefficients and a smooth function g. Subtracting a local smooth solution of $Lh = -g$ reduces the assertion to the classical

fact that the maximum of two continuous distributional subsolutions of L is again a subsolution.

For a bounded upper semicontinuous function v, admissibility means that every *upper test* q at a point x, that is, every C^2 function q near x with $q \geq v$ near x and $q(x) = v(x)$, satisfies

$$\omega(x) + \mathrm{dd}^{\mathrm{c}} q(x) \in \overline{\Gamma_m(\omega(x))}. \tag{13.2}$$

For continuous v, this agrees with (13.1): Gårding duality reduces both conditions to subsolution inequalities for smooth strictly elliptic linear operators, and we use the classical equivalence of viscosity and distributional subsolutions for such operators. In the bounded class, pairing (13.2) with ω^{n-1} makes v, up to a local smooth function, a viscosity subsolution, and hence a subharmonic function, for such an operator. As for ordinary subharmonic functions, $v(x)$ is then the limit of the essential suprema of v on small coordinate balls centered at x. Thus v is the *canonical representative* of its class in $L^1(X)$, and this representative is determined by the distribution defined by v.

The first estimate controls a scalar shift of the cone. It will give lower bounds for every coefficient of the linearized Hessian operator.

Lemma 13.1.1 *For $\lambda \in \Gamma_m$, put*

$$e_m = \frac{\sigma_m(\lambda)}{m\sigma_{m-1}(\lambda)}.$$

Then $\lambda - t\mathbf{1} \in \Gamma_m$ for $0 \leq t < e_m$. If $m \geq 2$, the conclusion also holds for $t = e_m$. For $m = 1$, the endpoint lies on the boundary.

Proof The polynomial

$$P(x) = \sigma_m(\lambda + x\mathbf{1}) = \sum_{j=0}^{m} \binom{m}{j} \sigma_j(\lambda) x^{m-j}$$

is a positive multiple of the $(n-m)$th derivative of $\prod_i (x + \lambda_i)$. Hence all its roots are real, by Rolle's theorem. Its coefficients are positive, so all its roots are strictly negative. Write $P(x) = \prod_{i=1}^{m} (x + r_i)$ with $r_i > 0$. Logarithmic differentiation at zero gives

$$\sum_{i=1}^{m} r_i^{-1} = \frac{m\sigma_{m-1}(\lambda)}{\sigma_m(\lambda)} = e_m^{-1}.$$

Thus $\min_i r_i \geq e_m$, with strict inequality if $m \geq 2$. For $x > -\min_i r_i$, every derivative of P is positive: differentiating its product expansion produces sums of products of positive factors. Since

$$P^{(m-j)}(x) = \frac{m!}{j!} \sigma_j(\lambda + x\mathbf{1}), \qquad 1 \leq j \leq m,$$

all the required symmetric polynomials are positive at $x = -t$. When $m = 1$, the sole root is $-e_1$. □

Lemma 13.1.2 *If $\alpha, \gamma \in \Gamma_m(\omega)$, then*

$$\frac{\alpha^{m-1} \wedge \gamma \wedge \omega^{n-m}}{\omega^n} \frac{\alpha \wedge \omega^{n-1}}{\omega^n} \geq \frac{1}{m!} \frac{\alpha^m \wedge \omega^{n-m}}{\omega^n} \frac{\gamma \wedge \omega^{n-1}}{\omega^n}.$$

The assertion extends to $\gamma \in \overline{\Gamma_m(\omega)}$ by continuity.

Proof Write $\sigma_k = \alpha^k \wedge \omega^{n-k}/\omega^n$, and $M_k = \alpha^{k-1} \wedge \gamma \wedge \omega^{n-k}/\omega^n$. Let $1 \leq k < m$. For $0 \leq t < \sigma_{k+1}/((k+1)\sigma_k)$, Lemma 13.1.1, with $k+1$ in place of m, puts $\alpha - t\omega$ in $\Gamma_{k+1}(\omega)$. Gårding positivity for the $k+1$ forms $\alpha - t\omega, \alpha, \dots, \alpha, \gamma$ of $\Gamma_{k+1}(\omega)$ gives $(\alpha - t\omega) \wedge \alpha^{k-1} \wedge \gamma \wedge \omega^{n-k-1} \geq 0$, that is,

$$M_{k+1} \geq t M_k \quad \left(0 \leq t < \frac{\sigma_{k+1}}{(k+1)\sigma_k}\right).$$

Let t increase to the displayed endpoint. Iterating, starting with $M_1 = \gamma \wedge \omega^{n-1}/\omega^n$, and using $\prod_{k=1}^{m-1} \sigma_{k+1}/((k+1)\sigma_k) = \sigma_m/(m!\sigma_1)$, yields

$$M_m \geq \frac{\sigma_m}{m!\sigma_1} M_1.$$

All denominators are positive. Approximating a form in the closed cone by $\gamma + \delta\omega$, $\delta \downarrow 0$, gives the last assertion. □

At a diagonal Hermitian matrix Λ in the interior of $C_{n,m}$, order its eigenvalues as $\lambda_1 \geq \cdots \geq \lambda_n$ and write

$$f = F(\Lambda), \quad F_i = \partial_{\lambda_i} F, \quad S = \operatorname{tr} \Lambda, \quad \mathcal{F} = \sum_i F_i, \quad Q = \sum_i F_i \lambda_i^2.$$

The next inequalities use these derivatives of the normalized root, rather than derivatives of the polynomial S_m.

Lemma 13.1.3 *If $M \in C_{n,m}$, then*

$$W_M := \sum_i F_i M_{i\bar{i}} \geq \frac{f}{m!S} \operatorname{tr} M, \qquad F_i \geq \frac{f}{m!S}, \qquad |M_{1\bar{1}}| \leq \operatorname{tr} M.$$

Proof In the notation of the proof of Lemma 13.1.2, with α and γ the forms with matrices Λ and M, differentiation gives $DF_\Lambda(M) = f M_m/\sigma_m$. The estimate $M_m \geq \sigma_m M_1/(m!\sigma_1)$, with $M_1 = \operatorname{tr} M/n$ and $\sigma_1 = S/n$, proves the first inequality. The positive semidefinite rank-one coordinate projections belong to $C_{n,m}$; taking M to be each projection proves the second inequality. Since $m \geq 2$, $C_{n,m}$ is contained in $C_{n,2}$, so the eigenvalues μ of M satisfy

$$\operatorname{tr} M \geq 0, \qquad \sum_i \mu_i^2 = (\operatorname{tr} M)^2 - 2S_2(\mu) \leq (\operatorname{tr} M)^2.$$

Thus the operator norm of M is at most its trace, which proves the last inequality. The wedge pairing computes the derivative in the diagonal frame of Λ, regardless of the eigenvectors of M. □

Lemma 13.1.4 *If $0 < f_0 \le f \le f_1$, then*

$$Q := \sum_i F_i \lambda_i^2 \ge \frac{f_0}{m} S - \frac{n-m}{m} f_1^2, \qquad \mathcal{F} \ge 1.$$

Proof Write $(\lambda|i)$ for λ with its ith coordinate omitted. The derivative of the normalized root is $F_i = f S_{m-1}(\lambda|i)/(m S_m)$. The splitting identity $S_k(\lambda) = \lambda_i S_{k-1}(\lambda|i) + S_k(\lambda|i)$, applied with $k = m$ and $k = m+1$, gives

$$Q = \frac{f}{m} S - \frac{(m+1) f}{m S_m} S_{m+1}.$$

For $m = n$, set $S_{n+1} = 0$. If $S_{m+1} \le 0$, the second term is nonnegative. Otherwise $\lambda \in \Gamma_{m+1}$ and Maclaurin gives $S_{m+1} \le \binom{n}{m+1} f^{m+1}$. Since $S_m = \binom{n}{m} f^m$, the subtracted term is at most $(n-m) f^2/m$. This proves the first assertion. Summing the derivatives gives $\mathcal{F} = f \sigma_{m-1}/\sigma_m \ge 1$, again by Maclaurin. □

When $m < n$, admissible forms need not be semipositive. Positivity of a mixed Hessian measure therefore does not directly bound every coefficient of its factors. For degrees below m, deleted-coordinate positivity supplies the required absolute bound: if $S_{j;I}(\lambda) = S_j(\lambda|I)$ denotes S_j with the coordinates indexed by I omitted, then $S_{j;I}(\lambda) > 0$ for $\lambda \in \Gamma_m$ and $j + |I| \le m$, by polarized Gårding positivity. By continuity, $S_{j;I} \ge 0$ on $\overline{\Gamma_m}$ in the same range.

Lemma 13.1.5 *If $0 \le r < m$, $\alpha_1, \dots, \alpha_r \in \Gamma_m(\omega)$, and Ξ is a smooth form of bidegree $(n-r, n-r)$, then*

$$|\alpha_1 \wedge \cdots \wedge \alpha_r \wedge \Xi| \le C_{n,m,\Xi} (\alpha_1 + \cdots + \alpha_r)^r \wedge \omega^{n-r}.$$

For $r = 0$ the right-hand side means $C_{n,m,\Xi} \omega^n$. The same bound holds for $\alpha_1, \dots, \alpha_r \in \overline{\Gamma_m(\omega)}$.

Proof First take a single form α, diagonal in an ω-unitary frame, with ordered eigenvalues $\lambda \in \Gamma_m$. For $1 \le r < m$, repeated splitting and deleted-coordinate positivity give $S_r(\lambda) \ge \lambda_1 \cdots \lambda_r$. The sum of the last $n-r$ eigenvalues is $S_{1;I}(\lambda) > 0$, with $I = \{1, \dots, r\}$, so each negative eigenvalue has modulus at most $(n-r)\lambda_{r+1}$. Thus every product of r distinct eigenvalues has modulus at most

$$(n-r)^r \lambda_1 \cdots \lambda_r \le (n-r)^r S_r(\lambda).$$

Indeed, replace its negative factors by this bound and its positive factors by the corresponding largest eigenvalues. In the diagonal frame only complementary diagonal coefficients of Ξ contribute to $\alpha^r \wedge \Xi$. This proves $|\alpha^r \wedge \Xi| \le C_{n,m,\Xi} \alpha^r \wedge \omega^{n-r}$.

Apply this bound to the polynomial $(\alpha_1 + t_2\alpha_2 + \cdots + t_r\alpha_r)^r \wedge \Xi$, with $0 \le t_i \le 1$. Since $\beta = \alpha_1 + t_2\alpha_2 + \cdots + t_r\alpha_r$ lies in $\Gamma_m(\omega)$, its absolute value is at most $C_{n,m,\Xi}\beta^r \wedge \omega^{n-r}$. Expanding $(\alpha_1 + \cdots + \alpha_r)^r = (\beta + \sum_{i\ge 2}(1-t_i)\alpha_i)^r$ and using polarized positivity bounds this by the stated right-hand side. A fixed invertible interpolation matrix on this finite-dimensional polynomial space bounds each coefficient by its supremum on the cube. The coefficient of $t_2 \cdots t_r$ is a fixed positive multiple of $\alpha_1 \wedge \cdots \wedge \alpha_r \wedge \Xi$. This proves the mixed bound. The closed-cone case follows by adding $\varepsilon\omega$ to each form and taking the limit. □

The next coefficient bound also controls a torsion form whose diagonal coefficients can have either sign. It compares each coefficient $S_{q;ij}$ with two deleted coordinates with the nonnegative coefficients $S_{q;i}$ and $S_{q;j}$ of the associated gradient pairing.

Lemma 13.1.6 *If $\lambda \in \overline{\Gamma_m}$, $1 \le q \le \min\{m-1, n-2\}$, and $i \ne j$, then, with $N = n-2$ and*

$$c_{N,q} = \frac{q(N-q)}{(q+1)(N-q+1)} < 1,$$

one has

$$|S_{q;ij}(\lambda)|^2 \le \frac{S_{q;i}(\lambda)S_{q;j}(\lambda)}{1 - c_{N,q}}.$$

Proof First prove the unnormalized Newton inequality for an arbitrary real list μ of length N. For $1 \le q < N$, put $R(x) = \prod_{a=1}^{N}(x + \mu_a)$ and $G = R^{(N-q-1)}$. Repeated Rolle's theorem makes G a real-rooted polynomial of degree $d = q + 1$. Away from its roots r_a, logarithmic differentiation and Cauchy–Schwarz give

$$G'^2 - GG'' = G^2 \sum_{a=1}^{d} \frac{1}{(x - r_a)^2} \ge \frac{1}{d} G^2 \left(\sum_{a=1}^{d} \frac{1}{x - r_a} \right)^2 = \frac{1}{d} G'^2.$$

Continuity extends this polynomial inequality to every real x. At zero,

$$G = (N-q-1)!S_{q+1}(\mu), \quad G' = (N-q)!S_q(\mu), \quad G'' = (N-q+1)!S_{q-1}(\mu).$$

Substituting these values in $(1 - 1/d)G'^2 \ge GG''$ gives $S_{q-1}(\mu)S_{q+1}(\mu) \le c_{N,q}S_q(\mu)^2$. No sign restriction on μ or on these coefficients was used. For $q = N$, $S_{N+1}(\mu) = 0$, so the same inequality holds with $c_{N,N} = 0$.

Apply this to the list with i, j deleted. Put $A = S_{q-1;ij}$, $B = S_{q;ij}$, $C = S_{q+1;ij}$ and $D_i = S_{q;i}$, $D_j = S_{q;j}$. Deleted-coordinate positivity gives $A, D_i, D_j \ge 0$: the required degrees are $q + 1 \le m$. Splitting in the two omitted coordinates gives the exact identity

$$D_iD_j = AS_{q+1}(\lambda) + B^2 - AC.$$

Since $S_{q+1}(\lambda) \ge 0$ and $AC \le c_{N,q}B^2$, this is at least $(1 - c_{N,q})B^2$. □

A smooth maximum preserves the cone condition and can be chosen monotone in its smoothing scale as well as in its two arguments.

Lemma 13.1.7 *If smooth uniformly bounded admissible $u_j \downarrow u$ and $v_j \downarrow v$ are given, there are smooth uniformly bounded admissible $w_j \downarrow \max\{u, v\}$. For any chosen $\epsilon_j > 0$ decreasing to zero, they can be chosen to satisfy $w_j = u_j$ on $\{u_j - v_j > \epsilon_j\}$ and $w_j = v_j$ on $\{v_j - u_j > \epsilon_j\}$. The pointwise maximum of two bounded admissible functions is bounded admissible. For smooth functions f, g on an open subset of X with $\mathrm{dd^c} f, \mathrm{dd^c} g \in \overline{\Gamma_m(\omega)}$, the regularized maximum $M_\epsilon(f, g)$ constructed below satisfies the same unshifted cone condition.*

Proof Choose a nonnegative even smooth probability density ζ supported in $[-1, 1]$, and set

$$r_\epsilon(t) = \int |t - \epsilon s| \zeta(s)\, \mathrm{d}s, \qquad M_\epsilon(x, y) = \tfrac{1}{2}(x + y + r_\epsilon(x - y)).$$

The function r_ϵ is smooth and convex, $|r_\epsilon'| \le 1$, and $r_\epsilon(t) = |t|$ for $|t| \ge \epsilon$. Jensen's inequality and the triangle inequality give

$$\max\{x, y\} \le M_\epsilon(x, y) \le \max\{x, y\} + \epsilon/2.$$

Its partial derivatives in x, y are nonnegative. For fixed t, $r_\epsilon(t)$ is an even convex function of ϵ, since ζ is even. It is therefore nondecreasing for $\epsilon \ge 0$. Consequently $w_j = M_{\epsilon_j}(u_j, v_j)$ decreases pointwise to $\max\{u, v\}$ and has the asserted equalities and bounds.

Put $h_j = u_j - v_j$ and $a_j = (1 + r_{\epsilon_j}'(h_j))/2$. The chain rule gives

$$\omega + \mathrm{dd^c} w_j = a_j(\omega + \mathrm{dd^c} u_j) + (1 - a_j)(\omega + \mathrm{dd^c} v_j) + \tfrac{1}{2} r_{\epsilon_j}''(h_j)\mathrm{d}h_j \wedge \mathrm{d^c} h_j.$$

Here $0 \le a_j \le 1$, and the last form is semipositive. Convexity of the cone and its stability under semipositive addition prove admissibility. For smooth local functions satisfying the unshifted cone condition, omit ω from this chain rule; the same argument proves preservation of that condition.

Finally, let u, v be bounded admissible. Their pointwise maximum is bounded and upper semicontinuous, and an upper test for it at a point is an upper test for whichever of u, v attains the maximum there. Hence $\max\{u, v\}$ satisfies (13.2); in particular, it is its own canonical representative. □

Lemma 13.1.8 *If bounded admissible u, v satisfy $u \le v$ almost everywhere with respect to ω^n, then $u \le v$ everywhere.*

Proof By Lemma 13.1.7, $w = \max\{u, v\}$ is bounded admissible. Since $w = v$ almost everywhere, w and v define the same distribution. As bounded admissible functions, both are canonical representatives, and a canonical representative is determined by its distribution. Thus $w = v$ everywhere, that is, $u \le v$. □

13.2 Liouville and gradient estimates

The gradient estimate comes from rescaling a sequence at points of large gradient. The background then disappears, leaving a maximal function for a constant Euclidean cone. We first establish the Liouville theorem that rules out such a nonconstant limit; its proof is that of Theorem 10.15.1 for another cone.

For this Euclidean argument allow $1 \le m \le n$. Write $H(v) = (v_{i\bar{j}})$ for the complex Hessian. A continuous real function v on a domain in $\mathbb{C}^n$ is admissible for $C_{n,m}$ if

$$\operatorname{tr}(PH(v)) \ge 0 \quad \text{in distributions for every } P \in C^*_{n,m},$$

where $C^*_{n,m} = \{P : \operatorname{tr}(PA) \ge 0 \text{ for all } A \in C_{n,m}\}$ is the dual cone of Hermitian matrices. For smooth functions this says $H(v) \in C_{n,m}$. A *lower test* for v at z_0 is a smooth function ψ near z_0 with $\psi \le v$ near z_0 and $\psi(z_0) = v(z_0)$. Call v *maximal* if $H(\psi)(z_0)$ lies outside the interior of $C_{n,m}$ for every lower test ψ at every point z_0. Admissibility implies subharmonicity and is preserved by nonnegative convolution. The cone polynomial S_m increases under nonnegative matrix addition in the interior.

Lemma 13.2.1 *Local uniform limits of admissible maximal functions are admissible and maximal. Suppose v is admissible and maximal on a bounded open set D, and h is smooth and admissible there. If both are continuous on $\overline{D}$ and $h \le v$ on ∂D, then $h \le v$ on D.*

Proof The proof of Lemma 10.15.1 applies with $C_{n,m}$ in place of the cone $\{P_n(H) \ge 0\}$. It uses only that distributional admissibility passes to local uniform limits, that the interior of the cone is open, and that $H + \delta I$ lies in the interior for H in the cone and $\delta > 0$. □

Theorem 13.2.1 *Every bounded globally Lipschitz admissible function on $\mathbb{C}^n$ that is maximal for $C_{n,m}$ is constant.*

Proof For $m = 1$, the dual cone consists of the nonnegative multiples of I. Distributional subharmonicity and the lower-test condition then say that the function is a viscosity solution of the Laplace equation. The elementary equivalence with harmonicity and bounded harmonic Liouville give the conclusion. For $m = n$, admissibility is distributional plurisubharmonicity. The continuous representative is plurisubharmonic, and bounded plurisubharmonic Liouville applies. For a fixed $m \ge 2$, we argue by induction on $n \ge m$, the case $n = m$ being the plurisubharmonic one.

For a Hermitian $(n-1)$-by-$(n-1)$ matrix A and $m < n$,

$$S_j(A \oplus 0) = S_j(A), \qquad 1 \le j \le m.$$

Consequently $A \oplus 0$ belongs to $C_{n,m}$, respectively its interior, exactly when A belongs to $C_{n-1,m}$, respectively its interior. For the closed cones, the nonnegative inequalities characterize the closure: the expansion of $S_j(\lambda + \delta \mathbf{1})$ has nonnegative coefficients and a strictly positive δ^j term. If an admissible function is independent of z_n, convolving

it and using the block identity shows that the function in the remaining variables is admissible. Extending its lower tests independently of z_n shows that maximality also passes to dimension $n-1$.

With this reduction, the proof of Theorem 10.15.1 applies word for word, with $C_{n,m}$ in place of the cone $\{P_n(H) \geq 0\}$, the induction starting at $n = m$, and Lemma 13.2.1 in place of Lemma 10.15.1. Apart from these two inputs, that proof uses only properties shared by $C_{n,m}$: the cone is closed, convex and stable under adding nonnegative matrices; admissibility and maximality are preserved by translations, unitary changes of coordinates and $v \mapsto av + b$ with $a > 0$; and admissibility implies subharmonicity and is preserved by nonnegative convolution. For instance, for $v \geq 0$ the convolution at a fixed radius of $2v_\epsilon H(v_\epsilon)$ lies in $C_{n,m}$ by convexity. □

A trace bound quadratic in the gradient now bounds the gradient. A bound linear in the gradient would suffice by Lemma 10.4.2, since forms in $\Gamma_m(\omega)$ have positive trace; the quadratic bound keeps its form under rescaling, and Theorem 13.2.1 excludes the rescaled limits.

Proposition 13.2.1 *Let θ be a smooth real $(1,1)$-form on X, let $2 \leq m \leq n$, and let $M, A, B > 0$. There is a uniform bound for $\sup_X |\partial u|_\omega$ over all smooth real u such that $\theta_u = \theta + \mathrm{dd^c} u \in \Gamma_m(\omega)$ and*

$$|u| \leq M, \qquad \frac{\theta_u^m \wedge \omega^{n-m}}{\omega^n} \leq B, \qquad \operatorname{tr}_\omega \theta_u \leq A\bigl(1 + \sup_X |\partial u|_\omega^2\bigr).$$

Proof In local holomorphic coordinates write $\omega = \mathrm{i} g_{i\bar{\jmath}}\, \mathrm{d}z_i \wedge \mathrm{d}\bar{z}_j$ and $\theta_u = \mathrm{i}(b_{i\bar{\jmath}} + \kappa u_{i\bar{\jmath}})\, \mathrm{d}z_i \wedge \mathrm{d}\bar{z}_j$, where $\kappa = 1/(2\pi)$ comes from the normalization of $\mathrm{dd^c}$. Since $m \geq 2$, every eigenvalue of a form in $\Gamma_m(\omega)$ has modulus at most its trace, by the proof of Lemma 13.1.3. Thus the assumed trace estimate bounds every eigenvalue of θ_u relative to ω in absolute value by $A(1 + \sup |\partial u|^2)$.

Suppose, contrary to the claim, that a sequence u_j has maxima $R_j = \sup |\partial u_j|_\omega \to \infty$, attained at x_j. Passing to a subsequence, let $x_j \to x_\infty$, and work in a fixed chart, changing coordinates linearly so that $g(x_\infty) = I$. If y_j is the coordinate of x_j, set

$$v_j(z) = u_j(y_j + z/R_j), \qquad g_j(z) = g(y_j + z/R_j),$$
$$E_j(z) = R_j^{-2} b(y_j + z/R_j) + \kappa H(v_j)(z).$$

These are defined on balls with radii tending to infinity. On each fixed ball $g_j \to I$, the smooth background term tends to zero, and E_j is uniformly bounded by the eigenvalue estimate. In particular $H(v_j)$ and its Euclidean trace are uniformly bounded. Interior estimates for the fixed Euclidean Laplacian give local uniform $W^{2,q}$ bounds for every finite q. A diagonal subsequence converges in C^1 on compact subsets to a bounded globally Lipschitz v, with $|\partial v(0)| = 1$.

The Hermitian matrices $g_j^{-1/2} E_j g_j^{-1/2}$ belong to the interior of $C_{n,m}$ and differ from $\kappa H(v_j)$ by a term tending uniformly to zero on fixed balls. Pairing with any fixed matrix in the dual cone and passing in distributions proves that v is admissible. By homogeneity of S_m, the rescaled Hessian densities satisfy

$$\frac{S_m(g_j^{-1/2}E_j g_j^{-1/2})}{\binom{n}{m}} = R_j^{-2m}\frac{\theta_{u_j}^m \wedge \omega^{n-m}}{\omega^n}(y_j + z/R_j) \le BR_j^{-2m}.$$

We prove maximality with lower tests. If a smooth lower test ψ at z_0 had $H(\psi)(z_0)$ in the interior, choose a small $\delta > 0$ so that $H(\psi) - \delta I$ remains there near z_0. Strictifying the contact by subtracting $\delta|z - z_0|^2$ gives interior minimum points $z_j \to z_0$ for v_j minus this test. At those points

$$E_j \ge R_j^{-2}b + \kappa(H(\psi) - \delta I).$$

The matrix on the right, measured with g_j, is uniformly inside the cone. Monotonicity of S_m under nonnegative addition gives a fixed positive lower bound for the left side of the preceding equation. This contradicts $BR_j^{-2m} \to 0$. Hence v is maximal. Theorem 13.2.1 makes it constant, contradicting $|\partial v(0)| = 1$. □

13.3 A trace estimate with a prescribed potential

In a coordinate chart write $\omega = (\mathrm{i}/(2\pi))g_{i\bar{j}}\,\mathrm{d}z_i \wedge \mathrm{d}\bar{z}_j$. Covariant derivatives use the Chern connection of g. For a smooth strictly admissible u, put

$$H_{i\bar{j}} = g_{i\bar{j}} + u_{i\bar{j}}, \qquad W = g^{-1}H, \qquad L = \sum_i F_i \nabla_{i\bar{i}}$$

in a unitary frame diagonalizing H. Let $\lambda_1 \ge \cdots \ge \lambda_n$ be the eigenvalues of W, and use $f, \mathcal{F}, S, Q$ from Section 13.1. Since $m \ge 2$, one has $|\lambda_i| < S \le n\lambda_1$.

We shall estimate equations whose density involves another admissible potential v. In an approximation argument the derivatives of v may diverge. A logarithmic weight χ with $\chi'' = 12(\chi')^2$ absorbs these derivatives using only uniform closeness to one fixed smooth comparison function; in the next lemma q stands for χ'.

Lemma 13.3.1 *Suppose $f \geq f_0 > 0$, $C_H, C_G \geq 0$, and q satisfies*

$$q \geq 8, \qquad q \geq \frac{2nm!C_H}{f_0}, \qquad q^2 \geq \frac{4m!C_G}{f_0}.$$

For $0 \leq r \leq 1/2$, $a, b \in \mathbb{C}^n$ and $M \in C_{n,m}$, put $W_M = \sum_i F_i M_{i\bar{i}}$ and

$$E = q(W_M - \mathcal{F}) - \frac{C_H}{\lambda_1}|M_{1\bar{1}}| - \frac{C_G}{S}\sum_i |a_i|^2 + \sum_i F_i\big(12q^2|a_i - b_i|^2 - (9q^2 + 2q + rq^2)|a_i|^2 - q|a_i|\big).$$

Then

$$E \geq \frac{q}{2}W_M + \frac{q^2}{4}\sum_i F_i|a_i|^2 - (q + 1 + 84q^2 \max_i |b_i|^2)\mathcal{F}.$$

Proof Young's inequality gives $|a_i - b_i|^2 \geq (7/8)|a_i|^2 - 7|b_i|^2$, so $12q^2|a_i - b_i|^2 - 9q^2|a_i|^2 \geq (3/2)q^2|a_i|^2 - 84q^2|b_i|^2$. Since $q \geq 8$, one has $2q \leq q^2/4$, and $q|a_i| \leq q^2|a_i|^2/4 + 1$. Hence

$$12q^2|a_i - b_i|^2 - (9q^2 + 2q)|a_i|^2 - q|a_i| \geq q^2|a_i|^2 - 1 - 84q^2|b_i|^2.$$

Subtracting $rq^2|a_i|^2$, with $r \leq 1/2$, leaves $q^2|a_i|^2/2$ in place of $q^2|a_i|^2$. Since $S \leq n\lambda_1$, Lemma 13.1.3 and the hypothesis on q imply

$$\frac{C_H}{\lambda_1}|M_{1\bar{1}}| \leq \frac{nC_H}{S}\operatorname{tr} M \leq \frac{q}{2}W_M.$$

The lower bound for each F_i similarly gives

$$\frac{C_G}{S}\sum_i |a_i|^2 \leq \frac{q^2}{4}\sum_i F_i|a_i|^2.$$

Combining these inequalities yields the assertion. □

The two cases for the smallest eigenvalue will reduce to the following numerical estimate.

Proposition 13.3.1 *Fix $a, \delta > 0$ and $b, d, e \geq 0$. Let $K \geq 1$, and let $\lambda \in \Gamma_m$ be ordered decreasingly, with $0 < f_0 \leq f \leq f_1$. Suppose the following inequality holds according to the sign of $\lambda_n + \delta\lambda_1$:*

(1) *If $\lambda_n < -\delta\lambda_1$, then $0 \geq aQ/K - bK\mathcal{F} - d\lambda_1\mathcal{F} - e$.*
(2) *If $\lambda_n \geq -\delta\lambda_1$, then $0 \geq aQ/K - bKF_1 - e$.*

Then $S \leq CK$ and $\lambda_1 \leq CK$, with C depending only on the fixed displayed data and n, m.

Proof In the first case, symmetry and concavity of F imply $F_n \geq F_i$ for every i, so $Q \geq \delta^2 \lambda_1^2 \mathcal{F}/n$. To see the derivative ordering, apply monotonicity of the gradient of a concave function to λ and the vector obtained by exchanging its ith and nth coordinates. Divide the assumed inequality by $K\mathcal{F}$, use $\mathcal{F} \geq 1$, and set $t = \lambda_1/K$. It yields $a\delta^2 t^2/n \leq b + dt + e$, which bounds t. Since $S \leq n\lambda_1$, the trace bound follows. In the second case, since $Q \geq F_1 \lambda_1^2$, either $\lambda_1^2 \leq 2bK^2/a$, which gives the same trace bound directly, or $aQ/(2K) \geq bKF_1$. The assumed inequality then gives $Q \leq 2eK/a$. Lemma 13.1.4 bounds S by a multiple of K. Finally the Γ_2 identity gives $\lambda_1 \leq (\sum_i \lambda_i^2)^{1/2} < S$. □

The geometric computation uses a lower support for the largest eigenvalue, as in the second-order argument for the Gauduchon equation in Section 10.14.

Lemma 13.3.2 *At a point with $\lambda_1 > 1$, the largest eigenvalue has a smooth local lower support $\widetilde{\lambda}$, equal to λ_1 at the point, for which, at that point,*

$$L \log \widetilde{\lambda} \geq \frac{f_{1\bar{1}}}{\lambda_1} + N - \sum_i F_i |Z_i|^2 - C\mathcal{F} - \frac{C}{\lambda_1^2} \sum_i F_i |H_{1\bar{1};i}|,$$

where

$$N = -\frac{D^2 F_W(\nabla_1 W, \nabla_{\bar{1}} W)}{\lambda_1}, \qquad Z_i = \frac{H_{i\bar{1};1}}{\lambda_1}.$$

Here C depends only on the fixed metric, and the derivatives of F are evaluated at W.

Proof Choose a smooth unitary frame parallel for the Chern connection at the point and diagonalizing H there. The positive semidefinite tensor B, diagonal in this frame with entries $0, \beta, \ldots, \beta$, satisfies $\nabla B = 0$ at the point and $|\nabla^2 B| \leq C\beta$. For $0 < \beta < 1$, the largest eigenvalue of $W - B$ is simple, lies below that of W, and agrees with it at the point. This gives the required smooth support $\widetilde{\lambda}$. Differentiating the normalized eigenvector equation gives

$$\widetilde{\lambda}_{;k} = H_{1\bar{1};k}, \qquad \widetilde{\lambda}_{;k\bar{k}} \geq H_{1\bar{1};k\bar{k}} + \sum_{p>1} \frac{|H_{p\bar{1};k}|^2 + |H_{1\bar{p};k}|^2}{\lambda_1 - \lambda_p + \beta} - C\beta.$$

The denominators are at most $(n+1)\lambda_1$. Let $U = \partial\bar{\partial} u$, and take $T^p_{ki} = \Gamma^p_{ki} - \Gamma^p_{ik}$. The first commutations are

$$U_{i\bar{j};k} = U_{k\bar{j};i} - T^p_{ki} U_{p\bar{j}}, \qquad U_{i\bar{j};\bar{k}} = U_{i\bar{k};\bar{j}} - \overline{T^p_{kj}} U_{i\bar{p}}.$$

Apply the first identity, commute one mixed pair of covariant derivatives, and apply the conjugate identity. The result has the form

$$U_{1\bar{1};k\bar{k}} = U_{k\bar{k};1\bar{1}} - 2\,\mathrm{Re} \sum_p U_{k\bar{p};1} \overline{T^p_{k1}} + E_k, \qquad |E_k| \leq C|U|.$$

Indeed the mixed commutator contracts curvature with U, differentiating the first torsion coefficient contracts ∇T with U, and the second use of these first-order

identities produces contractions of two torsion coefficients with U. Thus all the omitted terms satisfy the displayed bound. Since g is parallel, the same derivative relations apply to H apart from the explicit g terms in the first commutation. At the diagonal point $|U| \le C\lambda_1$. For $p > 1$, rewrite $U_{k\bar{p};1} = U_{1\bar{p};k} + T^{j}_{k1} U_{j\bar{p}}$. Young's inequality absorbs its third derivative into the positive quotient $|H_{1\bar{p};k}|^2/(\lambda_1 - \lambda_p + \beta)$, at the cost of $C\lambda_1$. Consequently

$$\widetilde{\lambda}_{;k\bar{k}} \ge H_{k\bar{k};1\bar{1}} - 2\,\mathrm{Re}(H_{k\bar{1};1}\overline{T^{1}_{k1}}) - C\lambda_1.$$

Differentiating $F(W) = f$ twice in the first frame direction gives $\sum_k F_k H_{k\bar{k};1\bar{1}} = f_{1\bar{1}} + \lambda_1 N$. Sum the last inequality with weights F_k, and subtract $\sum_k F_k |\widetilde{\lambda}_{;k}|^2/\lambda_1^2$ to pass to $L \log \widetilde{\lambda}$. Write $D_k = H_{1\bar{1};k}$ and $t_k = T^{1}_{k1}$. The first commutation, using $U_{1\bar{1}} = \lambda_1 - 1$, gives

$$D_k = \lambda_1 Z_k - (\lambda_1 - 1) t_k.$$

The negative logarithmic square and the remaining torsion term therefore satisfy

$$-\frac{|D_k|^2}{\lambda_1^2} - 2\,\mathrm{Re}(Z_k \bar{t}_k) \ge -|Z_k|^2 - C - C\lambda_1^{-2}|D_k|.$$

For example, expanding the left side leaves $-|Z_k|^2 - 2\lambda_1^{-1}\,\mathrm{Re}(Z_k\bar{t}_k) - (1-\lambda_1^{-1})^2|t_k|^2$, and $\lambda_1|Z_k| \le |D_k| + C\lambda_1$. This proves the claim, with C independent of $\beta \in (0,1)$.□

Lemma 13.3.3 *Put $s = |\partial u|_g^2$ and $K = 1 + \sup_X s$. For every $\eta > 0$ one has*

$$Ls \ge \sum_{i,k} F_i(|u_{ki}|^2 + |u_{k\bar{i}}|^2) + 2\,\mathrm{Re}\sum_k f_k u_{\bar{k}} - \eta Q - C_\eta K\mathcal{F}.$$

The constant depends only on the metric and η.

Proof Differentiate the squared norm using metric compatibility. Its quadratic part is the displayed sum. Commuting the remaining third derivatives gives

$$u_{ki\bar{i}} = u_{i\bar{i};k} - T^{p}_{ik} u_{p\bar{i}} + R^{p}_{i\bar{i}k} u_p, \qquad u_{\bar{k}i\bar{i}} = u_{i\bar{i};\bar{k}} - \overline{T^{p}_{ik}}\, u_{i\bar{p}}.$$

The first differentiated equation is $\sum_i F_i u_{i\bar{i};k} = f_k$, because the background g is parallel. At the diagonal point the torsion errors are bounded by $C\sqrt{K}\sum_i F_i(|\lambda_i| + 1)$, and the curvature errors by $CK\mathcal{F}$. Use

$$C\sqrt{K} F_i |\lambda_i| \le \eta F_i \lambda_i^2 + C_\eta K F_i,$$

and sum over i. This gives the asserted bound. □

Proposition 13.3.2 *Fix $M, B > 0$ and a smooth real $h(x,t)$ on a neighborhood of $X \times [-B-1, B+1]$. There is $0 < \varepsilon \le 1$, depending only on M, B, h, ω, n, m, with the following property. For every smooth real φ with $\|\varphi\|_\infty \le B$, there is $C < \infty$, depending only on these data and on $\|\varphi\|_{C^2}$, such that if a smooth strictly admissible u and a smooth admissible v satisfy*

$$\|u\|_\infty \le M, \qquad \|v-\varphi\|_\infty \le \varepsilon, \qquad F(W) = \exp\bigl((u + h(x,v))/m\bigr),$$

then

$$\operatorname{tr}_\omega(\omega + \mathrm{dd}^{\mathrm{c}} u) \le C\bigl(1 + \sup_X |\partial u|_g^2\bigr).$$

Proof For $\varepsilon \le 1$ the root $f = F(W)$ of the density has fixed bounds $0 < f_0 \le f \le f_1$. Its first derivative is

$$f_k = \frac{f}{m}(u_k + h_k(x,v) + h_t(x,v)v_k).$$

In Lemma 13.3.3, the u_k terms contribute $2fs/m \ge 0$ to $2\operatorname{Re}\sum_k f_k u_{\bar k}$, so they can be dropped. At a point with $\lambda_1 > 1$, differentiating once more gives

$$f_{1\bar 1} \ge -C(1 + |\partial v|^2 + |1 + v_{1\bar 1}|).$$

Here $u_{1\bar 1} = \lambda_1 - 1 > 0$, the square from differentiating the exponential is nonnegative, and all derivatives of h are bounded on the fixed interval.

Set

$$\Phi(s) = -\tfrac{1}{2}\log(1 - s/(2K)), \qquad \Psi(u) = -2Au + (A\tau/2)u^2,$$
$$\chi(t) = -\tfrac{1}{12}\log(2\varepsilon - t), \qquad \tau = (M+1)^{-1}.$$

Then $(4K)^{-1} \le \Phi' \le (2K)^{-1}$ and $\Phi'' = 2(\Phi')^2$. Since $|u| \le M$ and $\tau M < 1$, one has $-3A \le \Psi' \le -A$, and $\Psi'' = A\tau$. Finally $q = \chi'$ lies between $1/(36\varepsilon)$ and $1/(12\varepsilon)$, with $\chi'' = 12q^2$. Take a maximum point of $G = \log\lambda_1 + \Phi(s) + \Psi(u) + \chi(v-\varphi)$. Replacing λ_1 by the smooth lower support $\widetilde{\lambda}$ of Lemma 13.3.2 gives a smooth function $\widetilde{G} \le G$ with a local maximum at this point. There put

$$a_i = v_i, \quad b_i = \varphi_i, \quad P_v = I + v_{i\bar j}, \quad W_v = \sum_i F_i(P_v)_{i\bar i}, \quad x_i = \Phi' s_i, \quad y_i = \Psi' u_i.$$

The first-order condition at the maximum, and the identity $D_i = \lambda_1 Z_i - (\lambda_1 - 1)T^1_{i1}$ from the proof of Lemma 13.3.2, give

$$H_{1\bar 1;i}/\lambda_1 = -x_i - y_i - q(a_i - b_i), \qquad Z_i = -x_i - y_i - qa_i + e_i, \quad e_i = qb_i + (1 - \lambda_1^{-1})T^1_{i1}.$$

Thus $|e_i| \le E_0$, with E_0 independent of A and of derivatives of v.

We first bound the term $\Phi' Ls$ of $L\widetilde{G}$. Lemma 13.3.3 and the identity $u_{i\bar j} = (\lambda_i - 1)\delta_{ij}$ give, after multiplication by Φ',

$$\Phi' Ls \geq \frac{1}{8K}\sum_{i,k} F_i(|u_{ki}|^2 + |u_{k\bar{i}}|^2) + \frac{3}{64K}Q - C\mathcal{F} - \frac{C'_G}{S}\sum_i |a_i|^2.$$

To justify the constants, keep half of the positive quadratic term, and bound the other half below by its diagonal terms $|u_{i\bar{i}}|^2$, using $(\lambda_i - 1)^2 \geq \lambda_i^2/2 - 1$. Choose η in that lemma so that the term $\eta\Phi' Q$ costs at most $Q/(128K)$. The remaining density-gradient error is at most $C|a|/\sqrt{K}$ plus a fixed constant. Young's inequality bounds $C|a|/\sqrt{K}$ by $C'_G|a|^2/S + cS/K$. Choose C'_G so large that, by Lemma 13.1.4, cS/K is at most $Q/(128K)$ plus a fixed constant. This leaves $3Q/(64K)$; fixed constants are at most $C\mathcal{F}$, since $\mathcal{F} \geq 1$. The last error term of Lemma 13.3.2, linear in $H_{1\bar{1};i}$, is estimated using the first-order condition. Its x_i term costs at most half of the kept quadratic sum and $C\mathcal{F}$. Its y_i term costs at most $\Psi''\sum_i F_i|u_i|^2/4 + \mathcal{F}$ whenever $\lambda_1 \geq L_0(A)$, where $L_0(A)$ is fixed independently of K, v. Its qa_i term costs at most $q\sum_i F_i|a_i|$ when λ_1 exceeds a fixed metric constant, which we take at least 1; its qb_i term costs $C\mathcal{F}$. If λ_1 lies below either threshold at the maximum point, the final comparison of this proof gives the required bound directly. Otherwise, with fixed C_H, C_G, C_0 independent of A, the second-order condition $L\widetilde{G} \leq 0$ gives

$$\begin{aligned} 0 \geq N &- \sum_i F_i|Z_i|^2 + 2\sum_i F_i|x_i|^2 + \tfrac{3}{4}\Psi''\sum_i F_i|u_i|^2 + \frac{Q}{32K} + \Psi'(f - \mathcal{F}) - C_0\mathcal{F} \\ &+ q(W_v - \mathcal{F}) - \frac{C_H}{\lambda_1}|(P_v)_{1\bar{1}}| - \frac{C_G}{S}\sum_i |a_i|^2 + \sum_i F_i\big(12q^2|a_i - b_i|^2 - q|a_i|\big). \end{aligned}$$

The fixed $qL\varphi$ term is included in C_0. Choose ε so small that the lower bound $q \geq 1/(36\varepsilon)$ meets all thresholds of Lemma 13.3.1. These thresholds depend on M, B, h and the metric, and not on derivatives of φ. This choice precedes that of A.

Let $0 < \delta < 1$; it is fixed at the end of the proof. First suppose $\lambda_n < -\delta\lambda_1$. Concavity gives $N \geq 0$, and

$$-|Z_i|^2 + 2|x_i|^2 \geq -6|y_i|^2 - 6q^2|a_i|^2 - 6E_0^2.$$

After decreasing the right side further by $(3q^2 + 2q)|a_i|^2$, apply Lemma 13.3.1 to the matrix P_v, with $r = 0$. Drop the nonnegative terms $qW_v/2$ and $q^2\sum_i F_i|a_i|^2/4$ that it provides, and the Ψ'' term. Since $\Psi' f \geq -3Af_1$ and $-\Psi'\mathcal{F} \geq 0$, this gives

$$0 \geq \frac{Q}{32K} - (54A^2 + C_*)K\mathcal{F} - 3Af_1,$$

where C_* is fixed independently of A, δ.

Now suppose $\lambda_n \geq -\delta\lambda_1$, and let $I = \{i : F_i > \delta^{-1}F_1\}$. For $i \in I$, $\lambda_i < \lambda_1$ and

$$\frac{F_i - F_1}{\lambda_1 - \lambda_i} \geq \frac{(1-\delta)F_i}{(1+\delta)\lambda_1} \geq \frac{(1-2\delta)F_i}{\lambda_1}.$$

The spectral formula for D^2F at the original eigenvalues, including its nonpositive diagonal part, therefore gives $N \geq (1 - 2\delta)\sum_{i\in I} F_i|Z_i|^2$; the difference quotients

used here do not degenerate, since $\lambda_i < \lambda_1$ for $i \in I$. Using $x_i = -Z_i - y_i - qa_i + e_i$ and $|z+w|^2 \geq \delta|z|^2 - \delta|w|^2/(1-\delta)$, one obtains

$$N - \sum_{i\in I} F_i|Z_i|^2 + 2\sum_{i\in I} F_i|x_i|^2 \geq -\frac{4\delta(\Psi')^2}{1-\delta}\sum_{i\in I} F_i|u_i|^2 - rq^2\sum_{i\in I} F_i|a_i|^2 - rE_0^2\mathcal{F}, \qquad r = \frac{8\delta}{1-\delta}.$$

Choose $\delta \leq \min\{1/17, \tau/(72A+\tau)\}$. Then the first coefficient is at most $\Psi''/2$, and $r \leq 1/2$. On the complementary indices the preceding elementary bound gives

$$-\sum_{i\notin I} F_i|Z_i|^2 + 2\sum_{i\notin I} F_i|x_i|^2 \geq -\frac{54A^2 + 6nE_0^2}{\delta}KF_1 - 6q^2\sum_{i\notin I} F_i|a_i|^2.$$

Apply Lemma 13.3.1 to the matrix P_v, with penalty r. To match its input, add the negative terms $-(9q^2+2q)|a_i|^2$ for $i \in I$ and $-(3q^2+2q+rq^2)|a_i|^2$ for $i \notin I$; this only decreases the right side. Its fixed-comparison error and the error $rE_0^2\mathcal{F}$ from I have constants bounded independently of A, δ. Enlarge C_* to include them, and drop the nonnegative terms provided by the lemma and the remaining Ψ'' term. We obtain

$$0 \geq \frac{Q}{32K} - \frac{54A^2 + 6nE_0^2}{\delta}KF_1 + (A - C_*)\mathcal{F} - 3Af_1.$$

Take $A \geq \max\{1, C_* + 1\}$, then fix $\delta = \min\{1/17, \tau/(72A+\tau)\}$ and the threshold $L_0(A)$. Both case inequalities now satisfy the hypotheses of Proposition 13.3.1, with fixed data. Hence $\lambda_1 \leq CK$ at the maximum point. Finally Φ, Ψ, χ have oscillations bounded independently of K and of derivatives of v. Comparing G at any other point with its maximum gives the same bound for λ_1 everywhere. Since $S \leq n\lambda_1$, this proves the trace estimate. □

Corollary 13.3.1 *Under the hypotheses of Proposition 13.3.2, one has*

$$\sup_X |\partial u|_g + \sup_X |\partial\bar\partial u|_g \leq C,$$

with a constant depending on the same fixed data and no derivatives of v.

Proof The Hessian density is $F(W)^m \leq f_1^m$. Apply Proposition 13.2.1 to the fixed background $\theta = \omega$, with the trace estimate of Proposition 13.3.2. It gives the uniform gradient bound. The trace is then uniformly bounded, and the Γ_2 identity bounds the absolute value of every eigenvalue of W by its trace. Subtracting the identity matrix gives the complex-Hessian bound. □

The bound controls the complex Hessian. An estimate for all entries of the real Hessian is not needed: the Euclidean Laplacian gives finite L^p estimates for those derivatives (Lemma 13.4.1).

13.4 Bounded complex Hessians and elliptic extensions

Uniform complex-Hessian bounds give enough compactness both to pass the nonlinear products to a limit and to make the Hessian root uniformly elliptic. In coordinates use the metric matrix g of the preceding section and write $H(u) = (u_{i\bar{j}})$, as in Section 13.2. Coefficient products with bounded measurable factors will be identified with currents by convolution.

Lemma 13.4.1 *Suppose u_j is smooth,*

$$\sup_j \|\mathrm{dd^c} u_j\|_{L^\infty} < \infty, \qquad u_j \longrightarrow u \quad \textit{uniformly on } X.$$

Then $u \in W^{2,p}(X)$ for every $1 < p < \infty$, its distributional complex Hessian has bounded coefficients, and $u_j \to u$ in $C^{1,\alpha}(X)$ for every $0 < \alpha < 1$. The same hypotheses give uniform $W^{2,p}$ bounds for the sequence.

Proof The uniform convergence makes the u_j uniformly bounded. On a coordinate ball the Euclidean Laplacian is $\Delta u_j = 4\sum_i (u_j)_{i\bar{i}}$, so it is uniformly bounded. The interior Laplace estimate on two nested balls gives

$$\|u_j\|_{W^{2,p}(B')} \le C_p\big(\|\Delta u_j\|_{L^p(B)} + \|u_j\|_{L^p(B)}\big).$$

A finite cover gives the global bound. Weak $W^{2,p}$ limits coincide with the uniform limit. The bounded complex Hessians have weak-star limits in L^∞; testing against compactly supported smooth functions identifies them with $H(u)$. Choose p with $1 - 2n/p > \alpha$. The compact Sobolev embedding into $C^{1,\alpha}$ gives convergence of subsequences. Every such limit is u, hence the whole sequence converges. □

Lemma 13.4.2 *Let $1 \le k \le n$. For $1 \le a \le k$, suppose that u_j^a and u^a satisfy the hypotheses of Lemma 13.4.1. Let θ_a be fixed smooth real $(1,1)$-forms. Then*

$$\bigwedge_{a=1}^{k} (\theta_a + \mathrm{dd^c} u_j^a) \longrightarrow \bigwedge_{a=1}^{k} (\theta_a + \mathrm{dd^c} u^a) \quad \textit{as currents.}$$

The current on the right is the product of the almost-everywhere coefficients. For each smooth test form η of bidegree $(n-k, n-k)$, the difference of the pairings with η is bounded by $C_\eta \max_a \|u_j^a - u^a\|_\infty$.

Proof Set $D_j^a = \mathrm{dd^c} u_j^a$ and $D^a = \mathrm{dd^c} u^a$. On a smaller coordinate ball, convolution gives smooth u_ϵ^a whose complex Hessians are uniformly bounded and converge to $H(u^a)$ in every finite L^p space. Their coefficient products consequently converge in finite L^p spaces. In particular the products of the D^a are closed currents, since their smooth approximants are closed.

Expand the smooth background factors and telescope each remaining product. Every difference is then a sum of terms of the form

$$\mathrm{dd^c}(u_j^a - u^a) \wedge T_j,$$

where T_j is a product of D_j^b and D^b with $b \neq a$. It has uniformly bounded coefficients and is closed. Local convolution, followed by strong convergence of its finite coefficient products, justifies integration by parts:

$$\left\langle \mathrm{dd^c}(u_j^a - u^a) \wedge T_j, \xi \right\rangle = \int (u_j^a - u^a)\, T_j \wedge \mathrm{dd^c}\xi.$$

Here ξ includes the chosen smooth background factors and the test form, so their derivatives are kept. The right-hand side has the asserted uniform bound. A finite partition of unity proves the claim on X. □

Lemma 13.4.3 *For $f_0 > 0$ and $L > 0$, assume the set*

$$K = \{A \in C_{n,m} : \operatorname{tr} A \le L,\ \sigma_m(A) \ge f_0^m\}$$

is nonempty. It is a compact subset of the interior, and there are $\vartheta, \Theta > 0$, depending only on n, m, f_0, L, such that

$$\vartheta I \le DF_A \le \Theta I \qquad (A \in K).$$

The function on all Hermitian matrices defined by

$$\widehat{F}(B) = \inf_{A \in K} \big(F(A) + \operatorname{tr}(DF_A(B - A))\big)$$

is concave, agrees with F on K, and satisfies, for $P \ge 0$,

$$\vartheta \operatorname{tr} P \le \widehat{F}(B + P) - \widehat{F}(B) \le \Theta \operatorname{tr} P.$$

Proof Since $m \ge 2$, the eigenvalues of $A \in C_{n,m}$ have modulus at most $\operatorname{tr} A$, by the proof of Lemma 13.1.3. Thus K is bounded and closed. Maclaurin gives $\sigma_j(A) \ge \sigma_m(A)^{j/m} \ge f_0^j$ for $j \le m$, so every matrix in K is in the interior. In a unitary eigenframe,

$$F_i = \frac{F^{1-m} S_{m-1}(\lambda|i)}{m\binom{n}{m}}, \qquad \frac{f_0}{m!L} \le F_i \le \frac{L^{m-1}}{n f_0^{m-1}}.$$

The lower bound is Lemma 13.1.3; the upper bound uses $|\lambda_i| \le L$ in the polynomial numerator and its positivity. These bounds give possible values of ϑ and Θ. Concavity places each tangent plane above F on the cone; the plane at B gives equality when $B \in K$. An infimum of affine functions is concave. Each plane increases under $P \ge 0$ by a number between the two displayed trace bounds; taking infima preserves these bounds. The same estimates give global Lipschitz continuity of $\widehat{F}$. □

To interpret this as a real elliptic operator, let $C(R)$ be the complex Hessian of the real quadratic function $q(x) = \frac{1}{2} x^{\mathsf{T}} R x$, where $x \in \mathbb{R}^{2n}$. For $R \ge 0$, one has $C(R) \ge 0$

and $\operatorname{tr} C(R) = \operatorname{tr} R/4$. If $\mu I \le g(x) \le MI$, then

$$\mathcal{G}(x,R) = \widehat{F}\big(I + g(x)^{-1/2} C(R) g(x)^{-1/2}\big)$$

is uniformly elliptic on real symmetric matrices, with constants $\vartheta/(4M)$ and $\Theta/(4\mu)$. Its dependence on x satisfies on compact coordinate balls

$$|\mathcal{G}(x,R) - \mathcal{G}(y,R)| \le C|x-y|(1+\|R\|).$$

The ellipticity follows from the trace identity for C, and the x-dependence from the Lipschitz continuity of $\widehat{F}$ and the smooth dependence of the positive matrix square root of the fixed metric.

We use the classical interior Evans–Krylov–Caffarelli estimate for a concave operator on all real symmetric matrices with fixed positive ellipticity constants and coefficient oscillation bounded by $C|x-y|^{\gamma}(1+\|R\|)$, where $\gamma > 0$. A Hölder right-hand side then gives an interior $C^{2,\beta}$ estimate, for some $\beta > 0$. This estimate applies also to continuous viscosity solutions. The real extension just constructed satisfies these conditions. Once its solution is $C^{2,\beta}$ and the attained complex Hessians remain in a compact subset of the open cone, the original smooth Hessian operator and Schauder estimates give the higher derivatives.

We next identify the limit of the proper auxiliary equations that will be used for regularity.

Proposition 13.4.1 *Suppose $v_j \to \varphi$ uniformly on X, $h(x,t)$ is smooth on a neighborhood of $X \times J$ for a compact interval J containing all their values, and smooth admissible u_j satisfy*

$$(\omega + \mathrm{dd^c} u_j)^m \wedge \omega^{n-m} = e^{u_j + h(x,v_j)} \omega^n.$$

Assume $u_j \to u$ uniformly and the complex Hessians of u_j are uniformly bounded. Then $\omega + \mathrm{dd^c} u \in \overline{\Gamma_m(\omega)}$ almost everywhere, and its coefficient product satisfies

$$(\omega + \mathrm{dd^c} u)^m \wedge \omega^{n-m} = e^{u + h(x,\varphi)} \omega^n.$$

On every coordinate ball u also solves the corresponding uniformly elliptic concave extension in the viscosity sense.

Proof Lemma 13.4.1 gives $u \in W^{2,p}$ for every finite p and a bounded complex Hessian. The matrices $A_j = I + g^{-1/2} H(u_j) g^{-1/2}$ converge weakly in L^2 to the analogous matrix A for u. Mazur's lemma gives convex combinations of tails converging strongly in L^2, hence almost everywhere along a subsequence. The pointwise closed cone is convex, so $A \in C_{n,m}$ almost everywhere. Lemma 13.4.2 passes the equation to the limit as currents, since its right-hand side converges uniformly. Equality of the bounded densities then gives the equation almost everywhere.

The right-hand sides are bounded above and away from zero. The complex Hessian bound gives a uniform upper bound L for $\operatorname{tr} A_j$. Choose $f_0 > 0$ below all their normalized roots. Thus every A_j, and almost every A, belongs to the compact set K

of Lemma 13.4.3. Every u_j is a classical solution of

$$\mathcal{G}(x, D^2 u_j) = r_j(x), \qquad r_j(x) = e^{(u_j + h(x,v_j))/m}.$$

Let a smooth test q touch u from above at an interior point x_0. Replacing q by $q+|x-x_0|^4$ makes the local contact strict and does not change $D^2 q(x_0)$. Local maximum points x_j of $u_j - q$ then tend to x_0. At these points $D^2 u_j \le D^2 q$, and ellipticity gives $\mathcal{G}(x_j, D^2 q(x_j)) \ge r_j(x_j)$. Passing to the limit gives $\mathcal{G}(x_0, D^2 q(x_0)) \ge e^{(u+h(x,\varphi))/m}$ at x_0. A test from below gives the reverse inequality. The extension is defined on every real Hessian, so no admissibility of a lower test has been assumed. This proves the viscosity statement. □

13.5 Proper equations and admissible approximation

For a smooth potential write $H_m(w) = (\omega + \mathrm{dd^c} w)^m \wedge \omega^{n-m}$. The positive exponential coefficient controls additive constants by the maximum principle. The resulting smooth equations will also construct admissible approximations of bounded potentials, before any continuous Hessian product is defined.

Lemma 13.5.1 *For every $a > 0$ and smooth real H on X, there is a unique smooth strictly admissible solution of*

$$H_m(w) = e^{aw+H} \omega^n.$$

It satisfies

$$-\frac{\max_X H}{a} \le w \le -\frac{\min_X H}{a}.$$

Proof At a maximum of w, the admissible form $\omega + \mathrm{dd^c} w$ is at most ω, hence $e^{aw+H} \le 1$ there. At a minimum it is at least ω, and the reverse inequality holds. These give the bounds. For two solutions, a positive maximum of their difference would imply $H_m(w_1) \le H_m(w_2)$ there, contradicting the exponential factor. This proves uniqueness.

First take $a = 1$ and use the path $H_m(w_t) = e^{w_t + tH} \omega^n$, starting with $w_0 = 0$. By the bounds just proved, $\|w_t\|_\infty \le \|H\|_\infty$. The trace estimate of Proposition 13.3.2 applies with $h(x, \tau) = \tau H(x)$, the constant prescribed potential $v = t$, and a constant comparison function $\varphi = t_i$. One such constant serves for all t with $|t - t_i| \le \varepsilon$, so finitely many give constants uniform for $t \in [0, 1]$. Corollary 13.3.1, which combines this trace estimate with Proposition 13.2.1, then gives a uniform gradient and complex-Hessian bound. Lemma 13.4.1 gives uniform $W^{2,p}$ and hence $C^{1,\alpha}$ bounds, for any fixed $\alpha < 1$. Consequently $r_t = e^{(w_t + tH)/m}$ has a uniform Hölder bound.

Take f_0 and L in Lemma 13.4.3 from the uniform lower bound for r_t and the uniform trace bound. The resulting extension is uniformly elliptic and concave on all real Hessians, with coefficient oscillation bounded by $C|x - y|(1 + \|R\|)$. The

analytic estimate recalled in Section 13.4 therefore gives uniform $C^{2,\beta}$ bounds on a finite cover. The original smooth Hessian operator is uniformly elliptic near the attained compact cone set. Differentiating and applying Schauder estimates gives bounds for every higher derivative.

The linearization of $\log(H_m(w)/\omega^n) - w - tH$ is $L_w - 1$, where

$$L_w\xi = \frac{m\mathrm{dd}^{\mathrm{c}}\xi \wedge (\omega + \mathrm{dd}^{\mathrm{c}}w)^{m-1} \wedge \omega^{n-m}}{H_m(w)}.$$

Its principal coefficients are positive definite. The maximum principle gives zero kernel; smooth elliptic Fredholm theory gives surjectivity and Schauder inversion. The implicit-function theorem proves openness. The uniform estimates prove closedness, so the path reaches $t = 1$. For general a, apply the case $a = 1$ to the metric $a\omega$ and the potential $\widehat{w} = aw$. Both sides of the resulting equation acquire the factor a^n; after division by it, this is exactly the stated equation. □

An ordinary smooth approximation need not be admissible. To obtain an admissible one, choose a smooth obstacle above the potential and penalize the solution's distance from that obstacle.

Lemma 13.5.2 *Every continuous admissible φ on X is the uniform limit of smooth strictly admissible functions.*

Proof Continuous admissible functions are bounded admissible (Section 13.1). Let u be bounded admissible, fix a smooth $h \geq u$ and a constant

$$b > \max\left\{1, \|H_m(h)/\omega^n\|_\infty\right\}.$$

For each positive integer N solve, by the preceding lemma,

$$H_m(w_N) = be^{N(w_N - h)}\omega^n.$$

At a positive maximum of $w_N - h$, the form $\omega + \mathrm{dd}^{\mathrm{c}}h$ lies above the strictly admissible form $\omega + \mathrm{dd}^{\mathrm{c}}w_N$, hence is admissible there. Monotonicity of σ_m on the cone would give $be^{N(w_N-h)} \leq H_m(h)/\omega^n < b$, which is impossible since $w_N > h$ there. Thus $w_N \leq h$. The zero-order estimate of Lemma 13.5.1, with $a = N$ and $H = \log b - Nh$, also gives

$$\|w_N\|_\infty \leq M_N := \|h\|_\infty + \frac{\log b}{N}.$$

The upper semicontinuous function $u - (1 + N^{-1})w_N$ attains its maximum at some x_N, and $(1 + N^{-1})w_N$ plus this maximum is an upper test for u there. Put

$$T_N = (\omega + \mathrm{dd}^{\mathrm{c}}w_N)^{m-1} \wedge \omega^{n-m}, \qquad d_N = H_m(w_N)/\omega^n.$$

At x_N, condition (13.2) puts $\omega + (1 + N^{-1})\mathrm{dd}^{\mathrm{c}}w_N = (1 + N^{-1})(\omega + \mathrm{dd}^{\mathrm{c}}w_N) - N^{-1}\omega$ in the closed cone. Its wedge product with T_N is therefore nonnegative, by polarized Gårding positivity; this is the first inequality below. Maclaurin's inequality $\sigma_{m-1} \geq \sigma_m^{(m-1)/m}$ for $\omega + \mathrm{dd}^{\mathrm{c}}w_N$ gives the second:

$$(N+1)d_N \geq \omega \wedge T_N/\omega^n \geq d_N^{(m-1)/m}.$$

Therefore $d_N \geq (N+1)^{-m}$. Since $d_N = be^{N(w_N - h)}$, at x_N

$$w_N - h \geq -E_N, \qquad E_N = \frac{m\log(N+1) + \log b}{N}.$$

Since $u \leq h$, the maximum of $u - (1+N^{-1})w_N$ is at most $E_N + M_N/N$. It follows on all of X that

$$u - E_N - 2M_N/N \leq w_N \leq h. \tag{13.3}$$

Given $\delta > 0$, apply this to $u = \varphi$ and a smooth h with $\varphi \leq h \leq \varphi + \delta$, and choose N so large that $E_N + 2M_N/N \leq \delta$. The resulting w_N is strictly admissible and within δ of φ. Let $\delta \downarrow 0$. □

13.5.1 Decreasing approximation of bounded potentials

Upper semicontinuity gives smooth obstacles decreasing to a bounded potential. The same penalty equation gives admissible functions between the potential and each obstacle, up to an error tending to zero. Choosing successive obstacles below the preceding smooth approximation makes the admissible approximations decrease as well.

Lemma 13.5.3 *Every bounded upper semicontinuous real function u on X is the pointwise decreasing limit of smooth functions $h_j > u$.*

Proof Fix a Riemannian distance d on X and put

$$c_j(x) = \sup_{y\in X}\{u(y) - jd(x,y)\}.$$

Each c_j is Lipschitz, $c_j \geq u$, and $c_{j+1} \leq c_j$. The supremum is attained. If y_j attains it at a fixed x, then $jd(x,y_j) \leq \sup_X u - u(x)$, so $y_j \to x$. Upper semicontinuity gives $\limsup_j c_j(x) \leq u(x)$, and hence $c_j \downarrow u$ pointwise.

Set $\delta_j = 4^{-j}$. Smooth approximation of continuous functions on a compact manifold gives $s_j \in C^\infty(X)$ with $\|s_j - c_j\|_\infty \leq \delta_j/4$. Take $h_j = s_j + \delta_j$. Then

$$u + \tfrac{3}{4}\delta_j \leq c_j + \tfrac{3}{4}\delta_j \leq h_j \leq c_j + \tfrac{5}{4}\delta_j, \qquad h_{j+1} \leq c_j + \tfrac{5}{16}\delta_j < c_j + \tfrac{3}{4}\delta_j \leq h_j.$$

The bounds also imply $h_j(x) \to u(x)$ at every point. □

Lemma 13.5.4 *Let u be bounded admissible and $h \geq u$ smooth. Choose*

$$b > \max\{1, \|H_m(h)/\omega^n\|_\infty\}.$$

For each positive integer N, let w_N be the smooth strictly admissible solution of

$$H_m(w_N) = be^{N(w_N - h)}\omega^n.$$

With

$$M_N = \|h\|_\infty + \frac{\log b}{N}, \qquad E_N = \frac{m\log(N+1) + \log b}{N},$$

one has $\|w_N\|_\infty \leq M_N$ and

$$u - E_N - 2M_N/N \leq w_N \leq h.$$

Proof Existence follows from Lemma 13.5.1. The bounds are (13.3) and the zero-order bound in the proof of Lemma 13.5.2, which use only that u is bounded admissible and $u \leq h$. □

Theorem 13.5.1 *Every bounded admissible potential u on X is the pointwise decreasing limit of smooth strictly admissible potentials. The sequence may be chosen uniformly bounded.*

Proof Choose smooth $h_k \downarrow u$ by Lemma 13.5.3 and put $\varepsilon_j = 8^{-j}$. We construct recursively smooth strictly admissible u_j with $u_j \geq u + \varepsilon_j$. For $j = 1$, take $k_1 = 1$. At a subsequent step choose $k_j \geq \max\{j, k_{j-1} + 1\}$ so large that

$$h_{k_j} < u_{j-1} - \tfrac{1}{2}\varepsilon_{j-1} \quad \text{on } X.$$

Such an index exists: the open sets $\{h_k < u_{j-1} - \varepsilon_{j-1}/2\}$ increase with k and cover X, since $h_k \downarrow u \leq u_{j-1} - \varepsilon_{j-1}$. Compactness gives a single index for all of X.

For this fixed obstacle h_{k_j} choose b_j as in Lemma 13.5.4, and then choose N_j so large that its error $E_{N_j} + 2M_{N_j}/N_j$ is at most ε_j. The resulting smooth solution w_j satisfies

$$u - \varepsilon_j \leq w_j \leq h_{k_j}.$$

Put $u_j = w_j + 2\varepsilon_j$. Adding a constant preserves strict admissibility, and $u_j \geq u + \varepsilon_j$. For $j \geq 2$,

$$u_j \leq h_{k_j} + 2\varepsilon_j < u_{j-1} - \tfrac{1}{2}\varepsilon_{j-1} + 2\varepsilon_j = u_{j-1} - \tfrac{1}{4}\varepsilon_{j-1}.$$

Finally, $u \leq u_j \leq h_{k_j} + 2\varepsilon_j$ and $k_j \geq j$ give pointwise convergence to u. The bounds $\inf_X u \leq u_j \leq \sup_X u_1$ give a common uniform bound. □

13.6 Continuous products, envelopes and domination

Uniform admissible approximation allows us to define the mixed Hessian products. Their local bounds keep the derivatives of the Hermitian background; closedness of the background is not required.

Proposition 13.6.1 *For continuous admissible $u_1, \dots, u_k$, $1 \le k \le m$, the current*

$$(\omega + \mathrm{dd^c} u_1) \wedge \cdots \wedge (\omega + \mathrm{dd^c} u_k) \wedge \omega^{n-m}$$

is well defined as the limit of uniform smooth admissible approximations. It is positive, local, and continuous when its potentials converge uniformly. Its local masses are bounded in terms of uniform bounds on the potentials. In particular the Hessian measure $H_m(u)$, the product with every potential equal to u, is a positive Radon measure.

Proof Lemma 13.5.2 provides the smooth approximations. For a smooth admissible u put $\sigma = \omega + \mathrm{dd^c} u$. Since $\mathrm{d}\sigma = \mathrm{d}\omega$, the coefficients of $\mathrm{dd^c}(\chi \sigma^{r-1} \wedge \omega^{n-r})$ are sums of smooth test factors paired with powers σ^q, $q \le r-1$, where χ is a fixed cutoff. Lemma 13.1.5 and integration by parts give the Chern–Levine–Nirenberg bound by induction on $r \le m$: for a compact set K in an open set U,

$$\int_K \sigma^r \wedge \omega^{n-r} \le C_{K,U,\omega} (1 + \|u\|_{L^\infty(U)})^r .$$

At the induction step write $\sigma^r = \omega \wedge \sigma^{r-1} + \mathrm{dd^c} u \wedge \sigma^{r-1}$, integrate the second term against χ, and move $\mathrm{dd^c}$ to $\chi \sigma^{r-1} \wedge \omega^{n-r}$. The resulting powers have degrees $r-1, r-2, r-3$, with negative powers absent, and the induction applies on nested slightly larger compact sets. Mixed masses reduce to the mass of the average potential, since every term in the expansion of its power is positive. Positive mixed currents have total variation controlled by these trace masses.

To prove independence, telescope the difference between two smooth mixed products whose potentials have a common uniform bound. Each term paired with a fixed smooth complementary test is

$$\int \mathrm{dd^c}(u_a - v_a) \wedge T \wedge \eta = \int (u_a - v_a) \mathrm{dd^c}(T \wedge \eta),$$

where T is a product of the remaining admissible factors and ω^{n-m}. The identity $\mathrm{d}(\omega + \mathrm{dd^c} u) = \mathrm{d}\omega$ and its $\mathrm{d^c}$ analogue show that derivatives of T introduce only fixed metric derivatives; every term has at most $k - 1 < m$ admissible factors. The absolute mixed-form bound and the mass estimate just proved bound the variation of $\mathrm{dd^c}(T \wedge \eta)$ uniformly. Consequently the pairing difference is at most $C_\eta \max_a \|u_a - v_a\|_\infty$. The same construction with an arbitrary complementary smooth test also defines the signed products of fewer than m factors without the factor ω^{n-m}: their variation is bounded by the absolute-form lemma. These products give a meaning to the metric-derivative terms above when the potentials are only continuous. This proves

that all approximations have the same limit and that the products are continuous in the uniform topology. Positivity passes from the smooth mixed products. Using a test supported in an open set on which two limiting potentials coincide proves locality by the same estimate on a smaller neighborhood of its support. □

Locality and uniform continuity of these products give comparison on a contact set.

Lemma 13.6.1 *If $u \le v$ are continuous admissible functions and the remaining $m-1$ factors are fixed continuous admissible functions, then on $\{u = v\}$ the mixed measure with first factor $\omega + \mathrm{dd}^{\mathrm{c}} u$ is at most the one with first factor $\omega + \mathrm{dd}^{\mathrm{c}} v$. In particular $\mathbf{1}_{\{u=v\}} H_m(u) \le \mathbf{1}_{\{u=v\}} H_m(v)$.*

Proof Write $M(w)$ for the mixed measure with first potential w. Positivity and locality give, for $t > 0$,

$$M(\max(v, u+t)) \ge \mathbf{1}_{\{v>u+t\}} M(v) + \mathbf{1}_{\{u\ge v\}} M(u).$$

The first indicators increase to $\mathbf{1}_{\{v>u\}}$ as $t \downarrow 0$, and the potentials on the left converge uniformly to v. Integrate against nonnegative continuous tests and pass to the limit. Subtract the first restricted measure to obtain the claimed contact inequality. Replacing one factor at a time proves the pure-power claim and the analogous contact inequalities for all mixed powers. □

For a continuous real obstacle h, define $P(h)$ to be the supremum of continuous admissible functions bounded above by h. Constants make the family nonempty.

Lemma 13.6.2 *The function $P(h)$ is continuous and admissible, and $H_m(P(h))$ is supported on $\{P(h) = h\}$. Moreover $\|P(h_1) - P(h_2)\|_\infty \le \|h_1 - h_2\|_\infty$.*

Proof For smooth h, let w_N be the penalty solutions of Lemma 13.5.4. They are competitors, and that lemma applies to every competitor with an error depending only on h and N. Taking the supremum over the competitors gives

$$P(h) - E_N - 2M_N/N \le w_N \le P(h),$$

where the error tends to zero. Thus $P(h)$ is their uniform limit and is continuous and admissible. On every compact subset of $\{P(h) < h\}$ the penalty densities $be^{N(w_N - h)}$ tend uniformly to zero. Product continuity implies the contact support assertion. Adding constants to competitors gives the stated Lipschitz bound. Approximate a continuous h uniformly by smooth obstacles. Their envelopes converge uniformly to $P(h)$, proving continuity and admissibility. Any compact subset of $\{P(h) < h\}$ also lies outside the contact sets of all sufficiently close smooth obstacles; product continuity gives the support assertion for h. □

Proposition 13.6.2 *For every smooth real obstacle h, the envelope $P(h)$ equals the pointwise supremum of all bounded admissible functions below h.*

Proof The solutions w_N of Lemma 13.5.4 are continuous admissible competitors, so $w_N \le P(h)$. For each bounded admissible $u \le h$, the same lemma gives

$$u \le w_N + E_N + 2M_N/N \le P(h) + E_N + 2M_N/N.$$

Let $N \to \infty$ for this fixed obstacle and fixed b. Thus $P(h)$ dominates every bounded competitor. Since $P(h)$ itself is a bounded competitor, it equals their supremum. □

The trace test $\gamma_1 = \cdots = \gamma_{m-1} = \omega$ in (13.1) gives $L_\omega u \ge -n$ in distributions, where $L_\omega u = n\mathrm{dd}^{\mathrm{c}} u \wedge \omega^{n-1}/\omega^n$. For smooth admissible u, the Green-kernel argument in the proof of Proposition 10.11.1, applied to the operator L_ω, gives

$$\int_X (\sup_X u - u)\omega^n \le C_\omega. \tag{13.4}$$

By Lemma 13.5.2, this estimate extends to every continuous admissible u.

To use envelopes in domination, we must first exclude a potential with zero total Hessian mass.

Lemma 13.6.3 *Every continuous admissible u satisfies $\int_X H_m(u) > 0$.*

Proof Suppose this integral is zero, so that $H_m(u) = 0$. For $s > 0$ put $w = \max(u, \min_X u + s)$ and $U = \{u < \min_X u + s\}$. The open set U is nonempty, $0 \le w - u \le s$, and $w = u$ outside it. Set $\sigma_u = \omega + \mathrm{dd}^{\mathrm{c}} u$, $\sigma_w = \omega + \mathrm{dd}^{\mathrm{c}} w$, and $\mu_q = \sigma_u^q \wedge \sigma_w^{m-q} \wedge \omega^{n-m}$, so that $\mu_m = H_m(u) = 0$. For $q < m$ the telescoping identity is

$$\mu_q - \mu_m = \mathrm{dd}^{\mathrm{c}}(w-u) \wedge T_q, \qquad T_q = \sigma_u^q \wedge \sum_{r=0}^{m-q-1} \sigma_w^r \wedge \sigma_u^{m-q-1-r} \wedge \omega^{n-m}.$$

Integration by parts on X therefore gives

$$\int_X \mu_q = \int_X (w-u)\mathrm{dd}^{\mathrm{c}} T_q.$$

This identity follows first for smooth approximants and then by uniform product continuity. The derivatives of T_q contain only fixed metric derivatives and products of at most $m-1$ admissible factors, so the absolute-form lemma makes $\mathrm{dd}^{\mathrm{c}} T_q$ a measure of uniformly controlled variation. Uniform convergence also justifies its continuous weight. On U, locality gives $\sigma_w = \omega$ and thus

$$|\mathrm{dd}^{\mathrm{c}} T_q| \le C \sum_{r=0}^{m-1} \mu_r.$$

The constant depends only on the metric and m, n, and not on u or s. Since $0 \le w - u \le s$ and $w - u$ vanishes outside U,

$$\int_U \mu_q \le \int_X \mu_q \le Cs \sum_{r=0}^{m-1} \int_U \mu_r \qquad (0 \le q < m).$$

Summing over $q < m$ gives $(1 - mCs) \sum_{r=0}^{m-1} \int_U \mu_r \le 0$. For $mCs < 1$ these nonnegative integrals vanish. But $\mu_0 = \omega^n$ on the nonempty open set U, a contradiction. □

The envelope of a stretched difference now converts vanishing of a restricted measure into an order relation. The linear bound (13.4) prevents these envelopes from tending to $-\infty$ on an open set.

Proposition 13.6.3 *If continuous admissible u, v satisfy $\mathbf{1}_{\{u<v\}} H_m(u) = 0$, then $u \ge v$.*

Proof Replace v by $\max(u, v)$, so $u \le v$. For $b > 1$ put $p_b = P(bu - (b-1)v)$ and $z_b = b^{-1} p_b + (1 - b^{-1})v \le u$. On the contact set $C_b = \{p_b = bu - (b-1)v\}$ one has $z_b = u$; Lemma 13.6.1 and polarized positivity give

$$\mathbf{1}_{C_b} b^{-m} H_m(p_b) \le \mathbf{1}_{C_b} H_m(z_b) \le \mathbf{1}_{C_b} H_m(u).$$

By Lemmas 13.6.2 and 13.6.3, the envelope measure $H_m(p_b)$ is supported on C_b and has positive mass. It vanishes on $C_b \cap \{u < v\}$ by the hypothesis, so some point of $C_b \cap \{u = v\}$ exists. Hence $\sup_X p_b \ge \inf_X v$, while $p_b \le u$. We apply (13.4) to p_b. On the open set $E_a = \{u < v - a\}$, for $a > 0$, one has $p_b \le v - ba$, and therefore

$$C_\omega \ge \int_X (\sup_X p_b - p_b)\omega^n \ge (ba - \operatorname{osc}_X v) \int_{E_a} \omega^n.$$

Letting $b \to \infty$ gives zero volume for E_a. Since it is open it is empty, and $a \downarrow 0$ gives $u \ge v$. □

Proposition 13.6.4 *If continuous admissible u, v satisfy*

$$\mathbf{1}_{\{u<v\}} H_m(u) \le c \mathbf{1}_{\{u<v\}} H_m(v), \qquad 0 \le c < 1,$$

then $u \ge v$.

Proof Choose $b > 1$ so that $(1 - b^{-1})^m > c$, and put $p = P(bu - (b-1)v)$ and $z = b^{-1} p + (1 - b^{-1})v \le u$. On $C = \{p = bu - (b-1)v\}$ the contact comparison and polarized positivity give

$$\mathbf{1}_C \big(b^{-m} H_m(p) + (1 - b^{-1})^m H_m(v) \big) \le \mathbf{1}_C H_m(u).$$

Restricting to $\{u < v\}$ and using the hypothesis shows that $H_m(p)$ vanishes on $C \cap \{u < v\}$. Its support lies in C, where $p < v$ is equivalent to $u < v$. Thus $H_m(p)$ vanishes on $\{p < v\}$. Proposition 13.6.3 gives $p \ge v$. Since $p \le bu - (b-1)v$, it follows that $u \ge v$. □

Strict domination yields the comparison needed for the proper equations.

Lemma 13.6.4 *For continuous admissible u, v and $a > 0$, if*

$$e^{-av} H_m(v) \geq e^{-au} H_m(u),$$

then $u \geq v$.

Proof Fix $\delta > 0$. On $\{u < v - \delta\}$, the assumed inequality gives

$$H_m(u) \leq e^{a(u-v)} H_m(v) \leq e^{-a\delta} H_m(v - \delta).$$

The constant $e^{-a\delta}$ is strictly smaller than one, so Proposition 13.6.4 applied to u and $v - \delta$ gives $u \geq v - \delta$. Letting $\delta \downarrow 0$ proves the assertion. □

13.7 Continuous equations and regularity

The proper equation of Lemma 13.5.1 is stable in the uniform norm, so its smooth solutions extend to continuous data. This also identifies the limit of the auxiliary equations with a prescribed continuous weak solution.

Proposition 13.7.1 *Let $a > 0$ and let smooth H_j converge uniformly to a continuous H. The smooth solutions of $H_m(w_j) = e^{aw_j + H_j} \omega^n$ converge uniformly to the unique continuous admissible solution of $H_m(w) = e^{aw+H} \omega^n$. For two continuous data $H, \widetilde{H}$, the corresponding solutions $w, \widetilde{w}$ satisfy*

$$\|w - \widetilde{w}\|_\infty \leq a^{-1} \|H - \widetilde{H}\|_\infty.$$

They also satisfy $-\max_X H/a \leq w \leq -\min_X H/a$.

Proof Smooth solutions exist by Lemma 13.5.1. For smooth data, at a maximum of $w - \widetilde{w}$, monotonicity of the smooth Hessian operator gives $e^{aw+H} \leq e^{a\widetilde{w}+\widetilde{H}}$. Hence $a(w - \widetilde{w}) \leq \widetilde{H} - H$ there. Exchanging the functions gives the stated norm estimate for smooth data. Thus a uniformly convergent sequence of smooth data gives a uniformly Cauchy sequence of solutions. Its limit is admissible because the linear inequalities (13.1) pass to distributional limits, and Proposition 13.6.1 passes the equation to the limit. The bounds follow from those for the smooth solutions. Taking limits in the smooth norm estimate proves it for continuous data. Finally, Lemma 13.6.4 in both directions gives uniqueness among continuous admissible solutions. □

For smooth positive densities depending on the potential, the estimate of Section 13.3 gives complex-Hessian bounds independent of derivatives of the prescribed approximation. The elliptic extension then upgrades the continuous solution to a classical one.

Theorem 13.7.1 *If φ is continuous and admissible on X, $f(x,t)$ is smooth on a neighborhood of the compact set $X \times \varphi(X)$, and*

$$H_m(\varphi) = e^{f(x,\varphi)}\omega^n,$$

then φ is smooth.

Proof Choose smooth admissible $v_j \to \varphi$ uniformly, by Lemma 13.5.2. After discarding finitely many terms, their values lie in a fixed compact interval I such that f is smooth on a neighborhood of $X \times I$. Put $h(x,t) = -t + f(x,t)$ and solve

$$H_m(u_j) = e^{u_j + h(x,v_j)}\omega^n.$$

Proposition 13.7.1, with $a = 1$, shows that u_j converges uniformly to a continuous admissible u satisfying

$$e^{-u}H_m(u) = e^{-\varphi + f(x,\varphi)}\omega^n = e^{-\varphi}H_m(\varphi).$$

Lemma 13.6.4, applied in both directions, gives $u = \varphi$.

The zero-order bounds of Proposition 13.7.1 for u_j are uniform in j. Fix $M \geq 1$ bounding them and choose $B > \|\varphi\|_\infty + 1$. Modify h outside a neighborhood of $X \times \varphi(X)$ so that it is smooth on a neighborhood of $X \times [-B-1, B+1]$; this does not change the equations for large j. The tolerance ε in Proposition 13.3.2 depends on M, B and h, but not on derivatives of the smooth comparison function chosen next. Choose a fixed smooth $\widetilde{\varphi}$ with $\|\widetilde{\varphi} - \varphi\|_\infty < \varepsilon/2$ and $\|\widetilde{\varphi}\|_\infty \leq B$. For large j, $\|v_j - \widetilde{\varphi}\|_\infty < \varepsilon$. Corollary 13.3.1 then gives uniform gradient and complex-Hessian bounds for u_j, independent of derivatives of v_j. Lemma 13.4.1 yields $\varphi \in W^{2,p}$ for every finite p, and $\varphi \in C^{1,\alpha}$ for every $\alpha < 1$.

Proposition 13.4.1 makes φ a continuous viscosity solution of the concave uniformly elliptic extension, with right-hand side $e^{f(x,\varphi)/m}$. This right-hand side is Hölder continuous, since $\varphi \in C^{1,\alpha}$. The analytic estimate in Section 13.4, which applies directly to this viscosity limit, therefore gives $\varphi \in C^{2,\beta}$, for some $\beta > 0$. In the proof of Proposition 13.4.1, the matrix A of $\omega + \mathrm{dd}^c\varphi$ lies almost everywhere in the compact set K of Lemma 13.4.3; it is now continuous, so it lies in K everywhere. The extension agrees with the Hessian root F on K. The equation is therefore classical, depends smoothly on x, φ and the complex Hessian of φ, and is uniformly elliptic near K. Difference quotients and interior Schauder estimates give $C^{3,\beta}$; iteration proves smoothness. □

The proper equation can now be solved when its exponent depends smoothly on the unknown potential, provided that constant barriers bound the desired range and the dependence is strictly increasing.

Theorem 13.7.2 *Let $A < B$ and let f be smooth on a neighborhood of $X \times [A, B]$, with $\partial_t f > 0$ on $X \times [A, B]$ and, for every $x \in X$,*

$$f(x, A) \le 0 \le f(x, B).$$

There is a unique continuous admissible solution with values in $[A, B]$ of

$$H_m(w) = e^{f(x,w)}\omega^n.$$

This solution is smooth and strictly admissible.

Proof By compactness there are $0 < \kappa \le L$ with $\kappa \le \partial_t f \le L$ on $X \times [A, B]$. Choose $a \ge L$ and let $C = \{v \in C(X) : A \le v \le B\}$, a complete subset of $C(X)$ in the uniform norm. For $v \in C$ put $H_v(x) = f(x, v(x)) - av(x)$ and let $T(v)$ be the continuous proper solution with exponent $aT(v)+H_v$, furnished by Proposition 13.7.1. The function $f(x, t) - at$ is nonincreasing in t, so

$$-aB \le f(x, B) - aB \le H_v(x) \le f(x, A) - aA \le -aA.$$

The zero-order bounds in that proposition give $A \le T(v) \le B$. For $v, z \in C$, the mean-value theorem and the stability estimate give

$$\|H_v - H_z\|_\infty \le (a - \kappa)\|v - z\|_\infty, \qquad \|T(v) - T(z)\|_\infty \le (1 - \kappa/a)\|v - z\|_\infty.$$

Since $0 \le 1 - \kappa/a < 1$, the contraction mapping theorem gives a unique fixed point $w \in C$. It is admissible because it equals $T(w)$, and its equation is exactly the stated equation. Conversely, a continuous admissible solution $\widetilde{w}$ with values in $[A, B]$ solves the continuous proper equation with data $H_{\widetilde{w}}$. Uniqueness for that equation gives $T(\widetilde{w}) = \widetilde{w}$, so $\widetilde{w} = w$. This proves uniqueness in the stated class. Theorem 13.7.1 makes w smooth. The form $\omega + \mathrm{dd}^c w$ lies in $\overline{\Gamma_m(\omega)}$ and has positive mth symmetric polynomial; Maclaurin's inequalities then put it in $\Gamma_m(\omega)$. Thus w is strictly admissible. □

13.8 Bounded potentials and recursive Hessian currents

The decreasing approximation of Theorem 13.5.1 gives limits that uniform approximation cannot reach. A bounded potential can multiply a current with measure coefficients, so one can apply dd^c to the resulting current. Repeating this operation requires bounds on every coefficient, rather than only on the trace of a positive mixed measure. The absolute-form estimate of Lemma 13.1.5 provides these bounds below degree m; at degree m, positivity is obtained after wedging with ω^{n-m}.

Bounded admissible functions are canonical representatives (Section 13.1); their pointwise values are used when multiplying measure coefficients. For a real current T of order zero, write $\|T\|_K$ for the sum of the total variations of its coefficients on a

compact coordinate set K. Fix $K \Subset U$ in a coordinate chart. Different fixed coordinate frames give equivalent local bounds.

Lemma 13.8.1 *If u is bounded admissible, then $\mathrm{dd^c}u$ has signed Radon measure coefficients. For $K \Subset U$ contained in a coordinate chart,*

$$\|\mathrm{dd^c}u\|_K \le C_{K,U,\omega}(1 + \|u\|_{L^\infty(X)}).$$

Proof Choose smooth strictly admissible $u_j \downarrow u$ by Theorem 13.5.1, and put $\sigma_j = \omega + \mathrm{dd^c}u_j$. After discarding finitely many terms, $\|u_j\|_\infty \le \|u\|_\infty + 1$. Indeed, the increasing open sets $\{u_j < \sup_X u + 1\}$ cover X, and compactness gives a single index whose open set is all of X; the lower bound follows from $u_j \ge u$.

Since $m \ge 2$, the eigenvalues of σ_j relative to ω have modulus at most its trace, by the proof of Lemma 13.1.3. Thus every coefficient of σ_j is bounded by a fixed multiple of its trace. The degree-one Chern–Levine–Nirenberg estimate in the proof of Proposition 13.6.1, applied to these smooth functions, bounds the integral of that trace on K by $C(1 + \|u\|_\infty)$. Subtracting the fixed smooth ω gives the same bound for the coefficient variations of $\mathrm{dd^c}u_j$. Bounded convergence gives $u_j \to u$ in local L^1, and therefore $\mathrm{dd^c}u_j \to \mathrm{dd^c}u$ in distributions. The uniform coefficient bounds show that this limit has measure coefficients with the asserted bound. □

Set $D_0 = 1$. For an ordered list $u_1, \dots, u_p$ of bounded admissible potentials, define recursively

$$D_p(u_1, \dots, u_p) = \mathrm{dd^c}\big(u_p D_{p-1}(u_1, \dots, u_{p-1})\big). \tag{13.5}$$

The next result justifies the bounded multiplication at every step up to $p = m$.

Proposition 13.8.1 *For $1 \le p < m$, the recursion defines a real closed current of order zero and bidegree (p, p). If $M = \max_a \|u_a\|_{L^\infty(X)}$, then*

$$\|D_p(u_1, \dots, u_p)\|_K \le C_{K,U,\omega,p}(1 + M)^p.$$

For $p = m$, it defines a real closed distributional current. For $1 \le p \le m$ and any smooth admissible sequences $u_a^{j_a} \downarrow u_a$ with a common uniform bound, one has

$$D_p(u_1, \dots, u_p) = \lim_{j_p \to \infty} \cdots \lim_{j_1 \to \infty} \mathrm{dd^c}u_1^{j_1} \wedge \cdots \wedge \mathrm{dd^c}u_p^{j_p} \tag{13.6}$$

in distributions. The index j_1 tends to infinity first. Below degree m, all these limits have locally uniformly bounded coefficient variations.

Proof The case $p = 1$ is the preceding lemma. Suppose the assertions hold through degree $p - 1 < m$. Since u_p is bounded Borel and D_{p-1} has measure coefficients, their product is well defined. Applying $\mathrm{dd^c}$ gives a real current that is both d- and $\mathrm{d^c}$-closed.

For a fixed smooth complementary test form η, dominated convergence against the fixed signed measures gives

$$\begin{aligned}\langle D_p, \eta\rangle &= \langle u_p D_{p-1}, \mathrm{dd^c}\eta\rangle \\ &= \lim_{j_p\to\infty} \langle u_p^{j_p} D_{p-1}, \mathrm{dd^c}\eta\rangle.\end{aligned}$$

With j_p fixed, the inductive iterated limit for D_{p-1} can be paired with the smooth form $u_p^{j_p}\,\mathrm{dd^c}\eta$. Integration by parts for the smooth closed factors identifies the resulting pairing with the smooth product on the right of (13.6). This proves that identity also when $p = m$.

For $p < m$, expand the smooth product using $\mathrm{dd^c}u_a^{j_a} = \sigma_a^{j_a} - \omega$. A term with q admissible factors, $0 \le q \le p$, has the form

$$\sigma_{a_1}^{j_{a_1}} \wedge \cdots \wedge \sigma_{a_q}^{j_{a_q}} \wedge \omega^{p-q}.$$

Every such q is strictly less than m. Apply Lemma 13.1.5 to each coefficient test, and then the smooth mass estimate from Proposition 13.6.1 to the potential sum. The term $q = 0$ is the fixed smooth form ω^p. For $q \ge 1$, the sum of its admissible forms is $q(\omega + \mathrm{dd^c}\bar{u})$, where $\bar{u}$ is their average and $\|\bar{u}\|_\infty \le M + 1$. The result is a coefficient variation bound $C(1 + M)^p$, after choosing the approximations with norm at most $M + 1$ as in the preceding proof. For arbitrary uniformly bounded approximations the same argument uses their common bound. The bounds survive each distributional limit, proving that D_p has order zero and the stated estimate. This completes the induction. □

For a subset $I = \{i_1 < \cdots < i_q\} \subset \{1, \ldots, p\}$, write $D_I = D_q(u_{i_1}, \ldots, u_{i_q})$, with $D_\varnothing = 1$. Define the shifted current and its wedge with ω^{n-m} by

$$R_p(u_1, \ldots, u_p) = \sum_{I\subset\{1,\ldots,p\}} \omega^{p-|I|} \wedge D_I, \qquad Q_p = R_p \wedge \omega^{n-m}. \tag{13.7}$$

Below degree m, R_p has measure coefficients. At degree m it is a priori only a distributional current, but its wedge Q_m is a positive measure by the next theorem.

Put $M = \max_a \|u_a\|_{L^\infty(X)}$.

Theorem 13.8.1 *For $1 \le p \le m$, Q_p is a positive current of order zero, and*

$$\int_K Q_p \wedge \omega^{m-p} \le C_{K,U,\omega,p}(1 + M)^p.$$

For approximations as in Proposition 13.8.1, it is the iterated limit, in the order of (13.6), of

$$(\omega + \mathrm{dd^c}u_1^{j_1}) \wedge \cdots \wedge (\omega + \mathrm{dd^c}u_p^{j_p}) \wedge \omega^{n-m}.$$

In particular Q_m is a positive Radon measure. These currents are local in the canonical potentials and independent of the chosen approximations for the fixed ordering. When all potentials are continuous, Q_p equals the continuous mixed product of Proposition 13.6.1.

Proof Expand the smooth shifted product over its subsets. Applying (13.6) to each subset, in its inherited ordering, gives (13.7). The absent indices have no effect on a subset's limit.

Each smooth product is positive: pairing with a decomposable strongly positive complementary test appends $m - p$ nonnegative rank-one $(1,1)$-forms, and polarized Gårding positivity applies. The smooth mixed mass bound is uniform in all indices. Distributional limits preserve positivity and the local trace bound at every step. Positive currents have order zero, so this gives the assertions for Q_p, including the top-degree measure.

The recursion uses only bounded multiplication of canonical values and local distributional differentiation. It is therefore local. Its definition contains no approximation choices, so the proved limit is independent of those choices while the ordering is fixed. For continuous potentials, the approximations converge uniformly by Dini's theorem. The telescoping estimate in Proposition 13.6.1 is uniform in all remaining indices and shows that their iterated limit is the same continuous mixed product. This also proves permutation invariance in the continuous case. □

The shifted currents are not closed. Their derivatives can be stated without multiplying a discontinuous function by a distribution of positive order. This is the torsion identity needed before considering integration by parts with bounded weights.

Proposition 13.8.2 *Write $R^{\widehat{a}}_{p-1}$ for the shifted current obtained by omitting u_a, in the inherited order, and use two hats for two omissions. Put $R_0 = 1$. For $1 \le p \le m$,*

$$\begin{aligned}
\mathrm{d}R_p &= \mathrm{d}\omega \wedge \sum_{a=1}^{p} R^{\widehat{a}}_{p-1},\\
\mathrm{d}^{\mathrm{c}}R_p &= \mathrm{d}^{\mathrm{c}}\omega \wedge \sum_{a=1}^{p} R^{\widehat{a}}_{p-1},\\
\mathrm{dd}^{\mathrm{c}}R_p &= \mathrm{dd}^{\mathrm{c}}\omega \wedge \sum_{a=1}^{p} R^{\widehat{a}}_{p-1} + 2\,\mathrm{d}\omega \wedge \mathrm{d}^{\mathrm{c}}\omega \wedge \sum_{a<b} R^{\widehat{a}\widehat{b}}_{p-2}.
\end{aligned}$$

The last sum is absent for $p = 1$. All currents on the right have order zero. If all but at most one potential are smooth, R_p is independent of the ordering and is the ordinary product of the smooth factors with $\omega + \mathrm{dd}^{\mathrm{c}}u$, where u is the remaining potential.

Proof Every D_I is both d- and d^{c}-closed. Differentiating (13.7) therefore gives

$$\mathrm{d}R_p = \mathrm{d}\omega \wedge \sum_{I} (p - |I|)\omega^{p-|I|-1} \wedge D_I,$$

with zero terms omitted. The coefficient $p - |I|$ counts the omitted indices outside I, which proves the first formula; the second is identical with d^{c} in place of d. Apply d to the second formula, and use the first formula for each $R^{\widehat{a}}_{p-1}$. Each pair of omitted indices occurs twice, and $-\mathrm{d}^{\mathrm{c}}\omega \wedge \mathrm{d}\omega = \mathrm{d}\omega \wedge \mathrm{d}^{\mathrm{c}}\omega$ because both forms have odd degree

three. This gives the displayed $\mathrm{dd^c}$ formula, in which every background derivative appears.

If all but one potential are smooth, applying $\mathrm{dd^c}$ to the bounded potential times a smooth closed product gives its $\mathrm{dd^c}$ wedged with that product. At later smooth steps, closedness again makes the recursion ordinary smooth multiplication. This is independent of the position of the bounded potential, and the subset expansion gives the last assertion. □

Local capacity arguments naturally concern functions whose unshifted Hessian lies in the cone. On a coordinate ball U, call a bounded function f, taken in its canonical upper semicontinuous representative, *m-subharmonic relative to ω* if

$$\mathrm{dd^c} f \wedge \gamma_1 \wedge \cdots \wedge \gamma_{m-1} \wedge \omega^{n-m} \geq 0$$

in distributions on U, for all smooth $\gamma_i \in \Gamma_m(\omega)$. Equivalently, its local upper tests q satisfy $\mathrm{dd^c} q \in \overline{\Gamma_m(\omega)}$. The same Hermitian form ω is used here and in the chapter's shifted admissibility.

Lemma 13.8.2 *Let U be a coordinate ball compactly contained in a holomorphic chart. There is a smooth function ρ on a neighborhood of $\overline{U}$ such that $\mathrm{dd^c}\rho \geq 2\omega$ and, for every bounded admissible u on X, $u + \rho$ is m-subharmonic relative to ω on U. Adding ρ to the approximations in Theorem 13.5.1 gives uniformly bounded smooth decreasing approximations in this local class.*

Proof Take $\rho = C|z|^2$ in the chart. On the fixed compact set $\overline{U}$, a sufficiently large C gives $\mathrm{dd^c}\rho \geq 2\omega$. An upper test q for $u + \rho$ at x gives the upper test $q - \rho$ for u there. Hence

$$\mathrm{dd^c} q = (\omega + \mathrm{dd^c}(q - \rho)) + (\mathrm{dd^c}\rho - \omega) \in \overline{\Gamma_m(\omega)},$$

since the second summand is positive and the cone is preserved by adding nonnegative forms. Smooth addition preserves the canonical representative. The same identity applies pointwise to every smooth admissible approximation. Adding a fixed smooth function preserves pointwise decrease and convergence, and its bounded norm on $\overline{U}$ preserves the common local bound. □

The correction is a local inclusion of classes. It leaves the metric in the cone tests unchanged. Although $\tau = \mathrm{dd^c}\rho$ is closed, $\tau - \omega$ has derivative $-\mathrm{d}\omega$, so the background terms in the preceding torsion identity remain present. In particular, $\omega + \mathrm{dd^c} u = \mathrm{dd^c}(u + \rho) + (\omega - \tau)$. Comparison of the unshifted Hessians of the corrected functions must therefore account for this smooth remainder.

Example 13.8.1 Take $n = 3$ and $m = 2$. On the unit ball in $\mathbb{C}^3$, put $\beta_j = \mathrm{dd^c}|z_j|^2$, $\beta = \sum_{j=1}^3 \beta_j$ and

$$u = -\tfrac{1}{2}\sum_{j=1}^{3} |z_j|^2, \qquad v = -\tfrac{4}{5}|z_1|^2.$$

Both $\beta + \mathrm{dd^c}u$ and $\beta + \mathrm{dd^c}v$ are strictly positive, but

$$(\beta + \mathrm{dd^c}v)^2 \wedge \beta > (\beta + \mathrm{dd^c}u)^2 \wedge \beta, \qquad (\mathrm{dd^c}v)^2 \wedge \beta < (\mathrm{dd^c}u)^2 \wedge \beta.$$

Thus shifted Hessian order does not imply unshifted Hessian order, even for smooth potentials with positive perturbed forms.

Proof The eigenvalues of the two perturbed forms relative to β are $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ and $(\frac{1}{5}, 1, 1)$. For a diagonal form α with relative eigenvalues $\lambda_1, \lambda_2, \lambda_3$, its second Hessian $\alpha^2 \wedge \beta$ is $(\lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3)\beta^3/3$. Consequently the shifted measures of u, v are $\frac{1}{4}\beta^3$ and $\frac{7}{15}\beta^3$, while their unshifted expressions are $\frac{1}{4}\beta^3$ and 0. □

The iterated limit fixes the order of the potentials. Symmetry and simultaneous convergence require control of discontinuous weights against varying coefficient measures. Sobolev capacity controls first Hessians but not products of two of them (Section 13.9); the local Hessian capacities of Section 13.10 give the control in all degrees.

13.9 Sobolev capacity and two Hessian factors

Weak convergence of measures is not enough to pass to the limit in their products with bounded potentials; one needs a capacity that dominates the measures and makes the potentials quasicontinuous. Sobolev capacity does this for the first Hessian of a bounded admissible potential, through its linear trace, but a product of two Hessians can charge a set of zero Sobolev capacity. The trace is taken against a $\mathrm{dd^c}$-closed conformal multiple of ω^{n-1}: an admissible form need not be positive in every tangent direction, so only its trace relative to ω is known to be nonnegative.

Choose $\rho > 0$ by Theorem 4.1.1 for ω and put

$$T = \rho\omega^{n-1}, \qquad \mathrm{dd^c}T = 0, \qquad A = \int_X \omega \wedge T > 0.$$

Write $H^1(X) = W^{1,2}(X, \mathbb{R})$ and use the equivalent Sobolev norm

$$\|f\|_{H^1}^2 = \int_X f^2\omega^n + \int_X \mathrm{d}f \wedge \mathrm{d^c}f \wedge T.$$

For an open set $G \subset X$ define

$$\mathrm{Cap}_1(G) = \inf\{\|f\|_{H^1}^2 : f \in H^1(X),\ f \geq 1 \text{ almost everywhere on } G\},$$

and for an arbitrary set E take the infimum over open $G \supset E$. It is distinct from the local Hessian capacities of Section 13.10. Truncation allows $0 \le f \le 1$ in the infimum for open sets.

Lemma 13.9.1 *The capacity* Cap_1 *is countably subadditive. For a smooth real* f *and* $a > 0$*,*

$$\mathrm{Cap}_1(\{|f| > a\}) \le a^{-2}\|f\|_{H^1}^2.$$

If a positive Radon measure μ *satisfies* $|\int_X \phi \, \mathrm{d}\mu| \le C\|\phi\|_{H^1}$ *for every smooth real* ϕ*, then for every Borel set* E*,*

$$\mu(E) \le C\,\mathrm{Cap}_1(E)^{1/2}.$$

Proof Let G_i be open sets with $\sum_i \mathrm{Cap}_1(G_i) < \infty$; if the sum is infinite there is nothing to prove. Fix $\epsilon > 0$ and choose Sobolev functions $0 \le f_i \le 1$ equal to one almost everywhere on G_i, with $\|f_i\|_{H^1}^2 \le \mathrm{Cap}_1(G_i) + \epsilon 2^{-i}$. The Sobolev chain rule gives

$$\|\max_{i \le N} f_i\|_{H^1}^2 \le \sum_{i \le N} \|f_i\|_{H^1}^2.$$

Indeed, almost everywhere the gradient of the maximum is the gradient of an active function, and the squared value is bounded by the sum of the squared values. These maxima increase with N, converge in L^2, are bounded in H^1, and have a weak H^1 limit equal to one almost everywhere on $\bigcup_i G_i$. Lower semicontinuity of the norm gives $\mathrm{Cap}_1(\bigcup_i G_i) \le \sum_i \mathrm{Cap}_1(G_i) + \epsilon$. This proves subadditivity for open sets, and taking open supersets proves it for arbitrary sets. The level-set estimate follows by testing with $\min\{1, |f|/a\}$ and the same Sobolev chain rule.

Fix a compact $K \subset G$ and a Sobolev competitor $0 \le f \le 1$ for the open set G. Mollify f in finitely many coordinate charts by nonnegative kernels and glue the results with a fixed partition of unity. This gives smooth nonnegative $f_\epsilon \to f$ in H^1 that satisfy $f_\epsilon = 1$ on K for small ϵ: every convolution ball centered on K then lies in G, where $f = 1$ almost everywhere. The assumed estimate gives $\mu(K) \le C \lim_\epsilon \|f_\epsilon\|_{H^1} = C\|f\|_{H^1}$. Take the infimum over f and then the supremum over compact $K \subset G$ to obtain the estimate for G. For a Borel set E, apply it to every open $G \supset E$ and use $\mu(E) \le \mu(G)$. □

Lemma 13.9.2 *Every bounded admissible* u *belongs to* $H^1(X)$*. Its trace*

$$\mu_u = (\omega + \mathrm{dd^c} u) \wedge T$$

is a positive Radon measure of total mass A*. If* $\|u\|_\infty \le M$*, then*

$$\|u\|_{H^1} \le C(1 + M), \qquad \mu_u(E) \le C(1 + M)\,\mathrm{Cap}_1(E)^{1/2}$$

for every Borel E*. On each fixed relatively compact coordinate chart, the total variation of every coefficient of* $\mathrm{dd^c} u$ *is also bounded on* E *by* $C(1 + M)\,\mathrm{Cap}_1(E)^{1/2}$*.*

Proof First let u be smooth admissible. The first-cone inequality gives $\mu_u \ge 0$, since T is conformal to ω^{n-1}, and $\mathrm{dd^c} T = 0$ gives $\mu_u(X) = A$. Stokes' theorem and

$\mathrm{dd^c}T = 0$ give $\int_X \mathrm{dd^c}(u^2) \wedge T = 0$. Since $\mathrm{dd^c}(u^2) = 2u\mathrm{dd^c}u + 2\mathrm{d}u \wedge \mathrm{d^c}u$, this yields the energy identity

$$\int_X \mathrm{d}u \wedge \mathrm{d^c}u \wedge T = -\int_X u\,\mathrm{dd^c}u \wedge T = -\int_X u\,\mathrm{d}\mu_u + \int_X u\omega \wedge T \le 2MA.$$

Together with the bound $M^2 \int_X \omega^n$ for the L^2 term, this gives $\|u\|_{H^1} \le C(1+M)$. For bounded admissible u, choose smooth admissible $u_j \downarrow u$ by Theorem 13.5.1, discard a finite initial segment, and take $\|u_j\|_\infty \le M+1$ as in Lemma 13.8.1. Bounded convergence gives strong L^2 convergence. The energy bound gives weak H^1 compactness, and every weak cluster point agrees with u in L^2; thus the entire sequence converges weakly to u in H^1. Lower semicontinuity of the norm proves the Sobolev bound. The trace measures converge weakly to μ_u, preserving positivity and total mass.

For any smooth real ϕ, the first-order Stokes identity is

$$\int_X \phi\,\mathrm{dd^c}u \wedge T = -\int_X \mathrm{d}\phi \wedge \mathrm{d^c}u \wedge T + \int_X \phi\,\mathrm{d^c}u \wedge \mathrm{d}T.$$

It remains valid for $u \in H^1$ by smooth Sobolev approximation. Both terms are bounded by $C\|u\|_{H^1}\|\phi\|_{H^1}$; the background term $\int \phi\omega \wedge T$ is bounded by $C\|\phi\|_{H^1}$. Hence $|\int_X \phi\,\mathrm{d}\mu_u| \le C(1+M)\|\phi\|_{H^1}$, and Lemma 13.9.1 gives the bound for μ_u.

For a smooth admissible approximant, the eigenvalues of $\omega + \mathrm{dd^c}u_j$ relative to ω have modulus at most its trace, by the proof of Lemma 13.1.3, since $m \ge 2$. Consequently every coefficient of this form is dominated by a fixed multiple of its trace. On a fixed coordinate chart this bounds the coefficient variations of $\mathrm{dd^c}u_j$ by $C(\mu_{u_j} + \omega^n)$. Passing to the weak measure limits of Lemma 13.8.1 gives the same domination by $C(\mu_u + \omega^n)$. Finally $\int_E \omega^n \le \mathrm{Cap}_1(E)$ and $\mathrm{Cap}_1(E) \le \|1\|_{H^1}^2$ absorb the smooth term into the claimed square-root estimate. □

Lemma 13.9.3 *Every bounded admissible potential is continuous on the complement of an open set of arbitrarily small* Cap_1. *The same open set can be chosen for any countable family of such potentials.*

Proof Choose uniformly bounded smooth admissible $u_j \downarrow u$ by Theorem 13.5.1. By the preceding lemma these functions converge weakly to u in H^1. Mazur's lemma gives finite convex combinations w_N of terms with $j \ge N$ such that

$$\|w_N - u\|_{H^1} \le 8^{-N}.$$

Because $u \le w_N \le u_N$, they converge pointwise to the canonical function u, not just to an unspecified almost-everywhere representative. Put

$$G_{N_0} = \bigcup_{N \ge N_0} \{|w_{N+1} - w_N| > 2^{-N}\}.$$

Lemma 13.9.1 implies

$$\operatorname{Cap}_1(G_{N_0}) \le \sum_{N \ge N_0} 4^N \|w_{N+1} - w_N\|^2_{H^1} \le C \sum_{N \ge N_0} 16^{-N} \longrightarrow 0.$$

Outside G_{N_0}, the successive differences form a uniformly summable series. Thus w_N converge uniformly there and their limit u is continuous on that closed set. For a countable family, choose the exceptional open sets with capacities less than $\epsilon 2^{-i}$ and take their union. Countable subadditivity proves the last assertion. □

The capacity estimate depends on using a first Hessian. A product of two Hessians can already charge a set of zero Sobolev capacity.

Lemma 13.9.4 *Let E be a compact subset of a smooth real submanifold of X of codimension two. Then* $\operatorname{Cap}_1(E) = 0$.

Proof It suffices to treat a compact subset of a submanifold chart, since finitely many such subsets cover E and the capacity is subadditive. Choose real coordinates (s, t) there, with $t \in \mathbb{R}^2$ and the submanifold given by $t = 0$. Choose $0 \le \eta \le 1$ smooth, supported in this chart and equal to one near the compact subset. Put $r = |t|$. For $0 < \epsilon < 1/2$ define

$$h_\epsilon(r) = \begin{cases} 1, & 0 \le r \le \epsilon^2, \\ \dfrac{\log(\epsilon/r)}{\log(1/\epsilon)}, & \epsilon^2 < r < \epsilon, \\ 0, & r \ge \epsilon. \end{cases}$$

The Sobolev function $f_\epsilon = \eta h_\epsilon$ equals one on an open neighborhood of that subset. On the fixed compact coordinate set the Sobolev norm is equivalent to the Euclidean one. Integration in the two normal variables gives

$$\|f_\epsilon\|^2_{H^1} \le C \left(\epsilon^2 + \int_{\epsilon^2}^{\epsilon} \frac{\mathrm{d}r}{r \log(1/\epsilon)^2} \right) = C \left(\epsilon^2 + \frac{1}{\log(1/\epsilon)} \right) \longrightarrow 0.$$

The derivatives of η contribute at most $C\epsilon^2$. Since f_ϵ is a competitor for an open neighborhood of the subset, the capacity of the subset is at most $\|f_\epsilon\|^2_{H^1}$ for every ϵ, which proves the assertion. □

Example 13.9.1 There are bounded continuous ω-plurisubharmonic potentials u, v and a compact set E such that

$$\operatorname{Cap}_1(E) = 0, \qquad \int_E Q_2(u, v) \wedge \omega^{m-2} > 0.$$

Consequently the first-Hessian capacity estimate of Lemma 13.9.2 does not extend to this positive two-factor measure, even within the continuous ω-plurisubharmonic subclass.

Proof Fix a holomorphic chart with coordinates $z_j = x_j + \mathrm{i} y_j$ and a smooth nonnegative cutoff χ supported in the chart and equal to one on a smaller coordinate box. Extend

$$u = a\chi|x_1|, \qquad v = a\chi|x_2|$$

by zero outside the chart. These functions are globally continuous and bounded. To check their curvature, replace $|x_j|$ by $g_\delta(x_j) = (x_j^2 + \delta^2)^{1/2}$, with $0 < \delta \le 1$. The Hessian of χg_δ has the nonnegative term $\chi \mathrm{dd^c} g_\delta$; all its remaining terms are uniformly bounded forms, since g_δ and g'_δ are uniformly bounded on the fixed support. Thus $\mathrm{dd^c}(\chi g_\delta) \ge -C\omega$ with C independent of δ. Choosing $a > 0$ sufficiently small gives $\omega + \mathrm{dd^c}(a\chi g_\delta) \ge \omega/2$ for both coordinates. Their decreasing limits u, v as $\delta \downarrow 0$ are therefore ω-plurisubharmonic and admissible.

On the box where $\chi = 1$, set $\alpha_j = \frac{\mathrm{i}}{4\pi} \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j$. The book's normalization gives

$$\mathrm{dd^c} g_\delta(x_j) = \tfrac{1}{2} g''_\delta(x_j)\alpha_j, \qquad g''_\delta(x_j) = \frac{\delta^2}{(x_j^2 + \delta^2)^{3/2}} \longrightarrow 2\delta_{\{x_j=0\}}.$$

The measures $g''_\delta(t)\,\mathrm{d}t$ have mass two, while their mass outside any fixed neighborhood of zero tends to zero. Thus $\Theta = \mathrm{dd^c} u = a\delta_{\{x_1=0\}}\alpha_1$ on the box. This current is d- and $\mathrm{d^c}$-closed and depends only on x_1, while $v = a|x_2|$ there. Hence $\mathrm{dd^c}(g_\delta(x_2)\Theta) = \mathrm{dd^c} g_\delta(x_2) \wedge \Theta$, and since $g_\delta \to |\cdot|$ uniformly, (13.5) gives on the box

$$D_2(u, v) = \mathrm{dd^c}(v\Theta) = \lim_{\delta \downarrow 0} a\, \mathrm{dd^c} g_\delta(x_2) \wedge \Theta = a^2 \delta_{\{x_1=0\}}\delta_{\{x_2=0\}}\alpha_1 \wedge \alpha_2;$$

the two factors depend on distinct real variables, so their product converges to the tensor-product measure. Choose E to be a compact piece of $\{x_1 = x_2 = 0\}$ inside this box with nonempty relative interior. The preceding lemma gives $\mathrm{Cap}_1(E) = 0$. On this box, the positive measure $D_2(u, v) \wedge \omega^{n-2}$ has strictly positive mass on E, since ω is strictly positive. The smooth term ω^n and the two single-Hessian terms in the expansion

$$Q_2(u, v) \wedge \omega^{m-2} = \omega^n + \omega^{n-1} \wedge (\mathrm{dd^c} u + \mathrm{dd^c} v) + D_2(u, v) \wedge \omega^{n-2}$$

have zero mass on E. Indeed, on the box $\omega^{n-1} \wedge \mathrm{dd^c} u$ is a smooth multiple of the hypersurface measure of $\{x_1 = 0\}$, which vanishes on E, and similarly for v. Hence the asserted mass is strictly positive. A bound by $C\,\mathrm{Cap}_1(F)^b$ for all Borel F, with any $b > 0$, would contradict this choice of E. □

The preceding Sobolev-capacity estimate therefore cannot be iterated to control higher-degree coefficient measures. A local Hessian capacity gives that control in the next section.

13.10 Local capacities and decreasing products

The codimension-two example Example 13.9.1 shows why Sobolev capacity cannot control a product of Hessians. Local capacities built from smooth Hessian measures give the stronger control needed for the recursive currents. Convergence in these

capacities, and hence quasicontinuity, comes from an energy variation that retains the derivatives of the Hermitian background.

Work on a coordinate ball U with compact closure in a holomorphic chart, with the restriction of ω. A smooth local m-subharmonic function means a smooth function f on U that is m-subharmonic relative to ω in the sense defined before Lemma 13.8.2, that is, $\mathrm{dd^c} f \in \overline{\Gamma_m(\omega)}$ pointwise. For such f put

$$\Lambda_s(f) = (\mathrm{dd^c} f)^s \wedge \omega^{n-s}, \qquad 0 \le s \le m,$$

where $\Lambda_0(f) = \omega^n$. The local mass and gradient estimates needed for the capacity use ordinary smooth products.

Lemma 13.10.1 *Fix $K \Subset U$ and $M < \infty$. For smooth local m-subharmonic $f_1, \dots, f_q$ with $|f_i| \le M$, $0 \le q \le m$, the mass on K of their mixed trace $\mathrm{dd^c} f_1 \wedge \cdots \wedge \mathrm{dd^c} f_q \wedge \omega^{n-q}$ is at most a constant depending only on K, U, ω, m, M. If also f is smooth local m-subharmonic with $|f| \le M$, then, for $q \le m-1$ and a constant C depending on the same data,*

$$\int_K \mathrm{d}f \wedge \mathrm{d^c} f \wedge \mathrm{dd^c} f_1 \wedge \cdots \wedge \mathrm{dd^c} f_q \wedge \omega^{n-q-1} \le C.$$

Proof For one potential f, the form $\omega + \mathrm{dd^c} f$ lies in $\Gamma_m(\omega)$, and $(\omega + \mathrm{dd^c} f)^q \wedge \omega^{n-q}$ expands into nonnegative terms, one of which is $(\mathrm{dd^c} f)^q \wedge \omega^{n-q}$. The Chern–Levine–Nirenberg bound in the proof of Proposition 13.6.1, whose proof uses only the values of the potential on U, therefore bounds the mass on K by $C(1+M)^q$. For mixed factors, positivity of every term in the expansion of $(\sum_i \mathrm{dd^c} f_i)^q \omega^{n-q}$ bounds the mixed trace by a fixed multiple of the single trace for $\sum_i f_i$. This proves the first assertion, including its degree-zero convention.

Replace f by $f + M$, so that $0 \le f \le 2M$ without changing its gradient or Hessian. The identity

$$\mathrm{dd^c}(f^2) = 2f\,\mathrm{dd^c} f + 2\mathrm{d}f \wedge \mathrm{d^c} f$$

also shows that f^2 is smooth local m-subharmonic. Wedge it with the remaining q Hessians and ω^{n-q-1}. The first term on the right then becomes nonnegative, since $f \ge 0$ and the degree is $q+1 \le m$. Hence the gradient term is at most the mixed trace of $f^2, f_1, \dots, f_q$, whose mass on K is bounded by the first assertion with $\max\{M, 4M^2\}$ in place of M. □

Definition 13.10.1 For Borel $E \subset U$ define the smooth capacity

$$c_s(E) = \sup\left\{\int_E \Lambda_s(a) : a \in C^\infty(U),\ -1 \le a \le 0, \mathrm{dd^c} a \in \overline{\Gamma_m(\omega)}\right\},$$

for $0 \le s \le m$. Its outer-open version is

$$c_s^*(E) = \inf\{c_s(G) : E \subset G \subset U,\ G \text{ open}\}.$$

A function on a compact $K \Subset U$ is quasicontinuous for this capacity if, for every $\epsilon > 0$, it is continuous on $K \setminus G$ for some open $G \subset U$ with $c_s(G) < \epsilon$.

In particular $c_0(E) = \int_E \omega^n$. Each c_s is monotone, countably subadditive and continuous under increasing unions. For the last assertion, the supremum over competitors commutes with the increasing supremum over sets. By Lemma 13.10.1, c_s is finite on compact subsets of U. The outer capacity c_s^* is monotone and countably subadditive and agrees with c_s on open sets.

The capacities control mixed traces for smooth local m-subharmonic functions $f_1, \dots, f_q$ with a common bound. Choose a fixed smooth strictly plurisubharmonic b with $\mathrm{dd^c} b \ge \omega$ on U. For $q \le s$, choose constants B and $L \ge 1$ so that $(f_1 + \cdots + f_q + b - B)/L$ lies in $[-1, 0]$. Both constants depend only on the common bound and the fixed data. Positivity of the smooth binomial expansion gives

$$(\mathrm{dd^c} f_1 + \cdots + \mathrm{dd^c} f_q)^q \omega^{n-q} \le C\Lambda_s\big((\textstyle\sum_i f_i + b - B)/L\big). \tag{13.8}$$

Indeed, the expansion of the right side contains the term

$$(\mathrm{dd^c} f_1 + \cdots + \mathrm{dd^c} f_q)^q (\mathrm{dd^c} b)^{s-q} \omega^{n-s}$$

with a positive coefficient, and replacing any of its $s - q$ factors $\mathrm{dd^c} b$ by ω subtracts a nonnegative mixed trace. For $q = 0$ the left side is ω^n, and the same argument applies. Thus every such mixed trace, and every coefficient of a smooth product of $q < m$ Hessians, is dominated on Borel sets by Cc_s. For the coefficients, this follows from the absolute mixed-form estimate Lemma 13.1.5.

Lemma 13.10.2 *Let $0 \le q \le m-1$ and let α be a smooth real $(1,1)$-form in $\overline{\Gamma_m(\omega)}$ pointwise. Put $T = \alpha^q \omega^{n-q-2}$. For smooth real f and nonnegative Borel weights a, supported on a fixed compact set,*

$$\left| \int a \, \mathrm{d}f \wedge \mathrm{d}^{\mathrm{c}} \omega \wedge T \right| \le C \left(\int a \, \mathrm{d}f \wedge \mathrm{d}^{\mathrm{c}} f \wedge \omega \wedge T \right)^{1/2} \left(\int a \, \omega^2 \wedge T \right)^{1/2},$$

where C depends only on ω and the compact set. The corresponding inequality with $\mathrm{d}^{\mathrm{c}} f \wedge \mathrm{d}\omega$ holds as well. For a mixed product of q smooth forms in $\overline{\Gamma_m(\omega)}$ in place of α^q, the left side is bounded by the sum of these bounds over the finitely many nonempty subset sums of its factors.

Proof At one point choose an ω-unitary coframe e_i that diagonalizes α, with eigenvalues $\lambda \in \overline{\Gamma_m}$. Put $\epsilon_i = \mathrm{i} e_i \wedge \bar{e}_i$ and $\Omega = \epsilon_1 \cdots \epsilon_n$, so that $\omega = \sum_i \epsilon_i$ and $\alpha = \sum_i \lambda_i \epsilon_i$. Writing $B_{ij} = S_{q;ij}(\lambda)$ and $D_i = S_{q;i}(\lambda)$, ordinary wedge expansion gives

$$\alpha^q \omega^{n-q-2} = q!(n-q-2)! \sum_{i<j} B_{ij} \prod_{k \notin \{i,j\}} \epsilon_k .$$

Only the remaining two-plane contributes to its wedge with $\mathrm{d}f \wedge \mathrm{d}^{\mathrm{c}} \omega$. If $\partial f = \sum_i f_i e_i$, its absolute density relative to Ω is bounded by

$$C \sum_{i<j} (|f_i| + |f_j|) |B_{ij}|.$$

The constant bounds the tensor coefficients of $\mathrm{d}^{\mathrm{c}} \omega$ in all unitary frames on the fixed compact set; no derivative of this pointwise frame is taken.

For $q \ge 1$, Lemma 13.1.6 applies, since $q \le m-1 \le n-2$, and gives $|B_{ij}| \le C (D_i D_j)^{1/2}$; for $q = 0$, $B_{ij} = D_i = 1$. Finite Cauchy–Schwarz therefore bounds this sum by

$$C \sum_i |f_i| \sqrt{D_i} \sum_j \sqrt{D_j} \le C \left(\sum_i |f_i|^2 D_i \right)^{1/2} \left(\sum_i D_i \right)^{1/2}.$$

Here $\sum_i D_i = (n-q) S_q(\lambda)$. Moreover

$$\alpha^q \omega^{n-q} = q!(n-q)! S_q(\lambda) \Omega,$$
$$\mathrm{d}f \wedge \mathrm{d}^{\mathrm{c}} f \wedge \alpha^q \omega^{n-q-1} = \kappa q!(n-q-1)! \sum_i |f_i|^2 D_i \, \Omega,$$

where $\kappa = 1/(2\pi)$ by the book's d^{c} normalization. These two densities are nonnegative. Thus the pointwise torsion density is bounded by a constant times their geometric mean. Multiplication by a and integral Cauchy–Schwarz give the assertion, including the cases where either density vanishes. The other wedge is conjugate up to a fixed sign, which does not affect its absolute value.

For the mixed assertion with $q \ge 1$, use the exact polarization identity

$$q!\,\alpha_1 \wedge \cdots \wedge \alpha_q = \sum_{J \subset \{1,\dots,q\}} (-1)^{q-|J|} \left(\sum_{i \in J} \alpha_i \right)^q .$$

Every nonempty subset sum remains in the cone. Applying the single power estimate to each term and taking absolute values proves the mixed assertion. At $q = 0$ use the single term $\alpha^0 = 1$. □

Lemma 13.10.3 *Let f_j be uniformly bounded smooth local m-subharmonic functions with $f_j \downarrow f$, and fix a smaller concentric ball $V \Subset U$. There are uniformly bounded smooth local m-subharmonic functions $v_j \downarrow v$ which equal f_j on a neighborhood of $\overline{V}$ and equal one fixed strictly plurisubharmonic ψ on one fixed collar $\{R-\delta_0 < |z| < R\}$ of the boundary of $U = \{|z| < R\}$. Moreover $v = f$ near $\overline{V}$ and $v = \psi$ on that collar.*

Proof Let $|f_j| \le M$. Take $\psi = A(|z|^2 - R^2) + M + 3$, with A so large that $\psi < -M - 2$ on a neighborhood of $\overline{V}$. Since $\psi = M + 3$ on ∂U, it exceeds $M + 2$ on a sufficiently thin collar $\{R - \delta_0 < |z| < R\}$.

Use the regularized maximum M_ϵ constructed in Lemma 13.1.7. Choose $0 < \epsilon_j \downarrow 0$ with $\epsilon_j < 1$, and put $v_j = M_{\epsilon_j}(f_j, \psi)$. That lemma shows that v_j is smooth local m-subharmonic. The monotonicity of M_ϵ in its arguments and in ϵ gives $v_j \downarrow \max(f, \psi) = v$. Since $f_j - \psi > 2 > \epsilon_j$ near $\overline{V}$ and $\psi - f_j > 2 > \epsilon_j$ on the collar, the two asserted regions of exact equality follow. The bounds

$$\max(f_j, \psi) \le v_j \le \max(f_j, \psi) + \epsilon_j/2$$

give a common bound. Since every v_j equals ψ on the collar, all differences $v_k - v_j$ have support in the fixed compact set $\{|z| \le R - \delta_0\}$. □

Proposition 13.10.1 *For the sequence $v_j \downarrow v$ of Lemma 13.10.3 and $0 \le s \le m$,*

$$c_s\{v_k - v > \delta\} \longrightarrow 0 \quad (k \to \infty) \qquad \text{for every } \delta > 0.$$

Consequently the pointwise limit of a uniformly bounded decreasing sequence of smooth local m-subharmonic functions is quasicontinuous on every compact subset of U, for each c_s^* as well as c_s, $0 \le s \le m$.

Proof The assertion for $s = 0$ follows by bounded convergence for volume. Induct on s, assuming it for $s-1$, and put $q = s-1$ and $\theta = \omega^{n-s}$. For $j \ge k$ let $h = v_k - v_j \ge 0$. The support of h lies in one fixed compact set $K_0 \Subset U$. The capacity bound (13.8) and the induction hypothesis give

$$\int h\, dd^c g_1 \cdots dd^c g_q \omega^{n-q} \longrightarrow 0 \quad (k \to \infty), \tag{13.9}$$

uniformly in $j \ge k$ and in the smooth local m-subharmonic $g_1, \dots, g_q$ with a fixed common bound. To see this, split at $h = \delta$. On $\{h \le \delta\}$ the mass bound of Lemma 13.10.1 gives $C\delta$. On the remaining set use $h \le v_k - v$, the capacity bound

with $s-1$ in place of s, and the induction hypothesis. Let k tend to infinity before $\delta \downarrow 0$. For $q = 0$ this is the volume estimate.

Consider the real smooth energy

$$E_s(w) = \int_U w(\mathrm{dd^c} w)^s \theta, \qquad w_t = v_j + th, \quad 0 \le t \le 1.$$

Each $w_t = (1-t)v_j + tv_k$ is smooth local m-subharmonic, with the common bound of the v_j. The energies $E_s(v_j)$ are bounded: on the fixed collar the integrand is fixed, and on the remaining compact set Lemma 13.10.1 gives a uniform mass bound. Since h vanishes near the boundary, integration by parts and closedness of $\mathrm{dd^c} w_t$ give the exact identity

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} E_s(w_t) = (s+1) \int h(\mathrm{dd^c} w_t)^s \theta \\ + s \int h(\mathrm{dd^c} w_t)^q \left(\mathrm{d}w_t \wedge \mathrm{d^c}\theta - \mathrm{d^c} w_t \wedge \mathrm{d}\theta + w_t \mathrm{dd^c}\theta \right). \end{aligned} \tag{13.10}$$

Here

$$\begin{aligned} \mathrm{d}\theta &= (n-s)\mathrm{d}\omega \wedge \omega^{n-s-1}, \\ \mathrm{dd^c}\theta &= (n-s)\mathrm{dd^c}\omega \wedge \omega^{n-s-1} \\ &\quad + (n-s)(n-s-1)\mathrm{d}\omega \wedge \mathrm{d^c}\omega \wedge \omega^{n-s-2}, \end{aligned}$$

and the analogous $\mathrm{d^c}$ formula holds. A zero scalar coefficient means that its term is absent. As $k \to \infty$, the two first-order terms in (13.10) tend to zero uniformly in t and $j \ge k$, by Lemma 13.10.2, the gradient bound of Lemma 13.10.1, and (13.9). The last term has the same property by the absolute-form estimate at degree $q < m$ and (13.9). More explicitly, the first torsion integral is bounded by

$$C \left(\int h \, \mathrm{d}w_t \wedge \mathrm{d^c} w_t \wedge (\mathrm{dd^c} w_t)^q \omega^{n-q-1} \right)^{1/2} \left(\int h (\mathrm{dd^c} w_t)^q \omega^{n-q} \right)^{1/2}.$$

The first factor is bounded since h has a common bound and Lemma 13.10.1 applies; the second tends to zero by (13.9). Conjugation handles the other first-order term. Integrating (13.10) over $0 \le t \le 1$ therefore gives, for $j \ge k$,

$$E_s(v_k) - E_s(v_j) = (s+1)\Delta_{kj} + r_{kj}, \quad \Delta_{kj} = \int_0^1 \int h \Lambda_s(w_t) \, \mathrm{d}t \ge 0, \quad |r_{kj}| \le e_k,$$

where $e_k \to 0$ does not depend on j. It follows that $E_s(v_j) \le E_s(v_k) + e_k$ for $j \ge k$, so $\limsup_j E_s(v_j) \le E_s(v_k) + e_k$ for every k. Letting k run through a subsequence along which $E_s(v_k)$ tends to its lower limit shows that the bounded sequence $E_s(v_j)$ converges. Hence $\Delta_{kj} \to 0$ as $k \to \infty$, uniformly for $j \ge k$. Positivity of the binomial expansion $\Lambda_s(w_t) \ge (1-t)^s \Lambda_s(v_j)$ then gives

$$\int h \Lambda_s(v_j) \longrightarrow 0 \quad (k \to \infty), \text{ uniformly for } j \ge k. \tag{13.11}$$

It remains to replace each Hessian factor by a capacity competitor. For any smooth local m-subharmonic a with $-1 \le a \le 0$, put

$$I_\ell = \int h(\mathrm{dd^c} v_j)^\ell (\mathrm{dd^c} a)^{s-\ell}\theta, \quad 0 \le \ell \le s,$$
$$\eta_\ell = (\mathrm{dd^c} v_j)^\ell (\mathrm{dd^c} a)^{s-\ell-1}, \qquad 0 \le \ell < s.$$

Stokes' theorem for the compactly supported form $h\,\mathrm{d^c} a \wedge \eta_\ell\theta$, with every metric derivative kept, gives

$$I_\ell = -\int \mathrm{d}h \wedge \mathrm{d^c} a \wedge \eta_\ell\theta + \int h\,\mathrm{d^c} a \wedge \eta_\ell \mathrm{d}\theta.$$

The second term is $o(1)$ as $k \to \infty$, uniformly in a and $j \ge k$. Indeed, apply the mixed form of Lemma 13.10.2 with $f = a$ and weight h; each subset sum has a uniform gradient bound by Lemma 13.10.1, and its lower trace is controlled by (13.9).

The form $\eta_\ell\theta$ is positive of bidegree $(n-1, n-1)$: a complementary nonnegative rank-one form completes a polarized product of $s \le m$ cone forms. Its associated Hermitian gradient pairing is nonnegative, so Cauchy–Schwarz and the uniform gradient bound for a give

$$I_\ell \le C F_\ell(h)^{1/2} + o(1), \qquad F_\ell(h) = \int \mathrm{d}h \wedge \mathrm{d^c} h \wedge \eta_\ell\theta.$$

Stokes' theorem for $h\,\mathrm{d^c} h \wedge \eta_\ell\theta$ gives

$$F_\ell(h) = I_{\ell+1} - \int h\mathrm{dd^c} v_k \wedge \eta_\ell\theta + \int h\,\mathrm{d^c} h \wedge \eta_\ell \mathrm{d}\theta \le I_{\ell+1} + o(1).$$

The dropped term is nonnegative. For the last term the same polarized torsion estimate applies, with $f = h$ and weight h. Its first gradient factor is uniformly bounded because $0 \le h \le C$ and $h = v_k - v_j$: for every positive single-power gradient pairing, the gradient energy of the difference is at most twice the sum of those of v_k and v_j. Its second factor tends to zero by (13.9). Starting with (13.11) at I_s, finite backward induction gives $I_0 \to 0$ uniformly in a and $j \ge k$.

Since $I_0 = \int h\Lambda_s(a)$, Chebyshev's inequality now yields $c_s\{v_k - v_j > \delta\} \le \delta^{-1} \sup_a I_0$. These sets increase to $\{v_k - v > \delta\}$ as $j \to \infty$. Continuity of c_s under increasing unions proves the induction.

For quasicontinuity, fix $0 \le s \le m$ and $\epsilon > 0$, and choose k_N so that the sets $G_N = \{v_{k_N} - v > 2^{-N}\}$ satisfy $c_s(G_N) < \epsilon 2^{-N}$, $N \ge 1$. They are open because v is upper semicontinuous and v_{k_N} is smooth. On $U \setminus \bigcup_N G_N$ these smooth functions converge uniformly to the pointwise limit v. Hence v is continuous there. The union has capacity less than ϵ. Every compact subset of U lies in a smaller concentric ball V, and $v = f$ near $\overline{V}$ in Lemma 13.10.3; this transfers the conclusion to f. Open sets have the same outer capacity, which also proves the assertion for c_s^*. □

Lemma 13.10.4 *Every bounded admissible potential on X is locally quasicontinuous for each smooth capacity c_s^*, $0 \le s \le m$. For $0 \le p < m$, each coefficient measure ν of the ordered D_p satisfies*

$$|\nu|(E) \le C c_{m-1}^*(E), \qquad E \subset K \Subset U,$$

where C depends only on K, U, ω and a common bound for the potentials. For $p = 0$ the assertion concerns the volume measure, the coefficient of $D_0 = 1$.

Proof Let u be bounded admissible. The global decreasing approximation of Theorem 13.5.1 and Lemma 13.8.2 provide uniformly bounded smooth local m-subharmonic functions decreasing to $u+\rho$ on U. By Proposition 13.10.1, this pointwise limit, formed with the canonical values of u, is quasicontinuous. Subtracting the smooth ρ preserves quasicontinuity.

For smooth admissible potentials u_i write $\mathrm{dd^c} u_i = \mathrm{dd^c}(u_i + \rho) - \mathrm{dd^c}\rho$ and expand the product. Each term has $q \le p < m$ local cone factors and fixed smooth complementary factors. The absolute-form estimate and (13.8), with $s = m-1$, bound each coefficient variation by $C c_{m-1}$ on Borel sets. The constants depend only on the common bound and local data.

Take the iterated limits of Proposition 13.8.1 on a slightly larger compact neighborhood of K. Lower semicontinuity of signed measure variation on open sets preserves these estimates at each limit. For any open $G \supset E$, intersect with a fixed open neighborhood of K having compact closure in U; this gives $|\nu|(E) \le C c_{m-1}(G)$. Taking the infimum proves the outer estimate. □

Lemma 13.10.5 *Fix $0 \le s \le m$ and the outer smooth capacity c_s^* on U. Let signed Radon measures $\nu_j \to \nu$ weakly have uniformly bounded local variations, with all variations dominated on compact subsets by $C c_s^*$. Suppose, for $1 \le i \le \ell$, that bounded Borel functions $b_i^j \downarrow b_i$ have one common bound, and that every b_i^j and every limit b_i is locally quasicontinuous for this capacity. For any continuous real F on a compact box containing their ranges,*

$$F(b_1^j, \dots, b_\ell^j)\nu_j \longrightarrow F(b_1, \dots, b_\ell)\nu$$

weakly as signed Radon measures.

Proof Fix $\phi \in C_c(U)$, a compact $K \Subset U$ containing its support, and $\epsilon > 0$. For each of the countably many functions b_i^j and b_i, choose an exceptional open set for quasicontinuity on K, with capacities summing to less than ϵ. Their union G is open, $c_s(G) < \epsilon$, and every function is continuous on the compact set $A = K \setminus G$. Dini's theorem gives $b_i^j \to b_i$ uniformly on A for each i. Uniform continuity of F on the compact box therefore gives $f_j \to f$ uniformly on A, where $f_j = F(b_1^j, \dots, b_\ell^j)$ and $f = F(b_1, \dots, b_\ell)$. Put $M = \|F\|_\infty$. Extend $f|_A$ to a continuous g on U with $|g| \le M$; if A is empty, take $g = 0$. Then

$$\left|\int \phi f_j \,\mathrm{d}\nu_j - \int \phi f \,\mathrm{d}\nu\right| \le \|\phi\|_\infty \|f_j - f\|_A |\nu_j|(K) + \left|\int \phi g \,\mathrm{d}(\nu_j - \nu)\right| + 2M\|\phi\|_\infty \big(|\nu_j|(K \cap G) + |\nu|(K \cap G)\big).$$

For fixed G, the first two terms tend to zero as $j \to \infty$: the first by uniform convergence on A, the second by weak convergence applied to $\phi g \in C_c(U)$. Since $|\nu_j|(K \cap G) \le C c_s(G)$, and similarly for ν, the last term is at most $4MC\|\phi\|_\infty \epsilon$. Let ϵ tend to zero. □

Theorem 13.10.1 *For uniformly bounded admissible potentials $u_i^j \downarrow u_i$ on X, the currents $D_p(u_1^j, \dots, u_p^j)$ converge to $D_p(u_1, \dots, u_p)$ in distributions for $p \le m$, and weakly with measure coefficients for $p < m$. These currents are symmetric in their potentials. For $p < m$ the coefficient convergence also holds after multiplication by a uniformly bounded decreasing admissible weight. The positive shifted currents Q_p converge weakly for $p \le m$; Q_m converges as a positive Radon measure.*

Proof Work on a compact coordinate test set, and induct on p; the case $p = 0$ is trivial since $D_0 = 1$. Suppose the result holds for D_{p-1}, which has measure coefficients since $p - 1 < m$. Lemmas 13.10.4 and 13.10.5, applied coefficientwise with $s = m - 1$ and weight u_p^j, give weak convergence of $u_p^j D_{p-1}(u_1^j, \dots, u_{p-1}^j)$ to $u_p D_{p-1}(u_1, \dots, u_{p-1})$. Applying dd^c proves distributional convergence at degree p. For $p < m$, the uniform coefficient bounds in Proposition 13.8.1 upgrade this to weak convergence of measures, by smooth approximation of compactly supported continuous tests. Lemma 13.10.5, now with a decreasing weight $a_j \downarrow a$, gives the additional weighted conclusion.

For each fixed list choose simultaneous smooth decreasing global approximations. Their smooth raw Hessian wedges are symmetric. The convergence just proved identifies their limit with D_p in any ordering. This proves symmetry without exchanging the iterated limits of (13.6).

The subset expansion for R_p uses fixed smooth powers of ω, so its distributional convergence and symmetry follow from those of D_I. Its wedge Q_p is positive with uniform local masses by Theorem 13.8.1. Distributional convergence and these mass bounds imply weak convergence as an order-zero positive current, including the Radon measure conclusion at $p = m$. A partition of unity on a finite coordinate cover proves the global assertions. □

The coefficient bound for the raw currents D_p, $p < m$, uses only c_{m-1}. For the positive shifted products, c_m controls the traces in every degree $p \le m$, including the top degree, where D_m itself has no coefficient bound. This allows a bounded decreasing weight in every Hessian degree.

Lemma 13.10.6 *On $K \Subset U$, the positive shifted currents $Q_p(u_1, \dots, u_p)$, $1 \le p \le m$, for globally bounded admissible potentials have trace and coefficient variations bounded on Borel $E \subset K$ by $Cc_m^*(E)$. The constants are uniform when all potentials have a common bound.*

Proof First take smooth potentials, put $\sigma_i = \omega + \mathrm{dd}^c u_i$, and let ρ be the fixed local correction of Lemma 13.8.2, with $\tau = \mathrm{dd}^c\rho \ge 2\omega$. Then $\beta_i = \mathrm{dd}^c(u_i+\rho) = \sigma_i+(\tau-\omega)$ lies in $\overline{\Gamma_m(\omega)}$. Since $\tau - \omega \ge 0$, mixed positivity gives

$$0 \le \int_E Q_p \wedge \omega^{m-p} \le \int_E \beta_1 \cdots \beta_p \omega^{n-p} \le C c_m(E), \qquad E \subset K.$$

Here $Q_p \wedge \omega^{m-p} = \sigma_1 \cdots \sigma_p \omega^{n-p}$. The last bound is the domination of mixed traces by c_m that follows (13.8), applied with $s = m$ to the corrected potentials $u_i + \rho$, which are bounded on U. Positivity of Q_p bounds each of its coefficient variations on the compact coordinate set by a fixed multiple of its trace measure.

For bounded potentials choose uniformly bounded smooth decreasing approximations by Theorem 13.5.1. The joint convergence in Theorem 13.10.1 identifies their positive-current limit with the previously constructed Q_p. On a compact neighborhood of K inside U, lower semicontinuity on open sets preserves trace and coefficient variation estimates. For an arbitrary open $G \supset E$ intersect with a fixed relatively compact neighborhood of K before taking this limit; its capacity is at most $c_m(G)$. Therefore every such trace or variation measure μ satisfies $\mu(E) \le C c_m(G)$. Infimize over G to obtain the asserted outer-open bound. The constants of the smooth estimates are independent of the approximation index. □

Theorem 13.10.2 *Suppose $a_j \downarrow a$ and $u_i^j \downarrow u_i$ are bounded admissible potentials on X with common bounds. Then, for $1 \le p \le m$,*

$$a_j Q_p(u_1^j, \dots, u_p^j) \longrightarrow a Q_p(u_1, \dots, u_p)$$

weakly as currents with measure coefficients. At $p = m$ this is weak convergence of weighted Radon measures.

Proof Work on a compact coordinate test set. By Theorem 13.10.1, every coefficient measure of the positive shifted currents converges weakly and has a uniform local variation bound. Lemma 13.10.6 strengthens this to domination by Cc_m^* on Borel sets. Lemma 13.10.4 makes the weights a_j and a quasicontinuous for this capacity. Apply Lemma 13.10.5 with $s = m$ to each coefficient, and then use a finite partition of unity. At $p = m$, Q_m is a positive Radon measure, so this also gives the stated weighted measure convergence. □

13.11 Locality and maxima of bounded potentials

Where one bounded admissible potential is strictly larger than another, the maximum has the same positive Hessian current as the larger potential. These strict comparison sets need not be open. Quasicontinuity gives cutoffs whose products with the approximating currents converge, allowing smooth locality to pass to these sets.

Proposition 13.11.1 *Let u, v be bounded admissible, put $w = \max\{u, v\}$, and fix $1 \le p \le m$. For any bounded admissible $t_2, \dots, t_p$,*

$$\mathbf{1}_{\{u>v\}} Q_p(w, t_2, \dots, t_p) = \mathbf{1}_{\{u>v\}} Q_p(u, t_2, \dots, t_p).$$

The equality is coefficientwise as currents with measure coefficients. The corresponding equality on $\{v > u\}$ follows by exchanging u, v.

Proof Choose smooth uniformly bounded decreasing approximations of all the potentials by Theorem 13.5.1. Construct w_j by Lemma 13.1.7. Work on one compact coordinate test set. Fix $\delta > 0$ and a smooth $\chi_\delta : \mathbb{R} \to [0, 1]$ that vanishes for $x \le \delta/2$ and equals one for $x \ge \delta$. For j large enough that $\epsilon_j < \delta/2$, ordinary smooth locality gives

$$\chi_\delta(u_j - v_j) Q_p(w_j, t_2^j, \dots, t_p^j) = \chi_\delta(u_j - v_j) Q_p(u_j, t_2^j, \dots, t_p^j).$$

Indeed, wherever the weight is nonzero, $w_j = u_j$ on an open neighborhood. The equality elsewhere follows from its zero weight.

Both sequences of positive currents converge by Theorem 13.10.1; every coefficient variation is uniformly dominated by Cc_m^* by Lemma 13.10.6. The potentials and their limits are quasicontinuous by Lemma 13.10.4. Apply Lemma 13.10.5 with $s = m$ and $F(x, y) = \chi_\delta(x - y)$ to the real and imaginary parts of each coefficient. Passing to the limit gives the same identity with $u, v, w, t_2, \dots, t_p$. As $\delta \downarrow 0$, $\chi_\delta(x) \to \mathbf{1}_{\{x>0\}}$ for every real x. Dominated convergence against the fixed finite coefficient variations on the test set gives the asserted restriction. A partition of unity proves it globally.□

On the contact set, a positive constant shift gives a strict comparison before the limit is taken. Decreasing the shift then recovers that set by dominated convergence for a fixed measure.

Theorem 13.11.1 *Let u, v be bounded admissible. Fix $1 \le r \le m$ and bounded admissible $z_1, \dots, z_{m-r}$. Define the positive Radon measure*

$$\mu_r(t) = Q_m(\underbrace{t, \dots, t}_{r}, z_1, \dots, z_{m-r}).$$

For $w = \max\{u, v\}$,

$$\mu_r(w) \ge \mathbf{1}_{\{u \ge v\}} \mu_r(u) + \mathbf{1}_{\{u < v\}} \mu_r(v).$$

The inequality with u, v exchanged also holds.

Proof Symmetry and Proposition 13.11.1, applied to the r slots one at a time, give $\mu_r(w) = \mu_r(v)$ on $\{u < v\}$ and $\mu_r(w) = \mu_r(u)$ on $\{u > v\}$. To include the contact set, put $w_\epsilon = \max\{u + \epsilon, v\}$. Strict locality and invariance under addition of constants give

$$\mu_r(w_\epsilon) \ge \mathbf{1}_{\{u+\epsilon > v\}} \mu_r(u).$$

The invariance follows from the smooth products and their joint decreasing limits. As $\epsilon \downarrow 0$, $w_\epsilon \downarrow w$, so the left side converges weakly to $\mu_r(w)$ by Theorem 13.10.1. Against any nonnegative continuous compactly supported test function, the right side converges by dominated convergence for the fixed measure $\mu_r(u)$ to $\mathbf{1}_{\{u \ge v\}} \mu_r(u)$. This proves the measure inequality on the contact set as well as on $\{u > v\}$. On its disjoint complement $\{u < v\}$, the equality proved first gives $\mu_r(w) = \mu_r(v)$. Restricting the positive measures to these two Borel sets and adding gives the assertion. Exchanging u, v gives its other form. □

For bounded admissible v, write $H_m(v) = Q_m(v, \dots, v)$ for the positive Radon measure of Theorem 13.8.1 with all potentials equal to v. For continuous v this is the measure $H_m(v)$ of Section 13.6. Convex rescaling gives a positive background term in this measure.

Lemma 13.11.1 *For bounded admissible v, $0 \le \alpha \le 1$ and a real constant b, the function $v_\alpha = \alpha v + b$ is bounded admissible and*

$$H_m(v_\alpha) \ge \alpha^m H_m(v) + (1 - \alpha)^m \omega^n.$$

Proof Take smooth decreasing uniformly bounded admissible $v_j \downarrow v$ by Theorem 13.5.1. Cone convexity makes $\alpha v_j + b$ admissible because

$$\omega + \mathrm{dd}^{\mathrm{c}}(\alpha v_j + b) = \alpha(\omega + \mathrm{dd}^{\mathrm{c}} v_j) + (1 - \alpha)\omega.$$

For $\alpha > 0$, an upper test q for $\alpha v + b$ gives the upper test $(q - b)/\alpha$ for v. The same convex combination of forms therefore proves admissibility of the limit. For $\alpha = 0$ the potential is constant. The pointwise decreasing limit has the displayed canonical affine values. Expand the smooth top product. Every mixed term is positive by polarized Gårding positivity. Keeping the two extreme terms gives

$$H_m(\alpha v_j + b) \geq \alpha^m H_m(v_j) + (1-\alpha)^m \omega^n.$$

Joint decreasing convergence from Theorem 13.10.1 passes this inequality against each nonnegative continuous test to the asserted inequality of positive Radon measures. □

The weighted limits also pass integration by parts against metric derivatives.

Lemma 13.11.2 *Put $k = m-1$ and $\ell = n-m$. For bounded admissible $t_1, \dots, t_k$, let $T = R_k(t_1, \dots, t_k) \wedge \omega^\ell$. Then $\mathrm{dd^c} T$ is a signed Radon measure. If bounded admissible $t_i^j \downarrow t_i$ have a common bound, these measures converge weakly, also after multiplication by any continuous function of finitely many uniformly bounded decreasing admissible potentials.*

Proof The subset definition and the torsion formulas in Proposition 13.8.2 give, in distributions,

$$\begin{aligned}
\mathrm{dd^c} T = \mathrm{dd^c}\omega \wedge \left(\sum_{a=1}^{k} R_{k-1}^{\widehat{a}} \omega^\ell + \ell R_k \omega^{\ell-1} \right) \\
+ 2\,\mathrm{d}\omega \wedge \mathrm{d^c}\omega \wedge \sum_{a<b} R_{k-2}^{\widehat{a}\widehat{b}} \omega^\ell \\
+ 2\ell\,\mathrm{d}\omega \wedge \mathrm{d^c}\omega \wedge \sum_{a=1}^{k} R_{k-1}^{\widehat{a}} \omega^{\ell-1} \\
+ \ell(\ell-1)\,\mathrm{d}\omega \wedge \mathrm{d^c}\omega \wedge R_k \omega^{\ell-2}.
\end{aligned}$$

The sum with two hats is absent when $k = 1$, and the last term is absent when $\ell = 1$. Every factor R_r has degree $r \leq k < m$, so its coefficients are signed measures. Expanding each R_r over subsets expresses the displayed measure as fixed smooth forms wedged with raw currents D_p of degrees $p \leq m-1$. Lemma 13.10.4 therefore bounds its variation locally by $C c_{m-1}^*$, uniformly under common bounds on the potentials. Joint convergence follows from Theorem 13.10.1 applied to this expansion. All the canonical weights are quasicontinuous for c_{m-1}^*, by Lemma 13.10.4. Lemma 13.10.5, with $s = m-1$, thus proves the weighted assertion coefficientwise on a finite coordinate cover. A partition of unity gives it on X. □

13.12 Hessian mass and global comparison

The maximum inequality of Theorem 13.11.1 compares measures on complementary Borel sets. Integrating it leaves a difference of total masses. The recursive currents express this difference through the complex Hessians of the background powers, extending the mass calculation of Chapter 2 to bounded admissible potentials. Their vanishing characterizes the unmodified comparison principle. When they do not vanish, a small sublevel set limits the change made by the maximum; the resulting torsion error can still be absorbed after rescaling the upper potential.

For bounded admissible $t_1, \dots, t_m$, put

$$\mathcal{M}_m(t_1,\dots,t_m)=\int_X Q_m(t_1,\dots,t_m),$$

the total mass of a mixed positive measure by Theorem 13.8.1. Recall from Section 13.11 that $H_m(t)=Q_m(t,\dots,t)$ is the finite positive Radon measure with all potentials equal to t. For a nonempty subset $I=\{i_1<\cdots<i_q\}\subset\{1,\dots,m\}$, put $i(I)=i_q$ and $I^-=I\setminus\{i_q\}$. The currents D_I and D_{I^-} are the recursive currents of the list $t_1,\dots,t_m$ introduced before (13.7); in particular $D_\varnothing=1$.

Lemma 13.12.1 *For every bounded admissible list,*

$$\mathcal{M}_m(t_1,\dots,t_m)=\int_X\omega^n+\sum_{\varnothing\neq I\subset\{1,\dots,m\}}\int_X t_{i(I)}D_{I^-}\wedge \mathrm{dd^c}(\omega^{n-|I|}). \tag{13.12}$$

Proof Expand Q_m using (13.7). The term for I pairs D_I with the smooth form $\omega^{n-|I|}$. By (13.5),

$$\langle D_I,\omega^{n-|I|}\rangle=\langle t_{i(I)}D_{I^-},\mathrm{dd^c}(\omega^{n-|I|})\rangle.$$

Since $|I^-|\le m-1$, Proposition 13.8.1 gives measure coefficients for D_{I^-}; their bounded product with $t_{i(I)}$ is well defined. Thus the displayed pairing is exactly the integral in the sum, and is finite on compact X. The empty subset contributes $\int_X\omega^n$.□

Let μ_r be the repeated-slot mixed measure from Theorem 13.11.1, for fixed r and $z_1,\dots,z_{m-r}$. The total masses of $\mu_r(w)$ and $\mu_r(u)$ in the next proposition are values of $\mathcal{M}_m$, so (13.12) for the two corresponding lists expresses their difference explicitly.

Proposition 13.12.1 *If u,v are bounded admissible and $w=\max\{u,v\}$, then*

$$\int_{\{u<v\}}\mathrm{d}\mu_r(v)\le\int_{\{u<v\}}\mathrm{d}\mu_r(u)+\int_X\mathrm{d}\mu_r(w)-\int_X\mathrm{d}\mu_r(u).$$

Proof Set $E=\{u<v\}$. Theorem 13.11.1, whose first term includes the contact set $\{u=v\}$, gives

$$\mu_r(w)\ge\mathbf{1}_{X\setminus E}\mu_r(u)+\mathbf{1}_E\mu_r(v).$$

All these measures are finite and positive. Integrating on X and using $\mu_r(u)(X\setminus E)=\mu_r(u)(X)-\mu_r(u)(E)$ gives exactly the assertion. □

The unmodified global comparison principle holds precisely when the powers occurring in this mass formula are $\mathrm{dd^c}$-closed. Only the powers $n-m,\dots,n-1$ occur at Hessian degree m.

Theorem 13.12.1 *The following conditions are equivalent:*

(1) $\mathrm{dd^c}(\omega^{n-q}) = 0$ *for every* $1 \le q \le m$;
(2) $\int_X H_m(u)$ *is independent of the smooth admissible potential* u;
(3) $\mathcal{M}_m(t_1, \dots, t_m) = \int_X \omega^n$ *for every bounded admissible list;*
(4) *every pair of bounded admissible* u, v *satisfies*

$$\int_{\{u<v\}} H_m(v) \le \int_{\{u<v\}} H_m(u).$$

Under these conditions, for every $1 \le r \le m$, *every fixed bounded admissible list* $z_1, \dots, z_{m-r}$ *and all bounded admissible* u, v, *one also has*

$$\int_{\{u<v\}} \mathrm{d}\mu_r(v) \le \int_{\{u<v\}} \mathrm{d}\mu_r(u).$$

Proof The first condition implies the third by (13.12). The third implies the second by taking all inputs equal to a smooth potential.

To prove that the second implies the first, fix any smooth real f. For all sufficiently small real t of either sign, $\omega + t\mathrm{dd^c} f$ is positive, so tf is strictly admissible. The polynomial

$$\int_X (\omega + t\mathrm{dd^c} f)^m \wedge \omega^{n-m} = \sum_{q=0}^{m} \binom{m}{q} t^q \int_X (\mathrm{dd^c} f)^q \wedge \omega^{n-q}$$

is constant on an interval. Consequently every coefficient with $q \ge 1$ vanishes. For $1 \le q \le m$, Step 1 of the proof of Theorem 2.2.2, with $j = n - q$, derives $\mathrm{dd^c}(\omega^{n-q}) = 0$ from the vanishing of $\int_X (\mathrm{dd^c} f)^q \wedge \omega^{n-q}$ for every smooth real f.

Under the third condition, $\mu_r(w)$ and $\mu_r(u)$ both have total mass $\int_X \omega^n$. Proposition 13.12.1 proves the mixed comparison assertion, and $r = m$ gives the fourth condition. Finally, suppose the fourth condition holds and fix smooth admissible u, v. Choose C so large that $u < v + C$ everywhere. Constant invariance gives $H_m(v + C) = H_m(v)$, so comparison gives $\int_X H_m(v) \le \int_X H_m(u)$. Exchanging u, v gives equality, which is the second condition. □

The varying background of Example 2.3.1, which obstructs Monge–Ampère comparison, also obstructs comparison at each intermediate Hessian degree.

Example 13.12.1 Let $X = (\mathbb{C}/(\mathbb{Z}+\mathrm{i}\mathbb{Z}))^n$, with $z_j = x_j+\mathrm{i}y_j$, β_j and $b = 2+\cos(2\pi x_1)$ as in Example 2.3.1, and put

$$\omega = \beta_1 + b\beta_2 + \sum_{j=3}^{n} \beta_j .$$

For $|t| < \pi^{-2}$ the potential $u_t = t\cos(2\pi x_1)$ is smooth strictly admissible, and, with $B_n = \int_X \beta_1 \wedge \cdots \wedge \beta_n$,

$$\int_X H_m(u_t) = n! \left(2 - \frac{m\pi^2 t}{2n}\right) B_n .$$

Thus every $-\pi^{-2} < t < 0$, together with a constant $C > |t|$, gives a pair $u = 0$, $v = u_t + C$ that violates the unmodified global comparison inequality.

Proof As in Example 2.3.1, $\mathrm{dd^c} u_t = -\pi^2 t\cos(2\pi x_1)\beta_1$. The coefficient of β_1 in $\omega + \mathrm{dd^c} u_t$ is strictly positive in the stated range, and all the other coefficients are positive. Since $(\mathrm{dd^c} u_t)^2 = 0$, expansion gives

$$H_m(u_t) = \omega^n + m\mathrm{dd^c} u_t \wedge \omega^{n-1} = n! b \left(1 - \frac{m}{n}\pi^2 t\cos(2\pi x_1)\right) \beta_1 \wedge \cdots \wedge \beta_n .$$

The period averages of b and $b\cos(2\pi x_1)$ are respectively 2 and $1/2$, proving the mass formula. For $t < 0$ this mass is strictly larger than $\int_X H_m(0) = 2n! B_n$. Meanwhile $C > |t|$ gives $\{0 < u_t + C\} = X$, so the comparison inequality for this pair would assert the opposite ordering of these masses. □

13.12.1 Modified comparison

On a sublevel set just above the infimum of a difference, the maximum changes the lower potential by at most a prescribed small number. Integration by parts then expresses the mass defect through this small difference and a torsion measure. Mixed lower-degree traces control its variation. Rescaling the upper potential adds a positive background term that relates these traces to the top Hessian measure.

For bounded admissible u, V, use the notation

$$\sigma_u = \omega + \mathrm{dd^c} u, \qquad M_{p,q} = \sigma_u^p \sigma_V^q \omega^{n-p-q} \quad (0 \le p+q \le m).$$

These expressions denote the positive mixed measure Q_m with p slots equal to u, q slots equal to V and the remaining slots equal to the constant zero. In particular $M_{0,0} = \omega^n$.

For $p+q \le m-1$, put

$$T_{p,q} = \sigma_u^p \sigma_V^q \omega^{n-p-q-1}.$$

This denotes R_{m-1} with the same nonconstant slots and zero in its other slots, wedged with ω^{n-m}.

Lemma 13.12.2 *Let u, V be bounded admissible, b a real constant and $s > 0$. Suppose $0 \le \max\{u, V+b\} - u \le s$. Set $E = \{u < V+b\}$ and $b_{p,q} = M_{p,q}(E)$. There is a constant C, depending only on X, ω, m, n, such that*

$$b_{p,q+1} \le b_{p+1,q} + Cs \sum_{a+d \le p+q} b_{a,d} \qquad (p+q \le m-1).$$

For $q = 0$, the error can instead be bounded by $Cs(b_{p,0} + b_{p-1,0} + b_{p-2,0})$, with negative indices zero.

Proof First estimate the torsion of $T_{p,q}$, writing $r = p+q$, for smooth admissible potentials. Since $\mathrm{d}\sigma_u = \mathrm{d}\sigma_V = \mathrm{d}\omega$, $\mathrm{d}^{\mathrm{c}}\sigma_u = \mathrm{d}^{\mathrm{c}}\sigma_V = \mathrm{d}^{\mathrm{c}}\omega$ and $\mathrm{dd}^{\mathrm{c}}\sigma_u = \mathrm{dd}^{\mathrm{c}}\sigma_V = \mathrm{dd}^{\mathrm{c}}\omega$, the product rule expresses $\mathrm{dd}^{\mathrm{c}}T_{p,q}$ as a sum of fixed smooth metric forms wedged with products of σ_u, σ_V of degrees $r, r-1, r-2$. Every term contains either $\mathrm{dd}^{\mathrm{c}}\omega$ or $\mathrm{d}\omega \wedge \mathrm{d}^{\mathrm{c}}\omega$, with its product-rule multiplicity and the appropriate remaining background power; the terms of degree r come from differentiating the background power. Terms with negative numbers of factors are absent.

All variable degrees are at most $r < m$. Apply Lemma 13.1.5 to each term. Expanding the resulting positive power of the sum of its variable factors gives the pointwise bound

$$|\mathrm{dd}^{\mathrm{c}}T_{p,q}| \le C \sum_{a+d \le r} M_{a,d}.$$

For $q = 0$ every variable factor is σ_u, so the sharper bound is $C(M_{p,0} + M_{p-1,0} + M_{p-2,0})$. The constant bounds fixed metric coefficients on a finite coordinate cover and the finitely many multiplicities; it is independent of the potentials and their bounds.

Choose uniformly bounded smooth decreasing approximations to u, V. By Lemma 13.11.2, the torsion measures converge weakly; by Theorem 13.10.1, all positive measures in these bounds also converge weakly. Against nonnegative continuous tests, lower semicontinuity of total variation preserves both bounds. They are consequently inequalities of measures and may be restricted to the Borel set E.

Put $w = \max\{u, V+b\}$ and $h = w - u$. Apply the mixed maximum theorem Theorem 13.11.1 with $r = 1$, with $V+b$ in place of v, and with the $m-1$ slots defining $T_{p,q}$ as the fixed list. Since adding the constant b does not change the measures, it gives

$$(\omega + \mathrm{dd}^{\mathrm{c}}w)T_{p,q} \ge \mathbf{1}_E \sigma_V T_{p,q} + \mathbf{1}_{X\setminus E}\sigma_u T_{p,q}.$$

All products here are the corresponding positive mixed measures. After integration, this implies

$$b_{p,q+1} \le b_{p+1,q} + \int_X \big((\omega + \mathrm{dd^c} w) - \sigma_u\big) T_{p,q}.$$

Regularized maxima of the smooth decreasing approximations, from Lemma 13.1.7, give a uniformly bounded decreasing smooth list $w_j \downarrow w$. Stokes' theorem on compact X gives for these lists

$$\int_X \big((\omega + \mathrm{dd^c} w_j) - \sigma_{u_j}\big) T^j_{p,q} = \int_X (w_j - u_j) \mathrm{dd^c} T^j_{p,q}.$$

Joint decreasing convergence passes the positive mixed measures on the left to their limits. The continuous two-variable weight $F(x, y) = y - x$ in Lemma 13.11.2 passes the right side to $\int_X h \mathrm{dd^c} T_{p,q}$. Since h is zero outside E and $0 \le h \le s$,

$$\int_X h \mathrm{dd^c} T_{p,q} \le s |\mathrm{dd^c} T_{p,q}|(E).$$

Together with the torsion bounds restricted to E, the last three displays prove both assertions. □

Theorem 13.12.2 *There is $C \ge 1$, depending only on X, ω, m, n, with the following property. For bounded admissible u, v and $0 < \alpha < 1$, put $\eta = 1 - \alpha$, $V = \alpha v$ and $S = \inf_X(u - V)$. For $0 < s < \eta^3/C$, let $E_s = \{u < V + S + s\}$. Then*

$$\int_{E_s} H_m(V) \le \left(1 + \frac{Cs}{\eta^m}\right) \int_{E_s} H_m(u).$$

Proof Use the mixed measures above on E_s, writing $b_{p,q} = M_{p,q}(E_s)$ and $a_k = b_{k,0}$. The definition of S gives $0 \le \max\{u, V + S + s\} - u \le s$, so Lemma 13.12.2 applies. Repeatedly moving a V slot to a u slot by this lemma gives, for $p + q = r$,

$$b_{p,q} \le a_r + C_1 s \sum_{d=0}^{r-1} A_d, \qquad A_d = \max_{i+j=d} b_{i,j}, \qquad 1 \le r \le m.$$

Here there are at most m moves, and the multiplicity $d + 1$ in the lower-degree sum is absorbed in C_1, which depends only on X, ω, m, n. For $0 < s \le 1$, induction on r shows

$$A_r \le a_r + K_r s \sum_{k<r} a_k.$$

In fact $A_0 = a_0$, and one may take $K_r = C_1(1 + \sum_{1 \le d < r} K_d)$: insert the earlier bounds in the displayed recurrence and use $s^2 \le s$. Thus, for a constant C_2 depending only on X, ω, m, n,

$$b_{0,m} \le a_m + C_2 s \sum_{k<m} a_k.$$

It remains to control the lower diagonal masses by a_m. The identity $\sigma_V = \alpha \sigma_v + \eta\omega$ and mixed positivity give $b_{k,1} \ge \eta a_k$ for $0 \le k < m$; this identity and inequality

pass from smooth approximations by joint convergence. The assertion for $q = 0$ in Lemma 13.12.2, with C_3 the constant of that lemma, gives

$$\eta a_k \le a_{k+1} + C_3 s(a_k + a_{k-1} + a_{k-2}) \qquad (0 \le k < m),$$

with $a_{-1} = a_{-2} = 0$. Set $B_k = \eta^{m-k} a_k$ for $0 \le k \le m$, and set $B_{-1} = B_{-2} = 0$. Multiplying by η^{m-k-1} and using $0 < \eta < 1$ gives

$$B_k \le B_{k+1} + \delta(B_k + B_{k-1} + B_{k-2}), \qquad \delta = C_3 s \eta^{-3}.$$

For $\bar{B} = \max_{0 \le k < m} B_k$, summing from any $j < m$ through $m - 1$ gives $B_j \le B_m + 3m\delta\bar{B}$. If $s < \eta^3/(6mC_3)$, it follows that $\bar{B} \le 2B_m$. Consequently

$$a_k \le 2\eta^{-(m-k)} a_m \quad (k < m), \qquad \sum_{k<m} a_k \le 2m\eta^{-m} a_m.$$

Choose $C \ge \max\{1, 6mC_3, 2mC_2\}$. Then the asserted range for s meets every preceding restriction, and substitution in the bound for $b_{0,m}$ proves the theorem. □

13.12.2 Domination and proper equations

The weak modified comparison estimate extends the strict domination of Proposition 13.6.4 to bounded potentials on an arbitrary Hermitian background. A convex rescaling gives a positive volume term, while choosing a sufficiently small sublevel set absorbs the torsion error. Canonical representatives show that this sublevel set has positive volume even when it is not open.

Theorem 13.12.3 *Let u, v be bounded admissible. If, for some $0 \le c < 1$,*

$$\mathbf{1}_{\{u<v\}} H_m(u) \le c\mathbf{1}_{\{u<v\}} H_m(v),$$

then $u \ge v$ everywhere. In particular, $H_m(u)(\{u < v\}) = 0$ implies $u \ge v$.

Proof Add the same constant to u, v so that $v \ge 0$. This preserves the measure hypothesis and the desired conclusion. Choose $0 < \alpha < 1$ with $\alpha^m > c$ and put $V = \alpha v$, $\eta = 1 - \alpha$. Suppose $S = \inf_X(u - V) < 0$. For $0 < s < -S$, the set $E_s = \{u < V + S + s\}$ is contained in $\{u < V\}$ and hence in $\{u < v\}$. It has positive volume. Indeed, if its volume were zero, then $u \ge V + S + s$ almost everywhere. Both functions are bounded admissible, so Lemma 13.1.8 would give this order everywhere, contradicting the definition of S.

Choose also $s < \eta^3/C$, where C is from Theorem 13.12.2. By Lemma 13.11.1, Theorem 13.12.2 and the measure hypothesis on $E_s \subset \{u < v\}$,

$$\begin{aligned}\alpha^m H_m(v)(E_s) + \eta^m \int_{E_s} \omega^n &\le H_m(V)(E_s)\\ &\le (1 + Cs\eta^{-m}) H_m(u)(E_s)\\ &\le c(1 + Cs\eta^{-m}) H_m(v)(E_s).\end{aligned}$$

If $c > 0$, take s still smaller so that $c(1 + Cs\eta^{-m}) < \alpha^m$. If $c = 0$, this strict inequality already holds. All measures are finite and positive, so the display forces the positive-volume term to be zero, a contradiction. Hence $S \ge 0$ and $u \ge \alpha v$. Let α increase to one, with $\alpha^m > c$. Since v is bounded, $\alpha v \to v$ uniformly, and therefore $u \ge v$. The final assertion is $c = 0$. □

Proposition 13.12.2 *Every bounded admissible potential u has*

$$\int_X H_m(u) > 0.$$

Proof Otherwise $H_m(u) = 0$, and the domination principle Theorem 13.12.3, applied with $c = 0$ and the constant $v = \sup_X u + 1$, gives $u \ge \sup_X u + 1$, a contradiction. □

Corollary 13.12.1 *Fix $a > 0$. If bounded admissible u, v satisfy*

$$e^{-av} H_m(v) \ge e^{-au} H_m(u),$$

then $u \ge v$. Consequently, for any fixed finite positive Radon measure ν, the equation $H_m(u) = e^{au}\nu$ has at most one bounded admissible solution.

Proof The proof of Lemma 13.6.4 applies verbatim, with Theorem 13.12.3 in place of Proposition 13.6.4: multiplying the assumed measure inequality by the positive bounded Borel function e^{au} gives $H_m(u) \le e^{-a\delta} H_m(v - \delta)$ on $\{u < v - \delta\}$, by invariance of H_m under constants. Two solutions of the stated equation have equal weighted measures $e^{-au} H_m(u) = \nu$, and the order assertion in both directions proves uniqueness. □

A strictly increasing exponent also controls changes in logarithmic data for a fixed Radon right-hand measure.

Proposition 13.12.3 *Let $A < B$, $\kappa > 0$, and let f be continuous on $X \times [A, B]$ with $f(x,t) - f(x,t') \geq \kappa(t - t')$ for $A \leq t' \leq t \leq B$. Let ν be a finite positive Radon measure. Suppose bounded admissible u, v have values in $[A, B]$ and satisfy*

$$H_m(u) = e^{f(x,u)+h_1}\nu, \qquad H_m(v) = e^{f(x,v)+h_2}\nu,$$

where h_1, h_2 are bounded Borel functions. Then

$$\|u - v\|_\infty \leq \kappa^{-1}\|h_1 - h_2\|_{L^\infty(\nu)}.$$

Proof Put $K = \|h_1 - h_2\|_{L^\infty(\nu)}$ and fix $s > K/\kappa$. On $E_s = \{v + s < u\}$, the pointwise values of u, v lie in $[A, B]$, so $f(x,u) - f(x,v) \geq \kappa s$. The equations and the essential bound on $h_2 - h_1$ therefore give, as restricted measures,

$$\mathbf{1}_{E_s} H_m(v + s) = \mathbf{1}_{E_s} H_m(v) \leq e^{K-\kappa s}\mathbf{1}_{E_s} H_m(u).$$

The constant is smaller than one. Applying Theorem 13.12.3 to $v + s, u$ gives $u \leq v + s$ pointwise. Let $s \downarrow K/\kappa$ and exchange u, v to obtain the norm bound. The exponent is evaluated only at u, v, so $v + s$ need not take its values in $[A, B]$. □

Corollary 13.12.2 *Under the hypotheses of Theorem 13.7.2, every bounded admissible solution with values in $[A, B]$ coincides with the smooth solution given there. Moreover, for every $a > 0$ and smooth real H, the equation $H_m(u) = e^{au+H}\omega^n$ has exactly one bounded admissible solution, and that solution is smooth and strictly admissible.*

Proof By Theorem 13.8.1, the bounded and the continuous Hessian measures agree on continuous potentials, so the smooth solution of Theorem 13.7.2 solves the bounded equation as well. By compactness $\partial_t f \geq \kappa$ on $X \times [A, B]$ for some $\kappa > 0$, and the mean value theorem gives the monotonicity hypothesis of Proposition 13.12.3. That proposition, with $\nu = \omega^n$ and $h_1 = h_2 = 0$, identifies any other bounded solution with values in $[A, B]$ with the smooth one. For the final assertion, smooth existence is Lemma 13.5.1, and Corollary 13.12.1 with $\nu = e^H\omega^n$ gives uniqueness among all bounded solutions. □

13.13 Stability for concave equations

We now consider fully nonlinear equations defined by a concave symmetric function of the eigenvalues of $\omega^{-1}(\chi + \mathrm{dd}^c u)$. For such equations, a fixed admissible reference with a strict asymptotic gap replaces the determinant bounds used for the Monge–Ampère equation in Chapter 11: it bounds the positive matrix perturbations of the reference along which the operator stays below a given level. An auxiliary Monge–Ampère equation, whose density is concentrated where the difference of two solutions is nearly maximal, then shows that this set cannot be small. The resulting stability estimate remains uniform when the positive right-hand sides tend to zero.

Throughout this section and the next, X is a connected compact complex manifold of dimension $n \geq 1$, ω is a Hermitian form and χ is a fixed smooth real $(1,1)$-form on X; the degree m fixed at the beginning of the chapter plays no role. In these two sections, f, F and admissibility refer to the function of eigenvalues, the operator and the cone Γ introduced below, not to $\sigma_m^{1/m}$ and the cone of Section 13.1. For a smooth real function w and an endomorphism A of $T^{1,0}X$ that is Hermitian with respect to ω, put

$$A_w = \omega^{-1}(\chi + \mathrm{dd^c} w), \qquad F(A) = f(\lambda(A)),$$

where $\lambda(A) \in \mathbb{R}^n$ is the vector of eigenvalues of A. Let Γ be an open convex symmetric cone with $\Gamma_n \subseteq \Gamma \subseteq \Gamma_1$, where Γ_n is the positive orthant and $\Gamma_1 = \{\sum_i \lambda_i > 0\}$. Assume that $f \in C^\infty(\Gamma)$ is symmetric, concave and positive, and satisfies

$$f_i > 0, \qquad \lim_{t\to\infty} f(t\lambda) = +\infty \quad (\lambda \in \Gamma),$$

where $f_i = \partial f / \partial \lambda_i$. We call w *admissible* if $\lambda(A_w) \in \Gamma$ everywhere. Write $e_1, \dots, e_n$ for the standard basis of $\mathbb{R}^n$ and $\mathbf{1} = (1, \dots, 1)$. For $\mu \in \Gamma$, define

$$f_{\infty,i}(\mu) = \lim_{t\to\infty} f(\mu + t e_i), \qquad f_\infty(\mu) = \min_i f_{\infty,i}(\mu),$$

allowing the value $+\infty$. Fix a smooth admissible function $\underline{u}$ and positive constants H_0, δ_0. We consider smooth admissible solutions u of

$$F(A_u) = h, \qquad 0 < h \leq H_0, \qquad f_\infty(\lambda(A_{\underline{u}}(x))) \geq h(x) + \delta_0,$$

where the last condition is the strict asymptotic gap at the reference $\underline{u}$. The right-hand sides satisfying these two conditions form the *family* of right-hand sides, and the solutions are normalized by $\inf_X u = 0$. We call $X, \omega, \chi, f, \Gamma, \underline{u}, H_0, \delta_0$ the *fixed data*; all constants in this section depend only on them. Norms are taken with respect to ω^n, and $\langle a \rangle_\omega = (\int_X \omega^n)^{-1} \int_X a\omega^n$ denotes the average of a function a.

Lemma 13.13.1 *There are a positive continuous function H on X and constants $r, R, \delta > 0$ with the following properties. Every right-hand side in the family satisfies $h \leq H$; on X,*

$$\mu - 2r\mathbf{1} \in \Gamma, \qquad f_\infty(\mu - 2r\mathbf{1}) \geq H + 3\delta, \qquad \mu = \lambda(A_{\underline{u}});$$

and at every point of X, every positive semidefinite Hermitian endomorphism P satisfies

$$F(A_{\underline{u}} - r\,\mathrm{Id} + P) \leq H + 2\delta \quad \Longrightarrow \quad P \leq R\,\mathrm{Id}\,.$$

Proof Each $f_{\infty,i}$ is an increasing limit of positive concave functions, hence concave with values in $(0, +\infty]$. Suppose it is infinite at one point and finite at another point of Γ. Extend the segment joining them slightly beyond the second point, inside the open cone. The second point is then an interior point of the longer segment, so concavity forces the value there to be infinite, a contradiction. Thus $f_{\infty,i}$ is either

finite everywhere, and then continuous, or infinite everywhere. By symmetry, the same alternative holds for every i.

Take $\delta = \frac{1}{8}\min\{\delta_0, \min_X f(\mu)\} > 0$. In the finite case set $H = \min\{H_0, f_\infty(\mu) - 4\delta\}$; otherwise set $H = H_0$. Since $f_\infty(\mu) \geq f(\mu) \geq 8\delta$, this function is positive. The conditions on the family give $h \leq H$ and $f_\infty(\mu) \geq H + 4\delta$. Since $\mu(X)$ is a compact subset of Γ and the functions $f_{\infty,i}$ are continuous (or identically infinite), a sufficiently small $r > 0$ gives $\mu - 2r\mathbf{1} \in \Gamma$ and $f_\infty(\mu - 2r\mathbf{1}) \geq H + 3\delta$ on X.

If the matrix bound failed, there would be points $x_j \to x_*$ and $P_j \geq 0$ with unbounded norms and $F(A_{\underline{u}}(x_j) - r\,\mathrm{Id} + P_j) \leq H(x_j) + 2\delta$. By monotonicity of eigenvalues, the ordered eigenvalues $\lambda^{(j)}$ of $A_{\underline{u}}(x_j) - r\,\mathrm{Id} + P_j$ satisfy $\lambda_i^{(j)} \geq \mu_i(x_j) - r$. Their sum tends to infinity, so some component k tends to infinity along a subsequence. For every fixed $t > 0$, eventually

$$\lambda^{(j)} \geq \mu(x_*) - 2r\mathbf{1} + te_k.$$

Since f is increasing in each variable, letting $t \to \infty$ shows that the lower limit of $F(A_{\underline{u}}(x_j) - r\,\mathrm{Id} + P_j)$ is at least $f_\infty(\mu(x_*) - 2r\mathbf{1}) \geq H(x_*) + 3\delta$, which contradicts the upper bound $H(x_j) + 2\delta$. Adding positive semidefinite matrices preserves the cone, so all these comparisons take place in Γ. □

Proposition 13.13.1 *Every solution satisfies* $0 \leq u \leq C$.

Proof Put $\theta = \chi + \mathrm{dd^c}\underline{u}$ and $w = u - \underline{u} - \sup_X(u - \underline{u}) \leq 0$. Since $\Gamma \subseteq \Gamma_1$, we have $\mathrm{tr}_\omega\,\mathrm{dd^c} w \geq -\mathrm{tr}_\omega\,\theta \geq -C$. The Green-kernel argument in the proof of Proposition 10.11.1, applied to the operator $\mathrm{tr}_\omega\,\mathrm{dd^c}$, therefore bounds $\|w\|_{L^1}$ uniformly.

Center a holomorphic coordinate ball of fixed size at a minimum point of w, and rescale it to the real unit ball. For a fixed sufficiently small $\varepsilon > 0$, set $q = w + \varepsilon|x|^2$ and $w_0 = \min_X w$. The real contact-set construction of (10.15) gives a compact set E on which

$$D^2 q \geq 0, \qquad q \leq w_0 + \varepsilon/2, \qquad |B_{\varepsilon/2}| \leq \int_E \det D^2 q\,\mathrm{d}x.$$

On E, $\mathrm{dd^c} w \geq -C\varepsilon\omega$. Choose ε so that $C\varepsilon < r$. Then $P = \omega^{-1}(\mathrm{dd^c} w + r\omega) \geq 0$ and

$$F(A_{\underline{u}} - r\,\mathrm{Id} + P) = F(A_u) = h \leq H.$$

Lemma 13.13.1 bounds P, and hence the complex trace of the Hessian of q. The nonnegative real Hessian of q on E therefore has bounded trace, and hence bounded determinant, so the contact inequality implies $|E| \geq c > 0$. Since $w \leq w_0 + \varepsilon/2$ on E, the L^1 bound gives $c(-w_0 - \varepsilon/2) \leq C$. This bounds the oscillation of w, hence that of u; with the normalization $\inf_X u = 0$, this is the assertion. □

Theorem 13.13.1 *There is a constant C, depending only on the fixed data, such that any two normalized solutions u_1, u_2 with right-hand sides h_1, h_2 in the family satisfy*

$$\|u_1 - u_2\|_\infty \le C\big(\|h_1 - h_2\|_\infty + \|u_1 - u_2\|_{L^1}\big).$$

Proof Set $a = u_1 - \underline{u}$, $b = u_2 - \underline{u}$ and $\theta = \chi + \mathrm{dd^c}\underline{u}$. By Proposition 13.13.1, $|a|, |b| \le C_0$ for a fixed $C_0 \ge 1$. It suffices to bound $s = \max_X(b - a)$ when it is positive; the opposite difference is handled by interchanging the solutions.

Fix r, R, δ from Lemma 13.13.1 and put $\alpha = r\omega$. For every smooth positive ρ with

$$\rho \ge \tfrac{1}{2}, \qquad \langle \rho \rangle_\alpha = 1, \qquad \langle \rho^2 \rangle_\alpha \le 4,$$

Theorem 10.1.1 gives a smooth solution

$$(\alpha + \mathrm{dd^c}\psi)^n = c\rho\alpha^n, \qquad \alpha + \mathrm{dd^c}\psi > 0, \qquad \sup_X \psi = 0.$$

At a maximum of ψ, $c\rho \le 1$, so $c \le 2$. Apply Theorem 11.3.1 with $p = 2$, background α and volume form α^n. Its potential and lower-density bounds give fixed $c_0, C_\psi > 0$ with

$$c \ge c_0, \qquad -C_\psi \le \psi \le 0.$$

These constants depend only on the three bounds for ρ, and not on the cutoff function chosen below.

Choose $\kappa > 0$ so that $2C_0\kappa \le 1/4$, $\kappa C_0 \le 1/16$ and $\kappa C_\psi/2 \le 1/16$. Next choose $M \ge 1$ with $2^{-n}c_0 M r^n > R^n$, and then $\varepsilon_0 > 0$ with

$$\tfrac{16}{7}\varepsilon_0 \le \min\{(2M)^{-1}, M^{-2}\}.$$

If one of the inequalities

$$\|h_1 - h_2\|_\infty \le \kappa\delta s, \qquad \langle (b - a)_+ \rangle_\omega \le \varepsilon_0 s$$

fails, then $s \le C(\|h_1 - h_2\|_\infty + \|u_1 - u_2\|_{L^1})$, with C depending only on $\kappa, \delta, \varepsilon_0$ and the volume of X. We assume both and derive a contradiction. Put $t = \kappa s$; since $s \le 2C_0$, we have $t \le 1/4$. On $\Omega = \{(1 - t)b - a > s/2\}$, we have $b - a > 7s/16$, so by Chebyshev's inequality the relative volume of Ω is at most $16\varepsilon_0/7$.

Take a smooth $\zeta\colon \mathbb{R} \to [0, 1]$, zero on $(-\infty, 1/2]$ and one on $[3/4, \infty)$, and set

$$\rho = b_0 + M\zeta\big(((1 - t)b - a)/s\big), \qquad b_0 = 1 - M\big\langle \zeta\big(((1 - t)b - a)/s\big)\big\rangle_\omega.$$

The second term is supported in Ω, so $1/2 \le b_0 \le 1$ and $\langle \rho^2 \rangle_\alpha \le 2 + 2M^2(16\varepsilon_0/7) \le 4$; here we use that averages with respect to $\alpha = r\omega$ and ω agree. Thus the auxiliary solution for this ρ satisfies the bounds with the same constants c_0, C_ψ.

At a point where $b - a = s$, the function $\Phi = (1 - t)b - a + t\psi/2$ is at least $7s/8$, by the choice of κ. At a maximum point x_0 of Φ, therefore, $(1 - t)b - a \ge 7s/8$ because $\psi \le 0$, and so $\rho(x_0) \ge M$. The second derivative test gives, at x_0,

$$P = \alpha + \mathrm{dd}^{\mathrm{c}}\big((a - (1-t)b)/t\big) \geq \tfrac{1}{2}(\alpha + \mathrm{dd}^{\mathrm{c}}\psi) + \tfrac{1}{2}\alpha > 0.$$

The identity

$$\theta + \mathrm{dd}^{\mathrm{c}} a = t\big((\theta - \alpha) + P\big) + (1-t)(\theta + \mathrm{dd}^{\mathrm{c}} b)$$

and concavity imply

$$F\big(\omega^{-1}((\theta - \alpha) + P)\big) \leq h_2 + (h_1 - h_2)/t \leq H + \delta.$$

Both forms in the convex combination are admissible, since $\mu - r\mathbf{1} \in \Gamma$ and $P > 0$. By Lemma 13.13.1, $P \leq R\omega$. On the other hand, $P \geq \frac{1}{2}(\alpha + \mathrm{dd}^{\mathrm{c}}\psi)$, so the auxiliary equation gives

$$R^n \geq P^n/\omega^n \geq 2^{-n} c\rho r^n \geq 2^{-n} c_0 M r^n > R^n,$$

a contradiction. Hence one of the two smallness inequalities fails, which gives the required bound for s. Interchanging the solutions proves the theorem. □

Corollary 13.13.1 *Let $\mathcal{H}$ be a set of right-hand sides in the family that is equicontinuous. Then the normalized solutions with right-hand sides in $\mathcal{H}$ form a precompact subset of $C^0(X)$ and have a common modulus of continuity.*

Proof Admissibility and $\Gamma \subseteq \Gamma_1$ give $\mathrm{tr}_\omega\, \mathrm{dd}^{\mathrm{c}} u \geq -C$. By Theorem 4.1.1, there is a smooth function $\rho_0 > 0$ with $\mathrm{dd}^{\mathrm{c}}(\rho_0 \omega^{n-1}) = 0$ (for $n = 1$, take $\rho_0 = 1$), so that $\int_X \mathrm{dd}^{\mathrm{c}}\varphi \wedge \rho_0 \omega^{n-1} = 0$ for every smooth φ. Integrating the product rule $\mathrm{dd}^{\mathrm{c}}(u^2) = 2u\mathrm{dd}^{\mathrm{c}} u + 2\mathrm{d}u \wedge \mathrm{d}^{\mathrm{c}} u$ against $\rho_0 \omega^{n-1}$, and using $0 \leq u \leq C$ from Proposition 13.13.1, we obtain a uniform bound for $\int_X |\mathrm{d}u|^2_\omega \omega^n$, because ρ_0 is bounded above and below. Rellich's theorem therefore gives precompactness in L^1.

Given a sequence of such solutions, the Arzelà–Ascoli theorem gives a subsequence whose right-hand sides converge uniformly, since they are equicontinuous and bounded by H_0. A further subsequence of the solutions is Cauchy in L^1, and Theorem 13.13.1 makes it uniformly Cauchy. This proves precompactness in $C^0(X)$. A common modulus of continuity follows from a finite uniform $\varepsilon/3$-net, since each of its finitely many members is uniformly continuous. □

13.14 Gradient bounds for concave equations

We keep the fixed data, the family of right-hand sides and the normalization of Section 13.13, and assume in addition that $\|\partial h\|_{L^\infty(\omega)} \le H_1$; throughout the section, u denotes a solution with such a right-hand side h. The gradient bound is obtained by comparison with a nearby smooth admissible function V. A maximum principle for a weighted gradient quantity bounds the gradient of u in terms of that of V, provided $\|u - V\|_\infty$ is below a threshold that does not depend on the derivatives of V. The C^0 compactness of Corollary 13.13.1 then shows that finitely many such comparison functions suffice for all solutions.

For the derivative calculation, write $U = u/(2\pi)$ and $\underline{U} = \underline{u}/(2\pi)$. Thus $\mathrm{dd^c} u = \mathrm{i}\partial\bar\partial U$. For any smooth real V, put $\mathcal{A}_V = \omega^{-1}(\chi + \mathrm{i}\partial\bar\partial V)$ and call V admissible if $\lambda(\mathcal{A}_V) \in \Gamma$. We use the Chern connection of ω, and define the endomorphism $B = DF(\mathcal{A}_U)$ by $\frac{\mathrm{d}}{\mathrm{d}t} F(\mathcal{A}_U + tM)|_{t=0} = \operatorname{tr}(BM)$. Write $\lambda = (\lambda_1, \dots, \lambda_n) = \lambda(\mathcal{A}_U)$. At a point where $\mathcal{A}_U$ is diagonal in a unitary frame, set

$$B_i = f_i(\lambda), \qquad \mathcal{L}q = \sum_i B_i q_{i\bar i}, \qquad T = \sum_i B_i, \qquad |\partial q|_B^2 = \sum_i B_i |q_i|^2.$$

Here subscripts denote frame derivatives, $U_{ki} = \nabla_i U_k$ and $U_{i\bar j} = \partial_i \partial_{\bar j} U$; the second derivatives U_{ki} need not be symmetric. The trace expressions below do not depend on the unitary frame.

Lemma 13.14.1 *There are fixed $\tau, \kappa > 0$ such that*

$$T \ge \tau, \qquad 0 \le \operatorname{tr}(B\mathcal{A}_U) \le h,$$
$$\mathcal{L}(U - V) \le h \quad (V \text{ admissible}), \qquad \mathcal{L}(\underline{U} - U) \ge -h.$$

Moreover, at every point at least one of the alternatives

$$\mathcal{L}(\underline{U} - U) \ge \kappa T, \qquad B \ge \kappa T \operatorname{Id}$$

holds.

Proof If $\lambda(P) \in \Gamma$, then $\mathcal{A}_U + tP$ is admissible for $t \ge 0$. The function $t \mapsto F(\mathcal{A}_U + tP)$ is positive and concave on $[0, \infty)$, so its derivative at $t = 0$ cannot be negative; thus $\operatorname{tr}(BP) \ge 0$. For $0 < t < 1$, positivity of F and the tangent inequality at $\mathcal{A}_U$, evaluated at $t\mathcal{A}_U$, give $0 < F(t\mathcal{A}_U) \le h - (1-t)\operatorname{tr}(B\mathcal{A}_U)$; letting $t \downarrow 0$ gives $\operatorname{tr}(B\mathcal{A}_U) \le h$. Choose $a_* > 0$ with $F(a_* \operatorname{Id}) > H_0 + 1$. Concavity gives

$$H_0 + 1 < F(a_* \operatorname{Id}) \le h + a_* T - \operatorname{tr}(B\mathcal{A}_U) \le H_0 + a_* T.$$

Thus $T \ge a_*^{-1}$. Since $\mathcal{L}(U - V) = \operatorname{tr}(B\mathcal{A}_U) - \operatorname{tr}(B\mathcal{A}_V)$ and $\mathcal{L}(\underline{U} - U) = \operatorname{tr}(B\mathcal{A}_{\underline{U}}) - \operatorname{tr}(B\mathcal{A}_U)$, the remaining two bounds follow from the nonnegativity of $\operatorname{tr}(B\mathcal{A}_V)$ and $\operatorname{tr}(B\mathcal{A}_{\underline{U}})$.

We derive the alternative from the strict gap. Fix r, H, δ from Lemma 13.13.1, and put $\mu = \lambda(\mathcal{A}_{\underline{U}})$. Compactness and the gap give one $S > r$ such that

$$f(\mu - 2r\mathbf{1} + Se_i) > H + 2\delta \qquad (1 \le i \le n)$$

everywhere. We first prove a scalar alternative. At a point of X, let $\lambda \in \Gamma$ satisfy $f(\lambda) = \sigma$, where $\sigma = h$ at that point, and consider the set

$$K = \{v \ge \mu - r\mathbf{1} : f(v) \le \sigma\}.$$

It is contained in Γ because $\mu - r\mathbf{1} \in \Gamma$, it is closed, and the matrix bound of Lemma 13.13.1, applied to diagonal matrices, makes it bounded; thus K is compact. When nonempty, it contains $\mu - r\mathbf{1}$ and is star-shaped about that point: moving along a segment toward it decreases each coordinate, hence decreases f. Thus K is connected. Let $N = \nabla f(\lambda)$, whose entries are positive, and let Π be the tangent hyperplane $N \cdot (v - \lambda) = 0$.

If $f(\mu - r\mathbf{1}) > \sigma$, concavity gives $N \cdot (\mu - \lambda) > r \sum_i N_i$. Otherwise $\mu - r\mathbf{1} \in K$. The diagonal ray $\mu - r\mathbf{1} + t\mathbf{1}$, $t \ge 0$, meets $\{f = \sigma\}$, since f diverges along it; at the meeting point v, the tangent inequality gives $N \cdot (v - \lambda) \ge 0$. If Π misses K, connectedness makes $N \cdot (y - \lambda)$ strictly positive for every $y \in K$, and in particular for $y = \mu - r\mathbf{1}$; this again gives $N \cdot (\mu - \lambda) > r \sum_i N_i$.

If $v \in \Pi \cap K$, put $z_i = v - r\mathbf{1} + Se_i$. Then $z_i \ge \mu - 2r\mathbf{1} + Se_i$, so $f(z_i) > \sigma$. The tangent inequality at λ gives

$$0 < N \cdot (z_i - v) = SN_i - r \sum_j N_j.$$

Thus $N_i > (r/S) \sum_j N_j$ for every i. We have proved that $N \cdot (\mu - \lambda) > r \sum_i N_i$ or $N_i > (r/S) \sum_j N_j$ for all i, at every positive level σ, with constants independent of σ.

To pass to matrices, take $\lambda = \lambda(\mathcal{A}_U)$, and order λ and μ decreasingly. By concavity and symmetry, the numbers $f_i(\lambda)$ are then ordered increasingly. In the eigenframe of $\mathcal{A}_U$, the diagonal of $\mathcal{A}_{\underline{U}}$ is a doubly stochastic average of permutations of μ, so the rearrangement inequality gives

$$\operatorname{tr}\big(B(\mathcal{A}_{\underline{U}} - \mathcal{A}_U)\big) \ge \sum_i f_i(\lambda)(\mu_i - \lambda_i).$$

Since $N = (f_i(\lambda))_i = (B_i)_i$, taking κ smaller than $\min\{r, r/S\}$ proves the matrix alternative. □

Lemma 13.14.2 *Put $Q = 1 + |\partial U|^2_\omega$. For any smooth real ξ, one has*

$$\begin{aligned} \mathrm{e}^{-\xi}\mathcal{L}(\mathrm{e}^{\xi}Q) \geq \sum_{i,k} B_i |U_{ki} + U_k\xi_i|^2 + \frac{1}{4}\sum_i B_i\lambda_i^2 + Q\mathcal{L}\xi + |\partial\xi|^2_B \\ + 2\,\mathrm{Re}\sum_i B_i\lambda_i U_i\xi_{\bar\imath} - 2\,\mathrm{Re}\sum_{i,k} B_i\chi_{i\bar k}U_k\xi_{\bar\imath} - C_gTQ - 2H_1|\partial U|_\omega, \end{aligned}$$

where C_g depends only on ω and χ.

Proof Write $\mathscr{T}^p_{ik}$ and $R^p_{i\bar jk}$ for the torsion and curvature of the Chern connection. The commutation formula and the differentiated equation read

$$\nabla_{\bar\jmath}\nabla_i U_k = \nabla_k U_{i\bar\jmath} + \sum_p \big(R^p_{i\bar\jmath k}U_p - \mathscr{T}^p_{ik}U_{p\bar\jmath}\big),$$

$$\sum_i B_i\nabla_k U_{i\bar\imath} = h_k - \sum_i B_i\nabla_k\chi_{i\bar\imath}.$$

The product rule for $|\partial U|^2$ gives both squares $\sum B_i|U_{ki}|^2$ and $\sum B_i|U_{k\bar\imath}|^2$, the term $2\,\mathrm{Re}\sum_k h_kU_{\bar k}$, a curvature error bounded by $CT|\partial U|^2$, and the errors from $\nabla\chi$ and torsion. The latter are bounded by

$$CT|\partial U| + CT|\partial U|^2 + C|\partial U|T^{1/2}\Big(\sum_{i,k} B_i|U_{k\bar\imath}|^2\Big)^{1/2}.$$

Absorb the last term into half of the mixed square. Since $U_{i\bar\jmath} = \lambda_i\delta_{ij} - \chi_{i\bar\jmath}$, the remaining half is at least $\frac{1}{4}\sum_i B_i\lambda_i^2 - CT$. Hence

$$\mathcal{L}|\partial U|^2 \geq \sum_{i,k} B_i|U_{ki}|^2 + \frac{1}{4}\sum_i B_i\lambda_i^2 - C_gTQ - 2H_1|\partial U|.$$

Finally,

$$\partial_i|\partial U|^2 = \sum_k U_{ki}U_{\bar k} + \lambda_iU_i - \sum_k \chi_{i\bar k}U_k.$$

In the product rule for $\mathrm{e}^{\xi}Q$, combine the cross term coming from the first sum with $|\partial U|^2|\partial\xi|^2_B$ and with the square $\sum B_i|U_{ki}|^2$. This gives $\sum B_i|U_{ki} + U_k\xi_i|^2$, while the remaining weight term is $|\partial\xi|^2_B$. □

Proposition 13.14.1 *There are constants $\eta, C > 0$, depending only on the fixed data and H_1, such that every smooth admissible V with $\|U - V\|_\infty \le \eta$ satisfies*

$$\|\partial U\|_\infty \le C\big(1 + \|\partial V\|_\infty^2\big).$$

Proof Let $a \ge 1$ and $\beta > 1$ be constants, to be chosen below. Assuming $\|U - V\|_\infty \le \beta^{-1}$, put

$$w = U - V - \beta^{-1}, \qquad \xi = a(\underline{U} - U) + \beta w^2, \qquad K_V = 1 + \|\partial \underline{U}\|_\infty + \|\partial V\|_\infty.$$

Then $-2\beta^{-1} \le w \le 0$ and

$$\xi_i = a(\underline{U}_i - V_i) + (-a + 2\beta w)(U_i - V_i), \qquad |-a + 2\beta w| \le a + 4.$$

Writing $D = |\partial(U - V)|_B^2$, the Cauchy–Schwarz inequality gives

$$\Big|2\,\mathrm{Re}\sum_i B_i \lambda_i U_i \xi_{\bar\imath}\Big| \le \frac{1}{8}\sum_i B_i \lambda_i^2 + C_a K_V^4 T + C_a(Q + K_V^2)D,$$
$$\Big|2\,\mathrm{Re}\sum_{i,k} B_i \chi_{i\bar k} U_k \xi_{\bar\imath}\Big| \le \tfrac{1}{2} Q D + C_a T + C_a K_V Q^{1/2} T.$$

For the first line, split ξ_i as above and use $\sum B_i |U_i|^2 \le 2D + 2K_V^2 T$; then apply Young's inequality to the resulting three products, each absorbing at most $\frac{1}{24}\sum B_i\lambda_i^2$. For the second line, use $|\partial\xi|_B \le a K_V T^{1/2} + (a+4)D^{1/2}$ and the bound for χ. In both lines, C_a depends on a but not on β or on the derivatives of V.

By Lemma 13.14.1 and $w \le 0$,

$$\mathcal{L}\xi \ge a\mathcal{L}(\underline{U} - U) - 4H_0 + 2\beta D.$$

Choose a so large that $(a\kappa - C_g - 2)\tau > 4H_0 + 2H_1 + 1$, and then $\beta \ge 2C_a + 2$. Combining the preceding bounds with Lemma 13.14.2, we obtain, at every point where $Q \ge C'(1 + K_V^4)$ for a suitable constant C',

$$\mathrm{e}^{-\xi}\mathcal{L}(\mathrm{e}^{\xi} Q) \ge \frac{1}{8}\sum_i B_i \lambda_i^2 + Q\big[a\mathcal{L}(\underline{U} - U) - (C_g + 2)T + \beta D - (4H_0 + 2H_1)\big].$$

Indeed, the coefficient of D before absorption is $(2\beta - C_a - 1/2)Q - C_a K_V^2$, which is at least βQ once C' is large enough. The other errors satisfy $C_a K_V^4 \le Q$ and $C_a + C_a K_V Q^{1/2} \le Q$, and they account for the additional term $-2T$ in the bracket.

Consider a maximum point of $\mathrm{e}^{\xi} Q$ at which $Q \ge C'(1 + K_V^4)$. In the first alternative of Lemma 13.14.1, the bracket is positive by the choice of a, since $T \ge \tau$. In the second, $D \ge \kappa T |\partial(U - V)|^2$. For $Q \ge 4(1 + K_V^2)$, the triangle inequality gives $|\partial(U - V)|^2 \ge c_* Q$, where $c_* = (\sqrt{3} - 1)^2/4$. Since $\mathcal{L}(\underline{U} - U) \ge -H_0$, the bracket

is then at least

$$(\beta\kappa c_* Q - C_g - 2)T - (a+4)H_0 - 2H_1.$$

This is positive once C' is large enough, because $T \geq \tau$. In both cases $\mathcal{L}(\mathrm{e}^{\xi} Q) > 0$ at a maximum point, which contradicts the second derivative test. Thus $Q \leq C(1 + K_V^4)$ at the maximum point. The weight has uniformly bounded oscillation:

$$\operatorname{osc}_X \xi \leq a(\operatorname{osc}_X \underline{U} + \operatorname{osc}_X U) + 4/\beta.$$

By the potential bound in Proposition 13.13.1, this oscillation is bounded independently of V, so the estimate at the maximum point gives $Q \leq C(1 + K_V^4)$ everywhere. Taking square roots and setting $\eta = \beta^{-1}$ proves the assertion. □

Theorem 13.14.1 *There is a constant C, depending only on the fixed data and H_1, such that $\|\mathrm{d}u\|_{L^\infty(\omega)} \leq C$ for every normalized solution u whose right-hand side h lies in the family and satisfies $\|\partial h\|_\infty \leq H_1$.*

Proof The right-hand sides in the family with $\|\partial h\|_\infty \leq H_1$ form an equicontinuous set. By Corollary 13.13.1, the corresponding functions U form a precompact subset of $C^0(X)$. Choose a finite η-net in this set, consisting of smooth admissible functions $V_1, \ldots, V_N$. Each solution is within η of some V_j, so Proposition 13.14.1 bounds its gradient by $C(1 + \max_j \|\partial V_j\|_\infty^2)$. The set of solutions, and hence the chosen finite net, depends only on the fixed data and H_1. Finally, $u = 2\pi U$, so the same conclusion holds for $\mathrm{d}u$. □

Part III
Relative singularities and equations

Chapter 14
Envelopes and volumes of singularities

Throughout this chapter, X is a connected compact complex manifold of dimension $n \geq 1$, with an auxiliary Hermitian metric ω and a smooth positive volume form $\mathrm{d}V$, and θ denotes a smooth real $(1,1)$-form on X whose positivity is specified in each section.

On a compact Kähler manifold, and more generally on a compact complex manifold with the bounded mass property of Definition 7.5.1, the non-pluripolar Monge–Ampère mass of a potential in a closed background depends only on its singularity type; this is what makes mass a tool for comparing singularities. In a nonclosed background, a bounded change of the potential can change the mass, and a singularity type may contain potentials of different, even infinite, masses. This chapter opens the part of the book on relative singularities and the equations they govern, and introduces two kinds of invariants that take this into account.

The first records the variation of mass: the lower and upper volumes $\underline{\mathrm{vol}}_\theta(u)$ and $\overline{\mathrm{vol}}_\theta(u)$ of a potential u are the infimum and the supremum of the masses of all potentials of the same singularity type. The second replaces a mass condition by a condition on envelopes. Fix a reference potential φ. Stretching a potential u away from φ by a factor $b \geq 1$ gives the obstacle $bu - (b-1)\varphi$, which is more singular than u where u lies below φ. The *relative envelope class* $\mathcal{E}_\varphi$ consists of the potentials u at least as singular as φ for which every such stretched obstacle lies above some θ-plurisubharmonic function. This class is convex and stable under finite rooftops, its members have the same singularity envelope as φ, and its definition involves no mass.

The main tool is the contact property of envelopes: the Monge–Ampère measure of an envelope is carried by the set where it touches its obstacle. Together with ordered contact, it gives the domination principle for bounded potentials in a semipositive background in which every bounded potential has positive mass, and it bounds the measures of rooftop envelopes. For envelopes of logarithmic growth on $\mathbb{C}^n$, which correspond to envelopes on $\mathbb{P}^n$, we also examine how the contact of the regularized envelope with its obstacle depends on the growth allowance.

In a semipositive big background, the contact property also produces positive mass. Let φ be a model reference, that is, a fixed point of the singularity envelope that admits a competitor in a smaller background. Every member u of $\mathcal{E}_\varphi$ then has a

competitor in a background $a\theta$ with $0 < a < 1$, and an envelope in the remaining big background $(1-a)\theta$ has positive mass, which it transfers to u on the contact set. The argument uses only that each bounded potential has positive mass, so it applies even when the infimum of these masses is zero. On the other hand, if X does not have the bounded mass property of Definition 7.5.1, the relative envelope class of the zero potential in the background ω contains potentials of infinite mass. Envelope conditions and mass bounds are therefore different requirements, and this distinction is essential for the equations with prescribed singularities in the next chapter.

We use the notation of Chapter 1: the order $u \preceq v$, the equivalence $u \sim v$ and singularity types (Definition 1.12.1); the envelopes $P_\theta(f)$, which may be identically $-\infty$, the rooftop envelopes $P_\theta(u,v) = P_\theta(\min(u,v))$ and $V_\theta = P_\theta(0)$ (Section 1.19); and the singularity envelopes $P_\theta[u]$, $P_{\theta,F}[u]$ and envelope fixed points (Definition 1.19.1). We write $\theta_u = \theta + \mathrm{dd^c}u$ and $\omega_u = \omega + \mathrm{dd^c}u$. For $u \in \mathrm{PSH}(X,\theta)$, $\mathrm{MA}_\theta(u)$ denotes the extended Monge–Ampère measure $\mathrm{MA}^{\mathrm{ext}}_\theta(u)$ of Theorem 7.1.1. It is the Bedford–Taylor measure $(\theta + \mathrm{dd^c}u)^n$ when u is bounded, it puts no mass on pluripolar sets, and its total mass, the *extended mass* of u, may be infinite.

14.1 Contact sets

The Monge–Ampère measure of an envelope is concentrated on the contact set, where the envelope meets its obstacle. For a smooth obstacle and a Hermitian background, the envelope is the uniform limit of solutions of exponential equations whose densities tend to zero wherever the envelope stays strictly below the obstacle. Continuity of bounded products then carries this property to quasi-continuous obstacles, first bounded ones, through monotone approximation, and then general ones, through truncation.

Proposition 14.1.1 *Let $f \in C^\infty(X,\mathbb{R})$. For $\beta > 0$, denote by u_β the smooth strictly ω-plurisubharmonic solution of*

$$\omega_{u_\beta}^n = \mathrm{e}^{\beta(u_\beta - f)}\omega^n.$$

As $\beta \to +\infty$, these solutions converge uniformly to $p = P_\omega(f)$. Moreover,

$$\mathbf{1}_{\{p<f\}}\omega_p^n = 0.$$

Proof The solutions exist by Theorem 10.6.1. Choose $A \geq 0$ with $\mathrm{dd^c}f \leq A\omega$. At a maximum point of $u_\beta - f$, the second derivative test gives

$$\omega_{u_\beta} \leq \omega + \mathrm{dd^c}f \leq (1+A)\omega.$$

Evaluating the equation at this point, we obtain

$$u_\beta \le f + \frac{C_f}{\beta}, \qquad C_f = n \log(1 + A).$$

Thus $u_\beta - C_f/\beta$ is a competitor for $P_\omega(f)$, and $u_\beta \le p + C_f/\beta$.

For the lower bound, note that p is continuous by Proposition 1.19.1, and choose smooth strictly ω-plurisubharmonic functions $v_j \downarrow p$ by Theorem 1.11.1; by Dini's theorem the convergence is uniform. For any $\delta > 0$, choose j so that

$$p \le v_j \le p + \delta/2.$$

Then $v_j - \delta \le f$, since $p \le f$. Since $\omega_{v_j} > 0$ and X is compact, there is $c_j > 0$ such that $\omega_{v_j}^n \ge c_j \omega^n$. Put $C_j = \max(0, -\log c_j)$. The function

$$w_{j,\beta} = v_j - \delta - C_j/\beta$$

satisfies

$$\omega_{w_{j,\beta}}^n = \omega_{v_j}^n \ge \mathrm{e}^{-C_j} \omega^n \ge \mathrm{e}^{\beta(w_{j,\beta} - f)} \omega^n.$$

Both $w_{j,\beta}$ and u_β are smooth and strictly ω-plurisubharmonic, so the smooth comparison principle, Proposition 10.6.1 with $\lambda = \beta$ and $F = -\beta f$, gives $w_{j,\beta} \le u_\beta$. Consequently

$$p - \delta - C_j/\beta \le u_\beta \le p + C_f/\beta.$$

Letting first $\beta \to \infty$ with j fixed, so that no lower bound for c_j uniform in j is needed, and then $\delta \downarrow 0$, we obtain uniform convergence. Bounded Bedford–Taylor continuity then gives weak convergence of the measures.

On a compact subset of the open set $\{p < f\}$, uniform convergence gives $u_\beta - f \le -\epsilon$ for some $\epsilon > 0$ and all large β. There the measures $\omega_{u_\beta}^n$ have densities at most $\mathrm{e}^{-\beta\epsilon}$ with respect to ω^n. Testing against continuous functions with compact support in $\{p < f\}$ and passing to the limit gives $\omega_p^n = 0$ on this open set. □

Proposition 14.1.2 *If f is bounded and lower semicontinuous, then $P_\omega(f)$ is bounded, lies below f quasi-everywhere, and its Monge–Ampère measure is concentrated on $\{P_\omega(f) = f\}$.*

Proof Suppose first that f is continuous. Choose smooth functions $f_j \to f$ uniformly. The Lipschitz bound for envelopes in Proposition 1.19.1 gives $p_j = P_\omega(f_j) \to p = P_\omega(f)$ uniformly. Every compact subset of the open set $\{p < f\}$ has a neighborhood on which $p_j < f_j$ for all large j. The preceding proposition and convergence of bounded products then give $\omega_p^n = 0$ on $\{p < f\}$.

For lower semicontinuous f, choose continuous functions $f_j \uparrow f$ with common upper and lower bounds. For example, for a fixed distance d inducing the topology of X, one may take

$$f_j(x) = \inf_{y \in X} \big(f(y) + j d(x, y) \big).$$

These functions are Lipschitz and increase to f: if convergence failed at x, approximate minimizers y would approach x, contradicting lower semicontinuity of f at x.

The functions $p_j = P_\omega(f_j)$ increase and stay bounded, so the regularized limit $p_\infty = (\lim_j p_j)^*$ is a bounded ω-plurisubharmonic function. It agrees with $\lim_j p_j$ outside a pluripolar set, and since each p_j lies below $f_j \le f$ outside a pluripolar set, we obtain $p_\infty \le f$ quasi-everywhere. Thus $p_\infty \le P_\omega(f)$.

For fixed k, the set $\{p_\infty < f_k\}$ is open, since p_∞ is upper semicontinuous and f_k is continuous. If $j \ge k$, then $p_j \le p_\infty < f_k \le f_j$ there, so the continuous case gives $\omega_{p_j}^n = 0$ on that open set. Bedford–Taylor continuity along increasing sequences applies after adding a fixed smooth local function whose Hessian dominates ω and expanding with respect to the resulting smooth form. Together with the lower semicontinuity of the mass of open sets under weak convergence, it yields

$$\omega_{p_\infty}^n(\{p_\infty < f_k\}) = 0.$$

Taking the union over k gives zero measure on $\{p_\infty < f\}$, hence also on $\{p_\infty < P_\omega(f)\}$ outside a pluripolar set, since $P_\omega(f) \le f$ quasi-everywhere. Bounded measures of this kind do not charge pluripolar sets, so the domination principle, Theorem 11.2.2, gives $p_\infty \ge P_\omega(f)$. The two functions are therefore equal, and the assertion follows. □

Theorem 14.1.1 *Let $\theta \ge 0$ be smooth and let f be bounded and quasi-continuous. Then $p = P_\theta(f)$ is bounded, $p \le f$ quasi-everywhere, and*

$$\mathbf{1}_{\{p<f\}}\mathrm{MA}_\theta(p) = 0.$$

Proof First suppose that $\theta = \omega$ is Hermitian. Choose bounded lower semicontinuous functions $f_j \downarrow f$ quasi-everywhere, with common bounds, as in Lemma 1.8.5. The bounded functions $p_j = P_\omega(f_j)$ decrease to p. Indeed, their limit is a competitor for p, since it lies below f quasi-everywhere, and the reverse inequality follows from monotonicity.

For fixed k, put $H_k = \min\{1, (f - p_k)_+\}$, a bounded quasi-continuous function. Where $H_k > 0$ and $j \ge k$, we have $p_j \le p_k < f \le f_j$ outside a pluripolar set. The lower semicontinuous case, Proposition 14.1.2, therefore gives

$$\int_X H_k\, \omega_{p_j}^n = 0.$$

The convergence of bounded products against quasi-continuous weights in Section 1.8 gives $\int_X H_k\, \omega_p^n = 0$. Since the sets $\{p_k < f\}$ increase to $\{p < f\}$, this proves the assertion for Hermitian θ.

For general semipositive θ, set $\theta_j = \theta + j^{-1}\omega$. The functions $p_j = P_{\theta_j}(f)$ have the common bounds $\inf_X f \le p_j \le \sup_X f$ and decrease. Their limit is θ-plurisubharmonic by distributional convergence and lies below f quasi-everywhere, so it is at most $p = P_\theta(f)$; since $p \le p_j$ for all j, it equals p. For each fixed k, the same coefficient $H_k = \min\{1, (f - p_k)_+\}$ gives

$$\int_X H_k \, \mathrm{MA}_{\theta_j}(p_j) = 0 \qquad (j \geq k).$$

All p_j are bounded potentials for the fixed Hermitian form $\theta + \omega$. Expanding the products with background θ_j with respect to this fixed background, we may use convergence of bounded products and uniform capacity bounds to pass the last identity to the limit, with the quasi-continuous weight H_k; this gives $\int_X H_k \, \mathrm{MA}_\theta(p) = 0$. Exhausting $\{p < f\}$ as before completes the proof. □

Theorem 14.1.2 *Let θ be smooth and semipositive, and let f be a quasi-continuous obstacle. If $p = P_\theta(f)$ is not identically $-\infty$, then $p \leq f$ quasi-everywhere and*

$$\mathbf{1}_{\{p<f\}} \mathrm{MA}_\theta(p) = 0.$$

Proof The local envelope theorem, applied to a countable family realizing the supremum (Choquet's lemma), gives $p \leq f$ outside a pluripolar set. Choose $M > \sup_X p$ and replace f by $\min(f, M)$. This does not change the envelope, since p itself is a competitor for the smaller obstacle. Since $p < M$, the contact set is also unchanged, so it suffices to prove the theorem for the smaller obstacle; that is, we may assume that f is bounded above.

Set $f_j = \max(f, -j)$ and $p_j = P_\theta(f_j)$. The constants $-j$ are competitors, so p_j is bounded. Theorem 14.1.1 gives $p_j \leq f_j$ quasi-everywhere and

$$\mathbf{1}_{\{p_j<f_j\}} \mathrm{MA}_\theta(p_j) = 0.$$

The sequence p_j decreases and stays above p. Its limit is therefore θ-plurisubharmonic, it lies below f quasi-everywhere, and so it is a competitor for p. Hence $p_j \downarrow p$.

Fix an integer $k \geq 1$ and $C > 0$, and define the bounded nonnegative quasi-continuous function

$$H_{k,C} = \min\{1, (f - p_k)_+\} \, \min\{1, (p + C)_+\}.$$

Its values on pluripolar sets are immaterial. Where this coefficient is positive, $p_k < f$ and $p > -C$. For $j \geq k$, we then have $p_j \leq p_k < f \leq f_j$ and $p_j \geq p > -C$ there. Plurifine locality and Theorem 14.1.1 therefore imply

$$\int_X H_{k,C} \, \mathrm{MA}_\theta(\max(p_j, -C)) = 0.$$

The truncated potentials are uniformly bounded and decrease to $\max(p, -C)$, so their measures converge weakly by bounded Bedford–Taylor continuity. The uniform capacity bounds and the weighted convergence in Section 1.8 also allow testing against $H_{k,C}$. Hence the last identity holds with p in place of p_j. By plurifine locality, this says that the extended measure of p vanishes on $\{p_k < f\} \cap \{p > -C\}$. Off the pole set of p, these sets increase to $\{p < f\}$ as first k and then C tend to infinity. Continuity of positive measures along increasing sequences proves the assertion, even if the total extended mass is infinite. □

For a smooth big background, constants need not be competitors, and a strictly θ-plurisubharmonic barrier with poles replaces the constant lower truncations. The same contact property for quasi-continuous obstacles finite almost everywhere is proved in this setting in Theorem 15.7.2. Combined with ordered contact, either contact theorem controls the measures of rooftop envelopes. In the following bound, the strict inequality in the first term assigns the common contact set $\{p = u = v\}$ to the second term.

Proposition 14.1.3 *Let θ be a smooth real $(1,1)$-form that is semipositive or big, and let $u, v \in \mathrm{PSH}(X,\theta)$. If $p = P_\theta(u,v)$ is not identically $-\infty$, then*

$$\mathrm{MA}_\theta(p) \le \mathbf{1}_{\{p=u<v\}}\mathrm{MA}_\theta(u) + \mathbf{1}_{\{p=v\}}\mathrm{MA}_\theta(v).$$

Proof The envelope lies below both obstacles quasi-everywhere, hence everywhere, since the obstacles are upper semicontinuous. By the contact theorem (Theorem 14.1.2 or Theorem 15.7.2), its measure is carried by $\{p = \min(u,v)\}$. Partition this set into $\{p = u < v\}$ and $\{p = v\}$, and apply Proposition 7.2.1 on each part. All measures vanish on the pluripolar pole sets, which can therefore be discarded throughout. □

Proposition 14.1.4 *Let θ be smooth and semipositive. Every envelope fixed point φ satisfies*

$$\mathrm{MA}_\theta(\varphi) \le \mathbf{1}_{\{\varphi=0\}}\theta^n.$$

In particular, its extended mass is finite.

Proof Since $\varphi = P_\theta[\varphi] \le 0$, the function φ is a competitor for $P_\theta(\varphi+1, 0)$. This rooftop is also a competitor for $P_\theta[\varphi] = \varphi$, being at most $\varphi + 1$ and at most zero, so it equals φ. Now apply Proposition 14.1.3 with $u = \varphi + 1$ and $v = 0$. The contact set $\{\varphi = \varphi + 1 < 0\}$ is contained in the pole set of φ, which the extended measure does not charge. Only contact with zero contributes, and this gives the asserted bound. □

14.1.1 Affine growth and projective envelopes

The contact property concerns the regularized envelope. For the classical weighted extremal functions on $\mathbb{C}^n$, the obstacle is imposed at every point of a compact set, and one may ask when the regularized envelope still touches the obstacle there. Compactifying $\mathbb{C}^n$ to $\mathbb{P}^n$ turns the logarithmic growth allowance into a semipositive background. This correspondence distinguishes contact of the raw supremum with the obstacle from contact of its upper semicontinuous regularization, and it shows how both depend on the growth allowance.

In the rest of this section we work on $\mathbb{C}^n$ and on $\mathbb{P}^n$ rather than on X. Write $\chi(z) = \log(1 + \|z\|^2)$ on $\mathbb{C}^n$, and let ω_{FS} be the Fubini–Study form on $\mathbb{P}^n$ with this local potential, so that $\int_{\mathbb{P}^n} \omega_{\mathrm{FS}}^n = 1$ with our normalization of dd^{c}. For $\delta > 0$ put

$$\theta_\delta = \tfrac{\delta}{2}\omega_{\mathrm{FS}}, \qquad \mathcal{L}_\delta = \{u \in \mathrm{PSH}(\mathbb{C}^n) : u(z) \le \delta \log^+\|z\| + C_u\}.$$

Here $\log^+ = \max(\log, 0)$, and C_u is a constant depending on u. Thus $\mathcal{L}_\delta$ is the class of plurisubharmonic functions of logarithmic growth at most δ. Let $\mathcal{L}^+_\delta$ consist of the members of $\mathcal{L}_\delta$ that are also bounded below by $\delta \log^+\|z\| - C$ for some constant C.

Lemma 14.1.1 *The map $u \mapsto u - \delta\chi/2$, followed by the canonical extension across the hyperplane at infinity, is a bijection from $\mathcal{L}_\delta$ onto* $\mathrm{PSH}(\mathbb{P}^n, \theta_\delta)$.

Proof Use homogeneous coordinates $[Z_0 : \cdots : Z_n]$ and the affine chart $Z_0 \neq 0$. On a chart $Z_i \neq 0$ meeting $H = \{Z_0 = 0\}$, the local plurisubharmonic representative of $p = u - \delta\chi/2$ is

$$p + \frac{\delta}{2} \log \frac{\sum_{j=0}^n |Z_j|^2}{|Z_i|^2} = u + \delta \log |Z_0/Z_i|.$$

This function is locally bounded above near H, by the growth bound on u, because near each point of H the norm of the affine coordinates is at most a constant divided by $|Z_0/Z_i|$. By Lemma 1.3.7, it therefore extends canonically across H. Since the local potentials of θ_δ differ by smooth functions on overlaps, the extensions agree there.

Conversely, every $p \in \mathrm{PSH}(\mathbb{P}^n, \theta_\delta)$ is bounded above, since $\mathbb{P}^n$ is compact. The function $u = p + \delta\chi/2$ is plurisubharmonic on the affine chart, and

$$0 \leq \tfrac{1}{2}\chi - \log^+\|z\| \leq \tfrac{1}{2}\log 2$$

gives the required growth bound. Neither function is identically $-\infty$ on the affine chart. Uniqueness of the canonical extension shows that the two maps are inverse to each other. □

Fix a non-pluripolar compact set $K \subset \mathbb{C}^n$ and a bounded real function ψ on K. Define the raw and the regularized envelope

$$V_\delta = \sup\{u \in \mathcal{L}_\delta : u|_K \leq \psi\}, \qquad F_\delta = V^*_\delta,$$

where the obstacle condition is imposed at every point of K. On $\mathbb{P}^n$ put

$$G_\delta = \sup\{p \in \mathrm{PSH}(\mathbb{P}^n, \theta_\delta) : p|_K \leq \psi - \delta\chi/2\}.$$

Proposition 14.1.5 *The regularized envelope G^*_δ on $\mathbb{P}^n$ is bounded and θ_δ-plurisubharmonic. On the affine chart,*

$$V_\delta = \tfrac{\delta}{2}\chi + G_\delta, \qquad F_\delta = \tfrac{\delta}{2}\chi + G^*_\delta \in \mathcal{L}^+_\delta. \tag{14.1}$$

Moreover, $V_\delta = F_\delta$ outside a pluripolar set, in particular almost everywhere.

Proof By Lemma 14.1.1, the map $u \mapsto u - \delta\chi/2$ is a bijection between the two families of competitors, which proves the identity for V_δ. Constants below $\inf_K(\psi - \delta\chi/2)$ are competitors for G_δ.

Since all competitors for G_δ lie below the constant $\sup_K(\psi - \delta\chi/2)$ on K, Lemma 1.19.1 gives a common upper bound for them. Together with a constant competitor, the upper-envelope theorem shows that G_δ^* is bounded and θ_δ-plurisubharmonic. Addition of the smooth function $\delta\chi/2$ in (14.1) commutes with regularization, and the two-sided bounds for G_δ^* give $F_\delta \in \mathcal{L}_\delta^+$.

Locally, adding a potential of θ_δ makes the competitors plurisubharmonic. The upper-envelope theorem recalled in Chapter 1 shows that regularization changes their supremum only on a pluripolar set. Thus $G_\delta = G_\delta^*$ outside a pluripolar set, and adding $\delta\chi/2$ gives the same assertion for V_δ. □

Corollary 14.1.1 *The envelope F_δ is unchanged if $u|_K \le \psi$ is imposed only quasi-everywhere, or only on $K \setminus E$ for any pluripolar $E \subseteq K$.*

Proof By Lemma 14.1.1, the map $u \mapsto u - \delta\chi/2$ is a bijection from the competitors for each of these weakened constraints onto the θ_δ-plurisubharmonic functions satisfying the same constraint with the obstacle $\psi - \delta\chi/2$. Apply Proposition 1.19.2 on $\mathbb{P}^n$ to the Kähler, hence big, background θ_δ and this bounded obstacle on K, and add $\delta\chi/2$ as in the proof of Proposition 14.1.5. □

These equalities concern regularized envelopes. The raw envelope V_δ is still defined with the obstacle imposed at every point of K.

Changing δ changes both the background θ_δ and the obstacle $\psi - \delta\chi/2$. Nevertheless, the set of parameters for which contact holds has a simple structure. Put

$$\Lambda = \{\delta > 0 : V_\delta|_K = \psi\}, \qquad \Lambda_{\mathrm{reg}}(a) = \{\delta \in \Lambda : F_\delta(a) = \psi(a)\} \quad (a \in K).$$

The parameters $\delta \in \Lambda$ are those for which the raw envelope touches the obstacle at every point of K; we say that *raw contact* holds for them. For $a \in K$, the parameters in $\Lambda_{\mathrm{reg}}(a)$ are those for which, in addition, the regularized envelope touches the obstacle at a; we say that *regular contact* holds at a.

Theorem 14.1.3 *The set Λ is empty, $(0,\infty)$, or $[\lambda,\infty)$ for some $\lambda > 0$. If $\Lambda = [\lambda,\infty)$, each $\Lambda_{\mathrm{reg}}(a)$ is empty, $\{\lambda\}$, or Λ. If $\Lambda = (0,\infty)$, each $\Lambda_{\mathrm{reg}}(a)$ is empty or Λ. The same alternatives hold for $\bigcap_{a\in K}\Lambda_{\mathrm{reg}}(a)$.*

Proof Let $m = \inf_K \psi$. Inclusion of the growth classes and scaling of competitors give, for $0 < \alpha < \beta$,

$$0 \le V_\beta - V_\alpha \le \frac{\beta - \alpha}{\alpha}(V_\alpha - m). \tag{14.2}$$

Indeed, if u is a competitor for V_β, then $(\alpha/\beta)(u - m) + m$ is a competitor for V_α, since $\psi \ge m$. The constant m is also a competitor, so all terms are finite and the right-hand side is nonnegative. The estimate shows that $V_\delta(z)$ depends continuously on $\delta > 0$ at every point z. Since $V_\delta \le \psi$ on K and V_δ increases with δ, the set Λ is closed upwards, and by continuity it is relatively closed in $(0,\infty)$. This proves the three possible forms of Λ.

For $0 \le t \le 1$, convex combinations of competitors give

$$V_{(1-t)\alpha+t\beta} \geq (1-t)V_\alpha + tV_\beta.$$

By Proposition 14.1.5, the same inequality holds for the regularized envelopes almost everywhere, and hence everywhere by Lemma 1.3.3. Thus $F_\delta(z)$ is a concave nondecreasing function of δ at every point z.

For $a \in K$ and $\delta \in \Lambda$, the defect $d_\delta = F_\delta(a) - \psi(a)$ is nonnegative, concave and nondecreasing in δ. Suppose $d_\alpha = 0$, and that there is $\eta \in \Lambda$ with $\eta < \alpha$. For every $\beta > \alpha$, write α as a convex combination of η and β with positive weights. Concavity then forces both nonnegative defects d_η and d_β to vanish, and monotonicity gives $d_\delta = 0$ for all $\delta \in \Lambda$ with $\delta < \alpha$. Hence a zero of the defect at a parameter other than the smallest element of Λ forces every defect to vanish. This proves the alternatives for each $\Lambda_{\mathrm{reg}}(a)$, and intersecting them gives the alternatives for $\bigcap_{a\in K} \Lambda_{\mathrm{reg}}(a)$. □

Thus raw contact persists as the growth allowance increases, whereas regular contact at a point may hold only at the smallest contact parameter. The correspondence with $\mathbb{P}^n$ keeps the background and the obstacle matched throughout this variation.

14.1.1.1 Growth slopes and supporting probabilities

By concavity, the rate at which F_δ grows when the growth allowance δ is increased is at least its rate of growth for large δ, and the latter is the classical extremal function of K, without weight. For continuous weights, supporting probabilities express the initial rate of growth through tests by functions of the fixed class $\mathcal{L}_1$.

When the weight varies, we write $V_{\delta,q}$ and $F_{\delta,q} = V^*_{\delta,q}$ for the raw and regularized envelopes defined as above, with a bounded obstacle q on the fixed compact set K in place of ψ. Thus $F_\delta = F_{\delta,\psi}$. Set

$$V_K = V_{1,0}, \qquad V^*_K = F_{1,0}, \qquad m = \inf_K \psi, \qquad M = \sup_K \psi.$$

Thus V^*_K is the extremal function of K. Since the constant 0 is a competitor for V_K, and since competitors can be scaled and translated by constants, we have $V^*_K \geq 0$ and

$$\delta V^*_K + m \leq F_\delta \leq \delta V^*_K + M. \tag{14.3}$$

Lemma 14.1.2 *For $0 < \alpha < \beta$, one has on $\mathbb{C}^n$*

$$(\beta-\alpha)V^*_K \leq F_\beta - F_\alpha \leq \frac{\beta-\alpha}{\alpha}(F_\alpha - m). \tag{14.4}$$

Proof If u is a competitor for $V_\alpha = V_{\alpha,\psi}$ and v is a competitor for $V_K = V_{1,0}$, then $u + (\beta-\alpha)v$ is a competitor for V_β. Taking the suprema over u and v independently gives

$$V_\beta \geq V_\alpha + (\beta-\alpha)V_K.$$

By Proposition 14.1.5, the raw envelopes agree almost everywhere with their regularizations, so this inequality holds almost everywhere for the regularizations, and

hence everywhere by Lemma 1.3.3. This is the first inequality. The same argument applied to (14.2) gives the second. □

Proposition 14.1.6 *For every $a \in \mathbb{C}^n$ and $\tau > 0$, the right derivative*

$$D_\tau(a) = \lim_{h \downarrow 0} \frac{F_{\tau+h}(a) - F_\tau(a)}{h}$$

exists and is finite. The difference quotients decrease as $h > 0$ increases, and

$$V_K^*(a) \le \frac{F_{\tau+h}(a) - F_\tau(a)}{h} \le D_\tau(a) \le \frac{F_\tau(a) - m}{\tau}. \tag{14.5}$$

As $h \to \infty$, they converge to $V_K^*(a)$, with the error bound

$$0 \le \frac{F_{\tau+h}(a) - F_\tau(a)}{h} - V_K^*(a) \le \frac{M - m}{h}. \tag{14.6}$$

Moreover, $D_\tau(a) = V_K^*(a)$ if and only if

$$F_\delta(a) = F_\tau(a) + (\delta - \tau) V_K^*(a) \qquad (\delta \ge \tau). \tag{14.7}$$

Proof For $0 < h_1 < h_2$, concavity of $F_\delta(a)$ gives

$$F_{\tau+h_1}(a) - F_\tau(a) \ge \frac{h_1}{h_2}\big(F_{\tau+h_2}(a) - F_\tau(a)\big).$$

Thus the difference quotients decrease with h. Lemma 14.1.2 bounds them between $V_K^*(a)$ and $(F_\tau(a) - m)/\tau$, which proves the existence and finiteness of $D_\tau(a)$ and (14.5). Combining the lower bound for the increments with (14.3) gives (14.6), with a bound independent of a. If $D_\tau(a) = V_K^*(a)$, then by (14.5) every difference quotient equals $V_K^*(a)$, which gives (14.7). Conversely, that formula gives $D_\tau(a) = V_K^*(a)$.□

Corollary 14.1.2 *Let $a \in K$ and $\tau \in \Lambda_{\mathrm{reg}}(a)$. Regular contact at a holds for every $\delta > \tau$ if and only if $D_\tau(a) = 0$. If $D_\tau(a) > 0$, regular contact at a fails for every $\delta > \tau$, and τ is the smallest element of Λ.*

Proof At τ one has $F_\tau(a) = \psi(a)$, and raw contact holds for all $\delta \ge \tau$ by Theorem 14.1.3. If $D_\tau(a) = 0$, then by (14.5) every difference quotient at τ vanishes, so $F_\delta(a) = \psi(a)$ for all $\delta > \tau$. If $D_\tau(a) > 0$, then $F_\delta(a) > \psi(a)$ for all δ slightly larger than τ, and by monotonicity for all $\delta > \tau$. In this case $\Lambda_{\mathrm{reg}}(a)$ contains τ but no larger parameter, so by Theorem 14.1.3 it equals $\{\lambda\}$, where $\lambda = \tau$ is the smallest element of Λ. □

Assume now that the weights are continuous. If the obstacle is imposed only quasi-everywhere on K, the regularized envelope itself becomes a competitor, so the supremum needs no regularization; a constraint imposed at every point would miss the values created by regularization. Indeed, for $q \in C(K, \mathbb{R})$,

$$F_{\delta,q}(z) = \sup\{u(z) : u \in \mathcal{L}_\delta,\ u \le q \text{ q.e. on } K\}. \tag{14.8}$$

The envelope $F_{\delta,q}$ itself is a competitor on the right, since it agrees with the raw envelope off a pluripolar set by Proposition 14.1.5. Conversely, by Corollary 14.1.1, every competitor on the right lies below $F_{\delta,q}$.

For $a \in \mathbb{C}^n$, let $\mathcal{J}_a(K)$ be the set of *supporting pairs* (μ, c), where μ is a Borel probability measure on K and $c \ge 0$, such that

$$u(a) \le \int_K f \, d\mu + c \tag{14.9}$$

whenever $u \in \mathcal{L}_1$, $f \in C(K, \mathbb{R})$ and $u \le f$ quasi-everywhere on K. We call μ a *supporting probability* at a.

Theorem 14.1.4 *For every $a \in \mathbb{C}^n$, $q \in C(K, \mathbb{R})$ and $\delta > 0$,*

$$F_{\delta,q}(a) = \min_{(\mu,c)\in\mathcal{J}_a(K)} \left(\int_K q \, d\mu + \delta c \right). \tag{14.10}$$

In particular, $\mathcal{J}_a(K)$ is nonempty and

$$\min_{(\mu,c)\in\mathcal{J}_a(K)} c = V_K^*(a).$$

Proof On the real Banach space $C(K)$, put $T(f) = F_{1,f}(a)$. By (14.8), T is concave, since convex combinations of competitors are competitors. Comparison of obstacles and translation by constants show that T is nondecreasing and satisfies

$$T(f + t) = T(f) + t, \qquad |T(f) - T(g)| \le \|f - g\|_\infty.$$

Fix $f_0 \in C(K)$. The directional derivative of the convex Lipschitz function $-T$ at f_0 is finite and sublinear. By the Hahn–Banach theorem, there is a continuous linear functional dominated by it, and since difference quotients of a convex function are monotone, this functional supports $-T$ at f_0. Changing the sign, we obtain a continuous linear functional ℓ with

$$T(f) \le T(f_0) + \ell(f - f_0) \qquad (f \in C(K)).$$

Testing $f = f_0 + th$ with $h \ge 0$ and $t > 0$, and using monotonicity, gives $\ell(h) \ge 0$. Testing with $f = f_0 \pm t$ gives $\ell(1) = 1$. By the Riesz representation theorem, $\ell(f) = \int_K f \, d\mu$ for a Borel probability measure μ on K. Set $c = T(f_0) - \int_K f_0 \, d\mu$. The supporting inequality at $f = 0$ gives $c \ge T(0) = V_K^*(a) \ge 0$. Together with (14.8), it shows that $(\mu, c) \in \mathcal{J}_a(K)$ and

$$T(f_0) = \int_K f_0 \, d\mu + c.$$

Conversely, applying (14.9) to $u = F_{1,f_0}$, which lies below f_0 quasi-everywhere on K, shows that $T(f_0) \le \int_K f_0 \, d\mu + c$ for every pair in $\mathcal{J}_a(K)$. This proves (14.10) for

$\delta = 1$, with the minimum attained. The identity $F_{\delta,q} = \delta F_{1,q/\delta}$ gives the general case, and the zero obstacle gives the last assertion. □

Theorem 14.1.5 *If $\psi \in C(K,\mathbb{R})$, then for every $a \in \mathbb{C}^n$ and $\tau > 0$,*

$$D_\tau(a) = \min\left\{c : (\mu, c) \in \mathcal{J}_a(K),\ \int_K \psi\, \mathrm{d}\mu + \tau c = F_\tau(a)\right\}. \tag{14.11}$$

Consequently, $D_\tau(a) = V_K^*(a)$ if and only if there is a Borel probability measure μ on K such that

$$(\mu, V_K^*(a)) \in \mathcal{J}_a(K), \qquad \int_K \psi\, \mathrm{d}\mu = F_\tau(a) - \tau V_K^*(a).$$

Proof By Corollary 14.1.3 below, applied with $q = \psi$, the right derivative $D_\tau(a)$ is the minimum of c over the pairs that are active at (τ, ψ), which is (14.11). Every pair in $\mathcal{J}_a(K)$ has $c \geq V_K^*(a)$ by Theorem 14.1.4, so this minimum equals $V_K^*(a)$ exactly when some active pair has the constant $V_K^*(a)$, which is the stated criterion. □

At a point a of regular contact, $F_\tau(a) = \psi(a)$, so the criterion involves only the original weight and tests by functions in $\mathcal{L}_1$. If the defining tests instead required $u \leq f$ at every point of K, the pair formed by the unit point mass at $a \in K$ and the constant 0 would always be allowed. The resulting infimum would then be at most $\psi(a)$, so it could not detect a defect created by regularization.

14.1.1.2 Variation of continuous weights

Supporting probabilities also describe how the envelope responds to changes of the weight, jointly with changes of the growth allowance. Keep the non-pluripolar compact set $K \subset \mathbb{C}^n$ fixed. For $a \in \mathbb{C}^n$, $\delta > 0$ and $q \in C(K,\mathbb{R})$, let

$$\mathcal{J}_a^{\mathrm{act}}(\delta, q) = \left\{(\mu, c) \in \mathcal{J}_a(K) : \int_K q\, \mathrm{d}\mu + \delta c = F_{\delta,q}(a)\right\} \tag{14.12}$$

be the set of pairs that are active at (δ, q). We give the weights the uniform norm, the probability measures the weak topology, and the constants the usual topology of $\mathbb{R}$.

Lemma 14.1.3 *The set $\mathcal{J}_a^{\mathrm{act}}(\delta, q)$ is nonempty, convex and compact. For $\delta, \eta \geq \epsilon > 0$ and $p, q \in C(K,\mathbb{R})$,*

$$|F_{\eta,p}(a) - F_{\delta,q}(a)| \leq \|p - q\|_\infty + |\eta - \delta|\left(V_K^*(a) + \frac{\operatorname{osc}_K q}{\epsilon}\right). \tag{14.13}$$

If $\delta_j \to \delta > 0$, $q_j \to q$ uniformly and $(\mu_j, c_j) \in \mathcal{J}_a^{\mathrm{act}}(\delta_j, q_j)$, then these pairs have a convergent subsequence, and every subsequential limit belongs to $\mathcal{J}_a^{\mathrm{act}}(\delta, q)$.

Proof Nonemptiness follows from Theorem 14.1.4. Since the test functions are continuous, $\mathcal{J}_a(K)$ is closed and convex, and the equality defining active pairs is a closed affine condition. For an active pair, the defining equality and $\int_K q\,\mathrm{d}\mu \geq \inf_K q$ give the first upper bound below, and (14.3), applied with the weight q, gives the second:

$$0 \leq c \leq \frac{F_{\delta,q}(a) - \inf_K q}{\delta} \leq V_K^*(a) + \frac{\operatorname{osc}_K q}{\delta}.$$

Since the Borel probability measures on the compact metric space K form a weakly compact set, the pairs in $\mathcal{J}_a(K)$ with bounded constants form a compact set; hence the set of active pairs is compact.

With q fixed, Lemma 14.1.2 and (14.3), applied with the weight q, bound $|F_{\eta,q}(a) - F_{\delta,q}(a)|$ by $|\eta - \delta|(V_K^*(a) + \operatorname{osc}_K q/\epsilon)$. Comparison of obstacles and translation by constants bound $|F_{\eta,p}(a) - F_{\eta,q}(a)|$ by $\|p - q\|_\infty$. Adding these bounds gives (14.13).

For the sequence in the statement, the constants c_j have a common bound, by the bound for the constants of active pairs proved above, because the weights q_j are uniformly bounded and the parameters δ_j are bounded below by a positive constant. Weak compactness of probability measures gives a convergent subsequence of pairs, and its limit belongs to $\mathcal{J}_a(K)$, which is closed. The estimate (14.13) gives $F_{\delta_j,q_j}(a) \to F_{\delta,q}(a)$, while uniform convergence of the q_j and weak convergence of the μ_j give $\int_K q_j\,\mathrm{d}\mu_j \to \int_K q\,\mathrm{d}\mu$. Passing to the limit in the equality defining active pairs proves the last assertion. □

Proposition 14.1.7 *For $r \in C(K,\mathbb{R})$ and $s \in \mathbb{R}$, one has*

$$\lim_{h\downarrow 0} \frac{F_{\delta+hs,q+hr}(a) - F_{\delta,q}(a)}{h} = \min_{(\mu,c)\in\mathcal{J}_a^{\mathrm{act}}(\delta,q)} \left(\int_K r\,\mathrm{d}\mu + sc \right). \tag{14.14}$$

The same limit holds along any sequence $h_j \downarrow 0$ if s, r in the difference quotient are replaced by s_j, r_j, where $s_j \to s$ and $r_j \to r$ uniformly.

Proof By compactness of the set of active pairs, the minimum on the right is attained. For a fixed pair (μ, c) active at (δ, q), the representation (14.10) at the perturbed parameters gives

$$F_{\delta+hs,q+hr}(a) \leq F_{\delta,q}(a) + h\left(\int_K r\,\mathrm{d}\mu + sc \right).$$

Choosing an active pair that attains the minimum on the right, we see that the upper limit of the difference quotients is at most this minimum. Conversely, choose a pair (μ_h, c_h) active at $(\delta + hs, q + hr)$. The representation at (δ, q) gives

$$\frac{F_{\delta+hs,q+hr}(a) - F_{\delta,q}(a)}{h} \geq \int_K r\,\mathrm{d}\mu_h + sc_h.$$

The estimate (14.13) bounds the difference quotients. Along a sequence of values of h realizing their lower limit, Lemma 14.1.3 gives a subsequence of pairs converging

to a pair active at (δ, q). For this pair, $\int_K r\,\mathrm{d}\mu + sc$ is at least the minimum on the right. This proves (14.14).

For convergent directions, the same two inequalities hold. The fixed active pair attaining the minimum gives an upper bound with the right limit. The active pairs at the perturbed parameters still have uniformly bounded constants, and their subsequential limits are active at (δ, q). Uniform convergence of the r_j controls the integrals $\int_K r_j\,\mathrm{d}\mu_{h_j}$, and $s_j \to s$ controls the terms $s_j c_{h_j}$. The lower bound therefore has the same limit. Note that all perturbed growth allowances $\delta + h_j s_j$ are positive for small h_j. □

Corollary 14.1.3 *For fixed a and q, the function $\delta \mapsto F_{\delta,q}(a)$ has finite left and right derivatives $D^-_{\delta,q}(a)$ and $D^+_{\delta,q}(a)$ at every positive parameter, with*

$$D^-_{\delta,q}(a) = \max_{\mathcal{J}_a^{\mathrm{act}}(\delta,q)} c, \qquad D^+_{\delta,q}(a) = \min_{\mathcal{J}_a^{\mathrm{act}}(\delta,q)} c. \tag{14.15}$$

It is differentiable exactly when all active pairs have the same constant c. For each fixed a, q, the set of positive parameters where it fails to be differentiable is at most countable.

Proof Apply (14.14) with $r = 0$ and $s = \pm 1$. The direction $s = -1$ gives minus the left derivative, which proves (14.15). Concavity gives

$$D^+_{\alpha,q}(a) \geq D^-_{\beta,q}(a) \qquad (0 < \alpha < \beta).$$

Hence the open intervals $(D^+_{\delta,q}(a), D^-_{\delta,q}(a))$, which are nonempty exactly at the parameters where differentiability fails, are pairwise disjoint. Choosing a rational number in each of them proves countability. □

Corollary 14.1.4 *For fixed a and $\delta > 0$, the map $q \mapsto F_{\delta,q}(a)$ on $C(K, \mathbb{R})$ is Gâteaux differentiable at q if and only if all pairs active at (δ, q) have the same probability measure μ. In that case its derivative is*

$$r \longmapsto \int_K r\,\mathrm{d}\mu.$$

Proof With $s = 0$, (14.14) shows that the right derivative in the direction r is the minimum of $\int_K r\,\mathrm{d}\mu$ over active pairs, and the direction $-r$ shows that the left derivative is the maximum. If all active pairs have the same probability measure, these values agree and define a continuous linear derivative. Conversely, if the map is Gâteaux differentiable, the minimum and the maximum agree for every continuous r. Continuous functions separate Borel probability measures on K, so all active pairs have the same probability measure; their constant is then determined by (14.12), since $\delta > 0$. □

Example 14.1.1 Let $D \subset \mathbb{C}$ be the open unit disk, let $K = \overline{D}$, and fix $\delta > 0$. For $q_t(z) = -t|z|^2$,

$$F_{\delta,q_t}(0) = \begin{cases} -t, & t \geq 0, \\ 0, & -\delta/2 \leq t \leq 0. \end{cases} \tag{14.16}$$

Thus the map $q \mapsto F_{\delta,q}(0)$ is not Gâteaux differentiable at $q = 0$, and there are active pairs at $(\delta, 0)$ with different probability measures.

Proof For $t \geq 0$, each competitor is at most $-t$ on the unit circle, so the maximum principle for subharmonic functions bounds it by $-t$ on D. This bound persists after regularization on D, and the constant $-t$ is a competitor. Hence $F_{\delta,q_t}(0) = -t$.

For $t = -b$ with $0 \leq b \leq \delta/2$, define

$$u_b(z) = \begin{cases} b|z|^2, & |z| \leq 1, \\ b + \delta \log |z|, & |z| > 1. \end{cases}$$

This function is continuous and subharmonic: its Laplacian has density $4b$ inside the disk, density zero outside, and a nonnegative contribution on the circle coming from the jump $\delta - 2b \geq 0$ of the radial derivative. It belongs to $\mathcal{L}_\delta$ and equals q_{-b} on K. Hence $F_{\delta,q_{-b}}(0) \geq u_b(0) = 0$. On D the raw envelope is at most the continuous function $b|z|^2$, so its regularization at zero is at most zero. This proves (14.16). In particular, the right and left derivatives in t at $t = 0$ are -1 and 0, and Corollary 14.1.4, applied in the direction $r = -|z|^2$, gives the final assertion. □

14.1.1.3 Local regularity and strict weights

Whether the regularized envelope touches the obstacle at a point $a \in K$ depends only on the parts of K near a. Without weight, this local regularity is detected by the relative extremal function $h_{E,B}$ of Definition 1.6.1, for compact pieces E of K near a and small balls B centered at a. If the weight extends to a global function whose complex Hessian has a strict lower bound near a, a failure of this regularity produces a quantitative defect of the weighted envelope.

For a non-pluripolar compact set $E \subset \mathbb{C}^n$, write V_E and V_E^* for the raw and regularized envelopes of growth one with zero obstacle on E; thus $V_K = V_{1,0}$ as before. For other weights and growth allowances, write $F_{\delta,E,q}$ for the regularized envelope with growth allowance δ and obstacle q imposed at every point of E.

The pluripolar case of Lemma 14.1.5 below uses functions of logarithmic growth with poles on a given bounded pluripolar set.

Lemma 14.1.4 *For every bounded pluripolar set $E \subset \mathbb{C}^n$, there is $b \in \mathcal{L}_1$ with $b = -\infty$ on E.*

Proof The set E is pluripolar as a subset of $\mathbb{P}^n$: it is locally pluripolar near every affine point, and a neighborhood of the hyperplane at infinity does not meet it. By Josefson's theorem on compact manifolds, [Xia26, Theorem 1.1.5], there is a quasi-plurisubharmonic function on $\mathbb{P}^n$, not identically $-\infty$, that equals $-\infty$ on E. A small positive multiple p of it belongs to $\mathrm{PSH}(\mathbb{P}^n, \theta_1)$, and $b = p + \chi/2$ belongs to $\mathcal{L}_1$ by Lemma 14.1.1. □

Lemma 14.1.5 *Let $a \in \mathbb{C}^n$, $0 < r < R$ and $B = B(a, R)$, and let $E \subseteq \overline{B}(a, r)$ be compact. If E is pluripolar, then $h^*_{E,B} = 0$ on B. If E is non-pluripolar, then*

$$h^*_{E,B}(a) = -1 \quad \Longleftrightarrow \quad V^*_E(a) = 0.$$

Proof In the pluripolar case, choose $b \in \mathcal{L}_1$ with $b = -\infty$ on E by Lemma 14.1.4, and a finite upper bound C for b on B. The functions

$$u_\epsilon = \max\{-1, \epsilon(b - C)\}$$

are competitors for $h_{E,B}$, and they tend to zero almost everywhere as $\epsilon \downarrow 0$. Hence $h^*_{E,B} \geq 0$ almost everywhere, and then everywhere by Lemma 1.3.3. Since $h^*_{E,B} \leq 0$, it vanishes.

Now suppose that E is non-pluripolar. Proposition 14.1.5 gives a finite upper bound $M \geq 1$ for V^*_E on B. For each competitor v for V_E, the function $\max(-1, v/M - 1)$ is a competitor for $h_{E,B}$. Taking suprema and regularizing gives

$$h^*_{E,B}(a) \geq \max\{-1, V^*_E(a)/M - 1\}.$$

As $V^*_E \geq 0$, the equality $h^*_{E,B}(a) = -1$ forces $V^*_E(a) = 0$.

For the converse, choose $r < R_0 < R$ and $k > 0$ with $k \log(R_0/r) > 1$. The plurisubharmonic function

$$L(z) = k \log\big(\|z - a\|/r\big)$$

belongs to $\mathcal{L}_k$ and equals $-\infty$ at a. For a competitor u for $h_{E,B}$, define $W = \max(u + 1, L)$ on B and $W = L$ outside B. On the collar $R_0 < \|z - a\| < R$ we have $L > 1 \geq u + 1$, so the two definitions agree there, and W is plurisubharmonic by Lemma 1.3.2. On E one has $u = -1$ and $L \leq 0$, so $W = 0$; on the ball $B(a, r)$ one has $L < 0 \leq u + 1$, so $W = u + 1$. Hence W/k is a competitor for V_E, with k independent of u. Taking the supremum over u and regularizing at a gives

$$V^*_E(a) \geq (1 + h^*_{E,B}(a))/k.$$

Thus $V^*_E(a) = 0$ forces $h^*_{E,B}(a) = -1$. □

Let $K \subset \mathbb{C}^n$ again be a non-pluripolar compact set. For $a \in K$ and $r > 0$, put $K_r = K \cap \overline{B}(a, r)$.

Definition 14.1.1 The compact set K is *locally pluriregular* at $a \in K$ if every K_r, $r > 0$, is non-pluripolar and $V^*_{K_r}(a) = 0$. For $\delta > 0$ and a bounded function q on K, it is *locally (δ, q)-regular* at a if $F_{\delta,K_r,q}(a) = q(a)$ for every $r > 0$.

For local pluriregularity it suffices to check arbitrarily small radii, since envelopes decrease as the compact set increases. For local (δ, q)-regularity, monotonicity gives only $F_{\delta,K_s,q}(a) \leq F_{\delta,K_r,q}(a)$ for $r < s$, and the definition requires every radius. By Lemma 14.1.5, K is locally pluriregular at a if and only if $h^*_{K_r,B(a,R)}(a) = -1$ for all small $0 < r < R$.

Lemma 14.1.6 *Let q be bounded and upper semicontinuous on K, and suppose that raw contact holds for the parameter $\delta > 0$ and the obstacle q on K, that is, $V_{\delta,q} = q$ on K. If K is locally pluriregular at a, then K is locally (η, q)-regular at a for every $\eta \geq \delta$; in particular, $F_{\eta,K,q}(a) = q(a)$.*

Proof Raw contact also holds for every $\eta \geq \delta$, since $V_{\eta,q}$ increases with η and is at most q on K; so it suffices to treat $\eta = \delta$. Fix $s > 0$. The compact set K_s inherits raw contact: on K_s, its raw envelope is at least the raw envelope for K, which equals q, and at most the obstacle q. For $0 < r < s$ put $M_r = \sup_{K_r} q$. Raw contact on K_s, monotonicity of envelopes with respect to the compact set, and (14.3) for K_r give

$$q(a) \leq F_{\delta,K_s,q}(a) \leq F_{\delta,K_r,q}(a) \leq M_r + \delta V^*_{K_r}(a) = M_r.$$

Since q is upper semicontinuous and $a \in K_r$, we have $M_r \downarrow q(a)$ as $r \downarrow 0$, which gives $F_{\delta,K_s,q}(a) = q(a)$. □

Theorem 14.1.6 *Let $a \in K$, $\tau > 0$ and $\Psi \in \mathcal{L}_\tau \cap C(\mathbb{C}^n)$, and put $q = \Psi|_K$. Suppose that, for some $R, c > 0$,*

$$\mathrm{dd^c}\Psi \geq c\omega_0 \quad \text{on } B(a,R), \qquad \omega_0 = \mathrm{dd^c}\|z\|^2.$$

For $0 < r < R$ and $\delta \geq \tau$,

$$F_{\delta,K,q}(a) - \Psi(a) \geq cr^2\big(1 + h^*_{K_r,B(a,R)}(a)\big). \tag{14.17}$$

Regular contact at a, that is, $F_{\delta,K,q}(a) = q(a)$, holds for some $\delta \geq \tau$ if and only if K is locally pluriregular at a. In that case it holds for every $\delta \geq \tau$.

Proof Set $B = B(a,R)$ and $E = K_r$. For a competitor u for $h_{E,B}$, the function

$$H_u(z) = \Psi(z) - c\|z-a\|^2 + cr^2(u(z)+1)$$

is plurisubharmonic on B, because the assumption on $\mathrm{dd^c}\Psi$ makes $\Psi - c\|z-a\|^2$ plurisubharmonic there. Since $u \leq 0$, we have $H_u \leq \Psi$ where $\|z-a\| \geq r$. Define $W = \max(\Psi, H_u)$ on B and $W = \Psi$ outside B. The two definitions agree near ∂B, so W is plurisubharmonic on $\mathbb{C}^n$, and it belongs to $\mathcal{L}_\tau$ because it equals Ψ outside B. On E, the equality $u = -1$ gives $H_u \leq \Psi$, and on $K \setminus E$ the distance bound gives the same inequality. Thus $W = q$ on K, so W is a competitor for $F_{\delta,K,q}$ for each $\delta \geq \tau$. Since $W \geq H_u$ near a, taking the supremum over u, regularizing, and using the continuity of Ψ, we obtain (14.17).

The function Ψ itself is a competitor, so raw contact holds for every $\delta \geq \tau$. If regular contact holds for one such δ, the left-hand side of the estimate vanishes, so $h^*_{K_r,B(a,R)}(a) = -1$ for all $0 < r < R$. By Lemma 14.1.5, every K_r with $r < R$ is then non-pluripolar and satisfies $V^*_{K_r}(a) = 0$; for $r \geq R$ this follows by monotonicity. Conversely, local pluriregularity and Lemma 14.1.6 give regular contact at a for every $\delta \geq \tau$. □

In this regular case, $V^*_K(a) \le V^*_{K_r}(a) = 0$ by monotonicity, and the right derivative $D_\tau(a)$ with respect to the growth allowance, for the weight q, vanishes by Corollary 14.1.2, since regular contact persists.

Corollary 14.1.5 *Let $\tau > 0$, $\Psi \in \mathcal{L}_\tau \cap C^2(\mathbb{C}^n)$ and $a \in K$. If $F_{\tau,K,\Psi|_K}(a) = \Psi(a)$, but $F_{\delta,K,\Psi|_K}(a) > \Psi(a)$ for some $\delta > \tau$, then*

$$(\mathrm{dd}^{\mathrm{c}}\Psi)^n(a) = 0.$$

Proof The complex Hessian of Ψ is semipositive. If its top power did not vanish at a, it would be positive definite at a, and by continuity $\mathrm{dd}^{\mathrm{c}}\Psi \ge c\omega_0$ on a neighborhood of a for some $c > 0$. Theorem 14.1.6 would then give regular contact at a for every $\delta \ge \tau$, a contradiction. □

14.1.1.4 A regularization defect along a complex disk

Raw contact with a constant obstacle can hold for every positive growth parameter while regularization raises the value of the envelope at a point of the compact set. In the following example, a complex disk through the point, transverse to the planar direction, prevents any local plurisubharmonic function from separating the point from the rest of the compact set, but it does not remove this defect.

In the complex plane, let D be the open unit disk and put

$$p_j = 2 + 2^{-j}, \qquad r_j = \exp(-16\,2^j), \qquad E = \overline{D} \cup \{2\} \cup \bigcup_{j\ge 1} \overline{D}(p_j, r_j).$$

Lemma 14.1.7 *The set E is compact, and every neighborhood of each of its points contains an open disk contained in E. The series*

$$h(\zeta) = \sum_{j\ge 1} 2^{-j} \log|\zeta - p_j| + 2\log 2$$

defines a subharmonic function on $\mathbb{C}$ of logarithmic growth at most one; it satisfies $h(2) = 0$ and is at most $-16 + 2\log 2$ on every small disk $\overline{D}(p_j, r_j)$. The function

$$v(\zeta) = \frac{8\log^+|\zeta| + h(\zeta) - 4\log 2}{9}$$

is subharmonic of logarithmic growth at most one, is nonpositive on $E \setminus \{2\}$, and satisfies $v(2) = 4\log 2/9$.

Proof Since $r_j < 2^{-j-3}$, the small disks are pairwise disjoint and disjoint from $\overline{D}$, and they accumulate only at 2. This proves compactness. Every neighborhood of 2 contains one of the small disks, and every other point of E lies in the closure of the interior of one of the disks.

Fix $R > 0$. On the disk $\{|\zeta| < R\}$ every term satisfies $\log|\zeta - p_j| < \log(R+3)$, so the partial sums of $\sum_j 2^{-j}(\log|\zeta - p_j| - \log(R+3))$ decrease there. Their limit is therefore subharmonic or identically $-\infty$ on this disk, and it is finite at $\zeta = 0$. Since $\sum_j 2^{-j} = 1$, this limit is $h - 2\log 2 - \log(R+3)$, so h is subharmonic on $\mathbb{C}$. The bound $|\zeta - p_j| \le |\zeta| + 3$ gives $h(\zeta) \le \log(|\zeta|+3) + 2\log 2$, hence the growth bound, and $\sum_j j2^{-j} = 2$ gives $h(2) = 0$.

On the jth small disk, all distances to the other centers are less than one, so the corresponding terms are negative, and

$$h \le 2^{-j}\log r_j + 2\log 2 = -16 + 2\log 2.$$

On $\overline{D}$ all distances to the centers are less than four, hence $h \le 4\log 2$ and $v \le 0$. On each small disk $|\zeta| < 4$, so $9v \le 14\log 2 - 16 < 0$. The growth bound for v and its value at 2 follow directly from the formula. □

Now let $n \ge 2$ and $K = E \times \overline{D}^{n-1} \subset \mathbb{C}^n$. For a non-pluripolar compact set $L \subset \mathbb{C}^n$, write $V_{\delta,L}$ and $F_{\delta,L}$ for the raw envelope with growth allowance δ and zero obstacle on L, and for its regularization; thus $F_{\delta,L} = F_{\delta,L,0}$ in the notation of Section 14.1.1.3.

Proposition 14.1.8 *For every point of K and every neighborhood W of it, the set $K \cap W$ is non-pluripolar. For $\delta > 0$ and $a = (2, w)$ with $w \in \overline{D}^{n-1}$,*

$$V_{\delta,K}|_K = 0, \qquad F_{\delta,K}(a) \ge \frac{4\delta}{9}\log 2 > 0. \tag{14.18}$$

For every $r > 0$, the same lower bound holds for $F_{\delta,K\cap\overline{B}(a,r)}(a)$. At $a_0 = (2, 0, \dots, 0)$, every plurisubharmonic function f defined near a_0 satisfies

$$f(a_0) \le \limsup_{K\setminus\{a_0\}\ni z\to a_0} f(z). \tag{14.19}$$

Thus no plurisubharmonic function near a_0 separates a_0 from $K \setminus \{a_0\}$, that is, K is not plurithin at a_0. Nevertheless, the regularized envelope does not touch the obstacle at a_0 for any positive growth parameter.

Proof By the preceding lemma, every neighborhood of a point of K contains a nonempty open subset of K, namely a product of open disks. This proves the first assertion, which applies in particular to each $K \cap \overline{B}(a, r)$, $r > 0$.

The plurisubharmonic function $z \mapsto v(z_1)$ belongs to $\mathcal{L}_1$ and is nonpositive on K outside the pluripolar set $K \cap \{z_1 = 2\}$. By Corollary 14.1.1 with zero obstacle, $v(z_1) \le F_{1,K}(z)$ for all z. The value $v(2) = 4\log 2/9$ gives the lower bound for $\delta = 1$. Since 0 is a competitor, $V_{\delta,K}|_K = 0$, and scaling of competitors gives $F_{\delta,K} = \delta F_{1,K}$. Replacing K by a smaller compact set only increases the envelope, which proves the local assertion.

Finally, let f be plurisubharmonic near a_0, and restrict it to the complex line through a_0 in which only the second coordinate varies. Small circles in this line centered at a_0 lie in $K \setminus \{a_0\}$, and the submean inequality on these circles gives (14.19). □

14.2 Bounded domination in a semipositive background

We return to the compact manifold X.

The domination principle for Hermitian backgrounds, Theorem 11.2.2, extends to a smooth semipositive background θ provided every bounded θ-plurisubharmonic potential has positive mass. For bounded potentials u, v satisfying the measure inequality below, contact support bounds $\sup_X P_\theta(bu - (b-1)v)$ from below as the stretching parameter $b > 1$ increases. A fixed gap between u and v would then force the L^1 norms of these envelopes, normalized by their suprema, to diverge, contradicting compactness.

We call a smooth semipositive form θ *non-collapsing on bounded potentials* if

$$\int_X \mathrm{MA}_\theta(w) > 0 \quad \text{for every } w \in \mathrm{PSH}(X,\theta) \cap L^\infty(X).$$

Every Hermitian form has this property by Corollary 11.2.1. Bigness also implies it, by the following envelope argument.

Proposition 14.2.1 *If a smooth real $(1,1)$-form θ is big, every bounded $u \in \mathrm{PSH}(X,\theta)$ has strictly positive Monge–Ampère mass.*

Proof Choose $\rho \in \mathrm{PSH}(X,\theta)$ and $a > 0$ with $\theta_\rho \geq a\omega$, normalize $\rho \leq 0$, and put

$$p = P_{a\omega}(u - \rho).$$

The obstacle is finite almost everywhere and bounded below, so Lemma 1.19.2 shows that p is bounded. The obstacle is quasi-continuous off the pluripolar pole set of ρ. The contact theorem, Theorem 14.1.2, applied in the Hermitian background $a\omega$, shows that $\mathrm{MA}_{a\omega}(p)$ is carried by $D = \{p + \rho = u\}$, outside the pole set of ρ.

By Lemma 1.3.3, $p + \rho \leq u$ everywhere, and

$$\theta + \mathrm{dd^c}(p+\rho) = (a\omega + \mathrm{dd^c}p) + (\theta_\rho - a\omega) \geq a\omega + \mathrm{dd^c}p \geq 0.$$

On common finite truncation levels, positivity of the mixed products and ordered contact give

$$\mathbf{1}_D \mathrm{MA}_{a\omega}(p) \leq \mathbf{1}_D \mathrm{MA}_\theta(p+\rho) \leq \mathbf{1}_D \mathrm{MA}_\theta(u).$$

Exhausting by these levels proves the same inequalities for the extended measures. Since p is bounded, its Monge–Ampère mass with respect to the Hermitian form $a\omega$ is positive by Corollary 11.2.1. Integrating the inequalities proves the assertion. □

Theorem 14.2.1 (Guedj–Lu) *Let θ be non-collapsing on bounded potentials. Suppose that $u, v \in \mathrm{PSH}(X,\theta) \cap L^\infty(X)$ and $0 \le c < 1$. If*

$$\mathbf{1}_{\{u<v\}}\mathrm{MA}_\theta(u) \le c\,\mathbf{1}_{\{u<v\}}\mathrm{MA}_\theta(v),$$

then $u \ge v$.

Proof If the conclusion failed, we could choose $a > 0$ such that $E = \{u < v - a\}$ is nonempty. This set has positive volume by Lemma 1.3.4. Fix C with $|u|, |v| \le C$.

For $b > 1$ define the bounded envelope

$$w_b = P_\theta\big(bu - (b-1)v\big), \qquad D_b = \{w_b = bu - (b-1)v\}.$$

Since $\theta \ge 0$, constants below the obstacle are competitors. By the bounded contact theorem, Theorem 14.1.1, the measure $\mathrm{MA}_\theta(w_b)$ is concentrated on D_b. The potential $b^{-1}w_b + (1-b^{-1})v$ lies below u quasi-everywhere, hence everywhere by Lemma 1.3.3, and it equals u on D_b outside a pluripolar set. Ordered contact and positivity of the mixed terms give

$$\mathbf{1}_{D_b}\left[b^{-n}\mathrm{MA}_\theta(w_b) + (1-b^{-1})^n\mathrm{MA}_\theta(v)\right] \le \mathbf{1}_{D_b}\mathrm{MA}_\theta(u).$$

Choose b so large that $(1-b^{-1})^n > c$. Then the hypothesis forces $\mathrm{MA}_\theta(w_b) = 0$ on $D_b \cap \{u < v\}$. Since θ is non-collapsing, w_b has positive total mass, so some of this mass is carried by $D_b \cap \{u \ge v\}$. At a point of this set outside the exceptional pluripolar set,

$$w_b = bu - (b-1)v = u + (b-1)(u-v) \ge -C.$$

Consequently $\sup_X w_b \ge -C$ for all large b.

On the other hand, on the fixed set E the obstacle bound gives $w_b \le C - (b-1)a$ quasi-everywhere. Hence

$$\int_X |w_b - \sup_X w_b|\,\mathrm{d}V \ge \big((b-1)a - 2C\big)\,\mathrm{vol}_{\mathrm{d}V}(E)$$

for all sufficiently large b. This contradicts the uniform L^1 bound for normalized potentials in Theorem 1.4.1. □

The exponential comparison principle follows as in Corollary 11.2.2. For unbounded potentials, the stretched envelopes must also be kept in the required singularity class; this is the role of the relative classes introduced below.

14.3 Upper and lower volumes

Let θ be any smooth real $(1,1)$-form. Taking the infimum and the supremum of the mass over a singularity type gives two invariants with values in $[0,+\infty]$. They can differ, because a bounded change of potential need not preserve the mass.

Definition 14.3.1 For $u \in \mathrm{PSH}(X,\theta)$, define

$$\underline{\mathrm{vol}}_\theta(u) = \inf_{v\in\mathrm{PSH}(X,\theta),\ v\sim u} \int_X \mathrm{MA}_\theta(v),$$
$$\overline{\mathrm{vol}}_\theta(u) = \sup_{v\in\mathrm{PSH}(X,\theta),\ v\sim u} \int_X \mathrm{MA}_\theta(v).$$

These are the *lower* and *upper volumes* of the singularity type of u in the background θ. When $\mathrm{PSH}(X,\theta)$ is nonempty, we write $\underline{\mathrm{vol}}(\theta)$ and $\overline{\mathrm{vol}}(\theta)$ for their values at the potential V_θ with minimal singularities.

For semipositive θ, the singularity type of the zero potential consists of the bounded potentials. Extending the lower mass of Definition 7.5.1 to this background, we put

$$v_-(\theta) = \underline{\mathrm{vol}}_\theta(0) = \inf_{v\in\mathrm{PSH}(X,\theta)\cap L^\infty(X)} \int_X \mathrm{MA}_\theta(v).$$

The condition $v_-(\theta) > 0$ is stronger than the non-collapsing property used in bounded domination: it requires a single positive lower bound for the masses of all bounded potentials. Singularity volumes under a bounded mass hypothesis are compared in Chapter 31.

Both volumes are invariant under bounded changes of u, by definition. They are also invariant under the smooth change of decomposition $(\theta,u) \mapsto (\theta + \mathrm{dd}^{\mathrm{c}} f, u - f)$, for $f \in C^\infty(X,\mathbb{R})$: the map $v \mapsto v - f$ identifies the competitors, their currents and their measures. Scaling gives

$$\underline{\mathrm{vol}}_{t\theta}(tu) = t^n \underline{\mathrm{vol}}_\theta(u), \qquad \overline{\mathrm{vol}}_{t\theta}(tu) = t^n \overline{\mathrm{vol}}_\theta(u) \quad (t>0).$$

Pullback by a modification maps the singularity type of u onto that of its pullback. Together with the following identity of measures, this shows that both volumes are preserved when the background is pulled back as well.

Proposition 14.3.1 *Let $\pi : Y \to X$ be a modification of connected compact complex manifolds, let θ be a smooth real $(1,1)$-form on X, and let $u \in \mathrm{PSH}(X,\theta)$. Then*

$$\pi_* \mathrm{MA}_{\pi^*\theta}(u \circ \pi) = \mathrm{MA}_\theta(u).$$

Both singularity volumes satisfy

$$\underline{\mathrm{vol}}_{\pi^*\theta}(u \circ \pi) = \underline{\mathrm{vol}}_\theta(u), \qquad \overline{\mathrm{vol}}_{\pi^*\theta}(u \circ \pi) = \overline{\mathrm{vol}}_\theta(u),$$

including infinite values.

Proof Choose a proper analytic set $S \subset X$ such that π restricts to a biholomorphism from $Y \setminus D$ onto $X \setminus S$, where $D = \pi^{-1}(S)$. On these complements, the truncations satisfy $\max(u \circ \pi, -t) = \max(u, -t) \circ \pi$, and Bedford–Taylor products of bounded potentials are preserved by the biholomorphic change of coordinates. Passing to the increasing limits in (7.1) gives the measure identity there. By Theorem 7.1.1, the two measures give no mass to the pluripolar sets S and D, which proves the global identity and the equality of total masses.

Pullback is a bijection between the potentials in the two backgrounds by Proposition 1.22.4. Since π is surjective, an inequality $v \le u + C$ holds on X if and only if $v \circ \pi \le u \circ \pi + C$ holds on Y. Hence this bijection maps the singularity type of u onto that of $u \circ \pi$, which identifies the competitors in the definitions of the two pairs of volumes. Their masses agree by the first assertion. Taking their infima and suprema proves the volume identities. □

The background on Y is the pullback $\pi^*\theta$ itself, which may be degenerate even when $\theta > 0$. These identities do not compare volumes for an independently chosen Hermitian form on Y.

14.4 Relative envelope classes

Fix a reference potential $\varphi \in \mathrm{PSH}(X,\theta)$. For $b \ge 1$, the stretched obstacle $bu - (b-1)\varphi = \varphi + b(u - \varphi)$ magnifies the difference between u and φ by the factor b, and so becomes more singular where u lies below φ. Requiring a competitor below every finite stretch defines a relative class that is stable under convex combinations and finite rooftops.

For $u \in \mathrm{PSH}(X,\theta)$ and $b \ge 1$, define the family of competitors

$$\mathcal{A}_b(u,\varphi) = \left\{ h \in \mathrm{PSH}(X,\theta) : b^{-1}h + (1 - b^{-1})\varphi \le u \right\}.$$

For $b = 1$ the inequality means $h \le u$. Away from the poles of u and φ, the inequality means $h \le bu - (b-1)\varphi$. The convex combination on the left is defined everywhere, so the definition does not require a value of $bu - (b-1)\varphi$ at common poles. When $\mathcal{A}_b(u,\varphi)$ is nonempty, we write

$$Q_b(u,\varphi) = P_\theta(bu - (b-1)\varphi)$$

for its upper envelope. Since the obstacle is finite almost everywhere, the competitors have a common upper bound by Section 1.19, so $Q_b(u,\varphi)$ is then a θ-plurisubharmonic function; in particular, it is not identically $-\infty$. The upper-envelope theorem gives its defining inequality almost everywhere, and Lemma 1.3.3 makes it pointwise; thus $Q_b(u,\varphi)$ is the largest member of $\mathcal{A}_b(u,\varphi)$.

Definition 14.4.1 The *relative envelope class* of φ is

$$\mathcal{E}_{\mathrm{env}}(X,\theta,\varphi) = \{u \in \mathrm{PSH}(X,\theta) : u \preceq \varphi,\ \mathcal{A}_b(u,\varphi) \neq \varnothing \text{ for every } b \geq 1\}.$$

When the background θ is fixed, we abbreviate $\mathcal{E}_\varphi = \mathcal{E}_{\mathrm{env}}(X,\theta,\varphi)$.

Proposition 14.4.1 *The class $\mathcal{E}_{\mathrm{env}}(X,\theta,\varphi)$ is convex, contains all potentials of the singularity type of φ, and is stable under bounded changes of its members. If u belongs to it, $v \in \mathrm{PSH}(X,\theta)$, $u \leq v$ and $v \preceq \varphi$, then v belongs to it. In particular, it is stable under maxima of two members. Replacing φ by another member of $\mathrm{PSH}(X,\theta)$ of the same singularity type does not change the class. For every smooth real function f,*

$$u \in \mathcal{E}_{\mathrm{env}}(X,\theta,\varphi) \quad\Longleftrightarrow\quad u - f \in \mathcal{E}_{\mathrm{env}}(X,\theta + \mathrm{dd^c} f, \varphi - f).$$

Proof If $u \geq \varphi - C$, then $h = \varphi - bC$ belongs to $\mathcal{A}_b(u,\varphi)$ for each b; hence every $u \sim \varphi$ belongs to the class. More generally, if $u' \geq u - C$ and $h \in \mathcal{A}_b(u,\varphi)$, then $h - bC \in \mathcal{A}_b(u',\varphi)$. Since the condition $u \preceq \varphi$ is also unchanged by bounded changes of u, the class is stable under such changes. A convex combination of two competitors is a competitor for the same convex combination of the two potentials, which proves convexity. Finally, a competitor for u is a competitor for any larger v, and the assumption $v \preceq \varphi$ gives the remaining condition.

If $\varphi' \in \mathrm{PSH}(X,\theta)$ and $\varphi' \sim \varphi$, choose $C \geq 0$ with $|\varphi' - \varphi| \leq C$ off their common pole set. For $h \in \mathcal{A}_b(u,\varphi)$, the function $h - (b-1)C$ belongs to $\mathcal{A}_b(u,\varphi')$: the required inequality holds off the poles, and it extends everywhere by Lemma 1.3.3. The reverse implication follows by interchanging the references, and $u \preceq \varphi$ is equivalent to $u \preceq \varphi'$.

For the last assertion, note that $h \mapsto h - f$ is a bijection from $\mathrm{PSH}(X,\theta)$ onto $\mathrm{PSH}(X,\theta + \mathrm{dd^c} f)$. It preserves the order $\preceq$ and transforms each stretching inequality into

$$b^{-1}(h - f) + (1 - b^{-1})(\varphi - f) \leq u - f.$$

This inequality is equivalent to the original one also at poles, because f is smooth and finite. Thus the corresponding families of competitors are in bijection for every $b \geq 1$. □

Lemma 14.4.1 *If $u \in \mathcal{E}_\varphi$ and $u \leq \varphi$, then $Q_b(u,\varphi) \in \mathcal{E}_\varphi$ for every $b \geq 1$. More generally, if $u, w \in \mathcal{E}_\varphi$ and $u \leq w$, then $Q_b(u,w) \in \mathcal{E}_\varphi$.*

Proof Fix $b, s \geq 1$ and put $h = Q_{bs}(u,\varphi)$. Its defining inequality gives, off the pole sets,

$$s^{-1}h + (1 - s^{-1})\varphi \le bu - (b-1)\varphi.$$

The left-hand side is θ-plurisubharmonic, so it is at most $Q_b(u,\varphi)$. Hence $h \in \mathcal{A}_s(Q_b(u,\varphi),\varphi)$ for every s. Also $Q_b(u,\varphi) \le u \le \varphi$, so $Q_b(u,\varphi) \preceq \varphi$. The inequalities between convex combinations extend across the poles by Lemma 1.3.3, which proves the first assertion.

For the second assertion, choose a constant C such that $u - C \le \varphi$ and $w - C \le \varphi$. Since $Q_b(u-C, w-C) = Q_b(u,w) - C$ and class membership is unchanged by translation, we may assume $u, w \le \varphi$. Because $u \le w \le \varphi$, the obstacle for $Q_b(u,w)$ lies above that for $Q_b(u,\varphi)$ and below u. Thus

$$Q_b(u,\varphi) \le Q_b(u,w) \le u \le \varphi.$$

Since $Q_b(u,\varphi) \in \mathcal{E}_\varphi$ by the first assertion, the stability under increase proved in Proposition 14.4.1 gives the conclusion. □

Proposition 14.4.2 *If $u_1, \dots, u_N \in \mathcal{E}_\varphi$, $N \ge 1$, then the rooftop envelope $P_\theta(\min_i u_i)$ is not identically $-\infty$ and belongs to $\mathcal{E}_\varphi$.*

Proof Adding a common constant to all u_i, we may assume $u_i \le \varphi$. By Lemma 14.4.1 and Proposition 14.4.1,

$$h = \frac{1}{N}\sum_{i=1}^{N} Q_N(u_i,\varphi) \in \mathcal{E}_\varphi.$$

Each summand is at most φ. For every i, the defining inequality of $Q_N(u_i,\varphi)$ therefore gives

$$h \le N^{-1}Q_N(u_i,\varphi) + (1 - N^{-1})\varphi \le u_i.$$

These inequalities between convex combinations also hold at common poles. Thus h is a competitor for $p = P_\theta(\min_i u_i)$, so p is not identically $-\infty$, and $h \le p \le u_1 \preceq \varphi$. Stability under increase gives $p \in \mathcal{E}_\varphi$, and undoing the translation preserves this conclusion. □

The finiteness of the family is essential: every $\varphi - j$ belongs to $\mathcal{E}_\varphi$, whereas $P_\theta(\inf_{j\ge1}(\varphi - j)) \equiv -\infty$. The constructions of common envelopes for sequences in Chapter 33 require additional estimates.

Proposition 14.4.3 *If $u \in \mathcal{E}_\varphi$, then for every smooth real function F,*

$$P_{\theta,F}[u] = P_{\theta,F}[\varphi].$$

In particular, $P_\theta[u] = P_\theta[\varphi]$.

Proof The relation $u \preceq \varphi$ gives $P_{\theta,F}[u] \le P_{\theta,F}[\varphi]$. For the reverse inequality, let $a \in (0,1)$ and choose $h_a \in \mathcal{A}_{1/(1-a)}(u,\varphi)$, so that $a\varphi + (1-a)h_a \le u$. Combining competitors below F gives

$$aP_{\theta,F}[\varphi] + (1-a)P_{\theta,F}[h_a] \le P_{\theta,F}[u].$$

Indeed, if g_1 and g_2 are competitors for $P_{\theta,F}[\varphi]$ and $P_{\theta,F}[h_a]$, then $ag_1 + (1-a)g_2$ lies below F and is at least as singular as $a\varphi + (1-a)h_a \le u$. Taking suprema over g_1 and g_2 gives the inequality for the unregularized envelopes. Each of these agrees with its regularization almost everywhere, so the inequality holds almost everywhere, and hence everywhere by Lemma 1.3.3. Write $m = \min_X F$ and $M = \max_X F$. Translating competitors gives

$$P_\theta[h_a] + m \le P_{\theta,F}[h_a] \le P_\theta[h_a] + M.$$

Every $P_\theta[h_a]$ has supremum zero, since it lies above $h_a - \sup_X h_a$ and below zero. The uniform L^1 bound for normalized potentials in Theorem 1.4.1 therefore bounds the L^1 norms of the $P_{\theta,F}[h_a]$ uniformly in a. Letting $a \uparrow 1$ in the first inequality gives $P_{\theta,F}[\varphi] \le P_{\theta,F}[u]$ almost everywhere, and by Lemma 1.3.3 the inequality holds everywhere. □

14.5 Model references and positive mass

In this section θ is semipositive and big. Suppose that a potential $u \in \mathrm{PSH}(X,\theta)$ has a competitor in a smaller background $a\theta$, $0 < a < 1$, that is, $P_{a\theta}(u) \not\equiv -\infty$. Then the big form $(1-a)\theta$ remains available: an envelope in this background has positive mass, and the contact property transfers this mass to u. To obtain such competitors for all members of a relative class, we use a reference φ satisfying

$$P_\theta[\varphi] = \varphi, \qquad P_{\varepsilon\theta}(\varphi) \not\equiv -\infty \quad \text{for some } 0 < \varepsilon < 1. \tag{14.20}$$

We call such a φ a *model reference* and fix one such ε. Note that $\varphi = P_\theta[\varphi] \le 0$. For $u \in \mathcal{E}_\varphi$, the envelope of $u - \varphi$ belongs to the relative envelope class $\mathcal{E}_0 = \mathcal{E}_{\mathrm{env}}(X,\theta,0)$ of the zero potential, and stretching realizes it in an arbitrarily small background. Under a bounded mass hypothesis, Chapter 31 uses instead the condition (31.5), in which a positive lower volume replaces the condition on the smaller background; the two conditions are compared there.

Lemma 14.5.1 *Let φ be a model reference. If $u \in \mathcal{E}_\varphi$ and $u \le \varphi$, then*

$$w = P_\theta(u - \varphi) \in \mathcal{E}_0, \qquad w \le 0, \qquad w + \varphi \le u.$$

Proof Since $\varphi \le 0$, the potential u lies below the obstacle $u - \varphi$ away from the pole set of φ, so it is a competitor. Hence the envelope w is not identically $-\infty$. It is at most zero, since $u - \varphi \le 0$, and it satisfies $w + \varphi \le u$ off the poles, hence everywhere by Lemma 1.3.3. For $b \ge 1$, let $h_b = Q_b(u,\varphi)$. Then, off the poles,

$$h_b \le bu - (b-1)\varphi \le b(u - \varphi).$$

Since $\theta \geq 0$, the function $b^{-1}h_b$ is θ-plurisubharmonic, and so it is a competitor for w. Hence $h_b \leq bw$, so $\mathcal{A}_b(w,0)$ is nonempty for every b. Since also $w \leq 0$, this means exactly that $w \in \mathcal{E}_0$. □

Lemma 14.5.2 *Let $0 < a < 1$ and $u \in \mathrm{PSH}(X,\theta)$. If $q = P_{a\theta}(u)$ is not identically $-\infty$, then*

$$\int_X \mathrm{MA}_\theta(u) \geq (1-a)^n v_-(\theta),$$

and the mass on the left is strictly positive, even when $v_-(\theta) = 0$.

Proof The obstacle $u - q$ is nonnegative away from the poles, so by Lemma 1.19.2 the envelope $p = P_{(1-a)\theta}(u-q)$ is bounded. By the contact theorem, Theorem 14.1.2, in the semipositive background $(1-a)\theta$, its measure is carried by $D = \{p+q = u\}$, and $p + q \leq u$ everywhere. Since $p+q$ is θ-plurisubharmonic, positivity of mixed products and ordered contact imply

$$\mathrm{MA}_{(1-a)\theta}(p) \leq \mathbf{1}_D \mathrm{MA}_\theta(p+q) \leq \mathbf{1}_D \mathrm{MA}_\theta(u),$$

where all assertions are first restricted to the complement of the poles. Writing $p = (1-a)\widetilde{p}$ with $\widetilde{p} \in \mathrm{PSH}(X,\theta) \cap L^\infty(X)$ and integrating, we obtain the lower bound from the definition of $v_-(\theta)$. Finally, the bounded potential p has strictly positive mass by Proposition 14.2.1, since $(1-a)\theta$ is big. □

Theorem 14.5.1 *Let φ be a model reference, with ε as in (14.20). For every $u \in \mathcal{E}_\varphi$ and every $0 < \delta < 1-\varepsilon$, the envelope $P_{(\varepsilon+\delta)\theta}(u)$ is not identically $-\infty$. Consequently,*

$$\int_X \mathrm{MA}_\theta(u) \geq (1-\varepsilon)^n v_-(\theta), \qquad \int_X \mathrm{MA}_\theta(u) > 0.$$

Proof After adding a constant to u, we may assume $u \leq \varphi$. Let $w \in \mathcal{E}_0$ be as in Lemma 14.5.1. Since $w \in \mathcal{E}_0$, the envelope $P_\theta(w/\delta)$ is not identically $-\infty$, and $\delta P_\theta(w/\delta) \leq w$. The potential

$$P_{\varepsilon\theta}(\varphi) + \delta P_\theta(w/\delta) \leq \varphi + w \leq u$$

belongs to $\mathrm{PSH}(X,(\varepsilon+\delta)\theta)$, so it is a competitor for $P_{(\varepsilon+\delta)\theta}(u)$. Applying Lemma 14.5.2 with $a = \varepsilon + \delta$ and letting $\delta \downarrow 0$ gives the uniform estimate. The positivity follows from the same lemma with any one such δ. □

14.6 Infinite mass inside the envelope class

We now take the Hermitian background $\theta = \omega$ and the reference $\varphi = 0$. If X does not have the bounded mass property, that is, if the upper mass $v_+(\omega)$ of Definition 7.5.1 is infinite, then the series constructed in Theorem 7.6.1 has infinite mass. It splits into

a smooth finite sum and a tail whose curvature is bounded below by an arbitrarily small multiple of $-\omega$. Multiplying a sufficiently late tail by the stretching factor gives a competitor for that stretch, so the series lies in the relative envelope class of zero.

Proposition 14.6.1 *If $v_+(\omega) = +\infty$, the potential U in Theorem 7.6.1 can be chosen so that, in addition,*

$$U \in \mathcal{E}_{\mathrm{env}}(X, \omega, 0), \qquad P_\omega[U] = 0.$$

The same conclusions hold for εU, $0 < \varepsilon \le 1$. Each of these potentials has infinite extended mass and trivial multiplier ideals at every positive weight.

Proof Use the smooth nonpositive ω-plurisubharmonic functions u_j, $j \ge 1$, of Theorem 7.6.1 and their convergent series $U = \sum_{j\ge1} 2^{-j} u_j$. For each integer $N \ge 1$, write

$$U = A_N + V_N, \qquad A_N = \sum_{j\le N} 2^{-j} u_j, \qquad V_N = \sum_{j>N} 2^{-j} u_j, \qquad \delta_N = 2^{-N}.$$

The head A_N is smooth, and since the finite partial sums of the tail decrease, $V_N \in \mathrm{PSH}(X, \delta_N\omega)$; the tail is not identically $-\infty$, because $U = A_N + V_N$ is not.

Fix $b > 1$ and choose N so large that $b\delta_N \le 1$. Then

$$h_b = bV_N + \min_X(bA_N)$$

is ω-plurisubharmonic, because $\mathrm{dd^c} h_b \ge -b\delta_N\omega \ge -\omega$. Moreover, $h_b \le bU$. Thus $h_b \in \mathcal{A}_b(U, 0)$ for every b, and since $U \le 0$, this proves $U \in \mathcal{E}_{\mathrm{env}}(X, \omega, 0)$. For εU, use the tails εV_N, and choose N with $b\varepsilon\delta_N \le 1$. By Proposition 14.4.3, the singularity envelope of each of these potentials is $P_\omega[0] = 0$. Their infinite mass and trivial multiplier ideals are given by Corollary 7.6.1. □

The bound on the mass in Proposition 14.1.4 concerns the envelope fixed point itself. This example shows that potentials with the same singularity envelope as a fixed point, even with trivial multiplier ideals at every positive weight, need not have finite mass.

Corollary 14.6.1 *The following conditions on a Hermitian metric ω are equivalent:*

(1) $v_+(\omega) < +\infty$;
(2) *every member of $\mathcal{E}_{\mathrm{env}}(X, \omega, 0)$ has finite extended mass;*
(3) *every $u \in \mathrm{PSH}(X, \omega)$ with $P_\omega[u] = 0$ has finite extended mass.*

Proof If $v_+(\omega) < +\infty$, all ω-plurisubharmonic functions have finite extended mass by Proposition 7.5.2. If $v_+(\omega) = +\infty$, Proposition 14.6.1 contradicts each of the other two conditions. □

Chapter 15
Equations with prescribed singularities

Throughout this chapter we fix a Hermitian metric ω on X and a smooth positive probability volume form $\mathrm{d}V$. Unless another measure is named, L^p norms, volumes and the phrase "almost everywhere" refer to $\mathrm{d}V$, while capacity and convergence in capacity refer to Cap_ω from Section 1.8. The background θ is a smooth real $(1,1)$-form whose positivity is specified in each section. For a smooth real $(1,1)$-form α and $u \in \mathrm{PSH}(X,\alpha)$ we write $\alpha_u = \alpha + \mathrm{dd^c}u$, and we call $\{u = -\infty\}$ the *pole set* of u; it is pluripolar and has zero volume. We write $\mathrm{MA}_\theta(u)$ for the extended non-pluripolar Monge–Ampère measure $\mathrm{MA}^{\mathrm{ext}}_\theta(u)$ of Theorem 7.1.1. It equals $(\theta + \mathrm{dd^c}u)^n$ when u is bounded, charges no pluripolar set, and may have infinite total mass; we keep the superscript where this possibility matters. Finally, $s_+ = \max(s,0)$ for real s.

We use the singularity order $\preceq$, the equivalence $\sim$ and the ordinary singularity types of Definition 1.12.1; the envelopes $P_\theta(f)$, the rooftop envelopes $P_\theta(u,v)$, the envelope $V_\theta = P_\theta(0)$ and minimal singularities of Section 1.19; and the ordinary envelope $P_\theta[u]$ and its fixed points of Definition 1.19.1. A potential $\rho \in \mathrm{PSH}(X,\theta)$ with $\theta_\rho \geq \delta\omega$ for some $\delta > 0$ is called a *strict barrier*; it exists exactly when θ is big, and by Theorem 1.15.2 it can be chosen with analytic singularities.

We seek potentials u in the ordinary singularity type of a given reference $\varphi \in \mathrm{PSH}(X,\theta)$ solving

$$\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu \quad (\lambda > 0) \qquad \text{or} \qquad \mathrm{MA}_\theta(u) = c\mu,$$

where μ is a given measure, typically $f\,\mathrm{d}V$ with $f \in L^p$ and $p > 1$, and the constant $c > 0$ is part of the unknown. When φ is unbounded, the comparison principles for bounded potentials no longer apply, and a limit of approximate solutions may leave the prescribed type. For a semipositive big background, both difficulties are overcome by a smaller background. If $P_{\delta\theta}(\varphi) \not\equiv -\infty$ for some $0 < \delta < 1$, then the part $\delta\theta$ of the background can carry the singularity of φ, while bounded envelopes in the remaining big background $(1-\delta)\theta$ have Monge–Ampère measures of positive mass. This gives a domination principle for singular potentials (Theorem 15.1.1), and hence uniqueness. Existence starts from bounded equations with L^p densities, solved with a subsolution built from a strict barrier. A compactness criterion with a common

lower barrier identifies limits of Monge–Ampère measures, and an addition theorem for weighted measures in a fixed type removes a possible deficit of mass in a limit. Together with a uniform estimate relative to the reference (Theorem 15.11.2), these results solve both equations in the type of an envelope fixed point that admits such a smaller-background envelope (Theorem 15.11.1). At exponent zero, comparison determines the constant c but, in general, not the potential.

If θ is big but not semipositive, V_θ may be unbounded and the smaller-background construction is not available; a strict analytic barrier ρ takes over its role in comparison. Resolving the singularities of analytic approximants of V_θ reduces the equations to semipositive big backgrounds. Stability above the barrier then produces solutions with minimal singularities which are locally Hölder continuous off the poles of ρ (Theorems 15.15.1, 15.15.2 and 15.16.1). They are smooth off these poles when the density is smooth and strictly positive on the whole complement (Theorem 15.17.1).

The same barrier controls smooth equations as a semipositive background degenerates (Theorem 15.18.1). For small resolutions of threefold nodes with a closed background, the potentials converge smoothly away from the exceptional curves, and comparing the limiting and approximating metrics reduces to controlling the diameters of neighborhoods of these curves. In the projective example of Theorem 12.6.1, where the normalized potential is not unique, symmetry selects the zero-exponent limit, while local examples show that a positive density on a small open set does not by itself give regularity. Resolved logarithmic poles and minimal types on blowups give explicit equations whose data are determined by the residual form and the Jacobian of the modification.

The last part of the chapter concerns a Hermitian background h and general right-hand sides. If the masses of bounded h-plurisubharmonic functions have no positive lower bound, some probability measure μ with density bounded below by a positive constant admits no solution of $\mathrm{MA}_h^{\mathrm{ext}}(u) = c\mu$; with such a bound, every nonzero finite non-pluripolar measure admits one (Corollary 33.1.1 in Chapter 33). Without such a bound, a measure with bounded density with respect to the Monge–Ampère measure of a Hölder potential still admits a normalized Hölder solution of the homogeneous equation (Theorem 15.24.1). For singular data supported on shrinking coordinate spheres, a bounded-density test against a fixed probability supplies a strict Hölder reference, a normalized Hölder solution of the homogeneous equation and bounded exponential solutions (Proposition 15.25.1, Corollary 15.25.1, and Theorem 15.25.1), even when the lower mass vanishes. For an individual nonnegative density $f \in L^1$, bounded truncations and Gauduchon tests determine the largest coefficient a for which $\mathrm{MA}_h(u) \geq a f \, \mathrm{d}V$ has a solution.

Passing from this threshold to an exact equation at exponent zero requires control of the potentials as well as of their measures. On products $X_0 \times Y$, exact threshold equations are obtained, also when the lower mass vanishes, for the product of a fixed smooth measure on X_0 with an integrable density on a closed Kähler factor Y, or with a measure of continuous logarithmic potential on $Y = \mathbb{P}^1$ (Theorems 15.29.1, 15.29.2 and 15.30.2). The roots of the closed-factor densities form a closed subspace of the density roots approximable by bounded ones (Proposition 15.31.5). A density whose measure is dominated by a power $\mathrm{Cap}_h^{1+\alpha}$ of the capacity, $\alpha > 0$, also admits an exact

threshold equation (Theorem 15.32.1), and such densities exist outside every L^p with $p > 1$ (Theorem 15.32.2). At a positive exponent, exponential decay controls the mass on deep sublevels (Theorem 15.33.1); this control is lost as the exponent tends to zero. For an individual density, the greatest solutions of the exponential equations have one-sided limits in the exponent and jump at no more than countably many exponents (Propositions 15.34.5 to 15.34.7). For the sublogarithmic densities on products with a projective space, a radial flux identifies the limit of the supremum-normalized greatest solutions as the exponent tends to zero (Theorem 15.35.1). On projective space itself, the radial singularity types of these densities separate the regularity of the residual from the pole, both when the density is prescribed (Theorem 15.36.1) and when an envelope of a singular radial obstacle selects the measure (Theorem 15.37.1).

15.1 Domination with a smaller background

Throughout this section θ is semipositive and big. Let $0 < \delta < 1$. When a singular potential lies above a $\delta\theta$-plurisubharmonic function, the gap between them can be measured by an envelope in the remaining background $(1-\delta)\theta$. This background is again semipositive and big, so its bounded potentials have Monge–Ampère measures of positive mass by Proposition 14.2.1, and bounded domination in this background bounds the envelope from below. A family of stretched envelopes then converts a strict measure inequality into a pointwise inequality, as in the bounded case of Theorem 14.2.1.

We say that $u \in \mathrm{PSH}(X,\theta)$ *admits a smaller-background envelope* if $P_{\delta\theta}(u) \not\equiv -\infty$ for some $0 < \delta < 1$. This property depends only on the ordinary singularity type of u: if $u' \geq u - C$, then $P_{\delta\theta}(u) - C$ is a competitor for $P_{\delta\theta}(u')$.

Lemma 15.1.1 *Suppose $u, v \in \mathrm{PSH}(X,\theta)$, $v \preceq u$, and $\mathrm{MA}_\theta^{\mathrm{ext}}(u)(\{u < v\}) = 0$. If $0 < \delta < 1$ and $v_\delta = P_{\delta\theta}(v) \not\equiv -\infty$, then $u \geq v_\delta$.*

Proof Choose C with $v - C \leq u$. Since $v_\delta \leq v$, the obstacle $u - v_\delta$ is bounded below by $-C$ off the pole sets, and it is finite almost everywhere. Hence the envelope

$$p = P_{(1-\delta)\theta}(u - v_\delta)$$

is bounded by Lemma 1.19.2. The obstacle is quasi-continuous off the pluripolar pole sets, so contact support (Theorem 14.1.2) shows that the measure of p is carried by $D = \{p + v_\delta = u\}$. Moreover $p + v_\delta \leq u$ everywhere, and

$$\theta + \mathrm{dd^c}(p + v_\delta) = ((1-\delta)\theta + \mathrm{dd^c} p) + (\delta\theta + \mathrm{dd^c} v_\delta)$$

is a sum of two positive currents. Positivity of mixed products and ordered contact therefore give

$$\mathbf{1}_D \mathrm{MA}_{(1-\delta)\theta}(p) \leq \mathbf{1}_D \mathrm{MA}_\theta^{\mathrm{ext}}(u);$$

this inequality is first proved on common finite truncation levels and then extended by exhaustion. Off the pole sets, every point of $D \cap \{p < 0\}$ satisfies $u = p + v_\delta < v_\delta \le v$. The hypothesis therefore gives $\mathrm{MA}_{(1-\delta)\theta}(p)(\{p < 0\}) = 0$. Since $(1-\delta)\theta$ is semipositive and big, bounded domination (Theorem 14.2.1, with Proposition 14.2.1), applied to p and the constant 0 with coefficient zero, yields $p \ge 0$. Hence $u \ge p + v_\delta \ge v_\delta$. □

Theorem 15.1.1 *Let $u \in \mathrm{PSH}(X,\theta)$ satisfy $P_{\delta\theta}(u) \not\equiv -\infty$ for some $0 < \delta < 1$. Suppose $v \in \mathrm{PSH}(X,\theta)$, $v \preceq u$, and $0 \le c < 1$. If*

$$\mathbf{1}_{\{u<v\}}\mathrm{MA}_\theta^{\mathrm{ext}}(u) \le c\,\mathbf{1}_{\{u<v\}}\mathrm{MA}_\theta^{\mathrm{ext}}(v),$$

then $u \ge v$.

Proof Since $\max(u,v) = v$ on the plurifine open set $\{u < v\}$, plurifine locality shows that replacing v by $\max(u,v)$ preserves the hypothesis; it does not change the conclusion. We may therefore assume $u \le v$, and since $v \preceq u$ there is $C \ge 0$ with $u \le v \le u + C$. For $b > 1$ consider the stretched envelopes

$$p_b = P_\theta(bu - (b-1)v), \qquad D_b = \{p_b = bu - (b-1)v\}.$$

The function $u - (b-1)C$ is a competitor, so $p_b \not\equiv -\infty$; and since $bu - (b-1)v \le u$, we have $u - (b-1)C \le p_b \le u$. By contact support, the measure of p_b is concentrated on D_b. The potential $b^{-1}p_b + (1-b^{-1})v$ lies below u and agrees with it on D_b, so ordered contact and positivity of mixed products give

$$\mathbf{1}_{D_b}\left[b^{-n}\mathrm{MA}_\theta^{\mathrm{ext}}(p_b) + (1-b^{-1})^n\mathrm{MA}_\theta^{\mathrm{ext}}(v)\right] \le \mathbf{1}_{D_b}\mathrm{MA}_\theta^{\mathrm{ext}}(u).$$

Choose b so large that $(1-b^{-1})^n > c$. On $D_b \cap \{u < v\}$ the hypothesis then forces $\mathrm{MA}_\theta^{\mathrm{ext}}(p_b)$ to vanish. Since the masses may be infinite, we justify this subtraction as follows. On each set where $u, v, p_b > -k$, every measure involved is finite, being bounded by the mass of the corresponding bounded truncation. We subtract the finite terms on each such set and then take the union over k, so no infinite masses are ever subtracted.

It follows that the measure of p_b is carried by $D_b \cap \{u = v\} \subset \{p_b = u\}$. We apply Lemma 15.1.1 to the pair (p_b, u): their singularities are equivalent, $\mathrm{MA}_\theta^{\mathrm{ext}}(p_b)(\{p_b < u\}) = 0$, and $P_{\delta\theta}(u) \not\equiv -\infty$. We obtain

$$p_b \ge P_{\delta\theta}(u)$$

for all sufficiently large b. Suppose now that $u \ge v$ fails. By Lemma 1.3.3, the set $\{u < v\}$ has positive volume. Since both potentials are finite almost everywhere, some set $E = \{u < v - a\}$ with $a > 0$ has positive volume and both functions are finite on E. On E the obstacle inequality gives

$$P_{\delta\theta}(u) \le p_b \le v - ba$$

quasi-everywhere. Letting b tend to infinity through the integers shows that the quasi-plurisubharmonic function $P_{\delta\theta}(u) \not\equiv -\infty$ equals $-\infty$ on a set of positive volume, a contradiction. Hence $u \geq v$. □

Corollary 15.1.1 *Let $u, v \in \mathrm{PSH}(X, \theta)$ satisfy $v \preceq u$, and suppose that u admits a smaller-background envelope. If $\lambda > 0$ and*

$$\mathrm{e}^{-\lambda u}\mathrm{MA}_\theta^{\mathrm{ext}}(u) \leq \mathrm{e}^{-\lambda v}\mathrm{MA}_\theta^{\mathrm{ext}}(v)$$

off the pole sets, then $u \geq v$. In particular, for every $\lambda > 0$ and every positive measure μ, the equation $\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu$ has at most one solution in an ordinary singularity type admitting a smaller-background envelope.

Proof Fix $t > 0$. On $\{u < v - t\}$ the assumed inequality gives $\mathrm{MA}_\theta^{\mathrm{ext}}(u) \leq \mathrm{e}^{-\lambda t}\mathrm{MA}_\theta^{\mathrm{ext}}(v - t)$. The theorem, applied to u and $v - t$ with $c = \mathrm{e}^{-\lambda t}$, gives $u \geq v - t$; now let $t \downarrow 0$. For two solutions of the same equation in the stated type, apply this result in both directions. The weighted measures are compared off the common pole sets, which these measures do not charge, so no value of an infinite weight is needed.□

15.2 Bounded equations in a semipositive big background

In this section θ is again semipositive and big, and we solve the exponential and the normalized equations with L^p densities by bounded potentials. The basic tool is a bounded subsolution. A strict barrier, corrected by the solution of a Hermitian equation, is composed with a bounded convex function; the result is a bounded θ-plurisubharmonic function whose Monge–Ampère measure dominates a prescribed density. The derivative of the convex function decays polynomially near the poles of the barrier, and we compensate for this decay by multiplying the auxiliary density by the reciprocal weight.

Fix a strict barrier $\rho \in \mathrm{PSH}(X, \theta)$ and $0 < a \leq 1$ such that

$$\theta + \mathrm{dd}^{\mathrm{c}}\rho \geq a\omega, \qquad \rho \leq -1.$$

No property of ρ other than these two inequalities will be used.

Lemma 15.2.1 *Fix $p > 1$ and $L > 0$. There is $b > 0$ such that, for every $f \geq 0$ with $\|f\|_{L^p(\mathrm{d}V)} \leq L$, one can find $s \in \mathrm{PSH}(X, \theta)$ satisfying*

$$-1 \leq s \leq 0, \qquad \mathrm{MA}_\theta(s) \geq b f \, \mathrm{d}V.$$

The constant b depends only on the fixed geometry and on ρ, a, p, L.

Proof Choose $1 < q < p$. Since a fixed quasi-plurisubharmonic function belongs to every finite L^r space, Hölder's inequality gives a uniform L^q bound for

$$g = (-\rho)^{2n} f.$$

By Theorem 11.1.1 there are $c > 0$ and a continuous ω-plurisubharmonic v, normalized by $\sup_X v = -1$, such that

$$\omega_v^n = c(1 + g) \, \mathrm{d}V.$$

The construction in that theorem gives a uniform upper bound for c, because in its Gauduchon trace estimate the root integral of $1 + g$ is bounded below by the root integral of 1. The lower-density estimate of Theorem 11.3.1 gives $c\|1+g\|_q \geq c_q > 0$, so c is also bounded below. The potential estimate of the same theorem then gives $-C \leq v \leq -1$ with a uniform C.

Put $h = \rho + av \leq -1$. Then $\theta_h \geq a\omega_v$. On $(-\infty, -1]$ define

$$\chi(t) = -t^{-1} - 1, \qquad \chi'(t) = t^{-2} \in (0, 1], \qquad \chi''(t) = -2t^{-3} > 0,$$

and set $s = \chi(h)$, with value -1 at the poles of h. To justify both the positivity of $\theta + \mathrm{dd^c} s$ and the computation of its Monge–Ampère measure, we first work with the bounded functions $h_k = \max(h, -k)$, $k \geq 1$. The chain rule for bounded potentials and $\theta \geq 0$ give

$$\theta + \mathrm{dd^c}\chi(h_k) = (1 - \chi'(h_k))\theta + \chi'(h_k)\theta_{h_k} + \chi''(h_k) \, \mathrm{d}h_k \wedge \mathrm{d^c} h_k \geq 0.$$

These bounded potentials decrease to s, so $s \in \mathrm{PSH}(X, \theta)$ and $-1 \leq s \leq 0$. On $\{h > -k\}$, locality and positivity of mixed products give

$$\mathrm{MA}_\theta(s) \geq a^n |h|^{-2n} \omega_v^n \geq a^n c |h|^{-2n} (-\rho)^{2n} f \, \mathrm{d}V.$$

Letting $k \to \infty$, we obtain this inequality on the set where h is finite; its complement is pluripolar and has zero volume. Since $|h| \leq (-\rho) + aC \leq (1 + aC)(-\rho)$, the assertion follows with any common lower bound for $a^n c(1 + aC)^{-2n}$. □

The lemma gives a subsolution for each density in an L^p-bounded family. It does not give a lower bound for the masses of all bounded θ-plurisubharmonic functions.

15.2.1 The exponential equation

We approximate θ from above by the Hermitian forms $\theta+\varepsilon\omega$. The bounded subsolution provides a common lower barrier for the corresponding solutions, and exponential comparison shows that they decrease as ε decreases. Their limit solves the equation in the original background.

Theorem 15.2.1 *Let $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, be nonzero, and let $\lambda > 0$. There is a unique bounded $v \in \mathrm{PSH}(X,\theta)$ such that*

$$\mathrm{MA}_\theta(v) = \mathrm{e}^{\lambda v} f \,\mathrm{d}V.$$

For fixed λ, these solutions are uniformly bounded when $\|f\|_p$ is bounded above and $\int_X f^{1/n}\,\mathrm{d}V$ is bounded away from zero.

Proof For $0 < \varepsilon \le 1$, set $\theta_\varepsilon = \theta + \varepsilon\omega$. By the Hermitian existence theorem Theorem 11.1.2, there is a continuous solution v_ε of

$$\mathrm{MA}_{\theta_\varepsilon}(v_\varepsilon) = \mathrm{e}^{\lambda v_\varepsilon} f \,\mathrm{d}V.$$

Take s, b from Lemma 15.2.1, and put $w = s + \lambda^{-1}\log b$. Positivity of bounded products gives

$$\mathrm{MA}_{\theta_\varepsilon}(w) \ge \mathrm{MA}_\theta(s) \ge b f \,\mathrm{d}V \ge \mathrm{e}^{\lambda w} f \,\mathrm{d}V,$$

and bounded exponential comparison (Corollary 11.2.2) yields $w \le v_\varepsilon$. If $0 < \varepsilon' < \varepsilon$, then $v_{\varepsilon'}$ is also θ_ε-plurisubharmonic and

$$\mathrm{MA}_{\theta_\varepsilon}(v_{\varepsilon'}) \ge \mathrm{e}^{\lambda v_{\varepsilon'}} f \,\mathrm{d}V,$$

so the same comparison gives $v_{\varepsilon'} \le v_\varepsilon$. Hence v_ε decreases to a bounded θ-plurisubharmonic function v with $w \le v \le v_1$.

To pass to the limit in the equation, we use the fixed Hermitian background $\Omega = \theta + \omega$ and the expansion

$$(\theta_\varepsilon + \mathrm{dd^c} v_\varepsilon)^n = \sum_{k=0}^{n} \binom{n}{k} (\Omega + \mathrm{dd^c} v_\varepsilon)^k \wedge \big(-(1-\varepsilon)\omega\big)^{n-k}.$$

Continuity of bounded products along monotone sequences applies to every term, and the smooth coefficients converge. Dominated convergence applies to the right-hand sides. This proves the equation. Uniqueness follows from Corollary 15.1.1, since bounded potentials admit a smaller-background envelope when $\theta \ge 0$.

For the uniform assertion, note first that the lower barrier w is uniform by Lemma 15.2.1. The upper barrier v_1 solves an equation in the fixed Hermitian background Ω. The proof of Lemma 11.4.1 bounds its supremum using only an upper bound for the density norm and a positive lower bound for the root integral: these two bounds control, respectively, the Hölder term and the denominator in (11.10),

and they survive the smooth approximation used there. Thus v_1, and hence v, has a uniform upper bound. □

Local continuity of this bounded solution will follow from Theorem 15.5.1.

15.2.2 The normalized equation

For the normalized equation the approximating potentials need not be monotone. Upper envelopes and lower envelopes of their tails provide monotone sequences on either side of an L^1 cluster limit. Comparison identifies the two limits, and hence the equation, while the constant is controlled by the Gauduchon trace and the bounded subsolution.

Theorem 15.2.2 *For every nonzero $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, there are a bounded $\varphi \in \mathrm{PSH}(X, \theta)$ and a uniquely determined $c > 0$ such that*

$$\mathrm{MA}_\theta(\varphi) = c f \, \mathrm{d}V, \qquad \sup_X \varphi = 0.$$

The construction gives uniform bounds for $\|\varphi\|_\infty$, c and c^{-1} when $\|f\|_p$ is bounded above and $\int_X f^{1/n} \, \mathrm{d}V$ is bounded away from zero.

Only the constant c is asserted to be unique; the normalized potential need not be unique, as Theorem 12.6.1 shows already for a Hermitian background.

Proof *Step 1: approximating equations and their constants.* For integers $j \ge 1$, put $\omega_j = \theta + 2^{-j}\omega$, and use Theorem 11.1.1, with the Hermitian form ω_j in place of ω, to choose continuous normalized solutions

$$\mathrm{MA}_{\omega_j}(\varphi_j) = c_j f \, \mathrm{d}V, \qquad \sup_X \varphi_j = 0.$$

We first bound c_j. Fix a Gauduchon metric γ_0; when $n = 1$ any Hermitian form will do, and the trace estimate reduces to Stokes' theorem. The smooth approximation in the proof of Theorem 11.1.1, together with the Gauduchon trace, gives a common upper bound: the numerator $\int_X \omega_j \wedge \gamma_0^{n-1}$ is bounded above by $\int_X (\theta + \omega) \wedge \gamma_0^{n-1}$, and the root integral of f is positive and fixed. Here the estimate is applied to the smooth approximating equations before passing to their continuous limits.

For a lower bound, take s, b from Lemma 15.2.1. If $c_j < b$, then

$$\mathrm{MA}_{\omega_j}(\varphi_j) \le (c_j/b)\mathrm{MA}_{\omega_j}(s + A)$$

for every constant A, and strict domination in the Hermitian background ω_j (Theorem 11.2.2) would give $\varphi_j \ge s + A$ for every A, which is absurd. Thus $c_j \ge b$. In fact the constants decrease with j: if $k > j$, positivity gives $\mathrm{MA}_{\omega_j}(\varphi_k) \ge c_k f \, \mathrm{d}V$, and the same argument excludes $c_j < c_k$. Hence $c_j \downarrow c > 0$.

Step 2: a uniform bound for the potentials. For the Hermitian form $\Omega = \theta + \omega$, all φ_j are normalized Ω-plurisubharmonic functions. Using uniform exponential integrability (Lemma 1.5.1) and Hölder's inequality, choose $\gamma > 0$ small and $1 < q < p$ such that

$$g_j = \mathrm{e}^{-\gamma\varphi_j} f$$

is bounded in L^q. Since $\varphi_j \le 0$, we have $g_j \ge f$. Theorem 15.2.1, applied with q in place of p and γ in place of λ, therefore gives uniformly bounded $w_j \in \mathrm{PSH}(X, \theta)$ with

$$\mathrm{MA}_\theta(w_j) = \mathrm{e}^{\gamma w_j} g_j \, \mathrm{d}V.$$

Choose $C_0 \ge \max\{1, \sup_j c_j\}$ and set $z_j = w_j - \gamma^{-1} \log C_0$. Then

$$\mathrm{MA}_{\omega_j}(z_j) \ge \mathrm{e}^{\gamma z_j} c_j g_j \, \mathrm{d}V, \qquad \mathrm{MA}_{\omega_j}(\varphi_j) = \mathrm{e}^{\gamma\varphi_j} c_j g_j \, \mathrm{d}V,$$

and exponential comparison gives $z_j \le \varphi_j \le 0$. Thus all φ_j take values in a common bounded interval.

Step 3: the upper envelopes. Extract a subsequence converging in L^1 and almost everywhere to the canonical θ-plurisubharmonic representative φ. We keep the indices j; the coefficients of ω in ω_j still decrease to zero. Define

$$\Phi_j^+ = \left(\sup_{\ell \ge j} \varphi_\ell \right)^*, \qquad h_j = \inf_{\ell \ge j} \varphi_\ell, \qquad p_j = P_\theta(h_j).$$

All these functions take values in a common bounded interval. The upper envelopes are ω_j-plurisubharmonic and decrease to φ, since the regularized upper envelope agrees almost everywhere with the L^1 limit; this also follows from the local Hartogs lemma applied to the bounded tails.

For $\ell \ge j$, positivity gives $\mathrm{MA}_{\omega_j}(\varphi_\ell) \ge c_\ell f \, \mathrm{d}V \ge c f \, \mathrm{d}V$. By ordered contact (Lemma 1.7.1), finite maxima of these functions satisfy the same lower bound. Continuity of bounded products along increasing sequences, followed by the decreasing limit in j and the expansion in the smooth background, gives

$$\mathrm{MA}_{\omega_j}(\Phi_j^+) \ge c f \, \mathrm{d}V, \qquad \mathrm{MA}_\theta(\varphi) \ge c f \, \mathrm{d}V. \tag{15.1}$$

Step 4: the lower envelopes. For the reverse inequality we use lower envelopes, which carry a useful exponential weight. For $N \ge j$ put

$$p_{j,N} = P_\theta \left(\min_{j \le \ell \le N} \varphi_\ell \right).$$

The obstacle is continuous and bounded, so the measure of $p_{j,N}$ is carried by its contact set. Partition this set into disjoint Borel pieces according to the first index ℓ at which $p_{j,N} = \varphi_\ell$. On such a piece, positivity and ordered contact in the background ω_ℓ give

$$\mathrm{MA}_\theta(p_{j,N}) \le \mathrm{MA}_{\omega_\ell}(p_{j,N}) \le c_\ell f \, \mathrm{d}V.$$

Moreover $p_{j,N} = \varphi_\ell \geq h_j$ there, and $c_\ell \leq c_j$. Summing over the pieces therefore yields the global inequality

$$\mathrm{MA}_\theta(p_{j,N}) \leq \mathrm{e}^{p_{j,N}-h_j} c_j f \, \mathrm{d}V.$$

As $N \to \infty$, the envelopes decrease to p_j. Indeed, their limit lies below each tail obstacle quasi-everywhere and is thus a competitor for $P_\theta(h_j)$, while the reverse inequality is immediate. Continuity of bounded products along monotone sequences and dominated convergence give

$$\mathrm{MA}_\theta(p_j) \leq \mathrm{e}^{p_j-h_j} c_j f \, \mathrm{d}V.$$

The functions p_j increase. Let $\varphi' = (\lim_j p_j)^*$. Since $h_j \to \varphi$ almost everywhere, we have $\varphi' \leq \varphi$ for the canonical representatives. Passing to the increasing limit and using (15.1), we obtain

$$\mathrm{MA}_\theta(\varphi') \leq \mathrm{e}^{\varphi'-\varphi} c f \, \mathrm{d}V \leq \mathrm{e}^{\varphi'-\varphi} \mathrm{MA}_\theta(\varphi).$$

Since bounded potentials admit a smaller-background envelope, Corollary 15.1.1 with $\lambda = 1$ forces $\varphi' \geq \varphi$. Thus $\varphi' = \varphi$, and the first inequality of the last display, together with (15.1), proves the equation. If necessary, subtracting $\sup_X \varphi$ gives the required normalization.

Step 5: uniformity and uniqueness of c. All bounds used above depend only on the two stated density bounds and the fixed data, which proves the uniform assertion for the constructed solution. Finally, two different positive constants for the same f would contradict strict domination after an arbitrary translation of one of the bounded solutions, exactly as in the lower bound for c_j; the domination theorem is now the one for the semipositive big background θ (Theorem 14.2.1 and Proposition 14.2.1). This proves the uniqueness of c. □

For $\lambda > 0$, let v_λ be the bounded solution of $\mathrm{MA}_\theta(v_\lambda) = \mathrm{e}^{\lambda v_\lambda} f \, \mathrm{d}V$, and let (φ, c) be a normalized solution given by Theorem 15.2.2. The two exponential barriers used in Proposition 11.6.1 give

$$\|\lambda v_\lambda - \log c\|_\infty \leq \lambda \operatorname{osc}_X \varphi$$

also in this semipositive background.

15.3 A quantitative stability estimate

If a θ-plurisubharmonic function exceeds a bounded solution only by a small amount in L^1, the excess is also uniformly small. To prove this, we restrict the density of the solution to a strict superlevel set of the difference and construct a subsolution for the restricted density. Domination bounds the height of the superlevel set in terms of the

norm of the restricted density, and Hölder's inequality converts its volume into a power of the L^1 error.

Throughout this section θ is semipositive and big, and $p > 1$ is fixed.

Theorem 15.3.1 *Fix $L, C_0 > 0$. Let $u \in \mathrm{PSH}(X,\theta)$ satisfy*

$$-C_0 \le u \le 0, \qquad \mathrm{MA}_\theta(u) = f\,\mathrm{d}V, \qquad 0 \le f \in L^p(\mathrm{d}V), \quad \|f\|_p \le L.$$

For every

$$0 < \eta < \frac{p-1}{np+p-1}$$

there is a constant C, depending only on the fixed data and η, such that every $v \in \mathrm{PSH}(X,\theta)$ with $v \le 0$ satisfies

$$\sup_X (v-u)_+ \le C\left(\int_X (v-u)_+\,\mathrm{d}V\right)^{\eta}.$$

The comparison potential v need not be bounded.

Proof Replacing v by $\max(v,u)$ leaves both sides unchanged, so we may assume $-C_0 \le u \le v \le 0$. Put $I = \int_X (v-u)\,\mathrm{d}V$. If $I = 0$, the two canonical potentials agree everywhere and there is nothing to prove, so we assume $I > 0$.

Choose $q \in (1,p)$, to be specified at the end. For $\delta > 0$ put

$$E_\delta = \{u < v - \delta\}, \qquad g_\delta = \mathbf{1}_{E_\delta} f, \qquad m_\delta = \|g_\delta\|_q.$$

If $m_\delta > 0$, apply Lemma 15.2.1 to g_δ/m_δ. It gives $h \in \mathrm{PSH}(X,\theta)$ and $b > 0$ such that

$$-1 \le h \le 0, \qquad \mathrm{MA}_\theta(h) \ge b\, g_\delta/m_\delta\,\mathrm{d}V.$$

The constant b does not depend on δ, f, u, v, since the auxiliary density has L^q norm one.

First suppose $0 < m_\delta < b/2$, and take

$$t = (2m_\delta/b)^{1/n} \in (0,1), \qquad w = (1-t)v + th - \delta - C_0 t.$$

Because $h - v \le C_0$, we have $w \le v - \delta$, and hence $\{u < w\} \subset E_\delta$. Positivity of bounded mixed products gives

$$\mathrm{MA}_\theta(w) \ge t^n \mathrm{MA}_\theta(h) \ge 2g_\delta\,\mathrm{d}V.$$

On $\{u < w\}$ the last measure equals $2\mathrm{MA}_\theta(u)$. Bounded strict domination in the semipositive big background (Theorem 14.2.1, with Proposition 14.2.1), applied with coefficient $1/2$, therefore implies $u \ge w$. Consequently

$$v - u \le v - w = \delta + C_0 t + t(v-h) \le \delta + (C_0+1)t.$$

If $m_\delta = 0$, the measure of u vanishes on E_δ, and the same domination theorem with coefficient zero, applied to u and $v - \delta$, gives $v - u \le \delta$. If $m_\delta \ge b/2$, the bound

$v - u \le C_0$ gives the following inequality after an increase of its constant. Thus in every case

$$\sup_X (v - u) \le \delta + C_1 m_\delta^{1/n}. \tag{15.2}$$

Hölder's inequality and Chebyshev's inequality give

$$m_\delta \le \|f\|_p \, \mathrm{d}V(E_\delta)^{1/q-1/p} \le L(I/\delta)^{(p-q)/(pq)}.$$

Set $a = (p-q)/(npq) > 0$. For $0 < I \le 1$, choose $\delta = I^{a/(1+a)}$. Substituting in (15.2) yields

$$\sup_X (v - u) \le C_2 I^{\eta_q}, \qquad \eta_q = \frac{p-q}{npq + p - q}.$$

As $q \downarrow 1$, the exponent η_q tends to $(p-1)/(np+p-1)$. For the prescribed η we choose $q > 1$ so close to 1 that $\eta_q > \eta$; then $I^{\eta_q} \le I^\eta$ on $(0, 1]$. When $I > 1$, the bound $v - u \le C_0$ gives the same conclusion after increasing C. This proves the theorem. □

Corollary 15.3.1 *Fix $C_0 > 0$. Let $u_j, u \in \mathrm{PSH}(X, \theta)$ satisfy $-C_0 \le u_j, u \le 0$ and*

$$\mathrm{MA}_\theta(u_j) = f_j \, \mathrm{d}V, \qquad \mathrm{MA}_\theta(u) = f \, \mathrm{d}V, \qquad \sup_j \|f_j\|_p + \|f\|_p < +\infty.$$

If $u_j \to u$ in L^1, then $u_j \to u$ uniformly. More precisely, for every exponent η as in the theorem,

$$\|u_j - u\|_\infty \le C \|u_j - u\|^\eta_{L^1(\mathrm{d}V)}.$$

Proof Apply the theorem first with solution u and comparison potential u_j, and then with solution u_j and comparison potential u. The constants are uniform because of the common range and the common density bound. The larger of the two suprema is the uniform norm, and each integral of a positive part is at most the full L^1 difference.□

The corollary assumes that the limit u already solves an equation with an L^p density. In an existence proof, this must be established before the corollary can be applied.

15.4 Hölder continuity from bounded stability

Bounded stability also gives a Hölder modulus. We apply it to Demailly's Chern-connection regularization of a solution: the L^1 error of the regularization controls its pointwise error, and a gradient estimate for the smooth averages then gives the modulus. We use the averaging operators R_t and the transforms $\Phi_{c,\delta}$ of Lemma 1.21.1, with the constants A, K given there, together with the estimates of Lemma 1.21.2. Interpolating between the smoothing error and the gradient bound yields the exponent below.

Theorem 15.4.1 *Let $p > 1$ and let $u \in \mathrm{PSH}(X,\omega) \cap L^\infty(X)$ solve*

$$\mathrm{MA}_\omega(u) = f\,\mathrm{d}V, \qquad 0 \le f \in L^p(\mathrm{d}V).$$

Then u is Hölder continuous with every exponent

$$0 < \beta < \frac{2(p-1)}{np + 3(p-1)}.$$

The Hölder constants are uniform for a fixed bound $-C_0 \le u \le 0$ and a fixed bound for $\|f\|_p$. In particular, the continuous solutions constructed in Theorem 11.1.1 have this regularity.

Proof After adding a constant we may assume $-C_0 \le u \le 0$. Choose

$$0 < \eta < \frac{p-1}{np + p - 1}, \qquad \alpha = 2\eta < 1.$$

For small $\delta > 0$ set

$$c = \delta^\alpha, \qquad e = Ac + 2K\delta, \qquad r = -C_0 - 1, \qquad v_\delta = \frac{\Phi_{c,\delta} + er}{1+e}.$$

The curvature estimate of Lemma 1.21.1 gives $v_\delta \in \mathrm{PSH}(X,\omega)$. Evaluating the infimum defining $\Phi_{c,\delta}$ at $t = \delta$ gives $\Phi_{c,\delta} \le R_\delta u \le 0$. On the other hand,

$$\Phi_{c,\delta} \ge u - K(\delta^2 + \delta),$$

since $R_t u + Kt^2 \ge u$, $Kt \ge 0$ and $-c\log(t/\delta) \ge 0$ for $0 < t \le \delta$. For small δ this lower bound is at least r, and therefore

$$v_\delta \le \Phi_{c,\delta} \le R_\delta u \le 0.$$

The Hermitian form ω is semipositive and big, so Theorem 15.3.1 applies in the background ω. Together with the L^1 estimate of Lemma 1.21.2, it gives

$$\sup_X (v_\delta - u)_+ \le C\left(\int_X (v_\delta - u)_+\,\mathrm{d}V\right)^\eta \le C\|R_\delta u - u\|_1^\eta \le C\delta^\alpha.$$

Since $e \le C\delta^\alpha$ and $u - r \le C_0 + 1$, it follows that

$$\Phi_{c,\delta} - u = (1+e)(v_\delta - u) + e(u - r) \le C\delta^\alpha. \tag{15.3}$$

We now bound the minimizing radius from below. Fix z and a radius $t = t(z) \in (0,\delta]$ at which the infimum is attained. The lower estimate at this radius and (15.3) give

$$-c\log(t/\delta) \le \Phi_{c,\delta}(z) - u(z) + K(\delta^2 + \delta) \le C\delta^\alpha = Cc.$$

Thus $t(z) \ge a\delta$ with the constant $a = \mathrm{e}^{-C} > 0$, which is independent of z and δ. Since $R_s u + Ks^2$ is increasing in s,

$$\begin{aligned} R_{a\delta}u + K(a\delta)^2 &\le R_t u + Kt^2 \\ &= \Phi_{c,\delta} + K\delta^2 + K\delta - Kt + c\log(t/\delta) \\ &\le \Phi_{c,\delta} + K\delta^2 + K\delta. \end{aligned}$$

Combining this with (15.3) and the lower bound $R_s u - u \ge -Ks^2$, and setting $s = a\delta$, we obtain

$$\|R_s u - u\|_\infty \le Cs^\alpha$$

for every sufficiently small $s > 0$.

Finally, let x, y be points at a small distance d. The gradient estimate $\|\nabla R_s u\|_\infty \le C/s$ gives

$$|u(x) - u(y)| \le 2Cs^\alpha + Cd/s.$$

With $s = d^{1/(1+\alpha)}$, both terms are bounded by a constant times $d^{\alpha/(1+\alpha)}$, and larger distances are controlled by the bound on the oscillation. This proves Hölder continuity with exponent $\alpha/(1+\alpha)$. Letting η approach its upper endpoint gives every β in the theorem; smaller exponents follow on compact X by changing the constant. □

15.4.1 Regularity for a measure with a Hölder reference

The same smoothing argument applies when the Monge–Ampère measure of the solution is singular but dominated by the Monge–Ampère measure of a Hölder potential. The square-capacity bound for such a measure supplies stability, and the weighted Chern estimate supplies the smoothing error.

Theorem 15.4.2 *Let $u \in \mathrm{PSH}(X,\omega) \cap C^{0,\alpha}(X)$, $0 < \alpha \le 1$, and suppose $\varphi \in \mathrm{PSH}(X,\omega) \cap C(X)$ satisfies*

$$\omega_\varphi^n = \mu \le A\omega_u^n, \qquad A > 0.$$

Then φ is Hölder continuous with exponent

$$\beta = \frac{\alpha}{(\alpha + 2n)(6n+1) + \alpha} > 0.$$

Its Hölder constant is uniform for fixed bounds on A, $\|u\|_{C^{0,\alpha}}$ and the oscillation of φ.

Proof Normalize $\sup_X \varphi = 0$ and write $M = \|\varphi\|_\infty$. By Proposition 1.8.1, the measure μ has a square-capacity bound with a fixed constant. Put

$$\eta = \frac{2\alpha}{\alpha + 2n}, \qquad \gamma = \frac{1}{6n+1}, \qquad q = \eta\gamma/2 < 1.$$

The weighted estimate of Lemma 1.21.5 gives $\|R_\delta\varphi - \varphi\|_{L^1(\mu)} \le C\delta^\eta$.

Let A_0 and K denote the constants A and K of Lemma 1.21.1 for $C_0 = M$; we may take $A_0 \ge 1$. Define the unshifted transform

$$F = \inf_{0<t\le\delta}\{R_t\varphi + Kt^2 + Kt - b\log(t/\delta)\}.$$

It differs from $\Phi_{b,\delta}$ by the constant $K(\delta^2+\delta)$, so

$$\omega + \mathrm{dd}^{\mathrm{c}}F \ge -(A_0 b + 2K\delta)\omega, \qquad \varphi \le F \le R_\delta\varphi + 2K\delta$$

when $\delta \le 1$. Choose $\varepsilon = \delta^q$ and $b = (\varepsilon - 2K\delta)/A_0$. For small δ, $\varepsilon \le 1/2$ and $\varepsilon/(2A_0) \le b \le \varepsilon/A_0$.

Set $T = (1-\varepsilon)F$. Then $\omega + \mathrm{dd}^{\mathrm{c}}T \ge \varepsilon^2\omega$. The transform is continuous: its positive logarithmic penalty excludes radii near zero uniformly in the base point for each fixed δ, b, and the minimum of its continuous defining function on the resulting compact interval depends continuously on the base point. Take $B \ge M + 2K + 1$ and set $\psi = T - B\varepsilon$. Since $R_\delta\varphi \le 0$, we have $-M - B \le \psi \le 0$. Moreover,

$$\psi - \varphi \le (1-\varepsilon)(R_\delta\varphi - \varphi + 2K\delta) + \varepsilon(M - B) \le (1-\varepsilon)(R_\delta\varphi - \varphi + 2K\delta).$$

The weighted smoothing estimate and the fixed bound for $\mu(X)$ give $\int(\psi-\varphi)_+\,\mathrm{d}\mu \le C\delta^\eta$. Thus Proposition 11.3.1, applied to the pair φ, ψ with the fixed lower bound $H = M + B + 1$ for ψ, gives

$$\sup_X(T - \varphi) \le C\delta^{\eta\gamma} + B\varepsilon \le C_1\varepsilon.$$

Since $\varphi \le 0$,

$$0 \le F - \varphi = \frac{T - \varphi + \varepsilon\varphi}{1-\varepsilon} \le 2C_1\varepsilon.$$

At a minimizing radius $t_0 \in (0, \delta]$, monotonicity of $R_t\varphi + Kt^2$ implies

$$b\log(\delta/t_0) \le F - \varphi \le 2C_1\varepsilon, \qquad t_0 \ge \kappa\delta, \quad \kappa = \exp(-4A_0C_1) > 0.$$

The constant κ is independent of δ and the point. Applying monotonicity again gives $R_{\kappa\delta}\varphi - \varphi \le 2C_1\delta^q$. With the lower bound $R_r\varphi - \varphi \ge -Kr^2$ and $r = \kappa\delta$, this proves $\|R_r\varphi - \varphi\|_\infty \le Cr^q$.

Finally Lemma 1.21.2 gives $\|\nabla R_r\varphi\|_\infty \le C/r$. For points a distance s apart,

$$|\varphi(x) - \varphi(y)| \le 2Cr^q + Cs/r.$$

Take $r = s^{1/(1+q)}$ for small s; boundedness handles larger distances. The exponent is $q/(1+q) = \beta$. □

15.5 Local continuity in a semipositive big background

We return to a semipositive big background θ. Where θ degenerates, regularization loses some positivity, and this curvature error can be repaired by adding a small

multiple of a strict barrier. Bounded stability then shows that a bounded solution is continuous wherever the barrier is locally bounded.

Fix $\rho \in \mathrm{PSH}(X,\theta)$ with $\theta + \mathrm{dd}^{\mathrm{c}}\rho \geq \kappa\omega$ for some $\kappa > 0$. For bounded $u \in \mathrm{PSH}(X,\theta)$ we use the averages $R_t u$, the transforms $\Phi_{c,\delta}$ and the curvature modulus $\epsilon(\delta)$, with the constants A, K, from Section 1.21.1 and Lemma 1.21.3.

Theorem 15.5.1 *Let $p > 1$ and let $u \in \mathrm{PSH}(X,\theta) \cap L^{\infty}(X)$ satisfy*

$$\mathrm{MA}_{\theta}(u) = f\,\mathrm{d}V, \qquad 0 \leq f \in L^{p}(\mathrm{d}V).$$

Then u is continuous on every open set $U \subseteq X$ on which ρ is locally bounded.

Proof After adding constants we may assume $-C_0 \leq u \leq 0$ and $\rho \leq -C_0 - 1$. Choose

$$0 < \eta < \frac{p-1}{np+p-1}, \qquad \alpha = 2\eta < 1,$$

and set, for small $\delta > 0$,

$$c = \delta^{\alpha} + \epsilon(\delta), \qquad e = Ac + \epsilon(\delta), \qquad s = e/\kappa, \qquad v_{\delta} = \frac{\Phi_{c,\delta} + s\rho}{1+s}.$$

Since $s\kappa = e$, the curvature bounds for $\Phi_{c,\delta}$ and ρ give $v_{\delta} \in \mathrm{PSH}(X,\theta)$. Moreover

$$\rho \leq -C_0 - 1 \leq \Phi_{c,\delta} \leq R_{\delta}u \leq 0$$

for small δ: the lower bound follows from $\Phi_{c,\delta} \geq u - K(\delta^2 + \delta)$, exactly as in the proof of Theorem 15.4.1. Hence $v_{\delta} \leq \Phi_{c,\delta} \leq R_{\delta}u \leq 0$.

Theorem 15.3.1 allows an unbounded comparison potential, so it applies to v_{δ}. Together with the L^1 smoothing estimate stated after Lemma 1.21.3, it gives

$$\sup_X (v_{\delta} - u)_+ \leq C\left(\int_X (v_{\delta} - u)_+\,\mathrm{d}V\right)^{\eta} \leq C\delta^{\alpha}.$$

Fix M and consider the points where $\rho \geq -M$. Since $s \leq Cc$ and $c \to 0$, the identity

$$\Phi_{c,\delta} - u = (1+s)(v_{\delta} - u) + s(u - \rho)$$

gives $\Phi_{c,\delta} - u \leq C_M c$ there.

At a radius t where the infimum defining $\Phi_{c,\delta}$ is attained, the same lower estimate yields

$$-c\log(t/\delta) \leq \Phi_{c,\delta} - u + K(\delta^2 + \delta) \leq C_M c.$$

Thus $t \geq a_M \delta$ with $a_M > 0$ independent of the point and of δ. Since $R_t u + Kt^2$ is increasing in t, we obtain

$$R_{a_M\delta}u + K(a_M\delta)^2 \leq R_t u + Kt^2 \leq \Phi_{c,\delta} + K\delta^2 + K\delta.$$

This upper bound and the lower bound $R_s u \geq u - Ks^2$ imply

$$\sup_{\{\rho \geq -M\}} |R_{a_M \delta} u - u| \leq C_M \big(\delta^\alpha + \epsilon(\delta)\big) \longrightarrow 0.$$

On every compact subset of U, the function ρ is bounded below. The smooth functions $R_{a_M \delta} u$ therefore converge uniformly to u on each such compact set, which proves that u is continuous on U. □

The proof also gives a modulus of continuity on each compact subset of U, obtained by combining the uniform smoothing error with the gradient bound C/t. A Hölder exponent would require a power bound for the modulus $\epsilon(\delta)$. For potentials that are locally bounded off the poles of a strict analytic barrier, Lemma 1.21.6 refines the classical expansion and gives a curvature error linear in δ. Combined with stability above a strict barrier, this yields local Hölder continuity even for solutions with minimal singularities (Theorem 15.16.1). Neither argument gives continuity across the pole set of the barrier.

15.6 Homogeneous replacement on a coordinate ball

To show that the Monge–Ampère measure of an envelope vanishes off the contact set, we shall replace the envelope on small balls by a solution of the homogeneous equation with the same boundary values. We first recall the smooth Dirichlet problem with a smooth positive density; the replacement is obtained by letting a constant density tend to zero, and continuous boundary values are then handled by uniform approximation.

In this section we work on the ball $B = B_R = \{|z| < R\} \subset \mathbb{C}^n$, put $\beta = \mathrm{dd^c}|z|^2$, and let θ be any smooth real $(1,1)$-form defined near $\overline{B}$. Distances and the Laplacian Δ are Euclidean.

We use smooth comparison on subdomains D of the ball. Let a, b be smooth functions on $\overline{D}$ with $\theta + \mathrm{dd^c} a > 0$ and $\theta + \mathrm{dd^c} b > 0$. If $(\theta + \mathrm{dd^c} a)^n \geq (\theta + \mathrm{dd^c} b)^n$ in D and $a \leq b$ on ∂D, then $a \leq b$ in D. Indeed, integrating the derivative of the logarithm of the determinant along the segment between the two positive matrices gives a linear elliptic inequality for $a - b$, and the maximum principle proves the assertion. The boundary of D need not be smooth for this argument.

Lemma 15.6.1 *Let $\varphi \in C^\infty(\partial B)$, and let F be a smooth strictly positive function on $\overline{B}$. There is a unique smooth function v on $\overline{B}$ such that*

$$\theta + \mathrm{dd^c} v > 0, \qquad (\theta + \mathrm{dd^c} v)^n = F\beta^n, \qquad v|_{\partial B} = \varphi.$$

Proof Extend φ to a smooth function $\widetilde{\varphi}$ on $\overline{B}$. For sufficiently large A, the function $\underline{v} = \widetilde{\varphi} + A(|z|^2 - R^2)$ is a smooth strict subsolution with the prescribed boundary values: $\theta + \mathrm{dd^c}\underline{v}$ is positive and its determinant is at least $(\max_{\overline{B}} F)\beta^n$. For $n \geq 2$, the classical smooth Dirichlet theorem of Guan–Li therefore applies on the ball, with

the arbitrary smooth background $\chi = \theta$ and the smooth positive right-hand side F. Their operator is $(\mathrm{i}/2)\partial\bar\partial$, so our potential is π times theirs. No closedness condition is needed here; such conditions enter only in their separate theorems on compact manifolds. For $n = 1$ the lemma is the Dirichlet problem for the Poisson equation. Uniqueness follows from smooth comparison, applied in both directions. □

Theorem 15.6.1 *For every $\varphi \in C(\partial B)$ there exists a continuous θ-plurisubharmonic function H_φ on $\overline{B}$ such that*

$$(\theta + \mathrm{dd^c} H_\varphi)^n = 0 \quad \text{on } B, \qquad H_\varphi|_{\partial B} = \varphi.$$

It dominates every continuous θ-plurisubharmonic function p on $\overline{B}$ with $p|_{\partial B} \le \varphi$. For $\varphi, \psi \in C(\partial B)$, the construction satisfies

$$\|H_\varphi - H_\psi\|_{C(\overline{B})} \le \|\varphi - \psi\|_{C(\partial B)}.$$

For smooth boundary data the constructed function is Lipschitz on $\overline{B}$.

Proof *Step 1: boundary barriers.* We first take φ smooth and $0 < \varepsilon \le 1$, and let v_ε be the solution of Lemma 15.6.1 with the constant density $F \equiv \varepsilon$. The strict subsolution $\underline{v}$ constructed there for $F \equiv 1$ lies below every v_ε, by smooth comparison. Choose a smooth function q with $q = \varphi$ on the boundary and $\Delta q = -C$, where C is so large that $\theta + \mathrm{dd^c} v_\varepsilon \ge 0$ implies $\Delta v_\varepsilon \ge -C$. The maximum principle for the Laplacian gives $v_\varepsilon \le q$. Both barriers are smooth and equal $\widetilde{\varphi}$ on the boundary. Consequently

$$|v_\varepsilon(z) - \widetilde{\varphi}(z)| \le C_1 \operatorname{dist}(z, \partial B), \tag{15.4}$$

with C_1 independent of ε.

Step 2: a Lipschitz bound independent of ε. Let $h \in \mathbb{C}^n$ be small, and set $D_h = \{z \in B : z + h \in B\}$. Since θ is smooth, there is $C_2 > 0$, independent of the translation h, such that

$$\theta(z) - \theta(z + h) + C_2|h|\beta \ge 0.$$

For a constant $C_3 > 0$ to be fixed, define on D_h

$$w(z) = v_\varepsilon(z + h) + C_2|h|(|z|^2 - R^2) - C_3|h|.$$

Then $\theta + \mathrm{dd^c} w \ge \theta(z + h) + \mathrm{dd^c} v_\varepsilon(z + h) > 0$, and the determinant of $\theta + \mathrm{dd^c} w$ is at least $\varepsilon\beta^n$, because the Euclidean volume form is translation invariant. On ∂D_h, one of the points z and $z + h$ lies on ∂B, and the other has distance at most $|h|$ from ∂B. Estimate (15.4) and the Lipschitz bound for $\widetilde{\varphi}$ therefore give $w \le v_\varepsilon$ on ∂D_h if C_3 is chosen sufficiently large, and smooth comparison extends this inequality to all of D_h. Together with the same argument for $-h$, this gives

$$|v_\varepsilon(z + h) - v_\varepsilon(z)| \le C_4|h|,$$

with constants independent of ε. The same bound for arbitrary pairs of points follows by subdividing the segment joining them inside the convex ball, and it extends to $\overline{B}$ because each v_ε is continuous there.

Step 3: the limit $\varepsilon \to 0$. Smooth comparison also shows that v_ε increases as ε decreases. By the barriers and the common Lipschitz bound, the limit H_φ is Lipschitz and the convergence is uniform on $\overline{B}$. The inequality $\theta + \mathrm{dd^c} v_\varepsilon > 0$ of currents passes to the limit, and continuity of local bounded products under uniform convergence gives $(\theta + \mathrm{dd^c} H_\varphi)^n = 0$. For smooth boundary values φ, ψ, denote the solutions of Lemma 15.6.1 with $F \equiv \varepsilon$ by $v_{\varepsilon,\varphi}$ and $v_{\varepsilon,\psi}$. Smooth comparison gives $\|v_{\varepsilon,\varphi} - v_{\varepsilon,\psi}\|_\infty \le \|\varphi - \psi\|_\infty$, and passing to the limit proves the contraction for smooth data.

Step 4: continuous boundary values. For continuous φ, choose smooth $\varphi_j \to \varphi$ uniformly on ∂B. By the contraction, H_{φ_j} is uniformly Cauchy. Its limit H_φ is continuous and θ-plurisubharmonic, has the prescribed boundary values, and satisfies the homogeneous equation, again by continuity of bounded products under uniform convergence. Passing to the limit also proves the contraction for all continuous boundary data.

Step 5: domination. It remains to prove the domination property, which does not follow from a comparison of total masses. First take φ smooth and p as in the statement. For $\delta > 0$ put $p_\delta = p + \delta(|z|^2 - R^2)$, so that $\theta + \mathrm{dd^c} p_\delta \ge \delta\beta$. By Lemma 1.3.9, applied with $\theta - \delta\beta$ in place of θ, every C^2 function a touching p_δ from above at an interior point z_0 satisfies

$$\theta(z_0) + \mathrm{dd^c} a(z_0) \ge \delta\beta.$$

Suppose that $p_\delta - v_\varepsilon$ had a positive maximum. Since this difference is nonpositive on the boundary, the maximum would be attained at an interior point, where v_ε plus the maximum touches p_δ from above. The determinant of $\theta + \mathrm{dd^c} v_\varepsilon$ at that point would then be at least $\delta^n \beta^n$, contradicting the equation for v_ε when $\varepsilon < \delta^n$. Hence $p_\delta \le v_\varepsilon$ for small ε. Letting first ε and then δ decrease to zero, we obtain $p \le H_\varphi$.

Finally, for continuous φ, take the smooth approximants φ_j above and set $a_j = \|\varphi_j - \varphi\|_\infty$. The boundary values of $p - a_j$ are at most φ_j, so the result for smooth data gives $p - a_j \le H_{\varphi_j}$. Uniform convergence proves $p \le H_\varphi$. □

15.7 Contact support in a big background

Throughout this section, θ is an arbitrary smooth big background, not necessarily semipositive. For a continuous obstacle, regularization shows that the envelope is continuous off the poles of a strict analytic barrier. On a small ball where the envelope stays below the obstacle, a homogeneous replacement then shows that the envelope already solves the homogeneous equation, so its Monge–Ampère measure vanishes away from the contact set.

Applying Theorem 1.15.2 to a strict barrier and taking an approximant with a sufficiently small curvature error, we may fix $\rho \in \mathrm{PSH}(X,\theta)$ with analytic singularities and a smooth remainder after resolution such that

$$\theta + \mathrm{dd}^{\mathrm{c}}\rho \geq \kappa\omega, \qquad \kappa > 0.$$

Its pole set $Z = \{\rho = -\infty\}$ is an analytic, hence pluripolar, subset of X.

Theorem 15.7.1 *Let $h \in C(X)$. Then $p = P_\theta(h)$ is a θ-plurisubharmonic function, continuous on $X \setminus Z$, whose Monge–Ampère measure satisfies*

$$\mathbf{1}_{\{p<h\}}\mathrm{MA}_\theta(p) = 0.$$

Proof Subtract a constant from ρ so that $\rho + 2 \leq h$. Then $\rho + 2$ is a competitor, and since the obstacle is continuous, the envelope lies below the obstacle everywhere; thus

$$\rho + 2 \leq p \leq h.$$

In particular, p is locally bounded off Z. We first prove that p is continuous there; this part does not use the contact assertion.

Apply Lemma 1.21.6 to $p - \sup_X p$ and add back the constant. The transforms $\Phi_{c,\delta}$ of that lemma are defined on X and, with fixed constants $A, K, C > 0$ and $\epsilon(\delta) = C\delta$, satisfy

$$\theta + \mathrm{dd}^{\mathrm{c}}\Phi_{c,\delta} \geq -(Ac + \epsilon(\delta))\omega. \tag{15.5}$$

Off Z the infimum defining $\Phi_{c,\delta}$ is attained at a positive radius, and

$$p - K(\delta^2 + \delta) \leq \Phi_{c,\delta} \leq R_\delta p \leq R_\delta h.$$

For small δ the first inequality gives $\Phi_{c,\delta} \geq \rho + 1$, hence $\rho \leq \Phi_{c,\delta}$ on X.

Set

$$m(\delta) = \sup_{0<t\leq\delta} \|R_t h - h\|_\infty, \qquad c = m(\delta) + \epsilon(\delta) + \delta, \qquad s = \frac{Ac + \epsilon(\delta)}{\kappa}.$$

The averaging kernels are positive, have integral one and are supported in balls of radius comparable to t, so uniform continuity of h gives $m(\delta) \to 0$. Define

$$v_\delta = \frac{\Phi_{c,\delta} + s\rho}{1+s}.$$

By the choice of s, the strict barrier and (15.5) show that v_δ is θ-plurisubharmonic. Since $\rho \leq \Phi_{c,\delta}$,

$$v_\delta \leq \Phi_{c,\delta} \leq R_\delta h \leq h + m(\delta).$$

Thus $v_\delta - m(\delta)$ is a competitor for the envelope, and $v_\delta \leq p + m(\delta)$.

Let $L \Subset X \setminus Z$ be compact. On L the difference $p - \rho$ is bounded above. Since $s \leq Cc$, the identity

$$\Phi_{c,\delta} - p = (1+s)(v_\delta - p) + s(p - \rho)$$

therefore gives $\Phi_{c,\delta} - p \le C_L c$ on L. At a radius t where the infimum is attained,

$$-c\log(t/\delta) \le \Phi_{c,\delta} - p + K(\delta^2 + \delta) \le C_L c.$$

Since $c \ge \delta$, this yields $t \ge a_L \delta$ with a fixed $a_L > 0$. Because $R_t p + Kt^2$ is increasing in t, it follows that

$$p - K(a_L\delta)^2 \le R_{a_L\delta} p \le \Phi_{c,\delta} + K\delta^2 + K\delta.$$

Consequently $R_{a_L\delta} p \to p$ uniformly on L. These averages are smooth for positive radius, even though p is unbounded on X. Applying this to compact neighborhoods of the points of $X \setminus Z$ proves that p is continuous on $X \setminus Z$.

We turn to the contact assertion. By the continuity just proved, the set $\{p < h\} \setminus Z$ is open. Let x be one of its points, and choose a small coordinate ball B_r centered at x whose closure lies in this set. Let H be the homogeneous replacement of Theorem 15.6.1 with boundary values $p|_{\partial B_r}$. Its domination property gives $H \ge p$ on the ball.

We claim that $H \le h$ on B_r if r is small enough. Indeed, in the chosen Euclidean coordinates z, with Laplacian Δ, the inequality $\theta + \mathrm{dd}^{\mathrm{c}} H \ge 0$ implies a fixed bound $\Delta H \ge -C$. Comparison with the function $\sup_{\partial B_r} p + C'(r^2 - |z-x|^2)$, with C' fixed large enough, gives

$$H \le \sup_{\partial B_r} p + C' r^2.$$

As $r \to 0$, the right-hand side tends to $p(x)$, whereas $\inf_{B_r} h \to h(x) > p(x)$. This proves the claim for sufficiently small r.

Now replace p by H on the ball. The resulting function is θ-plurisubharmonic. To see this, fix $a > 0$ and take $\max\{p, H - a\}$ inside the ball and p outside. Since H and p are continuous with equal boundary values, $H - a < p$ on a collar of ∂B_r, so this is an ordinary local maximum construction. Letting $a \downarrow 0$ gives the function equal to H on B_r and to p outside, with locally uniform convergence near the boundary of the ball. This function is bounded above by h on all of X, so the definition of the envelope forces it to be at most p. Thus $p = H$ on B_r, and $\mathrm{MA}_\theta(p)$ vanishes there.

These balls cover $\{p < h\} \setminus Z$. There the extended measure agrees with the local bounded product, and it does not charge the pluripolar set Z. This proves the contact assertion on X. □

15.7.1 Quasi-continuous obstacles

We now extend the contact property to discontinuous obstacles. Continuous approximations of an obstacle need not preserve every competitor of its envelope pointwise. For lower semicontinuous obstacles, the approximation of Lemma 1.19.3 repairs this, and quasi-continuity then allows us to pass to limits of weighted measures. An unbounded obstacle is truncated at finite levels; in an arbitrary smooth background this produces signed products, whose limits are controlled by Lemma 1.8.7 before

we restrict to the region where the product is positive. The strict analytic barrier ρ and its pole set Z are those fixed above.

Proposition 15.7.1 *If f is bounded and lower semicontinuous, then $p = P_\theta(f)$ is locally bounded on $X \setminus Z$, and*

$$\mathbf{1}_{\{p<f\}}\mathrm{MA}_\theta(p) = 0.$$

Proof Using the infimum construction in the proof of Proposition 14.1.2, take continuous $f_j \uparrow f$ with common bounds $m \le f_j \le M$, and shift ρ below m. The envelopes $p_j = P_\theta(f_j)$ increase and satisfy

$$\rho \le p_j \le M.$$

Their regularized limit $q = (\lim_j p_j)^*$ is therefore θ-plurisubharmonic. Outside a pluripolar set it equals the pointwise limit, which is at most f; hence $q \le p = P_\theta(f)$.

We prove the reverse inequality without using a domination theorem. Let u be any competitor for $P_\theta(f)$, and let $u_t \le f$ be the potentials given by Lemma 1.19.3, which satisfy this inequality everywhere. For fixed $t, \eta > 0$, the sets

$$\{u_t < f_j + \eta\}, \qquad j \ge 1,$$

are open, increase with j, and cover X. By compactness, $u_t \le f_j + \eta$ everywhere for all large j. Thus $u_t - \eta$ is a competitor for p_j, and $q \ge u_t - \eta$. Letting first $\eta \downarrow 0$ and then $t \downarrow 0$, we obtain $q \ge u$ almost everywhere, hence everywhere by Lemma 1.3.3. Taking the supremum over all competitors u gives $q \ge p$, so $q = p$.

For fixed k, the set $U_k = \{p < f_k\}$ is open. If $j \ge k$, then $p_j \le p < f_k \le f_j$ on U_k, so Theorem 15.7.1 gives $\mathrm{MA}_\theta(p_j) = 0$ there. On every relatively compact open subset of $X \setminus Z$ the functions p_j have common bounds. After a common smooth correction and a finite expansion, the local Bedford–Taylor continuity along increasing sequences shows that their measures converge to $\mathrm{MA}_\theta(p)$ there. Since the measure of an open set is at most the lower limit of the measures of the approximating sequence, $\mathrm{MA}_\theta(p)$ vanishes on $U_k \setminus Z$. The union of these sets is $\{p < f\} \setminus Z$, and the extended measure does not charge the pluripolar set Z. □

Proposition 15.7.2 *The conclusion of Proposition 15.7.1 holds for every bounded quasi-continuous obstacle f.*

Proof By Lemma 1.8.5, there are bounded lower semicontinuous f_j with common bounds and $f_j \downarrow f$ quasi-everywhere. The envelopes $p_j = P_\theta(f_j)$ decrease to $p = P_\theta(f)$. Indeed, the common lower barrier shows that the limit is not identically $-\infty$; the limit lies below f quasi-everywhere and is thus a competitor for $P_\theta(f)$, while monotonicity gives the other inequality. All these functions have common local bounds off Z.

For fixed k, put $H_k = \min\{1, (f - p_k)_+\}$ on $X \setminus Z$, a bounded quasi-continuous function there. If $j \ge k$, then on $\{H_k > 0\}$ we have $p_j \le p_k < f \le f_j$ outside a pluripolar set, so Proposition 15.7.1 gives $H_k\mathrm{MA}_\theta(p_j) = 0$ off Z.

Let g be continuous with compact support in $X \setminus Z$. The continuity of bounded products against quasi-continuous weights from Section 1.6 applies to the fixed quasi-continuous coefficient gH_k. Consequently $H_k \mathrm{MA}_\theta(p) = 0$ off Z. As $k \to \infty$, the sets $\{H_k > 0\}$ exhaust $\{p < f\} \setminus Z$, and Z is not charged because it is pluripolar.□

For an unbounded obstacle, we truncate and use the convergence of signed products in Lemma 1.8.7.

Theorem 15.7.2 *Let θ be smooth and big. Let f be a quasi-continuous obstacle, finite almost everywhere, such that $p = P_\theta(f) \not\equiv -\infty$. Then $p \le f$ quasi-everywhere and*

$$\mathbf{1}_{\{p<f\}} \mathrm{MA}_\theta(p) = 0.$$

Proof As in the proof of Theorem 14.1.2, the local envelope theorem and a countable Choquet family give $p \le f$ quasi-everywhere. Set $f_j = \max(f, -j)$ and $p_j = P_\theta(f_j)$. Then p is a competitor for every f_j. Since $f_j \le f_1$ and f_1 is finite almost everywhere, the finite-sublevel argument in Section 1.19 gives a common upper bound for all competitors. Thus the envelopes p_j are θ-plurisubharmonic and uniformly bounded above. Choose M strictly larger than this bound.

The obstacle $g_j = \min(f_j, M)$ is bounded and quasi-continuous. Since $p_j < M$ and $p_j \le f_j$ quasi-everywhere, p_j is a competitor for $P_\theta(g_j)$; the reverse inequality follows from $g_j \le f_j$. Thus $p_j = P_\theta(g_j)$, and Proposition 15.7.2 gives contact support for p_j with respect to g_j. Because $p_j < M$, no contact occurs at the artificial upper cutoff, so

$$\mathbf{1}_{\{p_j<f_j\}} \mathrm{MA}_\theta(p_j) = 0. \tag{15.6}$$

Each p_j is locally bounded off Z, with a lower barrier of the form $\rho - C_j$. The sequence decreases to p: its limit is not identically $-\infty$ because it dominates p, and it lies below f quasi-everywhere, so it is itself a competitor for $P_\theta(f)$.

Fix a Hermitian form $\Omega \ge \theta$. For integers $C, k \ge 1$, consider the bounded Ω-plurisubharmonic functions

$$v_{j,C} = \max(p_j, -C), \qquad v_C = \max(p, -C),$$

and set

$$H_{k,C} = \min\{1, (f - p_k)_+\} \min\{1, (p + C)_+\} \quad \text{on } X \setminus Z,$$

extended by zero on Z. This function is bounded and quasi-continuous. Indeed, off Z the function p_k is a finite quasi-plurisubharmonic function, truncation by bounded continuous functions preserves quasi-continuity, and the pluripolar set Z can be removed by an open set of arbitrarily small capacity. At a pole of p the second factor vanishes.

For $j \ge k$, on $\{H_{k,C} > 0\}$ we have, outside pluripolar sets,

$$p_j \ge p > -C, \qquad p_j \le p_k < f \le f_j.$$

By plurifine locality, the signed measure $\nu_{j,C} = (\theta + \mathrm{dd}^{\mathrm{c}} v_{j,C})^n$ therefore agrees there with $\mathrm{MA}_\theta(p_j)$, and (15.6) gives

$$H_{k,C}\nu_{j,C} = 0.$$

For fixed C, the potentials $v_{j,C}$ have common global bounds and decrease to v_C, so Lemma 1.8.7 yields

$$\int_X H_{k,C}(\theta + \mathrm{dd^c} v_C)^n = 0.$$

On the set where $H_{k,C}$ is positive we have $p > -C$, so the restricted product in Theorem 7.1.1 identifies $H_{k,C}(\theta + \mathrm{dd^c} v_C)^n$ with the positive measure $H_{k,C}\mathrm{MA}_\theta(p)$; on the complement the coefficient is zero. Hence $\mathrm{MA}_\theta(p)$ vanishes on $\{H_{k,C} > 0\}$.

Finally, if $p(x)$ is finite and $p(x) < f(x)$, then $p_k(x) < f(x)$ for large k and $p(x) > -C$ for large C. The countable union of the sets $\{H_{k,C} > 0\}$ therefore covers $\{p < f\}$ outside $Z \cup \{p = -\infty\}$. Both excluded sets are pluripolar, and the theorem follows. □

15.8 Compactness with a common lower barrier

We return to a semipositive big background θ. Convergence of potentials in L^1 does not by itself imply convergence of their Monge–Ampère measures. We give a criterion for convergence in capacity when the potentials are bounded above and stay above a reference admitting a smaller-background envelope, up to a fixed constant. This common lower barrier ensures that all finite and infinite rooftop envelopes of the sequence are θ-plurisubharmonic. A small error integral of each potential against its own Monge–Ampère measure bounds the corresponding error of the rooftops by a summable tail. Weighted lower semicontinuity passes this bound to the limit, and singular domination identifies the limit of the lower envelopes with the L^1 limit. Squeezing the sequence between lower and upper envelopes then gives convergence in capacity, and a domination of the measures also controls their mass on deep sublevels.

Theorem 15.8.1 *Let $\theta \geq 0$ be big. Suppose $\varphi \in \mathrm{PSH}(X,\theta)$ satisfies $P_{\delta\theta}(\varphi) \not\equiv -\infty$ for some $0 < \delta < 1$. Let $u_j \in \mathrm{PSH}(X,\theta)$ satisfy, for a fixed C,*

$$\varphi - C \leq u_j \leq C, \qquad u_j \longrightarrow u \text{ in } L^1(X), \qquad u \preceq \varphi.$$

If

$$I_j := \int_X |\mathrm{e}^{u_j} - \mathrm{e}^{u}|\,\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) \longrightarrow 0, \tag{15.7}$$

then $u_j \to u$ in capacity.

The condition $u \preceq \varphi$ holds if $u_j \leq \varphi + C$ for all j. The approximants themselves need not have the singularity type of φ; they may be bounded while φ is unbounded.

Proof The canonical representative of the limit satisfies $\varphi - C \leq u \leq C$, hence $u \sim \varphi$. It suffices to show that every subsequence has a further subsequence converging in capacity. We may therefore assume that $I_j \leq 2^{-j}$ and that $u_j \to u$ almost everywhere. For $N \geq j$ define

$$p_{j,N} = P_\theta(\min\{u_j, \dots, u_N\}), \qquad p_j = P_\theta\left(\inf_{\ell \geq j} u_\ell\right).$$

All these envelopes lie between $\varphi - C$ and C. The contact theorem and ordered contact give an inequality for the finite rooftops. Partition the contact set of $p_{j,N}$ according to the first index ℓ for which $p_{j,N} = u_\ell$. On the corresponding piece, the measure of $p_{j,N}$ is bounded by that of u_ℓ, and $\mathrm{e}^{p_{j,N}} = \mathrm{e}^{u_\ell}$. Therefore

$$\int_X |\mathrm{e}^{p_{j,N}} - \mathrm{e}^u| \, \mathrm{MA}_\theta^{\mathrm{ext}}(p_{j,N}) \leq \sum_{\ell=j}^{N} I_\ell \leq 2^{1-j}.$$

Only positive measures enter this argument, so it applies also when their total masses are infinite.

As $N \to \infty$, $p_{j,N}$ decreases to p_j. Indeed, the limit lies below every finite minimum, hence below the infimum quasi-everywhere, and every competitor for p_j lies below every finite rooftop. The sequence p_j increases, and its canonical limit p is θ-plurisubharmonic. By Lemma 1.8.9, both monotone limits hold in capacity. Applying Lemma 7.3.1 to the bounded nonnegative weights $|\mathrm{e}^{p_{j,N}} - \mathrm{e}^u|$, first in N and then in j, we obtain

$$\int_X |\mathrm{e}^{p} - \mathrm{e}^u| \, \mathrm{MA}_\theta^{\mathrm{ext}}(p) = 0.$$

The inequalities $p_j \leq u_\ell$ for all $\ell \geq j$ imply $p \leq u$, first almost everywhere and then for the canonical representatives. Since $\varphi - C \leq p \leq u$, also $p \sim u \sim \varphi$. Off the pole sets, $\mathrm{e}^p < \mathrm{e}^u$ exactly on $\{p < u\}$, so $\mathrm{MA}_\theta^{\mathrm{ext}}(p)(\{p < u\}) = 0$. Since $P_{\delta\theta}(\varphi) - C$ is a competitor for $P_{\delta\theta}(p)$, the latter is not identically $-\infty$. Theorem 15.1.1, applied with coefficient zero, gives $p \geq u$, and hence $p = u$.

Finally, by Hartogs' lemma and the almost-everywhere convergence, $U_j = (\sup_{\ell \geq j} u_\ell)^*$ decreases to u. The inequalities $p_j \leq u_j \leq U_j$, together with the convergence in capacity of both monotone sequences, prove the claim. □

Corollary 15.8.1 *In the setting of Theorem 15.8.1, suppose either*

(1) $\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) \leq \nu$ *for a fixed finite non-pluripolar measure* ν*; or*
(2) $\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) \leq A\omega_{w_j}^n$*, where* A *is fixed and* $w_j \in \mathrm{PSH}(X, \omega)$ *are uniformly bounded and converge in capacity to a bounded* w*.*

Then $u_j \to u$ *in capacity, all the measures* $\mathrm{MA}_\theta^{\mathrm{ext}}(u_j)$ *and* $\mathrm{MA}_\theta^{\mathrm{ext}}(u)$ *have finite mass, and*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) \rightharpoonup \mathrm{MA}_\theta^{\mathrm{ext}}(u).$$

Proof In the first case, Lemma 1.23.1, applied to the exponentials e^{u_j}, proves (15.7). In the second case we use Lemma 1.23.3 instead, since these exponentials are bounded and have a common lower bound for their curvature. In both cases the criterion gives convergence in capacity.

The masses are uniformly bounded by those of the dominating measures. For the deep sublevels we use $u_j \geq \varphi - C$, which gives $\{u_j < -T\} \subset \{\varphi < C - T\}$. In the first case the masses of these sublevels are at most $\nu(\{\varphi < C - T\})$, which tends to zero as $T \to \infty$ because ν does not charge the pole set of φ. In the second case Lemma 1.8.6 bounds them by $C' \operatorname{Cap}_\omega(\{\varphi < C - T\})$, which also tends to zero. The tail criterion Theorem 8.4.1, in the general-background form of Theorem 8.6.1, now gives finiteness of the limiting measure and weak convergence. □

When no common lower barrier is available, one has to be constructed in a relative envelope class; this is done in Section 33.6.

15.9 A bounded subsolution theorem

The bounded exponential equation remains solvable for right-hand sides that are singular with respect to smooth volume, provided they have a bounded density with respect to the Monge–Ampère measure of a bounded potential. We regularize this potential to obtain equations with bounded densities with respect to smooth volume; their solutions are controlled by a moving dominating measure. When the density is allowed to vanish, the normalizations are bounded by the mixed density inequality of Lemma 1.7.3 and the trace estimate for a fixed reference in Lemma 1.8.10.

Theorem 15.9.1 *Let θ be semipositive and big. Suppose*

$$\mu = f \operatorname{MA}_\theta(\psi), \qquad \psi \in \operatorname{PSH}(X,\theta) \cap L^\infty(X), \qquad 0 \leq f \in L^\infty(\operatorname{MA}_\theta(\psi)),$$

and $\mu(X) > 0$. For every $\lambda > 0$ there is a unique bounded $u \in \operatorname{PSH}(X,\theta)$ satisfying

$$\operatorname{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu.$$

Proof Write $\nu = \operatorname{MA}_\theta(\psi)$, a finite non-pluripolar measure.

Step 1: smooth strictly positive density. First suppose f is smooth and strictly positive. Choose a constant $A \geq 1$ with $\Omega = A\omega \geq \theta$ and, by Theorem 1.11.1, smooth Ω-plurisubharmonic functions $\psi_j \downarrow \psi$. The envelopes

$$q_j = P_\theta(\psi_j)$$

lie between ψ and ψ_j, so they are uniformly bounded and decrease to ψ. Contact support and ordered contact in the background Ω give

$$\operatorname{MA}_\theta(q_j) \leq \mathbf{1}_{\{q_j = \psi_j\}} \Omega^n_{\psi_j}.$$

In particular, each $\operatorname{MA}_\theta(q_j)$ has a bounded density with respect to smooth volume. Moreover $\operatorname{MA}_\theta(q_j)(X) > 0$ by Proposition 14.2.1. Theorem 15.2.1 therefore gives bounded $u_j \in \operatorname{PSH}(X,\theta)$ with

$$\operatorname{MA}_\theta(u_j) = \mathrm{e}^{\lambda u_j} f \operatorname{MA}_\theta(q_j).$$

Since f is bounded above and away from zero, we can choose C such that $e^{\lambda(q_j-C)} f \le 1 \le e^{\lambda(q_j+C)} f$ for all j. Then $q_j - C$ and $q_j + C$ are a subsolution and a supersolution, and bounded exponential comparison gives $q_j - C \le u_j \le q_j + C$.

Extract an L^1 limit u, which is again bounded. The moving domination

$$\mathrm{MA}_\theta(u_j) \le C' \Omega^n_{\psi_j}$$

and Corollary 15.8.1, applied with the reference zero and with the uniformly bounded ω-plurisubharmonic functions $w_j = \psi_j / A$, for which $\Omega^n_{\psi_j} = A^n \omega^n_{w_j}$, give convergence in capacity. Continuity of bounded products under convergence in capacity, with the quasi-continuous weights $e^{\lambda u_j} f$, then gives

$$\mathrm{MA}_\theta(u) = e^{\lambda u} f \nu.$$

Step 2: bounded measurable density. Choose smooth strictly positive functions f_j with a common upper bound $L > 0$ such that $f_j \to f$ in $L^1(\nu)$. Such an approximation exists. Continuous functions are dense in $L^1(\nu)$, since ν is a finite Borel measure on the compact manifold X, and continuous functions are uniform limits of smooth ones. Truncating to a fixed bounded interval and adding a positive constant tending to zero preserves the bounds and the convergence. Step 1 gives bounded u_j with

$$\mathrm{MA}_\theta(u_j) = e^{\lambda u_j} f_j \nu.$$

For a fixed C we have $e^{\lambda(\psi - C)} f_j \le 1$ for all j, so $\psi - C$ is a subsolution and $u_j \ge \psi - C$. The upper bound requires a separate argument, because $\inf_X f_j$ may tend to zero.

The construction in Step 1 also gives the mixed inequality

$$\theta_{u_j} \wedge \theta_\psi^{n-1} \ge e^{\lambda u_j / n} f_j^{1/n} \nu. \tag{15.8}$$

To prove it without a mixed inequality for arbitrary singular measures, we use the approximants $u_{j,k}$ and q_k of Step 1, for which

$$\mathrm{MA}_\theta(u_{j,k}) = e^{\lambda u_{j,k}} f_j \mathrm{MA}_\theta(q_k),$$

and both Monge–Ampère measures have densities with respect to smooth volume. Lemma 1.7.3 applies to the pair $(u_{j,k}, q_k)$. For fixed j, both sequences are uniformly bounded and converge in capacity to (u_j, ψ), so continuity of bounded products against quasi-continuous weights passes the mixed inequality to the limit and yields (15.8).

Put $M_j = \sup_X u_j$, $v_j = u_j - M_j$ and $c_j = e^{\lambda M_j}$. Suppose the M_j were unbounded above. After extraction, $M_j \to +\infty$ and $v_j \to v$ in $L^1(X)$, where $v \in \mathrm{PSH}(X, \theta)$ is normalized. By Lemma 1.23.1, $e^{\lambda v_j / n} \to e^{\lambda v / n}$ in $L^1(\nu)$. Together with the inequalities

$$|f_j^{1/n} - f^{1/n}|^n \le |f_j - f|, \qquad 0 \le f_j \le L,$$

this gives

$$\int_X \mathrm{e}^{\lambda v_j/n} f_j^{1/n}\,\mathrm{d}\nu \longrightarrow \int_X \mathrm{e}^{\lambda v/n} f^{1/n}\,\mathrm{d}\nu > 0;$$

the last integral is positive because $\mu(X) > 0$ and ν does not charge the pole set of v. Integrating (15.8) and applying Lemma 1.8.10, we find some $a > 0$ such that, along the extraction,

$$a c_j^{1/n} \le \int_X \theta_{v_j} \wedge \theta_\psi^{n-1} \le C_\psi,$$

contradicting $M_j \to +\infty$. Thus the u_j are uniformly bounded above as well as below.

Extract an L^1 limit u. Now $\mathrm{MA}_\theta(u_j) \le C''\nu$, so Corollary 15.8.1 gives convergence in capacity and weak convergence of the Monge–Ampère measures. The right-hand sides converge in total variation, by the convergence of exponentials against the fixed measure ν and the estimate

$$\int_X |\mathrm{e}^{\lambda u_j} f_j - \mathrm{e}^{\lambda u} f|\,\mathrm{d}\nu \le C\|f_j - f\|_{L^1(\nu)} + \|f\|_\infty \int_X |\mathrm{e}^{\lambda u_j} - \mathrm{e}^{\lambda u}|\,\mathrm{d}\nu.$$

Consequently $\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu$. Uniqueness follows from bounded exponential comparison. □

15.10 Adding weighted Monge–Ampère measures

Let u and v be potentials of the same ordinary singularity type. We show that the sum of their weighted measures $\mathrm{e}^{-\lambda u}\mathrm{MA}_\theta(u) + \mathrm{e}^{-\lambda v}\mathrm{MA}_\theta(v)$ is again of the form $\mathrm{e}^{-\lambda a}\mathrm{MA}_\theta(a)$ for a potential a of that type. Truncating u and v gives bounded equations, solved by the bounded subsolution theorem, whose solutions have a common lower barrier. Their exponential error tends to zero even though the approximating masses may grow, so the common-barrier criterion gives convergence in capacity. This addition step will remove a deficit of mass in the existence proof of Section 15.11.

Theorem 15.10.1 *Let $\theta \geq 0$ be big, and suppose $\varphi \in \mathrm{PSH}(X,\theta)$ satisfies $P_{\delta\theta}(\varphi) \not\equiv -\infty$ for some $0 < \delta < 1$. Let $u, v \in \mathrm{PSH}(X,\theta)$ satisfy $u, v \sim \varphi$, and let $\lambda > 0$. If*

$$\mu = \mathrm{e}^{-\lambda u}\mathrm{MA}_\theta^{\mathrm{ext}}(u) + \mathrm{e}^{-\lambda v}\mathrm{MA}_\theta^{\mathrm{ext}}(v)$$

is a finite measure, then there exists $a \in \mathrm{PSH}(X,\theta)$, with $a \sim \varphi$, such that

$$\mathrm{MA}_\theta^{\mathrm{ext}}(a) = \mathrm{e}^{\lambda a}\mu.$$

Here the weights are evaluated off the pole sets. The solution a is unique in the ordinary singularity type of φ.

Proof *Step 1: truncated equations and barriers.* Choose $S \geq \max(\sup_X u, \sup_X v)$. A common translation reduces the theorem to the case $u, v \leq 0$: translating u and v by $-S$ multiplies μ by $\mathrm{e}^{\lambda S}$, and the resulting solution is then translated back by S.

For integers $j \geq 1$, set $u_j = \max(u, -j)$, $v_j = \max(v, -j)$ and

$$\mu_j = \mathrm{e}^{-\lambda u_j}\mathrm{MA}_\theta(u_j) + \mathrm{e}^{-\lambda v_j}\mathrm{MA}_\theta(v_j).$$

For $s_j = (u_j + v_j)/2$, positivity of mixed products gives

$$\mu_j \leq 2^n \mathrm{e}^{\lambda j}\mathrm{MA}_\theta(s_j).$$

Thus μ_j has a bounded density with respect to the Monge–Ampère measure of the bounded potential s_j, and $\mu_j \neq 0$ by Proposition 14.2.1. The bounded subsolution theorem Theorem 15.9.1 therefore gives bounded $a_j \in \mathrm{PSH}(X,\theta)$ with $\mathrm{MA}_\theta(a_j) = \mathrm{e}^{\lambda a_j}\mu_j$, and bounded exponential comparison yields

$$a_j \leq \min(u_j, v_j) \leq 0.$$

Choose C with $|u - v| \leq C$ off the common pole set; then $|u_j - v_j| \leq C$. Put

$$D = C/2 + n\lambda^{-1}\log 2, \qquad b_j = s_j - D.$$

Because $|u_j - v_j| \leq C$, positivity of mixed products gives

$$\mathrm{e}^{-\lambda b_j}\mathrm{MA}_\theta(b_j) \geq 2^{-n}\mathrm{e}^{\lambda D - \lambda s_j}\big(\mathrm{MA}_\theta(u_j) + \mathrm{MA}_\theta(v_j)\big) \geq \mu_j.$$

Hence $a_j \geq b_j \geq w$, where

$$w = (u + v)/2 - D \sim \varphi$$

is a fixed θ-plurisubharmonic function. Extract a subsequence with $a_j \to a$ in $L^1(X)$. The upper barriers decrease to $\min(u, v)$, so $w \leq a \leq \min(u, v)$ and $a \sim \varphi$.

Step 2: convergence in capacity. Write $E_j = \{\min(u, v) > -j\}$. By locality, $\mu_j = \mu$ on E_j; since $a_j \leq 0$, it follows that

$$\int_{E_j} |\mathrm{e}^{a_j} - \mathrm{e}^{a}|\,\mathrm{MA}_\theta(a_j) \le \int_X |\mathrm{e}^{a_j} - \mathrm{e}^{a}|\,\mathrm{d}\mu \longrightarrow 0$$

by Lemma 1.23.1. On $X \setminus E_j$ we have $a_j \le -j$ and $a \le -j$, whence $|\mathrm{e}^{a_j} - \mathrm{e}^{a}| \le \mathrm{e}^{-j}$. Moreover

$$\mathrm{MA}_\theta(a_j) = \mathrm{e}^{\lambda(a_j-u_j)}\mathrm{MA}_\theta(u_j) + \mathrm{e}^{\lambda(a_j-v_j)}\mathrm{MA}_\theta(v_j) \le \mathrm{MA}_\theta(u_j) + \mathrm{MA}_\theta(v_j).$$

Choose a Hermitian form $\Omega \ge \theta$. Since $\mathrm{MA}_\theta(u_j) \le \mathrm{MA}_\Omega(u_j)$ and $-j \le u_j \le 0$, and similarly for v_j, the range estimate of Proposition 8.1.1, with $r_n = \max(n-2, 0)$, gives

$$\int_{X\setminus E_j} |\mathrm{e}^{a_j} - \mathrm{e}^{a}|\,\mathrm{MA}_\theta(a_j) \le C'\mathrm{e}^{-j}(1+j)^{r_n} \longrightarrow 0.$$

Since $w \le a_j \le 0$ and $a \sim \varphi$, Theorem 15.8.1, applied with the reference w, now gives $a_j \to a$ in capacity.

Step 3: the limiting equation. We identify the equation on sets where w is bounded below, without first proving weak convergence of the measures $\mathrm{MA}_\theta(a_j)$. For $T > 0$, let g_T be the continuous cutoff from the proof of Lemma 7.3.1, and put $\zeta_T = g_T(w)$. Where $\zeta_T > 0$, we have $w > -2T$, hence every a_j and a is above $-2T$. If $j > 2T$ and $x \notin E_j$, the inequalities $w(x) \le a_j(x) \le -j$ force $\zeta_T(x) = 0$. Consequently, for all $j > 2T$,

$$\zeta_T\mathrm{MA}_\theta(a_j) = \zeta_T\mathrm{e}^{\lambda a_j}\mu.$$

We test this identity against a continuous function. On the left we may replace a_j by $\max(a_j, -2T)$ and use continuity of bounded products under convergence in capacity, with the fixed bounded quasi-continuous weight ζ_T. On the right, the exponentials converge against the fixed measure μ, which gives convergence in total variation. Therefore

$$\zeta_T\mathrm{MA}_\theta^{\mathrm{ext}}(a) = \zeta_T\mathrm{e}^{\lambda a}\mu.$$

As $T \to \infty$, the cutoffs increase to one off the pole set of w, which no measure involved charges. This proves the equation and, in particular, shows that $\mathrm{MA}_\theta^{\mathrm{ext}}(a)$ has finite mass. Uniqueness follows from Corollary 15.1.1. □

15.11 Equations in a prescribed ordinary type

We return to a semipositive big background θ and now require the solution to have the ordinary singularity type of a fixed reference φ. Throughout this section, $\varphi \in \mathrm{PSH}(X, \theta)$ is a model reference in the sense of (14.20): an envelope fixed point, $P_\theta[\varphi] = \varphi$, that admits a smaller-background envelope. We fix $0 < \varepsilon < 1$ with $P_{\varepsilon\theta}(\varphi) \not\equiv -\infty$. Note that $\varphi \le 0$, since every competitor for $P_\theta[\varphi]$ is nonpositive. For a normalized solution u, the fixed-point property gives the upper bound $u \le \varphi$, while the smaller-background envelope $P_{\varepsilon\theta}(\varphi)$ yields a uniform lower bound $u \ge \varphi - C$. A supersolution and the addition theorem then prove existence in this type.

Theorem 15.11.1 *Let θ and φ be as above, and let $\mu = f\,\mathrm{d}V$, where $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, and $\mu(X) > 0$. Then:*

(1) *For every $\lambda > 0$ there is a unique $u \in \mathrm{PSH}(X,\theta)$ with $u \sim \varphi$ such that*

$$\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu.$$

(2) *There are $u \in \mathrm{PSH}(X,\theta)$ with $u \sim \varphi$ and a uniquely determined constant $c > 0$ such that*

$$\mathrm{MA}_\theta(u) = c\mu.$$

The potential in part (2) need not be unique up to an additive constant, even for a Hermitian background and the reference $\varphi = 0$; see the counterexample in Theorem 12.6.1.

The proof uses two estimates in the prescribed type. The first is a lower bound for the norm of a Monge–Ampère density.

Lemma 15.11.1 *Fix $q > 1$. There is $b_q > 0$ such that every $u \in \mathrm{PSH}(X,\theta)$ with $u \sim \varphi$ and*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) = F\,\mathrm{d}V, \qquad 0 \le F \in L^q(\mathrm{d}V),$$

satisfies $\|F\|_q \ge b_q$. For fixed θ and ε, the constant b_q is independent of φ.

Proof Since $u \sim \varphi$, the smaller-background envelope $P_{\varepsilon\theta}(u)$ is not identically $-\infty$, so $\mathrm{MA}_\theta^{\mathrm{ext}}(u) \ne 0$ by Lemma 14.5.2; hence $N = \|F\|_q > 0$.

Apply Lemma 15.2.1 in the semipositive big background $(1-\varepsilon)\theta$ to F/N, whose L^q norm is one. It gives a bounded s with $-1 \le s \le 0$ and a constant $b_q > 0$, depending only on this fixed background and on q, such that

$$\mathrm{MA}_{(1-\varepsilon)\theta}(s) \ge (b_q/N)F\,\mathrm{d}V.$$

Put $h = P_{\varepsilon\theta}(\varphi) + s$. This is a θ-plurisubharmonic function with $h \preceq u$. Since θ_h is the sum of the positive currents $\varepsilon\theta + \mathrm{dd^c}P_{\varepsilon\theta}(\varphi)$ and $(1-\varepsilon)\theta + \mathrm{dd^c}s$, the same density bound holds for $\mathrm{MA}_\theta^{\mathrm{ext}}(h)$. To justify this, we truncate $P_{\varepsilon\theta}(\varphi)$ at finite levels, use positivity of bounded mixed products, and exhaust the set where $P_{\varepsilon\theta}(\varphi)$ is finite; its pole set has zero volume.

If $N < b_q$, we would have

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) \le (N/b_q)\mathrm{MA}_\theta^{\mathrm{ext}}(h+t) \quad \text{for every } t \in \mathbb{R}.$$

Since $N/b_q < 1$, Theorem 15.1.1 would then imply $u \ge h + t$ for every t, which is impossible at a point where both potentials are finite. Thus $N \ge b_q$. The construction of s used no property of φ, which proves the last assertion. □

The lemma bounds the L^q norm of the density from below; it gives no positive lower bound for the total mass $\int_X F\,\mathrm{d}V$.

The second estimate bounds a normalized solution from below by φ minus a uniform constant. It is what keeps limits of solutions in the ordinary singularity type of φ.

Theorem 15.11.2 *Let θ and φ be as above. Suppose $u \in \mathrm{PSH}(X,\theta)$, $u \sim \varphi$, $\sup_X u = 0$, and*

$$\mathrm{MA}_\theta(u) \le f\,\mathrm{d}V, \qquad 0 \le f \in L^p(\mathrm{d}V), \quad p > 1.$$

There is a constant C, depending only on the fixed geometry, φ, p, and an upper bound for $\|f\|_{L^p}$, such that

$$\varphi - C \le u \le \varphi.$$

Proof The upper bound follows directly from the fixed-point property: since $u \le 0$ and $u \preceq \varphi$, the function u is a competitor for $P_\theta[\varphi] = \varphi$.

For the lower bound we construct a subsolution ψ of the type of φ whose measure dominates a multiple of $\mathrm{e}^{-\gamma u}\mathrm{MA}_\theta(u)$, and then apply singular domination. Put $\varphi_\varepsilon = P_{\varepsilon\theta}(\varphi)$, so that $\varphi_\varepsilon \le \varphi \le 0$. Uniform exponential integrability (Lemma 1.5.1) for the normalized potentials u and exponential integrability for the fixed function φ_ε allow us to choose $\gamma > 0$ and $1 < q < p$ such that

$$g = \mathrm{e}^{-\gamma(\varphi_\varepsilon + u)} f$$

has a uniform L^q bound. Explicitly, we first choose $q < p$ and apply Hölder's inequality with exponents p/q and $p/(p-q)$; then we choose γ so small that the exponential factor is controlled by the Cauchy–Schwarz inequality and the two exponential integrability estimates.

Theorem 15.2.2, applied in the semipositive big background $(1-\varepsilon)\theta$ to the density $1+g$, gives a bounded v with $\sup_X v = 0$ and a constant $c > 0$ such that

$$((1-\varepsilon)\theta + \mathrm{dd^c} v)^n = c(1+g)\,\mathrm{d}V.$$

Both $\|v\|_\infty$ and c^{-1} are uniformly bounded. Adding the constant one to g makes the root integral of the density at least one, so no lower bound for the integral of f is needed.

Put $h = \varphi_\varepsilon + v \le \varphi$, which is θ-plurisubharmonic, and define

$$\chi(t) = \frac{1}{1-t} \quad (t \le 0), \qquad \psi = \varphi + \chi(h - \varphi),$$

where the formula for ψ is evaluated off the pole sets. Then $0 \le \chi \le 1$, $0 < \chi' \le 1$, and $\chi'' > 0$. For bounded potentials with $h \le \varphi$, Lemma 1.7.6, applied with $G(x,y) = x + \chi(y - x)$, gives

$$\theta + \mathrm{dd^c} G(\varphi, h) \ge (1 - \chi'(h-\varphi))\theta_\varphi + \chi'(h-\varphi)\theta_h \ge 0.$$

For unbounded potentials we apply this bounded assertion to $\varphi_k = \max(\varphi, -k)$ and $h_k = \max(h, -k) \le \varphi_k$. Since G is nondecreasing in both variables, $G(\varphi_k, h_k)$ decreases as k increases. Its limit is θ-plurisubharmonic and lies between φ and $\varphi + 1$; hence it is not identically $-\infty$, and it equals ψ. Locality on common finite levels and positivity of bounded products give

$$\mathrm{MA}^{\mathrm{ext}}_\theta(\psi) \ge (\varphi - h + 1)^{-2n}\mathrm{MA}^{\mathrm{ext}}_\theta(h) \ge c(\varphi - h + 1)^{-2n}\mathrm{e}^{-\gamma\varphi_\varepsilon}\mathrm{e}^{-\gamma u} f\,\mathrm{d}V,$$

where the second inequality uses $\mathrm{MA}_\theta^{\mathrm{ext}}(h) \geq ((1-\varepsilon)\theta + \mathrm{dd^c}v)^n \geq cg\,\mathrm{d}V$. These inequalities are first obtained off the pole sets; since pole sets have zero volume, they extend by exhaustion to all Borel sets.

Write $T = -\varphi_\varepsilon \geq 0$. If $C_1 \geq 1 + \|v\|_\infty$, then $\varphi - h + 1 \leq T + C_1$. Since

$$A := \inf_{T\geq 0} \frac{\mathrm{e}^{\gamma T}}{(T+C_1)^{2n}} > 0,$$

we obtain, with a uniform $d > 0$,

$$\mathrm{MA}_\theta^{\mathrm{ext}}(\psi) \geq d\mathrm{e}^{-\gamma u}\mathrm{MA}_\theta^{\mathrm{ext}}(u).$$

Choose $K = 2+\gamma^{-1}|\log d|$. On $\{u < \psi - K\}$ we have $u < 1-K$, because $\psi \leq \varphi+1 \leq 1$; hence

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) \leq d^{-1}\mathrm{e}^{\gamma(1-K)}\mathrm{MA}_\theta^{\mathrm{ext}}(\psi - K) \leq \mathrm{e}^{-\gamma}\mathrm{MA}_\theta^{\mathrm{ext}}(\psi - K).$$

The potentials u and $\psi - K$ have the same ordinary singularity type. Moreover, $u \geq \varphi - C_u$ for some constant C_u, so $P_{\varepsilon\theta}(u) \geq \varphi_\varepsilon - C_u$ is not identically $-\infty$. Theorem 15.1.1 therefore yields $u \geq \psi - K \geq \varphi - K$. All constants used to choose K are uniform over the family in the theorem; the possibly nonuniform constant C_u serves only to show that the smaller-background envelope of u is not identically $-\infty$, and it enters no estimate. □

The existence proof starts from a supersolution in the prescribed type. It is obtained from a bounded solution by a rooftop envelope with φ.

Lemma 15.11.2 *For nonzero $0 \leq f \in L^p(\mathrm{d}V)$, $p > 1$, and $\lambda > 0$, there exists $w \in \mathrm{PSH}(X,\theta)$ with $w \sim \varphi$ such that*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(w) \leq \mathrm{e}^{\lambda w} f\,\mathrm{d}V.$$

Proof Let v be the bounded solution of $\mathrm{MA}_\theta(v) = \mathrm{e}^{\lambda v} f\,\mathrm{d}V$ given by Theorem 15.2.1, and write $M = \sup_X v$ and $a = v - M \leq 0$. The rooftop envelope $r = P_\theta(a,\varphi)$ satisfies $\varphi - \|a\|_\infty \leq r \leq \varphi$, so $r \sim \varphi$. By the rooftop inequality of Proposition 14.1.3,

$$\mathrm{MA}_\theta^{\mathrm{ext}}(r) \leq \mathbf{1}_{\{r=a\leq\varphi\}}\mathrm{MA}_\theta(a) + \mathbf{1}_{\{r=\varphi<a\}}\mathrm{MA}_\theta^{\mathrm{ext}}(\varphi).$$

The second term vanishes: by Proposition 14.1.4, the measure of φ is carried by $\{\varphi = 0\}$, whereas $\varphi < a \leq 0$ on the set in question. On the first contact set, $a = r$ and $\mathrm{MA}_\theta(a) = \mathrm{e}^{\lambda(a+M)} f\,\mathrm{d}V$. Thus $\mathrm{MA}_\theta^{\mathrm{ext}}(r) \leq \mathrm{e}^{\lambda(r+M)} f\,\mathrm{d}V$, and $w = r + M$ has the required properties. □

15.11.1 Existence for a positive exponent

We prove part (1) of Theorem 15.11.1 by maximizing a weighted mass. Among the potentials $z \sim \varphi$ whose weighted measure $\mathrm{e}^{-\lambda z}\mathrm{MA}_\theta(z)$ lies below $\mu = f\,\mathrm{d}V$, we find

one of maximal weighted mass; Theorem 15.11.2 provides the compactness needed for this. If its weighted measure were strictly smaller than μ, the addition theorem would produce a potential of larger weighted mass.

Proof (of Theorem 15.11.1, part (1)) *Step 1: a maximizing sequence.* Fix $\lambda > 0$. By Lemma 15.11.2, there is $w \sim \varphi$ such that

$$\nu_0 := \mathrm{e}^{-\lambda w}\mathrm{MA}_\theta^{\mathrm{ext}}(w) \le \mu = f\,\mathrm{d}V.$$

Consider all potentials $z \sim \varphi$ for which

$$\nu_0 \le \nu_z := \mathrm{e}^{-\lambda z}\mathrm{MA}_\theta^{\mathrm{ext}}(z) \le \mu,$$

and let B be the supremum of their weighted masses $\nu_z(X)$. This family is nonempty, since it contains w, and $B \le \mu(X) < +\infty$. Choose u_j in the family with $\nu_{u_j}(X) \to B$. Since $\nu_0 \le \nu_{u_j}$, Corollary 15.1.1 gives $u_j \le w$.

Step 2: uniform bounds. We need a lower bound for the suprema of the u_j. Put $\varphi_\varepsilon = P_{\varepsilon\theta}(\varphi) \le 0$, and let ψ be the bounded solution of

$$((1-\varepsilon)\theta + \mathrm{dd}^{\mathrm{c}}\psi)^n = \mathrm{e}^{\lambda\psi} f\,\mathrm{d}V$$

given by Theorem 15.2.1. For $h = \psi + \varphi_\varepsilon$, positivity of the splitting $\theta_h = (\varepsilon\theta + \mathrm{dd}^{\mathrm{c}}\varphi_\varepsilon) + ((1-\varepsilon)\theta + \mathrm{dd}^{\mathrm{c}}\psi)$ gives

$$\mathrm{e}^{-\lambda h}\mathrm{MA}_\theta^{\mathrm{ext}}(h) \ge \mathrm{e}^{-\lambda\varphi_\varepsilon} f\,\mathrm{d}V \ge \mu \ge \nu_{u_j}.$$

Here $h \preceq u_j$, because $h \preceq \varphi$ and $u_j \sim \varphi$, and each u_j admits a smaller-background envelope because $u_j \sim \varphi$. Corollary 15.1.1 therefore gives $u_j \ge h$. Consequently

$$\sup_X h \le M_j := \sup_X u_j \le \sup_X w,$$

and the lower bound is finite.

We now apply Theorem 15.11.2 to $v_j = u_j - M_j$. Their measures satisfy

$$\mathrm{MA}_\theta^{\mathrm{ext}}(v_j) = \mathrm{MA}_\theta^{\mathrm{ext}}(u_j) \le \mathrm{e}^{\lambda \sup_X w} f\,\mathrm{d}V,$$

so $\varphi - C \le v_j \le \varphi$. Together with the bounds for M_j, this shows that all u_j lie within a fixed bounded distance of φ.

Step 3: the limit. Extract an L^1 limit u. By Corollary 15.8.1, applied with the dominating measure $\mathrm{e}^{\lambda \sup_X w}\mu$, the convergence holds in capacity, $u \sim \varphi$, and

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) \rightharpoonup \mathrm{MA}_\theta^{\mathrm{ext}}(u).$$

After a further extraction we may also assume $\nu_{u_j} \rightharpoonup \nu$. Since $\nu_0 \le \nu_{u_j} \le \mu$, the limit satisfies the same inequalities. We claim that

$$\mathrm{e}^{\lambda u_j}\nu_{u_j} \rightharpoonup \mathrm{e}^{\lambda u}\nu.$$

Indeed, $e^{\lambda u_j} \to e^{\lambda u}$ in $L^1(\mu)$ by Lemma 1.23.1, and all the measures ν_{u_j} are at most μ; this controls the error caused by changing the weight. The fixed bounded weight $e^{\lambda u}$ can be approximated in $L^1(\mu)$ by continuous functions, and since both ν_{u_j} and ν are dominated by μ, this approximation is uniform in the integrals. The weak convergence $\nu_{u_j} \rightharpoonup \nu$ then gives the claim, and the same argument applies after multiplication by any continuous test function.

Since $e^{\lambda u_j}\nu_{u_j} = \mathrm{MA}_\theta^{\mathrm{ext}}(u_j)$, it follows that

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) = e^{\lambda u}\nu, \qquad e^{-\lambda u}\mathrm{MA}_\theta^{\mathrm{ext}}(u) = \nu, \qquad \nu(X) = B.$$

The second identity holds off the pole set of u, which has zero μ-measure. Hence u belongs to the family and attains the supremum B. Write $\nu = g\,dV$, with $0 \le g \le f$.

Step 4: no deficit. Suppose that $g \neq f$ in $L^1(dV)$. Applying Lemma 15.11.2 to $f - g$, we find $v \sim \varphi$ with

$$\nu_v := e^{-\lambda v}\mathrm{MA}_\theta^{\mathrm{ext}}(v) \le (f - g)\,dV.$$

Since $v \sim \varphi$ admits a smaller-background envelope, $\mathrm{MA}_\theta^{\mathrm{ext}}(v) \neq 0$ by Lemma 14.5.2. Thus $\nu_v(X) > 0$.

The sum $\nu + \nu_v$ is finite, so Theorem 15.10.1 gives $a \sim \varphi$ with

$$e^{-\lambda a}\mathrm{MA}_\theta^{\mathrm{ext}}(a) = \nu + \nu_v.$$

This measure still lies between ν_0 and μ, but its mass is $B + \nu_v(X) > B$, a contradiction. Therefore $g = f$, and u solves the required equation. Uniqueness in the prescribed ordinary type follows from Corollary 15.1.1. □

15.11.2 The normalized equation

For the normalized equation we let the positive exponent tend to zero. Bounded comparison controls the normalizing constants from above, while the lower density estimate of Lemma 15.11.1 keeps them away from zero. The common barriers then give convergence of the Monge–Ampère measures and keep the limit in the prescribed type.

Proof (of Theorem 15.11.1, part (2)) For each $j \ge 1$, part (1) gives $u_j \sim \varphi$ with

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) = e^{u_j/j} f\,dV.$$

Write $M_j = \sup_X u_j$, $v_j = u_j - M_j$ and $c_j = e^{M_j/j}$. Let h be a bounded normalized solution of $\mathrm{MA}_\theta(h) = \alpha f\,dV$, with $\alpha > 0$, given by Theorem 15.2.2.

For $H_j = h + \|h\|_\infty + j \log \alpha$ we have

$$\mathrm{e}^{-H_j/j} \mathrm{MA}_\theta(H_j) = \mathrm{e}^{-(h+\|h\|_\infty)/j} f \, \mathrm{d}V \le f \, \mathrm{d}V = \mathrm{e}^{-u_j/j} \mathrm{MA}_\theta^{\mathrm{ext}}(u_j).$$

The bounded function H_j admits a smaller-background envelope, and $u_j \preceq H_j$, so Corollary 15.1.1 gives $u_j \le H_j$. In particular,

$$0 < c_j \le \alpha \exp(2\|h\|_\infty / j).$$

The lower density bound of Lemma 15.11.1, applied to v_j with $q = p$, gives

$$b_p \le \|c_j \mathrm{e}^{v_j/j} f\|_p \le c_j \|f\|_p,$$

so the c_j are also bounded away from zero.

Since $\sup_X v_j = 0$ and $\mathrm{MA}_\theta^{\mathrm{ext}}(v_j) = c_j \mathrm{e}^{v_j/j} f \, \mathrm{d}V \le C f \, \mathrm{d}V$, Theorem 15.11.2 gives $\varphi - C' \le v_j \le \varphi$. Extract a subsequence with $v_j \to v$ in $L^1(X)$ and $c_j \to c > 0$. By Corollary 15.8.1, the convergence holds in capacity, $v \sim \varphi$, and the Monge–Ampère measures converge weakly. Meanwhile, Lemma 1.23.2 gives

$$\int_X |\mathrm{e}^{v_j/j} - 1| f \, \mathrm{d}V \longrightarrow 0.$$

Thus the right-hand sides converge in total variation to $c f \, \mathrm{d}V$, and $\mathrm{MA}_\theta^{\mathrm{ext}}(v) = c f \, \mathrm{d}V$.

For the uniqueness of the constant, suppose that $a, b \sim \varphi$ solve the normalized equation with constants $c_a < c_b$. For every real t,

$$\mathrm{MA}_\theta^{\mathrm{ext}}(a) = \frac{c_a}{c_b} \mathrm{MA}_\theta^{\mathrm{ext}}(b + t),$$

so Theorem 15.1.1 gives $a \ge b + t$. This is impossible as $t \to +\infty$, at any point where both potentials are finite. Interchanging the roles of the two constants proves that c is unique. □

15.12 Comparison above a strict singular barrier

From now on, θ is an arbitrary smooth big real $(1,1)$-form. Then V_θ may be unbounded, and the smaller-background construction used above is no longer available. Its role is taken by a strict analytic barrier. When two potentials bounded below by the barrier are compared, adding a small multiple of the barrier forces a negative infimum of their difference to be attained away from its pole set, and there a local comparison principle rules it out.

Since θ is big, Theorem 1.15.2 provides $\rho \in \mathrm{PSH}(X, \theta)$ with analytic singularities and a smooth remainder after resolution such that $\theta_\rho \ge \kappa \omega$ for some $\kappa > 0$. Replacing ω by a small positive multiple and subtracting a constant from ρ, we fix such a ρ with $\rho \le 0$ and $\theta_\rho \ge \omega$. Put $Z = \{\rho = -\infty\}$ and $\Omega = X \setminus Z$. From now on until the end of

Section 15.17, these data remain fixed, and Ω always denotes this open set rather than an auxiliary Hermitian form. A potential bounded below by $\rho - C$ is locally bounded on Ω, and there its extended Monge–Ampère measure agrees with the local bounded product.

Lemma 15.12.1 *In this lemma, θ may be any smooth real $(1,1)$-form on X. Let u and v be bounded θ-plurisubharmonic functions on a coordinate ball $B \subset X$, and assume*

$$\liminf_{x \to \partial B} (u - v)(x) \geq 0, \qquad \mathbf{1}_{\{u<v\}} \mathrm{MA}_\theta(u) \leq \mathbf{1}_{\{u<v\}} \mathrm{MA}_\theta(v).$$

Then $u \geq v$ on B.

Proof Fix $\varepsilon > 0$ and set $v_\varepsilon = \max(u, v - \varepsilon)$, which equals u near the boundary. We claim that

$$\mathrm{MA}_\theta(u) \leq \mathrm{MA}_\theta(v_\varepsilon)$$

throughout B. On $\{u > v - \varepsilon\}$ this follows from locality, since $v_\varepsilon = u$ there. On $\{u \leq v - \varepsilon\}$ it follows from ordered contact and from the assumed inequality, which applies because this set is contained in $\{u < v\}$.

Let q be the plurisubharmonic envelope on B of the obstacle $u - v_\varepsilon$, in the sense of (1.5). This obstacle is bounded, quasi-continuous and zero near the boundary. Hence it lies between Ah and zero for a sufficiently large A, where $h < 0$ is a smooth strictly plurisubharmonic defining function for the ball. Thus q tends to zero at the boundary, and (1.5) shows that $(\mathrm{dd}^{\mathrm{c}} q)^n$ is carried by $D = \{q = u - v_\varepsilon\}$.

Since $q \leq u - v_\varepsilon$ quasi-everywhere, the canonical representatives satisfy $q + v_\varepsilon \leq u$. Ordered contact, followed by the measure inequality above, yields

$$\mathbf{1}_D \mathrm{MA}_\theta(q + v_\varepsilon) \leq \mathbf{1}_D \mathrm{MA}_\theta(u) \leq \mathbf{1}_D \mathrm{MA}_\theta(v_\varepsilon).$$

The positive expansion of $(\theta_{v_\varepsilon} + \mathrm{dd}^{\mathrm{c}} q)^n$ contains both $\mathrm{MA}_\theta(v_\varepsilon)$ and $(\mathrm{dd}^{\mathrm{c}} q)^n$. Thus $\mathbf{1}_D (\mathrm{dd}^{\mathrm{c}} q)^n = 0$, and since $(\mathrm{dd}^{\mathrm{c}} q)^n$ is carried by D, we get $(\mathrm{dd}^{\mathrm{c}} q)^n = 0$. The classical Bedford–Taylor comparison principle for bounded plurisubharmonic functions with zero boundary values implies $q = 0$. Consequently $u \geq v_\varepsilon$, first quasi-everywhere and then everywhere. Finally let $\varepsilon \downarrow 0$. □

Theorem 15.12.1 *Suppose $u, v \in \mathrm{PSH}(X, \theta)$ satisfy, for constants $C_0, C_1 \geq 0$,*

$$\min(u, v) \geq \rho - C_0, \qquad v \leq u + C_1.$$

If, for some $0 \leq c < 1$,

$$\mathbf{1}_{\{u<v\}} \mathrm{MA}_\theta(u) \leq c\, \mathbf{1}_{\{u<v\}} \mathrm{MA}_\theta(v),$$

then $u \geq v$.

Proof We first modify the barrier. Let $r = a\rho - A$, where $a > 1$ is sufficiently close to one and A is sufficiently large. Since

$$\theta + \mathrm{dd}^{\mathrm{c}}(a\rho) = a\theta_\rho + (1-a)\theta,$$

we can arrange $\theta_r \geq \omega/2$ and $r - v \leq -1$, and moreover $u - r \to +\infty$ at Z, because $u - r \geq (a-1)(-\rho) + A - C_0$.

Choose $0 < \delta < 1$ with $(1-\delta)^n > c$, and put $v_\delta = (1-\delta)v + \delta r$. Then

$$u - v_\delta \geq \delta(u - r) - (1-\delta)C_1.$$

Suppose that the infimum m of $u - v_\delta$ on Ω were negative. Since $u - v_\delta \to +\infty$ at Z, a sequence x_j along which $u - v_\delta$ tends to m has a subsequence converging to a point $x_0 \in \Omega$. Choose a coordinate ball $B \Subset \Omega$ centered at x_0, identify it with $\{|z| < 1\}$, and take

$$\eta(z) = b(1 - |z|^2), \qquad 0 < b < 1, \qquad \omega/2 + \mathrm{dd}^{\mathrm{c}}\eta \geq 0.$$

The function $w = v_\delta + \delta\eta + m$ is θ-plurisubharmonic on B, and $\liminf_{\partial B}(u - w) \geq 0$ because η vanishes on ∂B. Also

$$w - v = \delta(r - v + \eta) + m < 0.$$

Thus $\{u < w\} \subset \{u < v\}$, and positivity of mixed products gives on this set

$$\mathrm{MA}_\theta(u) \leq c\mathrm{MA}_\theta(v) \leq (1-\delta)^n \mathrm{MA}_\theta(v) \leq \mathrm{MA}_\theta(w).$$

By Lemma 15.12.1, $u \geq w$ on B. Evaluating at x_j gives $u(x_j) - v_\delta(x_j) \geq m + \delta\eta(x_j)$, which contradicts $\eta(x_0) > 0$ in the limit. Hence $u \geq v_\delta$ on Ω. Letting $\delta \downarrow 0$ along values with $(1-\delta)^n > c$, we obtain $u \geq v$ on Ω, and Lemma 1.3.3 extends this inequality to X. □

Corollary 15.12.1 *Let $u, v \in \mathrm{PSH}(X,\theta)$ satisfy $u, v \geq \rho - C$ for some $C \geq 0$ and $u \sim v$.*

(1) *If $\mathrm{MA}_\theta(u) \leq c\mathrm{MA}_\theta(v)$ for some $c \geq 0$, then $c \geq 1$.*
(2) *If $\mathrm{MA}_\theta(u) \leq \mathrm{e}^{\lambda(u-v)}\mathrm{MA}_\theta(v)$ for some $\lambda > 0$, then $u \geq v$.*

In particular, let μ be a positive measure. Among potentials bounded below by ρ minus a constant, each ordinary singularity type contains at most one solution of $\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu$, and the constant $c > 0$ in $\mathrm{MA}_\theta(u) = c\mu$ is uniquely determined within each type.

Proof If $c < 1$, apply the theorem to u and $v + A$ for every constant $A > 0$; the constants in its hypotheses may depend on A. The resulting inequality $u \geq v + A$ for all A is impossible. For the second assertion, fix $a > 0$ and restrict the measure inequality to $\{u < v - a\}$. There the factor is at most $\mathrm{e}^{-\lambda a} < 1$, so the theorem gives $u \geq v - a$; now let $a \downarrow 0$. For two solutions of the exponential equation, apply the second assertion in both orders; for two solutions of the normalized equation, apply the first. □

15.13 Subsolutions and the Gauduchon trace

The strict barrier also yields subsolutions within a bounded distance of any reference between ρ and zero. The curvature estimate comes from composing a convex function with the difference between the reference and a barrier-based subsolution, which produces a convex combination of positive currents. Pairing the equation with a Gauduchon metric then bounds the normalizing constants through a linear trace.

Lemma 15.13.1 *Fix $p > 1$ and $L > 0$. There is $m > 0$ such that, whenever*

$$\rho \leq \eta \leq 0, \quad \eta \in \mathrm{PSH}(X,\theta), \qquad g \geq 0, \quad \|g\|_{L^p(\mathrm{d}V)} \leq L,$$

there exists $v \in \mathrm{PSH}(X,\theta)$ with

$$\eta \leq v \leq \eta + 1, \qquad \mathrm{MA}_\theta(v) \geq mg\,\mathrm{d}V.$$

The constant m depends only on the fixed geometry and on ρ, p, L.

Proof Choose $1 < q < p$. By Hölder's inequality and the finiteness of all polynomial moments of the fixed quasi-plurisubharmonic function ρ, the functions $h = g(2 - \rho)^{2n}$ have uniformly bounded L^q norms. Lemma 15.2.1, applied in the Hermitian background ω with the constant strict barrier -1, gives $s \in \mathrm{PSH}(X,\omega)$ with $-1 \leq s \leq 0$ and

$$\omega_s^n \geq mh\,\mathrm{d}V$$

for a fixed $m > 0$. Put $u_1 = \rho + s$. Then $\theta_{u_1} \geq \omega_s$ and $u_1 \leq \rho \leq \eta$. On Ω define

$$t = 1 + \eta - u_1, \qquad v = \eta + t^{-1},$$

so that $1 \le t \le 2-\rho$. Lemma 1.7.6, applied on relatively compact subsets of Ω with $a = \eta$, $h = u_1$ and $\chi(x) = (1-x)^{-1}$, gives

$$\theta_v = (1-t^{-2})\theta_\eta + t^{-2}\theta_{u_1} + 2t^{-3}\,\mathrm{d}(\eta - u_1) \wedge \mathrm{d}^{\mathrm{c}}(\eta - u_1) \ge t^{-2}\theta_{u_1}.$$

Since $\eta \le v \le \eta + 1$, Lemma 1.3.7 extends v across Z, and the bounds extend by Lemma 1.3.3. On Ω, positivity of the local products gives

$$\mathrm{MA}_\theta(v) \ge t^{-2n}\theta_{u_1}^n \ge (2-\rho)^{-2n}\omega_s^n \ge mg\,\mathrm{d}V.$$

Neither side charges Z, so the inequality holds on X. □

Proposition 15.13.1 *Let γ be a Gauduchon metric on X, write $\gamma^n = G\,\mathrm{d}V$, and put*

$$T_\gamma = \int_X \theta \wedge \gamma^{n-1}.$$

Then $T_\gamma > 0$. If $u \in \mathrm{PSH}(X,\theta)$ satisfies $u \ge \rho - C$ and $\mathrm{MA}_\theta(u) \ge F\,\mathrm{d}V$ with $F \ge 0$, then

$$\int_X F^{1/n} G^{1-1/n}\,\mathrm{d}V \le T_\gamma. \tag{15.9}$$

Consequently, let $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, be nonzero, and let $\lambda > 0$. Every solution of $\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u} f\,\mathrm{d}V$ that is bounded below by ρ minus a constant satisfies $\sup_X u \le C'$, where C' depends only on θ, γ, f and λ. Moreover, if $v_j \in \mathrm{PSH}(X,\theta)$, $j \ge 1$, and constants $C_j \ge 0$ and $c_j > 0$ satisfy

$$\sup_X v_j = 0, \qquad \mathrm{MA}_\theta(v_j) = c_j \mathrm{e}^{v_j/j} f\,\mathrm{d}V, \qquad v_j \ge \rho - C_j,$$

then the constants c_j are uniformly bounded above.

Proof The current $\theta_u \wedge \gamma^{n-1}$ is linear in θ_u, so it is a globally defined positive current, even at Z. Since $\mathrm{dd}^{\mathrm{c}}\gamma^{n-1} = 0$, Stokes' theorem in the sense of currents gives

$$\int_X \theta_u \wedge \gamma^{n-1} = \int_X \theta \wedge \gamma^{n-1}.$$

Applied to ρ, this identity proves $T_\gamma \ge \int_X \omega \wedge \gamma^{n-1} > 0$.

On Ω, Lemma 1.7.3, applied to θ_u and γ, gives

$$\theta_u \wedge \gamma^{n-1} \ge F^{1/n} G^{1-1/n}\,\mathrm{d}V.$$

Exhausting Ω and using positivity of the global trace measure, we obtain (15.9), since the complement Z has zero volume.

For the exponential equation, write $a = \sup_X u$ and $w = u - a$. The estimate becomes

$$\mathrm{e}^{\lambda a/n} \int_X \mathrm{e}^{\lambda w/n} f^{1/n} G^{1-1/n} \, \mathrm{d}V \le T_\gamma .$$

The integral has a positive lower bound over all normalized $w \in \mathrm{PSH}(X, \theta)$. Otherwise, normalized compactness would give a sequence of such w with integrals tending to zero and converging almost everywhere to a θ-plurisubharmonic function; since $w \le 0$ and $f^{1/n} G^{1-1/n}$ is integrable, dominated convergence would give a strictly positive limit integral, a contradiction. This bounds a from above.

For the last assertion, the same estimate contains the factors $c_j^{1/n}$ and $\mathrm{e}^{v_j/(jn)}$. If the integrals $\int_X \mathrm{e}^{v_j/(jn)} f^{1/n} G^{1-1/n} \, \mathrm{d}V$ tended to zero along a subsequence with $j \to \infty$, normalized compactness and almost-everywhere convergence would give $v_j/j \to 0$ almost everywhere along a further subsequence. Dominated convergence would then give the positive limit $\int_X f^{1/n} G^{1-1/n} \, \mathrm{d}V$. This contradiction bounds c_j for all large j, and each of the finitely many remaining constants is finite. Neither argument uses the total Monge–Ampère mass. □

15.14 Stability above a strict barrier

The subsolutions of Lemma 15.13.1 stay within a bounded distance of a given reference. With them, the proof of Theorem 15.3.1 extends to unbounded potentials lying above ρ, and the resulting estimate is uniform over all references between ρ and zero. For a fixed reference, it upgrades L^1 compactness of solutions to uniform convergence of their differences on Ω.

Theorem 15.14.1 *Fix $C_0, L > 0$, $p > 1$, and*

$$0 < \gamma < \frac{p-1}{np + p - 1}.$$

Let $\eta \in \mathrm{PSH}(X, \theta)$ satisfy $\rho \le \eta \le 0$, and suppose that $u \in \mathrm{PSH}(X, \theta)$ satisfies

$$\eta - C_0 \le u \le \eta, \qquad \mathrm{MA}_\theta(u) = f \, \mathrm{d}V, \qquad 0 \le f \in L^p(\mathrm{d}V), \quad \|f\|_p \le L.$$

There is a constant C, depending only on the fixed geometry and on ρ, C_0, L, p, γ, such that every $v \in \mathrm{PSH}(X, \theta)$ with $v \le \eta$ satisfies

$$\sup_\Omega (v - u)_+ \le C \left(\int_\Omega (v - u)_+ \, \mathrm{d}V \right)^\gamma . \tag{15.10}$$

The comparison potential v need not have the ordinary singularity type of η. Since Z is pluripolar and has zero volume, the integral may equally be taken over X, and the resulting inequality $v \le u + C\big(\int_\Omega (v-u)_+ \, \mathrm{d}V\big)^\gamma$ on Ω extends to X.

Proof Replacing v by $\max(v,u)$, we may assume $\eta - C_0 \le u \le v \le \eta$, so that $0 \le v - u \le C_0$ on Ω. We repeat the proof of Theorem 15.3.1, with $1 < q < p$, $E_\delta = \{u < v - \delta\}$, $g_\delta = \mathbf{1}_{E_\delta} f$ and $m_\delta = \|g_\delta\|_q$, and with suprema taken over Ω. Three ingredients change. First, when $m_\delta > 0$, Lemma 15.13.1, applied to g_δ/m_δ and followed by subtracting one from its potential, gives $h \in \mathrm{PSH}(X,\theta)$ and a fixed $b > 0$ such that

$$\eta - 1 \le h \le \eta, \qquad \mathrm{MA}_\theta(h) \ge b\, g_\delta/m_\delta \, \mathrm{d}V;$$

the constant b is uniform in η, because the auxiliary density has L^q norm one and the strict barrier is fixed. These bounds give $h - v \le C_0$ and $v - h \le 1$, the two properties of h used there. Second, all potentials involved are bounded below by ρ minus a constant, so positivity of mixed products holds on Ω, where they are locally bounded, and Z is not charged. Third, bounded domination is replaced by Theorem 15.12.1: the potentials $v - \delta$ and $w = (1-t)v + th - \delta - C_0 t$ lie below $u + C_0$, so this theorem applies with the coefficients zero and $1/2$. The rest of the argument is unchanged. It gives (15.10) with the exponent $\gamma_q = (p-q)/(npq + p - q)$, and we choose $q > 1$ so close to one that $\gamma_q > \gamma$. □

Corollary 15.14.1 *Fix $\eta \in \mathrm{PSH}(X,\theta)$ with $\rho \le \eta \le 0$, constants $C_0, L > 0$, and $p > 1$. Consider the family of potentials $u \in \mathrm{PSH}(X,\theta)$ satisfying*

$$\eta - C_0 \le u \le \eta, \qquad \mathrm{MA}_\theta(u) = f_u \, \mathrm{d}V, \qquad 0 \le f_u \in L^p(\mathrm{d}V), \quad \|f_u\|_p \le L.$$

This family is compact for uniform convergence of differences on Ω. If u_j belong to the family and $u_j \to u$ in L^1, then u belongs to the family and

$$\sup_\Omega |u_j - u| \le C \|u_j - u\|^\gamma_{L^1(\mathrm{d}V)} \longrightarrow 0$$

for every exponent γ as in Theorem 15.14.1. Moreover, f_{u_j} converges weakly to f_u in L^p.

Proof Applying the theorem in both orders to u_j and u_k, with the common reference η, we see that an L^1-Cauchy sequence in the family is uniformly Cauchy in differences on Ω. Its uniform limit on Ω is locally θ-plurisubharmonic and agrees almost everywhere, hence everywhere for the canonical representatives, with the L^1 limit u. The bounds $\eta - C_0 \le u \le \eta$ hold on all of X. Letting $k \to \infty$ in the pairwise estimate proves the quantitative bound, before any equation for u is used.

The densities form a bounded set in the reflexive space L^p. Let f be the weak limit of a subsequence $f_{u_{j_k}}$. Since the potentials are locally bounded and converge locally uniformly on Ω, Bedford–Taylor continuity, which holds also for the smooth nonclosed background, gives $\mathrm{MA}_\theta(u) = f \, \mathrm{d}V$ on Ω. Neither measure charges the analytic set Z, so this equality holds on X. Positivity and weak lower semicontinuity of the norm give $f \ge 0$ and $\|f\|_p \le L$. Thus every weak cluster point equals the density f_u of $\mathrm{MA}_\theta(u)$, which proves weak convergence of the whole sequence. Finally, the common bounds by η and $\eta - C_0$ give L^1 compactness of the family, by

the compactness of quasi-plurisubharmonic functions, and the preceding argument upgrades each convergent subsequence to uniform convergence of differences on Ω.□

For solutions with minimal singularities, normalized by $\sup_X u = 0$, we take $\eta = V_\theta$. Then $\rho \le V_\theta \le 0$, and every normalized comparison potential $v \in \mathrm{PSH}(X,\theta)$ satisfies $v \le V_\theta$, as the theorem requires.

15.15 Minimal singularities in an arbitrary big background

We now solve the equations in the ordinary singularity type of V_θ. Demailly's regularization approximates V_θ by strict potentials with analytic singularities. On a resolution of these singularities, the corresponding equations become equations in a semipositive big background, which we have already solved. The solutions satisfy bounds relative to the approximating singularities that are uniform in the approximation, and upper and lower envelopes then produce a solution with minimal singularities. For the normalized equation, the normalized exponential solutions lie in a common interval below V_θ, so the compactness theorem of the preceding section applies directly.

Since $\rho \le 0$ is θ-plurisubharmonic, it is a competitor for V_θ, and hence $\rho \le V_\theta \le 0$. We use the analytic regularization of Theorem 1.15.2 in its resolved form.

Lemma 15.15.1 *Let $\psi \in \mathrm{PSH}(X,\theta)$ satisfy $\theta_\psi \ge a\omega$ for some $a > 0$, and suppose that ψ has a smooth remainder after resolution, as in Theorem 1.15.2. For every nonzero $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, and every $\lambda > 0$, there exists $u \in \mathrm{PSH}(X,\theta)$ with $u \sim \psi$ such that*

$$\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u} f \,\mathrm{d}V.$$

Proof Let $\pi : Y \to X$ be a modification resolving the singularities of ψ, obtained by a finite sequence of blowups with smooth centers. The pulled-back current has the form

$$\pi^*\theta + \mathrm{dd^c}\pi^*\psi = \beta + [D],$$

where D is an effective real divisor, $[D]$ is its current of integration, and β is a smooth real $(1,1)$-form on Y. Off D we have $\beta \ge a\pi^*\omega$, and by continuity this inequality holds everywhere. In particular, β is semipositive.

We check that β is big. Blowups along smooth hypersurfaces are isomorphisms, so Lemma 1.15.1, applied to π and ω, gives $C_0 > 0$ and an effective real exceptional divisor F such that $\kappa = C_0\pi^*\omega - \eta_F$ is a Hermitian form. With $\zeta = C_0^{-1}\sigma_F$, the Poincaré–Lelong formula gives

$$\pi^*\omega + \mathrm{dd^c}\zeta = C_0^{-1}\big(\kappa + [F]\big).$$

Hence $\beta + a\mathrm{dd^c}\zeta \ge aC_0^{-1}\kappa$, so β is big.

Fix a smooth positive volume form $\mathrm{d}V_Y$ on Y, and write $\pi^*\mathrm{d}V = J\,\mathrm{d}V_Y$. The function J is smooth, nonnegative and bounded, and it is positive off the exceptional

locus. Define

$$H = \mathrm{e}^{\lambda \pi^* \psi} (f \circ \pi) J.$$

This function is nonzero and belongs to $L^p(\mathrm{d}V_Y)$. Indeed, ψ is bounded above, and the change of variables formula gives

$$\int_Y |(f \circ \pi) J|^p \, \mathrm{d}V_Y \le \|J\|_\infty^{p-1} \int_X |f|^p \, \mathrm{d}V;$$

the analytic exceptional sets do not affect these integrals. Theorem 15.2.1, applied on Y in the semipositive big background β, gives a bounded $s \in \mathrm{PSH}(Y, \beta)$ with

$$\mathrm{MA}_\beta(s) = \mathrm{e}^{\lambda s} H \, \mathrm{d}V_Y.$$

The function $\pi^*\psi + s$ is $\pi^*\theta$-plurisubharmonic, because the corresponding current is $\beta_s + [D] \ge 0$, where $\beta_s = \beta + \mathrm{dd^c} s$. By Theorem A.2.1 it descends to some $u \in \mathrm{PSH}(X, \theta)$, and $u \sim \psi$ because s is bounded. Off the exceptional set and the pole set of ψ, the change of variables formula gives the required equation. Neither the extended measure on the left nor the measure $\mathrm{e}^{\lambda u} f \, \mathrm{d}V$ on the right charges these analytic sets, so the equation holds on all of X. □

Theorem 15.15.1 *Let θ be any smooth big real $(1,1)$-form and let $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, be nonzero. For every $\lambda > 0$ there is a unique $u \in \mathrm{PSH}(X, \theta)$ such that*

$$u \sim V_\theta, \qquad \mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u} f \, \mathrm{d}V.$$

Proof *Step 1: approximating singularity types.* Apply Theorem 1.15.2 to V_θ. Set $\delta_j = 2^{-j}$ and pass to a subsequence whose curvature errors are at most δ_j. Denote its potentials by $\widetilde{\psi}_j$; thus $\widetilde{\psi}_j \downarrow V_\theta$ and $\theta_{\widetilde{\psi}_j} \ge -\delta_j \omega$. Set

$$a_j = (1 - \delta_j) \sup_X \widetilde{\psi}_j \ge 0, \qquad \psi_j = (1 - \delta_j) \widetilde{\psi}_j + \delta_j \rho - a_j.$$

By Hartogs' lemma, $\sup_X \widetilde{\psi}_j \to \sup_X V_\theta = 0$, so $a_j \to 0$. A common resolution of the singularities of $\widetilde{\psi}_j$ and ρ shows that ψ_j has a smooth remainder after resolution. Moreover,

$$\theta_{\psi_j} \ge \delta_j^2 \omega, \qquad (1 - \delta_j) V_\theta + \delta_j \rho - a_j \le \psi_j \le V_\theta. \tag{15.11}$$

The upper inequality holds because ψ_j is θ-plurisubharmonic and nonpositive.

Step 2: uniform bounds. By Lemma 15.15.1 there are $u_j \sim \psi_j$ satisfying

$$\mathrm{MA}_\theta(u_j) = \mathrm{e}^{\lambda u_j} f \, \mathrm{d}V.$$

Each u_j is bounded below by ρ minus a constant depending on j, so Proposition 15.13.1 gives a uniform upper bound for $\sup_X u_j$.

There is also a uniform lower bound. We apply Lemma 15.13.1 with the barrier ρ replaced by $\rho - 1 - \sup_j a_j$ and with the reference $\psi_j - 1$, which lies between this barrier and zero. The lemma gives v_j with $\psi_j - 1 \le v_j \le \psi_j \le 0$ and $\mathrm{MA}_\theta(v_j) \ge m f \, \mathrm{d}V$,

where $m > 0$ is independent of j. Put $w_j = v_j + \lambda^{-1}\log m$. Then

$$\mathrm{MA}_\theta(w_j) \geq \mathrm{e}^{\lambda w_j} f\,\mathrm{d}V, \qquad u_j \sim w_j \sim \psi_j,$$

and the exponential order in Corollary 15.12.1 gives $u_j \geq w_j$. Thus, for a uniform constant C,

$$\psi_j - C \leq u_j \leq V_\theta + C. \tag{15.12}$$

After extraction, u_j converges in L^1 and almost everywhere to a θ-plurisubharmonic function u, and (15.11) and (15.12) give $V_\theta - C \leq u \leq V_\theta + C$.

Step 3: the equation for the limit. We identify the equation using envelopes. Put $A_j = \sup_{k\geq j} a_k$ and, for $N \geq j$, define

$$p_{j,N} = P_\theta(\min\{u_j, \dots, u_N\}), \qquad t_{j,N} = \max_{j\leq k\leq N} u_k.$$

Both functions lie above the θ-plurisubharmonic function

$$b_j = (1-\delta_j)V_\theta + \delta_j\rho - C - A_j,$$

which is a competitor for every $p_{j,N}$; in particular, no envelope is identically $-\infty$. All these functions are locally bounded on Ω, uniformly in j and N. Contact support (Theorem 15.7.2), together with a partition of the contact set according to the first index of contact, gives

$$\mathrm{MA}_\theta(p_{j,N}) \leq \mathrm{e}^{\lambda p_{j,N}} f\,\mathrm{d}V.$$

No factor N appears here: on each piece the envelope equals a single u_k, and ordered contact bounds its measure by the measure of that u_k. Similarly, locality and ordered contact for a finite maximum give

$$\mathrm{MA}_\theta(t_{j,N}) \geq \mathrm{e}^{\lambda t_{j,N}} f\,\mathrm{d}V.$$

Now let $N \to \infty$, and set $p_j = \lim_N p_{j,N}$ and $t_j = (\lim_N t_{j,N})^*$. Continuity of local bounded products along monotone sequences passes both inequalities to these limits. Next, p_j increases to a regularized limit p, while t_j decreases to u by Hartogs' lemma. The bound by b_j gives $p \sim V_\theta$, and $p \leq u$. A second application of the monotone continuity gives

$$\mathrm{MA}_\theta(p) \leq \mathrm{e}^{\lambda p} f\,\mathrm{d}V, \qquad \mathrm{MA}_\theta(u) \geq \mathrm{e}^{\lambda u} f\,\mathrm{d}V.$$

These inequalities hold on Ω, and they extend across Z because no measure involved charges Z. Thus $\mathrm{MA}_\theta(p) \leq \mathrm{e}^{\lambda(p-u)}\mathrm{MA}_\theta(u)$, and the exponential order in Corollary 15.12.1 gives $p \geq u$. Hence $p = u$, and the equation follows. Applying the same order in both directions proves uniqueness. □

Lemma 15.15.2 *Under the hypotheses of Theorem 15.15.1, let u_λ be the solution given there, and put*

$$v_\lambda = u_\lambda - \sup_X u_\lambda, \qquad c_\lambda = \mathrm{e}^{\lambda \sup_X u_\lambda}.$$

There are constants $B, C, m > 0$, independent of $0 < \lambda \le 1$, such that

$$V_\theta - B \le v_\lambda \le V_\theta, \qquad m \le c_\lambda \le C. \tag{15.13}$$

Proof The normalized potential satisfies

$$\mathrm{MA}_\theta(v_\lambda) = c_\lambda \mathrm{e}^{\lambda v_\lambda} f \, \mathrm{d}V.$$

Fix a Gauduchon metric γ and write $\gamma^n = G \, \mathrm{d}V$. Since $v_\lambda \le 0$ and $0 < \lambda \le 1$, (15.9) gives

$$c_\lambda^{1/n} \int_X \mathrm{e}^{v_\lambda/n} f^{1/n} G^{1-1/n} \, \mathrm{d}V \le \int_X \theta \wedge \gamma^{n-1}.$$

The integrals on the left have a positive lower bound independent of λ. Otherwise, the compactness of normalized quasi-plurisubharmonic functions would give an almost everywhere convergent sequence of the normalized v_λ whose integrals tend to zero, with a θ-plurisubharmonic limit; since f is nonzero, dominated convergence would give a positive limit integral. Thus $c_\lambda \le C$.

To control the singularity type, we use uniform exponential integrability (Lemma 1.5.1) and Hölder's inequality to choose $\varepsilon > 0$ and $1 < q < p$ such that $\|\mathrm{e}^{-\varepsilon v_\lambda} f\|_q$ is uniformly bounded. Lemma 15.13.1, applied with the reference $V_\theta - 1$ and the barrier $\rho - 1$, gives $\eta_\lambda \in \mathrm{PSH}(X, \theta)$ with

$$V_\theta - 1 \le \eta_\lambda \le V_\theta, \qquad \mathrm{MA}_\theta(\eta_\lambda) \ge m \mathrm{e}^{-\varepsilon v_\lambda} f \, \mathrm{d}V \ge m f \, \mathrm{d}V$$

for a fixed $m > 0$. The scalar order in Corollary 15.12.1, together with $\mathrm{MA}_\theta(v_\lambda) \le c_\lambda f \, \mathrm{d}V$, gives $c_\lambda \ge m$. Set $D = \varepsilon^{-1} \max\{0, \log(C/m)\}$ and $w_\lambda = \eta_\lambda - D$. Then

$$\mathrm{e}^{\varepsilon(w_\lambda - v_\lambda)} \mathrm{MA}_\theta(v_\lambda) \le m \mathrm{e}^{-\varepsilon v_\lambda} f \, \mathrm{d}V \le \mathrm{MA}_\theta(w_\lambda),$$

since $\lambda v_\lambda \le 0$ and $\eta_\lambda \le 0$. The exponential order in Corollary 15.12.1 gives $v_\lambda \ge w_\lambda$, which proves the uniform bounds with $B = 1 + D$. □

Theorem 15.15.2 *Under the assumptions of Theorem 15.15.1, there are $v \in \mathrm{PSH}(X, \theta)$, a constant $C > 0$ and a uniquely determined $c > 0$ such that*

$$\sup_X v = 0, \qquad V_\theta - C \le v \le V_\theta, \qquad \mathrm{MA}_\theta(v) = c f \, \mathrm{d}V.$$

As in the semipositive case, only the constant c is asserted to be unique, not the normalized potential.

Proof Apply Lemma 15.15.2 with $\lambda = 1/j$, and write $v_j = v_{1/j}$ and $c_j = c_{1/j}$. Extract a subsequence with $v_j \to v$ in L^1 and almost everywhere, and with $c_j \to c \in [m, C]$. The limit is normalized and θ-plurisubharmonic, and (15.13) gives $V_\theta - B \le v \le V_\theta$, which is the asserted two-sided bound. The densities $c_j \mathrm{e}^{v_j/j} f$ have a common L^p bound. Hence Corollary 15.14.1, applied with the reference V_θ, upgrades the convergence to uniform convergence of differences on Ω and identifies the weak L^p limit of these densities with the density of $\mathrm{MA}_\theta(v)$.

At almost every point, v_j converges to a finite value, so $v_j/j \to 0$. Since $v_j \le 0$ and $c_j \le C$, dominated convergence gives $c_j e^{v_j/j} f \to cf$ in L^p. Therefore $\mathrm{MA}_\theta(v) = cf\,\mathrm{d}V$. The uniqueness of the constant follows from Corollary 15.12.1.□

15.16 Local Hölder continuity in a big background

We now combine the stability estimate above ρ with the regularization of Lemma 1.21.6, whose curvature error is linear in the regularization parameter. As in the proof of Theorem 15.4.1, a small multiple of the strict barrier absorbs the curvature error of the transforms, and a lower bound for the minimizing radius converts the estimate for the transforms into an estimate for the smoothing on compact subsets of Ω. This gives a local Hölder modulus for solutions with minimal singularities.

Theorem 15.16.1 *Fix $C_0 > 0$. Let $u \in \mathrm{PSH}(X,\theta)$ satisfy*

$$\sup_X u = 0, \qquad V_\theta - C_0 \le u \le V_\theta, \qquad \mathrm{MA}_\theta(u) = f\,\mathrm{d}V,$$

where $0 \le f \in L^p(\mathrm{d}V)$ and $p > 1$. Then u is locally Hölder continuous on Ω, with every exponent

$$0 < \beta < \frac{2(p-1)}{np + 3(p-1)}.$$

For fixed bounds on C_0 and $\|f\|_p$, the Hölder constants are uniform on each fixed compact subset of Ω.

Proof Since $u \ge V_\theta - C_0 \ge \rho - C_0$, we may subtract a constant from the strict barrier so that $\rho \le u - 1$; it still satisfies $\theta + \mathrm{dd}^{\mathrm{c}}\rho \ge \kappa\omega$ for a fixed $\kappa > 0$. Choose $0 < \gamma < (p-1)/(np+p-1)$ and set $\alpha = 2\gamma < 1$. Apply Lemma 1.21.6 with $a = u$, and let A, K, C and $\Phi_{c,\delta}$ be the constants and transforms given there. For small $\delta > 0$ set

$$c = \delta^\alpha, \qquad e = Ac + C\delta, \qquad s = e/\kappa, \qquad v_\delta = \frac{\Phi_{c,\delta} + s\rho}{1+s}.$$

By the curvature bound of Lemma 1.21.6, $v_\delta \in \mathrm{PSH}(X,\theta)$. The lower bound for $\Phi_{c,\delta}$ in that lemma implies $\rho \le \Phi_{c,\delta}$ when δ is small. Thus

$$v_\delta \le \Phi_{c,\delta} \le R_\delta u \le 0.$$

In particular $v_\delta \le V_\theta$, so Theorem 15.14.1 applies with the reference V_θ and the comparison potential v_δ. Together with the L^1 smoothing estimate, it gives

$$\sup_\Omega (v_\delta - u)_+ \le C\|(R_\delta u - u)_+\|_1^\gamma \le C\delta^\alpha.$$

Fix a compact subset of Ω on which $\rho \ge -M$. Since $s \le C\delta^\alpha$ and $u \le 0$, the identity

$$\Phi_{c,\delta} - u = (1+s)(v_\delta - u) + s(u - \rho)$$

gives $\Phi_{c,\delta} - u \le C_M \delta^\alpha$ there.

At a radius t where the infimum defining $\Phi_{c,\delta}$ is attained, the monotone lower bound yields

$$-c \log(t/\delta) \le \Phi_{c,\delta} - u + K(\delta^2 + \delta) \le C_M c.$$

Hence $t \ge a_M \delta$ for a fixed $a_M > 0$. As in the proof of Theorem 15.4.1, monotonicity of the smoothing now gives

$$R_{a_M \delta} u - u \le C_M \delta^\alpha, \qquad R_{a_M \delta} u - u \ge -K(a_M \delta)^2$$

on the chosen compact set. Since every small radius has the form $a_M \delta$, we obtain $|R_t u - u| \le C_M t^\alpha$ there for all small $t > 0$, with constants uniform on that compact set.

We apply this to a slightly larger compact neighborhood, on which also $\|\nabla R_t u\| \le C_M / t$. For nearby points x, y at distance $d > 0$, the smooth function $R_t u$ and a short path in that neighborhood give

$$|u(x) - u(y)| \le C_M (t^\alpha + d/t).$$

The choice $t = d^{1/(1+\alpha)}$ gives the exponent $\alpha/(1+\alpha)$, and letting γ approach its upper endpoint gives the asserted range. Pairs of points farther apart are handled by the boundedness of u on the compact neighborhood. □

It suffices that the equation hold on Ω, where $u \ge \rho - C_0$ is locally bounded and $\mathrm{MA}_\theta(u)$ is the local bounded product: since Z has zero volume and is not charged by $\mathrm{MA}_\theta(u)$, the equation then holds on X.

15.16.1 Limits of normalized exponential solutions

For the normalized exponential family (v_λ, c_λ) of Lemma 15.15.2, the reference interval $[V_\theta - B, V_\theta]$ is uniform in $0 < \lambda \le 1$. Stability therefore controls differences on all of Ω, and the local Hölder estimates strengthen the convergence on compact subsets. We fix the nonzero density $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, and use the constants B, C, m of that lemma.

Proposition 15.16.1 *For the family (v_λ, c_λ) of Lemma 15.15.2, the normalized potentials are relatively compact for uniform convergence of differences on Ω. As $\lambda \downarrow 0$, every cluster limit v is normalized, has minimal singularities, and satisfies*

$$\mathrm{MA}_\theta(v) = c f \, \mathrm{d}V,$$

where $c > 0$ is the unique constant of Theorem 15.15.2. Moreover,

$$c_\lambda \longrightarrow c, \qquad c_\lambda \mathrm{e}^{\lambda v_\lambda} f \longrightarrow c f \quad \text{in } L^p(\mathrm{d}V).$$

Every sequence of parameters in $(0, 1]$ has a subsequence for which the potentials converge in C^β on each compact subset of Ω, for all

$$0 < \beta < \frac{2(p-1)}{np + 3(p-1)}.$$

Proof By Lemma 15.15.2, all v_λ lie in a common reference interval, and their densities are bounded in L^p by $C\|f\|_p$. We may therefore apply Corollary 15.14.1 with $\eta = V_\theta$. Given any sequence $\lambda_j \downarrow 0$, we pass to a subsequence with $v_{\lambda_j} \to v$ uniformly in differences on Ω and in $L^1(X)$, and with $c_{\lambda_j} \to c_* \in [m, C]$. The limit satisfies the same reference bounds and is normalized. Off the pole set,

$$0 \le 1 - \mathrm{e}^{\lambda v_\lambda} \le 1 - \mathrm{e}^{\lambda(V_\theta - B)} \longrightarrow 0$$

as $\lambda \downarrow 0$. Multiplying by f and using dominated convergence, we obtain $\mathrm{e}^{\lambda v_\lambda} f \to f$ in L^p for the whole family. The compactness corollary therefore identifies the limit equation as $\mathrm{MA}_\theta(v) = c_* f \, \mathrm{d}V$, and the uniqueness of the constant gives $c_* = c$. Since every subsequential limit of c_λ as $\lambda \downarrow 0$ equals c, we conclude that $c_\lambda \to c$, and the stated convergence of the densities follows.

Finally, Theorem 15.16.1 gives uniform local C^α bounds for every α below its endpoint, and the uniform limit inherits these bounds by pointwise convergence. For $0 < \beta < \alpha$, uniform convergence and the elementary interpolation inequality on smaller coordinate balls give

$$[v_{\lambda_j} - v]_{C^\beta} \le C \|v_{\lambda_j} - v\|_\infty^{1-\beta/\alpha} [v_{\lambda_j} - v]_{C^\alpha}^{\beta/\alpha} \longrightarrow 0,$$

where the Hölder bounds are taken on slightly larger compact neighborhoods. The uniformly convergent subsequence given by compactness thus converges in C^β on every compact subset of Ω and for every exponent below the endpoint; no further extraction depending on the exponent is needed. □

Corollary 15.16.1 *Suppose a group G of biholomorphisms of X preserves θ and the measure $f \, \mathrm{d}V$. Then every v_λ and every cluster limit as $\lambda \downarrow 0$ is G-invariant. If the normalized zero-exponent equation has only one G-invariant solution with minimal singularities, then the whole family v_λ converges to it as $\lambda \downarrow 0$, uniformly in differences on Ω and in each local C^β topology of the proposition.*

Proof For $g \in G$, pullback by g preserves θ-plurisubharmonicity, and $V_\theta \circ g = V_\theta$ by the definition of the envelope. Pullback also preserves the positive-exponent equation, so its uniqueness gives $u_\lambda \circ g = u_\lambda$, and the normalization preserves this equality. Pullback by a fixed smooth biholomorphism is continuous on $L^1(X)$, since its Jacobian relative to the fixed smooth volume is bounded. Hence every cluster limit is invariant almost everywhere, and therefore everywhere for the canonical representatives. This argument does not require g to preserve the barrier ρ or the set Ω.

If the invariant solution is unique, every cluster limit is the same potential. Relative compactness then implies convergence of the whole family for uniform convergence of differences on Ω: a sequence staying at a fixed positive distance from that potential would have a convergent subsequence, a contradiction. The interpolation argument above upgrades this to the asserted local C^β convergence. □

The proposition gives compactness and identifies the equation of every cluster limit, and the symmetry criterion is one way to identify the limit itself. In general, Theorem 16.12.1 shows that the whole family converges, and it describes the limit through the greatest subsolution of the zero-exponent equation that satisfies a family of integral normalizations determined by how the constant c depends on the density. This holds even when the zero-exponent equation has several normalized solutions.

15.17 Interior smoothness on the regular locus

Throughout this section, θ is a smooth big real $(1,1)$-form, and $\rho \le 0$, $Z = \{\rho = -\infty\}$ and $\Omega = X \setminus Z$ are the strict analytic barrier, its pole set and its regular locus fixed in Section 15.12; thus $\theta + \mathrm{dd^c}\rho \ge \omega$. A potential with minimal singularities is bounded below by $\rho - C$, hence locally bounded on Ω, and its Monge–Ampère measure there is the bounded-potential product.

For such a potential, Theorem 15.16.1 gives local Hölder continuity on Ω when the density belongs to L^p for some $p > 1$. We show that a smooth strictly positive density on the whole of Ω then makes the potential smooth on Ω. Local information alone cannot give this: Example 15.20.2 below is a locally Lipschitz plurisubharmonic function with constant Monge–Ampère density that is not C^1. The argument therefore uses the barrier globally.

Theorem 15.17.1 *Suppose $\varphi \in \mathrm{PSH}(X, \theta)$ has minimal singularities and is locally Hölder continuous on Ω. Let f be smooth and strictly positive on Ω, and assume*

$$(\theta + \mathrm{dd^c}\varphi)^n = f\,\omega^n \quad \text{on } \Omega.$$

Then $\varphi \in C^\infty(\Omega)$ and $\theta + \mathrm{dd^c}\varphi > 0$ there.

By Theorem 15.16.1, the Hölder hypothesis holds when $f \in L^p(X, \omega^n)$ for some $p > 1$ (Corollary 15.17.1).

The proof is by contradiction and occupies the rest of the section. Let $\mathcal{S} \subset \Omega$ be the closed set of points near which φ is not smooth, suppose that $\mathcal{S} \neq \varnothing$, and let $p \in \mathcal{S}$ minimize $\varphi - \rho$ on $\mathcal{S}$; it exists by the proper gap of Lemma 1.19.4, and strict positivity of $\theta + \mathrm{dd^c}\rho$ turns this minimum into a quadratic separation along $\mathcal{S}$. On a small ball centered at p the argument has four steps.

(1) A comparison principle for continuous solutions (Section 15.17.1) and the smooth Dirichlet problem give smooth solutions u_j of the equation that converge uniformly to φ at a polynomial rate in j, while the traces of $\theta + \mathrm{dd^c}u_j$, measured in Euclidean coordinates, grow at most polynomially in j (Section 15.17.3).
(2) Near p, the quadratic separation, together with uniform trace bounds on compact sets where φ is already smooth, improves the trace growth to $O(j^\gamma)$ for every $\gamma > 0$. Interpolation with the convergence rate then gives uniform $C^{1,\alpha}$ bounds for every $\alpha < 1$ (Section 15.17.4).
(3) A third-order estimate whose constants depend polynomially on the ellipticity turns uniform $C^{1,\alpha}$ bounds, for α close to one, into a uniform trace bound (Section 15.17.5).
(4) A continuous solution with bounded trace is smooth (Section 15.17.2). Hence φ is smooth near p, contradicting $p \in \mathcal{S}$ (Section 15.17.6).

15.17.1 Local comparison

The comparison principle below, for continuous sub- and supersolutions on a bounded domain, identifies the given continuous potential with limits of smooth Dirichlet solutions. Since the background need not be closed, the classical Bedford–Taylor comparison does not apply directly. We freeze the background at a point, apply the closed-current comparison formula (1.4) to quadratic modifications, and control the error by the mixed inequalities of Lemmas 1.7.3 and 1.7.4.

The local arguments of this section take place in coordinate balls, always with compact closure in a larger coordinate chart. There we put $\beta = \mathrm{dd^c}|z|^2$, write $\partial_i = \partial/\partial z_i$ and $\partial_{\bar\imath} = \partial/\partial \bar z_i$, and take coefficient matrices of real $(1,1)$-forms relative to β. For such a form A, $\mathrm{tr}_\beta A$ and $\det_\beta A$ denote the trace and the determinant of its coefficient matrix, so that $A^n = (\det_\beta A)\beta^n$, and $\Delta_\beta = \mathrm{tr}_\beta\,\mathrm{dd^c}$. For a positive form T, tr_T is the trace computed with the inverse of the coefficient matrix of T. Indices run from 1 to n, repeated indices in tensor expressions are summed, and I denotes

the identity matrix. In local statements, ϑ denotes a smooth real $(1,1)$-form, not necessarily closed, which plays the role of θ.

Proposition 15.17.1 *Let $D \Subset \mathbb{C}^n$ be a bounded domain and let ϑ be a smooth real $(1,1)$-form near $\overline{D}$. Suppose $u, v \in C(\overline{D})$ are ϑ-plurisubharmonic,*

$$(\vartheta + \mathrm{dd^c} u)^n \geq F\beta^n, \qquad (\vartheta + \mathrm{dd^c} v)^n \leq F\beta^n,$$

where $F > 0$ is continuous on D. If $u \leq v$ on ∂D, then $u \leq v$ in D.

Proof *Step 1: v admits no strict smooth lower test.* Suppose, to the contrary, that a C^2 function q touches v from below at a point $p \in D$ and that, at p,

$$\vartheta + \mathrm{dd^c} q > 0, \qquad \det_{\beta}(\vartheta + \mathrm{dd^c} q) > F.$$

Choose a sufficiently small $a > 0$. After replacing q by $q - a|z-p|^2$, which makes the contact strict, and restricting to a small ball $B_r(p)$, there are fixed $\lambda, \eta > 0$ such that

$$A := \vartheta + \mathrm{dd^c} q \geq \lambda\beta, \qquad A^n \geq (F + \eta)\beta^n.$$

To compare with closed currents, freeze the background at p: put $\kappa = \sup_{B_r} \|\vartheta - \vartheta(p)\|_\beta$ and $B_0 = \vartheta(p) + \kappa\beta$. Then $S = B_0 - \vartheta$ satisfies $0 \leq S \leq 2\kappa\beta$. For small $s > 0$ the set $E = \{v < q + s\}$ is nonempty and relatively compact in B_r. If Q_0 is a quadratic potential for the constant form B_0, then

$$P = \mathrm{dd^c}(v + Q_0) = \vartheta + \mathrm{dd^c} v + S, \qquad Q = \mathrm{dd^c}(q + s + Q_0) = A + S \geq \lambda\beta$$

are closed positive Hessians. Applying (1.4), expanding P, and using $S \leq 2\kappa\beta \leq 2\kappa\lambda^{-1}Q$, we obtain

$$\int_E P^n \leq \int_E F\beta^n + e_\kappa \int_E P^n, \qquad e_\kappa = \left(1 + \frac{2\kappa}{\lambda}\right)^n - 1.$$

On the other hand, (1.4) with $j = n$ gives

$$\int_E P^n \geq \int_E Q^n \geq \int_E (F + \eta)\beta^n.$$

If M bounds F on the ball, the two estimates give $(1 - e_\kappa)\eta \leq e_\kappa M$. Since $\kappa \to 0$, and hence $e_\kappa \to 0$, as $r \downarrow 0$, this is impossible for small r. Thus no such lower test exists.

Step 2: comparison. Suppose that $\max_D(u - v) > 0$. We regularize u to produce a forbidden lower test for v. Let χ_δ be a nonnegative smooth mollifier of mass one and radius $\delta > 0$. For small $t > 0$ and $C \geq \sup_D |z|^2$, set

$$u_t = u + t(|z|^2 - C).$$

Then $u_t \le v$ on ∂D, the positive maximum persists, and $T_t = \vartheta + \mathrm{dd^c} u_t \ge t\beta$. Lemma 1.7.3 gives

$$T_t^n \ge (F^{1/n} + t)^n \beta^n.$$

Choose a relatively compact neighborhood W of the positive maximum set. By Lemma 1.7.4,

$$(T_t * \chi_\delta)^n \ge \big((F^{1/n} + t) * \chi_\delta\big)^n \beta^n.$$

Since $T_t \ge t\beta$, the convolution satisfies $T_t * \chi_\delta \ge t\beta$. Moreover,

$$\vartheta + \mathrm{dd^c}(u_t * \chi_\delta) = T_t * \chi_\delta + (\vartheta - \vartheta * \chi_\delta),$$

so if $\epsilon_\delta = \|\vartheta - \vartheta * \chi_\delta\|_{C^0(W)}$, then

$$\vartheta + \mathrm{dd^c}(u_t * \chi_\delta) \ge (1 - \epsilon_\delta/t)(T_t * \chi_\delta).$$

For fixed $t > 0$, uniform convergence of the convolved density and $\epsilon_\delta \to 0$ show that the left-hand side is positive and has determinant strictly larger than F on W for small δ. Since $u_t * \chi_\delta \to u_t$ uniformly on W, the difference $u_t * \chi_\delta - v$ still attains its maximum over $\overline{W}$ at an interior point of W for small δ. Subtracting this maximum from $u_t * \chi_\delta$ produces a smooth function touching v from below with the properties excluded in Step 1, a contradiction. □

15.17.2 Bounded traces and smoothness

We begin with the criterion used in the last step of the proof. A bounded trace of a positive current bounds all its coefficient measures, so a continuous solution with bounded trace is a strong solution with $W^{2,q}$ regularity for every finite q. The equation then bounds the eigenvalues from below, so it becomes uniformly elliptic. Since the solution is not yet known to be C^2, the Evans–Krylov theory is applied to it as a viscosity solution of a concave extension of the operator.

Proposition 15.17.2 *Let $B \subset \mathbb{C}^n$ be a coordinate ball, let ϑ be a smooth real $(1,1)$-form there and let $F \in C^\infty(B)$ be strictly positive. Suppose u is continuous, $T = \vartheta + \mathrm{dd^c} u \ge 0$, and*

$$T^n = F\beta^n. \tag{15.14}$$

If $\mathrm{tr}_\beta T$ has a bounded measure density near a point $p \in B$, then u is smooth near p.

Proof Say $\mathrm{tr}_\beta T \le L$ on a ball around p. Positivity makes every coefficient of the matrix-valued current T a bounded density. Hence the Euclidean Laplacian of u is locally bounded, and interior Poisson estimates give $u \in W^{2,q}_{\mathrm{loc}}$ for every finite q. Since T has bounded coefficients, $T = T_{\mathrm{ac}}$, and Lemma 1.7.5 gives $\det_\beta T = F$ almost everywhere. On a smaller ball with $F \ge m > 0$, all eigenvalues of T are at most L and their product is at least m, so

$$\frac{m}{L^{n-1}} I \le T \le L I$$

almost everywhere.

It remains to show that this strong solution is smooth. Let P be the real linear map from real symmetric $2n \times 2n$ Hessian matrices to Hermitian $n \times n$ matrices, sending $D^2 u$ to the coefficient matrix of $\mathrm{dd}^{\mathrm{c}} u$ relative to β, and let $\Theta(x)$ be the matrix of ϑ. With $\delta = m L^{1-n}$, define the concave extension

$$G(x, M) = \inf_{\delta I \le A \le L I} \left\{ \log\det A + \operatorname{tr}\left[A^{-1} \big(\Theta(x) + P(M) - A \big) \right] \right\} - \log F(x).$$

For fixed A, write $\mathcal{E}_A(x, M)$ for the expression in braces minus $\log F(x)$. The tangent-plane formula for $\log\det$ gives

$$G(x, D^2 u) = 0 \quad \text{almost everywhere.}$$

The operator G is concave and uniformly elliptic on real symmetric matrices. The affine operators $\mathcal{E}_A$ have constant principal matrices with eigenvalues bounded above and below in terms of δ and L. Their x-dependence is smooth uniformly in A, so G has a uniform local Lipschitz modulus in x.

We claim that u is a viscosity solution of $G(x, D^2 u) = 0$. Suppose that a smooth function ψ touches u from above at x_0 and that $G(x_0, D^2\psi) < 0$. Some fixed matrix A then satisfies $\mathcal{E}_A(x, D^2\psi) < -\eta$ on a small ball. Since $G(x, D^2 u) = 0$ almost everywhere, $\mathcal{E}_A(x, D^2 u) \ge 0$ there. Thus $u - \psi$ has an interior maximum and satisfies

$$\operatorname{tr}\Big(A^{-1} P(D^2(u - \psi)) \Big) \ge \eta \quad \text{almost everywhere.}$$

After a small quadratic perturbation makes the contact strict, this contradicts the Aleksandrov maximum principle [GT01, Chapter 9, Theorem 9.1]. If instead ψ touches u from below and $G(x_0, D^2\psi) > 0$, then $\mathcal{E}_A(x, D^2\psi) > \eta$ for every admissible A on a small ball. At almost every point choose an admissible $A(x)$ with $\mathcal{E}_{A(x)}(x, D^2 u) < \eta/2$. A measurable choice is obtained by taking the first qualifying element of a countable dense family. The function $\psi - u$ then violates the same maximum principle for the measurable uniformly elliptic coefficients $A(x)^{-1}$. This proves the claim.

The classical variable-coefficient Evans–Krylov theorem gives $u \in C^{2,\alpha}_{\mathrm{loc}}$ for some $0 < \alpha < 1$. The original equation is now uniformly elliptic with smooth data. Difference quotients satisfy linear equations whose principal coefficients are averages of inverse matrices along positive definite segments. Interior Schauder estimates first give $u \in C^{3,\alpha}_{\mathrm{loc}}$ and then, by iteration, $u \in C^{\infty}$. $\square$

15.17.3 Smooth exact approximants

We now construct the approximants of the first step. The smooth Dirichlet theorem, Lemma 15.6.1, applies to every smooth positive density F. Approximating the Hölder boundary values of φ by smooth functions therefore gives smooth solutions, and the point is to control them quantitatively. The estimates below bound the trace of a smooth solution by a power of the C^4 norm of its boundary value. The C^4 norms of the boundary approximations grow polynomially while their uniform errors decay polynomially, so both the convergence and the trace growth have polynomial rates.

In this subsection balls are centered at the origin. On the boundary sphere, ν is the outward Euclidean unit normal and $\nabla_{\tan}$ denotes tangential differentiation. For a positive form T, inner products $\langle \cdot, \cdot \rangle_T$ and norms $|\cdot|_T$ of $(1,0)$-covectors are computed with the inverse of the coefficient matrix of T.

Lemma 15.17.1 *Let B be a coordinate ball, let ϑ and $F > 0$ be smooth near $\overline{B}$, and let q be smooth with $R = \vartheta + \mathrm{dd^c} q \geq \varepsilon\beta$ for a fixed $\varepsilon > 0$. Fix a finite $L \geq 0$. If $u \in C^\infty(B)$ satisfies*

$$T = \vartheta + \mathrm{dd^c} u > 0, \qquad T^n = F\beta^n, \qquad |u - q| \leq L,$$

put $S = \mathrm{tr}_\beta T$ and $v = u - q$. For $A > 0$, define

$$Q_A = \log S - Av + \mathrm{e}^{-v}.$$

There are A_0, C, depending only on the fixed data and L, such that at every interior local maximum of Q_A, with $A \geq A_0$,

$$S \leq C\sqrt{A}. \tag{15.15}$$

Proof This is the maximum-principle computation behind the classical second-order estimate. The non-closedness of ϑ produces a first-order error term, which is absorbed by the convex term e^{-v}.

At the point in question make a constant unitary change of coordinates so that the matrix of T is diagonal, with eigenvalues λ_i. Write

$$\Delta_T = \sum_i \lambda_i^{-1} \partial_i \partial_{\bar{\imath}}, \qquad H = \mathrm{tr}_T \beta.$$

Differentiating $\log\det T = \log F$ and commuting derivatives gives

$$\Delta_T S = \Delta_\beta \log F + P + E_0, \quad P = \sum_{k,i,j} \frac{|\partial_k T_{i\bar{\jmath}}|^2}{\lambda_i \lambda_j}, \quad |E_0| \leq C_0 H. \tag{15.16}$$

The non-closedness of ϑ enters through the bounded vector

$$E_p = \sum_k (\partial_p \vartheta_{k\bar{k}} - \partial_k \vartheta_{p\bar{k}}).$$

Since $\partial_p S - E_p = \sum_k \partial_k T_{p\bar{k}}$, weighted Cauchy–Schwarz gives $|\partial S - E|_T^2 \le S P$. From (15.16),

$$\Delta_T \log S \ge -\frac{C_0(1+H)}{S} - \frac{2}{S}\,\mathrm{Re}\langle \partial \log S, E\rangle_T . \tag{15.17}$$

Set $g_A(s) = -As + \mathrm{e}^{-s}$. At a local maximum of Q_A, $\partial \log S = -g_A'(v)\partial v$ and $\Delta_T v = n - \mathrm{tr}_T\, R$. Combining these identities with (15.17), then completing the square in ∂v, yields

$$0 \ge \left[\varepsilon A - \frac{C_2}{S} - \frac{C_2 (A+1)^2}{S^2} \right] H - C_3 (A+1) - \frac{C_2}{S}. \tag{15.18}$$

Here $|E|_T^2 \le CH$ and the bounds for v have been used. If $S \ge K\sqrt{A}$ and K is large, the bracket is at least $\varepsilon A/2$. Thus $H \le C$ independently of A and u. The determinant upper bound then gives $S \le C'$. Enlarging K makes this incompatible with $S \ge K\sqrt{A}$, proving (15.15). $\square$

We shall also use the following variant, in which a gradient bound replaces the convex term. If $|\nabla(u-q)|_\beta$ is bounded, then at an interior maximum of $\log S - A(u-q)$ the trace is bounded independently of A for all large A. Indeed, at the maximum $\partial \log S = A\partial(u-q)$, and (15.17) gives

$$0 \ge \left[A\varepsilon - \frac{C}{S} - \frac{CA}{S} \right] H - nA - \frac{C}{S}.$$

For S larger than a fixed constant the bracket is at least $A\varepsilon/2$; the determinant again controls S.

A polynomial boundary trace estimate

Proposition 15.17.3 *Let B be the unit ball. Fix smooth ϑ, smooth $F > 0$ near $\overline{B}$, and $M < \infty$. There are $C, N < \infty$, depending only on these fixed data, such that every smooth solution u with boundary value $g \in C^\infty(\partial B)$ of*

$$T = \vartheta + \mathrm{dd}^{\mathrm{c}} u > 0, \qquad T^n = F\beta^n, \qquad \|u\|_{L^\infty(B)} \le M, \qquad u|_{\partial B} = g,$$

satisfies

$$\sup_{\overline{B}} \operatorname{tr}_\beta T \le C\big(1 + \|g\|_{C^4(\partial B)}\big)^N. \tag{15.19}$$

Proof We follow the classical boundary estimates for the Dirichlet problem on a ball and keep track of the dependence on $L = 1 + \|g\|_{C^4}$. Barriers bound the first derivatives, and differentiation along rotations bounds the mixed normal–tangential second derivatives on the boundary. A lower bound for the tangential block of T then bounds the double normal derivative through the equation, and Lemma 15.17.1 reduces the interior bound to the boundary bound.

Step 1: first derivatives. Fix a sufficiently large constant $K > 0$, depending only on the fixed data. Extend g to G on $\overline{B}$ with $\|G\|_{C^4} \le CL$. For $A = CL$ large enough,

$$\underline{u} = G + A(|z|^2 - 1), \qquad \underline{T} = \vartheta + \mathrm{dd}^{\mathrm{c}} \underline{u} \ge K\beta, \qquad \underline{T}^n \ge F\beta^n.$$

Proposition 15.17.1 gives $\underline{u} \le u$. If h solves $\Delta_\beta h = -C_\vartheta$ with $h = g$ on ∂B and $C_\vartheta \ge \sup \operatorname{tr}_\beta \vartheta$, then $u \le h$. Boundary differentiation of these barriers gives

$$|\partial_\nu u| + |\nabla_{\tan} u| \le CL \quad \text{on } \partial B. \tag{15.20}$$

Let $H = \operatorname{tr}_T \beta$, let $\mathcal{L} = \operatorname{tr}_T \mathrm{dd}^{\mathrm{c}}$ be the linearized operator, and let $w = u - \underline{u} \ge 0$ be the barrier function used below. Since $H \ge nF^{-1/n}$ is bounded below, taking K fixed and sufficiently large gives

$$\mathcal{L} w = n - \operatorname{tr}_T \underline{T} \le -(1 + H). \tag{15.21}$$

Let $E = \sum_k (z_k \partial_{z_k} + \bar{z}_k \partial_{\bar{z}_k})$ be the real radial field. Write Θ for the coefficient matrix of ϑ. Differentiating $\log\det T = \log F$ and commuting E with complex second derivatives yields

$$\mathcal{L}(Eu) = E(\log F) + 2n - \operatorname{tr}_T(E\Theta + 2\Theta).$$

Hence $|\mathcal{L}(Eu)| \le C(1+H)$. The same estimate holds for Zu when Z is an infinitesimal unitary rotation. Applying the minimum principle to $CL + C'w \pm Yu$, with $Y = E$ or Z, gives

$$|Eu| + |Zu| \le CL \quad \text{on } B. \tag{15.22}$$

Step 2: mixed second derivatives on the boundary. For a rotation Z, the function $R_Z = Z(u - G)$ vanishes on ∂B and satisfies $|\mathcal{L} R_Z| \le CL(1 + H)$. The barriers $CLw \pm R_Z$ and (15.21) imply $|\partial_\nu R_Z| \le CL^2$ on the boundary. Rotations span the

real tangent space of ∂B, so all real mixed normal–tangential second derivatives are $O(L^2)$. Pure tangential second derivatives are $O(L)$ by differentiating $u = g$ twice and using (15.20). Thus all complex Hessian entries except the complex normal–normal entry are $O(L^2)$.

Step 3: the double normal derivative. It remains to bound the tangential block of T from below. For $n \geq 2$, let

$$\lambda_0 = \min\{T_p(\xi, \bar{\xi}) : p \in \partial B,\ \xi \in T_p^{1,0}(\partial B),\ |\xi|_\beta = 1\}.$$

Choose a minimizing pair (p, ξ) and extend ξ to a smooth unit tangential field near p, with uniformly bounded derivatives. On the sphere, any smooth function vanishing there satisfies

$$\mathrm{dd}^{\mathrm{c}}(u - G)(\xi, \bar{\xi}) = \frac{1}{2} E(u - G).$$

Define

$$a(z) = (\vartheta + \mathrm{dd}^{\mathrm{c}} G)_z(\xi(z), \overline{\xi(z)}) - \frac{1}{2} EG(z), \qquad P = a + \frac{1}{2} Eu - \lambda_0.$$

Then $P \geq 0$ on the nearby spherical boundary and $P(p) = 0$. The boundary first-derivative bound also gives $\lambda_0 \leq CL$ by the tangential identity above. Consequently

$$|P| \leq CL, \qquad |\mathcal{L}P| \leq CL(1 + H).$$

On a fixed cap $D = B \cap B_r(p)$ consider

$$\Psi = P + C_1 L w + C_2 L |z - p|^2.$$

Choose C_2 so that $\Psi \geq 0$ on the interior spherical part of ∂D, and then C_1 so that $\mathcal{L}\Psi \leq 0$ by (15.21). The minimum principle yields $\Psi \geq 0$. At p the normal derivative of $|z - p|^2$ vanishes, and therefore

$$\partial_\nu P + C_1 L \partial_\nu w \leq 0.$$

Equations (15.20)–(15.22) now give $D^2 u(\nu, \nu)(p) \leq CL^2$, hence $\mathrm{tr}_\beta T(p) \leq CL^2$. Positivity and $\det_\beta T = F$ imply

$$\lambda_0 \geq c L^{-2(n-1)}. \tag{15.23}$$

At any boundary point write

$$T = \begin{pmatrix} B_0 & b \\ b^* & d \end{pmatrix}$$

in tangential–normal blocks, with B_0 an $(n-1) \times (n-1)$ Hermitian matrix, $b \in \mathbb{C}^{n-1}$ and $d \in \mathbb{R}$. We have $|b| \leq CL^2$ and $B_0 \geq \lambda_0 I$. The determinant equation and the Schur complement give

$$d = \frac{F}{\det B_0} + b^* B_0^{-1} b \le (\sup F)\lambda_0^{-(n-1)} + CL^4\lambda_0^{-1},$$

which is polynomial in L. For $n = 1$, the trace is F and this argument is unnecessary.

Step 4: the interior. Choose a fixed quadratic q with $\vartheta + \mathrm{dd^c} q \ge \varepsilon\beta$. Since $|u - q| \le M + \sup_B |q|$, Lemma 15.17.1 applies with this bound in place of its constant L; we use it with one fixed A. At an interior maximum of its auxiliary function the trace is uniformly bounded, while at a boundary maximum the polynomial boundary bound of Steps 2 and 3 applies. Since the remaining terms are uniformly bounded, (15.19) follows. □

Exact approximation

We apply these estimates to smooth approximations of the boundary values of φ on a ball in Ω. Recall that θ, ω and f are the data of Theorem 15.17.1.

Proposition 15.17.4 *Let $B \Subset \Omega$ be a coordinate ball, identified with $\{|z| < r_0\}$, with closure in a larger coordinate chart, and put*

$$F = \frac{f\,\omega^n}{\beta^n}.$$

There are $a > 0$, $b, C < \infty$, and $u_j \in C^\infty(\overline{B})$ such that

$$\theta + \mathrm{dd^c} u_j > 0, \qquad (\theta + \mathrm{dd^c} u_j)^n = F\beta^n, \tag{15.24}$$

and

$$\|u_j - \varphi\|_{L^\infty(B)} \le Cj^{-a}, \qquad \sup_{\overline{B}} \mathrm{tr}_\beta(\theta + \mathrm{dd^c} u_j) \le Cj^b. \tag{15.25}$$

Proof Since φ is Hölder continuous near $\overline{B}$, mollification at scale $1/j$ gives smooth boundary approximations g_j with

$$\|g_j - \varphi\|_{L^\infty(\partial B)} \le Cj^{-a}, \qquad \|g_j\|_{C^4(\partial B)} \le Cj^4.$$

Lemma 15.6.1, applied with the background θ, the density F and the boundary value g_j, gives $u_j \in C^\infty(\overline{B})$ satisfying (15.24) with $u_j = g_j$ on ∂B. Comparing u_j with $\varphi \pm \|g_j - \varphi\|_\infty$ by Proposition 15.17.1 proves the first bound in (15.25). The functions u_j are therefore uniformly bounded. Proposition 15.17.3, applied after rescaling B to the unit ball, and the polynomial C^4 boundary norm give the second bound. □

15.17.4 Localization near the nonregular set

In the auxiliary function of Lemma 15.17.1, the term $-A(u_j - q)$ penalizes points where $u_j - q$ is large. A quadratic gap of $u - q$ along the nonregular set therefore

suppresses the contribution of the boundary of a neighborhood of that set. Taking A proportional to $\log j$ makes this suppression beat the polynomial growth of the boundary trace, and near the point of minimal $u - q$ the trace then grows more slowly than every positive power of j. Interpolation with the polynomial approximation rate gives uniform $C^{1,\alpha}$ bounds.

Distances and norms below are Euclidean. By a smooth exact solution for (ϑ, F) we mean a smooth function u_j with $\vartheta + \mathrm{dd^c} u_j > 0$ and $(\vartheta + \mathrm{dd^c} u_j)^n = F\beta^n$.

Proposition 15.17.5 *For $R > 0$, let $B = B_R(0)$, let ϑ and $F > 0$ be smooth near $\overline{B}$, and let $u \in C(\overline{B})$. Suppose q is smooth and $\vartheta + \mathrm{dd^c} q \geq \varepsilon\beta$ for some $\varepsilon > 0$. Put $v = u - q$ and $m = v(0)$, and fix $\kappa > 0$. Let $\mathcal{S} \subset \overline{B}$ be closed, contain 0, and satisfy*

$$v(0) = m, \qquad v(z) \geq m + \kappa|z|^2 \quad (z \in \mathcal{S}). \tag{15.26}$$

Assume that smooth exact solutions u_j, $j \geq 2$, satisfy, for fixed constants $a > 0$, $b \geq 0$ and $C > 0$,

$$\|u_j - u\|_\infty \leq C j^{-a}, \tag{15.27}$$

$$\sup_{\overline{B}} \mathrm{tr}_\beta(\vartheta + \mathrm{dd^c} u_j) \leq C j^b, \tag{15.28}$$

$$\sup_j \sup_K \mathrm{tr}_\beta(\vartheta + \mathrm{dd^c} u_j) < \infty \quad \text{for every } K \Subset \overline{B} \setminus \mathcal{S}. \tag{15.29}$$

Then for every $\gamma > 0$ there are a neighborhood U_γ of 0 and C_γ such that

$$\mathrm{tr}_\beta(\vartheta + \mathrm{dd^c} u_j) \leq C_\gamma j^\gamma \quad \text{on } U_\gamma. \tag{15.30}$$

For every $0 < \alpha < 1$, there is also a neighborhood U of 0 on which $u_j \to u$ in $C^{1,\alpha}$. In particular the approximants are uniformly bounded in that norm and $u \in C^{1,\alpha}(U)$.

Proof *Step 1: the trace bound* (15.30). Write $e_j = C j^{-a}$, fix $d = \kappa R^2/2$, and choose $D > (b+1)/d$. Choose $\eta > 0$ so that $2D\eta < \gamma/2$. For small $\delta > 0$, the neighborhood

$$W_\delta = \{z \in B : \mathrm{dist}(z, \mathcal{S}) < \delta\}$$

contains 0, has $v \geq m - \eta$ on its closure, and satisfies $v \geq m + d$ on $\partial B \cap \overline{W}_\delta$ with the fixed d above. These properties follow from (15.26), continuity, and compactness. The remaining part of the boundary of W_δ is a compact subset of $\overline{B} \setminus \mathcal{S}$, where (15.29) applies.

Set $A_j = D \log j$. Apply the maximum principle on $\overline{W}_\delta$ to

$$Q_j = \log \mathrm{tr}_\beta(\vartheta + \mathrm{dd^c} u_j) - A_j(u_j - q) + \mathrm{e}^{-(u_j - q)}.$$

At an interior maximum, Lemma 15.17.1 gives $\log \mathrm{tr}_\beta(\vartheta + \mathrm{dd^c} u_j) \leq C + \frac{1}{2}\log A_j$. On the part of ∂W_δ inside B, which stays away from $\mathcal{S}$, the trace has a bound independent of j. On the outer part $\partial B \cap \overline{W}_\delta$, (15.28) and the gap d give

$$Q_j \leq C + b \log j - A_j(m + d - e_j),$$

and the choice $Dd > b + 1$ makes this smaller than the bound below for large j. Hence throughout W_δ,

$$Q_j \le C_\delta + \tfrac{1}{2}\log A_j - A_j(m - \eta - e_j).$$

On a small ball about 0 where $v \le m + \eta$, this implies

$$\log \operatorname{tr}_\beta(\vartheta + \mathrm{dd}^{\mathrm{c}} u_j) \le C_\delta + \tfrac{1}{2}\log A_j + A_j(2\eta + 2e_j).$$

By the choice of η, the logarithmic factor and $A_j e_j$ are absorbed into the remaining power. This proves (15.30).

Step 2: $C^{1,\alpha}$ convergence. On a smaller ball, the real Laplacian of u_j is bounded by Cj^γ. Interior Poisson estimates give a W^{2,q_0} bound for every finite q_0, and Sobolev embedding with q_0 large then gives, for every $\sigma < 1$,

$$\|u_j\|_{W^{2,q_0}} + \|u_j\|_{C^{1,\sigma}} \le Cj^\gamma.$$

For $d_k = u_{2^{k+1}} - u_{2^k}$,

$$\|d_k\|_\infty \le C2^{-ka}, \qquad \|d_k\|_{C^{1,\sigma}} \le C2^{k\gamma}.$$

If $\alpha < \sigma$ and $t = (1+\alpha)/(1+\sigma)$, Hölder interpolation gives

$$\|d_k\|_{C^{1,\alpha}} \le C2^{k[-a(1-t)+\gamma t]}.$$

Since γ may be arbitrarily small, the exponent can be made negative. The dyadic series converges in $C^{1,\alpha}$ to its uniform limit u. If $2^k \le j < 2^{k+1}$, interpolation applied to $u_j - u_{2^k}$ gives the same decaying bound. Hence the full sequence converges to u in $C^{1,\alpha}$ on a smaller ball. □

Proposition 15.17.6 *Suppose u_j solve the same smooth equation as in Proposition 15.17.5, satisfy (15.27)–(15.28), and u is smooth with $\vartheta + \mathrm{dd}^{\mathrm{c}} u > 0$ on an open set $O \subset B$. Then, for every $K \Subset O$,*

$$\sup_j \sup_K \operatorname{tr}_\beta(\vartheta + \mathrm{dd}^{\mathrm{c}} u_j) < \infty. \tag{15.31}$$

Proof Where u is smooth and strictly ϑ-plurisubharmonic, it provides its own quadratic separation, so the argument of Proposition 15.17.5 applies around every point of O. A first application gives trace growth of order $\sqrt{\log j}$; the resulting gradient convergence then allows the variant of Lemma 15.17.1 with a gradient bound, which removes this growth.

For p in a fixed compact subset of O, take a uniformly sized ball $B_r(p) \Subset O$ and set

$$q_p(z) = u(z) - \kappa|z - p|^2$$

with $\kappa > 0$ small, so $\vartheta + \mathrm{dd}^{\mathrm{c}} q_p \ge \varepsilon\beta$. If $e_j = Cj^{-a}$ and $v_{j,p} = u_j - q_p$, then

$$v_{j,p} \ge -e_j, \quad v_{j,p}(p) \le e_j, \quad v_{j,p} \ge \kappa r^2 - e_j \text{ on } \partial B_r(p).$$

With $A_j = D\log j$ and $D\kappa r^2 > b+1$, the maximum argument of Step 1 in the proof of Proposition 15.17.5 gives

$$\operatorname{tr}_\beta(\vartheta + \mathrm{dd}^{\mathrm{c}} u_j)(p) \le C\sqrt{\log j}. \tag{15.32}$$

The constants are uniform on compact subsets of O.

Put $T_j = \vartheta + \mathrm{dd}^{\mathrm{c}} u_j$ and $w_j = u_j - u$. On slightly larger compact sets, (15.32) bounds $|\Delta w_j|$ by $C\sqrt{\log j}$. Poisson estimates and Sobolev embedding give

$$\|w_j\|_{C^{1,\sigma}} \le C_\sigma \sqrt{\log j}, \qquad \sigma < 1.$$

Interpolating with $\|w_j\|_\infty \le C j^{-a}$ shows $\nabla w_j \to 0$ uniformly on smaller compact sets. The gradients of $v_{j,p}$ are therefore uniformly bounded. Repeat the maximum argument with $\log \operatorname{tr}_\beta T_j - A_j v_{j,p}$, now using the variant of Lemma 15.17.1 with a gradient bound. Its interior trace bound is now independent of A_j, while the boundary term is suppressed as before. Evaluation at p gives

$$\operatorname{tr}_\beta T_j(p) \le C\exp(2A_j e_j) \le C,$$

proving (15.31). □

15.17.5 Derivative estimates and bounded traces

Uniform $C^{1,\alpha}$ bounds control averages of the trace $\operatorname{tr}_\beta(\vartheta + \mathrm{dd}^{\mathrm{c}} u_j) = \operatorname{tr}_\beta \vartheta + \Delta_\beta u_j$ over small balls. To pass from averages to pointwise values we need a bound for the gradient of the trace, that is, a third-order estimate, whose dependence on the ellipticity constants is polynomial. We prove it by Calabi's maximum-principle argument, computed with the Chern connection of the solution so that the torsion of the nonclosed background appears explicitly. The resulting averaging inequality is nonlinear in the trace, and it closes under iteration when α is sufficiently close to one.

In this subsection B is the unit ball and $B_r(p)$ is the Euclidean ball of radius r centered at p. For a Hermitian metric g, ∂g denotes the ordinary coordinate derivatives of its coefficient matrix, while tensor derivatives in the proofs are taken with respect to the Chern connection of g.

Proposition 15.17.7 *Fix smooth ϑ and smooth $F > 0$ near the closed unit ball. There are $C, N < \infty$, depending only on bounds for finitely many derivatives of ϑ and F and for $1/F$, such that any smooth solution on B of*

$$g = \vartheta + \mathrm{dd}^{\mathrm{c}} u, \qquad g^n = F\beta^n, \qquad M^{-1}\beta \le g \le M\beta, \qquad M \ge 1,$$

satisfies

$$\sup_{B_{1/2}} |\partial g|_\beta \le C M^N. \tag{15.33}$$

If, instead, the data are given near the closure of a ball $B_r(p)$ with $r \le 1$, satisfy the same bounds there, and $\operatorname{tr}_\beta g \le L$ *on* $B_r(p)$*, then*

$$\sup_{B_{r/2}(p)} |\nabla \operatorname{tr}_\beta g|_\beta \le C r^{-1}(1+L)^N. \tag{15.34}$$

Proof The quantity $\mathcal{R}$ below is the squared norm of the difference between the Chern connection of g and the flat connection. We derive a differential inequality for $\mathcal{R}$ in which the torsion enters through fixed derivatives of ϑ, and combine it with the trace in a maximum principle; every constant is kept a power of M.

We compare the flat connection of β with the Chern connection ∇ of g. Write $g^{k\bar\ell}$ for the coefficients of the inverse matrix and $\bar\nabla$ for covariant differentiation in $(0,1)$ directions; tensor norms and inner products without subscript are computed with g. The difference of the two connections and its squared norm are

$$\Psi^k_{ij} = g^{k\bar\ell}\partial_i g_{j\bar\ell}, \qquad \mathcal{R} = |\Psi|^2_g.$$

The torsion of ∇ is determined by the fixed background, because the third derivatives of u cancel in it. Its lowered and raised components are

$$C_{ij\bar\ell} = \partial_i \vartheta_{j\bar\ell} - \partial_j \vartheta_{i\bar\ell}, \qquad \mathcal{T}^k_{ij} = g^{k\bar\ell} C_{ij\bar\ell}. \tag{15.35}$$

Consequently, for a fixed integer N,

$$\begin{aligned} |\mathcal{T}| &\le C M^N, \\ |\nabla\mathcal{T}| + |\bar\nabla\mathcal{T}| &\le C M^N (1+\sqrt{\mathcal{R}}), \\ |\nabla^2\mathcal{T}| + |\nabla\bar\nabla\mathcal{T}| + |\bar\nabla^2\mathcal{T}| &\le C M^N (1+\mathcal{R}+|\nabla\Psi|+|\bar\nabla\Psi|). \end{aligned} \tag{15.36}$$

Every factor here is either a fixed derivative of ϑ, a connection factor Ψ, or a factor of $g^{\pm1}$; this explains the polynomial dependence.

For tensor fields, use $\Delta_g = g^{p\bar q}\nabla_p\nabla_{\bar q}$. We need the following curvature identities. With

$$\operatorname{Rm}^k{}_{i\bar j l} = -\partial_{\bar j}\Psi^k_{il}, \qquad \operatorname{Rm}_{i\bar j l\bar k} = g_{m\bar k}\operatorname{Rm}^m{}_{i\bar j l},$$

set

$$\operatorname{Ric}^{(2)}_{l\bar k} = g^{i\bar j}\operatorname{Rm}_{i\bar j l\bar k}, \qquad \operatorname{Ric}^{(1)}_{l\bar k} = g^{i\bar j}\operatorname{Rm}_{l\bar k i\bar j} = -\partial_l\partial_{\bar k}\log F.$$

The two curvature interchanges are

$$\begin{aligned} \operatorname{Rm}_{i\bar j l\bar k} - \operatorname{Rm}_{l\bar j i\bar k} &= -\nabla_{\bar j} C_{il\bar k}, \\ \operatorname{Rm}_{i\bar j l\bar k} - \operatorname{Rm}_{i\bar k l\bar j} &= -\nabla_i \overline{C}_{\bar j\bar k l}. \end{aligned}$$

They follow by differentiating the torsion identity and its conjugate. Successive interchange and contraction give

$$\operatorname{Ric}^{(2)}_{l\bar k} = \operatorname{Ric}^{(1)}_{l\bar k} - g^{i\bar j}\big(\nabla_l \overline{C}_{\bar j\bar k i} + \nabla_{\bar j} C_{il\bar k}\big). \tag{15.37}$$

Metric compatibility, (15.36), and the fixed smooth formula for $\mathrm{Ric}^{(1)}$ imply

$$|\mathrm{Ric}^{(2)}| \le CM^N(1+\sqrt{\mathcal{R}}), \qquad |\nabla \mathrm{Ric}^{(2)}| \le CM^N(1+\mathcal{R}+|\nabla\Psi|+|\bar\nabla\Psi|). \tag{15.38}$$

Moreover,

$$\nabla_{\bar q}\Psi^k_{ij} = -\mathrm{Rm}^k{}_{i\bar q j},$$

and the Chern–Bianchi identity, including its torsion term, is

$$\nabla_p \mathrm{Rm}^k{}_{i\bar q j} - \nabla_i \mathrm{Rm}^k{}_{p\bar q j} = -\mathcal{T}^r_{pi}\,\mathrm{Rm}^k{}_{r\bar q j}.$$

Consequently,

$$\Delta_g \Psi^k_{ij} = -\nabla_i(\mathrm{Ric}^{(2)})^k{}_j + g^{p\bar q}\mathcal{T}^r_{pi}\,\mathrm{Rm}^k{}_{r\bar q j}. \tag{15.39}$$

If $A^k{}_l = (\mathrm{Ric}^{(2)})^k{}_l$, define its action on a tensor of type $(1,2)$ by

$$(A\cdot\Psi)^k_{ij} = A^k{}_l\Psi^l_{ij} - A^l{}_i\Psi^k_{lj} - A^l{}_j\Psi^k_{il}.$$

Differentiating the squared norm and using metric compatibility gives

$$\Delta_g\mathcal{R} = |\nabla\Psi|^2 + |\bar\nabla\Psi|^2 + 2\,\mathrm{Re}\langle\Delta_g\Psi,\Psi\rangle - \mathrm{Re}\langle A\cdot\Psi,\Psi\rangle. \tag{15.40}$$

There is no mixed torsion term because the Chern torsion has type $(2,0)$. Using (15.36), (15.38), and (15.39) in (15.40), Young's inequality absorbs the products of $\sqrt{\mathcal{R}}$ with $|\nabla\Psi| + |\bar\nabla\Psi|$. We obtain

$$\Delta_g\mathcal{R} \ge -CM^N(\mathcal{R}^{3/2}+1). \tag{15.41}$$

Let $s = \mathrm{tr}_\beta\, g$. Direct differentiation as in (15.16) gives

$$\Delta_g s \ge M^{-1}\mathcal{R} - CM, \qquad |\partial s|^2_g \le nM^2\mathcal{R}. \tag{15.42}$$

Take a cutoff ζ supported in B, equal to one on $B_{1/2}$, set $\eta = \zeta^2$ and $h = 2nM - s$, so $nM \le h \le 2nM$, and maximize

$$Q = \frac{\eta\mathcal{R}}{h}.$$

At a positive interior maximum of Q, the identity $\partial\mathcal{R}/\mathcal{R} = \partial h/h - \partial\eta/\eta$ gives

$$0 \ge \frac{\Delta_g Q}{Q} = \frac{\Delta_g\mathcal{R}}{\mathcal{R}} + \frac{\Delta_g\eta}{\eta} - \frac{\Delta_g h}{h} - \frac{2|\partial\eta|^2_g}{\eta^2} + \frac{2\,\mathrm{Re}\langle\partial\eta,\partial h\rangle_g}{\eta h}.$$

We have $|\partial\eta|^2_g \le CM\eta$, $|\Delta_g\eta| \le CM$ and $|\partial h|_g \le \sqrt{n}M\sqrt{\mathcal{R}}$. Insert (15.41) and (15.42) and multiply by ηQ. The positive term is $M^{-1}Q^2$. The two terms involving $\sqrt{\mathcal{R}}$ are bounded in absolute value by a fixed power of M times $Q^{3/2}$, since $\mathcal{R} = hQ/\eta$ and $h \le 2nM$. The remaining cutoff terms are bounded by $CM^N(Q+1)$. Therefore

$$0 \ge M^{-1}Q^2 - CM^N(Q^{3/2}+Q+1).$$

Thus Q, and then $\mathcal{R}$ on $B_{1/2}$, are bounded by a fixed polynomial in M. Since $|\partial g|_\beta^2 \le M^3\mathcal{R}$, this proves (15.33).

If $s \le L$, the determinant lower bound gives $\lambda_{\min}(g) \ge (\inf F)L^{1-n}$, so M is a fixed power of $1+L$. For a ball $B_r(p)$, set $u_r(z) = r^{-2}u(p+rz)$ and use the coefficient matrices $\vartheta_r(z) = \vartheta(p+rz)$, $g_r(z) = g(p+rz)$ and $F_r(z) = F(p+rz)$. Then $g_r = \vartheta_r + \mathrm{dd^c}u_r$ in the unit coordinates and $\det g_r = F_r$. For $r \le 1$ the derivatives of the fixed data remain uniformly bounded. The unit-ball estimate for the ordinary coefficient derivatives ∂g_r and their conjugates gives the factor r^{-1} in (15.34). □

Proposition 15.17.8 *There is $\alpha_* < 1$, depending only on n and the exponent N in Proposition 15.17.7, with the following property. Let B_0 be a Euclidean ball, fix a smooth real $(1,1)$-form ϑ and a smooth function $F > 0$ near $\overline{B_0}$, and let u_j be smooth exact solutions of $(\vartheta + \mathrm{dd^c}u_j)^n = F\beta^n$ with $\vartheta + \mathrm{dd^c}u_j > 0$. If*

$$\sup_j \|u_j\|_{C^{1,\alpha}(B_0)} < \infty \qquad \text{for some } \alpha_* < \alpha < 1,$$

then $\mathrm{tr}_\beta(\vartheta + \mathrm{dd^c}u_j)$ is uniformly bounded on every smaller ball.

Proof Fix a center for concentric balls $B_r \Subset B_0$, put $s_j = \mathrm{tr}_\beta(\vartheta + \mathrm{dd^c}u_j)$, and set

$$M_j(r) = 1 + \sup_{B_r} s_j.$$

For concentric balls $B_r \Subset B_R \Subset B_0$, $d = R - r \le 1$, Proposition 15.17.7 gives

$$|\nabla s_j| \le Cd^{-1}M_j(R)^N \quad \text{on } B_{r+d/2}. \tag{15.43}$$

Fix $p \in B_r$ and let χ_h be a nonnegative mollifier of radius $h \le d/4$. Since

$$s_j = \mathrm{tr}_\beta\,\vartheta + \Delta_\beta u_j,$$

integration by parts, after subtracting the constant vector $\nabla u_j(p)$, and the uniform $C^{1,\alpha}$ bound give

$$(s_j * \chi_h)(p) \le C(1 + h^{\alpha-1}).$$

Equation (15.43) gives

$$|s_j(p) - (s_j * \chi_h)(p)| \le Cd^{-1}M_j(R)^N h.$$

Choose

$$h = \frac{d}{4}M_j(R)^{-N/(2-\alpha)}.$$

Taking the supremum over $p \in B_r$ and absorbing fixed powers of d^{-1}, we obtain

$$M_j(r) \le Cd^{-1}M_j(R)^q, \qquad q = \frac{N(1-\alpha)}{2-\alpha}. \tag{15.44}$$

Choose α_* so close to one that $q < 1$ for $\alpha > \alpha_*$. Choose $0 < r_0 < R_0$ with $B_{R_0} \Subset B_0$. Let $r_k \uparrow R_0$ with $r_{k+1} - r_k = (R_0 - r_0)2^{-k-1}$. Iterating (15.44) gives

$$M_j(r_0) \le \prod_{k=0}^{m-1} \left[C(r_{k+1} - r_k)^{-1}\right]^{q^k} M_j(r_m)^{q^m}.$$

The infinite product converges because $\sum q^k(1+k) < \infty$. For each fixed j, the final factor tends to one as $m \to \infty$. The resulting bound is independent of j. □

15.17.6 Proof of the main theorem

We now assemble the four steps outlined after Theorem 15.17.1. The barrier enters only once, but globally: it locates a point of the nonsmooth set at which the quadratic separation required in Proposition 15.17.5 holds.

Proof (of Theorem 15.17.1) If $n = 1$, the equation is the linear identity

$$\mathrm{dd^c}\varphi = f\,\omega - \theta \quad \text{on } \Omega.$$

Its right-hand side is smooth, so interior elliptic regularity gives $\varphi \in C^\infty(\Omega)$, and the original equation gives $\theta + \mathrm{dd^c}\varphi = f\,\omega > 0$. We henceforth assume $n \ge 2$.

Let

$$\mathcal{S} = \{p \in \Omega : \varphi \text{ is not smooth on any neighborhood of } p\}.$$

This is closed in Ω. Suppose $\mathcal{S} \neq \varnothing$. By Lemma 1.19.4, the restriction of $\varphi - \rho$ to $\mathcal{S}$ attains its minimum m at some $p \in \mathcal{S}$.

Choose nested coordinate balls

$$B \Subset B_0 \Subset \Omega$$

centered at p. On B_0, Proposition 15.17.4 gives smooth exact Dirichlet solutions u_j converging uniformly to φ at a polynomial rate and having polynomial global trace growth.

On the open set $\Omega \setminus \mathcal{S}$ the function φ is smooth, and $\theta + \mathrm{dd^c}\varphi$ is semipositive with positive determinant, hence positive definite. Proposition 15.17.6 therefore gives a uniform trace bound for the u_j on every compact subset of $B_0 \setminus \mathcal{S}$.

In coordinates centered at p, take $\kappa > 0$ small and set

$$q(z) = \rho(z) - \kappa|z|^2.$$

After shrinking B, the lower bound $\theta + \mathrm{dd^c}\rho \ge \omega$ gives $\theta + \mathrm{dd^c}q > 0$ on $\overline{B}$. If $v = \varphi - q$, then the minimizing property of p gives

$$v(p) = m, \qquad v(z) = (\varphi - \rho)(z) + \kappa|z|^2 \ge m + \kappa|z|^2 \quad (z \in \mathcal{S} \cap \overline{B}).$$

Thus Proposition 15.17.5 applies on B with the closed set $\mathcal{S} \cap \overline{B}$: the approximation rate and global polynomial trace bound come from Proposition 15.17.4, and the bound away from $\mathcal{S}$ comes from Proposition 15.17.6. Its second assertion gives, for every $0 < \alpha < 1$, a neighborhood of p on which the approximants have a uniform $C^{1,\alpha}$ bound.

Choose $\alpha > \alpha_*$ as in Proposition 15.17.8, and apply that proposition on a ball centered at p inside this neighborhood. It yields

$$\sup_j \operatorname{tr}_\beta(\theta + \mathrm{dd}^{\mathrm{c}} u_j) < \infty$$

on a still smaller neighborhood of p. Since $u_j \to \varphi$ uniformly, $\mathrm{dd}^{\mathrm{c}} u_j \to \mathrm{dd}^{\mathrm{c}} \varphi$ as currents. The trace bound passes to the limit, so $\operatorname{tr}_\beta(\theta + \mathrm{dd}^{\mathrm{c}} \varphi)$ has a bounded density near p. Proposition 15.17.2 now implies that φ is smooth near p, contradicting $p \in \mathcal{S}$.

Hence $\mathcal{S} = \varnothing$ and $\varphi \in C^\infty(\Omega)$. The form $\theta + \mathrm{dd}^{\mathrm{c}} \varphi$ is smooth and semipositive there, while

$$(\theta + \mathrm{dd}^{\mathrm{c}} \varphi)^n = f\, \omega^n > 0.$$

Its nonnegative eigenvalues have positive product, so every eigenvalue is positive. Therefore $\theta + \mathrm{dd}^{\mathrm{c}} \varphi > 0$ on Ω. □

Corollary 15.17.1 *Let $0 \le f \in L^p(X, \omega^n)$, $p > 1$, be smooth and strictly positive on Ω, and let $c > 0$. Every $\varphi \in \mathrm{PSH}(X, \theta)$ with minimal singularities and $(\theta + \mathrm{dd}^{\mathrm{c}} \varphi)^n = c f\, \omega^n$ on Ω is smooth and strictly θ-plurisubharmonic on Ω.*

Proof Neither $\mathrm{MA}_\theta(\varphi)$ nor $f\, \omega^n$ charges the pluripolar set Z, so $\mathrm{MA}_\theta(\varphi) = c f\, \omega^n$ on X. After adding a constant, $\sup_X \varphi = 0$, and then $V_\theta - C \le \varphi \le V_\theta$. Theorem 15.16.1, applied with the density $c f\, \omega^n / \mathrm{d}V \in L^p(\mathrm{d}V)$, gives local Hölder continuity on Ω, and Theorem 15.17.1 applies with the density $c f$. □

By Theorem 15.15.2, such a solution exists, with a uniquely determined $c > 0$. For a smooth closed big representative on a compact Kähler manifold, Theorem 29.10.1 applies this conclusion to the classical full-mass solution and identifies the regular open set with the ample locus.

15.18 Degenerating semipositive backgrounds

Adding a small multiple of a Hermitian form makes a semipositive background strictly positive, and the smooth theory then solves the normalized equation with a smooth positive density. We study these smooth solutions as the added multiple tends to zero. Bounded subsolutions give uniform bounds for the potentials and the normalizing constants, and a strict barrier gives uniform bounds for the trace on compact sets away from its poles. Smooth local convergence then identifies the limiting equation, without any global continuity property of the Monge–Ampère operator.

The main example is a small resolution of a threefold with nodes (Section 15.18.1). There the background is the pullback of a positive form from the singular threefold,

and it degenerates exactly along the exceptional curves. When this background is closed, the limit is unique and the whole family converges smoothly away from the exceptional curves. Convergence of the associated metric spaces then reduces to controlling the diameters of neighborhoods of these curves, in the limiting metric and in the approximating metrics (Section 15.18.3).

The background θ, the barrier ρ and its pole set Z are new data, independent of those fixed in Section 15.12. Let $\theta \geq 0$ be a smooth real $(1,1)$-form, and fix a Hermitian form η and a smooth positive probability volume form μ on X. Suppose that there are $\delta > 0$ and $\rho \in \mathrm{PSH}(X,\theta)$ such that

$$\theta + \mathrm{dd}^{\mathrm{c}}\rho \geq \delta\eta, \qquad \sup_X \rho = 0, \tag{15.45}$$

the pole set $Z = \{\rho = -\infty\}$ is closed, and ρ is smooth on $X \setminus Z$. For $0 < \varepsilon \leq 1$, the form $\theta + \varepsilon\eta$ is Hermitian, and Theorem 10.1.1 gives $u_\varepsilon \in C^\infty(X)$ and $c_\varepsilon > 0$ with

$$\omega_\varepsilon := \theta + \varepsilon\eta + \mathrm{dd}^{\mathrm{c}}u_\varepsilon > 0, \qquad \omega_\varepsilon^n = c_\varepsilon\mu, \qquad \sup_X u_\varepsilon = 0. \tag{15.46}$$

Lemma 15.18.1 *There are constants $b, B, C > 0$, independent of ε, such that*

$$b \leq c_\varepsilon \leq B, \qquad -C \leq u_\varepsilon \leq 0.$$

Proof Write $\mu = f\,\mathrm{d}V$, where f is smooth and positive. The strict barrier makes θ big. By Lemma 15.2.1, there is a bounded $s \in \mathrm{PSH}(X,\theta)$ with $-1 \leq s \leq 0$ and $\mathrm{MA}_\theta(s) \geq b\mu$ for some fixed $b > 0$. If $c_\varepsilon < b$, positivity gives

$$\mathrm{MA}_{\theta+\varepsilon\eta}(u_\varepsilon) \leq (c_\varepsilon/b)\mathrm{MA}_{\theta+\varepsilon\eta}(s + A)$$

for every constant A. Since $c_\varepsilon/b < 1$, strict bounded domination in the Hermitian background $\theta + \varepsilon\eta$ (Theorem 11.2.2) would imply $u_\varepsilon \geq s + A$ for every A, which is absurd. Hence $c_\varepsilon \geq b$.

For an upper bound, choose a fixed Gauduchon form γ. The arithmetic–geometric mean inequality and Stokes give

$$c_\varepsilon^{1/n}\int_X \left(\frac{\mu}{\gamma^n}\right)^{1/n}\gamma^n \leq \int_X \omega_\varepsilon \wedge \gamma^{n-1} = \int_X (\theta + \varepsilon\eta)\wedge\gamma^{n-1} \leq \int_X (\theta+\eta)\wedge\gamma^{n-1}.$$

The first integral is fixed and positive, giving $c_\varepsilon \leq B$.

For the lower bound on u_ε, note that all u_ε are normalized $(\theta+\eta)$-plurisubharmonic functions. Uniform exponential integrability, Lemma 1.5.1, gives $a > 0$ such that $g_\varepsilon = \mathrm{e}^{-au_\varepsilon}f$ has a common L^2 bound. Also $g_\varepsilon \geq f$. By Theorem 15.2.1, the solutions

$$w_\varepsilon \in \mathrm{PSH}(X,\theta), \qquad \mathrm{MA}_\theta(w_\varepsilon) = \mathrm{e}^{aw_\varepsilon}g_\varepsilon\,\mathrm{d}V$$

are uniformly bounded. Put $B_0 = \max\{1, B\}$ and $z_\varepsilon = w_\varepsilon - a^{-1}\log B_0$. Then

$$\mathrm{MA}_{\theta+\varepsilon\eta}(z_\varepsilon) \geq \mathrm{e}^{az_\varepsilon}c_\varepsilon g_\varepsilon\,\mathrm{d}V, \qquad \mathrm{MA}_{\theta+\varepsilon\eta}(u_\varepsilon) = \mathrm{e}^{au_\varepsilon}c_\varepsilon g_\varepsilon\,\mathrm{d}V.$$

The comparison principle for the exponential equation in the background $\theta + \varepsilon\eta$ (Corollary 11.2.2) gives $z_\varepsilon \le u_\varepsilon \le 0$, and z_ε is bounded below uniformly in ε. □

Since θ may degenerate along Z, we do not estimate the trace by local arguments. Instead we apply the maximum principle on all of $X \setminus Z$ to the logarithm of the trace minus a multiple of $u_\varepsilon - \rho$. This function tends to $-\infty$ at Z, so its maximum is attained off Z, where $\theta + \mathrm{dd^c}\rho \ge \delta\eta$ provides positivity. An additional logarithmic term in the weight absorbs the torsion term without a gradient bound for ρ.

Lemma 15.18.2 *For every compact $K \Subset X \setminus Z$, there is $C_K > 1$, independent of ε, such that*

$$C_K^{-1}\eta \le \omega_\varepsilon \le C_K\eta \quad \text{on } K.$$

Proof For the differentiated calculation write $U = u_\varepsilon/(2\pi)$ and $P = \rho/(2\pi)$, so that $\mathrm{dd^c}u_\varepsilon = \mathrm{i}\partial\bar\partial U$. Put

$$S = \operatorname{tr}_\eta \omega_\varepsilon, \quad R = \operatorname{tr}_{\omega_\varepsilon} \eta, \quad D = \omega_\varepsilon^n/\eta^n.$$

The previous lemma and the smooth positive density give $0 < d_1 \le D \le d_2$. Choose a fixed C_0 so that $v = U - P + C_0 > 1$, and set

$$H = \log S - \gamma(v), \qquad \gamma(v) = Av + \log v,$$

where A will be fixed below. For each ε, S is smooth on X and v tends to infinity at Z. Thus $H \to -\infty$ there, and H attains its maximum on $X \setminus Z$. We bound this maximum independently of ε.

In local holomorphic coordinates, h and g denote the coefficient matrices of η and ω_ε with respect to $\mathrm{i}\,\mathrm{d}z_j \wedge \mathrm{d}\bar z_k$, and commas denote coordinate derivatives, as in $h_{i\bar j,k} = \partial_k h_{i\bar j}$ and $h_{i\bar j,\bar k} = \partial_{\bar k} h_{i\bar j}$. At a maximum point choose holomorphic coordinates in which

$$h_{i\bar j} = \delta_{ij}, \qquad h_{i\bar i,k} = 0, \qquad g_{i\bar j} = \lambda_i \delta_{ij}.$$

A linear change first normalizes h and diagonalizes g. In these coordinates, the quadratic change $z_i = w_i + \frac{1}{2}\sum_{j,k} a^i_{jk} w_j w_k$, with symmetric coefficients chosen so that $a^i_{ik} = -h_{i\bar i,k}$, then kills $h_{i\bar i,k}$. Its coefficients are uniformly bounded on a fixed finite cover. Write

$$\Delta_g = \sum_i \lambda_i^{-1}\partial_i\partial_{\bar i}, \quad T_{ij\bar k} = g_{j\bar k,i} - g_{i\bar k,j}, \quad E_i = \sum_j T_{ij\bar j}.$$

The lowered torsion coefficients $T_{ij\bar k}$ and their first derivatives are uniformly bounded: the potential Hessian cancels, leaving only derivatives of $\theta + \varepsilon\eta$.

Let μ_{loc} be the density of μ with respect to the coordinate volume form. Differentiating $S = h^{k\bar\ell} g_{k\bar\ell}$ and the equation $\log\det g = \log c_\varepsilon + \log\mu_{\mathrm{loc}}$ gives

$$\Delta_g S \ge Q - CRS - CR, \qquad Q = \sum_{i,j} \frac{|g_{i\bar j,j}|^2}{\lambda_i\lambda_j}. \tag{15.47}$$

The second derivatives of g contribute $\sum_{i,j,k} |g_{i\bar{j},k}|^2/(\lambda_i\lambda_j)$, up to $-CR - C$, by commuting the potential derivatives. The first derivatives of h contribute $-2\,\mathrm{Re}\sum_{i,j,k} \lambda_i^{-1} h_{j\bar{k},\bar{i}} g_{k\bar{j},i}$. Its terms with $j = k$ vanish. Replace $g_{k\bar{j},i}$ by $g_{i\bar{j},k} + T_{ik\bar{j}}$ and apply weighted Cauchy–Schwarz. This consumes exactly the terms of the preceding sum of squares with $j \neq k$ and costs at most $CRS + CR$. The second derivatives of h^{-1} cost CRS. Finally the fixed density term $-C$ is absorbed using $SR \geq n^2$.

Since $\partial_i S = \sum_j g_{i\bar{j},j} + E_i$ at the maximum point, weighted Cauchy–Schwarz also gives

$$|\partial S|_g^2 \leq SQ + 2\,\mathrm{Re}\langle E, \partial S\rangle_g.$$

Indeed expand around $\partial S - E$ and discard the resulting term $-|E|_g^2$. Combining with (15.47) yields

$$\Delta_g \log S \geq -CR - CR/S - \frac{2}{S}\,\mathrm{Re}\langle E, \partial \log S\rangle_g.$$

At the maximum, $\partial \log S = \gamma' \partial v$. Moreover (15.45) gives $\Delta_g v \leq n - \delta R$, and $|E|_g^2 \leq CR$. Young's inequality with the positive coefficient $-\gamma'' = v^{-2}$ therefore gives

$$0 \geq \Delta_g H \geq (\delta\gamma' - C)R - n\gamma' - CR/S - \frac{CR}{S^2}\frac{\gamma'^2}{-\gamma''}.$$

Since $\gamma' = A + v^{-1}$, choose A with $\delta A - C \geq 1$. The resulting inequality is

$$0 \geq R - C\left(1 + \frac{R}{S} + \frac{R(1+v^2)}{S^2}\right). \tag{15.48}$$

All constants are independent of ε.

If the maxima of H were unbounded, then at a sequence of maximizing points

$$\log(S/v) = H + Av \longrightarrow +\infty.$$

Thus $S/v \to \infty$ and $S \to \infty$. Inequality (15.48) would imply $R \leq 2C$ along this sequence. But

$$S = D\sum_i \prod_{j\neq i} \lambda_j^{-1} \leq nDR^{n-1} \leq nd_2(2C)^{n-1},$$

a contradiction. Hence H is uniformly bounded. On K, both U and P are bounded, so $S \leq C_K$. The determinant lower bound then gives $\lambda_i \geq d_1/C_K^{n-1}$, completing the proof. □

Theorem 15.18.1 *In the preceding setting, c_ε decreases to a positive number c_0 as $\varepsilon \downarrow 0$. This number is independent of the perturbing Hermitian form η. For every sequence $\varepsilon_j \downarrow 0$, a subsequence satisfies*

$$u_{\varepsilon_j} \longrightarrow u_0 \quad \text{in } L^1(X) \cap C^\infty_{\mathrm{loc}}(X \setminus Z),$$

where $u_0 \in \mathrm{PSH}(X,\theta) \cap L^\infty(X)$, $\sup_X u_0 = 0$, and

$$\mathrm{MA}_\theta(u_0) = c_0 \mu.$$

If θ is closed, then $c_0 = \int_X \theta^n$.

Proof *Monotonicity of the constants.* Let $\alpha \le \beta$ be Hermitian forms, and let (u_α, c_α) and (u_β, c_β) be the normalized smooth solutions of $(\alpha + \mathrm{dd^c} u_\alpha)^n = c_\alpha \mu$ and $(\beta + \mathrm{dd^c} u_\beta)^n = c_\beta \mu$. At a maximum point of $u_\alpha - u_\beta$,

$$\alpha + \mathrm{dd^c} u_\alpha \le \beta + \mathrm{dd^c} u_\beta,$$

so $c_\alpha \le c_\beta$. Hence c_ε decreases as $\varepsilon \downarrow 0$, and Lemma 15.18.1 gives $c_0 \ge b > 0$. Write $c_\eta(\varepsilon)$ for the constant c_ε defined with the perturbation η. For two perturbations with $M^{-1}\eta_1 \le \eta_2 \le M\eta_1$, the same comparison gives

$$c_{\eta_1}(\varepsilon/M) \le c_{\eta_2}(\varepsilon) \le c_{\eta_1}(M\varepsilon).$$

Letting $\varepsilon \downarrow 0$ shows that c_0 does not depend on η.

Local estimates. The trace bound of Lemma 15.18.2 gives uniform ellipticity on compact subsets of $X \setminus Z$. The smooth coefficients of $\theta + \varepsilon\eta$ and all derivatives of $\log \mu_{\mathrm{loc}}$ have common bounds there. Interior complex Evans–Krylov and Schauder estimates, as in the ellipticity argument of Proposition 15.17.2, give uniform C^k bounds on smaller compact subsets for every k.

Convergence and the limiting equation. Normalized compactness in the fixed background $\theta + \eta$, followed by a diagonal extraction over an exhaustion of $X \setminus Z$, gives the stated convergence. The uniform bound $-C \le u_\varepsilon \le 0$ passes to the canonical limit, and normalized compactness preserves the supremum zero. Current convergence gives $\theta + \mathrm{dd^c} u_0 \ge 0$, and smooth convergence identifies the equation on $X \setminus Z$. The bounded non-pluripolar measure $\mathrm{MA}_\theta(u_0)$ and the smooth measure μ both vanish on the pluripolar set Z. This proves the equation on all of X. For closed θ, bounded closed-background Stokes gives $\int_X \mathrm{MA}_\theta(u_0) = \int_X \theta^n$, proving the last assertion. □

Since $\theta \ge 0$ is nef, Theorem 27.9.1 identifies c_0 with the Gauduchon variational volume $\mathcal{V}_\mu(\theta)$. If the bounded normalized limiting equation has a unique solution, all cluster limits coincide, and the whole family converges in $L^1(X) \cap C^\infty_{\mathrm{loc}}(X \setminus Z)$.

15.18.1 Small resolutions of nodes

The pullback of a positive form from a singular space loses positivity along the contracted curves. In the standard small resolution of a node, the logarithm of the norm in the normal bundle of the exceptional curve provides the missing curvature along the curve. Cutting this logarithm off where the pullback is already positive gives a global strict barrier, so the results above apply.

Assume now $n = 3$. Let Y be a connected compact normal complex threefold, smooth outside a finite set Σ of nodes. A node is locally the hypersurface

$$\{x_1x_4 - x_2x_3 = 0\} \subset \mathbb{C}^4.$$

Let $\pi\colon X \to Y$ be a proper holomorphic map that is biholomorphic over $Y \setminus \Sigma$. Assume each local resolution is the standard small resolution: a neighborhood of its exceptional curve is identified with a neighborhood of the zero section of

$$\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1),$$

and the contraction, in bundle coordinates (z, y_1, y_2), is

$$\pi(z, y_1, y_2) = (y_1, zy_1, y_2, zy_2). \tag{15.49}$$

Interchanging the two rulings gives the other standard model. Write $E = \pi^{-1}(\Sigma)$; its components E_j are disjoint copies of $\mathbb{P}^1$.

Fix a positive form ω_Y on Y in the local embedding sense: in each ambient chart it is induced by a smooth strictly positive real $(1,1)$-form. Set $\theta = \pi^*\omega_Y$, and let η be an arbitrary Hermitian form on X.

Lemma 15.18.3 *The form θ is smooth and semipositive, positive on $X \setminus E$. There are $\delta > 0$ and $\rho \in \mathrm{PSH}(X, \theta)$ with analytic poles precisely on E, smooth on $X \setminus E$, such that*

$$\sup_X \rho = 0, \qquad \theta + \mathrm{dd}^{\mathrm{c}}\rho \geq \delta\eta.$$

Proof Smoothness and positivity off E follow from the pullback and the biholomorphism. In one standard model put

$$r^2 = (1 + |z|^2)(|y_1|^2 + |y_2|^2), \qquad b = \mathrm{dd}^{\mathrm{c}} \log(1 + |z|^2).$$

The function r^2 is the sum of the squared absolute values of the four contracted coordinates. It is therefore globally defined on the bundle; b is the pullback of the Fubini–Study form from its base. In the chart (15.49),

$$\mathrm{dd}^{\mathrm{c}} \log r^2 = b + \mathrm{dd}^{\mathrm{c}} \log(|y_1|^2 + |y_2|^2) \geq b. \tag{15.50}$$

This is an inequality of currents across the zero section as well as a smooth inequality off it.

Ambient positivity gives $\theta \geq c\mathrm{dd}^{\mathrm{c}} r^2$ on a sufficiently small tube, for some $c > 0$. The smooth form $\mathrm{dd}^{\mathrm{c}} r^2 + b$ is positive definite there. Indeed the contraction is an immersion off the zero section, and on that section $\mathrm{dd}^{\mathrm{c}} r^2$ is positive in both fiber directions while b is positive along the base. Compactness of the zero section and a smaller closed tube give a uniform lower bound by a positive multiple of η. Thus, for every fixed $a > 0$,

$$\theta + a\mathrm{dd}^{\mathrm{c}} \log r^2 \geq c\mathrm{dd}^{\mathrm{c}} r^2 + ab \geq \delta_a \eta$$

near the zero section, with $\delta_a > 0$.

Choose disjoint model tubes with nonempty complement and numbers $0 < t_1 < t_2 < 1$ in each. Let $0 \leq \chi \leq 1$ be a smooth cutoff of r^2, equal to one for $r^2 \leq t_1$ and zero for $r^2 \geq t_2$. Extend $\chi(r^2) \log r^2$ by zero outside its tube. On the compact transition annuli, the pullback θ is uniformly positive and the second derivatives of this cutoff potential are bounded. A sufficiently small positive rational a therefore makes

$$\rho = a \sum_j \chi_j(r_j^2) \log r_j^2$$

satisfy $\theta + \mathrm{dd}^{\mathrm{c}} \rho \geq \delta\eta$ there and near every E_j. On the remaining compact set, $\rho = 0$ and θ is positive, so decreasing δ proves the global inequality. Near E_j the function is $a \log(|y_1|^2 + |y_2|^2) + a \log(1 + |z|^2)$, hence has analytic singularities with rational coefficient and pole set exactly E_j. It is smooth elsewhere and nonpositive; its zero extension gives supremum zero. If E is empty, take $\rho = 0$ and use the positivity of θ on compact X. □

Corollary 15.18.1 *For the preceding geometric data and every smooth positive probability volume form μ on X, the solutions of* (15.46) *have the bounds and subsequential convergence of Theorem 15.18.1, with $Z = E$. Moreover, for each exceptional curve,*

$$\int_{E_j} \omega_\varepsilon = \varepsilon \int_{E_j} \eta \longrightarrow 0.$$

Proof Apply the theorem using Lemma 15.18.3. Since π contracts E_j, the restriction of θ to it is zero. Stokes on the compact smooth curve gives $\int_{E_j} \mathrm{dd}^{\mathrm{c}} u_\varepsilon = 0$, proving the area identity. □

When K_X is torsion, the density can be chosen so that these metrics are Chern–Ricci flat. Recall that the Chern–Ricci form $\mathrm{Ric}^{\mathrm{Ch}}(\omega)$ of a Hermitian metric ω is $-\mathrm{i}\partial\bar{\partial}$ of the logarithm of the density of ω^n with respect to a coordinate volume form. Choose a nowhere vanishing holomorphic section of $K_X^{\otimes m}$ for some $m \geq 1$. If its local expression is $f(\mathrm{d}z_1 \wedge \mathrm{d}z_2 \wedge \mathrm{d}z_3)^{\otimes m}$, use $|f|^{2/m}$ as the density relative to the standard coordinate volume form. The coordinate Jacobians cancel, so this defines a smooth positive volume form on X; normalize it to total mass one and take it as μ. Its logarithmic density is pluriharmonic. Applying $-\mathrm{i}\partial\bar{\partial} \log$ to $\omega_\varepsilon^3 = c_\varepsilon \mu$ therefore gives $\mathrm{Ric}^{\mathrm{Ch}}(\omega_\varepsilon) = 0$, and the same holds for the smooth limiting metric on $X \setminus E$.

15.18.2 Measures and bounded potentials at a node

We keep the standard small resolution $\pi\colon X \to Y$ and a smooth positive probability volume form μ on X. Write $\mu_Y = \pi_*\mu$, and let ν_Y be ω_Y^3 on the regular locus, extended by zero at the nodes. The pullback of the ambient volume form ω_Y^3 vanishes along the exceptional curves, and its order of vanishing determines the integrability of the density of μ_Y with respect to ν_Y.

Lemma 15.18.4 *The measures μ_Y and ν_Y are finite and give the nodes zero mass. There is a positive density F with $\mu_Y = F\nu_Y$. If Σ is nonempty, then for $1 \le p < \infty$,*

$$F \in L^p(Y, \nu_Y) \quad \Longleftrightarrow \quad p < 3.$$

At each node, F is comparable to r^{-2}, where $r^2 = \sum_{j=1}^4 |x_j|^2$ in the local model $\{x_1x_4 - x_2x_3 = 0\}$ of the node. There are $c, C, T_0 > 0$ such that

$$cT^{-3} \le \nu_Y(\{F > T\}) \le CT^{-3} \qquad (T \ge T_0).$$

Thus the endpoint is weak L^3, with $F \notin L^3$. If Σ is empty, F is smooth and bounded.

Proof The map is biholomorphic off E, and both smooth measures μ and θ^3 give E zero mass. Hence $\nu_Y = \pi_*\theta^3$ is finite, μ_Y has no node atom, and on the regular locus

$$F \circ \pi = \mu/\theta^3.$$

In the chart (15.49), put $h = 1 + |z|^2$ and $s = |y_1|^2 + |y_2|^2$, so $r^2 = hs$. The fiber block of the complex Hessian of r^2 is h times the identity, and its Schur complement is s/h. Consequently

$$\det\bigl((r^2)_{j\bar{k}}\bigr) = hs = r^2, \qquad \frac{(\mathrm{dd^c} r^2)^3}{3!} = \frac{r^2}{\pi^3}\,\mathrm{d}\lambda_6, \tag{15.51}$$

where λ_6 is Euclidean volume in (z, y_1, y_2). Ambient positivity bounds θ above and below by positive multiples of $\mathrm{dd^c} r^2$ on a smaller tube. The smooth density of μ is bounded above and below there. On compact subcharts covering the base $\mathbb{P}^1$, it follows that

$$\theta^3 \asymp r^2\,\mathrm{d}\lambda_6, \qquad F \circ \pi \asymp r^{-2}.$$

The other bundle chart gives the same comparison, since r is the norm of the contracted coordinates.

On each bounded base subchart, r is comparable to the Euclidean radius R in the two complex fiber variables. Up to fixed positive constants, the contribution to $\int F^p\,\mathrm{d}\nu_Y$ is therefore

$$\int_0^{R_0} R^{2-2p} R^3\,\mathrm{d}R = \int_0^{R_0} R^{5-2p}\,\mathrm{d}R.$$

Two compact base subcharts give the upper estimate; a fixed open base subchart gives the lower estimate. The integral is finite precisely when $p < 3$, and diverges logarithmically at $p = 3$. Away from the finitely many nodes the density is smooth. The same comparisons give $\nu_Y(\{r < t\}) \asymp t^6$ near a node. The two bounds for F sandwich $\{F > T\}$ between such tubes of radii comparable to $T^{-1/2}$, proving the endpoint estimate. □

The descent theorem Theorem A.2.1 applies to the normal space Y and its resolution π, also when the ambient background is not closed. Every bounded θ-plurisubharmonic function is therefore constant on each exceptional curve and is the pullback of a unique bounded ω_Y-plurisubharmonic function. In particular each cluster limit has the form $u_0 = \psi_0 \circ \pi$, with $\sup_Y \psi_0 = 0$. Let $\mathrm{MA}^{\mathrm{ext}}_{Y,\omega_Y}$ be the extended Monge–Ampère measure on Y, defined in (A.7) as the pushforward of the extended measure of the pullback to a resolution; it does not depend on the resolution. Then

$$\mathrm{MA}^{\mathrm{ext}}_{Y,\omega_Y}(\psi_0) = c_0\mu_Y = c_0 F \nu_Y. \tag{15.52}$$

The equality holds also at the nodes, since both measures give them zero mass. Thus ψ_0 is a bounded solution of an equation on the singular space; nothing is claimed about its continuity at a node.

Theorem 15.18.2 *Suppose additionally that θ is closed. The equation*

$$\mathrm{MA}_\theta(u) = c\mu, \qquad u \in \mathrm{PSH}(X,\theta) \cap L^\infty(X), \qquad \sup_X u = 0,$$

has a unique pair (c, u) with $c > 0$. Its scalar is $c = c_0 = \int_X \theta^3$. The entire family u_ε converges to u in $L^1(X)$ and smoothly on compact subsets of $X \setminus E$. The limit is independent of η, and its descended potential is the unique bounded normalized solution of (15.52).

Proof Existence follows from Corollary 15.18.1. For uniqueness we use an energy identity, which is available because θ is closed. Fix one cluster limit u_0 and put $A = \theta + \mathrm{dd^c} u_0$. This is a positive closed current with bounded local potentials, smooth and strictly positive on $X \setminus E$ by the trace bounds. If v is another bounded solution with scalar c, bounded closed-background Stokes gives $c = \int_X \theta^3 = c_0$. Put $B = \theta + \mathrm{dd^c} v$ and $w = v - u_0$. The bounded integration-by-parts calculus of Section 1.6 gives

$$0 = \int_X w(B^3 - A^3) = -\sum_{j=0}^{2} \int_X \mathrm{d}w \wedge \mathrm{d^c} w \wedge B^j \wedge A^{2-j}.$$

All three energies are nonnegative, so each of them vanishes, in particular the one with A^2. On every relatively compact chart of $X \setminus E$, the smooth positive factor A^2 controls the weak gradient of w, so $\mathrm{d}w = 0$ there. The complement of a proper complex analytic subset in the connected manifold X is connected. Thus w is constant almost everywhere off E. Since E has ordinary volume zero, canonical almost-everywhere

equality of quasi-plurisubharmonic functions, Lemma 1.3.3, gives $v = u_0 + C$ on all of X. Normalization fixes $C = 0$.

The compactness and smooth local estimates of Theorem 15.18.1 now make every subsequential limit equal to u_0, proving convergence of the whole family. The equation and its unique normalized solution do not depend on η. Descent and the zero exceptional mass identify bounded solutions upstairs and downstairs, proving the final assertion. □

15.18.3 Distances across finitely many contracting regions

On the small resolutions above, the potential limit for a closed background gives smooth convergence of the metrics away from the exceptional curves. Global distances, however, are also affected by paths through neighborhoods of these curves, so their convergence requires control of the diameters of such neighborhoods. The criterion of Proposition 15.18.1 separates the two controls: uniform convergence on compact subsets of the regular part, and small diameters of neighborhoods of the contracted sets, in the limiting metric and uniformly in the approximating metrics. We first recall the Gromov–Hausdorff distance.

Definition 15.18.1 For nonempty subsets E, F of a metric space (W, d_W), their Hausdorff distance is

$$d_{\mathrm{H}}^{W}(E,F) = \max\left\{ \sup_{x\in E}\inf_{y\in F} d_W(x,y), \sup_{y\in F}\inf_{x\in E} d_W(x,y) \right\}.$$

For nonempty bounded metric spaces E, F, put

$$d_{\mathrm{GH}}(E,F) = \inf_{W,\iota,\jmath} d_{\mathrm{H}}^{W}(\iota(E),\jmath(F)),$$

where $\iota, \jmath$ are isometric embeddings into a common metric space W. A correspondence is a subset $R \subset E \times F$ whose two coordinate projections are surjective. Its distortion is

$$\operatorname{dis} R = \sup_{(x,y),(x',y')\in R} |d_E(x,x') - d_F(y,y')|.$$

Lemma 15.18.5 *For any correspondence R between nonempty bounded metric spaces,*

$$\frac{1}{2}|\operatorname{diam} E - \operatorname{diam} F| \le d_{\mathrm{GH}}(E,F) \le \frac{1}{2}\operatorname{dis} R.$$

Proof Choose $\varepsilon > 0$ with $2\varepsilon \ge \operatorname{dis} R$. Give the disjoint union $E \sqcup F$ its original distances within each space, and use the symmetric cross distance

$$D(x,y) = \inf_{(x_0,y_0)\in R}\bigl(d_E(x,x_0) + \varepsilon + d_F(y_0,y)\bigr), \qquad x \in E,\ y \in F.$$

It is finite and at least ε. The triangle inequalities involving one cross distance follow from the triangle inequality within the corresponding space and the infimum. For two cross distances, take $(x_0,y_0), (x_1,y_1) \in R$. The distortion bound gives

$$\begin{aligned} d_E(x,x') &\le d_E(x,x_0) + d_F(y_0,y_1) + 2\varepsilon + d_E(x_1,x') \\ &\le d_E(x,x_0) + d_F(y_0,y) + \varepsilon + \varepsilon + d_F(y,y_1) + d_E(x_1,x'). \end{aligned}$$

Taking the two infima gives $d_E(x,x') \le D(x,y) + D(y,x')$. The same argument with E, F exchanged treats the other case. Hence D is a metric extending both metrics. Each point has a correspondent at cross distance ε, so the Hausdorff distance between E and F in $(E \sqcup F, D)$ is at most ε. Letting ε decrease to half the distortion proves the upper bound, including zero distortion.

For the lower bound, suppose two isometric images have Hausdorff distance h. For every $\delta > 0$, two points in the first image have points in the second within $h + \delta$. Their distance is at most $\operatorname{diam} F + 2h + 2\delta$. Taking the supremum and then $\delta \downarrow 0$ gives $\operatorname{diam} E \le \operatorname{diam} F + 2h$. Exchanging the images and taking the infimum over embeddings proves the lower bound. □

We now work in a purely metric setting, which contains the nodal one. Let Y be a compact metrizable space and let $S = \{s_1, \dots, s_N\} \subset Y$ be finite. Assume $Y^\circ = Y \setminus S$ is a dense connected smooth manifold. Let g_0 be a smooth Riemannian metric on Y°, and denote its intrinsic length distance by d_0.

Lemma 15.18.6 *Suppose every s_a has a neighborhood basis of open sets $V_{a,r}$ such that*

$$\lim_{r\downarrow 0} \operatorname{diam}_{d_0}(V_{a,r} \setminus S) = 0.$$

Then the metric completion $(\widehat{Y}, \widehat{d_0})$ of (Y°, d_0) is compact, and the identity on Y° extends to a homeomorphism $\iota\colon Y \to \widehat{Y}$.

Proof The neighborhoods can be chosen nested, with closures avoiding the other points of S. Every sequence of regular points converging to s_a is d_0-Cauchy: its tail lies in each prescribed neighborhood, whose punctured diameter tends to zero. Interleaving two such sequences shows that they define the same completion point, denoted by $\widehat{s}_a$. Density of Y° guarantees the existence of these sequences.

For each prescribed positive diameter, remove sufficiently small neighborhoods of all the points of S. Their complement is a compact subset of Y° and hence is totally

bounded for d_0. The removed punctured neighborhoods each have the prescribed small diameter. Thus Y° is totally bounded, and its completion is compact.

The points $\widehat{s}_a$ are distinct from one another and from the regular points. To see this, fix s_a and a different point $y \in Y$. Choose an open neighborhood V of s_a whose closure avoids y and $S \setminus \{s_a\}$, and choose $W \Subset V$ containing s_a. There is a smooth function χ on Y°, equal to one on $W \setminus S$ and zero outside V, whose derivative has compact support in Y°. Indeed the transition takes place in the compact regular set $\overline{V} \setminus W$, where smooth cutoffs exist. Therefore $|\mathrm{d}\chi|_{g_0} \le C$ for some finite C. Integration along curves gives

$$|\chi(p) - \chi(q)| \le C d_0(p,q), \qquad p, q \in Y^\circ.$$

The Lipschitz extension of χ to the completion has value one at $\widehat{s}_a$ and zero at the regular point y, or at $\widehat{s}_b$ if $y = s_b$. This proves the assertion.

Define ι by the identity on the regular locus and by $\iota(s_a) = \widehat{s}_a$. The intrinsic distance induces the smooth topology at each regular point. At s_a, adjoining $\widehat{s}_a$ to a punctured neighborhood does not increase its diameter, so the diameter hypothesis proves continuity of ι. Every completion point is represented by a Cauchy sequence in Y°. Compact metrizability of Y gives a subsequence converging topologically either to a regular point or to some s_a; the preceding arguments give its metric limit. Hence ι is surjective as well as injective. A continuous bijection from compact Y to Hausdorff $\widehat{Y}$ is a homeomorphism. □

Let X be a compact connected smooth manifold and let $\pi\colon X \to Y$ be a continuous surjection, with $E = \pi^{-1}(S)$ and with $\pi\colon X \setminus E \to Y^\circ$ a diffeomorphism. Assume that the identity of Y° extends to a homeomorphism $\iota\colon Y \to \widehat{Y}$, for instance under the hypothesis of Lemma 15.18.6, and write $G = \iota \circ \pi$. For a smooth Riemannian metric g on X and an open subset U, let $\operatorname{diam}^{\mathrm{int}}(U, g)$ denote the intrinsic diameter of U, computed with the lengths of piecewise smooth curves contained in U.

Proposition 15.18.1 *Let g_j be smooth Riemannian metrics on X such that $g_j \to \pi^* g_0$ uniformly on every compact subset of $X \setminus E$. Suppose each s_a has a nested neighborhood basis $V_{a,r}$, with $U_{a,r} = \pi^{-1}(V_{a,r})$, satisfying*

$$\lim_{r \downarrow 0} \operatorname{diam}_{\widehat{d}_0}(\iota(V_{a,r})) = 0, \qquad \lim_{r \downarrow 0} \limsup_{j \to \infty} \operatorname{diam}^{\mathrm{int}}(U_{a,r}, g_j) = 0.$$

Then, for the global length distances d_j of g_j,

$$\lim_{j \to \infty} \sup_{x, x' \in X} |d_j(x,x') - \widehat{d}_0(G(x), G(x'))| = 0.$$

Consequently (X, d_j) converges to $(\widehat{Y}, \widehat{d}_0)$ in Gromov–Hausdorff distance.

Proof If S is empty, uniform metric comparison on compact X proves the distance convergence. Assume $N \ge 1$. Fix $\tau > 0$. Choose pairwise disjoint neighborhoods V_a from the specified bases with

$$\operatorname{diam}_{\widehat{d}_0}(\iota(V_a)) < \tau, \qquad \operatorname{diam}^{\mathrm{int}}(U_a, g_j) < \tau$$

for all sufficiently large j, where $U_a = \pi^{-1}(V_a)$. Pick $q_a \in V_a \setminus S$, and then choose open neighborhoods W_a of s_a with $\overline{W}_a \subset V_a$ and $q_a \notin \overline{W}_a$. The regular cores

$$K_Y = Y \setminus \bigcup_a W_a, \qquad K_X = \pi^{-1}(K_Y)$$

are compact and identified by π. Uniform convergence and positivity of g_0 on this core give numbers $A_j \geq 1$, with $A_j \to 1$, such that lengths of curves contained in the cores compare in both directions by the factor A_j.

Consider a piecewise smooth path in Y° joining two points of K_Y. The preimages of the closed sets $\overline{W}_a$ are compact subsets of its parameter interval. At its first visit to any such set, delete the portion from that visit through its last visit to the same set. Resume with the remaining tail. That set is never visited again; thus there are at most N deleted portions. Each remaining segment lies in K_Y, and each deleted portion has endpoints in a single $\overline{W}_a \subset V_a$. The lifts of those endpoints can be joined in U_a with g_j-length less than τ, up to an arbitrarily small error. Applying this procedure to almost minimizing paths, comparing the lengths of the remaining segments, and letting the errors tend to zero gives, for $x, x' \in K_X$,

$$d_j(x, x') \leq A_j \widehat{d}_0(G(x), G(x')) + N\tau.$$

Conversely start with an almost minimizing g_j-path in X and apply the same deletion procedure to the closed sets $\pi^{-1}(\overline{W}_a)$. Project its remaining segments, which avoid E, to K_Y. The two images of each deleted portion have completion distance less than τ. The triangle inequality in $\widehat{Y}$ therefore gives

$$\widehat{d}_0(G(x), G(x')) \leq A_j d_j(x, x') + N\tau, \qquad x, x' \in K_X.$$

For any $x \notin K_X$ with $\pi(x) \in W_a$, replace x by $p = \pi^{-1}(q_a) \in U_a \cap K_X$. Then $d_j(x, p) < \tau$ and $\widehat{d}_0(G(x), G(p)) < \tau$. Leave points of K_X fixed. Replacing both endpoints extends the two inequalities to all $x, x' \in X$, with the same error

$$C_j = (N + 2 + 2A_j)\tau.$$

If $D = \operatorname{diam}(\widehat{Y}, \widehat{d}_0)$, the first inequality gives $\operatorname{diam}(X, d_j) \leq A_j D + C_j$. The positive and negative distance errors are consequently bounded by

$$(A_j - 1)D + C_j, \qquad (A_j - 1)(A_j D + C_j) + C_j,$$

respectively. Their upper limits are at most $(N + 4)\tau$. Letting $\tau \downarrow 0$ proves uniform convergence.

The graph of G is a correspondence between X and $\widehat{Y}$, and its distortion b_j for the distances d_j and $\widehat{d}_0$ is the supremum in the statement. By Lemma 15.18.5,

$$d_{\mathrm{GH}}((X,d_j),(\widehat{Y},\widehat{d}_0)) \le b_j/2 \longrightarrow 0.$$

For the standard nodal small resolutions of Section 15.18.1 with closed θ, the smooth convergence of the whole family off E in Theorem 15.18.2 gives the uniform convergence on compact subsets of $X \setminus E$ required in Proposition 15.18.1. What remains are the two conditions on the diameters of neighborhoods. The next estimates control the approximating neighborhoods by metric bounds along and normal to the exceptional curves.

15.18.3.1 Curve areas and radial tube lengths

The approximating metrics give each exceptional curve an area tending to zero, by Corollary 15.18.1. Along the curve, small area forces small diameter as long as the metrics are bounded above; in the normal directions, an integrable bound for the speed of radial rays has the same effect.

Let C be a connected compact Riemann surface. Fix a smooth Hermitian metric h of area one on C, and use the associated Riemannian metrics for lengths and areas.

Lemma 15.18.7 *There is $K_h > 0$ such that every smooth Hermitian metric $g \le Mh$, with $M > 0$ and $A = \mathrm{Area}_g(C)$, satisfies*

$$\mathrm{diam}(C,g) \le K_h\sqrt{A\big(1+\log(M/A)\big)}.$$

Here $0 < A \le M$. In particular, for a fixed upper bound M, area tending to zero implies diameter tending to zero.

Proof Replace g by g/M. Its area is $a = A/M \le 1$, and it is bounded above by h. We prove the estimate for this normalized metric.

Choose a finite cover by holomorphic coordinate disks U_i represented by $\{|z| < 1\}$, with each coordinate defined on the larger disk $\{|z| < 2\}$. Write $g = \rho_i|\mathrm{d}z|^2$ there. There is a common $L > 0$ such that $0 < \rho_i \le L$ on all the closed unit disks, and

$$\int_{U_i} \rho_i \,\mathrm{d}\lambda_2 \le a,$$

where λ_2 is Euclidean area. For $p, z \in U_i$, let $\ell_p(z)$ be the g-length of their straight coordinate segment, which stays in U_i. Polar coordinates about p and Fubini give

$$\int_{U_i} \ell_p(z)\,\mathrm{d}\lambda_2(z) \le 2\int_{U_i} \frac{\sqrt{\rho_i(w)}}{|w-p|}\,\mathrm{d}\lambda_2(w).$$

Indeed each ray meets the convex unit disk in an interval of length $R \le 2$. Interchanging the radial integrals multiplies the length integrand at radius s by $(R^2 - s^2)/2 \le 2$.

For $0 < a \le 1/4$, put $t = \sqrt{a}$. The integral on the right, without its factor two, is bounded by

$$\sqrt{L}\int_{\{|w-p|<t\}}\frac{\mathrm{d}\lambda_2(w)}{|w-p|}+\left(\int_{U_i}\rho_i\,\mathrm{d}\lambda_2\right)^{1/2}\left(\int_{\{t<|w-p|<2\}}\frac{\mathrm{d}\lambda_2(w)}{|w-p|^2}\right)^{1/2}$$
$$\le 2\pi\sqrt{L}\,t+\sqrt{2\pi a\log(2/t)}\le K\sqrt{a\big(1+\log(1/a)\big)}.$$

Thus for any $p,q\in U_i$, averaging $\ell_p(z)+\ell_q(z)$ over $z\in U_i$ gives a two-segment path of length at most $K\sqrt{a(1+\log(1/a))}$. The constant is common to the finite cover.

The overlap graph of this cover is connected because C is connected. Choose points in consecutive overlaps along a simple path in this finite graph. The local estimates join any two points of C through at most the number of covering disks, proving the global bound for $a\le 1/4$. For $1/4\le a\le 1$, the comparison $g\le h$ gives $\operatorname{diam}(C,g)\le\operatorname{diam}(C,h)$, and increasing K_h gives the same estimate. Restoring the factor M scales lengths by $\sqrt{M}$ and proves the assertion. □

Example 15.18.1 On every such (C,h) there are smooth Hermitian metrics $g_\varepsilon\le h$, for $\varepsilon\downarrow 0$, such that

$$A_\varepsilon:=\operatorname{Area}_{g_\varepsilon}(C)\asymp\varepsilon^2\log(1/\varepsilon),$$
$$\operatorname{diam}(C,g_\varepsilon)\asymp\varepsilon\log(1/\varepsilon).$$

Consequently the diameter bound of Lemma 15.18.7 has sharp order under its uniform upper bound. Without that bound, area can tend to zero while diameter stays bounded below by a positive constant.

Proof Choose a coordinate disk $\{|z|<R\}$ with $0<R<1$, on whose closure $h=\lambda(z)|\mathrm{d}z|^2$ with $\lambda>0$ smooth. Choose a smooth cutoff $0\le\chi\le 1$, supported in this disk and equal to one for $|z|\le R/2$, and a constant $c>0$ with $c\le\min\lambda$. For sufficiently small $\varepsilon>0$, set

$$g_\varepsilon=\left(\chi\frac{c\varepsilon^2}{\lambda(z)(\varepsilon^2+|z|^2)}+(1-\chi)\varepsilon^2\right)h$$

in the disk, and set $g_\varepsilon=\varepsilon^2 h$ outside it. The cutoff makes this a smooth positive metric, and both coefficients in its convex combination are at most one.

On $|z|\le R/2$ its metric density is $c\varepsilon^2/(\varepsilon^2+|z|^2)$, whose area is

$$\pi c\varepsilon^2\log\left(1+\frac{R^2}{4\varepsilon^2}\right).$$

The remaining area is $O(\varepsilon^2)$, proving the asserted area comparison. Any path from the origin to a point outside this smaller disk must first cross its boundary. Radial variation is bounded by Euclidean speed, so that initial portion has length at least

$$\sqrt{c}\,\varepsilon\int_0^{R/2}\frac{\mathrm{d}r}{\sqrt{\varepsilon^2+r^2}}=\sqrt{c}\,\varepsilon\operatorname{arsinh}(R/(2\varepsilon))\asymp\varepsilon\log(1/\varepsilon).$$

This gives the diameter lower bound. The upper bound follows from Lemma 15.18.7, since $A_\varepsilon(1+\log(1/A_\varepsilon))\asymp\varepsilon^2\log^2(1/\varepsilon)$. Finally rescale by

$$\widetilde{g}_\varepsilon = \frac{g_\varepsilon}{\varepsilon^2 \log^2(1/\varepsilon)}.$$

Its area is comparable to $1/\log(1/\varepsilon)$ and tends to zero, whereas its diameter is comparable to one. At the origin its ratio to h tends to infinity, so it has no uniform upper bound. □

Let $V \to C$ be a holomorphic vector bundle of positive rank with a fixed smooth Hermitian fiber norm. Write $U_\delta = \{v \in V : |v| < \delta\}$ and identify its zero section with C.

Proposition 15.18.2 *Let g_j be smooth Hermitian metrics on U_{δ_0}. Suppose their restrictions to C satisfy*

$$g_j|_C \le Mh, \qquad A_j = \operatorname{Area}_{g_j|_C}(C) \longrightarrow 0$$

for a fixed $M > 0$. Suppose also that there are $B > 0$ and $\alpha > 0$ such that, for every unit fiber vector v over any $x \in C$, the ray $\gamma_{x,v}(s) = sv$ satisfies

$$g_j(\gamma'_{x,v}(s), \gamma'_{x,v}(s)) \le Bs^{2\alpha-2}, \qquad 0 < s < \delta_0,$$

uniformly in j, x, v. Then

$$\operatorname{diam}^{\mathrm{int}}(U_\delta, g_j) \le \frac{2\sqrt{B}}{\alpha}\delta^\alpha + K_h\sqrt{A_j\bigl(1 + \log(M/A_j)\bigr)}, \qquad 0 < \delta < \delta_0.$$

In particular,

$$\lim_{\delta \downarrow 0} \limsup_{j \to \infty} \operatorname{diam}^{\mathrm{int}}(U_\delta, g_j) = 0.$$

Proof A point of fiber radius $r < \delta$ joins its base point on the zero section along the radial segment, entirely inside U_δ. Its length is at most

$$\sqrt{B}\int_0^r s^{\alpha-1}\,\mathrm{d}s = \frac{\sqrt{B}}{\alpha}r^\alpha.$$

The metric is smooth at the zero section for each j; the integrable uniform bound controls the whole segment. Join the two base points by an almost minimizing path in $(C, g_j|_C)$. This path also lies in U_δ, and up to an arbitrarily small excess its length is bounded by Lemma 15.18.7. Together with the two radial segments, this gives the asserted bound for the intrinsic diameter. Since $A_j(1 + \log(M/A_j)) \to 0$, taking $j \to \infty$ and then $\delta \downarrow 0$ proves the final assertion. □

Return to the standard nodal small resolutions of Section 15.18.1. On a nodal tube put $r^2 = \sum_{k=1}^4 |x_k \circ \pi|^2$, the squared bundle norm used in Lemma 15.18.3. Let $V_\delta \subset Y$ be the nodal coordinate ball $\sum_k |x_k|^2 < \delta^2$. For sufficiently small δ, properness gives $\pi^{-1}(V_\delta) = \{r < \delta\}$. Write g_ε for the Riemannian metric associated to ω_ε. Its area on E_a is $\varepsilon\int_{E_a}\eta$ by Corollary 15.18.1. Fix an area-one Hermitian metric h_a on E_a.

If $g_\varepsilon|_{E_a} \le M h_a$ uniformly and the radial bound of Proposition 15.18.2 holds with $\alpha = 1/2$, substitution in that proposition gives

$$\operatorname{diam}^{\mathrm{int}}(\{r < \delta\}, g_\varepsilon) \le K\delta^{1/2} + K\sqrt{\varepsilon\bigl(1 + \log(1/\varepsilon)\bigr)}$$

for sufficiently small ε and δ, with K independent of both. Under these two pointwise metric bounds, the condition on approximating neighborhoods in Proposition 15.18.1 therefore holds. The condition on punctured neighborhoods in the limiting metric does not follow from it, as the next example shows.

In the following surface example the volume form is fixed and the approximating tubes shrink, but the punctured neighborhoods of the contracted point keep a positive diameter in the limiting metric.

Example 15.18.2 There are a compact complex surface X, a holomorphic map $\pi\colon X \to Y = \mathbb{P}^2$ contracting a curve E to a point p, and smooth Hermitian metrics g_j, g_* on X with the following properties. The map is biholomorphic from $X \setminus E$ to $Y \setminus \{p\}$, and $g_j = g_*$ eventually on each compact subset of $X \setminus E$. Their associated Hermitian forms satisfy

$$\omega_j^2 = \omega_*^2.$$

The curve areas tend to zero, and the metrics have a uniform integrable radial bound near E. For a neighborhood basis V_δ of p, with $U_\delta = \pi^{-1}(V_\delta)$,

$$\lim_{\delta \downarrow 0} \limsup_{j \to \infty} \operatorname{diam}^{\mathrm{int}}(U_\delta, g_j) = 0.$$

Nevertheless, for the intrinsic distance d_0 of $g_0 = (\pi^{-1})^* g_*$ on $Y \setminus \{p\}$, there is $c > 0$ such that

$$\operatorname{diam}_{d_0}(V_\delta \setminus \{p\}) \ge c$$

for every sufficiently small δ.

Proof Put $p = [1 : 0 : 0]$ and take the incidence surface

$$X = \bigl\{([z_0 : z_1 : z_2], [u : v]) \in \mathbb{P}^2 \times \mathbb{P}^1 : z_1 v = z_2 u\bigr\},$$
$$\pi([z], [u : v]) = [z].$$

Off p, the second coordinate is $[z_1 : z_2]$, so π is biholomorphic there. Its exceptional curve is $E = \{p\} \times \mathbb{P}^1$. In the affine chart $z_0 = 1$, a chart of X is

$$(t, w) \longmapsto ([1 : w : tw], [1 : t]).$$

The other chart is $([1 : t'w' : w'], [t' : 1])$, with $t' = 1/t$ and $w' = tw$ on their overlap. These are the charts of the tautological line bundle $L = \mathcal{O}_{\mathbb{P}^1}(-1)$. They also show smoothness along E; smoothness elsewhere follows from the biholomorphism. Thus X is a compact smooth complex surface. Its connected dense complement $X \setminus E$ makes it connected.

Let $q\colon L \to \mathbb{P}^1$ be the bundle projection. Give L the norm induced by its tautological inclusion into $\mathbb{P}^1 \times \mathbb{C}^2$. In the first chart its squared norm is

$$r^2 = (1 + |t|^2)|w|^2 = |z_1|^2 + |z_2|^2.$$

Fix an area-one Hermitian metric h on $\mathbb{P}^1$. The Chern connection of the fiber norm defines a smooth semipositive Hermitian tensor k on L by the squared norm of the covariant fiber differential. Explicitly, with $\lambda = 1 + |t|^2$,

$$k = \lambda \left| \mathrm{d}w + w\partial \log \lambda \right|^2 .$$

Under $t' = 1/t$, $w' = tw$, the differential in this formula multiplies by t and λ divides by $|t|^2$, so the tensor is globally defined. It has complex rank one, is positive on nonzero vertical tangent vectors, and restricts to zero on TE. Consequently $q^*h + k$ is a smooth positive Hermitian metric on L.

Choose $0 < \delta_0 < 1$ and extend this metric from $U_{\delta_0} = \{r < \delta_0\}$ to a smooth Hermitian metric g_* on X. Such an extension follows by a smooth cutoff supported in a slightly larger bundle tube and a convex combination with any global Hermitian metric. It agrees exactly with $q^*h + k$ on U_{δ_0}.

Choose $0 < \varepsilon_j < 1$ tending to zero, with $2\varepsilon_j^2 < \delta_0$, and a smooth function $0 \le \chi \le 1$ on $[0, \infty)$, equal to zero on $[0, 1]$ and to one on $[4, \infty)$. Set

$$\begin{aligned} f_j(r) &= \varepsilon_j^2 + (1 - \varepsilon_j^2)\chi(r^2/\varepsilon_j^4), \\ g_j &= f_j q^*h + f_j^{-1} k, \end{aligned}$$

on U_{δ_0}, and set $g_j = g_*$ elsewhere. The function f_j is smooth at E, is at least ε_j^2, and equals one for $r \ge 2\varepsilon_j^2$. Hence the metrics glue smoothly and are positive. They equal g_* eventually on every compact subset of $X \setminus E$.

Let β, κ be the Hermitian forms of q^*h, k. Their complex rank one gives $\beta^2 = \kappa^2 = 0$. On the tube,

$$(f_j\beta + f_j^{-1}\kappa)^2 = 2\beta \wedge \kappa = (\beta + \kappa)^2.$$

Outside it the metrics coincide. This proves the global volume identity. On E, the restriction is $g_j|_E = \varepsilon_j^2 h$, so its area is ε_j^2.

For any unit fiber vector a, a radial ray $\gamma(s) = sa$ has k-squared speed one and zero base component. Thus its g_j-squared speed is $f_j(s)^{-1}$. If $s < 2\varepsilon_j^2$, this is at most $\varepsilon_j^{-2} \le 2/s$; otherwise it is one, still at most $2/s$ for $s < \delta_0$. This is the uniform integrable radial bound with exponent $1/2$.

A radial segment of radius $r < \delta$ has length at most $r + 2\varepsilon_j$: its portion below $2\varepsilon_j^2$ has speed at most ε_j^{-1} and length at most $2\varepsilon_j$, and its remaining speed is one. Connect two points by their radial segments and an almost minimizing path on E. All three segments stay inside U_δ, giving

$$\operatorname{diam}^{\mathrm{int}}(U_\delta, g_j) \le 2\delta + \varepsilon_j\big(4 + \operatorname{diam}(\mathbb{P}^1, h)\big).$$

Taking $j \to \infty$ before $\delta \downarrow 0$ proves the asserted shrinking condition. The affine balls

$$V_\delta = \{[1 : z_1 : z_2] : |z_1|^2 + |z_2|^2 < \delta^2\}$$

have precisely these tubes as their preimages.

Let d_* be the global distance of the fixed smooth metric g_*. Choose distinct points $a, b \in E$, and put $c = d_*(a, b) > 0$. Choose points a_r, b_r of fiber norm $r > 0$ over these base points. They tend to a, b as $r \downarrow 0$. Every path in $Y \setminus \{p\}$ lifts to a path in $X \setminus E$ with the same g_0-length as its g_*-length. Therefore

$$d_0(\pi(a_r), \pi(b_r)) \geq d_*(a_r, b_r) \longrightarrow c.$$

For any fixed $\delta > 0$, these two images lie in V_δ when $r < \delta$. Taking the supremum and then $r \downarrow 0$ gives its punctured diameter at least c. □

15.19 Symmetry and the selected projective limit

Let ω be a Hermitian form and $0 \leq f \in L^p(\mathrm{d}V)$, $p > 1$, a nonzero density. When the normalized equation $\mathrm{MA}_\omega(u) = c f \, \mathrm{d}V$ has several solutions, the positive-exponent equations $\mathrm{MA}_\omega(u) = \mathrm{e}^{\lambda u} f \, \mathrm{d}V$ still have unique solutions, and their normalized limits as $\lambda \downarrow 0$ single out particular solutions of the normalized equation. We study this selection in the projective example of Theorem 12.6.1, whose density vanishes on an open band. The starting point is a general observation: uniqueness at positive exponent preserves the symmetries of the data, and uniform compactness carries them to every normalized cluster limit. For the projective data this already gives a solution distinct from the two smooth solutions exchanged by the involution. The radial form of the equation then identifies the selected limit (Theorem 15.19.1) and shows that this limit and all the positive-exponent solutions fail to be C^2 along the same two hypersurfaces (Proposition 15.19.2).

Proposition 15.19.1 *Let ω be a Hermitian form on X, let $0 \leq f \in L^p(\mathrm{d}V)$, $p > 1$, be nonzero, and let $c > 0$ be the scalar of Theorem 11.1.1 for f. For $\lambda > 0$ denote by u_λ the unique bounded solution of*

$$\mathrm{MA}_\omega(u_\lambda) = \mathrm{e}^{\lambda u_\lambda} f \, \mathrm{d}V, \qquad a_\lambda = u_\lambda - \sup_X u_\lambda.$$

The family $\{a_\lambda : 0 < \lambda \leq 1\}$ is relatively compact in $C(X)$. Every cluster limit as $\lambda \downarrow 0$ is a normalized bounded solution of $\mathrm{MA}_\omega(a) = c f \, \mathrm{d}V$.

Suppose a group G of biholomorphisms of X preserves ω and the measure $f \, \mathrm{d}V$. Then every a_λ and every such cluster limit is G-invariant. In particular, the zero-exponent equation has a G-invariant normalized solution.

Proof The barriers of Proposition 11.6.1 give a uniform oscillation bound for u_λ and uniform convergence $\mathrm{e}^{\lambda u_\lambda} \to c$. Thus the equation densities are uniformly bounded in

L^p for $0 < \lambda \leq 1$ and tend to $c f$ in L^p as $\lambda \downarrow 0$, and Theorem 11.3.2, together with the weak continuity of bounded Monge–Ampère measures under uniform convergence, gives the first paragraph, as noted after Proposition 11.6.1.

For the second, ω is big with $V_\omega = 0$, and $\rho = 0$ is a strict barrier for it as in Section 15.12, with empty pole set; the solutions with minimal singularities are then the bounded ones. Hence Corollary 15.16.1 applies to $\theta = \omega$ and gives the invariance of every a_λ and of every cluster limit. A cluster limit exists by the first paragraph. □

Corollary 15.19.1 *On $\mathbb{P}(\mathbb{C}^2 \oplus \mathbb{C}^2)$, with block homogeneous coordinates $[A : B]$, for the data in Theorem 12.6.1, the zero-exponent equation has at least three distinct normalized bounded solutions. Two are smooth, and a third is Hölder continuous and fixed by the involution $\iota[A : B] = [B : A]$.*

Proof Use the radial coordinate r, rank-one forms P, Q, R, Fubini–Study form ω_X, and scalar functions F, x, y, z from Sections 12.6.1 and 12.6.2. Set $w = F(r)$, $\omega_0 = xP + yQ + zR$, $H = \mathrm{dd^c} w/2$, $A_0 = \omega_0 - H$ and $B_0 = \omega_0 + H$ as in Section 12.6.3. The Hermitian form is $\omega = \omega_0 + \mathrm{dd^c}\rho$, with $\rho = \varepsilon(r^2 - 1)\kappa(r)$ from Section 12.6.4. The cutoff κ is even, so ρ is even, while $w = F(r)$ is odd. The involution sends r to $-r$, interchanges P, Q, and fixes R. The coefficient identities $x(-r) = y(r)$ and $z(-r) = z(r)$ show that $\iota^*\omega_0 = \omega_0$. Thus $\iota^*\omega = \omega$, and $\iota^*\omega_X = \omega_X$.

Since $\iota^* H = -H$, the involution interchanges A_0 and B_0. Their cubes are equal, so $f\omega_X^3$ is invariant. It also interchanges $\widetilde{u} = -w/2 - \rho$ and $\widetilde{v} = w/2 - \rho$. These functions therefore have the same supremum, and the normalized solutions u, v are interchanged. They are distinct, so neither is invariant.

Apply Proposition 15.19.1 to the group generated by ι. Here $c = 1$, as the two constructed solutions already have that scalar. An invariant cluster solution a cannot equal either u or v. Its bounded equation with smooth density gives Hölder regularity by Theorem 15.4.1. This gives a third solution. Its regularity is determined in Sections 15.19.1 and 15.19.2: it is $C^{1,1}$ but not C^2. □

To identify the whole normalized limit for these symmetric data, we now use the radial form of the equation. It determines the invariant solution even among bounded potentials whose differentiability is not known in advance.

15.19.1 The invariant projective solution

We keep the symmetric projective data of Corollary 15.19.1. Thus $X = \mathbb{P}^3$ with block coordinates $[A : B]$ and Fubini–Study form ω_X; on $U = \{A \neq 0,\ B \neq 0\}$ we use the radial coordinate r, the rank-one forms P, Q, R and the coefficients x, y, z, p, z_0 of Sections 12.6.1 and 12.6.2. The function $\rho(r)$ is the even correction of Section 12.6.4, not a barrier, and $\omega = \omega_0 + \mathrm{dd^c}\rho$ is the resulting Hermitian form, with $\omega_0 = xP + yQ + zR$; the density f is that of Theorem 12.6.1. Let K be the group generated by the block unitary group $U(2) \times U(2)$ and the exchange $\iota[A : B] = [B : A]$. Both preserve ω and $f\omega_X^3$. Denote by u_* the normalized

ι-invariant solution of Proposition 12.6.1, given by the symmetric transition pair $(-t_*, t_*)$.

Theorem 15.19.1 *For the symmetric projective data, u_* is the unique normalized bounded K-invariant solution of*

$$\mathrm{MA}_\omega(u) = f\omega_X^3.$$

If u_λ is the unique bounded solution of

$$\mathrm{MA}_\omega(u_\lambda) = \mathrm{e}^{\lambda u_\lambda} f\omega_X^3, \qquad \lambda > 0,$$

then

$$u_\lambda - \sup_X u_\lambda \longrightarrow u_* \quad \textit{uniformly as } \lambda \downarrow 0.$$

The selected limit u_* is $C^{1,1}$.

Uniqueness holds only within the K-invariant class: by Proposition 12.6.1, the same equation has uncountably many normalized solutions. Since a cluster limit is a priori only bounded, the proof needs the radial form of the equation for bounded invariant solutions, which we derive first, for every exponent $\lambda \geq 0$. For a real scalar s, define the coefficient polynomial $A(r, s)$ and its primitive $\Phi(r, s)$ by

$$A(r, s) = (x(r) + s)(y(r) - s), \qquad \Phi(r, s) = \int_0^s A(r, t)\,\mathrm{d}t.$$

Partial derivatives of Φ with respect to r below keep s fixed.

Lemma 15.19.1 *Let $\lambda \geq 0$, and let u be a bounded block-unitary-invariant solution of $\mathrm{MA}_\omega(u) = \mathrm{e}^{\lambda u} f \omega_X^3$. On U write $u + \rho = G(r)$. Then G is C^1, and its derivative $h = G'$ is continuous, has locally bounded variation, and satisfies*

$$-x(r) \leq h(r) \leq y(r), \qquad \mathrm{d}h + z\,\mathrm{d}r \geq 0.$$

With

$$k(r) = \begin{cases} 0, & |r| \leq 1, \\ x(r)y(r)z(r), & |r| > 1, \end{cases} \qquad k_\lambda(r) = \mathrm{e}^{\lambda(G(r)-\rho(r))}\, k(r),$$

h satisfies the distributional identity on $\mathbb{R}$

$$\mathrm{d}\big(\Phi(r, h(r))\big) = \big(k_\lambda(r) - A(r, h(r))z(r) + \partial_r \Phi(r, h(r))\big)\,\mathrm{d}r. \tag{15.53}$$

On every open interval where $A(r, h(r)) > 0$, h is smooth and

$$h' = \frac{k_\lambda(r)}{A(r, h)} - z.$$

Proof *Reduction to one variable.* Block-unitary orbits in U are exactly the level sets of r. Indeed, once the homogeneous coordinates are normalized so that the squared norms of the two blocks add up to one, the value of r determines both norms, and $U(2) \times U(2)$ acts transitively on pairs of vectors with prescribed nonzero norms. Thus invariance gives a function G of one real variable.

On a compact r-interval, the radial direction annihilates P, Q, while R is strictly positive in that direction. Testing positivity of $\omega_0 + \mathrm{dd}^{\mathrm{c}} G(r)$ in this direction gives $G'' + z \geq 0$ as a one-dimensional distribution. One may obtain this first by smooth approximation of the scalar distribution and then by testing with nonnegative functions. Adding a twice integrated primitive of z makes G convex. A finite bounded convex function on an interval is locally Lipschitz, and its derivative is locally of bounded variation. The quasi-plurisubharmonic representative is the corresponding continuous representative. The horizontal directions give $x + h \geq 0$, $y - h \geq 0$ almost everywhere. Continuity of x, y and the one-sided limits of a function of bounded variation give these bounds for both one-sided representatives. The radial inequality is precisely $\mathrm{d}h + z\,\mathrm{d}r \geq 0$; in particular jumps of h can only be upward.

The radial form of the measure. We next justify the formula for the measure, including possible jumps. Identify U smoothly with $\mathbb{R} \times Y$, where $Y = \{r = 0\}$, by real scaling of one block. The forms P, Q are independent of r in this identification. So is $\mathrm{d}^{\mathrm{c}} r$: in local coordinates it is the angular differential of the relative complex scale plus fixed base terms. Therefore

$$P \wedge Q \wedge R = \mathrm{d}r \wedge \nu,$$

where ν is a fixed smooth positive volume form on the compact orbit Y, with the orientation chosen by this equality.

For smooth G, the rank-one wedge formula gives the radial coefficient of its Monge–Ampère measure, after dividing by 6ν, as

$$A(r,h)(z+h')\,\mathrm{d}r = A(r,h)z\,\mathrm{d}r + \mathrm{d}\big(\Phi(r,h)\big) - \partial_r\Phi(r,h)\,\mathrm{d}r.$$

This identity passes to a locally Lipschitz G with the preceding lower Hessian bound. Indeed, convolve G on a slightly larger compact interval. The derivatives h_ϵ are uniformly bounded and converge to h almost everywhere and in every finite L^q space. Their second derivatives have a uniform lower bound. Consequently the functions $G_\epsilon(r)$ have a common quasi-plurisubharmonic lower Hessian bound on compact subsets of U, and converge uniformly to $G(r)$. A fixed smooth local correction and bounded Bedford–Taylor continuity give convergence of their signed products. On the right-hand side of the formula, the two ordinary terms converge in L^1, and $\Phi(r,h_\epsilon)\to\Phi(r,h)$ in L^1, giving distributional convergence of its derivative. This proves the same formula for the bounded product of G.

Continuity of h. After the same division by 6ν, the right-hand side $\mathrm{e}^{\lambda u} f\,\omega_X^3$ has radial coefficient $k_\lambda(r)\,\mathrm{d}r$, and k_λ is continuous because G is. This gives (15.53), initially for the almost-everywhere representative of h. If h jumped at r_0, the atom of its left side would be

$$\Phi(r_0,h(r_0+)) - \Phi(r_0,h(r_0-)) = \int_{h(r_0-)}^{h(r_0+)} (x(r_0)+s)(y(r_0)-s)\,\mathrm{d}s.$$

The right side has no atom. An upward jump of positive size lies in $[-x(r_0),y(r_0)]$, where the integrand is strictly positive in the interior. Since $x+y>0$ everywhere, such an integral is strictly positive. Thus there are no jumps. A locally bounded-variation function whose jumps all vanish has a continuous representative here: its distributional derivative consists of an absolutely continuous measure and a nonnegative singular measure, so its one-sided representatives coincide at every point. We henceforth use this representative. In particular G is C^1 and $G'=h$ everywhere.

All coefficients on the right of (15.53) are now continuous, so $\Phi(r,h(r))$ is C^1. Where $\partial_s\Phi = A > 0$, the implicit function theorem gives $h\in C^1$ and the stated ordinary differential equation. The function k is smooth, since the original density and its flat joins are smooth. Smoothness of h follows, since (G,h) solves the system $G'=h$, $h' = \mathrm{e}^{\lambda(G-\rho)}k/A(r,h) - z$ with smooth coefficients. □

Proof (of Theorem 15.19.1) We first show that a normalized bounded K-invariant solution u coincides with u_*, by computing its slope h on the band $|r|\le 1$ and outside it. Apply Lemma 15.19.1 with $\lambda=0$, so that $k_0=k$. Because ρ and u are invariant under block exchange, G is even, so h is odd and $h(0)=0$.

The slope on the band. On $0<r<1$ we have $x=p>0$ and $y=p+b\ge p>0$. Starting from zero, as long as $-p<h<y$, the equation is $h'=-z_0$. Thus $h(r) = -\int_0^r z_0(s)\,\mathrm{d}s$. This decreasing function cannot meet the positive upper boundary y. Its first meeting with the lower boundary $-p$ occurs at the unique $t_*\in(0,1)$ satisfying

$$\int_0^{t_*} z_0(s)\,\mathrm{d}s = p(t_*),$$

whose existence and uniqueness were proved in Proposition 12.6.1.

After that meeting, h must stay on $-p$ until $r = 1$. To prove it, suppose a connected component (a_0, b_0) of $\{h + p > 0\}$ lies to its right. Its left endpoint has $a_0 \geq t_*$ and $h(a_0) = -p(a_0) < 0$. Near that endpoint, $h < 0 < y$; hence on the component the interior equation applies for as long as $h < 0$. It forces $h' = -z_0 < 0$, so h cannot first reach zero. The equation therefore holds throughout the component. But then

$$(h + p)' = p' - z_0 < 0,$$

contradicting positivity of $h + p$ immediately to the right of its zero at a_0. Thus $h = -p$ on $[t_*, 1]$. Oddness determines the entire negative half of the band. The slope on $[-1, 1]$ is exactly that of the symmetric explicit solution.

The slope outside the band. Here boundedness of u at the projective ends matters. On $r > 1$, $k = xyz > 0$. Neither horizontal coefficient can vanish at a finite point of this interval. For the lower one define

$$\Psi(r) = \int_{-x(r)}^{h(r)} A(r, s)\,\mathrm{d}s \geq 0.$$

Equation (15.53) and the smooth lower endpoint show that Ψ is C^1. If $h(r_0) = -x(r_0)$, its derivative at that point is $k(r_0) > 0$: the Az term vanishes, and the two partial-r terms agree at the common endpoint. This contradicts that the nonnegative C^1 function Ψ has an interior minimum zero there. The upper-endpoint integral $\int_{h(r)}^{y(r)} A(r, s)\,\mathrm{d}s$ gives the same contradiction, with derivative $-k(r_0)$, when $h = y$. Thus $-x < h < y$ everywhere on $r > 1$, and the smooth ordinary differential equation of the lemma applies.

For $r \geq 2$ the background is Fubini–Study:

$$x = \sigma, \qquad y = 1 - \sigma, \qquad z = \sigma', \qquad \sigma = \frac{\mathrm{e}^r}{1 + \mathrm{e}^r}.$$

Set $a = \sigma + h \in (0, 1)$. Its equation is

$$a(1 - a)a' = \sigma(1 - \sigma)\sigma' > 0.$$

Hence a is increasing and has a limit in $(0, 1]$ at $+\infty$. If that limit were less than one, $h = a - \sigma$ would approach a negative constant and its primitive G would be unbounded below, contrary to boundedness of u and ρ. The limit is therefore one. Integrating the equation gives

$$\frac{a^2}{2} - \frac{a^3}{3} = \frac{\sigma^2}{2} - \frac{\sigma^3}{3} + C.$$

Taking the limit at infinity gives $C = 0$. The scalar function $s^2/2 - s^3/3$ is strictly increasing on $(0, 1)$, so $a = \sigma$ and $h = 0$ for $r \geq 2$. Ordinary uniqueness propagates this zero solution throughout $r > 1$, since the coefficients are smooth and the two horizontal coefficients stay positive there. Oddness gives $h = 0$ on $r < -1$ as well.

We have proved that G' agrees everywhere with the symmetric explicit slope. Thus u differs from u_* by a constant on U. The two quasi-plurisubharmonic functions then agree up to that constant on X by their canonical representatives. Their supremum normalization removes the constant. This proves uniqueness within the bounded K-invariant class.

Convergence. Finally Proposition 15.19.1 gives uniform precompactness of the normalized positive-exponent family. Every cluster limit solves the zero-exponent equation and is K-invariant, because the data are K-invariant. The scalar is one for these data. The uniqueness just proved identifies every cluster limit with u_*. If convergence of the whole family failed, a sequence staying a fixed positive uniform distance from u_* would have a convergent subsequence, contradicting that identification. This proves uniform convergence. Finally, u_* is $C^{1,1}$ by Proposition 12.6.1. □

15.19.2 Second derivatives at positive exponent

On the band where the density vanishes, the exponential factor plays no role, and the radial equation determines the slope of the solution independently of the exponent. Consequently the positive-exponent solutions have the same failure of second differentiability as the selected limit u_*. We keep the notation A, Φ, k_λ and the transition radius t_* of Section 15.19.1.

Proposition 15.19.2 *For the symmetric projective data of Theorem 15.19.1, let u_λ be the bounded solution of*

$$\mathrm{MA}_\omega(u_\lambda) = \mathrm{e}^{\lambda u_\lambda} f \omega_X^3, \qquad \lambda > 0.$$

On the open band $\{-1 < r < 1\}$, $u_\lambda - u_*$ is constant. Both potentials are $C^{1,1}$ there and fail to be C^2 at every point of the two hypersurfaces $r = \pm t_*$. In particular, none of these positive-exponent solutions is a globally C^2 potential.

Proof By Proposition 15.19.1, u_λ is K-invariant, so Lemma 15.19.1 applies to it; write $u_\lambda + \rho = G_\lambda(r)$ on U and $h_\lambda = G'_\lambda$. On $-1 < r < 1$ the coefficient k_λ is identically zero, and exchange invariance makes h_λ odd. On the positive half-band, the first-contact and no-departure arguments in the proof of Theorem 15.19.1, which use only these two facts and the lemma, apply verbatim: the first branch solves $h' = -z_0$, reaches $-p$ at t_*, and cannot leave that lower boundary to its right because $(h + p)' = p' - z_0 < 0$ on every proposed departure interval. Therefore

$$h_\lambda(r) = \begin{cases} p(r), & -1 < r \le -t_*, \\ -\displaystyle\int_0^r z_0(s)\,\mathrm{d}s, & -t_* \le r \le t_*, \\ -p(r), & t_* \le r < 1. \end{cases}$$

This is the derivative of $u_* + \rho$. Its primitive differs from G_λ by one constant on the connected band. The claimed local $C^{1,1}$ regularity follows from the explicit slope.

At $t_* \in (0,1)$ the two one-sided derivatives of this slope are

$$h'_\lambda(t_*-) = -z_0(t_*) < 0, \qquad h'_\lambda(t_*+) = -p'(t_*) > 0.$$

These unequal derivatives prevent the scalar primitive from having a second derivative at t_*. Through every point of that level set, the submersion r provides a smooth curve parametrized by r, transferring the failure to u_λ. Smooth ρ cannot cancel the jump. Oddness gives the conclusion at $-t_*$, and the identical slope proves it for u_*. □

The zero set of f contains the whole band. This example therefore does not concern the smooth strictly positive densities in the Hermitian Calabi–Yau theorem. It shows that smoothness of a nonnegative density alone does not give second-order regularity, even when the equation includes a strictly increasing exponential factor.

15.20 Local density and boundary control

Theorem 15.17.1 requires a smooth positive density on the whole regular locus of a global strict barrier, and the global minimum of the difference between the solution and the barrier drives its proof. If the density is smooth and positive only on a smaller open set, smoothness can fail there. A bounded solution with minimal singularities can be discontinuous on a ball where its density is constant (Theorem 15.20.2), and a locally Lipschitz plurisubharmonic function with constant density need not be C^1 (Example 15.20.2). Boundary data give a different route to regularity: if the solution has smooth boundary values on small balls, uniqueness for the bounded Dirichlet problem identifies it with the smooth Dirichlet solution.

In this section θ is a smooth real $(1,1)$-form on X, not necessarily closed, and $\Omega \subset X$ is an open set.

15.20.1 Smooth boundary traces

Let $B \Subset \Omega$ be a coordinate ball. A function u on B has *boundary trace* g, a function on ∂B, if $u(z) \to g(\xi)$ as $B \ni z \to \xi$, for every $\xi \in \partial B$.

Theorem 15.20.1 *Suppose that $u \in \mathrm{PSH}(X,\theta)$ is locally bounded on Ω and*

$$\mathrm{MA}_\theta(u) = f\,\mathrm{d}V \quad \text{on } \Omega, \qquad f \in C^\infty(\Omega), \quad f > 0.$$

Assume that Ω is covered by coordinate Euclidean balls $B_j \Subset \Omega$ such that u has a smooth boundary trace on ∂B_j. Then $u \in C^\infty(\Omega)$ and $\theta + \mathrm{dd^c}u > 0$ there.

Proof Fix one of the balls B, and let g be the boundary trace of u on ∂B. Lemma 15.6.1, applied with the background θ, the density $f\,\mathrm{d}V$ and the boundary value g, gives $V \in C^\infty(\overline{B})$ with

$$\theta + \mathrm{dd^c}V > 0, \qquad (\theta + \mathrm{dd^c}V)^n = f\,\mathrm{d}V \text{ on } B, \qquad V|_{\partial B} = g.$$

The functions u and V are bounded and θ-plurisubharmonic on B, they have the same Monge–Ampère measure, and $u - V$ tends to zero at every boundary point. The local comparison principle of Lemma 15.12.1 gives $u \geq V$ and $V \geq u$. Therefore $u = V$ is smooth on B, and $\theta + \mathrm{dd^c}u > 0$ there. Since the balls cover Ω, the theorem follows.□

If some $\rho \in \mathrm{PSH}(X,\theta)$, for instance a strict barrier, is locally bounded on Ω, then every $u \in \mathrm{PSH}(X,\theta)$ with minimal ordinary singularities is locally bounded on Ω, since $\rho - \sup_X \rho \leq V_\theta \leq u + C$.

Example 15.20.1 For every $n \geq 1$, the trace criterion applies to a bounded continuous solution whose global density belongs to L^1 but to no L^p with $p > 1$, and which is not Hölder continuous at the points of $X \setminus \Omega$.

Let $E = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$, $X = E^n$, and

$$\omega_j = \frac{\mathrm{i}}{2\pi}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j, \qquad \theta = \sum_{j=1}^{n} \omega_j, \qquad D = \{0\} \times E^{n-1}.$$

Use the volume form θ^n in place of $\mathrm{d}V$. Choose a nonnegative smooth cutoff χ supported in a coordinate disk $|z| < r_0 < 1$ of E and equal to one near zero. For $\beta > 0$, define

$$\psi(z) = \chi(z)(-\log|z|^2)^{-\beta} \quad (z \neq 0), \qquad \psi(0) = 0,$$

and extend by zero. On the smaller punctured disk put $t = |z|^2$ and $L = -\log t$. Differentiation gives

$$\psi_{z\bar{z}} = \frac{\beta(\beta+1)}{tL^{\beta+2}}, \qquad t\frac{\mathrm{d}}{\mathrm{d}t}L^{-\beta} = \beta L^{-\beta-1} \longrightarrow 0. \tag{15.54}$$

The first expression is integrable and the second is the radial flux. Integration by parts on punctured disks, using also $\psi \to 0$, shows that the distributional Hessian

has no atom at zero. On the cutoff annulus its negative part is bounded. Hence, for a sufficiently small fixed $\eta > 0$, $u = \eta\psi(z_1)$ satisfies $\theta + \mathrm{dd}^{\mathrm{c}} u \geq \theta/2$.

Only one coordinate contributes to $\mathrm{dd}^{\mathrm{c}} u$. Locally approximate the bounded subharmonic potential in that coordinate from above and use Theorem 1.6.1; the smooth diagonal determinant identity passes to the limit and yields

$$\mathrm{MA}_\theta(u) = f\theta^n, \qquad f = 1 + \eta\psi_{z\bar{z}}(z_1).$$

Give f any positive value on the null set D. It is smooth and positive on $\Omega = X \setminus D$, where u is smooth and has smooth traces on all relatively compact coordinate balls. Near D its singular term is comparable to $|z_1|^{-2}(-\log|z_1|^2)^{-\beta-2}$. Its radial L^1 integral converges, whereas

$$\int_0^{r_0} s^{1-2p}(-\log s^2)^{-p(\beta+2)}\,\mathrm{d}s = +\infty \qquad (p > 1).$$

Also $(-\log|z|^2)^{-\beta}/|z|^\alpha \to +\infty$ for every $\alpha > 0$, so u is not Hölder at D.

15.20.2 A bounded discontinuous potential

We construct a bounded plurisubharmonic function on a bidisk whose Monge–Ampère measure is a constant multiple of Lebesgue measure, but which is discontinuous. The construction places logarithmic poles with summable weights at points accumulating at the origin; the resulting weight b vanishes at the poles but not at their limit. It is the local model for Theorem 15.20.2.

For a smooth function V on an open subset of $\mathbb{C}^n$ and the Euclidean volume $\mathrm{d}\Lambda$,

$$(\mathrm{dd}^{\mathrm{c}} V)^n = \frac{n!}{\pi^n}\det(V_{j\bar{k}})\,\mathrm{d}\Lambda. \tag{15.55}$$

Let $U = W = \{|z| < 1/4\} \subset \mathbb{C}$, with coordinates z and w on the two factors. Write $\mathrm{d}A$ for planar Lebesgue measure and set

$$a_k = \alpha_k = 2^{-k-4}, \qquad A = \sum_{k\geq 1}\alpha_k = \frac{1}{16}, \qquad S = \{0\} \cup \{a_k : k \geq 1\}.$$

Define on U

$$h(z) = \sum_{k\geq 1}\alpha_k \log|z - a_k|^2, \qquad b(z) = \mathrm{e}^{h(z)}, \tag{15.56}$$

with $b = 0$ where $h = -\infty$.

Lemma 15.20.1 *The function*

$$a(z) = \frac{1}{\pi} \int_U \log|z-\zeta|^2 \, b(\zeta)^{-1} \, \mathrm{d}A(\zeta)$$

is bounded and continuous on U. The bounded function

$$P(z,w) = a(z) + b(z)|w|^2$$

is plurisubharmonic on $U \times W$, is smooth off $S \times W$, and satisfies

$$(\mathrm{dd}^{\mathrm{c}} P)^2 = \frac{2}{\pi^2} \, \mathrm{d}A(z) \, \mathrm{d}A(w). \tag{15.57}$$

It is discontinuous at every $(0, w_0)$ with $w_0 \neq 0$.

Proof The summands defining h are negative on U, so their partial sums decrease. Since $h(0) = 2\sum_k \alpha_k \log a_k > -\infty$, the limit is subharmonic. On compact subsets of $U \setminus S$, the series and its derivatives converge uniformly, making h harmonic there. Moreover $0 \leq b \leq 1$, $b(a_k) = 0$, and $b(0) > 0$.

Weighted arithmetic–geometric mean gives, almost everywhere,

$$b(z)^{-2} \leq \sum_{k \geq 1} \frac{\alpha_k}{A} |z - a_k|^{-4A}. \tag{15.58}$$

For the countable inequality, first use the first N weights and assign the remaining weight to the factor 1; then let N increase. Since $4A = 1/4 < 2$, the integrals of the terms $|z - a_k|^{-4A}$ on U are bounded uniformly in k. Thus $b^{-1} \in L^2(U)$.

The logarithmic kernel has uniformly bounded L^2 norm on the translates of U that occur in the definition of a. Cauchy–Schwarz bounds a uniformly. Choose a compactly supported $L^2(\mathbb{C})$ kernel agreeing with $\log|z|^2$ on $U - U$. Translation continuity in L^2, paired with $b^{-1}\mathbf{1}_U \in L^2$, proves continuity of a. The distribution identity $\partial_z \partial_{\bar{z}} \log|z|^2 = \pi\delta_0$ gives $a_{z\bar{z}} = b^{-1}$. In particular a is subharmonic and, by local elliptic regularity, smooth off S.

The second summand of P is $\exp(h(z) + \log|w|^2)$, a convex increasing composition of a plurisubharmonic function, extended by zero at its poles. Thus P is bounded and plurisubharmonic. At smooth points the harmonicity of $h = \log b$ gives $b b_{z\bar{z}} = |b_z|^2$, and hence

$$\det(P_{j\bar{k}}) = b a_{z\bar{z}} + |w|^2 \big(b b_{z\bar{z}} - |b_z|^2 \big) = 1.$$

The set $S \times W$ is contained in the pole set of $h(z) + \log|z|$, a plurisubharmonic function not identically $-\infty$. It is also Lebesgue-null. Pluripolar nullity in Theorem 1.6.1 and (15.55) prove (15.57) throughout the bidisk. Finally,

$$P(a_k, w_0) = a(a_k) \longrightarrow a(0) < a(0) + b(0)|w_0|^2 = P(0, w_0)$$

when $w_0 \neq 0$. □

15.20.3 Compact gluing

We now glue the local example into a compact torus by a regularized maximum with a quadratic function. The result is a bounded solution with minimal singularities whose density is constant on a ball, where the solution is discontinuous; in the transition region the density acquires singularities that destroy every L^p bound with $p > 1$.

We work on the flat complex torus $X = (\mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z}))^2$, identifying a small coordinate ball with a ball in $U \times W$. For a smooth function v on the ball write $H_v = (v_{j\bar{k}})$ for its complex Hessian and I for the two-dimensional identity matrix.

Theorem 15.20.2 *There are a positive flat Kähler form θ on X, a bounded $u \in \mathrm{PSH}(X,\theta)$ with minimal ordinary singularities, and a density $F > 0$ almost everywhere such that*

$$\mathrm{MA}_\theta(u) = F\theta^2, \qquad F \in L^1(X,\theta^2), \qquad F \notin L^p(X,\theta^2) \quad (p > 1).$$

On an open coordinate ball Ω, F is a positive constant and u is discontinuous. The constant function $\rho = 0$ is a smooth strict barrier on X.

Proof Set $R = 1/8$ and choose M so that $P_0 = P + M$ satisfies $0 \le P_0 \le L$ on $B_R \Subset U \times W$ for some finite L. Choose $C > 0$ with

$$\frac{3CR^2}{16} \ge 2, \qquad \frac{5CR^2}{16} > L + 2, \tag{15.59}$$

and put $Q(Z) = C(|Z|^2 - R^2/4)$ for $Z = (z,w)$. Let χ be the convolution of $|t|$ with an even nonnegative smooth probability kernel supported in $(-1,1)$. Then χ is smooth and convex, $|\chi'| \le 1$, $\chi'(0) = 0$, and $\chi(t) = |t|$ for $|t| \ge 1$. The function

$$m(s,t) = \frac{s + t + \chi(s-t)}{2}, \qquad v = m(P_0, Q)$$

is bounded and plurisubharmonic, because m is convex and nondecreasing in each variable. The collar inequalities imply

$$v = P_0 \quad \text{on } B_{R/4}, \qquad v = Q \quad \text{when } 3R/4 \le |Z| < R.$$

Indeed, $Q \le -2 \le P_0 - 2$ on the first ball and $Q > L + 2 \ge P_0 + 2$ on the collar. Define

$$\theta = \frac{\mathrm{i}C}{2\pi}(\mathrm{d}z \wedge \mathrm{d}\bar{z} + \mathrm{d}w \wedge \mathrm{d}\bar{w}), \qquad u = \begin{cases} v - Q & \text{on } B_R, \\ 0 & \text{outside } B_R. \end{cases}$$

The collar identity makes u identically zero near ∂B_R and gives a global bounded potential with $\theta + \mathrm{dd}^{\mathrm{c}} u = \mathrm{dd}^{\mathrm{c}} v$ on the ball and $\theta + \mathrm{dd}^{\mathrm{c}} u = \theta$ outside. Thus $u \in \mathrm{PSH}(X,\theta)$, and, being bounded, u has minimal ordinary singularities.

Outside the pluripolar set $(S \times W) \cap B_R$, set $\lambda = (1 + \chi'(P_0 - Q))/2$. Differentiating the regularized maximum gives

$$H_v = \lambda H_P + (1-\lambda)CI + \frac{\chi''(P_0-Q)}{2}\partial(P_0-Q)\otimes\overline{\partial(P_0-Q)}. \tag{15.60}$$

Both H_P and CI are positive definite there; the final term is positive semidefinite. Hence the classical density is positive. Bounded products charge no pluripolar sets, so there is no further measure on the exceptional set. Their mass on compact X is finite by Proposition 1.7.1. This proves $\mathrm{MA}_\theta(u) = F\theta^2$ for an almost-everywhere positive $F \in L^1$. On $\Omega = B_{R/4}$, (15.57) gives the representative $F = C^{-2}$. There $u = P + M - Q$ is discontinuous at $(0, w_0)$ for $0 < |w_0| < R/4$.

We prove failure of every global L^p directly in the transition region. For large k, set

$$m_k = a(a_k)+M \in [0,L], \qquad w_k = \left(R^2/4+m_k/C-|a_k|^2\right)^{1/2}, \qquad Z_k = (a_k, w_k).$$

Then $w_k > 0$, $P_0(Z_k) = Q(Z_k) = m_k$, and (15.59) gives $|Z_k|^2 < 9R^2/16$. Near the isolated pole a_k,

$$h(z) = \alpha_k \log|z-a_k|^2 + H_k(z),$$

where H_k is smooth and harmonic. Consequently b and P_0 are continuous near Z_k. There is a product neighborhood $V_k \Subset B_R$ of Z_k, with $|w|$ bounded below by a positive number, on which $1/4 \le \lambda \le 3/4$.

At smooth points of V_k, positivity and the two-dimensional determinant expansion give

$$\begin{aligned}\det H_v &\ge \det\bigl(\lambda H_P + (1-\lambda)CI\bigr)\\ &\ge \lambda(1-\lambda)C\,\mathrm{tr}\,H_P \ge c_k b_{z\bar z},\end{aligned}$$

where $c_k > 0$ may depend on k and C. Here $P_{z\bar z} = b^{-1} + |w|^2 b_{z\bar z}$. With $\zeta = z - a_k$, shrink the first factor of V_k so that

$$b_{z\bar z} = |\zeta|^{2\alpha_k} \mathrm{e}^{H_k}\left|\frac{\alpha_k}{\zeta} + \partial_z H_k\right|^2 \ge c_k'|\zeta|^{2\alpha_k-2}$$

for some $c_k' > 0$. Since $F = C^{-2}\det H_v$, this lower bound implies $F \notin L^p$ whenever $p(1-\alpha_k) \ge 1$: the w-disk has positive area and the radial integral in z contains

$$\int_0^{\delta_k} r^{1-2p(1-\alpha_k)}\,\mathrm{d}r = +\infty.$$

For each $p > 1$ such a k exists because $\alpha_k \downarrow 0$. □

The bounded-product capacity estimate in Section 1.6 also holds for the discontinuous potential P. Thus a first-order capacity bound, even together with a constant positive density, cannot supply the missing boundary control.

Example 15.20.2 On the bidisk in $\mathbb{C}^2$ with coordinates (z, w), the function

$$H(z,w) = 2|z|(1+|w|^2)$$

is locally Lipschitz, plurisubharmonic and not C^1 along $\{z=0\}$. Plurisubharmonicity follows by writing its two summands as convex exponential compositions of $\log|z|$ and $\log|z| + \log|w|^2$. At $z \neq 0$,

$$H_{z\bar z} = \frac{1+|w|^2}{2|z|}, \qquad H_{w\bar w} = 2|z|, \qquad |H_{z\bar w}|^2 = |w|^2.$$

Hence $\det(H_{j\bar k}) = 1$, and pluripolar nullity gives $(\mathrm{dd^c} H)^2 = 2\pi^{-2}\,\mathrm{d}\Lambda$ on the entire bidisk.

The last example rules out a purely local implication from Hölder continuity and a smooth positive density to smoothness; being local, it says nothing about the global hypotheses of Theorem 15.17.1. Nor does Theorem 15.20.2 contradict that theorem: there the solution is not locally Hölder continuous on the ball Ω, and its density is not smooth on the rest of X, which is the regular locus of the barrier $\rho = 0$. The trace criterion of Theorem 15.20.1 avoids both global hypotheses by identifying the bounded solution with a smooth Dirichlet solution on each ball.

15.21 Equations with resolved logarithmic poles

We now prescribe a singularity type with logarithmic poles. If the poles become a divisor on a modification, the singular part can be removed there, and what remains is a smooth residual form. When this residual form is strictly positive, the bounded Hermitian theory on the modification provides the remaining bounded potential, and pushing forward solves the equation in the prescribed type.

In this section the background is a Hermitian form h on X. Let ψ be a quasi-plurisubharmonic function that is smooth outside a nonempty analytic set A, its pole set. Assume that there is a modification $\pi\colon Y \to X$, biholomorphic over $X \setminus A$, such that $\pi^{-1}(A)$ is the support $|D|$ of a simple normal crossing divisor $D = \sum_j D_j$ and

$$\psi\circ\pi = \sum_j c_j \log|s_j|^2 + b, \qquad \beta = \pi^*h + \mathrm{dd^c} b - \sum_j c_j \Theta_j > 0, \tag{15.61}$$

where $c_j > 0$, b is smooth on Y, s_j is a holomorphic section of $\mathcal{O}(D_j)$ defining D_j, and Θ_j is the curvature form of a smooth Hermitian metric on $\mathcal{O}(D_j)$, so that $\mathrm{dd^c}\log|s_j|^2 = [D_j] - \Theta_j$. Thus β is the smooth part of the current $\pi^*h + \mathrm{dd^c}(\psi\circ\pi)$, which we call the *residual form*.

Theorem 15.21.1 *For every nonzero nonnegative $f \in L^p(X, h^n)$, $p > 1$, there are $\kappa > 0$ and a normalized $u \in \mathrm{PSH}(X,h)$ such that*

$$\mathrm{MA}_h^{\mathrm{ext}}(u) = \kappa f h^n, \qquad u = \psi + O(1). \tag{15.62}$$

The lifted remainder $v = (u - \psi) \circ \pi$ extends Hölder continuously over Y, so u is locally Hölder on $X \setminus A$. Its current is

$$\pi^* h + \mathrm{dd}^{\mathrm{c}}(u \circ \pi) = \sum_j c_j [D_j] + \beta + \mathrm{dd}^{\mathrm{c}} v. \tag{15.63}$$

The positive residual current $\beta + \mathrm{dd}^{\mathrm{c}} v$ has zero Lelong numbers. If h is Kähler, then $\kappa = \int_Y \beta^n / \int_X f h^n$, and the normalized solution is unique among all normalized solutions u' of (15.62) with $u' = \psi + O(1)$. If the density $g = (f \circ \pi)\, \pi^* h^n / \beta^n$ is smooth and strictly positive on Y, the residual current $\beta + \mathrm{dd}^{\mathrm{c}} v$ may be chosen smooth and strictly positive.

Proof Define $J = \pi^* h^n / \beta^n$ and $g = (f \circ \pi) J$ off the exceptional set, assigning any value to g on that volume-zero set. The smooth nonnegative J is bounded. Change of variables on the biholomorphic locus gives

$$\int_Y g \beta^n = \int_X f h^n, \qquad \int_Y g^p \beta^n \le \|J\|_\infty^{p-1} \int_X f^p h^n. \tag{15.64}$$

Thus g is a nonzero L^p density. Theorems 11.1.1 and 15.4.1 give a bounded Hölder β-plurisubharmonic v and $\kappa > 0$ with $(\beta + \mathrm{dd}^{\mathrm{c}} v)^n = \kappa g \beta^n$. If h is Kähler, then β is Kähler and the mass identity fixes the stated scalar. For smooth positive g, Theorem 10.1.1 gives a smooth solution instead.

Set $u = \psi + v \circ \pi^{-1}$ on $X \setminus A$ and $u = -\infty$ on A. Push forward the positive current $\sum c_j [D_j] + \beta + \mathrm{dd}^{\mathrm{c}} v$. Pushforward commutes with dd^{c}, $\pi_* \pi^* h = h$, and the pushed locally integrable potential is u almost everywhere. Its quasi-plurisubharmonic representative equals $\psi + v \circ \pi^{-1}$ off A and tends to $-\infty$ at A because v is bounded. Hence $h + \mathrm{dd}^{\mathrm{c}} u \ge 0$ and $u = \psi + O(1)$ in the two-sided ordinary sense. The pulled-back identity is (15.63); bounded residuals have zero Lelong number. The equation holds off A because $g \beta^n = (f \circ \pi)\, \pi^* h^n$ off the exceptional set, and Theorem 7.1.1 extends it as (15.62). Subtracting $\sup_X u$ changes neither the equation nor the ordinary type.

For uniqueness when h is Kähler, let u' be a normalized solution with $u' = \psi + O(1)$. On $Y \setminus D$, $w = (u' - \psi) \circ \pi$ is bounded and β-plurisubharmonic. Since w is bounded, Lemma 1.3.7 extends it across D to a bounded β-plurisubharmonic function on Y. On each relatively compact subset of $Y \setminus D$, sufficiently deep truncations of u' equal u'. Thus locality in Theorem 7.1.1 gives the bounded equation $\kappa g \beta^n$ there; both sides give D zero mass. The global equation follows, its mass fixes κ, and closed full-mass uniqueness gives $w = v$+constant as recalled in Section 36.21. Normalization fixes that constant. □

The Hölder constants depend on the fixed geometry and divisor metrics, the c_j, a lower bound for the positivity of β, p, and the L^p norm of the density κg itself. In particular, the argument gives no estimate that is uniform as β degenerates or as κ becomes unbounded. In simple normal crossing coordinates a density comparable to $\prod_j |z_j|^{-2a_j}$ belongs locally to L^p exactly when

$$pa_j < 1 \quad \text{for every } j, \tag{15.65}$$

by the factors $\int_0^\epsilon r_j^{1-2pa_j}\,\mathrm{d}r_j$. A smooth positive density f on X need not give a strictly positive g on Y, since the Jacobian factor $J = \pi^* h^n/\beta^n$ of the proof can vanish on the exceptional set.

Example 15.21.1 Take $X = \mathbb{P}^n$ with $n \geq 2$, $h = \lambda\omega_{\mathrm{FS}}$, $\int_X \omega_{\mathrm{FS}}^n = 1$, and $0 < c < \lambda$. Put

$$\psi_c = c \log \frac{\sum_{j=1}^n |Z_j|^2}{\sum_{j=0}^n |Z_j|^2}.$$

Its pole is $P = [1 : 0 : \cdots : 0]$. On the point blowup $\pi\colon Y \to X$, let $q\colon Y \to \mathbb{P}^{n-1}$ be the direction map. In affine blowup coordinates $x = (z, zw)$, Poincaré–Lelong gives

$$\pi^* h + \mathrm{dd}^{\mathrm{c}}(\psi_c \circ \pi) = c[E] + (\lambda - c)\pi^*\omega_{\mathrm{FS}} + cq^*\omega_{\mathrm{FS}}.$$

The residual form $\beta = (\lambda - c)\pi^*\omega_{\mathrm{FS}} + cq^*\omega_{\mathrm{FS}}$ is positive: $\pi^*\omega_{\mathrm{FS}}$ is positive in the directions normal to E, and $q^*\omega_{\mathrm{FS}}$ in the directions tangent to E. Write $H = \pi^*[\omega_{\mathrm{FS}}]$. Since $H|_E = 0$ and $E^n = (-1)^{n-1}$, its mass is

$$\int_Y \bigl((\lambda - c)\pi^*\omega_{\mathrm{FS}} + cq^*\omega_{\mathrm{FS}}\bigr)^n = \lambda^n - c^n. \tag{15.66}$$

Hence, for every nonzero nonnegative $f \in L^p$ with $\int_X f h^n = \lambda^n - c^n$, Theorem 15.21.1 gives $\kappa = 1$ and a solution of $\mathrm{MA}_h^{\mathrm{ext}}(u) = f h^n$ with the nonminimal type $u = \psi_c + O(1)$.

For an explicit solution, choose a smooth function v on Y with $v|_E$ nonconstant, small enough that $S = \beta + \mathrm{dd}^{\mathrm{c}} v$ is still positive, and put $u = \psi_c + v \circ \pi^{-1}$ off P. The density f of its measure, given by $f \circ \pi = S^n/\pi^* h^n$ off E, lies in every L^p with $1 < p < n/(n-1)$: $\pi^* h^n$ vanishes to order $|z|^{2(n-1)}$ along E, so the local integral of f^p is finite precisely when $(n-1)(p-1) < 1$. Different directions of approach to P give different limits of $u - \psi_c$. The divisor $c[E]$ of the current upstairs leaves no divisor downstairs, since E is contracted to a point. On $X \setminus \{P\}$, where u is smooth and strictly h-plurisubharmonic, the Hermitian metric $h + \mathrm{dd}^{\mathrm{c}} u$ is isometric to $(Y \setminus E, S)$. Its metric completion is Y with the metric S, so the whole divisor E is restored: removing the smooth real codimension-two submanifold E does not change the infimum of path lengths in Y. Finally, if h is replaced by a sufficiently small nonclosed perturbation, both h and the residual form remain positive, so the existence part of Theorem 15.21.1 still applies.

15.21.1 Residual mass

When h is not Kähler, the scalar in Theorem 15.21.1 is still determined by masses if h satisfies the power condition $\mathrm{dd}^{\mathrm{c}}(h^k) = 0$, $1 \leq k \leq n-1$, of Theorem 2.2.2, which

fixes the mass of every bounded solution. For $n = 2$ the condition is simply $\mathrm{dd}^{\mathrm{c}} h = 0$, so it includes the nonclosed Gauduchon metric $\alpha + \zeta$ on the Hopf surface X_2 of Section 21.6. The residual form then has strictly smaller mass than h, the difference being carried by the divisor.

Proposition 15.21.1 *Under* (15.61)*, assume* $\mathrm{dd}^{\mathrm{c}}(h^k) = 0$ *for* $1 \le k \le n-1$*, and put* $H = \pi^* h$*. Then*

$$\int_X h^n - \int_Y \beta^n = \sum_j c_j \sum_{i=0}^{n-1} \int_{D_j} H^i \wedge \beta^{n-1-i} > 0. \tag{15.67}$$

Thus the constructed solution of Theorem 15.21.1 has $\kappa = \int_Y \beta^n / \int_X f h^n$*. If* $\int_X f h^n = \int_X h^n$*, its scalar is strictly less than one.*

Proof Let $C = \sum_j c_j \Theta_j - \mathrm{dd}^{\mathrm{c}} b$, so $\beta = H - C$ with C closed. The mixed forms $Q_i = H^i \wedge \beta^{n-1-i}$ satisfy $\mathrm{dd}^{\mathrm{c}} Q_i = 0$: expand $\beta = H - C$ and apply the power condition to each H^k. With $\Psi = \psi \circ \pi$, $\mathrm{dd}^{\mathrm{c}} \Psi = \sum_j c_j [D_j] - C$. Distributional integration by parts against the smooth dd^{c}-closed Q_i gives

$$\int_Y (H^n - \beta^n) = \sum_i \int_Y C \wedge Q_i = \sum_j c_j \sum_i \int_{D_j} Q_i .$$

Change of variables gives $\int_Y H^n = \int_X h^n$. Every integrand on the right is nonnegative, since H is semipositive and β positive; the $i = 0$ term on each nonempty D_j is strictly positive. The same expansion of $\beta = H - C$ gives $\mathrm{dd}^{\mathrm{c}}(\beta^k) = 0$ for $1 \le k \le n-1$, so by Theorem 2.2.2 the residual solution has mass $\int_Y \beta^n$, giving the scalar. □

15.22 Minimal singularities on blowups

On a blowup, the background itself can force an exceptional divisor into every positive current. The coefficient of this divisor is then part of the minimal ordinary type, while the rest of the potential descends to the target of the blowup, so the equation reduces to one on the target. The density changes by the Jacobian of the blowup, which determines the loss of integrability.

In this section the roles of the letters differ from those in Section 15.21: X is a blowup and Y its target. Let (Y, h) be a connected compact Hermitian n-fold, let $Z \subset Y$ be a connected smooth center of codimension $r \ge 2$, and write $\pi\colon X = \mathrm{Bl}_Z Y \to Y$ for the blowup, with exceptional divisor E. Let s_E be the canonical section of $\mathcal{O}(E)$, choose a smooth Hermitian metric on $\mathcal{O}(E)$ with curvature form Θ_E, so that $\mathrm{dd}^{\mathrm{c}} \log |s_E|^2 = [E] - \Theta_E$, put $L = \log |s_E|^2$, and fix $c > 0$. The smooth background

$$\theta = \pi^* h + c \Theta_E \tag{15.68}$$

can be indefinite, and it is nonclosed when h is nonclosed. Positivity is imposed on $\theta + \mathrm{dd}^{\mathrm{c}} u$, not on θ.

Proposition 15.22.1 *Every $u \in \mathrm{PSH}(X, \theta)$ has, up to an additive constant, a unique h-plurisubharmonic target potential q such that*

$$u = cL + q \circ \pi + C, \qquad \theta + \mathrm{dd}^{\mathrm{c}} u = \pi^*(h + \mathrm{dd}^{\mathrm{c}} q) + c[E]. \tag{15.69}$$

The potentials with minimal ordinary singularities are exactly those of the form $cL + O(1)$, and u is one of them if and only if q is bounded. There is a strict θ-plurisubharmonic barrier with logarithmic coefficient $c + \delta$ for some $\delta > 0$.

Proof Put $T = \theta + \mathrm{dd}^{\mathrm{c}} u$ and $q = \pi_*(u - cL)$ as a scalar distribution. The distribution q is locally integrable: off Z it is the function $(u - cL) \circ \pi^{-1}$, and a smooth target volume pulls back with bounded Jacobian, so its absolute integral is controlled by that of $u - cL$ on X. Since $\pi_*[E] = 0$,

$$S = \pi_* T = h + \mathrm{dd}^{\mathrm{c}} q \geq 0.$$

A smooth local strictly plurisubharmonic shift gives a quasi-plurisubharmonic representative of q. Its pullback $q \circ \pi$ is not identically $-\infty$ and satisfies $\pi^*(h + \mathrm{dd}^{\mathrm{c}} q) \geq 0$, by Lemma 1.22.1: since π is dominant, the composition of a local plurisubharmonic function with π is again plurisubharmonic and not identically $-\infty$.

Off E, $T = \pi^* S$. Hence the difference $R = T - \pi^* S$ is an order-zero current supported on E, and it is closed: $\mathrm{d}T$ and $\mathrm{d}(\pi^* S)$ both equal $\pi^* \mathrm{d}h$. In a chart where $E = \{z = 0\}$, order zero gives $zR = \bar{z}R = 0$. Closedness then gives $\mathrm{d}z \wedge R = \mathrm{d}\bar{z} \wedge R = 0$. Thus only its normal $(1,1)$ coefficient remains, a measure on E; the tangential equations in $\mathrm{d}R = 0$ make that measure a constant multiple of integration on E. Connectedness of E identifies one constant a globally, so $R = a[E]$. Since $R = c\Theta_E + \mathrm{dd}^{\mathrm{c}}(u - q \circ \pi)$, its cohomology class is $c[E]$. On a line in an exceptional fiber, $O(E)$ has degree -1, so $[E]$ is nonzero and $a = c$. The difference between the two scalar potentials in (15.69) has dd^{c} zero. Local elliptic regularity makes it pluriharmonic; compactness and connectedness make it constant.

Since $h > 0$, $q = 0$ is admissible, and every h-plurisubharmonic q is bounded above on compact Y. Hence cL is minimal and two-sided bounded comparison with it is exactly boundedness of q. For the barrier choose another divisor metric, with logarithmic norm $L' = L + \sigma$ and curvature $\Theta'_E = \Theta_E - \mathrm{dd}^{\mathrm{c}}\sigma$ for a smooth real function σ, so that $-\Theta'_E$ is positive on the vertical tangent bundle of $E \to Z$, using the tautological $O(-1)$. In adapted coordinates

$$\pi(y, z, w) = (y, z, zw), \tag{15.70}$$

$\pi^* h$ controls the nonvertical directions and $|z|^2 |\mathrm{d}w|^2$. For small $\delta > 0$, the vertical positivity of $-\delta\Theta'_E$ dominates its mixed errors by completing the square; away from E, compactness allows the same δ. Thus $\pi^* h - \delta\Theta'_E > 0$. For the fixed background θ in (15.68), take

$$\rho = (c + \delta)L' - c\sigma = (c + \delta)L + \delta\sigma.$$

Then $\theta + \mathrm{dd^c}\rho = (c+\delta)[E] + \pi^*h - \delta\Theta'_E$ dominates a positive Hermitian form. The smooth correction $\delta\sigma$ preserves the stated logarithmic coefficient. □

Theorem 15.22.1 *Let $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, be nonzero. There exist $\kappa > 0$ and a potential u with minimal ordinary singularities such that*

$$\mathrm{MA}^{\mathrm{ext}}_\theta(u) = \kappa f \,\mathrm{d}V, \qquad u = cL + q \circ \pi + C, \tag{15.71}$$

where q is Hölder on Y and u is normalized by $\sup_X u = 0$. The divisor coefficient is exactly c. In (15.70), $u - cL = q(y,0) + C + O(|z|^\alpha)$ on smaller charts for some $\alpha > 0$, so its limit is constant along each exceptional fiber.

If $\pi_*(f\,\mathrm{d}V) = Fh^n$, then $F \in L^s(Y, h^n)$ for every

$$1 < s < \frac{rp}{(r-1)p+1}. \tag{15.72}$$

For a point center and data bounded above and below by positive constants near E, the density F is L^s-integrable near Z exactly for $s < n/(n-1)$, and for finite p no strictly larger universal exponent than $np/((n-1)p+1)$ can be guaranteed. If h is Kähler, the class $[\theta]$ is big and non-nef, has volume $\int_Y h^n$, and $\kappa = \int_Y h^n / \int_X f\,\mathrm{d}V$; the normalized solution is unique among all normalized solutions of (15.71) with minimal ordinary singularities. Under $\mathrm{dd^c}(h^j) = 0$, $1 \le j \le n-1$, the scalar of the constructed solution has the same integral formula.

Proof *Integrability and existence.* In (15.70) the complex Jacobian is z^{r-1}. Put $J = \pi^*h^n/\mathrm{d}V$, so $J \asymp |z|^{2(r-1)}$ on nested charts and $F \circ \pi = f/J$ off E. For $1 < s < p$, change of variables and Hölder's inequality give

$$\int_Y F^s h^n = \int_X f^s J^{1-s}\,\mathrm{d}V \tag{15.73}$$

$$\le \|f\|^s_{L^p} \left(\int_X J^{-(s-1)p/(p-s)}\,\mathrm{d}V \right)^{(p-s)/p}. \tag{15.74}$$

The normal complex integral is finite precisely if $(r-1)(s-1)p/(p-s) < 1$, which is (15.72). Theorems 11.1.1 and 15.4.1 on (Y,h) supply q and κ with $(h + \mathrm{dd^c}q)^n = \kappa F h^n$. Use (15.69) off E and Theorem 7.1.1 across it. Hölder continuity gives $q(\pi(y,z,w)) = q(y,0) + O(|z|^\alpha)$ uniformly on smaller charts.

Sharpness of the exponent. For f bounded above and below by positive constants near the exceptional divisor of a point center, F is comparable to $\mathrm{dist}_h(\cdot, Z)^{-2(n-1)}$ near Z, giving the strict threshold. More generally, $f \asymp |s_E|^{-2a}$ with $0 \le a < 1/p$ belongs to L^p upstairs, while its pushforward has threshold $s < n/(n-1+a)$. Letting $a \uparrow 1/p$ proves the stated limitation; it makes no finite-p endpoint assertion.

The Kähler case. When h is Kähler, the strict barrier current $(c+\delta)[E] + \pi^*h - \delta\Theta'_E$ constructed in Proposition 15.22.1 makes $[\theta]$ big. Its degree on an exceptional-fiber line is $-c$, so it is not nef. The minimal current $c[E] + \pi^*h$ has non-pluripolar top mass $\int_Y h^n$, and the minimal mass theorem identifies this with $\mathrm{vol}([\theta])$ as

recalled in Section 36.16. Full-mass uniqueness and mass integration fix the scalar and normalized current as recalled in Section 36.21. Another normalized solution with minimal ordinary singularities, possibly discontinuous, still has bounded target potential by Proposition 15.22.1. It is bounded on every relatively compact subset of $X \setminus E$, so its deep truncations coincide with it there, and locality in Theorem 7.1.1 gives the bounded target equation off Z. Both target measures give Z zero mass, so closed uniqueness applies globally. If only the power conditions hold, Theorem 2.2.2 fixes the scalar of the constructed solution. □

For the prescribed pullback measure $\pi^*(h^n)$, choose $q = 0$ and $\kappa = 1$. The current is $c[E] + \pi^*h$. Its smooth part π^*h is degenerate on E exactly in the directions tangent to the exceptional fibers, and its density relative to a smooth volume form is $|z|^{2(r-1)}$ times a smooth positive factor. On $X \setminus E$, where the solution $u = cL$ is smooth and strictly θ-plurisubharmonic, the metric $\theta + \mathrm{dd}^{\mathrm{c}}u = \pi^*h$ is isometric to $(Y \setminus Z, h)$; its metric completion is Y, so each exceptional fiber collapses to a point of Z. The distance to E is comparable to $|z|$ on bounded adapted charts. Neither the density order nor this distance exponent depends on c.

Example 15.22.1 If $n = 2$ and Z is a point on a compact Kähler surface, $[\theta] = \pi^*[h] + c[E]$ has Zariski positive part $\pi^*[h]$ and negative part cE: $E^2 = -1$ and $\pi^*[h] \cdot E = 0$. Its volume is $\int_Y h^2$, although it is non-nef. The solution with $\kappa = 1$ for the measure $\pi^*(h^2)$ has logarithmic coefficient exactly c, its density vanishes to order $|z|^2$ along E, and the completion of its metric on $X \setminus E$ collapses E to the point Z. The stronger barrier from Proposition 15.22.1 has coefficient $c + \delta$.

Theorem 15.22.2 *Let $\pi\colon Y_m \to Y_0$ be a finite nonempty sequence of blowups $\pi_j\colon Y_j \to Y_{j-1}$ of connected smooth centers of codimensions $r_j \geq 2$ over a compact Hermitian (Y_0, h_0). Let E_j be the exceptional divisor of π_j, let $\pi_{>j}\colon Y_m \to Y_j$ denote the composition of the subsequent blowups, choose $a_j > 0$, and set*

$$\Delta = \sum_j a_j \pi_{>j}^* E_j, \qquad \mathrm{dd^c} L_\Delta = [\Delta] - \Theta_\Delta, \qquad \theta = \pi^* h_0 + \Theta_\Delta,$$

where $L_\Delta = \sum_j a_j \log |s_j|^2$, with s_j the canonical section of $\pi_{>j}^* O(E_j)$ for a smooth Hermitian metric, and Θ_Δ is the corresponding combination of curvature forms. Every θ-positive current is $\pi^*(h_0 + \mathrm{dd^c} q) + [\Delta]$ and every potential is $L_\Delta + q \circ \pi + C$, with $q \in \mathrm{PSH}(Y_0, h_0)$. Minimal ordinary type is exactly $L_\Delta + O(1)$, equivalent to bounded q. For potentials of this type, every prime exceptional coefficient is the one prescribed by the effective total transform Δ. There is a strict barrier with divisor $\Delta + B$ for an effective exceptional B positive on every exceptional component. For nonzero nonnegative upstairs L^p data, $p > 1$, a minimal solution has Hölder q, with pushforward density in every L^s satisfying

$$1 < s < \frac{Rp}{(R-1)p+1}, \qquad R = \prod_j r_j. \tag{15.75}$$

For Kähler h_0 the mass, scalar, and closed full-mass uniqueness are those of the target; the power conditions on h_0 fix the scalar for the constructed nonclosed solution. For the pullback measure $\pi^*(h_0^n)$, the metric $\pi^* h_0$ on the complement of the exceptional set has metric completion Y_0, each fiber collapsing to a point, and the zero divisor of the density is that of the squared modulus of the holomorphic Jacobian, independently of the a_j.

Proof Apply Proposition 15.22.1 successively, allowing an intermediate smooth background to be indefinite. The difference between each current and the pullback of its pushforward is closed of order zero, and exceptional-fiber degree fixes its coefficient. Pullback of earlier divisor currents produces the total transforms, proving the representation without an SNC requirement. The initial $q = 0$ is admissible; upper bounds on compact targets prove minimality and the bounded-q criterion. At SNC charts this is the usual sum of logarithms.

At each stage choose an auxiliary divisor metric as in Proposition 15.22.1, with curvature Θ'_{E_j}, so that $h_j = \pi_j^* h_{j-1} - \delta_j \Theta'_{E_j} > 0$ for a sufficiently small $\delta_j > 0$. Expanding yields $h_m = \pi^* h_0 - \Theta_B$ for $B = \sum_j \delta_j \pi_{>j}^* E_j$, effective with a positive coefficient on every exceptional component. If $\mathrm{dd^c} L_B = [B] - \Theta_B$ uses these auxiliary metrics, then the fixed background satisfies $\theta + \mathrm{dd^c}(L_\Delta + L_B) = h_m + [\Delta + B]$, which is the strict barrier current. The Jacobian calculation and Hölder estimate apply stagewise. If $s = rp/((r-1)p+1)$, then $1 - 1/s = (1 - 1/p)/r$; iteration over the stages gives (15.75), with strict inequality. The target Hölder equation, mass and uniqueness are obtained as in Theorem 15.22.1. For the pullback measure take $q = 0$ and use the target path comparison away from the exceptional set. The Jacobian fixes

the density, while $[\Delta]$ fixes the divisor current; these are independent pieces of the equation. □

15.23 Representing measures in an ordinary type

We return to the general setting of the chapter, with the Hermitian form ω fixed at its beginning. Locally, every finite non-pluripolar measure is absolutely continuous with respect to the Monge–Ampère measure of a bounded plurisubharmonic function (Lemma 1.6.2). We globalize this statement in a prescribed ordinary type. The local potentials are glued to a strictly θ-plurisubharmonic potential lying below the reference and are then truncated relative to the reference. A decreasing convex sum of the resulting countable family has a Monge–Ampère measure that dominates every local piece in the sense of absolute continuity. Its potential has the prescribed ordinary type, and the given measure has an integrable density with respect to its Monge–Ampère measure.

Proposition 15.23.1 *Let θ be a smooth big real $(1,1)$-form, and let $\varphi \in \mathrm{PSH}(X,\theta)$ be such that the envelope $P_{\epsilon\,\theta}(\varphi)$ of Section 1.19 is not identically $-\infty$ for some $0 < \epsilon < 1$. For every finite non-pluripolar measure μ there are*

$$\psi \in \mathrm{PSH}(X,\theta), \qquad \psi \sim \varphi, \qquad 0 \le f \in L^1\big(\mathrm{MA}_\theta^{\mathrm{ext}}(\psi)\big)$$

such that

$$\mu = f\,\mathrm{MA}_\theta^{\mathrm{ext}}(\psi).$$

Proof Translate φ below zero. Put $q = P_{\epsilon\,\theta}(\varphi) \le \varphi$, and choose a strict barrier $\rho \le 0$ with $\theta + \mathrm{dd^c}\rho \ge a\omega$, where $a > 0$. Then

$$s = q + (1-\epsilon)\rho \le \varphi, \qquad \theta + \mathrm{dd^c} s \ge (1-\epsilon)a\omega.$$

In particular s is θ-plurisubharmonic and not identically $-\infty$. Its pole set is pluripolar, as is that of φ, so both are negligible for μ.

Cover X by finitely many closed coordinate balls K_i, each contained in a larger coordinate ball B_i. Apply Lemma 1.6.2 to $\mu_i = \mathbf{1}_{K_i}\mu$. This gives bounded local plurisubharmonic v_i, normalized by $1 \le v_i \le 2$, with

$$\mu_i \ll \nu_i := (\mathrm{dd^c} v_i)^n.$$

Take a smooth cutoff χ_i which is zero near K_i and -3 on a boundary collar of B_i. Choose $\delta_i > 0$ so small that

$$\theta + \mathrm{dd^c}\big(s + \delta_i(v_i + \chi_i)\big) \ge \tfrac{1}{2}(1-\epsilon)a\omega + \delta_i \mathrm{dd^c} v_i.$$

Let S_i equal $\max(s, s + \delta_i(v_i + \chi_i))$ on B_i and s elsewhere. The boundary collar makes this a global θ-plurisubharmonic function. Since $\chi_i \le 0$ may be arranged,

$$s \le S_i \le s + 2\delta_i \le \varphi + 2\delta_i.$$

For every positive integer m, set

$$q_{i,m} = \max(\varphi - m, S_i) - 2\delta_i.$$

These potentials have the same ordinary type as φ, and

$$\varphi - m - 2\delta_i \le q_{i,m} \le \varphi. \tag{15.76}$$

On a neighborhood of K_i, $S_i = s + \delta_i v_i$ away from the pole set of s. Let

$$D_{i,m} = \{s + \delta_i v_i > \varphi - m\}$$

in that neighborhood, restricting to points where the potentials are finite. This is a plurifinely open set there. Locality and positive mixed products give

$$\mathbf{1}_{D_{i,m}} \mathrm{MA}_\theta^{\mathrm{ext}}(q_{i,m}) \ge \delta_i^n \mathbf{1}_{D_{i,m}} \nu_i. \tag{15.77}$$

Here s need not be bounded. After a local smooth correction, the positive product inequality holds at every common finite truncation level of s and the bounded potential v_i. We then exhaust the complement of the pole set of s; the non-pluripolar measure ν_i loses no mass in this exhaustion. Restricting the resulting inequality by plurifine locality proves (15.77) for the extended measures.

For μ_i-almost every point of K_i, both s and φ are finite, and hence that point belongs to $D_{i,m}$ for some m. Consequently, a Borel set which is null for every $\mathrm{MA}_\theta^{\mathrm{ext}}(q_{i,m})$ is null for every μ_i, and therefore for μ.

Enumerate the countable family $q_{i,m}$ as q_k, and write its bounds as $\varphi - C_k \le q_k \le \varphi$ with $C_k < \infty$. Choose positive t_k such that

$$\sum_k t_k < 1, \qquad B := \sum_k t_k C_k < \infty.$$

The finite convex combinations

$$\psi_N = \sum_{k=1}^{N} t_k q_k + \left(1 - \sum_{k=1}^{N} t_k\right) \varphi$$

decrease and lie between $\varphi - B$ and φ. Their limit ψ is θ-plurisubharmonic, satisfies $\psi \ge \varphi - B$, and hence $\psi \sim \varphi$. For any fixed k, separate its summand to write

$$\psi = t_k q_k + (1 - t_k) w_k, \qquad w_k \in \mathrm{PSH}(X, \theta), \qquad w_k \sim \varphi.$$

The remainder is obtained by the same decreasing convex-sum construction. Positive non-pluripolar mixed products give

$$\mathrm{MA}_\theta^{\mathrm{ext}}(\psi) \ge t_k^n \mathrm{MA}_\theta^{\mathrm{ext}}(q_k).$$

This is valid also for infinite measures: first restrict away from the common pole set, which is the pole set of φ by the ordinary-type bounds, and then exhaust by finite truncations. All measures ignore that pole set. The preceding null-set conclusion now gives $\mu \ll \mathrm{MA}_\theta^{\mathrm{ext}}(\psi)$.

By Theorem 7.1.1, the measure $\mathrm{MA}_\theta^{\mathrm{ext}}(\psi)$, which may have infinite mass, is σ-finite. The Radon–Nikodym theorem applies to this σ-finite measure and the finite μ. It gives a nonnegative density f with

$$\int_X f \,\mathrm{MA}_\theta^{\mathrm{ext}}(\psi) = \mu(X) < \infty,$$

as claimed. □

Lemma 15.23.1 *Let $\theta \geq 0$ be a smooth big real $(1,1)$-form, and let μ be a finite non-pluripolar measure on X. There are a bounded $\psi \in \mathrm{PSH}(X,\theta)$ and a nonnegative $f \in L^1(\mathrm{MA}_\theta(\psi))$ such that*

$$\mu = f\,\mathrm{MA}_\theta(\psi).$$

Proof Apply Proposition 15.23.1 with $\varphi = 0$. Semipositivity of θ gives $P_{\epsilon\theta}(0) = 0$ for every $0 < \epsilon < 1$, and $\psi \sim 0$ means that ψ is bounded. □

These representations give only an L^1 density, whereas Theorem 15.9.1 requires a bounded density with respect to the measure of a bounded potential; for an unbounded prescribed type, moreover, the representing potential is itself unbounded. The existence argument for general measures in Section 33.6 therefore applies Lemma 15.23.1 only to the Hermitian background ω, and then approximates the resulting bounded potential and integrable density, passing to the limit by compactness and continuity theorems.

15.24 The homogeneous equation with a Hölder reference

A measure with a bounded density with respect to the Monge–Ampère measure of a Hölder potential admits an exact homogeneous equation, whose coefficient is obtained together with the solution. The density must be approximated by continuous functions before the reference measure is replaced by smooth ones, since weak convergence of measures does not preserve integrals of a fixed Borel function.

Theorem 15.24.1 *Let h be a Hermitian form on X. Suppose $u \in \mathrm{PSH}(X,h)\cap C^{0,\alpha}(X)$ for some $0<\alpha\le 1$, and let $\nu \neq 0$ be a positive measure with*

$$\nu \le A\mathrm{MA}_h(u), \qquad A>0.$$

There are a constant $c>0$ and a Hölder continuous $w\in \mathrm{PSH}(X,h)$ such that

$$\sup_X w = 0, \qquad \mathrm{MA}_h(w) = c\nu.$$

No bound for the masses of bounded h-plurisubharmonic functions is assumed.

Proof *Step 1: approximate the density and reference.* Write $\sigma = \mathrm{MA}_h(u)$ and $\nu = Ag\sigma$, with $0\le g\le 1$. Since σ is a finite Borel measure on the compact manifold X, there are continuous $g_l\colon X\to[0,1]$ tending to g in $L^1(\sigma)$. The inequality $|s^{1/n}-t^{1/n}|^n\le |s-t|$ on $[0,1]$, followed by Hölder's inequality, gives $g_l^{1/n}\to g^{1/n}$ in $L^1(\sigma)$.

By Lemma 1.21.4, there are smooth $v_j\in\mathrm{PSH}(X,h)$ tending uniformly to u with a common Hölder norm. Their Monge–Ampère measures $\sigma_j = h_{v_j}^n$ tend weakly to σ. Choose a dense countable family of continuous tests, including the constant one. For each l, choose $j(l)\ge l$ so that integration of $g_l^{1/n}$, and of g_l multiplied by the first l tests, against $\sigma_{j(l)}-\sigma$ has absolute value at most $1/l$. Write $\widehat{v}_l = v_{j(l)}$ and

$$\nu_l = Ag_l h_{\widehat{v}_l}^n.$$

The masses of the measures $h_{\widehat{v}_l}^n$ are uniformly bounded, since the potentials $\widehat{v}_l$ are; as $0\le g_l\le 1$, the masses of the ν_l are uniformly bounded as well. The choice of $j(l)$ and the convergence of g_l and $g_l^{1/n}$ in $L^1(\sigma)$ imply

$$\nu_l \rightharpoonup \nu, \qquad \lim_l \int_X g_l^{1/n} h_{\widehat{v}_l}^n = \int_X g^{1/n}\,\mathrm{d}\sigma > 0.$$

After discarding finitely many terms, each ν_l has positive mass and a continuous density with respect to smooth volume. Thus Theorem 11.1.1 gives continuous normalized h-plurisubharmonic φ_l and $c_l>0$ with $h_{\varphi_l}^n = c_l\nu_l$.

Step 2: control the coefficients and potentials. Since both right-hand sides have smooth-volume densities, Lemma 1.7.3 gives

$$h_{\varphi_l}\wedge h_{\widehat{v}_l}^{n-1} \ge (c_lA)^{1/n} g_l^{1/n} h_{\widehat{v}_l}^n.$$

By Lemma 1.8.10, the trace bound is uniform for references with a common bound on their supremum norms. Integrating, and using the positive limit of $\int_X g_l^{1/n} h_{\widehat{v}_l}^n$ from Step 1, therefore bounds c_l above. By Proposition 1.8.1, the solution measures $\mu_l = h_{\varphi_l}^n \le C h_{\widehat{v}_l}^n$ have a common square-capacity bound.

In Lemma 11.3.1, take φ_l as the first potential, the zero function as the second, $H=1$ and $\varepsilon = 1/2$, and fix $D>0$ with $2D<\delta_{1/2}$. The backwards estimate of Lemma 11.3.2 gives

$$\operatorname{Cap}_h\{\varphi_l < \inf_X \varphi_l + D\} \geq a_0 > 0.$$

The normalized tail estimate (1.15) then bounds $-\inf_X \varphi_l$ uniformly. Applying (11.5) with $s = t = D$ also gives

$$D^n a_0 \leq C\mu_l(X) = Cc_l\nu_l(X).$$

Since the masses of ν_l are uniformly bounded above, c_l is bounded away from zero.

Step 3: take a uniform limit. Extract a subsequence with $c_l \to c > 0$ and φ_l converging in $L^1(\mathrm{d}V)$, using Theorem 1.4.1. All potentials now have common upper and lower bounds. The common Hölder norms of $\widehat{v}_l$ and Corollary 1.8.2 give, uniformly in j, k,

$$\int_X |\varphi_j - \varphi_k|\,\mathrm{d}\mu_j \leq C\|\varphi_j - \varphi_k\|_{L^1(\mathrm{d}V)}^{\alpha/(\alpha+2n)} \longrightarrow 0.$$

Use Proposition 11.3.1 in both orders to see that this subsequence is uniformly Cauchy. Its continuous limit w is h-plurisubharmonic and has supremum zero. Continuity of bounded Monge–Ampère products under uniform convergence and $\nu_l \rightharpoonup \nu$ yield $\mathrm{MA}_h(w) = c\nu$. Finally Theorem 15.4.2, with reference u and domination constant cA, gives the Hölder assertion. □

15.25 Singular measures on shrinking spheres

A measure with bounded density with respect to the Monge–Ampère measure of a bounded potential admits exponential equations by Theorem 15.9.1. The following construction turns this analytic hypothesis into a test against a probability measure fixed by a coordinate ball. It applies to singular data on every Hermitian background, including backgrounds with zero lower mass (Definition 7.5.1).

Fix a Hermitian form h on X, a holomorphic coordinate ball centered at a point p, and a smooth cutoff $0 \leq \chi \leq 1$ supported in that ball and equal to one near a smaller closed ball. Choose $r_0 > 0$ so small that all spheres of radii $r_j = r_0 2^{-j}$, $j \geq 1$, lie in the smaller ball. Let σ_j denote the normalized Euclidean surface measures of these spheres, which are probability measures, extended by zero to X, and put

$$S_n = \sum_{j\geq 1} r_j^{n/2}, \qquad \rho = \frac{1}{S_n}\sum_{j\geq 1} r_j^{n/2}\sigma_j. \tag{15.78}$$

For a finite positive measure ν, define

$$M_\rho(\nu) = \left\| \frac{\mathrm{d}\nu}{\mathrm{d}\rho} \right\|_{L^\infty(\rho)},$$

with value $+\infty$ when $\nu \not\ll \rho$. The measure ρ is chosen before ν or the exponent in the equation, and the test depends on the chart and the radii only through ρ.

Proposition 15.25.1 *The measure ρ is a singular non-pluripolar probability. There are $k > 0$ and $u_\rho \in \mathrm{PSH}(X,h) \cap C^{0,1/2}(X)$ with $\sup_X u_\rho = 0$ such that*

$$h + \mathrm{dd^c} u_\rho \geq h/2, \qquad \mathrm{MA}_h(u_\rho) \geq k\rho.$$

Consequently, every finite positive measure ν with $M_\rho(\nu) < \infty$ is non-pluripolar and satisfies $\nu \leq k^{-1} M_\rho(\nu) \mathrm{MA}_h(u_\rho)$.

Proof In the coordinate ball write $R = |z|$ and set

$$\psi_j = \chi \log \max\{R^2, r_j^2\} + (1 - \chi),$$

extending ψ_j by one outside it. On the transition region of χ, R is bounded away from zero and exceeds all r_j. Hence all ψ_j agree there with the same smooth function. Where $\chi = 1$ they are plurisubharmonic, and outside its support they are constant. One fixed $K \geq 1$ therefore satisfies $\mathrm{dd^c}\psi_j \geq -Kh$ for every j. Put

$$Z = 2K \sum_{j \geq 1} \sqrt{r_j}, \qquad d_j = \frac{\sqrt{r_j}}{Z}, \qquad u_0 = \sum_{j \geq 1} d_j \psi_j.$$

The estimate $\|\psi_j\|_\infty \leq C(1 + |\log r_j|)$ gives uniform convergence, and $\sum d_j = 1/(2K)$ yields $h + \mathrm{dd^c} u_0 \geq h/2$.

For two points of the inner ball at Euclidean distance t, the reverse triangle inequality gives

$$|\log \max\{|z|^2, r_j^2\} - \log \max\{|z'|^2, r_j^2\}| \leq 2 \log(1 + t/r_j).$$

The contribution of $r_j \geq t$ is at most $Ct \sum_{r_j \geq t} r_j^{-1/2} \leq C\sqrt{t}$. For $r_j < t$, summing the geometric tail gives at most $C\sqrt{t} \sum_{\ell \geq 0} 2^{-\ell/2}(1 + \ell)$. On the cutoff transition and outside, the series is a fixed smooth function. These estimates give a local $1/2$-Hölder bound near every point. A finite cover and the bound on the oscillation give the global bound.

On the smaller ball, the currents $\mathrm{dd^c}\psi_j$ are closed and positive. Expand the nth power of $h + \sum_{j \leq N} d_j \mathrm{dd^c}\psi_j$ there. All terms are positive, and we keep the diagonal terms $d_j^n (\mathrm{dd^c}\psi_j)^n$, which equal $d_j^n \sigma_j$ by Lemma 1.6.7. The partial sums $\sum_{j \leq N} d_j \psi_j$ are h-plurisubharmonic and converge uniformly to u_0, so their Monge–Ampère measures converge weakly to $\mathrm{MA}_h(u_0)$. Testing against nonnegative functions, and keeping first any fixed finite number of diagonal terms, therefore gives

$$\mathrm{MA}_h(u_0) \geq \sum_{j \geq 1} d_j^n \sigma_j = \frac{S_n}{Z^n} \rho.$$

Here the right-hand side is supported strictly inside the smaller ball, so the inequality extends to X. Each σ_j is non-pluripolar by the same lemma; countable additivity

gives that property for ρ. The countable union of the spheres has zero smooth volume, which proves singularity. Set $k = S_n/Z^n$ and subtract $\sup_X u_0$ to obtain u_ρ. The last assertion follows from $\nu \le M_\rho(\nu)\rho$. □

Corollary 15.25.1 *Let ν be a finite nonzero positive measure with $M_\rho(\nu) < \infty$. There are $c > 0$ and a Hölder continuous $w \in \mathrm{PSH}(X, h)$ satisfying*

$$\sup_X w = 0, \qquad \mathrm{MA}_h(w) = c\nu.$$

Proof By Proposition 15.25.1, $\nu \le k^{-1} M_\rho(\nu)\mathrm{MA}_h(u_\rho)$, where u_ρ is Hölder continuous with exponent $1/2$. Apply Theorem 15.24.1. □

Theorem 15.25.1 *Let ν be a finite nonzero positive measure with $M_\rho(\nu) < \infty$. For every $\lambda > 0$ there is a unique bounded $U_\lambda \in \mathrm{PSH}(X, h)$ satisfying*

$$\mathrm{MA}_h(U_\lambda) = \mathrm{e}^{\lambda U_\lambda}\nu.$$

Proof By Proposition 15.25.1, $\nu = f\,\mathrm{MA}_h(u_\rho)$ with $0 \le f \le k^{-1}M_\rho(\nu)$. The form $h > 0$ is semipositive and big, so Theorem 15.9.1 gives the unique bounded solution.□

The theorem does not assert uniqueness among unbounded potentials. Its hypothesis holds, for example, for $\nu = g\sigma_j$ with $j \ge 1$ and $g \ge 0$ bounded, $\nu \ne 0$. The test is only a sufficient condition. For the homogeneous equation, Corollary 15.25.1 gives some coefficient $c > 0$, but the construction identifies neither a largest admissible coefficient nor a unique solution.

15.26 Lower mass and solvability for all measures

Throughout the rest of the chapter, h denotes a Hermitian form on X, used as the background; the sections that use another background say so explicitly. No singularity type is prescribed: the unknown potentials range over all of $\mathrm{PSH}(X, h)$. Constants belong to $\mathrm{PSH}(X, h)$, and every $u \in \mathrm{PSH}(X, h)$ is bounded above, so $u \preceq v$ for every bounded $v \in \mathrm{PSH}(X, h)$.

The masses of bounded potentials are measured by the lower and upper masses $v_\pm(h)$ of Definition 7.5.1, the infimum and the supremum of $\int_X (h + \mathrm{dd^c} v)^n$ over the bounded $v \in \mathrm{PSH}(X, h)$; recall that h has positive lower mass if $v_-(h) > 0$. If h is Kähler, Stokes' theorem gives $v_-(h) = v_+(h) = \int_X h^n$; on a non-Kähler manifold the lower mass can vanish (Theorem 27.18.1).

For a nonzero nonnegative $q \in L^p(\mathrm{d}V)$ with $p > 1$, Theorem 11.1.1 gives a constant $c > 0$ and a continuous $u \in \mathrm{PSH}(X, h)$ with $\mathrm{MA}_h(u) = cq\,\mathrm{d}V$, and Proposition 11.6.1 shows that c is determined by q. We call $c_h(q) = c$ the *normalization scalar* of q, and for the measure $\nu = q\,\mathrm{d}V$ we also write $c_h(\nu) = c_h(q)$. We say that a finite measure ν admits a *positive scaled subsolution* if there are $u \in \mathrm{PSH}(X, h)$ and a constant $a > 0$ such that $\mathrm{MA}_h^{\mathrm{ext}}(u) \ge a\nu$.

If ν admits a solution of $\mathrm{MA}_h^{\mathrm{ext}}(u) = c\nu$ with $c > 0$, then it admits in particular a positive scaled subsolution. We show that, when h has zero lower mass, some probability density admits no positive scaled subsolution. Conversely, if $\nu_-(h) > 0$, every nonzero finite non-pluripolar measure admits such a solution (Corollary 33.1.1 in Chapter 33). The key estimate is a consequence of strict domination: if $\mathrm{MA}_h^{\mathrm{ext}}(u) \geq aq\,\mathrm{d}V$, then $a \leq c_h(q)$. When $\nu_-(h) = 0$, there are smooth probability densities with arbitrarily small normalization scalars, and a weighted sum of such densities, with scalars decreasing fast enough, is a single probability density admitting no positive scaled subsolution.

Lemma 15.26.1 *Let $0 \leq q \in L^p(\mathrm{d}V)$, $p > 1$, be nonzero. If $u \in \mathrm{PSH}(X,h)$ and $a > 0$ satisfy*

$$\mathrm{MA}_h^{\mathrm{ext}}(u) \geq aq\,\mathrm{d}V,$$

then $a \leq c_h(q)$. Consequently, for nonzero nonnegative $q, r \in L^p(\mathrm{d}V)$,

$$q \leq r \implies c_h(q) \geq c_h(r), \qquad c_h(tq) = t^{-1}c_h(q) \quad (t > 0).$$

Proof Let v be a continuous solution of $\mathrm{MA}_h(v) = c_h(q)q\,\mathrm{d}V$, and suppose, to reach a contradiction, that $a > c_h(q)$. Then $\tau = c_h(q)/a < 1$ and, for every real A,

$$\mathrm{MA}_h(v + A) = c_h(q)q\,\mathrm{d}V \leq \tau\,\mathrm{MA}_h^{\mathrm{ext}}(u).$$

The translate $v + A$ is bounded, so $P_{h/2}(v + A) \geq \inf_X v + A$ is not identically $-\infty$, and $u \preceq v + A$. Theorem 15.1.1, applied with $\delta = 1/2$ and $c = \tau$ to $v + A$ in place of u and to u in place of v, therefore gives $v + A \geq u$ for every real A. Letting $A \to -\infty$ at a point where u is finite gives a contradiction. Hence $a \leq c_h(q)$.

If $q \leq r$, a continuous solution of $\mathrm{MA}_h(w) = c_h(r)r\,\mathrm{d}V$ satisfies $\mathrm{MA}_h(w) \geq c_h(r)q\,\mathrm{d}V$, so the first assertion gives $c_h(r) \leq c_h(q)$. Finally, the equation $\mathrm{MA}_h(v) = c_h(q)q\,\mathrm{d}V$ can be written as $\mathrm{MA}_h(v) = t^{-1}c_h(q)\,(tq)\,\mathrm{d}V$, and uniqueness of the scalar gives $c_h(tq) = t^{-1}c_h(q)$. □

Lemma 15.26.2 *Fix $p > 1$. If nonzero nonnegative densities $q_j, q \in L^p(\mathrm{d}V)$ satisfy $q_j \to q$ in L^p, then $c_h(q_j) \to c_h(q)$.*

Proof We first show that the scalars $c_h(q_j)$ lie in a compact subset of $(0, +\infty)$. Fix a Gauduchon metric γ on X (Definition 4.1.1; in dimension one, any Hermitian metric). Pairing the equation for q_j with γ^{n-1}, as in Proposition 15.13.1 with $\theta = h$ and barrier $\rho = 0$, gives

$$c_h(q_j)^{1/n} \int_X \left(\frac{q_j\,\mathrm{d}V}{\gamma^n}\right)^{1/n} \gamma^n \leq \int_X h \wedge \gamma^{n-1}.$$

Since $q_j^{1/n} \to q^{1/n}$ in L^n, the integrals on the left converge to the corresponding integral for q, which is positive. Hence the scalars are bounded above. The lower-density estimate of Theorem 11.3.1 gives $c_h(q_j)\|q_j\|_{L^p} \geq c_p > 0$, so they are also bounded away from zero.

Consequently, the normalized continuous solutions u_j of $\mathrm{MA}_h(u_j) = c_h(q_j)q_j\,\mathrm{d}V$ have uniformly bounded L^p densities. By Theorem 11.3.2, every subsequence has a further subsequence along which u_j converges uniformly to some u and $c_h(q_j)$ converges to some $c > 0$. Continuity of bounded Monge–Ampère products under uniform convergence gives $\mathrm{MA}_h(u) = cq\,\mathrm{d}V$, and uniqueness of the scalar gives $c = c_h(q)$. Since every subsequence of $(c_h(q_j))$ has a further subsequence converging to $c_h(q)$, the whole sequence converges to $c_h(q)$. □

Theorem 15.26.1 *Suppose $v_-(h) = 0$. There is a probability measure $\mu = f\,\mathrm{d}V$, with $f \in L^1(\mathrm{d}V)$ and $f \geq m$ almost everywhere for a constant $m > 0$, such that no $u \in \mathrm{PSH}(X, h)$ and $a > 0$ satisfy*

$$\mathrm{MA}_h^{\mathrm{ext}}(u) \geq a\mu. \tag{15.79}$$

In particular, the equation $\mathrm{MA}_h^{\mathrm{ext}}(u) = c\mu$ has no solution $u \in \mathrm{PSH}(X, h)$ for any $c > 0$; neither a singularity type nor finiteness of the mass of u is imposed.

Proof By Theorems 10.8.1 and 10.8.2, $v_-(h)$ is the infimum of the normalization scalars $c_h(\nu)$ over smooth strictly positive probability measures ν. Since $v_-(h) = 0$, we can choose such measures $\nu_j = f_j\,\mathrm{d}V$ with

$$c_h(\nu_j) \leq 2^{-2j} \qquad (j \geq 1).$$

Set

$$\mu = \sum_{j=1}^{\infty} 2^{-j}\nu_j, \qquad f = \sum_{j=1}^{\infty} 2^{-j} f_j.$$

By monotone convergence, $\int_X f\,\mathrm{d}V = 1$, so $f \in L^1$. Moreover $f \geq \frac{1}{2}\min_X f_1 =: m > 0$, and μ is non-pluripolar, being absolutely continuous with respect to $\mathrm{d}V$.

Suppose that (15.79) held for some u and $a > 0$. Since $\mu \geq 2^{-j}\nu_j$, we would have

$$\mathrm{MA}_h^{\mathrm{ext}}(u) \geq a2^{-j}\nu_j$$

for every j, and Lemma 15.26.1 would give $a2^{-j} \leq c_h(\nu_j) \leq 2^{-2j}$. Thus $a \leq 2^{-j}$ for every j, contradicting $a > 0$. □

Corollary 15.26.1 *Assume $v_-(h) = 0$, and let $\mu = f\,\mathrm{d}V$ be the measure of Theorem 15.26.1. Then $f \notin L^p(\mathrm{d}V)$ for every $p > 1$. Moreover, for every probability density $g \in L^1(\mathrm{d}V)$ and every $0 < t < 1$, the probability density*

$$g_t = (1-t)g + tf$$

admits no positive scaled subsolution, although $\|g_t - g\|_{L^1} \leq 2t$. In particular, the probability densities admitting no positive scaled subsolution are dense among all probability densities in $L^1(\mathrm{d}V)$.

Proof If $f \in L^p$ for some $p > 1$, Theorem 11.1.1 would give a continuous solution of $\mathrm{MA}_h(u) = c f\,\mathrm{d}V$ with $c > 0$, contradicting the theorem. Since $g_t\,\mathrm{d}V \geq t\mu$, a positive scaled subsolution for $g_t\,\mathrm{d}V$ would also give one for μ. Finally, $\|g_t - g\|_{L^1} = t\|f - g\|_{L^1} \leq 2t$. □

Smooth strictly positive probability densities are also dense in $L^1(\mathrm{d}V)$, and each of them admits a smooth solution of the normalized equation. Thus, when the lower mass vanishes, solvability is not preserved under L^1 approximation. The obstruction comes from the way the density concentrates, not from its size: the density in Theorem 15.26.1 is bounded below by a positive constant. It occurs, for instance, on the compact threefold of Theorem 27.18.1, where every Hermitian form has zero lower mass.

Theorem 15.26.1 does not exclude unbounded solutions of the exponential equation $\mathrm{MA}_h^{\mathrm{ext}}(u) = \mathrm{e}^{\lambda u}\mu$ with $\lambda > 0$, because the factor $\mathrm{e}^{\lambda u}$ need not have a positive lower bound. It does exclude bounded solutions, since a bounded solution would satisfy $\mathrm{MA}_h^{\mathrm{ext}}(u) \geq \mathrm{e}^{\lambda \inf_X u}\mu$. On suitable products, Theorem 15.34.3 chooses the concentrating components of such a density so that its exponential equation has a solution for every positive exponent. We do not know whether this holds for every density admitting no positive scaled subsolution, nor which individual measures admit a solution of the normalized equation when $v_-(h) = 0$.

15.27 Subsolutions for an individual density

When $v_-(h) = 0$, some densities admit no positive scaled subsolution. For one fixed density f, we now determine the largest coefficient a for which $\mathrm{MA}_h^{\mathrm{ext}}(u) \geq a f \, \mathrm{d}V$ has a solution. The bounded truncations of f have normalization scalars, and their limit is the answer. Moreover, the largest coefficient is attained: the normalized continuous solutions for the truncations have an L^1 cluster limit, and a lower bound for Monge–Ampère densities passes to L^1 limits of potentials by means of monotone upper envelopes.

Let $0 \leq f \in L^1(\mathrm{d}V)$ have positive integral. Its truncations are bounded and nonzero, and we put

$$f_k = \min(f, k), \qquad b_h(f) = \lim_{k \to \infty} c_h(f_k). \tag{15.80}$$

By Lemma 15.26.1, the scalars $c_h(f_k)$ decrease as k increases, so the limit exists in $[0, c_h(f_1)]$. We call $b_h(f)$ the *subsolution threshold* of f; Theorem 15.27.1 below justifies the name. The same lemma gives

$$b_h(f) = \inf\{c_h(q) : 0 \leq q \leq f, \ q \in L^\infty, \ \textstyle\int_X q \, \mathrm{d}V > 0\}. \tag{15.81}$$

Indeed, every such q satisfies $q \leq f_k$ for $k \geq \|q\|_{L^\infty}$, hence $c_h(q) \geq c_h(f_k) \geq b_h(f)$, while the truncations f_k are themselves admissible.

The Monge–Ampère operator is not continuous under L^1 convergence of potentials, but a lower bound for a Monge–Ampère density survives such convergence. The proof replaces the sequence by monotone upper envelopes, which converge in capacity, and uses the continuity of damped measures, Theorem 8.3.1, applied with the Hermitian form h and the weight $t \mapsto \mathrm{e}^{-t}$. For nonpositive $v \in \mathrm{PSH}(X, h)$, the corresponding damped measure is $\mathrm{e}^v \mathrm{MA}_h^{\mathrm{ext}}(v)$, with weight zero on the pole set; it has finite mass and depends weakly continuously on v under convergence in capacity.

Lemma 15.27.1 *Let $s_j, s \in \mathrm{PSH}(X, h)$ satisfy $s_j \to s$ in $L^1(X)$. Suppose $g_j, g \geq 0$, $g_j \to g$ in $L^1(\mathrm{d}V)$, and*

$$\mathrm{MA}_h^{\mathrm{ext}}(s_j) \geq g_j \, \mathrm{d}V \quad \textit{for every } j.$$

Then $\mathrm{MA}_h^{\mathrm{ext}}(s) \geq g \, \mathrm{d}V$.

Proof If x_j is a maximum point of s_j, Lemma 1.4.1 and compactness of X give $\limsup_j \sup_X s_j \leq \sup_X s$, so the potentials have a common upper bound. Subtracting it, which does not change the extended measures, we may assume that all potentials are nonpositive. Pass to a subsequence along which $s_j \to s$ and $g_j \to g$ almost everywhere, and set

$$a_j = \inf_{\ell \geq j} g_\ell, \qquad w_j = \left(\sup_{\ell \geq j} s_\ell \right)^*.$$

Then $0 \le a_j \uparrow g$ almost everywhere, and $w_j \in \mathrm{PSH}(X,h)$ decreases to s by Lemma 1.4.1.

Step 1: $\mathrm{MA}_h^{\mathrm{ext}}(w_j) \ge a_j\,\mathrm{d}V$. Fix j. For $N \ge j$, the finite maximum $m_N = \max_{j\le\ell\le N} s_\ell$ satisfies $\mathrm{MA}_h^{\mathrm{ext}}(m_N) \ge a_j\,\mathrm{d}V$: on the set $\{m_N = s_\ell\}$, Proposition 7.2.1 gives $\mathrm{MA}_h^{\mathrm{ext}}(m_N) \ge \mathrm{MA}_h^{\mathrm{ext}}(s_\ell) \ge g_\ell\,\mathrm{d}V \ge a_j\,\mathrm{d}V$, and these sets cover X. As $N \to \infty$, the functions m_N increase, are bounded above by zero, and have regularized limit w_j. By Lemma 1.8.9, $m_N \to w_j$ in capacity, so the damped measures $\mathrm{e}^{m_N}\mathrm{MA}_h^{\mathrm{ext}}(m_N)$ converge weakly to $\mathrm{e}^{w_j}\mathrm{MA}_h^{\mathrm{ext}}(w_j)$. Since $m_N \to w_j$ almost everywhere and $a_j \le g$, dominated convergence gives $\mathrm{e}^{m_N}a_j \to \mathrm{e}^{w_j}a_j$ in $L^1(\mathrm{d}V)$. Passing to the limit in $\mathrm{e}^{m_N}\mathrm{MA}_h^{\mathrm{ext}}(m_N) \ge \mathrm{e}^{m_N}a_j\,\mathrm{d}V$ gives

$$\mathrm{e}^{w_j}\mathrm{MA}_h^{\mathrm{ext}}(w_j) \ge \mathrm{e}^{w_j}a_j\,\mathrm{d}V.$$

The pole set of w_j has zero volume and is not charged by $\mathrm{MA}_h^{\mathrm{ext}}(w_j)$. Dividing by the positive weight off this set gives $\mathrm{MA}_h^{\mathrm{ext}}(w_j) \ge a_j\,\mathrm{d}V$.

Step 2: passage to s. Since w_j decreases to $s \in \mathrm{PSH}(X,h)$, Lemma 1.8.9 gives $w_j \to s$ in capacity, and the damped measures of w_j converge weakly to $\mathrm{e}^{s}\mathrm{MA}_h^{\mathrm{ext}}(s)$. Moreover $0 \le \mathrm{e}^{w_j}a_j \le g$ and $\mathrm{e}^{w_j}a_j \to \mathrm{e}^{s}g$ almost everywhere, so dominated convergence and Step 1 give

$$\mathrm{e}^{s}\mathrm{MA}_h^{\mathrm{ext}}(s) \ge \mathrm{e}^{s}g\,\mathrm{d}V.$$

Dividing by e^{s} on $\{s > -N\}$ and letting $N \to \infty$ gives $\mathrm{MA}_h^{\mathrm{ext}}(s) \ge g\,\mathrm{d}V$ off the pole set of s. Both measures give zero mass to the pole set, which proves the lemma. □

Theorem 15.27.1 *For every nonnegative $f \in L^1(\mathrm{d}V)$ with positive integral,*

$$b_h(f) = \max\{a \ge 0 : \textit{there exists } u \in \mathrm{PSH}(X,h) \textit{ with } \mathrm{MA}_h^{\mathrm{ext}}(u) \ge a f\,\mathrm{d}V\}.$$

In particular, $f\,\mathrm{d}V$ admits a positive scaled subsolution if and only if $b_h(f) > 0$.

Proof If $\mathrm{MA}_h^{\mathrm{ext}}(u) \ge a f\,\mathrm{d}V$ with $a > 0$, then also $\mathrm{MA}_h^{\mathrm{ext}}(u) \ge a f_k\,\mathrm{d}V$ for every k, and Lemma 15.26.1 gives $a \le c_h(f_k)$. Letting $k \to \infty$ gives $a \le b_h(f)$. If $b_h(f) = 0$, the coefficient zero is attained by the zero potential.

Suppose now that $b = b_h(f) > 0$, and let u_k be continuous solutions of

$$\mathrm{MA}_h(u_k) = c_h(f_k)f_k\,\mathrm{d}V, \qquad \sup_X u_k = 0.$$

By Theorem 1.4.1, a subsequence u_{k_j} converges in L^1 to some $u \in \mathrm{PSH}(X,h)$ with $\sup_X u = 0$. The densities $c_h(f_{k_j})f_{k_j}$ are dominated by $c_h(f_1)f$ and converge to bf almost everywhere, hence in L^1. Lemma 15.27.1 now gives $\mathrm{MA}_h^{\mathrm{ext}}(u) \ge bf\,\mathrm{d}V$, so the maximum is attained at $a = b$. □

When $b_h(f) > 0$, a function $u \in \mathrm{PSH}(X,h)$ with $\mathrm{MA}_h^{\mathrm{ext}}(u) \ge b_h(f)f\,\mathrm{d}V$ is called a *threshold subsolution* for f, and a threshold subsolution with $\mathrm{MA}_h^{\mathrm{ext}}(u) = b_h(f)f\,\mathrm{d}V$ is called a *threshold solution*; the theorem shows that threshold subsolutions exist. Its proof preserves only a lower bound for the Monge–Ampère density. It

shows neither that the normalized solutions u_k converge in capacity, nor that their limit has finite mass, nor that equality holds in the limit; these are the additional difficulties of the exact equation for an individual density. The threshold does not depend on the choice of truncations. If (q_k) is any increasing sequence of nonzero bounded densities converging to f almost everywhere, then $c_h(q_k) \geq b_h(f)$ by (15.81); conversely, the second half of the proof, applied to normalized solutions for q_k, gives a subsolution with coefficient $\lim_k c_h(q_k)$, which is therefore at most $b_h(f)$. Hence $\lim_k c_h(q_k) = b_h(f)$.

15.27.1 The greatest normalized threshold subsolution

Suppose $b_h(f) > 0$. The threshold subsolutions with supremum zero form a family that is stable under finite maxima and under L^1 limits. Its upper envelope is therefore again a normalized threshold subsolution, and it dominates all the others.

Proposition 15.27.1 *Suppose $b = b_h(f) > 0$. The family*

$$\mathcal{S}_f = \{s \in \mathrm{PSH}(X,h) : \sup_X s = 0,\ \mathrm{MA}_h^{\mathrm{ext}}(s) \geq b f\, dV\}$$

is nonempty and compact in $L^1(X)$. Its upper envelope

$$\Phi_f = \left(\sup_{s\in\mathcal{S}_f} s\right)^*$$

belongs to $\mathcal{S}_f$ and is therefore its greatest member.

Proof The family is nonempty by Theorem 15.27.1. By Theorem 1.4.1, an L^1 limit of members of $\mathcal{S}_f$ is h-plurisubharmonic with supremum zero, and Lemma 15.27.1, with $g_j = g = bf$, shows that it satisfies the measure inequality. Thus $\mathcal{S}_f$ is a closed subset of the compact set of normalized h-plurisubharmonic functions, hence compact. It is also stable under finite maxima: if $s_1, s_2 \in \mathcal{S}_f$, then $\max(s_1, s_2)$ has supremum zero, and Proposition 7.2.1, applied on the sets where the maximum equals s_1 or s_2, gives $\mathrm{MA}_h^{\mathrm{ext}}(\max(s_1,s_2)) \geq bf\, dV$.

By Choquet's lemma there is a sequence $(s_i)_{i\geq 1}$ in $\mathcal{S}_f$ with $(\sup_i s_i)^* = \Phi_f$. Fix $s_0 \in \mathcal{S}_f$. The finite maxima $t_N = \max(s_0, s_1, \dots, s_N)$ belong to $\mathcal{S}_f$, increase with N, and satisfy $s_0 \leq t_N \leq 0$; their regularized limit is Φ_f. Hence $t_N \to \Phi_f$ almost everywhere and, by dominated convergence, in L^1. Lemma 15.27.1 gives $\mathrm{MA}_h^{\mathrm{ext}}(\Phi_f) \geq bf\, dV$, and $s_0 \leq \Phi_f \leq 0$ forces $\sup_X \Phi_f = 0$. Thus $\Phi_f \in \mathcal{S}_f$, and by construction it dominates every member of $\mathcal{S}_f$. □

The proposition gives a greatest normalized threshold subsolution, but not the equation $\mathrm{MA}_h^{\mathrm{ext}}(\Phi_f) = bf\, dV$; in particular, it does not show that Φ_f has finite mass.

Proposition 15.27.2 *Let $0 \le f \in L^1(\mathrm{d}V)$ have positive integral. If $f \in L^p(\mathrm{d}V)$ for some $p > 1$, then $b_h(f) = c_h(f)$. In general,*

$$b_h(f) \ge \frac{v_-(h)}{\int_X f \,\mathrm{d}V}.$$

If h is Kähler, then $b_h(f) = \int_X h^n / \int_X f \,\mathrm{d}V$.

Proof If $f \in L^p$ with $p > 1$, then $f_k \to f$ in L^p, and Lemma 15.26.2 gives $b_h(f) = \lim_k c_h(f_k) = c_h(f)$.

For general f, the continuous solution for f_k has total mass $c_h(f_k) \int_X f_k \,\mathrm{d}V$, which is at least $v_-(h)$ by the definition of the lower mass. Letting $k \to \infty$ and using monotone convergence gives the lower bound for $b_h(f)$. If h is Kähler, Stokes' theorem for bounded potentials gives instead the identity $c_h(f_k) \int_X f_k \,\mathrm{d}V = \int_X h^n$, and the same limit proves the last assertion. □

15.27.2 Smooth comparison tests

The infimum (15.81) runs over bounded measurable densities below f. One may ask whether smooth densities suffice to compute the threshold. A smooth test g is in general not below f, so we compare it with f through the ratio $\|g/f\|_{L^\infty}$. Thus let f be a probability density bounded below by a positive constant, and define

$$C_h^{\mathrm{sm}}(f) = \inf_g c_h(g) \|g/f\|_{L^\infty},$$

where g ranges over smooth strictly positive probability densities. For each such g, with $R = \|g/f\|_{L^\infty}$, a subsolution with coefficient a for $f \,\mathrm{d}V$ is a subsolution with coefficient a/R for $g \,\mathrm{d}V$, so $a \le R c_h(g)$ by Lemma 15.26.1. Hence $b_h(f) \le C_h^{\mathrm{sm}}(f)$, with equality if f is smooth (take $g = f$). When the lower mass vanishes, the two quantities can differ even in their positivity.

Theorem 15.27.2 *If $v_-(h) = 0$, there is a probability density $f \in L^1(\mathrm{d}V)$ bounded below by a positive constant such that*

$$b_h(f) = 0 < C_h^{\mathrm{sm}}(f).$$

Thus $f \,\mathrm{d}V$ admits no positive scaled subsolution, although $c_h(g)\|g/f\|_{L^\infty}$ is bounded below by a positive constant for all smooth strictly positive probability densities g.

Proof We construct $f = (1 + Q)/M$, where $Q \ge 0$ concentrates on closed nowhere dense sets in such a way that it has essential infimum zero on every nonempty open set.

Step 1: the pieces. Let $(U_i)_{i \ge 1}$ be a countable basis of nonempty open subsets of X. We construct inductively numbers $\alpha_j > 0$, nonzero bounded densities q_j, and closed nowhere dense sets $K_j \subset X$ such that

$$\int_X q_j \, dV \le 1, \qquad q_j = 0 \text{ outside } K_j, \qquad c_h(q_j) \le \alpha_j 2^{-j}.$$

Start with $\alpha_1 = 1/4$. Once α_j has been chosen, Theorems 10.8.1 and 10.8.2 and $v_-(h) = 0$ give a smooth strictly positive probability density g_j with $c_h(g_j) < \alpha_j 2^{-j}$. Removing from X an open dense set of small volume, we obtain a closed nowhere dense set K_j for which $q_j = g_j \mathbf{1}_{K_j}$ is nonzero and, since g_j is bounded, arbitrarily close to g_j in L^2. By Lemma 15.26.2 with $p = 2$, we can therefore choose K_j so that $c_h(q_j) \le \alpha_j 2^{-j}$.

After $K_1, \dots, K_N$ have been chosen, put

$$m_N = \min_{1 \le i \le N} dV\left(U_i \setminus \bigcup_{j=1}^{N} K_j \right) > 0;$$

each set inside the minimum is open and nonempty, since a finite union of closed nowhere dense sets is closed and nowhere dense. Choose

$$0 < \alpha_{N+1} \le \min\{\alpha_N/2,\ 2^{-N-2} m_N\},$$

and continue. In particular,

$$\sum_{j>N} \alpha_j \le 2\alpha_{N+1} \le 2^{-N-1} m_N.$$

Step 2: the concentrating part. The function $Q = \sum_j \alpha_j q_j$ is nonnegative and integrable. We claim that its essential infimum on every nonempty open set is zero. Since every nonempty open set contains some U_i, it suffices to consider U_i. Take $N \ge i$ and remove $K_1, \dots, K_N$ from U_i. On the remaining set, whose volume is at least m_N, the first N summands of Q vanish. By Markov's inequality, the set where the tail $\sum_{j>N} \alpha_j q_j$ exceeds 2^{-N} has volume at most

$$2^N \sum_{j>N} \alpha_j \int_X q_j \, dV \le m_N/2.$$

Hence $Q \le 2^{-N}$ on a subset of U_i of positive volume. Letting $N \to \infty$ proves the claim.

Step 3: $b_h(f) = 0$. Set $M = \int_X (1+Q)\, dV$ and $f = (1+Q)/M$, a probability density with $f \ge 1/M$. If $\mathrm{MA}_h^{\mathrm{ext}}(u) \ge a f \, dV$ with $a > 0$, then $\mathrm{MA}_h^{\mathrm{ext}}(u) \ge (a\alpha_j/M) q_j \, dV$, and Lemma 15.26.1 gives

$$\frac{a\alpha_j}{M} \le c_h(q_j) \le \alpha_j 2^{-j} \quad \text{for every } j.$$

This is impossible. Hence $b_h(f) = 0$ by Theorem 15.27.1.

Step 4: $C_h^{\mathrm{sm}}(f) > 0$. Let g be a smooth strictly positive probability density and put $R = \|g/f\|_{L^\infty}$. Then $g \le R(1+Q)/M$ almost everywhere, and this forces $g \le R/M$

everywhere. Otherwise, by continuity, $g > R(1+\varepsilon)/M$ on a nonempty open set for some $\varepsilon > 0$, while $Q < \varepsilon$ on a subset of positive volume of this set by Step 2, a contradiction. The order and homogeneity of the scalar in Lemma 15.26.1 now give $c_h(g) \geq c_h(R/M) = (M/R)c_h(1)$, that is,

$$Rc_h(g) \geq Mc_h(1).$$

The constant density $g = 1$ attains this value, since $\operatorname{ess\,inf}_X Q = 0$ gives $\|1/f\|_{L^\infty} = M$. Thus $C_h^{\mathrm{sm}}(f) = Mc_h(1) > 0$, as required. □

The point is that a smooth test g with $\|g/f\|_{L^\infty} \leq R$ is controlled by the essential infimum of f on open sets. It therefore cannot detect an increase of f that leaves this essential infimum unchanged on every open set, whereas bounded measurable tests below f do detect it. The theorem concerns only the existence of subsolutions. Exact solutions of $\mathrm{MA}_h^{\mathrm{ext}}(u) = b_h(f)f\,\mathrm{d}V$, the singularity class in which they should be sought, and compactness of their normalized solutions are further problems; see Problem 35.1.1.

15.28 Gauduchon tests for integrable densities

For a smooth strictly positive probability density, the nth root of the normalization scalar is an infimum of Gauduchon quotients by Theorem 10.8.1. We extend this dual formula to the threshold of an arbitrary integrable density. A uniform bound for all Gauduchon tests in a single L^r norm allows the passage from smooth to bounded densities, and monotone truncation then reaches all integrable densities, including those with zero threshold. The formula is most naturally written for the nth root of the density.

Recall that a Hermitian form γ is *Gauduchon* if $\mathrm{dd^c}(\gamma^{n-1}) = 0$ (Definition 4.1.1); in dimension one every Hermitian form is Gauduchon. For a Gauduchon form γ and a nonnegative measurable function p, put

$$L_{h,\gamma}(p) = \frac{\displaystyle\int_X p(\mathrm{d}V/\gamma^n)^{1/n}\gamma^n}{\displaystyle\int_X h\wedge\gamma^{n-1}} = \int_X p\,a_\gamma\,\mathrm{d}V, \qquad a_\gamma = \frac{(\gamma^n/\mathrm{d}V)^{(n-1)/n}}{\displaystyle\int_X h\wedge\gamma^{n-1}}.$$

Since the coefficient a_γ is smooth and strictly positive, each individual test $L_{h,\gamma}$ is finite on L^n; the supremum over γ may be infinite. In dimension one the test is $\int_X p\,\mathrm{d}V/\int_X h$. For real or complex $p \in L^n(\mathrm{d}V)$, define the *root functional*

$$\rho_h(p) = b_h(|p|^n)^{-1/n} \quad (p \neq 0), \qquad \rho_h(0) = 0,$$

with the convention $0^{-1/n} = +\infty$.

Theorem 15.28.1 *For every real or complex $p \in L^n(\mathrm{d}V)$,*

$$\rho_h(p) = \sup_{\gamma \text{ Gauduchon}} L_{h,\gamma}(|p|). \tag{15.82}$$

Equivalently, for every nonzero nonnegative $f \in L^1(\mathrm{d}V)$,

$$b_h(f)^{1/n} = \inf_{\gamma \text{ Gauduchon}} \frac{\displaystyle\int_X h \wedge \gamma^{n-1}}{\displaystyle\int_X (f\,\mathrm{d}V/\gamma^n)^{1/n}\gamma^n}. \tag{15.83}$$

Both sides of (15.82) *may be $+\infty$, and both sides of* (15.83) *may vanish.*

Proof It suffices to prove (15.82) for $p \geq 0$, since both sides depend only on $|p|$; the case $p = 0$ is immediate.

Step 1: smooth strictly positive p. Apply Theorem 10.8.1, with background h, to the probability measure $\mu = p^n \mathrm{d}V/Z$, where $Z = \int_X p^n\,\mathrm{d}V$. By homogeneity (Lemma 15.26.1) its scalar is $c_h(\mu) = Z c_h(p^n)$, while the denominator of each Gauduchon quotient is $Z^{-1/n}\int_X p(\mathrm{d}V/\gamma^n)^{1/n}\gamma^n$. The factors of Z cancel, and the dual formula becomes

$$c_h(p^n)^{-1/n} = \sup_\gamma L_{h,\gamma}(p).$$

Since $b_h(p^n) = c_h(p^n)$ by Proposition 15.27.2, this proves (15.82) for such p.

Step 2: a uniform bound for the tests. The lower-density estimate of Theorem 11.3.1, with exponent two, gives a constant $m_h > 0$ such that

$$c_h(p^n)\|p^n\|_{L^2} \geq m_h$$

for smooth strictly positive p. By Step 1, therefore,

$$L_{h,\gamma}(p) \leq m_h^{-1/n}\|p\|_{L^{2n}} \quad \text{for every } \gamma. \tag{15.84}$$

Approximating nonnegative functions in L^{2n} by smooth strictly positive ones extends this bound to all nonnegative $p \in L^{2n}$. Since $a_\gamma > 0$, duality gives $\|a_\gamma\|_{L^{2n/(2n-1)}} \leq m_h^{-1/n}$ for every γ. In particular, $p \mapsto \sup_\gamma L_{h,\gamma}(p)$ is Lipschitz on the cone of nonnegative functions in L^{2n}, with the same constant.

Step 3: bounded p. Let $p \geq 0$ be bounded and nonzero, and approximate it in L^{2n} by smooth strictly positive functions p_j. By Hölder's inequality, $p_j^n \to p^n$ in L^2, so Lemma 15.26.2 gives

$$c_h(p_j^n) \longrightarrow c_h(p^n) > 0.$$

By the Lipschitz bound derived from (15.84),

$$\sup_\gamma L_{h,\gamma}(p_j) \longrightarrow \sup_\gamma L_{h,\gamma}(p).$$

Passing to the limit in the identity of Step 1 for p_j, and using $b_h(p^n) = c_h(p^n)$ (Proposition 15.27.2), we obtain (15.82) for bounded p.

Step 4: general p. For nonnegative $p \in L^n$, put $p_k = \min(p, k)$. Then $p_k^n = \min(p^n, k^n)$ are truncations of p^n, so the definition of b_h and Step 3 give

$$\rho_h(p) = \lim_{k\to\infty} c_h(p_k^n)^{-1/n} = \sup_k \sup_\gamma L_{h,\gamma}(p_k).$$

For each fixed γ, monotone convergence gives $\sup_k L_{h,\gamma}(p_k) = L_{h,\gamma}(p)$. Interchanging the two suprema proves (15.82). Finally, every denominator in (15.83) is positive; taking reciprocals in (15.82) with $p = f^{1/n}$ proves (15.83), including the case $b_h(f) = 0$. □

For a nonsmooth density we do not claim that the infimum in (15.83) is attained by a Gauduchon form; attainment by Gauduchon currents is studied in Section 27.8.

For $n \geq 2$, the coefficient of the test has norm

$$\|a_\gamma\|_{L^{n/(n-1)}} = \frac{(\int_X \gamma^n)^{(n-1)/n}}{\int_X h \wedge \gamma^{n-1}}$$

in the exponent dual to n. By (10.12) in Theorem 10.8.2, positive lower mass is therefore equivalent to a uniform bound for these norms, that is, for all tests on L^n simultaneously. Without positive lower mass, Step 2 of the proof still bounds the coefficients uniformly in the weaker norm of $L^{2n/(2n-1)}$, and this suffices for the proof.

Proposition 15.28.1 *Let f be a nonzero nonnegative function in $L^1(\mathrm{d}V)$ and let $p \in L^n(\mathrm{d}V)$ be real or complex. If h_1, h_2 are Hermitian forms on X and $m h_1 \leq h_2 \leq M h_1$ for positive constants m, M, then*

$$m^n b_{h_1}(f) \leq b_{h_2}(f) \leq M^n b_{h_1}(f), \qquad M^{-1}\rho_{h_1}(p) \leq \rho_{h_2}(p) \leq m^{-1}\rho_{h_1}(p).$$

If $h_2 - h_1 = \mathrm{dd}^{\mathrm{c}} a$ for a smooth real function a, then $b_{h_1}(f) = b_{h_2}(f)$ and $\rho_{h_1}(p) = \rho_{h_2}(p)$.

Proof For every Gauduchon form γ, the numerators in (15.83) satisfy

$$m \int_X h_1 \wedge \gamma^{n-1} \leq \int_X h_2 \wedge \gamma^{n-1} \leq M \int_X h_1 \wedge \gamma^{n-1},$$

while the denominators do not depend on the background. Taking the infimum over γ and then nth powers proves the bounds for the thresholds; taking reciprocal nth roots gives the bounds for ρ, with the stated conventions at zero and infinity. If $h_2 - h_1 = \mathrm{dd}^{\mathrm{c}} a$, the numerators agree by Stokes' theorem, since $\mathrm{dd}^{\mathrm{c}}(\gamma^{n-1}) = 0$. □

Any two Hermitian metrics on the compact manifold X are comparable. Hence the positivity of $b_h(f)$, and the set of real or complex $p \in L^n$ with $\rho_h(p) < +\infty$, do not depend on the choice of h. The dependence on the reference volume is equally

simple. If $\mathrm{d}V$ is replaced by $\mathrm{d}V' = \sigma\,\mathrm{d}V$ with a smooth function $\sigma > 0$, the measure $f\,\mathrm{d}V$ has density $f' = f/\sigma$ with respect to $\mathrm{d}V'$, and Theorem 15.27.1 shows that the threshold of f' relative to $\mathrm{d}V'$ equals $b_h(f)$. Correspondingly, the map $p \mapsto \sigma^{-1/n} p$ from $L^n(\mathrm{d}V)$ to $L^n(\mathrm{d}V')$ preserves both the L^n norm and the root functional.

15.29 Exact equations on products with zero lower mass

Positive lower mass guarantees solvability for all measures, but an individual density can have a positive threshold and an exact equation even when the lower mass vanishes. We construct such examples on products $X_0 \times Y$ of a compact Hermitian manifold (X_0, h_0) with $\nu_-(h_0) = 0$ and a closed Kähler factor Y. On $Y = \mathbb{P}^k$, a potential with a sublogarithmic pole, pulled back from the projective factor, solves an explicit equation whose density is integrable but belongs to no L^p with $p > 1$. The coefficient is computed from the normalization scalars of bounded truncations, the equation follows from the product structure of the Monge–Ampère measure, and the product background again has zero lower mass.

Throughout this section, (X_0, h_0) is a connected compact Hermitian manifold of dimension $m \geq 1$, (Y, κ) is a connected compact Kähler manifold of dimension $k \geq 1$ normalized by $\int_Y \kappa^k = 1$, and $X = X_0 \times Y$, a compact complex manifold of dimension $N = m + k$ with the Hermitian background $H = h_0 + \kappa$. Forms and functions on the factors are identified with their pullbacks to X. Put

$$V_0 = \int_{X_0} h_0^m, \qquad C = \binom{N}{m}, \qquad \mu = V_0^{-1} h_0^m \wedge \kappa^k, \qquad R = \frac{\mu}{\mathrm{d}V},$$

where $\mathrm{d}V$ is any smooth positive probability volume form on X. Then μ is a smooth positive probability volume form, and R is smooth and bounded above and below by positive constants. The normalization scalar c_H and the threshold b_H are those of Sections 15.26 and 15.27, now on the N-dimensional manifold X with background H.

Lemma 15.29.1 (1) *For every $v \in \mathrm{PSH}(Y, \kappa)$,*

$$\mathrm{MA}_H^{\mathrm{ext}}(v) = C h_0^m \wedge \mathrm{MA}_\kappa^{\mathrm{ext}}(v). \tag{15.85}$$

(2) *For every nonzero nonnegative $g \in L^1(Y, \kappa^k)$, with $M_g = \int_Y g\kappa^k$,*

$$b_H(Rg) = \frac{CV_0}{M_g},$$

and $c_H(Rg) = CV_0/M_g$ if g is bounded.

(3) *If $\nu_-(h_0) = 0$, then $\nu_-(H) = 0$ and H is not closed.*

Proof (1) Forms and currents of bidegree (j, j) pulled back from X_0 vanish for $j > m$, and those pulled back from Y vanish for $j > k$. For bounded v, Proposition 1.7.1

and Lemma 1.7.2 and the binomial expansion therefore give $\mathrm{MA}_H(v) = Ch_0^m \wedge \mathrm{MA}_\kappa(v)$. For general v, apply this to $\max(v, -j)$, multiply by the pullback of $\mathbf{1}_{\{v>-j\}}$, and let $j \to \infty$ as in the definition (7.1) of the extended measure.

(2) Suppose first that g is bounded. Theorem 11.1.1 on Y, with reference volume κ^k, gives a continuous $v \in \mathrm{PSH}(Y, \kappa)$ and $c > 0$ with $\mathrm{MA}_\kappa(v) = cg\kappa^k$. Since κ is closed, Stokes' theorem gives $\int_Y \mathrm{MA}_\kappa(v) = 1$, hence $c = 1/M_g$, and (1) gives

$$\mathrm{MA}_H(v) = \frac{C}{M_g} h_0^m \wedge g\kappa^k = \frac{CV_0}{M_g} Rg \, \mathrm{d}V.$$

Uniqueness of the scalar gives $c_H(Rg) = CV_0/M_g$. For general g, write $a = \min_X R$ and $A = \max_X R$. The truncations of Rg that define $b_H(Rg)$ in (15.80) satisfy

$$R \min(g, j/A) \le \min(Rg, j) \le R \min(g, j/a).$$

By the bounded case and the monotonicity of the scalar in Lemma 15.26.1,

$$\frac{CV_0}{M_{\min(g,j/a)}} \le c_H(\min(Rg, j)) \le \frac{CV_0}{M_{\min(g,j/A)}}.$$

By monotone convergence, both bounds tend to CV_0/M_g.

(3) For bounded $w \in \mathrm{PSH}(X_0, h_0)$, the same expansion gives $\mathrm{MA}_H(w) = C\,\mathrm{MA}_{h_0}(w) \wedge \kappa^k$, a measure of mass $C \int_{X_0} \mathrm{MA}_{h_0}(w)$. Pulling back bounded potentials whose masses on X_0 tend to zero gives $v_-(H) = 0$. If h_0 were closed, Stokes' theorem would give every bounded h_0-plurisubharmonic function the mass $V_0 > 0$, contradicting $v_-(h_0) = 0$. Hence h_0, and therefore H, is not closed. □

Until Section 15.29.1, let $Y = \mathbb{P}^k$, with κ the Fubini–Study form normalized by $\int_Y \kappa^k = 1$. For $0 < \alpha < 1$ define, on Y,

$$\psi([z_0 : \cdots : z_k]) = \log \frac{\sum_{i=1}^k |z_i|^2}{\sum_{i=0}^k |z_i|^2}, \qquad u_\alpha = 1 - (1 - \psi)^\alpha.$$

The function ψ is smooth on $Y \setminus \{o\}$, where $o = [1 : 0 : \cdots : 0]$, and tends to $-\infty$ at o; moreover $\psi \le 0$, with equality exactly on the hyperplane $\{z_0 = 0\}$. Thus $u_\alpha \le 0$, and the pole set of u_α is $\{o\}$.

Lemma 15.29.2 *The function u_α is κ-plurisubharmonic with supremum zero, and*

$$\mathrm{MA}_\kappa^{\mathrm{ext}}(u_\alpha) = q_\alpha \kappa^k, \qquad \int_Y q_\alpha \kappa^k = 1.$$

The density q_α is smooth and strictly positive on $Y \setminus \{o\}$, belongs to $L^1(\kappa^k)$, and belongs to no $L^p(\kappa^k)$ with $p > 1$. If we set $q_\alpha(o) = +\infty$, then every truncation $\min(q_\alpha, s)$, $s > 0$, is strictly positive and continuous on Y.

Proof In homogeneous coordinates, $\kappa + \mathrm{dd^c}\psi = \mathrm{dd^c} \log \sum_{i=1}^k |z_i|^2 \ge 0$ on $Y \setminus \{o\}$. The function $\chi(t) = 1 - (1 - t)^\alpha$, $t \le 0$, is increasing and convex, with $0 < \chi' \le \alpha$.

Hence, off o,

$$\kappa + \mathrm{dd^c} u_\alpha \geq (1 - \chi'(\psi))\kappa + \chi'(\psi)(\kappa + \mathrm{dd^c}\psi) \geq (1-\alpha)\kappa.$$

Since u_α is bounded above, it extends across o as a κ-plurisubharmonic function, with value $-\infty$ at o. On the hyperplane $\{z_0 = 0\}$ we have $\psi = 0$, hence $u_\alpha = 0$, so $\sup_Y u_\alpha = 0$.

To compute the measure, work in the affine chart $z_0 = 1$ and put $r = \sum_{i=1}^k |z_i|^2$. On this chart $\kappa = \mathrm{dd^c}\log(1+r)$ and $\kappa + \mathrm{dd^c} u_\alpha = \mathrm{dd^c} F$, where $s(r) = 1 - \log(r/(1+r))$ and

$$F(r) = \log(1+r) + 1 - s(r)^\alpha, \qquad Q(r) = rF'(r) = \frac{r + \alpha s(r)^{\alpha-1}}{1+r}.$$

For a radial function, the Levi form has the tangential eigenvalue $F'(r) = Q(r)/r$, with multiplicity $k-1$, and the radial eigenvalue

$$Q'(r) = \frac{1 - \alpha s(r)^{\alpha-1} + \alpha(1-\alpha) r^{-1} s(r)^{\alpha-2}}{(1+r)^2} > 0.$$

Integrating in the radial variable gives $\int_{\{a<r<b\}} (\mathrm{dd^c} F)^k = Q(b)^k - Q(a)^k$; the constant of proportionality is fixed by the case $F = \log(1+r)$, for which $(\mathrm{dd^c} F)^k = \kappa^k$ has mass one on the chart and $Q(r) = r/(1+r)$. Since again $Q(0) = 0$ and $Q(\infty) = 1$, the punctured chart has mass one. The complement of the chart is a hyperplane, which has zero volume, and the non-pluripolar measure does not charge o. Hence $\int_Y q_\alpha \kappa^k = 1$.

Relative to κ^k, whose tangential and radial eigenvalues are $1/(1+r)$ and $1/(1+r)^2$, the density on this chart is

$$q_\alpha(r) = (1+r)^{k+1} (Q(r)/r)^{k-1} Q'(r) \asymp r^{-k} (1 - \log r)^{-1-k(1-\alpha)} \quad (r \downarrow 0).$$

Near $r = 0$ the smooth volume is comparable to $r^{k-1}\mathrm{d}r$ in the radial variable, so q_α is integrable. For $p > 1$, the substitution $L = 1 - \log r$ turns the integral of q_α^p into the integral of $\mathrm{e}^{k(p-1)L}$ times a power of L, which diverges. The same asymptotics show that $q_\alpha(r) \to +\infty$ uniformly as $r \downarrow 0$, so every truncation $\min(q_\alpha, s)$ is continuous at o. Finally, $q_\alpha \geq (1-\alpha)^k > 0$ by the lower bound $\kappa + \mathrm{dd^c} u_\alpha \geq (1-\alpha)\kappa$. □

Theorem 15.29.1 *For every $0 < \alpha < 1$, the function $f = R q_\alpha$ is a probability density in $L^1(\mathrm{d}V)$, belonging to no $L^p(\mathrm{d}V)$ with $p > 1$, and*

$$b_H(f) = CV_0 > 0, \qquad \mathrm{MA}_H^{\mathrm{ext}}(u_\alpha) = b_H(f) f \, \mathrm{d}V.$$

The potential u_α, pulled back to X, has supremum zero and finite mass CV_0. It belongs to $\mathcal{E}_{\mathrm{env}}(X, H, 0)$, and $P_{\delta H}(u_\alpha) \not\equiv -\infty$ for every $0 < \delta < 1$. If $v_-(h_0) = 0$, then H is not closed and $v_-(H) = 0$.

Proof *Step 1: the equation.* By Lemma 15.29.2, $\int_X f\,\mathrm{d}V = \int_X q_\alpha\mu = \int_Y q_\alpha\kappa^k = 1$. Part (2) of Lemma 15.29.1, with $g = q_\alpha$ and $M_g = 1$, gives $b_H(f) = CV_0$, and part (1) gives

$$\mathrm{MA}_H^{\mathrm{ext}}(u_\alpha) = Ch_0^m \wedge q_\alpha\kappa^k = CV_0\, q_\alpha\mu = CV_0 f\,\mathrm{d}V,$$

a measure of mass CV_0. Since R is bounded above and below by positive constants, f has the integrability properties of q_α. The last assertion is part (3) of the lemma.

Step 2: the envelopes. Choose $\alpha < \beta < 1$. For $b \geq 1$ and $\varepsilon > 0$ with $\varepsilon\beta \leq 1$, the number

$$K_b = \sup_{s\geq 1}\{bs^\alpha - \varepsilon s^\beta\} - b$$

is finite, since $\beta > \alpha$. The function $g_b = -\varepsilon(1-\psi)^\beta - K_b$ on Y satisfies $\kappa + \mathrm{dd^c} g_b \geq (1-\varepsilon\beta)\kappa \geq 0$, by the chain-rule computation in the proof of Lemma 15.29.2, applied to the increasing convex function $t \mapsto -\varepsilon(1-t)^\beta$, whose derivative is at most $\varepsilon\beta$ for $t \leq 0$. With $L = 1-\psi \geq 1$, the inequality $g_b \leq bu_\alpha$ reads $bL^\alpha - \varepsilon L^\beta - b \leq K_b$, so it holds on $Y\setminus\{o\}$, and also at o, where both sides are $-\infty$. Thus the pullback of g_b belongs to $\mathrm{PSH}(X,H)$ and lies below bu_α. Since $b \geq 1$ is arbitrary, $u_\alpha \in \mathcal{E}_{\mathrm{env}}(X,H,0)$ by Definition 14.4.1. For $0 < \delta < 1$, take $b = 1$ and $\varepsilon\beta \leq \delta$. Then g_1 is δH-plurisubharmonic and lies below u_α, so $P_{\delta H}(u_\alpha) \geq g_1$ is not identically $-\infty$. □

For the elliptic threefold of Theorem 27.18.1 as X_0 and $Y = \mathbb{P}^2$, Corollary 27.18.6 computes the threshold $b_H(f) = 240\pi$. On these products, the radial flux also identifies the limit of the supremum-normalized exponential solutions as the exponent tends to zero (Theorem 15.35.1); a first moment of the limiting measure determines whether the solutions shifted by $-\lambda^{-1}\log b_H(f)$ have a finite limit (Proposition 15.35.1). Problem 35.1.1 asks for an exact equation for every integrable density with positive threshold.

15.29.1 Closed factors and arbitrary integrable densities

The preceding examples use an explicit potential on projective space. On a closed Kähler factor, Stokes' theorem instead bounds the non-pluripolar mass, and this bound makes every threshold subsolution on the factor exact. This gives equations for arbitrary integrable densities on that factor, with no assumption that they are bounded below.

In this subsection (Y,κ) is again an arbitrary connected compact Kähler manifold with $\int_Y \kappa^k = 1$, with the notation X, N, H, C, V_0, μ and R introduced at the beginning of the section, and we put $c_0 = CV_0$.

Theorem 15.29.2 *For every nonzero nonnegative $g \in L^1(Y,\kappa^k)$, with $M_g = \int_Y g\kappa^k$, one has*

$$b_H(Rg) = \frac{c_0}{M_g}.$$

There is $v \in \mathrm{PSH}(Y,\kappa)$ with $\sup_Y v = 0$ and $\int_Y \mathrm{MA}_\kappa^{\mathrm{ext}}(v) = 1$ whose pullback u satisfies

$$\mathrm{MA}_H^{\mathrm{ext}}(u) = \frac{c_0}{M_g} Rg\,\mathrm{d}V, \qquad \int_X \mathrm{MA}_H^{\mathrm{ext}}(u) = c_0.$$

Proof The formula for $b_H(Rg)$ is part (2) of Lemma 15.29.1. By Proposition 15.27.2, applied on Y with the Kähler form κ and the reference volume κ^k, $b_\kappa(g) = 1/M_g$. Thus Theorem 15.27.1 on Y, followed by subtraction of a constant, gives $v \in \mathrm{PSH}(Y,\kappa)$ with supremum zero and

$$\mathrm{MA}_\kappa^{\mathrm{ext}}(v) \geq (g/M_g)\kappa^k.$$

For $v_j = \max(v,-j)$, Stokes' theorem for the closed form κ gives $\int_Y \mathrm{MA}_\kappa(v_j) = 1$. Since $\mathrm{MA}_\kappa^{\mathrm{ext}}(v)$ is the increasing limit of the measures $\mathbf{1}_{\{v>-j\}}\mathrm{MA}_\kappa(v_j)$ in (7.1), it follows that $\int_Y \mathrm{MA}_\kappa^{\mathrm{ext}}(v) \leq 1$. The prescribed lower measure has mass one, so equality holds both in mass and in the measure inequality. Part (1) of Lemma 15.29.1 now gives

$$\mathrm{MA}_H^{\mathrm{ext}}(u) = Ch_0^m \wedge \mathrm{MA}_\kappa^{\mathrm{ext}}(v) = \frac{c_0}{M_g} Rg\,\mathrm{d}V,$$

a measure of mass c_0. □

When $v_-(h_0) = 0$, the background H has zero lower mass and is not closed, by part (3) of Lemma 15.29.1. For $Y = \mathbb{P}^k$ and $g = q_\alpha$, the potential u_α of Theorem 15.29.1 is such a solution. For the elliptic fivefold of Corollary 27.18.6, $c_0 = 240\pi$; thus, for every probability density g on $\mathbb{P}^2$, the density Rg has threshold 240π and admits a threshold solution of finite mass. The theorem concerns factor densities; it does not give an exact equation for an arbitrary density on X or identify limits of normalized exponential solutions.

15.30 Singular measures on products with a curve

The closed-factor equations of Section 15.29.1 concern integrable densities. A curve factor also allows a direct test on a measure: its logarithmic potential determines an exact equation, and continuity of that potential supplies the energy control needed for every positive exponent. The resulting product measure may be singular with respect to smooth volume, even when the solution is continuous.

Let $Y = \mathbb{P}^1$ and let κ be the Fubini–Study form of mass one. For the chordal distance $d_{\mathrm{ch}}([x_0 : x_1],[z_0 : z_1]) = |x_0z_1 - x_1z_0|/(|x|\,|z|)$, under which antipodal points have distance one, put

$$G(x,y) = \log d_{\mathrm{ch}}(x,y)^2 + 1, \qquad K(x,y) = 1 - G(x,y) \geq 0.$$

For a Borel probability μ on Y, define

$$Q_\mu(x) = \int_Y G(x,y)\,\mathrm{d}\mu(y). \tag{15.86}$$

We say that μ is non-polar if it gives every polar Borel set mass zero.

15.30.1 The logarithmic potential and its measure trace

Lemma 15.30.1 *For every probability μ, the function Q_μ is κ-plurisubharmonic, has κ-mean zero, and satisfies*

$$\kappa + \mathrm{dd}^{\mathrm{c}} Q_\mu = \mu, \qquad Q_\mu \le 1.$$

If μ is non-polar, then $\mathrm{MA}_\kappa^{\mathrm{ext}}(Q_\mu) = \mu$. Every $p \in \mathrm{PSH}(Y,\kappa)$ with $\mathrm{MA}_\kappa^{\mathrm{ext}}(p) \ge \mu$ is of the form $Q_\mu + C$.

Proof The Poincaré–Lelong formula gives $\mathrm{dd}^{\mathrm{c}}_x G(x,y) = \delta_y - \kappa$. After a rotation sending y to $[1:0]$, the variable $t = d_{\mathrm{ch}}(x,y)^2$ is uniformly distributed on $[0,1]$ for the measure κ. Thus

$$\int_Y G(x,y)\kappa(x) = \int_0^1 (\log t + 1)\,\mathrm{d}t = 0, \qquad \sup_y \int_Y |G(x,y)|\kappa(x) < \infty.$$

The continuous potentials $\int \max(G(x,y), -L)\,\mathrm{d}\mu(y)$ are κ-plurisubharmonic by Lemma 1.3.1 and averaging, and decrease to Q_μ. By Tonelli's theorem, Q_μ is integrable with κ-mean zero. As a decreasing limit of these potentials, it is κ-plurisubharmonic, and its values are the canonical ones. Fubini's theorem for distributions proves the curvature identity.

In degree one the extended measure of p is

$$\mathrm{MA}_\kappa^{\mathrm{ext}}(p) = \mathbf{1}_{\{p > -\infty\}}(\kappa + \mathrm{dd}^{\mathrm{c}} p).$$

The pole set of Q_μ is polar, so non-polarity gives its asserted measure. For any p, distributional Stokes gives the full current $\kappa + \mathrm{dd}^{\mathrm{c}} p$ mass one, and its extended measure has mass at most one. Domination of the probability μ therefore forces both measures to equal μ. The distribution $p - Q_\mu$ is harmonic on the compact connected curve, hence constant, and canonical equality gives the assertion everywhere. □

For real Sobolev functions on Y use the Dirichlet form

$$E(p,q) = \int_Y \mathrm{d}p \wedge \mathrm{d}^{\mathrm{c}} q.$$

It is symmetric and nonnegative on the diagonal. The space $H^1(Y)$ and its norm are taken for the Fubini–Study Riemannian metric.

Lemma 15.30.2 *Suppose $Q = Q_\mu$ is finite and continuous. Then μ is non-polar, $Q \in H^1(Y)$, and smooth restriction extends to a continuous linear map*

$$\operatorname{Tr}_\mu : H^1(Y) \longrightarrow L^1(\mu).$$

For every $g \in H^1(Y)$,

$$\int_Y \operatorname{Tr}_\mu g \, d\mu = \int_Y g\kappa - E(g, Q). \tag{15.87}$$

For $g \in \operatorname{PSH}(Y, \kappa) \cap H^1(Y)$ this trace agrees μ-almost everywhere with its canonical values. The trace commutes with Lipschitz real compositions.

Proof Continuity gives the uniform kernel bound

$$A_\mu := \sup_x \int_Y K(x, y) \, d\mu(y) = 1 - \inf_Y Q < \infty. \tag{15.88}$$

To prove non-polarity, let $s \not\equiv -\infty$ be subharmonic on a coordinate disk D, and let $D' \Subset D'' \Subset D$ be concentric subdisks; we show that $\{s = -\infty\} \cap D'$ has μ-mass zero. The logarithmic potential of the finite measure $\mathbf{1}_{D''} dd^c s$ has the same dd^c as s on D'', so their difference is harmonic there. This is the local Riesz decomposition; it also follows directly from the distributional fundamental solution and harmonic regularity. On compact subdisks the kernel $-\log|z - a|^2$ differs from $K(z, a)$ by a bounded smooth term. Symmetry of K, Tonelli's theorem and (15.88) make the logarithmic potential integrable against μ on D', and the harmonic remainder is bounded there. Consequently $\{s = -\infty\} \cap D'$ has μ-mass zero. A countable disk cover proves the claim for polar Borel sets.

The group $\mathrm{SU}(2)$ acts on Y by Fubini–Study isometries. Average with smooth nonnegative conjugation-invariant kernels of mass one supported near the identity. This gives smooth Q_ϵ, probability measures μ_ϵ, and

$$Q_\epsilon \longrightarrow Q \text{ uniformly}, \qquad dd^c Q_\epsilon = \mu_\epsilon - \kappa, \qquad \int_Y Q_\epsilon \kappa = 0.$$

For the smooth functions Q_ϵ, Stokes' theorem gives $E(Q_\epsilon, Q_\epsilon) = -\int Q_\epsilon \, d\mu_\epsilon \le \|Q\|_\infty$. Poincaré's inequality and weak Hilbert compactness show that $Q \in H^1$. Its distributional curvature identity gives (15.87) first for smooth g. Apply this identity to $\sqrt{g^2 + \epsilon^2}$, whose gradient has magnitude at most $|dg|$. Letting $\epsilon \downarrow 0$ yields

$$\int_Y |g| \, d\mu \le \|g\|_{L^1(\kappa)} + E(g, g)^{1/2} E(Q, Q)^{1/2} \le C_\mu \|g\|_{H^1}.$$

Completion defines the trace and extends the identity.

These group averages converge strongly in H^1: rotations act continuously and isometrically on that space, and the averaging kernels approach the identity. They also converge pointwise to every canonical quasi-subharmonic function. Indeed their induced averages about a point are radial by conjugation invariance. In a coordinate

centered at that point they are positive averages of circular means; after a smooth quadratic correction, the submean inequality and upper semicontinuity give the required limit. The correction error tends to zero, and upper semicontinuity gives the limit also at a pole. For a κ-plurisubharmonic H^1 function, strong convergence and the trace bound give $L^1(\mu)$ convergence, while a subsequence converges μ-almost everywhere to the trace. Non-polarity makes the canonical value finite there and identifies the two. Finally, the Sobolev chain rule and smooth approximation show that Lipschitz compositions are continuous in H^1; the corresponding $L^1(\mu)$ limits prove the composition assertion. □

15.30.2 Exponential equations and uniform selection on the curve

Theorem 15.30.1 *Suppose $Q = Q_\mu$ is finite and continuous. For every $\lambda > 0$ there is a unique bounded $w_\lambda \in \mathrm{PSH}(Y, \kappa)$ satisfying*

$$\kappa + \mathrm{dd}^{\mathrm{c}} w_\lambda = \mathrm{e}^{\lambda w_\lambda} \mu. \tag{15.89}$$

Since $\kappa + \mathrm{dd}^{\mathrm{c}} w_\lambda$ has mass one, $\int_Y \mathrm{e}^{\lambda w_\lambda}\, \mathrm{d}\mu = 1$. The solution is continuous and obeys

$$Q - \sup_Y Q \le w_\lambda \le Q - \inf_Y Q. \tag{15.90}$$

The entire family satisfies, as $\lambda \downarrow 0$,

$$\left\| w_\lambda - Q + \int_Y Q \, \mathrm{d}\mu \right\|_\infty = O(\lambda).$$

The limit is the greatest $p \in \mathrm{PSH}(Y, \kappa)$ with $\mathrm{MA}_\kappa^{\mathrm{ext}}(p) \ge \mu$ and $\int_Y p \, \mathrm{d}\mu \le 0$.

Proof *Variational existence.* On the real Hilbert space

$$\mathcal{H} = \{ q \in H^1(Y) : \int_Y q \kappa = 0 \}$$

consider the extended functional

$$J_\lambda(q) = \tfrac{1}{2} E(q, q) + \lambda^{-1} \log \int_Y \mathrm{e}^{\lambda \operatorname{Tr}_\mu q} \, \mathrm{d}\mu.$$

It is finite at zero. Jensen's inequality and (15.87) give

$$J_\lambda(q) \ge \tfrac{1}{2} E(q, q) - E(q, Q),$$

so it is coercive and bounded below. Strong H^1 convergence gives $L^1(\mu)$ convergence of traces. A subsequence realizing the lower limit converges almost everywhere for μ, and Fatou's lemma proves strong lower semicontinuity of the exponential term.

Jensen excludes a zero integral or a logarithm tending to $-\infty$. Hölder's inequality makes the logarithmic term convex, while the Dirichlet term is strictly convex on $\mathcal{H}$. Separation of a closed convex epigraph makes J_λ weakly lower semicontinuous. Weak compactness therefore gives a unique minimizer q_λ, with

$$0 < Z_\lambda := \int_Y \mathrm{e}^{\lambda \operatorname{Tr}_\mu q_\lambda}\, \mathrm{d}\mu < \infty.$$

Variation in a smooth mean-zero direction g is justified by its boundedness, which dominates the differentiated integrand by a constant times the integrable exponential. Put $\rho_\lambda = \mathrm{e}^{\lambda \operatorname{Tr}_\mu q_\lambda}/Z_\lambda$. The Euler equation is

$$E(q_\lambda, g) + \int_Y g\rho_\lambda \,\mathrm{d}\mu = 0, \qquad \mathrm{dd}^{\mathrm{c}} q_\lambda = \rho_\lambda \mu - \kappa.$$

The second identity follows for arbitrary smooth tests by subtracting their κ-mean; $\int \rho_\lambda \,\mathrm{d}\mu = 1$ fixes the constant term. The minimizer has a canonical κ-plurisubharmonic representative. Its mean-zero Green representation and Lemma 15.30.2 give

$$q_\lambda(x) = \int_Y G(x,y)\rho_\lambda(y)\,\mathrm{d}\mu(y) \le 1, \qquad \rho_\lambda \le \mathrm{e}^{\lambda}/Z_\lambda \quad \mu\text{-a.e.}$$

The kernel bound then gives $q_\lambda \ge 1 - (\mathrm{e}^{\lambda}/Z_\lambda)A_\mu$.

To obtain continuity, the continuous functions $\int \min(K(x,y), L)\,\mathrm{d}\mu(y)$ increase to the continuous function $1 - Q(x)$. Dini's theorem gives

$$\sup_x \int_Y (K(x,y) - L)_+ \,\mathrm{d}\mu(y) \longrightarrow 0. \tag{15.91}$$

Multiplication by the bounded density ρ_λ preserves this estimate. Its truncated Green potentials converge uniformly, so q_λ is continuous. Now

$$w_\lambda = q_\lambda - \lambda^{-1} \log Z_\lambda$$

solves (15.89), since $\rho_\lambda = \mathrm{e}^{\lambda w_\lambda}$.

Uniqueness and the barriers. Every bounded κ-plurisubharmonic function is in H^1, by the same smoothing and energy bound used above after subtracting its κ-mean. For two bounded solutions w_1, w_2, their right-hand sides have bounded densities relative to μ. The trace inequality extends distributional Stokes to their difference, giving

$$E(w_1 - w_2, w_1 - w_2) = -\int_Y (w_1 - w_2)(\mathrm{e}^{\lambda w_1} - \mathrm{e}^{\lambda w_2})\,\mathrm{d}\mu \le 0.$$

The difference is constant; the unit exponential integrals force that constant to be zero.

Set $L_0 = Q - \sup Q \le 0$, $R_0 = Q - \inf Q \ge 0$, and $z = (L_0 - w_\lambda)_+$. The H^1 chain rule, canonical trace and extended Stokes give

$$E(z,z) = -\int_Y z(1 - \mathrm{e}^{\lambda w_\lambda})\,\mathrm{d}\mu \le 0.$$

Indeed $z > 0$ implies $w_\lambda < L_0 \le 0$. Thus z is constant. If it were a positive constant, continuity would give $w_\lambda = L_0 - z \le -z$ everywhere, contradicting its unit exponential integral. This proves the lower barrier. Testing $(w_\lambda - R_0)_+$ gives the upper barrier with the same argument and the reversed exponential difference. In particular $\|w_\lambda\|_\infty \le \operatorname{osc} Q$.

Selection. Write $q_\lambda = w_\lambda - \int_Y w_\lambda \kappa$, consistent with the variational normalization. The barriers give $\mathrm{e}^{\lambda w_\lambda} = 1 + O(\lambda)$ uniformly, and the Green identity gives

$$\|q_\lambda - Q\|_\infty \le \|\mathrm{e}^{\lambda w_\lambda} - 1\|_\infty \sup_x \int_Y |G(x,y)|\,\mathrm{d}\mu(y) = O(\lambda).$$

The last kernel integral is bounded by $1 + A_\mu$. The q_λ are uniformly bounded, so Taylor's formula gives

$$\lambda^{-1} \log Z_\lambda = \int_Y q_\lambda\,\mathrm{d}\mu + O(\lambda) = \int_Y Q\,\mathrm{d}\mu + O(\lambda).$$

This proves the whole-family estimate. Finally, Lemma 15.30.1 shows that every $p \in \mathrm{PSH}(Y,\kappa)$ with $\mathrm{MA}^{\mathrm{ext}}_\kappa(p) \ge \mu$ has the form $Q + C$. Its μ-mean is at most zero exactly when $C \le -\int Q\,\mathrm{d}\mu$, proving the order characterization. □

15.30.3 The product equation and its maximal coefficient

We use the product $X = X_0 \times Y$ with background $H = h_0 + \kappa$ of Section 15.29, now with $Y = \mathbb{P}^1$, so that $k = 1$ and $\binom{N}{m} = N = m + 1$, and with $V_0 = \int_{X_0} h_0^m$. For $M > 0$ put

$$\nu = M\,\frac{h_0^m}{V_0} \otimes \mu, \qquad c = \frac{NV_0}{M}.$$

Thus $\nu(X) = M$. When $v_-(h_0) = 0$, the background H has zero lower mass and is not closed, by Lemma 15.29.1.

Proposition 15.30.1 *If μ is non-polar, then ν is non-pluripolar and*

$$u = (Q_\mu - \sup_Y Q_\mu) \circ \pi_Y, \qquad \mathrm{MA}_H^{\mathrm{ext}}(u) = c\nu, \qquad \int_X \mathrm{MA}_H^{\mathrm{ext}}(u) = NV_0.$$

This is the unique supremum-zero solution among pullbacks of κ-plurisubharmonic functions. It is bounded exactly when Q_μ is bounded, and continuous exactly when Q_μ is continuous.

Proof Let $s \not\equiv -\infty$ be plurisubharmonic on a product chart. By local integrability and Fubini's theorem, for almost every point of X_0 with respect to smooth volume, the restriction of s to the corresponding curve slice is subharmonic and not identically $-\infty$. Non-polarity of μ and a second application of Fubini give its pole set product measure zero. A countable cover proves that ν is non-pluripolar.

By part (1) of Lemma 15.29.1, every $p \in \mathrm{PSH}(Y, \kappa)$ satisfies

$$\mathrm{MA}_H^{\mathrm{ext}}(p \circ \pi_Y) = N h_0^m \otimes \mathrm{MA}_\kappa^{\mathrm{ext}}(p), \tag{15.92}$$

where $\mathrm{MA}_\kappa^{\mathrm{ext}}(p) = \kappa + \mathrm{dd^c} p$ when p is bounded. Lemma 15.30.1 now proves the equation, the finite mass and the uniqueness among pullbacks. Boundedness and continuity follow from the same constant-difference identity. □

For this fixed product measure define its subsolution threshold by

$$b_H(\nu) = \sup\{a \geq 0 : \exists p \in \mathrm{PSH}(X, H), \mathrm{MA}_H^{\mathrm{ext}}(p) \geq a\nu\}.$$

For $\nu = f\,\mathrm{d}V$ this agrees with the individual-density threshold of Theorem 15.27.1.

Theorem 15.30.2 *Suppose $Q = Q_\mu$ is finite and continuous. Then $b_H(\nu) = c$, attained by the bounded continuous u of Proposition 15.30.1. For every $\lambda > 0$, the continuous potential*

$$U_\lambda = \lambda^{-1} \log c + w_\lambda \circ \pi_Y$$

satisfies

$$\mathrm{MA}_H(U_\lambda) = \mathrm{e}^{\lambda U_\lambda}\nu, \qquad \int_X \mathrm{MA}_H(U_\lambda) = NV_0.$$

It dominates every $p \in \mathrm{PSH}(X, H)$ with $\mathrm{MA}_H^{\mathrm{ext}}(p) \geq \mathrm{e}^{\lambda p}\nu$, and it is the unique bounded solution. For $v_\lambda = U_\lambda - \sup_X U_\lambda$ and $c_\lambda = \mathrm{e}^{\lambda \sup_X U_\lambda}$,

$$\mathrm{MA}_H(v_\lambda) = c_\lambda \mathrm{e}^{\lambda v_\lambda}\nu, \qquad c_\lambda = c\,\mathrm{e}^{\lambda \sup_Y w_\lambda},$$
$$\|v_\lambda - u\|_\infty = O(\lambda), \qquad \lambda^{-1}\log(c_\lambda/c) \longrightarrow \sup_Y Q - \int_Y Q\,\mathrm{d}\mu.$$

All the constructed potentials belong to $\mathcal{E}_{\mathrm{env}}(X, H, 0)$. The same coefficient c is attained if the defining supremum for $b_H(\nu)$ is restricted to bounded potentials or to this envelope class.

Proof Lemma 15.30.2 gives non-polarity, and Theorem 15.30.1 constructs w_λ. Substituting it in (15.92) proves the exponential equation and its finite mass. Supremum normalization gives the displayed c_λ, and the uniform curve estimate gives both limits. The constructed potentials are bounded, hence of the singularity type of 0, and Proposition 14.4.1 proves the envelope-class assertion.

Greatestness. The potential U_λ is bounded and satisfies $\mathrm{MA}_H(U_\lambda) = \mathrm{e}^{\lambda U_\lambda}\nu$. Let $r \in \mathrm{PSH}(X,H)$ satisfy $\mathrm{MA}_H^{\mathrm{ext}}(r) \geq \mathrm{e}^{\lambda r}\nu$. Off the pole set of r, $\mathrm{e}^{-\lambda U_\lambda}\mathrm{MA}_H(U_\lambda) = \nu \leq \mathrm{e}^{-\lambda r}\mathrm{MA}_H^{\mathrm{ext}}(r)$. Since U_λ is bounded, $r \preceq U_\lambda$ and U_λ admits a smaller-background envelope, so Corollary 15.1.1 gives $r \leq U_\lambda$. Applied to two bounded solutions, this gives bounded uniqueness.

The maximal coefficient. Suppose $\mathrm{MA}_H^{\mathrm{ext}}(r) \geq a\nu$ with $a > 0$, and put $B = \sup_X r$. For every $\lambda > 0$ the potential $r + \lambda^{-1}\log a - B$ is an exponential subsolution, since its exponential is at most a. Greatestness gives

$$r + \lambda^{-1}\log a - B \leq \lambda^{-1}\log c + w_\lambda \circ \pi_Y .$$

Evaluate at a fixed point where r is finite, multiply by λ, and let $\lambda \downarrow 0$. The uniform barriers for w_λ give $a \leq c$. The equation for u attains c in each stated class. □

The order characterization of the zero-exponent limit in Theorem 15.30.1 concerns only potentials pulled back from the curve. Bounded uniqueness at a positive exponent and maximality of the coefficient do not by themselves determine all normalized solutions of $\mathrm{MA}_H^{\mathrm{ext}}(p) = c\nu$ on the nonclosed product.

Proposition 15.30.2 *Suppose, for some $C > 0$ and $\delta > 0$,*

$$\mu(B_{\mathrm{ch}}(x,r)) \leq Cr^{\delta} \qquad (x \in Y,\ 0 < r \leq 1).$$

Then Q_μ is finite and continuous, so Theorem 15.30.2 applies. In particular it applies to

$$\mathrm{d}\mu(\vartheta) = (1 + \epsilon\cos\vartheta)\frac{\mathrm{d}\vartheta}{2\pi}, \qquad 0 < \epsilon < 1,$$

on the circle $[1 : \mathrm{e}^{\mathrm{i}\vartheta}]$. This measure and its product ν are singular with respect to smooth volume, and Q_μ is nonconstant on its support.

Proof Since $K = -\log d_{\mathrm{ch}}^2$, layer-cake integration gives

$$\sup_x \int_{\{K(x,y)>L\}} K(x,y)\,\mathrm{d}\mu(y) \leq C\left(L + \frac{2}{\delta}\right)\mathrm{e}^{-\delta L/2}.$$

Consequently the continuous truncated-kernel potentials converge uniformly to Q_μ. Weighted arclength on the circle has the ball bound with $\delta = 1$. On that circle the kernel is $1 + \log\sin^2((\vartheta - \varphi)/2)$, whose first cosine coefficient is -2. Integrating its L^1 Fourier series against $1 + \epsilon\cos\varphi$ gives

$$Q_\mu([1 : \mathrm{e}^{\mathrm{i}\vartheta}]) = 1 - 2\log 2 - \epsilon\cos\vartheta.$$

This proves nonconstancy, and the support has smooth curve volume zero. Its product with X_0 also has smooth volume zero. □

15.31 Stability of the subsolution threshold

We now study how the threshold $b_h(f)$ depends on f. It is upper semicontinuous on L^1, but when $v_-(h) = 0$ it is continuous only where it vanishes. For density roots, (15.82) expresses ρ_h as a supremum of positive linear functionals. Hence ρ_h is a norm on the space where it is finite, this space is complete for a suitable norm, and the threshold depends continuously on the root in that norm.

Theorem 15.31.1 *Suppose f_j, f are nonzero nonnegative L^1 densities and $f_j \to f$ in $L^1(\mathrm{d}V)$. Then*

$$\limsup_{j\to\infty} b_h(f_j) \le b_h(f).$$

Moreover,

$$b_h(f) = \lim_{\delta\downarrow 0} \sup\{c_h(q) : q \in L^\infty,\ q \ge 0,\ q \not\equiv 0,\ \|q - f\|_{L^1} < \delta\}, \tag{15.93}$$

and the suprema are finite for sufficiently small δ.

Proof For a Gauduchon form γ on X, write $A_\gamma(f) = \int_X (f\,\mathrm{d}V/\gamma^n)^{1/n}\gamma^n$. Since $|s^{1/n} - t^{1/n}| \le |s-t|^{1/n}$ for $s, t \ge 0$, Hölder's inequality shows that A_γ is continuous on L^1; it is positive at every nonzero nonnegative density. By Theorem 15.28.1, $b_h^{1/n}$ is the infimum over γ of the continuous positive functions $f \mapsto \int_X h \wedge \gamma^{n-1}/A_\gamma(f)$. It is therefore upper semicontinuous, and testing a single γ bounds it above on an L^1 neighborhood of f.

For the second assertion, recall that $b_h(q) = c_h(q)$ for bounded q (Proposition 15.27.2). By the first assertion, $\limsup_j c_h(q_j) \le b_h(f)$ for every sequence of nonzero nonnegative bounded densities $q_j \to f$ in L^1. Hence the limit in (15.93) is at most $b_h(f)$, and the local upper bound makes the suprema finite for small δ. Conversely, the truncations f_k converge to f in L^1 and satisfy $c_h(f_k) \ge b_h(f)$, so every supremum in (15.93) is at least $b_h(f)$. □

Since $b_h \ge 0$, upper semicontinuity makes b_h continuous at every density where it vanishes. If $v_-(h) = 0$, these are its only points of continuity. Indeed, let $b_h(f) > 0$ and $g = f/\int_X f\,\mathrm{d}V$. By Corollary 15.26.1 and Theorem 15.27.1, the densities $(1-t)g + t f_0$, where f_0 is the density of Theorem 15.26.1, have zero threshold and converge to g in L^1 as $t \downarrow 0$; multiplying them by $\int_X f\,\mathrm{d}V$ gives densities with zero threshold converging to f.

15.31.1 Finite sums of densities

In the statements below, f and g are nonzero nonnegative L^1 densities, and p and q are real or complex functions in L^n.

Proposition 15.31.1 *The functional ρ_h satisfies $\rho_h(tp) = |t|\rho_h(p)$ for scalars $t \neq 0$, $\rho_h(p) \le \rho_h(q)$ whenever $|p| \le |q|$, and*

$$\rho_h(p+q) \le \rho_h(p) + \rho_h(q).$$

In particular, if $b_h(f), b_h(g) > 0$, then

$$b_h(f+g) \ge \big(b_h(f)^{-1/n} + b_h(g)^{-1/n}\big)^{-n} > 0.$$

Proof By (15.82), $\rho_h(p) = \sup_\gamma L_{h,\gamma}(|p|)$ is a supremum of positive linear functionals evaluated at $|p|$. This gives homogeneity and monotonicity, and, together with $|p+q| \le |p| + |q|$, the triangle inequality. Finally, $(f+g)^{1/n} \le f^{1/n} + g^{1/n}$ gives

$$b_h(f+g)^{-1/n} = \rho_h\big((f+g)^{1/n}\big) \le \rho_h(f^{1/n}) + \rho_h(g^{1/n}) = b_h(f)^{-1/n} + b_h(g)^{-1/n},$$

which is equivalent to the asserted lower bound for $b_h(f+g)$. □

Proposition 15.31.2 *The real or complex vector space*

$$\mathcal{D}_h = \{p \in L^n(\mathrm{d}V) : \rho_h(p) < +\infty\}$$

is complete for the norm $\|p\|_{\mathcal{D}_h} = \|p\|_{L^n} + \rho_h(p)$. It is an order ideal in L^n: if $q \in \mathcal{D}_h$ and $|p| \le |q|$, then $p \in \mathcal{D}_h$. It contains L^∞ and is therefore dense in L^n for the L^n norm.

Moreover, $\mathcal{D}_h = L^n$ if and only if $v_-(h) > 0$, and in that case the norm $\|\cdot\|_{\mathcal{D}_h}$ is equivalent to the L^n norm. If h is Kähler, then

$$\rho_h(p) = \left(\int_X h^n\right)^{-1/n} \|p\|_{L^n}.$$

Proof By Proposition 15.31.1, $\mathcal{D}_h$ is a vector space and an order ideal, and $\|\cdot\|_{\mathcal{D}_h}$ is a norm on it. Bounded functions belong to $\mathcal{D}_h$, because $b_h(|p|^n) = c_h(|p|^n) > 0$ for bounded nonzero p.

We next show that ρ_h is lower semicontinuous on L^n. If $p_j \to p$ in L^n, then $|p_j|^n \to |p|^n$ in L^1, and the upper semicontinuity of b_h in Theorem 15.31.1 gives

$$\rho_h(p) \le \liminf_j \rho_h(p_j).$$

This includes the case $b_h(|p|^n) = 0$, in which the right side must be infinite; for $p = 0$ the inequality is trivial. Now let (p_j) be a Cauchy sequence in $\mathcal{D}_h$. It converges in L^n to some p, and lower semicontinuity gives $\rho_h(p) \le \liminf_j \rho_h(p_j) < +\infty$, so $p \in \mathcal{D}_h$. Applied to $p_j - p_k$ as $k \to \infty$, lower semicontinuity also gives

$$\rho_h(p_j - p) \le \liminf_{k\to\infty} \rho_h(p_j - p_k),$$

which tends to zero as $j \to \infty$. Together with L^n convergence, this proves completeness.

If $v_-(h) > 0$, Proposition 15.27.2 gives $\rho_h(p) \le v_-(h)^{-1/n}\|p\|_{L^n}$, so $\mathcal{D}_h = L^n$ with an equivalent norm. If $v_-(h) = 0$, the density f of Theorem 15.26.1 has $b_h(f) = 0$ by Theorem 15.27.1; thus $f^{1/n}$ belongs to L^n but not to $\mathcal{D}_h$. The Kähler formula follows from the same proposition. □

Thus, when the lower mass vanishes, $\mathcal{D}_h$ is a proper dense subspace of L^n, complete for the stronger norm $\|\cdot\|_{\mathcal{D}_h}$; its nonnegative elements are the nth roots of the densities admitting a positive scaled subsolution. A finite sum of such densities again admits a positive scaled subsolution, but a convergent infinite sum need not: the smooth summands in Theorem 15.26.1 give an example. By Proposition 15.28.1, the space $\mathcal{D}_h$ does not depend on the Hermitian form h, and the norms $\|\cdot\|_{\mathcal{D}_h}$ for different Hermitian forms are equivalent. The closure of the bounded functions for this norm is studied in Section 15.31.2.

The restriction of ρ_h to $\mathcal{D}_h$ is itself a norm: if $p \neq 0$, then $\rho_h(p) \ge L_{h,\gamma}(|p|) > 0$ for any Gauduchon form γ. Whether this weaker norm is complete depends on the upper mass.

Proposition 15.31.3 *The norm ρ_h is complete on $\mathcal{D}_h$ if and only if $v_+(h) < +\infty$. When the upper mass is finite,*

$$\|p\|_{L^n} \le v_+(h)^{1/n}\rho_h(p) \qquad (p \in \mathcal{D}_h),$$

so ρ_h and $\|\cdot\|_{\mathcal{D}_h}$ are equivalent norms.

Proof Suppose $v_+(h) < +\infty$. A continuous solution for a bounded density q has mass $c_h(q)\int_X q\,\mathrm{d}V \le v_+(h)$. Applying this to the truncations of $|p|^n$ and letting the truncation level increase gives $b_h(|p|^n)\|p\|_{L^n}^n \le v_+(h)$, which is the asserted inequality. Hence ρ_h and $\|\cdot\|_{\mathcal{D}_h}$ are equivalent, and completeness follows from Proposition 15.31.2.

Suppose instead that $v_+(h) = +\infty$. By Theorems 10.8.1 and 10.8.2, the normalization scalars of smooth strictly positive probability densities are unbounded above. For each j, choose such a density f_j with

$$c_h(f_j) \ge (j2^j)^n, \qquad \text{and put} \qquad p_j = j f_j^{1/n}.$$

Then $\|p_j\|_{L^n} = j$ and $\rho_h(p_j) = j c_h(f_j)^{-1/n} \le 2^{-j}$, so the partial sums $P_N = \sum_{j\le N} p_j$ form a Cauchy sequence for ρ_h.

Fix a Gauduchon form γ. Its coefficient a_γ is bounded below by a positive constant, so (15.82) gives $\rho_h(p) \ge c\|p\|_{L^1}$ with a constant $c > 0$. Hence $\sum_j p_j$ converges in L^1 to a nonnegative function P. If P_N converged for ρ_h to an element of $\mathcal{D}_h$, that element would also be the L^1 limit, hence equal to P almost everywhere. But $P \ge p_j$ gives $\|P\|_{L^n} \ge j$ for every j, so $P \notin L^n$. Therefore ρ_h is not complete. □

The two mass properties play separate roles: positive lower mass makes every L^n function admissible, whereas finite upper mass makes the root functional control the L^n norm. The norm $\|\cdot\|_{\mathcal{D}_h}$ is complete without either assumption.

All four combinations occur. On a compact Kähler manifold both mass properties hold. The Hopf threefold has positive lower mass and infinite upper mass (Corollary 9.7.1). The elliptic threefold of Theorem 27.18.1 has zero lower mass, and finite upper mass for its balanced metric (Corollary 27.18.3). The product manifolds of Corollary 27.18.2 have zero lower mass and infinite upper mass. Since neither $\mathcal{D}_h$ nor the completeness of ρ_h depends on the choice of h (Propositions 7.5.1 and 15.28.1), we obtain the following table.

Manifold	Domain $\mathcal{D}_h$	Is ρ_h complete?
Kähler manifold	L^n	Yes
Hopf threefold	L^n	No
Elliptic threefold	$\mathcal{D}_h \subsetneq L^n$	Yes
Product example	$\mathcal{D}_h \subsetneq L^n$	No

15.31.2 Approximation by bounded density roots

Bounded functions are dense in $\mathcal{D}_h$ for the L^n norm, but not obviously for the norm $\|\cdot\|_{\mathcal{D}_h}$. Let $\mathcal{D}_h^0$ denote the closure of L^∞ in $\mathcal{D}_h$ for this norm. We show that a function belongs to $\mathcal{D}_h^0$ exactly when its tails are small uniformly over all Gauduchon tests. The basic tool is a subcritical estimate. For every $r > n$,

$$\rho_h(p) \le C_r \|p\|_{L^r} \qquad (p \in L^r), \tag{15.94}$$

where the constant C_r depends on h, the reference volume and r, but not on p. Indeed, for nonzero $p \in L^r$ the density $|p|^n$ belongs to $L^{r/n}$, so $b_h(|p|^n) = c_h(|p|^n)$ by Proposition 15.27.2, and the lower-density estimate of Theorem 11.3.1, with exponent r/n, gives $c_h(|p|^n)\||p|^n\|_{L^{r/n}} \ge c_{r/n} > 0$.

Proposition 15.31.4 *For $p \in \mathcal{D}_h$, the following conditions are equivalent:*

(1) $p \in \mathcal{D}_h^0$;
(2) $\rho_h(|p|\mathbf{1}_{\{|p|>K\}}) \to 0$ *as* $K \to \infty$;
(3) *whenever* $|q_j| \le |p|$ *and* $q_j \to 0$ *in measure, one has* $\|q_j\|_{\mathcal{D}_h} \to 0$.

By (15.82)*, the second condition can be written as*

$$\sup_{\gamma \text{ Gauduchon}} L_{h,\gamma}(|p|\mathbf{1}_{\{|p|>K\}}) \longrightarrow 0 \qquad (K \to \infty).$$

The space $\mathcal{D}_h^0$ is a closed order ideal in $\mathcal{D}_h$. It contains L^r for every $r > n$, and smooth functions are dense in it. If $v_-(h) > 0$, then $\mathcal{D}_h^0 = \mathcal{D}_h = L^n$.

Proof For $K > 0$, set $p_K = p \min(1, K/|p|)$ where $p \neq 0$, and $p_K = 0$ where $p = 0$. Then $|p - p_K| = (|p| - K)_+$ and

$$(|p| - K)_+ \le |p|\mathbf{1}_{\{|p|>K\}}, \qquad |p|\mathbf{1}_{\{|p|>2K\}} \le 2(|p| - K)_+.$$

Since the L^n norms of the tails tend to zero, these inequalities and the monotonicity of ρ_h show that the second condition is equivalent to $p_K \to p$ in $\mathcal{D}_h$; in particular, it implies the first. Conversely, if a is bounded and $K \ge \|a\|_{L^\infty}$, then $(|p|-K)_+ \le |p-a|$, so $\|p - p_K\|_{\mathcal{D}_h} \le \|p - a\|_{\mathcal{D}_h}$. Approximating p by bounded functions a shows that the first condition implies $p_K \to p$, hence the second.

Assume the second condition, and let q_j be as in the third. Fix K. On $\{|p| \le K\}$, the functions q_j are bounded by K and tend to zero in measure, hence in L^{2n}; by (15.94) with $r = 2n$, the root functional of $q_j\mathbf{1}_{\{|p|\le K\}}$ tends to zero. On the complementary set, $\rho_h(q_j\mathbf{1}_{\{|p|>K\}}) \le \rho_h(|p|\mathbf{1}_{\{|p|>K\}})$. Letting first j and then K tend to infinity gives $\rho_h(q_j) \to 0$, and dominated convergence gives $\|q_j\|_{L^n} \to 0$. Conversely, the third condition applied to $q_j = |p|\mathbf{1}_{\{|p|>j\}}$ gives the second.

Since the second condition passes from p to every function dominated by $|p|$, the space $\mathcal{D}_h^0$ is an order ideal; it is closed by definition. For $p \in L^r$ with $r > n$, the truncations p_K converge to p in L^r, hence in $\mathcal{D}_h$ by (15.94). Bounded functions can be approximated in L^{2n} by smooth functions, which then converge in $\mathcal{D}_h$ by the same estimate; hence smooth functions are dense in $\mathcal{D}_h^0$. Under positive lower mass, Proposition 15.31.2 shows that $\mathcal{D}_h = L^n$ with an equivalent norm, and bounded functions are dense in L^n; hence $\mathcal{D}_h^0 = \mathcal{D}_h = L^n$. □

By Proposition 15.28.1, the space $\mathcal{D}_h^0$ does not depend on the choice of the Hermitian form h. Convergence in $\mathcal{D}_h$ controls the threshold: if $p_j \to p \neq 0$ in $\mathcal{D}_h$, then

$$|\rho_h(p_j) - \rho_h(p)| \le \rho_h(p_j - p) \longrightarrow 0, \qquad b_h(|p_j|^n) \longrightarrow b_h(|p|^n) > 0.$$

For $p \in \mathcal{D}_h^0$, the uniform tail condition also gives a uniform modulus for exponential damping and error bounds for the scalars of the exponential equations; see Proposition 15.34.1. It does not by itself give capacity convergence of the solutions. We do not know whether $\mathcal{D}_h^0 = \mathcal{D}_h$ when $\nu_-(h) = 0$.

15.31.3 Density roots from a closed factor

The threshold formula of Lemma 15.29.1 identifies a complete subspace of density roots with uniformly vanishing tails even when the ambient lower mass is zero. Use the product data $X_0, Y, N, H, C, V_0, \mu, R$ of Section 15.29, for an arbitrary closed Kähler factor (Y, κ), and put $c_0 = CV_0$ and $A = \max_X R$. The spaces $\mathcal{D}_H$ and $\mathcal{D}_H^0$ are now taken on the N-dimensional product with reference volume $\mathrm{d}V$.

Proposition 15.31.5 *For real or complex $\xi \in L^N(Y, \kappa^k)$, put $J\xi = R^{1/N}\pi_Y^*\xi$. Then*

$$\|J\xi\|_{L^N(\mathrm{d}V)} = \|\xi\|_{L^N(\kappa^k)}, \qquad \rho_H(J\xi) = c_0^{-1/N}\|\xi\|_{L^N(\kappa^k)}.$$

The image of J is a closed linear subspace of $\mathcal{D}_H^0$. If $g = |\xi|^N$, then

$$\rho_H\bigl(J\xi\mathbf{1}_{\{|J\xi|>K\}}\bigr) \le \left(\frac{1}{c_0}\int_{\{g>K^N/A\}} g\kappa^k\right)^{1/N} \longrightarrow 0 \qquad (K \to \infty).$$

For $\mathrm{d}V = \mu$, so that $R = 1$, the exact tail formula is

$$\rho_H\bigl(\pi_Y^*\xi\mathbf{1}_{\{|\xi|>K\}}\bigr) = \left(\frac{1}{c_0}\int_{\{|\xi|>K\}} |\xi|^N\kappa^k\right)^{1/N}.$$

Proof Since $|J\xi|^N = R|\xi|^N$, integration against $\mathrm{d}V$ gives the first norm identity. For nonzero ξ, part (2) of Lemma 15.29.1, applied to $g = |\xi|^N$, gives the root-functional identity; for $\xi = 0$ it holds by definition. Applying the two identities to $\xi - \eta$ yields

$$\|J\xi - J\eta\|_{\mathcal{D}_H} = (1 + c_0^{-1/N})\|\xi - \eta\|_{L^N(\kappa^k)}.$$

Bounded approximations of ξ in L^N have bounded images because R is bounded, so $J\xi \in \mathcal{D}_H^0$. Completeness of $L^N(Y, \kappa^k)$ and the displayed identity make the image closed.

On $\{|J\xi| > K\}$ one has $g > K^N/A$, and hence

$$|J\xi|\mathbf{1}_{\{|J\xi|>K\}} \le \bigl(Rg\mathbf{1}_{\{g>K^N/A\}}\bigr)^{1/N}.$$

Monotonicity of the root functional and its identity for factor roots give the estimate, with the value zero when the tail vanishes. Integrability of g makes the bound tend to zero; the L^N tails also vanish. When $R = 1$, the amplitude cutoff itself is pulled back from Y, so the same norm identity gives equality. □

For $Y = \mathbb{P}^k$ and $\mathrm{d}V = \mu$, take $p_\alpha = \pi_Y^*(q_\alpha^{1/N})$ with q_α from Lemma 15.29.2. Then $p_\alpha = J(q_\alpha^{1/N}) \in \mathcal{D}_H^0$, but $p_\alpha \notin L^r(\mathrm{d}V)$ for every $r > N$. Moreover,

$$\rho_H\bigl(p_\alpha\mathbf{1}_{\{p_\alpha>K\}}\bigr) \asymp (\log K)^{-k(1-\alpha)/N} \qquad (K \to +\infty).$$

Indeed, in the radial variable r of the proof of that lemma, $q_\alpha \asymp r^{-k}(1 - \log r)^{-1-k(1-\alpha)}$ as $r \downarrow 0$. With $L = 1 - \log r$, the density level set $\{q_\alpha > K^N\}$ lies between two sets $\{L > c\log K\}$ for fixed positive constants c, once K is large. Since $q_\alpha\kappa^k$ has radial density comparable to $L^{-1-k(1-\alpha)}\,\mathrm{d}L$ near o, integrating over $\{L > c\log K\}$ and taking the Nth root of the exact tail formula gives the claimed tail. Finally, $p_\alpha \notin L^r(\mathrm{d}V)$ because $\int_X p_\alpha^r\,\mathrm{d}\mu = \int_Y q_\alpha^{r/N}\kappa^k$ and $q_\alpha \notin L^{r/N}(\kappa^k)$.

These roots enlarge the examples in $\mathcal{D}_H^0$ beyond the subcritical L^r classes. They give neither $\mathcal{D}_H^0 = \mathcal{D}_H$ nor a general criterion for an exact equation at zero lower mass.

15.32 Power capacity bounds and threshold equations

The Hermitian form h and the probability volume form $\mathrm{d}V$ are as before. The root-tail condition of Proposition 15.31.4 controls the scalar threshold. A power bound for one density's measure supplies the missing control of its approximating potentials: the capacity estimates give a common bound, and stability against the solution measure gives uniform convergence.

Theorem 15.32.1 *Let $0 \le f \in L^1(\mathrm{d}V)$ have positive integral. Suppose that $A, \alpha > 0$ satisfy*

$$\int_E f \,\mathrm{d}V \le A \operatorname{Cap}_h(E)^{1+\alpha} \qquad (E \subset X \text{ Borel}). \tag{15.95}$$

There is $C > 0$, depending only on $(X, h), A, \alpha$, such that every measurable $0 \le g \le f$ with positive integral satisfies

$$b_h(g) \ge C \left(\int_X g \,\mathrm{d}V \right)^{-\alpha/(1+\alpha)}, \qquad \rho_h(g^{1/n}) \le C^{-1/n} \left(\int_X g \,\mathrm{d}V \right)^{\alpha/(n(1+\alpha))}. \tag{15.96}$$

In particular, $f^{1/n} \in \mathcal{D}_h^0$. There is $\phi \in \mathrm{PSH}(X, h) \cap C(X)$, with $\sup_X \phi = 0$, such that

$$\mathrm{MA}_h(\phi) = b_h(f) f \,\mathrm{d}V, \tag{15.97}$$

where $b_h(f) > 0$.

Proof *Step 1: the coefficient and root tails.* Put $g_j = \min(g, j)$ and let ϕ_j be normalized continuous solutions of

$$\mathrm{MA}_h(\phi_j) = c_j g_j \,\mathrm{d}V, \qquad c_j = c_h(g_j), \qquad \sup_X \phi_j = 0,$$

given by Theorem 11.1.1. The definition (15.80) gives $c_j \downarrow b_h(g)$. Their measures satisfy a power capacity bound with constant $c_j A$.

Apply Lemmas 11.3.1 and 11.3.2, with h in place of ω, to $u = \phi_j$ and $v = 0$, with $H = 1$ and $\varepsilon = 1/2$; let C_1 be the resulting constant C_H, and C_α the constant in (11.6). Fix $D > 0$ with $2D < \delta_{1/2}$, and put $U_D = \{\phi_j < \inf_X \phi_j + D\}$. The backward estimate (11.6) and the thin-sublevel estimate (11.5) with $s = t = D$ give

$$D \le C_\alpha (C_1 c_j A)^{1/n} \operatorname{Cap}_h(U_D)^{\alpha/n}, \qquad D^n \operatorname{Cap}_h(U_D) \le C_1 c_j \int_X g_j \,\mathrm{d}V.$$

Eliminating $\operatorname{Cap}_h(U_D)$ yields

$$c_j^{1+\alpha}\left(\int_X g_j\,\mathrm{d}V\right)^{\alpha} \geq c > 0,$$

with c depending only on (X, h), A and α. Letting $j \to \infty$ proves the lower bound for $b_h(g)$. Since $\rho_h(g^{1/n}) = b_h(g)^{-1/n}$ by definition, this lower bound is equivalent to the root estimate; by (15.82), the root estimate bounds every Gauduchon test of $g^{1/n}$. Apply it to $g = f\mathbf{1}_{\{f>K^n\}}$, for which $g^{1/n} = f^{1/n}\mathbf{1}_{\{f^{1/n}>K\}}$, whenever this density is nonzero; the zero density has root functional zero. Since the mass of g tends to zero as $K \to \infty$, the tails of $f^{1/n}$ tend to zero in both L^n and ρ_h. The case $g = f$ shows that $f^{1/n} \in \mathcal{D}_h$, and Proposition 15.31.4 gives $f^{1/n} \in \mathcal{D}_h^0$.

Step 2: uniform convergence of a subsequence. Take $g = f$ in Step 1, so that $g_j = f_j$ and $b_h(f) > 0$. Since $c_j \leq c_1$, the measures satisfy

$$\mathrm{MA}_h(\phi_j)(E) \leq Ac_1\,\mathrm{Cap}_h(E)^{1+\alpha},$$

and the first estimate of Step 1 bounds $\mathrm{Cap}_h(U_D)$ below by a positive constant independent of j. Since $U_D = \{\phi_j < \inf_X \phi_j + D\}$, the tail estimate (1.15), applied with h in place of ω, gives one $M \geq 1$ with $-M \leq \phi_j \leq 0$ for all j.

By Theorem 1.4.1, extract a subsequence converging in $L^1(\mathrm{d}V)$ and almost everywhere. Since $f\,\mathrm{d}V$ is a fixed finite absolutely continuous measure and the potentials have a common bound, dominated convergence gives

$$\int_X |\phi_j - \phi_k|\,\mathrm{MA}_h(\phi_j) \leq c_1 \int_X |\phi_j - \phi_k| f\,\mathrm{d}V \longrightarrow 0$$

along this subsequence as both indices tend to infinity. The same bound holds with $\mathrm{MA}_h(\phi_k)$ in place of $\mathrm{MA}_h(\phi_j)$. Applying Proposition 11.3.1 with $H = M$ to the pairs $(u, v) = (\phi_j, \phi_k)$ and (ϕ_k, ϕ_j) shows that the subsequence is uniformly Cauchy. Its limit ϕ is continuous, h-plurisubharmonic by distributional positivity, and has supremum zero.

Step 3: the limit equation. Uniform convergence implies convergence in capacity, so the continuity of bounded products recalled in Sections 1.6 and 1.8 gives $\mathrm{MA}_h(\phi_j) \rightharpoonup \mathrm{MA}_h(\phi)$ along the subsequence. On the other hand, $c_j f_j \to b_h(f) f$ in $L^1(\mathrm{d}V)$. The two limits prove (15.97). □

The capacity assumption concerns the fixed datum and implies the root-tail condition. When $v_-(h) > 0$, which always holds for $n \leq 2$ by Corollary 27.18.1, the root tails and the lower bound $b_h(g) \geq v_-(h)/\int_X g\,\mathrm{d}V$ already follow from Propositions 15.27.2 and 15.31.4, and Corollary 33.7.1 gives an exact threshold equation for every integrable density; the capacity assumption then provides a continuous solution. Whether root tails alone suffice for the exact equation when $v_-(h) = 0$ remains part of Problem 35.1.1; whether every positive-threshold root has these tails is the question following Proposition 15.31.4.

15.32.1 A logarithmic density with threshold one

The power capacity condition can hold for a density outside every L^p class with $p > 1$. A bounded continuous potential with logarithmic modulus at one point gives such a datum on each fixed Hermitian background.

Theorem 15.32.2 *In the setting of Theorem 15.32.1, fix $\alpha > 0$. There is $u \in \mathrm{PSH}(X,h) \cap C(X)$ whose measure $\mathrm{MA}_h(u) = f\,\mathrm{d}V$ satisfies*

$$f \in L^1(\mathrm{d}V), \qquad f \notin L^p(\mathrm{d}V) \quad \text{for every } p > 1.$$

For some $A > 0$ it satisfies (15.95), and $b_h(f) = 1$; thus u is a threshold solution for f. For a constant C and all sufficiently large K,

$$\rho_h\bigl(f^{1/n}\mathbf{1}_{\{f>K\}}\bigr) \le C(\log K)^{-\alpha}. \tag{15.98}$$

Proof Fix a point $x_0 \in X$ and holomorphic coordinates z centered at x_0 on a neighborhood of the closed unit ball $\{|z| \le 1\}$, and write $R = |z|$ and $B_r = \{R < r\}$. Choose a smooth cutoff $0 \le \chi \le 1$ supported in B_1 and equal to one on a smaller ball. Put

$$q(s) = (1-s)^{-\alpha}, \qquad s = \log R^2, \qquad \psi = \chi q(\log R^2) + 1 - \chi.$$

Set $\psi(x_0) = 0$ and extend $\psi = 1$ outside the ball. In the smaller ball, $q' = \alpha(1-s)^{-\alpha-1} > 0$ and $q'' = \alpha(\alpha+1)(1-s)^{-\alpha-2} > 0$, so ψ is plurisubharmonic there, including its continuous extension at x_0. The transition is smooth and $\mathrm{dd^c}\psi \ge -K_0 h$ globally for a fixed K_0. For sufficiently small $\delta > 0$, the normalized potential $u = \delta\psi - \sup_X \delta\psi$ satisfies $h + \mathrm{dd^c}u \ge h/2$.

By Proposition 1.7.1, the bounded product $\mathrm{MA}_h(u)$ does not charge the pluripolar set $\{x_0\}$, and off x_0 the potential is smooth. Thus the measure, which has finite mass, is absolutely continuous; write it as $f\,\mathrm{d}V$ with $f \in L^1(\mathrm{d}V)$.

In the smaller ball, where $u = \delta q(\log R^2) - \delta$, the tangential and radial eigenvalues of the complex Hessian of $q(\log R^2)$ are q'/R^2 and q''/R^2. With the normalization $\mathrm{dd^c} = \frac{\mathrm{i}}{2\pi}\partial\bar\partial$, this gives

$$(\mathrm{dd^c}u)^n = \frac{\delta^n n!\alpha^n(\alpha+1)}{\pi^n} R^{-2n}(1-\log R^2)^{-n(\alpha+1)-1}\,\mathrm{d}\lambda, \tag{15.99}$$

where $\mathrm{d}\lambda$ is Euclidean volume.

For r smaller than the radius of the smaller ball, integration over B_r gives

$$\int_{B_r} (\mathrm{dd^c}u)^n = \delta^n\alpha^n(1-\log r^2)^{-n(\alpha+1)}.$$

In the expansion of $\mathrm{MA}_h(u)$ into the terms $\binom{n}{k}(\mathrm{dd^c}u)^k \wedge h^{n-k}$, all terms are positive in this ball. For $k < n$, their densities are bounded by $CR^{-2k}(1-\log R^2)^{-k(\alpha+1)}$, and their masses on B_r are $O(r^2)$. For $p > 1$, the pure top term gives a positive multiple

of the divergent integral

$$\int_0^r R^{2n(1-p)-1}(1-\log R^2)^{-p(n(\alpha+1)+1)}\,\mathrm{d}R$$

as a lower bound for $\int_{B_r} f^p\,\mathrm{d}V$, since smooth volume densities are comparable. Thus $f \notin L^p$. For all sufficiently small r,

$$\int_{B_r} f\,\mathrm{d}V \le C\big[(1-\log r^2)^{-n(\alpha+1)} + r^2\big], \qquad \operatorname*{ess\,sup}_{X\setminus B_r} f \le Cr^{-2n}. \tag{15.100}$$

By (1.14), applied with h in place of ω, every Borel set $E \subset X$ satisfies

$$\mathrm{d}V(E) \le C\exp\big(-a\,\mathrm{Cap}_h(E)^{-1/n}\big).$$

Let $s = \mathrm{Cap}_h(E) > 0$ be small and choose $r = \exp[-as^{-1/n}/(4n)]$. Splitting E inside and outside B_r and applying (15.100), we obtain

$$\int_E f\,\mathrm{d}V \le Cs^{1+\alpha} + C\exp[-as^{-1/n}/(2n)] + C\exp[-as^{-1/n}/2] \le C's^{1+\alpha}.$$

For larger s use the finite total mass; for $s = 0$ the volume estimate gives $\mathrm{d}V(E) = 0$. This proves (15.95) for every Borel set.

Suppose that $v \in \mathrm{PSH}(X,h)$ and $a > 1$ satisfy $\mathrm{MA}_h^{\mathrm{ext}}(v) \ge af\,\mathrm{d}V$. For every real A, the bounded potential $u + A$ satisfies $\mathrm{MA}_h(u+A) = f\,\mathrm{d}V \le a^{-1}\mathrm{MA}_h^{\mathrm{ext}}(v)$, $P_{h/2}(u+A) \ge \inf_X u + A$ and $v \preceq u + A$. Theorem 15.1.1, with $\delta = 1/2$ and $c = a^{-1}$, gives $v \le u + A$, which fails for sufficiently negative A at a point where v is finite. Hence every subsolution coefficient is at most 1, and u attains 1, so $b_h(f) = 1$ by Theorem 15.27.1.

Finally take $r_K = K^{-1/(4n)}$. The outside bound in (15.100) gives $f \le CK^{1/2} < K$ almost everywhere on $X \setminus B_{r_K}$ for large K. Therefore

$$\int_{\{f>K\}} f\,\mathrm{d}V \le C(\log K)^{-n(\alpha+1)} + CK^{-1/(2n)} \le C'(\log K)^{-n(\alpha+1)}.$$

Apply (15.96) to $g = f\mathbf{1}_{\{f>K\}}$, for which $g^{1/n} = f^{1/n}\mathbf{1}_{\{f>K\}}$. Since the exponent there is $\alpha/(n(1+\alpha))$, the last bound gives (15.98). □

The potential has logarithmic modulus at x_0 and is not Hölder there. Its root-tail estimate follows from the capacity bound despite the failure of higher L^p integrability. The construction applies to each fixed background, including one with $v_-(h) = 0$.

15.33 Limits of exponential equations

In this section θ is an arbitrary smooth real $(1,1)$-form and μ a finite non-pluripolar measure. We consider solutions $u_\lambda \in \mathrm{PSH}(X,\theta)$ of

$$\mathrm{MA}_\theta(u_\lambda) = \mathrm{e}^{\lambda u_\lambda}\mu \qquad (\lambda > 0).$$

As $\lambda \downarrow 0$, one expects suitably normalized solutions to converge to a solution of $\mathrm{MA}_\theta(u) = c\mu$ with a scalar $c > 0$. For an L^p density with $p > 1$ and a Hermitian background, the continuous solutions satisfy $\lambda u_\lambda \to \log c$ uniformly (Proposition 11.6.1). For solutions in a fixed unbounded ordinary type, no such uniform convergence is possible: by Proposition 1.23.1, the L^∞ distance from λu_λ to every finite constant is infinite for each $\lambda > 0$. The limit must instead be formulated through normalized potentials and their measures. Write

$$v_\lambda = u_\lambda - \sup_X u_\lambda, \qquad c_\lambda = \mathrm{e}^{\lambda \sup_X u_\lambda},$$

so that $\mathrm{MA}_\theta(v_\lambda) = c_\lambda \mathrm{e}^{\lambda v_\lambda}\mu$. By Lemma 1.23.2, $\mathrm{e}^{\lambda v_\lambda} \to 1$ in $L^1(\mu)$ as $\lambda \downarrow 0$; hence, if $c_\lambda \to c < +\infty$, the right-hand sides converge in total variation to $c\mu$. If $\mu = f\,\mathrm{d}V$ with $f \in L^p$ for some $p > 1$, Lemma 1.23.2 gives an error linear in λ.

To identify the Monge–Ampère measure of an L^1 limit of the v_λ, one needs in addition a continuity theorem for the Monge–Ampère operator. For potentials of a fixed ordinary type, Corollary 15.8.1 identifies the limiting measure under a common barrier and a domination hypothesis. In a relative envelope class, Theorem 33.3.1 instead constructs a common lower envelope under positive lower mass. For a semipositive big θ and a reference satisfying (14.20), it shows that an L^1-convergent sequence in the class whose measures are dominated by one finite non-pluripolar measure converges in capacity, and that its measures converge weakly. That argument, with its mass hypothesis, is developed in Chapter 33.

15.33.1 A positive limiting exponent

When the exponents converge to a positive limit, the factor $\mathrm{e}^{\lambda u}$ makes the Monge–Ampère mass of deep sublevels uniformly small. If the potentials converge in capacity, the tail criterion of Theorem 8.4.1 then identifies the limit of their Monge–Ampère measures, and convergence of the right-hand sides identifies the equation.

Theorem 15.33.1 *Let θ be any smooth real $(1,1)$-form and μ a finite non-pluripolar measure. Suppose $f_j \geq 0$, $f_j \to f$ in $L^1(\mu)$, and $\lambda_j > 0$ with $\lambda_j \to \lambda > 0$. Let $u_j, u \in \mathrm{PSH}(X,\theta)$ be nonpositive, suppose $u_j \to u$ in capacity, and assume*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) = \mathrm{e}^{\lambda_j u_j} f_j \, \mu.$$

Then $\mathrm{MA}_\theta^{\mathrm{ext}}(u)$ is finite and

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) = \mathrm{e}^{\lambda u} f \, \mu.$$

Moreover, $\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) \rightharpoonup \mathrm{MA}_\theta^{\mathrm{ext}}(u)$ weakly, and the right-hand sides converge in total variation.

Proof Discarding finitely many indices, we may assume that $\lambda_j \geq \lambda/2$. The measures $\mathrm{MA}_\theta^{\mathrm{ext}}(u_j)$ have masses at most $\|f_j\|_{L^1(\mu)}$, which are uniformly bounded, and since $u_j \leq 0$, their deep sublevels satisfy

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u_j)(\{u_j < -T\}) \leq \mathrm{e}^{-\lambda T/2} \|f_j\|_{L^1(\mu)}.$$

The tail criterion Theorem 8.4.1, which holds for an arbitrary smooth background by Theorem 8.6.1, therefore shows that $\mathrm{MA}_\theta^{\mathrm{ext}}(u)$ is finite and that $\mathrm{MA}_\theta^{\mathrm{ext}}(u_j) \rightharpoonup \mathrm{MA}_\theta^{\mathrm{ext}}(u)$.

It remains to identify this limit with $\mathrm{e}^{\lambda u} f \, \mu$. By Lemma 1.8.8, $u_j \to u$ in μ-measure. Since $t \mapsto \mathrm{e}^{\lambda t}$ is λ-Lipschitz for $t \leq 0$, the functions $\mathrm{e}^{\lambda u_j}$, which are bounded by one, converge in μ-measure to $\mathrm{e}^{\lambda u}$. The change of exponent is uniform for nonpositive arguments, because the mean value theorem gives

$$\sup_{t \leq 0} |\mathrm{e}^{\lambda_j t} - \mathrm{e}^{\lambda t}| \leq |\lambda_j - \lambda| \sup_{s \geq 0} s \mathrm{e}^{-(\lambda/2)s} \longrightarrow 0.$$

Thus $\mathrm{e}^{\lambda_j u_j} \to \mathrm{e}^{\lambda u}$ in μ-measure, and hence in $f\mu$-measure, since the finite measure $f\mu$ is absolutely continuous with respect to μ. As these functions are bounded by one, the convergence also holds in $L^1(f\mu)$. Therefore

$$\int_X |\mathrm{e}^{\lambda_j u_j} f_j - \mathrm{e}^{\lambda u} f| \, \mathrm{d}\mu \leq \|f_j - f\|_{L^1(\mu)} + \int_X |\mathrm{e}^{\lambda_j u_j} - \mathrm{e}^{\lambda u}| f \, \mathrm{d}\mu \longrightarrow 0.$$

This proves the total-variation convergence of the right-hand sides, and comparing the two limits gives the equation for u. □

Additive normalizations of the potentials can be absorbed into the densities: translating each u_j by a constant multiplies f_j by an exponential factor, and the theorem applies as long as the modified densities still converge in $L^1(\mu)$. Convergence in capacity is a hypothesis; the theorem says nothing about sequences that converge only in L^1, and it does not keep the limit in a prescribed envelope class.

The proof also shows where the difficulty lies as $\lambda_j \downarrow 0$: the tail bound $e^{-\lambda_j T}\|f_j\|_{L^1(\mu)}$ no longer tends to zero uniformly in j as $T \to \infty$. A zero-exponent limit therefore needs another source of uniform tail control, such as the integrability condition of Corollary 8.4.1; a uniform bound on the total masses does not by itself control these tails.

15.34 Exponential equations for an individual density

We return to the Hermitian background h and a fixed nonzero density $0 \le f \in L^1(dV)$, and consider, for $\lambda > 0$, the exponential equation

$$\mathrm{MA}_h^{\mathrm{ext}}(u) = e^{\lambda u} f\, dV, \qquad u \in \mathrm{PSH}(X,h). \tag{15.101}$$

An *exponential subsolution* at exponent λ is a function $s \in \mathrm{PSH}(X,h)$ with $\mathrm{MA}_h^{\mathrm{ext}}(s) \ge e^{\lambda s} f\, dV$. Bounded solutions of the truncated equations dominate every exponential subsolution, so as soon as one subsolution exists, their decreasing limit is the greatest solution. A positive threshold $b_h(f)$ provides exponential subsolutions for every exponent, and the normalized greatest solutions then recover $b_h(f)$ as $\lambda \downarrow 0$. The domination comes from the following case of Corollary 15.1.1, which we state for an arbitrary measure.

Lemma 15.34.1 *Let $\lambda > 0$ and let ν be a positive Borel measure on X. If $t \in \mathrm{PSH}(X,h)$ is bounded and $\mathrm{MA}_h(t) \le e^{\lambda t}\nu$, then every $s \in \mathrm{PSH}(X,h)$ with $\mathrm{MA}_h^{\mathrm{ext}}(s) \ge e^{\lambda s}\nu$ satisfies $s \le t$.*

Proof Since t is bounded, $s \preceq t$, and $P_{h/2}(t) \ge \inf_X t$, so t admits a smaller-background envelope. Off the pole set of s,

$$e^{-\lambda t}\mathrm{MA}_h(t) \le \nu \le e^{-\lambda s}\mathrm{MA}_h^{\mathrm{ext}}(s).$$

Corollary 15.1.1, applied with $\theta = h$, $u = t$ and $v = s$, gives $s \le t$. □

Theorem 15.34.1 *Fix $\lambda > 0$, and suppose that (15.101) has an exponential subsolution $s \in \mathrm{PSH}(X,h)$. For $k \ge 1$, let u_k be the continuous solution of*

$$\mathrm{MA}_h(u_k) = e^{\lambda u_k} f_k\, dV, \qquad f_k = \min(f,k),$$

given by Theorem 11.1.2. Then u_k decreases to a solution u_λ of (15.101), and $\mathrm{MA}_h^{\mathrm{ext}}(u_\lambda)$ has finite mass. Moreover, u_λ is the greatest exponential subsolution: every exponential subsolution v at exponent λ satisfies $v \le u_\lambda$. In particular, u_λ is the greatest solution of (15.101).

Proof Since $f_k \le f_{k+1}$, Lemma 11.4.2 gives $u_{k+1} \le u_k$. Since $\mathrm{MA}_h(u_k) = e^{\lambda u_k} f_k\, dV \le e^{\lambda u_k} f\, dV$, Lemma 15.34.1, with $\nu = f\, dV$, shows that every exponential subsolution v satisfies $v \le u_k$ for every k.

In particular $s \le u_k \le u_1$ for every k, so the decreasing limit $u_\lambda \ge s$ is not identically $-\infty$ and belongs to $\mathrm{PSH}(X,h)$. By Lemma 1.8.9, $u_k \to u_\lambda$ in capacity. Put $A = \sup_X u_1$. The nonpositive functions $u_k - A$ satisfy

$$\mathrm{MA}_h^{\mathrm{ext}}(u_k - A) = \mathrm{e}^{\lambda(u_k - A)}\, \mathrm{e}^{\lambda A} f_k \, \mathrm{d}V,$$

and $\mathrm{e}^{\lambda A} f_k \to \mathrm{e}^{\lambda A} f$ in $L^1(\mathrm{d}V)$. Theorem 15.33.1, applied with $\mu = \mathrm{d}V$ and constant exponents, shows that $\mathrm{MA}_h^{\mathrm{ext}}(u_\lambda)$ is finite and that u_λ solves (15.101). Finally, every exponential subsolution lies below all u_k, hence below u_λ, which is itself a subsolution. □

Being the greatest subsolution, u_λ does not depend on the approximation used to construct it. The equation may have other, smaller solutions with different singularities. If $f \in L^p$ for some $p > 1$, the continuous solution of Theorem 11.1.2 is a subsolution and therefore lies below u_λ. Thus u_λ is bounded, and bounded exponential comparison (Corollary 11.2.2) shows that it coincides with the continuous solution.

Corollary 15.34.1 *Suppose $b = b_h(f) > 0$. Then (15.101) has a greatest solution u_λ for every $\lambda > 0$. If $v \in \mathrm{PSH}(X,h)$ satisfies $\sup_X v = 0$ and $\mathrm{MA}_h^{\mathrm{ext}}(v) \ge b f \, \mathrm{d}V$, then*

$$u_\lambda \ge v + \lambda^{-1} \log b.$$

Put

$$M_\lambda = \sup_X u_\lambda, \qquad v_\lambda = u_\lambda - M_\lambda, \qquad c_\lambda = \mathrm{e}^{\lambda M_\lambda}.$$

Then, as $\lambda \downarrow 0$,

$$c_\lambda \longrightarrow b_h(f), \qquad \mathrm{e}^{\lambda u_\lambda} f \longrightarrow b_h(f) f \quad \text{in } L^1(\mathrm{d}V).$$

Proof By Theorem 15.27.1, a function v as in the statement exists. Since $v \le 0$, the function $s = v + \lambda^{-1} \log b$ satisfies

$$\mathrm{MA}_h^{\mathrm{ext}}(s) \ge b f \, \mathrm{d}V \ge \mathrm{e}^{\lambda s} f \, \mathrm{d}V,$$

so it is an exponential subsolution. Theorem 15.34.1 gives the greatest solution u_λ and the bound $u_\lambda \ge s$; in particular, $c_\lambda \ge b$.

Set $g_\lambda = \mathrm{e}^{\lambda v_\lambda} f$. By Lemma 1.23.2, applied to the finite measure $f \, \mathrm{d}V$, we have $g_\lambda \to f$ in L^1 as $\lambda \downarrow 0$. Each g_λ is nonzero, since v_λ is finite almost everywhere. The equation for u_λ reads $\mathrm{MA}_h^{\mathrm{ext}}(v_\lambda) = c_\lambda g_\lambda \, \mathrm{d}V$, so Theorem 15.27.1, applied to the density g_λ, gives

$$c_\lambda \le b_h(g_\lambda).$$

The upper semicontinuity of the threshold (Theorem 15.31.1) therefore gives

$$b \le \liminf_{\lambda \downarrow 0} c_\lambda \le \limsup_{\lambda \downarrow 0} c_\lambda \le b_h(f) = b.$$

This proves $c_\lambda \to b$. Since $\mathrm{e}^{\lambda u_\lambda} f = c_\lambda g_\lambda$ and $g_\lambda \to f$ in L^1, it also proves $\mathrm{e}^{\lambda u_\lambda} f \to b f$ in L^1. □

Corollary 15.34.2 *Suppose $b_h(f) > 0$, and let u_λ and v_λ be as in Corollary 15.34.1. Every L^1 limit of a sequence v_{λ_j} with $\lambda_j \downarrow 0$ belongs to the family $\mathcal{S}_f$ of Proposition 15.27.1, and hence lies below Φ_f.*

Proof Suppose $v_{\lambda_j} \to v$ in L^1 with $\lambda_j \downarrow 0$. By Theorem 1.4.1, v is h-plurisubharmonic with supremum zero. The measures $\mathrm{MA}_h^{\mathrm{ext}}(v_{\lambda_j}) = \mathrm{e}^{\lambda_j u_{\lambda_j}} f \, \mathrm{d}V$ have densities converging in L^1 to $b_h(f) f$, by Corollary 15.34.1. Applying Lemma 15.27.1 to these densities and potentials shows that $v \in \mathcal{S}_f$. □

Since $\mathrm{MA}_h^{\mathrm{ext}}(v_\lambda) = \mathrm{e}^{\lambda u_\lambda} f \, \mathrm{d}V$, the Monge–Ampère measures of the normalized greatest solutions converge in total variation to $b_h(f) f \, \mathrm{d}V$. This answers the questions about the scalar and the measure raised in Section 15.33; the potentials v_λ themselves are harder to control. Under positive lower mass, Corollary 33.7.1 shows that every L^1-convergent sequence v_{λ_j} with $\lambda_j \downarrow 0$ converges in capacity to a solution of $\mathrm{MA}_h^{\mathrm{ext}}(v) = b_h(f) f \, \mathrm{d}V$. For L^p data with $p > 1$, Proposition 15.19.1 gives uniform precompactness. In both cases, convergence of the whole family (v_λ) requires that all its cluster limits coincide. This holds when the dual problem for $f^{1/n}$ over Gauduchon currents, studied in Section 27.8, has a smooth strictly positive maximizer: the whole family then converges uniformly (Corollary 27.8.2). If the maximizer is smooth and semipositive, and strictly positive outside a proper analytic set, Proposition 27.8.5 gives L^1 convergence of the whole family and a fixed lower barrier, and Corollary 27.8.5 strengthens this to uniform convergence for L^p data with $p > 1$.

For a general positive threshold, the corollary gives the lower bound

$$v_\lambda \geq v - \lambda^{-1} \log(c_\lambda / b_h(f))$$

for every normalized threshold subsolution v. The next subsection estimates the scalar error $\log(c_\lambda / b_h(f))$ when $f^{1/n}$ can be approximated by bounded functions in the norm of $\mathcal{D}_h$. When $b_h(f) = 0$, Theorem 15.34.1 still applies whenever an exponential subsolution exists; examples are constructed in Section 15.34.2.

15.34.1 An error bound for the scalar limit

Suppose that $f^{1/n}$ belongs to the space $\mathcal{D}_h^0$ of Section 15.31.2. The uniform tail condition of Proposition 15.31.4 then gives a rate for the convergence $c_\lambda \to b_h(f)$ in Corollary 15.34.1, through a modulus for the damping of the root $f^{1/n}$ that is uniform over normalized potentials.

Proposition 15.34.1 *Let $0 \leq p \in \mathcal{D}_h^0$ be nonzero, and put $f = p^n$, $R = \rho_h(p)$ and $b = R^{-n} = b_h(f)$. There is a constant C_h, depending only on h and $\mathrm{d}V$, such that the function*

$$\eta_p(\lambda) = \inf_{K>0} \left(\| p \mathbf{1}_{\{p>K\}} \|_{\mathcal{D}_h} + C_h K \lambda \right)$$

tends to zero as $\lambda \downarrow 0$ and satisfies

$$\left\|p(1-\mathrm{e}^{\lambda v/n})\right\|_{\mathcal{D}_h} \le \eta_p(\lambda) \qquad \big(v \in \mathrm{PSH}(X,h),\ \sup_X v = 0\big). \tag{15.102}$$

Let u_λ, v_λ and c_λ be the greatest exponential solutions and their normalizations from Corollary 15.34.1. If $\eta_p(\lambda) < R$, then

$$b \le c_\lambda \le \big(R - \eta_p(\lambda)\big)^{-n}$$

and

$$\left\|\mathrm{MA}_h^{\mathrm{ext}}(v_\lambda) - b f\,\mathrm{d}V\right\|_{\mathrm{TV}} \le (c_\lambda - b)\|p\|_{L^n}^n + nb\|p\|_{L^n}^{n-1}\eta_p(\lambda).$$

If $p \in L^r$ for some finite $r > n$, the same bounds hold with $\eta_p(\lambda)$ replaced by $C_{h,r}\|p\|_{L^r}\lambda$, where $C_{h,r}$ depends only on h, $\mathrm{d}V$ and r. In particular, for each such p the scalar and total-variation errors are $O(\lambda)$.

Proof By Lemma 1.5.1, normalized h-plurisubharmonic functions are uniformly bounded in every L^s with $s < +\infty$. By (15.94) with $r = 2n$, their norms $\|v\|_{\mathcal{D}_h}$ are therefore uniformly bounded; let C_h be $1/n$ times this bound. Let v be normalized and $K > 0$. On $\{p \le K\}$ we have

$$0 \le p(1-\mathrm{e}^{\lambda v/n}) \le K\lambda(-v)/n,$$

since $1 - \mathrm{e}^{-x} \le x$ for $x \ge 0$, while on $\{p > K\}$ we use the bound $p(1-\mathrm{e}^{\lambda v/n}) \le p$. The monotonicity of $\|\cdot\|_{\mathcal{D}_h}$ and the triangle inequality give

$$\left\|p(1-\mathrm{e}^{\lambda v/n})\right\|_{\mathcal{D}_h} \le \|p\mathbf{1}_{\{p>K\}}\|_{\mathcal{D}_h} + C_h K\lambda,$$

and taking the infimum over K proves (15.102). By Proposition 15.31.4, and since the L^n tails of p tend to zero, $\|p\mathbf{1}_{\{p>K\}}\|_{\mathcal{D}_h} \to 0$ as $K \to \infty$. Choosing K first and then letting $\lambda \downarrow 0$ shows that $\eta_p(\lambda) \to 0$.

Now set $q_\lambda = p\,\mathrm{e}^{\lambda v_\lambda/n}$, so that $q_\lambda^n = g_\lambda = \mathrm{e}^{\lambda v_\lambda} f$, with g_λ as in the proof of Corollary 15.34.1. Since $0 \le q_\lambda \le p$ and $\rho_h(p - q_\lambda) \le \eta_p(\lambda)$ by (15.102), the triangle inequality and monotonicity give

$$R - \eta_p(\lambda) \le \rho_h(q_\lambda) \le R.$$

As in the proof of Corollary 15.34.1, $b \le c_\lambda \le b_h(g_\lambda) = \rho_h(q_\lambda)^{-n}$, which yields the scalar bound. For the measures, the factorization of $p^n - q_\lambda^n$ gives $0 \le p^n - q_\lambda^n \le np^{n-1}(p - q_\lambda)$, and Hölder's inequality implies

$$\|f - g_\lambda\|_{L^1} \le n\|p\|_{L^n}^{n-1}\|p - q_\lambda\|_{L^n} \le n\|p\|_{L^n}^{n-1}\eta_p(\lambda).$$

Since $\mathrm{MA}_h^{\mathrm{ext}}(v_\lambda) = c_\lambda g_\lambda\,\mathrm{d}V$, the identity $c_\lambda g_\lambda - bf = (c_\lambda - b)g_\lambda - b(f - g_\lambda)$, together with $\|g_\lambda\|_{L^1} \le \|f\|_{L^1} = \|p\|_{L^n}^n$, proves the total-variation bound.

Finally, suppose $p \in L^r$ with $n < r < +\infty$, and choose $n < s < r$. By (15.94) with exponent s, and since $\|\cdot\|_{L^n} \le \|\cdot\|_{L^s}$, we have $\|\cdot\|_{\mathcal{D}_h} \le (1 + C_s)\|\cdot\|_{L^s}$. Hölder's inequality and Lemma 1.5.1 then give, for every normalized v,

$$\|p(-v)\|_{\mathcal{D}_h} \le (1+C_s)\|p(-v)\|_{L^s} \le (1+C_s)\|p\|_{L^r}\|v\|_{L^{rs/(r-s)}} \le nC_{h,r}\|p\|_{L^r}.$$

Since $0 \le 1 - \mathrm{e}^{\lambda v/n} \le \lambda(-v)/n$, the left side of (15.102) is therefore at most $C_{h,r}\|p\|_{L^r}\lambda$, and the scalar and total-variation estimates can be repeated with this bound in place of $\eta_p(\lambda)$. For λ so small that $C_{h,r}\|p\|_{L^r}\lambda \le R/2$, the function $t \mapsto (R-t)^{-n}$ has bounded derivative on $[0, R/2]$, so the scalar error $c_\lambda - b$, and hence the total-variation error, is $O(\lambda)$. □

For a general $p \in \mathcal{D}_h^0$ the modulus $\eta_p(\lambda)$ tends to zero, but we do not know whether it is $O(\lambda)$. Combined with the lower bound $v_\lambda \ge v - \lambda^{-1}\log(c_\lambda/b)$ for normalized threshold subsolutions v, the proposition gives

$$0 \le \lambda^{-1}\log(c_\lambda/b) \le -\frac{n}{\lambda}\log\bigl(1-\eta_p(\lambda)/R\bigr),$$

which stays bounded as $\lambda \downarrow 0$ when $\eta_p(\lambda) = O(\lambda)$, in particular for $p \in L^r$ with $r > n$. The proposition estimates only scalars and measures; even with a uniform lower barrier for v_λ, passing to the limit in capacity requires the singularity-type or lower-mass hypotheses of a continuity theorem, and when the lower mass vanishes the proposition does not give an exact equation at the threshold.

15.34.2 Exponential equations at zero threshold

When $b_h(f) = 0$, no positive multiple of $f\,\mathrm{d}V$ is dominated by a Monge–Ampère measure. An exponential subsolution s only needs $\mathrm{MA}_h^{\mathrm{ext}}(s) \ge \mathrm{e}^{\lambda s} f\,\mathrm{d}V$, however, and the factor $\mathrm{e}^{\lambda s}$ is small where s is very negative. A density concentrating near the pole of a singular potential can therefore admit exponential subsolutions although its threshold vanishes. We construct such densities on the products of Section 15.29.

We keep the notation of that section: $X = X_0 \times Y$ with $Y = \mathbb{P}^k$, the background $H = h_0 + \kappa$, the constants N, C and V_0, the probability volume form $\mu = V_0^{-1} h_0^m \wedge \kappa^k$ with density $R = \mu/\mathrm{d}V$, and the potential u_α with density q_α from Lemma 15.29.2, for a fixed $0 < \alpha < 1$. On $Y \setminus \{o\}$ put $L = 1 - \psi \ge 1$, so that $u_\alpha = 1 - L^\alpha$, and $t = \mathrm{e}^{1-L}$. As functions of L, define

$$a = \alpha L^{\alpha-1}, \qquad Q = t + (1-t)a, \qquad D = t(1-a) + (1-t)\alpha(1-\alpha)L^{\alpha-2},$$

$$W = kQ^{k-1}D, \qquad B_\tau = q_\alpha \mathrm{e}^{\tau(L^\alpha - 1)} \quad (\tau > 0), \qquad b_0 = (1-\alpha)^k.$$

Here Q is the function $Q(r)$ of Lemma 15.29.2, written in terms of L, and $D = -\mathrm{d}Q/\mathrm{d}L$. The weight B_τ is chosen so that $\mathrm{e}^{\tau u_\alpha} B_\tau = q_\alpha$ off o.

Lemma 15.34.2 *As functions of L,*

$$q_\alpha = Q^{k-1} D\,\mathrm{e}^{k(L-1)}, \qquad q_\alpha \ge b_0,$$

and $W(L) \sim k\alpha^k(1-\alpha)L^{-1-k(1-\alpha)}$ *as* $L \to +\infty$. *For every* $\tau > 0$, $\mathrm{e}^{\tau(L^\alpha-1)}W(L) \to +\infty$ *as* $L \to +\infty$. *In particular,* $B_\tau\kappa^k$ *has infinite mass on every punctured neighborhood of* o.

Proof In the notation of the proof of Lemma 15.29.2, $t = r/(1+r)$ and $s(r) = L$. Since the mass of $q_\alpha\kappa^k$ on $\{r_1 < r < r_2\}$ is $Q(r_2)^k - Q(r_1)^k$, this measure has radial distribution $-\mathrm{d}(Q^k)/\mathrm{d}L = W(L)\,\mathrm{d}L$. The Fubini–Study measure κ^k corresponds to $Q = t$, so its radial distribution is $k\mathrm{e}^{-k(L-1)}\mathrm{d}L$. Both measures are invariant under the unitary group of the chart, so q_α is the ratio of the two distributions, which is the stated formula. The lower bound $q_\alpha \geq b_0$ follows from $\kappa + \mathrm{dd}^{\mathrm{c}}u_\alpha \geq (1-\alpha)\kappa$ off o. As $L \to +\infty$, t tends to zero exponentially, so $Q \sim \alpha L^{\alpha-1}$ and $D \sim \alpha(1-\alpha)L^{\alpha-2}$; this gives the asymptotics of W. Consequently,

$$\log\bigl(\mathrm{e}^{\tau(L^\alpha-1)}W(L)\bigr) = \tau L^\alpha - [1+k(1-\alpha)]\log L + O(1) \longrightarrow +\infty,$$

and the mass of $B_\tau\kappa^k$ on $\{L > T\}$, which equals $\int_T^\infty \mathrm{e}^{\tau(L^\alpha-1)}W(L)\,\mathrm{d}L$, is infinite for every T. □

Theorem 15.34.2 *Let* $f \geq 0$ *be a nonzero integrable function on* X, *and put* $F = f/R$.

(1) *If* $\lambda > 0$ *and* $F \leq AB_\lambda$ *almost everywhere for a constant* $A > 0$, *then*

$$s_\lambda = u_\alpha + \lambda^{-1}\log(CV_0/A)$$

is an exponential subsolution at exponent λ, *of mass* CV_0, *and the equation* $\mathrm{MA}_H^{\mathrm{ext}}(u) = \mathrm{e}^{\lambda u}f\,\mathrm{d}V$ *has a greatest solution, whose mass is finite and positive.*

(2) *Suppose that* F *is locally essentially bounded on* $X \setminus (X_0 \times \{o\})$ *and*

$$\lim_{T\to+\infty} \operatorname*{ess\,sup}_{x\in X_0,\ L(y)>T} \frac{\log^+(F(x,y)/q_\alpha(y))}{L(y)^\alpha} = 0,$$

where $\log^+ r = \log\max(1,r)$ *for* $r \geq 0$. *Then the hypothesis of (1), and hence its conclusion, holds for every* $\lambda > 0$.

Proof (1) By Theorem 15.29.1, $\mathrm{MA}_H^{\mathrm{ext}}(u_\alpha) = CV_0q_\alpha\mu$. Off the pole fiber $X_0 \times \{o\}$ we have $\mathrm{e}^{\lambda u_\alpha}B_\lambda = q_\alpha$ and $f\,\mathrm{d}V = F\mu \leq AB_\lambda\mu$, hence

$$\mathrm{e}^{\lambda s_\lambda}f\,\mathrm{d}V \leq (CV_0/A)\mathrm{e}^{\lambda u_\alpha}AB_\lambda\mu = CV_0q_\alpha\mu = \mathrm{MA}_H^{\mathrm{ext}}(s_\lambda),$$

and both measures vanish on the pole fiber. Thus s_λ is an exponential subsolution, and its mass is CV_0, since adding a constant does not change the Monge–Ampère measure. Theorem 15.34.1 now gives the greatest solution u and the finiteness of its mass. The mass $\int_X \mathrm{e}^{\lambda u}f\,\mathrm{d}V$ is positive because u is finite almost everywhere and f has positive integral.

(2) Fix $\lambda > 0$. By hypothesis, there is T such that $F/q_\alpha \leq \mathrm{e}^{(\lambda/2)L^\alpha}$ almost everywhere on $\{L > T\}$. Since $\mathrm{e}^{(\lambda/2)L^\alpha} \leq \mathrm{e}^\lambda\mathrm{e}^{\lambda(L^\alpha-1)}$ for $L \geq 1$, we have $F \leq \mathrm{e}^\lambda B_\lambda$ there. On the complement $\{L \leq T\}$, which is compact and disjoint from the pole fiber, F is essentially bounded, while $B_\lambda \geq q_\alpha \geq b_0 > 0$. Increasing the constant, we obtain $F \leq AB_\lambda$ almost everywhere on X. □

Theorem 15.34.3 *Let F_j, $j \geq 1$, be smooth strictly positive probability densities on X_0 with respect to $\mathrm{d}V_0 = V_0^{-1} h_0^m$, whose normalization scalars on X_0, relative to h_0 and $\mathrm{d}V_0$, satisfy $c_{h_0}(F_j) \leq 2^{-2j}/C$. There are smooth strictly positive probability densities g_j on Y with respect to κ^k such that the probability densities*

$$f_j = RF_j g_j, \qquad f = \sum_{j \geq 1} 2^{-j} f_j$$

on X satisfy $c_H(f_j) \leq 2^{-2j}$ and $b_H(f) = 0$. The function f is finite everywhere, smooth off the pole fiber $X_0 \times \{o\}$, has integral one and is bounded below by a positive constant. For every $\lambda > 0$, the equation $\mathrm{MA}_H^{\mathrm{ext}}(u) = \mathrm{e}^{\lambda u} f \, \mathrm{d}V$ has an exponential subsolution and a greatest solution, of finite mass, and every solution is unbounded.

Proof Put $w_j = 2^{-j}$ and $M_j = \max_{X_0} F_j$.

Step 1: a smooth baseline. Fix C^j norms $\| \cdot \|_{C^j}$ on X_0, computed in a fixed finite atlas, so that they increase with j. Choose $0 < \varepsilon_j \leq 1/2$ with

$$w_j \varepsilon_j M_j \leq b_0 2^{-j-2}, \qquad w_j \varepsilon_j \| F_j \|_{C^j} \leq 2^{-j}.$$

Then $E = \sum_j w_j \varepsilon_j F_j$ converges in every C^l norm, so E is smooth on X_0, and $0 < E \leq b_0/4$.

Step 2: bumps escaping to the pole. Choose a smooth function $\chi \colon \mathbb{R} \to [0,1]$, supported in $[0,1]$, positive on $(0,1)$ and flat at 0 and 1. Starting with $T_0 = 0$, choose inductively $T_j \geq 2$ with $T_j > T_{j-1} + 2$ and

$$Z_j = \int_{T_j}^{T_j+1} \chi(L - T_j) \mathrm{e}^{(L^\alpha - 1)/j} W(L) \, \mathrm{d}L \geq 2 w_j M_j.$$

This is possible because, by Lemma 15.34.2 with $\tau = 1/j$, the factor $\mathrm{e}^{(L^\alpha - 1)/j} W(L)$ is uniformly large on sufficiently distant intervals. On the annulus $\{T_j \leq L \leq T_j + 1\}$ put $\beta_j = \chi(L - T_j) B_{1/j} / Z_j$, extend β_j by zero, and set $g_j = \varepsilon_j + (1 - \varepsilon_j)\beta_j$. Since χ is flat at the endpoints, β_j is smooth; it vanishes near o and near the hyperplane $\{L = 1\}$. Its integral against κ^k is one, so g_j is a smooth strictly positive probability density with respect to κ^k. Moreover,

$$w_j M_j (1 - \varepsilon_j) \beta_j \leq B_{1/j}/2.$$

Step 3: the scalars $c_H(f_j)$. By Theorem 10.1.1 on Y, there is a smooth ϕ_j with $\kappa + \mathrm{dd}^{\mathrm{c}} \phi_j > 0$ and $(\kappa + \mathrm{dd}^{\mathrm{c}} \phi_j)^k = g_j \kappa^k$; the scalar is one because both sides have mass one. On X_0, the same theorem gives a smooth v_j with $h_0 + \mathrm{dd}^{\mathrm{c}} v_j > 0$ and $(h_0 + \mathrm{dd}^{\mathrm{c}} v_j)^m = c_{h_0}(F_j) F_j \, \mathrm{d}V_0$. Since only one term of the binomial expansion survives for dimensional reasons,

$$(H + \mathrm{dd}^{\mathrm{c}}(v_j + \phi_j))^N = C c_{h_0}(F_j) F_j g_j \mu = C c_{h_0}(F_j) f_j \, \mathrm{d}V.$$

Uniqueness of the scalar gives $c_H(f_j) = C c_{h_0}(F_j) \leq 2^{-2j}$.

Step 4: properties of f. By Tonelli's theorem, $\int_X f\,\mathrm{d}V = \sum_j w_j = 1$. The density of $f\,\mathrm{d}V$ with respect to μ is

$$F = f/R = \sum_j w_j F_j g_j = E + \sum_j w_j(1-\varepsilon_j)F_j\beta_j.$$

The annuli are disjoint and escape to the pole, so only finitely many bumps meet any compact subset of $Y \setminus \{o\}$; hence F, and therefore f, is smooth off the pole fiber. On the pole fiber all bumps vanish and $F = E$, so f is finite everywhere. Also $f \geq w_1 f_1 \geq \frac{1}{2}\min_X f_1 > 0$. Since $f \geq w_j f_j$ and $c_H(f_j) \leq 2^{-2j} = w_j^2$ for every j, the argument in the proof of Theorem 15.26.1 shows that no $u \in \mathrm{PSH}(X,H)$ and $a > 0$ satisfy $\mathrm{MA}_H^{\mathrm{ext}}(u) \geq a f\,\mathrm{d}V$. Hence $b_H(f) = 0$ by Theorem 15.27.1.

Step 5: exponential subsolutions. Fix $\lambda > 0$, and choose J with $1/j \leq \lambda$ for $j \geq J$. Since $L^\alpha - 1 \geq 0$, we have $B_{1/j} \leq B_\lambda$ for these j. As the bumps have disjoint supports, Step 2 bounds their total contribution $\sum_{j\geq J} w_j(1-\varepsilon_j)F_j\beta_j$ by $B_\lambda/2$. The baseline satisfies $E \leq b_0/4 \leq B_\lambda/4$, because $B_\lambda \geq q_\alpha \geq b_0$. The remaining finitely many bumps are bounded by the constant

$$K_\lambda = \max_Y \sum_{j<J} w_j(1-\varepsilon_j)M_j\beta_j,$$

and $K_\lambda \leq (K_\lambda/b_0)B_\lambda$. Hence $F \leq A_\lambda B_\lambda$ with $A_\lambda = 3/4 + K_\lambda/b_0$, and Theorem 15.34.2 gives the exponential subsolution $u_\alpha + \lambda^{-1}\log(CV_0/A_\lambda)$ and the greatest solution. Finally, a bounded solution u would satisfy $\mathrm{MA}_H^{\mathrm{ext}}(u) \geq \mathrm{e}^{\lambda \inf_X u} f\,\mathrm{d}V$, a positive scaled subsolution, contradicting $b_H(f) = 0$. Hence every solution is unbounded. □

The density just constructed also satisfies the criterion in part (2) of Theorem 15.34.2: on the jth annulus, $F/q_\alpha \leq 1/4 + \frac{1}{2}\mathrm{e}^{(L^\alpha-1)/j}$, and away from all annuli the ratio is at most $1/4$. Approaching the pole through the annuli forces $j \to \infty$, so the required limit is zero. Since $b_H(f) = 0$, Proposition 15.27.2 also shows that f belongs to no L^p with $p > 1$.

Densities F_j as in the theorem exist whenever $v_-(h_0) = 0$, by Theorems 10.8.1 and 10.8.2. By Theorem 27.18.1, this holds on the elliptic threefold, so the construction applies to this threefold times $\mathbb{P}^k$ for every $k \geq 1$. With the balanced metric of Corollary 27.18.3, $V_0 = 24\pi$, so the explicit subsolutions have mass CV_0, equal to 96π on the fourfold ($k = 1$) and 240π on the fivefold ($k = 2$). These values concern the subsolutions; the greatest solutions have their own finite positive masses.

In this construction the projective components g_j are chosen after the densities F_j, to fit them. We do not know whether the exponential equation is solvable for every prescribed concentration family of this kind, and we have no necessary criterion for solvability at zero threshold.

15.34.3 Solvable exponents and collapse at zero threshold

We return to a general Hermitian form h on X and a nonzero density $0 \le f \in L^1(\mathrm{d}V)$. Let $\Lambda_h(f)$ be the set of exponents $\lambda > 0$ for which (15.101) has a solution. By Theorem 15.34.1, $\lambda \in \Lambda_h(f)$ if and only if the equation has an exponential subsolution at exponent λ.

Proposition 15.34.2 *If $\lambda_0 \in \Lambda_h(f)$, then $[\lambda_0, +\infty) \subset \Lambda_h(f)$. Consequently, $\Lambda_h(f)$ is empty, or equals $(a, +\infty)$ for some $a \ge 0$, or equals $[a, +\infty)$ for some $a > 0$. If $b_h(f) > 0$, then $\Lambda_h(f) = (0, +\infty)$.*

Proof Let u be a solution at λ_0, and put $s = u - \max(0, \sup_X u)$. Then $s \le \min(u, 0)$, so for $\lambda \ge \lambda_0$,

$$\begin{aligned} \mathrm{MA}_h^{\mathrm{ext}}(s) &= \mathrm{e}^{\lambda_0 u} f \, \mathrm{d}V \\ &\ge \mathrm{e}^{\lambda_0 s} f \, \mathrm{d}V \ge \mathrm{e}^{\lambda s} f \, \mathrm{d}V. \end{aligned}$$

Thus s is an exponential subsolution at every $\lambda \ge \lambda_0$, and Theorem 15.34.1 gives a solution. Hence $\Lambda_h(f)$ contains $[\lambda_0, +\infty)$ for each of its points λ_0. A nonempty subset of $(0, +\infty)$ with this property is determined by its infimum a and by whether a belongs to it, which gives the stated forms. If $b_h(f) > 0$, Corollary 15.34.1 gives $\Lambda_h(f) = (0, +\infty)$. □

Proposition 15.34.3 *Suppose $b_h(f) = 0$. Let $\lambda_j > 0$ decrease to zero, and let u_j be any solution of (15.101) at exponent λ_j. Put*

$$M_j = \sup_X u_j, \qquad v_j = u_j - M_j, \qquad c_j = \mathrm{e}^{\lambda_j M_j}.$$

Then

$$\begin{gathered} c_j \longrightarrow 0, \qquad \lambda_j M_j \longrightarrow -\infty, \\ \|\mathrm{MA}_h^{\mathrm{ext}}(u_j)\|_{\mathrm{TV}} = \|\mathrm{MA}_h^{\mathrm{ext}}(v_j)\|_{\mathrm{TV}} \longrightarrow 0. \end{gathered}$$

Moreover, $\mathrm{e}^{\lambda_j u_j} \to 0$ uniformly on X, where the exponential is zero on the pole set, and $u_j \to -\infty$ uniformly, that is, $\sup_X u_j \to -\infty$.

Proof Set $g_j = \mathrm{e}^{\lambda_j v_j} f$. The argument for the upper bound in the proof of Corollary 15.34.1, which does not use maximality of the solutions, gives $g_j \to f$ in L^1 and $0 < c_j \le b_h(g_j)$, hence $\limsup_j c_j \le b_h(f) = 0$. Since $0 \le g_j \le f$, the total masses are at most $c_j \|f\|_{L^1}$ and tend to zero. Moreover $\lambda_j M_j = \log c_j \to -\infty$, and $0 \le \mathrm{e}^{\lambda_j u_j} \le \mathrm{e}^{\lambda_j M_j} = c_j$. Finally, once $\lambda_j < 1$ and $\log c_j < 0$, we have $M_j = \lambda_j^{-1} \log c_j \le \log c_j$, so $M_j \to -\infty$. □

For the density of Theorem 15.34.3, solutions exist for every positive exponent, so the collapse applies as $\lambda \downarrow 0$ to every choice of solutions, in particular to the greatest ones. By contrast, the subsolutions of Theorem 15.34.2 keep the mass CV_0

for every exponent, while the masses of the solutions above them tend to zero: on a nonclosed background, the order of two potentials does not give an order of their total Monge–Ampère masses. Whether the normalized potentials converge in capacity, and which equation a normalized cluster limit satisfies at exponent zero, are not decided by this argument. Positive endpoints of $\Lambda_h(f)$ are governed instead by the compactness criterion below.

15.34.4 Positive endpoints and one-sided limits

For $\lambda \in \Lambda_h(f)$, write U_λ for the greatest solution given by Theorem 15.34.1. When the exponents stay above a fixed $\tau > 0$, the solution of a bounded truncated equation at exponent τ is a common upper barrier for all subsolutions. Together with the maximality of greatest solutions, this barrier gives an order estimate between U_{λ_1} and U_{λ_2}, which controls the positive endpoint of $\Lambda_h(f)$ and the one-sided limits of U_λ.

Lemma 15.34.3 *For every $\tau > 0$, there is a nonnegative $t_\tau \in \mathrm{PSH}(X,h) \cap C(X)$ such that every exponential subsolution at an exponent $\lambda \geq \tau$ lies below t_τ. Put $K_\tau = \sup_X t_\tau$. If $\tau \leq \lambda_1 \leq \lambda_2$ belong to $\Lambda_h(f)$, then*

$$U_{\lambda_2} \geq U_{\lambda_1} - K_\tau \frac{\lambda_2 - \lambda_1}{\lambda_2}.$$

Consequently, $\lambda \mapsto U_\lambda + K_\tau \log(\lambda/\tau)$ is nondecreasing on $\Lambda_h(f) \cap [\tau, +\infty)$.

Proof Set $q = \min(f, 1)$, a nonzero bounded density. By Theorem 11.1.2, there is a continuous $w \in \mathrm{PSH}(X,h)$ with $\mathrm{MA}_h(w) = \mathrm{e}^{\tau w} q \, \mathrm{d}V$. Put $t_\tau = w + \max(0, -\inf_X w)$. Then $t_\tau \geq 0$ and $t_\tau \geq w$, so for $\lambda \geq \tau$,

$$\mathrm{MA}_h(t_\tau) = \mathrm{e}^{\tau w} q \, \mathrm{d}V \leq \mathrm{e}^{\lambda t_\tau} f \, \mathrm{d}V.$$

Lemma 15.34.1, with $\nu = f \, \mathrm{d}V$, therefore shows that every exponential subsolution at an exponent $\lambda \geq \tau$ lies below t_τ.

For the order estimate, put $\delta = K_\tau(\lambda_2 - \lambda_1)/\lambda_2$. By the first part, $U_{\lambda_1} \leq t_\tau \leq K_\tau$, so $(\lambda_2 - \lambda_1)U_{\lambda_1} \leq \lambda_2 \delta$, that is, $\lambda_1 U_{\lambda_1} \geq \lambda_2(U_{\lambda_1} - \delta)$. Since the Monge–Ampère measure is unchanged by adding constants,

$$\begin{aligned} \mathrm{MA}_h^{\mathrm{ext}}(U_{\lambda_1} - \delta) &= \mathrm{e}^{\lambda_1 U_{\lambda_1}} f \, \mathrm{d}V \\ &\geq \mathrm{e}^{\lambda_2 (U_{\lambda_1} - \delta)} f \, \mathrm{d}V. \end{aligned}$$

Thus $U_{\lambda_1} - \delta$ is an exponential subsolution at λ_2, and the maximality of U_{λ_2} gives $U_{\lambda_2} \geq U_{\lambda_1} - \delta$. Finally, $1 - x^{-1} \leq \log x$ for $x \geq 1$; with $x = \lambda_2/\lambda_1$ this gives $\delta \leq K_\tau \log(\lambda_2/\lambda_1)$, which proves the monotonicity. □

Proposition 15.34.4 *Suppose $\Lambda_h(f) \neq \varnothing$ and $a = \inf \Lambda_h(f) > 0$. Then $a \in \Lambda_h(f)$ if and only if there are exponents $\lambda_j > a$ decreasing to a and solutions u_j at exponent λ_j with $\inf_j \sup_X u_j > -\infty$.*

If $a \notin \Lambda_h(f)$, then every sequence of solutions u_j at exponents $\lambda_j > a$ decreasing to a satisfies

$$\sup_X u_j \longrightarrow -\infty, \qquad \mathrm{e}^{\lambda_j u_j} \longrightarrow 0 \text{ uniformly on } X,$$
$$\|\mathrm{MA}_h^{\mathrm{ext}}(u_j)\|_{\mathrm{TV}} \longrightarrow 0,$$

where the exponential is zero on the pole sets.

Proof Suppose first that $a \in \Lambda_h(f)$, let u_a be a solution at a, and put $s = u_a - \max(0, \sup_X u_a)$. As in the proof of Proposition 15.34.2, s is an exponential subsolution at every $\lambda \geq a$, so $U_\lambda \geq s$. Hence the greatest solutions at any exponents $\lambda_j \downarrow a$ have suprema at least $\sup_X s > -\infty$.

Conversely, let λ_j and u_j be as in the statement. The lemma with $\tau = a$ gives the common upper bound $u_j \leq K_a$. Write $M_j = \sup_X u_j$. After extraction, $M_j \to M \in \mathbb{R}$ and, by Theorem 1.4.1, $u_j - M_j \to v$ in L^1 for some $v \in \mathrm{PSH}(X,h)$. Then $u_j \to u = v + M$ in L^1 and, after a further extraction, almost everywhere. Since $a \leq \lambda_j \leq \lambda_1$, $u_j \leq K_a$ and $K_a \geq 0$, the densities $\mathrm{e}^{\lambda_j u_j} f$ are bounded by $\mathrm{e}^{\lambda_1 K_a} f$, and dominated convergence gives

$$\mathrm{e}^{\lambda_j u_j} f \longrightarrow \mathrm{e}^{au} f \quad \text{in } L^1(\mathrm{d}V).$$

Lemma 15.27.1 now gives $\mathrm{MA}_h^{\mathrm{ext}}(u) \geq \mathrm{e}^{au} f \, \mathrm{d}V$. Thus u is an exponential subsolution at a, and Theorem 15.34.1 gives a solution at a.

Finally, suppose $a \notin \Lambda_h(f)$, and let u_j be solutions at exponents $\lambda_j \downarrow a$. If $\sup_X u_j$ did not tend to $-\infty$, a subsequence would satisfy the condition just shown to imply $a \in \Lambda_h(f)$. Hence $M_j \to -\infty$. Once $M_j < 0$, since $\lambda_j \geq a > 0$,

$$0 \leq \mathrm{e}^{\lambda_j u_j} \leq \mathrm{e}^{\lambda_j M_j} \leq \mathrm{e}^{a M_j} \longrightarrow 0,$$

and the total mass $\int_X \mathrm{e}^{\lambda_j u_j} f \, \mathrm{d}V$ is at most $\mathrm{e}^{a M_j} \|f\|_{L^1}$. □

Proposition 15.34.5 *For every $a \in \Lambda_h(f)$, the greatest solutions satisfy*

$$U_\lambda \longrightarrow U_a \quad \text{in } L^1 \text{ and in capacity} \quad (\lambda \downarrow a),$$

and their Monge–Ampère measures converge in total variation to $\mathrm{e}^{aU_a} f \, \mathrm{d}V$.

Proof Let $K = K_a$ be the constant of the lemma with $\tau = a$, and let $\lambda_j > a$ decrease to a. By the lemma, the functions

$$V_j = U_{\lambda_j} + K \log(\lambda_j / a)$$

decrease and satisfy $V_j \geq U_a$. Their limit V therefore belongs to $\mathrm{PSH}(X,h)$ and satisfies $V \geq U_a$. Dominated convergence between the integrable bounds U_a and

V_1 gives $V_j \to V$ in L^1, and Lemma 1.8.9 gives convergence in capacity. Since the added constants tend to zero, $U_{\lambda_j} \to V$ in both senses.

The nonpositive functions $U_{\lambda_j} - K$ satisfy

$$\mathrm{MA}_h^{\mathrm{ext}}(U_{\lambda_j} - K) = \mathrm{e}^{\lambda_j (U_{\lambda_j} - K)}\, \mathrm{e}^{\lambda_j K} f \, \mathrm{d}V,$$

and $\mathrm{e}^{\lambda_j K} f \to \mathrm{e}^{aK} f$ in L^1. Theorem 15.33.1 therefore gives $\mathrm{MA}_h^{\mathrm{ext}}(V) = \mathrm{e}^{aV} f \, \mathrm{d}V$, together with total-variation convergence of the measures $\mathrm{MA}_h^{\mathrm{ext}}(U_{\lambda_j})$. Hence V is a solution at a, so $V \le U_a$ by maximality, while $V \ge U_a$ by construction. Thus $V = U_a$. Since every sequence of exponents converging to a from above has a decreasing subsequence, this proves the assertions. □

The monotonicity of the corrected family $U_\lambda + K_\tau \log(\lambda/\tau)$ in Lemma 15.34.3 also controls the approach from the left. The left limit solves the equation at the limiting exponent, but maximality only places it below the greatest solution there.

Proposition 15.34.6 *Suppose $\Lambda_h(f) \neq \varnothing$, and put $a_0 = \inf \Lambda_h(f)$. For every $a > a_0$, there is $L_a \in \mathrm{PSH}(X, h)$ such that*

$$U_\lambda \longrightarrow L_a \quad \text{in } L^1 \text{ and in capacity} \quad (\lambda \uparrow a).$$

It satisfies

$$L_a \le U_a, \qquad \mathrm{MA}_h^{\mathrm{ext}}(L_a) = \mathrm{e}^{aL_a} f \, \mathrm{d}V.$$

The equation measures converge in total variation to this finite measure. The left limit does not depend on the auxiliary exponent or on the sequence of exponents used to construct it in the proof.

Proof Since $a > a_0$, there is $\tau \in \Lambda_h(f)$ with $\tau < a$, and Proposition 15.34.2 gives $[\tau, +\infty) \subset \Lambda_h(f)$. Put $K = K_\tau$ from Lemma 15.34.3 and

$$V_\lambda = U_\lambda + K \log(\lambda/\tau) \qquad (\lambda \in \Lambda_h(f),\ \lambda \ge \tau).$$

By Lemma 15.34.3, this family is nondecreasing in λ. For a sequence $\lambda_j \uparrow a$ with $\tau \le \lambda_j < a$, its members lie between the integrable potentials V_τ and V_a. The upper-envelope construction in Section 1.19 gives

$$V_a^- = \left(\sup_{\tau \le \lambda < a} V_\lambda \right)^* = \left(\lim_j V_{\lambda_j} \right)^* \in \mathrm{PSH}(X, h).$$

Indeed, since the family is monotone, every such sequence has the same pointwise supremum, and regularization changes this supremum only on a pluripolar set. Dominated convergence between the integrable bounds gives L^1 convergence to V_a^-, and Lemma 1.8.9 gives convergence in capacity. Since the added constants converge, set

$$L_a = V_a^- - K \log(a/\tau).$$

Then $U_{\lambda_j} \to L_a$ in both topologies and $L_a \le U_a$; the latter inequality survives regularization because V_a is upper semicontinuous.

By the lemma, $U_{\lambda_j} \le t_\tau \le K$, so the potentials $U_{\lambda_j} - K$ and their limit $L_a - K$ are nonpositive. As in the proof of Proposition 15.34.5, Theorem 15.33.1, applied with exponents tending to $a > 0$ and densities $\mathrm{e}^{\lambda_j K} f \to \mathrm{e}^{aK} f$ in L^1, proves the equation for L_a and the total-variation convergence.

Every sequence of exponents converging to a from the left has an increasing subsequence. If convergence of the potentials in either topology, or total-variation convergence of the equation measures, failed, there would be such a sequence staying outside a fixed neighborhood of the proposed limit, and its increasing subsequence would contradict the preceding argument. This proves the full left limit and measure convergence. Repeating the construction with another τ gives the same L^1 limit, and quasi-plurisubharmonic functions that agree almost everywhere agree everywhere. Thus the auxiliary choice does not change L_a. □

Proposition 15.34.7 *Suppose $\Lambda_h(f) \neq \varnothing$, put $a_0 = \inf \Lambda_h(f)$, and let L_a be the left limit of Proposition 15.34.6. The set*

$$D = \{a > a_0 : L_a \neq U_a\}$$

is at most countable. At every $a > a_0$ with $a \notin D$, the greatest solutions are continuous in L^1 and in capacity as $\lambda \to a$, and their equation measures converge in total variation to $\mathrm{e}^{aU_a} f \, \mathrm{d}V$.

Proof Fix $a_0 < a_1 < a_2$, choose $\tau \in \Lambda_h(f)$ with $\tau < a_1$, and use the corrected family V_λ from the preceding proof on $[a_1, a_2]$. Its integrals

$$I(\lambda) = \int_X V_\lambda \, \mathrm{d}V$$

form a bounded nondecreasing real function on $[a_1, a_2]$. For $a \in (a_1, a_2]$, the L^1 convergence in Proposition 15.34.6 gives

$$I(a) - I(a-) = \int_X (U_a - L_a) \, \mathrm{d}V.$$

If $L_a \neq U_a$, this quantity is positive: the integrable nonnegative difference $U_a - L_a$ cannot vanish almost everywhere, since quasi-plurisubharmonic functions that agree almost everywhere agree everywhere. For finitely many such points in $(a_1, a_2]$, the sum of their jumps is at most $I(a_2) - I(a_1)$. Consequently, for each positive integer ℓ there are only finitely many jumps of size at least $1/\ell$ in $(a_1, a_2]$, and hence at most countably many positive jumps. Covering $(a_0, +\infty)$ by countably many intervals $(a_1, a_2]$ with rational endpoints proves the assertion about D.

If $a \notin D$, the left limit is U_a, and Proposition 15.34.5 supplies the same right limit in both topologies and for the equation measures. Splitting an arbitrary approaching sequence into its two sides proves the stated two-sided convergence. □

We do not know whether the set D can be nonempty. We also know neither an intrinsic criterion on f for $a \in \Lambda_h(f)$, nor whether a positive infimum $a \notin \Lambda_h(f)$

can occur. These one-sided limits concern greatest solutions at positive exponents; they do not identify limits of arbitrary solutions with other singularities or supply a zero-exponent limit.

15.35 Radial selection for integrable densities

The scalar and measure limits in Corollary 15.34.1 do not by themselves determine a limit potential. For the sublogarithmic densities of Lemma 15.29.2, the radial flux supplies the missing information. It also distinguishes convergence after subtracting the supremum from convergence after subtracting $\lambda^{-1}\log b$, where b is the subsolution threshold of the density: the latter depends on a first moment of the limiting measure.

Use the product $X = X_0 \times Y$ with $Y = \mathbb{P}^k$, the background $H = h_0 + \kappa$ and the probability volume $\mu = V_0^{-1} h_0^m \wedge \kappa^k$ of Section 15.29, and put $b = CV_0 = \binom{N}{m} V_0$. For $0 < \alpha < 1$, let $U_{\lambda,\alpha}$ be the greatest solution of

$$\mathrm{MA}_H^{\mathrm{ext}}(U_{\lambda,\alpha}) = \mathrm{e}^{\lambda U_{\lambda,\alpha}} q_\alpha \mu, \qquad \lambda > 0.$$

The threshold of this forcing measure is $b > 0$, by Theorem 15.29.1, so the solutions exist by Corollary 15.34.1.

15.35.1 Separation and the radial flux

Lemma 15.35.1 *There is a κ-plurisubharmonic function $w_{\lambda,\alpha}$ on Y such that*

$$U_{\lambda,\alpha} = \lambda^{-1}\log b + w_{\lambda,\alpha}, \qquad w_{\lambda,\alpha} \geq u_\alpha, \qquad \mathrm{MA}_\kappa^{\mathrm{ext}}(w_{\lambda,\alpha}) = \mathrm{e}^{\lambda w_{\lambda,\alpha}} q_\alpha \kappa^k.$$

The last measure has mass one. On the affine chart centered at o, set $s = \log|z|^2$ and $\rho(s) = \log(1+\mathrm{e}^s)$, so that $\kappa = \mathrm{dd}^{\mathrm{c}}\rho$ there, and write radial functions as functions of s. The function $w = w_{\lambda,\alpha}$ is radial and satisfies

$$\begin{aligned} Q_{\lambda,\alpha}(s) &= \int_{-\infty}^{s} \mathrm{e}^{\lambda w(\sigma)} q_\alpha(\sigma)\, \mathrm{d}(\rho'(\sigma)^k), \\ w'(s) &= Q_{\lambda,\alpha}(s)^{1/k} - \rho'(s), \qquad Q_{\lambda,\alpha}(-\infty) = 0, \quad Q_{\lambda,\alpha}(+\infty) = 1. \end{aligned} \tag{15.103}$$

Proof Choose a smooth nondecreasing concave function χ on $[0,+\infty)$, equal to the identity on $[0,1]$ and constant on $[2,+\infty)$, and put $q_j = j\chi(q_\alpha/j)$. Concavity gives $q_j \uparrow q_\alpha$ and

$$\min(q_\alpha, j) \leq q_j \leq \min(q_\alpha, 2j).$$

The functions q_j are smooth and positive on Y: near o they are constant, and elsewhere the density is smooth. Let w_j be the continuous solution of $\mathrm{MA}_\kappa(w_j) = \mathrm{e}^{\lambda w_j} q_j \kappa^k$ from Theorem 11.1.2. Since $u_\alpha \leq 0$ and $q_j \leq q_\alpha$, we have $\mathrm{MA}_\kappa^{\mathrm{ext}}(u_\alpha) =$

$q_\alpha \kappa^k \geq \mathrm{e}^{\lambda u_\alpha} q_j \kappa^k$, and Lemma 15.34.1 on Y gives $u_\alpha \leq w_j$. Since the q_j increase, Lemma 11.4.2 shows that the w_j decrease; let w be their limit.

By part (1) of Lemma 15.29.1, the pullback $W_j = \lambda^{-1} \log b + w_j$ is bounded and satisfies $\mathrm{MA}_H(W_j) = \mathrm{e}^{\lambda W_j} q_j \mu \leq \mathrm{e}^{\lambda W_j} q_\alpha \mu$. Lemma 15.34.1 on X, with $\nu = q_\alpha \mu$, gives $U_{\lambda,\alpha} \leq W_j$.

Monotone convergence in capacity and Theorem 15.33.1 on Y identify the equation for $w \geq u_\alpha$. Moreover, since κ is closed, Stokes' theorem gives $\int_Y \mathrm{e}^{\lambda w_j} q_j \kappa^k = 1$. The integrands are dominated by $\mathrm{e}^{\lambda \sup_Y w_1} q_\alpha$, so dominated convergence gives the same identity for w. By (15.85), the pullback of $\lambda^{-1} \log b + w$ solves the equation on X, and hence lies below $U_{\lambda,\alpha}$. Together with $U_{\lambda,\alpha} \leq \lim_j W_j$, this proves separation and all measure assertions.

Uniqueness makes each w_j invariant under the unitary action on the affine coordinates $z \in \mathbb{C}^k$, and the limit has the same invariance. A radial κ-plurisubharmonic function has a convex nondecreasing profile $\rho + w$. Symmetric convolution of this convex profile on compact s intervals, decreasing as the scale decreases, and bounded-product continuity give its radial measure $\mathrm{d}((\rho + w)'^k)$, as in Lemma 15.29.2. The bound $w \geq u_\alpha$, whose growth is $o(|s|)$ at $-\infty$, forces the initial derivative to vanish. The positive density makes the flux continuous and positive at finite s, yielding (15.103). Its final value is the mass one already proved. The profile is consequently smooth on finite s intervals. □

15.35.2 The normalized limit

Set

$$a_{\lambda,\alpha} = \sup_Y w_{\lambda,\alpha}, \qquad v_{\lambda,\alpha} = w_{\lambda,\alpha} - a_{\lambda,\alpha}, \qquad c_{\lambda,\alpha} = \mathrm{e}^{\lambda \sup_X U_{\lambda,\alpha}} = b\mathrm{e}^{\lambda a_{\lambda,\alpha}}.$$

In particular $a_{\lambda,\alpha} \geq 0$.

Theorem 15.35.1 *Let $\alpha_j \to \alpha \in (0,1)$ and $\lambda_j \to 0$ with $\lambda_j > 0$, at arbitrary relative rates. Then*

$$v_{\lambda_j,\alpha_j} \longrightarrow u_\alpha \quad \text{in } L^1(Y), \text{ in capacity, and locally uniformly on } Y \setminus \{o\}.$$

Their pullbacks converge in $L^1(X)$, in capacity on X, and locally uniformly off $X_0 \times \{o\}$. Furthermore $q_{\alpha_j} \to q_\alpha$ in $L^1(\kappa^k)$, $c_{\lambda_j,\alpha_j} \to b$, and the equation measures converge in total variation to $b q_\alpha \mu$. The limiting pullback has finite mass b and belongs to $\mathcal{E}_{\mathrm{env}}(X, H, 0)$.

Proof Choose $[\alpha_-, \alpha_+] \Subset (0,1)$ containing all late parameters. The density formula in Lemma 15.29.2 gives, uniformly for δ in this interval and $t = -s$ large,

$$q_\delta(s) \leq C\mathrm{e}^{kt}(1+t)^{-1-k(1-\alpha_+)}.$$

Away from o the densities are jointly smooth, including on the hyperplane at infinity. Thus one integrable density g dominates all q_δ, and dominated convergence gives $q_{\alpha_j} \to q_\alpha$ in L^1.

For every normalized κ-plurisubharmonic v, Theorem 1.4.1 gives $\int_Y (-v)\kappa^k \le C_k$. Absolute continuity of the integral of g therefore implies

$$\sup_{\sup_Y v=0} \int_{\{v<-T\}} g\kappa^k \longrightarrow 0 \quad (T \to \infty).$$

Taking $T = \lambda^{-1/2}$ shows that

$$\int_Y (1 - \mathrm{e}^{\lambda v}) q_\delta \kappa^k \le \lambda T + \int_{\{v<-T\}} g\kappa^k \longrightarrow 0$$

uniformly in δ and v. Write $v_j = v_{\lambda_j,\alpha_j}$ and $a_j = a_{\lambda_j,\alpha_j}$. The exact mass identity of the lemma is

$$1 = \mathrm{e}^{\lambda_j a_j} \int_Y \mathrm{e}^{\lambda_j v_j} q_{\alpha_j} \kappa^k. \tag{15.104}$$

It gives $\mathrm{e}^{\lambda_j a_j} \to 1$, hence $\lambda_j a_j \to 0$. The equation densities $\mathrm{e}^{\lambda_j a_j} \mathrm{e}^{\lambda_j v_j} q_{\alpha_j}$ converge in L^1 to q_α and are bounded by $2g$ for all late j.

Their cumulative radial fluxes Q_j consequently converge uniformly to the flux Q_α of u_α. By (15.103), $v_j' \to u_\alpha'$ uniformly on compact s intervals. The normalized L^1 bound controls one value of each profile on a fixed annulus; thus every sequence has a subsequence converging uniformly on compact s intervals to $u_\alpha + d$ for some finite constant d.

Near infinity the common bound for the densities is smooth and bounded. Hence $1 - Q_j(s) \le C\mathrm{e}^{-s}$ and $|v_j'(s)| \le C\mathrm{e}^{-s}$ for large s. Integrating gives uniform convergence up to the hyperplane at infinity. At the other end, monotonicity of $\rho + v_j$ gives, for $s < s_0$,

$$v_j(s) \le v_j(s_0) + \rho(s_0).$$

Choose s_0 so negative that the right side is negative for all late members of the subsequence. The supremum zero is then attained outside this inner ball. Uniform convergence on its compact complement gives $\sup_Y (u_\alpha + d) = 0$, so $d = 0$. Normalized compactness identifies every L^1 cluster limit with u_α; this proves convergence of the entire family and its local uniform convergence off o.

For capacity convergence, fix $T > 1$. The sublevel $\{u_\alpha < -T\}$ is a neighborhood of o, and its complement is a compact set on which convergence is uniform. By (1.15), the capacity of this sublevel is at most C/T. Let $j \to \infty$ and then $T \to \infty$. The same argument on X uses the normalized pullback of u_α and proves convergence in Hermitian capacity there; comparison sets omit pole sets. Pullback also preserves the L^1 and local uniform limits. Formula (15.85) proves total-variation convergence of the measures, and Theorem 15.29.1 supplies the exact limiting mass and envelope class. □

15.35.3 Scalar centering and its moment boundary

Proposition 15.35.1 *The first moment*

$$M_\alpha = \int_Y u_\alpha q_\alpha \kappa^k$$

is finite exactly when $\alpha < k/(k+1)$. In this range,

$$-M_\alpha = \alpha \int_1^\infty x^{\alpha-1} \left[\alpha x^{\alpha-1} + (1-\alpha x^{\alpha-1})\mathrm{e}^{1-x}\right]^k \, \mathrm{d}x. \tag{15.105}$$

For the joint limits of Theorem 15.35.1, if $\alpha < k/(k+1)$, then

$$a_{\lambda_j,\alpha_j} \to -M_\alpha, \qquad U_{\lambda_j,\alpha_j} - \lambda_j^{-1} \log b \to u_\alpha - M_\alpha$$

in L^1, in capacity, and locally uniformly away from the pole fiber. If $\alpha \geq k/(k+1)$, then $a_{\lambda_j,\alpha_j} \to +\infty$ and the latter translated potentials tend to $+\infty$ locally uniformly off the pole fiber, although $\lambda_j a_{\lambda_j,\alpha_j} \to 0$. In both cases the exact scalar error is

$$\lambda_j^{-1} \log(c_{\lambda_j,\alpha_j}/b) = a_{\lambda_j,\alpha_j}.$$

Proof As $t = -s \to \infty$, the radial profile of Lemma 15.29.2 satisfies

$$u_\alpha \sim -t^\alpha, \qquad Q'_\alpha(s) \sim k(1-\alpha)\alpha^k t^{-1-k(1-\alpha)}.$$

Thus the first moment is finite precisely when $(k+1)\alpha < k$. In that case, write $x = 1 - \psi$. By Lemma 15.34.2, the cumulative flux of u_α is

$$Q_\alpha = \left[\alpha x^{\alpha-1} + (1-\alpha x^{\alpha-1})\mathrm{e}^{1-x}\right]^k.$$

Integration by parts gives (15.105); the pole boundary term vanishes since $(k+1)\alpha - k < 0$.

Suppose $\alpha < k/(k+1)$ and choose the parameter interval in the preceding proof with $\alpha_+ < k/(k+1)$. Its dominating density g satisfies $(-u_{\alpha_+})g \in L^1$. Write $w_j = w_{\lambda_j,\alpha_j} = v_j + a_j$. Since $w_j \geq u_{\alpha_j} \geq u_{\alpha_+}$, the mass identity and $\mathrm{e}^r \geq 1 + r$ give $\int_Y w_j q_{\alpha_j} \kappa^k \leq 0$. On a fixed annulus A away from o, the normalized smooth-volume L^1 bound gives

$$\int_A w_j q_{\alpha_j} \kappa^k \geq a_j \int_A q_{\alpha_j} \kappa^k - C.$$

On the complement, use $w_j \geq u_{\alpha_+}$ and the integrable lower bound. The annulus has uniformly positive measure, so $0 \leq a_j \leq C_*$. Along any subsequence with $a_j \to a$, local uniform convergence gives $w_j \to u_\alpha + a$ off o. In the exact identity

$$\int_Y \frac{\mathrm{e}^{\lambda_j w_j} - 1}{\lambda_j} q_{\alpha_j} \kappa^k = 0,$$

the negative part is dominated by $(-u_{\alpha_+})g$ and the positive part by $C_* e^{C_*} g$. Dominated convergence gives $M_\alpha + a = 0$. Every subsequence therefore has the asserted scalar limit. Adding these convergent constants to the normalized potentials gives all three claimed topologies.

If $\alpha \geq k/(k+1)$, use instead

$$\int_Y \frac{1 - e^{\lambda_j v_j}}{\lambda_j} q_{\alpha_j} \kappa^k = \frac{1 - e^{-\lambda_j a_j}}{\lambda_j} \leq a_j.$$

The integrands are nonnegative and tend off o to $(-u_\alpha)q_\alpha$. Fatou's lemma forces $a_j \to +\infty$. The preceding theorem gives $\lambda_j a_j \to 0$; local uniform convergence of v_j gives the stated divergence of w_j. This includes sequences approaching the moment boundary from below. □

For the elliptic fivefold of Corollary 27.18.6, $b = 240\pi$ and the moment boundary is $\alpha = 2/3$. The choice $\alpha = 1/2$ has a finite centering constant. For $2/3 \leq \alpha < 1$, the supremum-normalized family still converges to the exact threshold solution, while its scalar-centered version diverges off the pole fiber. Thus convergence of the normalization scalars alone does not give a bounded first-order scalar error.

15.35.4 Order in the separated class

We use here the closed-class uniqueness theorem: on a compact connected Kähler manifold, two potentials whose non-pluripolar measures have full mass and are equal differ by a constant. This theorem is not proved in this book; its source is given in Section 36.11.

Corollary 15.35.1 *Every separated threshold subsolution $r = r(y)$ with $r \in \mathrm{PSH}(Y, \kappa)$ and $\mathrm{MA}_H^{\mathrm{ext}}(r) \geq b q_\alpha \mu$ equals $u_\alpha + d$ for some constant d. In particular, u_α is the only such subsolution with supremum zero. If $\alpha < k/(k+1)$, then $u_\alpha - M_\alpha$ is the greatest such subsolution whose q_α-mean on Y is nonpositive.*

Proof By part (1) of Lemma 15.29.1, the inequality reduces to $\mathrm{MA}_\kappa^{\mathrm{ext}}(r) \geq q_\alpha \kappa^k$. As in the proof of Theorem 15.29.2, Stokes' theorem on Y bounds the mass of $\mathrm{MA}_\kappa^{\mathrm{ext}}(r)$ by one, so equality holds and the measure has full mass. Closed-class uniqueness yields $r = u_\alpha + d$. In the finite-moment range its mean is $M_\alpha + d$, which is nonpositive exactly when $d \leq -M_\alpha$. □

At and beyond the moment boundary every constant translate has mean $-\infty$, so that mean condition no longer chooses a greatest translate. The order assertion concerns separated potentials. Selection for arbitrary integrable densities and uniqueness of arbitrary critical potentials on the nonclosed product remain the questions of Problem 35.1.1 and Section 15.33.

15.36 Residual regularity in radial singularity types

Subtracting a prescribed singularity separates regularity of the residual from the pole of the solution. On projective space, central radial mass gives an exact inverse for the equation. A relative density error then controls the residual through two derivatives, even when the density is merely integrable at the pole. The inverse also permits smooth crossing of density zeros away from it. The pole is the one used in Lemma 15.29.2; here its ordinary type is fixed while the density varies.

15.36.1 The prescribed ordinary type

Let $k \geq 1$, let κ be the Fubini–Study form on $\mathbb{P}^k$ with $\int \kappa^k = 1$, and put $o = [1 : 0 : \cdots : 0]$. On the chart $Z_0 \neq 0$ write $z_i = Z_i/Z_0$ and set

$$r = \sum_{j=1}^{k} |z_j|^2, \qquad P = \frac{r}{1+r}, \qquad T = 1 - \log P, \qquad u_\alpha = 1 - T^\alpha \quad (0 < \alpha < 1).$$

The ratio $P = \sum_{i=1}^{k} |Z_i|^2 / \sum_{i=0}^{k} |Z_i|^2$ is global. Give u_α its canonical value $-\infty$ at o and value zero on the hyperplane $Z_0 = 0$. Primes denote differentiation in r. Throughout this section $b_\alpha = k(1-\alpha) > 0$ and, near $r = 0$, $L = 1 - \log r$.

Since $\log P$ is the function ψ of Section 15.29, u_α is the potential of Lemma 15.29.2: it is κ-plurisubharmonic and smooth off o, and its non-pluripolar measure $q_\alpha \kappa^k$ has mass one, with a positive density q_α in L^1 and in no L^p, $p > 1$. In the notation of the proof of that lemma, the total potential and its radial flux are

$$\Phi_\alpha = \log(1+r) + u_\alpha, \qquad Q_\alpha = r\Phi_\alpha' = P + \alpha(1-P)T^{\alpha-1}. \tag{15.106}$$

The radial determinant integration of that proof applies to every smooth radial function Φ on an annulus $\{a < |z|^2 < b\}$:

$$\int_{a<|z|^2<b} (\mathrm{dd^c}\Phi)^k = Q(b)^k - Q(a)^k, \qquad Q = r\Phi'. \tag{15.107}$$

Since $Q_\alpha(0) = 0$ and $Q_\alpha(\infty) = 1$, the density is

$$q_\alpha = \frac{(Q_\alpha^k)'}{(P^k)'} > 0. \tag{15.108}$$

Near $r = 0$, direct differentiation gives

$$Q_\alpha \sim \alpha L^{\alpha-1}, \qquad Q'_\alpha \sim \frac{\alpha(1-\alpha)}{r} L^{\alpha-2}, \tag{15.109}$$
$$(Q_\alpha^k)' \sim \frac{k\alpha^k(1-\alpha)}{r} L^{-1-k(1-\alpha)}.$$

In particular $q_\alpha \sim \alpha^k(1-\alpha) r^{-k} L^{-1-b_\alpha}$.

Lemma 15.36.1 *Distinct α give distinct ordinary types, although every multiplier ideal $\mathcal{I}(tu_\alpha)$, $t > 0$, is trivial.*

Proof For $\alpha < \beta$, the difference $u_\alpha - u_\beta = T^\beta - T^\alpha$ is unbounded at o. Near o, integrability of e^{-tu_α} against the smooth volume, which is comparable to $r^{k-1}\,\mathrm{d}r$ in the radial variable, reduces to $\int^\infty \mathrm{e}^{-kL+tL^\alpha}\,\mathrm{d}L < \infty$. Ordinary type information here is strictly finer than this ideal information. □

15.36.2 Density criterion and exact inverse

Here "radial" means depending only on r in the displayed chart. The smoothness assumption along the hyperplane includes a smooth profile in $t = 1/r$ near $t = 0$.

Theorem 15.36.1 *Let $q \geq 0$ be radial, smooth on $\mathbb{P}^k \setminus \{o\}$, including the hyperplane at infinity, and satisfy $\int q\kappa^k = 1$. Suppose that for some $\delta, A, r_0 > 0$,*

$$\left| \frac{q(r)}{q_\alpha(r)} - 1 \right| \leq A r^\delta \quad (0 < r < r_0). \tag{15.110}$$

Then there is a radial κ-psh potential U, normalized to have value zero on the hyperplane, with

$$\mathrm{MA}_\kappa(U) = q\kappa^k, \qquad U \sim u_\alpha.$$

Its residual $R = U - u_\alpha$ off o extends continuously across o, with value R_0, and for small r satisfies

$$\begin{aligned} |R - R_0| &\leq C r^\delta L^{\alpha-2}, \\ |R'| \leq C r^{\delta-1} L^{\alpha-2}, &\qquad |R''| \leq C r^{\delta-2} L^{\alpha-2}. \end{aligned} \tag{15.111}$$

It is locally $C^{0,\min\{1,2\delta\}}$. For $\delta \geq \frac{1}{2}$ it extends C^1 with zero gradient at o; for $\delta \geq 1$ it extends C^2 with zero Hessian at o. The potential and its residual are smooth on $\mathbb{P}^k \setminus \{o\}$, including every density zero there.

No derivative of the error in (15.110) is controlled. Smoothness across density zeros comes from the positivity of the central mass of every ball around o, which (15.110) implies, not from the smoothness of the density alone.

Proof Let

$$\begin{aligned} M(r) &= \int_{|z|^2<r} q\kappa^k, \qquad Q(r) = M(r)^{1/k}, \\ U'(r) &= \frac{Q(r)-P(r)}{r}, \quad U(\infty) = 0. \end{aligned} \tag{15.112}$$

Condition (15.110) makes q positive near o, so $M(r) > 0$ for every $r > 0$. One has $M' = q(P^k)' \geq 0$, $Q' \geq 0$ and $M(\infty) = 1$. Thus the positive root defining Q is smooth for every finite $r > 0$, even where q vanishes. This proves smoothness across all finite density zeros without a vanishing-order assumption. In $t = 1/r$, the tail $1 - M$ is the integral from 0 to t of the smooth density profile times $-\mathrm{d}(1+t)^{-k}/\mathrm{d}t$. Thus M and Q are smooth in t near $t = 0$, with $Q = 1$ at $t = 0$. The equation $\mathrm{d}U/\mathrm{d}t = -(Q-P)/t$ is smooth there, since $Q - P$ vanishes at $t = 0$. This both defines the normalization and proves smoothness along the full hyperplane.

From (15.110) and (15.109),

$$|M - Q_\alpha^k| \leq C \int_0^r s^{\delta-1}(1-\log s)^{-1-b_\alpha}\,\mathrm{d}s \leq Cr^\delta L^{-1-b_\alpha}. \tag{15.113}$$

The last inequality follows by writing $s = r\mathrm{e}^{-y}$ and bounding $\int_0^\infty \mathrm{e}^{-\delta y}(L+y)^{-1-b_\alpha}\,\mathrm{d}y$. Dividing by $Q_\alpha^k \asymp L^{-b_\alpha}$ shows that Q/Q_α tends to one; the mean value theorem for the positive kth root then gives

$$|Q - Q_\alpha| \leq Cr^\delta L^{\alpha-2}. \tag{15.114}$$

Moreover the exact identity

$$\frac{Q'}{Q'_\alpha} = \frac{q}{q_\alpha}\left(\frac{Q_\alpha}{Q}\right)^{k-1}$$

gives $|Q' - Q'_\alpha| \leq Cr^{\delta-1}L^{\alpha-2}$. Since $R' = (Q - Q_\alpha)/r$, integration to the origin gives a finite R_0 and the first two inequalities in (15.111). Differentiating this formula gives the last inequality.

The local potential $\Phi = \log(1+r) + U$ has semipositive Hessian away from o because $Q/r > 0$ and $Q' \geq 0$. Also $U = u_\alpha + R_0 + o(1)$ near o, so Φ tends to $-\infty$. As a function of $\log r$, the profile of Φ has the nonnegative increasing slope Q, so the maxima $\max\{\Phi, \Phi(\varepsilon)\}$ are plurisubharmonic. They decrease to the locally integrable function Φ as $\varepsilon \downarrow 0$, and the local decreasing-limit property shows that Φ, with the value $-\infty$ at o, is plurisubharmonic. Thus U is globally κ-psh. Formula (15.107) or the local determinant ratio gives its measure $q\kappa^k$ away from o; the defining non-pluripolar truncations exclude o and are locally identical to this smooth product elsewhere. The measure has exactly total mass one. The bounded global residual proves the ordinary type.

In real coordinates, with $\rho = |z| = \sqrt{r}$,

$$|\nabla R| \leq C\rho^{2\delta-1}L^{\alpha-2}, \qquad |\nabla^2 R| \leq C\rho^{2\delta-2}L^{\alpha-2}.$$

For $2\delta < 1$, integration along radii bounds differences by $C|\rho_1 - \rho_2|^{2\delta}$ and hence by $C|z_1 - z_2|^{2\delta}$. For $2\delta \geq 1$ the gradient is bounded, giving local Lipschitz regularity. At $\delta = \frac{1}{2}$ its bound tends to zero and $R - R_0 = o(\rho)$, which give the claimed C^1 extension. At $\delta = 1$, the Hessian bound tends to zero and $\nabla R = o(\rho)$, giving the C^2 extension. Larger δ only improves these bounds. These arguments justify crossing the pole for R; they do not assign a finite value to $U(o)$. □

Among radial solutions with this density that are smooth off o, including the hyperplane, and have no central flux ($Q(r) \to 0$ as $r \to 0$), the potential is unique up to constants: its flux satisfies $Q^k = M$, and (15.112) then fixes its derivative. No nonsymmetric uniqueness is asserted.

Since $U \sim u_\alpha$, Proposition 14.4.1 gives $U \in \mathcal{E}_{\mathrm{env}}(\mathbb{P}^k, \kappa, u_\alpha)$.

15.36.3 Smooth residuals from flat errors

Proposition 15.36.1 *Under the hypotheses of Theorem 15.36.1, suppose* $g = q/q_\alpha - 1$ *satisfies*

$$(r\partial_r)^j g = O(r^N) \quad (r \to 0) \quad \text{for every } N, j \geq 0.$$

Then $R - R_0$ *extends smoothly across* o*, with every derivative zero there.*

Proof Put $x = -\log r$, $a(x) = Q_\alpha(\mathrm{e}^{-x})^k$ and $E(x) = M(\mathrm{e}^{-x}) - a(x)$. The exact central integral is

$$E(x) = \int_x^\infty g(\mathrm{e}^{-y})[-a'(y)]\,\mathrm{d}y, \qquad E'(x) = g(\mathrm{e}^{-x})a'(x).$$

Every derivative of a has at most polynomial growth in $1+x$, and $a \asymp (1+x)^{-b_\alpha}$. The integral bounds E by an exponential of arbitrarily large rate. Repeated differentiation of E' expresses each derivative as a finite Leibniz sum of flat derivatives of g and polynomially bounded derivatives of a; all derivatives of E are therefore exponentially flat. Every derivative of $1/a$ has at most polynomial growth, by (15.106). Thus E/a has the same flatness. Applying smooth composition near one to

$$Q - Q_\alpha = Q_\alpha\left[(1 + E/a)^{1/k} - 1\right]$$

proves flatness of every x derivative of $Q - Q_\alpha$. Since $\mathrm{d}R/\mathrm{d}x = -(Q - Q_\alpha)$, integration from x to infinity proves the same for $R - R_0$, including its zeroth derivative. An ordinary jth derivative in r is r^{-j} times a polynomial in ∂_x; arbitrarily strong exponential decay absorbs that factor. All ordinary derivatives extend continuously as zero at $r = 0$, and repeated one-dimensional integration verifies smoothness there. Composition with $r = |z|^2$ gives the conclusion. □

15.36.4 Examples with sharp derivative thresholds

Proposition 15.36.2 *The C^1 threshold $\delta = 1/2$ and the C^2 threshold $\delta = 1$ of Theorem 15.36.1 are sharp. When $2\delta < 1$, the stated Hölder exponent is also sharp. There are radial probability densities $q \neq q_\alpha$ to which Proposition 15.36.1 applies, so that $R - R_0$ is smooth and flat at o.*

Proof Choose smooth nonnegative radial cutoffs χ_0, χ_1 with disjoint supports: $\chi_0 = 1$ near $r = 0$ and vanishes outside a finite small ball; $\chi_1 \not\equiv 0$ is supported in a finite annulus. Put

$$B = \frac{\int r^\delta \chi_0 q_\alpha \kappa^k}{\int \chi_1 q_\alpha \kappa^k}, \qquad q_c = q_\alpha \big[1 + c(r^\delta \chi_0 - B\chi_1)\big]. \tag{15.115}$$

For sufficiently small $c > 0$, q_c is positive, smooth off o and along the hyperplane, and has total mass exactly one. It is comparable to q_α near o, hence is in no L^p, $p > 1$. Theorem 15.36.1 gives a solution on $\mathbb{P}^k$ in the prescribed ordinary type. Near o the difference $M_c - Q_\alpha^k$ of the central masses is exactly $c\int_0^r s^\delta (Q_\alpha^k)'\,\mathrm{d}s$. The substitution $s = r\mathrm{e}^{-y}$ and dominated convergence give

$$\int_0^r s^{\delta-1}(1 - \log s)^{-1-b_\alpha}\,\mathrm{d}s \sim \delta^{-1} r^\delta L^{-1-b_\alpha}.$$

Using (15.109) and the kth-root expansion yields

$$R_c'(r) \sim \frac{c\alpha(1-\alpha)}{\delta} r^{\delta-1} L^{\alpha-2},$$
$$R_c(r) - R_c(0) \sim \frac{c\alpha(1-\alpha)}{\delta^2} r^\delta L^{\alpha-2}.$$

For $\delta < \frac{1}{2}$, the radial gradient $2\sqrt{r}\,R_c'$ diverges, ruling out C^1 extension. For $\delta < 1$, a tangential real Hessian eigenvalue $2R_c'$ diverges, ruling out C^2 extension. The annular compensation does not change the central mass near o. For $2\delta < 1$ the displayed asymptotic also rules out any larger Hölder exponent. The critical derivative thresholds in Theorem 15.36.1 are therefore sharp for this family.

Replacing $r^\delta \chi_0$ in (15.115) by $\mathrm{e}^{-1/r}\chi_0$ and adjusting B by the same integral formula gives a nonzero mass-preserving perturbation satisfying every flat-error hypothesis. Proposition 15.36.1 then shows that $R - R_0$ is smooth and flat at the pole. □

15.36.5 Smooth residuals with an open density-zero annulus

Proposition 15.36.3 *There is a nonnegative radial probability density q_0, smooth off o and along the hyperplane, which equals q_α near o and vanishes on a nonempty*

open annulus. Its solution has ordinary type u_α and a globally smooth residual, constant near o.

Proof Choose a smooth radial h with $0 \le h \le 1$, equal to zero on a nonempty finite annulus and equal to one near o and outside a larger finite annulus. Choose a nonzero nonnegative smooth radial cutoff χ supported in a disjoint finite annulus where $h = 1$. Set

$$C = \frac{\int (1-h) q_\alpha \kappa^k}{\int \chi q_\alpha \kappa^k} > 0, \qquad q_0 = q_\alpha (h + C\chi).$$

This is nonnegative and has integral one. It is smooth at every point off o, including the hyperplane, and equals q_α near o. It vanishes on the chosen open annulus and is still in no L^p, $p > 1$. Theorem 15.36.1 applies, with zero relative density error on a neighborhood of o. There the exact central fluxes agree, so $Q = Q_\alpha$ and $R' = 0$. Hence R is constant near o and extends smoothly across it. It is smooth everywhere else by the positive-root inverse, including the entire zero annulus. The solution itself keeps its pole at o; the smoothness assertion concerns the residual. □

15.36.6 Products with a Hermitian factor

Use the product $X = X_0 \times Y$ with $Y = \mathbb{P}^k$, the background $H = h_0 + \kappa$, $N = m + k$ and $V_0 = \int_{X_0} h_0^m$ of Section 15.29. Let q and U be as in Theorem 15.36.1; in the product formulas below, U and R denote their pullbacks.

Corollary 15.36.1 *The pullback of U has the ordinary type of the pullback of u_α, belongs to its relative envelope class $\mathcal{E}_{\mathrm{env}}(X, H, u_\alpha)$ and has finite Monge–Ampère mass $\binom{N}{m} V_0$. Its residual has the regularity of Theorem 15.36.1 across $X_0 \times \{o\}$. If $v_-(h_0) = 0$, the background H is nonclosed and $v_-(H) = 0$.*

Proof By part (1) of Lemma 15.29.1,

$$\mathrm{MA}_H(U) = \binom{N}{m} h_0^m \wedge q\kappa^k, \qquad \int_X \mathrm{MA}_H(U) = \binom{N}{m} V_0.$$

The residual is the pullback of R, so its regularity extends across $X_0 \times \{o\}$, and it is bounded. Thus the pullback of U has the ordinary type of the pullback of u_α, and Proposition 14.4.1 gives the envelope class. The last assertion is part (3) of Lemma 15.29.1. □

15.37 Envelopes for singular radial obstacles

The radial inverse of Section 15.36 starts with a density. One can instead perturb the singular reference as an obstacle and let the envelope select its measure. The

resulting total profile is a convex minorant. Its affine intervals give open regions of zero density, while its contact with the reference at both ends controls the residual across the pole and the hyperplane at infinity.

Keep $0 < \alpha < 1$, the normalized Fubini–Study form κ on $\mathbb{P}^k$ and the potential u_α of Section 15.36. Write $r = |z|^2$ on the chart $Z_0 \neq 0$, put $x = \log r$, and define

$$A(x) = \log(1 + \mathrm{e}^x) + u_\alpha(x).$$

Here $u_\alpha(x)$ denotes its radial profile. Throughout this section primes denote derivatives in x. For $F \in C_c^2(\mathbb{R};\mathbb{R})$, extend $F(\log r)$ by zero near the pole o and the hyperplane at infinity, and set

$$E_F = \Big(\sup\{v \in \mathrm{PSH}(\mathbb{P}^k, \kappa) : v \le u_\alpha + F \text{ everywhere},\ v \preceq u_\alpha\}\Big)^*. \tag{15.116}$$

The obstacle takes the value $-\infty$ at o. Since F is bounded, its pointwise constraint already implies ordinary domination for each competitor.

Lemma 15.37.1 *The profile A is smooth and strictly convex, with $0 < A' < 1$, bounded A'' and asymptotic slopes $0, 1$. Its tangents tend uniformly to $-\infty$ on each compact interval as their point of tangency tends to $-\infty$, and uniformly to the line $y \mapsto y$ as that point tends to $+\infty$. Moreover, $A(x) - x \searrow 0$ at the right end.*

Proof As functions of x, the quantities P and T of Section 15.36 are $P = \mathrm{e}^x/(1+\mathrm{e}^x)$ and $T = 1 - \log P$; put $a = \alpha T^{\alpha-1}$. Since A is the profile of Φ_α and $\mathrm{d}/\mathrm{d}x = r\,\mathrm{d}/\mathrm{d}r$, (15.106) and one more differentiation give

$$A' = P + (1-P)a, \qquad A'' = P(1-P)(1-a) + \alpha(1-\alpha)(1-P)^2 T^{\alpha-2}.$$

Since $T \ge 1$ and $0 < a \le \alpha < 1$, these formulas give the asserted positivity and bounds. On the left, $A(x) = 1 - (1-x)^\alpha + o(1)$ and $A - xA'$ has leading term $-(1-\alpha)(-x)^\alpha$. On the right, $A = x + (1-\alpha)\mathrm{e}^{-x} + O(\mathrm{e}^{-2x})$ and $A' = 1 - (1-\alpha)\mathrm{e}^{-x} + O(\mathrm{e}^{-2x})$. The tangent at x is $y \mapsto A'(x)y + A(x) - xA'(x)$, so both uniform limits follow. Finally $A' - 1 < 0$ gives the decreasing right-end limit. □

15.37.1 The convex minorant and its residual

Put $g = A + F$, $K_F = \|F\|_\infty$, and let C_F be the supremum of all affine functions below g on $\mathbb{R}$. Since $A - K_F$ is convex and is the supremum of its supporting lines,

$$A - K_F \le C_F \le A + F \le A + K_F. \tag{15.117}$$

In particular C_F is a finite convex function. We call the open set $\{C_F < g\}$ the *free set*, its components *free intervals*, and $\{C_F = g\}$ the *contact set*.

Theorem 15.37.1 *The envelope* (15.116) *is*

$$E_F = C_F(\log r) - \log(1 + r)$$

on the punctured affine chart, with its canonical extensions at both ends. It has ordinary type u_α. *The residual* $R_F = E_F - u_\alpha$ *off* o *extends to* $C^{1,1}(\mathbb{P}^k)$, *is zero near* o *and is smooth near infinity. The function* C_F *is affine on each free interval, and, with* $L_2 = \sup_{\mathbb{R}} \max(g'', 0)$,

$$0 \le C_F'(b) - C_F'(a) \le L_2(b - a) \qquad (a < b). \tag{15.118}$$

Proof We first identify the full envelope, then examine contact and the two ends. Every affine minorant of g has slope in $[0, 1]$, because $g \to -\infty$ on the left and $g = x + o(1)$ on the right; hence every supporting slope of C_F lies in $[0, 1]$. The sandwich forces its asymptotic slopes to be $0, 1$.

The competitor family is invariant under the unitary group $U(k)$ acting on the affine coordinates. Its regularized supremum is therefore radial. Adding smooth local potentials of κ gives a locally upper-bounded family of plurisubharmonic functions, so the envelope is κ-plurisubharmonic. On the affine chart $\kappa = \mathrm{dd}^{\mathrm{c}} \log(1 + r)$, so a function v is κ-plurisubharmonic there exactly when its total potential $\log(1 + r) + v$ is plurisubharmonic. The competitor $u_\alpha - K_F$ prevents collapse, and upper semicontinuity of the obstacle preserves the everywhere bound after regularization, including at o. A radial plurisubharmonic total potential $B(\log r)$ has convex profile B: restriction to a complex line gives this from radial subharmonicity. Conversely a convex nondecreasing profile has nonnegative tangential and radial Hessian eigenvalues B'/r and B''/r, first for smooth profiles and then by convex smoothing on annuli. The envelope's total profile is consequently a convex minorant of g, and is at most C_F.

The same criterion makes $C_F(\log r)$ plurisubharmonic on the punctured chart. Its upper bound near o and (15.117) give a removable extension with value $-\infty$. Since $C_F' \le 1$, the function $C_F - x$ is nonincreasing and, by the sandwich, has a finite right-end limit. The proposed E_F is thus locally bounded above near infinity. Add a local smooth potential of κ and apply Lemma 1.3.7 across the hyperplane. This gives a global competitor, hence the formula. The sandwich proves ordinary equivalence without subtracting the infinite values at o.

On a compact interval in the free set the gap $g - C_F$ is positive. Replacing C_F on a sufficiently short subinterval by its chord preserves convexity at both endpoints and stays below g. Unless C_F is locally affine this would give a larger minorant. It follows that C_F is affine on each free interval. At a contact point y, domination by g gives $C_{F,-}'(y) \ge g'(y) \ge C_{F,+}'(y)$, and convexity reverses the outer inequality. Thus C_F is differentiable at contacts, with derivative g', and it is differentiable on free intervals as well. Its derivative is continuous. Let $a < b$. If both points lie in one free interval, then $C_F'(b) = C_F'(a)$. Otherwise, if a lies in a free interval, move it to the right endpoint of that interval, and if b lies in a free interval, move it to the left endpoint; these endpoints exist, since a free interval lacking the needed endpoint would contain both points. The resulting contact points $a' \le b'$ in $[a, b]$ have the

same respective derivatives, and

$$0 \le C_F'(b) - C_F'(a) = g'(b') - g'(a') \le L_2(b' - a') \le L_2(b - a),$$

which proves (15.118).

For all sufficiently negative x, the tangent to A at x lies below g on $\operatorname{supp} F$, by Lemma 15.37.1, and below $A = g$ elsewhere. Hence $C_F = A$ on a left tail. The residual is zero near o.

For the other end, put $L_F = \inf_{\mathbb{R}}(g(x) - x) \le 0$. If this infimum is attained at y, the affine minorant $x + L_F$ touches g there. The bound $C_F' \le 1$ forces $C_F = x + L_F$ for $x \ge y$. A negative infimum is always attained, since $g - x$ tends to $+\infty$ on the left and to zero on the right. With $R = 1/r$, the envelope on this tail is

$$E_F = L_F - \log(1 + R). \tag{15.119}$$

If $L_F = 0$ is not attained and $F \neq 0$, the continuous function $g(y) - y$ has a positive minimum on $\operatorname{supp} F$. The right tangents of A approach the line y uniformly there, so every sufficiently late tangent lies below g and gives $C_F = A$ at its contact point. Thus $C_F = A$ on a right tail; for $F = 0$ this holds everywhere. Both alternatives give a smooth residual near infinity, because $R = 1/r$ and u_α are smooth there.

On the intervening finite annulus,

$$|C_F - A| \le K_F, \qquad |C_F' - A'| \le 1, \qquad |(C_F - A)''| \le L_2 + \|A''\|_\infty \quad \text{a.e.}$$

Composing with the smooth coordinate function $x = \log r$ gives $C^{1,1}$ regularity. The exact end formulas extend it over the whole projective space. □

Corollary 15.37.1 *The non-pluripolar measure of E_F has mass one and satisfies*

$$\mathrm{MA}_\kappa(E_F)(\{a < \log r < b\}) = C_F'(b)^k - C_F'(a)^k \qquad (-\infty < a < b < \infty).$$

It vanishes on $\{\log r \in I\}$ for every free interval I. For any $F, G \in C_c^2(\mathbb{R};\mathbb{R})$,

$$\|R_F - R_G\|_{C^0(\mathbb{P}^k)} \le \|F - G\|_\infty. \tag{15.120}$$

Proof On an annulus, the radial flux formula (15.107), followed by convex smoothing and bounded Bedford–Taylor continuity, gives the displayed measure. Continuity of C_F' excludes boundary atoms. The non-pluripolar truncations discard o, and smoothness at infinity excludes hyperplane mass. The asymptotic slopes 0, 1 therefore give mass one. An affine profile has zero radial eigenvalue and hence zero top-degree measure. Finally, if $\delta = \|F - G\|_\infty$, subtracting δ from a competitor for F gives a competitor for G, and conversely. The resulting value bound for the envelopes gives (15.120) off o, and continuity of the residuals extends it across both ends. □

For example, take a nonnegative smooth compactly supported bump χ with $\chi(x_0) = 1$ and $F = -M\chi$, where $M > A(x_0) - x_0$. Then $L_F < 0$, and (15.119) gives an open free region of zero density. The envelope still has its sublogarithmic pole, while its residual is globally $C^{1,1}$.

15.37.2 Uniform bounds through moving contact

For a fixed interval $[a,b]$ and $M>0$, let $\mathcal{F}_{a,b,M}$ consist of real C^2 functions supported in $[a,b]$ with $\|F\|_\infty+\|F''\|_\infty\le M$. Take all norms in one fixed finite smooth projective atlas.

Theorem 15.37.2 *There is $K=K(k,\alpha,a,b,M)$ such that*

$$\|R_F\|_{C^{1,1}}\le K\qquad (F\in\mathcal{F}_{a,b,M}).$$

For every $0<\gamma<1$ and $F,G\in\mathcal{F}_{a,b,M}$ with $\delta=\|F-G\|_\infty\le 1$,

$$\|R_F-R_G\|_{C^{1,\gamma}}\le K_\gamma\delta^{(1-\gamma)/2},$$

where K_γ has the same fixed-parameter dependence. These bounds hold across changes of the infinity alternative in Theorem 15.37.1.

Proof The left tangents of A lie below $A-M$ on $[a,b]$ once their contact point is less than a common $a_0<a$. They are minorants of every $A+F$, so all residuals vanish on the same neighborhood of o. The sandwich and slope bound give uniform real Hessian bounds on any fixed finite annulus, since $L_2\le\|A''\|_\infty+M$. It remains to control infinity, where the right contact point need not stay bounded.

Put $B=\max(b,0)+1$. For $x>B$, the obstacle is A. At contact $C_F'=A'$. On a free interval with finite right contact endpoint d, its constant slope is $A'(d)\ge A'(x)$; an unbounded right free interval has slope one. Thus

$$0\le d_F(x):=1-C_F'(x)\le 1-A'(x)\le \mathrm{e}^{-x}.$$

The locally Lipschitz functions C_F',A' agree on the contact set, so their derivatives agree almost everywhere there. Indeed, a differentiable Lipschitz function has derivative zero at almost every density point of its zero set. On the free set $C_F''=0$. The reference formula consequently gives

$$0\le C_F''(x)\le A''(x)\le (1+\alpha(1-\alpha))\mathrm{e}^{-x}\quad\text{a.e. on } x>B. \tag{15.121}$$

In a chart $Z_j\neq 0$, $j\ge 1$, use $\zeta=Z_0/Z_j$ and $y=(Z_i/Z_j)_{i\ge1,i\neq j}$. Then

$$R=\frac{|\zeta|^2}{1+\sum|y_i|^2},\qquad x=-\log R.$$

On fixed relatively compact chart pieces, $|\nabla R|\le c\sqrt{R}$ and $|\nabla^2 R|\le c$. Write the envelope as $\mathcal{E}_F(R)=C_F(-\log R)+\log R-\log(1+R)$ for $R>0$. Its derivatives are

$$(\mathcal{E}_F)_R=\frac{d_F}{R}-\frac{1}{1+R},\qquad (\mathcal{E}_F)_{RR}=\frac{C_F''-d_F}{R^2}+\frac{1}{(1+R)^2}\quad\text{a.e.}$$

The tail bounds give $|(\mathcal{E}_F)_R|\le 2$ and $|(\mathcal{E}_F)_{RR}|\le c_\alpha/R+1$. The real chain rule therefore bounds its Hessian uniformly, and its gradient tends to zero as

$\zeta \to 0$. Also $L_F = \lim_{x\to\infty}(C_F(x) - x) \in [-M, 0]$, and integrating $d_F \le \mathrm{e}^{-x}$ gives $|C_F(x) - x - L_F| \le \mathrm{e}^{-x}$. Hence $\mathcal{E}_F - L_F = O(R)$, so the extended envelope is differentiable on the hyperplane with zero gradient. The Hessian bound makes the gradient Lipschitz off the hyperplane. Apply it along straight coordinate segments and use the continuous zero gradient at a crossing; on segments contained in the hyperplane the gradient is zero. This proves a uniform $C^{1,1}$ extension. Subtract the fixed smooth u_α near infinity and join this estimate to the common pole neighborhood and a fixed annulus.

For the stability estimate put $Z = R_F - R_G$. By (15.120), $\|Z\|_\infty \le \delta$, and its gradient has a uniform Lipschitz bound. On a smaller fixed chart, a real directional increment of length h gives $|\partial_e Z| \le 2\delta/h + Kh$. Taking h of order $\sqrt{\delta}$, or enlarging the constant when limited by the chart radius, yields $\|\nabla Z\|_\infty \le c\sqrt{\delta}$. For points ξ, ξ' at distance t,

$$|\nabla Z(\xi) - \nabla Z(\xi')| \le \min(2c\sqrt{\delta}, 2Kt).$$

Dividing by t^γ and splitting at $t = \sqrt{\delta}$ gives the estimate in a finite atlas. When $\delta = 0$, the value bound gives equality. This argument does not assert convergence in $C^{1,1}$.□

15.37.3 The full envelope on a Hermitian product

Let (Y, h_0) be a connected compact Hermitian manifold of dimension $m \ge 1$, and on $X = \mathbb{P}^k \times Y$ put $H = \mathrm{pr}_1^*\,\kappa + \mathrm{pr}_2^*\,h_0$, $N = k + m$ and $V_0 = \int_Y h_0^m$. Pull u_α and F back to X, and define E_F^H by (15.116) using the full family $\mathrm{PSH}(X, H)$.

Proposition 15.37.1 *One has $E_F^H = \mathrm{pr}_1^*\,E_F$ and*

$$\mathrm{MA}_H(E_F^H) = \binom{N}{k}\mathrm{MA}_\kappa(E_F) \otimes h_0^m, \qquad \mathrm{MA}_H(E_F^H)(X) = \binom{N}{k}V_0.$$

The pulled-back residual is globally $C^{1,1}$, and E_F^H belongs to $\mathcal{E}_{\mathrm{env}}(X, H, u_\alpha)$. On each fixed product the uniform bounds and derivative stability of Theorem 15.37.2 hold for these residuals.

Proof Every competitor restricts to a projective slice as a κ-plurisubharmonic function or as $-\infty$ identically. To check the exact lower bound when H is nonclosed, choose a smooth quadratic correction q with $\mathrm{dd^c}q \ge H$ on a coordinate polydisk and let $v_\epsilon = (v + q) * \eta_\epsilon - q$ on a smaller polydisk, where η_ϵ is the rescaling of a fixed smooth nonnegative radial probability kernel. Radial plurisubharmonic convolutions decrease to the canonical $v + q$, and

$$\mathrm{dd^c}v_\epsilon \ge (\mathrm{dd^c}q - H) * \eta_\epsilon - \mathrm{dd^c}q \ge -H - o(1)\omega_{\mathrm{Eucl}}.$$

Restrict this smooth inequality to a fixed slice. The corrected restrictions decrease either to $-\infty$ or to a plurisubharmonic function in local L^1; in the latter case the background error tends to zero distributionally. Subtracting q gives the exact κ bound,

and these canonical restrictions patch on the connected slice. The everywhere obstacle constraint now gives $v \le \mathrm{pr}_1^* E_F$ on every slice. Upper regularization preserves this bound. Conversely every base competitor lifts, since $h_0 > 0$, proving equality of the full envelopes.

Part (1) of Lemma 15.29.1, applied with the Hermitian factor Y in place of X_0 and the Kähler factor $\mathbb{P}^k$, gives the stated non-pluripolar measure and its finite mass. The residual regularity and uniform estimates pull back in a fixed smooth product atlas. Finally, E_F^H has the ordinary type of u_α by Theorem 15.37.1, so Proposition 14.4.1 gives the envelope class. □

Example 15.37.1 There is a nonclosed positive background on a non-Kähler fourfold for which the preceding full envelope formula and uniform residual estimates hold.

Proof Take $Y = (\mathbb{C}^2\setminus\{0\})/\langle \zeta \mapsto \mathrm{e}\zeta\rangle$. With $s = \log|\zeta|^2$, the form $h_0 = \mathrm{dd^c}s + \mathrm{d}s\wedge\mathrm{d^c}s$ is invariant under the quotient. Its two summands are positive on the tangential and radial complex directions, respectively, so it is Hermitian. Moreover $\mathrm{d}h_0 = -\mathrm{d}s\wedge\mathrm{dd^c}s \neq 0$. The Hopf surface is diffeomorphic to $S^3\times S^1$ and has odd first Betti number, so it is not Kähler. A Kähler form on $X = \mathbb{P}^2\times Y$ would restrict to one on each fiber Y. Thus X is a non-Kähler fourfold with $\mathrm{d}H \neq 0$, and Proposition 15.37.1 applies. □

Chapter 16
Normalization and weak adjoint measures

Fix a smooth positive volume form $\mathrm{d}V$ on X. Function spaces are real, and Lebesgue norms are taken with respect to $\mathrm{d}V$. Until Section 16.8, θ is a smooth semipositive big real $(1,1)$-form and $\varphi \in \mathrm{PSH}(X,\theta)$ is a model reference in the sense of (14.20): it is a fixed point of the ordinary singularity envelope $P_\theta[\cdot]$ of Definition 1.19.1, and its envelope $P_{\varepsilon\theta}(\varphi)$ in a smaller background is not identically $-\infty$ for some $0 < \varepsilon < 1$. We write $u \sim \varphi$ when u has the ordinary singularity type of φ (Definition 1.12.1), and $\mathrm{MA}_\theta^{\mathrm{ext}}$ for the extended non-pluripolar Monge–Ampère measure of Theorem 7.1.1; when its total mass is finite, it is the usual non-pluripolar measure, which we also denote by MA_θ.

In the ordinary type of φ, the equation $\mathrm{MA}_\theta^{\mathrm{ext}}(u) = c\,\mathrm{e}^F \mathrm{d}V$ determines the constant $c > 0$ uniquely, even when it has several normalized solutions. As a function of the logarithmic density F, the logarithm of this constant is concave. Its supporting functionals are probability measures that describe the first-order response of the constant to a change of density; for smooth Hermitian equations they are the Gauduchon adjoint probabilities of the linearized operator. For the zero reference in a Hermitian background, the supporting probability at a continuous F is still unique, and stationary for the linearized operator, when a bounded solution has a continuous positive form $\theta + \mathrm{d}\mathrm{d}^{\mathrm{c}}u$; the proof uses the Fredholm theory of the trace operator of such a form on Sobolev spaces. The convex dual of the functional controls the concentration of these measures through entropy, and this gives compactness when the density, the reference or the positive background varies. The same measures provide the integral normalizations that single out the limit of the exponential equations as the exponent tends to zero. The scalar functional does not, however, determine the singularities of the solutions: examples with moving divisors have identical functionals but different limiting potentials. In a Hermitian background, for a nonzero divisorial reference with a positive residual form, the subsolution threshold on the whole one-sided class is that of the residual background, and for smooth positive densities it is strictly smaller than the global threshold. On closed projective backgrounds, convex geometry gives quantitative criteria for varying references, including references with nonanalytic poles or degenerate residuals. On products

with a projective line, logarithmic references give explicit selected limits under simultaneous variation of the reference, background, density and exponent.

16.1 The scalar functional

The domination principle makes the constant decrease as the density increases, and mixed Monge–Ampère products strengthen this monotonicity to concavity in the logarithmic density. Compactness of the solutions then gives continuity under uniformly bounded L^1 perturbations.

For a bounded real measurable function F, part (2) of Theorem 15.11.1, applied to the bounded density e^F, gives $u \in \mathrm{PSH}(X,\theta)$ with $u \sim \varphi$ and a unique constant $c_\varphi(F) > 0$ such that

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) = c_\varphi(F)\mathrm{e}^F \,\mathrm{d}V. \tag{16.1}$$

The *normalization scalar* is

$$B_\varphi(F) = \log c_\varphi(F).$$

Since the constant is unique, B_φ is well defined; it depends on φ only through its ordinary singularity type.

Proposition 16.1.1 *Let $F, G \in L^\infty(\mathrm{d}V)$ be real. The functional B_φ is decreasing, and for every constant $a \in \mathbb{R}$,*

$$B_\varphi(F + a) = B_\varphi(F) - a.$$

Consequently

$$-\operatorname*{ess\,sup}_X(G - F) \le B_\varphi(G) - B_\varphi(F) \le -\operatorname*{ess\,inf}_X(G - F), \tag{16.2}$$

and $|B_\varphi(G) - B_\varphi(F)| \le \|G - F\|_\infty$.

Proof Suppose that $F \le G$ almost everywhere and, to reach a contradiction, that $c_\varphi(F) < c_\varphi(G)$. Let $u, v \sim \varphi$ solve the two equations. Since $F \le G$ and the extended measure is unchanged by adding constants, for every constant a,

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) \le \frac{c_\varphi(F)}{c_\varphi(G)}\mathrm{MA}_\theta^{\mathrm{ext}}(v + a),$$

and the ratio is strictly less than one. As $u \sim \varphi$, the envelope $P_{\varepsilon\theta}(u)$ dominates a translate of $P_{\varepsilon\theta}(\varphi)$ and is therefore not identically $-\infty$; moreover $v + a \preceq u$. The singular strict domination principle, Theorem 15.1.1, gives $u \ge v + a$ for every a, which is impossible at any point where both potentials are finite. Thus $B_\varphi(F) \ge B_\varphi(G)$.

Replacing F by $F + a$ and the constant by $\mathrm{e}^{-a}c_\varphi(F)$ leaves the right-hand side of (16.1) unchanged, so uniqueness of the constant proves the translation rule. Finally,

G lies between the constant translates $F + \operatorname{ess\,inf}_X(G - F)$ and $F + \operatorname{ess\,sup}_X(G - F)$; monotonicity and the translation rule give (16.2). □

The scalar does not depend on the smooth representative of the background: for a smooth real h, replacing θ by $\theta + \mathrm{dd^c} h$ and each potential u, including φ, by $u - h$ preserves the currents and the prescribed ordinary type, and hence leaves the equation and its constant unchanged.

Theorem 16.1.1 *The functional B_φ is concave on the real space $L^\infty(\mathrm{d}V)$: for F, G in that space,*

$$B_\varphi(tF + (1-t)G) \geq tB_\varphi(F) + (1-t)B_\varphi(G), \qquad 0 \leq t \leq 1.$$

Proof Let $u, v \sim \varphi$ solve the equations for F and G, and set $H = tF + (1-t)G$ and $D = c_\varphi(F)^t c_\varphi(G)^{1-t}$. Lemma 7.4.1 and the arithmetic–geometric mean inequality give

$$\begin{aligned} \mathrm{MA}_\theta^{\mathrm{ext}}(tu + (1-t)v) &\geq \bigl(t(c_\varphi(F)\mathrm{e}^F)^{1/n} + (1-t)(c_\varphi(G)\mathrm{e}^G)^{1/n}\bigr)^n \,\mathrm{d}V \\ &\geq D\mathrm{e}^H \,\mathrm{d}V. \end{aligned}$$

The averaged potential $tu + (1-t)v$ has the ordinary type of φ. If $c_\varphi(H) < D$, its measure dominates the measure of a solution $w \sim \varphi$ for H by the factor $D/c_\varphi(H) > 1$. Applying the singular strict domination principle to w and to every constant translate of the averaged potential, as in the proof of Proposition 16.1.1, leads to the same contradiction. Hence $c_\varphi(H) \geq D$, and taking logarithms proves concavity. □

For smooth Hermitian data, this concavity agrees with the second-variation formula of Theorem 10.7.1.

Theorem 16.1.2 *If F_j, F are uniformly bounded real measurable functions and $F_j \to F$ in $L^1(\mathrm{d}V)$, then*

$$B_\varphi(F_j) \longrightarrow B_\varphi(F).$$

Proof Choose solutions $u_j \sim \varphi$ normalized by $\sup_X u_j = 0$, and write $c_j = c_\varphi(F_j)$. If $|F_j| \leq M$, monotonicity and the translation rule of Proposition 16.1.1 give

$$\mathrm{e}^{-M} c_\varphi(0) \leq c_j \leq \mathrm{e}^M c_\varphi(0).$$

In particular, the Monge–Ampère measures of the u_j are dominated by a fixed multiple of $\mathrm{d}V$, and Theorem 15.11.2 gives a constant C such that

$$\varphi - C \leq u_j \leq \varphi.$$

From any subsequence, extract a further subsequence with $c_j \to c > 0$ and $u_j \to u$ in $L^1(X)$. By Corollary 15.8.1, the convergence also holds in capacity, $u \sim \varphi$, and the measures converge weakly. On the other hand, the exponential function is Lipschitz on the common range of the F_j, so $c_j \mathrm{e}^{F_j} \to c\mathrm{e}^F$ in $L^1(\mathrm{d}V)$. Therefore

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) = c\mathrm{e}^F \,\mathrm{d}V,$$

and uniqueness of the constant gives $c = c_\varphi(F)$. Since every subsequence has a further subsequence along which $c_j \to c_\varphi(F)$, the whole sequence converges, and so do the logarithms. □

16.2 Supporting probability measures

A supporting functional of the concave function B_φ describes the first-order response of the scalar to a change of logarithmic density. Monotonicity and the translation rule force such a functional to be minus the integral against a probability measure. We work first on the space $C(X)$ of continuous real functions, with the uniform norm. For $F \in C(X)$, let $\mathcal{A}_\varphi(F)$ be the set of Borel probability measures ν on X satisfying

$$B_\varphi(G) \le B_\varphi(F) - \int_X (G - F)\,\mathrm{d}\nu \qquad \text{for every } G \in C(X). \tag{16.3}$$

Thus $G \mapsto -\int_X G\,\mathrm{d}\nu$ is a supergradient of B_φ at F. We call the members of $\mathcal{A}_\varphi(F)$ the *supporting probabilities* at F.

Theorem 16.2.1 *For every $F \in C(X)$, the set $\mathcal{A}_\varphi(F)$ is nonempty, convex and weakly compact. Every one of its measures is absolutely continuous with respect to* $\mathrm{d}V$.

Proof The function $C_\varphi = -B_\varphi$ is convex and Lipschitz. Fix $F \in C(X)$. For a direction $h \in C(X)$, the limit

$$p(h) = \lim_{t \downarrow 0} \frac{C_\varphi(F + th) - C_\varphi(F)}{t}$$

exists and is finite: convexity makes the difference quotients monotone in $t > 0$, and their absolute values are at most $\|h\|_\infty$. Clearly p is positively homogeneous. Given another direction $k \in C(X)$, apply convexity at the midpoint of $F + 2th$ and $F + 2tk$ and let $t \downarrow 0$; this gives $p(h + k) \le p(h) + p(k)$. Thus p is sublinear.

By the Hahn–Banach theorem, the zero functional on $\{0\}$ extends to a linear functional $\ell \le p$. Since $p(h) \le \|h\|_\infty$, applying this bound to h and $-h$ shows that ℓ is continuous. Monotonicity of the convex difference quotients gives

$$C_\varphi(F + h) \ge C_\varphi(F) + \ell(h),$$

which is (16.3) once ℓ is represented by a probability. Testing the translation rule with positive and negative constants gives $\ell(1) = 1$. If $h \ge 0$, monotonicity of B_φ gives $C_\varphi(F - th) \le C_\varphi(F)$ for $t > 0$, whereas the inequality just proved gives the lower bound $C_\varphi(F) - t\ell(h)$. Hence $\ell(h) \ge 0$. By the Riesz representation theorem, $\ell(h) = \int_X h\,\mathrm{d}\nu$ for a probability measure ν, and $\nu \in \mathcal{A}_\varphi(F)$. Each inequality in (16.3)

defines a weakly closed half-space in the compact space of probability measures, so their intersection is closed and convex, hence weakly compact.

It remains to prove absolute continuity. Let K be compact with $\mathrm{d}V(K) = 0$, and choose continuous functions $0 \le \chi_j \le 1$, equal to one on K, whose supports have volumes tending to zero. For any $\nu \in \mathcal{A}_\varphi(F)$, the supporting inequality with $G = F + \chi_j$ gives

$$0 \le \nu(K) \le \int_X \chi_j \,\mathrm{d}\nu \le B_\varphi(F) - B_\varphi(F + \chi_j).$$

The right-hand side tends to zero by Theorem 16.1.2, so $\nu(K) = 0$. By inner regularity of finite Borel measures on X, the same holds for every Borel set of zero volume. The Radon–Nikodym theorem now gives $\mathrm{d}\nu = h_\nu \mathrm{d}V$ with $h_\nu \ge 0$ and $\int_X h_\nu \mathrm{d}V = 1$. □

Proposition 16.2.1 *Let $F_j, F \in C(X)$ with $F_j \to F$ uniformly, and let $\nu_j \in \mathcal{A}_\varphi(F_j)$. Then every weak cluster point of (ν_j) belongs to $\mathcal{A}_\varphi(F)$. In particular, if the latter set consists of a single measure, ν_j converges weakly to it.*

Proof Take a weakly convergent subsequence and fix $G \in C(X)$. In the inequality

$$B_\varphi(G) \le B_\varphi(F_j) - \int_X (G - F_j) \,\mathrm{d}\nu_j,$$

the Lipschitz continuity of B_φ, the uniform convergence of F_j and the weak convergence of the probabilities allow passage to the limit. The limiting inequality holds for every G, so the limit measure belongs to $\mathcal{A}_\varphi(F)$. If that set is a singleton, every convergent subsequence has the same limit, and compactness of the space of probability measures proves the final assertion. □

16.2.1 Directional derivatives

As ν runs through the supporting probabilities at F, the integral $-\int_X G \,\mathrm{d}\nu$ ranges exactly between the two one-sided derivatives of B_φ at F in the direction G. Consequently, differentiability is equivalent to uniqueness of the supporting probability. For $F, G \in C(X)$, write $D^+B_\varphi(F)[G]$ and $D^-B_\varphi(F)[G]$ for the limits of the difference quotient $(B_\varphi(F + tG) - B_\varphi(F))/t$ as $t \downarrow 0$ and $t \uparrow 0$, respectively.

Proposition 16.2.2 *For $F, G \in C(X)$, the one-sided derivatives exist, and*

$$D^+B_\varphi(F)[G] = -\max_{\nu \in \mathcal{A}_\varphi(F)} \int_X G \,\mathrm{d}\nu,$$

$$D^-B_\varphi(F)[G] = -\min_{\nu \in \mathcal{A}_\varphi(F)} \int_X G \,\mathrm{d}\nu.$$

Consequently B_φ is Gâteaux differentiable at F if and only if $\mathcal{A}_\varphi(F)$ is a singleton. Its derivative is then $G \mapsto -\int_X G \,\mathrm{d}\nu$.

Proof The one-sided limits exist and are finite by concavity and the Lipschitz bound. For $t > 0$, every supporting probability ν at F gives

$$\frac{B_\varphi(F + tG) - B_\varphi(F)}{t} \le -\int_X G \, \mathrm{d}\nu.$$

Conversely, choose $\nu_t \in \mathcal{A}_\varphi(F + tG)$ and apply its supporting inequality with the test function F; this bounds the same quotient below by $-\int_X G \, \mathrm{d}\nu_t$. Along a sequence $t \downarrow 0$, the measures ν_t have a weak limit point, which belongs to $\mathcal{A}_\varphi(F)$ by Proposition 16.2.1. The two bounds identify the right derivative with minus the maximum. Applying this identity to $-G$ gives the formula for the left derivative.

If $\mathcal{A}_\varphi(F)$ is a singleton $\{\nu\}$, the two derivatives coincide for every G and equal $-\int_X G \, \mathrm{d}\nu$, a continuous linear functional of G. Conversely, if the two one-sided derivatives agree for every G, all supporting probabilities have the same integral against every continuous function, and the Riesz representation theorem shows that they coincide. □

16.2.2 Generic uniqueness

On every affine line in $C(X)$, the concave function B_φ is differentiable outside a countable set, so the gap between its two directional derivatives along the line vanishes on a dense subset. Upper semicontinuity of this gap and separability of $C(X)$ turn the one-dimensional fact into uniqueness of the supporting probability on a dense G_δ subset of $C(X)$.

Theorem 16.2.2 *The set*

$$\mathcal{U}_\varphi = \{F \in C(X, \mathbb{R}) : \mathcal{A}_\varphi(F) \text{ is a singleton}\}$$

is a dense G_δ subset of $C(X)$ with the uniform norm. At every $F \in \mathcal{U}_\varphi$, the functional B_φ is Gâteaux differentiable, and its derivative is minus the integral against the unique supporting probability. If $F_j \to F$ uniformly and $\nu_j \in \mathcal{A}_\varphi(F_j)$, then ν_j converges weakly to that probability.

Proof For $h \in C(X, \mathbb{R})$ define the width

$$w_h(F) = \max_{\nu \in \mathcal{A}_\varphi(F)} \int_X h \, \mathrm{d}\nu - \min_{\nu \in \mathcal{A}_\varphi(F)} \int_X h \, \mathrm{d}\nu \ge 0.$$

The extrema exist by weak compactness. We claim that the width is upper semicontinuous in F. Indeed, let $F_j \to F$ and choose maximizing and minimizing probabilities at each F_j. Along a subsequence realizing the upper limit of the widths, compactness of the space of probability measures gives weak limits of both choices. By Proposition 16.2.1, both limits belong to $\mathcal{A}_\varphi(F)$, so the difference of their integrals against h is at most $w_h(F)$.

Consequently, for every $\epsilon > 0$ the set $\{F : w_h(F) < \epsilon\}$ is open. It is also dense. If $h = 0$ there is nothing to prove. Otherwise fix F and an open ball about it. The function

$$b(t) = B_\varphi(F + th)$$

is finite and concave on the real line, so it is differentiable outside an at most countable set: its one-sided derivatives are finite and monotone, and the nonempty open intervals between unequal one-sided derivatives at distinct points are disjoint, so each contains a different rational number. Choose a point of differentiability t so close to zero that $F + th$ lies in the given ball. The derivative formulas of Proposition 16.2.2 give $w_h(F + th) = 0$, which proves density.

The compact manifold X is metrizable, so $C(X)$ is separable. Choose a countable uniformly dense family (h_j) in $C(X)$. By the Baire category theorem in the complete metric space $C(X)$, the set

$$\bigcap_{j,k\geq 1} \{F : w_{h_j}(F) < 1/k\}$$

is a dense G_δ. At every point of this intersection, all supporting probabilities have the same integral against every h_j. Since they have unit mass, uniform approximation gives equal integrals against every continuous function, and the Riesz representation theorem shows that the probabilities coincide. Conversely, a singleton has zero width in every direction, so the intersection is exactly $\mathscr{U}_\varphi$. The formula for the derivative follows from Proposition 16.2.2, and the approximation assertion from Proposition 16.2.1. □

16.2.3 Gauduchon adjoints

For a smooth Hermitian equation, differentiating the scalar identifies the supporting probability with the adjoint probability of the linearized operator.

Theorem 16.2.3 *Let $\theta = \omega$ be a Hermitian form, let $\varphi = 0$, and let F be a smooth real function. Let u be the smooth solution of (16.1), unique up to an additive constant by Corollary 12.4.1, and let μ_u be its Gauduchon adjoint probability from Proposition 4.8.1; for $n = 1$, this is $\mu_u = (\omega + \mathrm{dd}^{\mathrm{c}} u)/\int_X \omega$. Then*

$$\mathscr{A}_0(F) = \{\mu_u\}.$$

Proof Suppose first that $n \geq 2$. By uniqueness of the constant, the normalization scalar of the smooth equation in Theorem 10.7.1 is B_0, and that theorem gives, for every smooth real G,

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} B_0(F + tG) = -\int_X G \,\mathrm{d}\mu_u.$$

Let $\nu \in \mathcal{A}_0(F)$. Testing the supporting inequality at $F + tG$ for both signs of t and letting $t \to 0$ shows that G has the same integral against ν and μ_u. Since smooth functions are uniformly dense in $C(X)$, it follows that $\nu = \mu_u$, and the assertion follows from nonemptiness of $\mathcal{A}_0(F)$. In dimension one, Stokes' theorem gives $c_0(F) = \int_X \omega / \int_X \mathrm{e}^F \mathrm{d}V$; differentiating its logarithm in the same way yields the stated measure, since $\omega + \mathrm{dd^c} u = c_0(F)\mathrm{e}^F \mathrm{d}V$. □

Consequently, in the Hermitian case the supporting probabilities at a continuous F can be approached through smooth data. If smooth F_j converge uniformly to F, then by Proposition 16.2.1 every weak cluster point of the Gauduchon adjoint probabilities of the solutions for F_j belongs to $\mathcal{A}_0(F)$, and is therefore absolutely continuous. If $\mathcal{A}_0(F)$ is a singleton, as it is for every F in the dense G_δ set $\mathcal{U}_0$ of Theorem 16.2.2 and under the continuous positive Hessian hypothesis of Theorem 16.2.4, these probabilities converge to its unique member, independently of the approximation. We do not know whether $\mathcal{A}_0(F)$ is a singleton for every continuous F, even in this Hermitian setting; the same question is open for the singular references considered above.

For $n \geq 2$, the Hermitian zero-reference case admits a quantitative strengthening: Corollary 27.7.2 bounds the densities of all supporting probabilities uniformly in L^q, for $1 < q < n/(n-1)$, when $\|F\|_\infty$ is bounded, and also for $q = n/(n-1)$ when ω has positive lower mass $v_-(\omega) > 0$ in the sense of Definition 7.5.1. The proof uses the Gauduchon-current formulation of Chapter 27, and we do not claim such a bound for an arbitrary singular reference.

A weak linearized operator requires more than a supporting probability. In the same Hermitian zero-reference setting, Theorem 27.6.1 represents each supporting probability by a determinant-weighted positive $\mathrm{dd^c}$-closed current. Identifying that current with a scalar multiple of a weak power $(\omega + \mathrm{dd^c} u)^{n-1}$ would require a well-defined product and its compatibility with approximation.

A similar representation holds for a functional defined by subsolutions rather than by an equation in a prescribed singularity type. Let $0 \leq f \in L^1(\mathrm{d}V)$ be nonzero with $f^{1/n} \in \mathcal{D}^0_\omega$, the space of densities with uniformly vanishing Gauduchon tails from Section 15.31.2. Then Proposition 27.8.3 describes the supporting probabilities of $F \mapsto \log b_\omega(\mathrm{e}^F f)$, where $b_\omega(\cdot)$ is the subsolution threshold of Theorem 15.27.1.

16.2.4 Fredholm theory for a continuous positive form

A continuous strictly positive $(1,1)$-form still defines a uniformly elliptic trace operator. Its range is described in Sobolev spaces: continuity of the coefficients suffices for the $W^{2,p}$ estimates, and the maximum principle identifies the kernel. The lemma is used for continuous positive complex Hessians in the next subsection.

Lemma 16.2.1 *Let σ be a continuous strictly positive real $(1,1)$-form on X, and let $\mathrm{d}V$ be a smooth positive volume form. For $p > 2n$, the real operator*

$$L_\sigma v = \operatorname{tr}_\sigma \mathrm{d}\mathrm{d}^{\mathrm{c}} v, \qquad L_\sigma \colon W^{2,p}(X) \longrightarrow L^p(\mathrm{d}V),$$

is Fredholm of index zero. Its kernel consists of the constants, and its range has a one-dimensional annihilator in $L^{p'}(\mathrm{d}V)$*, where* $p' = p/(p-1)$.

Proof Fix a smooth positive Hermitian form η and put $L_s = (1-s)L_\eta + sL_\sigma$, $0 \le s \le 1$. In local holomorphic coordinates, these are real second-order operators without lower-order terms, whose principal matrices are continuous, uniformly positive and have a common modulus of continuity.

Estimates. The interior L^p estimates for continuous coefficients [GT01, Theorem 9.11], applied on a finite cover by coordinate balls, give, uniformly in s,

$$\|v\|_{W^{2,p}} \le C\bigl(\|L_s v\|_p + \|v\|_p\bigr) \qquad (v \in W^{2,p}(X)). \tag{16.4}$$

Kernels. Since $p > 2n$, every $v \in W^{2,p}(X)$ is continuous and belongs to $W^{2,2n}$ on coordinate balls, so the strong maximum principle for strong solutions [GT01, Theorem 9.6] applies in coordinate charts. If $L_s v = 0$, then v attains its maximum, the set of maximum points is open by that principle and closed, and v is constant because X is connected. Thus $\ker L_s = \mathbb{R}$. If $(L_s - 1)v = 0$ and v has a positive maximum, the same principle for the operator $L_s - 1$, whose zeroth-order coefficient is negative, shows in the same way that v is a positive constant, which contradicts $L_s v = v$. Applying this to $-v$ gives $\ker(L_s - 1) = 0$.

Index and range. Compactness of $W^{2,p} \hookrightarrow L^p$ and (16.4) now give

$$\|v\|_{W^{2,p}} \le C\|(L_s - 1)v\|_p \tag{16.5}$$

uniformly in s. If this failed, we could normalize a contradicting sequence in $W^{2,p}$, pass to $s_j \to s$ and a strongly L^p convergent subsequence, and apply (16.4) to differences. Operator norm continuity in s gives strong $W^{2,p}$ convergence to a nonzero member of $\ker(L_s - 1)$, which is impossible. The smooth operator $L_\eta - 1$ has index zero and trivial kernel, hence is an isomorphism, by smooth elliptic theory. The method of continuity [GT01, Theorem 5.2] and (16.5) show that every $L_s - 1$ is an isomorphism. Finally, L_σ differs from $L_\sigma - 1$ by the compact inclusion $W^{2,p} \hookrightarrow L^p$, so it is Fredholm of index zero. Its one-dimensional kernel makes its range have codimension one. L^p duality gives the annihilator assertion. □

16.2.5 Continuous positive complex Hessians

The recovery theorem Theorem 16.2.3 extends to a bounded solution u for which the form $\omega + \mathrm{d}\mathrm{d}^{\mathrm{c}} u$ is continuous and everywhere positive. In this case one can vary the potential in both directions, and the resulting densities force every supporting probability to be *stationary* for the linearized operator L, that is, to satisfy $\int_X Lh \, \mathrm{d}\nu = 0$ for every smooth h. The description of the range of L in Sobolev spaces, Lemma 16.2.1, then gives uniqueness.

Theorem 16.2.4 *Let $n \geq 2$, let $\theta = \omega$ be a Hermitian form, and let $\varphi = 0$. Suppose that $F \in C(X)$ and that a bounded solution $u \in \mathrm{PSH}(X, \omega)$ of (16.1) has the property that the current*

$$\sigma = \omega + \mathrm{dd}^{\mathrm{c}} u$$

is a continuous form, positive everywhere on X. Then $\mathcal{A}_0(F)$ consists of one probability ν_F. With the operator $L_\sigma = \mathrm{tr}_\sigma \, \mathrm{dd}^{\mathrm{c}}$ of Lemma 16.2.1, it satisfies

$$\int_X L_\sigma h \, \mathrm{d}\nu_F = 0 \quad (h \in C^\infty(X)),$$
$$DB_0(F)[H] = -\int_X H \, \mathrm{d}\nu_F \quad (H \in C(X)).$$

The derivative is taken in the Gâteaux sense.

Proof In local coordinates, the Euclidean trace of $\mathrm{dd}^{\mathrm{c}} u$ is a constant multiple of the Laplacian of u, and it is continuous. Local constant-coefficient elliptic regularity gives $u \in W^{2,p}$ for every finite p. Indeed, subtract the Newtonian potential of a cutoff of this trace. The second derivatives of that potential are given by the order-zero Fourier multipliers $\xi_i \xi_j / |\xi|^2$, so they lie in every L^p, $p < \infty$; the remaining distribution is harmonic. Sobolev embedding gives a $C^{1,\alpha}$ representative for every $\alpha < 1$, which agrees with the quasi-psh representative.

On a coordinate ball choose smooth ρ with $\mathrm{dd}^{\mathrm{c}} \rho \geq \omega$. The radial regularizations of the bounded psh function $u + \rho$ decrease to it, and their complex Hessians converge uniformly on smaller balls. Continuity of bounded products and the finite smooth-background expansion identify $(\omega + \mathrm{dd}^{\mathrm{c}} u)^n$ with σ^n. For smooth real h, the same argument applies to $u + th$, which is admissible for all sufficiently small signed t by uniform positivity. Define

$$F_t = F + \log \frac{(\sigma + t\mathrm{dd}^{\mathrm{c}} h)^n}{\sigma^n}.$$

Then $F_t \in C(X)$ and $(\omega + \mathrm{dd}^{\mathrm{c}}(u + th))^n = c_0(F) \mathrm{e}^{F_t} \mathrm{d}V$, so uniqueness of the constant gives $B_0(F_t) = B_0(F)$. The supporting inequality (16.3) with $G = F_t$ gives $\int_X (F_t - F) \, \mathrm{d}\nu \leq 0$ for each $\nu \in \mathcal{A}_0(F)$. Since $F_t - F = tL_\sigma h + O(t^2)$ uniformly, using both signs of t yields $\int_X L_\sigma h \, \mathrm{d}\nu = 0$.

Choose $p = 4n + 1 > 2n$ and $q = p/(p-1) = 1 + 1/(4n)$. By Corollary 27.7.2, every supporting probability has an L^q density, since $q < n/(n-1)$. Smooth functions are dense in $W^{2,p}(X)$, and $L_\sigma \colon W^{2,p}(X) \to L^p(\mathrm{d}V)$ is bounded, so this identity extends to every $h \in W^{2,p}(X)$. Thus the densities of all supporting probabilities annihilate the range of L_σ, and by Lemma 16.2.1 they lie in a one-dimensional subspace of $L^q(\mathrm{d}V)$. Their integrals equal one, so they coincide. Nonemptiness is Theorem 16.2.1, and the derivative formula follows from Proposition 16.2.2. □

16.2.6 Products with a Hermitian factor

An explicit product construction shows that the hypothesis of Theorem 16.2.4 allows densities with an arbitrary continuous modulus. Let (Y,κ) be a connected compact Hermitian k-fold, $k \geq 1$, with $\int_Y \kappa^k = 1$. Denote by ν_κ the smooth stationary probability of $L_\kappa = \operatorname{tr}_\kappa \mathrm{dd}^{\mathrm{c}}$, given by the Gauduchon multiplier when $k \geq 2$ (Proposition 4.8.1); for $k = 1$ it is κ. Write ω_1 for the Fubini–Study form of mass one on $\mathbb{P}^1$. All factor forms and functions below are pulled back to $X = \mathbb{P}^1 \times Y$.

Proposition 16.2.3 *Let $a > 0$, let b be smooth and positive on $\mathbb{P}^1$, and let A be continuous and positive there, with $\int_{\mathbb{P}^1} A\omega_1 = a$. For*

$$\omega = a\omega_1 + b\kappa, \qquad \mathrm{d}V = \omega_1 \wedge \kappa^k, \qquad F = \log((k+1)Ab^k),$$

one has

$$c_0(F) = 1, \qquad \mathcal{A}_0(F) = \left\{ \frac{A}{a}\, \omega_1 \otimes \nu_\kappa \right\}.$$

Proof The Green solution of $\mathrm{dd}^{\mathrm{c}} u = (A-a)\omega_1$ on $\mathbb{P}^1$ has continuous complex Hessian and is bounded. Its pullback yields $\sigma = A\omega_1 + b\kappa > 0$, and $\sigma^{k+1} = (k+1)Ab^k\omega_1 \wedge \kappa^k$. Thus uniqueness of the constant gives $c_0(F) = 1$, and Theorem 16.2.4 gives a unique supporting probability. For every smooth real h on X, the block decomposition

$$L_\sigma h = A^{-1} L_{\omega_1} h + b^{-1} L_\kappa h,$$

in which L_{ω_1} and L_κ act along the two factors, shows that $(A/a)\omega_1 \otimes \nu_\kappa$ is stationary: the first term integrates to zero by Stokes' theorem on $\mathbb{P}^1$, and the second by stationarity on Y and Fubini's theorem. This candidate is a probability with a bounded density with respect to $\mathrm{d}V$, so its stationarity extends to $W^{2,p}$ tests as in the proof of Theorem 16.2.4. The one-dimensional annihilator of Lemma 16.2.1 and probability normalization therefore identify it with the supporting probability. □

Example 16.2.1 There is a non-Kähler threefold with a nonclosed Hermitian background and a positive continuous density whose logarithm is neither Dini continuous nor Hölder continuous of any positive order, but whose supporting probability is explicit and unique.

Proof Take the Hopf surface $Y = (\mathbb{C}^2 \setminus \{0\})/\langle z \mapsto 2z \rangle$ and its unit-volume metric $\kappa = c|z|^{-2}\mathrm{dd}^{\mathrm{c}}|z|^2$, where $c > 0$ is chosen so that $\int_Y \kappa^2 = 1$. Put $s = \log|z|^2$, $\alpha = \mathrm{dd}^{\mathrm{c}} s$ and $\beta = \mathrm{d}s \wedge \mathrm{d}^{\mathrm{c}} s$, so that $\kappa = c(\alpha + \beta)$. The computation of $\mathrm{dd}^{\mathrm{c}}\beta$ in (9.1) applies in this dimension and gives $\mathrm{dd}^{\mathrm{c}}\beta = -\alpha^2$. Here α has rank one, so $\mathrm{dd}^{\mathrm{c}}\kappa = -c\alpha^2 = 0$. Thus κ is Gauduchon and $\nu_\kappa = \kappa^2$. Polar coordinates identify Y with $S^3 \times S^1$, hence $H^2(Y,\mathbb{R}) = 0$. A closed positive form would be exact and have positive square integral, contradicting Stokes' theorem. Thus Y is non-Kähler, and so is $\mathbb{P}^1 \times Y$, by restriction.

Let $\tau \in [-1,1]$ be the height on $\mathbb{P}^1$, set $b = 2 + \tau$, and put

$$f(t) = \begin{cases} 1/\log(\mathrm{e}/|t|), & 0 < |t| \le 1, \\ 0, & t = 0, \end{cases}$$

$$A = a + \varepsilon \left(f(\tau) - \int_{\mathbb{P}^1} f(\tau)\omega_1 \right), \quad 0 < \varepsilon < a.$$

Since $0 \le f \le 1$, this is a positive continuous function of mean a. Along a normal segment to the equator, $|\tau|$ is comparable to the distance r from the equator. The oscillation of A is therefore bounded below by a positive multiple of $1/\log(\mathrm{e}/r)$. Its Dini integral $\int_0^{r_0} \mathrm{d}r/(r\log(\mathrm{e}/r))$ diverges, and this modulus is larger than every r^γ, $\gamma > 0$, as $r \downarrow 0$. The logarithm of A has the same lower bound, because A is uniformly positive and bounded. The smooth term $2\log b$ contributes only $O(r)$, so $F = \log(3Ab^2)$ has neither regularity. The mixed component $\mathrm{d}b \wedge \kappa$ of $\mathrm{d}\omega$ shows that $\omega = a\omega_1 + b\kappa$ is nonclosed. By Proposition 16.2.3, its supporting probability is $(A/a)\,\omega_1 \wedge \kappa^2$. □

16.2.7 Divisorial smoothing

For a singular reference, the supporting probability can be unique while smoothings of the solution have different limiting adjoint probabilities. The reason is that the non-pluripolar product discards the mass on a divisor that the smoothings keep. On a product of two projective lines, explicit formulas exhibit both the lost mass and its dependence on the smoothing.

In this subsection, $X = \mathbb{P}^1 \times \mathbb{P}^1$. Let ω_1 and ω_2 be the pullbacks of the Fubini–Study forms of the two factors, normalized to have integral one on their factors, and put

$$\theta = 2\omega_1 + 2\omega_2, \qquad \beta = \omega_1 + 2\omega_2, \qquad \omega = \omega_1 + \omega_2, \qquad \mathrm{d}V = \omega_1 \wedge \omega_2.$$

Thus $\mathrm{d}V$ is a probability volume form, $\theta^2 = 8\mathrm{d}V$ and $\beta^2 = 4\mathrm{d}V$. Let z be the affine coordinate on the first factor, and put

$$r = \frac{|z|^2}{1 + |z|^2}, \qquad \psi = \log r, \qquad D = \{0\} \times \mathbb{P}^1.$$

The function r extends smoothly to the first factor, with value one at infinity, and ψ is quasi-plurisubharmonic with value $-\infty$ exactly on D. The Poincaré–Lelong formula gives $\mathrm{d}\mathrm{d}^\mathrm{c}\psi = [D] - \omega_1$, so $\theta + \mathrm{d}\mathrm{d}^\mathrm{c}\psi = \beta + [D]$.

For a smooth positive $(1,1)$-form Ω on the surface X, the linearization $v \mapsto 2\mathrm{d}\mathrm{d}^\mathrm{c}v \wedge \Omega/\Omega^2$ of the logarithmic Monge–Ampère operator has the adjoint probability $\rho\Omega^2$, where the Gauduchon multiplier $\rho > 0$ satisfies $\mathrm{d}\mathrm{d}^\mathrm{c}(\rho\Omega) = 0$ and $\int_X \rho\Omega^2 = 1$ (Proposition 4.8.1). We call $Q = \rho\Omega$ the *adjoint cofactor* of Ω: it is $\mathrm{d}\mathrm{d}^\mathrm{c}$-closed, and $\Omega \wedge Q$ is the adjoint probability.

Lemma 16.2.2 *Write $x = \log|z|^2$ and $\chi(x) = \log(1+\mathrm{e}^x)$. The envelope $\varphi = P_\theta[\psi]$ is*

$$\varphi = W(x) - 2\chi(x), \qquad W(x) = \begin{cases} x + 2\log 2, & x \le 0, \\ 2\chi(x), & x \ge 0. \end{cases} \tag{16.6}$$

It satisfies $\varphi \le 0$, $0 \le \varphi - \psi \le 2\log 2$, and the model condition (14.20) with $\varepsilon = 3/4$.

Proof The automorphism $z \mapsto 1/z$ of the first factor preserves ω_1 and maps r to the function $\ell = 1/(1+|z|^2)$ of Section 16.13. After this change of coordinate, θ and ψ are the form θ and the function $\psi = d\log\ell$ of that section for $Y = \mathbb{P}^1$, $\kappa = \omega_2$, $a = d = 1$ and $b = 2$. Then $K = 2$, $R = 1$ and $C_* = 2\log 2$, (16.43) becomes (16.6) in the variable x, and Proposition 16.13.1 below gives the envelope identity, $\psi \le \varphi \le \psi + 2\log 2$, $\sup_X \varphi = 0$, and the smaller-background condition for every $\delta \in (1/2, 1)$, in particular for $\varepsilon = 3/4$. □

Proposition 16.2.4 *Every $v \in \mathrm{PSH}(X,\theta)$ with $v \sim \varphi$ is uniquely of the form $v = \psi + w$, where $w \in \mathrm{PSH}(X,\beta)$ is bounded. Its non-pluripolar measure is*

$$\mathrm{MA}_\theta(v) = (\beta + \mathrm{dd^c} w)^2, \qquad \int_X \mathrm{MA}_\theta(v) = 4. \tag{16.7}$$

For every real continuous F,

$$B_\varphi(F) = \log 4 - \log \int_X \mathrm{e}^F \mathrm{d}V, \qquad \mathcal{A}_\varphi(F) = \left\{ \frac{\mathrm{e}^F \mathrm{d}V}{\int_X \mathrm{e}^F \mathrm{d}V} \right\}.$$

At $v = \psi$, the non-pluripolar part of $\theta + \mathrm{dd^c}\psi$ is β, whose adjoint cofactor is $Q = \beta/4$, with adjoint probability $\beta \wedge Q = \mathrm{d}V$.

Proof On the first factor, $r = |s|^2_{h_{\mathrm{FS}}}$ for the section s of $\mathcal{O}(1)$ vanishing at $z = 0$ and the Fubini–Study metric h_{FS}, whose curvature form is ω_1. Since $\varphi \sim \psi$, Theorem 16.9.1 below, applied to the pullback of s with weight one and residual form $\theta - \omega_1 = \beta$, gives the decomposition, the first identity in (16.7) and $B_\varphi = B_\beta$; uniqueness of $w = v - \psi$ is clear. Put $T = \beta + \mathrm{dd^c} w$. The bounded-product construction gives

$$T^2 - \beta^2 = \mathrm{dd^c}\big(w(T+\beta)\big),$$

and Stokes' theorem against the constant function yields $\int_X T^2 = 4$. Integrating the equation in the background β therefore gives the formula for B_φ. This formula is Gâteaux differentiable on $C(X)$, with derivative $G \mapsto -\int_X G\,\mathrm{e}^F \mathrm{d}V / \int_X \mathrm{e}^F \mathrm{d}V$, so Proposition 16.2.2 identifies $\mathcal{A}_\varphi(F)$.

For a smooth real h and small $|t|$, the potential $\psi + th$ has the ordinary type of φ and residual measure $(\beta + t\mathrm{dd^c} h)^2$, whose derivative at $t = 0$ is $2\mathrm{dd^c} h \wedge \beta$. Since β is closed, the constant $1/4$ is its normalized Gauduchon multiplier by the uniqueness in Theorem 4.1.1; hence $Q = \beta/4$ and $\beta \wedge Q = \mathrm{d}V$. □

Theorem 16.2.5 *Let g be a smooth real function on the second factor with $\gamma = \mathrm{dd^c} g \geq -\omega_2$, regarded as a function on X. For $\delta > 0$ put*

$$u_{\delta,g} = \log(r + \delta \mathrm{e}^g), \qquad \Omega_{\delta,g} = \theta + \mathrm{dd^c} u_{\delta,g}, \qquad F_{\delta,g} = \log \frac{\Omega_{\delta,g}^2}{4\mathrm{d}V}.$$

The forms satisfy $\Omega_{\delta,g} \geq \omega$ and $\int_X \Omega_{\delta,g}^2 = 8$. For each fixed g, as $\delta \downarrow 0$ the smooth potentials $u_{\delta,g}$ decrease pointwise to ψ, converge in L^1 and in the capacity Cap_ω, and converge smoothly off D. Moreover $F_{\delta,g} \to 0$ in $L^p(\mathrm{d}V)$ for every $1 \leq p < \infty$. The adjoint cofactors and adjoint probabilities of the forms $\Omega_{\delta,g}$ are

$$Q_{\delta,g} = \Omega_{\delta,g}/8, \qquad \nu_{\delta,g} = \Omega_{\delta,g}^2/8,$$

and they have the weak limits

$$\begin{aligned} Q_{\delta,g} &\longrightarrow (\beta + [D])/8, \\ \nu_{\delta,g} &\longrightarrow \tfrac{1}{2}\mathrm{d}V + \tfrac{1}{2}[D] \wedge \omega_2 + \tfrac{1}{8}[D] \wedge \mathrm{dd^c} g. \end{aligned} \tag{16.8}$$

The limiting probability has mass $1/2$ on D and depends on g, although $\mathcal{A}_\varphi(0) = \{\mathrm{d}V\}$. The cofactors do not approach $\beta/4$, and their limit is not $\rho\beta$ for any locally integrable scalar ρ.

Proof Smoothness and monotone convergence follow directly from the defining formula, also on D, where $\psi = -\infty$. Off D, set

$$\lambda = \frac{r}{r + \delta \mathrm{e}^g}, \qquad h = \psi - g - \log \delta.$$

The chain rule gives $\Omega_{\delta,g} = A_\delta + R_\delta$, where

$$A_\delta = (2 - \lambda)\omega_1 + 2\omega_2 + (1 - \lambda)\gamma, \qquad R_\delta = \lambda(1 - \lambda)\mathrm{d}h \wedge \mathrm{d^c} h.$$

Thus $A_\delta \geq \omega_1 + \omega_2$ and $R_\delta \geq 0$, and the inequality $\Omega_{\delta,g} \geq \omega$ extends across D by continuity. Closedness of θ and Stokes' theorem give the mass eight. Since $\Omega_{\delta,g}$ is closed, the uniqueness in Theorem 4.1.1 makes its normalized Gauduchon multiplier the constant $1/8$, which gives the stated adjoint cofactors and probabilities.

The rank-one form R_δ has square zero, hence $\Omega_{\delta,g}^2 = A_\delta^2 + 2A_\delta \wedge R_\delta$. The bounded coefficients of A_δ converge off D to those of β, which gives $A_\delta^2 \to 4\mathrm{d}V$. Write $N = \mathrm{d}\psi \wedge \mathrm{d^c}\psi$. In $A_\delta \wedge R_\delta$, the part in which R_δ contributes only derivatives along the first factor (the normal part) is

$$\lambda(1 - \lambda)N \wedge \big(2\omega_2 + (1 - \lambda)\gamma\big).$$

Terms mixing a derivative along the first factor with one along the second vanish for degree reasons. The remaining tangential term is $\lambda(1 - \lambda)(2 - \lambda)\omega_1 \wedge \mathrm{d}g \wedge \mathrm{d^c} g$, which tends to zero by dominated convergence.

The normal coefficients can be computed in polar coordinates. Put $s = \delta \mathrm{e}^{g}$ and $t = |z|^2$. Then

$$\lambda = \frac{t}{s + (1+s)t}, \qquad N = \frac{\mathrm{i}\,\mathrm{d}z \wedge \mathrm{d}\bar{z}}{2\pi t (1+t)^2}.$$

After angular integration the radial measure of $\lambda(1-\lambda)N$ is

$$\frac{s\,\mathrm{d}t}{(1+t)(s+(1+s)t)^2} = \frac{\mathrm{d}y}{(1+sy)(1+(1+s)y)^2}, \qquad t = sy.$$

It is bounded by $\mathrm{d}y/(1+y)^2$ and tends to that measure, whose integral is one. With an additional factor $1-\lambda$, the rescaled measure becomes $\mathrm{d}y/(1+(1+s)y)^3$, which is bounded by $\mathrm{d}y/(1+y)^3$ and has limiting integral $1/2$. The mass of both measures outside every fixed tube around D tends to zero, uniformly in the point of the second factor because g is bounded. Testing against continuous functions therefore gives

$$\lambda(1-\lambda)N \longrightarrow [D], \qquad \lambda(1-\lambda)^2 N \longrightarrow \tfrac{1}{2}[D].$$

Combining the normal and the remaining terms proves

$$\Omega_{\delta,g}^2 \longrightarrow 4\mathrm{d}V + [D] \wedge (4\omega_2 + \gamma). \tag{16.9}$$

Dividing by eight gives the second line of (16.8). The limit is positive since $4\omega_2 + \gamma \geq 3\omega_2$, and its mass on D is $4/8 = 1/2$ by Stokes' theorem on the second factor. Taking $g = 0$ and a small nonconstant smooth g gives different limits, since $\mathrm{d}\mathrm{d}^{\mathrm{c}} g$ vanishes identically only for constant g.

The difference of the potentials satisfies

$$0 \leq u_{\delta,g} - \psi \leq \log\bigl(1 + \delta \mathrm{e}^{\max g}/r\bigr)$$

off D. The right-hand side tends to zero and is dominated by an integrable logarithmic pole, which proves L^1 convergence. Continuity of distributional differentiation then gives the limit of the cofactors. A locally integrable function times the smooth form β has absolutely continuous coefficients, so it cannot produce the singular coefficients of $[D]$.

For convergence in capacity, note that $\psi \in \mathrm{PSH}(X,\omega)$ and $\sup_X \psi = 0$, so (1.15) gives

$$\mathrm{Cap}_\omega(\{\psi < -t\}) \leq C/t, \qquad t > 1. \tag{16.10}$$

For $\eta > 0$, the set of points off D where $u_{\delta,g} - \psi > \eta$ is contained in

$$\left\{ r < \frac{\delta \mathrm{e}^{\max g}}{\mathrm{e}^{\eta} - 1} \right\}.$$

This is a sublevel set of ψ whose level tends to $-\infty$ as $\delta \downarrow 0$, so (16.10) proves convergence in capacity.

Finally, $\Omega_{\delta,g} \geq \omega$ gives $F_{\delta,g} \geq -\log 2$. In the affine chart $|\mathrm{d}\psi|_\omega^2 \leq C/r$, while g and its first two derivatives are bounded. The decomposition above yields

$\Omega_{\delta,g} \le C_g(1+r^{-1})\omega$, hence

$$|F_{\delta,g}| \le C_g + 2|\log r|.$$

Every finite power of this logarithmic bound is integrable. Smooth convergence off D gives $F_{\delta,g} \to 0$ there, and dominated convergence proves convergence in L^p. □

The exponential densities satisfy $\int_X \mathrm{e}^{F_{\delta,g}}\,\mathrm{d}V = 2$ for every δ, whereas their pointwise limit has integral one, so they are not uniformly integrable: the smoothings leave the ordinary type of φ and keep the mass eight of bounded potentials instead of the mass four of that type. Convergence in every L^p is weaker than the hypotheses of Proposition 16.2.1 and Theorem 16.1.2. In the fixed divisorial type, a common bound on the residual potentials gives a recovery estimate.

Corollary 16.2.1 *Let $w_j, w \in \mathrm{PSH}(X,\beta)$ satisfy $|w_j|, |w| \le M$ for a fixed $M \ge 1$, and suppose that $w_j \to w$ in capacity. Put*

$$T_j = \beta + \mathrm{dd^c} w_j, \qquad T = \beta + \mathrm{dd^c} w,$$
$$Q_j = T_j/4, \quad Q = T/4, \qquad \nu_j = T_j^2/4, \quad \nu = T^2/4.$$

Then $Q_j \rightharpoonup Q$ as currents and $\nu_j \rightharpoonup \nu$ as probability measures. More precisely, for real $f \in C^\infty(X)$ choose $C_f \ge 0$ with $-C_f\beta \le \mathrm{dd^c} f \le C_f\beta$. For every $\eta > 0$, writing $E_{j,\eta} = \{|w_j - w| > \eta\}$, one has

$$\left|\int_X f\,(\mathrm{d}\nu_j - \mathrm{d}\nu)\right| \le C_f\big(2\eta + 2M^2\,\mathrm{Cap}_\beta(E_{j,\eta})\big). \tag{16.11}$$

If T_j is smooth and strictly positive, then Q_j and ν_j are its adjoint cofactor and adjoint probability. If $\psi + w$ solves (16.1) for some $F \in C(X)$, the limit ν is the unique member of $\mathcal{A}_\varphi(F)$.

Proof The capacity is the one of Section 1.8, and $\beta \le 2\omega$ gives $\mathrm{Cap}_\beta \le 4\,\mathrm{Cap}_\omega$. Set $a_j = (w_j + M)/(2M)$ and $S_j = \beta + \mathrm{dd^c} a_j$. Then $0 \le a_j \le 1$ and

$$S_j = \left(1 - \frac{1}{2M}\right)\beta + \frac{1}{2M}T_j,$$
$$S_j^2 \ge \left(\frac{1}{M} - \frac{1}{2M^2}\right)\beta \wedge T_j \ge \frac{1}{2M}\beta \wedge T_j.$$

Positivity of bounded mixed products justifies these inequalities. Thus $\int_E \beta \wedge T_j \le 2M\,\mathrm{Cap}_\beta(E)$ for every Borel set E, and the same bound holds for T. By closedness and Stokes' theorem, $\int_X \beta \wedge (T_j + T) = 8$. Now test the identity

$$T_j^2 - T^2 = \mathrm{dd^c}\big((w_j - w)(T_j + T)\big)$$

against f. On $X \setminus E_{j,\eta}$ the potential difference has absolute value at most η, and on $E_{j,\eta}$ at most $2M$. Consequently

$$\left|\int_X f(T_j^2 - T^2)\right| \le C_f \left(8\eta + 2M \int_{E_{j,\eta}} \beta \wedge (T_j + T)\right)$$
$$\le C_f \left(8\eta + 8M^2 \operatorname{Cap}_\beta(E_{j,\eta})\right).$$

Dividing by four proves (16.11). Letting first $j \to \infty$ with η fixed and then $\eta \downarrow 0$ shows that $\int_X f \, \mathrm{d}\nu_j \to \int_X f \, \mathrm{d}\nu$. All ν_j and ν have mass one by Proposition 16.2.4, and smooth functions are uniformly dense in $C(X)$, so $\nu_j \rightharpoonup \nu$.

The constant test potential in the definition of capacity gives $4\mathrm{d}V(E) \le \operatorname{Cap}_\beta(E)$. Splitting the potential difference over the same sets yields

$$\|w_j - w\|_{L^1(\mathrm{d}V)} \le \eta + \tfrac{M}{2} \operatorname{Cap}_\beta(E_{j,\eta}).$$

Taking the limits in the same order proves L^1 convergence, and hence convergence of the currents Q_j by continuity of distributional differentiation.

If T_j is smooth and strictly positive, it is closed, so the uniqueness in Theorem 4.1.1 and the mass four make its normalized Gauduchon multiplier the constant $1/4$. Finally, if $\psi + w$ solves the equation at F, then $\nu = T^2/4 = \mathrm{e}^F \mathrm{d}V / \int_X \mathrm{e}^F \mathrm{d}V$, which is the unique supporting probability by Proposition 16.2.4. □

In the corollary, the potentials $\psi + w_j$ keep their pole along D, and the adjoint objects refer to the smooth residual forms T_j. We do not know whether a similar recovery holds for more general singular references, for unbounded residual sequences, or, beyond the product model of Section 16.13.4, when the Gauduchon multipliers are not constant.

16.3 A convex functional on probability measures

To pass to the limit in the supporting inequality, we need to control the mass that supporting probabilities can put on small sets. The convex dual of B_φ provides this control: the measures in each of its sublevel sets are uniformly absolutely continuous. This allows the logarithmic density and the supporting measure to vary simultaneously.

Let $\mathcal{P}(X)$ denote the set of Borel probability measures on X, with the weak topology. Define

$$H_\varphi(\nu) = \sup_{G \in C(X)} \left(B_\varphi(G) + \int_X G \, \mathrm{d}\nu\right), \qquad \nu \in \mathcal{P}(X). \tag{16.12}$$

The value $+\infty$ is allowed, while taking $G = 0$ shows that $H_\varphi \ge B_\varphi(0) > -\infty$.

Proposition 16.3.1 *The functional H_φ is convex and weakly lower semicontinuous. For every $F \in C(X)$,*

$$B_\varphi(F) = \min_{\nu \in \mathcal{P}(X)} \left(H_\varphi(\nu) - \int_X F \, \mathrm{d}\nu \right). \tag{16.13}$$

The minimizers are exactly $\mathcal{A}_\varphi(F)$.

Proof Each function of ν in the supremum defining H_φ is affine and weakly continuous, so their supremum is convex and lower semicontinuous. By definition, $H_\varphi(\nu) - \int_X F \, \mathrm{d}\nu \geq B_\varphi(F)$, with equality precisely when

$$B_\varphi(G) + \int_X G \, \mathrm{d}\nu \leq B_\varphi(F) + \int_X F \, \mathrm{d}\nu \quad \text{for every } G \in C(X).$$

This is the defining inequality of $\mathcal{A}_\varphi(F)$, and the minimum is attained because $\mathcal{A}_\varphi(F)$ is nonempty by Theorem 16.2.1. □

Theorem 16.3.1 *For every finite real* L*, the set*

$$\mathcal{K}_L = \{\nu \in \mathcal{P}(X) : H_\varphi(\nu) \leq L\}$$

is weakly compact. Its measures are uniformly absolutely continuous with respect to $\mathrm{d}V$*: for every* $\epsilon > 0$ *there is a* $\delta > 0$ *such that*

$$\mathrm{d}V(E) < \delta \quad \Longrightarrow \quad \sup_{\nu \in \mathcal{K}_L} \nu(E) < \epsilon \tag{16.14}$$

for every Borel set E*. In particular, writing* $\mathrm{d}\nu = h_\nu \mathrm{d}V$*, the densities in this set are uniformly integrable in* $L^1(\mathrm{d}V)$*. On* $\mathcal{K}_L$*, weak convergence of measures also implies weak convergence of their densities in* $L^1(\mathrm{d}V)$*.*

Proof Weak compactness follows from lower semicontinuity and compactness of $\mathcal{P}(X)$. We may assume that the sublevel set is nonempty, so that $L \geq B_\varphi(0)$. Fix $t > 0$. By Theorem 16.1.2, there is $\delta_t > 0$ such that

$$0 \leq \chi \leq 1, \quad \chi \in C(X), \quad \int_X \chi \, \mathrm{d}V < \delta_t \quad \Longrightarrow \quad B_\varphi(t\chi) > B_\varphi(0) - 1.$$

Indeed, otherwise there would be a sequence of such functions χ tending to zero in L^1 with $B_\varphi(t\chi) \leq B_\varphi(0) - 1$, contrary to that theorem. For $\nu \in \mathcal{K}_L$, the defining supremum gives

$$t \int_X \chi \, \mathrm{d}\nu \leq L - B_\varphi(t\chi) < L - B_\varphi(0) + 1.$$

Choose t so large that $(L - B_\varphi(0) + 1)/t < \epsilon/2$. If $\mathrm{d}V(E) < \delta_t/2$, outer regularity gives an open neighborhood O of E with $\mathrm{d}V(O) < \delta_t$. For any compact $K \subset E$, choose a continuous cutoff $0 \leq \chi \leq 1$ equal to one on K and supported in O. The last inequality gives $\nu(K) < \epsilon/2$, uniformly in $\nu \in \mathcal{K}_L$, and inner regularity yields $\nu(E) \leq \epsilon/2 < \epsilon$. This proves (16.14).

In particular, every measure in the sublevel set is absolutely continuous, with a density of integral one. Together with the uniform small-set estimate, this is exactly uniform integrability on the finite measure space $(X, \mathrm{d}V)$.

Finally, suppose that $\nu_j \to \nu$ weakly within the sublevel set. For a bounded measurable G, choose continuous G_k with $|G_k| \le \|G\|_\infty$ and $G_k \to G$ in $\mathrm{d}V$-measure; such approximations exist by Lusin's theorem. Uniform absolute continuity gives

$$\sup_{\sigma \in \mathcal{K}_L} \int_X |G_k - G| \, \mathrm{d}\sigma \longrightarrow 0.$$

To see this, split X into $\{|G_k - G| \le a\}$ and its complement; the integral over the first set is at most a, and the integral over the second is at most $2\|G\|_\infty$ times a uniformly small measure. For fixed k, weak convergence applies to G_k. Letting first $j \to \infty$ and then $k \to \infty$ proves convergence of the integrals against G, which is exactly weak $L^1(\mathrm{d}V)$ convergence of the densities. In particular, every sequence in the sublevel set has a subsequence whose densities converge weakly in L^1. □

16.4 Bounded measurable densities and stability

The supporting probabilities of continuous approximations of a bounded measurable logarithmic density lie in one sublevel set of H_φ, so their limits can be identified by Theorem 16.3.1.

Theorem 16.4.1 *For every $F \in L^\infty(\mathrm{d}V)$ there is a nonempty set of absolutely continuous probability measures, again denoted $\mathcal{A}_\varphi(F)$, characterized by*

$$B_\varphi(G) \le B_\varphi(F) - \int_X (G - F) \, \mathrm{d}\nu \qquad \text{for every } G \in L^\infty(\mathrm{d}V). \tag{16.15}$$

For continuous F, this set agrees with the one defined by (16.3). Formula (16.13) also holds for bounded measurable F, with these measures as its minimizers; there the objective is interpreted as $+\infty$ whenever $H_\varphi(\nu) = +\infty$, and otherwise ν is absolutely continuous, so that the integral of the L^∞ class F is well defined.

Suppose that $F_j, F \in L^\infty(\mathrm{d}V)$ are uniformly bounded and $F_j \to F$ in $L^1(\mathrm{d}V)$, and let $\nu_j \in \mathcal{A}_\varphi(F_j)$. Then the densities of the ν_j are uniformly integrable, and every weak L^1 cluster point belongs to $\mathcal{A}_\varphi(F)$. If the latter is a singleton, the whole sequence converges to it weakly in $L^1(\mathrm{d}V)$.

Proof First, if $H_\varphi(\nu) < \infty$, the measure ν is absolutely continuous by Theorem 16.3.1. A bounded measurable G admits uniformly bounded continuous approximations $G_k \to G$ in $L^1(\mathrm{d}V)$, and absolute continuity together with the common bound gives $\int_X G_k \, \mathrm{d}\nu \to \int_X G \, \mathrm{d}\nu$. Combined with Theorem 16.1.2, this proves

$$H_\varphi(\nu) = \sup_{G \in L^\infty(\mathrm{d}V)} \left(B_\varphi(G) + \int_X G \, \mathrm{d}\nu \right) \tag{16.16}$$

for such ν.

To construct supporting probabilities at F, choose continuous F_j with $|F_j| \le M = \|F\|_\infty$ and $F_j \to F$ in $L^1(\mathrm{d}V)$, and choose $\nu_j \in \mathcal{A}_\varphi(F_j)$. The minimum formula of Proposition 16.3.1 and the Lipschitz bound give

$$H_\varphi(\nu_j) = B_\varphi(F_j) + \int_X F_j \,\mathrm{d}\nu_j \le B_\varphi(0) + 2M.$$

By Theorem 16.3.1, a subsequence converges weakly in L^1 to a probability measure ν. Uniform absolute continuity gives

$$\int_X |F_j - F| \,\mathrm{d}\nu_j \longrightarrow 0,$$

by splitting again at a fixed small positive threshold: the sets where $|F_j - F|$ exceeds it have $\mathrm{d}V$-measure tending to zero. Since F is bounded and measurable, weak L^1 convergence applies to its integral. Lower semicontinuity of H_φ and continuity of B_φ yield

$$H_\varphi(\nu) \le \liminf_j H_\varphi(\nu_j) = B_\varphi(F) + \int_X F \,\mathrm{d}\nu.$$

The reverse inequality follows from (16.16) with $G = F$. Equality, combined with the same supremum formula, proves (16.15) for ν and the minimum formula.

For continuous F, every measure satisfying (16.3) has finite dual value by Proposition 16.3.1, so (16.16) extends its supporting inequality to all bounded measurable tests; the converse inclusion is trivial. Thus the two definitions agree. Finally, an absolutely continuous probability satisfying (16.15) has finite dual value and attains the minimum, so the characterization is exact.

For the stability assertion, we start from the given F_j and ν_j instead of an approximating construction. The common uniform bound again places all ν_j in a single finite sublevel set of H_φ, and the same compactness and limiting argument shows that every weak L^1 cluster point attains the minimum for F. If the set of minimizers is a singleton, compactness forces the whole sequence to converge to its member. □

16.5 Entropy and the closed case

In a closed background on a compact Kähler manifold, all potentials of an ordinary singularity type have the same mass, and integrating the equation determines the scalar explicitly; its convex dual is then the relative entropy. The variational formula for the entropy also gives estimates in nonclosed backgrounds.

For a Borel probability measure ν absolutely continuous with respect to $\mathrm{d}V$, let h be its nonnegative density and define the *relative entropy*

$$\mathrm{Ent}_{\mathrm{d}V}(\nu) = \int_X h \log h \, \mathrm{d}V, \qquad 0 \log 0 = 0,$$

and set $\mathrm{Ent}_{\mathrm{d}V}(\nu) = +\infty$ if ν is not absolutely continuous. The integral is well defined in $(-\infty, +\infty]$, since $h \log h \geq -\mathrm{e}^{-1}$. By Jensen's inequality it is at least $-\log \mathrm{d}V(X)$, so it can be negative only when $\mathrm{d}V(X) > 1$.

Lemma 16.5.1 *For every $\nu \in \mathcal{P}(X)$,*

$$\mathrm{Ent}_{\mathrm{d}V}(\nu) = \sup_{F \in C(X)} \left(\int_X F \, \mathrm{d}\nu - \log \int_X \mathrm{e}^F \, \mathrm{d}V \right).$$

Proof If ν is not absolutely continuous, choose a compact set K with $\mathrm{d}V(K) = 0$ and $\nu(K) > 0$. For $t > 0$, regularity gives a continuous cutoff $0 \leq \chi_t \leq 1$, equal to one on K, whose support has volume at most e^{-2t}. For $F = t\chi_t$, the expression in the supremum is at least

$$t\nu(K) - \log\bigl(\mathrm{d}V(X) + \mathrm{e}^{-t}\bigr),$$

which tends to infinity with t.

Suppose now that $\nu = h \, \mathrm{d}V$. If its entropy is finite, Jensen's inequality for the logarithm gives

$$\int_X F \, \mathrm{d}\nu - \mathrm{Ent}_{\mathrm{d}V}(\nu) \leq \log \int_{\{h>0\}} \mathrm{e}^F \, \mathrm{d}V \leq \log \int_X \mathrm{e}^F \, \mathrm{d}V.$$

For infinite entropy this upper bound is automatic. For the opposite inequality, set

$$F_k = \log\bigl(\max\{k^{-1}, \min\{h, k\}\}\bigr), \qquad k \geq 1.$$

Then $\int_X \mathrm{e}^{F_k} \, \mathrm{d}V \to 1$ and $\int_X F_k \, \mathrm{d}\nu \to \mathrm{Ent}_{\mathrm{d}V}(\nu)$, with the value $+\infty$ allowed: the positive parts of the integrands hF_k increase to that of $h \log h$, and their negative parts are dominated by the integrable negative part of $h \log h$. For each fixed k, approximate F_k in $L^1(\mathrm{d}V + \nu)$ by continuous functions with the same bounds; both integrals, including the exponential one, converge to those of F_k. Hence the supremum over continuous functions is at least the value of the expression at F_k, and letting $k \to \infty$ completes the proof. □

Suppose now that X is Kähler and θ is closed. By the closed-background theory of [Xia26], all potentials of the ordinary type of φ have the same non-pluripolar mass m_φ, which is positive under the standing hypotheses. For $F \in L^\infty(\mathrm{d}V)$, integrating the equation gives

$$B_\varphi(F) = \log m_\varphi - \log \int_X \mathrm{e}^F \, \mathrm{d}V, \qquad \mathcal{A}_\varphi(F) = \left\{ \frac{\mathrm{e}^F \, \mathrm{d}V}{\int_X \mathrm{e}^F \, \mathrm{d}V} \right\}. \tag{16.17}$$

For the second identity, write ν_F for the normalized exponential measure. By the first identity, the supporting inequality (16.15) for ν_F and $G \in L^\infty(\mathrm{d}V)$ is Jensen's inequality $\log \int_X \mathrm{e}^{G-F} \, \mathrm{d}\nu_F \geq \int_X (G - F) \, \mathrm{d}\nu_F$. Conversely, testing the supporting

inequality of any $\nu \in \mathcal{A}_\varphi(F)$ with $G = F \pm th$, $h \in L^\infty(\mathrm{d}V)$, and letting $t \downarrow 0$ gives $\int_X h\,\mathrm{d}\nu = \int_X h\,\mathrm{d}\nu_F$, so $\nu = \nu_F$.

Proposition 16.5.1 *If X is Kähler and θ is closed, then*

$$H_\varphi(\nu) = \log m_\varphi + \mathrm{Ent}_{\mathrm{d}V}(\nu).$$

Thus the minimum formula (16.13) *becomes the usual variational formula for the logarithmic integral in* (16.17)*.*

Proof Substitute (16.17) into the defining supremum and apply Lemma 16.5.1. □

The dual functional depends on the reference volume. If $\mathrm{d}V$ is replaced by $\mathrm{d}V' = \mathrm{e}^a \mathrm{d}V$ for a continuous real a, the corresponding functionals satisfy

$$B'_\varphi(F) = B_\varphi(F + a), \qquad H'_\varphi(\nu) = H_\varphi(\nu) - \int_X a\,\mathrm{d}\nu.$$

The first identity follows from the equation, and the second, which also holds when the values are infinite, from the change of variable $G \mapsto G + a$ in the defining supremum. In a nonclosed background, H_φ is still defined by the same supremum, but the total mass need not be constant on an ordinary type, so the entropy computation no longer determines it.

16.6 Entropy bounds

For a Hermitian background and the zero reference, a lower bound for the L^q norm of the density in the equation and an upper bound coming from a Gauduchon trace place B_0 between two logarithmic integrals. Dualizing gives two-sided entropy estimates, which identify the set where H_0 is finite.

Proposition 16.6.1 *Let $\theta = \omega$ be a Hermitian form, and let $\varphi = 0$. There are real constants A and, for every $q > 1$, a_q such that every Borel probability measure ν satisfies*

$$a_q + \frac{1}{q}\,\mathrm{Ent}_{\mathrm{d}V}(\nu) \le H_0(\nu) \le A + n\,\mathrm{Ent}_{\mathrm{d}V}(\nu).$$

The inequalities include the value $+\infty$. Thus $H_0(\nu) < \infty$ if and only if ν has finite entropy, and a family of probability measures has uniformly bounded values of H_0 if and only if its entropies are uniformly bounded.

Proof In dimension one, ω is a Kähler form, and Proposition 16.5.1 gives $H_0 = \log \int_X \omega + \mathrm{Ent}_{\mathrm{d}V}$; since $\mathrm{Ent}_{\mathrm{d}V} \ge -\log \mathrm{d}V(X)$, the assertions follow. Suppose that $n \ge 2$.

For $G \in C(X)$, apply the lower bound for $\|F\|_q$ in Theorem 11.3.1 to a bounded solution with density $F = \mathrm{e}^{B_0(G)+G}$. It gives a constant $c_q > 0$, independent of G, such that

$$B_0(G) \geq \log c_q - \frac{1}{q}\log \int_X \mathrm{e}^{qG}\,\mathrm{d}V.$$

For the upper bound, fix a smooth Gauduchon metric γ, rescaled so that $\int_X \omega\wedge\gamma^{n-1} = 1$, and write

$$k_\gamma = \left(\frac{\gamma^n}{\mathrm{d}V}\right)^{(n-1)/n}, \qquad m = \min_X k_\gamma > 0.$$

For smooth G, the trace bound (4.3) gives

$$\mathrm{e}^{B_0(G)/n}\int_X \mathrm{e}^{G/n}k_\gamma\,\mathrm{d}V \leq 1.$$

Uniform approximation and continuity of B_0 extend this inequality to continuous G. Consequently

$$B_0(G) \leq -n\log m - n\log\int_X \mathrm{e}^{G/n}\,\mathrm{d}V.$$

Taking the defining suprema in the two scalar bounds and applying Lemma 16.5.1 with $F = qG$ and $F = G/n$, respectively, proves both inequalities for every probability measure, including infinite values, with $a_q = \log c_q$ and $A = -n\log m$. The finite-domain and bounded-family assertions follow by taking $q = 2$. □

16.6.1 The mass bounds as entropy differences

In a closed background the difference between H_0 and the entropy is constant. In general it measures how much the total Monge–Ampère mass of bounded potentials varies: its infimum and supremum recover the lower and upper masses $v_-(\omega)$ and $v_+(\omega)$ of Definition 7.5.1.

Let $\mathcal{P}_{\mathrm{ent}}(X)$ denote the set of probability measures of finite entropy relative to $\mathrm{d}V$. By Proposition 16.6.1, the difference $H_0(\nu) - \mathrm{Ent}_{\mathrm{d}V}(\nu)$ is a finite real number for every ν in this set.

Proposition 16.6.2 *For every Hermitian background ω,*

$$\inf_{\nu\in\mathcal{P}_{\mathrm{ent}}(X)}\big(H_0(\nu) - \mathrm{Ent}_{\mathrm{d}V}(\nu)\big) = \log v_-(\omega),$$
$$\sup_{\nu\in\mathcal{P}_{\mathrm{ent}}(X)}\big(H_0(\nu) - \mathrm{Ent}_{\mathrm{d}V}(\nu)\big) = \log v_+(\omega).$$

The same extrema are obtained over smooth strictly positive probability measures. Here $\log 0 = -\infty$ and $\log(+\infty) = +\infty$. Thus the difference is bounded above if and only if X has the bounded mass property, and bounded below if and only if ω has positive lower mass.

Proof Write $Z_G = \int_X \mathrm{e}^G\,\mathrm{d}V$. A bounded solution for $G \in C(X)$ has mass $\mathrm{e}^{B_0(G)}Z_G$. If $v_-(\omega) > 0$, the lower mass bound gives

$$B_0(G) \geq \log \nu_-(\omega) - \log Z_G.$$

Taking the defining supremum for H_0 and using Lemma 16.5.1 gives $H_0 - \mathrm{Ent}_{\mathrm{d}V} \geq \log \nu_-(\omega)$ on $\mathcal{P}_{\mathrm{ent}}(X)$. In the same way, if $\nu_+(\omega) < \infty$, the upper mass bound gives $H_0 - \mathrm{Ent}_{\mathrm{d}V} \leq \log \nu_+(\omega)$. These bounds hold trivially when $\nu_-(\omega) = 0$ or $\nu_+(\omega) = +\infty$.

To prove sharpness, first take a smooth real G and its Gibbs probability $\nu_G = Z_G^{-1} \mathrm{e}^G \, \mathrm{d}V$, which has finite entropy. Testing the supremum defining H_0 with G gives

$$H_0(\nu_G) - \mathrm{Ent}_{\mathrm{d}V}(\nu_G) \geq B_0(G) + \log Z_G.$$

In the other direction, let σ_G be the smooth strictly positive Gauduchon adjoint probability of Theorem 16.2.3. It lies in $\mathcal{A}_0(G)$, so its dual value is $B_0(G) + \int_X G \, \mathrm{d}\sigma_G$, and Lemma 16.5.1 yields

$$H_0(\sigma_G) - \mathrm{Ent}_{\mathrm{d}V}(\sigma_G) \leq B_0(G) + \log Z_G.$$

As G runs over smooth real functions, the masses $\mathrm{e}^{B_0(G)} Z_G$ run over the masses $\int_X (\omega + \mathrm{dd}^{\mathrm{c}} u)^n$ of all smooth u with $\omega + \mathrm{dd}^{\mathrm{c}} u > 0$, since such a u solves the equation for $G = \log((\omega + \mathrm{dd}^{\mathrm{c}} u)^n / \mathrm{d}V)$. By Theorem 1.11.1, the mass of every bounded ω-plurisubharmonic function is a limit of such masses. Hence the infimum and supremum of $\mathrm{e}^{B_0(G)} Z_G$ over smooth G are $\nu_-(\omega)$ and $\nu_+(\omega)$. Since ν_G and σ_G are smooth and strictly positive, the last two inequalities prove sharpness on this smaller domain, including the cases $\nu_-(\omega) = 0$ and $\nu_+(\omega) = +\infty$. The two equivalences follow. □

The difference does not depend on the smooth positive reference volume. For smooth real a, replacing $\mathrm{d}V$ by $\mathrm{d}V' = \mathrm{e}^a \mathrm{d}V$ changes $H_0(\nu)$ by $-\int_X a \, \mathrm{d}\nu$, by the transformation formula preceding Proposition 16.6.1. Since $\mathrm{d}\nu/\mathrm{d}V' = \mathrm{e}^{-a} \, \mathrm{d}\nu/\mathrm{d}V$, the entropy changes by the same amount, and finiteness of the entropy is unchanged because a is bounded. Similarly, replacing ω by a Hermitian form $\omega + \mathrm{dd}^{\mathrm{c}} h$ preserves B_0 (Section 16.1), and hence this difference. Finally, the difference is constant on $\mathcal{P}_{\mathrm{ent}}(X)$ exactly when $\nu_-(\omega) = \nu_+(\omega)$, that is, when all bounded ω-plurisubharmonic functions have the same mass.

16.6.2 Entropy for a prescribed ordinary type

Return to the standing setting: θ is semipositive and big, and φ is a model reference, with $\varepsilon \in (0, 1)$ as in (14.20). The part $(1 - \varepsilon)\theta$ of the background left over by the smaller-background envelope gives a uniform lower bound for the density norm, while comparison in a larger Hermitian background gives the upper bound. Together they extend the entropy estimates to the prescribed type.

Corollary 16.6.1 *For every $q > 1$ there is a finite real a_q, and there is a finite real A, such that*

$$a_q + \frac{1}{q}\operatorname{Ent}_{\mathrm{d}V}(\nu) \le H_\varphi(\nu) \le A + n\operatorname{Ent}_{\mathrm{d}V}(\nu)$$

for every Borel probability measure ν, including infinite values. Thus $H_\varphi(\nu)$ is finite exactly when ν has finite entropy, and uniformly bounded dual values are equivalent to uniformly bounded entropies. For fixed θ and ε, the constants can be chosen independently of the model reference φ.

Proof For $G \in C(X)$, let $u \sim \varphi$ solve the equation with density $c\mathrm{e}^G$, where $c = \mathrm{e}^{B_\varphi(G)}$. The lower bound $b_q > 0$ of Lemma 15.11.1 for the L^q norm of the density gives $c\|\mathrm{e}^G\|_q \ge b_q$, hence

$$B_\varphi(G) \ge \log b_q - \frac{1}{q}\log\int_X \mathrm{e}^{qG}\,\mathrm{d}V.$$

For the upper bound, choose a Hermitian form $\Omega \ge \theta$, and let $B_{\Omega,0}$ and $H_{\Omega,0}$ denote its normalization scalar and dual functional for the zero reference. Let v be a bounded solution for G in this background; being bounded, it has a smaller-background envelope $P_{\delta\Omega}(v) \ge \inf_X v$ for every $0 < \delta < 1$. Moreover $u \in \mathrm{PSH}(X,\Omega)$, $u \preceq v$, and positivity gives

$$\mathrm{MA}^{\mathrm{ext}}_\Omega(u) \ge \mathrm{MA}^{\mathrm{ext}}_\theta(u) = c\mathrm{e}^G\,\mathrm{d}V.$$

If $c > \mathrm{e}^{B_{\Omega,0}(G)}$, the singular strict domination principle, applied in the background Ω to v and $u + t$ for every real t, would give $v \ge u + t$ for all t, which is impossible. Hence $B_\varphi \le B_{\Omega,0}$, and therefore $H_\varphi \le H_{\Omega,0}$; Proposition 16.6.1 now gives the upper bound.

The lower scalar estimate gives the lower bound with $a_q = \log b_q$, exactly as in the proof of Proposition 16.6.1, and the finite-domain and bounded-family assertions follow by taking $q = 2$. Finally, Lemma 15.11.1 is uniform in the reference for fixed θ and ε, and the upper bound uses only Ω, which proves the last assertion. □

This gives compactness when the reference varies. Let the model references φ_j satisfy the smaller-background condition with the same ε, let $G_j \in L^\infty(\mathrm{d}V)$ satisfy $\|G_j\|_\infty \le M$ for a fixed constant M, and choose $\nu_j \in \mathcal{A}_{\varphi_j}(G_j)$. With Ω as in the proof,

$$H_{\varphi_j}(\nu_j) = B_{\varphi_j}(G_j) + \int_X G_j\,\mathrm{d}\nu_j \le B_{\Omega,0}(0) + 2M.$$

The uniform lower estimate of Corollary 16.6.1 bounds their entropies, and the Hermitian upper estimate then places all ν_j in a single finite sublevel set of $H_{\Omega,0}$. Hence Theorem 16.3.1, applied in the background Ω, gives weak compactness and uniform integrability of their densities, and every weak limit has finite entropy. Identifying such a limit as a supporting probability for a limiting reference requires further information.

16.7 Changing the prescribed singularities

A more singular reference has a smaller normalization scalar. A competitor built from the smaller-background envelope also gives a quantitative comparison with the zero reference, uniform over all logarithmic densities. In this section the reference varies among the model potentials satisfying (14.20) for the fixed semipositive big background θ, and B_0, H_0 denote the functionals of the zero reference in the same background.

Proposition 16.7.1 *Let φ, ψ satisfy (14.20). If $\varphi \preceq \psi$, then $B_\varphi \le B_\psi$ and $H_\varphi \le H_\psi$. Moreover, if $0 < \varepsilon < 1$ and $P_{\varepsilon\theta}(\varphi) \not\equiv -\infty$, then*

$$B_0(G) + n\log(1-\varepsilon) \le B_\varphi(G) \le B_0(G) \qquad (G \in C(X)),$$

and

$$H_0(\nu) + n\log(1-\varepsilon) \le H_\varphi(\nu) \le H_0(\nu) \qquad (\nu \in \mathcal{P}(X)).$$

The dual inequalities include infinite values.

Proof Fix G, and let u_φ, u_ψ solve the equations in the ordinary types of φ and ψ, with constants c_φ, c_ψ. Then $u_\varphi \preceq u_\psi$. If $c_\varphi > c_\psi$, the singular strict domination principle applied to u_ψ and $u_\varphi + t$ would give $u_\psi \ge u_\varphi + t$ for every real t, which is impossible at a point where both are finite. This proves the monotonicity of B, and taking the defining supremum gives the monotonicity of H. Since $\varphi \preceq 0$ and the zero potential is itself a model reference, this also proves both upper bounds.

For the lower bound, let v be a bounded solution for G with the zero reference, with constant c_0, and put

$$h = P_{\varepsilon\theta}(\varphi) + (1-\varepsilon)v.$$

The two summands are $\varepsilon\theta$- and $(1-\varepsilon)\theta$-plurisubharmonic, respectively, so positivity and homogeneity give

$$\mathrm{MA}_\theta^{\mathrm{ext}}(h) \ge (1-\varepsilon)^n \mathrm{MA}_\theta(v) = (1-\varepsilon)^n c_0 \mathrm{e}^G \, \mathrm{d}V.$$

As in the proof of Lemma 15.11.1, this inequality is justified first on finite truncation levels and then off the pole set of $P_{\varepsilon\theta}(\varphi)$, which has zero smooth volume. Also $h \preceq u_\varphi$. If $c_\varphi < (1-\varepsilon)^n c_0$, strict domination would force $u_\varphi \ge h + t$ for every t. Thus $c_\varphi \ge (1-\varepsilon)^n c_0$. Taking logarithms proves the scalar bound, and taking the defining suprema proves the dual bound. □

16.7.1 Uniform comparison of the scalar functionals

When two scalar functionals are uniformly close, their supporting probabilities are close as well. To make this precise, define for two model references φ, ψ in the

preceding setting

$$\Delta(\varphi,\psi)=\sup_{G\in C(X)}|B_\varphi(G)-B_\psi(G)|. \tag{16.18}$$

The comparison with B_0 in Proposition 16.7.1 makes this number finite for every such pair. It is a pseudometric: symmetry, vanishing on the diagonal and the triangle inequality follow from the properties of the uniform norm. Distinct singularity types can have distance zero, as Example 16.10.1 shows.

Proposition 16.7.2 *For these model references,*

$$\Delta(\varphi,\psi)=\sup_{\mathrm{Ent}_{\mathrm{d}V}(\nu)<\infty}|H_\varphi(\nu)-H_\psi(\nu)|,$$

where the supremum is over probability measures of finite entropy. The same value is obtained if (16.18) *uses all bounded measurable G, and it does not depend on the smooth positive reference volume.*

Proof By Corollary 16.6.1, both duals are finite exactly on the probability measures of finite entropy. A uniform bound for $|B_\varphi - B_\psi|$ passes to the defining suprema, so the supremum in the statement is at most $\Delta(\varphi,\psi)$. Conversely, denote that supremum by D. If $\nu\in\mathcal{A}_\psi(G)$, then ν has finite entropy, and the dual minimum formula gives

$$B_\varphi(G)\le H_\varphi(\nu)-\int_X G\,\mathrm{d}\nu\le B_\psi(G)+D.$$

Interchanging the references gives the opposite inequality, so $\Delta(\varphi,\psi)\le D$.

For bounded measurable G, take continuous approximations with the same uniform bound converging in $L^1(\mathrm{d}V)$; the continuity theorem Theorem 16.1.2, applied to each of the two references, extends the scalar bound to G. Finally, a change of volume $\mathrm{d}V'=\mathrm{e}^a\mathrm{d}V$ replaces each $B(G)$ by $B(G+a)$, and translation by $a\in C(X)$ is a bijection of the space of tests, so the supremum is unchanged. □

Proposition 16.7.3 *Let φ_j,φ be model references in the same background with $\Delta(\varphi_j,\varphi)\to 0$. Let G_j,G be uniformly bounded measurable functions with $G_j\to G$ in $L^1(\mathrm{d}V)$, and choose $\nu_j\in\mathcal{A}_{\varphi_j}(G_j)$. Then the densities of the ν_j are uniformly integrable, every weak cluster point of (ν_j) belongs to $\mathcal{A}_\varphi(G)$, and a weakly convergent subsequence also converges weakly in L^1. If $\mathcal{A}_\varphi(G)$ is a singleton, the whole sequence converges to it in both senses.*

Proof Write $\delta_j=\Delta(\varphi_j,\varphi)$. The bounds $|B_{\varphi_j}-B_\varphi|\le\delta_j$ on bounded measurable tests, from Proposition 16.7.2, and $|H_{\varphi_j}-H_\varphi|\le\delta_j$, together with the identity $H_{\varphi_j}(\nu_j)=B_{\varphi_j}(G_j)+\int_X G_j\,\mathrm{d}\nu_j$, give

$$0\le H_\varphi(\nu_j)-\int_X G_j\,\mathrm{d}\nu_j-B_\varphi(G_j)\le 2\delta_j. \tag{16.19}$$

If $\|G_j\|_\infty\le M$, then $H_\varphi(\nu_j)\le B_\varphi(0)+2M+2\delta_j$, so all the measures lie in one finite sublevel set of H_φ. By Theorem 16.3.1, their densities are uniformly integrable, and a weakly convergent subsequence also converges weakly in L^1.

Let ν be the limit of such a subsequence. Uniform integrability, the common bound and convergence in measure give $\int_X |G_j - G|\,\mathrm{d}\nu_j \to 0$; weak L^1 convergence applies to the bounded function G, and Theorem 16.1.2 gives $B_\varphi(G_j) \to B_\varphi(G)$. Lower semicontinuity of H_φ in (16.19) yields

$$H_\varphi(\nu) - \int_X G\,\mathrm{d}\nu - B_\varphi(G) \le 0.$$

The minimum formula for bounded measurable densities in Theorem 16.4.1 gives the reverse inequality, and equality means exactly that $\nu \in \mathcal{A}_\varphi(G)$. When this set is a singleton, compactness gives convergence of the whole sequence. □

The smaller-background comparison gives an explicit family of references converging to the zero reference.

Corollary 16.7.1 *Let the model references φ_j satisfy $P_{\varepsilon_j\theta}(\varphi_j) \not\equiv -\infty$, where $0 < \varepsilon_j < 1$ and $\varepsilon_j \to 0$. Then $B_{\varphi_j} \to B_0$ uniformly on $C(X)$. Let $G_j, G \in C(X)$ with $G_j \to G$ uniformly, and let $\nu_j \in \mathcal{A}_{\varphi_j}(G_j)$. Then every weak cluster point of (ν_j) belongs to $\mathcal{A}_0(G)$, the densities of the ν_j are uniformly integrable, and a weakly convergent subsequence also converges weakly in $L^1(\mathrm{d}V)$. If $\mathcal{A}_0(G)$ is a singleton, the whole sequence converges to it in both senses.*

Proof Put $\delta_j = -n\log(1-\varepsilon_j) \to 0$. Proposition 16.7.1 gives $\Delta(\varphi_j, 0) \le \delta_j$, which is the uniform convergence of B_{φ_j} to B_0 on $C(X)$. The assertions about supporting probabilities follow from Proposition 16.7.3 with the zero reference as limit. □

For ordinary divisorial types, a comparison of positive residual forms gives a geometric criterion for this convergence of functionals, as the next two sections show. We do not know general convergence criteria for references with degenerate residuals or with arbitrary singularities; Section 16.11 treats special families on closed projective backgrounds.

16.8 Varying positive backgrounds

A two-sided inequality between two Hermitian forms compares their normalization scalars uniformly in the logarithmic density. This turns geometric convergence of backgrounds into the uniform comparison of functionals used above.

From this section on, each section specifies its own background and reference. Here we vary a Hermitian form β on X and use the zero reference. Write B_β, H_β and $\mathcal{A}_\beta$ for the normalization scalar, its dual functional and the supporting sets, defined as above with $\theta = \beta$ and $\varphi = 0$. For two such backgrounds, set

$$d(B_\beta, B_{\beta'}) = \sup_{F \in L^\infty(\mathrm{d}V)} |B_\beta(F) - B_{\beta'}(F)|.$$

The supremum ranges over all bounded measurable tests, with no common bound on their norms.

Theorem 16.8.1 *Suppose* $0 < \varepsilon < 1$ *and* $(1-\varepsilon)\beta \le \beta' \le (1+\varepsilon)\beta$. *For every bounded measurable* F,

$$n\log(1-\varepsilon) \le B_{\beta'}(F) - B_{\beta}(F) \le n\log(1+\varepsilon). \tag{16.20}$$

In particular, $d(B_\beta, B_{\beta'}) \le -n\log(1-\varepsilon)$. *The same conclusion holds if the two inequalities between forms are satisfied by positive representatives* $\beta + \mathrm{dd^c}h$ *and* $\beta' + \mathrm{dd^c}h'$, *with* h, h' *smooth real functions.*

Proof First let F be smooth, and let v, v' be smooth solutions in the two backgrounds. At a maximum point of $v' - (1+\varepsilon)v$,

$$0 < \beta' + \mathrm{dd^c}v' \le (1+\varepsilon)(\beta + \mathrm{dd^c}v).$$

Taking determinants and canceling $\mathrm{e}^F \mathrm{d}V$ gives the upper bound. At a minimum point of $v' - (1-\varepsilon)v$, the reverse inequality between forms gives the lower bound. For bounded measurable F, choose smooth uniformly bounded approximations converging in L^1 and apply Theorem 16.1.2 in each of the two backgrounds. Finally, $B_{\beta+\mathrm{dd^c}h} = B_\beta$ whenever both backgrounds are positive, since the scalar does not depend on the smooth representative of the background (Section 16.1). This proves the assertion about representatives. □

Corollary 16.8.1 *Let* β_j *be Hermitian forms with* $(1-\varepsilon_j)\beta \le \beta_j \le (1+\varepsilon_j)\beta$, *where* $0 < \varepsilon_j < 1$ *and* $\varepsilon_j \to 0$. *Suppose that* $F_j \to F$ *in* $L^1(\mathrm{d}V)$ *with a common* L^∞ *bound. Then* $B_{\beta_j}(F_j) \to B_\beta(F)$. *For any* $\nu_j \in \mathcal{A}_{\beta_j}(F_j)$, *the densities are uniformly integrable, and every subsequence has a further subsequence converging weakly in* L^1 *to an element of* $\mathcal{A}_\beta(F)$. *If this set is a singleton, which is the case in particular when* F *is smooth, the whole sequence converges to it weakly in* L^1 *and as probability measures.*

Proof Put $\delta_j = -n\log(1-\varepsilon_j)$, so that $|B_{\beta_j} - B_\beta| \le \delta_j$ on bounded measurable tests by (16.20). The scalar assertion follows from this bound and Theorem 16.1.2 in the fixed background β. The proof of Proposition 16.7.3 uses only such a bound, together with Theorems 16.1.2, 16.3.1 and 16.4.1 and Proposition 16.3.1 for each functional separately; with B_{β_j} and B_β in place of B_{φ_j} and B_φ, it gives the assertions about the ν_j. For smooth F, the set $\mathcal{A}_\beta(F)$ is a singleton by Theorem 16.2.3. □

Smooth changes of the background also control the potentials, because the inverse of the linearized equation used in Proposition 10.3.1 remains available when the form itself is included among the parameters.

Proposition 16.8.1 *Let β_j, β be Hermitian forms and F_j, F smooth real functions with $\beta_j \to \beta$ and $F_j \to F$ in C^∞. Then the solutions v_j for (β_j, F_j) normalized by $\int_X v_j \, \mathrm{d}V = 0$ converge in C^∞ to the similarly normalized solution v for (β, F), and their scalars converge as well.*

Proof Fix $k \geq 0$ and $0 < \alpha < 1$. On the open set where $\eta + \mathrm{dd^c} w > 0$, consider the smooth map

$$(\eta, G, w, a) \longmapsto \log \frac{(\eta + \mathrm{dd^c} w)^n}{\mathrm{d}V} - a - G$$

defined for $C^{k,\alpha}$ forms η and functions G, mean-zero $C^{k+2,\alpha}$ potentials w and $a \in \mathbb{R}$, with values in $C^{k,\alpha}$. At the limiting solution its derivative in (w, a) is $(h, q) \mapsto \Delta_\Omega h - q$, where $\Omega = \beta + \mathrm{dd^c} v$ and $\Delta_\Omega h = n \mathrm{dd^c} h \wedge \Omega^{n-1} / \Omega^n$. Given a target g, choose $q = -\int_X g \, \mathrm{d}\mu_v$, where μ_v is the adjoint probability of Proposition 4.8.1. The Fredholm alternative then gives a unique mean-zero h with $\Delta_\Omega h = g + q$, and the inverse is bounded as in Proposition 10.3.1. In dimension one this is the Fredholm alternative for the Poisson equation.

The implicit function theorem gives a local branch of solutions depending continuously on the parameters in these spaces. For smooth parameters, uniqueness of bounded solutions modulo constants, Corollary 12.4.1, identifies this branch with the global smooth solution. Thus $v_j \to v$ in $C^{k+2,\alpha}$ for every k, which proves the assertion. □

The smooth adjoint probabilities determine the functional up to its value at zero. If $\nu_{\beta,F}$ denotes the unique supporting probability at a smooth F, integrating the derivative of Theorem 16.2.3 along the path $s \mapsto sF$ gives

$$B_\beta(F) = B_\beta(0) - \int_0^1 \int_X F \, \mathrm{d}\nu_{\beta, sF} \, \mathrm{d}s. \tag{16.21}$$

Here the derivative is continuous along the path by the implicit function theorem. Consequently, two Hermitian backgrounds on the same X, with the same $\mathrm{d}V$, have the same functional on bounded measurable tests if and only if their values at zero and all their adjoint probabilities at smooth tests agree. One direction follows by differentiation; the other follows from (16.21) for smooth tests and from bounded L^1 approximation for the remaining tests.

16.9 Divisorial types and moving sections

For a reference given by logarithms of holomorphic sections, removing its divisorial part identifies its ordinary singularity type with the bounded potentials of the residual background. The normalization scalar therefore depends only on the residual form, whereas convergence of the full potentials also requires control of the sections defining the divisors.

In this section, θ is a smooth real $(1,1)$-form on X. Fix finitely many holomorphic line bundles L_r on X with smooth Hermitian metrics h_r, nonzero holomorphic sections s_r, and weights $t_r > 0$. Put

$$\varphi = \sum_r t_r \log |s_r|^2_{h_r}, \qquad \Gamma = \sum_r t_r \operatorname{div}(s_r), \qquad \beta = \theta - \sum_r t_r c_1(L_r, h_r),$$

where $c_1(L_r, h_r)$ is the Chern curvature form, and assume that the residual form β is positive. The empty family is allowed, with $\varphi = 0$ and $\beta = \theta$.

Theorem 16.9.1 *The ordinary singularity type of φ is described by*

$$\{u \in \mathrm{PSH}(X,\theta) : u \sim \varphi\} = \{\varphi + w : w \in \mathrm{PSH}(X,\beta) \cap L^\infty(X)\},$$

and for each such w,

$$\mathrm{MA}^{\mathrm{ext}}_\theta(\varphi + w) = (\beta + \mathrm{dd^c} w)^n. \tag{16.22}$$

For every bounded measurable F, the equation (16.1) in this ordinary type has a unique constant and a solution unique modulo constants. Its normalization scalar $B_{\theta,[\varphi]}$ satisfies

$$B_{\theta,[\varphi]}(F) = B_\beta(F),$$

so its dual functional and supporting sets coincide with H_β and $\mathcal{A}_\beta$.

Proof The Poincaré–Lelong formula gives $\theta + \mathrm{dd^c}\varphi = [\Gamma] + \beta$, so every function on the right is θ-plurisubharmonic and has the type of φ. Conversely, let $u \in \mathrm{PSH}(X,\theta)$ with $u \sim \varphi$. Then u has the same generic Lelong numbers as φ along the components of Γ, namely the coefficients of Γ, and Lemma 1.17.1 gives

$$\theta + \mathrm{dd^c} u - [\Gamma] \geq 0.$$

The locally integrable difference $w = u - \varphi$ therefore has a β-plurisubharmonic representative. It is bounded off the divisor because $u \sim \varphi$, and the local submean property preserves both bounds across the divisor. Since u and $\varphi + w$ agree almost everywhere, they are equal as quasi-plurisubharmonic functions.

Off the divisor, the currents $\theta + \mathrm{dd^c}(\varphi + w)$ and $\beta + \mathrm{dd^c} w$ agree, and locality identifies their Monge–Ampère products there. The extended measure on the left gives the divisor zero mass, and so does the bounded product on the right, which is finite and charges no analytic set by Section 1.7. This proves (16.22). In the background β, the equation with bounded density has a continuous solution and a unique constant by Theorem 11.1.1 and Proposition 11.6.1. The density e^F is bounded above and below by positive constants, so Corollary 12.4.1 gives uniqueness modulo constants. The correspondence transfers these assertions to the whole ordinary type and identifies the functionals. □

In particular, the comparison of backgrounds and the stability of supporting probabilities in Theorem 16.8.1 and Corollary 16.8.1 apply whenever the positive residual forms satisfy the two-sided inequalities there; the location of the divisors

does not enter these estimates. This statement concerns ordinary types. To interpret it in terms of model references, one also needs the envelope $P_\theta[\varphi]$ to differ from φ by a bounded function. For a Hermitian θ, Lemma 16.9.1 below proves this bounded difference and the smaller-background model condition.

For a fixed positive residual β, uniformly bounded $F_j \to F$ in L^1 give uniform convergence of the residual solutions normalized by $\sup_X v_j = 0$. Indeed, continuity of the scalar gives uniformly bounded densities $e^{B_\beta(F_j)+F_j}$ converging in every L^p, $p < \infty$. The compactness theorem Theorem 11.3.2 gives uniform convergence of subsequences, continuity of bounded products identifies the limiting equation, and Corollary 12.4.1 identifies every limit. Subtracting the means gives the same conclusion for solutions normalized to have mean zero.

Theorem 16.9.2 *Keep the line bundles and metrics fixed. Let nonzero sections s_{jr} converge to $s_r \neq 0$ in the finite-dimensional spaces of global sections, and let positive weights satisfy $t_{jr} \to t_r > 0$. Given Hermitian forms β_j, define φ_j and $\theta_j = \beta_j + \sum_r t_{jr} c_1(L_r, h_r)$ by the preceding formulas. Suppose either that $\beta_j \to \beta$ in C^∞ for a Hermitian form β and $F_j \to F$ in C^∞, or that $\beta_j = \beta$ for all j and the tests F_j converge to F in L^1 with a common L^∞ bound. Then $\varphi_j \to \varphi$ in L^1, and the solutions $u_j \sim \varphi_j$ for (θ_j, F_j), normalized by $\sup_X u_j = 0$, converge in L^1 to the similarly normalized solution $u \sim \varphi$. For the residual solutions v_j normalized to have mean zero, one also has*

$$\sup_X(\varphi_j + v_j) \longrightarrow \sup_X(\varphi + v).$$

The residuals converge smoothly in the first regime and uniformly in the second.

Proof The convergence of the residual solutions follows from Proposition 16.8.1 in the first case and from the preceding compactness argument in the second. The fixed metrics and bounded weights give a common smooth lower bound for $\mathrm{dd}^c\varphi_j$. The convergent sections give a common upper bound for φ_j and convergence at a point outside all limiting divisors, so the suprema of the φ_j are bounded above and below. By Theorem 1.4.1, every subsequence has a further subsequence converging in L^1. Off the limiting divisors, convergence of the sections and weights gives local uniform convergence to φ; since those divisors have zero volume, every L^1 cluster point equals φ. Thus $\varphi_j \to \varphi$ in $L^1(X)$.

Put $C_j = \sup_X(\varphi_j + v_j)$ and $C = \sup_X(\varphi + v)$. The sums converge in L^1 and have a common smooth lower bound for their complex Hessians, since $\mathrm{dd}^c v_j \geq -\beta_j$ and the residual forms are uniformly bounded. Applying Lemma 1.4.1 at maximum points gives $\limsup_j C_j \leq C$. A maximum point of $\varphi + v$ lies off the poles of φ, and evaluation there, using local uniform convergence, gives $\liminf_j C_j \geq C$.

It follows that $u_j = \varphi_j + v_j - C_j \to \varphi + v - C = u$ in L^1. For an empty family of divisors, the assertion reduces to the convergence of the residual solutions and of their suprema. □

16.9.1 One-sided divisorial classes and sharp thresholds

The ordinary-type correspondence of Theorem 16.9.1 extends to potentials with additional poles. It then identifies the largest subsolution coefficient in the whole one-sided class, even for merely integrable data. For smooth positive data, the prescribed divisor causes a strict loss relative to the global coefficient.

Keep the divisorial setting above, assume also that θ is a Hermitian form, and subtract a constant from φ so that $\sup_X \varphi = 0$. Write $S = \operatorname{Supp}\Gamma$ and $\psi = P_\theta[\varphi]$. For a Hermitian form h and a nonzero density $0 \le f \in L^1(\mathrm{d}V)$, we use the subsolution threshold $b_h(f)$ of Theorem 15.27.1 and, for $f \in L^p(\mathrm{d}V)$ with $p > 1$, the normalization scalar $c_h(f)$ of Section 15.26; thus $c_h(\mathrm{e}^F) = \exp B_h(F)$ for bounded F. For a nonempty class $C \subset \mathrm{PSH}(X,\theta)$, define

$$b_{\theta,C}(f) = \sup\{c \ge 0 : \text{some } u \in C \text{ satisfies } \mathrm{MA}_\theta^{\mathrm{ext}}(u) \ge c f\,\mathrm{d}V\}.$$

The three classes considered here are

$$\begin{aligned} C_{\preceq} &= \{u \in \mathrm{PSH}(X,\theta) : u \preceq \varphi\},\\ C_{\sim} &= \{u \in \mathrm{PSH}(X,\theta) : u \sim \varphi\},\\ C_{\mathrm{env}} &= \mathcal{E}_{\mathrm{env}}(X,\theta,\psi). \end{aligned}$$

The following lemma describes the reference ψ used in the last definition.

Lemma 16.9.1 *The map $w \mapsto \varphi + w$ is a bijection from $\mathrm{PSH}(X,\beta)$ onto $C_{\preceq}$, and*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(\varphi + w) = \mathrm{MA}_\beta^{\mathrm{ext}}(w).$$

The bounded members of $\mathrm{PSH}(X,\beta)$ correspond to $C_\sim$. The envelope ψ satisfies $\varphi \le \psi \le \varphi + K$ for some finite K, meets the model condition (14.20), and

$$C_\sim \subset C_{\mathrm{env}} \subset C_{\preceq}.$$

Proof If $u \preceq \varphi$, its generic Lelong numbers dominate the coefficients of Γ. By Lemma 1.17.1, $\theta + \mathrm{dd}^{\mathrm{c}} u - [\Gamma] \ge 0$. The locally integrable distribution $w = u - \varphi$ therefore has a canonical β-psh representative. Off S it equals the difference and is bounded above. The identity $u = \varphi + w$ holds almost everywhere, hence everywhere for the canonical quasi-psh functions. Conversely, $w \in \mathrm{PSH}(X,\beta)$ has finite supremum, and

$$\theta + \mathrm{dd}^{\mathrm{c}}(\varphi + w) = \beta + \mathrm{dd}^{\mathrm{c}} w + [\Gamma] \ge 0, \qquad \varphi + w \preceq \varphi.$$

These maps are inverse to each other. The bounded correspondence is Theorem 16.9.1. Off S the currents agree; locality of the extended products identifies their measures there. Both extended measures give the analytic set S zero mass by Theorem 7.1.1, which proves the measure identity.

Let v be a competitor in the definition of $\psi = P_\theta[\varphi]$, that is, $v \in \mathrm{PSH}(X,\theta)$ with $v \le 0$ and $v \preceq \varphi$. The corresponding $w = v - \varphi \in \mathrm{PSH}(X,\beta)$ satisfies $w \le -\varphi$ off S. The L^1 bound of Theorem 1.4.1, applied to $w - \sup_X w$, gives a constant C_β with

$$\mathrm{d}V(X)\sup_X w \le \int_X w\,\mathrm{d}V + C_\beta \le \int_X (-\varphi)\,\mathrm{d}V + C_\beta,$$

so $\sup_X w \le K$ for a constant K independent of v. Thus $v = \varphi + w \le \varphi + K$ everywhere. The function φ itself is a competitor, so the ordinary envelope construction gives a θ-psh function between φ and $\varphi + K$. The upper bound survives upper regularization because $\varphi + K$ is upper semicontinuous. Since $\psi \sim \varphi$ and the envelope depends only on the ordinary type, $P_\theta[\psi] = P_\theta[\varphi] = \psi$. For $\varepsilon < 1$ sufficiently close to one, $\beta - (1-\varepsilon)\theta > 0$, so φ is an $\varepsilon\theta$-psh competitor below ψ. Hence $P_{\varepsilon\theta}(\psi)$ is not identically $-\infty$, as required in (14.20).

If $u \sim \varphi \sim \psi$, choose $K_u \ge 0$ with $\psi - K_u \le u$. For each $t \ge 1$, the θ-psh function $a_t = \psi - tK_u$ obeys

$$t^{-1}a_t + (1 - t^{-1})\psi = \psi - K_u \le u.$$

This proves membership in the relative envelope class of Definition 14.4.1 (compare Proposition 14.4.1), including the convention at $t = 1$ and common poles. The other containment follows from its defining order $u \preceq \psi \sim \varphi$. □

Theorem 16.9.3 *For every nonzero nonnegative $f \in L^1(\mathrm{d}V)$,*

$$b_{\theta,C_\preceq}(f) = b_\beta(f),$$

and this coefficient is attained. If f is smooth and strictly positive, then

$$b_{\theta,C_\sim}(f) = b_{\theta,C_{\mathrm{env}}}(f) = b_{\theta,C_\preceq}(f) = c_\beta(f),$$

and the equation $\mathrm{MA}^{\mathrm{ext}}_\theta(u) = c_\beta(f) f\,\mathrm{d}V$ has a solution $u \sim \varphi$. If also $\Gamma \neq 0$, then

$$0 < c_\beta(f) < c_\theta(f) = b_\theta(f).$$

Proof The correspondence lemma and Theorem 15.27.1 for the Hermitian form β prove the first assertion, including attainment. For smooth positive f, Theorem 10.1.1 gives a smooth w with $\beta + \mathrm{dd^c}w > 0$ and $(\beta + \mathrm{dd^c}w)^n = c_\beta(f) f\,\mathrm{d}V$. The potential $\varphi + w$ belongs to the ordinary type and hence to all three classes, and by Lemma 16.9.1 it solves the asserted equation. The class containments and the first assertion give equality of their coefficients, and Proposition 15.27.2 gives $b_\beta(f) = c_\beta(f)$.

It remains to prove strictness. Let v be a smooth solution of $(\theta + \mathrm{dd^c}v)^n = c_\theta(f) f\,\mathrm{d}V$, given by Theorem 10.1.1, and put $\sigma = \theta + \mathrm{dd^c}v$, $\eta = \beta + \mathrm{dd^c}w$. The subsolution $\varphi + w$ and Theorem 15.27.1 and Proposition 15.27.2 give $c_\beta(f) \le b_\theta(f) = c_\theta(f)$. If equality held, then on $X \setminus S$ the smooth function $a = \varphi + w - v$ would satisfy

$$\mathrm{dd^c} a = \eta - \sigma, \qquad \mathrm{dd^c} a \wedge Q = 0, \qquad Q = \sum_{j=0}^{n-1} \eta^j \wedge \sigma^{n-1-j}.$$

The form Q is smooth and strictly positive (for $n = 1$, $Q = 1$), so $\mathrm{dd^c} a \wedge Q = 0$ is a linear elliptic equation for a without zeroth-order term. Also $a \to -\infty$ near S, so its maximum is attained inside a component U of $X \setminus S$. The strong maximum principle [GT01, Theorem 3.5], applied on coordinate balls, and connectedness of U make a constant on U. The boundary of U is nonempty and lies in S: otherwise U would be both open and closed in connected X, although $S \neq \varnothing$. Constancy then contradicts $a \to -\infty$ at that boundary. □

In dimension one, every Hermitian form is closed, and the theorem becomes explicit. By Proposition 15.27.2, $b_\beta(f) \int_X f \, \mathrm{d}V = \int_X \beta$ for every f as in the first assertion, and integrating $\theta + \mathrm{dd^c}\varphi = [\Gamma] + \beta$ gives $\int_X \theta = \deg \Gamma + \int_X \beta$, where $\deg \Gamma$ is the sum of the coefficients of Γ. Thus the loss caused by the divisor is $c_\theta(f) - c_\beta(f) = \deg \Gamma / \int_X f \, \mathrm{d}V$.

Corollary 16.9.1 *For smooth strictly positive f, no threshold subsolution $u \in \mathrm{PSH}(X, \theta)$ for f has a nonzero divisorial component in $\theta + \mathrm{dd^c} u$.*

Proof Suppose that $\theta + \mathrm{dd^c} u \geq \lambda_0 [E_0]$ for a prime divisor E_0 and some $\lambda_0 > 0$, and put $E = \lambda E_0$ with $0 < \lambda \leq \lambda_0$. Let s be the canonical section of the line bundle $\mathcal{O}(E_0)$, choose a smooth Hermitian metric h_E on it, and put $\ell_E = \lambda \log |s|^2_{h_E}$ and $\gamma_E = \lambda c_1(\mathcal{O}(E_0), h_E)$; then $[E] = \gamma_E + \mathrm{dd^c} \ell_E$ by the Poincaré–Lelong formula. For λ sufficiently small, $\theta - \gamma_E > 0$. Divisorial removal, Lemma 1.17.1, gives a $(\theta - \gamma_E)$-psh representative of $u - \ell_E$, whose finite supremum implies $u \preceq \ell_E$. The preceding theorem, applied to the single section s with weight λ, would then give

$$b_\theta(f) \leq c_{\theta - \gamma_E}(f) < c_\theta(f) = b_\theta(f),$$

a contradiction. □

For $n \geq 2$, the later rigidity result Corollary 27.8.1 strengthens this obstruction by determining every smooth-data threshold subsolution through the maximizing Gauduchon form.

Example 16.9.1 Let (X_0, h_0) be a connected compact Hermitian m-fold, let $k \geq 1$, and let κ be the mass-one Fubini–Study form on $\mathbb{P}^k$. On $X_0 \times \mathbb{P}^k$, choose a smooth positive volume form $\mathrm{d}V$. With pullbacks implicit, put

$$n = m + k, \quad C = \binom{n}{m}, \qquad V_0 = \int_{X_0} h_0^m,$$
$$H = h_0 + \kappa, \quad \mu = (h_0^m/V_0) \wedge \kappa^k, \quad R = \mu/\mathrm{d}V.$$

For $D = \{Z_0 = 0\}$ pulled back to the product, take $E = \lambda D$, $0 < \lambda < 1$, with

$$\ell_E = \lambda \log \frac{|Z_0|^2}{\sum_{i=0}^k |Z_i|^2}, \qquad H_E = h_0 + (1-\lambda)\kappa.$$

The global threshold of R is CV_0, while its ordinary, relative envelope and one-sided thresholds prescribed by E are all $CV_0(1-\lambda)^k$, attained by ℓ_E. If $v_-(h_0) = 0$, both H and H_E are nonclosed and have zero lower mass.

Proof The Poincaré–Lelong formula gives $[D] = \kappa + \mathrm{dd^c}(\ell_E/\lambda)$. Since $h_0^{m+1} = 0$ and $\kappa^{k+1} = 0$, the binomial expansion gives

$$H^n = CV_0 R\,\mathrm{d}V, \qquad H_E^n = CV_0(1-\lambda)^k R\,\mathrm{d}V.$$

Thus the zero residual solves each smooth positive equation and fixes its unique scalar. The theorem identifies all three restricted coefficients, and the correspondence gives $\mathrm{MA}_H^{\mathrm{ext}}(\ell_E) = H_E^n$. If $v_-(h_0) = 0$, pullbacks of bounded h_0-psh potentials with mass tending to zero have masses multiplied respectively by C and $C(1-\lambda)^k$. Bounded submersion pullback and the product expansion therefore give zero lower mass for both backgrounds. If h_0 were closed, Stokes' theorem would give every bounded potential the mass $V_0 > 0$; hence h_0 is nonclosed, and so are both backgrounds. □

16.10 The mass determined by the functional

In a closed background, the information lost by the scalar functional can be described exactly: for the model references below, two functionals agree precisely when the masses agree. This also gives examples in which the supporting probabilities stay fixed while the potentials have different limits.

In this section X is a compact Kähler manifold and θ is any smooth closed big real $(1,1)$-form. Let $\varphi \in \mathrm{PSH}(X,\theta)$ satisfy $P_\theta[\varphi] = \varphi$, have small unbounded locus in the sense of Section 1.12, and have positive mass

$$m_\varphi = \int_X \mathrm{MA}_\theta(\varphi) > 0.$$

Write $B_{\theta,\varphi}$ and $\mathcal{A}_{\theta,\varphi}$ for the normalization scalar and the supporting sets of the equation (16.1) in the ordinary type of φ, and put $Z_F = \int_X \mathrm{e}^F \, \mathrm{d}V$.

Theorem 16.10.1 *For every such pair (θ,φ) and every bounded measurable F, the equation in the ordinary type of φ has a unique constant and a solution unique modulo constants. Its normalization scalar is*

$$B_{\theta,\varphi}(F) = \log m_\varphi - \log Z_F.$$

For continuous real F, its supporting set is

$$\mathcal{A}_{\theta,\varphi}(F) = \{\mathrm{e}^F \mathrm{d}V / Z_F\}.$$

For two such pairs on the same X, with the same $\mathrm{d}V$ and possibly in different classes,

$$\sup_{F \in L^\infty} |B_{\theta,\varphi}(F) - B_{\theta',\psi}(F)| = |\log m_\varphi - \log m_\psi|. \tag{16.23}$$

Thus an invariant of these pairs is determined by the whole functional if and only if it is constant on every fiber of the map $(\theta,\varphi) \mapsto m_\varphi$.

Proof Existence follows from Corollary 34.8.1. Indeed, a Kähler manifold has the bounded mass property by Lemma 7.5.1, and since in a closed background all potentials of the ordinary type of φ have the same mass $m_\varphi > 0$, the reference satisfies the positive-volume model condition (31.5). Uniqueness modulo constants is the classical theorem on equations with prescribed singularities in closed big classes, which applies under the model and small-unbounded-locus hypotheses; see the notes in Section 36.11.

Every potential of the ordinary type of φ has mass m_φ, so integrating the equation forces $cZ_F = m_\varphi$, which is the scalar formula. The supporting set follows from it as for (16.17). Subtracting the scalar formulas gives (16.23). Finally, $m_\varphi = \mathrm{d}V(X) \exp B_{\theta,\varphi}(0)$, so the functional and the mass determine each other, and an invariant factors through either one exactly when it is constant on the fibers of the mass map. □

For an ordinary divisorial type whose residual form β is positive and closed, Theorem 16.9.1 gives the same formula with mass $\int_X \beta^n$, since Stokes' theorem for bounded products in a closed background fixes the mass of every bounded β-plurisubharmonic function. This computation does not require the real weighted sum of divisors to have a model representative.

Example 16.10.1 Let $X = \mathbb{P}^n$ with $n \geq 1$, let θ be the Fubini–Study form, normalized by $\int_X \theta^n = 1$, and let D be a hyperplane. Let s_D be a section of $\mathcal{O}(1)$ defining D, measured with the Fubini–Study metric and rescaled so that $a_D = \log|s_D|^2 \leq 0$, and choose $t \in (0,1)$. Then

$$\mathrm{dd^c} a_D = [D] - \theta, \qquad \theta + \mathrm{dd^c}(t a_D) = (1-t)\theta + t[D].$$

Put $\varphi_{t,D} = P_\theta[t a_D]$. By Lemma 16.9.1, applied to s_D with weight t and residual form $(1-t)\theta$, this envelope fixed point has the ordinary type of $t a_D$. Moreover $t a_D \in \mathrm{PSH}(X, t\theta)$ and $t a_D \leq \varphi_{t,D}$, so $\varphi_{t,D}$ satisfies (14.20) with $\varepsilon = t$. Off D, the potential $t a_D$ has Monge–Ampère measure $(1-t)^n\theta^n$, and its non-pluripolar measure puts no mass on D.

Since mass is invariant under bounded changes of the potential in a closed background on the Kähler manifold $\mathbb{P}^n$,

$$m_{\varphi_{t,D}} = (1-t)^n, \qquad B_{\varphi_{t,D}}(G) - B_0(G) = n\log(1-t) \quad (G \in C(X,\mathbb{R})).$$

Thus the constant in Proposition 16.7.1 is sharp.

Now let $n = 1$, and take for D two distinct points $p, q \in \mathbb{P}^1$. The corresponding references have distance zero, although $\varphi_{t,p}$ is unbounded near p and bounded near q, while $\varphi_{t,q}$ behaves in the opposite way; in particular they are not ordinarily equivalent. For every positive integer m,

$$\mathcal{I}(m\varphi_{t,p}) = \mathcal{O}(-\lfloor mt \rfloor p), \qquad \mathcal{I}(m\varphi_{t,q}) = \mathcal{O}(-\lfloor mt \rfloor q),$$

by Lemma 1.16.1, and for $mt \geq 1$ these ideals differ. Thus the distance Δ of (16.18) does not even separate multiplier-ideal types, and it determines neither the ordinary nor the ideal singularity type.

Fix a smooth real function F on $\mathbb{P}^1$, and let w_F be the solution of $\theta + \mathrm{dd^c} w_F = \mathrm{e}^F \mathrm{d}V / Z_F$ with $\int_X w_F \, \mathrm{d}V = 0$. The normalized solutions for $D = p$ and $D = q$ are

$$u_{t,D} = t a_D + (1-t) w_F - \sup_X \bigl(t a_D + (1-t) w_F\bigr).$$

Their difference satisfies $\mathrm{dd^c}(u_{t,p} - u_{t,q}) = t([p] - [q]) \neq 0$. Hence the sequence of solutions obtained by alternating between $D = p$ and $D = q$ has two different L^1 cluster points, although the scalar functional and all its supporting sets are the same at every step. Within this one-parameter family, the functional does recover t, since $t \mapsto (1-t)^n$ is injective.

Example 16.10.2 On $\mathbb{P}^n$, $n \geq 2$, let ω be the Fubini–Study form, normalized as in Example 16.10.1, and let f be a nonconstant smooth real function. Fix $0 < t < 1$ and put

$$\beta = \mathrm{e}^f \omega, \qquad \theta = \beta + t\omega, \qquad \varphi_D = t \log|s_D|^2.$$

Here s_D defines a hyperplane D as in Example 16.10.1, and $\theta + \mathrm{dd^c}\varphi_D = \beta + t[D]$, so every hyperplane has the same positive residual form β. This form is not closed: $\mathrm{d}\beta = \mathrm{e}^f \mathrm{d}f \wedge \omega \not\equiv 0$, since $\mathrm{d}f$ is nonzero somewhere and wedging a one-form

with ω^{n-1} is injective. By Theorem 16.9.1, the ordinary types of all φ_D have the same functional B_β on bounded tests, hence the same supporting sets. At a smooth test, the common supporting probability is the Gauduchon adjoint probability of Theorem 16.2.3.

If v_F is the common smooth residual solution, the normalized solutions are the functions $\varphi_D + v_F - \sup_X(\varphi_D + v_F)$. Two distinct hyperplanes D_1, D_2 give different poles, different divisorial coefficients and, for sufficiently large k, different multiplier ideals $\mathcal{O}(-\lfloor kt \rfloor D_i)$, by the calculation in normal coordinates of Lemma 1.16.1. The potentials differ by more than a constant, since the difference of their complex Hessians is $t([D_1] - [D_2])$, and alternating between the two hyperplanes again produces two distinct L^1 limits. These conclusions come from the common residual form; for general nonclosed backgrounds we do not claim that the functional is determined by a mass.

16.11 Convex-body criteria for varying references

When the singularities of a torus-invariant reference are described by a convex body, the closed mass formula of Theorem 16.10.1 turns convergence of the convex bodies into convergence of the functionals. A translated simplex contained in all the bodies keeps the limiting reference solvable, and the Hausdorff distance then controls the functionals uniformly over all continuous tests, even though the ordinary types vary.

Let ω_{FS} be the Fubini–Study form on $\mathbb{P}^n$, normalized by $\int_{\mathbb{P}^n} \omega_{\mathrm{FS}}^n = 1$. In homogeneous coordinates $[Z_0 : \cdots : Z_n]$, consider the torus $T = \{Z_0 \cdots Z_n \neq 0\}$, and put

$$x_i = \log |Z_i/Z_0|^2, \qquad L(x) = \log\bigl(1 + \textstyle\sum_i \mathrm{e}^{x_i}\bigr), \qquad \Sigma = \{p_i \geq 0 : \textstyle\sum_i p_i \leq 1\},$$

where $1 \leq i \leq n$ and $\Sigma \subset \mathbb{R}^n$ is the standard simplex. Functions of $x = (x_1, \dots, x_n)$ are regarded as functions on T; thus $\omega_{\mathrm{FS}} = \mathrm{dd}^{\mathrm{c}} L$ on T. Define the continuous function ℓ on Σ by

$$\ell(p) = \sum_i p_i \log p_i + (1 - \textstyle\sum_i p_i) \log(1 - \textstyle\sum_i p_i), \qquad 0 \log 0 = 0.$$

For a convex body $K \subset \Sigma$, that is, a compact convex set with nonempty interior, set

$$h_K(x) = \sup_{p \in K} p \cdot x, \qquad g_K(x) = \sup_{p \in K} \{p \cdot x - \ell(p)\}.$$

16.11.1 Canonical models and their mass

Lemma 16.11.1 *The functions $h_K - L$ and $g_K - L$ on T extend uniquely to ω_{FS}-plurisubharmonic functions ψ_K, Φ_K on $\mathbb{P}^n$. They satisfy*

$$\psi_K \le \Phi_K \le \psi_K + \log(n+1), \qquad \sup \Phi_K = 0,$$
$$P_{\omega_{\mathrm{FS}}}[\psi_K] = \Phi_K = P_{\omega_{\mathrm{FS}}}[\Phi_K].$$

If $p_* + \delta\Sigma \subset K$ for some $p_* \in \mathbb{R}^n$ and $0 < \delta < 1$, then $P_{(1-\delta)\omega_{\mathrm{FS}}}(\Phi_K) \not\equiv -\infty$.

Proof The Legendre transform of L is $L^* = \ell$ on Σ and $+\infty$ outside it. Since $-\log(n+1) \le \ell \le 0$,

$$h_K \le g_K \le h_K + \log(n+1), \qquad g_K \le L, \qquad h_K \le h_\Sigma \le L.$$

A finite convex function of the logarithmic moduli is plurisubharmonic: for a smooth convex function the complex Hessian is positive, and convolution gives the general case. On $T \cap \{Z_j \neq 0\}$, add the local Fubini–Study potential $\log \sum_i |Z_i/Z_j|^2$; the resulting functions are again convex functions of the logarithmic moduli of the affine coordinates, and they are bounded above by that smooth potential because $h_K, g_K \le L$. Lemma 1.3.7 therefore gives the extensions across the coordinate hyperplanes and the stated bounds. If $p \in \operatorname{int} K$, then $p \in \operatorname{int}\Sigma$, and at $x_i = \log(p_i/(1 - \sum_k p_k))$ the supremum defining $L = \ell^*$ is attained at $p \in K$, so $g_K(x) = L(x)$. This proves $\sup \Phi_K = 0$.

For $C \ge \log(n+1)$, let $v_C = P_{\omega_{\mathrm{FS}}}(\psi_K + C, 0)$, the envelope of the obstacle $\min(\psi_K + C, 0)$. Since Φ_K is a competitor, $\Phi_K \le v_C \le 0$. The obstacle is upper semicontinuous, so $v_C \le \psi_K + C$ everywhere. The compact torus acting by rotations of the coordinates preserves the obstacle, so v_C is invariant, and on T the function $f = v_C + L$ is a finite convex function of x with $f \le L$ and $f \le h_K + C$. The second inequality shows that the Legendre transform f^* is $+\infty$ outside K, and the first gives $f^* \ge \ell$. Thus

$$f(x) = \sup_{p \in \operatorname{dom} f^*} \{p \cdot x - f^*(p)\} \le g_K(x).$$

Hence $v_C \le \Phi_K$ on T, and since both functions are quasi-plurisubharmonic and T is the complement of an analytic set, $v_C = \Phi_K$ on $\mathbb{P}^n$. Letting $C \to \infty$ gives $P_{\omega_{\mathrm{FS}}}[\psi_K] = \Phi_K$. Since $\Phi_K \sim \psi_K$ and the envelope depends only on the ordinary type, also $P_{\omega_{\mathrm{FS}}}[\Phi_K] = \Phi_K$.

For the final assertion put $\varepsilon = 1 - \delta$, $b = \sum_i p_{*,i}$ and $\chi = p_* \cdot x - \varepsilon L$ on T. The inclusion $p_* + \delta\Sigma \subset \Sigma$ implies $p_{*,i} \ge 0$ and $b \le \varepsilon$, and the extension of χ to $\mathbb{P}^n$ satisfies

$$\varepsilon\omega_{\mathrm{FS}} + \mathrm{dd}^{\mathrm{c}}\chi = \sum_i p_{*,i}[Z_i = 0] + (\varepsilon - b)[Z_0 = 0] \ge 0.$$

Since $h_K \ge p_* \cdot x + \delta h_\Sigma$ and $L - h_\Sigma \le \log(n+1)$, one has $\chi - \delta\log(n+1) \le \Phi_K$. Thus $\chi - \delta\log(n+1)$ is an $\varepsilon\omega_{\mathrm{FS}}$-plurisubharmonic competitor for $P_{\varepsilon\omega_{\mathrm{FS}}}(\Phi_K)$, which proves the claim. □

In the rest of this section, either $X = \mathbb{P}^n$ and $\pi = \mathrm{id}$, or $n \ge 2$ and $\pi\colon X = \mathrm{Bl}_q \mathbb{P}^n \to \mathbb{P}^n$ is the blowup at a fixed point q, with exceptional divisor E_q; put $\Theta = \pi^*\omega_{\mathrm{FS}}$.

Lemma 16.11.2 *The manifold X is Kähler, and Θ is smooth, closed, semipositive and big. For $r > 0$, the reference $r\pi^*\Phi_K$ satisfies $P_{r\Theta}[r\pi^*\Phi_K] = r\pi^*\Phi_K$, has small unbounded locus, and satisfies*

$$\int_X \mathrm{MA}_{r\Theta}(r\pi^*\Phi_K) = r^n n!\,\mathrm{Vol}_n(K). \tag{16.24}$$

If $p_* + \delta\Sigma \subset K$ *with* $0 < \delta < 1$, *then* $P_{(1-\delta)r\Theta}(r\pi^*\Phi_K) \not\equiv -\infty$.

Proof On $\mathbb{P}^n$, the envelope identity is Lemma 16.11.1. In the blowup case, pullback is a bijection from $\mathrm{PSH}(\mathbb{P}^n, \omega_{\mathrm{FS}})$ onto $\mathrm{PSH}(X, \Theta)$ by Proposition 1.22.4; it preserves inequalities and hence envelopes, and positive scaling preserves them as well.

On projective space, $\Theta = \omega_{\mathrm{FS}}$ is Kähler. For bigness in the blowup case, choose unitary homogeneous coordinates $[W_0 : \cdots : W_n]$, used only in this paragraph, in which $q = [1 : 0 : \cdots : 0]$, set

$$\psi_q = \log \frac{\sum_{i=1}^n |W_i|^2}{\sum_{i=0}^n |W_i|^2},$$

and let $Q\colon X \to \mathbb{P}^{n-1}$ be the map induced by the linear projection from q. For $0 < t < 1$,

$$\Theta + \mathrm{dd^c}(t\pi^*\psi_q) = (1-t)\pi^*\omega_{\mathrm{FS}} + tQ^*\omega_{\mathrm{FS},n-1} + t[E_q].$$

The smooth part is a Kähler form, since (π, Q) embeds X as the incidence graph in $\mathbb{P}^n \times \mathbb{P}^{n-1}$ and both coefficients are positive. This proves the assertions about the manifold and the form Θ.

For a smooth convex g, the coordinates $z_i = \mathrm{e}^{x_i/2+\mathrm{i}\vartheta_i}$ give

$$\mathrm{d^c} g = \frac{1}{2\pi}\sum_i g_i\,\mathrm{d}\vartheta_i, \qquad \int_T (\mathrm{dd^c} g)^n = n! \int_{\mathbb{R}^n} \det(D^2 g)\,\mathrm{d}x.$$

For a finite convex g, convolution and the continuity of bounded Monge–Ampère products under local uniform convergence identify $(\mathrm{dd^c} g)^n$ on T in the same way: its image under $z \mapsto x$ is $n!M_g$, where M_g is the Alexandrov measure, which assigns to a Borel set of $\mathbb{R}^n$ the Euclidean volume of its subgradient image. For g_K, the bound $g_K \le h_K + \log(n+1)$ and a separation argument put every subgradient in K. Conversely, if $p \in \operatorname{int} K$ and $p + c\mathbb{B} \subset K$, where $\mathbb{B}$ is the closed unit ball of $\mathbb{R}^n$ and $c > 0$, then $g_K(x) - p \cdot x \ge c|x|$; this function attains a minimum, so p is a subgradient of g_K at some point. Consequently

$$\operatorname{int} K \subset \partial g_K(\mathbb{R}^n) \subset K, \qquad M_{g_K}(\mathbb{R}^n) = \mathrm{Vol}_n(K),$$

since the boundary of a convex body has zero volume. The analytic complement of T receives no non-pluripolar mass, so this is the full mass. Locality off the exceptional divisor proves the formula after pullback, since the divisor and its image carry no non-pluripolar mass, and homogeneity gives the factor r^n. The reference is locally bounded off the coordinate hyperplanes and their inverse image, an analytic set, hence has small unbounded locus. Finally, scaling by r and pulling back the competitor found at the end of the proof of Lemma 16.11.1 gives the last assertion. □

16.11.2 Hausdorff control of the functional

Theorem 16.11.1 *Let convex bodies $P_j, P \subset \Sigma$ contain the same translated simplex $p_* + \delta\Sigma$, with $0 < \delta < 1$, and suppose that their Hausdorff distance $d_j = d_H(P_j, P)$ tends to zero. Let $r_j > 0$ tend to $r > 0$. On the fixed X above, set*

$$(\theta_j, \varphi_j) = (r_j\Theta, r_j\pi^*\Phi_{P_j}), \qquad (\theta, \varphi) = (r\Theta, r\pi^*\Phi_P).$$

For every bounded real Borel function F, the equation (16.1) is solvable in the ordinary type of each φ_j and of φ; these types have masses $m_j = r_j^n n!\,\mathrm{Vol}_n(P_j)$ and $m = r^n n!\,\mathrm{Vol}_n(P) > 0$. Fix a smooth positive volume form $\mathrm{d}V$ on X, put $Z_F = \int_X \mathrm{e}^F\,\mathrm{d}V$, and let B_j, B and $\mathcal{A}_j, \mathcal{A}$ be the corresponding normalization scalars and supporting sets. Then

$$B_j(F) = \log m_j - \log Z_F, \qquad B(F) = \log m - \log Z_F.$$

Writing $s = \delta/(n + \sqrt{n})$, one has

$$\sup_{G \in C(X,\mathbb{R})} |B_j(G) - B(G)| \le n|\log(r_j/r)| + n\log(1 + d_j/s). \tag{16.25}$$

Moreover, $n!\,\mathrm{Vol}_n(P_j) \ge \delta^n$ and $n!\,\mathrm{Vol}_n(P) \ge \delta^n$, the backgrounds converge smoothly, and the references converge in $L^1(X)$ and locally uniformly on $\pi^{-1}(T)$. For continuous real F, every supporting set at F equals $\{\nu_F\}$, where $\nu_F = \mathrm{e}^F\mathrm{d}V/Z_F$. If continuous F_j converge uniformly to F and $\nu_j \in \mathcal{A}_j(F_j)$, then

$$\int_X \left|\frac{\mathrm{d}\nu_j}{\mathrm{d}V} - \frac{\mathrm{d}\nu_F}{\mathrm{d}V}\right| \mathrm{d}V \le \mathrm{e}^{2\|F_j - F\|_\infty} - 1. \tag{16.26}$$

Proof By the two preceding lemmas, each φ_j and φ is an envelope fixed point with small unbounded locus and the stated positive mass, and it satisfies the smaller-background condition. Theorem 16.10.1 therefore gives solvability and the formulas for the scalars and supporting sets.

The common simplex contains a ball $c + s\mathbb{B}$ of radius s, the inradius of $\delta\Sigma$, where $\mathbb{B}$ is the closed unit ball. The Hausdorff distance gives $P_j \subset P + d_j\mathbb{B}$, while $s\mathbb{B} \subset P - c$ and convexity give

$$P + d_j\mathbb{B} \subset c + (1 + d_j/s)(P - c).$$

Exchanging the roles of P and P_j bounds both volume ratios by $(1 + d_j/s)^n$. Subtracting the scalar formulas and including the factor r_j^n/r^n gives (16.25). The volume bound holds because $\delta\Sigma$ has volume $\delta^n/n!$.

Let ω_ℓ be the modulus of uniform continuity of ℓ on Σ. Matching the maximizers in the definitions of g_{P_j} and g_P by points at distance at most d_j gives

$$|g_{P_j}(x) - g_P(x)| \le d_j|x| + \omega_\ell(d_j).$$

This proves local uniform convergence on T. The common lower bound $\chi - \delta \log(n+1) \le \Phi_{P_j}$ obtained at the end of the proof of Lemma 16.11.1 is integrable on X, also after pullback, so boundedness of r_j and dominated convergence off the coordinate hyperplanes give L^1 convergence. Finally, the ratio of the normalized exponential densities lies in $[\mathrm{e}^{-2\eta}, \mathrm{e}^{2\eta}]$, where $\eta = \|F_j - F\|_\infty$. This proves (16.26) and hence convergence in total variation. □

16.11.3 A family with nonanalytic isolated poles

The hypothesis of a common simplex allows ordinary singularity types that contain no potential with analytic singularities. Assume $n \ge 2$, and let $o = [1 : 0 : \cdots : 0]$ be the origin of the affine chart $Z_0 \ne 0$, with affine coordinates $z_i = Z_i/Z_0$. For $0 < a \le 1$, define

$$P_a = \{p \in \Sigma : \textstyle\sum_i \sqrt{p_i} \ge \sqrt{a}\}, \qquad \kappa_n = \tfrac{2^n n!}{(2n)!} = \tfrac{1}{(2n-1)!!}.$$

Proposition 16.11.1 *The function Φ_{P_a} is locally bounded outside the single point o, where it has a pole, and it does not have the ordinary singularity type of any function with analytic singularities. Its generic Lelong number along every prime divisor of $\mathbb{P}^n$ is zero, and*

$$\int_{\mathbb{P}^n} \mathrm{MA}_{\omega_{\mathrm{FS}}}(\Phi_{P_a}) = 1 - \kappa_n a^n > 0. \tag{16.27}$$

The same assertions about the pole, nonanalyticity and generic Lelong numbers hold for $\pi^*\Phi_{P_a}$ on the blowup $\mathrm{Bl}_q \mathbb{P}^n$ at a fixed point $q \ne o$.

Proof The function $p \mapsto \sum_i \sqrt{p_i}$ is concave, so P_a is convex. Set

$$b = \frac{n+1}{2n}, \qquad p_{*,i} = \frac{b}{n}, \qquad \delta = \frac{n-1}{4n}.$$

Then $b + \delta < 1$ and $\sum_i \sqrt{p_{*,i}} = \sqrt{(n+1)/2} > 1$. Adding a point of $\delta\Sigma$ preserves the last inequality and keeps the sum of the coordinates below one, so $p_* + \delta\Sigma$ lies in every P_a.

All the standard basis vectors e_i belong to P_a, and therefore

$$\psi_{P_a} \ge \log \frac{\max_{1 \le i \le n} |Z_i|^2}{\sum_{i=0}^n |Z_i|^2}.$$

Hence the reference is locally bounded away from o. The Cauchy–Schwarz inequality gives $\sum_i p_i \ge a/n$ on P_a. When all $x_i < 0$, this yields $h_{P_a}(x) \le (a/n) \max_i x_i$, which tends to $-\infty$ as $z \to o$, and Lemma 16.11.1 gives the pole at o. Local boundedness outside one point implies that all generic Lelong numbers along prime divisors vanish, and pulling back to the blowup at a different point introduces no pole.

To test analyticity, fix positive $\lambda_1, \dots, \lambda_n$ and approach o along $|z_i|^2 = \mathrm{e}^{-R\lambda_i}$, $R \to \infty$. Writing $y_i = \sqrt{p_i}$, the Cauchy–Schwarz inequality gives

$$\min_{p \in P_a} \sum_i \lambda_i p_i = \frac{a}{\sum_i \lambda_i^{-1}} =: d_a(\lambda), \qquad y_i = \frac{\sqrt{a}}{\lambda_i \sum_k \lambda_k^{-1}}, \tag{16.28}$$

where the minimizer satisfies $\sum_i p_i \le a \le 1$, so that it lies in P_a. Since $L(-R\lambda) \to 0$ and $\Phi_{P_a} - \psi_{P_a}$ is bounded, $-\Phi_{P_a}/R \to d_a(\lambda)$ along these curves.

Suppose that the type were that of $c \log \sum_k |f_k|^2$ near o, with $c > 0$ and holomorphic germs $f_k = \sum_\alpha c_{k,\alpha} z^\alpha$. Exponentiating the two inequalities expressing this equivalence and averaging over the arguments of the z_i gives the positive series

$$\sum_{k,\alpha} |c_{k,\alpha}|^2 \mathrm{e}^{-R\alpha\cdot\lambda},$$

whose exponential decay rate in R must be $d_a(\lambda)/c$. That rate is the minimum of $\alpha \cdot \lambda$ over the monomials with nonzero coefficients: a minimizing term gives one bound, and convergence at a fixed R_0 bounds the series above by a constant times $\mathrm{e}^{-(R-R_0)\min \alpha\cdot\lambda}$. Near any λ with positive coordinates, only finitely many monomials can attain the minimum, because these coordinates have a common positive lower bound and any one nonzero coefficient bounds the degree of the minimizing monomials. The rate is therefore locally piecewise linear in λ. But

$$d_a(s, 1, \dots, 1) = \frac{as}{1 + (n-1)s}, \qquad \frac{\mathrm{d}^2}{\mathrm{d}s^2} d_a(s, 1, \dots, 1) = -\frac{2a(n-1)}{(1 + (n-1)s)^3} < 0.$$

This contradiction proves nonanalyticity. The assertion persists on a blowup at a point different from o, which does not change a neighborhood of the pole.

The removed corner $C_a = \{p \in \mathbb{R}^n : p_i \ge 0,\ \sum_i \sqrt{p_i} < \sqrt{a}\} = aC_1$ lies in Σ. The substitution $p_i = y_i^2$ gives

$$\mathrm{Vol}_n(C_1) = 2^n \int_{y_i \ge 0,\, \sum y_i < 1} \prod_i y_i \, \mathrm{d}y = \frac{2^n}{(2n)!}.$$

The last equality follows by iterated integration of monomials over a simplex. Removing the corner from Σ, whose volume is $1/n!$, and applying (16.24) gives (16.27). Since $n \ge 2$, one has $\kappa_n < 1$. □

Corollary 16.11.1 *Put $a_j = 1 - 1/(j+2)$ and $r_j = 1 + 1/(j+2)$ for $j \ge 1$. On $\mathbb{P}^n$, or on its blowup at a fixed point $q \ne o$, the references $r_j \pi^* \Phi_{P_{a_j}}$ satisfy the hypotheses of Theorem 16.11.1, with limit $\pi^* \Phi_{P_1}$. Their ordinary types and masses vary with j, and*

$$\sup_G |B_j(G) - B(G)| \le n |\log r_j| + \frac{n\kappa_n}{1 - \kappa_n} |1 - a_j| \longrightarrow 0.$$

The equation is solvable in the limiting ordinary type, whose mass is $1 - \kappa_n$; for $n = 2$ this mass is $2/3$.

Proof If $p \in P_{a_j} \setminus P_1$, then $u = (\sum_i \sqrt{p_i})^2 \in [a_j, 1)$ and $p/u \in P_1$. Since $|p| \le \sum_i p_i \le u$, one has $|p/u - p| \le 1 - u \le 1 - a_j$. Thus $d_H(P_{a_j}, P_1) \le 1 - a_j$, and the hypotheses of the theorem hold with the simplex found in the proof of Proposition 16.11.1. The mass formula (16.27) and the derivative bound $|\mathrm{d}\log(1-\kappa_n a^n)/\mathrm{d}a| \le n\kappa_n/(1-\kappa_n)$ give the estimate for the functionals. By (16.28), the directional coefficients of the jth reference are multiplied by $r_j a_j = 1 - (j+2)^{-2}$, which strictly increases with j, so the ordinary types are distinct, while the masses $r_j^n(1 - \kappa_n a_j^n)$ strictly decrease to $1 - \kappa_n > 0$. □

16.11.4 Positive mass with a degenerate smooth residual

A second family has a logarithmic pole whose resolution leaves a smooth residual form degenerating along a different exceptional divisor. Let $n \ge 2$ and $o = [1 : 0 : \cdots : 0]$ as in Section 16.11.3, and fix $q \ne o$ in $\mathbb{P}^n$. On $X = \mathrm{Bl}_q \mathbb{P}^n$, keep $\Theta = \pi^*\omega_{\mathrm{FS}}$. For $0 < t < 1$, put

$$\psi_t = t \log \frac{\sum_{i=1}^n |Z_i|^2}{\sum_{i=0}^n |Z_i|^2}, \qquad K_t = \{p \in \Sigma : \textstyle\sum_i p_i \ge t\}.$$

On T, the function $\psi_t + L$, a potential of $\omega_{\mathrm{FS}} + \mathrm{dd}^{\mathrm{c}}\psi_t$, equals $(1-t)L + t\log\sum_i \mathrm{e}^{x_i}$. It differs by a bounded function from

$$(1-t)h_\Sigma + t\max_i x_i = h_{K_t}, \qquad K_t = (1-t)\Sigma + t\,\mathrm{conv}(e_1, \ldots, e_n).$$

Thus $\psi_t \sim \psi_{K_t} \sim \Phi_{K_t}$, and $n!\,\mathrm{Vol}_n(K_t) = 1 - t^n$. We call ψ_t the prototype of this ordinary type.

Proposition 16.11.2 *Let $\tau\colon Z \to X$ be the blowup of the point of X above o, with exceptional divisor E_o, and denote the inverse image of E_q in Z again by E_q. There is a smooth semipositive closed form α_t on Z such that*

$$(\pi\circ\tau)^*\omega_{\mathrm{FS}} + \mathrm{dd}^{\mathrm{c}}(\psi_t\circ\pi\circ\tau) = \alpha_t + t[E_o]. \tag{16.29}$$

It is positive away from E_q, has rank exactly one at every point of E_q, and satisfies $\int_Z \alpha_t^n = 1 - t^n > 0$. Moreover, for every $u \in \mathrm{PSH}(X,\Theta)$ with the ordinary type of $\pi^*\Phi_{K_t}$, in particular for every solution of the equation in this type, the function $u\circ\tau - \psi_t\circ\pi\circ\tau$ extends from $Z\setminus E_o$ to a bounded α_t-plurisubharmonic function on Z.

Proof On $Y = \mathrm{Bl}_o\,\mathbb{P}^n$, let π_o be the blowdown map and $Q_o\colon Y \to \mathbb{P}^{n-1}$ the map induced by the linear projection from o. As in the proof of Lemma 16.11.2, the incidence graph embedding shows that

$$\beta_t = (1-t)\pi_o^*\omega_{\mathrm{FS}} + tQ_o^*\omega_{\mathrm{FS},n-1}$$

is a Kähler form, where $\omega_{\mathrm{FS},n-1}$ is the Fubini–Study form of $\mathbb{P}^{n-1}$. Since the two centers are distinct, Z is also the blowup $\sigma\colon Z \to Y$ of the point above q. Put $\alpha_t = \sigma^*\beta_t$. The Poincaré–Lelong formula gives (16.29); there is no term along E_q, since ψ_t is locally bounded near q.

In a blowup chart, σ is $(s, w_2, \dots, w_n) \mapsto (s, sw_2, \dots, sw_n)$. At $s = 0$ its differential has rank one and kills the $n-1$ tangent directions of E_q. Positivity of β_t therefore gives the assertion about the rank; off E_q the map is biholomorphic.

Write $H = \pi_o^*[\omega_{\mathrm{FS}}]$ and $E = [E_o]$ for the cohomology classes on Y. The current identity gives $[\beta_t] = H - tE$. Since $H|_{E_o} = 0$, all mixed top intersections vanish. The normal bundle of $E_o \cong \mathbb{P}^{n-1}$ is $\mathcal{O}_{\mathbb{P}^{n-1}}(-1)$, so $\int_Y E^n = (-1)^{n-1}$. Therefore

$$\int_Z \alpha_t^n = \int_Y (H - tE)^n = 1 - t^n.$$

The first equality holds because σ has degree one. Finally, if $u \sim \pi^*\Phi_{K_t} \sim \pi^*\psi_t$, then $u\circ\tau$ differs from the pulled-back prototype $\psi_t\circ\pi\circ\tau$ by a bounded function off E_o. By (16.29), after adding a local potential of α_t this bounded difference becomes plurisubharmonic off E_o, and it extends across E_o by Lemma 1.3.7. □

Corollary 16.11.2 *Let $0 < t_{\max} < 1$, let $t_j, t \in (0, t_{\max}]$ with $t_j \to t$, and let $r_j > 0$ tend to $r > 0$. The references $r_j\pi^*\Phi_{K_{t_j}}$ satisfy the hypotheses of Theorem 16.11.1, so in particular the equation is solvable in the limiting ordinary type, and*

$$\sup_G |B_j(G) - B(G)| \le n|\log(r_j/r)| + \frac{nt_{\max}^{n-1}}{1 - t_{\max}^n}|t_j - t|.$$

At continuous tests, all supporting sets consist of the normalized exponential probabilities of that theorem. At the endpoint $t = 1$, where the mass vanishes, no equation with a positive bounded density is solvable in the ordinary type of the prototype $\pi^*\psi_1$.

Proof Choose $p_{*,i} = t_{\max}/n$ and $\delta = (1 - t_{\max})/2$; then $p_* + \delta\Sigma$ lies in every K_t with $t \le t_{\max}$. For $s \ge t$, moving a point of K_t toward a vertex e_i until its coordinate sum reaches s gives

$$d_H(K_s, K_t) \le \frac{\sqrt{2}}{1 - t_{\max}} |s - t|.$$

Hence the stability theorem applies. Instead of its general bound, the bound $n t_{\max}^{n-1}/(1 - t_{\max}^n)$ for the derivative of $-\log(1 - t^n)$ on $(0, t_{\max}]$ and the mass formula give the sharper estimate for the functionals.

At $t = 1$, the resolved residual form α_1 is pulled back from $\mathbb{P}^{n-1}$, so $\int_Z \alpha_1^n = 0$. The argument of Proposition 16.11.2 shows that every potential of the ordinary type of $\pi^*\psi_1$ has a bounded α_1-plurisubharmonic remainder. Stokes' theorem for bounded products in a closed background gives this remainder mass zero, and positivity forces its Monge–Ampère measure to vanish. Thus no such potential can solve the equation with density $ce^F \mathrm{d}V$ for bounded F and $c > 0$. In contrast, the endpoint $a = 1$ in Corollary 16.11.1 has positive mass and remains solvable. □

Scaling by $r > 0$ multiplies both the resolved residual form and the divisor coefficient by r. Solvability is obtained on X through the criterion of Theorem 16.10.1, without requiring the resolved residual form to be strictly positive. These criteria use closed projective backgrounds and fixed modifications; we do not know analogous convergence criteria for general nonclosed Hermitian backgrounds, for moving modifications, or for unrestricted degenerations of the singularities. Moreover, the supporting probabilities here only identify the limiting functional: we make no claim about convergence of the solutions or of weighted adjoint products.

16.12 Selection by a vanishing exponential parameter

The equation with exponent zero can have several normalized solutions. Nevertheless, the solutions of the equations with positive exponent select one of them in the limit: the scalar functional provides a family of integral normalizations, and the selected potential is the greatest subsolution of the critical equation satisfying all of them. To prove that it is the greatest, we use supporting probabilities of a second functional, obtained by varying the forcing term in the subsolution inequality.

In this section, θ is an arbitrary smooth big real $(1,1)$-form on X, not necessarily semipositive. Fix a strict analytic barrier $\rho \le 0$ as in Section 15.12, and put $\Omega = X \setminus \{\rho = -\infty\}$. A potential has *minimal type* if it has the ordinary singularity type of the envelope $V_\theta = P_\theta(0)$, which has minimal singularities (Section 1.19). Fix a nonzero $0 \le f \in L^p(\mathrm{d}V)$, with $1 < p < \infty$. For a nonzero nonnegative density g in some L^q, $q > 1$, let $c(g) > 0$ be the unique constant of Theorem 15.15.2 for which the equation $\mathrm{MA}_\theta(v) = c(g) g \, \mathrm{d}V$ has a solution v of minimal type. We call $c(g)$ the *critical scalar* of g and such solutions *critical potentials* for g; they are continuous on Ω by Theorem 15.16.1. Here MA_θ is the extended measure $\mathrm{MA}_\theta^{\mathrm{ext}}$. On Ω it is the

locally bounded product for every potential of minimal type, and it gives zero mass to the analytic set $X \setminus \Omega$.

Put $c = c(f)$ and define

$$\begin{aligned} \kappa(r) &= \log c - \log c(\mathrm{e}^r f), \\ \mathcal{P}_f &= \left\{ \nu \in \mathcal{P}(X) : \int_X r \, \mathrm{d}\nu \le \kappa(r) \text{ for every } r \in C(X) \right\}. \end{aligned} \tag{16.30}$$

When θ is semipositive and $f = \mathrm{e}^F$ with $F \in C(X)$, we have $\kappa(r) = B_0(F) - B_0(F+r)$, and $\mathcal{P}_f$ is the set $\mathcal{A}_0(F)$ of supporting probabilities. The definition also applies when f vanishes on a set of positive measure and when θ is not semipositive.

Theorem 16.12.1 *For $\lambda > 0$, let u_λ be the solution of minimal type of*

$$\mathrm{MA}_\theta(u_\lambda) = \mathrm{e}^{\lambda u_\lambda} f \, \mathrm{d}V$$

given by Theorem 15.15.1 and unique by Corollary 15.12.1. The set $\mathcal{P}_f$ is nonempty and weakly compact, and every potential of minimal type is integrable against all its measures. As $\lambda \downarrow 0$,

$$w_\lambda = u_\lambda - \lambda^{-1} \log c$$

converges locally uniformly on Ω and in $L^1(X)$ to a potential w of minimal type, continuous on Ω. It is the greatest among the potentials s of minimal type, continuous on Ω, satisfying

$$\mathrm{MA}_\theta(s) \ge c f \, \mathrm{d}V, \qquad \int_X s \, \mathrm{d}\nu \le 0 \quad (\nu \in \mathcal{P}_f). \tag{16.31}$$

Moreover,

$$\mathrm{MA}_\theta(w) = c f \, \mathrm{d}V, \qquad \max_{\nu \in \mathcal{P}_f} \int_X w \, \mathrm{d}\nu = 0.$$

Consequently, $u_\lambda - \sup_X u_\lambda$ converges to $w - \sup_X w$ in the same two topologies. For a Hermitian θ, one may take $\Omega = X$, and both convergences of potentials are uniform.

16.12.1 A norm defined by the critical scalar

Mixed products turn the comparison of critical scalars into a triangle inequality for nth roots of densities, and the subsolution estimate associated with the strict barrier gives continuity in the corresponding Lebesgue norm.

Fix $1 < q < p$ and set $\ell = nq$. On real $L^\ell(\mathrm{d}V)$ define

$$N(a) = \begin{cases} c(|a|^n)^{-1/n}, & a \ne 0, \\ 0, & a = 0. \end{cases}$$

Proposition 16.12.1 *The function N is a lattice norm on $L^\ell(\mathrm{d}V)$, that is, a norm satisfying $N(a) \le N(b)$ whenever $|a| \le |b|$. There is $C_q < \infty$, depending only on $X, \theta, \mathrm{d}V$ and q, such that*

$$N(a) \le C_q \|a\|_\ell, \qquad |N(a) - N(b)| \le C_q \|a - b\|_\ell. \tag{16.32}$$

Consequently, for nonzero $g, h \ge 0$ in L^q,

$$|c(g)^{-1/n} - c(h)^{-1/n}| \le C_q \|g - h\|_q^{1/n}.$$

Proof If a potential v of minimal type satisfies $\mathrm{MA}_\theta(v) \ge dg\,\mathrm{d}V$, then $d \le c(g)$, since otherwise comparison with a critical potential for g contradicts part (1) of Corollary 15.12.1. Hence $g \le h$ implies $c(g) \ge c(h)$. Uniqueness of the critical scalar also gives $c(tg) = c(g)/t$ for $t > 0$. These facts prove that N is monotone and absolutely homogeneous, and positivity follows from $0 < c(g) < \infty$.

For the triangle inequality, let $a, b \ge 0$ be nonzero, put $A = N(a)$ and $B = N(b)$, and choose critical potentials v_a, v_b with measures $(a/A)^n \mathrm{d}V$ and $(b/B)^n \mathrm{d}V$. On every relatively compact subset of Ω, (1.12) gives

$$\mathrm{MA}_\theta\left(\frac{Av_a + Bv_b}{A + B}\right) \ge \left(\frac{a + b}{A + B}\right)^n \mathrm{d}V,$$

where both potentials are locally bounded. Exhausting Ω gives the inequality on X, since $X \setminus \Omega$ has zero smooth volume. The averaged potential has minimal type, so the first paragraph gives $N(a + b) \le A + B$. The cases where a or b vanishes are trivial, and for real functions we use $|a + b| \le |a| + |b|$ and monotonicity.

Next, apply Lemma 15.13.1 with $\eta = V_\theta$ to nonnegative densities of L^q norm at most one. It gives a uniform $m > 0$ and subsolutions of minimal type, hence $c(g) \ge m$ for each such nonzero density. For $a \ne 0$, take $g = |a|^n / \|a\|_\ell^n$, which has L^q norm one. Homogeneity gives $N(a) \le m^{-1/n} \|a\|_\ell$, which is the first bound in (16.32), and the reverse triangle inequality gives the second. Applying the second bound to $g^{1/n}, h^{1/n}$ and using $|g^{1/n} - h^{1/n}|^n \le |g - h|$ proves the last assertion. □

16.12.2 Norming probabilities and integral bounds

Set $a = f^{1/n} \in L^\ell(\mathrm{d}V)$ and $A = N(a) = c^{-1/n}$. The probabilities in (16.30) come from positive linear functionals that norm N at a, that is, functionals dominated by N and equal to $N(a)$ at a. This representation will control integrals of unbounded potentials as well as bounded logarithmic perturbations.

With $\ell' = \ell/(\ell - 1)$, define

$$\mathcal{H}_a = \left\{ \eta \in L^{\ell'}(\mathrm{d}V) : \eta \ge 0,\ \int_X a\eta\,\mathrm{d}V = A,\ \int_X b\eta\,\mathrm{d}V \le N(b) \text{ for every } b \in L^\ell \right\}. \tag{16.33}$$

Proposition 16.12.2 *The set $\mathcal{H}_a$ is nonempty and weakly compact in $L^{\ell'}$. The function κ of (16.30) is given by*

$$\kappa(r) = n \log \frac{N(a\mathrm{e}^{r/n})}{A},$$

and it is convex, increasing and 1-Lipschitz for the uniform norm on $C(X)$, with $\kappa(0) = 0$ and $\kappa(r + b) = \kappa(r) + b$ for constants b. Its subdifferential at zero is

$$\mathcal{P}_f = \{\nu_\eta : \eta \in \mathcal{H}_a\}, \qquad \mathrm{d}\nu_\eta = A^{-1} a\eta \,\mathrm{d}V. \tag{16.34}$$

If z is measurable and $az \in L^\ell$, then

$$\sup_{\nu \in \mathcal{P}_f} \int_X |z| \,\mathrm{d}\nu \le C_q A^{-1} \|az\|_\ell. \tag{16.35}$$

If also $a\mathrm{e}^{tz/n} \in L^\ell$ and $N(a\mathrm{e}^{tz/n}) = A$ for some $t > 0$, then $\int_X z \,\mathrm{d}\nu \le 0$ for every $\nu \in \mathcal{P}_f$.

Proof The formula for κ follows from $N(a\mathrm{e}^{r/n}) = c(\mathrm{e}^r f)^{-1/n}$. The Hahn–Banach theorem gives a linear functional that norms N at a. By (16.32) it is continuous on L^ℓ, and hence is represented by some $\eta_0 \in L^{\ell'}$. For $b \ge 0$, testing on $\operatorname{sgn}(\eta_0) b$ gives

$$\int_X b|\eta_0| \,\mathrm{d}V \le N(b).$$

Since $a \ge 0$ and $\int_X a\eta_0 \,\mathrm{d}V = A = N(a)$, the same equality holds with $|\eta_0|$ in place of η_0. Monotonicity of N extends the domination to real b, so $|\eta_0| \in \mathcal{H}_a$. The defining conditions of $\mathcal{H}_a$ are weakly closed and give $\|\eta\|_{\ell'} \le C_q$, so reflexivity proves weak compactness.

For nonzero nonnegative b_0, b_1, the weighted arithmetic–geometric mean inequality applied to $b_j / N(b_j)$, together with monotonicity and the triangle inequality, gives

$$N(b_0^{1-t} b_1^t) \le N(b_0)^{1-t} N(b_1)^t, \qquad 0 \le t \le 1.$$

Taking $b_j = a\mathrm{e}^{r_j/n}$ proves convexity of κ. Monotonicity and the translation rule follow from the corresponding properties of N, and together they imply the Lipschitz bound.

Every ν_η is a probability and satisfies

$$\frac{N(a\mathrm{e}^{r/n})}{A} \ge \int_X \mathrm{e}^{r/n} \,\mathrm{d}\nu_\eta.$$

Jensen's inequality gives $\kappa(r) \ge \int_X r \,\mathrm{d}\nu_\eta$, which proves one inclusion in (16.34). The map $\eta \mapsto \nu_\eta$ is weakly continuous, since $ar \in L^\ell$ for $r \in C(X)$.

Conversely, let $\nu \in \mathcal{P}_f$ and put $L(r) = \int_X r \,\mathrm{d}\nu$. Differentiating the curve $t \mapsto a\mathrm{e}^{tr/n}$ in L^ℓ at $t = 0$ gives

$$N(a\mathrm{e}^{tr/n}) = N(a) + \frac{t}{n} N'_+(a; ar) + o(t), \qquad t \downarrow 0.$$

Here the one-sided derivative $N'_+(a;\cdot)$ of the continuous convex function N exists, and the error estimate follows from its Lipschitz bound. The inequality $L \le \kappa$, applied to tr and $-tr$, therefore gives

$$|AL(r)| \le C_q \|ar\|_\ell, \qquad AL(r) \le N'_+(a; ar) \le N(a + ar) - A. \tag{16.36}$$

Thus $T(ar) = AL(r)$ defines a continuous linear functional on the space of products ar, $r \in C(X)$. These products are dense in

$$S_a = \{b \in L^\ell : b = 0 \text{ almost everywhere on } \{a = 0\}\} :$$

indeed, it suffices to approximate b/a in $L^\ell(a^\ell \,\mathrm{d}V)$ by continuous functions. If $ar \ge 0$, then $ar = ar_+$ almost everywhere, and the first inequality in (16.36) gives $L(r - r_+) = 0$, so $T(ar) \ge 0$. Approximating nonnegative elements of S_a by products ar with $r \ge 0$ continuous shows that the continuous extension of T to S_a is positive.

Represent this extension by some $\eta \ge 0$ in $L^{\ell'}$ vanishing on $\{a = 0\}$. Since $T(a) = A$, extending the second inequality in (16.36) to S_a and applying it to $b - a$ gives $T(b) \le N(b)$ for $b \in S_a$. For general $b \in L^\ell$,

$$\int_X b\eta \,\mathrm{d}V \le N(b\mathbf{1}_{\{a>0\}}) \le N(b).$$

Thus $\eta \in \mathcal{H}_a$, and ν_η agrees with ν on all continuous functions. This proves the representation (16.34), and weak compactness of $\mathcal{P}_f$ follows from that of $\mathcal{H}_a$.

The bound (16.35) follows from Hölder's inequality and $\|\eta\|_{\ell'} \le C_q$. Under the last hypothesis, the norming inequality gives $\int_X \mathrm{e}^{tz/n} \,\mathrm{d}\nu \le 1$. Since z is ν-integrable by (16.35), Jensen's inequality implies $\int_X z \,\mathrm{d}\nu \le 0$. □

These estimates apply uniformly to all $\nu \in \mathrm{PSH}(X, \theta)$ with $\sup_X \nu = 0$. Indeed, Lemma 1.5.1, applied in a Hermitian background larger than θ, gives uniform bounds for all finite moments of $|\nu|$, and Hölder's inequality gives

$$\int_X |\nu|^{nq} f^q \,\mathrm{d}V \le \|f\|_p^q \left(\int_X |\nu|^{nqp/(p-q)} \,\mathrm{d}V \right)^{(p-q)/p}. \tag{16.37}$$

Consequently, (16.35) gives a uniform bound for $\int_X |\nu| \,\mathrm{d}\nu$ over all such ν and all $\nu \in \mathcal{P}_f$; constant translates of such ν are integrable as well. The representation also shows that these probabilities charge neither $\{f = 0\}$ nor $X \setminus \Omega$.

16.12.3 Bounds for the centered potentials

We call the potentials w_λ of Theorem 16.12.1 the *centered potentials*. Choose a critical potential ψ for f with $\sup_X \psi = 0$, so that $\mathrm{MA}_\theta(\psi) = c f \, \mathrm{d}V$. The integral normalizations prevent the centered potentials from drifting upward by constants.

Lemma 16.12.1 *There is $K < \infty$, independent of $0 < \lambda \le 1$, such that*

$$\psi \le w_\lambda, \qquad 0 \le k_\lambda := \sup_X w_\lambda \le K, \qquad \psi - K \le v_\lambda := w_\lambda - k_\lambda \le V_\theta. \quad (16.38)$$

Moreover $\int_X w_\lambda \, \mathrm{d}\nu \le 0$ for every $\nu \in \mathcal{P}_f$. The family w_λ is relatively compact for locally uniform convergence on Ω, and every subsequence converging locally uniformly on Ω also converges in $L^1(X)$, to a potential of minimal type.

Proof The centered potential satisfies $\mathrm{MA}_\theta(w_\lambda) = c\mathrm{e}^{\lambda w_\lambda} f \, \mathrm{d}V$, while $\psi \le 0$ gives $\mathrm{MA}_\theta(\psi) \ge c\mathrm{e}^{\lambda\psi} f \, \mathrm{d}V$. Part (2) of Corollary 15.12.1 therefore gives $\psi \le w_\lambda$, hence $k_\lambda \ge 0$. For each fixed λ, the density $\mathrm{e}^{\lambda w_\lambda} f$ lies in L^p, and its critical scalar is c. Therefore

$$N(a\mathrm{e}^{\lambda w_\lambda/n}) = A.$$

The moment estimate above gives $a w_\lambda \in L^\ell$, so Proposition 16.12.2, with $z = w_\lambda$ and $t = \lambda$, gives the asserted integral inequalities. Fix one $\nu \in \mathcal{P}_f$. Then

$$k_\lambda \le -\int_X v_\lambda \, \mathrm{d}\nu \le C_q A^{-1} \| a v_\lambda \|_\ell \le K$$

by (16.37), with K independent of λ. The normalization $\sup_X v_\lambda = 0$ gives $v_\lambda \le V_\theta$, and subtracting k_λ from the lower bound $\psi \le w_\lambda$ proves (16.38).

The potentials $v_\lambda = u_\lambda - \sup_X u_\lambda$ form the normalized exponential family of Proposition 15.16.1, which gives subsequences converging locally uniformly on Ω and in $L^1(X)$ to potentials of minimal type. Extracting a convergent subsequence of $k_\lambda \in [0, K]$ gives the same compactness for $w_\lambda = v_\lambda + k_\lambda$. The bound $|w_\lambda| \le |\psi| + K$ also gives L^1 convergence along every subsequence converging locally uniformly on Ω, since $X \setminus \Omega$ has zero volume. □

In particular,

$$\mathrm{e}^{\lambda \sup_X u_\lambda} = c\mathrm{e}^{\lambda k_\lambda} \longrightarrow c.$$

Moreover, for $0 < \lambda \le 1$ the densities of the equations satisfy $\|ce^{\lambda w_\lambda} f\|_p \le ce^K \|f\|_p$. It matters that this bound includes the scalar factor: an L^p bound on the data alone, after an arbitrary rescaling, does not bound the normalized potentials. For example, on $\mathbb{P}^1$, with $\omega_{\mathrm{FS}} = \mathrm{dd}^{\mathrm{c}} \log(1 + |z|^2)$ and $0 < \epsilon < 1$, the function

$$\psi_\epsilon = \log \frac{|z|^2 + \epsilon^2}{1 + |z|^2}$$

is smooth and ω_{FS}-plurisubharmonic, with supremum zero and infimum $\log \epsilon^2$. Write $\mathrm{MA}_{\omega_{\mathrm{FS}}}(\psi_\epsilon) = g_\epsilon \omega_{\mathrm{FS}}$, and take L^p norms with respect to ω_{FS} in this example. The data $g_\epsilon / \|g_\epsilon\|_p$ have norm one, and ψ_ϵ solves the equation with these data and scalar $\|g_\epsilon\|_p$; but the densities g_ϵ of the equations themselves are not bounded in L^p, and the potentials are not uniformly bounded.

16.12.4 Scalar envelopes of subsolutions

To compare a cluster potential with every member of the class (16.31), fix a point $x \in \Omega$ and add a continuous forcing term r to the subsolution inequality. Let $\mathcal{S}$ be the convex set of potentials of minimal type that are continuous on Ω, and define

$$\mathcal{F}_{\lambda,x}(r) = \sup\{s(x) : s \in \mathcal{S},\ \mathrm{MA}_\theta(s) \ge ce^{\lambda s + r} f \, \mathrm{d}V\}. \tag{16.39}$$

Lemma 16.12.2 *The functional $\mathcal{F}_{\lambda,x}$ is finite, concave, decreasing and λ^{-1}-Lipschitz for the uniform norm on $C(X)$. It satisfies*

$$\mathcal{F}_{\lambda,x}(0) = w_\lambda(x), \qquad \mathcal{F}_{\lambda,x}(r + b) = \mathcal{F}_{\lambda,x}(r) - b/\lambda.$$

There is a probability measure $\nu_{\lambda,x}$ such that, for every $r \in C(X)$ and every s admissible for r in (16.39)*,*

$$\lambda s(x) + \int_X r \, \mathrm{d}\nu_{\lambda,x} \le \lambda w_\lambda(x). \tag{16.40}$$

Proof By part (2) of Corollary 15.12.1, w_λ is the greatest admissible potential for $r = 0$. If $m = \min_X r$ and $M = \max_X r$, then $w_\lambda - M/\lambda$ is admissible for r, and translating any potential admissible for r by m/λ makes it admissible for $r = 0$. Hence

$$w_\lambda(x) - M/\lambda \le \mathcal{F}_{\lambda,x}(r) \le w_\lambda(x) - m/\lambda.$$

The mixed density inequality on Ω and the arithmetic–geometric mean inequality show that a convex combination of potentials admissible for r_0 and r_1 is admissible for the same convex combination of the forcing terms. Approximating the two suprema at x proves concavity. Inclusion of the admissible classes proves monotonicity, and translation by constants proves the translation rule and the Lipschitz bound.

Now apply the Hahn–Banach construction in the proof of Theorem 16.2.1 to $-\lambda\mathcal{F}_{\lambda,x}$ at zero. This functional is finite, continuous, convex and increasing, and it satisfies $-\lambda\mathcal{F}_{\lambda,x}(r+b) = -\lambda\mathcal{F}_{\lambda,x}(r) + b$ for constants b. The same tests with nonnegative functions and constants show that its subgradient is a probability measure $\nu_{\lambda,x}$. The resulting supporting inequality, combined with $s(x) \le \mathcal{F}_{\lambda,x}(r)$, gives (16.40); it is not necessary that the supremum be attained. □

Proof (of Theorem 16.12.1) By Proposition 16.12.2 and the moment bound following it, $\mathcal{P}_f$ has the stated properties, and Lemma 16.12.1 gives precompactness of the centered potentials. It remains to identify every cluster potential.

Step 1: limits of the forcing probabilities. Fix $x \in \Omega$. For $r \in C(X)$, choose a critical potential ψ_r for $\mathrm{e}^r f$ with supremum zero. By the definition of κ,

$$\mathrm{MA}_\theta(\psi_r) = c\mathrm{e}^{-\kappa(r)+r} f \, \mathrm{d}V.$$

Since $\psi_r \le 0$, the potential $\psi_r - \kappa(r)/\lambda$ is admissible in (16.39). Thus

$$\int_X r \, \mathrm{d}\nu_{\lambda,x} \le \kappa(r) + \lambda\big(w_\lambda(x) - \psi_r(x)\big). \tag{16.41}$$

For fixed x and r, the values $w_\lambda(x)$ are uniformly bounded by Lemma 16.12.1, and $\psi_r(x)$ is finite. Hence every weak cluster point of $\nu_{\lambda,x}$ as $\lambda \downarrow 0$ belongs to $\mathcal{P}_f$.

Step 2: a pointwise comparison. Let s satisfy (16.31). The function $-\lambda s$ is lower semicontinuous and bounded below on X, with the value $+\infty$ allowed, so metric inf-convolution and truncation give continuous functions $r_j \uparrow -\lambda s$. Since $r_j \le -\lambda s$, the potential s is admissible for every r_j, and (16.40) bounds the integrals of the r_j from above. Monotone convergence, after subtracting a common lower bound, shows that s is integrable against $\nu_{\lambda,x}$ and gives

$$w_\lambda(x) \ge s(x) - \int_X s \, \mathrm{d}\nu_{\lambda,x}. \tag{16.42}$$

Step 3: cluster potentials belong to the class (16.31). Take a sequence $\lambda_j \downarrow 0$ with $w_{\lambda_j} \to z$ locally uniformly on Ω and in $L^1(X)$. The bounds (16.38) show that z has minimal type, and z is continuous on Ω. Since $c\mathrm{e}^{\lambda_j w_{\lambda_j}} f \to c f$ in L^p, by dominated convergence and the common bound $w_\lambda \le K$, continuity of local bounded products gives $\mathrm{MA}_\theta(z) = c f \, \mathrm{d}V$ on Ω, and then on X because neither side charges $X \setminus \Omega$.

For each $\nu \in \mathcal{P}_f$, the bound $|w_\lambda| \le |\psi| + K$ has a ν-integrable right-hand side, and ν charges only Ω. Dominated convergence in $\int_X w_\lambda \, \mathrm{d}\nu \le 0$ therefore gives $\int_X z \, \mathrm{d}\nu \le 0$. Hence z belongs to the class (16.31).

Step 4: z is the greatest element of the class. For any s in the class and fixed $x \in \Omega$, pass to a further subsequence with $\nu_{\lambda_j,x} \rightharpoonup \nu$, so that $\nu \in \mathcal{P}_f$ by Step 1. The potential s is upper semicontinuous and bounded above on X, so

$$\limsup_j \int_X s \, \mathrm{d}\nu_{\lambda_j,x} \le \int_X s \, \mathrm{d}\nu \le 0.$$

For an unbounded s, this follows from the portmanteau theorem applied to $\max(s,-M)$, followed by $M\to\infty$; the final integral is finite by (16.35). Passing to the limit in (16.42) gives $z(x)\ge s(x)$. This holds for every $x\in\Omega$, and hence on X by Lemma 1.3.3. Thus every cluster potential is the greatest element w of the class, and precompactness gives convergence of the whole family.

Step 5: the normalizations. The maximum of $\int_X w\,\mathrm{d}\nu$ over $\mathcal{P}_f$ is attained, because under (16.34) it is the maximum of the weakly continuous functional $\eta\mapsto A^{-1}\int_X aw\eta\,\mathrm{d}V$ on $\mathcal{H}_a$; note that $aw\in L^\ell$. If the maximum were negative, a small positive constant translate of w would still satisfy (16.31), contrary to greatestness. Thus it is zero.

Finally, $k_\lambda\to\sup_X w$. Indeed, from any subsequence choose a further one with $k_\lambda\to k$. Then $v_\lambda=w_\lambda-k_\lambda$ converges in L^1 to $w-k$, and Theorem 1.4.1 gives $\sup_X(w-k)=0$. Thus $k=\sup_X w$ for every cluster point of (k_λ). Subtracting the constants proves the convergence of $u_\lambda-\sup_X u_\lambda$. If θ is Hermitian, we may take $\rho=0$ and $\Omega=X$, so that local uniform convergence is uniform convergence. □

Together with Proposition 15.16.1, the theorem also gives convergence of the normalized potentials in the topology of uniform convergence of differences on Ω and in every local C^β topology stated there: compactness in those topologies and uniqueness of the L^1 limit force every subsequence to have the same limit. The fact that the limit is selected does not imply that the critical equation has a unique normalized solution.

16.13 Selection for logarithmic models on products

For a logarithmic reference with a positive residual background, removing the divisor turns the singular exponential equation into a bounded equation. On a product with a curve, the selected residual can then be determined by the Green operator. This also controls simultaneous changes of the reference, background and density, for which the centering constant must follow the changing data.

Let (Y,κ) be a connected compact Hermitian manifold of dimension $k\ge 1$, with $\int_Y\kappa^k=1$, and put $X=\mathbb{P}^1\times Y$, so that $n=k+1$. Write ω_1 for the mass-one Fubini–Study form on $\mathbb{P}^1$ and $\mathrm{d}W=\omega_1\wedge\kappa^k$; pullbacks to X are implicit. Fix $a,d>0$, a smooth positive function b on $\mathbb{P}^1$, and a continuous positive function H on $\mathbb{P}^1$. Set

$$\theta=(a+d)\omega_1+b\kappa,\qquad \eta=a\omega_1+b\kappa,$$
$$f=nb^kH,\qquad m=\int_{\mathbb{P}^1}H\omega_1,\qquad c_0=\frac{a}{m}.$$

Both forms are positive, and they are not closed when b is nonconstant: the component $\mathrm{d}b\wedge\kappa$ of horizontal degree one cannot cancel the vertical component $b\,\mathrm{d}\kappa$. For homogeneous coordinates $[Z_0:Z_1]$, put

$$\ell = \frac{|Z_0|^2}{|Z_0|^2 + |Z_1|^2}, \qquad D = \{Z_0 = 0\} \times Y, \qquad \psi = d \log \ell.$$

The Poincaré–Lelong formula gives $\theta + \mathrm{dd}^{\mathrm{c}}\psi = \eta + d[D]$.

16.13.1 The ordinary model and the greatest solutions

Put $K = a + d$, $R = a/d$ and $C_* = K \log(K/d) - a \log(a/d)$. In the coordinate $z = Z_1/Z_0$ define

$$\phi_{a,d} = \begin{cases} 0, & |z|^2 \le R, \\ a \log |z|^2 + C_* - K \log(1 + |z|^2), & |z|^2 > R, \end{cases} \qquad \phi_{a,d}|_D = -\infty. \quad (16.43)$$

Proposition 16.13.1 *The function $\phi_{a,d}$ is an ordinary model reference for θ, with*

$$P_\theta[\psi] = P_\theta[\phi_{a,d}] = \phi_{a,d}, \qquad \psi \le \phi_{a,d} \le \psi + C_*, \qquad \sup_X \phi_{a,d} = 0.$$

For $d/(a+d) < \delta < 1$, the envelope $P_{\delta\theta}(\phi_{a,d})$ is not identically $-\infty$. The model has finite positive Monge–Ampère measure

$$\mathrm{MA}_\theta(\phi_{a,d}) = n b^k K \mathbf{1}_{\{|z|^2 < R\}}\, \mathrm{d}W.$$

The difference $\phi_{a,d} - \psi$ extends continuously across D. If $(a_j, d_j) \to (a, d)$ in $(0, +\infty)^2$, these residual extensions converge uniformly.

Proof First take the ordinary envelope on the curve with background $K\omega_1$ and obstacle $\min(\psi + C, 0)$. Rotation of z preserves both the obstacle and the admissible family, so the envelope is rotation invariant. With $s = \log |z|^2$ and $F(s) = K \log(1+\mathrm{e}^s)$, its profile $u(s)$ has the property that $u + F$ is convex and lies below F. The comparison with $\psi + C$ bounds the final slope of $u + F$ by a. At $s_* = \log R$, one has $F'(s_*) = a$. Above s_*, convexity and the final slope bound give

$$u(s) + F(s) \le F(s_*) + a(s - s_*),$$

and below s_* the obstacle gives $u \le 0$. The function in (16.43) attains both bounds. Its convex profile, including its logarithmic value at infinity, gives a curve potential, and direct calculation gives $\psi \le \phi_{a,d} \le \psi + C_*$. It is therefore a competitor for every $C \ge C_*$, so the curve envelope is exactly $\phi_{a,d}$.

A product competitor restricts to each curve $\mathbb{P}^1 \times \{y\}$ as either the constant $-\infty$ or a curve competitor. It is thus bounded above by $\phi_{a,d}$. Conversely, the curve model is a product competitor because the vertical form $b\kappa$ is positive. Hence the product envelope is the same function. Ordinary-type invariance of the singularity envelope gives its model property. Moreover,

$$\delta\theta + \mathrm{dd}^{\mathrm{c}}\psi = (\delta K - d)\omega_1 + \delta b \kappa + d[D] \ge 0,$$

which proves the smaller-background assertion.

Off D, the horizontal curvature of $\phi_{a,d}$ is $K\mathbf{1}_{\{|z|^2<R\}}\omega_1$. The first derivatives of its convex profile agree at R, so there is no boundary-circle atom. Expanding the product off D gives the displayed measure, and the non-pluripolar product assigns D zero mass. Finally, in terms of ℓ the residual extension is

$$\phi_{a,d}-\psi=\begin{cases}-d\log\ell, & \ell\geq d/K,\\ a\log(1-\ell)+C_*, & \ell<d/K.\end{cases}$$

The formulas agree at the moving interface and extend to both endpoints. On compact positive parameter ranges each logarithm is used away from its singular endpoint. Joint continuity on the resulting compact set proves uniform variation. □

Lemma 16.13.1 *For every $\sigma>0$, the equation*

$$\mathrm{MA}_\theta(s_\sigma)=\mathrm{e}^{\sigma s_\sigma}f\,\mathrm{d}W \tag{16.44}$$

has a solution of ordinary type $\phi_{a,d}$, unique in that type. It is the greatest exponential subsolution among all $u\in\mathrm{PSH}(X,\theta)$ with $u\preceq\phi_{a,d}$, and its measure has finite mass.

Proof The one-sided ordinary class has the description

$$u\preceq\psi\quad\Longleftrightarrow\quad u=\psi+r,\quad r\in\mathrm{PSH}(X,\eta),\qquad \mathrm{MA}_\theta^{\mathrm{ext}}(\psi+r)=\mathrm{MA}_\eta^{\mathrm{ext}}(r). \tag{16.45}$$

This is Lemma 16.9.1 for the pullback of the hyperplane bundle of $\mathbb{P}^1$ with its Fubini–Study metric, the section Z_0 and the weight d; the residual form is η, and $\sup_X\psi=0$. Bounded r correspond to the ordinary type of ψ.

The function $g_\sigma=\mathrm{e}^{\sigma\psi}f$ extends continuously by zero on D and is positive elsewhere. Since $\eta>0$, Theorem 11.1.2 gives a continuous solution R_σ of

$$\mathrm{MA}_\eta(R_\sigma)=\mathrm{e}^{\sigma R_\sigma}g_\sigma\,\mathrm{d}W.$$

For this bounded density, the greatest solution of Theorem 15.34.1 is bounded and coincides with R_σ by Corollary 11.2.2. Thus it dominates every η-plurisubharmonic exponential subsolution. The correspondence (16.45) transfers this order to the entire one-sided ordinary class and gives $s_\sigma=\psi+R_\sigma$. Bounded residual uniqueness gives uniqueness in the ordinary type. The bounded residual equation has finite mass, so the singular equation does too. □

16.13.2 The selected limit with varying data

Let $G(x,z) = \log d_{\mathrm{ch}}(x,z)^2 + 1$ be the kernel on $\mathbb{P}^1$ defined by the chordal distance d_{ch} before (15.86), where the mass-one Fubini–Study form ω_1 was written κ. By Lemma 15.30.1, applied to the Dirac mass at z, $\mathrm{dd}^{\mathrm{c}}_x G = \delta_z - \omega_1$ and $\int G(x,z)\omega_1(x) = 0$. For $z = [0:1]$ one has $G(x,z) = \log \ell(x) + 1$. For a bounded density q, write $G * q(x) = \int G(x,z)q(z)\omega_1(z)$. Define

$$\begin{aligned} A_0 &= \frac{aH}{m}, \qquad w_0 = G * (A_0 - a), \qquad S_0 = \sup_X(\psi + w_0), \\ \mathrm{d}\nu_0 &= \frac{H}{m}\,\mathrm{d}W, \qquad J = \int_X (\psi + w_0)\,\mathrm{d}\nu_0. \end{aligned} \tag{16.46}$$

The function w_0 is continuous, has mean zero on the curve and satisfies $\mathrm{dd}^{\mathrm{c}} w_0 = (A_0 - a)\omega_1$. We use the probability ν_0 to normalize the separated potentials.

Theorem 16.13.1 *Let $a_j, d_j > 0$ converge to $a, d > 0$, let smooth positive functions b_j converge uniformly to b, and let continuous positive functions H_j converge uniformly to H. For any $\sigma_j > 0$ tending to zero, put*

$$\begin{aligned} &\theta_j = (a_j + d_j)\omega_1 + b_j\kappa, \quad \psi_j = d_j \log \ell, \qquad \phi_j = \phi_{a_j,d_j}, \\ &f_j = n b_j^k H_j, \qquad\qquad m_j = \int_{\mathbb{P}^1} H_j\omega_1, \quad c_{0,j} = a_j/m_j. \end{aligned}$$

Let s_j be the greatest solution of (16.44) for these data, and set $v_j = s_j - \sup_X s_j$, $c_j = \mathrm{e}^{\sigma_j \sup_X s_j}$. Then

$$v_j \longrightarrow v_0 = \psi + w_0 - S_0, \qquad s_j - \sigma_j^{-1}\log c_{0,j} \longrightarrow s_0 = \psi + w_0 - J$$

in $L^1(X)$, in capacity, and locally uniformly on $X \setminus D$. Moreover,

$$c_j \longrightarrow c_0, \qquad \sigma_j^{-1}\log(c_j/c_{0,j}) \longrightarrow S_0 - J, \tag{16.47}$$

and $\mathrm{MA}_{\theta_j}(v_j) \to c_0 f\,\mathrm{d}W$ in total variation. The limit has ordinary type $\phi_{a,d}$ and belongs to $\mathcal{E}_{\mathrm{env}}(X,\theta,\phi_{a,d})$, with

$$\mathrm{MA}_\theta(v_0) = c_0 f\,\mathrm{d}W.$$

There is no restriction on the relative rates of parameter and exponent convergence.

Proof *Uniform construction.* The kernel has uniformly integrable logarithmic tails and $C_G = \sup_x \int |G(x,z)|\omega_1(z) < \infty$. These facts follow by rotation invariance and the radial integral of $|\log \ell + 1|$ on $0 < \ell < 1$. Truncating the kernel shows that convolution of bounded densities is continuous.

Temporarily omit subscripts. On continuous mean-zero functions on the curve define

$$T_\sigma(w) = G * (A(w) - a), \qquad A(w) = \frac{aH\mathrm{e}^{\sigma(\psi+w)}}{Z(w)}, \qquad Z(w) = \int_{\mathbb{P}^1} H\mathrm{e}^{\sigma(\psi+w)}\omega_1.$$

For all late parameters choose fixed bounds $a \le a_{\max}$, $0 < d_{\min} \le d \le d_{\max}$ and $0 < h_{\min} \le H \le h_{\max}$. Put $B = 4a_{\max}h_{\max}/h_{\min}$ and choose $R_0 > C_G(B + a_{\max})$. On the ball $\|w\|_\infty \le R_0$, for sufficiently small σ,

$$Z(w) \ge \frac{h_{\min}\mathrm{e}^{-\sigma R_0}}{1 + \sigma d_{\max}} \ge h_{\min}/2, \qquad 0 \le A(w) \le B.$$

Here $\int \ell^{\sigma d}\omega_1 = (1 + \sigma d)^{-1}$. Thus T_σ maps the mean-zero ball into itself. Since $\int A(w)\omega_1 = a$, the derivative of $A(w)$ in the direction h is

$$\sigma A(w)\left(h - a^{-1}\int hA(w)\omega_1\right).$$

It follows that $\|T_\sigma(w) - T_\sigma(v)\|_\infty \le 2\sigma BC_G\|w - v\|_\infty$. The map is a contraction for all sufficiently small exponents, uniformly over the late parameters. Its fixed point w_j has mean zero and $\|w_j\|_\infty \le R_0$.

Write $Z_j = Z(w_j)$ and $A_j = A(w_j)$. The fixed-point equation gives $\mathrm{dd}^{\mathrm{c}}w_j = (A_j - a_j)\omega_1$, and the potential

$$\psi_j + w_j + \sigma_j^{-1}\log(a_j/Z_j) \tag{16.48}$$

has residual form $A_j\omega_1 + b_j\kappa$, continuous and semipositive. On coordinate balls, mollification of w_j converges together with its continuous complex Hessian. The local bounded product, expanded after a common strict smooth correction, therefore equals $nb_j^k A_j\,\mathrm{d}W$. Removing D identifies the singular product with the same measure. The identity

$$nb_j^k A_j = \mathrm{e}^{\sigma_j(\psi_j+w_j)}(a_j/Z_j)f_j$$

shows that (16.48) solves the exponential equation in the ordinary type. By Lemma 16.13.1, it is s_j.

Residual and normalized limits. Uniform boundedness and dominated convergence give $Z_j \to m$ and $A_j \to A_0$ in $L^1(\omega_1)$, with a common bound on A_j. For uniformly bounded densities $q_j \to 0$ in L^1, one has

$$\|G * q_j\|_\infty \le \|q_j\|_\infty \sup_x \int_{\{|G(x,z)|>M\}} |G(x,z)|\omega_1(z) + M\|q_j\|_1.$$

Applied to $q_j = A_j - a_j - A_0 + a$, first letting $j \to \infty$ and then $M \to \infty$ proves $w_j \to w_0$ uniformly. Although $A_j = 0$ on D and $A_0 > 0$ there, the L^1 convergence suffices for this argument.

Put $S_j = \sup_X(\psi_j + w_j)$. A point with $\ell = 1$ supplies a uniformly bounded competing value. Since $d_j \ge d_{\min} > 0$, the upper bound $d_{\min}\log\ell + R_0$ tends uniformly to $-\infty$ near D, so all maximizing points stay in a fixed compact subset of $X \setminus D$. Uniform convergence there gives $S_j \to S_0$. Hence $v_j = \psi_j + w_j - S_j$

has the asserted L^1 and local uniform limits. For capacity convergence, note that the complements of the sets $\{\psi < -T\}$ are compact subsets of $X \setminus D$, while (1.15), applied to $\psi \in \mathrm{PSH}(X,\theta)$ with $\sup_X \psi = 0$, bounds the capacity of these sets by C/T. Local uniform convergence, followed by $T \to \infty$, proves convergence in capacity, with the common poles omitted.

The measure formula and uniform convergence of b_j give total-variation convergence to $nb^k A_0\,\mathrm{d}W = c_0 f\,\mathrm{d}W$. The same bounded product calculation gives this equation for v_0. The residual extensions in Proposition 16.13.1 show directly that $v_0 \sim \phi_{a,d}$. Hence v_0 belongs to the relative envelope class $\mathcal{E}_{\mathrm{env}}(X,\theta,\phi_{a,d})$ of Definition 14.4.1, which contains the whole ordinary type of $\phi_{a,d}$ by Proposition 14.4.1.

Scalar centering. The exact formula is $c_j = a_j \mathrm{e}^{\sigma_j S_j}/Z_j$. In the difference quotient

$$\frac{Z_j - m_j}{\sigma_j} = \int H_j \frac{\mathrm{e}^{\sigma_j(\psi_j + w_j)} - 1}{\sigma_j}\omega_1,$$

the negative part is bounded by $h_{\max}(d_{\max}|\log \ell| + R_0)$ and the positive part by a constant. Dominated convergence gives a limit $\int H(\psi + w_0)\omega_1 = mJ$. Since $m_j \to m > 0$,

$$\sigma_j^{-1}\log(Z_j/m_j) \longrightarrow J.$$

The identity $c_j/c_{0,j} = m_j \mathrm{e}^{\sigma_j S_j}/Z_j$ proves (16.47). Subtracting $\sigma_j^{-1}\log c_{0,j}$ in (16.48) leaves $\psi_j + w_j - \sigma_j^{-1}\log(Z_j/m_j)$, proving the centered limit in all three topologies. □

16.13.3 The critical coefficient and separated order

Corollary 16.13.1 *For the fixed data, c_0 is the largest coefficient $c > 0$ for which some $u \in \mathrm{PSH}(X,\theta)$ with $u \preceq \phi_{a,d}$ satisfies $\mathrm{MA}_\theta^{\mathrm{ext}}(u) \geq c\, f\, \mathrm{d}W$. Among separated potentials $u = \psi + r(x) \preceq \phi_{a,d}$, the critical subsolutions, those with $\mathrm{MA}_\theta^{\mathrm{ext}}(u) \geq c_0 f\, \mathrm{d}W$, are exactly the translates of $\psi + w_0$. Thus v_0 is their unique supremum-zero member, and s_0 is their greatest member with $\int_X u\, \mathrm{d}\nu_0 \leq 0$.*

Proof The first assertion is the case of Theorem 16.9.3 with residual form η, since $\phi_{a,d} \sim \psi$: the largest coefficient on the one-sided class is attained and equals $b_\eta(f)$, which is $c_\eta(f)$ by Proposition 15.27.2. Finally, $c_\eta(f) = c_0$, because w_0 is a bounded solution of $\mathrm{MA}_\eta(w_0) = c_0 f\, \mathrm{d}W$.

For a separated critical subsolution, (16.45) and the product expansion give $\mathrm{MA}^{\mathrm{ext}}_{a\omega_1}(r) \geq A_0\omega_1$. Closed curve truncations have mass a, so the non-pluripolar measure has mass at most a. Since $\int A_0\omega_1 = a$, it must equal $A_0\omega_1$. The positive closed current $a\omega_1 + \mathrm{dd}^{\mathrm{c}} r$ also has mass a, so it has no remaining measure on the pole set. Consequently $\mathrm{dd}^{\mathrm{c}} r = (A_0 - a)\omega_1$ as distributions, and $r = w_0 + C$ because harmonic distributions on $\mathbb{P}^1$ are constant. Supremum zero gives $C = -S_0$, while the mean condition gives $C \leq -J$. □

The scalar in the centering follows the varying data. Replacing $c_{0,j}$ by c_0 adds $\sigma_j^{-1} \log(c_{0,j}/c_0)$ to the centered potential, and this term need not vanish. For example, take $a = 1$, $d = 1/4$, $H = 1$ and $b = 2 + t$, where t is the sphere height. Then $w_0 = 0$, $S_0 = 0$, $J = -1/4$ and $c_0 = 1$, so $s_0 = \psi + 1/4$ and the scalar-error limit is $1/4$. If $a_j = 1 + \sqrt{\sigma_j}$ and the other data are fixed, the appropriate centering uses $c_{0,j} = a_j$. Centering by 1 instead adds

$$\frac{\log(1 + \sqrt{\sigma_j})}{\sigma_j} \longrightarrow +\infty.$$

These examples can be taken on a non-Kähler manifold by choosing Y to be the Hopf surface with the unit-volume form κ of Example 16.2.1. The product cannot be Kähler, since a Kähler form would restrict to one on each slice $\{x\} \times Y$. The pole divisor remains present in the selected ordinary type.

16.13.4 Weak adjoints on a product surface

On the product of two projective lines, the probability ν_0 of (16.46) is the unique supporting probability of the limiting equation of Theorem 16.13.1. The continuous positive residual defines a weak linearization, and spherical harmonics show that this linearization has only one stationary probability, without any regularity assumption on that probability.

In this subsection take $Y = \mathbb{P}^1_y$, $k = 1$ and $\kappa = \omega_2$, the mass-one Fubini–Study form of the second factor. Keep $a, d, b, H, \theta, \eta, \psi$ and $\mathrm{d}W = \omega_1 \wedge \omega_2$ as above, with b nonconstant, and write $\phi = \phi_{a,d}$, $A = A_0 = c_0 H$ and $u = \psi + w_0$. Choose a smooth positive probability volume form $\mathrm{d}V = q\,\mathrm{d}W$ and put $F = \log(2bH/q)$. Then

$$\Omega = A\omega_1 + b\omega_2, \qquad \theta + \mathrm{dd^c}u = \Omega + d[D], \qquad \mathrm{MA}_\theta(u) = \Omega^2 = c_0 \mathrm{e}^F \mathrm{d}V.$$

The model is admissible by Proposition 16.13.1, and $u \sim \phi$ gives $B_\phi(F) = \log c_0$. Nonconstant b cannot be harmonic, so $\mathrm{dd^c}\theta = (\mathrm{dd^c}b) \wedge \omega_2 \neq 0$. Every smooth potential perturbation $\theta + \mathrm{dd^c}g$ has the same $\mathrm{dd^c}\theta$ and is therefore non-Gauduchon.

For a smooth function h on the product, let $\mathrm{dd}^{\mathrm{c}}_i h$ be the component of $\mathrm{dd^c}h$ along the ith factor, and define the partial Laplacians by $\mathrm{dd}^{\mathrm{c}}_i h = (\Delta_i h)\omega_i$. With the mass-one Fubini–Study normalization, Δ_i is the Laplacian of the unit round sphere.

Lemma 16.13.2 *Let $V > 0$ be continuous on $\mathbb{P}^1$, and let T be a finite signed measure such that*

$$\int_{\mathbb{P}^1} (\Delta f - Vf)\,\mathrm{d}T = 0 \qquad (f \in C^\infty(\mathbb{P}^1)).$$

Then $T = 0$.

Proof Let $\mathrm{e}^{t\Delta}$ be the heat operator of the sphere. On spherical harmonics of degree ℓ, it acts by $\mathrm{e}^{-t\ell(\ell+1)}$. The harmonic series defines a smooth heat solution for $t > 0$; for smooth initial functions, the maximum principle gives positivity and contraction in the uniform norm. Density of smooth functions extends $\mathrm{e}^{t\Delta}$ to a positive, strongly continuous contraction on $C(\mathbb{P}^1)$. Choose $M \geq \max V$ and define

$$R_M g = \int_0^\infty \mathrm{e}^{-Mt}\mathrm{e}^{t\Delta} g\,\mathrm{d}t.$$

Then $\|R_M\| \leq M^{-1}$ and $(M - \Delta)R_M g = g$ in distributions. For any $g \in C(\mathbb{P}^1)$, the map

$$f \longmapsto R_M((M - V)f - g)$$

is a contraction of $C(\mathbb{P}^1)$ with constant at most $1 - \min V/M < 1$. Its continuous fixed point satisfies $\Delta f = Vf + g$ in distributions, with a continuous right-hand side. The functions $f_\varepsilon = \mathrm{e}^{\varepsilon\Delta} f$ are smooth and satisfy

$$f_\varepsilon \longrightarrow f, \qquad \Delta f_\varepsilon = \mathrm{e}^{\varepsilon\Delta}(Vf + g) \longrightarrow Vf + g$$

uniformly. Thus $(\Delta - V)f_\varepsilon \to g$ uniformly. Testing against the finite measure T gives $\int g\,\mathrm{d}T = 0$ for every continuous g, hence $T = 0$. □

Theorem 16.13.2 *The function ρ and the current Q given by*

$$\rho = \frac{1}{2ab}, \qquad Q = \rho\Omega = \frac{A}{2ab}\omega_1 + \frac{1}{2a}\omega_2$$

satisfy $\mathrm{d}Q = 0$ and $\int_X \rho\Omega^2 = 1$. For smooth real h, the measure-valued and logarithmic linearizations at u are

$$D_u\mathrm{MA}_\theta(h) = 2\Omega \wedge \mathrm{dd^c}h, \qquad L_u h = \frac{2\Omega \wedge \mathrm{dd^c}h}{\Omega^2} = \frac{\Delta_1 h}{A} + \frac{\Delta_2 h}{b}.$$

Call a Borel probability μ stationary *for L_u if $\int_X L_u h\,\mathrm{d}\mu = 0$ for every smooth real h. The only stationary probability is*

$$\nu = \rho\Omega^2 = \frac{A}{a}\,\mathrm{d}W = \nu_0, \qquad \mathcal{A}_\phi(F) = \{\nu\}. \tag{16.49}$$

If $r \geq 0$ is measurable, $\int_X r\Omega^2 = 1$ and $\mathrm{dd^c}(r\Omega) = 0$, then $r = \rho$ almost everywhere.

Proof *The current and operator.* The coefficient ρ is smooth and Ω is continuous, so the product defining Q is well defined. Since A/b depends only on the first curve, $\mathrm{d}((A/b)\omega_1) = 0$ distributionally; this also follows by uniform smooth approximation of its coefficient. Thus Q is closed. Moreover, $\rho\Omega^2 = (A/a)\,\mathrm{d}W$ has mass $\int A\omega_1/a = 1$, and it equals ν_0 because $A = aH/m$. Uniform positivity of Ω gives $\Omega + t\mathrm{dd^c}h > 0$ for sufficiently small signed t. The bounded-product calculation used in Theorem 16.13.1, now with the smooth perturbation th, gives

$$\mathrm{MA}_\theta(u + th) = \Omega^2 + 2t\Omega \wedge \mathrm{dd^c}h + t^2(\mathrm{dd^c}h)^2.$$

This proves the two linearizations; the mixed components of $\mathrm{dd^c}h$ have zero wedge product with the diagonal form Ω. Distributional Stokes gives $\int L_u h\,\mathrm{d}\nu = 2\int \mathrm{dd^c}h \wedge Q = 0$, so ν is stationary.

Stationary probabilities. Choose real spherical harmonics e_j on the second sphere, with $e_0 = 1$ and $\Delta_2 e_j = -\lambda_j e_j$. Only the constant mode has eigenvalue zero. For a stationary probability μ, define finite signed measures T_j on the first sphere by

$$\int f\,\mathrm{d}T_j = \int_X \frac{f(x)e_j(y)}{A(x)}\,\mathrm{d}\mu.$$

Testing L_u on $f(x)e_j(y)$ gives

$$\int \left(\Delta_1 f - \lambda_j \frac{A}{b} f\right)\mathrm{d}T_j = 0.$$

For $j = 0$, testing on spherical harmonics f of positive degree on the first sphere shows that T_0 annihilates them, so $T_0 = C\omega_1$ with $C = T_0(1)$. Indeed, the finite span of the spherical harmonics is uniformly dense, because restrictions of polynomials decompose into harmonics and polynomials are dense on the sphere. For every nonconstant mode, Lemma 16.13.2 with $V = \lambda_j A/b > 0$ gives $T_j = 0$. Since the T_j are measures, their defining identities hold for every continuous f, and replacing f by fA gives

$$\int_X f(x)e_j(y)\,\mathrm{d}\mu = \int fA\,\mathrm{d}T_j = \begin{cases} C\int fA\omega_1, & j = 0, \\ 0, & j \neq 0. \end{cases}$$

For $j = 0$ and $f = 1$, probability normalization gives $C = 1/a$. The probability $(A/a)\omega_1 \wedge \omega_2$ has the same integrals, and finite sums of the products $f(x)e_j(y)$ are uniformly dense in $C(X)$. Hence $\mu = (A/a)\omega_1 \wedge \omega_2$.

Supporting probabilities. Let $\mu \in \mathcal{A}_\phi(F)$ and, for smooth real h, set

$$F_t = F + \log \frac{(\Omega + t\mathrm{dd}^{\mathrm{c}}h)^2}{\Omega^2}.$$

This is continuous, and $u + th$ solves the ordinary-type equation with density e^{F_t} and scalar c_0. Thus $B_\phi(F_t) = B_\phi(F)$, by uniqueness of the scalar. Uniform positivity gives $F_t - F = tL_u h + O(t^2)$ in the uniform norm. Apply (16.3) for both signs of t to obtain $\int L_u h\,\mathrm{d}\mu = 0$. Uniqueness of the stationary probability and nonemptiness of $\mathcal{A}_\phi(F)$ from Theorem 16.2.1 now prove (16.49).

Weights. Finally, $\Omega^2/\mathrm{d}W$ is bounded above and below by positive constants, so a normalized r as in the statement belongs to $L^1(\mathrm{d}W)$ and $r\Omega$ is an order-zero current. Distributional Stokes makes the probability $r\Omega^2$ stationary. The probability uniqueness just proved gives $r\Omega^2 = \rho\Omega^2$, hence $r = \rho$ almost everywhere. □

Proposition 16.13.2 *Let $w_j \in C^\infty(X)$ be real and suppose*

$$\Omega_j = \eta + \mathrm{dd}^{\mathrm{c}}w_j > 0, \qquad \Omega_j \longrightarrow \Omega \quad \textit{uniformly as } (1,1)\textit{-forms}.$$

Put $u_j = \psi + w_j$ and $F_j = \log(\Omega_j^2/(c_0\mathrm{d}V))$. Let ρ_j be the Gauduchon multiplier of Ω_j, normalized by

$$\mathrm{dd}^{\mathrm{c}}(\rho_j\Omega_j) = 0, \qquad \int_X \rho_j\Omega_j^2 = 1,$$

so that $\rho_j\Omega_j$ and $\rho_j\Omega_j^2$ are the adjoint cofactor and the adjoint probability of Ω_j in the sense of Section 16.2.7. Then $F_j \to F$ uniformly, and the entire sequence satisfies

$$\rho_j\Omega_j^2 \rightharpoonup \nu, \qquad \rho_j\Omega_j \rightharpoonup Q, \qquad \rho_j\,\mathrm{d}W \rightharpoonup \rho\,\mathrm{d}W.$$

Such approximations exist with $w_j \to w_0$ uniformly and can depend on both curve variables.

Proof For each j, Theorem 4.1.1, applied to the smooth positive form Ω_j on the surface, gives a smooth positive ρ_j with $\mathrm{dd}^{\mathrm{c}}(\rho_j\Omega_j) = 0$, unique after normalization. Put $\nu_j = \rho_j\Omega_j^2$ and $L_j h = 2\Omega_j \wedge \mathrm{dd}^{\mathrm{c}} h/\Omega_j^2$. Distributional Stokes gives $\int L_j h \, \mathrm{d}\nu_j = 0$. Uniform convergence to the positive form Ω supplies a common determinant lower bound and gives $L_j h \to L_u h$ uniformly for every fixed smooth h. Every weak cluster point of the probabilities ν_j is therefore stationary for L_u. By Theorem 16.13.2, it equals ν, proving convergence of the whole sequence. The same determinant convergence gives $F_j \to F$ uniformly.

For a smooth real $(1,1)$ test form ξ, the ratios $\Omega_j \wedge \xi/\Omega_j^2$ converge uniformly to $\Omega \wedge \xi/\Omega^2$. Consequently,

$$\int_X \rho_j \Omega_j \wedge \xi = \int_X \frac{\Omega_j \wedge \xi}{\Omega_j^2} \, \mathrm{d}\nu_j \longrightarrow \int_X \rho\Omega \wedge \xi.$$

Similarly, the positive densities $p_j = \Omega_j^2/\mathrm{d}W$ converge uniformly to $p = \Omega^2/\mathrm{d}W$. For every $g \in C(X)$,

$$\int_X g\rho_j \, \mathrm{d}W = \int_X (g/p_j) \, \mathrm{d}\nu_j \longrightarrow \int_X (g/p) \, \mathrm{d}\nu = \int_X g\rho \, \mathrm{d}W.$$

For existence, heat-smooth A on the first sphere to positive smooth functions A_j of the same mean a, converging uniformly to A. Green inversion gives smooth mean-zero $w_j(x)$ with $\mathrm{dd}^{\mathrm{c}} w_j = (A_j - a)\omega_1$ and $w_j \to w_0$ uniformly. The forms $\eta + \mathrm{dd}^{\mathrm{c}} w_j = A_j\omega_1 + b\omega_2$ converge uniformly to Ω. Adding $\varepsilon_j h$ to w_j, for a fixed smooth function $h(x,y)$ depending on both variables and sufficiently small $\varepsilon_j \to 0$, preserves positivity and gives examples depending on both variables. Each u_j keeps the fixed logarithmic divisor, and its non-pluripolar equation has scalar c_0 and logarithmic density F_j. □

Example 16.13.1 Take $a = 1$, $d = 1/4$, $q = 1$, and let t be the sphere height

$$t = \frac{|Z_0|^2 - |Z_1|^2}{|Z_0|^2 + |Z_1|^2}, \qquad b = 2 + t, \qquad H = A = \tfrac{1}{2} + |t|.$$

Here $c_0 = 1$ and $F = \log(2bA)$ is Lipschitz but not C^1. The residual has continuous positive coefficients with the same loss of differentiability. Its supporting probability differs from normalized Monge–Ampère volume:

$$\int_X \Omega^2 = 4, \qquad \int_X t \, \mathrm{d}\nu = 0, \qquad \int_X t \, \frac{\Omega^2}{4} = \frac{5}{24}.$$

Even for smooth constant H, approximation of the pole by globally smooth potentials need not recover the weak adjoint. For $H = a$, so that $w_0 = 0$ and $\nu = \mathrm{d}W$, put

$$\psi_\delta = d \log(\ell + \delta(1 - \ell)), \qquad P_\delta = \omega_1 + \mathrm{dd}^{\mathrm{c}}(\psi_\delta/d), \qquad \delta > 0.$$

As $\delta \downarrow 0$, the smooth potentials ψ_δ decrease pointwise to ψ and converge in L^1 and smoothly off D, but the adjoint probabilities of the smooth positive forms $\theta + \mathrm{dd}^{\mathrm{c}}\psi_\delta$ converge to

$$\frac{a\omega_1 + d\delta_p}{a+d} \wedge \omega_2, \qquad p = \{Z_0 = 0\}.$$

This limit assigns mass $d/(a+d) > 0$ to D, and the limit of the adjoint cofactors also contains a nonzero multiple of $[D]$.

Proof Under ω_1, the height t is uniformly distributed on $[-1, 1]$, so $\int A\omega_1 = 1$ and $\int tA\omega_1 = 0$. Since $\Omega^2 = 2bA\,\mathrm{d}W$, its mass is four and its normalized height moment is

$$\frac{1}{2}\int_{-1}^{1} t^2\left(\tfrac{1}{2} + |t|\right)\frac{\mathrm{d}t}{2} = \frac{5}{24}.$$

The function of Theorem 16.13.2 is $\rho = 1/(2b)$, and $\nu = A\,\mathrm{d}W$.

For the smoothing family, direct differentiation gives

$$\frac{P_\delta}{\omega_1} = \frac{4\delta}{((1+\delta) + (1-\delta)t)^2}.$$

It is positive of mass one and converges weakly to δ_p as $\delta \to 0$. Thus $\widehat{\Omega}_\delta = \theta + \mathrm{dd}^{\mathrm{c}}\psi_\delta = a\omega_1 + dP_\delta + b\omega_2$ is a smooth positive form. Its Gauduchon multiplier and adjoint probability are

$$\widehat{\rho}_\delta = \frac{1}{2(a+d)b}, \qquad \widehat{\nu}_\delta = \frac{a\omega_1 + dP_\delta}{a+d} \wedge \omega_2.$$

The cofactors $\widehat{\rho}_\delta\widehat{\Omega}_\delta$ are closed and $\widehat{\nu}_\delta$ has mass one, so these are the normalized multiplier and adjoint probability. Their limits follow from the weak convergence of P_δ to δ_p. In particular, the limit of the cofactors contains $d[D]/(2(a+d)b(p))$. These smoothings fill the pole and do not satisfy the fixed-divisor and uniform-residual hypotheses of Proposition 16.13.2. □

Chapter 17
Multiplier-ideal envelopes and $\mathcal{I}$-goodness

Throughout this chapter, X is a connected compact complex manifold of dimension n and θ is a smooth real $(1,1)$-form, which need not be closed. We fix a smooth positive probability volume form $\mathrm{d}V$ on X and take L^1 norms with respect to it. For $u \in \mathrm{PSH}(X,\theta)$, $\mathrm{MA}_\theta(u)$ denotes the extended non-pluripolar Monge–Ampère measure $\mathrm{MA}_\theta^{\mathrm{ext}}(u)$ of Theorem 7.1.1; its total mass may be infinite. We write $u \preceq v$ and $u \sim v$ for the ordinary singularity order and equivalence of Definition 1.12.1, and $u \preceq_{\mathcal{I}} v$ and $u \sim_{\mathcal{I}} v$ for inclusion and equality of the multiplier ideals $\mathcal{I}(ku)$ and $\mathcal{I}(kv)$ at every weight $k > 0$, as in Section 1.12. Finally, $P_\theta[u]$ and $P_{\theta,F}[u]$ denote the ordinary singularity envelopes of Definition 1.19.1, below zero and below a smooth obstacle F, and $V_\theta = P_\theta(0)$ is the potential with minimal singularities of Section 1.19.

Multiplier ideals record weighted holomorphic integrability. Two potentials can have the same multiplier ideals at every weight although their ordinary envelopes differ, so passing from the ordinary singularity type of a potential to its multiplier-ideal class loses information. Analytic approximation makes this loss visible: along dyadic Demailly approximations, the envelopes converge to the greatest nonpositive representative of the multiplier-ideal class, which may exceed the ordinary envelope of the original potential.

In the Kähler theory, equality of these two envelopes is also characterized by a mass condition. In a nonclosed background the equality remains meaningful even when the potential has infinite mass, and we therefore study both envelopes along analytic approximations. Examples on curves show that this still does not give continuity of mass or completeness, because diffuse polar curvature can disappear in the limit. The same examples explain why the obstacle has to be taken into account when the background changes. The final section adds to a divisorial potential a residual whose ordinary type has representatives of arbitrarily small negative curvature; multiplier ideals cannot detect such a residual. The ordinary type of the sum contains a strict current exactly when θ minus the smooth curvature form of the divisor has a smooth strictly positive representative (Theorem 17.7.1). Zero-volume models on products and fibrations, and polar currents of models on modifications, are treated in Chapter 32.

17.1 The Kähler characterization

First suppose that X is Kähler, that θ is a smooth closed big real $(1,1)$-form, and that $u \in \mathrm{PSH}(X,\theta)$ has positive non-pluripolar mass. Let $P_\theta[u]_{\mathcal{I}}$ be the multiplier-ideal envelope of Definition 17.2.1 below, and let d_S be the singularity pseudometric of [Xia26, Section 6.2]. The characterization of $\mathcal{I}$-good singularities in [Xia26, Theorem 7.1.1] states that the following three conditions are equivalent:

$$u \text{ lies in the } d_S\text{-closure of analytic singularities,}$$
$$\int_X \mathrm{MA}_\theta(u) = \int_X \mathrm{MA}_\theta(P_\theta[u]_{\mathcal{I}}),$$
$$P_\theta[u] = P_\theta[u]_{\mathcal{I}}.$$

The last condition involves only envelopes. It therefore offers a starting point for the Hermitian problem even where the metric and the mass characterizations are unavailable.

The integral on the right of the middle condition is the *multiplier-ideal volume* of [Xia26]. The upper and lower volumes of Definition 14.3.1, by contrast, vary the potential within its ordinary singularity type. The distinction matters in a nonclosed background, where even a bounded change of the potential can alter its total mass.

17.2 Ideal projection

We return to the general setting fixed at the beginning of the chapter. The multiplier-ideal envelope is the greatest nonpositive potential in a multiplier-ideal class. Taking a supremum over the class could a priori enlarge the multiplier ideals; the increasing-limit form of strong openness rules this out, and the envelope is then a projection onto the class.

Definition 17.2.1 For $u \in \mathrm{PSH}(X,\theta)$ define

$$P_\theta[u]_{\mathcal{I}} := \left(\sup\{v \in \mathrm{PSH}(X,\theta) : v \le 0,\ v \sim_{\mathcal{I}} u\}\right)^*.$$

As for ordinary envelopes, the members of the defining family are called *competitors*. The family is nonempty because it contains $u - \sup_X u$; bounded perturbations do not change multiplier ideals. Since it is bounded above by zero, its regularized supremum is a θ-plurisubharmonic function. The envelope depends only on the multiplier-ideal class of u.

Theorem 17.2.1 *For every $u \in \mathrm{PSH}(X,\theta)$,*

$$P_\theta[u]_{\mathcal{I}} \sim_{\mathcal{I}} u, \qquad P_\theta[P_\theta[u]_{\mathcal{I}}]_{\mathcal{I}} = P_\theta[u]_{\mathcal{I}}.$$

The regularization in Definition 17.2.1 does not change the pointwise supremum. Moreover,

$$\sup_X P_\theta[u]_{\mathcal{I}} = 0.$$

Proof Write $q = P_\theta[u]_{\mathcal{I}}$. By Choquet's lemma, some sequence from the defining family has regularized supremum q. Replacing its terms by successive finite maxima, we obtain by Lemma 1.16.3 an increasing sequence v_j that remains in the family and converges to q almost everywhere.

Fix $x \in X$ and $\lambda > 0$. On a coordinate ball about x, choose one smooth function r such that every $v_j + r$ and $q + r$ is plurisubharmonic; this is possible because all these functions have complex Hessian bounded below by the same smooth form $-\theta$. The increasing-limit assertion of Theorem 1.16.1 gives

$$\mathcal{I}(\lambda(q+r))_x = \mathcal{I}(\lambda(v_j+r))_x = \mathcal{I}(\lambda u)_x$$

for all large j; the second equality holds because smooth additions do not change multiplier ideals. Hence $q \sim_{\mathcal{I}} u$.

Since $q \le 0$, it belongs to its own defining family. The pointwise supremum is therefore at least q, and it is at most its regularization q; this proves the assertion about regularization. The defining families for $P_\theta[q]_{\mathcal{I}}$ and $P_\theta[u]_{\mathcal{I}}$ coincide, which proves idempotence. Finally, if $\sup_X q < 0$, then $q - \sup_X q$ would be a larger competitor, which is impossible. □

Proposition 17.2.1 *Let $\pi : Y \to X$ be a modification of connected compact complex manifolds. For every smooth real $(1,1)$-form θ on X and every $u \in \mathrm{PSH}(X,\theta)$,*

$$P_{\pi^*\theta}[u \circ \pi]_{\mathcal{I}} = P_\theta[u]_{\mathcal{I}} \circ \pi.$$

Proof Pullback gives a bijection from $\mathrm{PSH}(X,\theta)$ onto $\mathrm{PSH}(Y,\pi^*\theta)$, by Proposition 1.22.4. Since π is surjective, pullback preserves and reflects pointwise inequalities, in particular the constraint $v \le 0$, and Lemma 1.22.12 shows that it preserves and reflects all-weight multiplier-ideal equivalence. Hence pullback is an order-preserving bijection between the two competitor families. By Theorem 17.2.1, each ideal envelope is itself a competitor, hence the greatest member of its family. The pullback of $P_\theta[u]_{\mathcal{I}}$ is therefore the greatest competitor upstairs, which proves the identity. □

Proposition 17.2.2 *For every $u \in \mathrm{PSH}(X,\theta)$,*

$$u - \sup_X u \le P_\theta[u] \le P_\theta[u]_{\mathcal{I}} \le V_\theta.$$

Moreover,

$$P_\theta[u] \sim_{\mathcal{I}} u, \qquad P_\theta[P_\theta[u]]_{\mathcal{I}} = P_\theta[u]_{\mathcal{I}},$$
$$P_\theta[P_\theta[u]_{\mathcal{I}}] = P_\theta[u]_{\mathcal{I}}.$$

Proof Let $h \le 0$ be a competitor for the ordinary envelope, so that $h \preceq u$. The function $\max(h, u - \sup_X u)$ is nonpositive, ordinarily equivalent to u, and dominates h. Since ordinarily equivalent potentials have the same multiplier ideals, it is a competitor for the ideal envelope. Taking regularized suprema gives $P_\theta[u] \le P_\theta[u]_{\mathcal{I}}$; the two outer inequalities follow directly from the defining families.

Projection and monotonicity of local integrability now give, for every $\lambda > 0$,

$$\mathcal{I}(\lambda u) \subseteq \mathcal{I}(\lambda P_\theta[u]) \subseteq \mathcal{I}(\lambda P_\theta[u]_{\mathcal{I}}) = \mathcal{I}(\lambda u).$$

Thus the ordinary envelope preserves all multiplier ideals, and it has the same ideal envelope as u. For $q = P_\theta[u]_{\mathcal{I}}$, the inequalities already proved and idempotence give $q \le P_\theta[q] \le P_\theta[q]_{\mathcal{I}} = q$. □

We call $u \in \mathrm{PSH}(X,\theta)$ *normalized* if $\sup_X u = 0$. Recall from Definition 1.19.1 that u is an ordinary model, or envelope fixed point, if $P_\theta[u] = u$; we call u an *ideal model* if $P_\theta[u]_{\mathcal{I}} = u$. Models of either kind are normalized, and by the last identity in Proposition 17.2.2 every ideal model is an ordinary model. The map $u \mapsto P_\theta[u]_{\mathcal{I}}$ is called *ideal projection*.

Corollary 17.2.1 *If θ is semipositive and big, then $P_\theta[u]_{\mathcal{I}}$ has finite extended Monge–Ampère mass for every $u \in \mathrm{PSH}(X,\theta)$.*

Proof By the last identity in Proposition 17.2.2, $P_\theta[u]_{\mathcal{I}}$ is an ordinary model, that is, an envelope fixed point. The mass estimate for envelope fixed points in Proposition 14.1.4 applies. □

Proposition 17.2.3 *For $u, v \in \mathrm{PSH}(X, \theta)$,*

$$u \preceq_{\mathcal{I}} v \quad \Longrightarrow \quad P_\theta[u]_{\mathcal{I}} \leq P_\theta[v]_{\mathcal{I}}.$$

For $a > 0$ one has $P_{a\theta}[au]_{\mathcal{I}} = aP_\theta[u]_{\mathcal{I}}$. For smooth real $(1,1)$-forms θ_i and potentials $u_i \in \mathrm{PSH}(X, \theta_i)$, $i = 1, 2$,

$$P_{\theta_1}[u_1]_{\mathcal{I}} + P_{\theta_2}[u_2]_{\mathcal{I}} \leq P_{\theta_1+\theta_2}[u_1 + u_2]_{\mathcal{I}}.$$

Proof Let h be a competitor for $P_\theta[u]_{\mathcal{I}}$. Since $h \sim_{\mathcal{I}} u \preceq_{\mathcal{I}} v$, the Lelong-number criterion in Theorem 1.16.1 shows that $\max(h, v - \sup_X v)$ is multiplier-ideal equivalent to v. This function is nonpositive and dominates h, so it is a competitor for $P_\theta[v]_{\mathcal{I}}$; taking the supremum over h proves that the envelope preserves the order. Multiplication by a is a bijection between the defining families in the scaling identity. For the sum, $P_{\theta_1}[u_1]_{\mathcal{I}} + P_{\theta_2}[u_2]_{\mathcal{I}}$ is nonpositive and $(\theta_1 + \theta_2)$-plurisubharmonic, and it is multiplier-ideal equivalent to $u_1 + u_2$ by projection and Lemma 1.16.3. It is therefore a competitor for the envelope on the right. □

17.2.1 Analytic and divisorial singularities

For a potential a with analytic singularities, the multiplier-ideal order controls the ordinary order: by Lemma 1.17.10, every quasi-plurisubharmonic v with $v \preceq_{\mathcal{I}} a$ satisfies $v \preceq a$. The ideal envelope of a is then a competitor for its ordinary envelope, and the two envelopes coincide. This extends [Xia26, Proposition 3.2.10] to smooth backgrounds.

Proposition 17.2.4 *Let $a \in \mathrm{PSH}(X, \theta)$ have analytic singularities. Then*

$$a \sim P_\theta[a] = P_\theta[a]_{\mathcal{I}}.$$

In particular the common envelope has analytic singularities with the same coefficient and ideal as a, up to a bounded remainder.

Proof Set $q = P_\theta[a]_{\mathcal{I}}$. Projection gives $q \sim_{\mathcal{I}} a$, so Lemma 1.17.10 gives $q \preceq a$. Since $q \leq 0$, it is a competitor for $P_\theta[a]$. Together with Proposition 17.2.2, this yields

$$a - \sup_X a \leq P_\theta[a] \leq q \leq P_\theta[a].$$

Hence the two envelopes are equal and ordinarily equivalent to a. Since they differ from a by a bounded function, they inherit its analytic presentation. □

Corollary 17.2.2 *Let S be a closed positive divisorial current of finite mass on a connected compact complex manifold Z, and choose a smooth residual decomposition $S = \beta + \mathrm{dd}^{\mathrm{c}} v$ as in Corollary 1.17.1. For every Hermitian form Ω with $v \in \mathrm{PSH}(Z, \Omega)$,*

$$P_\Omega[v] = P_\Omega[v]_{\mathcal{I}} \sim v.$$

Proof The proof of Proposition 17.2.4 applies, with Lemma 1.17.5 in place of Lemma 1.17.10. □

17.3 Changes of background and envelope mass

A smooth change of representative $(\theta, u) \mapsto (\theta + \mathrm{dd^c} f, u - f)$ preserves the current $\theta + \mathrm{dd^c} u$ but moves the zero obstacle. The resulting envelopes remain ordinarily equivalent after transport. In a nonclosed background, however, their masses can differ.

Proposition 17.3.1 *Let $u \in \mathrm{PSH}(X, \theta)$ and $f \in C^\infty(X, \mathbb{R})$. Set $\theta' = \theta + \mathrm{dd^c} f$, $u' = u - f$, $m = \min_X f$ and $M = \max_X f$. Then*

$$P_\theta[u]_{\mathcal{I}} + m \le P_{\theta'}[u']_{\mathcal{I}} + f \le P_\theta[u]_{\mathcal{I}} + M.$$

The same inequalities hold for the ordinary envelopes.

Proof If v is a competitor for $P_\theta[u]_{\mathcal{I}}$, then $v - f + m$ is θ'-plurisubharmonic, nonpositive and multiplier-ideal equivalent to u'. Therefore $v \le P_{\theta'}[u']_{\mathcal{I}} + f - m$, and taking the envelope gives the first inequality. Conversely, a competitor v' in the new background gives the competitor $v' + f - M$ in the old one, which proves the second inequality. For the ordinary envelopes, the same calculation applies with the ordinary singularity order in place of multiplier-ideal equivalence. □

For $u \in \mathrm{PSH}(X, \theta)$, put

$$W_\theta(u) = \int_X \mathrm{MA}_\theta(P_\theta[u]_{\mathcal{I}}) \in [0, +\infty].$$

This quantity depends on the decomposition of the current, even for smooth potentials.

Proposition 17.3.2 *There are two smooth decompositions of the same positive current for which the corresponding values of W are different.*

Proof Use the Hermitian form ω and the smooth function $g = u_t$ of Example 2.3.1, with $t \neq 0$ sufficiently small. Both ω and $\omega' = \omega + \mathrm{dd^c} g$ are Hermitian, and

$$\omega + \mathrm{dd^c} 0 = \omega' + \mathrm{dd^c}(-g).$$

For a smooth potential in a Hermitian background, all multiplier ideals are trivial and zero is a competitor for its multiplier-ideal envelope, so that envelope is zero. Consequently

$$W_\omega(0) = \int_X \omega^2, \qquad W_{\omega'}(-g) = \int_X (\omega + \mathrm{dd^c} g)^2.$$

By the explicit formula (2.3), the two integrals differ. □

The underlying torus is Kähler; the dependence comes from the chosen nonclosed background. In a closed background on a manifold with the bounded mass property, in particular on a compact Kähler manifold, mass is invariant under bounded changes of the potential. Since $\mathrm{MA}_{\theta'}(P_{\theta'}[u']_{\mathcal{I}}) = \mathrm{MA}_{\theta}(P_{\theta'}[u']_{\mathcal{I}} + f)$, the bounds of Proposition 17.3.1 then give $W_{\theta'}(u') = W_{\theta}(u)$.

Singularity volumes, on the other hand, range over the entire ordinary type, and the gauge change $v \mapsto v - f$ maps the members of the type bijectively onto those of the transported type without changing their measures; hence it preserves both volumes. Under bounded mass, Proposition 31.10.1 accordingly transports ordinary models of positive volume and preserves their upper-volume comparison.

17.3.1 Envelope equality and infinite mass

For each $u \in \mathrm{PSH}(X, \theta)$, the equality

$$P_{\theta}[u] = P_{\theta}[u]_{\mathcal{I}} \tag{17.1}$$

is meaningful. We call it *envelope goodness in the fixed background*. Its relation to mass differs from the Kähler characterization: it can hold when the potential has infinite mass while its envelope has finite mass.

Proposition 17.3.3 *If X fails the bounded mass property of Definition 7.5.1 and ω is a Hermitian form on X, there exists $U \in \mathrm{PSH}(X, \omega)$ such that*

$$P_{\omega}[U] = P_{\omega}[U]_{\mathcal{I}} = 0, \qquad U \in \mathcal{E}_{\mathrm{env}}(X, \omega, 0), \qquad \mathrm{MA}_{\omega}^{\mathrm{ext}}(U)(X) = +\infty,$$

where $\mathcal{E}_{\mathrm{env}}(X, \omega, 0)$ is the relative envelope class of Definition 14.4.1.

Proof Take the potential U of Proposition 14.6.1, which has the stated ordinary envelope, class membership and infinite mass. Since $P_{\omega}[U] = V_{\omega} = 0$, Proposition 17.2.2 gives $P_{\omega}[U]_{\mathcal{I}} = 0$. □

17.4 L^1 geometry of ideal classes

Ideal projection selects a canonical representative of each multiplier-ideal class, so the L^1 distance between these representatives defines a metric on the classes. Since the representatives have supremum zero, normalized compactness makes this metric totally bounded and its completion compact. The points added in the completion, however, need not be ideal models. On a curve this happens as badly as possible: approximating curvature measures by finite atomic measures shows that the completion contains every normalized potential.

Let

$$\mathfrak{S}_{\mathcal{I}}(X,\theta) = \mathrm{PSH}(X,\theta)/\!\sim_{\mathcal{I}}, \qquad \mathfrak{M}_{\mathcal{I}}(X,\theta) = \{q \in \mathrm{PSH}(X,\theta) : P_\theta[q]_{\mathcal{I}} = q\}$$

be the set of multiplier-ideal classes and the set of ideal models, and write $[u]_{\mathcal{I}}$ for the multiplier-ideal class of u. By Theorem 17.2.1, the map $[u]_{\mathcal{I}} \mapsto P_\theta[u]_{\mathcal{I}}$ identifies these two sets, and every ideal envelope has supremum zero.

Proposition 17.4.1 *The formula*

$$d_{\mathcal{I},\theta,\mathrm{d}V}([u]_{\mathcal{I}},[v]_{\mathcal{I}}) = \int_X |P_\theta[u]_{\mathcal{I}} - P_\theta[v]_{\mathcal{I}}|\,\mathrm{d}V \tag{17.2}$$

defines a finite metric on $\mathfrak{S}_{\mathcal{I}}(X,\theta)$. This metric space is totally bounded. Its metric completion is canonically the L^1 closure of $\mathfrak{M}_{\mathcal{I}}(X,\theta)$ inside the compact space $\mathcal{K}_\theta$ of normalized θ-plurisubharmonic functions in Theorem 1.4.1. Changing $\mathrm{d}V$ *to another smooth positive probability volume form gives a bi-Lipschitz equivalent metric.*

Proof The ideal envelopes are integrable and depend only on the ideal classes, so the formula is well defined and finite; symmetry and the triangle inequality are those of the L^1 norm. If the distance is zero, the two envelopes agree almost everywhere, hence everywhere by Lemma 1.3.3, and projection shows that the two ideal classes coincide. Conversely, equal ideal classes have the same envelope.

By the normalized compactness theorem, $\mathfrak{M}_{\mathcal{I}}$ lies in the compact space $\mathcal{K}_\theta$. Every subset of a compact metric space is totally bounded, and its completion is its closure in that space. Finally, the ratio of two smooth positive volume forms on the compact manifold X is bounded above and below by positive constants, and multiplying the integral by these bounds proves the last assertion. □

17.4.1 Completion on a curve

On a curve, a normalized potential is determined by its curvature measure. Replacing that measure by finite atomic approximations produces ideal models converging to the potential, even when the potential itself is not an ideal model.

Theorem 17.4.1 *Let Y be a compact connected Riemann surface and θ a smooth positive form of total mass one. Fix a smooth positive probability volume form on Y, again denoted by $\mathrm{d}V$, and use $\mathcal{K}_\theta$, the envelopes and the ideal metric on Y in place of X. Then the ideal models with finitely many logarithmic poles and zero non-pluripolar mass are dense in $\mathcal{K}_\theta$ for the L^1 topology. The logarithmic coefficients can be chosen rational, so these approximants have analytic singularities in the convention of [Xia26]. Consequently the completion in Proposition 17.4.1 is all of $\mathcal{K}_\theta$, and the metric on ideal classes is not complete.*

Proof We first show that the normalized Green potentials of finite atomic probability measures are ideal models, and then approximate the curvature measure of an arbitrary $u \in \mathcal{K}_\theta$ by such measures.

Let δ_p denote the Dirac mass at p. For distinct points $p_1, \dots, p_N \in Y$, consider a finitely supported probability measure

$$\nu = \sum_{j=1}^{N} a_j \delta_{p_j}, \qquad a_j > 0, \quad \sum_j a_j = 1.$$

The Green-kernel construction gives a potential q_ν with

$$\theta + \mathrm{dd}^{\mathrm{c}} q_\nu = \nu, \qquad \sup_Y q_\nu = 0.$$

Indeed, choose a kernel with $\mathrm{dd}^{\mathrm{c}}_x G(x,p) = \delta_p - \theta$. The sum $\sum_j a_j G(x,p_j)$ has curvature ν, and subtracting its finite supremum gives q_ν. In a coordinate z centered at p_j, the function q_ν is $a_j \log|z|^2$ plus a smooth function, and it is smooth away from the finite support of ν. Since all of its curvature lies on its pole set, its non-pluripolar measure is zero.

We claim that $P_\theta[q_\nu]_{\mathcal{I}} = q_\nu$. Let $w \in \mathrm{PSH}(Y,\theta)$ satisfy $w \sim_{\mathcal{I}} q_\nu$. The Lelong-number criterion of Theorem 1.16.1 gives $\nu(w,p_j) = a_j$, where $\nu(w,p_j)$ is the Lelong number of w at p_j. On a curve, this Lelong number is exactly the mass of the atom at p_j of the positive measure $\theta + \mathrm{dd}^{\mathrm{c}} w$. By Stokes' theorem for currents, that measure has total mass one, so the prescribed atoms already exhaust its mass and $\theta + \mathrm{dd}^{\mathrm{c}} w = \nu$. Thus $\mathrm{dd}^{\mathrm{c}}(w - q_\nu) = 0$, and the difference is a harmonic distribution, hence a smooth function that is constant on the compact connected curve. Among these translates, the greatest nonpositive one is q_ν, which proves the claim. By Proposition 17.2.2, every such q_ν is also an ordinary model.

Now let $u \in \mathcal{K}_\theta$. Its curvature $\nu = \theta + \mathrm{dd}^{\mathrm{c}} u$ is a positive probability measure. Choose finitely supported probability measures ν_j converging weakly to ν. For instance, partition the compact metric space Y into finitely many Borel sets of diameter tending to zero, place the mass of each set at a chosen point of it, and omit the sets of mass zero; uniform continuity of each continuous test function gives weak convergence.

The weights can be made rational. For a partition with weights $a_1, \dots, a_N$, choose a large integer M, replace a_i by $b_i = \lfloor M a_i \rfloor / M$ for $i < N$, put $b_N = 1 - \sum_{i<N} b_i$, and discard zero weights. The total absolute change is at most $2(N-1)/M$, which

can be made to tend to zero along the partitions, so weak convergence is preserved. Writing $b_i = m_i/M$, the corresponding Green potential has the analytic type

$$\left(\frac{1}{M}, O_Y\left(-\sum_i m_i p_i\right)\right)$$

with smooth local remainders.

Let $q_j = q_{\nu_j}$. By normalized compactness, every subsequence has a further subsequence converging in L^1 to some $q \in \mathcal{K}_\theta$. Passing to the limit in the curvature identity in the sense of distributions gives

$$\theta + \mathrm{dd^c} q = \nu = \theta + \mathrm{dd^c} u.$$

By the same harmonic argument, $q - u$ is constant, and the constant is zero because both suprema vanish. Every cluster value is therefore u, so the whole sequence converges to u in L^1. This proves the density assertion and identifies the completion.

To see incompleteness concretely, choose a nonconstant smooth real function f. For sufficiently small $\varepsilon > 0$, one has $\theta + \varepsilon \mathrm{dd^c} f > 0$, so $u = \varepsilon f - \sup_Y(\varepsilon f)$ is a smooth, normalized and nonconstant element of $\mathcal{K}_\theta$. Its multiplier ideals are all trivial, so zero is a competitor and $P_\theta[u]_{\mathcal{I}} = 0$. Thus $u \notin \mathfrak{M}_{\mathcal{I}}(Y, \theta)$, although u lies in the completion. The set of ideal models is therefore not closed in its completion, and the metric is not complete. □

17.4.2 Mass and capacity convergence

The approximation in Theorem 17.4.1 loses all non-pluripolar mass. For the nonconstant smooth u at the end of its proof, the approximants q_j have zero non-pluripolar measure, whereas $\mathrm{MA}_\theta(u) = \theta + \mathrm{dd^c} u > 0$.

After normalization to mean zero, the same sequence converges in the Gauduchon trace metric of Corollary 6.2.1. Nevertheless, the sequence q_j does not converge to u in capacity. Otherwise, since $q_j, u \le 0$, Theorem 8.3.1 would give

$$0 = \mathrm{e}^{q_j} \mathrm{MA}_\theta(q_j) \rightharpoonup \mathrm{e}^{u}(\theta + \mathrm{dd^c} u),$$

contradicting the strict positivity of the limiting measure. Thus weak convergence of traces and L^1 convergence do not imply convergence in capacity, even on a curve.

The same sequence shows that ideal projection is not continuous for the L^1 topology: $P_\theta[q_j]_{\mathcal{I}} = q_j \to u$, whereas $P_\theta[u]_{\mathcal{I}} = 0$. The completion therefore distinguishes smooth normalized potentials that the quotient by multiplier-ideal equivalence identifies.

The loss of mass also occurs along sequences of fixed positive mass.

Proposition 17.4.2 *In the curve setting above, for every rational $0 < \tau < 1$ there are ideal models r_j with analytic singularities such that*

$$d_{I,\theta,\mathrm{d}V}([r_j]_I, [0]_I) \longrightarrow 0,$$
$$\mathrm{MA}_\theta(r_j)(Y) = \tau, \qquad \mathrm{MA}_\theta(r_j) \rightharpoonup \tau\theta.$$

In particular, non-pluripolar mass is not continuous for this metric, even on a family whose masses have a fixed positive lower bound.

Proof Choose finitely supported probability measures ν_j with rational weights converging weakly to θ, and let q_j be their normalized Green potentials; the proof of Theorem 17.4.1 gives $q_j \to 0$ in L^1. Put $a = 1 - \tau$ and $r_j = P_\theta[aq_j]_I$. The scaled potential lies in the same background, since

$$\theta + \mathrm{dd}^{\mathrm{c}}(aq_j) = (1-a)\theta + a\nu_j \geq 0.$$

Since aq_j is a competitor, $aq_j \leq r_j \leq 0$, and hence $r_j \to 0$ in L^1. By projection, r_j is an ideal model. By Proposition 17.2.4, it has exactly the logarithmic coefficients of aq_j, which are rational, with bounded local remainders, and it is locally bounded away from the finite pole set. On the curve, removing the polar part of the curvature therefore gives

$$\mathrm{MA}_\theta(r_j) = \theta + \mathrm{dd}^{\mathrm{c}} r_j - a\nu_j.$$

By Stokes' theorem for currents, this measure has total mass $1 - a = \tau$. The positive curvature measures $\theta + \mathrm{dd}^{\mathrm{c}} r_j$ have mass one and converge as distributions, hence weakly, to θ. Since $\nu_j \rightharpoonup \theta$, the last identity gives $\mathrm{MA}_\theta(r_j) \rightharpoonup \tau\theta$. Finally, $P_\theta[0]_I = 0$, so the distance in the statement is $\int_Y |r_j|\,\mathrm{d}V$, which tends to zero. The limiting potential 0 has non-pluripolar measure θ, of mass one. □

For comparison, let $d_{S,\theta}$ denote the singularity pseudometric d_S recalled at the beginning of the chapter, computed in the background θ. Since $r_j \leq 0$, the ordered-distance formula of [Xia26, Lemma 6.2.2] applies and gives

$$d_{S,\theta}(r_j, 0) = \frac{1}{2}\left(\int_Y \theta - \int_Y \mathrm{MA}_\theta(r_j)\right) = \frac{1-\tau}{2} > 0.$$

Thus $d_{S,\theta}$ keeps the fixed mass gap that disappears in the elementary ideal metric.

17.4.3 Comparison with scalar normalization

We return to X of arbitrary dimension, with the fixed volume form $\mathrm{d}V$. The scalar pseudometric Δ of (16.18) compares the normalization scalars B_φ of Section 16.1 uniformly over all continuous logarithmic densities. When an analytic singularity is shrunk to zero, the estimate in a smaller background makes this scalar distance tend to zero. At a fixed mass, however, the normalization scalar in a closed background does not see the location of the poles.

Example 17.4.1 Let θ be semipositive and big, and let $a \in \mathrm{PSH}(X,\theta)$ have analytic singularities with $a \le 0$. Use the normalization scalars B_φ and the sets $\mathcal{A}_\varphi(G)$ of supporting probabilities from Sections 16.1 and 16.2, defined with the fixed volume form $\mathrm{d}V$. Set

$$\varphi_t = P_\theta[ta], \qquad t \in \mathbb{Q} \cap (0,1).$$

By Proposition 17.2.4, $\varphi_t \sim ta$; since the ordinary envelope depends only on the ordinary type, $P_\theta[\varphi_t] = \varphi_t$. Moreover $ta \in \mathrm{PSH}(X, t\theta)$ and $ta \le \varphi_t$, so the envelope $P_{t\theta}(\varphi_t)$ in the smaller background is not identically $-\infty$. By Proposition 16.7.1,

$$\sup_{G \in C(X)} |B_{\varphi_t}(G) - B_0(G)| \le -n \log(1-t) \longrightarrow 0 \qquad (t \downarrow 0).$$

When θ is Hermitian and G is fixed and smooth, every choice in $\mathcal{A}_{\varphi_t}(G)$ converges weakly in L^1 to the smooth Gauduchon adjoint probability, by Corollary 16.7.1 and Theorem 16.2.3.

By contrast, the projective references in Example 16.10.1 have equal mass and identical normalization functionals, although their poles and multiplier ideals differ.

17.5 Limits of analytic envelopes

By the curve construction of Theorem 17.4.1, convergence of potentials alone does not determine the limit of their ideal envelopes. Two further conditions do: a common lower bound in the multiplier-ideal order, and convergence of all divisorial Lelong numbers. Together they force every cluster value of the envelopes to have the multiplier ideals of the limiting potential, and maximality of the envelope then identifies the cluster value. The dyadic analytic approximants satisfy both conditions.

We fix $u \in \mathrm{PSH}(X,\theta)$ and a Hermitian form h on X, which will absorb the curvature errors of the approximants, and we use the local divisors E over X and their generic Lelong numbers ν_E of Section 1.17.

Proposition 17.5.1 *Let $\epsilon_j \geq 0$ tend to zero, and let $a_j \in \mathrm{PSH}(X, \theta + \epsilon_j h)$ satisfy*

$$u \preceq_{\mathcal{I}} a_j, \qquad \nu_E(a_j) \longrightarrow \nu_E(u) \quad \text{for every local divisor } E \text{ over } X.$$

Then

$$P_{\theta+\epsilon_j h}[a_j]_{\mathcal{I}} \longrightarrow P_\theta[u]_{\mathcal{I}} \quad \text{in } L^1.$$

If the a_j have analytic singularities, their ordinary envelopes converge to $P_\theta[u]$ in L^1 if and only if $P_\theta[u] = P_\theta[u]_{\mathcal{I}}$.

Proof Set $p = P_\theta[u]_{\mathcal{I}}$ and $q_j = P_{\theta+\epsilon_j h}[a_j]_{\mathcal{I}}$. Projection and the ideal order give

$$p \leq P_{\theta+\epsilon_j h}[p]_{\mathcal{I}} \leq q_j \leq 0.$$

The first inequality holds because p is a competitor in the larger background, and the second follows from Proposition 17.2.3, since $p \sim_{\mathcal{I}} u \preceq_{\mathcal{I}} a_j$. All q_j have supremum zero and belong to $\mathrm{PSH}(X, \theta + Ch)$ for a fixed $C \geq \sup_j \epsilon_j$. By normalized compactness, every subsequence therefore has a further subsequence converging in L^1. Let q be such a limit. Passing to the limit in the current inequalities gives $q \in \mathrm{PSH}(X, \theta)$, and Lemma 1.3.3 gives $p \leq q \leq 0$.

Fix a local modification $\pi\colon Z \to U$ and a prime divisor $E \subset Z$. By Lemma 1.22.8, the chosen subsequence still converges in L^1_{loc} after pullback. Since $q_j \sim_{\mathcal{I}} a_j$, the two functions have the same divisorial Lelong numbers. At every point of E, the pullback of each q_j has Lelong number at least $\nu_E(q_j)$; applying the pointwise semicontinuity of Lemma 1.17.6 at a point of E where the pullback of q has Lelong number $\nu_E(q)$ yields

$$\nu_E(q) \geq \limsup_j \nu_E(q_j) = \lim_j \nu_E(a_j) = \nu_E(u).$$

On the other hand, $p \leq q$ gives $\nu_E(q) \leq \nu_E(p) = \nu_E(u)$. Thus q and u have the same Lelong numbers along all local divisors. Blowing up a point of any local modification recovers the pointwise Lelong number there as a generic divisorial number; on a curve the point is already a divisor. By Theorem 1.16.1, $q \sim_{\mathcal{I}} u$.

Consequently q is a competitor for p, so $q \leq p$. Every cluster value is therefore p, which proves convergence of the whole sequence. For analytic a_j, Proposition 17.2.4 identifies the ordinary and ideal envelopes, and the last assertion follows from uniqueness of L^1 limits and Lemma 1.3.3. □

The divisorial convergence follows from inclusions of multiplier ideals at fixed weights. The local Bergman estimate bounds the error in terms of the log discrepancy, and this error tends to zero after division by the weight.

Corollary 17.5.1 *Let $\epsilon_j \geq 0$ tend to zero, and let $a_j \in \mathrm{PSH}(X, \theta + \epsilon_j h)$ satisfy $u \preceq_{\mathcal{I}} a_j$. Suppose that, for every $\delta > 0$ and every sufficiently large integer m,*

$$\mathcal{I}((1+\delta)ma_j) \subseteq \mathcal{I}(mu) \quad \text{for all sufficiently large } j.$$

Then the conclusions of Proposition 17.5.1 hold.

Proof Fix a local divisor E over a coordinate neighborhood U, and write $A = A_U(E)$ for its log discrepancy. By Lemma 1.17.8, the assumed inclusion implies

$$(1+\delta)m\nu_E(a_j) \geq \mathrm{ord}_E\, \mathcal{I}((1+\delta)ma_j) \geq \mathrm{ord}_E\, \mathcal{I}(mu) \geq m\nu_E(u) - A$$

for all large j. Taking the lower limit in j first, and then letting $m \longrightarrow \infty$ and $\delta \downarrow 0$, we obtain $\liminf_j \nu_E(a_j) \geq \nu_E(u)$. The opposite inequality $\nu_E(a_j) \leq \nu_E(u)$ holds for every j because $u \preceq_{\mathcal{I}} a_j$. Hence Proposition 17.5.1 applies. □

The common ideal bound cannot be omitted, even when all divisorial Lelong numbers converge. In the atomic approximation of Theorem 17.4.1, every individual atom can tend to zero while the limiting potential is not an ideal model.

Example 17.5.1 On a compact connected curve Y with a positive form θ of total mass one, there are normalized $a_j \in \mathrm{PSH}(Y, \theta)$ with analytic singularities and a smooth $u \in \mathrm{PSH}(Y, \theta)$ such that

$$a_j \longrightarrow u \text{ in } L^1, \qquad \nu_E(a_j) \longrightarrow \nu_E(u) \text{ for every local divisor } E,$$

but $P_\theta[a_j] = P_\theta[a_j]_{\mathcal{I}} \to u \neq P_\theta[u] = P_\theta[u]_{\mathcal{I}}$ in L^1.

Proof Choose u smooth, nonconstant and normalized, with $\theta + \mathrm{dd}^{\mathrm{c}} u > 0$, and apply the construction in Theorem 17.4.1 to the smooth probability measure $\nu = \theta + \mathrm{dd}^{\mathrm{c}} u$. Since ν has no atoms, the largest mass of a cell tends to zero along partitions of vanishing diameter. Choose the rational approximations of the weights so that their total absolute error tends to zero. The resulting atomic probability measures ν_j converge weakly to ν, and their largest atoms tend to zero.

By the same theorem, their normalized Green potentials a_j have analytic singularities, satisfy $P_\theta[a_j] = P_\theta[a_j]_{\mathcal{I}} = a_j$, and converge to u in L^1. At every point $y \in Y$,

$$0 \leq \nu(a_j, y) = \nu_j(\{y\}) \leq \max_{z \in Y} \nu_j(\{z\}) \longrightarrow 0 = \nu(u, y).$$

Since local modifications of a smooth curve are isomorphisms, this gives all the required divisorial limits. As u is bounded and $\theta > 0$, both envelopes of u are zero. Since u is normalized and not constant, $u \neq 0$, so the limit u of the envelopes of the a_j differs from the envelopes of u. Each a_j has a logarithmic pole, so $u \preceq_{\mathcal{I}} a_j$ fails.□

17.5.1 Dyadic regularization

Demailly regularization provides the common ideal bound, since its approximants lie above u, and the Bergman estimate gives convergence of each divisorial Lelong number. Along dyadic indices, subadditivity of multiplier ideals also orders the ordinary singularity types of the approximants, which makes their envelopes decrease pointwise. This is the mechanism used in [Xia26, Theorem 7.1.1, proof of (3) $\Rightarrow$ (1), Step 1].

Apply Theorem 1.15.1 to u, with $\gamma = -\theta$, along the indices $m_j = 2^j$, starting at a sufficiently large index. Denote the resulting potentials by a_j, and choose numbers $\epsilon_j > 0$ decreasing to zero such that

$$a_j \in \mathrm{PSH}(X, \theta + \epsilon_j h), \qquad a_j \geq u, \qquad a_j \text{ has type } (1/m_j, \mathcal{I}(m_j u)). \tag{17.3}$$

These approximants satisfy $a_j \to u$ in L^1. Their values need not decrease, but their ordinary singularity types do: the subadditivity property in (1.23) gives

$$\mathcal{I}(2m_j u) \subseteq \mathcal{I}(m_j u)^2, \qquad a_{j+1} \preceq a_j,$$

where the ordinary comparison follows by taking logarithms of local generators, with a constant that may depend on j.

Theorem 17.5.1 *For every $u \in \mathrm{PSH}(X, \theta)$ and every choice in (17.3), put*

$$q_j = P_{\theta+\epsilon_j h}[a_j].$$

Then q_j has analytic singularities, and

$$q_j = P_{\theta+\epsilon_j h}[a_j]_{\mathcal{I}} \searrow P_\theta[u]_{\mathcal{I}}$$

pointwise and in L^1.

Proof By Proposition 17.2.4, $q_j = P_{\theta+\epsilon_j h}[a_j]_{\mathcal{I}} \sim a_j$. Since the backgrounds decrease and $a_{j+1} \preceq a_j$, the function q_{j+1} is a competitor for $P_{\theta+\epsilon_j h}[a_j]$. Hence $q_{j+1} \leq q_j$.

The inequality $a_j \geq u$ gives $u \preceq_{\mathcal{I}} a_j$. For every local divisor E, Lemma 1.17.8 gives

$$\nu_E(a_j) = \frac{1}{m_j} \operatorname{ord}_E \mathcal{I}(m_j u) \longrightarrow \nu_E(u).$$

By Proposition 17.5.1, $q_j \to p := P_\theta[u]_{\mathcal{I}}$ in L^1, and its proof gives $p \leq q_j \leq 0$. The decreasing limit of the q_j is therefore bounded below by p; it is a θ-plurisubharmonic function that agrees with p almost everywhere, hence everywhere. □

Corollary 17.5.2 *For the sequence above, the following are equivalent:*

$$P_\theta[u] = P_\theta[u]_{\mathcal{I}}, \qquad q_j \longrightarrow P_\theta[u] \text{ in } L^1, \qquad q_j \searrow P_\theta[u] \text{ pointwise}.$$

For the fixed volume form $\mathrm{d}V$*,*

$$\lim_j \int_X (q_j - P_\theta[u])\,\mathrm{d}V = \int_X (P_\theta[u]_{\mathcal{I}} - P_\theta[u])\,\mathrm{d}V \geq 0,$$

and the vanishing of this limit is a further equivalent condition.

Proof The theorem identifies both limits of the q_j as $P_\theta[u]_{\mathcal{I}}$, and its L^1 convergence gives the integral formula. By Proposition 17.2.2, the limiting difference is nonnegative. Its integral therefore vanishes exactly when the two envelopes agree almost everywhere, hence everywhere by Lemma 1.3.3. This proves all the equivalences. □

17.5.2 Absorbing the curvature error

When $\theta = \omega$ is Hermitian, we can take $h = \omega$, and dividing the approximants by $1+\epsilon_j$ absorbs the curvature error. If the error bounds are rational, the analytic coefficients remain rational, so the whole approximation takes place in the fixed class $\mathrm{PSH}(X,\omega)$.

Corollary 17.5.3 *Let* ω *be Hermitian and let* $u \in \mathrm{PSH}(X,\omega)$*. There are* $b_j \in \mathrm{PSH}(X,\omega)$ *with analytic singularities and decreasing ordinary singularity types, such that* $b_j \to u$ *in* L^1 *and*

$$P_\omega[b_j] = P_\omega[b_j]_{\mathcal{I}} \searrow P_\omega[u]_{\mathcal{I}}.$$

Consequently the classes of potentials with analytic singularities are dense in the whole metric space of ideal classes of Proposition 17.4.1. The equality $P_\omega[u] = P_\omega[u]_{\mathcal{I}}$ *is equivalent to convergence of these ordinary envelopes to* $P_\omega[u]$ *in* L^1*.*

Proof Use $h = \omega$ in (17.3). Enlarge the error bounds to positive rational numbers $\epsilon_j \downarrow 0$, and set

$$c_j = \frac{1}{1+\epsilon_j}, \qquad b_j = c_j a_j.$$

Then $c_j \uparrow 1$ is rational and

$$\omega + \mathrm{dd^c} b_j = c_j\big((1+\epsilon_j)\omega + \mathrm{dd^c} a_j\big) \geq 0.$$

The analytic coefficient of b_j is c_j/m_j. Moreover $b_j \to u$ in L^1, since $a_j \to u$ there and $c_j \to 1$. Because $a_{j+1} \preceq a_j$, $c_{j+1} \geq c_j$, and each a_j is bounded above, one has $b_{j+1} \preceq b_j$.

Put $q_j = P_{(1+\epsilon_j)\omega}[a_j] = P_{(1+\epsilon_j)\omega}[a_j]_{\mathcal{I}}$. Theorem 17.5.1 gives $q_j \searrow P_\omega[u]_{\mathcal{I}}$ pointwise and in L^1. Since b_j has analytic singularities, Proposition 17.2.4 and the homogeneity in Proposition 17.2.3 give

$$r_j := P_\omega[b_j] = P_\omega[b_j]_{\mathcal{I}} = c_j q_j.$$

Here $q_j \le 0$ decreases and $c_j \uparrow 1$, so r_j also decreases, with pointwise limit $P_\omega[u]_{\mathcal{I}}$. Moreover

$$P_\omega[u]_{\mathcal{I}} \le q_j \le r_j \le 0.$$

Since the lower bound is integrable, dominated convergence gives convergence in L^1.

The distance in Proposition 17.4.1 from $[b_j]_{\mathcal{I}}$ to $[u]_{\mathcal{I}}$ is exactly the L^1 distance from r_j to $P_\omega[u]_{\mathcal{I}}$, so it tends to zero. The last assertion follows because the limit of the r_j has been identified. □

17.6 A topology on ordinary singularity types

The ordinary and the ideal envelope of a potential record different parts of its singularity. If a topology on ordinary singularity types requires both envelopes to converge, their equality becomes a closed condition, which every limit of analytic singularity types satisfies. For the converse, the dyadic approximants of a potential with equal envelopes must converge to it. We therefore add coordinates that these approximants control in a fixed Hermitian background: controlled inclusions of multiplier ideals, and the weighted total variation of divisorial Lelong numbers on each fixed modification. The closure of the analytic types is then exactly the set of types with equal envelopes, on the whole space of ordinary types.

This topology depends on the zero obstacle hidden in the envelopes, and a smooth change of background moves that obstacle. Testing all smooth obstacles removes this dependence and raises the question whether equality of the envelopes below zero implies equality below every obstacle. We answer it here for potentials with prescribed countable pole sets and strictly positive curvature (Section 17.6.5); models of zero volume built from divisors in fibrations and products over curves, on which polar curvature gives complete metrics, are treated in Chapter 32. Some remarks compare these topologies with the curve models of Chapter 31; those results do not depend on the present section.

Fix a Hermitian form ω on X and write

$$\mathfrak{S}(X,\omega) = \mathrm{PSH}(X,\omega)/\!\sim.$$

In this section $[u]$ denotes the ordinary class of u. For each smooth compact modification $\pi\colon Y \to X$, fix a Hermitian form γ_π on Y. For a prime divisor $E \subset Y$, set $w_E = \int_E \gamma_\pi^{n-1}$, and let $\nu_E(u) = \nu(u \circ \pi, E)$ be the generic Lelong number of Section 1.17.

Lemma 17.6.1 *For $u, v \in \mathrm{PSH}(X,\omega)$, the sum*

$$d_\pi([u],[v]) = \sum_{E \subset Y \text{ prime}} w_E |\nu_E(u) - \nu_E(v)| \tag{17.4}$$

is finite and defines a pseudometric on $\mathfrak{S}(X,\omega)$. Changing γ_π gives an equivalent pseudometric for the fixed modification π.

Proof The current $T = \pi^*\omega + \mathrm{dd}^{\mathrm{c}}(u \circ \pi)$ is positive. Divisorial removal in Lemma 1.17.1 gives, for every finite family of prime divisors,

$$\sum_E w_E \nu_E(u) \le \int_Y T \wedge \gamma_\pi^{n-1} < \infty.$$

Taking the supremum over finite families shows that the sum converges and that only countably many coefficients are positive. The same bound holds for v. Bounded changes of u or v do not affect the coefficients, and the triangle inequality holds term by term. Finally, two Hermitian forms on the compact manifold Y bound one another by fixed positive multiples, and so do the corresponding divisor weights. □

For $t > 0$ and $a > 1$, let

$$\mathcal{U}(t,a) = \{([u],[v]) : \mathcal{I}(atu) \subseteq \mathcal{I}(tv), \quad \mathcal{I}(atv) \subseteq \mathcal{I}(tu)\}. \tag{17.5}$$

These are relations between ordinary classes, since a bounded change leaves every multiplier ideal unchanged. Let d_I be the pullback of the metric (17.2), in background ω, under $[u] \mapsto [u]_{\mathcal{I}}$, and put

$$d_O([u],[v]) = \|P_\omega[u] - P_\omega[v]\|_{L^1(\mathrm{d}V)}.$$

Both are pseudometrics on ordinary classes: d_I compares the ideal envelopes, whereas d_O also detects differences of ordinary envelopes within a single multiplier-ideal class.

Proposition 17.6.1 *Finite intersections of the relations $d_O < r$, $d_I < r$, $d_\pi < r$ and $\mathcal{U}(t,a)$, with $r,t > 0$ and $a > 1$, form a base of entourages for a uniformity on $\mathfrak{S}(X,\omega)$. Denote its topology by $\tau_{\omega,0}$, where the subscript 0 records the zero obstacle in the envelopes. A net $[u_j]$ converges to $[u]$ if and only if both envelopes converge in L^1, $d_\pi([u_j],[u]) \to 0$ for every fixed π, and $([u_j],[u]) \in \mathcal{U}(t,a)$ eventually for every fixed t,a. Smooth positive choices of $\mathrm{d}V$ and of the γ_π do not change the uniformity.*

Proof All generators are symmetric. They contain the diagonal; for $\mathcal{U}(t,a)$ this holds because multiplier ideals decrease as the weight increases, as one sees from local integrability after subtracting a local upper bound from the potential. For $b = \sqrt{a}$, put $\mathcal{V} = \mathcal{U}(t,b) \cap \mathcal{U}(bt,b)$. If $([x],[y]),([y],[z]) \in \mathcal{V}$, then

$$\mathcal{I}(atx) \subseteq \mathcal{I}(bty) \subseteq \mathcal{I}(tz),$$

and the same argument with x and z exchanged shows that $\mathcal{V} \circ \mathcal{V} \subseteq \mathcal{U}(t,a)$. For the pseudometric relations, halving the radius gives the corresponding refinement under

composition. Taking finite intersections of these refinements proves the axioms of a uniformity, and convergence in it means eventual membership in each generating entourage. Bounded ratios of smooth positive volume forms and Lemma 17.6.1 show that the auxiliary choices do not matter. □

Each entourage involves only finitely many modifications, so no cofinal family of modifications is needed.

17.6.1 Analytic closure

We now show that the dyadic approximants of Corollary 17.5.3 converge in $\tau_{\omega,0}$ exactly when the two envelopes agree. Let $\mathcal{A}_\omega$ be the set of ordinary classes with analytic singularities, with rational coefficients as in Section 1.13; the rational curvature correction in Corollary 17.5.3 keeps the approximants in this set. In the controlled ideal relations, the scale factor of that correction has to be taken into account.

Theorem 17.6.1 *The closure and the sequential closure of $\mathcal{A}_\omega$ in $\tau_{\omega,0}$ are both equal to*

$$\mathcal{G}_\omega = \{[u] : P_\omega[u] = P_\omega[u]_{\mathcal{I}}\}. \tag{17.6}$$

For every $u \in \mathrm{PSH}(X,\omega)$, the dyadic sequence b_j of Corollary 17.5.3 satisfies

$$[b_j] \longrightarrow [u] \text{ in } \tau_{\omega,0} \quad \Longleftrightarrow \quad [u] \in \mathcal{G}_\omega.$$

Proof We use the notation of Corollary 17.5.3: $m_j = 2^j$, a_j has type $(1/m_j, \mathcal{I}(m_j u))$, $c_j = (1+\epsilon_j)^{-1} \uparrow 1$ is rational, and $b_j = c_j a_j$. The ordinary and ideal envelopes of b_j agree and decrease to $P_\omega[u]_{\mathcal{I}}$. Thus the two envelope coordinates of $[b_j]$ converge to those of $[u]$ exactly when $[u] \in \mathcal{G}_\omega$. It remains to show that the other coordinates always converge.

Fix $t > 0$ and $a > 1$, and choose $1 < s < a$. For large j, one has $ac_j > s$ and $m_j \geq st/(s-1)$, so Lemma 1.14.1 gives

$$\mathcal{I}(atb_j) = \mathcal{I}(atc_j a_j) \subseteq \mathcal{I}(sta_j) \subseteq \mathcal{I}(tu).$$

Since $u \preceq a_j \preceq b_j$, the reverse relation $\mathcal{I}(atu) \subseteq \mathcal{I}(tb_j)$ holds for every j.

For a divisor E on a fixed modification π, the Bergman divisorial estimate gives

$$c_j \max\left\{\nu_E(u) - \frac{A(E)}{m_j}, 0\right\} \leq \nu_E(b_j) \leq c_j \nu_E(u) \leq \nu_E(u),$$

where $A(E)$ is the log discrepancy, computed over a coordinate neighborhood of a point of the center of E. Each coefficient therefore converges to $\nu_E(u)$, and the summability in Lemma 17.6.1 gives $d_\pi([b_j],[u]) \to 0$ by dominated convergence. This proves convergence of all the remaining coordinates.

Finally, analytic classes have equal envelopes by Proposition 17.2.4. The map $[u] \mapsto (P_\omega[u], P_\omega[u]_{\mathcal{I}})$ is continuous into $L^1 \times L^1$, whose diagonal is closed. Thus every limit of a net of analytic classes lies in $\mathcal{G}_\omega$. The sequence b_j gives the converse and the sequential assertion. □

17.6.2 Separation and mass

The topology $\tau_{\omega,0}$ sees an ordinary class only through its envelopes, its multiplier ideals and its divisorial Lelong numbers. Distinct ordinary types can have the same ordinary envelope, and such types cannot be separated.

Proposition 17.6.2 *Two classes have the same $\tau_{\omega,0}$ neighborhoods if and only if their ordinary envelopes coincide. This topology is not T_0, hence not metrizable.*

Proof The coordinate d_O distinguishes different ordinary envelopes. Conversely, potentials with equal ordinary envelopes have equal multiplier ideals at every weight by Proposition 17.2.2. Their ideal envelopes and all their divisorial coefficients therefore agree, so every generating entourage has identical sections at the two classes.

To produce distinct classes of this kind, choose a cutoff logarithmic pole $w \le 0$ supported in a coordinate ball, plurisubharmonic near its pole and smooth on the annulus where the cutoff varies. After multiplication by a small positive constant, $w \in \mathrm{PSH}(X, \omega)$. Put $\chi(s) = 1 - \sqrt{1 - s}$ for $s \le 0$ and $u = \chi(w)$. The inequalities $0 < \chi' \le 1/2$ and $\chi'' \ge 0$, applied first to bounded truncations, give $u \in \mathrm{PSH}(X, \omega)$. In local coordinates z centered at the pole, u has size comparable to $\sqrt{-\log|z|^2}$, so it is unbounded and e^{-tu} is integrable for every $t > 0$. Hence all its multiplier ideals are trivial and $P_\omega[u]_{\mathcal{I}} = 0$.

For every $0 < \delta \le 1$, the function $\delta s - \chi(s)$ is bounded above on $(-\infty, 0]$. Hence $\delta w \preceq u$, and δw is a competitor for the ordinary envelope of u. Letting $\delta \downarrow 0$ off the pole gives $P_\omega[u] = 0$. Thus $[u]$ and $[0]$ are distinct classes with the same neighborhoods. □

By Theorem 7.1.1, the two envelopes have the extended masses

$$\int_X \mathrm{MA}_\omega(P_\omega[u]), \qquad \int_X \mathrm{MA}_\omega(P_\omega[u]_{\mathcal{I}}).$$

In the present positive background, both are at most $\int_X \omega^n$. For the ideal envelope this follows from Propositions 14.1.4 and 17.2.2. For the ordinary envelope, normalize $u \le 0$ and put $p_j = P_\omega(u + j, 0)$ and $p = P_\omega[u]$, with the obstacle envelopes of Section 1.19. Then $p_j \to p$ in capacity by Lemma 1.8.9. For $k > 0$, set $\eta_k = \min\{1, (u+k)_+\}$. If $j > k$, the contact set of p_j with $u + j$ lies outside $\{\eta_k > 0\}$, so Proposition 14.1.3 gives $\eta_k \mathrm{MA}_\omega(p_j) \le \eta_k \omega^n$. Testing against a nonnegative continuous function and applying Lemma 7.3.1, we obtain the same inequality for

p. Letting $k \to \infty$ proves $\mathrm{MA}_\omega(p) \le \omega^n$, since the remaining set $\{u = -\infty\}$ is pluripolar.

The potential itself can nevertheless have infinite mass: when X fails the bounded mass property, Proposition 17.3.3 gives such a u whose two envelopes are both zero. Its ordinary class belongs to $\mathcal{G}_\omega$. Thus the topology and its analytic-closure criterion need no mass condition on the representative, whereas the two canonical envelopes never have infinite mass in a Hermitian background.

17.6.3 Transport with an obstacle

The identification $u \mapsto u - \chi$ of $\mathrm{PSH}(X,\omega)$ with $\mathrm{PSH}(X,\omega')$, $\omega' = \omega + \mathrm{dd}^{\mathrm{c}}\chi$, in Proposition 1.2.1 moves the zero obstacle to $-\chi$. To compare the topologies in different backgrounds, we therefore need envelopes below an arbitrary smooth obstacle.

For a smooth real obstacle F, we use the ordinary envelope $P_{\omega,F}[u]$ of Definition 1.19.1, and we write $P_{\omega,F}[u]_{\mathcal{I}}$ for the ideal envelope defined with $v \le F$ in place of $v \le 0$. The competitor family of the latter contains a translate of u. Replacing a competitor by its maximum with this translate shows that the ideal envelope may equivalently be defined with the condition $v \preceq_{\mathcal{I}} u$. The same notation applies to any smooth real background that admits u. Define $\tau_{\omega,F}$ by replacing the two envelope coordinates in Proposition 17.6.1 by these obstacle envelopes.

Proposition 17.6.3 *Let $\chi \in C^\infty(X,\mathbb{R})$ and $\omega' = \omega + \mathrm{dd}^{\mathrm{c}}\chi > 0$. The map $[u] \mapsto [u-\chi]$ is a uniform isomorphism*

$$(\mathfrak{S}(X,\omega), \tau_{\omega,F}) \longrightarrow (\mathfrak{S}(X,\omega'), \tau_{\omega',F-\chi}).$$

For $s > 0$, $[u] \mapsto [su]$ is a uniform isomorphism from $\tau_{\omega,F}$ to $\tau_{s\omega,sF}$. Both maps transport the ordinary and the ideal envelopes exactly.

Proof For either envelope operator P, the correspondence of competitors gives

$$P_{\omega',F-\chi}[u-\chi] = P_{\omega,F}[u] - \chi, \qquad P_{s\omega,sF}[su] = sP_{\omega,F}[u].$$

Smooth additions leave all multiplier ideals and divisorial Lelong numbers unchanged. Scaling multiplies the divisorial and envelope distances by s and changes the ideal weights by $t \mapsto st$. Each map and its inverse therefore transport the generating entourages to generating entourages. □

For two smooth obstacles, translation of competitors gives

$$P_{\omega,F}[u] + \min_X(G-F) \le P_{\omega,G}[u] \le P_{\omega,F}[u] + \max_X(G-F),$$

and the same inequalities hold for the ideal envelopes. If instead the zero obstacle is kept in both backgrounds, the transport gives $P_{\omega',0}[u-\chi] = P_{\omega,\chi}[u] - \chi$, so the

change of background compares the obstacles 0 and χ. The bounded comparison of obstacles does not make this change continuous, even on a curve.

Proposition 17.6.4 *Let Y be a compact connected Riemann surface and $\theta > 0$ a smooth form. If F is a nonconstant smooth function with $\theta' = \theta + \mathrm{dd^c} F > 0$, then the bijection*

$$\Gamma_F \colon (\mathfrak{S}(Y,\theta), \tau_{\theta,0}) \longrightarrow (\mathfrak{S}(Y,\theta'), \tau_{\theta',0}), \qquad [u] \longmapsto [u - F],$$

and its inverse are both discontinuous. Equivalently, $\tau_{\theta,0}$ and $\tau_{\theta,F}$ are incomparable topologies on the same space of ordinary types.

Proof For $s > 0$, the uniform isomorphisms $[u] \mapsto [su]$ of Proposition 17.6.3 conjugate Γ_F to Γ_{sF} and map $\tau_{\theta,F}$ to $\tau_{s\theta,sF}$, so we may assume $\int_Y \theta = 1$. The smooth function θ'/θ has integral one against the probability measure θ, and it is not constant: otherwise $\mathrm{dd^c} F = 0$, and the harmonic function F would be constant on Y. Choose c with $\min_Y(\theta'/\theta) < c < 1$, so that $\theta' \not\geq c\theta$.

Next choose probability measures μ_j without atoms, each carried by a polar set, that converge weakly to θ. To construct them, start with finite atomic probability measures converging weakly to θ, as in Theorem 17.4.1, and replace each atom by a nonatomic probability measure carried by a compact polar set in a small coordinate disk about that atom, keeping its weight; such measures exist in every disk by Lemma 1.6.5. If the diameters of the disks tend to zero uniformly, uniform continuity of continuous test functions preserves weak convergence, and the finite union of the polar carriers is polar.

Let q_j be the normalized Green potentials with $\theta + \mathrm{dd^c} q_j = \mu_j$ and $\sup_Y q_j = 0$. By normalized compactness, the sequence has L^1 cluster values, and each of them has curvature θ and supremum zero, hence vanishes. Thus $q_j \to 0$ in L^1. Put $u_j = cq_j$. Then

$$\theta + \mathrm{dd^c} u_j = (1-c)\theta + c\mu_j, \qquad u_j \leq P_\theta[u_j] \leq 0.$$

Since μ_j has no atoms, all point Lelong numbers of u_j vanish. Because modifications of a smooth curve are isomorphisms, Theorem 1.16.1 gives trivial multiplier ideals at every weight. Thus $P_\theta[u_j]_{\mathcal{I}} = 0$, all controlled ideal relations with $[0]$ hold, and every divisorial distance to $[0]$ vanishes. The bounds above give convergence of the ordinary envelopes to zero in L^1. Consequently $[u_j] \to [0]$ in $\tau_{\theta,0}$.

Set $v_j = P_{\theta,F}[u_j]$. Every competitor $h \preceq u_j$ satisfies $\theta + \mathrm{dd^c} h \geq c\mu_j$, by Corollary 1.6.1. The family of competitors below F contains a translate of u_j and is stable under finite maxima, so Choquet selection and distributional convergence of increasing maxima show that v_j satisfies the same inequality. If $[u_j] \to [0]$ in $\tau_{\theta,F}$, the ordinary-envelope coordinate would give $v_j \to P_{\theta,F}[0] = F$ in L^1, because F is a bounded θ-plurisubharmonic competitor. Passing to the limit in the current inequality would then give $\theta' \geq c\theta$, a contradiction.

By Proposition 17.6.3, $[u] \mapsto [u - F]$ is a uniform isomorphism from $\tau_{\theta,F}$ onto $\tau_{\theta',0}$, so the sequence above shows that Γ_F is discontinuous. The same argument applied to θ' and $-F$ shows that the inverse is discontinuous. Under this transport,

continuity of Γ_F and of its inverse are precisely the two comparisons between $\tau_{\theta,0}$ and $\tau_{\theta,F}$, which proves incomparability. □

The sequence u_j has trivial multiplier ideals, and its polar curvature $\mathbf{1}_{\{u_j=-\infty\}}(\theta + \mathrm{dd^c} u_j)$ (Section 1.6.1) is $c\mu_j$. By the classification of curve models through their polar curvature in Lemma 31.13.1 and the bounded obstacle comparison, its ordinary envelopes in both backgrounds have non-pluripolar mass $1 - c$ when $\int_Y \theta = 1$. The discontinuity therefore persists among models whose masses are bounded below by a fixed positive constant. On the same curve, the total variation of polar curvature is a complete metric on ordinary models, by Theorem 31.13.1 and Corollary 31.13.1, and after ordinary projection it is invariant under smooth changes of background; it keeps the diffuse curvature that the sequence above loses.

17.6.4 All smooth obstacles

We return to the Hermitian background ω on X. By Proposition 17.6.4, a smooth change of background need not preserve convergence when the obstacle is kept at zero. Requiring convergence below every smooth obstacle gives a topology that is stable under such changes. The dyadic approximation still determines its analytic closure, which is now described by equality of the two envelopes below every obstacle.

Let $\tau_\omega^{\mathrm{all}}$ be the topology of the uniformity generated by all the uniformities $\tau_{\omega,F}$ with $F \in C^\infty(X, \mathbb{R})$. Thus an entourage involves finitely many distances between obstacle envelopes, together with finitely many of the controlled ideal relations and divisorial distances defined above.

Proposition 17.6.5 *A countable family $\{F_k\}_{k\geq 1} \subset C^\infty(X, \mathbb{R})$ that is dense for the uniform norm suffices for the obstacle coordinates of $\tau_\omega^{\mathrm{all}}$. For smooth χ with $\omega' = \omega + \mathrm{dd^c}\chi > 0$, the map $[u] \mapsto [u - \chi]$ is a uniform isomorphism between $\tau_\omega^{\mathrm{all}}$ and $\tau_{\omega'}^{\mathrm{all}}$. For $s > 0$, positive scaling gives the corresponding uniform isomorphism onto $\tau_{s\omega}^{\mathrm{all}}$. This topology is not T_0 on ordinary classes.*

Proof For either envelope operator, the obstacle comparison gives

$$\|P_{\omega,F}[u] - P_{\omega,G}[u]\|_{L^1(\mathrm{d}V)} \leq \|F - G\|_\infty.$$

Hence the two envelope distances for a pair $[u]$, $[v]$ differ by at most $2\|F - G\|_\infty$, uniformly in the pair. The space $C^\infty(X, \mathbb{R})$ contains a countable subset that is dense for the uniform norm: approximate each member of a countable dense subset of $C(X)$ by smooth functions with errors $1/m$, $m \geq 1$. Given an entourage, we approximate each of its finitely many obstacles by a member of the family and shrink the radii accordingly; this proves that the family generates the same uniformity. The divisorial coordinates still range over all fixed compact modifications.

The transport in Proposition 17.6.3 relabels the entire family of smooth obstacles by $F \mapsto F - \chi$, or by $F \mapsto sF$. These are bijections of the family, which proves both isomorphisms.

For nonseparation, use the unbounded function $u = 1 - \sqrt{1-w}$ of Proposition 17.6.2. It has trivial multiplier ideals and zero divisorial Lelong numbers, and $\delta w \preceq u$ for $0 < \delta \leq 1$. Fix F and put $q = P_\omega(F)$, the envelope of the obstacle F in the sense of Section 1.19. Since $\omega > 0$, the constant $\min_X F$ is a competitor, so q is bounded and $q \leq F$. The function

$$h_\delta = (1-\delta)q + \delta w - \delta\|F\|_\infty$$

is ω-plurisubharmonic, lies below F, and satisfies $h_\delta \preceq u$. Thus $h_\delta \leq P_{\omega,F}[u] \leq q$. Letting $\delta \downarrow 0$ off the pole set of w and applying Lemma 1.3.3 gives $P_{\omega,F}[u] = q = P_{\omega,F}[0]$. Since u and 0 have the same multiplier ideals, their ideal envelopes below F also agree. Every generating entourage therefore has identical sections at $[u]$ and $[0]$, although these ordinary classes differ. □

Below a smooth obstacle, the ideal envelope is the ordinary envelope of the ideal model obtained with the zero obstacle. Equality of the two envelopes below every obstacle therefore reduces to a question about ordinary envelopes alone.

Proposition 17.6.6 *Let θ be a smooth real $(1,1)$-form, let $u \in \mathrm{PSH}(X,\theta)$ and put $q = P_\theta[u]_{\mathcal{I}}$. For every smooth real F,*

$$P_{\theta,F}[q] = P_{\theta,F}[u]_{\mathcal{I}} \sim q.$$

If $u \in \mathcal{E}_{\mathrm{env}}(X,\theta,q)$, then also

$$P_{\theta,F}[u] = P_{\theta,F}[u]_{\mathcal{I}} \quad \textit{for every smooth real } F.$$

Proof Set $q_F = P_{\theta,F}[u]_{\mathcal{I}}$. Translation of competitors gives

$$q + \min_X F \leq q_F \leq q + \max_X F,$$

so $q_F \sim q$. The identity $q_F - F = P_{\theta+\mathrm{dd^c}F}[u-F]_{\mathcal{I}}$ and Theorem 17.2.1 give $q_F \sim_{\mathcal{I}} u$. Since $q_F \leq F$ and $q_F \sim q$, it is a competitor for $P_{\theta,F}[q]$. Conversely, the maximum of a competitor for $P_{\theta,F}[q]$ with a sufficiently low translate of q is a competitor for $P_{\theta,F}[q]_{\mathcal{I}}$, and this ideal envelope equals q_F because $q \sim_{\mathcal{I}} u$. These bounds give

$$q_F \leq P_{\theta,F}[q] \leq P_{\theta,F}[q]_{\mathcal{I}} = q_F.$$

The last assertion follows from Proposition 14.4.3, which gives $P_{\theta,F}[u] = P_{\theta,F}[q]$.□

Theorem 17.6.2 *The closure and the sequential closure of $\mathcal{A}_\omega$ in $\tau_\omega^{\mathrm{all}}$ are both equal to*

$$\mathcal{G}_\omega^{\mathrm{all}} = \{[u] : P_{\omega,F}[u] = P_{\omega,F}[u]_{\mathcal{I}} \text{ for every smooth } F\}.$$

It suffices to test equality on the countable dense family above. The dyadic sequence b_j of Corollary 17.5.3 converges to $[u]$ in this topology if and only if $[u] \in \mathcal{G}_\omega^{\mathrm{all}}$.

Proof Fix a smooth obstacle F and put $\theta_F = \omega + \mathrm{dd}^{\mathrm{c}} F$. Translation by F identifies the competitor families, so that

$$P_{\omega,F}[v] = P_{\theta_F}[v - F] + F, \qquad P_{\omega,F}[v]_{\mathcal{I}} = P_{\theta_F}[v - F]_{\mathcal{I}} + F$$

for every $v \in \mathrm{PSH}(X, \omega)$. If v has analytic singularities, so has $v - F$, and Proposition 17.2.4, applied to $v - F$ in the background θ_F, gives $P_{\omega,F}[v] = P_{\omega,F}[v]_{\mathcal{I}}$. In particular, analytic classes lie in $\mathcal{G}_\omega^{\mathrm{all}}$.

The functions $b_j - F$ belong to $\mathrm{PSH}(X, \theta_F)$ and satisfy $u - F \preceq b_j - F$, hence $u - F \preceq_{\mathcal{I}} b_j - F$. By the divisorial estimate in the proof of Theorem 17.6.1, which holds for every local divisor E over X, $\nu_E(b_j) \to \nu_E(u)$. Proposition 17.5.1, applied in the background θ_F with all $\epsilon_j = 0$, therefore gives

$$P_{\omega,F}[b_j] = P_{\omega,F}[b_j]_{\mathcal{I}} \longrightarrow P_{\omega,F}[u]_{\mathcal{I}} \quad \text{in } L^1.$$

The controlled ideal relations and the divisorial distances on fixed modifications do not involve F, and they converge by the proof of Theorem 17.6.1. Consequently the dyadic sequence converges in $\tau_\omega^{\mathrm{all}}$ exactly when $P_{\omega,F}[u] = P_{\omega,F}[u]_{\mathcal{I}}$ for every F, by uniqueness of L^1 limits and Lemma 1.3.3.

For each F, equality of the two L^1 envelope coordinates is a closed condition. Since analytic classes satisfy all these equalities, the closure of the analytic classes under nets lies in $\mathcal{G}_\omega^{\mathrm{all}}$, and the dyadic sequence proves the converse. Finally, the uniform obstacle bound in the proof of Proposition 17.6.5 extends equality from a dense family to every smooth obstacle. □

Taking $F = 0$ gives $\mathcal{G}_\omega^{\mathrm{all}} \subseteq \mathcal{G}_\omega$. For the reverse inclusion, let $q = P_\omega[u]_{\mathcal{I}}$. By Proposition 17.6.6, it suffices to show that $P_{\omega,F}[u] = P_{\omega,F}[q]$ for every F, and membership $u \in \mathcal{E}_{\mathrm{env}}(X, \omega, q)$ is sufficient for this. Under bounded mass and positive lower volume of q, this membership is equivalent to equality of the envelopes below zero, by Proposition 31.8.1. On curves the two closures coincide even at zero mass, by Proposition 31.13.3. We do not know whether the reverse inclusion holds in general. The next subsection proves equality for potentials with prescribed countable poles, and Chapter 32 proves it for explicit classes of zero volume obtained by reduction along a divisor in a fibration, including one on a manifold without the bounded mass property. Although $\tau_\omega^{\mathrm{all}}$ removes the dependence on a fixed obstacle, it still has incomplete curve families and is strictly weaker than polar total variation on ordinary curve models, by Proposition 31.13.4. On the product and fibration model loci of Chapter 32, complete metrics come instead from polar curvature.

17.6.5 Countable poles and all obstacles

If a potential w is at least as singular as every multiple bu, $b \geq 1$, of a potential u, then convex combinations with w supply ordinary competitors for u below every continuous obstacle; this is the mechanism of the single-pole example in the proof of Proposition 17.6.5. We construct such a pair with poles on any prescribed countable set. The potentials have trivial multiplier ideals even when their poles are dense, so both envelopes lose this ordinary singularity information.

We keep the Hermitian form ω and the volume form $\mathrm{d}V$. Envelopes below a continuous obstacle F are defined by the competitor families of Definitions 1.19.1 and 17.2.1 with the bound $v \leq F$.

Theorem 17.6.3 *For every nonempty finite or countably infinite subset $\{x_j\} \subset X$, there are nonpositive $u, w \in \mathrm{PSH}(X, \omega)$ such that*

$$\omega + \mathrm{dd^c} u,\ \omega + \mathrm{dd^c} w \geq \tfrac{3}{4}\omega, \qquad u(x_j) = w(x_j) = -\infty.$$

The same w satisfies $tw \preceq u$ for every $0 < t \leq 1$, and

$$u \in \mathcal{E}_{\mathrm{env}}(X, \omega, 0), \qquad \mathrm{e}^{-su},\ \mathrm{e}^{-sw} \in L^1(X, \mathrm{d}V) \quad (s > 0).$$

Thus $\mathcal{I}(su) = \mathcal{I}(sw) = \mathcal{O}_X$ for every $s > 0$. If the prescribed set is dense, both potentials are unbounded on every nonempty open set.

Proof Fix $0 < \beta < \beta' < 1$, and index the prescribed points by integers $j \geq 1$. Choose a finite atlas of relatively compact charts with larger coordinate neighborhoods, on which the coefficients of $\mathrm{d}V$ are uniformly bounded. For each j, let z_j be the coordinate of x_j in a chart of the atlas containing it, and choose a coordinate ball of radius $r_j \leq 1$ centered at z_j and contained in the corresponding larger neighborhood. Let χ be a smooth cutoff supported inside the unit ball, with $0 \leq \chi \leq 1$ and $\chi = 1$ near zero. On this ball set

$$\chi_j(z) = \chi((z - z_j)/r_j), \qquad L_j = 1 - \log(|z - z_j|^2/r_j^2).$$

The functions $-\chi_j L_j^{\sigma}$, for $\sigma = \beta, \beta'$, extend by zero outside the ball. Near the center, where $\chi_j = 1$, they are plurisubharmonic off z_j: for $s = \log(|z - z_j|^2/r_j^2)$ the profile $q_\sigma(s) = -(1 - s)^{\sigma}$ is convex and increasing, since

$$q'_\sigma = \sigma(1 - s)^{\sigma - 1} > 0, \qquad q''_\sigma = \sigma(1 - \sigma)(1 - s)^{\sigma - 2} > 0.$$

Being nonpositive, they extend across z_j with the value $-\infty$ by Lemma 1.3.7. On the annulus where the cutoff varies they are smooth, so their complex Hessians have a finite lower bound.

Put $e_j = 2^{-j-2}$. Scale both profiles by the same $0 < \delta_j \leq 2^{-j}$ so that the nonpositive functions

$$a_j = -\delta_j \chi_j L_j^{\beta}, \qquad c_j = -\delta_j \chi_j L_j^{\beta'}, \qquad \mathrm{dd^c} a_j,\ \mathrm{dd^c} c_j \geq -e_j \omega \tag{17.7}$$

have these lower bounds globally; only the cutoff annulus contributes a negative Hessian. Coordinate rescaling and the finite atlas give

$$\|a_j\|_{L^1} + \|c_j\|_{L^1} \le C\delta_j.$$

Consequently the nonpositive partial sums decrease with uniformly bounded L^1 norms. Since $\sum_{j\ge1} e_j = \frac14$, the partial sums $\sum_{j\le N} a_j$ and $\sum_{j\le N} c_j$ have complex Hessians at least $-\frac14\omega$. By Corollary 1.4.1, their pointwise limits

$$u = \sum_j a_j, \qquad w = \sum_j c_j$$

belong to $\mathrm{PSH}(X,\omega)$, and the lower bound $\frac34\omega$ for their currents passes to the limits. Each prescribed point is a pole of its own summand, and the other summands are nonpositive, so $u(x_j) = w(x_j) = -\infty$. This also proves the assertion when the prescribed set is dense.

For $b \ge 1$ let

$$S = \sum_j \delta_j > 0, \qquad M_b = \max\{0, \sup_{L\ge1}(bL^\beta - L^{\beta'})\} < \infty.$$

Multiplying the inequality $-L^{\beta'} \le -bL^\beta + M_b$, valid for $L \ge 1$, at $L = L_j$ by $\delta_j\chi_j \ge 0$ and summing over j gives

$$w \le bu + M_bS. \tag{17.8}$$

The inequality first holds for finite sums and then for their decreasing limits, including common poles. Taking $b = 1/t$ gives $tw \preceq u$; here $tw = tw + (1-t)\cdot 0$ is ω-plurisubharmonic by convexity, since $\omega > 0$. Also, $b^{-1}(w - M_bS) \le u$, so $w - M_bS$ belongs to the family $\mathcal{A}_b(u,0)$ of stretching competitors for every $b \ge 1$, and Definition 14.4.1 gives $u \in \mathcal{E}_{\mathrm{env}}(X,\omega,0)$.

It remains to prove integrability at every positive exponential scale. Fix $s > 0$ and use Jensen's inequality with weights δ_j/S, assigning the unused weight in each finite sum to zero:

$$\mathrm{e}^{-s\sum_{j\le N} a_j} \le 1 - \sum_{j\le N}\frac{\delta_j}{S} + \sum_{j\le N}\frac{\delta_j}{S}\mathrm{e}^{sS\chi_jL_j^\beta}.$$

The integrals on the right are bounded uniformly in j, since coordinate rescaling reduces the contribution inside the ball to a constant times

$$\int_0^1 \exp\bigl(sS(1-\log R^2)^\beta\bigr)R^{2n-1}\,\mathrm{d}R.$$

With $x = -\log R^2$, its exponent is $-nx + sS(1+x)^\beta$, which is at most $-nx/2$ for large x. Integrating the Jensen inequality and using monotone convergence proves $\mathrm{e}^{-su} \in L^1$. The same calculation with β' proves the assertion for w. Every holomorphic germ is

bounded on a smaller coordinate neighborhood, so Definition 1.12.2 gives the stated trivial ideals. □

Proposition 17.6.7 *For the potentials of Theorem 17.6.3 and every $F \in C(X, \mathbb{R})$,*

$$P_{\omega,F}[u] = P_{\omega,F}[u]_{\mathcal{I}} = P_\omega(F).$$

In particular, $P_\omega[u] = P_\omega[u]_{\mathcal{I}} = 0$.

Proof The proof of the nonseparation assertion of Proposition 17.6.5 uses only that $w \le 0$ is ω-plurisubharmonic with $\delta w \preceq u$ for $0 < \delta \le 1$, that u has trivial multiplier ideals, and that F is continuous. By Theorem 17.6.3, these conditions hold here, so that proof gives $P_{\omega,F}[u] = P_{\omega,F}[0] = P_\omega(F)$ and $P_{\omega,F}[u]_{\mathcal{I}} = P_{\omega,F}[0]_{\mathcal{I}}$. The latter equals $P_\omega(F)$ because $P_{\omega,F}[0] \le P_{\omega,F}[0]_{\mathcal{I}} \le P_\omega(F)$. For $F = 0$, zero is a competitor, so $P_\omega(0) = 0$. □

For a dense prescribed set, the ordinary type of u differs from the bounded type, although their envelopes agree below every continuous obstacle. The construction applies to every Hermitian form ω, including one with unbounded upper mass or zero lower mass.

17.7 Small curvature and strict ordinary types

Multiplier ideals can miss an unbounded residual. To decide whether an ordinary type containing such a residual admits a strict current, that is, a current $\theta + dd^c v$ bounded below by a positive multiple of a Hermitian form, we need representatives in that same ordinary type with a controlled curvature bound. Fix a Hermitian form h on X and a quasi-plurisubharmonic function q such that

$$\text{for every } \delta > 0 \text{ there is } q_\delta \sim q \quad \text{with} \quad dd^c q_\delta \ge -\delta h. \tag{17.9}$$

This property does not depend on the choice of h, since any two Hermitian forms are uniformly comparable on X.

Lemma 17.7.1 *If (17.9) holds, then q has zero point Lelong numbers and zero polar curvature. The pullback of q to every smooth modification has the same small-curvature property, and its point Lelong numbers and polar curvature also vanish.*

Proof The function q_δ/δ belongs to $\mathrm{PSH}(X, h)$. The uniform Lelong bound of Lemma 1.17.7, and invariance under bounded changes, give

$$0 \le \nu(q, x) = \nu(q_\delta, x) = \delta \nu(q_\delta/\delta, x) \le C_h \delta.$$

Letting δ tend to zero proves the first assertion.

For polar curvature, put $S = \{q = -\infty\}$ and let $R = \mathbf{1}_S \mathrm{dd}^{\mathrm{c}} q$, a closed positive current by Proposition 1.25.3. Choose a Gauduchon metric γ when $n \geq 2$, using Theorem 4.1.1; when $n = 1$ take any Hermitian form and use $\gamma^0 = 1$. Bounded-type invariance in Proposition 1.25.1 gives $R = \mathbf{1}_S(\delta h + \mathrm{dd}^{\mathrm{c}} q_\delta)$. The positive current $\delta h + \mathrm{dd}^{\mathrm{c}} q_\delta$ has total γ-trace $\delta \int_X h \wedge \gamma^{n-1}$ by distributional Stokes. Consequently

$$0 \leq \int_X R \wedge \gamma^{n-1} \leq \delta \int_X h \wedge \gamma^{n-1}.$$

Thus $R = 0$ by positivity and $\delta \downarrow 0$.

Now let $\pi \colon X' \to X$ be a smooth modification, and fix a Hermitian form h' on X'. The pullbacks $\pi^* q = q \circ \pi$ and $\pi^* q_\delta$ are quasi-plurisubharmonic by Proposition 1.22.4. Their ordinary inequalities hold pointwise, including at poles, and $\pi^* h \leq C h'$ for some fixed C. Hence $\pi^* q_\delta \sim \pi^* q$ and $\mathrm{dd}^{\mathrm{c}} \pi^* q_\delta \geq -C\delta h'$. Since $C\delta$ can be arbitrarily small, the preceding argument on X' proves both vanishing assertions. □

The point Lelong statement and the polar statement use different parts of the current: an isolated logarithmic pole in dimension at least two has a positive Lelong number, but the complex Hessian of its potential gives it no mass. Small curvature controls both.

Lemma 17.7.2 *Let B be a smooth real $(1,1)$-form. If q' is quasi-plurisubharmonic, all its point Lelong numbers vanish, and $B + \mathrm{dd}^{\mathrm{c}} q' \geq \epsilon h$ with $\epsilon > 0$, then there is a smooth real f such that $B + \mathrm{dd}^{\mathrm{c}} f \geq \epsilon h/2$.*

Proof Apply the smooth regularization of closed currents recalled in Section 1.15.1 to $T = \mathrm{dd}^{\mathrm{c}} q'$, with smooth representative zero and the lower bound $\gamma = \epsilon h - B$, which need not be closed. Since X is compact, the Hermitian form used there is bounded by a constant multiple of h, and since all Lelong numbers of q' vanish, the loss of positivity tends to zero uniformly. The regularization therefore gives smooth functions f_k on X and constants $\delta_k \to 0$ with $\mathrm{dd}^{\mathrm{c}} f_k \geq \gamma - \delta_k h$. Choose k with $\delta_k \leq \epsilon/2$ and set $f = f_k$; then $B + \mathrm{dd}^{\mathrm{c}} f \geq \epsilon h/2$. □

Let $\Delta = \sum_{j=1}^{N} a_j D_j$ be a finite effective real divisor, where the D_j are distinct prime divisors and $a_j > 0$. Choose smooth Hermitian metrics on the line bundles $\mathcal{O}_X(D_j)$, with canonical sections s_j, and put

$$\sigma_\Delta = \sum_j a_j \log \|s_j\|^2, \qquad \mathrm{dd}^{\mathrm{c}} \sigma_\Delta = [\Delta] - \chi_\Delta,$$

where χ_Δ is smooth and closed. The case $\Delta = 0$ means $\sigma_\Delta = \chi_\Delta = 0$.

Theorem 17.7.1 *Suppose (17.9) holds and put $B = \theta - \chi_\Delta$. The ordinary type of $\sigma_\Delta + q$ contains a strict θ-current, that is, some $v \sim \sigma_\Delta + q$ satisfies $\theta + \mathrm{dd}^{\mathrm{c}} v \geq \epsilon h$ with $\epsilon > 0$, if and only if B has a smooth strictly positive representative: $B + \mathrm{dd}^{\mathrm{c}} f > 0$ for some smooth real f.*

Proof Suppose first that $v \sim \sigma_\Delta + q$ and $\theta + \mathrm{dd^c} v \geq \epsilon h$ with $\epsilon > 0$. Lelong numbers are additive and invariant under bounded changes, so Lemma 17.7.1 gives $\nu(v, D_j) = a_j$. Apply Lemma 1.17.1 with background $\theta - \epsilon h$ to obtain

$$\theta + \mathrm{dd^c} v - [\Delta] \geq \epsilon h.$$

For $\Delta = 0$ this is the original strict inequality. The L^1 difference $v - \sigma_\Delta$ consequently has a global quasi-plurisubharmonic representative q' with $B + \mathrm{dd^c} q' \geq \epsilon h$. Off the divisor it satisfies $q' \sim q$; these inequalities extend everywhere by Lemma 1.3.3. Thus q' has zero point Lelong numbers, and Lemma 17.7.2 gives $B + \mathrm{dd^c} f > 0$ for a smooth f.

Conversely, compactness gives $B + \mathrm{dd^c} f \geq \epsilon h$ for some $\epsilon > 0$. Choose q_δ in (17.9) with $\delta < \epsilon/2$. Then $v = \sigma_\Delta + q_\delta + f$ is in the specified ordinary type, and Poincaré–Lelong gives

$$\theta + \mathrm{dd^c} v = [\Delta] + B + \mathrm{dd^c} f + \mathrm{dd^c} q_\delta \geq \epsilon h/2.$$

Part IV
Interpolation and restriction

Chapter 18
Perron interpolation and the geodesic equation

Let X be a connected compact complex manifold of dimension $n \geq 1$, and let θ be a smooth real $(1,1)$-form. Fix a smooth positive volume form $\mathrm{d}V$ for the $L^1(X)$ topology.

This part studies two operations on singular potentials: interpolation between two potentials along an auxiliary real parameter, and restriction to a complex submanifold. A one-parameter family of θ-plurisubharmonic functions can be regarded as a single function on the product of X with a strip in $\mathbb{C}$, and positivity of its curvature current on the product is a convexity condition in the parameter. The largest such family with prescribed endpoints, the Perron envelope, interpolates between two potentials; in Kähler geometry it gives the weak geodesics of the space of potentials. A Legendre transform in the parameter relates rays made of Perron segments to concave decreasing families of envelope fixed points, the test curves of Chapter 19. Restriction behaves differently: the restriction of a potential to a submanifold can acquire poles that are invisible to its multiplier ideals, and Chapter 20 works instead with the information carried by the restricted multiplier ideals. All these constructions use only positivity of currents, so they apply to nonclosed backgrounds.

In Kähler geometry, the smooth geodesic equation for the Mabuchi L^2 metric on potentials is a homogeneous Monge–Ampère equation on such a product. Positivity on the product still defines Perron interpolation for a nonclosed background, but the two constructions can now differ: integration by parts in the L^2 action produces torsion terms. Even a smooth Perron segment given by translation by constants can fail to be stationary for that action. On the scalar Hopf manifolds $(\mathbb{C}^n \setminus \{0\})/\langle z \mapsto \mathrm{e}^{-1} z \rangle$, $n \geq 2$, the finite-energy potentials pulled back from the projective base form a complete CAT(0) space whose geodesics are Perron segments, while translations by constants, though stationary, do not minimize the L^2 action.

The Perron construction itself rests on convexity and barriers. Convexity in the parameter bounds every admissible path by the chord between its endpoints, while a lower barrier ensures that the envelope attains the endpoints. For two potentials of the same ordinary singularity type, two affine barriers suffice and give a Lipschitz bound in the parameter. Maximality on each finite subinterval is the property that links Perron segments with Legendre duality in Chapter 19.

18.1 Subgeodesics and endpoint bounds

As in Lemma 1.20.1, for a nonempty open interval $I \subset \mathbb{R}$ write

$$S_I = \{z \in \mathbb{C} : \operatorname{Re} z \in I\}, \qquad \pi \colon X \times S_I \longrightarrow X,$$

where π is the projection. A family $u_t \in \mathrm{PSH}(X,\theta)$, $t \in I$, is a *subgeodesic* if its complexification $U(x,z) = u_{\operatorname{Re} z}(x)$ is quasi-plurisubharmonic and satisfies $\pi^*\theta + \mathrm{dd}^{\mathrm{c}} U \geq 0$ on $X \times S_I$. For finite $a < b$ and $u_a, u_b \in \mathrm{PSH}(X,\theta)$, a subgeodesic on (a,b) *runs from* u_a *to* u_b if $u_t \to u_a$ in $L^1(X)$ as $t \downarrow a$ and $u_t \to u_b$ as $t \uparrow b$.

The elementary operations below produce lower barriers. Convexity in t, given by Lemma 1.20.1, and the minimum principle of Section 1.20 then give the continuity and upper bounds needed for the Perron envelope.

Proposition 18.1.1 *Subgeodesics are preserved by adding an affine function of t, by a positive affine reparametrization of time, by finite maxima, and by convex combinations. For $h \in C^\infty(X,\mathbb{R})$, replacing θ by $\theta + \mathrm{dd}^{\mathrm{c}} h$ and u_t by $u_t - h$ also preserves them.*

Proof Affine functions of $\operatorname{Re} z$ have zero complex Hessian, and a positive affine change of time is induced by a holomorphic affine change of the strip coordinate. Convex combinations preserve the inequality between currents, and the change $(\theta, u_t) \mapsto (\theta + \mathrm{dd}^{\mathrm{c}} h, u_t - h)$ preserves the current itself. Maxima are covered by Lemma 1.3.1. □

Proposition 18.1.2 *A subgeodesic is continuous in $L^1(X)$ on the interior of its parameter interval. If the interval is (a,b) with $a < b$ finite, and the subgeodesic runs from u_a to u_b, where $u_a, u_b \in \mathrm{PSH}(X,\theta)$, then*

$$u_t \leq \frac{b-t}{b-a} u_a + \frac{t-a}{b-a} u_b, \qquad a < t < b. \tag{18.1}$$

Moreover, the following upper limits, taken jointly in space and time, hold at the endpoints:

$$\limsup_{\substack{y \to x \\ t \downarrow a}} u_t(y) \leq u_a(x), \qquad \limsup_{\substack{y \to x \\ t \uparrow b}} u_t(y) \leq u_b(x).$$

Proof For a compact interval $J \Subset I$, choose $a' < \inf J \leq \sup J < b'$ in I. By Lemma 1.20.1, the path lies below the chord through $u_{a'}$ and $u_{b'}$, hence below a constant on J, and by Proposition 1.20.1 the function $\inf_{a'<t<b'} u_t$ is an integrable lower bound. For almost every x, the function $t \mapsto u_t(x)$ is finite and convex, hence continuous, and dominated convergence proves continuity in L^1 on J.

For the chord bound, choose sequences of times tending to a and b along which the assumed L^1 convergence holds almost everywhere. Applying convexity between these times and passing to the limit proves (18.1) almost everywhere in x, for each t, and hence everywhere by Lemma 1.3.3.

For the joint upper limit at a, note that in (18.1) the coefficient of u_a tends to one, while the term involving u_b has a vanishing coefficient and a uniform upper bound.

Upper semicontinuity of u_a gives the assertion when $u_a(x)$ is finite. At a pole of u_a, bound u_a near x above by an arbitrary real number, take the joint upper limit, and let that number tend to $-\infty$. The argument at b is the same. □

18.2 Perron segments

The chord bound survives taking a regularized supremum, and a single competitor attaining the endpoints bounds that supremum from below and forces the same endpoint limits. Fix $u_0, u_1 \in \mathrm{PSH}(X,\theta)$ and call a subgeodesic on $(0,1)$ *admissible* if its joint upper limits at $t = 0$ and $t = 1$, in the sense of Proposition 18.1.2, are at most u_0 and u_1, respectively. Define the *Perron envelope*

$$U = \left(\sup\{V : V \text{ is an admissible complexification}\}\right)^*,$$

where the upper semicontinuous regularization is taken on $X \times S_{(0,1)}$, and denote the slices of U by u_t^{P}. When these slices form a subgeodesic running from u_0 to u_1, they define the *Perron segment* from u_0 to u_1, and we set $u_0^{\mathrm{P}} = u_0$ and $u_1^{\mathrm{P}} = u_1$. An affine change of parameter gives the same construction on any finite interval.

Proposition 18.2.1 *If an admissible subgeodesic exists, the Perron envelope is the largest admissible subgeodesic. If some admissible subgeodesic runs from u_0 to u_1, so does the envelope, which is then the Perron segment. In that case, for every $[a,b] \subset [0,1]$ with $a < b$, its restriction to $[a,b]$ is the Perron segment between u_a^{P} and u_b^{P}.*

Proof Convexity in t and the upper limits at the endpoints bound every competitor by $(1-t)u_0 + tu_1$. This bound is jointly upper semicontinuous on the interior and locally bounded above, so it persists after regularization. The local envelope theorem of Section 1.19 shows that U is quasi-plurisubharmonic with $\pi^*\theta + \mathrm{dd}^{\mathrm{c}}U \geq 0$, and U is still independent of $\Im z$. Since it dominates an admissible competitor, each slice belongs to $\mathrm{PSH}(X,\theta)$. The chord bound gives the joint upper limits at the endpoints, by the argument at the end of the proof of Proposition 18.1.2. The envelope therefore belongs to its defining family and is its largest member.

If a competitor w_t runs from u_0 to u_1, then

$$w_t \leq u_t^{\mathrm{P}} \leq (1-t)u_0 + tu_1.$$

In particular, almost everywhere,

$$|u_t^{\mathrm{P}} - u_0| \leq |w_t - u_0| + t|u_1 - u_0|,$$

which proves $u_t^{\mathrm{P}} \to u_0$ in L^1 as $t \downarrow 0$. The argument at $t = 1$ is the same.

For $0 \leq a < b \leq 1$, let v_t be the Perron envelope on (a,b) between u_a^{P} and u_b^{P}. The original path restricted to (a,b) runs between these endpoints and is admissible

by Proposition 18.1.2. Thus v_t also runs between them and dominates u_t^{P}. The joint upper limits of v_t at a and b allow us to glue its complexification to U outside the smaller strip, by Lemma 1.3.2; if $a = 0$ or $b = 1$, the same upper limits show that the glued family is still admissible for u_0, u_1. The resulting competitor cannot exceed U, which proves $v_t = u_t^{\mathrm{P}}$ on (a, b). □

When the endpoints differ by a bounded function, an explicit lower barrier is available.

Theorem 18.2.1 *Suppose that $u_0 \sim u_1$, and choose $C \geq 0$ with $u_0 - C \leq u_1 \leq u_0 + C$. Then the Perron segment from u_0 to u_1 exists, every slice has the ordinary singularity type of u_0 and u_1, and*

$$u_0 - Ct \leq u_t^{\mathrm{P}} \leq u_0 + Ct, \qquad u_1 - C(1-t) \leq u_t^{\mathrm{P}} \leq u_1 + C(1-t).$$

Moreover, away from the common pole set,

$$|u_t^{\mathrm{P}} - u_s^{\mathrm{P}}| \leq C|t - s|, \qquad 0 \leq s, t \leq 1.$$

Proof Consider the barrier

$$w_t = \max\{u_0 - Ct,\ u_1 - C(1-t)\}.$$

It is a subgeodesic by Proposition 18.1.1, and the bounded difference gives $w_0 = u_0$ and $w_1 = u_1$. Upper semicontinuity gives the joint upper limits at the endpoints. Off the common poles, $|w_t - u_0| \leq Ct$ and $|w_t - u_1| \leq C(1-t)$, so w_t runs from u_0 to u_1, and Proposition 18.2.1 applies.

The lower bounds follow from $w_t \leq u_t^{\mathrm{P}}$, and the upper bounds from the chord estimate; together they give the asserted singularity type. Off the common poles, put $f(t) = u_t^{\mathrm{P}}(x)$. For $0 < s < t < 1$, monotonicity of the slopes of secants of the convex function f gives

$$-C \leq \frac{f(s) - f(0)}{s} \leq \frac{f(t) - f(s)}{t - s} \leq \frac{f(1) - f(t)}{1 - t} \leq C.$$

The two-sided estimates at the endpoints cover the cases $s = 0$ and $t = 1$, which proves the Lipschitz bound on $[0, 1]$. □

For endpoints of different types, the constant path at a common lower potential $v \leq \min(u_0, u_1)$, $v \in \mathrm{PSH}(X, \theta)$, is still admissible, so the envelope exists whenever such a potential exists. Attaining the endpoints requires a barrier approaching the prescribed values; Proposition 18.2.1 separates this requirement from the existence of the envelope.

18.3 The smooth product equation

For a smooth path, positivity on the product compares the acceleration with the spatial gradient of the velocity. Let u_t be smooth in $(x,t) \in X \times I$, suppose that $\eta_t = \theta + \mathrm{dd^c} u_t > 0$, and write, in local holomorphic coordinates $(x_1, \dots, x_n)$,

$$\theta = \frac{\mathrm{i}}{2\pi} \sum_{i,j=1}^{n} g_{i\bar{j}}\, \mathrm{d}x_i \wedge \mathrm{d}\bar{x}_j, \qquad H = (g_{i\bar{j}} + \partial_i \partial_{\bar{j}} u), \qquad \xi = (\partial_i \dot{u})_{i=1}^{n}.$$

Here dots denote derivatives in t, ξ is a column vector, and ξ^* is its conjugate transpose. Since $t = \operatorname{Re} z$, the form $\pi^*\theta + \mathrm{dd^c} U$ on the product has, in the coordinates $(x_1, \dots, x_n, z)$, the coefficient matrix

$$\begin{pmatrix} H & \frac{1}{2}\xi \\ \frac{1}{2}\xi^* & \frac{1}{4}\ddot{u} \end{pmatrix}.$$

Proposition 18.3.1 *The form $\pi^*\theta + \mathrm{dd^c} U$ is semipositive exactly when*

$$\ddot{u} \geq \xi^* H^{-1} \xi.$$

Its $(n+1)$st power vanishes exactly when equality holds. Equivalently, the homogeneous Monge–Ampère equation on the product is

$$\ddot{u}\, \eta_t^n = n\, \mathrm{d}\dot{u} \wedge \mathrm{d^c}\dot{u} \wedge \eta_t^{n-1}. \tag{18.2}$$

Proof The Schur complement of H in the block matrix is $(\ddot{u} - \xi^* H^{-1} \xi)/4$. Since $H > 0$, the block matrix is semipositive exactly when this complement is nonnegative, and its determinant, which is $\det H$ times the complement, vanishes exactly when the complement vanishes.

To rewrite the equation in terms of forms, note that

$$\mathrm{d}\dot{u} \wedge \mathrm{d^c}\dot{u} = \frac{\mathrm{i}}{2\pi} \partial\dot{u} \wedge \bar{\partial}\dot{u}, \qquad n \frac{\mathrm{d}\dot{u} \wedge \mathrm{d^c}\dot{u} \wedge \eta_t^{n-1}}{\eta_t^n} = \operatorname{tr}(H^{-1}\xi\xi^*) = \xi^* H^{-1} \xi.$$

Multiplying the second identity by $\eta_t^n > 0$ gives (18.2). □

This calculation identifies the homogeneous equation for smooth paths. Whether nonsmooth Perron segments solve a weak form of this equation is a separate question, which we do not address.

18.4 The L^2 action and torsion

The homogeneous equation on the product involves the spatial form η_t only pointwise. In the L^2 action, integration by parts also differentiates the powers of η_t, which produces terms absent from that equation. Fix a Hermitian form ω, write $\omega_u = \omega + dd^c u$, and recall from Chapter 2 the space $\mathcal{H}_\omega$ of smooth real functions u with $\omega_u > 0$. On $\mathcal{H}_\omega$ consider the Riemannian metric

$$G_u(v,w) = \int_X vw\,\omega_u^n, \qquad v, w \in C^\infty(X, \mathbb{R}).$$

For a smooth path $u_t \in \mathcal{H}_\omega$, $0 \le t \le 1$, its action is

$$\mathcal{S}(u) = \frac{1}{2}\int_0^1 \int_X \dot{u}_t^2\,\omega_{u_t}^n\,dt.$$

Stationarity under smooth variations fixing both endpoints is the formal geodesic equation of this metric.

Theorem 18.4.1 *Let $u_t \in \mathcal{H}_\omega$, $0 \le t \le 1$, be a smooth path, and put $\eta_t = \omega_{u_t}$ and $T_t = \eta_t^{n-1}$. The path is stationary for $\mathcal{S}$ if and only if*

$$\begin{aligned} \ddot{u}_t\,\eta_t^n &= n\,d\dot{u}_t \wedge d^c\dot{u}_t \wedge T_t \\ &\quad + n\dot{u}_t\big(d\dot{u}_t \wedge d^c T_t - d^c\dot{u}_t \wedge dT_t\big) + \frac{n}{2}\dot{u}_t^2 dd^c T_t. \end{aligned} \tag{18.3}$$

All differentials on the right act only in the X variable.

Proof Vary the path as $u_t + sv_t$, where v_t is smooth in (x,t) and $v_0 = v_1 = 0$, and put $f = \dot{u}_t$. Differentiating the action at $s = 0$, using $\partial_s(\omega^n_{u_t+sv_t})|_{s=0} = n dd^c v_t \wedge T_t$, gives

$$\delta\mathcal{S} = \int_0^1 \left(\int_X f\dot{v}_t\,\eta_t^n + \frac{n}{2}\int_X f^2 dd^c v_t \wedge T_t \right) dt.$$

We integrate by parts in t in the first term, using $\partial_t(\eta_t^n) = n dd^c f \wedge T_t$, and on X in the second, using Lemma 2.1.1 with the whole form $f^2 T_t$. Thus

$$\delta\mathcal{S} = \int_0^1 \int_X v_t \left(-\ddot{u}_t\eta_t^n - nf\,dd^c f \wedge T_t + \frac{n}{2} dd^c(f^2 T_t) \right) dt.$$

The product rule gives

$$\begin{aligned} dd^c(f^2 T_t) &= 2f\,dd^c f \wedge T_t + 2\,df \wedge d^c f \wedge T_t \\ &\quad + 2f\big(df \wedge d^c T_t - d^c f \wedge dT_t\big) + f^2 dd^c T_t. \end{aligned}$$

After the term $nf\,dd^c f \wedge T_t$ cancels, the integrand is v_t times the difference between the right-hand and left-hand sides of (18.3). Since $\eta_t > 0$ on the compact set $X \times [0,1]$,

every smooth v_t gives an admissible variation for small $|s|$, and the fundamental lemma of the calculus of variations gives both implications. □

For a Kähler background, $\mathrm{d}T_t = \mathrm{d}^{\mathrm{c}}T_t = 0$, and the equation reduces to (18.2). The Gauduchon condition $\mathrm{dd}^{\mathrm{c}}T_t = 0$ on η_t removes only the last term. Translations by constants isolate that term and give an exact test for stationarity.

Corollary 18.4.1 *Fix $u \in \mathcal{H}_\omega$ and a nonzero constant a. The path $u_t = u + at$, $0 \le t \le 1$, is a Perron segment and solves the homogeneous equation (18.2). It is stationary for the L^2 action if and only if*

$$\mathrm{dd}^{\mathrm{c}}(\omega_u^{n-1}) = 0.$$

For $n \ge 2$, this means that ω_u is a Gauduchon metric.

Proof The path is admissible and equals the chord between its endpoints, so by the chord bound no admissible competitor can be larger; it is therefore the Perron segment. Its acceleration and the spatial derivatives of its velocity vanish, which proves (18.2). In (18.3), the only remaining term is $\frac{n}{2}a^2\mathrm{dd}^{\mathrm{c}}(\omega_u^{n-1})$, which vanishes exactly under the stated condition. □

The three-dimensional torus in Example 4.8.1 carries a Gauduchon form ω and a smooth potential $u \in \mathcal{H}_\omega$ for which ω_u is not Gauduchon. Thus $u_t = u + t$ is a smooth Perron segment that is not stationary for the L^2 action. On a surface, the Gauduchon condition persists under perturbation of the potential, since $\mathrm{dd}^{\mathrm{c}}\omega_u = \mathrm{dd}^{\mathrm{c}}\omega$ (compare Proposition 4.8.2), but it does not cancel the terms in (18.3) involving first derivatives of T_t.

18.5 Basic finite energy potentials on a Hopf manifold

The Hopf geometry of Section 9.6 gives a nonclosed background in which the potentials pulled back from the projective base carry a complete geodesic metric. The Perron envelope between two such potentials uses all subgeodesics, whereas the metric measures lengths only along paths of pulled-back potentials. The distinction matters: allowing additional smooth paths can decrease the length.

Fix $n \ge 2$ and put $d = n - 1$. Write

$$X = H_n, \qquad Y = \mathbb{P}^d, \qquad p = p_n, \qquad \omega = \omega_n = \alpha_n + \beta_n.$$

Let κ be the mass-one Fubini–Study form on Y, so $\alpha_n = p^*\kappa$. On the cover, write $r = \log|z|^2$ for the function denoted s in Section 9.6; it descends to X as a function defined modulo two, and $\beta_n = \mathrm{d}r \wedge \mathrm{d}^{\mathrm{c}}r$. The compact elliptic group

$$E = \mathbb{C}/(\mathbb{Z} + 2\pi\mathrm{i}\mathbb{Z})$$

acts by scalar multiplication, preserves ω, and acts transitively on each fiber. A potential is *basic* if it is invariant under this action; the basic potentials are exactly the pullbacks p^*v of functions v on Y.

These manifolds are not products of positive-dimensional compact complex manifolds. Indeed, H_n is diffeomorphic to $S^{2n-1} \times S^1$, whose intermediate positive even-degree real cohomology vanishes. A product with a compact complex factor of dimension k, $0 < k < n$, would have a nonzero degree-$2k$ class: pullback of that factor's top class is injective because projection has a smooth section. Thus the fiber reduction below takes place on a nonproduct manifold.

By Lemma 9.6.1 with $a = b = 1$, a quasi-plurisubharmonic function V on $Y \times S_I$, for a nonempty open interval $I \subset \mathbb{R}$, satisfies

$$\mathrm{pr}_X^*\,\omega + \mathrm{dd}^{\mathrm{c}}(p \times \mathrm{id})^*V \geq 0 \quad \Longleftrightarrow \quad \mathrm{pr}_Y^*\,\kappa + \mathrm{dd}^{\mathrm{c}}V \geq 0,$$

and every $v \in \mathrm{PSH}(Y, \kappa)$ satisfies

$$\mathrm{MA}_\omega^{\mathrm{ext}}(p^*v) = n\beta_n \wedge p^*\,\mathrm{MA}_\kappa^{\mathrm{ext}}(v), \tag{18.4}$$

with the fiber measure $\beta_n \wedge p^*\nu$ of that lemma. In particular, for every nonnegative Borel function F on Y,

$$\int_X p^*F\ \mathrm{MA}_\omega^{\mathrm{ext}}(p^*v) = 2n \int_Y F\ \mathrm{MA}_\kappa^{\mathrm{ext}}(v). \tag{18.5}$$

Proposition 18.5.1 *If $v_0, v_1 \in \mathrm{PSH}(Y, \kappa)$ admit an admissible subgeodesic on Y, then the Perron envelope on X between p^*v_0 and p^*v_1, taken over all admissible subgeodesics on X, is the pullback of the Perron envelope on Y between v_0 and v_1.*

Proof The family of all admissible subgeodesics on X between p^*v_0 and p^*v_1 is preserved by E, so its supremum and the upper semicontinuous regularization are invariant. By the positivity equivalence above, a pulled-back admissible subgeodesic of Y belongs to this family, so the envelope is the largest admissible subgeodesic by Proposition 18.2.1. By Lemma 9.6.1(2) and the same equivalence, this invariant envelope descends to an admissible subgeodesic on Y, since its joint upper limits at the endpoints descend as well. Conversely, every admissible subgeodesic on Y pulls back to one on X. The two envelopes consequently agree; in submersion charts, upper semicontinuous regularization commutes with pullback. □

Let $\mathcal{E}^2(Y, \kappa)$ denote the set of $v \in \mathrm{PSH}(Y, \kappa)$ of non-pluripolar mass one with finite squared energy $\int_Y |v|^2\,\mathrm{MA}_\kappa^{\mathrm{ext}}(v)$. Define

$$\mathcal{B}_\infty = \{p^*v : v \in C^\infty(Y),\ \kappa + \mathrm{dd}^{\mathrm{c}}v > 0\}, \qquad \mathcal{B}_2 = p^*\mathcal{E}^2(Y, \kappa).$$

Thus $\mathcal{B}_2$ consists precisely of the basic potentials of extended mass $2n$ with $\int_X |u|^2\,\mathrm{MA}_\omega^{\mathrm{ext}}(u) < \infty$. On $\mathcal{B}_\infty$, let d_B be the infimum of the lengths of smooth paths in $\mathcal{B}_\infty$ for the L^2 metric G of the previous section.

A geodesic metric space (Z, δ) is CAT(0) if every constant-speed geodesic $\gamma\colon [0,1] \to Z$ satisfies

$$\delta(z,\gamma_t)^2 \le (1-t)\delta(z,\gamma_0)^2 + t\delta(z,\gamma_1)^2 - t(1-t)\delta(\gamma_0,\gamma_1)^2$$

for $z \in Z$ and $0 \le t \le 1$. In particular, its geodesics are unique.

We use the classical Kähler completion theorem: the length metric of the L^2 metric $\int_Y \xi_1\xi_2(\kappa + \mathrm{dd^c}v)^d$ has completion $\mathcal{E}^2(Y,\kappa)$, the completed metric is CAT(0), and its geodesic between two points is their Perron segment. That segment attains its endpoints in capacity, and all its slices lie in $\mathcal{E}^2$. We also use the full-mass characterization $P_\kappa[v] = 0$ for $v \in \mathcal{E}^2$. These are closed-background inputs on Y; their exact source and operator normalization are specified in Section 36.12.

Theorem 18.5.1 *The completion of $(\mathcal{B}_\infty, d_B)$ is $\mathcal{B}_2$. It is a CAT(0) space, and its unique constant-speed geodesic between two points is their full Hermitian Perron segment. This segment attains both endpoints in $L^1(X, \mathrm{d}V)$ for every smooth positive volume form $\mathrm{d}V$. For every $u \in \mathcal{B}_2$ and $c \in \mathbb{R}$,*

$$d_B(u, u+c) = \sqrt{2n}|c|.$$

Proof Fiber integration gives the exact restricted metric

$$G_{p^*v}(p^*\xi_1, p^*\xi_2) = 2n\int_Y \xi_1\xi_2(\kappa + \mathrm{dd^c}v)^d. \tag{18.6}$$

The Kähler completion theorem therefore gives the stated completion and CAT(0) property after multiplying base distances by $\sqrt{2n}$. If $\widetilde{d}_D$ denotes the completed metric written with operator $\mathrm{i}\partial\bar\partial$ and background κ, the normalization is explicitly

$$d_B(p^*v, p^*w) = 2\pi\sqrt{2n}\,\widetilde{d}_D(v/(2\pi), w/(2\pi)), \tag{18.7}$$

because $\kappa + \mathrm{i}\partial\bar\partial(v/(2\pi)) = \kappa + \mathrm{dd^c}v$. The geodesic between two points of $\mathcal{B}_2$ is the pullback of their Perron segment on Y, and Proposition 18.5.1 identifies it with their Perron segment on X, taken over all admissible subgeodesics. The base segment attains its endpoints in capacity, hence in $L^1(Y)$. Since

$$\int_X |p^*(v-w)|\frac{\omega^n}{2n} = \int_Y |v-w|\kappa^d,$$

the same holds on X; every smooth positive $\mathrm{d}V$ has bounded density ratio with ω^n on compact X. For smooth v, the constant translation is a base geodesic with speed $|c|$. Smooth metric approximation and translation by the same constant extend the distance formula to every $v \in \mathcal{E}^2$. □

For an absolutely continuous curve $\gamma\colon [0,1] \to \mathcal{B}_2$, define its metric action by

$$\mathcal{A}_B(\gamma) = \frac{1}{2}\int_0^1 |\dot\gamma|_{d_B}^2 \,\mathrm{d}t.$$

Here the metric derivative is the almost everywhere defined limit

$$|\dot\gamma_t|_{d_B} = \lim_{h\to 0} \frac{d_B(\gamma_{t+h},\gamma_t)}{|h|}.$$

The action is defined even when the potentials are unbounded; it does not require a Riemannian tensor at a singular potential. To distinguish them from $\mathcal{A}_B$, we call the action $\mathcal{S}$ of a smooth path in $\mathcal{H}_\omega$, defined in the previous section, its *raw action*, and its length for G its *raw length*.

Proposition 18.5.2 *The unique action minimizer between two points of $\mathcal{B}_2$ is their constant-speed Perron segment, and its action is half their squared distance. On every smooth path in $\mathcal{B}_\infty$, $\mathcal{A}_B$ equals the raw action $\mathcal{S}$. Every smooth basic Perron segment with strictly positive spatial forms is stationary for $\mathcal{S}$ under all smooth fixed-endpoint variations on X.*

Proof The length inequality and Cauchy–Schwarz give

$$d_B(\gamma_0,\gamma_1)^2 \le \left(\int_0^1 |\dot\gamma|_{d_B}\,\mathrm{d}t\right)^2 \le 2\mathcal{A}_B(\gamma).$$

Equality holds for the constant-speed geodesic. Equality for a connecting curve forces constant speed and equality between length and endpoint distance, hence forces a geodesic; uniqueness follows from the CAT(0) property.

For a smooth basic path p^*v_s, put $h = v_{s+\delta} - v_s$. On the base Perron segment from v_s to $v_{s+\delta}$, parametrized by $0 \le \tau \le 1$, the chord $v_s + \tau h$ bounds above, and

$$v_s + \tau h - C\delta^2\tau(1-\tau)$$

bounds below. Indeed, the spatial forms along the chord have a uniform positive lower bound, $\mathrm{d}h = O(\delta)$, and Proposition 18.3.1 makes the displayed path a subgeodesic for a fixed sufficiently large C. The right initial velocity of the Perron segment is therefore $h + O(\delta^2)$ uniformly. The classical smooth-endpoint Kähler distance formula gives

$$\lim_{\delta\to 0}\frac{d_B(p^*v_s,p^*v_{s+\delta})^2}{\delta^2} = 2n\int_Y \dot v_s^{\,2}(\kappa + \mathrm{dd^c}v_s)^d.$$

Integration proves $\mathcal{A}_B = \mathcal{S}$ along this path.

For stationarity, let $u_t = p^*v_t$ be a smooth basic Perron segment with strictly positive spatial forms; by Proposition 18.5.1, v_t is the Perron segment on Y between its endpoints. Let $L(w)$ be the derivative of $\mathcal{S}(u+\varepsilon w)$ at $\varepsilon = 0$. Uniform spatial positivity allows every smooth endpoint-zero variation w. Group invariance of the action and of the path gives $L(g^*w) = L(w)$ for $g \in E$. Average over Haar probability measure to obtain

$$L(w) = L\left(\int_E g^* w \, \mathrm{d}g\right) = L(p^* q)$$

for a smooth endpoint-zero base variation q. Fiber integration reduces this to $2n$ times the closed-base first variation. With $\eta_t = \kappa + \mathrm{dd}^{\mathrm{c}} v_t$, integration by parts gives that variation as

$$\int_0^1 \int_Y q_t \left[-\ddot{v}_t \eta_t^d + d \, \mathrm{d}\dot{v}_t \wedge \mathrm{d}^{\mathrm{c}} \dot{v}_t \wedge \eta_t^{d-1}\right] \mathrm{d}t.$$

The bracket vanishes because the base segment solves the homogeneous equation (18.2) on Y. Indeed, if the $(d+1)$st power of its semipositive product form were positive at some point, that form would be strictly positive near the point, and adding a small nonzero nonnegative bump function of y and of the real part of the strip variable, supported near the point, would give a larger admissible subgeodesic. Thus $L(w) = 0$ for every variation on X. □

The metric action is minimized only among curves in $\mathcal{B}_2$. Stationarity holds under all smooth variations on X, but it does not imply minimization among all smooth Hermitian paths, and Section 18.6 shows that minimization fails. Likewise, the initial-velocity formula used above requires smooth endpoints; it is not a definition of the metric on all of $\mathcal{B}_2$.

18.6 Negative directions of the raw action

On this Hopf background, translations by constants are stationary Perron segments, and radial variations show that they do not minimize the raw action. Every smooth two-periodic function of r defines a smooth function on X, and for a radial path $u(t, r)$,

$$\omega + \mathrm{dd}^{\mathrm{c}} u = (1 + u_r)\alpha_n + (1 + u_{rr})\beta_n.$$

For $\langle F \rangle = \frac{1}{2}\int_0^2 F(r) \, \mathrm{d}r$, fiber integration therefore gives

$$\mathcal{S}(u) = n \int_0^1 \left\langle u_t^2 (1 + u_r)^d (1 + u_{rr}) \right\rangle \mathrm{d}t. \tag{18.8}$$

Changing the local base section shifts the start of the full period and leaves the average unchanged.

Theorem 18.6.1 *For every nonzero constant c, the full Perron segment $u_t = ct$, $0 \le t \le 1$, is stationary for the raw action, and the Hessian of the raw action at this segment is negative definite on an infinite-dimensional space of smooth fixed-endpoint variations. The infimum of raw lengths over all smooth strictly positive paths from 0 to c is strictly smaller than $d_B(0,c) = \sqrt{2n}|c|$.*

Proof The path is basic, equals the chord, and is the full Perron segment by Corollary 18.4.1. Since ω is Gauduchon, the same corollary shows that it is stationary; this also follows from Proposition 18.5.2. Let $w(t,r)$ be smooth, two-periodic in r, and zero at $t = 0, 1$. The paths $ct + \varepsilon w$ have positive spatial forms for sufficiently small $|\varepsilon|$. Differentiating (18.8) twice in ε at $\varepsilon = 0$ gives

$$2n \int_0^1 \left\langle w_t^2 + 2cw_t(dw_r + w_{rr}) + c^2 \left(\frac{d(d-1)}{2} w_r^2 + dw_r w_{rr} \right) \right\rangle \mathrm{d}t.$$

The integral of $w_r w_{rr}$ over a period vanishes, while $\int_0^1 \langle w_t w_{rr} \rangle \mathrm{d}t = -\frac{1}{2}[\langle w_r^2 \rangle]_0^1 = 0$. Completing the square gives the Hessian

$$2n \int_0^1 \left\langle (w_t + dcw_r)^2 - \frac{d(d+1)}{2} c^2 w_r^2 \right\rangle \mathrm{d}t. \tag{18.9}$$

For a positive integer k, take

$$w_k(t,r) = \sin(\pi t) \cos\bigl(\pi k (r - dct)\bigr).$$

Then $w_{k,t} + dcw_{k,r} = \pi \cos(\pi t) \cos(\pi k(r - dct))$, and (18.9) is

$$\frac{n\pi^2}{2} \left(1 - \frac{d(d+1)}{2} c^2 k^2 \right).$$

It is negative for all sufficiently large k. For distinct k, the functions $w_{k,t} + dcw_{k,r}$ are orthogonal under the r average, and so are the derivatives $w_{k,r}$. Hence (18.9) is negative on every nonzero finite combination w of the modes w_k with k this large. For sufficiently small $\varepsilon \neq 0$, the path $ct + \varepsilon w$ is strictly positive and has raw action strictly below $\mathcal{S}(ct) = nc^2$. By the Cauchy–Schwarz inequality on the unit interval, its raw length ℓ satisfies $\ell^2 \le 2\mathcal{S}(ct + \varepsilon w) < 2nc^2$, so $\ell < \sqrt{2n}|c|$. □

These competing paths need not be subgeodesics. The strict length inequality does not establish definiteness or completeness of the unrestricted length infimum. For $n = 3$ and $c = 1$, the mode $k = 1$ gives the concrete path

$$u_\varepsilon(t,r) = t + \varepsilon \sin(\pi t) \cos\bigl(\pi(r - 2t)\bigr).$$

Its spatial coefficients are at least $1 - \pi|\varepsilon|$ and $1 - \pi^2|\varepsilon|$. Its exact action is

$$\mathcal{S}(u_\varepsilon) = 3 - \frac{3\pi^2}{2}\varepsilon^2 + \frac{111\pi^4}{64}\varepsilon^4. \tag{18.10}$$

Indeed, put $f = \sin(\pi t)$, $g = f'$, and $\vartheta = \pi(r - 2t)$. Substitute $u_t = 1 + \varepsilon(g\cos\vartheta + 2\pi f \sin\vartheta)$, $u_r = -\varepsilon\pi f \sin\vartheta$, and $u_{rr} = -\varepsilon\pi^2 f\cos\vartheta$ into (18.8). Odd powers have zero periodic average. The averaged quadratic coefficient is $(g^2 - 3\pi^2 f^2 - 2\pi^2 fg)/2$, and the quartic coefficient is $\pi^2 f^2(g^2 + 6\pi^2 fg + 12\pi^2 f^2)/8$. The identities

$$\int_0^1 fg\,\mathrm{d}t = \int_0^1 f^3 g\,\mathrm{d}t = 0,$$
$$\int_0^1 f^2\,\mathrm{d}t = \frac{1}{2}, \quad \int_0^1 g^2\,\mathrm{d}t = \frac{\pi^2}{2},$$
$$\int_0^1 f^4\,\mathrm{d}t = \frac{3}{8}, \quad \int_0^1 f^2 g^2\,\mathrm{d}t = \frac{\pi^2}{8}$$

give (18.10). Taking $\varepsilon = 1/100$ gives a positive path with action $2.9985212487\ldots < 3$.

18.7 Explicit basic geodesics and singular potentials

Diagonal projective automorphisms give smooth nonconstant geodesics. For real $a_0, \ldots, a_{n-1}$, set

$$v_t([z]) = \log \frac{\sum_{j=0}^{n-1} \mathrm{e}^{t a_j}|z_j|^2}{\sum_{j=0}^{n-1}|z_j|^2}.$$

The family is spatially nonconstant when the a_j are not all equal.

Proposition 18.7.1 *The pullback p^*v_t, $0 \le t \le 1$, is the full Hermitian Perron segment from 0 to p^*v_1, and it is stationary under all smooth fixed-endpoint variations. The distance between its endpoints and its action are*

$$d_B(0, p^*v_1)^2 = \frac{2[(\sum_j a_j)^2 + \sum_j a_j^2]}{n+1},$$
$$\mathcal{S}(p^*v) = \mathcal{A}_B(p^*v) = \frac{(\sum_j a_j)^2 + \sum_j a_j^2}{n+1}.$$

Proof Let s be the coordinate on the strip $S_{(0,1)}$, and let $V([z], s) = v_{\operatorname{Re} s}([z])$ be the complexification. Since $|\mathrm{e}^{a_j s/2}|^2 = \mathrm{e}^{a_j \operatorname{Re} s}$, the holomorphic map $F([z], s) = [\mathrm{e}^{a_0 s/2} z_0 : \cdots : \mathrm{e}^{a_{n-1} s/2} z_{n-1}]$ satisfies $\mathrm{pr}_Y^*\,\kappa + \mathrm{dd}^{\mathrm{c}} V = F^*\kappa$. The form $F^*\kappa$ has rank d, strictly positive spatial slices, and zero $(d+1)$st power. The classical uniqueness for the bounded Kähler Dirichlet problem identifies v_t with the base Perron segment from 0 to v_1; Propositions 18.5.1 and 18.5.2 give the full segment and its stationarity.

For $x_j = |z_j|^2/\sum|z_j|^2$, the initial velocity is $\sum_j a_j x_j$. A standard complex Gaussian vector has independent exponential squared coordinates, and its projective direction has distribution κ^d. Dividing those coordinates by their sum gives the uniform simplex distribution, with moments

$$\int_Y x_i^2 \kappa^d = \frac{2}{n(n+1)}, \qquad \int_Y x_i x_j \kappa^d = \frac{1}{n(n+1)} \quad (i \neq j).$$

The smooth distance formula and (18.6) give the claimed squared distance; the action is half of it. □

For instance, $n = 3$ and $(a_0, a_1, a_2) = (1, -1, 0)$ give squared distance one and action $1/2$.

The completed space also contains potentials with different ordinary singularity types. Put $o = [1 : 0 : \cdots : 0] \in Y$ and

$$\psi = \log \frac{\sum_{j=1}^d |z_j|^2}{\sum_{j=0}^d |z_j|^2}, \qquad v_a = 1 - (1 - \psi)^a, \qquad 0 < a < 1,$$

with the canonical value $v_a(o) = -\infty$.

Proposition 18.7.2 *Every v_a has non-pluripolar mass one, and*

$$v_a \in \mathcal{E}^2(Y, \kappa) \quad \Longleftrightarrow \quad a < \frac{d}{d+2}.$$

At equality its squared energy diverges logarithmically. The pullback has mass $2n$ and pole set $p^{-1}(o)$. If $0 < a < b < d/(d+2)$, the two pullbacks have different ordinary singularity types and finite positive d_B distance.

Proof In the chart $z_0 = 1$, write $z' = (z_1, \ldots, z_d)$, $s = \log |z'|^2$, $L = \log(1 + \mathrm{e}^s)$, $P = \mathrm{e}^s/(1 + \mathrm{e}^s)$, and $T = 1 + L - s \geq 1$. The local potential $\Phi = L + 1 - T^a$ has derivative

$$A = P + a(1 - P)T^{a-1}, \qquad A' = P'(1 - aT^{a-1}) + a(1 - a)(1 - P)^2 T^{a-2} > 0.$$

Both A and A' are positive, so $\mathrm{dd^c}\Phi = A\,\mathrm{dd^c}s + A'\,\mathrm{d}s \wedge \mathrm{d^c}s$ is strictly positive off o. Its bounded-above local potential extends plurisubharmonically across o by Lemma 1.3.7. The function extends smoothly across the hyperplane at infinity. Its bounded plurisubharmonic truncations decrease to a nontrivial canonical potential: near o, $|v_a| = O((1 - s)^a)$ is integrable against radial smooth volume $O(\mathrm{e}^{ds})\,\mathrm{d}s$.

The radial mass flux is $Q(s) = A(s)^d$. Its sphere coefficient is one, since $\mathrm{d^c} \log |z'|^2$ integrates to one on the Hopf circle and $\mathrm{dd^c} \log |z'|^2$ induces mass-one Fubini–Study on $\mathbb{P}^{d-1}$. Stokes' theorem on the annulus $\{s_1 < s < s_2\}$ gives mass $Q(s_2) - Q(s_1)$. Since $Q(-\infty) = 0$ and $Q(+\infty) = 1$, finite-level locality and annular exhaustion give non-pluripolar mass one. The pole has no atom, and smoothness gives no mass on the hyperplane at infinity.

As $s \to -\infty$,

$$|v_a|^2 \sim T^{2a}, \qquad Q'(s) \sim da^d(1 - a)T^{-d(1-a)-1}, \qquad T \sim 1 - s.$$

The squared energy therefore converges exactly when $2a - d(1 - a) < 0$. At equality the integrand has order T^{-1}, giving logarithmic divergence. All other compact regions

have smooth potentials and measures. Fiber integration gives the pullback assertions. Finally, $v_a - v_b = T^b - T^a \to +\infty$ at o, so mutual bounded comparison fails. Both points lie in the completed metric space and are distinct, giving finite positive distance. □

18.8 Model projection and Perron segments

The distance just constructed retains information about the potentials that their ordinary model projection forgets. This distinction connects the example with the singularity spaces studied earlier; the compatibility of pullback with the Legendre transforms of the next chapter is Corollary 19.1.1.

Proposition 18.8.1 *For every $v \in \mathrm{PSH}(Y, \kappa)$,*

$$P_\omega[p^* v] = p^* P_\kappa[v].$$

Consequently $P_\omega[u] = 0$ for every $u \in \mathcal{B}_2$. Let $(v_t)_{t \geq 0}$ be a family in $\mathrm{PSH}(Y, \kappa)$ whose restriction to every finite interval is a Perron segment. If every v_t, including v_0, lies in $\mathcal{E}^2(Y, \kappa)$, then each finite segment of its pullback is a d_B geodesic and minimizes the metric action with its prescribed endpoints.

Proof For finite C, the obstacle $p^* \min(v + C, 0)$ and the family of all competitors for its envelope on X are invariant under E. The same argument as in Proposition 18.5.1 gives

$$P_\omega(p^* \min(v + C, 0)) = p^* P_\kappa(\min(v + C, 0)).$$

Increasing limits and their upper semicontinuous regularizations commute with pullback in submersion charts, proving the model identity. The classical full-mass characterization on Y gives $P_\kappa[v] = 0$ for every $v \in \mathcal{E}^2$, hence the assertion for $\mathcal{B}_2$.

Proposition 18.5.1 identifies the pullback of every finite segment of (v_t) with the full Hermitian Perron segment; its finite-energy endpoints and Theorem 18.5.1 and Proposition 18.5.2 give the geodesic and minimizing assertions. □

For endpoints of the same ordinary singularity type, Theorem 18.2.1 already gives their common type throughout the segment. The mass formula (18.5) computes the mass of a chosen basic potential; it does not compute the lower or upper volume of its entire ordinary type, which also includes nonbasic potentials. A separating complete metric on all ordinary models, with compatibility under modifications and changes of background, remains open.

Chapter 19
Test curves and Legendre duality

We keep the setting of Chapter 18 and fix $\varphi \in \mathrm{PSH}(X,\theta)$ with $P_\theta[\varphi] = \varphi$, where $P_\theta[\cdot]$ is the ordinary singularity envelope of Definition 1.19.1; thus $\varphi \le 0$ and $\sup_X \varphi = 0$.

Legendre duality in the parameter exchanges two maximality conditions: each member of a test curve is a fixed point of the singularity envelope, while every finite segment of the corresponding ray is a Perron segment in the sense of Chapter 18. Replacing a finite segment of the ray by the Perron segment with the same endpoints changes each inverse transform only by a bounded amount, so the envelope condition forces Perron maximality. Conversely, comparison with affine paths built from obstacle envelopes shows that the inverse transforms of a ray of Perron segments are envelope fixed points. The argument uses only convexity, the minimum principle and the Perron construction, and therefore applies to nonclosed backgrounds as well as to Kähler forms. On the Hopf manifolds of Section 18.5, pullback from the projective base commutes with both transforms.

The transforms are the pointwise Legendre transforms of Section 1.20.1, and the proofs follow the envelope arguments for the Ross–Witt Nyström correspondence in [Xia26].

19.1 Envelope test curves and rays

The members of a test curve become more singular as the parameter increases. The cutoff of the curve becomes the slope of the transformed ray, while its limit as the parameter tends to $-\infty$ gives the initial potential.

Definition 19.1.1 An *envelope test curve* with initial potential φ consists of a real number c and potentials $\Gamma_\tau \in \mathrm{PSH}(X,\theta)$ for $\tau < c$ such that

(1) $\tau \mapsto \Gamma_\tau$ is pointwise concave and decreasing;
(2) $P_\theta[\Gamma_\tau] = \Gamma_\tau$ for every $\tau < c$;
(3) $(\sup_{\tau<c} \Gamma_\tau)^* = \varphi$.

Its cutoff is c. We extend it by

$$\Gamma_c = \lim_{\tau \uparrow c} \Gamma_\tau, \qquad \Gamma_\tau \equiv -\infty \quad (\tau > c).$$

Each Γ_τ is an envelope fixed point and hence has supremum zero, so Theorem 1.4.1 and Corollary 1.4.1 give $\Gamma_c \in \mathrm{PSH}(X,\theta)$.

Definition 19.1.2 An *envelope ray* with initial potential φ and slope c is a subgeodesic u_t, $t > 0$, such that

(1) $u_t \to \varphi$ in $L^1(X)$ as $t \downarrow 0$, and, with $u_0 = \varphi$, the restriction of the ray to every interval (a,b) with $0 \le a < b < \infty$ is the Perron segment from u_a to u_b;
(2) $P_\theta[u_t] = \varphi$ for every $t > 0$;
(3) $\sup_X u_t = ct$ for every $t > 0$.

For an envelope test curve, write U_t for its forward transform (1.40), the pointwise supremum of $\Gamma_\tau + t\tau$ over $\tau < c$. For a family $(u_t)_{t>0}$, write $\widehat{u}_\tau$ for its inverse transform (1.41), the pointwise infimum of $u_t - t\tau$ over $t > 0$. The forward transform is a subgeodesic by Theorem 1.20.1; it remains to show that its finite segments are Perron segments.

On the Hopf manifold of Section 18.5, pullback from the projective base preserves these notions.

Corollary 19.1.1 *In the notation of Section 18.5, let $\psi \in \mathrm{PSH}(Y,\kappa)$ satisfy $P_\kappa[\psi] = \psi$. Pullback by p maps envelope test curves on (Y,κ) with initial potential ψ to envelope test curves on (X,ω) with initial potential $p^*\psi$, and it commutes with the forward and backward Legendre transforms* (1.40) *and* (1.41).

Proof Pullback preserves the concavity and monotonicity in the parameter and the cutoff of an envelope test curve on Y, and, by the model identity of Proposition 18.8.1, the envelope condition and the initial potential. It commutes with pointwise infima, and the submersion property makes it commute with upper semicontinuous regularization of suprema. Thus it commutes with both Legendre transforms. □

19.2 Perron maximality

The slope and the singularity envelope of the transformed ray follow directly from the normalization of the test curve. To prove Perron maximality, we replace the transform on a finite interval by the Perron segment with the same endpoints. The estimate

(19.1) keeps each inverse transform in its original ordinary singularity type, where the envelope condition forces equality.

Theorem 19.2.1 *The forward transform of an envelope test curve with cutoff c is an envelope ray with the same initial potential and slope c. Moreover, $U_t - ct$ decreases as t increases.*

Proof By Theorem 1.20.1, U_t is a subgeodesic. The inequalities

$$\Gamma_\tau + t\tau \le U_t \le \varphi + ct$$

give $\sup_X U_t \ge t\tau$ for each $\tau < c$, since $\sup_X \Gamma_\tau = 0$, and $\sup_X U_t \le ct$. Letting $\tau \uparrow c$ proves $\sup_X U_t = ct$. The formula

$$U_t - ct = \sup_{\tau<c}\big(\Gamma_\tau - t(c-\tau)\big)$$

shows that $U_t - ct$ is decreasing in t.

Since $\Gamma_\tau \preceq U_t \preceq \varphi$, monotonicity of the singularity envelope gives

$$\Gamma_\tau = P_\theta[\Gamma_\tau] \le P_\theta[U_t] \le P_\theta[\varphi] = \varphi.$$

Taking the regularized supremum in τ proves $P_\theta[U_t] = \varphi$.

The local envelope theorem in Section 1.19 gives $\sup_{\tau<c}\Gamma_\tau = \varphi$ almost everywhere, and at those points the bounds $\Gamma_\tau + t\tau \le U_t \le \varphi + ct$ imply $U_t \to \varphi$ as $t \downarrow 0$. For any fixed $\tau_0 < c$ and $0 < t \le 1$, they also give

$$\Gamma_{\tau_0} - |\tau_0| \le U_t \le \varphi + |c|.$$

Both bounds are integrable, so dominated convergence gives $U_t \to \varphi$ in L^1 as $t \downarrow 0$.

It remains to prove Perron maximality. Fix $0 \le a < b$ and replace U_t on (a,b) by the Perron segment from U_a to U_b, with $U_0 = \varphi$ when $a = 0$; it exists by Proposition 18.2.1, because the restriction of U_t itself runs between these endpoints. The joint upper limits of the Perron segment at a and b allow gluing by Lemma 1.3.2 and Proposition 18.1.2. The resulting subgeodesic v_t dominates U_t and equals it outside (a,b), where therefore $v_t \le ct$. On (a,b), the chord bound (18.1) together with $U_a \le ca$ and $U_b \le cb$ gives $v_t \le ct$ as well.

For $\tau < c$, put $\chi_\tau = \inf_{t>0}(v_t - t\tau)$. Since $v_t - t\tau \ge U_t - t\tau \ge \Gamma_\tau$, Proposition 1.20.1 gives $\chi_\tau \in \mathrm{PSH}(X,\theta)$. We claim that

$$\Gamma_\tau \le \chi_\tau \le \Gamma_\tau + (b-a)(c-\tau). \tag{19.1}$$

The first inequality was just noted. For the second, let $s > 0$. If $s \in [a,b]$, then $v_b = U_b$, and since $U_t - ct$ is decreasing,

$$\chi_\tau \le U_b - b\tau \le U_s - s\tau + (b-s)(c-\tau).$$

If $s \notin [a,b]$, then $\chi_\tau \le v_s - s\tau = U_s - s\tau$. In both cases $\chi_\tau \le U_s - s\tau + (b-a)(c-\tau)$. Taking the infimum over $s > 0$ and using Lemma 1.20.3 proves (19.1).

Moreover, $v_t \le ct$ gives $\chi_\tau \le t(c-\tau)$ for every $t>0$, hence $\chi_\tau \le 0$. By (19.1), χ_τ has the ordinary type of Γ_τ, so it is a competitor for $P_\theta[\Gamma_\tau]=\Gamma_\tau$. Thus $\chi_\tau=\Gamma_\tau$. Recovering v_t from its inverse transforms by Lemma 1.20.2 gives $v_t = U_t$, which proves Perron maximality on every finite interval. □

19.3 Recovering the singularity envelopes

The minimum principle shows that the inverse transforms are θ-plurisubharmonic, but their envelope property uses the whole ray. For an inverse transform g and $C>0$, the envelope $P_\theta(g+C,0)$ of the obstacle $\min(g+C,0)$ gives an admissible affine path on every interval $[0,M]$, lying below the ray translated by an affine function of t. Letting the length M vary, and then letting the relative time t/M tend to zero, removes the additive constant C.

Theorem 19.3.1 *If u_t is an envelope ray of slope c with initial potential φ, its inverse transform is an envelope test curve with cutoff c and initial potential φ. The forward and inverse transforms are mutually inverse bijections between the envelope test curves of Definition 19.1.1 and the envelope rays of Definition 19.1.2 with initial potential φ.*

Proof By Lemma 1.20.1, for each x the function $t\mapsto u_t(x)-ct$ is either finite and convex or identically $-\infty$. It is bounded above by zero, so it is decreasing. Since $\sup_X(u_t-ct)=0$, Theorem 1.4.1 and Corollary 1.4.1 give

$$\widehat{u}_c = \inf_{t>0}(u_t-ct) \in \mathrm{PSH}(X,\theta).$$

For $\tau<c$, we have $\widehat{u}_c \le \widehat{u}_\tau \le u_1-\tau$, so Proposition 1.20.1 gives $\widehat{u}_\tau\in\mathrm{PSH}(X,\theta)$. Also

$$\widehat{u}_\tau \le \inf_{t>0} t(c-\tau) = 0 \quad (\tau\le c),$$

while the bound $\widehat{u}_\tau \le t(c-\tau)$ with $t\to\infty$ shows $\widehat{u}_\tau\equiv-\infty$ for $\tau>c$. Concavity and monotonicity in τ hold because $\widehat{u}_\tau$ is an infimum of affine functions of τ with negative slopes $-t$.

Fix $\tau<c$ and write $g=\widehat{u}_\tau$. The inequality $g\le u_M-M\tau$, valid for every $M>0$, gives

$$P_\theta[g] \le P_\theta[u_M] = \varphi.$$

For $C>0$ put $q_C=P_\theta(g+C,0)$. Since $g\le 0$, the function g is a competitor, so $q_C\not\equiv-\infty$, and

$$g \le q_C \le P_\theta[g] \le \varphi, \qquad q_C - C \le g \le u_M - M\tau.$$

The affine path $q_C - Ct/M$ is admissible for the endpoints φ and $u_M-M\tau$ on $[0,M]$. Adding $-t\tau$ gives a bijection between the competitor families for the endpoints φ, u_M

and $\varphi, u_M - M\tau$, by Proposition 18.1.1, so the Perron segment for the latter endpoints is $u_t - t\tau$. For $0 < s \le 1$ this yields

$$q_C - Cs \le u_{sM} - sM\tau.$$

Taking the infimum over $M > 0$ gives $q_C - Cs \le g$, and letting $s \downarrow 0$ gives $q_C \le g$. Thus $q_C = g$ for every $C > 0$, and letting $C \to \infty$ in the definition of $P_\theta[g]$ proves $P_\theta[g] = g$.

Let $\psi = (\sup_{\tau<c} \widehat{u}_\tau)^*$; since $\widehat{u}_\tau = P_\theta[\widehat{u}_\tau] \le \varphi$, we have $\psi \le \varphi$. Because $u_t - ct$ is decreasing in t, the limit of $u_t(x)$ as $t \downarrow 0$ exists at every x, and an almost everywhere convergent sequence given by the L^1 convergence $u_t \to \varphi$ identifies it with φ almost everywhere. At these points, the endpoint identity in Lemma 1.20.2 gives

$$\sup_{\tau<c} \inf_{t>0} (u_t(x) - t\tau) = \varphi(x).$$

Therefore $\psi = \varphi$ by Lemma 1.3.3.

As $\tau \uparrow c$, the transforms decrease to $\widehat{u}_c$, since the infima over $t > 0$ and over $\tau < c$ commute; this is the convention for the value at the cutoff. Finally, Lemma 1.20.2 recovers u_t from its inverse transforms, also at points where $t \mapsto u_t(x)$ is identically $-\infty$, and Lemma 1.20.3 recovers each test curve from its forward transform. Hence the correspondence is a bijection. □

19.4 A logarithmic pole on the projective line

Let $X = \mathbb{P}^1$ and $\varphi = 0$. The envelope of a multiple of a logarithmic pole is obtained by replacing part of a convex function by a tangent line. The corresponding ray is smooth, and all its slices are bounded; the pole appears only in the behavior of the ray as $t \to \infty$. In the affine coordinate z, take

$$\theta = \mathrm{dd^c} \log(1 + |z|^2), \qquad g(z) = \log \frac{|z|^2}{1 + |z|^2}.$$

The form θ extends to the Fubini–Study form of $\mathbb{P}^1$, which has integral one. The potential g is θ-plurisubharmonic, has a logarithmic pole at zero, and extends smoothly across infinity, with value zero there.

Put $H(\lambda) = -\lambda \log \lambda - (1-\lambda)\log(1-\lambda)$ for $0 < \lambda < 1$, with $H(0) = H(1) = 0$. For $0 < \lambda < 1$, define

$$\Gamma_\lambda(z) = \begin{cases} \lambda \log|z|^2 - \log(1+|z|^2) + H(\lambda), & |z|^2 \le \lambda/(1-\lambda), \\ 0, & |z|^2 \ge \lambda/(1-\lambda). \end{cases} \tag{19.2}$$

Set $\Gamma_\lambda = 0$ for $\lambda \le 0$, $\Gamma_1 = g$, and $\Gamma_\lambda = -\infty$ for $\lambda > 1$.

Proposition 19.4.1 *The family (Γ_λ) is an envelope test curve with initial potential zero and cutoff one. Its forward transform is*

$$U_t(z) = \log \frac{1 + e^t|z|^2}{1+|z|^2}, \qquad t > 0. \tag{19.3}$$

At $z = 0$ its value is zero, and at infinity its value is t.

Proof Fix $0 < \lambda < 1$. For $s = \log|z|^2$, let $q_\lambda(s) = \Gamma_\lambda(z) + \log(1+e^s)$. In the first region of (19.2) this is the affine function $\lambda s + H(\lambda)$, which is tangent to $\log(1+e^s)$ at $s_\lambda = \log(\lambda/(1-\lambda))$, and in the second region it is $\log(1+e^s)$. Its derivative is continuous at the junction and nondecreasing, so q_λ is convex. By Corollary 1.20.1, Γ_λ is θ-plurisubharmonic away from zero, and it extends across its logarithmic pole by Lemma 1.3.7. It vanishes near infinity and has the ordinary type of λg.

To prove $\Gamma_\lambda = P_\theta[\lambda g]$, let $v \le 0$ be a competitor with $v \preceq \lambda g$. On the punctured disk $0 < |z|^2 < \lambda/(1-\lambda)$, the function

$$v - \Gamma_\lambda = v + \log(1+|z|^2) - \lambda \log|z|^2 - H(\lambda)$$

is subharmonic, since $v + \log(1+|z|^2)$ is subharmonic and $\log|z|^2$ is harmonic there. Since $v \preceq \lambda g$, this difference is bounded above near zero, so it extends subharmonically across zero by Lemma 1.3.7. Its upper limits at the boundary circle are at most zero, since $v \le 0$ and $\Gamma_\lambda = 0$ there. The maximum principle gives $v \le \Gamma_\lambda$ on the disk (at zero both have the value $-\infty$), and outside the disk the same inequality follows from $\Gamma_\lambda = 0$. Thus every competitor lies below Γ_λ, which is itself a competitor, so $\Gamma_\lambda = P_\theta[\lambda g]$. Since the envelope depends only on the ordinary type, also $P_\theta[\Gamma_\lambda] = \Gamma_\lambda$. The zero potential is an envelope fixed point as well.

For a fixed $0 < |z| < \infty$, put $p = |z|^2/(1+|z|^2)$. The function $\lambda \mapsto \Gamma_\lambda(z)$ is zero for $\lambda \le p$. For $p < \lambda < 1$ its derivative is

$$\log|z|^2 + \log\frac{1-\lambda}{\lambda}.$$

It vanishes at $\lambda = p$ and decreases in λ, which proves that $\lambda \mapsto \Gamma_\lambda(z)$ is concave and decreasing. At $z = 0$, the function is zero for $\lambda \le 0$ and $-\infty$ for $\lambda > 0$; at $z = \infty$, it is zero for $\lambda \le 1$. These functions are concave and decreasing as well. Finally, $\sup_{\lambda<1} \Gamma_\lambda = 0$, and the limit as $\lambda \uparrow 1$ is g, so the family is an envelope test curve with cutoff one.

For finite nonzero z and $t > 0$, the supremum defining $U_t(z)$ is attained at a parameter $\lambda > p$, namely at

$$\lambda = \frac{e^t |z|^2}{1 + e^t |z|^2},$$

and substituting this value into the first line of (19.2) gives (19.3). At $z = 0$, only the parameters $\lambda \le 0$ give finite values, and the supremum is zero; at $z = \infty$, the supremum of $t\lambda$ over $\lambda < 1$ is t. In the coordinate $w = 1/z$, the formula reads $\log((|w|^2 + e^t)/(|w|^2 + 1))$, so the ray is smooth in both charts. □

Each slice of this ray is bounded and has envelope zero, whereas its inverse transforms for $0 < \lambda \le 1$ have logarithmic poles. At $z = 0$, the parameters $\lambda \le 0$ determine the transform: omitting them would give the wrong value $U_t(0) = -\infty$.

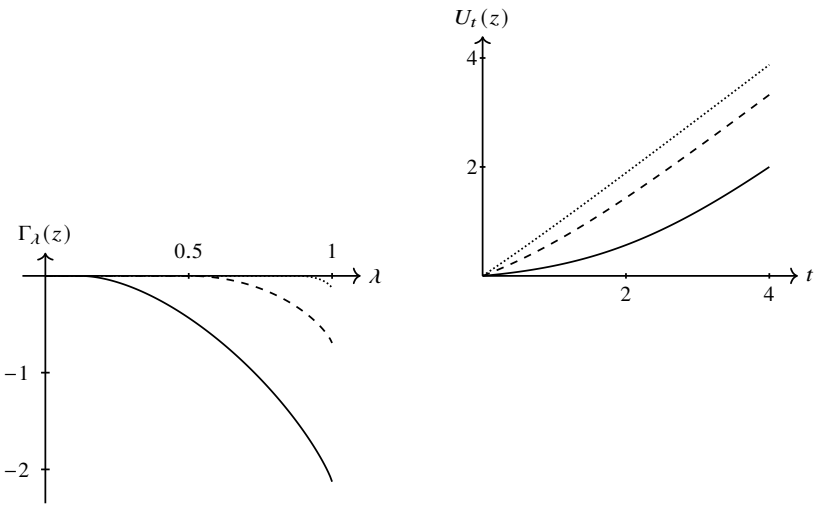


Fig. 19.1 The explicit test curve and its ray, at points with $\log |z|^2 = -2$ (solid), 0 (dashed), and 2 (dotted). The parameter functions on the left are concave; their transforms on the right are convex.

Chapter 20
Restriction and intrinsic trace data

Let X be a connected compact complex manifold, let $Y \subset X$ be a connected compact complex submanifold of positive dimension, and fix a quasi-plurisubharmonic function $u \in \mathrm{QPSH}(X)$ whose generic Lelong number along Y vanishes: $\nu(u, Y) = \inf_{y \in Y} \nu(u, y) = 0$.

Even when all multiplier ideals of u are trivial, its restriction to Y can have logarithmic poles. To keep only information determined by the multiplier ideals of u on X, we restrict the ideals themselves and take their asymptotic vanishing orders on modifications of Y. These orders form the divisorial trace data. Subadditivity of multiplier ideals and analytic approximation make the trace data additive under sums of potentials. For a logarithmic divisor not containing Y plus a residual of arbitrarily small negative curvature, the trace data is that of the divisor alone, even when the restriction of the residual to Y is identically $-\infty$.

The next question is whether a potential on Y realizes these numbers. On a curve, they form an atomic measure, whose Green potential has exactly the prescribed Lelong numbers. In higher dimension, restricted analytic envelopes converge to the largest potential satisfying the trace inequalities, but Lelong semicontinuity gives only a lower bound for its divisorial Lelong numbers. When Y is Kähler, convergence in a sufficiently large closed background gives the reverse bound. Pullback from a curve gives another exact realization, also on non-Kähler targets. On the scalar Hopf manifolds H_n of Section 9.6, elliptic symmetry and extraction of divisors on its projective base extend this construction to projective pullback data of arbitrary dimension. The product examples show that exact trace data still does not determine the Monge–Ampère mass. The final ruled family computes both mass extremes within one unbounded ordinary type, and a blowup separates positivity of its lower volume from existence of a strict current in that type.

Choose a smooth real $(1,1)$-form θ with $u \in \mathrm{PSH}(X, \theta)$ and a Hermitian form ω on X. We use the analytic approximants u_m of Corollary 1.15.1: they have analytic type $(1/m, \mathcal{I}(mu))$ in the sense of Section 1.13 and satisfy $\theta + \mathrm{dd}^{\mathrm{c}} u_m \geq -\epsilon_m \omega$ with $\epsilon_m \downarrow 0$. By Corollary 1.22.2, their restrictions to Y are quasi-plurisubharmonic, in particular not identically $-\infty$. As in Chapter 7, $\mathrm{MA}^{\mathrm{ext}}$ denotes the extended

non-pluripolar Monge–Ampère measure, also written MA when its total mass is finite.

20.1 Restricted ideals and trace data

The restriction of a multiplier ideal to Y records the vanishing of its local generators along Y. Its vanishing order along a divisor on a modification of Y can grow with the weight, even when the ideal is generically trivial on Y.

For $t > 0$ put

$$\mathfrak{a}_t(u;Y) = \mathcal{I}(tu)\mathcal{O}_Y.$$

Here $\mathcal{I}(tu)\mathcal{O}_Y$ denotes the image of $\mathcal{I}(tu)$ under the restriction map $\mathcal{O}_X \to \mathcal{O}_Y$. This is a coherent ideal sheaf on Y, and it is nonzero: since $\nu(u,Y) = 0$, there is $y \in Y$ with $\nu(u,y) < 1/t$, and Skoda's integrability theorem gives $\mathcal{I}(tu)_y = \mathcal{O}_{X,y}$.

In this chapter, a *divisor over Y* is a prime divisor E on a connected smooth compact modification $\pi\colon Z \to Y$. For a nonzero coherent ideal $\mathfrak{a}$ on Y, let $\operatorname{ord}_E(\mathfrak{a})$ be the generic vanishing order along E of the ideal generated by its pullback to Z; it is a nonnegative integer.

Theorem 20.1.1 *For every divisor E over Y, the limit*

$$\tau_{Y,u}(E) := \lim_{t\to\infty} \frac{\operatorname{ord}_E \mathfrak{a}_t(u;Y)}{t} = \sup_{t>0} \frac{\operatorname{ord}_E \mathfrak{a}_t(u;Y)}{t} \tag{20.1}$$

exists and is finite. These numbers depend only on the multiplier-ideal equivalence class of u on X.

Proof Fix the modification $\pi\colon Z \to Y$ carrying E, and write $b(t) = \operatorname{ord}_E \mathfrak{a}_t(u;Y)$ for $t > 0$. For $s, t > 0$, the subadditivity of multiplier ideals (the second inclusion in (1.23)), followed by restriction to Y, gives

$$\mathfrak{a}_{s+t} \subseteq \mathfrak{a}_s\mathfrak{a}_t, \qquad b(s+t) \geq b(s) + b(t).$$

Moreover, $b(t) \geq 0$. For fixed $s > 0$ and $t = ks + r$ with k a nonnegative integer and $0 \leq r < s$, repeated superadditivity gives $b(t) \geq kb(s)$. Hence

$$\liminf_{t\to\infty} \frac{b(t)}{t} \geq \frac{b(s)}{s}.$$

Taking the supremum over s proves the existence of the limit and the supremum formula, with the value $+\infty$ allowed at this stage.

For every sufficiently large integer m, the restriction $u_m|_Y$ has analytic type $(1/m, \mathfrak{a}_m(u;Y))$ by Proposition 1.13.1, and therefore

$$\frac{b(m)}{m} = \nu\big((u_m|_Y) \circ \pi, E\big).$$

These numbers are uniformly bounded by Corollary 1.22.2, so the limit along integers, and hence the supremum, is finite. Finally, the ideals $\mathfrak{a}_t(u;Y)$ are unchanged when u is replaced by a potential with the same multiplier ideals, which proves the last assertion. □

We call the function $E \mapsto \tau_{Y,u}(E)$, defined on all divisors over Y, the *divisorial trace data* of u along Y.

Inclusions of ideals give monotonicity and homogeneity of the trace data directly. For sums, subadditivity of multiplier ideals gives one inequality; for the reverse inequality, an analytic approximant of the sum controls sufficiently fine approximants of both summands.

Proposition 20.1.1 *Let $v \in \mathrm{QPSH}(X)$ also have zero generic Lelong number along Y, and recall that $u \preceq_{\mathcal{I}} v$ means $\mathcal{I}(ku) \subseteq \mathcal{I}(kv)$ for every real $k > 0$ (Section 1.12). Then, as functions on divisors over Y,*

(1) *$u \preceq_{\mathcal{I}} v$ implies $\tau_{Y,u} \geq \tau_{Y,v}$;*
(2) *$\tau_{Y,au} = a\tau_{Y,u}$ for every $a > 0$;*
(3) *$\tau_{Y,u+v} = \tau_{Y,u} + \tau_{Y,v}$;*
(4) *$\tau_{Y,\max(u,v)} \leq \min\{\tau_{Y,u}, \tau_{Y,v}\}$.*

Proof The inclusions of ideals defining $\preceq_{\mathcal{I}}$ persist after restriction and reverse the vanishing orders, which proves (1). Replacing t by at in (20.1) proves (2). Moreover,

$$\mathcal{I}(tu) + \mathcal{I}(tv) \subseteq \mathcal{I}(t\max(u,v)).$$

Since the order of a sum of ideals is the minimum of their orders, restriction and passage to the limit prove (4); note that $\max(u,v)$ also has zero generic Lelong number along Y.

For (3), note first that generic Lelong numbers of sums add, so $u+v$ also has zero generic Lelong number along Y and $\tau_{Y,u+v}$ is defined. Subadditivity of multiplier ideals gives $\tau_{Y,u+v} \geq \tau_{Y,u} + \tau_{Y,v}$. For the reverse inequality, fix a sufficiently large integer k and the analytic approximant S_k of $u+v$ given by Theorem 1.15.1, of analytic type $(1/k, \mathcal{I}(k(u+v)))$; that theorem gives $u+v \leq S_k$. Let U_m, V_m be the corresponding approximants of u and v. Lemma 1.17.11 gives

$$U_m + V_m \leq (1-\delta_m)S_k + O(1), \qquad \delta_m \downarrow 0.$$

Here both δ_m and the bounded error may depend on the fixed k. The restrictions of U_m, V_m and S_k to Y are not identically $-\infty$ by Corollary 1.22.2. Restricting, pulling back to a fixed modification and taking the generic Lelong number along a divisor E on it, we obtain

$$\frac{\mathrm{ord}_E\,\mathfrak{a}_m(u;Y)}{m} + \frac{\mathrm{ord}_E\,\mathfrak{a}_m(v;Y)}{m} \geq (1-\delta_m)\frac{\mathrm{ord}_E\,\mathfrak{a}_k(u+v;Y)}{k}.$$

Letting first $m \to \infty$, with k fixed, and then $k \to \infty$, the finite limits of Theorem 20.1.1 give the reverse inequality. □

20.2 Comparison with ordinary restriction

When the restriction $u|_Y$ is not identically $-\infty$, the restriction theorem for multiplier ideals bounds the trace data by its divisorial Lelong numbers. For analytic singularities, comparison with the approximants gives the reverse inequality.

Proposition 20.2.1 *If $v = u|_Y$ is not identically $-\infty$, then*

$$\tau_{Y,u}(E) \le \nu(v \circ \pi, E)$$

for every divisor E over Y. If u has analytic singularities, equality holds.

Proof The restriction theorem, which is the first inclusion in (1.23), and Lemma 1.17.8, applied on Y, give for every $t > 0$

$$\operatorname{ord}_E \mathfrak{a}_t(u;Y) \le \operatorname{ord}_E \mathcal{I}(tv) \le t\nu(v \circ \pi, E).$$

Divide by t and use Theorem 20.1.1.

If u has analytic singularities, apply Lemma 1.17.11 to the pair $(u, 0)$, with $s = u$, $U_m = u_m$ and $V_m = 0$. It gives

$$u_m \le (1 - \delta_m)u + O(1), \qquad \delta_m \downarrow 0.$$

Restricting to Y, pulling back by π and taking the generic Lelong number along E, we obtain, since u_m has analytic type $(1/m, \mathcal{I}(mu))$,

$$\frac{1}{m} \operatorname{ord}_E \mathfrak{a}_m(u;Y) \ge (1 - \delta_m)\nu((u|_Y) \circ \pi, E).$$

Letting $m \to \infty$ proves the reverse inequality, hence equality. All the restrictions involved are quasi-plurisubharmonic, because $\nu(u, Y) = 0$. □

The inequality can be strict, as the following local example shows. On a small bidisk in $\mathbb{C}^2$ with coordinates (z, w), write

$$u(z, w) = \max\{-\log(-\log|z|^2),\ \log|w|^2\}, \qquad Y = \{z = 0\}.$$

The bidisk is chosen with $|z| < \mathrm{e}^{-1}$, and the first function is extended by $-\infty$ along $z = 0$. It is plurisubharmonic, since $s \mapsto -\log(-s)$ is increasing and convex for $s < 0$, and hence so is the maximum. For every $t > 0$,

$$\mathrm{e}^{-tu} \le (-\log|z|^2)^t,$$

which is locally integrable. Thus all multiplier ideals of u are trivial and the trace data vanishes, whereas $u|_Y = \log|w|^2$ has Lelong number one at the origin. This example is taken from [Xia26, Example 8.1.2].

20.2.1 Logarithmic divisors with a small-curvature residual

An ambient residual can be invisible to every multiplier ideal even when its restriction to Y is identically $-\infty$. The restricted ideals then recover the imposed logarithmic divisor directly.

Proposition 20.2.2 *Let $q \in \mathrm{QPSH}(X)$ have representatives of arbitrarily small negative curvature in its ordinary type, as in (17.9), and let $\Delta = \sum_j a_j D_j$ be a finite effective real divisor with logarithmic potential σ_Δ, as defined before Theorem 17.7.1. For every $t > 0$,*

$$\mathcal{I}\big(t(\sigma_\Delta + q)\big) = \mathcal{I}(t\sigma_\Delta).$$

If Y is contained in none of the components D_j, then for every divisor E over Y, on a modification $\pi\colon Z \to Y$,

$$\tau_{Y,\sigma_\Delta+q}(E) = \sum_j a_j \operatorname{ord}_E \pi^*(D_j|_Y).$$

Proof The first equality follows from Lemmas 1.16.4 and 17.7.1. Let s_j be local equations of the D_j, so that $\sigma_\Delta = \sum_j a_j \log|s_j|^2 + O(1)$ locally. If $f = g\prod_j s_j^{\lceil ta_j\rceil}$, then $|f|^2 e^{-t\sigma_\Delta}$ is locally bounded, so $f \in \mathcal{I}(t\sigma_\Delta)$. Conversely, near a general point of D_j, where $\sigma_\Delta = a_j \log|s_j|^2 + O(1)$ with s_j a coordinate, Lemma 1.16.1 in codimension one shows that every germ of $\mathcal{I}(t\sigma_\Delta)$ vanishes to order at least $\lfloor ta_j\rfloor$ along D_j. Such a germ is therefore divisible by $\prod_j s_j^{\lfloor ta_j\rfloor}$, and

$$\mathcal{O}_X\Big(-\sum_j \lceil ta_j\rceil D_j\Big) \subseteq \mathcal{I}(t\sigma_\Delta) \subseteq \mathcal{O}_X\Big(-\sum_j \lfloor ta_j\rfloor D_j\Big).$$

Since Y is contained in no D_j, each restricted equation $s_j|_Y$ is a nonzero holomorphic germ on Y. Its pullback to Z is nonzero and has a finite order $N_{j,E} = \operatorname{ord}_E \pi^*(D_j|_Y)$ along E. Restricting the inclusions to Y and pulling back to Z give

$$\sum_j \lfloor ta_j\rfloor N_{j,E} \le \operatorname{ord}_E \mathfrak{a}_t(\sigma_\Delta + q; Y) \le \sum_j \lceil ta_j\rceil N_{j,E}.$$

Moreover $\nu(\sigma_\Delta + q, Y) = 0$, since at points of Y outside the finitely many divisors both summands have zero Lelong number. Thus these orders are those of Theorem 20.1.1, and dividing by t and letting $t \to \infty$ proves the trace formula. □

20.3 Realization on a compact curve

We return to the compact setting, and until Section 20.6 we assume that Y is a curve. Since every modification of a compact smooth curve is an isomorphism, the divisors over Y are its points, and the trace data prescribes an atomic curvature measure. Stokes' theorem bounds its total mass by the degree of the background on Y, and filling the remaining mass with a smooth measure gives a positive current whose potential realizes the data.

Set

$$D_u = \sum_{p \in Y} \tau_{Y,u}(p)\, \delta_p,$$

where the sum is understood as the supremum of its finite subsums.

Theorem 20.3.1 *Put $M = \int_Y \theta|_Y$. Then*

$$0 \le D_u(Y) \le M. \tag{20.2}$$

The finite atomic measure D_u determines a canonical ordinary singularity type of quasi-plurisubharmonic functions on Y: for every smooth positive form η on Y with $\int_Y \eta = 1$, this type has a representative $v \in \mathrm{PSH}(Y, \theta|_Y)$ with

$$\theta|_Y + \mathrm{dd}^{\mathrm{c}} v = D_u + (M - D_u(Y))\eta. \tag{20.3}$$

The extended Monge–Ampère measure of v is $(M - D_u(Y))\eta$. The ordinary type does not depend on η, on the choice of the smooth form θ with $u \in \mathrm{PSH}(X, \theta)$, or on the choice of u within its multiplier-ideal class.

Proof For each positive integer m, let

$$D_m = \frac{1}{m} \sum_{p \in Y} \mathrm{ord}_p\, \mathfrak{a}_m(u; Y)\, \delta_p.$$

A nonzero coherent ideal on a compact smooth curve has finitely many zeros, so this sum is finite. Since $u_m|_Y$ has analytic type $(1/m, \mathfrak{a}_m(u; Y))$, the measure D_m is the atomic part of the positive current $(\theta + \epsilon_m \omega)|_Y + \mathrm{dd}^{\mathrm{c}}(u_m|_Y)$, and Stokes' theorem on the curve gives

$$D_m(Y) \le M + \epsilon_m \int_Y \omega|_Y.$$

For any finite subset of Y, passing to the limit in the sum of the corresponding coefficients by Theorem 20.1.1, and then taking the supremum over finite subsets, gives (20.2). In particular, the positive coefficients have at most countable support and define a finite measure. Write $d = D_u(Y)$.

Choose a Green kernel G for dd^{c} on the curve, normalized so that

$$\mathrm{dd}^{\mathrm{c}}_x G(x, p) = \delta_p - \eta, \qquad G(x, p) \le 0.$$

Such a kernel exists: the Green kernel of the scalar Laplacian on the compact Riemann surface has, in a local coordinate z, the singularity $\log|z(x)-z(p)|^2$, is uniformly bounded above, and satisfies $\sup_p \int_Y |G(x,p)|\eta(x) < \infty$; subtracting a constant gives the sign condition. By Fubini's theorem,

$$w(x) = \int_Y G(x,p)\,\mathrm{d}D_u(p)$$

is integrable, and it is quasi-subharmonic with $\mathrm{dd}^{\mathrm{c}} w = D_u - d\,\eta$. Its Lelong number at every point p is the mass of D_u at p, namely $\tau_{Y,u}(p)$.

Since $M\eta - \theta|_Y$ is smooth with integral zero, the Poisson equation gives a smooth s with $\mathrm{dd}^{\mathrm{c}} s = M\eta - \theta|_Y$. Then $v = w + s$ satisfies (20.3), and the right-hand side is positive because $M \geq d$. At every atom of D_u, the positive Lelong number forces $v = -\infty$. The extended measure of v is the restriction of the current in (20.3) to the set where $v > -\infty$; this set contains no atom of D_u, and its complement is polar and therefore not charged by η. Hence the extended measure is $(M-d)\eta$.

If another smooth form η or another smooth background θ is used, the two constructed representatives have the same atomic curvature, and all other terms in their equations are smooth. Their difference therefore has smooth dd^{c}, so it is smooth by elliptic regularity, and in particular bounded. This proves independence of these choices at the level of ordinary types. Independence of the choice of u within its multiplier-ideal class holds because the numbers defining D_u depend only on that class. □

We denote this ordinary type by $\mathrm{Tr}^{\mathrm{at}}_Y(u)$ and call it the *atomic trace* of u on the curve Y; the same symbol also denotes any of its representatives. All potentials on a curve with these Lelong numbers have the same multiplier ideals, but they need not have the same ordinary type; the construction selects the type whose curvature, apart from the atoms, is a smooth measure. Additivity of the trace data (Proposition 20.1.1) and linearity of the Green operator give

$$\mathrm{Tr}^{\mathrm{at}}_Y(u+v) \sim \mathrm{Tr}^{\mathrm{at}}_Y(u) + \mathrm{Tr}^{\mathrm{at}}_Y(v), \qquad \mathrm{Tr}^{\mathrm{at}}_Y(au) \sim a\,\mathrm{Tr}^{\mathrm{at}}_Y(u) \quad (a>0).$$

Here $v \in \mathrm{QPSH}(X)$ is any function that also has zero generic Lelong number along Y.

Keep D_m, G and w from the proof of Theorem 20.3.1, and put $w_m = \int_Y G(\cdot,p)\,\mathrm{d}D_m(p)$.

Proposition 20.3.1 *The measures D_m satisfy $D_m \leq D_u$ and converge to D_u in total variation, and they increase along dyadic indices. For a fixed nonpositive Green kernel, the potentials w_{2^j} decrease to w, and $w_m \to w$ in $L^1(Y)$. Each w_m has the ordinary singularity type of $u_m|_Y$.*

If the restriction $u|_Y$ is not identically $-\infty$, it is at least as singular as the atomic trace:

$$u|_Y \preceq \mathrm{Tr}^{\mathrm{at}}_Y(u).$$

If u has analytic singularities, the two are ordinarily equivalent.

Proof The supremum formula in (20.1) gives $D_m \le D_u$ coefficient by coefficient. Since the coefficients converge and are dominated by those of the finite measure D_u, dominated convergence gives $D_m(Y) \to D_u(Y)$, and hence convergence in total variation. Superadditivity with $s = t = m$ gives $D_{2m} \ge D_m$, and since $G \le 0$ the potentials w_{2^j} decrease. Monotone convergence for $-G$ identifies their pointwise limit with w, including at poles. The uniform L^1 bound for the kernel also gives

$$\|w_m - w\|_{L^1(\eta)} \le C\|D_m - D_u\|_{\mathrm{TV}} \longrightarrow 0.$$

Each D_m is finitely supported, and locally both w_m and $u_m|_Y$ are the sum of the logarithmic singularities prescribed by D_m and a bounded function. Thus their ordinary types agree.

If $a = u|_Y$ is not identically $-\infty$, Proposition 20.2.1 gives $\nu(a, p) \ge \tau_{Y,u}(p)$ at every point. Since $\mathrm{dd^c} w = D_u - D_u(Y)\eta$, Lemma 1.17.5 implies $a \preceq w$. If u has analytic singularities, Proposition 20.2.1 identifies the finitely many logarithmic coefficients of $u|_Y$ with those of D_u, and the bounded local remainders give ordinary equivalence. □

20.4 The Poisson equation in the atomic trace type

Write $d = D_u(Y)$, and let w be the Green representative constructed in the proof of Theorem 20.3.1. For every $\theta|_Y$-plurisubharmonic potential in the ordinary type of w, the part of its curvature measure carried by polar sets is D_u, and its extended measure is the remaining part, of total mass $M - d$. Prescribing the extended measure therefore amounts to solving a Poisson equation for the bounded difference from w.

Theorem 20.4.1 *Let* $\mathrm{d}V$ *be a smooth positive volume form on* Y*, and let* $0 \le f \in L^p(\mathrm{d}V)$*,* $p > 1$*, with* $\int_Y f\,\mathrm{d}V > 0$*. There is a solution* v *in the atomic trace type of*

$$\mathrm{MA}^{\mathrm{ext}}_{\theta|_Y}(v) = c f\,\mathrm{d}V, \qquad c > 0,$$

if and only if $M > d$*. In that case*

$$c = \frac{M - d}{\int_Y f\,\mathrm{d}V},$$

and v *is unique up to an additive constant. If* $M = d$*, there is instead a potential in this type with extended measure zero, unique up to an additive constant.*

Proof Assume that $M > d$, take the value of c given above, and solve

$$\mathrm{dd^c} s = d\,\eta + c f\,\mathrm{d}V - \theta|_Y.$$

The right-hand side has integral zero, so the elliptic L^p theory gives a solution $s \in W^{2,p}(Y)$, unique up to constants. In real dimension two, the Sobolev embedding

for $p > 1$ makes s continuous and bounded. The potential $v = w + s$ has the atomic trace type and satisfies

$$\theta|_Y + \mathrm{dd}^{\mathrm{c}} v = D_u + c\, f\, \mathrm{d}V \geq 0.$$

Its extended measure is $c\, f\, \mathrm{d}V$, since this absolutely continuous measure does not charge the polar set of poles, while the atoms lie in it. For $M = d$, the same construction with $c = 0$ has a smooth right-hand side and gives the asserted potential with zero extended measure.

To prove necessity and uniqueness, let a be any $\theta|_Y$-plurisubharmonic potential in the ordinary type of w, and put $\mu = \theta|_Y + \mathrm{dd}^{\mathrm{c}} a$. By Corollary 1.6.1, ordinary equivalence with w makes the polar part of μ equal to D_u. Consequently

$$\mathrm{MA}^{\mathrm{ext}}_{\theta|_Y}(a) = \mu - D_u, \qquad \int_Y \mathrm{MA}^{\mathrm{ext}}_{\theta|_Y}(a) = M - d.$$

Integrating the equation therefore forces the stated value of c, so a positive c requires $M > d$. If two potentials solve the equation with the same density and constant, their curvature currents both equal $D_u + c\, f\, \mathrm{d}V$; their difference has zero dd^{c} and is therefore constant on the compact connected curve. The same argument proves uniqueness in the case of zero extended measure. □

20.5 Singularity distance on a curve

On the curve Y, the singularity pseudometric d_S of [Xia26, Definition 6.2.1] measures the difference of the masses of two ordered potentials. For the Green representatives w and w_m, it is half the atomic mass missing from D_m. When X is Kähler, this identifies the atomic trace with the Kähler trace of [Xia26, Section 8.1], up to P-equivalence, that is, up to potentials with the same singularity envelope; the distance calculation itself uses only the geometry of the curve.

Proposition 20.5.1 *Choose a smooth positive form η of integral one on Y and choose $A > D_u(Y)$ so large that every restricted approximant $u_m|_Y$ is $A\eta$-plurisubharmonic. Define w and w_m as before Proposition 20.3.1, with this form η and a fixed nonpositive Green kernel. Then*

$$d_{S,A\eta}(w, u_m|_Y) = d_{S,A\eta}(w, w_m) = \frac{1}{2}\big(D_u(Y) - D_m(Y)\big) \longrightarrow 0. \tag{20.4}$$

In particular, the atomic trace has $\mathcal{I}$-good singularities on the curve, in the sense recalled in Chapter 17. If X is Kähler, the atomic trace represents the P-equivalence class of the Kähler trace.

Proof The curvature bounds of Corollary 1.15.1, restricted to Y, are dominated by a single form $A\eta$ once A is large. Since $D_m \leq D_u$ and $G \leq 0$, we have $w \leq w_m$. Also

$$A\eta + \mathrm{dd}^{\mathrm{c}} w = D_u + (A - D_u(Y))\eta, \qquad A\eta + \mathrm{dd}^{\mathrm{c}} w_m = D_m + (A - D_m(Y))\eta.$$

Hence their extended masses are $A - D_u(Y)$ and $A - D_m(Y)$, both positive. For $a, b \in \mathrm{PSH}(Y, A\eta)$ with $a \le b$, the formula for the distance of ordered potentials [Xia26, Lemma 6.2.2] reads

$$d_{S,A\eta}(a,b) = \frac{1}{2}\left(\int_Y \mathrm{MA}_{A\eta}(b) - \int_Y \mathrm{MA}_{A\eta}(a)\right).$$

Substituting $a = w$ and $b = w_m$ proves the second equality in (20.4). Since $u_m|_Y$ and w_m are ordinarily equivalent, their distance vanishes, which proves the first equality. The convergence of D_m in total variation proves the limit.

Each w_m has finitely many logarithmic poles with rational coefficients, hence analytic singularities. The characterization in [Xia26, Theorem 7.1.1], applied on the compact Kähler curve Y, shows that their d_S-limit w is $\mathcal{I}$-good.

Finally, suppose that X is Kähler. The Kähler trace is defined in [Xia26, Definition 8.1.1 and Proposition 8.1.2] through a quasi-equisingular approximation of u with analytic types $(2^{-j}, \mathcal{I}(2^j u))$, whose restriction to Y has the ordinary type of w_{2^j}. The same distance calculation shows that these restrictions converge to w for $d_{S,A\eta}$, so w represents the Kähler trace up to P-equivalence. □

20.5.1 Total variation of the atomic data

The distance calculation extends to all multiplier-ideal envelopes $P_\beta[\cdot]_{\mathcal{I}}$ of Definition 17.2.1 on the curve Y. Such an envelope is determined by its atomic curvature, and the maximum of two of them has as atomic curvature the common part of the two atomic measures. The formula for the distance of ordered potentials can therefore compare envelopes whose poles are unrelated. Throughout this subsection, fix a smooth positive form β on Y, put $A = \int_Y \beta$, and choose a nonpositive Green kernel G for $\eta = \beta/A$. All atomic measures below have countable support and finite total mass.

For a positive atomic measure D with $D(Y) \le A$, set

$$w_D(x) = \int_Y G(x,p)\,\mathrm{d}D(p), \qquad q_D = P_\beta[w_D]_{\mathcal{I}}.$$

As in the proof of Theorem 20.3.1, w_D is a quasi-plurisubharmonic function, not identically $-\infty$, with $\mathrm{dd}^{\mathrm{c}} w_D = D - D(Y)\eta$.

Lemma 20.5.1 *The function w_D is β-plurisubharmonic, and*

$$q_D \sim w_D, \qquad \int_Y \mathrm{MA}_\beta(q_D) = \int_Y \mathrm{MA}_\beta(w_D) = A - D(Y).$$

The map $D \mapsto q_D$ is a bijection from the positive atomic measures of mass at most A onto the set of all envelopes $P_\beta[a]_I$, $a \in \mathrm{PSH}(Y,\beta)$, and q_D has positive mass exactly when $D(Y) < A$. The measure D is the atomic part of $\beta + \mathrm{dd}^c q_D$.

Proof One has

$$\beta + \mathrm{dd}^c w_D = D + (A - D(Y))\eta \geq 0,$$

and, as for the extended measure in the proof of Theorem 20.3.1, $\mathrm{MA}_\beta(w_D) = (A - D(Y))\eta$.

Put $q = q_D$. By Theorem 17.2.1, q has the same multiplier ideals, and hence the same Lelong numbers at points, as w_D. Applied to $\mathrm{dd}^c w_D = D - D(Y)\eta$, Lemma 1.17.5 gives $q \preceq w_D$, and the envelope inequality $q \geq w_D - \sup_Y w_D$ proves $q \sim w_D$. By Corollary 1.6.1, the polar part of $\beta + \mathrm{dd}^c q$ is therefore also D, and Stokes' theorem gives the non-pluripolar mass $A - D(Y)$.

For any $a \in \mathrm{PSH}(Y,\beta)$, its Lelong numbers at points give an atomic measure D_a of mass at most A. On a smooth curve these numbers determine all multiplier ideals, by Theorem 1.16.1. Thus $a \sim_I w_{D_a}$, so $P_\beta[a]_I = q_{D_a}$. If $D_a(Y) = A$, the atoms exhaust the whole curvature measure, and the non-pluripolar mass is zero. This proves surjectivity and the description of the potentials of positive mass. Distinct atomic measures give distinct Lelong numbers, which proves injectivity. □

Theorem 20.5.1 *For positive atomic measures D, E with masses less than A,*

$$\frac{1}{96}\|D - E\|_{\mathrm{TV}} \leq d_{S,\beta}(q_D, q_E) \leq \frac{1}{2}\|D - E\|_{\mathrm{TV}}.$$

Thus the singularity distance is a metric on these ideal envelopes, and convergence for it is exactly convergence of the atomic measures in total variation.

Proof Write $d_S = d_{S,\beta}$ and $a \vee b = \max(a,b)$. For potentials $a, b \in \mathrm{PSH}(Y,\beta)$ of positive mass, besides the formula for ordered potentials we use the inequality [Xia26, Lemma 6.2.1]

$$d_S(a,b) \leq d_S(a, a \vee b) + d_S(b, a \vee b) \leq 48 d_S(a,b)$$

on a curve; here $48 = 3(1+1)2^{1+2}$ is the constant of that lemma in dimension one.

Put $z = \max(w_D, w_E)$. By Lemma 20.5.1, the polar parts of the curvature measures of w_D and w_E are D and E. By (1.8), the polar part of the curvature measure of z is $D \wedge E$, the atomic measure with the smaller coefficient at each point. Stokes' theorem therefore gives

$$\mathrm{MA}_\beta(z)(Y) = A - (D \wedge E)(Y).$$

The formula for ordered potentials yields

$$\begin{aligned} d_S(w_D, z) + d_S(w_E, z) &= \frac{1}{2}\big(D(Y) + E(Y) - 2(D \wedge E)(Y)\big) \\ &= \frac{1}{2}\|D - E\|_{\mathrm{TV}}. \end{aligned}$$

Combining this equality with the two inequalities above proves both bounds, since replacing w_D, w_E by the ordinarily equivalent q_D, q_E does not change the singularity distance. The lower bound and the injectivity in Lemma 20.5.1 make the pseudometric a metric on these envelopes. □

Corollary 20.5.1 *For $0 < \varepsilon < A$, the ideal envelopes q satisfying $\mathrm{MA}_\beta(q)(Y) \geq \varepsilon$ form a complete space for $d_{S,\beta}$. Analytic ideal envelopes are dense in this space. The space is not totally bounded.*

Proof By Lemma 20.5.1, the space corresponds to the positive atomic measures of mass at most $A - \varepsilon$. This set is complete for total variation. Indeed, a Cauchy sequence has as limit a positive finite measure with the same mass bound. The union of the countable supports of the sequence is countable, and total-variation convergence shows that the limit gives zero mass to the complement of that union, so the limit is again atomic. The two metric estimates of Theorem 20.5.1 transfer completeness to the ideal envelopes.

Finite truncation, followed by rational approximation of each coefficient from below, preserves the mass bound and converges in total variation. The resulting Green potentials have finitely many logarithmic poles with rational coefficients, and their ideal envelopes have the same ordinary types by Lemma 20.5.1. These envelopes therefore have analytic singularities, and the upper metric estimate proves density.

Finally, fix $0 < a < A - \varepsilon$. The measures $a\delta_p$, for distinct $p \in Y$, have pairwise total-variation distance $2a$, so their ideal envelopes have pairwise d_S distance at least $a/48$. This infinite separated family rules out total boundedness. □

This topology differs from the one given by the L^1 metric on ideal envelopes of Proposition 17.4.1, since that metric is totally bounded. The sequence in Proposition 17.4.2 makes the difference quantitative: it converges for that L^1 metric while remaining at a fixed positive d_S distance from its limit.

20.6 The trace envelope

We return to the general setting, in which $Y \subset X$ has arbitrary positive dimension. The curvature errors of the restricted approximants tend to zero, so normalized compactness gives a limit potential in the original background $\theta|_Y$, whose divisorial Lelong numbers dominate the trace data. These inequalities define an envelope, and comparison with the resolved restricted ideals identifies it as the limit of the restricted analytic envelopes.

Proposition 20.6.1 *There is a potential $v \in \mathrm{PSH}(Y, \theta|_Y)$ such that*

$$v(v \circ \pi, E) \geq \tau_{Y,u}(E) \quad \text{for every divisor } E \text{ over } Y.$$

In particular, $v \in \mathrm{PSH}(Y, \Omega_0)$ for every Hermitian form $\Omega_0 \geq \theta|_Y$.

Proof The restrictions $u_m|_Y$ have a common smooth lower bound for their curvature. After subtracting their suprema, normalized compactness gives a subsequence converging in L^1 to some v. The inequalities $(\theta + \epsilon_m\omega)|_Y + \mathrm{dd}^{\mathrm{c}}(u_m|_Y) \geq 0$ pass to the limit as $\epsilon_m \to 0$, giving $\theta|_Y + \mathrm{dd}^{\mathrm{c}} v \geq 0$.

For each fixed compact modification $\pi\colon Z \to Y$, Lemma 1.22.8 gives L^1 convergence of the pullbacks along the same subsequence, and Lemma 1.17.6 gives, at every point z of a prime divisor $E \subset Z$,

$$v(v \circ \pi, z) \geq \limsup_m v((u_m|_Y) \circ \pi, z) \geq \lim_m \frac{\mathrm{ord}_E\, \mathfrak{a}_m(u; Y)}{m} = \tau_{Y,u}(E).$$

Taking the generic value along E proves the assertion. No diagonal extraction over divisors is needed, since the pullback lemma applies to each modification once the subsequence on Y has been fixed. □

For a smooth real $(1,1)$-form Ω on Y for which the following family is nonempty, define

$$\mathcal{T}_\Omega(u; Y) := \Big(\sup\{a \in \mathrm{PSH}(Y, \Omega) : a \leq 0, \\ v(a \circ \pi, E) \geq \tau_{Y,u}(E) \ \text{ for every } E\}\Big)^*. \tag{20.5}$$

Here E ranges over all divisors over Y, and we call the members of this family *trace competitors* for Ω. By the preceding proposition, $v - \sup_Y v$ is a trace competitor for $\Omega = \theta|_Y$ and for every larger background. The envelope $\mathcal{T}_\Omega(u; Y)$ depends only on Ω and on the multiplier-ideal class of u on X.

20.6.1 Limits of restricted analytic envelopes

On a log resolution of $\mathfrak{a}_m(u; Y)$, the trace inequalities dominate every coefficient of the restricted analytic singularity, so each trace competitor is at least as singular as $u_m|_Y$. This gives a common lower barrier for the analytic envelopes. Semicontinuity of Lelong numbers puts every cluster point back into the family of trace competitors, where maximality identifies it.

Theorem 20.6.1 *Let Ω be a smooth real $(1,1)$-form on Y, let h be a Hermitian form on Y, and let $\delta_m \geq 0$ be numbers tending to zero such that, for all sufficiently large m,*

$$a_m := u_m|_Y \in \mathrm{PSH}(Y, \Omega + \delta_m h).$$

Then the family of trace competitors for Ω is nonempty, $T = \mathcal{T}_\Omega(u;Y)$ belongs to $\mathrm{PSH}(Y,\Omega)$, and

$$q_m := P_{\Omega+\delta_m h}[a_m] = P_{\Omega+\delta_m h}[a_m]_{\mathcal{I}} \longrightarrow T \quad \text{in } L^1(Y). \tag{20.6}$$

If δ_{2^j} decreases in j, then $q_{2^j} \searrow T$ pointwise. The limit depends only on Ω and the multiplier-ideal class of u on X. Moreover,

$$P_\Omega[T]_{\mathcal{I}} = P_\Omega[T] = T.$$

Proof *Step 1: T is the largest trace competitor.* The argument of Proposition 20.6.1, applied with the assumed curvature bounds, gives a normalized Ω-plurisubharmonic limit v of a subsequence of the a_m, whose divisorial Lelong numbers dominate $\tau_{Y,u}$. Hence the family defining T is nonempty, and its supremum lies between v and zero. The family is stable under finite maxima, since the Lelong number of a maximum is the minimum of the two Lelong numbers, also after pullback. By Choquet's lemma and finite maxima, there is an increasing sequence of trace competitors whose limit has upper semicontinuous regularization T. Convergence of the pullbacks and semicontinuity of Lelong numbers show that T satisfies every defining inequality. Thus T is itself a trace competitor, and it is the largest one.

Step 2: a common lower barrier. Fix m, and let $\pi\colon Z \to Y$ be a log resolution of $\mathfrak{a}_m(u;Y)$. For each prime component E of the resulting divisor, (20.1) gives

$$\nu(T\circ\pi, E) \geq \tau_{Y,u}(E) \geq \frac{1}{m}\operatorname{ord}_E \mathfrak{a}_m(u;Y) = \nu(a_m\circ\pi, E).$$

The criterion of Lemma 1.17.10, which involves only these finitely many divisors, gives $T \preceq a_m$. Since $T \leq 0$ and $\delta_m \geq 0$, the function T is a competitor for q_m. Moreover, Proposition 17.2.4 identifies the two envelopes in (20.6) and gives $q_m \sim a_m$. Therefore

$$T \leq q_m \leq 0, \qquad \nu(q_m\circ\pi, E) = \frac{1}{m}\operatorname{ord}_E \mathfrak{a}_m(u;Y)$$

for every divisor over Y.

Step 3: convergence. All q_m have supremum zero and a common smooth lower bound for their curvature. Let q be any L^1 cluster point. It belongs to $\mathrm{PSH}(Y,\Omega)$, lies between T and zero, and satisfies

$$\nu(q\circ\pi, E) \geq \limsup_m \nu(q_m\circ\pi, E) = \tau_{Y,u}(E)$$

along the convergent subsequence, by the pullback and semicontinuity argument of Proposition 20.6.1 on each fixed compact modification. Thus q is a trace competitor,

so $q \le T$, and hence $q = T$. Normalized compactness now gives L^1 convergence of the whole sequence.

Step 4: monotonicity and the fixed-point identities. Subadditivity gives $\mathfrak{a}_{2m}(u;Y) \subseteq \mathfrak{a}_m(u;Y)^2$, hence $a_{2m} \preceq a_m$. When the backgrounds $\Omega + \delta_{2^j} h$ decrease, this implies $q_{2^{j+1}} \le q_{2^j}$. The lower bound by T shows that the pointwise limit is not identically $-\infty$, and since it agrees with T almost everywhere by L^1 convergence, it equals T everywhere. The family defining T depends only on Ω and on the multiplier ideals of u, which proves the asserted independence.

Finally, $P_\Omega[T]_{\mathcal{I}}$ has the same multiplier ideals as T, hence the same divisorial Lelong numbers, so it is a trace competitor. Maximality and the inequality $P_\Omega[T]_{\mathcal{I}} \ge T$ give $P_\Omega[T]_{\mathcal{I}} = T$, and the identity $P_\Omega[T] = T$ follows from Proposition 17.2.2. □

Taking $\Omega = \theta|_Y$, $h = \omega|_Y$ and $\delta_m = \epsilon_m$ gives the limit in the original restricted background, monotone along dyadic indices since $\epsilon_m \downarrow 0$. In a fixed, sufficiently large Hermitian background one can take $\delta_m = 0$.

Inclusion of the families of trace competitors also orders the trace envelopes: increasing the prescribed data decreases $\mathcal{T}_\Omega$, whereas enlarging the background increases it whenever both envelopes are defined.

20.6.2 Envelopes on modifications

Let $\pi\colon Z \to Y$ be a modification of connected compact complex manifolds. Extend the restricted ideals to Z by taking the ideals $\mathfrak{a}_t(u;Y)\mathcal{O}_Z$ generated by their pullbacks. For a divisor F on a further modification $\rho\colon W \to Z$, their asymptotic order is $\tau_{Y,u}(F)$, computed on $\pi \circ \rho\colon W \to Y$. For a smooth real $(1,1)$-form Γ on Z, write $\mathcal{T}_\Gamma^{\pi}(u;Y)$ for the envelope on Z defined as in (20.5), with background Γ and the inequalities $\nu(a \circ \rho, F) \ge \tau_{Y,u}(F)$ for all divisors F over Z. Thus the trace data still comes from the multiplier ideals of u on X.

For the pulled-back background $\pi^*\Omega$, every competitor descends to Y, and a common refinement of two models identifies the divisorial inequalities on them.

Theorem 20.6.2 *Suppose Ω is a smooth real $(1,1)$-form on Y for which $T = \mathcal{T}_\Omega(u;Y)$ is defined, and let h be a Hermitian form on Z. Then*

$$\mathcal{T}^{\pi}_{\pi^*\Omega}(u;Y) = T \circ \pi, \qquad \mathcal{T}^{\pi}_{\pi^*\Omega+\varepsilon h}(u;Y) \searrow T \circ \pi \quad (\varepsilon \downarrow 0). \tag{20.7}$$

The convergence is pointwise and in $L^1(Z)$.

Let $\delta_m \geq 0$ tend to zero and suppose, for all sufficiently large m, that

$$b_m := (u_m|_Y) \circ \pi \in \mathrm{PSH}(Z, \pi^*\Omega + \delta_m h).$$

Then

$$P_{\pi^*\Omega+\delta_m h}[b_m] = P_{\pi^*\Omega+\delta_m h}[b_m]_{\mathcal{I}} \longrightarrow T \circ \pi \quad \text{in } L^1(Z).$$

If δ_{2^j} decreases, these envelopes decrease pointwise to $T \circ \pi$ along the dyadic indices.

Proof For any nonempty family of trace competitors, the argument with finite maxima and semicontinuity in Step 1 of the proof of Theorem 20.6.1 shows that the envelope is its largest member. By Proposition 1.22.4, every $\pi^*\Omega$-plurisubharmonic function is uniquely of the form $a \circ \pi$ with $a \in \mathrm{PSH}(Y, \Omega)$, and since π is surjective, $a \leq 0$ if and only if $a \circ \pi \leq 0$. If a satisfies all trace inequalities over Y, then $a \circ \pi$ satisfies those over Z, since every modification of Z, composed with π, is a modification of Y.

Conversely, suppose that $a \circ \pi$ satisfies the inequalities over Z. Let E be a prime divisor on a smooth compact modification $V \to Y$. Resolving the graph of the bimeromorphic map between V and Z gives a smooth compact modification W dominating both. The map $W \to V$ is an isomorphism near the generic point of E, so the strict transform F of E has the same ideal orders and generic Lelong numbers as E, and the inequality at F over Z is exactly the required inequality at E. Thus the two families of competitors correspond under pullback, and the largest member on Z is $T \circ \pi$. This proves the first identity.

For $\varepsilon > 0$, the pullback $T \circ \pi$ is still a competitor for $\mathcal{T}^{\pi}_{\pi^*\Omega+\varepsilon h}(u;Y)$. These envelopes lie between $T \circ \pi$ and zero and decrease as $\varepsilon \downarrow 0$, so their limit S satisfies $S \geq T \circ \pi$ and is $\pi^*\Omega$-plurisubharmonic. The L^1 convergence of pullbacks and semicontinuity of Lelong numbers preserve every trace inequality, so S belongs to the unperturbed family. Maximality gives $S \leq T \circ \pi$, hence equality. Since the lower bound $T \circ \pi$ is integrable on Z, dominated convergence gives L^1 convergence.

For the final assertion, denote the analytic envelope in the statement by q_m. The proof of Theorem 20.6.1 applies to the global coherent ideals $\mathfrak{a}_m(u;Y)\mathcal{O}_Z$: resolving each of them and applying Lemma 1.17.10 gives $T \circ \pi \preceq b_m$. Hence

$$T \circ \pi \leq q_m \leq 0, \qquad \nu(q_m \circ \rho, F) = \frac{1}{m} \operatorname{ord}_F \mathfrak{a}_m(u;Y)$$

for every divisor F over Z, where ordinary and ideal envelopes agree by Proposition 17.2.4. Every L^1 cluster point is $\pi^*\Omega$-plurisubharmonic and satisfies all trace inequalities, so the first identity in (20.7) identifies it with $T \circ \pi$, and compactness gives convergence of the whole sequence. Subadditivity orders the dyadic analytic types, decreasing dyadic backgrounds then order the envelopes, and their pointwise limit is the same L^1 limit. □

For $\Omega = \theta|_Y$, the last hypothesis holds with $\delta_m = C\epsilon_m$, where the constant C satisfies $\pi^*(\omega|_Y) \le Ch$. Thus the restricted analytic envelopes converge to the same pulled-back trace envelope on every compact model. Formula (20.7) also gives compatibility under successive modifications.

Equality of the ordinary types of trace envelopes in different backgrounds would require a reverse bound on their singularities. For curve fibrations, the fixed divisorial potential of Theorem 20.7.1 gives this bound in every sufficiently large background.

20.6.3 Exact realization on a Kähler submanifold

By Theorem 20.6.1, semicontinuity of Lelong numbers gives $\nu(\mathcal{T}_\Omega(u;Y)\circ\pi, E) \ge \tau_{Y,u}(E)$. When Y is Kähler, positive mass in a sufficiently large closed background strengthens this to equality: for decreasing sequences of envelope fixed points whose limit has positive mass, the divisorial Lelong numbers converge. The envelope obtained in this way then gives the reverse inequality in every larger background.

Theorem 20.6.3 *Suppose Y is Kähler. Choose a Hermitian form $\Omega_0 \ge \theta|_Y$ and a Kähler form $\kappa \ge 2\Omega_0$ on Y. For every Hermitian form $\Omega \ge \kappa$,*

$$\nu\big(\mathcal{T}_\Omega(u;Y)\circ\pi, E\big) = \tau_{Y,u}(E)$$

for every prime divisor E on every smooth compact modification $\pi\colon Z \to Y$.

Proof The curvature errors ϵ_m tend to zero, so $u_m|_Y \in \mathrm{PSH}(Y,\kappa)$ for all sufficiently large m. By Theorem 20.6.1, applied with $\Omega = \kappa$ and $\delta_m = 0$, the envelope fixed points

$$q_j = P_\kappa[u_{2^j}|_Y]$$

decrease to $T = \mathcal{T}_\kappa(u;Y)$, after discarding finitely many indices. The normalized potential v of Proposition 20.6.1 belongs to $\mathrm{PSH}(Y,\Omega_0) \subset \mathrm{PSH}(Y,\kappa/2)$ and satisfies $v \le T$, so $P_{\kappa/2}(T) \not\equiv -\infty$. Thus Lemma 14.5.2 and Stokes' theorem give, with $r = \dim Y$,

$$\int_Y \mathrm{MA}_\kappa(T) \ge 2^{-r}\int_Y \kappa^r > 0.$$

For a decreasing sequence of envelope fixed points in a closed background, positive mass of the limit implies convergence of every divisorial Lelong number [Xia26, Proposition 3.1.12]. Applied on Y, this yields

$$\nu(T\circ\pi, E) = \lim_j \nu(q_j\circ\pi, E) = \lim_j 2^{-j}\,\mathrm{ord}_E\,\mathfrak{a}_{2^j}(u;Y) = \tau_{Y,u}(E).$$

Here the middle equality uses the ordinary equivalence of each q_j with $u_{2^j}|_Y$.

If $\Omega \geq \kappa$, then T is a trace competitor for Ω, so $T \leq T_\Omega = \mathcal{T}_\Omega(u;Y)$. Together with the defining inequalities of T_Ω, this gives

$$\tau_{Y,u}(E) \leq \nu(T_\Omega \circ \pi, E) \leq \nu(T \circ \pi, E) = \tau_{Y,u}(E).$$

This proves the assertion for every such Ω. □

Only Y is required to be Kähler here; the closed theory is applied on Y. If Y has the bounded mass property, Corollary 31.5.2 gives the same exactness in a sufficiently large Hermitian background with positive lower mass. For a general non-Kähler Y, the envelope limit gives only the lower bound on its divisorial Lelong numbers.

20.6.4 Positive mass and prescribed equations

If a Hermitian background admits a trace competitor, then in the doubled background the same competitor lies in the half-sized background. This gives the trace envelope positive mass and the model condition needed for equations in its ordinary type.

Theorem 20.6.4 *Choose Ω_0 and ν as in Proposition 20.6.1 and set $\Omega = 2\Omega_0$. Then $T = \mathcal{T}_\Omega(u;Y)$ belongs to $\mathrm{PSH}(Y,\Omega)$ and satisfies all the divisorial inequalities in (20.5). Moreover,*

$$P_\Omega[T]_I = T, \qquad P_\Omega[T] = T, \qquad P_{\Omega/2}(T) \not\equiv -\infty.$$

Writing $r = \dim Y$, its extended measure satisfies

$$0 < \int_Y \mathrm{MA}^{\mathrm{ext}}_\Omega(T) < +\infty, \qquad \mathrm{MA}^{\mathrm{ext}}_\Omega(T) \leq \mathbf{1}_{\{T=0\}}\Omega^r.$$

Proof Since $\theta|_Y + \epsilon_m\omega|_Y \leq \Omega$ for all sufficiently large m, the restricted approximants belong to $\mathrm{PSH}(Y,\Omega)$. Theorem 20.6.1, applied with $\delta_m = 0$, gives T, its defining inequalities and both fixed-point identities.

Normalize v so that $\sup_Y v = 0$. Since $v \in \mathrm{PSH}(Y,\Omega/2)$ is a trace competitor, it lies below T and is a competitor for $P_{\Omega/2}(T)$. The contact estimate for envelope fixed points, Proposition 14.1.4, applied on Y, gives the bound for the measure and its finiteness, and Lemma 14.5.2 gives strict positivity. □

Corollary 20.6.1 *Let Ω and T be as in Theorem 20.6.4, let $\mathrm{d}V$ be a smooth positive volume form on Y, and let $0 \le f \in L^p(\mathrm{d}V)$, $p > 1$, with $\int_Y f\,\mathrm{d}V > 0$. For every $\lambda > 0$ there is a unique $a \in \mathrm{PSH}(Y,\Omega)$ with $a \sim T$ and*

$$\mathrm{MA}^{\mathrm{ext}}_{\Omega}(a) = \mathrm{e}^{\lambda a} f\,\mathrm{d}V.$$

There are also a constant $c > 0$ and $a \in \mathrm{PSH}(Y,\Omega)$ with $a \sim T$ and $\mathrm{MA}^{\mathrm{ext}}_{\Omega}(a) = c f\,\mathrm{d}V$, and the constant c is unique.

Proof By Theorem 20.6.4, T satisfies both parts of the model condition (14.20) for the Hermitian background Ω on Y, so Theorem 15.11.1 applies. □

On a curve, Lemma 1.17.5 shows that the trace inequalities are equivalent to domination by the Green representative w in the ordinary sense. Thus

$$\mathcal{T}_{A\eta}(u;Y) = P_{A\eta}[w] = P_{A\eta}[w]_{\mathcal{I}}$$

for every $A > 0$ with $A \ge D_u(Y)$. The last equality, and the fact that the envelope has exactly the prescribed Lelong numbers, follow from Lemma 20.5.1, also when the residual mass $A - D_u(Y)$ is zero.

20.7 Exact traces over a curve

An exact representative on a curve also gives higher-dimensional examples by pullback. Multiplier ideals commute with pullback by a submersion, so the trace data on a modification is given by the trace data on the curve, multiplied by the multiplicities of the pulled-back fibers. The pulled-back Green potential has exactly these divisorial Lelong numbers, and its domination property identifies the ordinary type of the trace envelope.

In this section, $p\colon X \to M$ is a surjective holomorphic submersion of connected compact complex manifolds with connected fibers. Let $C \subset M$ be a connected compact smooth curve, put $Y = p^{-1}(C)$, and write $q = p|_Y\colon Y \to C$; then Y is a connected compact submanifold of X. Take $U \in \mathrm{QPSH}(M)$ with $\nu(U,C) = 0$, and set $u = U \circ p$. The generic Lelong number of u along Y is zero, since in local product coordinates it equals that of U along C. Write $\tau_{C,U}$ for the trace data of U along C, defined as in Section 20.1 with (M,C,U) in place of (X,Y,u), and let

$$D = \sum_{b\in C} \tau_{C,U}(b)\,\delta_b, \qquad \delta = D(C) < +\infty.$$

Choose a smooth positive form η of integral one on C and a nonpositive Green kernel for η. The construction in the proof of Theorem 20.3.1 gives $w \in \mathrm{QPSH}(C)$, not identically $-\infty$, with

$$\mathrm{dd}^{\mathrm{c}} w = D - \delta\eta.$$

For $b \in C$, put $Y_b = q^{-1}(b)$, and let $[Y_b]$ be its current of integration. Set $v = w \circ q$. Pullback of distributions by a submersion is defined by duality with integration along the fibers; it commutes with differentiation and, in the present situation, with positive sums, so

$$\mathrm{dd}^{\mathrm{c}} v = \sum_{b \in C} \tau_{C,U}(b)[Y_b] - \delta q^* \eta. \tag{20.8}$$

Each fiber is a smooth connected prime divisor, and the sum is a positive closed current of finite mass, as also follows locally from the positivity of $\mathrm{dd}^{\mathrm{c}}(w \circ q) + \delta q^* \eta$.

Theorem 20.7.1 *The potential v realizes the entire divisorial trace data of u along Y:*

$$\nu(v \circ \pi, E) = \tau_{Y,u}(E)$$

for every prime divisor E on every smooth compact modification $\pi\colon Z \to Y$. More explicitly, write $h = q \circ \pi$. If $h(E) = C$, both sides are zero. If $h(E) = \{b\}$, choose a local coordinate t_b on C vanishing at b; then both sides equal

$$\mathrm{ord}_E(t_b \circ h)\, \tau_{C,U}(b).$$

Choose any Hermitian form Ω_0 with $v \in \mathrm{PSH}(Y, \Omega_0)$, and set $\Omega = 2\Omega_0$. Then

$$\mathcal{T}_\Omega(u; Y) = P_\Omega[v] = P_\Omega[v]_{\mathcal{I}} \sim v. \tag{20.9}$$

The envelope has positive finite extended mass, and the conclusions of Corollary 20.6.1 on equations in its ordinary type hold. Its ordinary type does not depend on the choice of the Green kernel, of the form η, or of the sufficiently large background Ω.

Proof Lemma 1.16.5 and restriction to Y give

$$\mathfrak{a}_t(u; Y) = \mathfrak{a}_t(U; C)\mathcal{O}_Y.$$

The nonzero coherent ideal $\mathfrak{a}_t(U; C)$ on the smooth curve is generated near b by $t_b^{k_t(b)}$, where $k_t(b) = \mathrm{ord}_b\, \mathfrak{a}_t(U; C)$. For a divisor E with $h(E) = \{b\}$, the order along E of the extended ideal is therefore $k_t(b)\, \mathrm{ord}_E(t_b \circ h)$. For a divisor dominating C, the generic order is zero, since the ideal on the curve has only finitely many zeros. Dividing by t and letting $t \to \infty$ gives exactly the stated formula for $\tau_{Y,u}(E)$.

For the Lelong numbers of $v \circ \pi$, suppose first that $h(E) = \{b\}$. At a general point of E, a local equation $z_1 = 0$ of E satisfies $t_b \circ h = z_1^k g$ with g holomorphic and nonvanishing, and $k = \mathrm{ord}_E(t_b \circ h)$. Replacing z_1 by $z_1 g^{1/k}$ for a local holomorphic kth root, we may assume $t_b \circ h = z_1^k$. The Lelong number of the one-variable function $w(z_1^k)$ at 0 is $k\nu(w, b) = k\tau_{C,U}(b)$, as follows from the definition through circular means, and this is also the generic Lelong number of $v \circ \pi$ along E in these product coordinates. If $h(E) = C$, the regular locus of E contains a point mapped to a point where w is finite, because the nonconstant holomorphic map from that locus to C has open image; the pullback is finite there, so its generic Lelong number along E is zero. This proves exact realization on every compact model.

By Lemma 1.17.5 and (20.8), a quasi-plurisubharmonic function a on Y satisfies $a \preceq v$ if and only if $\nu(a, Y_b) \geq \tau_{C,U}(b)$ for every b with $\tau_{C,U}(b) > 0$. These fiber inequalities are among the trace inequalities, and conversely domination by v implies all trace inequalities on modifications, by the exact formula just proved. Thus the trace competitors for Ω are exactly the competitors for the ordinary envelope of v, and $\mathcal{T}_\Omega(u; Y) = P_\Omega[v]$.

For $\psi = P_\Omega[v]_I$, the Lelong numbers along the fibers are those of v, since ψ and v have the same multiplier ideals, and the same domination lemma gives $\psi \preceq v$. Hence

$$v - \sup_Y v \leq P_\Omega[v] \leq \psi \preceq v.$$

Since $\psi \leq 0$, it is a competitor for the ordinary envelope of v, so $\psi \leq P_\Omega[v]$. This proves (20.9).

The potential $v - \sup_Y v$ is a competitor for $P_{\Omega/2}(P_\Omega[v])$. The contact estimate for envelope fixed points, Proposition 14.1.4, gives finite mass, and Lemma 14.5.2 gives strict positivity. The two envelope properties give the model condition (14.20), so Theorem 15.11.1 gives the stated equations.

Two Green constructions on C differ by a smooth function, by the Poisson argument in the proof of Theorem 20.3.1, so their pullbacks have the same ordinary type. Every background in the statement gives an envelope ordinarily equivalent to this pullback. This proves the asserted independence, at the level of ordinary types rather than of normalized potentials. □

The same construction respects sums and multiplication by positive real numbers at the level of ordinary types. Indeed, the trace measures on the curve add by Proposition 20.1.1, their Green representatives add for a fixed kernel, and pullback preserves sums. Each envelope in a sufficiently large background represents the corresponding ordinary type.

20.8 Exact traces on Hopf manifolds

The projective base of a scalar Hopf manifold has higher dimension than the curve in Section 20.7. A divisor on a modification of the Hopf manifold can therefore map to a set of codimension greater than one in the base. Extracting its image as a divisor on a modified base reduces the trace constraint to a closed-background problem. The elliptic action then identifies the full trace envelope with the pullback of the base envelope. Thus exact realization is possible for these data even though the target has unbounded mass.

Fix $n \geq 3$ and put

$$X = H_{n+1}, \qquad Y = H_n \subset X, \qquad p \colon X \longrightarrow \mathbb{P}^n, \qquad q = p|_Y \colon Y \longrightarrow B = \mathbb{P}^{n-1},$$

where Y is defined by $z_n = 0$ and $B \subset \mathbb{P}^n$ is the same hyperplane. Both maps are holomorphic submersions with connected elliptic fibers, and $q = p_n$ in the notation

of Section 9.6. Set $d = n - 1$, let κ be the Fubini–Study form on B normalized by $\int_B \kappa^d = 1$, and write $\alpha = \alpha_n = q^*\kappa$ and $\beta = \beta_n$ for the forms of that section. For $a, b > 0$ write $\Omega_{a,b} = a\alpha + b\beta$; this is a Hermitian form on Y.

Take $U \in \mathrm{QPSH}(\mathbb{P}^n)$ with $\nu(U, B) = 0$, set $u = U \circ p$, and choose $C > 0$ with $C\omega_{\mathrm{FS}} + \mathrm{dd}^{\mathrm{c}} U \geq 0$, where ω_{FS} is the Fubini–Study form on $\mathbb{P}^n$ with $\omega_{\mathrm{FS}}|_B = \kappa$. Throughout the construction assume $a \geq 2C$ and $b > 0$. The submersion identity of Lemma 1.16.5, followed by restriction, gives

$$\mathfrak{b}_t = \mathcal{I}(tU)\mathcal{O}_B, \qquad \mathfrak{a}_t(u; Y) = \mathfrak{b}_t \mathcal{O}_Y \quad (t > 0). \tag{20.10}$$

Local product coordinates also give $\nu(u, Y) = 0$. The ideals $\mathfrak{b}_t$ are nonzero, and their trace data $\tau_{B,U}$ is defined as in (20.1), with $(\mathbb{P}^n, B, U)$ in place of (X, Y, u). We do not require $U|_B \not\equiv -\infty$.

20.8.1 Divisors under a dominant map

Lemma 20.8.1 *Let $f\colon Z \to M$ be a dominant holomorphic map of connected smooth compact complex manifolds, and let D be a prime divisor on Z. If $f(D) \neq M$, there are a smooth compact modification $\sigma\colon M' \to M$, a prime divisor $F \subset M'$ dominated by D, and an integer $e \geq 1$ such that f lifts holomorphically near a general point of D and*

$$\mathrm{ord}_D(\mathfrak{b}\mathcal{O}_Z) = e\,\mathrm{ord}_F(\mathfrak{b}\mathcal{O}_{M'}), \qquad \nu(w \circ f, D) = e\nu(w \circ \sigma, F) \tag{20.11}$$

for every nonzero coherent ideal $\mathfrak{b}$ on M and every $w \in \mathrm{QPSH}(M)$. If $f(D) = M$, both quantities on the left are zero.

Proof We first extract a divisor on the base and then check the effect of ramification on generic Lelong numbers. Put $r = \dim M$. The extraction replaces the base, step by step, by smooth compact modifications of M. At each step, the closure of the graph of the induced meromorphic map from Z to the present base is a compact analytic set, proper over Z and isomorphic to Z near a general point of D; it maps onto the present base and may be singular away from this neighborhood. By the proper mapping theorem, the image S of D in the present base is an irreducible analytic subset. It is a proper subset, since its image in M is $f(D) \neq M$.

Near a general point of D, where the map to the present base, still denoted f, is holomorphic, let $J_D(f)$ be the least vanishing order of the coefficients of $f^*(\mathrm{d}y_1 \wedge \cdots \wedge \mathrm{d}y_r)$ in a smooth source frame. Dominance makes it a finite nonnegative integer. Changes of smooth coordinates multiply the target volume form by a unit and change the source frame by an invertible matrix, so this integer is independent of the coordinates.

Suppose S has codimension $c \geq 2$. Blow up its reduced coherent ideal. A general point of D is mapped to the smooth part of S; near its image, use coordinates with $S = \{y_1 = \cdots = y_c = 0\}$ and put

$$m = \min_{i \le c} \operatorname{ord}_D(f^* y_i) \ge 1.$$

Choose y_1 attaining this minimum. After removing a proper analytic subset of D, the leading coefficient of $f^* y_1$ along D is a unit, and all ratios $f^* y_i / f^* y_1$ are holomorphic. The map therefore lifts near these points to the smooth blowup chart

$$(y_1, \dots, y_c) = (s, st_2, \dots, st_c).$$

Resolve the target blowup by a proper analytic resolution which is an isomorphism over its whole smooth locus; the resolution input is recalled in Section 36.13. The resulting base is smooth and compact, and the lift near the general point of D is preserved. Taking the closure of the graph of the resulting meromorphic lift again gives a compact graph proper over Z, unchanged in this neighborhood.

In these coordinates,

$$\mathrm{d}y_1 \wedge \cdots \wedge \mathrm{d}y_r = s^{c-1} \mathrm{d}s \wedge \mathrm{d}t_2 \wedge \cdots \wedge \mathrm{d}t_c \wedge \mathrm{d}y_{c+1} \wedge \cdots \wedge \mathrm{d}y_r.$$

Consequently the new Jacobian order is $J_D(f) - (c-1)m \ge 0$. It strictly decreases at every step, so after finitely many steps the image S has codimension one; as noted above, it is never the whole base. Call the final base M' and this prime divisor F. This proves extraction without resolving the source graph in a way that changes the general point of D.

Let e be the order along D of the pullback of a local equation of the extracted divisor F. Near a general point of F, the ideal $\mathfrak{b}\mathcal{O}_{M'}$ is a power of that equation times a unit ideal. Pullback gives the ideal-order identity in (20.11).

The map $D \to F$ has tangential rank $r-1$ at general smooth points. Use the tangential base coordinates as source coordinates and absorb the unit in the normal pullback into a local holomorphic eth root. The resulting local form of the map is

$$(z_1, z_2, \dots, z_r, z') \longmapsto (z_1^e, z_2, \dots, z_r).$$

Siu's divisorial removal from Section 1.17 (Lemma 1.17.1) writes, near a general point of F,

$$w \circ \sigma = c_0 \log |y_1|^2 + R, \qquad c_0 = \nu(w \circ \sigma, F),$$

with R quasi-plurisubharmonic and $\nu(R, F) = 0$. By Siu's analyticity theorem for the level sets $\{\nu(R, \cdot) \ge c\}$, recalled in Section 1.13.1, the point Lelong number of R is zero at a very general point of F. Skoda's theorem then makes e^{-tR} locally integrable for every $t > 0$ at this point.

For $e > 1$, choose $1 < p < e/(e-1)$ and let p' be its conjugate exponent. The density $|y_1|^{2/e-2}$ belongs locally to L^p. Change of variables and Hölder's inequality, using integrability of $\mathrm{e}^{-tp'R}$, give

$$\int \mathrm{e}^{-tR(z_1^e, z_2, \dots, z_r)} \, \mathrm{d}V_z = c_e \int \mathrm{e}^{-tR(y)} |y_1|^{2/e-2} \, \mathrm{d}V_y < +\infty$$

on sufficiently small corresponding polydisks. Free source variables contribute a finite product factor; for $e = 1$ Fubini's theorem suffices. A positive generic divisor coefficient c' of the pulled-back residual would instead force nonintegrability for $tc' \geq 1$. The residual coefficient is therefore zero, and the logarithmic term has coefficient ec_0. This proves the Lelong identity even if $R|_F \equiv -\infty$.

Finally, if $f(D) = M$, the map is a submersion near a very general point of D. A nonzero ideal is a unit at a general image point, and Skoda's theorem at an image point of zero Lelong number, followed by Fubini's theorem in product coordinates, gives zero generic Lelong number on D. □

20.8.2 The full trace envelope

Let $w_a = \mathcal{T}_{a\kappa}(U; B)$. The closed-background construction of Theorem 20.6.3 applies to U along $B \subset \mathbb{P}^n$, with background $C\omega_{\mathrm{FS}}$, with $\Omega_0 = C\kappa$, and with the Kähler form $a\kappa \geq 2\Omega_0$ in place of the Kähler form there. It gives

$$\begin{aligned} \nu(w_a \circ \sigma, F) &= \tau_{B,U}(F) \quad \text{for every divisor } F \text{ over } B, \\ (a/2)^d \leq \int_B \mathrm{MA}_{a\kappa}(w_a) &\leq a^d. \end{aligned} \tag{20.12}$$

The lower bound is the mass bound in the proof of that theorem, which comes from the trace majorant in the half-sized background; the upper bound comes from closed mass monotonicity. In particular, the mass is positive, as required for convergence of the divisorial Lelong numbers of the decreasing base models.

Theorem 20.8.1 *For the projective pullback data above,*

$$\mathcal{T}_{\Omega_{a,b}}(u; Y) = q^* w_a. \tag{20.13}$$

For every smooth real $(1,1)$*-form* $\Omega \geq a\alpha$ *on* Y*, every smooth compact modification* $\pi\colon Z \to Y$*, and every prime divisor* $D \subset Z$*,*

$$\nu\big(\mathcal{T}_{\Omega}(u; Y) \circ \pi, D\big) = \tau_{Y,u}(D).$$

Proof Apply Lemma 20.8.1 to $f = q \circ \pi$. If D dominates B, the last assertion of the lemma, applied to the ideals $\mathfrak{b}_t$ and to w_a, shows that $\tau_{Y,u}(D)$ and $\nu(q^* w_a \circ \pi, D)$ are both zero. Otherwise the ideal identity (20.10) and the extracted divisor F give

$$\tau_{Y,u}(D) = e\tau_{B,U}(F) = e\nu(w_a \circ \sigma, F) = \nu(q^* w_a \circ \pi, D).$$

Thus the single potential $q^* w_a$ has exactly the prescribed divisorial Lelong numbers on every compact model.

For a basic potential, that is, a potential pulled back from B, positivity is equivalent to positivity on the base: by Lemma 9.6.1(1), with S a point, every quasi-plurisubharmonic function v on B satisfies

$$\Omega_{a,b} + \mathrm{dd}^{\mathrm{c}} q^{*} v \geq 0 \quad \Longleftrightarrow \quad a\kappa + \mathrm{dd}^{\mathrm{c}} v \geq 0, \tag{20.14}$$

and the same holds with $a\alpha$ in place of $\Omega_{a,b}$. Since $w_a \in \mathrm{PSH}(B, a\kappa)$ and $w_a \leq 0$, the potential $q^{*} w_a$ is therefore a trace competitor for every smooth real $(1,1)$-form $\Omega \geq a\alpha$, in particular for $\Omega_{a,b}$.

The compact elliptic group $\mathbb{C}^{*}/\langle \mathrm{e}^{-1} \rangle$ acts by scalar multiplication on Y. It preserves α, β and all ideals in (20.10). It also preserves the family of trace competitors for $\Omega_{a,b}$: for a competitor g and a translation ρ, the constraint for $g \circ \rho$ on π is the constraint for g on the modification $\rho \circ \pi$, with the same ideal order. By Theorem 20.6.1, the envelope is the largest member of this family, so it is invariant under the group. The action is transitive on each fiber, so Lemma 9.6.1(2), with S a point, gives $\mathcal{T}_{\Omega_{a,b}}(u; Y) = q^{*} v$ for a quasi-plurisubharmonic function $v \leq 0$ on B, and $v \in \mathrm{PSH}(B, a\kappa)$ by (20.14).

Every smooth compact modification $\sigma \colon B' \to B$ lifts to the smooth compact modification $Y \times_B B' \to Y$. The inverse image of a prime divisor $F \subset B'$ is prime, because the fibers are connected, and has generic multiplicity one. The trace constraint on this divisor says exactly that v satisfies the constraint at F on B'. Thus $v \leq w_a$. The opposite inequality follows from the competitor $q^{*} w_a$, proving (20.13).

For $\Omega \geq a\alpha$, the competitor $q^{*} w_a$ lies below $T_\Omega = \mathcal{T}_\Omega(u; Y)$. The defining lower bound and the order of the two potentials give

$$\tau_{Y,u}(D) \leq \nu(T_\Omega \circ \pi, D) \leq \nu(q^{*} w_a \circ \pi, D) = \tau_{Y,u}(D),$$

which proves exactness in every such background. □

20.8.3 Mass and the ideal class

The mass of a basic potential is computed on the closed base: by Lemma 9.6.1(3) with $F = 1$, every $v \in \mathrm{PSH}(B, a\kappa)$ satisfies

$$\int_Y \mathrm{MA}_{\Omega_{a,b}}(q^{*} v) = 2nb \int_B \mathrm{MA}_{a\kappa}(v), \tag{20.15}$$

and both masses are finite. This concerns basic potentials only; the target still fails the bounded mass property by Corollary 9.6.1.

Proposition 20.8.1 *The trace envelope in* (20.13) *has mass between* $2nb(a/2)^d$ *and* $2nba^d$.

Proof Apply (20.15) to $v = w_a$ and use (20.12). □

Example 20.8.1 Set

$$U([z_0 : \cdots : z_n]) = \log \frac{\sum_{i=0}^{n-2} |z_i|^2}{\sum_{i=0}^{n} |z_i|^2}, \qquad P = [0 : \cdots : 0 : 1 : 0] \in B.$$

Then $\nu(U, B) = 0$ and, for every $t > 0$,

$$\mathfrak{b}_t = \mathfrak{m}_P^{\max\{\lfloor t-d+1 \rfloor, 0\}}, \qquad \tau_{Y,u}(D) = \operatorname{ord}_D(\mathfrak{m}_P \mathcal{O}_Z)$$

for every divisor D on every smooth compact modification $Z \to Y$, with the ideal extended through q. The potential $q^*(U|_B)$ is an exact realizer, singular along the elliptic fiber above P. For this explicit potential it suffices to take $a \geq 1$, and

$$\int_Y \mathrm{MA}_{\Omega_{a,b}}(q^*(U|_B)) = 2nb(a^d - 1).$$

It need not equal the normalized trace envelope.

Proof The pole set of U is the projective line L defined by $z_0 = \cdots = z_{n-2} = 0$, of codimension d, and U is smooth off L. Near L, in affine coordinates, U is $\log \sum_{i=0}^{n-2} |z_i|^2$ up to a bounded term, so Lemma 1.16.1 gives $\mathcal{I}(tU) = \mathcal{I}_L^{\max\{\lfloor t \rfloor - d+1, 0\}}$. The line L meets B transversally at the single point P, so restriction to B gives the displayed powers of $\mathfrak{m}_P$; in particular $\nu(U, B) = 0$. Dividing their orders by t proves the trace formula. Exactness of the ordinary restriction follows from Proposition 20.2.1, since the ambient pullback has analytic singularities.

On the point blowup $\sigma\colon \widetilde{B} \to B$ at P, let F be the exceptional divisor, $H = [\sigma^*\kappa]$, and let η be the pullback of the Fubini–Study form by the resolved projection to $\mathbb{P}^{d-1}$. The logarithm defining $U|_B$ gives

$$\sigma^*(a\kappa) + \mathrm{dd}^{\mathrm{c}}(U|_B \circ \sigma) = [F] + (a-1)\sigma^*\kappa + \eta.$$

The residual form is smooth semipositive for $a \geq 1$, with class $aH - F$. Its smooth top product off F computes the non-pluripolar mass, and the smooth measure does not charge F. Since $H|_F = 0$, all mixed intersections involving both H and F vanish; $H^d = 1$ and $F^d = (-1)^{d-1}$ give $(aH - F)^d = a^d - 1$. Formula (20.15) now supplies the elliptic factor $2nb$. □

Corollary 20.8.1 *The ideal singularity class of $\mathcal{T}_{\Omega_{a,b}}(u; Y)$ is independent of $a \geq 2C$ and $b > 0$, and depends only on the multiplier-ideal class of U. These classes respect addition and positive scaling of the ambient data: for $U_1, U_2 \in \mathrm{QPSH}(\mathbb{P}^n)$ with zero generic Lelong number along B, $u_i = U_i \circ p$ and $\lambda > 0$, in sufficiently large invariant backgrounds,*

$$\mathcal{T}(u_1 + u_2; Y) \sim_{\mathcal{I}} \mathcal{T}(u_1; Y) + \mathcal{T}(u_2; Y), \qquad \mathcal{T}(\lambda u_1; Y) \sim_{\mathcal{I}} \lambda \mathcal{T}(u_1; Y),$$

where the backgrounds on the individual envelopes are omitted from the notation.

Proof All sufficiently large base backgrounds give the same divisorial Lelong numbers (20.12). On the compact Kähler base, equality of these numbers on all compact models is equivalent to equality of the multiplier ideals at every real positive weight; see the closed criterion recalled in Section 36.13. Hence all w_a have the same ideal class. The submersion identity of Lemma 1.16.5 transfers the equality to q^*w_a, which does not depend on b. Equal ambient multiplier ideals also give equal trace data.

By Proposition 20.1.1, the trace data of $U_1 + U_2$ is the sum of those of U_1 and U_2. The sum of two exact base realizers has precisely these divisorial Lelong numbers, so the same closed criterion identifies its ideal class with that of the base trace envelope for $U_1 + U_2$. Positive scaling is identical. Pullback gives both assertions on Y. □

The corollary concerns the invariant backgrounds $\Omega_{a,b}$ only. Exactness in the larger backgrounds of Theorem 20.8.1 does not by itself identify an ordinary or ideal class there; the closed criterion in the proof is applied on B.

20.8.4 Decreasing analytic models

Let U_m be the analytic approximants of U on $\mathbb{P}^n$ given by Corollary 1.15.1, as in Theorem 20.6.1. After discarding finitely many dyadic indices, their restrictions belong to $\mathrm{PSH}(B, a\kappa)$. Put

$$w_{a,j} = P_{a\kappa}[U_{2^j}|_B], \qquad v_j = q^*w_{a,j}.$$

Proposition 20.8.2 *The potentials v_j are the ordinary envelopes of the pulled-back analytic approximants,*

$$v_j = P_{\Omega_{a,b}}[q^*(U_{2^j}|_B)].$$

They decrease to $\mathcal{T}_{\Omega_{a,b}}(u;Y)$ in $L^1(Y)$. For every smooth compact modification $\pi\colon Z \to Y$, their pullbacks converge in $L^1(Z)$, and for every prime divisor $D \subset Z$,

$$\nu(v_j \circ \pi, D) \longrightarrow \tau_{Y,u}(D).$$

Proof Let $h \in \mathrm{PSH}(B, a\kappa)$ and $A > 0$. The family of competitors for the envelope $P_{\Omega_{a,b}}(q^*h + A, 0)$ is invariant under the elliptic action, hence so is the envelope, and as in the proof of Theorem 20.8.1 it is pulled back from the base. By (20.14), a basic potential q^*g is $\Omega_{a,b}$-plurisubharmonic exactly when g is $a\kappa$-plurisubharmonic, and since q is surjective, $q^*g \le \min(q^*h + A, 0)$ exactly when $g \le \min(h + A, 0)$. Hence

$$P_{\Omega_{a,b}}(q^*h + A, 0) = q^*P_{a\kappa}(h + A, 0).$$

Letting $A \to +\infty$ and taking upper semicontinuous regularizations, we obtain

$$P_{\Omega_{a,b}}[q^*h] = q^*P_{a\kappa}[h].$$

For $h = U_{2j}|_B$ this is the asserted identity. By Proposition 17.2.4, $w_{a,j}$ has the analytic type of $U_{2j}|_B$, so v_j has the analytic type of $q^*(U_{2j}|_B)$.

The base models $w_{a,j}$ decrease to w_a by Theorem 20.6.1, applied on B with $\Omega = a\kappa$ and $\delta_m = 0$. Their pullbacks therefore decrease to q^*w_a, and they converge in $L^1(Y)$ by monotone convergence, since $q^*w_a \le v_j \le 0$ and q^*w_a is integrable. The divisorial Lelong numbers of the $w_{a,j}$ converge on every compact model of B, by the positive-mass argument in the proof of Theorem 20.6.3. The extracted-divisor formula (20.11) transfers this convergence to every D over Y; divisors dominating B have coefficient zero throughout. Finally, Lemma 1.22.8 gives L^1 convergence on each compact modification. □

These results answer the trace question for all projective data pulled back to the Hopf pair (H_{n+1}, H_n), including nonanalytic data. They neither require bounded mass on H_n nor give convergence of the lower and upper singularity volumes there. The unrestricted realization problem of Problem 35.8.1 still concerns data without this projective pullback structure.

20.9 Product realizations and their measures

For a sum of potentials pulled back from the two factors of a product with a curve, dimension reasons force every term of the Monge–Ampère expansion to contain exactly one factor from the curve. The resulting product formula computes the measure of the pulled-back Green representative and solves equations with separated densities. Adding a potential from the other factor with trivial multiplier ideals can keep every trace coefficient while changing finite mass to infinite mass.

In this section, C is a compact connected curve with a smooth positive form η of integral one, and (Z, ω_Z) is a compact connected Hermitian manifold of dimension $k \ge 1$. On $Y = C \times Z$, let $q\colon Y \to C$ and $r\colon Y \to Z$ be the projections. Choose $A > 0$ and set

$$\Omega = q^*(A\eta) + r^*\omega_Z.$$

For positive Borel measures μ on C and ν on Z, $\mu \otimes \nu$ denotes their product measure. The extended measures below are σ-finite by Theorem 7.1.1, so this notation is meaningful also when a total mass is infinite.

Proposition 20.9.1 *For $w \in \mathrm{PSH}(C, A\eta)$ and $h \in \mathrm{PSH}(Z, \omega_Z)$, the potential $a = w \circ q + h \circ r$ is Ω-plurisubharmonic and*

$$\mathrm{MA}^{\mathrm{ext}}_{\Omega}(a) = (k+1)\,\mathrm{MA}^{\mathrm{ext}}_{A\eta}(w) \otimes \mathrm{MA}^{\mathrm{ext}}_{\omega_Z}(h). \tag{20.16}$$

Proof The curvature current of a is the sum of the pullbacks of the two positive currents $A\eta + \mathrm{dd^c}w$ and $\omega_Z + \mathrm{dd^c}h$, which proves positivity.

Suppose first that both potentials are bounded. Expanding the bounded product by multilinearity, only the terms with one factor from C and k factors from Z survive: two factors from C vanish for dimension reasons, and so do $k+1$ factors from Z. The

binomial coefficient is $k+1$, which gives the formula for bounded potentials once the exterior product $q^*(A\eta + \mathrm{dd^c} w) \wedge r^*(\omega_Z + \mathrm{dd^c} h)^k$ is identified with the product measure $(A\eta + \mathrm{dd^c} w) \otimes (\omega_Z + \mathrm{dd^c} h)^k$. To justify this identification at the present regularity, add local plurisubharmonic corrections on each factor and regularize by decreasing convolution on smaller product charts. For the smooth approximants the identity is the exterior product of forms. Local Bedford–Taylor convergence, together with the finite expansion in the smooth background, gives weak convergence of the factor measures, with uniformly bounded variation on compact subcharts. The product measures then converge weakly: one first tests against products of continuous functions on the two factors and then uses uniform approximation by finite sums of such products. This proves the identity in the limit.

For general potentials, subtract constants so that $w, h \le 0$, and set $w_j = \max(w, -j)$ and $h_j = \max(h, -j)$. On

$$R_j = \{w > -j\} \times \{h > -j\}$$

the sum agrees with the bounded potential $w_j \circ q + h_j \circ r$. This set is plurifine open and Borel, and it lies in $\{a > -2j\}$. Locality of the products of bounded truncations therefore identifies the restriction to R_j of the extended measure of a with that of the bounded sum. By the bounded case, this is $k+1$ times the product of the restrictions of the truncated measures, which are exactly the restrictions of the two extended measures of the factors.

The sets R_j increase to the product of the complements of the two pole sets, and both sides of (20.16) give zero mass to the remaining set. For the left-hand side, this set is the pole set of a. For the right-hand side, the claim holds because each factor measure charges no pluripolar set and the product measures are σ-finite. Monotone convergence of positive measures proves the formula, even when the total mass is infinite. □

Suppose now that D is a finite atomic measure on C, for instance the trace measure of Section 20.7, put $\delta = D(C)$, and let w be the Green potential with $\mathrm{dd^c} w = D - \delta\eta$. For $A > \delta$, the potential $v = w \circ q$ satisfies

$$\mathrm{MA}^{\mathrm{ext}}_{\Omega}(v) = (k+1)(A-\delta)\,\eta \otimes \omega_Z^k. \tag{20.17}$$

Thus its measure is smooth and positive, although v may have infinitely many singular fibers. This identity concerns the extended measure, whose definition discards the divisorial part of the curvature current.

The product formula also solves equations with separated densities in this ordinary type. Fix a smooth positive volume form $\mathrm{d}V_Z$ on Z and $p > 1$, and let $0 \le f \in L^p(C, \eta)$ and $0 \le g \in L^p(Z, \mathrm{d}V_Z)$ have positive integrals. The proof of Theorem 20.4.1 uses only that D is a finite positive atomic measure. Applied on C in the background $A\eta$, it gives $w_f \sim w$ with

$$\mathrm{MA}^{\mathrm{ext}}_{A\eta}(w_f) = c_C f \eta, \qquad c_C = \frac{A-\delta}{\int_C f\eta}.$$

By Theorem 11.1.1, there are a bounded $h_g \in \mathrm{PSH}(Z, \omega_Z)$ and $c_Z > 0$ with

$$\mathrm{MA}_{\omega_Z}(h_g) = c_Z g \, \mathrm{d}V_Z.$$

The sum $a = w_f \circ q + h_g \circ r$ is ordinarily equivalent to v, and

$$\mathrm{MA}^{\mathrm{ext}}_{\Omega}(a) = (k+1) c_C c_Z \, (f\eta) \otimes (g \, \mathrm{d}V_Z).$$

The product density belongs to L^p by Fubini's theorem. This is an explicit construction for separated densities; existence for general L^p densities, in a sufficiently large background, follows from Corollary 20.6.1.

Proposition 20.9.2 *Suppose that Z does not have the bounded mass property, that is, $v_+(\omega_Z) = +\infty$ in the notation of Definition 7.5.1, and that $A > \delta$. Then there is $a \in \mathrm{PSH}(Y, \Omega)$ such that*

$$a \sim_{\mathcal{I}} v, \qquad \int_Y \mathrm{MA}^{\mathrm{ext}}_{\Omega}(v) < +\infty, \qquad \int_Y \mathrm{MA}^{\mathrm{ext}}_{\Omega}(a) = +\infty.$$

Both potentials have the same divisorial Lelong numbers on every modification; when D is the trace measure of Section 20.7, these are the trace data computed in Theorem 20.7.1. The product manifold Y does not have the bounded mass property.

Proof Choose the nonpositive potential H of Theorem 7.6.1 on Z; it has infinite extended mass, and e^{-tH} is integrable for every $t > 0$. Put $a = w \circ q + H \circ r$. The product formula and $A - \delta > 0$ show that the mass of a is infinite, while (20.17) gives the finite positive mass of v.

We prove equality of the multiplier ideals locally, for every $t > 0$. Since $H \le 0$, integrability against e^{-ta} implies integrability against e^{-tv}, so $\mathcal{I}(ta) \subseteq \mathcal{I}(tv)$. Conversely, Lemma 1.16.5 gives $\mathcal{I}(tv) = \mathcal{I}(tw)\mathcal{O}_Y$. Local generators pulled back from C are square integrable against e^{-tw}, so by Fubini's theorem they are also square integrable against $\mathrm{e}^{-tw}\mathrm{e}^{-tH} = \mathrm{e}^{-ta}$; multiplication by bounded holomorphic functions and finite sums preserve this local integrability. Thus $\mathcal{I}(tv) \subseteq \mathcal{I}(ta)$, which proves equality at every positive weight. Theorem 1.16.1 then gives equality of all divisorial Lelong numbers after modification.

Finally, since a has infinite mass, Proposition 7.5.2 shows that Y does not have the bounded mass property. □

For a concrete ambient realization, take a connected compact complex manifold M containing the curve C, a potential $U \in \mathrm{QPSH}(M)$ with $\nu(U, C) = 0$, and $X = M \times Z$. Then $Y = C \times Z$ and $u = U \circ \mathrm{pr}_M$, where $\mathrm{pr}_M : X \to M$ is the projection, satisfy the hypotheses of Theorem 20.7.1 with $p = \mathrm{pr}_M$. The factor Z can be the Hopf threefold of Theorem 9.0.1. Small logarithmic poles of U at finitely many points of C already give nonzero trace data, and the construction on the curve also allows any countable trace data arising from the multiplier ideals of U.

The mass formula of Section 29.7 uses more information: the restriction itself in a closed background, with a finite divisorial part and a bounded remainder on a Kähler

model. Its residual class is nef and computes the mass of that restriction. Divisorial trace data alone does not provide such a remainder, as the preceding example shows.

20.10 Ordinary trace volumes on a nonclosed ruled surface

The preceding product formula computes the mass of a separated representative. When the background itself varies along the curve, even bounded changes of the representative can change its mass. On a product of projective lines we can optimize over the entire ordinary singularity type, including potentials depending on both factors. The potential will be a logarithmic pole along a divisor plus an unbounded residual with zero Lelong number. Its multiplier ideals see only the imposed divisor, whereas its ordinary type also retains the residual pole.

In this section let $Y = \mathbb{P}^1 \times \mathbb{P}^1$, let w be the probability Fubini–Study form on $\mathbb{P}^1$, and denote its pullbacks by w_1, w_2. Fix $p \in \mathbb{P}^1$, and let $G_p \le 0$ be the Green potential satisfying $\mathrm{dd^c} G_p = [p] - w$ and $\sup G_p = 0$. Functions on the first factor are also viewed as functions on Y. Put

$$D = \{p\} \times \mathbb{P}^1, \qquad U = 1 - (1 - G_p)^{\alpha}, \qquad r = bG_p + U, \qquad 0 < \alpha < 1,$$

where $b \ge 0$. The term bG_p is identically zero when $b = 0$. Choose $c > 0$ and a smooth positive function a on the first curve, and set

$$\theta = cw_1 + aw_2. \tag{20.18}$$

Thus θ is Hermitian and is nonclosed precisely when a is nonconstant. The function r specifies an ordinary type; whether that type has θ-plurisubharmonic representatives is decided in Theorem 20.10.1.

Lemma 20.10.1 *There are $U_L \sim U$ with $\sup U_L = 0$, $\mathrm{dd^c} U_L \ge -\epsilon_L w$ and $\epsilon_L \downarrow 0$, such that $U_L \to 0$ in $L^1(\mathbb{P}^1)$. Each has zero Lelong number at p and is locally bounded off p.*

Proof For $t \le 0$ put $\chi(t) = 1 - (1 - t)^{\alpha}$. Then

$$\chi'(t) = \alpha(1 - t)^{\alpha - 1} > 0, \qquad \chi''(t) = \alpha(1 - \alpha)(1 - t)^{\alpha - 2} > 0.$$

For $L > 0$ set $\epsilon_L = \chi'(-L)$, $C_L = -\epsilon_L L - \chi(-L)$, and define

$$\chi_L(t) = \begin{cases} \chi(t) + C_L, & t < -L, \\ \epsilon_L t, & -L \le t \le 0. \end{cases}$$

The values and first derivatives match, and χ_L is convex and increasing with derivative at most ϵ_L. Off p, the chain rule gives $\mathrm{dd^c}\chi_L(G_p) \ge -\epsilon_L w$. Canonical extension across p retains this bound by the removable-singularity theorem, so $U_L = \chi_L(G_p)$

has the required curvature. Near p it differs from U by C_L, and elsewhere both are bounded. Its growth is sublogarithmic, hence its Lelong number is zero.

Writing $x = 1 + L$, we have $C_L = (1-\alpha)x^{\alpha} + \alpha x^{\alpha-1} - 1 \geq 0$. Also $\chi(t) \leq \epsilon_L t$ for $-L \leq t \leq 0$, by integrating $\chi' \geq \epsilon_L$ on this interval. It follows that $U \leq U_L \leq 0$. Since $\epsilon_L \to 0$, the functions tend to zero off p, and dominated convergence gives the L^1 limit. □

Theorem 20.10.1 *The ordinary type of r has θ-plurisubharmonic representatives if and only if $b < c$. In this range it contains a strict current: some $v \sim r$ satisfies $\theta + \mathrm{dd}^{\mathrm{c}} v \geq \delta(w_1 + w_2)$ with $\delta > 0$. Every $v \in \mathrm{PSH}(Y, \theta)$ with $v \sim r$ has positive finite extended mass, given by*

$$\int_Y \mathrm{MA}_{\theta}^{\mathrm{ext}}(v) = \int_Y \theta^2 - 2ba(p) + 2\int_Y v\,\mathrm{dd}^{\mathrm{c}}\theta. \tag{20.19}$$

The volumes of Definition 14.3.1 are

$$\underline{\mathrm{vol}}_{\theta}(v) = 2(c-b)\min a, \qquad \overline{\mathrm{vol}}_{\theta}(v) = 2(c-b)\max a. \tag{20.20}$$

Both values are limits of masses of representatives of this ordinary type.

Proof We first remove the logarithmic divisor. If $v \sim r$, then $\nu(v, D) = b$, so Lemma 1.17.1 gives $\theta + \mathrm{dd}^{\mathrm{c}} v - b[D] \geq 0$. Taking $q = v - bG_p$ in L^1 and then its quasi-plurisubharmonic representative gives

$$q \in \mathrm{PSH}(Y, \beta), \qquad q \sim U, \qquad \beta = kw_1 + aw_2, \qquad k = c - b.$$

Indeed the ordinary inequalities hold off D and extend canonically across it. Conversely such a q gives $v = bG_p + q$ with

$$\theta + \mathrm{dd}^{\mathrm{c}} v = b[D] + \beta + \mathrm{dd}^{\mathrm{c}} q.$$

Their locally bounded products agree off D, and both extended measures give D zero mass. Thus their extended measures agree everywhere.

If $k > 0$, choose L with $\epsilon_L < k/2$. The potential $bG_p + U_L$ belongs to the specified type and satisfies

$$\theta + \mathrm{dd}^{\mathrm{c}}(bG_p + U_L) \geq b[D] + (k - \epsilon_L)w_1 + aw_2 \geq \min(k/2, \min a)(w_1 + w_2).$$

For necessity, average q over the unitary group acting transitively on the second projective line, with its Haar probability measure. The background β is invariant, so the average is β-plurisubharmonic and invariant, hence a function q_0 on the first factor, with $q_0 \sim U$ and $kw + \mathrm{dd}^{\mathrm{c}} q_0 \geq 0$. One can first average in L^1 and then take the canonical representative; the curvature inequality and the ordinary bounds pass to it. The positive curve measure has total mass k, excluding $k < 0$. When $k = 0$, q_0 is harmonic on the compact curve and therefore constant, contradicting $q_0 \sim U$.

It remains to compute the mass for every competitor, before using the average. Choose a closed Kähler form $K = C(w_1 + w_2) \geq \beta$ and a fixed $u_0 = U_L$ as above.

The closed positive current $S_0 = K + \mathrm{dd^c} u_0$ has no coefficient mass on D, since its first-factor curvature has no atom at p. The product formula of Proposition 20.9.1 gives

$$\langle S_0 \rangle = S_0, \qquad \langle S_0^2 \rangle = 2C(Cw_1 + \mathrm{dd^c} u_0) \wedge w_2, \qquad \int_Y \langle S_0^2 \rangle = \int_Y K^2.$$

For $S = K + \mathrm{dd^c} q$, both S and S_0 are closed positive currents in the same Kähler class, with bounded difference of potentials and small unbounded locus D. By Proposition 1.25.1, S and S_0 put the same mass on D, namely zero, so $\langle S \rangle = S$. The classical closed-class comparison of non-pluripolar products, applied in both singularity orders, gives

$$\int_Y \langle S^2 \rangle = \int_Y \langle S_0^2 \rangle = \int_Y K^2.$$

The closed comparison is used only for K.

Put $R = \beta - K$. Bedford–Taylor expansion off D, followed by extension by zero, now yields

$$\mathrm{MA}_\beta^{\mathrm{ext}}(q) = \langle S^2 \rangle + 2R \wedge S + R^2.$$

The first term is finite by closed comparison, the second has finite variation because S has finite trace, and the last is smooth. Since none has coefficient mass on D, this is also an identity on Y. Distributional Stokes therefore gives

$$\int_Y \mathrm{MA}_\beta^{\mathrm{ext}}(q) = \int_Y \beta^2 + 2 \int_Y q \, \mathrm{dd^c} \beta = 2 \int_{\mathbb{P}^1} a(kw + \mathrm{dd^c} q_0). \tag{20.21}$$

The pairing uses only $q \in L^1$. Averaging preserves it, since $\mathrm{dd^c} \beta = \mathrm{dd^c} a \wedge w_2$. The last measure is positive of total mass k, proving the bounds in (20.20) for all original competitors. Substituting $q = v - bG_p$ and $\int G_p \mathrm{dd^c} a = a(p) - \int aw$ proves (20.19).

To approach either bound, choose a point x where a takes the corresponding extreme value, smooth strictly positive measures ζ_j of mass k tending weakly to $k\delta_x$, and smooth g_j with $\mathrm{dd^c} g_j = \zeta_j - kw$. Choose L_j so that ϵ_{L_j} is less than half the minimum density of ζ_j relative to w and $|\int U_{L_j} \mathrm{dd^c} a| < 1/j$. Then

$$v_j = bG_p + g_j + U_{L_j} \sim r, \qquad \beta + \mathrm{dd^c}(g_j + U_{L_j}) \geq \tfrac{1}{2}\zeta_j + aw_2 > 0.$$

Their masses tend to $2ka(x)$ by (20.21). □

No finite first moment is used here. For the reference $u_0 = U_L$, the closed first moment $\int_Y (-u_0)\langle (K + \mathrm{dd^c} u_0)^2 \rangle$ is finite exactly when $\alpha < 1/2$. Near p put $t = -G_p$. Then $-u_0 \asymp t^\alpha$ and $\chi''(G_p) \asymp t^{\alpha-2}$, while angular integration of $\mathrm{d}G_p \wedge \mathrm{d^c} G_p$ is comparable to $\mathrm{d}t$. The singular contribution is thus comparable to $\int^\infty t^{2\alpha-2} \, \mathrm{d}t$. The smooth contributions are integrable, and the mass formula includes the divergent range.

20.10.1 Ambient ideal orders and ordinary type

To realize these data as intrinsic traces, set $X = Y \times \mathbb{P}^1$ and identify Y with $Y \times \{z_0\}$ for a fixed z_0. Write w_3 for the third-factor form and let $\mathrm{pr}_Y \colon X \to Y$ be the projection. Assume $b < c$, and choose L with $\epsilon_L < (c-b)/2$, as in the proof of Theorem 20.10.1. Put

$$\Theta = \mathrm{pr}_Y^{*}\,\theta + w_3, \qquad u_X = \mathrm{pr}_Y^{*}(bG_p + U_L), \qquad D_X = D \times \mathbb{P}^1.$$

Then $u_X \in \mathrm{PSH}(X, \Theta)$ and $\nu(u_X, Y) = 0$.

Proposition 20.10.1 *For every real $s > 0$,*

$$\mathcal{I}(su_X) = \mathcal{O}_X(-\lfloor sb \rfloor D_X), \qquad \mathcal{I}(su_X)\mathcal{O}_Y = \mathcal{O}_Y(-\lfloor sb \rfloor D).$$

For every prime divisor E on every smooth compact modification $\pi \colon Y' \to Y$, the intrinsic trace data are

$$\tau_{Y,u_X}(E) = b\,\mathrm{ord}_E(\pi^{*}D).$$

Nevertheless the ordinary type of $bG_p + U_L$ has no analytic representative.

Proof The function $G_p \circ \mathrm{pr}_Y$ is the logarithm of the squared norm of the canonical section of $\mathcal{O}_X(D_X)$ for a smooth metric. By Lemma 20.10.1, the potentials $U_{L'} \circ \mathrm{pr}_Y \sim U_L \circ \mathrm{pr}_Y$ show that $U_L \circ \mathrm{pr}_Y$ satisfies (17.9). Apply Proposition 20.2.2 with $\Delta = bD_X$, $\sigma_\Delta = bG_p \circ \mathrm{pr}_Y$ and $q = U_L \circ \mathrm{pr}_Y$; the divisor D_X is smooth and meets Y transversally along D. Its first identity and $\mathcal{I}(s\sigma_\Delta) = \mathcal{O}_X(-\lfloor sb \rfloor D_X)$, which is Lemma 1.16.1 in codimension one near each point of D_X when $b > 0$ and is trivial when $b = 0$, give the ambient ideals, restriction gives the ideals on Y, and its trace formula gives $\tau_{Y,u_X}(E) = b\,\mathrm{ord}_E(\pi^{*}D)$.

The residual U_L has trivial multiplier ideals at every weight, but is unbounded. Near a generic point of D, a potential with analytic singularities and generic Lelong number b along D is $b \log |z_1|^2 + O(1)$. Our residual tends to $-\infty$ like $-t^{\alpha}$, excluding ordinary equivalence. This also covers $b = 0$. □

20.10.2 Pullback and strict representatives

The lower volume above is positive, and its type contains a strict current on Y. A modification preserves both optimized masses but can destroy this last property when the ordinary type is fixed.

Proposition 20.10.2 *For every modification $\pi\colon Y' \to Y$ and $v \in \mathrm{PSH}(Y,\theta)$ with $v \sim r$,*

$$\underline{\mathrm{vol}}_{\pi^*\theta}(\pi^*v) = \underline{\mathrm{vol}}_{\theta}(v), \qquad \overline{\mathrm{vol}}_{\pi^*\theta}(\pi^*v) = \overline{\mathrm{vol}}_{\theta}(v).$$

These masses and the ideal orders are also unchanged by a smooth gauge $(\theta, v, r) \mapsto (\theta + \mathrm{dd}^{\mathrm{c}} f, v - f, r - f)$.

Proof The identities are Proposition 14.3.1 applied to v. Smooth gauge preserves the curvature currents and the measures; it preserves multiplier ideals because e^{sf} is locally bounded above and below. □

Proposition 20.10.3 *Suppose $b < c$ and let $\pi\colon Y' \to Y$ be a nonempty finite composition of point blowups. There is a strict current $\pi^*\theta + \mathrm{dd}^{\mathrm{c}} v'$ with $v' \le \pi^* r + C$, but none with $v' \sim \pi^* r$. Both ordinary-type volumes retain the values in (20.20). Blowup centers may lie on D or on earlier exceptional curves.*

Proof Put $\beta = (c-b)w_1 + a w_2$. The representatives of Lemma 20.10.1 show that U satisfies (17.9); by Lemma 17.7.1 so does π^*U. The logarithmic divisor is $b\pi^*D$, whose smooth curvature is $b\pi^* w_1$, so its residual background is $\pi^*\beta$. If a strict current existed in the ordinary type of $\pi^* r$, Theorem 17.7.1 would give a smooth f with $\pi^*\beta + \mathrm{dd}^{\mathrm{c}} f > 0$. The exceptional curve E of the last blowup is contracted by the composite map. Restriction to this smooth curve would give $\mathrm{dd}^{\mathrm{c}}(f|_E) > 0$, contrary to its zero integral by Stokes. Only the smooth function f is restricted, so this argument applies also when the original potential has a pole along E.

For the one-sided assertion, write $\pi = \pi_1 \circ \cdots \circ \pi_N$ with point blowups $\pi_j\colon Y_j \to Y_{j-1}$ and $Y_0 = Y$, and start with the strict representative $v_0 = bG_p + U_L$, $\epsilon_L < (c-b)/2$, from the proof of Theorem 20.10.1. Suppose v_j is strict on Y_j, with lower bound $\delta_j h_j$ for a Hermitian form h_j and $\delta_j > 0$, and let π_{j+1} be the next point blowup. For its exceptional divisor E_{j+1}, choose a smooth metric on $\mathcal{O}(E_{j+1})$ whose curvature η_{j+1} is negative along E_{j+1}, which is possible since this bundle has degree -1 on E_{j+1}, and write $\mathrm{dd}^{\mathrm{c}}\ell_{j+1} = [E_{j+1}] - \eta_{j+1}$. For all sufficiently small $t_j > 0$, $\pi_{j+1}^* h_j - t_j\eta_{j+1} > 0$. Indeed, $\pi_{j+1}^* h_j$ is semipositive, positive definite off E_{j+1}, and at a point of E_{j+1} its kernel is the tangent line of E_{j+1}, on which $-\eta_{j+1}$ is positive. If unit tangent vectors w_t satisfied $(\pi_{j+1}^* h_j - t\eta_{j+1})(w_t, w_t) \le 0$ along a sequence $t \downarrow 0$, then $\pi_{j+1}^* h_j(w_t, w_t) \le t\eta_{j+1}(w_t, w_t) \to 0$, so a limit of a subsequence would lie in this kernel. Then $-\eta_{j+1}(w_t, w_t) > 0$ for small t along the subsequence, and semipositivity of $\pi_{j+1}^* h_j$ would give $(\pi_{j+1}^* h_j - t\eta_{j+1})(w_t, w_t) > 0$, a contradiction. Thus

$$v_{j+1} = \pi_{j+1}^* v_j + \delta_j t_j \ell_{j+1}$$

is strict. The pullbacks are quasi-plurisubharmonic and not identically $-\infty$, even when a center is a pole. The logarithmic terms are bounded above, so induction gives $v' \le \pi^* r + C$. Finally Proposition 20.10.2 preserves both volumes in the pulled-back ordinary type. The additional exceptional poles of the one-sided witness are not competitors in that fixed type. □

For example take $c = 1$, $b = 1/4$, $\alpha = 3/4$ and $a = 2 + \tau$, where $\tau = (|z|^2 - 1)/(|z|^2 + 1)$ is the height on the first sphere. The lower and upper volumes are $3/2$ and $9/2$. The choice $L = 31$ has $\epsilon_L = (3/4)32^{-1/4} < 3/8$ and gives a strict representative with infinite closed first moment. The intrinsic trace data are $\frac{1}{4}\,\mathrm{ord}_E(\pi^*D)$. Every nonempty finite composition of point blowups, including centers on D, preserves

both volumes and permits one-sided strictness while excluding a strict representative in the same ordinary type.

Part V
Holomorphic sections and geometric models

Chapter 21
Holomorphic sections and Hermitian mass

On a projective manifold, holomorphic sections of line bundles connect singularities with geometry: a singular weight selects the sections whose vanishing compensates for its poles, and the growth of the selected section spaces is governed by Monge–Ampère masses. On a compact Hermitian manifold the curvature of a Hermitian line bundle is still closed, whereas the background form of the Monge–Ampère operator need not be. This part of the book asks what section spaces, and the geometric models attached to sections and singularities, can still detect. Gauduchon degrees often take the place of intersection numbers: they bound the vanishing orders of sections, the valuations along a flag and the divisorial parts of positive currents on modifications.

Two kinds of geometric models appear. A complete flag turns vanishing orders into lattice points and section growth into the volume of a convex body, and the elliptic fibration of a Hopf manifold over projective space reduces its sections and its descended potentials to the base. Compact modifications carry the divisorial part of a singularity: on each model, the divisorial Lelong numbers or trace numbers sum to a closed positive current, and subtracting it from the pulled-back background gives residual classes that form b-divisor classes. Finally, positive Hermitian perturbations of a background organize multiplier-ideal envelopes into compatible systems; the limit of such a system in the original background need not recover the system.

In this chapter, X is a connected compact complex manifold of dimension n, L is a holomorphic line bundle on X with a smooth Hermitian metric h, $u \in \mathrm{QPSH}(X)$ is a fixed weight, and $\mathrm{d}V$ is a smooth positive volume form. In a local holomorphic frame e of L, write $|e|_h^2 = \mathrm{e}^{-\chi}$. The form $\beta = \mathrm{dd^c}\chi$ does not depend on the frame; it is the smooth closed curvature form of h.

The weight selects the global sections of L^m that are square integrable against $h^m \mathrm{e}^{-mu}$, and the Bergman kernels of these finite-dimensional Hilbert spaces produce closed positive currents in the curvature class of L. Pairing the Poincaré–Lelong formula with a Gauduchon metric bounds the vanishing that a nonzero section can carry; on a submanifold, this gives a necessary condition for realizing the trace data of Chapter 20 by restricted sections. When X is a curve, $\beta + \mathrm{dd^c}u \geq 0$ and u has finitely many logarithmic poles, Riemann–Roch identifies the leading growth of the section spaces with the extended Monge–Ampère mass of u. In higher dimensions

this identification depends on the background. Homogeneous sections on a Hopf manifold grow only like the sections on its projective base, and a potential of infinite Hermitian mass with trivial multiplier ideals leaves every section space unchanged. Section counts therefore cannot recover a mass computed in a Hermitian background unrelated to the curvature of L.

21.1 Weighted sections and Bergman currents

The multiplier ideals of mu determine which sections of L^m have finite weighted L^2 norm. Evaluation at points of the resulting Hilbert space defines a Bergman kernel, and its extremal characterization turns comparisons of norms into comparisons of the associated potentials.

For every positive integer m, set

$$\mathcal{H}_m(u) = H^0(X, L^m \otimes \mathcal{I}(mu)), \qquad \|s\|_{m,u}^2 = \int_X |s|_{h^m}^2 \mathrm{e}^{-mu}\, \mathrm{d}V.$$

The integrand is defined off the pole set $\{u = -\infty\}$, which has volume zero. A section $s \in H^0(X, L^m)$ belongs to $\mathcal{H}_m(u)$ if and only if $\|s\|_{m,u} < \infty$. Indeed, if the local coefficients of s lie in $\mathcal{I}(mu)$, the integrand is locally integrable, and a finite covering of the compact manifold X gives a finite integral; conversely, a finite integral gives local integrability on every coordinate ball. Thus $\mathcal{H}_m(u)$, a subspace of the finite-dimensional space $H^0(X, L^m)$, is a Hilbert space for this norm. Write $N_m = \dim \mathcal{H}_m(u)$.

Let $v \in \mathrm{QPSH}(X)$. If $u \preceq_{\mathcal{I}} v$, that is, $\mathcal{I}(ku) \subseteq \mathcal{I}(kv)$ for every real $k > 0$ (Section 1.12), then $\mathcal{H}_m(u) \subseteq \mathcal{H}_m(v)$ for every m. In particular, weights of the same ordinary singularity type (Definition 1.12.1) have the same section spaces, by Proposition 1.12.2. When the two weights differ by a bounded function, their Hilbert norms are also comparable. Below, $\|s\|_{m,v,\mathrm{d}V'}$ denotes the norm defined with v and $\mathrm{d}V'$ in place of u and $\mathrm{d}V$.

Proposition 21.1.1 *Let $v \in \mathrm{QPSH}(X)$, let $D, E \geq 0$, and let $F \in C^\infty(X, \mathbb{R})$. Suppose that $|u - v| \leq D$ outside the union of the pole sets of u and v, and that $\mathrm{d}V' = \mathrm{e}^F \mathrm{d}V$ with $|F| \leq E$. Then u and v have the same section spaces, and for every positive integer m and every s in the common space $\mathcal{H}_m(u) = \mathcal{H}_m(v)$,*

$$\mathrm{e}^{-mD-E}\|s\|_{m,u}^2 \leq \|s\|_{m,v,\mathrm{d}V'}^2 \leq \mathrm{e}^{mD+E}\|s\|_{m,u}^2.$$

Proof Off the pole sets, the integrand of $\|s\|_{m,v,\mathrm{d}V'}^2$ is that of $\|s\|_{m,u}^2$ multiplied by $\mathrm{e}^{-m(v-u)+F}$, a function with values between e^{-mD-E} and e^{mD+E}. Hence one norm is finite if and only if the other is, so the section spaces coincide, and integration gives both estimates. □

Fix $m \geq 1$ and choose an orthonormal basis $s_1, \dots, s_{N_m}$ of $\mathcal{H}_m(u)$. Define $K_{m,u}$ on X and $B_{m,u}$ off the pole set by

$$K_{m,u}(x) = \sum_{j=1}^{N_m} |s_j(x)|^2_{h^m}, \qquad B_{m,u}(x) = K_{m,u}(x)\mathrm{e}^{-mu(x)},$$

and set $B_{m,u} = 0$ on the pole set. When $N_m = 0$, both functions are zero. Since two orthonormal bases differ by a unitary matrix, $K_{m,u}$ does not depend on the choice of basis. We call $\Phi_m = m^{-1} \log K_{m,u}$ the *Bergman potential*; when $N_m > 0$, $\beta + \mathrm{dd^c}\Phi_m$ is its *Bergman current*. For $N_m = 0$, we have $\Phi_m \equiv -\infty$.

Proposition 21.1.2 *If $N_m > 0$, then*

$$K_{m,u}(x) = \sup_{0 \neq s \in \mathcal{H}_m(u)} \frac{|s(x)|^2_{h^m}}{\|s\|^2_{m,u}}, \qquad \int_X B_{m,u}\, \mathrm{d}V = N_m.$$

Moreover, the Bergman potential $\Phi_m = m^{-1} \log K_{m,u}$ is not identically $-\infty$, has analytic singularities, and satisfies

$$\beta + \mathrm{dd^c}\Phi_m \geq 0.$$

In particular, the Bergman current $\beta + \mathrm{dd^c}\Phi_m$ is a closed positive current.

Proof Write a section as $s = \sum_j a_j s_j$ and trivialize the fiber of L^m at x by a unit vector. By the Cauchy–Schwarz inequality, $|s(x)|^2_{h^m} \leq \|s\|^2_{m,u} K_{m,u}(x)$, with equality when the coefficient vector (a_j) is proportional to the complex conjugate of $(s_j(x))$. If $K_{m,u}(x) = 0$, both sides of the extremal formula vanish. This proves the extremal formula. Integrating the finite sum defining $B_{m,u}$ gives $\sum_j \|s_j\|^2_{m,u} = N_m$.

Locally $s_j = f_j e^{\otimes m}$ with holomorphic functions f_j, so

$$\Phi_m = \frac{1}{m} \log \sum_j |f_j|^2 - \chi, \qquad \beta + \mathrm{dd^c}\Phi_m = \frac{1}{m}\mathrm{dd^c} \log \sum_j |f_j|^2 \geq 0.$$

Since X is connected, a nonzero section cannot vanish identically on an open set; hence the sum $\sum_j |f_j|^2$ does not vanish identically. The local expression therefore shows that Φ_m is a quasi-plurisubharmonic function with analytic singularities, not identically $-\infty$. It also shows directly that $\beta + \mathrm{dd^c}\Phi_m$ is closed. □

Write $\Phi_m(u, \mathrm{d}V)$ when the dependence on the weight and the volume form matters. Under the hypotheses of Proposition 21.1.1, and provided that the common section space is nonzero, the extremal formula gives

$$|\Phi_m(u, \mathrm{d}V) - \Phi_m(v, \mathrm{d}V')| \leq D + E/m$$

outside the common base locus. Both potentials equal $-\infty$ on that locus. In particular, the two potentials have the same ordinary singularity type, and for a fixed weight a change of the smooth volume form changes the Bergman potential by $O(m^{-1})$.

A simultaneous change of metric and weight leaves the weighted norm unchanged. Let $a \in C^\infty(X, \mathbb{R})$, replace the metric by $h' = h\mathrm{e}^{-a}$, so that $|\cdot|^2_{h'} = |\cdot|^2_h \mathrm{e}^{-a}$, and replace the weight by $u' = u - a$. Then $h'^m \mathrm{e}^{-mu'} = h^m \mathrm{e}^{-mu}$, so the Hilbert space, its norm and the density $B_{m,u}\,\mathrm{d}V$ are unchanged, while

$$\beta' = \beta + \mathrm{dd^c} a, \qquad \Phi'_m = \Phi_m - a.$$

Consequently the Bergman current $\beta + \mathrm{dd^c}\Phi_m$ does not change.

21.2 Gauduchon degree and vanishing

The zero divisor of a holomorphic section is effective, and the Poincaré–Lelong formula expresses its integration current as the curvature form of the line bundle plus $\mathrm{dd^c}$ of a global function. Pairing with a Gauduchon metric removes the $\mathrm{dd^c}$ term, so the size of the divisor is bounded by a degree of the line bundle. Recall that a Hermitian form γ is Gauduchon if $\mathrm{dd^c}(\gamma^{n-1}) = 0$ (Definition 4.1.1); every Hermitian form on a curve is Gauduchon. Write $h^0(X, L^m) = \dim H^0(X, L^m)$. For a Gauduchon metric γ on X, the degree

$$\deg_\gamma L = \int_X \beta \wedge \gamma^{n-1}$$

is the degree $\deg_{\gamma^{n-1}} L$ of Section 5.1, with $\gamma^0 = 1$ when $n = 1$; as there, Stokes' theorem shows that it does not depend on the smooth metric on L.

Theorem 21.2.1 *If $\deg_\gamma L < 0$, then $H^0(X, L^m) = 0$ for every $m \geq 1$. If $\deg_\gamma L = 0$ and $H^0(X, L^m) \neq 0$, then L^m is holomorphically trivial and $h^0(X, L^m) = 1$. At degree zero, each nonzero weighted subspace $\mathcal{H}_m(u)$ is also one-dimensional.*

Proof Let s be a nonzero section of L^m and let D be its zero divisor. The function $\log|s|^2_{h^m}$ is locally integrable, and the Poincaré–Lelong formula gives

$$\mathrm{dd^c} \log|s|^2_{h^m} = [D] - m\beta.$$

Since γ is Gauduchon, integration by parts against γ^{n-1} annihilates the left-hand side, and therefore

$$m \deg_\gamma L = \int_D \gamma^{n-1} \geq 0,$$

where the components of D are counted with multiplicities. If $D \neq 0$, the right-hand side is strictly positive, because γ is positive on the regular locus of each component. Hence negative degree is impossible. At degree zero we obtain $D = 0$, so s has no zeros and trivializes L^m. Every other section of L^m is then a holomorphic function times s, and holomorphic functions on the connected compact manifold X are

constant. Thus $h^0(X, L^m) = 1$, and every nonzero subspace $\mathcal{H}_m(u) \subseteq H^0(X, L^m)$ is one-dimensional. □

The degree constrains positive curvature currents in the same way. If $T = \beta + \mathrm{dd}^{\mathrm{c}} u \geq 0$, the trace identity of Proposition 4.2.1 gives

$$\deg_\gamma L = \int_X T \wedge \gamma^{n-1} \geq 0.$$

If the degree is zero, the positivity of T and the strict positivity of γ imply $T = 0$.

21.3 Restricted sections and trace data

Let $Y \subset X$ be a connected compact complex submanifold of dimension $r \geq 1$ with $\nu(u, Y) = 0$, and let $\mathfrak{a}_m = \mathfrak{a}_m(u; Y) = \mathcal{I}(mu)\mathcal{O}_Y$ and $\tau_{Y,u}$ be the restricted ideals and the trace data of Section 20.1. Put

$$V_m = \operatorname{im}\bigl(\mathcal{H}_m(u) \longrightarrow H^0(Y, L^m|_Y)\bigr).$$

The local coefficients of a restricted section lie in $\mathfrak{a}_m$, so $V_m \subseteq H^0(Y, L^m|_Y \otimes \mathfrak{a}_m)$; the right-hand space can contain sections that do not extend to X.

On a modification of Y, the pullback of a nonzero restricted section vanishes along each prime divisor at least to the order imposed by the image ideal. Subtracting finitely many of these orders still leaves an effective divisor, which the Poincaré–Lelong formula turns into a positive current. If such sections exist in arbitrarily large degrees, normalized compactness produces a positive current in the class obtained by subtracting the trace data from the curvature class, and pairing it with a Gauduchon metric gives a degree inequality. Below, c_1 denotes the first Chern class in real Bott–Chern cohomology (Section 24.4); it is represented by the curvature form of any smooth Hermitian metric.

Theorem 21.3.1 *Suppose that $V_m \neq 0$ for arbitrarily large integers m. Let $\pi\colon Z \to Y$ be any smooth compact modification, let $E_1, \dots, E_q$ be distinct prime divisors on Z, and let γ be a Gauduchon metric on Z. Then*

$$\deg_\gamma(\pi^*(L|_Y)) \geq \sum_{i=1}^{q} \tau_{Y,u}(E_i) \int_{E_i} \gamma^{r-1}.$$

Moreover, the real closed $(1,1)$ class

$$c_1(\pi^*(L|_Y)) - \sum_{i=1}^{q} \tau_{Y,u}(E_i) c_1(\mathcal{O}_Z(E_i))$$

is pseudoeffective: it contains a closed positive current.

Proof Choose an unbounded set of degrees m and nonzero sections $s_m \in V_m$, and put $b_i(m) = \operatorname{ord}_{E_i} \mathfrak{a}_m$. The pullback $\pi^* s_m$ is nonzero, and its local coefficients belong to the pulled-back ideal $\mathfrak{a}_m \mathcal{O}_Z$, so it vanishes to order at least $b_i(m)$ along E_i. Thus

$$D_m := \operatorname{div}(\pi^* s_m) - \sum_{i=1}^{q} b_i(m) E_i \geq 0.$$

Choose smooth Hermitian metrics on the line bundles $\mathcal{O}_Z(E_i)$, with curvature forms β_i, and set

$$\Theta_m = \pi^*(\beta|_Y) - \sum_{i=1}^{q} \frac{b_i(m)}{m} \beta_i, \qquad \Theta = \pi^*(\beta|_Y) - \sum_{i=1}^{q} \tau_{Y,u}(E_i) \beta_i.$$

Dividing $\pi^* s_m$ by the corresponding powers of the canonical sections of the bundles $\mathcal{O}_Z(E_i)$ gives a nonzero holomorphic section of $\pi^*(L^m|_Y) \otimes \mathcal{O}_Z(-\sum_i b_i(m) E_i)$ with divisor D_m. Equip this line bundle with the metric induced by h and the chosen metrics. One mth of the logarithm of the squared norm of this section is a quasi-plurisubharmonic function w_m on Z, and the Poincaré–Lelong formula gives

$$\Theta_m + \mathrm{dd}^{\mathrm{c}} w_m = \frac{1}{m} [D_m] \geq 0.$$

By Theorem 20.1.1, $b_i(m)/m \to \tau_{Y,u}(E_i)$ as $m \to \infty$ through all integers, hence also along the chosen degrees, so $\Theta_m \to \Theta$ smoothly. Consequently a single Hermitian form Λ on Z dominates all the forms Θ_m, and every w_m belongs to $\mathrm{PSH}(Z, \Lambda)$. Normalize $\sup_Z w_m = 0$. By Theorem 1.4.1, a subsequence converges in L^1, and its limit w satisfies

$$\Theta + \mathrm{dd}^{\mathrm{c}} w \geq 0.$$

Since Θ is closed and represents the class in the statement, this positive current proves the asserted pseudoeffectivity.

Finally, pair this current with γ^{r-1}, where $\gamma^{r-1} = 1$ if $r = 1$. Integration by parts removes $\mathrm{dd}^{\mathrm{c}} w$ because γ is Gauduchon, and the Poincaré–Lelong formula for the

canonical section of $\mathcal{O}_Z(E_i)$ gives $\int_Z \beta_i \wedge \gamma^{r-1} = \int_{E_i} \gamma^{r-1}$. The nonnegativity of the pairing is exactly the stated degree inequality. □

Equivalently, if one of these inequalities fails for some choice of $Z, E_1, \dots, E_q$ and γ, then $V_m = 0$ for all sufficiently large m.

The converse of the theorem fails in general. Let $C = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$ be the square elliptic curve, choose an irrational real number a, and let L be the flat line bundle on C with multipliers $\mathrm{e}^{2\pi \mathrm{i} a}$ and 1 along the two generators: its sections lift to holomorphic functions f on $\mathbb{C}$ with $f(z+1) = \mathrm{e}^{2\pi \mathrm{i} a} f(z)$ and $f(z+\mathrm{i}) = f(z)$. The constant metric on L has curvature zero. Take $u = 0$ and $Y = X = C$. Then every trace number is zero, and the class in the theorem is zero, hence pseudoeffective. Nevertheless no positive tensor power of L has a nonzero holomorphic section. Indeed, a nonzero section of L^m would have a zero divisor of degree zero and would therefore be nowhere zero. The logarithm of its norm would be harmonic on C, hence constant, so its lift f would be a holomorphic function of constant modulus on $\mathbb{C}$, and thus constant. Since $ma \notin \mathbb{Z}$, the relation $f(z+1) = \mathrm{e}^{2\pi \mathrm{i} m a} f(z)$ then forces $f = 0$, a contradiction.

21.4 Bounded weighted sections

The spaces $\mathcal{H}_m(u)$ need not be closed under multiplication: the product of two sections with integrable weighted squared norms need not have an integrable weighted squared norm, as the following logarithmic example shows. Requiring the weighted pointwise norm to be bounded rather than integrable gives spaces that do multiply. They form the graded algebra used in Chapter 22.

Example 21.4.1 Let $X = \mathbb{P}^1$ with affine coordinate z, and let δ_0 be the Dirac mass at the origin. Take $L = \mathcal{O}(1)$ with its Fubini–Study metric, whose curvature form is β, and let

$$u(z) = \frac{3}{5} \log \frac{|z|^2}{1 + |z|^2}.$$

Then $\beta + \mathrm{dd^c} u = \frac{2}{5}\beta + \frac{3}{5}\delta_0 \geq 0$, $\mathcal{I}(u) = \mathcal{O}$ and $\mathcal{I}(2u) = \mathcal{O}(-[0])$. A section of L that does not vanish at 0 belongs to $\mathcal{H}_1(u)$, but its square does not belong to $\mathcal{H}_2(u)$.

Proof The curvature formula follows from the Poincaré–Lelong formula $\mathrm{dd^c} \log |z|^2 = \delta_0$ and from $\beta = \mathrm{dd^c} \log(1 + |z|^2)$ on $\mathbb{C}$. The weight is bounded near infinity and equals $\frac{3}{5} \log |z|^2 + O(1)$ near zero, its only singularity. The one-variable calculation of Example 1.12.1 therefore gives the two stated ideals: $\mathcal{I}(u)$ is trivial, while $\mathcal{I}(2u)$ consists of the functions vanishing at the origin. A section of L with $s(0) \neq 0$ thus lies in $\mathcal{H}_1(u)$, whereas s^2 does not vanish at the origin and so does not lie in $\mathcal{H}_2(u)$.□

We return to the general data (X, L, h, u). Define $\mathcal{W}_0 = \mathbb{C}$ and, for $m \geq 1$,

$$\mathcal{W}_m(u) = \{ s \in H^0(X, L^m) : |s|^2_{h^m} \mathrm{e}^{-mu} \text{ is bounded off } \{u = -\infty\} \}.$$

These are vector spaces, since $|s+t|^2 \le 2|s|^2 + 2|t|^2$; they are finite-dimensional and do not depend on the smooth metric, because the ratio of two smooth metrics is bounded above and below on X. Since X has finite volume, they are contained in $\mathcal{H}_m(u)$. For positive integers m, k, $s \in \mathcal{W}_m(u)$ and $t \in \mathcal{W}_k(u)$, the identity

$$|st|^2_{h^{m+k}} \mathrm{e}^{-(m+k)u} = \big(|s|^2_{h^m} \mathrm{e}^{-mu}\big)\big(|t|^2_{h^k} \mathrm{e}^{-ku}\big)$$

proves $\mathcal{W}_m(u)\mathcal{W}_k(u) \subseteq \mathcal{W}_{m+k}(u)$. Thus $\bigoplus_m \mathcal{W}_m(u)$ is a graded subalgebra of the section ring $\bigoplus_{m\ge 0} H^0(X, L^m)$ of L.

On a smooth projective variety, for big line bundles with metrics of positive curvature current and positive multiplier-ideal volume, the almost-semigroup construction of [Xia26, Section 10.3] works with the integrable section spaces themselves.

21.5 Section growth on a curve

On a curve, a weight with logarithmic poles imposes vanishing orders at the poles. Integrability rounds the required order down and boundedness rounds it up, so the two resulting divisors differ by a bounded amount, and Riemann–Roch gives the same leading growth for both section spaces.

Throughout this section, X is a compact Riemann surface and $\beta + \mathrm{dd}^{\mathrm{c}} u \ge 0$. Let $p_1, \dots, p_r$ be distinct points such that u is locally bounded on $X \setminus \{p_1, \dots, p_r\}$ and, in coordinates z_i centered at p_i,

$$u = c_i \log |z_i|^2 + O(1), \qquad c_i > 0.$$

Set $\ell = \deg L = \int_X \beta$ and $d = \sum_i c_i$. Recall that $\mathrm{MA}^{\mathrm{ext}}_\beta(u)$ denotes the extended non-pluripolar Monge–Ampère measure of Theorem 7.1.1.

Theorem 21.5.1 *In this situation $0 \le d \le \ell$, and*

$$\mathcal{H}_m(u) = H^0\left(X, L^m\Big(-\sum_i \lfloor mc_i \rfloor p_i\Big)\right),$$
$$\mathcal{W}_m(u) = H^0\left(X, L^m\Big(-\sum_i \lceil mc_i \rceil p_i\Big)\right).$$

Moreover,

$$\lim_{m\to\infty} \frac{\dim \mathcal{H}_m(u)}{m} = \lim_{m\to\infty} \frac{\dim \mathcal{W}_m(u)}{m} = \ell - d = \int_X \mathrm{MA}^{\mathrm{ext}}_\beta(u).$$

Proof *Step 1: the section spaces.* A germ of a section of L^m vanishing to order k at p_i has weighted squared norm comparable to $|z_i|^{2(k-mc_i)}$. As in Example 1.12.1, it is locally integrable if and only if $k - mc_i > -1$, that is, $k \ge \lfloor mc_i \rfloor$; it is bounded if and only if $k \ge mc_i$, that is, $k \ge \lceil mc_i \rceil$. Away from the points p_i the weight is locally

bounded and imposes no condition. Since X is compact, this gives the displayed global identifications.

Step 2: the mass. The positive measure $\beta + \mathrm{dd^c} u$ has total mass ℓ, and it has an atom of mass c_i at p_i. Subtracting these atoms leaves a positive measure, which proves $d \le \ell$. This remaining measure does not charge polar sets: after subtraction of the logarithmic singularity, its local potential has a bounded quasi-plurisubharmonic representative, and by Proposition 1.7.1 its curvature gives polar sets zero mass. Thus the remaining measure is exactly the extended measure, and its mass is $\ell - d$.

Step 3: growth. If $\ell > d$, the degrees of both line bundles in the display equal $m(\ell - d) + O(1)$ and tend to infinity. Let g be the genus of X. By Riemann–Roch and Serre duality, $h^0 = \deg + 1 - g$ once the degree exceeds $2g - 2$. Dividing by m proves both limits.

If $\ell = d$, both degrees remain bounded. A line bundle of negative degree has no nonzero section, and one of nonnegative degree k has at most $k + 1$ independent sections: the sections vanishing at $k + 1$ given distinct points are sections of a line bundle of negative degree, hence zero. Therefore both dimensions are uniformly bounded, and both limits vanish. These arguments include the case $r = 0$. □

21.6 Homogeneous sections on a Hopf manifold

On a scalar Hopf manifold, a holomorphic section of a line bundle lifts to a function on $\mathbb{C}^n \setminus \{0\}$ with an exact homogeneity property under the contraction. Hartogs extension reduces the section spaces to spaces of homogeneous polynomials, and the invariance of the metric makes the Bergman kernel constant. The resulting growth is of order m^{n-1}, reflecting the dimension of the projective base rather than that of the manifold.

Let $n \ge 2$. We write $X_n = (\mathbb{C}^n \setminus \{0\})/\langle z \mapsto \mathrm{e}^{-1} z \rangle$ for the scalar Hopf manifold H_n of Section 9.6, now with covering coordinates $z = (z_1, \dots, z_n)$, and $\pi = p_n \colon X_n \to \mathbb{P}^{n-1}$ for its projection, whose fibers are elliptic curves. As there, put

$$\alpha = \alpha_n = \mathrm{dd^c} \log |z|^2, \qquad \zeta = \beta_n, \qquad \omega = \omega_n = \alpha + \zeta.$$

By that section, α is the pullback of the Fubini–Study form of total mass one, ω is a Gauduchon form, and

$$\alpha^n = 0, \qquad \zeta^2 = 0, \qquad \int_{X_n} \alpha^{n-1} \wedge \zeta = 2, \qquad \int_{X_n} \omega^n = 2n.$$

Set $\mathrm{d}V = \omega^n/(2n)$, a probability measure.

For an integer $k \ge 0$, let L_k be the line bundle on X_n obtained from $(\mathbb{C}^n \setminus \{0\}) \times \mathbb{C}$ through the identification

$$(z, t) \sim (\mathrm{e}^{-1} z, \mathrm{e}^{-k} t).$$

Give it the metric $|t|^2_{h_k} = |t|^2|z|^{-2k}$. Under the identification the two factors change by e^{-2k} and e^{2k}, so this metric descends. In the frame $t = 1$ the squared norm is $|z|^{-2k}$, and the curvature form is $k\alpha$. We apply the constructions of Section 21.1 with $X = X_n$, $L = L_k$, $h = h_k$ and this probability volume form.

Proposition 21.6.1 *For $m \geq 1$ and the weight $u = 0$,*

$$N_m := h^0(X_n, L_k^m) = \binom{km + n - 1}{n - 1}, \qquad B_{m,0} = N_m.$$

The Bergman potential is the constant $m^{-1} \log N_m$, and the Bergman current is $k\alpha$. Moreover,

$$\deg_\omega L_k = 2k(n-1), \qquad \lim_{m\to\infty} m^{-n} N_m = 0, \qquad (k\alpha)^n = 0,$$

whereas $\int_{X_n} \omega^n = 2n > 0$.

Proof A section of L_k^m lifts to a holomorphic function f on $\mathbb{C}^n \setminus \{0\}$ satisfying

$$f(\mathrm{e}^{-1} z) = \mathrm{e}^{-km} f(z).$$

Since $n \geq 2$, Hartogs' theorem extends f holomorphically to $\mathbb{C}^n$. In its Taylor series, a monomial of total degree j is multiplied by e^{-j} under the contraction, so comparing coefficients shows that only monomials of degree km occur. Conversely, every homogeneous polynomial of that degree has the required equivariance. Counting monomials gives the binomial coefficient; when $k = 0$, only the constant polynomials remain.

The inner product defined by h_k^m and $\mathrm{d}V$ is invariant under the unitary group $U(n)$, so the basis-independent function $K_{m,0}$ is $U(n)$-invariant. It is also invariant under positive radial dilations: every section is homogeneous of degree km, and the factor $|z|^{-2km}$ in its squared norm cancels the effect of a dilation. Unitary transformations and positive radial dilations act transitively on $\mathbb{C}^n \setminus \{0\}$, so $K_{m,0}$ is constant. By Proposition 21.1.2 its integral is N_m, and $\mathrm{d}V$ has mass one. Thus $K_{m,0} = B_{m,0} = N_m$.

The statements about the Bergman potential and current follow immediately. Since $\alpha^n = 0$ and $\zeta^2 = 0$,

$$\int_{X_n} k\alpha \wedge \omega^{n-1} = k(n-1) \int_{X_n} \alpha^{n-1} \wedge \zeta = 2k(n-1).$$

The binomial coefficient grows at most like m^{n-1}, which proves the normalized limit, and $\alpha^n = 0$ proves the vanishing of the curvature mass. □

For $n = 3$, X_3 is the Gauduchon threefold of Theorem 9.0.1, on which Hermitian Monge–Ampère masses are unbounded; for $k > 0$, the line bundles L_k nevertheless have positive degree and quadratic section growth. In the notation of Section 5.1.11, L_k is the line bundle E_a on X_q with $q = \mathrm{e}^{-1}$ and $a = \mathrm{e}^{-k}$, so Example 5.1.2 also gives the convexity indices of the complements $\mathbb{P}(\mathcal{O} \oplus L_k) \setminus \mathbb{P}(\mathcal{O})$.

21.7 Section counts and infinite Hermitian mass

We return to a connected compact complex manifold X of dimension n. For a weight u with trivial multiplier ideals, every weighted section space $\mathcal{H}_m(u)$ is the full space $H^0(X, L^m)$. When X fails the bounded mass property of Definition 7.5.1, such a weight can have infinite extended mass, so section growth in a closed curvature class does not determine the mass in an unrelated Hermitian background.

Proposition 21.7.1 *Let ω be a Hermitian metric on X with $v_+(\omega) = +\infty$. There is a potential $U \in \mathrm{PSH}(X, \omega)$ such that, with $L = \mathcal{O}_X$ and its constant metric,*

$$\dim \mathcal{H}_m(U) = 1 \quad \textit{for every } m \geq 1, \qquad \int_X \mathrm{MA}_\omega^{\mathrm{ext}}(U) = +\infty.$$

All Bergman currents of U are identically zero.

Proof Take U from Theorem 7.6.1; its extended mass is infinite, and $\mathcal{I}(tU) = \mathcal{O}_X$ for every $t > 0$. Global holomorphic functions on X are constant, and the constant function 1 lies in $\mathcal{H}_m(U)$ because $\mathcal{I}(mU) = \mathcal{O}_X$ and X is compact. Hence every weighted section space is one-dimensional, and its orthonormal basis consists of a nonzero constant. Each Bergman potential Φ_m is therefore constant. Since the trivial metric has curvature zero, every Bergman current is zero. □

Chapter 22
Flags, convex bodies and section growth

Let X be a connected compact complex manifold of dimension n, and let L be a holomorphic line bundle on X. Along a complete flag of compact complex submanifolds, successive vanishing orders turn each section of L^m into a lattice point, and the Gauduchon degree bounds of Chapter 21 keep the normalized points in a compact convex body. For a graded algebra of sections with some nonzero piece of positive degree, counting these points gives the growth exponent and the leading coefficient of its graded pieces. The exponent is the transcendence degree of the field of section ratios, and the coefficient is an intrinsic lattice volume of the body. A smooth metric on L gives an additional bound: the maximal rank of its curvature on the locus where the curvature is semipositive bounds the growth exponent, even if the curvature has no global sign.

Not every compact complex manifold carries such a flag, but the blowup of a point does, and Hartogs extension identifies the sections on the blowup with those on X; the count therefore applies to every compact complex manifold. Growth of the maximal order m^n forces X to admit a modification from a projective manifold, and hence to have the bounded mass property. On the Hopf threefold, where Hermitian masses are unbounded, the homogeneous line bundles of positive degree from Section 21.6 still have bodies that are triangles: their area detects quadratic section growth, although their three-dimensional volume vanishes.

22.1 Flag valuations and degree bounds

Successive vanishing orders along a flag compute the lexicographically smallest exponent in a local Taylor expansion. The coefficient of that exponent is a linear functional on sections, and this shows that the number of distinct values on a space of sections equals its dimension. At each step, the effective divisor that remains gives, through a Gauduchon degree, a linear bound for the next order.

Until Section 22.5, suppose that X carries a *complete flag*, that is, a chain

$$X = Y_0 \supset Y_1 \supset \cdots \supset Y_n = \{p\},$$

where p is a point and each Y_i, $1 \le i \le n$, is a smooth connected compact prime divisor in Y_{i-1}. For a nonnegative integer m and a nonzero section $s \in H^0(X, L^m)$, let $\nu_1(s)$ be its vanishing order along Y_1. Dividing s by the corresponding power of the canonical section of $\mathcal{O}_X(Y_1)$ and restricting to Y_1 gives a nonzero section of $L^m(-\nu_1(s)Y_1)|_{Y_1}$. Repeating this procedure along the flag defines $\nu_2(s), \ldots, \nu_n(s)$, and the vector $\nu(s) = (\nu_1(s), \ldots, \nu_n(s)) \in \mathbb{N}^n$ is the *flag valuation* of s.

Near p, choose holomorphic coordinates in which $Y_i = \{z_1 = \cdots = z_i = 0\}$, and a nonvanishing local frame of L. Then $\nu(s)$ is the lexicographically smallest exponent occurring in the Taylor expansion of the local coefficient of s. Consequently, with respect to the lexicographic order on $\mathbb{N}^n$,

$$\nu(st) = \nu(s) + \nu(t), \qquad \nu(s+t) \ge \min\{\nu(s), \nu(t)\},$$

where the first identity holds for nonzero sections s, t of arbitrary degrees, and the second when s, t have the same degree and $s + t \neq 0$. If $\nu(s) \neq \nu(t)$, equality holds in the second formula.

Lemma 22.1.1 *For every finite-dimensional vector subspace $V \subset H^0(X, L^m)$,*

$$\#\{\nu(s) : 0 \neq s \in V\} = \dim V.$$

Proof Sections with distinct valuations are linearly independent: in a nontrivial finite linear combination of them, the smallest valuation occurs in exactly one summand and cannot cancel. Hence there are at most $\dim V$ values.

For the reverse inequality, we argue by induction on $\dim V$, the case $\dim V = 0$ being immediate. The nonempty set of values has a smallest element a in the lexicographic order. The Taylor coefficient of the monomial z^a is a nonzero linear functional on V. Its kernel has codimension one, and every nonzero element of the kernel has valuation strictly larger than a. By the induction hypothesis the kernel contributes $\dim V - 1$ distinct values, all different from a; together with a, these are $\dim V$ values. □

The valuations of sections of L^m grow at most linearly in m. To see this, pair the residual divisor at each stage of the construction with a Gauduchon metric on the corresponding member of the flag.

Theorem 22.1.1 *There are constants $C_1, \ldots, C_n \ge 0$, depending only on L and the chosen flag, such that*

$$0 \le \nu_i(s) \le C_i m \quad (1 \le i \le n)$$

for every $m \ge 1$ and every nonzero $s \in H^0(X, L^m)$.

Proof For each i, equip Y_{i-1} with a Gauduchon metric γ_i; when Y_{i-1} is a curve, any Hermitian metric will do. Put $d_i = \dim Y_{i-1} = n - i + 1$ and

$$a_i = \int_{Y_i} \gamma_i^{d_i - 1} > 0, \qquad A_i = \deg_{\gamma_i}(L|_{Y_{i-1}}).$$

For $i = n$ the integral defining a_i is the integral of 1 over the point p, so $a_n = 1$.

For $j < i$, also put

$$B_{ij} = \deg_{\gamma_i}\left(\mathcal{O}_{Y_{j-1}}(Y_j)|_{Y_{i-1}}\right).$$

All these numbers depend only on L, the flag and the chosen metrics.

At the ith stage of the construction of $\nu(s)$, the residual section is a section of the line bundle

$$M_{i-1} = L^m|_{Y_{i-1}} \otimes \bigotimes_{j<i} \mathcal{O}_{Y_{j-1}}(-\nu_j(s)Y_j)|_{Y_{i-1}}.$$

Its zero divisor contains $\nu_i(s)Y_i$. The Poincaré–Lelong formula, tested against $\gamma_i^{d_i-1}$ as in the proof of Theorem 21.2.1, therefore gives

$$a_i\nu_i(s) \le mA_i - \sum_{j<i} B_{ij}\nu_j(s). \tag{22.1}$$

Define recursively

$$C_i = \frac{|A_i| + \sum_{j<i} |B_{ij}|C_j}{a_i}.$$

Induction on i in (22.1) gives $\nu_i(s) \le C_i m$. □

22.2 The body of a graded section algebra

The valuations of the nonzero homogeneous elements of a graded algebra of sections form a semigroup whose part in degree m has exactly as many elements as the dimension of the degree-m piece. The linear bound of Theorem 22.1.1 places this semigroup in a cone with compact height-one slice, and a semigroup counting theorem relates the asymptotic count to the volume of this slice.

Let $W_\bullet = \bigoplus_{m\ge 0} W_m$ be a graded subalgebra of the section ring $\bigoplus_{m\ge 0} H^0(X, L^m)$, with $W_0 = \mathbb{C}$. For a weight $u \in \mathrm{QPSH}(X)$, the bounded weighted section algebra $\mathcal{W}_\bullet(u)$ of Section 21.4 is an example.

Set

$$S = \{(0,0)\} \cup \{(\nu(s), m) : m \ge 1,\ 0 \ne s \in W_m\}.$$

This is an additive semigroup, because the product of two nonzero sections on the connected manifold X is nonzero and valuations add. Choose constants C_i as in Theorem 22.1.1. Then S lies in the closed cone

$$\{(x,t) \in \mathbb{R}^n \times \mathbb{R} : t \ge 0,\ 0 \le x_i \le C_i t\},$$

whose only point at height zero is the origin. Define

$$\Delta(W_\bullet) = \overline{\mathrm{conv}}\{m^{-1}\nu(s) : m \geq 1,\ 0 \neq s \in W_m\},$$

where $\overline{\mathrm{conv}}$ denotes the closed convex hull. If some positive-degree piece of $W_\bullet$ is nonzero, this is a compact convex subset of $\mathbb{R}^n$; equivalently, it is the height-one slice of the closed convex cone generated by S. If every positive-degree piece is zero, $\Delta(W_\bullet) = \varnothing$. Write $\mathrm{vol}_{\mathbb{R}^n}$ for Euclidean volume.

Proposition 22.2.1 *For every such algebra,*

$$\limsup_{m\to\infty} m^{-n} \dim W_m \leq \mathrm{vol}_{\mathbb{R}^n} \Delta(W_\bullet).$$

In particular, if $\Delta(W_\bullet)$ *has empty interior, then* $\dim W_m = o(m^n)$.

Proof The empty case is immediate. Otherwise, let $S_m = \{\nu(s) : 0 \neq s \in W_m\}$. By Lemma 22.1.1, S_m has $\dim W_m$ elements, and $m^{-1}S_m \subseteq \Delta(W_\bullet)$. Attach to each $a \in S_m$ the half-open cube $m^{-1}(a + [0,1)^n)$. These cubes are disjoint, their total volume is $m^{-n}\dim W_m$, and their union is contained in

$$\Delta(W_\bullet) + [0, m^{-1}]^n.$$

These compact sets decrease to $\Delta(W_\bullet)$ as $m \to \infty$, so their volumes converge to its volume. This proves the inequality. A compact convex set has positive n-dimensional volume exactly when it has nonempty interior, which gives the last statement. □

We use the following semigroup counting theorem, recalled in [Xia26, Theorem C.2.1]. Let $r \geq 0$, and let $\Sigma \subset \mathbb{Z}^{r+1}$ be an additive semigroup that generates $\mathbb{Z}^{r+1}$ as a group and lies in a closed convex cone in $\mathbb{R}^r \times \mathbb{R}_{\geq 0}$ whose height-one slice is compact and whose only point at height zero is the origin. Then the number of points of Σ at height m, divided by m^r, converges to the Euclidean volume of the height-one slice of the closed convex cone generated by Σ.

Corollary 22.2.1 *If the subgroup generated by* S *is all of* $\mathbb{Z}^{n+1}$*, then*

$$\lim_{m\to\infty} m^{-n} \dim W_m = \mathrm{vol}_{\mathbb{R}^n} \Delta(W_\bullet).$$

Proof The semigroup S satisfies the cone conditions established above and, by hypothesis, generates $\mathbb{Z}^{n+1}$. The semigroup counting theorem therefore applies to it. By Lemma 22.1.1, the number of points of S at height m is $\dim W_m$, and the height-one slice of the closed convex cone generated by S is $\Delta(W_\bullet)$. □

22.3 Intrinsic dimension, lattice and growth

A body of zero volume in $\mathbb{R}^n$ can still detect growth of lower order. The group generated by the valuation semigroup determines a degree period and a lattice in which the body is full-dimensional, and counting in this lattice gives the growth

exponent and the leading coefficient. Ratios of sections identify the exponent with a transcendence degree, and since the coefficient is a limit of graded dimensions, it does not depend on the flag.

Throughout this section, some positive-degree piece of $W_\bullet$ is nonzero. Let $G \subset \mathbb{Z}^{n+1}$ be the subgroup generated by S, and let pr_{n+1} be the projection to the last coordinate. Define the *degree period* $e \geq 1$ by

$$\mathrm{pr}_{n+1}(G) = e\mathbb{Z}.$$

Thus e is the greatest common divisor of the degrees m with $W_m \neq 0$. Set

$$\Lambda = G \cap (\mathbb{R}^n \times \{0\}), \qquad d = \operatorname{rank} \Lambda.$$

We regard Λ as a lattice in its real span, a d-dimensional subspace of $\mathbb{R}^n \times \{0\}$. Its lattice measure assigns volume one to a fundamental parallelepiped. On each translate of this span we use the translated measure, denoted by vol_Λ; when $d = 0$, a point has volume one.

Theorem 22.3.1 *The affine dimension of $\Delta(W_\bullet)$ is d. Its lattice volume is finite and strictly positive, and*

$$\lim_{q\to\infty} \frac{\dim W_{eq}}{(eq)^d} = \mathrm{vol}_\Lambda\, \Delta(W_\bullet).$$

If e does not divide m, then $W_m = 0$. In particular, $W_m \neq 0$ for all sufficiently large multiples m of e.

Proof Choose $g_0 = (b, e) \in G$ and an integral basis $\lambda_1, \ldots, \lambda_d$ of Λ. Every element of G has a unique expression

$$\sum_{i=1}^{d} a_i \lambda_i + q g_0, \qquad (a, q) \in \mathbb{Z}^d \times \mathbb{Z}.$$

These coordinates identify G with $\mathbb{Z}^{d+1}$. Let S' be the image of S under this identification. It generates $\mathbb{Z}^{d+1}$, and the height of the image of $(x, m) \in S$ is $q = m/e \geq 0$.

The points $(m^{-1}x, 1)$, for $(x, m) \in S$ with $m > 0$, span the $(d+1)$-dimensional real vector space $\mathrm{span}_{\mathbb{R}}\, G$. Since they all have height one, their affine span has dimension d. Taking the convex hull and the closure does not change this affine span, which proves the dimension assertion.

Intersect the cone containing S with $\mathrm{span}_{\mathbb{R}}\, G$. The linear change of coordinates carries this intersection to a closed cone with compact height-one slice whose only point at height zero is the origin. Indeed, a vector (x, m) with $|x| \leq Cm$ has horizontal coordinates bounded by a constant times m, hence by a constant times q. The height-one body of S' is

$$\Delta' = e\big(\Delta(W_\bullet) - b/e\big),$$

where the right-hand side is expressed in the basis $\lambda_1, \dots, \lambda_d$. It is full-dimensional in $\mathbb{R}^d$, and therefore

$$\operatorname{vol}_{\mathbb{R}^d} \Delta' = e^d \operatorname{vol}_\Lambda \Delta(W_\bullet) > 0.$$

Apply the semigroup counting theorem of Section 22.2 to S', which generates its full lattice. The points of S' at height q correspond bijectively to the valuations of the nonzero elements of W_{eq}, and by Lemma 22.1.1 there are $\dim W_{eq}$ of them. Hence

$$\lim_{q\to\infty} \frac{\dim W_{eq}}{q^d} = \operatorname{vol}_{\mathbb{R}^d} \Delta'.$$

Dividing by e^d gives the asserted limit, and its positivity gives eventual nonvanishing. For $d = 0$ the body is a point of volume one, so the integer dimensions converge to one and are eventually equal to one. Finally, every degree with a nonzero section lies in the projection $e\mathbb{Z}$ of G, which proves the vanishing outside these multiples. □

The growth dimension also has a description independent of the flag. Write $\mathcal{M}(X)$ for the field of meromorphic functions on X and define the *section-ratio field*

$$K(W_\bullet) = \{s/t : s, t \in W_m \text{ for some } m \geq 0,\ t \neq 0\} \subset \mathcal{M}(X).$$

It is a field: sums and products of two such ratios, with denominators $t_1 \in W_{m_1}$ and $t_2 \in W_{m_2}$, can be written with the common denominator $t_1 t_2$ in degree $m_1 + m_2$, and inversion exchanges numerator and denominator. It contains $\mathbb{C}$ because $W_0 = \mathbb{C}$.

Proposition 22.3.1 *The transcendence degree of $K(W_\bullet)$ over $\mathbb{C}$ is exactly d. If $\Delta(W_\bullet)$ has nonempty interior in $\mathbb{R}^n$, then X has n algebraically independent meromorphic functions. Moreover, d, the period e and the intrinsic volume in Theorem 22.3.1 are the same for any two complete flags on X.*

Proof Suppose first that $d > 0$. Choose sections $s_0, \dots, s_d$ of positive degrees $m_0, \dots, m_d$ whose normalized valuations $\nu(s_i)/m_i$ are affinely independent. Let M be a common multiple of their degrees and put $t_i = s_i^{M/m_i} \in W_M$. The ratios $f_i = t_i/t_0$ belong to $K(W_\bullet)$. The flag valuation extends to quotients by subtraction, giving

$$\nu(f_i) = M\left(\frac{\nu(s_i)}{m_i} - \frac{\nu(s_0)}{m_0}\right), \qquad 1 \leq i \leq d.$$

These vectors are linearly independent. Distinct monomials in the f_i therefore have distinct valuations, so the term of smallest valuation in a nonzero polynomial expression cannot cancel, and the ratios are algebraically independent. Thus the transcendence degree is at least d; this is trivial when $d = 0$.

Conversely, let $f_1, \dots, f_r$, with $r \geq 1$, be algebraically independent elements of the field. Using the product of their denominators, write $f_i = A_i/B$ with $A_i, B \in W_M$ and $B \neq 0$ for a common positive degree M. For every integer $q \geq 1$ and every multi-index $a = (a_1, \dots, a_r) \in \mathbb{N}^r$ with $|a| = a_1 + \cdots + a_r \leq q$, the sections

$$B^{q-|a|} A_1^{a_1} \cdots A_r^{a_r} \in W_{Mq}$$

are linearly independent, because division by B^q turns a linear relation among them into a polynomial relation among the f_i. There are $\binom{q+r}{r}$ such sections. Since $W_M \neq 0$, the degree M is a multiple of e, and Theorem 22.3.1 gives $\dim W_{Mq} = O(q^d)$. Comparing with the binomial lower bound forces $r \leq d$, which proves equality of the transcendence degree with d. A body with nonempty interior has $d = n$, and this gives the stated consequence for X.

The field and the set of degrees with nonzero sections do not depend on the flag, so neither d nor e does. By Theorem 22.3.1, the lattice volume is, for either flag, the limit of the same dimensions with the same normalization. It is therefore independent of the flag as well, although the bodies and their lattices may differ. □

22.4 Moishezon manifolds and bounded mass

Recall from Lemma 7.5.1 that a Moishezon manifold, that is, a compact complex manifold of dimension n with n algebraically independent meromorphic functions, has the bounded mass property.

Theorem 22.4.1 *If X fails the bounded mass property, then for every complete flag on X, every holomorphic line bundle L and every graded subalgebra $W_\bullet$ of its section ring as above,*

$$\operatorname{vol}_{\mathbb{R}^n} \Delta(W_\bullet) = 0, \qquad \dim W_m = o(m^n).$$

Proof If $\Delta(W_\bullet)$ had nonempty interior, X would be Moishezon by Proposition 22.3.1, contradicting Lemma 7.5.1. The body therefore has empty interior, and Proposition 22.2.1 gives both conclusions. □

22.5 Flags on a point blowup

We now remove the assumption that X carries a complete flag. A curve always does, since any point completes the flag. In dimension at least two, the exceptional divisor of the blowup of a point is a projective space and carries a complete flag of linear subspaces, while Hartogs extension identifies the sections on the blowup with those on X. The growth formula on the blowup therefore counts the original section spaces.

Lemma 22.5.1 *Every connected compact complex manifold X of dimension $n \geq 1$ admits a smooth compact modification $\pi\colon \widetilde{X} \to X$ such that $\widetilde{X}$ carries a complete flag and, for every holomorphic line bundle L on X and every $m \geq 0$, pullback is an isomorphism*

$$H^0(X, L^m) \simeq H^0(\widetilde{X}, \pi^* L^m).$$

These isomorphisms are compatible with products.

Proof If $n = 1$, take π to be the identity and any point as the last member of the flag; the remaining assertions are then immediate. If $n \geq 2$, blow up a point $p \in X$. The local description of the blowup of $\mathbb{C}^n$ at the origin shows that $\widetilde{X}$ is smooth and that its exceptional divisor is $E \simeq \mathbb{P}^{n-1}$. A nested sequence of linear subspaces of E, together with $\widetilde{X}$, gives a flag

$$\widetilde{X} \supset E \supset \mathbb{P}^{n-2} \supset \cdots \supset \mathbb{P}^0.$$

Each member is smooth, compact and connected, and is a prime divisor in the preceding member. The modification is an isomorphism off E.

Let s be a section of $\pi^* L^m$. Off E, it gives a section of L^m on $X \setminus \{p\}$. In a trivialization of L^m on a coordinate ball centered at p, its coefficient is holomorphic on the punctured ball, and by Hartogs' extension theorem, valid since $n \geq 2$, it extends holomorphically across p. The extension glues with the section away from p to a section of L^m on X. Its pullback agrees with s off E, hence everywhere by the identity theorem. This proves surjectivity; injectivity and compatibility with products follow because pullback is an isomorphism on the dense open set $\widetilde{X} \setminus E$. □

Given a graded subalgebra $W_\bullet$ of the section ring of L, choose a complete flag on the modification of Lemma 22.5.1 and apply the constructions of Section 22.2 on $\widetilde{X}$ to the pulled-back algebra $\widetilde{W}_m = \pi^* W_m \subseteq H^0(\widetilde{X}, \pi^* L^m)$. The shape of the resulting body may depend on the point and on the flag, but its dimension and its normalized count are determined by $W_\bullet$.

Theorem 22.5.1 *Assume that some positive-degree piece of $W_\bullet$ is nonzero. There are an integer $0 \leq d \leq n$, a degree period $e \geq 1$, and a number $V(W_\bullet) > 0$ such that*

$$\lim_{q \to \infty} \frac{\dim W_{eq}}{(eq)^d} = V(W_\bullet), \qquad W_m = 0 \quad \text{if } e \nmid m.$$

Here d is the transcendence degree of the section-ratio field $K(W_\bullet)$, e is the greatest common divisor of the degrees m with $W_m \neq 0$, and $V(W_\bullet)$ is the intrinsic lattice volume of every body constructed on such a modification. In particular, these three quantities do not depend on the choice of the modification or the flag.

Proof Pullback preserves the graded dimensions and the degrees with nonzero sections. Applying Theorems 22.1.1 and 22.3.1 on $\widetilde{X}$ gives the displayed limit and the positivity of the lattice volume. Pullback also identifies the section-ratio fields, since their elements are the corresponding ratios and pullback is injective on the dense set where π is an isomorphism. Thus Proposition 22.3.1 identifies d with the transcendence degree of the original field. The unchanged degrees with nonzero sections determine e, and the limit of the unchanged dimensions determines $V(W_\bullet)$. None of the three depends on the auxiliary choices. □

Corollary 22.5.1 *For every holomorphic line bundle L on a connected compact complex manifold X of dimension n,*

$$h^0(X, L^m) = O(m^n).$$

If X is not Moishezon, then, for each L, either all positive-degree section spaces of L vanish, or there is an integer $d < n$ for which

$$h^0(X, L^m) = O(m^d).$$

In particular $m^{-n} h^0(X, L^m) \to 0$. These conclusions apply when X fails the bounded mass property.

Proof If all positive-degree section spaces vanish, there is nothing to prove. Otherwise, apply the theorem to the full section ring. Since $W_m = 0$ for $e \nmid m$, the limit gives $h^0(X, L^m) = O(m^d)$ for all degrees, with $d \le n$. If $d = n$, the section-ratio field contains n algebraically independent meromorphic functions on X, so X is Moishezon. Thus $d < n$ when X is not Moishezon. Finally, a Moishezon manifold has the bounded mass property by Lemma 7.5.1, which gives the last assertion. □

Since the multiplier-ideal section spaces $\mathcal{H}_m(u)$ of Section 21.1 are contained in $H^0(X, L^m)$, the same upper bounds hold for them. Identifying weighted section spaces on a modification, however, requires a correction by the Jacobian; Lemma 1.22.10 gives the corresponding formula involving the relative canonical bundle.

22.6 Curvature rank and section growth

The transcendence degree of a section-ratio field can be bounded by a smooth curvature form, even when that form changes rank and has no global sign. The key is to arrange that a maximum of a section norm occurs where the section map has its maximal rank. This gives a pointwise restriction on the growth counted by the preceding sections.

Keep the connected compact complex manifold X and line bundle L, and choose a smooth Hermitian metric h on L. In a local frame write $|e|_h^2 = \mathrm{e}^{-\chi}$ and put $\beta = \mathrm{dd}^{\mathrm{c}}\chi$, as in Chapter 21. Let $r = \max_X \operatorname{rank} \beta$. No positivity of β, constant-rank condition or Kähler metric is assumed. Set $S_h = \{x : \beta_x \ge 0\}$; when S_h is nonempty, let $r_h^+ = \max_{S_h} \operatorname{rank} \beta$.

Proposition 22.6.1 *For a nonzero subspace $W \subset H^0(X, L^p)$, $p \ge 1$, let B_W be its common zero set and let $\Phi_W \colon X \setminus B_W \to \mathbb{P}(W^*)$ be its section map. If d is the maximal rank of its differential, there is $x_* \notin B_W$ such that*

$$\operatorname{rank} \mathrm{d}\Phi_W|_{x_*} = d, \qquad \beta_{x_*} \ge 0, \qquad \operatorname{rank} \beta_{x_*} \ge d.$$

Proof Choose a basis $s_0, \dots, s_N$ of W. Principalizing the ideal of their local coefficients gives a smooth compact modification $\pi\colon Y \to X$, isomorphic over $X \setminus B_W$, and an effective divisor D with

$$\pi^* s_j = s_D v_j, \qquad v_j \in H^0(Y, \pi^* L^p(-D)),$$

where the v_j have no common zero. The resolved map $f = [v_0 : \cdots : v_N]\colon Y \to \mathbb{P}^N$ has maximal differential rank d, since it agrees with $\Phi_W \circ \pi$ on a dense open set.

The critical set $C = \{y : \operatorname{rank} \mathrm{d}f_y < d\}$ is analytic. Proper mapping and the image-dimension theorem imply that $Z = f(Y)$ and $T = f(C)$ are compact analytic sets, with $\dim Z = d$ and $\dim T \le d - 1$ when T is nonempty. For the last inequality apply the rank bound to the smooth locus of every irreducible component of C. By Chow's theorem T is algebraic. Choose a homogeneous polynomial P of degree $a \ge 1$ that vanishes on T and does not vanish identically on Z. If T is empty, take $P = 1$ and $a = 0$, which also covers $d = 0$.

Then $\sigma = P(s_0, \dots, s_N)$ is a nonzero section of L^{pa}. Indeed, the set where $P(v_0, \dots, v_N) \neq 0$ meets the dense locus where π is an isomorphism. The smooth function

$$\Psi = \sum_j |\sigma s_j|^2_{h^{p(a+1)}}$$

has a positive maximum x_*. At that point $\sigma \neq 0$ and some $s_j \neq 0$, so $x_* \notin B_W$ and its lift is outside C. The tuples (σs_j) and (s_j) define the same projective map near x_*. The complex Hessian of $\log \Psi$ is seminegative at its maximum point x_*. With $\omega_{\mathrm{FS}} = \mathrm{dd^c} \log(1 + |w|^2)$ in an affine chart, this gives

$$0 \ge \mathrm{dd^c} \log \Psi|_{x_*} = \Phi_W^* \omega_{\mathrm{FS}}|_{x_*} - p(a+1)\beta_{x_*}.$$

The first form on the right is semipositive of rank d. Thus β_{x_*} is semipositive and its kernel is contained in the kernel of that form, proving the assertion. □

If some positive power of L has a nonzero section, the *Kodaira–Iitaka dimension* $\kappa(L)$ is the transcendence degree of the section-ratio field of the full section ring; otherwise put $\kappa(L) = -\infty$.

Corollary 22.6.1 *If S_h is empty, then $H^0(X, L^p) = 0$ for every $p \ge 1$. Otherwise*

$$\kappa(L) \le r_h^+ \le r, \qquad h^0(X, L^p) = O(p^{r_h^+}).$$

In this second case, every graded section algebra with a nonzero positive-degree piece has growth exponent at most r_h^+, and all its flag bodies, including those constructed on a point blowup, have dimension at most r_h^+. For every fixed weight $u \in \mathrm{QPSH}(X)$, the integrable spaces satisfy $\dim \mathcal{H}_p(u) = O(p^{r_h^+})$.

Proof The proposition applied to a single section excludes every positive power if S_h is empty. If all positive-degree section spaces vanish, the remaining section bounds are immediate and $\kappa(L) = -\infty$. Otherwise, suppose a graded algebra has a nonzero positive-degree piece. Write finitely many algebraically independent elements of its

ratio field with a common denominator in one positive degree, as in the proof of Proposition 22.3.1. They are ratios of coordinates of the section map of the span W of the numerators and the denominator. The image of the resolved section map is a projective variety, by proper mapping and Chow's theorem, and the given elements are pullbacks of rational functions on it; its dimension is therefore at least their number. This dimension is the maximal rank of $\mathrm{d}\Phi_W$, which the proposition bounds by r_h^+. Hence the transcendence degree of the ratio field is at most r_h^+.

Apply Theorem 22.5.1 to this algebra and to the full section ring. The degree-period formula gives the growth bound in every degree, including the zero spaces outside the period. By Theorem 22.3.1, the dimension of every flag body equals the growth exponent. Finally, $\mathcal{H}_p(u) \subseteq H^0(X, L^p)$, so the same upper bound holds even though the integrable spaces need not form a graded algebra. □

Lemma 22.6.1 *For every fixed holomorphic line bundle A on X,*

$$h^0(X, L^p \otimes A) = O(p^r).$$

Proof Induct on the dimension, simultaneously for all connected compact complex manifolds and all integer upper bounds on curvature rank. Dimension zero is immediate. If $H^0(X, L^p \otimes A) = 0$ for every $p \geq 1$, there is nothing to prove. Otherwise fix a nonzero section of $L^a \otimes A$ for some $a \geq 1$, and resolve its zero divisor on a smooth compact modification $\mu\colon Y \to X$. With $M = \mu^* L$, the pulled-back twist is $M^{-a}(D)$ for a fixed effective divisor D whose components are smooth. Pullback is injective on sections, and the pulled-back curvature has rank at most r.

Write $D = H_1 + \cdots + H_s$, allowing repeated components, and put $D_j = H_1 + \cdots + H_j$. For $m = p - a > 0$, the divisor sequences give

$$h^0(Y, M^m(D)) \leq h^0(Y, M^m) + \sum_{j=1}^{s} h^0\big(H_j, (M|_{H_j})^m \otimes \mathcal{O}_Y(D_j)|_{H_j}\big).$$

The first term is $O(m^r)$ by the preceding corollary; every other term is $O(m^r)$ by induction, since curvature rank cannot increase on restriction. The number of terms and their twists are fixed. Absorb the finitely many $p \leq a$ in the constant. □

Corollary 22.6.2 *For every fixed holomorphic vector bundle E on X,*

$$h^0(X, L^p \otimes E) = O(p^r), \qquad h^n(X, L^p \otimes E) = O(p^r).$$

Proof For E of positive rank, let $\pi\colon Y = \mathbb{P}(E^*) \to X$ be the bundle of lines in E^*, and let $\mathcal{O}_Y(1)$ be the dual of its tautological line bundle. Fiberwise linear forms give $\pi_* \mathcal{O}_Y(1) = E$, hence

$$H^0(X, L^p \otimes E) = H^0(Y, (\pi^* L)^p \otimes \mathcal{O}_Y(1)).$$

The pulled-back curvature has rank at most r, so the fixed-line-twist lemma, proved in all dimensions, gives the first bound. Serre duality identifies the dual of the top cohomology with

$$H^0(X, (L^*)^p \otimes E^* \otimes K_X).$$

The dual curvature is $-\beta$ and has the same rank; apply the bound for h^0 to L^*, with $E^* \otimes K_X$ in place of E. A bundle of rank zero has zero cohomology and satisfies both bounds. □

Proposition 22.6.2 *For every fixed holomorphic vector bundle E, the Euler characteristic $\chi(X, L^p \otimes E)$ is a polynomial in p of degree at most r.*

Proof The Hirzebruch–Riemann–Roch theorem, which holds on every compact complex manifold, gives

$$\chi(X, L^p \otimes E) = \int_X \mathrm{e}^{p\beta} \operatorname{ch}(E) \operatorname{td}(T^{1,0}X) = \sum_{j=0}^{r} \frac{p^j}{j!} \int_X \beta^j \wedge \operatorname{ch}(E) \wedge \operatorname{td}(T^{1,0}X).$$

Here the integral selects real degree $2n$, and the Chern character and Todd forms are computed with fixed Hermitian metrics. The curvature normalization represents $c_1(L)$, and $\beta^{r+1} = 0$ pointwise by its rank bound. Every coefficient is independent of p. □

Corollary 22.6.3 *If $\dim X = 2$, then $h^q(X, L^p \otimes E) = O(p^r)$ for every fixed holomorphic vector bundle E and $0 \le q \le 2$.*

Proof The endpoints have already been bounded. Since $\chi = h^0 - h^1 + h^2$, the middle dimension is at most $h^0 + h^2 + |\chi| = O(p^r)$ by the Euler polynomial. □

Example 22.6.1 Let $0 \le r \le n$, let Y be any connected compact complex manifold of dimension $n - r$, and take $X = \mathbb{P}^r \times Y$ and $L = \mathrm{pr}_1^* O(1)$ with the pulled-back Fubini–Study metric. Its curvature has rank r, and

$$h^0(X, L^p) = \binom{p + r}{r}.$$

Thus the section-growth exponent cannot be reduced, including when Y is non-Kähler.

Proof A section, expressed in a line-bundle frame on the projective factor, is a holomorphic function of Y and hence is constant there. Restriction to any slice $\mathbb{P}^r \times \{y_0\}$ is therefore inverse to pullback on sections. Counting degree-p homogeneous monomials in $r + 1$ variables gives the formula; for $r = 0$ it equals one. □

These bounds concern the closed curvature of a line bundle. They do not estimate the mass of a potential in an unrelated nonclosed Hermitian background, as Proposition 21.7.1 shows. In dimension at least three, the endpoint bounds and the Euler polynomial alone do not control the individual intermediate cohomology dimensions. The Hopf body computed next realizes the rank bound $n - 1$ for the pulled-back positive line bundles of Proposition 22.7.1.

22.7 The Hopf flag and its lower-dimensional body

On a scalar Hopf manifold, the coordinate subspaces give a complete flag ending at a point of an elliptic fiber. Along this flag, a homogeneous section reduces to a nowhere-vanishing section on the elliptic fiber, so the last coordinate of its valuation is zero; the remaining coordinates fill a simplex whose intrinsic volume gives the section growth.

Let X_n and L_k, with $n \geq 2$ and $k \geq 0$, be as in Section 21.6. In the covering coordinates $z_1, \dots, z_n$, the image of $\{z_1 = \cdots = z_{n-1} = 0\} \setminus \{0\}$ in X_n is an elliptic curve; choose a point p on it. Consider the flag $Y_0 = X_n$,

$$Y_i = \{z_1 = \cdots = z_i = 0\}/\langle \mathrm{e}^{-1} \rangle \quad (1 \leq i \leq n-1), \qquad Y_n = \{p\} \subset Y_{n-1},$$

where $/\langle \mathrm{e}^{-1} \rangle$ denotes the quotient of the punctured linear subspace by the contraction $z \mapsto \mathrm{e}^{-1} z$. Each member is smooth, compact and connected, and is a prime divisor in the preceding member. Take $W_m = H^0(X_n, L_k^m)$.

Proposition 22.7.1 *For this flag,*

$$\{\nu(s) : 0 \neq s \in W_m\} = \{(a_1, \dots, a_{n-1}, 0) : a_i \in \mathbb{N},\ \textstyle\sum_i a_i \leq km\}.$$

Consequently

$$\Delta(W_\bullet) = \{(x_1, \dots, x_{n-1}, 0) : x_i \geq 0,\ \textstyle\sum_i x_i \leq k\}.$$

Its n-dimensional volume is zero. When $k > 0$, its volume in the coordinate hyperplane $x_n = 0$, with the Euclidean measure for which the coordinate lattice has covolume one, is

$$\frac{k^{n-1}}{(n-1)!} = \lim_{m \to \infty} \frac{\dim W_m}{m^{n-1}}.$$

For $k = 0$ the body is a point and $\dim W_m = 1$.

Proof By Proposition 21.6.1, sections of L_k^m are homogeneous polynomials of degree km. Dividing successively by the highest powers of $z_1, \dots, z_{n-1}$ that divide the section, and restricting at each step, leaves a nonzero multiple of $z_n^{km - a_1 - \cdots - a_{n-1}}$, where a_i is the ith valuation. This residual section has no zeros on Y_{n-1}, so the last valuation is zero, and the preceding ones are nonnegative with sum at most km.

Conversely, every such tuple is the valuation of the monomial

$$z_1^{a_1} \cdots z_{n-1}^{a_{n-1}} z_n^{km - a_1 - \cdots - a_{n-1}}.$$

The normalized values lie in the displayed simplex and, for $m = 1$, include all its vertices, so their closed convex hull is the simplex. For $k > 0$, its coordinate-lattice volume is $k^{n-1}/(n-1)!$, the leading coefficient of the binomial count in Proposition 21.6.1. For $k = 0$, only the constant polynomials remain. □

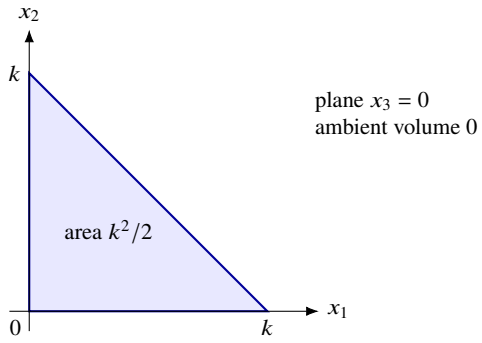


Fig. 22.1 For the Hopf threefold and $k > 0$, the body is a triangle in a coordinate plane. Its area gives the quadratic section growth, while its ambient volume vanishes.

Chapter 23
Hopf fibrations and toric potentials

Let $n \geq 2$ and let X_n be the scalar Hopf manifold of Section 21.6, which fibers over $\mathbb{P}^{n-1}$ with compact elliptic fibers. A potential in a pulled-back closed class is constant on each fiber and therefore descends to the base. For such a weight, the weighted section integrands also descend, and fiber integration identifies the section norms and Bergman kernels on the two spaces.

A potential of the standard Gauduchon form may vary along the fibers. For one that descends, the vertical part of this form supplies a single factor in the Hermitian Monge–Ampère measure, and fiber integration reduces the equation to the projective base. This measure identity is a special case of Lemma 9.6.1. For bounded descended potentials, closedness on the base determines the normalizing constant and gives uniqueness up to additive constants. On the Hopf threefold, the descended masses are at most six, so unbounded masses require variation along the fibers.

On the base, invariance under separate rotations of the homogeneous coordinates gives a real convex description in logarithmic coordinates. Strict positivity of the curvature makes the gradient a parametrization of the relative interior of a simplex.

The projection $\pi\colon X_n \to \mathbb{P}^{n-1}$, the forms α, ζ, ω and the probability volume form $\mathrm{d}V = \omega^n/(2n)$ are as in Section 21.6. Write σ for the Fubini–Study form on $\mathbb{P}^{n-1}$ normalized by $\int_{\mathbb{P}^{n-1}} \sigma^{n-1} = 1$; it is the form ω_{FS} of Section 9.6. We write $\mathrm{MA}^{\mathrm{ext}}$ for the extended non-pluripolar Monge–Ampère measure of Theorem 7.1.1. The fiber calculations use the following identities, where $y \in \mathbb{P}^{n-1}$:

$$\alpha = \pi^*\sigma, \qquad \omega = \alpha + \zeta, \qquad \zeta^2 = 0, \qquad \mathrm{d}V = \tfrac{1}{2}\alpha^{n-1} \wedge \zeta, \qquad \int_{\pi^{-1}(y)} \zeta = 2.$$

23.1 Weighted sections on the projective base

For an integer $k \geq 0$, the line bundle L_k of Section 21.6, with its metric h_k, is the pullback of $\mathcal{O}(k)$ with its Fubini–Study metric. By the proof of Proposition 21.6.1, sections of L_k^m are represented by homogeneous polynomials f of degree km in the

covering coordinates $z \in \mathbb{C}^n \setminus \{0\}$, exactly as sections of $\mathcal{O}(km)$ on $\mathbb{P}^{n-1}$, and the pointwise squared norm $|f(z)|^2|z|^{-2km}$ is invariant under every nonzero complex scalar. If the weight also descends, the weighted integrand is a function on the base, and fiber integration identifies the norms and Bergman kernels of Section 21.1 on the two spaces.

Theorem 23.1.1 *Let $k \geq 0$ be an integer and $u \in \mathrm{PSH}(X_n, k\alpha)$. Then $u = \pi^* v$ for a unique $v \in \mathrm{PSH}(\mathbb{P}^{n-1}, k\sigma)$. For every positive integer m, pullback identifies the weighted section spaces isometrically: for a section s in the space on the base,*

$$H^0\big(\mathbb{P}^{n-1}, \mathcal{O}(km) \otimes \mathcal{I}(mv)\big) \simeq H^0\big(X_n, L_k^m \otimes \mathcal{I}(mu)\big),$$
$$\|\pi^* s\|^2_{X_n,m,u} = \|s\|^2_{\mathbb{P}^{n-1},m,v}.$$

Here the norm on the base is defined with the Fubini–Study metric and the volume form σ^{n-1}. The kernels K and, off their pole sets, B on X_n are the pullbacks of the corresponding kernels on the base. Pullback also identifies the bounded weighted section algebras $\mathcal{W}_\bullet(v)$ and $\mathcal{W}_\bullet(u)$.

Proof The descent of the potential follows from Proposition 1.22.1. By the proof of Proposition 21.6.1, the unweighted section spaces on both manifolds consist of the same homogeneous polynomials of degree km, and the pointwise norms of corresponding sections agree because the metrics correspond. For every nonnegative Borel function F on the base, fiber integration gives

$$\int_{X_n} (F \circ \pi)\, \mathrm{d}V = \frac{1}{2} \int_{\mathbb{P}^{n-1}} F\sigma^{n-1} \int_{\pi^{-1}(y)} \zeta = \int_{\mathbb{P}^{n-1}} F\sigma^{n-1}.$$

Applied to $F = |s|^2 \mathrm{e}^{-mv}$, with arbitrary values on the pole set, this formula gives equality of the two possibly infinite integrals. Since finiteness of the weighted norm characterizes the multiplier-ideal section spaces, this proves the asserted isometry. An orthonormal basis therefore pulls back to an orthonormal basis, which proves the kernel identities. Finally, a nonnegative function on the base is bounded if and only if its pullback is bounded, so pullback also identifies the bounded weighted sections, compatibly with multiplication. □

23.2 Descended measures and equations

The pulled-back curvature of a descended potential has no component in the fiber direction; the form ζ supplies it. Since $\zeta^2 = 0$, the Hermitian top power of a descended potential consists of a single mixed term, and fiber integration of this term determines the mass and reduces the Monge–Ampère equation for descended potentials to the base.

For a positive Borel measure ν on $\mathbb{P}^{n-1}$, we write $\pi^*\nu \wedge \zeta$ for the measure on X_n defined, on nonnegative Borel functions F, by

$$\int_{X_n} F\,\pi^*\nu \wedge \zeta = \int_{\mathbb{P}^{n-1}} \left(\int_{\pi^{-1}(y)} F\zeta \right) \mathrm{d}\nu(y).$$

When ν is given by a smooth top-degree form, this agrees with the pullback of the form wedged with ζ. It is the measure $\beta_n \wedge p_n^*\nu$ of Lemma 9.6.1(3). With S a point and $a = b = 1$, parts (1) and (3) of that lemma say that, for a quasi-plurisubharmonic function v on $\mathbb{P}^{n-1}$, the pullback $u = \pi^*v$ belongs to $\mathrm{PSH}(X_n, \omega)$ if and only if $v \in \mathrm{PSH}(\mathbb{P}^{n-1}, \sigma)$, and that then

$$\mathrm{MA}_\omega^{\mathrm{ext}}(u) = n\,\pi^*\mathrm{MA}_\sigma^{\mathrm{ext}}(v) \wedge \zeta, \qquad \int_{X_n} \mathrm{MA}_\omega^{\mathrm{ext}}(u) = 2n \int_{\mathbb{P}^{n-1}} \mathrm{MA}_\sigma^{\mathrm{ext}}(v) \le 2n,$$

with equality for bounded v. With $a = k$ and $b = 0$, part (3) gives $\mathrm{MA}_{k\alpha}^{\mathrm{ext}}(\pi^*v) = 0$ for every integer $k \ge 0$ and $v \in \mathrm{PSH}(\mathbb{P}^{n-1}, k\sigma)$.

If an ω-plurisubharmonic potential is constant on the fibers, it descends to a quasi-plurisubharmonic function on the base by Lemma 9.6.1(2), so its Hermitian mass is at most $2n$. On the Hopf threefold this bound is six, so the unbounded masses of Chapter 9 require potentials that vary along the fibers.

For the equation, let $p > 1$ and let $f \ge 0$ belong to $L^p(\mathbb{P}^{n-1}, \sigma^{n-1})$, with $\int f\sigma^{n-1} > 0$. For potentials $u = \pi^*v$ with bounded $v \in \mathrm{PSH}(\mathbb{P}^{n-1}, \sigma)$, the equation

$$(\omega + \mathrm{dd^c}u)^n = c(f \circ \pi)\,\mathrm{d}V, \qquad c > 0, \tag{23.1}$$

is equivalent to the base equation

$$(\sigma + \mathrm{dd^c}v)^{n-1} = \frac{c}{2n} f\sigma^{n-1}.$$

Indeed, the base equation implies (23.1) by the measure formula above and the formula for $\mathrm{d}V$. Conversely, pushing (23.1) forward to the base gives $2n(\sigma + \mathrm{dd^c}v)^{n-1} = cf\sigma^{n-1}$.

Integrating the base equation determines the constant:

$$c = \frac{2n}{\int_{\mathbb{P}^{n-1}} f\sigma^{n-1}}.$$

Since σ is closed, it satisfies the Guan–Li condition (12.35). Applied to $(\mathbb{P}^{n-1}, \sigma)$, Theorem 12.5.1 gives a continuous solution of the base equation with this constant, unique up to additive constants among bounded σ-plurisubharmonic functions. Consequently (23.1) has a continuous solution, unique up to additive constants among bounded descended potentials. For smooth positive f, Theorem 10.1.1 on the base gives a smooth solution v, and the constant is again determined by Stokes' theorem.

For a real parameter λ, the same calculation shows that the equation with right-hand side $ce^{\lambda u}(f \circ \pi)\mathrm{d}V$ is equivalent to the base equation with right-hand side $(c/(2n))\mathrm{e}^{\lambda v} f \sigma^{n-1}$.

23.3 Logarithmic coordinates and the gradient simplex

For potentials invariant under rotations of the individual coordinates, the complex Hessian becomes the real Hessian of a function of the logarithmic coordinates. Invariance under complex scaling gives this Hessian a null direction; transverse to it, positivity becomes convexity, and boundedness of the potential confines the gradient to a simplex. Logarithmic growth makes every interior point of the simplex a gradient value, and strict positivity on the base makes the gradient injective modulo the null direction.

In this section, $k > 0$ is a real number. Let v be a smooth real function on $\mathbb{P}^{n-1}$ that is invariant under the torus action

$$[z_1 : \cdots : z_n] \longmapsto [\mathrm{e}^{\mathrm{i}t_1} z_1 : \cdots : \mathrm{e}^{\mathrm{i}t_n} z_n], \qquad t \in \mathbb{R}^n,$$

and put $u = \pi^* v$. On the locus where all covering coordinates z_i are nonzero, set $x_i = \log |z_i|^2$. By this invariance, u is a function of $x = (x_1, \ldots, x_n)$, which we denote by $u(x)$, and we set

$$F(x) = k \log \left(\sum_{i=1}^{n} \mathrm{e}^{x_i} \right) + u(x).$$

Write $\nabla F = (F_i)$ and $D^2 F = (F_{ij})$ for its Euclidean gradient and Hessian, and let $\mathbf{1} = (1, \ldots, 1)$. Since u is invariant under complex scaling, for every $t \in \mathbb{R}$,

$$F(x + t\mathbf{1}) = F(x) + kt. \tag{23.2}$$

Proposition 23.3.1 *On this locus, $k\alpha + \mathrm{dd}^{\mathrm{c}} u \geq 0$ if and only if F is convex. If F is convex, its gradient lies in*

$$\Sigma_k = \{p \in \mathbb{R}^n : p_i \geq 0,\ \textstyle\sum_i p_i = k\},$$

and $D^2 F\, \mathbf{1} = 0$. If $k\sigma + \mathrm{dd}^{\mathrm{c}} v$ is positive definite on $\mathbb{P}^{n-1}$, the gradient induces a bijection from $\mathbb{R}^n / \mathbb{R}\mathbf{1}$ onto the relative interior of Σ_k.

Proof On the covering space, $k\alpha + \mathrm{dd^c} u = \mathrm{dd^c} F$. In the coordinates z_i, the Hermitian coefficient matrix of this form is a positive constant times

$$\left(\frac{F_{ij}}{z_i \overline{z}_j} \right)_{i,j} .$$

This matrix is congruent to the real Hessian (F_{ij}) through the invertible diagonal matrix with entries $1/z_i$, which proves the first equivalence. Differentiating (23.2) gives $\sum_i F_i = k$ and $D^2 F\, \mathbf{1} = 0$.

Since v is bounded on the compact base, there is a constant C such that

$$\left| F(x) - k \log \sum_i \mathrm{e}^{x_i} \right| \le C.$$

Fix x and a coordinate vector e_i. Since $n \ge 2$, the convex function $t \mapsto F(x + te_i)$ is bounded above as $t \to -\infty$. Its derivative cannot be negative anywhere: otherwise the supporting line at such a point would tend to $+\infty$ as $t \to -\infty$. Thus $F_i \ge 0$, which proves $\nabla F \in \Sigma_k$.

Now assume that $k\sigma + \mathrm{dd^c} v$ is positive definite. The kernel of the differential of the projection $\mathbb{C}^n \setminus \{0\} \to \mathbb{P}^{n-1}$ is exactly the complex scalar direction, so the Hessian of F is positive definite transverse to $\mathbf{1}$. In particular F is strictly convex on $H = \{x : \sum_i x_i = 0\}$. Along each line $x + te_i$, the derivative of F is strictly increasing and nonnegative everywhere, hence positive. Thus every gradient lies in the relative interior of Σ_k.

Conversely, take $p \in \mathbb{R}^n$ with $p_i > 0$ and $\sum_i p_i = k$. The function $F(x) - p \cdot x$ is invariant under translation along $\mathbf{1}$, and we claim that it tends to $+\infty$ on H as $|x| \to \infty$. Indeed,

$$k \max_i x_i - p \cdot x = \sum_i p_i (\max_j x_j - x_i)$$

is strictly positive at every nonzero vector of H, and hence has a positive minimum on the unit sphere of H. Since F differs from $k \log \sum_i \mathrm{e}^{x_i}$ by a bounded function and $\log \sum_i \mathrm{e}^{x_i} \ge \max_i x_i$, the claim follows. Hence $F(x) - p \cdot x$ has a unique minimizer on H. At this point $\nabla F - p$ is a multiple of $\mathbf{1}$; both vectors have coordinate sum k, so the multiple is zero. This proves surjectivity, and strict convexity on H proves injectivity. □

Chapter 24
Divisorial currents on compact modifications

Keep the setting of Chapter 20: $Y \subset X$ is a connected compact complex submanifold of dimension $r \geq 1$, θ is a smooth real $(1,1)$-form and ω a Hermitian form on X, and $u \in \mathrm{PSH}(X,\theta)$ has $\nu(u,Y) = 0$.

The divisorial trace data of u along Y (Chapter 20) assign a number to every prime divisor on every smooth compact modification of Y. A single model sees only part of these data, since blowing up can reveal vanishing that had higher codimension on Y. Summing the coefficients visible on one model gives a closed positive current, whose mass is controlled by a Gauduchon degree. These currents agree under pushforward, whereas on a higher model the trace current exceeds the pullback of the trace current from a lower model by an effective exceptional divisor. The trace envelope, which imposes the trace inequalities on all models at once, pulls back exactly. This discrepancy explains why the ordinary singularity type of the divisorial part on one model need not realize the full trace data.

The divisorial currents are closed even when the background is not. In a closed restricted background, subtracting them gives residual Bott–Chern classes that are compatible between models. To identify the exceptional correction for an infinite divisorial sum, we need pullback to be continuous on closed positive currents; this continuity comes from choosing, in each Bott–Chern class, the representative of constant Gauduchon trace. Residual classes with a nonzero prescribed exterior derivative belong to the affine spaces of Chapter 25.

Let u_m and ϵ_m, defined for all sufficiently large integers m, be the analytic approximants and curvature errors of Corollary 1.15.1: u_m has analytic type $(1/m, \mathcal{I}(mu))$, and $\theta + \mathrm{dd}^{\mathrm{c}} u_m \geq -\epsilon_m \omega$ with $\epsilon_m \to 0$. For a smooth compact modification $\pi\colon Z \to Y$, also called a *model* of Y, put

$$\theta_Z = \pi^*(\theta|_Y), \qquad \omega_Z^0 = \pi^*(\omega|_Y), \qquad b_m(E) = \mathrm{ord}_E\big(\mathcal{I}(mu)\mathcal{O}_Y\big) \quad (m \in \mathbb{Z}_{\geq 1}).$$

Here E is a prime divisor on Z, $[E]$ denotes its integration current, and $b_m(E)$ is the vanishing order along E of the image ideal $\mathfrak{a}_m(u;Y) = \mathcal{I}(mu)\mathcal{O}_Y$ pulled back to Z. By Theorem 20.1.1, the trace numbers satisfy

$$\frac{b_m(E)}{m} \longrightarrow \tau_{Y,u}(E), \qquad \tau_{Y,u}(E) = \sup_{m\geq 1} \frac{b_m(E)}{m}.$$

By Corollary 1.22.2, the restrictions $u_m|_Y$ are not identically $-\infty$, and the curvature bound pulls back to $\mathrm{dd^c}((u_m|_Y)\circ\pi) \geq -\theta_Z - \epsilon_m\omega_Z^0$.

24.1 Divisorial trace currents

On a model Z, the curvature current of the pulled-back analytic approximant dominates the divisor of the image ideal of $\mathcal{I}(mu)$. Pairing with a Gauduchon metric removes the potential from this inequality, and the resulting bound passes to the limit in each coefficient and controls the sum of all trace coefficients. For a Gauduchon metric γ on Z, we call $\int_Z T\wedge\gamma^{r-1}$ the *trace mass* of a positive $(1,1)$-current T; when $r=1$, $\gamma^{r-1}=1$ and γ may be any Hermitian metric.

For a positive integer m, with E ranging over the prime divisors on Z, define the finite divisorial current

$$D_{m,Z} = \frac{1}{m}\sum_E b_m(E)[E].$$

The sum is finite: the pulled-back image ideal is a nonzero coherent ideal on the compact manifold Z, so its zero set has only finitely many components of codimension one. We also consider the possibly infinite sum

$$S_Z(u;Y) = \sum_E \tau_{Y,u}(E)[E],$$

which a priori is defined only through its finite subsums.

Theorem 24.1.1 *The sum $S_Z(u;Y)$ defines a closed positive $(1,1)$-current of finite mass on Z. For every Gauduchon metric γ on Z,*

$$\int_Z S_Z(u;Y)\wedge\gamma^{r-1} \leq \int_Z \theta_Z\wedge\gamma^{r-1}, \tag{24.1}$$

where $\gamma^{r-1}=1$ if $r=1$. Moreover,

$$0 \leq D_{m,Z} \leq S_Z(u;Y), \qquad \int_Z \big(S_Z(u;Y) - D_{m,Z}\big)\wedge\gamma^{r-1} \longrightarrow 0.$$

In particular, $D_{m,Z}$ converges weakly to $S_Z(u;Y)$ as currents, and the dyadic sequence $D_{2^j,Z}$ is increasing.

Proof Set $w_m = (u_m|_Y)\circ\pi$. The restriction $u_m|_Y$ has analytic type $(1/m, \mathfrak{a}_m(u;Y))$ by Proposition 1.13.1, as in the proof of Corollary 1.22.2, so its pullback w_m has generic Lelong number $b_m(E)/m$ along E. Thus

$$T_m = \theta_Z + \epsilon_m\omega_Z^0 + \mathrm{dd^c}w_m \geq 0, \qquad \nu(w_m,E) = b_m(E)/m.$$

Lemma 1.17.1 gives $D_{m,Z} \leq T_m$. Integration by parts for currents and the Gauduchon condition therefore give

$$\int_Z D_{m,Z} \wedge \gamma^{r-1} \leq \int_Z \theta_Z \wedge \gamma^{r-1} + \epsilon_m \int_Z \omega_Z^0 \wedge \gamma^{r-1}.$$

For any fixed finite set of prime divisors, let $m \to \infty$ in the coefficients of these divisors, using Theorem 20.1.1. This proves (24.1) for every finite subsum of $S_Z(u;Y)$.

Every prime divisor has strictly positive γ-area. Hence at most countably many coefficients are positive: for each positive integer k, only finitely many terms can have trace mass at least $1/k$, and every positive term is among these for some k. Enumerate the corresponding divisors. The finite partial sums are increasing positive closed currents with uniformly bounded trace mass. They therefore converge, in the sense of coefficient measures and weakly as currents, to a positive current of finite mass, which is closed because differentiation is continuous on currents. This constructs $S_Z(u;Y)$ and proves the integral inequality.

The supremum formula for $\tau_{Y,u}(E)$ gives $b_m(E)/m \leq \tau_{Y,u}(E)$ for every m and E, and the limit along the integers gives coefficientwise convergence. Dominated convergence for the countable series with summable weights $\tau_{Y,u}(E) \int_E \gamma^{r-1}$ shows that the trace mass of $S_Z - D_{m,Z}$ tends to zero. Since a fixed smooth test form is bounded in absolute value, on positive $(1,1)$-currents, by a constant times the trace mass, this gives weak convergence. Finally, the subadditivity inclusion in (1.23) gives $\mathcal{I}(2mu) \subseteq \mathcal{I}(mu)^2$, hence $b_{2m}(E) \geq 2b_m(E)$ and $D_{2m,Z} \geq D_{m,Z}$. □

24.2 Compatibility between models

Under a further modification, a prime divisor and its strict transform define the same vanishing order on the image ideals, while exceptional divisors disappear under pushforward. Hence the trace currents are compatible with pushforward. Under pullback, exceptional divisors can carry additional vanishing, and divisibility of the pulled-back ideals shows that this additional vanishing is nonnegative.

Theorem 24.2.1 *Let $g\colon Z' \to Z$ be a smooth compact modification, and regard Z' as a model over Y through $\pi \circ g$. Then*

$$g_* S_{Z'}(u;Y) = S_Z(u;Y).$$

The current $S_Z(u;Y)$ is additive and positively homogeneous in u. For every finite set $\mathcal{F}$ of prime divisors on Z, the finite effective real divisor

$$A_{\mathcal{F}} = \sum_{E \in \mathcal{F}} \tau_{Y,u}(E) E$$

satisfies

$$[g^* A_{\mathcal{F}}] \le S_{Z'}(u;Y).$$

Proof Every prime divisor on Z' is either the strict transform of a prime divisor on Z or exceptional for g. If E' is the strict transform of a prime divisor $E \subset Z$, then g is biholomorphic near generic points of E'. Hence E' and E define the same vanishing order on the image ideals from Y, so $\tau_{Y,u}(E') = \tau_{Y,u}(E)$, and $g_*[E'] = [E]$.

The image of an exceptional prime divisor F has codimension at least two. The pullback of a smooth $(r-1, r-1)$ test form therefore restricts to zero on the regular locus of F, where the restriction of g has complex rank at most $r-2$; thus $g_*[F] = 0$. In dimension one, a modification between smooth compact curves is an isomorphism, and there are no exceptional divisors. Pushforward under the proper map g is continuous on currents. Applying it to the increasing finite partial sums, or equivalently testing their summable positive coefficient measures, proves the displayed compatibility. Additivity and homogeneity follow coefficientwise from Proposition 20.1.1.

For the inequality, fix a positive integer m and let $\mathfrak{a}_{m,Z} = \mathcal{I}(mu)\mathcal{O}_Z$ denote the image ideal on Z. Its divisorial vanishing gives

$$\mathfrak{a}_{m,Z} \subseteq \mathcal{O}_Z\left(-\sum_{E \in \mathcal{F}} b_m(E) E\right).$$

Indeed, on a manifold, a holomorphic function vanishing to the indicated generic orders is divisible by the corresponding powers of the local equations of the divisors. Pulling back this inclusion and taking the order along a prime divisor $F \subset Z'$ gives

$$b_m(F) \ge \sum_{E \in \mathcal{F}} b_m(E) \operatorname{ord}_F(g^* E). \tag{24.2}$$

Divide by m and let $m \to \infty$; the sum is finite. Thus every coefficient of $g^* A_{\mathcal{F}}$ is at most the corresponding coefficient $\tau_{Y,u}(F)$ of $S_{Z'}(u;Y)$. Since both sides are positive divisorial currents, this proves the asserted inequality of currents. □

24.3 The trace envelope as a common majorant

The trace envelope of Section 20.6 imposes the trace inequalities along all divisors over Y, including those that do not appear on a given model. It therefore gives a single potential on Y, in the original background $\theta|_Y$, whose pulled-back curvature current dominates the trace current on every model.

Proposition 24.3.1 *The trace envelope $v = \mathcal{T}_{\theta|_Y}(u;Y)$ of (20.5) satisfies, for every smooth compact modification $\pi\colon Z \to Y$,*

$$\pi^*(\theta|_Y) + \mathrm{dd^c}(v \circ \pi) \geq S_Z(u;Y).$$

Proof By Theorem 20.6.1, applied with $\Omega = \theta|_Y$, $h = \omega|_Y$ and $\delta_m = \epsilon_m$, the envelope v is defined, belongs to $\mathrm{PSH}(Y, \theta|_Y)$ and satisfies $\nu(v \circ \pi, E) \geq \tau_{Y,u}(E)$ for every prime divisor $E \subset Z$. On a fixed model Z, Lemma 1.17.1 therefore allows us to subtract every finite partial sum of $S_Z(u;Y)$ from $\theta_Z + \mathrm{dd^c}(v \circ \pi)$ while preserving positivity. Passing to the limit of these partial sums, constructed in Theorem 24.1.1, preserves positivity and proves the assertion. □

This potential is canonical: by Theorem 20.6.2, $v \circ \pi$ is the trace envelope on Z in the background θ_Z for the same trace data. Its generic Lelong numbers may nevertheless exceed the prescribed trace numbers. Exact realization of the trace data requires the reverse inequality, which Chapter 20 establishes for analytic singularities, for Kähler Y in sufficiently large Hermitian backgrounds, and for pullbacks from curves. By contrast, the ordinary singularity type attached to S_Z by Corollary 1.17.1 records only the divisorial part on the particular model Z.

24.4 Bott–Chern classes and closed currents

Bott–Chern cohomology records a closed $(1,1)$-current modulo $\mathrm{dd^c}$ of global functions, and so separates the smooth background of the current from its potential. When the restricted background $\theta|_Y$ is closed, subtracting the trace current from the pulled-back background gives a residual class on each model, and compatibility with pushforward organizes these classes into a b-divisor class in the sense of [Xia26, Section 11.1].

Recall from Lemma 1.17.4 that the Bott–Chern group $H^{1,1}_{\mathrm{BC}}(Z,\mathbb{R})$ of a complex manifold Z, the space of smooth closed real $(1,1)$-forms modulo $\mathrm{dd^c}$ of smooth real functions, is also the quotient of closed real $(1,1)$-currents by $\mathrm{dd^c}$ of real distributions.

From now on, Z is connected and compact. Thus every closed real $(1,1)$-current T has a Bott–Chern class, written $\{T\}_{\mathrm{BC}}$; the class is *pseudoeffective* if it contains a closed positive current. Pullback of classes is defined on smooth representatives. Pushforward of classes under a modification is defined on current representatives; it is well defined because pushforward commutes with $\mathrm{dd^c}$, and by Lemma 1.17.4 the resulting class is again a smooth Bott–Chern class. For a modification g and

every smooth real $(1,1)$-form β on its target, $g_*g^*\beta = \beta$ as currents, by the change of variables formula for the degree-one map g.

A Gauduchon metric γ defines a linear functional on this cohomology:

$$\ell_\gamma(\{T\}_{\mathrm{BC}}) = \int_Z T \wedge \gamma^{\dim Z - 1}.$$

Integration by parts for currents shows that it does not depend on the representative. On a positive representative it is nonnegative, and it vanishes only if that representative is zero, by the strict positivity of the metric. In particular, the pseudoeffective cone contains no line: if $T_1, T_2 \geq 0$ represent opposite classes, their sum has zero Gauduchon pairing, hence $T_1 = T_2 = 0$.

Gauduchon pairings detect closed curvature information that is invisible in de Rham cohomology. Consider the Hopf manifold X_n, $n \geq 2$, with its Gauduchon form ω and covering coordinates $z \in \mathbb{C}^n \setminus \{0\}$, as in Section 21.6. The form $\alpha = \mathrm{dd^c} \log |z|^2$ is exact as a de Rham form, because $\mathrm{d^c} \log |z|^2$ descends to the quotient: the logarithm changes only by a constant under the contraction. Nevertheless, by the calculation in Proposition 21.6.1,

$$\ell_\omega(\{\alpha\}_{\mathrm{BC}}) = \int_{X_n} \alpha \wedge \omega^{n-1} = 2(n-1) > 0,$$

so the Bott–Chern class of α is nonzero.

We return to the smooth compact modifications of the manifold Y fixed at the beginning of the chapter. Two such modifications $\pi\colon Z \to Y$ and $\pi'\colon Z' \to Y$ are compared through modifications $g\colon Z' \to Z$ with $\pi' = \pi \circ g$; we then say that Z' dominates Z.

Definition 24.4.1 A *Bott–Chern b-divisor class* over Y is a family

$$\mathbb{A} = (\mathbb{A}_Z)_{\pi\colon Z \to Y}, \qquad \mathbb{A}_Z \in H^{1,1}_{\mathrm{BC}}(Z, \mathbb{R}),$$

indexed by the smooth compact modifications of Y and compatible with pushforward: $g_*\mathbb{A}_{Z'} = \mathbb{A}_Z$ whenever $g\colon Z' \to Z$ is a modification over Y. It is *pseudoeffective* if every $\mathbb{A}_Z$ is pseudoeffective. It is *Cartier* if there are a model Z and a class $a_Z \in H^{1,1}_{\mathrm{BC}}(Z, \mathbb{R})$ such that $\mathbb{A}_{Z'} = g^*a_Z$ for every model $g\colon Z' \to Z$ dominating Z; the pair (Z, a_Z) is then called a *Cartier realization* of $\mathbb{A}$.

When the models are Kähler, the $\partial\bar\partial$-lemma identifies this cohomology with the real $(1,1)$-cohomology used in [Xia26, Section 11.1], and the definition agrees with the one given there.

We now return to the trace data fixed at the beginning of the chapter. When the restricted background $\theta|_Y$ is closed, the trace envelope of Proposition 24.3.1 provides positive representatives of the residual classes.

Proposition 24.4.1 *If the restricted background $\theta|_Y$ is closed (for instance, if θ is closed), the classes*

$$\mathbb{A}_Z(u;Y,\theta) = \{\pi^*(\theta|_Y) - S_Z(u;Y)\}_{\mathrm{BC}}$$

form a pseudoeffective Bott–Chern b-divisor class. They depend only on the ambient multiplier-ideal class of u and the Bott–Chern class of $\theta|_Y$.

Proof By Proposition 24.3.1, the trace envelope $v = \mathcal{T}_{\theta|_Y}(u;Y)$ satisfies

$$R_Z = \pi^*(\theta|_Y) + \mathrm{dd^c}(v \circ \pi) - S_Z(u;Y) \geq 0$$

on every model. Since $\theta|_Y$ and S_Z are closed, R_Z is closed, and its class is the class in the statement. This proves pseudoeffectivity. Compatibility with pushforward follows from Theorem 24.2.1 together with $g_*g^*(\theta_Z) = \theta_Z$. Finally, the trace data do not change under ambient multiplier-ideal equivalence, and replacing $\theta|_Y$ by $\theta|_Y + \mathrm{dd^c} f$ with f smooth changes no class; this proves the last assertion. □

If $\theta|_Y$ is not closed, the same expression R_Z is still a positive current, but

$$\mathrm{d}R_Z = \pi^*(\mathrm{d}\theta|_Y).$$

It is therefore not closed in general and defines no Bott–Chern class. Classes whose representatives have a prescribed nonzero exterior derivative are treated in Chapter 25.

24.5 The b-divisor class of a closed positive current

For a closed positive current, the construction can be carried out with its own potential in place of the restricted ambient ideals. The trace coefficients then become the generic Lelong numbers of the pulled-back potential along divisors, and divisorial removal gives the positive residual currents directly.

In this section, T is a closed positive $(1,1)$-current on Y, written $T = \beta + \mathrm{dd^c} w$ as in Lemma 1.17.4. On a smooth compact modification $\pi\colon Z \to Y$, the pullback π^*T is the positive current defined by composing local plurisubharmonic potentials with π; since π is surjective, these compositions are not identically $-\infty$. With E ranging over the prime divisors on Z, set

$$N_Z(T) = \sum_E \nu(w \circ \pi, E)[E], \qquad \mathrm{Reg}_Z(T) = \pi^*T - N_Z(T).$$

Divisorial removal (Lemma 1.17.1) bounds every finite partial sum of $N_Z(T)$ by π^*T. Hence these partial sums converge to a closed positive current, and $\mathrm{Reg}_Z(T)$ is positive and closed. The following lemma identifies the coefficients of $N_Z(T)$ with trace numbers when the ambient manifold is Y itself.

Lemma 24.5.1 *Let w be a quasi-plurisubharmonic function on Y, and take the ambient manifold in (20.1) to be Y itself. For every prime divisor E on a smooth compact modification $\pi\colon Z \to Y$, the resulting self-trace satisfies*

$$\tau_{Y,w}(E) = \nu(w \circ \pi, E).$$

In particular, for the potential w of the closed positive current T, one has $N_Z(T) = S_Z(w; Y)$.

Proof Since w is not identically $-\infty$, its Lelong number vanishes at almost every point; hence $\nu(w, Y) = 0$, and the self-trace is defined. Fix a divisor E on a compact modification and apply Lemma 1.17.8 on a coordinate neighborhood of a point of its image in Y. With $\nu_E = \nu(w \circ \pi, E)$ and A_E the log discrepancy of E, it gives

$$\max\{\nu_E - A_E/m, 0\} \le \frac{1}{m} \operatorname{ord}_E \mathcal{I}(mw) \le \nu_E.$$

The limit along the integers defining $\tau_{Y,w}(E)$ is therefore ν_E. Since all coefficients agree, the two divisorial currents are equal. □

Theorem 24.5.1 *The family*

$$\mathbb{D}(T)_Z = \{\operatorname{Reg}_Z(T)\}_{\mathrm{BC}}$$

is a pseudoeffective Bott–Chern b-divisor class. For closed positive currents T_1, T_2 and $a \ge 0$,

$$\mathbb{D}(T_1 + T_2) = \mathbb{D}(T_1) + \mathbb{D}(T_2), \qquad \mathbb{D}(aT_1) = a\mathbb{D}(T_1).$$

On a Kähler manifold, with Kähler models, it agrees with the b-divisor class of the same current in [Xia26, Section 11.1.2].

Proof By the lemma, $N_Z(T) = S_Z(w; Y)$, so Theorem 24.2.1 gives $g_* N_{Z'}(T) = N_Z(T)$ for every modification $g\colon Z' \to Z$ over Y. Moreover, $g_* g^*(\pi^* T) = \pi^* T$. To see this, write the current as a smooth form plus dd^{c} of a potential; the change of variables formula for the degree-one map g applies both to the smooth form and to the locally integrable pulled-back potential, and pushforward commutes with dd^{c}. Therefore

$$g_* \operatorname{Reg}_{Z'}(T) = \operatorname{Reg}_Z(T).$$

Taking classes proves compatibility, and positivity of the regular parts proves pseudoeffectivity. Generic Lelong numbers are additive under addition of positive currents and scale under multiplication by nonnegative constants. Hence so are the divisorial parts and the regular parts, which proves the two identities. In the Kähler case, the regular parts and, by the $\partial\bar\partial$-lemma, their cohomology classes are exactly those of the construction cited in the statement. □

24.6 Nef Cartier realizations

If T has analytic singularities, a log resolution splits its pullback into a finite divisor and a current with bounded potentials. The bounded part acquires no divisorial component on higher models, so its class determines the whole b-divisor class there, and smooth regularization of its bounded potential shows that this class is nef.

A class $a \in H^{1,1}_{\mathrm{BC}}(Z,\mathbb{R})$ is *nef* if, for one Hermitian metric γ, it has for every $\epsilon > 0$ a smooth representative $\beta_\epsilon \geq -\epsilon\gamma$. Since any two Hermitian metrics are comparable, this does not depend on the metric. A Cartier b-divisor class is called *nef* if it has a Cartier realization (Z, a_Z) with a_Z nef.

Lemma 24.6.1 *On a connected compact complex manifold Z, let β be a smooth real $(1,1)$-form and v a bounded quasi-plurisubharmonic function. If $R = \beta + \mathrm{dd}^{\mathrm{c}} v \geq 0$ is closed, then $\{R\}_{\mathrm{BC}}$ is nef.*

Proof Since R is closed, so is β. Fix a Hermitian metric γ. By Proposition 1.10.1, there are smooth functions v_j and numbers $\epsilon_j \downarrow 0$ such that

$$\beta + \mathrm{dd}^{\mathrm{c}} v_j \geq -\epsilon_j \gamma.$$

These forms are closed representatives of $\{R\}_{\mathrm{BC}}$. Given $\epsilon > 0$, choosing j with $\epsilon_j \leq \epsilon$ gives the estimate required in the definition of nefness. □

Theorem 24.6.1 *If the closed positive current T has analytic singularities, then $\mathbb{D}(T)$ is Cartier and has a nef Cartier realization on a smooth compact modification.*

Proof Resolve the analytic singularity ideal, using the log-resolution theorem in [Xia26, Theorem 1.6.1]. On a suitable smooth compact model $\pi\colon Z \to Y$,

$$\pi^* T = [D] + R,$$

where D is a finite effective rational divisor and R is a closed positive current with locally bounded potentials. To obtain a global bounded potential, choose smooth metrics on the line bundles of the components of D and subtract from $w \circ \pi$ the logarithms of the squared norms of their canonical sections, weighted by the coefficients of D. The remainder v is globally bounded, and

$$R = \beta_Z + \mathrm{dd}^{\mathrm{c}} v$$

for a smooth closed form β_Z. By the lemma, the class of R is nef.

On every further compact modification $g\colon Z' \to Z$, the current g^*R still has bounded local potentials and therefore zero Lelong numbers. Hence the divisorial part of $g^*\pi^*T$ is exactly $[g^*D]$, and its regular part is g^*R. It follows that $\mathbb{D}(T)_{Z'} = g^*\{R\}_{\mathrm{BC}}$. Thus $(Z, \{R\}_{\mathrm{BC}})$ is a nef Cartier realization. □

For a general current, analytic regularization gives closed approximants bounded below by $-\epsilon_m\gamma$. Adding $\epsilon_m\gamma$ makes them positive but, when γ is not closed, changes

their exterior derivative. The approximation then takes place in the larger spaces of Chapter 25, and does not by itself approximate the original Bott–Chern b-divisor class by nef Cartier classes.

24.7 Continuous representatives and pullback

A finite divisorial sum is pulled back through the local equations of its divisors. For an infinite sum we need pullback to be continuous on closed positive currents whose Bott–Chern classes vary. Solving a scalar equation for the Gauduchon trace selects a smooth representative in each class; the complementary normalized potentials then converge in L^1, and Lemma 1.22.8 shows that their pullbacks converge as well.

In this section, Y denotes an arbitrary connected compact complex manifold of dimension $n \geq 1$, and γ is a Gauduchon metric on Y; in dimension one, any Hermitian metric will do. Fix smooth norms on tensor fields induced by γ. Put $V_\gamma = \int_Y \gamma^n$ and, for a real $(1,1)$-form or current B, define its *scalar trace* $\sigma_\gamma(B)$ by

$$\sigma_\gamma(B)\,\gamma^n = B \wedge \gamma^{n-1}.$$

For a current this is an identity of distributions. As in the proof of Theorem 4.1.1, we use the classical elliptic Fredholm theorem, elliptic regularity, and the continuity on distributions of the normalized elliptic inverse.

Lemma 24.7.1 *Every closed real $(1,1)$-current T has a unique decomposition*

$$T = B_\gamma(T) + \mathrm{dd^c} v_\gamma(T), \qquad \int_Y v_\gamma(T)\,\gamma^n = 0,$$

where $v_\gamma(T)$ is a real distribution and $B_\gamma(T)$ is a smooth closed real form with constant scalar trace. Both assignments are linear and continuous from weak convergence of currents to convergence of distributions. The forms $B_\gamma(T)$ lie in a finite-dimensional space, and their convergence takes place in C^∞.

This space maps isomorphically onto $H^{1,1}_{\mathrm{BC}}(Y,\mathbb{R})$. If $T \geq 0$, then $v_\gamma(T)$ is quasi-plurisubharmonic. On the cone of positive closed currents, $v_\gamma(T)$ depends continuously on T in L^1, and for every nonnegative integer l,

$$\|B_\gamma(T)\|_{C^l} + \|v_\gamma(T)\|_{L^1(\gamma^n)} \leq C_l \int_Y T \wedge \gamma^{n-1}, \tag{24.3}$$

where the constants C_l depend only on the fixed metric and norms.

Proof Set $L_\gamma f = \sigma_\gamma(\mathrm{dd^c} f)$. We use the inverse G_γ of (6.1), normalized with the volume form γ^n/V_γ. It acts on top-degree distributions of total mass zero and is continuous by the elliptic estimates for the operator and its transpose.

Step 1: construction. Define

$$c_T = \frac{\int_Y T \wedge \gamma^{n-1}}{V_\gamma}, \qquad v_\gamma(T) = G_\gamma\big((\sigma_\gamma(T) - c_T)\gamma^n\big), \qquad B_\gamma(T) = T - \mathrm{dd^c} v_\gamma(T).$$

The current $B_\gamma(T)$ is closed and has scalar trace c_T. We now prove its smoothness and the finite-dimensionality assertion without using any Kähler identity.

Step 2: an elliptic system. Let J be the complex structure acting on real covectors. On real $(1,1)$-forms consider the first-order operator

$$\mathscr{P} B = (\mathrm{d}B, \mathrm{d}\sigma_\gamma(B)).$$

Its principal symbol at a nonzero real covector ξ is $B \mapsto (\xi \wedge B, \xi\, \sigma_\gamma(B))$, and we claim that it is injective. If $\xi \wedge B = 0$, then the $(1,1)$ condition forces B to be a real multiple of $\xi \wedge J\xi$. To check this, choose unitary real coordinates with $\xi = \mathrm{d}x_1$: every term of B contains $\mathrm{d}x_1$, and invariance under J forces every term to contain $\mathrm{d}y_1$ as well. If moreover $\xi\, \sigma_\gamma(B) = 0$, then $\sigma_\gamma(B) = 0$; since the scalar trace of $\xi \wedge J\xi$ is nonzero, the multiple is zero.

Let $\mathscr{P}^*$ be the formal adjoint of $\mathscr{P}$ for the fixed metric and volume. Since the symbol of $\mathscr{P}$ is injective, $\mathscr{P}^*\mathscr{P}$ is elliptic. By elliptic regularity and compactness, its distributional kernel is a finite-dimensional space of smooth forms. Since $\mathscr{P} B_\gamma(T) = 0$, the form $B_\gamma(T)$ belongs to this space. Conversely, the kernel consists exactly of the closed forms with constant scalar trace: integrating $\langle \mathscr{P}^*\mathscr{P} B, B\rangle$ gives $\mathscr{P} B = 0$, and connectedness makes the trace constant. Denote this kernel by $\mathscr{H}_\gamma$.

Step 3: Bott–Chern representatives. The construction is linear and continuous on distributions. Convergence of the components $B_\gamma(T)$ therefore takes place in the finite-dimensional space $\mathscr{H}_\gamma$, hence in every smooth norm. The decomposition is unique: if two decompositions are given, L_γ of the difference of the two potentials is a constant, integration forces this constant to be zero, and the normalized kernel of L_γ is zero. A smooth closed form β has the representative $B_\gamma(\beta) \in \mathscr{H}_\gamma$ of its Bott–Chern class, and two elements of $\mathscr{H}_\gamma$ in the same class differ by $\mathrm{dd^c} f$ with f smooth, which has the decomposition $0 + \mathrm{dd^c}(f - V_\gamma^{-1} \int_Y f\gamma^n)$. By uniqueness, $\mathscr{H}_\gamma$ contains exactly one representative of each Bott–Chern class.

Step 4: positive currents. Suppose now that $T \geq 0$. The inequality $\mathrm{dd^c} v_\gamma(T) \geq -B_\gamma(T)$ gives a quasi-plurisubharmonic representative of the distribution $v_\gamma(T)$. Let positive closed currents T_j converge weakly to T. Their components $B_\gamma(T_j)$ converge smoothly, so the potentials $v_\gamma(T_j)$ have a common smooth lower curvature bound. By Theorem 1.4.1, after subtracting their suprema they form an L^1-precompact family. If $a_j = \sup_Y v_\gamma(T_j)$, the zero-mean normalization gives

$$0 \leq a_j V_\gamma = \int_Y \big(a_j - v_\gamma(T_j)\big)\, \gamma^n,$$

and the right-hand side is uniformly bounded. Hence the potentials $v_\gamma(T_j)$ themselves are L^1-precompact. Every cluster point equals their distributional limit $v_\gamma(T)$, so the whole sequence converges in L^1.

Finally, the positive closed currents with trace mass $\int_Y T \wedge \gamma^{n-1} = 1$ form a weakly compact set: positivity bounds their coefficient measures, weak limits remain positive and closed, and testing with γ^{n-1} preserves the trace mass. The smooth and L^1 continuity just proved bounds both norms on this set, and positive homogeneity gives (24.3). A positive current of trace mass zero is zero, so this case is included.□

The smooth representatives give a common curvature bound for converging normalized potentials. This is exactly the hypothesis needed to pull back their L^1 convergence.

Proposition 24.7.1 *Let $g\colon Z \to Y$ be a smooth compact modification, and fix a Gauduchon metric γ' on Z. Pullback of positive closed $(1,1)$-currents, defined through their local potentials, is weakly continuous. Moreover, there is a constant C_g, depending only on the modification and the fixed metrics, such that*

$$\int_Z g^*T \wedge (\gamma')^{n-1} \le C_g \int_Y T \wedge \gamma^{n-1} \qquad (T \ge 0,\ \mathrm{d}T = 0). \tag{24.4}$$

Proof Write $T = B_\gamma(T) + \mathrm{dd}^{\mathrm{c}} v_\gamma(T)$. The pullback is

$$g^*T = g^*B_\gamma(T) + \mathrm{dd}^{\mathrm{c}}(v_\gamma(T) \circ g).$$

This current is positive and closed, since it is locally dd^{c} of plurisubharmonic functions, and the composed potential is not identically $-\infty$ because g is surjective. The formula agrees with the definition through local potentials, so it does not depend on the decomposition.

If $T_j \to T$, the smooth parts converge in C^∞, and the potentials converge in $L^1(Y)$ with a common lower curvature bound. By Lemma 1.22.8, their compositions with g converge in $L^1(Z)$, and taking dd^{c} proves the weak convergence of the pullbacks.

Integration by parts on Z gives

$$\int_Z g^*T \wedge (\gamma')^{n-1} = \int_Z g^*B_\gamma(T) \wedge (\gamma')^{n-1}.$$

The absolute value of the right-hand side is at most a fixed constant times $\|B_\gamma(T)\|_{C^0}$. Since the left-hand side is nonnegative, (24.3) gives (24.4). □

Corollary 24.7.1 *Let $g\colon Z \to Y$, γ and γ' be as in Proposition 24.7.1. Let D_j be prime divisors on Y, with integration currents $[D_j]$, and let $c_j > 0$. If $S = \sum_j c_j [D_j]$ is a positive closed current of finite mass, then*

$$g^* S = \sum_j c_j [g^* D_j]$$

as positive currents of finite mass. The trace masses on Z of the positive tails of this series tend to zero. In particular, g^*S is again purely divisorial.

Proof Apply the proposition to the increasing finite partial sums S_N and to their positive closed tails $S - S_N$. Pullback is additive on positive closed currents, by addition of local potentials. Thus $g^*S - g^*S_N = g^*(S - S_N) \geq 0$, and its trace mass is at most C_g times the trace mass of $S - S_N$, which tends to zero. The pullback of a finite divisor is the usual pullback of its local equations, which identifies g^*S_N with the corresponding partial sum of the series, and hence g^*S with its limit. Its terms are integration currents of prime divisors with nonnegative coefficients, and regrouping this countable positive sum of finite mass does not change it. □

24.8 Exceptional divisors between trace models

We return to the trace data fixed at the beginning of the chapter, with $\dim Y = r$, and write $S_Z = S_Z(u; Y)$. By Corollary 24.7.1, the full trace current can now be pulled back under a modification $g\colon Z' \to Z$. The pullback is bounded above by the trace current on Z', with equal coefficients along strict transforms, so the difference is a finite effective divisor supported on the exceptional locus. Its logarithmic potential measures the change in the ordinary singularity types of the divisorial parts.

Theorem 24.8.1 *For a smooth compact modification $g\colon Z' \to Z$ over Y, there is a unique finite effective real divisor E_g supported on the exceptional locus of g such that*

$$S_{Z'} = g^* S_Z + [E_g]. \tag{24.5}$$

For a further smooth compact modification $h\colon Z'' \to Z'$, these divisors satisfy

$$E_{g\circ h} = E_h + h^* E_g. \tag{24.6}$$

Their coefficients are additive and positively homogeneous in the ambient potential u.

For every m, the analytic-ideal divisors on the two models differ by a finite effective divisor $E_{m,g}$ supported on the exceptional locus:

$$[E_{m,g}] = D_{m,Z'} - g^* D_{m,Z}.$$

For every Gauduchon metric γ' on Z', the measures

$$[E_{m,g}] \wedge (\gamma')^{r-1} \longrightarrow [E_g] \wedge (\gamma')^{r-1}$$

converge in total variation.

Proof *Step 1: the exceptional divisor.* Enumerate the prime divisors on Z with positive trace coefficient. By Theorem 24.2.1, the pullback of every finite partial sum of S_Z is at most $S_{Z'}$, and Corollary 24.7.1 identifies the increasing limit of these pullbacks with $g^* S_Z$. Thus $S_{Z'} - g^* S_Z$ is a positive closed divisorial current. The two currents have equal coefficients along the strict transform of every prime divisor on Z, because these coefficients are computed by the same valuation. Every other prime divisor on Z' is exceptional for g, and the exceptional analytic set has only finitely many irreducible components on the compact manifold Z'. Hence the difference is the integration current of a finite effective real divisor E_g supported on its divisorial components, and the coefficients of integration currents determine E_g uniquely.

Step 2: the cocycle identity. Pullback of positive closed $(1,1)$-currents is functorial, since it is defined by composing local plurisubharmonic potentials. Using (24.5) for h and for g gives

$$S_{Z''} = h^* S_{Z'} + [E_h] = (g \circ h)^* S_Z + [h^* E_g] + [E_h],$$

and uniqueness gives (24.6). Additivity and positive homogeneity follow from the same properties of the trace coefficients and the linearity of pullback.

Step 3: approximation. For fixed m, apply (24.2) to the finitely many prime divisors in the support of the image ideal on Z. After division by m, it gives $D_{m,Z'} - g^{*}D_{m,Z} \geq 0$. The coefficients along strict transforms vanish, so this difference is the integration current of an effective real divisor $E_{m,g}$ supported on the exceptional locus.

Choose also a Gauduchon metric γ on Z. By Theorem 24.1.1, the currents $S_Z - D_{m,Z}$ and $S_{Z'} - D_{m,Z'}$ are positive, and their trace masses tend to zero. Subtracting the two identities for the exceptional currents gives

$$[E_{m,g}] - [E_g] = -(S_{Z'} - D_{m,Z'}) + g^{*}(S_Z - D_{m,Z}),$$

where both $S_{Z'} - D_{m,Z'}$ and $g^{*}(S_Z - D_{m,Z})$ are positive. By (24.4), the total variation of the signed measure $([E_{m,g}] - [E_g]) \wedge (\gamma')^{r-1}$ is therefore at most

$$\int_{Z'} (S_{Z'} - D_{m,Z'}) \wedge (\gamma')^{r-1} + C_g \int_{Z} (S_Z - D_{m,Z}) \wedge \gamma^{r-1}.$$

Both terms tend to zero, which proves convergence in total variation. □

For a modification $g\colon Z' \to Z$ as in the theorem, the coefficient of E_g along an exceptional prime divisor $F \subset Z'$ is

$$\tau_{Y,u}(F) - \sum_{D \subset Z} \tau_{Y,u}(D)\, \mathrm{ord}_F(g^{*}D), \tag{24.7}$$

where D ranges over the prime divisors on Z with positive trace coefficient. This countable sum converges: it is the coefficient of $g^{*}S_Z$ along F in the positive divisorial series of Corollary 24.7.1, and it is at most $\tau_{Y,u}(F)$. The same current identity compares the ordinary singularity types attached to S_Z and $S_{Z'}$ by Corollary 1.17.1.

Corollary 24.8.1 *Choose decompositions $S_Z = B_Z + \mathrm{dd}^{\mathrm{c}} v_Z$ and $S_{Z'} = B_{Z'} + \mathrm{dd}^{\mathrm{c}} v_{Z'}$ with smooth closed forms $B_Z, B_{Z'}$, as in Corollary 1.17.1. Let ρ_{E_g} be a global logarithmic potential of E_g, that is, a quasi-plurisubharmonic function on Z' such that $\mathrm{dd}^{\mathrm{c}} \rho_{E_g} - [E_g]$ is smooth. Then*

$$v_{Z'} - (v_Z \circ g) - \rho_{E_g}$$

is represented by a smooth function. In particular,

$$v_{Z'} \preceq v_Z \circ g$$

in the sense of Definition 1.12.1, and the two ordinary singularity types are equal if and only if $E_g = 0$.

Proof Global logarithmic potentials exist: choose smooth Hermitian metrics on the line bundles of the prime components of E_g, let α_{E_g} be the corresponding weighted sum of their curvature forms, and let ρ_{E_g} be the weighted sum of the logarithms of the squared norms of the canonical sections. Then

$$\mathrm{dd}^{\mathrm{c}} \rho_{E_g} = [E_g] - \alpha_{E_g},$$

and any two global logarithmic potentials differ by a smooth function, by Lemma 1.17.3. We may therefore use this choice. The function ρ_{E_g} is quasi-plurisubharmonic and bounded above on the compact manifold Z'.

The chosen decompositions and (24.5) give

$$\mathrm{dd}^{\mathrm{c}} \left(v_{Z'} - v_Z \circ g - \rho_{E_g} \right) = g^* B_Z + \alpha_{E_g} - B_{Z'},$$

which is smooth. By Lemma 1.17.3, this difference of locally integrable functions is represented by a smooth function. The resulting equality of quasi-plurisubharmonic functions extends from almost everywhere to everywhere by Lemma 1.3.3. Since ρ_{E_g} is bounded above, this gives the ordinary domination.

If $E_g = 0$, the smooth difference gives ordinary equivalence. Conversely, ordinary equivalence would give equal generic Lelong numbers along every prime divisor of Z', whereas the identity just proved shows that along a prime component of E_g these numbers differ by its positive coefficient. Hence ordinary equivalence forces $E_g = 0$. □

When the restricted background $\theta|_Y$ is closed, the positive residual currents R_Z constructed in the proof of Proposition 24.4.1, and their Bott–Chern classes, satisfy

$$\begin{aligned} R_{Z'} &= g^* R_Z - [E_g], \\ \mathbb{A}_{Z'}(u; Y, \theta) &= g^* \mathbb{A}_Z(u; Y, \theta) - \{[E_g]\}_{\mathrm{BC}}. \end{aligned}$$

Indeed, the smooth background and the common trace potential pull back, while (24.5) gives the change in the subtracted divisorial current. The correction $\{[E_g]\}_{\mathrm{BC}}$ accounts for the additional divisorial vanishing prescribed on the higher model. This

is consistent with the exact pullback of the trace envelope in Theorem 20.6.2, since the envelope imposes the trace inequalities on all models at once. The following example shows that the correction can be nonzero even when the trace data are realized exactly by a potential.

Example 24.8.1 Let $Y = \mathbb{P}^2$ with homogeneous coordinates $[t_0 : t_1 : t_2]$, let σ be the Fubini–Study form normalized by $\int_Y \sigma^2 = 1$, and choose $a \in \mathbb{Q}_{>0}$. Put

$$u = a \log \frac{|t_1|^2 + |t_2|^2}{|t_0|^2 + |t_1|^2 + |t_2|^2},$$

and take the ambient manifold to be Y itself. Then $u \in \mathrm{PSH}(Y, a\sigma)$. It has analytic singularities, with its only pole at $p = [1 : 0 : 0]$, and zero generic Lelong number on Y. By Proposition 20.2.1, its trace data are given by the generic Lelong numbers of its pullbacks: $\tau_{Y,u}(E) = \nu(u \circ \pi, E)$ for every prime divisor E on a smooth compact modification π of Y. Since u has no divisorial poles on Y, we have $S_Y = 0$, whose canonical ordinary type is that of a smooth function. If $\pi \colon \widetilde{Y} \to Y$ is the blowup at p with exceptional curve F, then

$$S_{\widetilde{Y}} = a[F], \qquad E_\pi = aF.$$

Indeed, on either standard chart of the blowup, the numerator is the squared modulus of the exceptional coordinate times a smooth positive function, so the pole order is a along F and zero along every other prime divisor. Thus the trace data are realized exactly by u, although $E_\pi \neq 0$. The exceptional difference does not obstruct exact realization; it prevents the identification of the realizing potential with the canonical ordinary type of the divisorial part on the original model, which here is smooth.

The same phenomenon occurs on a manifold without the bounded mass property. Let (W, ω_W) be a connected compact Hermitian manifold of dimension k that fails the bounded mass property, such as the Hopf threefold of Theorem 9.0.1, and pull u back to $Y \times W$; take both the ambient manifold and the submanifold to be $Y \times W$. The pole set of the pullback is $\{p\} \times W$, of codimension two, and the modification $\widetilde{Y} \times W \to Y \times W$ has exceptional coefficient a along $F \times W$. Since the pullback of u has analytic singularities, Proposition 20.2.1 again shows that it realizes the trace data exactly. Let pr_Y and pr_W be the two product projections.

Choose smooth ω_W-plurisubharmonic functions H_j whose masses tend to infinity. For the Hermitian form $\Omega = \mathrm{pr}_Y^*\,\sigma + \mathrm{pr}_W^*\,\omega_W$, the pullbacks of the H_j have masses

$$\binom{k+2}{2} \left(\int_Y \sigma^2 \right) \left(\int_W (\omega_W + \mathrm{dd^c} H_j)^k \right) \longrightarrow +\infty,$$

by the binomial expansion of smooth forms and the dimensions of the two factors. Hence the product also fails the bounded mass property, by Proposition 7.5.1. The pulled-back potential belongs to $\mathrm{PSH}(Y \times W, C\Omega)$ for every $C \geq a$, since its curvature is bounded below by $-a\,\mathrm{pr}_Y^*\,\sigma$.

Chapter 25
Classes with a nonclosed background

Let Y be a connected compact complex manifold of dimension $n \geq 1$, let θ be a smooth real $(1,1)$-form on Y, and let $\pi\colon Z \to Y$ denote a smooth compact modification.

When the background θ is not closed, subtracting a closed divisorial current from its pullback leaves the exterior derivative unchanged, so the residual currents of Chapter 24 are not closed and define no Bott–Chern class. They do define classes in an affine space over Bott–Chern cohomology, since two representatives with the same exterior derivative differ by a closed current. The trace envelope still gives positive residual representatives, and the exceptional divisors of Section 24.8 describe how the classes change between models.

Analytic singularities give nef Cartier realizations in these affine spaces. For a general potential, analytic regularization gives such realizations in slightly larger backgrounds, and on each fixed model the positive residual currents converge weakly. The error in the background tends to zero smoothly, but removing it can destroy nefness. Nor does this linear convergence control Monge–Ampère mass: for a Hermitian background with unbounded upper mass, a potential of infinite mass can have the same residual classes as the zero potential, whose classes are realized by a smooth positive form.

25.1 Affine classes with prescribed derivative

Two real $(1,1)$-currents on Z with exterior derivative $\pi^*\mathrm{d}\theta$ differ by a closed current, and modulo dd^{c} of global distributions such differences are Bott–Chern classes. Let $\mathcal{D}'(Z,\mathbb{R})$ denote the space of real distributions on Z, and set

$$\mathscr{A}_Z(\theta) = \frac{\{\pi^*\theta + C : C \text{ is a closed real } (1,1)\text{-current on } Z\}}{\mathrm{dd}^{\mathrm{c}}\mathcal{D}'(Z,\mathbb{R})}.$$

Write $\{T\}$ for the class of T in this space. By Lemma 1.17.4, every class has a smooth representative, and translation by Bott–Chern classes gives

$$\mathscr{A}_Z(\theta) = \{\pi^*\theta\} + H^{1,1}_{\mathrm{BC}}(Z, \mathbb{R}).$$

A smooth real $(1,1)$-form η represents an element of $\mathscr{A}_Z(\theta)$ exactly when $\mathrm{d}\eta = \pi^*\mathrm{d}\theta$. As for Bott–Chern classes, pullback of a class is defined on a smooth representative and pushforward on current representatives.

Proposition 25.1.1 *If $g\colon Z' \to Z$ is a smooth compact modification over Y, pushforward and pullback induce maps*

$$g_*\colon \mathscr{A}_{Z'}(\theta) \to \mathscr{A}_Z(\theta), \qquad g^*\colon \mathscr{A}_Z(\theta) \to \mathscr{A}_{Z'}(\theta), \qquad g_*g^* = \mathrm{id}.$$

Pushforward preserves classes with positive current representatives.

Proof Write $\pi' = \pi \circ g$. Translation by the pulled-back background identifies the required maps with the Bott–Chern maps of Section 24.4. Indeed, for a closed current C' on Z' and a smooth closed form β on Z,

$$g_*(\pi'^*\theta + C') = \pi^*\theta + g_*C', \qquad g^*(\pi^*\theta + \beta) = \pi'^*\theta + g^*\beta,$$

where the first equality uses the change of variables formula for the degree-one map g on the smooth background. Since the Bott–Chern maps are well defined on the closed terms, these identities define the affine maps independently of the representatives. Their composition is the identity because g_*g^* is the identity both on the background and on Bott–Chern classes. Finally, pushforward under a proper map preserves positive currents. □

The prescribed derivative is essential for this pushforward: $g_*\pi'^*\mathrm{d}\theta = \pi^*\mathrm{d}\theta$ is smooth, whereas the pushforward of an arbitrary smooth exact form can be a singular current.

We call a family $(\mathbb{A}_Z)$, with $\mathbb{A}_Z \in \mathscr{A}_Z(\theta)$ for every model Z and compatible with pushforward, a *b-divisor class over the background θ*. Pseudoeffectivity and Cartier realizations are defined as in Definition 24.4.1, with affine classes in place of Bott–Chern classes. A class in $\mathscr{A}_Z(\theta)$ is *nef* if, for a Hermitian form γ on Z, it has for every $\epsilon > 0$ a smooth representative bounded below by $-\epsilon\gamma$, as in Section 24.6.

25.2 Residual trace classes

Since the trace envelope dominates the trace currents on every model, subtracting them from its curvature current gives positive residual representatives, with exterior derivative $\pi^*\mathrm{d}\theta$. The classes are determined by the prescribed trace currents, even when the envelope has larger Lelong numbers.

In this section, Y is a complex submanifold of a connected compact complex manifold X. Let Θ be a smooth real $(1,1)$-form on X with $\Theta|_Y = \theta$, and let

$u \in \mathrm{PSH}(X, \Theta)$ have generic Lelong number $\nu(u, Y) = 0$. We use the trace currents $S_Z(u; Y)$ of Section 24.1, now with Θ in the role of the ambient background.

Theorem 25.2.1 *The family defined by*

$$\mathbb{B}_Z(u; Y, \theta) = \{\pi^*\theta - S_Z(u; Y)\} \in \mathscr{A}_Z(\theta)$$

is a pseudoeffective b-divisor class over θ. It is unchanged by ambient multiplier-ideal equivalence and, after the resulting affine spaces are identified, by a smooth dd^{c} change of the restricted background.

Proof The currents S_Z are closed, so the classes belong to the indicated affine spaces, and their compatibility follows from Theorem 24.2.1 and Proposition 25.1.1. By Proposition 24.3.1, with Θ in place of θ, the trace envelope $v = \mathcal{T}_\theta(u; Y)$ satisfies

$$\pi^*\theta + \mathrm{dd}^{\mathrm{c}}(v \circ \pi) - S_Z(u; Y) \geq 0$$

on every model, which proves pseudoeffectivity in the original restricted background. The currents S_Z depend only on the ambient multiplier ideals. For a smooth real function f on Y, the change $\theta \mapsto \theta + \mathrm{dd}^{\mathrm{c}} f$ adds $\mathrm{dd}^{\mathrm{c}}(f \circ \pi)$ to every representative, which is zero in the quotient. This proves both invariances. □

When θ is closed, this family is the Bott–Chern family of Proposition 24.4.1.

On a higher model, the trace current can exceed the pullback of the trace current from a lower model. The effective exceptional difference of Theorem 24.8.1 therefore translates the residual class, and the translation preserves the prescribed exterior derivative.

Corollary 25.2.1 *Let $g\colon Z' \to Z$ be a smooth compact modification over Y, and let E_g be the effective exceptional divisor of Theorem 24.8.1. The trace classes of Theorem 25.2.1 satisfy*

$$\mathbb{B}_{Z'}(u; Y, \theta) = g^*\mathbb{B}_Z(u; Y, \theta) - \{[E_g]\}_{\mathrm{BC}},$$

where the Bott–Chern class on the right acts by translation on the affine space $\mathscr{A}_{Z'}(\theta)$.

Proof Let $\pi\colon Z \to Y$ be the structure map and put $\pi' = \pi \circ g$. Abbreviate $S_Z = S_Z(u; Y)$, and use Lemma 1.17.3 to write $S_Z = B_Z + \mathrm{dd}^{\mathrm{c}} v_Z$ with B_Z smooth and closed and v_Z quasi-plurisubharmonic. The affine class $\mathbb{B}_Z$ has the smooth representative $\pi^*\theta - B_Z$. Its pullback is therefore the class of $\pi'^*\theta - g^*S_Z$, since the two representatives differ by $\mathrm{dd}^{\mathrm{c}}(v_Z \circ g)$. Substituting $S_{Z'} = g^*S_Z + [E_g]$ gives the asserted identity. Both representatives have exterior derivative $\pi'^*\mathrm{d}\theta$, because the subtracted currents are closed, so the subtraction takes place within $\mathscr{A}_{Z'}(\theta)$. □

The cocycle identity (24.6) makes these translations compatible with composition of modifications, and by Theorem 24.8.1 their analytic-ideal approximants converge in total variation of the trace measures. As in the closed case of Section 24.8, the translation can be nonzero even when the trace data are realized exactly (Example 24.8.1).

For the rest of the chapter, the ambient manifold is Y itself, and $u \in \mathrm{PSH}(Y,\theta)$. For a smooth compact modification $\pi\colon Z \to Y$, we write $N_Z(u)$ for the divisorial current of $u\circ\pi$ and $\mathbb{D}_\theta(u)$ for the resulting residual family. By the self-trace identity of Lemma 24.5.1,

$$N_Z(u) = \sum_E \nu(u\circ\pi, E)[E] = S_Z(u;Y), \qquad \mathbb{D}_\theta(u)_Z = \{\pi^*\theta - N_Z(u)\},$$

where E ranges over the prime divisors on Z. By Theorem 25.2.1, $\mathbb{D}_\theta(u)$ is pseudoeffective. In this case divisorial removal gives the positive representative $\pi^*\theta + \mathrm{dd^c}(u\circ\pi) - N_Z(u) \geq 0$ directly, by Lemma 1.17.1 and passage to the limit of increasing finite sums.

25.3 Nef realizations and residual-current limits

On a resolution of an analytic singularity, the bounded residual potential has smooth approximants with arbitrarily small curvature error, and these prove nefness within the same affine class. Boundedness also prevents new divisorial terms on further modifications, which gives the Cartier realization.

Proposition 25.3.1 *If $u \in \mathrm{PSH}(Y,\theta)$ has analytic singularities, then $\mathbb{D}_\theta(u)$ has a nef Cartier realization on a smooth compact modification.*

Proof Resolve the analytic singularity ideal as in the proof of Theorem 24.6.1. On a suitable model $\pi\colon Z \to Y$ we obtain

$$\pi^*\theta + \mathrm{dd^c}(u\circ\pi) = [D] + R, \qquad R = \beta + \mathrm{dd^c}\nu \geq 0,$$

where D is a finite effective rational divisor, β is smooth with $\mathrm{d}\beta = \pi^*\mathrm{d}\theta$, and ν is globally bounded after the choice of metrics on the line bundles of the components of D. Positivity of R follows from divisorial removal.

Fix a Hermitian form γ on Z and apply Proposition 1.10.1 to ν in the smooth background β. It gives smooth functions ν_j with $\beta + \mathrm{dd^c}\nu_j \geq -\epsilon_j\gamma$ and $\epsilon_j \downarrow 0$. These forms represent the same affine class and have exterior derivative $\pi^*\mathrm{d}\theta$, which proves nefness.

The Cartier property follows as in the proof of Theorem 24.6.1: on every further modification $g\colon Z' \to Z$, the bounded potential $\nu\circ g$ has zero Lelong numbers, so the divisorial part of the full pullback is exactly g^*D, and the residual affine class on Z' is $g^*\{R\}$. □

For general u, add twice the curvature error of analytic regularization to the background. This gives positive currents with analytic singularities, to which the proposition applies, while the ideal orders identify the limiting residual family.

Theorem 25.3.1 *Let $u \in \mathrm{PSH}(Y,\theta)$ and fix a Hermitian form ω on Y. There are potentials u_m with analytic singularities, converging to u in $L^1(Y)$, and positive numbers $\epsilon_m \downarrow 0$, indexed by sufficiently large positive integers m, such that the backgrounds $\theta_m = \theta + 2\epsilon_m\omega$ satisfy*

$$\theta_m + \mathrm{dd^c} u_m \geq \epsilon_m \omega.$$

For each m, the family $\mathbb{D}_{\theta_m}(u_m)$ has a nef Cartier realization. On every fixed compact model $\pi\colon Z \to Y$, its class has the current representative

$$C_{m,Z} = \pi^*\theta + 2\epsilon_m\pi^*\omega - D_{m,Z}, \qquad D_{m,Z} = \frac{1}{m}\sum_E \mathrm{ord}_E\, \mathcal{I}(mu)[E],$$

where E ranges over the prime divisors on Z and the order is computed after pulling the ideal back to Z. These representatives satisfy

$$C_{m,Z} \longrightarrow \pi^*\theta - N_Z(u) \quad \textit{weakly as currents},$$

and the positive residual representatives satisfy

$$R_{m,Z} := C_{m,Z} + \mathrm{dd^c}(u_m \circ \pi) \longrightarrow \pi^*\theta + \mathrm{dd^c}(u\circ\pi) - N_Z(u) \geq 0 \quad \textit{weakly as currents}.$$

The exterior derivatives of both $C_{m,Z}$ and $R_{m,Z}$ converge smoothly to $\pi^*\mathrm{d}\theta$.

Proof Take u_m from Corollary 1.15.1; they converge to u in L^1. If $e_m \to 0$ are their curvature errors, put $\epsilon_m = \sup_{\ell\geq m} e_\ell + 1/m$. These numbers are positive and decrease to zero, and the indices m are unchanged. Adding $2\epsilon_m\omega$ to the background turns the lower bound $-\epsilon_m\omega$ into $\epsilon_m\omega$. Since each u_m has analytic type $(1/m, \mathcal{I}(mu))$, Proposition 25.3.1 gives a nef Cartier realization in the background θ_m.

On a fixed model, the generic divisorial Lelong numbers of $u_m \circ \pi$ are exactly $m^{-1}\,\mathrm{ord}_E\,\mathcal{I}(mu)$, so $N_Z(u_m) = D_{m,Z}$ and $C_{m,Z} = \pi^*\theta_m - N_Z(u_m)$ represents the class of $\mathbb{D}_{\theta_m}(u_m)$ on Z. The self-trace identity and Theorem 24.1.1 give $D_{m,Z} \to N_Z(u)$ in the trace mass of Section 24.1, hence weakly as currents. Since the smooth error $2\epsilon_m\pi^*\omega$ tends to zero with all its derivatives, $C_{m,Z}$ converges as stated.

Divisorial removal gives $R_{m,Z} \geq 0$. All the u_m and u have a common smooth lower curvature bound: choose $A > 0$ with $\theta + \epsilon_m\omega \leq A\omega$ for every index m considered. By Lemma 1.22.8, the convergence in $L^1(Y)$ implies $u_m\circ\pi \to u\circ\pi$ in $L^1(Z)$. Applying $\mathrm{dd^c}$ and adding the convergent currents $C_{m,Z}$ proves the convergence of the residual currents, and positivity persists in the weak limit. Finally, since each $D_{m,Z}$ is closed,

$$\mathrm{d}R_{m,Z} = \mathrm{d}C_{m,Z} = \pi^*\mathrm{d}\theta + 2\epsilon_m\pi^*\mathrm{d}\omega,$$

which proves the smooth convergence of the exterior derivatives. □

The classes $\mathbb{D}_{\theta_m}(u_m)_Z$ lie in $\mathscr{A}_Z(\theta_m)$. Translation by $-2\epsilon_m\pi^*\omega$ identifies this space with $\mathscr{A}_Z(\theta)$, but subtracting this form need not preserve nefness. Thus the convergence of currents above does not give an approximation by nef Cartier classes with exactly the prescribed exterior derivative.

25.4 Divisorial classes and infinite mass

Now let the background be a Hermitian form ω on Y, with upper mass $v_+(\omega)$ (Definition 7.5.1) and extended Monge–Ampère measure $\mathrm{MA}_\omega^{\mathrm{ext}}$ (Theorem 7.1.1). When $v_+(\omega) = +\infty$, the potential of Theorem 7.6.1 has the same multiplier ideals as a smooth potential. The two potentials therefore have the same residual classes on every model, while their extended masses differ.

Proposition 25.4.1 *If $v_+(\omega) = +\infty$, there is $U \in \mathrm{PSH}(Y, \omega)$ such that*

$$\mathbb{D}_\omega(U) = \mathbb{D}_\omega(0), \qquad \int_Y \mathrm{MA}_\omega^{\mathrm{ext}}(U) = +\infty.$$

The common family has the nef Cartier realization $(Y, \{\omega\})$.

Proof Take U from Theorem 7.6.1. All its multiplier ideals at positive weights are trivial, so every divisorial order in $D_{m,Z}$ vanishes on every model. The self-trace identity gives $N_Z(U) = 0$, which is also the divisorial part of the zero potential. Both families therefore consist of the pullbacks of the class of ω. Since ω is smooth and positive, its affine class is nef and realizes the family on Y itself. The extended mass of U is infinite by construction. □

Chapter 26
Envelope systems over Hermitian perturbations

Let X be a connected compact complex manifold of dimension $n \geq 1$, let θ be a smooth real $(1,1)$-form on X, and fix real numbers $a < c$.

For $u \in \mathrm{PSH}(X,\theta)$, the multiplier-ideal envelope $P_\theta[u]_{\mathcal{I}}$ of Definition 17.2.1 is the regularized supremum of the nonpositive θ-plurisubharmonic functions with the multiplier ideals of u. It depends on the background θ. When θ is perturbed by Hermitian forms, the envelopes of one multiplier-ideal class in the perturbed backgrounds form a compatible family, and as the perturbation tends to zero these normalized representatives decrease to a potential in the original background. Projecting the limit back to the perturbed backgrounds recovers the family exactly when the limit keeps the multiplier ideals of the family; otherwise it gives the greatest family below it that comes from the original background.

We apply this to test curves that are constant for parameters $\tau \leq a$ and have cutoff c. Recovery can fail even in a Kähler background: on a ruled surface, at zero volume, the limit acquires an additional divisorial pole. On the other hand, for big backgrounds on manifolds with the bounded mass property, recovery holds by Theorem 31.5.3 when the lower volumes $\underline{\mathrm{vol}}$ of the initial slices (Definition 14.3.1) have a positive limit. On the Hopf threefold, whose upper mass is unbounded, recovery holds instead for systems whose components over scalar multiples of the standard Gauduchon form are invariant along the elliptic fibers, provided that the limiting initial slice on the projective base has positive lower volume; both recovery theorems are proved in Chapter 31. Smooth changes of the background commute with recovery, whereas the limiting Monge–Ampère mass can change when the background is not closed.

We also use the ordinary singularity envelope $P_\theta[u]$ of Definition 1.19.1, the ordinary and multiplier-ideal singularity orders of Section 1.12, and the extended Monge–Ampère measure $\mathrm{MA}^{\mathrm{ext}}$ of Theorem 7.1.1. A potential u with $P_\theta[u]_{\mathcal{I}} = u$ is called $P_\theta[\cdot]_{\mathcal{I}}$-fixed.

26.1 Curves constant on a left half-line

We restrict the envelope test curves of Definition 19.1.1 by requiring that each slice be fixed by the multiplier-ideal envelope and that the curve be constant for $\tau \le a$. Constancy identifies the initial potential explicitly as Γ_a, and it confines the associated ray between the translates of this potential by at and by ct.

Definition 26.1.1 Let $\mathcal{T}_{a,c}(\theta)$ be the set of families $\Gamma = (\Gamma_\tau)_{\tau<c}$ such that

(1) $\Gamma_\tau \in \mathrm{PSH}(X,\theta)$ and $P_\theta[\Gamma_\tau]_I = \Gamma_\tau$ for every $\tau < c$;
(2) the family is pointwise concave and decreasing in τ;
(3) $\Gamma_\tau = \Gamma_a$ for $\tau \le a$.

The cutoff of Γ is c, and we extend Γ by $\Gamma_c = \lim_{\tau\uparrow c}\Gamma_\tau$ and $\Gamma_\tau = -\infty$ for $\tau > c$. Such curves are ordered by the pointwise order of their slices Γ_τ, $\tau < c$.

Every slice has supremum zero by Theorem 17.2.1. Hence, by Theorem 1.4.1 and Corollary 1.4.1, the decreasing limit Γ_c belongs to $\mathrm{PSH}(X,\theta)$. It is also $P_\theta[\cdot]_I$-fixed. Indeed, $\Gamma_c \le 0$ is a competitor for its own envelope, and $\Gamma_c \le \Gamma_\tau$, so the order preservation of Proposition 17.2.3 gives

$$\Gamma_c \le P_\theta[\Gamma_c]_I \le \Gamma_\tau \quad (\tau < c).$$

Taking the infimum over τ proves equality.

Proposition 26.1.1 *A curve $\Gamma \in \mathcal{T}_{a,c}(\theta)$ is an envelope test curve with initial potential $\varphi = \Gamma_a$. Its forward transform U_t, $t > 0$, of (1.40) is an envelope ray with initial potential φ and slope c, in the sense of Definition 19.1.2, and satisfies*

$$\varphi + at \le U_t \le \varphi + ct \qquad (t > 0).$$

In particular $U_t \sim \varphi$ for every $t > 0$.

Proof By Proposition 17.2.2, a $P_\theta[\cdot]_I$-fixed potential is also fixed by the ordinary envelope P_θ. The regularized supremum of the curve is Γ_a, since the curve is decreasing and constant for $\tau \le a$. Hence all the conditions of Definition 19.1.1 hold, and Theorem 19.2.1 applies. The term $\tau = a$ in the transform gives the lower bound, and the inequalities $\Gamma_\tau \le \varphi$ for $\tau < c$ give the upper bound. □

26.2 Transition maps and envelope systems

Curves over different backgrounds are related by projecting each slice to the ideal envelope in the larger background. For these projections to form a consistent system, projecting in two steps must agree with projecting directly. This follows because each projection preserves multiplier ideals, while in each background a multiplier-ideal class has a unique representative fixed by the ideal envelope. The same fact gives

injectivity: equal projected slices have equal multiplier ideals, hence come from equal fixed representatives.

Proposition 26.2.1 *Let $\beta \geq \theta$ be a smooth real $(1,1)$-form on X. Slicewise projection, $\Gamma \mapsto (P_\beta[\Gamma_\tau]_{\mathcal{I}})_{\tau<c}$, defines an injective and order-preserving map*

$$P_\beta[\cdot]_{\mathcal{I}} : \mathscr{T}_{a,c}(\theta) \longrightarrow \mathscr{T}_{a,c}(\beta).$$

For every smooth real $(1,1)$-form β' with $\beta \geq \beta' \geq \theta$, these maps satisfy

$$P_\beta\big[P_{\beta'}[\Gamma]_{\mathcal{I}}\big]_{\mathcal{I}} = P_\beta[\Gamma]_{\mathcal{I}}.$$

Proof Since $\beta \geq \theta$, every slice is β-plurisubharmonic, and by Theorem 17.2.1 its projection is a β-plurisubharmonic function fixed by $P_\beta[\cdot]_{\mathcal{I}}$. Decreasingness, constancy for $\tau \leq a$, and order preservation follow from Proposition 17.2.3.

For concavity, let $r, t < c$ and $0 < s < 1$, and put $\tau = (1-s)r + st$. By the same proposition and the concavity of the original curve,

$$\begin{aligned}(1-s)P_\beta[\Gamma_r]_{\mathcal{I}} + sP_\beta[\Gamma_t]_{\mathcal{I}} &\leq P_\beta[(1-s)\Gamma_r + s\Gamma_t]_{\mathcal{I}} \\ &\leq P_\beta[\Gamma_\tau]_{\mathcal{I}}.\end{aligned}$$

Here the first inequality follows by combining the scaling and addition formulas of that proposition, with backgrounds $(1-s)\beta$ and $s\beta$.

Each projection preserves the multiplier ideals at every weight. Hence the two sides of the transition identity are $P_\beta[\cdot]_{\mathcal{I}}$-fixed representatives of the same multiplier-ideal class, and they are equal. If two projected curves agree, their original slices have the same multiplier ideals; since those slices are $P_\theta[\cdot]_{\mathcal{I}}$-fixed, they agree as well. This proves injectivity. □

Let $\mathscr{H}(X)$ denote the set of Hermitian forms on X; for $\eta \in \mathscr{H}(X)$ we also write $\eta > 0$.

Definition 26.2.1 An *envelope system over* θ, with parameters $a < c$, is a family

$$\mathbf{\Gamma} = (\Gamma^\eta)_{\eta \in \mathscr{H}(X)}, \qquad \Gamma^\eta \in \mathscr{T}_{a,c}(\theta + \eta),$$

such that

$$\Gamma^\eta = P_{\theta+\eta}[\Gamma^{\eta'}]_{\mathcal{I}} \quad \text{whenever } \eta \geq \eta'. \tag{26.1}$$

Write $\mathscr{S}_{a,c}(\theta)$ for the set of these systems, ordered componentwise. For $\Gamma \in \mathscr{T}_{a,c}(\theta)$, set

$$\mathsf{E}(\Gamma)^\eta = P_{\theta+\eta}[\Gamma]_{\mathcal{I}} \qquad (\eta \in \mathscr{H}(X)).$$

A system is *realized in the background* θ if it equals $\mathsf{E}(\Gamma)$ for some $\Gamma \in \mathscr{T}_{a,c}(\theta)$.

By Proposition 26.2.1, $\mathsf{E}(\Gamma)$ is an envelope system, and the map E is injective. All components of an envelope system have the same multiplier ideals at each $\tau < c$: compare two components Γ^η and $\Gamma^{\eta'}$ with the component indexed by $\eta + \eta'$.

If $u \in \mathrm{PSH}(X, \theta)$, the constant family $\Gamma_\tau = P_\theta[u]_{\mathcal{I}}$, $\tau < c$, with cutoff c, lies in $\mathscr{T}_{a,c}(\theta)$, and its projections form the envelope system $\mathsf{E}(\Gamma)$.

26.3 The limiting curve and recovery

Every Hermitian form dominates all sufficiently small multiples of a fixed Hermitian form. A single decreasing sequence of components therefore determines the limit of the whole system. Projecting the resulting curve back gives the greatest realized system below the original one, and strict inequality means that recovery fails.

Theorem 26.3.1 *Fix a Hermitian form h on X, and let j range over the positive integers. For every $\mathbf{\Gamma} \in \mathscr{S}_{a,c}(\theta)$ and every $\tau < c$, the sequence $\Gamma_\tau^{h/j}$ decreases, and its limit*

$$(\mathsf{R}\mathbf{\Gamma})_\tau = \lim_{j\to\infty} \Gamma_\tau^{h/j}$$

is θ-plurisubharmonic. These limits define an element $\mathsf{R}\mathbf{\Gamma}$ of $\mathscr{T}_{a,c}(\theta)$, which does not depend on h. The maps E and R satisfy

$$\mathsf{R}\mathsf{E} = \mathrm{id}, \qquad \mathsf{E}\mathsf{R}(\mathbf{\Gamma}) \leq \mathbf{\Gamma}.$$

Both maps preserve order. More precisely,

$$\mathsf{E}(\Gamma) \leq \mathbf{\Gamma} \quad \Longleftrightarrow \quad \Gamma \leq \mathsf{R}\mathbf{\Gamma}. \tag{26.2}$$

Thus $\mathsf{E}\mathsf{R}(\mathbf{\Gamma})$ is the greatest system below $\mathbf{\Gamma}$ that is realized in the original background.

Proof *Step 1: the limit.* Fix $\tau < c$ and put $q_j = \Gamma_\tau^{h/j}$. By compatibility, $q_j = P_{\theta+h/j}[q_{j+1}]_{\mathcal{I}}$, and $q_{j+1} \leq 0$ is a competitor for this envelope; hence $q_j \geq q_{j+1}$. The

functions q_j have supremum zero and a common smooth lower curvature bound. By Theorem 1.4.1 and Corollary 1.4.1, their decreasing limit q is not identically $-\infty$ and has supremum zero. Passing to the limit in the distributional inequalities

$$\theta + h/j + \mathrm{dd}^{\mathrm{c}} q_j \geq 0$$

gives $q \in \mathrm{PSH}(X, \theta)$.

The defining family of an ideal envelope grows when the background increases. Together with $q \leq q_j$ and order preservation, this gives

$$q \leq P_\theta[q]_{\mathcal{I}} \leq P_{\theta+h/j}[q]_{\mathcal{I}} \leq P_{\theta+h/j}[q_j]_{\mathcal{I}} = q_j.$$

Taking the infimum over j proves $P_\theta[q]_{\mathcal{I}} = q$. Concavity, decreasingness and constancy on the left half-line pass to the pointwise limits, so these limits form a curve in $\mathscr{T}_{a,c}(\theta)$.

Step 2: independence of h. For every $\eta > 0$ there is j with $h/j \leq \eta$, and compatibility then gives $\Gamma_\tau^\eta \geq \Gamma_\tau^{h/j}$. Consequently

$$(\mathsf{R}\boldsymbol{\Gamma})_\tau = \inf_{\eta \in \mathscr{H}(X)} \Gamma_\tau^\eta,$$

which proves that $\mathsf{R}\boldsymbol{\Gamma}$ does not depend on h. This formula also shows that R preserves order.

Step 3: the two compositions. Write $q = (\mathsf{R}\boldsymbol{\Gamma})_\tau$. Since $q \leq \Gamma_\tau^\eta$ and Γ_τ^η is fixed by its envelope, order preservation gives $P_{\theta+\eta}[q]_{\mathcal{I}} \leq \Gamma_\tau^\eta$. This proves $\mathsf{ER} \leq \mathrm{id}$.

Now start with a curve $\Gamma \in \mathscr{T}_{a,c}(\theta)$. Its slice Γ_τ is a competitor in every larger background, so

$$\Gamma_\tau \leq (\mathsf{R}\mathsf{E}\Gamma)_\tau \leq P_{\theta+\eta}[\Gamma_\tau]_{\mathcal{I}} \quad (\eta > 0).$$

The two outer terms have the same multiplier ideals, so the two ideal inclusions given by these inequalities show that the middle term has those ideals as well. Since both the middle term and Γ_τ are $P_\theta[\cdot]_{\mathcal{I}}$-fixed, they are equal. This proves $\mathsf{RE} = \mathrm{id}$.

Step 4: the adjunction. By Proposition 26.2.1, E preserves order. If $\mathsf{E}(\Gamma) \leq \boldsymbol{\Gamma}$, applying R gives $\Gamma = \mathsf{RE}(\Gamma) \leq \mathsf{R}\boldsymbol{\Gamma}$. Conversely, if $\Gamma \leq \mathsf{R}\boldsymbol{\Gamma}$, applying E and using $\mathsf{ER} \leq \mathrm{id}$ gives $\mathsf{E}(\Gamma) \leq \boldsymbol{\Gamma}$. This proves (26.2), and the final maximality assertion follows from it. □

Proposition 26.3.1 *A system $\mathbf{\Gamma} \in \mathcal{S}_{a,c}(\theta)$ is realized in θ if and only if, for every $\tau < c$, its limiting slice $(\mathsf{R}\mathbf{\Gamma})_\tau$ is multiplier-ideal equivalent to one, and hence every, slice Γ^η_τ.*

Proof For a realized system, this follows from $\mathsf{RE} = \mathrm{id}$ and the preservation of multiplier ideals by each projection. Conversely, the assumed equivalence and the fact that Γ^η_τ is fixed by its envelope in the larger background give

$$P_{\theta+\eta}[(\mathsf{R}\mathbf{\Gamma})_\tau]_I = \Gamma^\eta_\tau.$$

Thus the system equals $\mathsf{ER}(\mathbf{\Gamma})$. □

By Theorem 1.16.1, multiplier-ideal equivalence means equality of all Lelong numbers on all local modifications. Recovery therefore holds exactly when the limiting slices keep these Lelong numbers. They are the same for all components of a system, whereas the limiting slice can have larger values: in the ruled-surface example of Section 26.5, one of them increases from zero to one. For big backgrounds on manifolds with the bounded mass property, Theorem 31.5.3 excludes such an increase when the initial slices have positive limiting lower volume.

When X is Kähler, the Kähler forms are cofinal among the Hermitian forms: a sufficiently small positive multiple of a fixed Kähler form lies below any given Hermitian form. The compatibility condition therefore determines a system by its components indexed by Kähler forms. Conversely, every compatible family indexed by Kähler forms extends uniquely to a system: for $\eta > 0$, choose a Kähler form $\kappa \le \eta$ and set $\Gamma^\eta = P_{\theta+\eta}[\Gamma^\kappa]_I$. Two choices of κ give the same result, as one sees by passing to a common smaller Kähler form and using the transition identity.

26.4 Changes of representative and limiting mass

Subtracting a smooth function transports a curve to a new background, and projection restores the envelope normalization. The gauge estimate is uniform in the perturbation, so this transport also respects the limiting curve. The limiting Monge–Ampère mass behaves differently: in a nonclosed background, it can change even for the system all of whose slices are zero.

Proposition 26.4.1 *Let $f \in C^\infty(X,\mathbb{R})$ and $\theta' = \theta + \mathrm{dd}^{\mathrm{c}} f$. For $\Gamma \in \mathcal{T}_{a,c}(\theta)$ and $\mathbf{\Gamma} \in \mathcal{S}_{a,c}(\theta)$, the formulas*

$$(\mathsf{G}_f\Gamma)_\tau = P_{\theta'}[\Gamma_\tau - f]_I, \qquad (\mathsf{G}_f\mathbf{\Gamma})^\eta_\tau = P_{\theta'+\eta}[\Gamma^\eta_\tau - f]_I$$

define bijections $\mathsf{G}_f\colon \mathcal{T}_{a,c}(\theta) \to \mathcal{T}_{a,c}(\theta')$ and $\mathsf{G}_f\colon \mathcal{S}_{a,c}(\theta) \to \mathcal{S}_{a,c}(\theta')$. They satisfy the cocycle law $\mathsf{G}_g\mathsf{G}_f = \mathsf{G}_{f+g}$, where G_g is formed in the background θ', and they commute with E and R.

Proof Subtracting f changes the background from θ to θ' and preserves concavity, decreasingness and constancy on the left half-line. Slicewise projection preserves these properties by the argument of Proposition 26.2.1. Since f is smooth, every new slice has the multiplier ideals of the original one. Applying the construction with $-f$ therefore gives a $P_\theta[\cdot]_I$-fixed slice with the same ideals as the original slice, hence the original slice itself. This proves that G_{-f} is the inverse of G_f; the same uniqueness of fixed representatives proves the cocycle law.

Compatibility of the transformed system and commutation with E follow by comparing multiplier ideals in the common final background. For commutation with R, the gauge estimate of Proposition 17.3.1, applied in each background $\theta+\eta$, gives

$$\Gamma^\eta_\tau + \min_X f \le (\mathsf{G}_f\mathbf{\Gamma})^\eta_\tau + f \le \Gamma^\eta_\tau + \max_X f,$$

with constants independent of η. Pass to the decreasing limits along a cofinal sequence. The new limit has the same ordinary singularity type as the old limit minus f, and hence the same multiplier ideals. It is $P_{\theta'}[\cdot]_I$-fixed by Theorem 26.3.1, so it equals $P_{\theta'}[(\mathsf{R}\mathbf{\Gamma})_\tau - f]_I$. This proves commutation with R. □

For closed backgrounds on a compact Kähler manifold, [Xia26, Section 13.1] indexes such systems by Kähler perturbations, as above, and defines their volume as a limit of the masses of the initial potentials. The components of those systems are general test curves fixed by the ideal envelope, with initial potentials of positive mass, whereas our curves are constant on a prescribed left half-line and our system spaces impose no mass condition. For nonclosed positive backgrounds, the corresponding limit of masses is not invariant under the gauge bijections.

Proposition 26.4.2 *There are a compact complex surface X, a Hermitian form ω on X, and a smooth real function f with $\omega' = \omega + \mathrm{dd^c} f > 0$, with the following property. For every Hermitian form h on X and all $a < c$, the system $\mathbf{\Gamma} \in \mathcal{S}_{a,c}(\omega)$ all of whose slices are zero is such that the limits*

$$\lim_{\epsilon \downarrow 0} \int_X \mathrm{MA}^{\mathrm{ext}}_{\omega+\epsilon h}(\Gamma_a^{\epsilon h}), \qquad \lim_{\epsilon \downarrow 0} \int_X \mathrm{MA}^{\mathrm{ext}}_{\omega'+\epsilon h}((\mathsf{G}_f \mathbf{\Gamma})_a^{\epsilon h})$$

both exist and are different.

Proof Take the torus X and the smooth perturbation of Proposition 17.3.2, so that $\omega' = \omega + \mathrm{dd^c} f$ is Hermitian and $\int_X (\omega')^2 \neq \int_X \omega^2$. The zero family is an envelope system for every $a < c$, since zero is the multiplier-ideal envelope of the unit-ideal class in a positive background. Its first limit is

$$\lim_{\epsilon \downarrow 0} \int_X (\omega + \epsilon h)^2 = \int_X \omega^2.$$

The gauge bijection sends every slice to $P_{\omega'+\eta}[-f]_{\mathcal{I}} = 0$, since f is smooth and the new background is positive. The second limit is therefore $\int_X (\omega')^2$, which is different. □

26.5 Failure of recovery in a positive background

We construct a Kähler background on a ruled surface over an elliptic curve and an envelope system that is not realized in this background. The obstruction is a section D of the ruling whose class is nef but contains only one closed positive current, namely $[D]$. Prescribing a logarithmic pole along one fiber in the Kähler background leaves the class of D as the residual class. In every positive perturbation of the background, this class has smooth positive representatives, so the perturbed envelopes have only the fiber pole. In the limit, however, the only positive representative left is $[D]$, so the limiting envelope acquires D as an additional pole, and the residual volume tends to zero.

In this section, $E = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$, and X is the compact ruled surface

$$X = (\mathbb{C}_z \times \mathbb{P}^1_w)/\!\sim, \qquad (z, w) \sim (z + 1, w), \qquad (z, w) \sim (z + \mathrm{i}, w + 1).$$

These commuting transformations define a holomorphic $\mathbb{P}^1$-bundle $\pi\colon X \to E$. Let D be its section $w = \infty$. The function $v = w - \operatorname{Im} z$ is invariant under both transformations and descends to a smooth function on $X \setminus D$.

Lemma 26.5.1 *Let $\sigma_0 = (\mathrm{i}/2)\,\mathrm{d}z \wedge \mathrm{d}\bar{z}$ be the flat probability form on E. There is a smooth closed real $(1,1)$-form $\theta \in c_1(\mathcal{O}_X(D))$ whose class is nef and for which*

$$\kappa = \theta + \pi^*\sigma_0$$

is a Kähler form. The only closed positive current of the form $\theta + \mathrm{dd}^{\mathrm{c}}u$ is $[D]$.

Proof Let s_D be the canonical section of $\mathcal{O}_X(D)$, with zero divisor D. For a real parameter $t > 0$, define a smooth Hermitian metric on this line bundle by prescribing the norm of s_D:

$$|s_D|_t^2 = (1 + t|v|^2)^{-1} \quad \text{on } X \setminus D.$$

To check that it extends smoothly, use the coordinate $\xi = 1/w$ near D. The squared norm of a local nonvanishing frame is

$$\bigl(|\xi|^2 + t|1 - (\operatorname{Im} z)\xi|^2\bigr)^{-1},$$

which is smooth and positive. The prescribed norm is invariant under the transformations on the complement of D, so these local extensions define a global metric. Its curvature form is

$$\theta_t = \mathrm{dd}^{\mathrm{c}} \log(1 + t|v|^2) \quad \text{on } X \setminus D.$$

Set $\theta = \theta_1$, and let $\sigma = \sigma_0/\pi$ on E, where the denominator is the number π.

We now estimate the curvature from below. Put $r = |v|^2$ and let H be the complex Hessian of $\log(1 + tr)$ in the coordinates (z, w). Direct differentiation gives

$$\begin{aligned} H_{w\bar{w}} &= \frac{t}{(1+tr)^2}, \\ H_{z\bar{w}} &= \frac{\mathrm{i}t(1 - tv^2)}{2(1+tr)^2}, \\ H_{z\bar{z}} &= \frac{t}{2(1+tr)} - \frac{t^2(\operatorname{Re} v)^2}{(1+tr)^2}. \end{aligned}$$

Consequently

$$H_{z\bar{z}} - \frac{|H_{z\bar{w}}|^2}{H_{w\bar{w}}} = \frac{t(1-tr)}{4(1+tr)} \geq -\frac{t}{4}.$$

Completing the square gives $\theta_t \geq -(t/4)\pi^*\sigma$ on $X \setminus D$, and by continuity this inequality extends across D. The restriction of θ_t to every fiber is a positive Fubini–Study form, including at infinity. Thus $\theta_t + A\pi^*\sigma$ is strictly positive off D whenever $A > t/4$. At $\xi = 0$, its coefficient matrix, with the common factor $\mathrm{i}/(2\pi)$ omitted, is

$$\begin{pmatrix} A & \mathrm{i}/2 \\ -\mathrm{i}/2 & 1/t \end{pmatrix}.$$

Its vertical coefficient is positive and its Schur complement is $A - t/4 > 0$, so the form is strictly positive along D as well. In particular, $\kappa = \theta + \pi^*\sigma_0$ is Kähler.

The function

$$f_t = \log \frac{1 + t|v|^2}{1 + |v|^2}$$

extends smoothly across D and satisfies $\theta_t = \theta + \mathrm{dd}^{\mathrm{c}} f_t$. Letting $t \downarrow 0$ in the curvature lower bound shows that the class of θ is nef in the sense of Section 24.6.

Write $\psi_D = \log |s_D|_1^2$. By the Poincaré–Lelong formula, $\theta + \mathrm{dd}^{\mathrm{c}}\psi_D = [D]$. Let $u \in \mathrm{PSH}(X, \theta)$. Then $u - \psi_D$ is plurisubharmonic on $X \setminus D$. Its lift $\Phi(z, w)$ to $\mathbb{C}^2$ is invariant under the two deck transformations and, since u is bounded above on the compact surface X, satisfies

$$\Phi(z, w) \le C + \log\bigl(1 + |w - \operatorname{Im} z|^2\bigr)$$

for some real constant C. For each fixed w, the function $z \mapsto \Phi(z, w)$ is either identically $-\infty$ or subharmonic. It is 1-periodic and has logarithmic upper growth, so Lemma 1.3.6 makes it constant. Thus $\Phi(z, w) = \phi(w)$, where ϕ is a subharmonic function of w, not identically $-\infty$. The second deck transformation gives $\phi(w + 1) = \phi(w)$, and the bound at $z = 0$ gives logarithmic growth, so the same lemma makes ϕ constant. Therefore $u = \psi_D + C$ off D, and Lemma 1.3.3 extends this equality across D. Hence $\theta + \mathrm{dd}^{\mathrm{c}} u = [D]$. □

Fix a point $p \in E$ and let $F = \pi^{-1}(p)$. The Green-kernel construction in the proof of Theorem 20.3.1 gives a quasi-plurisubharmonic function g on E with $\sigma_0 + \mathrm{dd}^{\mathrm{c}} g = \delta_p$. Its pullback $\psi_F = g \circ \pi$ has a logarithmic pole of coefficient one along F, and

$$\pi^* \sigma_0 + \mathrm{dd}^{\mathrm{c}} \psi_F = [F].$$

Theorem 26.5.1 *For the Kähler background κ of Lemma 26.5.1 and every $a < c$, there is a system $\mathbf{\Gamma} \in \mathcal{S}_{a,c}(\kappa)$ for which*

$$\mathsf{ER}(\mathbf{\Gamma}) \neq \mathbf{\Gamma}.$$

Every slice Γ_τ^η, $\tau < c$, of every component has multiplier ideals

$$\mathcal{I}\,(t\Gamma_\tau^\eta) = \mathcal{O}_X(-\lfloor t \rfloor F) \qquad (t > 0),$$

whereas its limiting slices have multiplier ideals

$$\mathcal{I}\,\bigl(t(\mathsf{R}\mathbf{\Gamma})_\tau\bigr) = \mathcal{O}_X(-\lfloor t \rfloor (F + D)).$$

In particular, on the system space $\mathcal{S}_{a,c}(\kappa)$, which carries no mass condition, E is not surjective and R is not injective.

Proof For each Hermitian form η, the nef approximation of Lemma 26.5.1 gives a smooth function f_η such that $\theta + \eta + \mathrm{dd}^{\mathrm{c}} f_\eta \ge 0$. Then

$$u_\eta = \psi_F + f_\eta \in \mathrm{PSH}(X, \kappa + \eta).$$

Let every slice Γ^η_τ, $\tau < c$, of the component Γ^η be $q_\eta = P_{\kappa+\eta}[u_\eta]_{\mathcal{I}}$, with cutoff c. Since u_η has analytic singularities, Proposition 17.2.4 gives $q_\eta \sim u_\eta \sim \psi_F$; in particular q_η has the multiplier ideals of ψ_F. Hence the compatibility condition follows by comparing these ideals in the larger background, and the constant curves form a system with the required parameters.

Let q be its limiting slice under R, and let $\nu(q, F)$ be its generic Lelong number along F. Since $q \le q_\eta$ and $q_\eta \sim \psi_F$, we have $\nu(q, F) \ge \nu(q_\eta, F) = 1$. By Lemma 1.17.1,

$$\kappa + \mathrm{dd^c} q - [F] \ge 0.$$

By the Poincaré–Lelong identity for ψ_F, this current equals $\theta + \mathrm{dd^c}(q - \psi_F)$, where the difference $q - \psi_F$ is first taken as a locally integrable function. Since the current is positive, this difference has a canonical θ-plurisubharmonic representative, and by Lemma 26.5.1 it differs from ψ_D by a constant. Hence

$$q = \psi_F + \psi_D + C.$$

The divisors F and D are smooth and meet transversely, since one is a fiber and the other a section. A direct integrability calculation in local product coordinates therefore gives the stated ideals for every real $t > 0$. In particular, the Lelong number at a generic point of D has increased from zero to one, and Proposition 26.3.1 shows that recovery fails.

Finally, since $\mathsf{RE} = \mathrm{id}$, the system $\mathbf{\Gamma}$ and the realized system $\mathsf{ER}\mathbf{\Gamma} \ne \mathbf{\Gamma}$ have the same image under R. Thus R is not injective, and $\mathbf{\Gamma}$ is not in the image of E. □

Proposition 26.5.1 *Let q_η be the common slice of the component Γ^η constructed in the preceding proof. Along the closed perturbations $\eta = \epsilon\kappa$, with real $\epsilon > 0$,*

$$\int_X \mathrm{MA}^{\mathrm{ext}}_{\kappa+\epsilon\kappa}(q_{\epsilon\kappa}) = 2\epsilon \int_X \theta \wedge \kappa + \epsilon^2 \int_X \kappa^2 \longrightarrow 0.$$

The limiting slice has zero non-pluripolar Monge–Ampère measure.

Proof By the proof of Theorem 26.5.1, $q_\eta \sim \psi_F$. Removing the divisorial component $[F]$ gives a positive current in the background $\theta + \eta$, whose global potential $w_\eta = q_\eta - \psi_F$ is bounded by this ordinary equivalence.

Off F, plurifine locality identifies the Monge–Ampère measure of q_η with $(\theta + \eta + \mathrm{dd^c} w_\eta)^2$. This bounded product gives no mass to F by Proposition 1.7.1. For closed $\eta = \epsilon\kappa$, Stokes' theorem gives its total mass as $\int_X (\theta + \epsilon\kappa)^2$.

The normal bundle of D is trivial. Indeed, the transformation $w \mapsto w + 1$ becomes $\xi \mapsto \xi/(1 + \xi)$, whose derivative at $\xi = 0$ is one, and the other transformation fixes w. Hence $\int_X \theta^2 = D^2 = 0$, and expanding the square gives the formula. The limiting potential has curvature current $[F] + [D]$, which vanishes away from the pluripolar set $F \cup D$; its non-pluripolar measure is therefore zero. □

Part VI
Gauduchon duality and finite volumes

Chapter 27
A Gauduchon variational volume

Let X be a connected compact complex manifold of dimension $n \geq 1$, let θ be a smooth real $(1,1)$-form, which we call the *background*, and let μ be a smooth strictly positive probability form on X. If $\theta + \mathrm{dd}^{\mathrm{c}} v > 0$ for some smooth real function v, there is exactly one constant $c_\theta(\mu) > 0$ for which the Monge–Ampère equation $(\theta + \mathrm{dd}^{\mathrm{c}} u)^n = c_\theta(\mu)\mu$ has a smooth solution u with $\theta + \mathrm{dd}^{\mathrm{c}} u > 0$. When θ is closed, Stokes' theorem gives $c_\theta(\mu) = \int_X \theta^n$, independently of μ. For a nonclosed background the total mass $\int_X (\theta + \mathrm{dd}^{\mathrm{c}} u)^n$ can depend on the potential, and the constant is the one number that the equation attaches to the pair (θ, μ). By Theorem 10.8.1, its nth root is the infimum of the quotients $\int_X \theta \wedge \gamma^{n-1} / D_\mu(\gamma)$ over Gauduchon metrics γ. For every pseudoeffective background this infimum is the root volume $\lambda_\mu(\theta)$, and $\mathcal{V}_\mu(\theta) = \lambda_\mu(\theta)^n$ is the Gauduchon variational volume of Definition 10.8.1. We keep the notation D_μ, $\mathscr{P}$, $\Omega_\theta = \theta + \mathrm{dd}^{\mathrm{c}} u_\theta$ and $c_\theta(\mu)$ of Section 10.8, where the elementary properties of λ_μ, its dependence on μ and its extrema over μ are proved. This chapter studies the volume as a function of the background.

The root volume is concave. On the positive cone, a minimizing Gauduchon metric computes its derivative, and the Hessian of its negative logarithm defines a Riemannian metric on background classes in each finite-dimensional family. Invariant forms on compact quotients of complex Lie groups and warped products with a curve show that the unit-volume hypersurface of this metric may be complete or may acquire a boundary at finite distance.

For $n \geq 2$, the same quotient computes the subsolution threshold $b_\omega(f)$ for a Hermitian background ω and an integrable density f. Allowing positive dd^{c}-closed currents as tests makes this dual problem compact. For continuous densities the maximizing currents represent exactly the supporting probabilities of the normalization functional; for integrable densities, a uniform tail condition passes the determinant integral to weak limits and again gives a maximizing current. A maximizing test that is a smooth strictly positive form gives more: equality in the trace inequality determines the solution current and forces an exact smooth equation. The exact equation persists when the maximizing form degenerates along a proper analytic set, whereas a threshold solution with a point pole can admit no smooth maximizing test at all. If the solution forms $\omega + \mathrm{dd}^{\mathrm{c}} v$ are uniformly comparable to ω, their Gauduchon

multipliers and the determinant densities of the corresponding exact maximizing currents are bounded above and below uniformly, without bounds on derivatives of the complex Hessian. The determinant measures of Gauduchon currents do not depend on the auxiliary Hermitian form. When a Hermitian form β satisfies $\mathrm{dd}^{\mathrm{c}}(\beta^j) = 0$ for $1 \le j \le n-1$, the normalized Gauduchon currents of every $\beta + \mathrm{dd}^{\mathrm{c}} u > 0$, their determinant measures and a sharp bound for them depend only on β; examples on a flat torus show that these data cannot detect a loss of ellipticity, and that exact maximizers for varying densities can concentrate.

The last part of the chapter concerns the boundary of the positive cone and the lower and upper Monge–Ampère masses, which are the extrema of the volume over μ. At a nef background the variational volume is the limit of the smooth equation constants of nearby positive backgrounds. On a compact Kähler manifold, along a path of backgrounds ending at a nef point, a closed nef core of positive top intersection keeps the root volume positive up to the endpoint, and the decrease of the root volume is measured by a subtraction cost; this cost is also computed on warped products whose endpoint is not big and for a residual on a ruled surface. A projective radial model gives positive variational volume without bigness, even at the finite maximal time of a Chern–Ricci flow; along tests approaching the bigness threshold its determinant denominators collapse, as they must for every such background. On threefolds, curvature conditions characterize the smooth potentials of extremal mass, and they give the upper mass of invariant backgrounds on compact Lie-group quotients. A threefold with zero lower mass shows that positivity for each smooth density need not be uniform over densities; it is a primitive elliptic bundle, and on every such bundle the upper mass is computed exactly and separated potentials keep a positive mass. On an elliptic threefold, by contrast, a nonnegative sum of curvature squares gives a bounded cubic energy identity and hence a positive lower bound for every potential, even when the manifold admits no pluriclosed metric. Finally, nef degeneration gives a further contrast with closed Kähler classes: intersection polynomials are replaced by fractional, logarithmic or oscillating decay, and a single blowup can change its order.

Pseudoeffective and big backgrounds are those of Section 10.8. Throughout the chapter, θ is called *nef* if, for a Hermitian form h and every $\epsilon > 0$, there is a smooth real f_ϵ with $\theta + \mathrm{dd}^{\mathrm{c}} f_\epsilon \ge -\epsilon h$; this is the condition of Section 24.6 for the affine class of θ in Section 25.1. Like pseudoeffectivity and bigness, nefness does not require θ to be closed and is unchanged by a smooth dd^{c} change of θ.

27.1 The equality case of concavity and the first variation

Since λ_μ is an infimum of linear functions, each minimizing Gauduchon metric at θ defines a linear functional that supports λ_μ at θ. If equality holds in the concavity inequality along a segment, a metric minimizing at an interior point must minimize at both endpoints, and Theorem 10.8.1 then makes the two endpoint solution forms proportional. Differentiating the smooth equation shows that the supporting functional is the derivative of λ_μ. For $\theta \in \mathscr{P}$ we write γ_θ for a minimizing Gauduchon metric, which exists by Theorem 10.8.1; variation directions are smooth real $(1,1)$-forms.

Theorem 27.1.1 *Let $\theta_0, \theta_1 \in \mathscr{P}$ and $0 < s < 1$. Equality holds in the concavity inequality*

$$\lambda_\mu((1-s)\theta_0 + s\theta_1) \geq (1-s)\lambda_\mu(\theta_0) + s\lambda_\mu(\theta_1)$$

if and only if, for some $r > 0$ and some smooth real f,

$$\theta_1 = r\theta_0 + \mathrm{dd}^{\mathrm{c}} f.$$

Proof Suppose first that $n \geq 2$ and that equality holds. Take a minimizing Gauduchon metric for the middle background $(1-s)\theta_0 + s\theta_1$. Its quotient at the middle background is the same convex combination of its quotients at θ_0 and θ_1, and each of these is at least the corresponding infimum. Equality therefore forces this metric to minimize both endpoint quotients.

By Theorem 10.8.1, the solution forms $\Omega_i = \Omega_{\theta_i}$, with constants $c_i = c_{\theta_i}(\mu)$ for $i = 0, 1$, are then both conformal to this metric. Their volume forms are $c_0\mu$ and $c_1\mu$, so their pointwise proportionality factor is the constant $r = (c_1/c_0)^{1/n}$. Hence $\Omega_1 = r\Omega_0$, which gives the stated smooth dd^{c} relation. Conversely, if that relation holds, invariance under smooth dd^{c} changes and homogeneity give equality.

When $n = 1$, the function is $\theta \mapsto \int_X \theta$. Take $r = \int_X \theta_1 / \int_X \theta_0$. The difference $\theta_1 - r\theta_0$ has integral zero, so the Green operator of the compact curve writes it as $\mathrm{dd}^{\mathrm{c}} f$ with f smooth. Thus the same equivalence holds. □

Proposition 27.1.1 *The function λ_μ is smooth along smooth finite-dimensional families in $\mathscr{P}$. For a smooth real $(1,1)$-form χ and $n \geq 2$, its derivative at θ is*

$$(\mathrm{d}\lambda_\mu)_\theta(\chi) = \frac{\displaystyle\int_X \chi \wedge \gamma_\theta^{n-1}}{D_\mu(\gamma_\theta)},$$

where γ_θ is any minimizing Gauduchon metric. In dimension one the derivative is $\int_X \chi$.

Proof Choose a smooth potential u_* with $\theta + \mathrm{dd}^{\mathrm{c}} u_* > 0$. Backgrounds θ_t near θ then have the positive representatives $\theta_t + \mathrm{dd}^{\mathrm{c}} u_*$. The implicit-function argument in the proof of Proposition 16.8.1 applies to this family and shows that the solution and its constant depend smoothly on the parameters. Adding u_* to these solution potentials and fixing the mean gives the normalized solutions for θ_t, so $\lambda_\mu(\theta_t) = c_{\theta_t}(\mu)^{1/n}$ is smooth.

To compute the derivative, write $\lambda = \lambda_\mu(\theta)$ and $\Omega = \Omega_\theta$, and write the minimizing metric as $\gamma_\theta = \mathrm{e}^g\Omega$ with g smooth and real, as Theorem 10.8.1 permits. Along $\theta_t = \theta + t\chi$ for small real t, put $\Omega_t = \theta_t + \mathrm{dd^c}u_t$ and $\lambda_t = \lambda_\mu(\theta_t)$, and let dots denote derivatives at $t = 0$. Differentiating $\Omega_t^n = \lambda_t^n\mu$ gives

$$(\chi + \mathrm{dd^c}\dot{u}) \wedge \Omega^{n-1} = \lambda^{n-1}\dot{\lambda}\,\mu.$$

We multiply by $\mathrm{e}^{(n-1)g}$, so that Ω^{n-1} becomes γ_θ^{n-1}, and integrate. The term with $\mathrm{dd^c}\dot{u}$ vanishes by the Gauduchon condition, and $\int_X \chi \wedge \gamma_\theta^{n-1} = \lambda^{n-1}\dot{\lambda}\int_X \mathrm{e}^{(n-1)g}\mu$. On the other hand, $\gamma_\theta^n = \mathrm{e}^{ng}\lambda^n\mu$, so that

$$D_\mu(\gamma_\theta) = \lambda^{n-1}\int_X \mathrm{e}^{(n-1)g}\mu.$$

Dividing the two identities proves the derivative formula. In dimension one the formula follows directly from the definition. □

27.2 Second variation on background classes

Differentiating the smooth equation twice along an affine family of backgrounds computes the second derivative of the root volume. After integration against the adjoint probability of the solution form, the second derivative of the potential disappears, and what remains is the integrated squared norm of the linearized solution form. Consequently the Hessian of $-\log\lambda_\mu$ is positive definite once smooth $\mathrm{dd^c}$ directions are factored out. On the cone of closed Kähler classes, where $\lambda_\mu(\theta) = (\int_X \theta^n)^{1/n}$, it is $1/n$ times the Hessian of $-\log\int_X \theta^n$.

Fix $\theta \in \mathscr{P}$ and write $\Omega = \Omega_\theta$, $u = u_\theta$ and $\lambda = \lambda_\mu(\theta)$. Let ν_Ω be the adjoint probability of Ω from Proposition 4.8.1, that is, the smooth positive probability measure, given by the Gauduchon multiplier of Ω, which satisfies $\int_X \mathrm{tr}_\Omega(\mathrm{dd^c}v)\,\mathrm{d}\nu_\Omega = 0$ for every smooth real v. In dimension one this probability is $\Omega/\int_X \Omega = \mu$.

Let $\theta_t = \theta + t\chi$, where t is real and χ is a smooth real $(1,1)$-form on X. For sufficiently small $|t|$, the background θ_t stays in $\mathscr{P}$. We normalize the solution potentials u_t by one fixed condition on their mean, and dots denote derivatives of u_t and of $\lambda_\mu(\theta_t)$ at $t = 0$. Define the Hermitian endomorphism

$$A_\chi = \Omega^{-1}(\chi + \mathrm{dd^c}\dot{u})$$

of $T^{1,0}X$, where Ω^{-1} means raising an index with the positive form Ω; traces of powers of A_χ are independent of coordinates. We write I for the identity endomorphism of $T^{1,0}X$ and $\|\cdot\|_\Omega$ for the Hilbert–Schmidt norm on its endomorphisms induced by Ω.

Proposition 27.2.1 *Along this affine variation,*

$$\frac{\ddot{\lambda}}{\lambda} = -\frac{1}{n}\int_X \left\| A_\chi - \frac{\dot{\lambda}}{\lambda} I \right\|_\Omega^2 \, \mathrm{d}\nu_\Omega, \tag{27.1}$$

$$-\frac{\mathrm{d}^2}{\mathrm{d}t^2} \log \lambda_\mu(\theta_t) = \frac{1}{n}\int_X \mathrm{tr}(A_\chi^2)\, \mathrm{d}\nu_\Omega. \tag{27.2}$$

The first expression vanishes exactly when $\chi = r\theta + \mathrm{dd}^{\mathrm{c}} f$ for a real constant r and a smooth real f. The second vanishes exactly when $\chi = \mathrm{dd}^{\mathrm{c}} f$ for a smooth real f.

Proof Smooth dependence on t was proved in Proposition 27.1.1. We differentiate the logarithmic form $\log(\Omega_t^n/\mu) = n \log \lambda_\mu(\theta_t)$ of the equation. Since μ is fixed,

$$\mathrm{tr}\, A_\chi = n\frac{\dot{\lambda}}{\lambda}.$$

In particular this trace is constant on X. Differentiating once more along the affine path θ_t gives

$$\mathrm{tr}_\Omega(\mathrm{dd}^{\mathrm{c}}\ddot{u}) - \mathrm{tr}(A_\chi^2) = n\left(\frac{\ddot{\lambda}}{\lambda} - \frac{\dot{\lambda}^2}{\lambda^2}\right).$$

Integration against the probability ν_Ω, which is fixed since everything is evaluated at $t = 0$, removes the first term. This proves (27.2).

Put $r = \dot{\lambda}/\lambda$. Since A_χ is Hermitian and $\mathrm{tr}\, A_\chi = nr$,

$$\|A_\chi - rI\|_\Omega^2 = \mathrm{tr}(A_\chi^2) - nr^2.$$

Substituting this into (27.2) gives (27.1). Since the adjoint measure has a strictly positive smooth density, the right side of (27.1) vanishes exactly when $A_\chi = rI$ everywhere, that is, when

$$\chi + \mathrm{dd}^{\mathrm{c}}\dot{u} = r(\theta + \mathrm{dd}^{\mathrm{c}} u).$$

This is the stated relation with $f = ru - \dot{u}$. Conversely, if $\chi = r\theta + \mathrm{dd}^{\mathrm{c}} f$, then θ_t differs from $(1 + rt)\theta$ by a smooth dd^{c} term for small t. Homogeneity gives $\lambda_\mu(\theta_t) = (1 + rt)\lambda$, whose second derivative is zero.

The integrand in (27.2) is the squared norm of the Hermitian endomorphism A_χ. Its integral therefore vanishes exactly when $A_\chi = 0$, that is, when $\chi = -\mathrm{dd}^{\mathrm{c}}\ddot{u}$. Conversely, $\lambda_\mu(\theta + t\mathrm{dd}^{\mathrm{c}} f) = \lambda$ for all t. This proves both equality statements. □

The linearized solution and the derivative of the constant depend linearly on χ. For smooth real $(1,1)$-forms χ_1, χ_2 on X, polarization therefore gives a symmetric bilinear form

$$\mathfrak{g}_\theta(\chi_1, \chi_2) = -D^2 \log \lambda_\mu|_\theta(\chi_1, \chi_2) = \frac{1}{n}\int_X \mathrm{tr}(A_{\chi_1} A_{\chi_2})\, \mathrm{d}\nu_\Omega. \tag{27.3}$$

We call the classes of smooth real $(1,1)$-forms modulo $\mathrm{dd}^{\mathrm{c}}C^{\infty}(X,\mathbb{R})$ *background classes*. By Proposition 27.2.1, the form $\mathfrak{g}_\theta$ descends to background classes of directions and is positive definite there. A change of representative of θ leaves its solution form Ω unchanged, and a change of representative of a direction is absorbed by the derivative of the solution potential. Thus $\mathfrak{g}$ is intrinsically defined on the background classes in $\mathscr{P}$. On every finite-dimensional affine family whose directions are linearly independent modulo smooth dd^{c} terms, it is a smooth Riemannian metric.

These variations change the background class. Directions obtained by varying a potential inside one fixed class are exactly the directions annihilated by (27.3). Thus this construction concerns a different space from the potential-space geometry of Chapter 3.

27.3 Recovering a positive class from its Gauduchon derivative

By Proposition 27.1.1, the derivative of λ_μ at $\theta \in \mathscr{P}$ is represented by a minimizing Gauduchon metric, which is determined up to a constant factor. The solution form is conformal to this metric and has a prescribed volume form, so the value and the derivative of λ_μ together reconstruct the solution form, and hence the background class. Fix $\theta \in \mathscr{P}$ and a minimizing metric γ_θ.

For $n \geq 2$ define the smooth positive $(n-1,n-1)$-form

$$\Xi_\theta = \frac{\gamma_\theta^{n-1}}{D_\mu(\gamma_\theta)}.$$

Since numerator and denominator are both homogeneous of degree $n-1$ in γ_θ, this form does not depend on the choice of the minimizing metric. It is dd^{c}-closed, and by Proposition 27.1.1 it represents the derivative in every smooth real $(1,1)$-direction χ:

$$(\mathrm{d}\lambda_\mu)_\theta(\chi) = \int_X \chi \wedge \Xi_\theta.$$

Proposition 27.3.1 *Let $n \geq 2$ and $\theta \in \mathscr{P}$. The value $\lambda_\mu(\theta)$ and the derivative of λ_μ at θ on all smooth real $(1,1)$-forms together determine the background class of θ, and the derivative alone determines this class up to multiplication by a positive constant. More precisely, if $\widehat{\gamma}$ is the positive $(1,1)$-form with $\widehat{\gamma}^{n-1} = \Xi_\theta$, then*

$$D_\mu(\widehat{\gamma}) = 1, \qquad \Omega_\theta = \lambda_\mu(\theta)\left(\frac{\mu}{\widehat{\gamma}^n}\right)^{1/n}\widehat{\gamma}.$$

Moreover, assigning to a background class in $\mathscr{P}$ with $\lambda_\mu = 1$ the ray of its minimizing Gauduchon metrics is a bijection onto the set of rays of Gauduchon metrics.

Proof The values of the derivative on all smooth $(1,1)$-forms determine the smooth form Ξ_θ. Its positive root exists and is unique by the positive-root construction

preceding the proof of Theorem 4.3.1. If $D = D_\mu(\gamma_\theta)$, this root is $\widehat{\gamma} = D^{-1/(n-1)}\gamma_\theta$, and the homogeneity of D_μ gives $D_\mu(\widehat{\gamma}) = 1$.

By Theorem 10.8.1, the solution form Ω_θ is conformal to $\widehat{\gamma}$, and its volume form is $\lambda_\mu(\theta)^n\mu$. These two requirements determine the conformal factor pointwise and give the formula for Ω_θ. Thus the value and the derivative recover Ω_θ, a representative of the class of θ. Without the value, the same formula recovers Ω_θ up to a positive constant factor. Conversely, multiplying the class by a positive constant multiplies Ω_θ by the same constant and leaves its minimizing Gauduchon ray, and hence Ξ_θ, unchanged; so the derivative cannot determine more than the ray of the class.

For the surjectivity in the last assertion, take any Gauduchon metric γ and set

$$\Omega_\gamma = \left(\frac{\mu}{\gamma^n}\right)^{1/n}\gamma.$$

This Hermitian form is unchanged by constant rescaling of γ and satisfies $\Omega_\gamma^n = \mu$. The zero potential therefore solves the equation for the background Ω_γ with constant one, so its class has $\lambda_\mu = 1$, and γ is a minimizing Gauduchon metric by Theorem 10.8.1. Injectivity follows from the recovery formula: two classes with $\lambda_\mu = 1$ and the same Gauduchon ray have the same solution form. This proves the bijection. □

In dimension one, a background class is determined by its integral, by the Green operator of the curve. The derivative is then the fixed functional $\chi \mapsto \int_X \chi$, and the value alone determines the class.

27.4 The Hessian metric and its completion

We now study the metric $\mathfrak{g}$ of (27.3) on finite-dimensional cones of backgrounds, where it defines an ordinary Riemannian path distance. Homogeneity of λ_μ separates the scale of a positive class from its position on a level set of λ_μ, and the logarithm of this scale turns out to be a Euclidean coordinate orthogonal to the level sets. The metric is therefore a product, and its completeness reduces to that of the unit-volume hypersurface.

Let V be a finite-dimensional real vector space of smooth real $(1,1)$-forms such that

$$V \cap \mathrm{dd^c}C^\infty(X,\mathbb{R}) = \{0\}, \qquad \mathscr{C} = V \cap \mathscr{P} \neq \varnothing.$$

Then $\mathscr{C}$ is an open convex cone in V. Write $\lambda = \lambda_\mu|_{\mathscr{C}}$, and let $g = -D^2 \log \lambda$ be the restriction to $\mathscr{C}$ of the metric (27.3); it is a smooth Riemannian metric on $\mathscr{C}$ because V meets $\mathrm{dd^c}C^\infty(X,\mathbb{R})$ only in zero. Put

$$\mathscr{S} = \{\theta \in \mathscr{C} : \lambda(\theta) = 1\},$$

and equip this smooth hypersurface with the induced metric $g_{\mathscr{S}}$.

Proposition 27.4.1 *The map*

$$\Phi\colon \mathbb{R}\times\mathcal{S}\longrightarrow\mathscr{C},\qquad (r,q)\longmapsto \mathrm{e}^{r}q$$

is an isometry from the product metric $\mathrm{d}r^2+g_{\mathcal{S}}$ *onto* g*. In particular, if* d_g *and* $d_{\mathcal{S}}$ *are the intrinsic distances, then*

$$d_g(\mathrm{e}^{r_0}q_0,\mathrm{e}^{r_1}q_1)^2=|r_1-r_0|^2+d_{\mathcal{S}}(q_0,q_1)^2.$$

The metric on $\mathscr{C}$ *is complete if and only if the induced metric on* $\mathcal{S}$ *is complete. Every path for which* λ *tends to zero or to infinity has infinite length.*

Proof Homogeneity gives $\lambda(s\theta)=s\lambda(\theta)$ for $s>0$. Hence $r=\log\lambda(\theta)$ and $q=\theta/\lambda(\theta)$ define the smooth inverse of Φ. The level set is smooth because the Euler identity gives $D\lambda_\theta(\theta)=\lambda(\theta)>0$.

Differentiating the Euler identity $D\log\lambda_\theta(\theta)=1$ in a direction $\chi\in V$ gives

$$g_\theta(\theta,\chi)=D\log\lambda_\theta(\chi),\qquad g_\theta(\theta,\theta)=1.$$

For directions $\chi,\eta\in V$, differentiating homogeneity twice also gives

$$g_{s\theta}(s\chi,s\eta)=g_\theta(\chi,\eta).$$

Let $q\in\mathcal{S}$, let $\xi,\eta\in T_q\mathcal{S}$, and let a,b,r be real numbers. Since ξ and η are tangent to a level set of λ, the first of these identities shows that both are g_q-orthogonal to the radial vector q. Using also the second identity and the scaling rule, we obtain

$$g_{\mathrm{e}^r q}\big(\mathrm{e}^r(aq+\xi),\mathrm{e}^r(bq+\eta)\big)=ab+g_q(\xi,\eta),$$

which proves the metric identity.

It remains to prove the distance formula and its consequences. A path $(r(t),q(t))$ in the product has length $\int\sqrt{|r'|^2+|q'|_{g_{\mathcal{S}}}^2}\,\mathrm{d}t$. The triangle inequality in $\mathbb{R}^2$ (applied to the integral of the vector $(|r'|,|q'|_{g_{\mathcal{S}}})$) bounds this below by the square root of the sum of the squared endpoint difference in r and the squared length of the projected path in $\mathcal{S}$. For the opposite bound, choose a path in $\mathcal{S}$ of length within ϵ of the distance of its endpoints, parametrize it with constant speed, and let r vary linearly; then let ϵ decrease to zero. This proves the distance formula. It identifies Cauchy sequences in $\mathscr{C}$ with pairs of Cauchy sequences in the two factors; since $\mathbb{R}$ is complete, the completeness assertion follows. Finally, the length of any path is at least the total change of $\log\lambda$ along it, and this change is infinite when λ tends to zero or to infinity. □

27.4.1 Product and blowup examples

The following two examples of closed classes show that the unit-volume hypersurface may be complete or may have a boundary at finite distance. The first also shows why the derivative is needed in Proposition 27.3.1: knowing the volume for every smooth probability measure need not identify the class.

For the first example take $X = \mathbb{P}^1 \times \mathbb{P}^1$, and let α, β be the pulled-back Fubini–Study forms of integral one on their respective factors. They satisfy

$$\alpha^2 = \beta^2 = 0, \qquad \int_X \alpha \wedge \beta = 1.$$

For $a, b > 0$ the form $\theta_{a,b} = a\alpha + b\beta$ is Kähler. Since it is closed, integrating the smooth equation gives, for every smooth positive probability measure μ,

$$\mathcal{V}_\mu(\theta_{a,b}) = 2ab, \qquad \lambda_\mu(\theta_{a,b}) = \sqrt{2ab}.$$

Consequently, for real t, all the forms $\theta_{e^t, e^{-t}}$ have volume 2 for every μ. Their background classes are nevertheless distinct: for any fixed point p of the second factor, integration over $\mathbb{P}^1 \times \{p\}$ gives e^t, and this integral is unchanged by the addition of $dd^c f$.

By direct differentiation, the restriction of (27.3) to this two-dimensional cone is

$$\mathfrak{g} = \frac{1}{2}\left(\frac{da^2}{a^2} + \frac{db^2}{b^2}\right).$$

The coordinates $((\log a)/\sqrt{2}, (\log b)/\sqrt{2})$ identify this metric with the Euclidean metric on $\mathbb{R}^2$, so the metric on this family is complete. The curve $a = e^t$, $b = e^{-t}$ has unit speed. All distances here are intrinsic to this two-dimensional family.

In the second example, the blowup of $\mathbb{P}^2$ at a point, the cone is incomplete, and the boundary added by the completion consists of classes of positive volume.

Example 27.4.1 Let $\pi\colon X \to \mathbb{P}^2$ be the blowup at one point. Write H for the pullback of the hyperplane class and E for the exceptional divisor class. In the two-dimensional space they span, the cone of classes with a positive smooth representative is

$$\mathscr{C} = \{aH - bE : a > b > 0\}.$$

These are closed classes, and we represent them by fixed smooth forms in order to apply Proposition 27.4.1. Their intersection numbers are $H^2 = 1$, $H \cdot E = 0$, $E^2 = -1$. Consequently, for every smooth positive probability measure μ,

$$\lambda_\mu(aH - bE) = \sqrt{a^2 - b^2}.$$

In the coordinates

$$a = e^r \cosh s, \qquad b = e^r \sinh s, \qquad r \in \mathbb{R}, \quad s > 0,$$

the Hessian metric is $\mathrm{d}r^2 + \mathrm{d}s^2$. Its completion is therefore the Euclidean closed half-plane. The added boundary $s = 0$ consists of the nef classes $\mathrm{e}^r H$, each of positive volume e^{2r}.

Proof The projection from the blown-up point extends to a morphism $f\colon X \to \mathbb{P}^1$ with class $L = H - E$. The incidence map $(\pi, f)\colon X \to \mathbb{P}^2 \times \mathbb{P}^1$ is an embedding. If h and ℓ are pullbacks of Fubini–Study forms of integral one, then

$$(a - b)h + b\ell$$

is positive for $a > b > 0$, by the product embedding. Conversely, a positive representative has positive integrals over the exceptional curve and the strict transform of a line through the blown-up point. These integrals are respectively b and $a - b$, proving the cone description. The same curves show that no nonzero linear combination in this two-dimensional space is smoothly dd^{c}-exact.

The intersection numbers follow from the hyperplane pullback and the normal bundle $O_{\mathbb{P}^1}(-1)$ of the exceptional curve; equivalently $H^2 = H \cdot L = 1$ and $L^2 = 0$. Since the classes are closed, integrating the smooth equation gives $\mathcal{V}_\mu = a^2 - b^2$, independently of μ. Direct differentiation of $-\log\sqrt{a^2 - b^2}$ gives

$$g = \frac{(a^2 + b^2)(\mathrm{d}a^2 + \mathrm{d}b^2) - 4ab\,\mathrm{d}a\,\mathrm{d}b}{(a^2 - b^2)^2}.$$

Substituting the stated coordinates gives $\mathrm{d}r^2 + \mathrm{d}s^2$. The coordinate domain is exactly $\mathbb{R} \times (0, \infty)$, so the intrinsic distance and completion are those of that Euclidean half-plane. Finally, H has a semipositive pullback representative and is a limit of positive classes, hence is nef. Its square is one, but its integral over E is zero, so it has no positive representative. This proves all assertions about the added boundary.□

27.4.2 Invariant backgrounds on compact Lie-group quotients

On a compact quotient of a complex Lie group, the backgrounds with constant coefficients in an invariant coframe form a cone of nonclosed backgrounds on which everything can be computed explicitly. With respect to the normalized invariant volume, the root volume is the nth root of a determinant, the minimizing metric is the background itself, and the Hessian metric is the trace metric on positive Hermitian matrices. Its unit-volume hypersurface is complete, so the boundary of the cone of coefficient matrices lies at infinite distance after normalization.

Let G be a connected complex Lie group of dimension $n \geq 2$, and let $X = \Gamma\backslash G$ be compact, where the discrete subgroup Γ acts on the left. Choose a left-invariant holomorphic coframe $\alpha_1, \ldots, \alpha_n$, which descends to X. For a Hermitian matrix A

put

$$\theta_A = \frac{\mathrm{i}}{2\pi}\sum_{j,k} A_{jk}\alpha_j \wedge \bar{\alpha}_k, \qquad h_0 = \theta_I,$$

$$H = \int_X h_0^n, \qquad \mu = \frac{h_0^n}{H}.$$

Here $\mathrm{Herm}(n)$ is the real vector space of Hermitian $n \times n$ matrices, and $A > 0$ (respectively $A \geq 0$) means that A is positive definite (respectively semidefinite). Fix a real linear subspace $W \subset \mathrm{Herm}(n)$ containing a positive matrix, and write $V_W = \{\theta_A : A \in W\}$. Throughout this subsection, the probability form is the normalized invariant volume μ just defined.

Proposition 27.4.2 *The backgrounds in $V_W \cap \mathscr{P}$ are exactly the forms θ_A with $A \in W$ and $A > 0$. Moreover,*

$$V_W \cap \mathrm{dd^c} C^\infty(X, \mathbb{R}) = \{0\}.$$

Every positive θ_A is balanced. Its root volume and its unique minimizing Gauduchon metric normalized by $D_\mu = 1$ are given by

$$\lambda_\mu(\theta_A) = (H \det A)^{1/n}, \qquad \gamma_A = \frac{\theta_A}{\lambda_\mu(\theta_A)}, \qquad \gamma_A^n = \mu. \tag{27.4}$$

On this coefficient cone the Hessian metric is

$$g_A(U, Z) = \frac{1}{n}\operatorname{tr}(A^{-1}UA^{-1}Z), \qquad U, Z \in W. \tag{27.5}$$

For $A \in W$, there is a real distribution u with $\theta_A + \mathrm{dd^c} u \geq 0$ exactly when $A \geq 0$. The closure of $\{A \in W : A > 0\}$ is $W \cap \{A \geq 0\}$, and each of its points A is represented by the smooth semipositive form θ_A. For $A = 0$, the only positive current $\mathrm{dd^c} u$ is zero. If G is nonabelian, every positive background in this cone is nonclosed, and adding a smooth $\mathrm{dd^c}$ term never makes it closed.

Proof *Invariant averages and balancedness.* Let $Z_1, \dots, Z_n$ be the invariant holomorphic vector fields dual to $\alpha_1, \dots, \alpha_n$, and set $\Upsilon = \alpha_1 \wedge \cdots \wedge \alpha_n$. This holomorphic top-degree form and its conjugate are closed. For an invariant holomorphic vector field Z, evaluation on the invariant frame gives

$$\mathcal{L}_Z \Upsilon = -\operatorname{tr}(\operatorname{ad} Z)\Upsilon, \qquad \mathcal{L}_Z \bar{\Upsilon} = 0.$$

Since h_0^n is a positive constant multiple of the appropriately oriented form $\Upsilon \wedge \bar{\Upsilon}$, Stokes' theorem on compact X gives

$$0 = \int_X \mathcal{L}_Z h_0^n = -H \operatorname{tr}(\operatorname{ad} Z).$$

Thus the fields Z_j and their conjugates have zero divergence with respect to μ. In particular,

$$\int_X Z_j f\,\mu = \int_X \bar{Z}_j f\,\mu = 0 \qquad (f \in C^\infty(X,\mathbb{C})). \tag{27.6}$$

Cartan's formula also gives $\mathrm{d}(\iota_{Z_j}\Upsilon) = 0$. These contractions span the invariant $(n-1,0)$-forms. The form θ_A^{n-1} is a constant linear combination of their wedges with conjugates, so $\mathrm{d}\theta_A^{n-1} = 0$.

The holomorphic fields commute with the antiholomorphic fields, and $\mathrm{d}\alpha_j$ has type $(2,0)$. The coefficient matrix of $\mathrm{dd^c}u$ is therefore $(Z_j\bar{Z}_k u)$. For smooth u its entries have μ-mean zero by (27.6), so the matrix of μ-means of the coefficients of $\theta_A + \mathrm{dd^c}u$ is A. If $\theta_A + \mathrm{dd^c}u > 0$, averaging the positive coefficient matrix gives $A > 0$. The converse follows by taking $u = 0$. The same average shows that V_W contains no nonzero smooth $\mathrm{dd^c}$ direction.

For a distribution u, the adjoints of Z_j and $\bar{Z}_k$ with respect to μ annihilate constants, so the mean of each $Z_j\bar{Z}_k u$ is still zero. The coefficients of a positive current in this global frame form a positive matrix of measures, whose matrix of total masses is semipositive; for $\theta_A + \mathrm{dd^c}u \geq 0$ this gives $A \geq 0$. Conversely, θ_A itself is a positive current when $A \geq 0$. If $A = 0$, the positive trace measure of $\mathrm{dd^c}u$ has total mass zero, and polarization then makes the entire current zero. Finally, for a fixed positive $A_+ \in W$, the matrices $A + \varepsilon A_+$ with $\varepsilon \downarrow 0$ approximate every semipositive $A \in W$ from the positive cone. This proves the closure assertion.

The quotient and its minimizers. Put $Q = H \det A$, so that $\theta_A^n = Q\mu$. Thus θ_A is its own solution form, with constant $c_{\theta_A}(\mu) = Q$, and it is balanced, hence Gauduchon. By Theorem 10.8.1, $\lambda_\mu(\theta_A) = Q^{1/n}$, and the minimizing Gauduchon metrics are the constant multiples $c\theta_A$, $c > 0$. Since $D_\mu(c\theta_A) = c^{n-1}Q^{(n-1)/n}$, the normalization $D_\mu = 1$ gives $c = \lambda_\mu(\theta_A)^{-1}$, and then $\gamma_A^n = \theta_A^n/Q = \mu$.

Differentiating $-\log\lambda_\mu(\theta_A) = -(\log H + \log\det A)/n$ twice gives (27.5). The corresponding squared norm of U is $\|A^{-1/2}UA^{-1/2}\|_{\mathrm{F}}^2/n$, which is positive for $U \neq 0$; here $\|B\|_{\mathrm{F}}^2 = \mathrm{tr}(B^*B)$ denotes the squared Frobenius norm.

Finally,

$$\partial\theta_A = \frac{\mathrm{i}}{2\pi}\sum_{j,k} A_{jk}\,\mathrm{d}\alpha_j \wedge \bar{\alpha}_k.$$

For invertible A, the vanishing of this form forces every $\mathrm{d}\alpha_j = 0$, by independence of the conjugate coframe, and the Lie algebra would then be abelian. Since adding $\mathrm{dd^c}u$ does not change $\mathrm{d}\theta_A$, this proves the last assertion. □

Lemma 27.4.1 *On the cone of all positive definite Hermitian $n \times n$ matrices, the Riemannian distance of the metric (27.5) is*

$$d_{\text{full}}(A, B) = \frac{1}{\sqrt{n}} \| \log(A^{-1/2} B A^{-1/2}) \|_{\text{F}}. \tag{27.7}$$

For endpoints with equal determinant, this is also the intrinsic distance in the hypersurface of all positive definite matrices with that determinant.

Proof By the trace formula, the congruence $M \mapsto P^* M P$ by a fixed invertible matrix P is an isometry. Taking $P = A^{-1/2}$, we may assume that the endpoints are I and $C = A^{-1/2} B A^{-1/2} > 0$. The curve $\exp(t \log C)$, $0 \le t \le 1$, has constant speed $\| \log C \|_{\text{F}} / \sqrt{n}$, which gives the upper bound for the distance.

For the reverse inequality, diagonalization gives the convergent resolvent formula for the matrix logarithm and its derivative

$$\log M = \int_0^\infty \big((1+s)^{-1} I - (M + sI)^{-1} \big) \, \mathrm{d}s,$$
$$D \log_M [U] = \int_0^\infty (M + sI)^{-1} U (M + sI)^{-1} \, \mathrm{d}s.$$

On a neighborhood of M where the spectrum is bounded below by a positive constant, differentiation under the integral sign is justified by an integrable bound. In a unitary basis diagonalizing M, with eigenvalues $d_i > 0$, the derivative has entries $c_{ij} U_{ij}$, where

$$c_{ij} = \frac{\log d_i - \log d_j}{d_i - d_j} \quad (d_i \ne d_j), \qquad c_{ij} = \frac{1}{d_i} \quad (d_i = d_j).$$

Writing $d_i / d_j = \mathrm{e}^{2\tau}$ gives $\sqrt{d_i d_j}\, |c_{ij}| = |\tau / \sinh \tau| \le 1$, including the limiting value at $\tau = 0$. Hence

$$\| D \log_M [U] \|_{\text{F}} \le \| M^{-1/2} U M^{-1/2} \|_{\text{F}}.$$

Applying this along a piecewise C^1 path of positive matrices from I to C and integrating, we bound its length below by $\| \log C \|_{\text{F}} / \sqrt{n}$. This proves the distance formula. In terms of the original endpoints, the length-minimizing curve is

$$M(t) = A^{1/2} (A^{-1/2} B A^{-1/2})^t A^{1/2}. \tag{27.8}$$

Its determinant is $(\det A)^{1-t} (\det B)^t$, so for endpoints with equal determinant it stays in the fixed-determinant hypersurface. □

Write $\lambda(A) = \lambda_\mu(\theta_A)$ and

$$S_W = \{ A \in W : A > 0, \ H \det A = 1 \}.$$

The radial derivative of λ is positive, so S_W is a smooth hypersurface of the cone $\{ A \in W : A > 0 \}$. We write d_W for the intrinsic path distance of S_W with the metric induced by (27.5).

Theorem 27.4.1 *The metric space (S_W, d_W) is complete and proper, and a sequence in S_W is Cauchy exactly when its coefficient matrices converge to a positive matrix in S_W. For $A, B \in S_W$,*

$$d_W(A,B) \geq \frac{1}{\sqrt{n}} \| \log(A^{-1/2} B A^{-1/2}) \|_{\mathrm{F}}, \tag{27.9}$$

$$d_W(A,B) \leq \frac{\|A - B\|_{\mathrm{F}}}{\sqrt{n}\, \min\{\lambda_{\min}(A), \lambda_{\min}(B)\}}. \tag{27.10}$$

Here $\lambda_{\min}$ denotes the smallest eigenvalue. The normalized minimizing metrics of (27.4) satisfy, pointwise on X,

$$\mathrm{e}^{-\sqrt{n} d_W(A,B)} \gamma_A \leq \gamma_B \leq \mathrm{e}^{\sqrt{n} d_W(A,B)} \gamma_A. \tag{27.11}$$

In particular, the completion of S_W adds no point, and no point of the boundary of the coefficient cone is a limit of a Cauchy sequence.

If $A_j > 0$ in W converges to a nonzero singular semipositive A_0, then $\widehat{A}_j = A_j / \lambda(A_j)$ lies in S_W and, for fixed $A_\star \in S_W$,

$$d_W(A_\star, \widehat{A}_j) \geq \frac{|\log \det A_j|}{n\sqrt{n}} + O(1) \longrightarrow \infty. \tag{27.12}$$

By contrast, a sequence of coefficient matrices tending to zero can normalize to a point of S_W.

Proof *Paths in W.* Every path in S_W is a path in the hypersurface of all positive matrices with $H \det A = 1$, so Lemma 27.4.1 gives (27.9). To obtain an upper bound by paths inside W, put

$$U = B - A, \qquad C(t) = A + tU, \qquad N(t) = \frac{C(t)}{\lambda(C(t))}.$$

Since the positive cone is convex and W is linear, $N(t)$ stays in S_W and joins A to B. With $K = C^{-1/2} U C^{-1/2}$, differentiation gives

$$g_N(N', N') = \frac{1}{n} \operatorname{tr}(K^2) - \frac{1}{n^2} (\operatorname{tr} K)^2 \leq \frac{\|U\|_{\mathrm{F}}^2}{n m^2},$$
$$m = \min\{\lambda_{\min}(A), \lambda_{\min}(B)\}.$$

Indeed $C(t) \geq mI$ and $(\log \lambda(C))' = \operatorname{tr}(C^{-1} U)/n$. Integrating proves (27.10), and shows in particular that S_W is connected.

Put $d = d_W(A,B)$. Every eigenvalue of $A^{-1/2} B A^{-1/2}$ has a logarithm of absolute value at most $\| \log(A^{-1/2} B A^{-1/2}) \|_{\mathrm{F}} \leq \sqrt{n}\, d$, by (27.9). Hence

$$\mathrm{e}^{-\sqrt{n} d} A \leq B \leq \mathrm{e}^{\sqrt{n} d} A. \tag{27.13}$$

Since $\gamma_A = \theta_A$ on S_W, this proves (27.11).

Cauchy sequences and bounded balls. For a Cauchy sequence (A_j) in (S_W, d_W), comparison with one fixed element of the sequence, using (27.13), gives $mI \le A_j \le MI$ with fixed positive m, M. Comparing two elements then gives

$$\|A_j - A_k\|_{\mathrm{op}} \le M\bigl(\mathrm{e}^{\sqrt{n}\,d_W(A_j,A_k)} - 1\bigr),$$

where $\|\cdot\|_{\mathrm{op}}$ is the operator norm. The coefficient matrices therefore form a Cauchy sequence. Their limit lies in the closed linear space W, is positive, and satisfies $H \det A = 1$, and the upper bound (27.10) proves convergence to that point in d_W. Conversely, coefficient convergence to a positive point gives convergence in d_W by the same bound. This proves completeness and the Cauchy criterion. The minimizing forms then converge in C^∞, because their coefficients multiply a fixed smooth coframe.

By these estimates, d_W induces the topology of coefficient convergence. A closed ball of radius R around $A_\star$ is contained in the compact matrix interval

$$\mathrm{e}^{-\sqrt{n}R} A_\star \le A \le \mathrm{e}^{\sqrt{n}R} A_\star,$$
$$A \in W, \qquad H \det A = 1.$$

The ball is closed in that interval, hence compact. This proves properness. In particular, distances from a fixed reference point are bounded exactly when the eigenvalues relative to that point are bounded above and away from zero.

The coefficient boundary. Write $\lambda_{\max}$ for the largest eigenvalue. If $A_j \to A_0$ with $A_0 \ge 0$ singular and nonzero, then $\lambda(A_j) \to 0$ while $\lambda_{\max}(A_j)$ has a positive limit, so the largest eigenvalue of $\widehat{A}_j$ tends to infinity. Applying (27.13) to $A_\star, \widehat{A}_j$ gives

$$\begin{aligned} d_W(A_\star, \widehat{A}_j) &\ge \frac{1}{\sqrt{n}} \log \frac{\lambda_{\max}(A_j)}{\lambda(A_j)\lambda_{\max}(A_\star)} \\ &= \frac{|\log\det A_j|}{n\sqrt{n}} + O(1). \end{aligned}$$

The error term is bounded because $A_0 \ne 0$. This proves (27.12). For comparison, $A_j = t_j A_\star$ with $t_j \downarrow 0$ normalizes to $A_\star$ for every j; its coefficient limit, the zero matrix, gives no new point of the completion. □

By Proposition 27.4.1, the whole cone $\{\theta_A : A \in W,\ A > 0\}$ is then complete as well. The argument uses the invariant probability and the linearity of W; in other geometries the unit-volume hypersurface can have a boundary at finite distance, as in Example 27.4.1.

The length-minimizing path (27.8) need not stay in a proper subspace W. For $n = 3$, take

$$W = \mathrm{span}_{\mathbb{R}}\{I, \mathrm{diag}(0,1,2)\}, \qquad a = H^{-1/3}.$$

The points aI and $a6^{-1/3}\,\mathrm{diag}(1,2,3)$ of S_W have, on the path (27.8), the midpoint $a6^{-1/6}\,\mathrm{diag}(1,\sqrt{2},\sqrt{3})$. The diagonal entries fail the relation $x_1 - 2x_2 + x_3 = 0$, since $1 - 2\sqrt{2} + \sqrt{3} \ne 0$. Thus this path leaves W, whereas the normalized segment $N(t)$ used in the proof remains in it.

Example 27.4.2 On $G = \mathbb{C}^3$ take the multiplication

$$z \cdot w = (z_1 + w_1, z_2 + w_2, z_3 + w_3 + z_1 w_2),$$

the Gaussian lattice $\Gamma = \mathbb{Z}[\mathrm{i}]^3$, and $X = \Gamma\backslash G$. This is a compact non-Kähler manifold of dimension three, with invariant coframe

$$\alpha_1 = \mathrm{d}z_1, \qquad \alpha_2 = \mathrm{d}z_2, \qquad \alpha_3 = \mathrm{d}z_3 - z_1 \mathrm{d}z_2.$$

For its full nine-dimensional cone $W = \mathrm{Herm}(3)$ and the normalized invariant probability,

$$H = \frac{6}{\pi^3}, \qquad \mathcal{V}_\mu(\theta_A) = \frac{6}{\pi^3} \det A, \qquad S = \{A > 0 : \det A = \pi^3/6\}.$$

Every positive background is nonclosed. The eight-dimensional unit-volume hypersurface is complete and proper, and its intrinsic distance is

$$d_S(A, B) = \frac{1}{\sqrt{3}} \| \log(A^{-1/2} B A^{-1/2}) \|_{\mathrm{F}}. \tag{27.14}$$

Proof The bilinear cross term makes the multiplication associative, with identity zero and inverse $(-z_1, -z_2, -z_3 + z_1 z_2)$. These operations preserve the Gaussian lattice. Its action by left multiplication is free and properly discontinuous: for a compact set $K \subset G$, elements γ with $\gamma K \cap K \neq \varnothing$ belong to the compact set KK^{-1}, containing only finitely many lattice elements.

Choose Gaussian integers γ_1, γ_2 that move z_1, z_2 into the half-open unit square, and then γ_3 that moves $z_3 + \gamma_1 z_2$ into it. This gives a representative in the real six-dimensional unit cube. If two points of the half-open cube are equivalent, then $\gamma_1 = \gamma_2 = 0$ and then $\gamma_3 = 0$. The cube is therefore a measurable fundamental domain, and X is compact, being the image of its closure.

This coframe is left-invariant, descends to X, and has

$$\mathrm{d}\alpha_3 = -\alpha_1 \wedge \alpha_2, \qquad \alpha_1 \wedge \alpha_2 \wedge \alpha_3 = \mathrm{d}z_1 \wedge \mathrm{d}z_2 \wedge \mathrm{d}z_3.$$

The density of h_0^3 is $6/\pi^3$ times real Lebesgue volume on the fundamental cube, giving the stated H. The coefficient of $\alpha_1 \wedge \alpha_2 \wedge \bar{\alpha}_3$ in $\partial\theta_A$ is $-\mathrm{i}A_{33}/(2\pi)$, so positivity makes every such background nonclosed.

To see that X admits no Kähler metric at all, suppose that a positive closed $(1,1)$-form κ existed, and average its coefficient functions in this coframe against μ. Their mean is a positive matrix A. In the coefficients of $\partial\kappa$, derivative terms have mean zero by (27.6); the remaining means are exactly the coefficients of $\partial\theta_A$. Closedness would force them to vanish, contradicting $A_{33} > 0$. Thus X is non-Kähler.

The assertions about the volume, the minimizers and completeness follow from Proposition 27.4.2 and Theorem 27.4.1. As W is the full matrix space, the equal-determinant exponential path stays in S, and Lemma 27.4.1 gives (27.14). For $A_\star = aI$,

$a = (\pi^3/6)^{1/3}$, and with $\lambda_1(A), \lambda_2(A), \lambda_3(A)$ the eigenvalues of A, this becomes

$$d_S(A_\star, A)^2 = \frac{1}{3}\sum_{j=1}^{3}\left(\log\frac{\lambda_j(A)}{a}\right)^2.$$

The three logarithms sum to zero.

For a concrete degeneration, take $A_\varepsilon = \operatorname{diag}(1,1,\varepsilon)$, $0 < \varepsilon \le 1$. As $\varepsilon \to 0$, the coefficients converge to the nonzero smooth semipositive form $\theta_{\operatorname{diag}(1,1,0)}$. The normalized matrices and their distances, on the other hand, are

$$\begin{aligned} \widehat{A}_\varepsilon &= a\operatorname{diag}(\varepsilon^{-1/3}, \varepsilon^{-1/3}, \varepsilon^{2/3}), \\ d_S(A_\star, \widehat{A}_\varepsilon) &= \frac{\sqrt{2}}{3}|\log\varepsilon|. \end{aligned} \tag{27.15}$$

Thus, after normalization, the boundary form $\theta_{\operatorname{diag}(1,1,0)}$ lies at infinite distance. □

27.4.3 Warped product cones and flat completion

The invariant cones above have complete unit-volume hypersurfaces. On the product of a compact Kähler manifold with a curve, a nonconstant coefficient in the fiber direction gives cones of nonclosed backgrounds whose unit-volume hypersurfaces are flat and may instead have a Euclidean strip or half-plane as their metric completion. The volume is the reciprocal of an integral of the reciprocal fiber coefficient, and the variance of that reciprocal coefficient measures the distance to the boundary. The probability measure in this variance is a trace of a minimizing balanced metric.

Let (B, κ) be a connected compact Kähler manifold of dimension $m \ge 1$, with $\int_B \kappa^m = 1$. Let C be a connected compact complex curve with a smooth positive $(1,1)$-form w of integral one. Fix a smooth strictly positive probability form σ on B and a smooth nonconstant real function a on B with $\min a = 0$ and $\max a = M > 0$. On $X = B \times C$, of dimension $n = m + 1$, put $\mu = \sigma \wedge w$; we omit pullbacks from the notation. For real parameters b, c, e, the backgrounds in the real span of κ, aw, w are

$$\theta_{b,c,e} = b\kappa + Lw, \qquad L = ca + e.$$

Theorem 27.4.2 *This span contains no nonzero smooth* $\mathrm{dd^c}$ *direction, and its intersection with* $\mathscr{P}$ *is the cone*

$$\mathscr{C} = \{b > 0,\ e > 0,\ e + cM > 0\}.$$

For $\theta = \theta_{b,c,e} \in \mathscr{C}$, *let* $H(c,e) = \int_B \sigma/L$. *Then*

$$\mathcal{V}_\mu(\theta) = \frac{nb^m}{H(c,e)}. \tag{27.16}$$

The minimizing Gauduchon metrics are the constant multiples of the balanced metric

$$\gamma = L^{-1/m}(A + Lw), \qquad A = b\kappa + \mathrm{dd}^{\mathrm{c}}\phi > 0, \qquad A^m = \frac{b^m}{H}\frac{\sigma}{L},$$

where ϕ is a smooth real function on B. Backgrounds with $c \neq 0$ are nonclosed.

Proof Pairing a smooth positive representative $\theta + \mathrm{dd}^{\mathrm{c}}u$ with the closed semipositive form $\kappa^{m-1} \wedge w$ gives $b > 0$, and its integral over the compact fiber $\{y\} \times C$ is $L(y) > 0$, by Stokes' theorem. These are exactly the stated inequalities; conversely, they make θ itself positive. The same tests applied to a smooth dd^{c} direction give $b = 0$ and $ca + e = 0$ everywhere, and since a is nonconstant, $c = e = 0$. Also $\mathrm{d}\theta = c\,\mathrm{d}a \wedge w$, which is not identically zero when $c \neq 0$.

Next we apply Theorem 10.1.1 on B to the closed positive background $b\kappa$ and the density $(b^m/H)\sigma/L$. Since this density has integral $b^m = \int_B (b\kappa)^m$, Stokes' theorem forces the constant in the equation to be one, and we obtain A. Put $\Omega = A + Lw$. Since $A^{m+1} = 0$ and $(Lw)^2 = 0$, the binomial expansion gives

$$\Omega^n = \frac{nb^m}{H}\mu, \qquad \gamma^m = L^{-1}A^m + mA^{m-1} \wedge w. \tag{27.17}$$

The first summand is the pullback of a top-degree form on B, and the second is closed. Hence γ is balanced, in particular Gauduchon, and it is conformal to the solution form $\Omega = \theta + \mathrm{dd}^{\mathrm{c}}\phi$ of θ. Since $n \geq 2$, Theorem 10.8.1 proves (27.16) and the description of the minimizers. By Stokes' theorem and (27.17), the quotient of γ has numerator $\int_X \Omega \wedge \gamma^m = n\int_B A^m = nb^m$, and hence denominator $D_\mu(\gamma) = nb^m/\lambda_\mu(\theta) = (nb^m/H)^{m/n}H$. □

The minimizing metric gives the probability identity

$$\frac{Lw \wedge \gamma^m}{b^m} = \frac{\sigma \wedge w}{HL}. \tag{27.18}$$

Its marginal $\sigma/(HL)$ on B is the probability measure that governs the Hessian metric. To compute that metric we use the ratio $v = c/e$ and set

$$v = c/e \in (-1/M, \infty), \qquad Q(v) = \int_B \frac{\sigma}{1 + va},$$
$$\mathrm{d}\nu_v = \frac{\sigma}{Q(v)(1 + va)}, \qquad \ell_v = \frac{a}{1 + va}.$$

Proposition 27.4.3 *On the unit-volume hypersurface $\mathcal{S} = \{\theta \in \mathscr{C} : \mathcal{V}_\mu(\theta) = 1\}$, the functions $U = \log b$ and v are global coordinates, in which*

$$g_{\mathcal{S}} = m\,\mathrm{d}U^2 + \rho(v)\,\mathrm{d}v^2,$$
$$\rho(v) = \frac{Q''/Q - 2(Q'/Q)^2}{n} = \frac{2}{n}\operatorname{Var}_{\nu_v}(\ell_v) > 0. \tag{27.19}$$

For any $v_* \in (-1/M, \infty)$, the coordinates

$$Y = \sqrt{m}\, U, \qquad R(v) = \int_{v_*}^{v} \sqrt{\rho(t)}\, \mathrm{d}t$$

define a global isometry onto $\mathbb{R} \times I$, where I is an open interval with the Euclidean metric. The intrinsic distance is the Euclidean distance in (Y, R).

Proof Consider an affine variation $\delta = (\delta b, \delta c, \delta e)$ of the coefficients, write $F = \delta L / L$, and note that $\sigma/(HL) = \mathrm{d}\nu_v$ because $HL = Q(v)(1 + va)$. Differentiating under the integral sign gives

$$D^2 \log H(\delta, \delta) = 2\mathbb{E}_{\nu_v} F^2 - (\mathbb{E}_{\nu_v} F)^2,$$

where $\mathbb{E}_{\nu_v}$ denotes expectation with respect to ν_v. This is positive unless $\delta L = 0$ everywhere, which forces $\delta c = \delta e = 0$. Since $\log \lambda_\mu = (\log n + m \log b - \log H)/n$,

$$g = -D^2 \log \lambda_\mu = \frac{m}{n} \left(\frac{\mathrm{d}b}{b} \right)^2 + \frac{1}{n} D^2 \log H$$

is positive definite. On the unit-volume hypersurface, the relation $H(c, e) = Q(v)/e = nb^m$ gives

$$e = \frac{Q(v)}{n \exp(mU)}, \qquad c = ve.$$

To pull back the Hessian, write $t = \mathrm{d}e/e$ and $z = Q'/Q$. Along an affine path of coefficients one has $v'' = -2tv'$, so differentiating $\log H = -\log e + \log Q(v)$ twice gives

$$D^2 \log H = t^2 - 2zt\, \mathrm{d}v + (Q''/Q - z^2)\, \mathrm{d}v^2.$$

On $\mathcal{S}$ we have $t = z\, \mathrm{d}v - m\, \mathrm{d}U$; substituting this and using $n = m + 1$ gives (27.19). Since $Q'/Q = -\mathbb{E}_{\nu_v} \ell_v$ and $Q''/Q = 2\mathbb{E}_{\nu_v} \ell_v^2$, the variance formula follows, and nonconstancy of a makes this variance positive. In the coordinates (Y, R) the metric is Euclidean on the convex set $\mathbb{R} \times I$, so straight segments prove the distance assertion. □

27.4.3.1 Lengths of the two ends

The interval I has two ends, corresponding to $v \to \infty$ and $v \to -1/M$. Whether they lie at finite distance depends on how a attains its extreme values. Assume that each of the extreme sets $\{a = 0\}$ and $\{a = M\}$ is a finite disjoint union of smooth compact real submanifolds Z_i of positive codimension, and that near each component, respectively,

$$a \asymp \mathrm{dist}(Z_i)^{k_i}, \qquad M - a \asymp \mathrm{dist}(Z_i)^{k_i}, \qquad k_i > 0,$$

where $\asymp$ means comparability up to positive constant factors and $\mathrm{dist}(Z_i)$ is the distance to Z_i for a fixed Riemannian metric on B. The function a, respectively $M - a$,

is then bounded below by a positive constant outside small disjoint tubes around these components. Define, with minima over the components Z_i of the respective extreme set,

$$r_0 = \min_{\{a=0\}} \frac{\operatorname{codim}_{\mathbb{R}} Z_i}{k_i}, \qquad r_M = \min_{\{a=M\}} \frac{\operatorname{codim}_{\mathbb{R}} Z_i}{k_i}.$$

Lemma 27.4.2 *The end $v \to \infty$ of I has finite length exactly when $r_0 > 1$, and the end $v \to -1/M$ exactly when $r_M > 1$. More precisely, at the end $v \to \infty$ put $f = a$, $\epsilon = 1/v$ and $r = r_0$; at the end $v \to -1/M$ put $f = M - a$, $\epsilon = -1/v - M$ and $r = r_M$. Let*

$$H_f = \int_B \frac{\sigma}{f+\epsilon}, \quad J_f = \int_B \frac{\sigma}{(f+\epsilon)^2}, \quad K_f = \int_B \frac{\sigma}{(f+\epsilon)^3}.$$

Then $\epsilon \downarrow 0$ at the end, and

$$\rho(v)\,\mathrm{d}v^2 = G_f(\epsilon)\,\mathrm{d}\epsilon^2, \quad G_f = \frac{2}{n}\left[\frac{K_f}{H_f} - \left(\frac{J_f}{H_f}\right)^2\right]. \tag{27.20}$$

Moreover, $G_f \asymp \epsilon^{-2}$ for $r < 1$ and $G_f \asymp \epsilon^{-2}/\log(1/\epsilon)$ for $r = 1$, while for $r > 1$ the function $\sqrt{G_f}$ is integrable near $\epsilon = 0$.

Proof At the end $v \to \infty$, where $\epsilon = 1/v$, we have

$$\ell_v = \epsilon - \frac{\epsilon^2}{a+\epsilon}, \qquad \mathrm{d}v = -\epsilon^{-2}\mathrm{d}\epsilon.$$

At the end $v \to -1/M$, where $\epsilon = -1/v - M$, we obtain

$$\ell_v = \frac{(M+\epsilon)^2}{M-a+\epsilon} - (M+\epsilon), \qquad \mathrm{d}v = (M+\epsilon)^{-2}\mathrm{d}\epsilon.$$

In both cases ℓ_v is an affine function of $1/(f+\epsilon)$ with coefficients depending only on ϵ, and (27.20) expresses $\rho(v)\,\mathrm{d}v^2$ through the variance of $1/(f+\epsilon)$ under the probability $\sigma/[H_f(f+\epsilon)]$. This probability is ν_v, the marginal in (27.18), because L is a positive multiple of $f+\epsilon$ and positive scaling of L does not change that marginal.

Integration over tubes around the components Z_i, splitting the normal radius at ϵ^{1/k_i}, gives

$$H_f \asymp \begin{cases} \epsilon^{r-1} & r < 1, \\ \log(1/\epsilon) & r = 1, \\ 1 & r > 1, \end{cases} \qquad K_f \asymp \begin{cases} \epsilon^{r-3} & r < 3, \\ \log(1/\epsilon) & r = 3, \\ 1 & r > 3. \end{cases}$$

The analogous threshold for J_f is two. For a finite sum of tube integrals, the smallest ratio governs.

If $r < 1$, choose a component attaining r, and two normal annuli at fixed multiples of ϵ^{1/k_i}, so far apart that the two-sided bounds separate their ranges of f/ϵ. Each

annulus has probability bounded below, since its weighted mass and H_f both have order ϵ^{r-1}, and the values of $1/(f+\epsilon)$ on the two annuli differ by at least $c\epsilon^{-1}$. The identity

$$\operatorname{Var}_\nu T = \frac{1}{2}\iint (T(x)-T(y))^2 \,\mathrm{d}\nu(x)\mathrm{d}\nu(y)$$

applied to $T = 1/(f+\epsilon)$ gives the lower bound, and $T \le \epsilon^{-1}$ gives the upper bound.

If $r = 1$, the near annulus has probability at least $c/\log(1/\epsilon)$. In the same tube, the normal radii between $\epsilon^{1/(2k_i)}$ and a fixed small radius carry probability bounded below, by the logarithmic integral. There $1/(f+\epsilon) \le C\epsilon^{-1/2}$, while $1/(f+\epsilon) \ge c\epsilon^{-1}$ on the near annulus. The variance identity and the bound $G_f \le 2K_f/(nH_f)$ prove the claimed logarithmic estimate. In both cases $r < 1$ and $r = 1$, the integral of $\sqrt{G_f}$ diverges at $\epsilon = 0$.

For $r > 1$, H_f tends to a finite positive integral and $G_f \le CK_f$. For $1 < r < 3$, $\sqrt{G_f}$ is therefore bounded by $C\epsilon^{(r-3)/2}$, whose exponent exceeds -1. At $r = 3$ the bound is $C\sqrt{\log(1/\epsilon)}$, and for $r > 3$ it is a constant. All these bounds are integrable.□

Theorem 27.4.3 *Under the hypotheses of Lemma 27.4.2 on the extreme sets, the metric completion of $\mathcal{S}$ is a closed Euclidean strip if $r_0, r_M > 1$, a closed Euclidean half-plane if exactly one of them exceeds one, and the whole Euclidean plane otherwise. Each boundary line of the completion is parametrized by $b > 0$, and its points are represented by the forms*

$$\theta_+(b) = b\kappa + \frac{H_a}{nb^m}aw, \qquad H_a = \int_B \frac{\sigma}{a} \quad (r_0 > 1),$$

$$\theta_-(b) = b\kappa + \frac{H_-}{nb^m}(1-a/M)w, \qquad H_- = \int_B \frac{\sigma}{1-a/M} \quad (r_M > 1).$$

These forms are smooth, semipositive, nef, nonclosed and not big, and their variational volume is one. The completion has no other boundary points.

Proof The completion of the convex product $\mathbb{R}\times I$ of Proposition 27.4.3 is its Euclidean closure. Along a Cauchy sequence, Y converges, so b has a finite positive limit, and R converges to a point of I or to a finite endpoint of I. Lemma 27.4.2 decides which endpoints are finite and gives the three alternatives.

On $\mathcal{S}$ we have $e = Q(v)/(nb^m)$ and $c = ve$. At the end $v \to \infty$, $vQ(v) \to H_a$ by monotone convergence, which gives the coefficient limit $\theta_+(b)$. At the end $v \to -1/M$, write $1 + va = 1 - a/M + (v+1/M)a$; dominated convergence gives $Q(v) \to H_-$, and hence the limit $\theta_-(b)$. These are the coefficient limits at all finite ends. The forms are semipositive, hence nef, and they are nonclosed. Their fiber coefficients $L_\pm$ vanish somewhere, so they are not big by Proposition 27.4.4 below. By the definitions of H_a and H_-, $\int_B \sigma/L_\pm = nb^m$. For $\epsilon > 0$, the form $\theta_\pm(b) + \epsilon(\kappa + w)$ lies in $\mathcal{C}$, and by (27.16) and monotone convergence its volume

$$\frac{n(b+\epsilon)^m}{\int_B \sigma/(L_\pm+\epsilon)}$$

tends to one as $\epsilon \downarrow 0$. Theorem 27.9.1, with the Hermitian form $\kappa + w$, shows that the variational volume of $\theta_{\pm}(b)$ is one. □

27.4.3.2 Current cones and endpoint equations

We keep the product $X = B \times C$, the backgrounds $\theta = b\kappa + (ca + e)w$ with real parameters b, c, e, and the probability $\mu = \sigma \wedge w$ of Section 27.4.3, and write $h = \kappa + w$ for the product Hermitian form. Pullbacks of nonnegative top-degree forms on B are closed, and testing against them determines which backgrounds have positive current representatives, not only which have smooth positive ones. At a finite end of the unit-volume hypersurface, the reciprocal of the fiber coefficient is integrable with an exponent greater than one, so the equation on the base still has a continuous solution.

Proposition 27.4.4 *In this real span, the pseudoeffective and nef cones both equal*

$$\{b \geq 0,\ e \geq 0,\ e + cM \geq 0\}.$$

A background in this span is big, that is, admits a representative $\theta + \mathrm{dd}^{\mathrm{c}} u \geq \delta h$ with $\delta > 0$, if and only if $b > 0$, $e > 0$ and $e + cM > 0$. Thus the big backgrounds in this span form exactly the cone $\mathscr{C}$ of Theorem 27.4.2.

Proof Put $L = ca + e$, and suppose that a quasi-plurisubharmonic u satisfies $\theta + \mathrm{dd}^{\mathrm{c}} u \geq \delta h$ for some $\delta \geq 0$. Pairing with the closed semipositive form $\kappa^{m-1} \wedge w$ and using Stokes' theorem for currents gives $b \geq \delta$. For every smooth nonnegative top-degree form η on B, its pullback to X is closed for degree reasons, and

$$\int_X (\theta + \mathrm{dd}^{\mathrm{c}} u - \delta h) \wedge \eta = \int_B (L - \delta)\eta \geq 0.$$

If $L < \delta$ at some point, a nonzero form η supported in the open set where $L < \delta$ contradicts this inequality. Hence $L \geq \delta$ everywhere. For $\delta = 0$ these conditions are exactly $b \geq 0$, $e \geq 0$ and $e + cM \geq 0$; conversely, they make θ semipositive, and such forms are nef because $\theta + \varepsilon h > 0$ for every $\varepsilon > 0$. Conversely, if θ is nef, applying the same tests to smooth positive representatives of $\theta + \varepsilon h$ and letting ε tend to zero gives these inequalities. For $\delta > 0$ the conditions force the three strict inequalities, and conversely these make θ itself strictly positive. □

Proposition 27.4.5 *Under the hypotheses of Lemma 27.4.2 on the extreme sets, let $\theta = b\kappa + L_0 w$, with $b > 0$, be one of the boundary forms $\theta_+(b)$ or $\theta_-(b)$ of Theorem 27.4.3 at a finite end. There is a continuous θ-plurisubharmonic function u, pulled back from B, such that*

$$(\theta + \mathrm{dd}^{\mathrm{c}} u)^n = \mu. \tag{27.21}$$

Here the product is the usual product of bounded potentials. No solution of this equation is of class $C^{1,1}$ on X.

Proof Let $f = a$ for $\theta_+(b)$ and $f = 1 - a/M$ for $\theta_-(b)$, and let $r > 1$ be the corresponding ratio r_0 or r_M. The coefficient L_0 is a positive constant multiple of f. In each tube around an extreme component, the integral of f^{-p} converges exactly when

$$\int_0^{s_0} s^{\operatorname{codim}_{\mathbb{R}} Z_i - 1 - k_i p}\, \mathrm{d}s$$

converges, where s is the normal radius and $s_0 > 0$ is small. Thus $1/L_0 \in L^p(B, \sigma)$ for every $1 < p < r$. The zero set of L_0 has volume zero, so the value assigned to the density there is irrelevant. The formulas for $\theta_\pm(b)$ give

$$H_0 := \int_B \frac{\sigma}{L_0} = n b^m.$$

The closed form $b\kappa$ satisfies the Guan–Li condition, so Theorem 12.5.1 on B, applied to the density of σ/L_0 relative to $(b\kappa)^m$, gives a continuous ϕ with $A = b\kappa + \mathrm{dd}^{\mathrm{c}}\phi \geq 0$ and $A^m = c_0 \sigma/L_0$, where $c_0 = \int_B (b\kappa)^m/H_0 = b^m/H_0$. Set $u = \phi \circ p_B$, where $p_B \colon X \to B$ is the projection. Then $\theta + \mathrm{dd}^{\mathrm{c}} u = A + L_0 w \geq 0$.

The pullback identity of Lemma 1.7.2 gives $A^{m+1} = 0$ on X, and $(L_0 w)^2 = 0$. The multilinear expansion of Proposition 1.7.1 therefore gives

$$(A + L_0 w)^n = n L_0 A^m \wedge w = \frac{n b^m}{H_0} \sigma \wedge w = \mu.$$

The cancellation $L_0(\sigma/L_0) = \sigma$ holds as measures, since the zero set of L_0 has volume zero and A^m is absolutely continuous with density c_0/L_0 relative to σ.

Suppose that some solution q, not necessarily pulled back from B, were of class $C^{1,1}$ on X. Since μ/h^n is smooth and strictly positive on compact X, Proposition 1.7.3 would give $\theta + \mathrm{dd}^{\mathrm{c}} q \geq \delta h$ for some $\delta > 0$. This contradicts Proposition 27.4.4, because L_0 vanishes somewhere. □

27.4.3.3 Limits of the base potentials

The base potentials used in Theorem 27.4.2 to construct the minimizing balanced metrics converge uniformly at each finite end, and the limit solves the endpoint equation (27.21). Both the compactness and the uniqueness needed here concern only the equation on B, whose background is closed.

Proposition 27.4.6 *Under the hypotheses of Theorem 27.4.3, let $\theta_j = b_j\kappa + L_j w \in \mathcal{S}$ converge in coefficients to a boundary form $\theta_0 = b\kappa + L_0 w$ at a finite end. Normalize the smooth base potentials ϕ_j of Theorem 27.4.2 by $\sup_B \phi_j = 0$ and put $u_j = \phi_j \circ p_B$. Then the u_j converge uniformly on X to a continuous solution u_0 of (27.21) pulled back from B. For every $1 \le k \le n$,*

$$(\theta_j + \mathrm{dd^c} u_j)^k \rightharpoonup (\theta_0 + \mathrm{dd^c} u_0)^k.$$

Moreover, u_0 is the unique bounded solution of (27.21) that is pulled back from B and has supremum zero.

Proof Write $v_j = c_j/e_j$ and $\psi_j = \phi_j/b_j$. On $\mathcal{S}$ the equation for ϕ_j on B becomes

$$(\kappa + \mathrm{dd^c}\psi_j)^m = F_{v_j}\sigma, \qquad F_v = \frac{1}{Q(v)(1+va)}, \qquad \sup_B \psi_j = 0.$$

At the end where $\theta_0 = \theta_+(b)$, $v_j \to \infty$. Set $\epsilon = 1/v$ and $H_\epsilon = \int_B \sigma/(a+\epsilon)$. Then $F_v = [H_\epsilon(a+\epsilon)]^{-1}$ and $H_\epsilon \to H_a$. At the end where $\theta_0 = \theta_-(b)$, $v_j \to -1/M$. With $f = 1 - a/M$ and $t = v + 1/M > 0$, the denominator of F_v is $Q(v)(f+ta)$, and $Q(v) \to H_-$. Thus the respective limits of F_{v_j} are

$$F_+ = \frac{1}{H_a a}, \qquad F_- = \frac{1}{H_-(1-a/M)}.$$

Choose $1 < p < r$, where $r > 1$ is the ratio r_0 or r_M of that end. Integration over tubes as in the proof of Proposition 27.4.5 gives $a^{-1} \in L^p(B,\sigma)$, respectively $f^{-1} \in L^p(B,\sigma)$. Near the first end $F_v \le C/a$, and near the second $F_v \le C/f$. Since the extreme sets have volume zero, dominated convergence gives $F_{v_j} \to F_\pm$ in $L^p(B,\sigma)$, and each limiting density has integral one. Since σ/κ^m is smooth and strictly positive on compact B, the densities of the equations also have a common bound in $L^p(B,\kappa^m)$.

We apply Theorem 11.3.2 to the fixed background κ: every subsequence of (ψ_j) has a uniformly convergent further subsequence. Continuity of bounded products under uniform convergence identifies its limit with a normalized solution of $(\kappa + \mathrm{dd^c}\psi_0)^m = F_\pm\sigma$. Since κ is closed, it satisfies the Guan–Li condition, so Theorem 12.5.1 makes this solution unique. Every cluster limit is consequently the same, which proves uniform convergence of the whole sequence. Since $b_j \to b > 0$, we obtain $u_j \to u_0 = (b\psi_0) \circ p_B$ uniformly.

To prove convergence of the products, note that on each coordinate ball the convergence of the background coefficients provides a common smooth function q, a quadratic polynomial, such that all $u_j + q$ are plurisubharmonic. These corrected potentials are uniformly bounded and converge uniformly. The continuity of bounded products under convergence in capacity (Section 1.6) gives weak convergence of all their products, with uniform local mass bounds. Expanding the products as in (1.10), and using smooth convergence of the remaining background factors, we obtain the asserted convergence of currents. In top degree, $(\theta_j + \mathrm{dd^c}u_j)^n = \mu$ for all j, so the limit solves the endpoint equation.

For uniqueness, let $v = \chi \circ p_B$ be a bounded solution pulled back from B, with $\sup_B \chi = 0$. The current $A_\chi = b\kappa + \mathrm{dd^c}\chi$ on B is positive, as one sees by pairing the pulled-back current $\theta_0 + \mathrm{dd^c}v$ with w and with pullbacks of positive forms on B. The product expansion gives $nL_0 A_\chi^m \wedge w = \sigma \wedge w$, which determines $A_\chi^m = \sigma/(nL_0)$ off the extreme set. That density already has total mass $H_0/n = b^m$, whereas mass invariance for the closed background $b\kappa$ gives $\int_B A_\chi^m = b^m$. Positivity therefore excludes any mass of A_χ^m on the extreme set. Thus χ/b satisfies the same normalized equation on B as ψ_0, and Theorem 12.5.1 proves the asserted uniqueness. □

The proposition concerns the limit of the base potentials constructed above. It does not describe solutions of the endpoint equation that are not pulled back from B, nor does it give a pointwise limit of the balanced metrics $\gamma_j = L_j^{-1/m}(b_j\kappa + \mathrm{dd^c}\phi_j + L_j w)$.

27.4.3.4 Traces of minimizing metrics

Uniform convergence of the base potentials does not control the minimizing balanced metrics themselves: by the cofactor formula (27.17), their $(n-1)$st powers contain the square of the reciprocal fiber coefficient. Depending on the ratio at the end, the traces of these powers stay bounded or blow up, and in the second case the trace-normalized powers concentrate on fibers over the extreme set.

Keep the sequence $\theta_j \to \theta_0$ of Proposition 27.4.6, and let ϕ_0 be the uniform limit of its base potentials ϕ_j. Put $A_j = b_j\kappa + \mathrm{dd^c}\phi_j$, and write $G_j = \gamma_j^m$, where γ_j is the balanced minimizing metric of Theorem 27.4.2 for θ_j. Traces are taken with respect to $h = \kappa + w$.

Proposition 27.4.7 *Let $r > 1$ be the ratio r_0 or r_M of the end. After discarding finitely many terms, write $L_j = s_j(f + \epsilon_j)$ with $s_j > 0$, $s_j \to s_0 > 0$ and $\epsilon_j \to 0^+$. Here $f = a$, $s_j = c_j$ at the first face, and $f = M - a$, $s_j = -c_j$ at the second. Set $J_j = \int_B \sigma/L_j^2$. Then*

$$G_j = \frac{\sigma}{nL_j^2} + mA_j^{m-1} \wedge w, \qquad t_j := \int_X h \wedge G_j = \frac{J_j}{n} + mb_j^{m-1}. \tag{27.22}$$

For $1 < r < 2$, $t_j \asymp \epsilon_j^{r-2}$; for $r = 2$, $t_j \asymp \log(1/\epsilon_j)$. For $r > 2$, the entire sequence converges weakly to the finite positive closed current

$$G_0 = \frac{\sigma}{nL_0^2} + mA_0^{m-1} \wedge w, \qquad A_0 = b\kappa + \mathrm{dd^c}\phi_0,$$

and $t_j \to t_0 = \int_X h \wedge G_0 > 0$.

Proof On $\mathcal{S}$, $H(c_j, e_j) = nb_j^m$, so $A_j^m = \sigma/(nL_j)$. Substituting this in (27.17) gives the formula for G_j. Stokes' theorem on B, for the closed form κ, gives $\int_B \kappa \wedge A_j^{m-1} = b_j^{m-1}$, which proves the trace identity. The factorization of L_j uses $\epsilon_j = e_j/c_j$ at the first end and $\epsilon_j = e_j/(-c_j) - M$ at the second, and the coefficient limits give $s_j \to s_0 > 0$.

On each tube around an extreme component, $f \asymp \operatorname{dist}(Z_i)^{k_i}$, so the integral of f^{-2} is comparable to

$$\int_0^{\varrho_0} \frac{\varrho^{\operatorname{codim}_{\mathbb{R}} Z_i - 1}}{(\varrho^{k_i} + \epsilon_j)^2}\, \mathrm{d}\varrho,$$

where ϱ is normal radius, $\varrho_0 > 0$ is small and $r_i = \operatorname{codim}_{\mathbb{R}} Z_i / k_i$. Splitting the integral at $\varrho = \epsilon_j^{1/k_i}$ gives order $\epsilon_j^{r_i - 2}$ for $r_i < 2$, order $\log(1/\epsilon_j)$ for $r_i = 2$, and a finite limit for $r_i > 2$. The finite sum over components is governed by $r = \min_i r_i$. Since b_j and s_j have positive finite limits, these estimates prove the three alternatives for t_j.

If $r > 2$, then f^{-2} is integrable and dominates L_j^{-2} up to a fixed constant factor, so $\sigma/L_j^2 \to \sigma/L_0^2$ in total variation. Since $w \wedge w = 0$, we have $A_j^{m-1} \wedge w = (\theta_j + \mathrm{dd}^{\mathrm{c}} u_j)^{m-1} \wedge w$, which converges weakly by Proposition 27.4.6. This proves $G_j \rightharpoonup G_0$, and G_0 is positive and closed as a weak limit of positive closed forms. Pairing with h gives the limit of the traces, which is positive because of the term $\sigma/(nL_0^2)$. □

Proposition 27.4.8 *Suppose $1 < r \le 2$, and put $T_j = G_j / t_j$. Every weak cluster limit of (T_j) has the form $p_B^* \nu$, where ν is a probability measure on B supported on the extreme components with ratio r. Such a limit is positive and closed, has trace one relative to h, and satisfies $\theta_0 \wedge p_B^* \nu = 0$. If those components reduce to a single point p, the whole sequence converges to the current of integration $[\{p\} \times C]$.*

The trace-normalized smooth metrics $\eta_j = t_j^{-1/m} \gamma_j$ satisfy

$$\int_X \theta_j \wedge \eta_j^m = D_\mu(\eta_j) = \frac{n b_j^m}{t_j} \longrightarrow 0.$$

Both quantities have order ϵ_j^{2-r} when $1 < r < 2$, and order $1/\log(1/\epsilon_j)$ when $r = 2$. Their quotient remains one.

Proof Let $\nu_j = \sigma/(J_j L_j^2)$. These are probability measures on B, and

$$T_j = \frac{J_j}{n t_j} p_B^* \nu_j + \frac{m}{t_j} A_j^{m-1} \wedge w.$$

The first coefficient tends to one. The second current is positive with trace $m b_j^{m-1}/t_j \to 0$, hence converges weakly to zero. Compactness of probability measures on B and weak continuity of pullback under the submersion p_B therefore show that every cluster limit is $p_B^* \nu$ for a cluster limit ν of (ν_j). Away from small tubes around the extreme set, the density $1/L_j^2$ is uniformly bounded, while $J_j \to \infty$; that region receives no mass in the limit. The tube integrals for components with ratio greater than r are $o(J_j)$ by the preceding estimates, so only components with ratio r keep mass.

For a limit $T = p_B^* \nu$, $\kappa \wedge T = 0$ for degree reasons on B, while $L_0 w \wedge T = 0$ because L_0 vanishes on the support of ν. Its trace is one because $\int_C w = 1$. A top-degree

measure on B is closed, so its pullback is closed as well. If the support is the single point p, every cluster limit of (ν_j) is δ_p; compactness then gives convergence of the whole sequence, and $p_B^*\delta_p$ is the current of integration over the fiber $\{p\} \times C$.

By the proof of Theorem 27.4.2, the quotient of γ_j has numerator nb_j^m and, since $\lambda_\mu(\theta_j) = 1$ on $\mathcal{S}$, also denominator nb_j^m. Both are multiplied by t_j^{-1} under the trace normalization, and the trace estimates give their limits and rates. □

In the projective strip of Example 27.4.3, both ratios are two and each extreme set is a single point. At each of the two ends, the normalized minimizing tests therefore converge to the current of integration over the corresponding fiber $\{p\} \times \mathbb{P}^1$, and the numerator and the denominator of the quotient both decay like $1/\log(1/\epsilon_j)$. This is convergence of currents; it does not give a pointwise limit of the metrics.

27.4.3.5 Projective and torus examples

Example 27.4.3 On $B = \mathbb{P}^2$, take normalized Fubini–Study κ, $\sigma = \kappa^2$, $C = \mathbb{P}^1$, and

$$a([Z]) = \frac{|Z_1|^2/2 + |Z_2|^2}{|Z_0|^2 + |Z_1|^2 + |Z_2|^2}.$$

Then $M = 1$, $r_0 = r_M = 2$, and the completion is a closed strip with finite faces

$$b\kappa + \frac{4\log 2}{3b^2} aw, \qquad b\kappa + \frac{4\log 2}{3b^2}(1-a)w.$$

For $v \neq 0$,

$$Q(v) = \frac{4}{v^2}\big[(v+1)\log(1+v) - (v+2)\log(1+v/2)\big], \qquad Q(0) = 1.$$

The width of the strip is the finite positive integral

$$\int_{-1}^{\infty} \sqrt{\frac{Q''/Q - 2(Q'/Q)^2}{3}}\, \mathrm{d}v.$$

Proof The extreme sets are the points $[1:0:0]$ and $[0:0:1]$. In affine coordinates, a and $1-a$ vanish there to exactly second order, so both real codimensions are four and both orders are two. To compute the law of a under σ, polar integration in the chart $Z_0 = 1$ gives $2\,\mathrm{d}s_1\mathrm{d}s_2/(1+s_1+s_2)^3$, where $s_j = |z_j|^2$. The transformation $u_j = s_j/(1+s_1+s_2)$ gives the uniform density two on the triangle $u_1, u_2 \geq 0$, $u_1 + u_2 \leq 1$, and $a = u_1/2 + u_2$. For $0 \leq s \leq 1/2$, the triangle $u_1/2 + u_2 \leq s$ has area s^2, so the law of a has density $4s$ on $[0, 1/2]$. With $u_0 = 1 - u_1 - u_2$, exchanging u_0 and u_2 preserves the uniform measure and replaces a by $1-a$; hence the law has density $4(1-s)$ on $[1/2, 1]$. Integrating against these two densities gives $H_a = H_- = 4\log 2$ and the formula for Q, and the integral definition of Q shows that $Q(0) = 1$ is its

continuous extension. Theorem 27.4.3 gives the boundary lines and the finite width, with Euclidean coordinates $Y = \sqrt{2}\log b$ and $R(v)$. □

Example 27.4.4 Let $E = \mathbb{C}/(\mathbb{Z} + i\mathbb{Z})$, $B = E^m$, and write $z_j = u_j + i\tau_j$. Put

$$\eta_j = \frac{i}{2}\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j, \qquad s = (m!)^{-1/m}, \qquad \kappa = s\sum_j \eta_j, \qquad \sigma = \kappa^m.$$

For $a = \sin^2(\pi u_1)$, the unit-volume hypersurface is itself a complete Euclidean plane, with Euclidean coordinates

$$Y = \sqrt{m}\log b, \qquad R = \frac{\log(1+v)}{2\sqrt{n}}, \qquad v > -1.$$

In this example the minimizing balanced metrics of Theorem 27.4.2 can be written down by integrating a periodic function twice.

Proof The forms descend to B, and $\sigma = \eta_1 \wedge \cdots \wedge \eta_m$ is a probability form. Both extreme sets are real hypersurfaces along which a, respectively $1 - a$, vanishes to second order, so $r_0 = r_M = 1/2$. Splitting the period and substituting $t = \tan(\pi u)$ gives

$$\begin{aligned} Q(v) &= \int_0^1 \frac{\mathrm{d}u}{1 + v\sin^2(\pi u)} \\ &= \frac{2}{\pi}\int_0^\infty \frac{\mathrm{d}t}{1 + (1+v)t^2} = \frac{1}{\sqrt{1+v}}. \end{aligned}$$

Let $F_v(u) = [Q(v)(1 + v\sin^2(\pi u))]^{-1}$, which has mean one over a period. Integrating the equation

$$\phi_v'' = 4\pi b s(F_v - 1)$$

twice, whose right side has mean zero, gives a smooth periodic potential ϕ_v of $u = u_1$; the first integration constant is chosen so that ϕ_v' has mean zero. In our normalization $\mathrm{d}\mathrm{d}^{\mathrm{c}}\phi_v = \phi_v''\eta_1/(4\pi)$, hence

$$A = bs\left(F_v\eta_1 + \sum_{j=2}^m \eta_j\right) > 0, \qquad A^m = b^m F_v \sigma = \frac{b^m}{H}\frac{\sigma}{L}.$$

Thus the minimizer of Theorem 27.4.2, with its cofactor formula (27.17), is explicit in this example, without an appeal to Theorem 10.1.1. Differentiating Q gives

$$\rho(v) = \frac{1}{4n(1+v)^2}, \qquad g_{\mathcal{S}} = m\,\mathrm{d}U^2 + \frac{1}{4n}\left(\frac{\mathrm{d}v}{1+v}\right)^2.$$

As U and v range over $\mathbb{R}$ and $(-1, \infty)$, the coordinates (Y, R) range over all of $\mathbb{R}^2$. Both boundary faces of the coefficient cone are therefore at infinite distance on the unit-volume hypersurface. □

These distances are those of the fixed three-parameter cone with the product probability $\mu = \sigma \wedge w$. The endpoint criterion of Lemma 27.4.2 uses the hypothesis that the extreme sets are disjoint unions of submanifolds along which a vanishes to finite power order; intersecting extreme strata or vanishing of infinite order would require further estimates. The variances are computed from the minimizing metrics, but we do not assert that these metrics have a pointwise limit at a finite end.

27.5 A compact dual problem over Gauduchon currents

We now fix a Hermitian background and vary the density. For a smooth positive density, Theorem 10.8.1 provides a minimizing Gauduchon metric; for a density that is only continuous or integrable, there is no smooth equation to produce one. The extremal problem becomes compact when positive $\mathrm{dd^c}$-closed $(n-1,n-1)$-currents are allowed as tests, provided the denominator D_μ is extended to currents. For this we take the determinant of the matrix density of a current relative to its trace measure. The integral of its nth root is an infimum of continuous linear functionals, hence upper semicontinuous under weak convergence, and its maximum over normalized currents is attained. The smooth equation identifies this maximum with the supremum over smooth tests.

From here until Section 27.9, we assume $n \geq 2$, and we fix a Hermitian form ω and a smooth positive probability volume form $\mathrm{d}V$ on X; integrability and convergence in measure refer to $\mathrm{d}V$. As in Chapter 7, $\mathrm{MA}_\omega(u) = (\omega + \mathrm{dd^c}u)^n$ for bounded $u \in \mathrm{PSH}(X,\omega)$, and $\mathrm{MA}_\omega^{\mathrm{ext}}(u)$ is the extended non-pluripolar Monge–Ampère measure of Theorem 7.1.1. Let $\mathscr{G}_\omega$ be the compact convex set of normalized Gauduchon currents of Section 4.4, that is, of positive $(n-1,n-1)$-currents T with $\mathrm{dd^c}T = 0$ and $\int_X \omega \wedge T = 1$. Finally, let $F \in C(X,\mathbb{R})$.

For $T \in \mathscr{G}_\omega$, the trace measure $\sigma_T = \omega \wedge T$ is a probability measure, and all coefficient measures of T are absolutely continuous with respect to it. In a unitary frame for ω, let B be the matrix of a smooth real $(1,1)$-form β relative to ω. The coefficient densities of T form a measurable positive semidefinite Hermitian matrix P_T, determined σ_T-almost everywhere by

$$\beta \wedge T = \frac{1}{n}\operatorname{tr}(BP_T)\,\mathrm{d}\sigma_T, \qquad \operatorname{tr} P_T = n \quad \sigma_T\text{-almost everywhere.}$$

The matrix P_T transforms by unitary conjugation when the frame changes, so its determinant is well defined. Define

$$f_F = \left(\frac{\mathrm{e}^F\,\mathrm{d}V}{\omega^n}\right)^{1/n}, \qquad \Gamma_F(T) = f_F(\det P_T)^{1/n}\sigma_T, \qquad \mathfrak{D}_F(T) = \Gamma_F(T)(X). \tag{27.23}$$

We call $\Gamma_F(T)$ the *determinant measure* of T. These quantities are finite, since the arithmetic–geometric mean inequality gives $(\det P_T)^{1/n} \leq 1$; the measure $\Gamma_F(T)$ may vanish for some currents. The same definitions apply to every positive $(n-1,n-1)$-

current T of finite mass, with the finite measure $\sigma_T = \omega \wedge T$ and $\operatorname{tr} P_T = n$, and Γ_F is positively homogeneous of degree one in T. If the coefficient measures of T are written instead as a matrix density P' relative to another positive measure σ' with respect to which they are absolutely continuous, then $(\det P_T)^{1/n}\sigma_T = (\det P')^{1/n}\sigma'$, by homogeneity of the determinant.

For a smooth positive $T = \gamma^{n-1}$, we may therefore use the reference measure ω^n. The coefficient matrix of T relative to ω^n is the cofactor matrix of the matrix of γ, and its determinant is $(\gamma^n/\omega^n)^{n-1}$. The definition therefore gives

$$\mathfrak{D}_F(\gamma^{n-1}) = \int_X \left(\frac{\mathrm{e}^F \,\mathrm{d}V}{\gamma^n}\right)^{1/n} \gamma^n.$$

Thus $\mathfrak{D}_F$ extends the denominator D_μ of Definition 10.8.1 to currents, with μ replaced by the possibly unnormalized density $\mathrm{e}^F \,\mathrm{d}V$.

Lemma 27.5.1 *For every positive $(n-1,n-1)$-current T of finite mass on X,*

$$\mathfrak{D}_F(T) = \inf_{\substack{\beta>0 \text{ continuous} \\ \beta^n = \mathrm{e}^F \,\mathrm{d}V}} \int_X \beta \wedge T.$$

In particular, $\mathfrak{D}_F$ is concave, positively homogeneous and weakly upper semicontinuous on $\mathcal{G}_\omega$, and its value does not depend on the auxiliary frame or on the metric used to express the matrix density.

Proof Let I be the identity matrix of size n. For any positive semidefinite Hermitian matrix P of that size,

$$(\det P)^{1/n} = \inf_{Q>0,\ \det Q=1} \frac{1}{n} \operatorname{tr}(QP).$$

The inequality $\le$ is the arithmetic–geometric mean inequality. For the reverse inequality, take, for real $\varepsilon > 0$,

$$Q_\varepsilon = \det(P + \varepsilon I)^{1/n} (P + \varepsilon I)^{-1}.$$

Its determinant is one. If P is positive definite, the trace expression $\frac{1}{n}\operatorname{tr}(Q_\varepsilon P)$ converges to $(\det P)^{1/n}$ as $\varepsilon \downarrow 0$. If P is singular, it converges to zero: each ratio of an eigenvalue of P to that eigenvalue plus ε is bounded, while the prefactor tends to zero. When $\operatorname{tr} P = n$, the trace expression is at most $1 + \varepsilon$.

We apply this formula to P_T and multiply by f_F. Every continuous positive β in the asserted infimum has matrix $f_F Q$ with $\det Q = 1$, so it gives an upper bound for $\mathfrak{D}_F(T)$. Conversely, for fixed $\varepsilon > 0$ the measurable field Q_ε formed with $P = P_T$ is bounded above and below by positive constants. It can be approximated in $L^1(\sigma_T)$ by continuous positive definite Hermitian fields with bounds of the same type. One way to construct them is to apply Lusin's theorem in finitely many bundle charts, extend the continuous fields from the resulting compact sets, and glue with a partition of unity; spectral truncation to a fixed positive interval preserves the bounds. Finally we

divide each approximating field by the nth root of its determinant. This normalization is uniformly continuous on fields with spectrum in a fixed compact positive interval, so the normalized fields still converge in $L^1(\sigma_T)$ and have determinant one. Their trace integrals converge to that of Q_ε.

Now let $\varepsilon \downarrow 0$. The bound $1+\varepsilon$ permits dominated convergence, and we obtain equality of the infimum and the integral. For a current of arbitrary positive finite trace mass we use homogeneity, and the zero current is immediate. The infimum is taken over functionals $T \mapsto \int_X \beta \wedge T$ that are linear and weakly continuous, which proves concavity and upper semicontinuity. It also expresses $\mathfrak{D}_F(T)$ without a matrix frame or an auxiliary metric, which proves the final assertion. □

Let $B_0(F)$ be the normalization scalar of Section 16.1 for the background ω and the reference potential zero. Thus the equation $\mathrm{MA}_\omega(u) = \mathrm{e}^{B_0(F)+F}\mathrm{d}V$ has a bounded solution $u \in \mathrm{PSH}(X,\omega)$, and $B_0(F)$ is the only constant with this property.

Theorem 27.5.1 *For every $F \in C(X,\mathbb{R})$,*

$$M(F) := \max_{T \in \mathscr{G}_\omega} \mathfrak{D}_F(T) = \mathrm{e}^{-B_0(F)/n} > 0. \tag{27.24}$$

The maximum equals the supremum over the smooth strictly positive members of $\mathscr{G}_\omega$. The set $\mathscr{M}(F)$ of maximizing currents is nonempty, compact and convex.

If F is smooth, there is exactly one maximizing current, namely the normalized $(n-1)$st power of the Gauduchon metric conformal to the smooth solution form $\omega + \mathrm{d}\mathrm{d}^{\mathrm{c}}u$ of $\mathrm{MA}_\omega(u) = \mathrm{e}^{B_0(F)+F}\mathrm{d}V$.

Proof Upper semicontinuity of $\mathfrak{D}_F$ on the compact set $\mathscr{G}_\omega$ gives attainment, and testing a smooth positive member gives a positive value. Concavity makes the set of maximizers convex, and upper semicontinuity makes it closed.

Suppose first that F is smooth, and let $\Omega = \omega + \mathrm{d}\mathrm{d}^{\mathrm{c}}u > 0$ be the smooth solution form with $\Omega^n = \mathrm{e}^{B_0(F)+F}\mathrm{d}V$. The form $\beta = \mathrm{e}^{-B_0(F)/n}\Omega$ is an admissible test in the preceding lemma, so Stokes' theorem for currents gives, for every $T \in \mathscr{G}_\omega$,

$$\mathfrak{D}_F(T) \le \int_X \beta \wedge T = \mathrm{e}^{-B_0(F)/n} \int_X \omega \wedge T = \mathrm{e}^{-B_0(F)/n}.$$

If γ is the Gauduchon metric conformal to Ω, equality holds in the pointwise matrix inequality for T proportional to γ^{n-1}; normalizing T so that $\int_X \omega \wedge T = 1$, we obtain the formula and its attainment by a smooth test.

For uniqueness, let T be a maximizer. Equality in the integrated inequalities forces equality in the matrix arithmetic–geometric mean inequality σ_T-almost everywhere. Since the matrix of β is positive definite and $\operatorname{tr} P_T = n$, this means that the coefficient measures of T are a nonnegative scalar measure times the smooth coefficients of γ^{n-1}. Thus $T = a\gamma^{n-1}$, where a is a positive measure, regarded as a distribution. The equation $\mathrm{d}\mathrm{d}^{\mathrm{c}}(a\gamma^{n-1}) = 0$ is the adjoint of a scalar elliptic equation, so elliptic regularity makes a smooth. Since γ is Gauduchon, Proposition 4.8.1 identifies the kernel of this adjoint operator with the constants: the trace operator $v \mapsto n\mathrm{d}\mathrm{d}^{\mathrm{c}}v \wedge \gamma^{n-1}/\gamma^n$ has the constants as kernel and index zero, and integration

against γ^n annihilates its range. Thus a is constant, and the normalization fixes the constant. This proves uniqueness.

For continuous F, G, the pointwise definition gives

$$\mathrm{e}^{-\|F-G\|_\infty/n}\mathfrak{D}_G(T) \le \mathfrak{D}_F(T) \le \mathrm{e}^{\|F-G\|_\infty/n}\mathfrak{D}_G(T).$$

The same bounds hold for the maxima, and for the suprema over smooth positive members. We approximate F uniformly by smooth functions. Since B_0 is 1-Lipschitz for the uniform norm by Proposition 16.1.1, the formula just proved for smooth data gives (27.24) in the limit, including the equality with the smooth supremum. Note that only F is approximated here: some normalized Gauduchon currents are not limits of smooth positive ones, as Example 4.4.1 shows. □

27.6 Supporting probabilities represented by maximizing currents

By Theorem 27.5.1, $B_0 = -n\log M$, and $M(F)$ is the maximum of the functionals $T \mapsto \mathfrak{D}_F(T) = \int_X \mathrm{e}^{F/n}\,\mathrm{d}\Gamma_0(T)$, each of which is linear in $\mathrm{e}^{F/n}$. The supporting probabilities of B_0 should therefore be the normalized determinant measures of maximizing currents, and we show that they are exactly these. The point that needs proof is the continuity of the determinant measures: on the set of maximizers, upper semicontinuity of the total determinant integral becomes equality. Recall from Section 16.2 that the set $\mathcal{A}_0(F)$ of *supporting probabilities* of B_0 at $F \in C(X)$ consists of the Borel probability measures ν with $B_0(G) \le B_0(F) - \int_X (G-F)\,\mathrm{d}\nu$ for all $G \in C(X)$; by Theorem 16.2.1 it is nonempty, convex and weakly compact.

For $T \in \mathscr{M}(F)$ put

$$\nu_{T,F} = \Gamma_F(T)/M(F).$$

This is a probability measure.

Lemma 27.6.1 *Suppose $T_j \to T$ weakly in $\mathscr{G}_\omega$ and $\mathfrak{D}_F(T_j) \to M(F)$. Then $T \in \mathscr{M}(F)$ and $\Gamma_F(T_j) \rightharpoonup \Gamma_F(T)$. On $\mathscr{M}(F)$ the map $T \mapsto \Gamma_F(T)$ is affine and continuous.*

Proof Upper semicontinuity gives $\mathfrak{D}_F(T) \ge M(F)$, hence equality. The measures $\Gamma_F(T_j)$ have bounded mass; let Γ' be a weak cluster point, whose mass is $M(F)$. For a continuous real function h, choose $\varepsilon > 0$ with $1 \pm \varepsilon h > 0$. Since $\Gamma_{F+n\log(1\pm\varepsilon h)} = (1\pm\varepsilon h)\Gamma_F$, upper semicontinuity of $\mathfrak{D}$ for these two continuous logarithmic densities gives

$$\int_X (1\pm\varepsilon h)\,\mathrm{d}\Gamma' \le \int_X (1\pm\varepsilon h)\,\mathrm{d}\Gamma_F(T).$$

Since both measures have mass $M(F)$, the two signs give equality of the integrals of h. Thus every cluster point equals $\Gamma_F(T)$, which proves the convergence.

For affineness, take $T_1, T_2 \in \mathscr{G}_\omega$ and $t \in [0,1]$, and express the coefficient measures of both currents through densities with respect to the sum of their trace measures.

Concavity and homogeneity of the nth root of the determinant, applied to these matrix densities, give

$$\Gamma_F(tT_1 + (1-t)T_2) \geq t\Gamma_F(T_1) + (1-t)\Gamma_F(T_2), \qquad 0 \leq t \leq 1.$$

If T_1 and T_2 maximize, so does the current on the left, by concavity of $\mathfrak{D}_F$. Both measures in the inequality then have mass $M(F)$, so equality holds. This proves affineness, and continuity is the special case of the first assertion in which every T_j is a maximizer. □

Theorem 27.6.1 *For every $F \in C(X, \mathbb{R})$,*

$$\mathcal{A}_0(F) = \{\nu_{T,F} : T \in \mathscr{M}(F)\}.$$

Consequently the determinant measures $\Gamma_F(T)$ of all maximizing currents T are absolutely continuous with respect to $\mathrm{d}V$.

Proof For a maximizing T and any $G \in C(X)$,

$$M(G) \geq \mathfrak{D}_G(T) = M(F) \int_X \mathrm{e}^{(G-F)/n}\, \mathrm{d}\nu_{T,F}.$$

Since $B_0 = -n \log M$, taking $-n$ times the logarithm and applying Jensen's inequality gives

$$B_0(G) \leq B_0(F) - n \log \int_X \mathrm{e}^{(G-F)/n}\, \mathrm{d}\nu_{T,F} \leq B_0(F) - \int_X (G-F)\, \mathrm{d}\nu_{T,F}.$$

Hence every probability $\nu_{T,F}$ with $T \in \mathscr{M}(F)$ belongs to $\mathcal{A}_0(F)$.

Denote the set of these probabilities by $\mathcal{N}(F)$; by the preceding lemma it is compact and convex. To prove the reverse inclusion we compute the right derivative $(B_0)'_+(F; h) = \lim_{t \downarrow 0} (B_0(F + th) - B_0(F))/t$ for $h \in C(X)$. Since $\mathfrak{D}_{F+th}(T) = \int_X \mathrm{e}^{th/n}\, \mathrm{d}\Gamma_F(T)$ and the masses of the measures $\Gamma_F(T)$ are bounded on $\mathscr{G}_\omega$, a Taylor expansion of the exponential gives, as $t \downarrow 0$,

$$\mathfrak{D}_{F+th}(T) = \mathfrak{D}_F(T) + \frac{t}{n} \int_X h\, \mathrm{d}\Gamma_F(T) + O(t^2),$$

with the error bounded uniformly in T. Testing maximizers at F gives

$$\liminf_{t \downarrow 0} \frac{M(F+th) - M(F)}{t} \geq \frac{M(F)}{n} \max_{\nu \in \mathcal{N}(F)} \int_X h\, \mathrm{d}\nu.$$

For the reverse inequality, choose a maximizer T_t at $F + th$. The multiplicative bounds in the proof of Theorem 27.5.1 imply $\mathfrak{D}_F(T_t) \to M(F)$. Every sequence $t \downarrow 0$ has a subsequence along which T_t converges, and the preceding lemma identifies the limit of $\Gamma_F(T_t)$ with the determinant measure of a maximizer at F. Since $\mathfrak{D}_F(T_t) \leq M(F)$, the Taylor expansion gives the matching upper bound for the difference quotients of

M. Since $B_0 = -n \log M$, therefore

$$(B_0)'_+(F; h) = -\max_{\nu \in \mathcal{N}(F)} \int_X h \, \mathrm{d}\nu.$$

If $\nu \in \mathcal{A}_0(F)$, its supporting inequality in direction th implies

$$\int_X h \, \mathrm{d}\nu \le \max_{\widetilde{\nu} \in \mathcal{N}(F)} \int_X h \, \mathrm{d}\widetilde{\nu} \quad \text{for every } h \in C(X).$$

If ν were not in the weakly compact convex set $\mathcal{N}(F)$, the Hahn–Banach separation theorem in the dual of $C(X)$ would give some $h \in C(X)$ violating this inequality. Hence $\nu \in \mathcal{N}(F)$, and the two sets are equal. The absolute continuity of their members was proved in Theorem 16.2.1. □

For continuous F, the supporting probability is therefore unique exactly when all maximizing currents have the same normalized determinant measure, which is weaker than uniqueness of the maximizing current. Whether a maximizing current can be written as a conformal multiple of a power $(\omega + \mathrm{dd^c} u)^{n-1}$ of a weak solution form is a further question, discussed in Section 36.15; the construction above works directly with the coefficient measures of the currents.

27.7 Integrability of Gauduchon determinant measures

A lower bound for the constants of smooth equations in terms of an L^p norm of the density gives, by duality, a uniform L^q bound for the determinant measures of all normalized Gauduchon currents, for every exponent $q < n/(n-1)$. At the endpoint $q = n/(n-1)$, the optimal constant is exactly $v_-(\omega)^{-1/n}$, where $v_-(\omega)$ is the lower Monge–Ampère mass of Definition 7.5.1. Thus a uniform bound for the determinant densities of the dual tests at the critical exponent is equivalent to positive lower mass of the background.

Write $\Gamma_0(T)$ for the determinant measure $\Gamma_F(T)$ of (27.23) with $F = 0$.

Theorem 27.7.1 *For every $T \in \mathcal{G}_\omega$, the measure $\Gamma_0(T)$ is absolutely continuous with respect to $\mathrm{d}V$; we write $\Gamma_0(T) = k_T \, \mathrm{d}V$ and call $k_T \ge 0$ the* determinant density *of T. For every $1 < q < n/(n-1)$ there is $C_q > 0$ such that*

$$\|k_T\|_{L^q(\mathrm{d}V)} \le C_q \qquad (T \in \mathcal{G}_\omega).$$

Proof Fix $p > 1$ and a smooth strictly positive function h. By Theorem 10.1.1 there are a smooth form $\Omega_h = \omega + \mathrm{dd^c} u_h > 0$ and a constant $c_h > 0$ such that

$$\Omega_h^n = c_h h^n \, \mathrm{d}V.$$

The lower-density estimate of Theorem 11.3.1 gives a constant $c_p > 0$, independent of h, such that

$$c_h \|h^n\|_{L^p(\mathrm{d}V)} \geq c_p.$$

The pointwise matrix arithmetic–geometric mean inequality, applied to Ω_h and the coefficient density of T, gives the inequality of measures

$$c_h^{1/n} h \, \Gamma_0(T) \leq \Omega_h \wedge T.$$

Integrating and using $\mathrm{d}\mathrm{d}^{\mathrm{c}} T = 0$ yields

$$\int_X h \, \mathrm{d}\Gamma_0(T) \leq c_h^{-1/n} \int_X \Omega_h \wedge T = c_h^{-1/n} \leq c_p^{-1/n} \|h\|_{L^{np}(\mathrm{d}V)}.$$

Approximation by smooth strictly positive functions extends this bound to every continuous nonnegative h, and applied to $|h|$ it bounds the integral of every continuous function against $\Gamma_0(T)$ by its L^{np} norm. Since continuous functions are dense in L^{np}, the duality between L^{np} and $L^{np/(np-1)}$ shows that $\Gamma_0(T)$ has a density in $L^{np/(np-1)}$, with norm at most $c_p^{-1/n}$. As p ranges over $(1, +\infty)$, the exponent $np/(np-1)$ ranges over $(1, n/(n-1))$. This proves the theorem. □

In particular, the determinant of the matrix density vanishes on the singular part of the trace measure: if $\sigma_T = \sigma_{T,\mathrm{ac}} + \sigma_{T,\mathrm{s}}$ is its Lebesgue decomposition with respect to $\mathrm{d}V$, then

$$\det P_T = 0 \quad \sigma_{T,\mathrm{s}}\text{-almost everywhere.}$$

Indeed, $f_0 = (\mathrm{d}V/\omega^n)^{1/n}$ is everywhere positive and $\Gamma_0(T) = f_0 (\det P_T)^{1/n} \sigma_T$ is absolutely continuous. This says nothing about a possible singular part of the current T itself.

Theorem 27.7.2 *With the extended value $+\infty$ allowed, one has*

$$\sup_{T \in \mathcal{G}_\omega} \|k_T\|_{L^{n/(n-1)}(\mathrm{d}V)} = v_-(\omega)^{-1/n},$$

where the right side is interpreted as $+\infty$ when $v_-(\omega) = 0$. The supremum does not change if T is restricted to the smooth strictly positive members of $\mathcal{G}_\omega$. Thus a uniform estimate at the endpoint exponent holds if and only if $v_-(\omega) > 0$.

Proof Suppose first that $v_-(\omega) > 0$. For the smooth equation $\Omega_h^n = c_h h^n \, \mathrm{d}V$ in the proof of Theorem 27.7.1, the definition of v_- gives

$$c_h \int_X h^n \, \mathrm{d}V = \int_X \Omega_h^n \geq v_-(\omega).$$

The same pointwise inequality and Stokes' theorem therefore give

$$\int_X h \, \mathrm{d}\Gamma_0(T) \leq v_-(\omega)^{-1/n} \|h\|_{L^n(\mathrm{d}V)}.$$

By duality, $\|k_T\|_{L^{n/(n-1)}} \le v_-(\omega)^{-1/n}$ for every $T \in \mathcal{G}_\omega$.

For the reverse bound, let g be any smooth positive probability density relative to $\mathrm{d}V$, and solve

$$\Omega_g^n = c(g) g \, \mathrm{d}V.$$

Let T_g be the trace-normalized $(n-1)$st power of the Gauduchon metric conformal to Ω_g. Equality in the pointwise matrix inequality gives

$$\int_X g^{1/n} \, \mathrm{d}\Gamma_0(T_g) = c(g)^{-1/n}.$$

Since $\|g^{1/n}\|_{L^n} = 1$, Hölder's inequality gives $\|k_{T_g}\|_{L^{n/(n-1)}} \ge c(g)^{-1/n}$. By Theorems 10.8.1 and 10.8.2,

$$\inf_{g>0 \text{ smooth}, \, \int_X g \, \mathrm{d}V = 1} c(g) = v_-(\omega).$$

Taking the supremum over these smooth currents T_g proves the matching lower bound; if $v_-(\omega) = 0$, the same argument makes this supremum infinite. Since the currents T_g are smooth, this also proves that allowing nonsmooth currents does not change the supremum. □

The formula shows that the optimal endpoint constant does not depend on the choice of $\mathrm{d}V$, whereas the subcritical constants C_q may depend on it.

Since the lower-mass quotient (10.12) is monotone and homogeneous in the background, the endpoint bound passes to every larger background. For a Hermitian form h, define $\mathcal{G}_h$ and the determinant densities k_T, $T \in \mathcal{G}_h$, as above with h in place of ω.

Corollary 27.7.1 *Let g and h be Hermitian forms with $h \ge cg$ for some $c > 0$. Then*

$$v_-(h) \ge c^n v_-(g), \qquad \sup_{T \in \mathcal{G}_h} \|k_T\|_{L^{n/(n-1)}(\mathrm{d}V)} \le c^{-1} v_-(g)^{-1/n}, \tag{27.25}$$

where the right side of the second inequality is $+\infty$ when $v_-(g) = 0$.

Proof Since $h \ge cg$, each quotient in (10.12) for h is at least c times the corresponding quotient for g, so $v_-(h) \ge c^n v_-(g)$. Theorem 27.7.2, applied to h, gives $\sup_{T \in \mathcal{G}_h} \|k_T\|_{L^{n/(n-1)}(\mathrm{d}V)} = v_-(h)^{-1/n} \le c^{-1} v_-(g)^{-1/n}$. □

Corollary 27.7.2 *For every $R > 0$ and $1 < q < n/(n-1)$ there is a constant $C_{R,q}$ with the following property. If $F \in C(X)$ and $\|F\|_\infty \le R$, every supporting probability $\nu \in \mathcal{A}_0(F)$ of Theorem 27.6.1 has a density a satisfying*

$$\nu = a\,\mathrm{d}V, \qquad \|a\|_{L^q(\mathrm{d}V)} \le C_{R,q}.$$

If $\nu_-(\omega) > 0$, this holds also for $q = n/(n-1)$.

Proof Recall that $M(F) = \exp(-B_0(F)/n)$, and represent ν by a maximizing current $T \in \mathscr{M}(F)$, using Theorem 27.6.1. Since $\Gamma_F(T) = \mathrm{e}^{F/n}\Gamma_0(T)$ and B_0 is 1-Lipschitz,

$$\nu = \frac{\mathrm{e}^{F/n} k_T}{M(F)}\,\mathrm{d}V, \qquad M(F) \ge \mathrm{e}^{-R/n} M(0) > 0.$$

Theorem 27.7.1 therefore gives the subcritical bound with constant $\mathrm{e}^{2R/n} M(0)^{-1} C_q$. When $\nu_-(\omega) > 0$, the same argument with the endpoint constant $\nu_-(\omega)^{-1/n}$ of Theorem 27.7.2 gives the case $q = n/(n-1)$. □

On the threefold of Theorem 27.18.1, every Hermitian form has zero lower mass, so the endpoint estimate fails for the whole family $\mathscr{G}_\omega$, while all the subcritical estimates above remain valid. For a fixed smooth density, Theorem 27.5.1 gives a unique smooth maximizing current, and its determinant density is bounded. Sequences of currents whose values approach the maximum need not, however, have uniformly bounded higher moments, as Proposition 27.17.1 shows.

On the other hand, the endpoint estimate can hold on a manifold with infinite upper mass. On the Hopf threefold of Chapter 9, take the Gauduchon metric ω of that chapter, for which $\nu_-(\omega) = 6$ and $\nu_+(\omega) = +\infty$, and set $\mathrm{d}V = \omega^3/6$. The normalized Gauduchon form $T_0 = \omega^2/6$ has $\sigma_{T_0} = \mathrm{d}V$ and $P_{T_0} = I$. Hence $k_{T_0} = 6^{-1/3}$, and

$$\sup_{T \in \mathscr{G}_\omega} \|k_T\|_{L^{3/2}(\mathrm{d}V)} = 6^{-1/3}.$$

Thus an endpoint determinant estimate, with its optimal constant attained, can coexist with infinite upper Monge–Ampère mass.

27.7.1 Uniform bounds near a smooth density

The endpoint estimate can fail for the family of all tests while the maximizing tests for a family of densities remain uniformly controlled. For smooth densities close in $C^{2,\alpha}$ to the normalized volume form $\omega^n / \int_X \omega^n$ of the background, a fixed-point argument keeps the solution forms close to ω and their Gauduchon multipliers close to that of ω. This gives uniform bounds for both the determinant density of the maximizing current and the density of the supporting probability.

Fix $0 < \alpha < 1$. All Hölder norms are defined using a fixed finite atlas and fixed smooth connections. Put

$$H = \frac{\omega^n}{\mathrm{d}V}, \qquad V = \int_X \omega^n, \qquad F_* = \log(H/V).$$

Thus $\mathrm{e}^{F_*}\mathrm{d}V = \omega^n/V$ is a probability measure. By Theorem 4.1.1, there is a unique smooth positive function χ_0, the normalized Gauduchon multiplier of ω, such that

$$\mathrm{dd}^{\mathrm{c}}(\chi_0\omega^{n-1}) = 0, \qquad \int_X \chi_0\omega^n = 1.$$

Write $m = \min_X \chi_0$, $a = \max_X \chi_0$ and $C_0 = \|\chi_0\|_{C^{2,\alpha}}$.

We first specify the constants that determine the size of the neighborhood. For a smooth positive $(1,1)$-form Ω, set

$$L_\omega v = \frac{n\mathrm{dd}^{\mathrm{c}} v \wedge \omega^{n-1}}{\omega^n}, \qquad A_\Omega \eta = \frac{\mathrm{dd}^{\mathrm{c}}(\eta\Omega^{n-1})}{\mathrm{d}V}.$$

The compatibility argument of Propositions 4.8.1 and 10.3.1 shows that

$$\mathcal{L}(v,c) = L_\omega v - c, \qquad \int_X v\,\mathrm{d}V = 0,$$

is an isomorphism from the space of pairs $(v,c) \in C^{4,\alpha} \times \mathbb{R}$ with this normalization onto $C^{2,\alpha}$. Let $K \geq 1$ bound its inverse for the norm $\|v\|_{C^{4,\alpha}} + |c|$. The formal adjoint of A_ω with respect to $\mathrm{d}V$ is $(H/n)L_\omega$, whose kernel consists of the constants and whose index is zero. Consequently A_ω has kernel $\mathbb{R}\chi_0$ and range

$$Z_0 = \left\{ g \in C^{0,\alpha} : \int_X g\,\mathrm{d}V = 0 \right\}.$$

Its restriction to

$$E_\chi = \left\{ \eta \in C^{2,\alpha} : \int_X \eta\omega^n = 0 \right\}$$

has a bounded inverse $\mathcal{R}\colon Z_0 \to E_\chi$. Fix $G \geq 1$ bounding this inverse.

We normalize the Hölder norms of forms and of endomorphism fields so that they dominate pointwise operator norms and are submultiplicative under composition of endomorphisms; the subscript alg below refers to this norm on endomorphism fields. Choose $D \geq 1$ so that

$$\|\mathrm{dd}^{\mathrm{c}} v\|_{C^{2,\alpha}} \leq D\|v\|_{C^{4,\alpha}}, \qquad \|\omega^{-1}\mathrm{dd}^{\mathrm{c}} v\|_{C^{2,\alpha},\mathrm{alg}} \leq D\|v\|_{C^{4,\alpha}}.$$

Let $C_{\mathrm{tr}} \geq 1$ be a bound for the trace map on endomorphism fields, and put $Q = 2C_{\mathrm{tr}}D^2$. Expanding $\Omega^{n-1} - \omega^{n-1}$ gives a constant $C_A \geq 1$ such that

$$\|A_\Omega - A_\omega\|_{C^{2,\alpha}\to C^{0,\alpha}} \leq C_A\|\Omega - \omega\|_{C^{2,\alpha}} \quad \text{if } \|\Omega - \omega\|_{C^{2,\alpha}} \leq 1.$$

Only two derivatives of Ω enter this polynomial estimate. Define

$$r = \min\left\{\frac{1}{2D}, \frac{1}{4KQ}, \frac{m}{4GC_A DC_0}\right\}, \qquad \varepsilon = \frac{r}{2K}. \tag{27.26}$$

These constants depend only on ω, $\mathrm{d}V$, α and the chosen norms.

Lemma 27.7.1 *Let F be smooth with $\|F - F_*\|_{C^{2,\alpha}} \le \varepsilon$. The solution form $\Omega_F = \omega + \mathrm{dd^c} u_F$ of $\mathrm{MA}_\omega(u_F) = \mathrm{e}^{B_0(F)+F}\mathrm{d}V$ and its Gauduchon multiplier χ_F, normalized as below, satisfy*

$$\tfrac{1}{2}\omega \le \Omega_F \le \tfrac{3}{2}\omega, \qquad \frac{m}{2} \le \chi_F \le \frac{3a}{2},$$
$$\mathrm{dd^c}(\chi_F \Omega_F^{n-1}) = 0, \qquad \int_X \chi_F \omega^n = 1.$$

The potential can be normalized by $\int_X u_F \,\mathrm{d}V = 0$ with $\|u_F\|_{C^{4,\alpha}} + |B_0(F) - \log V| \le r$.

Proof For $\|v\|_{C^{4,\alpha}} \le 1/(2D)$ put

$$N(v) = \log\det(I + \omega^{-1}\mathrm{dd^c} v) - L_\omega v.$$

The logarithm series starts at degree two. If the endomorphism norms of $\omega^{-1}\mathrm{dd^c} v$ and $\omega^{-1}\mathrm{dd^c} w$ are at most $\rho \le 1/2$, telescoping their powers bounds the difference series by $C_{\mathrm{tr}}\rho/(1-\rho)$ times the endomorphism difference. Hence

$$\|N(v)\|_{C^{2,\alpha}} \le Q\|v\|^2_{C^{4,\alpha}},$$
$$\|N(v) - N(w)\|_{C^{2,\alpha}} \le Q(\|v\|_{C^{4,\alpha}} + \|w\|_{C^{4,\alpha}})\|v - w\|_{C^{4,\alpha}}.$$

On the closed radius-r ball of normalized pairs (v, c), the map

$$(v, c) \longmapsto \mathcal{L}^{-1}(F - F_* - N(v))$$

has image norm at most $K\varepsilon + KQr^2 \le 3r/4$ and Lipschitz constant at most $2KQr \le 1/2$. Its fixed point (u, c) satisfies

$$\log\frac{(\omega + \mathrm{dd^c} u)^n}{\omega^n} - c = F - F_*.$$

The estimate $Dr \le 1/2$ gives the form comparison. The scalar is $b = \log V + c$. Differentiating the local equation gives a uniformly elliptic equation for each first derivative of u, with coefficients initially in $C^{2,\alpha}$. Interior Schauder estimates improve the regularity one order at a time, so u is smooth. Uniqueness in Theorem 10.1.1 identifies it with u_F and gives $b = B_0(F)$. No uniform higher-derivative bounds on F enter the estimates above.

Put $\Delta A = A_{\Omega_F} - A_\omega$. Stokes' theorem makes its image lie in Z_0. On the complete space E_χ, the affine map

$$\eta \longmapsto -\mathcal{R}\Delta A(\chi_0 + \eta)$$

has Lipschitz constant $\kappa = GC_A Dr \le m/(4C_0) \le 1/4$. Its fixed point obeys

$$\|\eta\|_{C^{2,\alpha}} \le \frac{\kappa C_0}{1-\kappa} \le 2\kappa C_0 \le m/2.$$

Since $A_\omega \chi_0 = 0$ and $A_\omega \mathcal{R}$ is the identity on Z_0, the function $\chi_F = \chi_0 + \eta$ satisfies $A_{\Omega_F} \chi_F = 0$, and it has the stated normalization and bounds. Elliptic regularity makes it smooth. Every Gauduchon multiplier of Ω_F with the same normalization satisfies the same affine fixed-point equation, so this multiplier is the one given by Theorem 4.1.1. □

Theorem 27.7.3 *For every smooth F in the ball of Lemma 27.7.1, let T_F be the maximizing current of Theorem 27.5.1 and ν_F the supporting probability of B_0 at F; both are unique. With*

$$H_{\max} = \max_X H, \qquad C_\nu = \frac{3^{n+1}}{4} a H_{\max}, \qquad C_k = \frac{3^n}{2} a H_{\max}^{(n-1)/n},$$

their densities satisfy

$$\nu_F = q_F \, \mathrm{d}V, \qquad \|q_F\|_{L^\infty} \le C_\nu, \qquad \|k_{T_F}\|_{L^\infty} \le C_k.$$

In particular, for $p = n/(n-1)$,

$$\|q_F\|_{L^p} \le C_\nu^{1/n}, \qquad \|k_{T_F}\|_{L^p} \le C_k.$$

The constants do not involve $v_-(\omega)$; in particular, these bounds hold also when $v_-(\omega) = 0$.

Proof Set $\Omega = \Omega_F$, $\chi = \chi_F$ and

$$Z = \int_X \chi \omega \wedge \Omega^{n-1}.$$

The form comparison and $\int_X \chi \omega^n = 1$ give

$$2^{-(n-1)} \le Z \le (3/2)^{n-1}. \tag{27.27}$$

Since $\mathrm{dd}^{\mathrm{c}}(\chi \Omega^{n-1}) = 0$, Stokes' theorem also gives $Z = \int_X \chi \Omega^n$. The form $T = (\chi/Z)\Omega^{n-1}$ is normalized and is a constant multiple of the $(n-1)$st power of the Gauduchon metric conformal to Ω. By Theorem 27.5.1, it is the unique maximizing current, so $T = T_F$. The equality case in the proof of that theorem gives $\nu_F = \Omega \wedge T_F$,

and hence

$$\begin{aligned} T_F &= \frac{\chi}{Z}\Omega^{n-1}, \\ \nu_F &= \Omega \wedge T_F = \frac{\chi}{Z}\Omega^n, \\ q_F &= \frac{\chi}{Z}\frac{\Omega^n}{\mathrm{d}V}, \\ k_{T_F} &= \frac{\chi}{Z}\left(\frac{\Omega^n}{\mathrm{d}V}\right)^{(n-1)/n}. \end{aligned} \tag{27.28}$$

The last identity uses the positive root $\gamma = (\chi/Z)^{1/(n-1)}\Omega$, for which $T_F = \gamma^{n-1}$. Uniqueness of the supporting probability follows from Theorem 27.6.1.

Now $\chi/Z \le (3a/2)2^{n-1}$ and $\Omega^n/\mathrm{d}V \le (3/2)^n H_{\max}$, and multiplying gives exactly $q_F \le C_\nu$ and $k_{T_F} \le C_k$. Since $\int_X q_F \,\mathrm{d}V = 1$, we have $\int_X q_F^p \,\mathrm{d}V \le C_\nu^{p-1}$, which gives the sharper L^p estimate for q_F. The L^p estimate for k_{T_F} follows from $\int_X \mathrm{d}V = 1$. □

Corollary 27.7.3 *The ball of Lemma 27.7.1 contains smooth injectively parametrized families, of arbitrarily large finite dimension, of data F normalized by $\int_X \mathrm{e}^F \mathrm{d}V = 1$. It also contains a smooth path of such data whose maximizing currents are pairwise distinct.*

Proof For a small smooth function f, put

$$F(f) = F_* + f - \log \int_X \mathrm{e}^{F_*+f} \,\mathrm{d}V.$$

If $F(f) = F(g)$, then $f - g$ is constant, so the map $f \mapsto F(f)$ is injective on functions of zero $\mathrm{d}V$-mean. By its continuity in $C^{2,\alpha}$, it maps a sufficiently small neighborhood of zero in any fixed finite-dimensional space of such smooth functions into the ball, and these spaces can have arbitrary dimension.

For the path, choose a smooth real ψ of zero $\mathrm{d}V$-mean whose complex Hessian at some point x_0 has, in an ω-unitary frame, a matrix Q_0 that is not a multiple of the identity. A cutoff of $|z_1|^2 - |z_2|^2$ in local coordinates, minus its mean, is such a function. For small real t put

$$\Omega_t = \omega + t\mathrm{dd}^{\mathrm{c}}\psi, \qquad V_t = \int_X \Omega_t^n, \qquad F_t = \log \frac{\Omega_t^n}{V_t \,\mathrm{d}V}.$$

These normalized data tend to F_* in $C^{2,\alpha}$ as $t \to 0$, and by uniqueness in Theorem 10.1.1 their solution forms are the forms Ω_t. If the maximizing currents for F_s and F_t agree, (27.28) makes Ω_s^{n-1} and Ω_t^{n-1} proportional at x_0. Positive cofactor matrices determine positive forms up to scale, so $I + sQ_0$ and $I + tQ_0$ are proportional. Since I and Q_0 are linearly independent, comparing coefficients gives $s = t$. □

The small $C^{2,\alpha}$ neighborhood gives a condition directly on the prescribed density. The next estimate instead assumes uniform bounds for the solution forms and covers a larger smooth family without derivative bounds on its densities. Whether such uniform

bounds hold for the maximizers of arbitrary bounded families of continuous densities remains open when $v_-(\omega) = 0$. The concentrating sequences of Proposition 27.17.1 consist of near-maximizers, so these estimates do not apply to them.

27.7.2 Uniformly controlled complex Hessians

The preceding construction bounds the solution form and its Gauduchon multiplier by controlling derivatives of the logarithmic density. A bound on the solution form itself leads to a different estimate. If $\Omega = \omega + \mathrm{dd}^{\mathrm{c}} v$, then $\mathrm{d}\Omega = \mathrm{d}\omega$, so the exterior derivative of the cofactor Ω^{n-1} depends only on the fixed torsion and the size of Ω. This permits uniform bounds for the multiplier even when derivatives of the varying complex Hessian are unbounded.

For $n = 2$, every Hermitian form has positive lower mass by Corollary 27.18.1, so the critical integrability in Theorem 27.7.4 below, as in Theorem 27.7.3, also follows from Theorem 27.7.2 and Corollary 27.7.2; in that dimension the point of these theorems is the pointwise form of their bounds. We first give the real elliptic estimate used below. A cube denoted by tQ has the same center as Q and t times its side length, and $\fint_Q$ denotes its Euclidean average.

Lemma 27.7.2 *Let $Q \subset \mathbb{R}^m$, $m > 2$, be a cube. Suppose that u is a smooth strictly positive solution on a neighborhood of $3Q$ of*

$$\operatorname{div}(A\nabla u + Bu) = 0, \qquad \lambda I \le A \le \Lambda I, \qquad |B| \le K,$$

where A is a smooth symmetric matrix field and B is a smooth vector field. Then

$$\sup_{Q/2} u \le C \inf_{Q/2} u,$$

where C depends only on m, λ, Λ, K and the side length of Q. Derivatives of A and B do not enter C.

Proof We first estimate positive and negative powers, then compare their small moments by an estimate for $\log u$.

For $p \neq 0, 1$, test the weak equation by $\eta^2 u^{p-1}$, where η is a smooth cutoff. The principal term is

$$(p-1)\int \eta^2 u^{p-2}\langle A\nabla u, \nabla u\rangle.$$

The other terms contain respectively $2\eta u^{p-1} A\nabla u \cdot \nabla\eta$, $(p-1)\eta^2 u^{p-1} B \cdot \nabla u$, and $2\eta u^p B \cdot \nabla\eta$. Taking absolute values after moving the principal term to one side and applying Young's inequality absorbs half of its energy. Thus, for either sign of p,

$$\int \eta^2 |\nabla u^{p/2}|^2 \le C_1 p^2 (1 + |p-1|^{-2}) \int (|\nabla\eta|^2 + K^2\eta^2) u^p, \tag{27.29}$$

with C_1 depending only on the ellipticity bounds. Each test is legitimate since this individual smooth positive solution is bounded above and away from zero on the cutoff support.

Put $\beta = m/(m-2) > 1$ and choose $p_0 = \beta^{-k-1/2}$, with k to be fixed later. Apply the Euclidean Sobolev inequality to $\eta u^{p/2}$ on nested cubes shrinking from Q to $Q/2$. Choose the cutoff gaps proportional to 2^{-j} and the powers $p_j = p_0\beta^j$. The positive powers never equal one: the closest two are $\beta^{-1/2}$ and $\beta^{1/2}$. Consequently the constants in (27.29), together with the cutoff terms from $\nabla(\eta u^{p/2})$, grow at most polynomially in p_j and exponentially in j. Since $\sum_j (j+1)\beta^{-j} < \infty$, their iteration product converges. Negative powers give the same calculation for u^{-1}. We obtain

$$\sup_{Q/2} u \le C_2 \left(\fint_Q u^{p_0} \right)^{1/p_0}, \qquad \inf_{Q/2} u \ge C_2^{-1} \left(\fint_Q u^{-p_0} \right)^{-1/p_0}. \tag{27.30}$$

Here C_2 also depends on p_0 and the fixed cube size.

Testing instead by η^2/u gives

$$\int \eta^2 |\nabla \log u|^2 \le C_3 \int (|\nabla \eta|^2 + K^2 \eta^2).$$

For every subcube $S \subset Q$ of side length ℓ, a cutoff supported in $2S \subset 3Q$, equal to one on S, has gradient bounded by a dimensional constant times $1/\ell$. Poincaré's inequality and Cauchy–Schwarz therefore give

$$\fint_S |\log u - (\log u)_S| \le C_4(1 + K\ell) \le K_0,$$

where $K_0 \ge 1$ is uniform and $(\log u)_S$ denotes the mean. We spell out the exponential consequence of this mean oscillation bound.

Write $f = \log u$ and fix $a > 2$. Select the maximal dyadic subcubes S of Q on which $\fint_S |f - f_Q| > aK_0$. Their total measure is at most $|Q|/a$. The parent of each selected cube has average at most aK_0, hence $|f_S - f_Q| \le 2^m a K_0$. Outside their union, differentiation of dyadic averages gives $|f - f_Q| \le aK_0$ almost everywhere. Repeat this selection inside each chosen cube, subtracting its own mean. The union at generation j has measure at most $a^{-j}|Q|$, and outside generation $j+1$ the accumulated mean changes and last oscillation give

$$|f - f_Q| \le (j 2^m a + a) K_0.$$

The intersection of all generations is null. Choosing $c > 0$ so that $\exp(c 2^m a) < a$ and summing over the generations yields

$$\fint_Q \exp\left(\frac{c|f - f_Q|}{K_0} \right) \le C_5,$$

with dimensional constants. Now fix k large enough that $p_0 < c/K_0$. Multiplying the estimates for the positive and negative exponentials cancels f_Q and gives

$$\left(\fint_Q u^{p_0}\right)\left(\fint_Q u^{-p_0}\right) \le C_5^2.$$

Combining this with (27.30) proves the result. Every constant comes from the stated coefficient bounds and the fixed cube. □

Proposition 27.7.1 *Fix $0 < \delta \le \Lambda$. For every smooth real v with*

$$\delta\omega \le \Omega_v := \omega + \mathrm{d}\mathrm{d}^{\mathrm{c}} v \le \Lambda\omega,$$

let ρ_v be its Gauduchon multiplier, normalized by

$$\mathrm{d}\mathrm{d}^{\mathrm{c}}(\rho_v \Omega_v^{n-1}) = 0, \qquad \int_X \rho_v \,\mathrm{d}V = 1.$$

There is $C_H \ge 1$, depending only on $X, \omega, \mathrm{d}V, \delta, \Lambda$, such that $C_H^{-1} \le \rho_v \le C_H$ on X for every such v.

Proof Existence and uniqueness of ρ_v follow from Theorem 4.1.1. Fix a real reference metric on X, and put $Q_v = \Omega_v^{n-1}$ and $J_v = \Omega_v^n/\mathrm{d}V$. Since $\mathrm{d}\Omega_v = \mathrm{d}\omega$,

$$\mathrm{d}Q_v = (n-1)\mathrm{d}\omega \wedge \Omega_v^{n-2}. \tag{27.31}$$

Both Q_v and $\mathrm{d}Q_v$ have uniformly bounded coefficients in fixed charts.

Define a real tensor A_v and vector field b_v by

$$\langle A_v \mathrm{d}\phi, \mathrm{d}\psi\rangle \,\mathrm{d}V = n\,\mathrm{d}\psi \wedge \mathrm{d}^{\mathrm{c}}\phi \wedge Q_v, \qquad (b_v \cdot \mathrm{d}\phi)\,\mathrm{d}V = -n\,\mathrm{d}^{\mathrm{c}}\phi \wedge \mathrm{d}Q_v$$

for smooth real functions ϕ, ψ. Only the $(1,1)$ part of $\mathrm{d}\psi \wedge \mathrm{d}^{\mathrm{c}}\phi$ contributes to the first identity; it is

$$\frac{\mathrm{i}}{4\pi}(\partial\psi \wedge \bar\partial\phi + \partial\phi \wedge \bar\partial\psi).$$

Thus A_v is symmetric and uniformly positive definite, with a uniform upper bound. The second identity and (27.31) give a uniform bound for b_v.

For $L_{\Omega_v}\phi = n\mathrm{d}\mathrm{d}^{\mathrm{c}}\phi \wedge Q_v/\Omega_v^n$, Stokes' theorem gives

$$\begin{aligned}\int_X \psi J_v L_{\Omega_v}\phi \,\mathrm{d}V &= -n\int_X \mathrm{d}\psi \wedge \mathrm{d}^{\mathrm{c}}\phi \wedge Q_v + n\int_X \psi \mathrm{d}^{\mathrm{c}}\phi \wedge \mathrm{d}Q_v \\ &= -\int_X \langle A_v \mathrm{d}\phi, \mathrm{d}\psi\rangle \,\mathrm{d}V - \int_X \psi b_v \cdot \mathrm{d}\phi \,\mathrm{d}V.\end{aligned}$$

For $\psi = \rho_v$ the left side vanishes for every ϕ, since $\mathrm{d}\mathrm{d}^{\mathrm{c}}(\rho_v Q_v) = 0$. As A_v is symmetric, the smooth function ρ_v therefore solves

$$\mathrm{div}_{\mathrm{d}V}(A_v \mathrm{d}\rho_v + b_v \rho_v) = 0.$$

In each coordinate chart, multiplying the coefficients by the fixed smooth volume density converts this to Euclidean divergence form while preserving uniform ellipticity and a uniform bound for the vector coefficient. Lemma 27.7.2 applies, with $m = 2n \geq 4$. Cover X by the half cubes $Q/2$ of finitely many coordinate cubes Q for which $3Q$ lies in a chart. Since X is connected, the overlap graph of this cover is connected, and chaining the local estimates gives $\sup_X \rho_v \leq C_H \inf_X \rho_v$. The normalization and $\mathrm{d}V(X) = 1$ imply the claimed two-sided bound. □

For a fixed $0 < \varepsilon < 1$, consider the full family

$$\mathcal{U}_\varepsilon = \{v \in C^\infty(X, \mathbb{R}) : \sup_X v = 0, -\varepsilon\omega \leq \mathrm{dd^c} v \leq \varepsilon\omega\}. \tag{27.32}$$

With $J_v = \Omega_v^n/\mathrm{d}V$ as in the preceding proof, set $c_v = \int_X \Omega_v^n$ and $F_v = \log(J_v/c_v)$, so that $\int_X \mathrm{e}^{F_v}\mathrm{d}V = 1$. Here the density F_v is read off from a given solution form instead of being prescribed in advance.

Theorem 27.7.4 *Let*

$$j_- = (1-\varepsilon)^n \min_X \frac{\omega^n}{\mathrm{d}V}, \qquad j_+ = (1+\varepsilon)^n \max_X \frac{\omega^n}{\mathrm{d}V},$$

and take C_H from Proposition 27.7.1 with $\delta = 1-\varepsilon$ and $\Lambda = 1+\varepsilon$. For every $v \in \mathcal{U}_\varepsilon$, the exact maximizing current $T_v \in \mathcal{G}_\omega$ for F_v is unique. Its determinant density k_{T_v} and the density $r_v = \mathrm{d}\nu_{T_v,F_v}/\mathrm{d}V$ of its supporting probability satisfy

$$\frac{j_-}{C_H j_+} \leq r_v \leq \frac{C_H j_+}{j_-}, \qquad j_+^{-1/n}\frac{j_-}{C_H j_+} \leq k_{T_v} \leq j_-^{-1/n}\frac{C_H j_+}{j_-}. \tag{27.33}$$

In particular both densities are uniformly bounded at the critical exponent $n/(n-1)$, without an assumption on the lower mass. Moreover,

$$\|F_v - F_0\|_\infty \leq n \log\frac{1+\varepsilon}{1-\varepsilon},$$

where $F_0 = F_* = \log(\omega^n/(c_0\,\mathrm{d}V))$ is the density for $v = 0$, and this normalized family contains infinitely many distinct nonconstant densities on the fixed manifold X.

Proof Use the normalization of Proposition 27.7.1. The equation $\Omega_v^n = c_v \mathrm{e}^{F_v}\mathrm{d}V$ gives $B_0(F_v) = \log c_v$, so Ω_v is the solution form for F_v. The identification of the maximizing current and of its densities (27.28) in the proof of Theorem 27.7.3 uses only that Ω_F is the solution form of a smooth datum and that $\chi_F > 0$ is a Gauduchon multiplier of Ω_F. Applied with Ω_v and ρ_v in place of Ω_F and χ_F, and with

$$N_v = \int_X \rho_v \Omega_v^n = \int_X J_v \rho_v \,\mathrm{d}V$$

in place of Z, it shows that $T_v = (\rho_v/N_v)\Omega_v^{n-1}$ is the unique maximizing current and that

$$k_{T_v} = \frac{\rho_v}{N_v} J_v^{(n-1)/n}, \qquad r_v = \frac{J_v \rho_v}{N_v}. \tag{27.34}$$

Here $j_- \le J_v \le j_+$ and, since $\int_X \rho_v \, \mathrm{d}V = 1$, also $j_- \le N_v \le j_+$. The bounds for ρ_v in Proposition 27.7.1 give the first pair of inequalities in (27.33); the identity $k_{T_v} = J_v^{-1/n} r_v$ gives the second. The critical norms are bounded by the respective upper constants because $\mathrm{d}V$ is a probability volume.

The inequalities

$$(1-\varepsilon)^n \le \frac{J_v}{J_0} \le (1+\varepsilon)^n, \qquad (1-\varepsilon)^n \le \frac{c_v}{c_0} \le (1+\varepsilon)^n$$

give the bound for $F_v - F_0$ by subtracting logarithms. To prove nontriviality, choose a nonconstant smooth real ϕ. For small $|t|$, the potential $v_t = t\phi - \sup_X(t\phi)$ belongs to $\mathcal{U}_\varepsilon$. If $F_{v_s} = F_{v_t}$, then v_s and v_t solve the same smooth equation with the same normalization, so $v_s = v_t$ by the uniqueness in Theorem 10.1.1; hence $(s-t)\phi$ is constant and $s = t$. Thus the densities F_{v_t} are pairwise distinct. Since $\int_X \mathrm{e}^{F_v} \mathrm{d}V = 1$ and $\mathrm{d}V$ is a probability volume, the only constant density in the family is zero, so at most one F_{v_t} is constant. □

Uniform bounds for the solution forms need not give derivative bounds, or even equicontinuity, for their normalized logarithmic densities.

Example 27.7.1 For every fixed $0 < \varepsilon < 1$, there are $v_j \in \mathcal{U}_\varepsilon$ with $\|v_j\|_{C^0} \to 0$ such that their normalized densities F_{v_j} are pairwise distinct, bounded in C^0, and unbounded in C^1. This family is not equicontinuous. Nevertheless, the determinant densities of the maximizing currents and the densities of the supporting probabilities satisfy the uniform bounds of Theorem 27.7.4.

Proof Choose holomorphic coordinates centered at a point p, put $x = \operatorname{Re} z_1$, and take a smooth real cutoff ζ supported in the chart and equal to one near p. We may assume $0 \le \zeta \le 1$. On the chart set

$$\beta = \mathrm{d}x \wedge \mathrm{d}^{\mathrm{c}}x = \mathrm{d}\mathrm{d}^{\mathrm{c}}(x^2/2) = \frac{\mathrm{i}}{8\pi} \mathrm{d}z_1 \wedge \mathrm{d}\bar{z}_1.$$

This is a nonzero semipositive form of rank one. For a constant $a > 0$ to be chosen, extend

$$w_j = \frac{a}{j^2} \zeta \sin(jx)$$

by zero outside the chart, and put $v_j = w_j - \sup_X w_j$. The product rule gives

$$\begin{aligned} \mathrm{d}\mathrm{d}^{\mathrm{c}} w_j = & -a\zeta \sin(jx)\beta + \frac{a}{j} \cos(jx)(\mathrm{d}\zeta \wedge \mathrm{d}^{\mathrm{c}}x + \mathrm{d}x \wedge \mathrm{d}^{\mathrm{c}}\zeta) \\ & + \frac{a}{j^2} \sin(jx) \mathrm{d}\mathrm{d}^{\mathrm{c}}\zeta. \end{aligned}$$

All fixed forms in this expression have compact support after multiplication by the indicated cutoff or its derivatives. Thus a fixed $C > 0$ bounds the operator norm of $\mathrm{dd}^{\mathrm{c}} w_j$ relative to ω by aC for every $j \geq 1$. Choose a so small that $aC \leq \varepsilon/2$. Then $v_j \in \mathcal{U}_\varepsilon$, and $\|v_j\|_{C^0} \leq 2a/j^2$.

Write $s = \mathrm{tr}_{\omega(p)} \beta > 0$. Since $x(p) = 0$ and $\zeta = 1$ near p, we have $\Omega_{v_j}(p) = \omega(p)$ and

$$\partial_x \Omega_{v_j}(p) = \partial_x \omega(p) - aj\beta.$$

Differentiating the logarithm of the determinant, and noting that the normalizing constant c_{v_j} does not depend on the point, gives the identity

$$\partial_x F_{v_j}(p) = \partial_x \log \frac{\omega^n}{\mathrm{d}V}(p) - ajs.$$

These derivatives are pairwise different and their absolute values tend to infinity. Hence the densities are pairwise distinct and their C^1 norms are unbounded. Their uniform C^0 bound is part of Theorem 27.7.4.

To check failure of equicontinuity, take points $p_j^{\pm}$ with $x = \pm\pi/(2j)$ and all other real coordinates zero. For large j they lie where $\zeta = 1$, and $\Omega_{v_j}(p_j^{\pm}) = \omega(p_j^{\pm}) \mp a\beta$. The integral normalizing F_{v_j} cancels from the difference, so

$$F_{v_j}(p_j^-) - F_{v_j}(p_j^+) \longrightarrow \log \frac{1+as}{1-as} > 0.$$

Indeed, $\beta^2 = 0$ gives $(\omega(p) \pm a\beta)^n = (1 \pm as)\omega(p)^n$, and $as < 1$ by the chosen form comparison. Both points tend to p, which proves the failure of equicontinuity. The bounds of Theorem 27.7.4 still apply because the entire sequence lies in $\mathcal{U}_\varepsilon$. □

The same estimate applies to the family with $\delta\omega \leq \Omega_v \leq \Lambda\omega$, with C_H taken for these δ, Λ and with j_- and j_+ replaced by $\delta^n \min_X(\omega^n/\mathrm{d}V)$ and $\Lambda^n \max_X(\omega^n/\mathrm{d}V)$. Applying the estimate to a prescribed family of densities requires uniformly comparable solution forms.

27.8 Integrable densities in the compact Gauduchon problem

We keep the Hermitian background ω and replace the continuous logarithmic density by an integrable density. For an unbounded density, weak compactness of the test currents alone does not pass the determinant integral to the limit; uniform control of the weighted tails provides the missing upper semicontinuity.

We use the following notation from Sections 15.27 and 15.28, with ω in place of h. For a nonzero nonnegative $f \in L^1(\mathrm{d}V)$, the subsolution threshold $b_\omega(f) \geq 0$ is, by Theorem 15.27.1, the largest $b \geq 0$ for which some $u \in \mathrm{PSH}(X,\omega)$ satisfies $\mathrm{MA}^{\mathrm{ext}}_\omega(u) \geq b f \,\mathrm{d}V$. For $p \in L^n$, the root functional is $\rho_\omega(p) = b_\omega(|p|^n)^{-1/n}$ if $p \neq 0$, and $\rho_\omega(0) = 0$. The space $\mathcal{D}_\omega$ of Proposition 15.31.2 consists of the $p \in L^n$ with $\rho_\omega(p) < +\infty$, and $\mathcal{D}^0_\omega$ is the closure of the bounded functions in $\mathcal{D}_\omega$, characterized by vanishing tails in Proposition 15.31.4. For a Gauduchon metric γ, $L_{\omega,\gamma}(p)$ denotes the test quotient of Section 15.28, and Theorem 15.28.1 gives $\rho_\omega(p) = \sup_\gamma L_{\omega,\gamma}(|p|)$. Finally, k_T is the determinant density of $T \in \mathcal{G}_\omega$ from Theorem 27.7.1.

If $p \geq 0$ is measurable, define

$$D_p(T) = \int_X p k_T \,\mathrm{d}V \qquad (T \in \mathcal{G}_\omega),$$

allowing the value $+\infty$. If $T = \gamma^{n-1}/\int_X \omega \wedge \gamma^{n-1}$ for a Gauduchon metric γ, then $D_p(T) = L_{\omega,\gamma}(p)$.

Theorem 27.8.1 *For every nonnegative $p \in L^n$,*

$$\sup_{T \in \mathcal{G}_\omega} D_p(T) = \rho_\omega(p), \tag{27.35}$$

with the value $+\infty$ allowed. If $p \in \mathcal{D}^0_\omega$, the functional D_p is finite, concave and weakly upper semicontinuous on $\mathcal{G}_\omega$, its maximum is attained, and the densities $p k_T$, for $T \in \mathcal{G}_\omega$, are uniformly integrable.

Proof For a smooth strictly positive p, we have $D_p = \mathfrak{D}_{n \log p}$ and $\mathrm{e}^{-B_0(n\log p)/n} = \rho_\omega(p)$, so Theorem 27.5.1, applied to $F = n \log p$, proves (27.35). By Hölder's inequality and Theorem 27.7.1, for every real $r > n$ there is a constant C_r such that, for every $q \in L^r$,

$$\sup_{T \in \mathcal{G}_\omega} \int_X |q| k_T \,\mathrm{d}V \leq C_r \|q\|_{L^r}. \tag{27.36}$$

Next, approximate a bounded nonnegative p in L^{2n} by smooth strictly positive functions. This estimate and (15.94) give uniform convergence on $\mathcal{G}_\omega$ of the corresponding functionals D and convergence of the root functionals, so the supremum formula holds for bounded p. For such p, the functional D_p is also upper semicontinuous and concave, by Lemma 27.5.1 and uniform convergence. For general $p \in L^n$, we use the truncations $\min(p, K)$ at positive integer levels K. Monotone convergence

and the definition of b_ω through truncations give

$$\sup_T D_p(T) = \sup_T \sup_K D_{\min(p,K)}(T) = \sup_K \rho_\omega(\min(p,K)) = \rho_\omega(p).$$

Now suppose $p \in \mathcal{D}^0_\omega$. The supremum formula applied to the tail gives

$$0 \le D_p(T) - D_{\min(p,K)}(T) \le \rho_\omega(p\mathbf{1}_{\{p>K\}}) \longrightarrow 0$$

uniformly in T, by Proposition 15.31.4. Hence D_p is finite and is a uniform limit of upper semicontinuous concave functionals, and compactness of $\mathcal{G}_\omega$ gives a maximizer. Finally, fix $1 < q < n/(n-1)$. By Hölder's inequality and Theorem 27.7.1, for every measurable set E,

$$\sup_T \int_E p k_T \, \mathrm{d}V \le \rho_\omega(p\mathbf{1}_{\{p>K\}}) + K C_q \, \mathrm{d}V(E)^{1-1/q}.$$

Choosing first K large and then the volume of E small proves uniform absolute continuity. Since the total masses are bounded by $\rho_\omega(p)$, the densities are uniformly integrable. □

For a nonzero density, a maximizing current gives a supporting probability for the threshold, as we show below. Whether the threshold is attained by an exact equation depends on the regularity of the maximizing current, which enters through the trace inequalities of Section 27.8.3.

27.8.1 The tail condition and perturbations of the density

Vanishing tails are equivalent to continuity of the threshold under uniformly bounded multiplicative perturbations of the density that converge in measure. This explains the role of the tail condition: if it fails, a perturbation supported on sets of shrinking volume can lower the threshold by a fixed factor.

Proposition 27.8.1 *Let $0 \le p \in \mathcal{D}_\omega$ be nonzero. The following are equivalent:*

(1) $p \in \mathcal{D}^0_\omega$;
(2) *the densities pk_T, for $T \in \mathcal{G}_\omega$, are uniformly integrable;*
(3) *whenever F_j are smooth real functions, uniformly bounded and tending to zero in measure, one has*
$$b_\omega(\mathrm{e}^{F_j} p^n) \longrightarrow b_\omega(p^n).$$

If the first condition holds, the convergence in the third also holds for bounded measurable F_j with the same uniform bound and convergence in measure. If it fails, there are smooth $F_j \ge 0$, uniformly bounded and tending to zero in measure, such that

$$\mathrm{e}^{F_j} p^n \longrightarrow p^n \quad \text{in } L^1, \qquad b_\omega(\mathrm{e}^{F_j} p^n) \le 2^{-n} b_\omega(p^n).$$

Proof The first condition implies the second by Theorem 27.8.1. Conversely, $\mathrm{d}V(\{p > K\}) \to 0$ as $K \to \infty$, so uniform integrability and (27.35) give

$$\rho_\omega(p\mathbf{1}_{\{p>K\}}) = \sup_{T\in\mathcal{G}_\omega} \int_{\{p>K\}} pk_T \,\mathrm{d}V \longrightarrow 0.$$

This is the first condition by Proposition 15.31.4.

If $p \in \mathcal{D}^0_\omega$ and $|F_j| \le A$, then $p(\mathrm{e}^{F_j/n} - 1)$ tends to zero in measure and its absolute value is at most $(\mathrm{e}^{A/n} + 1)p$. The dominated-convergence criterion of Proposition 15.31.4, applied to $(\mathrm{e}^{A/n} + 1)p \in \mathcal{D}^0_\omega$, makes its root functional tend to zero, and the triangle inequality gives $\rho_\omega(p\mathrm{e}^{F_j/n}) \to \rho_\omega(p) > 0$. Taking negative nth powers proves the third condition, also for bounded measurable F_j.

Suppose now that the tails do not tend to zero, and put $R = \rho_\omega(p) > 0$. The formula of Theorem 15.28.1, which uses only smooth tests, yields $\delta > 0$, numbers $K_j \to \infty$ and smooth strictly positive $T_j \in \mathcal{G}_\omega$ such that

$$\int_{\{p>K_j\}} pk_{T_j} \,\mathrm{d}V > \delta.$$

Since the finite measure $pk_{T_j}\,\mathrm{d}V$ is regular, we can choose a compact set $C_j \subset \{p > K_j\}$ of mass greater than $\delta/2$ for this measure. By outer regularity of $\mathrm{d}V$, choose an open set $O_j \supset \{p > K_j\}$ with $\mathrm{d}V(O_j) \le \mathrm{d}V(\{p > K_j\}) + 1/j$, and a smooth cutoff $0 \le \chi_j \le 1$ that equals one on C_j and has support in O_j. Fix $A > 0$ with $\mathrm{e}^{A/n}\delta/2 > 2R$, and set $F_j = A\chi_j$. Then $F_j \to 0$ in measure, while

$$\rho_\omega(p\mathrm{e}^{F_j/n}) \ge \int_X p\mathrm{e}^{F_j/n} k_{T_j} \,\mathrm{d}V > 2R.$$

Since $b_\omega(\mathrm{e}^{F_j}p^n) = \rho_\omega(p\mathrm{e}^{F_j/n})^{-n}$, this gives the asserted uniform gap. Finally, $|\mathrm{e}^{F_j} - 1|p^n \le (\mathrm{e}^A - 1)p^n\mathbf{1}_{O_j}$, whose integral tends to zero by absolute continuity of the integral. This proves the last assertion and the converse to the third condition. □

For $f = p^n$, the perturbed densities satisfy $\mathrm{e}^{-A}f \le \mathrm{e}^{F_j}f \le \mathrm{e}^A f$ with a fixed A, so the continuity statement concerns only such order intervals. The instability under general L^1 approximation in Section 15.31 involves approximations without these bounds.

For attainment at one density, uniform integrability along a single maximizing sequence is enough, and this weaker condition also identifies the limit of its weighted measures.

Proposition 27.8.2 *Let $0 \le p \in \mathcal{D}_\omega$ be nonzero and put $R = \rho_\omega(p)$. Suppose $T_j \in \mathcal{G}_\omega$ converge weakly to T and satisfy $D_p(T_j) \to R$. If the densities pk_{T_j} are uniformly integrable, then*

$$D_p(T) = R, \qquad pk_{T_j}\,\mathrm{d}V \rightharpoonup pk_T\,\mathrm{d}V.$$

Consequently, a maximizing sequence for D_p with uniformly integrable densities pk_{T_j} has a maximizing current among its weak cluster limits.

Proof For positive integers K, put $p_K = \min(p, K)$. Since p is finite almost everywhere, uniform integrability gives

$$\varepsilon_K := \sup_j \int_{\{p>K\}} pk_{T_j}\,\mathrm{d}V \longrightarrow 0.$$

For every nonnegative continuous function h, the weight hp_K is bounded and measurable, so the upper semicontinuity of D_{hp_K} proved in Theorem 27.8.1 gives

$$\limsup_j \int_X hpk_{T_j}\,\mathrm{d}V \le \int_X hp_K k_T\,\mathrm{d}V + \|h\|_\infty \varepsilon_K.$$

Now let $K \to \infty$. For $h = 1$ this yields $R \le D_p(T)$, and the supremum formula (27.35) gives the reverse inequality. If Λ is a weak cluster limit of the measures $pk_{T_j}\,\mathrm{d}V$, the same estimate for all nonnegative continuous h gives $\Lambda \le pk_T\,\mathrm{d}V$. Both measures have mass R, so they agree. Since these measures have bounded masses, they form a weakly relatively compact family, and every cluster limit has now been identified. The final assertion follows from the compactness of $\mathscr{G}_\omega$. □

27.8.2 Supporting probabilities for an integrable density

Under the vanishing-tail condition, we can also perturb the density by continuous multiplicative factors e^F. The logarithm of the threshold is then a concave function of F, and its supporting probabilities are given by the maximizing currents, just as for the continuous densities in Section 27.6.

Let $f \in L^1(\mathrm{d}V)$ be nonzero and nonnegative, and assume $p = f^{1/n} \in \mathcal{D}^0_\omega$. For a continuous real F, set

$$B_f(F) = \log b_\omega(\mathrm{e}^F f) = -n \log \rho_\omega(p\mathrm{e}^{F/n}), \qquad M_f(F) = \rho_\omega(p\mathrm{e}^{F/n}).$$

Both quantities are finite and $M_f(F) > 0$, since $\mathcal{D}^0_\omega$ is an order ideal. Let $\mathscr{M}_f(F)$ be the set of currents in $\mathscr{G}_\omega$ maximizing $D_{p\mathrm{e}^{F/n}}$, which is nonempty by Theorem 27.8.1, and put

$$\nu_{T,F} = \frac{p\mathrm{e}^{F/n}k_T}{M_f(F)}\,\mathrm{d}V \quad (T \in \mathscr{M}_f(F)).$$

These are probability measures. Let $\mathscr{A}_f(F)$, the set of *supporting probabilities* of B_f at F, consist of all Borel probability measures ν satisfying

$$B_f(G) \le B_f(F) - \int_X (G - F)\,\mathrm{d}\nu \qquad (G \in C(X, \mathbb{R})). \tag{27.37}$$

Proposition 27.8.3 *The functional B_f is concave, decreasing and* 1*-Lipschitz for the uniform norm, and $B_f(F + a) = B_f(F) - a$ for constants a. Moreover,*

$$\mathcal{A}_f(F) = \{\nu_{T,F} : T \in \mathscr{M}_f(F)\}.$$

This set is nonempty, convex and weakly compact, and its measures are absolutely continuous. For each $R > 0$, the densities of all these measures, for all F with $\|F\|_\infty \le R$, are uniformly integrable. For $H \in C(X, \mathbb{R})$, the one-sided directional derivatives of B_f at F are

$$(B_f)'_+(F; H) = -\max_{\nu \in \mathcal{A}_f(F)} \int_X H \, \mathrm{d}\nu, \qquad (B_f)'_-(F; H) = -\min_{\nu \in \mathcal{A}_f(F)} \int_X H \, \mathrm{d}\nu.$$

Proof For each T, Hölder's inequality makes $F \mapsto \int_X p\mathrm{e}^{F/n} k_T \, \mathrm{d}V$ logarithmically convex, and taking suprema preserves the inequality defining logarithmic convexity. Thus M_f is logarithmically convex and $B_f = -n \log M_f$ is concave. Monotonicity and homogeneity of ρ_ω give the translation rule and, by comparison with constant translates, the Lipschitz estimate.

For fixed F, the weight $p\mathrm{e}^{F/n}$ belongs to $\mathcal{D}^0_\omega$, so the densities $p\mathrm{e}^{F/n} k_T$, $T \in \mathcal{G}_\omega$, are uniformly integrable by Theorem 27.8.1. If $T_j \to T$ and $D_{p\mathrm{e}^{F/n}}(T_j) \to M_f(F)$, Proposition 27.8.2 makes T a maximizer and gives convergence of the weighted measures. For $T_1, T_2 \in \mathcal{G}_\omega$ and $0 \le t \le 1$, the proof of Lemma 27.6.1 gives $\Gamma_F(tT_1 + (1-t)T_2) \ge t\Gamma_F(T_1) + (1-t)\Gamma_F(T_2)$, where $\Gamma_F(T) = \mathrm{e}^{F/n} k_T \, \mathrm{d}V$. Multiplying by $p \ge 0$ shows, as there, that convex combinations of maximizing currents maximize and have the corresponding convex combinations of weighted measures. The set $\mathcal{N}_f(F) = \{\nu_{T,F} : T \in \mathscr{M}_f(F)\}$ is therefore nonempty, compact and convex.

A maximizing T gives

$$M_f(G) \ge M_f(F) \int_X \mathrm{e}^{(G-F)/n} \, \mathrm{d}\nu_{T,F}.$$

Taking $-n$ times the logarithm and applying Jensen's inequality proves (27.37), so $\mathcal{N}_f(F) \subset \mathcal{A}_f(F)$. For the reverse inclusion, a Taylor expansion of the exponential gives

$$D_{p\mathrm{e}^{(F+tH)/n}}(T) = D_{p\mathrm{e}^{F/n}}(T) + \frac{t}{n} \int_X H p \mathrm{e}^{F/n} k_T \, \mathrm{d}V + O(t^2),$$

where the error is uniform in T since $D_{p\mathrm{e}^{F/n}}(T) \le M_f(F)$. Maximizers at $F + tH$ are near-maximizers at F, by the multiplicative bounds $\mathrm{e}^{-t\|H\|_\infty/n} \le M_f(F + tH)/M_f(F) \le \mathrm{e}^{t\|H\|_\infty/n}$. The convergence of their weighted measures proved above and this uniform expansion are exactly the two inputs of the one-sided differentiation argument in the proof of Theorem 27.6.1, which gives

$$(B_f)'_+(F; H) = -\max_{\nu \in \mathcal{N}_f(F)} \int_X H \, \mathrm{d}\nu.$$

The separation argument of that proof then identifies all supporting probabilities with $\mathcal{N}_f(F)$, and replacing H by $-H$ gives the left derivative.

Finally, $M_f(F) \geq \mathrm{e}^{-R/n}\rho_\omega(p)$ when $\|F\|_\infty \leq R$, so the densities of the probabilities $\nu_{T,F}$ are at most $\mathrm{e}^{2R/n}\rho_\omega(p)^{-1}pk_T$. The uniform integrability of the densities pk_T in Theorem 27.8.1 proves the remaining assertions. □

When $f = 1$, we have $B_f = B_0$, and the proposition reduces to Theorem 27.6.1. For general f, the functional B_f is defined through the subsolution threshold and does not presuppose that the corresponding equation has a solution. If $v_-(\omega) > 0$, every nonzero nonnegative $f \in L^1$ satisfies $f^{1/n} \in \mathcal{D}^0_\omega$ by Proposition 15.31.4, and Corollary 33.7.1 identifies $b_\omega(f)$ as the unique constant for which the equation $\mathrm{MA}^{\mathrm{ext}}_\omega(v) = b\,f\,\mathrm{d}V$ has a solution v in the envelope class $\mathcal{E}_{\mathrm{env}}(X,\omega,0)$ of that corollary. Without positive lower mass, the uniform tail condition is an explicit sufficient hypothesis for this compact dual description.

27.8.3 Trace inequalities at the subsolution threshold

For a nonzero $0 \leq f \in L^1(\mathrm{d}V)$ with $b = b_\omega(f) > 0$, recall from Section 15.27 that a threshold subsolution is a function $u \in \mathrm{PSH}(X,\omega)$ with $\mathrm{MA}^{\mathrm{ext}}_\omega(u) \geq b f\,\mathrm{d}V$, and a threshold solution is one for which equality holds; threshold subsolutions exist by Theorem 15.27.1. A smooth Gauduchon test can be paired with the full current $\omega + \mathrm{dd}^{\mathrm{c}}u$ of a threshold subsolution, and by Lemma 7.4.2 the resulting trace measure is bounded below by the determinant-weighted density. The nonnegative difference has a mass that depends only on the test. Along a maximizing sequence this mass tends to zero, uniformly over all threshold subsolutions. We write $\|\cdot\|_{\mathrm{TV}}$ for the total variation of a finite signed measure.

Proposition 27.8.4 *Let $0 \leq f \in L^1$ be nonzero with $b = b_\omega(f) > 0$, write $p = f^{1/n}$, and let u be a threshold subsolution. For every smooth strictly positive $T \in \mathcal{G}_\omega$, the measure*

$$R_{u,T} = (\omega + \mathrm{dd}^{\mathrm{c}}u) \wedge T - b^{1/n}pk_T\,\mathrm{d}V$$

is positive and has mass

$$R_{u,T}(X) = 1 - b^{1/n}D_p(T).$$

In particular, if (T_j) is a sequence of smooth strictly positive members of $\mathcal{G}_\omega$ with $D_p(T_j) \to \rho_\omega(p)$, then $\|R_{u,T_j}\|_{\mathrm{TV}} \to 0$, uniformly over all threshold subsolutions u.

If also $p \in \mathcal{D}^0_\omega$ and $T_j \to T$ as currents, then T maximizes D_p and

$$(\omega + \mathrm{dd}^{\mathrm{c}}u) \wedge T_j \rightharpoonup b^{1/n}pk_T\,\mathrm{d}V.$$

Proof A smooth strictly positive $T \in \mathcal{G}_\omega$ is γ^{n-1} for a Gauduchon metric γ normalized by $\int_X \omega \wedge \gamma^{n-1} = 1$. Lemma 7.4.2 proves positivity of $R_{u,T}$, and Stokes' theorem for

currents gives $((\omega + \mathrm{dd}^{\mathrm{c}} u) \wedge T)(X) = \int_X \omega \wedge T = 1$. This proves the mass formula. Since $\rho_\omega(p) = b^{-1/n}$, the nonnegative mass tends to zero along a maximizing sequence; it equals the total variation norm and does not depend on u.

Under the tail hypothesis, Theorem 27.8.1 gives uniform integrability of the densities pk_{T_j}. Thus Proposition 27.8.2 makes T a maximizer and gives $pk_{T_j}\,\mathrm{d}V \rightharpoonup pk_T\,\mathrm{d}V$. Adding the error measures R_{u,T_j}, which tend to zero in total variation, proves the limit formula. □

The limit formula concerns smooth maximizing sequences. It does not define the product $(\omega + \mathrm{dd}^{\mathrm{c}} u) \wedge T$ for a singular current T, nor does it show that the limit is the same for different maximizing currents.

If a maximizing test is itself a smooth strictly positive form, the trace inequality becomes an equality of currents, and since $n \geq 2$ this forces an exact smooth equation. The local regularity needed for this comes from Lemma 1.1.1.

Corollary 27.8.1 *Let $0 \leq f \in L^1(\mathrm{d}V)$ be nonzero, and put $b = b_\omega(f)$. If $D_{f^{1/n}}$ has a smooth strictly positive maximizer $T \in \mathcal{G}_\omega$, then $b > 0$, $f^{1/n}$ agrees almost everywhere with a smooth nonnegative function, and every threshold subsolution u is smooth and satisfies*

$$\mathrm{MA}_\omega(u) = b f\,\mathrm{d}V.$$

In particular a smooth threshold solution exists, and any two threshold subsolutions differ by a constant.

Proof The determinant density k_T of the smooth form T is bounded, so $\rho_\omega(f^{1/n}) = D_{f^{1/n}}(T) < \infty$ by (27.35), that is, $b > 0$. Let u be any threshold subsolution; one exists by Theorem 15.27.1. Write $T = \gamma^{n-1}$, where γ is the smooth Gauduchon metric given by the positive $(n-1)$st root of T, and set

$$a = \left(\frac{b f\,\mathrm{d}V}{\gamma^n}\right)^{1/n}.$$

Put $A = \omega + \mathrm{dd}^{\mathrm{c}} u \geq 0$. Since T maximizes, the positive measure $R_{u,T}$ of Proposition 27.8.4 has mass zero, which gives the equality

$$A \wedge \gamma^{n-1} = a\gamma^n.$$

The coefficient measures of the positive current A are controlled by this trace measure, which is absolutely continuous. Thus A has a matrix density that is positive semidefinite almost everywhere, with coefficients in L^n because $a \in L^n$.

For every smooth Hermitian form β, Lemma 7.4.2 also gives

$$A \wedge \beta^{n-1} \geq \left(\frac{b f\,\mathrm{d}V}{\beta^n}\right)^{1/n} \beta^n.$$

The resulting pointwise matrix inequalities hold almost everywhere, simultaneously for all positive definite matrices: on each coordinate chart, first use a countable dense

family of constant positive matrices, realized on smaller charts by global smooth Hermitian forms, and then use continuity in the matrix. The variational formula for $(\det P)^{1/n}$ in the proof of Lemma 27.5.1 therefore yields

$$\left(\frac{A^n}{\gamma^n}\right)^{1/n} \geq a \quad \text{almost everywhere.}$$

Here A^n denotes only the determinant of the measurable coefficient matrix of A; no nonlinear product of currents is used. The trace equality gives the reverse inequality, by the arithmetic–geometric mean inequality. Equality then forces all eigenvalues of A relative to γ to equal a, also where $a = 0$. Consequently $A = a\gamma$ as currents.

By Lemma 1.1.1, applied with $\theta = \omega$, both a and u are smooth, and $f^{1/n} = b^{-1/n} a (\gamma^n / \mathrm{d}V)^{1/n}$ has a smooth nonnegative representative. Since u is smooth, we may now take the top power: $(\omega + \mathrm{dd^c} u)^n = a^n \gamma^n = b f \, \mathrm{d}V$. Finally, the function a was defined from f, b and γ alone, so two threshold subsolutions u_1, u_2 satisfy $\omega + \mathrm{dd^c} u_1 = a\gamma = \omega + \mathrm{dd^c} u_2$. Their smooth difference is therefore pluriharmonic, hence constant on the connected compact manifold X. □

If f is smooth and strictly positive, Theorem 27.5.1, applied to $F = \log f$, provides a smooth strictly positive maximizer of $D_{f^{1/n}} = \mathfrak{D}_F$. The corollary therefore applies, and shows in particular that all threshold subsolutions, bounded or not, differ from the smooth solution by constants.

Corollary 27.8.2 *Under the hypotheses of Corollary 27.8.1, let u be the threshold solution with $\sup_X u = 0$. If u_λ is the unique bounded solution of*

$$\mathrm{MA}_\omega(u_\lambda) = \mathrm{e}^{\lambda u_\lambda} f \, \mathrm{d}V, \qquad \lambda > 0,$$

then $u_\lambda - \sup_X u_\lambda \to u$ uniformly as $\lambda \downarrow 0$.

Proof By Corollary 27.8.1, f agrees almost everywhere with a smooth function, so it belongs to L^p for every finite p. By Proposition 15.27.2, the constant c for which $\mathrm{MA}_\omega(v) = c f \, \mathrm{d}V$ has a bounded solution is $b_\omega(f)$. Thus Proposition 15.19.1, with the trivial symmetry group, shows that the normalized family $u_\lambda - \sup_X u_\lambda$ is uniformly precompact as $\lambda \downarrow 0$ and that every cluster limit is a normalized solution with this constant. Each cluster limit is therefore a normalized threshold subsolution and equals u by Corollary 27.8.1. If the whole family did not converge uniformly, a sequence staying at a fixed positive uniform distance from u would have a uniformly convergent subsequence, a contradiction. □

Corollary 27.8.3 *Suppose ω is Kähler. For a nonzero nonnegative $f \in L^1(\mathrm{d}V)$, the functional $D_{f^{1/n}}$ has a smooth strictly positive maximizer in $\mathscr{G}_\omega$ if and only if f agrees almost everywhere with a smooth strictly positive function.*

Proof If a smooth strictly positive maximizer exists, the proof of Corollary 27.8.1 gives

$$\omega + \mathrm{dd^c} u = a\gamma, \qquad a \geq 0,$$

with a, u smooth and $a \not\equiv 0$. Since $\mathrm{d}\omega = 0$, the first-order equation in Lemma 1.1.1 becomes $\mathrm{d}a = -aC$ for a smooth one-form C. Along any smooth path σ, this is the linear scalar equation

$$(a \circ \sigma)' = -(a \circ \sigma)C(\dot{\sigma}),$$

whose solution is the initial value multiplied by a positive exponential. Since X is connected and $a \not\equiv 0$, it follows that $a > 0$ everywhere, and $f = b_\omega(f)^{-1} a^n \gamma^n / \mathrm{d}V$ is smooth and strictly positive. Conversely, if f is smooth and strictly positive, Theorem 27.5.1, applied to $F = \log f$, for which $\mathfrak{D}_F = D_{f^{1/n}}$, provides a smooth strictly positive maximizer. □

For nonclosed backgrounds, the strict positivity of f in Corollary 27.8.3 can fail on every compact complex manifold of dimension at least two, including Kähler manifolds.

Example 27.8.1 On every connected compact complex manifold X of dimension $n \geq 2$, with a fixed smooth strictly positive probability volume $\mathrm{d}V$, there is a nonclosed Hermitian form ω and a nonzero smooth nonnegative density f vanishing on an open set such that $b_\omega(f) = 1$ and $D_{f^{1/n}}$ has a smooth strictly positive maximizer.

Proof Fix a Gauduchon metric γ and a point x. Choose a smooth real function ψ such that $\mathrm{dd}^{\mathrm{c}}\psi \leq -\gamma$ on a coordinate neighborhood U of x; in holomorphic coordinates $z = (z_1, \ldots, z_n)$ centered at x, such a function is obtained by multiplying $-M|z|^2$, with M large enough, by a cutoff that equals one near x, and then shrinking U.

Choose a smooth function χ with $0 \leq \chi \leq 1$ that vanishes on a smaller neighborhood of x and equals one outside U. Put

$$a = L\chi, \qquad \omega = a\gamma - \mathrm{dd}^{\mathrm{c}}\psi.$$

Since X is compact, ω is strictly positive for sufficiently large $L > 0$: on U the term $-\mathrm{dd}^{\mathrm{c}}\psi$ is at least γ, and outside U the term $L\gamma$ dominates $\mathrm{dd}^{\mathrm{c}}\psi$. Set $f = a^n\gamma^n/\mathrm{d}V$. The smooth function ψ satisfies $\omega + \mathrm{dd}^{\mathrm{c}}\psi = a\gamma \geq 0$ and $\mathrm{MA}_\omega(\psi) = f\,\mathrm{d}V$, so $b_\omega(f) \geq 1$.

Stokes' theorem and $\mathrm{dd}^{\mathrm{c}}\gamma^{n-1} = 0$ give

$$\int_X \omega \wedge \gamma^{n-1} = \int_X a\gamma^n = \int_X (f\,\mathrm{d}V/\gamma^n)^{1/n}\gamma^n.$$

Testing (15.83) with γ therefore gives the opposite inequality $b_\omega(f) \leq 1$. Hence $b_\omega(f) = 1$, the metric γ attains the infimum, and the normalized current $\gamma^{n-1}/\int_X \omega \wedge \gamma^{n-1}$ is a smooth strictly positive maximizer of $D_{f^{1/n}}$. If ω were closed, Corollary 27.8.3 would force f to be strictly positive, contrary to its vanishing on an open set. Thus ω is nonclosed. □

In Example 27.8.1, Corollaries 27.8.1 and 27.8.2 show that every threshold subsolution is a constant translate of ψ and that the normalized solutions of the exponential equations converge uniformly to $\psi - \sup_X \psi$. Thus vanishing of the

density on an open set does not by itself force nonuniqueness or the failure of this approximation.

The restriction $n \geq 2$ in Corollary 27.8.1 is necessary. On a compact connected Riemann surface X with a Hermitian form ω, a normalized Gauduchon current has bidegree $(0,0)$, and the equation $\mathrm{dd}^{\mathrm{c}} T = 0$ makes it a harmonic function, hence the constant $(\int_X \omega)^{-1}$. This single smooth strictly positive current maximizes for every nonzero nonnegative L^1 density, which need not have a smooth representative.

27.8.4 A smooth maximizing form with an analytic degeneracy set

We now allow the smooth maximizing form to degenerate. Where it is strictly positive, the argument of Corollary 27.8.1 still determines the solution current and gives the equation. If the degeneracy set is a proper analytic subset, neither side of the equation charges it, so the equation holds on all of X. Uniqueness follows by passing to the maximum of two threshold subsolutions and applying Lemma 1.17.2.

Theorem 27.8.2 *Let $0 \leq f \in L^1(\mathrm{d}V)$ be nonzero, and put $b = b_\omega(f)$. Suppose that $D_{f^{1/n}}$ has a smooth semipositive maximizer $T \in \mathscr{G}_\omega$ that is strictly positive on $X \setminus Z$, where Z is a proper analytic subset of X. Then $b > 0$, and every threshold subsolution u is smooth on $X \setminus Z$ and satisfies*

$$\mathrm{MA}_\omega^{\mathrm{ext}}(u) = b f \, \mathrm{d}V$$

on X. In particular a threshold solution exists. The function $f^{1/n}$ has a smooth nonnegative representative on $X \setminus Z$, and any two threshold subsolutions differ by a constant.

Proof Put $p = f^{1/n}$. For a smooth semipositive form T with coefficient matrix P' relative to ω^n, (27.23) gives $k_T \, \mathrm{d}V = f_0 (\det P')^{1/n} \omega^n$, so that $k_T = f_0^{1-n} (\det P')^{1/n}$ is a continuous function of that matrix. In particular k_T is bounded, so $D_p(T) < +\infty$, and since T maximizes, (27.35) gives $\rho_\omega(p) = D_p(T) < +\infty$, that is, $b > 0$. Let u be any threshold subsolution, which exists by Theorem 15.27.1, and put $A = \omega + \mathrm{dd}^{\mathrm{c}} u$. Fix a smooth strictly positive $T_0 \in \mathscr{G}_\omega$, and for real $\varepsilon > 0$ set $T_\varepsilon = (T + \varepsilon T_0)/(1+\varepsilon)$. Each T_ε is a smooth strictly positive member of $\mathscr{G}_\omega$, so Proposition 27.8.4 gives

$$A \wedge T_\varepsilon \geq b^{1/n} p k_{T_\varepsilon} \, \mathrm{d}V.$$

By the formula for k_T above, $k_{T_\varepsilon} \to k_T$ uniformly on the compact manifold X. Since $p \in L^1$, passing to the limit gives

$$A \wedge T \geq b^{1/n} p k_T \, \mathrm{d}V.$$

The left side has mass one by Stokes' theorem and $\mathrm{dd}^{\mathrm{c}} T = 0$, and the right side has mass $b^{1/n} D_p(T) = 1$ because T maximizes. Their difference is positive and has mass

zero, hence

$$A \wedge T = b^{1/n} p k_T \, \mathrm{d}V. \tag{27.38}$$

On $U = X \setminus Z$ write $T = \gamma^{n-1}$, where γ is smooth and strictly positive, and set $a = (b f \, \mathrm{d}V/\gamma^n)^{1/n}$. The trace equality becomes $A \wedge \gamma^{n-1} = a\gamma^n$ on U. On relatively compact subsets of U, it controls every coefficient measure of A by an L^n density. The trace inequalities for all Hermitian forms and the matrix arithmetic–geometric mean inequality, used exactly as in the proof of Corollary 27.8.1, force $A = a\gamma$ on U; this uses only the positivity of γ on U, since the Hermitian test forms needed there are global forms obtained by extension from smaller coordinate charts.

The local regularity lemma Lemma 1.1.1 makes u and a smooth on U. Taking the top power proves the equation on U, and $f^{1/n} = b^{-1/n} a (\gamma^n/\mathrm{d}V)^{1/n}$ is smooth and nonnegative there. Both $\mathrm{MA}^{\mathrm{ext}}_{\omega}(u)$ and $f \, \mathrm{d}V$ put no mass on the pluripolar set Z. Hence the equality holds on X.

It remains to prove uniqueness. Let v be another threshold subsolution and put $w = \max(u, v)$. By Proposition 7.2.1,

$$\mathrm{MA}^{\mathrm{ext}}_{\omega}(w) \geq \mathbf{1}_{\{u \geq v\}} \mathrm{MA}^{\mathrm{ext}}_{\omega}(u) + \mathbf{1}_{\{u < v\}} \mathrm{MA}^{\mathrm{ext}}_{\omega}(v) \geq b f \, \mathrm{d}V.$$

Here the pole sets carry no mass for any of the measures involved. Thus w is also a threshold subsolution, and the preceding argument gives the same smooth current $\omega + \mathrm{dd}^{\mathrm{c}} u = \omega + \mathrm{dd}^{\mathrm{c}} v = \omega + \mathrm{dd}^{\mathrm{c}} w = a\gamma$ on U.

We apply Lemma 1.17.2 to $u \leq w$ and to $v \leq w$: their complex Hessians agree on $X \setminus Z$, so $w - u$ and $w - v$ are constant. Hence $u - v$ is constant. □

27.8.5 The limit of the exponential equations

The uniqueness just proved identifies the limit, as $\lambda \downarrow 0$, of the normalized solutions of the exponential equations $\mathrm{MA}_\omega^{\mathrm{ext}}(u) = \mathrm{e}^{\lambda u} f \, \mathrm{d}V$. To control their normalization, we use the probability measure obtained from the maximizing form and the density. Normalized potentials have bounded moments with respect to this measure, and Jensen's inequality converts this bound into an estimate for the constants and a common lower barrier for the potentials.

Proposition 27.8.5 *Assume the hypotheses of Theorem 27.8.2, and let u_0 be its unique threshold solution with $\sup_X u_0 = 0$. For real $\lambda > 0$, let u_λ be the greatest solution of the exponential equation (15.101), with ω in place of h, given by Corollary 15.34.1, and put*

$$v_\lambda = u_\lambda - \sup_X u_\lambda, \qquad c_\lambda = \mathrm{e}^{\lambda \sup_X u_\lambda}.$$

There is a constant C independent of $\lambda > 0$ such that

$$b \le c_\lambda \le b\, \mathrm{e}^{C\lambda}, \qquad u_0 - C \le v_\lambda \le 0.$$

Moreover $v_\lambda \to u_0$ in $L^1(X)$ as $\lambda \downarrow 0$.

Proof Write $p = f^{1/n}$ and introduce the probability measure

$$\mathrm{d}\nu = b^{1/n} p k_T \, \mathrm{d}V.$$

It has mass one because T maximizes. Its density belongs to L^n, since $p \in L^n$ and the smooth semipositive form T has bounded determinant density k_T. By Lemma 1.5.1, ω-plurisubharmonic functions with supremum zero are uniformly exponentially integrable, so their moments of every finite order are uniformly bounded. Hölder's inequality with the exponents n and $n/(n-1)$ therefore gives

$$\int_X (-v_\lambda) \, \mathrm{d}\nu \le C.$$

Since $\mathrm{MA}_\omega^{\mathrm{ext}}(v_\lambda) = c_\lambda \mathrm{e}^{\lambda v_\lambda} f \, \mathrm{d}V$, the trace inequality of Lemma 7.4.2, applied with the positive regularizations of T and passed to the limit as in the proof of Theorem 27.8.2, gives $(\omega + \mathrm{dd}^{\mathrm{c}} v_\lambda) \wedge T \ge (c_\lambda/b)^{1/n} \mathrm{e}^{\lambda v_\lambda / n} \, \nu$. Integrating, we obtain

$$1 \ge (c_\lambda/b)^{1/n} \int_X \mathrm{e}^{\lambda v_\lambda / n} \, \mathrm{d}\nu \ge (c_\lambda/b)^{1/n} \mathrm{e}^{-C\lambda/n},$$

where the last step is Jensen's inequality. This proves the upper bound for c_λ. The lower bound and the comparison

$$v_\lambda \ge u_0 - \lambda^{-1} \log(c_\lambda/b)$$

follow from Corollary 15.34.1 applied with $v = u_0$, and together with the upper bound they give the common lower barrier $u_0 - C$.

Take any sequence $\lambda_j \downarrow 0$. By Theorem 1.4.1, a subsequence of (v_{λ_j}) converges in L^1 to some $v \in \mathrm{PSH}(X,\omega)$ with supremum zero, and Corollary 15.34.2 makes v a threshold subsolution. By the uniqueness in Theorem 27.8.2, $v - u_0$ is constant, and since both functions have supremum zero, $v = u_0$. Since every sequence tending to zero has a subsequence with this same L^1 limit, the whole family converges. □

Under the additional envelope condition of the next corollary, the lower barrier upgrades this to convergence in capacity. If the maximizing form is strictly positive everywhere, Corollary 27.8.2 already gives uniform convergence.

Corollary 27.8.4 *Under the hypotheses of Proposition 27.8.5, suppose in addition that the obstacle envelope $P_{\delta\omega}(u_0)$ of Section 1.19 is not identically $-\infty$ for some real $0 < \delta < 1$. Then $v_\lambda \to u_0$ in capacity as $\lambda \downarrow 0$. In particular this holds if the threshold solution u_0 is bounded.*

Proof Fix a sequence $\lambda_j \downarrow 0$. Proposition 27.8.5 gives $u_0 - C \le v_{\lambda_j} \le 0$ and $v_{\lambda_j} \to u_0$ in L^1. For all j with $\lambda_j \le 1$,

$$\mathrm{MA}_\omega^{\mathrm{ext}}(v_{\lambda_j}) = c_{\lambda_j} \mathrm{e}^{\lambda_j v_{\lambda_j}} f \, \mathrm{d}V \le b\mathrm{e}^{C} f \, \mathrm{d}V.$$

The bounded functions $|\mathrm{e}^{v_{\lambda_j}} - \mathrm{e}^{u_0}|$ tend to zero in $\mathrm{d}V$-measure. Since $f \in L^1$, their integrals against $f \, \mathrm{d}V$ tend to zero, so the error term in (15.7) tends to zero. We apply Theorem 15.8.1 with $\theta = \omega$ and $\varphi = u_0$; its hypothesis $u \preceq \varphi$ on the L^1 limit u holds since $u = u_0$. This proves convergence in capacity along every sequence, and hence for the family. Finally, if u_0 is bounded, a constant below it is a competitor for $P_{\delta\omega}(u_0)$, so this envelope is not identically $-\infty$. □

In general, the condition on $P_{\delta\omega}(u_0)$ is an additional hypothesis. The next subsection shows that it holds when the Monge–Ampère mass of ω is invariant, or when the singularity of u_0 satisfies a quantitative bound.

27.8.6 Envelope criteria and extension across the degeneracy set

The capacity criterion above requires the obstacle envelope of u_0 in a smaller background $\delta\omega$ not to be identically $-\infty$. We give two sufficient conditions of different origin: invariance of the Monge–Ampère mass controls the contact measures of envelopes of stretched truncations, while a sublinear bound on the singularity provides explicit competitors. Both conditions also give zero Lelong numbers. In the setting of Theorem 27.8.2, if the maximizing form has a globally smooth root of rank at least two, zero Lelong numbers then allow the threshold solution to be extended smoothly across the analytic degeneracy set.

Inequalities between quasi-plurisubharmonic functions refer to their canonical representatives. We will use the following consequence of Lemma 1.17.7, where C_ω is the bound for the Lelong numbers of ω-plurisubharmonic functions given there: if

$u \in \mathrm{PSH}(X,\omega)$ and, for every $0 < \delta < 1$, some $v_\delta \in \mathrm{PSH}(X,\delta\omega)$ satisfies $v_\delta \le u$, then

$$0 \le \nu(u,x) \le \nu(v_\delta,x) = \delta\nu(v_\delta/\delta,x) \le \delta C_\omega.$$

Thus u has zero Lelong numbers; the bound is uniform in δ because $v_\delta/\delta \in \mathrm{PSH}(X,\omega)$.

Theorem 27.8.3 *Suppose* $\mathrm{dd}^{\mathrm{c}}(\omega^j) = 0$ *for* $1 \le j < n$, *and put* $V = \int_X \omega^n$. *For every nonzero* $0 \le f \in L^1(\mathrm{d}V)$, *with* $M = \int_X f\,\mathrm{d}V$, *one has*

$$b_\omega(f) = \frac{V}{M},$$

and every threshold subsolution u *satisfies* $\mathrm{MA}_\omega^{\mathrm{ext}}(u) = (V/M)f\,\mathrm{d}V$, *so it has mass* V. *Every* $u \in \mathrm{PSH}(X,\omega)$ *with* $\int_X \mathrm{MA}_\omega^{\mathrm{ext}}(u) = V$ *has zero Lelong numbers, and* $P_{\delta\omega}(u)$ *is not identically* $-\infty$ *for every* $0 < \delta < 1$.

Proof By Theorem 2.2.2, every bounded ω-plurisubharmonic function has mass V, and truncation therefore bounds every extended mass by V. For $f_k = \min(f,k)$, integration of the bounded equation gives $c_\omega(f_k) = V/\int_X f_k\,\mathrm{d}V$, so the definition (15.80) gives $b_\omega(f) = V/M$. A threshold subsolution has mass at least $b_\omega(f)M = V$. Its mass is consequently V, and the nonnegative difference between its measure and $b_\omega(f)f\,\mathrm{d}V$ has mass zero.

Now let $u \in \mathrm{PSH}(X,\omega)$ have extended mass V. Normalize $u \le 0$, put $\mu_u = \mathrm{MA}_\omega^{\mathrm{ext}}(u)$, and fix $a > 1$. For $u_j = \max(u,-j)$, let $w_j = P_\omega(au_j)$. Comparison with constants gives $-aj \le w_j \le 0$. The bounded contact theorem Theorem 14.1.1 shows that $\mathrm{MA}_\omega(w_j)$ is concentrated on the contact set $C_j = \{w_j = au_j\}$. Since the two functions are canonical representatives and $w_j \le au_j$ almost everywhere, this inequality holds everywhere. Both functions are $a\omega$-plurisubharmonic, so positivity and Proposition 7.2.1 give

$$\mathrm{MA}_\omega(w_j) \le \mathbf{1}_{C_j}\mathrm{MA}_{a\omega}(w_j) \le a^n\mathbf{1}_{C_j}\mathrm{MA}_\omega(u_j).$$

For integers $j \ge N$, locality of the products on $\{u > -N\}$ and the mass identity for bounded functions yield

$$\int_{\{u_j \le -N\}} \mathrm{MA}_\omega(u_j) = V - \mu_u(\{u > -N\}) = \mu_u(\{u \le -N\}).$$

Choose N with $a^n\mu_u(\{u \le -N\}) < V$; this is possible because μ_u is finite and puts no mass on the pole set of u. If $\sup_X w_j < -aN$, the contact set C_j lies in $\{u_j < -N\}$, and the preceding inequalities contradict $\int_X \mathrm{MA}_\omega(w_j) = V$. Hence $\sup_X w_j \ge -aN$ for $j \ge N$, and for smaller j the bound $\sup_X w_j \ge -aj$ suffices.

The envelopes w_j decrease with j, and by compactness of normalized quasi-plurisubharmonic functions this uniform lower bound for their suprema prevents their limit from being identically $-\infty$. Their limit $w \in \mathrm{PSH}(X,\omega)$ satisfies $w \le au$, also at the poles of u. With $\delta = 1/a$, the function $\delta w \in \mathrm{PSH}(X,\delta\omega)$ is therefore a competitor

for $P_{\delta\omega}(u)$, so this envelope is not identically $-\infty$. Since $a > 1$ was arbitrary, this holds for every $0 < \delta < 1$, and the Lelong estimate preceding the theorem completes the proof. The normalization $u \le 0$ does not affect any conclusion. □

Proposition 27.8.6 *Let $\psi \le 0$ be a quasi-plurisubharmonic function with $\mathrm{dd^c}\psi \ge -C\omega$ for some $C > 0$, and let $u \in \mathrm{PSH}(X, \omega)$ satisfy, everywhere on X,*

$$u \ge -A(1-\psi)^{\alpha} - B, \qquad A > 0, \quad B \in \mathbb{R}, \quad 0 < \alpha < 1.$$

Then $P_{\delta\omega}(u)$ is not identically $-\infty$ for every $0 < \delta < 1$, and u has zero Lelong numbers.

Proof Choose $\alpha < \beta < 1$ and $\varepsilon > 0$ with $\varepsilon\beta C \le \delta$. The number

$$K = B + \sup_{t \ge 1}(At^{\alpha} - \varepsilon t^{\beta})$$

is finite, and $v_\delta = -\varepsilon(1-\psi)^{\beta} - K \le u$. The function $t \mapsto -\varepsilon(1-t)^{\beta}$ is increasing and convex on $t \le 0$, with derivative at most $\varepsilon\beta$. Applying the chain inequality for quasi-plurisubharmonic functions to $\max(\psi, -j)$ and passing to the decreasing limit as $j \to \infty$, we obtain $\mathrm{dd^c} v_\delta \ge -\varepsilon\beta C\omega \ge -\delta\omega$. The limit is not identically $-\infty$ because ψ is finite almost everywhere; at the poles of ψ it takes the value $-\infty$, where the comparison with u still holds. Thus $v_\delta \in \mathrm{PSH}(X, \delta\omega)$ is a competitor for $P_{\delta\omega}(u)$ for each δ, and the Lelong estimate preceding Theorem 27.8.3 proves the last assertion. □

27.8.6.1 A globally smooth root of rank at least two

Theorem 27.8.4 *Assume the hypotheses of Theorem 27.8.2, with analytic degeneracy set Z, and suppose that $T = \gamma^{n-1}$ for a smooth semipositive $(1,1)$-form γ on X whose rank is at least two everywhere. If the threshold solution u_0 with $\sup_X u_0 = 0$ has zero Lelong numbers, then u_0 is smooth on X, and hence so is every threshold subsolution.*

Proof Put $U = X \setminus Z$, $b = b_\omega(f)$ and $A = \omega + \mathrm{dd^c} u_0$. On U the form γ is positive, since $\gamma^{n-1} = T$ is, so it is the positive root of T used in the proof of Theorem 27.8.2. That proof shows that $A = r\gamma$ on U, where $r = (bf\,\mathrm{dV}/\gamma^n)^{1/n}$ is smooth and nonnegative on U.

Since $\mathrm{dd^c} u_0$ is closed,

$$\mathrm{d}r \wedge \gamma + r\,\mathrm{d}\gamma = \mathrm{d}\omega \quad \text{on } U.$$

For a semipositive form γ of rank at least two, the real-linear map $W_\gamma : \xi \mapsto \xi \wedge \gamma$ from real one-forms to three-forms is injective at every point. Indeed, write $\gamma = \sum_k \lambda_k\, \mathrm{d}x_k \wedge \mathrm{d}y_k$ at the point in suitable complex linear coordinates $z_k = x_k + \mathrm{i}y_k$, with $\lambda_k \ge 0$ and at least two λ_k positive. For each index j choose $k \ne j$ with $\lambda_k > 0$.

The coefficient of $\mathrm{d}x_j \wedge \mathrm{d}x_k \wedge \mathrm{d}y_k$ in $\xi \wedge \gamma$ is λ_k times the coefficient of $\mathrm{d}x_j$ in ξ, and similarly for $\mathrm{d}y_j$. Hence $\xi \wedge \gamma = 0$ forces $\xi = 0$.

Choose an auxiliary Riemannian metric. The left inverse $L_\gamma = (W_\gamma^* W_\gamma)^{-1} W_\gamma^*$ is smooth and uniformly bounded on compact X. Thus

$$\mathrm{d}r = \alpha - r\beta, \qquad \alpha = L_\gamma(\mathrm{d}\omega), \quad \beta = L_\gamma(\mathrm{d}\gamma), \tag{27.39}$$

with smooth one-forms on X. In particular $|\mathrm{d}\log(1+r)| \le C$ on U.

Points of U can be joined by short paths avoiding Z. In a coordinate ball, stratify Z into countably many smooth strata of real dimension at most $2n-2$. For a fixed point outside Z, the endpoints of segments from it that meet one stratum lie in a cone of dimension at most $2n-1$, which is Lebesgue-null. Given two nearby points outside Z, choose a generic intermediate point in a ball of radius comparable to their distance; both connecting segments then avoid Z, and their combined length is bounded by a fixed multiple of that distance. Using a finite connected cover by coordinate balls, with intermediate points chosen in the overlaps, we also obtain a uniform bound for the lengths of connecting paths in U.

The bound on $\mathrm{d}\log(1+r)$ now bounds r on U, and (27.39) bounds its derivative. The short paths therefore give a unique locally Lipschitz extension of r across Z. Since Z is Lebesgue-null, the weak derivative of the extension satisfies (27.39) on X. The right side is continuous, so r is C^1, and iterating makes it smooth.

Set $S = r\gamma$. It is smooth and semipositive, and $\mathrm{d}S = \mathrm{d}\omega$ on X by density of U. The current $R = A - S$ is therefore closed, and it is supported on Z. It is also positive: since $A = S$ off Z and S puts no mass on Z, the current R is the restriction of the positive current A to Z. The support theorem recalled in Section 1.17 expresses R as a nonnegative combination of currents of integration along the components of Z of codimension one. Locally $S - \omega$ is smooth and closed, so it has a smooth local dd^{c}-potential, and the coefficients of R are the generic Lelong numbers of u_0 along these components. All of them vanish, so $\mathrm{dd}^{\mathrm{c}} u_0 = S - \omega$ is smooth. Taking a Euclidean trace and applying elliptic regularity gives a smooth function equal to u_0 almost everywhere, and hence everywhere, since both functions are quasi-plurisubharmonic. Threshold subsolutions differ from u_0 by constants. □

By Theorem 27.8.3 and Proposition 27.8.6, the condition on the Lelong numbers holds under either of the conditions above. The smooth root and its rank bound are needed: a smooth semipositive $(n-1, n-1)$-form need not have a smooth $(1,1)$-root across its degeneracy set. On a surface the rank condition already forces $\gamma > 0$ everywhere.

27.8.6.2 Two attained instances

Example 27.8.2 On the Hopf surface $X = (\mathbb{C}^2 \setminus \{0\})/\langle z \mapsto 2z \rangle$, the background $\omega = |z|^{-2}\mathrm{dd}^{\mathrm{c}}|z|^2$ satisfies $\mathrm{dd}^{\mathrm{c}}\omega = 0$, so its Monge–Ampère mass is invariant. There are smooth densities whose threshold solutions are smooth and nonconstant and whose maximizing tests are smooth strictly positive forms.

Proof The form is invariant under $z \mapsto 2z$, so it descends to X, and it is positive. For $s = |z|^2$ and $F(s) = s^{-1}$, the identity $sF'' + 2F' = 0$ gives $\mathrm{dd}^{\mathrm{c}}\omega = \mathrm{dd}^{\mathrm{c}}(s^{-1}) \wedge \mathrm{dd}^{\mathrm{c}} s = 0$, while ω is not closed. Choose a nonconstant smooth real q and small nonzero ε such that $A = \omega + \mathrm{dd}^{\mathrm{c}}(\varepsilon q) > 0$. Put $V = \int_X \omega^2$, $\mathrm{d}V = \omega^2/V$, $f = A^2/\mathrm{d}V$ and $u = \varepsilon q - \sup_X \varepsilon q$. Invariance of the mass gives $\int_X f \, \mathrm{d}V = \int_X A^2 = V$, hence $b_\omega(f) = 1$ by Theorem 27.8.3.

The form $T = A/V$ satisfies $\mathrm{dd}^{\mathrm{c}} T = 0$, $\int_X \omega \wedge T = 1$ and $D_{\sqrt{f}}(T) = 1$. For every smooth strictly positive $Q \in \mathscr{G}_\omega$, the arithmetic–geometric mean inequality and Stokes' theorem give $D_{\sqrt{f}}(Q) \leq \int_X A \wedge Q = 1$. Thus T is a smooth strictly positive maximizer. Since u is a threshold solution, Corollary 27.8.1 shows that every threshold subsolution is u plus a constant. □

Example 27.8.3 Let $n \geq 3$, and view the smooth quadric of dimension n as a double cover of $\mathbb{P}^n$. For the restricted Fubini–Study form and a suitable smooth density, the maximizing test is a smooth semipositive form whose smooth root has rank at least two everywhere and degenerates along the ramification divisor.

Proof Let

$$X = \{[w : z_0 : \cdots : z_n] : w^2 = \textstyle\sum_{i=0}^n z_i^2\} \subset \mathbb{P}^{n+1}, \qquad F([w : z]) = [z].$$

The quadratic form is nonsingular, and F is finite of degree two, with smooth ramification divisor $Z = \{w = 0\}$. Local branch coordinates show that $\mathrm{d}F$ has rank $n - 1$ on Z and n elsewhere. Let ω be the restriction to X of the normalized Fubini–Study form of $\mathbb{P}^{n+1}$, and let $A = F^*\omega_{\mathrm{FS}}$, where ω_{FS} is the normalized Fubini–Study form of $\mathbb{P}^n$. Then

$$u = \log \frac{\sum_i |z_i|^2}{|w|^2 + \sum_i |z_i|^2}$$

is a smooth function on X with $-\log 2 \leq u \leq 0$, $\sup_X u = 0$ and $A = \omega + \mathrm{dd}^{\mathrm{c}} u$. The change of variables for the branched cover gives $\int_X A^n = 2$, and Stokes' theorem gives $V = \int_X \omega^n = 2$.

Set $\mathrm{d}V = \omega^n/V$, $f = A^n/\mathrm{d}V$, $T = A^{n-1}/V$ and $\gamma = A/V^{1/(n-1)}$. Since ω is closed, Theorem 27.8.3 gives $b_\omega(f) = V/\int_X f \, \mathrm{d}V = 1$. Stokes' theorem gives $\mathrm{dd}^{\mathrm{c}} T = 0$ and $\int_X \omega \wedge T = 1$, and computing the determinant density gives $D_{f^{1/n}}(T) = 1$. The arithmetic–geometric mean inequality, which remains valid for semipositive A, bounds $D_{f^{1/n}}(Q)$ by $\int_X A \wedge Q = 1$ for every smooth strictly positive $Q \in \mathscr{G}_\omega$. Thus T is a maximizer, strictly positive off the proper analytic set Z, and by Theorem 27.8.2 the threshold solution with supremum zero is u. Its root γ has rank $n - 1 \geq 2$ on Z and is positive elsewhere, as required. □

The invariant-mass and sublinear conditions settle the envelope condition only in special situations. We do not know whether general moment or entropy bounds imply either of them, or the hypothesis on the smooth root.

Corollary 27.8.5 *Under the hypotheses of Theorem 27.8.2, suppose in addition that $f \in L^p(\mathrm{d}V)$ for some $p > 1$. Then the normalized threshold solution u_0 is continuous, and the normalized greatest exponential solutions converge uniformly to u_0 as the exponent tends to zero.*

Proof By Proposition 15.27.2, the constant c for which $\mathrm{MA}_\omega(v) = c f \,\mathrm{d}V$ has a continuous solution is $b_\omega(f)$. The continuous normalized solution is therefore a threshold solution, and it equals u_0 by the uniqueness in Theorem 27.8.2. For L^p data the greatest exponential solution coincides with the bounded solution, as explained after Theorem 15.34.1. By Proposition 15.19.1, applied with the trivial group, the normalized family is uniformly precompact, and every cluster limit solves the threshold equation. Uniqueness identifies every cluster limit with u_0, which proves uniform convergence of the whole family. □

27.8.7 A full-mass point pole without a smooth maximizing test

The envelope criteria above do not require L^p integrability of the density. In the following example on projective space, with a closed background, an explicit threshold solution is unbounded, while its envelopes $P_{\delta\omega}(u)$ in smaller backgrounds are not identically $-\infty$. The dual supremum is approached by smooth tests but attained by no smooth semipositive test.

Proposition 27.8.7 *Let $X = \mathbb{P}^n$, $n \geq 2$, and let ω be the normalized Fubini–Study form, with $\int_X \omega^n = 1$ and $\mathrm{d}V = \omega^n$. For $0 < \alpha < 1$ put*

$$\psi = \log \frac{\sum_{i=1}^n |z_i|^2}{\sum_{i=0}^n |z_i|^2}, \qquad u = 1 - (1-\psi)^\alpha, \qquad o = [1 : 0 : \cdots : 0].$$

The function u belongs to $\mathrm{PSH}(X,\omega)$, satisfies $\sup_X u = 0$, has o as its only pole, and has extended mass one. For every $0 < \delta < 1$, the envelope $P_{\delta\omega}(u)$ is not identically $-\infty$. The measure $\mathrm{MA}_\omega^{\mathrm{ext}}(u)$ is $f\,\mathrm{d}V$ with $f \in L^1(\mathrm{d}V)$ and $f \notin L^p(\mathrm{d}V)$ for every $p > 1$.

Moreover, $b_\omega(f) = 1$ and $\rho_\omega(f^{1/n}) = 1$, so u is a threshold solution for f. No smooth semipositive $T \in \mathscr{G}_\omega$ attains $D_{f^{1/n}}(T) = 1$.

Proof The numerator in the definition of ψ vanishes only at o, and locally $\omega + \mathrm{dd}^{\mathrm{c}}\psi = \mathrm{dd}^{\mathrm{c}} \log \sum_{i=1}^n |z_i|^2 \geq 0$. The function $\chi(t) = 1 - (1-t)^\alpha$ is increasing and convex on $t \leq 0$, with $0 \leq \chi' \leq \alpha$. Applying the chain inequality to bounded truncations of ψ and taking the decreasing limit, we obtain $u = \chi \circ \psi \in \mathrm{PSH}(X,\omega)$, with

$$A = \omega + \mathrm{dd}^{\mathrm{c}} u \geq (1-\alpha)\omega.$$

The function u is smooth off o, and it attains its supremum zero on $\{z_0 = 0\}$. Since $u = -(1-\psi)^\alpha + 1$, Proposition 27.8.6 shows that all the envelopes $P_{\delta\omega}(u)$ are not identically $-\infty$.

To compute the mass, use the chart $z_0 = 1$ and write $s = \sum_{i=1}^{n} |z_i|^2$. A local potential F for A, as a function of s, and the function $q = sF'$ are

$$\begin{aligned} F(s) &= \log(1+s) + 1 - \Bigl(1 - \log\frac{s}{1+s}\Bigr)^{\alpha}, \\ q(s) &= sF'(s) = \frac{s + \alpha(1 - \log(s/(1+s)))^{\alpha-1}}{1+s}. \end{aligned} \tag{27.40}$$

The eigenvalues of A are q/s in the tangential directions and q' in the radial direction. Polar integration gives the radial top-degree measure $nq^{n-1}q'\,\mathrm{d}s$, where the angular normalization follows from the Fubini–Study case $q = s/(1+s)$, whose radial density is $ns^{n-1}(1+s)^{-n-1}\,\mathrm{d}s$ and whose total mass is one. Consequently, for $0 < s_1 < s_2$,

$$\int_{\{s_1<|z|^2<s_2\}} A^n = q(s_2)^n - q(s_1)^n.$$

Here $q(s) \to 0$ as $s \downarrow 0$ and $q(s) \to 1$ as $s \to \infty$. On each compact annulus the extended measure equals the smooth top power, and it puts no mass on o or on the hyperplane at infinity, since both are pluripolar. Exhausting the affine chart by annuli therefore gives extended mass exactly one.

Near o write $L = 1 - \log s$. Differentiating (27.40) gives

$$\frac{q}{s} \sim \alpha s^{-1} L^{\alpha-1}, \qquad q' \sim \alpha(1-\alpha) s^{-1} L^{\alpha-2},$$
$$f(s) \asymp s^{-n} L^{-1-n(1-\alpha)}.$$

The smooth volume form, written in terms of s, is comparable to $s^{n-1}\,\mathrm{d}s$, so the contribution of a neighborhood of o to $\int_X f\,\mathrm{d}V$ is comparable to $\int^{\infty} L^{-1-n(1-\alpha)}\,\mathrm{d}L$, which is finite. For every $p > 1$, the corresponding integral of f^p contains an exponentially growing factor $\exp(n(p-1)L)$ and diverges. The measure is smooth off o and puts no mass on o, so it is $f\,\mathrm{d}V$ on all of X. Since its total mass is one and ω is closed, Theorem 27.8.3 gives $b_\omega(f) = 1$, and hence $\rho_\omega(f^{1/n}) = b_\omega(f)^{-1/n} = 1$.

Suppose that a smooth semipositive $T \in \mathscr{G}_\omega$ attained this value. The positive current A satisfies $\int_X A \wedge T = 1$ by Stokes' theorem for currents. On $X \setminus \{o\}$, the matrix arithmetic–geometric mean inequality gives

$$A \wedge T \geq f^{1/n} k_T\,\mathrm{d}V.$$

The right side has total mass one by attainment, so the nonnegative difference of the two sides vanishes off o. Since A is smooth and strictly positive there, the equality case gives $T = tA^{n-1}$ on $X \setminus \{o\}$. Indeed, the arithmetic–geometric mean inequality for the positive semidefinite coefficient matrix of T conjugated by $A^{1/2}$ becomes an equality only when all its eigenvalues agree; at points where T is degenerate they therefore all vanish. The function $t = (T \wedge A)/A^n$ is smooth and nonnegative on $X \setminus \{o\}$.

Since A is closed, the equation $\mathrm{dd}^{\mathrm{c}} T = 0$ becomes $\mathrm{dd}^{\mathrm{c}} t \wedge A^{n-1} = 0$, a locally strictly elliptic scalar equation on $X \setminus \{o\}$. All eigenvalues of A tend to infinity

at o, and since $n \geq 2$, so do all eigenvalues of A^{n-1}, which are products of $n-1$ eigenvalues of A. Since the coefficient matrix of the smooth form T is bounded, it follows that $t \to 0$ at o.

Set $t(o) = 0$. The resulting continuous nonnegative function on compact X is not identically zero, by the normalization of T, so it attains a positive maximum away from o. The strong maximum principle on the connected manifold $X \setminus \{o\}$ forces it to be constant there, contrary to its zero limit at o. This excludes every smooth semipositive maximizing test; no assumption on the degeneracy set of T was needed. □

Corollary 27.8.6 *For the density and potential of Proposition 27.8.7,*

$$\int_X f \log f \, \mathrm{d}V < +\infty \quad \Longleftrightarrow \quad \alpha < 1 - \frac{1}{n}.$$

For every real $r > 0$,

$$\int_X (-u)^r f \, \mathrm{d}V < +\infty \quad \Longleftrightarrow \quad \alpha r < n(1-\alpha).$$

In particular, finite entropy of the density can coexist with an unbounded threshold solution of full mass and with the absence of a smooth semipositive maximizing test.

Proof Only a neighborhood of o can contribute a divergence. There the estimates in the proof of Proposition 27.8.7 give $f \, \mathrm{d}V \asymp L^{-1-n(1-\alpha)} \, \mathrm{d}L$, $\log f \sim nL$ and $-u \asymp L^{\alpha}$. Hence the entropy integral converges exactly when $n(1-\alpha) > 1$, and the weighted integral exactly when $\alpha r < n(1-\alpha)$. In the equality cases both integrals are comparable to $\int^{\infty} L^{-1} \, \mathrm{d}L$ and diverge. The negative part of $f \log f$ is bounded, hence integrable for the probability volume, so the entropy criterion is unambiguous. □

These are integrals of the single density f, not uniform moment or entropy bounds for the normalized determinant measures of a maximizing sequence as in Theorem 27.8.5. Since the example has no smooth semipositive maximizer, it lies outside the setting of Theorem 27.8.2, and the proposition does not identify u with the greatest normalized threshold subsolution of Proposition 15.27.1.

27.8.8 Quantitative convergence of the currents

The bound for the constants c_λ in Proposition 27.8.5 also bounds the trace error linearly in λ. Away from the degeneracy set, the deficit in the arithmetic–geometric mean inequality converts this bound into a rate of convergence for all coefficients of the currents. The total variation of a current of order zero is measured with respect to a fixed smooth Hermitian metric; on a relatively compact chart it is equivalent to the sum of the total variations of its coefficients.

Proposition 27.8.8 *Under the hypotheses and notation of Proposition 27.8.5, put*

$$A_\lambda = \omega + \mathrm{dd^c} v_\lambda, \qquad A_0 = \omega + \mathrm{dd^c} u_0.$$

There is a constant C such that

$$\|A_\lambda \wedge T - A_0 \wedge T\|_{\mathrm{TV}} \le C\lambda \qquad (\lambda > 0).$$

For each compact set $K \Subset X \setminus Z$, there is a constant C_K such that

$$\|A_\lambda - A_0\|_{\mathrm{TV}(K)} \le C_K \sqrt{\lambda} \qquad (0 < \lambda \le 1).$$

Proof Put $p = f^{1/n}$, use the probability measure $\nu = A_0 \wedge T = b^{1/n} p k_T \, \mathrm{d}V$ from the proof of Proposition 27.8.5, where the equality is (27.38), and set

$$q_\lambda = (c_\lambda / b)^{1/n} \mathrm{e}^{\lambda v_\lambda / n}, \qquad R_\lambda = A_\lambda \wedge T - q_\lambda \nu \ge 0.$$

Here $R_\lambda \ge 0$ by the trace inequality used in that proof, and the same proof gives $c_\lambda \ge b$ and $\int_X (-v_\lambda)\, \mathrm{d}\nu \le C_0$ for a constant C_0 independent of λ. Write $s_+ = \max(s, 0)$ for the positive part of a real number s. Since $q_\lambda \ge \mathrm{e}^{\lambda v_\lambda / n}$ and $1 - \mathrm{e}^x \le -x$, we obtain

$$(1 - q_\lambda)_+ \le 1 - \mathrm{e}^{\lambda v_\lambda / n}, \qquad \int_X (1 - q_\lambda)_+ \, \mathrm{d}\nu \le C_0 \lambda / n.$$

Since $A_\lambda \wedge T$ and ν both have mass one,

$$\|A_\lambda \wedge T - \nu\|_{\mathrm{TV}} \le R_\lambda(X) + \int_X |q_\lambda - 1| \, \mathrm{d}\nu = 2 \int_X (1 - q_\lambda)_+ \, \mathrm{d}\nu.$$

This proves the first estimate, with $C = 2C_0/n$, and also gives $R_\lambda(X) \le C_0 \lambda / n$ and $\int_X |q_\lambda - 1| \, \mathrm{d}\nu \le 2C_0 \lambda / n$.

On $U = X \setminus Z$, write $T = \gamma^{n-1}$ and $A_0 = a\gamma$ as in the proof of Theorem 27.8.2, and put $\alpha_\lambda = q_\lambda a$. Decompose the positive current A_λ into its absolutely continuous part, with matrix density B_λ relative to γ, and its positive singular part. Since $\mathrm{MA}^{\mathrm{ext}}_\omega(v_\lambda) = c_\lambda \mathrm{e}^{\lambda v_\lambda} f \, \mathrm{d}V$, the trace inequalities for all Hermitian forms imply, as in the proof of Corollary 27.8.1,

$$(\det B_\lambda)^{1/n} \ge \alpha_\lambda \quad \text{almost everywhere on } U.$$

This is a statement about determinants of matrices; no nonlinear product of coefficient measures is involved. Put $s_\lambda = \operatorname{tr} B_\lambda / n$ and $\delta_\lambda = s_\lambda - \alpha_\lambda \geq 0$. On U, the absolutely continuous part of the trace error R_λ is $\delta_\lambda \gamma^n$, and its singular part is the trace measure of the singular part of A_λ. Thus

$$\int_U s_\lambda \gamma^n \leq 1, \qquad \int_U \delta_\lambda \gamma^n \leq C_0 \lambda / n,$$

and the singular trace mass on U has the same upper bound.

For nonnegative numbers $\ell_1, \dots, \ell_n$ with geometric mean at least $\alpha \geq 0$ and arithmetic mean s, the arithmetic–geometric mean inequality and the Cauchy–Schwarz inequality give

$$\sum_i (\sqrt{\ell_i} - \sqrt{\alpha})^2 \leq n(s - \alpha), \qquad \sum_i |\ell_i - \alpha| \leq 2n\sqrt{s(s-\alpha)}.$$

Indeed $\sum_i \sqrt{\ell_i} \geq n\sqrt{\alpha}$, while $\sum_i (\sqrt{\ell_i} + \sqrt{\alpha})^2 \leq 4ns$. Applying this to the eigenvalues of B_λ relative to γ and integrating, with the Cauchy–Schwarz inequality once more, we see that the absolutely continuous part of $A_\lambda - \alpha_\lambda \gamma$ has integrated trace norm at most $2n\sqrt{C_0 \lambda / n}$ on U. Its singular part has norm at most $C_0 \lambda$ in the same metric. Finally

$$\int_U |\alpha_\lambda - a| \gamma^n = \int_U |q_\lambda - 1| \, \mathrm{d}\nu \leq 2 C_0 \lambda / n.$$

Since γ is comparable to the fixed metric on the compact set $K \subset U$, these estimates and $\lambda \leq \sqrt{\lambda}$ for $0 < \lambda \leq 1$ give the second conclusion. □

The coefficient estimate is local on $X \setminus Z$. For convergence in capacity across Z we use the additional envelope condition of Corollary 27.8.4.

27.8.9 The classes of the divisorial components

Uniqueness constrains the divisors along which the threshold solution can have positive Lelong numbers. A linear relation between their Bott–Chern classes would produce a global potential that changes the coefficients of these divisors while preserving the equation off the degeneracy set, contradicting uniqueness. We write $\{D\} \in H^{1,1}_{\mathrm{BC}}(X, \mathbb{R})$ for the Bott–Chern class of the current of integration $[D]$ along a prime divisor D, and $\nu(u, D)$ for the generic Lelong number of u along D.

Corollary 27.8.7 *Under the hypotheses of Theorem 27.8.2, let u be a threshold solution and let $D_1, \dots, D_k$ be the prime divisors with $\nu(u, D_j) > 0$. Then the classes $\{D_j\}$ are linearly independent in $H^{1,1}_{\mathrm{BC}}(X, \mathbb{R})$. In particular their number is at most* $\dim_{\mathbb{R}} H^{1,1}_{\mathrm{BC}}(X, \mathbb{R})$.

Proof Since u is smooth off Z, these divisors are among the finitely many components of Z of codimension one. Put $a_j = \nu(u, D_j) > 0$. By Lemma 1.17.1,

$$R = \omega + \mathrm{dd^c} u - \sum_j a_j [D_j] \geq 0.$$

Suppose that real numbers $c_1, \ldots, c_k$ satisfy $\sum_j c_j \{D_j\} = 0$. Fix a Gauduchon metric η and apply Lemma 24.7.1 to the currents of integration $[D_j]$, with the integrable potentials $v_\eta([D_j])$ given there. Since the smooth representative in that lemma depends linearly on the Bott–Chern class, we obtain

$$h = \sum_j c_j v_\eta([D_j]) \in L^1(X), \qquad \mathrm{dd^c} h = \sum_j c_j [D_j].$$

Choose $t \neq 0$ sufficiently small that $a_j + t c_j \geq 0$ for every j. The distribution $u_t = u + t h$ satisfies

$$\omega + \mathrm{dd^c} u_t = R + \sum_j (a_j + t c_j)[D_j] \geq 0,$$

so it is represented by a function $u_t \in \mathrm{PSH}(X, \omega)$. On $X \setminus Z$, the function h is smooth and pluriharmonic, so u_t has the same Monge–Ampère measure as u there. Since neither extended measure charges the pluripolar set Z, u_t is another threshold solution. Uniqueness makes $t h = u_t - u$ constant. Hence $\sum_j c_j [D_j] = 0$, and testing near a generic point of each of the distinct divisors D_j gives $c_j = 0$. □

27.8.10 A branched projective example

The squaring map of projective space gives a smooth threshold solution whose maximizing form degenerates along the coordinate hyperplanes. The density is smooth, but its nth root is not differentiable along these hyperplanes.

For this example take $X = \mathbb{P}^n$, $n \geq 2$, with homogeneous coordinates $[Z_0 : \cdots : Z_n]$, and let ω be the Fubini–Study form normalized by $\int_X \omega^n = 1$. Set $\mathrm{d}V = \omega^n$.

Example 27.8.4 There is a smooth nonnegative density f on $\mathbb{P}^n$, vanishing precisely on the coordinate hyperplanes, such that $b_\omega(f) = 1$ and $D_{f^{1/n}}$ has a smooth semipositive maximizer strictly positive off those hyperplanes. The threshold solution is smooth, but $f^{1/n}$ is not differentiable at a general point of any coordinate hyperplane. There is no smooth strictly positive maximizing form. The normalized greatest exponential solutions converge uniformly to the normalized threshold solution.

Proof Define the holomorphic map $F \colon \mathbb{P}^n \to \mathbb{P}^n$ and the semipositive form A by

$$F[Z_0 : \cdots : Z_n] = [Z_0^2 : \cdots : Z_n^2], \qquad A = \tfrac{1}{2} F^* \omega.$$

The homogeneous formula

$$u = \tfrac{1}{2}\log\left(\sum_{i=0}^{n} |Z_i|^4\right) - \log\left(\sum_{i=0}^{n} |Z_i|^2\right)$$

defines a smooth function on projective space, and $A = \omega + \mathrm{dd}^{\mathrm{c}} u$. The form A is semipositive, and it is strictly positive exactly where every homogeneous coordinate is nonzero. Define

$$f = A^n/\omega^n, \qquad T = A^{n-1}.$$

Since ω and A are closed and cohomologous, Stokes' theorem gives $\int_X f\omega^n = \int_X \omega \wedge T = 1$, and $\mathrm{dd}^{\mathrm{c}} T = 0$; thus $T \in \mathscr{G}_\omega$. Since ω is closed, Theorem 27.8.3 gives $b_\omega(f) = 1$.

The determinant formula for smooth semipositive forms gives

$$k_T = f^{(n-1)/n}, \qquad D_{f^{1/n}}(T) = \int_{\mathbb{P}^n} f\omega^n = 1.$$

By (27.35) and $\rho_\omega(f^{1/n}) = b_\omega(f)^{-1/n} = 1$, the form T is a maximizer. It is strictly positive away from the coordinate hyperplanes because A is; on these hyperplanes A has deficient rank, and hence so does the coefficient matrix of $T = A^{n-1}$. Since $\mathrm{MA}_\omega(u) = f\omega^n$, the smooth function u is a threshold solution, and the uniqueness in Theorem 27.8.2, with Z the union of the coordinate hyperplanes, shows that every threshold subsolution is u plus a constant.

For the density root, use the chart $Z_0 = 1$ with coordinates $z_1, \ldots, z_n$. Pulling back the Fubini–Study volume by $(z_1, \ldots, z_n) \mapsto (z_1^2, \ldots, z_n^2)$ and including the factor 2^{-n} gives

$$f(z) = 2^n \left(\prod_{i=1}^{n} |z_i|^2\right) \frac{(1 + \sum_i |z_i|^2)^{n+1}}{(1 + \sum_i |z_i|^4)^{n+1}}.$$

Near a point of $\{z_1 = 0\}$ at which all the other coordinates are nonzero, $f^{1/n}$ is $|z_1|^{2/n}$ times a smooth strictly positive function. Restricting to a real line transverse to that hyperplane proves nondifferentiability, also in the case $n = 2$, where $|z_1|^{2/n} = |z_1|$. By symmetry of the coordinates, the same holds on every coordinate hyperplane. The absence of a smooth strictly positive maximizer follows from Corollary 27.8.3, since the continuous function f has zeros. Finally, f is bounded, so Corollary 27.8.5 gives the uniform convergence of the normalized exponential solutions. □

27.8.11 A moment condition on maximizing tests

For a general Hermitian background, a uniform bound on the moments of normalized potentials with respect to the probabilities of one smooth maximizing sequence can replace the regularity of the maximizing current. Together with the envelope

condition in a smaller background, it shows that the cluster limits of the normalized solutions of the exponential equations are exact threshold solutions.

We use the class $\mathcal{S}_f$ of threshold subsolutions with supremum zero and its greatest member Φ_f from Proposition 15.27.1, and, as in Proposition 27.8.5, the greatest solutions u_λ of the exponential equations $\mathrm{MA}_\omega^{\mathrm{ext}}(u) = \mathrm{e}^{\lambda u} f\,\mathrm{d}V$, with $v_\lambda = u_\lambda - \sup_X u_\lambda$ and $c_\lambda = \mathrm{e}^{\lambda \sup_X u_\lambda}$.

Theorem 27.8.5 *Let $0 \le f \in L^1(\mathrm{d}V)$ be nonzero with $b = b_\omega(f) > 0$, and put $p = f^{1/n}$ and $R = \rho_\omega(p) = b^{-1/n}$. Suppose that there are smooth strictly positive $T_j \in \mathcal{G}_\omega$, with associated probability measures ν_j, such that*

$$d_j = D_p(T_j) \longrightarrow R, \qquad \mathrm{d}\nu_j = d_j^{-1} p k_{T_j}\,\mathrm{d}V,$$

and a constant C satisfying

$$\int_X (-v)\,\mathrm{d}\nu_j \le C \quad \text{for all } j \text{ and all } v \in \mathrm{PSH}(X,\omega) \text{ with } \sup_X v = 0. \tag{27.41}$$

Then, for every $\lambda > 0$,

$$b \le c_\lambda \le b\mathrm{e}^{C\lambda}, \qquad \Phi_f - C \le v_\lambda \le 0.$$

Suppose in addition that $P_{\delta\omega}(\Phi_f)$ is not identically $-\infty$ for some $0 < \delta < 1$. Whenever $\lambda_k \downarrow 0$ and $v_{\lambda_k} \to v$ in $L^1(X)$, the convergence holds in capacity and

$$\mathrm{MA}_\omega^{\mathrm{ext}}(v) = b f\,\mathrm{d}V, \qquad \Phi_f - C \le v \le \Phi_f.$$

In particular an exact threshold solution exists.

Proof Since $\mathrm{MA}_\omega^{\mathrm{ext}}(v_\lambda) = c_\lambda \mathrm{e}^{\lambda v_\lambda} f\,\mathrm{d}V$, the trace inequality of Lemma 7.4.2 and Stokes' theorem give, for each j and $\lambda > 0$,

$$1 \ge c_\lambda^{1/n} d_j \int_X \mathrm{e}^{\lambda v_\lambda/n}\,\mathrm{d}\nu_j \ge c_\lambda^{1/n} d_j \mathrm{e}^{-C\lambda/n}.$$

The second inequality follows from Jensen's inequality and (27.41). Letting $j \to \infty$ and using $R = b^{-1/n}$, we obtain the upper bound for c_λ. The lower bound and the comparison with every normalized threshold subsolution were proved in Corollary 15.34.1; applying that comparison to Φ_f, we obtain

$$v_\lambda \ge \Phi_f - \lambda^{-1}\log(c_\lambda/b) \ge \Phi_f - C.$$

Now let $v_{\lambda_k} \to v$ in L^1. By Corollary 15.34.2, the limit v lies in $\mathcal{S}_f$, hence below Φ_f. The lower bound passes to the limit, so $\Phi_f - C \le v \le \Phi_f$. The proof of Corollary 27.8.4, with Φ_f in place of u_0 and v as the L^1 limit, now gives convergence in capacity: the bounds $\Phi_f - C \le v_{\lambda_k} \le 0$ and $c_{\lambda_k} \le b\mathrm{e}^{C\lambda_k}$ are those used there, the condition on $P_{\delta\omega}(\Phi_f)$ is assumed, and the hypothesis $v \preceq \Phi_f$ of Theorem 15.8.1 holds since $v \le \Phi_f$.

The continuity of damped measures under convergence in capacity, Theorem 8.3.1, identifies the equation without a mass-conservation argument. Write $g_k = c_{\lambda_k} \mathrm{e}^{\lambda_k v_{\lambda_k}} f = \mathrm{e}^{\lambda_k u_{\lambda_k}} f$. By Corollary 15.34.1, $g_k \to bf$ in L^1, and $0 \le g_k \le b\mathrm{e}^C f$ for large k. Hence

$$\mathrm{e}^{v_{\lambda_k}} \mathrm{MA}_\omega^{\mathrm{ext}}(v_{\lambda_k}) = \mathrm{e}^{v_{\lambda_k}} g_k \, \mathrm{d}V \longrightarrow \mathrm{e}^v b f \, \mathrm{d}V$$

in total variation, while Theorem 8.3.1 gives the weak limit $\mathrm{e}^v \mathrm{MA}_\omega^{\mathrm{ext}}(v)$. Dividing by e^v off the pole set of v, which neither measure charges, gives the equation and the finiteness of the mass. Finally, compactness of normalized ω-plurisubharmonic functions provides an L^1 cluster limit along any sequence $\lambda_k \downarrow 0$, which proves existence. □

The moment condition holds if the densities of the ν_j are uniformly bounded in $L^q(\mathrm{d}V)$ for some $q > 1$, by Hölder's inequality and the uniform bounds for the moments of normalized ω-plurisubharmonic functions. More generally, a uniform entropy bound suffices. Write $a_j = \mathrm{d}\nu_j/\mathrm{d}V$ and suppose $\int_X a_j \log a_j \, \mathrm{d}V \le H$ for all j. (By Proposition 16.6.1, a uniform entropy bound is equivalent to a uniform bound for the dual entropy $H_0(\nu_j)$ with zero reference defined there.) Choose $\alpha > 0$ and A from Lemma 1.5.1, so that $\int_X \mathrm{e}^{-\alpha v} \, \mathrm{d}V \le A$ for every $v \in \mathrm{PSH}(X, \omega)$ with $\sup_X v = 0$. The elementary entropy inequality, applied first to the truncations $v_N = \max(v, -N)$ at levels $N > 0$, gives

$$\alpha \int_X (-v_N) a_j \, \mathrm{d}V \le \int_X a_j \log a_j \, \mathrm{d}V + \log \int_X \mathrm{e}^{-\alpha v_N} \, \mathrm{d}V \le H + \log A.$$

To prove this inequality, apply $t \log t - t + 1 \ge 0$ to $t = a_j/\varrho_N$, integrate against $\varrho_N \, \mathrm{d}V$, where $\varrho_N = \mathrm{e}^{-\alpha v_N} / \int_X \mathrm{e}^{-\alpha v_N} \, \mathrm{d}V$ is a probability density, and rearrange. Monotone convergence as $N \to \infty$ proves (27.41) with $C = (H + \log A)/\alpha$.

The same entropy bound also gives attainment of the dual supremum along this sequence. Indeed, $t \log t \ge -\mathrm{e}^{-1}$ and, for $K > 1$,

$$\int_{\{a_j > K\}} a_j \, \mathrm{d}V \le \frac{H + \mathrm{e}^{-1}}{\log K}.$$

This proves uniform integrability of the a_j, and since $d_j \to R > 0$, also of the densities $pk_{T_j} = d_j a_j$. By Proposition 27.8.2, every weak cluster limit T of (T_j) maximizes D_p, and along a subsequence converging to T the probabilities ν_j converge to $R^{-1} pk_T \, \mathrm{d}V$. For each normalized v, the function $-v$ is nonnegative and lower semicontinuous, so the lower semicontinuity of its integral under weak convergence of measures passes the moment bound to this limiting probability. These conclusions give no regularity of T.

The envelope condition holds if Φ_f is bounded. In general, it and the moment bound are additional hypotheses; the tail condition on $f^{1/n}$ alone does not provide them in this argument. The theorem identifies the equation satisfied by each cluster limit. Uniqueness, which gave convergence of the whole family in Proposition 27.8.5, is not available here.

27.9 Nef limits and nonattainment

We return to arbitrary dimension $n \geq 1$. If θ is nef and h is a Hermitian form, then the backgrounds $\theta + \epsilon h$ lie in $\mathscr{P}$ for every $\epsilon > 0$, directly from the definition of nefness; their volumes are therefore scalars of smooth equations. As ϵ decreases to zero, the infimum over Gauduchon metrics commutes with the decreasing limit of the background. Thus the boundary value is determined by smooth equations, even when no metric attains it.

Theorem 27.9.1 *For every nef background θ and every Hermitian form h,*

$$\lambda_\mu(\theta) = \lim_{\epsilon \downarrow 0} \lambda_\mu(\theta + \epsilon h), \qquad \mathcal{V}_\mu(\theta) = \lim_{\epsilon \downarrow 0} c_{\theta+\epsilon h}(\mu).$$

Both quantities on the right decrease as $\epsilon \downarrow 0$, and their limits do not depend on h.

Proof For $0 < \epsilon' < \epsilon$ the difference of the two backgrounds is the positive form $(\epsilon - \epsilon')h$, so the monotonicity in Proposition 10.8.1 shows that $\lambda_\mu(\theta + \epsilon h)$ decreases as $\epsilon \downarrow 0$. Its limit is therefore the infimum over $\epsilon > 0$. Infima over $\epsilon > 0$ and over Gauduchon metrics commute, and for a fixed metric the quotient is affine in ϵ. Hence

$$\begin{aligned} \inf_{\epsilon>0} \lambda_\mu(\theta + \epsilon h) &= \inf_{\gamma} \inf_{\epsilon>0} \frac{\int_X \theta \wedge \gamma^{n-1} + \epsilon \int_X h \wedge \gamma^{n-1}}{D_\mu(\gamma)} \\ &= \lambda_\mu(\theta). \end{aligned}$$

Here the second equality holds because the coefficient of ϵ is positive, so the inner infimum is the value at $\epsilon = 0$. This proves the first identity, and its right side is the definition of $\lambda_\mu(\theta)$, which does not involve h. The second identity follows by raising the first to the nth power, since $\lambda_\mu(\theta + \epsilon h)^n = c_{\theta+\epsilon h}(\mu)$ by Theorem 10.8.1. □

If X is Kähler and θ is closed and nef, choose h Kähler. Each perturbed solution form is then closed and cohomologous to $\theta + \epsilon h$, so integrating its equation and applying Stokes' theorem gives

$$c_{\theta+\epsilon h}(\mu) = \int_X (\theta + \epsilon h)^n.$$

Letting $\epsilon \downarrow 0$, we obtain $\mathcal{V}_\mu(\theta) = \int_X \theta^n$ for every closed nef θ on a compact Kähler manifold and every μ. We refer to this as the *closed nef formula*.

Proposition 27.9.1 *Suppose θ is pseudoeffective and $\lambda_\mu(\theta) = 0$. The infimum in Definition 10.8.1 is attained if and only if $\theta = \mathrm{dd}^{\mathrm{c}} f$ for a smooth real function f.*

Proof Since $D_\mu(\gamma) > 0$, a metric γ attaining the value zero satisfies $\int_X \theta \wedge \gamma^{n-1} = 0$. By Lemma 4.3.1, including its dimension-one case, θ is then smoothly dd^{c}-exact. Conversely, if $\theta = \mathrm{dd}^{\mathrm{c}} f$, Stokes' theorem makes every quotient zero, so every admissible metric attains the infimum. □

For example, let X be the ruled surface and D the section of Lemma 26.5.1. The class $c_1(\mathcal{O}_X(D))$ is nef, has zero square, and contains the nonzero positive current $[D]$. By the closed nef formula its volume is zero. A smooth dd^{c}-exact representative would make $[D]$ itself dd^{c}-exact, which is impossible because $\int_D \kappa > 0$ for a Kähler form κ. By Proposition 27.9.1, no Gauduchon metric minimizes the defining quotient.

27.10 Subtraction with a closed nef core

Consider a path of backgrounds that starts in $\mathscr{P}$ and reaches its boundary at a nef point. If the path contains a closed nef class of positive top intersection, this closed core keeps the root volume positive up to and including the boundary, and adding a semipositive form, closed or not, preserves this lower bound. By Proposition 27.1.1, the total decrease of the root volume along the path is the integral of an explicit ratio formed with the minimizing Gauduchon metrics; we call it the subtraction cost. Two test metrics chosen at the initial point bound this cost from above, with explicit dependence on the density and on the derivatives of the added form.

Throughout this section, X is a compact Kähler manifold of dimension $n \geq 2$ and Ω is a fixed Kähler form on X. Let α be a smooth closed nef real $(1,1)$-form, let $B \geq 0$ be a smooth real $(1,1)$-form, and let $t > 0$. For real s, put

$$a_s = \alpha + (t-s)\Omega, \qquad \theta_s = a_s + B, \qquad V_s = \int_X a_s^n.$$

The class of a_0 is Kähler. Choose a Kähler representative κ_0 of $[a_0]$, and let u_0 be the smooth solution of the Calabi–Yau equation $\gamma_0 = \kappa_0 + \mathrm{dd}^{\mathrm{c}} u_0 > 0$, $\gamma_0^n = V_0 \mu$. Define the fixed quantities

$$\begin{aligned} I_B &= \int_X B \wedge \kappa_0^{n-1}, \qquad U_\mu = \frac{V_0 + I_B}{D_\mu(\kappa_0)}, \\ \Delta_\mu(B) &= V_0^{-(n-1)/n} \int_X B \wedge \gamma_0^{n-1}. \end{aligned} \tag{27.42}$$

Theorem 27.10.1 *Suppose that $V := V_t > 0$ and that there is a smooth compact complex curve $C \subset X$ with $\int_C \alpha = \int_C B = 0$. Then θ_s has a smooth positive representative for $s < t$, θ_t is nef but has no smooth positive representative, and θ_s is not nef for $s > t$. Moreover,*

$$\lambda_\mu(a_s) = V_s^{1/n} \quad (0 \leq s \leq t), \qquad \lambda_\mu(\theta_t) \geq V^{1/n} > 0.$$

For every choice of minimizing Gauduchon metrics g_s for θ_s, $0 \le s < t$, the ratio $\int_X \Omega \wedge g_s^{n-1}/D_\mu(g_s)$ is a continuous function of s, and the subtraction cost satisfies

$$\begin{aligned} 0 \le C &:= \int_0^t \frac{\int_X \Omega \wedge g_s^{n-1}}{D_\mu(g_s)}\, \mathrm{d}s = \lambda_\mu(\theta_0) - \lambda_\mu(\theta_t) \\ &\le \min\{U_\mu, V_0^{1/n} + \Delta_\mu(B)\} - V^{1/n} < \infty, \end{aligned} \tag{27.43}$$

where the integral is improper at t. If $B \le M\kappa_0$ for a constant $M \ge 0$, then

$$\Delta_\mu(B) \le M V_0^{1/n}, \qquad C \le (1+M)V_0^{1/n} - V^{1/n}.$$

Finally, there are a quasi-plurisubharmonic function ρ and a constant $\varepsilon > 0$ with $\alpha + \mathrm{dd^c}\rho \ge \varepsilon\Omega$ and $\theta_t + \mathrm{dd^c}\rho \ge \varepsilon\Omega$.

Proof We first locate the boundary of $\mathscr{P}$ along the path. Let $s < t$. Since α is nef, for every $\delta > 0$ there is a smooth function v_δ with $\alpha + \mathrm{dd^c} v_\delta \ge -\delta\Omega$. Taking $\delta < t - s$ makes $a_s + \mathrm{dd^c} v_\delta$ Kähler, and adding $B \ge 0$ gives a positive representative of θ_s. The same approximations show that θ_t is nef. On the other hand, the hypotheses on C and Stokes' theorem on the compact curve give

$$\int_C (\theta_s + \mathrm{dd^c} v) = (t-s)\int_C \Omega$$

for every smooth v. At $s = t$ the right side vanishes, which excludes a positive representative of θ_t. For $s > t$, a representative with $\theta_s + \mathrm{dd^c} v \ge -\delta\Omega$ and $\delta < s - t$ would contradict this identity, so θ_s is not nef.

Since every a_s with $s \le t$ is closed and nef, the closed nef formula following Theorem 27.9.1 gives $\lambda_\mu(a_s) = V_s^{1/n}$, including at $s = t$. The difference $\theta_t - a_t = B$ is semipositive, so monotonicity gives $\lambda_\mu(\theta_t) \ge V^{1/n}$. Since $\theta_s = \theta_t + (t-s)\Omega$, Theorem 27.9.1 with the Hermitian form Ω identifies $\lambda_\mu(\theta_t)$ with the limit of $\lambda_\mu(\theta_s)$ as $s \uparrow t$.

Write $f(s) = \lambda_\mu(\theta_s)$ for $s < t$. The path has constant velocity $-\Omega$, so Proposition 27.1.1 shows that f is smooth and

$$H(s) := \frac{\int_X \Omega \wedge g_s^{n-1}}{D_\mu(g_s)} = -f'(s).$$

In particular, the ratio H is continuous and independent of the choice of minimizing metrics. For $r < t$, the fundamental theorem of calculus gives

$$\int_0^r H(s)\, \mathrm{d}s = f(0) - f(r).$$

Since $H \ge 0$, letting $r \uparrow t$ and using the limit just established yields the identity in (27.43).

It remains to bound $f(0) = \lambda_\mu(\theta_0)$ from above by two test metrics. The Kähler form κ_0 is Gauduchon, and since the closed forms κ_0 and a_0 are cohomologous, Stokes' theorem gives $\int_X \theta_0 \wedge \kappa_0^{n-1} = V_0 + I_B$; hence $\lambda_\mu(\theta_0) \le U_\mu$. For the second test γ_0, the equation $\gamma_0^n = V_0\mu$ and the cohomology class of γ_0 give

$$D_\mu(\gamma_0) = V_0^{(n-1)/n}, \qquad \int_X a_0 \wedge \gamma_0^{n-1} = V_0,$$

so its quotient for θ_0 is $V_0^{1/n} + \Delta_\mu(B)$. Combined with the lower bound $\lambda_\mu(\theta_t) \ge V^{1/n}$, these two tests prove the upper bound in (27.43). If $B \le M\kappa_0$, positivity and Stokes' theorem yield $\int_X B \wedge \gamma_0^{n-1} \le M \int_X \kappa_0 \wedge \gamma_0^{n-1} = MV_0$, which gives the remaining cost bounds.

Finally, $V = \int_X \alpha^n > 0$, so the closed nef class of α is big by Demailly and Păun's criterion for nef classes on compact Kähler manifolds. Hence it contains a Kähler current $\alpha + \mathrm{dd^c}\rho \ge \varepsilon\Omega$. Adding $B \ge 0$ gives the last assertion. Unlike the first current, the second is not closed when $\mathrm{d}B \ne 0$, since its exterior derivative is $\mathrm{d}B$. □

Such data arise from a bimeromorphic holomorphic map $p\colon X \to Y$ onto a compact Kähler manifold Y that contracts a smooth curve C to a point. Take $\alpha = p^*\omega_Y$, with ω_Y Kähler, and any smooth $B \ge 0$ vanishing on TC. Since p has degree one, $\int_X \alpha^n = \int_Y \omega_Y^n > 0$, while both curve integrals vanish.

The second initial test separates the closed intersection number I_B from a correction that involves only $\mathrm{dd^c}B$.

Proposition 27.10.1 *Set $S = \sum_{j=0}^{n-2} \gamma_0^j \wedge \kappa_0^{n-2-j}$. Then*

$$\Delta_\mu(B) = V_0^{-(n-1)/n} \left(I_B + \int_X u_0 \, \mathrm{dd^c}B \wedge S \right). \tag{27.44}$$

If $\mathrm{dd^c}B \ge -C_B\kappa_0^2$ in the sense of weak positivity of $(2,2)$-forms, then

$$\Delta_\mu(B) \le V_0^{-(n-1)/n} I_B + (n-1)C_B \operatorname{osc}_X(u_0) V_0^{1/n}.$$

Such a finite constant C_B exists for every fixed smooth B.

Proof The factors of S are closed, and $\gamma_0^{n-1} - \kappa_0^{n-1} = \mathrm{dd^c}u_0 \wedge S$. Integration by parts therefore gives (27.44). Since $\int_X \mathrm{dd^c}B \wedge S = 0$, the right side does not change when a constant is added to u_0, so we may assume $\sup_X u_0 = 0$. Then $-\operatorname{osc}_X(u_0) \le u_0 \le 0$, and the lower bound for $\mathrm{dd^c}B$ together with the positivity of the summands of S bounds the correction term above by

$$C_B \operatorname{osc}_X(u_0) \int_X \kappa_0^2 \wedge S = (n-1)C_B \operatorname{osc}_X(u_0) V_0.$$

The equality holds because the cohomologous Kähler forms γ_0, κ_0 have all mixed top intersections equal to V_0. Finally, restrict $\mathrm{dd^c}B$ to the complex two-planes of TX and

normalize by κ_0^2; compactness of the bundle of these planes bounds the restrictions and gives C_B. □

The potential u_0 solves a single complex Monge–Ampère equation, with density $V_0\mu/\kappa_0^n$ relative to κ_0^n. Its oscillation therefore carries the dependence on μ, whereas the derivatives $\mathrm{d}\theta_s = \mathrm{d}B$ and $\mathrm{dd}^{\mathrm{c}}\theta_s = \mathrm{dd}^{\mathrm{c}}B$ do not change along the path.

For the closed core alone the cost is explicit. Put $P_s = \int_X \Omega \wedge a_s^{n-1}$. Since $V_s' = -nP_s$,

$$-\frac{\mathrm{d}}{\mathrm{d}s}\lambda_\mu(a_s) = \frac{P_s}{V_s^{(n-1)/n}}, \qquad \int_0^t \frac{P_s}{V_s^{(n-1)/n}}\,\mathrm{d}s = V_0^{1/n} - V^{1/n}.$$

Moreover $P_s' = -(n-1)\int_X \Omega^2 \wedge a_s^{n-2} \le 0$, since all classes on the segment are nef. Integrating $V_s' = -nP_s$ and using $P_s \le P_0$ gives

$$V \ge V_0 - nt \int_X \Omega \wedge a_0^{n-1}. \tag{27.45}$$

This argument does not use $V > 0$, so a positive right side guarantees the hypothesis $V > 0$ of Theorem 27.10.1. The following example shows that this sufficient condition is not necessary.

Example 27.10.1 On the blowup of $\mathbb{P}^2$ at a point there are data as above with $t = 1$, $\mathrm{d}B \neq 0$,

$$\lambda_\mu(\theta_1) \ge \frac{3}{2}, \qquad 0 \le C \le \frac{11}{\sqrt{8}} - \frac{3}{2},$$

and an explicit current $\theta_1 + \mathrm{dd}^{\mathrm{c}}\rho \ge \frac{1}{2}\Omega$ at the nef endpoint.

Proof Realize the blowup as $X = \{(x, [u : v]) \in \mathbb{P}^2 \times \mathbb{P}^1 : x_1 v = x_2 u\}$, with projections p, q and exceptional curve $E = p^{-1}([1 : 0 : 0])$. Let $L = p^*\omega_{\mathrm{FS},\mathbb{P}^2}$, $Q = q^*\omega_{\mathrm{FS},\mathbb{P}^1}$ and $\Omega = L + Q$, with the Fubini–Study forms normalized to have integral one on projective lines. The incidence embedding makes Ω Kähler. The degree-one blowdown, a fiber of q, and the isomorphism $q|_E \colon E \to \mathbb{P}^1$ give

$$\int_X L^2 = \int_X L \wedge Q = 1, \quad Q^2 = 0, \quad L|_{TE} = 0, \quad \int_E Q = 1.$$

Take $t = 1$, $\alpha = L$, $\kappa_0 = 2L + Q$ and $\mu = \kappa_0^2/8$. With $r = 1 - s$, the core volumes are $V_s = 1 + 4r + 3r^2$; in particular $V_0 = 8$ and $V = V_1 = 1$. Choose a nonconstant smooth function $0 \le \chi \le 1$ supported away from E, and put

$$b = \frac{1}{3}\int_X \chi L \wedge \kappa_0, \qquad f = \chi - b, \qquad B = (1 + f/2)L.$$

Since $\int_X L \wedge \kappa_0 = 3$, we have $|f| \le 1$ and $\int_X fL \wedge \kappa_0 = 0$. Thus $B \ge L/2$ and $B|_{TE} = 0$, so the curve $C = E$ satisfies the hypotheses of Theorem 27.10.1. Where

$\mathrm{d}\chi \neq 0$, the form L is symplectic, and wedging with L is injective on real one-forms in real dimension four. Hence $\mathrm{d}B = \frac{1}{2}\mathrm{d}f \wedge L \neq 0$.

At the endpoint $\theta_1 = (2+f/2)L \geq 3L/2$, and $\lambda_\mu(L) = 1$ by the closed nef formula; this gives the lower bound for $\lambda_\mu(\theta_1)$. Since $\kappa_0^2 = 8\mu$, the initial solution is $\gamma_0 = \kappa_0$ with $u_0 = 0$. Therefore $D_\mu(\kappa_0) = \sqrt{8}$, $I_B = 3$, $\Delta_\mu(B) = 3/\sqrt{8}$ and $U_\mu = 11/\sqrt{8}$, so $\lambda_\mu(\theta_0) \leq 11/\sqrt{8}$. The cost identity and the lower bound at the endpoint give the asserted estimate for C.

The pullback of $\{x_1 = 0\}$ is E plus a fiber of q, so $\mathcal{O}(E) = p^*\mathcal{O}_{\mathbb{P}^2}(1) \otimes q^*\mathcal{O}_{\mathbb{P}^1}(-1)$. Give this bundle its product metric h_E, with curvature $L - Q$, and let s_E be its canonical section. The Poincaré–Lelong formula gives, for $\rho = \frac{1}{2}\log|s_E|^2_{h_E}$,

$$\theta_1 + \mathrm{dd}^{\mathrm{c}}\rho = B + \tfrac{1}{2}\Omega + \tfrac{1}{2}[E] \geq \tfrac{1}{2}\Omega.$$

Finally, $\int_X \Omega \wedge a_0 = 5$, so the right side of (27.45) is $8 - 10 = -2$, although $V_1 = 1$.□

27.11 Subtraction on warped products

The closed core in Section 27.10 keeps the endpoint big. On a warped product, the subtraction cost can instead be computed all the way to an endpoint that is not big. Its rate is determined by two inverse moments of the fiber coefficient, and the total cost remains finite even when the endpoint volume vanishes.

Let (B, α) be a connected compact Kähler manifold of dimension $m \geq 1$, put $W = \int_B \alpha^m$, and let C be a connected compact complex curve. Fix smooth strictly positive probability forms σ on B and w on C. On $X = B \times C$, of dimension $n = m+1$, use $\mu = \sigma \wedge w$ and omit pullbacks. For a smooth function $a \geq 0$ on B and a real number $t > 0$, set

$$\theta_s = \alpha + (a + t - s)w \qquad (s \leq t).$$

The endpoint $\theta_t = \alpha + aw$ is semipositive, hence nef. For $\epsilon > 0$, write

$$H(\epsilon) = \int_B \frac{\sigma}{a+\epsilon}, \qquad J(\epsilon) = \int_B \frac{\sigma}{(a+\epsilon)^2}, \qquad H(0) = \int_B \frac{\sigma}{a},$$

allowing $H(0) = +\infty$, with $1/a = +\infty$ at a zero of a.

Theorem 27.11.1 *For $s < t$, the infimum defining $\lambda_\mu(\theta_s)$ is attained by a balanced metric, and*

$$\mathcal{V}_\mu(\theta_s) = \frac{nW}{H(t-s)}, \qquad \mathcal{V}_\mu(\theta_t) = \frac{nW}{H(0)},$$

where the last value is zero if $H(0) = +\infty$. Every minimizing Gauduchon metric g_s for θ_s gives the same cost rate

$$\frac{\int_X w \wedge g_s^{n-1}}{D_\mu(g_s)} = C(s) := n^{-(n-1)/n} W^{1/n} H(t-s)^{-(n+1)/n} J(t-s). \tag{27.46}$$

For every selection of these metrics, the improper integral satisfies

$$\int_0^t C(s)\,\mathrm{d}s = \lambda_\mu(\theta_0) - \lambda_\mu(\theta_t) < \infty. \tag{27.47}$$

Proof Put $\epsilon = t-s$ and $L = a+\epsilon$. The second paragraph of the proof of Theorem 27.4.2 uses neither the normalization $\int_B \kappa^m = 1$ nor the conditions on a imposed there. Applied with α in place of $b\kappa$, so that b^m becomes W, and with the fiber coefficient L, it gives a smooth function ϕ_ϵ on B with

$$A_\epsilon = \alpha + \mathrm{dd^c}\phi_\epsilon > 0, \qquad A_\epsilon^m = \frac{W}{H(\epsilon)}\frac{\sigma}{L}, \qquad \Omega_\epsilon = A_\epsilon + Lw = \theta_s + \mathrm{dd^c}\phi_\epsilon,$$

and shows that $\Omega_\epsilon^n = (nW/H(\epsilon))\mu$ and that the balanced metric $\gamma_s = L^{-1/m}\Omega_\epsilon$ minimizes the quotient, with numerator nW and denominator

$$D_\mu(\gamma_s) = (nW/H(\epsilon))^{m/n} H(\epsilon).$$

This proves the volume formula for $s < t$.

For a fixed Gauduchon metric, the quotient at $\theta_t + \epsilon w$ is affine in ϵ with nonnegative slope, so it decreases to the quotient at θ_t as $\epsilon \downarrow 0$. The two infima, over $\epsilon > 0$ and over the metrics, commute, as in Theorem 27.9.1. Hence $\lambda_\mu(\theta_t)$ is the limit of $\lambda_\mu(\theta_s)$ as $s \uparrow t$. Monotone convergence of $H(\epsilon)$ gives the asserted value, including when the integral is infinite.

For the balanced metric γ_s, the formula $\gamma_s^m = L^{-1}A_\epsilon^m + mA_\epsilon^{m-1} \wedge w$ of (27.17) shows that the numerator of the cost rate is $\int_X w \wedge \gamma_s^m = WJ(\epsilon)/H(\epsilon)$. Dividing by $D_\mu(\gamma_s)$ gives (27.46). Also $\mathrm{d}H(t-s)/\mathrm{d}s = J(t-s)$, so differentiation of the volume formula gives $-\mathrm{d}\lambda_\mu(\theta_s)/\mathrm{d}s = C(s)$. By Proposition 27.1.1, applied in the direction $-w$, this is the cost rate of every minimizing metric, regardless of the regularity of the selection. On each interval $[0, r]$ with $r < t$, integrate this identity. Since $C \geq 0$ and the root volume has the finite endpoint limit just proved, letting $r \uparrow t$ gives (27.47).□

The identity (27.47) applies to every smooth nonnegative coefficient, including coefficients with flat zeros. Finite-order zeros give more precise rates. Fix a Riemannian metric on B for the following distance comparisons.

Proposition 27.11.1 *Suppose that $Z = \{a = 0\}$ is a nonempty smooth compact real submanifold of positive codimension d, that $a \asymp \operatorname{dist}(\cdot, Z)^k$ in a fixed tube for a real number $k > 0$, and that a is bounded below by a positive constant outside that tube. Put $r = d/k$. Then, as $\epsilon \downarrow 0$,*

$$C(t-\epsilon) \asymp \begin{cases} \epsilon^{(1-r)/n-1}, & r < 1, \\ \epsilon^{-1}(\log(1/\epsilon))^{-(n+1)/n}, & r = 1, \\ \epsilon^{r-2}, & 1 < r < 2, \\ \log(1/\epsilon), & r = 2, \\ 1, & r > 2. \end{cases} \tag{27.48}$$

Each rate is integrable, and the endpoint volume is positive exactly when $r > 1$, that is, when $d > k$. For $r > 2$, the cost rate tends to

$$n^{-(n-1)/n} W^{1/n} H(0)^{-(n+1)/n} J(0) > 0, \qquad J(0) = \int_B \frac{\sigma}{a^2}.$$

For finitely many disjoint zero components satisfying these hypotheses with codimensions d_i and orders k_i, the same assertions hold with $r = \min_i d_i/k_i$.

Proof The positive density and the tubular Jacobians are bounded above and below by positive constants. For $p = 1, 2$, radial integration therefore gives

$$\int_B \frac{\sigma}{(a+\epsilon)^p} \asymp 1 + \int_0^R \frac{u^{d-1}}{(u^k+\epsilon)^p}\,\mathrm{d}u = 1 + \frac{\epsilon^{r-p}}{k} \int_0^{R^k/\epsilon} \frac{v^{r-1}}{(1+v)^p}\,\mathrm{d}v,$$

with the equality applying to the radial expression. For $r < p$ the last integral tends to a finite positive constant; for $r = p$ it grows logarithmically; for $r > p$ the unregularized moment is finite, and monotone convergence gives its positive limit. Applying this to H and J in (27.46) proves the five rates and the criterion $H(0) < \infty$ exactly when $r > 1$.

The power $(1-r)/n - 1$ exceeds -1 for $r < 1$, as does $r - 2$ for $1 < r < 2$. At $r = 1$, substitution of $\log(1/\epsilon)$ gives an integrable tail because $(n+1)/n > 1$. The logarithmic and bounded rates are also integrable. When $r > 2$, both inverse moments have finite positive limits, which gives the displayed limit. With finitely many disjoint components, the moments are finite sums of positive tube contributions and a bounded complement term; their smallest ratio determines the divergence, without cancellation. □

Proposition 27.11.2 *The endpoint θ_t is big exactly when a is positive everywhere. If a has a zero, there is no decomposition*

$$\theta_t + \mathrm{dd^c} f = \chi + \beta$$

with f smooth and real, χ a smooth closed real $(1,1)$-form with nef class and $\int_X \chi^n > 0$, and $\beta \geq 0$ a smooth $(1,1)$-form.

Proof Every Hermitian form on X is comparable to $\alpha + w$. Suppose therefore that $\theta_t + \mathrm{dd^c} u \geq \delta(\alpha + w)$ for a quasi-plurisubharmonic u and $\delta > 0$. Pull back an arbitrary smooth nonnegative top-degree form ζ on B. It is $\mathrm{dd^c}$-closed by dimension, so Stokes' theorem for distributions gives

$$0 \leq \int_X (\theta_t + \mathrm{dd^c} u - \delta(\alpha + w)) \wedge \zeta = \int_B (a - \delta)\zeta.$$

If a has a zero, a nonzero ζ supported where $a < \delta$ contradicts this inequality, so θ_t is not big. Conversely, if $a > 0$ everywhere, compactness makes θ_t itself Hermitian, so it is big.

Suppose that a has a zero and that such a decomposition exists. The manifold X is compact Kähler, and the closed nef class of χ has positive top self-intersection, so it is big by the criterion of Demailly and Păun used in the proof of Theorem 27.10.1. Hence there are a quasi-plurisubharmonic v and $\delta > 0$ with $\chi + \mathrm{dd^c} v \geq \delta(\alpha + w)$. Then

$$\theta_t + \mathrm{dd^c}(f + v) = \chi + \mathrm{dd^c} v + \beta \geq \delta(\alpha + w),$$

and $f + v$ is quasi-plurisubharmonic, contradicting the first part. □

In the closed case $a \equiv a_0 \geq 0$, put $P = \alpha + (a_0 + t)w$ and $Q = tw$. These closed nef forms satisfy

$$\int_X P^n - n \int_X P^{n-1} \wedge Q = nW a_0 = \mathcal{V}_\mu(\theta_t). \tag{27.49}$$

This is the lower bound in the transcendental Morse inequality for the difference of the nef classes of P and Q, discussed in Section 36.22; the equality follows from $w^2 = 0$, $\alpha^{m+1} = 0$ and $\int_C w = 1$. For nonconstant a, the exterior derivative $\mathrm{d}\theta_s = \mathrm{d}a \wedge w$ is not identically zero, and the endpoint value depends on the probability σ through $H(0)$. The comparison in (27.49) therefore concerns the constant closed case.

For example, take $m \geq 2$, $B = \mathbb{P}^m$, normalized Fubini–Study α, $\sigma = \alpha^m$, and $a([Z]) = \sum_{j=1}^m |Z_j|^2 / \sum_{j=0}^m |Z_j|^2$. The push-forward of σ under a is $m x^{m-1}\,\mathrm{d}x$ on $[0,1]$, by the polar integration formula (27.54). Hence $W = 1$, $H(0) = m/(m-1)$ and $\mathcal{V}_\mu(\theta_t) = (m+1)(m-1)/m > 0$, while the endpoint is not big and admits no closed nef core of positive top intersection. Its sole zero has real codimension $2m$ and order two, so $r = m$. The cost rate is therefore logarithmic when $m = 2$ and tends to a positive finite value when $m > 2$, by Proposition 27.11.1. Replacing a by $ma/(m-1)$ gives $H(0) = 1$ and recovers the value $\mathcal{V}_\mu = n$ of Theorem 27.13.1.

27.12 Volume and the bigness threshold

In this section X is again an arbitrary connected compact complex manifold of dimension $n \geq 1$. The root volume and the threshold of the Gauduchon criterion for bigness divide the same pairing by different quantities. Positive volume can therefore fail to imply bigness only if these denominators separate along tests approaching the threshold, and the following proposition isolates this degeneration. For the radial background of the next section it occurs, by Proposition 27.13.2.

Fix a Hermitian form h on X and a smooth strictly positive probability form μ, and let θ be a smooth pseudoeffective real $(1,1)$-form. For a Gauduchon metric γ, abbreviate the pairings by

$$A_\theta(\gamma) = \int_X \theta \wedge \gamma^{n-1}, \qquad A_h(\gamma) = \int_X h \wedge \gamma^{n-1}.$$

The threshold $\kappa_h(\theta)$ of Corollary 4.3.1 is the infimum of the quotient A_θ/A_h over Gauduchon metrics, and it is positive exactly when θ is big; we call it the *bigness threshold*. The root volume $\lambda_\mu(\theta)$ is the infimum of A_θ/D_μ, with the determinant denominator $D_\mu(\gamma)$ of Definition 10.8.1.

Proposition 27.12.1 *Let h, θ and μ be as above, and let (γ_j) be a sequence of Gauduchon metrics.*

(1) *Suppose $n \geq 2$, $\lambda_\mu(\theta) > 0$, and θ is not big. If*

$$A_h(\gamma_j) = 1, \qquad A_\theta(\gamma_j) \longrightarrow 0,$$

then $D_\mu(\gamma_j) \to 0$. Such sequences exist.

(2) *More generally, fix a real number $\delta > 0$. If $A_h(\gamma_j) = 1$ and*

$$A_\theta(\gamma_j) \longrightarrow \kappa_h(\theta), \qquad \liminf_j D_\mu(\gamma_j) \geq \delta > 0,$$

then

$$\kappa_h(\theta) \geq \delta \lambda_\mu(\theta).$$

In particular, positive root volume then implies bigness.

Proof Suppose first that θ is not big. Since θ is pseudoeffective, the threshold criterion of Corollary 4.3.1 gives $\kappa_h(\theta) = 0$, so there are Gauduchon metrics whose quotients A_θ/A_h tend to zero. Multiplying a Gauduchon metric by a positive constant multiplies A_θ, A_h and D_μ by the same factor; for $n \geq 2$ this factor can be chosen to make $A_h = 1$, without changing either quotient. This gives the sequences in the first assertion. By the definition of the root volume, every Gauduchon metric satisfies

$$\lambda_\mu(\theta) D_\mu(\gamma) \leq A_\theta(\gamma). \tag{27.50}$$

Dividing by the positive root volume proves the first assertion. Applying the same inequality to the sequence in the second assertion and taking lower limits gives

$\delta\lambda_\mu(\theta) \le \kappa_h(\theta)$. If the root volume is positive, then $\kappa_h(\theta) > 0$, and the threshold criterion gives bigness. □

The quotient D_μ/A_h is always bounded above: (27.50) with $\theta = h$ gives

$$D_\mu(\gamma) \le \lambda_\mu(h)^{-1} A_h(\gamma),$$

where $\lambda_\mu(h) > 0$ by Corollary 10.8.1.

27.13 Positive volume without bigness

By Corollary 10.8.1, every big background has positive volume. For a closed nef class on a compact Kähler manifold the converse also holds: the volume is then $\int_X \theta^n$, and a closed nef class with positive top self-intersection is big. The converse fails for nonclosed nef backgrounds, even on a projective manifold: such a background can have positive volume without containing a Kähler current. The background θ defined below is the endpoint of the example following Theorem 27.11.1, taken with $C = \mathbb{P}^1$ and with its coefficient multiplied by $m/(m-1)$. The coefficient of its fiber direction vanishes at one point of the base. A radial potential on the base compensates for this vanishing: it is bounded, and the determinant of the resulting current bounds every Gauduchon quotient from below.

Throughout this section fix an integer $m \ge 2$, and put $n = m+1$, $t = (m-1)/m$ and $B = \mathbb{P}^m$. Give B and $\mathbb{P}^1$ their Fubini–Study forms ω and w, normalized by $\int_B \omega^m = \int_{\mathbb{P}^1} w = 1$. On B define

$$x([Z_0 : \cdots : Z_m]) = \frac{\sum_{j=1}^m |Z_j|^2}{\sum_{j=0}^m |Z_j|^2}, \qquad a = x/t, \qquad p = [1 : 0 : \cdots : 0].$$

Thus x is a smooth function on B with values in $[0,1]$; it vanishes only at p and equals one on the hyperplane $\{Z_0 = 0\}$. On $X = B \times \mathbb{P}^1$, with pullbacks understood, set

$$h = \omega + w, \qquad \mu = \omega^m \wedge w, \qquad \theta = \omega + aw.$$

For $\epsilon > 0$ put

$$H_\epsilon = m \int_0^1 \frac{y^{m-1}}{y/t + \epsilon}\, \mathrm{d}y. \tag{27.51}$$

Lemma 27.13.1 *There are smooth real functions φ_ϵ on B such that $A_\epsilon = \omega + \mathrm{dd}^{\mathrm{c}}\varphi_\epsilon > 0$ and*

$$A_\epsilon^m = \frac{\omega^m}{H_\epsilon(a+\epsilon)}. \tag{27.52}$$

As $\epsilon \downarrow 0$ they converge uniformly to a bounded continuous function $\varphi \in \mathrm{PSH}(B,\omega)$ that is smooth on $B \setminus \{p\}$. On $B \setminus \{p\}$ the current $A = \omega + \mathrm{dd}^{\mathrm{c}}\varphi$ is a smooth strictly positive form satisfying

$$A^m = a^{-1}\omega^m. \tag{27.53}$$

Moreover $H_\epsilon \to 1$ as $\epsilon \downarrow 0$.

Proof In the affine chart $Z_0 = 1$, write $s = |z|^2$. Then $x = s/(1+s)$ and $\omega = \mathrm{dd^c}\log(1+s)$. Polar integration gives

$$\int_B b(x)\omega^m = m\int_0^1 b(y)y^{m-1}\,\mathrm{d}y \tag{27.54}$$

for every integrable function b on $[0,1]$, since the normalized radial density of ω^m in the variable s is $ms^{m-1}(1+s)^{-m-1}\,\mathrm{d}s$. In particular $\int_B a^{-1}\omega^m = mt/(m-1) = 1$, and monotone convergence in (27.51) gives $H_\epsilon \to 1$.

Define $q_\epsilon \geq 0$ and φ_ϵ on $[0,1]$ by

$$q_\epsilon(x)^m = \frac{m}{H_\epsilon}\int_0^x \frac{y^{m-1}}{y/t+\epsilon}\,\mathrm{d}y, \qquad \varphi_\epsilon(x) = \int_0^x \frac{q_\epsilon(y)-y}{y(1-y)}\,\mathrm{d}y.$$

Near zero,

$$q_\epsilon(x) = x\left(\frac{m}{H_\epsilon}\int_0^1 \frac{u^{m-1}}{xu/t+\epsilon}\,\mathrm{d}u\right)^{1/m},$$

so q_ϵ is x times a smooth positive function; moreover $q_\epsilon(1) = 1$. Hence both apparent singularities of φ'_ϵ, at 0 and at 1, are removable, and φ_ϵ is smooth on $[0,1]$. Composing with the smooth function x gives a smooth function on B, again denoted φ_ϵ.

To compute A_ϵ, recall that for a local potential of the form $F(s)$, with $q = sF'(s)$, the complex Hessian has the tangential eigenvalue q/s, of multiplicity $m-1$, and the radial eigenvalue $\mathrm{d}q/\mathrm{d}s$. For $F_\epsilon = \log(1+s) + \varphi_\epsilon(x)$ one finds $sF'_\epsilon = q_\epsilon(x)$. Differentiating the integral defining q_ϵ shows that both eigenvalues are positive, also at $s = 0$ by the smooth factorization above. Their product relative to the Fubini–Study determinant is

$$\frac{A_\epsilon^m}{\omega^m} = \frac{q_\epsilon^{m-1}q'_\epsilon}{x^{m-1}} = \frac{1}{H_\epsilon(a+\epsilon)}.$$

At a point of the hyperplane $\{Z_0 = 0\}$, the smooth form A_ϵ is semipositive, as a limit of the positive matrices in the affine chart. Since the determinant just computed extends there with a strictly positive value, every eigenvalue is positive. This proves strict positivity on all of B and (27.52).

We now pass to the limit. For $x > 0$, $q_\epsilon(x)$ tends to x^t. To bound q_ϵ uniformly, note that q_ϵ^m is the distribution function of the probability density $\frac{m}{H_\epsilon}\frac{y^{m-1}}{y/t+\epsilon}$ on $[0,1]$. Relative to the probability density my^{m-1}, this density has a decreasing weight; relative to $(m-1)y^{m-2}$, its weight is proportional to $y/(y+t\epsilon)$ and hence increasing. If one probability density on $[0,1]$ has a decreasing weight relative to another, its average weight on $[0,x]$ is at least one and its distribution function is the larger of the two; an increasing weight reverses the inequality. Consequently

$$x^m \leq q_\epsilon(x)^m \leq x^{m-1}, \qquad \text{that is,} \qquad x \leq q_\epsilon(x) \leq x^t.$$

The function

$$\varphi(x) = \int_0^x \frac{y^t - y}{y(1-y)}\,\mathrm{d}y$$

is bounded and continuous on $[0,1]$, since $t > 0$, and smooth on $(0,1]$; its derivative extends smoothly at one. The integrable function $(y^t - y)/[y(1-y)]$ dominates the derivatives of the functions φ_ϵ, so dominated convergence gives uniform convergence $\varphi_\epsilon \to \varphi$ on B. Distributional convergence shows that A is a closed positive current. On each coordinate chart, a smooth local potential of ω plus φ is a uniform limit of the corresponding plurisubharmonic functions built with φ_ϵ, including at p. Hence φ is ω-plurisubharmonic. Off p, the limit $q = x^t$ gives

$$\frac{A^m}{\omega^m} = \frac{(x^t)^{m-1} t x^{t-1}}{x^{m-1}} = \frac{t}{x} = \frac{1}{a}.$$

Smoothness of φ off p and the same determinant argument at the hyperplane $\{Z_0 = 0\}$ prove (27.53). □

Theorem 27.13.1 *The background θ on X is smooth, semipositive, nef and nonclosed. It is not big, but*

$$\lambda_\mu(\theta) = n^{1/n}, \qquad \mathcal{V}_\mu(\theta) = n. \tag{27.55}$$

For $\theta_\epsilon = \omega + (a+\epsilon)w$, one has

$$\mathcal{V}_\mu(\theta_\epsilon) = \frac{n}{H_\epsilon},$$

and the smooth positive metrics

$$\gamma_\epsilon = (a+\epsilon)^{-1/m}\big(A_\epsilon + (a+\epsilon)w\big) \tag{27.56}$$

are balanced and minimize the defining quotient at θ_ϵ.

Proof The coefficient a is nonnegative, and $\theta + \eta h > 0$ for every $\eta > 0$; this proves semipositivity and nefness. Since a is nonconstant, $\mathrm{d}\theta = \mathrm{d}a \wedge w \neq 0$. Since $a(p) = 0$, Proposition 27.11.2, applied with $(B, \alpha) = (\mathbb{P}^m, \omega)$, $C = \mathbb{P}^1$ and $\sigma = \omega^m$, shows that θ is not big.

For the lower bound on the volume, pull back the bounded potential φ of Lemma 27.13.1. The positive current $\Omega = A + aw = \theta + \mathrm{dd}^{\mathrm{c}}\varphi$ is a smooth strictly positive form off $Z = \{p\} \times \mathbb{P}^1$, where

$$\Omega^n = naA^m \wedge w = n\mu.$$

Let γ be a Gauduchon metric. Since φ is bounded, Stokes' theorem for currents lets us replace θ by Ω in the numerator of its quotient. The set Z has zero smooth volume and $\Omega \wedge \gamma^{n-1}$ is a positive measure, so the pointwise arithmetic–geometric mean inequality off Z gives

$$\int_X \theta \wedge \gamma^{n-1} \geq n^{1/n} \int_X (\mu/\gamma^n)^{1/n} \gamma^n = n^{1/n} D_\mu(\gamma).$$

This lower bound holds for every test in Definition 10.8.1; no power of the current Ω along Z is needed.

For the upper bound, note that θ_ϵ is the background $\theta_{1,1,\epsilon}$ of Theorem 27.4.2, with $\kappa = \omega$, $C = \mathbb{P}^1$, $\sigma = \omega^m$ and the coefficient a, which is nonconstant with $\min a = 0$. By (27.54), the integral $H(1,\epsilon)$ there is $\int_B \omega^m/(a+\epsilon) = H_\epsilon$, so the form A_ϵ of Lemma 27.13.1 satisfies the conditions imposed on A there. That theorem therefore gives $\mathcal{V}_\mu(\theta_\epsilon) = n/H_\epsilon$ and shows that γ_ϵ is balanced and minimizes the quotient at θ_ϵ. Since $\theta \le \theta_\epsilon$, monotonicity gives $\lambda_\mu(\theta) \le (n/H_\epsilon)^{1/n}$, and letting $\epsilon \downarrow 0$ proves (27.55). By Theorem 27.9.1, this value is also the limit of the volumes of the Hermitian perturbations $\theta + \epsilon h$. □

The value n refers to the probability measure μ fixed above, but positivity holds for every smooth positive probability measure by Proposition 10.8.2. The construction needs $m \ge 2$, so that $t > 0$ and the endpoint potential is bounded; it says nothing about surfaces, or about closed nef classes on general compact complex manifolds.

27.13.1 Volume tests and threshold tests

The volume quotient and the bigness threshold $\kappa_h(\theta)$ of Section 27.12 divide the same pairing $\int_X \theta \wedge \gamma^{n-1}$ by the determinant integral $D_\mu(\gamma)$ and by the trace $\int_X h \wedge \gamma^{n-1}$, respectively. For the pseudoeffective but nonbig background θ of Theorem 27.13.1, the threshold vanishes. The same metrics can approach both infima, but need not do so. In the radial model, the volume-minimizing metrics approach the bigness threshold only in dimension three, while a second family of smooth balanced metrics approaches it in every dimension by concentrating on the fiber over p.

Keep the data of Theorem 27.13.1, and put

$$T_\epsilon = \gamma_\epsilon^m, \qquad I_\epsilon = \int_B (a+\epsilon)^{-2}\omega^m, \qquad M_\epsilon = m + I_\epsilon/H_\epsilon, \qquad g_\epsilon = M_\epsilon^{-1/m}\gamma_\epsilon.$$

Write $Z = \{p\} \times \mathbb{P}^1$. Its integration current $[Z]$ satisfies $\int_Z h = 1$ and annihilates θ, since $\theta|_Z = a(p)w = 0$.

Proposition 27.13.1 *The metrics g_ϵ are balanced, satisfy $\int_X h \wedge g_\epsilon^m = 1$, and*

$$\int_X \theta \wedge g_\epsilon^m = \frac{n - \epsilon I_\epsilon / H_\epsilon}{M_\epsilon}, \qquad D_\mu(g_\epsilon) = \frac{(n/H_\epsilon)^{m/n} H_\epsilon}{M_\epsilon}. \tag{27.57}$$

Their volume quotients tend to $n^{1/n}$ as $\epsilon \downarrow 0$. If $n = 3$, then g_ϵ^2 converges weakly to $[Z]$, and

$$\int_X \theta \wedge g_\epsilon^2 \sim \frac{6}{\log(1/\epsilon)}, \qquad D_\mu(g_\epsilon) \sim \frac{2\,3^{2/3}}{\log(1/\epsilon)}. \tag{27.58}$$

If $n \geq 4$, put $M_0 = m + (m-1)^2/[m(m-2)]$. Then

$$\int_X \theta \wedge g_\epsilon^m \longrightarrow \frac{n}{M_0} > 0, \qquad D_\mu(g_\epsilon) \longrightarrow \frac{n^{m/n}}{M_0} > 0.$$

Thus this family approaches the bigness threshold, which is zero, exactly when $n = 3$.

Proof By (27.17) and (27.52),

$$T_\epsilon = \frac{\omega^m}{H_\epsilon (a+\epsilon)^2} + m A_\epsilon^{m-1} \wedge w.$$

Since the closed forms A_ϵ and ω are cohomologous, Stokes' theorem gives $\int_B \omega \wedge A_\epsilon^{m-1} = 1$. Consequently the two terms have h-traces I_ϵ / H_ϵ and m, which proves the trace normalization of $g_\epsilon^m = T_\epsilon / M_\epsilon$. Subtracting ϵw from θ_ϵ subtracts $\epsilon I_\epsilon / H_\epsilon$ from the pairing $\int_X \theta_\epsilon \wedge T_\epsilon = n$, which is the numerator computed in the proof of Theorem 27.4.2. Rescaling γ_ϵ by $M_\epsilon^{-1/m}$ divides both the linear pairing and D_μ by M_ϵ, proving (27.57).

Polar integration gives

$$I_\epsilon = m t^2 \int_0^1 \frac{x^{m-1}}{(x + t\epsilon)^2}\, \mathrm{d}x.$$

When $m = 2$, put $q = \epsilon/2$. Direct integration yields

$$I_\epsilon = \tfrac{1}{2}\left(\log \frac{1+q}{q} + \frac{q}{1+q} - 1 \right), \qquad H_\epsilon = 1 - q \log \frac{1+q}{q}.$$

Hence $I_\epsilon \sim \tfrac{1}{2}\log(1/\epsilon)$ and $\epsilon I_\epsilon \to 0$. Since $H_\epsilon \to 1$, the asymptotics in (27.58) follow from (27.57).

For the weak limit when $n = 3$, note that the mixed positive term $2A_\epsilon \wedge w$ has h-trace two; divided by $M_\epsilon \to \infty$, it tends weakly to zero. The horizontal probability measure

$$I_\epsilon^{-1} (a+\epsilon)^{-2} \omega^2$$

on B concentrates at p: outside any fixed neighborhood of p its unnormalized density stays bounded, while $I_\epsilon \to \infty$. For a smooth $(1,1)$-test form on X, integration along the fibers $\{b\} \times \mathbb{P}^1$ gives a smooth function on B. The convergence of these probability

measures to the Dirac mass at p therefore shows that the horizontal term of g_ϵ^2 converges to $[Z]$, because its coefficient $I_\epsilon/(H_\epsilon M_\epsilon)$ tends to one.

For $m \geq 3$, monotone convergence instead gives

$$I_\epsilon \longrightarrow mt^2 \int_0^1 x^{m-3}\,\mathrm{d}x = \frac{(m-1)^2}{m(m-2)}.$$

In both cases $\epsilon I_\epsilon \to 0$. Using $H_\epsilon \to 1$ in (27.57) proves the remaining limits, including that of the volume quotients. □

For $n \geq 4$ the metrics g_ϵ still minimize the perturbed volumes, and their quotients converge to $\lambda_\mu(\theta)$, but their normalized pairings with θ stay bounded away from zero. This does not contradict nonbigness: the bigness threshold is approached by a different family.

Proposition 27.13.2 *There are smooth balanced metrics b_ρ, $0 < \rho < 1$, with*

$$\int_X h \wedge b_\rho^m = 1, \qquad b_\rho^m \rightharpoonup [Z], \qquad \int_X \theta \wedge b_\rho^m \longrightarrow 0, \qquad D_\mu(b_\rho) \longrightarrow 0$$

as $\rho \downarrow 0$.

Proof Choose a nonzero smooth function $f_\rho \geq 0$ on B, supported in $\{x < \rho\}$ and normalized by $\int_B f_\rho \omega^m = 1$. The form

$$\Gamma_\rho = \frac{f_\rho \omega^m + \rho h^m}{1 + n\rho} \tag{27.59}$$

is smooth, strictly positive and closed: the first summand has top degree on the base, and h is Kähler. The positive-root construction preceding the proof of Theorem 4.3.1 gives a unique smooth metric b_ρ with $b_\rho^m = \Gamma_\rho$, and b_ρ is balanced because Γ_ρ is closed. Since $\int_X h^{m+1} = n$, its h-trace is one.

Put $C_\theta = \int_X \theta \wedge h^m$. Since $a < \rho/t$ on the support of f_ρ,

$$0 \leq \int_X \theta \wedge \Gamma_\rho \leq \frac{\rho/t + \rho C_\theta}{1 + n\rho} \longrightarrow 0.$$

Since $\lambda_\mu(\theta) = n^{1/n}$ by (27.55), testing the infimum with b_ρ gives $D_\mu(b_\rho) \leq n^{-1/n} \int_X \theta \wedge \Gamma_\rho$, so the determinant integrals also tend to zero. The probability measures $f_\rho \omega^m$ converge to the Dirac mass at p because their supports shrink to that point. Fiber integration, as in the proof of Proposition 27.13.1, gives the current limit $[Z]$, while the term ρh^m tends to zero. □

Thus the normalized current $[Z]$ is a weak limit of smooth balanced metrics normalized by their h-traces. It annihilates θ, and its determinant measure vanishes by Proposition 27.20.1. The two families give explicit instances of the separation between the volume and threshold denominators described in Proposition 27.12.1, without any general smoothing result for Gauduchon currents.

27.14 The lower-mass quotient

From now on X is again an arbitrary connected compact complex manifold of dimension $n \geq 1$, and ω is a Hermitian form on X. By Theorem 10.8.2, the lower mass $v_-(\omega)$ is the infimum of the volumes $\mathcal{V}_\mu(\omega)$ over the set $\mathscr{M}$ of smooth strictly positive probability measures, and (10.12) expresses it through Gauduchon metrics alone. We relate attainment of this infimum to smooth minimizers of the mass, and then extend the quotient to nef backgrounds.

Proposition 27.14.1 *The infimum in* (10.12) *is attained if and only if the infimum of the masses* $\int_X (\omega + \mathrm{dd^c} v)^n$ *over smooth functions* v *with* $\omega + \mathrm{dd^c} v > 0$ *is attained. For every such minimizing potential* v*, the metric* $\omega + \mathrm{dd^c} v$ *is Gauduchon.*

Proof Suppose that a Gauduchon metric γ attains the infimum in (10.12), and put $m = v_-(\omega)^{1/n}$. For the measure $\mu_\gamma = \gamma^n / \int_X \gamma^n$, equality holds in the Hölder inequality of the proof of Theorem 10.8.2, so the test γ gives $\lambda_{\mu_\gamma}(\omega) \leq m$. Theorem 10.8.2 gives the opposite inequality. Hence the solution of the smooth equation for μ_γ has mass $m^n = v_-(\omega)$, which gives a minimizing smooth potential. For $n \geq 2$, γ also minimizes the quotient for μ_γ, so by Theorem 10.8.1 the solution form is conformal to γ. Since both have volume forms proportional to μ_γ, the conformal factor is constant, and the solution metric is Gauduchon.

Conversely, let v be a smooth potential minimizing the mass. For every smooth real f, the potentials $v \pm t f$ are admissible for sufficiently small $t > 0$. The vanishing of the first variation of the mass therefore gives

$$\int_X f \, \mathrm{dd^c}\big((\omega + \mathrm{dd^c} v)^{n-1}\big) = 0 \quad \text{for every smooth real } f.$$

Thus $\Omega = \omega + \mathrm{dd^c} v$ is Gauduchon. Taking $\gamma = \Omega$ in the quotient and using Stokes' theorem for the Gauduchon metric Ω gives

$$\frac{\int_X \omega \wedge \Omega^{n-1}}{\big(\int_X \Omega^n\big)^{(n-1)/n}} = \Big(\int_X \Omega^n\Big)^{1/n} = v_-(\omega)^{1/n},$$

so the infimum in (10.12) is attained. In dimension one all metrics are Gauduchon and all masses equal $\int_X \omega$. □

The infimum over densities also makes sense beyond the positive cone. For a pseudoeffective smooth real $(1,1)$-form θ on X, put

$$\lambda_*(\theta) = \inf_{\gamma \text{ Gauduchon}} \frac{\int_X \theta \wedge \gamma^{n-1}}{\big(\int_X \gamma^n\big)^{(n-1)/n}}.$$

The Hölder argument in the proof of Theorem 10.8.2 proves

$$\lambda_*(\theta) = \inf_{\mu \in \mathscr{M}} \lambda_\mu(\theta).$$

Like each λ_μ, the function λ_* is finite, nonnegative, concave, positively homogeneous, and invariant under smooth dd^{c} changes of θ. Suppose θ is nef, fix a Hermitian form h, and for real $\epsilon > 0$ choose a Hermitian representative ω_ϵ of $\theta + \epsilon h$. Interchanging the infima over the test metric and over $\epsilon > 0$, as in Theorem 27.9.1, gives

$$\lambda_*(\theta) = \lim_{\epsilon \downarrow 0} \lambda_*(\theta + \epsilon h) = \lim_{\epsilon \downarrow 0} v_-(\omega_\epsilon)^{1/n}.$$

By invariance under smooth dd^{c} changes, the last expression does not depend on the choice of the representatives ω_ϵ. This extends the lower-mass formula to the nef boundary. We do not compare λ_* with other notions of lower volume for singular classes.

27.15 Curvature conditions for mass extrema

We now ask when the lower and upper masses are attained by smooth potentials. At a smooth local extremum of the mass, the first variation forces the metric $\Omega = \omega + \mathrm{dd}^{\mathrm{c}} v$ to be Gauduchon, and the second variation gives a sign for $\mathrm{dd}^{\mathrm{c}}(\Omega^{n-2})$. On a threefold these necessary conditions are also sufficient, because the mass is then a polynomial of degree two in the variation. The last subsection applies this criterion to invariant backgrounds on compact quotients of three-dimensional complex Lie groups, where it gives the upper mass. Fix a Hermitian form ω, and recall from Chapter 2 the set $\mathcal{H}_\omega$ of smooth real functions u with $\omega + \mathrm{dd}^{\mathrm{c}} u > 0$ and the mass functional $\mathrm{M}_\omega(u) = \int_X (\omega + \mathrm{dd}^{\mathrm{c}} u)^n$.

Recall that a smooth real $(n-1, n-1)$-form T is nonnegative, written $T \geq 0$, if $\mathrm{i}\alpha \wedge \bar{\alpha} \wedge T \geq 0$ for every $(1,0)$-covector α; in this bidegree this is the usual notion of positivity. The second variation of the mass produces integrals of the form $\int_X \mathrm{d}f \wedge \mathrm{d}^{\mathrm{c}} f \wedge T$, and the following elementary lemma shows that their signs detect the sign of T.

Lemma 27.15.1 *Suppose a smooth real $(n-1, n-1)$-form T satisfies*

$$\int_X \mathrm{d}f \wedge \mathrm{d}^{\mathrm{c}} f \wedge T \geq 0 \quad \textit{for every smooth real } f.$$

Then $T \geq 0$.

Proof Suppose instead that $\mathrm{i}\alpha \wedge \bar{\alpha} \wedge T < 0$ at a point x_0 for some $(1,0)$-covector α. Every $(1,0)$-covector is the $(1,0)$-part of a real covector, so we can choose a real linear function ℓ in local real coordinates around x_0 whose differential has $(1,0)$-part α at x_0. Then $\mathrm{d}\ell \wedge \mathrm{d}^{\mathrm{c}}\ell$ is a positive multiple of $\mathrm{i}\alpha \wedge \bar{\alpha}$ at x_0, and after shrinking the coordinate ball, $\mathrm{d}\ell \wedge \mathrm{d}^{\mathrm{c}}\ell \wedge T$ is strictly negative on it. Choose a nonzero smooth cutoff function χ supported in this ball and, for positive integers k, set

$$f_k = k^{-1} \chi \sin(k\ell),$$

extended by zero to X. Since $\mathrm{d}\ell$ is a constant nonzero covector in these coordinates,

$$\mathrm{d}f_k \wedge \mathrm{d}^{\mathrm{c}} f_k = \chi^2 \cos^2(k\ell)\, \mathrm{d}\ell \wedge \mathrm{d}^{\mathrm{c}}\ell + O(k^{-1}),$$

where k times the remainder has uniformly bounded coefficients. In $\cos^2(k\ell) = (1+\cos(2k\ell))/2$, the oscillatory part contributes integrals tending to zero, as one sees by integrating by parts in a coordinate direction in which ℓ has nonzero derivative. Hence

$$\lim_{k\to\infty} \int_X \mathrm{d}f_k \wedge \mathrm{d}^{\mathrm{c}} f_k \wedge T = \frac{1}{2} \int_X \chi^2 \mathrm{d}\ell \wedge \mathrm{d}^{\mathrm{c}}\ell \wedge T < 0,$$

which contradicts the hypothesis. □

Proposition 27.15.1 *Let $n \geq 2$, and let $v \in \mathcal{H}_\omega$ be a local minimum of M_ω for the C^2 topology. Then $\Omega = \omega + \mathrm{dd}^{\mathrm{c}} v$ satisfies*

$$\mathrm{dd}^{\mathrm{c}}(\Omega^{n-1}) = 0, \qquad \mathrm{dd}^{\mathrm{c}}(\Omega^{n-2}) \leq 0.$$

At a local maximum the first condition still holds and the second inequality is reversed.

Proof For every smooth real f, the potentials $v + tf$ lie in $\mathcal{H}_\omega$ for all sufficiently small $|t|$. The vanishing of the first variation therefore gives the Gauduchon condition. The second variation is

$$\begin{aligned} \left.\frac{\mathrm{d}^2}{\mathrm{d}t^2}\right|_{t=0} \mathrm{M}_\omega(v+tf) &= n(n-1) \int_X (\mathrm{dd}^{\mathrm{c}} f)^2 \wedge \Omega^{n-2} \\ &= -n(n-1) \int_X \mathrm{d}f \wedge \mathrm{d}^{\mathrm{c}} f \wedge \mathrm{dd}^{\mathrm{c}}(\Omega^{n-2}), \end{aligned}$$

where the last equality follows by integrating by parts twice, using that $\mathrm{dd}^{\mathrm{c}}(\Omega^{n-2})$ is both d-closed and d^{c}-closed. At a local minimum this second variation is nonnegative for every f, and Lemma 27.15.1, applied to $-\mathrm{dd}^{\mathrm{c}}(\Omega^{n-2})$, gives $\mathrm{dd}^{\mathrm{c}}(\Omega^{n-2}) \leq 0$. The case of a maximum is identical, with the opposite sign. □

In dimension three these necessary conditions are also sufficient, and every local extremum is global.

Theorem 27.15.1 *Assume $n = 3$ and $v \in \mathcal{H}_\omega$. The following are equivalent:*

(1) *v is a local minimum of M_ω;*
(2) *v is a global minimum of M_ω on $\mathcal{H}_\omega$;*
(3) *$\Omega = \omega + \mathrm{dd^c} v$ is Gauduchon and $\mathrm{dd^c}\omega \leq 0$.*

In this case $v_-(\omega) = \int_X \Omega^3$, and the infimum in (10.12) is attained. The corresponding statements for maxima hold with $\mathrm{dd^c}\omega \geq 0$, and then $v_+(\omega) = \int_X \Omega^3 < +\infty$.

Proof Since $\mathrm{dd^c}\Omega = \mathrm{dd^c}\omega$, the preceding proposition shows that the first condition implies the third. Conversely, if Ω is Gauduchon, expanding the cube and integrating by parts gives, for every smooth real f,

$$\int_X (\Omega + \mathrm{dd^c} f)^3 - \int_X \Omega^3 = -3 \int_X \mathrm{d}f \wedge \mathrm{d^c} f \wedge \mathrm{dd^c}\omega. \tag{27.60}$$

The form $\mathrm{d}f \wedge \mathrm{d^c} f$ is nonnegative, so the right side is nonnegative if $\mathrm{dd^c}\omega \leq 0$; then v is a global minimum, and in particular a local one. By Theorem 10.8.2, regularization identifies the infimum of the masses of smooth strictly ω-plurisubharmonic potentials with $v_-(\omega)$, and Proposition 27.14.1 proves attainment in (10.12). Reversing the sign in the same identity proves the statements for maxima, including the equality of $\int_X \Omega^3$ with the supremum over bounded potentials. □

These conditions characterize smooth minimizers when they exist; neither argument produces a Gauduchon metric of the form $\omega + \mathrm{dd^c} v$ in an arbitrary background.

For the Hopf metric ω and the pulled-back Fubini–Study form α of Chapter 9, one has $\mathrm{dd^c}\omega = -\alpha^2 \leq 0$, and ω is Gauduchon. Thus the minimum of the mass is attained by the constant potential and equals 6, while the upper mass is infinite. On a surface, by contrast, the mass $\int_X \omega^2 + 2\int_X v\,\mathrm{dd^c}\omega$ is affine in the potential, so a local extremum exists precisely when ω is Gauduchon, and then all masses are equal. In dimension $n \geq 4$ the sign condition of Proposition 27.15.1 does not control the terms of order three and higher in the mass polynomial, and the argument of Theorem 27.15.1 does not show that the conditions are sufficient.

27.15.1 Invariant backgrounds in dimension three

In dimension three, an invariant positive background has a nonnegative complex Hessian. Together with Theorem 27.15.1, this determines its upper Monge–Ampère mass and shows that the singular semipositive coefficient limits have zero top mass. On the Iwasawa quotient, the same calculation identifies exactly which smooth reference probabilities keep the determinant formula for the volume.

We use the compact quotients $X = \Gamma\backslash G$ of complex Lie groups, the invariant coframe, h_0, H and θ_A of Section 27.4.2, and assume $n = 3$. Write $\mu_0 = h_0^3/H$ to distinguish this invariant probability from other reference measures, and let $\mathscr{M}$ be the set of smooth strictly positive probability forms on X. Recall that v_+ denotes the upper Monge–Ampère mass of Definition 7.5.1.

Proposition 27.15.2 *For $A > 0$ on any such compact complex Lie-group quotient,*

$$\nu_+(\theta_A) = H \det A = \max_{\nu \in \mathscr{M}} \mathcal{V}_\nu(\theta_A). \tag{27.61}$$

The invariant probability μ_0 attains this maximum. In particular the quotient has the bounded mass property.

Proof Since the forms $\mathrm{d}\alpha_j$ are holomorphic, we have

$$\mathrm{dd^c}\theta_A = \frac{1}{4\pi^2} \sum_{j,k} A_{jk}\, \mathrm{d}\alpha_j \wedge \mathrm{d}\bar{\alpha}_k \geq 0. \tag{27.62}$$

Indeed $\bar{\partial}(\alpha_j \wedge \bar{\alpha}_k) = -\alpha_j \wedge \mathrm{d}\bar{\alpha}_k$, and applying ∂ and multiplying by $(\mathrm{i}/(2\pi))^2$ gives the identity in (27.62). Diagonalizing the positive coefficient matrix expresses the result as a sum of nonnegative multiples of $\beta \wedge \bar{\beta}$ for invariant $(2,0)$-forms β. In dimension three these are nonnegative $(2,2)$-forms: the wedge product with $\mathrm{i}\xi \wedge \bar{\xi}$, for a $(1,0)$-form ξ, has the squared modulus of $\xi \wedge \beta$ as coefficient.

By Proposition 27.4.2, θ_A is balanced, hence Gauduchon. The maximum case of Theorem 27.15.1 therefore gives $\nu_+(\theta_A) = \int_X \theta_A^3 = H \det A$. The supremum of $\mathcal{V}_\nu(\theta_A)$ over $\nu \in \mathscr{M}$ equals this upper mass by Theorem 10.8.2, and (27.4) shows that it is attained at μ_0. Finally, Proposition 7.5.1 makes the finiteness of the upper mass a property of the manifold. □

Corollary 27.15.1 *If $A \geq 0$ is singular, then every bounded $u \in \mathrm{PSH}(X, \theta_A)$ satisfies*

$$\int_X (\theta_A + \mathrm{dd^c}u)^3 = 0.$$

For every $u \in \mathrm{PSH}(X, \theta_A)$, bounded or not, the non-pluripolar top power of $\theta_A + \mathrm{dd^c}u$ is also zero.

Proof For $\varepsilon > 0$, the same bounded u is $\theta_{A+\varepsilon I}$-plurisubharmonic, so the preceding proposition bounds its mass by $H \det(A + \varepsilon I)$, which tends to zero with ε. By the multilinear construction of bounded products in Proposition 1.7.1,

$$(\theta_{A+\varepsilon I} + \mathrm{dd^c}u)^3 = \left(\theta_A + \mathrm{dd^c}u + \varepsilon h_0\right)^3$$

is a polynomial in ε whose coefficients are mixed products of finite mass. Its total mass therefore converges to $\int_X (\theta_A + \mathrm{dd^c}u)^3$ as $\varepsilon \downarrow 0$, so this mass is zero, and positivity makes the measure zero.

For unbounded u, we subtract a constant so that $u \leq 0$. Since $\theta_A \geq 0$, the truncations $u_k = \max(u, -k)$ are bounded and θ_A-plurisubharmonic, so their top products vanish by the bounded case. Restricting them to $\{u > -k\}$ and passing to the limit that defines the non-pluripolar product gives zero as well. □

We now specialize to the Iwasawa quotient of Example 27.4.2. Put

$$E = \mathbb{C}/\mathbb{Z}[\mathrm{i}], \qquad B = E^2, \qquad p\colon X \longrightarrow B, \quad [z] \longmapsto ([z_1], [z_2]).$$

This is a holomorphic submersion with connected elliptic fibers, and the coordinate description of the fundamental cube gives $p_*\mu_0 = \mu_B$, where μ_B is the Haar probability on B. A smooth probability is called *fiber invariant* if it has the form

$$\mu_f = (f \circ p)\mu_0, \qquad f \in C^\infty(B), \quad f > 0, \quad \int_B f\,\mu_B = 1.$$

Thus a smooth probability is fiber invariant when its density relative to μ_0 is constant along the elliptic fibers of p; no invariance in the base directions is required.

Theorem 27.15.2 *For every $A > 0$ and fiber-invariant μ_f,*

$$\mathcal{V}_{\mu_f}(\theta_A) = H \det A, \qquad H = \frac{6}{\pi^3}. \tag{27.63}$$

The unique minimizing Gauduchon metric of θ_A normalized by $D_{\mu_f} = 1$ is

$$\gamma_{A,f} = \frac{\theta_A + \mathrm{dd^c}(p^* v_{A,f})}{(H \det A)^{1/3}}, \qquad \gamma_{A,f}^3 = \mu_f, \tag{27.64}$$

where $v_{A,f}$ is the solution of the Monge–Ampère equation (27.65) on the torus B, with a closed background, constructed in the proof. Conversely, for a fixed $A > 0$, the probabilities $\nu \in \mathscr{M}$ maximizing $\mathcal{V}_\nu(\theta_A)$ are exactly the fiber-invariant ones.

For each fixed real linear $W \subset \mathrm{Herm}(3)$ containing a positive matrix, the Hessian metric for μ_f is

$$g_A(U, Z) = \frac{1}{3}\operatorname{tr}(A^{-1}UA^{-1}Z).$$

Its unit-volume hypersurface $S_W = \{A \in W : A > 0,\ H \det A = 1\}$ is complete and proper, and the Cauchy criterion and the distance bounds of Theorem 27.4.1 hold with $n = 3$.

Proof The map p is well defined because multiplication by lattice elements adds Gaussian integers to the first two coordinates. It is a submersion, and fixing representatives of those coordinates identifies each fiber with the central quotient E. Integrating first in the third coordinate over the fundamental cube proves the pushforward identity.

The closed equation on the base. Write the positive Hermitian matrix in block form and take its Schur complement:

$$A = \begin{pmatrix} C & b \\ b^* & a \end{pmatrix},$$
$$a = A_{33} > 0, \qquad S = C - a^{-1}bb^* > 0.$$

On B, let

$$\omega_S = \frac{\mathrm{i}}{2\pi}\sum_{j,k=1}^{2} S_{jk}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k, \qquad Q_B = \int_B \omega_S^2 = \frac{2}{\pi^2}\det S.$$

We apply Theorem 10.1.1 to this closed background with the density $Q_B f \mu_B$. Since ω_S is closed and the density has total mass Q_B, Stokes' theorem forces the constant in the equation to be one. We obtain a smooth real $v = v_{A,f}$, unique up to an additive constant, such that

$$(\omega_S + \mathrm{dd^c} v)^2 = Q_B f \mu_B, \qquad \omega_S + \mathrm{dd^c} v > 0. \tag{27.65}$$

If K is the coefficient matrix of $\mathrm{dd^c} v$ in the base coframe, this says $\det(S + K) = (\det S) f$. The form

$$\Omega = \theta_A + \mathrm{dd^c}(p^* v)$$

has coefficient matrix $\left(\begin{smallmatrix} C+K & b \\ b^* & a \end{smallmatrix}\right)$ in the invariant coframe. Since $a > 0$ and the Schur complement $S+K$ is positive, $\Omega > 0$; its determinant is $a \det(S+K) = (\det A)(f \circ p)$. Consequently $\Omega^3 = (H \det A)\mu_f$.

This solution form is also balanced. Put $\beta = \mathrm{dd^c}(p^* v)$, which is closed and uses only α_1, α_2 and their conjugates. Since $\mathrm{d}\alpha_3 = -\alpha_1 \wedge \alpha_2$, every term of $\partial\theta_A$ contains $\alpha_1 \wedge \alpha_2$, and every conjugate term contains $\bar\alpha_1 \wedge \bar\alpha_2$. Thus $\mathrm{d}\theta_A \wedge \beta = 0$. Together with $\mathrm{d}\theta_A^2 = 0$, this proves

$$\mathrm{d}\Omega^2 = \mathrm{d}\theta_A^2 + 2\mathrm{d}\theta_A \wedge \beta = 0.$$

Theorem 10.8.1 now gives (27.63). Its characterization of the minimizers, the balancedness of Ω and the homogeneity of degree two of D_{μ_f} give the unique normalized minimizer (27.64).

All maximizing densities. Let $Z_3 = \partial/\partial z_3$ be dual to α_3, and set $\tau_j = (\mathrm{i}/(2\pi))\alpha_j \wedge \bar\alpha_j$. Directly from (27.62),

$$\mathrm{dd^c}\theta_A = A_{33}\tau_1 \wedge \tau_2, \qquad h_0^3 = 6\tau_1 \wedge \tau_2 \wedge \tau_3.$$

Since θ_A is Gauduchon, the identity (27.60) therefore becomes, for every smooth real u,

$$\int_X (\theta_A + \mathrm{dd^c} u)^3 = H \det A - \frac{A_{33} H}{2} \int_X |Z_3 u|^2 \mu_0. \tag{27.66}$$

Here $\mathrm{d}u \wedge \mathrm{d^c}u = (\mathrm{i}/(2\pi))\partial u \wedge \bar\partial u$, and wedging with $\tau_1 \wedge \tau_2$ selects its coefficient on $\alpha_3 \wedge \bar\alpha_3$; this also accounts for the factor $1/2$.

For an arbitrary $\nu \in \mathscr{M}$, let u_ν be the smooth solution of the equation for θ_A and ν. Its mass equals $\mathcal{V}_\nu(\theta_A)$ by Theorem 10.8.1. By (27.66), this mass equals $H \det A$ exactly when $Z_3 u_\nu = 0$. As u_ν is real, its derivatives in both real directions of each elliptic fiber then vanish, and connectedness of the fibers gives $u_\nu = p^* v$ for a smooth v on B. The density of $(\theta_A + \mathrm{dd^c} u_\nu)^3$ relative to μ_0 is then a function on B, so the equation makes ν fiber invariant. The converse was proved by the construction on the base.

Completion in the background parameters. For fixed f, formula (27.63) holds on the entire cone of positive matrices, and differentiating minus the logarithm of its cube root twice gives the stated Hessian metric. Since this is the same metric as in Theorem 27.4.1, the matrix-path and coefficient-limit arguments of that proof apply

unchanged on each fixed W. They give the distance bounds, completeness, properness and the Cauchy criterion; they do not use that the minimizing forms have constant coefficients. □

For the minimizing metrics $\gamma_{A,f}$, the pointwise estimate (27.11) is replaced by an estimate for coefficient means. For a form η whose coefficient matrix in the invariant coframe is (η_{jk}), write $\overline{\eta}$ for the matrix of means $\int_X \eta_{jk}\,\mu_0$. The mean-zero identity (27.6) gives

$$\overline{\gamma}_{A,f} = \frac{A}{(H \det A)^{1/3}}.$$

Thus on S_W the distance bounds control these means, and along a normalized sequence whose coefficients converge to a nonzero singular semipositive matrix the largest eigenvalue of the means diverges. The corresponding minimizing metrics then cannot be bounded above by a fixed multiple of h_0. For $f \neq 1$, however, we do not obtain the pointwise bound (27.11) for the nonconstant forms $\gamma_{A,f}$.

Reference densities that vary along the fibers give a strictly smaller volume. Choose a smooth real χ supported in a coordinate ball with $Z_3\chi \not\equiv 0$, and let $t \neq 0$ be sufficiently small that $\theta_A + t\mathrm{dd}^{\mathrm{c}}\chi > 0$. Put

$$M(t) = \int_X (\theta_A + t\mathrm{dd}^{\mathrm{c}}\chi)^3, \qquad \nu_t = \frac{(\theta_A + t\mathrm{dd}^{\mathrm{c}}\chi)^3}{M(t)}.$$

Then $t\chi$ solves the equation for θ_A and ν_t with constant $M(t)$, so Theorem 10.8.1 and (27.66) give

$$\mathcal{V}_{\nu_t}(\theta_A) = H \det A - \frac{A_{33}H}{2}t^2 \int_X |Z_3\chi|^2 \mu_0 < H \det A.$$

Here ν_t is constructed from the fixed background θ_A. The example therefore shows that the volume depends on the reference measure, but it does not determine the Hessian completion for a fixed density that varies along the fibers.

27.16 Curvature pairs and positive lower mass

On a threefold, the lower mass can be bounded below by a curvature condition. Consider a principal elliptic bundle over a Kähler surface, with a connection form η whose curvature has real and imaginary parts α and β. If $\alpha^2 + \beta^2$ is semipositive, the split metric g built from the base metric and η satisfies $\mathrm{dd}^{\mathrm{c}}(g^2) = 0$ and $\mathrm{dd}^{\mathrm{c}} g \leq 0$. A cubic energy identity for bounded potentials then shows that no bounded potential has mass below $\int_X g^3$, and the determinant estimate of Theorem 27.7.2 transfers this lower bound to every Hermitian form on the manifold. The same curvature sign excludes pluriclosed metrics on these manifolds.

In this section X is a connected compact complex threefold and $\mathrm{d}V$ is a smooth positive probability volume form. Products of bounded potentials are those of

Proposition 1.7.1. For a Hermitian form h, $\mathcal{G}_h$ denotes the compact convex set of normalized Gauduchon currents of Section 4.4, and k_T is the determinant density of $T \in \mathcal{G}_h$ relative to $\mathrm{d}V$, as in Theorem 27.7.1.

27.16.1 The bounded cubic identity

When the square of a Hermitian metric g is dd^{c}-closed, the mass of a bounded g-plurisubharmonic function differs from $\int_X g^3$ by a term that is linear in $\mathrm{dd}^{\mathrm{c}} g$. Its coefficient is a positive current playing the role of $\mathrm{d}u \wedge \mathrm{d}^{\mathrm{c}} u$.

Lemma 27.16.1 *Let g be a Hermitian form with $\mathrm{dd}^{\mathrm{c}}(g^2) = 0$, and let $u \in \mathrm{PSH}(X, g)$ be bounded. The current*

$$E_u = \tfrac{1}{2}\mathrm{dd}^{\mathrm{c}}(u^2) - u\mathrm{dd}^{\mathrm{c}} u$$

is positive, and

$$\int_X (g + \mathrm{dd}^{\mathrm{c}} u)^3 = \int_X g^3 - 3\int_X E_u \wedge \mathrm{dd}^{\mathrm{c}} g. \tag{27.67}$$

If $-\mathrm{dd}^{\mathrm{c}} g \geq 0$, then $v_-(g) = \int_X g^3$.

Proof Locally write $u = p - a$, with p bounded plurisubharmonic and a smooth. Expanding in p and a and using multilinearity, we obtain closed currents $C_r = (\mathrm{dd}^{\mathrm{c}} u)^r$, not necessarily positive, with $C_0 = 1$. The recursive definition of bounded products, expanded in the same way, gives

$$\mathrm{dd}^{\mathrm{c}}(uC_r) = C_{r+1}, \qquad r = 0, 1, 2.$$

By Lemma 1.6.1 and invariance under smooth corrections, the weighted product $u\mathrm{dd}^{\mathrm{c}} u$ does not depend on the local decomposition. On a smaller ball choose smooth bounded plurisubharmonic p_j decreasing to p, and put $u_j = p_j - a$. The same lemma gives $u_j \mathrm{dd}^{\mathrm{c}} u_j \to u\mathrm{dd}^{\mathrm{c}} u$, while boundedness gives $u_j^2 \to u^2$ in L^1. Hence

$$\mathrm{d}u_j \wedge \mathrm{d}^{\mathrm{c}} u_j = \tfrac{1}{2}\mathrm{dd}^{\mathrm{c}}(u_j^2) - u_j \mathrm{dd}^{\mathrm{c}} u_j \longrightarrow E_u.$$

The smooth currents on the left are positive, so their weak limit E_u is positive and has order zero.

Now expand the cube. Repeated use of Stokes' theorem for currents gives

$$\int_X C_3 = 0, \qquad \int_X g^2 \wedge C_1 = \int_X u\mathrm{dd}^{\mathrm{c}}(g^2) = 0, \qquad \int_X g \wedge C_2 = \int_X uC_1 \wedge \mathrm{dd}^{\mathrm{c}} g.$$

Since $\mathrm{dd}^{\mathrm{c}} g$ is closed, $\int_X \mathrm{dd}^{\mathrm{c}}(u^2) \wedge \mathrm{dd}^{\mathrm{c}} g = 0$, so the last integral equals $-\int_X E_u \wedge \mathrm{dd}^{\mathrm{c}} g$. This proves (27.67). If $-\mathrm{dd}^{\mathrm{c}} g \geq 0$, the correction term is nonnegative, and $u = 0$ attains the lower bound $\int_X g^3$. □

27.16.2 The curvature criterion

We now describe threefolds on which the hypotheses of Lemma 27.16.1 hold. Let $\pi\colon X \to B$ be a holomorphic principal elliptic bundle over a compact connected Kähler surface (B,ω), with fiber the elliptic curve $E = \mathbb{C}/\Lambda$ and fiber coordinate z. Assume that there is a smooth $(1,0)$-form η on X, invariant under the action of E and restricting to $\mathrm{d}z$ on every fiber, such that

$$\mathrm{d}\eta = \pi^* F, \qquad F = \alpha + \mathrm{i}\beta,$$

where α, β are smooth closed real $(1,1)$-forms on B. Put

$$Q = \alpha^2 + \beta^2, \qquad e = \mathrm{i}\eta\wedge\bar\eta, \qquad g = \pi^*\omega + se, \quad s > 0, \qquad A_E = \int_E \mathrm{i}\,\mathrm{d}z\wedge\mathrm{d}\bar z.$$

We call g the split metric.

Theorem 27.16.1 *Suppose $Q \geq 0$ and $Q \not\equiv 0$. The split form g is Hermitian, and its lower mass is*

$$v_-(g) = V = 3sA_E\int_B \omega^2. \tag{27.68}$$

Every bounded $u \in \mathrm{PSH}(X,g)$ satisfies

$$\int_X (g + \mathrm{dd^c}u)^3 = V + \frac{3s}{2\pi}\int_X E_u \wedge \pi^* Q \geq V.$$

Every Hermitian form $h \geq cg$, with $c > 0$, satisfies (27.25). Thus the endpoint estimate of Theorem 27.7.2 holds for every Hermitian form on X, with a constant depending on the form. The manifold X admits no pluriclosed Hermitian metric.

Proof We suppress pullbacks in the curvature calculation. Comparing types in $\mathrm{d}\eta = F$ gives $\partial\eta = 0$, $\bar\partial\eta = F$, $\partial\bar\eta = \bar F$ and $\bar\partial\bar\eta = 0$. Hence

$$\partial e = -\mathrm{i}\eta\wedge\bar F, \qquad \bar\partial e = \mathrm{i}F\wedge\bar\eta, \qquad \mathrm{dd^c}e = -\frac{1}{2\pi}F\wedge\bar F = -\frac{1}{2\pi}Q.$$

Since $e^2 = 0$ and the base has dimension two,

$$\mathrm{dd^c}g = -\frac{s}{2\pi}\pi^*Q, \qquad \mathrm{dd^c}(g^2) = -\frac{s}{\pi}\pi^*(\omega\wedge Q) = 0, \qquad g^3 = 3s\pi^*(\omega^2)\wedge e.$$

The form g is positive, because $\pi^*\omega$ is positive on directions transverse to the fibers and e is positive on the vertical line. Fiber integration of g^3 gives V. Lemma 27.16.1 gives the mass identity, and Corollary 27.7.1, with $n = 3$, gives the endpoint bound. By compactness, every Hermitian form h satisfies $h \geq cg$ for some $c > 0$.

Finally, $S = -\mathrm{dd}^{\mathrm{c}} g$ is smooth, semipositive and not identically zero. If a Hermitian form b were pluriclosed, Stokes' theorem would give

$$0 = \int_X g \wedge \mathrm{dd}^{\mathrm{c}} b = \int_X b \wedge \mathrm{dd}^{\mathrm{c}} g = -\int_X b \wedge S < 0,$$

where the last inequality holds because b is strictly positive and S is semipositive and nonzero on an open set. □

Proposition 27.16.1 *The split metric g is balanced if and only if $\omega \wedge \alpha = \omega \wedge \beta = 0$. For such a primitive pair one has $Q \le 0$, with strict inequality wherever the pair is nonzero. In particular, a balanced split metric never satisfies the hypothesis $Q \ge 0$, $Q \not\equiv 0$ of Theorem 27.16.1.*

Proof One has $\mathrm{d}(g^2) = 2s\pi^*\omega \wedge \mathrm{d}e$, and $\mathrm{d}e$ has one component containing η and one containing $\bar\eta$. These two vertical types are independent, so $\mathrm{d}(g^2) = 0$ if and only if $\omega \wedge F = 0$, which is equivalent to the two stated real identities. A primitive real $(1,1)$-form on a surface has eigenvalues $\lambda, -\lambda$ in an ω-unitary frame, so its square is $-\lambda^2\omega^2$. Applying this to α and to β gives the sign of Q. In this case $\mathrm{dd}^{\mathrm{c}} g \ge 0$, so (27.67) bounds the masses of bounded potentials from above rather than from below, and gives no information on the lower mass. □

27.16.3 Negative-line-bundle quotients

The simplest examples of the curvature criterion come from negative line bundles. Dividing the complement of the zero section of a negative line bundle over $\mathbb{P}^1 \times \mathbb{P}^1$ by a dilation produces a principal elliptic bundle, and the logarithm of the fiber norm gives a connection form whose curvature is a multiple of a Kähler form on the base.

Take $B = \mathbb{P}^1 \times \mathbb{P}^1$, let ω_1, ω_2 be the pullbacks of the Fubini–Study forms of the two factors, normalized to have integral one, and put

$$\Omega = a\omega_1 + b\omega_2, \quad a, b \in \mathbb{N}_{>0}, \qquad L = \mathcal{O}(-a, -b),$$
$$X = L^{\times}/\langle v \mapsto \rho v \rangle, \quad 0 < \rho < 1,$$

where $L^{\times}$ is the total space of L with the zero section removed. Equip L with the Hermitian norm induced by the standard norms on the tautological bundles of the two factors. Let $r = \log \|v\|^2$ and set $\eta = \partial r$.

Proposition 27.16.2 *The quotient X is a compact complex threefold, and the projection to B is a principal elliptic fibration. For $g = \pi^*\Omega + si\eta \wedge \bar\eta$, $s > 0$,*

$$v_-(g) = 24\pi sab \log(1/\rho). \tag{27.69}$$

Every Hermitian form on X satisfies the endpoint bound of Theorem 27.16.1 over its full set of normalized Gauduchon currents. The manifold has $H^2(X, \mathbb{R}) \neq 0$ and

admits no balanced or pluriclosed metric. In particular, it is not diffeomorphic to a Hopf threefold $(\mathbb{C}^3 \setminus \{0\})/\langle z \mapsto \lambda z\rangle$, $0 < |\lambda| < 1$.

Proof In product coordinates, the squared norm of the standard local frame of L is $(1+|z_1|^2)^a(1+|z_2|^2)^b$. Therefore

$$\mathrm{d}\mathrm{d}^{\mathrm{c}} r = \pi^*\Omega, \qquad \mathrm{d}\eta = 2\pi\mathrm{i}\pi^*\Omega.$$

Dilation changes r by a constant, so η descends to X. The function r changes by the nonzero constant $\log\rho^2$ under the generator, so the action is free and properly discontinuous. The product of the compact unit circle bundle with a fundamental interval for r maps onto X, which is therefore compact. Each fiber is $\mathbb{C}^*/\rho^{\mathbb{Z}}$. Its logarithmic coordinate has periods $\log\rho$ and $2\pi\mathrm{i}$, and the curvature $F = 2\pi\mathrm{i}\Omega$ has real part 0 and imaginary part $2\pi\Omega$; hence

$$A_E = 4\pi\log(1/\rho), \qquad Q = 4\pi^2\Omega^2 > 0, \qquad \int_B \Omega^2 = 2ab.$$

Theorem 27.16.1 now gives (27.69) and the endpoint bound, and excludes pluriclosed metrics.

To prove $H^2(X,\mathbb{R}) \neq 0$, put $A = \eta + \bar\eta$ and $B_1 = \mathrm{i}(\eta - \bar\eta)$. Then $\mathrm{d}A = 0$ and $\mathrm{d}B_1 = -4\pi\pi^*\Omega$. Suppose that $\pi^*(\omega_1 - \omega_2) = \mathrm{d}\zeta$ for a real one-form ζ. Averaging ζ over the compact group of elliptic translations, we may assume ζ invariant, so that

$$\zeta = \pi^*\xi + (\pi^* f)A + (\pi^* j)B_1$$

for a one-form ξ and functions f, j on B. The components of $\mathrm{d}\zeta$ with one vertical and one horizontal direction must vanish, which gives $\mathrm{d}f = \mathrm{d}j = 0$. Thus

$$\omega_1 - \omega_2 = \mathrm{d}\xi - 4\pi j\Omega.$$

Integrating over the two rulings gives $1 = -4\pi ja$ and $-1 = -4\pi jb$, which is impossible because $a, b > 0$. A Hopf threefold as above is diffeomorphic to $S^5 \times S^1$, whose second real cohomology vanishes.

Finally, $\pi^*\Omega = -\mathrm{d}B_1/(4\pi)$ is a nonzero semipositive exact form. Its integral against the closed positive square of a balanced metric would vanish by Stokes' theorem and be strictly positive by positivity. Hence no balanced metric exists. □

27.16.4 A Calabi–Eckmann family without LCK metrics

A Hermitian metric Ω is *locally conformally Kähler*, or LCK, if $\mathrm{d}\Omega = \vartheta \wedge \Omega$ for a closed real one-form ϑ, its Lee form; such metrics are studied in Chapter 30. The quotients of the preceding subsection resemble Hopf manifolds. A second family shows that the curvature criterion also applies on manifolds that carry no LCK metric at all.

Let $\tau = A + \mathrm{i}B_0$, where $A < 0$ and $B_0 > 0$, and let $\mathbb{C}$ act holomorphically on $(\mathbb{C}^2 \setminus \{0\})^2$ by

$$t \cdot (z, w) = (\mathrm{e}^t z, \mathrm{e}^{\tau t} w).$$

Write X_τ for the quotient and put $r_1 = \log |z|^2$, $r_2 = \log |w|^2$. The map $(z, w) \mapsto ([z], [w])$ descends to a projection $\pi\colon X_\tau \to \mathbb{P}^1 \times \mathbb{P}^1$; let ω_1, ω_2 be the pullbacks to the base of the Fubini–Study forms of the two factors, normalized to have integral one.

Proposition 27.16.3 *The quotient X_τ is a compact complex threefold diffeomorphic to $S^3 \times S^3$, and π is a principal elliptic fibration over $\mathbb{P}^1 \times \mathbb{P}^1$. The form $\eta = \tau \partial r_1 - \partial r_2$ descends to X_τ, and for*

$$g = \pi^*(\omega_1 + \omega_2) + s\mathrm{i}\eta \wedge \bar\eta, \qquad s > 0,$$

one has

$$v_-(g) = 48\pi^2 s B_0. \tag{27.70}$$

Every Hermitian form $h \geq cg$, with $c > 0$, satisfies

$$\sup_{T \in \mathscr{G}_h} \|k_T\|_{L^{3/2}(\mathrm{d}V)} \leq c^{-1}(48\pi^2 s B_0)^{-1/3}.$$

There is no balanced, pluriclosed or LCK metric on X_τ.

Proof Every orbit contains a unique point with $|z| = |w| = 1$: the required parameter is determined by

$$\operatorname{Re} t = -r_1/2, \qquad \operatorname{Im} t = (r_2 - A r_1)/(2B_0).$$

These smooth formulas identify $(\mathbb{C}^2 \setminus \{0\})^2$ with $\mathbb{C} \times (S^3 \times S^3)$, so the action is free and proper, and X_τ is diffeomorphic to $S^3 \times S^3$. To obtain holomorphic charts, choose a nonzero coordinate z_k of z and a branch of its logarithm, and use the action to normalize $z_k = 1$; the functions z_l / z_k, $l \neq k$, and $w_j z_k^{-\tau}$ are then local holomorphic coordinates on the quotient. Changing the branch of the logarithm accounts for the periods along the fibers. Thus X_τ is a compact complex threefold.

The invariant form η vanishes on the generator $z\partial_z + \tau w \partial_w$ of the action, so it descends. On an elliptic fiber it equals $\mathrm{d}\chi$, where $\chi = \tau \log v_1 - \log v_2$ in terms of fiber coordinates $(v_1 z, v_2 w)$; its period lattice is generated by $2\pi\mathrm{i}\tau$ and $-2\pi\mathrm{i}$. Hence $A_E = 8\pi^2 B_0$. Direct differentiation gives

$$F = 2\pi\mathrm{i}(\tau\omega_1 - \omega_2), \quad \alpha = -2\pi B_0 \omega_1,$$
$$\beta = 2\pi(A\omega_1 - \omega_2), \quad Q = -8\pi^2 A\, \omega_1 \wedge \omega_2 > 0.$$

Since $\int_B (\omega_1 + \omega_2)^2 = 2$, Theorem 27.16.1 gives the lower mass, the endpoint bound and the absence of pluriclosed metrics.

Moreover $\mathrm{d}\operatorname{Re}\eta = -2\pi B_0 \pi^* \omega_1$, so $\pi^*\omega_1$ is a nonzero semipositive exact form. It excludes every balanced metric by the Stokes argument at the end of the proof

of Proposition 27.16.2, and hence also every Kähler metric. The quotient is simply connected, so the Lee form of a hypothetical LCK metric would be exact, equal to $\mathrm{d}f$; multiplying the metric by e^{-f} would then give a Kähler metric, a contradiction. In particular X_τ carries no Vaisman metric, that is, no LCK metric with parallel Lee form. □

The curvature criterion controls bounded potentials and the full set of normalized Gauduchon currents in these threefolds. Curvature pairs of changing sign, primitive curvature pairs and higher-dimensional analogues are not covered by this argument.

27.17 Geometric bounds and determinant concentration

By Theorem 27.7.2, the determinant densities of all normalized Gauduchon currents are bounded in the critical space $L^{n/(n-1)}$ exactly when the lower mass is positive. Together with the mass-invariance criterion of Theorem 2.2.2, this gives geometric sufficient conditions for the critical bound. The critical exponent cannot be raised, even for currents that are powers of Kähler forms in one fixed class. The same concentrating family shows that currents approaching the maximum of the determinant functional for a fixed smooth density need not satisfy any uniform higher integrability, although the maximizer itself is smooth. In the subsections that follow, we compute the complete determinant profile of a background that satisfies the conditions of Theorem 2.2.2, show that this profile does not control the ellipticity of the background, and construct exact maximizers for varying densities whose weak limit has zero determinant measure.

Throughout this section $n \geq 2$ and $\mathrm{d}V$ is a smooth positive probability volume form. For a Hermitian form h, k_T denotes the determinant density relative to $\mathrm{d}V$ of a current $T \in \mathscr{G}_h$, as in Section 27.7. Put $q = n/(n-1)$.

Corollary 27.17.1 *Suppose g is a Hermitian form with $\mathrm{dd}^{\mathrm{c}} g = \mathrm{dd}^{\mathrm{c}}(g^2) = 0$, and put $V_g = \int_X g^n$. For every Hermitian form $h \geq cg$, $c > 0$,*

$$v_-(h) \geq c^n V_g, \qquad \sup_{T \in \mathscr{G}_h} \|k_T\|_{L^q(\mathrm{d}V)} \leq c^{-1} V_g^{-1/n}.$$

Equality holds in the determinant bound when $h = g$ and $c = 1$. In particular, on a compact Kähler manifold or a compact complex surface, the critical norm is uniformly bounded over the normalized Gauduchon currents of each fixed Hermitian form.

Proof The identity $\mathrm{dd}^{\mathrm{c}}(g^2) = 2g \wedge \mathrm{dd}^{\mathrm{c}} g + 2\mathrm{d}g \wedge \mathrm{d}^{\mathrm{c}} g$ shows that $\mathrm{d}g \wedge \mathrm{d}^{\mathrm{c}} g = 0$, so Theorem 2.2.2 gives $v_-(g) = V_g$. Corollary 27.7.1 then gives both bounds, and Theorem 27.7.2 gives equality in the determinant bound when $h = g$.

A Kähler form satisfies both differential identities. On a surface, Theorem 4.1.1 gives a metric g with $\mathrm{dd}^{\mathrm{c}} g = 0$, and $\mathrm{dd}^{\mathrm{c}}(g^2) = 0$ for degree reasons. By compactness, each fixed h satisfies $h \geq cg$ for some $c > 0$. □

For a fixed smooth F, Theorem 27.5.1 gives a unique maximizing current for the determinant functional $\mathfrak{D}_F$ of (27.23), and this current is smooth and strictly positive, so its determinant density is bounded. The following construction shows that currents approaching this maximum satisfy no uniform bound on higher moments of their determinant densities.

Proposition 27.17.1 *Let $X = E^n$, where $E = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$ carries its flat area form of total area one, and write ω_i for the pullback of this form from the ith factor. Set*

$$h = \sum_{i=1}^{n} \omega_i, \qquad \mathrm{d}V = \omega_1 \wedge \cdots \wedge \omega_n, \qquad V = n!, \qquad \alpha = \frac{n-1}{n}.$$

For $F = 0$, the unique maximizing current is $T_* = h^{n-1}/V$, with $k_{T_*} = V^{-1/n}$ and $\mathfrak{D}_0(T_*) = V^{-1/n}$. There are smooth strictly positive closed forms $S_\varepsilon \in \mathcal{G}_h$, $0 < \varepsilon < 1/4$, such that

$$\|k_{S_\varepsilon}\|_{L^q} = V^{-1/n}, \qquad \mathfrak{D}_0(S_\varepsilon) \longrightarrow V^{-1/n},$$

while for every fixed finite $p > q$,

$$\int_X k_{S_\varepsilon}^p \, \mathrm{d}V \geq C_{n,p} [\log(1/\varepsilon)]^{-\alpha p} \varepsilon^{2-2\alpha p} \longrightarrow +\infty. \tag{27.71}$$

Their L^∞ norms also diverge. Each S_ε is the $(n-1)$st power of a Kähler form in the fixed class $V^{-1/(n-1)}[h]$, and has the same Bott–Chern, de Rham and Aeppli classes as T_*. No S_ε attains the maximum of $\mathfrak{D}_0$.

Proof Put $H_i = \bigwedge_{j \neq i} \omega_j$. Expanding the wedge powers gives $h^{n-1} = (n-1)! \sum_i H_i$ and $h \wedge H_i = \mathrm{d}V$. Thus $T_* = (1/n) \sum_i H_i$ is normalized, and the determinant formula gives $k_{T_*} = V^{-1/n}$. The smooth equation for $F = 0$ is solved by the zero potential, since $h^n = V \, \mathrm{d}V$; Theorem 27.5.1 therefore gives the maximal value and the uniqueness of the maximizer.

To construct S_ε, choose a nonnegative smooth function β compactly supported in the unit disk in $\mathbb{C}$, of integral one, with $\beta \geq \beta_0 > 0$ on a disk of radius $r_0 \in (0, 1)$. On an embedded disk in the first elliptic factor put $b_\varepsilon(z) = \varepsilon^{-2} \beta(z/\varepsilon)$, extended by zero; it has integral one. For $0 < t < 1$, set

$$f_{\varepsilon,t} = 1 - t + t b_\varepsilon, \qquad a = V^{-1/(n-1)},$$
$$\gamma_{\varepsilon,t} = a(f_{\varepsilon,t}\omega_1 + \omega_2 + \cdots + \omega_n), \qquad T_{\varepsilon,t} = \gamma_{\varepsilon,t}^{n-1}.$$

The form $\gamma_{\varepsilon,t}$ is strictly positive, and it is closed because $f_{\varepsilon,t}$ depends only on the first factor, which has real dimension two. Direct expansion gives

$$T_{\varepsilon,t} = \frac{1}{n} \left(H_1 + f_{\varepsilon,t} \sum_{i=2}^{n} H_i \right), \qquad k_{T_{\varepsilon,t}} = V^{-1/n} f_{\varepsilon,t}^{\alpha}. \tag{27.72}$$

Since $\int_X f_{\varepsilon,t}\,\mathrm{d}V = 1$, the trace integral is one, so $T_{\varepsilon,t} \in \mathscr{G}_h$, and $\|k_{T_{\varepsilon,t}}\|_{L^q} = V^{-1/n}$ because $\alpha q = 1$.

The mean-zero function $f_{\varepsilon,t} - 1$ has a smooth real potential ψ on E with $\mathrm{dd^c}\psi = (f_{\varepsilon,t} - 1)\omega_1$; for example, the mean-zero Fourier series solving $\Delta\psi = 4\pi(f_{\varepsilon,t} - 1)$ in the area-one coordinates. Consequently $\gamma_{\varepsilon,t} - ah = \mathrm{dd^c}(a\psi)$. Since $T_* = (ah)^{n-1}$, the closed form $R = \sum_{j=0}^{n-2} \gamma_{\varepsilon,t}^{n-2-j} \wedge (ah)^j$ satisfies

$$T_{\varepsilon,t} - T_* = \mathrm{dd^c}(a\psi R).$$

This proves all the assertions about classes, including the case $n = 2$, where $R = 1$.

Jensen's inequality and $f_{\varepsilon,t} \geq 1 - t$ give

$$V^{-1/n}(1-t)^{\alpha} \leq \mathfrak{D}_0(T_{\varepsilon,t}) = V^{-1/n} \int_X f_{\varepsilon,t}^{\alpha}\,\mathrm{d}V < V^{-1/n},$$

where the inequality is strict because $f_{\varepsilon,t}$ is nonconstant and $0 < \alpha < 1$. On the product of the disk of radius $r_0\varepsilon$ with the other factors, we have $f_{\varepsilon,t} \geq t\beta_0\varepsilon^{-2}$, and this set has volume $\pi r_0^2\varepsilon^2$. Hence

$$\int_X k_{T_{\varepsilon,t}}^p\,\mathrm{d}V \geq V^{-p/n}\pi r_0^2 \beta_0^{\alpha p} t^{\alpha p} \varepsilon^{2-2\alpha p},$$
$$\|k_{T_{\varepsilon,t}}\|_{\infty} \geq V^{-1/n}\beta_0^{\alpha} t^{\alpha} \varepsilon^{-2\alpha}.$$

Now take $t = 1/\log(1/\varepsilon)$, which lies in $(0,1)$ because $\varepsilon < 1/4 < \mathrm{e}^{-1}$, and $S_\varepsilon = T_{\varepsilon,t}$. Then $\mathfrak{D}_0(S_\varepsilon)$ converges to its maximum, while both lower bounds diverge for $p > q$ because $\alpha p > 1$. All constants are independent of ε. □

If instead $t \in (0,1)$ is kept fixed in the same construction, then $\int_X k_{T_{\varepsilon,t}}^p\,\mathrm{d}V \geq C_{n,p,t}\varepsilon^{2-2\alpha p}$ and $\|k_{T_{\varepsilon,t}}\|_{\infty} \geq C_{n,t}\varepsilon^{-2\alpha}$. Thus there are no uniform bounds on higher moments, either over the full normalized cone or along a maximizing sequence for one smooth density. Every member of these families attains the supremum $V^{-1/n}$ of the critical norm in Corollary 27.17.1, whereas the maximum of $\mathfrak{D}_0$ is attained only by T_*.

27.17.1 Intrinsic determinant measures and the complete profile

For a positive $(n-1,n-1)$-current T and a Hermitian form h, let $\Gamma_{0,h}(T)$ denote the determinant measure of (27.23) with $F = 0$, computed with h in place of ω and with the fixed $\mathrm{d}V$. By Lemma 27.5.1, its total mass does not depend on h. The following coordinate formula shows that the measure itself does not depend on h, also when the coefficient measures of T are singular.

Lemma 27.17.1 *For every positive $(n-1,n-1)$-current T on X, the measure $\Gamma_{0,h}(T)$ is the same for all Hermitian forms h.*

Proof In local holomorphic coordinates put $\lambda = \bigwedge_{j=1}^n (\mathrm{i}\,\mathrm{d}z_j \wedge \mathrm{d}\bar z_j)$, $h = \mathrm{i}\sum H_{ij}\,\mathrm{d}z_i \wedge \mathrm{d}\bar z_j$ and $\mathrm{d}V = \nu\lambda$. The coefficients of T are measures. Choose a positive Radon measure ν with respect to which they are all absolutely continuous. There is a ν-measurable field Q of positive semidefinite Hermitian matrices such that $\xi \wedge T = \operatorname{tr}(\Xi Q)\nu$ for every smooth real $(1,1)$-form $\xi = \mathrm{i}\sum \Xi_{ij}\,\mathrm{d}z_i \wedge \mathrm{d}\bar z_j$. In an h-unitary frame, the trace measure $\sigma_h = h \wedge T$ and the matrix density P_h of Section 27.5, formed with h in place of ω, are

$$\sigma_h = \operatorname{tr}(HQ)\nu, \qquad P_h = \frac{nH^{1/2}QH^{1/2}}{\operatorname{tr}(HQ)}.$$

The second formula holds where $Q \neq 0$; the set where $Q = 0$ has zero σ_h-measure. Since $h^n = n!\det(H)\lambda$, the density in (27.23) is $(\nu/(n!\det H))^{1/n}$, and $\det P_h = n^n \det(H)\det(Q)/\operatorname{tr}(HQ)^n$. Hence

$$\Gamma_{0,h}(T) = n(n!)^{-1/n}\nu^{1/n}(\det Q)^{1/n}\nu. \tag{27.73}$$

The right side does not involve H. Replacing ν by $L\nu$, where $L > 0$ almost everywhere, replaces Q by Q/L, leaving the right side unchanged; a common dominating measure compares any two choices. Thus two Hermitian forms give the same measure on every coordinate chart, and hence on X. □

We write $\Gamma_0(T)$ for this measure; taking $h = \omega$ shows that it agrees with the measure $\Gamma_0(T)$ of Section 27.7.

Theorem 27.17.1 *Let β be a Hermitian form with $\mathrm{dd}^{\mathrm{c}}(\beta^j) = 0$ for $1 \le j \le n-1$, put $V = \int_X \beta^n$, and let $h = \beta + \mathrm{dd}^{\mathrm{c}}u > 0$ for a smooth real function u on X. Then*

$$\mathscr{G}_h = \mathscr{G}_\beta, \qquad \{\Gamma_{0,h}(T) : T \in \mathscr{G}_h\} = \{\Gamma_{0,\beta}(T) : T \in \mathscr{G}_\beta\}.$$

The density k_T of every measure $\Gamma_0(T) = k_T\,\mathrm{d}V$ in this common family satisfies

$$\|k_T\|_{L^{n/(n-1)}(\mathrm{d}V)} \le V^{-1/n},$$

and for every real or complex $p \in L^n(\mathrm{d}V)$,

$$\sup_{T\in\mathscr{G}_h} \int_X |p|\,\Gamma_0(T) = V^{-1/n}\|p\|_{L^n(\mathrm{d}V)}. \tag{27.74}$$

Proof For every positive $(n-1,n-1)$-current T with $\mathrm{dd}^{\mathrm{c}}T = 0$, distributional integration by parts gives $\int_X h \wedge T = \int_X \beta \wedge T + \int_X u\,\mathrm{dd}^{\mathrm{c}}T = \int_X \beta \wedge T$. Hence $\mathscr{G}_h = \mathscr{G}_\beta$, and Lemma 27.17.1 gives $\Gamma_{0,h}(T) = \Gamma_{0,\beta}(T)$ for every such T.

By Theorem 2.2.2, every bounded β-plurisubharmonic function has mass V, so $v_-(\beta) = V$. Hence Theorem 27.7.2, applied to β, gives the bound for k_T, and Hölder's

inequality gives, for every $T \in \mathscr{G}_\beta$ and every smooth $r \geq 0$,

$$\int_X r\,\Gamma_0(T) \leq V^{-1/n}\|r\|_{L^n(\mathrm{d}V)}. \tag{27.75}$$

The same inequality for $|p|$ in place of r proves the upper bound in (27.74).

For the reverse inequality, let $r > 0$ be smooth and put $M = \int_X r^n\,\mathrm{d}V$. By Theorem 10.1.1, applied to the background β and the density r^n, there are a smooth function w with $A = \beta + \mathrm{dd^c} w > 0$ and a constant $c > 0$ with $A^n = cr^n\,\mathrm{d}V$. By Theorem 2.2.2, $\int_X A^n = \int_X \beta^n = V$. This determines $c = V/M$, so

$$A^n = \frac{Vr^n}{M}\,\mathrm{d}V. \tag{27.76}$$

Put $T_r = A^{n-1}/V$. It is smooth and positive. Write $E = \mathrm{dd^c} w$; this form is d-closed and $\mathrm{d^c}$-closed. For $1 \leq j \leq n-1$, the binomial expansion gives

$$\mathrm{dd^c}(A^j) = \sum_{l=0}^{j} \binom{j}{l} \mathrm{dd^c}(\beta^{j-l}) \wedge E^l = 0,$$

since the term with $l = j$ contains $\mathrm{dd^c}(1) = 0$ and the others vanish by hypothesis. Thus $\mathrm{dd^c} T_r = 0$, and smooth integration by parts gives

$$\int_X \beta \wedge A^{n-1} = \int_X A^n - \int_X \mathrm{dd^c} w \wedge A^{n-1} = V.$$

Consequently $T_r \in \mathscr{G}_\beta$. We have $T_r = \gamma^{n-1}$ with $\gamma = V^{-1/(n-1)} A$, and the determinant measure of γ^{n-1} is $(\mathrm{d}V/\gamma^n)^{1/n}\gamma^n$, as computed in Section 27.5. Hence, by (27.76),

$$k_{T_r} = (\gamma^n/\mathrm{d}V)^{(n-1)/n} = V^{-1/n}\frac{r^{n-1}}{\|r\|_{L^n}^{n-1}}.$$

Thus $\int_X r k_{T_r}\,\mathrm{d}V = V^{-1/n}\|r\|_{L^n}$, which is equality in (27.75). For a nonnegative $p \in L^n$, take smooth positive $r_j \to p$ in L^n. By the bound for k_T, the error $|\int_X (p - r_j) k_{T_{r_j}}\,\mathrm{d}V|$ is at most $V^{-1/n}\|p - r_j\|_{L^n} \to 0$, which proves the lower bound in (27.74). For real or complex p, apply this to $|p|$. □

27.17.2 Fixed determinant data with degenerating ellipticity

By Theorem 27.17.1, if β satisfies its hypothesis and $h = \beta + \mathrm{dd^c} u > 0$, then the normalized Gauduchon currents of h, their determinant measures and the sharp constant in (27.74) depend only on β. These data therefore cannot detect a loss of ellipticity of h. The next two theorems make this explicit on a flat torus. As in Definition 7.5.1, ν_- and ν_+ denote the lower and upper masses.

Theorem 27.17.2 *Let $X = (\mathbb{C}/(\mathbb{Z}+\mathrm{i}\mathbb{Z}))^n$, with coordinates $z_j = x_j + \mathrm{i}y_j$ and the flat Kähler form $\beta = \sum_j \mathrm{d}x_j \wedge \mathrm{d}y_j$. There are Kähler forms $\omega_\varepsilon = \beta + \mathrm{dd}^{\mathrm{c}} u_\varepsilon$, $0 < \varepsilon \le 1$, in the class of β, such that the sets $\mathcal{G}_{\omega_\varepsilon}$ and their families of determinant measures do not depend on ε, and (27.74) holds for $h = \omega_\varepsilon$ with $V = n!$ and every smooth positive probability volume $\mathrm{d}V$. The minimum over X of the least eigenvalue of ω_ε relative to β equals ε. All C^k norms of the coefficients of ω_ε and of the potentials u_ε are bounded uniformly in ε, and*

$$v_-(\omega_\varepsilon) = v_+(\omega_\varepsilon) = n! > 0.$$

The manifold has a global holomorphic volume form, so $c_1(X) = 0$. Every ω_ε is balanced and pluriclosed.

Proof Write $\beta_j = (\mathrm{i}/2)\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j = \mathrm{d}x_j \wedge \mathrm{d}y_j$, so that $\beta = \sum_j \beta_j$, and put $V = \int_X \beta^n = n!$. Set

$$u_\varepsilon = \frac{1-\varepsilon}{4\pi}\cos(4\pi x_1), \qquad a_\varepsilon = 1 - (1-\varepsilon)\cos(4\pi x_1).$$

For a function of x_1, $\mathrm{dd}^{\mathrm{c}} u = (u''/(4\pi))\beta_1$. Hence

$$\omega_\varepsilon = \beta + \mathrm{dd}^{\mathrm{c}} u_\varepsilon = a_\varepsilon \beta_1 + \sum_{j=2}^{n} \beta_j.$$

These are smooth positive closed forms with $\varepsilon \le a_\varepsilon \le 2-\varepsilon$ and $\omega_\varepsilon \le 2\beta$. Their least eigenvalue relative to β is $\min(a_\varepsilon, 1)$, whose minimum over X is ε. As $\varepsilon \to 0$ they converge smoothly to a semipositive form losing rank on $\{x_1 \in \frac{1}{2}\mathbb{Z}\}$. All C^k norms of the coefficients and potentials are bounded uniformly in ε. Theorem 27.17.1, applied with this closed β and $h = \omega_\varepsilon$, gives $\mathcal{G}_{\omega_\varepsilon} = \mathcal{G}_\beta$, the equality of the families of determinant measures, and (27.74) with $V = n!$, for every ε.

Since ω_ε is closed, Theorem 2.2.2 gives mass $\int_X \omega_\varepsilon^n$ for every bounded ω_ε-plurisubharmonic potential. The explicit formula gives $\int_X \omega_\varepsilon^n = V$, since a_ε has mean one. Thus both extrema over bounded potentials equal V. The global form $\mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n$ trivializes the canonical bundle, and closed positive forms are balanced and pluriclosed. □

27.17.3 Nonclosed pluriclosed metrics with fixed determinant data

Theorem 27.17.3 *On the torus X of Theorem 27.17.2, with its flat Kähler form β, fix $0 < \tau < 1/2$. There are a nonclosed pluriclosed Hermitian metric h_τ and smooth functions u_ε, $0 < \varepsilon \le 1/2$, such that the forms $\Omega_\varepsilon = h_\tau + \mathrm{dd^c} u_\varepsilon$ are positive, and the sets $\mathcal{G}_{\Omega_\varepsilon}$ and their families of determinant measures do not depend on ε. For every smooth positive probability volume $\mathrm{d}V$ and every $p \in L^n(\mathrm{d}V)$,*

$$\sup_{T \in \mathcal{G}_{\Omega_\varepsilon}} \int_X |p|\, \Gamma_0(T) = V_\tau^{-1/n} \|p\|_{L^n(\mathrm{d}V)}, \qquad V_\tau = n!(1 - \tau^2/2).$$

The lower and upper masses of every Ω_ε equal V_τ. All C^k norms of the coefficients of Ω_ε and of u_ε are bounded uniformly in ε, while

$$\min_X \lambda_{\min}(\Omega_\varepsilon \mid \beta) = \varepsilon, \qquad \mathrm{d}\Omega_\varepsilon = \mathrm{d}h_\tau \ne 0,$$

where $\lambda_{\min}(\Omega_\varepsilon \mid \beta)$ is the least eigenvalue of Ω_ε relative to β. Each Ω_ε is pluriclosed, and β is a balanced metric on the same manifold, which has $c_1(X) = 0$.

Proof With the coordinates and the forms β_j of the proof of Theorem 27.17.2, set $s = \sin(2\pi x_1)$ and

$$\xi = \frac{\mathrm{i}}{2}(\mathrm{d}z_1 \wedge \mathrm{d}\bar z_2 + \mathrm{d}z_2 \wedge \mathrm{d}\bar z_1) = \mathrm{d}x_1 \wedge \mathrm{d}y_2 + \mathrm{d}x_2 \wedge \mathrm{d}y_1, \qquad h_\tau = \beta + \tau s \xi.$$

Relative to β, the matrix of h_τ in the first two coordinates is $\left(\begin{smallmatrix} 1 & \tau s \\ \tau s & 1 \end{smallmatrix}\right)$, and its remaining diagonal entries are one, so positivity follows from $|\tau s| < 1$. Moreover

$$\mathrm{d}h_\tau = 2\pi\tau \cos(2\pi x_1)\, \mathrm{d}x_1 \wedge \mathrm{d}x_2 \wedge \mathrm{d}y_1 \ne 0.$$

Since ξ has constant coefficients and $\mathrm{dd^c} s$ is a multiple of $\mathrm{d}z_1 \wedge \mathrm{d}\bar z_1$, we have $\mathrm{dd^c}(s\xi) = \mathrm{dd^c} s \wedge \xi = 0$. Thus the form $h' = \beta_1 + \beta_2 + \tau s \xi$ in the first two coordinates is pluriclosed. Its square is a multiple of $\beta_1 \wedge \beta_2$ with a coefficient depending only on x_1, hence closed, and its higher powers vanish. Since $h_\tau = h' + \sum_{j \ge 3} \beta_j$ and the forms β_j are closed with constant coefficients, expanding the powers of h_τ proves

$$\mathrm{dd^c}(h_\tau^j) = 0 \quad (1 \le j \le n-1).$$

Moreover $h_\tau^n = n!(1 - \tau^2 s^2)\beta_1 \wedge \cdots \wedge \beta_n$, so

$$\int_X h_\tau^n = n! \int_0^1 (1 - \tau^2 \sin^2(2\pi x_1))\, \mathrm{d}x_1 = V_\tau.$$

Use the same u_ε and a_ε as in Theorem 27.17.2, now with $\varepsilon \le 1/2$. Then

$$\Omega_\varepsilon = a_\varepsilon \beta_1 + \sum_{j \ge 2} \beta_j + \tau s \xi, \qquad a_\varepsilon = \varepsilon + 2(1 - \varepsilon)s^2.$$

The block of $\Omega_\varepsilon - \varepsilon\beta$ in the first two coordinates has diagonal entries $2(1-\varepsilon)s^2$ and $1-\varepsilon$, and determinant

$$\left[2(1-\varepsilon)^2 - \tau^2\right]s^2 \geq 0.$$

It is positive semidefinite, and the remaining diagonal entries of $\Omega_\varepsilon - \varepsilon\beta$ are $1-\varepsilon$. Hence $\Omega_\varepsilon \geq \varepsilon\beta$, and at points with $s = 0$ the least eigenvalue of Ω_ε relative to β equals ε. Also $\Omega_\varepsilon \leq 3\beta$, all C^k norms are bounded uniformly in ε, and the smooth semipositive limit as $\varepsilon \to 0$ loses rank where $s = 0$. Since $\mathrm{ddd^c} u_\varepsilon = 0$, the exterior derivative of Ω_ε is the fixed nonzero form $\mathrm{d}h_\tau$ computed above.

The binomial expansion in the proof of Theorem 27.17.1 applies to $\Omega_\varepsilon = h_\tau + \mathrm{dd^c} u_\varepsilon$; it gives $\mathrm{dd^c}(\Omega_\varepsilon^j) = 0$ for $1 \leq j \leq n-1$, in particular pluriclosedness. The volume of Ω_ε is V_τ, either by integrating the determinant $a_\varepsilon - \tau^2 s^2$ of its block in the first two coordinates, whose mean is $1 - \tau^2/2$, or by Theorem 2.2.2. That theorem also gives mass V_τ for every bounded Ω_ε-plurisubharmonic potential, so both mass extrema equal V_τ. Theorem 27.17.1, applied with h_τ in place of β and $h = \Omega_\varepsilon$, gives $\mathcal{G}_{\Omega_\varepsilon} = \mathcal{G}_{h_\tau}$, the equality of the families of determinant measures, and the asserted formula for every $p \in L^n(\mathrm{d}V)$. The balanced trace is fixed as well:

$$\int_X \Omega_\varepsilon \wedge \beta^{n-1} = n!,$$

since the off-diagonal term has zero trace and the mean of a_ε equals one. The global holomorphic volume form gives $c_1(X) = 0$. □

These nonclosed metrics show that the complete determinant profile, the balanced trace and positive mass extrema can stay fixed while the least relative eigenvalue tends to zero. The torus is Kähler, however, so the examples do not settle Problem 35.3.4, which remains open.

27.17.4 Concentration of exact maximizers for varying densities

In the proof of Theorem 27.17.1, each current T_r attains the sharp bound (27.75) for its own function r. Such exact maximizers, taken for varying r, need not have uniformly integrable critical powers $k_T^{n/(n-1)}$, and their weak limits can have zero determinant measure.

Theorem 27.17.4 *On $X = (\mathbb{C}/(\mathbb{Z}+\mathrm{i}\mathbb{Z}))^n$, with coordinates $z_j = x_j + \mathrm{i}y_j$, let $\beta_j = \mathrm{d}x_j \wedge \mathrm{d}y_j$, $\beta = \sum_j \beta_j$, $V = n!$ and $\mathrm{d}V = \beta^n/V$. There are Kähler forms $A_\varepsilon = \beta + \mathrm{dd^c} w_\varepsilon$, $0 < \varepsilon < 1$, such that the currents $T_\varepsilon = A_\varepsilon^{n-1}/V$ belong to $\mathscr{G}_\beta$ and satisfy*

$$\int_X k_{T_\varepsilon}^{n/(n-1)}\,\mathrm{d}V = V^{-1/(n-1)}.$$

Each T_ε attains the supremum in (27.74), with $h = \beta$, for some smooth positive p_ε with $\|p_\varepsilon\|_{L^n(\mathrm{d}V)} = 1$. The potentials w_ε are uniformly bounded and converge uniformly to a continuous β-plurisubharmonic function w_0. The currents converge weakly to $T_0 = (\beta + \mathrm{dd^c} w_0)^{n-1}/V \in \mathscr{G}_\beta$, but $\Gamma_0(T_0) = 0$. The functions $k_{T_\varepsilon}^{n/(n-1)}$ are not uniformly integrable. Nevertheless the Monge–Ampère masses $\int_X (\beta + \mathrm{dd^c} w_\varepsilon)^n$, $0 \le \varepsilon < 1$, all equal V.

Proof Put $r = 1 - \varepsilon$ and

$$P_r(x) = \frac{1 - r^2}{1 - 2r\cos(2\pi x) + r^2}, \qquad f_\varepsilon = \varepsilon + (1-\varepsilon)P_r.$$

The positive periodic Poisson kernel has mean one and Fourier series

$$P_r = 1 + 2\sum_{m\ge 1} r^m \cos(2\pi m x).$$

Define

$$w_\varepsilon = -\frac{2(1-\varepsilon)}{\pi}\sum_{m\ge 1}\frac{r^m\cos(2\pi m x_1)}{m^2}.$$

For fixed $\varepsilon > 0$ this is smooth. Differentiation gives $w_\varepsilon'' = 4\pi(f_\varepsilon - 1)$, and hence

$$A_\varepsilon = f_\varepsilon(x_1)\beta_1 + \sum_{j=2}^n \beta_j = \beta + \mathrm{dd^c} w_\varepsilon > 0.$$

It is closed, and $\int_X A_\varepsilon^n = V\int_X f_\varepsilon\,\mathrm{d}V = V$. Since $A_\varepsilon^n = V f_\varepsilon\,\mathrm{d}V$, the computation of k_{T_r} in the proof of Theorem 27.17.1, with $f_\varepsilon^{1/n}$ in place of r, gives

$$k_{T_\varepsilon} = V^{-1/n} f_\varepsilon^{(n-1)/n}, \qquad k_{T_\varepsilon}^{n/(n-1)} = V^{-1/(n-1)} f_\varepsilon.$$

Thus the integral of $k_{T_\varepsilon}^{n/(n-1)}$ equals $V^{-1/(n-1)}$, which by Theorem 27.17.1 is its largest value over *all* currents in $\mathscr{G}_\beta$. Moreover, $p_\varepsilon = f_\varepsilon^{1/n}$ has L^n norm one, and $\int_X p_\varepsilon k_{T_\varepsilon}\,\mathrm{d}V = V^{-1/n}$, so T_ε attains the supremum in (27.74) for $p = p_\varepsilon$.

Since $\sum_{m\ge1} m^{-2} = \pi^2/6$, the Fourier series gives $|w_\varepsilon| \le \pi/3$, and, as $\varepsilon \to 0$, it converges uniformly to

$$w_0 = -\frac{2}{\pi}\sum_{m\ge1}\frac{\cos(2\pi m x_1)}{m^2}.$$

This limit is continuous, and it is β-plurisubharmonic because $A_0 = \beta + \mathrm{dd^c} w_0$ is a positive current, as we now show. The probability measures $f_\varepsilon(x_1)\,\mathrm{d}x_1$ converge weakly to the periodic Dirac mass at $x_1 = 0$, as follows from their Fourier coefficients, which equal 1 for $m = 0$ and $(1-\varepsilon)^{|m|+1}$ for $m \neq 0$. Thus

$$A_0 = \delta_{x_1=0}\beta_1 + \sum_{j=2}^{n} \beta_j,$$

where the Dirac measure is tensored with Lebesgue measure in the remaining real coordinates. Since w_0 is bounded, the powers of A_0 are Bedford–Taylor products. Locally $\beta_1 + \mathrm{dd^c} w_0 = \mathrm{dd^c}\psi$ for a bounded subharmonic function ψ of z_1 alone, so $(\beta_1 + \mathrm{dd^c} w_0)^2 = \mathrm{dd^c}(\psi\,\mathrm{dd^c}\psi) = 0$ by the inductive definition of these products. The form $B = \sum_{j\geq 2} \beta_j$ is smooth and closed. Multilinear expansion therefore gives $A^{n-1} = B^{n-1} + (n-1)C \wedge B^{n-2}$, both for $A = A_\varepsilon$ with $C = f_\varepsilon \beta_1$ and for $A = A_0$ with $C = \beta_1 + \mathrm{dd^c} w_0$. Since $f_\varepsilon \beta_1 \to \beta_1 + \mathrm{dd^c} w_0$ weakly, we obtain $T_\varepsilon \to T_0 = A_0^{n-1}/V$ weakly. This current is positive, closed and normalized by $\int_X \beta \wedge T_0 = 1$.

To compute $\Gamma_0(T_0)$, take for ν in Lemma 27.17.1 the trace measure $\beta \wedge T_0$. The coefficient matrix Q of T_0 is then a positive multiple of the first matrix below on $\{x_1 \neq 0\}$, which carries the absolutely continuous part, and of the second on $\{x_1 = 0\}$, which carries the singular part:

$$\mathrm{diag}(1, 0, \ldots, 0), \qquad \mathrm{diag}(0, 1, \ldots, 1).$$

Their ranks are respectively one and $n-1$, both less than n. Hence $\det Q = 0$ almost everywhere for ν, and formula (27.73) gives $\Gamma_0(T_0) = 0$. This is not the Monge–Ampère measure of w_0, which has total mass V by Theorem 2.2.2.

Away from $x_1 = 0$, $f_\varepsilon \to 0$ pointwise, while its integral against $\mathrm{d}V$ remains one. Thus the functions $k_{T_\varepsilon}^{n/(n-1)} = V^{-1/(n-1)} f_\varepsilon$ are not uniformly integrable: almost-everywhere convergence to zero and uniform integrability would force their fixed positive integrals to vanish. Since $\Gamma_0(T_0) = 0$, the functional $T \mapsto \int_X k_T^{n/(n-1)}\,\mathrm{d}V$ is not continuous along $T_\varepsilon \to T_0$, although the potentials converge uniformly. The measures themselves satisfy $\Gamma_0(T_\varepsilon) \to 0$ in total mass: on the complement of a neighborhood of $\{x_1 = 0\}$ the densities converge uniformly to zero, and on the neighborhood Hölder's inequality and the uniform $L^{n/(n-1)}$ bound make their integrals small when the neighborhood is thin. Thus the measures $\Gamma_0(T_\varepsilon)$ converge to $\Gamma_0(T_0)$; only the integrals of the critical powers of their densities fail to converge.□

The function p_ε for which T_ε is maximizing varies with ε. The construction concerns the complete profile (27.74) and the critical power integral; it does not give a sequence of exact maximizers for one fixed smooth density.

27.18 Vanishing lower mass

By (10.12), the lower mass $\nu_-(h)$ vanishes as soon as there are Gauduchon metrics η whose degrees $\int_X h \wedge \eta^{n-1}$ stay bounded while their volumes $\int_X \eta^n$ tend to infinity. We construct such metrics on a compact threefold. Writing $\Gamma = \eta^2$, the volume of η is a determinant integral of the coefficient matrix of the positive $\mathrm{dd^c}$-closed form Γ. The construction first concentrates the trace of a normalized form Γ in one direction near a hypersurface, and then adds an average of compact rational curves that fills the two complementary directions. The degree stays bounded, while the determinant, and hence the volume of the positive root, becomes large. The smooth equation then converts these metrics into potentials whose masses tend to zero.

Theorem 27.18.1 *There is a connected compact complex threefold X such that, for every Hermitian form h on X, there are smooth real functions u_j satisfying*

$$h + \mathrm{dd^c} u_j > 0, \qquad \sup_X u_j = 0, \qquad \int_X (h + \mathrm{dd^c} u_j)^3 \longrightarrow 0.$$

In particular, $\nu_-(h) = 0$ for every Hermitian form h on this manifold.

27.18.1 The elliptic quotient

Write points of $\mathbb{P}^3$ as $[z : w]$, with $z, w \in \mathbb{C}^2$, and set

$$\Omega = \{[z : w] : z \neq 0,\ w \neq 0\}, \qquad q = \mathrm{e}^{2\pi}, \qquad g[z : w] = [z : qw].$$

Thus g is an automorphism of Ω; let $\langle g \rangle$ be the cyclic group it generates. This group acts freely. Indeed, if g^k fixed a point, the same scalar would have to equal 1 on the nonzero block z and q^k on the nonzero block w, which forces $k = 0$.

The function

$$t = \log \frac{\|w\|^2}{\|z\|^2}$$

is well defined on Ω and satisfies $t \circ g = t + 4\pi$. Consequently the action is properly discontinuous: t is bounded on compact subsets, so only finitely many translates of one compact set can meet another. Moreover, the strip $\{0 \leq t \leq 4\pi\}$ is compact. In homogeneous coordinates of total norm one, its defining inequalities bound both blocks away from zero, so it is a closed subset of $\mathbb{P}^3$ contained in Ω. Since the strip meets every orbit,

$$X = \Omega / \langle g \rangle$$

is a connected compact complex threefold.

The map $[z : w] \mapsto ([z], [w])$ induces a holomorphic submersion $X \to \mathbb{P}^1 \times \mathbb{P}^1$. Using a logarithm of the scalar relating two points with the same image, one sees that its fibers are the elliptic curves $E = \mathbb{C}/(2\pi\mathbb{Z} + 2\pi\mathrm{i}\mathbb{Z})$. Let α, β be the pullbacks

of the Fubini–Study forms of the two factors, each normalized to have integral one. Although t does not descend to X, its differentials $\mathrm{d}t$ and $\mathrm{d}^{\mathrm{c}}t$ do. The form

$$\omega = \alpha + \beta + \mathrm{d}t \wedge \mathrm{d}^{\mathrm{c}}t \tag{27.77}$$

is Hermitian: $\alpha + \beta$ is semipositive with kernel the vertical line, while $\mathrm{d}t \wedge \mathrm{d}^{\mathrm{c}}t$ is semipositive and strictly positive on the vertical line.

On the open set $\{z_0 w_0 \neq 0\}$ we use the coordinates

$$x = \frac{z_1}{z_0}, \qquad y = \frac{w_1}{w_0}, \qquad \zeta = \log \frac{w_0}{z_0} \in E.$$

Near $0 \in E$, the last coordinate is represented by an ordinary complex number. For any one of these coordinates, denoted by v, put $\sigma_v = \frac{\mathrm{i}}{2\pi}\mathrm{d}v \wedge \mathrm{d}\bar{v}$. Choose a smooth probability density p relative to σ_v that is positive on $\{|v| < 1\}$ and vanishes on $\{|v| \geq 1\}$. Choose also a smooth function χ_0 with values in $[0, 1]$, supported in $\{|v| < 1\}$ and equal to one on $\{|v| \leq 1/2\}$, and write $\chi(x, y) = \chi_0(x)\chi_0(y)$. For $0 < \delta < 1/100$ define

$$B_\delta = \chi(x, y)\delta^{-2} p(\zeta/\delta)\sigma_\zeta,$$

extended by zero. This is a smooth semipositive $(1, 1)$-form on X, because the chosen functions are smooth with compact support in their coordinate domains. As $\delta \downarrow 0$ it concentrates on the hypersurface $\{\zeta = 0\}$.

The first step shows that normalized positive dd^{c}-closed $(2, 2)$-forms can pair arbitrarily strongly with B_δ.

Lemma 27.18.1 *For $\delta > 0$ put*

$$C_\delta = \sup\left\{\int_X B_\delta \wedge \Theta : \Theta > 0,\ \mathrm{dd}^{\mathrm{c}}\Theta = 0,\ \int_X \omega \wedge \Theta = 1\right\},$$

the supremum being taken over smooth real $(2, 2)$-forms Θ. These numbers are finite and tend to infinity as $\delta \downarrow 0$.

Proof For fixed δ, choose a constant $M_\delta > 0$ with $B_\delta \leq M_\delta \omega$; this shows that $C_\delta \leq M_\delta$. Suppose instead that $C_{\delta_l} \leq C$ along a sequence $\delta_l \to 0$. Then $\int_X (C\omega - B_{\delta_l}) \wedge \Theta \geq 0$ for every admissible Θ, and in particular for the squares of Gauduchon metrics. By Theorem 4.3.1, there are quasi-plurisubharmonic functions v_l such that

$$C\omega + \mathrm{dd}^{\mathrm{c}} v_l \geq B_{\delta_l}, \qquad \sup_X v_l = 0.$$

By Theorem 1.4.1, a subsequence converges in L^1 to a $C\omega$-plurisubharmonic function v, not identically $-\infty$. Since $\delta^{-2}p(\zeta/\delta)\sigma_\zeta$ is an approximate identity in the variable ζ,

$$B_{\delta_l} \rightharpoonup \chi[\zeta = 0], \qquad C\omega + \mathrm{dd}^{\mathrm{c}} v \geq \chi[\zeta = 0].$$

On a coordinate ball where $\chi = 1$, add to v a smooth strictly plurisubharmonic function whose complex Hessian dominates $C\omega$. The resulting plurisubharmonic

function has a current that dominates $[\zeta = 0]$, so its Lelong number is at least one at every point of an open subset of this hypersurface. Smooth corrections do not change Lelong numbers, so the same holds for v. Siu's analyticity theorem, recalled in Section 1.13.1, shows that $Z = \{\nu(v, \cdot) \geq 1\}$ is an analytic subset of X. It is a proper subset because v is not identically $-\infty$.

Let $\widetilde{Z}$ be the inverse image of Z in Ω. It contains an open subset of $H_0 \cap \Omega$, where $H_0 = \{w_0 = z_0\}$. This surface is a projective plane with two points removed, so it is connected and irreducible, and the identity principle gives $H_0 \cap \Omega \subset \widetilde{Z}$. Since $\widetilde{Z}$ is invariant under g, also

$$H_k \cap \Omega \subset \widetilde{Z}, \qquad H_k = \{w_0 = q^k z_0\}, \qquad k \in \mathbb{Z}.$$

Now use the chart $z_1 = 1$ near the point $[0 : 1 : 1 : 0] \in \Omega$. There H_k is the graph $z_0 = q^{-k} w_0$. For each fixed (w_0, w_1) near $(1, 0)$, every local holomorphic defining function of $\widetilde{Z}$ vanishes at the distinct values $z_0 = q^{-k} w_0$, which tend to zero as $k \to +\infty$. The one-variable identity theorem makes each defining function vanish identically on a polydisk. Thus $\widetilde{Z}$ contains a nonempty open subset of the connected manifold Ω, contradicting properness. Hence no subsequence of C_δ stays bounded as $\delta \downarrow 0$. □

By the lemma we can choose $\delta_j \downarrow 0$ and then, by the definition of the supremum, smooth strictly positive dd^{c}-closed $(2,2)$-forms Θ_j with

$$\int_X \omega \wedge \Theta_j = 1, \qquad J_j = \int_X B_{\delta_j} \wedge \Theta_j > j^4. \tag{27.78}$$

27.18.2 Averaging the trace and compact curves

A large pairing with B_{δ_j} means that Θ_j has a large coefficient in one direction, but possibly only on a small set. Averaging by translations spreads this coefficient over a whole product region of size comparable to δ_j in the vertical direction. A second average, this time of compact rational curves, fills the two complementary directions, which is needed for a lower bound on a determinant. Fix j and write $\delta = \delta_j$.

For complex parameters a, b, τ, the automorphisms

$$T_{a,b,\tau}[z_0 : z_1 : w_0 : w_1] = [z_0 : z_1 + a z_0 : \mathrm{e}^{\tau} w_0 : \mathrm{e}^{\tau}(w_1 + b w_0)]$$

of Ω commute with g, and in the product chart they act by $(x, y, \zeta) \mapsto (x + a, y + b, \zeta + \tau)$.

Average $T^*_{a,b,\tau}\Theta_j$ against the probability density

$$3^{-4}p(a/3)p(b/3)(3\delta)^{-2}p(\tau/(3\delta))\sigma_a\wedge\sigma_b\wedge\sigma_\tau,$$

and denote the result by $\widehat{\Theta}_j$. It is smooth, strictly positive and dd^{c}-closed. The parameters range in a fixed compact set of automorphisms, so $(T^{-1}_{a,b,\tau})^*\omega\le C\omega$ with one constant C, and a change of variables gives

$$\int_X \omega\wedge\widehat{\Theta}_j\le C. \tag{27.79}$$

Let F_j be the coefficient of $\sigma_x\wedge\sigma_y$ in the restriction of Θ_j to the slices $\{\zeta=\text{constant}\}$, and let $\widehat{F}_j$ be the corresponding coefficient of $\widehat{\Theta}_j$. Positivity gives $F_j\ge 0$, and since the translations preserve $\sigma_x\wedge\sigma_y$, $\widehat{F}_j$ is an average of translates of F_j. At a point of

$$V_j=\{|x|,|y|<1/4,\ |\zeta|<\delta/4\},$$

every translation carrying it into $\operatorname{supp}B_\delta$ has $|a|,|b|\le 5/4$ and $|\tau|\le 5\delta/4$. On this range the first two averaging factors are bounded below by a positive constant, and the third by a positive constant times δ^{-2}. On the other hand, $\chi\le 1$ and p is bounded, so the number J_j of (27.78) is at most a constant times δ^{-2} times the integral of F_j over $\operatorname{supp}B_\delta$. Comparing the two integrals gives

$$\widehat{F}_j\ge cJ_j\quad\text{on } V_j, \tag{27.80}$$

with $c>0$ independent of j.

The second average fills the two horizontal directions. Put

$$f(a,b)=p(4(a-1))p(b/2),\qquad \rho_j(\tau)=4\delta^{-2}p(2\tau/\delta).$$

For parameters in the supports of these functions and $|u|\le 1/16$, consider the holomorphic maps $\iota_{\tau,u,a,b}\colon\mathbb{P}^1\to X$ defined by

$$\iota_{\tau,u,a,b}[z_0:z_1]=[z_0:z_1:\mathrm{e}^\tau(z_0+uz_1):\mathrm{e}^\tau(az_1+bz_0)]\in X. \tag{27.81}$$

For $0<\varepsilon\le 1/16$, average their integration currents against

$$\rho_j(\tau)f(a,b)\varepsilon^{-2}p(u/\varepsilon)\sigma_\tau\wedge\sigma_a\wedge\sigma_b\wedge\sigma_u,$$

and call the result $T_{j,\varepsilon}$. Define $T_{j,0}$ in the same way, but with $u=0$ and without the integral over u.

Lemma 27.18.2 *For $\varepsilon > 0$, the form $T_{j,\varepsilon}$ is smooth on X, semipositive and closed, and*

$$\int_X \omega \wedge T_{j,\varepsilon} \le C'$$

with C' independent of j and ε. For each fixed j, $T_{j,\varepsilon}$ converges smoothly to $T_{j,0}$ on compact subsets of the product chart as $\varepsilon \downarrow 0$. In this chart

$$T_{j,0} = \rho_j(\zeta)\sigma_\zeta \wedge K_f, \qquad K_f = \int_{\mathbb{C}} f(a, y - ax)\frac{\mathrm{i}}{2\pi}(\mathrm{d}y - a\,\mathrm{d}x) \wedge (\mathrm{d}\bar{y} - \bar{a}\,\mathrm{d}\bar{x})\,\sigma_a, \tag{27.82}$$

and on $\{|x|, |y| \le 1/4\}$ the form K_f is bounded below by a positive multiple of $\sigma_x + \sigma_y$.

Proof On the parameter supports, $|a - bu| \ge 3/4 - 2/16 = 5/8$. Hence the matrix $\left(\begin{smallmatrix} 1 & u \\ b & a \end{smallmatrix}\right)$ is uniformly invertible, and the projective line (27.81) lies in Ω. Distinct deck translates of this line are disjoint: equality at a point fixes $[z_0 : z_1]$, and invertibility of the matrix then forces the deck scalar to be one. Thus each map is an embedding onto a compact rational curve in X.

The evaluation map from $\mathbb{P}^1$ times the open parameter domain to X is a holomorphic submersion. Indeed, after fixing a projective gauge, variations of u and a span the two w-coordinates at points with $z_1 \neq 0$, variations of τ and b span them at points with $z_1 = 0$, and the coordinate on $\mathbb{P}^1$ supplies the remaining direction. For $\varepsilon > 0$, the average is the integral along the fibers of this submersion of a smooth compactly supported parameter density, which proves smoothness. As an average of closed positive integration currents, $T_{j,\varepsilon}$ is closed and semipositive.

All curve parameters lie in one compact subset of the domain where the matrix is invertible, so the ω-areas of the curves are uniformly bounded. The weights in τ and u have mass one, and the weight in (a, b) has a fixed finite mass. This proves the degree bound.

When $u = 0$, the curve is given in the product chart by $\zeta = \tau$ and $y = ax + b$. Integrating in b and τ yields (27.82); the $(1,1)$-factor there is the integration current of the graph of $y = ax + b$. On $\{|x|, |y| \le 1/4\}$, the density $f(a, y - ax)$ is bounded below by a positive constant for a in a fixed small disk about 1. The covectors $\mathrm{d}y - a\,\mathrm{d}x$, for a in that disk, have no common nonzero annihilator, so integrating their squared absolute values and using compactness of the unit sphere in the (x, y)-directions gives the asserted lower bound.

For general u, the curve is given by

$$\zeta = \tau + \log(1 + ux), \qquad y = \frac{ax + b}{1 + ux}.$$

On a fixed compact subset of the chart, $1 + ux$ stays away from zero for all sufficiently small u. Solving for the parameters gives $\tau = \zeta - \log(1 + ux)$ and $b = (1 + ux)y - ax$. Regarding ρ_j as a smooth function on E, this change of variables also applies near the boundary of its support. The Jacobians, the density coefficients and any fixed number of their derivatives depend smoothly on u. Averaging against the approximate

identity in u therefore gives local C^∞ convergence. This convergence is needed only for each fixed j; it is not uniform in j. $\square$

27.18.3 The volume of the positive root

The two averages contribute complementary blocks of the coefficient matrix of a $(2,2)$-form: the translated forms give a large vertical entry, and the curves give a positive horizontal block. A Schur-complement estimate combines them into a lower bound for the volume of the positive root. In the coframe (x, y, ζ) put $\mathrm{d}V_0 = \sigma_x \wedge \sigma_y \wedge \sigma_\zeta$. For a real $(1,1)$-form ξ, let H_ξ be its coefficient matrix in this normalization. A real $(2,2)$-form Θ corresponds to a Hermitian matrix Q_Θ, which we call its cofactor matrix, by

$$\xi \wedge \Theta = \operatorname{tr}(H_\xi Q_\Theta)\, \mathrm{d}V_0.$$

Strict positivity of Θ is equivalent to $Q_\Theta > 0$.

If $\eta^2 = \Theta$ and $H = H_\eta$, expanding the wedge products gives

$$Q_\Theta = 2(\det H)H^{-1}, \qquad \eta^3 = \frac{3}{\sqrt{2}}(\det Q_\Theta)^{1/2}\, \mathrm{d}V_0.$$

Conversely, these formulas uniquely recover a smooth positive H from $Q_\Theta > 0$; this is the three-dimensional case of the root construction preceding the proof of Theorem 4.3.1. We write $\Theta^{3/2} = \eta^3$.

Lemma 27.18.3 *There are smooth strictly positive* dd^{c}*-closed* $(2,2)$*-forms* Γ_j *such that*

$$\int_X \omega \wedge \Gamma_j \le C_2, \qquad R_j = \int_X \Gamma_j^{3/2} \longrightarrow +\infty.$$

Proof Write the cofactor matrix of $\widehat{\Theta}_j$ in block form, with a 2×2 Hermitian block A_j in the (x, y)-directions and a column $\kappa_j \in \mathbb{C}^2$:

$$Q_j = \begin{pmatrix} A_j & \kappa_j \\ \kappa_j^* & \widehat{F}_j \end{pmatrix}.$$

The cofactor matrix of $T_{j,0}$ has the form $\operatorname{diag}(B_j, 0)$. Let I_2 denote the 2×2 identity matrix. The lower bound for K_f and the definition of ρ_j give a constant $c_0 > 0$ such that $B_j \ge c_0 \delta_j^{-2} I_2$ on V_j. Since $Q_j > 0$, the Schur complement $A_j - \kappa_j \kappa_j^* / \widehat{F}_j$ is positive definite, and

$$\det\bigl(Q_j + \operatorname{diag}(B_j, 0)\bigr) = \widehat{F}_j \det\left(B_j + A_j - \frac{\kappa_j \kappa_j^*}{\widehat{F}_j}\right) \ge \widehat{F}_j \det B_j,$$

by monotonicity of the determinant on positive Hermitian matrices. Combining the root formula with (27.80), we obtain

$$(\widehat{\Theta}_j + T_{j,0})^{3/2} \geq c\delta_j^{-2}\sqrt{J_j}\,\mathrm{d}V_0 \quad \text{on } V_j.$$

The smaller region $W_j = \{|x|, |y| < 1/8,\ |\zeta| < \delta_j/8\}$ has $\mathrm{d}V_0$-volume equal to a fixed positive constant times δ_j^2. Consequently

$$I_j = \int_{W_j} (\widehat{\Theta}_j + T_{j,0})^{3/2} \geq c_1\sqrt{J_j} > c_1 j^2.$$

For this fixed j, Lemma 27.18.2 gives smooth convergence of $T_{j,\varepsilon}$ to $T_{j,0}$ on the closure of W_j. The cofactor matrix of $\widehat{\Theta}_j + T_{j,0}$ is positive definite there, so the integrand of the root volume depends continuously on these smooth coefficients. We can therefore choose $0 < \varepsilon_j \leq 1/16$ so small that

$$\int_{W_j} (\widehat{\Theta}_j + T_{j,\varepsilon_j})^{3/2} \geq I_j/2.$$

Set $\Gamma_j = \widehat{\Theta}_j + T_{j,\varepsilon_j}$. It is smooth, strictly positive and dd^{c}-closed. Its degree is at most $C_2 = C + C'$, by (27.79) and Lemma 27.18.2, whereas its root volume is at least $c_1 j^2/2$. □

Proof (of Theorem 27.18.1) Fix a Hermitian form h on X, and choose $A > 0$ with $h \leq A\omega$. Let η_j be the positive root of Γ_j, so that $\eta_j^2 = \Gamma_j$. These metrics are Gauduchon, and

$$P_j = \int_X h \wedge \eta_j^2 \leq AC_2, \qquad R_j = \int_X \eta_j^3 \longrightarrow +\infty.$$

Set $\mu_j = \eta_j^3/R_j$ and solve the smooth equation in the background h:

$$h_j = h + \mathrm{dd}^{\mathrm{c}} u_j > 0, \qquad h_j^3 = c_h(\mu_j)\mu_j, \qquad \sup_X u_j = 0.$$

Since $D_{\mu_j}(\eta_j) = R_j^{2/3}$, using η_j as a test in Theorem 10.8.1 gives

$$0 < \int_X h_j^3 = c_h(\mu_j) = \mathcal{V}_{\mu_j}(h) \leq \frac{P_j^3}{R_j^2} \longrightarrow 0.$$

The potentials u_j are smooth, hence bounded, so $v_-(h) = 0$. □

27.18.4 Consequences for dimension and equations

The example occurs in the smallest possible dimension: on a surface, comparison with a Gauduchon metric gives positive lower mass. Products with projective spaces carry the threefold degeneration to every higher dimension, and a product with the Hopf threefold produces, in addition, unbounded upper mass.

Corollary 27.18.1 *Every Hermitian form on a compact complex manifold of dimension at most two has positive lower mass. In every dimension $n \geq 3$ there is a connected compact complex manifold on which every Hermitian form has zero lower mass.*

Proof In dimension one the mass is $\int_X h > 0$, and the surface case is Corollary 27.17.1. In higher dimensions, Corollary 27.7.1 passes from a split metric to every Hermitian metric.

For $n = 3$ this is Theorem 27.18.1. For $n > 3$, take the product of the threefold X with $\mathbb{P}^{n-3}$, let κ be a Fubini–Study form on the projective factor, and consider the split metric $h_0 = \omega + \kappa$, with pullbacks understood. For the potentials u_j given by the theorem for the background ω,

$$\int_{X\times\mathbb{P}^{n-3}} (h_0 + \mathrm{dd^c} u_j)^n = \binom{n}{3} \left(\int_X (\omega + \mathrm{dd^c} u_j)^3 \right) \left(\int_{\mathbb{P}^{n-3}} \kappa^{n-3} \right) \longrightarrow 0.$$

For another Hermitian form h, choose A with $h \leq A h_0$. Corollary 27.7.1, applied to $h_0 \geq A^{-1} h$, gives $v_-(h) \leq A^n v_-(h_0) = 0$. □

Corollary 27.18.2 *In every dimension $n \geq 6$ there is a connected compact complex manifold with*

$$v_-(h) = 0, \qquad v_+(h) = +\infty$$

for every Hermitian form h.

Proof Take the product of the threefold above with the Hopf threefold of Chapter 9, with a split metric. Potentials pulled back from the first factor make the masses tend to zero, while the smooth potentials of Theorem 9.0.1, pulled back from the second factor, make them tend to infinity. In each case the product integral is $\binom{6}{3}$ times the product of the two factor masses. Corollary 27.7.1 handles the lower mass for every Hermitian form, and the metric independence in Proposition 7.5.1 handles the upper mass. Products with projective spaces give every higher dimension by the same expansion of wedge powers. □

Finally, fix a smooth positive probability measure μ on the threefold X. The smooth equation still gives $\mathcal{V}_\mu(h) > 0$ for each Hermitian form h. With $\mathscr{M}$ as in Section 10.8.2 and λ_* as in Section 27.14, Theorem 10.8.2 gives

$$\inf_{\nu\in\mathscr{M}} \mathcal{V}_\nu(h) = 0, \qquad \lambda_*(h) = 0.$$

Thus even a smooth strictly positive background can have vanishing infimum over densities. For the metrics η_j and numbers R_j, P_j in the proof of Theorem 27.18.1, the measures $\mu_j = \eta_j^3/R_j$ satisfy $\mathcal{V}_{\mu_j}(h) \le P_j^3/R_j^2 \to 0$. Hence the densities μ_j/μ cannot be uniformly bounded above: a bound $\mu_j \le M\mu$ would give $\mathcal{V}_{\mu_j}(h) \ge M^{-1}\mathcal{V}_{\mu}(h) > 0$, by the comparison of denominators in the proof of Proposition 10.8.2.

The example also shows that the endpoint estimate in Theorem 27.7.2 fails for the full family of normalized Gauduchon currents: the supremum of the $L^{3/2}$ norms of their determinant densities is infinite, although every subcritical L^q norm, $1 < q < 3/2$, remains uniformly bounded by Theorem 27.7.1. For the same fixed background, Theorem 27.7.3 and Corollary 27.7.3 give uniform L^∞ bounds for the exact maximizers and their adjoint probabilities in a small $C^{2,\alpha}$ family of smooth densities. A larger sufficient family is given by Theorem 27.7.4: uniformly comparable smooth solution forms yield uniform bounds for both densities, without derivative bounds on the varying Hessian. Endpoint estimates for arbitrary bounded families of continuous densities remain open.

27.18.5 Primitive elliptic bundles

The threefold of Theorem 27.18.1 belongs to a family of elliptic bundles over compact Kähler surfaces, built from line bundles with primitive curvature. On each of them the variation of the top-degree mass is a vertical Dirichlet integral. This gives the exact upper mass and a positive lower bound for separated potentials. Whether the lower mass vanishes on the other bundles of the family is open (Problem 35.3.3); over a base without curves, the averages of rational curves used in the proof of Theorem 27.18.1 are not available.

Let (B, η) be a connected compact Kähler surface and let $L \to B$ be a holomorphic line bundle such that

$$\int_B c_1(L) \wedge \eta = 0.$$

Choose a smooth Hermitian metric on L whose curvature is primitive with respect to η. Such a metric exists: for an arbitrary metric, the trace of the curvature with respect to η has mean zero by the condition above, and the scalar Poisson equation gives a smooth change of metric that cancels this trace. Let c be the resulting smooth closed curvature representative of $-c_1(L)$, so that $\eta \wedge c = 0$. Write $\pi\colon L \to B$ for the bundle projection and w for a nonzero vector in a fiber, with norm taken in the chosen metric. On the complement $L^\times$ of the zero section, put

$$t = \log |w|^2, \qquad \mathrm{dd^c} t = \pi^* c.$$

For $q > 1$, form the quotient

$$X = L^\times / \langle w \mapsto qw \rangle, \qquad \pi\colon X \longrightarrow B, \qquad T = 2\log q, \qquad A = \int_B \eta^2,$$

where we use the same symbol π for the induced projection. The action increases t by T. It is free and properly discontinuous, and the compact annulus $1 \le |w| \le q$ meets every orbit. Hence X is a connected compact complex threefold; in this subsection it replaces the general manifold of the chapter. Over a trivializing open set $U \subset B$, a local holomorphic frame of L identifies the fibration with $U \times (\mathbb{C}^*/q^{\mathbb{Z}})$.

Since $\mathrm{d}t$ and $\mathrm{d}^{\mathrm{c}}t$ are invariant under the action, they descend to X, and we set

$$D = \mathrm{d}t \wedge \mathrm{d}^{\mathrm{c}}t, \qquad \omega = \pi^*\eta + D.$$

The form D has rank one; it is semipositive and positive on the vertical line, and its kernel projects isomorphically onto the tangent space of B. Hence ω is Hermitian.

Proposition 27.18.1 *The metric ω is balanced and satisfies*

$$\mathrm{dd}^{\mathrm{c}}\omega = -\pi^*(c^2) \ge 0, \qquad \int_X \omega^3 = 3TA.$$

For every smooth real u,

$$\int_X (\omega + \mathrm{dd}^{\mathrm{c}}u)^3 = 3TA - 3\int_X \mathrm{d}u \wedge \mathrm{d}^{\mathrm{c}}u \wedge \bigl(-\pi^*(c^2)\bigr). \tag{27.83}$$

In particular $v_+(\omega) = 3TA$. The canonical bundles satisfy $K_X \simeq \pi^*K_B$. If $c_1(L) \neq 0$ in real cohomology, X admits neither a pluriclosed metric nor a closed Kähler current. If $c_1(L) = 0$ in real cohomology, the chosen ω is Kähler.

Proof Write $a = \mathrm{d}t$ and $b = \mathrm{d}^{\mathrm{c}}t$. Then $\mathrm{d}a = 0$, $\mathrm{d}b = \pi^*c$, $\mathrm{d}^{\mathrm{c}}a = -\pi^*c$ and $\mathrm{d}^{\mathrm{c}}b = 0$. Consequently

$$D^2 = 0, \qquad \mathrm{d}D = -a \wedge \pi^*c, \qquad \mathrm{dd}^{\mathrm{c}}D = -\pi^*(c^2),$$

$$\mathrm{d}(\omega^2) = -2\pi^*(\eta \wedge c) \wedge a = 0,$$

so ω is balanced. Since c is primitive, its eigenvalues relative to η are λ and $-\lambda$ for a function $\lambda \ge 0$, so $c^2 = -\lambda^2\eta^2$. This proves the asserted sign of $\mathrm{dd}^{\mathrm{c}}\omega$.

To compute the volume, fix an elliptic fiber $E = \pi^{-1}(b_0)$, where $b_0 \in B$. In a logarithmic fiber coordinate $x + \mathrm{i}y$, the real periods are $\log q$ and 2π, while t equals $2x$ up to an additive constant and $D|_E = \pi^{-1}\mathrm{d}x \wedge \mathrm{d}y$. Thus $\int_E D = T$. Since $\omega^3 = 3\pi^*(\eta^2) \wedge D$, the volume formula follows.

For the mass identity, expand the cube in (27.83). The terms with one and with three factors $\mathrm{dd}^{\mathrm{c}}u$ integrate to zero, because ω^2 is closed and $\mathrm{dd}^{\mathrm{c}}u$ is exact. For the remaining term, put $S = \mathrm{dd}^{\mathrm{c}}\omega$. Integration by parts gives

$$\int_X (\mathrm{dd}^{\mathrm{c}}u)^2 \wedge \omega = \int_X u\,\mathrm{dd}^{\mathrm{c}}u \wedge S = -\int_X \mathrm{d}u \wedge \mathrm{d}^{\mathrm{c}}u \wedge S,$$

where the last equality uses $\mathrm{d}S = 0$. This proves the identity. Its last integral is nonnegative, and $u = 0$ attains the upper bound $3TA$. Since smooth and bounded potentials have the same mass extrema, by Theorem 10.8.2, this proves $v_+(\omega) = 3TA$.

If $c_1(L) = 0$ in real cohomology, then c is exact and Stokes' theorem gives $\int_B c^2 = 0$. The pointwise identity $c^2 = -\lambda^2\eta^2$ then forces $c = 0$, and ω is closed. Otherwise $S = -\pi^*(c^2)$ is nonzero, smooth and semipositive. For a pluriclosed metric γ, Stokes' theorem would give

$$0 < \int_X \gamma \wedge S = \int_X \omega \wedge \mathrm{dd^c}\gamma = 0,$$

a contradiction. Likewise, a closed positive current $R \geq \varepsilon\omega$ with $\varepsilon > 0$ would give $0 < \int_X R \wedge S = \int_X \omega \wedge \mathrm{dd^c}R = 0$. All these pairings are pairings of currents with smooth forms.

Finally, let w be a local holomorphic fiber coordinate given by a frame of L, and let σ be a local holomorphic two-form on B. The form $\pi^*\sigma \wedge \mathrm{d}w/w$ is invariant under the quotient action. Changing the frame of L adds to $\mathrm{d}w/w$ a horizontal one-form, whose wedge product with $\pi^*\sigma$ vanishes. These local sections of K_X therefore have the transition functions of π^*K_B, which proves $K_X \simeq \pi^*K_B$. □

For separated potentials, positivity bounds both the amplitude and the downward slope of the vertical derivative. The sharp variance bound under these two constraints determines how much mass such a potential can lose. For $M, T > 0$, set

$$F(M,T) = \begin{cases} T^2/12, & T \leq 2M, \\ M^2 - 4M^3/(3T), & T \geq 2M. \end{cases}$$

Lemma 27.18.4 *If a smooth real T-periodic function p has mean zero, $|p| < M$ and $p' > -1$, then*

$$\frac{1}{T}\int_0^T p^2 \,\mathrm{d}s < F(M,T).$$

The supremum of the left side over this class is $F(M,T)$.

Proof For $p = 0$ the inequality holds because $F(M,T) > 0$. Otherwise set $a = \max p > 0$, $-b = \min p < 0$ and $r = a + b$. Integrating $-p' < 1$ over a forward interval I from a maximum to the next minimum gives $r < |I| \leq T$; also $r < 2M$, since $|p| < M$. The function $G(s) = (a - s)(s + b)$ is nonnegative on the range of p. Since p has mean zero and $-p' < 1$, integration over I gives

$$abT - \int_0^T p^2 \,\mathrm{d}s = \int_0^T G(p) \,\mathrm{d}s \geq -\int_I G(p)p' \,\mathrm{d}s = \frac{r^3}{6}.$$

As $ab \leq r^2/4$, the average of p^2 is at most $r^2/4 - r^3/(6T)$. This expression increases strictly for $0 < r < T$, and its value at $r = \min\{2M, T\}$ is $F(M,T)$. Since $r < \min\{2M, T\}$, the strict inequality follows.

For sharpness, consider the following T-periodic profiles p_0. When $T \geq 2M$, let p_0 be constant equal to M and to $-M$ on two intervals of length $(T - 2M)/2$ each, joined by a descent of slope -1 and length $2M$, and close the period by an upward jump. When $T \leq 2M$, let $p_0(s) = T/2 - s$ for $0 \leq s < T$, with its upward

jump at the end of the period. In both cases p_0 has mean zero, $|p_0| \le M$, $p_0' \ge -1$ in the sense of distributions, and the average of p_0^2 is $F(M,T)$; both constructions apply when $T = 2M$. Convolution with a nonnegative periodic approximate identity preserves the mean and the bound on the distributional derivative, and multiplication by $1-\varepsilon$ makes both inequalities strict. As the convolution scale and ε tend to zero, the resulting smooth functions converge to p_0 in L^2, which proves that the supremum is $F(M,T)$. □

Proposition 27.18.2 *Suppose $c_1(L) \neq 0$ in real cohomology, and put*

$$\kappa = -\frac{\int_B c^2}{A} > 0.$$

Every smooth potential $u = f(t) + \pi^* v$ with f of period T, $v \in C^\infty(B,\mathbb{R})$ and $\omega + \mathrm{dd^c} u > 0$ satisfies

$$\int_X (\omega + \mathrm{dd^c} u)^3 > 3TA\big(1 - \kappa F(\kappa^{-1/2}, T)\big) > 0.$$

Let $\lambda \ge 0$ be the nonnegative eigenvalue function of c relative to η, so $c^2 = -\lambda^2\eta^2$, and put $M_0 = 1/\max_B \lambda$. For the subclass $v = 0$, the exact mass infimum is

$$3TA\big(1 - \kappa F(M_0, T)\big) > 0.$$

It is approached by smooth potentials $f(t)$ with $\omega + \mathrm{dd^c} f(t) > 0$, and it is not attained by such potentials.

Proof Put $p = f'$. The chain rule gives

$$\omega + \mathrm{dd^c} u = \pi^*(\eta + pc + \mathrm{dd^c} v) + (1 + p')D.$$

Positivity on the vertical line implies $p' > -1$. Over every point of B, the function t takes every value modulo T, so restriction to $\ker \partial t$ shows that $\eta + pc + \mathrm{dd^c} v$ is a Kähler form on B for every value of p. Its square has integral

$$\int_B (\eta + pc + \mathrm{dd^c} v)^2 = A(1 - \kappa p^2) > 0,$$

since primitivity removes the mixed term and Stokes' theorem removes the exact terms. Hence $|p| < \kappa^{-1/2}$.

Only the vertical derivative enters (27.83), because $-\pi^*(c^2)$ has top degree in the base directions. For a T-periodic function F_0, integration over a fiber gives $\int_E F_0(t) D = \int_0^T F_0(s)\,\mathrm{d}s$, and therefore

$$\int_X (\omega + \mathrm{dd^c} u)^3 = 3TA\left(1 - \frac{\kappa}{T}\int_0^T p^2\,\mathrm{d}s\right).$$

Now apply Lemma 27.18.4 with $M = \kappa^{-1/2}$. The formula for F gives $F(M,T) < M^2$, which proves positivity of the bound.

For $v = 0$, diagonalizing c with respect to η shows that the horizontal positivity condition is exactly $|p| < M_0$; together with $p' > -1$ it also suffices for positivity of the full form. Moreover κ is the η^2-average of λ^2, so $\kappa \le M_0^{-2}$. Lemma 27.18.4 with $M = M_0$ therefore proves the claimed positive lower bound and its nonattainment. The smooth approximating functions in its proof have mean zero and hence periodic primitives f_ε, and the potentials $f_\varepsilon(t)$ prove that the bound is sharp. □

The proof of the first bound does not show that it is sharp for separated potentials over an arbitrary base: positivity of the classes $[\eta] + p[c]$ alone does not provide one function v that makes all the representatives $\eta + pc + \mathrm{dd^c} v$ positive. For the threefold of Theorem 27.18.1, which is the bundle over $\mathbb{P}^1 \times \mathbb{P}^1$ with $q = \mathrm{e}^{2\pi}$, the two bounds of Proposition 27.18.2 agree and equal 8, so the first bound is sharp there (Section 27.18.6).

A base without curves rules out the averages of horizontal curves used in the proof of Theorem 27.18.1. The following construction uses the period theorem for K3 surfaces and the description of their Kähler cone [Huy16, Theorems 7.4.1 and 8.5.2 of the online version].

Example 27.18.1 There is a K3 surface B with $\mathrm{Pic}(B) = \mathbb{Z}[L]$, $c_1(L)^2 = -4$, and a Kähler form η orthogonal to $c_1(L)$. It has no compact curves. The associated X has trivial canonical bundle, a balanced metric of finite upper mass, and no pluriclosed metric or closed Kähler current. It has no divisors, and every compact irreducible curve is an elliptic fiber.

Proof In a hyperbolic plane of the K3 lattice, choose a basis e, f with $e^2 = f^2 = 0$ and $e \cdot f = 1$, and put $\ell = e - 2f$. This vector is primitive and has square -4. Choose a period in $\ell^\perp$ outside the hyperplanes perpendicular to integral vectors not in $\mathbb{Z}\ell$. Such a period exists by Baire's theorem: the period domain in $\ell^\perp$, of signature $(3, 18)$, is nonempty and each excluded hyperplane is proper. The period theorem and the description of integral $(1,1)$-classes give the asserted Picard group. An irreducible curve would have nonzero class, because its Kähler area is positive, and hence square $-4m^2 < -2$ for some integer $m \neq 0$, contradicting adjunction. Thus there are no curves.

The orthogonal complement of ℓ in $H^{1,1}(B, \mathbb{R})$ has signature $(1, 18)$. Choose a class of positive square there, with sign chosen so that it lies in the component of the positive cone containing the Kähler classes. Since there are no rational curves, the Kähler-cone theorem shows that this class is Kähler; choose η in it. Proposition 27.18.1 then gives the assertions about the metric and the canonical bundle.

If a divisor $Z \subset X$ dominated B, the push-forward of its integration current would be $\pi_*[Z] = m$ for a positive generic degree m. Stokes' theorem and the projection formula would then give

$$0 = \int_X [Z] \wedge \mathrm{dd^c}\omega = m \int_B (-c^2) > 0.$$

Thus no divisor dominates B. The image of any other irreducible divisor would be a curve or a point; the base has no curves, and a fiber has dimension one, so both alternatives are impossible. Likewise, the image of an irreducible compact curve is a point, and the curve is therefore an entire elliptic fiber. □

Put $V = 3TA$. The root volumes of complementary tests can be computed explicitly for one radial family. For smooth real T-periodic functions a, g, consider

$$\Gamma = a(t)\eta^2 + g(t)\eta \wedge D.$$

Proposition 27.18.3 *The form Γ is strictly positive and dd^{c}-closed exactly when a is a positive constant and $g > 0$. For such a form put $\bar{g} = T^{-1}\int_0^T g(s)\,\mathrm{d}s$ and $d_\Gamma = \int_X \omega \wedge \Gamma$. Its positive square root and volume are*

$$\gamma = \sqrt{a}\,\eta + \frac{g(t)}{2\sqrt{a}}D, \qquad \int_X \gamma^3 = \frac{3}{2}TA\sqrt{a}\,\bar{g} \le \frac{d_\Gamma^{3/2}}{\sqrt{V}}.$$

Equality holds exactly when $\bar{g} = 2a$, without requiring g to be constant. In particular every degree-one test in this family has root volume at most $V^{-1/2}$.

Proof Since $\eta \wedge c = 0$, the form $g\eta \wedge D$ is closed. The chain rule gives

$$\mathrm{dd}^{\mathrm{c}}(a\eta^2) = a'' D \wedge \eta^2,$$

the $a'c \wedge \eta^2$ term having excess base degree. Thus $\mathrm{dd}^{\mathrm{c}}\Gamma = 0$ forces $a'' = 0$, and periodicity makes a constant. In a positive coframe with $\eta = \sigma_1 + \sigma_2$, $D = \sigma_3$, the complementary coefficient matrix of Γ is $\mathrm{diag}(g, g, 2a)$. Hence positivity is exactly $a, g > 0$. The displayed γ is positive and squares to Γ. Multiplication and fiber integration give

$$d_\Gamma = TA(a + \bar{g}), \qquad \gamma^3 = \tfrac{3}{2}\sqrt{a}\,g\eta^2 \wedge D.$$

For fixed $d_\Gamma > 0$ set $s = d_\Gamma/(TA)$. Then $0 < a < s$, $\bar{g} = s - a$, and $\sqrt{a}(s - a)$ has its unique maximum at $a = s/3$. Substitution proves the bound and its equality condition. □

For the divisor-free threefold of Example 27.18.1, Theorem 4.5.1 gives divergent transverse pairings against normalized Gauduchon tests. The radial family above has bounded root volume at fixed degree, even when g concentrates arbitrarily strongly. Large transverse pairings do not by themselves give unbounded root volumes or a sequence of potentials with masses tending to zero.

27.18.6 A balanced metric and a finite upper mass

Although every Hermitian form on the threefold of Theorem 27.18.1 has zero lower mass, the metric ω of (27.77) is balanced and has finite upper mass. The threefold is a primitive elliptic bundle, so the loss of mass under a potential is a vertical Dirichlet energy along the elliptic fibers, and the degeneration cannot be realized by potentials that are the sum of a periodic function of t and a function on the base: their masses stay above 8. Put $D = \mathrm{d}t \wedge \mathrm{d}^{\mathrm{c}}t$, so that $\omega = \alpha + \beta + D$.

Let $B = \mathbb{P}^1 \times \mathbb{P}^1$, let α_B, β_B be the Fubini–Study forms of the two factors, normalized to have integral one, so that α, β are their pullbacks, and put $\eta = \alpha_B + \beta_B$. On B consider the line bundle $\mathcal{O}(1) \boxtimes \mathcal{O}(-1)$, whose fiber at $([z], [w])$ is the space of linear maps from the line $\mathbb{C}z$ to the line $\mathbb{C}w$, with the metric induced by the standard norms. Sending $[z : w] \in \Omega$ to the linear map $z \mapsto w$ identifies Ω with the complement of the zero section, the function $\log(\|w\|^2/\|z\|^2)$ with the logarithm of the squared norm, and g with multiplication by $q = \mathrm{e}^{2\pi}$. The curvature representative of minus the first Chern class is $c = \beta_B - \alpha_B$, and $\eta \wedge c = 0$. Hence X is the primitive elliptic bundle of Section 27.18.5 for these data, with the same function t, and ω is its metric $\pi^*\eta + D$. The constants of that subsection are

$$T = 4\pi, \qquad A = 2, \qquad c^2 = -2\alpha_B \wedge \beta_B, \qquad \kappa = 1,$$

and the eigenvalues of c relative to η are ± 1, so that $M_0 = 1$.

Corollary 27.18.3 *On the threefold X of Theorem 27.18.1, the Hermitian form $\omega = \alpha + \beta + D$ satisfies*

$$\mathrm{d}(\omega^2) = 0, \qquad \mathrm{d}\mathrm{d}^{\mathrm{c}}\omega = 2\alpha \wedge \beta \geq 0, \qquad \int_X \omega^3 = 24\pi.$$

For every smooth real function u,

$$\int_X (\omega + \mathrm{d}\mathrm{d}^{\mathrm{c}}u)^3 = 24\pi - 6\int_X \mathrm{d}u \wedge \mathrm{d}^{\mathrm{c}}u \wedge \alpha \wedge \beta. \tag{27.84}$$

Consequently $v_+(\omega) = 24\pi$. In particular, X has the bounded mass property, while $v_-(h) = 0$ for every Hermitian form h on X.

Proof Proposition 27.18.1, with $T = 4\pi$, $A = 2$ and $-\pi^*(c^2) = 2\alpha \wedge \beta$, gives the balanced condition, the formula for $\mathrm{d}\mathrm{d}^{\mathrm{c}}\omega$, the volume $3TA = 24\pi$, the identity (27.84) and $v_+(\omega) = 24\pi$. The metric independence in Proposition 7.5.1 then gives the bounded mass property of X. The assertion about the lower mass is Theorem 27.18.1.□

For an admissible smooth u, equality in the upper-mass bound holds precisely when u is constant along every elliptic fiber. Indeed, $\alpha \wedge \beta$ is a positive volume form in the base directions, so the nonnegative integral in (27.84) vanishes exactly when the derivative of u along the fibers vanishes. The fibers are connected, and smooth local sections then show that u is the pullback of a smooth function on $\mathbb{P}^1 \times \mathbb{P}^1$. For

the potentials with masses tending to zero in Theorem 27.18.1, the same integral instead tends to 4π.

A function $f(t)$ descends to X when f is 4π-periodic, since $t \circ g = t + 4\pi$. Even after adding an arbitrary potential pulled back from the base, such functions leave a positive mass gap. The sharp gap follows from Lemma 27.18.4, through the bounds on the first two derivatives of f imposed by positivity.

Corollary 27.18.4 *Let $\pi\colon X \to B = \mathbb{P}^1 \times \mathbb{P}^1$ be the elliptic fibration above, and put $L = 4\pi$. Consider the smooth potentials*

$$u = f(t) + \pi^* v, \qquad \omega + \mathrm{dd^c} u > 0,$$

where $f\colon \mathbb{R} \to \mathbb{R}$ is smooth and L-periodic and $v \in C^\infty(B, \mathbb{R})$. The infimum of $\int_X (\omega + \mathrm{dd^c} u)^3$ over these potentials is 8. Every such mass is strictly greater than 8, and the infimum is approached already with $v = 0$.

Proof Here $L = T$ and $\kappa^{-1/2} = M_0 = 1$. Since $T \geq 2$, the definition of F gives $F(1, 4\pi) = 1 - 1/(3\pi)$, so both bounds in Proposition 27.18.2 equal $3TA\big(1 - F(1, 4\pi)\big) = 8$. That proposition therefore shows that every mass in the statement is strictly greater than 8, and that potentials $f(t)$ with $v = 0$ have masses tending to 8. In terms of $p = f'$, the mass formula in its proof reads

$$\int_X (\omega + \mathrm{dd^c} u)^3 = 6\left(L - \int_0^L p(s)^2 \, \mathrm{d}s\right). \tag{27.85}$$

Corollary 27.18.5 *The threefold X admits neither a pluriclosed Hermitian metric nor a closed Kähler current.*

Proof The first Chern class of $O(1) \boxtimes O(-1)$ is represented by $\alpha_B - \beta_B$, which is nonzero in real cohomology, so this is part of Proposition 27.18.1. □

Since a compact complex manifold carries a closed Kähler current exactly when it belongs to Fujiki's class $\mathcal{C}$, as recalled in Section 35.3, the threefold X lies outside this class. The example shows that a balanced metric does not guarantee positive lower mass, even together with bounded upper mass. The Hopf threefold, in contrast, has positive lower mass and infinite upper mass, so the two properties are logically independent, and the products in Corollary 27.18.2 show that both can fail at once. In particular, in a theorem asserting solvability for every nonzero finite non-pluripolar measure, bounded mass alone cannot replace the hypothesis of positive lower mass.

The exact equations of Section 15.29 on products $X_0 \times Y$ with $v_-(h_0) = 0$ apply to this threefold. We use the notation of that section with $Y = \mathbb{P}^2$: κ is the Fubini–Study form normalized by $\int_Y \kappa^2 = 1$, $\mathrm{d}V$ is a smooth positive probability volume form on $X_0 \times Y$, the functions R, q_α and u_α, for $0 < \alpha < 1$, are defined there, and b_H is the threshold of Section 15.27.

Corollary 27.18.6 *Let X_0 be the elliptic threefold of Theorem 27.18.1, and let h_0 be the balanced metric of Corollary 27.18.3. On $X_0 \times \mathbb{P}^2$, with background $H = h_0 + \kappa$, the densities $f = Rq_\alpha$ of Theorem 15.29.1 satisfy*

$$\nu_-(H) = 0, \qquad b_H(f) = 240\pi, \qquad \mathrm{MA}^{\mathrm{ext}}_H(u_\alpha) = 240\pi f \, \mathrm{d}V.$$

Proof Theorem 27.18.1 gives $\nu_-(h_0) = 0$, and Corollary 27.18.3 gives $V_0 = 24\pi$. Theorem 15.29.1 applies with $m = 3$, $k = 2$ and $C = \binom{5}{3} = 10$. □

27.19 Orders of degeneration at the nef boundary

If θ is nef and h is Hermitian, the volumes $\mathcal{V}_\mu(\theta + \epsilon h)$ are positive and, by Theorem 27.9.1, decrease to $\mathcal{V}_\mu(\theta)$ as $\epsilon \downarrow 0$. When this limit is zero, the rate of decay measures how degenerate θ is at the nef boundary. For a closed nef class on a compact Kähler manifold, choosing h Kähler makes the volume of the perturbation the intersection polynomial $\int_X (\theta + \epsilon h)^n$, so it decays like an integral power of ϵ determined by the numerical dimension. For a nonclosed nef background the logarithmic rate of decay can be fractional, and it need not converge. We therefore define lower and upper orders of degeneration as lower and upper limits of this rate. Changing the probability measure or the perturbing metric only introduces bounded factors, so the orders are invariants of the background.

Let X again be a connected compact complex manifold of dimension $n \geq 1$, and μ a smooth strictly positive probability form on X. Let θ be nef and let h be Hermitian. For $0 < \epsilon < 1$, put

$$a_{\mu,h}(\theta;\epsilon) = \frac{\log \mathcal{V}_\mu(\theta + \epsilon h)}{\log \epsilon}.$$

The logarithm is finite because $\theta + \epsilon h$ belongs to $\mathscr{P}$, so that its volume is positive.

Proposition 27.19.1 *The limits*

$$\underline{a}(\theta) = \liminf_{\epsilon \downarrow 0} a_{\mu,h}(\theta;\epsilon), \qquad \overline{a}(\theta) = \limsup_{\epsilon \downarrow 0} a_{\mu,h}(\theta;\epsilon)$$

are independent of μ and h, and satisfy

$$0 \leq \underline{a}(\theta) \leq \overline{a}(\theta) \leq n.$$

They are unchanged when θ is replaced by $\theta + \mathrm{dd}^{\mathrm{c}} f$, with f smooth, or by $r\theta$, with $r > 0$. If $\mathcal{V}_\mu(\theta) > 0$, both limits are zero.

Proof Since θ is pseudoeffective, monotonicity and homogeneity give

$$\epsilon \lambda_\mu(h) \leq \lambda_\mu(\theta + \epsilon h) \leq \lambda_\mu(\theta + h), \qquad 0 < \epsilon \leq 1.$$

Thus there are constants $c, C > 0$ such that

$$c\epsilon^n \le \mathcal{V}_\mu(\theta + \epsilon h) \le C, \qquad 0 < \epsilon \le 1.$$

Taking logarithms and dividing by the negative number $\log \epsilon$ shows that the quotients $a_{\mu,h}(\theta;\epsilon)$ stay bounded as $\epsilon \downarrow 0$, and that their lower and upper limits lie in $[0, n]$.

By Proposition 10.8.2, changing the smooth probability measure multiplies the volume by a factor between two fixed positive constants. The resulting error in the logarithm is bounded, and after division by $\log \epsilon$ it tends to zero.

For another Hermitian metric h', choose constants $c_2 \ge c_1 > 0$ with $c_1 h \le h' \le c_2 h$. Then

$$\mathcal{V}_\mu(\theta + c_1\epsilon h) \le \mathcal{V}_\mu(\theta + \epsilon h') \le \mathcal{V}_\mu(\theta + c_2\epsilon h).$$

For any fixed $t > 0$ and small ϵ,

$$\frac{\log \mathcal{V}_\mu(\theta + t\epsilon h)}{\log \epsilon} = a_{\mu,h}(\theta; t\epsilon)\frac{\log(t\epsilon)}{\log \epsilon}.$$

The last factor tends to one and the first is bounded, while the rescaling $\epsilon \mapsto t\epsilon$ does not change lower and upper limits as $\epsilon \downarrow 0$. Hence the left side has the same lower and upper limits as $a_{\mu,h}(\theta;\epsilon)$. Dividing the preceding inequalities by $\log \epsilon < 0$ reverses them, and taking lower and upper limits proves independence of the metric.

Invariance under smooth dd^{c} changes follows from that of $\mathcal{V}_\mu$. For scaling, homogeneity gives

$$\mathcal{V}_\mu(r\theta + \epsilon h) = r^n \mathcal{V}_\mu(\theta + (\epsilon/r)h).$$

The bounded error $n \log r$ and the same rescaling argument prove scale invariance. Finally, if $\mathcal{V}_\mu(\theta) > 0$, then by Theorem 27.9.1 the perturbed volumes lie between $\mathcal{V}_\mu(\theta)$ and $\mathcal{V}_\mu(\theta + h)$ for $0 < \epsilon \le 1$, so both limits are zero. □

We call $\underline{a}(\theta)$ and $\overline{a}(\theta)$ the *lower* and *upper orders of degeneration* of the nef background θ. When they agree, their common value is the *common order*, and we then also say that θ has this order. The two orders can differ, and a common order can be irrational. Even when both are zero, the volume can tend to zero, at a logarithmic rate. The elliptic examples in Section 27.19.1 exhibit each of these behaviors. Closed nef classes on Kähler manifolds, in contrast, degenerate polynomially.

Proposition 27.19.2 *Suppose X is Kähler, θ is closed and nef, and h is Kähler. Set*

$$r = \max\left\{0 \le k \le n : \int_X \theta^k \wedge h^{n-k} > 0\right\}.$$

Then

$$\underline{a}(\theta) = \overline{a}(\theta) = n - r.$$

Thus n minus the common order is the numerical dimension of the closed nef class of θ.

Proof All the intersection numbers in the definition of r are nonnegative. Indeed, the mixed integrals against h of smooth positive representatives of $\theta + \delta h$ are positive, agree with the cohomological integrals by Stokes' theorem, and converge to the stated numbers as $\delta \downarrow 0$. The number for $k = 0$ is $\int_X h^n > 0$, so r is well defined.

By the closed nef formula following Theorem 27.9.1,

$$\mathcal{V}_\mu(\theta + \epsilon h) = \int_X (\theta + \epsilon h)^n = \sum_{k=0}^{n} \binom{n}{k} \epsilon^{n-k} \int_X \theta^k \wedge h^{n-k}.$$

As $\epsilon \downarrow 0$, the leading nonzero term is a positive constant times ϵ^{n-r}, and taking the logarithmic quotient proves the assertion. By Proposition 27.19.1, the result holds for every Hermitian perturbation, not only for Kähler ones. □

The orders can also be computed on the non-Kähler Hopf threefold X of Chapter 9. Keep its forms $\omega = \alpha + \beta$, where $\alpha \ge 0$ is closed of rank two, $\beta \ge 0$ has rank one, $\alpha^3 = \beta^2 = 0$, and $\int_X \omega^3 = 6$. Take $\theta = \alpha$, $h = \omega$, and $\mu_0 = \omega^3/6$. The metric

$$\Omega_\epsilon = \alpha + \epsilon\omega = (1 + \epsilon)\alpha + \epsilon\beta$$

is strictly positive. Its volume is

$$\Omega_\epsilon^3 = \epsilon(1 + \epsilon)^2 \omega^3 = 6\epsilon(1 + \epsilon)^2 \mu_0.$$

Hence the zero potential solves the smooth normalized equation, and

$$\mathcal{V}_{\mu_0}(\alpha + \epsilon\omega) = 6\epsilon(1 + \epsilon)^2, \qquad \underline{a}(\alpha) = \overline{a}(\alpha) = 1.$$

By Proposition 27.19.1, the same orders are obtained for every smooth positive probability measure and every Hermitian perturbation.

Each perturbed metric nevertheless has infinite upper mass. If V_j are the smooth potentials of Theorem 9.0.1, then

$$\Omega_\epsilon + \mathrm{dd}^{\mathrm{c}}(\epsilon V_j) = \alpha + \epsilon(\omega + \mathrm{dd}^{\mathrm{c}} V_j) > 0,$$

and monotonicity of the determinant of positive matrices gives

$$\int_X \left(\Omega_\epsilon + \mathrm{dd}^{\mathrm{c}}(\epsilon V_j)\right)^3 \ge \epsilon^3 \int_X (\omega + \mathrm{dd}^{\mathrm{c}} V_j)^3 \longrightarrow +\infty.$$

Thus a finite order of degeneration at the boundary coexists with infinite upper mass at every positive perturbation.

27.19.1 Elliptic surface volumes and order spectra

On a product of two elliptic curves, a semipositive background whose coefficients depend on one factor only has an explicit volume. Solving the smooth equation reduces to a Poisson equation on that factor, and the volume of each Hermitian perturbation is governed by the harmonic mean of the perturbed vertical coefficient. The rate at which this coefficient vanishes therefore determines the order of degeneration. Zeros of finite order give fractional orders or logarithmic decay, while coefficients that vanish to infinite order can make the logarithmic rate oscillate.

Let $E = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$, write $X = E_z \times E_w$, and fix

$$\omega_1 = \frac{\mathrm{i}}{2}\mathrm{d}z \wedge \mathrm{d}\bar{z}, \qquad \omega_2 = \frac{\mathrm{i}}{2}\mathrm{d}w \wedge \mathrm{d}\bar{w}, \qquad \mu = \omega_1 \wedge \omega_2, \qquad h = \omega_1 + \omega_2.$$

Both area forms have integral one. Let a, q, r be smooth functions of z alone, with a, q real and nonnegative, r complex-valued, and $|r|^2 \le aq$. Set

$$\theta = \frac{\mathrm{i}}{2}\bigl(a\,\mathrm{d}z \wedge \mathrm{d}\bar{z} + r\,\mathrm{d}z \wedge \mathrm{d}\bar{w} + \bar{r}\,\mathrm{d}w \wedge \mathrm{d}\bar{z} + q\,\mathrm{d}w \wedge \mathrm{d}\bar{w}\bigr).$$

This is a smooth semipositive form, and in particular it is nef.

Proposition 27.19.3 *For $\epsilon > 0$, put $g = q + \epsilon$ and $I = \int_E g^{-1}\omega_1$. Then*

$$\mathcal{V}_\mu(\theta + \epsilon h) = \frac{2}{I}\left(\int_E (a+\epsilon)\omega_1 - \int_E \frac{|r|^2}{g}\omega_1\right), \tag{27.86}$$

and the infimum defining this volume is attained by a smooth positive Gauduchon metric. In particular, if a is a constant $b > 0$ and $r = 0$, then $\mathcal{V}_\mu(\theta+\epsilon h) = 2(b+\epsilon)/I$. If $|r|^2 = aq$, then

$$\mathcal{V}_\mu(\theta + \epsilon h) = \frac{2\epsilon(1+K)}{I}, \qquad K = \int_E \frac{a}{q+\epsilon}\omega_1. \tag{27.87}$$

Proof Define

$$c = I^{-1}\left(\int_E (a+\epsilon)\omega_1 - \int_E \frac{|r|^2}{g}\omega_1\right), \qquad A = \frac{c + |r|^2}{g}.$$

Since $(a+\epsilon)g - |r|^2 > 0$, we have $c > 0$, and the definition of c makes $A - a - \epsilon$ have zero mean with respect to ω_1. Dividing its nonconstant Fourier coefficients by the corresponding nonzero eigenvalues of the flat Laplacian produces a smooth real function u on E with $\mathrm{d}\mathrm{d}^{\mathrm{c}}u = (A - a - \epsilon)\omega_1$. Pull it back to X. The form

$\Omega = \theta + \epsilon h + \mathrm{dd^c} u$ has coefficient matrix $\left(\begin{smallmatrix} A & r \\ \bar{r} & g \end{smallmatrix}\right)$, which has determinant c and positive diagonal entries. Hence $\Omega > 0$ and $\Omega^2 = 2c\mu$, and Theorem 10.8.1 gives (27.86).

The metric $\gamma_* = \Omega/g$ has constant coefficient 1 in the fiber direction, and its remaining coefficients depend only on z. Applying $\mathrm{dd^c}$ to its base term or its mixed terms produces a repeated $\mathrm{d}z$ or $\mathrm{d}\bar{z}$, so $\mathrm{dd^c}\gamma_* = 0$. Thus γ_* is a smooth Gauduchon metric conformal to Ω, and by Theorem 10.8.1 it attains the infimum. The two special cases follow by direct substitution. □

Proposition 27.19.4 *For each integer $m > 1$ there is a smooth semipositive nonclosed form on X with common order $1 - 1/m$. For each $b > 0$ there is also one with*

$$\mathcal{V}_\mu(\theta + \epsilon h) \sim \frac{2b}{\pi \log(1/\epsilon)} \quad (\epsilon \downarrow 0);$$

it has volume zero and both orders zero.

Proof In Proposition 27.19.3 take $a = b > 0$ and $r = 0$. Fix a point $z_0 \in E$ and a flat coordinate $\zeta = z - z_0$ on a small disk about z_0. Choose q smooth, positive outside z_0, and equal to $|\zeta|^{2m}$ on that disk; a positive smooth extension exists away from the disk. For $m = 1$, polar coordinates give

$$I = \int_E \frac{\omega_1}{q + \epsilon} = \pi \log(1/\epsilon) + O(1),$$

and (27.86) gives the asymptotic formula of the statement. If $m > 1$, the substitution $s = |\zeta|^2 = \epsilon^{1/m} t$ gives

$$I \sim C_m \epsilon^{1/m - 1}, \qquad C_m = \pi \int_0^\infty \frac{\mathrm{d}t}{1 + t^m} > 0,$$

so the common order is $1 - 1/m$. In the logarithmic case, Theorem 27.9.1 gives $\mathcal{V}_\mu(\theta) = \lim_{\epsilon \downarrow 0} \mathcal{V}_\mu(\theta + \epsilon h) = 0$. Moreover, the quotient of the volume by ϵ^δ diverges for every $\delta > 0$, so both orders vanish. Finally, $\mathrm{d}\theta = \mathrm{d}q \wedge \omega_2 \neq 0$, so these backgrounds are not closed. □

Proposition 27.19.5 *For every $0 < \alpha < \beta < 1$ there is a smooth semipositive nonclosed background on X, with the same μ and h, such that*

$$\big(\underline{a}(\theta), \overline{a}(\theta)\big) = (1 + \alpha, 1 + \beta).$$

For every $0 < \sigma < 1$ there is such a background with common order $1 + \sigma$.

Proof We use the rank-one formula (27.87), with coefficients that vanish to infinite order at one point at a prescribed oscillating rate.

Step 1: the coefficients. For all sufficiently large t, define

$$p(t) = \frac{\alpha + \beta}{2} + \frac{\beta - \alpha}{2} \sin(\log \log t), \qquad P(t) = t p(t).$$

Choose T so large that $0 < c_0 \le P'(t) \le d_0 < 1$ for $t \ge T$; this is possible because

$$P'(t) = p(t) + \frac{\beta - \alpha}{2 \log t} \cos(\log \log t)$$

and $\alpha \le p \le \beta$. Fix $z_0 \in E$ and a flat coordinate ζ centered at z_0. On $E \setminus \{z_0\}$ choose a smooth function $t(z) \ge T$ that equals $|\zeta|^{-2}$ near z_0 and is bounded outside every neighborhood of z_0. Put

$$q = \mathrm{e}^{-t}, \qquad a = \mathrm{e}^{-P(t)}, \qquad r = \mathrm{e}^{-(P(t)+t)/2},$$

and set all three equal to zero at z_0. These functions are smooth and vanish to infinite order at z_0: we have $P(t) \ge \alpha t$, and every fixed derivative of P grows at most polynomially in t, so by the chain rule each coordinate derivative is bounded by a polynomial in $|\zeta|^{-1}$ times $\mathrm{e}^{-c/|\zeta|^2}$ for some $c > 0$. Since $|r|^2 = aq$, the resulting θ is semipositive. It is not closed: the component $\mathrm{d}q \wedge \omega_2$ of $\mathrm{d}\theta$ contains both fiber differentials, and derivatives of the remaining terms cannot cancel it.

Step 2: asymptotics of I and K. Write $\epsilon = \mathrm{e}^{-L}$, and let $T_0 \ge T$ be such that $t = |\zeta|^{-2}$ on the disk $\{|\zeta|^2 < 1/T_0\}$. Outside this disk the integrands of I and K in (27.87) are bounded, so their contributions are $O(1)$, while polar coordinates on the disk give

$$I_{\mathrm{loc}} = \pi \int_{T_0}^{\infty} \frac{t^{-2}\,\mathrm{d}t}{\mathrm{e}^{-t} + \mathrm{e}^{-L}}, \qquad K_{\mathrm{loc}} = \pi \int_{T_0}^{\infty} \frac{\mathrm{e}^{-P(t)} t^{-2}\,\mathrm{d}t}{\mathrm{e}^{-t} + \mathrm{e}^{-L}}.$$

The part of I_{loc} over $t \ge L$ is at least a fixed multiple of e^L/L. For an upper bound, use $(\mathrm{e}^{-t} + \mathrm{e}^{-L})^{-1} \le \mathrm{e}^t$ for $t \le L$ and $(\mathrm{e}^{-t} + \mathrm{e}^{-L})^{-1} \le \mathrm{e}^L$ for $t \ge L$, and split $[T_0, L]$ at $L/2$. Thus

$$\log I = L - \log L + O(1). \tag{27.88}$$

The derivatives of P and of $t - P(t)$ are positive and bounded away from zero. Splitting the integral for K at $t = L$ therefore gives $K \le C\mathrm{e}^{L-P(L)}$. On $[L, L+1]$ the denominator is at most $2\mathrm{e}^{-L}$ and $P(t) \le P(L) + 1$, so

$$c_1 L^{-2} \mathrm{e}^{L-P(L)} \le K \le c_2 \mathrm{e}^{L-P(L)};$$

the complementary $O(1)$ term is absorbed because $P(L) \le \beta L$. Consequently $\log K = L - P(L) + O(\log L)$, and $K \to \infty$.

Step 3: the orders. Combining these estimates with (27.87) and (27.88) yields

$$\log \mathcal{V}_\mu(\theta + \mathrm{e}^{-L} h) = -L - P(L) + O(\log L), \qquad \frac{\log \mathcal{V}_\mu(\theta + \epsilon h)}{\log \epsilon} = 1 + p(L) + o(1). \tag{27.89}$$

Since $\sin(\log \log L)$ oscillates between -1 and 1, the lower and upper limits of $p(L)$ are α and β, which proves the first assertion. If instead $P(t) = \sigma t$, the same smoothness and integral estimates give the common order $1 + \sigma$. In particular, $\sigma = \sqrt{2} - 1$ gives an irrational common order. $\square$

27.19.2 Comparison under an arbitrary modification

The orders of degeneration are defined on a fixed manifold. We now show that pulling a nef background back by a modification can only increase them. Recall that a modification $\pi\colon Y \to X$ is a proper surjective holomorphic map that restricts to a biholomorphism between dense Zariski open subsets. The proof compares Gauduchon tests on X with tests on Y. The pullback of a Gauduchon test is degenerate along the exceptional locus, and adding a small positive test on Y repairs this. To control the effect of the added test uniformly, we first bound the pairing of pulled-back tests with a Hermitian form on Y by their pairing with a Hermitian form on X, uniformly over the whole cone of tests.

In this subsection $\pi\colon Y \to X$ is a modification of connected compact complex manifolds of dimension $n \geq 2$. We use Hironaka's analytic elimination of indeterminacies in the following form: there are a finite composition $p\colon Z \to X$ of blowups along smooth centers and a modification $F\colon Z \to Y$ with $p = \pi \circ F$. Here p resolves the indeterminacies of the meromorphic map π^{-1} from X to Y, and F is the holomorphic extension of $\pi^{-1} \circ p$; the identity $p = \pi \circ F$ holds on the dense set where the maps are biholomorphic, hence everywhere. This classical input is discussed in Section 36.15. For an effective real divisor D, we write η_D for the curvature form of a smooth Hermitian metric on the corresponding real line bundle, as in Section 27.21.

Lemma 27.19.1 *Fix Hermitian forms h_X, h_Y for a modification $\pi\colon Y \to X$. There is a constant $K > 0$ such that*

$$\int_Y h_Y \wedge \pi^*\Psi \leq K \int_X h_X \wedge \Psi \tag{27.90}$$

for every smooth semipositive $(n-1,n-1)$-form Ψ on X with $\mathrm{dd}^{\mathrm{c}}\Psi = 0$.

Proof Use the factorization $p = \pi \circ F$ above and Lemma 1.15.1. By compactness there is $a > 0$ with $h_Z \geq aF^*h_Y$. Pulling back the semipositive form Ψ and integrating, we obtain

$$a \int_Y h_Y \wedge \pi^*\Psi \leq \int_Z h_Z \wedge p^*\Psi = C_0 \int_X h_X \wedge \Psi - \int_D p^*\Psi.$$

Here the changes of variables use that F and p have degree one, and the Poincaré–Lelong formula, together with $\mathrm{dd}^{\mathrm{c}}p^*\Psi = 0$, replaces the pairing with η_D by the integral over D. This last integral vanishes: on the regular locus of each component of D, the restriction of p has rank at most $n-2$, so the restriction of the $(n-1,n-1)$-form $p^*\Psi$ is zero. Taking $K = C_0/a$ proves the assertion. □

Proposition 27.19.6 *Let θ be a smooth nef background on X. Then $\pi^*\theta$ is nef. For fixed Hermitian forms h_X, h_Y and smooth positive probability forms μ_X, μ_Y, there are constants $C, K > 0$, independent of ϵ, such that*

$$\mathcal{V}_{\mu_Y}(\pi^*\theta + \epsilon h_Y) \le C\mathcal{V}_{\mu_X}(\theta + K\epsilon h_X) \qquad (\epsilon > 0). \tag{27.91}$$

Consequently

$$\underline{a}(\pi^*\theta) \ge \underline{a}(\theta), \qquad \overline{a}(\pi^*\theta) \ge \overline{a}(\theta).$$

Proof *Step 1: nefness.* Choose C' with $\pi^*h_X \le C'h_Y$. For $\epsilon > 0$, pull back a smooth representative $\theta + \mathrm{dd}^{\mathrm{c}}u_\tau \ge -\tau h_X$ with $\tau < \epsilon/C'$; this gives a positive representative of $\pi^*\theta + \epsilon h_Y$. Thus $\pi^*\theta$ is nef, and both volumes in (27.91) are positive.

Step 2: regularized tests. The ratio $\pi^*\mu_X/\mu_Y$ is smooth, nonnegative and bounded on the compact manifold Y, so there is $c > 0$ with $\mu_Y \ge c\pi^*\mu_X$. Fix a smooth Gauduchon metric γ_0 on Y, given by Theorem 4.1.1. For an arbitrary Gauduchon metric γ on X, write $\Psi = \gamma^{n-1}$, and for $\delta > 0$ define Γ_δ by

$$\Gamma_\delta^{n-1} = \pi^*\Psi + \delta\gamma_0^{n-1}.$$

The right side is a strictly positive $(n-1,n-1)$-form, and such a form has a unique smooth positive $(n-1)$st root, obtained by solving for the Hermitian matrix of the root from the positive cofactor matrix. Since the right side is dd^{c}-closed, Γ_δ is a smooth Gauduchon metric on Y.

Step 3: the numerator. For $\epsilon > 0$, Lemma 27.19.1 gives

$$\lim_{\delta\downarrow 0}\int_Y (\pi^*\theta+\epsilon h_Y)\wedge\Gamma_\delta^{n-1} = \int_X \theta\wedge\Psi + \epsilon\int_Y h_Y\wedge\pi^*\Psi \le \int_X (\theta+K\epsilon h_X)\wedge\Psi. \tag{27.92}$$

Step 4: the denominator. We claim that

$$\lim_{\delta\downarrow 0} D_{\mu_Y}(\Gamma_\delta) = \int_Y (\pi^*\gamma^n)^{(n-1)/n}\mu_Y^{1/n} \ge c^{1/n}D_{\mu_X}(\gamma) > 0. \tag{27.93}$$

To justify the limit on the exceptional locus, represent Γ_δ^{n-1} by its positive cofactor matrix. The density of $(\Gamma_\delta^n)^{(n-1)/n}$ is a fixed dimensional constant times the $1/n$th power of the determinant of that matrix, and this expression extends continuously to positive semidefinite matrices. Since Y is compact, dominated convergence applies. The inequality follows from $\mu_Y \ge c\pi^*\mu_X$ and the change of variables for the degree-one map π.

Step 5: conclusion. Test the infimum defining $\lambda_{\mu_Y}(\pi^*\theta+\epsilon h_Y)$ with Γ_δ, use (27.92) and (27.93), and let $\delta \downarrow 0$. For every Gauduchon metric γ on X this gives

$$\lambda_{\mu_Y}(\pi^*\theta + \epsilon h_Y) \le c^{-1/n}\frac{\int_X(\theta + K\epsilon h_X)\wedge\gamma^{n-1}}{D_{\mu_X}(\gamma)}.$$

Taking the infimum over γ and then the nth power proves (27.91) with $C = c^{-1}$.

Dividing the logarithm of (27.91) by $\log \epsilon < 0$ reverses the inequality. The bounded error $\log C$ and the rescaling $\epsilon \mapsto K\epsilon$ do not affect lower and upper limits, as in the proof of Proposition 27.19.1. This gives $\underline{a}(\pi^*\theta) \geq \underline{a}(\theta)$ and $\overline{a}(\pi^*\theta) \geq \overline{a}(\theta)$.□

In dimension one a modification is an isomorphism, so both orders are preserved.

27.19.3 Strict changes under one blowup

The inequalities of Proposition 27.19.6 can be strict. A blowup changes how much smooth volume lies near the zero set of a coefficient. Near a point of $\mathbb{P}^m$, the Fubini–Study volume of the ball $\{|z|^2 < s\}$ is of order s^m; after the point is blown up, the corresponding neighborhood of the exceptional divisor has volume of order s for a Kähler form upstairs. For product backgrounds whose fiber coefficient vanishes at the blown-up point, the smooth equation can be solved explicitly, and this change in the distribution of volume produces a strict increase of the order.

Throughout this subsection let $B = \mathbb{P}^m$ with $m \geq 2$, and let H be the Fubini–Study form normalized by $\int_B H^m = 1$. Let $\pi\colon \widetilde{B} \to B$ be the blowup of the origin of the affine chart $\{Z_0 \neq 0\}$. The projection $\rho\colon \widetilde{B} \to \mathbb{P}^{m-1}$ from the incidence description gives the form $L = \rho^*\omega_{\mathrm{FS}}$, with ω_{FS} normalized in the same way; we also write H for π^*H. The incidence embedding of $\widetilde{B}$ in $\mathbb{P}^m \times \mathbb{P}^{m-1}$ makes $\kappa = H + L$ a Kähler form.

Proposition 27.19.7 *For all integers $m \geq 2$ and $p, k \geq 1$, there are a single blowup, along a smooth center, of a projective manifold of dimension $m + p$ and a smooth semipositive nonclosed background on it such that the background and its pullback both have common orders, namely*

$$a_{\mathrm{down}} = \max\{p - m/k, 0\}, \qquad a_{\mathrm{up}} = p - 1/k.$$

Every positive integer occurs as the increase $a_{\mathrm{up}} - a_{\mathrm{down}}$, for a suitable dimension. In the same family, a background of positive volume can pull back to a background of volume zero, with both orders equal to zero.

Proof *Step 1: the backgrounds.* Put $W_0 = \mathbb{P}^p$, with Fubini–Study form ω normalized by $\int_{W_0} \omega^p = 1$. Write $s = |z|^2$ on the affine chart of B. Choose a smooth function $Q(s) \geq 0$ that equals s^k near zero, is positive for $s > 0$, and is constant for large s; it defines a smooth function q on B. Put $n = m + p$, choose $b > 0$, and define

$$\theta = bH + q\omega \quad \text{on } B \times W_0, \qquad \widetilde{\theta} = bH + (q \circ \pi)\omega \quad \text{on } \widetilde{B} \times W_0.$$

The map $\Pi = \pi \times \mathrm{id}$ is the blowup of the smooth center $\{0\} \times W_0$, and $\widetilde{\theta} = \Pi^*\theta$. Fix the Kähler perturbation directions $h = H + \omega$ and $\widetilde{h} = \kappa + \omega$, and the smooth positive probability densities $\mu = H^m \wedge \omega^p$ and $\widetilde{\mu} = \kappa^m \wedge \omega^p/(2^m - 1)$. Both backgrounds are smooth and semipositive, and they are not closed, since $\mathrm{d}\theta = \mathrm{d}q \wedge \omega \neq 0$.

Step 2: explicit solutions. Write $\nu_{\mathrm{down}} = H^m$ and $\nu_{\mathrm{up}} = \kappa^m/(2^m - 1)$ for the base forms; we show below that both are probability forms. For either product, write

the perturbed background as $\xi = A + g\omega$, where downstairs $A = (b+\epsilon)H$, upstairs $A = (b+\epsilon)H + \epsilon L$, and in both cases $g = q + \epsilon$, with pullback understood upstairs. The form A is Kähler on the relevant base. Let ν be the corresponding base probability form, and put $M = \int A^m$ and $J = \int g^{-p}\nu$. The smooth equation of Theorem 10.1.1, together with Stokes' theorem, gives a smooth u on the base with

$$\alpha = A + \mathrm{dd^c}u > 0, \qquad \alpha^m = (M/J)g^{-p}\nu.$$

Let $\Xi(s)$ denote the mass of ν on the set $\{|z|^2 < s\}$. These cumulative masses are

$$\begin{aligned} \Xi_{\mathrm{down}}(s) &= \left(\frac{s}{1+s}\right)^m, \\ \Xi_{\mathrm{up}}(s) &= \frac{\left(1+\frac{s}{1+s}\right)^m - 1}{2^m - 1}. \end{aligned} \tag{27.94}$$

To check these formulas and to compute M, write $t = \log|z|^2$ on the punctured affine chart. For a radial potential $F(t)$, polar integration of its complex Hessian gives

$$\int_{s_1<|z|^2<s_2} (\mathrm{dd^c}F(\log|z|^2))^m = F'(\log s_2)^m - F'(\log s_1)^m.$$

The potentials of H and L are $\log(1+\mathrm{e}^t)$ and t. Thus the potential of $xH+yL$ has slopes y and $x+y$ at the two ends, and the total mass of $xH+yL$ is $(x+y)^m - y^m$. The cases $(x,y) = (1,0)$ and $(1,1)$ give (27.94); the exceptional divisor and the hyperplane at infinity have zero smooth volume. In particular, $M = (b+\epsilon)^m$ downstairs and $M = (b+2\epsilon)^m - \epsilon^m$ upstairs. The form

$$\Omega = \alpha + g\omega = \xi + \mathrm{dd^c}u > 0, \qquad \Omega^n = \binom{n}{m}\frac{M}{J}\nu\wedge\omega^p \tag{27.95}$$

solves the normalized equation, since $\nu\wedge\omega^p$ is the chosen probability density, and Theorem 10.8.1 gives the exact identities

$$\mathcal{V}_\mu(\theta+\epsilon h) = \binom{n}{m}\frac{(b+\epsilon)^m}{J_{\mathrm{down}}}, \qquad \mathcal{V}_{\widetilde{\mu}}(\widetilde{\theta}+\epsilon\widetilde{h}) = \binom{n}{m}\frac{(b+2\epsilon)^m-\epsilon^m}{J_{\mathrm{up}}}, \tag{27.96}$$

where $J_{\mathrm{down}} = \int_B (q+\epsilon)^{-p}\nu_{\mathrm{down}}$ and $J_{\mathrm{up}} = \int_{\widetilde{B}}(q\circ\pi+\epsilon)^{-p}\nu_{\mathrm{up}}$.

Step 3: asymptotics of the denominators. Near zero, differentiation of (27.94) gives

$$\mathrm{d}\Xi_{\mathrm{down}} = (ms^{m-1} + O(s^m))\,\mathrm{d}s, \qquad \mathrm{d}\Xi_{\mathrm{up}} = \left(\frac{m}{2^m-1} + O(s)\right)\mathrm{d}s. \tag{27.97}$$

For a density $(as^{d-1} + O(s^d))\,\mathrm{d}s$ near zero and $q = s^k$, the integral of $(q+\epsilon)^{-p}$ over a fixed small interval $[0, s_0]$ has the following leading behavior as $\epsilon \downarrow 0$:

$$\begin{cases} \dfrac{a}{k}\mathrm{B}(d/k, p - d/k)\epsilon^{d/k-p} & kp > d, \\ \dfrac{a}{k}\log(1/\epsilon) & kp = d, \\ \text{a finite positive limit} & kp < d. \end{cases} \tag{27.98}$$

Here $\mathrm{B}(u,v) = \int_0^\infty t^{u-1}(1+t)^{-u-v}\,\mathrm{d}t$ is the Beta function. In the first case substitute $s = \epsilon^{1/k}v$ and then $t = v^k$. In the second case the integrand is asymptotic to a/s on the range $\epsilon^{1/k} \ll s \ll 1$. The error term $O(s^d)$ is negligible: on a sufficiently small fixed interval its relative size is arbitrarily small, while the integral over the complementary interval stays bounded. For $kp < d$, monotone convergence applies, since the limiting density is integrable. The numerators in (27.96) tend to $\binom{n}{m}b^m > 0$, and the integrals over the complement of a neighborhood of the origin stay bounded. Hence (27.97) and (27.98), with $d = m$ downstairs and $d = 1$ upstairs, give the stated orders. Downstairs the order is $p - m/k$ when $kp > m$ and zero when $kp \le m$. Upstairs $kp \ge 1 = d$, and the order $p - 1/k$ vanishes exactly in the logarithmic case $k = p = 1$.

Step 4: examples. For $m = k = 2$ and $p = 1$, the denominators satisfy $J_{\mathrm{down}} \sim \log(1/\epsilon)$ and $J_{\mathrm{up}} \sim (\pi/3)\epsilon^{-1/2}$, so the volumes are asymptotic to $3b^2/\log(1/\epsilon)$ and $(9b^2/\pi)\epsilon^{1/2}$, respectively. The order changes from 0 to $1/2$. For $m = 2$ and $k = p = 1$, the downstairs denominator has a finite positive limit, while $J_{\mathrm{up}} \sim (2/3)\log(1/\epsilon)$. Thus a background of positive volume downstairs pulls back to one whose perturbed volumes are asymptotic to $9b^2/[2\log(1/\epsilon)]$, and both orders are zero. Finally, for an integer $j \ge 1$, choose $m = j+2$, $p = j+1$ and $k = 1$. Then $kp < m$ gives a positive limit of the volumes downstairs, while the orders are 0 and j. □

27.19.4 Positive-current subtraction and divisorial residuals

Another way to change the order of degeneration is to subtract a positive current from the background. If $\theta - \beta$ differs from a positive current by a $\mathrm{dd^c}$-exact term, then every Gauduchon pairing of θ dominates the corresponding pairing of β. When the residual β is still nef, its perturbed volumes are therefore smaller, and its orders are at least those of θ. The main example is the subtraction of the curvature form of an effective divisor. Such a subtraction can preserve the order, raise it by an integer or by a fraction, or produce unequal lower and upper orders; and the residual need not be nef.

Proposition 27.19.8 *Let θ, β be smooth nef real $(1,1)$-forms on a compact complex manifold X of dimension n, and suppose that*

$$S = (\theta - \beta) + \mathrm{dd}^{\mathrm{c}}\varphi \geq 0$$

is a positive $(1,1)$-current for some integrable function φ. For every Hermitian form h, every smooth strictly positive probability form μ and every $\epsilon > 0$,

$$\mathcal{V}_\mu(\beta + \epsilon h) \leq \mathcal{V}_\mu(\theta + \epsilon h). \tag{27.99}$$

Consequently $\underline{a}(\beta) \geq \underline{a}(\theta)$ and $\overline{a}(\beta) \geq \overline{a}(\theta)$.

Proof For every Gauduchon metric γ, distributional integration by parts gives

$$\int_X (\theta - \beta) \wedge \gamma^{n-1} = \langle S, \gamma^{n-1} \rangle \geq 0.$$

Hence, for each test metric, the quotient for $\theta + \epsilon h$ is at least the quotient for $\beta + \epsilon h$, and taking infima gives $\lambda_\mu(\beta + \epsilon h) \leq \lambda_\mu(\theta + \epsilon h)$. Both root volumes are nonnegative, so raising to the nth power gives (27.99). Dividing the logarithms by the negative number $\log \epsilon$ reverses the inequality and proves both comparisons of orders. □

For an effective real divisor D with curvature form η_D, the Poincaré–Lelong formula gives $[D] = \eta_D + \mathrm{dd}^{\mathrm{c}}\varphi$ for a quasi-plurisubharmonic function φ. Thus Proposition 27.19.8 applies to $\beta = \theta - \eta_D$, with $S = [D]$, whenever β is nef. If $T = \theta + \mathrm{dd}^{\mathrm{c}}u$ is a positive current with $T - S \geq 0$, then $T - S = \beta + \mathrm{dd}^{\mathrm{c}}(u - \varphi)$, so β is the background of the positive current that remains after S is removed from T.

Proposition 27.19.9 *Subtracting the curvature form of an effective divisor from a smooth nef background, with a smooth nef residual, can preserve the common order, raise it by any integer between 1 and the dimension or by a fraction, or produce a residual whose lower and upper orders differ. The residual can also fail to be nef.*

Proof *Integer jumps.* On E^n, where E is an elliptic curve, let ω_i be the pullback of a flat area form of integral one from the ith factor, and set $h = \sum_i \omega_i$ and $\mu = \omega_1 \wedge \cdots \wedge \omega_n$. The divisor F_i obtained by fixing a point in the ith factor has ω_i as a smooth curvature form. Indeed, on that elliptic curve any curvature form of the corresponding line bundle differs from ω_i by dd^{c} of a solution of a zero-mean Poisson equation, so the metric can be adjusted on the curve and then pulled back. For $\theta = h$ and $D = \sum_{i=1}^{k} F_i$, the residual $\beta = \sum_{i=k+1}^{n} \omega_i$ is smooth and nef. The perturbed forms are closed, positive and of constant density with respect to μ, so the zero potential solves their equations, and Theorem 10.8.1 gives

$$\mathcal{V}_\mu(\theta + \epsilon h) = n!(1+\epsilon)^n, \qquad \mathcal{V}_\mu(\beta + \epsilon h) = n!\epsilon^k(1+\epsilon)^{n-k}. \tag{27.100}$$

These examples realize each integer jump $1, \ldots, n$. A nonzero divisor with jump zero is given by $\theta = 2h$ and $D = F_1$, since then the residual is still strictly positive.

A fractional jump. Use the background $\beta = b\omega_1 + q\omega_2$ of Proposition 27.19.4, where $q = |\zeta|^{2m}$ near its unique zero and $m > 1$. Choose $c > 0$ and put $\theta = \beta + c\omega_2 > 0$

and $D = c(E \times \{w_0\})$, whose smooth curvature form is $c\omega_2$. The original background is strictly positive, so both its orders are zero, whereas the residual has common order $1 - 1/m$.

Unequal orders. Let β_{res} be the oscillatory rank-one background of Proposition 27.19.5, and put

$$\theta = \beta_{\mathrm{res}} + ch > 0, \qquad D = c(\{z_0\} \times E) + c(E \times \{w_0\}), \qquad \eta_D = ch.$$

The background θ has both orders zero: since $\theta \geq ch$, every test quotient is bounded below by a positive constant, while one fixed test gives a finite upper bound. Its smooth nef residual is exactly β_{res}, whose lower and upper orders differ. In each of these examples, the residual plus $[D]$ is a positive current in the smooth $\mathrm{dd^c}$-class of θ, and its divisorial part is exactly $[D]$ because the residual is smooth. We do not need this current to have minimal singularities; that case is treated in Corollary 27.19.1.

A residual that is not nef. Blow up $\mathbb{P}^2$ at two distinct points. Let H be the pullback of the class of the normalized Fubini–Study form, let E_1, E_2 be the exceptional curves, and let L be the strict transform of the line through the two points. The pullback θ of the Fubini–Study form is smooth, semipositive and nef. Subtract curvature forms of $D = E_1 + E_2$. The residual β has class $H - E_1 - E_2$, which is the class of L, and the Poincaré–Lelong formula for the total transform of the line shows that it contains the positive current $[L]$. The intersection numbers $H^2 = 1$, $H \cdot E_i = 0$, $E_i^2 = -1$ and $E_1 \cdot E_2 = 0$ give

$$\int_L \beta = (H - E_1 - E_2)^2 = -1.$$

If β were nef, then, for a Hermitian form h, integrating smooth positive representatives $\beta + \epsilon h + \mathrm{dd^c} u_\epsilon$ over L would give $-1 + \epsilon \int_L h \geq 0$ for all $\epsilon > 0$, which is impossible.□

For a prime divisor D and $\delta > 0$, consider the infimum of the generic Lelong numbers along D of the positive currents $\theta + \delta h + \mathrm{dd^c} u$; its limit as $\delta \downarrow 0$ is the minimal multiplicity of θ along D. For a nef background all minimal multiplicities vanish, since every $\theta + \delta h$ has a smooth positive representative. Removing the divisor defined by these multiplicities therefore leaves a nef background unchanged. Removing the divisorial part of a particular positive current in the class of θ is a different operation, even when that current has minimal singularities.

Corollary 27.19.1 *On the ruled surface X of Lemma 26.5.1, keep its nef curvature form θ, its section D at infinity and its Kähler form $\kappa = \theta + \pi^*\sigma_0$. For every smooth positive probability form μ,*

$$\mathcal{V}_\mu(\theta + \epsilon\kappa) = 2\epsilon(1 + \epsilon), \qquad \mathcal{V}_\mu(\epsilon\kappa) = 2\epsilon^2.$$

The only positive current of the form $\theta + \mathrm{dd^c} u$ is $[D]$. It has minimal singularities, and removing its whole divisor leaves the zero background and changes the common order from one to two.

Proof Lemma 26.5.1 gives the uniqueness of $[D]$ and the nefness of θ, and the calculation in the proof of Proposition 26.5.1 gives $\int_X \theta^2 = 0$. Put $B = \pi^*\sigma_0$. Then

$B^2 = 0$, and

$$\int_X \theta \wedge B = \int_D B = 1,$$

by the Poincaré–Lelong formula, Stokes' theorem and the fact that the section D maps isomorphically onto the elliptic base. The closed nef formula following Theorem 27.9.1 now computes both volumes by expanding the squares, for every μ. Every θ-plurisubharmonic function u satisfies $\theta + \mathrm{dd}^{\mathrm{c}} u = [D]$, so any two such functions differ by a pluriharmonic function, hence by a constant; in particular each has minimal singularities. Since θ is itself a curvature form of $\mathcal{O}_X(D)$, removing the whole divisor leaves the zero background. The two volume polynomials give the orders one and two. □

27.19.5 Integer orders in fixed projective cones

In the examples of Section 27.19.1 the order is not an integer because a coefficient of the background vanishes at a point. We now consider backgrounds that are combinations, with strictly positive coefficient functions, of finitely many fixed closed semipositive generators on products of projective spaces and their linear blowups. Such backgrounds have integral orders even when they are not closed. The tool is a comparison with a closed semipositive form B. If θ lies between two positive multiples of B, then the Gauduchon quotients of $\theta + \epsilon h$ and of $B + \epsilon\Omega$, for a Kähler form Ω, are comparable, and the intersection polynomial of $B + \epsilon\Omega$ has order $n - R$, where R is the largest rank of B.

Lemma 27.19.2 *Let X be a compact Kähler manifold of dimension n, let B be a smooth closed semipositive $(1,1)$-form, and put $R = \max_X \operatorname{rank} B$. Let θ be a smooth real $(1,1)$-form with*

$$cB \le \theta \le CB$$

for constants $c, C > 0$. Then θ is nef, and for every Hermitian form h and every smooth positive probability form μ there are constants $L, U > 0$ such that

$$L\epsilon^{n-R} \le \mathcal{V}_\mu(\theta + \epsilon h) \le U\epsilon^{n-R}, \qquad 0 < \epsilon \le 1.$$

In particular, θ has common order $n - R$. The same holds, with the same exponent, for $\pi^*\theta$, for every modification $\pi\colon Y \to X$ from a compact Kähler manifold Y and every Hermitian form and smooth positive probability form on Y.

Proof Semipositivity of θ gives nefness. Fix a Kähler form Ω on X and constants $c_h, C_h > 0$ with $c_h\Omega \le h \le C_h\Omega$, and put $k = \min(c, c_h)$ and $K = \max(C, C_h)$. For $\epsilon > 0$ the forms $\theta + \epsilon h - k(B + \epsilon\Omega)$ and $K(B + \epsilon\Omega) - (\theta + \epsilon h)$ are semipositive, so the monotonicity and homogeneity in Proposition 10.8.1 give

$$k\lambda_\mu(B + \epsilon\Omega) \le \lambda_\mu(\theta + \epsilon h) \le K\lambda_\mu(B + \epsilon\Omega). \tag{27.101}$$

Since B is semipositive of rank at most R, the forms $B^j \wedge \Omega^{n-j}$ are semipositive and vanish for $j > R$. The closed nef formula following Theorem 27.9.1 therefore gives

$$\mathcal{V}_\mu(B + \epsilon\Omega) = \sum_{j=0}^{R} \binom{n}{j} \epsilon^{n-j} \int_X B^j \wedge \Omega^{n-j}, \tag{27.102}$$

with nonnegative coefficients. The form $B^R \wedge \Omega^{n-R}$ is positive at a point where B has rank R, so the coefficient of ϵ^{n-R} is positive. For $0 < \epsilon \le 1$ every power ϵ^{n-j} that occurs is at most ϵ^{n-R}, so the polynomial lies between its leading term and the sum of its coefficients times ϵ^{n-R}. Raising (27.101) to the nth power gives the two-sided bounds. Fixed positive factors do not affect the order, since their logarithms are divided by $\log \epsilon$.

For a modification $\pi\colon Y \to X$, the inequalities $c\pi^*B \le \pi^*\theta \le C\pi^*B$ hold on Y, and π^*B is smooth, closed and semipositive of rank at most R. The set where B has rank R is open and nonempty, so it meets the dense open set over which π is biholomorphic; hence π^*B has maximal rank R. Since $\dim Y = n$, the first part applies on Y. □

27.19.5.1 Linear blowups and the fixed generators

The manifolds are products of factors of two kinds. A *plain* factor is $Z_i = \mathbb{P}^{m_i}$ with $m_i \ge 1$; let H_i be its Fubini–Study form normalized to have integral one on lines. A *blown-up* factor is

$$Z_i = \mathrm{Bl}_{\mathbb{P}^{\ell_i}} \mathbb{P}^{m_i}, \qquad m_i \ge 2, \quad 0 \le \ell_i \le m_i - 2, \qquad r_i = m_i - \ell_i - 1,$$

the blowup of $\mathbb{P}^{m_i}$ along a linear subspace of dimension ℓ_i. We describe one such factor and suppress the index for the moment. In homogeneous coordinates $[z_0 : \cdots : z_\ell : w_0 : \cdots : w_r]$ on $\mathbb{P}^m$, with the center $\{w = 0\}$, the blowup has the incidence model

$$Z = \{(x, [t]) \in \mathbb{P}^m \times \mathbb{P}^r : w_a t_b = w_b t_a \text{ for all } a, b\}.$$

Let $p\colon Z \to \mathbb{P}^m$ and $q\colon Z \to \mathbb{P}^r$ be the projections. In the chart $t_j = 1$, the equations $w_a = t_a w_j$ are the usual charts of the blowup along $\{w_0 = \cdots = w_r = 0\}$. Writing $w = \lambda t$ shows that q is locally a $\mathbb{P}^{\ell+1}$-bundle, with fiber coordinates $[z : \lambda]$. Thus Z is smooth, q is a submersion, and the incidence inclusion is an embedding. With ω_m and ω_r the normalized Fubini–Study forms on $\mathbb{P}^m$ and $\mathbb{P}^r$, set

$$H = p^*\omega_m, \qquad Q = q^*\omega_r, \qquad \kappa = H + Q.$$

The form κ is Kähler, as the restriction of a product Kähler form, whereas Q is semipositive of constant rank r.

Let $E = \{w = 0\}$ be the exceptional divisor. The coordinate section w_j of $p^*\mathcal{O}(1)$ vanishes exactly on $E + q^{-1}\{t_j = 0\}$, so $\mathcal{O}(E) = p^*\mathcal{O}(1) \otimes q^*\mathcal{O}(-1)$. Give this line

bundle the product metric, whose curvature is $H - Q$. For its canonical section s_E, the function $\psi_E = \log |s_E|^2$ is locally integrable, and the Poincaré–Lelong formula, as in Section 27.21, gives

$$\mathrm{dd}^{\mathrm{c}}\psi_E = [E] - H + Q. \tag{27.103}$$

On $X = \prod_i Z_i$, put $n = \sum_i m_i$, and pull back all forms from the factors. Fix smooth strictly positive functions f_i on X, and, for each blown-up factor, a smooth strictly positive function g_i on X; these functions may depend on all coordinates of the product. The *fixed cone* consists of the backgrounds

$$\theta = \sum_i \big((a_i + b_i f_i)H_i + (c_i + d_i g_i)Q_i\big), \qquad a_i, b_i, c_i, d_i \geq 0, \tag{27.104}$$

where the terms involving Q_i, and the parameters c_i, d_i, are omitted for plain factors. All coefficient functions and generators are fixed, and only the parameters vary. Every member of the cone is smooth and semipositive, hence nef. We say that H_i *occurs* in θ if $a_i + b_i > 0$, and that Q_i occurs if $c_i + d_i > 0$. Define

$$R_i(\theta) = \begin{cases} m_i, & \text{if } H_i \text{ occurs,} \\ r_i, & \text{if } H_i \text{ does not occur and } Q_i \text{ occurs,} \\ 0, & \text{if neither occurs,} \end{cases} \qquad R(\theta) = \sum_i R_i(\theta). \tag{27.105}$$

The middle case can only occur on blown-up factors.

Theorem 27.19.1 *Every background θ in the fixed cone (27.104) has common order $n - R(\theta)$. More precisely, for every smooth positive probability form μ and Hermitian form h there are constants $L, U > 0$, depending on θ, μ, h, such that*

$$L\epsilon^{n-R(\theta)} \leq \mathcal{V}_\mu(\theta + \epsilon h) \leq U\epsilon^{n-R(\theta)}, \qquad 0 < \epsilon \leq 1. \tag{27.106}$$

Proof Let B_θ be the sum of the generators H_i and Q_i that occur in θ; it is closed and semipositive. The coefficient function of an occurring generator takes values in a compact subinterval of $(0, \infty)$, because f_i and g_i are strictly positive, so $cB_\theta \leq \theta \leq CB_\theta$ for some constants $c, C > 0$ (any constants if $\theta = 0$). Ranks of forms pulled back from different factors add pointwise. On a plain factor H_i has rank m_i. On a blown-up factor, H_i and $H_i + Q_i$ have rank m_i off E_i, where p_i is biholomorphic, while Q_i has constant rank r_i. Hence the maximal rank of B_θ, attained off all exceptional divisors, is $R(\theta)$, and Lemma 27.19.2 applies. □

Members of the fixed cone need not be closed: by construction

$$\mathrm{d}\theta = \sum_i b_i\, \mathrm{d}f_i \wedge H_i + \sum_{i \text{ blown up}} d_i\, \mathrm{d}g_i \wedge Q_i.$$

If only one of these terms has a nonzero coefficient and the corresponding form, say $\mathrm{d}f_i \wedge H_i$, is nonzero, then θ is not closed. On a blown-up factor, for instance, if

g is a small positive perturbation of a constant that varies along a fiber of q, then $\mathrm{d}g \wedge Q \neq 0$, since $r \geq 1$. On $\mathbb{P}^{m_1} \times \mathbb{P}^{m_2}$ with $m_1, m_2 \geq 1$, a positive function f that varies on the second factor gives the nonclosed background fH_1 of order m_2. Adding dd^{c} of a smooth function does not change the exterior derivative.

27.19.5.2 Equality of orders under modifications

By Proposition 27.19.6, pulling back by a modification can only increase the orders. For the fixed cone, the orders are preserved.

Proposition 27.19.10 *Let θ be in the fixed cone, and let $\pi\colon Y \to X$ be a modification from a compact Kähler manifold Y, for instance the product in which some plain factors are replaced by linear blowups as above. Then $\pi^*\theta$ has common order $n - R(\theta)$, and satisfies two-sided power bounds with this exponent for every smooth positive probability form and every Hermitian perturbation on Y.*

Proof By the proof of Theorem 27.19.1, $cB_\theta \leq \theta \leq CB_\theta$, where B_θ is closed and semipositive of maximal rank $R(\theta)$. The last assertion of Lemma 27.19.2 applies.□

27.19.5.3 Subtracting effective divisors

We now subtract effective divisors from members of the fixed cone. Let $\mathcal{O}(H_i)$ and $\mathcal{O}(Q_i)$ be the pullbacks of $\mathcal{O}(1)$ under the projections defining H_i and Q_i, with their Fubini–Study metrics, whose curvature forms are H_i and Q_i. Consider line bundles that are tensor products of integer powers of these bundles, negative powers included, and finite nonnegative real combinations of zero divisors of their nonzero holomorphic sections s_ν:

$$D = \sum_\nu t_\nu \operatorname{div}(s_\nu), \quad t_\nu \geq 0,$$

$$\eta_D = \sum_i (\delta_i H_i + \rho_i Q_i), \qquad \Phi_D = \sum_\nu t_\nu \log |s_\nu|^2.$$

Here η_D is the corresponding combination of the curvature forms of the fixed metrics, so that $[D] = \eta_D + \mathrm{dd}^{\mathrm{c}}\Phi_D$. The terms involving Q_i are omitted for plain factors.

Lemma 27.19.3 *The coefficients (δ_i, ρ_i) that occur for divisors D of this form are exactly those satisfying*

$$\delta_i \geq 0 \text{ on plain factors}; \qquad \delta_i \geq 0, \quad \delta_i + \rho_i \geq 0 \text{ on blown-up factors}.$$

Proof It suffices to treat one nonzero holomorphic section s, since nonnegative real combinations preserve the inequalities. Choose a point at which s is nonzero and which lies outside all exceptional divisors, and fix the coordinates of the other

factors. A projective line through that point inside a fiber of q_i has H_i-degree one and Q_i-degree zero. The restriction of s to it is a nonzero section, so the exponent of $O(H_i)$ is nonnegative. On a plain factor, a line through the point gives the same inequality.

On a blown-up factor, write the chosen point as $([z : w], [w])$ with $w \neq 0$, and choose a linear map A with $Aw = z$. The section of q_i given by $[t] \mapsto ([At : t], [t])$ passes through the point, and both H_i and Q_i have degree one on each projective line in its image. Such a line through the point exists because $r_i \geq 1$, and s restricts nontrivially to it. Its degree shows that the sum of the two exponents is nonnegative.

Conversely, let F_i be the pullback under q_i of a hyperplane. An admissible pair (δ_i, ρ_i) is realized by $\delta_i E_i + (\delta_i + \rho_i)F_i$, since E_i has curvature $H_i - Q_i$, and coefficients on plain factors are realized by hyperplanes. These are allowed nonnegative combinations and give every coefficient tuple in the stated range. □

Theorem 27.19.2 *Let θ be in the fixed cone, with parameters a_i, b_i, c_i, d_i, and let D be a divisor of the above form. Assume*

$$a_i \geq \delta_i \text{ for all } i, \qquad c_i \geq \rho_i \text{ for every blown-up factor.}$$

Then the residual $\beta = \theta - \eta_D$ lies in the fixed cone, and the common orders $a(\beta)$ and $a(\theta)$ satisfy

$$a(\beta) - a(\theta) = \sum_i \big(R_i(\theta) - R_i(\beta)\big) \geq 0.$$

The two orders are equal if and only if, for every factor,

$$\begin{aligned} a_i - \delta_i + b_i &> 0 \qquad \text{whenever } a_i + b_i > 0, \\ c_i - \rho_i + d_i &> 0 \qquad \text{whenever } a_i = b_i = 0 \text{ and } c_i + d_i > 0; \end{aligned}$$

the second condition concerns only blown-up factors.

Proof The residual parameters $(a_i - \delta_i, b_i, c_i - \rho_i, d_i)$ are nonnegative, so β lies in the fixed cone; in particular it is smooth, semipositive and nef. Apply Theorem 27.19.1 to θ and to β. If H_i occurs in θ but not in β, the rank R_i drops from m_i to $r_i < m_i$ or to zero. If H_i does not occur in θ, then $a_i = b_i = 0$, and the range of Lemma 27.19.3 together with $a_i \geq \delta_i$ forces $\delta_i = 0$ and hence $\rho_i \geq 0$; thus Q_i cannot appear in β if it was absent from θ. Therefore no rank R_i increases, each difference $R_i(\theta) - R_i(\beta)$ is nonnegative, and the sum vanishes exactly when the generator that determines $R_i(\theta)$ still occurs in β for every i. This is the stated pair of conditions. □

The parameter inequalities guarantee that the residual stays in the fixed cone; they do not describe all divisors whose residual is nef.

Example 27.19.1 On $Z = \mathrm{Bl}_{\mathbb{P}^\ell}\,\mathbb{P}^m$, choose a smooth function $g > 0$ with $\mathrm{d}g \wedge Q \neq 0$. For $t > 0$, the nonclosed positive background $\theta_t = tH + gQ$ has order zero. Removing sE, with its fixed curvature form $s(H - Q)$ and $0 \leq s \leq t$, gives

$$\beta_s = (t - s)H + (g + s)Q, \qquad \theta_t + \mathrm{dd}^{\mathrm{c}}(s\psi_E) = s[E] + \beta_s.$$

Every residual β_s is nef and not closed. Its order equals that of θ_t exactly for $s < t$, and at $s = t$ the order jumps by $\ell + 1$. These examples realize each integer jump $1, \ldots, m - 1$.

Proof The lower bound $\theta_t \geq \min(t, \min g)\kappa > 0$ shows that θ_t is positive. For $s < t$ the form H occurs in β_s and the rank is m; at $s = t$ only Q occurs, with rank $r = m - \ell - 1$. The order formula gives zero and $m - r = \ell + 1$, respectively. The exterior derivative of every residual is $\mathrm{d}g \wedge Q \neq 0$. Finally, let ℓ range from zero to $m - 2$. □

At the endpoint $s = t$, the current $t[E] + \beta_t$ in the class of θ_t has a divisorial part, although the minimal multiplicities of θ_t vanish because θ_t is nef, as explained before Corollary 27.19.1.

27.19.6 Variable multiples of semipositive generators

Lemma 27.19.2 applies to positive variable multiples of arbitrary closed semipositive generators on a compact Kähler manifold, and in particular to backgrounds that vanish on an open set.

Corollary 27.19.2 *On a compact Kähler manifold X of dimension n, fix finitely many smooth closed semipositive $(1,1)$-forms B_j and smooth strictly positive functions f_j. For $t_j \geq 0$, put*

$$\theta = \sum_j t_j f_j B_j, \qquad B = \sum_{j:t_j>0} B_j, \qquad R = \max_X \operatorname{rank} B.$$

The background θ is nef, has common order $n - R$, and satisfies the two-sided power bounds of Lemma 27.19.2. For every modification $\pi\colon Y \to X$ with Y a compact Kähler manifold, the pullback $\pi^*\theta$ has the same order and power bounds with the same exponent, for every smooth positive probability form and Hermitian perturbation on Y.

Proof If some t_j is positive, compactness and strict positivity of the functions f_j give $cB \leq \theta \leq CB$ with

$$c = \min_{j:t_j>0}\bigl(t_j \min_X f_j\bigr) > 0, \qquad C = \max_{j:t_j>0}\bigl(t_j \max_X f_j\bigr).$$

If all parameters vanish, then $B = \theta = 0$, and any $c, C > 0$ may be used. Lemma 27.19.2 gives all assertions. □

The constants depend on the parameters t_j and need not stay bounded as a positive parameter tends to zero. The rank of a sum records how the generators overlap, and it need not equal the sum of their ranks. Members of the family are not closed when $\sum_j t_j \, \mathrm{d}f_j \wedge B_j \neq 0$.

Example 27.19.2 Let $X = \mathbb{P}^1 \times \mathbb{P}^q$, $q \geq 1$, with projections p_1, p_2. Choose a smooth nonnegative function $b \not\equiv 0$ on $\mathbb{P}^1$ that vanishes on a nonempty open set, and a smooth strictly positive nonconstant function f on $\mathbb{P}^q$. With ω_1 the normalized Fubini–Study form on $\mathbb{P}^1$, set

$$B = p_1^*(b\omega_1), \qquad \theta = (f \circ p_2)B.$$

The smooth nef background θ is not closed, vanishes on a nonempty open set, and has common order q, with two-sided bounds by multiples of ϵ^q.

Proof The form $b\omega_1$ is closed, because the first factor is a curve, and semipositive. Its pullback has rank one where $b > 0$ and rank zero where $b = 0$, so its maximal rank is one. Since $n = q + 1$, Corollary 27.19.2 gives the order q. A nonconstant smooth function f on the connected second factor has $\mathrm{d}f \neq 0$ somewhere. At a point of the product where $b > 0$ and $\mathrm{d}f \neq 0$, the horizontal two-form B wedges nontrivially with the vertical one-form $p_2^*\mathrm{d}f$. Thus $\mathrm{d}\theta = p_2^*\mathrm{d}f \wedge B \neq 0$. □

These are geometric sufficient conditions for an integral common order, in the direction of Problem 27.19.1. They do not cover backgrounds without a closed semipositive comparison form, coefficient functions that are allowed to vanish, or the comparison of orders under residuals or non-Kähler modifications in general.

27.19.7 Flat anticanonical curves

The comparison criterion of Lemma 27.19.2 needs a closed semipositive comparison form. On a compact Kähler manifold S of dimension n, such a form of rank at most one arises from a smooth divisor D whose line bundle $\mathcal{O}_S(D)$ is *unitary flat* near D, that is, admits a Hermitian metric of zero curvature on a neighborhood of D. Cutting off the logarithm of the flat norm of the defining section gives a semipositive metric on $\mathcal{O}_S(D)$ whose curvature B vanishes near D and has rank at most one; every positive variable multiple fB then has order exactly $n - 1$, which is one on a surface. This holds whether or not some multiple of D moves in a linear system, so the order of degeneration does not detect semiampleness. We first show that a smooth anticanonical curve of self-intersection zero always has a unitary flat neighborhood, and then apply this to blowups of the plane at nine points of a cubic.

The construction of the flat metric rests on an estimate on an elliptic curve that is uniform in the holonomy of the twisting line bundle.

Lemma 27.19.4 *Let D be an elliptic curve with a flat Hermitian metric, let ϑ be a parallel $(1,0)$-form of unit length, and let v be its dual vector field. Let L be a unitary flat line bundle on D, with flat connection $\nabla = \partial_L + \bar\partial_L$. For every smooth L-valued $(0,1)$-form u, there is a unique smooth section f of L, orthogonal to the parallel sections, such that*

$$\bar\partial_L f = \iota_v \partial_L u.$$

If $\Pi_L^{0,1}$ denotes the orthogonal projection onto the parallel L-valued $(0,1)$-forms, then, for every Sobolev norm H^k defined by the flat structure,

$$\|f\|_{H^k} = \|(I - \Pi_L^{0,1})u\|_{H^k} \le \|u\|_{H^k}.$$

In particular, the bound does not depend on the holonomy of L.

Proof Put $P = \iota_v \partial_L$. Since the metric and L are flat, $P = \nabla_v$ and $P^* = -\nabla_{\bar v}$, and the Laplacian Δ_L satisfies $\Delta_L = P^*P = PP^*$ on sections and on $(0,1)$-forms. In degree $q \in \{0,1\}$, let G_q be the inverse of Δ_L on the orthogonal complement of its kernel, extended by zero on the kernel. The operators P and P^* commute with G_q. Set

$$f = \bar\partial_L^* G_1 P u.$$

The form Pu is orthogonal to the parallel forms, so $\bar\partial_L f = Pu$; also f is orthogonal to the parallel sections. Uniqueness follows because the kernel of $\bar\partial_L$ on sections consists of parallel sections. Moreover,

$$\|f\|_{L^2}^2 = \langle Pu, G_1 P u\rangle = \langle u, (I - \Pi_L^{0,1})u\rangle.$$

Covariant differentiation along a global parallel real frame commutes with all these operators, so applying the identity to every derivative proves the H^k assertion. The argument never divides by a lower bound for a nonzero eigenvalue of Δ_L: the derivative in Pu compensates for small eigenvalues. □

For the neighborhood argument, let S be a complex surface, not necessarily compact, and let $D \subset S$ be a smooth compact connected anticanonical divisor with $D^2 = 0$; thus D is the zero divisor of a holomorphic section s_D of the anticanonical bundle K_S^{-1}. By adjunction, $K_D = (K_S \otimes \mathcal{O}_S(D))|_D$ is trivial, so D is an elliptic curve. Its normal bundle $N = \mathcal{O}_S(D)|_D$ has degree zero, and therefore admits a unitary flat metric. Let $\pi\colon N \to D$ be the projection, $N^\times$ the complement of the zero section, and J_N the complex structure of N. The reciprocal s_D^{-1} is a meromorphic two-form Ω_S, nowhere zero on $S \setminus D$, with a simple pole along D; we denote its residue, a nowhere vanishing holomorphic one-form on D, by ϑ_D.

Choose parallel unitary local frames of N, with fiber coordinate w. On $N^\times$ put

$$\xi = w\partial_w, \qquad \Theta = \frac{\mathrm{d}w}{w}, \qquad \vartheta = \pi^*\vartheta_D, \qquad z = -\log|w|^2, \qquad \Omega_0 = \Theta \wedge \vartheta.$$

These objects do not depend on the frame, since parallel unitary frames differ by constant factors of modulus one. Give D the flat metric for which ϑ_D has unit length,

and let v be the parallel vector field dual to ϑ_D. Equip $N^\times$ with the flat cylindrical metric g for which the coframe ϑ, Θ is unitary, and let ∇ be its connection. The end $z \to +\infty$ of this cylinder is the end near the zero section.

Lemma 27.19.5 *There is a diffeomorphism Υ from a neighborhood of the zero section in N onto a neighborhood of D in S, equal to the identity on D, such that Υ maps each fiber disk of N to a holomorphic curve in S, and*

$$|\nabla^k(\Upsilon^*\Omega_S - \Omega_0)|_g = O(\mathrm{e}^{-z/2}) \qquad (k \geq 0).$$

Proof *Step 1: a first identification.* Start with the normal exponential map of a smooth Hermitian metric near D. Its differential along D is complex linear and induces the identity on the normal quotient. In a parallel local frame and a local coordinate z_0 on D with $\vartheta_D = \mathrm{d}z_0$, choose holomorphic coordinates Z, W on S with $D = \{W = 0\}$. Their pullbacks under the first map satisfy

$$Z = z_0 + b(z_0, \bar{z}_0)w + O(|w|^2), \qquad W = w + O(|w|^2).$$

The residue normalization gives $\Omega_S = (1 + Wa(Z, W))\,\mathrm{d}W/W \wedge \mathrm{d}Z$ with a holomorphic. Substituting, and noting that $\Theta \wedge \mathrm{d}w = w\Theta \wedge \Theta = 0$, we obtain $\Omega_S - \Omega_0 = O(|w|)$ in the coframe $\mathrm{d}z_0, \mathrm{d}\bar{z}_0, \Theta, \bar{\Theta}$. Every derivative by $\partial_{z_0}, \partial_{\bar{z}_0}, w\partial_w, \bar{w}\partial_{\bar{w}}$ preserves this bound. Since $|w| = \mathrm{e}^{-z/2}$, this proves the cylindrical estimate for the first map.

Step 2: holomorphic fiber disks. It remains to modify the first map so that the fiber disks become holomorphic, without changing its first jet along D. On a fiber disk $\Delta_{p,\rho} \subset N_p$ of radius ρ, perturb the map by $\Phi_Y(q) = \exp_q(Y_q)$, where Y is a vector field along the image of the disk. Identify the target tangent spaces with one fixed complex vector space over p, by Chern parallel transport along the radial paths and the short exponential paths. In these identifications, the antiholomorphic part of the differential of the perturbed disk is

$$\mathcal{F}_{p,\rho}(Y) = \tfrac{1}{2}\big(\mathrm{d}\Phi_Y + J_S \mathrm{d}\Phi_Y J_N\big)|_{T\Delta_{p,\rho}}.$$

Fix a Hölder exponent $0 < \nu < 1$, and use $C^{k,\nu}$ norms on the disks rescaled to unit size. The Cauchy–Green operator T_ρ of the disk gives the normalized right inverse

$$R_\rho a = T_\rho a - T_\rho a(0) - w(\partial_w T_\rho a)(0)$$

of $\bar{\partial}$. We restrict its domain to $(0,1)$-forms vanishing at the center; its image consists of fields whose value and first jet vanish at the center, and on this image $\bar{\partial}$ and R_ρ are inverse to each other. The differential of $\mathcal{F}_{p,\rho}$ at zero has the form $\bar{\partial} + K_{p,\rho}$, with

$$\|K_{p,\rho}R_\rho\| \leq C\rho, \qquad \|\mathcal{F}_{p,\rho}(0)\| \leq C\rho^2.$$

Indeed, the two complex structures agree along the zero section, and rescaling by $w \mapsto \rho w$ contributes an additional factor ρ to a pulled-back one-form. For small ρ, the operator $R_\rho(I + K_{p,\rho}R_\rho)^{-1}$ is a right inverse of the differential, and the quantitative

implicit function theorem gives a correction $Y = O(|w|^2)$ with vanishing first jet that makes the disk holomorphic.

Step 3: global smoothness. Compactness of D makes the choice of ρ uniform in p. The normalized Cauchy operator is equivariant under unitary changes of the fiber frame, so the intrinsic parallel transports and the uniqueness of the small solution make the corrections agree on overlaps. The parametric implicit function theorem gives smooth dependence on p, and interior elliptic regularity on slightly smaller fiber disks gives smoothness along the fibers. Since the first jet is unchanged, the correction changes the map by $O(|w|^2)$ together with its cylindrical derivatives. Composing the first map with this correction therefore preserves the estimate and proves the lemma.□

Theorem 27.19.3 *Let S be a complex surface and $D \subset S$ a smooth compact connected anticanonical divisor with $D^2 = 0$. Then the line bundle $\mathcal{O}_S(D)$ is unitary flat on a neighborhood of D.*

Proof We look for a fiber-preserving change of the identification of Lemma 27.19.5 after which the coordinate z becomes pluriharmonic. The logarithm of the flat norm of the defining section is then $-z$, up to a pluriharmonic correction.

Step 1: horizontal forms. Use the identification Υ of Lemma 27.19.5, and omit the pullback notation. A form on $N^\times$ is horizontal if its contractions with ξ and $\bar\xi$ vanish. For a horizontal form η define

$$\mathrm{d}_H\eta = \mathrm{d}\eta - \Theta \wedge \mathcal{L}_\xi\eta - \bar\Theta \wedge \mathcal{L}_{\bar\xi}\eta.$$

A form η_m on D with values in N^{-m} defines the horizontal form $w^m\eta_m$, homogeneous of degree m in w, and direct differentiation shows that d_H acts on it as the flat covariant exterior derivative. Since the fibers are holomorphic,

$$\iota_{\bar\xi}\Omega_S = 0, \qquad \Omega_S = \Theta \wedge \alpha + \beta, \qquad \alpha = \iota_\xi\Omega_S,$$

with α, β horizontal. Closedness gives $\mathcal{L}_{\bar\xi}\alpha = \mathcal{L}_{\bar\xi}\beta = 0$ and $\mathcal{L}_\xi\beta = \mathrm{d}_H\alpha$.

Step 2: a horizontal inverse. Fix $0 < \delta < 1/2$, and let $U_T = \{z \geq T\}$. For tensors define the weighted norms

$$\|\eta\|^2_{k,\delta,T} = \sum_{j=0}^{k} \int_{U_T} \mathrm{e}^{2\delta(z-T)} |\nabla^j\eta|^2_g \, \mathrm{d}V_g.$$

Let $\mathcal{H}^{k+1}_{\delta,T}$ be the space of functions f on U_T with $\|f\|_{k+1,\delta,T} < \infty$ and $\mathcal{L}_{\bar\xi}f = 0$, and let $\mathcal{D}^k_{\delta,T}$ be the space of complex two-forms γ on U_T with $\|\gamma\|_{k,\delta,T} < \infty$, $\mathrm{d}\gamma = 0$ and $\iota_{\bar\xi}\gamma = 0$. We construct an operator

$$Q_T \colon \mathcal{D}^k_{\delta,T} \longrightarrow \mathcal{H}^{k+1}_{\delta,T}, \qquad \|Q_T\gamma\|_{k+1,\delta,T} \leq C_{k,\delta}\|\gamma\|_{k,\delta,T},$$

with constant independent of T, such that

$$\mathrm{d}_H(Q_T\gamma\,\vartheta) = \gamma - \Theta \wedge \iota_\xi\gamma, \qquad Q_T(\mathrm{d}f \wedge \vartheta) = f. \tag{27.107}$$

Write $\gamma = \Theta \wedge \alpha + \beta$, with α, β horizontal. The conditions on γ give $\mathcal{L}_{\tilde{\xi}}\gamma = 0$ and, in the expansion in homogeneous components along the fibers, $m\beta_m = \mathrm{d}_H \alpha_m$. Only components with $m \geq 1$ occur, because the weighted norm is finite and $\delta > 0$ excludes the Laurent components of degree $m \leq 0$. Decompose

$$\alpha_m = -u_m \vartheta + v_m, \qquad u_m \in \Omega^0(D, N^{-m}), \quad v_m \in \Omega^{0,1}(D, N^{-m}).$$

Lemma 27.19.4 gives sections e_m with $\bar{\partial} e_m = \iota_v \partial v_m$ and $\|e_m\|_{H^k} \leq \|v_m\|_{H^k}$. Put $f_m = -(u_m + e_m)/m$. Then

$$\mathrm{d}_H(f_m \vartheta) = \beta_m, \qquad \|m f_m\|_{H^k} \leq C \|\alpha_m\|_{H^k}.$$

Also $\|\bar{\partial} f_m\|_{H^k} \leq C\|\beta_m\|_{H^k}$. Since D is flat, integration by parts gives the same bound for ∂f_m, which controls the horizontal derivatives of order $k+1$. A vertical derivative of a homogeneous component multiplies it by m, so every derivative involving a vertical direction is controlled by the bound for $m f_m$. The components are orthogonal on the circles $\{|w| = \text{constant}\}$, and the squared radial factor of the mth component is e^{-mz} with $m - 2\delta > 0$. Integrating the weight therefore gives the required estimate for $f = \sum_{m \geq 1} f_m$, where each f_m is identified with its homogeneous horizontal lift, and we put $Q_T \gamma = f$.

For forms with Sobolev coefficients, regularize each component by the heat operator of the flat base, which commutes with the differential equations above. The uniform estimates and orthogonality make the finite partial sums converge in the target space, and passing to the limit proves the first identity in (27.107). If $\gamma = \mathrm{d}f \wedge \vartheta$, then $\alpha_m = m f_m \vartheta$, so v_m and the normalized e_m vanish, and the construction returns f_m. This proves the second identity.

Step 3: a fixed point. Put $\eta = \Omega_S - \Omega_0$, and choose an integer $k \geq 4$. By the cylindrical estimate, the weighted H^{k+2} norm of η on U_{T-1} and the supremum norms there of its first $k+2$ covariant derivatives are bounded by a number a_T with $a_T \to 0$. For small $f \in \mathcal{H}^{k+1}_{\delta,T}$, the map $\Phi_f(\zeta) = \mathrm{e}^{f(\zeta)}\zeta$, which rescales each fiber point ζ, is a C^2 diffeomorphism onto its image and maps U_T into U_{T-1}. Its restriction to every fiber is holomorphic, so $\Phi_f^* \Omega_S - \Omega_0$ lies in the domain of Q_T. We look for f such that $\Phi_f^* \Omega_S$ has no horizontal component. Since

$$\Phi_f^* \Omega_0 - \Omega_0 = \mathrm{d}f \wedge \vartheta,$$

(27.107) reduces this to the equation

$$Q_T(\Phi_f^* \Omega_S - \Omega_0) = f + E_T(f) = 0, \qquad E_T(f) = Q_T(\Phi_f^* \eta).$$

On a fixed sufficiently small ball, multiplication and composition estimates in weighted Sobolev spaces on the flat cylinder give

$$\|E_T(f)\|_{k+1,\delta,T} \leq C a_T, \qquad \|E_T(f) - E_T(g)\|_{k+1,\delta,T} \leq C a_T \|f - g\|_{k+1,\delta,T}.$$

The constants are independent of T, because translation in the cylindrical direction is an isometry, the weight is normalized at T, and the shifts of z are uniformly bounded. Differentiating the pullback along the segment from f to g involves one Lie derivative of the fixed form η and at most one derivative of $f - g$, and the gain of one derivative in Q_T restores the norm. For T sufficiently large, $-E_T$ is therefore a contraction of the closed ball of radius $2Ca_T$ into itself. Let f be its fixed point.

Step 4: a pluriharmonic function. By the first identity in (27.107), the horizontal component of $\Phi_f^*\Omega_S$ vanishes, so $\Theta \wedge \Phi_f^*\Omega_S = 0$. The pulled-back two-form is nowhere zero and of type $(2,0)$ for $\Phi_f^* J_S$, so Θ is of type $(1,0)$ for this complex structure. Since $\mathrm{d}z = -(\Theta + \bar{\Theta})$ and $\mathrm{d}\Theta = 0$, the function z is pluriharmonic for $\Phi_f^* J_S$.

Step 5: the flat metric. Return to S through $A = \Upsilon \circ \Phi_f$. On each fiber, f is bounded and holomorphic on a punctured disk, so it extends holomorphically across the center, and its expansion in positive powers of w gives $f(0) = 0$. Since f is uniformly small, Rouché's theorem, comparing $w\mathrm{e}^{f(w)} - c$ with $w - c$ on a fixed smaller circle, shows that every sufficiently small value c is attained. Hence the image of A contains a punctured neighborhood of D. The function $r = z \circ A^{-1}$ is C^2 and pluriharmonic there, hence smooth by elliptic regularity. If w_S is a local holomorphic defining function for D, the unchanged first normal jet of Υ and the bounded factor e^f give

$$C^{-1}|w| \le |w_S \circ A| \le C|w|.$$

Thus $r + \log|w_S|^2$ is bounded and pluriharmonic off D. Both this function and its negative are bounded plurisubharmonic functions off D, so they extend plurisubharmonically across D; the sum of the extensions vanishes almost everywhere, hence everywhere, so the extension is pluriharmonic. It follows that $F = -r$, which has a logarithmic pole along D, satisfies $\mathrm{dd}^{\mathrm{c}} F = [D]$ near D. Locally, the metric for which the squared norm of the defining section equals e^F is smooth and flat, since $F - \log|w_S|^2$ is smooth and pluriharmonic. These local metrics agree on overlaps and give the required unitary flat metric. □

Proposition 27.19.11 *Let D be a smooth nonempty compact divisor in a complex manifold S, and suppose $\mathcal{O}_S(D)$ is unitary flat near D. Then $\mathcal{O}_S(D)$ has a smooth semipositive metric on S whose curvature B has rank at most one and vanishes both near D and outside a relatively compact neighborhood of D. If S is compact Kähler, then* $\max_S \operatorname{rank} B = 1$.

Proof Let s_D be the defining section and h_0 a flat metric on a neighborhood U of D. Put $F = \log|s_D|^2_{h_0}$, which is pluriharmonic on $U \setminus D$. Choose a relatively compact tubular neighborhood $U_0 \Subset U$ of D. Since F tends uniformly to $-\infty$ along D and is bounded on ∂U_0, after adding a constant to F we may assume $F > 0$ near ∂U_0. Choose a smooth concave function χ and a constant c with $\chi(t) = t + c$ for $t \le -1$ and $\chi(t) = 0$ for $t \ge 0$. Define $\widetilde{F} = \chi(F)$ on U_0, extended by zero outside U_0, and define a metric h on $S \setminus D$ by

$$|s_D|^2_h = \mathrm{e}^{\widetilde{F}}.$$

Near D this gives $h = \mathrm{e}^c h_0$, so h extends smoothly across D; it is also smooth near ∂U_0, where $\widetilde{F} = 0$. By the Poincaré–Lelong formula its curvature is $B = [D] - \mathrm{dd}^{\mathrm{c}}\widetilde{F}$,

and off D, since F is pluriharmonic,

$$B = -\chi''(F)\,\mathrm{d}F \wedge \mathrm{d}^{\mathrm{c}}F \geq 0.$$

This smooth closed form has rank at most one and vanishes where asserted. If S is compact, of dimension n, with Kähler form κ, then

$$\int_S B \wedge \kappa^{n-1} = \int_D \kappa^{n-1} > 0,$$

so B is not identically zero and its maximal rank is one. □

Corollary 27.19.3 *Let S be a compact Kähler manifold of dimension $n \geq 2$ and D a smooth nonempty divisor whose line bundle is unitary flat near D, and let B be the curvature form of Proposition 27.19.11. For every smooth function $f > 0$, every Hermitian form h and every smooth positive probability form μ, there are constants $L, U > 0$ such that*

$$L\epsilon^{n-1} \leq \mathcal{V}_\mu(fB + \epsilon h) \leq U\epsilon^{n-1}, \qquad 0 < \epsilon \leq 1.$$

For a Kähler form κ, the closed background B itself satisfies

$$\mathcal{V}_\mu(B + \epsilon\kappa) = n\epsilon^{n-1}\int_D \kappa^{n-1} + \epsilon^n \int_S \kappa^n.$$

For suitable f, the background fB is not closed.

Proof The inequalities $(\min f)B \leq fB \leq (\max f)B$ and Lemma 27.19.2, with $R = 1$, give the two-sided bounds. The polynomial follows from the closed nef formula, $B^2 = 0$ and the integral in the proof of Proposition 27.19.11. At a point where B has rank one, choose a smooth positive f whose differential has a nonzero component along the kernel of B. Then $\mathrm{d}(fB) = \mathrm{d}f \wedge B \neq 0$ there. □

Recall that a line bundle is *semiample* if some positive power of it is generated by global sections, and that $|kD|$ denotes the linear system of effective divisors linearly equivalent to kD. A line bundle on a curve is *torsion* if some positive power is trivial; the least such power is its exact order.

Proposition 27.19.12 *Let S be obtained from $\mathbb{P}^2$ by nine successive point blowups, each centered on the strict transform of a fixed smooth cubic. The final strict transform D of the cubic is a smooth elliptic anticanonical curve with $D^2 = 0$, and the anticanonical bundle has a smooth semipositive metric whose curvature has rank at most one, as in Proposition 27.19.11. The anticanonical bundle is semiample exactly when $N = \mathcal{O}_D(D)$ is torsion. If N has exact order m, then $|mD|$ is basepoint-free and $|kD| = \{kD\}$ for $1 \le k < m$. If N is not torsion, then $|kD| = \{kD\}$ for every $k \ge 1$. There are configurations of nine distinct centers for which N is not torsion.*

Proof A blowup at a point of the cubic replaces the class of the cubic by its pullback minus the exceptional curve, exactly as it does for the anticanonical class. The strict transform remains isomorphic to the cubic, and its self-intersection decreases by one. After nine steps, D is anticanonical and $D^2 = 9 - 9 = 0$. Theorem 27.19.3 and Proposition 27.19.11 give the metric.

For $k \ge 1$ we use the restriction sequences

$$0 \longrightarrow \mathcal{O}_S((k-1)D) \longrightarrow \mathcal{O}_S(kD) \longrightarrow N^k \longrightarrow 0.$$

A nontrivial line bundle of degree zero on an elliptic curve has no nonzero section, since such a section would have a zero divisor of degree zero and would trivialize the bundle. If N is not torsion, induction on k gives $h^0(S, \mathcal{O}_S(kD)) = h^0(S, \mathcal{O}_S) = 1$. This proves that every $|kD|$ is the single divisor kD, and in particular that the anticanonical bundle is not semiample.

Suppose now that N has exact order m. For $1 \le k < m$ we have $H^0(D, N^k) = 0$, and $H^1(D, N^k) = 0$ by Serre duality on the elliptic curve. Since S is rational, $H^1(S, \mathcal{O}_S) = 0$. The same sequences give $h^0(S, \mathcal{O}_S(kD)) = 1$ and $H^1(S, \mathcal{O}_S(kD)) = 0$ for $0 \le k < m$. For $k = m$ the restriction map to $H^0(D, N^m)$ is therefore surjective, and the constant section of $N^m \simeq \mathcal{O}_D$ lifts to a section of $\mathcal{O}_S(mD)$ that does not vanish on D. Together with s_D^m, this section generates $\mathcal{O}_S(mD)$ everywhere. Thus $|mD|$ is basepoint-free, while no smaller positive multiple of D moves.

Finally, fix eight distinct points on the cubic and vary the ninth. Under the identification of D with the cubic, the normal bundle is $\mathcal{O}_D(3) \otimes \mathcal{O}_D(-p_1 - \cdots - p_9)$, where $\mathcal{O}_D(3)$ is the restriction of $\mathcal{O}_{\mathbb{P}^2}(3)$. As p_9 varies, these bundles run through the whole group of line bundles of degree zero, by the Abel map of the elliptic curve. The torsion subgroup is countable, whereas the curve is uncountable, so even after excluding the first eight points there are non-torsion choices. For them, Corollary 27.19.3 gives nonclosed backgrounds of order one, although the anticanonical bundle is not semiample. □

27.19.8 Flat divisors and Hartogs extension

All the nine-point blowups of Proposition 27.19.12 carry backgrounds of order one, so the order of degeneration does not distinguish the torsion case, in which a multiple of D moves in a basepoint-free pencil, from the non-torsion case, in which no multiple

of D moves. The difference is visible in the complex geometry near D. On the tube where the curvature form of Proposition 27.19.11 vanishes, the logarithm of the flat norm of the defining section is pluriharmonic. Its regular level sets are therefore Levi-flat, that is, real hypersurfaces foliated by complex hypersurfaces. These complex leaves are governed by the unitary holonomy of the normal bundle: finite holonomy gives compact fibers of a local holomorphic function, whereas infinite holonomy makes every leaf dense and forces every holomorphic function on the punctured tube to be constant. In both cases the complement of D is not Stein.

In this subsection S is a complex manifold of dimension $n \geq 2$ and $D \subset S$ is a smooth compact connected divisor. Suppose that $L = \mathcal{O}_S(D)$ has a unitary flat metric h on a neighborhood of D. Write s_D for its defining section and $N = L|_D$ for the normal bundle, with the induced metric and flat connection, and let $\chi \colon \pi_1(D) \to \mathrm{U}(1)$ be the holonomy character of N, which describes parallel transport around closed loops. In parallel unit frames, s_D is represented by holomorphic functions W_j satisfying $W_j = t_{jk} W_k$ with locally constant $t_{jk} \in \mathrm{U}(1)$. The flat connection defines on each circle bundle $\{v \in N : |v| = r\}$ a foliation whose leaves are given locally, in parallel frames, by $\{w = \text{constant}\}$; we call it the flat horizontal foliation.

Lemma 27.19.6 *There are $R > 0$, a neighborhood U of D and a smooth diffeomorphism*

$$\Upsilon \colon \{v \in N : |v| < R\} \longrightarrow U$$

that is the identity on the zero section and preserves the defining function: in each parallel normal frame, $W_j(\Upsilon(p, w)) = w$. In particular, $|s_D|_h \circ \Upsilon = |w|$. For $0 < r < R$, the set

$$M_r = \{x \in U : |s_D(x)|_h = r\}$$

is a compact connected Levi-flat hypersurface, and Υ maps the leaves of the flat horizontal foliation of $\{|w| = r\}$ onto the complex leaves of M_r.

Proof Choose a smooth normal exponential map Υ_0 whose first normal derivative is complex linear; the derivative of s_D identifies the normal bundle with $L|_D$, and we use this identification to fix that first derivative. Parallel transport along the normal rays identifies $\Upsilon_0^* L$ with the pullback of N. In compatible parallel frames one then has

$$W_j(\Upsilon_0(p, a)) = a + O(|a|^2).$$

For each p, this is a smooth map between real two-dimensional disks whose derivative at zero is the identity. The inverse function theorem with parameters, uniformly over the compact base D, gives a unique small $a = a(p, w)$ with $W_j(\Upsilon_0(p, a)) = w$. It depends smoothly on (p, w), and its first derivative in w at $w = 0$ is the identity. On an overlap of frames, a and w change by the same constant unitary factor, so uniqueness makes these corrections agree. After decreasing R, the map $(p, w) \mapsto (p, a(p, w))$ is a diffeomorphism of a tube onto its image, and its composition with Υ_0 is the required map Υ.

Off D, the function $F = \log |s_D|_h^2$ is pluriharmonic, and its radial derivative in the tube is nonzero. Hence every M_r is a smooth hypersurface; it is compact and

connected, being a circle bundle over D. Locally its complex tangent space is $\ker \mathrm{d}W_j$, so it is Levi-flat, with complex leaves the hypersurfaces $\{W_j = \text{constant}\}$. The identity $W_j \circ \Upsilon = w$ pulls this distribution back to $\{\mathrm{d}w = 0\}$ in parallel frames, which is the flat horizontal foliation of the circle bundle. The map Υ is only a smooth tubular identification; its base projection need not be holomorphic. □

Proposition 27.19.13 *In this setting, N is torsion as a holomorphic line bundle if and only if its unitary holonomy has finite image. If N has exact order m, then a sufficiently small tube U admits a proper surjective holomorphic map*

$$\phi \colon U \longrightarrow \{z \in \mathbb{C} : |z| < R^m\}$$

with connected fibers and $\mathrm{div}(\phi) = mD$. Its fibers over nonzero values are smooth compact complex hypersurfaces.

Proof If $\chi^m = 1$, a parallel nowhere-zero section trivializes N^m holomorphically. Conversely, if a holomorphic section e trivializes N^m, then $\log |e|^2$ is pluriharmonic on the compact connected manifold D, hence constant by the maximum principle. In parallel holomorphic frames the coefficient of e is a holomorphic function of constant modulus, hence constant. Therefore e is parallel and $\chi^m = 1$. Consequently the least such m is both the order of N as a torsion line bundle and the order of the finite holonomy group.

By Lemma 27.19.6, the tube U retracts onto D. Hence the flat bundle L^m on U has trivial holonomy and a global parallel section e_U of unit length. Define $\phi = s_D^m / e_U$. Then ϕ is holomorphic, $\mathrm{div}(\phi) = mD$ and $|\phi| = |s_D|_h^m$. The inverse image of a compact subset of the target lies in a smaller closed normal disk bundle and is closed there, which proves properness, and the tubular model shows that every value in the disk is attained.

Choose local parallel unit frames e_j of L with $e_j^m = e_U$, so that $\phi = W_j^m$. In the tubular model, the fiber over a nonzero value consists, over each point of D, of the m roots of that value. Parallel transport acts on these roots through χ, whose image has exact order m and therefore acts transitively. Thus this fiber is a connected covering of D, as a real manifold. Since $\mathrm{d}\phi = mW_j^{m-1}\mathrm{d}W_j$, it is a smooth complex hypersurface, and it is compact because ϕ is proper. The fiber over zero has connected support D, which completes the proof that all fibers are connected. □

Theorem 27.19.4 *Suppose N is not torsion. Then every complex leaf of M_r is dense in M_r, and every holomorphic function on $U \setminus D$ is constant:*

$$H^0(U \setminus D, \mathcal{O}_S) = \mathbb{C}.$$

Suppose moreover that S is connected. For every compact set $K \subset S \setminus D$ such that $(S \setminus D) \setminus K$ is connected, every holomorphic function on $(S \setminus D) \setminus K$ is constant, and in particular extends holomorphically to $S \setminus D$. Similarly, for every connected open neighborhood V of D, every holomorphic function on $V \setminus D$ is constant and extends to $S \setminus D$.

Proof By Proposition 27.19.13, $\chi(\pi_1(D))$ is an infinite subgroup of $\mathrm{U}(1)$. Such a subgroup is dense, since a proper closed subgroup of the circle is finite, as one sees by lifting it to a closed subgroup of $\mathbb{R}$ containing $2\pi\mathbb{Z}$. The circle bundle with its flat horizontal foliation is the suspension of χ, that is, the quotient of $\widetilde{D} \times S^1$ by the diagonal action of $\pi_1(D)$, acting on S^1 through χ. A leaf projects onto D, and its intersection with each fiber contains the holonomy orbit of any one of its points, which is dense in the fiber. In a local trivialization, horizontal transport maps the orbits in nearby fibers to each other by a fixed rotation, so the leaf is dense in the whole circle bundle. By Lemma 27.19.6, every complex leaf is dense in M_r.

Let f be holomorphic on $U \setminus D$. Its modulus attains a maximum on the compact set M_r. On the complex leaf through a maximum point, $|f|$ is bounded by this maximum and attains it, so the maximum modulus principle makes f constant on the connected leaf. By density and continuity, f equals the same constant c on M_r. Near any point of M_r, choose holomorphic coordinates (z, W_j) with $z \in \mathbb{C}^{n-1}$. For each fixed z, the holomorphic function $W_j \mapsto f(z, W_j) - c$ vanishes on an arc of the circle $|W_j| = r$, so it vanishes on the coordinate neighborhood by the one-variable identity theorem. Analytic continuation on the connected punctured tube gives $f \equiv c$.

A compact set $K \subset S \setminus D$ is disjoint from a sufficiently small tube around the compact divisor D. A holomorphic function on $(S \setminus D) \setminus K$ is therefore constant on this punctured tube, and by the identity theorem on the connected set $(S \setminus D) \setminus K$ it is constant everywhere; the constant function is the required extension. Similarly, every open neighborhood V of D contains a full tube, and $V \setminus D$ is connected, because paths in the connected manifold V can be perturbed off the real codimension-two submanifold D. The identity theorem gives the assertion for V. The same path argument shows that $S \setminus D$ is connected. □

Recall that a complex manifold Y is *holomorphically separable* if its holomorphic functions separate points, and *holomorphically convex* if the holomorphic hull

$$\widehat{K}_Y = \{y \in Y : |f(y)| \le \max_K |f| \text{ for every } f \in \mathcal{O}(Y)\}$$

of every compact set $K \subset Y$ is compact. A Stein manifold has both properties.

Corollary 27.19.4 *For any holonomy of N, the punctured tube $U \setminus D$ admits no smooth strictly plurisubharmonic function, and $S \setminus D$ is not Stein. In the non-torsion nine-point blowups of Proposition 27.19.12, every holomorphic function on the complement of D is constant, and the Hartogs extension of Theorem 27.19.4 across compact holes holds.*

Proof The restriction of a smooth strictly plurisubharmonic function on $U \setminus D$ to a complex leaf of M_r is strictly plurisubharmonic. At a maximum point on the compact set M_r, this restriction has a local maximum on the leaf, which has positive dimension because $n \ge 2$; this contradicts the maximum principle.

In the finite-holonomy case, each fiber of the map of Proposition 27.19.13 over a nonzero value is a connected compact complex hypersurface of positive dimension. By the maximum modulus principle, every holomorphic function on $S \setminus D$ is constant

on it, so two distinct points of the fiber cannot be separated. In the infinite-holonomy case, Theorem 27.19.4 makes every holomorphic function constant on the punctured tube, which again contradicts holomorphic separability. In both cases $S \setminus D$ is not Stein. Finally, Theorem 27.19.3 gives the flat neighborhood for the nine-point blowups, whose normal bundle is non-torsion by assumption, and Theorem 27.19.4, also with $K = \varnothing$, gives the last assertion. □

The nine-point blowups show that a background with a well-defined order of degeneration can have very different holomorphic geometry near its null locus; Appendix B gives a second instance of this contrast, for Kähler classes on a ruled surface over an elliptic curve.

Problem 27.19.1 Which geometric conditions on a nonclosed nef background force a common degeneration order, or an integral one? When is equality attained in the modification and residual order comparisons?

27.20 Current tests for smooth positivity

Lamari's duality, Theorem 4.3.1, characterizes pseudoeffective backgrounds by their pairings with Gauduchon metrics. Smooth positivity, and hence nefness, is characterized in the same way by pairing with all positive dd^{c}-closed $(n-1,n-1)$-currents. By the Hahn–Banach theorem, the minimum of these pairings over normalized currents is exactly the supremum of the numbers a such that $\theta+\mathrm{dd}^{\mathrm{c}}u > ah$ for some smooth function u, and the nef boundary is where this minimum vanishes. The full set of currents is needed here: the closure of the smooth tests used in Lamari's duality detects pseudoeffectivity and bigness instead. The minimizing currents at the nef boundary appear again at the maximal time of the Chern–Ricci flow, and positive variational volume forces their determinant measures to vanish.

Let X again be a connected compact complex manifold of dimension $n \geq 1$. Fix a Hermitian metric h, and let $\mathscr{G}_h$ be the set of normalized Gauduchon currents of Section 4.4, that is, positive dd^{c}-closed $(n-1,n-1)$-currents T with $\int_X h \wedge T = 1$. This set is weakly compact for $n \geq 2$. For $n = 1$, elliptic regularity and the maximum principle give $\mathscr{G}_h = \{1/\int_X h\}$, so compactness holds in this case as well.

For a smooth real $(1,1)$-form θ, define

$$m_h(\theta) = \min_{T \in \mathscr{G}_h} \int_X \theta \wedge T.$$

The minimum exists by weak compactness, and it is finite since $-Ch \leq \theta \leq Ch$ for some constant C.

Theorem 27.20.1 *The background θ has a smooth positive representative if and only if $m_h(\theta) > 0$. Moreover, θ is nef if and only if $m_h(\theta) \geq 0$, and*

$$m_h(\theta) = \sup\{a \in \mathbb{R} : \ \exists u \in C^\infty(X,\mathbb{R}),\ \theta + \mathrm{dd}^{\mathrm{c}}u > ah\}.$$

Proof If $\theta + \mathrm{dd^c} u \geq ah$ with $a > 0$ and u smooth, then integration by parts gives $\int_X \theta \wedge T \geq a$ for every $T \in \mathscr{G}_h$. Since a smooth positive representative satisfies such a bound for some $a > 0$ by compactness, this proves one implication.

For the converse, work in the real Fréchet space of smooth real $(1,1)$-forms. Let $\mathcal{K}$ be its open convex cone of strictly positive forms, and let $\mathcal{L} = \mathrm{dd^c} C^\infty(X, \mathbb{R})$. If the affine subspace $\theta + \mathcal{L}$ is disjoint from $\mathcal{K}$, the Hahn–Banach separation theorem for an open convex set and a disjoint affine subspace gives a nonzero continuous real linear functional T with

$$T(\theta + \ell) \leq T(\eta) \qquad (\ell \in \mathcal{L},\ \eta \in \mathcal{K}).$$

Replacing ℓ by arbitrary real multiples shows that $T(\mathcal{L}) = 0$. Scaling a fixed $\eta \in \mathcal{K}$ to zero shows that $T(\theta) \leq 0$, while scaling it to infinity shows that $T(\eta) \geq 0$. Since every smooth semipositive form is a limit of strictly positive forms, T is a positive current, and positivity makes it a current of order zero. The equality $T(\mathcal{L}) = 0$ says exactly that $\mathrm{dd^c} T = 0$. Since $T \neq 0$ and $h > 0$, the trace $T(h)$ is positive, and normalizing T gives an element of $\mathscr{G}_h$ with nonpositive pairing against θ. This contradicts $m_h(\theta) > 0$ and proves the converse.

Applying the equivalence just proved to $\theta - ah$, and using $m_h(\theta - ah) = m_h(\theta) - a$, shows that the set in the supremum is exactly $(-\infty, m_h(\theta))$. Finally, θ is nef if and only if $\theta + \varepsilon h$ has a smooth positive representative for every $\varepsilon > 0$, that is, if and only if $m_h(\theta) + \varepsilon > 0$ for all such ε, or equivalently $m_h(\theta) \geq 0$. □

For $n \geq 2$, let $\mathscr{C}_h \subseteq \mathscr{G}_h$ be the weak closure of the smooth strictly positive normalized Gauduchon forms, and let $\kappa_h(\theta)$ be the linear positivity threshold of Corollary 4.3.1, which is the minimum of $\int_X \theta \wedge T$ over $T \in \mathscr{C}_h$, by Proposition 4.4.1. The inclusion $\mathscr{C}_h \subseteq \mathscr{G}_h$ gives $m_h(\theta) \leq \kappa_h(\theta)$. The larger test set detects smooth positivity and nefness, whereas the weak closure of the smooth tests detects the existence of positive and of strictly positive currents in the class, that is, pseudoeffectivity and bigness. The inequality can be strict. Let X, C, β_C and T_C be as in Example 4.4.1, with the Kähler form there taken as h. Since $T_C \in \mathscr{G}_h$ and $[C]$ is a positive representative of the class of β_C, the two thresholds satisfy

$$m_h(\beta_C) \leq -\frac{1}{\int_C h} < 0 \leq \kappa_h(\beta_C).$$

Proposition 27.20.1 *Let θ be nef with $\lambda_\mu(\theta) > 0$. If $T \in \mathscr{G}_h$ satisfies $\int_X \theta \wedge T = 0$, then the determinant measure of T, defined in (27.23) with reference metric h and with μ in place of $\mathrm{e}^F \mathrm{d}V$, vanishes.*

Proof In dimension one the hypotheses cannot hold, since $\int_X \theta \wedge T = \lambda_\mu(\theta) / \int_X h > 0$. Suppose $n \geq 2$, and write $D_\mu(T)$ for the total mass of this determinant measure; for $T = \gamma^{n-1}$ it is the denominator $D_\mu(\gamma)$ of Definition 10.8.1. If $\eta \in \mathscr{P}$, its solution form Ω satisfies $\Omega^n = \lambda_\mu(\eta)^n \mu$, so $\Omega / \lambda_\mu(\eta)$ is an admissible test in the infimum of

Lemma 27.5.1. Together with Stokes' theorem, this gives

$$\int_X \eta \wedge T = \int_X \Omega \wedge T \geq \lambda_\mu(\eta) D_\mu(T).$$

Apply this to $\eta = \theta + \varepsilon h$. The left side equals ε, while Theorem 27.9.1 gives $\lambda_\mu(\eta) \to \lambda_\mu(\theta) > 0$. Hence $D_\mu(T) = 0$. The determinant measure is positive and has total mass zero, so it vanishes. □

Thus, at a nef background of positive volume, every annihilating current in $\mathcal{G}_h$ has a coefficient matrix of deficient rank almost everywhere with respect to its trace measure. This alone does not force the trace measure to be supported on an analytic subset.

27.20.1 Maximal time of the Chern–Ricci flow

The Chern–Ricci flow deforms a Hermitian metric by $\partial_t \omega = -\operatorname{Ric}(\omega)$. Since the Chern–Ricci forms of any two Hermitian metrics differ by a dd^{c}-exact form, a solution starting at ω_0 stays in the affine family $\omega_0 - t \operatorname{Ric}(\omega_0)$ modulo dd^{c} of smooth functions. It can therefore exist only as long as this family has smooth positive representatives. Conversely, a positive representative at a later time provides a uniformly positive reference path, and parabolic estimates exclude an earlier singularity. The current criterion of Theorem 27.20.1 then computes the lifetime.

Let ω_0 be the initial Hermitian metric. For a Hermitian form ω on X we use the Chern–Ricci form

$$\operatorname{Ric}(\omega) = -\mathrm{i}\partial\bar{\partial} \log \omega^n = -2\pi \mathrm{dd}^{\mathrm{c}} \log \omega^n,$$

where ω^n is identified with its coefficient in local holomorphic coordinates; a change of coordinates changes $\log \omega^n$ by a pluriharmonic function and does not affect the form. This normalization fixes the time scale of the flow $\partial_t \omega = -\operatorname{Ric}(\omega)$. Put $\chi = \operatorname{Ric}(\omega_0)$ and $\theta_t = \omega_0 - t\chi$.

Theorem 27.20.2 *The Chern–Ricci flow starting from ω_0 has a unique maximal smooth solution on $[0, \tau)$, where*

$$\tau = \sup\{t \geq 0 : \ \theta_t \text{ has a smooth positive representative}\}.$$

Proof Call a time $t \geq 0$ admissible if θ_t has a smooth positive representative. The admissible times form an interval containing zero, since convex combinations of positive representatives at two times are positive representatives at the intermediate times, and the set is open because positivity is an open condition. We show that the flow exists on $[0, t_1)$ for every admissible $t_1 > 0$ and that it cannot exist beyond τ.

Step 1: the scalar equation. Fix an admissible $t_1 > 0$, and choose a smooth real function a with $\theta_{t_1} + \mathrm{dd}^{\mathrm{c}} a > 0$. The reference forms

$$\gamma_t = \theta_t + \frac{t}{t_1}\mathrm{dd}^{\mathrm{c}} a = \left(1 - \frac{t}{t_1}\right)\omega_0 + \frac{t}{t_1}(\theta_{t_1} + \mathrm{dd}^{\mathrm{c}} a), \qquad 0 \le t \le t_1,$$

are uniformly positive, with all derivatives uniformly bounded. Put

$$\Omega = \mathrm{e}^{a/(2\pi t_1)}\omega_0^n.$$

Then $\dot\gamma_t = -\operatorname{Ric}(\Omega)$, where the Ricci form of a volume form is defined by the same local expression. Consider the scalar equation

$$\dot u = 2\pi \log \frac{(\gamma_t + \mathrm{dd}^{\mathrm{c}} u)^n}{\Omega}, \qquad \gamma_t + \mathrm{dd}^{\mathrm{c}} u > 0, \qquad u(0) = 0. \tag{27.108}$$

Applying dd^{c} shows that $\omega(t) = \gamma_t + \mathrm{dd}^{\mathrm{c}} u(t)$ solves the metric flow. Conversely, for a solution $\omega(t)$ of the metric flow, the function $u(t) = 2\pi \int_0^t \log(\omega(s)^n/\Omega)\,\mathrm{d}s$ satisfies $\omega(t) = \gamma_t + \mathrm{dd}^{\mathrm{c}} u(t)$ and solves (27.108). The linearization of (27.108) is $\partial_t - 2\pi \operatorname{tr}_{\omega(t)} \mathrm{dd}^{\mathrm{c}}$, a strictly parabolic operator, so the classical short-time existence theory for smooth strictly parabolic equations gives a solution. The difference of two solutions satisfies a linear parabolic equation whose principal coefficients, the averages of the inverse metrics along the segment between the two solutions, are positive; the maximum principle therefore proves uniqueness.

Suppose that the maximal time t_* of this solution satisfies $t_* < t_1$. We derive estimates independent of $t < t_*$ that allow us to continue the solution.

Step 2: bounds for u and $\dot u$. For a large constant C, the functions Ct and $-Ct$ are a supersolution and a subsolution of (27.108), respectively: at a spatial extremum of u the complex Hessian has the appropriate sign, and γ_t^n/Ω is bounded above and away from zero. The maximum principle gives $|u| \le Ct_1$.

Set $v = \dot u$ and $\mathcal{L} = 2\pi \operatorname{tr}_{\omega(t)} \mathrm{dd}^{\mathrm{c}}$. Differentiation gives

$$(\partial_t - \mathcal{L})v = 2\pi \operatorname{tr}_{\omega(t)} \dot\gamma_t, \qquad (\partial_t - \mathcal{L})u = v - 2\pi n + 2\pi \operatorname{tr}_{\omega(t)} \gamma_t.$$

Since γ_t is affine in time, these identities imply

$$\begin{aligned}(\partial_t - \mathcal{L})(tv - u - 2\pi n t) &= -2\pi \operatorname{tr}_{\omega(t)} \omega_0 < 0,\\ (\partial_t - \mathcal{L})((t_1 - t)v + u + 2\pi n t) &= 2\pi \operatorname{tr}_{\omega(t)} \gamma_{t_1} > 0.\end{aligned}$$

By the maximum principle, the first expression is bounded above by its initial value zero, and the second is bounded below by $\min_X t_1 v(0)$. The bound for u and $t_1 - t \ge t_1 - t_* > 0$ give a lower bound for v. The first inequality gives an upper bound for v on $[t_*/2, t_*)$, and the smooth solution bounds it on $[0, t_*/2]$. In particular, $\omega(t)^n/\Omega = \mathrm{e}^{v/(2\pi)}$ is bounded above and away from zero.

Step 3: a parabolic trace inequality. We next bound the trace, keeping track of the time derivative of the reference metric. In the coordinates of Lemma 10.4.1, write $g(t)$ for the matrix of γ_t and, as in that lemma, $h = g + (u_{i\bar{j}})$ for the matrix of $\omega(t)$. Put

$$S = \operatorname{tr}_g h, \qquad B = \operatorname{tr}_h g, \qquad F = \log \frac{\Omega}{\gamma_t^n},$$

so that $\log(\det h/\det g) = v/(2\pi) + F$. The family $g(t)$ has uniformly bounded Chern torsion and curvature, so at each fixed time Lemma 10.4.1 applies with uniform constants. Direct differentiation of S gives

$$\partial_t \log S = \frac{\Delta_g v + \operatorname{tr}_g \dot{g}}{S} - \frac{\operatorname{tr}(g^{-1}\dot{g}\, g^{-1} h)}{S}.$$

In $(\partial_t - \mathcal{L}) \log S$, the term $\Delta_g v/S$ cancels the corresponding term in the lower bound for $\mathcal{L} \log S$ given by the lemma. With $\tau_k = \sum_i T^i_{ik}$ the torsion trace of $g(t)$, as in the lemma, we obtain on $\{S \geq 1\}$

$$(\partial_t - \mathcal{L}) \log S \leq -\frac{4\pi}{S^2} \operatorname{Re}(h^{k\bar{q}} \tau_k S_{\bar{q}}) + CB + C. \tag{27.109}$$

Indeed, $\Delta_g F$ and $\operatorname{tr}_g \dot{g}$ are uniformly bounded, $S^{-1} \leq B/n^2$, and $-Cg \leq \dot{g} \leq Cg$ bounds the absolute value of the last quotient by C. These bounds account for all the terms coming from the time dependence of the reference metric.

Step 4: the trace bound. Choose a constant C_0 such that $1 \leq s = u + C_0 \leq D$ on the whole time interval, and maximize

$$Q = \log S - Au + s^{-1}$$

on $X \times [0, t_0]$, where $t_0 < t_*$ and the constant $A \geq 1$ will be fixed independently of t_0. If the maximum is attained at time zero, the initial data bound Q. At a maximum with positive time, $\partial \log S = (A + s^{-2}) \partial u$. Since $|\tau|^2_h \leq CB$, Young's inequality bounds the absolute value of the first term on the right of (27.109) by

$$2\pi s^{-3} |\partial u|^2_h + \frac{C(A+1)^2 s^3}{S^2} B.$$

If $S^2 \leq (A+1)^2 D^3$, the trace is already bounded at the maximum. Otherwise the last coefficient is at most C, and

$$\begin{aligned} 0 \leq (\partial_t - \mathcal{L}) Q &\leq CB + C - (A + s^{-2})(v - 2\pi n + 2\pi B) - 2\pi s^{-3} |\partial u|^2_h \\ &\leq (C - 2\pi A) B + C(A+1). \end{aligned}$$

Choosing A with $2\pi A > C + 1$ bounds B at the maximum. In terms of the eigenvalues, $S \leq n(\det h/\det g) B^{n-1}$ for $n \geq 2$, and $S = \det h/\det g$ for $n = 1$, so the determinant bound of Step 2 bounds S at the maximum in every case. The bound for u transfers this estimate from Q to S everywhere, uniformly in t_0, and the lower bound for the

determinant then bounds the smallest eigenvalue away from zero. Thus $\omega(t)$ and γ_t are uniformly equivalent for $t < t_*$.

Step 5: higher estimates and continuation. Uniform ellipticity now gives the higher estimates by classical local parabolic regularity. In a coordinate ball, write the logarithmic determinant as a concave function of the complex part of the real Hessian. On the compact set of positive matrices just obtained, its supporting affine functions have uniformly positive and bounded slopes, and their infimum extends the operator to a concave, uniformly elliptic function of all real symmetric matrices. The dependence on space and time enters only through the smooth family $g(t)$ and the fixed density Ω. The parabolic Evans–Krylov estimate gives uniform $C^{2+\alpha,1+\alpha/2}$ bounds away from the initial time, and the smooth solution covers a fixed initial interval. Differentiating the equation and applying parabolic Schauder estimates then bounds all spatial and time derivatives uniformly. This is the parabolic counterpart of the local argument used in Corollary 10.4.1.

The bounds for the time derivatives show that $u(t)$ converges in every spatial C^k norm as $t \uparrow t_*$. By uniform ellipticity, the smooth limit still has a strictly positive solution form, and short-time existence with this limit as initial datum extends the solution beyond t_*, a contradiction. Hence the flow exists up to every admissible time t_1, and by uniqueness these solutions patch together on $[0, \tau)$.

Step 6: maximality. Integrating the metric flow gives

$$\omega(t) = \theta_t + \mathrm{dd^c}\left(2\pi \int_0^t \log \frac{\omega(s)^n}{\omega_0^n}\, \mathrm{d}s\right).$$

Hence every time at which a smooth positive solution exists is admissible, which proves maximality. □

Combining the continuation theorem with the current criterion gives a numerical description of the lifetime.

Corollary 27.20.1 *Let*

$$b = \max_{T \in \mathcal{G}_{\omega_0}} \int_X \chi \wedge T.$$

Then the maximal smooth existence time is

$$\tau = \begin{cases} 1/b, & b > 0, \\ +\infty, & b \le 0. \end{cases}$$

If $\tau < \infty$, the limiting background θ_τ is nef and there is a current $T_* \in \mathcal{G}_{\omega_0}$ with $\int_X \theta_\tau \wedge T_* = 0$. If moreover $\lambda_\mu(\theta_\tau) > 0$, every such annihilating current has zero determinant measure.

Proof For $t \geq 0$, the normalization of the test currents gives the identity

$$m_{\omega_0}(\theta_t) = 1 - tb.$$

Theorems 27.20.1 and 27.20.2 give the formula for τ. When $b > 0$, the maximum defining b is attained by weak compactness, and a maximizer T_* annihilates θ_τ. The same identity gives $m_{\omega_0}(\theta_\tau) = 0$, hence nefness. The last assertion is Proposition 27.20.1. □

Along the flow, $\omega(t) = \theta_t + \mathrm{dd}^{\mathrm{c}}\varphi_t$ with a smooth potential φ_t. Consequently, for every fixed $T \in \mathcal{G}_{\omega_0}$,

$$\int_X \omega(t) \wedge T = 1 - t \int_X \chi \wedge T.$$

These traces describe the obstruction at the maximal time without any bound on the total Monge–Ampère mass.

27.20.2 A positive-volume endpoint that is not big

Backgrounds reached at the maximal time of a Chern–Ricci flow are not better behaved, as far as the relation between positive volume and bigness is concerned. The radial background of Section 27.13 occurs, up to a smooth dd^{c} change, as the limiting background of a Chern–Ricci flow at its maximal time. The normalized currents annihilating it can also be described explicitly.

We keep the notation of Section 27.13: $X = \mathbb{P}^m \times \mathbb{P}^1$ with $n = m + 1 \geq 3$, the normalized Fubini–Study forms ω and w pulled back from the two factors, the nonnegative function $a = x/t$ on the first factor, which vanishes only at p, the background $\theta = \omega + aw$ and the probability form $\mu = \omega^m \wedge w$. Put $Z = \{p\} \times \mathbb{P}^1$, and start the flow from the Hermitian metric

$$\omega_0 = (n+1)\omega + (a+2)w. \tag{27.110}$$

Theorem 27.20.3 *With the normalization of the Ricci form in Section 27.20.1, the maximal smooth existence time of the Chern–Ricci flow starting from ω_0 is $\tau = 1/(2\pi)$. The limiting background $\alpha_\tau = \omega_0 - \tau\,\mathrm{Ric}(\omega_0)$ is nef and not big, and*

$$\alpha_\tau = \theta + \mathrm{dd}^{\mathrm{c}} \log(a+2), \qquad \mathcal{V}_\mu(\alpha_\tau) = n.$$

Its variational volume is positive for every smooth strictly positive probability form on X.

Proof The initial metric is strictly positive, and

$$\omega_0^n = n(n+1)^m (a+2)\mu.$$

With our normalization, the Ricci forms of the normalized Fubini–Study metrics are $2\pi n\omega$ and $4\pi w$. Consequently

$$\mathrm{Ric}(\omega_0) = 2\pi\big(n\omega + 2w - \mathrm{dd^c}\log(a+2)\big).$$

Write s for the flow time and put $r = 2\pi s$. The affine background is

$$\alpha_s = (n+1-nr)\omega + (a+2-2r)w + r\,\mathrm{dd^c}\log(a+2). \tag{27.111}$$

For $0 \le r < 1$, adding $\mathrm{dd^c}$ of the smooth function $-r\log(a+2)$ gives a strictly positive representative: the horizontal coefficient is at least one and the vertical coefficient is at least $2(1-r) > 0$. At $r = 1$ we obtain the identity $\alpha_\tau = \theta + \mathrm{dd^c}\log(a+2)$. Nefness, bigness and the variational volume are unchanged by this smooth $\mathrm{dd^c}$ change, by the conventions of Chapter 27 and Proposition 10.8.1, so Theorem 27.13.1 shows that the endpoint background is nef, not big, and of volume n.

For $r > 1$, choose a nonzero smooth nonnegative function f on the base supported in $\{a+2-2r < 0\}$, a neighborhood of p. Its pullback $S = f\omega^m$ is a closed semipositive $(n-1, n-1)$-form. For every quasi-plurisubharmonic function v, Stokes' theorem for currents gives

$$\int_X (\alpha_s + \mathrm{dd^c}v) \wedge S = \int_{\mathbb{P}^m} f(a+2-2r)\omega^m < 0,$$

so α_s is not even pseudoeffective for $r > 1$. At $r = 1$, a smooth positive representative would contradict nonbigness. Thus $r = 1$ is the supremum of the admissible times, and Theorem 27.20.2 gives the smooth flow on $[0, \tau)$ with exactly this lifetime; the reference representatives used above need not solve the flow equation. Finally, Proposition 10.8.2 gives positivity of the volume for every smooth positive probability form. □

Proposition 27.20.2 *The current $[Z]/2$ belongs to $\mathscr{G}_{\omega_0}$ and annihilates α_τ. Every $R \in \mathscr{G}_{\omega_0}$ with $\int_X \alpha_\tau \wedge R = 0$ has its trace probability $\sigma_R = \omega_0 \wedge R$ supported on Z. In the coefficient-matrix convention of Section 27.5, its matrix is*

$$P_R = n\Pi_{\mathrm{vert}} \qquad \sigma_R\text{-almost everywhere},$$

where Π_{vert} is the ω_0-orthogonal projection onto the tangent line of the fiber $\{p\}\times\mathbb{P}^1$. In particular, P_R has rank one, and the determinant measure of R vanishes for every smooth positive density.

Proof Only smooth forms need to be restricted to Z, and their restrictions satisfy $\omega_0|_Z = 2w$ and $\alpha_\tau|_Z = 0$. This proves that $[Z]/2$ is normalized and annihilates α_τ.

Let R be a general annihilator. Since α_τ and θ differ by $\mathrm{dd^c}$ of a smooth function, Stokes' theorem gives $\int_X \theta \wedge R = 0$. The measure $\theta \wedge R$ is nonnegative, so it vanishes. On every compact set $K \subset X \setminus Z$, the explicit form θ satisfies $\theta \ge c_K\omega_0$ for some $c_K > 0$. Thus $\sigma_R(K) = 0$, and a countable exhaustion by compact sets shows that σ_R is supported on Z.

Relative to ω_0, the matrix of R is characterized, for smooth test forms β with matrix B_β, by

$$\beta \wedge R = \frac{1}{n}\operatorname{tr}(B_\beta P_R)\sigma_R, \qquad P_R \geq 0, \qquad \operatorname{tr} P_R = n.$$

At points of Z, the form θ is positive on the horizontal m-plane and vanishes on the vertical line. The vanishing of $\theta \wedge R$ therefore forces all horizontal diagonal entries of P_R to vanish, and the inequality $|(P_R)_{ij}|^2 \leq (P_R)_{ii}(P_R)_{jj}$ for positive matrices removes their rows and columns. The remaining vertical entry is n, which proves the formula and the vanishing of the determinant. □

For each such annihilator R, the trace identity at the end of Section 27.20.1 becomes

$$\int_X \omega(s) \wedge R = 1 - 2\pi s, \qquad 0 \leq s < 1/(2\pi).$$

The support statement uses the explicit semipositive representative θ of the endpoint class; the vanishing of the determinant alone would not give it.

27.20.3 Twisting and distance bounds

A twisted Chern–Ricci flow is the equation $\partial_t \omega = -\operatorname{Ric}(\omega) + \zeta$, where the twist ζ is a fixed smooth closed real $(1,1)$-form. Diameter bounds along such flows cannot depend only on the maximal time and on the initial Monge–Ampère density: the twist and the length of the time interval enter the geometry, even for flat Kähler metrics.

Example 27.20.1 Let $E = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$, let $X = E^n$ with $n \geq 2$, and let $\eta_j = (\mathrm{i}/2)\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j$ be the flat forms pulled back from the factors. Put $\omega_0 = \sum_{j=1}^n \eta_j$. All the flows below start at ω_0, with potential zero and initial density one relative to ω_0^n. In particular, hypotheses on the initial data expressed through a decomposition $\mathrm{e}^{\psi^+ - \psi^-}$ of the density, with quasi-plurisubharmonic functions $\psi^\pm$ subject to fixed L^p bounds, hold uniformly for all these flows, with $\psi^+ = \psi^- = 0$.

For each $M > 0$, the twist $\zeta_M = M\eta_1 - \eta_2$ gives the explicit solution

$$\omega_M(t) = (1 + Mt)\eta_1 + (1 - t)\eta_2 + \sum_{j=3}^{n} \eta_j, \qquad 0 \leq t < 1,$$

of $\partial_t \omega = -\operatorname{Ric}(\omega) + \zeta_M$. Its maximal smooth existence time is exactly one, while

$$\operatorname{diam}(X, \omega_M(1/2)) \geq \sqrt{1 + M/2}\, \operatorname{diam}(E, \eta_1).$$

For the single fixed twist $\zeta = \omega_0$, the solution is $(1 + t)\omega_0$ for all $t \geq 0$, and its diameter grows as $\sqrt{1 + t}\, \operatorname{diam}(X, \omega_0)$.

Proof Each of these metrics has constant coefficients, so its Chern–Ricci form vanishes; differentiation in t proves the flow equations, and positivity of the coefficients gives the stated existence intervals. In the first family, pair a hypothetical positive representative at a time $t \geq 1$ with the closed positive form $\eta_1 \wedge \eta_3 \wedge \cdots \wedge \eta_n$, where the empty product is 1 when $n = 2$. The potential term integrates to zero, and the pairing is a positive fixed multiple of $1 - t$, so it cannot be strictly positive. Integrating the flow equation shows that every smooth solution differs from the affine background by dd^{c} of a smooth function, so no smooth positive continuation beyond $t = 1$ is possible.

Projection to the first elliptic factor does not increase lengths once the metric of that factor is scaled by $1 + Mt$. Taking two points that differ only in the first factor proves the lower bound for the diameter. For the second flow, every path length is multiplied by $\sqrt{1+t}$, which proves the exact formula for the diameter. □

On a fixed finite time interval, a common Dini modulus for the potentials and a uniform bound for the affine backgrounds do give control of distances and diameters, by Corollary 1.18.2. The example shows why such bounds must be imposed when the twist varies or the time interval grows.

27.21 Divisorial constraints and residual volume

Suppose we require a θ-plurisubharmonic function to have generic Lelong numbers at least a_j along prescribed prime divisors D_j. Removing the divisor $D = \sum_j a_j D_j$ leaves the residual background $\theta - \eta_D$, where η_D is a curvature form of D, and in each Gauduchon quotient this removal subtracts the Gauduchon degree of D from the numerator. Lamari's duality therefore decides when such functions exist, and the volume of the residual background measures what remains. By the correspondence of Theorem 16.9.1, equations in the ordinary singularity type of the logarithmic potential of D reduce to equations in the residual background. When the residual background has a smooth positive representative, the smooth maximizer of the dual problem forces every subsolution with the prescribed Lelong numbers and the residual volume as scalar to be the logarithmic potential of D plus a smooth function, up to a constant.

Fix a smooth real $(1,1)$-form θ on X. Let $D = \sum_{j=1}^{N} a_j D_j$ be an effective real divisor, with distinct prime divisors D_j and coefficients $a_j > 0$. Choose a smooth Hermitian metric on each line bundle $\mathcal{O}_X(D_j)$, write s_j for its canonical section, and let η_j be its Chern curvature form. Set

$$\sigma_D = \sum_j a_j \log |s_j|^2, \qquad \eta_D = \sum_j a_j \eta_j, \qquad \beta_D = \theta - \eta_D.$$

The Poincaré–Lelong formula reads

$$\mathrm{dd}^{\mathrm{c}} \sigma_D = [D] - \eta_D.$$

For a Gauduchon metric γ on X, define the Gauduchon degree

$$\deg_\gamma D = \int_X [D] \wedge \gamma^{n-1} = \sum_j a_j \int_{D_j} \gamma^{n-1},$$

where integration over a singular divisor means integration over its regular locus, or equivalently against its current of integration. This degree is finite, positive for $D \neq 0$, and equal to $\int_X \eta_D \wedge \gamma^{n-1}$ by Stokes' theorem for currents. We write $\nu(u, D_j)$ for the generic Lelong number of u along D_j, as in Section 1.13.

Proposition 27.21.1 *The following are equivalent:*

(1) *there is $u \in \mathrm{PSH}(X, \theta)$ with $\nu(u, D_j) \geq a_j$ for every j;*
(2) *the residual background β_D is pseudoeffective;*
(3) *for every Gauduchon metric γ,*

$$\int_X \theta \wedge \gamma^{n-1} \geq \deg_\gamma D.$$

The first condition does not require $u - \sigma_D$ to be bounded, nor the Lelong numbers of u to equal the prescribed ones.

Proof Suppose first that u satisfies the first condition. By Lemma 1.17.1, these Lelong numbers can be subtracted: $\theta + \mathrm{dd^c} u - [D] \geq 0$. Consequently

$$\beta_D + \mathrm{dd^c}(u - \sigma_D) = \theta + \mathrm{dd^c} u - [D] \geq 0,$$

where the difference $u - \sigma_D$ is first taken in L^1 and then replaced by its quasi-plurisubharmonic representative. This proves the second condition. Conversely, if $v \in \mathrm{PSH}(X, \beta_D)$, then $u = \sigma_D + v$ satisfies

$$\theta + \mathrm{dd^c} u = [D] + \beta_D + \mathrm{dd^c} v \geq 0,$$

so its generic Lelong numbers along the D_j are at least a_j. For $n \geq 2$, Lamari's duality, Theorem 4.3.1, says that a smooth background is pseudoeffective exactly when all its Gauduchon integrals are nonnegative; applied to β_D, together with the degree identity, it gives the equivalence of the second and third conditions. For $n = 1$ the same criterion reads $\int_X \beta_D \geq 0$: necessity follows from Stokes' theorem, and for sufficiency one solves $\mathrm{dd^c} f = (\int_X \beta_D)\mu - \beta_D$ for a smooth real function f with the scalar Green operator. □

For a nonzero effective real divisor D and a pseudoeffective θ, the exact threshold is therefore

$$\tau_\theta(D) = \inf_{\gamma \text{ Gauduchon}} \frac{\int_X \theta \wedge \gamma^{n-1}}{\deg_\gamma D}. \tag{27.112}$$

It is finite and nonnegative, and a function $u \in \mathrm{PSH}(X, \theta)$ with $\nu(u, D_j) \geq t a_j$ for all j exists if and only if $0 \leq t \leq \tau_\theta(D)$. The endpoint is included, because the

non-strict Gauduchon inequalities still hold there and duality gives a positive current. This is a threshold for the existence of a positive current, not for the existence of a bounded residual potential.

Whenever β_D is pseudoeffective, define the residual volume

$$\mathcal{V}_\mu(\theta; D) = \mathcal{V}_\mu(\beta_D).$$

The definition and the degree identity give the exact formula

$$\mathcal{V}_\mu(\theta; D)^{1/n} = \inf_{\gamma \text{ Gauduchon}} \frac{\int_X \theta \wedge \gamma^{n-1} - \deg_\gamma D}{D_\mu(\gamma)}, \tag{27.113}$$

and the residual volume is finite and nonnegative. Changing the auxiliary metrics on the line bundles changes η_D by a smooth dd^{c}-exact term, so the residual volume does not depend on these metrics, nor on a smooth dd^{c} change of θ. It depends on the prescribed divisor and on the smooth probability measure μ.

Here $\mathrm{MA}_\theta^{\mathrm{ext}}$ denotes the extended non-pluripolar Monge–Ampère measure of Theorem 7.1.1.

Theorem 27.21.1 *Suppose that β_D has a smooth positive representative, and put $c = \mathcal{V}_\mu(\theta; D)$. There is a θ-plurisubharmonic function $u_0 = \sigma_D + v$, with v smooth, satisfying*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u_0) = c\mu.$$

Every $u \in \mathrm{PSH}(X, \theta)$ such that

$$\nu(u, D_j) \geq a_j \quad (1 \leq j \leq N), \qquad \mathrm{MA}_\theta^{\mathrm{ext}}(u) \geq c\mu$$

differs from u_0 by a constant. In particular, a solution of a normalized equation $\mathrm{MA}_\theta^{\mathrm{ext}}(u) = c'\mu$ in the ordinary singularity type of σ_D has scalar $c' = c$, has a smooth remainder $u - \sigma_D$, and is unique up to constants.

Proof *Existence.* The smooth equation and Theorem 10.8.1 give a smooth v with

$$\Omega = \beta_D + \mathrm{dd}^{\mathrm{c}} v > 0, \qquad \Omega^n = c\mu.$$

Then $\theta + \mathrm{dd}^{\mathrm{c}}(\sigma_D + v) = [D] + \Omega \geq 0$. Off D, the extended measure of $u_0 = \sigma_D + v$ is Ω^n, and both measures give zero mass to D. This proves existence.

Reduction to the residual background. Let u satisfy the two inequalities. Removing the divisor as in the proof of Proposition 27.21.1, we obtain

$$w = u - \sigma_D - v \in \mathrm{PSH}(X, \Omega), \qquad \Omega + \mathrm{dd}^{\mathrm{c}} w = \theta + \mathrm{dd}^{\mathrm{c}} u - [D] \geq 0,$$

where the difference is first taken in L^1 and then replaced by its canonical representative. On $X \setminus D$, the two currents differ only by smooth changes in the decomposition of their local potentials, so their extended measures agree there, and both measures vanish on D. Hence

$$\mathrm{MA}_\Omega^{\mathrm{ext}}(w) = \mathrm{MA}_\theta^{\mathrm{ext}}(u) \geq c\mu.$$

Rigidity. Take $dV = \mu$. The zero function solves the smooth equation $\Omega^n = c\mu$ in the background Ω, so its subsolution threshold is $b_\Omega(1) = c$, by Proposition 15.27.2. For $n \geq 2$, the normalized $(n-1)$st power of the Gauduchon metric conformal to Ω is a smooth strictly positive maximizer of the dual problem. By Corollary 27.8.1, any two threshold subsolutions differ by a constant; comparing w with zero shows that w is constant. For $n = 1$, the positive current $\Omega + dd^c w$ dominates its extended measure, hence dominates $c\mu = \Omega$, while both have total mass $\int_X \Omega = c$ by Stokes' theorem. Thus $dd^c w = 0$, and w is again constant.

Solutions in the ordinary type. Finally, let u be in the ordinary singularity type of σ_D with $\mathrm{MA}_\theta^{\mathrm{ext}}(u) = c'\mu$ for some $c' \geq 0$. Then $\nu(u, D_j) = a_j$, and the same reduction gives a bounded $w \in \mathrm{PSH}(X, \Omega)$ with $\mathrm{MA}_\Omega(w) = c'\mu$. Suppose $c' < c$. For every constant C, the bounded functions w and C satisfy $\mathrm{MA}_\Omega(w) = (c'/c)\mathrm{MA}_\Omega(C)$, so Theorem 11.2.2 gives $w \geq C$, which is impossible for a bounded w. The case $c' > c$ is symmetric, with the roles of w and the constants exchanged. Therefore $c' = c$, and the rigidity just proved gives $u = u_0 + \text{constant}$. □

When θ is Hermitian and $n \geq 2$, write $\mu = f\,dV$ relative to a smooth strictly positive probability volume dV. For every $u \in \mathrm{PSH}(X, \theta)$ with $\theta + dd^c u \geq [D]$ and $\mathrm{MA}_\theta^{\mathrm{ext}}(u) \geq cf\,dV$, in particular for the potential u_0 of Theorem 27.21.1 with $c = \mathcal{V}_\mu(\theta; D)$, Proposition 29.8.1 bounds the coefficient c in terms of the subsolution threshold $b_\theta(f)$ of Theorem 15.27.1 and the linear threshold of the class of the divisor. The residual problem prescribes the divisor, whereas the threshold $b_\theta(f)$ allows all θ-plurisubharmonic functions.

Along a family $D_t = tD$, $t \geq 0$, in the feasible interval, the root of (27.113) is concave and nonincreasing in t: every test quotient is affine in t with nonpositive slope, and both properties pass to the infimum. On a subinterval where $\theta - t\eta_D$ has a smooth positive representative, choose a minimizing Gauduchon metric γ_t for that background. Proposition 27.1.1 then gives the sharper formula

$$\frac{d}{dt}\mathcal{V}_\mu(\theta; tD)^{1/n} = -\frac{\deg_{\gamma_t} D}{D_\mu(\gamma_t)} < 0. \tag{27.114}$$

In dimension one the derivative is $-\deg D$, since the denominator is one.

27.21.1 Varying divisorial coefficients

For a fixed support, the residual equation depends smoothly on the coefficients as long as the residual background stays in $\mathscr{P}$. The smooth remainders then give convergence in capacity of the solutions, and they determine all generic Lelong numbers of the limit on every modification. Keep D_j, s_j and η_j as above, and put $\sigma_j = \log|s_j|^2$.

For $a = (a_1, \ldots, a_N) \in \mathbb{R}^N_{\geq 0}$ put

$$D_a = \sum_j a_j D_j, \qquad \sigma_a = \sum_{a_j > 0} a_j \sigma_j, \qquad \beta_a = \theta - \sum_j a_j \eta_j,$$

where zero coefficients are omitted from the logarithmic sum. Let $\mathcal{A}$ be the set of such a for which β_a has a smooth positive representative.

Proposition 27.21.2 *There are unique smooth functions v_a, $a \in \mathcal{A}$, such that*

$$\int_X v_a \, \mathrm{d}\mu = 0, \qquad \Omega_a = \beta_a + \mathrm{dd}^{\mathrm{c}} v_a > 0, \qquad \Omega_a^n = c(a)\mu, \quad c(a) = \mathcal{V}_\mu(\beta_a) > 0.$$

The maps $a \mapsto v_a$ and $a \mapsto c(a)$ are smooth, and every smooth norm of v_a is bounded on compact subsets of $\mathcal{A}$. Here smoothness at a parameter with a zero coefficient means that the family is the restriction of a smooth family defined on an open subset of $\mathbb{R}^N$.

Set $u_a = \sigma_a + v_a$. If $a^{(k)} \in \mathcal{A}$ and $a^{(k)} \to a \in \mathcal{A}$, then $u_{a^{(k)}} \to u_a$ in L^1 and in capacity, and smoothly on compact subsets of $X \setminus \bigcup_j D_j$. Moreover,

$$\left\| \mathrm{MA}_\theta^{\mathrm{ext}}(u_{a^{(k)}}) - \mathrm{MA}_\theta^{\mathrm{ext}}(u_a) \right\|_{\mathrm{TV}} = |c(a^{(k)}) - c(a)| \longrightarrow 0,$$

where $\| \cdot \|_{\mathrm{TV}}$ is the total variation norm. For every modification $\pi \colon Y \to X$ with Y smooth and every prime divisor $E \subset Y$,

$$\nu(u_a \circ \pi, E) = \sum_j a_j \, \mathrm{ord}_E(\pi^* D_j),$$

where $\mathrm{ord}_E(\pi^* D_j)$ is the vanishing order along E of the pullback of a local equation of D_j. In particular, the limit u_a has exactly the ordinary singularity type of σ_a, and no further singularity appears in this finite family.

Proof The set of parameters in $\mathbb{R}^N$ for which β_a has a smooth positive representative is open, and the smooth equation gives v_a and $c(a)$ on it. The argument in the proof of Proposition 27.1.1 gives smooth dependence of both quantities on the parameter, with the fixed mean normalization; uniqueness makes the local families agree. Covering a compact set of parameters by finitely many such neighborhoods gives the asserted bounds.

The Poincaré–Lelong formula gives

$$\theta + \mathrm{dd}^{\mathrm{c}} u_a = [D_a] + \Omega_a,$$

so $u_a \in \mathrm{PSH}(X, \theta)$. Its extended measure equals $\Omega_a^n = c(a)\mu$ off the union of the divisors, and both measures vanish on that union; this also covers the case $D_a = 0$. Each fixed σ_j is locally integrable, so the convergence of the coefficients and the smooth convergence of v_a give convergence in L^1. Off the union of the divisors every σ_j is smooth, which gives smooth convergence there. For convergence in capacity, remove a neighborhood of this analytic set of arbitrarily small Hermitian capacity; on its compact complement the convergence is uniform. Since the union of the divisors has capacity zero, this proves convergence in capacity. The identity for the total variation distances follows from the equations and $\mu(X) = 1$.

Finally, after pullback the logarithmic potential $\sigma_j \circ \pi$ has generic Lelong number $\mathrm{ord}_E(\pi^* D_j)$ along E, whereas $v_a \circ \pi$ is smooth. Additivity of Lelong numbers for

the finite sum proves the formula on every modification. Since v_a is smooth also at the limiting parameter, u_a has the ordinary singularity type of σ_a. □

The smooth bounds above use that the limiting parameter lies in $\mathcal{A}$. When the sections and weights move, Theorem 16.9.2 gives L^1 convergence of the potentials normalized by $\sup_X = 0$; when the residual background is fixed, it also allows uniformly bounded measurable right-hand sides converging in L^1. On a modification, the pullback of a positive probability density can vanish along the exceptional divisor, so the smooth equation with positive density no longer applies directly. For a minimal exceptional pole, Proposition 29.11.3 gives a vanishing order of the density that is necessary for a smooth residual, but this order alone does not ensure smoothness.

Multiplier ideals need not stabilize in this family. At a smooth point of D_j outside the other divisors, take a local coordinate z_1 defining D_j. There $u_a = a_j \log|z_1|^2 + O(1)$, so

$$\mathcal{I}(u_a) = (z_1^{\lfloor a_j \rfloor})$$

by the one-variable integral in Example 1.12.1 and Fubini's theorem. Thus, if the parameter stays in $\mathcal{A}$ while a_j increases to an integer from below, this local ideal jumps at the limit. Convergence of the Lelong numbers to their limiting values does not imply that the multiplier ideals are eventually equal to those of the limit.

27.21.2 A ruled residual and its subtraction cost

On the ruled surface of Section 20.10, the Gauduchon variational volume depends on the data differently from the lower and upper volumes of the ordinary singularity type computed in Theorem 20.10.1; in particular, it depends on the measure μ. We keep the notation of that section: $Y = \mathbb{P}^1 \times \mathbb{P}^1$ with the forms w, w_1, w_2, the divisor $D = \{p\} \times \mathbb{P}^1$, the background $\theta = cw_1 + aw_2$ of (20.18), and $0 \le b < c$. Put $\beta = (c-b)w_1 + aw_2$. Let $m > 0$ be a smooth function on the first factor, with $\int_{\mathbb{P}^1} mw = 1$, and put

$$\mu = mw_1 \wedge w_2, \qquad k = c - b, \qquad H = \int_{\mathbb{P}^1} \frac{m}{a} w.$$

Proposition 27.21.3 *The infimum in Definition 10.8.1 for β is attained, and*

$$\mathcal{V}_\mu(\theta; bD) = \mathcal{V}_\mu(\beta) = \frac{2(c-b)}{H}, \qquad \mathcal{V}_\mu(\theta) = \frac{2c}{H}. \tag{27.115}$$

Writing $V = 2k/H$ and $g = Vm/(2a)$, a minimizing test is the closed positive form

$$T = \frac{g}{a} w_1 + w_2.$$

Proof Since $\int_{\mathbb{P}^1} gw = VH/2 = k$, there is a smooth function f on the first factor with $\mathrm{dd}^{\mathrm{c}} f = (g-k)w$. Then $\Omega = \beta + \mathrm{dd}^{\mathrm{c}} f = gw_1 + aw_2 > 0$ and $\Omega^2 = 2ga\, w_1 \wedge w_2 = V\mu$, so Theorem 10.8.1 gives $\mathcal{V}_\mu(\beta) = V$, and the infimum is attained by the Gauduchon metrics conformal to Ω. The displayed form $T = a^{-1}\Omega$ is such a metric, since it is closed: g/a depends only on the first factor. With the Fubini–Study metric on $\mathcal{O}_Y(D)$,

the curvature form of D is w_1, so the residual background $\theta - bw_1$ of bD is β. The case $b = 0$ gives the formula for θ. □

Along a subtraction path, the rate at which the root volume decreases is the integrand of the subtraction cost of Section 27.10. In the present family this rate is explicit. Fix a real constant d and a smooth real function e on the first factor. On an interval of values of s where $c_s = c - sd > b$ and $a_s = a - se > 0$, keep b, m fixed and put

$$k_s = c_s - b, \qquad H_s = \int_{\mathbb{P}^1} \frac{m}{a_s} w, \qquad V_s = \frac{2k_s}{H_s}, \qquad T_s = \frac{V_s m}{2a_s^2} w_1 + w_2.$$

By Proposition 27.21.3, applied with c_s and a_s in place of c and a, V_s is the variational volume of $\beta_s = k_s w_1 + a_s w_2 = \beta - s(dw_1 + ew_2)$, and T_s is a minimizing Gauduchon metric for it.

Corollary 27.21.1 *For the residual background $\beta_s = k_s w_1 + a_s w_2$,*

$$\frac{\mathrm{d}}{\mathrm{d}s}\sqrt{V_s} = -\frac{\int_Y (dw_1 + ew_2) \wedge T_s}{D_\mu(T_s)}. \tag{27.116}$$

Consequently, for every compact segment $[s_0, s_1]$ in that interval,

$$\sqrt{V_{s_0}} - \sqrt{V_{s_1}} = \int_{s_0}^{s_1} \frac{\int_Y (dw_1 + ew_2) \wedge T_s}{D_\mu(T_s)}\, \mathrm{d}s.$$

Proof The forms $\beta_s = \beta - s(dw_1 + ew_2)$ are strictly positive on the interval, and T_s is a minimizing Gauduchon metric for β_s. Hence Proposition 27.1.1, in the direction $-(dw_1 + ew_2)$, gives (27.116), and integration proves the second identity. □

Let $\tau = (|z|^2 - 1)/(|z|^2 + 1)$ be the height function on the first sphere. Its distribution under w is $\mathrm{d}\tau/2$ on $[-1, 1]$, as radial integration shows. For $c = 1$, $b = 1/4$, $a = 2 + \tau$ and $m = 1$, as in the example at the end of Section 20.10, we obtain $H = (\log 3)/2$, and the residual and original variational volumes are $3/\log 3$ and $4/\log 3$. The ordinary-type volumes from (20.20) are instead $3/2$ and $9/2$, independent of m. The subtraction identity of Corollary 27.21.1 differentiates the variational volume, which depends on the density. The invariance of the ordinary-type volumes under modifications in Proposition 20.10.2 has no direct counterpart here: on a modification, the pullback of μ degenerates along the exceptional divisor, so it is not one of the smooth positive probability densities used to define the variational volume upstairs.

The distinction persists along a segment that ends at a nef background. In the following example the subtraction cost is computed along such a segment, together with the ordinary-type volumes of a type that has no analytic representative, in the range of exponents in which its closed first moment is infinite.

Example 27.21.1 Let $\mu = w_1 \wedge w_2$, let τ be the height function on the first sphere, as above, and set

$$\theta_s = w_1 + (2 + \tau - s)w_2, \qquad 0 \le s < 1.$$

Fix the ordinary type of $U = 1-(1-G_p)^{\alpha}$ with $1/2 \le \alpha < 1$, and write $\sigma_-(s), \sigma_+(s)$ for the lower and upper volumes of Definition 14.3.1 of this type in the background θ_s. Then

$$V_s := \mathcal{V}_\mu(\theta_s) = \frac{4}{\log((3-s)/(1-s))}, \qquad \sigma_-(s) = 2(1-s), \qquad \sigma_+(s) = 2(3-s). \tag{27.117}$$

Let T_s be minimizing Gauduchon metrics for θ_s. The improper subtraction cost

$$\int_0^1 \frac{\int_Y w_2 \wedge T_s}{D_\mu(T_s)}\,\mathrm{d}s$$

is finite and equals $2/\sqrt{\log 3}$, while $\sigma_-(s) \to 0$ and $\sigma_+(s) \to 4$ as $s \to 1$. The endpoint $\theta_1 = w_1 + (1+\tau)w_2$ is nef, has zero variational volume, and is not big.

Proof Take $c = 1$, $b = 0$, $a_s = 2 + \tau - s$ and $m = 1$ in Proposition 27.21.3. Since τ has the law $\mathrm{d}\tau/2$ on $[-1, 1]$,

$$H_s = \frac{1}{2}\int_{-1}^{1} \frac{\mathrm{d}t}{2 + t - s} = \frac{1}{2}\log\frac{3-s}{1-s}.$$

Thus $V_s = 2/H_s$. Since $b = 0 < c$, Theorem 20.10.1, applied with a_s in place of a, shows that every θ_s-plurisubharmonic function of this ordinary type has finite positive mass, and gives the lower and upper volumes $2\min a_s = 2(1-s)$ and $2\max a_s = 2(3-s)$. The type has zero Lelong numbers by Lemma 20.10.1, and trivial multiplier ideals but no analytic representative by Proposition 20.10.1 with $b = 0$. Its closed first moment is infinite for $\alpha \ge 1/2$, as shown after the proof of Theorem 20.10.1; neither extremum uses such a moment.

For the minimizing metrics $T_s = V_s(2 + \tau - s)^{-2}w_1/2 + w_2$, Corollary 27.21.1 with $d = 0$ and $e = 1$ gives

$$\frac{\int_Y w_2 \wedge T_s}{D_\mu(T_s)} = -\frac{\mathrm{d}}{\mathrm{d}s}\sqrt{V_s} = \frac{2}{(1-s)(3-s)[\log((3-s)/(1-s))]^{3/2}}.$$

Integrating over $[0, s_1]$ with $s_1 < 1$ gives $\sqrt{V_0} - \sqrt{V_{s_1}}$. Since $V_{s_1} \to 0$ as $s_1 \to 1$, the improper integral is $\sqrt{V_0} = 2/\sqrt{\log 3}$. By Theorem 10.8.1, every minimizing metric is a constant multiple of T_s, and the quotient does not change under such multiples, so the cost does not depend on this choice.

The endpoint is smooth and semipositive, hence nef. Monotonicity in Proposition 10.8.1 gives

$$0 \le \lambda_\mu(\theta_1) \le \lambda_\mu(\theta_s) \longrightarrow 0,$$

so its variational volume vanishes. To exclude bigness, note that every Hermitian form dominates a positive multiple of $w_1 + w_2$. If θ_1 were big, there would be a quasi-plurisubharmonic q and $\epsilon > 0$ with $\theta_1 + \mathrm{dd}^{\mathrm{c}} q \geq \epsilon(w_1 + w_2)$. Choose a nonzero smooth nonnegative $(1,1)$-form ζ on the first sphere supported where $1 + \tau < \epsilon$. Its pullback is closed, so distributional Stokes gives

$$0 \leq \int_Y (\theta_1 + \mathrm{dd}^{\mathrm{c}} q - \epsilon(w_1 + w_2)) \wedge \zeta = \int_{\mathbb{P}^1} (1 + \tau - \epsilon)\zeta < 0,$$

a contradiction. Thus the positive segment ends outside the big cone, although its upper ordinary-type volume has a positive limit. □

27.22 A divisorial threshold on the Hopf threefold

On the Hopf threefold, the pullback of a hyperplane has an explicit divisorial threshold. The residual metric solves its normalized equation with zero potential, so the residual volume can be computed directly, also at the nef endpoint.

In this section X is the Hopf threefold of Chapter 9, with its projection $\pi\colon X \to \mathbb{P}^2$, and we write ζ for its vertical form, so that $\omega = \alpha + \zeta$. These forms satisfy

$$\alpha^3 = 0, \quad \zeta^2 = 0, \quad \mathrm{dd}^{\mathrm{c}} \zeta = -\alpha^2, \quad \int_X \alpha^2 \wedge \zeta = 2.$$

Let H be the inverse image under π of a projective line in $\mathbb{P}^2$. The pulled-back Fubini–Study metric gives $\eta_H = \alpha$ and a logarithmic potential σ_H with $\mathrm{dd}^{\mathrm{c}} \sigma_H = [H] - \alpha$.

Proposition 27.22.1 *The divisorial threshold is $\tau_\omega(H) = 1$. For $\mu_0 = \omega^3/6$ and $0 \leq a \leq 1$,*

$$\mathcal{V}_{\mu_0}(\omega; aH) = 6(1 - a)^2.$$

For $a < 1$ this is the scalar of the equation solved, as in Theorem 27.21.1, by $a\sigma_H$ plus a smooth function. At $a = 1$ the residual background is nef, and the logarithmic potential σ_H has zero non-pluripolar Monge–Ampère measure. For $a < 1$, every residual background has infinite upper mass.

Proof The residual background is

$$R_a = (1 - a)\alpha + \zeta.$$

It is positive for $0 \leq a < 1$ and semipositive for $a = 1$, so every coefficient $a \leq 1$ is feasible. For the reverse inequality, consider the metrics $\gamma_t = \alpha + t\zeta$, $t > 0$. They are Gauduchon, since

$$\mathrm{dd}^{\mathrm{c}}(\gamma_t^2) = -2t\alpha^3 = 0.$$

Using $\zeta^2 = 0$, we obtain

$$\frac{\int_X \omega \wedge \gamma_t^2}{\deg_{\gamma_t} H} = \frac{1+2t}{2t} \longrightarrow 1 \qquad (t \to \infty),$$

and the threshold formula gives $\tau_\omega(H) = 1$.

For $a < 1$,

$$R_a^3 = 3(1-a)^2\alpha^2 \wedge \zeta = (1-a)^2\omega^3 = 6(1-a)^2\mu_0.$$

Hence the zero potential solves the residual smooth equation, and uniqueness of the scalar gives the volume. At $a = 1$, the form $R_1 = \zeta$ is nef, because $\zeta + \epsilon\omega > 0$ for $\epsilon > 0$, and the same computation gives

$$\mathcal{V}_{\mu_0}(\zeta + \epsilon\omega) = 6\epsilon^2(1+\epsilon) \longrightarrow 0.$$

By Theorem 27.9.1, the residual volume at $a = 1$ is zero. Moreover $\omega + \mathrm{dd}^{\mathrm{c}}\sigma_H = [H] + \zeta$, so the non-pluripolar measure of σ_H is $\zeta^3 = 0$.

Finally, fix $a < 1$ and choose $0 < \delta \le 1 - a$, so that $R_a - \delta\omega \ge 0$. For the smooth functions V_j of Theorem 9.0.1,

$$R_a + \mathrm{dd}^{\mathrm{c}}(\delta V_j) = (R_a - \delta\omega) + \delta(\omega + \mathrm{dd}^{\mathrm{c}}V_j) > 0.$$

Monotonicity of the determinant on positive matrices bounds the mass of this form below by $\delta^3 \int_X (\omega + \mathrm{dd}^{\mathrm{c}}V_j)^3$, which tends to infinity. Thus a finite residual volume and an explicit divisorial threshold coexist with unbounded upper mass. □

Part VII
Partial kernels and equilibrium

Chapter 28
Partial kernels and quotient equilibrium

Let X be a connected compact complex manifold of dimension $n \geq 1$. Throughout this chapter, L is a holomorphic line bundle on X with a smooth Hermitian metric h. In a local holomorphic frame e write $|e|_h^2 = \mathrm{e}^{-\chi}$; the curvature form $c_1(L,h) = \mathrm{dd^c}\chi$ is a smooth closed real $(1,1)$-form, independent of the frame, as in Chapter 21.

A quasi-plurisubharmonic function φ on X, the selecting potential, selects in each tensor power L^m the holomorphic sections s for which $|s|_{h^m}^2 \mathrm{e}^{-m\varphi}$ is locally integrable. A partial Bergman kernel is the Bergman kernel of this space of sections for a Hilbert norm defined by a continuous weight: the singular function φ decides which sections occur, but it does not enter their norms. On a compact Kähler manifold, if φ is $c_1(L,h)$-plurisubharmonic with positive non-pluripolar Monge–Ampère mass, the normalized kernel measures converge to the Monge–Ampère measure of a partial envelope. The competitors defining this envelope have multiplier ideals contained in those of φ at every positive real weight, and lie below the continuous weight on the set where the norm is computed. The envelope is the largest of these $c_1(L,h)$-plurisubharmonic functions. This chapter studies how much of this equilibrium theory survives when X is not Kähler, and when the selecting potential is plurisubharmonic only with respect to a Hermitian form unrelated to the curvature of L.

One direction of the Kähler argument is local and works on every compact complex manifold. A Bernstein–Markov estimate, which follows from the submean inequality, keeps the partial Fubini–Study potentials below the weight, and Lelong-number estimates on local modifications keep their limits at least as singular as φ in the sense of multiplier ideals. Together these bound every limit of the potentials by the partial envelope. A matching lower bound requires global sections approximating the envelope, and this is where the Kähler hypothesis enters. If X admits a modification from a compact Kähler manifold, the relative canonical bundle with a Jacobian metric makes the selected spaces upstairs and downstairs isometric, and the Kähler theorem gives convergence of the kernel measures on X itself.

Fibrations show what can happen without such a modification. On a holomorphic submersion with compact connected fibers, every section of a pulled-back line bundle is constant along the fibers and descends to the base. Probability measures on the fibers make this descent an isometry, so the kernel measures on the total space are

lifts of kernel measures on the base. Their limit, the quotient equilibrium measure, is the lift of the equilibrium measure of the base, and it appears at the growth rate $m^{\dim B}$ of the section spaces rather than at $m^{\dim X}$. On the Hopf manifold it is a Hermitian Monge–Ampère measure; in dimension three this gives a nonzero limit on a manifold without the bounded mass property.

A selecting potential that is plurisubharmonic only with respect to a Hermitian form need not descend, but the sections it selects still do, so its singularities act only through a vanishing condition on the base. For isolated logarithmic poles, for poles along a section over a base divisor, and for anisotropic singularities controlled by integrability along the fibers, this condition differs from a multiplier-ideal condition on the base by a bounded number of vanishing orders. Such shifts cost only $O(m^{\dim B-1})$ sections, and for point poles and for poles along a section over a divisor the kernel measures still converge to a quotient equilibrium measure. The price is a loss of information: moving a pole along its fiber changes the multiplier ideals on X but leaves every kernel unchanged. Explicit point and divisor models on projective bases compute the limits through binomial tails, and a change of the integration volume separates the base equilibrium measure from the conditional fiber probabilities carried by its lift.

Two mechanisms control the explicit models that follow: the local metric weight determines the scale of evaluation, and the number of selected sections determines the mass of the kernel measure. When vanishing orders grow sublinearly in m, an entire-function model gives a Gaussian transition on a shrinking scale. Transferring it to a compact curve requires a uniform local comparison of norms and a global L^2 correction preserving the vanishing order; the remainder of the smooth weight moves the center of the transition. In two dimensions, competing monomials at a fixed curvature zero can already introduce a logarithmic factor into the scale of the full kernel.

The section count also distinguishes selections with similar potential limits. A Newton polytope gives a graded monomial family whose growth constraint determines an envelope and whose lattice points determine the limiting mass. For a lower-dimensional polytope, concentration of the monomial norms gives a positive limit carried by real tori, although the higher curvature powers of the envelope vanish. Multiplier-ideal spaces need not form a graded family: rounding at critical divisors can make the leading count oscillate between even and odd degrees, while at the base points of a pencil three neighboring monomial bands contribute to a single positive limit. Convergence of the potentials therefore needs to be separated from convergence of the normalized measures.

We use the singularity orders $\preceq$ and $\sim$ of Definition 1.12.1, their multiplier-ideal versions $\preceq_{\mathcal{I}}$ and $\sim_{\mathcal{I}}$ of Section 1.12, the multiplier-ideal envelope $P_\theta[u]_{\mathcal{I}}$ of Definition 17.2.1, the envelope V_θ with minimal singularities of Section 1.19, and the non-pluripolar Monge–Ampère measures MA_θ and $\mathrm{MA}_\theta^{\mathrm{ext}}$ of Theorem 7.1.1.

28.1 Selected spaces and partial Bergman kernels

Fix a quasi-plurisubharmonic function φ on X, the *selecting potential*, and a smooth positive probability measure ν on X. For each integer $m \geq 1$, the *selected space* of degree m and its dimension are

$$V_m = H^0\big(X, L^m \otimes \mathcal{I}(m\varphi)\big), \qquad N_m = \dim V_m.$$

Instead of φ, the Hilbert norm on V_m uses a real continuous weight $v \in C(X)$:

$$\|s\|_{m,v,\nu}^2 = \int_X |s|_{h^m}^2 \mathrm{e}^{-mv}\, \mathrm{d}\nu \quad (s \in V_m).$$

Since v is bounded, this integral is finite for every global section. It vanishes only for $s = 0$, because ν has a positive density and a holomorphic section vanishing almost everywhere vanishes identically. This separation of the selecting potential from the norm follows [Xia26, Section 14.3].

Fix also an integer $r \geq 1$, which determines the normalization below; it will be chosen as the growth degree of N_m, which may be smaller than n. Let $s_1, \dots, s_{N_m}$ be an orthonormal basis of V_m for the norm $\|\cdot\|_{m,v,\nu}$. The *partial Bergman kernel* and the *partial kernel measure* are

$$B_{m,v,\varphi,\nu}(x) = \mathrm{e}^{-mv(x)} \sum_j |s_j(x)|_{h^m}^2 \quad (x \in X), \qquad \beta_m^{(r)} = \frac{r!}{m^r} B_{m,v,\varphi,\nu}\, \nu.$$

By unitary invariance they do not depend on the orthonormal basis, and both are zero when $V_m = \{0\}$. The evaluation argument of Proposition 21.1.2 gives the extremal formula and the total mass:

$$B_{m,v,\varphi,\nu}(x) = \sup_{\substack{s \in V_m \\ \|s\|_{m,v,\nu} \leq 1}} |s(x)|_{h^m}^2 \mathrm{e}^{-mv(x)}, \qquad \int_X \beta_m^{(r)} = \frac{r! N_m}{m^r}.$$

The selecting potential enters only through V_m. In particular, for fixed h, v and ν, if the multiplier ideals $\mathcal{I}(m\varphi)$ are contained in those of another selecting potential, then so are the selected spaces, and the extremal formula shows that the kernels are ordered in the same way.

Replacing h by $h\mathrm{e}^{-a}$ and v by $v - a$, for a smooth real function a, leaves $h^m \mathrm{e}^{-mv}$, and hence the norm and the kernel measure, unchanged.

28.2 A local Bernstein–Markov estimate

A Bernstein–Markov estimate compares the supremum norm of sections on a weighted compact set with an integral norm, up to an exponential loss in the degree whose rate can be made arbitrarily small. It is what keeps partial Fubini–Study potentials below

the weight in the limit. For a smooth positive volume and a continuous weight on the whole manifold, it follows from the submean inequality.

For a compact set $K \subseteq X$ and $w \in C(K)$, write

$$\|s\|_{m,w,K} = \sup_K |s|_{h^m} \mathrm{e}^{-mw/2}.$$

A probability measure μ on K is *Bernstein–Markov* for (K, w) if, for every $\epsilon > 0$, there is a constant C_ϵ such that

$$\|s\|_{m,w,K} \le C_\epsilon \mathrm{e}^{\epsilon m} \left(\int_K |s|^2_{h^m} \mathrm{e}^{-mw} \, \mathrm{d}\mu \right)^{1/2}$$

for all $m \ge 1$ and all $s \in H^0(X, L^m)$. If K is non-pluripolar, the supremum norm is a norm: the zero set of a nonzero section is an analytic hypersurface, hence pluripolar, and cannot contain K. The Bernstein–Markov inequality then shows that the integral

$$\|s\|^2_{m,w,\mu} = \int_K |s|^2_{h^m} \mathrm{e}^{-mw} \, \mathrm{d}\mu$$

is also a norm on $H^0(X, L^m)$, and in particular on every selected space V_m. The definitions of Section 28.1 therefore apply with (X, v, ν) replaced by (K, w, μ). We write $B_{m,w,\varphi,\mu}$ for the resulting partial Bergman kernel on K and $\beta_m^{(r)} = (r!/m^r) B_{m,w,\varphi,\mu} \, \mu$ for the partial kernel measure; the extremal formula and the total mass $r! N_m / m^r$ hold as before.

Proposition 28.2.1 *Let $v \in C(X)$. Every smooth positive probability measure on X is Bernstein–Markov for (X, v).*

Proof Let ν be a smooth positive probability measure, and fix $\epsilon > 0$. Cover X by finitely many coordinate charts on which L has holomorphic frames, and choose relatively compact open subsets of these charts that still cover X. In a frame e write $|e|^2_h = \mathrm{e}^{-\chi}$ and $s = f e^{\otimes m}$, so that $|s|^2_{h^m} \mathrm{e}^{-mv} = |f|^2 \mathrm{e}^{-m(\chi+v)}$. By compactness and uniform continuity of the finitely many functions $\chi + v$, there is a radius $\delta > 0$ with the following property: every point x lies in one of the smaller sets, the coordinate ball $B(x, \delta)$ lies in the corresponding chart, and the oscillation of $\chi + v$ on $B(x, \delta)$ is at most 2ϵ. On these balls the density of ν with respect to coordinate Lebesgue measure is bounded below by a positive constant independent of x.

The submean inequality for the holomorphic function f on $B(x, \delta)$ therefore gives

$$\begin{aligned} |s(x)|^2_{h^m} \mathrm{e}^{-mv(x)} &= |f(x)|^2 \mathrm{e}^{-m(\chi+v)(x)} \\ &\le C_\delta \mathrm{e}^{2\epsilon m} \int_{B(x,\delta)} |f|^2 \mathrm{e}^{-m(\chi+v)} \, \mathrm{d}\nu \\ &\le C_\delta \mathrm{e}^{2\epsilon m} \|s\|^2_{m,v,\nu}, \end{aligned}$$

with C_δ independent of x and m. Taking the supremum over x and then the square root gives the Bernstein–Markov inequality with $C_\epsilon = C_\delta^{1/2}$. □

For a general compact set K and weight $w \in C(K)$, the Bernstein–Markov property is not automatic, and we assume it explicitly whenever it is needed.

28.3 Partial envelopes on a weighted compact set

Two constraints act on limits of partial Fubini–Study potentials. The continuous weight is an obstacle on the set where sections are measured, and the selecting potential imposes a multiplier-ideal constraint on the whole manifold. The partial envelope is the largest potential satisfying both. Since an obstacle on a non-pluripolar compact set bounds every competitor globally, the partial envelope lies below a translate of the multiplier-ideal envelope of the selecting potential; conversely, a translate of that envelope lying below the obstacle is a competitor. The partial envelope therefore differs from the multiplier-ideal envelope by a bounded function.

Let θ be a smooth real $(1,1)$-form on X, let $\varphi \in \mathrm{PSH}(X,\theta)$, let $K \subseteq X$ be compact and non-pluripolar, and let $w \in C(K,\mathbb{R})$. Recall that Theorem 1.16.1 characterizes the order $\preceq_{\mathcal{I}}$ by Lelong numbers on local modifications. As in Section 1.19, an inequality holds quasi-everywhere if it holds outside a pluripolar set. The *partial envelope* of φ with obstacle w on K is

$$E_{\theta,K}[\varphi]_{\mathcal{I}}(w) = \Big(\sup\big\{a \in \mathrm{PSH}(X,\theta) : a \preceq_{\mathcal{I}} \varphi, \\ a \le w \text{ quasi-everywhere on } K\big\}\Big)^{*}. \tag{28.1}$$

Theorem 28.3.1 *The function $E = E_{\theta,K}[\varphi]_{\mathcal{I}}(w)$ is θ-plurisubharmonic, and it is the largest member of its defining family. Let $q = P_\theta[\varphi]_{\mathcal{I}}$, and let C be a constant as in Lemma 1.19.1, so that $\sup_X a \le C$ for every $a \in \mathrm{PSH}(X,\theta)$ with $a \le w$ quasi-everywhere on K. Then*

$$q + \min_K w \le E \le q + C.$$

In particular, $E \sim q$ and

$$E \sim_{\mathcal{I}} \varphi.$$

Proof By Theorem 17.2.1, $q \le 0$ and $q \sim_{\mathcal{I}} \varphi$. Hence $q + \min_K w$ belongs to the defining family. Lemma 1.19.1 bounds the entire family above by C, so its regularized supremum E is θ-plurisubharmonic, and $E \ge q + \min_K w$.

For the upper bound, let a be a member of the family. Since $\sup_X a \le C$, Propositions 17.2.2 and 17.2.3 give

$$a - C \le a - \sup_X a \le P_\theta[a]_{\mathcal{I}} \le P_\theta[\varphi]_{\mathcal{I}} = q.$$

Taking the regularized supremum over a proves $E \le q + C$.

The two bounds show that $E - q$ is bounded. Hence E and q have the same multiplier ideals, and these are the multiplier ideals of φ.

It remains to show that E itself satisfies the obstacle condition, which is not automatic after regularization. By Choquet's lemma, E is the regularization of the supremum of a countable subfamily. Replace its members by their successive finite maxima. Finite maxima are again members of the family: the multiplier-ideal order is preserved by Theorem 1.16.1, and a finite union of pluripolar exceptional sets is pluripolar. We obtain an increasing sequence a_j in the family whose limit has regularization E. Outside a countable union of pluripolar subsets of K, every a_j is at most w, and hence so is $\lim_j a_j$. By the local upper-envelope theorem recalled in Section 1.19, E differs from $\lim_j a_j$ only on a pluripolar set. Thus $E \le w$ quasi-everywhere on K.

We have shown that E belongs to its defining family. It dominates every member by construction, so it is the largest member. In particular, the supremum in (28.1) equals E at every point even before regularization. □

Proposition 28.3.1 *The partial envelope depends only on the multiplier-ideal type of φ, and it is increasing in the obstacle and in φ with respect to $\preceq_{\mathcal{I}}$. For $w, w' \in C(K)$ and $c \in \mathbb{R}$,*

$$E_{\theta,K}[\varphi]_{\mathcal{I}}(w + c) = E_{\theta,K}[\varphi]_{\mathcal{I}}(w) + c,$$
$$\left|E_{\theta,K}[\varphi]_{\mathcal{I}}(w) - E_{\theta,K}[\varphi]_{\mathcal{I}}(w')\right| \le \|w - w'\|_{C(K)}.$$

The second line stands for the two corresponding pointwise inequalities, which make sense also at the poles. Let θ_0, θ_1 be smooth real $(1,1)$-forms on X, and let $\varphi_i \in \mathrm{PSH}(X, \theta_i)$ and $w_i \in C(K)$ for $i = 0, 1$. For $0 < t < 1$ put $\theta_t = (1-t)\theta_0 + t\theta_1$, and define φ_t and w_t in the same way. Then

$$(1-t)E_{\theta_0,K}[\varphi_0]_{\mathcal{I}}(w_0) + tE_{\theta_1,K}[\varphi_1]_{\mathcal{I}}(w_1) \le E_{\theta_t,K}[\varphi_t]_{\mathcal{I}}(w_t).$$

Finally, for a smooth real function f, changing the background and the obstacle gives the identity

$$E_{\theta+\mathrm{dd}^c f,K}[\varphi - f]_{\mathcal{I}}(w - f|_K) = E_{\theta,K}[\varphi]_{\mathcal{I}}(w) - f.$$

Proof The first two assertions hold because the defining families are equal or nested. Adding a constant c is a bijection between the families for w and $w + c$, which proves the translation rule. Applying monotonicity and translation to $w' - \|w - w'\|_{C(K)} \le w \le w' + \|w - w'\|_{C(K)}$ gives the two inequalities in the second line.

For concavity, the left-hand side is θ_t-plurisubharmonic, and it lies below w_t outside the union of two pluripolar exceptional sets. By Theorem 28.3.1, its two summands have the multiplier-ideal types of φ_0 and φ_1, so the addition and scaling rules of Lemma 1.16.3 show that the left-hand side has the multiplier-ideal type of φ_t. It is therefore a member of the family defining the right-hand side. Finally, $a \mapsto a - f$ is a bijection between the two families in the last identity, and regularization commutes with subtraction of a continuous function. □

In the next section the partial envelope bounds every limit of partial Fubini–Study potentials. The point is to show that both constraints, the multiplier-ideal order and the obstacle on K, survive L^1 limits.

28.4 An upper bound for partial Fubini–Study limits

The Bernstein–Markov inequality keeps the partial Fubini–Study potentials below the weight on K, up to an error that becomes arbitrarily small as the degree grows. The multiplier-ideal constraint survives in the limit because, on a fixed local modification, the Lelong numbers of the potentials fall short of those of the selecting potential by at most $O(m^{-1})$. To pass the obstacle condition to L^1 limits when K has zero smooth volume, we compare the potentials with the envelope of the obstacle alone, which is defined on all of X.

Now take $\theta = c_1(L,h)$, assume $\varphi \in \mathrm{PSH}(X,\theta)$, and let μ be a Bernstein–Markov probability measure for (K,w), with K and w as in Section 28.3. Use the selected spaces V_m and their dimensions N_m from Section 28.1. Choose an orthonormal basis $s_1,\dots,s_{N_m}$ of V_m for the norm $\|\cdot\|_{m,w,\mu}$ of Section 28.2, and define the *partial Fubini–Study potential*

$$U_m = \frac{1}{m}\log\sum_{j=1}^{N_m}|s_j|_{h^m}^2 \qquad (N_m>0).$$

Set $U_m \equiv -\infty$ if $N_m = 0$. When $N_m > 0$, the local expression in a holomorphic frame, as in the proof of Proposition 21.1.2, shows that $U_m \in \mathrm{PSH}(X,\theta)$ and that U_m has analytic singularities. On K it satisfies $\mathrm{e}^{m(U_m-w)} = B_{m,w,\varphi,\mu}$.

Let $\pi\colon Y\to W$ be a local smooth modification of an open set $W\subseteq X$, let $y\in Y$, and write J_π for the local holomorphic Jacobian determinant of π. Each s_j lies in V_m, so Lemma 1.17.9 applies to its local coefficient. Since the Lelong number of the logarithm of a sum of squared moduli of holomorphic functions is the minimum of their vanishing orders, and the smooth metric contributes nothing, we obtain

$$\nu(U_m\circ\pi,y)\ge \nu(\varphi\circ\pi,y)-\frac{n+\operatorname{ord}_y J_\pi}{m} \qquad (N_m>0).$$

Theorem 28.4.1 *The potentials U_m with $N_m>0$ form a relatively compact subset of $L^1(X)$. Let U be the L^1 limit of a sequence U_{m_j} with $N_{m_j}>0$ and $m_j\to\infty$. Then $U\in\mathrm{PSH}(X,\theta)$,*

$$U \preceq_{\mathcal{I}} \varphi,$$

and $U\le w$ quasi-everywhere on K. Consequently

$$\Big(\limsup_{m\to\infty} U_m\Big)^{*} \le E_{\theta,K}[\varphi]_{\mathcal{I}}(w).$$

If only finitely many selected spaces are nonzero, the left-hand side is identically $-\infty$.

Proof *Step 1: compactness.* The extremal formula for the kernel and the Bernstein–Markov inequality give, for every $\epsilon > 0$ and with $C_\epsilon \geq 1$,

$$U_m \leq w + \delta_{m,\epsilon} \quad \text{on } K, \qquad \delta_{m,\epsilon} = 2\epsilon + \frac{2\log C_\epsilon}{m}.$$

No factor depending on N_m occurs, because the extremal formula is a supremum over single sections of unit norm. Fix one $\epsilon > 0$. Since $\delta_{m,\epsilon} \leq 2\epsilon + 2\log C_\epsilon$, Lemma 1.19.1 gives an upper bound for $\sup_X U_m$ that is uniform over all m with $N_m > 0$. On the other hand, the total mass identity gives

$$\int_K \mathrm{e}^{m(U_m - w)}\, \mathrm{d}\mu = N_m \geq 1.$$

Since μ is a probability measure, $U_m - w \geq 0$ somewhere on K, and therefore $\sup_X U_m \geq \min_K w$. The normalized compactness theorem (Theorem 1.4.1) now gives relative compactness in $L^1(X)$, with limits in $\mathrm{PSH}(X,\theta)$.

Step 2: the multiplier-ideal order. Let $U_{m_j} \to U$ in $L^1(X)$, with $m_j \to \infty$. Fix a local smooth modification $\pi\colon Y \to W$ of an open set $W \subseteq X$, and a point $y \in Y$. By Lemma 1.22.8, $U_{m_j} \circ \pi \to U \circ \pi$ in L^1_{loc}. Upper semicontinuity of Lelong numbers (Lemma 1.17.6) and the order bound before the theorem give

$$\nu(U \circ \pi, y) \geq \limsup_j \nu(U_{m_j} \circ \pi, y) \geq \nu(\varphi \circ \pi, y).$$

Since π and y are arbitrary, the local valuative characterization in Theorem 1.16.1 gives $U \preceq_{\mathcal{I}} \varphi$.

Step 3: the obstacle. Since K may have zero smooth volume, the inequality on K in Step 1 does not pass directly to L^1 limits. We therefore use the envelope of the obstacle alone. Set

$$G = E_{\theta,K}[V_\theta]_{\mathcal{I}}(w).$$

Every $a \in \mathrm{PSH}(X,\theta)$ satisfies $a \leq V_\theta + \sup_X a$, and hence $a \preceq_{\mathcal{I}} V_\theta$. Thus the multiplier-ideal constraint in the definition of G is no restriction, and G is the largest θ-plurisubharmonic function lying below w quasi-everywhere on K. Since $U_m - \delta_{m,\epsilon}$ is such a function, Step 1 gives, on all of X,

$$U_m \leq G + \delta_{m,\epsilon}.$$

Passing to the L^1 limit gives $U \leq G + 2\epsilon$ almost everywhere, and hence everywhere by Lemma 1.3.3. Letting $\epsilon \downarrow 0$ gives $U \leq G$. Since $G \leq w$ quasi-everywhere on K by Theorem 28.3.1, the same holds for U. Thus U satisfies both defining conditions of the partial envelope, and $U \leq E_{\theta,K}[\varphi]_{\mathcal{I}}(w)$.

Step 4: the upper limit. By the local Hartogs lemma, if $U_{m_j} \to U$ in L^1 and $x_j \to x$, then $\limsup_j U_{m_j}(x_j) \leq U(x)$. Given a point x, a sequence of indices $m_j \to \infty$ with $N_{m_j} > 0$ and points $x_j \to x$, pass to a subsequence along which $U_{m_j}(x_j)$ converges

to $\limsup_j U_{m_j}(x_j)$ and U_{m_j} converges in L^1. By Steps 2 and 3 the L^1 limit is at most the partial envelope, so $\limsup_j U_{m_j}(x_j) \le E_{\theta,K}[\varphi]_I(w)(x)$. This proves the inequality for the regularized upper limit, also at the poles of the envelope. Indices with $N_m = 0$ contribute only $-\infty$. □

28.5 The determinant functional

As the continuous weight varies, the partial kernel measures are the derivatives of a single functional, the logarithm of a Gram determinant. Orthogonal projection expresses its second derivative as minus a sum of squared norms, which gives concavity, and the total mass identity gives a Lipschitz bound in terms of the dimension of the selected space.

We return to the norm $\|\cdot\|_{m,v,\nu}$ on the whole manifold from Section 28.1, with selecting potential $\varphi \in \mathrm{QPSH}(X)$. Fix m with $N_m > 0$ and a basis of V_m, and let $G_m(v)$ be the Gram matrix of this basis for $\|\cdot\|_{m,v,\nu}$, a positive definite Hermitian matrix. The *determinant functional* is

$$\mathcal{F}_m^{(r)}(v) = -\frac{r!}{m^{r+1}} \log \frac{\det G_m(v)}{\det G_m(0)}.$$

A change of basis multiplies both determinants by the same positive factor, so $\mathcal{F}_m^{(r)}$ does not depend on the basis. If $N_m = 0$, put $\mathcal{F}_m^{(r)} = 0$.

Proposition 28.5.1 *For $v, f \in C(X,\mathbb{R})$, the measure $\beta_m^{(r)}$ at the weight v satisfies*

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathcal{F}_m^{(r)}(v + tf) = \int_X f \,\mathrm{d}\beta_m^{(r)}.$$

The functional $\mathcal{F}_m^{(r)}$ is concave on $C(X,\mathbb{R})$, and for $v, w \in C(X,\mathbb{R})$ and $c \in \mathbb{R}$,

$$\mathcal{F}_m^{(r)}(v + c) = \mathcal{F}_m^{(r)}(v) + \frac{r!N_m}{m^r} c, \qquad |\mathcal{F}_m^{(r)}(v) - \mathcal{F}_m^{(r)}(w)| \le \frac{r!N_m}{m^r} \|v - w\|_\infty.$$

If $N_m > 0$, the second derivative of $t \mapsto \mathcal{F}_m^{(r)}(v + tf)$ is strictly negative unless f is constant.

Proof Choose a basis $s_1, \dots, s_{N_m}$ that is orthonormal for the weight v, so that $G_m(v) = I$ is the identity matrix of size N_m; the indices i, j below range from 1 to N_m. Since the weights are bounded, we may differentiate under the integral sign. With pointwise inner products taken in h^m, put

$$A_{ij} = \int_X f \langle s_i, s_j \rangle_{h^m} \mathrm{e}^{-mv} \,\mathrm{d}v, \qquad C_{ij} = \int_X f^2 \langle s_i, s_j \rangle_{h^m} \mathrm{e}^{-mv} \,\mathrm{d}v.$$

The first and second derivatives of $t \mapsto G_m(v + tf)$ at $t = 0$ are $-mA$ and m^2C. The identity $(\log\det G)' = \operatorname{tr}(G^{-1}G')$ gives the first derivative of $\mathcal{F}_m^{(r)}$, and differentiating it once more yields

$$\left.\frac{\mathrm{d}^2}{\mathrm{d}t^2}\right|_{t=0} \mathcal{F}_m^{(r)}(v + tf) = -\frac{r!}{m^{r-1}}\left(\int_X f^2 B_{m,v,\varphi,\nu}\,\mathrm{d}\nu - \sum_{i,j}|A_{ij}|^2\right).$$

Let P_v be the orthogonal projection of the weighted L^2 space of measurable sections of L^m onto V_m. Since $\int_X f^2 B_{m,v,\varphi,\nu}\,\mathrm{d}\nu = \sum_j \|fs_j\|^2_{m,v,\nu}$ and $P_v(fs_j) = \sum_i A_{ji}s_i$, Pythagoras' theorem shows that the quantity in parentheses equals

$$\sum_j \|(I - P_v)(fs_j)\|^2_{m,v,\nu} \geq 0.$$

This proves concavity. Multiplying every Gram entry by e^{-mc} shows that $\det G_m(v + c) = \mathrm{e}^{-mcN_m}\det G_m(v)$, which is the translation rule. By the first-variation formula and the total mass identity, the derivative of $\mathcal{F}_m^{(r)}$ along the segment from w to v is bounded by $(r!N_m/m^r)\|v - w\|_\infty$, which gives the Lipschitz estimate.

Suppose finally that $N_m > 0$ and that the second derivative vanishes. Then $(I - P_v)(fs_j) = 0$ for every j. Thus $s = s_1$ and $\widetilde{s} = P_v(fs_1)$ are elements of V_m with $s \neq 0$ and $fs = \widetilde{s}$ almost everywhere. Both sides are continuous and ν has a positive density, so $fs = \widetilde{s}$ everywhere. On a ball where $s \neq 0$, the quotient $\widetilde{s}/s = f$ is a real-valued holomorphic function, hence a constant c. The identity theorem gives $\widetilde{s} = cs$ on the connected manifold X, so $f = c$ wherever $s \neq 0$, and by continuity everywhere. Conversely, for constant f the translation rule shows that the second derivative vanishes. □

Thus $\mathcal{F}_m^{(r)}$ is a primitive of the partial kernel measures on every compact complex manifold. This does not affect the obstruction proved in Theorem 3.1.1 to a primitive of the Hermitian Monge–Ampère operator, since the derivative here is a partial kernel measure and not a Monge–Ampère measure.

Corollary 28.5.1 *If $N_m = o(m^n)$, then $\beta_m^{(n)} \to 0$ in total variation, and $\mathcal{F}_m^{(n)}(v) \to 0$ uniformly for v in any bounded subset of $C(X)$.*

Proof The measures $\beta_m^{(n)}$ are positive with total masses $n!N_m/m^n \to 0$. The second assertion follows from the Lipschitz estimate, since $\mathcal{F}_m^{(n)}(0) = 0$. □

For example, suppose that X is not Moishezon. By Corollary 22.5.1, either every positive-degree section space of L is zero, or $h^0(X, L^m) = O(m^d)$ for some $d < n$. Since $V_m \subseteq H^0(X, L^m)$, every sequence of selected spaces satisfies the hypothesis of Corollary 28.5.1. By the last assertion of Corollary 22.5.1, this applies in particular whenever X fails the bounded mass property of Definition 7.5.1.

28.5.1 Dependence on the continuous weight

When only the continuous weight changes, the selected space stays the same, and the two norms are related by a multiplication operator compressed to that space. Inverting this operator expresses the new reproducing vectors through the old ones. As a consequence the kernel itself, and not only the determinant functional, depends analytically on the weight.

Keep $m \geq 1$, the selecting potential φ, the metric h and the probability measure ν fixed as in Section 28.1, and write $B_v = B_{m,v,\varphi,\nu}$ and $\beta_v = \beta_m^{(r)}$ for the kernel and the kernel measure at the weight v, let P_v be the orthogonal projection onto V_m from the proof of Proposition 28.5.1, and let

$$Z_m = \{x \in X : s(x) = 0 \text{ for every } s \in V_m\}$$

be the common zero set of the selected sections. At each $x \in X$ choose a covector of unit length on the fiber $(L^m)_x$; it defines a complex-linear evaluation $\ell_x : V_m \to \mathbb{C}$. Let $k_{v,x} \in V_m$ be its Riesz vector, so that $\ell_x(s) = \langle s, k_{v,x}\rangle_{m,v,\nu}$ for all $s \in V_m$. Then

$$B_v(x) = \mathrm{e}^{-mv(x)}\ell_x(k_{v,x}) = \mathrm{e}^{-mv(x)}\|k_{v,x}\|^2_{m,v,\nu}. \tag{28.2}$$

These values do not depend on the phase of the chosen covector.

Proposition 28.5.2 *The map $v \mapsto B_v$ from $C(X, \mathbb{R})$ to $C(X, \mathbb{R})$ is real analytic. The map $v \mapsto \beta_v$ is real analytic with values in the finite signed measures on X, equipped with the total variation norm $\|\cdot\|_{\mathrm{TV}}$. If $V_m = \{0\}$, both maps vanish identically.*

Proof If $V_m = \{0\}$ there is nothing to prove. Fix a real weight v_0. For $f \in C(X, \mathbb{C})$ define an operator on V_m by

$$T(f)s = P_{v_0}(\mathrm{e}^{-mf}s).$$

Since the exponential series converges in the uniform norm, T is an entire function of f with values in the endomorphisms of V_m. Since P_{v_0} is a contraction,

$$\|T(f) - I\| \leq \mathrm{e}^{m\|f\|_\infty} - 1.$$

Hence, for $m\|f\|_\infty < \log(3/2)$, we have $\|T(f) - I\| < 1/2$, so the Neumann series for $T(f)^{-1}$ converges in operator norm and $\|T(f)^{-1}\| \leq 2$. For real f, the operator

$T(f)$ is selfadjoint and represents the new inner product:

$$\langle s,u\rangle_{m,v_0+f,\nu}=\langle T(f)s,u\rangle_{m,v_0,\nu}.$$

The Riesz identity therefore gives

$$k_{v_0+f,x}=T(f)^{-1}k_{v_0,x}.$$

For complex f the right-hand side is still a holomorphic function of f. Consequently

$$f\longmapsto \mathrm{e}^{-m(v_0(x)+f(x))}\ell_x\big(T(f)^{-1}k_{v_0,x}\big)$$

is a holomorphic extension of $f\mapsto B_{v_0+f}(x)$. Evaluations on the finite-dimensional space V_m are uniformly bounded over the compact manifold X, so the linear map $A\mapsto[x\mapsto\ell_x(Ak_{v_0,x})]$ from the endomorphisms of V_m to $C(X)$ is bounded; it is well defined because the phases of ℓ_x and $k_{v_0,x}$ cancel. Hence the extension is holomorphic as a map with values in $C(X)$. Finally, $B\mapsto(r!/m^r)B\nu$ is a bounded linear map into finite measures with the total variation norm, which proves the assertion about β_ν. At points of Z_m the Riesz vector is zero, so the argument includes the common zero set. □

Now let $I\subset\mathbb{R}$ be an open interval and let $t\mapsto v_t$ be a smooth curve in $C(X,\mathbb{R})$. The complete exponential Bell polynomials are

$$\mathcal{B}_j(x_1,\dots,x_j)=\sum_{\sum_{\ell=1}^{j}\ell n_\ell=j} j!\prod_{\ell=1}^{j}\frac{(x_\ell/\ell!)^{n_\ell}}{n_\ell!},\qquad j\ge 1,$$

where the sum runs over nonnegative integers $n_1,\dots,n_j$. Write $\mathcal{B}_j(-mv_t)$ for $\mathcal{B}_j$ evaluated at the first j derivatives of $-mv_t$ with respect to t, so that $\partial_t^{\,j}\mathrm{e}^{-mv_t}=\mathrm{e}^{-mv_t}\mathcal{B}_j(-mv_t)$ by Faà di Bruno's formula. Define operators on V_m by

$$\mathcal{T}_{t,j}u=-P_{v_t}\big(\mathcal{B}_j(-mv_t)u\big),\qquad u\in V_m.$$

For $k\ge 1$, let Λ_k be the set of compositions of k, that is, of ordered tuples $\alpha=(\alpha_1,\dots,\alpha_p)$ of positive integers with sum k, and set $\alpha!=\prod_{j=1}^{p}\alpha_j!$.

Theorem 28.5.1 *For every $k \geq 1$,*

$$\partial_t^k k_{v_t,x} = \sum_{\alpha \in \Lambda_k} \frac{k!}{\alpha!} \mathcal{T}_{t,\alpha_p} \cdots \mathcal{T}_{t,\alpha_1} k_{v_t,x}. \tag{28.3}$$

Suppose moreover that $s \geq 1$, that $J \Subset I$ is a compact interval, and that constants $A, R > 0$ satisfy

$$\sup_{t \in J} \|\partial_t^j v_t\|_\infty \leq A R^j (j!)^s \qquad (j \geq 1).$$

Then there are constants $C, L > 0$, depending only on the fixed data, on m, r, J, A, R, s and on $\sup_{t \in J} \|v_t\|_\infty$, such that

$$\sup_{t \in J} \left(\|\partial_t^k B_{v_t}\|_\infty + \|\partial_t^k \beta_{v_t}\|_{\mathrm{TV}} \right) \leq C L^k (k!)^s \qquad (k \geq 0). \tag{28.4}$$

Proof *The derivative formula.* By the proof of Proposition 28.5.2, $k_{v_t,x}$ depends smoothly on t, for any fixed norm on V_m. Fix a real parameter t and set

$$H(a) = P_{v_t}\left(\mathrm{e}^{-m(v_{t+a}-v_t)} \cdot\right)|_{V_m}.$$

Then $H(0) = I$, $H^{(j)}(0) = -\mathcal{T}_{t,j}$ by Faà di Bruno's formula, and $k_{v_{t+a},x} = H(a)^{-1} k_{v_t,x}$ as in the proof of Proposition 28.5.2. Since $H(a) - I = O(|a|)$, truncating the Neumann series of $H(a)^{-1}$ after k powers leaves a remainder $O(|a|^{k+1})$. Expanding the truncated series to order k in a gives (28.3), with its ordered products and coefficients $k!/\alpha!$.

Gevrey bounds. For nonnegative integers n_ℓ with $\sum \ell n_\ell = j$,

$$\prod_{\ell=1}^{j} (\ell!)^{(s-1)n_\ell} \leq (j!)^{s-1}.$$

The sum $\sum \prod_\ell (n_\ell!)^{-1}$ over these tuples is the coefficient of z^j in $\exp(z/(1-z))$, and evaluation at $z = 1/2$ shows that it is at most $\mathrm{e}\, 2^j$. Together with the assumed bounds this gives

$$\sup_{t \in J} \|\mathcal{B}_j(-m v_t)\|_\infty \leq L_0^j (j!)^s$$

for some $L_0 > 0$. Since P_{v_t} is a contraction, each ordered product in (28.3) has operator norm at most $L_0^k (\alpha!)^s$. There are 2^{k-1} compositions of k, and $\alpha! \leq k!$. Since v_t is uniformly bounded for $t \in J$, the norms $\|\cdot\|_{m,v_t,\nu}$ are uniformly equivalent, and the evaluations and Riesz vectors are bounded uniformly on $J \times X$. Hence the t-derivatives of the unweighted diagonal $x \mapsto \ell_x(k_{v_t,x})$ are bounded in $C(X)$ by $C(2L_0)^k (k!)^s$. The factor $\mathrm{e}^{-m v_t}$ satisfies bounds of the same Gevrey order, by the estimate for $\mathcal{B}_j(-m v_t)$. Leibniz's formula and $j!(k-j)! \leq k!$ then give (28.4) for B_{v_t}, and the bounded linear map $B \mapsto (r!/m^r) B\nu$ gives the bound in total variation.□

Corollary 28.5.2 *Let $v_t = v + tq$ with $v, q \in C(X, \mathbb{R})$. For $x \notin Z_m$, the probability measure*

$$\mathrm{d}\sigma_{t,x}(y) = \frac{|k_{v_t,x}(y)|^2_{h^m} \mathrm{e}^{-mv_t(y)}}{\|k_{v_t,x}\|^2_{m,v_t,\nu}} \, \mathrm{d}\nu(y) \tag{28.5}$$

satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t} \log B_{v_t}(x) = m \left(\int_X q \, \mathrm{d}\sigma_{t,x} - q(x) \right). \tag{28.6}$$

On Z_m the kernel and all its t-derivatives vanish. Moreover, $\int_X \partial_t B_{v_t} \, \mathrm{d}\nu = 0$.

Proof Since $\mathcal{T}_{t,1} u = m P_{v_t}(qu)$, the case $k = 1$ of (28.3) gives $\partial_t k_{v_t,x} = m P_{v_t}(q k_{v_t,x})$. Applying the fixed evaluation ℓ_x and the Riesz identity gives

$$\partial_t \ell_x(k_{v_t,x}) = m \int_X q |k_{v_t,x}|^2_{h^m} \mathrm{e}^{-mv_t} \, \mathrm{d}\nu.$$

Dividing by the positive number $\ell_x(k_{v_t,x}) = \|k_{v_t,x}\|^2_{m,v_t,\nu}$ and differentiating the factor $\mathrm{e}^{-mv_t(x)}$ in (28.2) proves (28.6); the density in (28.5) has integral one by the definition of the norm. At points of Z_m the evaluations and Riesz vectors vanish for every weight. Finally, differentiating the identity $\int_X B_{v_t} \, \mathrm{d}\nu = N_m$ in $C(X)$ proves the last assertion. □

The uniform norm on the weights is essential in these results, because it controls the weight on the whole region of integration. The next example shows that real-analytic dependence of the weight on a parameter at each interior point of a domain does not suffice.

Example 28.5.1 On the unit disk $D \subset \mathbb{C}$ with Euclidean area $\mathrm{d}A$, let K_t be the Bergman kernel of the Hilbert space of holomorphic functions f on D with the norm $(\int_D |f|^2 \rho_t \, \mathrm{d}A)^{1/2}$, where

$$\rho_t(z) = 1 + \frac{t^2}{(1 - |z|^2)^2 + t^2}, \qquad t \in \mathbb{R}.$$

The weights $-\log \rho_t$ are bounded and depend real analytically on $(z, t) \in D \times \mathbb{R}$, but

$$K_t(0,0) = \begin{cases} [\pi(1 + |t| \arctan(1/|t|))]^{-1}, & t \neq 0, \\ \pi^{-1}, & t = 0. \end{cases} \tag{28.7}$$

In particular, $K_t(0,0) = \pi^{-1} - |t|/2 + O(t^2)$, so $K_t(0,0)$ is not differentiable at $t = 0$.

Proof The density ρ_t is radial and lies between one and two. By radial symmetry, the constant functions are orthogonal to the holomorphic functions vanishing at the origin, so the reproducing value at the center is the reciprocal of $\int_D \rho_t \, \mathrm{d}A$. Polar coordinates and the substitution $a = 1 - r^2$ give, for $t \neq 0$,

$$\int_D \rho_t \, \mathrm{d}A = \pi \int_0^1 \left(1 + \frac{t^2}{a^2 + t^2} \right) \mathrm{d}a = \pi \big(1 + |t| \arctan(1/|t|) \big).$$

For $t = 0$ the integral is π. This proves (28.7) and the expansion. For every $t \neq 0$ we have $\| \log \rho_t \|_{L^\infty(D)} = \log 2$, whereas $\rho_0 = 1$. Thus the curve $t \mapsto \log \rho_t$ is not continuous at $t = 0$ for the uniform norm, and the example does not contradict Proposition 28.5.2. □

Throughout these estimates the degree m and the selected space were fixed, and the constants depend on m. A limit envelope need not even be differentiable in the weight, as the corner in Example 14.1.1 shows; differentiating limits would require estimates uniform in m.

28.5.2 Stability of normalized kernels

The total mass of $B_v \nu$ is the dimension of the selected space, so dividing by this dimension gives probability measures whose dependence on the weight can be controlled independently of the dimension. The squared reproducing kernel defines a symmetric coupling of such a probability measure with itself, and the positivity of the associated Markov operator sharpens the resulting variation bound to an optimal one.

Keep the fixed data of Section 28.5.1, and assume $N_m > 0$. For each real $v \in C(X)$, the *normalized kernel probability* is

$$\mu_v = \frac{B_v}{N_m} \nu. \tag{28.8}$$

Choose an orthonormal basis $s_1, \dots, s_{N_m}$ of V_m for the weight v, and use the unit evaluations ℓ_x of Section 28.5.1 to set

$$F_i(x) = \mathrm{e}^{-mv(x)/2} \ell_x(s_i), \qquad C_v(x, y) = \sum_i F_i(x) \overline{F_i(y)}.$$

The function C_v depends on the phases of the evaluations, but $|C_v|$ does not. For $x \notin Z_m$ let $\sigma_{v,x}$ be the probability measure (28.5) at the weight v, and define

$$Q_v f(x) = \int_X f(y) \, \mathrm{d}\sigma_{v,x}(y);$$

on Z_m, which is a μ_v-null set, put $Q_v f = 0$.

Lemma 28.5.1 *The measure*

$$\mathrm{d}J_\nu(x,y) = \frac{|C_\nu(x,y)|^2}{N_m}\,\mathrm{d}\nu(x)\,\mathrm{d}\nu(y) \tag{28.9}$$

on $X \times X$ is a symmetric probability measure whose two marginals are equal to μ_ν. The operator Q_ν is a selfadjoint, positive semidefinite contraction of $L^2(\mu_\nu)$; it fixes the constant functions μ_ν-almost everywhere and preserves integrals against μ_ν.

Proof Orthonormality gives $\int_X |C_\nu(x,y)|^2\,\mathrm{d}\nu(y) = B_\nu(x)$, so the first marginal of J_ν is μ_ν. By symmetry so is the second, and the total mass identity shows that J_ν has mass one. Off Z_m, dividing $|C_\nu(x,\cdot)|^2\,\mathrm{d}\nu$ by $B_\nu(x)$ gives the squared Riesz-vector density in (28.5). Thus the probabilities $\sigma_{\nu,x}$ are the conditional probabilities of J_ν given the first coordinate, and Q_ν is the associated Markov operator. Since these conditional probabilities have mass one and the marginal of J_ν is μ_ν, the operator Q_ν fixes constants and preserves integrals; Jensen's inequality shows that it is a contraction of $L^2(\mu_\nu)$, and the symmetry of J_ν shows that it is selfadjoint. For $f \in L^2(\mu_\nu)$, expanding the finite sum gives

$$\langle Q_\nu f, f\rangle_{L^2(\mu_\nu)} = \frac{1}{N_m}\sum_{i,j}\left|\int_X f F_i \overline{F_j}\,\mathrm{d}\nu\right|^2 \geq 0.$$

Each of these integrals is independent of the phases, and it exists because $|F_i\overline{F_j}| \leq B_\nu$ and $f \in L^1(\mu_\nu)$. This proves positivity. □

For $\nu, q \in C(X,\mathbb{R})$ we write $D\mu_\nu[q]$ for the derivative of $t \mapsto \mu_{\nu+tq}$ at $t = 0$ in total variation, which exists by Proposition 28.5.2.

Proposition 28.5.3 *For $q, \nu, w \in C(X,\mathbb{R})$, write $\operatorname{osc}_X q = \sup_X q - \inf_X q$. Then*

$$D\mu_\nu[q] = m(Q_\nu q - q)\mu_\nu, \qquad \|D\mu_\nu[q]\|_{\mathrm{TV}} \leq \frac{m}{2}\operatorname{osc}_X q. \tag{28.10}$$

Consequently

$$\|\mu_w - \mu_\nu\|_{\mathrm{TV}} \leq 2\tanh\left(\frac{m}{4}\operatorname{osc}_X(w-\nu)\right). \tag{28.11}$$

In particular, let $\nu_m, w_m \in C(X,\mathbb{R})$ along a sequence of degrees with nonzero selected spaces. If $m\operatorname{osc}_X(w_m - \nu_m) \to 0$, then the two sequences of normalized kernel probabilities have the same weak, or total variation, limit whenever one of them converges in that topology.

Proof The formula for $D\mu_\nu[q]$ follows from Corollary 28.5.2, since N_m does not depend on the weight. For the bound, put $g = q - Q_\nu q$, $A = \{g > 0\}$ and $p = \mu_\nu(A)$. Since Q_ν preserves integrals, g has mean zero, and selfadjointness gives

$$\int_A g\,\mathrm{d}\mu_\nu = \langle q, (I - Q_\nu)\mathbf{1}_A\rangle_{L^2(\mu_\nu)}.$$

The function $(I - Q_v)\mathbf{1}_A$ is nonnegative on A and nonpositive on its complement, almost everywhere, and its integral is zero. Subtracting $\inf_X q$ from q does not change the pairing, which is therefore at most

$$\operatorname{osc}_X q\big(p - \langle \mathbf{1}_A, Q_v \mathbf{1}_A\rangle\big).$$

Positivity of Q_v, applied to $\mathbf{1}_A - p$, gives $\langle \mathbf{1}_A, Q_v \mathbf{1}_A\rangle \geq p^2$: since Q_v fixes constants and preserves integrals, $\langle \mathbf{1}_A - p, Q_v(\mathbf{1}_A - p)\rangle = \langle \mathbf{1}_A, Q_v \mathbf{1}_A\rangle - p^2$. As g has mean zero, it follows that

$$\|g\|_{L^1(\mu_v)} = 2\int_A g \, d\mu_v \leq 2 \operatorname{osc}_X q \; p(1-p) \leq \tfrac{1}{2} \operatorname{osc}_X q.$$

The total variation norm of a signed measure is the integral of the absolute value of its density, so $\|D\mu_v[q]\|_{\mathrm{TV}} = m\|g\|_{L^1(\mu_v)}$, and (28.10) follows.

For the global estimate, take $q = w - v$, $v_t = v + tq$, $\mu_t = \mu_{v_t}$ and $Q_t = Q_{v_t}$ for $0 \leq t \leq 1$, and put $a = m \operatorname{osc}_X q$. For a Borel set E, the function $p_t = \mu_t(E)$ is differentiable, because integration against $\mathbf{1}_E$ is a bounded linear functional for the total variation norm, and selfadjointness gives

$$p_t' = -m\langle q, (I - Q_t)\mathbf{1}_E\rangle_{L^2(\mu_t)}.$$

The function $(I - Q_t)\mathbf{1}_E$ has positive and negative parts whose integrals both equal $p_t - \langle \mathbf{1}_E, Q_t \mathbf{1}_E\rangle$. Bounding q above and below on the supports of these parts, and applying positivity to $\mathbf{1}_E - p_t$ as before, we obtain

$$|p_t'| \leq a\big(p_t - \langle \mathbf{1}_E, Q_t \mathbf{1}_E\rangle\big) \leq a p_t(1 - p_t).$$

All the kernels B_{v_t} are positive exactly off the common zero set Z_m, so the probabilities μ_t are mutually absolutely continuous. Hence, if p_0 is zero or one, then $p_t = p_0$ for all t. Otherwise $0 < p_t < 1$ for all t, and integration gives

$$\left|\log \frac{p_1}{1 - p_1} - \log \frac{p_0}{1 - p_0}\right| \leq a.$$

Let $\sigma(z) = \mathrm{e}^z/(1 + \mathrm{e}^z)$ be the logistic function. If $p_1 \geq p_0$ and $z = \log(p_0/(1 - p_0))$, then

$$p_1 - p_0 \leq \sigma(z + a) - \sigma(z) = \frac{\sinh(a/2)}{\cosh(z + a/2) + \cosh(a/2)} \leq \tanh(a/4).$$

The case $p_1 \leq p_0$ is symmetric. Since $\mu_w - \mu_v$ has total mass zero, its total variation norm is $2 \sup_E |\mu_w(E) - \mu_v(E)|$, which proves (28.11). Under the hypothesis of the last assertion the right-hand side tends to zero, which gives the final statement. □

Example 28.5.2 For each $a > 0$ and $m \geq 1$, the bound (28.11) is optimal as a supremum over smooth weights, already for a selected space of dimension one. Indeed, let $X = \mathbb{P}^1$ with the trivial line bundle and its trivial metric, take the selecting potential zero, and let ν be the Fubini–Study area normalized to mass one. Then there are smooth real weights w_j with $m \operatorname{osc}_X w_j = a$ such that

$$\|\mu_{w_j} - \mu_0\|_{\mathrm{TV}} \longrightarrow 2\tanh(a/4).$$

Proof The selected space consists of the constant functions, so the normalized kernel probability at a weight u is $\mathrm{e}^{-mu}\nu / \int_X \mathrm{e}^{-mu}\,\mathrm{d}\nu$. Choose a spherical cap E with $\nu(E) = p = (1 + \mathrm{e}^{a/2})^{-1}$; its boundary circle has zero area. Choose smooth functions $0 \leq \chi_j \leq 1$, equal to one on an inner cap and zero outside an outer cap, converging almost everywhere to $\mathbf{1}_E$; each has oscillation one. Put $w_j = -a\chi_j/m$. The densities $\mathrm{e}^{a\chi_j}$ converge boundedly almost everywhere, and their normalizing constants converge to a positive limit, so μ_{w_j} converges in total variation to

$$\frac{\mathrm{e}^{a\mathbf{1}_E}}{Z}\nu, \qquad Z = 1 - p + \mathrm{e}^a p = \mathrm{e}^{a/2}.$$

This limit gives E the mass $1 - p$, and its density with respect to ν is greater than one on E and less than one outside E. Its total variation distance from $\nu = \mu_0$ is therefore $2(1 - 2p) = 2\tanh(a/4)$. The triangle inequality transfers this limit to the measures μ_{w_j}. This shows that the bound is optimal as a supremum; we do not claim that it is attained by a smooth weight. □

At the scale of perturbations of size $1/m$, the analytic dependence on the weight is also controlled independently of the dimension of the selected space.

Proposition 28.5.4 *Put $\rho = \log(3/2)$. The map $f \mapsto \mu_{v+f/m}$ extends holomorphically to the ball $\{f \in C(X, \mathbb{C}) : \|f\|_\infty < \rho\}$, with values in complex measures of total variation at most three. For real $q \in C(X)$ and $k \geq 1$,*

$$\left\| \frac{\mathrm{d}^k}{\mathrm{d}t^k}\bigg|_{t=0} \mu_{v+tq} \right\|_{\mathrm{TV}} \leq 3k! \left(\frac{m \operatorname{osc}_X q}{2\rho} \right)^k . \tag{28.12}$$

Proof On V_m use $T(f) = P_v(\mathrm{e}^{-f}\,\cdot\,)$, the operator in the proof of Proposition 28.5.2 for the perturbation f/m of the weight v. That proof gives $\|T(f)^{-1}\| \leq 2$ on the ball. The extended measure has the density

$$\frac{\mathrm{e}^{-mv(x)-f(x)}}{N_m} \ell_x\big(T(f)^{-1} k_{v,x}\big)$$

with respect to ν. By the Riesz identity at the weight v, its absolute value is at most $2\mathrm{e}^{\|f\|_\infty} B_v(x)/N_m \leq 3B_v(x)/N_m$, so its total variation is at most three. Holomorphy follows from the same finite-dimensional construction and the bounded linear passage from densities to measures. For a complex constant c we have $T(f + c) = \mathrm{e}^{-c}T(f)$, and the resulting factor e^c cancels the factor e^{-c} in the density; thus adding constants

to f does not change the extension. For nonconstant real q, replace q by $q - c$ with $c = (\sup_X q + \inf_X q)/2$, so that $\|q - c\|_\infty = \operatorname{osc}_X q/2$. Then $t \mapsto \mu_{\nu+tq}$ extends holomorphically to the disk $|t| < \rho/(m\|q - c\|_\infty)$, with total variation at most three. The Cauchy estimates for Banach-valued holomorphic functions, applied on circles of radius less than $\rho/(m\|q - c\|_\infty)$, followed by letting the radius increase to that value, prove (28.12). If q is constant, the measure does not depend on t, and all derivatives vanish. □

The hypothesis $m \operatorname{osc}_X(w_m - \nu_m) \to 0$ in Proposition 28.5.3 cannot in general be replaced by uniform convergence $w_m - \nu_m \to 0$: a perturbation of order $1/m$ along the fibers of a product survives in a nonzero measure limit.

Example 28.5.3 Let Y be a connected compact complex manifold of positive dimension with a smooth positive probability volume ν_Y. On $X = \mathbb{P}^1 \times Y$, use the pullback of $O(1)$ with its Fubini–Study metric, the selecting potential zero, and the volume $\nu = \nu_{\mathrm{FS}} \otimes \nu_Y$, where ν_{FS} is the Fubini–Study area of mass one. Choose a nonconstant $q \in C(Y, \mathbb{R})$ and put

$$\nu_m(b, y) = q(y)/m, \qquad Z_q = \int_Y \mathrm{e}^{-q}\, \mathrm{d}\nu_Y, \qquad \nu_q = Z_q^{-1}\mathrm{e}^{-q}\nu_Y.$$

Then $N_m = m + 1$ and, for every m, the kernel and the normalized kernel probability at the weight ν_m are

$$B_{m,\nu_m}(b, y) = (m + 1)\frac{\mathrm{e}^{-q(y)}}{Z_q}, \qquad \mu_{m,\nu_m} = \nu_{\mathrm{FS}} \otimes \nu_q. \tag{28.13}$$

Thus $\nu_m \to 0$ uniformly, but the kernel measures with $r = 1$ converge in total variation to $\nu_{\mathrm{FS}} \otimes \nu_q$, whereas at the weight zero the limit is $\nu_{\mathrm{FS}} \otimes \nu_Y$. In both cases the potentials $m^{-1} \log B_{m,\nu}$ tend uniformly to zero. With the normalization by the dimension of X, both sequences of kernel measures tend to zero in total variation.

Proof Every section descends to $\mathbb{P}^1$, because the fibers $\{b\} \times Y$ are compact and connected, as in Proposition 28.7.1. Hence the selected space is $H^0(\mathbb{P}^1, O(m))$, of dimension $m + 1$. Its unweighted Bergman kernel for ν_{FS} is constant, since the unitary group acts transitively on $\mathbb{P}^1$ and preserves the metric and the volume; its integral $m + 1$ determines the constant. For a section pulled back from $\mathbb{P}^1$, the norm on X at the weight ν_m is Z_q times the norm on $\mathbb{P}^1$, which proves (28.13). At the weight zero the normalized kernel probability is $\nu_{\mathrm{FS}} \otimes \nu_Y$. These probabilities differ: equality would force $\mathrm{e}^{-q}/Z_q = 1$ almost everywhere, and then everywhere, because q is continuous and every nonempty open set has positive volume; this contradicts the choice of q. The kernel measures with $r = 1$ are $(m + 1)/m$ times these probabilities, and the normalization in dimension $1 + \dim Y \geq 2$ gives measures tending to zero in total variation. Finally, $m^{-1} \log B_{m,\nu_m} = m^{-1}\bigl(\log(m + 1) - q - \log Z_q\bigr)$ and $m^{-1} \log B_{m,0} = m^{-1} \log(m + 1)$ both tend uniformly to zero. □

The same phenomenon, in which a fiber probability survives in the limit, reappears in Section 28.11, where the limit measure on the total space carries the conditional fiber probabilities of the integration volume.

28.6 Partial kernels under a modification

A modification identifies holomorphic sections, but it changes the integrability condition that defines multiplier ideals; the correction is a twist by the relative canonical bundle, through the Jacobian. If the relative canonical bundle carries the metric determined by two volume forms, the same Jacobian factor makes the identification of section spaces an isometry. The partial kernel measures downstairs are then exactly the pushforwards of those upstairs, and the convergence theorem on compact Kähler manifolds descends to every compact complex manifold that admits a Kähler modification.

Let $\pi\colon Y \to X$ be a modification of connected compact complex manifolds. We use the exceptional set $S \subset X$, its inverse image $D = \pi^{-1}(S) \subset Y$, the relative canonical bundle R_π and its Jacobian section J_π from Section 1.22.7, together with the transformation of multiplier ideals in Lemma 1.22.10 and the invariance of the multiplier-ideal order in Lemma 1.22.11.

Proposition 28.6.1 *Let θ be a smooth real $(1,1)$-form on X and let $\varphi \in \mathrm{PSH}(X,\theta)$. For $v \in C(X)$ let $E_X(v) = E_{\theta,X}[\varphi]_{\mathcal{I}}(v)$ be the partial envelope with obstacle v on the whole of X. Then*

$$E_{\pi^*\theta,Y}[\varphi\circ\pi]_{\mathcal{I}}(v\circ\pi) = E_X(v)\circ\pi.$$

Moreover, for every $u \in \mathrm{PSH}(X,\theta)$ the extended non-pluripolar measures satisfy

$$\pi_*\mathrm{MA}^{\mathrm{ext}}_{\pi^*\theta}(u\circ\pi) = \mathrm{MA}^{\mathrm{ext}}_{\theta}(u).$$

Proof By Proposition 1.22.4, $a \mapsto a\circ\pi$ is a bijection from $\mathrm{PSH}(X,\theta)$ onto $\mathrm{PSH}(Y,\pi^*\theta)$. For a continuous obstacle on the whole manifold, the quasi-everywhere and the pointwise obstacle conditions are equivalent, as recalled in Section 1.19, and since π is surjective, $a \le v$ on X if and only if $a\circ\pi \le v\circ\pi$ on Y. Together with Lemma 1.22.11, this identifies the two families of competitors. By Theorem 28.3.1, each partial envelope is the largest member of its family, which proves the envelope identity; no interchange of pullback and regularized supremum is needed.

The map π is a biholomorphism from $Y\setminus D$ onto $X\setminus S$, so locality of the non-pluripolar product identifies the two measures there. Neither measure charges the analytic sets D and S. Testing against nonnegative Borel functions, or exhausting the complements by increasing compact sets, proves the pushforward identity, including the case of infinite total mass. □

Choose smooth positive probability volumes ν_X on X and ν_Y on Y. They determine a smooth Hermitian metric h_R on R_π such that

$$|J_\pi|^2_{h_R}\,\nu_Y = \pi^*\nu_X. \tag{28.14}$$

Explicitly, let f_X and f_Y be the densities of ν_X and ν_Y in local coordinates. The local frame of R_π given by the quotient of the coordinate canonical frames is given the squared norm $(f_X \circ \pi)/f_Y$. This function is smooth and strictly positive, also where J_π vanishes, and the transformation rules of the coordinate densities show that it defines a metric. Equation (28.14) is an identity of smooth top-degree forms; it does not define the pullback of an arbitrary measure.

Proposition 28.6.2 *Let the line bundle L with metric h, the selecting potential $\varphi \in \mathrm{QPSH}(X)$ and the continuous weight $v \in C(X)$ be as in Section 28.1, and use the volume ν_X on X. On Y, use the line bundles $\pi^*L^m \otimes R_\pi$ with the metrics $\pi^*h^m \otimes h_R$, the selecting potential $\varphi \circ \pi$, the continuous weight $v \circ \pi$ and the volume ν_Y, and define the kernels and kernel measures as in Section 28.1, with the fixed factor (R_π, h_R) added. Then the identification $s \mapsto J_\pi\pi^*s$,*

$$H^0(X, L^m \otimes \mathcal{I}(m\varphi)) \simeq H^0(Y, \pi^*L^m \otimes R_\pi \otimes \mathcal{I}(m\varphi \circ \pi)),$$

is an isometry for the continuous weighted norms. The kernels and kernel measures on Y and X satisfy

$$B_{Y,m} = |J_\pi|^2_{h_R}(B_{X,m} \circ \pi), \qquad \pi_*\beta^{(r)}_{Y,m} = \beta^{(r)}_{X,m}$$

for the same normalization integer r, also when the section spaces are zero.

Proof The identification is Lemma 1.22.10 with $M = L^m$ and $a = m\varphi$. By (28.14) and the change of variables formula,

$$\int_Y |J_\pi\pi^*s|^2_{\pi^*h^m \otimes h_R} \mathrm{e}^{-mv\circ\pi}\,\mathrm{d}\nu_Y = \int_X |s|^2_{h^m}\mathrm{e}^{-mv}\,\mathrm{d}\nu_X.$$

The selecting potential has already determined the section spaces and does not enter the norms. Hence the identification maps an orthonormal basis to an orthonormal basis, and summing the squared pointwise norms of the basis vectors gives the kernel formula, also at the zeros of J_π. Multiplying by $r!/m^r$ and integrating a continuous function pulled back from X proves the measure formula. □

28.6.1 Convergence with a Kähler modification

Through the Jacobian isometry, the equilibrium theorem on a compact Kähler manifold can be pushed down along a modification. The limit does not depend on the modification, because both the partial envelope and its non-pluripolar Monge–Ampère measure descend.

Theorem 28.6.1 *Suppose that X admits a modification from a compact Kähler manifold. Let $L \to X$ be a holomorphic line bundle with a smooth Hermitian metric h, put $\theta = c_1(L,h)$, and let $\varphi \in \mathrm{PSH}(X,\theta)$ satisfy*

$$\mathrm{MA}_\theta(\varphi)(X) > 0.$$

Then, for every $v \in C(X)$ and every smooth positive probability volume ν_X, the partial kernel measures on X satisfy

$$\frac{n!}{m^n} B_{m,v,\varphi,\nu_X}\, \nu_X \rightharpoonup \mathrm{MA}_\theta(E_X(v)).$$

The limit measure is finite and nonzero. Its total mass is

$$M = \lim_{m\to\infty} \frac{n!}{m^n} h^0(X, L^m \otimes \mathcal{I}(m\varphi)).$$

Proof Choose a modification $\pi\colon Y \to X$ from a compact Kähler manifold Y, and a smooth positive probability volume ν_Y on Y. The pullback $\varphi \circ \pi$ belongs to $\mathrm{PSH}(Y, \pi^*\theta)$, and by Proposition 28.6.1 its non-pluripolar mass equals the given positive mass of φ. This mass is finite, by the theory of closed classes on compact Kähler manifolds.

Apply [Xia26, Theorem 14.3.1] on Y, with the line bundle π^*L, the auxiliary twisting line bundle R_π, the continuous obstacle $v \circ \pi$ on Y and the probability volume ν_Y. Its Bernstein–Markov hypothesis follows from the proof of Proposition 28.2.1, since a fixed smooth auxiliary bundle only changes the local constants. The multiplier ideals and the continuous weighted norms in that theorem agree exactly with those of Proposition 28.6.2. The theorem gives

$$\beta^{(n)}_{Y,m} \rightharpoonup \mathrm{MA}_{\pi^*\theta}\big(E_{\pi^*\theta,Y}[\varphi\circ\pi]_{\mathcal{I}}(v\circ\pi)\big),$$

and the limit measure has finite positive mass.

Now push this convergence forward to X, testing against continuous functions on X. By the kernel identity of Proposition 28.6.2 and the envelope and pushforward identities of Proposition 28.6.1, this gives the asserted limit on X. Testing the constant function one shows that $n!N_m/m^n$ converges to the total mass of the limit, which proves the formula for M. □

Since $M > 0$, the sections of L grow at the maximal rate m^n, so X is Moishezon by Corollary 22.5.1. A Kähler modification also forces the bounded mass property, by Lemma 7.5.1; the Jacobian identities, by contrast, hold for every modification of

compact complex manifolds. When the section spaces grow more slowly than m^n, the total mass identity shows that the kernel measures must instead be normalized by the growth degree of N_m, as in Section 28.7.

Corollary 28.6.1 *Under the hypotheses of Theorem 28.6.1, let M be the limit of the normalized dimensions in that theorem. The determinant functionals $\mathcal{F}_m^{(n)}$ of Section 28.5 converge pointwise on $C(X)$ to*

$$\mathcal{F}(v) = \int_0^1 \left(\int_X v \, \mathrm{dMA}_\theta(E_X(tv)) \right) \mathrm{d}t.$$

This functional is concave and vanishes at zero, and for $v, w \in C(X, \mathbb{R})$ and $c \in \mathbb{R}$,

$$\mathcal{F}(v + c) = \mathcal{F}(v) + Mc, \qquad |\mathcal{F}(v) - \mathcal{F}(w)| \le M \|v - w\|_\infty.$$

Proof Integrate the first-variation formula of Proposition 28.5.1 along the segment $t \mapsto tv$, $0 \le t \le 1$. For each fixed t, the integrand converges by the theorem, and its absolute value is bounded by

$$\|v\|_\infty \frac{n! N_m}{m^n},$$

which is bounded in m because it converges to $M\|v\|_\infty$. Dominated convergence on $[0, 1]$ gives the formula; the limit integrand is measurable in t as the pointwise limit of continuous functions. Concavity, the translation identity and the Lipschitz bound pass from $\mathcal{F}_m^{(n)}$ to the pointwise limit, since $n! N_m / m^n \to M$. □

This argument uses only weak convergence of measures along a segment of weights; it gives no lower bound for the partial Fubini–Study potentials complementing Theorem 28.4.1.

28.7 Kernels on a fibration

On a fibration with compact connected fibers, every section of a pulled-back line bundle is constant along the fibers and descends to the base. If probability measures are chosen on the fibers, this descent becomes an isometry, and the kernel measures on the total space are obtained from those on the base by integration along the fibers.

Let $\pi \colon X \to B$ be a surjective holomorphic submersion of connected compact complex manifolds, with connected fibers. Write $n = \dim X$ and $q = \dim B \ge 1$. Let A be a holomorphic line bundle on B with a smooth Hermitian metric h_A, choose a quasi-plurisubharmonic function ψ on B, and set

$$(L, h) = (\pi^* A, \pi^* h_A), \qquad \varphi = \pi^* \psi.$$

For $y \in B$ write $X_y = \pi^{-1}(y)$. Choose a family of smooth positive probability measures ν_y on the fibers X_y that varies smoothly with y; for instance, restrict a smooth positive vertical volume density to the fibers and divide by its fiber integrals. For $F \in C(X)$ set

$$\mathcal{A}F(y) = \int_{X_y} F \, \mathrm{d}\nu_y .$$

The function $\mathcal{A}F$ is continuous: a proper submersion is locally a smooth fiber bundle, the densities vary smoothly in a local trivialization, and F is uniformly continuous on compact sets. For a finite positive Borel measure μ on B, define its *lift* $\mathcal{L}\mu$ by

$$\int_X F \, \mathrm{d}\mathcal{L}\mu = \int_B \mathcal{A}F \, \mathrm{d}\mu \qquad (F \in C(X)).$$

Since $\mathcal{A}$ is positive and $\mathcal{A}1 = 1$, the Riesz representation theorem makes $\mathcal{L}\mu$ a finite positive measure with the same total mass as μ. Since $\mathcal{A}F$ is continuous, weak convergence of finite measures on B implies weak convergence of their lifts.

Proposition 28.7.1 *Pullback of sections induces an isomorphism*

$$H^0\big(B, A^m \otimes \mathcal{I}(m\psi)\big) \simeq H^0\big(X, L^m \otimes \mathcal{I}(m\varphi)\big).$$

Let $K \subseteq B$ be closed and non-pluripolar, let $w \in C(K)$, and let μ be a Bernstein–Markov probability measure for (K, w) on B. Set $K_X = \pi^{-1}(K)$, $w_X = w \circ \pi$ and $\mu_X = \mathcal{L}\mu$. Then pullback is an isometry from the norm $\|\cdot\|_{m,w,\mu}$ on B to the norm $\|\cdot\|_{m,w_X,\mu_X}$ on X, the measure μ_X is Bernstein–Markov for (K_X, w_X), and the partial kernel measures on B and X, normalized with the same integer r, satisfy

$$\beta^{(r)}_{X,m} = \mathcal{L}\big(\beta^{(r)}_{B,m}\big).$$

Proof Let s be a holomorphic section of $\pi^* A^m$. Over an open set of B on which A^m is trivial, the coefficient of s is a holomorphic function on the inverse image, and it is constant on each fiber, since the fibers are compact and connected. It is therefore the pullback of a function on the base, which is holomorphic because it is the composition of the coefficient with a local holomorphic section of the submersion. These local functions transform by the transition functions of A^m, so they define a section of A^m whose pullback is s. This proves the identification without multiplier ideals.

By Lemma 1.16.5, the multiplier ideal $\mathcal{I}(m\varphi)$ on X is generated by the pullbacks of the germs in $\mathcal{I}(m\psi)$. A pulled-back germ belongs to it exactly when the base germ belongs to $\mathcal{I}(m\psi)$: pullback gives one implication, and restriction to a local holomorphic section of π gives the other. This identifies the selected spaces.

Pointwise norms of corresponding sections agree and are constant along the fibers. Since each ν_y has mass one,

$$\int_{K_X} |\pi^* s|^2_{h^m} \mathrm{e}^{-mw_X} \, \mathrm{d}\mu_X = \int_K |s|^2_{h^m_A} \mathrm{e}^{-mw} \, \mathrm{d}\mu .$$

The supremum norms on K_X and K also agree, because π is surjective. Thus the Bernstein–Markov inequality transfers from (K, w, μ) to (K_X, w_X, μ_X), and the integral norms on X are nondegenerate. Pulling back an orthonormal basis gives $B_{X,m} = B_{B,m} \circ \pi$ on K_X, where each kernel is set to zero outside its weighted set. Integrating a continuous test function first along the fibers proves the identity of measures. □

28.7.1 The quotient equilibrium measure

On a Kähler base, the partial kernel measures converge to the Monge–Ampère measure of the partial envelope. Since lifting is continuous, the kernel measures on the total space converge to the lift of this measure, with the normalization given by the dimension of the base.

Keep the submersion and the weighted compact set (K, w, μ) above. Assume that B is Kähler, put $\theta = c_1(A, h_A)$, and take $\psi \in \mathrm{PSH}(B, \theta)$ with positive non-pluripolar Monge–Ampère mass. Write $B_{B,m}$ and $B_{X,m}$ for the partial Bergman kernels on K and K_X, put $\mu_X = \mathcal{L}\mu$, and let

$$E = E_{\theta,K}[\psi]_I(w)$$

be the partial envelope of Section 28.3. We call $\mathcal{L}(\mathrm{MA}_\theta(E))$ the *quotient equilibrium measure* of these data on X.

Theorem 28.7.1 *Under these hypotheses,*

$$\frac{q!}{m^q} B_{X,m}\,\mu_X \rightharpoonup \mathcal{L}\big(\mathrm{MA}_\theta(E)\big).$$

If $n > q$, then $(n!/m^n)B_{X,m}\,\mu_X \to 0$ in total variation.

Proof Apply [Xia26, Theorem 14.3.1] on the compact Kähler manifold B, with trivial auxiliary twisting line bundle. Its hypotheses are exactly our assumptions: ψ has positive mass, K is closed and non-pluripolar, w is continuous, and μ is a Bernstein–Markov probability measure. It gives

$$\frac{q!}{m^q} B_{B,m}\,\mu \rightharpoonup \mathrm{MA}_\theta(E).$$

By Proposition 28.7.1, the corresponding measures on X are the lifts of these measures, and lifting preserves weak convergence; this proves the first assertion. In particular the q-normalized total masses are bounded, and multiplying them by $(n!/q!)m^{q-n} \to 0$ proves the last assertion when $n > q$. □

28.7.2 The Hopf equilibrium measure

On the Hopf manifold, the quotient equilibrium measure is a Hermitian Monge–Ampère measure: the vertical part of the Hopf metric supplies the missing fiber direction. The constant relating the two measures combines the fiber probability with a coefficient of the top power.

Use the Hopf manifold X_n, $n \geq 2$, its projection $\pi\colon X_n \to \mathbb{P}^{n-1}$, the Fubini–Study form σ on $\mathbb{P}^{n-1}$ with $\int \sigma^{n-1} = 1$, and the forms $\alpha = \pi^*\sigma$, ζ and $\omega = \alpha + \zeta$ of Chapter 23; recall that $\zeta^2 = 0$ and that ζ has integral two on each fiber. Take $A = O(1)$ with its Fubini–Study metric, so that $\theta = \sigma$, and keep the selecting potential ψ and the weighted data (K, w, μ) above. On each elliptic fiber put $\nu_y = \zeta|_{X_y}/2$. The lift of a base measure τ is then

$$\mathcal{L}\tau = \tfrac{1}{2}\pi^*\tau \wedge \zeta,$$

in the fiber-integration notation of Chapter 23. Since $E \in \mathrm{PSH}(\mathbb{P}^{n-1}, \sigma)$, Lemma 9.6.1 gives $\mathcal{L}(\mathrm{MA}_\sigma(E)) = (2n)^{-1}\mathrm{MA}^{\mathrm{ext}}_\omega(\pi^*E)$, and Theorem 28.7.1 becomes

$$\frac{(n-1)!}{m^{n-1}} B_{X_n,m}\,\mu_X \rightharpoonup \frac{1}{2n}\mathrm{MA}^{\mathrm{ext}}_\omega(\pi^*E). \tag{28.15}$$

The factor $1/(2n)$ combines the fiber probability $1/2$ with the coefficient n of $\alpha^{n-1}\wedge\zeta$ in $\omega^n = (\alpha+\zeta)^n$. The closed curvature form α alone gives nothing: by Lemma 9.6.1, descended potentials have zero Monge–Ampère measure with respect to α, and the nonzero limit comes from the vertical form ζ. For $n = 3$ this gives a nonzero limit, with the normalization $2/m^2$, on a manifold without the bounded mass property (Theorem 9.0.1).

For a positive integer k, every $k\alpha$-plurisubharmonic function on X_n is pulled back from $\mathbb{P}^{n-1}$, by Theorem 23.1.1. An ω-plurisubharmonic function, by contrast, can have an isolated pole, and then it does not descend. Its selected sections still descend, so the pole enters only through a vanishing condition on the base. The next two sections determine this condition and its effect on the kernels.

28.8 Isolated logarithmic poles on a fibration

Return to a fibration with compact connected fibers, and let the selecting potential have isolated logarithmic poles on the total space. Every selected section descends to the base, and in large tensor powers it must vanish to a certain order at the images of the poles. This order is computed from integrability in all variables of the total space, so it differs from the order imposed by a pole of the same coefficient on the base by at most the fiber dimension. The jets in between span $O(m^{q-1})$ dimensions on a q-dimensional base. This is negligible after normalization by m^q, so the kernel measures converge to the same quotient equilibrium measure as for the corresponding poles on the base.

The comparison rests on the following elementary fact: for nested section spaces, the total variation of the difference of their kernel measures is the normalized codimension.

Lemma 28.8.1 *Let B be a compact complex manifold of dimension q, let $A \to B$ be a holomorphic line bundle with a smooth Hermitian metric, let $\nu \in C(B)$, and let ν be a smooth positive probability volume on B. For every integer $m \geq 1$ let $W_m \subseteq V_m \subseteq H^0(B, A^m)$ be subspaces, both with the norm $\|\cdot\|_{m,\nu,\nu}$, and put $d_m = \dim V_m - \dim W_m$. Write $B_{V_m,\nu}$, $B_{W_m,\nu}$ for their kernels and $\beta^{(q)}_{V_m,\nu}$, $\beta^{(q)}_{W_m,\nu}$ for their kernel measures, defined as in Section 28.1 with the space indicated by the subscript. Then*

$$B_{V_m,\nu} - B_{W_m,\nu} \geq 0,$$

$$\|\beta^{(q)}_{V_m,\nu} - \beta^{(q)}_{W_m,\nu}\|_{\mathrm{TV}} = \frac{q!\,d_m}{m^q}.$$

The determinant functionals of the two spaces, each normalized to vanish at the zero weight, satisfy

$$|\mathcal{F}^{(q)}_{V_m}(\nu) - \mathcal{F}^{(q)}_{W_m}(\nu)| \leq \frac{q!\,d_m}{m^q}\|\nu\|_\infty.$$

These statements include the case of zero-dimensional spaces.

Proof Extend an orthonormal basis of W_m to an orthonormal basis of V_m for the norm at the weight ν. The difference of the kernels is the sum of the squared weighted evaluations of the d_m added basis vectors. It is nonnegative, and its integral against ν is d_m, which gives the total variation formula. For the determinant functionals, apply the first-variation formula of Proposition 28.5.1 to each of the two fixed spaces, and integrate the difference along $t \mapsto t\nu$, $0 \leq t \leq 1$. By what we have just proved, at each t the difference of the kernel measures is positive with total mass $q!\,d_m/m^q$, so its integral against ν is at most $q!\,d_m\|\nu\|_\infty/m^q$ in absolute value. Since both functionals vanish at zero, this proves the last assertion. □

Return to a submersion $\pi\colon X \to B$ and a pulled-back line bundle $L = \pi^*A$ as in Section 28.7. Equip A with a smooth metric h_A and put $\theta = c_1(A, h_A)$. Choose a smooth positive probability volume ν_B on B, smooth fiber probabilities as in Section 28.7, and the smooth positive volume $\nu_X = \mathcal{L}\nu_B$ on X. Use the pullback metric on L, and for $\nu \in C(B)$ use the weight $\nu \circ \pi$ in the norm on X. For a selecting potential $\varphi \in \mathrm{QPSH}(X)$, we write $\beta^{(q)}_{X,m,\nu\circ\pi,\varphi}$ and $\mathcal{F}^{(q)}_{X,m}$ for the kernel measure and the determinant functional on X defined by these data, and we regard the selected spaces, by descent as in Proposition 28.7.1, as subspaces

$$V_m = H^0\big(X, L^m \otimes \mathcal{I}(m\varphi)\big) \subseteq H^0(B, A^m).$$

The following proposition reduces kernel limits for φ to the equilibrium theorem on the base, once the selected spaces are close to those of a potential on the base.

Proposition 28.8.1 *Assume that B is Kähler, let $\psi \in \mathrm{PSH}(B,\theta)$ have positive non-pluripolar mass, and suppose that*

$$W_m = H^0\big(B, A^m \otimes \mathcal{I}(m\psi)\big) \subseteq V_m, \qquad d_m = \dim V_m - \dim W_m = o(m^q).$$

Let $E(v) = E_{\theta,B}[\psi]_{\mathcal{I}}(v)$ for $v \in C(B)$. Then, for every $v \in C(B)$, the total variation distance between $\beta^{(q)}_{X,m,v\circ\pi,\varphi}$ and the lift of the kernel measure of W_m on B is $q!\,d_m/m^q$, and

$$\beta^{(q)}_{X,m,v\circ\pi,\varphi} \rightharpoonup \mathcal{L}\big(\mathrm{MA}_\theta(E(v))\big), \qquad \mathcal{F}^{(q)}_{X,m}(v\circ\pi) \longrightarrow \int_0^1 \left(\int_B v\,\mathrm{dMA}_\theta(E(tv))\right)\mathrm{d}t.$$

The normalized dimensions $q!N_m/m^q$ converge to the total mass of the limit measure, which is finite and positive. If $n > q$, the kernel measures with the normalization $n!/m^n$ tend to zero in total variation.

Proof For $s \in V_m$, the pointwise norm of π^*s and the weight $v\circ\pi$ are constant on the fibers, and every fiber probability has mass one, so the norm of π^*s on X equals the norm of s on B. As in Proposition 28.7.1, pulling back an orthonormal basis shows that $\beta^{(q)}_{X,m,v\circ\pi,\varphi}$ is the lift of the kernel measure of V_m on B, and the equality of the Gram matrices shows that $\mathcal{F}^{(q)}_{X,m}(v\circ\pi)$ is the determinant functional of V_m at v. By Lemma 28.8.1, the kernel measures of V_m and W_m differ by a positive measure of total mass $q!\,d_m/m^q$, and so do their lifts, since lifting preserves positivity and mass; the same lemma bounds the difference of the determinant functionals by $q!\,d_m\|v\|_\infty/m^q$. Apply Theorem 28.6.1 and Corollary 28.6.1 on B, with the identity as Kähler modification, to the spaces W_m selected by ψ. They give the convergence of the kernel measures and determinant functionals of W_m, and the total mass of the limit measure is finite and positive and equals the limit of $q!\dim W_m/m^q$. Lifting preserves weak convergence and $d_m = o(m^q)$, which proves the two limits and the assertion about the dimensions. Since the q-normalized masses are bounded, multiplying them by $(n!/q!)m^{q-n}$ proves the last assertion. □

We now let the selecting potential have isolated poles. Write $f = n - q$ for the fiber dimension. Fix distinct points $p_1,\dots,p_N$ of X and positive real numbers $a_1,\dots,a_N$. Suppose that $\varphi \in \mathrm{QPSH}(X)$ is locally bounded away from these points and that, in local coordinates x centered at p_j,

$$\varphi(x) = a_j \log|x|^2 + O(1). \tag{28.16}$$

Write

$$Z = \{\pi(p_1),\dots,\pi(p_N)\}, \qquad a_b = \max_{\pi(p_j)=b} a_j \quad (b \in Z),$$

and, for an integer $m \geq 1$,

$$r_{m,b} = \max\{\lfloor ma_b\rfloor - n + 1, 0\}, \qquad s_{m,b} = \max\{\lfloor ma_b\rfloor - q + 1, 0\}.$$

Let $\mathfrak{m}_b$ denote the ideal sheaf of a point $b \in B$; its zeroth power is $\mathcal{O}_B$.

Proposition 28.8.2 *The selected space of φ is*

$$V_m = H^0\left(B, A^m \otimes \bigcap_{b\in Z} \mathfrak{m}_b^{r_{m,b}}\right).$$

The space

$$W_m = H^0\left(B, A^m \otimes \bigcap_{b\in Z} \mathfrak{m}_b^{s_{m,b}}\right)$$

satisfies $W_m \subseteq V_m$ and

$$0 \le \dim V_m - \dim W_m \le \sum_{b\in Z} \sum_{j=r_{m,b}}^{s_{m,b}-1} \binom{j+q-1}{q-1} = O(m^{q-1}),$$

where an empty inner sum is zero. The implied constant depends only on the points and their coefficients, not on m.

Proof Lemma 1.16.1, applied with a point as center, gives

$$\mathcal{I}(m\varphi)_{p_j} = \mathfrak{m}_{p_j}^{\max\{\lfloor ma_j\rfloor - n+1,\,0\}};$$

away from the points p_j the ideal is the full structure sheaf.

Near p_j choose submersion coordinates (z,τ) with $\pi(z,\tau) = z$. A pulled-back section has a local coefficient $g(z)$ that does not depend on τ, and its vanishing order at $(0,0)$ equals the vanishing order of g at $z = 0$. The conditions imposed by the points of one fiber are therefore vanishing conditions at the same base point, and the strongest of them is vanishing to order $r_{m,b}$. This proves the identification.

For the dimension bound, map each section in V_m to its jets modulo $\mathfrak{m}_b^{s_{m,b}}$ at the finitely many points $b \in Z$. The kernel of this map is W_m, so local trivializations of A^m give an injection

$$V_m/W_m \hookrightarrow \bigoplus_{b\in Z} \mathfrak{m}_b^{r_{m,b}}/\mathfrak{m}_b^{s_{m,b}}.$$

In q variables there are $\binom{j+q-1}{q-1}$ monomials of degree j, which gives the dimensions of these quotients. Finally $0 \le s_{m,b} - r_{m,b} \le f$ and $s_{m,b} = O(m)$, which proves the bound, also when $q = 1$ or $f = 0$. □

Theorem 28.8.1 *Assume that B is Kähler and that there is $\psi \in \mathrm{PSH}(B,\theta)$ with positive non-pluripolar mass, locally bounded off Z, such that*

$$\psi(z) = a_b \log|z-b|^2 + O(1) \quad \text{near each } b \in Z$$

in local coordinates z on B. Then the selecting potential φ of (28.16) satisfies the conclusions of Proposition 28.8.1, and the total variation distance there is $O(m^{-1})$. In particular,

$$\beta^{(q)}_{X,m,v\circ\pi,\varphi} \rightharpoonup \mathcal{L}\big(\mathrm{MA}_\theta(E_{\theta,B}[\psi]_{\mathcal{I}}(v))\big) \qquad (v \in C(B)).$$

Proof By Lemma 1.16.1 in dimension q, the space of sections of A^m selected by ψ on B is the space W_m of Proposition 28.8.2. That proposition gives $W_m \subseteq V_m$ and $\dim V_m - \dim W_m = O(m^{q-1})$, so Proposition 28.8.1 applies. □

If $\theta > 0$, Lemma 1.16.2, applied on B with the Hermitian form θ, constructs the required ψ whenever the coefficients a_b are sufficiently small. Off the poles, $\theta + \mathrm{dd^c}\psi$ then dominates a positive multiple of θ, so the non-pluripolar mass of ψ is positive; it is finite because θ is closed and B is Kähler. The same lemma, applied on X with a Hermitian form, constructs φ. If the fibers have positive dimension, such a function φ does not descend to B, since it is finite at the other points of a fiber containing a pole.

The section formula of Proposition 28.8.2 also shows that the kernels cannot recover the singularities of the selecting potential. Moving an isolated pole within its fiber does not change any selected space, and neither does adding further poles in that fiber whose coefficients do not exceed the largest one. For the fixed pullback metric $h = \pi^* h_A$, continuous weight $v \circ \pi$ and lifted volume v_X, all kernels and determinant functionals are therefore the same in every degree. The multiplier ideals on X are nevertheless different: for large m, the ideal $\mathcal{I}(m\varphi)$ of one potential is a proper ideal at a point where the corresponding ideal of the other is the full local ring. Under the hypotheses of Theorem 28.8.1, this loss of information occurs together with a positive limit at the growth rate m^q. In the closed theory, by contrast, the limit envelope has the multiplier-ideal type of the selecting potential (Theorem 28.3.1); this property is lost when the selecting potential is plurisubharmonic only with respect to a Hermitian form.

28.9 A point pole on the Hopf manifold

On the Hopf manifold the comparison of Section 28.8 can be made completely explicit. A single point pole selects the homogeneous polynomials whose degree in affine coordinates centered at the image point is bounded below, and summing the squared norms of the normalized monomials expresses the kernel as a binomial tail. The exponential rate of this tail determines the limit of the Fubini–Study potentials, and its concentration determines the limit of the measures. The limit potential has

a pole along the entire fiber through the original pole, which makes the loss of the point singularity explicit.

We keep the notation of Section 28.7.2, with $q = n-1$, so that $\pi\colon X_n \to \mathbb{P}^q$. Use $L = L_1 = \pi^* O(1)$ with its pulled-back Fubini–Study metric and the probability volume $\nu_X = \omega^n/(2n)$, which is the lift of σ^q for the fiber probabilities $\zeta|_{X_y}/2$. Let $p \in X_n$ lie over $b = [1:0:\cdots:0]$. Suppose that φ has a single logarithmic pole at p, with coefficient $0 < a < 1$, as in (28.16), and is locally bounded elsewhere. The continuous weight is zero. In homogeneous coordinates $[Z_0 : \cdots : Z_q]$ on $\mathbb{P}^q$, set

$$t([Z_0:\cdots:Z_q]) = \frac{|Z_1|^2+\cdots+|Z_q|^2}{|Z_0|^2+\cdots+|Z_q|^2},$$
$$r_m = \max\{\lfloor ma\rfloor - q, 0\}, \qquad D_m = \binom{m+q}{q}.$$

Thus $t = 0$ exactly at b. Define the function e_a on $[0,1]$ by

$$e_a(t) = \begin{cases} a\log(t/a) + (1-a)\log((1-t)/(1-a)), & 0 < t < a, \\ 0, & a \le t \le 1, \\ -\infty, & t = 0, \end{cases} \tag{28.17}$$

and put $E_a = e_a \circ t$ on $\mathbb{P}^q$. For $q = 1$ this is the function Γ_a of (19.2); here it arises from a point pole on the total space of a fibration. We write $B_{X_n,m} = B_{m,0,\varphi,\nu_X}$ for the partial Bergman kernel.

Proposition 28.9.1 *The kernel and the dimension of the selected spaces are*

$$B_{X_n,m} = D_m \left(\sum_{j=r_m}^{m} \binom{m}{j} t^j (1-t)^{m-j} \right) \circ \pi, \qquad N_m = D_m - \binom{r_m+q-1}{q},$$

where the last binomial coefficient is zero when $r_m = 0$. In particular, all selected spaces are nonzero, and

$$\lim_{m\to\infty} \frac{q! N_m}{m^q} = 1 - a^q > 0.$$

Proof Since $n = q+1$, Proposition 28.8.2 identifies the selected space with the space of homogeneous polynomials of degree m on $\mathbb{C}^{q+1}$ that vanish to order at least r_m at b. For $\ell = (\ell_0, \ldots, \ell_q) \in \mathbb{Z}^{q+1}_{\ge 0}$ write $|\ell| = \sum_i \ell_i$ and $Z^\ell = \prod_i Z_i^{\ell_i}$. In the affine chart $Z_0 = 1$, the vanishing order of Z^ℓ at b is $\ell_1 + \cdots + \ell_q$, so the selected space is spanned by the monomials Z^ℓ with $|\ell| = m$ and $\ell_1 + \cdots + \ell_q \ge r_m$.

Distinct monomials are orthogonal, by invariance under the coordinate circle actions. Their squared norms with respect to σ^q are

$$\int_{\mathbb{P}^q} \frac{|Z^\ell|^2}{|Z|^{2m}} \sigma^q = \frac{q!\,\ell_0!\cdots\ell_q!}{(m+q)!}, \qquad |\ell| = m. \tag{28.18}$$

To verify this, integrate $|z^\ell|^2 \mathrm{e}^{-|z|^2}$ over $\mathbb{C}^{q+1}$ in two ways. As a product of one-variable integrals it equals $\pi^{q+1} \prod_i \ell_i!$, while in polar coordinates it is the average of $|z^\ell|^2$ over the unit sphere times a radial integral that contributes the factor $(m+q)!$. Dividing by the same computation for $\ell = 0$ gives the right-hand side of (28.18) as the normalized sphere average, and the pushforward of the normalized sphere measure to $\mathbb{P}^q$ is σ^q, since both are the probability measures invariant under the unitary group. This proves (28.18).

The reciprocal of the squared norm is $D_m m!/(\ell_0! \cdots \ell_q!)$. Summing first over the monomials with $\ell_1 + \cdots + \ell_q = j$, the multinomial formula gives the contribution

$$D_m \binom{m}{j} t^j (1-t)^{m-j}$$

to the kernel on $\mathbb{P}^q$. Since ν_X is the lift of σ^q, the norms and kernels lift exactly to X_n, which proves the kernel formula. There are $\binom{j+q-1}{q-1}$ monomials of affine degree j, and summing over the excluded degrees $0 \le j < r_m$ gives $\binom{r_m+q-1}{q}$. Since $0 \le r_m \le m$ and $r_m/m \to a$, this proves the dimension formula, the nonvanishing and the limit.□

Theorem 28.9.1 *Let $U_m = m^{-1} \log B_{X_n,m}$ be the partial Fubini–Study potential for the zero weight. Then*

$$U_m \longrightarrow E_a \circ \pi$$

in $L^1(X_n)$, pointwise as extended-real functions, and locally uniformly off the fiber $\pi^{-1}(b)$. The normalized kernel measures converge in total variation:

$$\frac{q!}{m^q} B_{X_n,m}\, \nu_X \longrightarrow \mathbf{1}_{\{t \circ \pi > a\}} \nu_X.$$

Proof Write $P_m(t)$ for the binomial sum in Proposition 28.9.1. For fixed $0 < t < 1$, $P_m(t)$ is the probability that a binomial random variable with parameters (m, t) is at least r_m. Its variance is $mt(1-t)$, so Chebyshev's inequality and $r_m/m \to a$ give

$$P_m(t) \longrightarrow \begin{cases} 0, & t < a, \\ 1, & t > a. \end{cases}$$

The pushforward of σ^q under t is $q t^{q-1}\, \mathrm{d}t$. To see this, note that in the affine chart the Fubini–Study measure has radial density $2q r^{2q-1}(1+r^2)^{-q-1}\, \mathrm{d}r$, and $t = r^2/(1+r^2)$. In particular, the level set $\{t = a\}$ has measure zero. Since $0 \le P_m \le 1$ and $q! D_m/m^q \to 1$, dominated convergence gives convergence of the densities in $L^1(\sigma^q)$, hence in $L^1(\nu_X)$ after lifting; this is the asserted convergence in total variation. The limit has mass $\int_a^1 q t^{q-1}\, \mathrm{d}t = 1 - a^q$.

To identify the limit of the potentials, put

$$H(s) = -s \log s - (1-s) \log(1-s), \qquad 0 \le s \le 1,$$

with $0 \log 0 = 0$. Then

$$\frac{\mathrm{e}^{mH(j/m)}}{m+1} \le \binom{m}{j} \le \mathrm{e}^{mH(j/m)} \quad (0 \le j \le m).$$

Indeed, $\binom{m}{j}\mathrm{e}^{-mH(j/m)}$ is the jth term of the binomial distribution with parameters $(m, j/m)$. It is at most one, and it is at least $1/(m+1)$ because, as the ratio of neighboring terms shows, it is a largest of the $m+1$ terms, which sum to one. For $j = 0$ and $j = m$ it equals one.

The logarithm of a sum of at most $m+1$ nonnegative terms differs from the largest of their logarithms by at most $\log(m+1)$. Hence $m^{-1} \log P_m(t)$ differs by at most $\log(m+1)/m$ from

$$\max_{r_m \le j \le m} \{H(j/m) + (j/m) \log t + (1 - j/m) \log(1-t)\}.$$

On compact subintervals of $(0, 1)$, uniform continuity and $r_m/m \to a$ show that these maxima converge uniformly to the maximum of the same expression over $a \le s \le 1$. Maximizing in s, we find that this maximum is zero for $t \ge a$, attained at $s = t$, and equals the first expression in (28.17) for $t < a$, attained at $s = a$. Thus the limit is $e_a(t)$, also at $t = a$.

The convergence is also uniform near $t = 1$. Choose c with $a < c < 1$. The same variance estimate gives $P_m(t) \ge 1 - C/m$ uniformly for $c \le t \le 1$ and large m, so $m^{-1} \log P_m \to 0$ uniformly there. Together with the argument on compact subintervals and $m^{-1} \log D_m \to 0$, this proves locally uniform convergence off $\{t = 0\}$. At $t = 0$ we have $r_m > 0$ for large m, so $P_m(0) = 0$ and the pointwise limit is $-\infty$.

Finally, $t^m \le P_m(t) \le 1$, because the term $j = m$ occurs in every binomial sum. Since $m^{-1} \log D_m$ is nonnegative and bounded, it follows that $\log(t \circ \pi) \le U_m \le C$ off the fiber over b. In local submersion coordinates near this fiber, $\log(t \circ \pi) = \log |z|^2 + O(1)$, which is integrable against a smooth volume. The pointwise convergence and these bounds give convergence in L^1, by dominated convergence. □

For rational a, the function E_a is the partial envelope of a point pole on the base. Put $Z' = (Z_1, \dots, Z_q)$ and $\psi_a = a \log t$ on $\mathbb{P}^q$. For $0 < a < 1$ this function is σ-plurisubharmonic. Indeed, away from b we have $\sigma + \mathrm{dd}^{\mathrm{c}}\psi_a = (1-a)\sigma + a\sigma'$, where σ' is the semipositive pullback of the Fubini–Study form under the projection $[Z_0 : Z'] \mapsto [Z']$ (and $\sigma' = 0$ if $q = 1$); near b, the function ψ_a is $a \log |z|^2$ plus a smooth function, so the positivity extends across b.

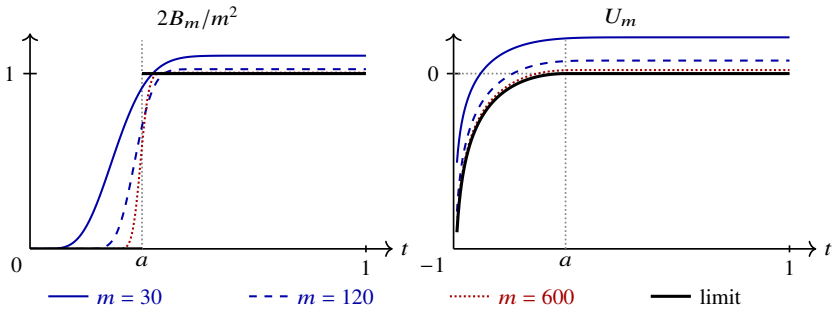


Fig. 28.1 The one-point model with $q = 2$ and $a = 1/3$. The normalized kernel density (left) and the Fubini–Study potential (right) are plotted against t. The black curves are their limits. The potential panel omits $t < 0.01$, where the limit tends to $-\infty$.

Proposition 28.9.2 *The function E_a belongs to* $\mathrm{PSH}(\mathbb{P}^q, \sigma)$*, and*

$$\mathrm{MA}_\sigma(E_a) = \mathbf{1}_{\{t>a\}}\sigma^q, \qquad \mathbf{1}_{\{t\circ\pi>a\}}\nu_X = \frac{1}{2n}\mathrm{MA}^{\mathrm{ext}}_\omega(E_a \circ \pi).$$

If a is rational, then $E_a = E_{\sigma,\mathbb{P}^q}[\psi_a]_{\mathcal{I}}(0)$, and $E_a \circ \pi$ is the largest $u \in \mathrm{PSH}(X_n, \alpha)$ with $u \le 0$ and $u \preceq_{\mathcal{I}} \varphi$.

Proof In the chart $Z_0 = 1$ put $z = Z'/Z_0$ and $R^2 = a/(1-a)$. For $|z| < R$, the local potential of $\sigma + \mathrm{dd^c}E_a$ is

$$\log(1 + |z|^2) + E_a = a\log|z|^2 - a\log a - (1-a)\log(1-a).$$

Its complex Hessian is semipositive and, away from $z = 0$, of rank $q - 1$. For $|z| > R$ we have $E_a = 0$, so the form is σ. The two expressions for E_a have equal values and equal first derivatives on the sphere $|z| = R$, so they glue to a $C^{1,1}$ function near this sphere, whose distributional Hessian has no part carried by the sphere. Near b, E_a has a logarithmic pole of coefficient a. This proves that E_a is σ-plurisubharmonic and that $E_a \sim \psi_a$.

Off b the local potentials of $\sigma + \mathrm{dd^c}E_a$ are $C^{1,1}$. By Proposition 1.7.2, their Bedford–Taylor top products are given by the determinants of their complex Hessians almost everywhere. By the rank computation, this determinant vanishes where $t < a$, while for $t > a$ the measure is σ^q. The sphere $\{t = a\}$ has volume zero, and a $C^{1,1}$ local potential puts no mass on it. Finally, the non-pluripolar measure does not charge the point b. This proves the measure formula. Since $E_a \in \mathrm{PSH}(\mathbb{P}^q, \sigma)$, Lemma 9.6.1 and the measure formula on the base give the identity on X_n.

Now let a be rational. For the envelope identity, let $w \in \mathrm{PSH}(\mathbb{P}^q, \sigma)$ satisfy $w \le 0$ and $w \preceq_{\mathcal{I}} \psi_a$. Since ψ_a has analytic singularities, Lemma 1.17.10 gives $w \le \psi_a + C$. Fix a complex line through 0 in the affine chart. If the restriction of $w + \log(1 + |z|^2)$ to this line is identically $-\infty$, there is nothing to prove on the line; otherwise the restriction is subharmonic. On the punctured disk of radius R in the line, subtract $a\log|z|^2$. The difference is subharmonic and, since $w \le \psi_a + C$, bounded above

near 0, so it extends subharmonically across 0. On the boundary circle it is at most $\log(1+R^2) - a\log R^2 = -a\log a - (1-a)\log(1-a)$, because $w \le 0$. By the maximum principle and the formula for $\log(1+|z|^2) + E_a$ on $\{|z| < R\}$, $w \le E_a$ on the disk. Every point with $|z| < R$ lies on such a line, and for $|z| \ge R$ we have $w \le 0 = E_a$. Since E_a is itself a competitor, it is the envelope. For the continuous obstacle zero, the quasi-everywhere and the pointwise obstacle conditions agree, as recalled in Section 1.19.

For the assertion on X_n, every α-plurisubharmonic function has the form $u = w \circ \pi$ with $w \in \mathrm{PSH}(\mathbb{P}^q, \sigma)$, by Proposition 1.22.1. Since φ has analytic singularities, $u \preceq_{\mathcal{I}} \varphi$ implies $u \preceq \varphi$ by Lemma 1.17.10. Restricting this inequality to $\tau = 0$ in product coordinates (z, τ) at p gives $w(z) \le a\log|z|^2 + C$, and hence $w \preceq \psi_a$ on $\mathbb{P}^q$. Conversely, if $w \preceq \psi_a$, then

$$w(z) \le a\log|z|^2 + C \le a\log(|z|^2 + |\tau|^2) + C$$

near p. Away from p, the function φ is locally bounded and w is bounded above, so by compactness the inequality $w \circ \pi \le \varphi + C$ holds globally. Thus $w \circ \pi \preceq \varphi$, which implies $w \circ \pi \preceq_{\mathcal{I}} \varphi$. Since Lemma 1.17.10 also applies on the base, the map $u = w \circ \pi \mapsto w$ identifies the competitors on X_n with the competitors for E_a considered above. Their largest members therefore correspond under pullback. □

For sufficiently small a, Lemma 1.16.2 gives a function $\varphi \in \mathrm{PSH}(X_n, \omega)$ whose only pole is at p. In particular, Theorem 28.9.1 applies on the Hopf threefold X_3, which does not have the bounded mass property, and gives a nonzero limit measure with the normalization $2/m^2$. The limit potential $E_a \circ \pi$ has a pole along the whole elliptic fiber over b, whereas φ is locally bounded at the other points of this fiber, so the multiplier ideals $\mathcal{I}(mE_a \circ \pi)$ are proper there for large m while those of φ are not. This makes explicit the loss of information described at the end of Section 28.8.

The limit measure also differs from the Hermitian Monge–Ampère measure of φ itself. In the construction of Lemma 1.16.2, choose φ to vanish outside a small neighborhood of p. A neighborhood of another point of the fiber over b is then an open set outside this neighborhood on which $t \circ \pi < a$. There $\varphi = 0$, so $\mathrm{MA}_\omega^{\mathrm{ext}}(\varphi) = \omega^n$ by locality, whereas the limit of the kernel measures vanishes. The identity in Proposition 28.9.2 concerns the envelope $E_a \circ \pi$, not the selecting potential φ.

28.10 Singularities along a section over a divisor

We now replace the isolated poles by a pole along the image of a holomorphic section of the fibration over a divisor of the base. The selected sections must then vanish to a certain order along that divisor. As for point poles, the normal variables in the fiber shift the integrability threshold only by a bounded amount, so this order differs from the order imposed by a divisorial pole on the base by a bounded number. Restriction sequences along the divisor show that each unit of vanishing order costs $O(m^{q-1})$ sections, and kernel convergence again reduces to the equilibrium theorem on the base.

For the restriction sequences we need a bound for the dimensions of section spaces that is uniform over a range of twists growing linearly with m.

Lemma 28.10.1 *Let M be a compact complex manifold of dimension $d \geq 0$, and let $E, F_1, \dots, F_N$ be holomorphic line bundles on M. For each $C_0 > 0$ there is a constant C_1 such that*

$$h^0\Big(M, E^m \otimes \bigotimes_{i=1}^{N} F_i^{j_i}\Big) \leq C_1 m^d \qquad (m \geq 1,\ j_i \in \mathbb{Z},\ |j_i| \leq C_0 m).$$

Proof For $d = 0$, the manifold M is a finite set and the dimension is at most its number of points. Let $d \geq 1$. Fix smooth Hermitian metrics on E and the F_i, a smooth positive volume form $\mathrm{d}V$, and finitely many coordinate charts with holomorphic frames, together with smaller subcharts that still cover M. For $|j_i| \leq C_0 m$, the local weight of the metric on $G = E^m \otimes \bigotimes_i F_i^{j_i}$ has second derivatives bounded by Cm. At a point p of a smaller subchart, multiply the frame by the exponential of a holomorphic affine function to remove the constant and real-linear terms of this weight at p. The new weight vanishes at p and is bounded by $Cm|z - p|^2$ on a fixed neighborhood. A ball of radius $\varepsilon m^{-1/2}$ about p lies in the chart, with $\varepsilon > 0$ independent of p, m and the j_i, and on this ball the weight is bounded. The submean inequality for the holomorphic coefficient of a section s on this ball gives

$$|s(p)|^2 \leq C m^d \int_M |s|^2 \,\mathrm{d}V,$$

where all constants depend only on the fixed charts, metrics, volume and C_0. Evaluation at p has a one-dimensional target, so for an L^2-orthonormal basis $s_1, \dots, s_D$ of $H^0(M, G)$ its squared norm is $\sum_{\ell=1}^{D} |s_\ell(p)|^2 \leq C m^d$. Integrating over M gives $D \leq C \operatorname{Vol}(M) m^d$. □

Let $\pi \colon X \to B$ be a surjective holomorphic submersion of connected compact complex manifolds, with connected fibers of dimension r, and let $q = \dim B$. Fix a holomorphic line bundle A on B with a smooth Hermitian metric h_A, and put $\theta = c_1(A, h_A)$. Let $Z \subset B$ be a nonempty smooth divisor, and suppose that π has a holomorphic section $\sigma \colon Z \to X$ over Z; put $Y = \sigma(Z)$. Fix $c > 0$ and a smooth metric h_Z on $\mathcal{O}_B(Z)$, with curvature Θ_Z, such that $\alpha := \theta - c\Theta_Z > 0$. Let s_Z be the canonical section of $\mathcal{O}_B(Z)$, and set $\psi = c \log |s_Z|^2_{h_Z}$.

Fix a Hermitian form ω on X, and suppose that $\varphi \in \mathrm{PSH}(X,\omega)$ is locally bounded on $X \setminus Y$ and that, in submersion coordinates $(x, z', w_1, \dots, w_r)$ near each point of Y,

$$\begin{aligned} \varphi &= c \log\left(|x|^2 + \sum_{j=1}^{r} |w_j|^2 \right) + O(1), \\ Z &= \{x = 0\}, \qquad Y = \{x = w_1 = \cdots = w_r = 0\}, \end{aligned} \tag{28.19}$$

with a locally bounded remainder. Here x and z' are coordinates on the base, and the vector z' of the $q-1$ tangential coordinates is absent when $q = 1$.

Theorem 28.10.1 *The function ψ is θ-plurisubharmonic, with positive non-pluripolar mass*

$$\int_B \mathrm{MA}_\theta(\psi) = \int_B \alpha^q > 0.$$

For every integer $m \geq 1$ put $\ell_m = \lfloor mc \rfloor$ and $k_m = \max\{\ell_m - r, 0\}$. Under holomorphic descent, the spaces selected by φ on X and by ψ on B are

$$\begin{aligned} V_m &:= H^0\big(X, \pi^* A^m \otimes \mathcal{I}(m\varphi)\big) = H^0\big(B, A^m(-k_m Z)\big), \\ W_m &:= H^0\big(B, A^m \otimes \mathcal{I}(m\psi)\big) = H^0\big(B, A^m(-\ell_m Z)\big) \subseteq V_m. \end{aligned}$$

There is a constant C, depending only on the fixed data, such that

$$0 \leq \dim V_m - \dim W_m \leq C m^{q-1}.$$

For $q = 1$ the difference is at most $r\,\#Z$ for every m.

Proof By the Poincaré–Lelong formula, $\theta + \mathrm{dd^c}\psi = \alpha + c[Z] \geq 0$. The divisorial measure identity (16.22), applied with the residual form $\alpha > 0$ and the bounded part zero, gives $\mathrm{MA}_\theta(\psi) = \alpha^q$, which proves the mass formula.

Every section of $\pi^* A^m$ descends by Proposition 28.7.1. In the $r+1$ normal variables (x, w), Lemma 1.16.1 gives

$$\mathcal{I}(m\varphi) = (x, w_1, \dots, w_r)^{k_m}.$$

The local coefficient $g(x, z')$ of a pulled-back section does not depend on w. Expanding g in powers of x, we see that g lies in this ideal exactly when the coefficients of x^j vanish for $j < k_m$, that is, when x^{k_m} divides g. This is an identity of germs in all tangential variables, so it holds at every point of Z. Outside Y the function φ is locally bounded. On the base, the same lemma in the single normal variable x gives $\mathcal{I}(m\psi) = \mathcal{O}_B(-\ell_m Z)$. These facts prove the two identities and the inclusion.

For $k_m \leq j < \ell_m$, the restriction sequence along Z,

$$\begin{aligned} 0 \longrightarrow A^m(-(j+1)Z) &\longrightarrow A^m(-jZ) \\ &\longrightarrow \big(A^m(-jZ)\big)|_Z \longrightarrow 0, \end{aligned}$$

shows that passing from vanishing order j to order $j+1$ along Z loses at most $h^0(Z,(A^m(-jZ))|_Z)$ dimensions. Apply Lemma 28.10.1 on Z with $E=A|_Z$ and $F_1=O_B(-Z)|_Z$; since $0\le j\le cm$, each loss is at most Cm^{q-1}, uniformly in j. There are at most r steps, which proves the estimate. When $q=1$, the divisor Z is a finite set and each restriction space has dimension $\#Z$, which gives the last assertion directly. □

We call ψ the *reference potential*, W_m the *reference spaces*, and $\dim V_m-\dim W_m$ the *defect*; the same terms are used for the analogous comparisons below. The following example shows that the order m^{q-1} in Theorem 28.10.1 is sharp.

Example 28.10.1 For all integers $q,r,k\ge 1$ and every real $0<c<k$, the hypotheses above are satisfied on $X=\mathbb{P}^q\times\mathbb{P}^r$ with $B=\mathbb{P}^q$ and $A=O_{\mathbb{P}^q}(k)$, for suitable choices made in the proof. For $q\ge 2$ the defect is

$$\dim V_m-\dim W_m=\frac{r(k-c)^{q-1}}{(q-1)!}m^{q-1}+O(m^{q-2}),$$

and for $q=1$ it equals $\min\{r,\lfloor mc\rfloor\}$ for every m.

Proof Use homogeneous coordinates $[Z_0:\cdots:Z_q]$ on the base and $[W_0:\cdots:W_r]$ on the fiber. Let $Z=\{Z_0=0\}$ and $Y=Z\times\{[1:0:\cdots:0]\}$, and let ω_B,ω_F be the Fubini–Study forms normalized to have top integrals one. Define

$$R_B=\frac{|Z_0|^2}{\sum_{i=0}^{q}|Z_i|^2},\qquad R_F=\frac{\sum_{j=1}^{r}|W_j|^2}{\sum_{j=0}^{r}|W_j|^2},\qquad \varphi=c\log(R_B+R_F),\quad \psi=c\log R_B.$$

The function φ is smooth off Y and satisfies (28.19). Over a common denominator, the numerator of R_B+R_F is

$$\sum_{j=0}^{r}|Z_0W_j|^2+\sum_{i=0}^{q}\sum_{j=1}^{r}|Z_iW_j|^2,$$

whose logarithm is plurisubharmonic in each affine chart. Hence $\mathrm{dd}^{\mathrm{c}}\varphi\ge -c(\omega_B+\omega_F)$. Choose an integer $b\ge c$ and set $\omega=k\omega_B+b\omega_F$. This is a Hermitian form, and $\omega+\mathrm{dd}^{\mathrm{c}}\varphi\ge(k-c)\omega_B+(b-c)\omega_F\ge 0$. On the base, $\theta=k\omega_B$, $\Theta_Z=\omega_B$ and $\alpha=(k-c)\omega_B>0$.

Put $\ell_m=\lfloor mc\rfloor$. Divisibility of homogeneous polynomials by powers of Z_0 gives the exact dimensions

$$\dim W_m=\binom{km-\ell_m+q}{q},\qquad \dim V_m=\binom{km-\max\{\ell_m-r,0\}+q}{q}.$$

For large m, the two residual degrees differ by r and equal $(k-c)m+O(1)$. Subtracting the two binomial coefficients, which are polynomials of degree q in the residual degrees, gives the asserted leading coefficient and remainder. For $q=1$ the difference is $\ell_m-\max\{\ell_m-r,0\}=\min\{r,\ell_m\}$ for every m. □

The bounded shift of divisorial order thus has negligible normalized codimension. To compare the kernels, use the volumes, metrics and continuous weights $v \circ \pi$ of Section 28.8, together with the notation $\beta^{(q)}_{X,m,v\circ\pi,\varphi}$ and $\mathcal{F}^{(q)}_{X,m}$ introduced there, and put $E(v) = E_{\theta,B}[\psi]_{\mathcal{I}}(v)$.

Corollary 28.10.1 *Then φ satisfies the conclusions of Proposition 28.8.1, with $n = q + r$, and the total variation distance between $\beta^{(q)}_{X,m,v\circ\pi,\varphi}$ and the lift of the kernel measure selected by ψ on B is $O(m^{-1})$. In particular,*

$$\beta^{(q)}_{X,m,v\circ\pi,\varphi} \rightharpoonup \mathcal{L}\big(\mathrm{MA}_\theta(E(v))\big) \qquad (v \in C(B)).$$

Proof Since α is a Kähler form, B is Kähler. By Theorem 28.10.1, ψ has positive non-pluripolar mass, and the reference spaces W_m satisfy $W_m \subseteq V_m$ and $\dim V_m - \dim W_m = O(m^{q-1})$. Hence Proposition 28.8.1 applies. □

Thus the selecting potentials of (28.19) admit the approximation asked for in Problem 35.9.2, whenever the curvature margin $\theta - c\Theta_Z$ is positive.

28.10.1 A bounded divisorial band from fiber integrability

The comparison of Theorem 28.10.1 does not need the exact local form (28.19). It suffices to bound the selecting potential from below by divisorial poles on all of X, and from above by an anisotropic model in a single chart over each divisor. The lower bound gives the inclusion of the reference spaces into the selected spaces. The upper bound forces vanishing along each divisor to an order that falls short of the reference order by a bounded amount: separate powers of the normal variables replace the shift by the fiber dimension with a weighted shift. The selecting potential may then be singular along a positive-dimensional subset of a fiber, and it need not descend to the base. Several divisors can be treated at once if they cross normally, since the integrability test on the base then separates in normal coordinates.

Let $\pi\colon X \to B$ be a surjective holomorphic submersion of connected compact complex manifolds with connected fibers of dimension f, and let $q = \dim B$. Fix a Hermitian form ω on X, a holomorphic line bundle A on B with a smooth Hermitian metric h, and $\varphi \in \mathrm{PSH}(X,\omega)$. Write $\theta = c_1(A,h)$, and use holomorphic descent to regard the selected spaces as subspaces of $H^0(B, A^m)$:

$$V_m = H^0\big(X, \pi^*A^m \otimes \mathcal{I}(m\varphi)\big) \subseteq H^0(B, A^m), \qquad m \geq 1.$$

Let $D_1, \dots, D_N$ be distinct nonempty smooth connected reduced divisors on B whose union has simple normal crossings. For each i, choose a smooth metric H_i on $\mathcal{O}_B(D_i)$, and write s_i for the canonical section and η_i for the curvature.

Lemma 28.10.2 *Let $r \geq 1$ and $\ell \geq 0$ be integers, let $p, b_1, \dots, b_r$ be positive integers, and let $P > 0$. The function*

$$|z|^{2\ell}\left(|z|^{2p} + \sum_{j=1}^{r} |w_j|^{2b_j}\right)^{-P}$$

of $(z, w) \in \mathbb{C} \times \mathbb{C}^r$ is locally integrable near the origin if and only if

$$\frac{\ell + 1}{p} + \sum_{j=1}^{r} \frac{1}{b_j} > P.$$

In the case of equality, the integral diverges logarithmically.

Proof After integrating over the angles, substitute $x_0 = |z|^{2p}$ and $x_j = |w_j|^{2b_j}$. Up to a positive constant, the integral becomes

$$\int (x_0 + \cdots + x_r)^{-P} x_0^{(\ell+1)/p-1} \prod_{j=1}^{r} x_j^{1/b_j-1} \, \mathrm{d}x_0 \cdots \mathrm{d}x_r$$

over a box near the origin. This box contains a small simplex $\{\sum_j x_j < \delta\}$ and is contained in a larger one. On either simplex use the coordinates $\rho = \sum_j x_j$ and $x_j = \rho y_j$. The integral over y is finite and positive, because all its exponents exceed -1, and the integral over ρ is

$$\int_0^{\delta} \rho^{(\ell+1)/p+\sum b_j^{-1}-P-1} \, \mathrm{d}\rho.$$

This gives the threshold and the logarithmic divergence at equality. Bounded factors in the weight, holomorphic units and additional tangential variables do not affect this local test. □

Theorem 28.10.2 *Suppose that $c_1, \dots, c_N > 0$ and $u \in C^\infty(B, \mathbb{R})$ satisfy*

$$\beta = \theta + \mathrm{dd^c} u - \sum_{i=1}^N c_i \eta_i \geq 0, \qquad \int_B \beta^q > 0, \qquad \varphi \geq \sum_{i=1}^N c_i \log \|\pi^* s_i\|^2_{H_i} - C \quad \text{on } X$$

for a constant C. Suppose further that for each i there is one submersion polydisk over an open subset of D_i, with coordinates

$$\pi(z, t, w, v) = (z, t), \quad D_i = \{z = 0\}, \quad t \in \mathbb{C}^{q-1}, \quad w \in \mathbb{C}^{r_i}, \quad v \in \mathbb{C}^{f-r_i},$$

where $1 \leq r_i \leq f$, on which

$$\varphi \leq \frac{c_i}{p_i} \log \left(|z|^{2p_i} + \sum_{j=1}^{r_i} |w_j|^{2b_{ij}} \right) + C_i$$

for positive integers $p_i, b_{i1}, \dots, b_{ir_i}$ and a constant C_i. Put

$$a_i = p_i \sum_{j=1}^{r_i} b_{ij}^{-1}, \quad L_{m,i} = \lfloor m c_i \rfloor, \quad K_{m,i} = \max\{\lfloor m c_i - a_i \rfloor, 0\},$$

$$\psi = u + \sum_{i=1}^N c_i \log \|s_i\|^2_{H_i}.$$

Then $\psi \in \mathrm{PSH}(B, \theta)$, $\mathrm{MA}_\theta(\psi) = \beta^q$, its mass is $\int_B \beta^q \in (0, \infty)$, and for every $m \geq 1$,

$$W_m := H^0\big(B, A^m \otimes \mathcal{I}(m\psi)\big) = H^0\left(B, A^m \left(-\sum_i L_{m,i} D_i\right)\right) \subseteq V_m,$$

$$V_m \subseteq H^0\left(B, A^m \left(-\sum_i K_{m,i} D_i\right)\right).$$

Moreover $\dim V_m - \dim W_m = O(m^{q-1})$. *For $q = 1$ this difference is at most $\sum_i \lceil a_i \rceil$ for every m.*

Proof By the Poincaré–Lelong formula, $\theta + \mathrm{dd^c}\psi = \beta + \sum_i c_i [D_i] \geq 0$. Off the divisors, locality identifies the non-pluripolar top power with β^q; the non-pluripolar product puts no mass on the divisors, and neither does the smooth measure β^q. Hence $\mathrm{MA}_\theta(\psi) = \beta^q$, whose mass is finite because β is smooth and positive by assumption.

Near a point of B, the divisors through that point are distinct coordinate hyperplanes $\{z_i = 0\}$, and $\psi = \sum_i c_i \log |z_i|^2 + O(1)$, the sum running over these divisors. If a holomorphic germ is divisible by each corresponding $z_i^{L_{m,i}}$, then it is divisible by their product; its residual coefficient is bounded, and since $L_{m,i} + 1 > m c_i$, Fubini's theorem gives integrability. Conversely, at generic nearby points of each coordinate hyperplane only one divisor passes, and the one-variable integrability condition

$\ell + 1 > mc_i$ forces divisibility by $z_i^{L_{m,i}}$; the divisor of the germ then gives divisibility throughout the chart. This proves $\mathcal{I}(m\psi) = \mathcal{O}_B(-\sum_i L_{m,i} D_i)$, also when some mc_i is an integer. Since u is bounded, the global lower bound implies $\varphi \geq \psi \circ \pi - C'$. Hence integrability on the base gives integrability of the pullback in every bounded submersion chart, which proves $W_m \subseteq V_m$.

For the upper inclusion, take a nonzero section in V_m and let ℓ be its vanishing order along D_i; note that D_i is irreducible, being smooth and connected. In the polydisk over D_i on which the upper bound holds, the coefficient of the section is $z^\ell g(z,t)$, where $g(0,t)$ does not vanish identically on the open subset of D_i, since otherwise the identity theorem would increase the order along D_i. Shrink the polydisk about a point with $w = 0$ at which $g(0,t) \neq 0$, so that g becomes a holomorphic unit. The upper bound on φ and Lemma 28.10.2 then force $\ell + 1 > mc_i - a_i$. For an integer $\ell \geq 0$ this is equivalent to $\ell \geq K_{m,i}$. Imposing these orders along all the divisors proves the upper inclusion.

Since $\lfloor mc_i \rfloor - \lfloor mc_i - a_i \rfloor \leq \lceil a_i \rceil$, we have $0 \leq L_{m,i} - K_{m,i} \leq \lceil a_i \rceil$. Raise the orders from $(K_{m,i})_i$ to $(L_{m,i})_i$, one divisor and one unit at a time. As in the proof of Theorem 28.10.1, the restriction sequence along D_i shows that each step loses at most

$$h^0\left(D_i,\, A^m\left(-\sum_j n_j D_j\right)\Big|_{D_i}\right), \qquad 0 \leq n_j \leq mc_j,$$

dimensions. By Lemma 28.10.1 with $E = A|_{D_i}$ and $F_j = \mathcal{O}_B(-D_j)|_{D_i}$, this is $O(m^{q-1})$ uniformly over all steps, and there are at most $\sum_i \lceil a_i \rceil$ steps. When $q = 1$, the distinct connected divisors are points, and each restriction space has dimension one. □

For a single divisor, that is, for $N = 1$, we drop the index 1 and write D, s_D, η, c, p, r, b_j, a, L_m and K_m.

Corollary 28.10.2 *Let $N = 1$, and let c and u satisfy the conditions on β in Theorem 28.10.2. Suppose that a nonempty compact smooth submanifold $Z \subset \pi^{-1}(D)$ projects submersively onto D, and that near every point of Z there are coordinates as in that theorem in which*

$$Z = \{z = w_1 = \cdots = w_r = 0\}, \qquad \varphi = \frac{c}{p} \log\left(|z|^{2p} + \sum_{j=1}^{r} |w_j|^{2b_j}\right) + O(1),$$

with fixed c, p, b_j. Suppose also that φ is locally bounded outside Z. Then φ satisfies the hypotheses of Theorem 28.10.2, so its conclusions hold, and moreover

$$V_m = H^0(B, A^m(-K_m D)) \qquad (m \geq 1).$$

Proof The local model is bounded below by $c \log |z|^2$ up to a constant. Near points of $\pi^{-1}(D)$ outside Z, the function φ is locally bounded, so the same lower bound holds there; away from $\pi^{-1}(D)$ both functions are locally bounded. By compactness, finitely many charts give the global lower bound. The upper bound holds in any chart

of the model. Conversely, if a section is divisible by z^{K_m}, its residual coefficient is bounded and holomorphic in each chart of the model, and the strict inequality $K_m + 1 > mc - a$ together with Lemma 28.10.2 gives integrability there. Outside Z the function φ is locally bounded, which proves the reverse inclusion. □

Such exact models exist with pole sets of positive dimension inside the fibers. Let B be projective, let A be ample, and let $X = B \times (\mathbb{P}^1)^f$ with π the first projection. Mark a point o_j in each of the first r factors $\mathbb{P}^1$ of the fiber, and let $E_j \subset X$ be the inverse image of o_j. With smooth metrics on the corresponding line bundles, put

$$Q = \|\pi^* s_D\|^{2p} + \sum_{j=1}^{r} \|s_{E_j}\|^{2b_j}, \qquad \varphi = \lambda \log Q, \qquad c = p\lambda.$$

The common zero set of the terms of Q is $D \times \{o_1\} \times \cdots \times \{o_r\} \times (\mathbb{P}^1)^{f-r}$. Near it, the positive factors of the metrics give exactly the local model of Corollary 28.10.2, and φ is smooth elsewhere. The inequality $Q \geq \|\pi^* s_D\|^{2p}$ gives the global lower bound. Locally, $\mathrm{dd^c} \log Q$ is bounded below by a convex combination of the negatives of the smooth curvature forms of the terms. Hence $\mathrm{dd^c} \log Q \geq -C_0\omega$ for a constant C_0 off the common zero set, and, after a quadratic correction, the extension theorem for plurisubharmonic functions carries this bound across it. Since A is ample, there is u with $\theta + \mathrm{dd^c} u > 0$. For sufficiently small $\lambda > 0$ we then have $\lambda C_0 \leq 1$, so that $\varphi \in \mathrm{PSH}(X, \omega)$, and $\theta + \mathrm{dd^c} u - p\lambda\eta > 0$. Thus the exact series of the corollary also occurs when $r < f$.

Example 28.10.2 Let $B = \mathbb{P}^q$ with a hyperplane D, let $A = \mathcal{O}_B(1)$, and let $X = B \times \mathbb{P}^1$. Let ω_B, ω_F be the Fubini–Study forms normalized to have total mass one, and use the corresponding metrics on the divisor line bundles. For a point $o \in \mathbb{P}^1$ set

$$\varphi = \tfrac{1}{4} \log\bigl(\|s_D\|^4 + \|s_o\|^6\bigr), \qquad \omega = \omega_B + \omega_F.$$

Then $p = 2$, $b_1 = 3$, $c = 1/2$, $a = 2/3$, and the reference potential $\psi = \frac{1}{2} \log \|s_D\|^2$ has mass 2^{-q}. The spaces V_1 and W_1 agree, and for every $m \geq 2$ the defect is

$$\dim V_m - \dim W_m = \binom{\lceil m/2 \rceil + q}{q-1} \sim \frac{2^{1-q}}{(q-1)!} m^{q-1}.$$

For $q = 1$ the defect is exactly one for every $m \geq 2$.

Proof Locally, the complex Hessian of the logarithm of a sum of two squared norms of holomorphic sections is bounded below by a convex combination of the negatives of their curvature forms. Hence $\mathrm{dd^c}\varphi \geq -\frac{1}{2}\omega_B - \frac{3}{4}\omega_F$ off the pole set, and across it by the extension theorem after a quadratic correction. Thus $\varphi \in \mathrm{PSH}(X, \omega)$. The pole set is $D \times \{o\}$, along which φ has the exact local model of Corollary 28.10.2. Here $u = 0$, $\beta = \frac{1}{2}\omega_B$, and $K_m = L_m - 1$ for $m \geq 2$, while $K_1 = L_1 = 0$. The spaces V_m and W_m therefore correspond to homogeneous polynomials of degrees $m - L_m + 1$ and $m - L_m$, and since $m - L_m = \lceil m/2 \rceil$, the difference of their dimensions is the displayed binomial coefficient. □

In this example φ is finite on most of each fiber over D and equals $-\infty$ on a proper subset of it, so φ differs from every pullback of a function on B by an unbounded amount. For $p = b_j = 1$ and $r = f$, the shift $a = f$ is the shift by the fiber dimension found in Theorem 28.10.1. Theorem 28.10.2 thus gives sufficient conditions for the approximation asked for in Problem 35.9.2, and when B is Kähler, Proposition 28.8.1 turns them into limits of the kernel measures. We do not know which other fiber singularities admit such an approximation.

28.10.2 Several crossing base divisors

In Theorem 28.10.2 the upper bounds are required only on one polydisk over each divisor, which can be chosen away from the crossings; integrability at the crossings is controlled by the global lower bound. Exact models can nevertheless have fiber poles over the crossings, as in the following example, where different divisors use separate fiber coordinates.

Example 28.10.3 Let $B = (\mathbb{P}^1)^q$, $X = B \times (\mathbb{P}^1)^q$ and $A = \mathcal{O}_B(1, \dots, 1)$. For each i, let $D_i \subset B$ be the inverse image of a marked point in the ith factor of B, and let $E_i \subset X$ be the inverse image of a marked point in the ith factor of the fiber. Use normalized Fubini–Study metrics on the corresponding line bundles, and write $\omega_{B,i}, \omega_{F,i}$ for the pullbacks to X of the normalized Fubini–Study forms on the ith factors of B and of the fiber. Put

$$R_i = \|\pi^* s_i\|^2, \quad T_i = \|s_{E_i}\|^2, \quad \varphi = \frac{1}{4} \sum_{i=1}^{q} \log(R_i^2 + T_i^3),$$
$$\omega = \sum_i (\omega_{B,i} + \omega_{F,i}), \qquad \psi = \frac{1}{2} \sum_i \log \|s_i\|^2.$$

Then $\varphi \in \mathrm{PSH}(X, \omega)$, and Theorem 28.10.2 applies with $N = q$, $u = 0$, $c_i = 1/2$, $p_i = 2$, $b_{i1} = 3$ and $a_i = 2/3$. The reference potential ψ has mass $q!/2^q$. With $L_m = \lfloor m/2 \rfloor$ and $K_m = \max\{\lfloor m/2 - 2/3 \rfloor, 0\}$, the selected and reference spaces are exactly

$$V_m = H^0\big(B, \mathcal{O}_B(m - K_m, \dots, m - K_m)\big),$$
$$W_m = H^0\big(B, \mathcal{O}_B(m - L_m, \dots, m - L_m)\big).$$

They agree for $m = 1$, and for $m \geq 2$ the defect is

$$\dim V_m - \dim W_m = (\lceil m/2 \rceil + 2)^q - (\lceil m/2 \rceil + 1)^q.$$

For $q \geq 2$ this is $q 2^{1-q} m^{q-1} + O(m^{q-2})$, and for $q = 1$ it is exactly one.

Proof By the calculation in Example 28.10.2, the complex Hessian of each summand of φ is bounded below by $-\frac{1}{2}\omega_{B,i} - \frac{3}{4}\omega_{F,i}$, also across its pole set by the extension theorem after a quadratic correction. Summing proves $\varphi \in \mathrm{PSH}(X, \omega)$. The global lower bound follows from $R_i^2 + T_i^3 \geq R_i^2$. For the upper bound over D_i, away from the other divisors, place the ith fiber coordinate at its marked point; all the other summands are smooth there. The residual is $\beta = \frac{1}{2} \sum_i \omega_{B,i} > 0$, whose top integral is $q!/2^q$.

The theorem gives the required generic orders along every D_i. Conversely, near a point where several of the pairs (R_i, T_i) vanish, the density $\mathrm{e}^{-m\varphi}$ is comparable to the product of the factors $(|z_i|^4 + |w_i|^6)^{-m/4}$ in disjoint pairs of variables. A descended coefficient divisible by every $z_i^{K_m}$ has a bounded holomorphic residual. Each factor is integrable because $K_m + 1 > m/2 - 2/3$, and Fubini's theorem gives integrability at the intersections; elsewhere the same argument uses fewer factors. This proves the identity for V_m, and the theorem gives W_m. A line bundle $\mathcal{O}_B(d, \dots, d)$ with $d \geq 0$ has $(d+1)^q$ independent sections, the products of monomials in the separate factors. Since $K_m = L_m - 1$ for $m \geq 2$ and $m - L_m = \lceil m/2 \rceil$, subtraction gives the displayed defect and its leading term. □

As before, the selecting potential takes finite values and the value $-\infty$ in the fibers over the divisors, so it does not descend.

28.10.3 Fiber integrability on a modified base

Keep $\pi\colon X \to B$, (A,h), θ, φ and V_m as in Section 28.10.1. A singularity of higher codimension on the base can become divisorial after a modification of the base. By Lemma 1.22.10, the integrability test upstairs then applies to the pulled-back section multiplied by the Jacobian. This fixed factor shifts the divisorial orders, while the restriction estimate still bounds the number of lost sections.

Let $\rho\colon Y \to B$ be a modification of connected compact complex manifolds, and form the fiber product and its projections

$$X' = X \times_B Y, \qquad \pi'\colon X' \to Y, \qquad \mu\colon X' \to X.$$

In local submersion coordinates one sees that X' is smooth, that π' is a submersion with the same compact connected fibers as π, and that μ is a modification. Let R_ρ and J_ρ be the relative canonical bundle and the Jacobian section of ρ, as in Section 1.22.7. The coordinate Jacobian matrix of μ is that of ρ, extended by the identity in the fiber variables, so

$$R_\mu = \pi'^{*} R_\rho, \qquad J_\mu = \pi'^{*} J_\rho.$$

Put $\varphi' = \varphi \circ \mu$ and $\theta' = \rho^{*}\theta$. Choose distinct nonempty smooth connected reduced divisors $D_1, \dots, D_N$ on Y whose union has simple normal crossings, together with smooth metrics H_i on $\mathcal{O}_Y(D_i)$, canonical sections s_i and curvatures η_i.

Theorem 28.10.3 *Suppose that $c_i > 0$ and $u \in C^\infty(Y, \mathbb{R})$ satisfy*

$$\beta = \theta' + \mathrm{dd^c} u - \sum_i c_i \eta_i \geq 0, \qquad \int_Y \beta^q > 0, \qquad \varphi' \geq \sum_i c_i \log \|\pi'^* s_i\|^2_{H_i} - C.$$

Suppose that for each i the function φ' satisfies an anisotropic upper bound, with respect to π', on a submersion polydisk over an open subset of D_i, as in Theorem 28.10.2, with fixed p_i, b_{ij}. Define a_i, $L_{m,i}$ and $K_{m,i}$ by the formulas of that theorem, and put

$$\psi' = u + \sum_i c_i \log \|s_i\|^2_{H_i}, \qquad M_m = \rho^* A^m \otimes R_\rho.$$

Then there is a unique $\psi \in \mathrm{PSH}(B, \theta)$ with $\psi \circ \rho = \psi'$, and

$$\mathrm{MA}_\theta(\psi) = \rho_*(\beta^q), \qquad \int_B \mathrm{MA}_\theta(\psi) = \int_Y \beta^q \in (0, \infty).$$

The map $T_m(s) = J_\rho \rho^* s$ identifies $H^0(B, A^m)$ with $H^0(Y, M_m)$, and for $W_m = H^0(B, A^m \otimes \mathcal{I}(m\psi))$,

$$T_m(W_m) = H^0\left(Y, M_m\left(-\sum_i L_{m,i} D_i\right)\right),$$
$$T_m(W_m) \subseteq T_m(V_m) \subseteq H^0\left(Y, M_m\left(-\sum_i K_{m,i} D_i\right)\right).$$

Consequently $W_m \subseteq V_m$ and $\dim V_m - \dim W_m = O(m^{q-1})$. If $q = 1$, the difference is at most $\sum_i \lceil a_i \rceil$ for every m.

Proof By the Poincaré–Lelong formula, $\theta' + \mathrm{dd^c}\psi' = \beta + \sum_i c_i [D_i]$. Thus $\psi' \in \mathrm{PSH}(Y, \rho^*\theta)$, and Proposition 1.22.4 gives the unique potential ψ on B with $\psi \circ \rho = \psi'$. As in the proof of Theorem 28.10.2, $\mathrm{MA}_{\theta'}(\psi') = \beta^q$, whose mass is finite and, by assumption, positive. The measure identity in Proposition 28.6.1 then gives the pushforward formula and the mass.

Lemma 1.22.10, applied first with weight zero and then with weight $m\psi$, gives the identification of the full section spaces and the formula for $T_m(W_m)$; here $\mathcal{I}(m\psi') = \mathcal{O}_Y(-\sum_i L_{m,i} D_i)$ by the integrability computation for simple normal crossings in the proof of Theorem 28.10.2. Since u is bounded, the global lower bound gives $\varphi' \geq \psi' \circ \pi' - C'$. As $\rho \circ \pi' = \pi \circ \mu$ and μ is surjective, this descends to $\varphi \geq \psi \circ \pi - C'$, and integrability in bounded submersion charts proves $W_m \subseteq V_m$.

For $s \in V_m$, the transformation of multiplier ideals under μ gives

$$J_\mu \mu^* s = \pi'^* T_m(s) \in H^0\big(X', \pi'^* M_m \otimes \mathcal{I}(m\varphi')\big).$$

Hence, in each polydisk where the upper bound holds, the descended coefficient of $T_m(s)$ vanishes to order at least $K_{m,i}$ along D_i, by Lemma 28.10.2 and the argument

with a unit coefficient in the proof of Theorem 28.10.2. This is the vanishing order of $T_m(s)$, including the Jacobian factor, and it proves the upper inclusion.

The chain of restrictions along the divisors on Y has at most $\sum_i \lceil a_i \rceil$ steps. On D_i, each target is $(\rho^* A|_{D_i})^m$ twisted by powers of the bundles $\mathcal{O}_Y(-D_j)|_{D_i}$ in a linear range and by the fixed bundle $R_\rho|_{D_i}$. Including $R_\rho|_{D_i}$ in Lemma 28.10.1 with exponent one, we see that every target has dimension $O(m^{q-1})$, uniformly. Summing gives the estimate. For $q = 1$ each divisor is a point and every target has dimension one. □

The theorem requires the modification and the divisorial data to be given; it does not say which selecting potentials admit such data. In the following model with a point pole on the base, both the residual and the Jacobian correction are explicit.

Example 28.10.4 Let $q \geq 2$, $B = \mathbb{P}^q$, $A = \mathcal{O}_B(1)$, $p = [1 : 0 : \cdots : 0]$ and $X = B \times \mathbb{P}^1$. Use the normalized Fubini–Study forms ω_B, ω_F and the corresponding metrics, and for a point $o \in \mathbb{P}^1$ put

$$R = \frac{\sum_{j=1}^{q} |Z_j|^2}{\sum_{j=0}^{q} |Z_j|^2}, \qquad T = \|s_o\|^2, \qquad \varphi = \tfrac{1}{4}\log(R^2 + T^3), \qquad \psi = \tfrac{1}{2}\log R.$$

Then $\varphi \in \mathrm{PSH}(X, \omega_B + \omega_F)$, and $\psi \in \mathrm{PSH}(B, \omega_B)$ has mass $1 - 2^{-q}$. Let $\mathfrak{m}_p$ be the ideal sheaf of p. The selected and reference spaces are exactly

$$V_m = H^0\big(B, A^m \otimes \mathfrak{m}_p^{k_m}\big), \qquad W_m = H^0\big(B, A^m \otimes \mathfrak{m}_p^{\ell_m}\big),$$
$$k_m = \max\{\lfloor m/2 - 2/3 \rfloor - q + 1, 0\}, \qquad \ell_m = \max\{\lfloor m/2 \rfloor - q + 1, 0\}.$$

Their defect vanishes for $m < 2q$, and for $m \geq 2q$ it is

$$\dim V_m - \dim W_m = \binom{\lfloor m/2 \rfloor - 1}{q - 1} \sim \frac{2^{1-q}}{(q-1)!} m^{q-1}.$$

Proof Over a common denominator, $R^2 + T^3$ is a sum of squared moduli of holomorphic coefficients of sections of $\mathcal{O}_X(2, 3)$, divided by the corresponding metric factor. Hence $\mathrm{dd}^{\mathrm{c}}\varphi \geq -\frac{1}{2}\omega_B - \frac{3}{4}\omega_F$, also across the common zero set by the extension theorem.

Let $\rho : Y \to B$ be the blowup of p, with exceptional divisor E, and let $\tau : Y \to \mathbb{P}^{q-1}$ be the extension to Y of the projection $[Z_0 : \cdots : Z_q] \mapsto [Z_1 : \cdots : Z_q]$. Since (ρ, τ) is an embedding, the forms

$$\alpha = \rho^* \omega_B, \qquad \gamma = \tau^* \omega_{\mathbb{P}^{q-1}}, \qquad \beta = \tfrac{1}{2}(\alpha + \gamma)$$

satisfy $\beta > 0$. The divisor bundle is $\mathcal{O}_Y(E) = \rho^* \mathcal{O}_B(1) \otimes \tau^* \mathcal{O}_{\mathbb{P}^{q-1}}(-1)$; its quotient metric has curvature $\alpha - \gamma$ and satisfies $R \circ \rho = \|s_E\|^2$. Thus the theorem applies with the single divisor E, $u = 0$, $c = 1/2$, $a = 2/3$ and residual β. Along $E \times \{o\}$ the pulled-back selecting potential has the exact local model $\frac{1}{4}\log(|z|^4 + |w|^6) + O(1)$, and it is locally bounded elsewhere. If $T_m(s)$ is divisible by z^{K_m}, where $K_m =$

$\max\{\lfloor m/2 - 2/3\rfloor, 0\}$, then Lemma 28.10.2 gives integrability of $\pi'^{*}T_m(s) = J_\mu \mu^* s$ in every chart of this model. By Lemma 1.22.10 for μ, this gives the inclusion opposite to the upper inclusion of the theorem.

In the blowup coordinates $Z_1/Z_0 = z$ and $Z_j/Z_0 = zt_j$ for $j \geq 2$, the Jacobian of ρ is z^{q-1}. Hence, if a section s of A^m vanishes to order d at p, then $T_m(s)$ vanishes to order $d + q - 1$ generically along E. The two divisorial thresholds $\lfloor m/2 \rfloor$ and K_m therefore give exactly the orders ℓ_m and k_m above. Homogeneous polynomials of degree m realize every jet of total degree at most m at p. For $m \geq 2q$ we have $k_m = \ell_m - 1$, and the additional jets of degree $\ell_m - 1$ are spanned by $\binom{\ell_m+q-2}{q-1}$ monomials, which gives the defect. For $m < 2q$ both orders vanish.

Finally, in cohomology $[\beta] = \rho^* c_1(A) - \frac{1}{2}[E]$. The intersection numbers containing both $\rho^* c_1(A)$ and $[E]$ vanish, because $\rho^* c_1(A)$ restricts to zero on E. The normal bundle of $E \simeq \mathbb{P}^{q-1}$ is $\mathcal{O}_E(-1)$, so $\int_Y [E]^q = (-1)^{q-1}$, while $\int_Y (\rho^* c_1(A))^q = 1$. Expanding gives $\int_Y \beta^q = 1 - 2^{-q}$, which is the mass of ψ by the theorem. □

The selecting potential is finite at most points of the fiber over p and equals $-\infty$ at the point (p, o), so again it does not descend, even up to a bounded function. The reference potential ψ has a pole of codimension q at p, and the Jacobian factor is what converts the divisorial orders along E into vanishing orders at p.

28.10.4 Semipositive residuals with positive volume

In Theorems 28.10.2 and 28.10.3 and Corollary 28.10.2 the residual β is only required to be semipositive with positive top integral. It enters only through the reference potential ψ, which it makes plurisubharmonic with positive non-pluripolar mass; the estimates for the divisorial orders do not involve β. A semipositive residual with positive top integral need not become strictly positive after any smooth change of gauge, that is, after adding $\mathrm{dd}^{\mathrm{c}} u$ for a smooth real function u. The following example shows this, with an exact computation of the selected spaces.

Example 28.10.5 Let $q \geq 2$, let $\rho\colon B \to \mathbb{P}^q$ be the blowup of $p = [1 : 0 : \cdots : 0]$, and let E be its exceptional divisor. Let $\tau\colon B \to \mathbb{P}^{q-1}$ be the extension to B of the projection from p, and put

$$\alpha = \rho^* \omega_{\mathbb{P}^q}, \qquad \gamma = \tau^* \omega_{\mathbb{P}^{q-1}}, \qquad \kappa = \alpha + \gamma > 0,$$

where $\omega_{\mathbb{P}^q}$ and $\omega_{\mathbb{P}^{q-1}}$ are the normalized Fubini–Study forms. Equip $\mathcal{O}_B(E) = \rho^* \mathcal{O}_{\mathbb{P}^q}(1) \otimes \tau^* \mathcal{O}_{\mathbb{P}^{q-1}}(-1)$ with the quotient metric, whose curvature is $\eta = \alpha - \gamma$, and let s_E be its canonical section. For an integer $k \geq 1$, equip

$$A = \rho^* \mathcal{O}_{\mathbb{P}^q}(k) \otimes \mathcal{O}_B(E) \quad \text{with curvature} \quad \theta = k\alpha + \eta.$$

Let $X = B \times \mathbb{P}^1$, with projection π to B. For a point $o \in \mathbb{P}^1$, use the normalized Fubini–Study metric on $\mathcal{O}_{\mathbb{P}^1}(o)$, and set

$$R = \|\pi^* s_E\|^2, \quad T = \|s_o\|^2, \quad \varphi = \tfrac{1}{2}\log(R^2 + T^3), \quad \omega = \pi^*\kappa + 2\omega_{\mathbb{P}^1},$$
$$\psi = \log \|s_E\|^2.$$

Then $\varphi \in \mathrm{PSH}(X, \omega)$, $\psi \in \mathrm{PSH}(B, \theta)$ and

$$\mathrm{MA}_\theta(\psi) = (k\alpha)^q, \qquad \int_B \mathrm{MA}_\theta(\psi) = k^q.$$

The residual $k\alpha$ cannot be made strictly positive by a smooth change of gauge. Nevertheless, for every $m \geq 1$ the selected and reference spaces coincide as subspaces of $H^0(B, A^m)$:

$$V_m = W_m = H^0(B, A^m) = s_E^m \rho^* H^0(\mathbb{P}^q, \mathcal{O}(km)), \qquad \dim V_m = \binom{km+q}{q}.$$

Proof Since (ρ, τ) is an embedding, $\kappa > 0$. The quotient metric satisfies

$$\|s_E\|^2 = \rho^* \left(\frac{\sum_{j=1}^q |Z_j|^2}{\sum_{j=0}^q |Z_j|^2} \right),$$

as in Example 28.10.4. The same calculation over a common denominator, now with coefficient $1/2$, gives

$$\mathrm{dd}^{\mathrm{c}} \varphi \geq -\pi^*\alpha - \tfrac{3}{2}\omega_{\mathbb{P}^1}$$

off the pole set $E \times \{o\}$, and across it by the extension theorem. Hence $\omega + \mathrm{dd}^{\mathrm{c}}\varphi \geq \pi^*\gamma + \frac{1}{2}\omega_{\mathbb{P}^1} \geq 0$. Along the smooth pole set $E \times \{o\}$, the function φ has the exact local model $\frac{1}{2}\log(|z|^4 + |w|^6) + O(1)$, and it is locally bounded elsewhere. It also satisfies the global lower bound $\varphi \geq \log R$.

We may therefore take $c = 1$, $u = 0$, $p = 2$, $b_1 = 3$ and $a = 2/3$, with residual $\beta = \theta - \eta = k\alpha \geq 0$. Its top integral is k^q, since the normalized Fubini–Study form has top integral one and ρ is a modification. Corollary 28.10.2 therefore gives

$$W_m = H^0(B, A^m(-mE)), \qquad V_m = H^0(B, A^m(-(m-1)E)).$$

Let $t \geq 0$ be an integer. A section of $\rho^*\mathcal{O}(km) \otimes \mathcal{O}_B(tE)$, divided by s_E^t, gives a holomorphic section of $\mathcal{O}(km)$ on $\mathbb{P}^q \setminus \{p\}$, which extends across p by Hartogs' theorem since $q \geq 2$. Pulling the extension back and multiplying by s_E^t recovers the original section off E, hence everywhere by the identity theorem. Thus every section of A^m has the form $s_E^m \rho^* s$, and the natural embeddings of W_m and V_m into $H^0(B, A^m)$ both have this image. Its dimension is the number $\binom{km+q}{q}$ of monomials of degree km.

Finally, let $C \subset E \simeq \mathbb{P}^{q-1}$ be a projective line; it is contracted by ρ. For every smooth real function u, Stokes' theorem gives $\int_C (k\alpha + \mathrm{dd}^{\mathrm{c}} u) = 0$, whereas a strictly

positive form has positive integral over C. Changing the smooth metric on the divisor bundle changes the residual only by dd^{c} of a smooth function, so this proves the obstruction. Moreover, φ is finite on most of each fiber over E and equals $-\infty$ at the point o of that fiber, so it does not descend, even up to a bounded function. □

Thus the selected and reference spaces agree, although the selecting potential does not descend and the reference has positive mass. The reason is that E is a fixed component of every section of A^m: all global sections vanish along E to order m, more than the local integrability threshold requires.

28.10.5 Fixed exceptional orders and coefficient thresholds

In Example 28.10.5 the fixed divisor E absorbs the local integrability conditions. If the coefficient of the selecting potential increases, the selected sections must eventually also vanish at the point that was blown up. The following family shows when this happens, and when the mass of the reference potential vanishes.

Keep the blowup $\rho\colon B \to \mathbb{P}^q$ with $q \geq 2$, its exceptional divisor E, the extended projection τ, the forms α, γ, κ, and the quotient metric on $\mathcal{O}_B(E)$ with curvature η from that example, and write $x = \rho(E)$ for the point blown up. Let $k \geq 1$ and $e \geq 0$ be integers and equip

$$A_e = \rho^{*}\mathcal{O}_{\mathbb{P}^q}(k) \otimes \mathcal{O}_B(eE), \qquad \theta = k\alpha + e\eta$$

with the product metric. Let $X = B \times \mathbb{P}^1$ with its projection π, let $o \in \mathbb{P}^1$, and let $R = \|\pi^{*}s_E\|^2$ and $T = \|s_o\|^2$ as before. For positive integers p, b and a real number $c > 0$, put

$$\varphi_c = \frac{c}{p}\log(R^p + T^b), \quad \omega_c = c\pi^{*}\kappa + (cb/p + 1)\omega_{\mathbb{P}^1}, \quad a = \frac{p}{b},$$

and let $V_m(c) \subseteq H^0(B, A_e^m)$ be the space selected by $\mathcal{I}(m\varphi_c)$, regarded on B by descent.

Proposition 28.10.1 *For every $c>0$ we have $\varphi_c\in \mathrm{PSH}(X,\omega_c)$, and the selected space is*

$$V_m(c)=s_E^{em}\rho^*H^0\big(\mathbb{P}^q,\mathcal{O}(km)\otimes\mathfrak{m}_x^{v_m}\big),$$
$$v_m=\max\{\lfloor m(c-e)-a\rfloor,0\},\qquad m\ge 1,$$

where $\mathfrak{m}_x$ is the ideal sheaf of x. For $0<c\le e+k$ put

$$d=\max\{c,e\},\quad \delta=d-e,\quad \psi_d=d\log\|s_E\|^2,\quad \ell_m=\lfloor m\delta\rfloor.$$

Then $\psi_d\in\mathrm{PSH}(B,\theta)$ and

$$\mathrm{MA}_\theta(\psi_d)=\big((k-\delta)\alpha+\delta\gamma\big)^q,\qquad \int_B\mathrm{MA}_\theta(\psi_d)=k^q-\delta^q.$$

The reference spaces satisfy

$$\begin{aligned}W_m(c)&:=H^0\big(B,A_e^m\otimes\mathcal{I}(m\psi_d)\big)\\&=s_E^{em}\rho^*H^0\big(\mathbb{P}^q,\mathcal{O}(km)\otimes\mathfrak{m}_x^{\ell_m}\big)\subseteq V_m(c).\end{aligned}$$

If $0<c\le e$, both spaces equal $H^0(B,A_e^m)$ for every m. If $e<c<e+k$, the defect and the leading term of the dimension are

$$\dim V_m(c)-\dim W_m(c)=\binom{\ell_m+q-1}{q}-\binom{v_m+q-1}{q}=O(m^{q-1}),$$
$$\dim V_m(c)=\frac{k^q-(c-e)^q}{q!}m^q+O(m^{q-1}),$$

where $\binom{q-1}{q}=0$. At $c=e+k$ the reference ψ_d has zero mass and

$$\dim V_m(e+k)=\frac{(\lceil a\rceil+1)k^{q-1}}{(q-1)!}m^{q-1}+O(m^{q-2}).$$

If $c>e+k$, the selected space is zero exactly when

$$m(c-e-k)\ge a+1;$$

in particular, it is zero for every $m\ge\lceil (a+1)/(c-e-k)\rceil$.

Proof As in Example 28.10.5, the norm identity writes R^p+T^b over the common denominator for $\mathcal{O}(p,b)$ on $\mathbb{P}^q\times\mathbb{P}^1$, pulled back to X, and its numerator is a sum of squared moduli of holomorphic coefficients. The lower bound for the complex Hessian of the logarithm of such a sum, together with the extension theorem across the pole set, gives

$$\mathrm{dd^c}\varphi_c\ge -c\pi^*\alpha-(cb/p)\omega_{\mathbb{P}^1}.$$

Thus $\omega_c+\mathrm{dd^c}\varphi_c\ge c\pi^*\gamma+\omega_{\mathbb{P}^1}\ge 0$.

Hartogs' theorem, as in that example, shows that every global section of A_e^m can be written uniquely as $s_E^{em}\rho^*s$ with $s\in H^0(\mathbb{P}^q,\mathcal{O}(km))$. If s vanishes to order j at

x, this section vanishes to order $em + j$ generically along E. In a chart of the pole set, $\varphi_c = (c/p)\log(|z|^{2p} + |w|^{2b}) + O(1)$. The argument with a unit coefficient in the proof of Theorem 28.10.2 and Lemma 28.10.2 require

$$em + j + 1 > mc - a, \quad \text{equivalently} \quad j \geq \max\{\lfloor m(c-e) - a \rfloor, 0\}.$$

Conversely, this divisorial order gives integrability in every chart of the pole set, and φ_c is locally bounded elsewhere. The vanishing order of s at x equals the generic vanishing order of $\rho^* s$ along E, as the blowup coordinates of Example 28.10.4 show. This proves the formula for $V_m(c)$.

For $0 \leq \delta \leq k$, the Poincaré–Lelong formula gives

$$\theta + \mathrm{dd}^{\mathrm{c}}\psi_d = (k-\delta)\alpha + \delta\gamma + d[E] \geq 0.$$

The non-pluripolar measure does not charge E, and it equals the top power of the smooth residual $(k-\delta)\alpha + \delta\gamma$. The class of this residual is $k\rho^* c_1(\mathcal{O}(1)) - \delta[E]$. The mixed intersection numbers vanish and $\int_B [E]^q = (-1)^{q-1}$, as computed in Example 28.10.4, so its mass is $k^q - \delta^q$. The divisorial ideal is $\mathcal{I}(m\psi_d) = \mathcal{O}_B(-\lfloor md \rfloor E)$. Since em is an integer, $\lfloor md \rfloor = em + \ell_m$, and the condition on s is exactly $j \geq \ell_m$. This gives the formula for $W_m(c)$, and the inclusion follows from $\ell_m \geq v_m$. If $c \leq e$, both point orders are zero.

For an integer $0 \leq j \leq km + 1$, homogeneous polynomials of degree km realize every jet of total degree less than j at x, and there are $\binom{j+q-1}{q}$ such jets. Hence

$$h^0\big(\mathbb{P}^q, \mathcal{O}(km) \otimes \mathfrak{m}_x^j\big) = \binom{km+q}{q} - \binom{j+q-1}{q}.$$

For $e < c < e + k$ we have $0 \leq v_m \leq \ell_m \leq km$ and $\ell_m - v_m \leq \lceil a \rceil$, and subtraction gives the exact defect. Each additional jet degree contributes $O(m^{q-1})$ monomials, which gives the bound, and $v_m = m(c-e) + O(1)$ gives the leading term of $\dim V_m(c)$.

At $c = e + k$ we have $v_m = \max\{km - \lceil a \rceil, 0\}$. For large m, the allowed monomials have vanishing orders $km - \lceil a \rceil, \ldots, km$ at x, and each of these $\lceil a \rceil + 1$ degrees has $k^{q-1}m^{q-1}/(q-1)! + O(m^{q-2})$ monomials, which proves the asymptotic formula. The residual is then $k\gamma$, whose top power vanishes. For $c > e + k$, a nonzero polynomial of degree km vanishes at x to order at most km. Hence $V_m(c) = 0$ exactly when $v_m > km$, which is equivalent to $m(c - e - k) - a \geq 1$. □

For $0 < c < e + k$ the reference ψ_d has positive mass and the defect is $O(m^{q-1})$, so this family admits the approximation asked for in Problem 35.9.2. At $c = e + k$ the reference mass vanishes and the growth degree of the selected spaces drops to $q - 1$; beyond this threshold the selected spaces eventually vanish.

28.10.6 A semipositive residual with nonzero defect

Blowing up two points instead of one, we obtain a residual that is semipositive, cannot be made strictly positive, and still has positive top integral. Unlike the example with one blown-up point, the gap between the two divisorial thresholds now produces a nonzero defect of the largest possible order m^{q-1}. Counting the spaces requires the following elementary interpolation of jets at several points.

Lemma 28.10.3 *Let $x_1, \dots, x_N$ be distinct points of $\mathbb{P}^q$, $q \geq 1$, and let $d, j_1, \dots, j_N$ be nonnegative integers with $\sum_i j_i \leq d + 1$. Then the map from $H^0(\mathbb{P}^q, \mathcal{O}(d))$ to the jets of order less than j_i at the points x_i is surjective. In particular,*

$$h^0\left(\mathbb{P}^q, \mathcal{O}(d) \otimes \bigcap_i \mathfrak{m}_{x_i}^{j_i}\right) = \binom{d+q}{q} - \sum_i \binom{j_i + q - 1}{q},$$

where $\binom{q-1}{q} = 0$.

Proof We may discard the points with $j_i = 0$; if no point remains, the target is zero. Choose an affine chart containing all the remaining points, and use the corresponding trivialization of $\mathcal{O}(d)$. For each pair $i \neq k$ choose an affine linear function L_{ik} vanishing at x_k and nonzero at x_i, and set

$$Q_i = \prod_{k \neq i} L_{ik}^{j_k}.$$

Given a jet b_i at x_i, divide it by the invertible jet of Q_i at x_i, and let T_i be the Taylor polynomial of degree at most $j_i - 1$ of the quotient. Then $Q_i T_i$ has the jet b_i at x_i and vanishes to order at least j_k at every other point x_k, and its degree is at most $\sum_k j_k - 1 \leq d$. Hence $\sum_i Q_i T_i$, homogenized to degree d, realizes all the prescribed jets. In q variables there are $\binom{j_i+q-1}{q}$ monomials of total degree less than j_i, which gives the dimension formula. □

Example 28.10.6 Let $q \geq 2$ and let $\rho\colon B \to \mathbb{P}^q$ be the blowup of the two points $x_1 = [1 : 0 : \cdots : 0]$ and $x_2 = [0 : 1 : 0 : \cdots : 0]$, with exceptional divisors E_1, E_2. Let $\tau_i\colon B \to \mathbb{P}^{q-1}$ be the extension to B of the projection from x_i. With the normalized Fubini–Study forms, put

$$\alpha = \rho^*\omega_{\mathbb{P}^q}, \qquad \gamma_i = \tau_i^*\omega_{\mathbb{P}^{q-1}}, \qquad \kappa = \alpha + \gamma_1 + \gamma_2 > 0.$$

Equip $\mathcal{O}_B(E_i) = \rho^*\mathcal{O}(1) \otimes \tau_i^*\mathcal{O}(-1)$ with the quotient metric, whose curvature is $\eta_i = \alpha - \gamma_i$, and let s_i be its canonical section. Equip

$$A = \rho^*\mathcal{O}(2) \otimes \mathcal{O}_B(E_1 + E_2), \qquad \theta = 2\alpha + \eta_1 + \eta_2$$

with the product metric. On $X = B \times \mathbb{P}^1$, with projection π and a point $o \in \mathbb{P}^1$, set

$$\begin{gathered} R_i = \|\pi^* s_i\|^2, \qquad T = \|s_o\|^2, \\ \varphi = \log(R_1^2 + T^3) + \log(R_2^2 + T^3), \\ \omega = 4\pi^*\kappa + 7\omega_{\mathbb{P}^1}, \\ \psi = 2\log\|s_1\|^2 + 2\log\|s_2\|^2, \qquad \beta = \gamma_1 + \gamma_2. \end{gathered}$$

Then $\varphi \in \mathrm{PSH}(X, \omega)$, $\psi \in \mathrm{PSH}(B, \theta)$, and

$$\mathrm{MA}_\theta(\psi) = \beta^q, \qquad \int_B \beta^q = 2^q - 2 > 0.$$

No smooth change of gauge makes β strictly positive. Nevertheless, writing V_m and W_m for the selected and reference spaces as before, their images in $H^0(B, A^m)$ are, for every $m \geq 1$,

$$\begin{aligned} V_m &= s_1^m s_2^m \rho^* H^0\left(\mathbb{P}^q, \mathcal{O}(2m) \otimes \mathfrak{m}_{x_1}^{m-1}\mathfrak{m}_{x_2}^{m-1}\right), \\ W_m &= s_1^m s_2^m \rho^* H^0\left(\mathbb{P}^q, \mathcal{O}(2m) \otimes \mathfrak{m}_{x_1}^{m}\mathfrak{m}_{x_2}^{m}\right). \end{aligned}$$

The defect and the common leading term of the dimensions are

$$\begin{gathered} \dim V_m - \dim W_m = 2\binom{m+q-2}{q-1} \sim \frac{2}{(q-1)!}m^{q-1}, \\ \dim V_m = \frac{2^q - 2}{q!}m^q + O(m^{q-1}), \qquad \dim W_m = \frac{2^q - 2}{q!}m^q + O(m^{q-1}). \end{gathered}$$

Proof The map (ρ, τ_1, τ_2) is an embedding: away from the exceptional divisors ρ is already an embedding, and along E_i the map τ_i separates the tangent directions of E_i while ρ detects the normal direction. Hence $\kappa > 0$. The identity for the quotient metric is the one-point identity of Example 28.10.5, applied at each center. For each i, the calculation over a common denominator writes $R_i^2 + T^3$ as a sum of squared moduli of holomorphic coefficients over the metric denominator for $\mathcal{O}(2, 3)$. Together

with the extension theorem across the pole set, this gives

$$\mathrm{dd}^{\mathrm{c}}\varphi \geq -4\pi^{*}\alpha - 6\omega_{\mathbb{P}^1}.$$

It follows that $\omega + \mathrm{dd}^{\mathrm{c}}\varphi \geq 4\pi^{*}\beta + \omega_{\mathbb{P}^1} \geq 0$.

By the Poincaré–Lelong formula,

$$\theta + \mathrm{dd}^{\mathrm{c}}\psi = \beta + 2[E_1] + 2[E_2] \geq 0.$$

The non-pluripolar product does not charge the divisors and equals β^q. The class of β is $2H - [E_1] - [E_2]$, where $H = \rho^{*}c_1(\mathcal{O}(1))$. As in Example 28.10.4, $H^q = 1$, the mixed intersection numbers involving H and some $[E_i]$ vanish, and $[E_i]^q = (-1)^{q-1}$; moreover, the two exceptional divisors are disjoint. Expanding gives $\int_B \beta^q = 2^q - 2$.

Let C be the strict transform of the line through x_1 and x_2. Both maps τ_i are constant on C, so $\int_C \beta = 0$. By Stokes' theorem, $\int_C (\beta + \mathrm{dd}^{\mathrm{c}}u) = 0$ for every smooth real function u, whereas a strictly positive form has positive integral over C. This proves the obstruction, although the top integral is positive.

The selecting potential satisfies $\varphi \geq \pi^{*}\psi$. Its pole set is the disjoint union of the sets $E_i \times \{o\}$. Near either of them the other summand is smooth, and

$$\varphi = \log(|z|^4 + |w|^6) + O(1), \qquad E_i = \{z = 0\}, \quad o = \{w = 0\}.$$

Thus $c_i = p_i = 2$, $b_{i1} = 3$ and $a_i = 2/3$. By Theorem 28.10.2, $W_m = H^0(B, A^m(-2m(E_1+E_2)))$, and V_m is contained in $H^0(B, A^m(-(2m-1)(E_1+E_2)))$; for an integer ℓ, the strict local integrability condition $\ell + 1 > 2m - 2/3$ means $\ell \geq 2m - 1$. Conversely, by Lemma 28.10.2, divisibility by these orders gives integrability in every chart of the pole set, and off the pole set the selecting potential is locally bounded. This proves equality for V_m.

Dividing a section of A^m by $s_1^m s_2^m$ gives a section of $\mathcal{O}(2m)$ away from the two centers, which extends across them by Hartogs' theorem since $q \geq 2$. The pullback of the extension, multiplied by $s_1^m s_2^m$, agrees with the original section everywhere, by the identity theorem. If the extended section vanishes to order j_i at x_i, the corresponding section of A^m vanishes to order $m + j_i$ generically along E_i. The two divisorial conditions therefore give exactly the vanishing orders at x_1 and x_2 displayed in the statement. Since the two point ideals have disjoint supports, their product equals their intersection.

Apply Lemma 28.10.3 with $d = 2m$ and orders (m, m) or $(m-1, m-1)$; in both cases their sum is at most $d + 1$. This gives

$$\dim W_m = \binom{2m+q}{q} - 2\binom{m+q-1}{q},$$
$$\dim V_m = \binom{2m+q}{q} - 2\binom{m+q-2}{q}.$$

Pascal's identity gives the exact defect, and the leading terms give both growth laws. For $m = 1$ the vanishing orders for V_1 are zero, and the defect is already two. □

In this example the residual β has zero integral over the curve C, so no smooth change of gauge makes it positive, and yet its top integral is positive. Thus the bound $O(m^{q-1})$ of Theorem 28.10.2 cannot be improved to $o(m^{q-1})$ for semipositive residuals either. As in the previous examples, the selecting potential is finite on most of each fiber over E_i and equals $-\infty$ at the point o of that fiber, so it does not descend, even up to a bounded function.

28.11 Horizontal poles over projective space

When the base divisor is a hyperplane in projective space, the descended vanishing conditions of Theorem 28.10.1 turn the kernels into binomial tail probabilities. The large-deviation rate of these tails is the domination envelope of the original singularity upstairs, and the concentration of evaluation sections identifies the limiting measure for every smooth integration volume. In this setting the weak limit of Corollary 28.10.1 improves to convergence in total variation.

We specialize Section 28.10 to $B = \mathbb{P}^q$ with homogeneous coordinates $[Z_0 : \cdots : Z_q]$, $A = O_B(k)$ and $Z = \{Z_0 = 0\}$, where q, r, k are positive integers and $0 < c < k$. Let ω_B be the Fubini–Study form on B, normalized by $\int_B \omega_B^q = 1$, and give A the Fubini–Study metric, whose curvature is $\theta = k\omega_B$. We keep the submersion $\pi\colon X \to B$ with connected r-dimensional fibers, and the selecting potential φ with the singularity (28.19) along the section Y over Z. Write

$$R = \frac{|Z_0|^2}{\sum_{i=0}^q |Z_i|^2}, \qquad a = \frac{c}{k}, \qquad D_c = \{R \geq a\}, \qquad C_c = \frac{(k-c)^q}{q!}.$$

Fix a smooth positive probability volume ν_X on X. Fiber integration gives $\nu_B = \pi_* \nu_X = w\, \omega_B^q$ with a smooth function $w > 0$; put $M = \max_B w$. Let B_m^B be the kernel of the descended space V_m for the norm on B with zero continuous weight and volume ν_B, and let B_m^X be the kernel of the pulled-back space for the norm on X with zero continuous weight and volume ν_X. We shall see that $B_m^X = B_m^B \circ \pi$.

The selecting potential φ need not be $\pi^*\theta$-plurisubharmonic. Its domination envelope in this curvature background is

$$\mathcal{P}(\varphi) = \left(\sup\{u \in \mathrm{PSH}(X, \pi^*\theta) : u \leq 0,\ u \preceq \varphi\}\right)^*,$$

where $u \preceq \varphi$ means that $u \leq \varphi + C$ for some constant C (Definition 1.12.1). The limit will be described by the function of R defined by

$$u_c(R) = \begin{cases} -k\left(a \log \dfrac{a}{R} + (1-a) \log \dfrac{1-a}{1-R}\right), & 0 < R < a, \\ 0, & a \leq R \leq 1, \\ -\infty, & R = 0. \end{cases} \tag{28.20}$$

Theorem 28.11.1 *The envelope satisfies $\mathcal{P}(\varphi) = u_c \circ \pi$. With $\ell_m = \lfloor mc \rfloor$ and $t_m = \max\{\ell_m - r, 0\}$,*

$$N_m = \dim V_m = \binom{km - t_m + q}{q} = C_c m^q + O(m^{q-1}).$$

For every $m \geq 1$, in the extended-value sense,

$$\frac{1}{m} \log B_m^X \geq \mathcal{P}(\varphi) - \frac{\log M}{m}. \tag{28.21}$$

The functions $m^{-1} \log B_m^X$ converge to $\mathcal{P}(\varphi)$ in L^1 for every smooth positive volume on X, and locally uniformly on $X \setminus \pi^{-1}(Z)$. In the normalization of Section 28.1, the kernel measures converge in total variation:

$$\begin{aligned} \frac{q!}{m^q} B_m^B \nu_B &\longrightarrow \mu_c := \mathbf{1}_{D_c} \theta^q, \\ \frac{q!}{m^q} B_m^X \nu_X &\longrightarrow \Lambda_{\nu_X} := \frac{k^q \mathbf{1}_{D_c} \circ \pi}{w \circ \pi} \nu_X. \end{aligned} \tag{28.22}$$

Both limits have mass $(k-c)^q$, and $\pi_* \Lambda_{\nu_X} = \mu_c$. Dividing the unnormalized kernel measures by N_m instead gives the probability limits $\mu_c/(k-c)^q$ and $\Lambda_{\nu_X}/(k-c)^q$, while normalization by m^{q+r} gives zero in total variation.

Let W_m be the reference spaces of Theorem 28.10.1. For every sequence of intermediate spaces $W_m \subseteq U_m \subseteq V_m$, equipped with the restricted norms, the lower bound and all these convergences remain valid. The dimensions of these spaces are $C_c m^q + O(m^{q-1})$, and the dimension normalization then uses $\dim U_m$.

28.11.1 The domination envelope

We first compute the envelope on the base, using the affine chart $Z_0 \neq 0$. Set $z_i = Z_i/Z_0$, $s = |z|^2$, $l = k - c$, $s_0 = l/c$ and $C = k \log k - c \log c - l \log l$. In this chart the reference weight of A is $\phi_A = k \log(1+s)$. Put

$$F_0(z) = \begin{cases} k \log(1+s), & s \leq s_0, \\ l \log s + C, & s \geq s_0. \end{cases}$$

As a function of $\log s$, F_0 is increasing and convex: its derivative increases to l and then stays constant. Hence F_0 is plurisubharmonic. The two branches match to first order at $s = s_0$, and the tangent-line inequality gives $F_0 \leq \phi_A$. On a chart $Z_j \neq 0$ with $j \geq 1$, write $x = Z_0/Z_j$ and denote the remaining ratios by ζ. Near Z the corresponding weight is

$$F_j = c \log |x|^2 + l \log(1 + |\zeta|^2) + C.$$

Therefore F_0 and the F_j define a global plurisubharmonic metric weight F on A, with a divisor pole of coefficient c along Z and a smooth remainder.

Let $v \le \phi_A$ be a competing plurisubharmonic weight, that is, $v = \phi_A + u$ with $u \in \mathrm{PSH}(B,\theta)$ and $u \le 0$, and suppose that $v \le c\log|x|^2 + O(1)$ near Z. Write v_0 for its expression in the chart $Z_0 \neq 0$. The obstacle gives $v_0 \le F_0$ on $\{s \le s_0\}$. On each complex line $z = t\xi$ with $|\xi| = 1$, either $v_0(t\xi) \equiv -\infty$, or $v_0(t\xi) - l\log|t|^2$ is subharmonic on $|t| > \sqrt{s_0}$ and bounded above near infinity. In the coordinate $1/t$ the puncture at infinity is then removable, and the maximum principle bounds this difference by C, its bound on the circle $|t| = \sqrt{s_0}$ given by the obstacle. Varying ξ proves $v \le F$ on B. Consequently $F = \phi_A + u_c$ is the largest such weight, and its contact set $\{F = \phi_A\}$ is D_c.

We now return to X. By Proposition 1.22.1, every $\pi^*\theta$-plurisubharmonic function has the form $u \circ \pi$ for a θ-plurisubharmonic function u on B. Restriction to the local section $w_1 = \cdots = w_r = 0$ shows that $u \circ \pi \preceq \varphi$ implies $u \le c\log|x|^2 + O(1)$ near Z. Conversely, this divisorial bound and the inequality $|x|^2 \le |x|^2 + \sum_j |w_j|^2$ give the domination $u \circ \pi \le \varphi + O(1)$ near Y. A finite cover of Y gives one constant there. On the compact complement of a smaller neighborhood of Y, the function φ is bounded below and $u \le 0$, so one constant works on all of X. The two competitor families therefore correspond to each other, which proves $\mathcal{P}(\varphi) = u_c \circ \pi$. The pole of this envelope fills $\pi^{-1}(Z)$, although φ is finite on $\pi^{-1}(Z) \setminus Y$.

28.11.2 Binomial kernels

For a fixed integer $j \ge 0$, set $d = km$, $t_{m,j} = \max\{\ell_m - j, 0\}$ and $\mathcal{H}_m^{(j)} = H^0(B, O(d)(-t_{m,j}Z))$. The choices $j = 0$ and $j = r$ give W_m and V_m. Give these spaces the reference norm defined by the Fubini–Study metric, zero continuous weight and the volume $\nu_0 = \omega_B^q$. For a binomial random variable $\mathrm{Bin}(d, R)$ with d trials and success probability R, their kernels are

$$B^0_{m,j}(R) = A_d Q_{m,j}(R), \qquad A_d = \binom{d+q}{q}, \qquad Q_{m,j}(R) = \mathbb{P}\{\mathrm{Bin}(d,R) \ge t_{m,j}\}. \tag{28.23}$$

To see this, apply the monomial norm formula (28.18) in degree d. Invariance under the coordinate circle actions makes the monomials orthogonal, and the selection keeps exactly those in which Z_0 has exponent at least $t_{m,j}$. Summing over the other exponents with the multinomial identity gives (28.23). Counting the selected monomials gives $\dim \mathcal{H}_m^{(j)} = \binom{d-t_{m,j}+q}{q} = C_c m^q + O(m^{q-1})$.

We next bound the tails from below. Write $D(v \mid R) = v\log(v/R) + (1-v)\log((1-v)/(1-R))$ for the relative entropy, and put $b = \lfloor da \rfloor = \ell_m$. If $R \ge b/d$, the binomial mode $\min\{d, \lfloor (d+1)R \rfloor\}$ is at least b. Since $t_{m,j} \le b$, the tail above $t_{m,j}$ contains this mode and is therefore at least $1/(d+1)$. If $0 < R < b/d$, the mass at b for the parameter $v = b/d$ is a largest mass, hence at least $1/(d+1)$. For the parameter R,

the same mass is at least $(d+1)^{-1}\exp(-dD(v \mid R))$. Since

$$\frac{\partial D(v \mid R)}{\partial v} = \log \frac{v(1-R)}{R(1-v)} \geq 0 \quad (v \geq R)$$

and $v \leq a$, it is at least $(d+1)^{-1}\exp(-dD(a \mid R))$. Together with $u_c \leq 0$, these cases give $Q_{m,j} \geq (d+1)^{-1}\exp(mu_c)$, including the cases $b = 0$ and $R = 1$. Since $A_d \geq d+1$, the reference kernels satisfy $m^{-1}\log B^0_{m,j} \geq u_c$; at $R = 0$ this holds in the extended sense.

For the matching upper asymptotics, fix $0 < R < a$. Stirling's formula at the first selected index $t_{m,j}$ gives the logarithmic rate $-kD(a \mid R)$. Since $t_{m,j}/d \to a > R$, the ratios of successive binomial masses above that index are eventually bounded by a number less than one, so the whole tail has the same logarithmic rate. For $R > a$, the binomial variance estimate gives $Q_{m,j} \to 1$. At $R = a$, the lower bound by the mode and $Q_{m,j} \leq 1$ give the logarithmic limit zero. The functions $m^{-1}\log Q_{m,j}(R)$ increase with R, and their limit u_c is continuous on each interval $[\varepsilon, 1]$ with $\varepsilon > 0$. On a finite partition of such an interval on which the oscillation of u_c is small, monotonicity upgrades pointwise convergence to uniform convergence.

The event that the first $t_{m,j}$ trials succeed gives

$$c\log R \leq \frac{t_{m,j}}{m}\log R \leq \frac{1}{m}\log Q_{m,j}(R) \leq 0.$$

Since $\log R$ is integrable against every smooth volume, dominated convergence proves global L^1 convergence of the functions $m^{-1}\log Q_{m,j}$, and the additional term $m^{-1}\log A_d$ tends to zero. For the measures, the variance estimate gives $Q_{m,j} \to \mathbf{1}_{D_c}$ away from $R = a$. The pushforward of ν_0 by R is $q(1-R)^{q-1}\,\mathrm{d}R$, by the radial computation in the proof of Theorem 28.9.1 applied to $1-R$. In particular the level set $\{R = a\}$ has zero volume. Since $q!A_d/m^q \to k^q$ and these constants are bounded, dominated convergence proves total-variation convergence of $q!m^{-q}B^0_{m,j}\nu_0$ to μ_c, and

$$\mu_c(B) = k^q \int_a^1 q(1-R)^{q-1}\,\mathrm{d}R = (k-c)^q.$$

28.11.3 Changing the integration volume

Put $\delta = \min_B w > 0$, and write $B^w_{m,j}$ for the kernel of $\mathcal{H}^{(j)}_m$ with volume $w\nu_0$. Comparing the two norms gives

$$M^{-1} B^0_{m,j} \le B^w_{m,j} \le \delta^{-1} B^0_{m,j}.$$

Together with the preceding subsection, this proves the lower bound (28.21) on the base and the convergence of the functions $m^{-1}\log B^w_{m,j}$. Comparability alone does not identify the measure limit, however.

Let Π_m be the orthogonal projection onto $\mathcal{H}^{(j)}_m$ for the reference L^2 norm. In a unit frame at x, write $F_{m,x}$ for the evaluation vector of $H^0(B, \mathcal{O}(d))$ at x, and put $K_{m,x} = \Pi_m F_{m,x}$. Formula (28.23) gives

$$\|F_{m,x}\|_0^2 = A_d, \qquad \|K_{m,x}\|_0^2 = A_d Q_{m,j}(R(x)),$$
$$\|F_{m,x} - K_{m,x}\|_0^2 = A_d(1 - Q_{m,j}(R(x))).$$

The full evaluation section satisfies

$$|F_{m,x}(y)|^2_{h_A^m} = A_d^2 \tau(x,y)^d, \qquad \tau([Z],[W]) = \frac{|\langle Z, W\rangle|^2}{|Z|^2|W|^2}.$$

Hence $A_d^{-1}|F_{m,x}|^2\nu_0$ is a probability measure concentrating at x: outside any neighborhood of x we have $\tau \le \rho$ for some $\rho < 1$, so its mass there is at most $A_d\rho^d \to 0$. Continuity of w therefore gives $\|(w - w(x))F_{m,x}\|_0 = o(\sqrt{A_d})$ for each fixed x.

On $\mathcal{H}^{(j)}_m$, let $T_m s = \Pi_m(ws)$. The quadratic form of T_m is the weighted norm, so $\delta I \le T_m \le MI$, and the Riesz identity, as in the proof of Proposition 28.5.2, gives

$$B^w_{m,j}(x) = \langle T_m^{-1} K_{m,x}, K_{m,x}\rangle_0.$$

For $R(x) > a$, the preceding norm identities give $\|F_{m,x} - K_{m,x}\|_0 = o(\sqrt{A_d})$. Since $(T_m - w(x)I)K_{m,x} = \Pi_m\big((w - w(x))K_{m,x}\big)$, it follows that $\|(T_m - w(x)I)K_{m,x}\|_0 = o(\sqrt{A_d})$. Using $\|T_m^{-1}\| \le \delta^{-1}$, we obtain

$$B^w_{m,j}(x) = w(x)^{-1} B^0_{m,j}(x) + o(A_d) \qquad (R(x) > a).$$

For $R(x) < a$, the upper comparison of kernels gives $m^{-q} B^w_{m,j}(x) \to 0$. Finally, $0 \le q!m^{-q} w B^w_{m,j} \le (M/\delta) q! A_d/m^q$ is uniformly bounded. Dominated convergence away from the null set $\{R = a\}$ proves that $q!m^{-q}B^w_{m,j} w\nu_0 \to \mu_c$ in total variation.

Proof (Proof of Theorem 28.11.1) The identification of the selected spaces is Theorem 28.10.1, and the dimension formula follows from the monomial count. The envelope computation and the kernel calculations for the two endpoint spaces W_m, V_m prove the assertions on the base. Isometric descent gives $B^X_m = B^B_m \circ \pi$. For

$f_c = k^q \mathbf{1}_{D_c}/w$, fiber integration gives

$$\int_X \left| \frac{q!}{m^q} B_m^X - f_c \circ \pi \right| \nu_X = \int_B \left| \frac{q!}{m^q} B_m^B - f_c \right| \nu_B.$$

This proves the total-variation limit on X and the identity $\pi_* \Lambda_{\nu_X} = \mu_c$. The same fiber integration transfers the L^1 convergence, and every other smooth positive volume on the compact manifold X is comparable to ν_X. Local uniform convergence pulls back to $X \setminus \pi^{-1}(Z)$. The dimension asymptotics give the probability limits, and $N_m/m^{q+r} = O(m^{-r})$ proves the zero limit in the ambient degree.

For $W_m \subseteq U_m \subseteq V_m$, monotonicity of the kernels gives $B_{W,m} \le B_{U,m} \le B_{V,m}$. By Lemma 28.8.1, the positive difference of the q-normalized U_m and W_m measures has total mass at most $q!(\dim V_m - \dim W_m)/m^q = O(m^{-1})$. The lower bound and the local uniform convergence for U_m follow from those for the two endpoint spaces by sandwiching. The L^1 errors of the endpoint functions bound the error of the intermediate one, which proves global convergence as well. □

The limit on the base is independent of ν_X, but its lift is not. If $\nu_X(\mathrm{d}b, \mathrm{d}x) = \nu_B(\mathrm{d}b)\lambda_b(\mathrm{d}x)$ is the disintegration of ν_X along the fibers, then $\Lambda_{\nu_X}(\mathrm{d}b, \mathrm{d}x) = \mu_c(\mathrm{d}b)\lambda_b(\mathrm{d}x)$. Thus the lifted limit keeps exactly the conditional fiber probabilities over D_c. Comparing with Corollary 28.10.1 for $\nu = 0$, we see that $\mu_c = \mathrm{MA}_\theta(E(0))$, so Λ_{ν_X} is the quotient equilibrium measure of Section 28.7.1 for the fiber probabilities λ_b. Conversely, equality of two lifts implies equality of these probabilities for μ_c-almost every b: test against $(g \circ \pi) f$, with $g \in C(B)$ and f in a countable dense subset of $C(X)$, and take one common exceptional null set. The reverse implication follows directly by integration.

This dependence occurs in the product example of Example 28.10.1. Keep its selecting potential $\varphi = c \log(R_B + R_F)$ and its integer $b \ge c$, and let $\eta_f = k\omega_B + bf\omega_F$, where $f > 0$ is a smooth function on $\mathbb{P}^r$ with $b \min f > c$. Then $\eta_f + \mathrm{dd^c}\varphi \ge (k-c)\omega_B + (bf - c)\omega_F > 0$. The normalized volume of η_f and the corresponding limit are

$$\nu_{X,f} = \omega_B^q \otimes \frac{f^r \omega_F^r}{\int_{\mathbb{P}^r} f^r \omega_F^r}, \qquad \Lambda_{\nu_{X,f}} = \mu_c \otimes \frac{f^r \omega_F^r}{\int_{\mathbb{P}^r} f^r \omega_F^r}.$$

A nonconstant f changes the lift. For $r \ge 2$, it also makes η_f nonclosed: $\mathrm{d}\eta_f = b\,\mathrm{d}f \wedge \omega_F$, and if this form vanished, wedging with ω_F^{r-2} would give $\mathrm{d}f \wedge \omega_F^{r-1} = 0$, hence $\mathrm{d}f = 0$ by the injectivity of $\xi \mapsto \xi \wedge \omega_F^{r-1}$ on one-forms. For $r = 1$ this derivative vanishes for degree reasons.

The spaces, the envelope and the kernels do not change when the admissible Hermitian background varies while π, A, h_A, φ and ν_X stay fixed. As for the point poles of Section 28.8, moving the fiber point of $Y = Z \times \{[1 : 0 : \cdots : 0]\}$ to another point of $\mathbb{P}^r$ does not change them either, although for $mc \ge r + 1$ the constant germ fails to be integrable at the old center and becomes integrable there after the move.

28.12 Degenerate curvature and growing vanishing orders

In the compact models above, the vanishing orders are proportional to the tensor degree. To see the transition when the order grows more slowly, we pass to a model space of entire functions. There the local metric fixes the scale of the full kernel, while the selecting ideal creates a second, larger radius. Grouping the monomials according to the residues of their degrees expresses the transition at that radius through Poisson tails.

In this section r and q denote new integers. Fix an integer $r \geq 1$. On $\mathbb{C}$, with coordinate $z = x + \mathrm{i}y$ and area measure $\mathrm{d}A = \mathrm{d}x\,\mathrm{d}y$, put $\phi(z) = |z|^{2r}$. For a real $m > 0$, let $H_{m,r}$ be the Hilbert space of entire functions with norm

$$\|f\|_{m,r}^2 = \int_{\mathbb{C}} |f|^2 \mathrm{e}^{-m\phi}\,\mathrm{d}A.$$

For an integer $q \geq 0$, let $H_{m,r}^{(q)}$ consist of the functions vanishing to order at least q at zero. Since the weight is bounded below on each fixed disk, the local submean and Cauchy inequalities bound point and jet evaluations, so this subspace is closed. Write $K_{m,r}^{(q)}(z, w)$ for its reproducing kernel, holomorphic in z and antiholomorphic in w, and put

$$B_{m,r}^{(q)}(z) = \mathrm{e}^{-m\phi(z)} K_{m,r}^{(q)}(z, z).$$

Thus the ideal (z^q) selects the functions, while the smooth weight determines the norm. We use $\Gamma(a) = \int_0^\infty t^{a-1}\mathrm{e}^{-t}\,\mathrm{d}t$ for $a > 0$.

Proposition 28.12.1 *For every $m > 0$ and integer $q \geq 0$,*

$$K_{m,r}^{(q)}(z, w) = \frac{r}{\pi} \sum_{k=q}^{\infty} \frac{m^{(k+1)/r} (z\bar{w})^k}{\Gamma((k+1)/r)}. \tag{28.24}$$

The series and all its derivatives converge uniformly on compact sets. For $\zeta \in \mathbb{C}$,

$$m^{-1/r} B_{m,r}^{(q)}(m^{-1/(2r)}\zeta) = \frac{r}{\pi} \mathrm{e}^{-|\zeta|^{2r}} \sum_{k=q}^{\infty} \frac{|\zeta|^{2k}}{\Gamma((k+1)/r)}. \tag{28.25}$$

Consequently, if m tends to infinity through positive integers and $q = q_m \to \infty$, the left side tends to zero uniformly on compact ζ-sets.

Proof The monomials are orthogonal by rotational invariance. For every integer $k \geq 0$, polar integration with radial variable ρ and the substitution $t = m\rho^{2r}$ give

$$\|z^k\|_{m,r}^2 = 2\pi \int_0^\infty \rho^{2k+1} \mathrm{e}^{-m\rho^{2r}}\,\mathrm{d}\rho = \frac{\pi}{r} m^{-(k+1)/r} \Gamma((k+1)/r).$$

For an entire function $f(z) = \sum_{k\geq 0} a_k z^k$, Parseval's identity on circles and Tonelli's theorem yield $\|f\|_{m,r}^2 = \sum_{k\geq 0} |a_k|^2 \|z^k\|_{m,r}^2$. Thus the normalized monomials form

an orthonormal basis, and those with $k \geq q$ form an orthonormal basis of the selected subspace. This proves (28.24). By Stirling's formula, the reciprocal Gamma coefficients decay faster than every fixed geometric progression, even after multiplication by any fixed power of k; this gives uniform convergence on compact sets together with all derivatives. Substituting $z = w = m^{-1/(2r)}\zeta$ gives (28.25). On a disk $|\zeta| \leq R$ of fixed radius, the series in (28.25) is bounded by the tail, from the index q on, of the convergent series with $|\zeta|$ replaced by R. This tail tends to zero as $q \to \infty$, proving the last assertion. □

For $r = 1$ and $q = 0$ the density is the constant m/π. For general r and $q = 0$ its value at zero is $rm^{1/r}/(\pi\Gamma(1/r))$, whereas it vanishes at zero when $q > 0$. The scale $m^{-1/(2r)}$ therefore describes fixed vanishing orders, but a growing order requires a further rescaling.

28.12.1 A uniform tail estimate

Grouping the monomial degrees by their residues modulo r expresses the kernel through weighted Poisson tails. The estimate of Gamma ratios below removes those weights, with an error independent of the vanishing threshold.

Write $\mathbb{P}$ and $\mathbb{E}$ for probability and expectation. For $T > 0$, let N_T be a Poisson random variable with parameter T, so that $\mathbb{P}(N_T = \ell) = \mathrm{e}^{-T}T^{\ell}/\ell!$ for integers $\ell \geq 0$. For a real $0 < \alpha \leq 1$, define

$$w_{\alpha,T}(\ell) = T^{\alpha-1}\frac{\Gamma(\ell+1)}{\Gamma(\ell+\alpha)}.$$

All constants in the next lemma may depend on α, but not on T.

Lemma 28.12.1 *There is a constant $C_\alpha > 0$ such that, for $T \geq 1$,*

$$\mathbb{E}\left|w_{\alpha,T}(N_T) - 1\right| \leq C_\alpha T^{-1/2}.$$

Proof Recall that $\mathbb{E}N_T = T$ and $\mathrm{Var}(N_T) = T$. Stirling's formula, applied to the ratio of Gamma functions, gives

$$\begin{aligned}
\frac{\Gamma(\ell+1)}{\Gamma(\ell+\alpha)} &= \ell^{1-\alpha}(1 + O_\alpha(\ell^{-1})) && (\ell \to \infty),\\
\frac{\Gamma(\ell+1)}{\Gamma(\ell+\alpha)} &\leq C_\alpha(\ell+1)^{1-\alpha} && (\ell \geq 0).
\end{aligned}$$

The second bound follows from the first after the constant is increased to account for finitely many small integers. Since $0 \leq 1 - \alpha < 1$, it implies

$$w_{\alpha,T}(\ell) \leq C_\alpha\left(1 + \frac{\ell+1}{T}\right), \qquad \mathbb{E}(w_{\alpha,T}(N_T)+1)^2 \leq C_\alpha \quad (T \geq 1).$$

For $T \geq 2$, consider the event $E_T = \{|N_T - T| \leq T/2\}$. On this event, the expansion of the Gamma ratio and the bounded derivative of $s \mapsto s^{1-\alpha}$ on $[1/2, 3/2]$ give

$$|w_{\alpha,T}(N_T) - 1| \leq C_\alpha \left(\frac{|N_T - T|}{T} + \frac{1}{T} \right).$$

By the Cauchy–Schwarz inequality, the expectation of the right side is at most $C_\alpha T^{-1/2}$. On the complement, Chebyshev's inequality gives $\mathbb{P}(E_T^c) \leq 4/T$, and the preceding second-moment bound together with the Cauchy–Schwarz inequality gives the same estimate for the contribution of E_T^c. Enlarging the constant handles $1 \leq T \leq 2$. □

For integers $q \geq 0$ and $0 \leq s < r$, put

$$L_s(q) = \left\lceil \frac{q - s}{r} \right\rceil \geq 0.$$

If a monomial degree is written as $k = r\ell + s$, the condition $k \geq q$ becomes $\ell \geq L_s(q)$.

Proposition 28.12.2 *There is a constant $C_r > 0$, depending only on r, such that for $m > 0$, $z \neq 0$ and $T = m|z|^{2r} \geq 1$,*

$$\left| \frac{\pi B_{m,r}^{(q)}(z)}{m r^2 |z|^{2r-2}} - \frac{1}{r} \sum_{s=0}^{r-1} \mathbb{P}\{N_T \geq L_s(q)\} \right| \leq C_r T^{-1/2} \quad (q \geq 0). \tag{28.26}$$

Proof Let $\rho = |z| > 0$ and put $\alpha_s = (s+1)/r$ for integers $0 \leq s < r$. Grouping the positive series (28.24) by the residues s and using $T = m\rho^{2r}$, we obtain the exact identity

$$\begin{aligned} B_{m,r}^{(q)}(z) &= \frac{r}{\pi \rho^2} \sum_{s=0}^{r-1} \mathrm{e}^{-T} \sum_{\ell \geq L_s(q)} \frac{T^{\ell + \alpha_s}}{\Gamma(\ell + \alpha_s)} \\ &= \frac{rT}{\pi \rho^2} \sum_{s=0}^{r-1} \mathbb{E}\big[w_{\alpha_s,T}(N_T) \mathbf{1}_{\{N_T \geq L_s(q)\}} \big]. \end{aligned}$$

Divide by $m r^2 \rho^{2r-2}/\pi = r^2 T/(\pi \rho^2)$ and apply Lemma 28.12.1 to each of the finitely many residues. Since the indicators are bounded by one independently of their thresholds, the resulting error is uniform in q. □

28.12.2 The limiting envelope and the transition scale

A vanishing order proportional to m removes the kernel density inside a fixed circle. Inside that circle the limit of the potentials is logarithmic, and outside it agrees with the smooth weight. Throughout this subsection, m tends to infinity through positive

integers. For $\lambda > 0$, set $q_m = \lfloor m\lambda \rfloor$ and $R_\lambda = (\lambda/r)^{1/(2r)}$. Define

$$V_\lambda(z) = \begin{cases} |z|^{2r}, & |z| \geq R_\lambda, \\ \lambda \log(|z|^2/R_\lambda^2) + \lambda/r, & 0 < |z| < R_\lambda, \\ -\infty, & z = 0. \end{cases}$$

In the following theorem, δ_0 denotes the unit point mass at zero. Recall that in dimension one the non-pluripolar measure of a subharmonic function discards the mass carried by its polar set.

Theorem 28.12.1 *One has*

$$\frac{1}{m} \log K_{m,r}^{(q_m)}(z,z) \longrightarrow V_\lambda(z)$$

pointwise with extended values and in $L^1_{\mathrm{loc}}(\mathbb{C})$. *The kernel measures converge locally in total variation:*

$$\frac{1}{m} B_{m,r}^{(q_m)} \, \mathrm{d}A \longrightarrow \mathbf{1}_{\{|z|>R_\lambda\}} \mathrm{dd^c}\phi. \tag{28.27}$$

The function V_λ *is the largest subharmonic function* v *on* $\mathbb{C}$ *such that* $v \leq \phi$ *and* $v(z) \leq \lambda \log|z|^2 + C$ *near zero for some constant* C. *Its curvature current is*

$$\mathrm{dd^c} V_\lambda = \lambda\delta_0 + \mathbf{1}_{\{|z|>R_\lambda\}} \mathrm{dd^c}\phi. \tag{28.28}$$

Thus the limit in (28.27) *is the non-pluripolar measure of this envelope, rather than its full distributional Laplacian.*

Proof For fixed $z \neq 0$, put $\rho = |z|$ and $T = m\rho^{2r}$. Then $L_s(q_m)/T \to \lambda/(r\rho^{2r})$, and this limit differs from one when $\rho \neq R_\lambda$. Chebyshev's inequality and (28.26) therefore give

$$\frac{1}{m} B_{m,r}^{(q_m)}(z) \longrightarrow \begin{cases} 0, & 0 < \rho < R_\lambda, \\ r^2\rho^{2r-2}/\pi, & \rho > R_\lambda. \end{cases} \tag{28.29}$$

The selected density is at most the full density, that is, the density with $q = 0$. When $T \geq 1$, the same comparison with $q = 0$ bounds it by $C_r m\rho^{2r-2}$; when $T \leq 1$, (28.25) with $q = 0$ bounds it by $C_r m^{1/r} \leq C_r m$. Thus $m^{-1} B_{m,r}^{(q_m)}$ is uniformly bounded on each compact set. Since the circle $\rho = R_\lambda$ and the origin have zero area, dominated convergence and $\mathrm{dd^c}\phi = (r^2/\pi)|z|^{2r-2}\mathrm{d}A$ prove (28.27).

To compute the limit of the potentials inside the circle, fix $0 < \rho < R_\lambda$ and write

$$d_{m,k} = \frac{m^{(k+1)/r}\rho^{2k}}{\Gamma((k+1)/r)} \quad (k \geq q_m).$$

The estimate of Gamma ratios from Stirling's formula gives

$$\frac{d_{m,k+1}}{d_{m,k}} \leq \left(\frac{rm\rho^{2r}}{q_m+1}\right)^{1/r} \left(1 + \frac{C_r}{q_m+1}\right) \quad (k \geq q_m)$$

for all sufficiently large m. The right side tends to $(r\rho^{2r}/\lambda)^{1/r} < 1$. Hence the sum of these terms lies between d_{m,q_m} and a fixed multiple of that term, and Stirling's formula yields

$$\frac{1}{m}\log K_{m,r}^{(q_m)}(z,z) \longrightarrow \lambda\log\rho^2 + \frac{\lambda}{r}\bigl(1-\log(\lambda/r)\bigr) = V_\lambda(z).$$

For $\rho > R_\lambda$, the limit of the densities in (28.29) is positive. Hence $m^{-1}\log B_{m,r}^{(q_m)}(z) \to 0$, and therefore $m^{-1}\log K_{m,r}^{(q_m)}(z,z)$ tends to $\phi(z)$.

At $\rho = R_\lambda$, put $k_m = \max\{q_m, \lfloor rm\rho^{2r}\rfloor\}$. Then $k_m/m \to r\rho^{2r} = \lambda$. The single monomial of degree k_m and Stirling's formula give $\liminf m^{-1}\log K_{m,r}^{(q_m)}(z,z) \geq \phi(z)$, and the bound for the full kernel already proved gives the reverse inequality for the upper limit. At zero the selected kernel vanishes for all sufficiently large m. This proves pointwise convergence everywhere.

For local L^1 convergence, the bounds for the full kernel give a uniform upper bound for $m^{-1}\log K_{m,r}^{(q_m)}(z,z)$ on each compact set. The single selected monomial of degree q_m gives the lower bound

$$\frac{1}{m}\log K_{m,r}^{(q_m)}(z,z) \geq c_m + \frac{q_m}{m}\log|z|^2,$$

where c_m is bounded as $m \to \infty$, again by Stirling's formula. Since $0 \leq q_m/m \leq \lambda$, the negative part on a fixed compact set is bounded by a constant plus $2\lambda|\log|z||$ near zero, which is locally integrable. Dominated convergence proves the L^1_{loc} assertion.

We turn to the envelope. The two radial expressions for V_λ and their first radial derivatives agree at R_λ, because $rR_\lambda^{2r} = \lambda$. Away from zero and that circle, direct differentiation proves (28.28). Since the first derivatives match, the circle carries no measure, and $\mathrm{dd^c}\log|z|^2 = \delta_0$ gives the atom at zero. The right side is positive, so V_λ is subharmonic. For $t = |z|^{2r}/(\lambda/r) < 1$, the inequality $\log t + 1 \leq t$ proves $V_\lambda \leq \phi$ inside the circle; outside it there is equality.

Finally, let v be subharmonic with $v \leq \phi$ and $v \leq \lambda\log|z|^2 + C$ near zero. On the punctured disk $0 < |z| < R_\lambda$, the function $v - \lambda\log|z|^2$ is subharmonic and bounded above near zero, so it extends subharmonically across zero. On the boundary circle it is at most $\lambda/r - \lambda\log R_\lambda^2$, since $v \leq \phi = \lambda/r$ there. The maximum principle gives $v \leq V_\lambda$ inside the disk, and outside the disk this follows from $v \leq \phi$. Since V_λ itself satisfies the conditions, it is the largest such function. Removing the atom at the pole gives the last assertion. □

The measures in this theorem have infinite total mass on $\mathbb{C}$, so their convergence is asserted only on compact sets. The weighted potentials $m^{-1}\log B_{m,r}^{(q_m)}$ converge to $V_\lambda - \phi$. The radius R_λ is created by the selecting ideal even when $r = 1$, whereas $r > 1$ also reflects the curvature zero at the origin.

For any integer sequence $q_m \to \infty$, the transition at the boundary circle is governed by the central limit theorem for the same Poisson tails. Let Φ be the standard Gaussian

distribution function,

$$\Phi(\tau) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\tau} \mathrm{e}^{-s^2/2}\,\mathrm{d}s \quad (\tau \in \mathbb{R}).$$

Proposition 28.12.3 *Let (q_m) be a sequence of nonnegative integers with $q_m \to \infty$, and fix a real τ. Choose $z_m \neq 0$ with*

$$m|z_m|^{2r} = \frac{q_m}{r} + \tau\sqrt{\frac{q_m}{r}},$$

which is positive for all sufficiently large m. Then

$$\frac{\pi B_{m,r}^{(q_m)}(z_m)}{mr^2|z_m|^{2r-2}} \longrightarrow \Phi(\tau).$$

Proof We first recall the central limit theorem for Poisson variables in the form used here. For a fixed real $M > 0$, Stirling's formula for $\ell!$ gives, uniformly over integers ℓ with $|\ell - T| \le M\sqrt{T}$,

$$\mathbb{P}(N_T = \ell) = \frac{1}{\sqrt{2\pi T}} \exp\left(-\frac{(\ell - T)^2}{2T}\right)(1 + O_M(T^{-1/2})) \quad (T \to \infty).$$

Indeed, insert the expansion of the factorial in $\mathrm{e}^{-T}T^{\ell}/\ell!$ and expand the logarithm in powers of $(\ell - T)/T$; the cubic remainder is $O_M(T^{-1/2})$. Riemann sums over bounded intervals therefore converge to Gaussian integrals, and Chebyshev's bound $\mathbb{P}(|N_T - T| > M\sqrt{T}) \le M^{-2}$ then allows $M \to \infty$. Consequently, if real thresholds a_T satisfy $(a_T - T)/\sqrt{T} \to a \in \mathbb{R}$, then $\mathbb{P}(N_T \ge a_T) \to 1 - \Phi(a)$; rounding changes the normalized threshold by at most $T^{-1/2}$, and Φ is continuous.

Set $T_m = m|z_m|^{2r}$. Then $T_m \to \infty$ and, for every integer $0 \le s < r$,

$$\frac{L_s(q_m) - T_m}{\sqrt{T_m}} \longrightarrow -\tau.$$

Apply the preceding Poisson limit in (28.26). The uniform error there tends to zero, and each of the finitely many probabilities tends to $1 - \Phi(-\tau) = \Phi(\tau)$. □

For a sequence $q_m \to \infty$, put $R_m = (q_m/(rm))^{1/(2r)}$. At fixed τ, the radii in Proposition 28.12.3 satisfy

$$|z_m| = R_m\left(1 + \frac{\tau}{2\sqrt{rq_m}} + O_{r,\tau}(q_m^{-1})\right).$$

If $q_m = o(m)$, these radii tend to zero, while their ratio to the scale $m^{-1/(2r)}$ of the full kernel tends to infinity; when $r > 1$, they shrink to a curvature zero. To transfer this transition to a compact manifold, both the local comparison of norms and the

error of the global extension must be uniform in the growing vanishing order. The next section establishes both estimates on a curve.

28.13 Localization on curves

We now transfer the transition of Proposition 28.12.3 to a compact curve. The selected sections concentrate on a shrinking disk around the distinguished point. When the remainder of the metric weight is negligible on that disk, the norm there can be compared directly with the model norm, and restriction to the disk gives an upper bound for the kernel. For the lower bound, positivity of the curvature away from the distinguished point permits a global L^2 correction of localized model sections that preserves the growing vanishing order.

Let C be a connected compact Riemann surface, let $x \in C$, and let (L, h) be a holomorphic line bundle on C with a smooth Hermitian metric. Fix an integer $r \geq 1$. Suppose that in a holomorphic coordinate z centered at x and a holomorphic frame e,

$$|e|_h^2 = \mathrm{e}^{-\phi}, \qquad \phi(z) = |z|^{2r} + O(|z|^{2r+1}).$$

Let ν be a smooth positive volume form on C, written as $a(z)\,\mathrm{d}A$ in this coordinate, where $\mathrm{d}A$ is Euclidean area, and assume that $a(0) = 1$. For positive integers m, choose nonnegative integers q_m such that

$$q_m \longrightarrow \infty, \qquad q_m = o(m^{1/(2r+1)}).$$

Write $\mathfrak{m}_x$ for the ideal sheaf of x, and equip

$$V_m = H^0(C, L^m \otimes \mathfrak{m}_x^{q_m})$$

with the norm $\|s\|_m^2 = \int_C |s|_{h^m}^2 \,\mathrm{d}\nu$. Its density is $B_m = \sum_j |s_{m,j}|_{h^m}^2$ for any orthonormal basis $(s_{m,j})$, and $B_m = 0$ if $V_m = 0$. As in Section 28.12, Φ denotes the standard Gaussian distribution function.

Theorem 28.13.1 *For a fixed real τ, let x_m lie in the coordinate neighborhood and satisfy*

$$T_m := m|z(x_m)|^{2r} = q_m/r + \tau\sqrt{q_m/r}.$$

Then

$$\limsup_{m\to\infty} \frac{\pi B_m(x_m)}{mr^2|z(x_m)|^{2r-2}} \le \Phi(\tau). \tag{28.30}$$

If the curvature of (L,h) is semipositive on C and strictly positive on $C\setminus\{x\}$, this ratio converges to $\Phi(\tau)$.

Proof We compare B_m with the model kernels of Section 28.12 on a shrinking disk, which gives the upper bound. For the lower bound, we cut off model polynomials concentrated near x and correct them to global sections by an L^2 estimate. Put $z_m = z(x_m)$, $\rho_m = |z_m|$, $R_m = (q_m/(rm))^{1/(2r)}$ and

$$S_m = \frac{mr^2\rho_m^{2r-2}}{\pi}.$$

Then $R_m \to 0$ and $\rho_m/R_m \to 1$, and $T_m > 0$ for all sufficiently large m. Positive constants denoted by M, with subscripts when needed, may change between estimates. They are independent of m and of the argument of z_m; they may depend on the fixed geometric data, r, τ, and fixed auxiliary cutoffs or radii. We write $q = q_m$ when no confusion is possible. The model densities $B_{m,r}^{(q)}$ are those of Section 28.12.

Step 1: the kernel on a shrinking disk. Fix a real $A > 1$ and set $D_m = \{|z| < AR_m\}$. On D_m,

$$m|\phi(z) - |z|^{2r}| \le M_A m R_m^{2r+1} = M_A \frac{q_m R_m}{r} = o(1), \qquad \sup_{D_m} |a-1| = o(1).$$

The first limit follows from $q_m R_m = r^{-1/(2r)}(q_m^{2r+1}/m)^{1/(2r)} \to 0$. Hence, on D_m, the norm defined by h^m and ν and the radial norm with weight $\mathrm{e}^{-m|z|^{2r}}\,\mathrm{d}A$ differ by a factor $1 + o(1)$, uniformly over all holomorphic functions on D_m. The same comparison holds for the corresponding pointwise weights at z_m.

For integers $k \ge 0$, let $I_{m,k}$ and $J_{m,k}$ be the squared norms of z^k on D_m and on the whole plane, respectively, with this radial weight. Polar integration gives

$$I_{m,k} = \frac{\pi}{r} m^{-(k+1)/r}\gamma((k+1)/r, U_m), \qquad U_m = m(AR_m)^{2r},$$
$$J_{m,k} = \frac{\pi}{r} m^{-(k+1)/r}\Gamma((k+1)/r), \qquad \gamma(b,U) = \int_0^U t^{b-1}\mathrm{e}^{-t}\,\mathrm{d}t.$$

Here $b, U > 0$. By Parseval's identity on circles and Tonelli's theorem, the normalized monomials form a complete orthonormal system in the space of holomorphic functions on D_m with finite radial norm. Indeed, the squared norm of a Taylor series is the sum of the squared norms of its monomials, so its partial sums converge in that norm.

Therefore the density of the subspace of functions vanishing to order at least q is

$$B_m^D(z_m) = \mathrm{e}^{-T_m} \sum_{k \ge q} \frac{\rho_m^{2k}}{I_{m,k}}.$$

Point evaluation is continuous by the local submean inequality, so this sum is also the supremum of $\mathrm{e}^{-T_m}|f(z_m)|^2$ over the unit ball of that subspace.

Choose fixed real numbers $1 < c < d < A^{2r}$ and put $k_m = \lfloor rcT_m \rfloor$. For large m, $k_m \ge q$ and $U_m \ge dT_m$. Uniformly for $0 \le k \le k_m + 1$,

$$\frac{I_{m,k}}{J_{m,k}} = 1 + O(\mathrm{e}^{-\eta T_m}) \tag{28.31}$$

for some fixed $\eta > 0$. Indeed, let G_b be a random variable with Gamma density $t^{b-1}\mathrm{e}^{-t}/\Gamma(b)$ on $(0,\infty)$. A change of variable gives $\mathbb{E}\mathrm{e}^{sG_b} = (1-s)^{-b}$ for $0 < s < 1$. With $b = (k+1)/r \le cT_m + O_r(1)$, Markov's inequality yields

$$1 - \frac{\gamma(b, U_m)}{\Gamma(b)} \le \mathrm{e}^{-sU_m}(1-s)^{-b}.$$

Since $d > c$, choose $s > 0$ small enough that $sd + c\log(1-s) > 0$. The right side is at most $M\mathrm{e}^{-\eta T_m}$, as required.

The Cauchy–Schwarz inequality for the positive radial measure gives $I_{m,k+1}^2 \le I_{m,k} I_{m,k+2}$, so $I_{m,k+1}/I_{m,k}$ increases with k. By (28.31) and the asymptotics of Gamma ratios,

$$\rho_m^2 \frac{I_{m,k_m}}{I_{m,k_m+1}} = (1+o(1))T_m^{1/r} \frac{\Gamma((k_m+1)/r)}{\Gamma((k_m+2)/r)} \longrightarrow c^{-1/r} < 1.$$

Choose $\kappa \in (c^{-1/r}, 1)$. For all large m, the ratio of consecutive terms $\rho_m^{2k}/I_{m,k}$ is at most κ for every $k \ge k_m$. Consequently,

$$\mathrm{e}^{-T_m} \sum_{k > k_m} \frac{\rho_m^{2k}}{I_{m,k}} \le \frac{\kappa}{1-\kappa} \mathrm{e}^{-T_m} \frac{\rho_m^{2k_m}}{I_{m,k_m}} = o(S_m).$$

For the last equality, replace I_{m,k_m} by J_{m,k_m} using (28.31). The resulting term is one of the terms of the series for $B_{m,r}^{(k_m)}(z_m)$. In (28.26) for this kernel, each residue threshold divided by T_m tends to $c > 1$, so Chebyshev's inequality makes all the Poisson tails tend to zero, and the comparison error is $O(T_m^{-1/2})$. Hence $B_{m,r}^{(k_m)}(z_m) = o(S_m)$.

For the terms with $q \le k \le k_m$, we use (28.31). Since $I_{m,k} \le J_{m,k}$ for every k, we obtain

$$B_{m,r}^{(q)}(z_m) \le B_m^D(z_m) \le (1+o(1))B_{m,r}^{(q)}(z_m) + o(S_m).$$

By Proposition 28.12.3,

$$B_m^D(z_m)/S_m \longrightarrow \Phi(\tau). \tag{28.32}$$

The global norm of a selected section dominates its norm on D_m. Restriction to D_m and the comparison of norms on D_m therefore give $B_m(x_m) \le (1+o(1))B_m^D(z_m)$, also when $V_m = 0$. This proves (28.30).

Step 2: polynomials with almost all their norm near the point. From now on we assume the curvature hypotheses of the last assertion. Fix a real $c > 1$ and put $K_m = \lfloor cq_m \rfloor$. Define

$$C_m = \sum_{k=q_m}^{K_m} \frac{\rho_m^{2k}}{J_{m,k}}, \qquad p_m(z) = C_m^{-1/2} \sum_{k=q_m}^{K_m} \frac{\bar z_m^k z^k}{J_{m,k}}.$$

For large m these are well defined, $\|p_m\|_{m,r} = 1$, and

$$\mathrm{e}^{-T_m}|p_m(z_m)|^2 = B_{m,r}^{(q_m)}(z_m) - B_{m,r}^{(K_m+1)}(z_m) = (\Phi(\tau)+o(1))S_m. \tag{28.33}$$

Here the first equality holds because p_m is the normalized evaluation vector at z_m of the span of $z^{q_m},\dots,z^{K_m}$. The omitted kernel $B_{m,r}^{(K_m+1)}(z_m)$ is $o(S_m)$ by (28.26), since its Poisson thresholds divided by T_m tend to $c > 1$; the second equality then follows from Proposition 28.12.3.

Choose $\delta \in (0,1/2)$ so that $\phi(z) \ge \beta|z|^{2r}$ on the coordinate disk $|z| < \delta$ for a fixed $\beta \in (1/2,1)$, and a is bounded above there. For a sufficiently large fixed $A > 1$,

$$\int_{AR_m<|z|<\delta} |p_m|^2 \mathrm{e}^{-m\phi} a \, \mathrm{d}A = o(1). \tag{28.34}$$

To see this, write $p_m(z) = \sum_{k=q_m}^{K_m} b_{m,k} z^k$, so that $\sum_k |b_{m,k}|^2 J_{m,k} = 1$. The lower bound for ϕ, the orthogonality of monomials on circles and polar integration bound the left side by a fixed constant times

$$\sum_{k=q_m}^{K_m} |b_{m,k}|^2 J_{m,k}\, \beta^{-(k+1)/r} \mathbb{P}\{G_{(k+1)/r} \ge \beta A^{2r} q_m/r\}.$$

The exponential moment of G_b at $s = 1/2$ gives, for $b, U > 0$,

$$\beta^{-b}\mathbb{P}\{G_b \ge U\} \le \exp\bigl(-U/2 + b\log(2/\beta)\bigr).$$

Choose A so large that $\beta A^{2r}/2 > c\log(2/\beta)+1$. Since $(k+1)/r \le (cq_m+1)/r$, the product of the last two factors in the sum is $O(\mathrm{e}^{-q_m/r})$, uniformly in k. This proves (28.34). On $|z| < AR_m$, the norm defined by h^m and ν and the radial model norm differ by a factor $1+o(1)$, as shown in Step 1, and the model norm of p_m on this disk is at most 1.

Fix a smooth cutoff function χ with $0 \le \chi \le 1$, compactly supported in $|z| < \delta$ and equal to one near zero, and extend $\chi p_m e^{\otimes m}$ by zero to C. The estimate (28.34) and the comparison on $|z| < AR_m$ give

$$\|\chi p_m e^{\otimes m}\|_m^2 \le 1 + o(1). \tag{28.35}$$

The numbers δ, β, A and the cutoff χ are fixed before m tends to infinity.

Step 3: a global correction with the same vanishing condition. Let $f_m = \bar\partial(\chi p_m e^{\otimes m})$. Its support is contained in a fixed annulus on which ϕ has a positive lower bound. The Cauchy–Schwarz inequality and the identity $\sum_k |b_{m,k}|^2 J_{m,k} = 1$ give, for $|z| < \delta < 1$,

$$|p_m(z)|^2 \le \sum_{k=q_m}^{K_m} \frac{|z|^{2k}}{J_{m,k}} \le M(K_m+1)m^{(K_m+1)/r}.$$

Here we used that $\Gamma(b)$ has a positive lower bound for $b \ge 1/r$. Since $q_m \log m = o(m)$, the factor $\mathrm{e}^{-m\phi}$ on the annulus dominates this growth of the coefficients.

Regard ν as a Kähler form on the curve; it defines pointwise norms of forms and the induced metric $h_{K_C^{-1}}$ on the anticanonical bundle K_C^{-1}. Choose a smooth cutoff function η with $0 \le \eta \le 1$, supported in the coordinate chart and equal to one near x; shrink its support if necessary. For a sufficiently small fixed $t_0 > 0$, the global smooth functions

$$\ell_t = \eta \log(|z|^2 + t^2), \qquad 0 < t \le t_0,$$

extended by zero outside the chart, are nonpositive. Their complex Hessians are nonnegative where $\eta = 1$, and elsewhere they are bounded below independently of t, because $\eta = 1$ near x.

Choose a fixed smooth real function ψ that equals a sufficiently large positive multiple of $|z|^2$ near x and is cut off inside the chart, so that

$$\mathrm{i}\Theta(K_C^{-1}) + \mathrm{i}\partial\bar\partial\psi \ge 2\nu$$

on a neighborhood of x where $\eta = 1$. Here $\mathrm{i}\Theta$ denotes the Chern curvature form, without the factor $1/(2\pi)$, so that the weight $\mathrm{e}^{-\psi}$ contributes $\mathrm{i}\partial\bar\partial\psi$. For all large m and all $0 < t \le t_0$, equip $E_m = L^m \otimes K_C^{-1}$ with the smooth metric

$$H_{m,t} = (h^m \otimes h_{K_C^{-1}})\, \mathrm{e}^{-\psi - q_m \ell_t}.$$

This metric satisfies

$$\mathrm{i}\Theta(E_m, H_{m,t}) \ge \nu. \tag{28.36}$$

Indeed, near x the two terms involving h and ℓ_t are nonnegative, and the other two together are at least 2ν. On the complement of a fixed such neighborhood, compactness and strict positivity give a lower bound $c_0 m\nu$ for $m\mathrm{i}\Theta(L,h)$, with $c_0 > 0$. Since $q_m = o(m)$, this term absorbs all remaining negative terms, which are bounded by $M(q_m+1)\nu$.

On the support of f_m, the functions ℓ_t are bounded below uniformly in t. The polynomial estimate, the bounded derivatives of χ, and $q_m \log m = o(m)$ therefore give a constant $a_0 > 0$ such that, for all large m,

$$\int_C |f_m|^2_{h^m,\nu} \mathrm{e}^{-\psi - q_m\ell_t}\, \mathrm{d}\nu \le \mathrm{e}^{-a_0 m} \qquad (0 < t \le t_0).$$

On a curve, $K_C = \Lambda^{1,0}$, so an L^m-valued $(0,1)$-form is the same as an E_m-valued $(1,1)$-form, and the induced metrics give it the same pointwise norm. The anti-canonical factor in E_m, whose curvature enters (28.36), accounts for this change of bidegree. In bidegree $(1,1)$ on a curve, (28.36) makes the curvature operator at least the identity. Since the curve is compact and every $(1,1)$-form on it is $\bar\partial$-closed, the L^2 existence theorem for complete Kähler manifolds [Dem11, Theorem 5.1] applies. It gives an L^m-valued weak solution $u_{m,t}$ with

$$\bar\partial u_{m,t} = f_m, \qquad \int_C |u_{m,t}|^2_{h^m} \mathrm{e}^{-\psi - q_m \ell_t}\, \mathrm{d}\nu \le \mathrm{e}^{-a_0 m}.$$

Now fix m and let $t \downarrow 0$. Since $\ell_t \le 0$ and ψ is bounded, the solutions $u_{m,t}$ are bounded in the original L^2 norm, and a subsequence converges weakly to some u_m. Continuity of $\bar\partial$ in the sense of distributions gives $\bar\partial u_m = f_m$. For fixed $t' > 0$ and $t < t'$, we have $\ell_t \le \ell_{t'}$, so the same bound holds in the smooth weighted space with weight $\mathrm{e}^{-\psi - q_m \ell_{t'}}$, whose norm is equivalent to the original one for this fixed m and t'. Weak lower semicontinuity, followed by monotone convergence as $t' \downarrow 0$, gives

$$\int_C |u_m|^2_{h^m} \mathrm{e}^{-\psi - q_m \ell}\, \mathrm{d}\nu \le \mathrm{e}^{-a_0 m}, \qquad \ell = \eta \log |z|^2.$$

In particular, $\|u_m\|^2_m \le M \mathrm{e}^{-a_0 m}$. Near x the right side f_m vanishes, so the Cauchy–Riemann equation, in the sense of distributions, makes u_m holomorphic in the frame $e^{\otimes m}$. The smooth factors in the weight are bounded above and below near x for each fixed m, so the coefficient of u_m is square integrable against $|z|^{-2q_m}$. By Lemma 1.16.1 in dimension one, this forces vanishing order at least q_m at x.

Since p_m also vanishes to order at least q_m at zero, it follows that $s_m = \chi p_m e^{\otimes m} - u_m$ is a global holomorphic section in V_m. By (28.35), $\|s_m\|_m \le 1 + o(1)$.

Step 4: evaluation at the moving point. For large m, the disk D_m lies in the region where $\chi = 1$ and $f_m = 0$. Apply the local comparison of norms and the extremal property of B^D_m to the holomorphic function representing u_m on D_m, which vanishes to order at least q_m. By (28.32),

$$|u_m(x_m)|^2_{h^m} \le (1 + o(1)) B^D_m(z_m) \|u_m\|^2_m \le M S_m \mathrm{e}^{-a_0 m}.$$

The pointwise factor $\mathrm{e}^{-m\phi(z_m)}$ differs from e^{-T_m} by a factor $1 + o(1)$. Since $\Phi(\tau) > 0$, (28.33) and the triangle inequality give

$$|s_m(x_m)|^2_{h^m} \ge (\Phi(\tau) + o(1)) S_m.$$

In particular, these sections are nonzero for large m. The extremal characterization of B_m and $\|s_m\|_m \le 1 + o(1)$ yield the matching lower bound, which proves the limit.□

The hypothesis $q_m = o(m^{1/(2r+1)})$ enters only through $m R_m^{2r+1} = o(1)$, which makes the first remainder term of the metric weight negligible throughout the disk carrying the norm. For larger orders, the curvature quadrature domains of Section 28.13.1 absorb the harmonic part of this error and determine where the

transition occurs. Theorem 28.13.2 gives the corrected transition for all sublinear orders, and Theorem 28.13.3 determines the optimal range in which the uncorrected law holds for every metric. On a fibration over a curve, the kernel identities of Section 28.7 transfer these transitions whenever the descended spaces carry the same vanishing conditions and the induced norm.

28.13.1 Curvature quadrature domains

The disk comparison above requires the error of the metric weight to be small throughout the disk carrying the norm. A larger range of orders is obtained by absorbing the harmonic part of this error into the holomorphic sections. The transition then takes place on the boundary of the region where the remaining obstacle potential detaches from the weight with first-order contact, and a quadrature identity for the curvature determines that boundary.

Keep $C, x, (L,h), r, z, e$ and $\nu = a(z)\,\mathrm{d}A$ from Section 28.13, and assume that the curvature of (L,h) is semipositive on C and strictly positive on $C \setminus \{x\}$. We now allow integers $q = q_m \to \infty$ with $q = o(m)$. Write $V_{m,q} = H^0(C, L^m \otimes \mathfrak{m}_x^q)$, and let $B_{m,q}$ be its density for the norm of Section 28.13. Put

$$\rho = \phi_{z\bar z}, \qquad t = q/m, \qquad R = (t/r)^{1/(2r)}.$$

Write $\mathbb{D}$ for the unit disk.

Lemma 28.13.1 *Let $b > 1$, and let $Q(R,\zeta)$ be a real function, jointly smooth for small real R and $|\zeta| < b$, with $Q(0,\zeta) = |\zeta|^{2r}$. Write $Q_R = Q(R,\cdot)$, and suppose that every Q_R is subharmonic and that $Q_{R,\zeta\bar\zeta} > 0$ on a fixed neighborhood of the unit circle. For small R, let U_R be the largest subharmonic function on $\{|\zeta| < b\}$ such that $U_R \le Q_R$ and $U_R \le 2r\log|\zeta| + O(1)$ near zero. Then the noncontact set $\Omega_R = \{U_R < Q_R\}$ is connected, and its boundary is a radial graph near the unit circle. Let $F_R \colon \mathbb{D} \to \Omega_R$ be the conformal map with $F_R(0) = 0$ and $F_R'(0) > 0$. For every fixed k and all sufficiently small R, depending on k, the map F_R and the harmonic function $(U_R - 2r\log|\zeta|)\circ F_R$ depend smoothly on R in the Hölder spaces $C^{k,\alpha}(\overline{\mathbb{D}})$, and they differ from their values at $R = 0$ by $O(R)$.*

The arguments below use only finitely many of these norms, so the dependence of the range of R on k is harmless.

Proof Fix $k \ge 3$ and $0 < \alpha < 1$. Let $A^{k,\alpha}$ be the real Banach space of functions that are holomorphic on $\mathbb{D}$, of class $C^{k,\alpha}$ on its closure, and real at zero. For $h \in A^{k,\alpha}$ near zero set $f(u) = u(1 + h(u))$. For small h, the map f is univalent, with $f(0) = 0$ and $f'(0) > 0$. On $|u| = 1$ define

$$K(R,h) = f(u)(Q_R)_\zeta(f(u)) - r, \qquad T(R,h) = (P_- K, \mathrm{Re}[K]_0),$$

where P_- is the projection onto the strictly negative Fourier modes and $[\cdot]_0$ is the constant Fourier coefficient.

The Cauchy projections are bounded on these Hölder spaces. This follows from the Hilbert kernel on the circle: subtract the value at the evaluation point, split the integral at twice the length of the increment, and use the Hölder bound for differences on the inner part and the bound for the differentiated kernel on the outer part. This gives the $C^{0,\alpha}$ bound, and commuting angular derivatives with the projections gives the $C^{k,\alpha}$ bound. The superposition operator $h \mapsto (Q_R)_\zeta \circ f$ involves no derivatives of h and is smooth. Thus T is a smooth map into the product of the space of negative-mode functions with $\mathbb{R}$. At $(R, h) = (0, 0)$ we have $T = 0$ and

$$D_h K(0,0)v = r^2(v + \bar{v}), \qquad D_h T(0,0)v = (r^2\overline{v - v(0)},\ 2r^2 v(0)).$$

The derivative $D_h T(0, 0)$ is a bounded real-linear isomorphism, so the implicit function theorem in Banach spaces gives a unique normalized branch f_R of solutions of $T(R, h) = 0$, depending smoothly on R in the fixed space, with $\|f_R - \mathrm{id}\| = O(R)$. The branches obtained for larger k agree with this one by uniqueness in the smaller space. For each fixed order, this gives the spatial estimates and the expansions in R needed below, on a parameter interval that depends on the order.

The equation $T = 0$ prescribes only the real part of $[K]_0$; we show that the imaginary part vanishes as well. Since $P_- K = 0$, the function K extends holomorphically to $\mathbb{D}$, and $K(0) = [K]_0$ is purely imaginary. On the other hand, Stokes' formula gives

$$\frac{1}{2\pi \mathrm{i}} \int_{|u|=1} (Q_R)_\zeta (f_R(u)) f_R'(u)\, \mathrm{d}u = \frac{1}{\pi} \int_{f_R(\mathbb{D})} (Q_R)_{\zeta\bar{\zeta}}\, \mathrm{d}A \in \mathbb{R}.$$

On the circle the integrand equals $(r + K) f_R'/f_R$, whose residue at zero is $r + K(0)$. Hence $K(0)$ is real as well, and therefore zero. Consequently

$$S_R(\zeta) = \frac{r + K(f_R^{-1}(\zeta))}{\zeta} = r/\zeta + H_R(\zeta)$$

with H_R holomorphic on $\Omega_R = f_R(\mathbb{D})$, and $S_R = (Q_R)_\zeta$ on the boundary of Ω_R. Let

$$U_R = 2r \log|\zeta| + 2\,\mathrm{Re} \int_0^\zeta H_R(w)\, \mathrm{d}w + C_R.$$

Its complex gradient agrees with that of Q_R on the boundary, so the difference $Q_R - U_R$ is constant there; choose the constant C_R to make it zero. Then the difference and its gradient vanish on the boundary. Since the tangential second derivative of the difference also vanishes there, its inward normal second derivative equals $\Delta Q_R > 0$. Hence $Q_R - U_R > 0$ in an inner collar of uniform width. At $R = 0$,

$$U_0 = 1 + 2r \log|\zeta|, \qquad |\zeta|^{2r} - 1 - 2r \log|\zeta| > 0 \quad (0 < |\zeta| < 1).$$

By uniform convergence, this gap keeps a positive minimum on the remaining compact subsets of the interior for small R, and the logarithmic term controls a neighborhood of zero.

Set $U_R = Q_R$ outside Ω_R. Since values and gradients match across the boundary, the Laplacian of the patched function carries no measure on the boundary. Its curvature is $r\delta_0$ inside Ω_R and $\mathrm{dd}^{\mathrm{c}} Q_R \geq 0$ outside, so it is an admissible minorant. If v is any competitor, then $v - U_R$ is subharmonic on $\Omega_R \setminus \{0\}$, bounded above near zero and nonpositive on the boundary. Removability of the singularity at zero and the maximum principle give $v \leq U_R$ in Ω_R, and outside Ω_R this follows from $v \leq Q_R$. This identifies the envelope and shows that there are no further noncontact components. The radial graph property and the estimates for the harmonic part follow from the normalized map and its dependence on the parameter. We henceforth write $F_R = f_R$. □

We apply the lemma to the given weight. By Taylor's formula with integral remainder, for small real R,

$$Q_R(\zeta) = R^{-2r}\phi(R\zeta) = |\zeta|^{2r} + RE(R,\zeta)$$

with a jointly smooth function E. Each Q_R is subharmonic, since the curvature of (L,h) is semipositive, and $(Q_R)_{\zeta\bar\zeta} = r^2|\zeta|^{2r-2} + O(R)$ is positive near the unit circle, so the lemma applies. By its proof, the envelope in Ω_R has the form

$$U_R = 2r\log|\zeta| + 2\operatorname{Re} G_R(\zeta), \qquad G_R = 1/2 + O(R),$$

where G_R is holomorphic on Ω_R and normalized by $G_R(0) \in \mathbb{R}$. Set $D_t = R\Omega_R$. In the original coordinate z, the obstacle potential is $\phi_t(z) = R^{2r}U_R(z/R)$ inside D_t and ϕ outside. Replacing ϕ by ϕ_t in this chart and keeping h elsewhere defines a global singular metric on L. Its local potential is C^1 across the boundary, bounded above by ϕ, and has the pole $2t\log|z| + O(1)$. On the boundary, $\phi_z = t/z + g_t'$ for a holomorphic function g_t. Green's formula therefore gives the quadrature identity

$$\int_{D_t} \rho(z) f(z)\,\mathrm{d}A = \pi t f(0) \tag{28.37}$$

for every function f holomorphic near $\overline{D_t}$. The curvature of the singular metric is $t\delta_x$ inside D_t and the original curvature outside. Finally, we write the boundary as $\Gamma_t = \{Rc_R(\theta)\mathrm{e}^{\mathrm{i}\theta}\}$ with a positive periodic function c_R, where $c_R - 1 = O(R)$ in each fixed norm used below.

28.13.2 Gaussian transitions at curvature boundaries

To prove the transition at the boundary Γ_t, we again correct localized sections by an L^2 estimate. The singular metric of Section 28.13.1, whose curvature at x is the point mass $t\delta_x$, enforces the growing vanishing order of the correction. Its curvature vanishes on the rest of D_t, so a bounded function varying on the scale R must first restore strict positivity there.

Lemma 28.13.2 *There are constants $c, C > 0$ with the following properties, for all sufficiently large m. There is a smooth function b_R, bounded independently of m, such that the metric on $L^m \otimes K_C^{-1}$ induced by the singular local potential $\psi_m = m\phi_t + b_R$ on L^m and by the anticanonical metric of ν has curvature at least $(c/R^2)\nu$ on C. On annuli outside D_t whose radii are comparable to R, the lower bound improves to $(cq/R^2)\nu$. If an L^m-valued $(0,1)$-form α is supported on such an annulus, the equation $\bar\partial v = \alpha$ has a solution v, depending linearly on α, with*

$$\|v\|^2_{h^m} \le C\frac{R^2}{q}\|\alpha\|^2_{h^m,\nu}, \tag{28.38}$$

where both sides are the L^2 norms defined by h^m and ν. If α vanishes near x, then v is holomorphic near x and vanishes to order at least q there.

Proof Take $b_R = |z|^2/R^2$ on a disk containing D_t, and cut it off before the radius $3R$, so that its negative curvature is supported outside D_t. This function is uniformly bounded and its Hessian is $O(R^{-2})$. On the support of its negative curvature, the curvature $m\rho$ is comparable to q/R^2; since $q \to \infty$, it absorbs the negative part. Inside D_t, the function b_R contributes the curvature c/R^2. Elsewhere in a fixed coordinate disk, the original curvature is bounded below by a fixed multiple of $m|z|^{2r-2}$, and off that disk it is at least cm. Since $mR^2 \to \infty$, all these bounds dominate c/R^2 and the bounded Ricci term of ν.

We use the L^2 existence theorem applied in the proof of Theorem 28.13.1, now with a pointwise lower bound for the curvature and a singular metric. As in the proof of Theorem 28.13.1, we regard L^m-valued $(0,1)$-forms as $(1,1)$-forms with values in $L^m \otimes K_C^{-1}$ and sections of L^m as $(1,0)$-forms with values in the same bundle; this identification preserves the pointwise norms and brings in the Ricci term of ν. The estimate then reads as follows: if H is a singular metric on L^m and the curvature of $L^m \otimes K_C^{-1}$ is at least $\lambda\nu$ for a continuous function $\lambda > 0$, then

$$\int_C |v|^2_H \, \mathrm{d}\nu \le \int_C \lambda^{-1}|\alpha|^2_{H,\nu} \, \mathrm{d}\nu.$$

On the support of α we have $\lambda^{-1} = O(R^2/q)$, $\phi_t = \phi$, and b_R is bounded. Elsewhere $\phi_t \le \phi$ gives $\mathrm{e}^{-m\phi} \le C\mathrm{e}^{-\psi_m}$. These comparisons yield (28.38). Choosing the solution of minimal norm in the singular weighted Hilbert space makes the correction linear in its source.

The estimate requires a continuous lower bound λ. We take λ equal to c/R^2 inside D_t and to cq/R^2 on the exterior annulus, interpolating across a thin exterior collar of Γ_t. For the cutoff sources used below, whose supports are separated from Γ_t, the collar can be chosen to miss the source. For an exterior source whose support accumulates on Γ_t, we let the collar shrink to Γ_t; since Γ_t has zero area, dominated convergence in the integrals of $\lambda^{-1}|\alpha|^2$ and weak compactness in the fixed weighted Hilbert space give the same estimate. The singular metric is handled by approximation, which gives the same estimate for a weak solution. Where the source vanishes, the Cauchy–Riemann equation in the sense of distributions makes the solution holomorphic. Near x, the

weight is $|z|^{-2q}$ times a smooth positive factor. A holomorphic term z^j is square integrable against it precisely when $j \geq q$, so the solution has the required vanishing order. □

Theorem 28.13.2 *Let D_t and Γ_t be as in Section 28.13.1. Let $d_t(z)$ be the signed Euclidean distance from z to Γ_t, positive outside D_t, and let $p_t(z)$ be the nearest point of Γ_t. For every fixed $M > 0$, uniformly over points z with $|d_t(z)| \leq MR/\sqrt{q}$,*

$$\frac{\pi a(z) B_{m,q}(z)}{m\rho(z)} - \Phi\big(2\sqrt{m\rho(p_t(z))}\, d_t(z)\big) \longrightarrow 0. \tag{28.39}$$

If $d_t(z) = o(R)$ and $\sqrt{q}\, d_t(z)/R$ tends to $-\infty$ or $+\infty$, the same normalized kernel tends to 0 or 1, respectively.

Proof We pass to logarithmic coordinates adapted to Γ_t, in which the norm of a section of order q becomes Gaussian in the distance to the boundary. Restriction to a thin ring gives the upper bound, and localized trial sections corrected by Lemma 28.13.2 give the lower bound. Finally we compare with the full kernel to obtain saturation. Throughout the proof, c and C denote positive constants independent of m, which may change from line to line.

Step 1: the effective potential. Define on the unit disk

$$A_R(w) = G_R(F_R(w)) + r \log(F_R(w)/w),$$
$$\Psi_R = Q_R(F_R(w)) - 2\,\mathrm{Re}\, A_R(w) - 2r \log|w|,$$

where log is the principal branch, which is defined because $F_R(w)/w$ is close to one. Then $A_R = 1/2 + O(R)$ and $F_R = w + O(R)$ in every fixed norm used below. The effective potential $\Psi_R = (Q_R - U_R) \circ F_R$ is positive inside the disk, and its value and gradient vanish on the boundary. For $w = \mathrm{e}^{v+\mathrm{i}\theta}$ it has the expansion

$$\Psi_R = (2r^2 + O(R))v^2 + O(v^3), \tag{28.40}$$

and the exact quadratic coefficient is $2(Q_R)_{\zeta\bar{\zeta}}(F_R(\mathrm{e}^{\mathrm{i}\theta}))|F_R'(\mathrm{e}^{\mathrm{i}\theta})|^2$.

The maps need not extend holomorphically across the circle. To obtain charts that also cover a neighborhood outside the circle, we truncate their Taylor series at the degree $K = \lfloor q^{1/16} \rfloor$ and denote the truncations by F_K, A_K. The Taylor coefficients of a function bounded in a sufficiently high fixed Hölder norm decay faster than a large fixed power of their index, so the truncation error is $O(q^{-10})$, together with the fixed number of derivatives needed here. Fix a small $c_0 > 0$. On $|w| \leq 1 + c_0/K$ the derivatives of the truncations remain bounded and $F_K - w, A_K - 1/2 = O(R)$, because the factor $(1 + c_0/K)^K \leq \mathrm{e}^{c_0}$ controls the sums. In particular, F_K is univalent there for small R. Put $\delta = c_0/K$ and

$$\Psi_K = Q_R(F_K) - 2\,\mathrm{Re}\, A_K - 2r \log|w|.$$

The errors in the value and the first derivative of Ψ_K on the circle are $O(q^{-10})$, and uniformly for $|v| \le \delta$,

$$\Psi_K = (2r^2 + O(R) + O(|v|))v^2 + O(q^{-10})(1 + |v|). \tag{28.41}$$

Inside the disk, the effective potential has uniform positive quadratic growth near the circle, a positive lower bound on compact annuli farther inside, and logarithmic growth at zero.

In the coordinate $z = RF_K(w)$, a local holomorphic section of order at least q at x is uniquely represented, in the original frame, as

$$s = w^q \exp((q/r)A_K(w))f(w), \qquad |s|^2_{h^m} = |f(w)|^2 \mathrm{e}^{-(q/r)\Psi_K(w)}, \tag{28.42}$$

with f holomorphic near zero. The volume in this chart is $R^2(1 + O(R))\,\mathrm{d}A_w$. Put $\eta = \sqrt{rq}\,v$. On each band $|\eta| \le N$ with N fixed,

$$(q/r)\Psi_K = 2\eta^2 + o(1).$$

Step 2: the upper bound. Let the observation point have coordinate η_0 in a fixed bounded interval. Restricting the norm to the band $|\eta| \le N$ bounds the kernel from above by the kernel of the restricted space. For $f(w) = \sum_{j \ge 0} b_j w^j$, the limiting diagonal norms of the monomials on this band are

$$n_{j,N} = \frac{2\pi R^2}{\sqrt{rq}} I_N(j/\sqrt{rq}), \qquad I_N(s) = \int_{-N}^{N} \mathrm{e}^{2s\eta - 2\eta^2}\,\mathrm{d}\eta.$$

After normalization by $rq/(\pi R^2)$, the limiting upper kernel is

$$\frac{1}{2}\mathrm{e}^{-2\eta_0^2} \int_0^\infty \frac{\mathrm{e}^{2s\eta_0}}{I_N(s)}\,\mathrm{d}s. \tag{28.43}$$

The passage from sums to integrals uses Riemann sums with mesh $1/\sqrt{rq}$. To control their tails, bound $I_N(s)$ from below by its integral over a fixed subinterval of $[-N, N]$ lying beyond every observation point η_0; this gives an exponential bound uniform in the mesh. Thus we may first pass to the Riemann integral and then let $N \to \infty$. Since $I_\infty(s) = \sqrt{\pi/2}\,\mathrm{e}^{s^2/2}$, the expression (28.43) tends to

$$\frac{1}{\sqrt{2\pi}} \int_0^\infty \mathrm{e}^{-(s-2\eta_0)^2/2}\,\mathrm{d}s = \Phi(2\eta_0). \tag{28.44}$$

Step 3: the lower bound. Fix $T > 0$, and use (28.42) with f in the span of the monomials w^j, $0 \le j \le T\sqrt{rq}$. For $f = \sum_j b_j w^j$, put $\|s\|^2_{\mathrm{trial}} = \sum_j |b_j|^2 n_j$, with n_j given by (28.46) below. Let χ be a radial cutoff function in the w-plane, supported in $|w| \le 1 + \delta$ and equal to one on $|w| \le 1 + \delta/2$. The annulus on which χ is not constant lies outside D_t. Equations (28.41) and (28.42) bound the squared norm of

the source, as a quadratic form on the entire span, by

$$\|\bar\partial(\chi s)\|^2_{h^m,\nu} \le C(R\delta)^{-2}\mathrm{e}^{-cq\delta^2}\|s\|^2_{\mathrm{trial}};$$

the linear term $2jv$ in the exponent is absorbed because $\sqrt{q}\,\delta \to \infty$. Lemma 28.13.2 gives a correction g_s, depending linearly on s, with $\bar\partial g_s = \bar\partial(\chi s)$ and

$$\|g_s\|^2_{h^m} \le C(q\delta^2)^{-1}\mathrm{e}^{-cq\delta^2}\|s\|^2_{\mathrm{trial}}. \tag{28.45}$$

The constants here depend on m only through R and q. The factor R^2 in (28.38) cancels the factor R^{-2} coming from the derivative of the cutoff; this cancellation is needed even when q grows more slowly than every power of m.

On this span, the Gram form of the global norm of χs converges, in the sense of relative operator norm, to the diagonal form

$$n_j = \frac{2\pi R^2}{\sqrt{rq}}\sqrt{\frac{\pi}{2}}\exp(j^2/(2rq)). \tag{28.46}$$

Indeed, the ratio of the weights is uniformly close to one on each fixed band in η. Outside such a band, the radial bound for the weight and Parseval's identity on circles bound arbitrary combinations by diagonal tails $\mathrm{e}^{-c\eta^2+2T|\eta|}$. We first pass to the limit on a fixed band and then enlarge it. The positive lower bound of the effective potential on compact annuli and its logarithmic singularity at zero control the remaining part of the disk. This argument controls the whole quadratic form, without a factor depending on the dimension of the span.

To evaluate the corrections we need a pointwise bound. Let u be a section holomorphic on a disk of radius $R/\sqrt{q}$ centered at a point z with $|z|$ comparable to R. Then a local submean estimate gives

$$|u(z)|^2_{h^m} \le C(q/R^2)\|u\|^2_{h^m}. \tag{28.47}$$

To prove it, multiply u by the exponential of a holomorphic polynomial that removes the harmonic part of the linear and quadratic Taylor terms of $m\phi$ at z. After rescaling the disk to unit size, the remaining quadratic coefficient is bounded and the third-order error is $O(q^{-1/2})$, which proves the estimate. It applies to the corrections, because their sources lie on the cutoff annulus, away from the observation points. Together with (28.45), it shows that the kernel of the corrected global trial sections differs by $o(q/R^2)$ from the local trial kernel. After normalization, the lower limit is the integral in (28.44) with upper limit T instead of infinity, and letting $T \to \infty$ gives the matching lower bound.

Step 4: return to the original coordinate. Let p be the nearest boundary point and $\mathrm{e}^{\mathrm{i}\theta}$ its preimage under F_R. Then

$$v = \frac{d_t(z)}{R|F_R'(\mathrm{e}^{\mathrm{i}\theta})|} + O(d_t(z)^2/R^2) + o(q^{-1/2}),$$
$$m\rho(p)R^2|F_R'(\mathrm{e}^{\mathrm{i}\theta})|^2 = rq(1+O(R)).$$

Moreover, $m\rho(z)/a(z) = (rq/R^2)(1 + o(1))$ in the band $|d_t(z)| \le MR/\sqrt{q}$. These identities convert $2\eta_0$ into the argument in (28.39) and give the normalization of the density there. This proves (28.39).

Step 5: the full density and saturation. Let B_m^{full} be the density of the full space $H^0(C, L^m)$. At points with $|z| \asymp R$, it satisfies

$$B_m^{\mathrm{full}}(z) = \frac{m\rho(z)}{\pi a(z)}(1 + o(1)), \tag{28.48}$$

uniformly in an $o(R)$-neighborhood of Γ_t. We give a direct localization argument, which also yields the saturation statements of the theorem. Remove the harmonic Taylor terms of $m\phi$ at the observation point and rescale by $(m\rho(z))^{-1/2}$. On every fixed rescaled disk, the remaining weight tends to $|u|^2$, where u is the rescaled coordinate, and the volume density tends to the constant $a(z)$. The Taylor estimate is uniform, because the third derivatives of ϕ at radius R are $O(R^{2r-3})$, so the normalized third-order error is $O(q^{-1/2})$. Restriction to such disks bounds the evaluation norm from above, and the Gaussian norm of the constant function on them tends to π as their radius grows. This proves the upper bound in (28.48).

For the lower bound, take the holomorphic exponential canceling the harmonic Taylor terms, and cut it off on a disk of radius εR centered at the observation point, where $\varepsilon \to 0$ and $\sqrt{q}\,\varepsilon \to \infty$. Its local squared norm is $(\pi a(z)/(m\rho(z)))(1 + o(1))$ times its squared weighted value at the center, which is one. The quadratic lower bound for the weight bounds the source on the cutoff annulus by an exponential factor $\mathrm{e}^{-cq\varepsilon^2}$. Add the same bounded function of scale R to the original metric and use the pointwise $\bar\partial$ estimate. The relative squared norm of the correction is bounded by $C(q\varepsilon^2)^{-1}\mathrm{e}^{-cq\varepsilon^2} = o(1)$, after decreasing c if necessary, and its value at the center is $o(1)$ by (28.47). This proves the matching lower bound.

For saturation inside D_t, use the polynomial chart $z = RF_K(w)$ of Step 1 and put $u = 1 - |w| > 0$. Restrict the norm in (28.42) to the exterior half-band $0 \le v \le c/\sqrt{q}$. On this half-band every monomial w^j, $j \ge 0$, has norm at least $cR^2/\sqrt{q}$. Parseval's identity on circles and the Cauchy–Schwarz inequality give

$$B_{m,q}(z) \le C\frac{\sqrt{q}}{R^2}\frac{\mathrm{e}^{-cqu^2}}{1 - |w|^2}.$$

If $u\sqrt{q} \to \infty$, this bound divided by q/R^2 tends to zero. The error $O(q^{-10})$ of the chart changes neither the hypothesis nor the conclusion.

For saturation outside D_t, cut off the localized Gaussian section at z on a disk of radius

$$\varepsilon R, \qquad \varepsilon = \min\{d_t(z)/(3R), q^{-1/4}\}.$$

This disk lies entirely outside D_t, and again $\varepsilon \to 0$ and $\sqrt{q}\,\varepsilon \to \infty$. Apply Lemma 28.13.2 with the singular obstacle metric, which equals the original metric on the support of the source. The correction has negligible norm and value, and it vanishes to order q at x. Thus a member of $V_{m,q}$ asymptotically attains the full

Gaussian density, and (28.48) gives the reverse inequality. This proves saturation to one. □

Thus, for every sublinear order, the transition is centered on Γ_t and has width $1/(2\sqrt{m\rho})$, which is comparable to $R/\sqrt{q}$.

28.13.3 Growth ranges and compact examples

In the coordinate z, the quadrature boundary Γ_t differs from the model circle $|z| = R$ by $O(R^2)$. Comparing this displacement with the transition width $R/\sqrt{q}$ gives the range of orders in which the uncorrected law holds for every metric. A holomorphic change of coordinate produces a compact example in which the displacement has exactly this size.

Theorem 28.13.3 *Keep the setting of Section 28.13.1, fix a real τ, and consider observation points with coordinates z_m and the law*

$$\frac{\pi B_{m,q}(z_m)}{mr^2|z_m|^{2r-2}} \longrightarrow \Phi(\tau), \qquad m|z_m|^{2r} = q/r + \tau\sqrt{q/r}, \tag{28.49}$$

If $q = o(m^{1/(r+1)})$, this law holds for every choice of the geometric data and every sequence of arguments of z_m. This range is optimal: for every sequence $q \to \infty$ with $q = o(m)$ that is not $o(m^{1/(r+1)})$, there is a compact example satisfying the same hypotheses in which the law fails along a subsequence.

For fixed geometric data, write Γ_t as $\{|z| = Rc_R(\theta)\}$. Then (28.49) holds for every fixed τ and every sequence of arguments if and only if

$$\sqrt{q}\,\|c_R - 1\|_\infty \longrightarrow 0. \tag{28.50}$$

Proof By Lemma 28.13.1, $c_R - 1 = O(R)$ in every fixed norm required here. The observation points in (28.49) satisfy

$$|z_m|/R = 1 + \tau/(2\sqrt{rq}) + O(q^{-1}).$$

Since the slope of the boundary graph tends to zero, the argument of Φ in Theorem 28.13.2 is $\tau + o(1)$ whenever $\sqrt{q}R \to 0$, and this condition is equivalent to $q = o(m^{1/(r+1)})$. Moreover, $a(z) = 1 + o(1)$ and $\rho(z) = r^2|z|^{2r-2}(1 + o(1))$ at these points, so the two normalizations agree. Thus Theorem 28.13.2 proves the law for all data in this range.

The same reasoning shows that (28.50) is sufficient. For necessity, suppose that (28.50) fails. On a subsequence, choose an angle realizing the maximal boundary displacement and observe at $|z| = R$ with $\tau = 0$. A nonzero finite subsequential limit of the scaled displacement gives a nonzero Gaussian argument, and an infinite one gives saturation to zero or one. Radial and normal distances have relative error

tending to zero, because the graph has vanishing slope. In either case the limit differs from $1/2$.

It remains to construct the compact examples. On $\mathbb{P}^1$ choose a radial metric on $\mathcal{O}(d)$ whose local potential $f(u)$, $u = |w|^2$, equals u^r near zero, such that $g(u) = uf'(u)$ is strictly increasing on $(0, \infty)$ and tends to d, and such that $g(u) = d - \epsilon/u$ for large u, with a constant $\epsilon > 0$. Such a smooth interpolation exists for a sufficiently large integer d: prescribe a positive function g', equal to r^2u^{r-1} near zero and to ϵ/u^2 near infinity, with total integral d. Then $f = d \log u + \text{constant} + \epsilon/u$ near infinity, so the potential in the frame at infinity is smooth with strictly positive curvature. Choose a smooth radial volume form equal to $\mathrm{d}A_w$ near zero.

For sufficiently small t, the domain D_t lies in the region where $f(u) = u^r$ and equals the disk $|w| < R$: rotational invariance and uniqueness make it a disk, and its curvature mass is $rR^{2r} = t$. Hence the transition and saturation statements of Theorem 28.13.2 apply to this compact example.

Now use instead the coordinate z defined by $w = z + cz^2$, with a constant $c > 0$. In this coordinate the same compact metric satisfies $\phi(z) = |z|^{2r} + O(|z|^{2r+1})$, the volume form still has $a(0) = 1$, and vanishing orders are unchanged. At the uncorrected observation point $z = R$ with $\tau = 0$,

$$2\sqrt{mr^2R^{2r-2}}\,(|w| - R) = 2c\sqrt{r}\,\sqrt{q}R,$$

and the normalizing factors computed in the two coordinates have ratio tending to one. If q is not $o(m^{1/(r+1)})$, there is a subsequence on which $\sqrt{q}R$ tends to a positive finite number or to infinity. Theorem 28.13.2, together with its saturation statement when the argument diverges, then gives a limit strictly larger than $1/2$. This proves that the range is sharp. □

28.13.4 Corrections from the smooth weight

The boundary Γ_t can be expanded in powers of R, and the leading terms of this expansion give explicit corrections to the uncorrected law. Write

$$\phi(z) = |z|^{2r} + P_{2r+1}(z, \bar{z}) + O(|z|^{2r+2}), \qquad P(\theta) = P_{2r+1}(\mathrm{e}^{\mathrm{i}\theta}, \mathrm{e}^{-\mathrm{i}\theta}),$$

where P_{2r+1} is a real homogeneous polynomial of degree $2r + 1$ in $(z, \bar{z})$. Let Λ be the Dirichlet-to-Neumann operator of the unit circle, $\Lambda \mathrm{e}^{\mathrm{i}n\theta} = |n|\mathrm{e}^{\mathrm{i}n\theta}$. The first boundary coefficient is given by

$$c_R = 1 + Rb_1 + O(R^2), \qquad b_1 = -\frac{((2r + 1) - \Lambda)P}{4r^2}. \tag{28.51}$$

To derive this, write the envelope U_R in Ω_R, to first order in R and in polar coordinates (s, θ) for ζ, as $1 + 2r \log s + RH$ with H harmonic. Matching values on the boundary $s = 1 + Rb_1$ gives $H|_{s=1} = P$; the terms involving b_1 cancel there, since the radial

model already matches first derivatives. Matching radial derivatives then gives $4r^2 b_1 + (2r+1)P - \Lambda P = 0$. If P_{2r+1} is harmonic, then $\Lambda P = (2r+1)P$ and $b_1 = 0$, as it must be, since such a term can be removed by a change of frame.

For $q = O(m^{1/(r+1)})$, the argument of Φ at the uncorrected observation point of (28.49) is

$$\tau + \frac{\sqrt{q}R}{2r^{3/2}}((2r+1) - \Lambda)P(\theta) + o(1), \tag{28.52}$$

uniformly in θ, and the normalized kernel differs from Φ of this argument by $o(1)$. The remainder is small because $\sqrt{q}R^2 \to 0$ in this range. For $w = z + cz^2$ we have $P = 2r\,\mathrm{Re}(ce^{i\theta})$, and the shift is $2\sqrt{rq}R\,\mathrm{Re}(ce^{i\theta})$, in agreement with the compact example in the proof of Theorem 28.13.3.

For every fixed integer $J \geq 1$, the parameter construction of Lemma 28.13.1 gives

$$c_R = 1 + \sum_{j=1}^{J} R^j b_j + O(R^{J+1})$$

in every fixed norm needed. The coefficients satisfy an explicit recursion, obtained by expanding the harmonic part of the envelope U_R in Ω_R in powers of R. At order j, matching values determines the boundary data of the new harmonic term from the Taylor term of degree $2r + j$ of ϕ and all preceding coefficients, and harmonic extension determines the term itself. Matching derivatives then gives an equation $4r^2 b_j + (\text{known terms}) = 0$; the new coefficient b_j does not enter the value equation of order j. For $j \geq 2$, nonlinear contributions of lower-order terms also appear.

Replacing Γ_t by the curve given by the truncated expansion moves the center of the transition by $O(R^{J+2})$. This curve can therefore be used in place of Γ_t whenever

$$\sqrt{q}R^{J+1} \to 0, \quad \text{equivalently} \quad q = o(m^{(J+1)/(r+J+1)}).$$

If $b_1 = \cdots = b_{j-1} = 0$ but $b_j \not\equiv 0$, the criterion (28.50) shows that, for this fixed weight, the uncorrected law holds exactly in the range $q = o(m^{j/(r+j)})$, and this range is sharp when all directions of observation are allowed. Indeed, the higher terms are relatively $O(R)$ at an angle where $b_j \neq 0$.

In the radial case $\phi = f(u)$ with $u = |z|^2$, the domain D_t is a disk by uniqueness and rotational invariance. By (28.37), its boundary circle $\{u = u_*\}$ is determined by

$$mu_* f'(u_*) = q.$$

With $\rho(u) = f'(u) + uf''(u)$, the asymptotic variance of the dominant monomial degree is $mu\rho(u)$, and the corresponding Gaussian argument is

$$\frac{muf'(u) - q}{\sqrt{mu\rho(u)}}. \tag{28.53}$$

For bounded arguments, Taylor expansion at u_* identifies it with $2\sqrt{m\rho(u_*)}(\sqrt{u} - \sqrt{u_*}) + o(1)$. Hence Theorem 28.13.2 proves this radial formula for all sublinear

orders. For $f = u^r + cu^{r+1} + \cdots$, the first shift at the uncorrected observation point is $c(r+1)r^{-3/2}\sqrt{q}R^2$, so the critical growth is $m^{2/(r+2)}$. This larger radial threshold does not apply in arbitrary coordinates.

No fixed finite Taylor jet of the weight determines the correction for every $q = o(m)$. For example, near zero take

$$f(u) = u^r + cu^r \mathrm{e}^{-1/\sqrt{u}}, \qquad c > 0,$$

with the second term extended by zero at $u = 0$, and interpolated globally as in the compact example in the proof of Theorem 28.13.3. It has the same Taylor jet as u^r at zero to every finite order, but at $u = R^2$ its relative contribution to $uf'(u)$ is $c\mathrm{e}^{-1/R}(1 + 1/(2rR))$. For $q = \lfloor m/\log m \rfloor$, we have $\sqrt{q}\mathrm{e}^{-1/R}/R \to \infty$. Hence the uncorrected observation point with $\tau = 0$ lies outside the true boundary circle by many transition widths, and the normalized kernel tends to one. The quadrature-domain correction includes this flat contribution.

28.14 A logarithmic kernel scale in two dimensions

In the curve models above, the metric weight has a single homogeneous leading term. In two dimensions, a mixed monomial in the weight can compete with the pure monomials and introduce a logarithm into the scale of the kernel. For an explicit metric on $\mathbb{P}^2$, the relevant global section norm can be computed directly.

Write $\mathrm{d}A$ for Euclidean area on $\mathbb{C}$. We first isolate the integral responsible for the logarithm.

Lemma 28.14.1 *Set $W(a, b) = ab + a^3 + b^3$ on $[0, \infty)^2$. As $t \to \infty$,*

$$J(t) := \int_0^\infty \int_0^\infty \mathrm{e}^{-tW(a,b)}\,\mathrm{d}a\,\mathrm{d}b = \frac{\log t}{3t} + O(t^{-1}),$$
$$\int_0^\infty \int_0^\infty (a+b)\mathrm{e}^{-tW(a,b)}\,\mathrm{d}a\,\mathrm{d}b = O(t^{-4/3}).$$

Proof Put $A = t^{-2/3}$ and $B = t^{-1/3}$; the logarithm comes from the interval $A \le a \le B$. For $a > 0$, the substitution $u = tab$ gives

$$H_t(a) := \int_0^\infty \mathrm{e}^{-t(ab+b^3)}\,\mathrm{d}b = \frac{1}{ta}\int_0^\infty \mathrm{e}^{-u-u^3/(t^2a^3)}\,\mathrm{d}u.$$

Since $0 \le 1 - \mathrm{e}^{-v} \le v$ for $v \ge 0$,

$$0 \le \frac{1}{ta} - H_t(a) \le \frac{6}{t^3a^4}.$$

Over $A \le a \le B$, the integral of this error is $O(t^{-1})$. Replacing e^{-ta^3} by one in $\int_A^B \mathrm{e}^{-ta^3}/(ta)\,\mathrm{d}a$ costs at most $\int_A^B a^2\,\mathrm{d}a = O(t^{-1})$ as well. Thus the middle interval

contributes

$$\int_A^B \mathrm{e}^{-ta^3} H_t(a)\,\mathrm{d}a = \frac{1}{t}\log\frac{B}{A} + O(t^{-1}) = \frac{\log t}{3t} + O(t^{-1}).$$

For $a < A$, the bound $H_t(a) \le \int_0^\infty \mathrm{e}^{-tb^3}\,\mathrm{d}b = O(t^{-1/3})$ gives a contribution $O(t^{-1})$. For $a > B$, use $H_t(a) \le 1/(ta)$ and substitute $v = ta^3$ to obtain the bound

$$\int_B^\infty \frac{\mathrm{e}^{-ta^3}}{ta}\,\mathrm{d}a = \frac{1}{3t}\int_1^\infty \frac{\mathrm{e}^{-v}}{v}\,\mathrm{d}v = O(t^{-1}).$$

Finally, $aH_t(a) \le 1/t$ gives

$$\int_0^\infty \int_0^\infty a\mathrm{e}^{-tW(a,b)}\,\mathrm{d}a\,\mathrm{d}b \le \frac{1}{t}\int_0^\infty \mathrm{e}^{-ta^3}\,\mathrm{d}a = O(t^{-4/3}).$$

Symmetry proves the same bound for b. □

On $L = O_{\mathbb{P}^2}(1)$, define a Hermitian metric by

$$|S|_h^2([Z]) = \frac{|S(Z)|^2}{\bigl(|Z_0|^6 + 3|Z_0Z_1Z_2|^2 + 3|Z_1|^6 + 3|Z_2|^6\bigr)^{1/3}} \qquad (S \text{ a linear form}).$$

The denominator is positive away from zero and homogeneous of degree two in $|Z|$, so the metric is well defined and smooth. Its local weight is one third of the logarithm of a sum of squared absolute values of holomorphic cubic functions, so its curvature is semipositive. In the chart $Z_0 \neq 0$, put $z_j = Z_j/Z_0$ and $p = [1:0:0]$. The frame $e = Z_0$ has

$$|e|_h^2 = \mathrm{e}^{-\phi}, \qquad \phi(z) = \tfrac{1}{3}\log\bigl(1 + 3W(|z_1|^2, |z_2|^2)\bigr). \tag{28.54}$$

In particular the curvature vanishes at p.

Proposition 28.14.1 *Let ν be a smooth positive volume form on $\mathbb{P}^2$, invariant under the rotations of z_1 and of z_2, and write*

$$\nu = \rho(z)\,\mathrm{d}A(z_1)\,\mathrm{d}A(z_2), \qquad \rho_0 = \rho(0) > 0$$

in the chart above. Give $H^0(\mathbb{P}^2, L^m)$ the norm $\int_{\mathbb{P}^2} |s|_{h^m}^2\,\mathrm{d}\nu$, and let B_m be its kernel density. Then

$$B_m(p) = \frac{3}{\pi^2\rho_0}\frac{m}{\log m}\left(1 + O\bigl((\log m)^{-1}\bigr)\right).$$

For the Fubini–Study probability volume, the leading constant is $3/2$.

Proof The homogeneous monomials form an orthogonal basis, because the metric and the volume are invariant under the two independent rotations and distinct monomials transform by distinct characters. At p, only Z_0^m has a nonzero value, equal to one in

the frame $e^{\otimes m}$. Consequently $B_m(p) = I_m^{-1}$, where

$$I_m = \int_{\mathbb{P}^2} |Z_0^m|_{h^m}^2 \, \mathrm{d}\nu = \int_{\mathbb{C}^2} (1 + 3W(|z_1|^2, |z_2|^2))^{-m/3} \rho(z) \, \mathrm{d}A(z_1) \, \mathrm{d}A(z_2).$$

Here the omitted projective line $\{Z_0 = 0\}$ has zero volume. By rotational invariance, the first derivatives of ρ vanish at zero, so $\rho(z) = \rho_0 + O(|z_1|^2 + |z_2|^2)$ near zero. In the variables $a = |z_1|^2$ and $b = |z_2|^2$, polar integration has Jacobian $\pi^2 \, \mathrm{d}a \, \mathrm{d}b$.

Choose a fixed small $\delta > 0$. On $W \le \delta$, set $\ell(W) = \frac{1}{3}\log(1 + 3W)$. For constants $c, C > 0$ independent of m,

$$cW \le \ell(W) \le W, \qquad 0 \le \mathrm{e}^{-m\ell(W)} - \mathrm{e}^{-mW} \le CmW^2 \mathrm{e}^{-cmW} \le \frac{C}{m} \mathrm{e}^{-cmW/2}.$$

The first estimate of Lemma 28.14.1 bounds the integral of this error in the weight by $O(m^{-2}\log m)$, and the second bounds the error from replacing ρ by ρ_0 by $O(m^{-4/3})$. On the complement $W > \delta$, the contribution to I_m is at most $\nu(\mathbb{P}^2)(1 + 3\delta)^{-m/3}$, and the corresponding tail of the exponential model is at most $\mathrm{e}^{-m\delta/2} J(m/2)$. Therefore

$$I_m = \pi^2 \rho_0 J(m) + O(m^{-4/3}) + O(m^{-2}\log m) = \frac{\pi^2 \rho_0}{3m} \log m + O(m^{-1}).$$

Inverting proves the first assertion. The Fubini–Study probability volume has the local expression

$$\nu = \frac{2}{\pi^2}(1 + |z_1|^2 + |z_2|^2)^{-3} \, \mathrm{d}A(z_1) \, \mathrm{d}A(z_2).$$

It has mass one, since $2\int_0^\infty \int_0^\infty (1 + a + b)^{-3} \, \mathrm{d}a \, \mathrm{d}b = 1$, and its density at zero is $2/\pi^2$. The leading constant is therefore $3/2$. □

Thus the full kernel, without any selection, already has a logarithmic factor at this curvature zero, with a coefficient depending on the volume density at p. A transition for growing vanishing ideals would require control of the norms of the selected monomials on the moving scale, as in the curve case.

28.15 Anisotropic point jets and isolated poles on fibrations

Vanishing to a given order at a point couples the variables through the total degree, even when the metric weight splits as a sum. On a compact product of curves, this coupling becomes a half-space condition on independent monomial indices, and the transition on each curve factor determines the local law in higher dimensions. An isolated pole on a submersion with connected fibers gives a further application: its multiplier ideal does not descend to the base, although the condition it imposes on pulled-back sections does.

Let q be a positive integer. For $1 \le j \le q$, choose integers $r_j \ge 2$ and numbers $\varepsilon_j > 0$. Put

$$Y = (\mathbb{P}^1)^q, \qquad x = (0, \dots, 0), \qquad d_j = r_j + 1, \qquad L = \boxtimes_j \mathcal{O}_{\mathbb{P}^1}(d_j).$$

In the affine coordinates z_j and the product frame e, put $t_j = |z_j|^2$ and define the smooth metric and volume by

$$|e|_h^2 = \mathrm{e}^{-\phi}, \qquad \phi = \sum_j \psi_j(t_j), \qquad \psi_j(t) = \log(1 + t^{r_j} + \varepsilon_j t^{r_j+1}),$$
$$\nu_Y = a(t) \prod_j \mathrm{d}A_j, \qquad a(t) = \prod_j a_j(t_j), \qquad a_j(t) = (1+t)^{-2},$$

where $\mathrm{d}A_j$ is Euclidean area in z_j. In particular $\nu_Y(Y) = \pi^q$ and $a(0) = 1$. Define

$$H_j(t) = t\psi_j'(t), \qquad G_j(t) = tH_j'(t), \qquad g_j(t) = G_j(t)/t, \qquad \theta = c_1(L, h).$$

The metric extends across infinity: there its weight, in terms of $u = |z_j|^{-2}$, is $\log(\varepsilon_j + u + u^{d_j})$, which has positive curvature at infinity. For $t > 0$, $H_j(t)$ and $G_j(t)$ are the mean and the variance of the exponents $0, r_j, d_j$ taken with probabilities proportional to $1, t^{r_j}, \varepsilon_j t^{d_j}$. Hence $G_j > 0$, and H_j increases from 0 to d_j. Near zero,

$$\begin{aligned} H_j(t) &= r_j t^{r_j} + (r_j+1)\varepsilon_j t^{r_j+1} + O(t^{2r_j}), \\ G_j(t) &= r_j^2 t^{r_j} + (r_j+1)^2 \varepsilon_j t^{r_j+1} + O(t^{2r_j}). \end{aligned} \tag{28.55}$$

Thus θ is semipositive, and its rank drops along the coordinate divisors $\{z_j = 0\}$.

For integers $k_m \to \infty$ with $k_m = o(m)$, let B_m be the density of

$$V_m = H^0(Y, L^m \otimes \mathfrak{m}_x^{k_m}), \qquad \|s\|^2 = \int_Y |s|_{h^m}^2 \nu_Y.$$

The ideal $\mathfrak{m}_x^{k_m}$ of the point x imposes vanishing to total order at least k_m, rather than separate orders in the individual variables. As before, Φ is the standard Gaussian distribution function.

Theorem 28.15.1 *Let $z = z^{(m)}$ be points of the affine chart, with $t_j = |z_j|^2$, such that $\mu_j := mH_j(t_j)$ satisfies $\mu_j/k_m \to \alpha_j > 0$. Set $V = \sum_j mG_j(t_j)$. If $(\sum_j \mu_j - k_m)/\sqrt{V} \to \tau \in \mathbb{R}$, then*

$$\frac{\pi^q a(t) B_m(z)}{m^q \prod_j g_j(t_j)} \longrightarrow \Phi(\tau). \tag{28.56}$$

For all sufficiently large m,

$$N_m := \dim V_m = \prod_j (d_j m + 1) - \binom{k_m + q - 1}{q}, \qquad N_m/m^q \longrightarrow D := \prod_j d_j. \tag{28.57}$$

Moreover $m^{-q} B_m \nu_Y \to \theta^q/q!$ in total variation.

Proof The proof reduces the kernel to a probability for a sum of independent variables, one for each factor, and applies the one-variable transition of Section 28.13 to each of them. In each variable, the affine coefficient of a global section of L^m is entire, and its expression in the frame at infinity must be holomorphic, so it is a polynomial of degree at most $d_j m$. Hence the monomials $z^\ell e^m$, $0 \le \ell_j \le d_j m$, form a basis of the global sections. A polynomial vanishes to order at least k_m at x exactly when all its monomials have total degree at least k_m, so the monomials with $\sum_j \ell_j \ge k_m$ form a basis of V_m. Once $k_m \le \min_j(d_j m) + 1$, which holds for large m, all monomials of total degree below k_m are global, and counting them gives (28.57).

Invariance under the rotation of each coordinate makes these monomials orthogonal. Their squared norms are $\prod_j J_{j,m,\ell_j}$, where

$$J_{j,m,\ell} = \pi \int_0^\infty t^\ell \mathrm{e}^{-m\psi_j(t)} a_j(t)\, \mathrm{d}t, \qquad w_{j,m,\ell}(t) = \frac{t^\ell \mathrm{e}^{-m\psi_j(t)}}{J_{j,m,\ell}}.$$

These integrals converge even for $\ell = d_j m$. Write $K_{j,m} = \sum_{\ell=0}^{d_j m} w_{j,m,\ell}$, and let $\mathsf{L}_1, \dots, \mathsf{L}_q$ be independent random variables with $\Pr\{\mathsf{L}_j = \ell\} = w_{j,m,\ell}(t_j)/K_{j,m}(t_j)$ for $0 \le \ell \le d_j m$. Then

$$B_m(z) = \prod_j K_{j,m}(t_j)\ \Pr\left\{ \sum_j \mathsf{L}_j \ge k_m \right\}. \tag{28.58}$$

Since $H_j(t_j) = \mu_j/m \to 0$ and H_j increases from $H_j(0) = 0$, we have $t_j \to 0$. Each factor $(\mathbb{P}^1, O(d_j))$, with the metric $\mathrm{e}^{-\psi_j(|z_j|^2)}$ and the volume $a_j\, \mathrm{d}A_j$, satisfies the hypotheses of Theorem 28.13.2, and $K_{j,m}$ is its full density. Put $V_j = mG_j(t_j)$. Since $\partial_{z_j}\partial_{\bar z_j}\psi_j(|z_j|^2) = g_j(t_j)$, the full density (28.48) gives $K_{j,m} = mg_j(t_j)/(\pi a_j(t_j))(1 + o(1))$. For any fixed $y \in \mathbb{R}$, apply the radial formula (28.53) to the integer order

$$b_{j,m} = \lfloor \mu_j + y\sqrt{V_j} \rfloor + 1.$$

This order tends to infinity and is sublinear. The selected kernel of the factor, divided by its full kernel, is $\Pr\{\mathsf{L}_j \ge b_{j,m}\}$, and it tends to $\Phi(-y)$. Hence

$$\frac{\mathsf{L}_j - \mu_j}{\sqrt{V_j}} \implies N(0,1).$$

Here μ_j and $\sqrt{V_j}$ serve only as centering and scale; we do not need the exact mean or variance of L_j. Since $V_j/V \to r_j\alpha_j/\sum_i r_i\alpha_i > 0$, independence gives a standard Gaussian limit for $(\sum_j \mathsf{L}_j - \sum_j \mu_j)/\sqrt{V}$. To see this, use the product of the limiting Gaussian laws on bounded rectangles, and then tightness and the fact that the boundary of a half-space has zero Gaussian measure. Consequently the probability in (28.58) tends to $\Phi(\tau)$, which proves the local assertion for every sequence $k_m \to \infty$ with $k_m = o(m)$.

For the global assertion we also need the full kernel at a fixed $t > 0$, where the formula for shrinking radii does not apply. We suppress the index of the factor and set $F(u) = \psi(\mathrm{e}^u)$. Then $F'(u) = H(\mathrm{e}^u)$ and $F''(u) = G(\mathrm{e}^u) > 0$. For ℓ/m in a compact subinterval of $(0, d)$, the saddle $s = H^{-1}(\ell/m)$ stays compact, and Laplace expansion gives uniformly

$$J_{m,\ell} = \pi a(s) s \sqrt{\frac{2\pi}{mG(s)}}\, \mathrm{e}^{\ell \log s - m\psi(s)}\,(1 + o(1)).$$

Indeed, the phase in the logarithmic variable has a uniformly positive second derivative there, and outside a fixed neighborhood of the saddle point, convexity and the slopes at the endpoints bound the tails exponentially. For $\ell = mH(t) + O(\sqrt{m})$, evaluation yields

$$w_{m,\ell}(t) = \frac{\sqrt{mG(t)}}{\pi a(t) t \sqrt{2\pi}}\, \mathrm{e}^{-(\ell - mH(t))^2/(2mG(t))}\,(1 + o(1)).$$

For indices with saddle points in a compact interval, the loss in the phase, as a function of ℓ, has second derivative $1/(mG(s)) \ge c/m$, which gives summable Gaussian tails. For the remaining indices, fix small numbers $\delta, \sigma > 0$, and bound the norm $J_{m,\ell}$ from below by the integral over an interval of length $1/m$ about $t\mathrm{e}^{-\sigma}$ or $t\mathrm{e}^{\sigma}$. If $|\ell/m - H(t)| \ge \delta$, the appropriate choice gains at least $m(\delta\sigma - C\sigma^2) \ge cm$ in the exponent. Each such weight $w_{m,\ell}(t)$ is then at most $Cm\mathrm{e}^{-cm}$, and there are $dm + 1$ indices. Gaussian summation therefore gives $K_m(t)/m \to g(t)/(\pi a(t))$.

In the product chart, $\theta^q/q! = \prod_j (g_j(t_j)/\pi)\mathrm{d}A_j$, and its mass is D because $\int_0^\infty g_j(t)\,\mathrm{d}t = H_j(\infty) - H_j(0) = d_j$. The normalized full density $m^{-q}\prod_j K_{j,m}$ therefore tends to the density of $\theta^q/q!$ with respect to ν_Y almost everywhere; the coordinate divisors have zero volume. The total mass of the normalized full kernel measure is $m^{-q}\prod_j (d_j m + 1) \to D$. For the nonnegative densities f_m, f of these measures with respect to ν_Y, dominated convergence applied to $\min(f_m, f)$ gives

$$\int |f_m - f| \nu_Y = \int f_m \nu_Y + \int f \nu_Y - 2\int \min(f_m, f)\nu_Y \longrightarrow 0.$$

Finally, the difference between the full and the selected kernel is nonnegative and has integral $\binom{k_m+q-1}{q} = o(m^q)$. This proves the total-variation convergence. □

28.15.1 Corrected and homogeneous centers

The transition in Theorem 28.15.1 is centered at the radii determined by the functions H_j. Replacing them by the radii of the homogeneous leading terms of the weight introduces a shift, which is negligible exactly in a range of orders determined by the largest exponent r_j.

Fix positive numbers α_j with sum one and a real τ, and set

$$R = \sum_j \alpha_j r_j, \qquad T_m = k_m + \tau\sqrt{Rk_m}, \qquad t_j = H_j^{-1}(\alpha_j T_m/m). \tag{28.59}$$

Then $\sum_j \mu_j = T_m$ and $V \sim Rk_m$, so (28.56) holds for every sublinear growing order. The radii $|z_j| = \sqrt{t_j}$ are asymptotic to $(\alpha_j k_m/(r_j m))^{1/(2r_j)}$.

Proposition 28.15.1 *Keep α_j, τ, R, T_m as above, but use instead the homogeneous radii given by $t_j = (\alpha_j T_m/(r_j m))^{1/r_j}$, and put $r_* := \max_j r_j$. For the metrics of this section, the law*

$$\frac{\pi^q B_m(z)}{m^q \prod_j r_j^2 |z_j|^{2r_j-2}} \longrightarrow \Phi(\tau)$$

holds for every fixed τ if and only if $k_m = o(m^{2/(r_*+2)})$. If $k_m/m^{2/(r_*+2)} \to c \in (0,\infty)$, the left side instead tends to

$$\Phi\big(\tau + C_\alpha c^{(r_*+2)/(2r_*)}\big), \qquad C_\alpha = \frac{1}{\sqrt{R}} \sum_{j: r_j = r_*} (r_*+1)\varepsilon_j (\alpha_j/r_*)^{1+1/r_*}.$$

If that ratio tends to infinity while $k_m = o(m)$, the limit is 1.

Proof By (28.55), $\sum_j \mu_j = T_m + \Delta_m$ with

$$\Delta_m = \sum_j (r_j+1)\varepsilon_j m t_j^{r_j+1} + O\left(\sum_j m t_j^{2r_j}\right), \qquad V \sim Rk_m.$$

Since $r_j \geq 2$, each error term is negligible compared with the corresponding positive correction. Consequently

$$\frac{\Delta_m}{\sqrt{Rk_m}} = (1+o(1)) \sum_j \frac{(r_j+1)\varepsilon_j (\alpha_j/r_j)^{1+1/r_j}}{\sqrt{R}} k_m^{1/2+1/r_j} m^{-1/r_j}.$$

At the critical scale only the indices with $r_j = r_*$ contribute in the limit. Below it every term tends to zero, and above it a positive term diverges. The Gaussian limit

of the normalized sum proved above applies in all three cases; in the last case its tightness gives probability one. Moreover, $a(t) \to 1$ and $g_j(t_j)/(r_j^2 t_j^{r_j-1}) \to 1$, so the two normalizations agree. If $k_m/m^{2/(r_*+2)}$ does not tend to zero, a subsequence has a positive finite or infinite limit, and the law fails at $\tau = 0$. This proves both directions of the equivalence for these metrics. □

28.15.2 An isolated selecting potential on a submersion

Let $\pi\colon X \to Y$ be a surjective holomorphic submersion from a compact complex manifold X, with connected fibers of dimension $f \geq 1$, and put $n = q + f = \dim X$. Fix a Hermitian form ω_X on X. Use $E = \pi^* L$ with the metric $h_E = \pi^* h$, and a smooth positive volume form ν_X satisfying

$$\pi_* \nu_X = \nu_Y. \tag{28.60}$$

Choose $p \in \pi^{-1}(x)$ and submersion coordinates (z, w) centered at p, in which $\pi(z, w) = z$. Let χ_p be a function equal to $\log(|z|^2 + |w|^2)$ near p, smooth on $X \setminus \{p\}$, and zero outside a larger coordinate ball. Define the selecting potentials, which depend on m, and their multiplier ideals,

$$\varphi_m = \frac{k_m + n - 1}{m} \chi_p, \qquad \mathcal{I}_m = \mathcal{I}(m\varphi_m).$$

The selected sections are measured with the smooth norm defined by h_E^m and ν_X.

Theorem 28.15.2 *For all large m, $\varphi_m \in \mathrm{PSH}(X, \omega_X)$ and $\mathcal{I}_m = \mathfrak{m}_p^{k_m}$. Neither φ_m nor $\mathcal{I}_m$ is pulled back from Y, but pullback gives an isometry of Hilbert spaces and an identity of densities*

$$H^0(X, E^m \otimes \mathcal{I}_m) = \pi^* V_m, \qquad B_{X,m} = B_m \circ \pi.$$

Thus the local laws of Theorem 28.15.1 and Proposition 28.15.1 hold for every sequence of points of X whose projections satisfy their hypotheses. Moreover $\log N_m / \log m \to q < \dim X$, and

$$m^{-q} B_{X,m} \nu_X \longrightarrow \left(\frac{\theta^q / q!}{\nu_Y} \circ \pi \right) \nu_X \quad \text{in total variation.} \tag{28.61}$$

Normalizing by N_m instead gives the limit divided by D, a probability measure.

Proof The function χ_p satisfies $\mathrm{dd}^{\mathrm{c}} \chi_p \geq -C\omega_X$, because it is plurisubharmonic near its pole and smooth elsewhere. Since its coefficient $(k_m + n - 1)/m$ tends to zero, φ_m is ω_X-plurisubharmonic for large m. Near p, $m\varphi_m = (k_m + n - 1)\log(|z|^2 + |w|^2)$, so Lemma 1.16.1, with the point p as center of codimension n, gives $\mathcal{I}_m = \mathfrak{m}_p^{k_m}$ there. Away from p the ideal is the unit ideal. The ideal is proper at p but trivial at

the other points of the same fiber, which is impossible for an ideal pulled back from Y; likewise, the pole of φ_m occurs only at p.

By Proposition 28.7.1 with zero selecting potential, every section of π^*L^m is the pullback of a section of L^m. In the submersion coordinates, a pulled-back coefficient depends only on z, so its order at p equals its order at x. This proves the identity of the selected spaces. Fiber integration and (28.60) give

$$\int_X |\pi^* s|^2_{h_E^m} \nu_X = \int_Y |s|^2_{h^m} \nu_Y.$$

Pulling back an orthonormal basis proves the identity of densities. If b_m and b are the densities, with respect to ν_Y, of the measures in the total-variation limit of Theorem 28.15.1, then the corresponding densities on X are $b_m \circ \pi$ and $b \circ \pi$, and their $L^1(\nu_X)$ distance equals $\int_Y |b_m - b| \nu_Y$. This proves (28.61). The dimension formula (28.57) gives $\log N_m / \log m \to q$ and the final normalization. □

Volume forms satisfying (28.60) exist on every such submersion: for a smooth family of fiber probability measures as in Section 28.7, the lift of ν_Y is one, since in a local trivialization its density is the product of the densities of ν_Y and of the fiber measures. On a product $X = Y \times F$, where F may be any compact connected complex manifold, Kähler or not, and ν_F is a smooth positive volume form on F, the volumes $W(y,u)\nu_Y \nu_F$ with $W > 0$ smooth and $\int_F W(y,u)\nu_F = 1$ satisfy (28.60).

For an arbitrary smooth positive volume ν'_X, the pulled-back norm becomes the norm on the base for the volume $\pi_* \nu'_X$, whereas the local laws above were proved for the specific volume ν_Y. Since the selecting potentials vary with m, Theorem 28.15.2 does not settle Problem 35.9.2 for fixed singularities. Weights with mixed or angular remainder terms, observation points with some $\alpha_j = 0$, and other ideals on the base are not treated here. The lifted measure keeps the chosen conditional fiber volume; it is not a Monge–Ampère measure in the dimension of X.

28.16 Newton selection and moment-shell limits

On a product of projective lines, selecting the monomials whose exponents lie in the dilates of a fixed polytope gives a graded family of section spaces for which the lower bound for the potentials can be proved directly, without a general approximation theorem. The limit of the kernel measures depends on the dimension of the polytope. When this dimension is smaller than that of the manifold, the limit is carried by a family of real tori and differs from the mixed curvature measures of the envelope.

Let $X = (\mathbb{P}^1)^n$ with $n \geq 1$, and let $L = \mathcal{O}(k_1, \dots, k_n)$, where each k_j is a positive integer. Equip L with its product Fubini–Study metric h. On the torus $T = (\mathbb{C}^*)^n$, in the standard frame e, write

$$r_j = \log |z_j|^2, \qquad \varphi_0(r) = \sum_j k_j \log(1 + \mathrm{e}^{r_j}), \qquad \theta = c_1(L,h) = \mathrm{dd^c}\varphi_0.$$

Put $P = \prod_j [0, k_j]$. Fix a nonempty rational polytope Q contained in the interior of P whose affine span contains a lattice point $b \in \mathbb{Z}^n$, and write

$$\operatorname{aff} Q = b + W;$$

the linear subspace W is rational because Q is. Set $d = \dim Q$ and $\Lambda = W \cap \mathbb{Z}^n$. Let $\mathrm{d}q_\Lambda$ be the Lebesgue measure on $b + W$ normalized so that a fundamental domain of the lattice Λ has volume one; when $d = 0$, it is the unit point mass at b. Write $\operatorname{vol}_\Lambda(Q)$ for the volume of Q with respect to $\mathrm{d}q_\Lambda$. In this section the selected spaces are defined directly by monomials, without a selecting potential: we select exactly the monomials with exponents in mQ,

$$V_m = \operatorname{span}\{z^\alpha e^m : \alpha \in mQ \cap \mathbb{Z}^n\}. \tag{28.62}$$

Each of these monomials is a global section of L^m, since $0 \le \alpha_j \le mk_j$. Since Q is convex, products of selected sections are selected, so these spaces form a graded family.

Choose a smooth real torus-invariant function w on X such that the function $F = \varphi_0 + w$ of r has positive definite Hessian $D^2F(r)$ for every $r \in \mathbb{R}^n$. Choose a smooth strictly positive torus-invariant probability measure ν on X. On T we then have $\mathrm{d}\nu = a(r)\,\mathrm{d}r\,\mathrm{d}\tau$, where $a > 0$, $\int a\,\mathrm{d}r = 1$, and $\mathrm{d}\tau$ is the Haar probability measure on $(S^1)^n$ in the angular variables. Give V_m the Hilbert norm $\int_X |s|^2_{h^m} \mathrm{e}^{-mw}\,\mathrm{d}\nu$, and for an orthonormal basis put

$$\mathcal{B}_m = \sum_j |s_{m,j}|^2_{h^m}, \qquad B_m = \mathrm{e}^{-mw}\mathcal{B}_m, \qquad U_m = m^{-1}\log \mathcal{B}_m.$$

As in Section 28.1, the density B_m includes the continuous weight.

28.16.1 The growth constraint and its envelope

Let $H_Q(r) = \max_{q \in Q} q \cdot r$ be the support function of Q, and define

$$C_Q = \{v \in \mathrm{PSH}(X, \theta) : v + \varphi_0 \le H_Q + C_v \text{ on } T \text{ for some constant } C_v\},$$
$$E_Q = \left(\sup\{v \in C_Q : v \le w \text{ quasi-everywhere on } X\}\right)^*.$$

This growth condition replaces the multiplier-ideal condition of Section 28.3; we do not identify E_Q with an ideal envelope.

Lemma 28.16.1 *The gradient $\nabla F \colon \mathbb{R}^n \to \operatorname{int} P$ is a smooth diffeomorphism. With $r(q) = (\nabla F)^{-1}(q)$ and $F^*(q) = q \cdot r(q) - F(r(q))$, set*

$$G_Q(r) = \max_{q \in Q}\{q \cdot r - F^*(q)\}.$$

Then $E_Q = G_Q - \varphi_0$ on T, and $E_Q = -\infty$ on $X \setminus T$. Moreover, a global section s of L^m belongs to V_m exactly when

$$|s/e^m|^2 \le C_s \mathrm{e}^{mH_Q(r)} \quad \text{on } T \quad \text{for some finite } C_s. \tag{28.63}$$

Proof In the charts at infinity one sees that every global section of L^m is a polynomial of degree at most mk_j in each variable z_j. If $s \in V_m$, the growth bound follows from the triangle inequality, since $|z^\alpha|^2 = \mathrm{e}^{\alpha\cdot r} \le \mathrm{e}^{mH_Q(r)}$ for $\alpha \in mQ$. Conversely, averaging the growth bound over the angles gives $\sum_\alpha |c_\alpha|^2 \mathrm{e}^{\alpha\cdot r} \le C_s \mathrm{e}^{mH_Q(r)}$ for the coefficients c_α of s. If some $c_\alpha \ne 0$ had $\alpha \notin mQ$, a vector u with $\alpha \cdot u > mH_Q(u)$, which exists by separation, would contradict this bound along the ray $r = tu$, $t \to \infty$. Thus the growth condition controls each monomial separately, even though the monomials of s may cancel pointwise.

For $p \in \operatorname{int} P$, put $\delta_p = \min_j (p_j, k_j - p_j) > 0$. Since w is bounded,

$$F(r) - p \cdot r \ge \delta_p \sum_j |r_j| - \|w\|_\infty. \tag{28.64}$$

By strict convexity, $F - p \cdot r$ therefore has a unique minimizer, at which $\nabla F = p$. Conversely, every gradient of F lies in P: compare the supporting hyperplanes of the convex function F with the asymptotic slopes of φ_0. By the inverse function theorem the image of ∇F is open, so it is contained in the interior of P. This proves that ∇F is a diffeomorphism onto $\operatorname{int} P$. On a compact neighborhood of Q in $\operatorname{int} P$, the inverse gradient and the first and second derivatives of F^* are bounded, and the Hessian of F^* is positive definite.

We next show that $G_Q - \varphi_0$ is a competitor. In a product chart in which some of the coordinates are inverted, the local potential of $G_Q - \varphi_0$ is

$$\max_{q \in Q} \left\{ \sum_j c_j(q) \log |\zeta_j|^2 - F^*(q) \right\}, \qquad c_j(q) = q_j \text{ or } k_j - q_j.$$

All c_j are bounded below by a fixed positive number. The expression is locally bounded above, tends uniformly to $-\infty$ at the coordinate hyperplanes when the other coordinates stay bounded, and is locally integrable by comparison with one fixed summand. At points where it is finite, a maximizing summand gives the submean inequality for plurisubharmonic functions; at points where it equals $-\infty$, this inequality is automatic. These local expressions glue. Boundedness of F^* on Q gives $G_Q - H_Q = O(1)$, and Fenchel's inequality gives $G_Q \le F$, that is, $G_Q - \varphi_0 \le w$. Thus $G_Q - \varphi_0$, extended by $-\infty$ on $X \setminus T$, belongs to C_Q and satisfies the obstacle condition.

Now let v be any competitor. The obstacle condition holds everywhere: the exceptional pluripolar set has Lebesgue measure zero, and submean values over shrinking balls carry the inequality with the continuous obstacle to every point. Consider the maximum over the angles, $M(r) = \max_\tau \{v(\mathrm{e}^{r/2}\tau) + \varphi_0(r)\}$, where $\mathrm{e}^{r/2}\tau$ denotes the point with coordinates $\mathrm{e}^{r_j/2}\tau_j$. By compactness, M is upper

semicontinuous. Rotating a point at which the maximum is attained gives the submean inequality for the invariant function $z \mapsto M(\log|z_1|^2, \dots, \log|z_n|^2)$, which is therefore plurisubharmonic on T. It extends across the affine coordinate hyperplanes by the local upper bound. If $M(r) = -\infty$ for some r, successive one-variable maximum principles would force this function to be $-\infty$ on the corresponding polydisk, contradicting $v \not\equiv -\infty$. Thus M is finite. In logarithmic coordinates this invariant function depends only on the real parts, so its positive complex Hessian shows that M is convex.

We have $M \le F$ and $M \le H_Q + C_v$. Comparing along rays shows that every supporting slope of M lies in Q. At any r, choose such a slope p. Since $M \le F$ gives $M^* \ge F^*$, convex duality yields $M(r) = p \cdot r - M^*(p) \le p \cdot r - F^*(p) \le G_Q(r)$. Hence $v \le G_Q - \varphi_0$ on T. Submean values over shrinking balls and the uniform decay of $G_Q - \varphi_0$ near $X \setminus T$ show that $v = -\infty$ on $X \setminus T$. Since $G_Q - \varphi_0$, extended by $-\infty$, is already upper semicontinuous, this proves the identity for the envelope after upper regularization. □

28.16.2 Norms, lattice growth and the lower bound

Theorem 28.16.1 *The norms are nondegenerate and satisfy the Bernstein–Markov inequality of Section 28.2 for the weight w. Writing $N_m = \dim V_m$, we have*

$$N_m = m^d \operatorname{vol}_\Lambda(Q) + O(m^{d-1}) \quad (d \ge 1), \qquad N_m = 1 \quad (d = 0).$$

For all sufficiently large m,

$$|U_m - E_Q| \le \frac{C}{m}\left(1 + \sum_j |r_j| + \log m\right) \quad \text{on } T. \tag{28.65}$$

Consequently $U_m \to E_Q$ locally uniformly on T and in L^p for every finite p with respect to smooth volume forms on X, and the closed positive currents $\theta + \mathrm{dd^c} U_m$ converge weakly to $\theta + \mathrm{dd^c} E_Q$. Both U_m and E_Q equal $-\infty$ on $X \setminus T$.

Proof Since the volume form is smooth and positive, the norm is positive on every nonzero section. Orthogonality of the angular Fourier modes diagonalizes it:

$$A_m(\alpha) = \|z^\alpha e^m\|_m^2 = \int_{\mathbb{R}^n} \mathrm{e}^{\alpha\cdot r - mF(r)} a(r)\,\mathrm{d}r,$$
$$\mathcal{B}_m(r) = \sum_{\alpha \in mQ \cap \mathbb{Z}^n} \frac{\mathrm{e}^{\alpha\cdot r - m\varphi_0(r)}}{A_m(\alpha)}.$$

For $q = \alpha/m$, the function $F(r) - q \cdot r$ has the unique minimum value $-F^*(q)$, attained at $r(q)$. Taylor's formula on a ball of radius $m^{-1/2}$ about $r(q)$, with uniform

bounds on a compact neighborhood of $r(Q)$, yields

$$c m^{-n/2} \mathrm{e}^{m F^*(q)} \le A_m(\alpha) \le \mathrm{e}^{m F^*(q)}. \tag{28.66}$$

The upper bound uses only the minimum and $\int a \, \mathrm{d}r = 1$.

Since $b \in \mathbb{Z}^n$, we have $(m \operatorname{aff} Q) \cap \mathbb{Z}^n = m b + \Lambda$. Counting fundamental cells of Λ in $m(Q - b)$ gives the dimension estimate: the cells meeting the relative boundary of $m(Q - b)$ lie in a boundary layer of fixed width, and there are $O(m^{d-1})$ of them. The same cells give convergence of the lattice Riemann sums on Q. Every $q \in Q$ lies within $O(m^{-1})$ of $Q_m = Q \cap (b + m^{-1}\Lambda)$: move q by $O(m^{-1})$ towards a fixed relative interior point so that a ball larger than the covering radius of the lattice fits inside Q, and then round to a lattice point inside that ball. This works for all sufficiently large m, and for a singleton it is exact.

Put $G_{Q,m} = \max_{q \in Q_m} (q \cdot r - F^*(q))$. The approximation by the grid gives $0 \le G_Q - G_{Q,m} \le C(1 + \sum_j |r_j|)/m$. The formula for $\mathcal{B}_m$ and (28.66) give

$$G_{Q,m} - \varphi_0 \le U_m \le G_{Q,m} - \varphi_0 + \frac{(n/2) \log m + \log N_m - \log c}{m}.$$

This proves (28.65). The selected monomials vanish on every component of $X \setminus T$. The functions $\log |z_j|^2$ have finite L^p norms for each finite p with respect to smooth volume forms, so (28.65) gives the global convergence, and differentiation in the sense of distributions gives the convergence of the currents.

Fenchel's inequality $\alpha \cdot r - m F(r) \le m F^*(\alpha/m)$ and (28.66) also give $\sup_X B_m \le C N_m m^{n/2}$. By the Cauchy–Schwarz inequality in the orthonormal basis, $\sup_X |s|_{h^m} \mathrm{e}^{-m w/2} \le C m^{d/2 + n/4} \|s\|_m$ for $s \in V_m$. The finitely many initial degrees can be absorbed in C, and a polynomial bound implies the Bernstein–Markov inequality with any positive exponential loss. $\square$

28.16.3 The limit measure and the ambient metric

For $q \in \operatorname{int} P$ and $\tau \in (S^1)^n$, let $\Psi(q,\tau) = \mathrm{e}^{r(q)/2}\tau$ be the point of T with coordinates $\mathrm{e}^{r_j(q)/2}\tau_j$. Thus $\Psi(q,\cdot)$ parametrizes the real torus $\{r = r(q)\}$, and we call the image of $Q \times (S^1)^n$ the *moment shell* of Q. Here and below, $\langle S^k \rangle$ denotes the non-pluripolar product of a closed positive $(1,1)$-current S; in top degree it is the non-pluripolar Monge–Ampère measure of Theorem 7.1.1.

Theorem 28.16.2 *Against every continuous test function on X,*

$$\frac{d!}{m^d} B_m \nu \rightharpoonup \mu_Q = \Psi_*(d!\, \mathrm{d}q_\Lambda\, \mathrm{d}\tau), \qquad \Psi(q,\tau) = \mathrm{e}^{r(q)/2}\tau. \tag{28.67}$$

The mass of μ_Q is $d!\operatorname{vol}_\Lambda(Q) > 0$. If $d = 0$, then μ_Q is the Haar probability measure on one real n-torus. If $d < n$, then μ_Q is singular with respect to smooth volume forms on X, and

$$\left\langle (\theta + \mathrm{dd^c} E_Q)^{d+1} \right\rangle = 0.$$

If $d = n$, then $\mu_Q = \langle (\theta + \mathrm{dd^c} E_Q)^n \rangle$.

Proof For $\alpha \in mQ \cap \mathbb{Z}^n$ and $q = \alpha/m$, the normalized monomial $z^\alpha e^m$ gives the probability measure $\mathrm{d}P_{m,q} = A_m(\alpha)^{-1}\mathrm{e}^{m(q\cdot r - F(r))} a(r)\, \mathrm{d}r\, \mathrm{d}\tau$ on T. For every $\eta > 0$ there is $c_\eta > 0$ such that $F(r) - q\cdot r + F^*(q) \geq c_\eta$ whenever $|r - r(q)| \geq \eta$, uniformly for $q \in Q$: on a large compact set this follows from the uniqueness of the minima, and outside it from (28.64), which is uniform on Q. Together with (28.66), this gives $P_{m,q}\{|r - r(q)| \geq \eta\} \leq C m^{n/2} \mathrm{e}^{-mc_\eta}$. Hence, for every continuous function on X, its integral against $P_{m,q}$ differs from its average over the torus $\{r = r(q)\}$ by $o(1)$, uniformly in $q \in Q$. The identity $d! m^{-d} B_m \nu = d! m^{-d} \sum_\alpha P_{m,\alpha/m}$ and the lattice Riemann sums prove (28.67). When $d < n$, the measure μ_Q is carried by the moment shell, which has real dimension $n + d < 2n$, so it is singular with respect to smooth volume forms.

For the curvature, note that $\theta + \mathrm{dd^c} E_Q = \mathrm{dd^c} G_Q$ on T. By uniform strong convexity of F^* on Q, the slope $q(r)$ maximizing $q \cdot r - F^*(q)$ over Q is unique. Adding the variational inequalities at r and s gives $(r - s)\cdot(q(r) - q(s)) \geq c|q(r) - q(s)|^2$. Thus q is Lipschitz, and comparison of the maximal values gives $\nabla G_Q = q$. Hence G_Q is $C^{1,1}$, and its Hessian has rank at most d almost everywhere, since $q(r) \in b + W$. In logarithmic coordinates, convex mollifications of G_Q converge in C^1, their Hessians stay bounded and converge in L^p_{loc} for every finite p, and their minors converge in L^1_{loc}. Induction on the degree in the Bedford–Taylor definition of products of bounded plurisubharmonic functions, together with integration by parts, identifies the powers of $\mathrm{dd^c} G_Q$ on T with the measures given by these minors. The non-pluripolar product agrees with them on T and puts no mass on the pluripolar set $X \setminus T$. The bound on the rank therefore proves the vanishing assertion.

When $d = n$, a maximizer in the interior of Q satisfies $r = \nabla F^*(q)$, so $G_Q = F$ on the set where $\nabla F(r) \in \operatorname{int} Q$. Elsewhere the maximizing slope lies in one of the finitely many facets of Q. On the set where the maximizing slope lies in a given facet, its component normal to that facet is constant, so the derivative of this component

vanishes at almost every point of the set. Indeed, at a point of differentiability and of density, a nonzero derivative would force a cone of positive density outside the set. Consequently $\det D^2 G_Q = \mathbf{1}_{\{\nabla F(r)\in Q\}} \det D^2 F$ almost everywhere. In our normalization, $(\mathrm{dd^c} F)^n = n! \det D^2 F(r)\, \mathrm{d}r\, \mathrm{d}\tau$. The change of variables $q = \nabla F(r)$ gives the asserted identity for the top product. □

For any Hermitian form ω on X, closed or not, the map Ψ embeds $\operatorname{relint} Q \times (S^1)^n$ onto a real $(n+d)$-dimensional submanifold $S_Q = \Psi(\operatorname{relint} Q \times (S^1)^n)$; for $d = 0$ this is a single torus. Let $g_\omega(u,v) = \omega(u, Jv)$ be the associated Riemannian metric, and define the function $\mathcal{J}_\omega > 0$ by $\Psi^* \mathrm{d}\operatorname{vol}_{\omega,S_Q} = \mathcal{J}_\omega\, \mathrm{d}q_\Lambda\, \mathrm{d}\tau$, where $\mathrm{d}\operatorname{vol}_{\omega,S_Q}$ is the Riemannian volume of S_Q for g_ω. Since the relative boundary has measure zero,

$$\mathrm{d}\mu_Q = \frac{d!}{\mathcal{J}_\omega \circ \Psi^{-1}}\, \mathrm{d}\operatorname{vol}_{\omega,S_Q}.$$

If $d < n$, the mixed measure $\langle (\theta + \mathrm{dd^c} E_Q)^d \rangle \wedge \omega^{n-d}$, where the zeroth power is interpreted as 1 when $d = 0$, has a locally bounded density on T and gives S_Q zero mass. Hence it differs from μ_Q. The limit μ_Q does not depend on the invariant probability measure ν, although the norms in each degree do.

28.16.4 A rank-one selection with nonclosed sampling

Example 28.16.1 On $(\mathbb{P}^1)^2$ take $L = O(4,4)$, $w = 0$, and $Q = \{(2+t, 2-t) : -1/2 \le t \le 1/2\}$. Then $\Lambda = \mathbb{Z}(1,-1)$, $\mathrm{d}q_\Lambda = \mathrm{d}t$, and

$$V_m = \operatorname{span}\{z_1^{2m+j} z_2^{2m-j} e^m : j \in \mathbb{Z},\ |j| \le m/2\}, \qquad N_m = 2\lfloor m/2 \rfloor + 1.$$

The limit μ_Q has mass one: it is the average, over $-1/2 \le t \le 1/2$, of the Haar probability measures on the real tori $\{|z_1|^2 = (2+t)/(2-t),\ |z_2|^2 = (2-t)/(2+t)\}$; indeed, these equations describe the inverse gradient map. The moment shell is a compact three-dimensional part of the hypersurface $|z_1 z_2| = 1$. Writing $c(p) = p\log(p/4) + (4-p)\log((4-p)/4)$, we have $F^*(2+t, 2-t) = 2c(2+t)$, which gives the envelope explicitly.

The quotient of two adjacent selected monomials is z_1/z_2. In homogeneous coordinates $z_j = v_j/u_j$, the corresponding map $[v_1 u_2 : u_1 v_2]$ to $\mathbb{P}^1$ is undefined at $(0,0)$ and (∞,∞). Thus the limit arises from radial concentration rather than from a globally defined submersion as in Section 28.7.

Let ω_0 be the sum of the two Fubini–Study forms, and set $f = \lambda|z_1|^2/(1+|z_1|^2)$ with a constant $\lambda \ne 0$. Then $\omega = \mathrm{e}^f \omega_0$ is a torus-invariant Hermitian form, and $\mathrm{d}\omega = \mathrm{e}^f \mathrm{d}f \wedge \omega_0 \ne 0$ wherever $\mathrm{d}f \ne 0$. Sampling with $\nu = \omega^2 / \int_X \omega^2$ gives the same limit of mass one, while the curvature remains $\theta = 4\omega_0$.

The assumption $b \in \mathbb{Z}^n$ excludes a parity obstruction: if $Q = \{b\}$ with b half-integral, then mQ contains no lattice point for odd m. Under our assumptions, Theorem 28.16.1 shows that $V_m \ne 0$ for all sufficiently large m. Polytopes touching

the boundary of P, weights or sampling measures that are not torus-invariant, and general multiplier-ideal families require additional arguments.

28.17 Rounding in multiplier-ideal kernel families

The growth degree $\lim_m \log N_m/\log m$ of the selected spaces does not by itself determine a limit of the kernel measures. At a critical logarithmic coefficient, the rounding in multiplier ideals can produce different leading section counts in even and odd degrees, while the potentials still converge to the ideal envelope. In the following product example the growth degree is positive.

Fix $q \geq 1$ and put $X = \mathbb{P}^1 \times Y$, where $Y = (\mathbb{P}^1)^q$. Write p_0 for the projection onto the first factor. For $0 \leq j \leq q$, let σ_j be the Fubini–Study form on the jth factor, normalized to have integral one, where $j = 0$ refers to the first factor. Use

$$L = \mathcal{O}(1, \dots, 1), \qquad \theta = c_1(L, h) = \sigma_0 + \theta_Y, \qquad \theta_Y = \sum_{j=1}^{q} \sigma_j,$$

where h is the product Fubini–Study metric and the forms σ_j are pulled back to X. We use the smooth positive probability measure $\nu = \sigma_0 \otimes \nu_Y$, where $\nu_Y = \sigma_1 \wedge \cdots \wedge \sigma_q = \theta_Y^q/q!$, and the continuous weight zero.

On the affine line of the first factor define

$$\psi_0(z) = \tfrac{1}{2} \log |z|^2 - \log(1 + |z|^2), \qquad \psi = \psi_0 \circ p_0.$$

In the coordinate $1/z$ at infinity, ψ_0 has the same expression. It equals $-\infty$ at 0 and ∞, and

$$\theta + \mathrm{dd^c}\psi = \tfrac{1}{2}[D_0] + \tfrac{1}{2}[D_\infty] + \theta_Y \geq 0, \qquad D_0 = \{0\} \times Y, \quad D_\infty = \{\infty\} \times Y. \tag{28.68}$$

We use ψ as selecting potential, so that the selected spaces are $V_m = H^0(X, L^m \otimes \mathcal{I}(m\psi))$, with the norm $\int_X |s|^2_{h^m} \, \mathrm{d}\nu$. For an orthonormal basis $(s_{m,j})$, write $\mathcal{B}_m = \sum_j |s_{m,j}|^2_{h^m}$ and $U_m = m^{-1} \log \mathcal{B}_m$. Let ς be the Haar probability measure on the circle $\{|z| = 1\} \subset \mathbb{P}^1$.

Theorem 28.17.1 *Put $a_m = 1$ for even m and $a_m = 2$ for odd m. Then*

$$N_m = \dim V_m = a_m (m+1)^q, \qquad \lim_{m\to\infty} \frac{\log N_m}{\log m} = q.$$

The partial envelope (28.1) with $K = X$ and obstacle zero is

$$E_{\theta,X}[\psi]_{\mathcal{I}}(0) = E = \psi + \log 2. \tag{28.69}$$

The norms satisfy the Bernstein–Markov inequality, and $U_m \to E$ locally uniformly on $X \setminus (D_0 \cup D_\infty)$ and in L^p for every finite p with respect to smooth volume forms; moreover, the currents $\theta + \mathrm{dd^c} U_m$ converge weakly to $\theta + \mathrm{dd^c} E$. However,

$$\frac{q!}{(2k)^q} \mathcal{B}_{2k} \nu \rightharpoonup q!\, \varsigma \otimes \nu_Y,$$
$$\frac{q!}{(2k+1)^q} \mathcal{B}_{2k+1} \nu \rightharpoonup 2q!\, \varsigma \otimes \nu_Y.$$

Thus the kernel measures normalized by $q!/m^q$ do not converge as $m \to \infty$ through all integers. Dividing instead by the dimension gives $N_m^{-1} \mathcal{B}_m \nu \rightharpoonup \varsigma \otimes \nu_Y$. The envelope has zero non-pluripolar top mass: $\langle (\theta + \mathrm{dd^c} E)^{q+1} \rangle = 0$.

Proof By Lemma 1.16.1,

$$\mathcal{I}(m\psi) = \mathcal{O}_X(-r_m D_0 - r_m D_\infty), \qquad r_m = \lfloor m/2 \rfloor.$$

In the product affine frame, global sections are polynomials of degree at most m in each variable. A global section belongs to V_m exactly when each of its monomials vanishes to order at least r_m along D_0 and along D_∞, that is, when the exponent α_0 of the first variable satisfies $r_m \le \alpha_0 \le m - r_m$; the other exponents range from 0 to m. There is one admissible value of α_0 when m is even and two when m is odd. This proves the dimension formula.

To identify the envelope, let $v \in \mathrm{PSH}(X, \theta)$ be a competitor, that is, $v \preceq_{\mathcal{I}} \psi$ in the multiplier-ideal order of Section 1.12 and $v \le 0$ quasi-everywhere. By Theorem 1.16.1, the Lelong numbers of v at the points of D_0 and D_∞ are at least $1/2$. Fix $y \in Y$ and restrict v to the line $\mathbb{P}^1 \times \{y\}$. The restriction is either identically $-\infty$ or σ_0-plurisubharmonic. In the latter case, its Lelong numbers at the two points over 0 and ∞ are at least those of v: the supremum over a ball in the line is at most the supremum over the corresponding ambient ball, and division by $\log r^2 < 0$ reverses the inequality. By the identification of Lelong numbers with atoms in Lemma 1.16.7, the curvature measure of the restriction has atoms of mass at least $1/2$ at these two points. By Stokes' theorem its total mass is one, so these atoms exhaust the measure. The difference between the restriction and ψ_0 is therefore harmonic on the compact projective line, hence constant.

It follows that $v(z, y) = \psi_0(z) + c(y)$ off $D_0 \cup D_\infty$, where $c(y) = -\infty$ on lines along which v is identically $-\infty$. Restriction to $z = 1$ shows that $c(y) = v(1, y) + \log 2$ is a θ_Y-plurisubharmonic function; it is not identically $-\infty$, since otherwise v would

be. Submean values over shrinking balls upgrade the obstacle condition to $v \le 0$ everywhere. Since $\max \psi_0 = -\log 2$, we get $c \le \log 2$, and hence $v \le E$. On $D_0 \cup D_\infty$ both v and E equal $-\infty$. Conversely, $E \le 0$ and E differs from ψ by a constant, so E is itself a competitor. This proves (28.69).

The norms of the monomials in one factor are given by (28.18) on $\mathbb{P}^1$, with $\ell = (m-\alpha, \alpha)$. With $x = |z|^2/(1+|z|^2)$, they are the beta integrals

$$\int_{\mathbb{P}^1} |z^\alpha e^m|^2_{h^m}\, \sigma_0 = \int_0^1 x^\alpha (1-x)^{m-\alpha}\, \mathrm{d}x = \frac{\alpha!(m-\alpha)!}{(m+1)!}. \tag{28.70}$$

Orthogonality under the rotations and integration over the product diagonalize the global norm. Summing over all monomials in one factor gives the constant kernel $m+1$, by the binomial formula. Therefore the full kernel on Y is $(m+1)^q$. With $C_k = (2k+1)\binom{2k}{k}$, the selected kernels are

$$\begin{aligned} \mathcal{B}_{2k} &= (2k+1)^q C_k \mathrm{e}^{2k\psi_0}, \\ \mathcal{B}_{2k+1} &= 2(2k+2)^q C_k \mathrm{e}^{2k\psi_0}. \end{aligned} \tag{28.71}$$

In the odd case the two monomials in the first factor have equal norms, and the sum of their squared moduli contributes the factor $1+|z|^2$, which gives the second identity. The full kernel on X is $(m+1)^{q+1}$, so evaluation gives $\sup_X |s|_{h^m} \le (m+1)^{(q+1)/2}\|s\|_m$ for every selected section. This polynomial bound proves the Bernstein–Markov inequality.

The central binomial coefficient is the largest in its row. Hence $4^k \le C_k \le (2k+1)4^k$, and $\log C_k = 2k\log 2 + O(\log(k+1))$. Taking logarithms in (28.71) gives

$$|U_m - E| \le \frac{C_q}{m}\bigl(1 + |\log|z|^2| + \log m\bigr) \quad \text{off } D_0 \cup D_\infty,$$

because $|\psi_0| = \log(2\cosh(\tfrac12 \log|z|^2))$ is bounded by $\log 2 + \tfrac12 |\log|z|^2|$. The function $\log|z|^2$ belongs to L^p for every finite p with respect to smooth volume forms. This proves the convergence of the potentials, and differentiating the L^1 limit proves the convergence of the currents. For every $m \ge 2$, both U_m and E are $-\infty$ on $D_0 \cup D_\infty$.

Let P_k be the probability measure on the first factor with density $C_k \mathrm{e}^{2k\psi_0}$ with respect to σ_0. Under P_k the angle of z is uniformly distributed, and x has the beta distribution with parameters $(k+1, k+1)$, whose mean is $1/2$ and whose variance is $1/[4(2k+3)]$. Chebyshev's inequality and continuity of the test functions in the angle therefore give $P_k \rightharpoonup \varsigma$. The measure identities

$$\mathcal{B}_{2k}\nu = (2k+1)^q P_k \otimes \nu_Y, \qquad \mathcal{B}_{2k+1}\nu = 2(2k+2)^q P_k \otimes \nu_Y$$

prove both limits along even and odd degrees and the limit after division by N_m. Since the two limits have different total masses, the q-normalized family does not converge. Normalization by the ambient degree, $(q+1)!/m^{q+1}$, instead gives zero in total variation.

Finally, off $D_0 \cup D_\infty$ the current in (28.68) is θ_Y, whose rank is q. The non-pluripolar product charges neither divisor, so the top product is zero. □

Unlike the Newton family of Section 28.16, this family of multiplier-ideal spaces is not multiplicative. For two odd degrees, the monomials of minimal order along D_0 are selected, but their product misses the condition in the even sum degree by one order. The Newton family has a leading term given by lattice counting in every sufficiently large degree; here the ideal envelope is identified, but the leading term oscillates. Testing against the constant function one shows that convergence of $q!N_m/m^q$ is necessary for the q-normalized kernel measures to converge. Since the envelope has zero non-pluripolar top mass, Theorem 28.6.1, whose hypotheses require positive mass, does not apply to this selecting potential.

28.18 Three monomial bands from a multiplier-ideal selection

For an isolated logarithmic pole of coefficient one on a complex surface, the multiplier ideal in degree m requires vanishing of order $m-1$, whereas domination by the pole would require order m. At the two base points of the pencil below, the multiplier-ideal conditions select three neighboring bands of monomials. All three contribute to the limit of the kernel measures, even though the competitor classes defined by multiplier ideals and by domination coincide.

Let $X = \mathbb{P}^1 \times \mathbb{P}^1$, and let $L = \mathcal{O}(1,1)$ carry its product Fubini–Study metric h, with $\theta = c_1(L,h)$. Let $[T_j : Z_j]$ be homogeneous coordinates on the jth factor, write $z_j = Z_j/T_j$, and set

$$\psi = \max\{\log|T_1T_2|_h^2, \log|Z_1Z_2|_h^2\}, \qquad p_+ = (0,\infty), \quad p_- = (\infty,0).$$

The two sections T_1T_2 and Z_1Z_2 of L vanish simultaneously only at $p_\pm$, so these are the only poles of ψ. In local frames the potentials of ψ are maxima of logarithms of squared moduli of holomorphic functions, so $\psi \in \mathrm{PSH}(X,\theta)$. On the torus $(\mathbb{C}^*)^2$ put

$$x_j = \log|z_j|^2, \qquad s = x_1 + x_2, \qquad \varphi_X = \log(1+\mathrm{e}^{x_1}) + \log(1+\mathrm{e}^{x_2}).$$

Then $\psi = \max(0,s) - \varphi_X$.

In this model the norm is computed on the compact set $K = \{x \in [-1,2]^2\} \subset (\mathbb{C}^*)^2$, which has nonempty interior. On K let

$$w = \frac{x_1^2 + x_2^2}{2} - \varphi_X, \qquad \mathrm{d}\nu = \frac{1}{9}\,\mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}\tau_1\,\mathrm{d}\tau_2,$$

where $\mathrm{d}\tau_j$ is the Haar probability measure in the angle of z_j. Thus ν is a probability measure supported on K, not a smooth positive volume form on X. With ψ as selecting

potential, set

$$V_m = H^0(X, L^m \otimes \mathcal{I}(m\psi)), \quad N_m = \dim V_m,$$
$$\|\sigma\|_m^2 = \int_K |\sigma|_{h^m}^2 \mathrm{e}^{-mw}\, \mathrm{d}\nu, \qquad \mathcal{B}_m = \sum_j |\sigma_{m,j}|_{h^m}^2,$$

where $(\sigma_{m,j})$ is an orthonormal basis. The norm is positive definite by the identity theorem on the interior of K. As in Section 28.1, we use $U_m = m^{-1}\log \mathcal{B}_m$ on X and $B_m = \mathrm{e}^{-mw}\mathcal{B}_m$ on K. Only the values of w on K enter the norm and the envelope.

Lemma 28.18.1 *For every $v \in \mathrm{PSH}(X,\theta)$,*

$$v \preceq_{\mathcal{I}} \psi \quad \Longleftrightarrow \quad v \le \psi + C_v \quad \textit{for some finite } C_v,$$

where $\preceq_{\mathcal{I}}$ is the multiplier-ideal order of Section 1.12. Consequently, for every obstacle, the envelopes defined with these two conditions agree. At either pole, for every real $t > 0$,

$$\mathcal{I}(t\psi) = \mathfrak{m}_{p_\pm}^{\max\{\lfloor t\rfloor - 1, 0\}}.$$

Away from the poles, $\mathcal{I}(t\psi)$ is the full structure sheaf.

Proof Near p_+ use the coordinates $\zeta_1 = z_1$, $\zeta_2 = 1/z_2$ and the frame $T_1 Z_2$. In this frame the two pencil sections T_1T_2 and Z_1Z_2 have coefficients ζ_2 and ζ_1, so

$$\psi = \log\max(|\zeta_1|^2, |\zeta_2|^2) - \log(1+|\zeta_1|^2) - \log(1+|\zeta_2|^2) = \log|\zeta|^2 + O(1).$$

The same calculation applies at p_-. The formula for the ideals is Lemma 1.16.1 with normal dimension two and coefficient one, including the case of equality in the integrability condition.

Now suppose that $v \preceq_{\mathcal{I}} \psi$. By Theorem 1.16.1, the Lelong numbers satisfy $\nu(v, p_\pm) \ge 1$; here $\nu(v,p)$ is the Lelong number of Section 1.13, not the measure ν. Let ϕ be the plurisubharmonic local potential of v in a frame near $p_\pm$, and consider its radial maximum

$$M(a) = \sup_{|\zeta| \le \mathrm{e}^{a/2}} \phi(\zeta)$$

inside a fixed coordinate ball. This function is finite and convex: by the three-circles theorem, the corresponding maximum on each complex line through the pole is convex, and M is the supremum of these functions; lines on which ϕ is identically $-\infty$ do not affect the supremum. The limiting slope of M as $a \to -\infty$ is the Lelong number. For $a < a_0$, convexity therefore gives

$$M(a) \le M(a_0) + \nu(v, p_\pm)(a - a_0) \le a + C.$$

Consequently $v \le \psi + C$ near both poles. After C is increased, the same bound holds on the compact complement, because v is bounded above and ψ is finite and continuous there. Conversely, domination $v \le \psi + C$ implies the inclusions of

multiplier ideals at all weights, by the definition through integrability. Thus the two competitor classes coincide before any obstacle is imposed. □

The pencil also describes the competitors that are not torus-invariant. Let ω_{FS} be the Fubini–Study form on $\mathbb{P}^1$, normalized to have integral one.

Lemma 28.18.2 *On $X \setminus \{p_+, p_-\}$ set*

$$f = [T_1T_2 : Z_1Z_2], \qquad \psi_{\mathrm{FS}} = \log(|T_1T_2|_h^2 + |Z_1Z_2|_h^2).$$

Every v in the class of Lemma 28.18.1 has the form

$$v = \psi_{\mathrm{FS}} + u \circ f, \qquad u \in \mathrm{PSH}(\mathbb{P}^1, \omega_{\mathrm{FS}}),$$

on $X \setminus \{p_+, p_-\}$, and $v(p_\pm) = -\infty$. Conversely, for every such u, the function $\psi_{\mathrm{FS}} + u \circ f$, extended by $-\infty$ at the two poles, belongs to that class.

Proof For $\xi \in \mathbb{C}^*$, the curve of the pencil

$$C_\xi = \{Z_1Z_2 = \xi T_1T_2\}$$

is a smooth rational curve of bidegree $(1,1)$ through both poles, and its θ-degree is two. The restriction of v to this connected curve is either identically $-\infty$ or $\theta|_{C_\xi}$-plurisubharmonic. In the latter case, its curvature measure has total mass two by Stokes' theorem. The local upper bounds of Lemma 28.18.1 give Lelong numbers at least one at the two poles, hence, by the identification of Lelong numbers with atoms in Lemma 1.16.7, atoms of mass at least one there. These atoms exhaust the measure. The curvature of $\psi|_{C_\xi}$ consists of the same two unit atoms. Thus $v - \psi$ is harmonic, in the sense of distributions, on the compact curve C_ξ, and therefore constant.

On the torus use the holomorphic coordinates $(z_1, \xi = z_1z_2)$. Local integrability and Fubini's theorem show that, for almost every ξ, the restriction of v to C_ξ is not identically $-\infty$; by the preceding paragraph it is then finite off the poles. Hence the restriction of v to the transverse curve $S = \{z_1 = 1\} \simeq \mathbb{P}^1$ is not identically $-\infty$. Put $u = v|_S + \log 2$. Since $\theta|_S = \omega_{\mathrm{FS}}$ and $\psi_{\mathrm{FS}}|_S = -\log 2$, the constants on the curves C_ξ give the asserted identity almost everywhere on the torus. Both sides have plurisubharmonic local potentials, so they agree everywhere off the poles. The logarithmic upper bounds give the value $-\infty$ at the poles.

Conversely, $\theta + \mathrm{dd^c}(\psi_{\mathrm{FS}} + u \circ f) = f^*(\omega_{\mathrm{FS}} + \mathrm{dd^c}u) \geq 0$ off the poles. In a chart at a pole, the local potential is bounded above by $\log(|\zeta_1|^2 + |\zeta_2|^2) + C$, since u is bounded above. It therefore extends across the pole as a plurisubharmonic function with value $-\infty$ there, which gives a global θ-plurisubharmonic function dominated by $\psi + C$. □

Theorem 28.18.1 *For these K and w, the partial envelope* (28.1) *is*

$$E_{\theta,K}[\psi]_I(w) = P = g(s) - \varphi_X,$$
$$g(s) = \sup_{0\le t\le 1}(ts - t^2) = \begin{cases} 0, & s \le 0, \\ s^2/4, & 0 \le s \le 2, \\ s - 1, & s \ge 2. \end{cases}$$

Here P is given by this formula on the torus and extends to a θ-plurisubharmonic function on X with $P(p_\pm) = -\infty$ and $\psi - 1 \le P \le \psi$.

Proof For an arbitrary competitor v, Lemma 28.18.2 gives, in terms of the affine coordinate ξ of $\mathbb{P}^1$,

$$\chi(\xi) = \log(1 + |\xi|^2) + u(\xi), \qquad v = \chi(z_1 z_2) - \varphi_X.$$

On the interior of K, the obstacle condition, which holds quasi-everywhere, becomes an inequality everywhere, by the submean inequality and the continuity of $w + \varphi_X$. Evaluating at $x_1 = x_2 = s/2$, with $s = \log|\xi|^2$, gives

$$\chi(\xi) \le s^2/4 \quad (-2 < s < 4)$$

for every argument of ξ. At $s = 0$ this bounds χ by zero on $|\xi| = 1$, and the maximum principle for the subharmonic function χ bounds it by zero on the whole disk $|\xi| \le 1$. At infinity use the local potential

$$\chi_\infty(\eta) = \chi(1/\eta) + \log|\eta|^2.$$

The obstacle at $s = 2$ bounds it by -1 on $|\eta| = \mathrm{e}^{-1}$, and the same maximum principle gives $\chi(\xi) \le s - 1$ for $s \ge 2$. Together with the obstacle for $0 \le s \le 2$, this proves $\chi \le g$, without any invariance assumption on the competitor.

Conversely, g is convex with slopes in $[0, 1]$, so $g(\log|\xi|^2) - \log(1 + |\xi|^2)$ is an ω_{FS}-plurisubharmonic function on $\mathbb{P}^1$. By Lemma 28.18.2, it defines the global function P. Directly,

$$\max(0, s) - 1 \le g(s) \le \max(0, s), \qquad g(s) \le s^2/4 \le (x_1^2 + x_2^2)/2.$$

Thus P belongs to the competitor class of Lemma 28.18.1 and satisfies the obstacle condition everywhere on K. Since it dominates every competitor, it is the envelope, including its values $-\infty$ at the poles. □

Proposition 28.18.1 *For every $m \geq 1$ the selected space is*

$$V_m = \operatorname{span}\{Z_1^a T_1^{m-a} Z_2^b T_2^{m-b} : 0 \leq a, b \leq m,\ |a-b| \leq 1\}, \qquad N_m = 3m+1.$$

In particular, N_m grows linearly in m. These spaces are not multiplicative. If the sections were instead required to satisfy $|\sigma|_{h^m}^2 \leq C\mathrm{e}^{m\psi}$, only the monomials with $a = b$ would remain. The measure ν is Bernstein–Markov for (K, w).

Proof At p_+ the monomial with exponents (a, b) has local exponents $(a, m-b)$, and at p_- it has local exponents $(m-a, b)$. By Lemma 28.18.1, both total orders must be at least $m-1$. Distinct exponent pairs cannot cancel, so these conditions select exactly the monomials with $|a-b| \leq 1$. The three bands $a = b$, $a = b+1$ and $a = b-1$ have $m+1$, m and m elements. The bound $|\sigma|_{h^m}^2 \leq C\mathrm{e}^{m\psi}$ would require total order m at both poles and leave only the diagonal. The product of two sections with offset $a - b = 1$ has offset two and lies outside the selected space of the sum degree; such products fail for every pair of positive degrees. Thus the linear growth of N_m is not the growth of a graded section ring.

Integration over the angles makes all global monomials orthogonal, including those outside the selected space. For $t \in [0, 1]$ set

$$I_m(t) = \int_{-1}^{2} \mathrm{e}^{m(tx - x^2/2)}\,\mathrm{d}x = \mathrm{e}^{mt^2/2} J_m(t), \qquad J_m(t) = \int_{-1}^{2} \mathrm{e}^{-m(x-t)^2/2}\,\mathrm{d}x.$$

The squared norm of the monomial with exponents (a, b) is

$$A_m(a, b) = \frac{1}{9} I_m(a/m) I_m(b/m). \tag{28.72}$$

With $c = \int_{-1}^{1} \mathrm{e}^{-y^2/2}\,\mathrm{d}y$, uniformly in $t \in [0, 1]$,

$$\frac{c}{\sqrt{m}} \leq J_m(t) \leq \sqrt{\frac{2\pi}{m}}, \qquad J_m(t) = \sqrt{\frac{2\pi}{m}}\bigl(1 + O(m^{-1/2}\mathrm{e}^{-m/2})\bigr).$$

For the lower bound, integrate over the interval of radius $m^{-1/2}$ centered at t; for the last assertion, note that each of the two omitted Gaussian tails is at most $\mathrm{e}^{-m/2}/m$. Hence

$$\frac{c^2}{9m} \mathrm{e}^{(a^2+b^2)/(2m)} \leq A_m(a, b) \leq \frac{2\pi}{9m} \mathrm{e}^{(a^2+b^2)/(2m)}. \tag{28.73}$$

For $x \in [-1, 2]^2$, completing squares bounds the weighted contribution of each normalized monomial by $9m/c^2$. Summing over all $(m+1)^2$ global monomials and using the evaluation inequality gives

$$\sup_K |\sigma|_{h^m} \mathrm{e}^{-mw/2} \leq C m^{3/2} \|\sigma\|_m \quad (\sigma \in H^0(X, L^m)).$$

This polynomial bound proves the Bernstein–Markov property. On the selected space the same argument gives the factor Cm. □

Theorem 28.18.2 *With values in $[-\infty,\infty)$ at the poles, the potentials satisfy*

$$U_m \geq P + \frac{\log(9m/(2\pi))}{m} - \frac{1}{4m^2}. \tag{28.74}$$

They converge to P locally uniformly on $X\setminus\{p_+,p_-\}$, in L^p with respect to a smooth volume form on X for every finite p, at the rate $O(\log m/m)$, and pointwise on X with values in $[-\infty,\infty)$. The currents $\theta+\mathrm{dd^c}U_m$ converge weakly to $\theta+\mathrm{dd^c}P$. Moreover, $\mathrm{e}^{U_m}\to\mathrm{e}^{P}$ uniformly on X, although for each fixed $m\geq 2$ the difference U_m-P is unbounded above near either pole.

Define a probability measure μ by

$$\int_X f\,\mathrm{d}\mu = \int_0^1 \int f(\mathrm{e}^{t/2}\tau_1,\mathrm{e}^{t/2}\tau_2)\,\mathrm{d}\tau_1\,\mathrm{d}\tau_2\,\mathrm{d}t.$$

Then, against every continuous test function on X,

$$N_m^{-1}B_m\nu \rightharpoonup \mu, \qquad m^{-1}B_m\nu \rightharpoonup 3\mu.$$

The measures $2m^{-2}B_m\nu$, normalized by the ambient dimension, tend to zero in total variation. The envelope satisfies $\langle(\theta+\mathrm{dd^c}P)^2\rangle=0$.

Proof The diagonal monomials belong to V_m. The upper bound for their norms in (28.73) gives

$$\mathcal{B}_m \geq \frac{9m}{2\pi}\mathrm{e}^{-m\varphi_X}\max_{0\leq j\leq m}\mathrm{e}^{js-j^2/m}.$$

For $0\leq s\leq 2$, the grid point j/m nearest to $s/2$ loses at most $1/(4m^2)$ compared with $g(s)$; outside this interval the maximizing endpoint lies on the grid. This proves (28.74) on the torus.

For a monomial with offset one, $(a,b)=(j+1,j)$, the lower bound in (28.73) shows that the exponent of its normalized contribution, apart from $-m\varphi_X$ and constants, is at most both

$$js-j^2/m+x_1, \qquad (j+1)s-(j+1)^2/m+1-x_2.$$

The band with $(a,b)=(j,j+1)$ is obtained by interchanging the indices. Put

$$R(x)=1+\min(\mathrm{e}^{x_1},\mathrm{e}^{1-x_2})+\min(\mathrm{e}^{x_2},\mathrm{e}^{1-x_1}).$$

The lower bound for the norms and the count of monomials give

$$U_m \leq P + \frac{C+2\log(m+1)+\log R(x)}{m}. \tag{28.75}$$

The function R extends continuously across the coordinate divisors away from the poles. In a chart at p_+, put $A=|\zeta_1|^2$ and $B=|\zeta_2|^2$. The possibly large term of R satisfies

$$\min(1/B,\mathrm{e}/A)\leq(\mathrm{e}+1)/(A+B),$$

and the other term is bounded; the same holds at p_-. Thus $R \le C(1 + r^{-2})$ near either pole, where $r = |\zeta|$ is the local radius, and $\log R$ belongs to L^p for every finite p with respect to smooth volume forms.

For $m \ge 2$, all selected sections vanish at the poles. The two diagonal sections $T_1^m T_2^m$ and $Z_1^m Z_2^m$ have no other common zeros, so $\mathcal{B}_m > 0$ off the poles. Away from the poles, P and U_m are continuous, so both inequalities extend from the torus to $X \setminus \{p_+, p_-\}$. They give the asserted local uniform convergence and convergence in L^p, and differentiation of the L^1 limit in the sense of distributions gives the weak convergence of the currents. At the poles both U_m and P equal $-\infty$ for $m \ge 2$, which also proves the pointwise statement on X.

For a fixed $m \ge 2$, the lowest-order terms of the selected sections at a pole include every binary monomial of degree $m - 1$, and the sum of their squared moduli is comparable to $r^{2(m-1)}$ in all directions. Hence

$$U_m = (1 - 1/m) \log r^2 + O_m(1), \qquad P = \log r^2 + O(1),$$

so the difference $U_m - P$ tends to $+\infty$ near each pole. Nevertheless, (28.75) gives, uniformly in $m \ge 2$ and $r < 1$,

$$\mathrm{e}^{U_m} \le C r^{2-2/m} \le C r, \qquad \mathrm{e}^{P} \le C r^2.$$

Combined with uniform convergence on compact subsets away from the poles, these bounds give uniform convergence of e^{U_m}.

The contribution of each normalized monomial to $B_m \nu$ is exactly the product of two truncated Gaussian probability measures in x_1, x_2, centered at $(a/m, b/m)$, with the Haar probability measures in the angles. Completing squares in (28.72) and using its uniform bounds proves concentration at these centers, uniformly over all three bands; for example, the probability of $|x_1 - a/m| \ge \eta$ is at most $C_\eta \sqrt{m}\, \mathrm{e}^{-m\eta^2/2}$. Uniform continuity of a test function then reduces the sum to three lattice Riemann sums. Each band contains asymptotically one third of the N_m monomials, and in each band the centers $(a/m, b/m)$ approach the diagonal point (t, t), $t \in [0, 1]$. This proves both positive limits. The total mass of $2m^{-2} B_m \nu$ is $2(3m + 1)/m^2 \to 0$, which proves the convergence to zero in total variation.

Off the poles, $\theta + \mathrm{dd^c} P$ is the pullback by f of the current on $\mathbb{P}^1$ defined by g. The function g is $C^{1,1}$, and the local complex Hessian of $g(s)$ has rank at most one. Smooth convex approximations and the product of locally bounded potentials show that its square vanishes, as in the Hessian argument of Section 28.16. The non-pluripolar product assigns no mass to the two poles, so the global square is zero. In particular, the positive-mass hypothesis of Theorem 28.6.1 fails in this model. □

The measure μ is carried by the real three-dimensional shell $S = \{(\mathrm{e}^{t/2}\tau_1, \mathrm{e}^{t/2}\tau_2) : 0 \le t \le 1\}$. As after Theorem 28.16.2, it has a positive density with respect to the Riemannian volume of S for every Hermitian form ω on X, which need be neither closed nor torus-invariant. The mixed measure $\langle \theta + \mathrm{dd^c} P \rangle \wedge \omega$ has a locally bounded density off the poles and gives S zero mass, so it cannot reproduce the positive limit of the kernel measures.

Part VIII
Closed classes and locally conformally Kähler geometry

Chapter 29
Bott–Chern classes and singular loci

This part of the book returns to closed backgrounds. A closed real $(1,1)$-form has a Bott–Chern class, and positivity can then be asked of the class rather than of a chosen representative. On a compact Kähler manifold, the positivity of a class is described by its non-nef and non-Kähler loci, by intersection numbers and by volumes. The present chapter asks which parts of this description survive on an arbitrary compact complex manifold, where no Kähler metric controls masses, and how closed classes interact with the Hermitian equations of the preceding parts. In Chapter 30, closed classes arise from the Lee form of a locally conformally Kähler metric.

Throughout the chapter, X is a connected compact complex manifold of dimension $n \geq 1$ as in the Conventions, and ω is a fixed Hermitian form on X; sections that require a Kähler manifold or a special geometry say so at their beginning. Without a Kähler class, Gauduchon metrics provide the substitute mass control. Pairing a closed current with the $(n-1)$st power of a Gauduchon metric eliminates its potential, so the currents of a fixed class whose negative part is bounded by a multiple of ω have uniformly bounded mass. With this compactness, regularization defines the non-nef locus through Lelong numbers of minimal currents and the non-Kähler locus through polar sets of Kähler currents. These loci separate the singularities forced by the class from those of a particular current. Restricted volumes, which describe the same loci on Kähler manifolds, behave differently: computed with a Hermitian error, they can be infinite even for the zero class. When X belongs to Fujiki's class $\mathcal{C}$, pulling a class back to a Kähler resolution of each subvariety gives instead a finite restricted mass; on a smooth subvariety, its positivity detects bigness of the restricted class. For a divisorial trace with bounded remainder, the non-pluripolar mass is the top intersection of a nef residual class; positive mass produces a Kähler current, but possibly with a larger singularity than the trace.

Closed classes also constrain the Hermitian Monge–Ampère equations of earlier chapters. A divisorial component of a positive current in a Hermitian background consumes Gauduchon trace but contributes no density to its non-pluripolar measure, so it can lower the largest coefficient of a subsolution. At the maximal coefficient, weak limits of maximizing Gauduchon tests annihilate the class of the divisor. On the elliptic threefold of Section 27.18, closedness forces every positive current to descend to the

base surface. No closed class is big there, although small Hermitian perturbations of the boundary classes have positive variational volume, whose degeneration can be computed explicitly.

Smooth solutions raise a metric question. For a big class on a compact Kähler manifold, the solution of the Monge–Ampère equation with a smooth positive density defines a Kähler metric on the ample locus, but its smoothness there does not determine the distance to the complement. Zeros of the density can come from ramification, from collapsed exceptional fibers or from a singular Hessian, and they give different normal lengths even when the determinant vanishes to the same order. For radial equations, boundedness of the potential and finiteness of the distance are governed by two different integrals: a continuous minimal remainder can have a complete end, whereas a Hölder remainder forces finite distance. For exponential equations with a divisorial pole, dividing the determinant by its tangential part gives a lower bound for normal length and a sharp logarithmic threshold for completeness. Explicit surfaces give exponential equations with unique finite-energy solutions and incomplete metrics. One solution has minimal singularities in a big non-nef class; another has a nef and big class but an unbounded cusp.

At the boundary of the nef cone, smooth semipositive representatives have a different metric behavior. On a complex torus, a constant semipositive form whose kernel leaves are dense is the only closed positive current in its class, and adding a small fixed Kähler form produces metrics whose diameters tend to zero. An explicit family of flat metrics collapses in the same way at a class with many semipositive representatives, so the collapse depends on how the class is approached. On the resolution of a Kummer quotient, an irrational nef boundary class has many smooth semipositive representatives. In a family of blown-up projective planes, degree-zero curves on the nearby fibers force a larger singularity along one component of the central curve, so a prescribed semipositive metric on the special fiber does not extend to the family. Finally, on Oeljeklaus–Toma manifolds the natural closed semipositive class contains positive currents supported on real hypersurfaces but no singular potentials, and in dimension three a natural Gauduchon metric has masses on both sides of its background mass.

29.1 Closed classes with Hermitian lower bounds

Testing a closed current against a Gauduchon form eliminates its potential. The resulting trace depends only on the class, and it bounds the mass of every almost positive representative.

Recall from Section 1.17 that $H^{1,1}_{\mathrm{BC}}(X,\mathbb{R})$ is the space of smooth closed real $(1,1)$-forms modulo dd^{c} of smooth real functions, and from Section 24.4 that a class is *pseudoeffective* if it contains a closed positive current. It is *nef* if, for every $\varepsilon > 0$, it has a smooth representative bounded below by $-\varepsilon\omega$ (Section 24.6). Fix a class $\alpha \in H^{1,1}_{\mathrm{BC}}(X,\mathbb{R})$ and a smooth closed representative θ. By Lemma 1.17.4, the closed real $(1,1)$-currents in α are exactly the currents $T = \theta + \mathrm{dd}^{\mathrm{c}}u$ with u a real

distribution, and the potential of a positive one is quasi-plurisubharmonic. A closed current T with $T \geq \delta\omega$ for some $\delta > 0$ is called a *Kähler current*; we also say that such a current is *strict*. The class α is *big* if it contains a Kähler current, that is, if θ is big in the sense of Section 1.19. None of these properties depends on the choice of θ, because replacing θ by $\theta + \mathrm{dd}^{\mathrm{c}} f$ with f smooth only changes the potentials by $-f$.

For $\varepsilon > 0$, let

$$\alpha[-\varepsilon\omega] = \{T \in \alpha : T \geq -\varepsilon\omega\}$$

be the set of closed currents in α whose negative part is bounded by $\varepsilon\omega$.

Proposition 29.1.1 *For each fixed $\varepsilon > 0$, the currents in $\alpha[-\varepsilon\omega]$ have uniformly bounded distributional mass.*

Proof Choose a Gauduchon metric γ. For $T = \theta + \mathrm{dd}^{\mathrm{c}} u$ in $\alpha[-\varepsilon\omega]$, the current $T + \varepsilon\omega$ is positive, and since $\mathrm{dd}^{\mathrm{c}}\gamma^{n-1} = 0$, Stokes' theorem for currents gives

$$\int_X (T + \varepsilon\omega) \wedge \gamma^{n-1} = \int_X (\theta + \varepsilon\omega) \wedge \gamma^{n-1}.$$

The right-hand side does not depend on T. Since γ is comparable to any fixed reference metric, this trace bounds all coefficient measures of the positive current $T + \varepsilon\omega$. Subtracting the fixed smooth form $\varepsilon\omega$ gives the asserted bound for T. In dimension one the same argument applies with $\gamma^{n-1} = 1$ and the ordinary Stokes theorem. □

Proposition 29.1.2 *The pseudoeffective cone in $H^{1,1}_{\mathrm{BC}}(X, \mathbb{R})$ is closed and convex and contains no nonzero line. For a fixed Gauduchon metric γ, the trace functional ℓ_γ of Section 24.4 is strictly positive on its nonzero elements, and the slice of pseudoeffective classes with $\ell_\gamma = 1$ is compact. This slice may be empty. Pullback and pushforward by a compact modification are continuous linear maps on Bott–Chern cohomology and preserve pseudoeffectivity.*

Proof The finite-dimensional space of smooth representatives in Lemma 24.7.1 identifies Bott–Chern cohomology with a finite-dimensional vector space and gives it its usual topology; the trace functional is continuous for this topology. Its positivity and the absence of lines in the cone were proved in Section 24.4, and convexity follows by adding positive currents.

Let pseudoeffective classes α_j converge to α, and choose positive closed representatives T_j. Their masses $\ell_\gamma(\alpha_j)$ are bounded, so by weak compactness a subsequence converges to a positive closed current T. Let B_γ be the continuous smooth representative map of Lemma 24.7.1. Along the subsequence, $B_\gamma(T_j)$ converges both to $B_\gamma(T)$ and to the representative of α, so T represents α. Thus the cone is closed. On its mass-one slice, (24.3) bounds $B_\gamma(T)$ in the finite-dimensional representative space. The slice is therefore closed and bounded there, hence compact.

The class maps of a modification are linear, and they are well-defined because classes can be represented equally by smooth forms and by currents. As linear maps between finite-dimensional spaces, they are continuous. Pullback preserves positive closed currents, since it can be computed on local plurisubharmonic potentials as

in Proposition 24.7.1, and proper pushforward preserves positivity and closedness. Applying these maps to positive representatives proves the assertions about the cones. □

This compactness is linear. The functional ℓ_γ pairs a closed current with a fixed smooth form; it is not a top-degree volume of the class, and it gives no bound for the Monge–Ampère masses of potentials.

The mass bound of Proposition 29.1.1 also shows that a nef class α is pseudo-effective. Indeed, choose smooth representatives $\theta_j \in \alpha$ with $\theta_j \geq -\varepsilon_j\omega$, where $\varepsilon_j \downarrow 0$. They have the same Gauduchon trace and hence uniformly bounded mass. A weak cluster value is a closed current, positive because the negative parts tend to zero, and it belongs to α because the smooth representative map of Lemma 24.7.1 is continuous.

29.2 Minimal multiplicities and the non-nef locus

In this section α is pseudoeffective. A current in $\alpha[-\varepsilon\omega]$ may have singularities that the class does not force. To isolate the forced ones, take in each family a current with the mildest singularities and let the allowed negative error tend to zero. The Lelong numbers that persist define the non-nef locus.

For $\varepsilon > 0$ put

$$V_\varepsilon = V_{\theta+\varepsilon\omega}, \qquad T_{\min,\varepsilon} = \theta + \mathrm{dd^c} V_\varepsilon,$$

where $V_{\theta+\varepsilon\omega} = P_{\theta+\varepsilon\omega}(0)$ is the normalized envelope of Section 1.19. Since α is pseudoeffective, $\mathrm{PSH}(X, \theta+\varepsilon\omega)$ is nonempty, so V_ε is $(\theta+\varepsilon\omega)$-plurisubharmonic and $T_{\min,\varepsilon} \in \alpha[-\varepsilon\omega]$. Moreover, V_ε has minimal singularities in this family: $u - \sup_X u \leq V_\varepsilon$ for every $u \in \mathrm{PSH}(X, \theta+\varepsilon\omega)$.

Definition 29.2.1 For $x \in X$, the *minimal multiplicity* of α at x is

$$\nu(\alpha, x) = \sup_{\varepsilon>0} \nu(T_{\min,\varepsilon}, x).$$

The *non-nef locus* is

$$E_{\mathrm{nn}}(\alpha) = \{x \in X : \nu(\alpha, x) > 0\}.$$

As ε decreases, the family $\alpha[-\varepsilon\omega]$ shrinks, so the Lelong numbers of its minimal current can only increase; the supremum is therefore a limit as $\varepsilon \downarrow 0$. Two Hermitian forms are comparable up to a constant factor, so the families defined by them are nested after rescaling ε; hence $\nu(\alpha, x)$ does not depend on ω. Replacing θ by $\theta + \mathrm{dd^c} f$ changes the minimal potentials by $-f$ up to a bounded normalization error, which does not affect Lelong numbers.

The smooth regularization of Section 1.15.1 turns vanishing Lelong numbers into an arbitrarily small loss of positivity. This gives the converse direction in the following characterization.

Proposition 29.2.1 *For a pseudoeffective class α, the set $E_{\mathrm{nn}}(\alpha)$ is a countable union of analytic subsets. Moreover,*

$$\alpha \text{ is nef} \quad \Longleftrightarrow \quad E_{\mathrm{nn}}(\alpha) = \varnothing.$$

Proof By the monotonicity just noted,

$$E_{\mathrm{nn}}(\alpha) = \bigcup_{k,\ell \geq 1} \{x : \nu(T_{\min,1/\ell}, x) \geq 1/k\},$$

and each set in this union is analytic by Siu's analyticity theorem (Section 1.13.1).

If α is nef, each family $\alpha[-\varepsilon\omega]$ contains a smooth current $\theta + \mathrm{dd}^{\mathrm{c}} f$. By minimality, $V_\varepsilon \geq f - \sup_X f$, so V_ε is bounded and all Lelong numbers of $T_{\min,\varepsilon}$ vanish. Conversely, if the non-nef locus is empty, every $T_{\min,\varepsilon}$ has zero Lelong numbers. The regularization just recalled then gives a smooth representative of α bounded below by $-2\varepsilon\omega$. Since $\varepsilon > 0$ is arbitrary, α is nef. □

The approximating families cannot be replaced by the single family of positive currents in α: even for a nef class, a positive current with minimal singularities in α itself may have nonzero Lelong numbers; see the example discussed in Section 36.16.

29.3 The non-Kähler locus

A big class contains Kähler currents, but in general none of them is smooth. The points at which every Kähler current in the class is singular form its non-Kähler locus. Analytic regularization, maxima of potentials and the descending chain condition for analytic sets produce a single Kähler current whose singular set is exactly this locus.

For an almost positive closed $(1,1)$-current T on X, write

$$E_+(T) = \{x \in X : \nu(T, x) > 0\}$$

for its *positive Lelong set*, where $\nu(T, x)$ is the Lelong number of a local potential of T at x.

Definition 29.3.1 The *non-Kähler locus* of a big class α is

$$E_{\mathrm{nK}}(\alpha) = \bigcap_{T} E_+(T),$$

where T ranges over the Kähler currents in α. For a pseudoeffective class that is not big, we set $E_{\mathrm{nK}}(\alpha) = X$.

Theorem 29.3.1 *Let α be big. There is a Kähler current in α with analytic singularities, smooth off its polar set, whose positive Lelong set is exactly $E_{\mathrm{nK}}(\alpha)$. In particular this locus is a proper analytic subset, and*

$$E_{\mathrm{nn}}(\alpha) \subset E_{\mathrm{nK}}(\alpha), \qquad E_{\mathrm{nK}}(\alpha) = \varnothing \quad \Longleftrightarrow \quad \alpha \text{ is Kähler.}$$

Proof *Step 1: reduction to analytic singularities.* By the analytic regularization of Section 1.15.1, a Kähler current $T \in \alpha$ can be approximated by currents in α with analytic singularities and smooth remainder away from their poles. After a sufficiently small loss of positivity these are still Kähler currents, and their positive Lelong sets are contained in $E_+(T)$. Hence the intersection defining $E_{\mathrm{nK}}(\alpha)$ may be taken over Kähler currents with analytic singularities and smooth remainder.

Step 2: intersections. Let $\theta + \mathrm{dd}^{\mathrm{c}} u_1$ and $\theta + \mathrm{dd}^{\mathrm{c}} u_2$ be two such currents. By Lemma 1.3.1, $\max(u_1, u_2)$ is the potential of a current dominating $\delta\omega$ for a common $\delta > 0$, and its Lelong number at each point is the minimum of those of u_1 and u_2. Regularizing this current as in Step 1 gives a Kähler current with analytic singularities whose positive Lelong set is contained in the intersection of the two previous sets.

Step 3: realization. The positive Lelong sets of Step 1 are analytic subsets of X. By the descending chain condition for analytic subsets of a compact complex manifold, some finite intersection of them is minimal, and it then equals the full intersection $E_{\mathrm{nK}}(\alpha)$. Repeated use of Step 2 gives a single Kähler current with analytic singularities whose polar set, which is also its positive Lelong set, is contained in this finite intersection. Since $E_{\mathrm{nK}}(\alpha)$ is contained in the positive Lelong set of every Kähler current in α, equality holds.

Step 4: the two consequences. For a positive current $T \in \alpha$ and $\varepsilon > 0$, we have $T \in \alpha[-\varepsilon\omega]$, so minimality gives $\nu(T_{\min,\varepsilon}, x) \leq \nu(T, x)$. Taking the supremum over ε and then the intersection over Kähler currents T gives $E_{\mathrm{nn}}(\alpha) \subset E_{\mathrm{nK}}(\alpha)$. If the non-Kähler locus is empty, the current of Step 3 has no poles and is therefore a Kähler form in α. Conversely, if α contains a Kähler form, using it in the defining intersection gives $E_{\mathrm{nK}}(\alpha) = \varnothing$. □

Proposition 29.3.1 *The non-Kähler locus of a big class has no isolated points.*

Proof Let $\theta + \mathrm{dd}^{\mathrm{c}}\Phi \geq \delta\omega$ be a current as in Theorem 29.3.1, and suppose that x is an isolated point of its polar set. Choose holomorphic coordinates $z = (z_1, \ldots, z_n)$ on a ball centered at x that contains no other pole. For a sufficiently large constant $A > 0$ and any real constant C, the local function $q(z) = A|z|^2 - C$ satisfies $\theta + \mathrm{dd}^{\mathrm{c}} q \geq \delta'\omega$ on a smaller ball, for some $\delta' > 0$.

Since Φ is smooth on a closed annular collar near the boundary of this smaller ball, we can choose C so that $q < \Phi$ on the collar. Since $\Phi \to -\infty$ at x, the function $\max(\Phi, q)$ agrees with q near x and with Φ on the collar. By Lemma 1.3.2, it glues with Φ outside the ball to the potential of a Kähler current in α that is smooth near x. Its Lelong number at x is zero, contradicting $x \in E_{\mathrm{nK}}(\alpha)$. □

Bigness of a closed class has a strong global consequence. A compact complex manifold carries a Kähler current if and only if it belongs to Fujiki's class $\mathcal{C}$; this is the classical modification criterion discussed in Section 36.16. Hence a compact complex manifold outside class $\mathcal{C}$ has no big Bott–Chern class at all, whereas the strictly positive nonclosed backgrounds studied in this book exist on every compact complex manifold.

29.4 Null loci and numerical restricted volumes

In this section X is Kähler and ω is a Kähler form. For a closed class α and an irreducible analytic subvariety $V \subset X$ of positive dimension, the intersection number $\int_V \alpha^{\dim V}$ is computed by pulling back a smooth closed representative to a resolution of V; by Stokes' theorem it does not depend on the representative. For a nef class α, the subvarieties on which this number vanishes form the classical *null locus*

$$\mathrm{Null}(\alpha) = \bigcup_{\substack{V \subset X \text{ irreducible},\ \dim V > 0 \\ \int_V \alpha^{\dim V} = 0}} V.$$

For a nef and big class, the classical theorem discussed in Section 36.16 gives

$$E_{\mathrm{nK}}(\alpha) = \mathrm{Null}(\alpha).$$

The same equality holds on compact manifolds in class $\mathcal{C}$.

For a pseudoeffective class that is not nef, intersection numbers are replaced by numerical restricted volumes. Let $V \subset X$ be an irreducible compact analytic subvariety of dimension $k > 0$, with regular locus V_{reg}. If $V \subset E_{\mathrm{nn}}(\alpha)$, the numerical restricted volume of α along V is defined to be zero. Otherwise, for real $\varepsilon > 0$, let T range over the closed currents in α with $T \geq -\varepsilon\omega$ that have analytic singularities

with smooth remainder and whose polar set does not contain V, and set

$$\langle \alpha^k \rangle_{X|V} = \lim_{\varepsilon \downarrow 0} \sup_T \int_{V_{\mathrm{reg}}} \big((T + \varepsilon\omega)|_{V_{\mathrm{reg}}}\big)^k_{\mathrm{ac}}, \tag{29.1}$$

where the subscript ac denotes the absolutely continuous part of a measure on V_{reg}. Because T has smooth remainder, the integrand is the extension by zero of a smooth measure defined away from an analytic subset of V.

The *numerical null locus* of α is the union of the positive-dimensional irreducible subvarieties on which (29.1) vanishes. For big classes, the trace theory of [Xia26, Section 8.3] expresses these restricted volumes as masses of trace currents on V; this is why singularity types and restrictions of multiplier ideals enter into the identification of this locus. Such a trace need not equal the pointwise restriction of a singular potential; see Chapter 20.

29.5 Restricted volumes and Hermitian perturbations

We return to an arbitrary compact complex manifold X with a Hermitian form ω. In (29.1), the negative part of T is measured by a closed Kähler form. If the closed perturbation is replaced by the Hermitian form ω, the supremum can be infinite even for the zero Bott–Chern class $\{0\}$ and $V = X$. Unbounded Monge–Ampère masses of smooth potentials, such as those on the Hopf threefold, already produce this failure.

Proposition 29.5.1 *Suppose there are smooth real functions v_j on X, for positive integers j, such that*

$$\omega + \mathrm{dd}^{\mathrm{c}} v_j \geq 0, \qquad \int_X (\omega + \mathrm{dd}^{\mathrm{c}} v_j)^n \longrightarrow +\infty.$$

Then for every $\varepsilon > 0$,

$$\sup_{\substack{T \in \{0\} \text{ smooth, closed} \\ T \geq -\varepsilon\omega}} \int_X (T + \varepsilon\omega)^n = +\infty. \tag{29.2}$$

This occurs for the Gauduchon form on the Hopf threefold in Theorem 9.0.1.

Proof Fix $\varepsilon > 0$ and take $T_j = \varepsilon \mathrm{dd}^{\mathrm{c}} v_j$. Each T_j is smooth and closed and represents the zero Bott–Chern class. Moreover,

$$T_j + \varepsilon\omega = \varepsilon(\omega + \mathrm{dd}^{\mathrm{c}} v_j) \geq 0,$$

and its top mass $\varepsilon^n \int_X (\omega + \mathrm{dd}^{\mathrm{c}} v_j)^n$ tends to $+\infty$. The functions v_j on the Hopf threefold are provided by Theorem 9.0.1. □

The currents T_j are smooth, so requiring analytic singularities with smooth remainder does not exclude them. Moreover, the supremum is infinite for each fixed

ε, so letting ε tend to zero afterwards cannot produce a finite limit. By contrast, if ω is Kähler and $T = \mathrm{dd}^{\mathrm{c}} v$ is smooth, Stokes' theorem gives

$$\int_X (T + \varepsilon\omega)^n = \varepsilon^n \int_X \omega^n \longrightarrow 0.$$

On the other hand, the zero class is nef, its non-nef locus is empty, and its only positive representative is the zero current. Indeed, a positive representative $\mathrm{dd}^{\mathrm{c}} u \geq 0$ would make u a global plurisubharmonic function on a compact manifold, hence constant. Thus (29.2) measures the nonlinear mass of the Hermitian error family rather than positivity of the closed class.

A numerical theory of restricted volumes in a Hermitian background must therefore either control the approximating family or use a different invariant. Two qualitative loci avoid this supremum. The non-Hermitian locus of a big background is defined through its strict currents, whose exterior derivative is prescribed by the background; Dang's null locus tests failure of bigness after restriction to subvarieties. The singularities allowed in the first definition, and the equality of the two loci for nef and big backgrounds, are recalled in Section 35.3.

29.6 Finite restricted masses of closed classes

The nonclosed error in Proposition 29.5.1 destroys the cohomological bound on masses. On a compact Kähler manifold, by contrast, the non-pluripolar mass of a positive current in a closed class is bounded by a top intersection number. Pulling a class back to a Kähler resolution of a subvariety therefore gives a finite invariant: the supremum of the non-pluripolar masses of positive currents in the pulled-back class. It measures positivity along every subvariety, including subvarieties along which a given modification of X fails to be an isomorphism.

In this section and the next, X belongs to Fujiki's class $\mathcal{C}$. Then every irreducible reduced subvariety $Y \subset X$ has a resolution $\pi\colon Z \to Y$ with Z a compact Kähler manifold: closed subspaces of a space in class $\mathcal{C}$ are again in class $\mathcal{C}$, and the Kähler modification theorem applies to Y. These classical facts are discussed in Section 36.16. In particular, a Kähler model of Y is obtained directly, not as a proper transform under one fixed modification of X.

Two such resolutions Z_1, Z_2 of Y admit a common Kähler refinement. Indeed, apply embedded resolution to the closure in $Z_1 \times Z_2$ of the graph of the induced bimeromorphic map. The resulting strict transform is smooth, and it is Kähler because the ambient resolution is a sequence of blowups of the compact Kähler product along smooth centers. For such a sequence p of blowups, after omitting the blowups along hypersurfaces, which are isomorphisms, Lemma 1.15.1, applied to a Kähler form h on $Z_1 \times Z_2$, gives the Kähler form $C_0 p^* h - \eta_D$ on the blown-up manifold.

For a closed positive $(1,1)$-current S on a compact Kähler manifold Z of dimension $k > 0$, write $\langle S^k \rangle$ for its non-pluripolar Monge–Ampère measure, that is, $\mathrm{MA}_\eta^{\mathrm{ext}}(u)$ for any decomposition $S = \eta + \mathrm{dd}^{\mathrm{c}} u$ with η smooth and closed (Theorem 7.1.1). For

a class $b \in H^{1,1}_{\mathrm{BC}}(Z,\mathbb{R})$, set

$$\mathcal{R}_Z(b) = \sup_{\substack{S \in b \\ S \geq 0}} \int_Z \langle S^k \rangle, \qquad \sup \varnothing = 0. \tag{29.3}$$

For a class $a \in H^{1,1}_{\mathrm{BC}}(X,\mathbb{R})$ and an irreducible reduced subvariety $Y \subset X$ of dimension $k > 0$, let $j\colon Y \hookrightarrow X$ be the inclusion, write $f = j \circ \pi$, and define

$$\mathcal{R}(a;Y) = \mathcal{R}_Z(f^* a). \tag{29.4}$$

Proposition 29.6.1 *The quantities in* (29.3) *and* (29.4) *are finite, and the latter is independent of the Kähler resolution. If $\eta \in b$ is smooth, κ is Kähler and $C > 0$ satisfies $C\kappa - \eta \geq 0$, then*

$$\mathcal{R}_Z(b + t\{\kappa\}) \leq (C+t)^k \int_Z \kappa^k \qquad (t \geq 0). \tag{29.5}$$

Let $\mu\colon X' \to X$ be a modification and Y' the proper transform of a subvariety Y that meets the locus where μ is an isomorphism. Then

$$\mathcal{R}(\mu^* a; Y') = \mathcal{R}(a;Y).$$

Proof *Step 1: Kähler forms.* Let λ be a Kähler form on Z. All bounded λ-plurisubharmonic functions have mass $\int_Z \lambda^k$, by Theorem 2.2.2, or by Stokes' theorem when $k = 1$, so that $v_+(\lambda) = \int_Z \lambda^k$. Hence Proposition 7.5.2, with $C = 1$, gives

$$\int_Z \langle (\lambda + \mathrm{dd^c} q)^k \rangle \leq \int_Z \lambda^k \qquad (q \in \mathrm{PSH}(Z,\lambda)).$$

The smooth current λ attains equality.

Step 2: the mass bound. Put $A = C\kappa - \eta$. For a positive current S in $b + t\{\kappa\}$, the current $S + A$ is positive and belongs to $(C+t)\{\kappa\}$. Since local non-pluripolar products are positive and multilinear, $\langle S^k \rangle \leq \langle (S+A)^k \rangle$, and Step 1 applied to the Kähler form $(C+t)\kappa$ yields (29.5). In particular $\mathcal{R}_Z(b)$ is finite. The definition involves only the class, so the choice of its smooth representative or of the comparison form does not change the supremum.

Step 3: modifications. Let $\rho\colon W \to Z$ be a modification between compact Kähler manifolds. By Proposition 1.22.4, every positive current in $\rho^* b$ is the pullback of a unique positive current in b. Their non-pluripolar measures correspond on the locus where ρ is an isomorphism, and neither charges the exceptional analytic sets. Hence their total masses agree and

$$\mathcal{R}_W(\rho^* b) = \mathcal{R}_Z(b).$$

Applying this identity on a common Kähler refinement shows that $\mathcal{R}(a;Y)$ is independent of the resolution. The same argument compares resolutions of Y and

Y', since their maps to X agree on a dense open set and therefore everywhere on a common refinement. □

On a compact Kähler manifold, positivity of this mass has a classical interpretation. We use two facts from Kähler pluripotential theory, discussed in Section 36.16. First, a class containing a positive current with positive non-pluripolar top mass is big. Second, in a big class a current with minimal singularities attains the volume, the largest non-pluripolar top mass, and for a nef and big class this volume is the top intersection number. For a nef class with positive top intersection number, a Kähler current in the class can also be obtained from the following normalized diagonal construction.

Proposition 29.6.2 *Let Z be a compact Kähler manifold of dimension $k > 0$ and $b \in H^{1,1}_{\mathrm{BC}}(Z,\mathbb{R})$. Then*

$$\mathcal{R}_Z(b) > 0 \quad \Longleftrightarrow \quad b \text{ is big}.$$

If b is nef, then

$$\mathcal{R}_Z(b) = \int_Z b^k. \tag{29.6}$$

Suppose that b is nef with $I = \int_Z b^k > 0$. Fix a Kähler form κ, let $p_1, p_2 \colon Z \times Z \to Z$ be the projections and $\Delta \subset Z \times Z$ the diagonal, and put

$$M = k \int_Z b^{k-1} \wedge \{\kappa\}.$$

Then there are $d > 0$ and a closed positive (k,k)-current Θ in $(p_1^* b + p_2^* b)^k$ with $\Theta \geq d[\Delta]$ such that

$$K_b = M^{-1}(p_1)_*(\Theta \wedge p_2^* \kappa) \in b, \qquad K_b \geq (d/M)\kappa. \tag{29.7}$$

Proof If $\mathcal{R}_Z(b) > 0$, some positive representative has positive non-pluripolar top mass, so b is big by the first fact above. Conversely, if $S \in b$ satisfies $S \geq \delta\kappa$ with $\delta > 0$, then $S - \delta\kappa$ is positive and closed, and expanding the non-pluripolar product gives $\langle S^k \rangle \geq \delta^k \kappa^k$. Hence $\mathcal{R}_Z(b) > 0$.

Suppose now that b is nef. For $t > 0$, the class $b + t\{\kappa\}$ is Kähler. Adding $t\kappa$ to an arbitrary positive representative of b and using the Kähler mass bound of Step 1 in the proof of Proposition 29.6.1 yields

$$\mathcal{R}_Z(b) \leq \int_Z (b + t\{\kappa\})^k.$$

Letting t decrease to zero gives the upper bound in (29.6). In particular, equality holds if $I = 0$.

Let $I > 0$, and choose $L > 0$ such that $L\{\kappa\} - b$ is Kähler. Since intersection numbers of nef classes are nonnegative,

$$0 \le \int_Z b^{k-1} \wedge (L\{\kappa\} - b) = LM/k - I,$$

so $M > 0$. The class $B = p_1^* b + p_2^* b$ is nef and satisfies

$$\int_{Z\times Z} B^{2k} = \binom{2k}{k} I^2 > 0.$$

The classical mass concentration theorem along the diagonal gives $\Theta \in B^k$ with $\Theta \ge d[\Delta]$ for some $d > 0$. When $B^k \wedge p_2^*\{\kappa\}$ is pushed forward by p_1, only the term containing exactly one factor $p_1^* b$ survives. Thus

$$(p_1)_*(B^k \wedge p_2^*\{\kappa\}) = Mb, \qquad (p_1)_*([\Delta] \wedge p_2^*\kappa) = \kappa.$$

Since $\Theta - d[\Delta]$ is positive, these identities prove (29.7); the $\partial\bar\partial$-lemma for currents on the compact Kähler manifold Z identifies the cohomology class computed in this way with the Bott–Chern class b. Therefore b is big, and by the second fact above a current with minimal singularities in b has non-pluripolar mass $\int_Z b^k$. This gives the remaining lower bound. □

When Y is smooth, the criterion descends to Y. Write $a_Y = j^* a$ for the restricted class, let $\pi\colon Z \to Y$ be a Kähler modification with Kähler form κ, and choose a Hermitian metric h on Y and $C > 0$ with $\pi^* h \le C\kappa$. If $K \in \pi^* a_Y$ satisfies $K \ge \delta\kappa$ with $\delta > 0$, then descent of potentials (Proposition 1.22.4) makes $\pi_* K$ a positive representative of a_Y, and

$$\pi_* K \ge (\delta/C)\pi_*\pi^* h = (\delta/C)h. \tag{29.8}$$

Conversely, let S be a strict positive representative of a_Y. On a coordinate ball in the locus where π is an isomorphism, $\pi^* S$ dominates a positive multiple of the closed form κ, so the local expansion of the non-pluripolar product shows that $\pi^* S$ has positive non-pluripolar mass on that ball. Consequently, $\mathcal{R}(a; Y) > 0$ if and only if the restricted class a_Y is big. The Hermitian metric h enters only through inequalities between currents; the products are compared with the closed form κ.

Proposition 29.6.3 *Let W be a connected compact Kähler manifold of dimension $k > 0$, let $g\colon W \to X$ be holomorphic, and suppose* $\dim g(W) < k$. *Then for every class $a \in H^{1,1}_{\mathrm{BC}}(X, \mathbb{R})$,*

$$\mathcal{R}_W(g^* a) = 0.$$

Proof If the image is a point, then $g^* a = 0$ and (29.6) gives the assertion. Otherwise, choose a Kähler resolution $r\colon Q \to g(W)$ and resolve the graph of the meromorphic lift of g to Q. This gives a Kähler modification $\rho\colon \widehat{W} \to W$ and a holomorphic map $F\colon \widehat{W} \to Q$ with $g \circ \rho = r \circ F$, where r is also viewed as a map into X.

Choose a smooth representative $\eta \in a$, a Kähler form κ_Q on Q, and $C > 0$ such that $C\kappa_Q - r^*\eta \geq 0$. Since $\rho^* g^* \eta = F^* r^* \eta$, the forms

$$P = CF^*\kappa_Q, \qquad A = P - \rho^* g^* \eta$$

are closed and semipositive. Since $\dim Q < k$, we have $P^k = 0$. For any positive current $S \in g^* a$, the current $\rho^* S + A$ is positive and lies in the nef class $\{P\}$. Hence

$$\int_{\widehat{W}} \langle (\rho^* S)^k \rangle \leq \int_{\widehat{W}} \langle (\rho^* S + A)^k \rangle \leq \mathcal{R}_{\widehat{W}}(\{P\}) = \int_{\widehat{W}} P^k = 0.$$

Modification invariance (Step 3 in the proof of Proposition 29.6.1) now gives the result. If the class has no positive representative, its invariant is zero by definition. □

Corollary 29.6.1 *Let a be a nef class on X. For every positive-dimensional irreducible reduced subvariety $Y \subset X$,*

$$\mathcal{R}(a; Y) = \int_Y a^{\dim Y}.$$

If $\int_X a^n > 0$, then

$$\bigcup_{\substack{Y \subset X \text{ irreducible, } \dim Y > 0 \\ \mathcal{R}(a;Y) = 0}} Y = \mathrm{Null}(a) = E_{\mathrm{nK}}(a). \tag{29.9}$$

Proof The pullback of a to a Kähler resolution of Y is nef: pull back smooth representatives with arbitrarily small negative part, and bound the pullback of ω by a multiple of a Kähler form on the resolution. Since the intersection number $\int_Y a^{\dim Y}$ is defined through the same pullback, (29.6) gives the first assertion. The first equality in (29.9) then follows from the definition of the null locus. For the second, the first assertion with $Y = X$ gives $\mathcal{R}(a; X) = \int_X a^n > 0$, so a is big by the criterion for smooth subvarieties preceding Proposition 29.6.3. The second equality is therefore the classical theorem on nef and big classes recalled earlier in this chapter. □

When X is Kähler and a is nef, the numerical restricted volume (29.1) along Y is also $\int_Y a^{\dim Y}$, so the two restricted invariants agree. Outside the nef case this comparison does not identify them: (29.4) is a supremum over all positive currents on a resolution of Y, and these currents need not extend to X.

29.7 Divisorial traces with bounded remainder

The invariant $\mathcal{R}(a; Y)$ is a supremum over all positive currents in a class. The trace of a given potential on Y, on the other hand, carries a specified singularity, and its non-pluripolar mass can be smaller. Suppose that on a resolution the trace becomes a divisorial logarithmic term plus a bounded remainder. The non-pluripolar product

then ignores the divisor and sees only the residual current, which has bounded potentials. The class of this residual current is nef, and its top intersection number computes the mass.

We keep the assumption that X belongs to class C. Let $Y \subset X$ be an irreducible reduced subvariety of dimension $k > 0$, let α_0 be a smooth closed real $(1,1)$-form on X, and let $u \in \mathrm{PSH}(X, \alpha_0)$ be such that its pointwise restriction $v = u|_Y$ is not identically $-\infty$. On a Kähler resolution $\pi\colon Z \to Y$, with $f = j \circ \pi\colon Z \to X$ as in Section 29.6, put $\alpha_Z = f^*\alpha_0$ and $v_\pi = u \circ f$. When Y is smooth, one may instead start from a smooth closed real $(1,1)$-form α_Y on Y and $v \in \mathrm{PSH}(Y, \alpha_Y)$, and put $\alpha_Z = \pi^*\alpha_Y$ and $v_\pi = v \circ \pi$. In both cases v_π is the composition of a given potential with a map; the intrinsic multiplier-ideal trace of Chapter 20 is a different construction.

Suppose that on one such resolution

$$v_\pi = \ell + w, \qquad \ell = \sum_{j=1}^{N} c_j \log |s_j|^2_{h_j}, \qquad w \text{ bounded}, \tag{29.10}$$

where $c_j > 0$, the D_j are prime divisors on Z, s_j is the canonical section of $\mathcal{O}(D_j)$, and h_j is a smooth Hermitian metric on $\mathcal{O}(D_j)$; the empty sum $\ell = 0$ is allowed. Write $\Gamma = \sum_j c_j D_j$ and $\gamma_j = c_1(\mathcal{O}(D_j), h_j)$ for the curvature forms, and define the *residual form* and the *residual class*

$$\eta_v = \alpha_Z - \sum_j c_j \gamma_j, \qquad b_v = \{\eta_v\} = \{\alpha_Z\} - \{[\Gamma]\}. \tag{29.11}$$

For singular Y, all assertions below are made on its resolutions.

Lemma 29.7.1 *The current $R_v = \eta_v + \mathrm{dd^c} w$ is positive and has bounded local potentials, and its class b_v is nef. Moreover,*

$$\int_Z \langle (\alpha_Z + \mathrm{dd^c} v_\pi)^k \rangle = \int_Z R_v^k = \int_Z b_v^k = \mathcal{R}_Z(b_v). \tag{29.12}$$

If $v_1, \dots, v_k$ satisfy the same conditions on Z, possibly for different closed backgrounds, then

$$\int_Z R_{v_1} \wedge \cdots \wedge R_{v_k} = \int_Z b_{v_1} \wedge \cdots \wedge b_{v_k}. \tag{29.13}$$

Proof Off the support $|\Gamma|$, we have $\mathrm{dd^c}\ell = -\sum_j c_j \gamma_j$, so the current $\alpha_Z + \mathrm{dd^c} v_\pi$ equals $\eta_v + \mathrm{dd^c} w$ there and is positive. After the addition of a smooth local potential for η_v, the bounded function w is plurisubharmonic off $|\Gamma|$. By Lemma 1.3.7, it extends across the divisor as a bounded plurisubharmonic function that agrees almost everywhere with the original one; we use this extension to represent w. Thus $R_v \geq 0$ on Z, and the Poincaré–Lelong formula gives

$$\alpha_Z + \mathrm{dd^c} v_\pi = [\Gamma] + R_v.$$

Since w is bounded, b_v is nef by Lemma 24.6.1. The non-pluripolar measure on the left of (29.12) agrees with R_v^k off the divisor, and neither measure charges the divisor. This gives the first equality.

Fix a Kähler form κ on Z. For all sufficiently large real t, the smooth closed form $\eta_v + t\kappa$ is Kähler, and w is a bounded potential for it. Therefore

$$\int_Z (R_v + t\kappa)^k = \int_Z (\eta_v + t\kappa)^k$$

by Theorem 2.2.2, or by Stokes' theorem when $k = 1$. Both sides are polynomials in t, so their constant coefficients agree; this gives $\int_Z R_v^k = \int_Z b_v^k$. The last equality in (29.12) is (29.6). Applying the same polynomial argument to positive linear combinations of the R_{v_i} and comparing coefficients gives (29.13). □

We denote the common value in (29.12) by $\mathfrak{r}(v;Y)$ and call it the *trace mass* of v on Y. When Y is smooth, it is also the non-pluripolar mass of the trace current on Y itself:

$$\mathfrak{r}(v;Y) = \int_Y \langle (\alpha_Y + \mathrm{dd^c} v)^k \rangle,$$

where $\alpha_Y = \alpha_0|_Y$ when v is a restriction. Indeed, the currents on Z and on Y correspond on the locus where π is an isomorphism, and their non-pluripolar measures give no mass to its analytic complement. A change of variables therefore identifies the total masses.

Theorem 29.7.1 *The number $\mathfrak{r}(v;Y)$ does not depend on the divisor metrics or on the Kähler resolution carrying* (29.10)*. If two potentials in the same closed background have the same ordinary singularity type and each has a decomposition of the form* (29.10)*, then their residual classes agree on a common refinement, and in particular their trace masses are equal. Residual classes add when the traces and their backgrounds are added, and their mixed masses are given by* (29.13)*.*

Proof Changing a divisor metric changes ℓ by a smooth function, w by its negative, and η_v by the corresponding $\mathrm{dd^c}$-term. Thus R_v is unchanged. The same cancellation occurs when a smooth change of the background is accompanied by the opposite change of the potential.

Let $\rho\colon W \to Z$ be a modification with W Kähler. Pulling back the decomposition, with the total divisor $\rho^*\Gamma$ including all exceptional coefficients, gives a decomposition on W with bounded remainder $w \circ \rho$, residual current $\rho^* R_v$ and nef residual class $\rho^* b_v$. Its mass is

$$\int_W (\rho^* b_v)^k = \int_Z b_v^k.$$

For two decompositions on different resolutions, pass to a common Kähler refinement and pull both back in this way. At a generic smooth point of a prime component of the union of the two divisors, the difference of the logarithmic terms is $(a_1 - a_2)\log|t|^2 + O(1)$, where t is a local defining coordinate of the component and a_1, a_2 are its two coefficients. The two potentials have bounded difference, and the resulting pair of

inequalities forces $a_1 = a_2$. Hence the total divisors coincide, and so do the residual classes in the common background. Finally, adding two decompositions adds both the divisors and the bounded remainders. The residual class of the sum is therefore the sum of the nef residual classes, and (29.13) computes its mixed terms. □

Proposition 29.7.1 *Suppose that $\mathfrak{r}(v;Y) > 0$. Then the class $\{\alpha_Z\}$ contains a Kähler current $\alpha_Z + \mathrm{dd}^{\mathrm{c}} p$ whose potential satisfies $p \le v_\pi + A$ for some constant A. If b_v has a smooth positive representative on a resolution carrying (29.10), then on that resolution p can be chosen with $p \sim v_\pi$, that is, with $p - v_\pi$ bounded. When Y is smooth, the Kähler current and the corresponding inequalities between potentials descend to Y.*

Proof Positive trace mass means $\int_Z b_v^k > 0$. Since b_v is nef, (29.7) gives a current $K = \eta_v + \mathrm{dd}^{\mathrm{c}} a \ge \delta\kappa$ in b_v, where κ is a Kähler form and $\delta > 0$. The current

$$[\Gamma] + K = \alpha_Z + \mathrm{dd}^{\mathrm{c}}(\ell + a)$$

is then a Kähler current. Since a is bounded above on Z and w is bounded below, its potential $p = \ell + a$ satisfies $p \le \ell + w + A = v_\pi + A$ for a constant A. If b_v has a smooth positive representative, a can be chosen smooth, and then $p - v_\pi = a - w$ is bounded.

For smooth Y, take $\alpha_Y = \alpha_0|_Y$ when v is a restriction. By Proposition 1.22.4, $p = q \circ \pi$ for some $q \in \mathrm{PSH}(Y, \alpha_Y)$. The current $\alpha_Y + \mathrm{dd}^{\mathrm{c}} q$ is strict by (29.8), and since π is surjective, each inequality between potentials descends. The hypothesis of a smooth positive representative refers to the specified resolution: the pullback of a Kähler class under a further blowup need not be Kähler. □

The trace mass can vanish even when the restricted class is big, because the non-pluripolar product ignores the divisorial part of the trace.

Example 29.7.1 Let $k \ge 1$, let $Y \simeq \mathbb{P}^k$ be a hyperplane in $X = \mathbb{P}^{k+1}$, and let another hyperplane $H_X \subset X$ cut out $H = H_X \cap Y$. Let ω_{FS} be the Fubini–Study form, normalized by $\int_Y \omega_{\mathrm{FS}}^k = 1$, and let s_{H_X} be a section of $\mathcal{O}(1)$ defining H_X, measured with the Fubini–Study metric. For $c > 0$ and $d \ge 0$, take $\alpha_0 = (c + d)\omega_{\mathrm{FS}}$ on X and $u = c \log |s_{H_X}|^2$. The restriction $v = u|_Y$ is not identically $-\infty$, and

$$\alpha_0|_Y + \mathrm{dd}^{\mathrm{c}} v = c[H] + d\omega_{\mathrm{FS}}|_Y, \qquad \mathfrak{r}(v;Y) = d^k, \qquad \mathcal{R}(\{\alpha_0\};Y) = (c + d)^k.$$

Thus for $d = 0$ the trace mass is zero, although the restricted invariant is positive. The divisor $c[H]$ is a nonzero part of the trace current, but the non-pluripolar top product discards it.

The next example shows that the inequality $p \le v_\pi + A$ in Proposition 29.7.1 cannot in general be improved to $p \sim v_\pi$, even for a bounded trace.

Example 29.7.2 Let $\mu\colon Y \to \mathbb{P}^2$ be the blowup of a point, with exceptional curve E, let ω_{FS} be the Fubini–Study form on $\mathbb{P}^2$ with total volume one, and put $\alpha_Y = \mu^*\omega_{\mathrm{FS}}$ and $v = 0$. The decomposition (29.10) holds on $Z = Y$ with no divisor, the residual class $\{\alpha_Y\}$ is nef and big, and

$$\mathfrak{r}(0;Y) = \mathcal{R}(\{\alpha_Y\};Y) = 1.$$

Nevertheless, there is no bounded $q \in \mathrm{PSH}(Y,\alpha_Y)$ for which $\alpha_Y + \mathrm{dd}^{\mathrm{c}} q \geq \delta\kappa$ with $\delta > 0$ and κ Kähler. Indeed, since q is bounded, its restriction to E is defined, and the disk criterion of Lemma 1.22.1 gives

$$\mathrm{dd}^{\mathrm{c}}(q|_E) \geq \delta\kappa|_E,$$

because $\alpha_Y|_E = 0$. Integration over the compact curve E contradicts Stokes' theorem. Thus positive trace mass gives the one-sided comparison of Proposition 29.7.1, but a Kähler current in the class may need additional singularities.

A subvariety contained in the null locus can nevertheless have positive restricted invariant; the union in (29.9) is needed to recover the locus.

Example 29.7.3 Let $\mu\colon X \to \mathbb{P}^3$ be the blowup along a line L, and set $a = \mu^*\{\omega_{\mathrm{FS}}\}$, where ω_{FS} has total volume one; the class a is nef and $\int_X a^3 = 1$. Since $N_{L/\mathbb{P}^3} \simeq \mathcal{O}(1) \oplus \mathcal{O}(1)$, the exceptional divisor is $E = \mathbb{P}(N_{L/\mathbb{P}^3}) \simeq \mathbb{P}^1 \times \mathbb{P}^1$, and on E the map μ is the projection to L. Hence $\int_E a^2 = 0$, whereas a section Y of $E \to L$ satisfies $\int_Y a = 1$. Every irreducible subvariety not contained in E has positive intersection number, because $\mu^*\omega_{\mathrm{FS}}$ is positive on its nonempty open part outside E. By Corollary 29.6.1,

$$E_{\mathrm{nK}}(a) = E, \qquad \mathcal{R}(a;E) = 0, \qquad \mathcal{R}(a;Y) = 1.$$

The union of the subvarieties with zero invariant is the whole null locus E, although the curve $Y \subset E$ has positive invariant. Positivity along one subvariety does not place it outside the null locus.

29.8 Divisorial components at the threshold

A divisorial component of a positive current $\omega + \mathrm{dd}^{\mathrm{c}} u$ consumes part of its Gauduchon trace but contributes no density to its non-pluripolar measure. It can therefore lower the largest coefficient in a Monge–Ampère subsolution inequality. At the maximal coefficient, every weak limit of a maximizing sequence of Gauduchon tests annihilates the class of the divisor, and uniform integrability along the sequence makes the limit a maximizer.

In this section X is again an arbitrary compact complex manifold, now of dimension $n \geq 2$, with the Hermitian form ω. Fix a smooth strictly positive probability volume form $\mathrm{d}V$ and a nonzero function $0 \leq f \in L^1(\mathrm{d}V)$, and write $p = f^{1/n}$ for its root

density. Let $b = b_\omega(f)$ be its subsolution threshold: by Theorem 15.27.1, it is the largest $a \geq 0$ for which some $u \in \mathrm{PSH}(X,\omega)$ satisfies $\mathrm{MA}^{\mathrm{ext}}_\omega(u) \geq a f \,\mathrm{d}V$, where $\mathrm{MA}^{\mathrm{ext}}_\omega$ is the extended non-pluripolar Monge–Ampère measure of Theorem 7.1.1. Such a u with $a = b$ is a threshold subsolution in the sense of Section 15.27. We also use the set $\mathscr{G}_\omega$ of normalized Gauduchon currents and the weak closure $\mathscr{C}_\omega$ of its smooth strictly positive members, the normalized powers of Gauduchon metrics (Section 4.4); the determinant density k_T of $T \in \mathscr{G}_\omega$, with $\Gamma_0(T) = k_T \,\mathrm{d}V$ (Theorem 27.7.1), and the functional $D_p(T) = \int_X p k_T \,\mathrm{d}V$, whose supremum over $\mathscr{G}_\omega$ is the root functional $\rho_\omega(p) = b^{-1/n}$; and the closure $\mathcal{D}^0_\omega$ of the bounded functions in the space of roots with finite root functional (Section 27.8). Finally, $\kappa_\omega(\beta)$ is the linear positivity threshold of a smooth real $(1,1)$-form β (Corollary 4.3.1), the infimum of $\int_X \beta \wedge T$ over the smooth strictly positive $T \in \mathscr{G}_\omega$.

Proposition 29.8.1 *Let $c > 0$ be real and let E be an effective real divisor on X. Suppose $u \in \mathrm{PSH}(X,\omega)$ satisfies*

$$\mathrm{MA}^{\mathrm{ext}}_\omega(u) \geq c f \,\mathrm{d}V, \qquad \omega + \mathrm{dd}^{\mathrm{c}} u \geq [E].$$

Let β_E be a smooth closed representative of the Bott–Chern class of $[E]$, and put $\kappa = \kappa_\omega(\beta_E)$. Then $b > 0$, $0 \leq \kappa < 1$ and

$$c \leq b(1-\kappa)^n. \tag{29.14}$$

Suppose that $c = b$, and let (T_j) be a smooth maximizing sequence *for D_p, that is, a sequence of smooth strictly positive forms $T_j \in \mathscr{G}_\omega$ with $D_p(T_j) \to \rho_\omega(p)$. Then*

$$\int_X \beta_E \wedge T_j = \int_X [E] \wedge T_j \longrightarrow 0.$$

Every weak cluster current T therefore satisfies $\int_X \beta_E \wedge T = 0$, and the class of E is not big. If the densities $p k_{T_j}$ of the chosen sequence are uniformly integrable, every such cluster current maximizes D_p. In particular, this holds for every smooth maximizing sequence when $p \in \mathcal{D}^0_\omega$.

Proof By Theorem 15.27.1, $b \geq c > 0$. Let $T \in \mathscr{G}_\omega$ be smooth and strictly positive. By Lemma 7.4.2, the measure

$$R = (\omega + \mathrm{dd}^{\mathrm{c}} u) \wedge T - c^{1/n} p k_T \,\mathrm{d}V$$

is positive, and its mass is $1 - c^{1/n} D_p(T)$. The absolutely continuous term gives no mass to the support of E, whereas $[E] \wedge T$ is concentrated there. Restricting R to the support of E and using $\omega + \mathrm{dd}^{\mathrm{c}} u \geq [E]$ therefore gives

$$\int_X [E] \wedge T \leq R(X) = 1 - c^{1/n} D_p(T). \tag{29.15}$$

Let ℓ_E be a global logarithmic potential, so that $[E] = \beta_E + \mathrm{dd}^{\mathrm{c}} \ell_E$ by the Poincaré–Lelong formula. Since ℓ_E is locally integrable and T is smooth with $\mathrm{dd}^{\mathrm{c}} T = 0$,

Stokes' theorem for currents identifies the left side with $\int_X \beta_E \wedge T$. This integral is nonnegative for every such T, so $\kappa \geq 0$. Since κ is the infimum of these integrals, (29.15) gives $\kappa \leq 1 - c^{1/n} D_p(T)$, and in particular $\kappa < 1$ because $D_p(T) > 0$. By Theorem 15.28.1, the supremum of $D_p(T)$ over these smooth T is $\rho_\omega(p) = b^{-1/n}$. Hence $c^{1/n} b^{-1/n} \leq 1 - \kappa$, which is (29.14).

Now let $c = b$, and let (T_j) be a smooth maximizing sequence. Then the right side of (29.15) for $T = T_j$ tends to zero. Since pairing with the fixed smooth form β_E is continuous under weak convergence of currents, every cluster current satisfies $\int_X \beta_E \wedge T = 0$. Moreover, (29.14) with $c = b$ forces $\kappa = 0$, so the class of E is not big by Corollary 4.3.1. If the densities along the sequence are uniformly integrable, Proposition 27.8.2 shows that every cluster current maximizes D_p. When $p \in \mathcal{D}^0_\omega$, Theorem 27.8.1 gives this uniform integrability for every sequence. □

Geometrically, at the threshold $c = b$ with $E \neq 0$, the equation $\int_X \beta_E \wedge T = 0$ defines a nonempty proper exposed face of $\mathscr{C}_\omega$. Indeed, $\mathscr{C}_\omega$ is compact and convex, and the pairing with β_E is continuous and nonnegative on it and strictly positive on every smooth strictly positive member. The proposition places every cluster current of a smooth maximizing sequence on this face. The statement concerns $\mathscr{C}_\omega$, which can be strictly smaller than $\mathscr{G}_\omega$ (Example 4.4.1). Moreover, for a singular T the smooth pairing $\int_X \beta_E \wedge T$ does not define a product $[E] \wedge T$.

The hypotheses of the next two corollaries on b and on the maximizing sequence hold whenever $p \in \mathcal{D}^0_\omega$, since this gives $b > 0$ and uniform integrability over the whole of $\mathscr{G}_\omega$ (Theorem 27.8.1).

Corollary 29.8.1 *Assume $b > 0$ and that D_p has a smooth maximizing sequence (T_j) with uniformly integrable densities pk_{T_j}. Let E be an effective real divisor whose class has a smooth semipositive representative β_E. If*

$$\int_{\{\beta_E = 0\}} f \, \mathrm{d}V = 0,$$

then no threshold subsolution u for f satisfies $\omega + \mathrm{dd}^{\mathrm{c}} u \geq [E]$.

Proof Suppose such a subsolution exists. By Proposition 29.8.1, there is a maximizing current $T \in \mathscr{G}_\omega$ with $\int_X \beta_E \wedge T = 0$. We use the matrix-density notation of Section 27.5: $\sigma_T = \omega \wedge T$ is the trace measure of T, P_T is its matrix density and B is the coefficient matrix of β_E. Since both matrices are positive semidefinite, the vanishing pairing gives $\operatorname{tr}(BP_T) = 0$ at σ_T-almost every point. Wherever $B \neq 0$, this forces P_T to be singular: a positive definite P_T would have strictly positive trace against the nonzero positive semidefinite matrix B. The determinant measure $\Gamma_0(T)$ vanishes where P_T is singular, so it is concentrated on $\{\beta_E = 0\}$. By Theorem 27.7.1, $\Gamma_0(T) = k_T \, \mathrm{d}V$, and the hypothesis on f gives $pk_T = 0$ almost everywhere. This contradicts $D_p(T) = \rho_\omega(p) > 0$. □

Corollary 29.8.2 *Assume $b > 0$ and that D_p has a smooth maximizing sequence (T_j) with uniformly integrable densities pk_{T_j}. Let $u \in \mathrm{PSH}(X,\omega)$ be a threshold subsolution for f, and suppose that $\omega + \mathrm{dd^c} u \geq [E]$ for an effective real divisor E whose Bott–Chern class is nef. Let β_E be any smooth closed representative of that class. Then for every smooth strictly positive probability measure μ, the Gauduchon root volume of Definition 10.8.1 satisfies*

$$\lambda_\mu(\beta_E) = 0.$$

Proof By Proposition 29.8.1, there is a maximizing current $T \in \mathscr{G}_\omega$ with $\int_X \beta_E \wedge T = 0$. If $\lambda_\mu(\beta_E) > 0$, then Proposition 27.20.1 shows that the determinant measure of T computed with μ vanishes. Since μ has a smooth positive density with respect to $\mathrm{d}V$, this measure is a positive continuous multiple of $\Gamma_0(T)$. Hence $\Gamma_0(T) = 0$ and $D_p(T) = 0$, contradicting the positivity of the maximal value $\rho_\omega(p)$. Since the nef class of β_E has $\lambda_\mu(\beta_E) \geq 0$, the conclusion follows for each μ. □

The semipositive criterion and the nef-volume criterion detect different obstructions. On the Hopf threefold, the class of a hyperplane divisor has zero variational volume, so Corollary 29.8.2 says nothing about it, but its nowhere vanishing semipositive representative excludes a divisorial component at the threshold.

To see this, let X be the Hopf threefold with its Hopf metric $\omega = \alpha + \zeta$ (Chapter 9 and Section 27.22), where α is the pullback of the Fubini–Study form under the elliptic fibration $X \to \mathbb{P}^2$, and let H be the inverse image of a projective hyperplane. The class of H is represented by α, which is semipositive of rank two everywhere. Its variational volume vanishes for every smooth positive volume form μ: the metrics $\gamma_t = \alpha + t\zeta$, $t > 0$, are Gauduchon by the proof of Proposition 27.22.1, and since $\alpha^3 = \zeta^2 = 0$ and $\omega^3 = 3\alpha^2 \wedge \zeta$, one has $\int_X \alpha \wedge \gamma_t^2 = \frac{2t}{3} \int_X \omega^3$ and $\gamma_t^3 = t\,\omega^3$. Hence the quotient in Definition 10.8.1 is $O(t^{1/3})$ as $t \downarrow 0$. By Corollary 9.7.1 and Proposition 15.31.4, one has $\mathcal{D}^0_\omega = L^3$ on this example. Hence, for every nonzero nonnegative L^1 density, a threshold subsolution has zero generic Lelong number along H. Otherwise the divisorial removal of Lemma 1.17.1 would give $\omega + \mathrm{dd^c} u \geq a[H]$, where $a > 0$ is this generic Lelong number, contradicting Corollary 29.8.1 applied to aH. This exclusion uses no bounded mass property. It is also stronger here than nonbigness of the class of H: the residual formulas of Proposition 27.22.1 allow a positive coefficient along H when the equation constant is smaller than the threshold.

These conditions concern the maximal global coefficient. Whether threshold subsolutions with divisorial components can be constructed from the resulting annihilating currents is open; the threshold within a prescribed singularity type is a different problem.

29.9 Currents on the elliptic threefold

On the elliptic threefold of Section 27.18, closed classes can be described completely. The product of the two horizontal forms is $\mathrm{dd^c}$-exact, so every positive $\mathrm{dd^c}$-closed

$(1,1)$-current has zero vertical component, and the fiberwise descent of Lemma 4.6.4 identifies it with a current on the base. Closedness descends as well. The closed positive cones, the singularity envelopes and the spaces of sections selected by multiplier ideals are therefore all computed on the base surface.

Let X be the compact threefold of Section 27.18, and let

$$\pi\colon X \longrightarrow B = \mathbb{P}^1 \times \mathbb{P}^1$$

be its elliptic fibration. Write σ_1, σ_2 for the Fubini–Study forms of the two factors, each of integral one, and put $\alpha = \pi^*\sigma_1$ and $\beta = \pi^*\sigma_2$. The product coordinates of that construction show that π is locally a holomorphic product with fiber the elliptic curve E. The logarithmic ratio t of that construction is defined only on the covering space, but its differentials descend to X. We take for ω the balanced metric $\omega = \alpha + \beta + D$, where $D = \mathrm{d}t \wedge \mathrm{d}^{\mathrm{c}}t$. By Corollary 27.18.3, it satisfies

$$\alpha \wedge \beta = \tfrac{1}{2}\mathrm{d}\mathrm{d}^{\mathrm{c}}\omega.$$

The arguments below use this identity, not the vanishing of the lower Monge–Ampère mass.

Lemma 29.9.1 *Every positive* $\mathrm{d}\mathrm{d}^{\mathrm{c}}$*-closed real* $(1,1)$*-current* T *on* X *has the form*

$$T = \pi^*S$$

for a unique positive $\mathrm{d}\mathrm{d}^{\mathrm{c}}$*-closed* $(1,1)$*-current* S *on* B*. Moreover,* T *is closed if and only if* S *is closed. Here* π^*S *is the pullback of currents under the holomorphic submersion* π*.*

Proof The measure $T \wedge \alpha \wedge \beta$ is positive. Since $\mathrm{d}\mathrm{d}^{\mathrm{c}}T = 0$, Stokes' theorem for currents gives

$$\int_X T \wedge \alpha \wedge \beta = \tfrac{1}{2}\int_X \omega \wedge \mathrm{d}\mathrm{d}^{\mathrm{c}}T = 0,$$

so this measure vanishes. All products in this identity pair a current with smooth forms. Now apply Lemma 4.6.4 to π, with the base Hermitian form $\eta = \sigma_1 + \sigma_2$. Since $\eta^2 = 2\sigma_1 \wedge \sigma_2$, we have $T \wedge \pi^*\eta^2 = 0$, and since the fiber is a compact connected elliptic curve, the lemma gives the unique positive $\mathrm{d}\mathrm{d}^{\mathrm{c}}$-closed current S. The same lemma shows that S is closed when T is; conversely, pullback commutes with d. □

This description concerns $(1,1)$-currents. It does not conflict with the Gauduchon forms of Lemma 27.18.3: those have bidegree $(2,2)$ and satisfy $\mathrm{d}\mathrm{d}^{\mathrm{c}}\Gamma_j = 0$, with bounded degree and divergent volume of their positive roots, and they belong to a different cone of currents.

Proposition 29.9.1 *Pullback*

$$\pi^*\colon H^{1,1}_{\mathrm{BC}}(B,\mathbb{R}) \longrightarrow H^{1,1}_{\mathrm{BC}}(X,\mathbb{R})$$

is injective. The pseudoeffective and nef cones on X coincide and are

$$\operatorname{Psef}(X) = \operatorname{Nef}(X) = \{a\{\alpha\}_{\mathrm{BC}} + b\{\beta\}_{\mathrm{BC}} : a, b \geq 0\}.$$

For the trace functional of the Gauduchon metric ω, the slice of this cone at trace one is the segment $a + b = (8\pi)^{-1}$. Every class in the cone has a smooth semipositive representative, and none is big. The real form $\beta - \alpha$ is de Rham exact but represents a nonzero Bott–Chern class.

Proof Suppose a smooth closed real form τ on B satisfies $\pi^*\tau = \mathrm{dd}^{\mathrm{c}}u$ with u smooth on X. The restriction of u to each compact connected elliptic fiber is harmonic, hence constant, and local holomorphic sections show that $u = v \circ \pi$ for a smooth function v on B. Thus $\pi^*(\tau - \mathrm{dd}^{\mathrm{c}}v) = 0$, which implies $\tau = \mathrm{dd}^{\mathrm{c}}v$ because π is a submersion. This proves injectivity.

The usual cohomology computation for $\mathbb{P}^1 \times \mathbb{P}^1$, together with the $\partial\bar{\partial}$-lemma on this Kähler surface, gives

$$H^{1,1}_{\mathrm{BC}}(B, \mathbb{R}) = \mathbb{R}\{\sigma_1\} \oplus \mathbb{R}\{\sigma_2\}, \qquad \int_B \sigma_1 \wedge \sigma_2 = 1, \qquad \sigma_1^2 = \sigma_2^2 = 0.$$

If S is a positive closed current on B, the two coefficients of its class are

$$a = \int_B S \wedge \sigma_2 \geq 0, \qquad b = \int_B S \wedge \sigma_1 \geq 0.$$

By Lemma 1.17.4 and Lemma 29.9.1, every pseudoeffective class on X is the pullback of the class of such a current, and it therefore lies in the cone of the statement. Conversely, every class in that cone has the smooth semipositive representative $a\alpha + b\beta$, so it is nef and pseudoeffective. This proves the equality of the cones; the whole Bott–Chern group of X is not needed.

Since $\omega^2 = 2\alpha \wedge \beta + 2(\alpha + \beta) \wedge D$ and $\int_E D = 4\pi$ on each fiber, integration along the fibers gives

$$\int_X (a\alpha + b\beta) \wedge \omega^2 = 8\pi(a + b).$$

This proves the assertion about the slice. By Corollary 27.18.5, X carries no closed Kähler current, so none of these classes is big. Finally, the one-form $\mathrm{d}^{\mathrm{c}}t$ is globally defined on X and satisfies

$$\mathrm{d}(\mathrm{d}^{\mathrm{c}}t) = \beta - \alpha.$$

The class of $\sigma_2 - \sigma_1$ on B is nonzero, and injectivity makes its pullback nonzero in Bott–Chern cohomology. Thus de Rham exactness does not imply dd^{c}-exactness in this example. □

Consequently, every pseudoeffective class on this threefold is nef and has empty non-nef locus, whereas its non-Kähler locus is all of X, since no class is big. All its closed positive representatives come from the two-dimensional base; in particular $(a\alpha + b\beta)^3 = 0$.

29.9.1 Potentials and envelopes from the base

Descent also applies to potentials and preserves both ordinary singularity types and multiplier ideals, so the singularity envelopes in these backgrounds are computed on the base. The three-dimensional Monge–Ampère measures of these potentials nevertheless vanish, because their currents have only two horizontal directions.

We use the ordinary singularity order $\preceq$ and equivalence $\sim$ of Definition 1.12.1, the ideal order $\preceq_{\mathcal{I}}$ and equivalence $\sim_{\mathcal{I}}$ of Section 1.12, and the ordinary and multiplier-ideal envelopes $P_\theta[u]$ and $P_\theta[u]_{\mathcal{I}}$ of Definitions 1.19.1 and 17.2.1.

Proposition 29.9.2 *Let $a, b \geq 0$, put $\theta_B = a\sigma_1 + b\sigma_2$ and $\theta = \pi^*\theta_B$. Pullback is a bijection*

$$\mathrm{PSH}(B, \theta_B) \longrightarrow \mathrm{PSH}(X, \theta), \qquad v \longmapsto v \circ \pi.$$

It preserves the ordinary singularity order and equivalence, as well as the ideal order and equivalence. For every $v \in \mathrm{PSH}(B, \theta_B)$,

$$P_\theta[v \circ \pi] = P_{\theta_B}[v] \circ \pi, \qquad P_\theta[v \circ \pi]_{\mathcal{I}} = P_{\theta_B}[v]_{\mathcal{I}} \circ \pi.$$

Nevertheless, every $u \in \mathrm{PSH}(X, \theta)$ satisfies

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) = 0.$$

Proof The form θ_B is closed, and π is a proper holomorphic submersion with connected fibers, so Proposition 1.22.1 gives the asserted bijection. Since π is surjective, an ordinary singularity inequality holds upstairs exactly when it holds downstairs, with the same constant.

By Lemma 1.16.5, multiplier ideals of potentials on B pull back to those of the composed potentials on X, at every positive weight. Restriction to local holomorphic sections shows conversely that an inclusion of multiplier ideals on X implies the corresponding inclusion on B. Hence the ideal orders and equivalences correspond as well.

The competitors defining both envelopes correspond under the bijection, including the bound by zero and the required ordinary or ideal condition, and their pointwise suprema commute with pullback. Upper semicontinuous regularization also commutes with pullback: in a product chart, the supremum of a pulled-back function over a sufficiently small product neighborhood equals its supremum over the corresponding base neighborhood. This proves the two envelope identities.

Finally, write $u = v \circ \pi$ and, for positive integers j, set $u_j = \max\{u, -j\}$, which descends to $v_j = \max\{v, -j\}$. Since θ_B is semipositive, these are bounded admissible potentials. By Lemma 1.7.2, $(\theta + \mathrm{dd^c} u_j)^3 = 0$ because the base has dimension two. The definition of the extended non-pluripolar measure by increasing truncations now proves the last assertion. □

Thus the ordinary and ideal envelopes coincide in these backgrounds exactly when they coincide on the base, while every three-dimensional Monge–Ampère measure vanishes. Criteria from the closed theory that require positive volume do not apply to

these classes, which are not big. The Hermitian background ω behaves differently: its Monge–Ampère equations with smooth positive densities are solvable, even though the infimum of its masses is zero.

29.9.2 Sections selected by multiplier ideals

Descent also identifies the spaces of holomorphic sections selected by multiplier ideals, together with their weighted L^2 norms. In particular, sections of pulled-back line bundles grow at most quadratically, as on the two-dimensional base. For integers a, b, write $\mathcal{O}_B(a,b)$ for the tensor product of the pullbacks of $\mathcal{O}_{\mathbb{P}^1}(a)$ and $\mathcal{O}_{\mathbb{P}^1}(b)$ from the two factors of B.

Proposition 29.9.3 *Let a, b be nonnegative integers, let $L_B = \mathcal{O}_B(a,b)$ and $L = \pi^* L_B$. For every quasi-plurisubharmonic v on B and positive integer m, pullback gives an isomorphism*

$$H^0\bigl(B, L_B^m \otimes \mathcal{I}(mv)\bigr) \longrightarrow H^0\bigl(X, L^m \otimes \mathcal{I}(mv \circ \pi)\bigr).$$

For $v = 0$ these spaces have dimension $(am+1)(bm+1)$.

Equip L_B with a smooth Hermitian metric and L with its pullback, choose a continuous real function φ on B, let $\mathrm{d}V_X$ be a smooth positive volume form on X, and put $\nu = \pi_* \mathrm{d}V_X$. Use the norms

$$\|s\|_B^2 = \int_B |s|^2 \mathrm{e}^{-m\varphi}\, \mathrm{d}\nu, \qquad \|\pi^* s\|_X^2 = \int_X |\pi^* s|^2 \mathrm{e}^{-m\varphi\circ\pi}\, \mathrm{d}V_X.$$

Then the isomorphism is isometric. Let $K_{m,B}$ and $K_{m,X}$ be the corresponding partial Bergman kernels, that is, the sums of the weighted pointwise squared norms $|s|^2\mathrm{e}^{-m\varphi}$ and $|\pi^* s|^2 \mathrm{e}^{-m\varphi\circ\pi}$ over orthonormal bases of the two spaces. Then

$$K_{m,X} = K_{m,B} \circ \pi, \qquad \pi_*(K_{m,X}\, \mathrm{d}V_X) = K_{m,B}\, \mathrm{d}\nu.$$

Proof The identification of the section spaces is Proposition 28.7.1, applied to L_B and v on the base. For $v = 0$, the usual basis of bihomogeneous monomials on $\mathbb{P}^1 \times \mathbb{P}^1$ has $(am+1)(bm+1)$ elements.

Integration along the fibers of the proper submersion shows that ν is a smooth positive volume form, and the defining identity $\int_B g\, \mathrm{d}\nu = \int_X (g\circ\pi)\, \mathrm{d}V_X$ proves equality of the two norms. Pulling back an orthonormal basis and comparing pointwise norms gives the first kernel identity, and the same integration identity gives the second. The assertions include the case of a zero selected space, for which both kernels vanish.□

Even without a singular selecting weight, the growth is therefore at most quadratic:

$$\lim_{m\to\infty} m^{-2} h^0(X, L^m) = ab, \qquad \lim_{m\to\infty} m^{-3} h^0(X, L^m) = 0.$$

The next calculation shows that the variational volume of a small Hermitian perturbation detects this missing dimension of growth.

29.9.3 The boundary volume and section growth

Although no class $a\{\alpha\}+b\{\beta\}$ is big, its small Hermitian perturbations have positive Gauduchon variational volume $\mathcal{V}_\mu$ (Definition 10.8.1). For the volume form of the balanced metric, the zero potential solves the normalized Monge–Ampère equation for every positive combination of α, β and D, so the volume is an explicit cubic polynomial in the perturbation parameter. Its order of vanishing at the nef boundary records how many of the two horizontal coefficients vanish.

Proposition 29.9.4 *On the elliptic threefold, let $a, b \geq 0$, put $\theta = a\alpha + b\beta$, and let $\mu_0 = \omega^3/(24\pi)$. Then for every $\epsilon > 0$,*

$$\mathcal{V}_{\mu_0}(\theta + \epsilon\omega) = 24\pi\epsilon(a+\epsilon)(b+\epsilon). \tag{29.16}$$

In particular, $\mathcal{V}_\mu(\theta) = 0$ for every smooth positive probability measure μ. Let q be the number of positive coefficients among a, b. Then the degeneration orders of Proposition 27.19.1 are

$$\underline{a}(\theta) = \overline{a}(\theta) = 3 - q,$$

for every smooth positive probability measure and every Hermitian perturbation direction.

Proof The forms α, β, D are semipositive of rank one, so their squares vanish, and their sum ω is strictly positive. Thus

$$\Omega_\epsilon = \theta + \epsilon\omega = (a+\epsilon)\alpha + (b+\epsilon)\beta + \epsilon D > 0.$$

Since $\omega^3 = 6\alpha \wedge \beta \wedge D$ and $\int_X \omega^3 = 24\pi$, the wedge expansion gives

$$\Omega_\epsilon^3 = 6\epsilon(a+\epsilon)(b+\epsilon)\alpha \wedge \beta \wedge D = 24\pi\epsilon(a+\epsilon)(b+\epsilon)\mu_0.$$

Hence the zero potential solves the smooth normalized equation for Ω_ϵ and μ_0. By Theorem 10.8.1, the scalar of this equation is $\mathcal{V}_{\mu_0}(\Omega_\epsilon)$, which proves (29.16). The continuity of the volume at nef classes in Theorem 27.9.1 and comparison of smooth densities then give $\mathcal{V}_\mu(\theta) = 0$.

As $\epsilon \downarrow 0$, the leading term of (29.16) is $24\pi ab\epsilon$ if a and b are both positive, $24\pi(a+b)\epsilon^2$ if exactly one of them is positive, and $24\pi\epsilon^3$ if both vanish. The logarithmic quotients defining the orders therefore have limits $1, 2, 3$, respectively. Their independence of the density and of the perturbing metric is part of Proposition 27.19.1. □

The same number q governs the growth of sections. For nonnegative integers a, b, the line bundle $L = \pi^* \mathcal{O}_B(a, b)$ satisfies

$$h^0(X, L^m) = (am + 1)(bm + 1)$$

by Proposition 29.9.3. The growth degree of these spaces is exactly q, whereas the boundary order of the smooth curvature form $a\alpha + b\beta$ of L is $3 - q$. In this explicit family, the boundary order thus records the loss of algebraic growth dimension, even though the three-dimensional Monge–Ampère measures in the unperturbed background vanish.

Formula (29.16) concerns one fixed smooth density; it does not assert that a total mass is conserved over all potentials. Indeed, by Theorem 27.18.1 every Hermitian form on X, in particular each Ω_ϵ, has zero lower mass, while its upper mass is finite because X has the bounded mass property (Definition 7.5.1 and Corollary 27.18.3). Whether comparable identifications hold beyond such explicit families is part of Problem 27.19.1.

29.9.4 Limits of the minimizing Gauduchon forms

The Gauduchon metrics that minimize the quotient defining the volume in (29.16) can be identified explicitly: they are the equation metrics themselves. Their normalized squares converge smoothly at the nef boundary, although the normalized metrics may diverge. Recall from Definition 10.8.1 that the root volume λ_{μ_0}, with $\lambda_{\mu_0}^3 = \mathcal{V}_{\mu_0}$, is the infimum over Gauduchon metrics γ of $\int_X \theta \wedge \gamma^2$ divided by the denominator $D_{\mu_0}(\gamma) = \int_X (\mu_0/\gamma^3)^{1/3} \gamma^3$. For positive functions of ϵ, we write $f \asymp g$ if f/g is bounded above and below by positive constants as $\epsilon \downarrow 0$.

Proposition 29.9.5 *Let $a, b \geq 0$, $\theta = a\alpha + b\beta$ and $\Omega_\epsilon = \theta + \epsilon\omega$. For every $\epsilon > 0$, the metric Ω_ϵ is Gauduchon, and it minimizes the quotient defining $\lambda_{\mu_0}(\Omega_\epsilon)$. Put*

$$I_\epsilon = 8\pi\big(ab + 2(a+b)\epsilon + 3\epsilon^2\big), \qquad T_\epsilon = \frac{\Omega_\epsilon^2}{I_\epsilon}, \qquad \gamma_\epsilon = \frac{\Omega_\epsilon}{\sqrt{I_\epsilon}}.$$

Then $\int_X \omega \wedge T_\epsilon = 1$, $\mathrm{dd^c} T_\epsilon = 0$ and $T_\epsilon = \gamma_\epsilon^2$. As $\epsilon \downarrow 0$, the forms T_ϵ converge smoothly to

$$T_0 = \begin{cases} (4\pi)^{-1}\alpha \wedge \beta, & a, b > 0, \\ (8\pi)^{-1}\alpha \wedge (\beta + D), & a > 0,\ b = 0, \\ (8\pi)^{-1}\beta \wedge (\alpha + D), & a = 0,\ b > 0, \\ (24\pi)^{-1}\omega^2, & a = b = 0. \end{cases}$$

In every case $\theta \wedge T_0 = 0$. If q is the number of positive coefficients among a, b, the denominator of the normalized minimizing metric satisfies

$$D_{\mu_0}(\gamma_\epsilon) = \frac{\big(24\pi\epsilon(a+\epsilon)(b+\epsilon)\big)^{2/3}}{I_\epsilon} \asymp \epsilon^{q/3}.$$

Proof Write $x = a + \epsilon$, $y = b + \epsilon$ and $z = \epsilon$. Since $\alpha^2 = \beta^2 = D^2 = 0$,

$$\Omega_\epsilon^2 = 2xy\alpha \wedge \beta + 2xz\alpha \wedge D + 2yz\beta \wedge D.$$

The forms α, β are closed and $\mathrm{dd^c} D = 2\alpha \wedge \beta$. Hence the first term is closed, and $\mathrm{dd^c}$ of each of the last two terms contains α^2 or β^2 and therefore vanishes. Thus $\mathrm{dd^c}(\Omega_\epsilon^2) = 0$. Since the zero potential solves the normalized equation for Ω_ϵ and μ_0, the characterization of minimizing metrics in Theorem 10.8.1 shows that Ω_ϵ is a minimizer.

Integration along the fibers gives $\int_X \alpha \wedge \beta \wedge D = 4\pi$. Hence

$$\int_X \omega \wedge \Omega_\epsilon^2 = 8\pi(xy + xz + yz) = I_\epsilon,$$

which is the trace normalization. Dividing the three coefficients in the wedge expansion of Ω_ϵ^2 by I_ϵ gives each of the stated limits; the convergence holds in every smooth norm, because the forms $\alpha \wedge \beta$, $\alpha \wedge D$ and $\beta \wedge D$ are fixed. The identities $\alpha^2 = \beta^2 = 0$ then give $\theta \wedge T_0 = 0$ in all four cases.

Set $c_\epsilon = 24\pi\epsilon(a+\epsilon)(b+\epsilon)$. The equation for Ω_ϵ gives

$$\gamma_\epsilon^3 = \frac{c_\epsilon}{I_\epsilon^{3/2}}\mu_0.$$

Substituting this in the definition of D_{μ_0} and using $\int_X \mu_0 = 1$ yields $D_{\mu_0}(\gamma_\epsilon) = c_\epsilon^{2/3}/I_\epsilon$. For fixed (a, b) one has $c_\epsilon \asymp \epsilon^{3-q}$ and $I_\epsilon \asymp \epsilon^{2-q}$, so the quotient is of order $\epsilon^{q/3}$. □

In a coframe adapted to the three rank-one forms α, β and D, the three complementary products in the proof show that the limiting positive $(2,2)$-forms have coefficient matrices of rank 1, 2 and 3 when $q = 2$, 1 and 0, respectively. If $a, b > 0$, the normalized metrics themselves converge smoothly to

$$\frac{a\alpha + b\beta}{\sqrt{8\pi ab}}.$$

If $a > 0$ and $b = 0$, however, the coefficient of α in γ_ϵ grows like a positive constant times $\epsilon^{-1/2}$, while its other two coefficients tend to zero. Thus the metrics γ_ϵ are unbounded relative to the fixed metric ω, although their squares converge smoothly; the same holds with α and β interchanged. This divergence is unavoidable, since a bounded subsequence of the γ_ϵ would converge to a semipositive $(1,1)$-form whose square is the rank-two form T_0. In dimension three, a positive $(2,2)$-form of rank two is not the square of a semipositive $(1,1)$-form: the coefficients of such a square are the pairwise products of the eigenvalues of the $(1,1)$-form, and the number of nonzero products is three, one or zero. At the vertex $a = b = 0$, the normalized metric instead stays equal to $\omega/\sqrt{24\pi}$.

For a nonzero boundary class, the smooth limit T_0 annihilates θ, and the denominators $D_{\mu_0}(\gamma_\epsilon)$ tend to zero; no concentration of the coefficient measures occurs in this family. This degeneration of the minimizing tests for a fixed density differs from the sequence of Gauduchon forms used in Lemma 27.18.3 to prove that the lower mass vanishes. At the vertex, the denominator stays positive, and the order three comes entirely from the scaling $\Omega_\epsilon = \epsilon\omega$ of the background.

29.10 Smooth equations on the ample locus

We now turn to the metric geometry of solutions of Monge–Ampère equations in big classes. In this section X is a compact Kähler manifold of dimension $n \geq 1$, and $\alpha \in H^{1,1}_{\mathrm{BC}}(X, \mathbb{R})$ is a big class with a smooth closed representative θ. Its *ample locus* is $\mathrm{Amp}(\alpha) = X \setminus E_{\mathrm{nK}}(\alpha)$. The solution of the Monge–Ampère equation with a smooth positive density has minimal singularities, and a Kähler current with analytic singularities provides a barrier that is smooth on the ample locus. The interior estimate of Corollary 15.17.1 then gives smoothness there.

Let $\mathrm{vol}(\alpha)$ be the volume of α, the non-pluripolar mass of a current with minimal singularities in α, and let $\mathcal{E}(X, \theta)$ be the class of potentials $\varphi \in \mathrm{PSH}(X, \theta)$ of *full mass*, that is, with $\int_X \langle (\theta + \mathrm{dd^c}\varphi)^n \rangle = \mathrm{vol}(\alpha)$.

Theorem 29.10.1 *Let μ be a smooth positive volume form with $\mu(X) = \mathrm{vol}(\alpha)$. The normalized full-mass solution $\varphi \in \mathcal{E}(X, \theta)$ of*

$$\left\langle (\theta + \mathrm{dd^c}\varphi)^n \right\rangle = \mu, \qquad \sup_X \varphi = 0, \tag{29.17}$$

is smooth on $\mathrm{Amp}(\alpha)$, and the restriction of $\theta + \mathrm{dd^c}\varphi$ to this locus is a Kähler form.

The existence and uniqueness of φ are part of the classical theory of Monge–Ampère equations in big classes, discussed in Section 36.21; only the regularity assertion needs proof.

Proof Fix a Kähler form ω and put $f = \mu/\omega^n$. Since X is compact and μ is smooth, $f \in L^p(X, \omega^n)$ for every finite $p > 1$. The classical estimate for this equation, discussed in Section 36.21, shows that φ has minimal singularities: there is a constant $C > 0$ such that

$$V_\theta - C \le \varphi \le V_\theta. \tag{29.18}$$

The upper bound follows from the normalization.

Let $T = \theta + \mathrm{dd^c}\rho$ be a Kähler current with analytic singularities as in Theorem 29.3.1. Its potential ρ has analytic singularities with smooth remainder, and for some constant $\varepsilon > 0$,

$$T = \theta + \mathrm{dd^c}\rho \ge \varepsilon\omega, \qquad \{\rho = -\infty\} = E_{\mathrm{nK}}(\alpha).$$

Thus ρ is smooth on $\Omega = \mathrm{Amp}(\alpha)$. After subtracting a constant we may assume $\rho \le 0$, hence $\rho \le V_\theta$, and (29.18) gives

$$\rho - C \le \varphi \le 0.$$

In particular φ is locally bounded on Ω, and by locality its non-pluripolar equation there is the equation for bounded potentials. The density f belongs to L^p on X and is smooth and strictly positive on Ω. Hence all hypotheses of Corollary 15.17.1 hold for the strict barrier ρ, so φ is smooth and $\theta + \mathrm{dd^c}\varphi > 0$ on Ω. Since θ is closed, this form is Kähler. □

The proof uses that the density is smooth and positive on the entire regular locus of the barrier. The theorem says nothing about the distance from the ample locus to its complement: the metric $\theta + \mathrm{dd^c}\varphi$ may reach the omitted set along paths of finite length. To determine that length, one must compare the normal and tangential directions of the metric. The following sections first examine zeros of the density; singular radial densities then separate boundedness of the potential from completeness of the metric.

29.11 Density zeros and metric degeneration

Where the density vanishes, the metric defined by a solution degenerates, and the rate of degeneration depends on how the zero arises. In complex dimension $n \ge 2$, a zero of the determinant can come from a ramified holomorphic map, from an exceptional fiber of a blowup, or from a singular Hessian. These mechanisms give different relations between the regularity of the potential and the distance to the zero set. Bounded Hölder solutions in a positive background exist by Theorems 11.1.1 and 15.4.1, but their existence gives no pointwise estimate for the metric at such a zero.

Proposition 29.11.1 *Let $F\colon X \to Y$ be a finite surjective holomorphic map between compact Kähler manifolds of dimension n, let ω_Y be a Kähler form on Y, and let θ be a Kähler form in the class $F^*[\omega_Y]$, which is a Kähler class. Then the full-mass solution in the background θ with measure $(F^*\omega_Y)^n$ has a smooth bounded potential with minimal singularities, and its current is the smooth semipositive form $F^*\omega_Y$. Fix a smooth positive local volume form $\mathrm{d}V$. In a chart at a generic point of the ramification divisor, away from its other components, in which $F(z,w) = (z^k, w)$ with $k \geq 2$,*

$$(F^*\omega_Y)^n = |z|^{2(k-1)} G \,\mathrm{d}V, \qquad F^*\omega_Y \asymp |z|^{2(k-1)}|\mathrm{d}z|^2 + |\mathrm{d}w|^2, \tag{29.19}$$

where G is smooth and positive and $\asymp$ means that each side is bounded by a constant multiple of the other. The distance for $F^*\omega_Y$ to the ramification divisor $\{z = 0\}$ is comparable to $|z|^k$. Thus the metric on the complement of the ramification divisor is incomplete, although the current has no divisorial component. Finally, there is a strict barrier with logarithmic pole along the global ramification divisor.

Proof The class $F^*[\omega_Y]$ is nef, with the semipositive representative $F^*\omega_Y$. For an irreducible subvariety $V \subset X$ of positive dimension, $F|_V$ is finite, so it has maximal rank at the generic smooth point of V, where $F^*\omega_Y$ is therefore positive on V; hence $\int_V (F^*\omega_Y)^{\dim V} > 0$. For $V = X$ this makes the class big, by Proposition 29.6.2, and its null locus is empty. By the equality of the non-Kähler locus and the null locus of a nef and big class on a compact Kähler manifold, recalled in Section 29.4, its non-Kähler locus is empty, so it is a Kähler class by Theorem 29.3.1.

By the $\partial\bar\partial$-lemma on the compact Kähler manifold X, the de Rham class $F^*[\omega_Y]$ determines the Bott–Chern class, as in Section 24.4. Thus $F^*\omega_Y = \theta + \mathrm{dd}^{\mathrm{c}} u$ for a smooth function u on X. Its mass is the full mass of the class, and uniqueness of full-mass solutions in the closed theory, discussed in Section 36.21, identifies u with the solution up to a constant. The Jacobian determinant of $(z,w) \mapsto (z^k, w)$ is kz^{k-1}, and comparison of ω_Y with the Euclidean form gives (29.19). Hence every path to $\{z = 0\}$ has length bounded below by the variation of a fixed multiple of $|z|^k$ along it, and a radial path with fixed w attains this order.

The determinant of the differential $\mathrm{d}F$ defines the global ramification divisor R. Choose a smooth metric on $\mathcal{O}(R)$, with curvature form Θ_R and canonical section s_R. Since θ is Kähler, $\theta - \epsilon\Theta_R > 0$ for sufficiently small $\epsilon > 0$, so by the Poincaré–Lelong formula $\epsilon \log |s_R|^2 - C$ is the potential of a strict positive θ-current. The barrier is built from the global divisor R; no local branch coordinate is treated as a global line bundle. □

Nonclosed backgrounds show the same behavior on manifolds that are not Kähler. On a Hopf manifold, the pullback of the standard invariant metric under a coordinate power map solves an equation in a nonclosed Hermitian background.

Example 29.11.1 Let $n \geq 2$, $q > 1$ and let $k \geq 2$ be an integer. Let $H_q = (\mathbb{C}^n \setminus \{0\})/(Z \sim qZ)$ be the Hopf manifold, and on the covering space put $R = \sum_j |Z_j|^2$, $S = \sum_j |Z_j|^{2k}$,

$$h_0 = \frac{\mathrm{i}}{2\pi} \frac{\sum_j \mathrm{d}Z_j \wedge \mathrm{d}\bar{Z}_j}{R}, \quad u_k = \log S - k \log R,$$

$$\theta_k = k\mathrm{dd}^{\mathrm{c}} \log R + \frac{\mathrm{i}}{2\pi} \partial \log S \wedge \bar{\partial} \log S.$$

These forms and functions are invariant under $Z \mapsto qZ$, so they descend to H_q; the form h_0 descends to every Hopf manifold $H_{q'}$. The coordinate power map $F_k([Z]) = [(Z_1^k, \ldots, Z_n^k)]$ defines a finite surjective map $H_q \to H_{q^k}$: indeed $F_k(qZ) = q^k F_k(Z)$, and the choices of roots are finite modulo scaling in the source. The form $\mathrm{dd}^{\mathrm{c}} \log R$ is positive transverse to the complex Euler direction, while $\partial \log S$ is nonzero on that direction; hence $\theta_k > 0$. Direct differentiation gives

$$\theta_k + \mathrm{dd}^{\mathrm{c}} u_k = F_k^* h_0 = \frac{\mathrm{i}k^2}{2\pi S} \sum_j |Z_j|^{2k-2} \mathrm{d}Z_j \wedge \mathrm{d}\bar{Z}_j. \tag{29.20}$$

The background θ_k is not closed, since $\mathrm{d}\theta_k = F_k^*(\mathrm{d}h_0)$ is not identically zero. Along a coordinate divisor $D = \{Z_j = 0\}$, at a generic point and in local coordinates (z, w) with $D = \{z = 0\}$, the determinant vanishes to order $2(k-1)$, the metric $F_k^* h_0$ is comparable to $|z|^{2(k-1)}|\mathrm{d}z|^2 + |\mathrm{d}w|^2$, and the distance to D is of order $|z|^k$, as in Proposition 29.11.1.

A small logarithmic potential along D gives a strict barrier. Choose a smooth metric on $\mathcal{O}(D)$ with curvature form Θ_D and canonical section s_D. For small $c > 0$, set $h_c = \theta_k + c\Theta_D$ and $U_c = c \log |s_D|^2 + u_k$. Then $h_c > 0$ and

$$h_c + \mathrm{dd}^{\mathrm{c}} U_c = c[D] + F_k^* h_0.$$

By Theorem 7.1.1, the extended Monge–Ampère measure of U_c in the background h_c is the measure $(F_k^* h_0)^n$, whose density vanishes along the coordinate divisors. The ordinary singularity type of U_c is exactly that of $c \log |s_D|^2$, which is not minimal, while its metric off D is the same smooth metric $F_k^* h_0$. Finally, $H_q \cong S^{2n-1} \times S^1$ has $H^2(H_q; \mathbb{R}) = 0$. A Kähler form would therefore be exact, and Stokes' theorem would make its positive top-degree integral zero; thus H_q is not Kähler.

In these constructions, the potential becomes bounded once its divisorial part is removed, and the zero set lies at finite distance. More generally, Theorem 1.18.1 bounds the distance by the square root of a Hölder modulus of the potential wherever the current is a smooth positive metric. Continuity alone has no such consequence, as the radial examples of Section 29.12 will show.

29.11.1 Determinant and normal distance

The order of vanishing of the determinant alone does not determine the normal distance. An upper bound for the trace bounds all eigenvalues from above; combined with a lower bound for the determinant, it bounds the smallest eigenvalue from below, and hence the normal length. We use the local length distance d_η^U of Theorem 1.18.1, the infimum of η-lengths of piecewise smooth paths in $U \setminus D$; the distance $d_\eta^U(x, D)$ to D is the infimum of the lengths of such paths starting at x and tending to D.

Proposition 29.11.2 *Let $n \geq 2$, let $U \subset \mathbb{C} \times \mathbb{C}^{n-1}$ be a polydisk with coordinates (z, w), and put $D = \{z = 0\}$. Let η be a smooth semipositive $(1,1)$-form on U whose Euclidean trace is bounded and whose determinant is comparable to $|z|^{2\ell}$ for an integer $\ell \geq 1$. Then on a smaller concentric polydisk,*

$$c|z|^{\ell+1} \leq d_\eta^U((z,w), D) \leq C|z| \tag{29.21}$$

for constants $c, C > 0$. Both bounds are attained, up to constants, by smooth closed forms with the same determinant order. If the block of η in the w-directions is uniformly positive, the distance is comparable to $|z|^{\ell+1}$.

Proof The least eigenvalue satisfies $\lambda_{\min} \geq \det\eta/(\operatorname{tr}\eta)^{n-1} \geq c|z|^{2\ell}$. Every path to D therefore has length at least a constant times the integral of $|z|^\ell|\mathrm{d}z|$ along it, which gives the lower bound; a radial path and the trace bound give the upper bound. To see that both bounds occur, let $n = 2$ first. For an integer $0 \leq a \leq \ell$, take $\eta_a = \mathrm{dd^c}|G_a|^2$ with $G_a(z,w) = (z^{a+1}/(a+1), z^{\ell-a}w)$; for $n > 2$, append Euclidean coordinates. The Jacobian determinant of G_a is z^ℓ, but at $(z, 0)$ the first component of G_a forces a distance of order $|z|^{a+1}$, attained along a radial path. The cases $a = \ell$ and $a = 0$ give the two endpoints.

Finally, write the Hermitian coefficient matrix of η in the normal and tangential directions as $\left(\begin{smallmatrix} h & b^* \\ b & B \end{smallmatrix}\right)$. When B is uniformly positive, its Schur complement $h - b^*B^{-1}b = \det\eta/\det B$ is comparable to $|z|^{2\ell}$. Completing the square gives the lower bound of order $|z|^{\ell+1}$, and solving $\mathrm{d}w/\mathrm{d}s = -B^{-1}b\,\mathrm{d}z/\mathrm{d}s$ along a radial normal curve, on nested charts, gives a path of matching length. $\square$

29.11.2 Smoothness of exceptional residuals

We return to the blowup $\pi\colon X = \mathrm{Bl}_Z Y \to Y$ of Section 15.22. Here (Y, h) is a compact Hermitian n-fold, the smooth center Z has codimension $r \geq 2$, E is the exceptional divisor with canonical section s_E, Θ_E is the curvature form of a smooth metric on $O(E)$, $L = \log|s_E|^2$, and the background is $\theta = \pi^* h + c\Theta_E$ with $c > 0$. By Proposition 15.22.1, every $u \in \mathrm{PSH}(X, \theta)$ has the form $u = cL + q \circ \pi + C$ with q an h-plurisubharmonic function, and u has minimal singularities exactly when q is bounded; we then call u a potential of *minimal type*. Fix a smooth positive volume form $\mathrm{d}V$ on X, and use adapted coordinates in which $\pi(y, z, w) = (y, z, zw)$ and $E = \{z = 0\}$.

Proposition 29.11.3 *Let $u \in \mathrm{PSH}(X, \theta)$ be of minimal type, and suppose that its residual current $S = (\theta + \mathrm{dd^c} u) - c[E]$ is a smooth form. Then, on smaller adapted charts, $0 \leq S^n/\mathrm{d}V \leq C|z|^{2(r-1)}$. Consequently, if the Monge–Ampère measure of u has density $|z|^{2\ell} g$ with g smooth and strictly positive, then $\ell \geq r - 1$. This order condition is necessary but not sufficient for a smooth residual. In particular, in this background a smooth strictly positive density is incompatible with a smooth residual for a potential of minimal type.*

Proof Write $u = cL + v$. Since $S = \pi^* h + \mathrm{dd^c} v$ is smooth, the distributional Laplacian of v is smooth, and local elliptic regularity makes v smooth. Off E the form S equals $\theta + \mathrm{dd^c} u \geq 0$, so by continuity it is semipositive. On each fiber of $E \to Z$, a compact projective space, $\pi^* h$ restricts to zero and $\mathrm{dd^c} v$ restricts to the semipositive form S; the maximum principle therefore makes v constant on the fiber. Hence every vertical diagonal entry of S is a smooth nonnegative function vanishing at $z = 0$, and is therefore $O(|z|^2)$. The other diagonal entries are bounded, so Hadamard's determinant inequality gives the stated bound. The pullback $\pi^* h$ attains this order, while the examples below show that the order condition is not sufficient. The argument does not apply to prescribed nonminimal singularities, whose residuals can be positive on the exceptional fibers. □

A singular Hessian, rather than a zero of the density, can also make a solution nonsmooth. The following local example has constant positive density and a potential that is not C^1, and it can be placed in a compact manifold.

Example 29.11.2 On a polydisk with coordinates $(z, w) \in \mathbb{C} \times \mathbb{C}^{n-1}$, $n \geq 2$, set $P(z, w) = |z|^{2/n}(1 + |w|^2)$. Off $z = 0$, its Hessian block in the w-directions is $|z|^{2/n} I$ and its normal Schur complement is $n^{-2}|z|^{2/n-2}$, so its Hessian determinant is the positive constant n^{-2}. Since P is bounded, it extends across $z = 0$ as a plurisubharmonic function whose Monge–Ampère measure has no additional mass on the divisor. Yet P is not C^1 along that divisor.

This local example extends to a compact Kähler model on $\mathbb{P}^n$. On an affine chart with coordinates $y = (z, w)$, the Fubini–Study form is $\omega_{\mathrm{FS}} = \mathrm{dd^c} \log(1 + |y|^2)$; for constants $\lambda > 0$ and B, put $H = \lambda \log(1 + |y|^2)$ and $Q = H - B$. Choose nested radii and the parameters so that $Q > P + 2\epsilon$ on an outer collar of a ball and $Q < P - 2\epsilon$ on

an inner ball. A smooth convex regularized maximum $U = M_\epsilon(P, Q)$ then equals P on the inner ball and Q on the collar. Put $v = U - H$ on the ball and $v = -B$ outside it, so that $\lambda\omega_{\mathrm{FS}} + \mathrm{dd^c} v$ equals $\mathrm{dd^c} U$ on the ball and $\lambda\omega_{\mathrm{FS}}$ outside. This current is positive and smooth off $\{z = 0\}$; on the gluing annulus its density is at most $C|z|^{2/n-2}$, so it belongs globally to L^p for every $1 < p < n/(n-1)$. On the inner ball the density is smooth and strictly positive, but the potential is not C^1. There the metric is comparable to $|z|^{2/n-2}|\mathrm{d}z|^2 + |z|^{2/n}|\mathrm{d}w|^2$, and the distance to the divisor is of order $|z|^{1/n}$.

The construction also works on a Hopf manifold. Choose a global Hermitian form equal to $\mathrm{dd^c}(\lambda|y|^2)$ on a coordinate ball and equal to a fixed nonclosed Hermitian form outside a slightly larger ball, using a convex interpolation of positive forms. Carry out the same regularized maximum with $H = \lambda|y|^2$ on the coordinate ball, and set $v = U - H$ there and $v = -B$ outside it. The background is still nonclosed on the outer region, and the local potential and density are unchanged. As in the projective case, the density is controlled globally only in L^p.

This example does not contradict Corollary 15.17.1. The density loses smoothness along the divisor $\{z = 0\}$ in the gluing region, so the polar set of an admissible barrier must contain that part of the divisor. Being analytic, it then also contains the continuation of the divisor into the inner ball. The corollary requires a smooth positive density on the entire regular locus of the barrier, and local smoothness of the density on the inner ball cannot replace this hypothesis.

On a blowup, a radial potential of power type produces densities with exceptional zeros or poles of every order. These examples show that the order condition in Proposition 29.11.3 is not sufficient.

Example 29.11.3 Let $\pi\colon X = \mathrm{Bl}_0 \mathbb{P}^n \to \mathbb{P}^n$ be the blowup at the origin of an affine chart with coordinates y, let E be its exceptional divisor, and write $s = |y|^2$. For every $\gamma, c > 0$, one can choose $\lambda > 0$ so that the background $\theta = \pi^*(\lambda\omega_{\mathrm{FS}}) + c\Theta_E$ admits a potential $u = cL + q \circ \pi$ with bounded Hölder remainder q, hence of minimal type, whose total local potential $\lambda \log(1 + s) + q$ equals $As^\gamma + B$ near the origin, for constants $A > 0$ and $B \in \mathbb{R}$. To construct q, choose a smooth function m with $m(s) > 0$ and $m'(s) > 0$, equal to $A\gamma s^\gamma$ near $s = 0$ and to $\lambda s/(1 + s)$ outside a slightly larger interval; taking A small puts the values at the two ends in the right order. Match the positive derivative germs on narrow collars, fill the remaining positive deficit in the integral of m' with a positive interior derivative and a bump, and integrate $U' = m/s$, choosing the constant so that $U(s) = \lambda \log(1 + s)$ for large s. Set $q(y) = U(|y|^2) - \lambda \log(1 + |y|^2)$ on the affine chart and extend it by zero outside. Since m and m' are positive, positivity holds throughout the compact interpolation region.

On the blowup, in coordinates with $y = (z, zw)$, the total local potential of u is $A|z|^{2\gamma}(1 + |w|^2)^\gamma + B$. The density of u is $|z|^{2\ell}G$ with G smooth and positive and $\ell = n\gamma - 1 > -1$, since the Hessian determinant of $A|y|^{2\gamma}$ is $A^n\gamma^{n+1}s^{n(\gamma-1)}$ and the squared Jacobian of the blowup is $|z|^{2(n-1)}$. The metric off E is comparable to

$$|z|^{2\gamma-2}|\mathrm{d}z|^2 + |z|^{2\gamma}|\mathrm{d}w|^2, \tag{29.22}$$

the distance to E is of order $|z|^\gamma$, and the metric completion collapses E to a point. If ℓ is a nonnegative integer, the density is smooth. At $\ell = 0$ the density is smooth and strictly positive, but the remainder has exact normal Hölder exponent $2/n$ for $n > 2$, is Lipschitz but not C^1 for $n = 2$, and the normal coefficient of the metric is unbounded. For $-1 < \ell < 0$, the density belongs to L^p exactly when $p\ell > -1$. Taking $\ell = n$ gives a smooth density with a zero of order $2n \geq 2(n-1)$, while the residual Hessian remains nonsmooth; this proves the insufficiency asserted in Proposition 29.11.3. A small nonclosed perturbation of the background supported away from the origin leaves these local calculations unchanged, and gives the same examples in a nonclosed background, with the Monge–Ampère measure of the perturbed potential as prescribed measure.

29.12 Radial densities and metric completeness

For a radially symmetric equation, the mass of the density inside a ball determines the radial derivative of the potential and hence both eigenvalues of its Hessian. Boundedness of the potential and finiteness of the distance to the center are then expressed by two different integrals of this derivative, and suitable singular densities separate the two properties. Throughout this section $n \geq 2$, the Fubini–Study form on $\mathbb{P}^n$ is normalized by $\int_{\mathbb{P}^n} \omega_{\mathrm{FS}}^n = 1$, and on the standard affine chart with coordinates $y \in \mathbb{C}^n$ we put $\omega_{\mathrm{E}} = \mathrm{dd^c}|y|^2$, so that $\int_{\{|y|^2<S\}} \omega_{\mathrm{E}}^n = S^n$ for $S > 0$. We write $s = |y|^2$.

Theorem 29.12.1 *Fix $\lambda > 0$ and a smooth positive function $D\colon (0,\infty) \to (0,\infty)$ such that $D(s) = \lambda^n/(1+s)^{n+1}$ for all sufficiently large s, and*

$$n \int_0^\infty s^{n-1} D(s)\, \mathrm{d}s = \lambda^n. \tag{29.23}$$

Let μ_D be the measure on $\mathbb{P}^n$ equal to $D(|y|^2)\omega_{\mathrm{E}}^n$ on the affine chart and to zero on the hyperplane at infinity; near that hyperplane it agrees with $(\lambda\omega_{\mathrm{FS}})^n$. Then the equation $\langle(\lambda\omega_{\mathrm{FS}} + \mathrm{dd^c}q)^n\rangle = \mu_D$ has a unique normalized full-mass solution q. It is smooth off $y = 0$, and on the affine chart its total local potential *$\lambda\log(1+|y|^2) + q(y)$ is a radial function $U(s)$ satisfying*

$$\begin{aligned} m(s)^n &= n\int_0^s t^{n-1} D(t)\, \mathrm{d}t, \qquad m(s) = sU'(s), \\ m'(s) &= \frac{s^{n-1}D(s)}{m(s)^{n-1}} > 0. \end{aligned} \tag{29.24}$$

The tangential and radial eigenvalues of its Hessian are m/s and m'. For any sufficiently small $s_0 > 0$, the solution is bounded, equivalently has minimal singularities, precisely when

$$\int_0^{s_0} \frac{m(s)}{s}\, \mathrm{d}s < \infty. \tag{29.25}$$

The distance from the origin for the metric $\lambda\omega_{\mathrm{FS}} + \mathrm{dd}^{\mathrm{c}} q$ on $\mathbb{P}^n \setminus \{0\}$ is finite precisely when

$$\int_0^{s_0} \frac{\sqrt{m'(s)}}{2\sqrt{s}}\,\mathrm{d}s < \infty. \tag{29.26}$$

If this integral diverges, the metric on $\mathbb{P}^n \setminus \{0\}$ is complete.

Let $\pi\colon X = \mathrm{Bl}_0 \mathbb{P}^n \to \mathbb{P}^n$ be the blowup at the origin, with exceptional divisor E, and let s_E and Θ_E be as in Section 15.22. For $\theta = \pi^*(\lambda\omega_{\mathrm{FS}}) + c\Theta_E$ with $c > 0$, the full-mass solution on X whose measure is the pullback of μ_D has potential $c \log |s_E|^2 + q \circ \pi + C$, generic Lelong number exactly c along E, and mass λ^n. In a blowup chart $\pi(z, w) = (z, zw)$, its density relative to the Euclidean volume form of the coordinates (z, w), normalized as above, is

$$|z|^{2(n-1)} D\big(|z|^2(1 + |w|^2)\big). \tag{29.27}$$

Its ordinary singularity type is minimal precisely under (29.25). In the unbounded cases, the residual $q \circ \pi$ grows sublogarithmically and has zero generic Lelong number, but the type is not minimal. The metric off E satisfies the same completeness criterion (29.26).

Proof Define m by the first equation in (29.24). It is positive for $s > 0$ and tends to zero at the origin. By (29.23) and the prescribed outer density, $m(s) = \lambda s/(1 + s)$ for large s: the two expressions for m^n have the same limit at infinity and the same derivative for large s. Integrate $U' = m/s$, choosing the additive constant so that $U(s) = \lambda \log(1 + s)$ for large s. The Hessian of $U(|y|^2)$ has eigenvalues m/s and m', and its determinant is $(m/s)^{n-1}m' = D$. Thus $q = U(|y|^2) - \lambda \log(1 + |y|^2)$ extends by zero outside the affine chart and defines a positive current that is smooth off the origin. Subtracting its supremum gives the normalization.

Since U is increasing, it is bounded above near zero, so Lemma 1.3.7 gives a plurisubharmonic extension across the origin, possibly with the value $-\infty$ there. If U is unbounded, then $m(s) \to 0$ implies $U(s) = o(|\log s|)$: for each $\epsilon > 0$, bound m by ϵ on a small interval and integrate m/s. The closed non-pluripolar product equals μ_D off the origin and puts no mass at the origin; the measure μ_D has no atom there and has mass λ^n by (29.23). Hence q is a full-mass solution, and it is unique by the uniqueness of full-mass solutions in the closed theory, discussed in Section 36.21. Integrating $U' = m/s$ shows that U is bounded near the origin precisely under (29.25).

Write $R = |y|$. For a path approaching the origin, the metric norm of its velocity is at least a fixed multiple of $\sqrt{m'(R^2)}|\dot{R}|$, and radial paths attain this bound up to a fixed constant. Since $\mathrm{d}R = \mathrm{d}s/(2\sqrt{s})$, this gives (29.26). The metric is smooth and positive on compact subsets away from the origin, so divergence of the integral is equivalent to completeness of the metric on $\mathbb{P}^n \setminus \{0\}$.

On the blowup, apply Proposition 15.22.1 and use the squared Jacobian $|z|^{2(n-1)}$. The closed non-pluripolar product is the extension by zero of the pullback density, has mass $\lambda^n = \mathrm{vol}([\theta])$, and is unique in its full-mass class. By Theorem 7.1.1, the extended operator agrees with it on potentials that are continuous off E, even when the residual tends to $-\infty$ along E. Sublogarithmic growth gives the residual zero

Lelong number, so the coefficient along E is exactly c, but it does not give a bounded difference from $c \log |s_E|^2$. The metric off E is isometric to the metric on $\mathbb{P}^n \setminus \{0\}$, which proves the same distance criterion. □

The inner asymptotics considered below can be glued to the prescribed outer density while keeping (29.23): take a small inner amplitude or radius, join through a smooth positive annulus, and add a nonnegative smooth annular bump to supply the remaining mass. Thus each of these asymptotic behaviors occurs for a density satisfying the hypotheses of Theorem 29.12.1. The assertions below concern the total local potential U; subtracting a smooth background potential can change a leading smooth term.

In the next theorem, $f \sim g$ means that $f/g \to 1$ as $s \downarrow 0$.

Theorem 29.12.2 *For the normalized family in Theorem 29.12.1, put $L = \log(1/s)$ for $0 < s < 1$.*

(1) *If $D(s) \sim Cs^{\ell-n+1}L^{-b}$ with $C > 0$, $\ell > -1$ and $b \in \mathbb{R}$, put $\gamma = (\ell+1)/n$ and $K = (C/\gamma)^{1/n}$. Then*

$$m \sim Ks^{\gamma}L^{-b/n}, \qquad U(s) - U(0) \sim \frac{K}{\gamma}s^{\gamma}L^{-b/n}, \qquad m' \sim \gamma K s^{\gamma-1}L^{-b/n}. \tag{29.28}$$

The residual is bounded and Hölder continuous. In bounded blowup charts, its metric is comparable to $L^{-b/n}(|z|^{2\gamma-2}|\mathrm{d}z|^2 + |z|^{2\gamma}|\mathrm{d}w|^2)$, and its distance to E is of order $|z|^{\gamma}(\log(1/|z|))^{-b/(2n)}$. In particular, the metric is incomplete near E.

(2) *If $D(s) \sim Cs^{-n}L^{-b}$, the local mass is finite precisely for $b > 1$. In that range put $a = (b-1)/n$ and $K = (nC/(b-1))^{1/n}$. Then*

$$m \sim KL^{-a}, \qquad m' \sim Kas^{-1}L^{-a-1}. \tag{29.29}$$

The potential has a finite limit at the origin, the solution has minimal singularities, and the distance is finite, each precisely when $b > n+1$; in that case $U(s) - U(0) \sim KL^{1-a}/(a-1)$. At $b = n+1$, U diverges as $-K\log L$, and for $1 < b < n+1$ as $-KL^{1-a}/(1-a)$. In these unbounded cases the solutions have full mass but not minimal singularities, and their metrics are complete near E. Their densities on the blowup lie in no L^p with $p > 1$.

(3) *If $D(s) \sim Cs^{-n}L^{-(n+1)}(\log L)^{-d}$ with $d \in \mathbb{R}$, put $C_* = C^{1/n}$. Then*

$$m \sim C_*L^{-1}(\log L)^{-d/n}, \qquad m' \sim C_*s^{-1}L^{-2}(\log L)^{-d/n}. \tag{29.30}$$

The potential is bounded precisely when $d > n$, with $U(s) - U(0)$ of order $(\log L)^{1-d/n}$. At $d = n$ it diverges as $-C_*\log\log L$, and for $d < n$ as $-C_*(\log L)^{1-d/n}/(1-d/n)$. The distance is finite precisely when $d > 2n$, with remaining length of order $(\log L)^{1-d/(2n)}$. Thus for $n < d \le 2n$ the residual is bounded and continuous but not Hölder, the solution has minimal singularities, and the metric is complete near E. All these critical densities have finite mass and lie in no L^p with $p > 1$.

In all cases the logarithmic coefficient of the solution along E is c, whereas by Proposition 15.22.1 a strict barrier can be chosen with coefficient $c+\delta$, for some $\delta>0$ independent of ℓ, b, d.

Proof We use the exact expressions (29.24). Each asymptotic formula follows by bounding D between $(1\pm\epsilon)$ times its model and integrating the model; no derivative of an unspecified error term is taken. In the first case $m^n \sim nCs^{\ell+1}L^{-b}/(\ell+1)$, which yields (29.28). Both Hessian eigenvalues are then comparable to $s^{\gamma-1}L^{-b/n}$, and the substitution of blowup coordinates gives the assertions about the metric and the distance. The positive power of s makes the potential Hölder continuous, after the logarithmic factor is absorbed by a slightly smaller exponent.

In the second case, $n\int_0^s t^{-1}C(\log(1/t))^{-b}\,\mathrm{d}t$ is finite precisely for $b>1$, and it is then asymptotic to $nCL^{1-b}/(b-1)$. This gives (29.29). Since $\mathrm{d}U=-m\,\mathrm{d}L$, boundedness of U is equivalent to convergence of $\int^\infty L^{-a}\,\mathrm{d}L$, that is, to $a>1$. The distance is a constant multiple of $\int^\infty L^{-(a+1)/2}\,\mathrm{d}L$ and has the same strict threshold. Explicit primitives give the stated finite and divergent behavior, including the logarithm at $a=1$. On the blowup the density is comparable to $|z|^{-2}(\log(1/|z|))^{-b}$; for every $p>1$, its pth power acquires an exponentially growing factor in logarithmic coordinates, so the density is not in L^p.

In the third case, $m^n\sim CL^{-n}(\log L)^{-d}$. The criteria for the potential and for the distance become convergence of

$$\int^\infty L^{-1}(\log L)^{-d/n}\,\mathrm{d}L \quad\text{and}\quad \int^\infty L^{-1}(\log L)^{-d/(2n)}\,\mathrm{d}L,$$

respectively. Their thresholds are $d>n$ and $d>2n$, and direct integration gives the stated tails and divergent behavior. A tail of order $(\log L)^{1-d/n}$, a negative power of $\log L$, tends to zero more slowly than every positive power of $|z|$, so the potential is not Hölder continuous. The mass integral is finite for every real d because the principal factor is $L^{-(n+1)}$ with $n\ge 2$, and the same exponential argument excludes L^p for $p>1$. In particular, all the endpoint cases occur within the prescribed family of Theorem 29.12.1. □

29.13 Equations on products with a curve

On the product of a curve with a Hermitian manifold, a density depending only on the curve variable reduces the Monge–Ampère equation to a Poisson equation on the curve, even when the second factor is not Kähler. Explicit primitives then allow a comparison with the radial families. For a single logarithmic factor, boundedness of the potential and finiteness of the distance have the same threshold, whereas the iterated logarithmic family of Theorem 29.12.2 separates them.

Theorem 29.13.1 *Let C be a compact Riemann surface with a positive $(1,1)$-form ω_C, let (N,ω_N) be a compact Hermitian manifold of dimension $n-1$, and put $h=\omega_C+\omega_N$ on $C\times N$, the two forms being pulled back by the projections. Let $p_0\in C$, and let $F\geq 0$ be an integrable function on C, smooth and positive away from p_0, with $\int_C F\omega_C=\int_C\omega_C$. Then there is a function v on C such that $v\circ\mathrm{pr}_C$ solves the equation $\mathrm{MA}_h^{\mathrm{ext}}(v\circ\mathrm{pr}_C)=(F\circ\mathrm{pr}_C)h^n$, and the current of this solution has no divisorial component along $\{p_0\}\times N$.*

Let z be a coordinate centered at p_0, write $r=|z|$ and $\omega_C=(\mathrm{i}/2\pi)g(z)\,\mathrm{d}z\wedge\mathrm{d}\bar z$, and call a local potential of $\omega_C+\mathrm{dd}^{\mathrm{c}}v$ near p_0 a total normal potential.

(1) *If $Fg=C_0r^{2\tau}$ near p_0, with $C_0>0$ and $\tau>-1$, then up to a smooth harmonic term the total normal potential is $C_0r^{2(\tau+1)}/(\tau+1)^2$. The solution is bounded and Hölder continuous, hence has minimal singularities, and the normal distance to $\{p_0\}\times N$ is of order $r^{\tau+1}$. If $\tau=-a$ with $0<a<1$, the density is locally L^p exactly when $pa<1$. If τ is a nonnegative integer, the potential is smooth.*

(2) *If $Fg=C_0r^{-2}t^{-b}$ near p_0, with $t=\log(1/r)$, the mass is finite exactly for $b>1$. The total normal potential, whose dd^{c} has no atom at p_0, is*

$$P_b(t)=\begin{cases}\dfrac{4C_0t^{2-b}}{(b-1)(b-2)}, & b\neq 2,\\ -4C_0\log t, & b=2,\end{cases}\tag{29.31}$$

up to constants and smooth harmonic terms. It is bounded and continuous, equivalently the solution has minimal singularities, precisely when $b>2$, and it is then not Hölder continuous. The normal end has finite distance precisely when $b>2$. For $1<b\leq 2$ the end is complete, and the potential is unbounded and does not have minimal singularities, but it has zero Lelong number and trivial multiplier ideals. No density in this logarithmic family belongs to L^p for $p>1$.

If ω_C and ω_N are Kähler, the unbounded solutions have full mass, and uniqueness in the closed theory excludes a different bounded solution for the same measure. A small strict logarithmic barrier along $\{p_0\}\times N$ exists also when the solutions are bounded and have minimal singularities.

Proof On the curve, the real current $F\omega_C-\omega_C$ has integral zero and is therefore $\mathrm{dd}^{\mathrm{c}}v$ for a global distribution v, unique up to a constant. Its quasi-subharmonic representative is smooth off p_0 by the local Poisson equation. Since $F\omega_C$ has no atom at p_0, neither does the current $\omega_C+\mathrm{dd}^{\mathrm{c}}v$. Pull v back to $C\times N$. Since every wedge product containing two factors from the curve vanishes, on the regular locus

$$(h+\mathrm{dd}^{\mathrm{c}}v)^n=n(F\omega_C)\wedge\omega_N^{n-1}=(F\circ\mathrm{pr}_C)h^n.\tag{29.32}$$

By Theorem 7.1.1, the extended product agrees with this smooth product off $\{p_0\}\times N$ and gives that analytic set zero mass. The right side has the same property and finite mass. This proves the global equation, including the case in which the curve potential has a finite value at p_0 but is discontinuous there.

For the power density, put $P_\tau(r) = C_0 r^{2(\tau+1)}/(\tau+1)^2$. Direct differentiation gives $\partial_z\bar\partial_z P_\tau = C_0 r^{2\tau}$, and the radial flux of P_τ through small circles vanishes as $r \downarrow 0$ when $\tau > -1$, so no Dirac term appears. The difference between the total potential and P_τ is therefore harmonic on the whole disk and smooth. To realize the model globally, cut off the local potential and solve the remaining smooth mean-zero equation on the curve. The normal line element of the metric is comparable to $r^\tau|\mathrm{d}r|$, which gives a distance of order $r^{\tau+1}$: a lower bound for all paths and an upper bound along a radial path. The L^p assertion is the convergence of $\int_0^\epsilon r^{1-2pa}\,\mathrm{d}r$.

For the logarithmic density, $P_b''(t) = 4C_0 t^{-b}$ and $\partial_z\bar\partial_z P_b = P_b''/(4r^2) = C_0 r^{-2} t^{-b}$. The flux tends to zero for $b > 1$, which excludes a logarithmic Dirac mass, and the same cutoff and smooth correction give a compact model. The mass of the density near p_0 is a constant multiple of $\int^\infty t^{-b}\,\mathrm{d}t$, finite exactly for $b > 1$, while the integral of its pth power, for $p > 1$, diverges after the logarithmic change of variables. The primitive has a finite limit exactly for $b > 2$, with tail of order t^{2-b}, which tends to zero more slowly than every power r^α. The normal length is a constant multiple of $\int^\infty t^{-b/2}\,\mathrm{d}t$, which has the same threshold. The projection to the curve bounds the length of every path from below by this normal integral, and a radial path with fixed point of N attains it.

For $1 < b \le 2$, $|P_b(t)|/t \to 0$, so the Lelong number is zero. For every weight $k > 0$, the integrand e^{-kP_b} in polar coordinates is bounded by $\exp(-2t + o(t))$ with respect to $\mathrm{d}t$, which is integrable; hence all multiplier ideals are trivial. These properties do not imply bounded ordinary type. In a Kähler product, the normalized curve integral gives full non-pluripolar mass, and uniqueness of full-mass solutions in the closed theory, discussed in Section 36.21, excludes another bounded solution. A bounded competitor for the extended operator coincides globally with its deep truncations, so the same exclusion applies there. Finally, for $D = \{p_0\} \times N$, a smooth metric on $\mathcal{O}(D)$ with curvature form Θ_D and canonical section s_D, and sufficiently small $\delta > 0$, we have $h - \delta\Theta_D > 0$, so $\delta \log|s_D|^2$ is a strict barrier. □

29.14 An incomplete metric in a big class

We now consider exponential Monge–Ampère equations whose reference measure has a nonintegrable pole along a divisor, and ask whether the solution defines a complete metric on the locus where its current is smooth and positive. A divisorial part of a solution with minimal singularities can cancel the pole and leave the deleted divisor at finite distance. On the blowup of the projective plane at a point, the resulting metric away from the exceptional divisor is the pullback of the Fubini–Study metric. The class is big but not nef, and the equation has a unique solution of finite energy.

Finite energy refers to the classical closed theory of [Xia26]. For a smooth closed real $(1,1)$-form θ with big class on a compact Kähler surface Y, and for $u \in \mathrm{PSH}(Y,\theta)$

with minimal singularities, the Monge–Ampère energy relative to the envelope V_θ is

$$E_\theta(u) = \frac{1}{3}\sum_{j=0}^{2}\int_Y (u - V_\theta)\left\langle (\theta + \mathrm{dd^c} u)^j \wedge (\theta + \mathrm{dd^c} V_\theta)^{2-j}\right\rangle .$$

For general $u \in \mathrm{PSH}(Y,\theta)$, $E_\theta(u)$ is the decreasing limit of $E_\theta(\max(u, V_\theta - k))$ as $k \to \infty$, and $\mathcal{E}^1(Y,\theta)$ is the set of potentials with $E_\theta(u) > -\infty$; they have full mass.

Theorem 29.14.1 *There exist a smooth projective surface Y, a smooth irreducible divisor E, a smooth closed real $(1,1)$-form θ with big class α, a Hermitian metric on $\mathcal{O}_Y(E)$, a defining section σ of $\mathcal{O}_Y(E)$, and a smooth positive volume form $\mathrm{d}V_Y$ with the following properties. The equation*

$$\left\langle (\theta + \mathrm{dd^c}\rho)^2\right\rangle = \mathrm{e}^{\rho}|\sigma|^{-2}\,\mathrm{d}V_Y \tag{29.33}$$

has a unique solution $\rho \in \mathcal{E}^1(Y,\theta)$, and this solution has minimal singularities. Moreover,

$$\mathrm{Amp}(\alpha) = Y \setminus E$$

is precisely the open set on which $\theta + \mathrm{dd^c}\rho$ is smooth and strictly positive. The restriction of this metric to every nonempty connected open subset of $Y \setminus E$ is incomplete. In particular, no dense Zariski open subset of $Y \setminus E$ carries a complete induced metric.

Let $p = [1:0:0] \in \mathbb{P}^2$, and let ω_2 be the Fubini–Study form on $\mathbb{P}^2$, normalized by $\int_{\mathbb{P}^2}\omega_2^2 = 1$.

Proof (Proof of Theorem 29.14.1) *The surface and the divisor.* Use the incidence model

$$Y = \{([Z_0 : Z_1 : Z_2], [W_1 : W_2]) \in \mathbb{P}^2 \times \mathbb{P}^1 : Z_1W_2 = Z_2W_1\}.$$

The projections $\pi\colon Y \to \mathbb{P}^2$ and $q\colon Y \to \mathbb{P}^1$ identify Y with the blowup of $\mathbb{P}^2$ at p, with exceptional divisor $E = \pi^{-1}(p)$. Let ω_1 be the Fubini–Study form on $\mathbb{P}^1$ with the same normalization as ω_2, and write

$$\beta = \pi^*\omega_2, \qquad \gamma = q^*\omega_1, \qquad \Omega = \beta + \gamma.$$

The form Ω is Kähler, being the restriction of a product Kähler metric.

The linear map $(Z_0, Z_1, Z_2) \mapsto (Z_1, Z_2)$ defines a section σ of

$$\pi^*\mathcal{O}_{\mathbb{P}^2}(1) \otimes q^*\mathcal{O}_{\mathbb{P}^1}(-1),$$

because on Y the vector (Z_1, Z_2) lies in the tautological line $[W_1 : W_2]$. The section vanishes simply along E, which identifies this line bundle with $\mathcal{O}_Y(E)$. For the product of the Fubini–Study metric and the metric induced on the tautological bundle, its squared norm is

$$s = |\sigma|^2 = \frac{|Z_1|^2 + |Z_2|^2}{|Z_0|^2 + |Z_1|^2 + |Z_2|^2}.$$

In a blowup chart $\pi(t,w) = (t, tw)$, $q(t,w) = [1 : w]$, one has

$$s = \frac{|t|^2(1+|w|^2)}{1+|t|^2(1+|w|^2)}, \qquad \mathrm{dd^c}\log s = [E] + \gamma - \beta. \tag{29.34}$$

The current identity follows either from this formula and the second blowup chart or from the Poincaré–Lelong formula.

The big class and its equation. Set

$$\theta = 3\beta - 2\gamma, \qquad \rho = 2\log s, \qquad T = \theta + \mathrm{dd^c}\rho = \beta + 2[E]. \tag{29.35}$$

Thus $\rho \in \mathrm{PSH}(Y,\theta)$. The class $\alpha = \{\theta\}$ is big, because the potential $\psi = (5/2)\log s$ satisfies

$$\theta + \mathrm{dd^c}\psi = \tfrac{1}{2}(\beta+\gamma) + \tfrac{5}{2}[E] \geq \tfrac{1}{2}\Omega. \tag{29.36}$$

It is not nef, since $\int_E \theta = -2$: the restriction of β to E vanishes and $q|_E$ is an isomorphism onto $\mathbb{P}^1$. A nef class has nonnegative degree on every curve, by its defining smooth approximations.

On $Y \setminus E$ define $\mathrm{d}V_Y = \beta^2/s$. To check that it extends across E, put

$$\mathrm{d}\lambda_{t,w} = \tfrac{\mathrm{i}}{2}\mathrm{d}t \wedge \mathrm{d}\bar{t} \wedge \tfrac{\mathrm{i}}{2}\mathrm{d}w \wedge \mathrm{d}\bar{w},$$

and let $\mathrm{d}\lambda_{z_1,z_2}$ be the analogous Euclidean volume in affine coordinates of $\mathbb{P}^2$. The normalized Fubini–Study volume is

$$\omega_2^2 = \frac{2}{\pi^2}\frac{\mathrm{d}\lambda_{z_1,z_2}}{(1+|z_1|^2+|z_2|^2)^3}.$$

Since the complex Jacobian determinant of $(t,w) \mapsto (t,tw)$ is t, we obtain

$$\beta^2 = \frac{2}{\pi^2}\frac{|t|^2\,\mathrm{d}\lambda_{t,w}}{[1+|t|^2(1+|w|^2)]^3},$$
$$\mathrm{d}V_Y = \frac{2}{\pi^2}\frac{\mathrm{d}\lambda_{t,w}}{(1+|w|^2)[1+|t|^2(1+|w|^2)]^2}.$$

The second expression is smooth and strictly positive at $t = 0$. Interchanging the two homogeneous coordinates gives the extension on the other chart. The two extensions agree off E, hence everywhere on their overlap.

The non-pluripolar product of T agrees with β^2 off E and gives no mass to E. The smooth measure β^2 also gives no mass to E, so

$$\langle T^2\rangle = \beta^2, \qquad \mathrm{e}^{\rho}|\sigma|^{-2}\,\mathrm{d}V_Y = s^2 s^{-1}\frac{\beta^2}{s} = \beta^2.$$

The right side of (29.33) is defined on $Y \setminus E$ and extended by zero on E, so no product of zero and infinity occurs. This proves the equation and gives $\int_Y \langle T^2\rangle = \int_{\mathbb{P}^2}\omega_2^2 = 1$.

Minimal singularities and energy. Let $u \in \mathrm{PSH}(Y,\theta)$. Since

$$\beta + \mathrm{dd^c}(u-\rho) = \theta + \mathrm{dd^c} u \geq 0 \quad \text{on } Y \setminus E,$$

the function $u-\rho$ on $Y \setminus E$ descends to an ω_2-plurisubharmonic function on $\mathbb{P}^2 \setminus \{p\}$. By Lemma 1.3.8, it is bounded above, so $u \leq \rho + C_u$ there. This inequality holds almost everywhere on Y, and Lemma 1.3.3 makes it hold everywhere. In particular $u = -\infty$ on E, and ρ has minimal singularities.

Let V_θ be the normalized envelope of Section 1.19. Since $s \leq 1$, we have $\rho \leq 0$, and minimality gives

$$\rho \leq V_\theta \leq \rho + C.$$

Thus $\rho - V_\theta$ is bounded off E, and the energy is

$$E_\theta(\rho) = \frac{1}{3}\sum_{j=0}^{2} \int_Y (\rho - V_\theta)\left\langle (\theta + \mathrm{dd^c}\rho)^j \wedge (\theta + \mathrm{dd^c} V_\theta)^{2-j} \right\rangle.$$

The mixed measures have finite mass in the closed Kähler theory and do not charge E. Hence $E_\theta(\rho) > -\infty$, so $\rho \in \mathcal{E}^1(Y,\theta)$. A potential with minimal singularities has full mass; in this example that mass is 1.

Uniqueness. Suppose $u \in \mathcal{E}^1(Y,\theta)$ is another solution, and let $\mathrm{d}\mu = s^{-1}\mathrm{d}V_Y$ on $Y \setminus E$, extended by zero on E. This measure is smooth and positive off E, but has infinite total mass. The classical comparison principle for full-mass potentials in the closed theory gives

$$\int_{\{u<\rho\}} \mathrm{e}^{\rho}\,\mathrm{d}\mu \leq \int_{\{u<\rho\}} \mathrm{e}^{u}\,\mathrm{d}\mu.$$

Both integrals are finite, since the two Monge–Ampère measures have mass 1. The integrand on the left is strictly larger, which forces $\mu(\{u<\rho\}) = 0$. Interchanging u and ρ gives $\mu(\{\rho<u\}) = 0$ as well. Therefore $u = \rho$ almost everywhere on $Y \setminus E$ with respect to a smooth volume form, hence almost everywhere on Y, since E has zero volume. By Lemma 1.3.3, $u = \rho$ everywhere.

The ample locus and incompleteness. The Kähler current in (29.36) is smooth off E. On the other hand, every $u \in \mathrm{PSH}(Y,\theta)$ satisfies $u \leq \rho + C_u$, so its Lelong numbers at points of E are at least those of $\rho = 2\log s$, which are positive. Thus every Kähler current in α is singular at every point of E, and Definition 29.3.1 gives

$$\mathrm{Amp}(\alpha) = Y \setminus E_{\mathrm{nK}}(\alpha) = Y \setminus E.$$

The current T equals β there, and it is strictly positive because π is a biholomorphism away from E. Its nonzero divisorial part $2[E]$ prevents smoothness at every point of E.

Let $U \subset Y \setminus E$ be a nonempty connected open set, and set $W = \pi(U)$. The map $\pi\colon (U, T|_U) \to (W, \omega_2|_W)$ is an isometry. Choose $x \in W$ and a smooth path $c\colon [0,1] \to \mathbb{P}^2$ from x to p. Its first exit time

$$t_0 = \min\{t \in [0,1] : c(t) \notin W\}$$

is positive, and $c([0,t_0)) \subset W$. The path has finite Fubini–Study length. For $t_k \nearrow t_0$, the lengths of its tails show that $c(t_k)$ is a Cauchy sequence for the intrinsic metric of W. It has no limit in W: intrinsic convergence implies ambient convergence, and the ambient limit is $c(t_0) \notin W$. Thus U is incomplete. A dense Zariski open subset of $Y \setminus E$ is the complement of a proper analytic subset in a connected manifold, hence connected, so the last assertion is a special case. □

The cancellation $\mathrm{e}^{\rho}|\sigma|^{-2} = s$ removes the pole, and the divisor E, which the metric collapses to the point p, stays at finite distance. Thus an exponential equation in a big class can have an incomplete metric even when its finite-energy solution is unique and has minimal singularities. In this example the class is not nef. The next example shows that nefness alone does not restore completeness; there the pole produces a complete cusp, but a different curve lies at finite distance.

29.14.1 A nef class with an incomplete null end

Nefness of the class does not prevent incompleteness. In the next construction, the pole of the measure along a divisor D does produce a complete cusp, but a disjoint curve E, on which the class has degree zero, lies at finite distance.

Theorem 29.14.2 *There are a smooth projective surface Y, disjoint smooth divisors E, D, a smooth closed semipositive form θ with nef and big class, a smooth Hermitian norm on $\mathcal{O}_Y(D)$, its defining section σ, and a smooth strictly positive volume form $\mathrm{d}V_Y$, such that the equation*

$$\left\langle (\theta + \mathrm{dd^c}\rho)^2 \right\rangle = \mathrm{e}^{\rho}|\sigma|^{-2}\,\mathrm{d}V_Y \tag{29.37}$$

has a unique solution $\rho \in \mathcal{E}^1(Y,\theta)$. The solution has zero Lelong numbers and is smooth on $Y \setminus (E \cup D)$, where its current $\Omega = \theta + \mathrm{dd^c}\rho$ is strictly positive. The curve E lies in the null locus of $\{\theta\}$. Every dense Zariski open set on which Ω is smooth and strictly positive is incomplete, although every path in $Y \setminus (E \cup D)$ approaching D has infinite length.

The measure $|\sigma|^{-2}\mathrm{d}V_Y$ has a nonintegrable pole along D with smooth strictly positive coefficient. The solution is unbounded near D and therefore does not have minimal singularities: since θ is semipositive, the envelope V_θ is zero.

The solution will be a radial function on $\mathbb{C}^2 \setminus \{0\}$, identified with $Y \setminus (E \cup D)$. Its current is $\mathrm{dd^c}F(\log|z|^2)$ for a convex function F whose derivative increases from 0 at E to 1 at D, and the volume form $\mathrm{d}V_Y$ is defined from F. The following lemma constructs the profile F; its two end identities make the volume form smooth and positive across E and D.

Lemma 29.14.1 *There is $F \in C^\infty(\mathbb{R})$ with $0 < p := F' < 1$ and $p' > 0$ such that, for $c = 1/100$, some $B \in \mathbb{R}$ and $K = (9c^2/8)\mathrm{e}^{-B} > 0$,*

$$F(t) = B - 3\log(1 - c\mathrm{e}^{t/2}), \qquad pp' = K\mathrm{e}^{F+t} \qquad (t \le -2), \tag{29.38}$$

$$F(t) = t + \log H(p(t)), \qquad pp' = \mathrm{e}^{F-t} \qquad (t \ge 2), \tag{29.39}$$

where $H(p) = (1-p)^2(1+2p)/6$. At the two ends,

$$F(t) = B + 3c\mathrm{e}^{t/2} + O(\mathrm{e}^t), \qquad p'(t) \sim (3c/4)\mathrm{e}^{t/2} \qquad (t \to -\infty),$$

$$1 - p(t) \sim 2/t, \qquad p'(t) \sim 2/t^2 \qquad (t \to +\infty),$$

and $F(t) - t = -2\log t + \log 2 + o(1)$ as $t \to +\infty$.

Proof For $t \le -2$ put $L(t) = -3\log(1 - c\mathrm{e}^{t/2})$. With $q = c\mathrm{e}^{t/2}$,

$$p_L = L' = \frac{3q}{2(1-q)}, \qquad p_L' = \frac{3q}{4(1-q)^2}, \qquad p_L p_L' = \frac{9c^2}{8}\mathrm{e}^t(1 - c\mathrm{e}^{t/2})^{-3}.$$

This proves the left identity for every additive constant B.

The increasing function

$$\mathcal{T}(p) = 2 + \int_{3/4}^{p} \frac{q}{H(q)}\,\mathrm{d}q$$

tends to $+\infty$ as $p \to 1$, because its integrand is asymptotic to $2/(1-p)^2$ there. Hence it has an inverse $p_R(t)$, defined on an open interval containing $[2, \infty)$, with $p_R(2) = 3/4$ and $p_R' = H(p_R)/p_R$. Since $H'(p) = (p-1)p$, the function $F_R = t + \log H(p_R)$ has derivative $1 + (p_R - 1)p_R p_R'/H(p_R) = p_R$ and satisfies the right identity. For $q = 1 - p_R$,

$$(1/q)' = \frac{3 - 2q}{6(1-q)} \longrightarrow \frac{1}{2}.$$

Hence $q \sim 2/t$ and $H(p_R) \sim 2/t^2$, and the asymptotics at the right end follow.

It remains to interpolate the derivatives. Set $d = p_R(2) - p_L(-2) > 0$. Choose disjoint short collars in $[-2, 2]$ at the two endpoints, and cutoffs α_L, α_R equal to one on smaller collars. For small $\varepsilon > 0$, the function

$$Q_0 = \alpha_L p_L' + \alpha_R p_R' + \varepsilon(1 - \alpha_L - \alpha_R)$$

is positive and agrees with the derivative germs at the endpoints. The collars and ε can be chosen so that $\int_{-2}^{2} Q_0 < d$. Adding a nonnegative smooth bump in the interior, with integral $d - \int_{-2}^{2} Q_0$, gives a positive function Q, and we put

$$p(t) = p_L(-2) + \int_{-2}^{t} Q(x)\,\mathrm{d}x.$$

This function matches both endpoint germs, is increasing and takes values in $(0,1)$. Integrating $F' = p$ with $F(2) = F_R(2)$ gives the required F and fixes the constant B at the left end. The left expansion follows directly. □

Proof (Proof of Theorem 29.14.2) *The background and the pole.* Use the incidence surface from the proof of Theorem 29.14.1, with blowup map $\pi\colon Y \to \mathbb{P}^2$ and exceptional curve E over $[1:0:0]$, and let β, γ be the pulled-back Fubini–Study forms as there. Put $D = \pi^{-1}(Z_0 = 0)$ and $A = \pi^* c_1(\mathcal{O}(1))$. Then $E \cap D = \varnothing$, $A^2 = 1$ and $A \cdot E = 0$. The class A is nef, since it has the smooth semipositive representative β, and $A \cdot E = 0$ places E in its null locus. Moreover $\{\gamma\} = A - [E]$, and $\beta + \varepsilon\gamma$ is Kähler for $\varepsilon > 0$. Thus

$$\frac{\beta + \varepsilon\gamma}{1+\varepsilon} + \frac{\varepsilon}{1+\varepsilon}[E]$$

is a Kähler current in A, which proves bigness.

Choose an even nonnegative smooth function ζ, supported in $[-1,1]$, positive on its interior and of integral one. The smooth convex function

$$U(t) = \int_{\mathbb{R}} \max(0, t-x)\zeta(x)\,\mathrm{d}x$$

satisfies $U = 0$ for $t \le -1$, $U = t$ for $t \ge 1$, and $U', U'' \ge 0$. On $Z_0 \neq 0$, write $z = (Z_1/Z_0, Z_2/Z_0)$, $s = |z|^2$ and $t = \log s$, and give the section Z_0 of $\mathcal{O}(1)$ the squared norm $\mathrm{e}^{-U(t)}$; this norm is constant near the affine origin. Near the line at infinity, use

$$w = Z_0/Z_1, \qquad v = Z_2/Z_1, \qquad h = 1 + |v|^2, \qquad s = h/|w|^2.$$

The section Z_1 then has squared norm $\mathrm{e}^{-U}/|w|^2 = 1/h$ near $w = 0$. These formulas define a smooth Hermitian metric on $\mathcal{O}(1)$. Pull it back to $\mathcal{O}_Y(D)$ and let $\sigma = \pi^* Z_0$. Its curvature form θ and the norm of σ are

$$\theta = \mathrm{dd^c} U(t), \quad |\sigma|^2 = \mathrm{e}^{-U(t)}, \qquad \theta = \mathrm{dd^c}\log h, \quad |\sigma|^2 = |w|^2/h \quad \text{near } D.$$

The radial Hessian shows that θ is semipositive. It vanishes near E and has rank one near D. In particular $V_\theta = 0$, and the nef class has volume $A^2 = 1$.

Extension of the potential. On $Y \setminus (E \cup D) \simeq \mathbb{C}^2 \setminus \{0\}$, set

$$\rho = F(t) - U(t), \qquad \Omega = \mathrm{dd^c} F(t),$$

so that $\theta + \mathrm{dd^c}\rho = \Omega$. The eigenvalues of the Hermitian matrix of Ω are p/s and p'/s, which are positive. Extend ρ to E by B and to D by $-\infty$.

Near the affine origin, the smooth functions $F(\log(|z|^2 + \varepsilon))$ are plurisubharmonic and decrease to the continuous extension of $F(\log|z|^2)$ as $\varepsilon \downarrow 0$. Indeed, their two Hessian eigenvalues are $p/(s+\varepsilon)$ and $(sp' + \varepsilon p)/(s+\varepsilon)^2$, with p, p' evaluated at $\log(s+\varepsilon)$. Since $\theta = 0$ there, pulling back to Y proves positivity near E.

Near D put $t_\varepsilon = \log h - \log(|w|^2 + \varepsilon)$ and $g(t) = F(t) - t$. The local potentials $g(t_\varepsilon) + \log h$ decrease to $\rho + \log h$, and their Hessians are

$$p(t_\varepsilon)\mathrm{dd^c}\log h + (1 - p(t_\varepsilon))\mathrm{dd^c}\log(|w|^2 + \varepsilon) + p'(t_\varepsilon)\frac{\mathrm{i}}{2\pi}\partial t_\varepsilon \wedge \bar\partial t_\varepsilon \geq 0.$$

Thus ρ is θ-plurisubharmonic on Y. Since ρ is bounded near E and grows like a double logarithm near D, all its Lelong numbers vanish.

The volume form and full mass. With Euclidean area $\mathrm{d}A$ in each complex variable, the radial determinant is

$$\Omega^2 = \frac{2}{\pi^2}\frac{pp'}{s^2}\mathrm{d}A_{z_1}\mathrm{d}A_{z_2}. \tag{29.40}$$

Define $\mathrm{d}V_Y = \mathrm{e}^{-F}\Omega^2$ off the two divisors. In an exceptional chart $z = (u, uv)$, the real Jacobian is $|u|^2$ and $s = |u|^2 h$, so the left profile identity gives

$$\mathrm{d}V_Y = \frac{2K}{\pi^2 h}\,\mathrm{d}A_u\mathrm{d}A_v.$$

At infinity, the map $z = (1/w, v/w)$ has real Jacobian $|w|^{-6}$, and the right profile identity gives

$$\mathrm{d}V_Y = \frac{2}{\pi^2 h^3}\,\mathrm{d}A_w\mathrm{d}A_v, \qquad |\sigma|^{-2}\mathrm{d}V_Y = \frac{2}{\pi^2|w|^2h^2}\,\mathrm{d}A_w\mathrm{d}A_v.$$

The analogous second charts cover each divisor. The extensions are smooth and strictly positive, and they agree on overlaps because they agree on the punctured overlaps. The last expression exhibits the nonintegrable pole with smooth positive coefficient.

On the complement of the two divisors, $\mathrm{e}^{\rho}|\sigma|^{-2}\mathrm{d}V_Y = \Omega^2$; we extend this density by zero on D. Both measures give zero mass to the two divisors. In polar coordinates on $\mathbb{C}^2$, the Euclidean volume is $\pi^2 s\,\mathrm{d}s$ times the normalized angular measure, so

$$\int_Y \langle\Omega^2\rangle = \int_{-\infty}^{\infty} 2pp'\,\mathrm{d}t = p(+\infty)^2 - p(-\infty)^2 = 1.$$

This proves the global equation and shows that ρ has full mass.

Bounded truncations and energy. Near D write $\alpha = \mathrm{dd^c}t = \mathrm{dd^c}\log h$ and $\beta_t = (\mathrm{i}/2\pi)\partial t \wedge \bar\partial t$. These forms have rank one, and

$$\theta = \alpha, \quad \Omega = p\alpha + p'\beta_t, \quad \Omega^2 = 2pp'\alpha \wedge \beta_t, \quad \Omega \wedge \theta = p'\alpha \wedge \beta_t, \quad \theta^2 = 0.$$

Integration of $\alpha \wedge \beta_t$ over the angular and projective variables gives $\mathrm{d}t$. Since $|\rho| = O(\log t)$ and $p' = O(t^{-2})$, the integrals of $|\rho|$ against Ω^2 and $\Omega \wedge \theta$ near D are finite. Near E the potential is bounded and $\theta = 0$, and the remaining region is compact with smooth data.

Choose M so that $\phi = \rho - M \leq 0$, and set $\phi_k = \max(\phi, -k)$. For large k the truncation changes ϕ only in the cusp near D, where $\phi = \log H(p(t)) - M$ is strictly

decreasing. If $\phi(T_k) = -k$, then $k = 2\log T_k + O(1)$. On $\mathbb{C}^2 \setminus \{0\}$, the radial local potential of $\theta + \mathrm{dd^c}\phi_k$ is

$$G_k(t) = \max(F(t) - M, t - k).$$

Near D, in the frame $\pi^* Z_1$, the local potential of $\theta + \mathrm{dd^c}\phi_k$ is $G_k(t) + \log|w|^2$, which equals the smooth function $\log h - k$ for $t \geq T_k$; the term $\log|w|^2$ cancels the divisorial part of $\mathrm{dd^c} G_k(t)$. At $t = T_k$, the derivative of G_k jumps upward from $p(T_k)$ to 1.

Convolving G_k in t with an even nonnegative smooth kernel gives convex functions decreasing to G_k as the scale decreases. The corresponding bounded local potentials converge uniformly on compact sets, so their Bedford–Taylor products converge. Integrating the smooth radial measures through the transition shows that the transition carries mass $1-p(T_k)^2$ for $(\theta+\mathrm{dd^c}\phi_k)^2$ and mass $1-p(T_k)$ for $(\theta+\mathrm{dd^c}\phi_k)\wedge\theta$. Beyond the transition, G_k is linear and both top-degree measures vanish. Since $\phi_k = -k$ on the transition, its energy contributions have absolute values $k(1 - p(T_k)^2)$ and $k(1 - p(T_k))$, and both tend to zero because $1 - p(T_k) \sim 2/T_k$. The contributions of the remaining region converge absolutely by the bounds above. At E, bounded potentials give no mass to the curve, in accordance with the vanishing radial boundary mass $p(-\infty)^2 = 0$. Consequently

$$E_\theta(\phi_k) = \frac{1}{3}\sum_{j=0}^{2} \int_Y \phi_k (\theta + \mathrm{dd^c}\phi_k)^j \wedge \theta^{2-j}$$

has a finite limit, and in particular is bounded below. This proves $\rho \in \mathcal{E}^1(Y, \theta)$.

Uniqueness follows from the full-mass comparison argument in the proof of Theorem 29.14.1, now with $\mathrm{d}\mu = |\sigma|^{-2}\mathrm{d}V_Y$ off D, extended by zero on D: for a second solution $u \in \mathcal{E}^1(Y, \theta)$, the measures $\mathrm{e}^{\rho}\,\mathrm{d}\mu$ and $\mathrm{e}^{u}\,\mathrm{d}\mu$ are Monge–Ampère measures of mass 1.

The two metric ends. Fix $v = v_0$ in an exceptional chart and write $h_0 = 1 + |v_0|^2$. The restriction of Ω to the normal line $\{v = v_0\}$ is

$$\Omega|_{v=v_0} = \frac{\mathrm{i}}{2\pi}\,\frac{p'(\log(|u|^2 h_0))}{|u|^2}\,\mathrm{d}u \wedge \mathrm{d}\bar{u}.$$

Its coefficient is asymptotic to $(3c\sqrt{h_0}/4)|u|^{-1}$, so the current has no smooth extension at any point of E. The radial path $u = r > 0$ has finite remaining length, bounded by $C\int_0^{r_0} r^{-1/2}\,\mathrm{d}r$. At D the normal coefficient is asymptotic to $2/(|w|^2 t^2)$ and also diverges. Thus every open set on which Ω is smooth and strictly positive misses both curves.

Let W be a dense Zariski open set on which Ω is smooth and strictly positive. Its complement $Y \setminus W$ is a proper analytic set containing E, and by local finiteness of irreducible components and compactness it has finitely many components, which are curves or points. Each curve other than E meets E in finitely many points. Choose a point of E that avoids all these intersections and all point components. In a sufficiently

small chart around it, the complement of W is exactly E, so a tail of the radial path above lies in W. Its points approaching E form a Cauchy sequence for the intrinsic metric with no limit in W: the smooth positive metric induces the manifold topology of W, and the only ambient limit is the removed point. Hence W is incomplete.

Finally, on the cusp $\Omega = p\,\mathrm{dd^c}t + p'\,\mathrm{d}t \wedge \mathrm{d^c}t$, both terms are nonnegative, and $p' \geq c_0/t^2$ for large t and some $c_0 > 0$. The length of any path with $t \to \infty$ is therefore bounded below by a positive multiple of $\int |\mathrm{d}t|/t$, which diverges. Thus the end at finite distance is the null curve E, disjoint from the pole. □

29.14.2 An incomplete metric with compatible flow data

Exponential equations of this kind arise as auxiliary elliptic equations in the smoothing of singular initial data by geometric flows; the completeness question for them is discussed in Section 36.17. In that setting the class, the divisor and the volume form are not arbitrary: they are determined by the initial potential through a log resolution. The following example shows that incompleteness persists under these compatibility conditions.

Let (X, ω) be a compact Kähler manifold and $\phi_0 \in \mathrm{PSH}(X, \omega)$ an initial potential with analytic singularities. *Compatible data* on a log resolution $\pi\colon Y \to X$ satisfy

$$\pi^*\phi_0 = \sum_j m_j \log|\sigma_j|^2 + \ell_0, \qquad \pi^*(\omega + \mathrm{dd^c}\phi_0) = \sum_j m_j [D_j] + R_0, \tag{29.41}$$

$$K_Y = \pi^*K_X + \sum_j b_j D_j, \qquad \pi^*\mathrm{d}V_X = \prod_j |\sigma_j|^{2b_j}\mathrm{d}V_Y. \tag{29.42}$$

Here $D = \sum_j D_j$ is a divisor with simple normal crossings, σ_j is a defining section of $\mathcal{O}_Y(D_j)$ with a smooth Hermitian metric, and the numbers m_j and b_j are the coefficients of the pulled-back singularity and of the relative canonical divisor. Moreover, ℓ_0 is smooth, R_0 is a smooth closed semipositive form, and $\mathrm{d}V_X$, $\mathrm{d}V_Y$ are smooth positive volume forms. If Θ_j denotes the curvature form of the metric on $\mathcal{O}_Y(D_j)$, put

$$\lambda = \max_j \frac{m_j}{1 + b_j}, \qquad \Theta = \sum_j (1 + b_j)\Theta_j, \qquad \theta = \lambda\Theta + R_0, \qquad |\sigma|^2 = \prod_j |\sigma_j|^2.$$

The number λ is the inverse of the exponential integrability index of ϕ_0: by (29.41) and (29.42), $\mathrm{e}^{-q\phi_0}$ is integrable exactly when $q\lambda < 1$. The metric in the next theorem is the solution of the auxiliary elliptic equation (29.43) in this background.

Theorem 29.14.3 *There are compatible data* (29.41)–(29.42) *on smooth projective surfaces, with* $\mathrm{d}V_X = \omega^2$, R_0 *Kähler and* D *a disjoint union of two smooth curves, such that the equation*

$$\langle(\theta + \mathrm{dd}^{\mathrm{c}}\rho)^2\rangle = \mathrm{e}^{\rho}|\sigma|^{-2}\,\mathrm{d}V_Y \tag{29.43}$$

has a unique solution in $\mathcal{E}^1(Y,\theta)$. *The solution has the Lelong numbers of a potential with minimal singularities, but every dense Zariski open set on which its current is smooth and strictly positive is incomplete. The measure* $|\sigma|^{-2}\mathrm{d}V_Y$ *has a nonintegrable pole with smooth strictly positive coefficient along both components of* D.

The class $\{\theta\}$ in this example is not nef. Thus this theorem and Theorem 29.14.2 show separately that neither the compatibility identities nor nefness of the class implies completeness. We do not know whether completeness holds when the class is nef and the compatibility identities hold. In neither example is the auxiliary metric identified with the metric of the maximal flow.

The cusp in the example comes from a rotation-invariant solution of an exponential equation on $\mathbb{P}^1$ whose measure has a simple pole.

Lemma 29.14.2 *On* $\mathbb{P}^1$, *let* $z = Z_1/Z_0$, $r = |z|^2$, $\alpha = \mathrm{dd}^{\mathrm{c}}\log(1+r)$ *and* $h = r/(1+r)$, *so that* $\int_{\mathbb{P}^1}\alpha = 1$. *For every* $c > 2$ *there is a* $c\alpha$*-plurisubharmonic function* u, *depending only on* $|z|$ *and smooth off* 0, *such that*

$$g_1 := c\alpha + \mathrm{dd}^{\mathrm{c}}u = \mathrm{e}^{u}h^{-1}c\alpha. \tag{29.44}$$

It has zero Lelong number at 0, *mass* $\int g_1 = c$, *and*

$$\int_{\mathbb{P}^1}|u|\alpha + \int_{\mathbb{P}^1}|u|g_1 < \infty.$$

Moreover, with $t = \log r$ *and* $s = -\log r$, *there is* $A \in \mathbb{R}$ *such that, as* $s \to \infty$,

$$u = \log(2/c) - 2\log(s+A) + O(\mathrm{e}^{-s}),$$
$$u_t = \frac{2}{s+A} + O(\mathrm{e}^{-s}), \qquad u_{tt} = \frac{2}{(s+A)^2} + O(\mathrm{e}^{-s}).$$

Proof *Barriers and the radial inverse.* Set $L = 1 - \log h = 1 + \log(1 + 1/r)$ and $w = -2\log L$. Differentiation gives

$$\mathrm{dd}^{\mathrm{c}}w = \left(-\frac{2}{L} + \frac{2}{rL^2}\right)\alpha, \qquad Q(r) := \frac{c\alpha + \mathrm{dd}^{\mathrm{c}}w}{\mathrm{e}^{w}h^{-1}c\alpha} = \frac{crL^2 - 2rL + 2}{c(1+r)}.$$

The function Q is positive, since $L \geq 1$ and $c > 2$. Its limits are $2/c$ at 0 and $1 - 2/c$ at infinity, so $0 < q_- \leq Q \leq q_+ < \infty$ for some constants $q_\pm$. Choose $a_0 \leq \min(0, \log q_-)$ and $b_0 \geq \max(0, \log q_+)$. Then $l = w + a_0$ and $H = w + b_0$ satisfy

$$c\alpha + \mathrm{dd}^{\mathrm{c}}l \geq \mathrm{e}^{l}h^{-1}c\alpha, \qquad c\alpha + \mathrm{dd}^{\mathrm{c}}H \leq \mathrm{e}^{H}h^{-1}c\alpha, \qquad l \leq w \leq H.$$

Exhaust $\mathbb{P}^1 \setminus \{0\}$ by the disks $|\zeta| < R$ in the coordinate $\zeta = 1/z$, and solve the equation on them with boundary value w. For $q = |\zeta|^2$, the equation becomes

$$\Delta u = f(q)\mathrm{e}^u - g(q), \qquad f(q) = \frac{4c}{1+q}, \qquad g(q) = \frac{4c}{(1+q)^2},$$

where Δ is the Euclidean real Laplacian.

For $k > 0$, the regular radial solution $v = V(q)$ of $(-\Delta + k)v = F(q)$ satisfies

$$V(q) = A_0 + \frac{1}{4}\int_0^q \log(q/t)\big(kV(t) - F(t)\big)\,\mathrm{d}t. \tag{29.45}$$

The j-fold iterate of the integral kernel applied to the constant 1 is $q^j/(4^j(j!)^2)$, so successive approximation converges uniformly on every finite interval. The coefficient of A_0 in the solution is $\sum_{j\ge0}(kq/4)^j/(j!)^2 > 0$, so exactly one value of A_0 gives any prescribed value at $q = R^2$. Moreover,

$$V'(q) = \frac{1}{4}\int_0^1 \big(kV(qt) - F(qt)\big)\,\mathrm{d}t,$$

and this formula gives smoothness at $q = 0$ by induction. The maximum principle for $-\Delta + k$ shows that this radial inverse is unique and preserves order.

Choose $k \ge \sup f\mathrm{e}^H$ on the fixed disk, and let $T(v)$ be the radial solution of $(-\Delta + k)T(v) = kv - f\mathrm{e}^v + g$ with boundary value w; the right side is nondecreasing in v on $[l, H]$. The barrier inequalities give $l \le T(l)$ and $T(H) \le H$, so the iterates of T starting from l increase and stay below H. The derivative formula gives a uniform bound for their first derivatives on the fixed interval. Compactness and monotonicity give uniform convergence; passing to the limit in (29.45) proves the equation, and the averaged derivative formula bootstraps the limit to smoothness.

The same barriers bound the solutions on every fixed disk in ζ as $R \to \infty$. For an exact solution $U(q)$ of the nonlinear equation,

$$U'(q) = \frac{1}{4}\int_0^1 \big(f(qt)\mathrm{e}^{U(qt)} - g(qt)\big)\,\mathrm{d}t,$$

and induction gives bounds for all derivatives on compact intervals. A diagonal limit is therefore smooth on the ζ-plane, including the point $z = \infty$, and satisfies $u = -2\log(-\log r) + O(1)$ at 0.

The asymptotics, mass and energy. In the variable $t = \log r$, the equation reads

$$u_{tt} = \frac{c\mathrm{e}^u}{1+\mathrm{e}^t} - \frac{c\mathrm{e}^t}{(1+\mathrm{e}^t)^2}.$$

Put $v(t) = u(t) + c\log(1+\mathrm{e}^t)$ and $V(s) = v(-s)$. Then

$$V'' = k_0(s)\mathrm{e}^V, \qquad k_0(s) = c(1+\mathrm{e}^{-s})^{-(c+1)}, \qquad V = -2\log s + O(1).$$

Convexity and this sublinear decay imply $V' < 0$ and $V' \to 0$ at infinity. The function $J = (V')^2 - 2k_0 \mathrm{e}^V$ satisfies $J' = -2k_0' \mathrm{e}^V$ and tends to zero. Since $k_0' = O(\mathrm{e}^{-s})$ and $\mathrm{e}^V \asymp s^{-2}$,

$$J(s) = 2\int_s^\infty k_0'(\tau)\mathrm{e}^{V(\tau)}\,\mathrm{d}\tau = O(\mathrm{e}^{-s}s^{-2}).$$

Thus $(\mathrm{e}^{-V/2})' = \sqrt{c/2}(1 + O(\mathrm{e}^{-s}))$, and integration gives $\mathrm{e}^{-V/2} = \sqrt{c/2}(s + A) + O(\mathrm{e}^{-s})$. This identity and the differential equation give all three asymptotic formulas.

The local potential $u+c\log(1+r)$ extends subharmonically across 0 by Lemma 1.3.7. Its radial slope tends to zero at the puncture, so the extension adds no point mass. The density of g_1 is asymptotic to

$$\frac{\mathrm{i}}{2\pi}\frac{2}{rs^2}\,\mathrm{d}z \wedge \mathrm{d}\bar z.$$

Stokes' theorem for currents gives mass c. The energy integral at the cusp is bounded by $C\int^\infty (\log s)/s^2\,\mathrm{d}s$, and the integral against the smooth form α is finite as well. □

Proof (Proof of Theorem 29.14.3) *The initial data and the resolution.* Take $X = \mathbb{P}^1_Z \times \mathbb{P}^1_W$, with the pulled-back normalized Fubini–Study forms α, β of the two factors, and put

$$\omega = 6(\alpha + \beta), \qquad \mathrm{d}V_X = \omega^2 = 72\alpha \wedge \beta.$$

Let $H = (Z_1 = 0)$ and $p = ([0:1],[1:0]) = (\infty, 0)$. Define

$$h_H = \frac{|Z_1|^2}{|Z_0|^2 + |Z_1|^2}, \qquad \eta = \frac{|Z_0|^2}{|Z_0|^2 + |Z_1|^2} + \frac{|W_1|^2}{|W_0|^2 + |W_1|^2},$$
$$\phi_0 = 2\log h_H + 2\log\eta.$$

The zero sets of h_H and η are H and $\{p\}$, which are disjoint. Let $\pi\colon Y \to X$ be the blowup at p, write E for the exceptional curve, and set $H_Y = \pi^{-1}H$ and $D = H_Y + E$. The surface Y is smooth, compact, connected and projective, and π is an isomorphism off p.

Use $|\sigma_H|^2 = \pi^* h_H$ and $|\sigma_E|^2 = \pi^*\eta$. Near p, put $x = Z_0/Z_1$ and $y = W_1/W_0$. In the blowup chart $x = a$, $y = ab$,

$$\pi^*\eta = |a|^2 A(a,b), \qquad A(a,b) = \frac{1}{1+|a|^2} + \frac{|b|^2}{1+|ab|^2} > 0.$$

Thus $\pi^*\eta$ defines a smooth Hermitian metric on $\mathcal{O}_Y(E)$ across E. The second chart is obtained by interchanging x and y, and the metric factors satisfy the transition rule.

The sections Z_0W_0, $\sqrt{2}Z_0W_1$, Z_1W_1 of $\mathcal{O}(1,1)$ have summed squared Fubini–Study norm η, and near p they generate the ideal $(x, \sqrt{2}xy, y) = (x, y)$. Dividing their pullbacks by σ_E gives a morphism $F\colon Y \to \mathbb{P}^2$ with

$$F^*\gamma = \pi^*(\alpha + \beta) - \Theta_E, \qquad \Theta_H = \pi^*\alpha, \tag{29.46}$$

where γ is the normalized Fubini–Study form on $\mathbb{P}^2$. In the first chart $F = [1 : \sqrt{2}ab : b]$, so $F|_E$ is an isomorphism onto a line.

Near p,

$$\eta = \frac{|x|^2 + |y|^2 + 2|xy|^2}{(1+|x|^2)(1+|y|^2)}.$$

The numerator is $N = |x|^2 + |y|^2 + |\sqrt{2}\,xy|^2$, so ϕ_0 has analytic singularities with smooth remainder and coefficient 2 in the sense of Section 1.13. The resolved identities are

$$\begin{aligned} \pi^*\phi_0 &= 2\log|\sigma_H|^2 + 2\log|\sigma_E|^2, \\ R_0 &= \pi^*\omega - 2\Theta_H - 2\Theta_E = 2\pi^*\alpha + 4\pi^*\beta + 2F^*\gamma. \end{aligned} \tag{29.47}$$

The first two terms of R_0 are positive away from E, and along E their kernel is TE, on which $F^*\gamma$ is positive. Hence R_0 is Kähler. The resolved current is $2[H_Y] + 2[E] + R_0$, which shows that ϕ_0 is ω-plurisubharmonic on $X \setminus \{p\}$. Its value $-\infty$ at p is also its upper limit there, so $\phi_0 \in \mathrm{PSH}(X,\omega)$ by Lemma 1.3.7. In the notation of (29.41), $m_H = m_E = 2$ and $\ell_0 = 0$.

The Jacobian determinant of the blowup in the chart is a, so $b_E = 1$, $b_H = 0$, $\lambda = \max\{2,1\} = 2$, and

$$\Theta = \Theta_H + 2\Theta_E, \qquad \theta = 2\Theta + R_0 = \pi^*\omega + 2\Theta_E. \tag{29.48}$$

By (29.46), $\Theta_E|_E = -F^*\gamma|_E$, and $F|_E$ is an isomorphism onto a line, so $\int_E \Theta_E = -1$. Hence the class $\{\theta\}$ has degree -2 on E, so it is not nef. It is big, since the potential $3\log|\sigma_E|^2$ gives the current

$$(\pi^*\omega - \Theta_E) + 3[E], \qquad \pi^*\omega - \Theta_E = 5\pi^*(\alpha+\beta) + F^*\gamma > 0.$$

The prescribed volume. Define $\mathrm{d}V_Y = \pi^*\omega^2/|\sigma_E|^2$ off E. In the chart above,

$$\mathrm{d}V_Y = \frac{72\,\mathrm{d}A_a\mathrm{d}A_b}{\pi^2 A(a,b)(1+|a|^2)^2(1+|ab|^2)^2},$$

since the factor $|a|^2$ from the Jacobian cancels exactly. The second chart gives the extension on the rest of E, and the extensions agree on overlaps. Thus $\mathrm{d}V_Y$ is a smooth strictly positive volume form with $\pi^*\mathrm{d}V_X = |\sigma_E|^2\mathrm{d}V_Y$, as required in (29.42). Since H_Y and E are disjoint, the measure $|\sigma_H\sigma_E|^{-2}\mathrm{d}V_Y$ has a nonintegrable pole with smooth positive coefficient along each component.

The solution and the minimal reference. Take u from Lemma 29.14.2 with $c = 6$, pulled back to X from the first factor, and put $g_X = (6\alpha + \mathrm{dd}^{\mathrm{c}}u) + 6\beta$. This form is smooth and strictly positive near p, and

$$g_X^2 = \mathrm{e}^u h_H^{-1}\omega^2.$$

Set $P = 2\log|\sigma_E|^2$ and $\rho = P + \pi^*u$. Then

$$\theta + \mathrm{dd^c}P = 2[E] + \pi^*\omega, \qquad \theta + \mathrm{dd^c}\rho = 2[E] + \pi^*g_X.$$

Off D, the equation follows from the identity

$$\mathrm{e}^{\rho}|\sigma_H\sigma_E|^{-2}\mathrm{d}V_Y = \mathrm{e}^{\pi^*u}|\sigma_H|^{-2}|\sigma_E|^2\mathrm{d}V_Y = \pi^*g_X^2.$$

Both measures extend by zero on D, which proves the global non-pluripolar equation.

Let ψ be θ-plurisubharmonic. Off E, the function $\psi - P$ descends to an ω-plurisubharmonic function on $X \setminus \{p\}$, which by Lemma 1.3.8 extends across p and is bounded above on X. Thus $\psi \le P + C_\psi$ off E, and Lemma 1.3.3 gives the inequality everywhere. Since P itself is θ-plurisubharmonic, it has minimal singularities.

The current of P has non-pluripolar mass $\int_X \omega^2 = 72$, which is therefore the volume of the class, whereas the cup-product square of the class is $72 + 4E^2 = 68$. The cusp has mass 6 and no atom, so the current of ρ also has mass $\int_X g_X^2 = 72$, and ρ has full mass. Near E, π^*u is smooth, so ρ has the Lelong numbers of P; along H_Y the cusp has zero Lelong number. The solution ρ does not itself have minimal singularities, because u is unbounded below near H.

Relative energy and uniqueness. Choose M with $u - M \le 0$ and use the potential $P + M$, which has minimal singularities, as reference. The canonical approximants of ρ are

$$\rho_k = \max(\rho, P + M - k) = P + M + \pi^*u_k, \qquad u_k = \max(u - M, -k),$$

and their difference from the reference off E is π^*u_k. On the curve let $\kappa(t) = 6\log(1 + \mathrm{e}^t)$ and $v(t) = \kappa(t) + u(t)$; these are radial local potentials of 6α and of g_1, as in the proof of Lemma 29.14.2. For large k there is a unique level $t = T_k$ in the cusp with $u(T_k) - M = -k$. If $S_k = -T_k$, the asymptotics of Lemma 29.14.2 give $S_k \asymp \mathrm{e}^{k/2}$.

The local potential of the truncated curve current is $\kappa + u_k = \max(v - M, \kappa - k)$. On $\{t < T_k\}$ its mass is $\kappa'(T_k)$, and at $t = T_k$ its derivative jumps upward by $u_t(T_k)$. Smooth convex approximation and continuity of Bedford–Taylor products of bounded potentials therefore give total mass $v'(T_k) = O(S_k^{-1})$ on the cutoff region $\{t \le T_k\}$. There $u_k = -k$, so the absolute energy contribution of this region is

$$kv'(T_k) = O(k\mathrm{e}^{-k/2}) \longrightarrow 0.$$

Outside it the density and the potential are unchanged, and $\int |u|g_1$ is finite. Consequently $\int |u_k|g_{1,k}$ has a finite limit, where $g_{1,k} = 6\alpha + \mathrm{dd^c}u_k$, and $\int |u_k|\alpha$ converges by dominated convergence.

The mixed measures of the product current and of its reference are

$$\omega^2 = 72\alpha\wedge\beta, \quad (g_{1,k}+6\beta)\wedge\omega = 6g_{1,k}\wedge\beta + 36\alpha\wedge\beta, \quad (g_{1,k}+6\beta)^2 = 12g_{1,k}\wedge\beta.$$

Their pullbacks are exactly the mixed non-pluripolar measures on Y, which do not charge E. All three relative energy integrals of the canonical approximants therefore have finite limits. This proves $\rho \in \mathcal{E}^1(Y,\theta)$; replacing $P+M$ by the normalized envelope V_θ changes the reference by a bounded amount and does not affect this conclusion.

Uniqueness follows from the full-mass comparison argument in the proof of Theorem 29.14.1, applied to the measure $|\sigma|^{-2}\mathrm{d}V_Y$ extended by zero on D; the two exponential densities being compared are Monge–Ampère measures of mass 72.

The missing metric boundary. The set on which the current is smooth and strictly positive is $Y \setminus (H_Y \cup E)$: the cusp coefficient diverges at H_Y, and the nonzero current $2[E]$ excludes smoothness along E. The smooth part $\pi^* g_X$ also degenerates along TE.

Let U be a dense Zariski open set on which the current is smooth and strictly positive. As in the proof of Theorem 29.14.2, choose a point of E avoiding every other component of the analytic set $Y \setminus U$; in a small chart around it the complement of U is exactly E. A transverse path $a = \xi > 0$, $b = b_0$ lies in U for small ξ and approaches that point. Since g_X is smooth near p and π is smooth, the coefficients of $\pi^* g_X$ are bounded in the chart, so the path has bounded speed and finite remaining length. Its tails give an intrinsic Cauchy sequence with no limit in U, since its only ambient limit is the removed point and a smooth positive Riemannian metric induces the manifold topology. Hence every such U is incomplete. □

29.15 Normal length near a divisor

To prove completeness near a divisor, one needs a lower bound for the metric in the normal direction. A lower bound for the determinant alone does not give it, because the tangential eigenvalues may absorb the volume. Dividing the determinant by the determinant of the tangential block gives the normal Schur complement. A lower bound for it by the inverse square of a defining function of the divisor, weakened by at most the square of the logarithm of that function, forces completeness. For solutions of exponential equations, this bound follows from bounds on the density, the potential and the tangential determinant, and an example on a torus shows that the logarithmic exponent 2 is sharp.

In this section X is again a connected compact complex manifold of dimension $n \geq 1$ with a Hermitian form ω. Let $D \subset X$ be a nonempty smooth effective divisor, and put $M = X \setminus D$. Let s_D be the canonical section of $\mathcal{O}_X(D)$, equipped with a smooth Hermitian norm, and set

$$R = |s_D|, \qquad \tau = -\log R^2.$$

Rescale the norm so that $\tau \geq 1$ on M. Write J for the complex structure. For a smooth positive real $(1,1)$-form T on M, we use the Riemannian metric $g_T(v,v) = \pi T(v, Jv)$ on real tangent vectors v. Thus, if $T = (\mathrm{i}/2\pi)\sum H_{j\bar{k}}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$, then

$g_T(v,v) = \sum H_{j\bar{k}}\, \mathrm{d}z_j(v)\overline{\mathrm{d}z_k(v)}$, and Hermitian norms below use this normalization of coefficients.

Near D, the form $\partial\tau$ does not vanish (see (29.50) below), and we define the tangential hyperplane field, the tangential block and the determinant

$$E_\tau = \ker \partial\tau, \qquad B_\tau = T|_{E_\tau}, \qquad \det_\omega T = \frac{T^n}{\omega^n}.$$

Similarly, $\det_\omega B_\tau$ denotes the determinant of B_τ relative to the restriction of ω to the complex hyperplane E_τ; when $n = 1$, it is 1.

Proposition 29.15.1 *Suppose that there are constants $c > 0$, $\tau_0 \geq 1$ and $p \leq 2$ such that, on $\{\tau > \tau_0\}$,*

$$\frac{\det_\omega T}{\det_\omega B_\tau} \geq c R^{-2} \tau^{-p}. \tag{29.49}$$

Then (M, g_T) is complete.

Proof In a coordinate neighborhood meeting D, write $s_D = z_1 e$, where e is a local holomorphic frame with $|e|^2 = \mathrm{e}^{-\psi}$ and ψ is smooth. Then

$$\tau = -\log|z_1|^2 + \psi, \qquad \partial\tau = -\frac{\mathrm{d}z_1}{z_1} + \partial\psi.$$

Covering the compact divisor by finitely many such neighborhoods and comparing with the smooth background, we obtain, after increasing τ_0,

$$c_1 R^{-2} \leq |\partial\tau|^2_\omega \leq C_1 R^{-2} \quad \text{on } \{\tau > \tau_0\} \tag{29.50}$$

for positive constants c_1, C_1. In particular, E_τ is a hyperplane there.

At a point of this set, choose an ω-unitary frame whose last $n-1$ vectors span E_τ. Let H be the Hermitian matrix that represents g_T as the quadratic form $w^* H w$ in the complex components w, and write

$$H = \begin{pmatrix} h & a^* \\ a & B \end{pmatrix}, \qquad S = h - a^* B^{-1} a = \frac{\det H}{\det B},$$

where a is a column vector and a^* its conjugate transpose. Completing the square gives, for a vector with components (ξ, η),

$$\begin{pmatrix} \bar{\xi} & \eta^* \end{pmatrix} H \begin{pmatrix} \xi \\ \eta \end{pmatrix} = S|\xi|^2 + (\eta + B^{-1}a\xi)^* B (\eta + B^{-1}a\xi) \geq S|\xi|^2.$$

For $n = 1$ the same identity holds without the second term. For the complex components of a real vector v we have $|\xi| = |\partial\tau(v)|/|\partial\tau|_\omega$, so (29.49) and (29.50) imply

$$g_T(v,v) \geq c_2 \tau^{-p} |\partial\tau(v)|^2 \geq \frac{c_2}{4} \tau^{-p} |\mathrm{d}\tau(v)|^2, \tag{29.51}$$

where $c_2 > 0$ is uniform. The last inequality uses $\mathrm{d}\tau(v) = 2\,\mathrm{Re}\,\partial\tau(v)$.

Define a continuous function Φ_p on $[1,\infty)$ by

$$\Phi_p(t) = \begin{cases} 0, & t \le \tau_0, \\ \displaystyle\int_{\tau_0}^{t} x^{-p/2}\,\mathrm{d}x, & t > \tau_0. \end{cases}$$

Integrating (29.51) along piecewise smooth curves shows that $\Phi_p \circ \tau$ is Lipschitz for the length distance of g_T; this also holds for curves crossing $\{\tau = \tau_0\}$, since Φ_p is locally Lipschitz and constant below that level. Since $p \le 2$, $\Phi_p(t)$ tends to infinity with t, so $\Phi_p \circ \tau$ is a proper function on M.

A Cauchy sequence for the length distance has bounded values of this Lipschitz function and hence lies in a compact subset of M. It therefore has a subsequence converging in the manifold topology. On a coordinate ball about the limit, the smooth positive form T is comparable to the Euclidean metric, so this subsequence also converges for the length distance. The Cauchy property then gives convergence of the whole sequence, which proves completeness. □

Corollary 29.15.1 *In the preceding setting, let θ be a smooth real $(1,1)$-form on X, and let $u \in C^\infty(M)$ satisfy $T = \theta + \mathrm{dd^c}u > 0$ on M. Suppose that, for some $\lambda > 0$ and some positive $f \in C^\infty(M)$,*

$$T^n = \mathrm{e}^{\lambda u} f \omega^n \quad \text{on } M.$$

Assume that, near D, there are constants $c, C > 0$ and $q, a, b \in \mathbb{R}$ such that

$$f \ge cR^{-2}\tau^{-q}, \qquad u \ge -\frac{a}{\lambda}\log\tau - C, \qquad \det_\omega B_\tau \le C\tau^b. \tag{29.52}$$

If $q + a + b \le 2$, then (M, g_T) is complete.

Proof The equation and the first two inequalities give $\det_\omega T \ge c\mathrm{e}^{-\lambda C} R^{-2}\tau^{-(q+a)}$. Dividing by the third bound gives (29.49) with $p = q + a + b$, and Proposition 29.15.1 applies. □

The relevant tangential determinant is that of the intrinsic hyperplane field E_τ, not that of an arbitrary block of coordinate directions.

29.15.1 The critical exponent on a compact torus

The exponent 2 in Corollary 29.15.1 is sharp. On a product torus, two explicit logarithmic profiles give a complete metric with $q + a + b = 2$ and incomplete metrics with $q + a + b = 2 + \beta$ for every $\beta > 0$. Both profiles have full mass, and the second one is bounded.

Let $E = \mathbb{C}/(\mathbb{Z}+\mathrm{i}\mathbb{Z})$, let $\omega_E = (\mathrm{i}/2\pi)\,\mathrm{d}z \wedge \mathrm{d}\bar{z}$, and put $X = E^n$, $D = \{0\} \times E^{n-1}$ and $M = X \setminus D$. Write $\pi_j \colon X \to E$ for the coordinate projections and $\omega_0 = \sum_{j=1}^n \pi_j^* \omega_E$.

Choose radii $0 < r_1 < r_2 < \mathrm{e}^{-1}$ and a smooth radial cutoff χ on the coordinate disk $\{|z| < r_2\} \subset E$, equal to 1 on $\{|z| \le r_1\}$ and to 0 near its outer boundary, with $0 \le \chi \le 1$. For fixed numbers $A, \eta, \beta > 0$, set $L = -\log|z|^2$ and define on E

$$v_{\mathrm{c}} = -A\chi \log L, \qquad v_\beta = \eta\chi L^{-\beta}. \tag{29.53}$$

Both functions are extended by zero outside the disk, and at its center we set $v_{\mathrm{c}}(0) = -\infty$ and $v_\beta(0) = 0$. The bounded profile also occurs in Example 15.20.1. We write $\mathrm{MA}_\theta(u)$ for the extended non-pluripolar Monge–Ampère measure $\mathrm{MA}_\theta^{\mathrm{ext}}(u)$ of Theorem 7.1.1.

Lemma 29.15.1 *For $C > 0$ sufficiently large, put $\theta = C\omega_0$. For either profile v in (29.53), the function $u = \pi_1^* v$ belongs to $\mathrm{PSH}(X, \theta)$, the form $T = \theta + \mathrm{dd^c} u$ is smooth and positive on $X \setminus D$, and*

$$\int_X \mathrm{MA}_\theta(u) = \int_X \theta^n.$$

The same C may be used for the two fixed profiles. The bounded profile has minimal ordinary singularities; the profile v_{c} does not.

Proof For a twice differentiable function G of $L = -\log|z|^2$, differentiation away from zero gives

$$\frac{\partial^2 G(L)}{\partial z \partial \bar z} = \frac{G''(L)}{|z|^2}.$$

On the inner disk this yields

$$(v_{\mathrm{c}})_{z\bar z} = \frac{A}{|z|^2 L^2}, \qquad (v_\beta)_{z\bar z} = \frac{\eta\beta(\beta+1)}{|z|^2 L^{\beta+2}}. \tag{29.54}$$

Both densities are integrable at zero. Both profiles are locally integrable, and their distributional Hessians have no additional atom at zero. To see this, integrate twice by parts outside $\{|z| = \varepsilon\}$: the boundary terms against a smooth test function are bounded by constant multiples of $\varepsilon|G(-\log\varepsilon^2)|$ and $|G'(-\log\varepsilon^2)|$, and both tend to zero for the two profiles. Thus (29.54) holds in the sense of distributions across the center. On the cutoff annulus all derivatives are bounded. Hence, for a sufficiently large common C,

$$\alpha_v = C\omega_E + \mathrm{dd^c} v$$

is a positive measure, smooth and strictly positive away from zero, with no atom at zero. Pairing $\mathrm{dd^c} v$ with the constant test function gives $\int_E \alpha_v = C\int_E \omega_E$.

Let $Y = E^{n-1}$, let $\pi_Y \colon X \to Y$ be the projection onto the remaining factors, and let θ_Y be C times the sum of the $n-1$ flat forms of Y. On $X \setminus D$ the smooth wedge product is

$$T^n = n\,\pi_1^*\alpha_v \wedge \pi_Y^*\theta_Y^{n-1};$$

for $n = 1$ this means just α_v, with Y a point. The right side defines a finite product measure on X that gives D zero mass. By Theorem 7.1.1, the non-pluripolar measure

agrees with this product off D and gives the pluripolar set D zero mass, so the two global measures are equal. Integrating the product proves the full-mass identity. Since $\theta > 0$, its envelope is $V_\theta = 0$ (Section 1.19). The bounded profile therefore has minimal ordinary singularities, whereas v_{c} is not bounded below by 0 minus a constant. □

We apply the preceding results with $\omega = \theta$. Choose a smooth norm on $\mathcal{O}_E(0)$, the line bundle of the point $0 \in E$, for which its canonical section has squared norm $|z|^2$ near zero; such a norm is obtained from any smooth norm by multiplying by a smooth positive function prescribed on a smaller disk. Pull it back to X and rescale it by a constant. For some fixed $K > 0$ we may then arrange

$$R^2 = \mathrm{e}^{-K}|z_1|^2, \qquad \tau = L + K$$

near D, and $\tau \geq 1$ on M. For both profiles, E_τ is exactly the tangent space of the remaining torus factors near D, so $\det_\theta B_\tau = 1$.

Example 29.15.1 Fix $\lambda > 0$ and $0 < A < 1/\lambda$, and take $u_{\mathrm{c}} = \pi_1^* v_{\mathrm{c}}$ with the background of Lemma 29.15.1. This gives a complete metric on M solving

$$\mathrm{MA}_\theta(u_{\mathrm{c}}) = \mathrm{e}^{\lambda u_{\mathrm{c}}} f_{\mathrm{c}} \theta^n, \qquad f_{\mathrm{c}} \in C^\infty(M) \cap L^1(X, \theta^n), \quad f_{\mathrm{c}} > 0,$$

where the right side is extended by zero on D. The three exponents in (29.52) can be taken to be

$$q = 2 - \lambda A > 1, \qquad a = \lambda A, \qquad b = 0,$$

so their sum equals the critical exponent 2.

Indeed, define $f_{\mathrm{c}} = \mathrm{e}^{-\lambda u_{\mathrm{c}}} \det_\theta(\theta + \mathrm{dd}^{\mathrm{c}} u_{\mathrm{c}})$ on M. On the inner disk the product structure gives

$$f_{\mathrm{c}} = L^{\lambda A} + \frac{A}{C|z_1|^2 L^{2-\lambda A}}.$$

The substitution $L = -\log r^2$ in the radial integral shows that the second term is integrable precisely when $\lambda A < 1$. The first term is integrable as well, and the density is smooth elsewhere. The identity of measures across D follows from the preceding lemma. Since $\tau = L + K$, all three bounds hold with the stated exponents, and completeness follows from Corollary 29.15.1. Along a radial normal segment the divergence of the length at the critical exponent is also visible directly:

$$\int_0^{r_1} \frac{\sqrt{A}}{r(-\log r^2)}\, \mathrm{d}r = \infty.$$

For arbitrary paths, however, completeness requires the corollary, not just this family of radial paths.

Example 29.15.2 Fix $\lambda, \eta, \beta > 0$ and take $u_\beta = \pi_1^* v_\beta$ with the background of Lemma 29.15.1. This is a continuous bounded solution of full mass with minimal ordinary singularities, and its metric on M is incomplete, although

$$\mathrm{MA}_\theta(u_\beta) = \mathrm{e}^{\lambda u_\beta} f_\beta \theta^n, \qquad f_\beta \in C^\infty(M) \cap L^1(X, \theta^n), \quad f_\beta > 0.$$

To see this, put $K_\beta = \eta\beta(\beta+1)$ and define the density by the equation on M. Near D it is

$$f_\beta = \mathrm{e}^{-\lambda\eta L^{-\beta}} \left(1 + \frac{K_\beta}{C|z_1|^2 L^{\beta+2}}\right).$$

It is integrable, and the global identity follows as before. Fix the other coordinates and let $z_1 = r \downarrow 0$ along a radial segment starting at r_0, with $0 < r_0 < r_1$. The length of this segment is at most

$$\sqrt{C}\, r_0 + \sqrt{K_\beta} \int_0^{r_0} \frac{\mathrm{d}r}{r(-\log r^2)^{1+\beta/2}} = \sqrt{C}\, r_0 + \frac{\sqrt{K_\beta}}{\beta}(-\log r_0^2)^{-\beta/2} < \infty.$$

Points along this segment approaching D form a Cauchy sequence with no limit in M.

Here the smallest admissible exponents in (29.52) are $q = \beta + 2$, $a = 0$ and $b = 0$: the density is asymptotic to a positive constant times $R^{-2}\tau^{-(\beta+2)}$, the potential is bounded, and the tangential determinant is 1. Their sum is $2 + \beta$. Since $\beta > 0$ is arbitrary, the threshold 2 in the criterion cannot be replaced by any larger universal number, even in a Kähler class with a bounded full-mass solution.

In these examples one may write $f = F/R^2$ with $F = R^2 f$ smooth and positive on M, but F does not extend smoothly and positively across D: the logarithmic factors make it tend to zero there. Thus the examples concern integrable divisorial poles weakened by logarithmic factors. They do not decide completeness for a nonintegrable pole with smooth positive coefficient across the divisor, nor the question of compatibility with the flow discussed in Section 36.17.

29.16 Rigid boundary classes on a torus

We now turn to the boundary of the nef cone. A nef class with zero top intersection number may have a smooth semipositive representative and yet leave little freedom to vary it. On a complex torus, the kernel of a constant semipositive form controls this freedom: when the leaves of the kernel are dense, the form is the only closed positive current in its class. The same condition decides whether adding a small fixed Kähler form produces metrics whose diameters tend to zero. For Kähler forms converging to the boundary in other ways, the diameters can tend to zero even at a class with many semipositive representatives.

Let $X = V/\Lambda$ be a complex torus, where $V \simeq \mathbb{C}^n$ and Λ is a full lattice. Let θ be a translation-invariant semipositive real $(1,1)$-form of rank r, and fix a translation-

invariant Kähler form κ. The class $\alpha = \{\theta\}_{\mathrm{BC}}$ is nef. If $r < n$, then $\int_X \theta^n = 0$ and α lies on the boundary of the nef cone: a Kähler form in α would have positive top intersection number, whereas Stokes' theorem gives every representative the top intersection number $\int_X \theta^n = 0$. We call α *rigid* if it contains exactly one closed positive current. Let $K = \ker\theta \subset V$, a complex subspace of dimension $n - r$, and let $H \subset X$ be the closure of its image in X. This is a connected real subtorus, which need not be complex. Diameters are computed for the Riemannian metric $g_\eta(v, w) = \eta(v, Jw)$ associated with a positive form η, where J is the complex structure.

Theorem 29.16.1 *The following are equivalent:*

(1) α *is rigid;*
(2) $H = X$*;*
(3) $\operatorname{diam}(X, \theta + t\kappa) \to 0$ *as* $t \downarrow 0$.

In the first case the unique current is θ*. The third condition is independent of the chosen translation-invariant Kähler form* κ*.*

Proof *Rigidity when* $H = X$. Suppose first that $H = X$. By Lemma 1.17.4, a positive current in α can be written $T = \theta + \mathrm{dd^c} u$ with u quasi-plurisubharmonic, in particular $u \in L^1(X)$. For a constant real vector $v \in K$, the coefficient $T(v, Jv)$ is a positive Radon measure. Its total integral against Haar measure is zero, because $\theta(v, Jv) = 0$ and the remaining part of the coefficient is a combination of constant-coefficient derivatives of u. Hence this coefficient vanishes. Writing D_v for differentiation along v and using that K is complex, we obtain

$$(D_v^2 + D_{Jv}^2)u = 0 \quad \text{as a distribution for every } v \in K.$$

For a real linear functional ℓ with $\ell(\Lambda) \subset \mathbb{Z}$, let $\widehat{u}(\ell)$ be the Fourier coefficient of u at the character $\exp(2\pi i\ell)$. The identity above gives

$$\big(\ell(v)^2 + \ell(Jv)^2\big)\widehat{u}(\ell) = 0.$$

Since the image of K is dense, a nonzero character cannot vanish identically on K. Thus every nonconstant Fourier coefficient of u is zero, and convolution with a smooth approximate identity shows that u is constant. Hence $T = \theta$.

Nonrigidity when $H \neq X$. If $H \neq X$, a nontrivial character of the real torus X/H gives a nonzero ℓ that vanishes on K. Put $\phi = \cos(2\pi\ell)$. Since K is complex, the form $\mathrm{dd^c}\phi$ vanishes whenever one of its arguments lies in K. On a fixed complement of K, θ is positive definite and $\mathrm{dd^c}\phi$ is uniformly bounded. Consequently, the forms $\theta \pm \varepsilon\mathrm{dd^c}\phi$ are semipositive for sufficiently small $\varepsilon > 0$. They represent the same class, and they are distinct because a nonconstant smooth function on a compact torus cannot have zero $\mathrm{dd^c}$. Hence α is not rigid.

Collapse when $H = X$. Suppose again that $H = X$, and fix $\delta > 0$. By density and compactness, there are finitely many vectors $k_1, \ldots, k_N \in K$ whose images form a δ-net in the flat metric of κ. Put $R = \max_j |k_j|_\kappa$. Since $\theta(k_j, Jk_j) = 0$, the straight path from 0 to the image of k_j has $(\theta + t\kappa)$-length $\sqrt{t}\,|k_j|_\kappa$. For $0 < t \leq 1$ these

metrics are bounded above by a fixed multiple of κ. Joining this path to a point within δ therefore gives

$$\operatorname{diam}(X, \theta + t\kappa) \leq \sqrt{t}\, R + C\delta,$$

since by translation invariance the diameter equals the supremum of the distances from 0. Letting first $t \downarrow 0$ for this fixed net and then $\delta \downarrow 0$ proves the third condition.

No collapse when $H \neq X$. If $H \neq X$, choose ℓ as above. Its differential vanishes on K, so $|\mathrm{d}\ell(v)| \leq L|v|_\theta$ for some $L > 0$, where $|v|_\theta$ is the seminorm defined by θ. Since $\theta + t\kappa \geq \theta$, the surjective map $\ell \bmod \mathbb{Z}$ onto the circle of length one is L-Lipschitz for every $t > 0$. Two points mapping to opposite points of the circle therefore have distance at least $1/(2L)$, and the diameters cannot tend to zero. This completes the proof of the equivalences. Since the second condition does not involve κ, neither does the third. □

On a two-dimensional square torus, the slope of the kernel line decides rigidity.

Example 29.16.1 On $X = (\mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z}))^2$, for $a \in \mathbb{R}$ set

$$\theta_a = \frac{\mathrm{i}}{2}(\mathrm{d}z_2 - a\,\mathrm{d}z_1) \wedge (\mathrm{d}\bar{z}_2 - a\,\mathrm{d}\bar{z}_1).$$

Its class is rigid exactly when a is irrational. In that case its ray is irrational, that is, no nonzero real multiple of its de Rham class is rational, and every family $\theta_a + t\kappa$ as above collapses to a point in diameter. For rational a the class is not rigid, and these diameters have a positive lower bound.

Proof The kernel is the complex line $z_2 = az_1$. Write $z_j = x_j + \mathrm{i}y_j$. An integral character has the form

$$\ell = m_1x_1 + m_2x_2 + n_1y_1 + n_2y_2, \qquad m_j, n_j \in \mathbb{Z},$$

and it vanishes on the kernel exactly when $m_1 + am_2 = n_1 + an_2 = 0$. For irrational a, only the zero character does so, hence the image of the kernel is dense. For rational a, a nonzero such character exists. Now apply Theorem 29.16.1. The periods of θ_a on the coordinate two-tori in the (x_2, y_2) and (x_1, y_2) directions are 1 and $-a$, respectively. If a nonzero multiple of its class were rational, the first period would make that multiple rational, and the second would then make a rational. This proves the assertion about the ray. □

The fixed positive direction κ in the theorem cannot be replaced by an arbitrary convergent family of Kähler forms, even when all the metrics are flat and Ricci-flat.

Example 29.16.2 On the same square torus let $\theta_0 = \frac{\mathrm{i}}{2}\mathrm{d}z_2 \wedge \mathrm{d}\bar{z}_2$ and, for $t > 0$, put

$$\beta_t = \frac{\mathrm{i}}{2}\left[(t\,\mathrm{d}z_1 + \mathrm{d}z_2) \wedge (t\,\mathrm{d}\bar{z}_1 + \mathrm{d}\bar{z}_2) + t^2 \mathrm{d}z_2 \wedge \mathrm{d}\bar{z}_2\right].$$

These are flat Ricci-flat Kähler forms, $\beta_t \to \theta_0$ smoothly, and

$$\operatorname{diam}(X, \beta_t) \le t.$$

The limiting class $\{\theta_0\}_{\mathrm{BC}}$ is not rigid.

Proof The coefficient matrix of β_t is $\left(\begin{smallmatrix} t^2 & t \\ t & 1+t^2 \end{smallmatrix}\right)$, with determinant $t^4 > 0$. Its coefficients are constant, which gives flatness and Ricci-flatness. Represent a point by (z_1, z_2) with real and imaginary parts in $[-1/2, 1/2]$, and choose a Gaussian integer k nearest to $-z_1 - z_2/t$. The displacement $(z_1 + k, z_2)$ represents the same point of the torus and has squared metric norm

$$|t(z_1 + k) + z_2|^2 + t^2|z_2|^2 \le t^2/2 + t^2/2 = t^2.$$

The straight path along this displacement proves the bound for the distance from zero, and translation invariance gives the diameter bound. For $0 < |s| < 1/\pi$, the form $\theta_0 + \mathrm{dd}^{\mathrm{c}}(s\cos(2\pi x_2)) = (1 - \pi s\cos(2\pi x_2))\theta_0$ is another semipositive representative, so the limiting class is not rigid. Finally, the coefficient matrix of $\beta_t - \theta_0$ has eigenvalues $t^2 \pm t$, so this difference is indefinite for $0 < t < 1$: the forms β_t do not approach θ_0 from above. □

29.17 Nonrigid classes on a quotient resolution

In contrast with Example 29.16.1, an irrational nef boundary ray can admit many smooth semipositive representatives. On a suitable two-dimensional torus, such representatives come from semipositive forms depending on a single real coordinate, with an irrational lattice direction making the ray irrational. If these forms vanish near the fixed points of the involution $z \mapsto -z$, they pass smoothly to the resolution of the quotient, a Kummer surface.

Let $A = \mathbb{C}^2/\Lambda$ be a complex two-torus, let $Y = A/\{\pm 1\}$, and let $r\colon X \to Y$ be the minimal resolution. The fixed points of the involution are the sixteen points of $A[2] = (\frac{1}{2}\Lambda)/\Lambda$, and their images are the sixteen singular points of Y.

Lemma 29.17.1 *The surface X is compact Kähler. Its exceptional set consists of sixteen disjoint projective lines with self-intersection -2.*

Proof Blow up the sixteen fixed points on A and take the quotient by the lifted involution. Near one exceptional line, a blowup chart has coordinates (u, v) with $z_1 = u$ and $z_2 = uv$, and the involution is $(u, v) \mapsto (-u, v)$. The quotient chart has smooth coordinates $(s, v) = (u^2, v)$. On the other chart, $v' = 1/v$ and $s' = v^2 s$. These charts describe the total space of $\mathcal{O}_{\mathbb{P}^1}(-2)$ near its zero section. The quotient is therefore smooth and resolves the double point, with the stated exceptional curve. Doing this at all fixed points gives the compact surface X.

Choose a flat Kähler form on A that equals $\mathrm{dd^c}\rho$, where $\rho = |z_1|^2 + |z_2|^2$, on disjoint balls centered at the fixed points. Away from those points it descends to Y. On the resolution chart,

$$\rho = |s|(1 + |v|^2), \qquad \rho^2 = |s|^2(1 + |v|^2)^2.$$

Choose a smooth function F on $[0, +\infty)$ with $F(\rho) = c\rho^2$ near zero, where $c > 0$, with $F' > 0$ on $(0, +\infty)$ and $F'' \geq 0$, and with $F(\rho) = \rho - C$ for $\rho \geq \delta$, for a sufficiently small $\delta > 0$. Such a function is obtained by choosing F' nondecreasing, linear near zero and equal to one beyond δ. Replace the flat form in each ball by $\mathrm{dd^c}F(\rho)$. This agrees with the flat form near the boundary of the ball and extends smoothly across the exceptional line. The resulting closed form η is positive off the exceptional set, while along it η is positive in the normal direction and zero in the tangent direction.

To restore positivity in the tangent direction, choose a smooth cutoff χ equal to one for $\rho \leq \delta$ and zero for $\rho \geq 2\delta$. For $s \neq 0$, set

$$q_v = \chi(\rho) \log \rho - \log |s|.$$

Near $s = 0$ this equals $\log(1 + |v|^2)$, so it extends smoothly. On the overlap of the two charts, $q_{v'} - q_v = -2 \log |v|$ is pluriharmonic, so their $\mathrm{dd^c}$ forms agree. They define a smooth closed form R supported in the resolved ball, positive in the tangent direction along the exceptional line and semipositive where $\chi = 1$. Hence $\eta + \varepsilon R$ is positive where $\chi = 1$, including along the line. On the compact transition annulus η is already positive, so $\eta + \varepsilon R$ remains positive there for sufficiently small $\varepsilon > 0$, and outside the ball $R = 0$. Adding these corrections in all the disjoint balls produces a Kähler form on X. □

Fix now $a \in \mathbb{R} \setminus \mathbb{Q}$ and take for Λ the lattice generated by

$$\lambda_1 = (1, 0), \quad \lambda_2 = (a, 1), \quad \lambda_3 = (\mathrm{i}, 0), \quad \lambda_4 = (0, \mathrm{i}).$$

Write $z_j = x_j + \mathrm{i}y_j$. Since y_1 takes integer values on Λ, the coordinate $y_1 \bmod \mathbb{Z}$ defines a map from A to the circle $\mathbb{R}/\mathbb{Z}$. Call a smooth even function $h\colon \mathbb{R}/\mathbb{Z} \to [0, +\infty)$ *admissible* if it has integral one and vanishes near $\frac{1}{2}\mathbb{Z}$; such functions are obtained from even pairs of nonnegative bumps supported away from that set. For an admissible

h, put

$$\theta_h = h(y_1)\,\mathrm{d}x_1 \wedge \mathrm{d}y_1.$$

Proposition 29.17.1 *The forms θ_h descend and extend by zero to smooth semipositive closed forms T_h on X. They all represent one nonzero nef Bott–Chern class α with $\alpha^2 = 0$. This class contains more than one closed positive current, and its image in real de Rham cohomology spans an irrational ray. Each T_h vanishes on a neighborhood of the exceptional set.*

Proof The form θ_h is closed, because $\mathrm{d}h(y_1)$ is proportional to $\mathrm{d}y_1$, and it is semipositive. Since h is even, θ_h is invariant under the involution. At every fixed point, y_1 belongs to $\frac{1}{2}\mathbb{Z}$ modulo $\mathbb{Z}$, so the form vanishes on a neighborhood of that point. Its descent to the complement of the singular points of Y therefore extends by zero across their resolved neighborhoods. This proves smoothness and the asserted vanishing.

Let h_1, h_2 be admissible and put $k = h_1 - h_2$. This function is even, has mean zero and vanishes near $\frac{1}{2}\mathbb{Z}$. Choose f with

$$f'(y) = 4\pi \int_0^y k(s)\,\mathrm{d}s, \qquad f'' = 4\pi k.$$

The derivative f' is periodic and odd, so its mean is zero, and f can be chosen periodic and even. Evenness and the zero mean of k give $\int_0^{1/2} k = 0$, hence $f' = 0$ near $\frac{1}{2}\mathbb{Z}$. Therefore $f(y_1)$ descends, and it extends by constants near each exceptional curve to a smooth function G on X. Since

$$\mathrm{dd}^{\mathrm{c}} f(y_1) = \frac{f''(y_1)}{4\pi}\,\mathrm{d}x_1 \wedge \mathrm{d}y_1 \quad \text{on } A, \qquad \text{we have} \qquad \mathrm{dd}^{\mathrm{c}} G = T_{h_1} - T_{h_2}.$$

Thus all the forms T_h have the same Bott–Chern class α. Distinct admissible functions give distinct forms, so α contains more than one closed positive current. The forms are semipositive and have square zero, so $\alpha^2 = 0$. By Lemma 29.17.1, adding any positive multiple of a Kähler form gives a Kähler representative of a nearby class; therefore α is nef.

By the mean-one condition, the periods of θ_h on the real two-tori generated by (λ_1, λ_3) and by (λ_2, λ_3) are 1 and a, respectively. Translate both cycles by $(0, \mathrm{i}/4)$. The translated cycles lie in $\{y_2 = 1/4\}$ modulo $\mathbb{Z}$ and avoid the sixteen fixed points, and their periods are unchanged because the form is closed. Their images in X lie in the complement of the exceptional curves, which is doubly covered without ramification by $A \setminus A[2]$. The first period shows that $\alpha \neq 0$. If a nonzero multiple $b\alpha$ had a rational de Rham class, then evaluating it on these cycles would make both b and ba rational, contradicting $a \notin \mathbb{Q}$. Finally, since $\alpha^2 = 0$, Stokes' theorem shows that α itself has no Kähler representative, so it lies on the boundary of the nef cone.□

29.18 Semipositive metrics in a four-point blowup family

The preceding constructions keep the manifold fixed. For a family of compact manifolds and a line bundle on the total space, one may ask whether a semipositive singular metric on the restriction of the bundle to a special fiber is the restriction of a semipositive metric over a neighborhood of that fiber. In a family of blowups of the projective plane at four points, the answer is negative. The obstruction comes from compact curves on the nearby fibers on which the bundle has degree zero: the curvature of a semipositive metric vanishes on such a curve, and this persists on a component of the limiting curve in the central fiber. Varying these curves identifies exactly which central metrics extend. Their weights have a logarithmic singularity of coefficient at least one along a fixed curve of the central fiber, whereas the prescribed metric has coefficient one half there.

Let Δ be a disk with coordinate t, and let $[x : y : z]$ be homogeneous coordinates on $\mathbb{P}^2$. Blow up $\mathbb{P}^2 \times \Delta$ along the four disjoint sections

$$p_1(t) = [1 : 0 : 0], \qquad p_2(t) = [0 : 1 : 0],$$
$$p_3(t) = [1 : 1 : t], \qquad p_4(t) = [0 : 0 : 1],$$

write $\pi \colon \mathcal{X} \to \Delta$ for the resulting family and $X_t = \pi^{-1}(t)$ for its fibers. Let H be the pullback of the hyperplane line bundle, let E_i be the exceptional divisor over p_i, and put

$$L = 2H - E_1 - E_2 - E_3 - E_4, \qquad L_t = L|_{X_t}.$$

We use additive notation for tensor products of line bundles and also for their divisor classes, and we write K_{X_t} for the canonical line bundle. A singular Hermitian metric h has squared norm $|e|_h^2 = \mathrm{e}^{-\varphi}$ in a local holomorphic frame e; the function φ is its local *weight*, and its curvature is $\mathrm{dd}^{\mathrm{c}}\varphi$. The metric is *semipositive* if its local weights are plurisubharmonic. For an effective divisor with local equation w, the *divisor metric* on the associated line bundle is the singular metric for which the canonical section has norm one; its weight is $\log |w|^2$ and its curvature is the integration current along the divisor.

Lemma 29.18.1 *The family is smooth and projective with rational surface fibers, and L is effective on $\mathcal{X}$. For every $t \neq 0$, L_t is generated by global sections, has square zero and is nef. For every integer $m \geq 1$,*

$$H^0(X_t, K_{X_t} + mL_t) = 0 \qquad (t \neq 0).$$

Proof Near one of the sections, coordinates centered on it identify the blowup with $\mathrm{Bl}_0(\mathbb{C}^2) \times \Delta$. Since the four centers are disjoint, these charts prove smoothness and the submersion property. The blowup of the projective morphism $\mathbb{P}^2 \times \Delta \to \Delta$ is projective and proper, and each fiber X_t is the blowup of $\mathbb{P}^2$ at the four points $p_i(t)$.

The quadratic forms

$$F_1 = x(z - ty), \qquad F_2 = y(z - tx)$$

vanish along all four sections and therefore lift to sections of L. They are nonzero, which proves that L is effective. For $t \neq 0$, their common zeros in $\mathbb{P}^2$ are exactly the four points. In suitable affine coordinates (u, v) centered at p_1, p_2, p_3, p_4, their linear initial terms are

$$(v - tu, -tu), \quad (-tu, v - tu), \quad (v - tu, v), \quad (u, v).$$

Each pair is linearly independent. Hence, after the exceptional divisors are subtracted, the transformed sections have no common zero on an exceptional curve. The two transformed sections therefore generate L_t, and L_t has nonnegative degree on every irreducible curve.

On a fiber, $H^2 = 1$, $H \cdot E_i = 0$, and $E_i \cdot E_j = -\delta_{ij}$. Hence $L_t^2 = 0$, and $K_{X_t} = -3H + \sum_i E_i$ gives $K_{X_t} \cdot L_t = -2$. A nonzero section of $K_{X_t} + mL_t$ would have an effective zero divisor whose intersection with the nef line bundle L_t is $-2 < 0$, a contradiction. This argument also excludes a nowhere-vanishing section, whose zero divisor would have intersection zero. □

On X_0, let ℓ be the strict transform of the line $\{z = 0\}$, which passes through $p_1(0)$, $p_2(0)$ and $p_3(0)$. In additive notation,

$$\ell = H - E_1 - E_2 - E_3, \qquad A = 2L_0 - \ell = 3H - E_1 - E_2 - E_3 - 2E_4.$$

Lemma 29.18.2 *The line bundle A is generated by global sections. There is a semipositive metric h_0 on L_0 whose weight along ℓ is*

$$\varphi_0 = \tfrac{1}{2} \log |w|^2 + \textit{a smooth function},$$

where w is a local equation of ℓ. It is smooth off ℓ and has analytic singularities.

Proof Consider the cubic forms

$$x^2 z, \qquad xyz, \qquad y^2 z, \qquad xy(x - y).$$

Their common zero set in $\mathbb{P}^2$ consists of the four central points. In affine coordinates at p_1, p_2 and p_3, their linear initial terms include (z, y), (z, x) and $(z, x-y)$, respectively, while at p_4 their quadratic initial terms include x^2, xy, y^2. These have no common zero on the corresponding exceptional projective lines. After the simple exceptional factors at p_1, p_2, p_3 and the double factor at p_4 are removed, the transformed cubics therefore generate A.

Their Fubini–Study metric h_A is smooth and semipositive. Let h_ℓ be the divisor metric on $O(\ell)$. Since $2L_0 = A + \ell$, the square root of $h_A \otimes h_\ell$ defines a metric h_0 on L_0, with

$$\Theta_{h_0} = \tfrac{1}{2}\Theta_{h_A} + \tfrac{1}{2}[\ell] \geq 0.$$

The stated form of the weight and of its singularities follows directly. □

29.18.1 The image of metric restriction

The degree-zero conics determine exactly which central metrics extend. We say that a semipositive metric h on L_0 *extends* if there is a semipositive metric on L over a neighborhood of X_0 whose local weights restrict to local weights of h; in particular, these restrictions are not identically $-\infty$. The following slice lemma passes the vanishing of curvature on nearby compact curves to the central fiber.

Lemma 29.18.3 *Let u be a plurisubharmonic function on a product of disks $D_\zeta \times D_t$ with $0 \in D_t$, and suppose that u is not identically $-\infty$. Suppose that $u(\cdot, t)$ is harmonic for almost every t. Then $u(\cdot, 0)$ is harmonic or identically $-\infty$.*

Proof Since u is locally integrable, Fubini's theorem gives $\partial_\zeta \partial_{\bar\zeta} u = 0$ in the sense of distributions. On smaller product disks, form the averages

$$u_\varepsilon(\zeta) = \frac{1}{\pi\varepsilon^2} \int_{|t|<\varepsilon} u(\zeta, t)\, \mathrm{d}A(t),$$

where $\mathrm{d}A$ is planar area. Local upper bounds for u and the reverse Fatou lemma give upper semicontinuity, and Fubini's theorem gives the submean inequality in ζ. The averages are locally integrable, since u is integrable on compact product disks. Thus each u_ε is subharmonic with zero distributional Laplacian, hence harmonic.

Since u is subharmonic in t, the disk averages decrease to $u(\zeta, 0)$ as $\varepsilon \downarrow 0$. A decreasing limit of harmonic functions is harmonic or identically $-\infty$: if the values are bounded below at one point, Harnack's inequality, applied to the differences from the first function, bounds the differences on compact subdisks, and compactness of harmonic functions gives a harmonic limit. This proves the alternative on every smaller disk, and hence on D_ζ. □

Let $f\colon X_0 \to \mathbb{P}^1$ be the morphism defined by the pencil of lines through $p_4 = p_4(0)$, that is, by $[x : y : z] \mapsto [x : y]$, let $a = x/y$ be the affine coordinate on $\mathbb{P}^1$, and put $P = f^* \mathcal{O}_{\mathbb{P}^1}(1)$.

Lemma 29.18.4 *The pencil f is a holomorphic map with connected fibers, $P = H - E_4$, and $L_0 = \mathcal{O}(\ell) + P$. Let s_ℓ be the canonical section of $\mathcal{O}(\ell)$ and h_ℓ its divisor metric, and let h_{FS} be the Fubini–Study metric on $\mathcal{O}_{\mathbb{P}^1}(1)$, with curvature ω_{FS}. Then the metric*

$$h_* = h_\ell \otimes f^* h_{\mathrm{FS}}$$

on L_0 extends to a semipositive metric on L over the whole family. Its curvature is $[\ell] + f^* \omega_{\mathrm{FS}}$.

Proof First blow up p_4 in $\mathbb{P}^2$. The map $[x : y]$ then becomes holomorphic: in the chart $x = r$, $y = rs$ with $z = 1$, it is $[1 : s]$, and the other chart covers the remaining direction. Its fibers are the strict transforms of the lines through p_4, and the pullback of $\mathcal{O}_{\mathbb{P}^1}(1)$ is $H - E_4$. Next blow up the other three central points, which lie in the directions $a = \infty, 0, 1$, respectively. Each affected fiber becomes the union of its

strict transform and one exceptional line meeting it, while every other fiber remains a projective line. This proves connectedness of the fibers and the formula for P. The equality $L_0 = \mathcal{O}(\ell) + P$ follows from the divisor classes.

In this identification, the restrictions of F_1, F_2 to X_0 are $s_\ell \otimes f^*x$ and $s_\ell \otimes f^*y$, where x and y are regarded as sections of $\mathcal{O}_{\mathbb{P}^1}(1)$. Indeed, at $t = 0$ the forms are xz and yz; the pullback of $\{z = 0\}$ is $\ell + E_1 + E_2 + E_3$, and removing the four exceptional factors from the transforms of xz and yz leaves exactly these sections. In a local frame of L, the weight $\log(|F_1|^2 + |F_2|^2)$ defines a semipositive metric, since a change of frame multiplies both coefficients by the same nonvanishing holomorphic function. On every fiber X_t with $t \neq 0$ the two sections have no common zero, so the weight is not identically $-\infty$ there. On the central fiber the weight is the sum of the divisor weight of ℓ and the pullback of the Fubini–Study weight, which proves the claim and the curvature formula. □

Lemma 29.18.5 *For $a \in \mathbb{C} \setminus \{0, 1\}$, the strict transforms of the conics*

$$F_1 - aF_2 = z(x - ay) + t(a - 1)xy = 0$$

have smooth rational fibers of L_t-degree zero for $t \neq 0$. Their central fiber is $\ell + q_a$, where $q_a = f^{-1}(a)$ is a smooth projective line of L_0-degree one. The two components meet transversely at $p_a = [a : 1 : 0]$, away from the blowup centers. Along $q_a \setminus \{p_a\}$, the surface formed by these conics is smooth and its projection to the disk is a submersion.

Proof The symmetric matrix of the quadratic form has determinant $-a(a - 1)t/4$, so the conic is smooth for $t \neq 0$; it passes simply through the four centers. Its strict transform has class L_t, so its degree is $L_t^2 = 0$. At $t = 0$ the conic splits into the two lines $\{z = 0\}$ and $\{x = ay\}$. Since a is finite and $a \notin \{0, 1\}$, the line $\{x = ay\}$ does not pass through the other three central points, so no additional exceptional component occurs, and $q_a = H - E_4$ has degree one against L_0. The two lines meet transversely at p_a, away from all centers.

Away from the centers, the central equation has nonzero derivative in the directions of $\mathbb{P}^2$ along $q_a \setminus \{p_a\}$. At the point of q_a on E_4, the chart $z = 1$, $x = r$, $y = rs$ gives, after the exceptional factor is removed,

$$1 - as + t(a - 1)rs = 0.$$

At $r = t = 0$ and $s = 1/a$, the derivative in s is $-a \neq 0$. These charts prove the final assertion. □

Theorem 29.18.1 *The semipositive metrics on L_0 that extend to semipositive metrics on L over a neighborhood of X_0 are exactly the metrics*

$$h_\ell \otimes f^* h_B,$$

where h_B ranges over all semipositive singular Hermitian metrics on $\mathcal{O}_{\mathbb{P}^1}(1)$. Every such central metric extends over the whole family.

Proof *Necessity.* Let h be a semipositive metric on L_0 that extends. By properness of π we may shrink the disk so that the extending metric is defined on the whole family. On the locus where f is smooth, local integrability and Fubini's theorem show that for almost every $a \in \mathbb{C} \setminus \{0, 1\}$ the restriction of the weights of h to q_a is not identically $-\infty$. Fix such an a. On each local product chart of the conic surface of Lemma 29.18.5 along $q_a \setminus \{p_a\}$, the weight of the extending metric is not identically $-\infty$, since its central slice is not. By Fubini's theorem, the metric restricts to almost every nearby compact conic, and on such a conic its curvature is a positive measure of degree zero, hence zero. Lemma 29.18.3 therefore makes the central slice of the weight harmonic on each chart; the alternative $-\infty$ is excluded by the choice of a.

Consequently, the curvature of $h|_{q_a}$ is supported at p_a. It is a positive measure of degree one, so it equals $[p_a]$, which is also the curvature of $h_*|_{q_a}$. The difference of the two weights is a globally defined harmonic function on the projective line q_a: near p_a, subtracting $\log |w|^2$ from either weight leaves a harmonic function, so the difference extends across p_a. It is therefore constant. Thus the weight difference u between h and h_* is constant along almost every fiber of f, away from ℓ.

Over every point of the base there are local holomorphic sections of f avoiding ℓ and the exceptional curves: over the finite chart use $[a : 1 : 1]$, and near infinity use $[1 : b : 1]$ with $b = 1/a$. If e_B is a local frame of $\mathcal{O}_{\mathbb{P}^1}(1)$, then $s_\ell \otimes f^* e_B$ is a nonvanishing holomorphic frame of L_0 along these sections. Composing the weight of h in this frame with a section defines a subharmonic weight on the base. It is not identically $-\infty$, because on almost every fiber the weight is harmonic away from p_a. Two choices of section give the same function at almost every base point, because u is constant on those fibers, and hence everywhere, since subharmonic functions that agree almost everywhere agree everywhere. Changes of the base frame give the transition law of a metric. These weights therefore define a semipositive metric h_B on $\mathcal{O}_{\mathbb{P}^1}(1)$. Fubini's theorem on local product charts, together with the fact that ℓ and the exceptional fibers have zero volume, shows that h and $h_\ell \otimes f^* h_B$ agree almost everywhere on X_0. By Lemma 1.3.3, their local plurisubharmonic weights agree everywhere, including on the three exceptional fibers. This proves necessity.

Sufficiency. Let h_B be a semipositive metric on $\mathcal{O}_{\mathbb{P}^1}(1)$, and represent it by a homogeneous weight V on $\mathbb{C}^2 \setminus \{0\}$, with coordinates (ξ_1, ξ_2) and $a = \xi_1/\xi_2$. In the chart $\xi_2 \neq 0$, if ϕ_B is the weight of h_B in the frame given by the section ξ_2, set

$$V(\xi_1, \xi_2) = \phi_B(\xi_1/\xi_2) + \log |\xi_2|^2.$$

The other chart gives the same function, by the transition law of the metric. The function V is plurisubharmonic and satisfies $V(\lambda\xi_1, \lambda\xi_2) = V(\xi_1, \xi_2) + \log |\lambda|^2$ for $\lambda \neq 0$. The weight difference between h_B and h_{FS} is bounded above on the compact base, so

$$V(\xi_1, \xi_2) \leq \log(|\xi_1|^2 + |\xi_2|^2) + C.$$

Hence V extends plurisubharmonically across the origin, with value $-\infty$ there, by Lemma 1.3.7. In each local frame of L, the function $V(F_1, F_2)$ is plurisubharmonic, and by homogeneity it transforms as the weight of a metric. Its restriction to each fiber X_t with $t \neq 0$ is not identically $-\infty$: there the two generating sections define a

nonconstant map onto $\mathbb{P}^1$, and the weight of h_B is not identically $-\infty$. On the central fiber, the factorization in Lemma 29.18.4 and homogeneity give exactly the divisor weight of ℓ plus the pullback of the weight of h_B. This is the required extension over $\mathcal{X}$. □

Corollary 29.18.1 *Among all semipositive metrics on L_0 that extend, h_* has the least singular weights: every such metric has weights ψ satisfying $\psi \le \psi_* + C$ in compatible local frames, where ψ_* is the weight of h_*; and h_* itself extends. The assertion concerns metrics that extend, not all semipositive metrics on L_0.*

Proof By Theorem 29.18.1, such a metric has the form $h_\ell \otimes f^* h_B$. Its weight difference from h_* is the pullback of the global quasi-plurisubharmonic difference between the weights of h_B and h_{FS}, which is bounded above on $\mathbb{P}^1$. This proves the inequality. That h_* extends is Lemma 29.18.4. □

Theorem 29.18.2 *There is no semipositive metric on L over an open neighborhood of X_0 whose local weights, restricted to X_0, satisfy*

$$\psi \ge \varphi_0 - C$$

in compatible local frames, for a constant C. In particular, h_0 is neither the restriction of a semipositive metric on such a neighborhood nor of the same singularity type as such a restriction.

Proof Suppose that such a metric exists. Since φ_0 is finite off ℓ, the lower bound shows that the restricted weights are not identically $-\infty$, so they are the weights of a semipositive metric on L_0 that extends. By Corollary 29.18.1, $\psi \le \psi_* + C'$ in compatible local frames. Near a point of ℓ, let w be a local equation of ℓ and e_B a local frame of $\mathcal{O}_{\mathbb{P}^1}(1)$, and use the frame $w^{-1} s_\ell \otimes f^* e_B$ of L_0. In this frame $\psi_* = \log|w|^2 + O(1)$, while $\varphi_0 = \frac{1}{2}\log|w|^2 + O(1)$ by Lemma 29.18.2. Hence $\frac{1}{2}\log|w|^2 \le \log|w|^2 + O(1)$ off ℓ, which fails as $w \to 0$. □

The obstruction concerns the prescribed singularity type: by Lemma 29.18.4, the metric h_* on L_0 extends over the whole family.

29.19 Oeljeklaus–Toma manifolds

The compact manifolds studied so far in this chapter carry many closed positive currents, coming from divisors, fibrations or blowups. Inoue surfaces and their higher-dimensional analogues, the Oeljeklaus–Toma manifolds, behave very differently. They are built from number fields, contain few or no complex subvarieties, and carry a natural closed semipositive $(1,1)$-form whose kernel is a holomorphic foliation by complex affine spaces. The leaves carry no nonconstant plurisubharmonic functions bounded above, and their closures are real tori. As a result, every potential in this class is, up to a fixed logarithmic term, a continuous convex function of the

transverse coordinates, so the class has no singular potentials at all, even though it contains positive currents supported on real hypersurfaces. In dimension three, a natural Gauduchon metric has an exactly quadratic mass functional of indefinite sign. Its masses lie on both sides of the background mass. These threefolds carry no Hermitian metric whose dd^{c} has a sign, so the known curvature criteria for bounded masses do not apply, and whether the masses remain bounded is an open question (Problem 35.3.6).

29.19.1 The construction

Let K be a number field of degree $s+2t$, with $s \geq 1$ real embeddings $\sigma_1, \dots, \sigma_s$ and $t \geq 1$ pairs of complex conjugate embeddings $\sigma_{s+k}, \overline{\sigma}_{s+k}$, $1 \leq k \leq t$. Write O_K for its ring of integers and $O_K^{*,+}$ for the group of totally positive units, that is, of units ε with $\sigma_j(\varepsilon) > 0$ for $1 \leq j \leq s$. Let $\mathbb{H} \subset \mathbb{C}$ be the upper half-plane. Write points of $\mathbb{H}^s \times \mathbb{C}^t$ as (w, z), with $w_j = x_j + \mathrm{i} y_j$, and put

$$\sigma_{\mathbb{R}}(a) = (\sigma_1(a), \dots, \sigma_s(a)), \qquad \sigma_{\mathbb{C}}(a) = (\sigma_{s+1}(a), \dots, \sigma_{s+t}(a)).$$

An element $a \in O_K$ acts by the translation $(w, z) \mapsto (w + \sigma_{\mathbb{R}}(a), z + \sigma_{\mathbb{C}}(a))$, and $\varepsilon \in O_K^{*,+}$ acts by

$$(w, z) \longmapsto \bigl(\sigma_1(\varepsilon) w_1, \dots, \sigma_s(\varepsilon) w_s,\ \sigma_{s+1}(\varepsilon) z_1, \dots, \sigma_{s+t}(\varepsilon) z_t\bigr).$$

A subgroup $U \subset O_K^{*,+}$ is *admissible* if its image

$$\Lambda_U = \{(\log \sigma_1(\varepsilon), \dots, \log \sigma_s(\varepsilon)) : \varepsilon \in U\}$$

is a lattice in $\mathbb{R}^s$. By Dirichlet's unit theorem, when $t = 1$ every subgroup of finite index in $O_K^{*,+}$ is admissible. For admissible U, Oeljeklaus and Toma showed that the semidirect product $\Gamma = O_K \rtimes U$ acts freely and properly discontinuously on $\mathbb{H}^s \times \mathbb{C}^t$ with compact quotient. The compact complex manifold

$$X = X(K, U) = (\mathbb{H}^s \times \mathbb{C}^t)/\Gamma, \qquad n = \dim X = s + t,$$

is the *Oeljeklaus–Toma manifold* of type (s, t). The same construction works if O_K is replaced by any lattice $L \subset K$ of rank $s + 2t$ that is stable under multiplication by U; for $s = t = 1$ this produces the Inoue surfaces S_M. The action is still free: for $\varepsilon \neq 1$, injectivity of σ_j gives $\sigma_j(\varepsilon) \neq 1$, so a point fixed by (a, ε) would have real coordinates w_j. It is still cocompact, because $(\sigma_{\mathbb{R}}, \sigma_{\mathbb{C}})(L)$ is again a lattice in $\mathbb{R}^s \times \mathbb{C}^t$. All results of this section other than Corollary 29.19.2 hold, with the same proofs, in this generality, with L in place of O_K; the only property of O_K they use is that it spans K over $\mathbb{Q}$. Corollary 29.19.2 also holds for every Inoue surface S_M, since these contain no curves. The sources are discussed in Section 36.16.

The functions $y_j = \operatorname{Im} w_j$ are invariant under the translations and are multiplied by $\sigma_j(\varepsilon)$ under the unit ε. Hence the closed real $(1,1)$-form

$$\theta = -\mathrm{dd}^{\mathrm{c}} \sum_{j=1}^{s} \log y_j = \sum_{j=1}^{s} \frac{\mathrm{i}\, \mathrm{d}w_j \wedge \mathrm{d}\overline{w}_j}{8\pi y_j^2}$$

on $\mathbb{H}^s \times \mathbb{C}^t$ is Γ-invariant and descends to a closed semipositive form on X, still denoted θ. It has rank s everywhere, and its kernel is spanned by the vectors $\partial/\partial z_k$. The potential $-\sum_j \log y_j$ itself is not invariant, since a unit adds $-\sum_j \log \sigma_j(\varepsilon)$ to it.

29.19.2 Potentials of the transverse form

Two properties force rigidity: the leaves $\{w\} \times \mathbb{C}^t$ carry no nonconstant plurisubharmonic functions bounded above, and the real parts of the translations are dense.

Lemma 29.19.1 *Since $t \geq 1$, the subgroup $\sigma_{\mathbb{R}}(\mathcal{O}_K)$ is dense in $\mathbb{R}^s$.*

Proof A subgroup G of $\mathbb{R}^s$ is dense unless there is a nonzero linear form ξ on $\mathbb{R}^s$ with $\xi(G) \subseteq \mathbb{Z}$. Indeed, the closure of G is the sum of a linear subspace V and a discrete subgroup of a complement of V; if $V \neq \mathbb{R}^s$, one can take ξ vanishing on V and integral on that discrete subgroup. Suppose $\xi(\sigma_{\mathbb{R}}(a)) = \sum_j \xi_j \sigma_j(a) \in \mathbb{Z}$ for every $a \in \mathcal{O}_K$. Since $\mathcal{O}_K$ spans K over $\mathbb{Q}$, the map $a \mapsto \sum_j \xi_j \sigma_j(a)$ is a $\mathbb{Q}$-linear map $K \to \mathbb{Q}$. The trace form of K is nondegenerate, so there is $b \in K$ with

$$\sum_{j=1}^{s} \xi_j \sigma_j(a) = \operatorname{Tr}_{K/\mathbb{Q}}(ba) = \sum_{\sigma} \sigma(b)\sigma(a) \qquad (a \in K),$$

where σ runs over all $s + 2t$ embeddings of K into $\mathbb{C}$. Distinct field embeddings are linearly independent over $\mathbb{C}$, by Dedekind's lemma on characters. Comparing the coefficients of a complex embedding σ_{s+1}, which occurs only on the right, gives $\sigma_{s+1}(b) = 0$. Hence $b = 0$, and then $\xi_j = \sigma_j(b) = 0$ for every j. □

For $\varepsilon \in U$ and $y \in (0,\infty)^s$, write $\sigma(\varepsilon)y = (\sigma_1(\varepsilon)y_1, \ldots, \sigma_s(\varepsilon)y_s)$.

Theorem 29.19.1 *Let $\mathcal{F}_U$ be the set of convex functions $f\colon (0,\infty)^s \to \mathbb{R}$ such that*

$$f(\sigma(\varepsilon)y) = f(y) - \sum_{j=1}^{s} \log \sigma_j(\varepsilon) \qquad (\varepsilon \in U). \tag{29.55}$$

For $f \in \mathcal{F}_U$, the function $u_f(w,z) = f(y) + \sum_j \log y_j$ is Γ-invariant, and $f \mapsto u_f$ is a bijection from $\mathcal{F}_U$ onto $\mathrm{PSH}(X,\theta)$. In particular, every θ-plurisubharmonic function on X is continuous and depends only on $y = \operatorname{Im} w$.

Proof We first note that a function g on $\mathbb{H}^s$ that depends only on y is plurisubharmonic if and only if, as a function on $(0,\infty)^s$, it is convex (with values in $[-\infty,+\infty)$ and not identically $-\infty$). For smooth g this follows from $\partial^2 g/\partial w_j \partial \overline{w}_k = \frac{1}{4}\partial^2 g/\partial y_j \partial y_k$. In general, convolution with a smooth radial kernel in w produces smooth functions of y alone. They decrease to g if g is plurisubharmonic, and they converge locally uniformly to g if g is convex, because a convex function that is not identically $-\infty$ on the open convex set $(0,\infty)^s$ is finite and continuous there. Since decreasing limits of convex functions are convex and locally uniform limits of plurisubharmonic functions are plurisubharmonic, the claim follows.

The identity (29.55) says exactly that u_f is invariant under U, and u_f is clearly invariant under the translations. On the universal cover, $\theta + \mathrm{dd}^{\mathrm{c}} u_f = \mathrm{dd}^{\mathrm{c}} f(y) \geq 0$. Hence $u_f \in \mathrm{PSH}(X,\theta)$, and f is recovered from u_f, so the map is injective.

Conversely, let $u \in \mathrm{PSH}(X,\theta)$, and let $\tilde{u}$ be its lift to $\mathbb{H}^s \times \mathbb{C}^t$. Since $\theta + \mathrm{dd}^{\mathrm{c}}\tilde{u} = \mathrm{dd}^{\mathrm{c}}(\tilde{u} - \sum_j \log y_j)$ there, the function $q = \tilde{u} - \sum_j \log y_j$ is plurisubharmonic on $\mathbb{H}^s \times \mathbb{C}^t$.

Step 1: constancy on the leaves. Fix $w^0 \in \mathbb{H}^s$. The restriction of q to the slice $\{w^0\} \times \mathbb{C}^t$ is plurisubharmonic or identically $-\infty$, and it is bounded above because u is upper semicontinuous on the compact manifold X. A plurisubharmonic function on $\mathbb{C}^t$ that is bounded above is constant by Lemma 1.3.5. Hence $\tilde{u}(w,z) = \tilde{u}(w,0)$ for all (w,z). In particular, the function $F(w) = q(w,0)$ is not identically $-\infty$, because $\tilde{u}$ is not, and it is plurisubharmonic on $\mathbb{H}^s$.

Step 2: invariance under real translations. The translations by O_K and Step 1 give $F(w + \sigma_{\mathbb{R}}(a)) = F(w)$ for every $a \in O_K$, since $\log y_j$ does not change under real translations. For $r > 0$, the averages F_r of F over the Euclidean balls of radius r are defined on $\{w : \min_j y_j > r\}$ and continuous there, because F is locally integrable, and they are invariant under the same translations. By Lemma 29.19.1 and continuity, each F_r is invariant under all translations by $\mathbb{R}^s$. Since F_r decreases to F as $r \downarrow 0$, the function F is invariant under all real translations as well, so $F(w) = f(y)$ for a function f on $(0,\infty)^s$.

Step 3: convexity and automorphy. Since $F = f(y)$ is plurisubharmonic, the first paragraph of the proof shows that f is convex, finite and continuous. Finally, $u = u_f$, and invariance of u under U is (29.55). □

Corollary 29.19.1 *Every $u \in \mathrm{PSH}(X,\theta)$ has the ordinary singularity type of 0, its Lelong numbers vanish, its multiplier ideals are trivial, and $P_\theta[u] = V_\theta = 0$. Moreover $\mathrm{MA}_\theta(u) = 0$, so θ is nef but not big.*

Proof By Theorem 29.19.1, the functions in $\mathrm{PSH}(X,\theta)$ are continuous, hence bounded on the compact manifold X, so any two of them are equivalent, and their Lelong numbers and multiplier ideals are those of a bounded function. Since $\theta \geq 0$, the constant 0 belongs to $\mathrm{PSH}(X,\theta)$, so $V_\theta = 0$, and $P_\theta[u] = P_\theta[0] = V_\theta$. By Theorem 29.19.1, locally $\theta + \mathrm{dd^c} u = \mathrm{dd^c} f(y)$ with f continuous and depending on the s real variables y_j; the Bedford–Taylor products of such a current vanish in degree $s+1 \leq n$, because they are limits of the corresponding products for smooth convex approximants of f, whose complex Hessians have rank at most s. Thus $(\theta + \mathrm{dd^c} u)^n = 0$. The form θ is smooth and semipositive, hence nef. Suppose that $\rho \in \mathrm{PSH}(X,\theta)$ satisfied $\theta + \mathrm{dd^c}\rho \geq \delta\omega$ with $\delta > 0$. The function ρ is continuous, and on a small coordinate ball with coordinates ζ we may write $\theta = \mathrm{dd^c} a$ with a smooth and $\delta\omega \geq \delta' \mathrm{dd^c}|\zeta|^2$ with $\delta' > 0$. Then $a + \rho - \delta'|\zeta|^2$ is plurisubharmonic, and expanding $(\mathrm{dd^c}(a+\rho))^n$ as a sum of positive mixed Bedford–Taylor products gives $\mathrm{MA}_\theta(\rho) \geq \delta'^n (\mathrm{dd^c}|\zeta|^2)^n \neq 0$ on the ball, a contradiction. Hence θ is not big.□

Singular currents nevertheless occur in this class. Let $s = t = 1$, so that X is an Inoue surface and U is generated by a unit ε_0 with $\lambda = \sigma_1(\varepsilon_0) > 1$. Fix $y_0 > 0$, put $y_k = \lambda^k y_0$ for $k \in \mathbb{Z}$, and let f be the continuous function that is affine on each interval $[y_k, y_{k+1}]$ with slope

$$a_k = -\frac{\log\lambda}{(\lambda-1)y_k}, \qquad f(y_0) = 0.$$

Then $a_k(y_{k+1} - y_k) = -\log\lambda$, so $f(\lambda y) = f(y) - \log\lambda$, and the slopes increase because $a_{k+1} = a_k/\lambda$ and $a_k < 0$. Thus $f \in \mathcal{F}_U$. On the universal cover,

$$\theta + \mathrm{dd^c} u_f = \mathrm{dd^c} f(y) = \frac{1}{4\pi} \sum_{k\in\mathbb{Z}} (a_k - a_{k-1})\, \delta(y - y_k)\, \mathrm{d}x \wedge \mathrm{d}y,$$

with $a_k - a_{k-1} > 0$. The closed positive current $\theta + \mathrm{dd^c} u_f$ therefore has a continuous potential but is supported on the compact real hypersurface of X formed by the images of the hyperplanes $\{y = y_k\}$, a three-dimensional torus that is a union of leaves of the foliation $\{w = \mathrm{const}\}$.

On an Inoue surface, θ represents the canonical Bott–Chern class. Indeed, the Tricerri metric $\omega_T = 8\pi\theta + y\,\mathrm{i}\,\mathrm{d}z \wedge \mathrm{d}\overline{z}$ has $\omega_T^2 = (2/y)\,\mathrm{i}\,\mathrm{d}w \wedge \mathrm{d}\overline{w} \wedge \mathrm{i}\,\mathrm{d}z \wedge \mathrm{d}\overline{z}$, so its normalized Chern–Ricci form, which represents $c_1^{\mathrm{BC}}(X)$, is $-\mathrm{dd^c}\log(2/y) = \mathrm{dd^c}\log y = -\theta$. Hence Theorem 29.19.1 describes all positive currents in $c_1^{\mathrm{BC}}(K_X)$, and none of them has a pole.

Arbitrary backgrounds behave differently, since a Hermitian form has potentials with logarithmic poles. Nevertheless, when $t = 1$ their singularities are confined to finite sets. By a theorem of Ornea and Verbitsky (Section 36.16), an Oeljeklaus–Toma manifold with $t = 1$ contains no complex subvarieties of positive dimension.

Corollary 29.19.2 *Let X be an Oeljeklaus–Toma manifold with $t = 1$, let η be a smooth real $(1,1)$-form on X, and let $u \in \mathrm{PSH}(X,\eta)$. For every $c > 0$, the set $\{x : \nu(u,x) \geq c\}$ is finite. Consequently, for every $k > 0$ the multiplier ideal $\mathcal{I}(ku)$ is trivial outside the finite set $\{x : \nu(u,x) \geq 1/k\}$, and it has finite colength.*

Proof By Siu's theorem (Section 1.13.1), the set $E_c = \{\nu(u,\cdot) \geq c\}$ is a compact analytic subset of X. It is a proper subset, because $\nu(u,x) = 0$ outside the pluripolar set $\{u = -\infty\}$, which has Lebesgue measure zero. Hence E_c is finite by the theorem of Ornea and Verbitsky. By Skoda's integrability criterion [Dem11, Lemma 5.6], $\mathcal{I}(ku)_x = \mathcal{O}_{X,x}$ whenever $k\nu(u,x) < 1$, so $\mathcal{I}(ku)$ is trivial outside the finite set $E_{1/k}$. A coherent ideal sheaf that is trivial outside a finite set has finite colength. □

29.19.3 Monge–Ampère masses on threefolds

Now let $s = 2$ and $t = 1$, so that X is a threefold. A totally positive unit has norm $\sigma_1(\varepsilon)\sigma_2(\varepsilon)|\sigma_3(\varepsilon)|^2 = 1$, so the form $y_1 y_2\, \mathrm{dd^c}|z|^2$ is Γ-invariant. The Hermitian form

$$\omega = \theta + \beta, \qquad \beta = y_1 y_2\, \mathrm{dd^c}|z|^2,$$

descends to X. Write $\mathrm{d}V_0 = (\mathrm{i}\,\mathrm{d}w_1 \wedge \mathrm{d}\overline{w}_1) \wedge (\mathrm{i}\,\mathrm{d}w_2 \wedge \mathrm{d}\overline{w}_2) \wedge (\mathrm{i}\,\mathrm{d}z \wedge \mathrm{d}\overline{z})$ on the universal cover. As in Chapter 2, $\mathrm{M}_\omega(u) = \int_X (\omega + \mathrm{dd^c}u)^3$; for bounded $u \in \mathrm{PSH}(X,\omega)$ the integrand is the Bedford–Taylor product.

If a unit ε maps a level set of y_1 to itself, then $\sigma_1(\varepsilon) = 1$, hence $\varepsilon = 1$ because σ_1 is injective. Hence the map $X \to T_U = \mathbb{R}^2/\Lambda_U$, $[w,z] \mapsto (\log y_1, \log y_2)$, has as fibers the real tori $(\mathbb{R}^2 \times \mathbb{C})/(\sigma_{\mathbb{R}}, \sigma_{\mathbb{C}})(\mathcal{O}_K)$. Since $\mathrm{d}V_0/(y_1 y_2)$ is Γ-invariant, there is a constant $c_X > 0$ such that, for every integrable Λ_U-periodic function g of $\varsigma = (\log y_1, \log y_2)$,

$$\int_X g(\varsigma)\, \frac{\mathrm{d}V_0}{y_1 y_2} = c_X \int_{T_U} g\, \mathrm{d}\varsigma. \tag{29.56}$$

We shall use an elementary property of lattices in the plane: every lattice $\Lambda \subset \mathbb{R}^2$ contains vectors in each of the four open quadrants. Indeed, let $\ell > 0$ be the length of a shortest nonzero vector of Λ, and let $0 < \delta < \ell$. By Minkowski's theorem, the symmetric convex set $\{k : |k_1 - k_2| < \delta,\ |k_1 + k_2| < N\}$ contains a nonzero lattice point once N is large. Such a point satisfies $(k_1+k_2)^2 = 2|k|^2 - (k_1-k_2)^2 \geq 2\ell^2 - \delta^2$, hence $4k_1k_2 = (k_1+k_2)^2 - (k_1-k_2)^2 > 0$; replacing k by $-k$ if necessary gives a point in the open first quadrant, and the line $\mathbb{R}(1,-1)$ gives the other two quadrants in the same way. If λ lies in the open first quadrant and $\mu \in \Lambda$ is independent of λ, then $N\lambda + \mu$ also lies there for large N, so the first quadrant contains two independent lattice vectors.

Proposition 29.19.1 *Let X be an Oeljeklaus–Toma threefold of type $(2,1)$ with the Hermitian form ω above.*

(1) *The form ω is Gauduchon, and $\mathrm{dd^c}\omega = \mathrm{dd^c}(y_1y_2)\wedge\mathrm{dd^c}|z|^2$ is neither semipositive nor seminegative.*
(2) *For every smooth real function u on X,*

$$\mathrm{M}_\omega(u) = \int_X \omega^3 + \frac{3}{16\pi^3}\int_X \mathrm{Re}\Big(\frac{\partial u}{\partial w_1}\,\overline{\frac{\partial u}{\partial w_2}}\Big)\mathrm{d}V_0.$$

(3) *Let $u\in\mathrm{PSH}(X,\omega)$ be bounded and depend only on y, and write $u = g(\varsigma)$ with g periodic under Λ_U. Then g is Lipschitz, $P = 1-\partial_1 g$ and $Q = 1-\partial_2 g$ are nonnegative almost everywhere, and*

$$\mathrm{M}_\omega(u) = \frac{\int_X\omega^3}{|T_U|}\int_{T_U} PQ\,\mathrm{d}\varsigma.$$

There is a constant $c_U\geq 1$, depending only on Λ_U, such that every such u satisfies $c_U^{-1}\int_X\omega^3 \leq \mathrm{M}_\omega(u)\leq c_U\int_X\omega^3$.
(4) *There are smooth $u\in\mathrm{PSH}(X,\omega)$, depending only on y, with $\mathrm{M}_\omega(u) > \int_X\omega^3$, and others with $\mathrm{M}_\omega(u) < \int_X\omega^3$. Consequently $v_-(\omega) < \int_X\omega^3 < v_+(\omega)$.*

Proof (1) Since $\mathrm{dd^c}|z|^2$ has constant coefficients and y_j is pluriharmonic,

$$\mathrm{dd^c}\beta = \mathrm{dd^c}(y_1y_2)\wedge\mathrm{dd^c}|z|^2,$$

where

$$\mathrm{dd^c}(y_1y_2) = \frac{\mathrm{i}}{2\pi}\big(\partial y_1\wedge\overline{\partial}y_2 + \partial y_2\wedge\overline{\partial}y_1\big) = \frac{\mathrm{i}}{8\pi}\big(\mathrm{d}w_1\wedge\mathrm{d}\overline{w}_2 + \mathrm{d}w_2\wedge\mathrm{d}\overline{w}_1\big).$$

The Hermitian matrix of $\mathrm{dd^c}(y_1y_2)$ in the basis $\mathrm{d}w_1,\mathrm{d}w_2$ is a positive multiple of $\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$, which is indefinite. Accordingly, for $\eta_\pm = \mathrm{d}w_1\pm\mathrm{d}w_2$ the $(3,3)$-forms $\mathrm{dd^c}\omega\wedge\mathrm{i}\eta_\pm\wedge\overline{\eta}_\pm$ are respectively negative and positive multiples of $\mathrm{d}V_0$, so $\mathrm{dd^c}\omega = \mathrm{dd^c}\beta$ is neither semipositive nor seminegative. Since θ is closed and $\beta^2=0$, we have $\mathrm{dd^c}(\omega^2) = 2\theta\wedge\mathrm{dd^c}\beta$, and this vanishes because every term of $\theta\wedge(\mathrm{d}w_1\wedge\mathrm{d}\overline{w}_2+\mathrm{d}w_2\wedge\mathrm{d}\overline{w}_1)$ repeats a differential.

(2) By Theorem 2.2.1 with $n=3$ and $\mathrm{dd^c}(\omega^2)=0$, $\mathrm{M}_\omega(u)-\int_X\omega^3 = 3\int_X u\,\mathrm{dd^c}u\wedge\mathrm{dd^c}\omega$. The form $\mathrm{dd^c}\omega$ is closed, so Stokes' theorem gives $\int_X u\,\mathrm{dd^c}u\wedge\mathrm{dd^c}\omega = -\int_X \mathrm{d}u\wedge\mathrm{d^c}u\wedge\mathrm{dd^c}\omega$. Now $\mathrm{d}u\wedge\mathrm{d^c}u = (\mathrm{i}/2\pi)\,\partial u\wedge\overline{\partial}u$, and a direct computation in the coordinates w_1,w_2,z gives

$$\mathrm{i}\,\partial u\wedge\overline{\partial}u\wedge\mathrm{i}\big(\mathrm{d}w_1\wedge\mathrm{d}\overline{w}_2+\mathrm{d}w_2\wedge\mathrm{d}\overline{w}_1\big)\wedge\mathrm{i}\,\mathrm{d}z\wedge\mathrm{d}\overline{z} = -2\,\mathrm{Re}\Big(\frac{\partial u}{\partial w_1}\,\overline{\frac{\partial u}{\partial w_2}}\Big)\mathrm{d}V_0.$$

Collecting the constants $(1/2\pi)(1/8\pi)(1/2\pi)$ gives the formula.

(3) Since $\mathrm{dd}^{\mathrm{c}}u$ has no components in the z-direction, $\omega+\mathrm{dd}^{\mathrm{c}}u = (\theta+\mathrm{dd}^{\mathrm{c}}u)+\beta$ splits into blocks, and $u \in \mathrm{PSH}(X,\omega)$ if and only if $u \in \mathrm{PSH}(X,\theta)$. By Theorem 29.19.1, $u = f(y) + \log(y_1 y_2)$ with f convex, so g is locally Lipschitz, hence Lipschitz by periodicity. In the variables $\varsigma_j = \log y_j$, convexity of f is equivalent to positivity of the symmetric matrix

$$H = I + D^2 g - \mathrm{diag}(\partial_1 g, \partial_2 g),$$

as one checks from $y_i y_j\, \partial^2 f/\partial y_i \partial y_j = \partial^2\Phi/\partial\varsigma_i\partial\varsigma_j - \delta_{ij}\,\partial\Phi/\partial\varsigma_i$ for $\Phi(\varsigma) = f(\mathrm{e}^{\varsigma}) = g(\varsigma) - \varsigma_1 - \varsigma_2$. This condition is linear in g with constant coefficients, so it is preserved by convolution with a smooth kernel on $\mathbb{R}^2$; the regularized functions are periodic, and they converge uniformly to g with gradients converging almost everywhere. By the continuity of bounded Monge–Ampère products under uniform convergence and dominated convergence, it suffices to prove all assertions for smooth g with $H \geq 0$.

For such g, the function u depends only on y, so $\partial u/\partial w_j = -\frac{\mathrm{i}}{2}\partial u/\partial y_j$ and the integrand in (2) is $\frac{1}{4}\partial_{y_1}u\,\partial_{y_2}u = \partial_1 g\,\partial_2 g/(4y_1y_2)$. A direct computation gives $\omega^3 = 3\theta^2\wedge\beta = \frac{3}{64\pi^3}\,\mathrm{d}V_0/(y_1y_2)$. By (29.56) and (2),

$$\mathrm{M}_\omega(u) = \frac{\int_X \omega^3}{|T_U|}\int_{T_U}\bigl(1+\partial_1 g\,\partial_2 g\bigr)\,\mathrm{d}\varsigma = \frac{\int_X\omega^3}{|T_U|}\int_{T_U} PQ\,\mathrm{d}\varsigma,$$

where the second equality uses $\int_{T_U}\partial_j g\,\mathrm{d}\varsigma = 0$.

The diagonal entry $H_{11} = P - \partial_1 P \geq 0$ says that $\mathrm{e}^{-\varsigma_1}P$ is nonincreasing in ς_1, that is,

$$P(\varsigma+\tau e_1) \leq \mathrm{e}^{\tau}P(\varsigma) \qquad (\tau \geq 0). \tag{29.57}$$

If $P(\varsigma) < 0$, the right side tends to $-\infty$ as $\tau\to\infty$, which is impossible for the bounded function P; hence $P \geq 0$, and likewise $Q \geq 0$. Thus $\partial_j g \leq 1$. The lattice Λ_U contains two independent vectors λ, λ' with positive coordinates, and every point of T_U has the form $\varsigma + \tau\lambda + \tau'\lambda'$ with $\tau,\tau' \in [0,1)$. Along $\tau \mapsto \varsigma+\tau\lambda$, the derivative of g is at most $|\lambda|_1 = \lambda_1+\lambda_2$, and its integral over $[0,1]$ vanishes by periodicity, so $|g(\varsigma+\tau\lambda) - g(\varsigma)| \leq |\lambda|_1$. Hence $\operatorname{osc} g \leq L$, with $L = |\lambda|_1 + |\lambda'|_1$ depending only on Λ_U.

Averaging (29.57) in the form $P(\varsigma-\tau e_1) \geq \mathrm{e}^{-\tau}P(\varsigma)$ over $\tau\in[0,1]$ gives

$$(1-\mathrm{e}^{-1})P(\varsigma) \leq \int_0^1 P(\varsigma-\tau e_1)\,\mathrm{d}\tau = 1 - g(\varsigma) + g(\varsigma-e_1) \leq 1+L.$$

On the other hand, averaging $P(\varsigma) \geq \mathrm{e}^{-\tau}P(\varsigma+\tau e_1)$ over $\tau\in[0,R]$ with $R = \max(2L,1)$ gives

$$P(\varsigma) \geq \frac{\mathrm{e}^{-R}}{R}\int_0^R P(\varsigma+\tau e_1)\,\mathrm{d}\tau = \frac{\mathrm{e}^{-R}}{R}\bigl(R - g(\varsigma+Re_1) + g(\varsigma)\bigr) \geq \frac{\mathrm{e}^{-R}}{2}.$$

The same bounds hold for Q. Therefore the average of PQ over T_U lies between $\mathrm{e}^{-2R}/4$ and $\big((1+L)/(1-\mathrm{e}^{-1})\big)^2$, which proves (3).

(4) Let k belong to the dual lattice $\Lambda_U^* = \{k \in \mathbb{R}^2 : \langle k, \lambda\rangle \in \mathbb{Z} \text{ for all } \lambda \in \Lambda_U\}$, and put $g_\epsilon(\varsigma) = \epsilon\cos(2\pi\langle k, \varsigma\rangle)$. For small $\epsilon > 0$, the matrix H of g_ϵ is positive definite, so $u_\epsilon = g_\epsilon(\varsigma)$ is a smooth function with $\omega + \mathrm{dd^c} u_\epsilon > 0$. By (3),

$$\frac{\mathrm{M}_\omega(u_\epsilon)}{\int_X \omega^3} = 1 + \frac{1}{|T_U|}\int_{T_U} \partial_1 g_\epsilon\, \partial_2 g_\epsilon \,\mathrm{d}\varsigma = 1 + 2\pi^2\epsilon^2 k_1 k_2.$$

Since the lattice Λ_U^* has vectors in every open quadrant, we may choose k with $k_1k_2 > 0$, respectively $k_1k_2 < 0$, which gives the two inequalities. □

For a general smooth potential, averaging over the fibers separates the invariant part from the fluctuations. For $u \in \mathrm{PSH}(X,\omega)$ let Au be the average of u over the fibers of $X \to T_U$, that is, over the translations $(w,z) \mapsto (w+r, z+\zeta)$ with $(r,\zeta) \in (\mathbb{R}^2\times\mathbb{C})/(\sigma_{\mathbb{R}},\sigma_{\mathbb{C}})(O_K)$. These translations preserve ω on the universal cover. They do not descend to X individually, but conjugation by a unit ε carries the translation by (r,ζ) to the translation by $(\sigma_1(\varepsilon)r_1, \sigma_2(\varepsilon)r_2, \sigma_3(\varepsilon)\zeta)$, and this automorphism of the translation torus preserves its Haar measure because ε has norm one. Hence Au is a well-defined ω-plurisubharmonic function depending only on y. As in the proof of Proposition 29.19.1(3), such a function is θ-plurisubharmonic, so it is continuous by Theorem 29.19.1, and Proposition 29.19.1(3) gives $c_U^{-1}\int_X\omega^3 \le \mathrm{M}_\omega(Au) \le c_U\int_X\omega^3$. For smooth u, the integral of a y-derivative of $u - Au$ over a fiber vanishes, and so does the integral of an x-derivative. The cross terms between Au and $u - Au$ in Proposition 29.19.1(2) therefore integrate to zero, and

$$\mathrm{M}_\omega(u) = \mathrm{M}_\omega(Au) + \frac{3}{16\pi^3}\int_X \mathrm{Re}\Big(\frac{\partial v}{\partial w_1}\,\overline{\frac{\partial v}{\partial w_2}}\Big)\mathrm{d}V_0, \qquad v = u - Au. \tag{29.58}$$

Since the extrema of the masses of bounded potentials equal the extrema over smooth strictly ω-plurisubharmonic potentials (Proposition 7.5.1 and Theorem 10.8.2), $v_+(\omega) < +\infty$ if and only if the fluctuation term in (29.58) is bounded above on smooth ω-plurisubharmonic functions.

29.19.4 Hermitian metrics whose dd^{c} has a sign

Proposition 29.19.2 *Let X be an Oeljeklaus–Toma manifold of type $(s,1)$ with $s \ge 2$. Then X admits no Hermitian form $\widetilde{\omega}$ such that $\mathrm{dd^c}\widetilde{\omega}$ is semipositive, or seminegative, on every complex 2-plane.*

Proof Suppose $\widetilde{\omega}$ is such a form, with $\mathrm{dd^c}\widetilde{\omega} \ge 0$ on every complex 2-plane; the other case is analogous. Average the lift of $\widetilde{\omega}$ over the translations $(w,z) \mapsto (w+r, z+\zeta)$, $(r,\zeta) \in (\mathbb{R}^s\times\mathbb{C})/(\sigma_{\mathbb{R}},\sigma_{\mathbb{C}})(O_K)$, as in the definition of Au above. The averaged form $\overline{\omega}$ is again Γ-invariant and Hermitian, its coefficients in the frame $\mathrm{d}w_1,\dots,\mathrm{d}w_s,\mathrm{d}z$

depend only on y, and $\mathrm{dd}^{\mathrm{c}}\overline{\omega} \geq 0$ on complex 2-planes, because translations are holomorphic and positivity on 2-planes is preserved by averaging.

Let $h(y) > 0$ be the coefficient of $\mathrm{i}\,\mathrm{d}z \wedge \mathrm{d}\overline{z}$ in $\overline{\omega}$. Invariance under a unit ε gives $h(\sigma(\varepsilon)y)|\sigma_{s+1}(\varepsilon)|^2 = h(y)$, that is, $h(\sigma(\varepsilon)y) = \sigma_1(\varepsilon)\cdots\sigma_s(\varepsilon)\,h(y)$, since ε has norm one. Hence $h/(y_1\cdots y_s)$ is invariant under U, and since Λ_U is a lattice it is bounded between two positive constants:

$$c\,y_1\cdots y_s \leq h(y) \leq C\,y_1\cdots y_s. \tag{29.59}$$

Let v be a nonzero vector in the span of $\partial/\partial w_1,\dots,\partial/\partial w_s$, and let P be the complex 2-plane spanned by v and $\partial/\partial z$. Write $\overline{\omega} = \sum_{a,b} g_{ab}(y)\,\mathrm{i}\,\xi_a \wedge \overline{\xi}_b$ with $\xi = (\mathrm{d}w_1,\dots,\mathrm{d}w_s,\mathrm{d}z)$. Since the forms ξ_a are closed, $\mathrm{dd}^{\mathrm{c}}\overline{\omega} = \sum_{a,b}\mathrm{dd}^{\mathrm{c}}g_{ab}\wedge\mathrm{i}\,\xi_a\wedge\overline{\xi}_b$, and each $\mathrm{dd}^{\mathrm{c}}g_{ab}$ has only w-components. A term restricted to P is nonzero only if it contains both $\mathrm{d}z$ and $\mathrm{d}\overline{z}$, which happens only for $\xi_a = \xi_b = \mathrm{d}z$. Hence the restriction of $\mathrm{dd}^{\mathrm{c}}\overline{\omega}$ to P equals the restriction of $\mathrm{dd}^{\mathrm{c}}h\wedge\mathrm{i}\,\mathrm{d}z\wedge\mathrm{d}\overline{z}$, and its sign is that of the value of $\mathrm{dd}^{\mathrm{c}}h$ on v. Thus $\mathrm{dd}^{\mathrm{c}}h \geq 0$ in the w-variables, and h is convex in y.

Let $R > 1$ and put $A = (R, R^{-1}, 1,\dots,1)$, $B = (R^{-1}, R, 1,\dots,1)$ and $M = (A+B)/2$. By (29.59), $h(A), h(B) \leq C$, whereas $h(M) \geq c(R+R^{-1})^2/4$, which contradicts convexity for large R. In the seminegative case, h is concave, so $t \mapsto h(t,\dots,t)$ grows at most linearly as $t \to \infty$; this contradicts the lower bound $h(t,\dots,t) \geq ct^s$ with $s \geq 2$. $\square$

For $s = t = 1$ the same averaging produces a coefficient h comparable to y, which may be affine; indeed the Tricerri metric is pluriclosed. In dimension three, the curvature criteria for finite upper and positive lower mass recalled in Section 36.16 require a metric with $\mathrm{dd}^{\mathrm{c}}\omega \geq 0$, respectively $\mathrm{dd}^{\mathrm{c}}\omega \leq 0$, and Proposition 29.19.2 excludes both on threefolds of type $(2,1)$.

Chapter 30
Locally conformally Kähler geometry

Let X be a connected compact complex manifold of dimension $n \geq 2$. A locally conformally Kähler (LCK) metric is a Hermitian metric that becomes Kähler after local conformal changes, and its Lee form measures the obstruction to making this change globally. On a covering on which the Lee form is exact, the Kähler metric and its admissible potentials change by the same character under deck transformations. Descending this potential calculus to X gives a twisted complex Hessian with a zeroth-order term, so that even adding a constant to a potential changes the metric. The standard Vaisman metric on the Hopf manifold illustrates the difference between the two potential theories: although it is locally conformally Kähler, its ordinary potentials have unbounded Monge–Ampère masses.

Prescribing the volume while keeping the Lee class and the Gauduchon gauge couples the determinant equation with an equation for the conformal factor. For a volume invariant under the Lee and anti-Lee flows of a Vaisman metric, uniqueness for the ordinary smooth equation gives a potential that is constant along these flows. Stokes' theorem in the transverse directions fixes the normalizing constant, and changing the Lee form reconstructs the required Vaisman metric. On scalar Hopf manifolds, radial symmetry reduces the coupled equations to a periodic ordinary differential equation. Its positive solutions are ordered by their Lee periods, so the metric is determined by its Lee period and its volume; for a nonconstant radial volume, this metric is not Vaisman. A different compatibility condition gives rigidity: when an LCK metric is both Gauduchon and Chern–Ricci flat, the Bianchi and Bochner identities force its harmonic Lee form to be parallel. On a Kodaira surface, however, the Chern–Ricci-flat metrics of a conformal class need not be Gauduchon.

The Lee form also produces a closed curvature class that can be nonzero in Bott–Chern cohomology while vanishing in de Rham cohomology. If the least eigenvalue of its natural representative relative to the LCK metric is constant, this eigenvalue vanishes and the class is pseudoeffective. Pairing these classes with all Gauduchon metrics defines the Lee–Gauduchon cone, which detects their pseudoeffectivity and is invariant under bimeromorphic maps. Strongly Gauduchon metrics exclude all nonzero positive de Rham-exact currents and bound the linear mass of closed positive currents by their de Rham classes. The Calabi–Eckmann example separates this

condition from the presence of zero in the Lee–Gauduchon cone, and a balanced elliptic fibration shows that it still allows zero lower Monge–Ampère mass.

30.1 The Lee form and the Kähler covering

A Hermitian form is locally conformally Kähler when its exterior derivative is controlled by a closed one-form.

Definition 30.1.1 A Hermitian form ω on X is *locally conformally Kähler*, or LCK, if there is a smooth real one-form λ on X such that

$$\mathrm{d}\omega = \lambda \wedge \omega, \qquad \mathrm{d}\lambda = 0.$$

The form λ is called its *Lee form*. The metric is *strictly LCK* if λ is not exact. A strictly LCK metric is *Vaisman* if its Lee form is parallel for the Levi–Civita connection of the associated Riemannian metric.

Fix an LCK metric ω with Lee form λ. On a compact manifold a nonzero parallel one-form cannot be exact: the differential of a function vanishes at a maximum, whereas a parallel form has constant length. Thus a nonzero parallel Lee form is automatically nowhere zero and not exact. A closed one-form whose metric dual is a Killing field is parallel: its covariant derivative is symmetric because the form is closed, and skew-symmetric by the Killing equation.

The Lee form is determined by ω, because the map $a \mapsto a \wedge \omega$ is injective on one-forms: after wedging with ω^{n-2}, this follows from the bijectivity of the Lefschetz map $a \mapsto a \wedge \omega^{n-1}$ on one-forms.

Locally write $\lambda = \mathrm{d}f$. Then

$$\Omega = \mathrm{e}^{-f}\omega > 0, \qquad \mathrm{d}\Omega = 0,$$

so Ω is a Kähler form. Choose a connected regular covering $\pi\colon \widetilde{X} \to X$ on which $\pi^*\lambda = \mathrm{d}F$, for example the universal covering. The quotient of $\widetilde{X}$ by its deck group is X. The same construction gives a Kähler form

$$\widetilde{\omega} = \mathrm{e}^{-F}\pi^*\omega. \tag{30.1}$$

For a deck transformation σ, the difference $\sigma^*F - F = c_\sigma$ is constant, and

$$\sigma^*\widetilde{\omega} = \chi(\sigma)\widetilde{\omega}, \qquad \chi(\sigma) = \mathrm{e}^{-c_\sigma}.$$

Thus χ is a homomorphism from the deck group to the multiplicative group of positive real numbers, which records how the Kähler metric is rescaled. Forms on $\widetilde{X}$ transforming by χ may equivalently be viewed as forms on X with values in the flat real line bundle defined by χ, the *weight bundle*. In this sense an LCK form is closed as a form with values in the weight bundle [OV24b, Section 3.4].

Let g be the Riemannian metric of ω, let J be the complex structure, and let $\lambda^\sharp$ be the vector field dual to λ with respect to g. On a chart where $\lambda = \mathrm{d}f$, the metric $\widetilde{g} = \mathrm{e}^{-f} g$ is Kähler. For real vector fields U, V, the Levi–Civita connections ∇ of g and $\widetilde{\nabla}$ of $\widetilde{g}$ are related by

$$\widetilde{\nabla}_U V = \nabla_U V - \tfrac{1}{2}\big(\lambda(U)V + \lambda(V)U - g(U,V)\lambda^\sharp\big).$$

Substituting JV for V and using $\widetilde{\nabla} J = 0$ gives

$$(\nabla_U J)V = \tfrac{1}{2}\big(\lambda(JV)U - \lambda(V)JU + g(U,V)J\lambda^\sharp + g(JU,V)\lambda^\sharp\big). \tag{30.2}$$

Both sides are global tensors, so this identity holds on all of X.

For $a \in C^\infty(X, \mathbb{R})$, the conformal change $\omega' = \mathrm{e}^a \omega$ replaces the Lee form by $\lambda + \mathrm{d}a$. In particular, the de Rham class $[\lambda]$ depends only on the conformal class of ω. If λ is exact, the construction above gives a Kähler form on X itself.

30.2 Gauduchon representatives and the Lee class

In the conformal class of an LCK metric, the Gauduchon representative is the one whose Lee form is harmonic. A closed $(1,1)$-form built from the Lee form pairs positively with this Gauduchon metric when the structure is strictly LCK, even though the form is de Rham exact. It therefore defines a nonzero Bott–Chern class, which obstructs Kähler modifications.

Set

$$\lambda^{\mathrm{c}} = \frac{\mathrm{i}}{4\pi}(\lambda^{0,1} - \lambda^{1,0}),$$

so that $\lambda^{\mathrm{c}} = \mathrm{d}^{\mathrm{c}} f$ wherever $\lambda = \mathrm{d}f$. The closed real $(1,1)$-form

$$\kappa_\lambda := \mathrm{d}^{\mathrm{c}}\lambda = -\mathrm{d}\lambda^{\mathrm{c}} \tag{30.3}$$

connects the Lee class with Bott–Chern cohomology. It is de Rham exact, but it need not be dd^{c}-exact.

Proposition 30.2.1 *An LCK metric is Gauduchon if and only if its Lee form is co-closed for that metric. Consequently, the Gauduchon representative in its conformal class has harmonic Lee form. Moreover, for every LCK metric,*

$$\int_X \kappa_\lambda \wedge \omega^{n-1} = (n-1) \int_X \lambda \wedge \lambda^{\mathrm{c}} \wedge \omega^{n-1}. \tag{30.4}$$

If $\lambda \not\equiv 0$, the right side is strictly positive.

Proof On a chart where ω is conformally Kähler,

$$\mathrm{d}^{\mathrm{c}}(\omega^{n-1}) = (n-1)\lambda^{\mathrm{c}} \wedge \omega^{n-1}.$$

Let $*$ be the real Hodge star of the Riemannian metric associated with ω, and let d^* be the formal adjoint of the exterior derivative. By the pointwise Hodge-star identities for Hermitian metrics, $\lambda^{\mathrm{c}} \wedge \omega^{n-1}$ is a fixed nonzero constant multiple of $*\lambda$. Consequently, the Gauduchon condition $\mathrm{dd}^{\mathrm{c}}(\omega^{n-1}) = 0$ is equivalent to $\mathrm{d} * \lambda = 0$, that is, to $\mathrm{d}^*\lambda = 0$. Since $\mathrm{d}\lambda = 0$, co-closedness is equivalent to harmonicity. The Gauduchon representative exists by Theorem 4.1.1, and conformal changes preserve the LCK condition.

Stokes' theorem and $\mathrm{d}(\omega^{n-1}) = (n-1)\lambda \wedge \omega^{n-1}$ give

$$\begin{aligned} -\int_X \mathrm{d}\lambda^{\mathrm{c}} \wedge \omega^{n-1} &= -\int_X \lambda^{\mathrm{c}} \wedge \mathrm{d}(\omega^{n-1}) \\ &= (n-1) \int_X \lambda \wedge \lambda^{\mathrm{c}} \wedge \omega^{n-1}. \end{aligned}$$

For a real one-form, $\lambda \wedge \lambda^{\mathrm{c}} = \mathrm{i}\lambda^{1,0} \wedge \lambda^{0,1}/(2\pi)$ is semipositive, and its trace with respect to ω is positive wherever $\lambda \neq 0$. This proves the last assertion. □

The same construction applies to every closed real one-form and defines a linear map

$$j_X \colon H^1_{\mathrm{dR}}(X, \mathbb{R}) \longrightarrow H^{1,1}_{\mathrm{BC}}(X, \mathbb{R}), \qquad [\lambda] \longmapsto \{\kappa_\lambda\}, \tag{30.5}$$

which we call the *Lee map*. It is well defined, since replacing λ by $\lambda + \mathrm{d}a$ replaces κ_λ by $\kappa_\lambda - \mathrm{dd}^{\mathrm{c}}a$. Consequently, the pairing of $\{\kappa_\lambda\}$ with any Gauduchon form γ^{n-1} is well defined. For a strictly LCK structure, the Gauduchon representative in its conformal class gives a strictly positive pairing by (30.4). In particular $\{\kappa_\lambda\} \neq 0$, although its image in de Rham cohomology is zero.

Corollary 30.2.1 *If X carries a strictly LCK metric, then no class in $H^{1,1}_{\mathrm{BC}}(X, \mathbb{R})$ is big. In particular, X does not belong to Fujiki's class $\mathcal{C}$.*

Proof If there were a big class, the modification criterion recalled in Chapter 29 would give a modification $\pi \colon Y \to X$ with Y compact Kähler. The form $\pi^*\kappa_\lambda$ is closed and de Rham exact, so the $\partial\bar\partial$-lemma on Y gives a smooth real function h on Y with $\pi^*\kappa_\lambda = \mathrm{dd}^{\mathrm{c}}h$. Since π has degree one, pushing forward gives the current identity $\kappa_\lambda = \mathrm{dd}^{\mathrm{c}}(\pi_* h)$. By Stokes' theorem, the pairing of this current with a Gauduchon

form is zero, which contradicts (30.4) for the Gauduchon representative of the LCK structure. □

Proposition 30.2.2 *Suppose that X carries a strictly LCK metric with Lee form λ and that $\{\kappa_\lambda\}$ is pseudoeffective. Then X admits no balanced metric.*

Proof Let $T = \kappa_\lambda + \mathrm{dd}^c u \geq 0$ be a closed positive current representing the class. It is nonzero, since the class is nonzero, and it is de Rham exact. If a Hermitian metric b were balanced, then $\mathrm{d}(b^{n-1}) = 0$, and Stokes' theorem for currents would give

$$\int_X T \wedge b^{n-1} = 0.$$

This is impossible, since the mass of a nonzero positive $(1,1)$-current against a smooth strictly positive $(n-1, n-1)$-form is strictly positive. □

By Proposition 30.2.2, the following conjecture would exclude balanced metrics on every compact complex manifold carrying a strictly LCK metric; its origin is recalled in Section 36.18.

Conjecture 30.2.1 If X carries a strictly LCK metric with Lee form λ, then the Bott–Chern class $\{\kappa_\lambda\}$ is pseudoeffective.

The positivity in (30.4) tests the class only against the Gauduchon representative of the LCK metric. By Theorem 4.3.1, pseudoeffectivity requires a nonnegative pairing with *every* Gauduchon metric.

30.2.1 The least eigenvalue of Lee curvature

Fix a strictly LCK metric ω with Lee form λ. The volume of $\omega + t\kappa_\lambda$ increases with t as long as this form is positive. If the least eigenvalue of κ_λ relative to ω were a negative constant, the first degenerate form on this ray would have a kernel everywhere, and hence total volume zero. This contradiction gives a pointwise criterion for pseudoeffectivity of $\{\kappa_\lambda\}$.

Set $Q = \lambda \wedge \lambda^c \geq 0$. The form Q is nonzero on an open set, since λ is not exact and hence is not identically zero.

Proposition 30.2.3 *Put $\eta_t = \omega + t\kappa_\lambda$ for $t \in \mathbb{R}$. The polynomial $V(t) = \int_X \eta_t^n$ has degree at most $n-1$, and on the open interval of those t for which $\eta_t > 0$,*

$$V'(t) = n(n-1)\int_X Q \wedge \omega \wedge \eta_t^{n-2} > 0.$$

Proof Since $\mathrm{d}\eta_t = \lambda \wedge \omega$ and $\mathrm{d}\lambda^{\mathrm{c}} = -\kappa_\lambda$, Stokes' theorem gives

$$0 = \int_X \mathrm{d}(\lambda^{\mathrm{c}} \wedge \eta_t^{n-1}) = -\int_X \kappa_\lambda \wedge \eta_t^{n-1} + (n-1)\int_X Q \wedge \omega \wedge \eta_t^{n-2}.$$

Since $V'(t) = n\int_X \kappa_\lambda \wedge \eta_t^{n-1}$, this proves the formula. When $\eta_t > 0$, the integrand $Q \wedge \omega \wedge \eta_t^{n-2}$ is positive wherever $\lambda \neq 0$ (for $n = 2$ it is $Q \wedge \omega$), so its integral is strictly positive. Finally, κ_λ is closed and de Rham exact, so

$$\int_X \kappa_\lambda^n = -\int_X \mathrm{d}(\lambda^{\mathrm{c}} \wedge \kappa_\lambda^{n-1}) = 0.$$

This removes the coefficient of degree n. □

Theorem 30.2.1 *If the least eigenvalue of κ_λ relative to ω is constant on X, then it is zero and κ_λ is semipositive. Its ordinary Bott–Chern class is nonzero and pseudoeffective.*

If all relative eigenvalues are constant, the rank of κ_λ is constant and lies between 1 and $n-1$. In particular, the conclusion holds on a compact quotient $\Gamma\backslash G$ of a Lie group with left-invariant complex structure and a left-invariant strictly LCK metric.

Proof Let r be the constant least relative eigenvalue. If $r < 0$, put $t_0 = -1/r$. Then $\eta_t > 0$ for $0 \leq t < t_0$, whereas η_{t_0} is semipositive with a kernel at every point. Thus $V(t_0) = 0$. The preceding proposition and polynomial continuity instead give $V(t_0) \geq V(0) > 0$. This proves $r \geq 0$ and semipositivity.

The nonnegative smooth form κ_λ^n has integral zero by the same proposition, hence vanishes everywhere. Thus the least eigenvalue is zero at every point. As in the discussion of the Lee map (30.5), the Lee degree (30.4) of the Gauduchon representative in the conformal class of ω shows that the class is nonzero. It is pseudoeffective because its representative κ_λ is semipositive.

Constant spectrum gives constant rank. The positive pairing (30.4) for ω excludes rank zero, and the vanishing top power excludes rank n. On the stated compact quotient, the matrix of κ_λ relative to ω is constant in a left-invariant frame, which proves the last assertion. □

The class is also nef: the same representative is bounded below by $-\epsilon h$ for every Hermitian form h and every $\epsilon > 0$. Thus Proposition 30.2.2 excludes balanced metrics under the least-eigenvalue hypothesis.

The hypothesis concerns the chosen LCK metric and only its least eigenvalue; the other eigenvalues may vary, as the second example below shows. The first example shows that, without the constant-minimum condition, the first degenerate form on the ray $\omega + t\kappa_\lambda$ can have positive total volume.

Example 30.2.1 For every $n \geq 2$, the scalar Hopf manifold H_n of Section 9.6 carries a strictly LCK metric with indefinite Lee curvature and a smooth semipositive representative of its ordinary Bott–Chern class. The first degenerate form on the ray $\omega + t\kappa_\lambda$, $t > 0$, has positive total volume.

Proof Fix $0 < \epsilon < 1$ and, in coordinates $z = (z_1, \dots, z_n)$ on $\mathbb{C}^n$, put

$$\rho = \sum_j |z_j|^2, \qquad \Phi = \rho + \epsilon \operatorname{Re}(z_1^2), \qquad \omega = \frac{\mathrm{dd^c}\rho}{\Phi}, \qquad \lambda = -\mathrm{d}\log\Phi.$$

The bounds $(1-\epsilon)\rho \leq \Phi \leq (1+\epsilon)\rho$ give positivity. Since $\mathrm{dd^c}\Phi = \mathrm{dd^c}\rho$ and both Φ and $\mathrm{dd^c}\rho$ scale by e^{-2} under $z \mapsto \mathrm{e}^{-1}z$, the metric and Lee form descend and satisfy the LCK equation. The Lee period from z to $\mathrm{e}^{-1}z$ is 2, which proves strictness.

Here $\kappa_\lambda = \mathrm{dd^c}\log\Phi$. With $F_i = \bar{z}_i + \epsilon z_1 \delta_{i1}$, its complex Hessian is

$$(\log\Phi)_{i\bar{j}} = \frac{\delta_{ij}}{\Phi} - \frac{F_i\overline{F_j}}{\Phi^2}.$$

Relative to ω, the eigenvalues are 1 with multiplicity $n-1$ and $h = 1 - |F|^2/\Phi$. Writing $z_1 = x + \mathrm{i}y$ and $B = \sum_{j>1}|z_j|^2$, one finds

$$h = \frac{\epsilon((1-\epsilon)y^2 - (1+\epsilon)x^2)}{(1+\epsilon)x^2 + (1-\epsilon)y^2 + B}, \qquad -\epsilon \leq h \leq \epsilon.$$

At $(1, 0, \dots, 0)$ the spectrum is $-\epsilon, 1, \dots, 1$, hence is indefinite. The bounded smooth function $u = \log\rho - \log\Phi$ is deck invariant, and

$$\kappa_\lambda + \mathrm{dd^c}u = \mathrm{dd^c}\log\rho \geq 0,$$

by the Cauchy–Schwarz inequality for the Hessian of $\log\rho$. The correction also gives the standard Hopf conformal representative: $\mathrm{e}^{-u}\omega = \mathrm{dd^c}\rho/\rho = \omega_n$.

Since the least value of h is $-\epsilon$, the first degenerate form occurs at $t_0 = 1/\epsilon$. At that value,

$$1 + h/\epsilon = \frac{2(1-\epsilon)y^2 + B}{\Phi}.$$

It vanishes only on the real z_1 rays with the remaining coordinates zero. The other relative eigenvalues of $\omega + t_0\kappa_\lambda$ equal $1 + 1/\epsilon > 0$. The degenerate form is therefore semipositive and its top power is positive on a nonempty open set, proving positive total volume. □

Example 30.2.2 For every $n \geq 2$, the same H_n carries a family of strictly LCK metrics whose least Lee-curvature eigenvalue is identically zero and whose remaining relative spectrum is nonconstant. Thus constancy of the least eigenvalue is strictly weaker than constancy of the whole spectrum.

Proof Put $q = |z_1|^2$ and, for $0 < \epsilon < 1$, define

$$\Psi = \rho + \epsilon \frac{q^2}{\rho}, \qquad \omega_\epsilon = \frac{\mathrm{dd^c}\Psi}{\rho}, \qquad \lambda = -\mathrm{d}\log\rho.$$

For $a = q/\rho \in [0,1]$, direct differentiation gives

$$\left(\frac{q^2}{\rho}\right)_{i\bar{j}} = \frac{2}{\rho}(q_i - a\rho_i)(q_{\bar{j}} - a\rho_{\bar{j}}) + 2a\delta_{i1}\delta_{j1} - a^2\delta_{ij}.$$

The first two terms are positive semidefinite and $a^2 \leq 1$, so $\mathrm{dd^c}\Psi \geq (1-\epsilon)\mathrm{dd^c}\rho > 0$. The potential is smooth on the punctured space and scales by e^{-2} under the deck map. Hence ω_ϵ descends, satisfies the LCK equation and has Lee period 2.

Its Lee curvature is $\mathrm{dd^c}\log\rho$, which is semipositive of rank $n-1$. The least relative eigenvalue is consequently zero everywhere. At $(1,0,\dots,0)$, the Hessian of q^2/ρ is $\mathrm{diag}(1,-1,\dots,-1)$, giving the spectrum

$$0, (1-\epsilon)^{-1}, \dots, (1-\epsilon)^{-1}.$$

At $(0,1,0,\dots,0)$ the term q^2/ρ vanishes to fourth order, so its Hessian is zero and the spectrum is $0,1,\dots,1$. These spectra differ for every $\epsilon > 0$. □

30.3 Ordinary and twisted potentials

Let ω be an LCK metric with Lee form λ.

The Kähler covering provides a potential calculus compatible with the LCK equation $\mathrm{d}\omega = \lambda \wedge \omega$. Conjugating the ordinary differential operators by the local conformal factor produces a twisted complex Hessian on X.

For the fixed closed one-form λ, define

$$\mathrm{d}_\lambda = \mathrm{d} - \lambda\wedge, \qquad \mathrm{d}^{\mathrm{c}}_\lambda = \mathrm{d}^{\mathrm{c}} - \lambda^{\mathrm{c}}\wedge, \qquad D_\lambda = \mathrm{d}_\lambda\mathrm{d}^{\mathrm{c}}_\lambda.$$

On a chart where $\lambda = \mathrm{d}f$, these operators satisfy

$$\mathrm{d}_\lambda = \mathrm{e}^f\mathrm{d}\mathrm{e}^{-f}, \qquad \mathrm{d}^{\mathrm{c}}_\lambda = \mathrm{e}^f\mathrm{d}^{\mathrm{c}}\mathrm{e}^{-f}, \qquad D_\lambda = \mathrm{e}^f\mathrm{dd^c}\mathrm{e}^{-f}. \tag{30.6}$$

These identities show that $D_\lambda u$ is a real $(1,1)$-form for every smooth real function u, and that $\mathrm{d}_\lambda D_\lambda u = 0$.

Proposition 30.3.1 *Let u be a smooth real function. Then*

$$\mathrm{d}(\omega + \mathrm{dd}^{\mathrm{c}}u) - \lambda \wedge (\omega + \mathrm{dd}^{\mathrm{c}}u) = -\lambda \wedge \mathrm{dd}^{\mathrm{c}}u,$$

whereas

$$\mathrm{d}(\omega + D_\lambda u) = \lambda \wedge (\omega + D_\lambda u).$$

Consequently, an ordinary potential preserves the LCK equation with the same Lee form precisely when $\lambda \wedge \mathrm{dd}^{\mathrm{c}}u = 0$. A twisted potential always preserves that equation, and it gives an LCK metric whenever $\omega + D_\lambda u > 0$.

Proof The first identity follows from $\mathrm{ddd}^{\mathrm{c}}u = 0$ and $\mathrm{d}\omega = \lambda \wedge \omega$. The second follows from $\mathrm{d}_\lambda \omega = 0$ and (30.6). □

On the Kähler covering, a twisted potential becomes the automorphic function $\widetilde{u} = \mathrm{e}^{-F}\pi^* u$, and

$$\mathrm{e}^{-F}\pi^*(\omega + D_\lambda u) = \widetilde{\omega} + \mathrm{dd}^{\mathrm{c}}\widetilde{u}. \tag{30.7}$$

The top power $(\widetilde{\omega} + \mathrm{dd}^{\mathrm{c}}\widetilde{u})^n$ transforms by χ^n under deck transformations, so its ratio to $\widetilde{\omega}^n$ is invariant and descends to X. A measure on X is obtained from this ratio only after a metric in the conformal class has been fixed; the automorphic top-degree form cannot simply be integrated over the noncompact cover.

The two kinds of potentials also differ in their normalization. Expanding the twisted Hessian gives

$$D_\lambda u = \mathrm{dd}^{\mathrm{c}}u - \mathrm{d}u \wedge \lambda^{\mathrm{c}} - \lambda \wedge \mathrm{d}^{\mathrm{c}}u + u\big(\kappa_\lambda + \lambda \wedge \lambda^{\mathrm{c}}\big). \tag{30.8}$$

In particular, $D_\lambda 1 = \kappa_\lambda + \lambda \wedge \lambda^{\mathrm{c}}$ need not vanish, so adding a constant to u can change the metric $\omega + D_\lambda u$. The usual normalization $\sup_X u = 0$ is therefore not harmless for twisted equations.

For a chosen Lee form λ in its de Rham class, the real twisted Bott–Chern group in bidegree $(1,1)$ is

$$H^{1,1}_{\mathrm{BC},\lambda}(X,\mathbb{R}) = \frac{\{\eta \in C^\infty(X, \Lambda^{1,1}_{\mathbb{R}}) : \mathrm{d}_\lambda \eta = 0\}}{D_\lambda C^\infty(X,\mathbb{R})}.$$

This is Bott–Chern cohomology with values in the weight bundle [OV24b, Section 18.5]. It differs both from ordinary Bott–Chern cohomology and from the quotient of all backgrounds by ordinary dd^{c}-terms considered in Chapter 25.

30.3.1 The Vaisman metric on the Hopf manifold

Return to the Hopf threefold and the forms s, α, β and $\omega = \alpha + \beta$ of Chapter 9; recall that $s = \log |z|^2$ on the covering $\mathbb{C}^3 \setminus \{0\}$. The radial description makes the Lee form explicit and shows that this Gauduchon metric is also Vaisman.

Proposition 30.3.2 *The Gauduchon form $\omega = \alpha + \beta$ of Chapter 9 is Vaisman, with Lee form $\lambda = -\mathrm{d}s$, and*

$$\kappa_\lambda = \alpha, \qquad D_\lambda 1 = \omega.$$

In particular, the bounded mass property of Definition 7.5.1 fails for the ordinary potential theory of a background that is simultaneously Gauduchon and Vaisman.

Proof Using $\alpha = \mathrm{d}\mathrm{d}^{\mathrm{c}} s$ and $\beta = \mathrm{d}s \wedge \mathrm{d}^{\mathrm{c}} s$, direct differentiation gives

$$\mathrm{d}\omega = -\mathrm{d}s \wedge \alpha = -\mathrm{d}s \wedge \omega, \qquad \mathrm{e}^s \omega = \mathrm{d}\mathrm{d}^{\mathrm{c}}(\mathrm{e}^s) = \mathrm{d}\mathrm{d}^{\mathrm{c}} |z|^2.$$

Although s is not a function on the quotient, $\mathrm{d}s$ descends to it. The integral of $\mathrm{d}s$ along the loop induced by $z \mapsto \mathrm{e}^{-1} z$ is -2, so the Lee form is not exact.

Let g_{S^5} be the round metric on the unit sphere. Up to a constant factor, the Riemannian metric associated with ω on the cover is

$$\frac{|\mathrm{d}z|^2}{|z|^2} = \mathrm{d}t^2 + g_{S^5}, \qquad t = \log |z|.$$

The one-form $\mathrm{d}t$ is parallel for this product metric, which proves the Vaisman condition. Finally, $\kappa_\lambda = \mathrm{d}\mathrm{d}^{\mathrm{c}} s = \alpha$ and $\lambda \wedge \lambda^{\mathrm{c}} = \beta$, so $D_\lambda 1 = \omega$ by (30.8). The unbounded masses are given by Theorem 9.0.1. □

Since $\kappa_\lambda = \alpha$ is smooth and semipositive, the class $\{\alpha\}$ is nef and its non-nef locus is empty. By Corollary 30.2.1 it is not big, so its non-Kähler locus is all of X by the convention for classes that are not big in Chapter 29. These statements concern the closed class $\{\alpha\}$; the Monge–Ampère equations on the Hopf manifold use instead the strictly positive, nonclosed form ω.

The unbounded masses arise in the ordinary family $\omega + \mathrm{d}\mathrm{d}^{\mathrm{c}} u$, whose members need not satisfy the LCK equation. The volume problems below impose this additional geometric constraint, since they prescribe the volume among LCK metrics with a fixed Lee class.

30.4 Volume equations in Gauduchon gauge

Prescribing the volume of an LCK metric in Gauduchon gauge couples two equations: one for the determinant and one for the conformal factor. To compare metrics with a fixed Lee class, we first identify the twisted groups for cohomologous Lee forms. For a closed real one-form λ, smooth real functions a, u and a smooth differential form η, multiplication by e^{-a} gives

$$\mathrm{d}_\lambda(\mathrm{e}^{-a}\eta) = \mathrm{e}^{-a}\mathrm{d}_{\lambda+\mathrm{d}a}\eta, \qquad D_\lambda(\mathrm{e}^{-a}u) = \mathrm{e}^{-a}D_{\lambda+\mathrm{d}a}u.$$

We use this identification when comparing classes; the additive constant in a corresponds to a constant rescaling of the weight bundle.

Let ω_1 be an LCK Gauduchon metric with Lee form λ, and let ω_2 be an LCK metric with Lee form $\lambda + \mathrm{d}a$. If their twisted classes agree under the chosen identification, then

$$\mathrm{e}^{-a}\omega_2 = \omega_1 + D_\lambda u =: A > 0$$

for a smooth real function u. Substituting $\omega_2 = \mathrm{e}^a A$ shows that ω_2 has the same volume form as ω_1 and is Gauduchon exactly when

$$\begin{cases} A^n = \mathrm{e}^{-na}\omega_1^n, \\ \mathrm{dd}^{\mathrm{c}}\left(\mathrm{e}^{(n-1)a}A^{n-1}\right) = 0. \end{cases} \tag{30.9}$$

The scalar Hermitian equation of Chapter 10 does not impose the second equation, and it uses $\mathrm{dd}^{\mathrm{c}}u$ in place of $D_\lambda u$. The prescribed-volume Gauduchon theorem, Theorem 10.10.1, does not assert that its solution is LCK or lies in the given twisted class.

Problem 30.4.1 Does a compact LCK manifold have at most one Gauduchon LCK metric with a given Lee class, a given twisted Bott–Chern class and a given smooth positive volume form?

This is the uniqueness problem posed in [OV24b, Question 47.56(3), p. 665], with the identification of classes made explicit. The question concerns metrics: uniqueness of a scalar potential u would additionally require control of the kernel of D_λ. The counterexamples with degenerate densities in Chapter 12 concern ordinary potentials and do not decide this question.

30.4.1 Invariant volumes on Vaisman manifolds

The flows of the Lee and anti-Lee vector fields preserve a Vaisman metric. For a prescribed volume invariant under these flows, uniqueness for the smooth Hermitian equation forces the potential to be constant along them. Stokes' theorem in the transverse directions then fixes the normalizing constant, and changing the Lee form

by the differential of the potential produces a Vaisman metric with the prescribed volume.

Theorem 30.4.1 *Let ω be a Vaisman metric on X with Lee form λ, and let μ be a smooth positive volume form invariant under the flows of both the Lee field $\lambda^\sharp$ and the anti-Lee field $J\lambda^\sharp$. Then there is a unique Vaisman metric ω' with Lee class $[\lambda]$ and $(\omega')^n = \mu$.*

We first describe the transverse geometry and the integration of transverse forms. Let g be the Riemannian metric of ω. Since λ is parallel and not identically zero, it has constant squared norm $s = |\lambda|_g^2 > 0$. Write

$$A = s^{-1}\lambda^\sharp, \qquad B = JA, \qquad \tau = \frac{s}{4\pi}, \qquad \eta = \kappa_\lambda.$$

Thus $\lambda(A) = 1$ and $\lambda(B) = 0$. Since $\nabla\lambda = 0$, the identity (30.2) gives

$$\begin{aligned} \tau\omega &= \eta + \lambda \wedge \lambda^{\mathrm{c}}, \\ \eta(U, JV) &= \frac{1}{4\pi}\big(sg(U,V) - \lambda(U)\lambda(V) - \lambda(JU)\lambda(JV)\big). \end{aligned} \tag{30.10}$$

To obtain these identities, use $\lambda^{\mathrm{c}} = -\lambda \circ J/(4\pi)$ and antisymmetrize $\nabla\lambda^{\mathrm{c}}$; the terms containing $\nabla\lambda$ vanish. In particular, η is a closed semipositive $(1,1)$-form whose kernel is the span $\mathcal{F}$ of A and B.

The field A is parallel, and the same connection formula yields

$$\nabla_U B = \tfrac{1}{2}\big(-JU + \lambda(U)B + \lambda(JU)A\big), \qquad \nabla_A J = 0.$$

The first identity shows that B is a Killing field and that $[A, B] = 0$. The second shows that A is holomorphic, and integrability of J then shows that $B = JA$ is holomorphic. Their flows preserve g, J, λ and η. Consequently $\mathcal{F}$ is a holomorphic foliation by complex curves, and η induces a Kähler form on each local transverse space. A form is called *horizontal* if its contractions with A and B vanish.

A smooth function v is *basic* for $\mathcal{F}$ if $Av = Bv = 0$. For such a function, $\mathrm{d}v$, $\mathrm{d}^{\mathrm{c}}v$ and $\mathrm{dd}^{\mathrm{c}}v$ are horizontal and invariant under both flows; for $\mathrm{dd}^{\mathrm{c}}v$ this follows from Cartan's formula. Put $m = n - 1$ and $\eta_v = \eta + \mathrm{dd}^{\mathrm{c}}v$. Then

$$\int_X \eta_v^m \wedge \lambda \wedge \lambda^{\mathrm{c}} = \int_X \eta^m \wedge \lambda \wedge \lambda^{\mathrm{c}}. \tag{30.11}$$

Indeed, $\eta_v^m - \eta^m = \mathrm{dd}^{\mathrm{c}}v \wedge T$, where $T = \sum_{j=0}^{m-1} \eta_v^j \wedge \eta^{m-1-j}$ is closed and horizontal. Since $\mathrm{d}(\lambda \wedge \lambda^{\mathrm{c}}) = \lambda \wedge \eta$, the differential of $\mathrm{d}^{\mathrm{c}}v \wedge T \wedge \lambda \wedge \lambda^{\mathrm{c}}$ is $\mathrm{dd}^{\mathrm{c}}v \wedge T \wedge \lambda \wedge \lambda^{\mathrm{c}}$ plus a term containing the horizontal form $\mathrm{d}^{\mathrm{c}}v \wedge T \wedge \eta$ of degree $2m + 1$, which vanishes because the transverse real dimension is $2m$. Stokes' theorem therefore proves (30.11).

The following transverse maximum principle identifies the metric once its scale has been fixed.

Proposition 30.4.1 *Let $\mathcal{F}$ be a foliation of a connected compact manifold M with a transverse holomorphic structure of complex dimension $m \geq 1$. Let η_0, η_1 be transverse Kähler forms, and let v be a smooth real basic function, with $\mathrm{dd^c} v$ computed in transverse holomorphic coordinates. Suppose*

$$\eta_1 = \eta_0 + \mathrm{dd^c} v, \qquad \eta_1^m = \eta_0^m.$$

Then v is constant and $\eta_1 = \eta_0$.

Proof The function v is constant on each local plaque. On every transverse coordinate chart it satisfies

$$\mathrm{dd^c} v \wedge \sum_{j=0}^{m-1} \eta_1^j \wedge \eta_0^{m-1-j} = 0.$$

This is a linear elliptic equation without zeroth-order term. Let $c = \max_M v$. If $v = c$ at some point, the strong maximum principle on a connected transverse chart shows that $v = c$ throughout that chart, and since v is basic, $v = c$ on the corresponding foliation chart. Hence $\{v = c\}$ is nonempty, open and closed, and connectedness of M proves the assertion. □

Proof (of Theorem 30.4.1) For a constant $c > 0$, the metric $c\omega$ is Vaisman with Lee form λ, its Lee and anti-Lee fields are $c^{-1}\lambda^\sharp$ and $c^{-1}J\lambda^\sharp$, and its constant τ is $c^{-1}\tau$. Replacing ω by $(\int_X \mu / \int_X \omega^n)^{1/n}\omega$, we may therefore assume $\int_X \mu = \int_X \omega^n$. Replacing then ω by $\tau\omega$ and μ by $\tau^n\mu$ preserves this equality, and $\omega' \mapsto \tau\omega'$ maps the solutions for μ bijectively onto those for $\tau^n\mu$. We may therefore also assume $\tau = 1$, so that $\omega = \eta + \lambda \wedge \lambda^\mathrm{c}$. Write $\mu = f\omega^n$; since μ and ω^n are invariant under both flows, the positive smooth function f is basic.

Existence. By Theorem 10.1.1, there are a unique smooth function u and a unique constant b such that

$$(\omega + \mathrm{dd^c} u)^n = \mathrm{e}^b f\omega^n, \qquad \omega + \mathrm{dd^c} u > 0, \qquad \sup_X u = 0.$$

Pullback by the flow of A or of B preserves the equation and the normalization, so uniqueness shows that u is basic. Hence $\eta_u = \eta + \mathrm{dd^c} u$ is a transverse Kähler form, and

$$(\omega + \mathrm{dd^c} u)^n = n\eta_u^m \wedge \lambda \wedge \lambda^\mathrm{c}.$$

Integrating over X and using (30.11) and $\int_X \mu = \int_X \omega^n$ gives $b = 0$.

Set $\lambda_u = \lambda - \mathrm{d}u$ and

$$\omega_u' = \eta_u + \lambda_u \wedge \lambda_u^\mathrm{c}. \tag{30.12}$$

Here $\eta_u = \kappa_{\lambda_u}$ has kernel $\mathcal{F}$, while $\lambda_u(A) = 1$ and $\lambda_u(B) = 0$; together these facts give $\omega_u' > 0$. Direct differentiation gives $\mathrm{d}\omega_u' = \lambda_u \wedge \omega_u'$. Since u is basic, A is a

holomorphic Killing field for the Riemannian metric g'_u of ω'_u, and (30.12) gives

$$g'_u(A,\cdot) = \frac{1}{4\pi}\lambda_u.$$

Since λ_u is closed and its metric dual $4\pi A$ is Killing, λ_u is parallel by Section 30.1, so ω'_u is Vaisman with Lee class $[\lambda]$.

Finally, η_u^m has top transverse degree, and every term of $\lambda_u \wedge \lambda_u^{\mathrm{c}} - \lambda \wedge \lambda^{\mathrm{c}}$ contains a horizontal one-form. Therefore

$$(\omega'_u)^n = n\eta_u^m \wedge \lambda \wedge \lambda^{\mathrm{c}} = (\omega + \mathrm{dd}^{\mathrm{c}} u)^n = \mu.$$

Uniqueness. Let $\widehat{\omega}$ be another Vaisman solution, with Lee form $\widehat{\lambda} = \lambda + \mathrm{d}v$, and denote by $\widehat{s}, \widehat{\tau}, \widehat{A}, \widehat{B}, \widehat{\eta}$ the corresponding objects for $\widehat{\omega}$. The form $\eta + \widehat{\eta}$ is de Rham exact and semipositive. Its top power is nonnegative and has zero integral, so it vanishes everywhere, and $\eta + \widehat{\eta}$ has a nonzero kernel at every point. The kernel of a sum of two semipositive Hermitian forms is the intersection of their kernels. Since both kernels are complex lines, they coincide, and both metrics define the same foliation $\mathcal{F}$.

The nowhere vanishing holomorphic vector field $A - \mathrm{i}B$ trivializes the holomorphic tangent bundle of $\mathcal{F}$. The analogous field $\widehat{A} - \mathrm{i}\widehat{B}$ is a holomorphic multiple of it, and the multiple is constant because X is compact. Write $\widehat{A} = aA + bB$ with $a, b \in \mathbb{R}$. Since $\widehat{\lambda}(\widehat{A}) = 1$ and $\widehat{\lambda}(\widehat{B}) = 0$,

$$\widehat{A}v = 1 - a, \qquad \widehat{B}v = b.$$

Both fields are complete and v is bounded, so v cannot grow linearly along their flows; hence these constants vanish, and $a = 1$, $b = 0$. Thus $\widehat{A} = A$, $\widehat{B} = B$, and v is basic.

From (30.10) we obtain the contraction formulas

$$\iota_B \iota_A \widehat{\omega}^n = \frac{n}{4\pi\widehat{\tau}^n}\widehat{\eta}^m, \qquad \iota_B \iota_A (\omega'_u)^n = \frac{n}{4\pi}\eta_u^m. \tag{30.13}$$

Since the two volume forms are equal, $\widehat{\eta}^m = \widehat{\tau}^n \eta_u^m$. Since $\widehat{\eta} = \eta - \mathrm{dd}^{\mathrm{c}} v$ and v is basic, integrating against $\lambda \wedge \lambda^{\mathrm{c}}$ and applying (30.11) twice gives $\widehat{\tau}^n = 1$. Thus $\widehat{\tau} = 1$ and $\widehat{\eta}^m = \eta_u^m$. Since $\widehat{\eta} - \eta_u = -\mathrm{dd}^{\mathrm{c}}(v + u)$, Proposition 30.4.1 shows that $v + u$ is constant. It follows that $\widehat{\lambda} = \lambda_u$, $\widehat{\eta} = \eta_u$ and $\widehat{\omega} = \omega'_u$. □

30.4.2 Radial volumes on scalar Hopf manifolds

Let $n \geq 2$ and $m = n - 1$. We use the scalar Hopf manifold H_n, its projection $p_n \colon H_n \to \mathbb{P}^m$ and the forms $\alpha = \alpha_n = \mathrm{dd}^{\mathrm{c}} r$ and $\beta = \beta_n = \mathrm{d}r \wedge \mathrm{d}^{\mathrm{c}} r$ of Section 9.6, where we write $r = \log|z|^2$ for the function denoted s there; it is well defined modulo 2 on H_n. Recall that α has rank m with kernel the complex radial line, β has rank one,

$\alpha^m \wedge \beta$ is a positive volume form, every fiber of p_n has β-area 2, and $\int_{H_n} \alpha^m \wedge \beta = 2$. We consider all metrics of the radial form

$$\omega = C(r)\alpha + D(r)\beta, \qquad C, D \in C^\infty(\mathbb{R}/2\mathbb{Z}), \quad C, D > 0. \tag{30.14}$$

The circle $\mathbb{R}/2\mathbb{Z}$ of values of r is oriented by increasing r. The *Lee period* of such a metric is the integral of its Lee form over a radial loop along which r increases from 0 to 2.

Lemma 30.4.1 *Every metric of the form* (30.14) *is strictly LCK. Its Lee form and Lee period are*

$$\lambda = l(r)\,\mathrm{d}r, \qquad l = \frac{C' - D}{C}, \qquad L = -\int_0^2 \frac{D}{C}\,\mathrm{d}r < 0.$$

The metric is Gauduchon if and only if $C^{m-1}(C' - D) = k$ for a constant k. In terms of the positive smooth 2-periodic function $F = C^m D$, this condition is equivalent to

$$C^m C' = kC + F, \qquad D = \frac{F}{C^m}. \tag{30.15}$$

Every positive periodic solution has $k < 0$, and

$$\begin{aligned} k\int_0^2 C\,\mathrm{d}r + \int_0^2 F\,\mathrm{d}r &= 0, \\ l = kC^{-m}, \qquad L(C) &= -\int_0^2 \frac{F}{C^{m+1}}\,\mathrm{d}r. \end{aligned} \tag{30.16}$$

Proof Besides $\alpha^{m+1} = \beta^2 = 0$ and $\mathrm{dd^c}\beta = -\alpha^2$ from Section 9.6, we use

$$\mathrm{d}\beta = -\mathrm{d}r \wedge \alpha, \qquad \mathrm{d^c}\beta = -\alpha \wedge \mathrm{d^c}r.$$

These identities give $\mathrm{d}\omega = (C' - D)\mathrm{d}r \wedge \alpha = \lambda \wedge \omega$. The closed form λ is the Lee form, since wedging one-forms with ω is injective. The integral of C'/C over a period is zero, which proves the formula for L; since $L \neq 0$, the Lee form is not exact. Two radial Lee forms have the same de Rham class exactly when their periods agree: a difference with zero period has a periodic primitive, and the radial loop detects the converse.

For a smooth function G of r, the chain rule gives $\mathrm{dd^c}G = G'\alpha + G''\beta$. Also $\mathrm{d^c}(G\beta) = -G\alpha \wedge \mathrm{d^c}r$, so

$$\mathrm{dd^c}(G\beta) = -G'\alpha \wedge \beta - G\alpha^2.$$

Expanding the powers of ω therefore gives

$$\begin{aligned} \omega^m &= C^m\alpha^m + mC^{m-1}D\alpha^{m-1} \wedge \beta, \\ \omega^n &= nC^m D\alpha^m \wedge \beta, \end{aligned}$$

and

$$\mathrm{dd}^{\mathrm{c}}(\omega^m) = \bigl[(C^m)'' - m(C^{m-1}D)'\bigr]\alpha^m \wedge \beta.$$

Since $(C^m)'' - m(C^{m-1}D)' = m\bigl(C^{m-1}(C' - D)\bigr)'$, this vanishes exactly when $C^{m-1}(C' - D)$ is constant. Multiplying the identity $C^{m-1}(C' - D) = k$ by C gives (30.15). Since $C^m C'$ has zero integral over a period, integrating (30.15) gives the first identity in (30.16), which forces $k < 0$. The formulas for l and L follow from the formula for the Lee form. □

Lemma 30.4.2 *For fixed $F > 0$ and each $k < 0$, there is at most one positive 2-periodic solution of (30.15). If $k_2 > k_1$, the corresponding solutions satisfy $C_2 > C_1$ everywhere and $L(C_2) > L(C_1)$.*

Proof Suppose $k_2 > k_1$, and put $R = C_2 - C_1$. At a zero of R, the equation gives $R' = (k_2 - k_1)C_1^{1-m} > 0$. Such zeros are isolated, so there are finitely many on the circle. A periodic differentiable function cannot cross zero only upwards, since returning around the circle would require a downward crossing. Hence R has no zero and a constant strict sign. The identity $\int_0^2 C_i \,\mathrm{d}r = -\int_0^2 F \,\mathrm{d}r/k_i$ shows that this sign is positive, and the formula for $L(C)$ in (30.16) then gives $L(C_2) > L(C_1)$.

For equal k, uniqueness for the smooth scalar equation $C' = kC^{1-m} + FC^{-m}$ on $C > 0$ shows that two solutions which meet coincide. Two solutions which never meet are strictly ordered, which contradicts the equality of their integrals. This proves uniqueness for fixed k. □

Theorem 30.4.2 *For every positive smooth 2-periodic function F and every $L_0 < 0$, there is exactly one metric of the form (30.14) which is Gauduchon, has Lee period L_0, and satisfies*

$$\omega^n = nF(r)\alpha^m \wedge \beta.$$

Proof Fix $k < 0$, put $c = -k$, and let $f_0 = \min F > 0$ and $f_1 = \max F$. The constant functions f_0/c and f_1/c are a subsolution and a supersolution, respectively, of $C' = C^{-m}(F - cC)$. By comparison, the forward flow of this equation preserves the compact interval $I_k = [f_0/c, f_1/c]$ of positive numbers, and existence and uniqueness for smooth ordinary differential equations give a continuous time-two map $I_k \to I_k$. Its difference from the identity is nonnegative at the left endpoint and nonpositive at the right one, so the intermediate value theorem gives a fixed point. Since the equation is periodic, the solution through a fixed point extends to a smooth positive periodic solution. It is unique by Lemma 30.4.2 and satisfies

$$\frac{f_0}{-k} \le C_k(r) \le \frac{f_1}{-k}. \tag{30.17}$$

By Lemma 30.4.2, the period $L(k) = -\int_0^2 F/C_k^{m+1} \,\mathrm{d}r$ is strictly increasing in k. It is also continuous. Indeed, for k in a compact neighborhood of any $k_0 < 0$, (30.17) gives uniform positive lower and finite upper bounds for C_k, and the equation then bounds all derivatives. Any sequence $k \to k_0$ therefore has smoothly convergent

subsequences, and the equation and uniqueness identify every limit with C_{k_0}. Thus $C_k \to C_{k_0}$ uniformly, which gives continuity of the integral.

The same bounds imply $L(k) \to 0$ as $k \uparrow 0$ and $L(k) \to -\infty$ as $k \to -\infty$. Hence exactly one k has period L_0. For this k, set $D = F/C_k^m$. The identities of Lemma 30.4.1 show that $C_k\alpha + D\beta$ has the required properties. Conversely, every metric of the form (30.14) with these properties gives a positive periodic solution of the same equation, so the ordering of the periods also proves uniqueness. □

Proposition 30.4.2 *Let ω be a metric as in Theorem 30.4.2, with Lee form λ. There is a positive smooth function U on H_n with $\omega = D_\lambda U$; in particular, the class of ω in $H^{1,1}_{\mathrm{BC},\lambda}(H_n, \mathbb{R})$ is zero. If ω_1, ω_2 are two such metrics, with positive potentials U_1, U_2 and cohomologous Lee forms λ_1 and $\lambda_2 = \lambda_1 + \mathrm{d}a$, then for every choice of the additive constant in a,*

$$\mathrm{e}^{-a}\omega_2 - \omega_1 = D_{\lambda_1}(\mathrm{e}^{-a}U_2 - U_1).$$

In particular, among metrics of the form (30.14), a solution of (30.9) is determined by its volume form and Lee class, even if the potential u in (30.9) is not assumed radial.

Proof Write $l = kC^{-m} < 0$ and $f(r) = \int_0^r l(s)\,\mathrm{d}s$ for real r. Then $f(r+2) = f(r) + L_0$, where $L_0 < 0$ is the Lee period of ω. The solution of $U' - lU = C$ given by

$$U(0) = \frac{\mathrm{e}^{L_0}}{1 - \mathrm{e}^{L_0}} \int_0^2 \mathrm{e}^{-f(s)} C(s)\,\mathrm{d}s,$$
$$U(r) = \mathrm{e}^{f(r)} \left(U(0) + \int_0^r \mathrm{e}^{-f(s)} C(s)\,\mathrm{d}s \right)$$

is positive on $[0, 2]$ and satisfies $U(2) = U(0)$. Since the equation is periodic, all derivatives also match at the endpoints, so U defines a positive smooth function on H_n. By (30.6), its twisted Hessian is

$$\begin{aligned} D_\lambda U &= (U' - lU)\alpha + \left[U'' - 2lU' + (l^2 - l')U\right]\beta \\ &= C\alpha + (C' - lC)\beta = \omega. \end{aligned}$$

This proves that the twisted class is zero. The identity $\mathrm{e}^{-a} D_{\lambda_2} U_2 = D_{\lambda_1}(\mathrm{e}^{-a} U_2)$ of Section 30.4 gives the formula for the difference. For two metrics of the form (30.14), equal volume forms give the same function F, and equal Lee classes give the same period, so Theorem 30.4.2 applies. This uniqueness concerns the metric, not the potentials representing it. □

Proposition 30.4.3 *If F is nonconstant, the metric of Theorem 30.4.2 is not Vaisman.*

Proof The radial complex line is orthogonal to $\ker \mathrm{d}r \cap \ker \mathrm{d}^{\mathrm{c}} r$. On a radial complex line, use a logarithmic coordinate $w = x + \mathrm{i}y$, so that $r = 2x + \text{constant}$ and $\beta = \pi^{-1}\mathrm{d}x \wedge \mathrm{d}y$. The Riemannian metric of ω restricts there to $D\pi^{-1}(\mathrm{d}x^2 + \mathrm{d}y^2)$, whence

$$|\mathrm{d}r|^2_\omega = \frac{4\pi}{D}, \qquad |\lambda|^2_\omega = 4\pi k^2 \frac{C^{-2m}}{D}.$$

A parallel Lee form has constant norm. Since $k \neq 0$, the Vaisman condition would therefore imply $D = QC^{-2m}$ for a constant $Q > 0$. At a maximum and at a minimum of C, where $C' = 0$, the Gauduchon identity would then give $kC^{m+1} + Q = 0$. This equation has only one positive root, so the maximum and the minimum of C coincide. Thus C, D and $F = C^m D$ would all be constant, a contradiction. □

Example 30.4.1 For

$$0 < \varepsilon < \min\left\{\frac{1}{2}, \frac{(3/2)^{1-m}}{4\pi}\right\},$$
$$C = 1 + \varepsilon\cos(\pi r), \qquad D = C' + C^{1-m},$$

the metric (30.14) is Gauduchon and non-Vaisman, with Lee form $\lambda = -C^{-m}\mathrm{d}r$ and density $F = C^m C' + C$. The probability volume $\mathrm{d}V = \frac{1}{2}\alpha^m \wedge \beta$ gives $\omega^n = 2nF\,\mathrm{d}V$ and $\int_{H_n} F\,\mathrm{d}V = 1$.

Proof Since $1/2 < C < 3/2$ and $m \geq 1$, we have $D \geq (3/2)^{1-m} - \pi\varepsilon > \frac{3}{4}(3/2)^{1-m} > 0$. The Gauduchon identity holds with $k = -1$, giving the Lee form and the positive density $F = C^m D$. Its values $F(0) = 1 + \varepsilon$ and $F(1) = 1 - \varepsilon$ differ, so Proposition 30.4.3 applies. The integral of $C^m C'$ over a period is zero and the average of C is one, so integration along the fibers of p_n gives $\int_{H_n} F\,\mathrm{d}V = 1$. □

The volume form in this example is not invariant under the flow of the Lee field of the Vaisman metric $\alpha + \beta$, which translates r, so Theorem 30.4.1 does not apply to it. For the new metric, the Gauduchon condition states that $lF/D = lC^m = k$ is constant. Measured with the volume form ω^n, the flux of its Lee field $\lambda^\sharp$ through each level set of r, oriented by increasing r, is $4\pi n\, lF/D = 4\pi nk$; here $k = -1$. The radial classification answers Problem 30.4.1 within the family (30.14); uniqueness among arbitrary LCK metrics remains open.

30.5 Chern–Ricci flatness and Vaisman rigidity

Return to an LCK metric ω on X with Lee form λ. Chern–Ricci flatness expresses the Riemannian Ricci tensor through λ and its covariant derivative. The Bianchi and Bochner identities then relate the part of this derivative that is not J-invariant to the divergence of the Lee field. When the metric is Gauduchon, this divergence vanishes by Proposition 30.2.1, and the identities force the Lee form to be parallel.

We write $\mathrm{Ric}^{\mathrm{Ch}} = \mathrm{Ric}/(2\pi)$ for the normalized Chern–Ricci form, which represents the first Bott–Chern class $c_1^{\mathrm{BC}}(X)$, with Ric as in Section 27.20.1.

Theorem 30.5.1 *Let ω be a Chern–Ricci-flat strictly LCK metric on X with Lee form λ, and let g be its associated Riemannian metric, with Levi–Civita connection ∇. Put $A = \lambda^\sharp$, $q = \mathrm{d}_g^*\lambda$ and*

$$S^-(U,V) = \tfrac{1}{2}\big((\nabla\lambda)(U,V) - (\nabla\lambda)(JU,JV)\big).$$

Then

$$\int_X |S^-|_g^2 \,\mathrm{d}\,\mathrm{vol}_g = \frac{1}{2}\int_X |q|^2 \,\mathrm{d}\,\mathrm{vol}_g\,.$$

For this metric the following conditions are equivalent: ω is Gauduchon; $q = 0$; A is real holomorphic; and ω is Vaisman.

Proof We compute the Ricci tensor without assuming that λ is co-closed, and compare the integrated Bianchi and Bochner identities; the Gauduchon condition enters only in Step 3. All norms, traces and contractions use g. Put

$$B = JA, \qquad S = \nabla\lambda, \qquad S^+(U,V) = \tfrac{1}{2}\big(S(U,V) + S(JU,JV)\big), \qquad u = |\lambda|_g^2.$$

Closedness of λ makes S symmetric, and $\mathrm{tr}_g\, S = -q$. The tensors S^+ and $S^- = S - S^+$ are pointwise orthogonal. Write $J\lambda = -\lambda\circ J$, so that $\lambda^{\mathrm{c}} = J\lambda/(4\pi)$.

Step 1: the Ricci tensor. We first show that

$$\mathrm{Ric}_g = nS^+ - (n-1)S + \tfrac{q-u}{2}g + \tfrac{1}{2}\lambda\otimes\lambda + \tfrac{n}{2}J\lambda\otimes J\lambda. \tag{30.18}$$

Locally write $\lambda = \mathrm{d}f$ and put $\widetilde{g} = \mathrm{e}^{-f}g$. The formula for the Ricci tensor under a conformal change in real dimension $2n$ gives

$$\mathrm{Ric}_{\widetilde{g}} = \mathrm{Ric}_g + (n-1)S + \tfrac{n-1}{2}\lambda\otimes\lambda - \tfrac{1}{2}\big(q + (n-1)u\big)g.$$

The local metric $\widetilde{g}$ is Kähler. Since the Chern–Ricci form of g vanishes, the Ricci form of $\widetilde{g}$, normalized as Ric, is $ni\partial\bar\partial f = (n/2)\mathrm{d}(J\lambda)$. Applying (30.2) to $(\nabla_U J\lambda)(V) = -S(U,JV) - \lambda((\nabla_U J)V)$ and antisymmetrizing gives

$$\mathrm{d}(J\lambda)(U,JV) = 2S^+(U,V) - ug(U,V) + \lambda(U)\lambda(V) + J\lambda(U)J\lambda(V).$$

Evaluating the Kähler Ricci form on (U, JV) and comparing with the conformal formula proves (30.18). Taking its trace gives the scalar curvature

$$s = (n-1)q - \tfrac{n-1}{2}u.$$

Step 2: the integral identity. Write $\|\cdot\|$ for the $L^2(g)$ norm and set

$$E_A = \int_X S(A,A)\,\mathrm{d}\,\mathrm{vol}_g, \qquad E_B = \int_X S(B,B)\,\mathrm{d}\,\mathrm{vol}_g\,.$$

Since $S(A,A) = A(u)/2$ and $\operatorname{div} A = -q$, integration by parts gives

$$\int_X uq\,\mathrm{d}\,\mathrm{vol}_g = 2E_A. \tag{30.19}$$

Pair the contracted Bianchi identity $\delta\,\mathrm{Ric}_g + \mathrm{d}s/2 = 0$ with λ and integrate. Here δ, minus the divergence, is the formal adjoint of the symmetrized covariant derivative. Thus

$$0 = \int_X \langle \mathrm{Ric}_g, S\rangle\,\mathrm{d}\,\mathrm{vol}_g + \tfrac{1}{2}\int_X sq\,\mathrm{d}\,\mathrm{vol}_g\,.$$

Substituting (30.18) and using $\langle S^+, S\rangle = |S^+|^2$, $\operatorname{tr}_g S = -q$, $(J\lambda)^\sharp = B$ and (30.19), we obtain

$$0 = n\|S^+\|^2 - (n-1)\|S\|^2 + \tfrac{n-2}{2}\|q\|^2 + \tfrac{4-n}{2}E_A + \tfrac{n}{2}E_B. \tag{30.20}$$

For the closed one-form λ, the integrated Bochner formula reads

$$\|q\|^2 = \|S\|^2 + \int_X \mathrm{Ric}_g(A,A)\,\mathrm{d}\,\mathrm{vol}_g\,.$$

The quartic terms in (30.18) cancel on (A,A), and $J\lambda(A) = 0$, so

$$\mathrm{Ric}_g(A,A) = (1-\tfrac{n}{2})S(A,A) + \tfrac{n}{2}S(B,B) + \tfrac{1}{2}qu.$$

Using (30.19), we obtain

$$\|q\|^2 = \|S\|^2 + \tfrac{4-n}{2}E_A + \tfrac{n}{2}E_B. \tag{30.21}$$

Substituting the value of $\frac{4-n}{2}E_A + \frac{n}{2}E_B$ given by (30.21) into (30.20) gives $n(\|S^+\|^2 - \|S\|^2) + n\|q\|^2/2 = 0$. Since $\|S\|^2 = \|S^+\|^2 + \|S^-\|^2$ by orthogonality, this is the asserted identity for S^-.

Step 3: rigidity. By Proposition 30.2.1, the Gauduchon condition is equivalent to $q = 0$. If $q = 0$, the integral identity forces $S = S^+$, hence $S(B,B) = S(JA,JA) = S(A,A)$. Equation (30.19) gives $E_A = E_B = 0$, and (30.21) implies $S = 0$. Since ω is strictly LCK, it is then Vaisman.

Equation (30.2) gives $\nabla_A J = 0$. Let T be the symmetric endomorphism defined by $g(TU, V) = S(U, V)$, so that $\nabla_U A = TU$. Since ∇ is torsion free,

$$(\mathcal{L}_A J)V = JTV - TJV.$$

Consequently A is real holomorphic exactly when T commutes with J, that is, when $S = S^+$. In that case the integral identity gives $q = 0$, hence $S = 0$ by the previous paragraph. Conversely, if ω is Vaisman, then $S = 0$, so $q = -\operatorname{tr}_g S = 0$ and $\mathcal{L}_A J = 0$.□

The rigidity in Theorem 30.5.1 requires Chern–Ricci flatness and the Gauduchon condition for the same metric. If $c_1^{\mathrm{BC}}(X) = 0$, each conformal class of Hermitian metrics contains a Chern–Ricci-flat metric: if a smooth real function h satisfies $\mathrm{Ric}^{\mathrm{Ch}}(\omega) = \mathrm{dd}^{\mathrm{c}} h$, then

$$\mathrm{Ric}^{\mathrm{Ch}}(\mathrm{e}^{h/n}\omega) = \mathrm{Ric}^{\mathrm{Ch}}(\omega) - \mathrm{dd}^{\mathrm{c}} h = 0.$$

By Gauduchon's theorem, the same conformal class also contains a Gauduchon metric. The rigidity theorem applies when these two metrics coincide. The following example shows that this coincidence cannot always be arranged within a conformal class. Conjecture 30.5.1, stated after it, asks instead for some Vaisman metric on the manifold.

30.5.1 A Chern–Ricci-flat conformal gauge that is not Gauduchon

Example 30.5.1 There is a Kodaira surface with holomorphically trivial canonical bundle that carries a strictly LCK Chern–Ricci-flat metric ω which is not Gauduchon. No metric conformal to ω is both Gauduchon and Chern–Ricci flat, although the same conformal class contains a Vaisman metric.

Proof On $G = \mathbb{R}^4$ use the multiplication

$$(x, y, z, t)(x', y', z', t') = (x + x', y + y', z + z' + xy', t + t').$$

The integer subgroup $\Gamma = \mathbb{Z}^4$ acts on the left by

$$(x, y, z, t) \longmapsto (x + a, y + b, z + c + ay, t + d).$$

Reducing x, y, t and then z to unit intervals gives a compact fundamental set, so $X = \Gamma\backslash G$ is connected and compact. The forms

$$e^1 = \mathrm{d}x, \quad e^2 = \mathrm{d}y, \quad e^3 = \mathrm{d}z - x\,\mathrm{d}y, \quad e^4 = \mathrm{d}t$$

are left-invariant and hence descend to X, with $\mathrm{d}e^3 = -e^1 \wedge e^2$ and the other derivatives zero. Declare $\phi^1 = e^1 + \mathrm{i}e^2$ and $\phi^2 = e^3 + \mathrm{i}e^4$ to be of type $(1, 0)$. Since

$$\mathrm{d}\phi^1 = 0, \qquad \mathrm{d}\phi^2 = -\tfrac{\mathrm{i}}{2}\phi^1 \wedge \overline{\phi^1},$$

there is no $(0,2)$ component and the complex structure is integrable. The form $\Sigma = \phi^1 \wedge \phi^2$ is closed and nowhere vanishing, hence is a holomorphic trivialization of the canonical bundle. In particular $c_1^{\mathrm{BC}}(X) = 0$. Projection to $x + \mathrm{i}y$ exhibits X as a principal elliptic bundle over the elliptic curve $\mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$; it is a primary Kodaira surface.

Choose a smooth unit-periodic function $f(x)$ with $C = 1 - f'' > 0$, and put

$$\lambda_1 = -e^4 + f' e^1, \qquad \eta_1 = J\lambda_1 = e^3 + f' e^2, \qquad \omega_1 = \lambda_1 \wedge \eta_1 + C e^1 \wedge e^2.$$

This is positive and Hermitian, with Riemannian metric

$$g_1 = \lambda_1 \otimes \lambda_1 + \eta_1 \otimes \eta_1 + C(e^1 \otimes e^1 + e^2 \otimes e^2).$$

Since C depends only on x, the identities $\mathrm{d}\lambda_1 = 0$ and $\mathrm{d}\eta_1 = -C e^1 \wedge e^2$ give $\mathrm{d}\omega_1 = \lambda_1 \wedge \omega_1$. The Lee period along the t-circle is -1, so this metric is strictly LCK. Its Lee dual is $-\partial_t$, which is Killing because the coefficients of g_1 in the coordinates (x, y, z, t) are independent of t. Since λ_1 is closed, it is parallel by Section 30.1, so ω_1 is Vaisman.

Write $e^{1234} = e^1 \wedge e^2 \wedge e^3 \wedge e^4$. The volume is $\omega_1^2/2 = C e^{1234}$, whereas $\Sigma \wedge \overline{\Sigma}$ has constant density in this coframe. For $h = C^{-1/2}$, the conformal metric $\omega = h\omega_1$ has volume e^{1234} and therefore is Chern–Ricci flat. Its Lee form is

$$\lambda = -e^4 + \mathrm{d}(f + \log h).$$

Let $g = h g_1$ be the Riemannian metric of ω, and let (E_i) be the frame dual to (e^i). Solving $g(A, \cdot) = \lambda$ for the Lee field A gives

$$A = h' E_1 + (f' h' - h^{-1}) E_4.$$

The Riemannian volume form of g is $e^{1234} = \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z \wedge \mathrm{d}t$, and the fields E_i have zero divergence for this volume. Hence $\operatorname{div} A = h''$ and $q = -h''$.

For the choice

$$f(x) = \frac{\cos(2\pi x)}{16\pi^2}, \qquad C(x) = 1 + \tfrac{1}{4}\cos(2\pi x) \geq \tfrac{3}{4},$$

one has

$$q(0) = -\frac{4\sqrt{5}\pi^2}{25}, \qquad q(1/2) = \frac{4\sqrt{3}\pi^2}{9}.$$

Thus ω is not Gauduchon, and Theorem 30.5.1 also excludes the Vaisman condition for this Chern–Ricci-flat metric.

Finally, let v be a smooth real function such that $\mathrm{e}^v\omega$ is also Chern–Ricci flat. Then $\mathrm{Ric}^{\mathrm{Ch}}(\mathrm{e}^v\omega) = \mathrm{Ric}^{\mathrm{Ch}}(\omega) - 2\mathrm{dd}^{\mathrm{c}} v$ gives $\mathrm{dd}^{\mathrm{c}} v = 0$. The real pluriharmonic function v is constant on the connected compact complex manifold by the maximum principle. For a constant $c > 0$, the metric $c\omega$ has the same Lee form λ, whose codifferential for cg is $c^{-1}q \neq 0$. Hence no Chern–Ricci-flat metric in this conformal class is

Gauduchon. The metric ω_1 already constructed is a Vaisman representative of the class. □

Conjecture 30.5.1 If X satisfies $c_1^{\mathrm{BC}}(X) = 0$ and admits a strictly LCK metric, then X admits a Vaisman metric.

The analytic problem is to achieve Chern–Ricci flatness and the Gauduchon condition simultaneously within a family of LCK metrics. The conjecture asks for one Vaisman metric; it does not assert that the Lee form of every LCK metric on X is parallel.

30.6 The Lee–Gauduchon cone

For the rest of the chapter, X is an arbitrary connected compact complex manifold of dimension $n \geq 2$, not necessarily LCK. By Theorem 4.3.1, a class in the image of the Lee map j_X of (30.5) is pseudoeffective exactly when it pairs nonnegatively with every Gauduchon metric. For a Gauduchon metric γ, the form $\mathrm{d}^{\mathrm{c}}(\gamma^{n-1})$ is closed, and these pairings depend only on its de Rham class. The classes obtained in this way lie in the annihilator of $\ker j_X$, where they form an open convex cone, and strict positivity against this cone detects every nonzero pseudoeffective class in the image of j_X.

Set $V_X = H^1_{\mathrm{dR}}(X, \mathbb{R})$ and

$$K_X = \ker j_X, \qquad W_X = K_X^{\perp} \subset H^{2n-1}_{\mathrm{dR}}(X, \mathbb{R}),$$

where the annihilator is taken with respect to the Poincaré pairing

$$\langle [\lambda], a \rangle = \int_X \lambda \wedge a \qquad \big(a \in H^{2n-1}_{\mathrm{dR}}(X, \mathbb{R})\big).$$

Let

$$C_X = \{[\lambda] \in V_X : j_X([\lambda]) \text{ is pseudoeffective}\}.$$

This is a convex cone, since pseudoeffective classes form a convex cone. The image of j_X lies in the kernel of the natural map from Bott–Chern to de Rham cohomology, and it can be strictly smaller than this kernel, as Example 30.7.1 will show.

Lemma 30.6.1 *The space K_X consists of the de Rham classes of real parts of closed holomorphic one-forms. Moreover,*

$$C_X \cap (-C_X) = K_X.$$

Proof If $j_X([\lambda]) = 0$, write $\mathrm{d}^{\mathrm{c}}\lambda = \mathrm{dd}^{\mathrm{c}} f$ for a smooth real function f. Since $\mathrm{d}^{\mathrm{c}}\mathrm{d} = -\mathrm{dd}^{\mathrm{c}}$, the form $\lambda + \mathrm{d}f$ is both d-closed and d^{c}-closed. Since the two equations separate the ∂- and $\bar\partial$-derivatives, its $(1,0)$-part is holomorphic and closed. Thus $\lambda + \mathrm{d}f$ is the real part of the closed holomorphic form $2(\lambda + \mathrm{d}f)^{1,0}$. Conversely, the

real part of a closed holomorphic one-form is closed for both operators, so its image under j_X is zero.

By Proposition 29.1.2, the pseudoeffective Bott–Chern cone contains no line. Thus $j_X(\xi)$ and $-j_X(\xi)$ are both pseudoeffective exactly when $j_X(\xi) = 0$, which is the identity $C_X \cap (-C_X) = K_X$. □

Definition 30.6.1 The *Lee–Gauduchon cone* is

$$\mathrm{LG}(X) = \bigl\{ [\mathrm{d}^{\mathrm{c}}(\gamma^{n-1})] : \gamma \text{ is a Gauduchon metric on } X \bigr\}.$$

For a closed real one-form λ and any smooth real $(n-1,n-1)$-form Γ with $\mathrm{dd}^{\mathrm{c}}\Gamma = 0$, integration by parts gives

$$\int_X \lambda \wedge \mathrm{d}^{\mathrm{c}}\Gamma = \int_X \kappa_\lambda \wedge \Gamma. \tag{30.22}$$

The form $\mathrm{d}^{\mathrm{c}}\Gamma$ is closed, and both sides depend only on the cohomology classes involved. In particular, the class $[\mathrm{d}^{\mathrm{c}}\Gamma]$ annihilates K_X, so $\mathrm{LG}(X) \subseteq W_X$.

Proposition 30.6.1 *The cone* $\mathrm{LG}(X)$ *is nonempty, convex and open in* W_X*. Its linear span is* W_X*.*

Proof Let $\mathcal{Z}_X$ be the real vector space of smooth $(n-1,n-1)$-forms Γ satisfying $\mathrm{dd}^{\mathrm{c}}\Gamma = 0$, and set

$$q_X(\Gamma) = [\mathrm{d}^{\mathrm{c}}\Gamma].$$

We first show that $q_X(\mathcal{Z}_X) = W_X$. The inclusion into W_X follows from (30.22). If $[\lambda]$ annihilates the image of q_X, then by (30.22) the class $\{\kappa_\lambda\}$ has zero pairing with every Gauduchon form. The generalized Lamari criterion, Theorem 4.3.1, implies that both $\{\kappa_\lambda\}$ and $-\{\kappa_\lambda\}$ are pseudoeffective, so $[\lambda] \in K_X$ by Lemma 30.6.1. Thus the image of q_X and W_X have the same annihilator in V_X, and since the Poincaré pairing between H^1_{dR} and H^{2n-1}_{dR} is nondegenerate, they are equal.

Let $\mathcal{Z}_X^+$ be the set of strictly positive forms in $\mathcal{Z}_X$. By the linear-algebra construction preceding the proof of Theorem 4.3.1, every such form is the $(n-1)$st power of a unique Hermitian form, which is then Gauduchon. Hence

$$\mathrm{LG}(X) = q_X(\mathcal{Z}_X^+).$$

By Gauduchon's theorem, $\mathcal{Z}_X^+$ is nonempty, and it is a convex cone; hence so is $\mathrm{LG}(X)$.

For openness, choose forms $\Gamma_1, \dots, \Gamma_r \in \mathcal{Z}_X$ whose images form a basis of W_X. If $\Gamma \in \mathcal{Z}_X^+$, then $\Gamma + \sum_i t_i \Gamma_i$ remains strictly positive for all sufficiently small real t_i, and the images of these forms cover a neighborhood of $q_X(\Gamma)$ in W_X. This includes the case $W_X = \{0\}$, with the empty basis. Since a nonempty open subset spans the ambient vector space, the assertion about the span follows. □

Theorem 30.6.1 *The cone C_X is closed, and*

$$C_X = \{\xi \in V_X : \langle \xi, a\rangle \geq 0 \text{ for every } a \in \mathrm{LG}(X)\}.$$

Conversely,

$$\mathrm{LG}(X) = \{a \in W_X : \langle \xi, a\rangle > 0 \text{ for every } \xi \in C_X \setminus K_X\}.$$

Proof The first equality is Theorem 4.3.1 applied to the smooth closed form κ_λ, together with (30.22). It expresses C_X as an intersection of closed half-spaces, so this cone is closed.

If $a = [\mathrm{d}^{\mathrm{c}}(\gamma^{n-1})] \in \mathrm{LG}(X)$ and $\xi = [\lambda] \in C_X \setminus K_X$, choose a closed positive current $T = \kappa_\lambda + \mathrm{dd}^{\mathrm{c}} u$. It is nonzero because it represents the nonzero class $j_X(\xi)$. Therefore

$$\langle \xi, a\rangle = \int_X T \wedge \gamma^{n-1} > 0.$$

Conversely, suppose that $a \in W_X \setminus \mathrm{LG}(X)$. Separating the point a from the nonempty open convex cone $\mathrm{LG}(X)$ in the finite-dimensional space W_X gives a nonzero linear functional ℓ on W_X such that

$$\ell(a) \leq 0, \qquad \ell(b) \geq 0 \quad (b \in \mathrm{LG}(X)).$$

Indeed, in the usual separating inequalities the constant can be taken to be zero, by scaling the elements of the cone. By Poincaré duality, $\ell = \langle \xi, \cdot\rangle$ for a class $\xi \in V_X$ that is unique modulo K_X, and $\xi \notin K_X$ because $\ell \neq 0$. The first equality of the theorem gives $\xi \in C_X$, and then $\langle \xi, a\rangle \leq 0$ shows that a does not belong to the right side of the second equality. This proves the second equality; if $\mathrm{LG}(X) = W_X$, there is nothing to prove. □

30.6.1 Bimeromorphic invariance

A modification induces an isomorphism on degree-one cohomology, and a class on the base is pseudoeffective exactly when its pullback is. These two facts identify the cones C_X, and duality then identifies the Lee–Gauduchon cones.

Lemma 30.6.2 *Let $\pi : Y \to X$ be a modification between connected compact complex manifolds, and let θ be a smooth closed real $(1,1)$-form on X. Then*

$$\{\theta\} \text{ is pseudoeffective} \iff \{\pi^*\theta\} \text{ is pseudoeffective}.$$

The pullback on real Bott–Chern cohomology in bidegree $(1,1)$ is injective.

Proof By Proposition 29.1.2, both pullback and pushforward preserve pseudoeffectivity; the pullback of potentials is the one used in Lemma 1.22.8. Since π has degree one,

$$\pi_*\pi^*\{\theta\} = \{\theta\}.$$

Indeed, the identity of currents $\pi_*\pi^*\theta = \theta$ follows by integration over the locus where π is an isomorphism, whose complement has volume zero. This left inverse proves injectivity and the reverse implication for pseudoeffectivity. □

Degree-one cohomology is compared by moving cycles away from the image of the exceptional set.

Lemma 30.6.3 *If $\pi\colon Y \to X$ is a modification between connected compact complex manifolds, then*

$$\pi^* : H^1_{\mathrm{dR}}(X,\mathbb{R}) \longrightarrow H^1_{\mathrm{dR}}(Y,\mathbb{R})$$

is an isomorphism. Its Poincaré dual is the isomorphism $\pi_*\colon H^{2n-1}_{\mathrm{dR}}(Y,\mathbb{R}) \to H^{2n-1}_{\mathrm{dR}}(X,\mathbb{R})$.

Proof For a proper analytic subset A of a connected complex manifold M, the restriction map $H^1(M,\mathbb{R}) \to H^1(M \setminus A,\mathbb{R})$ is injective, and it is an isomorphism if A has complex codimension at least two. Indeed, choose a Whitney stratification of A. General position moves every one-cycle off the strata, which have real codimension at least two. If those codimensions are at least four, it also moves every two-chain with boundary in $M \setminus A$ off the strata, relative to its boundary. Thus the map $H_1(M \setminus A,\mathbb{R}) \to H_1(M,\mathbb{R})$ is surjective in the first case and bijective in the second, and dualizing over $\mathbb{R}$ gives the assertions about restriction.

Since X is smooth, there is an analytic subset $A \subset X$ of complex codimension at least two such that π is an isomorphism over $U = X \setminus A$. Restriction from X to U is therefore an isomorphism, whereas restriction from Y to $\pi^{-1}U$ is injective. Every class on $\pi^{-1}U \simeq U$ extends to X and hence pulls back to a class on Y. The commutative diagram of restriction maps now shows that π^* is an isomorphism, and Poincaré duality and the projection formula give the assertion for π_*. □

Theorem 30.6.2 *Let X and Y be bimeromorphic connected compact complex manifolds. Under the natural identification of their first de Rham cohomology groups, and the Poincaré-dual identification of their de Rham cohomology groups of degree $2n-1$, the spaces K_X, W_X and the cones $C_X, \mathrm{LG}(X)$ correspond to those of Y.*

Proof First let $\pi\colon Y \to X$ be a modification. Holomorphic pullback commutes with d^{c}, so

$$j_Y(\pi^*\xi) = \pi^* j_X(\xi).$$

By Lemma 30.6.2, a class ξ lies in C_X if and only if $\pi^*\xi$ lies in C_Y, and likewise for K_X and K_Y, since the pullback is injective on Bott–Chern classes. By Lemma 30.6.3, every degree-one class on Y is of the form $\pi^*\xi$. Thus the isomorphism π^* in degree one identifies the two cones and the two kernels. Its dual π_* identifies W_Y with W_X, and Theorem 30.6.1 gives

$$\pi_*\,\mathrm{LG}(Y) = \mathrm{LG}(X).$$

For a bimeromorphic map between X and Y, resolve its graph to obtain modifications $p\colon Z \to X$ and $q\colon Z \to Y$ from a compact complex manifold Z. The preceding argument applies to both maps and identifies the cones and kernels via $(q^*)^{-1}p^*$

and its Poincaré dual. These identifications do not depend on the resolution: they are compatible with restriction to a common open set on which the original map is an isomorphism, and by the injectivity of degree-one restriction proved above, this determines each class. The dual identifications are therefore canonical as well. □

30.6.2 Balanced metrics and the Hopf cone

If zero belongs to the Lee–Gauduchon cone, then the image of j_X contains no nonzero pseudoeffective class; this is the case when X carries a balanced metric. On the Hopf threefold, by contrast, the Lee form spans a ray of classes with pseudoeffective image under j_X, and the Lee–Gauduchon cone is an open half-line.

Corollary 30.6.1 *The following conditions are equivalent:*

$$0 \in \mathrm{LG}(X), \qquad \mathrm{LG}(X) = W_X, \qquad C_X = K_X.$$

A balanced metric implies these conditions. On a manifold in Fujiki's class $\mathcal{C}$, one has $W_X = \{0\}$ and $\mathrm{LG}(X) = \{0\}$.

Proof An open cone in a finite-dimensional vector space that contains zero is the entire space, since scaling a neighborhood of zero reaches every point. The two dual descriptions in Theorem 30.6.1 now prove the equivalences, including the case $W_X = \{0\}$. If b is a balanced metric, the form b^{n-1} is closed and of pure type, hence also d^{c}-closed. Thus b is Gauduchon and gives the element $[\mathrm{d}^{\mathrm{c}}(b^{n-1})] = 0$ of $\mathrm{LG}(X)$.

On a compact Kähler manifold the $\partial\bar\partial$-lemma gives $j_X = 0$, since every κ_λ is closed and de Rham exact. Thus $K_X = V_X$ and $W_X = \{0\}$, and since $\mathrm{LG}(X)$ is nonempty, $\mathrm{LG}(X) = \{0\}$. By Theorem 30.6.2, this conclusion extends to Fujiki's class $\mathcal{C}$. □

Consequently, a nonzero pseudoeffective class in the image of j_X obstructs balanced metrics on every compact complex manifold bimeromorphic to X. For LCK manifolds, Conjecture 30.2.1 would provide such a class.

For the Hopf threefold of Chapter 9, with the Lee form λ of Proposition 30.3.2, the covering description identifies the underlying manifold with $S^1 \times S^5$, so $V_X = \mathbb{R}[\lambda]$. The class $j_X([\lambda]) = \{\alpha\}$ is nonzero by (30.4), and it is pseudoeffective because $\alpha \geq 0$. Its negative multiples are not pseudoeffective, since their pairing with the Gauduchon metric ω is negative. Thus

$$K_X = 0, \qquad C_X = \mathbb{R}_{\geq 0}[\lambda], \qquad \mathrm{LG}(X) = \{a \in H^5_{\mathrm{dR}}(X, \mathbb{R}) : \langle [\lambda], a \rangle > 0\}.$$

The same manifold has unbounded Monge–Ampère masses by Theorem 9.0.1. The positive ray detected here controls pairings with cohomology classes, which are linear and do not bound those nonlinear masses.

30.7 Strongly Gauduchon metrics

The closed form underlying a strongly Gauduchon metric pairs positively with every nonzero closed positive $(1,1)$-current, and this pairing depends only on the de Rham class of the current. It therefore bounds the mass of closed positive currents in terms of their de Rham classes and excludes nonzero positive de Rham-exact currents. The Lee–Gauduchon cone, by contrast, tests only the de Rham-exact Bott–Chern classes in the image of j_X. On the Calabi–Eckmann threefold this image vanishes while a nonzero positive exact current exists, so the two conditions differ.

Definition 30.7.1 A Hermitian metric γ is *strongly Gauduchon* if γ^{n-1} is the $(n-1,n-1)$-component of a smooth closed real $(2n-2)$-form.

Write the closed real form in the definition as

$$\Theta = \gamma^{n-1} + \beta + \overline{\beta}, \qquad \beta \text{ of type } (n, n-2).$$

Then $\partial(\gamma^{n-1}) = -\bar\partial\beta$. Conversely, this equation and its conjugate give $\mathrm{d}\Theta = 0$, because all other components of $\mathrm{d}\Theta$ have a type with an entry exceeding n. Thus γ is strongly Gauduchon if and only if $\partial(\gamma^{n-1})$ is $\bar\partial$-exact. Applying $\bar\partial$ shows that a strongly Gauduchon metric is Gauduchon. Every balanced metric is strongly Gauduchon, with $\Theta = \gamma^{n-1}$.

Proposition 30.7.1 *If γ is a strongly Gauduchon metric, then $[\mathrm{d}^{\mathrm{c}}(\gamma^{n-1})] = 0$; in particular $0 \in \mathrm{LG}(X)$. Moreover, X admits a strongly Gauduchon metric if and only if there is no nonzero positive $(1,1)$-current on X that is de Rham exact.*

Proof Put $\Gamma = \gamma^{n-1}$ and use β as above. Then

$$\mathrm{d}^{\mathrm{c}}\Gamma = \frac{\mathrm{i}}{4\pi}(\bar\partial\beta - \partial\overline{\beta}) = \mathrm{d}\left(\frac{\mathrm{i}}{4\pi}(\beta - \overline{\beta})\right).$$

Thus $[\mathrm{d}^{\mathrm{c}}\Gamma] = 0$.

Next, let S be a real current of degree one such that $T = \mathrm{d}S$ is a nonzero positive $(1,1)$-current, and let Θ be a closed real form with strictly positive $(n-1,n-1)$-part Γ. By type considerations and Stokes' theorem for currents,

$$\int_X T \wedge \Gamma = \int_X T \wedge \Theta = 0.$$

But the first integral is strictly positive because $T \neq 0$ and $\Gamma > 0$, a contradiction. Thus a strongly Gauduchon metric excludes nonzero positive exact currents.

Conversely, suppose that no strongly Gauduchon metric exists. In the real Fréchet space of smooth $(2n-2)$-forms, let $\mathcal{Z}$ be the closed subspace of closed forms, and let $\mathcal{P}$ be the open convex cone of forms with strictly positive $(n-1,n-1)$-part. This cone is nonempty, since it contains h^{n-1} for any Hermitian metric h. The two sets are disjoint, because the positive $(n-1)$st root of the $(n-1,n-1)$-part of a form in their

intersection would be a strongly Gauduchon metric. The Hahn–Banach separation theorem therefore gives a nonzero real current T of degree two such that

$$T|_{\mathcal{Z}} = 0, \qquad T(\Theta) \geq 0 \quad (\Theta \in \mathcal{P}).$$

Fix $\Theta_0 \in \mathcal{P}$. If a real form Ψ has zero $(n-1,n-1)$-part, then $\Theta_0 + t\Psi \in \mathcal{P}$ for every $t \in \mathbb{R}$, so $T(\Psi) = 0$. Thus T is of type $(1,1)$. For any semipositive $(n-1,n-1)$-form Γ, we have $\Gamma + \varepsilon\Theta_0 \in \mathcal{P}$ for $\varepsilon > 0$, and letting $\varepsilon \downarrow 0$ gives $T(\Gamma) \geq 0$; thus T is positive.

Since T annihilates every exact test form, it is closed. By the de Rham theorem for currents and Poincaré duality, its cohomology class is determined by its pairings with closed forms of complementary degree. All these pairings vanish, so $T = \mathrm{d}S$ for a real current S. Thus T is a nonzero positive exact current, which proves the converse. □

Equivalently, X admits a strongly Gauduchon metric if and only if every nonzero pseudoeffective Bott–Chern class has nonzero image in de Rham cohomology. Indeed, a closed positive current representing the zero Bott–Chern class has zero pairing with a Gauduchon metric and therefore vanishes, so the Bott–Chern classes of the nonzero positive de Rham-exact currents are exactly the nonzero pseudoeffective classes with zero de Rham image.

The same criterion characterizes the manifolds on which de Rham classes control the masses of closed positive currents.

Corollary 30.7.1 *Fix a Hermitian metric h on X and a norm on $H^2_{\mathrm{dR}}(X,\mathbb{R})$. Then X admits a strongly Gauduchon metric if and only if there is a constant $C > 0$ such that every closed positive $(1,1)$-current T satisfies*

$$\int_X T \wedge h^{n-1} \leq C\|[T]_{\mathrm{dR}}\|. \tag{30.23}$$

Under this condition, the family of closed positive $(1,1)$-currents whose de Rham classes belong to a fixed compact set is weakly compact, and the image of the pseudoeffective Bott–Chern cone in $H^2_{\mathrm{dR}}(X,\mathbb{R})$ is a closed convex cone containing no line.

Proof Suppose that γ is strongly Gauduchon, and choose a smooth closed real form Θ whose $(n-1,n-1)$-part is γ^{n-1}. By compactness, $h^{n-1} \leq A\gamma^{n-1}$ for some $A > 0$. Positivity and type considerations then give

$$\int_X T \wedge h^{n-1} \leq A \int_X T \wedge \gamma^{n-1} = A\langle [T]_{\mathrm{dR}}, [\Theta]_{\mathrm{dR}} \rangle.$$

The last pairing is a fixed linear functional on the finite-dimensional de Rham group, which proves (30.23). Conversely, this bound forces every closed positive de Rham-exact current to vanish, and Proposition 30.7.1 gives a strongly Gauduchon metric.

For the compactness assertion, the bound gives uniformly bounded trace masses. A weak limit of such currents is positive and closed, and its de Rham class is the limit

of their classes, as one sees by pairing with a finite basis of smooth closed forms of complementary degree and using Poincaré duality. Thus the family is weakly closed, and the usual weak compactness of positive currents with bounded mass proves the assertion.

Let c_j be a convergent sequence in the image cone, and choose closed positive currents T_j with $[T_j]_{\mathrm{dR}} = c_j$. By the compactness assertion, a subsequence converges to a closed positive current representing the limit of the classes. Its Bott–Chern class is pseudoeffective, so the image cone is closed. Convexity follows from addition and nonnegative scaling. Finally, if both c and $-c$ lie in the image, choose closed positive currents T and S representing them. Their sum is de Rham exact, so (30.23) gives $T + S = 0$, and positivity gives $T = S = 0$, hence $c = 0$. □

Example 30.7.1 Every Calabi–Eckmann threefold $X = X_\tau$ of Proposition 27.16.3 satisfies

$$\mathrm{LG}(X) = \{0\}, \qquad V_X = K_X = W_X = 0,$$

but has no strongly Gauduchon metric. Moreover, X has a nonzero pseudoeffective Bott–Chern class whose de Rham image is zero; this class is not in the image of j_X.

Proof By Proposition 27.16.3, X is a compact complex threefold diffeomorphic to $S^3 \times S^3$, and the projection $\pi\colon X \to \mathbb{P}^1 \times \mathbb{P}^1$ is a principal elliptic fibration. Hence X is connected and

$$H^1_{\mathrm{dR}}(X,\mathbb{R}) = H^2_{\mathrm{dR}}(X,\mathbb{R}) = H^5_{\mathrm{dR}}(X,\mathbb{R}) = 0.$$

It follows that $V_X = K_X = W_X = 0$ and $\mathrm{LG}(X) = \{0\}$; the latter cone is nonempty by Gauduchon's theorem and lies in the zero group $H^5_{\mathrm{dR}}(X,\mathbb{R})$.

With ω_1, ω_2 as in Proposition 27.16.3, the form $\alpha = \pi^*(\omega_1 + \omega_2)$ is closed and semipositive, nonzero because π is a submersion, and de Rham exact because $H^2_{\mathrm{dR}}(X,\mathbb{R}) = 0$. By Proposition 30.7.1, X admits no strongly Gauduchon metric.

The Bott–Chern class of α is nonzero: otherwise $\alpha = \mathrm{dd}^{\mathrm{c}} f$ for a smooth real function f, and the integral of α against any Gauduchon form would vanish, contradicting $\alpha \geq 0$ and $\alpha \neq 0$. The class is pseudoeffective, since α is a positive representative. Finally, j_X has zero domain, so this class is not in its image. □

Thus $0 \in \mathrm{LG}(X)$ does not imply the existence of a strongly Gauduchon metric, and hence does not imply the existence of a balanced metric.

On the threefold of Corollary 27.18.3, which carries a balanced and hence strongly Gauduchon metric, let α, β be the two pulled-back Fubini–Study forms. The nonzero class $\{\beta - \alpha\}$ has zero de Rham image, and neither it nor its negative is pseudoeffective, by Proposition 29.9.1. Every Hermitian form on this threefold has zero lower mass in the sense of Definition 7.5.1. Thus the existence of a strongly Gauduchon metric guarantees neither injectivity of the map from Bott–Chern to de Rham cohomology nor the positive lower mass needed for solvability with every finite non-pluripolar measure.

Part IX
Theory under mass bounds

Chapter 31
Volumes under bounded mass

In a Hermitian background, Monge–Ampère masses need not be bounded above or away from zero. On the standard Hopf threefold the masses of smooth admissible potentials of a Gauduchon form are unbounded (Theorem 9.0.1), while on the threefold of Theorem 27.18.1 every Hermitian form h admits smooth admissible potentials whose masses tend to zero. In this part such bounds are imposed explicitly. The upper bound is the bounded mass property of Definition 7.5.1: $v_+(\omega) < +\infty$ for one, and hence for every, Hermitian form ω (Proposition 7.5.1), where $v_+(\omega)$ and $v_-(\omega)$ denote the supremum and the infimum of the masses of bounded ω-plurisubharmonic functions. A lower bound takes one of two forms. In Chapter 33, which does not assume bounded mass, it is positive lower mass $v_-(\theta) > 0$ of a semipositive background θ. In this chapter and in Chapter 34, which assume bounded mass, it is usually a positive lower singularity volume of the particular potentials involved, rather than a bound for all bounded potentials.

Under the bounded mass property, in particular on a compact Kähler manifold, the non-pluripolar mass of a potential in a closed background depends only on its ordinary singularity type. In a nonclosed background, potentials of the same type can have different masses, so a singularity type carries two numbers, the infimum and the supremum of the masses of its members. These lower and upper singularity volumes keep the order information needed for comparison. Bounded mass makes the upper volume finite, and a positive lower volume prevents the auxiliary envelopes used below from collapsing. Together these bounds turn a small upper-volume deficit into a large admissible stretch, and this relates equality of volumes to the relative full-mass class and to ordinary envelopes.

The upper volume of the maximum of two potentials measures how far apart their ordinary envelopes are. On models with a uniform positive lower-volume bound, the stretching estimate controls three-point comparisons, monotone limits and convergence in capacity. The comparison itself need not satisfy the triangle inequality, even on the projective plane, and at zero volume it can identify distinct models, as product examples in Chapter 32 show. Finite chains nevertheless give a complete metric once the reciprocal of the lower volume is included to prevent escape to zero volume. Analytic regularization identifies the multiplier-ideal models

among the resulting limits. On the projective plane, a summable series of logarithmic poles along lines gives nonanalytic ideal models of positive volume in a nonclosed pluriclosed background. Through the Hopf fibration, positive lower volume on the projective plane forces recovery of envelope systems on the Hopf threefold whose scalar components are invariant along the elliptic fibers, although the threefold fails the bounded mass property. Strict barriers allow these comparisons to persist under small changes of background.

On a curve, the polar part of the curvature current determines the ordinary model, and the volume comparison is the total variation of polar measures. This description extends to the zero-volume boundary. Taking the atomic part gives the nearest ideal model, while diffuse polar curvature measures the failure of analytic approximation. Area rescaling and smooth gauges then describe changes of background, including the collapse of all normalized potentials when the backgrounds converge uniformly to a form of zero area. Pullback by a finite map of curves multiplies the polar metric by the degree, and averaging polar measures over the fibers retracts the model space onto the pulled-back models.

Throughout the chapter, X is a connected compact complex manifold of dimension $n \geq 1$ with the bounded mass property, and θ is a big smooth real $(1,1)$-form on X, not necessarily closed; bigness means that $\theta + \mathrm{dd}^{\mathrm{c}}\rho \geq \varepsilon\omega$ for some $\rho \in \mathrm{PSH}(X,\theta)$, some $\varepsilon > 0$ and a Hermitian form ω (Section 1.19). In Section 31.9, X is instead the Hopf threefold, which fails the bounded mass property, and from Section 31.13 on we work on a compact Riemann surface with the backgrounds specified there.

We use the singularity orders $\preceq$, $\preceq_{\mathcal{I}}$ and the equivalences $\sim$, $\sim_{\mathcal{I}}$ of Definition 1.12.1 and Section 1.12; the classes of $\sim$ are the ordinary singularity types. For $u \in \mathrm{PSH}(X,\theta)$, $\mathrm{MA}_\theta(u)$ is the non-pluripolar Monge–Ampère measure of Theorem 7.1.1, finite under bounded mass by Proposition 7.5.2, and $\underline{\mathrm{vol}}_\theta(u)$ and $\overline{\mathrm{vol}}_\theta(u)$ are the infimum and the supremum of $\int_X \mathrm{MA}_\theta(v)$ over $v \in \mathrm{PSH}(X,\theta)$ with $v \sim u$ (Definition 14.3.1). A statement about vol_θ, without a bar, is asserted for both volumes.

We use the envelopes $P_\theta(f)$ below obstacles, the rooftop envelopes $P_\theta(u,v) = P_\theta(\min(u,v))$ and $V_\theta = P_\theta(0)$ of Section 1.19, the ordinary envelope $P_\theta[u]$ of Definition 1.19.1 and the multiplier-ideal envelope $P_\theta[u]_{\mathcal{I}}$ of Definition 17.2.1. A potential u is an ordinary model if $P_\theta[u] = u$, and an ideal model if $P_\theta[u]_{\mathcal{I}} = u$; both kinds have supremum zero, and every ideal model is an ordinary model by Proposition 17.2.2. Finally, $\nu(u,x)$ denotes the Lelong number at x, and a potential is called *analytic* if it has analytic singularities in the rational-coefficient convention of Section 1.13.

31.1 Comparison of singularity volumes

Choose a Hermitian form $\Omega \geq \theta$. By Proposition 7.5.2 and Theorem 7.1.1, the mass of every θ-plurisubharmonic potential is at most $v_+(\Omega)$. Hence, for $u \in \mathrm{PSH}(X,\theta)$,

$$0 \leq \underline{\mathrm{vol}}_\theta(u) \leq \int_X \mathrm{MA}_\theta(u) \leq \overline{\mathrm{vol}}_\theta(u) < +\infty.$$

If θ is closed, both volumes equal $\int_X \mathrm{MA}_\theta(u)$, because under the bounded mass property the non-pluripolar mass in a closed background depends only on the singularity type; on compact Kähler manifolds this is classical [Xia26], and the source of the general statement is given in Section 36.19. For a nonclosed background the two volumes can differ even at $u = 0$, as Example 2.3.1 shows.

Both volumes increase when the background increases or the singularities become milder. In the proof, a rooftop envelope for the lower volume and a maximum for the upper volume produce competitors of the required singularity types, and locality of non-pluripolar products confines the error to sets whose mass tends to zero.

Theorem 31.1.1 *Let θ_1, θ_2 be smooth real $(1,1)$-forms on X. Suppose θ_1 is big, $\theta_1 \leq \theta_2$, and $u_i \in \mathrm{PSH}(X,\theta_i)$ satisfy $u_1 \preceq u_2$. Then*

$$\underline{\mathrm{vol}}_{\theta_1}(u_1) \leq \underline{\mathrm{vol}}_{\theta_2}(u_2), \qquad \overline{\mathrm{vol}}_{\theta_1}(u_1) \leq \overline{\mathrm{vol}}_{\theta_2}(u_2).$$

Proof We first treat the lower volume. After adding a constant to u_1 we may assume $u_1 \leq u_2$. Choose $v_2 \in \mathrm{PSH}(X,\theta_2)$ with $v_2 \sim u_2$ and, for real $t > 0$, consider the rooftop envelope

$$p_t = P_{\theta_1}(u_1 + t, v_2).$$

Since $u_1 \leq u_2 \leq v_2 + C$ for a constant C, the function $u_1 - C$ is a competitor; moreover $p_t \leq u_1 + t$. Hence $p_t \sim u_1$. By the contact theorem (Theorem 15.7.2), the measure $\mathrm{MA}_{\theta_1}(p_t)$ is carried by the union of $\{p_t = u_1 + t\} \subset \{u_1 \leq v_2 - t\}$ and $\{p_t = v_2\}$. On the first set, ordered contact in background θ_1 (Proposition 7.2.1) bounds it by $\mathrm{MA}_{\theta_1}(u_1)$. On the second set, p_t is also θ_2-plurisubharmonic and lies below v_2; positivity of the mixed products (Lemma 7.1.1) and ordered contact in background θ_2 bound it by $\mathrm{MA}_{\theta_2}(v_2)$. Therefore

$$\mathrm{MA}_{\theta_1}(p_t) \leq \mathbf{1}_{\{u_1 \leq v_2 - t\}} \mathrm{MA}_{\theta_1}(u_1) + \mathrm{MA}_{\theta_2}(v_2).$$

As $t \to +\infty$, the first measure on the right has total mass tending to zero: the sets $\{u_1 \leq v_2 - t\}$ decrease to a subset of the pole set of u_1, and $\mathrm{MA}_{\theta_1}(u_1)$ is a finite measure that puts no mass on pluripolar sets. Integrating and letting $t \to +\infty$ gives $\underline{\mathrm{vol}}_{\theta_1}(u_1) \leq \int_X \mathrm{MA}_{\theta_2}(v_2)$, and taking the infimum over v_2 proves the first inequality.

For the upper volume, choose any $v_1 \in \mathrm{PSH}(X,\theta_1)$ with $v_1 \sim u_1$, and put $w_t = \max(v_1, u_2 - t)$ for $t > 0$. This function is θ_2-plurisubharmonic and satisfies $w_t \sim u_2$. On the set $D_t = \{v_1 > u_2 - t\}$, locality and positivity of the mixed products imply

$$\mathbf{1}_{D_t} \mathrm{MA}_{\theta_1}(v_1) \leq \mathbf{1}_{D_t} \mathrm{MA}_{\theta_2}(w_t).$$

The sets D_t exhaust the complement of the pole set of v_1, which carries no mass. Integrating and letting $t \to +\infty$ therefore gives $\int_X \mathrm{MA}_{\theta_1}(v_1) \le \overline{\mathrm{vol}}_{\theta_2}(u_2)$, and taking the supremum over v_1 proves the second inequality. □

Adding a positive current also increases both volumes. For the lower volume, the potential of the added current is subtracted inside an envelope, which produces a competitor in the original background.

Proposition 31.1.1 *Let $u \in \mathrm{PSH}(X,\theta)$ and $v \in \mathrm{PSH}(X,\eta)$ for a smooth real $(1,1)$-form η on X. Then, for either the lower or the upper volume,*

$$\mathrm{vol}_\theta(u) \le \mathrm{vol}_{\theta+\eta}(u+v).$$

Proof The form $\theta+\eta$ is big: adding the positive current $\eta + \mathrm{dd}^{\mathrm{c}} v$ to a strictly positive current $\theta + \mathrm{dd}^{\mathrm{c}}\rho$ gives a strictly positive current $\theta + \eta + \mathrm{dd}^{\mathrm{c}}(\rho + v)$. For the upper volume, add v to each competitor $u' \sim u$ in $\mathrm{PSH}(X,\theta)$. Then $u'+v \sim u+v$, and positivity of the mixed products shows that $\int_X \mathrm{MA}_{\theta+\eta}(u'+v) \ge \int_X \mathrm{MA}_\theta(u')$.

For the lower volume, choose $w \in \mathrm{PSH}(X,\theta+\eta)$ with $w \sim u+v$, and let $p = P_\theta(w-v)$. Since $w - v = u + O(1)$ outside the pole sets, a translate of u is a competitor and $p \le u + C$ for a constant C. Thus $p \in \mathrm{PSH}(X,\theta)$ and $p \sim u$. By the contact theorem, $\mathrm{MA}_\theta(p)$ is carried by $D = \{p + v = w\}$.

Furthermore $p + v \le w$ everywhere by Lemma 1.3.3. Positivity of the mixed products and ordered contact therefore give

$$\mathrm{MA}_\theta(p) \le \mathbf{1}_D \mathrm{MA}_{\theta+\eta}(p+v) \le \mathbf{1}_D \mathrm{MA}_{\theta+\eta}(w).$$

Integrating and taking the infimum over w proves the assertion. □

31.2 Volume deficits and relative full mass

The upper-volume deficit measures how far one potential can be stretched away from another, and the positive lower volume of the less singular potential prevents the stretching envelopes from collapsing. We use the competitor families $\mathcal{A}_b(u,v)$, $b \ge 1$, of all $h \in \mathrm{PSH}(X,\theta)$ with $b^{-1}h + (1-b^{-1})v \le u$, their largest members $Q_b(u,v) = P_\theta(bu - (b-1)v)$ when they are nonempty, and the relative envelope classes $\mathcal{E}_{\mathrm{env}}(X,\theta,\varphi)$ of Definition 14.4.1.

Lemma 31.2.1 *Let $u, v \in \mathrm{PSH}(X,\theta)$ satisfy $u \preceq v$, and put*

$$\delta = \underline{\mathrm{vol}}_\theta(v) > 0, \qquad D = \overline{\mathrm{vol}}_\theta(v) - \overline{\mathrm{vol}}_\theta(u).$$

If $b > 1$ and $b^n D < \delta$, then $\mathcal{A}_b(u,v) \ne \varnothing$; that is, the stretching envelope $Q_b(u,v)$ is defined.

Proof Adding constants to u and v only translates the competitors, so we may assume $v \le 0$. Write $V = \overline{\mathrm{vol}}_\theta(v)$. Since $D < b^{-n}\delta$, the definition of the upper volume gives $w \in \mathrm{PSH}(X,\theta)$ with $w \sim u$ and

$$V - \int_X \mathrm{MA}_\theta(w) < b^{-n}\delta. \tag{31.1}$$

After a translation of w we may also assume $w \le v$. For positive integers j, put

$$w_j = \max(w, v - j), \qquad h_j = Q_b(w_j, v).$$

The function $v - bj$ is a competitor for h_j, and the defining convex inequality together with $w_j \le v$ gives $h_j \le v$. Thus $v - bj \le h_j \le v$, so $w_j \sim v$ and $h_j \sim v$, and h_j decreases as j increases. We claim that the decreasing limit h of the h_j is not identically $-\infty$. If it were, Theorem 1.4.1 would give $\sup_X h_j \to -\infty$.

By Theorem 15.7.2, the measure $\mathrm{MA}_\theta(h_j)$ is carried by the contact set $C_j = \{b^{-1}h_j + (1-b^{-1})v = w_j\}$. Positivity of the mixed products and ordered contact for $b^{-1}h_j + (1-b^{-1})v \le w_j$ give

$$\mathrm{MA}_\theta(h_j) \le b^n \mathbf{1}_{C_j} \mathrm{MA}_\theta(w_j).$$

Fix a positive integer k. If $h \equiv -\infty$, then $h_j \le -bk$ everywhere for all large $j > k$. Since $v \le 0$, every point of C_j then satisfies $w_j \le -k$, hence $w \le -k$. Moreover $w_j = w$ on $\{w > -k\} \subset \{w > v - j\}$, so locality gives

$$\begin{aligned}\delta \le \int_X \mathrm{MA}_\theta(h_j) &\le b^n \int_{\{w \le -k\}} \mathrm{MA}_\theta(w_j) \\ &\le b^n \left(V - \int_{\{w > -k\}} \mathrm{MA}_\theta(w) \right).\end{aligned}$$

Here the first inequality uses $h_j \sim v$, and the last uses $w_j \sim v$. Letting $k \to \infty$ contradicts (31.1), since the finite non-pluripolar measure $\mathrm{MA}_\theta(w)$ gives no mass to the pole set of w. This proves the claim. Consequently $h \in \mathrm{PSH}(X,\theta)$, and $b^{-1}h + (1-b^{-1})v \le w$. If $w \le u + C$, then $h - bC \in \mathcal{A}_b(u,v)$, which proves the lemma. □

Proposition 31.2.1 *Let $\pi\colon Z \to X$ be a smooth compact modification. There is a finite constant C_π, depending only on π and θ, such that, for u, v, δ, D as in Lemma 31.2.1,*

$$0 \le \nu(u \circ \pi, z) - \nu(v \circ \pi, z) \le C_\pi \min\left\{1, \left(\frac{D}{\delta}\right)^{1/n}\right\} \qquad (z \in Z).$$

The same estimate holds for generic Lelong numbers along every prime divisor on Z.

Proof Let C_π be the uniform bound for Lelong numbers of $\pi^*\theta$-plurisubharmonic functions on Z given by Lemma 1.17.7. Since $u \preceq v$, the difference of Lelong

numbers is nonnegative, and it is at most $\nu(u\circ\pi,z)\le C_\pi$. This proves the assertion when $D\ge\delta$.

If $D<\delta$, then by Lemma 31.2.1 every $b>1$ with $b^nD<\delta$ admits some $h\in\mathcal{A}_b(u,v)$. Pulling back the defining convex inequality and taking Lelong numbers gives

$$\nu(u\circ\pi,z)-\nu(v\circ\pi,z)\le b^{-1}\bigl(\nu(h\circ\pi,z)-\nu(v\circ\pi,z)\bigr)\le b^{-1}C_\pi.$$

For $D>0$, let $b\uparrow(\delta/D)^{1/n}$; for $D=0$, let $b\to\infty$. Taking generic values along a prime divisor proves the divisorial assertion. □

At zero deficit every finite stretch is possible. This leads to a second description of the relative envelope class. Fix a reference potential $\varphi\in\mathrm{PSH}(X,\theta)$.

Definition 31.2.1 The *relative full-mass class* of φ is

$$\mathcal{E}_{\mathrm{vol}}(X,\theta,\varphi)=\{u\in\mathrm{PSH}(X,\theta):u\preceq\varphi,\ \overline{\mathrm{vol}}_\theta(u)=\overline{\mathrm{vol}}_\theta(\varphi)\}.$$

Conversely, convex combinations turn the envelope condition into equality of volumes.

Theorem 31.2.1 *Suppose* $\underline{\mathrm{vol}}_\theta(\varphi)>0$. *Then*

$$\mathcal{E}_{\mathrm{env}}(X,\theta,\varphi)=\mathcal{E}_{\mathrm{vol}}(X,\theta,\varphi).$$

Every member u of this class also satisfies $\underline{\mathrm{vol}}_\theta(u)=\underline{\mathrm{vol}}_\theta(\varphi)$.

Proof First let $u\in\mathcal{E}_{\mathrm{env}}(X,\theta,\varphi)$. For each $a\in(0,1)$, a member of $\mathcal{A}_b(u,\varphi)$ with $b=(1-a)^{-1}$ gives $h\in\mathrm{PSH}(X,\theta)$ such that $a\varphi+(1-a)h\le u$. By Theorem 31.1.1 and Proposition 31.1.1 and homogeneity, for either volume,

$$a^n\,\mathrm{vol}_\theta(\varphi)\le\mathrm{vol}_\theta(a\varphi+(1-a)h)\le\mathrm{vol}_\theta(u)\le\mathrm{vol}_\theta(\varphi).$$

Letting $a\to1$ proves that both volumes of u equal those of φ; this direction does not use positive lower volume.

Conversely, let $u\in\mathcal{E}_{\mathrm{vol}}(X,\theta,\varphi)$. Then $D=0$ in Lemma 31.2.1 with $v=\varphi$, so $\mathcal{A}_b(u,\varphi)\neq\varnothing$ for every $b>1$. For $b=1$ the family contains u. This is the defining condition of $\mathcal{E}_{\mathrm{env}}(X,\theta,\varphi)$. □

31.3 Domination and ordinary envelopes

Since $u\preceq P_\theta[u]$, monotonicity gives one inequality between the volumes of a potential and those of its ordinary envelope. For the reverse inequality we identify a limit of obstacle envelopes by means of contact support and a domination principle. In the domination argument, positive lower volume guarantees that the stretching envelopes are defined. From now on we write $\mu_w=\mathrm{MA}_\theta(w)$ for $w\in\mathrm{PSH}(X,\theta)$.

Theorem 31.3.1 *Let $\varphi \in \mathrm{PSH}(X,\theta)$ have positive lower volume, let $u \in \mathcal{E}_{\mathrm{env}}(X,\theta,\varphi)$ and $v \in \mathrm{PSH}(X,\theta)$ with $v \preceq \varphi$, and let $0 \le c < 1$. If*

$$\mathbf{1}_{\{u<v\}}\mu_u \le c\,\mathbf{1}_{\{u<v\}}\mu_v,$$

then $u \ge v$.

Proof Put $\ell = \underline{\mathrm{vol}}_\theta(\varphi) > 0$ and $\mathcal{E}_\varphi = \mathcal{E}_{\mathrm{env}}(X,\theta,\varphi)$. We may replace v by $\max(u,v)$: by locality, the hypothesis is unchanged on $\{u < v\}$, and by upward stability (Proposition 14.4.1) the new v lies in $\mathcal{E}_\varphi$. Thus we may assume $u \le v$. Translate u, v by the same constant so that $v \le \varphi$, and then translate all three potentials so that $\varphi \le 0$. The idea is to stretch u away from v; the stretched potentials keep lower volume ℓ, and this is incompatible with a strict inequality $u < v$ on a set of positive volume. Set

$$q_b = Q_b(u,v), \qquad D_b = \{b^{-1}q_b + (1-b^{-1})v = u\}.$$

Lemma 14.4.1 gives $q_b \in \mathcal{E}_\varphi$ and $q_b \le u$. Since $b^{-1}q_b + (1-b^{-1})v \le u$ with contact set D_b, contact support and positivity of the mixed products give

$$b^{-n}\mu_{q_b} + (1-b^{-1})^n \mathbf{1}_{D_b}\mu_v \le \mathbf{1}_{D_b}\mu_u.$$

Here μ_{q_b} is carried by D_b. Choose b so large that $(1-b^{-1})^n > c$. On $D_b \cap \{u < v\}$ the hypothesis bounds the right-hand side by $c\mu_v$, and subtracting finite measures shows that μ_{q_b} vanishes there. Thus μ_{q_b} is carried by $D_b \cap \{u = v\} \subset \{q_b = u\}$. Since $q_b \le u$, ordered contact for this pair and Theorem 31.2.1, which gives $\underline{\mathrm{vol}}_\theta(q_b) = \ell$, now yield

$$\ell \le \mu_{q_b}(X) \le \mu_u(\{q_b = u\}). \tag{31.2}$$

Suppose now that $u < v$ somewhere. Then there is $\delta > 0$ such that $\{u < v - \delta\}$ has positive Lebesgue measure; otherwise $u \ge v$ almost everywhere, and hence everywhere by Lemma 1.3.3. On this set the convex inequality gives $q_b \le u - (b-1)\delta$. The family q_b decreases in b and is bounded above by u. Its limit is $-\infty$ on a set of positive measure, so it cannot be θ-plurisubharmonic and is therefore identically $-\infty$. Hence Theorem 1.4.1 gives $\sup_X q_b \to -\infty$. The sets $\{q_b = u\} \subset \{u \le \sup_X q_b\}$ then have μ_u-mass tending to zero, because μ_u is finite and puts no mass on the pole set of u. This contradicts (31.2). □

Corollary 31.3.1 *For u, v as in Theorem 31.3.1 and $\lambda > 0$,*

$$\mathrm{e}^{-\lambda u}\mu_u \le \mathrm{e}^{-\lambda v}\mu_v \quad \Longrightarrow \quad u \ge v.$$

If $\mu_u \le c\mu_v$ on X for a finite $c \ge 0$, then $c \ge 1$.

Proof Fix $\delta > 0$. On $\{u < v - \delta\}$ the weighted inequality gives $\mu_u \le \mathrm{e}^{-\lambda\delta}\mu_v = \mathrm{e}^{-\lambda\delta}\mu_{v-\delta}$. Theorem 31.3.1, applied to $v - \delta$ with $c = \mathrm{e}^{-\lambda\delta} < 1$, gives $u \ge v - \delta$; now let $\delta \downarrow 0$. If the second assertion failed, the theorem would apply with the same $c < 1$ to every translate $v + A$, $A \in \mathbb{R}$, and would give $u \ge v + A$ for all A. This is impossible at a point where u and v are both finite. □

The next lemma concerns the envelope obtained by replacing the obstacle 0 in Definition 1.19.1 by a potential w.

Lemma 31.3.1 *Let $u, w \in \mathrm{PSH}(X,\theta)$ with $P_\theta(u,w) \not\equiv -\infty$, and define the relative obstacle envelope*

$$P_\theta[u](w) = \left(\lim_{C\to\infty} P_\theta(u+C,w)\right)^*.$$

Then

$$\mu_{P_\theta[u](w)} \le \mathbf{1}_{\{P_\theta[u](w)=w\}}\mu_w.$$

Proof Set $p_C = P_\theta(u+C,w)$ for $C \ge 0$ and $p = P_\theta[u](w)$. The envelopes p_C increase with C, lie between p_0 and w, and converge to p in capacity by Lemma 1.8.9. By the contact theorem, μ_{p_C} is carried by $\{p_C = \min(u+C,w)\}$. Assigning the points where $u + C = w$ to the second term, ordered contact gives

$$\mu_{p_C} \le \mathbf{1}_{\{p_C=u+C<w\}}\mu_u + \mathbf{1}_{\{p_C=w\}}\mu_w.$$

The first measure has total mass at most

$$\varepsilon_C = \mu_u(\{u \le \sup_X w - C\}) \longrightarrow 0,$$

because μ_u is finite and non-pluripolar. Testing against nonnegative continuous functions and applying the lower semicontinuity of Lemma 7.3.1 gives $\mu_p \le \mu_w$.

The bounded quasi-continuous weight $\chi = \min(1,(w-p)_+)$, defined arbitrarily on the pluripolar set where $w-p$ is undefined, vanishes on $\{p_C = w\}$, since $p_C \le p \le w$. Thus $\int_X \chi\,\mu_{p_C} \le \varepsilon_C$. Weighted lower semicontinuity gives $\int_X \chi\,\mu_p = 0$, so μ_p is carried by $\{p = w\}$. Together with $\mu_p \le \mu_w$, this proves the lemma. □

Theorem 31.3.2 *For $u \in \mathrm{PSH}(X,\theta)$ and $p = P_\theta[u]$,*

$$\underline{\mathrm{vol}}_\theta(u) = \underline{\mathrm{vol}}_\theta(p).$$

If this volume is positive, then also

$$\overline{\mathrm{vol}}_\theta(u) = \overline{\mathrm{vol}}_\theta(p), \qquad u \in \mathcal{E}_{\mathrm{env}}(X,\theta,p), \qquad P_\theta[p] = p.$$

Proof Since $u \preceq p$, monotonicity gives $\underline{\mathrm{vol}}_\theta(u) \le \underline{\mathrm{vol}}_\theta(p)$ and $\overline{\mathrm{vol}}_\theta(u) \le \overline{\mathrm{vol}}_\theta(p)$. In particular, the lower volumes agree if $\underline{\mathrm{vol}}_\theta(p) = 0$. Suppose henceforth that $\underline{\mathrm{vol}}_\theta(p) > 0$. Normalize $u \le 0$, so that $u \le p$, and choose any $w \in \mathrm{PSH}(X,\theta)$ with $w \sim p$. Fix $K \ge 0$ with $p - K \le w \le p + K$. The rooftop envelope $p_C = P_\theta(u+C,w)$ satisfies $u - K \le p_C \le u + C$ for each $C \ge 0$, so it is θ-plurisubharmonic and equivalent to u. Its regularized increasing limit $r = P_\theta[u](w)$ satisfies

$$P_\theta(u + C + K, 0) - K \le p_C \le w, \qquad p - K \le r \le w \le p + K.$$

Indeed, the leftmost potential lies below $u + C$, and below $p - K \le w$ because $P_\theta(u + C + K, 0) \le P_\theta[u] = p$; it is therefore a competitor for the rooftop. Letting

$C \to \infty$ gives the lower bound for r. Thus $r \sim p$, and Proposition 14.4.1 puts r in $\mathcal{E}_{\mathrm{env}}(X, \theta, p)$.

By Lemma 31.3.1, μ_r vanishes on $\{r < w\}$. Theorem 31.3.1, with coefficient $c = 0$ and reference p, gives $r \geq w$; hence $r = w$. Since $p_C \to r$ in capacity, lower semicontinuity of the total mass (Lemma 7.3.1) and $p_C \sim u$ yield

$$\int_X \mu_w \leq \liminf_{C\to\infty} \int_X \mu_{p_C} \leq \overline{\mathrm{vol}}_\theta(u).$$

Taking the supremum over $w \sim p$ gives $\overline{\mathrm{vol}}_\theta(p) \leq \overline{\mathrm{vol}}_\theta(u)$, and hence equality of the upper volumes. Theorem 31.2.1 then gives $u \in \mathcal{E}_{\mathrm{env}}(X, \theta, p)$ and equality of the lower volumes. Finally, Proposition 14.4.3 gives $P_\theta[p] = P_\theta[u] = p$. □

Corollary 31.3.2 *If φ is an ordinary model of positive lower volume, then*

$$\mathcal{E}_{\mathrm{env}}(X, \theta, \varphi) = \mathcal{E}_{\mathrm{vol}}(X, \theta, \varphi) = \{u \in \mathrm{PSH}(X, \theta) : P_\theta[u] = \varphi\}.$$

Proof The first equality is Theorem 31.2.1. Membership in $\mathcal{E}_{\mathrm{env}}(X, \theta, \varphi)$ implies $P_\theta[u] = P_\theta[\varphi] = \varphi$ by Proposition 14.4.3. Conversely, if $P_\theta[u] = \varphi$, then Theorem 31.3.2 puts u in $\mathcal{E}_{\mathrm{env}}(X, \theta, \varphi)$. □

The ordinary envelope $P_\theta[u]$ is in general less singular than u. When the pole set of u is a compact set of Hausdorff dimension zero, however, the full local Monge–Ampère product of Theorem 1.6.3 is defined, and the ordinary envelope preserves its restriction to the pole set. Bounded mass controls the local approximations in every degree, and the products of lower degree put no mass on the pole set.

For potentials that are locally bounded off a compact set of Hausdorff dimension zero, the bounded truncations have full weak limits in every degree. This does not use the bounded mass property.

Lemma 31.3.2 *Let ω be a Hermitian form on X, let $S \subset X$ be a compact set of Hausdorff dimension zero, and let $v \in \mathrm{PSH}(X, \omega)$ be locally bounded on $X \setminus S$. Put $v_k = \max(v, -k)$. As $k \to \infty$, the currents $(\mathrm{dd}^{\mathrm{c}} v_k)^j$, $1 \leq j \leq n$, converge weakly to currents $W_j(v)$ of order zero, and the measures $(\omega + \mathrm{dd}^{\mathrm{c}} v_k)^n$ converge weakly to a finite positive measure Φ_v, which we call the* full Monge–Ampère measure *of v. One has $\int_X W_n(v) = 0$,*

$$J(v) := \Phi_v(X) = \int_X \omega^n + \sum_{j=1}^{n-1} \binom{n}{j} \int_X W_j(v) \wedge \omega^{n-j}, \tag{31.3}$$

and $v_-(\omega) \leq J(v) \leq v_+(\omega)$. The measure Φ_v equals $\mathrm{MA}_\omega(v)$ on $X \setminus S$. On each coordinate chart, $\mathbf{1}_S \Phi_v = \mathbf{1}_S (\mathrm{dd}^{\mathrm{c}}(v + q))^n_{\mathrm{full}}$ for every smooth function q making $v + q$ plurisubharmonic there; in particular, this restriction does not depend on q.

Proof On each relatively compact chart choose a smooth q_1 with $\mathrm{dd}^{\mathrm{c}} q_1 \geq \omega$, so that the functions $v_k + q_1$ are plurisubharmonic and decrease to $v + q_1$. Theorem 1.6.3 then gives the full weak limits in every degree. Indeed, $\mathrm{dd}^{\mathrm{c}} v_k = \mathrm{dd}^{\mathrm{c}}(v_k + q_1) - \mathrm{dd}^{\mathrm{c}} q_1$,

and the powers can be expanded in the closed positive powers of $\mathrm{dd^c}(v_k + q_1)$ and smooth forms. The local limits patch together because the approximating bounded currents are global. The limits are currents of order zero with finite mass on the compact manifold X, and Φ_v is positive. Stokes' theorem for the bounded potentials v_k gives $\int_X W_n(v) = 0$, and the binomial expansion of $(\omega + \mathrm{dd^c} v_k)^n$ gives (31.3). Every truncation v_k has mass between $v_-(\omega)$ and $v_+(\omega)$, which bounds $J(v)$. On every compact subset of $X \setminus S$, the truncations eventually agree with v, so Φ_v equals $\mathrm{MA}_\omega(v)$ there.

Every closed positive power of degree less than n in this expansion has coefficient measures vanishing on S, by Corollary 1.13.1. Thus only the top closed power contributes to $\mathbf{1}_S \Phi_v$, which identifies $\mathbf{1}_S \Phi_v$ with $\mathbf{1}_S (\mathrm{dd^c}(v + q_1))^n_{\mathrm{full}}$. If $v + q$ is plurisubharmonic for another smooth q, choose $C > 0$ with $\mathrm{dd^c}(q + C|z|^2) \geq \omega$ on a smaller chart. The same argument applies to $q + C|z|^2$, and by multilinearity the full top power of $\mathrm{dd^c}(v + q + C|z|^2)$ differs from that of $\mathrm{dd^c}(v + q)$ by full powers of lower degree wedged with smooth forms, which vanish on S. These assertions are local on each chart and do not require a single chart containing S. □

Theorem 31.3.3 *Let ω be a Hermitian form on X, and let $S \subset X$ be a compact set of Hausdorff dimension zero. Suppose $u \in \mathrm{PSH}(X, \omega)$ is locally bounded on $X \setminus S$ and has pole set $\{u = -\infty\} = S$. Put $p = P_\omega[u]$ and $p_m = P_\omega(u + m, 0)$ for real $m > 0$. Then p and p_m are locally bounded on $X \setminus S$, their full Monge–Ampère measures and that of u (Lemma 31.3.2) satisfy*

$$\mathbf{1}_S \Phi_p = \mathbf{1}_S \Phi_u,$$

and $\Phi_{p_m} \to \Phi_p$ weakly as $m \to \infty$.

Proof *Step 1: the rooftops.* Adding a constant to u changes neither p nor the family $(p_m)_{m>0}$, apart from a shift of the parameter m. We may therefore assume $u \leq 0$. Then

$$u \leq p_m \leq \min(u + m, 0), \qquad p_m \uparrow p,$$

where the increasing limit is understood after upper semicontinuous regularization. Thus $p_m \sim u$, and p_m and p are locally bounded off S. By Lemma 1.8.9, $p_m \to p$ in capacity. On compact subsets of $X \setminus S$ the potentials have a common bounded range, so all bounded local products converge there, by the bounded product theory of Section 1.6.

Step 2: full measures. By Step 1 and Lemma 31.3.2, the full Monge–Ampère measures Φ_v, the currents $W_j(v)$ and the total masses $J(v)$ of (31.3) are defined for $v = u$, $v = p$ and $v = p_m$.

Step 3: convergence of the lower-degree powers. We need local trace bounds for the closed powers of p_m that are uniform in m. Fix a relatively compact coordinate neighborhood and write $\kappa = \mathrm{dd^c}|z|^2$. Choose $B > 0$ with $B\kappa \geq \omega$ there, and $A > 1$ with $A\omega - B\kappa \geq \kappa$. For $p_{m,k} = \max(p_m, -k)$, the closed current $T_{m,k} = B\kappa + \mathrm{dd^c} p_{m,k}$ is positive and

$$Q_{m,k} := A\omega + \mathrm{dd^c} p_{m,k} \geq T_{m,k} + \kappa.$$

Expanding $Q_{m,k}^n$ with the smooth positive difference $Q_{m,k} - T_{m,k} \geq \kappa$ gives, for $1 \leq j \leq n$,

$$\binom{n}{j} T_{m,k}^j \wedge \kappa^{n-j} \leq Q_{m,k}^n, \qquad \int_X Q_{m,k}^n \leq A^n v_+(\omega).$$

The last bound holds because $Q_{m,k} = A(\omega + \mathrm{dd}^{\mathrm{c}}(p_{m,k}/A))$ and $p_{m,k}/A$ is a bounded ω-plurisubharmonic function. Letting $k \to \infty$ gives trace bounds, independent of m, for every full closed power of $\mathrm{dd}^{\mathrm{c}}(p_m + B|z|^2)$.

For $j < n$, every subsequential weak limit of these closed positive (j, j)-currents puts no mass on S, again by Corollary 1.13.1. Off S the bounded capacity convergence identifies the limit with the corresponding full power of $\mathrm{dd}^{\mathrm{c}}(p + B|z|^2)$, which also puts no mass on S. Hence the limit is unique. Expanding back in the smooth correction and using a finite coordinate cover yields

$$W_j(p_m) \longrightarrow W_j(p), \qquad 1 \leq j < n.$$

Hence (31.3) gives $J(p_m) \to J(p)$.

Step 4: the residual measures. Let $R = \mathbf{1}_S \Phi_u$. Since $p_m - u$ is bounded, the equality case of Lemma 1.6.6, applied on each chart after a common smooth correction, gives $\mathbf{1}_S \Phi_{p_m} = R$ for every m. The total masses $J(p_m)$ are bounded by $v_+(\omega)$, so along any sequence $m_i \to \infty$ a subsequence of $\Phi_{p_{m_i}}$ converges weakly to a measure η. Since $\Phi_{p_m} - R$ is positive, so is $\eta - R$. Off S, η equals $\mathrm{MA}_\omega(p)$, by the bounded convergence of Step 1, and its total mass is $\lim_m J(p_m) = J(p)$ by Step 3. The measure Φ_p has the same total mass and the same restriction to $X \setminus S$, so

$$\Phi_p(S) = \eta(S) \geq R(S).$$

On the other hand, $u \leq p$, so the inequality in Lemma 1.6.6 gives $\mathbf{1}_S \Phi_p \leq R$ on each chart. Together with the previous inequality, this forces $\Phi_p(S) = R(S)$, hence $\mathbf{1}_S \Phi_p = R$, which is the asserted equality of residual measures. It also gives $\mathbf{1}_S \eta = R$, since $\eta - R \geq 0$ and $\eta(S) = R(S)$. As η and Φ_p also agree off S, we conclude that $\eta = \Phi_p$. Every weak subsequential limit of Φ_{p_m} is therefore Φ_p, which proves the convergence $\Phi_{p_m} \to \Phi_p$. □

31.4 Strict barriers and background variation

Positive lower volume permits a small stretch away from any less singular potential, and the admissible stretch depends only on the lower volume and a mass bound. Mixing the stretched potential with a strict potential produces a strict potential lying below the original one. The same bound for mixed masses controls the change of the lower volume under a small smooth perturbation of the background. Throughout this section, ω is a fixed Hermitian form on X.

Lemma 31.4.1 *Let $u, r \in \mathrm{PSH}(X,\theta)$, with $u \le r$, and assume $\underline{\mathrm{vol}}_\theta(u) > 0$. For a Hermitian form $\Omega \ge \theta$, set*

$$b_0 = 1 + \frac{\underline{\mathrm{vol}}_\theta(u)}{2^{n+1} n\, v_+(\Omega)}.$$

Then $\mathcal{A}_b(u,r) \neq \varnothing$, so that $Q_b(u,r)$ is defined, for $1 \le b \le b_0$, and

$$\lim_{b\downarrow 1} \overline{\mathrm{vol}}_\theta(Q_b(u,r)) = \overline{\mathrm{vol}}_\theta(u).$$

Proof Put $K = v_+(\Omega)$. Since $0 < \underline{\mathrm{vol}}_\theta(u) \le K$, we have $1 < b_0 \le 2$. After a common translation we may assume $u \le r \le 0$. Suppose, to reach a contradiction, that $\mathcal{A}_b(u,r) = \varnothing$ for some $1 < b \le b_0$. Set

$$u_j = \max(u, r-j), \qquad v_j = u + (b-1)u_j, \qquad q_j = P_\theta(v_j - (b-1)r).$$

Here, as for stretching envelopes, a competitor g for q_j is required to satisfy $g + (b-1)r \le v_j$, which is meaningful also at common poles. Off the poles the obstacle $v_j - (b-1)r = u + (b-1)(u_j - r)$ lies between $u-(b-1)j$ and u. Hence q_j is θ-plurisubharmonic and $q_j \sim u$. As j increases, q_j decreases. A θ-plurisubharmonic decreasing limit would belong to $\mathcal{A}_b(u,r)$, so the limit is identically $-\infty$, and $\sup_X q_j \to -\infty$ by Theorem 1.4.1.

Let $C_j = \{q_j + (b-1)r = v_j\}$. By the contact theorem, μ_{q_j} is carried by C_j. The current $b\theta + \mathrm{dd^c}(q_j + (b-1)r)$ equals $(\theta + \mathrm{dd^c} q_j) + (b-1)(\theta + \mathrm{dd^c} r) \ge \theta + \mathrm{dd^c} q_j$. Positivity of the mixed products and ordered contact in background $b\theta$ therefore give

$$\mu_{q_j} \le \mathbf{1}_{C_j} \big((\theta + \mathrm{dd^c} u) + (b-1)(\theta + \mathrm{dd^c} u_j)\big)^n.$$

On C_j we have $q_j = u + (b-1)(u_j - r) \ge bu$, because $u_j \ge u$ and $r \le 0$. Fix $k > 0$. For all sufficiently large j, $q_j \le -bk$ everywhere, so $C_j \subset \{u \le -k\}$, and consequently

$$\underline{\mathrm{vol}}_\theta(u) \le \mu_{q_j}(X) \le \int_{\{u \le -k\}} \big((\theta + \mathrm{dd^c} u) + (b-1)(\theta + \mathrm{dd^c} u_j)\big)^n. \tag{31.4}$$

To estimate the right-hand side, note that for any $s, t \in \mathrm{PSH}(X,\theta)$ and $0 \le i \le n$, positivity of the non-pluripolar mixed products and Proposition 7.5.2 imply

$$\begin{aligned}
&\binom{n}{i} \int_X (\theta + \mathrm{dd^c} s)^{n-i} \wedge (\theta + \mathrm{dd^c} t)^i \\
&\quad \le \binom{n}{i} \int_X (\Omega + \mathrm{dd^c} s)^{n-i} \wedge (\Omega + \mathrm{dd^c} t)^i \\
&\quad \le 2^n \int_X \mathrm{MA}_\Omega((s+t)/2) \le 2^n K.
\end{aligned}$$

The second inequality follows by expanding $(\Omega + \mathrm{dd}^{\mathrm{c}}\frac{s+t}{2})^n$; for the possibly unbounded average, apply this expansion to its truncations at constant levels, which remain Ω-plurisubharmonic. Expanding the integrand in (31.4), and using $(b-1)^i \le b-1$ for $i \ge 1$, gives

$$\underline{\mathrm{vol}}_\theta(u) \le \mu_u(\{u \le -k\}) + n2^n K(b-1).$$

The measure μ_u is finite and puts no mass on the pole set of u, so the first term tends to zero as $k \to \infty$. By the choice of b_0, the remaining error is at most half of $\underline{\mathrm{vol}}_\theta(u) > 0$, a contradiction. Thus $\mathcal{A}_b(u,r) \neq \varnothing$ for $1 < b \le b_0$; for $b = 1$ the family contains u, and $Q_1(u,r) = u$.

Put $h_0 = Q_{b_0}(u,r)$ and, for $1 < b \le b_0$, $t = (b-1)/(b_0-1) \in (0,1]$. The inequality $h_0 \le b_0 u - (b_0-1)r$ shows that $(1-t)u + th_0$ is a competitor for $Q_b(u,r)$, and $Q_b(u,r) \le bu - (b-1)r \le u$. Hence

$$(1-t)u + th_0 \le Q_b(u,r) \le u.$$

Addition, homogeneity and volume monotonicity give

$$(1-t)^n \overline{\mathrm{vol}}_\theta(u) \le \overline{\mathrm{vol}}_\theta(Q_b(u,r)) \le \overline{\mathrm{vol}}_\theta(u).$$

Letting $b \downarrow 1$ proves the limit. Undoing the common translation changes neither the volumes nor the nonemptiness of the competitor families. □

Corollary 31.4.1 *If $u \in \mathrm{PSH}(X,\theta)$ has positive lower volume, then there are $w \in \mathrm{PSH}(X,\theta)$ and $\varepsilon > 0$ with*

$$w \le u, \qquad \theta + \mathrm{dd}^{\mathrm{c}} w \ge \varepsilon\omega.$$

Proof Translate u so that $u \le V_\theta$. By bigness, there are $\rho \in \mathrm{PSH}(X,\theta)$ and $\delta > 0$ with $\theta + \mathrm{dd}^{\mathrm{c}}\rho \ge \delta\omega$; after a translation, $\rho \le V_\theta$. By Lemma 31.4.1 with $r = V_\theta$, there is $b > 1$ for which $h = Q_b(u, V_\theta)$ is defined. Put

$$w = b^{-1}h + (1 - b^{-1})\rho.$$

Since $b^{-1}h + (1-b^{-1})V_\theta \le u$ and $\rho \le V_\theta$, we have $w \le u$; moreover $\theta + \mathrm{dd}^{\mathrm{c}} w \ge (1-b^{-1})\delta\omega$. Undoing the translation of u by the same translation of w gives the assertion with $\varepsilon = (1-b^{-1})\delta$. □

Proposition 31.4.1 *If $u \in \mathrm{PSH}(X,\theta)$ and $\underline{\mathrm{vol}}_\theta(u) > 0$, then $P_{\varepsilon\theta}(u) \not\equiv -\infty$ for some $\varepsilon \in (0,1)$.*

Proof By Corollary 31.4.1, choose $w \le u$ and $\delta > 0$ with $\theta + \mathrm{dd^c} w \ge \delta\omega$. Choose $C > 0$ with $\theta \le C\omega$, and $0 < t < \min(1, \delta/C)$. Then

$$(1-t)\theta + \mathrm{dd^c} w \ge (\delta - tC)\omega \ge 0.$$

Thus w is a competitor for $P_{(1-t)\theta}(u)$, and $\varepsilon = 1 - t$ has the required property. □

For a reference potential φ, the positive-volume model condition reads

$$P_\theta[\varphi] = \varphi, \qquad \underline{\mathrm{vol}}_\theta(\varphi) > 0. \tag{31.5}$$

When $\theta \ge 0$, Proposition 31.4.1 shows that (31.5) implies the smaller-background model condition (14.20) of Section 14.5. If moreover $v_-(\theta) > 0$, the converse holds. Indeed, if $P_{\varepsilon\theta}(\varphi) \not\equiv -\infty$, then a translate of the same competitor shows that $P_{\varepsilon\theta}(\psi) \not\equiv -\infty$ for every $\psi \sim \varphi$ in $\mathrm{PSH}(X,\theta)$, and Lemma 14.5.2 gives $\int_X \mathrm{MA}_\theta(\psi) \ge (1-\varepsilon)^n v_-(\theta)$. Hence

$$\underline{\mathrm{vol}}_\theta(\varphi) \ge (1-\varepsilon)^n v_-(\theta) > 0.$$

Without the lower bound $v_-(\theta) > 0$, the same lemma still shows that each potential equivalent to φ has positive mass, but not that these masses are bounded away from zero.

A small smooth increase of the background changes the lower volume by at most a linear error, and a strict potential lying below u gives a multiplicative comparison for both volumes.

Proposition 31.4.2 *There is $C > 0$, depending only on θ and ω, such that, for every $u \in \mathrm{PSH}(X,\theta)$, every $\varepsilon \in [0,1]$ and every smooth real $(1,1)$-form η with $\theta \le \eta \le \theta + \varepsilon\omega$,*

$$0 \le \underline{\mathrm{vol}}_\eta(u) - \underline{\mathrm{vol}}_\theta(u) \le C\varepsilon.$$

If $\underline{\mathrm{vol}}_\theta(u) > 0$, there is also $\delta > 0$, depending on u, such that, for every $\varepsilon \ge 0$ and every smooth η with $\theta \le \eta \le \theta + \varepsilon\omega$, both singularity volumes satisfy

$$\mathrm{vol}_\theta(u) \le \mathrm{vol}_\eta(u) \le (1 + \varepsilon/\delta)^n \,\mathrm{vol}_\theta(u).$$

Proof Choose a Hermitian form Ω with $\Omega \geq \theta$ and $\Omega \geq \omega$, and set $C = n2^n v_+(\Omega)$. The last two inequalities of the mixed-mass estimate in the proof of Lemma 31.4.1 hold for all $s, t \in \mathrm{PSH}(X, \Omega)$. For every $v \in \mathrm{PSH}(X, \theta)$ and $1 \leq k \leq n$, applying them to $s = v$ and $t = 0$, and using $\theta \leq \Omega$ and $\omega \leq \Omega$, gives

$$\binom{n}{k} \int_X (\theta + \mathrm{dd^c} v)^{n-k} \wedge \omega^k \leq 2^n v_+(\Omega).$$

Expanding in the smooth positive perturbation and using $\varepsilon^k \leq \varepsilon$ yields

$$\int_X \mathrm{MA}_\eta(v) \leq \int_X \mathrm{MA}_{\theta+\varepsilon\omega}(v) \leq \int_X \mathrm{MA}_\theta(v) + C\varepsilon.$$

Each $v \sim u$ in $\mathrm{PSH}(X, \theta)$ is also a competitor for the lower volume in background η. Taking the infimum over these v proves the upper bound for $\underline{\mathrm{vol}}_\eta(u) - \underline{\mathrm{vol}}_\theta(u)$. The lower bound is background monotonicity (Theorem 31.1.1).

Now assume that u has positive lower volume, and choose $w \leq u$ and $\delta > 0$ with $\theta + \mathrm{dd^c} w \geq \delta\omega$ by Corollary 31.4.1. The case $\varepsilon = 0$ is immediate. For $\varepsilon > 0$ put $t = \varepsilon/\delta$. The potential tw belongs to $\mathrm{PSH}(X, t(\theta - \delta\omega))$, and the form $\eta + t(\theta - \delta\omega)$ is big, since adding $\mathrm{dd^c}((1+t)w)$ to it gives a form at least $(1+t)\delta\omega - t\delta\omega = \delta\omega$. Addition and monotonicity give

$$\begin{aligned} \mathrm{vol}_\eta(u) &\leq \mathrm{vol}_{\eta + t(\theta - \delta\omega)}(u + tw) \\ &\leq \mathrm{vol}_{(1+t)\theta}((1+t)u) = (1+t)^n \,\mathrm{vol}_\theta(u). \end{aligned}$$

Here we used $\eta + t(\theta - \delta\omega) \leq (1+t)\theta$, which holds because $\eta \leq \theta + \varepsilon\omega$, and $u + tw \leq (1+t)u$. The lower bound is again background monotonicity. □

31.5 Monotone limits

We now show that ordinary envelopes and singularity volumes are continuous along monotone sequences whose limit has positive lower volume. For increasing sequences, contact support identifies the limit of the envelopes before its volume is known. For decreasing sequences of models, a uniform small stretch keeps a positive lower-volume bound. Normalized compactness then preserves the convex inequalities given by the deficit lemma, and strict barriers allow the background to decrease at the same time.

Theorem 31.5.1 *Let $u_j \in \mathrm{PSH}(X,\theta)$ be an increasing sequence such that $u = (\lim_j u_j)^*$ belongs to $\mathrm{PSH}(X,\theta)$ and has positive lower volume. Then*

$$\Big(\lim_j P_\theta[u_j]\Big)^* = P_\theta[u], \qquad P_\theta[u_j] \longrightarrow P_\theta[u] \quad \textit{in capacity},$$

and

$$\underline{\mathrm{vol}}_\theta(u_j) \nearrow \underline{\mathrm{vol}}_\theta(u), \qquad \overline{\mathrm{vol}}_\theta(u_j) \nearrow \overline{\mathrm{vol}}_\theta(u).$$

For every fixed smooth compact modification $\pi\colon Z \to X$,

$$\sup_{z\in Z}\big|\nu(u_j\circ\pi, z) - \nu(u\circ\pi, z)\big| \longrightarrow 0.$$

Proof *Step 1: the envelopes.* Translate so that $u \le 0$, and set $p_j = P_\theta[u_j]$, $p = P_\theta[u]$ and $s = (\lim_j p_j)^*$. The p_j increase, and $u \le s \le p \le V_\theta$. By Theorem 31.3.2, the lower volume of p is positive and $u \in \mathcal{E}_{\mathrm{env}}(X,\theta,p)$. Upward stability (Proposition 14.4.1) puts s in the same class.

The ordinary envelope can use V_θ in place of zero as obstacle (see the remark after Definition 1.19.1); that is, $p_j = P_\theta[u_j](V_\theta)$. Hence Lemma 31.3.1 gives

$$\mu_{p_j} \le \mathbf{1}_{\{p_j = V_\theta\}}\mu_{V_\theta}.$$

The weight $\min(1,(p-s)_+)$ vanishes on $\{p_j = V_\theta\}$, because $p_j \le s \le p \le V_\theta$, and $p_j \to s$ in capacity by Lemma 1.8.9. As in the proof of Lemma 31.3.1, weighted lower semicontinuity shows that μ_s vanishes on $\{s < p\}$, and Theorem 31.3.1, with reference p and coefficient $c = 0$, yields $s = p$. This proves the assertions about the envelopes.

Step 2: the upper volumes of the envelopes. Choose any $w \in \mathrm{PSH}(X,\theta)$ with $w \sim p$, and $K \ge 0$ with $p - K \le w \le p + K$. Define $r_j = P_\theta[u_j](w)$; this is meaningful because $u_j - K \le p - K \le w$, so that $P_\theta(u_j, w) \not\equiv -\infty$. For $C \ge 0$ the competitors compare as follows:

$$P_\theta(u_j + C + K, 0) - K \le P_\theta(u_j + C, w) \le p_j + K.$$

The leftmost potential lies below $u_j + C$ and below $p_j - K \le p - K \le w$. For the upper bound, note that the middle term minus K lies below $w - K \le p \le 0$ and is more singular than u_j, so it is at most p_j. Taking the increasing limit in C gives

$$p_j - K \le r_j \le p_j + K, \qquad r_j \le w.$$

The r_j increase with j and converge in capacity to $r = (\lim_j r_j)^*$, and $p - K \le r \le w \le p + K$. Thus $r \sim p$, so $r \in \mathcal{E}_{\mathrm{env}}(X,\theta,p)$. Lemma 31.3.1 gives $\mu_{r_j} \le \mathbf{1}_{\{r_j = w\}}\mu_w$, and weighted lower semicontinuity with the weight $\min(1,(w-r)_+)$ shows that μ_r vanishes on $\{r < w\}$. Domination with reference p gives $r = w$. Therefore

$$\int_X \mu_w \le \liminf_j \int_X \mu_{r_j} \le \lim_j \overline{\mathrm{vol}}_\theta(p_j),$$

where $r_j \sim p_j$ justifies the last inequality. Taking the supremum over $w \sim p$ and using volume monotonicity proves

$$\overline{\mathrm{vol}}_\theta(p_j) \nearrow \overline{\mathrm{vol}}_\theta(p).$$

Step 3: the volumes of u_j. Put $\delta = \underline{\mathrm{vol}}_\theta(p) > 0$ and $D_j = \overline{\mathrm{vol}}_\theta(p) - \overline{\mathrm{vol}}_\theta(p_j)$. Since $D_j \to 0$, we can choose $b_j > 1$ with $b_j \to \infty$ and $b_j^n D_j < \delta$ for all large j. With $a_j = 1 - b_j^{-1}$, Lemma 31.2.1 gives $h_j \in \mathrm{PSH}(X,\theta)$ such that

$$a_j p + (1-a_j)h_j \le p_j$$

for all sufficiently large j. Addition, homogeneity and monotonicity of the lower volume, together with Theorem 31.3.2, give

$$a_j^n \delta \le \underline{\mathrm{vol}}_\theta(p_j) = \underline{\mathrm{vol}}_\theta(u_j) \le \delta.$$

Since $a_j \to 1$, this proves the convergence of the lower volumes, and shows that $\underline{\mathrm{vol}}_\theta(u_j) > 0$ for all large j. For such j, Theorem 31.3.2 identifies the upper volumes of u_j and p_j, and also those of u and p; Step 2 therefore gives the asserted upper-volume limit. Finally, Proposition 31.2.1, applied to $u_j \preceq u$, gives the uniform convergence of Lelong numbers on each fixed modification. □

For the results on decreasing sequences, fix smooth real $(1,1)$-forms θ_j with $\theta \le \theta_{j+1} \le \theta_j$ that converge uniformly to θ; the constant sequence $\theta_j = \theta$ is allowed. The proofs first treat a fixed background, using normalized auxiliary potentials, and then handle the varying forms by means of corrected envelopes of tail suprema.

Theorem 31.5.2 *Suppose that $q_j \in \mathrm{PSH}(X,\theta_j)$ are ordinary models, $q_j = P_{\theta_j}[q_j]$, decreasing to $q \in \mathrm{PSH}(X,\theta)$. Then $P_\theta[q] = q$ and*

$$\underline{\mathrm{vol}}_{\theta_j}(q_j) \searrow \underline{\mathrm{vol}}_\theta(q).$$

If $\underline{\mathrm{vol}}_\theta(q) > 0$, then also

$$\overline{\mathrm{vol}}_{\theta_j}(q_j) \searrow \overline{\mathrm{vol}}_\theta(q),$$

and, for every fixed smooth compact modification $\pi\colon Z \to X$,

$$\sup_{z\in Z}\left|\nu(q_j\circ\pi,z) - \nu(q\circ\pi,z)\right| \longrightarrow 0.$$

In this case all generic divisorial Lelong numbers converge.

Proof *Step 1: the fixed point and positivity of the limit volume.* Since $q \le q_j \le 0$, monotonicity of the envelope in the background and in the potential gives $q \le P_\theta[q] \le P_{\theta_j}[q] \le P_{\theta_j}[q_j] = q_j$. Letting $j \to \infty$ proves $P_\theta[q] = q$. By Theorem 31.1.1, the lower volumes $\underline{\mathrm{vol}}_{\theta_j}(q_j)$ decrease; let ℓ be their limit. Monotonicity also gives $\underline{\mathrm{vol}}_\theta(q) \le \ell$, so the lower-volume assertion holds if $\ell = 0$.

Suppose $\ell > 0$. We show that then $\underline{\mathrm{vol}}_\theta(q) > 0$; the lower-volume assertion will then follow from the volume limits proved below for limits of positive lower volume.

Choose a Hermitian form $\Omega \geq \theta_1$, put $K = v_+(\Omega)$, and choose

$$b = 1 + \frac{\ell}{2^{n+1} n K}, \qquad a = 1 - b^{-1}.$$

Since $\Omega \geq \theta_j$ and $\underline{\mathrm{vol}}_{\theta_j}(q_j) \geq \ell$, Lemma 31.4.1, applied in background θ_j with $r = V_{\theta_j}$, shows that $Q_b(q_j, V_{\theta_j})$ is defined for every j, with the same b. Put $h_j = P_{\theta_j}[Q_b(q_j, V_{\theta_j})]$. Since $aV_{\theta_j} + (1-a)Q_b(q_j, V_{\theta_j}) \leq q_j$, $V_{\theta_j} \leq 0$ and $q_j = P_{\theta_j}[q_j]$, combining V_{θ_j} with the competitors for h_j, as in the proof of Proposition 14.4.3, gives

$$aV_{\theta_j} + (1-a)h_j \leq q_j.$$

The functions h_j have supremum zero and are Ω-plurisubharmonic. Normalized compactness and the uniform convergence of the backgrounds give a subsequential L^1 limit $h \in \mathrm{PSH}(X, \theta)$. Also $V_{\theta_j} \searrow V_\theta$, since the decreasing limit is θ-plurisubharmonic and lies between V_θ and zero. Passing to the limit in the convex inequality and using Lemma 1.3.3 gives

$$aV_\theta + (1-a)h \leq q.$$

Moreover, $v_-(\Omega) = \underline{\mathrm{vol}}_\Omega(0) \geq \underline{\mathrm{vol}}_{\theta_1}(q_1) \geq \ell > 0$ by volume monotonicity. By bigness, choose $\rho \leq V_\theta$ with $\theta + \mathrm{dd}^{\mathrm{c}}\rho \geq \kappa\Omega$ for some $\kappa > 0$. Proposition 31.1.1, applied to $0 \in \mathrm{PSH}(X, \kappa\Omega)$ and $\rho \in \mathrm{PSH}(X, \theta - \kappa\Omega)$, gives $\underline{\mathrm{vol}}_\theta(\rho) \geq \kappa^n v_-(\Omega)$. Monotonicity and addition applied to the convex combination above now yield

$$\underline{\mathrm{vol}}_\theta(q) \geq a^n \underline{\mathrm{vol}}_\theta(V_\theta) \geq (a\kappa)^n v_-(\Omega) > 0.$$

It remains to prove the volume limits when $\underline{\mathrm{vol}}_\theta(q) > 0$.

Step 2: a fixed background. Suppose first that $\theta_j = \theta$ for every j. Put

$$\delta = \underline{\mathrm{vol}}_\theta(q) > 0, \qquad L = \lim_j \overline{\mathrm{vol}}_\theta(q_j), \qquad e_k = \overline{\mathrm{vol}}_\theta(q_k) - L.$$

The limit exists because the upper volumes decrease, by Theorem 31.1.1, and are finite, by bounded mass; thus $e_k \searrow 0$. After discarding finitely many indices, choose $b_k > 1$ tending to infinity with $b_k^n e_k < \delta$, and set $a_k = 1 - b_k^{-1}$.

For every $j \geq k$, the upper-volume deficit $\overline{\mathrm{vol}}_\theta(q_k) - \overline{\mathrm{vol}}_\theta(q_j)$ is at most e_k, whereas the lower volume of q_k is at least δ. Thus Lemma 31.2.1 gives $h_{k,j} \in \mathrm{PSH}(X, \theta)$ with

$$a_k q_k + (1-a_k)h_{k,j} \leq q_j.$$

Since $q_k \leq 0$ and $q_j = P_\theta[q_j]$, combining q_k with the competitors for $P_\theta[h_{k,j}]$, as in the proof of Proposition 14.4.3, gives

$$a_k q_k + (1-a_k)P_\theta[h_{k,j}] \leq q_j.$$

The potentials $P_\theta[h_{k,j}]$ have supremum zero. For fixed k, normalized compactness gives a subsequential L^1 limit h_k as $j \to \infty$. Passing to the limit and applying

Lemma 1.3.3 gives

$$a_k q_k + (1 - a_k) h_k \le q.$$

For either volume, homogeneity, addition and monotonicity now yield

$$a_k^n \operatorname{vol}_\theta(q_k) \le \operatorname{vol}_\theta\big(a_k q_k + (1 - a_k) h_k\big) \le \operatorname{vol}_\theta(q) \le \operatorname{vol}_\theta(q_k).$$

Letting $k \to \infty$ proves both volume limits in the fixed background.

Step 3: varying backgrounds. Fix a Hermitian form ω. By Corollary 31.4.1, choose $w \le q$ and $d > 0$ with $\theta + \mathrm{dd}^{\mathrm{c}} w \ge d\omega$. Choose $\varepsilon_j > 0$ decreasing to zero with $\theta_j \le \theta + \varepsilon_j \omega$, and put

$$c_j = \frac{d}{d + \varepsilon_j}, \qquad f_j = c_j q_j + (1 - c_j) w,$$

$$F_j = \left(\sup_{k \ge j} f_k \right)^*, \qquad r_j = P_\theta[F_j].$$

The convex combination f_j is θ-plurisubharmonic, because its background is bounded above by θ:

$$c_j \theta_j + (1 - c_j)(\theta - d\omega) \le \theta.$$

Since $w \le q \le q_k$, we have $f_k \le q_k \le q_j$ for $k \ge j$, and upper regularization gives $F_j \le q_j$. Also $f_k \to q$ in L^1, so a subsequence converges almost everywhere, and $q \le F_j$ by Lemma 1.3.3.

Background monotonicity now gives

$$q \le r_j = P_\theta[F_j] \le P_{\theta_j}[F_j] \le P_{\theta_j}[q_j] = q_j.$$

The r_j decrease to q. Their lower volumes are at least $\underline{\operatorname{vol}}_\theta(q) > 0$, so Theorem 31.3.2 makes them ordinary models, and Step 2 applies to the sequence r_j in the fixed background θ. On the other hand, addition (with $\eta = (1 - c_j)(\theta - d\omega)$) and background monotonicity give, for either volume,

$$\operatorname{vol}_\theta(q) \le \operatorname{vol}_{\theta_j}(q_j) \le c_j^{-n} \operatorname{vol}_\theta(f_j) \le c_j^{-n} \operatorname{vol}_\theta(r_j).$$

Since $c_j \to 1$, Step 2 shows that the outer terms tend to the same limit. The volumes $\operatorname{vol}_{\theta_j}(q_j)$ decrease by monotonicity in both background and potential, which proves the asserted decreasing convergence.

Step 4: Lelong numbers. By Proposition 31.4.2, the volumes of the fixed potential q in background θ_j tend to its volumes in background θ. Hence the upper-volume deficit between q and q_j, measured in θ_j, tends to zero, while the lower volume of q_j is at least $\underline{\operatorname{vol}}_\theta(q) > 0$. Apply Proposition 31.2.1 in each background θ_j. Since $\theta_j \le \theta_1$, its proof works with one Lelong bound C_π for $\pi^* \theta_1$, independent of j. This proves the uniform convergence on each fixed modification, and taking generic values gives the divisorial assertion. □

For an arbitrary decreasing sequence, convergence of the ordinary envelopes is equivalent to continuity of the upper volumes.

Proposition 31.5.1 *Suppose that $u_j \in \mathrm{PSH}(X,\theta_j)$ decrease to $u \in \mathrm{PSH}(X,\theta)$, and that u has positive lower volume. Then*

$$P_{\theta_j}[u_j] \searrow P_\theta[u] \quad \Longleftrightarrow \quad \overline{\mathrm{vol}}_{\theta_j}(u_j) \searrow \overline{\mathrm{vol}}_\theta(u).$$

When these conditions hold, the lower volumes converge to that of u in background θ, and the Lelong numbers converge uniformly on every fixed smooth compact modification.

Proof Normalize $u_j \le 0$ by a common translation. Set $p_j = P_{\theta_j}[u_j]$, $p = P_\theta[u]$ and $q = \lim_j p_j$. Since $\underline{\mathrm{vol}}_{\theta_j}(u_j) \ge \underline{\mathrm{vol}}_\theta(u) > 0$, Theorem 31.3.2 in background θ_j shows that every p_j is an ordinary model with the same volumes as u_j. The envelopes decrease, and the uniform convergence of the backgrounds gives $q \in \mathrm{PSH}(X,\theta)$. Since $u \le p \le q$, the limit q has positive lower volume, and Theorem 31.5.2 gives, for either volume,

$$\mathrm{vol}_\theta(q) = \lim_j \mathrm{vol}_{\theta_j}(p_j) = \lim_j \mathrm{vol}_{\theta_j}(u_j), \qquad P_\theta[q] = q.$$

If $q = p$, these equalities and Theorem 31.3.2 show that both volumes of u_j converge to those of u; in particular the upper volumes do. Conversely, if the upper volumes converge, then u and q have the same upper volume, so $u \in \mathcal{E}_{\mathrm{vol}}(X,\theta,q)$. By Corollary 31.3.2, $P_\theta[u] = q$, that is, $q = p$. For the Lelong assertion, Proposition 31.4.2 shows that the upper-volume deficit between u and u_j in background θ_j tends to zero. Apply Proposition 31.2.1 with the common lower bound $\underline{\mathrm{vol}}_\theta(u) > 0$ and the uniform Lelong bound for $\pi^*\theta_1$. □

Corollary 31.5.1 *For every $u \in \mathrm{PSH}(X,\theta)$,*

$$\underline{\mathrm{vol}}_{\theta_j}(u) \searrow \underline{\mathrm{vol}}_\theta(u).$$

If this limiting volume is positive, then also

$$\overline{\mathrm{vol}}_{\theta_j}(u) \searrow \overline{\mathrm{vol}}_\theta(u), \qquad P_{\theta_j}[u] \searrow P_\theta[u].$$

Proof Fix a Hermitian form ω. Uniform convergence gives $\theta_j \le \theta + \varepsilon_j\omega$ with $\varepsilon_j \to 0$, so the volume limits follow from the two estimates in Proposition 31.4.2; monotonicity makes them decreasing. In the positive-lower-volume case, apply Proposition 31.5.1 to the constant sequence $u_j = u$. □

As an application, we show that trace envelopes realize their trace data exactly. Let $Y \subset X$ be a connected compact complex submanifold of positive dimension, and let $u \in \mathrm{PSH}(X,\theta)$ satisfy $\nu(u,Y) = 0$. We use the restricted multiplier ideals $\mathfrak{a}_m(u;Y) = \mathcal{I}(mu)\mathcal{O}_Y$, the trace data $\tau_{Y,u}(E) = \lim_m m^{-1}\operatorname{ord}_E \mathfrak{a}_m(u;Y)$ of Theorem 20.1.1, the analytic approximants u_m of u and the trace envelopes $\mathcal{T}_\Omega(u;Y)$ of (20.5) (Chapter 20). A majorant in a smaller background gives the trace envelope positive lower volume, so Theorem 31.5.2 applies on Y.

Corollary 31.5.2 *Suppose that Y has the bounded mass property, and let Ω be a Hermitian form on Y with $\Omega \geq 2\theta|_Y$ and $v_-(\Omega) > 0$. Then*

$$\nu\big(\mathcal{T}_\Omega(u;Y)\circ\pi, E\big) = \tau_{Y,u}(E)$$

for every prime divisor E on every smooth compact modification $\pi\colon Z \to Y$.

Proof Put $T = \mathcal{T}_\Omega(u;Y)$ and $r = \dim Y$. Normalize the majorant v of Proposition 20.6.1 by $\sup_Y v = 0$. It belongs to $\mathrm{PSH}(Y,\theta|_Y) \subset \mathrm{PSH}(Y,\Omega/2)$ and is a competitor for T, so $v \leq T$. For every $w \in \mathrm{PSH}(Y,\Omega)$ with $w \sim T$, a constant translate of v lies below w, so $P_{\Omega/2}(w) \not\equiv -\infty$. By Lemma 14.5.2, applied on Y with $a = 1/2$,

$$\int_Y \mathrm{MA}_\Omega(w) \geq 2^{-r} v_-(\Omega), \qquad \underline{\mathrm{vol}}_\Omega(T) \geq 2^{-r} v_-(\Omega) > 0.$$

Since $\Omega - \theta|_Y \geq \Omega/2$ is Hermitian and the curvature errors of the approximants tend to zero, the restrictions $u_m|_Y$ belong to $\mathrm{PSH}(Y,\Omega)$ for all sufficiently large m. By Theorem 20.6.1, their dyadic envelopes $q_j = P_\Omega[u_{2^j}|_Y] = P_\Omega[u_{2^j}|_Y]_{\mathcal{I}}$ decrease to T; they are ordinary models by Proposition 17.2.2. Theorem 31.5.2, applied on Y in the fixed background Ω, and the analytic type of $u_{2^j}|_Y$ give

$$\nu(T\circ\pi, E) = \lim_j \nu(q_j\circ\pi, E) = \lim_j 2^{-j}\operatorname{ord}_E \mathfrak{a}_{2^j}(u;Y) = \tau_{Y,u}(E),$$

as required. □

31.5.1 Recovery of envelope systems

We return to the big background θ on X and apply the monotone limits above to the envelope systems of Chapter 26. Fix real numbers $a < c$. We use the sets $\mathcal{T}_{a,c}(\theta)$ of concave decreasing curves $(\Gamma_\tau)_{\tau<c}$ of ideal models that are constant for $\tau \leq a$ (Definition 26.1.1), the envelope systems $\mathbf{\Gamma} \in \mathcal{S}_{a,c}(\theta)$ with components $\Gamma^\eta \in \mathcal{T}_{a,c}(\theta+\eta)$ indexed by the Hermitian forms η (Definition 26.2.1), and the maps E and R of Theorem 26.3.1, where $(\mathsf{R}\mathbf{\Gamma})_\tau = \lim_j \Gamma_\tau^{h/j}$ for any Hermitian form h. The system is recovered from its limit if $\mathbf{\Gamma} = \mathsf{E}\mathsf{R}\mathbf{\Gamma}$.

Compatibility fixes every local divisorial Lelong number of the slices before the perturbation tends to zero, and recovery means that these numbers do not increase

in the limit. A positive limiting lower volume rules out such an increase. Concavity propagates positivity from the constant left part to every finite parameter, so a single condition on the initial slice suffices for the whole system.

Theorem 31.5.3 *Let $\mathbf{\Gamma} \in \mathcal{S}_{a,c}(\theta)$, fix a Hermitian form h, and put $\Gamma = \mathsf{R}\mathbf{\Gamma}$. Then*

$$\lim_{j\to\infty} \underline{\mathrm{vol}}_{\theta+h/j}(\Gamma_a^{h/j}) = \underline{\mathrm{vol}}_{\theta}(\Gamma_a).$$

In particular this limit is independent of h. If it is positive, then $\mathbf{\Gamma} = \mathsf{E}(\Gamma)$, and for every $\tau < c$ both singularity volumes satisfy

$$\mathrm{vol}_{\theta+h/j}(\Gamma_\tau^{h/j}) \searrow \mathrm{vol}_\theta(\Gamma_\tau), \qquad \underline{\mathrm{vol}}_\theta(\Gamma_\tau) > 0.$$

Proof Every ideal model is an ordinary model by Proposition 17.2.2, and the retraction theorem gives $\Gamma_\tau^{h/j} \searrow \Gamma_\tau$ in the decreasing backgrounds $\theta + h/j$. Thus Theorem 31.5.2 proves the identity for the initial lower volumes, including the case where the limit is zero.

Suppose that this limit ℓ is positive. For $\tau \le a$, the slice is Γ_a. For $a < \tau < c$, choose $r \in (\tau, c)$ and put $s = (\tau - a)/(r - a)$, so that $\tau = (1 - s)a + sr$. Concavity gives

$$(1 - s)\Gamma_a + s\Gamma_r \le \Gamma_\tau.$$

Homogeneity, Proposition 31.1.1 and volume monotonicity imply

$$\underline{\mathrm{vol}}_\theta(\Gamma_\tau) \ge (1 - s)^n \ell > 0.$$

Theorem 31.5.2 now gives both volume limits for every fixed $\tau < c$.

It remains to show that the slices keep their multiplier ideals in the limit. Fix $\tau < c$ and write $q = \Gamma_\tau$, $q_j = \Gamma_\tau^{h/j}$ and $\theta_j = \theta + h/j$. By Proposition 31.4.2, the upper volumes of the fixed potential q in the backgrounds θ_j converge to its upper volume in θ, which is the limit of the upper volumes of the q_j. Consequently

$$D_j = \overline{\mathrm{vol}}_{\theta_j}(q_j) - \overline{\mathrm{vol}}_{\theta_j}(q) \longrightarrow 0, \qquad \underline{\mathrm{vol}}_{\theta_j}(q_j) \ge \underline{\mathrm{vol}}_\theta(q) > 0.$$

After discarding finitely many indices, choose $b_j > 1$ with $b_j \to +\infty$ and $b_j^n D_j < \underline{\mathrm{vol}}_{\theta_j}(q_j)$. Since $q \le q_j$, the stretching lemma Lemma 31.2.1 in background θ_j gives $v_j \in \mathrm{PSH}(X, \theta_j)$ such that

$$(1 - b_j^{-1})q_j + b_j^{-1} v_j \le q.$$

Choose a Hermitian form $\Omega \geq \theta + h$, so that all v_j are Ω-plurisubharmonic. By Lemma 1.5.1, there is $\alpha > 0$ such that $\mathcal{I}(\alpha v_j) = \mathcal{O}_X$ for all j; additive normalization does not change these ideals. Let E be a local divisor over a coordinate neighborhood U, with generic Lelong number ν_E and log discrepancy $A_U(E)$ as in Section 1.17. Lemma 1.17.8, with $m = \alpha$, gives $\nu_E(v_j) \leq A_U(E)/\alpha$. Taking Lelong numbers in the convex inequality therefore yields

$$0 \leq \nu_E(q) - \nu_E(q_j) \leq \frac{A_U(E)}{\alpha b_j} \longrightarrow 0.$$

Compatibility gives all components the same multiplier ideals, so $\nu_E(q_j)$ is independent of j, and hence equal to $\nu_E(q)$. Blowing up a point on a local modification converts its pointwise Lelong number into such a divisorial number; on a curve the point itself is a divisor. Thus all Lelong numbers of q and q_j on all local modifications agree, and Theorem 1.16.1 gives $q \sim_{\mathcal{I}} q_j$. This holds at every parameter $\tau < c$, so Proposition 26.3.1 proves recovery. □

The lower bound $(1 - s)^n \ell$ may tend to zero as $\tau \uparrow c$; the theorem gives no positive lower bound that is uniform up to the cutoff. The hypothesis is invariant under a smooth change of background $\theta' = \theta + \mathrm{dd}^{\mathrm{c}} f$, $f \in C^\infty(X, \mathbb{R})$, through the gauge bijection of Proposition 26.4.1: the estimate there makes the transported initial slice ordinarily equivalent to $\Gamma_a - f$, and the map $u \mapsto u - f$ preserves the curvature currents and hence both singularity volumes. The ruled-surface system of Theorem 26.5.1, which is not recovered from its limit, has zero initial limiting lower volume by Proposition 26.5.1, so the theorem does not apply to it.

31.6 Cauchy convergence of model potentials

The maximum of two potentials is a common upper bound for both, and the sum of the two upper-volume deficits from this maximum measures how far apart their ordinary envelopes are. A uniform positive lower-volume bound controls how this quantity changes when a third potential is introduced, and this replaces the triangle inequality in the Cauchy argument below. From now on we write $U(u) = \overline{\mathrm{vol}}_\theta(u)$ and $M = U(V_\theta)$; by monotonicity, $U \leq M$ on $\mathrm{PSH}(X, \theta)$. We also fix a smooth positive probability volume form $\mathrm{d}V$ for L^1 norms.

Definition 31.6.1 For $u, v \in \mathrm{PSH}(X, \theta)$, define

$$d_\theta(u, v) = 2U(\max(u, v)) - U(u) - U(v).$$

For $\delta > 0$, put

$$\mathcal{M}_\delta(\theta) = \{u \in \mathrm{PSH}(X, \theta) : P_\theta[u] = u,\ \underline{\mathrm{vol}}_\theta(u) \geq \delta\}.$$

We call d_θ the *volume comparison*. A sequence is d_θ-Cauchy if $d_\theta(u_j, u_k) \to 0$ as $j, k \to \infty$, and it converges to u for d_θ if $d_\theta(u_j, u) \to 0$.

The volume comparison is finite, nonnegative and symmetric. It need not satisfy the triangle inequality (Example 31.12.1), which is why we do not call it a distance. It depends only on the ordinary singularity types, since replacing either input by an equivalent one changes the maximum by a bounded amount. Monotonicity also gives

$$|U(u) - U(v)| \le d_\theta(u, v), \qquad u \le v \implies d_\theta(u, v) = U(v) - U(u). \tag{31.6}$$

In the separation result below, the restriction to positive lower volume is necessary, even for a closed positive background: at zero volume the comparison can vanish between distinct models (Example 32.2.4).

Proposition 31.6.1 *For potentials u, v of positive lower volume,*

$$d_\theta(u, v) = d_\theta(P_\theta[u], P_\theta[v]), \qquad d_\theta(u, v) = 0 \iff P_\theta[u] = P_\theta[v].$$

There is $C_\theta > 0$, depending on θ and $\mathrm{d}V$*, such that, for $u, v \in \mathcal{M}_\delta(\theta)$,*

$$\|u - v\|_{L^1} \le 2C_\theta \min\left\{1, \left(\frac{2d_\theta(u, v)}{\delta}\right)^{1/n}\right\}.$$

Proof Translate the inputs below zero, and put $p = P_\theta[u]$, $q = P_\theta[v]$, $m = \max(u, v)$ and $r = P_\theta[m]$. Since $u \le p$, $v \le q$ and the envelope is monotone,

$$m \le \max(p, q) \le r.$$

All these potentials have positive lower volume. Theorem 31.3.2 and monotonicity therefore give $U(m) = U(\max(p, q)) = U(r)$, while $U(u) = U(p)$ and $U(v) = U(q)$. This proves the first identity. If $d_\theta(u, v) = 0$, then $U(u) = U(v) = U(m)$, since both are at most $U(m)$. Thus $u, v \in \mathcal{E}_{\mathrm{vol}}(X, \theta, m)$, and Theorem 31.2.1 and Proposition 14.4.3 give $P_\theta[u] = P_\theta[m] = P_\theta[v]$. Conversely, if $p = q$, the first identity gives $d_\theta(u, v) = d_\theta(p, p) = 0$.

Now let $u, v \in \mathcal{M}_\delta(\theta)$ and set $s = d_\theta(u, v)$. By Theorem 1.4.1, there is C_θ bounding the L^1 norm of every θ-plurisubharmonic potential with supremum zero. Since ordinary models have supremum zero, the estimate holds if $s \ge \delta/2$. Suppose $0 < s < \delta/2$ and put $t = (2s/\delta)^{1/n}$. The upper-volume deficit from u to m is at most s, and the lower volume of m is at least δ. Thus Lemma 31.2.1, with $b = t^{-1}$, gives $h \in \mathrm{PSH}(X, \theta)$ such that

$$(1 - t)m + th \le u.$$

Since $m \le 0$ and $u = P_\theta[u]$, combining m with the competitors for $P_\theta[h]$, as in the proof of Proposition 14.4.3, gives

$$(1 - t)m + tP_\theta[h] \le u.$$

Hence $0 \le m - u \le t(m - P_\theta[h]) \le -tP_\theta[h]$, and $\int_X (m-u)\,\mathrm{d}V \le tC_\theta$ because $P_\theta[h]$ has supremum zero. The same argument applies to v, and adding the two estimates gives the assertion, since $|u-v| \le (m-u)+(m-v)$. If $s = 0$, the argument works for every $t \in (0,1)$; let $t \downarrow 0$. □

Although the triangle inequality fails, the volume comparison satisfies it up to an error controlled by a lower bound for the lower volume of the middle point.

Proposition 31.6.2 *Let $u, v, w \in \mathrm{PSH}(X,\theta)$ with $\underline{\mathrm{vol}}_\theta(v) \ge \delta > 0$. Then*

$$d_\theta(u,w) \le d_\theta(u,v) + d_\theta(v,w) + 2nM \min\left\{1, \left(\frac{2d_\theta(v,w)}{\delta}\right)^{1/n}\right\}. \tag{31.7}$$

Consequently every sequence in $\mathcal{M}_\delta(\theta)$ that converges for d_θ to an element of $\mathcal{M}_\delta(\theta)$ is d_θ-Cauchy, and this limit is unique.

Proof Put $s = d_\theta(v,w)$. If $s \ge \delta/2$, the inequality follows from $d_\theta(u,w) \le 2M$. If $0 < s < \delta/2$, set $t = (2s/\delta)^{1/n}$. Lemma 31.2.1, applied to $v \preceq \max(v,w)$, whose lower volume is at least δ, gives $h \in \mathrm{PSH}(X,\theta)$ with

$$(1-t)\max(v,w) + th \le v.$$

This inequality also gives $h \le v$. Set $r = \max(u,v)$ and $z = \max(u,v,w)$. Where $z = \max(v,w)$, the inequality above bounds $(1-t)z + th$ by $v \le r$; where $z = u$, both u and h lie below r. Thus

$$(1-t)z + th \le r.$$

Addition, homogeneity and volume monotonicity give

$$U(z) - U(r) \le \big(1 - (1-t)^n\big)U(z) \le nMt.$$

Since $\max(u,w) \le z$, it follows that

$$\begin{aligned} d_\theta(u,w) &\le 2U(r) + 2nMt - U(u) - U(w) \\ &= d_\theta(u,v) + U(v) - U(w) + 2nMt \\ &\le d_\theta(u,v) + s + 2nMt. \end{aligned}$$

Here the last step uses (31.6). For $s = 0$, the same proof works with every $t \in (0,1)$; let $t \downarrow 0$. This proves (31.7). Applying it with the limit as the middle point shows that a convergent sequence is Cauchy. If a sequence converges to two limits, applying it with a member of the sequence as the middle point shows that the two limits have volume comparison zero; by Proposition 31.6.1, they are equal. □

The same stretching controls the lower volume.

Proposition 31.6.3 *Let $p, q \in \mathrm{PSH}(X,\theta)$, one of which has lower volume at least $\delta > 0$. Then*

$$\left|\underline{\mathrm{vol}}_\theta(p) - \underline{\mathrm{vol}}_\theta(q)\right| \le nM \min\left\{1, \left(\frac{2d_\theta(p,q)}{\delta}\right)^{1/n}\right\}. \tag{31.8}$$

In particular, if $\underline{\mathrm{vol}}_\theta(p) > 0$ and $d_\theta(p_j, p) \to 0$, then $\underline{\mathrm{vol}}_\theta(p_j) \to \underline{\mathrm{vol}}_\theta(p)$.

Proof Write $L = \underline{\mathrm{vol}}_\theta$, and put $s = d_\theta(p,q)$ and $m = \max(p,q)$. If $s \ge \delta/2$, the bound follows from $|L(p) - L(q)| \le M$. Suppose $s < \delta/2$, and take $t = (2s/\delta)^{1/n}$ if $s > 0$ and any $t \in (0,1)$ if $s = 0$. The upper-volume deficit from either p or q to m is at most s, while $L(m) \ge \delta$. Thus Lemma 31.2.1 gives a potential $h \in \mathrm{PSH}(X,\theta)$ with

$$(1-t)m + th \le p.$$

Volume addition and monotonicity imply $L(p) \ge (1-t)^n L(m) \ge (1-t)^n L(q)$, so $L(q) - L(p) \le (1-(1-t)^n)M \le nMt$. Interchanging p and q, and letting $t \downarrow 0$ if $s = 0$, proves the bound. The last assertion is the case $\delta = \underline{\mathrm{vol}}_\theta(p)$. □

Theorem 31.6.1 *Every d_θ-Cauchy sequence in $\mathcal{M}_\delta(\theta)$ converges for d_θ to a unique element of $\mathcal{M}_\delta(\theta)$. It also converges to that element in $L^1(X, \mathrm{d}V)$.*

Proof The plan is to dominate a fast subsequence by the ordinary envelopes of its tail suprema; these decrease to the limit, and the stretching lemma controls the volume they add. Choose a subsequence a_j of the given sequence such that

$$d_\theta(a_j, a_{j+1}) < \delta\, 2^{-(j+1)n} \qquad (j \ge 1).$$

For $k \ge 0$, set

$$v_j^k = \max(a_j, \dots, a_{j+k}), \qquad v_j = \left(\sup_{l \ge j} a_l\right)^*.$$

For every $l \ge j$, Lemma 31.2.1 with $b = 2^l$, applied to $a_l \preceq \max(a_l, a_{l+1})$, gives $h_l \le a_l$ satisfying

$$(1 - 2^{-l})\max(a_l, a_{l+1}) + 2^{-l} h_l \le a_l.$$

The argument by cases in the proof of Proposition 31.6.2 gives

$$(1-2^{-l})v_j^{l+1-j} + 2^{-l}h_l \le v_j^{l-j}.$$

Consequently

$$U(v_j^{l+1-j}) - U(v_j^{l-j}) \le nM2^{-l}.$$

Summing from $l = j$ to $j + k - 1$ and then letting $k \to \infty$, using Theorem 31.5.1, yields

$$0 \le U(v_j) - U(a_j) \le nM2^{1-j}.$$

Here $v_j \in \mathrm{PSH}(X,\theta)$ has lower volume at least δ, because $a_j \le v_j \le 0$; this also justifies the use of that theorem.

Put $q_j = P_\theta[v_j]$. By Theorem 31.3.2, q_j is an ordinary model with the same two volumes as v_j. The sequence q_j decreases, and each q_j has supremum zero. By normalized compactness, its decreasing limit q is not identically $-\infty$, so $q \in \mathrm{PSH}(X,\theta)$. By Theorem 31.5.2,

$$q = P_\theta[q], \qquad \underline{\mathrm{vol}}_\theta(q) = \lim_j \underline{\mathrm{vol}}_\theta(q_j) \ge \delta, \qquad U(q_j) \longrightarrow U(q).$$

Both a_j and q lie below q_j. Hence

$$\begin{aligned} d_\theta(a_j, q) &\le \big(U(q_j) - U(a_j)\big) + \big(U(q_j) - U(q)\big) \\ &\le nM2^{1-j} + U(q_j) - U(q) \longrightarrow 0. \end{aligned}$$

To treat the whole sequence, apply (31.7) with a late member a_j of the subsequence as the middle point. The Cauchy property makes the first term small for all sufficiently late members of the original sequence, while $d_\theta(a_j, q)$ makes the other two terms small. Thus the original sequence converges to q. Uniqueness follows from Proposition 31.6.2, and the L^1 estimate in Proposition 31.6.1 proves the last assertion. □

The non-strict inequality $\underline{\mathrm{vol}}_\theta \ge \delta$ in the definition of $\mathcal{M}_\delta(\theta)$ is essential for completeness.

Example 31.6.1 Let θ be a smooth positive $(1,1)$-form of integral one on a compact Riemann surface, and let $0 < \delta < 1$. The ordinary models of mass strictly greater than δ do not form a complete space for d_θ.

Proof Fix a point x, and let $g_x \le 0$ be the Green function used in Theorem 20.3.1, with $\mathrm{dd^c} g_x = \delta_x - \theta$, where δ_x is the Dirac measure at x. For $0 \le a < 1$, put $q_a = P_\theta[a g_x]$. The current $\theta + \mathrm{dd^c}(a g_x)$ equals $a\delta_x + (1-a)\theta$, with non-pluripolar mass $1-a$. Since a Riemann surface is Kähler and the background is closed, both singularity volumes of $a g_x$ equal this mass, and Theorem 31.3.2 gives

$$q_a = P_\theta[q_a], \qquad \underline{\mathrm{vol}}_\theta(q_a) = U(q_a) = 1 - a.$$

Since $g_x \le 0$, the potentials q_a decrease as a increases, and (31.6) gives $d_\theta(q_a, q_b) = |a-b|$. Choose $a_j < 1-\delta$ with $a_j \uparrow 1-\delta$. The sequence q_{a_j} is d_θ-Cauchy, and every term has mass $1 - a_j > \delta$. By (31.6), any d_θ-limit has upper volume $\lim_j (1-a_j) = \delta$. In this closed background the two volumes coincide, so no such limit has mass strictly greater than δ. □

31.7 Rooftops and capacity convergence

A small upper-volume deficit also controls the rooftop envelope of two potentials. For a finite family it suffices that the sum of the deficits stays below the lower volume of a common upper bound. Applying this criterion to stretched potentials gives increasing lower bounds for a suitable subsequence of models. Together with the decreasing upper bounds of the preceding section, they turn convergence for the volume comparison into convergence in capacity.

Lemma 31.7.1 *Let $u_1, \dots, u_N, w \in \mathrm{PSH}(X, \theta)$, with $N \geq 1$ and $u_i \preceq w$. If*

$$\sum_{i=1}^{N} \big(U(w) - U(u_i)\big) < \underline{\mathrm{vol}}_\theta(w),$$

then the rooftop envelope $P_\theta(u_1, \dots, u_N) = P_\theta(\min_i u_i)$ is not identically $-\infty$.

Proof Whether a finite rooftop envelope is identically $-\infty$ does not change under bounded changes of its obstacles. By the definition of $U(u_i)$, we can choose $v_i \in \mathrm{PSH}(X, \theta)$ with $v_i \sim u_i$ whose masses satisfy

$$\sum_{i=1}^{N} \left(U(w) - \int_X \mu_{v_i} \right) < \underline{\mathrm{vol}}_\theta(w).$$

Translate so that $v_i \leq w \leq 0$, and put

$$v_{i,k} = \max(v_i, w - k), \qquad p_k = P_\theta(v_{1,k}, \dots, v_{N,k}).$$

The common competitor $w - k$ gives $w - k \leq p_k \leq w$, so $p_k \sim w$. The p_k decrease as k increases, and a θ-plurisubharmonic limit would be a competitor for $P_\theta(v_1, \dots, v_N)$. Suppose, to reach a contradiction, that $P_\theta(u_1, \dots, u_N) \equiv -\infty$, so that $P_\theta(v_1, \dots, v_N) \equiv -\infty$ as well. Then the p_k decrease to $-\infty$, and normalized compactness gives $\sup_X p_k \to -\infty$.

Fix $s > 0$ and choose $k > s$ so large that $p_k \leq -s$. Iterating Proposition 14.1.3, which bounds the measure of a rooftop by the measures of its obstacles on the contact sets, and using locality on $\{v_i > -s\}$, we obtain

$$\begin{aligned} \underline{\mathrm{vol}}_\theta(w) &\leq \int_X \mu_{p_k} \leq \sum_{i=1}^{N} \int_{\{v_{i,k} \leq -s\}} \mu_{v_{i,k}} \\ &\leq \sum_{i=1}^{N} \left(U(w) - \int_{\{v_i > -s\}} \mu_{v_i} \right). \end{aligned}$$

Here we used $v_{i,k} \sim w$, and $w - k < -s$, which gives $v_{i,k} = v_i$ on $\{v_{i,k} > -s\} = \{v_i > -s\}$. Let $s \to \infty$. The measures μ_{v_i} are finite and put no mass on pluripolar sets, so the resulting inequality contradicts the choice of the v_i. □

To control the volume of a rooftop envelope, we first stretch each obstacle away from the maximum of the two, and then move both stretched potentials a fixed fraction back toward this maximum. By the lemma, the resulting potentials have a rooftop envelope that is not identically $-\infty$.

Proposition 31.7.1 *Let $u_j, u \in \mathrm{PSH}(X,\theta)$, with $\underline{\mathrm{vol}}_\theta(u) > 0$, and suppose that $d_\theta(u_j,u) \to 0$. For all sufficiently large j, the rooftop envelope $p_j = P_\theta(u_j,u)$ is not identically $-\infty$, and*

$$d_\theta(p_j,u) \longrightarrow 0, \qquad \underline{\mathrm{vol}}_\theta(p_j) \longrightarrow \underline{\mathrm{vol}}_\theta(u).$$

Proof Put $\delta = \underline{\mathrm{vol}}_\theta(u)$, $m_j = \max(u_j,u)$ and $s_j = d_\theta(u_j,u)$. After discarding finitely many indices, choose $b_j \to \infty$ with $b_j > 1$ and $b_j^n s_j < \delta$. The upper-volume deficits of both u_j and u from m_j are at most s_j, while m_j has lower volume at least δ. Thus Lemma 31.2.1 shows that the stretching envelopes

$$G_j = Q_{b_j}(u_j,m_j), \qquad H_j = Q_{b_j}(u,m_j)$$

are defined. Choose $a \in (0,1)$ with $2nMa < \delta$, and set

$$g_j = (1-a)m_j + aG_j, \qquad h_j = (1-a)m_j + aH_j.$$

These potentials lie below m_j, because $G_j \le u_j$ and $H_j \le u$. Addition and monotonicity give

$$2U(m_j) - U(g_j) - U(h_j) \le 2\big(1-(1-a)^n\big)U(m_j) \le 2nMa < \delta.$$

Lemma 31.7.1, with $w = m_j$, therefore shows that $r_j = P_\theta(g_j,h_j)$ is θ-plurisubharmonic. For large j, put $t_j = (ab_j)^{-1} < 1$. Since $r_j \le g_j$ and $r_j \le h_j$, the defining inequalities for G_j and H_j show that

$$(1-t_j)m_j + t_j r_j \le u_j, u.$$

The left-hand side is therefore a competitor for p_j, so $p_j \not\equiv -\infty$. For either volume, addition and monotonicity give

$$(1-t_j)^n \,\mathrm{vol}_\theta(u) \le (1-t_j)^n \,\mathrm{vol}_\theta(m_j) \le \mathrm{vol}_\theta(p_j) \le \mathrm{vol}_\theta(u).$$

Since $t_j \to 0$, this proves $\underline{\mathrm{vol}}_\theta(p_j) \to \underline{\mathrm{vol}}_\theta(u)$; and since $p_j \le u$, (31.6) gives $d_\theta(p_j,u) = U(u) - U(p_j) \to 0$. □

For ordinary models, finite rooftop envelopes are again ordinary models. Their normalization prevents collapse when the number of obstacles tends to infinity.

Theorem 31.7.1 *Suppose $u_j, u \in \mathcal{M}_\delta(\theta)$ and $d_\theta(u_j,u) \to 0$. After passing to a subsequence and relabeling it, there are $v_j, w_j \in \mathcal{M}_{\delta/2}(\theta)$ such that*

$$w_j \le u_j \le v_j, \qquad w_j \le u \le v_j,$$

v_j decreases to u, w_j increases with $(\lim_j w_j)^* = u$, and

$$d_\theta(v_j, u) + d_\theta(w_j, u) \longrightarrow 0.$$

The original sequence converges to u in capacity.

Proof *Step 1: upper bounds.* For large j, the rooftop envelope $r_j = P_\theta(u_j, u)$ is θ-plurisubharmonic by Proposition 31.7.1. It is an ordinary model: since $r_j \le 0$ and the envelope is monotone, $r_j \le P_\theta[r_j] \le P_\theta[u_j] = u_j$ and likewise $P_\theta[r_j] \le u$, so $P_\theta[r_j]$ is a competitor for the rooftop and equals r_j. The sequence u_j is d_θ-Cauchy by Proposition 31.6.2, and $d_\theta(r_j, u) = U(u) - U(r_j) \to 0$ by Proposition 31.7.1. Choose a subsequence so that

$$d_\theta(u_j, u_{j+1}) < \delta\, 2^{-(j+1)n}, \qquad U(u) - U(r_j) < \delta\, 4^{-(j+1)n}.$$

The tail-maximum construction in the proof of Theorem 31.6.1 gives

$$v_j = P_\theta\left[\left(\sup_{l \ge j} u_l\right)^*\right] \searrow u, \qquad d_\theta(v_j, u) \longrightarrow 0.$$

Indeed, that construction produces the unique d_θ-limit, which here is u. The models v_j satisfy $v_j \ge u_j$ and $v_j \ge u$, and have lower volume at least δ.

Step 2: lower bounds. By Lemma 31.2.1 and the choice of the subsequence, the stretching envelopes

$$H_l = Q_{4^l}(r_l, u), \qquad K_l = Q_{2^l}(r_l, u)$$

are defined. Their defining inequalities show that $(1 - 2^{-l})u + 2^{-l}H_l$ is a competitor for K_l and that $K_l \le u$. Together with addition and monotonicity, this gives

$$(1 - 2^{-l})u + 2^{-l}H_l \le K_l \le u, \qquad U(u) - U(K_l) \le nM2^{-l}.$$

Choose j so large that $nM2^{1-j} < \delta$. For every $k \ge 0$, the sum of the deficits of $K_j, \dots, K_{j+k}$ from u is less than δ, so Lemma 31.7.1 shows that their rooftop envelope is not identically $-\infty$. Since $r_l \le u$, the stretching envelope $Q_b(r_l, u)$ decreases as b increases; hence, for $l \ge j$,

$$Q_{2^j}(r_l, u) \ge K_l.$$

Thus $R_j^k = P_\theta(Q_{2^j}(r_j, u), \dots, Q_{2^j}(r_{j+k}, u))$ is θ-plurisubharmonic. The convex inequality for each obstacle gives

$$(1 - 2^{-j})u + 2^{-j}R_j^k \le w_j^k := P_\theta(r_j, \dots, r_{j+k}) \le u.$$

For either volume,

$$(1 - 2^{-j})^n \operatorname{vol}_\theta(u) \le \operatorname{vol}_\theta(w_j^k) \le \operatorname{vol}_\theta(u). \tag{31.9}$$

The argument used for r_j shows that w_j^k is an ordinary model. For fixed j it decreases in k and has supremum zero, so normalized compactness shows that its limit w_j is θ-plurisubharmonic. The lower-volume conclusion of Theorem 31.5.2 passes the lower bound in (31.9) to w_j. This bound is positive, so the upper-volume conclusion of the same theorem passes the bound for the upper volume as well.

Step 3: convergence. The w_j increase, since each is defined by fewer obstacles than the previous one, and $w_j \le r_j \le u_j$, $w_j \le u$. After discarding finitely many indices, they have lower volume at least $\delta/2$, and (31.9) gives $d_\theta(w_j, u) \le nM2^{-j}$. By Proposition 31.6.1, $w_j \to u$ in L^1, so $(\lim_j w_j)^* = u$. The two monotone sequences v_j and w_j converge to u in capacity by Lemma 1.8.9, and the bounds $w_j \le u_j \le v_j$ give $u_j \to u$ in capacity along the chosen subsequence. Since every subsequence of the original sequence has a further subsequence of this kind, the original sequence converges to u in capacity. □

31.8 Volumes of multiplier-ideal envelopes

Analytic regularizations recover the multiplier-ideal envelope, whose volumes may exceed those of the original potential. Continuity along decreasing models computes this excess, and a strict analytic potential absorbs the curvature error of the regularization, so that the comparison can be carried out entirely in the fixed background. For $u \in \mathrm{PSH}(X, \theta)$, write

$$\mathcal{V}^-_{\mathcal{I},\theta}(u) = \underline{\mathrm{vol}}_\theta(P_\theta[u]_{\mathcal{I}}), \qquad \mathcal{V}^+_{\mathcal{I},\theta}(u) = \overline{\mathrm{vol}}_\theta(P_\theta[u]_{\mathcal{I}})$$

for the two volumes of its ideal envelope. They depend only on the multiplier-ideal class of u. By Proposition 17.3.1 and the gauge invariance of singularity volumes, they are also unchanged under a smooth change of decomposition $(\theta, u) \mapsto (\theta + \mathrm{dd^c} f, u - f)$, $f \in C^\infty(X, \mathbb{R})$. Moreover, Proposition 17.2.2 and Theorem 31.1.1 give

$$\underline{\mathrm{vol}}_\theta(u) \le \mathcal{V}^-_{\mathcal{I},\theta}(u), \qquad \overline{\mathrm{vol}}_\theta(u) \le \mathcal{V}^+_{\mathcal{I},\theta}(u).$$

The ideal envelope is an ordinary model by Proposition 17.2.2, and its volumes are limits of the volumes of the dyadic analytic approximations.

Corollary 31.8.1 *Let $u \in \mathrm{PSH}(X,\theta)$, fix a Hermitian form h, and let $a_j \in \mathrm{PSH}(X,\theta_j)$, $\theta_j = \theta + \epsilon_j h$, be the dyadic analytic regularizations of u of analytic type $(2^{-j}, \mathcal{I}(2^j u))$, with errors $\epsilon_j \downarrow 0$, as in (17.3). Then*

$$\underline{\mathrm{vol}}_{\theta_j}(a_j) \searrow \mathcal{V}^-_{\mathcal{I},\theta}(u).$$

If $\mathcal{V}^-_{\mathcal{I},\theta}(u) > 0$, then also

$$\overline{\mathrm{vol}}_{\theta_j}(a_j) \searrow \mathcal{V}^+_{\mathcal{I},\theta}(u).$$

Proof By Theorem 17.5.1, the analytic ordinary models $q_j = P_{\theta_j}[a_j]$ decrease to $P_\theta[u]_{\mathcal{I}}$, and the backgrounds θ_j decrease to θ. Thus Theorem 31.5.2 applies to the q_j. Since $q_j \sim a_j$ by Proposition 17.2.4, the volumes of q_j and a_j agree. □

When the ideal envelope has positive lower volume, the relative full-mass class detects whether u and its ideal envelope have the same upper volume. Here $P_{\theta,F}[u]$ and $P_{\theta,F}[u]_{\mathcal{I}}$ denote the ordinary and the multiplier-ideal envelopes with the constraint $h \le F$ in place of $h \le 0$ (Definition 1.19.1 and Section 17.6).

Proposition 31.8.1 *Let $u \in \mathrm{PSH}(X,\theta)$, put $q = P_\theta[u]_{\mathcal{I}}$, and suppose that $\underline{\mathrm{vol}}_\theta(q) > 0$. Then the following conditions are equivalent:*

(i) $P_\theta[u] = P_\theta[u]_{\mathcal{I}}$;
(ii) $\overline{\mathrm{vol}}_\theta(u) = \mathcal{V}^+_{\mathcal{I},\theta}(u)$;
(iii) $u \in \mathcal{E}_{\mathrm{env}}(X,\theta,q)$;
(iv) $P_{\theta,F}[u] = P_{\theta,F}[u]_{\mathcal{I}}$ *for every* $F \in C^\infty(X,\mathbb{R})$.

If they hold, then also $\underline{\mathrm{vol}}_\theta(u) = \mathcal{V}^-_{\mathcal{I},\theta}(u)$.

Proof Recall that q is an ordinary model and $u \preceq P_\theta[u] \le q$. If (i) holds, then $P_\theta[u] = q$ has positive lower volume, and Theorem 31.3.2 shows that both volumes of u equal those of q; in particular (ii) holds, together with the final assertion.

Conversely, (ii) and $u \preceq q$ give $u \in \mathcal{E}_{\mathrm{vol}}(X,\theta,q)$, and Corollary 31.3.2 gives $P_\theta[u] = q$, which is (i). The same corollary shows that (i) is equivalent to (iii). By Proposition 17.6.6, (iii) implies (iv), and (iv) with $F = 0$ is (i). □

Under the hypotheses of the proposition, Corollary 17.5.2 also identifies equality of the upper volumes with convergence of the dyadic ordinary envelopes $P_{\theta_j}[a_j]$ to $P_\theta[u]$. These envelopes live in the larger backgrounds θ_j. The next construction mixes the a_j with a strict analytic potential, which gives analytic approximants in the fixed background θ with the same limiting upper volume; their singularity types, however, need no longer be ordered.

Theorem 31.8.1 *Let $u \in \mathrm{PSH}(X,\theta)$, put $q = P_\theta[u]_{\mathcal{I}}$, and suppose $\underline{\mathrm{vol}}_\theta(q) > 0$. There are analytic $b_j \in \mathrm{PSH}(X,\theta)$ with $b_j \to u$ in L^1 such that the ordinary models $r_j = P_\theta[b_j]$ satisfy*

$$d_\theta(r_j, q) \longrightarrow 0, \qquad d_\theta(r_j, u) \longrightarrow \mathcal{V}^+_{\mathcal{I},\theta}(u) - \overline{\mathrm{vol}}_\theta(u). \tag{31.10}$$

Their lower volumes converge to $\underline{\mathrm{vol}}_\theta(q)$*, and* $r_j \to q$ *in* L^1 *and capacity. In particular* $d_\theta(r_j, u) \to 0$ *if and only if* $P_\theta[u] = P_\theta[u]_{\mathcal{I}}$.

Proof *Step 1: the approximants.* Fix a Hermitian form h. Bigness and Theorem 1.15.1 give an analytic $\rho \in \mathrm{PSH}(X, \theta)$ with $\rho \le 0$ and a rational $\kappa > 0$ such that $\theta + \mathrm{dd}^{\mathrm{c}}\rho \ge \kappa h$. Choose the dyadic regularizations a_j as in (17.3), enlarging their errors to positive rational numbers $\epsilon_j \downarrow 0$. Put

$$c_j = \frac{\kappa}{\kappa + \epsilon_j}, \qquad b_j = c_j a_j + (1 - c_j)\rho.$$

Then $\theta + \mathrm{dd}^{\mathrm{c}} b_j \ge ((1 - c_j)\kappa - c_j\epsilon_j)h = 0$. The coefficients are rational, so b_j has analytic singularities by Proposition 1.13.1. Since $a_j \to u$ in L^1 and $c_j \to 1$, also $b_j \to u$ in L^1.

Step 2: two-sided bounds. Choose $A > 0$ with $\theta \le Ah$, and set

$$\widehat{\theta}_j = \theta + (1 + A/\kappa)\epsilon_j h, \qquad t_j = P_{\widehat{\theta}_j}[a_j].$$

These backgrounds decrease to θ and dominate $\theta + \epsilon_j h$, so the a_j also satisfy (17.3) with the larger errors $(1 + A/\kappa)\epsilon_j$. Theorem 17.5.1 therefore gives $t_j \searrow q$. Moreover

$$c_j\widehat{\theta}_j - \theta = (1 - c_j)(Ah - \theta) + c_j\epsilon_j h \ge 0.$$

By Proposition 17.2.4, $r_j \sim b_j$. Hence r_j/c_j is a nonpositive $\widehat{\theta}_j$-plurisubharmonic function that is more singular than a_j, because $b_j \le c_j a_j$; it is therefore a competitor for t_j, and $r_j \le c_j t_j$. Also $q \le t_j \le c_j t_j$.

For a lower bound, recall that $q \sim_{\mathcal{I}} u$, and $u \preceq_{\mathcal{I}} a_j$ because $a_j \ge u$. Addition and scaling preserve this order by Theorem 1.16.1 and Lemma 1.16.3. Thus $c_j q + (1 - c_j)\rho \preceq_{\mathcal{I}} b_j$. This potential lies below zero, so it lies below its own ideal envelope, and the monotonicity of ideal envelopes in Proposition 17.2.3, together with Proposition 17.2.4, gives

$$c_j q + (1 - c_j)\rho \le P_\theta[b_j]_{\mathcal{I}} = r_j \le c_j t_j.$$

Step 3: volumes. For either singularity volume, homogeneity, addition and monotonicity now yield

$$c_j^n \,\mathrm{vol}_\theta(q) \le \mathrm{vol}_\theta(r_j) \le \mathrm{vol}_\theta(\max(r_j, q)) \le c_j^n \,\mathrm{vol}_{\widehat{\theta}_j}(t_j).$$

The last comparison uses both $\theta \le c_j\widehat{\theta}_j$ and $\max(r_j, q) \le c_j t_j$. By Theorem 31.5.2, the outer terms tend to $\mathrm{vol}_\theta(q)$. This proves the convergence of both volumes of r_j and the first limit in (31.10). In particular, after discarding finitely many indices, the ordinary models r_j and q have a common positive lower-volume bound, so $r_j \to q$ in L^1 and in capacity by Proposition 31.6.1 and Theorem 31.7.1.

For the second limit, translate u so that $u \le q$; this does not change $d_\theta(r_j, u)$. Then

$$\overline{\mathrm{vol}}_\theta(r_j) \le \overline{\mathrm{vol}}_\theta(\max(r_j, u)) \le \overline{\mathrm{vol}}_\theta(\max(r_j, q)).$$

Both bounds tend to $\overline{\mathrm{vol}}_\theta(q)$, so $d_\theta(r_j,u) = 2U(\max(r_j,u)) - U(r_j) - U(u)$ tends to $U(q) - U(u)$, as claimed. The last equivalence follows from Proposition 31.8.1.□

For a Hermitian background, the simpler rescaling in Corollary 17.5.3 gives decreasing analytic models, with the same limits by Theorem 31.5.2. In a general big background, the strict correction above may introduce additional poles, but the volume comparison still measures exactly the excess of the ideal envelope.

Theorem 31.8.2 *Let p be an ordinary model of positive lower volume. Then $p = P_\theta[p]_{\mathcal{I}}$ if and only if there are analytic ordinary models $p_j \in \mathrm{PSH}(X,\theta)$ with $d_\theta(p_j,p) \to 0$.*

More generally, if $p_j = P_\theta[p_j]_{\mathcal{I}}$ and $d_\theta(p_j,p) \to 0$, then $p = P_\theta[p]_{\mathcal{I}}$.

Proof If $p = P_\theta[p]_{\mathcal{I}}$, apply Theorem 31.8.1 with $u = p$. For the converse it suffices to prove the more general assertion, because analytic ordinary models are ideal models by Proposition 17.2.4.

Put $q = P_\theta[p]_{\mathcal{I}}$ and $\delta = \underline{\mathrm{vol}}_\theta(p) > 0$. Choose $s_j \in (0,1)$ tending to zero with $d_\theta(p_j,p) < \delta s_j^n$, after discarding finitely many indices. For $m_j = \max(p_j,p)$ one has

$$\underline{\mathrm{vol}}_\theta(m_j) \geq \delta, \qquad U(m_j) - U(p_j) \leq d_\theta(p_j,p).$$

By Lemma 31.2.1 with $b = s_j^{-1}$, there is $h_j \in \mathrm{PSH}(X,\theta)$ with $(1-s_j)m_j + s_j h_j \leq p_j$. Set $H_j = P_\theta[h_j]_{\mathcal{I}}$. Since $m_j \geq p$, monotonicity, addition and scaling of ideal envelopes (Proposition 17.2.3) give

$$(1-s_j)q + s_j H_j \leq P_\theta[(1-s_j)m_j + s_j h_j]_{\mathcal{I}} \leq P_\theta[p_j]_{\mathcal{I}} = p_j.$$

Consequently $\underline{\mathrm{vol}}_\theta(p_j) \geq (1-s_j)^n \underline{\mathrm{vol}}_\theta(q) \geq (1-s_j)^n\delta$ by addition and volume monotonicity, since $q \geq p$. The models p_j, p therefore lie in $\mathcal{M}_{\delta/2}(\theta)$ for large j, so Proposition 31.6.1 gives $p_j \to p$ in L^1.

The H_j have supremum zero, so normalized compactness bounds $\int_X (-H_j)\,\mathrm{d}V$ by a constant C_θ. Since $q \leq 0$, the convex inequality above gives $q - p_j \leq s_j(q - H_j) \leq -s_j H_j$, and hence

$$\int_X (q-p_j)_+\,\mathrm{d}V \leq s_j \int_X (-H_j)\,\mathrm{d}V \leq C_\theta s_j \longrightarrow 0.$$

Passing to the L^1 limit gives $q \leq p$ almost everywhere, and hence everywhere by Lemma 1.3.3. The opposite inequality $p = P_\theta[p] \leq P_\theta[p]_{\mathcal{I}} = q$ holds by Proposition 17.2.2, so $p = q$. □

Thus the ideal models of positive lower volume are exactly the d_θ-limits of analytic ordinary models. On each $\mathcal{M}_\delta(\theta)$ they form a closed set for this convergence and inherit Cauchy completeness from Theorem 31.6.1. The approximants constructed above have a common positive lower-volume bound, which need not equal the lower volume of the limiting model.

31.8.1 A countable divisorial positivity criterion

A prescribed divisorial part can determine the ordinary type of an ideal envelope even when it has infinitely many components. On the projective plane with a pluriclosed background, the smooth residual determines the mass of every representative of that type. This gives a positivity test directly from divisorial orders.

Let $X = \mathbb{P}^2$, and let κ be the Fubini–Study form with $\int_X \kappa^2 = 1$. Choose distinct hyperplanes H_j and coefficients $a_j > 0$ with $A = \sum_{j\ge1} a_j < \infty$. For unit defining sections s_j, put

$$\ell_j(Z) = \log \frac{|s_j(Z)|^2}{\|Z\|^2} \le 0, \qquad u_0 = \sum_{j\ge1} a_j \ell_j, \qquad u = u_0 - \sup_X u_0.$$

Let θ be a smooth real $(1,1)$-form satisfying $\mathrm{dd^c}\theta = 0$ and $R = \theta - A\kappa > 0$.

Lemma 31.8.1 *The series u_0 defines a quasi-plurisubharmonic function, and*

$$\mathrm{dd^c} u = S - A\kappa, \qquad S = \sum_{j\ge1} a_j [H_j], \qquad \theta + \mathrm{dd^c} u = R + S > 0.$$

For every smooth compact modification $\pi\colon Y \to X$ and every prime divisor E on Y, the series defining the order profile converges, and

$$b(E) := \sum_{j\ge1} a_j \operatorname{ord}_E(\pi^* H_j) = \nu(u\circ\pi, E).$$

Proof The Poincaré–Lelong formula gives $\mathrm{dd^c}\ell_j = [H_j] - \kappa$. Each ℓ_j is integrable, and by unitary invariance $c = -\int_X \ell_j \kappa^2$ does not depend on j. Since $\ell_j \le 0$, the partial sums of u_0 decrease, and they are $A\kappa$-plurisubharmonic. By monotone convergence, $\int_X u_0 \kappa^2 = -cA$ is finite, so the limit u_0 is not identically $-\infty$ and is $A\kappa$-plurisubharmonic. The partial sums converge in L^1, so their complex Hessians converge as currents; this gives the current identities, and $\theta + \mathrm{dd^c} u \ge R > 0$.

Fix π and E, and consider the tail $r_N = \sum_{j>N} a_j \ell_j$ of total coefficient $A_N = \sum_{j>N} a_j > 0$. As for u_0, it is $A_N\kappa$-plurisubharmonic, and its pullback $r_N \circ \pi$ is not identically $-\infty$ because π is biholomorphic over a dense open subset of X. Thus $A_N^{-1}(r_N\circ\pi) \in \mathrm{PSH}(Y, \pi^*\kappa)$, and Lemma 1.17.7 gives

$$0 \le \nu(r_N\circ\pi, E) \le A_N C_{\pi^*\kappa} \longrightarrow 0.$$

Lelong numbers are additive, and $\nu(\ell_j\circ\pi, E) = \operatorname{ord}_E(\pi^* H_j)$. Hence the Nth partial sum of $b(E)$ differs from $\nu(u\circ\pi, E)$ by $\nu(r_N\circ\pi, E)$. This proves convergence and the asserted equality. □

Proposition 31.8.2 *Let $v \in \mathrm{PSH}(X,\theta)$. The conditions*

$$\nu(v\circ\pi,E) = b(E) \text{ for every } \pi, E, \qquad v \sim_{\mathcal{I}} u$$

are equivalent. Under either condition,

$$q = P_\theta[v]_{\mathcal{I}} = P_\theta[u]_{\mathcal{I}} \sim u,$$

and every $w \in \mathrm{PSH}(X,\theta)$ with $w \sim q$ satisfies

$$\int_X \mathrm{MA}_\theta(w) = \underline{\mathrm{vol}}_\theta(q) = \overline{\mathrm{vol}}_\theta(q) = \int_X R^2 > 0. \tag{31.11}$$

Moreover, θ is big and X has the bounded mass property.

Proof *Ideal profile and envelope.* Choose an integer C with $C\kappa \geq \theta$. Then $C > A$, since $\theta > A\kappa$, and u and v are plurisubharmonic for the Kähler form $C\kappa$. By the first item of Theorem 1.16.1, $v \sim_{\mathcal{I}} u$ if and only if $\nu(v\circ\pi, y) = \nu(u\circ\pi, y)$ at every point y of the source of every local modification π of a neighborhood of a point of X. Over a smaller neighborhood, a local modification of a surface is a finite composition of point blowups, and these extend to a smooth compact modification of X. One further point blowup turns the Lelong number at a point into the generic Lelong number along the exceptional curve. Thus the stated profile gives equal Lelong numbers on all local modifications, and hence $v \sim_{\mathcal{I}} u$. Conversely, if $v \sim_{\mathcal{I}} u$, then $v\circ\pi \sim_{\mathcal{I}} u\circ\pi$ for every smooth compact modification π by Lemma 1.22.11. The first item of Theorem 1.16.1, applied with the identity of Y, then gives equal Lelong numbers at all points of Y, and hence equal generic Lelong numbers along every prime divisor E.

The potential u satisfies $\mathrm{dd}^{\mathrm{c}}u = S - A\kappa$, and $\theta = R + A\kappa$ is a Hermitian form. Hence Corollary 17.2.2, applied with the smooth closed form $A\kappa$ and the Hermitian form θ, shows that the ordinary and ideal envelopes of u agree and are ordinarily equivalent to u. Since $v \sim_{\mathcal{I}} u$, the families defining $P_\theta[v]_{\mathcal{I}}$ and $P_\theta[u]_{\mathcal{I}}$ coincide, which proves the assertion about q. The inequality $\theta + \mathrm{dd}^{\mathrm{c}}u \geq R > 0$ proves bigness. For any Hermitian form h on X, choose D with $D\kappa \geq h$. Bounded h-plurisubharmonic potentials are $D\kappa$-plurisubharmonic, so positivity of bounded mixed products and Stokes' theorem bound their masses by $\int_X (D\kappa)^2 = D^2$. This proves the bounded mass property.

The bounded residual. Fix $w \in \mathrm{PSH}(X,\theta)$ with $w \sim u$, and put $T = C\kappa + \mathrm{dd}^{\mathrm{c}}w$. Ordinary equivalence preserves generic Lelong numbers, so Lemma 31.8.1, applied with π the identity, gives $\nu(w,H_j) = \nu(u,H_j) = a_j$. Lemma 1.17.1, applied to the finite partial sums of S, and passage to the limit give

$$K = T - S = (C-A)\kappa + \mathrm{dd}^{\mathrm{c}}g \geq 0, \qquad g = w - u.$$

Since $w \sim u$, the function g is bounded off the poles of u. By the inequality just displayed, it agrees almost everywhere with a quasi-plurisubharmonic function, which is bounded by Lemma 1.3.3. Thus K has bounded local potentials, and its trace charges no pluripolar set. The current S is carried by the pole union $\bigcup_j H_j \subset \{w = -\infty\}$.

Thus $\langle T\rangle = K$. For every smooth real $(1,1)$-form L with $\mathrm{dd^c}L = 0$, distributional Stokes gives

$$\int_X L \wedge \langle T\rangle = (C-A)\int_X L\wedge\kappa.$$

Closed top mass and smooth correction. Put $T_u = C\kappa + \mathrm{dd^c}u = S + (C-A)\kappa$. For the finite partial sums $u_N = \sum_{j\le N} a_j\ell_j$, the non-pluripolar mass of $(C\kappa+\mathrm{dd^c}u_N)^2$ is $(C-\sum_{j\le N}a_j)^2$: outside $\bigcup_{j\le N}H_j$ the current is the smooth form $(C-\sum_{j\le N}a_j)\kappa$, and the non-pluripolar product gives that union zero mass. Since $u_0 \le u_N$, closed-background mass monotonicity, the closed case of Theorem 31.1.1, gives

$$\int_X \langle T_u^2\rangle \le \left(C - \sum_{j\le N} a_j\right)^2$$

for every N, and hence $\int_X\langle T_u^2\rangle \le (C-A)^2$. For the reverse inequality, locally write $S = \mathrm{dd^c}\psi$ with ψ plurisubharmonic. The currents $(C-A)\kappa + \mathrm{dd^c}\max(\psi,-k)$ have bounded local potentials and are at least $(C-A)\kappa$. On the plurifine open set $\{\psi > -k\}$, their squared products coincide with $\langle T_u^2\rangle$ and therefore dominate $(C-A)^2\kappa^2$. Letting k increase, and noting that κ^2 does not charge the pluripolar set $\{\psi=-\infty\}$, gives $\langle T_u^2\rangle \ge (C-A)^2\kappa^2$. The two bounds give $\int_X\langle T_u^2\rangle = (C-A)^2$. Since $C\kappa$ is a Kähler form and $w-u$ is bounded, invariance of closed masses under bounded changes then gives

$$\int_X \langle T^2\rangle = (C-A)^2.$$

Set $L = C\kappa - \theta \ge 0$, so $\mathrm{dd^c}L = 0$. Expanding the bounded products of $\max(w,-k)$ in the two backgrounds and restricting to $\{w > -k\}$ gives, in the limit,

$$\mathrm{MA}_\theta(w) = \langle T^2\rangle - 2L\wedge\langle T\rangle + L^2.$$

All three measures on the right are finite: the first has the mass just computed, $\langle T\rangle = K$, and the smooth measure L^2 gives the pole set zero mass. The same truncations are used in both backgrounds, so this identity is an equality of Borel measures. Integrating, and using the Stokes identity for $\int_X L\wedge\langle T\rangle$, yields

$$\int_X \mathrm{MA}_\theta(w) = (C-A)^2 - 2(C-A)\int_X L\wedge\kappa + \int_X L^2 = \int_X(\theta - A\kappa)^2.$$

This holds throughout the ordinary type, proving both singularity-volume identities and their strict positivity. □

Example 31.8.1 There is a nonclosed pluriclosed background on $\mathbb{P}^2$ to which Proposition 31.8.2 applies with a nonanalytic ideal type: no potential with analytic singularities is ideal equivalent to u. The prescribed hyperplanes are not contained in a single pencil, and both singularity volumes of the ideal model q are at least $1/16$.

Proof In a coordinate ball choose a smooth real cutoff χ equal to one near the origin, and extend $\alpha = \chi|z_2|^2 \mathrm{d}z_1$ by zero. Put

$$\tau = \partial\overline{\alpha} + \bar{\partial}\alpha.$$

It is real and smooth, with $\mathrm{dd}^{\mathrm{c}}\tau = 0$. Where $\chi = 1$,

$$\mathrm{d}\tau = \mathrm{d}z_2 \wedge \mathrm{d}\bar{z}_2 \wedge (\mathrm{d}z_1 - \mathrm{d}\bar{z}_1) \neq 0.$$

Choose $\varepsilon > 0$ small enough that $\theta = \kappa + \varepsilon\tau \geq \kappa/2$; then $\mathrm{dd}^{\mathrm{c}}\theta = 0$ and $\mathrm{d}\theta = \varepsilon\,\mathrm{d}\tau \neq 0$. Set $A = 1/4$, $a_j = A2^{-j}$, $t_j = 2^{-j}$ and

$$H_j = \{Z_0 + t_j Z_1 + t_j^2 Z_2 = 0\}.$$

The Vandermonde determinant shows that any three defining linear forms are independent; hence the hyperplanes are not in one pencil. Here $R = \theta - A\kappa \geq \kappa/4$, so $\int_X R^2 \geq 1/16$. Proposition 31.8.2 gives the mass throughout the ordinary type and both singularity volumes. By its equivalence, applied with π the identity and $E = H_j$, every potential ideal equivalent to u has generic Lelong number $a_j > 0$ along each of the infinitely many lines H_j. A potential with analytic singularities on a compact manifold has positive generic Lelong numbers only along the finitely many prime divisors in its pole set. Hence no analytic potential has this ideal type. □

Problem 31.8.1 For a big nonclosed background θ on a manifold with the bounded mass property, find general criteria for $\underline{\mathrm{vol}}_\theta(P_\theta[u]_{\mathcal{I}}) > 0$ in terms of restricted multiplier ideals or divisorial orders. When this lower volume vanishes, which implications between envelope equality and upper-volume equality, conditions (i) and (ii) of Proposition 31.8.1, remain valid?

One implication fails: when the ideal envelope has zero lower volume, equality of the upper volumes of a potential and of its ideal envelope does not force equality of the envelopes. In Example 32.2.3, the background Θ is positive and nonclosed on a projective manifold with the bounded mass property, and $q = P_\Theta[a]_{\mathcal{I}}$ satisfies $\overline{\mathrm{vol}}_\Theta(a) = \overline{\mathrm{vol}}_\Theta(q) = 0$ but $P_\Theta[a] \neq q$. The example leaves the reverse implication in Problem 31.8.1, from (i) to (ii), undecided.

31.9 Recovery through the Hopf fibration

Positive lower volume on a closed base can force recovery even when the ambient manifold has unbounded upper mass. We now take

$$X = (\mathbb{C}^3 \setminus \{0\})/\langle z \mapsto \mathrm{e}^{-1} z\rangle, \qquad B = \mathbb{P}^2, \qquad \pi\colon X \to B,$$

and use the geometry of Chapter 9. Write $\kappa = \omega_{\mathrm{FS}}$, with $\int_B \kappa^2 = 1$, and put

$$s = \log|z|^2, \qquad \alpha = \pi^*\kappa = \mathrm{dd^c} s, \qquad \beta = \mathrm{d}s \wedge \mathrm{d^c} s, \qquad h = \alpha + \beta.$$

In the notation of Section 9.6, $X = H_3$, $\pi = p_3$, $\alpha = \alpha_3$, $\beta = \beta_3$ and $h = \omega_3$. The derivatives of s descend to X. Scalar multiplication gives an action of the elliptic group $E = \mathbb{C}/(\mathbb{Z}+2\pi\mathrm{i}\mathbb{Z})$, transitive on each fiber of π and preserving h. By Theorem 9.0.1, h has unbounded upper mass, so X does not have the bounded mass property. Invariant potentials nevertheless reduce to potentials on B because the horizontal part of h is the pullback of κ. In this section the standing hypotheses of the chapter, bounded mass and bigness, are used only on the base B with the Kähler form κ. The envelope systems $\mathcal{S}_{a,c}$, for the real numbers $a < c$ fixed there, and the maps E and R are those of Section 31.5.1.

Proposition 31.9.1 *For every $b > 0$ and $v \in \mathrm{PSH}(B, b\kappa)$,*

$$P_{bh}[\pi^* v]_{\mathcal{I}} = \pi^* P_{b\kappa}[v]_{\mathcal{I}}.$$

The envelope on the left is taken over all competitors on X, not only over the E-invariant ones.

Proof First, since $bh = b\alpha + b\beta$, parts (1) and (2) of Lemma 9.6.1, with $n = 3$, S a point and both parameters equal to b, show that the E-invariant members of $\mathrm{PSH}(X, bh)$ are exactly the pullbacks $\pi^* w$ of the functions $w \in \mathrm{PSH}(B, b\kappa)$.

Now set $q = P_{bh}[\pi^* v]_{\mathcal{I}}$. The action of any $g \in E$ preserves the family of all competitors for this envelope, since it preserves the background, the bound zero, and the multiplier ideals of the invariant potential $\pi^* v$. The envelope is the greatest competitor by Theorem 17.2.1, so $q \circ g = q$. Descend $q = \pi^* w$ as above. By Lemma 1.16.5, its ideal equivalence to $\pi^* v$ implies $w \sim_{\mathcal{I}} v$: a germ pulled back from the base is weighted-integrable in a product chart if and only if the base germ is, by Fubini. Hence w is a base competitor. Every base competitor also lifts to an ambient competitor, by Lemma 1.16.5 and the first step. Taking the greatest members gives equality. □

For an envelope system $\mathbf{\Gamma} \in \mathcal{S}_{a,c}(h)$, we call the components $\Gamma^{\epsilon h}$, $\epsilon > 0$, its *scalar components*. If they are invariant under E, they alone determine an envelope system on the base, including its components over nonclosed perturbations of κ.

Lemma 31.9.1 *Let $\mathbf{\Gamma} \in \mathcal{S}_{a,c}(h)$, and suppose that $\Gamma_\tau^{\epsilon h}$ is E-invariant for every $\epsilon > 0$ and $\tau < c$. There is a unique envelope system $\mathbf{\Psi} \in \mathcal{S}_{a,c}(\kappa)$ such that*

$$\Gamma_\tau^{\epsilon h} = \pi^* \Psi_\tau^{\epsilon \kappa} \quad (\epsilon > 0,\ \tau < c).$$

Writing $\Gamma = \mathsf{R}\mathbf{\Gamma}$ and $\psi = \mathsf{R}\mathbf{\Psi}$, we have $\Gamma_\tau = \pi^* \psi_\tau$ for every $\tau < c$.

Proof The descent in Proposition 31.9.1 gives $\Gamma_\tau^{\epsilon h} = \pi^* \phi_\tau^\epsilon$ with $\phi_\tau^\epsilon \in \mathrm{PSH}(B, (1+\epsilon)\kappa)$. Normalization, decrease, concavity and constancy for $\tau \le a$ descend pointwise. By the fiber-envelope identity of Proposition 31.9.1, the envelope condition and the compatibility of the scalar components also descend:

$$\phi_\tau^{\epsilon_2} = P_{(1+\epsilon_2)\kappa}[\phi_\tau^{\epsilon_1}]_I \quad (\epsilon_2 \ge \epsilon_1 > 0).$$

In particular, the slices ϕ_τ^ϵ, $\epsilon > 0$, have the same multiplier ideals at every weight.

For a Hermitian form $\xi > 0$ on B, choose $\delta > 0$ with $\delta\kappa \le \xi$, fixed for the whole parameter interval, and define

$$\Psi_\tau^\xi = P_{\kappa+\xi}[\phi_\tau^\delta]_I.$$

The definition is independent of δ because the slices ϕ_τ^δ for different δ have the same ideals. For $\xi = \epsilon\kappa$ it gives $\Psi_\tau^{\epsilon\kappa} = \phi_\tau^\epsilon$, and Proposition 26.2.1 proves that each component is a curve in $\mathcal{T}_{a,c}(\kappa + \xi)$. Projection preserves the ideals, so comparing the complete competitor families proves compatibility for $\xi \ge \xi'$. Thus $\mathbf{\Psi}$ is an envelope system over κ. Every possible extension is forced by this formula, which proves uniqueness.

By Theorem 26.3.1, applied to both systems with the Hermitian forms h on X and κ on B, the decreasing limits in the identity $\Gamma_\tau^{h/j} = \pi^* \Psi_\tau^{\kappa/j}$ give $\Gamma_\tau = \pi^* \psi_\tau$. □

We call $\mathbf{\Psi}$ the *transverse system* of $\mathbf{\Gamma}$, and $\psi = \mathsf{R}\mathbf{\Psi}$ its *transverse limiting curve*.

Theorem 31.9.1 *Let $\mathbf{\Gamma} \in \mathcal{S}_{a,c}(h)$ have E-invariant scalar components, and let ψ be its transverse limiting curve. If*

$$\underline{\mathrm{vol}}_\kappa(\psi_a) > 0,$$

then for every $\tau < c$ and every Hermitian form $\eta > 0$ on X,

$$\Gamma_\tau^\eta = P_{h+\eta}[\pi^* \psi_\tau]_I.$$

Thus $\mathbf{\Gamma} = \mathsf{E}(\mathsf{R}\mathbf{\Gamma})$.

Proof The Kähler form κ is big, and B has the bounded mass property, since every bounded κ-plurisubharmonic function has mass $\int_B \kappa^2$ by Theorem 2.2.2. Apply Theorem 31.5.3 to the transverse system $\mathbf{\Psi} \in \mathcal{S}_{a,c}(\kappa)$. Its initial-volume hypothesis is exactly the displayed one, and it gives

$$\Psi_\tau^{\epsilon\kappa} = P_{(1+\epsilon)\kappa}[\psi_\tau]_I \quad (\epsilon > 0,\ \tau < c).$$

Since $\Gamma_\tau^{\epsilon h} = \pi^*\Psi_\tau^{\epsilon\kappa}$, Proposition 31.9.1 with $b = 1+\epsilon$ yields $\Gamma_\tau^{\epsilon h} = P_{(1+\epsilon)h}[\pi^*\psi_\tau]_{\mathcal{I}}$. For a general $\eta > 0$, choose $\epsilon h \le \eta$. Compatibility and the nested-background identity of Proposition 26.2.1 then give

$$\Gamma_\tau^{\eta} = P_{h+\eta}[\Gamma_\tau^{\epsilon h}]_{\mathcal{I}} = P_{h+\eta}[\pi^*\psi_\tau]_{\mathcal{I}}.$$

Since $(\mathsf{R}\boldsymbol{\Gamma})_\tau = \pi^*\psi_\tau$ by Lemma 31.9.1, this is the identity $\boldsymbol{\Gamma} = \mathsf{E}(\mathsf{R}\boldsymbol{\Gamma})$. □

The hypothesis concerns only the lower volume of one ordinary singularity type on the closed base, and the conclusion gives no uniform positive-volume bound as τ approaches c.

31.9.1 A nonanalytic family

The countable hyperplane series of Lemma 31.8.1 and Proposition 31.8.2 provides an explicit family satisfying the transverse condition. Choose infinitely many distinct hyperplanes $H_j \subset B$ and positive coefficients p_j with $\sum_j p_j = 1$, and let $\ell_j \le 0$ be the logarithm of the squared Fubini–Study norm of a unit defining section of H_j, as in Section 31.8.1. Put

$$u_0 = \sum_{j\ge1} p_j\ell_j - \sup_B \sum_{j\ge1} p_j\ell_j.$$

Then $u_0 \le 0$ is κ-plurisubharmonic and $\kappa + \mathrm{dd}^{\mathrm{c}}u_0 = \sum_j p_j[H_j]$.

Example 31.9.1 Fix $0 < \lambda_0 < 1$ and $r > 0$ with $\lambda_0 + r(c-a) < 1$. For $\tau < c$, set

$$\lambda(\tau) = \lambda_0 + r\max(\tau - a, 0), \qquad u_\tau = \lambda(\tau)u_0,$$

and, for every Hermitian form $\eta > 0$ on X, define

$$\Gamma_\tau^{\eta} = P_{h+\eta}[\pi^*u_\tau]_{\mathcal{I}}.$$

These components form an envelope system with E-invariant scalar slices. Its transverse limiting curve is $\psi_\tau = P_\kappa[u_\tau]_{\mathcal{I}}$, and

$$\underline{\mathrm{vol}}_\kappa(\psi_\tau) = \overline{\mathrm{vol}}_\kappa(\psi_\tau) = (1-\lambda(\tau))^2 > 0.$$

The system is recovered from its limit. Its ideal data cannot be represented by analytic singularities.

Proof The function λ is positive, increasing and convex, and $u_0 \le 0$, so u_τ decreases, is pointwise concave and is constant for $\tau \le a$. Moreover,

$$\kappa + \mathrm{dd}^{\mathrm{c}}u_\tau = (1-\lambda(\tau))\kappa + \lambda(\tau)\sum_j p_j[H_j] \ge 0, \qquad h + \mathrm{dd}^{\mathrm{c}}\pi^*u_\tau \ge \beta \ge 0.$$

Set $q_\tau = P_\kappa[u_\tau]_I$. The order, scaling and addition inequalities of Proposition 17.2.3 show that q_τ remains decreasing and concave; it is constant on the same left half-line. Thus $q \in \mathscr{T}_{a,c}(\kappa)$. Apply the same argument upstairs to $Q_\tau = P_h[\pi^* u_\tau]_I$. The fiber-envelope identity gives $Q_\tau = \pi^* q_\tau$, and the nested-background identity identifies the displayed family with $\mathsf{E}(Q)$. It is therefore an envelope system, and its limit is $Q = \pi^* q$ by $\mathsf{RE} = \mathrm{id}$. Its scalar components $P_{(1+\epsilon)h}[\pi^* u_\tau]_I = \pi^* P_{(1+\epsilon)\kappa}[u_\tau]_I$ are invariant, and Lemma 31.9.1 identifies the transverse limiting curve as $\psi = q$.

Apply Proposition 31.8.2 with background κ, coefficients $a_j = \lambda(\tau) p_j$ and residual $(1 - \lambda(\tau))\kappa$. It gives the asserted exact lower and upper volumes of q_τ. Recovery holds by construction, since the system is $\mathsf{E}(Q)$; the volume formula shows that Theorem 31.9.1 also applies.

For every j, the divisorial coefficient of u_τ along H_j is positive. Projection preserves all ideals and hence these coefficients. Analytic singularities on a compact manifold have only finitely many divisorial components, since their defining coherent ideals have this property. Thus no analytic singularity type has the same all-weight ideals. The same conclusion holds upstairs along the distinct divisors $\pi^{-1}(H_j)$, by submersion pullback. □

31.9.2 Exact pullback and smooth changes of background

Let $\mu\colon Y \to X$ be a modification and let $\mathbf{\Gamma}$ satisfy the hypotheses of Theorem 31.9.1. By Proposition 17.2.1, its recovery identity pulls back to

$$\Gamma^\eta_\tau \circ \mu = P_{\mu^*(h+\eta)}[\Gamma_\tau \circ \mu]_I.$$

Since $\mu^*\eta$ may be degenerate, these identities in the pulled-back backgrounds $\mu^*(h+\eta)$ do not define an envelope system indexed by the Hermitian forms on Y. For a modification $\nu\colon \widetilde{B} \to B$, Proposition 14.3.1 gives

$$\underline{\mathrm{vol}}_{\nu^*\kappa}(\psi_a \circ \nu) = \underline{\mathrm{vol}}_\kappa(\psi_a) > 0,$$

so the transverse positivity condition survives exact pullback.

Finally, the smooth transport G_f of Proposition 26.4.1 commutes with E and R, so recovery survives the change from h to $h + \mathrm{dd}^{\mathrm{c}} f$ for every smooth real f, although the transported scalar components need not be invariant under E and the limiting Monge–Ampère masses of the slices can change (Proposition 26.4.2). The recovery problem without the scalar symmetry or the transverse positive-volume condition remains open.

31.10 Model comparison under background changes

A smooth change of decomposition $(\theta, u) \mapsto (\theta + \mathrm{dd}^{\mathrm{c}} f, u - f)$ preserves the curvature current but changes the zero obstacle used to normalize models. Taking the ordinary envelope again gives an exact correspondence that preserves the volume comparison. An increase of the background changes the volumes; a common positive lower-volume bound controls this change uniformly, and this allows analytic approximants whose curvature error tends to zero.

Proposition 31.10.1 *Let $f \in C^\infty(X, \mathbb{R})$ and $\theta' = \theta + \mathrm{dd}^{\mathrm{c}} f$. For every $\delta > 0$, the map*

$$G_f \colon \mathcal{M}_\delta(\theta) \longrightarrow \mathcal{M}_\delta(\theta'), \qquad G_f(p) = P_{\theta'}[p - f],$$

is a bijection with inverse $q \mapsto P_\theta[q + f]$. It preserves both singularity volumes, and

$$d_{\theta'}(G_f(p), G_f(q)) = d_\theta(p, q).$$

It takes analytic models to analytic models and ideal models to ideal models.

Proof The bijection $u \mapsto u - f$ of Proposition 1.2.1 preserves curvature currents and maps ordinary types onto ordinary types; hence it preserves both singularity volumes. Also $\max(u - f, v - f) = \max(u, v) - f$, so

$$d_{\theta'}(u - f, v - f) = d_\theta(u, v).$$

By Proposition 17.3.1, $G_f(p) + f \sim p$, that is, $G_f(p) \sim p - f$. Thus $G_f(p)$ has the same volumes as p, and it is an ordinary model by Theorem 31.3.2; in particular $G_f(p) \in \mathcal{M}_\delta(\theta')$. Since the volume comparison depends only on ordinary types, the identity $d_{\theta'}(G_f(p), G_f(q)) = d_\theta(p, q)$ follows. Taking the ordinary envelope of $G_f(p) + f \sim p$ gives $P_\theta[G_f(p) + f] = P_\theta[p] = p$. The same argument with the two backgrounds interchanged proves that the map is bijective with the stated inverse.

Subtraction of a smooth function and bounded changes preserve analytic singularities. If $p = P_\theta[p]_{\mathcal{I}}$, Proposition 17.3.1 also gives

$$P_{\theta'}[p - f]_{\mathcal{I}} \sim p - f \sim G_f(p).$$

The ideal envelope is an ordinary model by Proposition 17.2.2, and ordinary models of the same ordinary type coincide; hence $P_{\theta'}[G_f(p)]_{\mathcal{I}} = P_{\theta'}[p - f]_{\mathcal{I}} = G_f(p)$. The inverse map gives the converse. □

For the rest of this section and the next, fix a Hermitian form h on X, a Hermitian form $\Omega \geq \theta + h$, a rational number $\kappa > 0$ and an analytic $\rho \in \mathrm{PSH}(X, \theta)$ with $\rho \leq 0$ and $\theta + \mathrm{dd}^{\mathrm{c}} \rho \geq \kappa h$; such κ and ρ exist by bigness and Theorem 1.15.1. Put $K = v_+(\Omega)$. For $\delta > 0$, set

$$b_\delta = 1 + \frac{\delta}{2^{n+1} n K}, \qquad \kappa_\delta = (1 - b_\delta^{-1})\kappa.$$

These constants stay fixed while the potentials and backgrounds vary.

Proposition 31.10.2 *Let β be a smooth real $(1,1)$-form with $\theta \le \beta \le \theta + h$, and let $u \in \mathrm{PSH}(X,\beta)$ satisfy $\underline{\mathrm{vol}}_\beta(u) \ge \delta > 0$. For every $\varepsilon \in [0,1]$ and every smooth real $(1,1)$-form γ with $\beta \le \gamma \le \beta + \varepsilon h$, both singularity volumes satisfy*

$$\mathrm{vol}_\beta(u) \le \mathrm{vol}_\gamma(u) \le (1 + \varepsilon/\kappa_\delta)^n \,\mathrm{vol}_\beta(u).$$

Proof Normalize $u \le 0$, so that $u \le V_\beta$, and note that $\rho \le V_\beta$. Since $\Omega \ge \beta$ and $\underline{\mathrm{vol}}_\beta(u) \ge \delta$, Lemma 31.4.1, applied in the big background β, shows that $H = Q_{b_\delta}(u, V_\beta)$ is defined. The potential

$$w = b_\delta^{-1} H + (1 - b_\delta^{-1})\rho$$

satisfies $w \le u$ and $\beta + \mathrm{dd^c} w \ge (1 - b_\delta^{-1})(\theta + \mathrm{dd^c}\rho) \ge \kappa_\delta h$. The addition argument in the proof of Proposition 31.4.2, with h and κ_δ in place of ω and δ, and hence with $t = \varepsilon/\kappa_\delta$, therefore gives the upper bound, with constants independent of β and u. The lower bound is volume monotonicity. □

Corollary 31.10.1 *Let $u, v \in \mathrm{PSH}(X,\theta)$ have lower volume at least $\delta > 0$, and let η be a smooth real $(1,1)$-form with $\theta \le \eta \le \theta + \varepsilon h$, where $0 \le \varepsilon \le 1$. Then*

$$\left| d_\eta(P_\eta[u], P_\eta[v]) - d_\theta(u,v) \right| \le 2K\left((1 + \varepsilon/\kappa_\delta)^n - 1\right). \tag{31.12}$$

Proof The preceding proposition, with $\beta = \theta$ and $\gamma = \eta$, applies to u, v and $m = \max(u,v)$, and their upper volumes in background θ are at most $v_+(\Omega) = K$. Thus, for $x = u, v, m$, the errors

$$e_x = \overline{\mathrm{vol}}_\eta(x) - \overline{\mathrm{vol}}_\theta(x)$$

lie between zero and $K((1 + \varepsilon/\kappa_\delta)^n - 1)$. The difference $d_\eta(u,v) - d_\theta(u,v)$ equals $2e_m - e_u - e_v$, whose absolute value is at most twice this bound. Finally, the projection invariance in Proposition 31.6.1, applied in background η, gives $d_\eta(u,v) = d_\eta(P_\eta[u], P_\eta[v])$, which is the comparison appearing in (31.12). □

In particular, for $p_j, p \in \mathcal{M}_\delta(\theta)$ and $\theta \le \eta_j \le \theta + \varepsilon_j h$ with $\varepsilon_j \to 0$,

$$d_\theta(p_j, p) \longrightarrow 0 \quad \Longleftrightarrow \quad d_{\eta_j}(P_{\eta_j}[p_j], P_{\eta_j}[p]) \longrightarrow 0.$$

Here the backgrounds η_j need not be monotone in j. The next result allows the analytic approximants themselves to belong only to the larger backgrounds η_j.

Theorem 31.10.1 *Let p be an ordinary model of positive lower volume, and let η_j be smooth real $(1,1)$-forms with $\theta \le \eta_j \le \theta + \varepsilon_j h$, where $\varepsilon_j \ge 0$ tends to zero. Then $p = P_\theta[p]_{\mathcal{I}}$ if and only if there are analytic $a_j \in \mathrm{PSH}(X, \eta_j)$ with*

$$d_{\eta_j}(a_j, p) \longrightarrow 0.$$

The a_j may be required to be ordinary models in their respective backgrounds. Given any such analytic approximants, convex combinations with the fixed potential ρ, with rational coefficients, produce analytic $b_j \in \mathrm{PSH}(X, \theta)$ whose ordinary envelopes satisfy

$$d_\theta(P_\theta[b_j], p) \longrightarrow 0, \qquad P_\theta[b_j] \longrightarrow p \quad \text{in } L^1 \text{ and capacity.}$$

Proof Suppose first that $p = P_\theta[p]_{\mathcal{I}}$. By Theorem 31.8.1, there are analytic ordinary models r_j in background θ with $d_\theta(r_j, p) \to 0$ and a common positive lower-volume bound. Corollary 31.10.1 and projection invariance in background η_j give $d_{\eta_j}(r_j, p) \to 0$, so $a_j = r_j$ works. One can also use the ordinary models $a_j = P_{\eta_j}[r_j]$, since $P_{\eta_j}[r_j] \sim r_j$ by Proposition 17.2.4.

Conversely, suppose that such a_j exist. Normalize them below zero, and put $m_j = \max(a_j, p)$ and $\delta = \underline{\mathrm{vol}}_\theta(p) > 0$. Inequality (31.6) and Proposition 31.10.2 give

$$\overline{\mathrm{vol}}_{\eta_j}(a_j), \quad \overline{\mathrm{vol}}_{\eta_j}(m_j) \longrightarrow \overline{\mathrm{vol}}_\theta(p).$$

Indeed, the upper volume of p in background η_j tends to its upper volume in background θ, and the differences of $\overline{\mathrm{vol}}_{\eta_j}(a_j)$ and $\overline{\mathrm{vol}}_{\eta_j}(m_j)$ from $\overline{\mathrm{vol}}_{\eta_j}(p)$ are bounded in absolute value by $d_{\eta_j}(a_j, p)$.

Enlarge the errors ε_j to positive rational numbers tending to zero, and put

$$c_j = \frac{\kappa}{\kappa + \varepsilon_j}, \qquad b_j = c_j a_j + (1 - c_j)\rho, \qquad r_j = P_\theta[b_j].$$

As in the proof of Theorem 31.8.1, $\theta + \mathrm{dd}^{\mathrm{c}} b_j \ge 0$ and b_j has analytic singularities. Choose $A > 0$ with $\theta \le Ah$, and set

$$\widehat{\theta}_j = \theta + (1 + A/\kappa)\varepsilon_j h, \qquad \widehat{m}_j = P_{\widehat{\theta}_j}[m_j].$$

Then $\widehat{\theta}_j \ge \eta_j$ and $c_j \widehat{\theta}_j \ge \theta$. Since $r_j \sim b_j$ and $\rho \le 0$, the potential r_j / c_j is a nonpositive $\widehat{\theta}_j$-plurisubharmonic function more singular than a_j, hence more singular than m_j. Therefore $r_j \le c_j \widehat{m}_j$. Also $p \le \widehat{m}_j \le c_j \widehat{m}_j$.

The lower volume of m_j in background η_j is at least δ. For large j, Proposition 31.10.2 applies with $\beta = \eta_j$, $\gamma = \widehat{\theta}_j$ and $\varepsilon = (1 + A/\kappa)\varepsilon_j$. Consequently

$$\overline{\mathrm{vol}}_{\widehat{\theta}_j}(m_j) \longrightarrow \overline{\mathrm{vol}}_\theta(p).$$

By Theorem 31.3.2, $\widehat{m}_j$ has the same upper volume in background $\widehat{\theta}_j$ as m_j. On the other hand,

$$c_j\eta_j + (1-c_j)(\theta-\kappa h) \le \theta.$$

Addition, applied to $c_j a_j \in \mathrm{PSH}(X, c_j\eta_j)$ and $(1-c_j)\rho \in \mathrm{PSH}(X,(1-c_j)(\theta-\kappa h))$, and background monotonicity therefore bound the upper volume of b_j below by $c_j^n\overline{\mathrm{vol}}_{\eta_j}(a_j)$. Since $r_j \sim b_j$, the two comparisons yield

$$\begin{aligned} c_j^n\overline{\mathrm{vol}}_{\eta_j}(a_j) &\le \overline{\mathrm{vol}}_\theta(r_j) \le \overline{\mathrm{vol}}_\theta(\max(r_j,p)) \\ &\le c_j^n\overline{\mathrm{vol}}_{\widehat{\theta}_j}(m_j). \end{aligned}$$

Here the last inequality uses $\theta \le c_j\widehat{\theta}_j$ and $\max(r_j,p) \le c_j\widehat{m}_j$. Both outer terms tend to $\overline{\mathrm{vol}}_\theta(p)$, which proves $d_\theta(r_j,p)\to 0$. By Proposition 17.2.4, $r_j = P_\theta[b_j]_{\mathcal{I}}$ is an ideal model, so Theorem 31.8.2 gives $p = P_\theta[p]_{\mathcal{I}}$.

By Proposition 31.6.3, $\underline{\mathrm{vol}}_\theta(r_j)\to\delta$. Thus Proposition 31.6.1 and Theorem 31.7.1 give the asserted L^1 and capacity convergence of the models r_j. □

Since the volume comparison depends only on singularity types, the theorem asserts convergence of the ordinary models $P_\theta[b_j]$, not of the analytic potentials b_j themselves. When the background errors may have either sign, p need not be admissible in the perturbed backgrounds; the next section constructs reference models there.

31.11 Two-sided background perturbations

A potential in the original background may cease to be admissible when the background decreases. Mixing it with the strict potential ρ restores positivity, and ordinary projection then gives a model in the new background. The volume comparison controls the error of this transport uniformly on models with a common positive lower-volume bound, and it shows that the resulting notion of analytic approximation does not depend on the strict potential used in the correction.

We first make the estimate of the preceding section uniform for pairs of forms near θ.

Lemma 31.11.1 *For every $\delta > 0$ there are $C_\delta > 0$ and $\varepsilon_0 > 0$ with the following property. Let β, γ be smooth real $(1,1)$-forms with*

$$\theta - \varepsilon h \le \beta \le \gamma \le \theta + \varepsilon h, \qquad 0 \le \varepsilon \le \varepsilon_0,$$

and let $u \in \mathrm{PSH}(X, \beta)$ satisfy $\underline{\mathrm{vol}}_\beta(u) \ge \delta$. Then both singularity volumes satisfy

$$0 \le \mathrm{vol}_\gamma(u) - \mathrm{vol}_\beta(u) \le C_\delta \varepsilon.$$

For two such potentials u, v, one also has

$$\bigl| d_\gamma(P_\gamma[u], P_\gamma[v]) - d_\beta(u, v) \bigr| \le 2 C_\delta \varepsilon.$$

Proof Put $\theta_0 = \theta - \kappa h/2$ and $h_0 = (1 + \kappa/2)h$. Since $\theta_0 + \mathrm{dd}^{\mathrm{c}}\rho \ge \kappa h/2$, the form θ_0 is big, and ρ is a strict analytic potential for it with the rational margin $\kappa/(2+\kappa)$ relative to h_0. Put $\varepsilon_0 = \min(\kappa/2, 1/2)$ and let $\varepsilon \le \varepsilon_0$. Then $\theta_0 \le \beta \le \theta_0 + h_0$ and $\beta \le \gamma \le \beta + 2\varepsilon h_0$, so Proposition 31.10.2 applies with θ_0, h_0, ρ and a fixed Hermitian form $\Omega_0 \ge \theta_0 + h_0$ in place of θ, h, ρ and Ω. These choices do not depend on β and γ. Since the volumes in background β are at most $v_+(\Omega_0)$, the multiplicative estimate of the proposition implies the additive estimate, with a constant C_δ independent of β, γ and u. Applying it to u, v and $\max(u, v)$, the three-error calculation in the proof of Corollary 31.10.1 and projection invariance in background γ give the second assertion. □

We now define the transport. Choose $A > 0$ with $\theta \le Ah$ and put $B = 2(1 + A/\kappa)$. For $p \in \mathrm{PSH}(X, \theta)$ with $p \le 0$, a number ε with $0 < \varepsilon < \min(\kappa/2, 1/B)$, and a smooth real $(1,1)$-form η with $\theta - \varepsilon h \le \eta \le \theta + \varepsilon h$, set

$$c_\varepsilon = 1 - \varepsilon/\kappa, \qquad T^\rho_{\eta,\varepsilon}(p) = P_\eta[c_\varepsilon p + (1 - c_\varepsilon)\rho], \qquad \widehat{\theta}_\varepsilon = \theta + B\varepsilon h. \quad (31.13)$$

The mixed potential $c_\varepsilon p + (1 - c_\varepsilon)\rho$ is η-plurisubharmonic, since $1 - c_\varepsilon = \varepsilon/\kappa$ and therefore $\eta + \mathrm{dd}^{\mathrm{c}}(c_\varepsilon p + (1 - c_\varepsilon)\rho) \ge \eta - \theta + \varepsilon h \ge 0$.

Theorem 31.11.1 *Let $p \in \mathcal{M}_\delta(\theta)$, and let ε and η be as above. The transport $T^\rho_{\eta,\varepsilon}(p)$ of (31.13) is an ordinary model in background η, of lower volume at least $c_\varepsilon^n \delta$, and its volumes satisfy*

$$c_\varepsilon^n \operatorname{vol}_\theta(p) \le \operatorname{vol}_\eta(T^\rho_{\eta,\varepsilon}(p)) \le c_\varepsilon^n \operatorname{vol}_{\widehat{\theta}_\varepsilon}(p). \tag{31.14}$$

For $p, q \in \mathcal{M}_\delta(\theta)$,

$$\begin{aligned} &\left| d_\eta(T^\rho_{\eta,\varepsilon}(p), T^\rho_{\eta,\varepsilon}(q)) - c_\varepsilon^n d_\theta(p,q) \right| \\ &\qquad \le 2c_\varepsilon^n K\left((1 + B\varepsilon/\kappa_\delta)^n - 1\right). \end{aligned} \tag{31.15}$$

For fixed p, these models tend to p in L^1 and in capacity as $\varepsilon \to 0$. Finally, suppose that two choices of the strict analytic potential ρ, of its rational margin κ and of error bounds $\varepsilon_j \to 0$ are admissible for the same backgrounds η_j. Then the volume comparison in background η_j between the two transports of p tends to zero.

Proof Write $c = c_\varepsilon$, $s = 1 - c$, $v = cp + s\rho$ and $T(p) = T^\rho_{\eta,\varepsilon}(p)$. Since $s\rho \in \mathrm{PSH}(X, s(\theta - \kappa h))$, addition gives $\operatorname{vol}_{\theta - s\kappa h}(v) \ge c^n \operatorname{vol}_\theta(p)$, and since $\theta - s\kappa h = \theta - \varepsilon h \le \eta$, monotonicity gives the lower bound in (31.14) for v. In particular v has positive lower volume in background η, so Theorem 31.3.2 shows that $T(p) = P_\eta[v]$ is an ordinary model with the same two volumes as v.

Since $\varepsilon < \kappa/2$, we have $c \ge 1/2$, and $\theta \le Ah$ together with the choice of B gives

$$c\widehat{\theta}_\varepsilon - (\theta + \varepsilon h) \ge \varepsilon(cB - 1 - A/\kappa)h \ge 0. \tag{31.16}$$

Since $\eta \le \theta + \varepsilon h \le c\widehat{\theta}_\varepsilon$, every competitor w for $T(p)$, that is, every nonpositive η-plurisubharmonic $w \preceq v$, satisfies $w/c \in \mathrm{PSH}(X, \widehat{\theta}_\varepsilon)$ and $w/c \preceq p$, because $\rho \le 0$. Taking envelopes, and using that $v \le 0$ is itself a competitor, yields

$$cp + s\rho \le T(p) \le cP_{\widehat{\theta}_\varepsilon}[p]. \tag{31.17}$$

Volume monotonicity, together with Theorem 31.3.2 in the larger background $\widehat{\theta}_\varepsilon$, proves the upper bound in (31.14).

Now let $q \in \mathcal{M}_\delta(\theta)$ and $m = \max(p, q)$. The same pointwise bounds give

$$cm + s\rho \le \max(T(p), T(q)) \le cP_{\widehat{\theta}_\varepsilon}[m].$$

Consequently the upper volumes of $T(p), T(q)$ and of their maximum lie between c^n times the respective upper volumes of p, q, m in the backgrounds θ and $\widehat{\theta}_\varepsilon$. The lower volumes of p, q and m are at least δ, and their upper volumes are at most K. By Proposition 31.10.2 with $\beta = \theta$ and $\gamma = \widehat{\theta}_\varepsilon$, each of the three errors $\overline{\operatorname{vol}}_\eta(T(x)) - c^n \overline{\operatorname{vol}}_\theta(x)$, $x = p, q, m$ (with $T(m)$ read as $\max(T(p), T(q))$), is nonnegative and at most $c^n K((1 + B\varepsilon/\kappa_\delta)^n - 1)$. As in the proof of Corollary 31.10.1, this proves (31.15).

For fixed p, Corollary 31.5.1 and Lemma 1.8.9 give $P_{\widehat{\theta}_\varepsilon}[p] \to p$ in L^1 and in capacity along every decreasing sequence $\varepsilon \downarrow 0$. An arbitrary sequence $\varepsilon_j \to 0$ is handled by comparing with the decreasing sequence $\sup_{k \ge j} \varepsilon_k$, since the envelopes

increase with the background and lie above p. Quasi-continuity and integrability give $cp + s\rho \to p$ in both senses. The bounds (31.17) prove the asserted limits.

To compare two choices along a sequence η_j, denote their coefficients by $c_j^{(i)}$ and their upper backgrounds by $\theta + B_i\varepsilon_j^{(i)}h$, $i = 0, 1$. Put $c_j = \min_i c_j^{(i)}$ and $a_j = \max_i B_i\varepsilon_j^{(i)}$. Since the envelopes are nonpositive, both transported models lie below $c_j P_{\theta+a_jh}[p]$. Moreover $\eta_j \le c_j(\theta + a_jh)$, by (31.16) for an index attaining the smaller coefficient. The upper volume of the maximum of the two transported models is therefore at most $c_j^n \overline{\mathrm{vol}}_{\theta+a_jh}(p)$, which tends to $\overline{\mathrm{vol}}_\theta(p)$. Each of their individual upper volumes is at least $(c_j^{(i)})^n \overline{\mathrm{vol}}_\theta(p)$ and at most the upper volume of the maximum, so it has the same limit. Hence their d_{η_j}-comparison tends to zero.□

For $p_j, p \in \mathcal{M}_\delta(\theta)$, (31.15) shows that $d_\theta(p_j, p) \to 0$ if and only if the comparisons of the transported models tend to zero along backgrounds whose errors tend to zero. The next criterion allows approximants chosen directly in those backgrounds.

Theorem 31.11.2 *Let p be an ordinary model of positive lower volume. Let ε_j be rational numbers with $0 < \varepsilon_j < \min(\kappa/2, 1/B)$ tending to zero, let η_j be smooth real $(1,1)$-forms with $\theta - \varepsilon_j h \le \eta_j \le \theta + \varepsilon_j h$, and put $p_j = T^\rho_{\eta_j,\varepsilon_j}(p)$. Then $p = P_\theta[p]_\mathcal{I}$ if and only if there are analytic $a_j \in \mathrm{PSH}(X, \eta_j)$ with*

$$d_{\eta_j}(a_j, p_j) \longrightarrow 0.$$

The approximants may be required to be ordinary models. This condition does not depend on the choice of the strict analytic potential and of the admissible error bounds used to transport p. For every such analytic approximation, the convex combinations with ρ of Theorem 31.10.1 give analytic potentials in the original background whose ordinary envelopes converge to p for d_θ, in L^1 and in capacity.

Proof Suppose that $p = P_\theta[p]_\mathcal{I}$. By Theorem 31.8.1, there are analytic ordinary models r_j in background θ with $d_\theta(r_j, p) \to 0$ and a common positive lower-volume bound. The transports $a_j = T^\rho_{\eta_j,\varepsilon_j}(r_j)$ have analytic singularities: the convex coefficients are rational, and ordinary projection preserves the singularity type of an analytic potential by Proposition 17.2.4. The estimate (31.15) gives $d_{\eta_j}(a_j, p_j) \to 0$.

Conversely, assume that such analytic approximants exist, and put $\delta = \underline{\mathrm{vol}}_\theta(p)$. By Theorem 31.11.1, the lower volume of p_j is at least $c_{\varepsilon_j}^n \delta \ge 2^{-n}\delta$. In background $\eta_j \le \Omega$ all upper volumes are at most $v_+(\Omega)$, so (31.8) in background η_j and $d_{\eta_j}(a_j, p_j) \to 0$ give $\underline{\mathrm{vol}}_{\eta_j}(a_j) \ge 2^{-n-1}\delta$ for all sufficiently large j.

Put $\beta_j = \theta + \varepsilon_j h$. Lemma 31.11.1, with $\beta = \eta_j$ and $\gamma = \beta_j$, and projection invariance in background β_j imply $d_{\beta_j}(a_j, p_j) \to 0$. Also $d_{\beta_j}(p_j, p) \to 0$. Indeed, (31.17) bounds $\max(p_j, p)$ above by $c_{\varepsilon_j} P_{\widehat{\theta}_{\varepsilon_j}}[p]$, and $c_{\varepsilon_j}\widehat{\theta}_{\varepsilon_j} \ge \beta_j$ by (31.16). Hence the upper volume of this maximum in background β_j is at most $c_{\varepsilon_j}^n \overline{\mathrm{vol}}_{\widehat{\theta}_{\varepsilon_j}}(p)$, which tends to $\overline{\mathrm{vol}}_\theta(p)$. The upper volumes of p_j and p in background β_j lie between $c_{\varepsilon_j}^n \overline{\mathrm{vol}}_\theta(p)$ and this bound, so they have the same limit.

In background β_j, the lower volume of p_j is at least $2^{-n}\delta$, and all upper volumes are at most $v_+(\Omega)$, since $\beta_j \le \Omega$. The three-point estimate (31.7) in background β_j,

with middle point p_j, therefore gives $d_{\beta_j}(a_j, p) \to 0$, with constants independent of j. Now Theorem 31.10.1, applied with the backgrounds β_j, shows that p is an ideal model and gives the stated approximation in the original background.

For another choice of transport, Theorem 31.11.1 shows that the two reference models in background η_j have volume comparison tending to zero. Both have lower volume at least $2^{-n}\delta$, so the three-point estimate in background η_j, with a reference model as middle point, transfers the convergence condition in both directions. This proves independence of the choice. □

31.12 A complete metric on positive-volume models

The three-point modulus controls convergence even though d_θ need not satisfy the triangle inequality; two lines in the projective plane already show the failure. A complete metric can nevertheless be constructed from finite chains, as in the metrization of uniform structures. Continuity of the lower volume controls the chain estimates locally, and the reciprocal of the lower volume prevents Cauchy sequences from escaping to the zero-volume boundary.

Example 31.12.1 Let θ be the Fubini–Study form on $\mathbb{P}^2$, normalized by $\int_{\mathbb{P}^2} \theta^2 = 1$. There are analytic ordinary models p, q, r of positive lower volume such that

$$d_\theta(p, q) > d_\theta(p, r) + d_\theta(r, q).$$

Proof Write $|Z|^2 = |Z_0|^2 + |Z_1|^2 + |Z_2|^2$ in homogeneous coordinates. Fix a rational $a \in (0, 1/2)$ and put

$$u = a \log \frac{|Z_0|^2}{|Z|^2}, \qquad v = a \log \frac{|Z_1|^2}{|Z|^2}, \qquad w = u + v.$$

With $H_i = \{Z_i = 0\}$, the corresponding currents are $a[H_0] + (1-a)\theta$, $a[H_1] + (1-a)\theta$ and $a[H_0] + a[H_1] + (1 - 2a)\theta$. The background is closed and $\mathbb{P}^2$ is Kähler, so each singularity volume equals the non-pluripolar mass, which is the mass of the smooth part:

$$U(u) = U(v) = (1 - a)^2, \qquad U(w) = (1 - 2a)^2.$$

The maximum $\max(u, v)$ has the ordinary type of

$$z = a \log \frac{|Z_0|^2 + |Z_1|^2}{|Z|^2}.$$

To compute its volume, blow up the common point $[0 : 0 : 1]$ of H_0 and H_1. Denote the blowup by $\pi \colon Y \to \mathbb{P}^2$, its exceptional divisor by E, and the pencil map $[Z_0 : Z_1]$ by $f \colon Y \to \mathbb{P}^1$. For the Fubini–Study form ω_1 of mass one on $\mathbb{P}^1$, the Poincaré–Lelong

formula gives

$$\pi^*\theta + \mathrm{dd}^{\mathrm{c}}(z \circ \pi) = a[E] + \beta, \qquad \beta = (1-a)\pi^*\theta + a f^*\omega_1.$$

Here β is smooth and semipositive. Also

$$\int_Y (\pi^*\theta)^2 = 1, \qquad \int_Y \pi^*\theta \wedge f^*\omega_1 = 1, \qquad (f^*\omega_1)^2 = 0.$$

The mixed integral is one because every fiber of the pencil maps onto a line. By Lemma 29.7.1 and change of variables off E,

$$U(\max(u,v)) = \int_Y \beta^2 = 1 - a^2.$$

Since $w \le u$ and $w \le v$, (31.6) and the definition of d_θ give

$$d_\theta(u,v) = 4a - 4a^2,$$
$$d_\theta(u,w) + d_\theta(w,v) = 4a - 6a^2.$$

Set $p = P_\theta[u]$, $q = P_\theta[v]$ and $r = P_\theta[w]$. By Proposition 17.2.4, these models have the same analytic singularity types as u, v and w, and their lower volumes are at least $(1-2a)^2$. By projection invariance (Proposition 31.6.1), they have the same comparisons, and the difference between the two sides is $2a^2 > 0$. □

We return to the big background θ on the n-dimensional manifold X. Write $L(p) = \underline{\mathrm{vol}}_\theta(p)$, and let

$$\mathcal{M}_+(\theta) = \bigcup_{\delta>0} \mathcal{M}_\delta(\theta)$$

be the space of ordinary models of positive lower volume. We keep the notation U and M of Section 31.6.

Theorem 31.12.1 *There is a complete metric $\mathfrak{d}_\theta$ on $\mathcal{M}_+(\theta)$ such that*

$$\mathfrak{d}_\theta(p_j, p) \longrightarrow 0 \quad \Longleftrightarrow \quad d_\theta(p_j, p) \longrightarrow 0. \tag{31.18}$$

A sequence p_j is $\mathfrak{d}_\theta$-Cauchy if and only if it is d_θ-Cauchy and the sequence $1/L(p_j)$ is Cauchy in $\mathbb{R}$. On each $\mathcal{M}_\delta(\theta)$, the metric $\mathfrak{d}_\theta$ and the comparison d_θ define the same uniform structure, and the restriction of $\mathfrak{d}_\theta$ is complete.

Proof *Step 1: a decreasing family of symmetric relations.* Set $C = 1 + M$ and $m(p) = L(p)/C$, so that $0 < m(p) < 1$. Put $a_1 = 1$, $a_{k+1} = na_k + 1$, and choose $0 < \varepsilon_1 < 1$. Recursively choose $\varepsilon_{k+1} > 0$ so small that, with $r_k = 4^{a_{k+1}}\varepsilon_{k+1}$,

$$3\varepsilon_{k+1} < \varepsilon_k, \qquad 3r_k + 4nM\left(\frac{2r_k}{C}\right)^{1/n} < \varepsilon_k.$$

Let $\mathcal{U}_0 = \mathcal{M}_+(\theta)^2$ and, for $k \geq 1$, set

$$\mathcal{U}_k = \left\{(p,q) : \left|\frac{1}{m(p)} - \frac{1}{m(q)}\right| < \varepsilon_k, \quad d_\theta(p,q) < \varepsilon_k \min\{m(p), m(q)\}^{a_k}\right\}.$$

These symmetric sets contain the diagonal and decrease with k, and their intersection is the diagonal by Proposition 31.6.1. We claim that

$$\mathcal{U}_{k+1} \circ \mathcal{U}_{k+1} \circ \mathcal{U}_{k+1} \subset \mathcal{U}_k \qquad (k \geq 0), \tag{31.19}$$

where composition of relations means joining successive pairs. For $k = 0$ this is immediate. For $k \geq 1$, suppose $(p_{i-1}, p_i) \in \mathcal{U}_{k+1}$ for $i = 1, 2, 3$, and put $m_* = \min_{0 \leq i \leq 3} m(p_i)$. The reciprocal differences along this chain have total less than $3\varepsilon_{k+1} < 1$, so $1/m(p_i) > 1/m_* - 1$. Hence $m(p_i) \leq 4m_*$ for every i: if $m_* \leq 3/4$, this follows from $1/m_* - 1 \geq 1/(4m_*)$, and otherwise from $m(p_i) < 1$. Each successive comparison is therefore less than

$$e = r_k m_*^{a_{k+1}}.$$

All four models have lower volume at least Cm_*. Applying (31.7) with $\delta = Cm_*$ twice, first with middle point p_2 and then with middle point p_1, yields

$$\begin{aligned} d_\theta(p_0, p_3) &\leq 3e + 4nM\left(\frac{2e}{Cm_*}\right)^{1/n} \\ &\leq \left(3r_k + 4nM\left(\frac{2r_k}{C}\right)^{1/n}\right) m_*^{a_k} \\ &< \varepsilon_k \min\{m(p_0), m(p_3)\}^{a_k}. \end{aligned}$$

Here $(a_{k+1} - 1)/n = a_k$, $a_{k+1} \geq a_k$ and $m_* < 1$. The endpoint reciprocal difference is less than $3\varepsilon_{k+1} < \varepsilon_k$, proving (31.19).

Step 2: the chain metric. Define

$$\mathfrak{d}_\theta(p,q) = \inf \sum_{i=1}^{N} 2^{-k_i}, \tag{31.20}$$

where the infimum runs over finite chains $p = p_0, p_1, \ldots, p_N = q$ and integers $k_i \geq 0$ with $(p_{i-1}, p_i) \in \mathcal{U}_{k_i}$; we call $\sum_i 2^{-k_i}$ the cost of the chain. A single edge in $\mathcal{U}_0$ gives $\mathfrak{d}_\theta \leq 1$. Reversing and concatenating chains gives symmetry and the triangle inequality, while diagonal edges with arbitrarily large indices give $\mathfrak{d}_\theta(p,p) = 0$.

We next prove the inclusions

$$\{\mathfrak{d}_\theta < 2^{-(k+1)}\} \subset \mathcal{U}_k \subset \{\mathfrak{d}_\theta \leq 2^{-k}\} \qquad (k \geq 1). \tag{31.21}$$

The second inclusion follows by using a single edge. For the first, we prove by induction on the number of edges that every chain of cost less than 2^{-k} has its

endpoints in $\mathcal{U}_{k-1}$, simultaneously for all $k \geq 1$. For one edge this follows because the $\mathcal{U}_k$ decrease. For a chain of cost $A < 2^{-k}$, choose $l \geq k$ with

$$2^{-(l+1)} \leq A < 2^{-l}.$$

Split the chain at the first edge at which the cumulative cost reaches $2^{-(l+1)}$. The subchains before and after this edge both have cost less than $2^{-(l+1)}$, so by induction their endpoint pairs lie in $\mathcal{U}_l$; an empty subchain is interpreted as a diagonal pair. Every edge cost is a power of two smaller than 2^{-l}, so the middle edge lies in $\mathcal{U}_{l+1} \subset \mathcal{U}_l$. By (31.19), the endpoints of the whole chain form a pair in $\mathcal{U}_{l-1} \subset \mathcal{U}_{k-1}$. This completes the induction and proves (31.21). Since the $\mathcal{U}_k$ intersect in the diagonal, $\mathfrak{d}_\theta$ separates points, so it is a metric.

Step 3: convergence and completeness. By (31.21), $\mathfrak{d}_\theta(p_j, p) \to 0$ if and only if (p_j, p) eventually lies in every $\mathcal{U}_k$. This implies $d_\theta(p_j, p) \to 0$. Conversely, if $d_\theta(p_j, p) \to 0$, then Proposition 31.6.3 gives $L(p_j) \to L(p) > 0$, so for each fixed k both inequalities defining $\mathcal{U}_k$ hold eventually. This proves (31.18).

The same inclusions show that a $\mathfrak{d}_\theta$-Cauchy sequence is d_θ-Cauchy and has Cauchy reciprocal lower volumes. These reciprocals are bounded, so a tail of the sequence lies in some $\mathcal{M}_\delta(\theta)$. By Theorem 31.6.1, the sequence converges there for d_θ, and hence for $\mathfrak{d}_\theta$. This proves completeness. Conversely, a d_θ-Cauchy sequence with Cauchy reciprocal lower volumes has a common positive lower-volume bound on a tail, and then both inequalities defining each $\mathcal{U}_k$ hold for all pairs of sufficiently late terms. This proves the Cauchy criterion.

On $\mathcal{M}_\delta(\theta)$, (31.8) controls the reciprocal differences uniformly in terms of d_θ, while $m(p) \geq \delta/C$ controls the second threshold in $\mathcal{U}_k$. Thus $\mathfrak{d}_\theta$ and d_θ define the same uniform structure there. Continuity of the lower volume makes $\mathcal{M}_\delta(\theta)$ closed for $\mathfrak{d}_\theta$, so the restriction of $\mathfrak{d}_\theta$ to it is complete. □

Corollary 31.12.1 *The ideal models in $\mathcal{M}_+(\theta)$ form a closed, hence complete, subspace of $(\mathcal{M}_+(\theta), \mathfrak{d}_\theta)$, and the analytic ordinary models of positive lower volume are dense in it. Equivalently, this subspace is the metric completion of the analytic ordinary models of positive lower volume with the restricted metric.*

Proof By Theorem 31.8.2, an ordinary model of positive lower volume that is a d_θ-limit of ideal models is again an ideal model. The same theorem gives, for every ideal model $p \in \mathcal{M}_+(\theta)$, analytic ordinary models p_j with $d_\theta(p_j, p) \to 0$, and Proposition 31.6.3 puts them in $\mathcal{M}_+(\theta)$ for large j. Now apply (31.18) and the completeness of $\mathfrak{d}_\theta$. □

31.13 Model potentials on curves

On a curve the preceding theory becomes explicit. The part of the curvature current carried by the pole set determines the ordinary envelope, and its atomic part determines the ideal envelope. Comparing these two measures describes both the zero-volume

boundary and the failure of analytic approximation for diffuse polar singularities. Total variation of the polar measures keeps this singularity data and at the same time controls convergence of the model potentials in capacity.

In this section, Y is a compact connected Riemann surface, θ is a smooth positive $(1,1)$-form on Y, and $A = \int_Y \theta$. By Stokes' theorem every bounded θ-plurisubharmonic function has mass A; thus Y has the bounded mass property, θ is big, and the preceding results apply with Y in place of X. For $u \in \mathrm{PSH}(Y,\theta)$ write $T_u = \theta + \mathrm{dd^c} u$, and let $\mu_u = \mathbf{1}_{\{u=-\infty\}} T_u = T_u^{\mathrm{pol}}$ be the polar curvature of u (Section 1.6.1); in this and the following sections, μ_u no longer denotes $\mathrm{MA}_\theta(u)$. A positive measure carried by a polar set will be called a polar measure. Since θ is closed and the Riemann surface Y is Kähler, the two singularity volumes of u coincide with

$$U(u) = \int_Y \mathrm{MA}_\theta(u) = A - \mu_u(Y).$$

Fix a nonpositive Green kernel G for $\eta = \theta/A$, as in Section 20.3, so that $\mathrm{dd}^{\mathrm{c}}_x G(x,y) = \delta_y - \eta$, where δ_y is the Dirac measure at y. We write $\|\cdot\|_{\mathrm{TV}}$ for the total variation norm of finite signed measures.

Lemma 31.13.1 *For a polar measure μ on Y with $\mu(Y) \le A$, put*

$$w_\mu(x) = \int_Y G(x,y)\,\mathrm{d}\mu(y), \qquad p_\mu = P_\theta[w_\mu].$$

Then $p_\mu \sim w_\mu$, $\mu_{p_\mu} = \mu$ and $U(p_\mu) = A - \mu(Y)$. Every $u \in \mathrm{PSH}(Y,\theta)$ satisfies

$$P_\theta[u] = p_{\mu_u}. \tag{31.22}$$

Consequently $\mu \mapsto p_\mu$ is a bijection from the polar measures of mass at most A onto the set of all ordinary models on Y.

Proof The bounds for the Green kernel give $w_\mu \in L^1(Y)$ and

$$T_{w_\mu} = \mu + (A - \mu(Y))\eta.$$

In particular w_μ is θ-plurisubharmonic, and its polar curvature is μ. We claim that, for every $v \in \mathrm{PSH}(Y,\theta)$,

$$v \preceq w_\mu \quad \Longleftrightarrow \quad T_v \ge \mu. \tag{31.23}$$

The forward implication follows from Corollary 1.6.1, which gives $T_v \ge \mu_v \ge \mu_{w_\mu} = \mu$. Conversely, if $T_v \ge \mu$, the distribution $v - w_\mu$ satisfies $\mathrm{dd^c}(v - w_\mu) \ge -(A - \mu(Y))\eta$, so its quasi-subharmonic representative is bounded above on the compact curve Y. By Lemma 1.3.3, $v \preceq w_\mu$.

Thus p_μ is the upper envelope of all $v \le 0$ with $T_v \ge \mu$. This family is closed under maxima by (1.8). By Choquet's lemma the envelope is the limit of an increasing sequence of such maxima, which converges in L^1; hence the current inequality $T_{p_\mu} \ge \mu$ passes to the limit. Also $p_\mu \ge w_\mu - \sup_Y w_\mu$. Hence $b = p_\mu - w_\mu$ has a

quasi-subharmonic representative that is bounded below, and compactness bounds it above. Therefore $p_\mu \sim w_\mu$. The positive measure

$$T_{p_\mu} - \mu = (A - \mu(Y))\eta + \mathrm{dd}^{\mathrm{c}} b$$

charges no polar set because b is bounded. This proves $\mu_{p_\mu} = \mu$ and $U(p_\mu) = A - \mu(Y)$, and ordinary equivalence gives $P_\theta[p_\mu] = P_\theta[w_\mu] = p_\mu$.

For arbitrary u, take $\mu = \mu_u$. Then $T_u \geq \mu$, so (31.23) gives $u \preceq p_\mu$, and $U(u) = A - \mu(Y) = U(p_\mu)$. If this mass is positive, Corollary 31.3.2 gives (31.22). If it is zero, then $\mu(Y) = A$ and $T_u = \mu = T_{w_\mu}$. The difference $u - w_\mu$ is then harmonic on Y and hence constant, which gives the same conclusion. Finally, every ordinary model p equals $P_\theta[p] = p_{\mu_p}$, and distinct polar measures give distinct models because $\mu_{p_\mu} = \mu$. □

Theorem 31.13.1 *For every $u, v \in \mathrm{PSH}(Y, \theta)$,*

$$d_\theta(u, v) = \|\mu_u - \mu_v\|_{\mathrm{TV}}. \tag{31.24}$$

In particular, d_θ is a metric on the set of ordinary models on Y. For positive atomic measures D, E of mass at most A, let q_D, q_E be the ideal models of Lemma 20.5.1, with polar curvatures D and E. Then

$$d_\theta(q_D, q_E) = \|D - E\|_{\mathrm{TV}}. \tag{31.25}$$

Proof By (1.8), the polar curvature of $\max(u, v)$ is the minimum $\mu_u \wedge \mu_v$ of the two measures. The formula $U = A - \mu(Y)$ therefore gives

$$\begin{aligned} d_\theta(u, v) &= 2U(\max(u, v)) - U(u) - U(v) \\ &= \mu_u(Y) + \mu_v(Y) - 2(\mu_u \wedge \mu_v)(Y) \\ &= \|\mu_u - \mu_v\|_{\mathrm{TV}}. \end{aligned}$$

The classification in Lemma 31.13.1 shows that d_θ separates ordinary models, and the triangle inequality follows from that of the total variation norm. By Lemma 20.5.1, the ideal models are exactly the models whose polar measures are atomic, which gives (31.25). □

Corollary 31.13.1 *The ordinary models on Y form a complete geodesic metric space for d_θ. The ideal models form a closed geodesically convex subspace, which is the metric completion of the analytic ordinary models of positive mass.*

Proof By Lemma 31.13.1 and Theorem 31.13.1, the space of ordinary models is isometric to the set of polar measures of mass at most A with the total variation norm. This set is closed in total variation: a limit is positive, satisfies the mass bound, and is carried by the countable union of polar carriers of the sequence. Completeness therefore follows from that of finite signed measures. For two such measures μ, ν, the measures $\mu_t = (1 - t)\mu + t\nu$, $0 \leq t \leq 1$, remain in this set and satisfy

$$\|\mu_t - \mu_s\|_{\mathrm{TV}} = |t - s| \, \|\mu - \nu\|_{\mathrm{TV}}.$$

Their models give a constant-speed geodesic.

The atomic measures form a closed convex subset, since a total-variation limit is carried by the countable union of the sets of atoms of the sequence. Restricting to finitely many atoms and approximating the coefficients from below by rational numbers, with an arbitrarily small further decrease if needed, approximates every such measure by finite atomic measures with rational coefficients and mass strictly less than A. Their models have analytic singularities by Lemma 20.5.1, and (31.25) gives the density. □

The models of positive mass do not form a complete space for d_θ: for a fixed $x \in Y$, the models of $tA\delta_x$ form a Cauchy family as $t \uparrow 1$, with a limit of zero mass. For the complete metric of Theorem 31.12.1, this family is not Cauchy, because its reciprocal lower volumes diverge. The two metrics induce the same topology on models of positive mass but have different Cauchy sequences near the boundary.

The metric d_θ also controls convergence of the potentials at the zero-mass boundary. There the polar part exhausts the limiting curvature current, so its total variation controls the whole current.

Theorem 31.13.2 *Let p_j, p be ordinary models on Y. If $d_\theta(p_j, p) \to 0$, then $p_j \to p$ in L^1 and in capacity. If $U(p) = 0$, then in addition*

$$\|T_{p_j} - T_p\|_{\mathrm{TV}} \le 2d_\theta(p_j, p) \longrightarrow 0. \tag{31.26}$$

Proof Write $\mu_j = \mu_{p_j}$ and $\mu = \mu_p$. By (31.24), $\mu_j \to \mu$ in total variation. Thus $U(p_j) = A - \mu_j(Y) \to U(p)$. If $U(p) > 0$, the models lie in a common $\mathcal{M}_\delta(\theta)$ for all large j, and Proposition 31.6.1 and Theorem 31.7.1 apply.

Suppose that $U(p) = 0$. Then $T_p = \mu$ and $\mu(Y) = A$, while $T_{p_j} - \mu_j$ is a positive measure of mass $A - \mu_j(Y) = \mu(Y) - \mu_j(Y)$. Consequently

$$\|T_{p_j} - T_p\|_{\mathrm{TV}} \le \|\mu_j - \mu\|_{\mathrm{TV}} + A - \mu_j(Y) \le 2\|\mu_j - \mu\|_{\mathrm{TV}},$$

which proves (31.26). By normalized compactness, each subsequence has a further subsequence converging in L^1 to a function q with supremum zero. Then $T_{p_j} \to T_q$ weakly, so $T_q = T_p$ and $\mathrm{dd}^{\mathrm{c}}(q - p) = 0$. Hence $q - p$ is constant, and the normalization gives $q = p$. This proves L^1 convergence of the whole sequence. Finally, apply Lemma 1.6.3 on finitely many nested coordinate disks: both terms on its right-hand side tend to zero, and the comparison between local and global capacities in Section 1.8 gives convergence in capacity. □

For potentials with supremum zero, weak convergence $T_{p_j} \rightharpoonup T_p$ is equivalent to $p_j \to p$ in L^1, by Theorem 6.1.1 and normalized compactness. It need not imply convergence in capacity, even when the sequence and the limit are ideal models. Indeed, scaling the area-one construction in Theorem 17.4.1 gives zero-mass ideal models converging to zero in L^1. Convergence in capacity would contradict Theorem 8.3.1, since their damped measures vanish while the limiting measure is θ.

Conversely, convergence in capacity does not imply convergence of the polar parts in total variation. Moving a single pole gives an example, even within the zero-mass boundary.

Example 31.13.1 On $\mathbb{P}^1$ with the Fubini–Study form θ of area one, there are zero-mass ideal models p_j, p with $p_j \to p$ in capacity but $d_\theta(p_j, p) = 2$ for every j.

Proof In the affine coordinate z, where $\theta = \mathrm{dd^c}\log(1+|z|^2)$, put for $a \in \mathbb{C}$

$$p_a(z) = \log \frac{|z-a|^2}{(1+|z|^2)(1+|a|^2)}.$$

This function extends to $\mathbb{P}^1$, has supremum zero by the Cauchy–Schwarz inequality, and satisfies $T_{p_a} = \delta_a$. Hence it is the ideal model with polar measure δ_a, by Lemma 20.5.1. Choose $a_j \neq 0$ tending to zero and set $p_j = p_{a_j}$, $p = p_0$. On the complement of any fixed disk about zero the convergence is uniform, including near infinity. Small disks about zero lie in sublevel sets $\{p_0 < -t\}$ with t large, so their capacities tend to zero as the radius tends to zero, by (1.15). Therefore $p_j \to p$ in capacity. Since $\|\delta_{a_j} - \delta_0\|_{\mathrm{TV}} = 2$, (31.24) gives the asserted distance. □

Return to the general curve Y. Write $\mu_u = D_u + \sigma_u$, where $D_u = \sum_{x\in Y} \nu(u,x)\delta_x$ is the atomic part and σ_u has no atoms. Set $\tau(u) = \sigma_u(Y) = A - U(u) - D_u(Y)$, the diffuse polar mass of u, and let $\mathcal{A}_+(Y,\theta)$ denote the set of analytic ordinary models of positive mass.

Proposition 31.13.1 *For every $u \in \mathrm{PSH}(Y,\theta)$,*

$$\inf_{a\in\mathcal{A}_+(Y,\theta)} d_\theta(u,a) = d_\theta(u, P_\theta[u]_{\mathcal{I}}) = \tau(u). \tag{31.27}$$

On ordinary models, $p \mapsto P_\theta[p]_{\mathcal{I}}$ is a 1-Lipschitz retraction onto the ideal models, and $P_\theta[p]_{\mathcal{I}}$ is the unique ideal model nearest to p. There are $a_j \in \mathcal{A}_+(Y,\theta)$ with $d_\theta(u,a_j) \to 0$ if and only if $P_\theta[u] = P_\theta[u]_{\mathcal{I}}$.

Proof The ideal envelope of u is q_{D_u}, by Lemma 20.5.1. For any positive atomic measure E of mass at most A, the nonatomic measure σ_u and the signed atomic measure $D_u - E$ are mutually singular. Thus (31.24) gives

$$d_\theta(u, q_E) = \tau(u) + \|D_u - E\|_{\mathrm{TV}}. \tag{31.28}$$

Among ideal models, the minimum is attained exactly at $E = D_u$. Since the analytic models of positive mass are dense among the ideal models (Corollary 31.13.1), this gives (31.27). Taking the atomic part does not increase total variation, which proves the Lipschitz assertion. Finally, Lemma 31.13.1 shows that $P_\theta[u] = p_{\mu_u}$ equals $q_{D_u} = p_{D_u}$ if and only if $\mu_u = D_u$, that is, $\tau(u) = 0$. □

Example 31.13.2 There is an ordinary model p on Y with zero mass whose multiplier ideals $\mathcal{I}(tp)$ are trivial for every $t > 0$. It satisfies

$$P_\theta[p]_{\mathcal{I}} = 0, \qquad \inf_{a\in\mathcal{A}_+(Y,\theta)} d_\theta(p,a) = A.$$

Proof By Lemma 1.6.5, there is a probability measure σ without atoms carried by a compact polar subset of Y. Let $p = p_{A\sigma}$ be the ordinary model of Lemma 31.13.1, so that $\mu_p = A\sigma$ and $U(p) = 0$. Since $T_p \geq \mu_p$ and both measures have mass A, we get $T_p = A\sigma$. The Lelong numbers of p vanish because σ has no atoms, so all its multiplier ideals are trivial and its ideal envelope is zero. Finally $\tau(p) = A$, which gives the distance by (31.27). □

31.13.1 The ordinary-type topology on a curve

We now compare the polar metric with the topology on ordinary singularity types introduced in Section 17.6. That topology also uses the ordinary envelope, but it measures the total variation only of the atomic part of the curvature. On ideal models the atomic part is the whole polar measure, so the two topologies agree there. On arbitrary ordinary models, diffuse polar mass can concentrate without being detected by the atomic total variation.

We keep Y, θ, A and the decomposition $\mu_u = D_u + \sigma_u$, and we use $\eta = \theta/A$ for L^1 norms. Let $\mathfrak{S}(Y,\theta) = \mathrm{PSH}(Y,\theta)/\sim$ be the space of ordinary singularity types, and write $[u]$ for the class of u. Recall from Proposition 17.6.1 that $\tau_{\theta,0}$ is the uniform topology on $\mathfrak{S}(Y,\theta)$ generated by the L^1 distances between ordinary envelopes and between ideal envelopes, by the total variation of divisorial Lelong numbers on each fixed modification, and by the controlled ideal relations $\mathcal{U}(t,a)$ of (17.5). For ordinary classes put

$$d^{\mathrm{ot}}_\theta([u],[v]) = \|P_\theta[u] - P_\theta[v]\|_{L^1(\eta)} + \|D_u - D_v\|_{\mathrm{TV}}. \tag{31.29}$$

Theorem 31.13.3 *The finite pseudometric d^{ot}_θ induces $\tau_{\theta,0}$ on $\mathfrak{S}(Y,\theta)$. For $0 \leq s \leq A$, the classes of ideal models*

$$C_s = \{[q_D] : D \text{ positive atomic},\ D(Y) \leq s\}$$

form a complete metric space both for d^{ot}_θ and for the distance $\|D - E\|_{\mathrm{TV}}$ between $[q_D]$ and $[q_E]$, and these two metrics induce the same topology. The space C_s is not totally bounded if $s > 0$, and $C_0 = \{[0]\}$.

Proof A smooth compact modification of a smooth curve is an isomorphism, and the divisor weight of a point is one. Hence the divisorial distance in Proposition 17.6.1 is exactly the atomic total variation, and the two summands of (31.29) are among the coordinates defining $\tau_{\theta,0}$.

Conversely, suppose that a net $[u_j]$ converges to $[u]$ for d^{ot}_θ. The ideal envelope of u is q_{D_u}, by Lemma 20.5.1. Convergence of the atomic parts in total variation implies L^1 convergence of these envelopes, by Theorems 31.13.1 and 31.13.2. These results also apply to nets: the map $D \mapsto q_D$ goes between metric spaces, so sequential continuity is equivalent to continuity. It remains to check the controlled ideal relations.

At a point x with local coordinate z, the multiplier ideals on the curve are

$$\mathcal{I}(tu)_x = (z^{\lfloor tD_u(\{x\})\rfloor}).$$

This follows from Theorem 1.16.1 and Lemma 1.16.1: on a smooth curve the Lelong number at x determines the ideal, and the logarithmic germ with that Lelong number has the displayed ideal. Write $D_j = D_{u_j}$ and $D = D_u$, and fix $t > 0$ and $a > 1$. The set $F = \{x : D(\{x\}) \geq 1/(2t)\}$ is finite. At its points, $aD_j(\{x\}) \geq D(\{x\})$ and $aD(\{x\}) \geq D_j(\{x\})$ eventually. Outside F, the bound

$$\sup_x |D_j(\{x\}) - D(\{x\})| \leq \|D_j - D\|_{\mathrm{TV}} \longrightarrow 0$$

makes both $\mathcal{I}(tu)_x$ and $\mathcal{I}(tu_j)_x$ unit ideals for large j. Hence both inclusions in (17.5) hold eventually at every point. This proves that d_θ^{ot} induces $\tau_{\theta,0}$.

The map $D \mapsto q_D$ is injective, so d_θ^{ot} is a metric on C_s. The positive atomic measures of mass at most s form a closed subset of the space of finite signed measures, as in the proof of Corollary 31.13.1, and hence a complete space for total variation. A d_θ^{ot}-Cauchy sequence in C_s is Cauchy in total variation; its limit is again atomic, and Theorem 31.13.2 gives convergence of the envelope coordinate. This proves completeness for d_θ^{ot} as well as the equality of the two topologies.

If $s > 0$, fix $0 < b \leq s$ and infinitely many distinct points x_j. The measures $b\delta_{x_j}$ have pairwise total-variation distance $2b$, so C_s is not totally bounded for either metric. For $s = 0$ only the zero measure remains. □

By Lemma 20.5.1, for $0 \leq \delta \leq A$ the set $C_{A-\delta}$ consists exactly of the classes of ideal models of mass at least δ. The whole set C_A, including the models of zero mass, is complete. Its subset of positive mass is incomplete: for a point x, the measures $D_j = A(1 - 1/j)\delta_x$ form a Cauchy sequence whose limit $A\delta_x$ has zero mass.

For general ordinary classes, the corresponding statement concerns a pseudometric. Put

$$\mathcal{G}_{\theta,\delta} = \left\{ [u] : P_\theta[u] = P_\theta[u]_{\mathcal{I}}, \quad \int_Y \mathrm{MA}_\theta(P_\theta[u]_{\mathcal{I}}) \geq \delta \right\}.$$

For $0 \leq \delta \leq A$ this set is complete for d_θ^{ot}, since every class $[u]$ in it is at distance zero from $[q_{D_u}] \in C_{A-\delta}$; limits need not be unique as ordinary classes. The set is not totally bounded when $\delta < A$. At $\delta = A$ the pseudometric vanishes identically on it, although it contains distinct classes by Proposition 17.6.2. The union of these sets over all $\delta > 0$ is incomplete, by the same one-atom sequence.

31.13.2 Concentration of diffuse polar curvature

The atomic total variation cannot be omitted from Theorem 31.13.3. On $\mathbb{P}^1$ with its Fubini–Study form θ of area one, fix a rational $0 < \tau < 1$ and take the atomic probability measures ν_j and the ideal models $r_j = P_\theta[(1 - \tau)q_j]_{\mathcal{I}}$ of Proposition 17.4.2, choosing the rational approximations so that the largest atom of ν_j tends to zero. The

ordinary and ideal envelopes of r_j both tend to zero in L^1, and for each $t > 0$ and $a > 1$, the pair $([r_j], [0])$ eventually lies in $\mathcal{U}(t, a)$. The currents $\theta + \mathrm{dd^c} r_j$ converge weakly to θ. Nevertheless,

$$d_\theta^{\mathrm{ot}}([r_j], [0]) \geq \|D_{r_j}\|_{\mathrm{TV}} = 1 - \tau > 0.$$

Thus these weaker conditions do not imply convergence for $\tau_{\theta,0}$, even at a fixed positive mass.

The opposite difficulty concerns diffuse polar mass: the pseudometric controls the ordinary envelope in L^1, but not the total variation of the diffuse measure itself.

Proposition 31.13.2 *For the Fubini–Study form θ of area one on $\mathbb{P}^1$, the space $(\mathfrak{S}(\mathbb{P}^1, \theta), d_\theta^{\mathrm{ot}})$ is incomplete. In fact, there is a d_θ^{ot}-Cauchy sequence without limit consisting of ordinary models of one fixed positive non-pluripolar mass, all with trivial multiplier ideals.*

Proof Choose a nonatomic polar probability measure σ with compact support in an affine chart, by Lemma 1.6.5. Fix $0 < c < 1$ and a point x in that chart. Translating and shrinking the support towards x gives nonatomic polar probability measures σ_j supported in disks about x with radii tending to zero. Let $p_j = p_{c\sigma_j}$ be the ordinary models of Lemma 31.13.1. Their polar measures are $c\sigma_j$, their masses are $1 - c$, and $D_{p_j} = 0$. Hence their Lelong numbers vanish, their multiplier ideals are trivial, and their ideal envelopes are zero.

We claim that $p_j \to q_{c\delta_x}$ in L^1. Write $q_y = q_{c\delta_y}$. Rotations of $\mathbb{P}^1$ preserving θ and sending x to y can be chosen to depend continuously on y near x. They transport q_x to q_y, so continuity of translations in L^1 gives $q_y \to q_x$ in L^1 as $y \to x$. Set

$$w_j = \int_{\mathbb{P}^1} q_y \, \mathrm{d}\sigma_j(y).$$

The common L^1 bound for normalized potentials makes this integral well defined. The function w_j is nonpositive, its curvature current dominates $c\sigma_j$, and $w_j \to q_x$ in L^1. By (31.23), p_j is the upper envelope of all $v \leq 0$ with $\theta + \mathrm{dd^c} v \geq c\sigma_j$, so $w_j \leq p_j$.

By normalized compactness, every subsequence of p_j has a further subsequence converging in L^1 to some $v \leq 0$. The inequalities $\theta + \mathrm{dd^c} p_j \geq c\sigma_j$ pass to the limit and give $\theta + \mathrm{dd^c} v \geq c\delta_x$. Since $q_x = p_{c\delta_x}$, the same envelope description gives $v \leq q_x$. The lower bound $w_j \leq p_j$ gives $v \geq q_x$ almost everywhere, hence everywhere. This identifies every cluster point and proves the claim.

Since $P_\theta[p_j] = p_j$ and $D_{p_j} = 0$, the classes $[p_j]$ form a d_θ^{ot}-Cauchy sequence. A limit $[v]$ would satisfy $D_v = 0$ and $P_\theta[v] = q_x$. This is impossible: the ordinary envelope preserves all multiplier ideals and hence all atoms, whereas q_x has an atom of mass c at x. The sequence therefore has no limit in $\mathfrak{S}(\mathbb{P}^1, \theta)$. □

On ordinary models, $\tau_{\theta,0}$ and the topology of the polar metric d_θ are also different. Choose the Cantor support of σ inside a positive real interval in the affine coordinate z, away from zero. The rotations $R_j(z) = \mathrm{e}^{\mathrm{i}\varepsilon_j} z$, with $0 < \varepsilon_j \to 0$, move this support

off itself. Put $p = p_{c\sigma}$ and $\widetilde{p}_j = p \circ R_j^{-1}$. By rotation invariance of the envelope, the $\widetilde{p}_j$ are ordinary models with polar measures $c(R_j)_*\sigma$, which have no atoms. Continuity of rotations in L^1 yields $d_\theta^{\mathrm{ot}}([\widetilde{p}_j], [p]) \to 0$, whereas the polar measures have disjoint supports, so $d_\theta(\widetilde{p}_j, p) = 2c$ by Theorem 31.13.1. Conversely, polar total variation controls both terms of (31.29), by Theorem 31.13.2. Thus $\tau_{\theta,0}$ is strictly weaker than the polar metric topology on ordinary models, although the two agree on ideal models. The failure of continuity under smooth gauge changes with a fixed zero obstacle (Proposition 17.6.4) gives a further difference: the polar metric is invariant under such changes, by Section 31.14.

31.13.3 Convergence for all obstacles

We keep the compact curve Y and its positive form θ of area A. The topology $\tau_\theta^{\mathrm{all}}$ of Section 17.6.4 is generated by the topologies $\tau_{\theta,F}$ for all smooth obstacles F, in which the ordinary and ideal envelopes below F replace those below zero. On a curve the envelopes below F are still determined by the polar measure, and their ideal counterparts by its atomic part. This gives a description of $\tau_\theta^{\mathrm{all}}$ by a single pseudometric and allows a comparison with polar total variation.

Choose, as in Proposition 17.6.5, a countable family $\{F_k\}_{k\geq 1} \subset C^\infty(Y, \mathbb{R})$ that is dense for the uniform norm, with $F_1 = 0$. Write $d_{O,F}$ and $d_{I,F}$ for the L^1 distances between the ordinary envelopes $P_{\theta,F}[\cdot]$ and between the ideal envelopes $P_{\theta,F}[\cdot]_I$, which define $\tau_{\theta,F}$ (Section 17.6), and put

$$\begin{aligned} d_\theta^{\mathrm{all}}([u],[v]) = \|D_u - D_v\|_{\mathrm{TV}} \\ + \sum_{k=1}^{\infty} 2^{-k} \big(\min\{1, d_{O,F_k}([u],[v])\} \\ + \min\{1, d_{I,F_k}([u],[v])\}\big). \end{aligned} \tag{31.30}$$

Proposition 31.13.3 *The pseudometric d_θ^{all} induces the uniformity of $\tau_\theta^{\mathrm{all}}$. Two classes have distance zero if and only if they have the same polar curvature. The closure of the analytic classes is $\{[u] : \mu_u = D_u\} = \mathcal{G}_\theta$, where $\mathcal{G}_\theta = \{[u] : P_\theta[u] = P_\theta[u]_I\}$ as in (17.6). On each set C_s, $0 \leq s \leq A$, the pseudometric d_θ^{all} is a complete metric inducing the topology of total variation of the atomic measures. On ordinary models, polar total variation implies convergence for d_θ^{all}.*

Proof *Step 1: envelopes below an obstacle.* For any smooth F, translating the competitors by F gives

$$P_{\theta,F}[u] - F = P_{\theta+\mathrm{dd^c}F}[u - F],$$

and likewise for the ideal envelope. The background $\theta + \mathrm{dd^c}F$ on the right is a smooth real form of area A, though it need not be positive. We claim that these two envelopes are the models of μ_u and of D_u, respectively, in that background, in the

sense of Proposition 31.14.1 in Section 31.14; the proof of that proposition does not use the present subsection. Set $\mu = \mu_u$ and let w_μ be its Green potential, and put $r = P_{\theta+\mathrm{dd^c}F}[w_\mu - F]$. By the gauge bounds of Proposition 17.3.1, $r + F$ differs from $P_\theta[w_\mu] = p_\mu \sim w_\mu$ by a bounded function, so $r \sim w_\mu - F$. Hence $r = P_{\theta+\mathrm{dd^c}F}[r]$ is an ordinary model, and its polar measure is μ. Moreover $u - F \preceq r$, and both have mass $A - \mu(Y)$. If this mass is positive, Corollary 31.3.2, applied in the big background $\theta + \mathrm{dd^c}F$, shows that $P_{\theta+\mathrm{dd^c}F}[u - F] = r$. If it is zero, the curvature measures of $u - F$ and $w_\mu - F$ both equal μ, so these functions differ by a constant, and again $P_{\theta+\mathrm{dd^c}F}[u - F] = r$. The assertion for the ideal envelope follows in the same way from ideal projection and the classification of atomic models. It also follows that the envelope $P_{\theta,F}[u]$ is the upper envelope of all $v \le F$ with $\theta + \mathrm{dd^c}v \ge \mu$.

Step 2: the uniformity. By Proposition 31.14.1, polar total variation gives L^1 convergence of these models for every fixed F. The same assertion for atomic measures gives continuity of every ideal-envelope coordinate with respect to the total variation of the atomic parts. The controlled ideal relations are also implied, uniformly, by atomic total variation: for fixed $t > 0$ and $a > 1$, the bound $\|D - E\|_{\mathrm{TV}} < (a-1)/(at)$ guarantees both inclusions in (17.5), by the formula for multiplier ideals on a curve. Indeed, if $tE(\{x\}) \ge 1$, the bound gives $aD(\{x\}) > E(\{x\})$; otherwise the target ideal is the unit ideal. Interchange D and E for the other inclusion. Each divisorial coordinate is the atomic total variation. Therefore the envelope terms in the series, together with the atomic term, generate exactly the uniformity of $\tau_\theta^{\mathrm{all}}$: finitely many coordinates control the sum up to its uniformly small tail, and the sum controls each fixed coordinate.

Step 3: zero distance and analytic closure. By Step 1 with $F = 0$, distance zero forces equal polar measures; the classification at every obstacle proves the converse. The same classification shows that the ordinary and ideal envelopes agree at every F if and only if $\mu_u = D_u$. Hence Theorem 17.6.2 identifies the closure of the analytic classes.

Step 4: the sets C_s. For a d_θ^{all}-Cauchy sequence in C_s, the atomic measures converge in total variation to a measure of mass at most s, which is again atomic. The convergence of models in Step 2 gives convergence of every obstacle coordinate, and the summable tail then gives convergence for d_θ^{all}. Conversely, convergence for d_θ^{all} implies convergence of the atomic parts, because of the first summand. This proves completeness and the equality of topologies on C_s. Finally, polar total variation controls the atomic part and all ordinary-envelope coordinates, which proves the last assertion. □

On a curve, membership in the relative envelope class of the ideal envelope has the same polar criterion, including for ideal envelopes of zero mass.

Corollary 31.13.2 *For $u \in \mathrm{PSH}(Y,\theta)$ put $q = P_\theta[u]_{\mathcal{I}}$. The following are equivalent:*

$$u \in \mathcal{E}_{\mathrm{env}}(Y,\theta,q), \qquad \mu_u = D_u,$$
$$P_\theta[u] = P_\theta[u]_{\mathcal{I}},$$
$$P_{\theta,F}[u] = P_{\theta,F}[u]_{\mathcal{I}} \quad \textit{for every smooth real } F.$$

Proof The classifications of ordinary and ideal models in Lemmas 20.5.1 and 31.13.1 show that $P_\theta[u] = P_\theta[u]_{\mathcal{I}}$ if and only if $\mu_u = D_u$. Step 1 of the proof of Proposition 31.13.3 shows that this is equivalent to equality of the envelopes at every smooth obstacle. Membership of u in $\mathcal{E}_{\mathrm{env}}(Y,\theta,q)$ implies the latter by Proposition 17.6.6.

Conversely, assume that $\mu_u = D_u$. By Lemma 20.5.1, $\mu_q = D_u$, so both singularity volumes of q equal $A - D_u(Y)$. If $D_u(Y) < A$, then $P_\theta[u] = P_\theta[u]_{\mathcal{I}}$, and Proposition 31.8.1 puts u in $\mathcal{E}_{\mathrm{env}}(Y,\theta,q)$.

If $D_u(Y) = A$, the positive curvature measures T_u and T_q both dominate D_u and both have total mass A. Hence $T_u = T_q = D_u$. The distribution $u - q$ is harmonic, and therefore constant on the compact connected curve. Thus $u = q + C$ for a constant C, and Proposition 14.4.1 gives $u \in \mathcal{E}_{\mathrm{env}}(Y,\theta,q)$ also in this case. □

Proposition 31.13.4 *For the Fubini–Study form θ of area one on $\mathbb{P}^1$, the pseudometric d_θ^{all} is incomplete on $\mathfrak{S}(\mathbb{P}^1,\theta)$: some Cauchy sequence of ordinary models of one fixed positive mass with trivial multiplier ideals has no limit. On ordinary models, its topology is strictly weaker than that of polar total variation and strictly stronger than $\tau_{\theta,0}$.*

Proof Use the diffuse polar probability measures σ_j concentrating at a point x and the models $p_j = p_{c\sigma_j}$ from the proof of Proposition 31.13.2, with $0 < c < 1$. Fix a smooth obstacle F and put $q_{y,F} = P_{\theta,F}[q_{c\delta_y}]$. Rotations preserving θ and sending x to y can be chosen to depend continuously on y near x; they transport the obstacle F to its composition with the rotation. Since envelopes below two obstacles differ by at most the uniform distance between the obstacles, continuity of rotations in L^1 gives $q_{y,F} \to q_{x,F}$ in L^1 as $y \to x$.

The uniform L^1 bound near x makes the average $w_{j,F} = \int q_{y,F}\,\mathrm{d}\sigma_j(y)$ well defined. It lies below F, its curvature current dominates $c\sigma_j$, and it converges to $q_{x,F}$ in L^1. The description of envelopes below obstacles in Step 1 of the proof of Proposition 31.13.3 gives $w_{j,F} \le P_{\theta,F}[p_j] \le F$. These bounds and normalized compactness give L^1 precompactness. Every cluster point of the envelopes has curvature at least $c\delta_x$, so it is at most $q_{x,F}$, and the lower bound gives the reverse inequality. Hence $P_{\theta,F}[p_j] \to q_{x,F}$ for every F. The ideal envelopes of the p_j below F all equal $P_\theta(F)$, since their multiplier ideals are trivial, and their atomic parts vanish. Thus $[p_j]$ is a d_θ^{all}-Cauchy sequence. Since $F_1 = 0$, a limit would also be a d_θ^{ot}-limit, which Proposition 31.13.2 excludes.

For the strict comparison with polar total variation, use the rotated diffuse models $\widetilde{p}_j = p \circ R_j^{-1}$ from the end of Section 31.13.1. For every smooth F,

$$P_{\theta,F}[\widetilde{p}_j] = P_{\theta,F\circ R_j}[p] \circ R_j^{-1}.$$

The same obstacle bound and the L^1 continuity of rotations give convergence to $P_{\theta,F}[p]$. All atomic parts vanish, so $d_\theta^{\mathrm{all}}([\widetilde{p}_j],[p]) \to 0$, whereas the polar total-variation distance remains $2c$. Proposition 31.13.3 gives the implication in the other direction.

Finally, the sequence in Proposition 17.6.4 converges for the zero obstacle but not for some smooth obstacle. By Step 1 of the proof of Proposition 31.13.3, replacing its members by their ordinary models does not change any obstacle envelope, so d_θ^{all} is strictly stronger than $\tau_{\theta,0}$ already on ordinary models. □

31.14 Background changes on curves

On a curve, the polar curvature separates the singularity data of a model from the area of the background. A smooth gauge change reduces any background of positive area to a positive form. Keeping the polar measure fixed preserves distances whenever both backgrounds admit it, while rescaling by the ratio of the areas identifies the entire model spaces; the gauge function controls the normalized representatives. At zero area the whole model space collapses to the unique normalized potential.

We keep the curve Y of Section 31.13, and we fix a smooth positive $(1,1)$-form η on Y with $\int_Y \eta = 1$, which we use for L^1 norms and capacities. In this section any smooth real $(1,1)$-form σ on Y with $A_\sigma = \int_Y \sigma \geq 0$ is allowed as a background. By Lemma 1.8.11, these are exactly the forms for which $\mathrm{PSH}(Y,\sigma)$ is nonempty, and $\sigma = A_\sigma \eta + \mathrm{dd^c} h$ for a smooth real function h. Let $\mathcal{M}(\sigma)$ be the set of ordinary models in $\mathrm{PSH}(Y,\sigma)$, and put $U_\sigma(p) = \int_Y \mathrm{MA}_\sigma(p)$. The volume comparison d_σ is defined by the formula of Definition 31.6.1; at zero area it vanishes identically. Polar curvatures, their atomic parts and ideal models are defined as before, with σ in place of θ.

Proposition 31.14.1 *Polar curvature identifies $\mathcal{M}(\sigma)$ with the set of polar measures of mass at most A_σ. Write $p_\sigma(\mu)$ for the model with polar measure μ. Then*

$$U_\sigma(p_\sigma(\mu)) = A_\sigma - \mu(Y), \qquad d_\sigma(p_\sigma(\mu), p_\sigma(\nu)) = \|\mu - \nu\|_{\mathrm{TV}},$$

and ideal projection corresponds to taking the atomic part. The space $(\mathcal{M}(\sigma), d_\sigma)$ is complete and geodesic, and convergence for d_σ implies convergence in L^1 and in capacity. If $A_\sigma = 0$, its only member is the smooth potential u_σ of Lemma 1.8.11.

Proof Suppose first that $A = A_\sigma > 0$, and write $\sigma = A\eta + \mathrm{dd^c} h$. For $v \in \mathrm{PSH}(Y, A\eta)$, the inequalities for ordinary envelopes in Proposition 17.3.1 give

$$P_{A\eta}[v] + \min_Y h \leq P_\sigma[v-h] + h \leq P_{A\eta}[v] + \max_Y h.$$

For a model p in background $A\eta$, the potential $q = P_\sigma[p-h]$ therefore satisfies $q + h \sim p$, and since the envelope depends only on the ordinary type, $P_\sigma[q] = P_\sigma[p-h] = q$. Conversely, for $q \in \mathcal{M}(\sigma)$, the same bounds with $v = q+h$ show that

$P_{A\eta}[q+h] \sim q+h$, and taking the envelope back recovers q. Thus $p \mapsto P_\sigma[p-h]$ is a bijection from $\mathcal{M}(A\eta)$ onto $\mathcal{M}(\sigma)$.

Smooth gauge changes preserve curvature currents, and ordinary equivalence preserves their polar parts by Corollary 1.6.1. Consequently the bijection preserves polar curvature. It also preserves singularity volumes, and subtracting the same smooth function commutes with maxima. The classification and the distance formula of Lemma 31.13.1 and Theorem 31.13.1, applied to the positive form $A\eta$, now give the first assertions. The same gauge bounds for ideal envelopes show that the transported ideal projections are ordinarily equivalent. Both are ordinary models by Proposition 17.2.2, and ordinary models of the same type are equal. Thus the bijection commutes with ideal projection, which still takes the atomic part. Completeness and the geodesics follow from the identification with polar measures, as in Corollary 31.13.1.

Since $\sigma + \mathrm{dd}^{\mathrm{c}}(-h) = A\eta > 0$, the form σ is big, and the proof of Theorem 31.13.2 applies with σ in place of θ; it shows that convergence for d_σ implies convergence in L^1 and in capacity.

If $A_\sigma = 0$, Lemma 1.8.11 leaves only u_σ after normalization. Its curvature current vanishes, and both envelope operations fix it. All assertions reduce to statements about the zero measure and the one-point metric space. □

The criterion for equality of the ordinary and ideal envelopes also depends only on the polar data, including at zero area.

Corollary 31.14.1 *For $u \in \mathrm{PSH}(Y,\sigma)$ with polar curvature μ_u and atomic part D_u, put $q = P_\sigma[u]_{\mathcal{I}}$. The following are equivalent:*

$$u \in \mathcal{E}_{\mathrm{env}}(Y,\sigma,q), \qquad \mu_u = D_u,$$
$$P_\sigma[u] = P_\sigma[u]_{\mathcal{I}},$$
$$P_{\sigma,F}[u] = P_{\sigma,F}[u]_{\mathcal{I}} \quad \textit{for every smooth real } F.$$

If $A_\sigma = 0$, all four conditions hold for every $u \in \mathrm{PSH}(Y,\sigma)$.

Proof Suppose first that $A = A_\sigma > 0$, and write $\sigma = A\eta + \mathrm{dd}^{\mathrm{c}}h$. Put $v = u + h$ and $q_+ = P_{A\eta}[v]_{\mathcal{I}}$. The gauge change preserves the curvature current and its pole set, hence both μ_u and D_u. By Proposition 17.3.1, $q + h \sim q_+$. The gauge invariance of the relative envelope class and its invariance under equivalent references (Proposition 14.4.1) give

$$u \in \mathcal{E}_{\mathrm{env}}(Y,\sigma,q) \quad \Longleftrightarrow \quad v \in \mathcal{E}_{\mathrm{env}}(Y,A\eta,q_+).$$

By Corollary 31.13.2, this is equivalent to $\mu_u = D_u$.

The gauge bounds for the ordinary and ideal envelopes, Corollary 1.6.1 and the classifications in the positive background $A\eta$ identify the polar curvatures of $P_\sigma[u]$ and $P_\sigma[u]_{\mathcal{I}}$ with μ_u and D_u, respectively. Both envelopes are ordinary models, the ideal one by Proposition 17.2.2. The classification of models in Proposition 31.14.1 therefore shows that they are equal if and only if $\mu_u = D_u$.

For a smooth real F, the bijection $a \mapsto a + h$ between competitors gives

$$P_{\sigma,F}[u] + h = P_{A\eta,F+h}[v],$$

and the same identity for the ideal envelopes. Indeed, this bijection preserves the obstacle bound, the ordinary order and multiplier-ideal equivalence, and addition of h commutes with upper regularization. Since $F \mapsto F + h$ is a bijection of the smooth obstacles, Corollary 31.13.2 gives the remaining equivalence.

If $A_\sigma = 0$, Lemma 1.8.11 gives $u = -h + c$ for a constant c, and the curvature current of u vanishes. Thus $\mu_u = D_u = 0$. All potentials in this background are smooth and ordinarily equivalent, so their ordinary and ideal competitor families below every obstacle coincide, and they are nonempty after a constant translation. The reference q is ordinarily equivalent to u, and Proposition 14.4.1 gives $u \in \mathcal{E}_{\mathrm{env}}(Y, \sigma, q)$. □

Theorem 31.14.1 *Let θ_0, θ_1 be smooth real $(1,1)$-forms on Y with positive areas $A_i = A_{\theta_i}$, and put $\lambda = A_1/A_0$. The map*

$$F_{10}\colon \mathcal{M}(\theta_0) \longrightarrow \mathcal{M}(\theta_1), \qquad F_{10}(p_{\theta_0}(\mu)) = p_{\theta_1}(\lambda\mu),$$

is a bijection and satisfies

$$\begin{aligned} d_{\theta_1}(F_{10}(p), F_{10}(q)) &= \lambda d_{\theta_0}(p, q), \\ U_{\theta_1}(F_{10}(p)) &= \lambda U_{\theta_0}(p), \\ P_{\theta_1}[F_{10}(p)]_{\mathcal{I}} &= F_{10}(P_{\theta_0}[p]_{\mathcal{I}}). \end{aligned} \tag{31.31}$$

For a third smooth background θ_2 of positive area, the same construction gives $F_{21}F_{10} = F_{20}$.

Proof Multiplication by λ is a bijection from the polar measures of mass at most A_0 onto those of mass at most A_1. By Proposition 31.14.1, it gives the asserted bijection, and the distances satisfy

$$\|\lambda\mu - \lambda\nu\|_{\mathrm{TV}} = \lambda\|\mu - \nu\|_{\mathrm{TV}}.$$

The volume identity follows from $A_1 - \lambda\mu(Y) = \lambda(A_0 - \mu(Y))$. Taking the atomic part commutes with multiplication by λ, which gives the ideal-envelope identity. The area ratios multiply, proving the composition rule. □

Thus, for $A_\sigma > 0$, the rescaled distances d_σ/A_σ make all these model spaces isometric. The unscaled spaces have diameter $2A_\sigma$: this is an upper bound for the total variation of the difference of two positive measures of mass at most A_σ, and the models of $A_\sigma\delta_x$ and $A_\sigma\delta_y$, $x \neq y$, attain it. Consequently the unscaled spaces can be isometric only when their areas agree.

Alternatively, $p_{\theta_0}(\mu) \mapsto p_{\theta_1}(\mu)$ is an isometry on the polar measures with $\mu(Y) \leq \min(A_0, A_1)$. The two models have the same ordinary type. Indeed, the Green representatives have the same polar term, so their difference has smooth dd^{c} and is smooth by elliptic regularity, while Lemma 31.13.1 and Proposition 31.14.1

show that each model differs from its Green representative, corrected by the smooth gauge function, by a bounded function. Under this correspondence the non-pluripolar volume changes by $A_1 - A_0$.

The rescaling map preserves analytic models when λ is rational, by the convention of Section 1.13: on a curve, the polar measures of analytic models are finite atomic measures with rational coefficients. If λ is irrational, an atom $a\delta_x$ with rational $0 < a < A_0$ acquires an irrational Lelong number, so its image is not analytic in this convention. Ideal models are preserved for every λ, by (31.31).

Proposition 31.14.2 *Choose a smooth real h with $\theta_1 = \lambda\theta_0 + \mathrm{dd}^{\mathrm{c}} h$. For $p \in \mathcal{M}(\theta_0)$ and $q = F_{10}(p)$,*

$$\begin{gathered} q = P_{\theta_1}[\lambda p - h], \\ \lambda p + \min_Y h \le q + h \le \lambda p + \max_Y h. \end{gathered} \tag{31.32}$$

If $U_{\theta_0}(p) = 0$, then

$$q = \lambda p - h - \sup_Y(\lambda p - h).$$

Proof The form $\theta_1 - \lambda\theta_0$ has integral zero, so the scalar Poisson equation gives h, as in Lemma 1.8.11. Write $p = p_{\theta_0}(\mu)$. Scaling the competitors in the ordinary envelope shows that λp is an ordinary model for $\lambda\theta_0$. The gauge construction in the proof of Proposition 31.14.1 therefore makes $P_{\theta_1}[\lambda p - h]$ an ordinary model in background θ_1. The curvature current of $\lambda p - h$ in this background is $\lambda(\theta_0 + \mathrm{dd}^{\mathrm{c}} p)$, so its polar measure is $\lambda\mu$. The construction preserves this measure, which gives the first identity in (31.32).

The function $\lambda p - h + \min_Y h$ is a competitor for q, which gives the lower bound. If $v \le 0$ is any competitor for q, then $v + h - \max_Y h$ is $\lambda\theta_0$-plurisubharmonic, nonpositive and more singular than λp, so it is at most λp. Taking the upper envelope gives the upper bound. These inequalities hold also at the poles, and off the common pole set they imply $|q - \lambda p| \le \operatorname{osc}_Y h$.

At zero mass, the polar measures exhaust the curvature currents of both p and q. Hence $\mathrm{dd}^{\mathrm{c}}(q - \lambda p + h) = 0$, and the uniqueness of harmonic functions together with the normalization $\sup_Y q = 0$ gives the last formula. □

Corollary 31.14.2 *Suppose that smooth real $(1,1)$-forms θ_j on Y of positive area converge uniformly to a smooth form θ_0 of positive area. Put $\lambda_j = A_{\theta_j}/A_{\theta_0}$, and let $F_{j0}\colon \mathcal{M}(\theta_0) \to \mathcal{M}(\theta_j)$ be the correspondence of Theorem 31.14.1. For every $t > 0$,*

$$\begin{aligned} \sup_{p \in \mathcal{M}(\theta_0)} \|F_{j0}(p) - p\|_{L^1(Y)} &\longrightarrow 0, \\ \sup_{p \in \mathcal{M}(\theta_0)} \operatorname{Cap}_\eta(\{|F_{j0}(p) - p| > t\}) &\longrightarrow 0. \end{aligned} \tag{31.33}$$

More generally, let $q_j \in \mathcal{M}(\theta_j)$ have polar measures ν_j, and let $p = p_{\theta_0}(\mu)$. If $\lambda_j^{-1}\nu_j \to \mu$ in total variation, then $q_j \to p$ in L^1 and in capacity.

Proof The inverse of the scalar Laplacian for η on functions of mean zero has a symmetric Green kernel G. Set

$$h_j(x) = \int_Y G(x,y)\,(\theta_j - \lambda_j\theta_0)(y).$$

The integrand form has integral zero, so $\mathrm{dd}^{\mathrm{c}} h_j = \theta_j - \lambda_j\theta_0$, and elliptic regularity makes h_j smooth. Symmetry and the uniform L^1 bound from Section 20.3 give

$$\|h_j\|_{L^\infty(Y)} \le C \left\| \frac{\theta_j - \lambda_j\theta_0}{\eta} \right\|_{L^\infty(Y)} \longrightarrow 0.$$

Here C depends only on η; the symmetric kernel may be made nonpositive by subtracting a constant, which does not change h_j.

By (31.32), applied with $h = h_j$, we have outside the common poles

$$|F_{j0}(p) - p| \le \operatorname{osc}_Y h_j + |\lambda_j - 1|\,|p|.$$

Normalized compactness bounds $\|p\|_{L^1}$ uniformly over $\mathcal{M}(\theta_0)$, which proves the first limit. Choose $C_0 > 0$ with $\theta_0 \le C_0\eta$. Then p/C_0 is η-plurisubharmonic with supremum zero, so the uniform sublevel estimate (1.15) bounds the capacity of $\{|\lambda_j - 1|\,|p| > t/2\}$, uniformly in p; since pole sets have zero capacity, this proves the second limit.

For the last assertion put $p_j = F_{0j}(q_j)$, so that $q_j = F_{j0}(p_j)$. The polar measure of p_j is $\lambda_j^{-1}\nu_j$, so Proposition 31.14.1 gives $p_j \to p$ in L^1 and in capacity. The two uniform limits in (31.33), applied to p_j, give $q_j - p_j \to 0$ in the same senses. The triangle inequality in L^1 and the subadditivity of capacity complete the proof. □

Finally, suppose that $\theta_j \to \theta_0$ uniformly, where $A_{\theta_j} \ge 0$ and $A_{\theta_0} = 0$. The limiting model space is $\{u_{\theta_0}\}$, and Theorem 1.8.1 gives L^1 and capacity convergence to u_{θ_0}, uniformly over all normalized potentials and hence over all models. The model spaces also collapse for their unscaled metrics, since their diameters are $2A_{\theta_j} \to 0$. On the other hand, for $A_{\theta_j} > 0$ the rescaled metric $d_{\theta_j}/A_{\theta_j}$ makes $\mathcal{M}(\theta_j)$ isometric to $\mathcal{M}(\eta)$, by Theorem 31.14.1. Thus the rescaled metric keeps singularity data that disappear in the limiting potentials.

31.14.1 Models pulled back by a finite curve map

Let $r\colon C' \to C$ be a finite surjective holomorphic map of degree d between connected compact Riemann surfaces. Fix a smooth real form β of area $A \geq 0$ on C and put $\sigma = r^*\beta$. The curve results of Section 31.13 and of this section apply on C and on C' in place of Y. The models pulled back from C occupy a proper subspace of the full model space on C' in general. The fiber maximum of Lemma 1.22.6 compares envelope competitors on the two curves. The weighted measure pullback then identifies the metric on this subspace and gives a retraction from the full space.

Proposition 31.14.3 *For $v \in \mathrm{PSH}(C,\beta)$ and every smooth real obstacle F on C,*

$$P_{\sigma,F\circ r}[v\circ r] = P_{\beta,F}[v]\circ r,$$
$$P_{\sigma,F\circ r}[v\circ r]_{\mathcal{I}} = P_{\beta,F}[v]_{\mathcal{I}}\circ r.$$

In particular, $J(v) = v\circ r$ embeds the ordinary model space $\mathcal{M}(\beta)$ into $\mathcal{M}(\sigma)$ and preserves ideal projection.

Proof Pullback sends every base competitor to an upstairs competitor: it preserves positivity, the obstacle inequality and the ordinary order. It also preserves and reflects the all-weight ideal order, because on a curve this order consists of the pointwise Lelong inequalities, as in the proof of Lemma 1.22.6, and $\nu(w\circ r,x) = e_x\nu(w,r(x))$ by Lemma 1.22.3, where e_x is the local degree of r at x. Conversely, let h be an upstairs competitor, with fiber maximum m_h. Then $h \leq m_h\circ r$, and $h \leq F\circ r$ implies $m_h \leq F$. By Lemma 1.22.6, $h \preceq v\circ r$ implies $m_h \preceq v$, and the same holds for the ideal order. For an ideal competitor, moreover, $v\circ r \preceq_{\mathcal{I}} h \preceq_{\mathcal{I}} m_h\circ r$, and reflection gives $v \preceq_{\mathcal{I}} m_h$. Thus m_h is a base competitor in both cases, and the two unregularized suprema are equal after pullback. A finite nonconstant holomorphic map of curves is open; upper regularization commutes with its pullback, as is seen in the coordinates $t = z^e$. This proves both identities. Take $F = 0$. An ordinary model is fixed by its ordinary envelope, so its pullback is a model; surjectivity of r gives injectivity. Applying the ideal identity to a model proves preservation of ideal projection. □

Let d_β and d_σ be the polar metrics from Proposition 31.14.1. If q is a model on C', denote its polar curvature by $\mu_q = \mathbf{1}_{\{q=-\infty\}}(\sigma + \mathrm{dd}^{\mathrm{c}}q)$. Write $p_\beta(\nu)$ for the base model with polar measure ν, using the same classification. Write $U_\beta(v) = \int_C \mathrm{MA}_\beta(v)$ and $U_\sigma(q) = \int_{C'} \mathrm{MA}_\sigma(q)$ for their non-pluripolar volumes.

Proposition 31.14.4 *For $u, v \in \mathcal{M}(\beta)$,*

$$d_\sigma(J(u), J(v)) = d\, d_\beta(u, v), \qquad U_\sigma(J(u)) = d\, U_\beta(u).$$

The image $J(\mathcal{M}(\beta))$ is closed and contains the linear polar-measure geodesic between each pair of its members. The map defined for $q \in \mathcal{M}(\sigma)$ by

$$R(q) = J\Big(p_\beta\Big(d^{-1} r_* \mu_q\Big)\Big)$$

is a 1-Lipschitz retraction onto this image. Its polar measure is

$$\mu_{R(q)} = d^{-1} r^* r_* \mu_q.$$

It preserves the polar mass and the non-pluripolar volume, and commutes with ideal projection on $\mathcal{M}(\sigma)$. At $A = 0$ both model spaces are smooth singletons.

Proof Change of variables off the finite branch sets gives $\int_{C'} \sigma = dA$. By Lemma 1.22.7, the polar measure of $J(v)$ is $r^* \mu_v$, with total mass $d\mu_v(C)$. The curve classification gives

$$d_\sigma(J(u), J(v)) = \|r^*(\mu_u - \mu_v)\|_{\mathrm{TV}} = d\|\mu_u - \mu_v\|_{\mathrm{TV}},$$
$$U_\sigma(J(v)) = dA - d\mu_v(C) = d\, U_\beta(v).$$

Completeness of the base model space proves closedness of the image: a convergent sequence in the image comes from a Cauchy sequence downstairs. Linear interpolation of polar measures commutes with r^*, giving the stated geodesics.

For any $q \in \mathcal{M}(\sigma)$, its polar measure has mass at most dA. Its pushforward is carried by the image of a polar carrier, which is polar by the image assertion of Lemma 1.22.7, so $d^{-1} r_* \mu_q$ has mass at most A and defines a base model. This proves that R is well defined and gives its displayed polar measure. The identity $r_* r^* = d$ makes $R \circ J = J$. For any finite signed measure τ on C', pushforward does not increase total variation, hence

$$\|d^{-1} r^* r_* \tau\|_{\mathrm{TV}} = \|r_* \tau\|_{\mathrm{TV}} \le \|\tau\|_{\mathrm{TV}}.$$

Apply this to the difference of two polar measures to obtain the Lipschitz estimate. The averaging operator $d^{-1} r^* r_*$ preserves the total mass of positive measures, so the formula $U_\sigma(q) = dA - \mu_q(C')$ gives the preservation of polar mass and volume. Both measure operations commute with taking the atomic part, which is ideal projection by the curve classification. This proves the commutation assertion. At zero area all polar measures vanish and both spaces are singletons. □

This retraction averages the singularity data over a fiber. Unlike the ideal retraction, it need not select a nearest model.

Example 31.14.1 For $r\colon \mathbb{P}^1 \to \mathbb{P}^1$, $r(z) = z^3$, and the Fubini–Study background β of area one, the retraction R above is not a nearest-point retraction, even on ideal models with analytic singularities.

Proof Put $\sigma = r^*\beta$, of area three, and let q be its model with polar measure δ_1. This is an analytic ideal model: in the affine coordinate, the global function

$$h(z) = \log|z-1|^2 - \tfrac{1}{3}\log(1+|z|^6)$$

extends smoothly across infinity and satisfies $\sigma + \mathrm{dd^c} h = \delta_1 + \frac{2}{3}\sigma$. Thus h is σ-plurisubharmonic with one logarithmic pole, and the curve classification identifies its envelope with q; Proposition 17.2.4 preserves its analytic type. Writing $\zeta = \exp(2\pi\mathrm{i}/3)$, one has

$$\mu_{R(q)} = \tfrac{1}{3}(\delta_1 + \delta_\zeta + \delta_{\zeta^2}).$$

The three atoms are distinct, so

$$d_\sigma(q, R(q)) = \tfrac{2}{3} + \tfrac{1}{3} + \tfrac{1}{3} = \tfrac{4}{3}.$$

The zero model is $J(0)$ and has zero polar measure, giving $d_\sigma(q, J(0)) = 1$. Hence the averaged model is strictly farther from q than a member of the same pullback subspace. □

31.15 A complete space over varying curve backgrounds

The polar description also compares models in different backgrounds. Using the background as an additional coordinate gives a complete space that includes the zero-area boundary. The singularity coordinate still records the diffuse polar curvature, and taking its atomic part still gives the nearest ideal model.

We keep the compact connected curve Y and the smooth positive form η of area one. Let

$$\mathcal{B}(Y) = \{\sigma \in C^\infty(Y, \Lambda^{1,1}_{\mathbb{R}}) : A_\sigma = \textstyle\int_Y \sigma \geq 0\}$$

be the set of admissible backgrounds. Choose C^k norms on smooth forms, defined by a fixed smooth metric on Y, and put

$$b(\sigma, \tau) = \sum_{k=0}^{\infty} 2^{-k-1} \min\{1, \|\sigma - \tau\|_{C^k}\}. \tag{31.34}$$

This is a complete metric for the C^∞ topology on $\mathcal{B}(Y)$. Indeed, a b-Cauchy sequence is Cauchy in every C^k norm; the limits agree and define a smooth form, whose area is nonnegative by C^0 convergence. Conversely, convergence in every C^k norm gives convergence for b: a finite part of the sum becomes small, and the tail is uniformly small.

Let $\mathcal{M}(\sigma)$ be the space of ordinary models of Proposition 31.14.1, and define

$$\mathscr{M}(Y) = \{(\sigma, p) : \sigma \in \mathcal{B}(Y), p \in \mathcal{M}(\sigma)\}.$$

For such a pair write $\mu_p^\sigma = \mathbf{1}_{\{p=-\infty\}}(\sigma + \mathrm{dd}^c p)$, and set

$$D((\sigma, p), (\tau, q)) = b(\sigma, \tau) + \|\mu_p^\sigma - \mu_q^\tau\|_{\mathrm{TV}}. \tag{31.35}$$

Let $\mathscr{I}(Y) \subset \mathscr{M}(Y)$ consist of the pairs (σ, p) with $P_\sigma[p]_{\mathcal{I}} = p$, and let $\mathscr{A}_+(Y)$ consist of the pairs (σ, p) for which p has analytic singularities and positive non-pluripolar mass in background σ. For a finite measure μ on Y, write $\mu = \mu^{\mathrm{at}} + \mu^{\mathrm{na}}$ for its decomposition into an atomic and a nonatomic part.

Theorem 31.15.1 *The metric space $(\mathscr{M}(Y), D)$ is complete. Its restriction to each fixed-background fiber is the polar metric d_σ. The map*

$$R(\sigma, p) = (\sigma, P_\sigma[p]_{\mathcal{I}})$$

is a 1-Lipschitz retraction onto the closed subspace $\mathscr{I}(Y)$, and $R(\sigma, p)$ is the unique ideal pair nearest to (σ, p). Moreover,

$$\overline{\mathscr{A}_+(Y)}^{\,D} = \mathscr{I}(Y), \qquad \operatorname{dist}_D((\sigma, p), \mathscr{I}(Y)) = (\mu_p^\sigma)^{\mathrm{na}}(Y). \tag{31.36}$$

If $(\sigma_j, p_j) \to (\sigma, p)$ for D, then $p_j \to p$ in $L^1(Y, \eta)$ and in capacity for η, including when $A_\sigma = 0$.

Proof By Proposition 31.14.1, polar curvature identifies $\mathscr{M}(Y)$ with the set of pairs (σ, μ) in which $\sigma \in \mathcal{B}(Y)$ and μ is a polar measure of mass at most A_σ. In these coordinates, (31.35) is the restriction of the product metric $b + \|\cdot\|_{\mathrm{TV}}$, so it separates pairs and restricts to d_σ on every fiber.

Suppose that (σ_j, μ_j) is a Cauchy sequence. Completeness of b gives $\sigma_j \to \sigma$ in the C^∞ topology, and completeness of the space of finite signed measures gives $\mu_j \to \mu$ in total variation. The limit μ is positive, satisfies $\mu(Y) \le A_\sigma$, and is a polar measure, as in the proof of Corollary 31.13.1. Thus it defines a model in background σ, which proves completeness.

Ideal projection takes the atomic part, by Proposition 31.14.1. Taking the atomic part does not increase total variation, so R is 1-Lipschitz. The positive atomic measures are closed in total variation, as in the proof of Corollary 31.13.1, so $\mathscr{I}(Y)$ is closed. Let μ be the polar measure of p. For every ideal pair (τ, q), whose polar measure ν is atomic, mutual singularity gives

$$D((\sigma, p), (\tau, q)) = b(\sigma, \tau) + \mu^{\mathrm{na}}(Y) + \|\mu^{\mathrm{at}} - \nu\|_{\mathrm{TV}}.$$

Its unique minimum is attained at $\tau = \sigma$ and $\nu = \mu^{\mathrm{at}}$, that is, at $R(\sigma, p)$. This proves the retraction and distance assertions.

For the analytic closure, let (σ, q) be an ideal pair. If $A_\sigma > 0$, finite atomic measures with rational coefficients and total mass strictly less than A_σ approximate

every positive atomic measure of mass at most A_σ in total variation. Their models in background σ are analytic, by the classification on curves and the gauge transport of Proposition 31.14.1, and they have positive non-pluripolar mass. If $A_\sigma = 0$, the polar measure of q is zero. The pairs with backgrounds $\sigma + j^{-1}\eta$ and zero polar measure converge to (σ, q) for D. Their models are bounded, so they have analytic singularities with trivial ideal, and their non-pluripolar masses are $j^{-1} > 0$. Hence every ideal pair lies in the closure of $\mathscr{A}_+(Y)$. Conversely, an analytic model is an ideal model by Proposition 17.2.4, and $\mathscr{I}(Y)$ is closed. This proves (31.36).

Finally, let $(\sigma_j, p_j) \to (\sigma, p)$, and write $\mu_j = \mu_{p_j}^{\sigma_j}$ and $\mu = \mu_p^\sigma$. If $A_\sigma > 0$, put $\lambda_j = A_{\sigma_j}/A_\sigma$. For all sufficiently large j these numbers are positive and tend to one, and

$$\|\lambda_j^{-1}\mu_j - \mu\|_{\mathrm{TV}} \le \lambda_j^{-1}\|\mu_j - \mu\|_{\mathrm{TV}} + |\lambda_j^{-1} - 1|\,\mu(Y) \longrightarrow 0.$$

Now apply Corollary 31.14.2. If $A_\sigma = 0$, the backgrounds converge uniformly and all areas are nonnegative, and the uniform collapse of Theorem 1.8.1 gives L^1 and capacity convergence to the unique normalized potential $u_\sigma = p$ in background σ.□

The C^∞ topology on backgrounds is needed for completeness: a sequence of smooth forms that is Cauchy for the uniform norm can have a limit that is not smooth. On a fixed background of positive area, D induces on models of positive mass the topology of Theorem 31.12.1, but it allows sequences to converge to models of zero mass. In particular, the Cauchy sequences for D need not satisfy the condition on reciprocal volumes in that theorem.

Smooth gauge changes and positive scaling act compatibly on all fibers of the joint space.

Corollary 31.15.1 *For $f \in C^\infty(Y, \mathbb{R})$, the map*

$$\mathcal{G}_f(\sigma, p) = (\sigma + \mathrm{dd^c} f, P_{\sigma + \mathrm{dd^c} f}[p - f])$$

is an isometry of $\mathscr{M}(Y)$, with inverse $\mathcal{G}_{-f}$ and $\mathcal{G}_g\mathcal{G}_f = \mathcal{G}_{f+g}$. It preserves analytic models and commutes with R.

For $\lambda > 0$, the bijection $\mathcal{H}_\lambda(\sigma, p) = (\lambda\sigma, \lambda p)$ commutes with R and satisfies

$$\min\{1, \lambda\} D(x, y) \le D(\mathcal{H}_\lambda x, \mathcal{H}_\lambda y) \le \max\{1, \lambda\} D(x, y).$$

Proof The gauge transport in Proposition 31.14.1 preserves the polar measure and, after subtraction of f, the ordinary type. In the coordinates (σ, μ) of the proof of Theorem 31.15.1, $\mathcal{G}_f$ is therefore the map $(\sigma, \mu) \mapsto (\sigma + \mathrm{dd^c} f, \mu)$. It does not change the area, and $b(\sigma + \mathrm{dd^c} f, \tau + \mathrm{dd^c} f) = b(\sigma, \tau)$. This proves that $\mathcal{G}_f$ is an isometry and the laws for inverses and compositions. Taking the atomic part commutes with this map, and subtraction of a smooth function and bounded changes preserve analytic singularities.

Scaling the competitors of the ordinary envelope shows that λp is an ordinary model for $\lambda\sigma$. In the coordinates (σ, μ), $\mathcal{H}_\lambda$ is the map $(\sigma, \mu) \mapsto (\lambda\sigma, \lambda\mu)$, with inverse $\mathcal{H}_{\lambda^{-1}}$, and taking the atomic part commutes with scaling. For $r \ge 0$,

$$\min\{1, \lambda\} \min\{1, r\} \le \min\{1, \lambda r\} \le \max\{1, \lambda\} \min\{1, r\}.$$

Apply this to every C^k norm in (31.34); the total-variation term is multiplied by λ. The two bounds follow. $\square$

Chapter 32
Polar curvature and zero-volume models

The preceding chapters give complete topologies on ordinary models in two settings. On a curve, the total variation of polar curvature is a complete metric on ordinary models, ideal projection keeps the atomic part of the polar curvature, and the distance to the closure of the analytic models is exactly the diffuse polar mass, by Corollary 31.13.1 and Proposition 31.13.1. This topology is stronger than the topology $\tau_{\omega,0}$ of Section 17.6 on ordinary models: Proposition 31.13.2 exhibits a concentration of diffuse polar curvature that is lost in the latter topology. Adding the smooth curve background as a further coordinate gives the complete space of Theorem 31.15.1. Its analytic closure is the locus of ideal models, also over backgrounds of zero area, and metric convergence controls the normalized potentials in L^1 and in capacity. This construction keeps the positive-volume topology but imposes a different Cauchy condition at the zero-volume boundary. In higher dimension, for a smooth big background with bounded mass, Theorem 31.12.1 gives a complete topology on ordinary models of positive lower volume. Its locus of ideal models is the completion of the analytic models of positive volume, by Corollary 31.12.1.

This chapter studies ordinary models in higher dimension outside the positive-volume theory. If a potential is the sum of a logarithmic potential with fixed divisorial poles and the pullback of a potential from the base of a submersion, both its envelopes reduce to envelopes on the base, and every representative of the induced ordinary types has zero top-degree mass. Equality of the two envelopes below every smooth obstacle, the question of Section 17.6.4, then reduces to the base as well. On products with curves, these zero-volume models keep information that top-degree mass cannot detect. A fixed divisor with unit coefficient forces every ordinary representative of the induced models to have zero mass, and the volume comparison vanishes on them. Polar curvature on the curve factors nevertheless distinguishes these models and gives a complete metric whose analytic closure is exactly the locus of ideal models, with ideal projection as the unique nearest-point retraction. These conclusions hold on the product loci of Propositions 32.2.2 and 32.3.7, including the nonclosed background with unbounded ambient mass in Example 32.3.1.

Changing the curve areas rescales the polar distances, and a zero area removes the corresponding coordinate. The comparisons in Proposition 32.3.4 quantify this

change even when different sets of coordinates survive. If the curve backgrounds converge uniformly to forms of area zero, the diameters tend to zero and the potentials converge in L^1, uniformly over the model tuples, and the limit still records the normalized smooth potentials of the limiting backgrounds, by Corollary 32.3.2. These loci thus describe explicit degenerations beyond the positive-volume theory.

A curve quotient can also describe the entire model space. If $p\colon X \to C$ is a surjective holomorphic map with connected fibers, including critical fibers, and β is a smooth real $(1,1)$-form of nonnegative area on the compact curve C, every potential in the background $\theta = p^*\beta$ descends to C. For $\dim X \geq 2$ all top-degree masses vanish, but the polar trace, normalized by the constant Gauduchon volume of the fiber divisor with its multiplicities, gives the complete curve metric on all ordinary θ-models by Proposition 32.4.1. The ideal models form the closure of the analytic models, and Corollary 32.4.1 characterizes equality of the ordinary and ideal envelopes below every smooth obstacle. For an arbitrary surjective map to a compact curve, Proposition 1.22.3 gives the finite factor used in Proposition 32.4.2, so disconnected fibers have the same complete model-space description. This description holds even on a threefold without the bounded mass property, by Example 32.4.2. The polar trace must be kept upstairs: in Example 32.4.1, two distinct analytic models have the same trace pushforward to the original base curve. The last section shows that polar curvature on all modifications does not extend these descriptions to general ordinary models in higher dimension.

Throughout the chapter, compact complex manifolds are connected, and the bounded mass property is not assumed; mass bounds, expressed through the lower and upper masses $v_\pm$ of Definition 7.5.1, are imposed only where they are stated. We keep the notation of Chapter 17. For a smooth real $(1,1)$-form θ and $u \in \mathrm{PSH}(X,\theta)$, $\mathrm{MA}_\theta(u)$ is the extended non-pluripolar measure of Theorem 7.1.1, whose total mass may be infinite; $\preceq$, $\sim$, $\preceq_{\mathcal{I}}$ and $\sim_{\mathcal{I}}$ are the ordinary and multiplier-ideal orders; and $P_{\theta,F}[u]$ and $P_{\theta,F}[u]_{\mathcal{I}}$ are the ordinary and ideal envelopes below a smooth or continuous obstacle F (Definitions 1.19.1 and 17.2.1 and Section 17.6.3), written $P_\theta[u]$ and $P_\theta[u]_{\mathcal{I}}$ when $F = 0$. A potential is normalized if its supremum is zero, and ordinary and ideal models are the fixed points of the two envelopes. We also use the singularity volumes of Definition 14.3.1, the relative envelope classes of Definition 14.4.1, and the volume comparison d_θ and the sets $\mathcal{M}_\delta(\theta)$ of Definition 31.6.1, where the comparison is used only when the upper volumes involved are finite. On curves we use the polar curvature of Section 1.6.1 and the classification of models in Sections 31.13 and 31.14.

32.1 Relative divisors and fiber obstacles

To test equality of the envelopes below all obstacles beyond curves, we need potentials whose envelopes can be computed. Consider a holomorphic submersion with compact connected fibers, and a potential that is the sum of a logarithmic potential Ψ with fixed weights along some divisors and the pullback of a potential w on the base.

Every competitor has at least the poles of Ψ, and removing them leaves a current pulled back from the base. Since the fibers are compact and connected, the remaining potential descends to the base, and minimizing the obstacle over the fibers reduces both envelopes to envelopes on the base. Over a fiber contained in the divisor the induced obstacle is infinite, but normalized compactness still bounds the competitors. Because the remaining current comes from a base of smaller dimension, all these potentials have zero top-degree mass. The reduction gives criteria on the base for membership in relative envelope classes and for equality of the envelopes below all obstacles. The divisors may be singular or ramified over the base; only the explicit product formula for the multiplier ideals requires relative normal crossings.

Let $p\colon M \to Y$ be a surjective holomorphic submersion between connected compact complex manifolds, with connected fibers, and put $d = \dim Y$ and $n = \dim M$, where $1 \le d < n$. Let $H_1, \dots, H_s$ be a finite family of distinct prime divisors on M and put $H = \bigcup_j H_j$. For positive real weights λ_j, choose smooth Hermitian metrics h_j on $\mathcal{O}_M(H_j)$, with canonical sections s_j and Chern curvatures κ_j. Set

$$A = \sum_{j=1}^{s} \lambda_j H_j, \qquad \Psi = \sum_{j=1}^{s} \lambda_j \log |s_j|^2_{h_j}, \qquad \kappa_A = \sum_{j=1}^{s} \lambda_j \kappa_j.$$

Thus $\kappa_A + \mathrm{dd}^{\mathrm{c}}\Psi = [A] = \sum_j \lambda_j [H_j]$. For a smooth real $(1,1)$-form β on Y and $w \in \mathrm{PSH}(Y, \beta)$, put

$$\Theta = p^*\beta + \kappa_A, \qquad u = \Psi + w \circ p.$$

The form β need not be closed or positive. Envelopes below continuous real obstacles are defined by the same competitor families as for smooth obstacles. Let

$$Z = \{y \in Y : p^{-1}(y) \subset H\}$$

be the set of points over which the whole fiber lies in H.

Proposition 32.1.1 *For every $F \in C(M, \mathbb{R})$, the function*

$$\widehat{F}(y) = \inf_{x \in p^{-1}(y) \setminus H} \{F(x) - \Psi(x)\}, \qquad \inf \varnothing = +\infty,$$

is continuous in the extended real sense, finite on $Y \setminus Z$ and equal to $+\infty$ on Z. Defining the base envelopes below this extended-real obstacle by the usual competitor families, one has

$$\begin{aligned} P_{\Theta,F}[u] &= \Psi + P_{\beta,\widehat{F}}[w] \circ p, \\ P_{\Theta,F}[u]_I &= \Psi + P_{\beta,\widehat{F}}[w]_I \circ p. \end{aligned} \tag{32.1}$$

For $F = 0$ and $L = -\sup_M \Psi$, there is a constant K such that

$$\begin{aligned} P_\beta[w] + L &\le P_{\beta,\widehat{0}}[w] \le P_\beta[w] + K, \\ P_\beta[w]_I + L &\le P_{\beta,\widehat{0}}[w]_I \le P_\beta[w]_I + K. \end{aligned} \tag{32.2}$$

Every representative of the ordinary type of u or of $P_\Theta[u]_I$ has zero extended mass. Consequently both singularity volumes of these types are zero.

If the ordinary and ideal envelopes of w agree below every smooth obstacle on Y, then those of u agree below every smooth obstacle on M.

Proof We first study the induced obstacle and bound the base competitors, then describe all competitors upstairs by divisorial removal and descent, and finally prove the mass assertion.

Step 1: the induced obstacle. Since p is open, $Y \setminus Z = p(M \setminus H)$ is open and nonempty. The function $F - \Psi$ is bounded below by $\min_M F - \sup_M \Psi$ and tends to $+\infty$ near H, because $\Psi \to -\infty$ there. On every fiber that meets $M \setminus H$, its infimum is therefore attained outside H. A local holomorphic section through a minimizer gives upper semicontinuity of $\widehat{F}$. Conversely, let $y_j \to y$ with bounded fiber minima. By compactness, a subsequence of the minimizers converges, and its limit lies outside H because $F - \Psi$ remains bounded along it; this gives lower semicontinuity. If $y \in Z$, the same limit would lie in H, a contradiction. Thus $\widehat{F} \to +\infty$ at Z, and $\widehat{F}$ is continuous in the extended sense.

Step 2: a uniform bound for the base competitors. Choose a relatively compact nonempty ball $U \subset Y \setminus Z$ and a holomorphic section s of p on a neighborhood of $\overline{U}$ with image outside H; such a pair exists because local sections pass through every point of the open set $M \setminus H$. There is a finite C_U with $\widehat{F} \le F \circ s - \Psi \circ s \le C_U$ on U. Fix a smooth probability volume form $\mathrm{d}V$ on Y. For every $b \in \mathrm{PSH}(Y, \beta)$ below $\widehat{F}$, normalized compactness in Theorem 1.4.1 gives

$$-C_\beta \le \int_U (b - \sup_Y b)\, \mathrm{d}V \le (C_U - \sup_Y b) \int_U \mathrm{d}V.$$

Hence every such b satisfies $\sup_Y b \le K_F = C_U + C_\beta / \int_U \mathrm{d}V$. A sufficiently low translate of w belongs to both base competitor families. Their regularized suprema

are therefore β-plurisubharmonic functions, not identically $-\infty$, and by the extended continuity of $\widehat{F}$ they still lie below $\widehat{F}$. For $F = 0$, one has $\widehat{0} \geq L$, so translates by L of the competitors below zero are competitors below $\widehat{0}$; conversely, every competitor b below $\widehat{0}$ gives the competitor $b - K_0$ below zero. This proves (32.2) with $K = K_0$.

Step 3: the competitors upstairs. The pullback of w is quasi-plurisubharmonic, so its generic Lelong numbers are nonnegative, and $\nu(u, H_j) = \lambda_j + \nu(w \circ p, H_j) \geq \lambda_j$. Let v be an ordinary competitor for u, or an ideal competitor with $v \sim_{\mathcal{I}} u$. By the ordinary order or by Theorem 1.16.1, its generic Lelong number along H_j is at least λ_j. Divisorial removal in Lemma 1.17.1 gives

$$\Theta + \mathrm{dd}^{\mathrm{c}} v - [A] \geq 0.$$

Hence $v - \Psi$ has a quasi-plurisubharmonic representative a satisfying $p^*\beta + \mathrm{dd}^{\mathrm{c}} a \geq 0$. Descent along the compact connected fibers, Proposition 1.22.1, gives $a = b \circ p$ for a unique $b \in \mathrm{PSH}(Y, \beta)$, and $v = \Psi + b \circ p$ for the canonical representatives. Conversely, every potential of this form is Θ-plurisubharmonic.

For such v, one has $v \preceq u$ if and only if $b \preceq w$. Indeed, canceling Ψ outside H gives the base inequality almost everywhere, by Fubini's theorem in product charts, hence everywhere by Lemma 1.3.3; the reverse implication follows by pullback and addition. Similarly, $v \sim_{\mathcal{I}} u$ if and only if $b \sim_{\mathcal{I}} w$: Lemma 1.16.3 permits both addition and cancellation of Ψ, the pullback formula in Lemma 1.16.5 then gives one direction, and restriction of the extended ideals along any local section gives the other. This comparison of ideals does not need a section outside H.

Step 4: the envelope identities. The bound $v \leq F$ is equivalent to $b \leq \widehat{F}$, with no constraint over Z. The competitor families upstairs and on the base therefore correspond to each other, which gives (32.1) outside H, since regularization commutes with pullback in product coordinates there. On H both sides equal $-\infty$: every competitor is at most $\Psi + K_F$, by the bound $b \leq K_F$, and this upper semicontinuous function equals $-\infty$ on H, so regularization does not change the values there.

Step 5: equality below all obstacles. Suppose that the two base envelopes agree below every smooth obstacle. Uniform obstacle comparison extends this equality to continuous real obstacles. The functions $G_j = \min(\widehat{F}, j)$ are continuous and real-valued. Every base competitor below $\widehat{F}$ has finite supremum and hence lies below G_j for every sufficiently large integer j. Each full competitor family is thus the union of its subfamilies below the G_j, and equality for all G_j gives equality below $\widehat{F}$ after taking regularized suprema. By the reduction, the envelopes upstairs agree below every smooth obstacle.

Step 6: zero mass. Take any $b \in \mathrm{PSH}(Y,\beta)$ and write $a = \Psi + b\circ p$. On a relatively compact chart outside H, the functions $b_j = \max(b,-j)$ are locally bounded and quasi-plurisubharmonic, and there

$$\Theta + \mathrm{dd^c}(\Psi + b_j\circ p) = p^*(\beta + \mathrm{dd^c} b_j).$$

Since $n > d$, the nth bounded product of this current is zero by Lemma 1.7.2. This identity does not require b_j to be β-plurisubharmonic: Proposition 1.7.1 defines the products for locally bounded quasi-plurisubharmonic potentials. By plurifine locality, this zero product agrees with $\mathrm{MA}^{\mathrm{ext}}_{\Theta}(a)$ on $\{b\circ p > -j\}$ in the chart. These sets cover the locus where $b\circ p$ is finite, while the remaining pole set and H receive no extended mass. Thus $\mathrm{MA}^{\mathrm{ext}}_{\Theta}(a) = 0$.

By divisorial removal and descent, every representative ordinarily equivalent to u has this form. By the identity already proved, the ideal model is also of this form, with base potential $P_{\beta,\widehat{0}}[w]_{\mathcal{I}}$, where $\widehat{0}$ is the induced obstacle for $F = 0$. The same removal and descent apply to every representative of its ordinary type. All these masses are zero, which proves both volume assertions. □

Recall from Definition 14.4.1 that $u \in \mathcal{E}_{\mathrm{env}}(X,\theta,\varphi)$ means that $u \preceq \varphi$ and that, for every $a \in (0,1)$, there is a *stretching competitor* $h \in \mathrm{PSH}(X,\theta)$ with $a\varphi + (1-a)h \le u$; here $a = 1 - b^{-1}$ in the notation of that definition.

Proposition 32.1.2 *For $\varphi \in \mathrm{PSH}(Y,\beta)$ put $v = \Psi + \varphi\circ p$. Then*

$$u \in \mathcal{E}_{\mathrm{env}}(M,\Theta,v) \quad\Longleftrightarrow\quad w \in \mathcal{E}_{\mathrm{env}}(Y,\beta,\varphi).$$

In particular, for $q_M = P_\Theta[u]_{\mathcal{I}}$ and $q_Y = P_\beta[w]_{\mathcal{I}}$,

$$u \in \mathcal{E}_{\mathrm{env}}(M,\Theta,q_M) \quad\Longleftrightarrow\quad w \in \mathcal{E}_{\mathrm{env}}(Y,\beta,q_Y).$$

Proof By Step 3 of the proof of Proposition 32.1.1, $u \preceq v$ if and only if $w \preceq \varphi$. Membership in the relative envelope class requires, for every $a \in (0,1)$, a stretching competitor $h \in \mathrm{PSH}(M,\Theta)$ with

$$av + (1-a)h \le u.$$

A base competitor $c \in \mathrm{PSH}(Y,\beta)$ with $a\varphi + (1-a)c \le w$ gives such an h, namely $h = \Psi + c\circ p$.

Conversely, put $\rho_j = \nu(w\circ p, H_j)$ and $\sigma_j = \nu(\varphi\circ p, H_j)$. The ordinary order $w \preceq \varphi$ gives $\rho_j \ge \sigma_j \ge 0$. Comparing generic Lelong numbers along H_j in the stretching inequality $av + (1-a)h \le u$ gives

$$a(\lambda_j + \sigma_j) + (1-a)\nu(h,H_j) \ge \lambda_j + \rho_j.$$

Consequently $\nu(h,H_j) \ge \lambda_j + (\rho_j - a\sigma_j)/(1-a) \ge \lambda_j$. Divisorial removal and descent along the fibers give $h = \Psi + c\circ p$ with $c \in \mathrm{PSH}(Y,\beta)$. Canceling Ψ outside H as in Step 3, we obtain $a\varphi + (1-a)c \le w$ on Y. This proves the first equivalence.

By (32.1) for $F = 0$, $q_M = \Psi + b \circ p$ with $b = P_{\beta,\widehat{0}}[w]_I$, and (32.2) gives $b \sim q_Y$. The first equivalence with reference b, together with the invariance of the class under ordinarily equivalent references in Proposition 14.4.1, proves the second equivalence. □

Corollary 32.1.1 *The ordinary and ideal envelopes of u agree below every smooth obstacle on M if and only if those of w agree below every smooth obstacle on Y.*

Put $q = P_\beta[w]_I$. If $w \in \mathcal{E}_{\mathrm{env}}(Y, \beta, q)$, then

$$P_{\Theta,F}[u] = P_{\Theta,F}[u]_I \quad \text{for every smooth real } F.$$

If Y is a curve, write

$$T_w = \beta + \mathrm{dd}^{\mathrm{c}} w, \qquad \mu_w = \mathbf{1}_{\{w=-\infty\}} T_w, \qquad D_w = \sum_{y \in Y} \nu(w, y)\delta_y$$

for the curvature measure of w, its polar part and its atomic part. Then equality of the two envelopes of u below zero, the equality $\mu_w = D_w$, and equality of the two envelopes of u below every smooth obstacle are equivalent. They are also equivalent to

$$u \in \mathcal{E}_{\mathrm{env}}(M, \Theta, P_\Theta[u]_I).$$

Proof That equality on Y implies equality on M is part of Proposition 32.1.1. For the converse, we approximate a smooth obstacle on the base by the induced obstacles of continuous obstacles upstairs. Suppose that the envelopes upstairs agree below every smooth obstacle. Uniform obstacle comparison extends this equality to continuous real obstacles. Fix a smooth real function B on Y and put $G = \widehat{0}$. The continuous obstacles

$$F_j = B \circ p + \max(\Psi, -j) \qquad (j \geq 1)$$

have induced base obstacles

$$G_j = B + \max(G - j, 0),$$

with value $+\infty$ on Z. The set Z has smooth volume zero: in each product chart of the submersion, the preimage of Z lies in H, which has smooth volume zero, and Fubini's theorem gives the assertion on the base. By the reduction and the assumed equality, the two base envelopes below G_j agree off Z, hence everywhere by Lemma 1.3.3:

$$q_j = P_{\beta,G_j}[w]_I = P_{\beta,G_j}[w].$$

Write $q_B = P_{\beta,B}[w]_I$. Then $q_B \leq q_j \leq q_1$, and the q_j decrease to a β-plurisubharmonic potential r, not identically $-\infty$, with convergence in L^1; indeed, their negative parts are bounded by that of q_B, and Corollary 1.4.1 applies. Off Z, the bounds $q_j \leq G_j \downarrow B$ give $r \leq B$, and almost-everywhere order in Lemma 1.3.3 extends the inequality to all of Y.

Each q_j is multiplier-ideal equivalent to w. To see this without assuming G_j finite, put $Q_0 = P_\Theta[u]_I$ and $Q_j = P_{\Theta,F_j}[u]_I$. Translation of the competitor families gives

$$Q_0 + \min_M F_j \le Q_j \le Q_0 + \max_M F_j.$$

Thus $Q_j \sim Q_0 \sim_{\mathcal{I}} u$ by Theorem 17.2.1. Since $Q_j = \Psi + q_j \circ p$, cancellation and reflection of multiplier ideals under submersions, Lemmas 1.16.3 and 1.16.5, give $q_j \sim_{\mathcal{I}} w$. The same bounded comparison with $P_\beta[w]_{\mathcal{I}}$ gives $q_B \sim_{\mathcal{I}} w$.

Let π be any smooth local modification of a coordinate neighborhood in the base. The bounds $q_B \circ \pi \le q_j \circ \pi \le q_1 \circ \pi$ give decreasing local L^1 convergence to $r \circ \pi$. At every point x of this modification, Lemma 1.17.6 and the ordinary order give

$$\nu(w \circ \pi, x) \le \nu(r \circ \pi, x) \le \nu(q_B \circ \pi, x) = \nu(w \circ \pi, x).$$

All these Lelong numbers are therefore equal, and Theorem 1.16.1 gives $r \sim_{\mathcal{I}} w$. Together with $r \le B$, this makes r a competitor for q_B. Hence $r = q_B$.

Put $\epsilon_j = \max\{0, \sup_Y (q_j - B)\}$. The upper semicontinuous functions $q_j - B$ decrease to $q_B - B \le 0$ on the compact manifold Y, so for every $\epsilon > 0$ the decreasing compact sets $\{q_j - B \ge \epsilon\}$ are eventually empty. Thus $\epsilon_j \to 0$. Every ordinary competitor b below G_j satisfies $b \le q_j$, so $b - \epsilon_j \le B$ and $b - \epsilon_j \preceq w$. Taking regularized suprema yields

$$q_j - \epsilon_j \le P_{\beta,B}[w].$$

Since $q_B \le q_j$, letting $j \to \infty$ proves $q_B \le P_{\beta,B}[w]$. The reverse inequality always holds, so the two envelopes agree below B. Since B was arbitrary, this proves the converse.

Under the relative-class hypothesis, Proposition 17.6.6 gives equality of the two base envelopes below every smooth obstacle, and the equivalence just proved gives the stated equality upstairs.

In the curve case, (32.2) shows that the two base envelopes below $\widehat{0}$ are ordinarily equivalent to $P_\beta[w]$ and q. By Proposition 31.14.1 and Corollary 1.6.1, their polar curvatures are therefore μ_w and D_w, respectively. Thus equality of the envelopes upstairs below zero forces $\mu_w = D_w$. Conversely, under this equality of measures, Corollary 31.14.1 gives equality below every smooth obstacle on the base, hence upstairs by the first assertion. Taking $F = 0$ proves the remaining implication. The same curve criterion identifies the equality of measures with membership of w in the base relative envelope class of q, and the equivalence for ideal references in Proposition 32.1.2 gives the last assertion upstairs. □

The reduction does not compute the multiplier ideals of u. These ideals have an explicit form when the divisors have relative simple normal crossings, that is, when near every point there are product coordinates (y, z) with $p(y, z) = y$ in which the components through the point are distinct coordinate hyperplanes in the fiber variables. Then

$$\mathcal{I}\big(t(\Psi + b \circ p)\big) = \mathcal{O}_M\left(-\sum_j \lfloor t\lambda_j \rfloor H_j\right)\big(\mathcal{I}(tb)\mathcal{O}_M\big) \qquad (t > 0,\ b \in \mathrm{PSH}(Y, \beta)). \tag{32.3}$$

To see this, work in such coordinates. Up to a smooth correction, Ψ is the sum of the weights $\lambda_j \log|z_j|^2$ over the components through the point. Since b is locally bounded above, integrability forces vanishing to order at least $\lfloor t\lambda_j \rfloor$ along each hyperplane. After this product of coordinate powers is factored out, the remaining fiber weight is

$$\prod_j |z_j|^{-2(t\lambda_j - \lfloor t\lambda_j \rfloor)}.$$

It is integrable and bounded below by a positive constant on a small polydisk. Integrability of the quotient against this weight therefore implies integrability against the pullback of e^{-tb} alone, and Lemma 1.16.5 puts the quotient in the extension of $\mathcal{I}(tb)$. Conversely, local generators of that ideal are weighted-integrable against the integrable product weight, by Fubini's theorem. Outside H the formula is the same pullback lemma. For one smooth divisor of weight one transverse to the fibers, the first factor is the ideal sheaf of the divisor raised to the power $\lfloor t \rfloor$. Without normal crossings the formula can fail, although the envelopes still reduce to the base.

Example 32.1.1 On $M = \mathbb{P}^1 \times \mathbb{P}^1$, let p, r be the projections and let σ be the probability Fubini–Study form. With homogeneous coordinates $[y_0 : y_1]$ and $[z_0 : z_1]$, the divisor

$$H = \{y_0 z_0^2 + y_1 z_1^2 = 0\}$$

is smooth, meets every fiber properly and ramifies over the first factor. Give $\mathcal{O}_M(H) = \mathcal{O}_M(1,2)$ its product metric, with logarithmic divisor potential Ψ and curvature $\kappa_H = p^*\sigma + 2r^*\sigma$. Put $\Theta = 2p^*\sigma + 2r^*\sigma$ and choose $w \in \mathrm{PSH}(\mathbb{P}^1, \sigma)$ with $\sigma + \mathrm{dd}^{\mathrm{c}} w = \delta_{[0:1]}$. Both obstacle envelopes of $u = \Psi + w \circ p$ reduce to the base, but the product formula for its multiplier ideals fails at a ramification point.

Proof For each $[z_0 : z_1]$ there is a unique point $[y_0 : y_1] = [-z_1^2 : z_0^2]$ on H, so H is a smooth graph over the second factor, and its projection to the base has degree two. In particular no fiber of p lies in H. Since $\Theta = p^*\sigma + \kappa_H$, Proposition 32.1.1 gives both envelope identities.

Near $([0:1],[1:0])$, use $y = y_0/y_1$ and $z = z_1/z_0$. Up to smooth additions,

$$\Psi = \log|y + z^2|^2, \qquad w = \log|y|^2, \qquad u = \log|y(y+z^2)|^2.$$

At $t = 3/4$, each of the two weights separately has trivial multiplier ideal, so the right side of (32.3) would be $\mathcal{O}_M$. However, $1 \notin \mathcal{I}(3u/4)$ at this point. Indeed, consider the region where $y = cz^2$ with $|c-1| < 1/4$ and $0 < |z| < \epsilon$; for fixed z, it has area comparable to $|z|^4$ in the y variable. The weighted integral of 1 is therefore bounded below by a positive constant times

$$\int_{0<|z|<\epsilon} |z|^{-6}|z|^4 \, \mathrm{d}\lambda_z = 2\pi \int_0^{\epsilon} \frac{\mathrm{d}r}{r} = +\infty,$$

where $\mathrm{d}\lambda_z$ is planar area. Thus the multiplier ideal is proper. □

If the divisor contains an entire fiber, the induced obstacle is infinite over that fiber. The next example still admits an explicit model for the zero obstacle.

Example 32.1.2 Let $M = \mathbb{P}^2 \times \mathbb{P}^1$, with projections p, r, and let σ_j be the probability Fubini–Study form on $\mathbb{P}^j$. In the positive closed background

$$\Theta = 2p^*\sigma_2 + r^*\sigma_1,$$

there is a logarithmic potential Ψ of a smooth prime divisor containing the fiber over $E = [0 : 0 : 1]$. Its fiber infimum for the zero obstacle is finite and continuous off E and tends to $+\infty$ at E. Nevertheless,

$$P_\Theta[\Psi] = P_\Theta[\Psi]_I = \Psi + q \circ p,$$

where $q \in \mathrm{PSH}(\mathbb{P}^2, \sigma_2)$ is bounded, $0 \le q \le \log 4$ and $q(E) = \log 4$. Every ordinary representative of this model has zero extended mass, so both singularity volumes vanish. The two envelopes also agree below every smooth obstacle.

Proof Use homogeneous coordinates $[y_0 : y_1 : y_2]$ and $[z_0 : z_1]$, and take

$$H = \{y_0 z_0 + y_1 z_1 = 0\}, \qquad \Psi = \log \frac{|y_0 z_0 + y_1 z_1|^2}{(|y_0|^2 + |y_1|^2 + |y_2|^2)(|z_0|^2 + |z_1|^2)}.$$

Outside E, the divisor is the graph $[z_0 : z_1] = [-y_1 : y_0]$. Its closure contains the whole fiber over E, so H is prime. It is smooth outside that fiber by the graph description, and smooth along it because the derivatives in the y_0 and y_1 directions are z_0 and z_1, which cannot both vanish. Its product metric has curvature $p^*\sigma_2 + r^*\sigma_1$, hence

$$\Theta + \mathrm{dd}^{\mathrm{c}}\Psi = p^*\sigma_2 + [H].$$

Writing $S = |y_0|^2 + |y_1|^2$, the Cauchy–Schwarz inequality gives the fiber obstacle

$$G(y) = -\max_{z \in \mathbb{P}^1} \Psi(y, z) = \log \frac{S + |y_2|^2}{S}, \qquad G(E) = +\infty.$$

It has no finite continuous extension at E.

We claim that the largest σ_2-plurisubharmonic potential below G is

$$q(y) = \begin{cases} \log \dfrac{4|y_2|^2}{S + |y_2|^2}, & S \le |y_2|^2, \\ \log \dfrac{S + |y_2|^2}{S}, & S \ge |y_2|^2. \end{cases}$$

To see that q is σ_2-plurisubharmonic, work on $y_2 \neq 0$ with $x = (y_0/y_2, y_1/y_2)$ and $t = \log |x|^2$. Adding the Fubini–Study potential $L(x) = \log(1 + |x|^2)$ gives

$$q + L = \begin{cases} \log 4, & t \le 0, \\ 2\log(1 + \mathrm{e}^t) - t, & t \ge 0. \end{cases}$$

This function and its first derivative are continuous at $t = 0$. For $t > 0$, its first two derivatives are $(\mathrm{e}^t - 1)/(\mathrm{e}^t + 1) \ge 0$ and $2\mathrm{e}^t/(1 + \mathrm{e}^t)^2 > 0$. It is therefore a convex

nondecreasing function of t, hence plurisubharmonic in x, across the unit sphere as well. The expression is constant near $x = 0$, and $q = G$ is smooth near $y_2 = 0$, so q is globally σ_2-plurisubharmonic and bounded.

If $b \in \mathrm{PSH}(\mathbb{P}^2, \sigma_2)$ lies below G, then $b + L \le \log 4$ on $|x| = 1$, and the maximum principle on the unit ball gives $b \le q$ inside it. Outside the ball, the obstacle already gives $b \le G = q$. For $0 < R = |x|^2 \le 1$, the inequality $q \le G$ reads $4R \le (1 + R)^2$, and at E it is automatic. The formula for q also gives the asserted bounds, which proves the claim.

By Proposition 32.1.1, applied with $\beta = \sigma_2$, $w = 0$ and $F = 0$, the two envelopes of Ψ are Ψ plus the pullbacks of the two envelopes of 0 below G. Every σ_2-plurisubharmonic function is at least as singular as 0, and q is bounded, hence multiplier-ideal equivalent to 0. Both base envelopes therefore equal q, which proves the envelope identities, and the same proposition gives the volume assertions. Finally, since the model is ordinarily equivalent to Ψ, Proposition 14.4.1 gives $\Psi \in \mathcal{E}_{\mathrm{env}}(M, \Theta, P_\Theta[\Psi]_{\mathcal{I}})$, and Proposition 17.6.6 gives equality below every smooth obstacle. □

Example 32.1.3 On a compact fourfold without the bounded mass property, there are a Hermitian background Θ and a potential u without an analytic ordinary type such that

$$P_\Theta[u] = P_\Theta[u]_{\mathcal{I}} = \Psi, \qquad u \nsim \Psi.$$

Both envelopes agree below every smooth obstacle, and their common model for the zero obstacle has both singularity volumes zero.

Proof Let (Y, ω) be the Hopf threefold of Theorem 9.0.1. On $\mathbb{P}^1$, let σ be the probability Fubini–Study form, choose a point D and a logarithmic potential ψ with $\sigma + \mathrm{dd}^{\mathrm{c}}\psi = [D]$ and $\sup \psi = 0$. On $M = Y \times \mathbb{P}^1$, with projections p, r, put

$$\Theta = p^*\omega + r^*\sigma, \qquad H = Y \times D, \qquad \Psi = \psi \circ r.$$

Take the potential $U \in \mathcal{E}_{\mathrm{env}}(Y, \omega, 0)$ of infinite mass from Proposition 14.6.1, which satisfies $P_\omega[U] = 0$. Its multiplier ideals are trivial, so also $P_\omega[U]_{\mathcal{I}} = 0$. Set $u = \Psi + U \circ p$. Here $\widehat{0} = 0$, so Proposition 32.1.1 and Corollary 32.1.1 give the envelope identities, the equality below all obstacles and the vanishing of the singularity volumes. Moreover $u \in \mathcal{E}_{\mathrm{env}}(M, \Theta, \Psi)$ by Proposition 32.1.2. Since U has infinite extended mass, it is unbounded; restriction to a section outside H therefore proves $u \nsim \Psi$. If a potential a with analytic singularities were ordinarily equivalent to u, Proposition 17.2.4 would give $u \sim a \sim P_\Theta[a] = P_\Theta[u] = \Psi$, a contradiction.

Finally, let V_j be the smooth potentials of Theorem 9.0.1. Their pullbacks are bounded and Θ-plurisubharmonic, and Proposition 20.9.1 gives

$$\int_M \mathrm{MA}_\Theta(V_j \circ p) = 4 \int_Y \mathrm{MA}_\omega(V_j) \longrightarrow +\infty.$$

Thus $v_+(\Theta) = +\infty$, and M fails the bounded mass property. □

The descent of potentials uses that the fibers are connected. Otherwise the residual potential can take independent values on different components of a fiber, as a finite isogeny shows.

Example 32.1.4 There is a surjective holomorphic submersion $p\colon M \to Y$ from a connected compact Kähler surface to an elliptic curve, with four components in every fiber, for which both identities in (32.1) fail. The pole divisor is smooth and maps onto Y, and the obstacle upstairs is smooth.

Proof Let $E = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$, let β be its flat probability form, and let $q\colon E \to E$ be multiplication by two. On $M = E \times \mathbb{P}^1$, denote the projections by a, r and put $p = q \circ a$. Then q is an unramified covering of degree four and $q^*\beta = 4\beta$. The manifold M is connected, whereas every fiber of p is the disjoint union of four projective lines.

On $\mathbb{P}^1$ take the probability Fubini–Study form σ, a point D and a logarithmic potential ψ with $\sigma + \mathrm{dd}^{\mathrm{c}}\psi = [D]$ and $\sup\psi = 0$. Put

$$H = E \times D, \qquad \Psi = \psi \circ r, \qquad \Theta = p^*\beta + r^*\sigma = 4a^*\beta + r^*\sigma.$$

Thus $\Theta + \mathrm{dd}^{\mathrm{c}}\Psi = p^*\beta + [H]$, which corresponds to the base potential $w = 0$. Choose $\epsilon > 0$ so small that

$$f(\zeta) = \epsilon \cos(2\pi \operatorname{Re}\zeta) \in \mathrm{PSH}(E, 4\beta),$$

and set $F = f \circ a$. The function f is well defined and smooth on E. Since it changes sign under translation by $1/2$, it is not constant on the fibers of q.

Apply Proposition 32.1.1 to the projection a, which has connected fibers, with base background 4β and base potential zero. Its induced obstacle is f, since $\sup\psi = 0$. The bounded potential f is a competitor for both base envelopes and equals their obstacle, so

$$P_{\Theta,F}[\Psi] = P_{\Theta,F}[\Psi]_I = \Psi + f \circ a.$$

For the map p, the fiber obstacle of Proposition 32.1.1 would instead be

$$\widehat{F}(y) = \min_{q(\zeta)=y} f(\zeta).$$

Every expression $\Psi + b \circ p$ is invariant under translation of the first factor by points of the kernel of q. The computed envelope is not invariant under translation by $1/2$, because Ψ is unchanged and f changes sign. It therefore cannot equal either right side of (32.1) for p. □

The correct formula takes the fiber minimum over each connected component of a fiber separately. These components are the points of the finite covering of the base in Lemma 1.22.2.

Envelopes and relative envelope classes also pass through this finite covering. Pullback of competitors and the maximum over the fibers of a finite unramified covering identify the envelopes upstairs and downstairs, and the same fiber maximum transfers stretching competitors from the covering space to the base.

Proposition 32.1.3 *Let $r\colon Y \to X$ be a finite unramified holomorphic covering between connected compact complex manifolds. Let θ be a smooth real $(1,1)$-form on X, let $u, \varphi \in \mathrm{PSH}(X,\theta)$, and put $\theta' = r^*\theta$ and $u' = u \circ r$. For every $F \in C(X,\mathbb{R})$,*

$$P_{\theta',F\circ r}[u'] = P_{\theta,F}[u] \circ r,$$
$$P_{\theta',F\circ r}[u']_{\mathcal{I}} = P_{\theta,F}[u]_{\mathcal{I}} \circ r.$$

Moreover,

$$u' \in \mathcal{E}_{\mathrm{env}}(Y,\theta',\varphi\circ r) \quad\Longleftrightarrow\quad u \in \mathcal{E}_{\mathrm{env}}(X,\theta,\varphi).$$

Proof For $v \in \mathrm{PSH}(Y,\theta')$, define

$$v_{\max}(x) = \max_{y\in r^{-1}(x)} v(y).$$

Over a sufficiently small ball in X, this is the finite maximum of the functions obtained by composing v with the local inverse branches of r. Each of them is θ-plurisubharmonic, so $v_{\max} \in \mathrm{PSH}(X,\theta)$ by Lemma 1.3.1. If $v \preceq u'$, then $v_{\max} \preceq u$. If $v \sim_{\mathcal{I}} u'$, a local biholomorphic change of variables makes each branch multiplier-ideal equivalent to u, and the maximum rule in Lemma 1.16.3 gives $v_{\max} \sim_{\mathcal{I}} u$. The obstacle bound $v \le F\circ r$ gives $v_{\max} \le F$.

Thus every competitor upstairs is dominated by the pullback of a corresponding competitor downstairs. Conversely, pullback of a competitor downstairs preserves the background inequality, the ordinary order or multiplier-ideal equivalence, and the obstacle bound. These two correspondences between the competitor families give the envelope identities; upper regularization commutes with r, since r is locally biholomorphic.

Since r is surjective, it reflects the ordinary order: $u' \preceq \varphi\circ r$ if and only if $u \preceq \varphi$. Fix $a \in (0,1)$. Pullback takes any stretching competitor h with $a\varphi + (1-a)h \le u$ to a stretching competitor upstairs. Conversely, if $v \in \mathrm{PSH}(Y,\theta')$ satisfies

$$a\varphi\circ r + (1-a)v \le u',$$

then taking the maximum over the fibers gives $a\varphi + (1-a)v_{\max} \le u$; this also holds at poles, because the convex sums are defined there. Applying the definition of the relative class for every a proves the last assertion. □

Corollary 32.1.2 *Keep the data $A, H, \Psi, \kappa_A, \beta, w, \Theta, u$ above, but allow the fibers of the submersion p to be disconnected. For the factorization $p = r \circ \pi$ of Lemma 1.22.2, set $\beta' = r^*\beta$ and $w' = w \circ r$. For $F \in C(M, \mathbb{R})$ define*

$$\widehat{F}_{\mathrm{comp}}(y') = \inf_{x \in \pi^{-1}(y') \setminus H} \{F(x) - \Psi(x)\}, \qquad \inf \varnothing = +\infty.$$

This obstacle is continuous in the extended real sense on Y', and

$$P_{\Theta,F}[u] = \Psi + P_{\beta',\widehat{F}_{\mathrm{comp}}}[w'] \circ \pi,$$
$$P_{\Theta,F}[u]_I = \Psi + P_{\beta',\widehat{F}_{\mathrm{comp}}}[w']_I \circ \pi.$$

Every representative of the ordinary type of u or of $P_\Theta[u]_I$ has zero extended mass, and both singularity volumes of these types are zero. The ordinary and ideal envelopes of u agree below every smooth obstacle on M if and only if those of w' agree below every smooth obstacle on Y'.

For $\varphi \in \mathrm{PSH}(Y, \beta)$, put $v = \Psi + \varphi \circ p$. Then

$$u \in \mathcal{E}_{\mathrm{env}}(M, \Theta, v) \quad \Longleftrightarrow \quad w \in \mathcal{E}_{\mathrm{env}}(Y, \beta, \varphi).$$

For $q_M = P_\Theta[u]_I$ and $q_Y = P_\beta[w]_I$, one also has

$$u \in \mathcal{E}_{\mathrm{env}}(M, \Theta, q_M) \quad \Longleftrightarrow \quad w \in \mathcal{E}_{\mathrm{env}}(Y, \beta, q_Y).$$

If Y is a curve, let μ_w, D_w be the polar and atomic measures of Corollary 32.1.1. Then equality of the envelopes of u below zero, the equality $\mu_w = D_w$, and equality of the envelopes of u below every smooth obstacle are equivalent. They are also equivalent to $u \in \mathcal{E}_{\mathrm{env}}(M, \Theta, q_M)$.

Proof The covering map r preserves the curvature inequality, so $w' \in \mathrm{PSH}(Y', \beta')$. Moreover $\dim Y' = \dim Y < \dim M$, and

$$\Theta = \pi^*\beta' + \kappa_A, \qquad u = \Psi + w' \circ \pi.$$

Applying Proposition 32.1.1 to the submersion π, which has connected fibers, gives the obstacle identities and the mass assertions, and Corollary 32.1.1 for the same submersion gives the equivalence for all obstacles. In Example 32.1.4, this factorization has $Y' = E$, $r = q$ and $\pi = a$, so it recovers the computed envelope.

For the assertions on relative classes, apply Proposition 32.1.2 to π, with reference $\varphi \circ r$ on Y', and then transfer membership through r by Proposition 32.1.3. That proposition, applied with the zero obstacle, also gives

$$P_{\beta'}[w']_I = q_Y \circ r.$$

Applying the same two equivalences with these ideal references proves the assertion for q_M and q_Y.

In the curve case, Y' is a curve. The covering is locally biholomorphic, so curvature measures and point Lelong numbers pull back without multiplicities. The pole set of

w' is $r^{-1}\{w = -\infty\}$, and therefore

$$\mu_{w'} = r^*\mu_w, \qquad D_{w'} = r^*D_w,$$

where the pullback of a measure is defined on each sheet through the local inverse of r. Such pullbacks reflect equality: over every sufficiently small ball in Y, the restriction of a measure is recovered from any sheet over that ball. Hence $\mu_{w'} = D_{w'}$ if and only if $\mu_w = D_w$. The curve case of Corollary 32.1.1, applied to π, identifies the former equality with the equality of the envelopes upstairs below zero, with their equality below every smooth obstacle and with membership in the relative envelope class upstairs. This proves the last assertion. □

32.2 Divisorial products and zero volume

The simplest case of the fibration reduction is a product of a curve C with a compact Kähler manifold Z, with the poles of Ψ along $C \times D$ for a smooth divisor $D \subset Z$. If the divisor potential on Z has supremum zero, the induced zero obstacle on C is zero, so the envelopes of $\Psi + w \circ p$ are determined by those of w on the curve. Projective products then give potentials without analytic ordinary type whose two envelopes agree, as well as distinct models with the same zero volumes. Since top-degree mass vanishes on all these models, volumes cannot distinguish them. Polar curvature on the curve can, and it gives a complete metric on the resulting class of product models.

Let C be a compact connected curve with a smooth positive form η, and let Z be a connected compact Kähler manifold of dimension $k \geq 1$. Suppose that $D \subset Z$ is a nonempty smooth reduced divisor, κ is a Kähler form on Z, and $\psi \in \mathrm{PSH}(Z, \kappa)$ is smooth off D and satisfies

$$\kappa + \mathrm{dd^c}\psi = [D], \qquad \sup_Z \psi = 0.$$

Thus, near D, ψ is $\log |z_1|^2$ plus a smooth function, where z_1 is a local equation of D. On $M = C \times Z$, write $p\colon M \to C$ and $r\colon M \to Z$ for the projections, and put

$$\Theta = p^*\eta + r^*\kappa, \qquad \Psi = \psi \circ r, \qquad H = C \times D.$$

For $w \in \mathrm{PSH}(C, \eta)$ set $u = \Psi + w \circ p$.

On the curve (C, η) we use the models of Section 31.13. The polar curvature of $q \in \mathrm{PSH}(C, \eta)$ is the restriction of $\eta + \mathrm{dd^c}q$ to the pole set $\{q = -\infty\}$ (Section 1.6.1). By Lemma 31.13.1, every positive measure of mass at most $\int_C \eta$ carried by a polar set is the polar curvature of exactly one ordinary model, and $P_\eta[q]$ is the model with the polar curvature of q. Ideal projection takes the atomic part of the polar curvature (Lemma 20.5.1 and Proposition 31.13.1), and the total variation of the difference of polar curvatures is a complete geodesic metric on ordinary models, equal to the volume comparison d_η of Definition 31.6.1 (Theorem 31.13.1 and Corollary 31.13.1).

Proposition 32.2.1 *For $F \in C(M,\mathbb{R})$, the function*

$$\widehat{F}(y) = \min_{z \in Z \setminus D} \{F(y,z) - \psi(z)\} \qquad (y \in C)$$

is finite and continuous. The two envelopes satisfy

$$\begin{aligned} P_{\Theta,F}[u] &= \Psi + P_{\eta,\widehat{F}}[w] \circ p, \\ P_{\Theta,F}[u]_{\mathcal{I}} &= \Psi + P_{\eta,\widehat{F}}[w]_{\mathcal{I}} \circ p. \end{aligned} \tag{32.4}$$

Proof Apply Proposition 32.1.1 to $p\colon M \longrightarrow C$, with $H = C \times D$, $\beta = \eta$ and $\Psi = \psi \circ r$. The local logarithmic form of ψ defines a smooth metric on $\mathcal{O}_M(H)$ with curvature $r^*\kappa$. Thus $\Theta + \mathrm{dd^c}\Psi = p^*\eta + [H]$, and H is smooth and transverse to the fibers. The induced fiber minimum is the function $\widehat{F}$ above. For $F = 0$, the normalization $\sup_Z \psi = 0$ gives $\widehat{0} = 0$. □

Example 32.2.1 Take $C = \mathbb{P}^1$ and $Z = \mathbb{P}^k$, $k \geq 1$, with their probability Fubini–Study forms, and let D be a hyperplane. In this product setting there is a potential u without an analytic ordinary type such that

$$P_\Theta[u] = P_\Theta[u]_{\mathcal{I}} = \Psi, \qquad u \nsim \Psi,$$

and the two envelopes of u agree below every smooth obstacle. Their common model for the zero obstacle has zero singularity volume.

Proof On the curve C, let w be the unbounded potential constructed in the proof of Proposition 17.6.2, whose two envelopes are zero, and let ψ be a logarithmic potential of the hyperplane D with supremum zero. For $u = \Psi + w \circ p$, Proposition 32.2.1 gives the envelope identities, Corollary 32.1.1 gives equality below every smooth obstacle, and Proposition 32.1.1 gives zero volumes. As in the proof of Example 32.1.3, restriction to a section outside H shows $u \nsim \Psi$, and Proposition 17.2.4 excludes an analytic ordinary type. □

Vanishing of the volumes does not force the two envelopes to agree.

Example 32.2.2 In the product setting, take $C = \mathbb{P}^1$ and $Z = \mathbb{P}^k$, $k \geq 1$, with probability Fubini–Study forms and a hyperplane D. The positive closed background Θ admits normalized ordinary models $a \leq b$ with

$$P_\Theta[a] = a, \qquad P_\Theta[a]_{\mathcal{I}} = P_\Theta[b] = P_\Theta[b]_{\mathcal{I}} = b,$$
$$a \sim_{\mathcal{I}} b, \qquad a \nsim b,$$
$$\underline{\mathrm{vol}}_\Theta(a) = \overline{\mathrm{vol}}_\Theta(a) = \underline{\mathrm{vol}}_\Theta(b) = \overline{\mathrm{vol}}_\Theta(b) = 0.$$

Thus equal singularity volumes of a type and of its ideal model do not imply equality of the envelopes at zero volume.

Proof Use the normalized diffuse polar curve model w of Example 31.13.2 on $\mathbb{P}^1$ with its probability Fubini–Study form. It satisfies $P_\eta[w] = w$ and $P_\eta[w]_{\mathcal{I}} = 0$, and it is unbounded. Take the normalized hyperplane potential on $\mathbb{P}^k$ and put

$$a = \Psi + w \circ p, \qquad b = \Psi.$$

The identities for the zero obstacle in Proposition 32.2.1 give all four envelope identities. Both functions have supremum zero, and $w \leq 0$ gives $a \leq b$. Ideal projection preserves the ideal type by Theorem 17.2.1, so $a \sim_{\mathcal{I}} b$. Restriction to $\mathbb{P}^1 \times \{z\}$, for any $z \notin D$, gives the unbounded difference w, so $a \nsim b$. Finally, Proposition 32.1.1 gives zero extended mass for every ordinary representative of either model, which proves all four volume identities. □

A semipositive term in the background that vanishes on the slices of a product does not change the envelopes of potentials pulled back from the other factor.

Lemma 32.2.1 *Let Y and T be connected compact complex manifolds, let $\pi : Y \times T \to Y$ be the projection, let γ be a smooth real $(1,1)$-form on Y, and let α be a smooth semipositive real $(1,1)$-form on $Y \times T$ whose restriction to every slice $Y \times \{t\}$ is zero. Put $\Gamma = \pi^*\gamma + \alpha$. For every $u \in \mathrm{PSH}(Y, \gamma)$, one has $u \circ \pi \in \mathrm{PSH}(Y \times T, \Gamma)$ and*

$$P_\Gamma[u \circ \pi] = P_\gamma[u] \circ \pi.$$

If $P_\gamma[u]_{\mathcal{I}}$ is ordinarily equivalent to a quasi-plurisubharmonic function g with $\mathrm{dd}^{\mathrm{c}} g = S - \beta'$, where S is a divisorial current on Y, possibly zero, as in Lemma 1.17.5 and β' is a smooth form, then also

$$P_\Gamma[u \circ \pi]_{\mathcal{I}} = P_\gamma[u]_{\mathcal{I}} \circ \pi.$$

Proof Since $\alpha \geq 0$, pullback maps $\mathrm{PSH}(Y, \gamma)$ into $\mathrm{PSH}(Y \times T, \Gamma)$. It preserves the bound $v \leq 0$, the ordinary order and, by Lemma 1.16.5, multiplier-ideal equivalence. Pullbacks of competitors are therefore competitors, and since upper regularization commutes with pullback by a projection, both envelopes upstairs are at least the pullbacks of the envelopes on Y.

Let v be an ordinary competitor for $u \circ \pi$, and replace it by the competitor $\max(v, u \circ \pi - \sup_Y u)$. For every $t \in T$, the restriction $v(\cdot, t)$ is γ-plurisubharmonic,

because Γ restricts to γ on $Y \times \{t\}$, and it is not identically $-\infty$, nonpositive and at least as singular as u. Hence $v(\cdot,t) \le P_\gamma[u]$, which proves the ordinary identity.

Under the additional hypothesis, put $q = P_\gamma[u]_{\mathcal{I}}$, so that $u \sim_{\mathcal{I}} q \sim_{\mathcal{I}} g$ by Theorem 17.2.1. The complex Hessian of $g \circ \pi$ is $\pi^*S - \pi^*\beta'$, and π^*S is the sum of the prime divisors $D_j \times T$ with the coefficients of the components D_j of S; it has finite mass for a product Hermitian metric. An ideal competitor v for $u \circ \pi$ satisfies $v \sim_{\mathcal{I}} g \circ \pi$, so Lemma 1.17.5 gives $v \preceq g \circ \pi \sim q \circ \pi$. Thus v is an ordinary competitor for $q \circ \pi$, and the ordinary identity and Proposition 17.2.2 give $v \le P_\gamma[q] \circ \pi = q \circ \pi$. □

The failure of Example 32.2.2 also occurs in a nonclosed background once a second curve factor is added.

Example 32.2.3 In every complex dimension $n \ge 3$, there is a projective manifold with the bounded mass property that admits a smooth positive nonclosed background Θ and normalized ordinary models $a \le b$ such that

$$P_\Theta[a] = a, \qquad P_\Theta[a]_{\mathcal{I}} = P_\Theta[b] = P_\Theta[b]_{\mathcal{I}} = b,$$
$$a \sim_{\mathcal{I}} b, \qquad a \nsim b,$$
$$\underline{\mathrm{vol}}_\Theta(a) = \overline{\mathrm{vol}}_\Theta(a) = \underline{\mathrm{vol}}_\Theta(b) = \overline{\mathrm{vol}}_\Theta(b) = 0.$$

In particular, equality of the upper volumes of a type and of its ideal model does not force equality of the envelopes when the ideal model has zero lower volume.

Proof Let $Y = \mathbb{P}^1 \times \mathbb{P}^1$, with projections π_1, π_2, and let σ be the probability Fubini–Study form on $\mathbb{P}^1$. Choose a positive nonconstant smooth function f on the first factor and put

$$\beta = \pi_1^*\sigma + (f \circ \pi_1)\pi_2^*\sigma.$$

This form is positive, and $\mathrm{d}\beta = \pi_1^*(\mathrm{d}f) \wedge \pi_2^*\sigma \ne 0$. Let w_0 be the normalized diffuse polar model of Example 31.13.2, which satisfies $P_\sigma[w_0] = w_0$ and $P_\sigma[w_0]_{\mathcal{I}} = 0$, and put $w = w_0 \circ \pi_1$. Lemma 32.2.1, applied to the first factor with $T = \mathbb{P}^1$, $\alpha = (f \circ \pi_1)\pi_2^*\sigma$ and $g = 0$, gives

$$P_\beta[w] = w, \qquad P_\beta[w]_{\mathcal{I}} = 0.$$

Set $k = n - 2$ and $M = Y \times \mathbb{P}^k$, with projections p, r. On the second factor take the probability Fubini–Study form κ, a hyperplane D, and its normalized potential ψ with $\kappa + \mathrm{dd}^{\mathrm{c}}\psi = [D]$. Put

$$\Theta = p^*\beta + r^*\kappa, \qquad \Psi = \psi \circ r, \qquad a = \Psi + w \circ p, \qquad b = \Psi.$$

The form Θ is positive and not closed. A sufficiently large multiple of the closed positive product form dominates Θ, so the bounded mass property follows from Stokes' theorem for that closed form and the metric independence in Proposition 7.5.1.

Apply Proposition 32.1.1 to $p\colon M \to Y$ and the divisor $Y \times D$, whose induced zero obstacle is zero because $\sup \psi = 0$. The two identities on the base give the four

identities upstairs. The remaining assertions follow as in the proof of Example 32.2.2; for the volumes, use $\dim Y = 2 < n$. □

Consequently, at zero lower volume the volume comparison of Definition 31.6.1 does not separate ordinary models, in contrast with Proposition 31.6.1.

Example 32.2.4 For every $n \geq 2$ there are a closed positive $(1,1)$-form θ on the projective manifold $\mathbb{P}^1 \times \mathbb{P}^{n-1}$ and two ordinary models $a \nsim b$ in $\mathrm{PSH}(\mathbb{P}^1 \times \mathbb{P}^{n-1}, \theta)$ such that

$$P_\theta[a]_I = b, \qquad \underline{\mathrm{vol}}_\theta(a) = \underline{\mathrm{vol}}_\theta(b) = 0, \qquad d_\theta(a,b) = 0.$$

For every $n \geq 3$, the same assertions hold for a suitable nonclosed positive form θ on a projective manifold of dimension n with the bounded mass property.

Proof Take the closed positive product background of Example 32.2.2 on $\mathbb{P}^1 \times \mathbb{P}^{n-1}$. Its models satisfy $a \leq b$ and $U(a) = U(b) = 0$, where $U = \overline{\mathrm{vol}}_\theta$ as in Definition 31.6.1. Thus $\max(a,b) = b$, and the defining formula gives

$$d_\theta(a,b) = 2U(b) - U(a) - U(b) = 0.$$

The other assertions are part of that example. For the nonclosed case use Example 32.2.3; its models are again ordered and have zero upper volumes, so the same calculation applies. □

We return to the product $M = C \times Z$ and its fixed divisor potential Ψ. All the models $\Psi + q \circ p$ have zero top-degree volumes, but the polar curvature of q on C still distinguishes them.

Definition 32.2.1 For a normalized ordinary model q in background η on C, put

$$\Phi(q) = \Psi + q \circ p, \qquad \mathcal{S}_\Psi = \{\Phi(q) : P_\eta[q] = q\}.$$

Write $\mu_q = \mathbf{1}_{\{q=-\infty\}}(\eta + \mathrm{dd}^{\mathrm{c}} q)$ for the polar curvature of q, and define

$$\rho_\Psi(\Phi(q_1), \Phi(q_2)) = \|\mu_{q_1} - \mu_{q_2}\|_{\mathrm{TV}}. \tag{32.5}$$

Proposition 32.2.2 *The set $\mathcal{S}_\Psi$ consists of normalized ordinary models in background Θ, and $(\mathcal{S}_\Psi, \rho_\Psi)$ is a complete geodesic metric space. The upper-volume comparison d_Θ, given by the formula of Definition 31.6.1 from the upper singularity volumes of a, b and $\max(a,b)$, which are finite here, vanishes identically on this set:*

$$d_\Theta(a,b) = 0 \qquad (a,b \in \mathcal{S}_\Psi).$$

The ideal models in $\mathcal{S}_\Psi$ form a closed geodesically convex subspace, equal to the ρ_Ψ-closure of the models with analytic singularities. Ideal projection is a 1-Lipschitz retraction onto this subspace and selects the unique nearest ideal model. For $a = \Phi(q)$, the distance from a to this subspace is the mass of the nonatomic part of μ_q. If $\rho_\Psi(a_j, a) \to 0$, then $a_j \to a$ in $L^1(M)$ for every smooth positive volume form.

Proof The identities for the zero obstacle in Proposition 32.2.1 give

$$P_\Theta[\Phi(q)] = \Phi(q), \qquad P_\Theta[\Phi(q)]_I = \Phi(P_\eta[q]_I).$$

Both summands of $\Phi(q)$ have supremum zero on their respective factors, so $\Phi(q)$ is normalized. Restriction to any section outside D shows that Φ is injective. The classification in Lemma 31.13.1 and the distance formula in Theorem 31.13.1 therefore make Φ an isometry from the space of curve models onto $\mathcal{S}_\Psi$. Completeness and the existence of geodesics follow from Corollary 31.13.1.

By Proposition 32.1.1, every ordinary representative of $\Phi(q)$ has zero extended mass. Moreover,

$$\max(\Phi(q_1), \Phi(q_2)) = \Psi + \max(q_1, q_2) \circ p,$$

and the same proposition, applied to the curve potential $\max(q_1, q_2)$, gives zero upper volume for the maximum. Thus all three upper volumes in the definition of d_Θ vanish.

Under Φ, ideal projection corresponds to taking the atomic part of μ_q. The assertions on closedness, convexity and nearest points therefore follow from Corollary 31.13.1 and Proposition 31.13.1. To prove analytic closure upstairs, approximate an atomic measure by finite atomic measures with rational coefficients and mass less than $\int_C \eta$. By Lemma 20.5.1, their curve models have the corresponding logarithmic poles up to bounded remainders. Adding Ψ preserves an analytic presentation: in the local product monomial, clear the denominators of the rational pole coefficients together with the unit coefficient of D. The resulting models in $\mathcal{S}_\Psi$ converge for ρ_Ψ. Conversely, by Proposition 17.2.4, every ordinary model with analytic singularities in this locus is equal to its ideal projection, so the closure of these models lies in the closed ideal subspace.

Finally, if $\rho_\Psi(\Phi(q_j), \Phi(q)) \to 0$, then Theorem 31.13.2 gives $q_j \to q$ in $L^1(C)$. Off the pole sets, which have smooth volume zero, the difference of the two product potentials is $(q_j - q) \circ p$. For a smooth positive volume form $\mathrm{d}V$ on M, integration along the compact fibers makes $p_* \mathrm{d}V$ a smooth positive density on C, and Fubini's theorem gives

$$\int_M |\Phi(q_j) - \Phi(q)|\,\mathrm{d}V \le C_{\mathrm{d}V}\|q_j - q\|_{L^1(C,\eta)} \longrightarrow 0,$$

where $C_{\mathrm{d}V}$ bounds the density of $p_*\mathrm{d}V$ with respect to η. □

The locus and its metric persist when an auxiliary curve factor with a variable nonnegative coefficient is added to the background, which can make it nonclosed or nonbig; see Corollary 32.3.1 with $m = 1$.

32.3 Several curve factors

A single curve factor produces a model space isometric to the space of models on one curve. Products of several curves give larger complete model spaces at zero volume. The envelopes of a sum of curve potentials are computed coordinatewise: the supremum of a competitor over all factors but one is a potential on the remaining curve, and restriction to slices treats the other factors by induction. The sum of the polar distances of the factors is then a complete metric on the induced product models.

Let $C_1, \dots, C_m$ be connected compact curves with smooth positive forms η_i, where $m \ge 1$, and put $A_i = \int_{C_i} \eta_i$. Set

$$Y = \prod_{i=1}^{m} C_i, \qquad \beta = \sum_{i=1}^{m} \pi_i^* \eta_i,$$

where $\pi_i \colon Y \to C_i$ are the projections.

The coordinatewise computation applies to arbitrary curve potentials, to curve backgrounds of nonnegative area that need not be positive, and to smooth obstacles that are sums of functions of the separate coordinates. Let σ_i be smooth real $(1,1)$-forms on C_i with $\mathrm{PSH}(C_i, \sigma_i) \ne \varnothing$, that is, with $\int_{C_i} \sigma_i \ge 0$ by Lemma 1.8.11, and choose $u_i \in \mathrm{PSH}(C_i, \sigma_i)$ and $F_i \in C^\infty(C_i, \mathbb{R})$. Put

$$\gamma = \sum_i \pi_i^* \sigma_i, \qquad u = \sum_i u_i \circ \pi_i, \qquad F = \sum_i F_i \circ \pi_i.$$

Proposition 32.3.1 *For the preceding data,*

$$\begin{aligned} P_{\gamma,F}[u] &= \sum_i P_{\sigma_i,F_i}[u_i]\circ\pi_i,\\ P_{\gamma,F}[u]_{\mathcal{I}} &= \sum_i P_{\sigma_i,F_i}[u_i]_{\mathcal{I}}\circ\pi_i. \end{aligned} \tag{32.6}$$

Moreover, $P_\gamma[u]_{\mathcal{I}}$ is ordinarily equivalent to a quasi-plurisubharmonic function g with $\mathrm{dd}^{\mathrm{c}} g = S - \beta'$, where β' is a smooth form and S is a divisorial current, possibly zero, as in Lemma 1.17.5, supported on coordinate fibers $\pi_i^{-1}(x)$.

Proof Write $e_i = P_{\sigma_i,F_i}[u_i]$. Sums of ordinary competitors in the separate coordinates are ordinary competitors on the product, and their regularized supremum is $\sum_i e_i\circ\pi_i$. This gives the lower bound in the first identity.

For the upper bound we use induction on the number of factors, the case of one factor being trivial. Let $v \le F$ be an ordinary competitor, so that $v \le u + C$ for some constant C. Choose a constant K with $u - K \le F$ and replace v by $\max(v, u - K)$. This preserves both bounds, after enlarging C if necessary, and ensures that the restrictions of v to the slices used below are not identically $-\infty$.

Write $Y = Y' \times C_m$, and on Y' put $u' = \sum_{i<m} u_i\circ\pi_i$ and $F' = \sum_{i<m} F_i\circ\pi_i$. The regularized partial supremum

$$a(t) = \left(\sup_{y'\in Y'}\big(v(y',t) - F'(y')\big)\right)^*$$

is σ_m-plurisubharmonic by the local upper-envelope theorem of Chapter 1: the functions $t \mapsto v(y',t) - F'(y')$ have the common upper bound F_m, and at least one of them is not identically $-\infty$, by the lower bound $v \ge u - K$. The bound $v \le u + C$ gives

$$a \le u_m + C + \sup_{Y'}(u' - F'), \qquad a \le F_m.$$

Thus a is a competitor for e_m, and $a \le e_m$. If $u_m(t)$ is finite, so is $e_m(t)$, since a sufficiently low translate of u_m is a competitor for e_m. The function $v(\cdot,t) - e_m(t)$ is then a potential in the product background on Y', not identically $-\infty$, lying below F' and at least as singular as u'. By induction, $v(\cdot,t) \le \sum_{i<m} e_i\circ\pi_i + e_m(t)$. At a pole of u_m, the bound $v \le u + C$ already gives $v = -\infty$. Every competitor is therefore bounded by $\sum_i e_i\circ\pi_i$, and taking the regularized supremum proves the ordinary identity.

For the ideal identity, put $B_i = \int_{C_i}\sigma_i \ge 0$ and $\widehat{\eta}_i = \eta_i/A_i$. By Lemma 1.8.11, choose smooth functions χ_i with

$$\sigma_i = B_i\widehat{\eta}_i + \mathrm{dd}^{\mathrm{c}}\chi_i.$$

Let E_i be the atomic part of $\sigma_i + \mathrm{dd}^{\mathrm{c}}u_i$. If $B_i > 0$, apply Lemma 20.5.1 to $u_i + \chi_i$ in the positive background $B_i\widehat{\eta}_i$. Subtracting χ_i from the resulting atomic Green potential gives a σ_i-plurisubharmonic function g_i such that

$$u_i \sim_{\mathcal{I}} g_i, \qquad P_{\sigma_i}[u_i]_{\mathcal{I}} \sim g_i,$$
$$\mathrm{dd}^{\mathrm{c}} g_i = E_i - E_i(C_i)\widehat{\eta}_i - \mathrm{dd}^{\mathrm{c}}\chi_i;$$

the second relation uses the smooth gauge bounds of Proposition 17.3.1. If $B_i = 0$, Lemma 1.8.11 gives $u_i = -\chi_i + c_i$ for a constant c_i, and $E_i = 0$; taking $g_i = -\chi_i$ gives the same conclusions. In either case,

$$\sigma_i + \mathrm{dd}^{\mathrm{c}} g_i = E_i + (B_i - E_i(C_i))\widehat{\eta}_i \geq 0.$$

Put $g = \sum_i g_i \circ \pi_i$. Pullback and addition, Lemmas 1.16.3 and 1.16.5, give $g \sim_{\mathcal{I}} u$. The complex Hessian of g is a countable sum of divisor currents supported on coordinate fibers, minus a smooth form. This divisor current has finite mass, because each coordinate fiber has constant mass for a fixed product metric and each E_i has finite mass. By Lemma 1.17.5, every ideal competitor $v \leq F$ for u satisfies $v \preceq g$, and is therefore bounded by $P_{\gamma,F}[g]$.

Set $r_i = P_{\sigma_i,F_i}[u_i]_{\mathcal{I}}$. Invariance of the ordinary envelope under ordinary equivalence and Proposition 17.6.6 give

$$P_{\sigma_i,F_i}[g_i] = P_{\sigma_i,F_i}[P_{\sigma_i}[u_i]_{\mathcal{I}}] = r_i.$$

By the ordinary identity already proved, applied to g, the ideal envelope of u is therefore at most $\sum_i r_i \circ \pi_i$. Conversely, each r_i lies below F_i and is multiplier-ideal equivalent to u_i, again by Proposition 17.6.6. Their sum is a γ-plurisubharmonic ideal competitor for u below F, which proves the reverse inequality. For $F = 0$, the ideal identity and $r_i \sim g_i$ give $P_\gamma[u]_{\mathcal{I}} \sim g$, which proves the last assertion. □

The proof of (32.6) uses that the obstacle is a sum of functions of the separate coordinates; for an arbitrary smooth obstacle on Y, the coordinatewise bounds are not available. Equality of the two envelopes below such mixed obstacles is treated in Section 32.3.3.

For normalized ordinary models q_i on (C_i, η_i), write

$$w = \sum_{i=1}^{m} q_i \circ \pi_i, \qquad q_{i,\mathcal{I}} = P_{\eta_i}[q_i]_{\mathcal{I}}.$$

By the case $\sigma_i = \eta_i$, $F = 0$ of Proposition 32.3.1, w is a normalized ordinary model on (Y, β) and $P_\beta[w]_{\mathcal{I}} = \sum_i q_{i,\mathcal{I}} \circ \pi_i$.

We now add a divisorial factor (Z, κ, D, ψ) as in Section 32.2. On $M = Y \times Z$, write p, r for the projections, set $h_i = \pi_i \circ p$, and put

$$\Theta = p^*\beta + r^*\kappa, \qquad \Psi = \psi \circ r, \qquad \Phi(q_1, \ldots, q_m) = \Psi + \sum_i q_i \circ h_i.$$

Let $\mathcal{S}_\Psi^{(m)}$ be the set of these potentials. For the polar curvatures $\mu_i = \mathbf{1}_{\{q_i = -\infty\}}(\eta_i + \mathrm{dd}^{\mathrm{c}} q_i)$ and the analogous measures ν_i of models b_i, define

$$\rho_\Psi^{(m)}(\Phi(q_1,\dots,q_m),\Phi(b_1,\dots,b_m)) = \sum_i \|\mu_i - \nu_i\|_{\mathrm{TV}}. \tag{32.7}$$

Proposition 32.3.2 *The set $\mathcal{S}_\Psi^{(m)}$ consists of normalized ordinary models and is a complete geodesic metric space for $\rho_\Psi^{(m)}$. Its ideal models form a closed geodesically convex subspace, equal to the metric closure of its models with analytic singularities. Ideal projection is the unique nearest-point retraction onto this subspace and is* 1*-Lipschitz, and the distance to the subspace is the sum of the nonatomic polar masses. Metric convergence implies $L^1(M)$ convergence for every smooth positive volume form. The upper-volume comparison d_Θ vanishes on every pair of points of this locus.*

Proof For $m = 1$ this is Proposition 32.2.2, and its proof applies coordinatewise. By the identities for w above and Proposition 32.1.1, whose induced zero obstacle is zero,

$$P_\Theta[\Phi(q_1,\dots,q_m)] = \Phi(q_1,\dots,q_m),$$
$$P_\Theta[\Phi(q_1,\dots,q_m)]_I = \Phi(q_{1,I},\dots,q_{m,I}),$$

and independence of the factors gives normalization. Restriction to sections on which the other potentials are finite determines each q_i up to a constant, which the normalization fixes. Thus Φ identifies the locus with the product of the spaces of curve models, with the sum of their metrics, and ideal projection takes the atomic part in each coordinate. Completeness, the linear interpolation of all polar measures as geodesics, and the assertions on the ideal subspace, its nearest points and the Lipschitz bound follow coordinate by coordinate from Corollary 31.13.1 and Proposition 31.13.1; the distances add. Finite rational atomic approximations in each coordinate give analytic models, as in the case $m = 1$, and every model with analytic singularities is ideal by Proposition 17.2.4. The L^1 convergence follows from Theorem 31.13.2 by integration along the compact fibers of each h_i. Finally, since $\dim Y < \dim M$, Proposition 32.1.1 gives zero upper volume for every model and every pairwise maximum, which is Ψ plus the pullback of a β-plurisubharmonic function on Y; hence d_Θ vanishes. □

An auxiliary curve factor with a variable coefficient makes the background nonclosed, and lets it degenerate, without changing this metric locus. Let C_0 be another connected compact curve with a smooth positive form τ. On $M = Y \times C_0 \times Z$, write p, s, r for the projections. For a smooth real function $f \geq 0$ on Y, put

$$\Theta_f = p^*\beta + (f \circ p)s^*\tau + r^*\kappa, \qquad \Psi = \psi \circ r,$$

and set $\Phi(q_1,\dots,q_m) = \Psi + \sum_i q_i \circ \pi_i \circ p$.

Corollary 32.3.1 *For every tuple of normalized ordinary curve models, one has*

$$P_{\Theta_f}[\Phi(q_1,\dots,q_m)] = \Phi(q_1,\dots,q_m),$$
$$P_{\Theta_f}[\Phi(q_1,\dots,q_m)]_{\mathcal{I}} = \Phi(q_{1,\mathcal{I}},\dots,q_{m,\mathcal{I}}).$$

On the locus of these potentials, use the sum of polar distances (32.7)*. All the conclusions of Proposition 32.3.2 hold, and the diameter is* $2\sum_i A_i$*. The locus, its metric, ideal projection and the analytic closure are independent of* f*, including at* $f = 0$*. The background* Θ_f *is positive when* $f > 0$ *and not closed when* f *is nonconstant, and* Θ_0 *is not big.*

Proof On $B = Y \times C_0$, let π be the projection to Y and put $\beta_f = \pi^*\beta + (f \circ \pi)s_0^*\tau$, where s_0 is the projection to C_0. By Proposition 32.3.1, $P_\beta[w] = w$, and $P_\beta[w]_{\mathcal{I}} = \sum_i q_{i,\mathcal{I}} \circ \pi_i$ satisfies the divisorial hypothesis of Lemma 32.2.1. That lemma, with $T = C_0$ and $\alpha = (f \circ \pi)s_0^*\tau$, gives

$$P_{\beta_f}[w \circ \pi] = w \circ \pi, \qquad P_{\beta_f}[w \circ \pi]_{\mathcal{I}} = \sum_i q_{i,\mathcal{I}} \circ \pi_i \circ \pi.$$

Proposition 32.1.1, applied to $M \to B$ and the divisor $B \times D$, whose induced zero obstacle is zero, gives the two identities upstairs, and normalization holds because the factors are independent.

The conclusions of Proposition 32.3.2 now follow by the same coordinatewise arguments, since restriction to sections determines the tuple and the identities show that ideal projection takes the atomic part in each coordinate. The analytic models used there pull back to analytic models under the projection forgetting C_0, and every model with analytic singularities is ideal by Proposition 17.2.4. Fiber integration, now including the factor C_0, gives L^1 convergence. Every model and every pairwise maximum is Ψ plus a pullback from B, and $\dim B < \dim M$, so Proposition 32.1.1 makes d_{Θ_f} vanish. None of these statements involves f. Each polar measure μ_i has mass at most A_i, so the diameter is at most $2\sum_i A_i$; by Lemma 31.13.1, the bound is attained by placing mass A_i at two distinct points of each C_i.

Finally, Θ_f is positive when $f > 0$, and $\mathrm{d}\Theta_f = p^*(\mathrm{d}f) \wedge s^*\tau$ is nonzero for nonconstant f. To see that Θ_0 is not big, put $n = m + 1 + k$ and use the smooth closed positive $(n-1, n-1)$-form T and the Hermitian form Ω given by

$$T = p^*\left(\bigwedge_i \pi_i^*\eta_i\right) \wedge r^*(\kappa^k), \qquad \Omega = p^*\beta + s^*\tau + r^*\kappa.$$

Then $\int_M \Theta_0 \wedge T = 0$, whereas $\int_M \Omega \wedge T > 0$. By Stokes' theorem, a lower bound $\Theta_0 + \mathrm{dd}^{\mathrm{c}}u \geq \varepsilon\Omega$ with u quasi-plurisubharmonic and $\varepsilon > 0$ would contradict these identities. □

32.3.1 Varying the curve coefficients

The coefficient f of the auxiliary curve in Corollary 32.3.1 does not affect the metric on the model locus. Scaling the curve forms themselves has a different effect: a positive coefficient rescales the polar distance in its coordinate, whereas a zero coefficient removes the coordinate altogether, so the diameter depends on the remaining curve areas. We describe the resulting family of model spaces and then compare its members.

Keep the product $M = Y \times C_0 \times Z$, its projections and the forms of that corollary. For $a = (a_1, \dots, a_m) \in [0, \infty)^m$ and a smooth function $f \geq 0$ on Y, put

$$J(a) = \{i : a_i > 0\}, \qquad \beta_a = \sum_i a_i \pi_i^* \eta_i, \qquad \Theta_{a,f} = p^* \beta_a + (f \circ p) s^* \tau + r^* \kappa.$$

The coordinates $i \in J(a)$ are called *active*. For normalized ordinary models q_i in the fixed curve backgrounds η_i, define

$$\Phi_a(q) = \Psi + \sum_{i \in J(a)} a_i q_i \circ \pi_i \circ p, \qquad \Psi = \psi \circ r.$$

Terms with zero coefficient are omitted, also on their pole sets, so $\Phi_a(q)$ depends only on the active coordinates. Write $\mathcal{S}_a$ for this locus of potentials, and $q_{i,I} = P_{\eta_i}[q_i]_I$.

Lemma 32.3.1 *For every a and f as above,*

$$P_{\Theta_{a,f}}[\Phi_a(q)] = \Phi_a(q), \qquad P_{\Theta_{a,f}}[\Phi_a(q)]_I = \Phi_a(q_I).$$

Both potentials have supremum zero. For $a = 0$, every tuple gives $\Phi_0(q) = \Psi$, so $\mathcal{S}_0 = \{\Psi\}$.

Proof For $a_i > 0$, put $\sigma_i = a_i \eta_i$ and $u_i = a_i q_i$; for $a_i = 0$, put $\sigma_i = 0$ and $u_i = 0$. Then $\beta_a = \sum_i \pi_i^* \sigma_i$ and $w_a = \sum_{i \in J(a)} a_i q_i \circ \pi_i = \sum_i u_i \circ \pi_i$. Multiplication by a_i is a bijection between the competitor families, so $P_{\sigma_i}[u_i] = a_i q_i$, and Proposition 17.2.3 gives $P_{\sigma_i}[u_i]_I = a_i q_{i,I}$; for $a_i = 0$ both envelopes of $u_i = 0$ are zero. By Proposition 32.3.1,

$$P_{\beta_a}[w_a] = w_a, \qquad P_{\beta_a}[w_a]_I = w_{a,I} = \sum_{i \in J(a)} a_i q_{i,I} \circ \pi_i,$$

and $w_{a,I}$ satisfies the divisorial hypothesis of Lemma 32.2.1. As in the proof of Corollary 32.3.1, that lemma on $B = Y \times C_0$, with the auxiliary term $(f \circ \pi) s_0^* \tau$, and Proposition 32.1.1, applied to $M \to B$ and the divisor $B \times D$, give the identities upstairs. Independence of the factors gives normalization. □

For two tuples $q, \widetilde{q}$, let μ_i, ν_i be their polar curvature measures in the fixed backgrounds η_i, and set

$$\rho_a(\Phi_a(q),\Phi_a(\widetilde{q})) = \sum_{i\in J(a)} a_i \|\mu_i - \nu_i\|_{\mathrm{TV}}. \tag{32.8}$$

On the singleton $\mathcal{S}_0$ this distance is zero.

Proposition 32.3.3 *For every a, the locus $\mathcal{S}_a$ is a complete geodesic metric space for ρ_a, independently of f. Its ideal subspace is closed, geodesically convex and equal to the metric closure of its models with analytic singularities. Ideal projection is the unique nearest-point retraction onto this subspace and is* 1*-Lipschitz, with distance*

$$\rho_a\big(\Phi_a(q),\Phi_a(q_{\mathcal{I}})\big) = \sum_{i\in J(a)} a_i \mu_i^{\mathrm{na}}(C_i),$$

where μ_i^{na} is the nonatomic part of μ_i. The diameter is $2\sum_i a_i A_i$, and metric convergence implies $L^1(M)$ convergence for every smooth positive volume form. The upper-volume comparison vanishes on this locus.

If a, b have the same nonempty active set J, the correspondence $T_{ba}\Phi_a(q) = \Phi_b(q)$ is well defined, commutes with ideal projection and satisfies, for $u, v \in \mathcal{S}_a$,

$$\min_{i\in J}\frac{b_i}{a_i}\,\rho_a(u,v) \le \rho_b(T_{ba}u, T_{ba}v) \le \max_{i\in J}\frac{b_i}{a_i}\,\rho_a(u,v).$$

For a fixed smooth positive volume form on M, there are constants K_i, independent of a, b and of the normalized model tuple q, such that

$$\|\Phi_a(q) - \Phi_b(q)\|_{L^1(M)} \le \sum_i K_i |a_i - b_i|.$$

Consequently, as all a_i tend to zero, the diameter tends to zero and every potential in $\mathcal{S}_a$ tends to Ψ in $L^1(M)$, uniformly over the normalized model tuples.

Proof Restriction to coordinate sections on which the other potentials are finite determines every active term $a_i q_i$ up to a constant, and normalization makes these constants zero. Thus (32.8) is well defined, and it identifies $\mathcal{S}_a$ with the product of the spaces of curve models in the active coordinates, with their distances multiplied by a_i. Coordinates with $a_i = 0$ do not appear, and for $J(a) = \varnothing$ the empty product is a single point.

The completeness, geodesic and atomic-projection results of Corollary 31.13.1 and Proposition 31.13.1 apply coordinate by coordinate, and the weighted distances add; this proves the metric assertions and the distance formula. The diameter bound is attained by choosing polar measures $A_i\delta_{x_i}$ and $A_i\delta_{y_i}$ with $x_i \neq y_i$ on each active curve. Fiber integration and Theorem 31.13.2 give L^1 convergence upstairs. The preceding lemma shows that ideal projection does not depend on f. Since every model and every pairwise maximum is Ψ plus a pullback from B, Proposition 32.1.1 gives zero upper volumes, so the upper-volume comparison vanishes.

For analytic density, approximate the atomic part $a_i\mu_i^{\mathrm{at}}$ of each scaled polar measure, where $\mu_i^{\mathrm{at}} = \mu_i - \mu_i^{\mathrm{na}}$, by finite atomic measures with rational coefficients and mass at most $a_i A_i$. These approximations converge in total variation. Their

curve models in the background $a_i\eta_i$ have finitely many logarithmic poles with rational coefficients, whose denominators, together with the unit coefficient of D, can be cleared in local product monomials. They therefore give analytic models in $\mathcal{S}_a$ converging to every ideal model. Conversely, every model in $\mathcal{S}_a$ with analytic singularities is an ideal model by Proposition 17.2.4, and closedness gives the analytic-closure assertion. The rational coefficients are needed for the scaled poles a_iq_i themselves: with the convention of Section 1.13, an irrational scaling need not preserve an analytic curve model.

For a common active set, the normalized tuple is determined by the potential. The two bounds for T_{ba} follow by comparing each positive weight b_i with a_i, and atomic projection in each coordinate commutes with this correspondence.

Finally, normalized compactness, Theorem 1.4.1, bounds the $L^1(C_i)$ norms of all the q_i uniformly. Integration along the compact fibers of $\pi_i \circ p$ gives constants K_i bounding the $L^1(M)$ norms of their pullbacks. Off the pole sets, which have smooth volume zero, the difference of the two potentials is the sum of the functions $(a_i - b_i)q_i \circ \pi_i \circ p$. The triangle inequality proves the uniform estimate, also when some coefficients vanish. Taking $b = 0$ gives uniform convergence to Ψ, and the diameter formula shows that the diameters tend to zero. □

32.3.1.1 Comparing bounded model spaces

When the active sets differ, the coordinatewise correspondence T_{ba} no longer defines a bijection between $\mathcal{S}_a$ and $\mathcal{S}_b$. We compare them as abstract metric spaces. They are bounded, but they are not compact as soon as some coefficient is positive. The Gromov–Hausdorff distance d_{GH} of bounded metric spaces from Definition 15.18.1, together with its estimate through correspondences in Lemma 15.18.5, applies without a compactness hypothesis.

Write $\mathcal{I}_a \subset \mathcal{S}_a$ for the ideal subspace of Proposition 32.3.3, with the restricted metric ρ_a.

Proposition 32.3.4 *For all nonnegative coefficient vectors a, b, including vectors with different active sets,*

$$\left|\sum_i (a_i - b_i)A_i\right| \le d_{\mathrm{GH}}(\mathcal{S}_a, \mathcal{S}_b) \le \sum_i |a_i - b_i|A_i. \tag{32.9}$$

The same bounds hold for $d_{\mathrm{GH}}(\mathcal{I}_a, \mathcal{I}_b)$. If $b_i \le a_i$ for every i, both distances equal $\sum_i (a_i - b_i)A_i$. In particular, the distance from $\mathcal{S}_a$ to the singleton $\mathcal{S}_0 = \mathcal{I}_0 = \{\Psi\}$ equals $\sum_i a_i A_i$.

For $b \le a$, the map $T_{ba}\Phi_a(q) = \Phi_b(q)$ is a surjective 1-Lipschitz map and commutes with ideal projection. Every fiber has diameter

$$2 \sum_{\{i : a_i > 0,\ b_i = 0\}} a_i A_i.$$

If some $a_i > 0$, then $\mathcal{S}_a$ and $\mathcal{I}_a$ are not totally bounded, and hence not compact.

Proof Use the correspondence

$$R_{ab} = \{(\Phi_a(q), \Phi_b(q)) : q_i \text{ are normalized ordinary curve models}\}.$$

Every potential in either locus arises from such a tuple, with arbitrary normalized models in its coordinates of zero coefficient, so both projections of R_{ab} are surjective. For any two tuples, put $d_i = \|\mu_i - \nu_i\|_{\mathrm{TV}}$. By (32.8), the difference of their two distances is $\sum_i (a_i - b_i)d_i$, with $0 \le d_i \le 2A_i$. Thus the distortion of this correspondence satisfies $\operatorname{dis} R_{ab} \le 2\sum_i |a_i - b_i|A_i$, and Lemma 15.18.5 gives the upper bound in (32.9). The diameter formula in Proposition 32.3.3 and the same lemma give the lower bound.

Restricting the correspondence to tuples of ideal curve models, we obtain a correspondence between the two ideal subspaces with the same distortion bound. The atomic measures $A_i\delta_{x_i}$ and $A_i\delta_{y_i}$ show that the ideal subspaces have the same diameters as the full spaces, so the lower bound also holds for them. When $b \le a$, the upper and lower bounds coincide, including when $b = 0$.

For the transition map, every active coordinate of b is active for a, and uniqueness of the normalized active tuple makes the map well defined. Every b-tuple extends to an a-tuple, so the map is surjective, and $\sum_i b_i d_i \le \sum_i a_i d_i$ proves the Lipschitz bound. Atomic projection in each coordinate commutes with discarding and rescaling coordinates. In a fiber, every coordinate with $b_i > 0$ is fixed, while the coordinates with $a_i > 0$ and $b_i = 0$ vary freely. The diameter of the fiber is therefore the sum above, by the calculation of the diameter on each curve.

Finally, fix an active coordinate i and all other coordinates at their zero models. Let q_i have polar measure $A_i\delta_x$, with x ranging over infinitely many distinct points of C_i. The resulting potentials are ideal models, and any two of them are at distance $2a_iA_i$. This infinite separated set excludes total boundedness of both spaces. In particular, the comparison above is between complete bounded spaces that are not compact. □

32.3.2 Arbitrary curve backgrounds

Scaling by coefficients is a special case of changing the curve backgrounds. By Proposition 32.3.1, the coordinatewise computation of the envelopes holds for arbitrary smooth curve backgrounds of nonnegative area, so the product metric depends on the areas of the curve backgrounds rather than on their positivity. A curve background of area zero has a unique normalized potential, which is smooth. Its coordinate no longer contributes to the metric, but its smooth potential remains a summand of the product potentials.

Keep $M = Y \times C_0 \times Z$, the projections p, s, r and Ψ from Corollary 32.3.1. On each C_i, choose a smooth real $(1,1)$-form σ_i with $\mathrm{PSH}(C_i, \sigma_i) \neq \varnothing$, and put $B_i = \int_{C_i} \sigma_i \geq 0$. For a smooth function $f \geq 0$ on Y, set

$$\gamma_\sigma = \sum_i \pi_i^* \sigma_i, \qquad \Theta_{\sigma,f} = p^* \gamma_\sigma + (f \circ p) s^* \tau + r^* \kappa.$$

For normalized ordinary models q_i in the curve backgrounds σ_i, write

$$\Phi_\sigma(q) = \Psi + \sum_i q_i \circ \pi_i \circ p, \qquad q_{i,\mathcal{I}} = P_{\sigma_i}[q_i]_{\mathcal{I}},$$

and let $\mathcal{S}_\sigma$ be the locus of these potentials. All coordinates are included: when $B_i = 0$, q_i is the unique normalized potential u_{σ_i} of Lemma 1.8.11, which is smooth.

Lemma 32.3.2 *Every $\Phi_\sigma(q)$ is a normalized ordinary model, and*

$$P_{\Theta_{\sigma,f}}[\Phi_\sigma(q)]_{\mathcal{I}} = \Phi_\sigma(q_{\mathcal{I}}).$$

Proof By Proposition 32.3.1, $w = \sum_i q_i \circ \pi_i$ satisfies $P_{\gamma_\sigma}[w] = w$, and $P_{\gamma_\sigma}[w]_{\mathcal{I}} = \sum_i q_{i,\mathcal{I}} \circ \pi_i$ satisfies the divisorial hypothesis of Lemma 32.2.1. As in the proof of Corollary 32.3.1, that lemma on $B = Y \times C_0$, with projections π to Y and s_0 to C_0 and the auxiliary term $(f \circ \pi) s_0^* \tau$, and Proposition 32.1.1, applied to $M \to B$ and the divisor $B \times D$, give the assertion; normalization follows from the independence of the factors. □

By Corollary 32.3.4, applied with $T = C_0$ and $\alpha = (f \circ \pi) s_0^* \tau$, the background $\Theta_{\sigma,f}$ is not big if $f = 0$ or some $B_i = 0$, it is big if $f > 0$ and all $B_i > 0$, and it is not closed if f is nonconstant.

For the polar curvatures μ_i of q_i and ν_i of another tuple $\widetilde{q}$, taken in the backgrounds σ_i, set

$$\rho_\sigma(\Phi_\sigma(q), \Phi_\sigma(\widetilde{q})) = \sum_i \|\mu_i - \nu_i\|_{\mathrm{TV}}. \tag{32.10}$$

Write $\mathcal{I}_\sigma \subset \mathcal{S}_\sigma$ for the ideal subspace and μ_i^{na} for the nonatomic part of μ_i.

Proposition 32.3.5 *The metric space $(\mathcal{S}_\sigma, \rho_\sigma)$ is complete and geodesic, with diameter $2\sum_i B_i$. The ideal subspace is closed, geodesically convex and equal to the metric closure of the models with analytic singularities. Ideal projection is the unique nearest-point retraction onto it and is 1-Lipschitz, and the distance to the subspace is $\sum_i \mu_i^{\mathrm{na}}(C_i)$. Metric convergence implies $L^1(M)$ convergence for every smooth positive volume form. These assertions are independent of f, and the upper-volume comparison vanishes on this locus.*

Let σ_i' be another tuple of smooth real forms with $\mathrm{PSH}(C_i, \sigma_i') \neq \varnothing$, with areas B_i'. Then

$$\left|\sum_i (B_i - B_i')\right| \leq d_{\mathrm{GH}}(\mathcal{S}_\sigma, \mathcal{S}_{\sigma'}) \leq \sum_i |B_i - B_i'|. \tag{32.11}$$

The same bounds hold for the ideal subspaces. If $B_i' \leq B_i$ for every i, both distances equal $\sum_i (B_i - B_i')$. In this case, rescaling the polar measures by the area ratios, with the unique normalized potential in each target coordinate of zero area, gives a surjective 1-Lipschitz map that commutes with ideal projection. Its fibers have diameter $2\sum_{\{i:B_i>0,\ B_i'=0\}} B_i$.

Both $\mathcal{S}_\sigma$ and $\mathcal{I}_\sigma$ are noncompact if some $B_i > 0$. If all $B_i = 0$, they coincide with the singleton $\{\Psi + \sum_i u_{\sigma_i} \circ \pi_i \circ p\}$.

Proof Restriction to coordinate sections on which the other potentials are finite determines each q_i up to a constant, and normalization fixes that constant. Put $a_i = B_i/A_i$, and for a polar measure μ on C_i of mass at most B_i write $p_{\sigma_i}(\mu)$ for the ordinary model in background σ_i with polar curvature μ, as in Proposition 31.14.1. Sending a curve model for η_i with polar measure μ_i, of mass at most A_i, to $p_{\sigma_i}(a_i\mu_i)$, which for $a_i = 0$ is the unique normalized potential, defines an isometry Λ_σ from the weighted locus $\mathcal{S}_a$ of Proposition 32.3.3 onto $\mathcal{S}_\sigma$. By the preceding lemma and Proposition 31.14.1, ideal projection takes the atomic part in each coordinate on both loci, so Λ_σ commutes with ideal projection and maps the ideal subspaces onto each other. Completeness, geodesics, the diameter, the assertions on the ideal subspace and its nearest-point retraction, the distance formula and noncompactness therefore follow from Propositions 32.3.3 and 32.3.4. Applying Proposition 32.3.4 with $a_i = B_i/A_i$ and $b_i = B_i'/A_i$ gives (32.11), the exact value for ordered areas, the contractions and the diameters of their fibers. In curve coordinates the contraction is

$$p_{\sigma_i}(\mu) \longmapsto \begin{cases} p_{\sigma_i'}((B_i'/B_i)\mu), & B_i > 0, \\ p_{\sigma_i'}(0), & B_i = 0. \end{cases}$$

In the second case $B_i' = 0$, because $B_i' \leq B_i$. Atomic projection commutes with both operations. As in Theorem 31.14.1, irrational area ratios need not preserve individual analytic models, but the ideal subspaces are preserved for every ratio. If all areas are zero, every curve coordinate is fixed at its unique normalized smooth potential, which proves the last assertion.

For analytic density, approximate each atomic polar measure by finite atomic measures with rational coefficients and mass at most B_i. By the atomic Green construction and transport under smooth gauge changes, their models in background

σ_i have the corresponding finitely many poles with rational coefficients, up to bounded remainders. Coordinates of zero area are smooth. Together with the unit coefficient of D, the poles in the coordinates of positive area allow the denominators to be cleared in local product monomials, which gives analytic models in the locus. They converge for the metric (32.10). Conversely, every model with analytic singularities is ideal by Proposition 17.2.4, and closedness proves the analytic-closure assertion.

Metric convergence gives L^1 convergence on the curves by Proposition 31.14.1, and fiber integration transfers it to M. By Proposition 32.1.1, applied to the divisor $B \times D$ with weight one, every model and every pairwise maximum in this locus has zero upper volume; for a maximum, use Ψ plus the pullback of the maximum of the two potentials on B. Thus all three terms of the upper-volume comparison vanish. None of these arguments involves f. □

Corollary 32.3.2 *For each i, let $\sigma_{i,j}$ be smooth real forms on C_i of nonnegative area that converge uniformly, as $j \to \infty$, to a smooth form $\sigma_{i,0}$ of area zero, and write $\sigma_j = (\sigma_{1,j}, \dots, \sigma_{m,j})$. For any nonnegative smooth auxiliary coefficients f_j, the diameters of $\mathcal{S}_{\sigma_j}$ tend to zero. For every fixed smooth positive volume form on M, the potentials in $\mathcal{S}_{\sigma_j}$ converge in $L^1(M)$ to*

$$\Psi + \sum_i u_{\sigma_{i,0}} \circ \pi_i \circ p,$$

uniformly over all normalized model tuples.

Proof The areas $B_{i,j}$ tend to zero, so the diameter formula applies. By Theorem 1.8.1, all normalized $\sigma_{i,j}$-plurisubharmonic potentials converge in $L^1(C_i)$ to $u_{\sigma_{i,0}}$, uniformly over this family. Integration along the fixed compact fibers bounds each pulled-back norm by a constant times the norm on the curve, and the finite sum gives the asserted uniform limit on M. The auxiliary coefficients do not enter these potentials or their metrics. □

Thus the metric comparison sees only the areas of the curve backgrounds, while the limiting potential still records their smooth gauges. These statements concern the separated product locus; extending the metric to general ordinary models is part of Problem 35.7.1.

32.3.3 Equality below all obstacles on separated products

We return to the question raised in Section 17.6.4: does equality of the two envelopes below zero imply their equality below every smooth obstacle? For sums of curve potentials the answer is yes, even for obstacles that mix the coordinates. By Proposition 17.6.6, membership in the relative envelope class of the ideal model implies equality below every obstacle, and this membership can be tested coordinatewise, because stretching competitors can be added before any obstacle is imposed.

Lemma 32.3.3 *Let $N \geq 1$, let X_j, $1 \leq j \leq N$, be connected compact complex manifolds with smooth real $(1,1)$-forms θ_j, and let $u_j, \varphi_j \in \mathrm{PSH}(X_j, \theta_j)$. On $X = \prod_j X_j$, with projections pr_j, put*

$$\vartheta = \sum_j \mathrm{pr}_j^* \theta_j, \qquad u = \sum_j u_j \circ \mathrm{pr}_j, \qquad \varphi = \sum_j \varphi_j \circ \mathrm{pr}_j .$$

Then

$$u \in \mathcal{E}_{\mathrm{env}}(X, \vartheta, \varphi) \quad \Longleftrightarrow \quad u_j \in \mathcal{E}_{\mathrm{env}}(X_j, \theta_j, \varphi_j) \quad \text{for every } j.$$

Proof The ordinary order between the sums is equivalent to the ordinary order in each coordinate. Indeed, the latter follows from the former by fixing the other coordinates at points where their potentials are finite and restricting the inequality, and the converse follows by adding the coordinate inequalities.

Suppose that each u_j belongs to its relative class. For $b > 1$, choose stretching competitors h_j as in Definition 14.4.1. The sum $h = \sum_j h_j \circ \mathrm{pr}_j$ is ϑ-plurisubharmonic and satisfies

$$b^{-1} h + (1 - b^{-1})\varphi \leq u.$$

This convex inequality is meaningful at poles. For $b = 1$, use $h = u$. Thus u belongs to the relative class on the product.

Conversely, fix $b > 1$ and a stretching competitor h on the product. Choose a point $x = (x_1, \ldots, x_N)$ at which h and all the coordinate potentials are finite; such points have full smooth volume. For each j, let $h^{(j)}$ be the restriction of h to the slice through x in the jth factor. It is θ_j-plurisubharmonic and not identically $-\infty$, since it is finite at x_j. Put

$$c_j = \sum_{\ell \neq j} u_\ell(x_\ell), \qquad d_j = \sum_{\ell \neq j} \varphi_\ell(x_\ell).$$

Restricting the product inequality to the slice gives

$$b^{-1}\big(h^{(j)} - b c_j + (b-1) d_j\big) + (1 - b^{-1})\varphi_j \leq u_j.$$

This gives a stretching competitor in the jth coordinate for every $b > 1$. Together with the ordinary order and the competitor u_j for $b = 1$, it proves the converse. □

Keep the curves C_i and the smooth forms σ_i with $\mathrm{PSH}(C_i, \sigma_i) \neq \varnothing$ from Section 32.3.2, and choose arbitrary $u_i \in \mathrm{PSH}(C_i, \sigma_i)$. Put

$$u = \sum_i u_i \circ \pi_i, \qquad q_i = P_{\sigma_i}[u_i], \qquad r_i = P_{\sigma_i}[u_i]_I, \qquad Q = \sum_i r_i \circ \pi_i.$$

Write μ_i and E_i for the polar and the atomic parts of the positive curvature measure $\sigma_i + \mathrm{dd}^c u_i$, respectively.

Proposition 32.3.6 *For these data, $Q = P_{\gamma_\sigma}[u]_I$. The following are equivalent:*

$$P_{\gamma_\sigma}[u] = P_{\gamma_\sigma}[u]_I, \qquad \mu_i = E_i \quad \textit{for every } i,$$
$$u \in \mathcal{E}_{\mathrm{env}}(Y, \gamma_\sigma, Q),$$
$$P_{\gamma_\sigma, F}[u] = P_{\gamma_\sigma, F}[u]_I \quad \textit{for every smooth real } F \textit{ on } Y.$$

Proof For the zero obstacle, Proposition 32.3.1 identifies the two envelopes with $\sum_i q_i \circ \pi_i$ and Q, respectively. Since the coordinates are independent and all these functions have supremum zero, the two sums are equal if and only if $q_i = r_i$ for every i. By Corollary 31.14.1, this is equivalent to $\mu_i = E_i$, and also to $u_i \in \mathcal{E}_{\mathrm{env}}(C_i, \sigma_i, r_i)$, for every i. The preceding lemma identifies these memberships with membership in the relative class on the product. Finally, Proposition 17.6.6 gives equality below every smooth obstacle, and taking $F = 0$ gives the reverse implication. This proves all four equivalences. □

The same criterion holds after adding a divisor of weight one and an auxiliary factor, by Corollary 32.3.3. These criteria give neither coordinatewise formulas for the envelopes below mixed obstacles nor the implication for general potentials of zero volume.

32.3.4 Auxiliary compact complex factors

So far, the auxiliary factor that makes the background nonclosed has been a curve. By Lemma 32.2.1, only the semipositivity of the auxiliary term and its vanishing on the slices parallel to the curve product matter. The auxiliary factor can therefore be any compact complex manifold of positive dimension, Kähler or not, and the potentials and their polar metric still depend only on the curve coordinates and the fixed divisor of weight one.

Keep Y, σ_i, B_i and γ_σ, as well as the data (Z, κ, D, ψ) of the divisor of weight one. Let T be a connected compact complex manifold of dimension $\ell \geq 1$, and put $B = Y \times T$, with projections π to Y and pr_T to T. Choose a smooth semipositive real $(1,1)$-form α on B whose restriction to every slice $Y \times \{t\}$ is zero. By semipositivity, the tangent spaces of these slices lie in the kernel of α, so in product coordinates α has only coefficients in the T-directions. On $M = B \times Z$, let ϖ, r be the projections, and put $p = \pi \circ \varpi$ and

$$\Gamma_\alpha = \pi^* \gamma_\sigma + \alpha, \qquad \Theta_\alpha = \varpi^* \Gamma_\alpha + r^* \kappa, \qquad \Psi = \psi \circ r.$$

We assume neither that α is closed nor that M has the bounded mass property. For $T = C_0$ and $\alpha = (f \circ \pi)\,\mathrm{pr}_T^*\,\tau$, Θ_α is the background $\Theta_{\sigma,f}$ of Section 32.3.2.

Lemma 32.3.4 *For $u_i \in \mathrm{PSH}(C_i, \sigma_i)$ put*

$$u = \sum_i u_i \circ \pi_i, \qquad U = \Psi + u \circ p, \qquad q_i = P_{\sigma_i}[u_i], \quad r_i = P_{\sigma_i}[u_i]_{\mathcal{I}}.$$

Then

$$P_{\Theta_\alpha}[U] = \Psi + \sum_i q_i \circ \pi_i \circ p,$$
$$P_{\Theta_\alpha}[U]_{\mathcal{I}} = \Psi + \sum_i r_i \circ \pi_i \circ p.$$

Every ordinary representative of either of these envelopes has zero extended mass, so their upper and lower singularity volumes are zero.

Proof By Proposition 32.3.1, $P_{\gamma_\sigma}[u] = \sum_i q_i \circ \pi_i$, and $P_{\gamma_\sigma}[u]_{\mathcal{I}} = \sum_i r_i \circ \pi_i$ satisfies the divisorial hypothesis of Lemma 32.2.1. That lemma gives

$$P_{\Gamma_\alpha}[u \circ \pi] = \sum_i q_i \circ \pi_i \circ \pi, \qquad P_{\Gamma_\alpha}[u \circ \pi]_{\mathcal{I}} = \sum_i r_i \circ \pi_i \circ \pi.$$

Proposition 32.1.1, applied to $M \to B$ and the divisor $B \times D$ with weight one, whose induced zero obstacle is zero because $\sup_Z \psi = 0$, gives both identities upstairs, and its mass assertion gives the volume assertions. □

For normalized ordinary curve models q_i, write

$$\Phi_\alpha(q) = \Psi + \sum_i q_i \circ \pi_i \circ p,$$

let $\mathcal{S}_\alpha$ be the locus of these potentials, and give it the metric

$$\rho_\alpha(\Phi_\alpha(q), \Phi_\alpha(\widetilde{q})) = \sum_i \|\mu_i - \widetilde{\mu}_i\|_{\mathrm{TV}},$$

where $\mu_i, \widetilde{\mu}_i$ are the polar curvatures of $q_i, \widetilde{q}_i$.

Proposition 32.3.7 *The map $\Phi_\alpha(q) \mapsto \Phi_\sigma(q)$ is an isometry from $\mathcal{S}_\alpha$ onto the locus $\mathcal{S}_\sigma$ of Proposition 32.3.5, and it maps the ideal subspaces and the models with analytic singularities onto each other. Consequently $\mathcal{S}_\alpha$ is complete and geodesic, with diameter $2\sum_i B_i$. Its ideal subspace is closed, geodesically convex and equal to the metric closure of the models with analytic singularities. Ideal projection is the unique nearest-point retraction onto this subspace and is 1-Lipschitz, and the distance to the subspace is the sum of the nonatomic polar masses.*

Changing T or α does not change this abstract metric space. All the area comparisons and the contractions commuting with ideal projection of Proposition 32.3.5 hold, even for different auxiliary data. The upper-volume comparison vanishes on the locus, and metric convergence implies $L^1(M)$ convergence for every smooth positive volume form. Suppose that T and Z are fixed and that the curve backgrounds converge uniformly to backgrounds of zero area. Then, for any choice of the semipositive terms α, the potentials converge in $L^1(M)$, uniformly over the model tuples, to the sum of Ψ and the pullbacks of the normalized smooth potentials of the limiting curve backgrounds, as in Corollary 32.3.2.

Proof Applied to model tuples, the lemma shows that every $\Phi_\alpha(q)$ is a normalized ordinary model and computes its ideal projection coordinatewise. Restrictions to curve sections on which the other potentials are finite determine the tuple up to constants, and the normalization of the coordinates determines these constants. Thus the map is well defined and bijective, and the formulas for the polar curvatures make it an isometry preserving ideal projection.

Each potential in either locus is pulled back from $Y \times Z$. Analytic presentations pull back under the projection that forgets the auxiliary factor. Conversely, by Proposition 1.13.1, the restriction of a potential with analytic singularities to a section with fixed auxiliary coordinate has analytic singularities and is not identically $-\infty$. Hence a potential in either locus has analytic singularities exactly when the common potential on $Y \times Z$ does, and the isometry maps the analytic models onto each other. The statements on the complete metric, the ideal closure, geodesics, nearest points and area comparisons now follow from Proposition 32.3.5.

A pairwise maximum is Ψ plus the pullback of the maximum of two Γ_α-plurisubharmonic functions on B. The reduction along the divisor of weight one gives zero upper volume for it and for both models, so the upper-volume comparison vanishes. For L^1 convergence, integration along the fixed compact fibers bounds the norm of each pulled-back curve potential by a constant times its norm on the curve. The same estimate and Theorem 1.8.1 give the asserted uniform limit at zero area. Neither estimate involves α. □

Corollary 32.3.3 *For the arbitrary curve potentials u_i of the lemma, put $R = \Psi + \sum_i r_i \circ \pi_i \circ p$. The following are equivalent: $P_{\Theta_\alpha}[U] = P_{\Theta_\alpha}[U]_{\mathcal{I}}$; for each i, the polar part of the curvature measure of u_i equals its atomic part; $U \in \mathcal{E}_{\mathrm{env}}(M, \Theta_\alpha, R)$; and the ordinary and ideal envelopes of U agree below every smooth real obstacle on M.*

Proof By the lemma and the normalization of the coordinates, equality of the envelopes below zero is equivalent to $q_i = r_i$ for every i. By Corollary 31.14.1, this is equivalent to equality of the polar and atomic measures, and to $u_i \in \mathcal{E}_{\mathrm{env}}(C_i, \sigma_i, r_i)$ for every i. In that case Lemma 32.3.3 gives $u \in \mathcal{E}_{\mathrm{env}}(Y, \gamma_\sigma, \sum_i r_i \circ \pi_i)$. Since $\alpha \geq 0$, pullbacks of stretching competitors are Γ_α-plurisubharmonic, so $u \circ \pi \in \mathcal{E}_{\mathrm{env}}(B, \Gamma_\alpha, \sum_i r_i \circ \pi_i \circ \pi)$, and Proposition 32.1.2 gives $U \in \mathcal{E}_{\mathrm{env}}(M, \Theta_\alpha, R)$. By the lemma, $R = P_{\Theta_\alpha}[U]_{\mathcal{I}}$, so Proposition 17.6.6 gives equality below every smooth obstacle. Taking the zero obstacle recovers the first condition, which proves the equivalence. □

Corollary 32.3.4 *If $\alpha = 0$ or some $B_i = 0$, the background Θ_α is not big. If all $B_i > 0$ and α is positive on every nonzero vector tangent to the factor T, it is big. In particular, for a Hermitian form ω_T on T and a smooth function $h \geq 0$ on Y, the choice $\alpha = (h \circ \pi)\, \mathrm{pr}_T^* \omega_T$ gives a background that is not closed when h is nonconstant, and big when $h > 0$ and all $B_i > 0$.*

Proof For $\alpha = 0$, the background is pulled back from $Y \times Z$, and by Proposition 1.22.1 every potential in it descends to $Y \times Z$. The corresponding currents have no component in the T-directions, so they cannot dominate a positive multiple of a Hermitian form on M. Thus the background is not big.

If $B_i = 0$, choose any Hermitian form ω_T on T and write $s \colon M \to T$ for the projection. The positive $(m + \ell + k - 1, m + \ell + k - 1)$-form

$$L_i = p^* \left(\bigwedge_{j \neq i} \pi_j^* \eta_j \right) \wedge s^*(\omega_T^\ell) \wedge r^*(\kappa^k)$$

is closed, because ω_T^ℓ has top degree on T. Its pairing with Θ_α is zero, since the only possible contribution contains the integral of σ_i, whereas its pairing with a positive product Hermitian form is strictly positive. By Stokes' theorem, no current in the background is bounded below by a positive multiple of a Hermitian form.

When all areas are positive, write $\sigma_i = B_i \eta_i / A_i + \mathrm{dd}^{\mathrm{c}} \chi_i$. Subtracting the smooth pullback of $\sum_i \chi_i \circ \pi_i$ gives the current $p^* \sum_i (B_i / A_i) \pi_i^* \eta_i + \varpi^* \alpha + r^* \kappa$. Positivity of α in the T-directions makes this form strictly positive, which proves bigness. For the stated choice of α, the exterior derivative of Θ_α has the component $\mathrm{d}h \wedge \omega_T$, with one direction in Y and two in T. It is nonzero for nonconstant h and cannot be canceled by the component with three directions in T. Hence $\mathrm{d}\Theta_\alpha \neq 0$. □

The proofs use no bound on the ambient masses, so the auxiliary manifold may fail the bounded mass property; an explicit example follows.

32.3.5 A polar metric in a background with unbounded mass

Taking the Hopf threefold of Theorem 9.0.1 as the auxiliary factor makes the ambient masses unbounded, while the complete metric on the product models is unchanged. Letting the coefficient of the Hopf factor vary over the curve product also makes the background nonclosed.

Example 32.3.1 There is a compact complex sixfold with a smooth positive background Θ that is not closed and has infinite upper mass $v_+(\Theta) = +\infty$ in the sense of Definition 7.5.1. Nevertheless, the locus of separated product models of Proposition 32.3.7, along a divisor of weight one, has a complete geodesic polar metric of diameter 4. The ideal subspace is closed and is the metric closure of the models with analytic singularities, and ideal projection is the unique nearest-point retraction onto it and is 1-Lipschitz. The upper-volume comparison vanishes on this locus. The locus contains distinct ideal models with analytic singularities at polar distance 1.

Proof Take $Y = C_1 \times C_2$ with $C_1 = C_2 = \mathbb{P}^1$ and normalized Fubini–Study forms ω_i of area one, and write $\pi_i \colon Y \to C_i$ for the projections. Let (T, ω_T, V_j) be the Hopf threefold, its Gauduchon form and the smooth potentials of Theorem 9.0.1. Take $Z = \mathbb{P}^1$ with its normalized Fubini–Study form κ, a point D and its normalized potential ψ satisfying $\kappa + \mathrm{dd^c}\psi = [D]$. On $M = Y \times T \times Z$, write p, s, r for the projections. Choose a positive nonconstant smooth function h on Y and put

$$\Theta = p^*(\pi_1^*\omega_1 + \pi_2^*\omega_2) + (h \circ p)s^*\omega_T + r^*\kappa, \qquad \Psi = \psi \circ r.$$

This form is strictly positive, and it is not closed by Corollary 32.3.4.

Put $c = \min_Y h > 0$. The smooth functions $W_j = cV_j \circ s$ are strictly Θ-plurisubharmonic, since their current in the T-directions is

$$h\omega_T + c\mathrm{dd^c}V_j = c(\omega_T + \mathrm{dd^c}V_j) + (h - c)\omega_T > 0.$$

Only the part of degree three in the T-directions contributes to the top-degree product. The three curve forms have integral one, so positivity of smooth mixed products and Fubini's theorem give

$$\begin{aligned}\int_M (\Theta + \mathrm{dd^c}W_j)^6 &= 120 \int_Y \pi_1^*\omega_1 \wedge \pi_2^*\omega_2 \int_T (h(y)\omega_T + c\mathrm{dd^c}V_j)^3 \\ &\geq 120c^3 \int_T (\omega_T + \mathrm{dd^c}V_j)^3 \longrightarrow +\infty.\end{aligned}$$

The factor $120 = 6!/3!$ counts the ways of choosing one factor for each curve direction and three for T. Each W_j is bounded, so Definition 7.5.1 gives $v_+(\Theta) = +\infty$, and the bounded mass property fails.

Apply Proposition 32.3.7 to this product, with both curve areas equal to one. On the locus of potentials $\Psi + q_1 \circ \pi_1 \circ p + q_2 \circ \pi_2 \circ p$, where each q_i ranges over the ordinary ω_i-models with supremum zero, it gives a complete metric ρ of diameter

4, the analytic closure of the ideal subspace, the nearest-point retraction and the vanishing of the upper-volume comparison.

For $x \in C_1$, let g_x be the normalized Green potential of the point x, with $\omega_1 + \mathrm{dd}^{\mathrm{c}} g_x = [x]$. By Lemma 32.3.4, the potentials

$$\Phi_0 = \Psi, \qquad \Phi_x = \Psi + g_x \circ \pi_1 \circ p$$

are ordinary and ideal models, and their divisorial poles with coefficient one give analytic singularities. Their tuples of polar curvatures are $(0,0)$ and $(\delta_x, 0)$, so $\rho(\Phi_0, \Phi_x) = \|\delta_x\|_{\mathrm{TV}} = 1$. Nevertheless, by the same lemma, every representative of either ordinary type has zero extended mass; in particular, all their upper and lower singularity volumes are zero. □

32.4 Models over a curve

The polar metric on a curve also describes models on a compact manifold that maps onto it with connected fibers, including critical fibers. A background pulled back from the curve has no vertical curvature, so every potential descends. Gauduchon volumes of the fiber divisors are constant, with their multiplicities included. The normalized polar trace therefore recovers the curve metric without choosing a product structure.

Let $p\colon X \to C$ be a surjective holomorphic map with connected fibers, where X is a connected compact complex manifold of dimension $n \geq 2$ and C is a connected compact Riemann surface. Fix a smooth real $(1,1)$-form β on C, put $A = \int_C \beta$ and $\theta = p^*\beta$, and choose a smooth positive area-one form η on C. For $v \in \mathrm{PSH}(C, \beta)$ write

$$T_v = \beta + \mathrm{dd}^{\mathrm{c}} v, \qquad \mu_v = \mathbf{1}_{\{v = -\infty\}} T_v.$$

Thus μ_v is the polar curvature of the curve potential.

Lemma 32.4.1 *The class* $\mathrm{PSH}(X, \theta)$ *is nonempty if and only if* $A \geq 0$. *For* $v \in \mathrm{PSH}(C, \beta)$ *and* $U = v \circ p$, *one has*

$$P_\theta[U] = P_\beta[v] \circ p, \qquad P_\theta[U]_I = P_\beta[v]_I \circ p.$$

Pullback is a bijection between the ordinary model spaces and between their ideal subspaces. Every potential in this background has zero extended Monge–Ampère measure, and every ordinary type has zero upper and lower singularity volumes.

If $A = 0$, *write* $\beta = \mathrm{dd}^{\mathrm{c}} h$ *with* h *smooth; then the only normalized potential is* $-h \circ p + \min_C h$, *and it is the unique ordinary and ideal model.*

Proof By Proposition 1.22.2, every potential descends, even for this possibly indefinite background. Thus nonemptiness is equivalent to the curve condition in Lemma 1.8.11. Descent preserves and reflects the ordinary order and the supremum, and Corollary 1.22.1 preserves and reflects the all-weight ideal order. Pullback is

therefore a bijection between the competitor families of the envelopes upstairs and on C. The map p is open: in coordinates centered at x and $p(x)$, its restriction to a complex line on which the first nonzero homogeneous term of p does not vanish is a nonconstant function of one variable. Upper regularization therefore commutes with pullback, proving the envelope identities and the model bijections.

Write $\beta = A\eta + \mathrm{dd}^{\mathrm{c}} h$ with h smooth, using Lemma 1.8.11. For $v \in \mathrm{PSH}(C, \beta)$, put $w = v + h \in \mathrm{PSH}(C, A\eta)$. Then

$$\theta + \mathrm{dd}^{\mathrm{c}}(v \circ p) = Ap^{*}\eta + \mathrm{dd}^{\mathrm{c}}(w \circ p).$$

When $A > 0$, put $W = w \circ p$. Its bounded truncations are $\max(W, -j) = \max(w, -j) \circ p$. Locally choose a smooth a with $\mathrm{dd}^{\mathrm{c}} a = A\eta$. The bounded subharmonic function $a + \max(w, -j)$ has smooth decreasing local approximants. Their pullbacks have complex Hessian rank at most one, including at critical points, so their nth powers vanish. Bounded product continuity makes each truncation have zero top product, and hence W has zero extended measure in the semipositive background $Ap^{*}\eta$. The current invariance in Theorem 7.1.1 gives zero extended measure in the original decomposition as well.

When $A = 0$, Lemma 1.8.11 gives $v = -h + c$. Its pullback is smooth with zero curvature current, so its top product is zero. Surjectivity gives the stated normalization. Every potential differs from this one by a constant, and both envelopes fix its normalized representative. Since descent applies to every representative of every ordinary type, both singularity volumes are zero in all cases. □

Henceforth assume $A \geq 0$. Fix a Gauduchon form γ on X. For a finite positive measure μ on C of mass M, let q_μ be the solution normalized by $\int_C q_\mu \, \eta = 0$ to

$$\mathrm{dd}^{\mathrm{c}} q_\mu = \mu - M\eta.$$

The curve Green kernel of Section 20.3 gives this quasi-subharmonic solution. Define the positive current

$$p^{*}\mu = Mp^{*}\eta + \mathrm{dd}^{\mathrm{c}}(q_\mu \circ p), \qquad L_\gamma(\mu) = p^{*}\mu \wedge \gamma^{n-1}.$$

Locally this is the pullback of a subharmonic potential of μ; changing that potential adds a harmonic function and does not change the current. Extend both operators to signed measures by linearity. Write $F_s = p^{-1}(s)$ for the fiber divisor with its scheme multiplicities.

Lemma 32.4.2 *There is a positive constant κ_γ such that, for every finite signed measure μ and every Borel set $B \subset C$,*

$$p_* L_\gamma(\mu) = \kappa_\gamma \mu, \qquad \|L_\gamma(\mu)\|_{\mathrm{TV}} = \kappa_\gamma \|\mu\|_{\mathrm{TV}},$$
$$L_\gamma(\mathbf{1}_B \mu) = \mathbf{1}_{p^{-1}(B)} L_\gamma(\mu), \qquad \int_{F_s} \gamma^{n-1} = \kappa_\gamma \quad (s \in C).$$

The last integral includes the multiplicities of the fiber.

Proof The positive $(0,0)$-current $S = p_*(\gamma^{n-1})$ on C satisfies $\mathrm{dd^c} S = p_* \mathrm{dd^c}(\gamma^{n-1}) = 0$. A harmonic distribution is smooth, so S is a constant κ_γ. On the complement of the critical values, smooth fiber integration identifies it with the positive integral of γ^{n-1} on a regular fiber. Thus $\kappa_\gamma > 0$.

First let μ be positive with mass $M > 0$. Choose smooth positive measures μ_j with the same mass and $\mu_j \rightharpoonup \mu$. Their mean-zero solutions q_j have a common lower curvature bound $-M\eta$. Normalized compactness bounds $\sup_C q_j$: their mean-zero condition makes this supremum the negative mean of their supremum-zero normalization. Hence (q_j) is precompact in $L^1(C)$. Every limit has mean zero and Hessian $\mu - M\eta$. Two such limits differ by a harmonic function, hence by a constant, which their means force to be zero. Therefore $q_j \to q_\mu$ in $L^1(C)$.

By Lemma 1.22.4, $q_j \circ p \to q_\mu \circ p$ in $L^1(X)$. Thus $p^*\mu_j \to p^*\mu$ as currents, and the measures $L_\gamma(\mu_j)$ converge weakly to $L_\gamma(\mu)$. For smooth μ_j, the definition of S gives $p_* L_\gamma(\mu_j) = \kappa_\gamma \mu_j$. Passing to the limit proves the identity for μ. The case $M = 0$ is zero, and linearity proves the signed identity.

For a positive μ, split it into $\mathbf{1}_B \mu$ and $\mathbf{1}_{C \setminus B} \mu$. Their traces are positive; the pushforward identity shows that they are respectively carried by $p^{-1}(B)$ and its complement. Additivity then gives the restriction identity. Apply the same argument to the positive and negative Jordan parts of a signed measure. Their disjoint Borel carriers pull back to disjoint carriers, so their traces are the Jordan parts upstairs. The pushforward identity gives the asserted total-variation norm.

Finally the Poincaré–Lelong formula, with our normalization of $\mathrm{dd^c}$, gives $p^*\delta_s = [F_s]$, including every zero multiplicity of a local equation of the fiber. Its trace has total mass κ_γ, by the pushforward identity with $\mu = \delta_s$. □

For $v \in \mathrm{PSH}(C, \beta)$, the definition of the pullback measure gives $p^*T_v = \theta + \mathrm{dd^c}(v \circ p)$: in the smooth decomposition $\beta = A\eta + \mathrm{dd^c} h$, both sides are $Ap^*\eta + \mathrm{dd^c}((v+h) \circ p)$. Let $\mathcal{M}(\theta)$ be the ordinary model space. For $U = v \circ p \in \mathcal{M}(\theta)$, its polar trace measure is

$$\lambda_U^\gamma = \mathbf{1}_{\{U = -\infty\}} (\theta + \mathrm{dd^c} U) \wedge \gamma^{n-1} = L_\gamma(\mu_v).$$

The last identity follows from $p^*T_v = \theta + \mathrm{dd^c} U$, the equality $\{U = -\infty\} = p^{-1}(\{v = -\infty\})$ and the Borel restriction formula of Lemma 32.4.2, including at critical fibers and for a nonatomic polar measure. Define

$$\rho_\theta(U, V) = \kappa_\gamma^{-1} \|\lambda_U^\gamma - \lambda_V^\gamma\|_{\mathrm{TV}}.$$

Proposition 32.4.1 *The metric ρ_θ is independent of the Gauduchon form γ. Pullback is an isometry from the complete geodesic curve model space $(\mathcal{M}(\beta), d_\beta)$ of Proposition 31.14.1 onto $(\mathcal{M}(\theta), \rho_\theta)$, which has diameter $2A$. The ideal models form a closed geodesically convex subspace equal to the metric closure of the models with analytic singularities. Ideal projection is its unique nearest-point retraction and is 1-Lipschitz. If $U = v \circ p$ is a model, its distance to that subspace is the nonatomic polar mass of v.*

Metric convergence implies $L^1(X)$ convergence for every smooth positive volume form. The upper-volume comparison on $\mathcal{M}(\theta)$ vanishes identically. At $A = 0$, $\mathcal{M}(\theta)$ is the one-point metric space of the unique smooth model.

Proof Apply the Borel restriction and total-variation identities of Lemma 32.4.2 to get

$$\lambda^\gamma_{v\circ p} = L_\gamma(\mu_v), \qquad \rho_\theta(v \circ p, w \circ p) = \|\mu_v - \mu_w\|_{\mathrm{TV}} = d_\beta(v, w).$$

The envelope bijection and Proposition 31.14.1 give the isometry, completeness and geodesics. Polar measures have mass at most A, so their mutual distance is at most $2A$. For $A > 0$, the models of $A\delta_s$ and $A\delta_t$, $s \neq t$, attain this bound. For $A = 0$, Lemma 32.4.1 gives the singleton.

Ideal projection corresponds to the atomic part of the polar measure by the same curve classification. If $\mu = \mu_{\mathrm{at}} + \mu_{\mathrm{diff}}$ is the decomposition of a polar measure into its atomic and nonatomic parts, and ν is atomic, the measures μ_{diff} and $\mu_{\mathrm{at}} - \nu$ are mutually singular. Therefore

$$\|\mu - \nu\|_{\mathrm{TV}} = \mu_{\mathrm{diff}}(C) + \|\mu_{\mathrm{at}} - \nu\|_{\mathrm{TV}}.$$

The minimum is attained exactly at $\nu = \mu_{\mathrm{at}}$. Taking the atomic part does not increase total variation, giving the Lipschitz estimate. Atomic polar measures of mass at most A form a closed convex subset of the polar-measure space.

For $A > 0$, the smooth-gauge model bijection in the proof of Proposition 31.14.1 preserves analytic singularities in both directions: its envelope differs by a bounded function from the smoothly transported analytic model, by Proposition 17.3.1. Thus the analytic density in Corollary 31.13.1 carries from $A\eta$ to β. The analytic descent equivalence of Lemma 1.22.5 and the envelope bijection then give the stated closure upstairs. At zero area the unique smooth model is already analytic and ideal.

Curve metric convergence implies $L^1(C)$ convergence by Proposition 31.14.1, and Lemma 1.22.4 gives convergence upstairs. Finally all top-degree masses, including those of pairwise maxima, vanish by Lemma 32.4.1. The upper-volume comparison therefore vanishes. □

Corollary 32.4.1 *For $v \in \mathrm{PSH}(C, \beta)$, let $U = v \circ p$ and $R = P_\theta[U]_I$. The following are equivalent: $P_\theta[U] = P_\theta[U]_I$; μ_v is atomic; $U \in \mathcal{E}_{\mathrm{env}}(X, \theta, R)$; and the ordinary and ideal envelopes of U agree below every smooth real obstacle on X. At zero area all four conditions hold.*

Proof Put $r = P_\beta[v]_I$. The envelope bijection gives $R = r \circ p$. By Corollary 31.14.1, base envelope equality is equivalent to atomic polar curvature and to $v \in \mathcal{E}_{\mathrm{env}}(C, \beta, r)$. For each stretching parameter $b \geq 1$, descent and surjectivity identify competitors g satisfying $b^{-1}g + (1 - b^{-1})r \leq v$ with their upstairs counterparts. This uses only convex combinations and remains valid at poles. Thus the relative-class conditions correspond. Apply Proposition 17.6.6 for equality below every smooth obstacle on X; the converse includes the zero obstacle. The zero-area assertion follows from the smooth singleton. □

The top-degree measure is zero for every potential, yet the first polar trace separates every pair of models. Here the one-dimensional quotient supplies all the singularity data.

32.4.1 Finite curve factors

Keep X and C as above, but now allow $p\colon X \to C$ to have disconnected fibers. By Proposition 1.22.3, it has the component factorization

$$X \xrightarrow{q} C' \xrightarrow{r} C, \qquad p = r \circ q,$$

where q is surjective and holomorphic with connected fibers, C' is a connected compact Riemann surface, and r is finite and holomorphic of degree $d \geq 1$. Let β be any smooth real $(1,1)$-form on C, put $A = \int_C \beta$, and set $\sigma = r^*\beta$ and $\theta = p^*\beta = q^*\sigma$.

Proposition 32.4.2 *The class $\mathrm{PSH}(X, \theta)$ is nonempty exactly when $A \geq 0$. For $A \geq 0$, the metric ρ_θ of Proposition 32.4.1, formed with q and normalized by the Gauduchon volume of its fibers, makes the entire ordinary θ-model space isometric to the curve model space in background σ, whose area is dA. Its diameter is $2dA$, and it has the analytic closure, ideal retraction, convergence and all-obstacle properties of Proposition 32.4.1 and Corollary 32.4.1. Every potential has zero extended top-degree measure and every ordinary type has zero upper and lower singularity volumes. At $A = 0$ the model space is a smooth singleton.*

If κ_q^γ and κ_p^γ are the constants defined by the pushforwards of γ^{n-1} to C' and C, then

$$\kappa_p^\gamma = d\kappa_q^\gamma.$$

For $A \geq 0$, normalizing the same polar-trace variation by κ_p^γ gives d^{-1} times the q-normalized metric, with diameter $2A$. This rescaled metric is also independent of γ. For a model $U = v \circ q$ with polar measure μ_v,

$$p_*\lambda_U^\gamma = \kappa_q^\gamma r_*\mu_v.$$

Proof Away from the finitely many critical values, r is a covering with d sheets. Change of variables on these sheets gives

$$\int_{C'} r^{*}\alpha = d \int_{C} \alpha$$

for every smooth real $(1,1)$-form α; the branch points and their inverse images have smooth area zero. In particular $\int_{C'} \sigma = dA$. The envelope lemma Lemma 32.4.1, applied to q, gives nonemptiness exactly for $dA \geq 0$, equivalent to $A \geq 0$, and proves the zero-mass and zero-area assertions. In this nonempty case, apply Proposition 32.4.1 and Corollary 32.4.1 to q and the smooth background σ, which may vanish at branch points or be indefinite.

Lemma 32.4.2, applied to q, gives $q_{*}(\gamma^{n-1}) = \kappa_{q}^{\gamma}$. For a smooth test form α on C, the projection formula and the covering calculation yield

$$\int_{X} p^{*}\alpha \wedge \gamma^{n-1} = \kappa_{q}^{\gamma} \int_{C'} r^{*}\alpha = d\kappa_{q}^{\gamma} \int_{C} \alpha.$$

Thus $p_{*}(\gamma^{n-1})$ is the constant $d\kappa_{q}^{\gamma}$. The ratio of the two normalizations is d^{-1}, giving every metric assertion. Finally $q_{*}\lambda_{U}^{\gamma} = \kappa_{q}^{\gamma}\mu_{v}$, so composing pushforwards proves the last identity. □

Example 32.4.1 Let $X = \mathbb{P}^{1} \times \mathbb{P}^{1}$, let q be its first projection and let $r\colon \mathbb{P}^{1} \to \mathbb{P}^{1}$ be $r([z_{0} : z_{1}]) = [z_{0}^{2} : z_{1}^{2}]$. Put $p = r \circ q$ and $\theta = p^{*}\omega_{\mathrm{FS}}$, with $\int_{\mathbb{P}^{1}} \omega_{\mathrm{FS}} = 1$. There are two distinct analytic ideal θ-models U_{+}, U_{-} which do not descend through p. Their q-normalized polar distance is four and their p-normalized distance is two, yet

$$p_{*}\lambda_{U_{+}}^{\gamma} = p_{*}\lambda_{U_{-}}^{\gamma}$$

for every Gauduchon form γ. Thus the pushforward of polar trace to the original base does not separate the models.

Proof The finite map r has degree two and is branched at zero and infinity. In the affine coordinate it is $r(z) = z^{2}$, so the smooth semipositive form $\sigma = r^{*}\omega_{\mathrm{FS}}$ has area two. For $s \in \{1, -1\}$, set

$$v_{s}(z) = 2\log|z - s|^{2} - \log(1 + |z|^{4}) - \log 8.$$

This extends smoothly across infinity and has curvature $\sigma + \mathrm{dd}^{\mathrm{c}} v_{s} = 2\delta_{s}$. The inequality $|z - s|^{4} \leq (|z| + 1)^{4} \leq 8(1 + |z|^{4})$, with equality at $z = -s$, shows that its supremum is zero. The model of $2\delta_{s}$ from Proposition 31.14.1 has precisely this total curvature, since its polar measure already has the full mass two. Harmonic uniqueness and normalization identify it with v_{s}. Write $v_{+} = v_{1}$ and $v_{-} = v_{-1}$. Both have analytic singularities, hence are ideal by Proposition 17.2.4. The pullback bijection of Lemma 32.4.1, applied to q, makes $U_{\pm} = v_{\pm} \circ q$ analytic ideal θ-models.

The function U_{+} is $-\infty$ on $q^{-1}(1)$ and finite on $q^{-1}(-1)$, although both divisors are mapped by p to the same base point 1. It therefore cannot descend through p; the

same argument with the divisors exchanged applies to U_-. The variation of $2\delta_1 - 2\delta_{-1}$ is four, and Proposition 32.4.2 gives the two distances and

$$p_* \lambda^{\gamma}_{U_\pm} = 2\kappa_q^{\gamma} \delta_1 .$$

The upstairs polar traces are distinct, while this common pushforward loses which component carries the pole. □

For the finite map $r : C' \to C$ of the component factorization, Propositions 31.14.3 and 31.14.4 embed the model space $\mathcal{M}(\beta)$ of Section 31.14 into $\mathcal{M}(\sigma)$ by $J(v) = v \circ r$, and give a 1-Lipschitz retraction R onto the image, which averages polar measures over the fibers of r.

Corollary 32.4.2 *Let $p : X \to C$ be a surjective holomorphic map from a connected compact complex manifold of dimension at least two, with component factorization $p = r \circ q$, $q : X \to C'$. For $\theta = p^*\beta$, the ordinary models that are constant on each p-fiber form a closed complete subspace. With the p-fiber normalization of Proposition 32.4.2, the map $v \mapsto v \circ p$ is an isometry from $(\mathcal{M}(\beta), d_\beta)$ onto this subspace.*

Every model has the form $U = u \circ q$, and

$$R_X(U) = R(u) \circ q$$

is a 1-Lipschitz retraction onto the subspace. It commutes with ideal projection. The subspace contains the linear polar-measure geodesic between each pair of its members.

Proof The full model correspondence for q is Lemma 32.4.1. A model $u \circ q$ is constant on p-fibers exactly when u is constant on r-fibers. In that case $u = m_u \circ r$, and Lemma 1.22.6 makes m_u a base potential. The envelope identity and injectivity of J in Proposition 31.14.3 then show that m_u is a model. Thus this subspace is exactly the pullback image of $\mathcal{M}(\beta)$.

By Proposition 32.4.2, the p-normalized metric is d^{-1} times the polar metric on C'. Proposition 31.14.4 proves the isometry, closedness, completeness, retraction and geodesic assertions. The ideal-envelope correspondence for q transfers the commutation with ideal projection. □

32.4.2 Hopf surface fibers and unbounded ambient mass

The complete model space of a curve submersion can occur on a manifold whose ambient Hermitian masses are unbounded. Blowing up the elliptic base locus of the projection $[z_0 : z_1]$ on the scalar Hopf threefold turns this projection into a submersion with Hopf surface fibers. The rank-one background then retains only the curve singularities, while the smooth potentials of the Hopf threefold still force unbounded mass in a positive ambient background.

Example 32.4.2 There is a connected compact complex threefold Z with a surjective holomorphic submersion $p\colon Z \to \mathbb{P}^1$ whose fibers are scalar Hopf surfaces. The manifold Z does not have the bounded mass property. For the pullback $\theta = p^*\omega_{\mathrm{FS}}$ of the Fubini–Study form of area one, the entire ordinary model space has the complete geodesic polar metric of Proposition 32.4.1, with diameter two. Its ideal subspace is the analytic closure, with the unique nearest-point, 1-Lipschitz ideal retraction. All potentials in this background have zero extended Monge–Ampère measure, yet the model space contains analytic ideal models at polar distance one.

Proof Put $\lambda = \mathrm{e}^{-1}$ and consider the vector bundle

$$\mathcal{V} = \mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}.$$

Let $\mathcal{V}^\times$ be its total space with the zero section removed, and define

$$Z = \mathcal{V}^\times / \langle v \mapsto \lambda v \rangle.$$

The bundle projection descends to a holomorphic map $p\colon Z \to \mathbb{P}^1$. A Hermitian norm on the bundle satisfies $\|\lambda^k v\| = \lambda^k \|v\|$. The action is free and proper: on compact subsets of $\mathcal{V}^\times$ the norms have positive lower and finite upper bounds, so only finitely many powers can move one such subset to meet another. Thus the quotient is a complex threefold. The annulus bundle $\{\lambda \le \|v\| \le 1\}$ is compact over $\mathbb{P}^1$ and meets every orbit, so Z is compact. Bundle charts identify $p^{-1}(U)$ with

$$U \times \big((\mathbb{C}^2 \setminus \{0\})/\langle w \mapsto \lambda w \rangle\big).$$

Hence p is a holomorphic submersion with connected compact Hopf surface fibers, and Z is connected.

The tautological inclusion into $\mathbb{C}^2$ identifies $\mathcal{V}^\times$ with the incidence manifold

$$\big\{([s:t], z) \in \mathbb{P}^1 \times (\mathbb{C}^3 \setminus \{0\}) : \ tz_0 = sz_1\big\}.$$

The last coordinate z_2 comes from the trivial summand. This is the blowup of $\mathbb{C}^3 \setminus \{0\}$ along $L = \{z_0 = z_1 = 0,\ z_2 \neq 0\}$: in the chart $s \neq 0$, write $a = t/s$ and $z_1 = az_0$, while in the chart $t \neq 0$ write $b = s/t$ and $z_0 = bz_1$. These are the two usual blowup charts of the ideal (z_0, z_1). The scalar contraction preserves the incidence relation and commutes with projection to z. It therefore descends to a modification

$$\pi\colon Z \longrightarrow H = (\mathbb{C}^3 \setminus \{0\})/\langle z \mapsto \lambda z\rangle.$$

The map is an isomorphism away from the elliptic curve $L/\langle\lambda\rangle$. Local quotient charts show that π is the blowup of H along this curve; in particular, π has degree one.

Take the Gauduchon form ω and smooth potentials V_j on H from Theorem 9.0.1. Since Z is compact, a sufficiently large multiple Ω of any Hermitian form satisfies $\Omega - \pi^*\omega > 0$. Put $W_j = V_j \circ \pi$. Then

$$\Omega + \mathrm{dd^c} W_j = \Omega - \pi^*\omega + \pi^*(\omega + \mathrm{dd^c} V_j) > 0.$$

Positivity of smooth mixed products and the degree-one change of variables off the exceptional divisor give

$$\begin{aligned}\int_Z (\Omega + \mathrm{dd^c} W_j)^3 &\geq \int_Z \pi^*(\omega + \mathrm{dd^c} V_j)^3 \\ &= \int_H (\omega + \mathrm{dd^c} V_j)^3 \longrightarrow +\infty.\end{aligned}$$

The exceptional divisor and its image have zero measure for smooth top-degree forms, so they do not affect the equality. Every W_j is smooth and bounded, so $v_+(\Omega) = +\infty$. By Proposition 7.5.1, Z does not have the bounded mass property.

Apply Lemma 32.4.1 and Proposition 32.4.1 to p and the area-one form ω_{FS}. This gives all the assertions about the full model space. To see the stated distance explicitly, fix $x \in \mathbb{P}^1$ and let g_x be its normalized Green model, with $\omega_{\mathrm{FS}} + \mathrm{dd^c} g_x = \delta_x$. The models 0 and $g_x \circ p$ are ordinary and ideal, and the latter has a logarithmic pole of coefficient one along the smooth reduced fiber $p^{-1}(x)$. Thus both have analytic singularities. Their base polar measures are 0 and δ_x, so their polar distance is one, although their top-degree measures vanish. □

32.5 Polar currents on modifications

We do not know how to extend the complete topology to general ordinary models while keeping analytic closure and compatibility across backgrounds. Since polar curvature is the coordinate that achieves this on a curve, it is natural to test polar curvature on all modifications in higher dimension. This section shows that this candidate fails in three ways: the first polar current misses isolated poles unless a modification exposes them; normalizing polar traces by Gauduchon forms makes the topology depend on the chosen forms; and diffuse poles of product type are invisible to polar curvature on every modification, although they lower the volume. Throughout the section, X is a compact complex manifold of dimension n.

32.5.1 Polar curvature in higher dimension

On a curve, the polar curvature records every logarithmic pole. In higher dimension, the restriction of the curvature current to the pole set no longer does: it vanishes for an isolated logarithmic pole, although such a pole changes the multiplier ideals. The curve metric therefore cannot be extended simply by replacing the polar measure with this restricted current.

Here X has dimension $n \geq 2$, and ω is a Hermitian form on X. For $p \in \mathrm{PSH}(X, \omega)$, the *first polar current* of p is the closed positive current

$$\mathrm{Pol}_1(p) = \mathrm{Pol}_\omega(p) = \mathbf{1}_{\{p=-\infty\}}(\omega + \mathrm{dd^c} p)$$

of Proposition 1.25.3.

Proposition 32.5.1 *For every $x \in X$, there is $a_0 > 0$ such that, for each rational $a \in (0, a_0)$, there is an ideal model p_a with analytic singularities and pole set $\{x\}$ satisfying*

$$p_a = a \log |z|^2 + O(1) \quad \text{near } x, \qquad \mathrm{Pol}_1(p_a) = 0.$$

The models p_a are pairwise ordinarily inequivalent and have pairwise distinct families of multiplier ideals $\mathcal{I}(t p_a)$, $t > 0$, whereas also $\mathrm{Pol}_1(0) = 0$. More precisely, with $\mathfrak{m}_x$ the maximal ideal at x,

$$\mathcal{I}(t p_a)_x = \mathfrak{m}_x^{\max\{\lfloor ta \rfloor - n + 1, 0\}} \qquad (t > 0). \tag{32.12}$$

Proof Choose the cutoff potentials u_a of Lemma 1.16.2 for the single point x, with a_0 smaller than the reciprocal of the constant in that lemma, and put $p_a = P_\omega[u_a]$. By Proposition 17.2.4, p_a is an ideal model with analytic singularities, ordinarily equivalent to u_a. It is locally bounded away from x, and its only pole has the stated logarithmic coefficient. Distinct coefficients give unbounded differences near x. Formula (32.12) follows from Lemma 1.16.1 with codimension n, and its exponents distinguish any two coefficients at sufficiently large weights.

It remains to show that the current has no coefficient mass at x. In coordinates centered at x, let $\beta = \mathrm{dd^c}|z|^2$ and let

$$\nu = (\omega + \mathrm{dd^c} p_a) \wedge \beta^{n-1}.$$

This is a positive measure. Choose smooth cutoff functions $0 \leq \chi_\varepsilon \leq 1$, equal to one on the ball of radius ε and supported in the ball of radius 2ε, with $-C\varepsilon^{-2}\beta \leq \mathrm{dd^c}\chi_\varepsilon \leq C\varepsilon^{-2}\beta$; their complex Hessians are supported in the annulus between the two balls. Since β is closed, distributional integration by parts gives

$$\int \chi_\varepsilon \, \mathrm{d}\nu = \int \chi_\varepsilon \omega \wedge \beta^{n-1} + \int p_a \, \mathrm{dd^c}\chi_\varepsilon \wedge \beta^{n-1}.$$

The logarithmic bound for p_a on the annulus and the Euclidean volume of the annulus therefore give

$$0 \le \nu(\{x\}) \le \int \chi_\varepsilon \, \mathrm{d}\nu \le C_a\big(\varepsilon^{2n} + \varepsilon^{2n-2}(1 + |\log \varepsilon|)\big) \longrightarrow 0.$$

This is where $n \ge 2$ is used. By positivity, the coefficient measures of $\omega + \mathrm{dd}^{\mathrm{c}} p_a$ are bounded by a constant multiple of its trace measure ν, so they have no mass at x. Hence the restriction to the pole set is zero, as asserted. The zero potential is an ordinary and an ideal model because $\omega > 0$, and its pole set is empty. □

Consequently, no comparison that depends only on the fixed background and on Pol_1 separates these analytic ideal models, and a complete topology in higher dimension must involve singularity data of higher codimension.

A modification can turn the isolated pole into a divisor. For a modification $\pi\colon Y \to X$, pullback maps $\mathrm{PSH}(X, \omega)$ to $\mathrm{PSH}(Y, \pi^*\omega)$ by Proposition 1.22.4, and we define

$$\mathrm{Pol}_\pi(p) = \mathbf{1}_{\{p \circ \pi = -\infty\}}(\pi^*\omega + \mathrm{dd}^{\mathrm{c}}(p \circ \pi)).$$

Corollary 32.5.1 *Fix x and the models p_a of Proposition 32.5.1. Let $\pi\colon Y \to X$ be the blowup at x, with exceptional divisor E, and let η be a Hermitian form on Y. Then*

$$\mathrm{Pol}_\pi(p_a) = a[E].$$

For $\mu_a = \mathrm{Pol}_\pi(p_a) \wedge \eta^{n-1}$ and any two admissible coefficients a, b,

$$\|\mu_a - \mu_b\|_{\mathrm{TV}} = |a - b| \int_E \eta^{n-1}.$$

Proof On relatively compact blowup charts, with $z = (s, sw)$, one has $p_a \circ \pi = a \log |s|^2 + O(1)$ along $E = \{s = 0\}$. Thus the pole set is E and the generic Lelong number along it is a. By the proof of Lemma 1.17.1, the restriction of $\pi^*\omega + \mathrm{dd}^{\mathrm{c}}(p_a \circ \pi)$ to E is $a[E]$. Consequently $\mu_a = a[E] \wedge \eta^{n-1}$. The positive measure $[E] \wedge \eta^{n-1}$ has mass $\int_E \eta^{n-1} > 0$, which proves the formula for the total variation. □

A fixed modification detects isolated poles only over the set where it is not an isomorphism, so countably many modifications still miss the poles at all points outside a countable union of nowhere dense analytic sets.

Corollary 32.5.2 *For every countable family of smooth modifications $\pi_j\colon Y_j \to X$, there are pairwise ordinarily inequivalent ideal ω-models p_a with analytic singularities and with pairwise distinct families of multiplier ideals, such that*

$$\mathrm{Pol}_{\pi_j}(p_a) = 0 \qquad \text{for every } j \text{ and every } a.$$

Proof For each j, choose a proper closed analytic subset $S_j \subset X$ over whose complement π_j is an isomorphism. Each S_j is nowhere dense, so the Baire category theorem gives a point $x \notin \bigcup_j S_j$. Apply Proposition 32.5.1 at this point.

Since $x \notin S_j$, the only pole of $p_a \circ \pi_j$ is the point $y_j = \pi_j^{-1}(x)$, and π_j maps a neighborhood of y_j biholomorphically onto a neighborhood of x. Near y_j, the current $\pi_j^*\omega + \mathrm{dd}^{\mathrm{c}}(p_a \circ \pi_j)$ is the pullback of $\omega + \mathrm{dd}^{\mathrm{c}} p_a$ by this biholomorphism, so

$\mathrm{Pol}_{\pi_j}(p_a)$ is the pullback of $\mathrm{Pol}_1(p_a) = 0$. The same proposition gives the ordinary inequivalence and the distinct multiplier ideals. □

32.5.2 Gauduchon polar trace comparisons

To use polar currents on all modifications at once, their sizes must be normalized compatibly. Pairing with Gauduchon forms does this: for a Gauduchon form η on a modification, the total η-trace of the curvature current does not depend on the potential. Taking the supremum, over a family of modifications, of the normalized total variations of the polar traces gives a comparison that separates analytic models without any bound on top-degree mass. The space of all possible values of these coordinates is complete; whether its limit points are realized by ordinary models is a separate question.

Here $n \geq 1$ is arbitrary, and ω is a Hermitian form on X. Fix a set $\mathscr{R}$ of smooth modifications $\pi\colon Y_\pi \to X$ that contains, for any two analytic singularity types, a common log resolution. On each Y_π, choose a Gauduchon form η_π, that is, a Hermitian form with $\mathrm{dd}^{\mathrm{c}}(\eta_\pi^{n-1}) = 0$ as in Definition 4.1.1. For $u \in \mathrm{PSH}(X, \omega)$, with the polar current $\mathrm{Pol}_\pi(u)$ of Section 32.5.1, set

$$\mu_\pi(u) = \mathrm{Pol}_\pi(u) \wedge \eta_\pi^{n-1},$$

and define

$$\begin{aligned} A_\pi &= \int_{Y_\pi} \pi^*\omega \wedge \eta_\pi^{n-1}, \\ \delta_\omega^{\mathrm{pol}}(u, v) &= \sup_{\pi \in \mathscr{R}} \frac{\|\mu_\pi(u) - \mu_\pi(v)\|_{\mathrm{TV}}}{A_\pi}. \end{aligned} \tag{32.13}$$

The Gauduchon forms are part of the data. Multiplying one of them by a positive constant does not change the corresponding term in the supremum.

Proposition 32.5.2 *Every A_π is positive. The comparison (32.13) is a pseudometric on $\mathrm{PSH}(X,\omega)$, bounded by two, and its restriction to ordinary models with analytic singularities is a metric. If X is a curve, then for all ordinary models p, q,*

$$\delta_\omega^{\mathrm{pol}}(p,q) = \frac{\|\operatorname{Pol}_\omega(p) - \operatorname{Pol}_\omega(q)\|_{\mathrm{TV}}}{\int_X \omega},$$

so $\delta_\omega^{\mathrm{pol}}$ induces the complete topology of the polar metric on curve models, in which the closure of the analytic models is exactly the set of ideal models.

Let f be a smooth real function and $\omega' = \omega + \mathrm{dd^c} f$. If $\delta_{\omega'}^{\mathrm{pol}}$ is defined by the same formula with the same Gauduchon forms, then its denominators are positive and

$$\delta_{\omega'}^{\mathrm{pol}}(u - f, v - f) = \delta_\omega^{\mathrm{pol}}(u,v) \qquad (u, v \in \mathrm{PSH}(X,\omega)).$$

Proof The form $\pi^*\omega$ is semipositive, and it is positive on the nonempty open set where π is an isomorphism, so $A_\pi > 0$. By Proposition 1.22.4, $u \circ \pi \in \mathrm{PSH}(Y_\pi, \pi^*\omega)$. Since $\mathrm{dd^c}(\eta_\pi^{n-1}) = 0$, Stokes' theorem for currents gives

$$\int_{Y_\pi} \big(\pi^*\omega + \mathrm{dd^c}(u \circ \pi)\big) \wedge \eta_\pi^{n-1} = A_\pi.$$

Thus $\mu_\pi(u)$ is a positive measure of mass at most A_π, and the normalized total variation distance of two such measures is at most two. Symmetry and the triangle inequality persist under the supremum, which proves the pseudometric assertion.

Let p, q be ordinary models with analytic singularities, choose a common log resolution π in $\mathscr{R}$, and suppose that $\delta_\omega^{\mathrm{pol}}(p,q) = 0$. Then their measures in this coordinate agree. Let E_i be the prime components of the resolved ideals on Y_π, and a_i, b_i the coefficients of p and q along them. The divisorial restriction argument of Lemma 1.17.1 gives

$$\mu_\pi(p) = \sum_i a_i [E_i] \wedge \eta_\pi^{n-1}, \qquad \mu_\pi(q) = \sum_i b_i [E_i] \wedge \eta_\pi^{n-1}.$$

Distinct components do not share trace mass on their intersections, and each component has strictly positive trace mass. Equality of these measures therefore implies $a_i = b_i$ for every i. The criterion of Lemma 1.17.10, which uses only the finitely many divisors E_i, applied in both directions gives $p \sim q$. An ordinary model is the ordinary envelope of every potential of its type, so ordinary models with the same type are equal. This proves separation.

On a curve, every smooth modification is an isomorphism and $\eta_\pi^{n-1} = 1$. Hence $A_\pi = \int_X \omega$, and pullback preserves the total variation of polar curvature, so every $\pi \in \mathscr{R}$ gives the same term in the supremum. This proves the formula on curves, and the assertions following it are consequences of Corollary 31.13.1 and Proposition 31.13.1.

Finally, $u - f$ has, in background ω', the same pulled-back curvature current and the same pole set as u in background ω, and Stokes' theorem gives $A_{\pi,\omega'} = A_{\pi,\omega}$.

The gauge formula follows. It holds for the potentials themselves, before any ordinary projection. □

The coordinates $\mu_\pi(u)$ take values in a complete space. For each π, let $\mathscr{B}_\pi$ be the set of positive finite Borel measures on Y_π of mass at most A_π.

Proposition 32.5.3 *The space $\prod_{\pi\in\mathscr{R}}\mathscr{B}_\pi$ is complete for the metric*

$$\widehat{\delta}((\lambda_\pi),(\nu_\pi)) = \sup_{\pi\in\mathscr{R}} \frac{\|\lambda_\pi - \nu_\pi\|_{\mathrm{TV}}}{A_\pi}.$$

The map $u \mapsto (\mu_\pi(u))_\pi$ takes values in this space and preserves the comparison $\delta^{\mathrm{pol}}_\omega$.

Proof The metric is finite, bounded by two, and separates points. A Cauchy sequence is Cauchy in total variation in each coordinate, and completeness of the space of finite signed measures gives a limit there; positivity and the mass bound pass to the limit. Given $\varepsilon > 0$, choose N such that every normalized coordinate distance is at most ε for all $j, k \geq N$. Letting k tend to infinity separately in each coordinate preserves this inequality, and taking the supremum proves convergence for $\widehat{\delta}$. The last assertion follows from the definition and the mass bound proved above. □

Since the coordinates μ_π are invariant under bounded changes of the potential, the gauge identity of Proposition 32.5.2, which concerns the potentials themselves, transfers to ordinary models.

Corollary 32.5.3 *The comparison $\delta^{\mathrm{pol}}_\omega$ descends to ordinary classes. For a smooth real function f and $\omega' = \omega + \mathrm{dd}^{\mathrm{c}} f$, the map*

$$G_f(p) = P_{\omega'}[p - f]$$

is a bijection between the ordinary models in the two backgrounds, with inverse $q \mapsto P_\omega[q + f]$. If the same Gauduchon forms are used in both backgrounds, G_f preserves δ^{pol}, analytic models and ideal models.

Proof Boundedly equivalent potentials have boundedly equivalent pullbacks on every modification, with the same pole set. Proposition 1.25.1 therefore gives equality of their polar currents and of their measures μ_π in every coordinate, which proves descent to ordinary classes.

If $p = P_\omega[p]$, the version of Proposition 17.3.1 for ordinary envelopes gives

$$p + \min_X f \leq G_f(p) + f \leq p + \max_X f.$$

Thus $G_f(p) \sim p - f$. Since ordinary envelopes depend only on the ordinary type, $P_{\omega'}[G_f(p)] = P_{\omega'}[p - f] = G_f(p)$, and similarly $P_\omega[G_f(p) + f] = P_\omega[p] = p$. The same argument applied to models in background ω' proves surjectivity and the formula for the inverse. Invariance of the polar coordinates and the gauge identity of Proposition 32.5.2 now give

$$\delta^{\mathrm{pol}}_{\omega'}(G_f(p), G_f(q)) = \delta^{\mathrm{pol}}_{\omega}(p, q).$$

Subtraction of a smooth function and bounded equivalence preserve analytic singularities. If $p = P_\omega[p]_I$, the gauge estimate for ideal envelopes gives $P_{\omega'}[p - f]_I \sim p - f \sim G_f(p)$. By Proposition 17.2.2, this ideal envelope is an ordinary model, and two ordinary models of the same ordinary type coincide. Hence it equals $G_f(p)$, which shows that G_f maps ideal models to ideal models; the inverse map gives the converse. □

The traces μ_π determine the whole polar currents, because these currents are closed by the local cutoff argument of Proposition 1.25.3, which also applies when the pole set is not closed.

Corollary 32.5.4 *Let $u, v \in \mathrm{PSH}(X, \omega)$. For every $\pi \in \mathscr{R}$, the currents $\mathrm{Pol}_\pi(u)$ and $\mathrm{Pol}_\pi(v)$ are closed, and they agree if and only if $\mu_\pi(u) = \mu_\pi(v)$. Consequently, $\delta^{\mathrm{pol}}_{\omega}(u, v) = 0$ exactly when all these polar currents agree.*

Proof By Proposition 1.22.4, the pullbacks belong to $\mathrm{PSH}(Y_\pi, \pi^*\omega)$, and Proposition 1.25.3, applied with background $\pi^*\omega$, shows that their polar currents are closed and positive. Both currents are carried by the union of the two Borel pole sets, which has Lebesgue measure zero. Equality of their η_π-traces therefore gives equality of the currents by Lemma 1.25.3, and the converse is immediate. Since every A_π is positive, the last assertion follows. □

For a sequence that is Cauchy for $\delta^{\mathrm{pol}}_{\omega}$, total-variation convergence of the traces keeps the limiting polar mass on the union of the pole sets of the sequence, and the recovery from traces then gives compatible limits of the full polar currents.

Proposition 32.5.4 *Let (p_j) be normalized functions in $\mathrm{PSH}(X, \omega)$ forming a Cauchy sequence for $\delta^{\mathrm{pol}}_{\omega}$. For every $\pi \in \mathscr{R}$, the currents $\mathrm{Pol}_\pi(p_j)$ converge weakly to a unique closed positive current R_π, with*

$$R_\pi \wedge \eta_\pi^{n-1} = \lim_{j\to\infty} \mu_\pi(p_j) \quad \text{in total variation.}$$

If $g : Y_{\pi'} \to Y_\pi$ is a modification with $\pi' = \pi \circ g$ and $\pi, \pi' \in \mathscr{R}$, then $g_* R_{\pi'} = R_\pi$. There is $h \in \mathrm{PSH}(X, \omega)$ such that every R_π is carried coefficientwise by $\{h \circ \pi = -\infty\}$.

Proof Fix π. By Proposition 32.5.3, the measures $\mu_\pi(p_j)$ converge in total variation to a positive measure λ_π. Put

$$B_\pi = \bigcup_{j\geq 1} \{p_j \circ \pi = -\infty\}.$$

This is a Borel set of Lebesgue measure zero. Every measure $\mu_\pi(p_j)$ vanishes on its complement, so $\lambda_\pi(Y_\pi \setminus B_\pi) = 0$. The polar currents are closed and positive by Corollary 32.5.4, and their η_π-trace masses are at most A_π. By compactness of positive currents of bounded mass, a subsequence converges weakly. Its limit R

is closed and positive, and $R \wedge \eta_\pi^{n-1} = \lambda_\pi$. Since positivity bounds all coefficient measures by the Hermitian trace, R is carried by B_π. Any two subsequential limits have the same null carrier and the same trace, so they agree by Lemma 1.25.3. As every subsequence has a further convergent subsequence, the whole sequence of currents converges weakly to this unique limit R_π.

Apply Proposition 1.25.2 to g, with background $\pi^*\omega$ and potential $p_j \circ \pi$. It gives $g_* \operatorname{Pol}_{\pi'}(p_j) = \operatorname{Pol}_\pi(p_j)$. Proper pushforward is weakly continuous, since a smooth test form downstairs pulls back to a smooth test form upstairs, and passing to the limit proves compatibility.

To obtain one pole carrier, put $h_m = \sum_{j=1}^m 2^{-j} p_j$. These functions decrease, and they are ω-plurisubharmonic because the coefficients have sum at most one and $\omega > 0$. Normalized compactness, Theorem 1.4.1, bounds their L^1 norms uniformly, so Corollary 1.4.1 gives a limit $h \in \operatorname{PSH}(X, \omega)$. Since all summands are nonpositive, each pole set of a p_j is contained in the pole set of h. Hence $B_\pi \subset \{h \circ \pi = -\infty\}$, which gives the last assertion. □

The common pole carrier does not identify R_π with the polar current of h. We do not know whether all the limits R_π are the polar currents of a single ordinary model.

The topology defined by $\delta_\omega^{\mathrm{pol}}$ also depends on the choice of the Gauduchon forms. On a surface, a Gauduchon form close to the pullback of a Kähler form assigns little trace mass to exceptional curves, so choosing the forms independently on the different modifications can make a sequence of moving point poles of fixed strength converge to zero. We compare $\delta_\omega^{\mathrm{pol}}$ with the upper-volume comparison d_ω and with the sets $\mathcal{M}_\delta(\omega)$ of ordinary models of lower volume at least δ, both introduced in Definition 31.6.1.

Proposition 32.5.5 *Let $X = \mathbb{P}^2$ and let ω be the Fubini–Study form normalized by $\int_X \omega^2 = 1$. For every set $\mathscr{R}$ as above and every rational $a \in (0,1)$, there are Gauduchon forms η_π and ideal models p_j with analytic singularities, whose pole sets are pairwise distinct single points with logarithmic coefficient a, such that*

$$\delta_\omega^{\mathrm{pol}}(p_j, 0) \le \frac{1}{j}, \qquad d_\omega(p_j, 0) = a^2.$$

All p_j and 0 belong to $\mathcal{M}_{(1-a^2)/2}(\omega)$. Thus, for these choices of the Gauduchon forms, $\delta_\omega^{\mathrm{pol}}$ does not induce the topology of Theorem 31.12.1 on models of positive volume.

Proof At $x = [1:0:0]$, put

$$u_x([Z_0 : Z_1 : Z_2]) = a \log \frac{|Z_1|^2 + |Z_2|^2}{|Z_0|^2 + |Z_1|^2 + |Z_2|^2}.$$

The homogeneous quotient defines a global nonpositive potential. Its curvature is positive, since $\log(|Z_1|^2 + |Z_2|^2)$ is plurisubharmonic in a local trivialization and the remaining background is $(1-a)\omega$. Its only pole is x, where it is $a \log |z|^2 + O(1)$. By Proposition 17.2.4, $p_x = P_\omega[u_x]$ is an ideal model with analytic singularities,

ordinarily equivalent to u_x. Unitary projective automorphisms preserve ω and commute with its envelopes for the zero obstacle. Transporting this construction to pairwise distinct points x_j, we obtain models p_j.

Fix $\pi\colon Y_\pi \to X$ and some Gauduchon form γ_π on Y_π. The closed polar current of $p_j \circ \pi$ is supported on the analytic fiber over x_j. The divisorial restriction in Lemma 1.17.1 and the support theorem, [Dem12, Chapter III, Corollaries 2.11 and 2.14, pp. 141–143], give

$$\operatorname{Pol}_\pi(p_j) = \sum_{E \mapsto x_j} b_{E,j}[E], \qquad b_{E,j} = a \operatorname{ord}_E(\mathfrak{m}_{x_j} \mathcal{O}_{Y_\pi}),$$

where the sum runs over the irreducible curves $E \subset Y_\pi$ with $\pi(E) = x_j$. Indeed, Corollary 32.5.4 shows that the polar current is closed, and after the curve components are removed, the remaining closed current of order zero is supported at finitely many points and therefore vanishes. The sum is finite. Every such curve is contracted by π, so the restriction of $\pi^*\omega$ to it is zero. Put

$$C_{\pi,j} = \sum_{E \mapsto x_j} b_{E,j} \int_E \gamma_\pi, \qquad J_\pi = \{j : C_{\pi,j} > 0\}.$$

A modification of a surface has finitely many exceptional curves, so there are only finitely many points over which π has exceptional curves. Since the points x_j are distinct, J_π is finite. Choose $\varepsilon_\pi > 0$ with $\varepsilon_\pi C_{\pi,j} \le 1/j$ for every $j \in J_\pi$; if J_π is empty, take $\varepsilon_\pi = 1$. Now fix

$$\eta_\pi = \pi^*\omega + \varepsilon_\pi \gamma_\pi.$$

This form is positive and Gauduchon, since on a surface the Gauduchon equation is $\mathrm{dd}^{\mathrm{c}}\eta_\pi = 0$. Moreover

$$A_\pi = 1 + \varepsilon_\pi \int_{Y_\pi} \pi^*\omega \wedge \gamma_\pi \ge 1, \qquad \|\mu_\pi(p_j)\|_{\mathrm{TV}} = \varepsilon_\pi C_{\pi,j} \le \frac{1}{j}.$$

These forms are fixed for the whole sequence, and the estimate holds for every modification. Since $\mu_\pi(0) = 0$, taking the supremum gives the asserted convergence.

To compute the singularity volumes, let $\rho\colon Y \to X$ be the blowup at x, with exceptional curve E, and let $F\colon Y \to \mathbb{P}^1$ be the morphism resolving the pencil $[Z_1 : Z_2]$. Write ω_1 for the normalized Fubini–Study form on $\mathbb{P}^1$. Then

$$\rho^*\omega + \mathrm{dd}^{\mathrm{c}}(u_x \circ \rho) = a[E] + H_a, \qquad H_a = (1-a)\rho^*\omega + aF^*\omega_1.$$

For every $v \sim u_x$, divisorial removal gives $H_a + \mathrm{dd}^{\mathrm{c}} w \ge 0$, where $w = v \circ \rho - u_x \circ \rho$ is bounded after extension across E. The bounded product of Proposition 1.7.1 assigns no mass to E, so change of variables off E identifies the non-pluripolar mass of v with $\int_Y (H_a + \mathrm{dd}^{\mathrm{c}} w)^2$. Since H_a is closed, integration by parts for bounded potentials gives

$$\int_Y (H_a + \mathrm{dd}^{\mathrm{c}} w)^2 = \int_Y H_a^2 = 1 - a^2.$$

For the last equality, note that H_a represents the class $H - aE$, where $H = \rho^*[\omega]$, and that $H^2 = 1$, $H \cdot E = 0$ and $E^2 = -1$. Hence both singularity volumes of p_x, and by unitary transport those of every p_j, are equal to $1 - a^2$. The zero model has both volumes equal to one. Since $p_j \le 0$, Definition 31.6.1 gives $d_\omega(p_j, 0) = 1 - (1 - a^2) = a^2$. This also proves the common positive lower bound for the volumes and the incompatibility of the topologies. □

Thus compatibility with the positive-volume topology, even on analytic models, requires a coordinated choice of the Gauduchon forms. The obstruction in the next subsection is of a different kind: it applies to every choice of the forms, on general ordinary models. The normalization by A_π also excludes curve backgrounds of zero area, for which $A_\pi = 0$.

32.5.3 Diffuse product poles

Even polar currents on all modifications, with any choice of Gauduchon forms, miss some ordinary models. A maximum of diffuse polar potentials on the factors of a polydisk has a joint pole set of Hausdorff dimension zero, so all its first polar currents vanish. Nevertheless, its bounded truncations keep a fixed positive amount of top-degree mass near that set. The non-pluripolar product loses this mass, so the volume of the model drops, although the polar comparison does not see any pole.

Lemma 32.5.1 *Let $\Delta \subset \mathbb{C}$ be the unit disk. There are a compact set $K \Subset \Delta$ of Hausdorff dimension zero and a negative subharmonic function g on Δ, smooth and harmonic off K, with*

$$\{g = -\infty\} = K, \qquad \nu(g, a) = 0 \quad (a \in \Delta), \qquad g|_{\partial\Delta} = 0.$$

Put $D = \Delta^n$, $S = K^n$ and

$$h(z) = \max_{1 \le j \le n} g(z_j), \qquad h_L = \max\{h, -L\} \quad (L > 0).$$

The function h is locally Lipschitz off S, and its bounded local Monge–Ampère measure $(\mathrm{dd}^{\mathrm{c}} h)^n$ vanishes there. The measures $(\mathrm{dd}^{\mathrm{c}} h_L)^n$ all have the same finite mass M. Their supports are compact and converge to S as $L \to \infty$, in the sense that they lie in any given neighborhood of S for all large L.

Proof Use the binary Cantor set K and the nonatomic probability measure σ constructed in the proof of Lemma 1.6.5, in the coordinate of Δ, with the interval of level zero contained in Δ. Its intervals of level k have length $r_k = \exp(-4^{k+1})$ and mass 2^{-k}. For every $s > 0$,

$$2^k r_k^s \longrightarrow 0, \qquad 2^{nk} (\sqrt{n} r_k)^s \longrightarrow 0.$$

Covering K by its 2^k intervals of level k, and S by the 2^{nk} products of such intervals, which have diameter at most $\sqrt{n}r_k$, shows that K and S have Hausdorff dimension zero. Define the Green potential of the disk

$$g(z) = \int_\Delta \log \left| \frac{z-y}{1-\bar{y}z} \right|^2 \, d\sigma(y).$$

Then $dd^c g = \sigma$, with the normalization of dd^c fixed in the conventions. The Green kernel is nonpositive, vanishes on the boundary and is smooth away from its pole. On a fixed smaller disk it is at most $\log |z-y|^2 + C$. For $z \in K$, integration over the interval of level k containing z, together with nonpositivity of the kernel elsewhere, gives

$$g(z) \le 2^{-k}(2 \log r_k + C) \longrightarrow -\infty.$$

Outside K the potential is finite, smooth and harmonic. Its Lelong number at a is $\sigma(\{a\}) = 0$. Since K stays away from the boundary, g is smooth near the boundary and tends uniformly to zero there.

Near a point $z^0 \notin S$, some coordinate z_j^0 lies outside K. Since g is upper semicontinuous and equal to $-\infty$ on K, every branch $g(z_i)$ with $z_i^0 \in K$ lies below the finite branch $g(z_j)$ near z^0. Hence h is locally the maximum of the at most n pluriharmonic branches $g(z_i)$ with $z_i^0 \notin K$, and in particular it is locally Lipschitz. For $m \le n$ pluriharmonic functions a_j, use the smooth approximations

$$\varepsilon \log \sum_{j=1}^m e^{a_j/\varepsilon},$$

which decrease to $\max_j a_j$ as $\varepsilon \downarrow 0$. Their complex Hessian is a positive multiple of the weighted covariance of the forms ∂a_j, with probability weights proportional to $e^{a_j/\varepsilon}$. The centered forms satisfy one weighted linear relation, so this Hessian has rank at most $m - 1 < n$, and its nth power vanishes. Bounded decreasing convergence in Theorem 1.6.1 gives the same conclusion for the maximum.

The bounded truncation h_L is continuous, since near a pole of any curve potential the corresponding branch lies below the constant $-L$. The same argument, applied to the $n+1$ branches $g(z_1), \dots, g(z_n), -L$, shows that $(dd^c h_L)^n$ is carried by

$$Q_L = \{z : g(z_j) = -L \text{ for every } j\}.$$

Indeed, off Q_L one branch lies strictly below the maximum. Discarding it, together with any branches near poles, leaves at most n pluriharmonic branches locally, so the measure vanishes there. Since Q_L is compact, the Chern–Levine–Nirenberg estimate (1.2) gives finite mass.

For $L, L' > 0$, the functions h_L and $h_{L'}$ agree near ∂D, since h tends uniformly to zero there. Integration by parts for bounded potentials gives

$$(\mathrm{dd^c} h_L)^n - (\mathrm{dd^c} h_{L'})^n = \mathrm{dd^c}\left((h_L - h_{L'}) \sum_{j=0}^{n-1} (\mathrm{dd^c} h_L)^j \wedge (\mathrm{dd^c} h_{L'})^{n-1-j} \right).$$

The current on the right has compact support, so its integral is zero. Finally, g is bounded on compact subsets of $\overline{\Delta} \setminus K$. Its sublevel sets $\{g \le -L\}$ therefore converge to K as $L \to \infty$, and Q_L converges to S. □

For a separable maximum, the full top-degree product of Theorem 1.6.3 can be computed exactly. Its restriction to the pole set, the residual measure of Section 1.6.2, depends only on the product of the weights of the branches, so different positive weights can give the same residual measure and different singularity types.

Lemma 32.5.2 *Use the Green potential g, the probability measure σ and the sets $D = \Delta^n$, $S = K^n$ of Lemma 32.5.1. For $s, L > 0$ and $a = (a_1, \dots, a_n) \in (0, +\infty)^n$, put*

$$\mu_s = \mathrm{dd^c} \max(g, -s), \qquad h_a(z) = \max_j a_j g(z_j),$$

and $h_{a,L} = \max(h_a, -L)$. Each μ_s is a nonatomic probability measure supported on $\{g = -s\}$, and $\mu_s \to \sigma$ weakly as $s \to \infty$. The bounded products satisfy

$$(\mathrm{dd^c} h_{a,L})^n = \left(\prod_{j=1}^{n} a_j \right) \mu_{L/a_1} \otimes \cdots \otimes \mu_{L/a_n}.$$

The function h_a has pole set S, is locally Lipschitz off S, and has zero bounded top-degree product there. Its full product in the sense of Theorem 1.6.3 is

$$(\mathrm{dd^c} h_a)^n_{\mathrm{full}} = \left(\prod_{j=1}^{n} a_j \right) \sigma^{\otimes n}.$$

In particular, the common mass M in Lemma 32.5.1 is 1.

Proof The truncation $\max(g, -s)$ agrees with g near the boundary of Δ. Their difference is a compactly supported integrable function, so its complex Hessian has integral zero, and $\mu_s(\Delta) = \sigma(\Delta) = 1$. The truncation is constant near K and harmonic off K wherever $g \ne -s$, which proves the support assertion. Since the truncation is bounded, μ_s has no point masses, by Theorem 1.6.1. Decreasing local L^1 convergence to g gives $\mu_s \to \mathrm{dd^c} g = \sigma$; for large s, the supports lie in one fixed compact subset of Δ.

The rank argument of Lemma 32.5.1 applies with the weights a_j. Away from S, the finite branches dominate the branches near poles, so h_a is locally a Lipschitz maximum of at most n pluriharmonic functions, with zero top-degree product. For the bounded maximum with the constant branch $-L$, the same argument shows that its top-degree product is supported on

$$Q_{a,L} = \{z : a_j g(z_j) = -L \text{ for every } j\}.$$

Indeed, off this set one branch lies strictly below the maximum, leaving at most n active pluriharmonic branches; this also applies near branches with poles, which lie below the constant branch.

Near a point of $Q_{a,L}$ at which all the differentials $\mathrm{d}g(z_j)$ are nonzero, set $x_j = a_j g(z_j) + L$. Each x_j is the real part of a local holomorphic coordinate in its own factor. For $\varepsilon > 0$, smooth the maximum locally by

$$f_\varepsilon(x) = \varepsilon \log\left(1 + \sum_{j=1}^{n} e^{x_j/\varepsilon}\right),$$

with weights

$$w_0 = \frac{1}{1 + \sum_j e^{x_j/\varepsilon}}, \qquad w_j = e^{x_j/\varepsilon} w_0.$$

Then $-L + f_\varepsilon$ decreases to $h_{a,L}$. The Hessian matrix of f_ε in the variables x_j is $\varepsilon^{-1}(\mathrm{diag}(w_j) - (w_j w_k)_{j,k})$, whose determinant is $\varepsilon^{-n} w_0 \prod_j w_j$. Hence

$$(\mathrm{dd^c} f_\varepsilon)^n = n! \varepsilon^{-n} w_0 \prod_{j=1}^{n} w_j \bigwedge_{j=1}^{n} (\mathrm{d}x_j \wedge \mathrm{d^c} x_j).$$

The scalar kernel on $\mathbb{R}^n$ is an approximate identity of mass 1. To check the constant, write $y = x/\varepsilon$. The map

$$y \longmapsto \left(\frac{e^{y_j}}{1 + \sum_k e^{y_k}}\right)_{j=1}^{n}$$

is a diffeomorphism onto the open simplex $\{w_j > 0,\ \sum_j w_j < 1\}$, with Jacobian $w_0 \prod_j w_j$. The simplex has volume $1/n!$, so the kernel above has integral 1. Integrability of the fixed kernel in y gives concentration at $x = 0$ as $\varepsilon \downarrow 0$.

Bounded decreasing convergence of products therefore identifies the local limit with the tensor product of the one-variable level measures. In the jth factor, the chain rule for $\max(x_j, 0)$ gives

$$\delta_0(x_j)\, \mathrm{d}x_j \wedge \mathrm{d^c} x_j = a_j \mathrm{dd^c} \max(g, -L/a_j),$$

where the Dirac notation denotes the level measure in the real normal coordinate x_j. The critical points of the nonconstant harmonic function g off K are isolated. In each coordinate neighborhood, their inverse images are analytic hypersurfaces, hence pluripolar. The bounded top-degree product gives them zero mass by Theorem 1.6.1, and so does the candidate tensor measure, because each μ_{L/a_j} has no point masses. Thus the formula extends across these points and, together with the support assertion, holds on all of D.

Tensor products of the curve probabilities converge to $\sigma^{\otimes n}$: this follows first for products of continuous functions of one variable, and then for all continuous test functions by uniform approximation on a common compact product. Finally, $h_{a,L} \downarrow h_a$, whose unbounded locus is the closed set S of Hausdorff dimension zero.

By Theorem 1.6.3, the weak limit is the full top-degree product of h_a. Taking $a_j = 1$ proves $M = 1$. □

From here on, $n \geq 2$ and ω is a Hermitian form on X. Place the polydisk D of Lemma 32.5.1 in a holomorphic coordinate chart of X, choose a smooth function $0 \leq \chi \leq 1$ with compact support in D and equal to one on a neighborhood of S, and extend $H = \chi h$ by zero. Thus $H \leq 0$, $\sup_X H = 0$, $\{H = -\infty\} = S$, and H is locally bounded off S. Neither the first polar currents nor the multiplier ideals of a potential bounded below by a multiple of H detect its poles.

Lemma 32.5.3 *There is $C > 0$ with $\mathrm{dd^c} H \geq -C\omega$; thus $\omega + \mathrm{dd^c}(tH) \geq (1 - Ct)\omega$ for $0 < t < 1/C$. If $p \in \mathrm{PSH}(X, \omega)$ satisfies $p \geq sH - c$ for constants $s > 0$ and c, then $\nu(p, x) = 0$ for every $x \in X$, $\mathcal{I}(\lambda p) = \mathcal{O}_X$ for every $\lambda > 0$, and $\mathrm{Pol}_\pi(p) = 0$ for every smooth modification $\pi \colon Y \to X$.*

Proof By Lemma 32.5.1, h is locally Lipschitz off S. In the distributional product rule

$$\mathrm{dd^c}(\chi h) = \chi \mathrm{dd^c} h + h \mathrm{dd^c} \chi + \mathrm{d}\chi \wedge \mathrm{d^c} h + \mathrm{d}h \wedge \mathrm{d^c} \chi,$$

the first term is positive, and the other terms have bounded coefficients, since they are supported on the compact set $\operatorname{supp} \mathrm{d}\chi$, which does not meet S. This gives C.

Let $p \geq sH - c$. Off S, the function p is locally bounded below, so its Lelong numbers vanish there. Near a point $a \in S$, where $\chi = 1$, we have $p \geq s\, g(z_1) - c$, and the function $g(z_1)$ has Lelong number $\nu(g, a_1) = 0$ at a; hence $\nu(p, a) = 0$. Skoda's local integrability theorem [Dem11, Lemma 5.6] then makes $\mathrm{e}^{-\lambda p}$ locally integrable for every $\lambda > 0$.

Fix π, let $Z \subset X$ be a proper analytic subset over whose complement π is an isomorphism, and let $E_1, \ldots, E_N$ be the irreducible components of dimension $n - 1$ of $\pi^{-1}(Z)$. The current $R = \mathrm{Pol}_\pi(p)$ is closed and positive by Proposition 1.25.3. Off $\pi^{-1}(Z)$, it is carried by the inverse image of S, which has Hausdorff dimension zero in each relatively compact chart; by Corollary 1.13.1, R vanishes there. The support theorem for closed positive currents [Dem12, Chapter III, Corollaries 2.11 and 2.14, pp. 141–143] therefore gives

$$R = \sum_i b_i [E_i], \qquad b_i \geq 0,$$

since analytic sets of dimension less than $n - 1$ carry no closed $(1,1)$-current of order zero. Since $R \leq \pi^* \omega + \mathrm{dd^c}(p \circ \pi)$, restriction to E_i and the proof of Lemma 1.17.1 give $b_i \leq \nu(p \circ \pi, E_i)$.

If $\pi(E_i) \not\subset S$, then $p \circ \pi$ is bounded below near some point of E_i, so $\nu(p \circ \pi, E_i) = 0$. Otherwise $\pi(E_i)$ is an irreducible analytic set contained in the set S of Hausdorff dimension zero, hence a point $a = (a_1, \ldots, a_n)$. At a general point of E_i, with a local equation ζ of E_i, the first coordinate satisfies

$$z_1 \circ \pi - a_1 = \zeta^m f, \qquad m \geq 1, \quad f \neq 0.$$

A local holomorphic mth root of the unit f turns this expression into ζ'^m for another local equation ζ' of E_i. Taking circular means gives

$$\nu(g(a_1 + \zeta'^m), E_i) = m\nu(g, a_1) = 0.$$

Since $p \circ \pi \geq s\, g(z_1 \circ \pi) - c$ there, again $\nu(p \circ \pi, E_i) = 0$. Thus $R = 0$. □

From now on, C is a fixed constant as in Lemma 32.5.3. In a nonclosed background, the mass of a potential also involves the terms of the mass expansion (2.1) that contain derivatives of the background, and these terms can vary within an ordinary type. Bounded truncations separate them from the loss of top-degree mass on S, which depends only on the ordinary type.

Proposition 32.5.6 *Let $u \in \mathrm{PSH}(X, \omega)$ be locally bounded off S, with $\{u = -\infty\} = S$, and put $A = \int_X \omega^n$. For $v \in \mathrm{PSH}(X, \omega)$ with $v \sim u$ and $k > 0$, put $v_k = \max(v, -k)$. As $k \to \infty$, the measures $(\omega + \mathrm{dd^c} v_k)^n$ converge weakly to the full Monge–Ampère measure Φ_v of Lemma 31.3.2, a finite positive measure such that*

$$\mathbf{1}_S \Phi_v = \mathbf{1}_S (\mathrm{dd^c}(u + \psi))^n_{\mathrm{full}} \tag{32.14}$$

on every coordinate neighborhood on which $u + \psi$ is plurisubharmonic for a smooth function ψ. The limits

$$c_j(v) = \lim_{k \to \infty} \int_X v_k (\mathrm{dd^c} v_k)^{j-1} \wedge \mathrm{dd^c}(\omega^{n-j}), \qquad 1 \leq j < n,$$

exist and are finite, and

$$\int_X \mathrm{MA}_\omega(v) = A + \sum_{j=1}^{n-1} \binom{n}{j} c_j(v) - \Phi_v(S).$$

In particular, $v_-(\omega) - \Phi_v(S) \leq \int_X \mathrm{MA}_\omega(v) \leq v_+(\omega) - \Phi_v(S)$, with the lower and upper masses of Definition 7.5.1; if $\mathrm{dd^c}(\omega^q) = 0$ for $1 \leq q < n$, then $\int_X \mathrm{MA}_\omega(v) = A - \Phi_v(S)$; and if $n = 2$, then $\int_X \mathrm{MA}_\omega(v) = A + 2\int_X v\mathrm{dd^c}\omega - \Phi_v(S)$.

For $0 < t < 1/C$ and $u = tH$, one has $\Phi_v(S) = Mt^n$ and $c_j(tH) = c_j t^j$, where $H_L = \max(H, -L)$ and

$$c_j = \lim_{L \to \infty} \int_X H_L (\mathrm{dd^c} H_L)^{j-1} \wedge \mathrm{dd^c}(\omega^{n-j}).$$

Thus $\int_X \mathrm{MA}_\omega(tH) = A + \sum_{j=1}^{n-1} \binom{n}{j} c_j t^j - Mt^n$.

Proof Since $v \sim u$, the function v is locally bounded off S and $\{v = -\infty\} = S$. By Lemma 31.3.2, the measures $(\omega + \mathrm{dd^c} v_k)^n$ and the currents $(\mathrm{dd^c} v_k)^j$, $1 \leq j \leq n$, converge weakly to Φ_v and to currents $W_j(v)$, and Φ_v equals $\mathrm{MA}_\omega(v)$ off S. On a relatively compact coordinate neighborhood, choose a smooth function ψ_1 with $\mathrm{dd^c}\psi_1 \geq \omega$; the same lemma gives $\mathbf{1}_S \Phi_v = \mathbf{1}_S (\mathrm{dd^c}(v + \psi_1))^n_{\mathrm{full}}$ there. If $u + \psi$ is

plurisubharmonic on this neighborhood, then $(u+\psi)-(v+\psi_1)$ is bounded, and Lemma 1.6.6 gives (32.14). Since $\mathrm{MA}_\omega(v)$ does not charge the pluripolar set S, we obtain $\int_X \mathrm{MA}_\omega(v) = \Phi_v(X) - \Phi_v(S)$.

For $1 \le j < n$, integration by parts for bounded potentials gives

$$\int_X (\mathrm{dd^c} v_k)^j \wedge \omega^{n-j} = \int_X v_k (\mathrm{dd^c} v_k)^{j-1} \wedge \mathrm{dd^c}(\omega^{n-j});$$

this follows by uniformly bounded decreasing regularization, as in Theorems 1.11.1 and 2.2.1. The left side converges to $\int_X W_j(v) \wedge \omega^{n-j}$, so $c_j(v)$ exists, is finite and equals this integral. The formula (31.3) for $\Phi_v(X)$ therefore proves the mass formula.

The bounds $v_-(\omega) \le \Phi_v(X) \le v_+(\omega)$ of Lemma 31.3.2 give the two bounds, the upper one in the extended real sense. If $\mathrm{dd^c}(\omega^q) = 0$ for $1 \le q < n$, every $c_j(v)$ vanishes. For $n = 2$, the convergence $v_k \to v$ in L^1 gives $c_1(v) = \int_X v \mathrm{dd^c}\omega$.

Finally, let $u = tH$, which belongs to $\mathrm{PSH}(X,\omega)$ by Lemma 32.5.3. Near S, the function $u = th$ is plurisubharmonic, so (32.14) with $\psi = 0$ and Lemma 32.5.2, with all weights equal to one, give $\mathbf{1}_S \Phi_v = t^n \sigma^{\otimes n}$, a measure of mass $t^n = Mt^n$. Since $\max(tH, -k) = tH_{k/t}$, the limit defining $c_j(tH)$ is t^j times the limit of the integrals defining c_j, taken along $L = k/t \to \infty$. Hence c_j exists and $c_j(tH) = c_j t^j$. □

Corollary 32.5.5 *Let $n = 2$, $0 < t < 1/C$ and $u = tH$. The non-pluripolar mass is constant on the ordinary type of u if and only if ω is Gauduchon. More precisely, for every smooth real f and every sufficiently small real ε,*

$$\int_X \mathrm{MA}_\omega(u + \varepsilon f) - \int_X \mathrm{MA}_\omega(u) = 2\varepsilon \int_X f \mathrm{dd^c}\omega.$$

Proof Choose $c_f > 0$ with $-c_f\omega \le \mathrm{dd^c} f \le c_f \omega$. By Lemma 32.5.3, $u + \varepsilon f$ is ω-plurisubharmonic for $|\varepsilon| < (1 - Ct)/c_f$. These potentials differ boundedly from u, so the surface formula of Proposition 32.5.6 gives the stated difference.

If $\mathrm{dd^c}\omega = 0$, the same formula gives mass $A - Mt^2$ for every potential ordinarily equivalent to u. Conversely, write $\mathrm{dd^c}\omega = \rho\omega^2$ with ρ smooth and real. If $\rho \not\equiv 0$, choose $f = \rho$. Then $\int_X f \mathrm{dd^c}\omega = \int_X \rho^2 \omega^2 > 0$, so the masses of the smooth perturbations above vary. Thus constancy of the mass forces $\mathrm{dd^c}\omega = 0$, which is the Gauduchon condition on a surface. □

Under the bounded mass property and positive lower mass, the bounds of Proposition 32.5.6 give a definite volume gap between the ordinary models of the potentials tH and zero, which the polar comparison does not see.

Proposition 32.5.7 *Assume*

$$0 < v_-(\omega) \le v_+(\omega) < +\infty;$$

this holds for every Kähler form, with $v_\pm(\omega) = \int_X \omega^n$, *and for every Hermitian form if* $n = 2$. *For* $0 < t < 1/C$ *with* $Mt^n < v_-(\omega)$, *the potential* $p_t = P_\omega[tH]$ *is an ordinary model with*

$$\begin{aligned} 0 < v_-(\omega) - Mt^n &\le \underline{\mathrm{vol}}_\omega(p_t) \le \overline{\mathrm{vol}}_\omega(p_t) \\ &\le v_+(\omega) - Mt^n. \end{aligned}$$

Its first polar currents vanish on every smooth modification, all its multiplier ideals are trivial, and

$$d_\omega(p_t, 0) \ge Mt^n > 0, \qquad \delta^{\mathrm{pol}}_\omega(p_t, 0) = 0$$

for every family of modifications and every choice of Gauduchon forms as in Section 32.5.2. Thus the polar comparison does not separate ordinary models of positive lower volume; p_t *is not an ideal model, and it belongs to the closure of the analytic models for the polar comparison. For any* $0 < \delta < v_-(\omega)$, *all sufficiently small positive parameters give* $p_t, 0 \in \mathcal{M}_\delta(\omega)$.

Proof By Proposition 32.5.6, every $v \sim tH$ has mass between $v_-(\omega) - Mt^n$ and $v_+(\omega) - Mt^n$, which gives these bounds for the singularity volumes of tH. The lower volume is positive, so by Theorem 31.3.2 the envelope p_t is an ordinary model with the same volumes. The type of zero consists of the bounded potentials, so its upper volume is $v_+(\omega)$. Since $p_t \le 0$, (31.6) gives $d_\omega(p_t, 0) = v_+(\omega) - \overline{\mathrm{vol}}_\omega(p_t) \ge Mt^n$. In particular $p_t \ne 0$.

Since $p_t \ge tH$, Lemma 32.5.3 shows that all first polar currents of p_t vanish and all its multiplier ideals are trivial. Hence the ideal envelope of p_t is the zero model, while $p_t \ne 0$, so p_t is not an ideal model. The zero model has analytic singularities and vanishing first polar currents, so $\delta^{\mathrm{pol}}_\omega(p_t, 0) = 0$, and p_t lies in the closure of the analytic models for the polar comparison. Given $0 < \delta < v_-(\omega)$, choose $t_0 < 1/C$ with $Mt_0^n < v_-(\omega) - \delta$. The lower bound for the volume then puts zero and every p_t with $0 < t \le t_0$ in $\mathcal{M}_\delta(\omega)$.

For a Kähler form, Stokes' theorem for bounded potentials gives $v_\pm(\omega) = \int_X \omega^n$. On a compact complex surface, a Gauduchon form has finite upper mass by Theorems 2.2.2 and 4.1.1. By the metric independence in Proposition 7.5.1, every Hermitian form then has finite upper mass, and Corollary 27.18.1 gives positive lower mass. □

The residual measure distinguishes the parameters of the family p_t even after ordinary projection, and volume addition controls the volumes over the whole range $0 < t < 1/C$. The ordered family is therefore a complete interval under the two mass bounds, although the terms $c_j(v)$ of Proposition 32.5.6 need not vanish.

Corollary 32.5.6 *Assume* $0 < v_-(\omega) \le v_+(\omega) < +\infty$, *and fix* $0 < t_0 < 1/C$. *Set* $p_t = P_\omega[tH]$ *for* $0 < t \le t_0$, *define* $p_0 = 0$ *separately, and write*

$$L(t) = \underline{\mathrm{vol}}_\omega(p_t), \qquad U(t) = \overline{\mathrm{vol}}_\omega(p_t).$$

For $0 \le t \le t_0$,

$$(1 - Ct)^n v_-(\omega) \le L(t) \le U(t) \le v_+(\omega) - Mt^n.$$

The models are pairwise distinct, and $p_s \ge p_t$ *for* $s \le t$. *For either* $V = L$ *or* $V = U$, *and* $0 \le s < t \le t_0$,

$$\left(\frac{1 - Ct}{1 - Cs}\right)^n V(s) \le V(t) \le V(s).$$

Consequently

$$d_\omega(p_s, p_t) = |U(s) - U(t)| \le \frac{nCv_+(\omega)}{1 - Ct_0}|s - t|, \qquad \delta^{\mathrm{pol}}_\omega(p_s, p_t) = 0.$$

With the restriction of d_ω, *the family is a complete ordered interval, isometric to* $[0, v_+(\omega) - U(t_0)]$. *Every model with positive parameter has trivial multiplier ideals and is not an ideal model. For any* $0 < \delta < v_-(\omega)$, t_0 *can be chosen so that the whole interval lies in* $\mathcal{M}_\delta(\omega)$.

Proof Put $u_t = tH$ for $t > 0$ and $u_0 = 0$. For $0 < t < 1/C$, the potential u_t is $Ct\omega$-plurisubharmonic by Lemma 32.5.3. Adding it to the zero potential in the background $(1 - Ct)\omega$, Proposition 31.1.1 and homogeneity show that either singularity volume of u_t is at least $(1 - Ct)^n$ times the corresponding volume of zero. In particular, the lower volume of u_t is positive on the whole range $0 < t < 1/C$, and ordinary projection preserves both volumes by Theorem 31.3.2. The upper bound follows from Proposition 32.5.6, and the zero endpoint has $L(0) = v_-(\omega)$ and $U(0) = v_+(\omega)$.

By Proposition 32.5.6 and Theorem 31.3.3, $\Phi_{p_t}(S) = \Phi_{u_t}(S) = Mt^n$ for $t > 0$, whereas $\Phi_0 = \omega^n$ gives S zero mass. Thus two models of the family coincide only if their parameters are equal.

For $s < t$, set $c = (1 - Ct)/(1 - Cs)$. Then $0 < c < 1$ and

$$u_t = cu_s + (t - cs)H, \qquad C(t - cs) = 1 - c.$$

The second summand is $(1 - c)\omega$-plurisubharmonic. Addition and homogeneity give $V(t) \ge c^n V(s)$, whereas monotonicity gives $V(t) \le V(s)$. The same computation holds for $s = 0$, with the separately defined zero potential. Monotonicity of the envelope gives $p_s \ge p_t$, so (31.6) gives $d_\omega(p_s, p_t) = U(s) - U(t)$. Since $U(s) \le v_+(\omega)$, the Lipschitz bound follows from

$$1 - c^n \le n(1 - c) = \frac{nC(t - s)}{1 - Cs} \le \frac{nC(t - s)}{1 - Ct_0}.$$

By this bound, U is continuous. Distinct ordinary models of positive lower volume are separated by Proposition 31.6.1, so U is strictly decreasing. Thus $t \mapsto v_+(\omega) - U(t)$ identifies the family isometrically with the closed interval in the statement, which proves completeness.

Since $p_t \geq u_t$, Lemma 32.5.3 shows that all first polar currents of p_t vanish and all its multiplier ideals are trivial. Hence the polar comparison vanishes on the family, and the ideal envelope of every p_t is zero. The deficit Mt^n of the upper volume separates every positive parameter from zero, so the models with positive parameter are not ideal models. Finally, choose $t_0 < 1/C$ with $(1 - Ct_0)^n v_-(\omega) > \delta$ to place the whole interval in $\mathcal{M}_\delta(\omega)$. □

If all bounded potentials have the same mass, the volumes of the family are explicit.

Corollary 32.5.7 *Assume*

$$\mathrm{dd}^{\mathrm{c}}(\omega^q) = 0, \qquad 1 \leq q < n;$$

for instance, ω is Kähler, or $n = 2$ and ω is Gauduchon. Put $A = \int_X \omega^n$ and fix $0 < t_0 < 1/C$. Every $v \in \mathrm{PSH}(X, \omega)$ with $v \sim tH$, $0 < t < 1/C$, satisfies $\int_X \mathrm{MA}_\omega(v) = A - Mt^n$. Set $p_t = P_\omega[tH]$ for $0 < t \leq t_0$ and $p_0 = 0$. Both singularity volumes of p_t equal $A - Mt^n$ for $0 \leq t \leq t_0$, and for all $s, t \in [0, t_0]$,

$$d_\omega(p_s, p_t) = M|s^n - t^n|, \qquad \delta^{\mathrm{pol}}_\omega(p_s, p_t) = 0.$$

Thus this family, with the restriction of d_ω, is a complete ordered interval, isometric to $[0, Mt_0^n]$. For $t > 0$, the model p_t has trivial multiplier ideals and vanishing first polar currents on every smooth modification, and it is not an ideal model. For any prescribed $0 < \delta < A$, one can choose $t_0 > 0$ so that the whole interval lies in $\mathcal{M}_\delta(\omega)$.

Proof By Theorem 2.2.2, every bounded ω-plurisubharmonic function has mass A, so $v_-(\omega) = v_+(\omega) = A$. The mass formula is the invariant-mass case of Proposition 32.5.6, and it shows that both singularity volumes of tH equal $A - Mt^n$. This number is positive by the lower bound in Corollary 32.5.6, so Theorem 31.3.2 shows that both volumes of p_t equal $A - Mt^n$, also for $t = 0$. The same corollary gives $d_\omega(p_s, p_t) = |U(s) - U(t)| = M|s^n - t^n|$; thus $p_t \mapsto Mt^n$ is an isometry onto $[0, Mt_0^n]$. The remaining assertions are those of Corollary 32.5.6 and Lemma 32.5.3, with $v_-(\omega) = A$. Finally, a Kähler form satisfies $\mathrm{dd}^{\mathrm{c}}(\omega^q) = 0$ for every q, and on a surface the only condition is $\mathrm{dd}^{\mathrm{c}}\omega = 0$, the Gauduchon condition; Gauduchon forms exist by Theorem 4.1.1. □

The residual measure distinguishes the parameters of one ordered family, but its total mass alone cannot distinguish poles placed in different regions. Cutoffs with disjoint supports give a pair whose maximum is exactly zero, so that the volume comparison remains explicit.

Proposition 32.5.8 *Assume $0 < v_-(\omega) \le v_+(\omega) < +\infty$, and fix $0 < \delta < v_-(\omega)$. For every sufficiently small $r > 0$, there are distinct ordinary models $p_1, p_2 \in \mathcal{M}_\delta(\omega)$ and disjoint compact sets S_1, S_2 of Hausdorff dimension zero such that the full Monge–Ampère measures Φ_{p_i} of Lemma 31.3.2 satisfy*

$$R_i := \mathbf{1}_{S_1 \cup S_2} \Phi_{p_i}, \qquad R_i(X) = r, \qquad \operatorname{supp} R_i \subset S_i.$$

Neither model is an ideal model, all their multiplier ideals are trivial, and their first polar currents vanish on every smooth modification. Moreover,

$$\max(p_1, p_2) = 0, \qquad d_\omega(p_1, p_2) \ge 2r, \qquad \delta_\omega^{\mathrm{pol}}(p_1, p_2) = 0.$$

If $\mathrm{dd^c}(\omega^q) = 0$ for $1 \le q < n$, both singularity volumes of both models equal $A - r$, where $A = \int_X \omega^n$, and their distance is exactly $2r$. Thus the first polar currents together with the total residual mass do not separate ordinary models, and when $\mathrm{dd^c}(\omega^q) = 0$ for $1 \le q < n$, they still do not separate them after both singularity volumes are added.

Proof Place the polydisk of Lemma 32.5.1 in two disjoint coordinate charts, and let H_1, H_2 be the corresponding cutoffs, with pole sets S_1, S_2 and constants C_1, C_2 as in Lemma 32.5.3. For small $r > 0$ put

$$t = (r/M)^{1/n}, \qquad u_i = tH_i, \qquad p_i = P_\omega[u_i].$$

Choose r so small that $C_i t < 1$ and $(1 - C_i t)^n v_-(\omega) > \delta$ for both i. By Corollary 32.5.6, the p_i are ordinary models in $\mathcal{M}_\delta(\omega)$; they are not ideal models, their multiplier ideals are trivial and their first polar currents vanish. Their upper volumes U_i satisfy $U_i \le v_+(\omega) - r$.

By Proposition 32.5.6 and Theorem 31.3.3, $\Phi_{p_i}(S_i) = \Phi_{u_i}(S_i) = Mt^n = r$. Outside the compact support of H_i, the bounds $u_i \le p_i \le 0$ force $p_i = 0$. In particular p_i vanishes near S_j for $j \ne i$, so $\Phi_{p_i} = \omega^n$ there, which gives S_j zero mass. Hence $R_i = \mathbf{1}_{S_i} \Phi_{p_i}$. These measures have the same total mass $r > 0$ and disjoint supports, so they differ, and so do p_1 and p_2.

Since the cutoffs have disjoint supports, at every point at least one p_i vanishes, and both are nonpositive; this proves $\max(p_1, p_2) = 0$. By Definition 31.6.1,

$$d_\omega(p_1, p_2) = 2v_+(\omega) - U_1 - U_2 \ge 2r.$$

All first polar coordinates of both models are zero, so their polar comparison vanishes. When $\mathrm{dd^c}(\omega^q) = 0$ for $1 \le q < n$, Corollary 32.5.7 gives both volumes equal to $A - Mt^n = A - r$, and $v_+(\omega) = A$. Substituting these values gives the exact distance $2r$ and the final assertion. □

Knowing the location and the size of the whole residual measure still does not determine the model. For anisotropic maxima on the same pole set, the residual measure depends only on the product of the weights, whereas the volume of the

maximum of two such potentials detects the individual weights. On a compact box of positive weights, this comparison has an explicit metric interpretation.

Proposition 32.5.9 *Assume* $\mathrm{dd^c}(\omega^q) = 0$ *for* $1 \le q < n$. *Put* $A = \int_X \omega^n$, *fix* $0 < \delta < A$ *and* $0 < \alpha < \beta$, *and write* $B = [\alpha, \beta]^n$. *There are a compact set* $S \subset X$ *of Hausdorff dimension zero, a nonatomic probability measure* ρ *supported on* S, *and* $t_0 > 0$ *with the following properties. For each fixed* $0 < t \le t_0$, *ordinary models* $p_a \in \mathcal{M}_\delta(\omega)$ *are defined for all* $a \in B$ *and satisfy*

$$\mathbf{1}_S \Phi_{p_a} = t^n \left(\prod_j a_j \right) \rho, \qquad \underline{\mathrm{vol}}_\omega(p_a) = \overline{\mathrm{vol}}_\omega(p_a) = A - t^n \prod_j a_j .$$

For $a, b \in B$, *set* $c_j = \min(a_j, b_j)$. *Then*

$$d_\omega(p_a, p_b) = t^n \left(\prod_j a_j + \prod_j b_j - 2 \prod_j c_j \right).$$

The restriction of d_ω *to this family is a metric, and*

$$t^n \alpha^{n-1} \max_j |a_j - b_j| \le d_\omega(p_a, p_b) \le t^n \beta^{n-1} \sum_j |a_j - b_j|.$$

Thus $a \mapsto p_a$ *identifies* B *homeomorphically with a compact complete metric space. No model in the family is an ideal model. Each model has trivial multiplier ideals, and its first polar currents vanish on every smooth modification. For* $\alpha^n < P < \beta^n$, *the subfamily defined by* $\prod_j a_j = P$ *is complete and homeomorphic to a convex polytope of real dimension* $n - 1$. *Throughout this subfamily, the residual measure on* S *is* $Pt^n \rho$ *and both singularity volumes are* $A - Pt^n$. *Hence these data, even together with all multiplier ideals and the first polar currents on all modifications, do not separate ordinary models.*

Proof Use the chart and the cutoff χ fixed before Lemma 32.5.3. For $a \in B$, extend $H_a = \chi h_a$ by zero and put $u_a = tH_a$, with h_a as in Lemma 32.5.2. We first prove a curvature bound that is uniform in a.

Choose a compact neighborhood Σ of the supports of the derivatives of χ, disjoint from S. The isotropic maximum $\max_j g(z_j)$ is bounded below on Σ by $-B_0$, for some $B_0 > 0$. Since $g \le 0$ and $a_j \le \beta$, it follows that $-\beta B_0 \le h_a \le 0$ there. If the jth branch $a_j g(z_j)$ attains the maximum, then $a_j g(z_j) \ge -\beta B_0$, so $g(z_j) \ge -\beta B_0/\alpha$. Within the fixed compact range of the coordinates, this superlevel set stays away from K, and g and its gradient are bounded on a neighborhood of it. Thus all maximizing branches have uniformly bounded gradients, and the maxima h_a satisfy one local Lipschitz bound on Σ, independent of $a \in B$.

In the distributional product rule for $\mathrm{dd^c}(\chi h_a)$, the term $\chi \mathrm{dd^c} h_a$ is positive. The remaining terms contain derivatives of χ together with h_a or its first derivatives, so the preceding bounds control their coefficients uniformly. As in the proof of Lemma 32.5.3, this gives one $C' > 0$ with $\mathrm{dd^c} H_a \ge -C'\omega$ for every $a \in B$. Choose

$t_0 < 1/C'$ with $t_0^n \beta^n < A - \delta$. Then all u_a, $0 < t \le t_0$, are normalized and satisfy $\omega + \mathrm{dd}^{\mathrm{c}} u_a \ge (1 - C't)\omega$.

By Theorem 2.2.2, all bounded ω-plurisubharmonic functions have mass A, so $v_-(\omega) = v_+(\omega) = A$; in particular, X has the bounded mass property. Near S, the function $u_a = t h_a$ is plurisubharmonic. Hence Proposition 32.5.6, with (32.14) for $\psi = 0$, and Lemma 32.5.2 show that every $v \in \mathrm{PSH}(X, \omega)$ with $v \sim u_a$ satisfies

$$\mathbf{1}_S \Phi_v = t^n \left(\prod_j a_j \right) \rho, \qquad \rho = \sigma^{\otimes n},$$

with ρ extended by zero from the chart, and

$$\int_X \mathrm{MA}_\omega(v) = A - t^n \prod_j a_j > \delta.$$

Hence, by Theorem 31.3.2, ordinary projection preserves both singularity volumes, and by Theorem 31.3.3 it preserves the residual measure. Setting $p_a = P_\omega[u_a]$ proves the formulas for the residual measure and the volumes simultaneously for all $a \in B$.

For $a, b \in B$ and $c_j = \min(a_j, b_j)$, positivity of the weights and $g \le 0$ give, pointwise and also at the poles,

$$\max(h_a, h_b) = h_c, \qquad \max(u_a, u_b) = u_c.$$

The invariance of the comparison under ordinary projection in Proposition 31.6.1 and the definition of the comparison give

$$\begin{aligned} d_\omega(p_a, p_b) &= d_\omega(u_a, u_b) \\ &= t^n \left(\prod_j a_j + \prod_j b_j - 2 \prod_j c_j \right). \end{aligned}$$

For the boxes $R_a = \prod_j [0, a_j] \subset \mathbb{R}^n$ and Lebesgue measure λ, the last expression is $t^n \lambda(R_a \triangle R_b)$, since $R_a \cap R_b = R_c$; equivalently, it is $t^n \|\mathbf{1}_{R_a} - \mathbf{1}_{R_b}\|_{L^1(\lambda)}$. This proves the triangle inequality within the family. A slab in the coordinate where $|a_j - b_j|$ is largest, with every other coordinate in $[0, \alpha]$, lies in the symmetric difference up to its boundary, and its volume gives the lower bound. The symmetric difference is covered by the n coordinate slabs between a_j and b_j, with every other coordinate in $[0, \beta]$, and their total volume gives the upper bound. In particular, the distance separates the parameters, and the two estimates show that it induces the Euclidean topology of B. A sequence that is Cauchy for the distance has Cauchy weights in the Euclidean sense, whose limit lies in B, and the upper bound gives convergence for the distance. Compactness follows from the same bounds.

Since $g \le 0$, we have $h_a \ge (\max_j a_j) h$, hence $p_a \ge u_a \ge t(\max_j a_j) H$. By Lemma 32.5.3, every first polar current of p_a vanishes and all its multiplier ideals

are trivial. The strict volume deficit $t^n \prod_j a_j$ shows that $p_a \neq 0$, whereas the ideal envelope of p_a is zero, so p_a is not an ideal model.

Finally, the constraint $\prod_j a_j = P$ defines a closed subset of B, so the subfamily is complete. Taking logarithms coordinatewise identifies it with

$$[\log \alpha, \log \beta]^n \cap \left\{ x \in \mathbb{R}^n : \sum_j x_j = \log P \right\}.$$

For $\alpha^n < P < \beta^n$, this convex polytope meets the interior of the cube at the point with $x_j = (\log P)/n$ for all j, and it has dimension $n - 1$. The formulas for the volume and the residual measure are constant on it, while the distance separates its members. For example, taking $\alpha = 1/2$, $\beta = 2$, $P = 1$, $a = (2, 1/2, 1, \ldots, 1)$ and $b = (1, 1, 1, \ldots, 1)$ gives two models with common residual measure $t^n \rho$, both volumes equal to $A - t^n$, and distance t^n. □

Finally, even the full residual measure on the pole set can miss anisotropy. In Proposition 32.5.9, distinct ordinary models of positive volume have the same residual measure, the same singularity volumes, the same multiplier ideals and the same first polar currents on every modification, while their volume comparison is positive. A general extension, as asked for in Problem 35.7.1, must therefore record more than these data.

Chapter 33
Relative equations under positive lower mass

Let X be a connected compact complex manifold of dimension $n \geq 1$, with a smooth semipositive big real $(1,1)$-form θ and a model reference $\varphi \in \mathrm{PSH}(X,\theta)$. Given a nonzero finite non-pluripolar measure μ, we seek solutions of $\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu$, $\lambda > 0$, and of $\mathrm{MA}_\theta(u) = c\mu$, $c > 0$, relative to φ. The main difficulty is to keep the singularity condition when approximating μ: a bounded difference from the reference may be lost in the limit.

Recall from Section 14.5 that, as a model reference, φ satisfies (14.20): $P_\theta[\varphi] = \varphi$ in the sense of Definition 1.19.1, and $P_{\varepsilon\theta}(\varphi) \not\equiv -\infty$ for some $\varepsilon \in (0,1)$, which we fix.

Throughout this chapter, unless explicitly stated otherwise, we assume that θ has positive lower mass:

$$v_-(\theta) > 0. \tag{33.1}$$

Here $v_-(\theta) = \underline{\mathrm{vol}}_\theta(0)$ is the infimum of $\int_X \mathrm{MA}_\theta(v)$ over all bounded $v \in \mathrm{PSH}(X,\theta)$; see Definition 7.5.1 and the discussion following Definition 14.3.1. The bounded mass property is not assumed. For a Hermitian background and the reference 0, positive lower mass is also necessary for solvability with all measures (Corollary 33.1.1).

Approximating μ by measures with L^p densities gives solutions of the ordinary singularity type of φ by Theorem 15.11.1, but the constants in their bounds $|u-\varphi| \leq C$ need not remain uniform, so a limit can leave this type. What survives the limit is membership in the relative envelope class of Definition 14.4.1: stretching the potential away from φ by any finite factor still leaves an envelope that is not identically $-\infty$.

Positive lower mass of θ prevents these stretched envelopes from collapsing. On disjoint contact sets, one finite measure controls the rooftop envelopes of arbitrarily many potentials. Consequently, a sequence of potentials whose measures are dominated by a fixed finite measure has, after extraction, a common lower envelope in the relative class. This envelope keeps limits in the class and controls the measures of deep sublevel sets, so convergence in capacity passes to the Monge–Ampère measures themselves. When the dominating measures are Monge–Ampère measures of bounded potentials that vary along the sequence, exponential comparison provides the common envelope instead.

For a positive exponent the solution is unique. At exponent zero the scalar is unique, but the normalized potential need not be: on projective manifolds with nonclosed Hermitian backgrounds, the same smooth measure can have several smooth normalized solutions (Corollary 33.7.2).

Since bounded mass is not assumed, members of the relative class may have infinite extended mass; the envelope constructions of Section 14.5 remain available for them. The solutions constructed below have finite mass because the right-hand sides of their equations are finite.

Write

$$\mathcal{E}_\varphi = \mathcal{E}_{\mathrm{env}}(X,\theta,\varphi), \qquad \mathcal{E}_0 = \mathcal{E}_{\mathrm{env}}(X,\theta,0), \qquad \mu_u = \mathrm{MA}_\theta^{\mathrm{ext}}(u).$$

Here $\mathcal{E}_{\mathrm{env}}(X,\theta,\psi)$ denotes the relative envelope class of a reference ψ (Definition 14.4.1), and $\mathrm{MA}_\theta^{\mathrm{ext}}$ is the extended non-pluripolar Monge–Ampère measure of Theorem 7.1.1, a positive measure whose total mass may be infinite. For $u, w \in \mathrm{PSH}(X,\theta)$ and $b \geq 1$, we use the stretching envelopes $Q_b(u,w) = P_\theta(bu-(b-1)w)$ introduced before Definition 14.4.1, and the rooftop envelopes $P_\theta(u_1,\dots,u_N) = P_\theta(\min_i u_i)$. Fix also a Hermitian form $\omega \geq \theta$ for capacity and a smooth positive volume form $\mathrm{d}V$ for L^1 convergence.

For $a \in \mathcal{E}_\varphi$, Proposition 14.4.3 gives $P_\theta[a] = P_\theta[\varphi] = \varphi$. Since $a - \sup_X a$ is one of the competitors defining $P_\theta[a]$, it follows that

$$a - \sup_X a \leq P_\theta[a] = \varphi. \tag{33.2}$$

33.1 Equations for finite measures

For a positive exponent, the right-hand side $e^{\lambda u}\mu$ changes when u is translated, and this fixes the additive constant. At exponent zero, translation leaves the equation unchanged; instead, a positive scalar in front of the prescribed measure is needed to match its mass.

Theorem 33.1.1 *Let μ be a nonzero finite non-pluripolar measure on X. Then:*

(1) *For every $\lambda > 0$ there is a unique $u \in \mathcal{E}_\varphi$ with $\mu_u = e^{\lambda u}\mu$.*
(2) *There are $u \in \mathcal{E}_\varphi$ and a unique $c > 0$ with $\mu_u = c\mu$.*

The theorem is proved in Section 33.6. The case of the reference zero is treated first in Theorem 33.5.1; its solutions then serve as comparison potentials for the reference φ. For general measures the theorem asserts membership in $\mathcal{E}_\varphi$, not a bounded difference from φ.

For a Hermitian form h on X and the reference 0, the theorem and Theorem 15.26.1 characterize positive lower mass of h; the next corollary does not assume (33.1). By Definition 14.4.1, $u \in \mathrm{PSH}(X, h)$ belongs to the relative envelope class $\mathcal{E}_{\mathrm{env}}(X, h, 0)$ if, for every $b \geq 1$, some $g \in \mathrm{PSH}(X, h)$ satisfies $g \leq bu$.

Corollary 33.1.1 *Let h be a Hermitian form on X. The following conditions are equivalent:*

(1) $v_-(h) > 0$.
(2) *For every nonzero finite non-pluripolar measure μ there are $c > 0$ and $u \in \mathcal{E}_{\mathrm{env}}(X, h, 0)$ such that $\mathrm{MA}_h^{\mathrm{ext}}(u) = c\mu$.*
(3) *For every probability density $f \in L^1(\mathrm{d}V)$ bounded below by a positive constant, there are $u \in \mathrm{PSH}(X, h)$ and $a > 0$ such that $\mathrm{MA}_h^{\mathrm{ext}}(u) \geq a f \,\mathrm{d}V$.*

Proof The Hermitian form h is semipositive and big, and the reference 0 satisfies (14.20), since $P_h[0] = 0$ and $P_{\varepsilon h}(0) = 0$ for every $0 < \varepsilon < 1$. Hence, if $v_-(h) > 0$, Theorem 33.1.1, applied with $\theta = h$ and $\varphi = 0$, gives (2). Condition (2) implies (3), because a finite measure that is absolutely continuous with respect to $\mathrm{d}V$ is non-pluripolar. Finally, if $v_-(h) = 0$, the probability measure of Theorem 15.26.1 has a density with respect to $\mathrm{d}V$ that is bounded below by a positive constant, and it violates (3); thus (3) implies (1). □

33.2 Domination

The domination principle compares a potential $u \in \mathcal{E}_\varphi$ with a potential v whose measure is larger on $\{u < v\}$. We first treat the case in which μ_u is finite. The argument is that of Theorem 31.3.1: stretching u away from $\max(u, v)$ produces envelopes whose measures are bounded by the fixed measure μ_u, and the small tails

of this finite measure exclude collapse. Here bounded mass is not assumed, and the lower mass bound for the stretched envelopes comes from Theorem 14.5.1 and (33.1):

$$\mu_a(X) \geq \kappa := (1-\varepsilon)^n v_-(\theta) > 0 \qquad (a \in \mathcal{E}_\varphi).$$

Theorem 33.2.1 *Let $u \in \mathcal{E}_\varphi$ have finite μ_u, and let $v \in \mathrm{PSH}(X,\theta)$ satisfy $v \preceq \varphi$. If, for some $0 \leq c < 1$,*

$$\mu_u \leq c\mu_v \quad \text{on } \{u < v\},$$

then $u \geq v$.

Proof We follow the proof of Theorem 31.3.1, with the extended measures μ_a in place of $\mathrm{MA}_\theta(a)$. Put $w = \max(u,v)$, which belongs to $\mathcal{E}_\varphi$ by upward stability (Proposition 14.4.1), and let $b > 1$ satisfy $(1-b^{-1})^n > c$. As in the first part of that proof, Lemma 14.4.1, contact support, positivity of mixed products and ordered contact give $q_b = Q_b(u,w) \in \mathcal{E}_\varphi$ with

$$q_b \leq u, \qquad \mu_{q_b} \leq \mathbf{1}_{\{q_b = u\}}\mu_u. \tag{33.3}$$

The measures subtracted there are finite, because on the contact set of q_b the measure μ_w is at most $(1-b^{-1})^{-n}\mu_u$. The collapse argument that ends that proof uses (33.3) for all such b, the lower bound for $\mu_{q_b}(X)$, here κ, and the finiteness of μ_u; it gives $u \geq w \geq v$. □

33.2.1 Domination for extended measures

We now remove the finiteness assumption on μ_u. The key case is an ordered pair $u \leq v$ in $\mathcal{E}_\varphi$ such that μ_u puts no mass on $\{u < v\}$. A difference envelope of u and a smaller-background envelope of v then has no mass where it is negative. Its measure is therefore finite, and Theorem 33.2.1, applied in a smaller background, proves that $u = v$. The general domination principle then follows from the stretching step in the proof of Theorem 33.2.1.

Lemma 33.2.1 *Let $u, v \in \mathcal{E}_\varphi$ satisfy $u \leq v$. If*

$$\mu_u(\{u < v\}) = 0,$$

then $u = v$.

Proof Translate both potentials by a common constant so that $u \leq v \leq \varphi \leq 0$. Fix δ with $\varepsilon < \delta < 1$, put $\beta = 1-\delta$, and set

$$q = P_{\delta\theta}(v).$$

This envelope is not identically $-\infty$ by Theorem 14.5.1, applied to v. By Lemma 14.5.1,

$$w = P_\theta(u-\varphi) \in \mathcal{E}_0, \qquad w + \varphi \leq u, \qquad w \leq 0.$$

The stretching lemma Lemma 14.4.1, applied to $w \le 0$ with reference zero, shows that $z = P_\theta(w/\beta) = Q_{1/\beta}(w, 0)$ belongs to $\mathcal{E}_0$. Scaling the background and each defining competitor gives

$$\beta z \in \mathcal{E}_{\mathrm{env}}(X, \beta\theta, 0).$$

Indeed, if $a \in \mathrm{PSH}(X, \theta)$ lies below tz, then $\beta a \in \mathrm{PSH}(X, \beta\theta)$ lies below $t\beta z$, for every $t \ge 1$.

Since $q \le v \le \varphi$, we have

$$\beta z \le w \le u - \varphi \le u - q$$

off the pole sets. Thus the envelope

$$k = P_{\beta\theta}(u - q)$$

is not identically $-\infty$. Here the difference obstacle is taken off the pluripolar pole sets; it is quasi-continuous and finite almost everywhere. Hence k is bounded above (Section 1.19), and $k \le u - q$ quasi-everywhere by Theorem 15.7.2; thus $k + q \le u$ everywhere by Lemma 1.3.3. Since $\beta z \le k$, upward stability gives $k \in \mathcal{E}_{\mathrm{env}}(X, \beta\theta, 0)$.

Write $\nu_k = \mathrm{MA}^{\mathrm{ext}}_{\beta\theta}(k)$. By contact support, ν_k is carried by

$$D = \{k + q = u\}$$

outside the pole sets. Positivity of mixed products and ordered contact give

$$\nu_k \le \mathbf{1}_D \mu_u.$$

These inequalities are first proved on common finite truncation levels and then pass to the limit by exhaustion. On $D \cap \{k < 0\}$ we have $u = k + q < q \le v$, so the hypothesis gives $\nu_k(\{k < 0\}) = 0$.

This vanishing implies that ν_k is finite. Indeed, $\widehat{k} = \max(k, -1)$ is a bounded $\beta\theta$-plurisubharmonic function, and locality gives

$$\nu_k(X) = \nu_k(\{k \ge 0\}) = \int_{\{k \ge 0\}} \mathrm{MA}_{\beta\theta}(\widehat{k}) \le \int_X \mathrm{MA}_{\beta\theta}(\widehat{k}) < \infty.$$

We now apply Theorem 33.2.1 in the background $\beta\theta$, with reference zero, to the potentials k and 0 and the coefficient zero. Its hypotheses hold: $\beta\theta$ is semipositive and big, the reference zero satisfies the model condition, ν_k is finite and vanishes on $\{k < 0\}$, and

$$v_-(\beta\theta) = \beta^n v_-(\theta) > 0.$$

The scaling identity follows from the bijection $a \mapsto \beta a$ between bounded potentials in the two backgrounds. The theorem yields $k \ge 0$, and hence $u \ge q$.

It remains to let $\delta \uparrow 1$. Fix $\delta_0 \in (\varepsilon, 1)$ and put $r = P_{\delta_0\theta}(v)$. For $\delta_0 < \delta < 1$, set $t = (\delta - \delta_0)/(1 - \delta_0)$. The potential $tv + (1 - t)r$ is $\delta\theta$-plurisubharmonic and lies below v. Therefore

$$u \ge P_{\delta\theta}(v) \ge tv + (1 - t)r.$$

Letting $t \uparrow 1$ at points where v and r are finite gives $u \geq v$ almost everywhere, and then everywhere by Lemma 1.3.3. Together with $u \leq v$, this proves equality. □

Theorem 33.2.2 *Let $u \in \mathcal{E}_\varphi$ and $v \in \mathrm{PSH}(X,\theta)$ with $v \preceq \varphi$. If $0 \leq c < 1$ and*

$$\mu_u \leq c\mu_v \quad \text{on } \{u < v\},$$

then $u \geq v$.

Proof Put $w = \max(u,v) \in \mathcal{E}_\varphi$, choose $b > 1$ with $(1-b^{-1})^n > c$, and let $p = Q_b(u,w) \in \mathcal{E}_\varphi$ with contact set $D = \{b^{-1}p + (1-b^{-1})w = u\}$. As in the first part of the proof of Theorem 31.3.1, $p \leq u$, μ_p is carried by D, and

$$b^{-n}\mu_p + (1-b^{-1})^n\mu_w \leq \mu_u \quad \text{on } D, \qquad \mu_u \leq c\mu_w \quad \text{on } \{u < w\}.$$

Now μ_u may be infinite, so we compare these measures on $D \cap \{u < w\} \cap \{w > -k\}$, where μ_w is finite by locality. Since $(1-b^{-1})^n > c$, the two inequalities force μ_w, and then μ_p, to vanish there. Letting $k \to \infty$ shows that μ_p is carried by $D \cap \{u = w\} \subseteq \{p = u\}$; that is, $\mu_p(\{p < u\}) = 0$.

Now Lemma 33.2.1, applied to $p \leq u$, both in $\mathcal{E}_\varphi$, gives $p = u$. Substituting this into the defining inequality $b^{-1}p + (1-b^{-1})w \leq u$ yields $w \leq u$ off the poles, hence everywhere by Lemma 1.3.3. Thus $v \leq w \leq u$. □

Corollary 33.2.1 *Let u, v be as in Theorem 33.2.2. Then, for every $\lambda > 0$,*

$$\mathrm{e}^{-\lambda u}\mu_u \leq \mathrm{e}^{-\lambda v}\mu_v \quad \Longrightarrow \quad u \geq v.$$

Moreover, if $\mu_u \leq c\mu_v$ on X for a finite constant $c \geq 0$, then $c \geq 1$.

Proof The proof of Corollary 31.3.1 applies, with Theorem 33.2.2 in place of Theorem 31.3.1; the exponential weights are evaluated off the pluripolar pole sets.□

In particular, applying the exponential implication in both directions shows that a solution of $\mu_u = \mathrm{e}^{\lambda u}\nu$ in $\mathcal{E}_\varphi$, for a fixed $\lambda > 0$, is unique. If $\mu_u = c_1\nu$ and $\mu_v = c_2\nu$ with $c_1, c_2 > 0$ and $\nu \neq 0$, the second assertion, applied in both directions, gives $c_1 = c_2$. Corollary 33.7.2 at the end of the chapter shows that this uniqueness of the scalar does not extend to the normalized potentials.

33.3 Common envelopes and dominated continuity

Let $u_j \in \mathcal{E}_\varphi$ converge in L^1, with measures dominated by one finite non-pluripolar measure ν. To keep the limit in $\mathcal{E}_\varphi$, we construct, after passing to a subsequence, a common lower envelope in this class. Partitioning the contact set of a finite rooftop according to the obstacle it touches shows that the rooftop measure is again dominated by ν, with no factor depending on the number of obstacles. A summable integration error, combined with the lower mass bound κ, then prevents the stretched rooftops from collapsing, uniformly in the number of obstacles.

Lemma 33.3.1 *Suppose $u_j \in \mathcal{E}_\varphi$, $u_j \to u \in \mathrm{PSH}(X,\theta)$ in $L^1(X)$, and $\mu_{u_j} \le \nu$ for a fixed finite non-pluripolar measure ν. There is a subsequence, denoted again by u_j, such that*

$$h = P_\theta\left(\inf_j u_j\right) \in \mathcal{E}_\varphi.$$

In particular $u \in \mathcal{E}_\varphi$.

Proof We first bound stretched rooftops of the tail of the sequence from below and then add the finitely many remaining terms.

The suprema of an L^1-convergent sequence are bounded, so (33.2) gives a constant C with $u_j \le \varphi + C$ for all j. After a common translation we may assume $u_j \le \varphi \le 0$, and then $u \le \varphi$. Since $\mu_{u_j} \le \nu$, Lemma 1.23.1 gives

$$I_j = \int_X |\mathrm{e}^{u_j} - \mathrm{e}^{u}|\,\mathrm{d}\mu_{u_j} \longrightarrow 0.$$

We extract, once and for all, a subsequence with $I_j \le 2^{-j}$ along which $u_j \to u$ almost everywhere for smooth volume.

For $N \ge j$ put $p_{j,N} = P_\theta(u_j,\dots,u_N)$. These rooftops belong to $\mathcal{E}_\varphi$ by Proposition 14.4.2. Partition their contact sets according to the first index at which the minimum is attained. Ordered contact on these disjoint pieces gives both

$$\mu_{p_{j,N}} \le \nu, \qquad \int_X |\mathrm{e}^{p_{j,N}} - \mathrm{e}^{u}|\,\mathrm{d}\mu_{p_{j,N}} \le \sum_{\ell=j}^{N} I_\ell \le 2^{1-j}. \tag{33.4}$$

The first bound has no factor N, because each piece is controlled by the restriction of the same measure ν to that piece.

Fix a real number $B \ge 1$ and set $r_{j,N} = Q_B(p_{j,N},\varphi)$. It belongs to $\mathcal{E}_\varphi$ by the stretching lemma, so $\mu_{r_{j,N}}(X) \ge \kappa$. With $D_{j,N} = \{B^{-1}r_{j,N} + (1-B^{-1})\varphi = p_{j,N}\}$, contact support and positivity give

$$\mu_{r_{j,N}} \le B^n \mathbf{1}_{D_{j,N}} \mu_{p_{j,N}} \le B^n \nu.$$

On this contact set, $\varphi \le 0$ gives $\mathrm{e}^{p_{j,N}} \le \mathrm{e}^{r_{j,N}/B}$. Consequently

$$\int_X \mathrm{e}^{u}\,\mathrm{d}\mu_{r_{j,N}} \le B^n 2^{1-j} + B^n \nu(X) \exp\Bigl(\frac{\sup_X r_{j,N}}{B}\Bigr). \tag{33.5}$$

On the other hand, choose $K > 0$ with $B^n\nu(\{u \le -K\}) < \kappa/2$; this is possible because $u \not\equiv -\infty$ and ν does not charge its pole set. The measure domination then gives $\mu_{r_{j,N}}(\{u \le -K\}) < \kappa/2$, and the lower mass bound yields

$$\int_X \mathrm{e}^{u}\,\mathrm{d}\mu_{r_{j,N}} \ge \mathrm{e}^{-K}\kappa/2.$$

Choose j so large that $B^n 2^{1-j} < \mathrm{e}^{-K}\kappa/4$. Then (33.5) bounds $\sup_X r_{j,N}$ from below, independently of N. Since the sequence decreases in N and is bounded above by φ, its limit R is not identically $-\infty$. It satisfies, for every $\ell \geq j$,

$$B^{-1}R + (1 - B^{-1})\varphi \leq u_\ell, \qquad R \leq \varphi. \tag{33.6}$$

The second inequality follows already from $r_{j,N} \leq p_{j,N} \leq \varphi$.

It remains to include the finitely many earlier terms and to verify every defining condition of the relative class. Fix $b \geq 1$, apply the preceding construction with $B = 2b$, and let j and R be as above. If $j > 1$, let $p_0 = P_\theta(u_1, \dots, u_{j-1})$ and $S = Q_{2b}(p_0, \varphi)$. By the finite-rooftop and stretching lemmas, neither is identically $-\infty$, and $S \leq \varphi$. If $j = 1$, set $S = \varphi$. The potential $H = (R+S)/2$ is θ-plurisubharmonic. For a tail index $\ell \geq j$ we use (33.6) and $S \leq \varphi$; for an earlier index we use the analogous inequality for S and $R \leq \varphi$. In either case,

$$b^{-1}H + (1 - b^{-1})\varphi \leq u_\ell \quad \text{for every } \ell \geq 1.$$

For $b = 1$ this shows that $h = P_\theta(\inf_\ell u_\ell)$ is not identically $-\infty$. For general b the left side is a competitor for h, so the same inequality holds with h in place of every u_ℓ; this is the defining condition of the relative class for b. Also $h \leq \varphi$. Thus $h \in \mathcal{E}_\varphi$.

Finally $h \leq u_j$ for all j, so the almost-everywhere convergence and Lemma 1.3.3 give $h \leq u \leq \varphi$. Upward stability proves $u \in \mathcal{E}_\varphi$. Undoing the common translation proves the lemma as stated. □

Once a common lower envelope is available, convergence in capacity follows from a single integral condition. The proof uses the zero-domination lemma, so infinite masses are allowed.

Proposition 33.3.1 *Let $h \in \mathcal{E}_\varphi$, let C be a constant, and let $u_j \in \mathrm{PSH}(X, \theta)$ satisfy*

$$h \leq u_j \leq \varphi + C, \qquad u_j \longrightarrow u \in \mathrm{PSH}(X, \theta) \quad \text{in } L^1(X).$$

If

$$I_j := \int_X |\mathrm{e}^{u_j} - \mathrm{e}^{u}|\, \mu_{u_j} \longrightarrow 0,$$

then $u \in \mathcal{E}_\varphi$ and $u_j \to u$ in capacity.

Proof The bounds pass to u almost everywhere along a subsequence, and then everywhere by Lemma 1.3.3; upward stability gives $u_j, u \in \mathcal{E}_\varphi$. After translating the entire family and its bounds by a common constant, we may assume that all potentials are nonpositive.

It suffices to prove convergence in capacity along a further subsequence of every subsequence. Extract so that $I_j \leq 2^{-j}$ and $u_j \to u$ almost everywhere for smooth volume, and form

$$p_{j,N} = P_\theta(u_j, \dots, u_N), \qquad p_j = P_\theta\left(\inf_{\ell \geq j} u_\ell\right).$$

All these envelopes lie above h and below $\varphi + C$; hence none of them is identically $-\infty$, and all belong to $\mathcal{E}_\varphi$. Partition the contact set of $p_{j,N}$ according to the first index ℓ with $p_{j,N} = u_\ell$. Ordered contact on each part gives

$$\int_X |e^{p_{j,N}} - e^u| \, \mu_{p_{j,N}} \le \sum_{\ell=j}^{N} I_\ell \le 2^{1-j}.$$

Only integrals of nonnegative functions occur here, so the estimate holds even if a rooftop has infinite mass.

As $N \to \infty$, $p_{j,N} \downarrow p_j$; as $j \to \infty$, the p_j increase, and their regularized limit p is not identically $-\infty$. Both monotone limits hold in capacity by Lemma 1.8.9. Moreover $h \le p \le u \le \varphi + C$, where $p \le u$ follows from the almost-everywhere convergence of the extracted u_j. Thus $p \in \mathcal{E}_\varphi$.

Apply Lemma 7.3.1, first as $N \to \infty$ and then as $j \to \infty$, with the bounded quasi-continuous weights in the preceding estimate. These weights converge in capacity because the exponential is Lipschitz on a common upper half-line. We obtain

$$\int_X |e^p - e^u| \, \mu_p = 0, \qquad \mu_p(\{p < u\}) = 0.$$

The second equality follows off the poles, which the extended measures ignore. The zero-domination lemma Lemma 33.2.1 now gives $p = u$; no finiteness of μ_p is needed.

Finally, the upper envelopes $U_j = (\sup_{\ell \ge j} u_\ell)^*$ decrease to u, by Hartogs' lemma (Lemma 1.4.1). Applying Lemma 1.8.9 to $p_j \uparrow u$ and $U_j \downarrow u$, the inequalities $p_j \le u_j \le U_j$ give convergence in capacity along this subsequence, and hence for the original sequence. □

33.3.1 Convergence under a fixed measure

The extraction lemma provides the lower envelope required by the integral criterion. The same fixed measure then controls the measures of deep sublevel sets, so convergence in capacity passes to the Monge–Ampère measures themselves.

Theorem 33.3.1 *Suppose $u_j \in \mathcal{E}_\varphi$, $u_j \to u \in \mathrm{PSH}(X, \theta)$ in $L^1(X)$, and $\mu_{u_j} \le \nu$ for a fixed finite non-pluripolar measure ν. Then*

$$u \in \mathcal{E}_\varphi, \qquad u_j \longrightarrow u \text{ in capacity}, \qquad \mu_{u_j} \rightharpoonup \mu_u \le \nu.$$

Proof Start with an arbitrary subsequence. By Lemma 33.3.1, a further subsequence has a common lower envelope $h \in \mathcal{E}_\varphi$, and $u \in \mathcal{E}_\varphi$. Inequality (33.2) and Hartogs' lemma give a common upper bound $u_j \le \varphi + C$.

After a common translation, assume $u_j, u \le 0$. Since $\mu_{u_j} \le \nu$, Lemma 1.23.1 gives

$$\int_X |e^{u_j} - e^u| \, d\mu_{u_j} \le \int_X |e^{u_j} - e^u| \, d\nu \longrightarrow 0.$$

Thus Proposition 33.3.1 gives convergence in capacity along this subsequence. For the measures themselves, the common lower bound gives

$$\mu_{u_j}(\{u_j \le -T\}) \le \nu(\{h \le -T\}) \longrightarrow 0 \quad \text{uniformly in } j,$$

since ν is finite and does not charge the pole set of $h \not\equiv -\infty$. The tail criterion for smooth backgrounds, Theorem 8.6.1, therefore gives $\mu_{u_j} \rightharpoonup \mu_u$, and weak limits preserve domination by the fixed finite measure ν. Every subsequence of the original sequence has a further subsequence with these conclusions and the same limit u, so both convergences hold for the original sequence. □

33.4 Moving bounded carrier measures

In the approximation scheme of Section 33.6, the measures of the approximating solutions are dominated by Monge–Ampère measures of uniformly bounded ω-plurisubharmonic potentials ψ_j, which vary with j. For a bounded ω-plurisubharmonic potential ψ, write $\omega_\psi = \omega + dd^c\psi$. A single dominating measure is then unavailable, but an exponential domination $\mu_{u_j} \le Ae^{\lambda u_j}\omega_{\psi_j}^n$ still provides a common lower envelope. A smaller-background potential and a strict barrier, in the sense of the introduction to Chapter 15, give comparison functions whose constants are uniform in j. If every u_j has the ordinary singularity type of φ, smaller-background comparison gives the same construction even when $v_-(\theta) = 0$. Convergence of the potentials ψ_j in capacity then identifies the limiting equation.

Proposition 33.4.1 *Fix $A, \lambda > 0$, $C \in \mathbb{R}$ and $L \geq 0$. For $j \geq 1$, suppose that $u_j \in \mathcal{E}_\varphi$, $u_j \leq \varphi + C$, and*

$$\mu_{u_j} \leq A e^{\lambda u_j} \omega_{\psi_j}^n, \qquad \psi_j \in \mathrm{PSH}(X, \omega), \qquad \|\psi_j\|_\infty \leq L. \tag{33.7}$$

Assume either $v_-(\theta) > 0$ or $u_j \sim \varphi$ for every j. Then

$$h = P_\theta \left(\inf_j u_j \right) \in \mathcal{E}_\varphi.$$

No convergence of ψ_j or u_j is assumed, and in the second alternative the constants in $u_j \sim \varphi$ need not be uniform in j.

Proof Translate all u_j by one constant so that $u_j \leq \varphi \leq 0$; this changes A by a fixed factor. Translate each ψ_j so that it takes values in $[-2L, 0]$; this does not change its measure. Set $q = P_{\varepsilon\theta}(\varphi)$ and choose a strict barrier $\rho \in \mathrm{PSH}(X, \theta)$ and $a > 0$ such that

$$q \leq \varphi, \qquad \rho \leq 0, \qquad \theta + \mathrm{dd}^{\mathrm{c}} \rho \geq a\omega.$$

Here $q \not\equiv -\infty$ by the model condition, and ρ exists because θ is big. Fix $0 < \delta < (1 - \varepsilon)a$.

For a fixed real $b \geq 1$, put $r_{j,b} = Q_b(u_j, \varphi)$. The stretching lemma puts it in $\mathcal{E}_\varphi$, and $r_{j,b} \leq u_j \leq 0$. Contact support and positivity imply

$$\mu_{r_{j,b}} \leq b^n \mathbf{1}_{D_{j,b}} \mu_{u_j}, \qquad D_{j,b} = \{ b^{-1} r_{j,b} + (1 - b^{-1})\varphi = u_j \}.$$

On this contact set, $\varphi \leq 0$ gives $u_j \leq r_{j,b}/b$. Thus (33.7) yields

$$\mathrm{e}^{-\lambda r_{j,b}/b} \mu_{r_{j,b}} \leq b^n A \omega_{\psi_j}^n, \tag{33.8}$$

where the weights are evaluated off the pluripolar pole sets.

Choose $C_b \geq 0$ so large that

$$\delta^n \exp(\lambda C_b / b) \geq b^n A,$$

and define

$$s_{j,b} = q + (1 - \varepsilon)\rho + \delta\psi_j - C_b.$$

This is a θ-plurisubharmonic function, not identically $-\infty$, since

$$\begin{aligned} \theta + \mathrm{dd}^{\mathrm{c}} s_{j,b} &= (\varepsilon\theta + \mathrm{dd}^{\mathrm{c}} q) + (1 - \varepsilon)(\theta + \mathrm{dd}^{\mathrm{c}} \rho) + \delta \mathrm{dd}^{\mathrm{c}} \psi_j \\ &\geq \delta \omega_{\psi_j}. \end{aligned}$$

Consequently $\mu_{s_{j,b}} \geq \delta^n \omega_{\psi_j}^n$: this inequality holds away from the poles by the local positive product inequality, and then globally because the bounded reference measure does not charge pluripolar sets. Also $s_{j,b} \leq -C_b$ and $s_{j,b} \preceq \varphi$. Together with (33.8), we conclude

$$\mathrm{e}^{-\lambda s_{j,b}/b} \mu_{s_{j,b}} \geq b^n A \omega_{\psi_j}^n \geq \mathrm{e}^{-\lambda r_{j,b}/b} \mu_{r_{j,b}}.$$

If $v_-(\theta) > 0$, apply Corollary 33.2.1 with first potential $r_{j,b}$ and second potential $s_{j,b}$. It gives $r_{j,b} \geq s_{j,b}$.

In the second alternative, $r_{j,b} \sim \varphi$: if $u_j \geq \varphi - K_j$, the potential $\varphi - bK_j$ is a competitor for $r_{j,b}$. Hence the smaller-background envelope $P_{\varepsilon\theta}(r_{j,b})$ is not identically $-\infty$, and $s_{j,b} \preceq r_{j,b}$. Therefore Corollary 15.1.1 also gives $r_{j,b} \geq s_{j,b}$, without a lower-mass hypothesis.

Since $\psi_j \geq -2L$, we now have a lower bound independent of j:

$$H_b = q + (1-\varepsilon)\rho - 2\delta L - C_b \leq r_{j,b}.$$

The potential H_b is θ-plurisubharmonic and not identically $-\infty$. By the definition of $r_{j,b}$, therefore,

$$b^{-1}H_b + (1-b^{-1})\varphi \leq u_j \quad \text{for every } j.$$

For $b = 1$ this shows that the infimum envelope h is not identically $-\infty$. For general b, the left side is a competitor for h, so it lies below h. Thus $b^{-1}H_b + (1-b^{-1})\varphi \leq h$, which is the defining condition of the relative class for b. Since $h \leq \varphi$, we obtain $h \in \mathcal{E}_\varphi$. Undoing the common translation finishes the proof. □

Corollary 33.4.1 *Under the hypotheses of Proposition 33.4.1, if $u_j \to u \in \mathrm{PSH}(X,\theta)$ in $L^1(X)$, then $u \in \mathcal{E}_\varphi$.*

Proof The common envelope gives $h \leq u_j \leq \varphi + C$. Along an almost-everywhere convergent subsequence, Lemma 1.3.3 gives $h \leq u \leq \varphi + C$, and upward stability gives the conclusion. □

33.4.1 Convergence with a common envelope

If the potentials ψ_j converge in capacity, the integration error in Proposition 33.3.1 tends to zero, and the common lower envelope controls the measures of deep sublevel sets. The dominating measure may also contain a fixed finite non-pluripolar part.

Theorem 33.4.1 *Let $h \in \mathcal{E}_\varphi$, let C be a constant, and let $u_j \in \mathrm{PSH}(X,\theta)$ satisfy*

$$h \le u_j \le \varphi + C, \qquad u_j \longrightarrow u \in \mathrm{PSH}(X,\theta) \quad \text{in } L^1(X).$$

Fix $A \ge 0$ and a finite non-pluripolar measure ν on X. Let $\psi_j \in \mathrm{PSH}(X,\omega)$ be uniformly bounded and converge in capacity to a bounded $\psi \in \mathrm{PSH}(X,\omega)$. Suppose

$$\mu_{u_j} \le A\omega_{\psi_j}^n + \nu.$$

Then $u \in \mathcal{E}_\varphi$, $u_j \to u$ in capacity, μ_u is finite, and

$$\mu_{u_j} \rightharpoonup \mu_u.$$

Proof Translate so that $u_j, u \le 0$. The functions $\mathrm{e}^{u_j}, \mathrm{e}^{u}$ are uniformly bounded quasi-plurisubharmonic functions with a common lower curvature bound. They converge in L^1 for smooth volume, since the exponential is Lipschitz on $(-\infty, 0]$. Hence Lemmas 1.23.1 and 1.23.3 give

$$\int_X |\mathrm{e}^{u_j} - \mathrm{e}^{u}|\,\omega_{\psi_j}^n \longrightarrow 0, \qquad \int_X |\mathrm{e}^{u_j} - \mathrm{e}^{u}|\,\mathrm{d}\nu \longrightarrow 0.$$

Thus the integration error in Proposition 33.3.1 tends to zero, and that proposition gives convergence in capacity and $u \in \mathcal{E}_\varphi$.

By Lemma 1.8.6, the bounded reference measures $\omega_{\psi_j}^n$ have uniformly bounded mass and satisfy a common capacity estimate. Hence, for a constant B independent of j and T,

$$\mu_{u_j}(\{u_j < -T\}) \le B\,\mathrm{Cap}_\omega(\{h < -T\}) + \nu(\{h < -T\}) \longrightarrow 0$$

uniformly in j. The first term tends to zero because h is a quasi-plurisubharmonic function not identically $-\infty$, and the second because ν is finite and does not charge the pole set of h. The total masses $\mu_{u_j}(X)$ are uniformly bounded by the assumed domination. The tail criterion for smooth backgrounds, Theorem 8.6.1, therefore gives finiteness of μ_u and the asserted weak convergence. □

33.5 General measures in the minimal envelope class

We first take the reference zero. It has minimal singularities because $\theta \ge 0$, and it satisfies (14.20) for every $\varepsilon \in (0,1)$; we call $\mathcal{E}_0$ the minimal envelope class. By Lemma 15.23.1, a finite non-pluripolar measure has an integrable density with respect to the Monge–Ampère measure of a bounded potential. Truncating that density gives bounded solutions whose measures are dominated by one fixed finite measure, and dominated continuity passes to the full density. Letting the exponent tend to zero then produces the normalized equation.

Theorem 33.5.1 *Let μ be a finite non-pluripolar measure with $\mu(X) > 0$. For every $\lambda > 0$ there is a unique $u \in \mathcal{E}_0$ satisfying*

$$\mathrm{MA}_\theta^{\mathrm{ext}}(u) = \mathrm{e}^{\lambda u}\mu.$$

There also exist $v \in \mathcal{E}_0$, normalized by $\sup_X v = 0$, and a unique positive scalar c such that

$$\mathrm{MA}_\theta^{\mathrm{ext}}(v) = c\mu.$$

Proof For the reference zero, $P_{\epsilon\theta}(0) = 0$ for every $\epsilon > 0$. Letting $\epsilon \downarrow 0$ in Theorem 14.5.1 therefore gives

$$\mathrm{MA}_\theta^{\mathrm{ext}}(w)(X) \geq v_-(\theta) \qquad (w \in \mathcal{E}_0). \tag{33.9}$$

Positive exponent. By Lemma 15.23.1, write $\mu = f\nu$ with $\nu = \mathrm{MA}_\theta(\psi)$, where $\psi \in \mathrm{PSH}(X,\theta)$ is bounded and $0 \leq f \in L^1(\nu)$. For positive integers j, let $f_j = \min(f, j)$. These densities are bounded, and the measures $f_j\nu$ are nonzero. By Theorem 15.9.1 there are bounded $u_j \in \mathrm{PSH}(X,\theta)$ with

$$\mathrm{MA}_\theta(u_j) = \mathrm{e}^{\lambda u_j} f_j \nu.$$

These bounded potentials belong to $\mathcal{E}_0$ by Proposition 14.4.1. Since f_j increases with j, exponential order (Corollary 33.2.1) shows that u_j decreases; in particular $u_j \leq u_1$. On the other hand, since $f_j \leq f$,

$$v_-(\theta) \leq \mathrm{MA}_\theta(u_j)(X) \leq \mathrm{e}^{\lambda \sup_X u_j}\mu(X),$$

so the suprema of the u_j have a common lower bound. Hence the decreasing limit u is a θ-plurisubharmonic function, not identically $-\infty$, and $u_j \to u$ in $L^1(X)$. Moreover

$$\mathrm{MA}_\theta(u_j) \leq \mathrm{e}^{\lambda \sup_X u_1}\mu.$$

Applying Theorem 33.3.1 with reference zero, we obtain $u \in \mathcal{E}_0$ and weak convergence of the Monge–Ampère measures to $\mathrm{MA}_\theta^{\mathrm{ext}}(u)$. The right-hand sides converge in total variation to $\mathrm{e}^{\lambda u} f\nu$, since the functions $\mathrm{e}^{\lambda u_j} f_j$ converge ν-almost everywhere and are bounded by $\mathrm{e}^{\lambda \sup_X u_1} f \in L^1(\nu)$. This proves the equation, and Corollary 33.2.1 gives uniqueness.

Zero exponent. For each positive integer j, let $u_j \in \mathcal{E}_0$ be the solution just constructed with exponent $1/j$. To bound the normalization, we compare with the solution $s \in \mathcal{E}_0$ for exponent one, so that $\mathrm{MA}_\theta^{\mathrm{ext}}(s) = \mathrm{e}^s\mu$. Set $H = \max(0, \sup_X s)$ and $w_j = s + (j-1)H$. Then $w_j/j \geq s$, and

$$\mathrm{e}^{-w_j/j}\mathrm{MA}_\theta^{\mathrm{ext}}(w_j) \leq \mathrm{e}^{-s}\mathrm{MA}_\theta^{\mathrm{ext}}(s) = \mu = \mathrm{e}^{-u_j/j}\mathrm{MA}_\theta^{\mathrm{ext}}(u_j).$$

Exponential order, now with exponent $1/j$, gives $u_j \leq w_j$. Put

$$M_j = \sup_X u_j, \qquad v_j = u_j - M_j, \qquad c_j = \mathrm{e}^{M_j/j}.$$

Then $\sup_X v_j = 0$ and $0 < c_j \le e^H$, because $M_j \le \sup_X w_j \le jH$. By (33.9),

$$v_-(\theta) \le c_j \int_X e^{v_j/j}\, d\mu \le c_j \mu(X),$$

so $c_j \ge v_-(\theta)/\mu(X) > 0$. After extraction, Theorem 1.4.1 gives $v_j \to v$ in $L^1(X)$, where $v \not\equiv -\infty$ and $\sup_X v = 0$, and $c_j \to c > 0$.

By Lemma 1.23.2, $a_j = e^{v_j/j} \to 1$ in $L^1(\mu)$. Consequently

$$\mathrm{MA}_\theta^{\mathrm{ext}}(v_j) = c_j a_j \mu \longrightarrow c\mu \quad \text{in total variation,} \qquad \mathrm{MA}_\theta^{\mathrm{ext}}(v_j) \le e^H \mu.$$

Applying Theorem 33.3.1 again, we obtain $v \in \mathcal{E}_0$ and identify the limit as $\mathrm{MA}_\theta^{\mathrm{ext}}(v)$; thus $\mathrm{MA}_\theta^{\mathrm{ext}}(v) = c\mu$. The scalar assertion of Corollary 33.2.1, applied in both directions to two solutions, proves that c is unique. □

The solutions for the reference zero also bound from above the approximating solutions for the reference φ. Indeed, every θ-plurisubharmonic potential is $\preceq 0$, so exponential order with reference zero compares them with arbitrary θ-plurisubharmonic potentials. This comparison controls the normalization in Section 33.6.

33.6 Equations with a model reference

We now prove Theorem 33.1.1. Write the given measure as $\mu = f\nu$, where ν is the Monge–Ampère measure of a bounded potential ψ. To keep the reference φ, we first smooth ψ: the prescribed ordinary-type theorem applies to the approximating data, exponential comparison provides a common lower envelope, and moving-measure continuity preserves the equation in the limit. With ψ fixed, we then approximate the integrable density f by bounded densities. Here the solutions for the reference zero give upper bounds, while positive lower mass prevents collapse.

Proof (of Theorem 33.1.1) The proof has four steps: continuous positive densities, bounded densities, integrable densities, and finally the zero exponent.

Let μ be the given nonzero finite non-pluripolar measure. Apply Lemma 15.23.1 to the Hermitian, hence semipositive and big, background ω to write

$$\mu = f\nu, \qquad \nu = \omega_\psi^n, \qquad \psi \in \mathrm{PSH}(X,\omega) \cap L^\infty(X), \qquad 0 \le f \in L^1(\nu).$$

Step 1: positive continuous density. Fix $\lambda > 0$ and suppose first that f is continuous and strictly positive. By Theorem 1.11.1, choose smooth strictly ω-plurisubharmonic $\psi_j \downarrow \psi$. These potentials are uniformly bounded; write $\|\psi_j\|_\infty \le L$ and $\nu_j = \omega_{\psi_j}^n$. Bounded Bedford–Taylor continuity gives $\nu_j \rightharpoonup \nu$. The measure $f\nu_j$ has a continuous positive density with respect to smooth volume, so it belongs to every finite L^p space. Hence the prescribed ordinary-type theorem Theorem 15.11.1 gives $u_j \sim \varphi$ with

$$\mu_{u_j} = e^{\lambda u_j} f \nu_j.$$

We first bound the u_j from above, uniformly in j. Choose K with $\mathrm{e}^{-\lambda(\psi_j+K)} \le \min_X f$ for all j. Then

$$\mathrm{e}^{-\lambda(\psi_j+K)}\omega^n_{\psi_j+K} \le f\nu_j = \mathrm{e}^{-\lambda u_j}\mu_{u_j} \le \mathrm{e}^{-\lambda u_j}\mathrm{MA}^{\mathrm{ext}}_\omega(u_j),$$

where the last inequality follows from $\omega \ge \theta$ and positivity of extended mixed products. Apply Corollary 15.1.1 in the background ω, with first potential $\psi_j + K$ and second potential u_j. Its hypotheses hold: the first potential is bounded, so it admits a smaller-background envelope, and $u_j \preceq \psi_j + K$. Thus $u_j \le \psi_j + K \le L + K =: C$.

By (33.2), $u_j \le \varphi + C$. Since

$$\mu_{u_j} \le \|f\|_\infty \mathrm{e}^{\lambda u_j}\omega^n_{\psi_j},$$

Proposition 33.4.1 gives a single $h \in \mathcal{E}_\varphi$ below all u_j. The bounds $h \le u_j \le \varphi + C$ and Theorem 1.4.1 give, after extraction, an L^1 limit $u \not\equiv -\infty$. The moving-measure theorem Theorem 33.4.1 now applies, because $\psi_j \to \psi$ in capacity and $\mu_{u_j} \le \mathrm{e}^{\lambda C}\|f\|_\infty \nu_j$. It gives $u \in \mathcal{E}_\varphi$, convergence in capacity and $\mu_{u_j} \rightharpoonup \mu_u$.

The right-hand sides converge weakly to $\mathrm{e}^{\lambda u} f\nu$. Indeed, $\mathrm{e}^{\lambda u_j} - \mathrm{e}^{\lambda u} \to 0$ in capacity with a common bound, so its integral against ν_j tends to zero by the uniform capacity estimate for these measures. The fixed function $\mathrm{e}^{\lambda u}$ is bounded and quasi-continuous. Approximating it by a continuous function outside an open set of arbitrarily small capacity, bounded product continuity and the same capacity estimate allow this fixed weight in $\nu_j \rightharpoonup \nu$. Multiplication by the continuous function f and by a continuous test function preserves both estimates. Hence $\mu_u = \mathrm{e}^{\lambda u} f\nu$.

Step 2: bounded density. Suppose next that $0 \le f \in L^\infty(\nu)$ and $\int_X f\,\mathrm{d}\nu > 0$. Choose positive continuous functions f_j, uniformly bounded by a constant F, with $f_j \to f$ almost everywhere for ν. Such a sequence is obtained from the density of continuous functions in $L^1(\nu)$ by clipping to a common interval, adding positive constants tending to zero, and extracting a summably convergent subsequence.

There is a bounded nonnegative function g with $g\nu \ne 0$ and $g \le f_j$ for all j. Indeed, choose $a > 0$ with $\nu(\{f \ge 2a\}) > 0$. By Egorov's theorem, there is a measurable subset $E \subseteq \{f \ge 2a\}$ of positive measure on which the convergence is uniform; hence $f_j \ge a$ on E for all $j \ge N$, for some N. The finitely many remaining functions $f_1, \dots, f_{N-1}$ are positive and continuous, so they have positive minima on X. If $b > 0$ is a lower bound for these minima and for a, then $g = b\mathbf{1}_E$ works.

By Step 1, let $u_j \in \mathcal{E}_\varphi$ solve $\mu_{u_j} = \mathrm{e}^{\lambda u_j} f_j \nu$. The theorem for the reference zero, Theorem 33.5.1, applied to the finite nonzero measure $g\nu$, gives $s \in \mathcal{E}_0$ satisfying

$$\mathrm{MA}^{\mathrm{ext}}_\theta(s) = \mathrm{e}^{\lambda s} g\nu.$$

Apply Corollary 33.2.1 with reference zero, first potential $s \in \mathcal{E}_0$ and second potential $u_j \preceq 0$. Since $g\nu \le f_j\nu$, it gives $u_j \le s$. With $C = \sup_X s$ we have

$$\mu_{u_j} \le \mathrm{e}^{\lambda C} F\nu, \qquad \kappa \le \mathrm{e}^{\lambda \sup_X u_j} F\nu(X),$$

where the second inequality uses $\mu_{u_j}(X) \geq \kappa$. These two bounds on the suprema give a subsequence converging in L^1 to some $u \not\equiv -\infty$. The fixed-measure continuity theorem Theorem 33.3.1 puts u in $\mathcal{E}_\varphi$ and identifies the weak limit of μ_{u_j} as μ_u.

By Lemma 1.23.1, $e^{\lambda u_j} \to e^{\lambda u}$ in $L^1(\nu)$, and $f_j \to f$ in $L^1(\nu)$ by dominated convergence. Since both factors are uniformly bounded,

$$e^{\lambda u_j} f_j \longrightarrow e^{\lambda u} f \quad \text{in } L^1(\nu).$$

Thus the right-hand sides converge in total variation, which proves the equation for f.

Step 3: integrable density. For general $f \in L^1(\nu)$, apply Step 2 to $f_j = \min(f, j)$ and obtain $u_j \in \mathcal{E}_\varphi$ with $\mu_{u_j} = e^{\lambda u_j} f_j \nu$. Since $f_j \leq f_{j+1}$, exponential order gives $u_{j+1} \leq u_j$. Furthermore

$$\kappa \leq e^{\lambda \sup_X u_j} \mu(X), \qquad \mu_{u_j} \leq e^{\lambda \sup_X u_1} \mu.$$

Hence the decreasing limit u is not identically $-\infty$, and the fixed-measure continuity theorem gives $u \in \mathcal{E}_\varphi$ and $\mu_{u_j} \rightharpoonup \mu_u$. Dominated convergence gives $e^{\lambda u_j} f_j \to e^{\lambda u} f$ in $L^1(\nu)$, so $\mu_u = e^{\lambda u} \mu$. Uniqueness follows from Corollary 33.2.1, applied in both directions.

Step 4: zero exponent. For each positive integer j, Step 3 with exponent $1/j$ gives $u_j \in \mathcal{E}_\varphi$ with $\mu_{u_j} = e^{u_j/j} \mu$. The zero-exponent part of the proof of Theorem 33.5.1 applies to these u_j, with the same comparison potential $s \in \mathcal{E}_0$, the solution of $\mathrm{MA}_\theta^{\mathrm{ext}}(s) = e^s \mu$. Indeed, the comparison $u_j \leq s + (j-1)\max(0, \sup_X s)$ made there uses exponential order with reference zero, first potential in $\mathcal{E}_0$ and second potential $u_j \preceq 0$; the lower mass bound $v_-(\theta)$ is replaced by κ; and Theorem 33.3.1 is applied with reference φ. We obtain $v \in \mathcal{E}_\varphi$ with $\sup_X v = 0$ and $c > 0$ with $\mu_v = c\mu$. Finally, the scalar assertion of Corollary 33.2.1, applied in both directions, shows that c is unique. □

33.7 Normalization at exponent zero

In this section the background is a Hermitian form h with $\nu_-(h) > 0$ and the reference is zero; thus $\theta = h$, $\varphi = 0$, and $\mathcal{E}_{\mathrm{env}}(X, h, 0)$ is the minimal envelope class. For nonzero $0 \le f \in L^1(\mathrm{d}V)$, recall from (15.80) that the subsolution threshold $b_h(f)$ is the limit of the normalizing constants of the truncations $\min(f, k)$; by Theorem 15.27.1, it is the largest $a \ge 0$ for which some $u \in \mathrm{PSH}(X, h)$ satisfies $\mathrm{MA}_h^{\mathrm{ext}}(u) \ge a f \, \mathrm{d}V$. We show that $b_h(f)$ is exactly the scalar of the normalized equation in the minimal envelope class: scalar comparison bounds every subsolution coefficient by this scalar. Dominated continuity also identifies the equation satisfied by every limit of normalized greatest exponential solutions; whether these limits coincide is a question about the potentials.

Corollary 33.7.1 *Let h be a Hermitian form on X with $\nu_-(h) > 0$, and let $0 \le f \in L^1(\mathrm{d}V)$ be nonzero. There is $v \in \mathcal{E}_{\mathrm{env}}(X, h, 0)$ such that*

$$\mathrm{MA}_h^{\mathrm{ext}}(v) = b_h(f) f \, \mathrm{d}V.$$

The number $b_h(f) > 0$ is the unique positive scalar for which this equation has a solution in that envelope class.

For $\lambda > 0$, let u_λ be the greatest solution of $\mathrm{MA}_h^{\mathrm{ext}}(u_\lambda) = \mathrm{e}^{\lambda u_\lambda} f \, \mathrm{d}V$ from Theorem 15.34.1, and set $v_\lambda = u_\lambda - \sup_X u_\lambda$. Each u_λ belongs to $\mathcal{E}_{\mathrm{env}}(X, h, 0)$. If $\lambda_j \downarrow 0$ and $v_{\lambda_j} \to v$ in L^1, then the convergence holds in capacity, and v belongs to the same envelope class and satisfies the equation above.

Proof By Theorem 33.5.1, applied with $\theta = h$, there are $c > 0$ and $w \in \mathcal{E}_{\mathrm{env}}(X, h, 0)$ with $\mathrm{MA}_h^{\mathrm{ext}}(w) = c f \, \mathrm{d}V$. If $s \in \mathrm{PSH}(X, h)$ satisfies $\mathrm{MA}_h^{\mathrm{ext}}(s) \ge a f \, \mathrm{d}V$ with $a > 0$, then

$$\mathrm{MA}_h^{\mathrm{ext}}(w) \le \frac{c}{a} \mathrm{MA}_h^{\mathrm{ext}}(s).$$

Every h-plurisubharmonic s satisfies $s \preceq 0$, so the scalar assertion of Corollary 33.2.1, with reference zero, gives $c/a \ge 1$. Taking the maximum over all subsolution coefficients a, Theorem 15.27.1 yields $b_h(f) \le c$. The reverse inequality holds because w itself is a subsolution with coefficient c. The same argument identifies the scalar of every solution in this envelope class with $b_h(f)$.

For $\lambda > 0$, Theorem 33.5.1 also gives a solution s_λ of the exponential equation in the envelope class. Since s_λ is in particular a subsolution, the greatest-solution property gives $s_\lambda \le u_\lambda$. Upward stability (Proposition 14.4.1) places u_λ and its translate v_λ in that class.

Write $c_\lambda = \exp(\lambda \sup_X u_\lambda)$. By Corollary 15.34.1, $c_\lambda \to b_h(f) > 0$ and

$$\mathrm{MA}_h^{\mathrm{ext}}(v_\lambda) = c_\lambda \mathrm{e}^{\lambda v_\lambda} f \, \mathrm{d}V \longrightarrow b_h(f) f \, \mathrm{d}V \quad \text{in total variation.}$$

Since $v_\lambda \le 0$ and c_λ stays bounded for small λ, these measures are dominated by one finite non-pluripolar measure $C f \, \mathrm{d}V$. Applying Theorem 33.3.1 to the given L^1-convergent sequence v_{λ_j}, we obtain class membership, convergence in capacity and the equation for its limit. □

By Theorem 1.4.1, every sequence $\lambda_j \downarrow 0$ has a subsequence along which v_{λ_j} converges in L^1. The corollary identifies the equation of every such limit; the examples below show that this equation alone need not determine the normalized potential.

33.7.1 Nonuniqueness in the minimal envelope class

In this subsection the manifold and the background are no longer fixed. The examples of Theorem 12.9.1 are smooth solutions on projective manifolds with nonclosed Hermitian backgrounds. Comparison with a Kähler form shows that these backgrounds have both positive lower mass and bounded mass, and smooth potentials belong to the minimal envelope class. Thus even both mass bounds together allow several normalized solutions with the same scalar.

Corollary 33.7.2 *In every complex dimension $n \geq 3$, there are a projective manifold X of dimension n, a smooth nonclosed Hermitian form ω on X, and a nonzero finite non-pluripolar measure ν on X such that*

$$0 < v_-(\omega) \leq v_+(\omega) < +\infty,$$

but the equation

$$\mathrm{MA}_\omega^{\mathrm{ext}}(u) = c\nu, \qquad u \in \mathcal{E}_{\mathrm{env}}(X, \omega, 0), \qquad \sup_X u = 0,$$

has at least two smooth solutions u for $c = 1$, and no solution for any other positive constant c. Relative to a Kähler volume form on X, the measure ν has a real analytic density, positive off two connected smooth real hypersurfaces and vanishing quadratically along them.

Proof Use the form and the two potentials of Theorem 12.9.1. Since the manifold is projective, Lemma 7.5.1 gives bounded mass. Fix a Kähler form Ω on X and choose $\varepsilon > 0$ with $\varepsilon\Omega \leq \omega$. Lower-volume monotonicity, Theorem 31.1.1, applied to the zero potential in both backgrounds, gives

$$v_-(\omega) \geq v_-(\varepsilon\Omega) = \varepsilon^n \int_X \Omega^n > 0.$$

Here the potentials of the same ordinary type as zero are exactly the bounded ones, so these lower volumes are the infima defining v_-.

The two smooth potentials belong to $\mathcal{E}_{\mathrm{env}}(X, \omega, 0)$ by Proposition 14.4.1, and their extended measures are the ordinary top powers. Their common measure ν is finite and does not charge pluripolar sets. The construction gives $c = 1$, and Corollary 33.2.1 shows that no other positive scalar is possible. The density and its zero set were computed in Theorem 12.9.1. □

Chapter 34
Big backgrounds under bounded mass

Let X be a connected compact complex manifold of dimension $n \geq 1$, let θ be a smooth big real $(1,1)$-form on X, not necessarily closed or semipositive, and let $\varphi \in \mathrm{PSH}(X,\theta)$. As in Chapter 33, we solve the equations $\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu$, $\lambda > 0$, and $\mathrm{MA}_\theta(u) = c\mu$, $c > 0$, for an arbitrary nonzero finite non-pluripolar measure μ, with u in the relative class of φ. Since θ need not be semipositive, the smaller-background model condition used there is replaced by upper and positive lower mass bounds.

Throughout this chapter we make two standing assumptions. First, X has the bounded mass property of Definition 7.5.1: $v_+(\omega) < +\infty$ for one, and hence every, Hermitian form ω on X (Proposition 7.5.1). Second, φ satisfies (31.5): it is an ordinary model, $P_\theta[\varphi] = \varphi$ (Definition 1.19.1), and its lower singularity volume $\underline{\mathrm{vol}}_\theta(\varphi)$ (Definition 14.3.1) is positive. Fix a Hermitian form ω for capacity and a smooth positive volume form $\mathrm{d}V$.

The two assumptions give the two bounds needed for approximation: every member of the relative class of φ has Monge–Ampère mass at least $\underline{\mathrm{vol}}_\theta(\varphi) > 0$ and at most $\overline{\mathrm{vol}}_\theta(\varphi) < \infty$. With these bounds, first moments relative to a reference prevent collapse and keep limits in the relative class, while higher moments control convergence against a smoothed measure. This first gives solutions in the class of the potential V_θ with minimal singularities.

To pass to the reference φ, we project a solution in the relative class of V_θ onto the relative class of φ. The projection keeps only part of the exponentially weighted measure, but weighted addition can fill any remaining deficit; maximizing the weighted mass therefore gives the required equation. The same contact estimates keep limits in the relative class when the model reference varies in the volume comparison of Section 31.6.

For general finite non-pluripolar data the solutions lie in the relative full-mass class; for L^p densities, a stronger moment estimate shows that they differ from φ by a bounded function. A positive exponent determines the solution uniquely; at exponent zero only the scalar is unique.

For $a \in \mathrm{PSH}(X,\theta)$, write

$$\mathcal{E}_\varphi = \mathcal{E}_{\mathrm{vol}}(X,\theta,\varphi) = \mathcal{E}_{\mathrm{env}}(X,\theta,\varphi), \qquad \theta_a = \theta + \mathrm{dd}^{\mathrm{c}} a, \qquad \mu_a = \mathrm{MA}_\theta(a).$$

Here $\mathcal{E}_{\mathrm{vol}}(X,\theta,\varphi)$ is the relative full-mass class of Definition 31.2.1, $\mathcal{E}_{\mathrm{env}}(X,\theta,\varphi)$ is the relative envelope class of Definition 14.4.1, and the two classes coincide by Theorem 31.2.1. The measure $\mathrm{MA}_\theta(a)$ is the non-pluripolar Monge–Ampère measure of Theorem 7.1.1; it is finite by Proposition 7.5.2. Put

$$\ell = \underline{\mathrm{vol}}_\theta(\varphi) > 0, \qquad M = \overline{\mathrm{vol}}_\theta(\varphi) < \infty.$$

By Theorem 31.2.1, every $u \in \mathcal{E}_\varphi$ satisfies

$$\ell \le \mu_u(X) \le M. \tag{34.1}$$

We use the stretching envelopes $Q_b(u,w) = P_\theta(bu-(b-1)w)$ introduced before Definition 14.4.1, the rooftop envelopes $P_\theta(u_1,\dots,u_N) = P_\theta(\min_i u_i)$, and $V_\theta = P_\theta(0)$ (Section 1.19). For $a \in \mathcal{E}_\varphi$, Proposition 14.4.3 gives $P_\theta[a] = \varphi$, and hence, as in (33.2),

$$a - \sup_X a \le \varphi.$$

We refer to this consequence of the identity $P_\theta[a] = \varphi$ as the envelope identity. Finally, *zero domination* refers to Theorem 31.3.1 with coefficient zero: if $u \in \mathcal{E}_\varphi$, $v \preceq \varphi$ and $\mu_u(\{u < v\}) = 0$, then $u \ge v$.

34.1 Equations for finite measures

A positive exponent determines a unique solution in the relative class. At exponent zero, the equation determines its positive scalar, but normalized potentials may still differ, as in Corollary 33.7.2.

Theorem 34.1.1 *For every nonzero finite non-pluripolar measure μ on X:*

(1) *There are $u \in \mathcal{E}_\varphi$ and a unique $c > 0$ such that $\mathrm{MA}_\theta(u) = c\mu$.*
(2) *For each $\lambda > 0$ there is a unique $u \in \mathcal{E}_\varphi$ such that $\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu$.*

If $\mu = f\,\mathrm{d}V$ with $0 \le f \in L^p(\mathrm{d}V)$ for some $p > 1$, the solutions in both parts may be taken with $u \sim \varphi$.

Part (1) is Theorem 34.7.2, part (2) is Theorem 34.7.1, and the assertion for L^p densities is Corollary 34.8.1.

34.2 Convergence under a fixed measure

As in the semipositive case, a fixed dominating measure gives a common lower envelope after extraction, and then convergence in capacity.

Theorem 34.2.1 *Suppose $u_j \in \mathcal{E}_\varphi$, $u_j \to u \in \mathrm{PSH}(X,\theta)$ in $L^1(X)$, and $\mu_{u_j} \le \nu$ for one finite non-pluripolar measure ν. Then there is a subsequence with*

$$h = P_\theta\left(\inf_j u_j\right) \in \mathcal{E}_\varphi.$$

Moreover, for the original sequence,

$$u \in \mathcal{E}_\varphi, \qquad u_j \longrightarrow u \text{ in capacity}, \qquad \mu_{u_j} \rightharpoonup \mu_u \le \nu.$$

Proof The proofs of Lemma 33.3.1, Proposition 33.3.1 and Theorem 33.3.1 apply, with three replacements: the envelope identity replaces (33.2), the lower bound ℓ of (34.1) replaces κ, and zero domination replaces Lemma 33.2.1. Their other ingredients do not use semipositivity of the background: finite rooftops and stretching envelopes of members of $\mathcal{E}_\varphi$ stay in $\mathcal{E}_\varphi$ (Proposition 14.4.2 and Lemma 14.4.1), contact support holds by Theorem 15.7.2, and Lemmas 1.4.1, 1.8.9, 1.23.1 and 7.3.1 and Theorem 8.6.1 hold for every smooth background.

In particular, after a common translation with $u_j, u \le \varphi \le 0$ and an extraction with $\int_X |\mathrm{e}^{u_j} - \mathrm{e}^{u}|\,\mu_{u_j} \le 2^{-j}$, the finite rooftops $p_{j,N} = P_\theta(u_j,\dots,u_N)$ satisfy, as in (33.4),

$$\mu_{p_{j,N}} \le \nu, \qquad \int_X |\mathrm{e}^{p_{j,N}} - \mathrm{e}^{u}|\,\mu_{p_{j,N}} \le 2^{1-j}, \tag{34.2}$$

and the stretched rooftops $Q_B(p_{j,N},\varphi) \in \mathcal{E}_\varphi$ have masses at least ℓ. The exponential test (33.5) then gives the common lower envelope $h \in \mathcal{E}_\varphi$ of the first assertion. In the

proof of Proposition 33.3.1, zero domination applies to the limit $p \in \mathcal{E}_\varphi$ of the tail envelopes, since $p \le u \preceq \varphi$ and $\mu_p(\{p < u\}) = 0$; it gives $p = u$. □

34.3 Relative moments

For $w \in \mathcal{E}_\varphi$ and $u \le w$, the relative first moment $\int_X (w-u)\,\mu_u$ measures how far u lies below w, weighted by the measure of u. A uniform bound on these moments prevents a sequence from collapsing and keeps its limits in $\mathcal{E}_\varphi$. The proof compares rooftop masses with the fixed measure of a potential of the same type as w whose mass is close to the upper volume M. For a real number t, write $t_+ = \max(t, 0)$.

Proposition 34.3.1 *Fix $w \in \mathcal{E}_\varphi$. Suppose $u_j \in \mathcal{E}_\varphi$, $u_j \le w$, and*

$$K := \sup_j \int_X (w - u_j)\,\mu_{u_j} < +\infty.$$

Then $\sup_X u_j$ is bounded below. If $u_j \to u \in \mathrm{PSH}(X,\theta)$ in $L^1(X)$, then $u \in \mathcal{E}_\varphi$.

The same conclusions hold if $u_j \le w$ is replaced by a common upper bound $u_j \le \varphi + C$ and the integral is replaced by $\int_X (w-u_j)_+\,\mu_{u_j}$.

Proof We first treat the case $u_j \le w$. Since $\overline{\mathrm{vol}}_\theta(w) = M$, we may choose $w' \sim w$ with $\mu_{w'}(X) > M - \ell/2$. Translate w' so that $w' \le w$, and put $p_j = P_\theta(u_j, w')$. These rooftops belong to $\mathcal{E}_\varphi$. Contact support and ordered contact imply

$$\int_X (w' - p_j)\,\mu_{p_j} \le \int_X (w - u_j)\,\mu_{u_j} \le K. \tag{34.3}$$

Indeed, the integrand vanishes on the contact part $\{p_j = w'\}$, and on the remaining contact part $p_j = u_j < w' \le w$.

We shall use the following estimate for maxima. If $a \in \mathcal{E}_\varphi$, $a \le w'$, and $t > 0$, then

$$\mu_a(\{a > w' - t\}) \le \mu_{w'}(\{a > w' - t\}) + M - \mu_{w'}(X). \tag{34.4}$$

To see this, put $m = \max(a, w' - t)$. It has the ordinary singularity type of w', so $\mu_m(X) \le M$. On $\{a > w' - t\}$, locality gives $\mu_m = \mu_a$. On the complementary set, where $m = w' - t$, ordered contact gives $\mu_m \ge \mu_{w'}$. Integrating over the two sets proves (34.4).

Suppose that a subsequence had $\sup_X u_j \to -\infty$; then also $\sup_X p_j \to -\infty$. Splitting the mass of p_j at the level $w' - s$, $s > 0$, and using (34.3) and (34.4), we obtain

$$\ell \le \mu_{p_j}(X) \le \frac{K}{s} + \mu_{w'}(\{p_j > w' - s\}) + M - \mu_{w'}(X).$$

For fixed s, the middle term tends to zero as $j \to \infty$, since w' is finite off a pluripolar set and $\mu_{w'}$ is finite and non-pluripolar. Letting $j \to \infty$ and then $s \to \infty$ gives $\ell \le M - \mu_{w'}(X) < \ell/2$, a contradiction. This proves the first assertion.

Now suppose $u_j \to u$ in L^1. Then $u \le w \preceq \varphi$. Fix $b > 1$ and suppose that $Q_b(u,w) \equiv -\infty$. Since $w - w'$ is bounded, translating competitors by a constant shows that $Q_b(u,w') \equiv -\infty$ as well. Keep the same p_j and set

$$h_j = Q_b(p_j, w') \in \mathcal{E}_\varphi.$$

We claim that $\sup_X h_j \to -\infty$. Otherwise, since $h_j \le p_j \le w'$, Theorem 1.4.1 would give a subsequence converging in L^1 to some $h \not\equiv -\infty$. Passing to the limit in

$$b^{-1}h_j + (1-b^{-1})w' \le p_j \le u_j,$$

first almost everywhere and then everywhere by Lemma 1.3.3, would give a competitor for $Q_b(u,w')$, a contradiction.

On the contact set of h_j one has $p_j = b^{-1}h_j + (1-b^{-1})w'$. Thus $h_j \le w' - bs$ implies $p_j \le w' - s$ there. Contact support and (34.3) therefore imply

$$\mu_{h_j}(\{h_j \le w' - bs\}) \le b^n K/s.$$

Applying (34.4) with $a = h_j$ and $t = bs$ on the complementary set, we obtain

$$\ell \le b^n K/s + \mu_{w'}(\{h_j > w' - bs\}) + M - \mu_{w'}(X).$$

Let first $j \to \infty$ and then $s \to \infty$. The choice of w' again gives $\ell < \ell/2$, a contradiction. Therefore $Q_b(u,w) \not\equiv -\infty$ for every $b > 1$. The resulting convex inequalities and Theorem 31.1.1 and Proposition 31.1.1 give

$$(1-b^{-1})^n M \le \overline{\mathrm{vol}}_\theta(u) \le M.$$

Letting $b \to \infty$ shows that the upper volumes of u and φ are equal; hence $u \in \mathcal{E}_\varphi$.

For the last assertion put $q_j = P_\theta(u_j, w)$. Contact gives

$$\int_X (w - q_j)\,\mu_{q_j} \le \int_X (w - u_j)_+ \,\mu_{u_j} \le K.$$

The case already treated bounds $\sup_X q_j$ from below, and therefore also $\sup_X u_j \ge \sup_X q_j$. If $u_j \to u$ in L^1, then, since $q_j \le w$, a subsequence of the q_j converges in L^1 to some $q \not\equiv -\infty$, and the case already treated gives $q \in \mathcal{E}_\varphi$. Since $q \le u \le \varphi + C$, upward stability gives $u \in \mathcal{E}_\varphi$ as well. □

A common lower bound in $\mathcal{E}_\varphi$ and a weighted first moment of the difference from the limit tending to zero give convergence in capacity: rooftops of the tails identify the limit.

Lemma 34.3.1 *Let $a_j \in \mathrm{PSH}(X,\theta)$ have a common upper bound and converge in $L^1(X)$ to $a \in \mathrm{PSH}(X,\theta)$ with $a \preceq \varphi$. Suppose there is $h \in \mathcal{E}_\varphi$ such that $h \le a_j$ for every j. If, for some $\lambda > 0$,*

$$\int_X |a_j - a| e^{\lambda a}\, \mu_{a_j} \longrightarrow 0,$$

then $a \in \mathcal{E}_\varphi$ and $a_j \to a$ in capacity.

Proof By Lemma 1.3.3, the lower bound passes to the limit, so $h \le a \preceq \varphi$, and upward stability gives $a \in \mathcal{E}_\varphi$. Start with any subsequence and extract a further subsequence along which the integrals in the hypothesis are at most 2^{-j} and $a_j \to a$ almost everywhere for smooth volume. For $N \ge j$ let $p_{j,N} = P_\theta(a_j, \dots, a_N)$; these rooftops are not identically $-\infty$ because h is a competitor. Partitioning the contact set according to the first obstacle touched gives

$$\int_X \min(1, |p_{j,N} - a|) e^{\lambda a}\, \mu_{p_{j,N}} \le \sum_{k=j}^{N} \int_X |a_k - a| e^{\lambda a}\, \mu_{a_k} \le 2^{1-j}.$$

The decreasing limits $p_j = P_\theta(\inf_{k \ge j} a_k)$ satisfy $h \le p_j \le a$, the second inequality by the almost-everywhere convergence, and hence so does their increasing regularized limit p. Upward stability puts the p_j and p in $\mathcal{E}_\varphi$, although the finite rooftops need not lie there.

Both $p_{j,N} \downarrow p_j$ and $p_j \uparrow p$ hold in capacity by Lemma 1.8.9, since h is a common lower bound. Apply Lemma 7.3.1 to this rooftop estimate, with the bounded weights $\min(1, |p_{j,N} - a|) e^{\lambda a}$ and then $\min(1, |p_j - a|) e^{\lambda a}$. These weights converge in capacity: discard the deep sublevel sets of h and a, and use continuity on the remaining finite ranges. The result is

$$\int_X \min(1, |p - a|) e^{\lambda a}\, \mu_p = 0.$$

Since the exponential is positive off a pluripolar set, which μ_p does not charge, $\mu_p(\{p < a\}) = 0$, and zero domination gives $p = a$. The upper envelopes $(\sup_{k \ge j} a_k)^*$ decrease to a. Squeezing the extracted a_j between these envelopes and p_j proves convergence in capacity. Since the initial subsequence was arbitrary, the conclusion holds for the original sequence. □

Without a given lower bound, the first-moment criterion produces one from the rooftops of the tails.

Theorem 34.3.1 *Let $u_j, u \in \mathcal{E}_\varphi$, with $u_j \to u$ in $L^1(X)$. If*

$$\int_X |u_j - u|\, \mu_{u_j} \longrightarrow 0,$$

then $u_j \to u$ in capacity. Moreover, after passing to a subsequence, $P_\theta(\inf_{k\geq j} u_k)$ belongs to $\mathcal{E}_\varphi$ for every j.

Proof Start with any subsequence and extract a further subsequence along which the integrals in the hypothesis are at most 2^{-j} and $u_j \to u$ almost everywhere for smooth volume. The envelope identity and Hartogs' lemma give a uniform bound $u_j, u \leq \varphi + C$. For $N \geq j$ let $p_{j,N} = P_\theta(u_j, \dots, u_N)$. Partitioning the contact set as in the proof of the preceding lemma gives

$$\int_X |p_{j,N} - u|\, \mu_{p_{j,N}} \leq 2^{1-j}. \tag{34.5}$$

By (34.5), Proposition 34.3.1, applied with $w = u$ in its one-sided form, bounds $\sup_X p_{j,N}$ from below, independently of N. Consequently the decreasing sequence has a limit

$$p_j = P_\theta\left(\inf_{k\geq j} u_k\right)$$

that is not identically $-\infty$, and the same proposition puts p_j in $\mathcal{E}_\varphi$. This proves the second assertion. Since $p_1 \leq u_k$ for every k and $e^u \leq e^{\sup_X u}$, the preceding lemma, applied to the extracted sequence with $h = p_1$ and $\lambda = 1$, shows that it converges to u in capacity. Since the initial subsequence was arbitrary, the conclusion holds for the original sequence. □

34.3.1 Higher moments in the minimal singularity type

In this subsection the reference is the potential $V = V_\theta = P_\theta(0)$ with minimal singularities. It is an ordinary model, and we write $\mathcal{E}_V = \mathcal{E}_{\mathrm{env}}(X, \theta, V)$ and

$$\ell_V = \underline{\mathrm{vol}}_\theta(V) \geq \ell > 0, \qquad M_V = \overline{\mathrm{vol}}_\theta(V) < \infty.$$

The first inequality follows from volume monotonicity, since $\varphi \preceq V$, and the last from bounded mass. Thus V satisfies the standing assumptions on φ, and the results above apply with φ replaced by V. For a probability measure ν, $A_\theta(\nu)$ denotes the supremum of $\int_X (-w)\, d\nu$ over the $w \in \mathrm{PSH}(X, \theta)$ with $\sup_X w = 0$, as in Section 1.23.4. We show that a first moment of $V - u$ against the measure μ_u of u itself controls a higher moment of $V - u$ against every fixed ν with $A_\theta(\nu) < \infty$. An envelope with a concave obstacle converts the distribution of $V - u$ under μ_u into this stronger integrability estimate. We state the envelope lemma for the reference φ; here it is applied with V in place of φ, and in Section 34.8 with φ itself.

Lemma 34.3.2 *Let $u \in \mathcal{E}_\varphi$ with $u \le \varphi$, and let $\chi\colon \mathbb{R}_{\le 0} \to \mathbb{R}_{\le 0}$ be increasing and concave, with $\chi(0) = 0$. Suppose that χ is $C^{1,1}$ on compact intervals and that, for a finite constant $L \ge 1$,*

$$1 \le \chi' \le L.$$

Then the envelope

$$p = P_\theta(\varphi + \chi(u - \varphi))$$

belongs to $\mathcal{E}_\varphi$, and

$$\mu_p \le \mathbf{1}_{\{p=\varphi+\chi(u-\varphi)\}}\chi'(u-\varphi)^n\mu_u.$$

Here the obstacle and the contact set are defined off the pluripolar set where $u - \varphi$ is undefined, which all these measures ignore.

Proof For $t \le 0$, the bounds on χ' give $Lt \le \chi(t) \le t$. Thus

$$Q_L(u, \varphi) \le p \le u \le \varphi.$$

The lower potential belongs to $\mathcal{E}_\varphi$ by Lemma 14.4.1, so p is not identically $-\infty$ and belongs to $\mathcal{E}_\varphi$ by upward stability. The obstacle is finite and quasi-continuous off the poles, so the contact theorem Theorem 15.7.2 applies.

Let $\tau = \chi^{-1}$. It is convex, with $L^{-1} \le \tau' \le 1$. We justify the transformation

$$w = \varphi + \tau(p - \varphi)$$

without assuming that φ is locally bounded on some open set. On a coordinate ball choose a smooth function q with $\theta + \mathrm{dd}^c q \ge 0$. The functions

$$p_m = \max(p, q - m), \qquad \varphi_m = \max(\varphi, q - m)$$

are locally bounded θ-plurisubharmonic functions on the ball, and $p_m \le \varphi_m$. Lemma 1.7.6 applies to $w_m = \varphi_m + \tau(p_m - \varphi_m)$ and gives

$$\theta_{w_m} \ge 0, \qquad \mu_{w_m} \ge \tau'(p_m - \varphi_m)^n \mu_{p_m}.$$

The function $(s, t) \mapsto t + \tau(s - t)$ on $\{s \le t\}$ is nondecreasing in each variable. Consequently w_m decreases as m increases. Since $p \le p_m \le w_m \le \varphi_m$, the limit is not identically $-\infty$ on the ball. Off the poles it equals w. The local limits therefore agree on overlaps after regularization and define a global θ-plurisubharmonic function.

On $D_m = \{p > q - m\}$, neither p nor φ is truncated. Thus $w_m = w$ there, and plurifine locality transfers the product inequality to D_m. These sets exhaust the complement of the poles of p in the ball. Since all products ignore the discarded pluripolar set,

$$\mu_w \ge \tau'(p - \varphi)^n \mu_p.$$

The obstacle bound $p \le \varphi + \chi(u - \varphi)$ gives $w \le u$, with equality on the contact set of p. Ordered contact, the identity $\tau'(p - \varphi) = 1/\chi'(u - \varphi)$ on that set, and contact support of μ_p prove the assertion. □

Proposition 34.3.2 *Set $N = 2n + 1$ and $r = 1 + (4N)^{-1}$. For $u \in \mathrm{PSH}(X, \theta)$ with $u \sim V$ and $\sup_X u = 0$, put $B = \int_X (V - u)\, \mu_u$. There is a constant C, depending only on n, ℓ_V, M_V, such that, for every non-pluripolar probability measure ν with $A = A_\theta(\nu) < \infty$,*

$$\int_X (V - u)^r \, \mathrm{d}\nu \le C(1 + A)(1 + B)^{1+1/N}.$$

In particular, normalized potentials $u \sim V$ with uniformly bounded B are uniformly integrable with respect to ν.

Proof From the distribution of $V - u$ under μ_u we build a concave weight adapted to u, apply the preceding lemma, and use A to bound the resulting envelope against ν. Throughout, c and C denote positive constants depending only on n, ℓ_V, M_V.

The normalization gives $u \le V$. Put $x = V - u \ge 0$ and

$$T = \sup\{t \ge 0 : \mu_u(\{x > t\}) > 0\},$$

with $T = 0$ if this set is empty. Since $u \sim V$, T is finite. Zero domination, applied to u and $V - t$ for $t > T$, shows that $u \ge V - T$; thus $x \le T$ off the pole set. If $T \le 2$ the claim is immediate. We may therefore suppose $T > 2$; in particular $B > 0$.

Fix a real number s with $0 < s < T - 2$. For $t \ge 0$, define

$$g_s(t) = \begin{cases} \big((1+t)^{3/2}\mu_u(\{x > t\})\big)^{-1}, & 0 \le t \le T - s, \\ (1+t)^{-3/2}, & t > T - s, \end{cases} \qquad F_s(t) = 1 + \int_0^t g_s(a)\, \mathrm{d}a,$$

and set

$$W_s(t) = \int_0^t F_s(a)^{1/N}\, \mathrm{d}a, \qquad \chi_s(-t) = -W_s(t).$$

The tail $\mu_u(\{x > t\})$ in the first denominator has a positive lower bound on $[0, T - s]$. Hence F_s is bounded, nondecreasing and locally Lipschitz, with $F_s \ge 1$. Since $u \le V$ and, by Proposition 14.4.1, $u \in \mathcal{E}_V$, the function χ_s satisfies the hypotheses of Lemma 34.3.2, with V in place of φ.

Tonelli's theorem gives

$$\int_X F_s(x)\, \mu_u = \mu_u(X) + \int_0^T g_s(t)\mu_u(\{x > t\})\, \mathrm{d}t \le 3\max(1, M_V), \tag{34.6}$$

since $\int_0^\infty (1+t)^{-3/2}\, \mathrm{d}t = 2$ and the tail is at most M_V on the second interval. Chebyshev's inequality gives $\mu_u(\{x > t\}) \le B/t$. Integrating first for F_s and then for W_s, we obtain

$$W_s(t) \ge c(1 + B)^{-1/N} t^\beta, \qquad 1 \le t \le T - s, \qquad \beta = 1 + \frac{1}{2N}. \tag{34.7}$$

Indeed, $g_s(t) \ge cB^{-1}t^{-1/2}$ for $t \ge 1$ in this interval, so $F_s(t) \ge c(1 + B)^{-1}t^{1/2}$. For $t \ge 2$ integrate this estimate over $[t/2, t]$; for $1 \le t \le 2$ use $W_s(t) \ge t$.

Let $p_s = P_\theta(V - W_s(x)) \in \mathcal{E}_V$, the envelope of Lemma 34.3.2 for $\chi = \chi_s$. The lemma, the bound $W_s(t) \le tF_s(t)^{1/N}$ and the Cauchy–Schwarz inequality give, since $N = 2n + 1$,

$$\begin{aligned}\int_X (V - p_s)^{1/2}\,\mu_{p_s} &\le \int_X x^{1/2} F_s(x)^{(n+1/2)/N}\,\mu_u \\ &\le B^{1/2}\left(\int_X F_s(x)\,\mu_u\right)^{1/2} \le CB^{1/2}.\end{aligned}$$

Write $a_s = \sup_X p_s \le 0$. By the definition of V, $p_s - a_s \le V$, so $V - p_s \ge -a_s$. Since $\mu_{p_s}(X) \ge \ell_V$, the last estimate yields $\ell_V\sqrt{-a_s} \le C\sqrt{B}$, and hence $-a_s \le CB$.

The definition of A now gives

$$\int_X W_s(x)\,\mathrm{d}\nu \le \int_X (V - p_s)\,\mathrm{d}\nu \le A - a_s \le A + CB.$$

Combining this with (34.7), and then letting $s \downarrow 0$ for each fixed $1 \le t < T$, we obtain

$$\nu(\{x > t\}) \le C(1 + B)^{1/N}(A + B)t^{-\beta}.$$

There is no tail for $t \ge T$. The layer-cake formula therefore gives

$$\int_X x^r\,\mathrm{d}\nu = r\int_0^\infty t^{r-1}\nu(\{x > t\})\,\mathrm{d}t \le 1 + C(1 + B)^{1/N}(A + B),$$

because $1 < r < \beta$. This implies the stated bound.

Finally $-u = -V + (V - u)$, where $-V \in L^1(\nu)$ because $A < \infty$. The first summand is fixed and integrable, and the second has a uniform $L^r(\nu)$ bound. This proves uniform integrability; no L^r bound for V itself is claimed. □

34.4 General measures in the minimal class

We now solve the equations in $\mathcal{E}_V$. For a measure ν with $A_\theta(\nu) < \infty$, the coordinate convolution of Section 1.23.4 gives approximating equations with smooth densities. The integral estimates for this smoothing, combined with the higher-moment bound, give convergence of the solutions against the original measure. The resulting self-measure error yields convergence in capacity, and a first-moment tail estimate identifies the limit of the Monge–Ampère measures.

Proposition 34.4.1 *Let ν be a non-pluripolar probability measure with $A_\theta(\nu) < \infty$, and let ν_j be its smooth approximations defined by* (1.44)*. Fix a constant $C > 0$ and let $v_j, v \in \mathrm{PSH}(X,\theta)$ satisfy*

$$v_j \sim V, \qquad \sup_X v_j = 0, \qquad \mu_{v_j} \le C\nu_j, \qquad v_j \longrightarrow v \text{ in } L^1(X).$$

Then $v \in \mathcal{E}_V$, the convergence also holds in $L^1(\nu)$ and in capacity, and

$$\mu_{v_j} \rightharpoonup \mu_v.$$

Proof The limit v is not identically $-\infty$ and satisfies $\sup_X v = 0$, and $v_j, v \le V \le 0$. By Lemma 1.23.5, $\sup_j A_\theta(v_j) < \infty$, which gives the uniform estimate

$$\int_X (-v_j)\,\mu_{v_j} \le C \int_X (-v_j)\,\mathrm{d}\nu_j \le C'. \tag{34.8}$$

In particular the relative first moments $\int_X (V - v_j)\mu_{v_j}$ are bounded, and Proposition 34.3.1, with fixed reference $w = V$, puts v in $\mathcal{E}_V$.

By Proposition 34.3.2, applied to the fixed measure ν, the v_j are uniformly integrable with respect to ν. For each fixed T, the bounded functions $\max(v_j, -T)$ have a common lower curvature bound in a Hermitian background $\Omega \ge \theta$ and converge in L^1 for smooth volume to $\max(v, -T)$; hence Lemma 1.23.1 gives their convergence in $L^1(\nu)$. These truncations need not be θ-plurisubharmonic; they are used only in the larger Hermitian background. Uniform integrability, followed by $T \to \infty$, gives $v_j \to v$ in $L^1(\nu)$, and the limit is integrable because $A_\theta(\nu) < \infty$.

The smoothing lemma, in the form (1.45), now yields

$$\int_X |v_j - v|\,\mu_{v_j} \le C \int_X |v_j - v|\,\mathrm{d}\nu_j \longrightarrow 0.$$

Therefore the self-measure convergence theorem Theorem 34.3.1 gives convergence in capacity. Finally, (34.8) gives the sublevel estimate

$$\sup_j \mu_{v_j}(\{v_j \le -T\}) \le C'/T,$$

and the masses of the μ_{v_j} are bounded by M_V. The tail criterion Theorem 8.6.1 therefore gives the asserted weak convergence. □

34.4.1 Exponential and normalized equations

The smoothing argument solves both equations for measures with finite A_θ. A bounded carrier representation reduces an arbitrary finite non-pluripolar measure to increasing bounded densities, whose solutions have measures dominated by one fixed finite measure; Theorem 34.2.1 then passes to the limit.

Theorem 34.4.1 *For every finite non-pluripolar measure μ with $\mu(X) > 0$, the following assertions hold.*

(1) *For every $\lambda > 0$ there is a unique $u \in \mathcal{E}_V$ such that*

$$\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu.$$

(2) *There are $v \in \mathcal{E}_V$ with $\sup_X v = 0$ and a unique scalar $c > 0$ such that*

$$\mathrm{MA}_\theta(v) = c\mu.$$

Proof First suppose that μ is a probability measure with $A_\theta(\mu) < \infty$, and let μ_j be its smooth probability approximations. The theorems for the minimal ordinary type, Theorems 15.15.1 and 15.15.2, apply to their smooth, nonzero densities.

For the normalized equation, they give $v_j \sim V$ with $\sup_X v_j = 0$ and $\mu_{v_j} = c_j\mu_j$. Since μ_j is a probability measure, the mass bounds $\ell_V \le \mu_{v_j}(X) \le M_V$ give $\ell_V \le c_j \le M_V$. After extraction, $c_j \to c > 0$ and $v_j \to v$ in $L^1(X)$. The preceding proposition applies, so $v \in \mathcal{E}_V$ and $\mu_{v_j} \rightharpoonup \mu_v$. The right-hand sides converge weakly to $c\mu$, which proves the equation.

For the positive-exponent equation, take $u_j \sim V$ satisfying $\mu_{u_j} = \mathrm{e}^{\lambda u_j}\mu_j$, and write $a_j = \sup_X u_j$ and $v_j = u_j - a_j$. The mass bounds and Jensen's inequality give

$$\ell_V \le \mathrm{e}^{\lambda a_j}, \qquad M_V \ge \mathrm{e}^{\lambda a_j}\int_X \mathrm{e}^{\lambda v_j}\,\mathrm{d}\mu_j \ge \mathrm{e}^{\lambda a_j - \lambda A_\theta(\mu_j)},$$

and $\sup_j A_\theta(\mu_j) < \infty$ by Lemma 1.23.5. Thus a_j is bounded above and below. After extraction, $a_j \to a$, $v_j \to v$ in $L^1(X)$, and $\mu_{v_j} \le C\mu_j$. The preceding proposition gives $v \in \mathcal{E}_V$, convergence in $L^1(\mu)$ and weak convergence of the Monge–Ampère measures. Set $u = v + a$. By the smoothing lemma, the right-hand sides converge weakly to $\mathrm{e}^{\lambda u}\mu$. This proves the equation, and $u \in \mathcal{E}_V$.

The same conclusions hold for every nonzero finite measure whose normalization to a probability measure has finite A_θ, since scaling the measure only changes the positive scalar, or translates the solution of the exponential equation by a constant.

Now let μ be an arbitrary finite non-pluripolar measure of positive mass. The bounded-carrier decomposition Lemma 15.23.1, applied to ω, writes

$$\mu = f\nu, \qquad \nu = \mathrm{MA}_\omega(\psi), \qquad \psi \in \mathrm{PSH}(X,\omega) \cap L^\infty(X), \qquad 0 \le f \in L^1(\nu).$$

For $\mu_k = \min(f,k)\nu$, we have $\mu_k(X) > 0$. By Lemma 1.23.4, the normalization of μ_k has finite A_θ for each fixed k.

For the normalized equation, the preceding case gives $v_k \in \mathcal{E}_V$ with $\sup_X v_k = 0$ and $\mu_{v_k} = c_k \mu_k$, where

$$\frac{\ell_V}{\mu(X)} \le c_k \le \frac{M_V}{\mu_1(X)}.$$

After extraction, $c_k \to c > 0$ and v_k converges in L^1 to some $v \not\equiv -\infty$. All the measures μ_{v_k} are bounded by the fixed finite non-pluripolar measure $M_V \mu / \mu_1(X)$. The fixed-dominator theorem Theorem 34.2.1 gives $v \in \mathcal{E}_V$ and weak convergence to μ_v. Since $c_k \mu_k \to c\mu$ in total variation, $\mu_v = c\mu$.

For the positive-exponent equation, let $u_k \in \mathcal{E}_V$ solve $\mu_{u_k} = \mathrm{e}^{\lambda u_k} \mu_k$. The argument that follows the construction of the approximating solutions in the positive-exponent part of the proof of Theorem 33.5.1 applies, with Corollary 31.3.1 and Theorem 34.2.1 and ℓ_V in place of Corollary 33.2.1 and Theorem 33.3.1 and $v_-(\theta)$. It shows that the u_k decrease to some $u \in \mathcal{E}_V$ with $\mu_u = \mathrm{e}^{\lambda u} \mu$.

Finally, Corollary 31.3.1 with reference V, applied in both directions, gives uniqueness of the exponential solution and of the positive scalar. □

Positive lower mass of a Hermitian form gives positive lower volume to the minimal reference of every smooth big background.

Corollary 34.4.1 *Assume only that X has the bounded mass property and that $v_-(\omega) > 0$. Then $\underline{\mathrm{vol}}_\theta(V_\theta) > 0$ for every smooth big real $(1,1)$-form θ on X; hence the preceding theorem applies to every such θ.*

Proof Choose a strict barrier $\rho \in \mathrm{PSH}(X, \theta)$, so that $\theta_\rho \ge a\omega$ for some $a > 0$. Then $\rho \in \mathrm{PSH}(X, \theta - a\omega)$. Volume addition (Proposition 31.1.1), followed by monotonicity in the singularities (Theorem 31.1.1), gives

$$0 < a^n v_-(\omega) = \underline{\mathrm{vol}}_{a\omega}(0) \le \underline{\mathrm{vol}}_\theta(\rho) \le \underline{\mathrm{vol}}_\theta(V_\theta).$$

In the addition step the big background is $a\omega$, and the added positive current is $(\theta - a\omega)_\rho$. Bounded mass gives finiteness of the upper volume. □

Conversely, the standing assumptions imply $v_-(\omega) > 0$. Indeed, choose $C > 0$ with $\theta \le C\omega$. Since $\varphi \preceq 0$, Theorem 31.1.1 gives

$$0 < \underline{\mathrm{vol}}_\theta(\varphi) \le \underline{\mathrm{vol}}_{C\omega}(0) = C^n v_-(\omega).$$

Thus, under bounded mass, some smooth big background admits a reference satisfying (31.5) exactly when $v_-(\omega) > 0$, and then V_θ satisfies it in every smooth big background θ.

34.5 Relative projection and supersolutions

To pass from the minimal class to the reference φ, we project a solution $z \in \mathcal{E}_V$ onto $\mathcal{E}_\varphi$. The projection is the increasing limit $P_\theta[\varphi](z)$ of the rooftops $P_\theta(\varphi + C, z)$ as $C \to \infty$, in the notation of Lemma 31.3.1. The small stretches of Lemma 31.4.1 first show that the rooftop $P_\theta(\varphi, z)$ is not identically $-\infty$ and has the full upper volume of φ. Contact support then bounds the measure of the projection by the measure of z on the set where the two coincide. The argument uses only that the reference of z lies above φ, so we formulate it for an arbitrary ordinary model above φ; this will also allow the reference to vary.

Let $\eta \in \mathrm{PSH}(X, \theta)$ satisfy $P_\theta[\eta] = \eta \geq \varphi$, and write $\mathcal{E}_\eta = \mathcal{E}_{\mathrm{env}}(X, \theta, \eta)$.

Proposition 34.5.1 *For every $z \in \mathcal{E}_\eta$, the rooftop $P_\theta(\varphi, z)$ is not identically $-\infty$ and belongs to $\mathcal{E}_\varphi$.*

Proof First suppose $z \leq \eta$; recall also that $\varphi \leq \eta \leq 0$. For $b > 1$ sufficiently close to one, Lemma 31.4.1 gives

$$h_b = Q_b(\varphi, \eta) \leq \varphi,$$

which is not identically $-\infty$. Set $a = 1 - b^{-1}$. The stretching lemma Lemma 14.4.1, now applied in $\mathcal{E}_\eta$, gives $z_{1/a} = Q_{1/a}(z, \eta) \in \mathcal{E}_\eta$. The potential

$$k_a = a z_{1/a} + (1 - a) h_b$$

lies below both φ and z. Indeed, $z_{1/a} \leq \eta$ and $(1-a)h_b + a\eta \leq \varphi$ give the first bound, while $h_b \leq \eta$ and $a z_{1/a} + (1 - a)\eta \leq z$ give the second. Thus $p = P_\theta(\varphi, z)$ is not identically $-\infty$. Volume addition and monotonicity give

$$(1 - a)^n \overline{\mathrm{vol}}_\theta(h_b) \leq \overline{\mathrm{vol}}_\theta(k_a) \leq \overline{\mathrm{vol}}_\theta(p) \leq \overline{\mathrm{vol}}_\theta(\varphi).$$

As $b \downarrow 1$, we have $a \downarrow 0$, and the first term tends to the last by Lemma 31.4.1. The upper volumes of p and φ are therefore equal. Since $p \leq \varphi$, Theorem 31.2.1 gives $p \in \mathcal{E}_\varphi$.

For general z, the envelope identity in $\mathcal{E}_\eta$ gives $z - \sup_X z \leq \eta$. Set $S = \max(\sup_X z, 0)$. The preceding case gives $P_\theta(\varphi, z - S) \in \mathcal{E}_\varphi$, and

$$P_\theta(\varphi, z - S) \leq P_\theta(\varphi, z) \leq \varphi.$$

Upward stability finishes the proof. □

Proposition 34.5.2 *Let $z \in \mathcal{E}_\eta$ and let $Q = P_\theta[\varphi](z) = \big(\lim_{C\to\infty} P_\theta(\varphi + C, z)\big)^*$ be its projection onto the relative class of φ. Then $Q \in \mathcal{E}_\varphi$, $Q \le z$, and*

$$\mu_Q \le \mathbf{1}_{\{Q=z\}}\mu_z.$$

Proof For $C \ge 0$ put $p_C = P_\theta(\varphi + C, z)$. The preceding proposition gives $p_0 \in \mathcal{E}_\varphi$. Since $p_0 \le p_C \preceq \varphi$, upward stability gives $p_C \in \mathcal{E}_\varphi$. The envelope identity, together with $p_C \le z$, yields the uniform bound

$$p_C \le \varphi + \sup_X p_C \le \varphi + \sup_X z.$$

Consequently the increasing regularized limit Q is not identically $-\infty$, belongs to $\mathcal{E}_\varphi$ and is at most z.

Finally, Lemma 31.3.1, applied with $u = \varphi$ and $w = z$, gives

$$\mu_Q \le \mathbf{1}_{\{Q=z\}}\mu_z. \tag{34.9}$$

Corollary 34.5.1 *For every nonzero finite non-pluripolar measure ν, there is $Q \in \mathcal{E}_\varphi$ such that*

$$0 \ne \mathrm{e}^{-Q}\mu_Q \le \nu.$$

Proof By Theorem 34.4.1, there is $z \in \mathcal{E}_V$ with $\mu_z = \mathrm{e}^z\nu$. Since V is an ordinary model and $V \ge \varphi$, the preceding proposition applies with $\eta = V$ and gives $Q = P_\theta[\varphi](z) \in \mathcal{E}_\varphi$. On the contact set $\{Q = z\}$,

$$\mu_Q \le \mathbf{1}_{\{Q=z\}}\mathrm{e}^z\nu \le \mathrm{e}^Q\nu.$$

The weights are finite off the pole sets, which carry no mass. Thus $\mathrm{e}^{-Q}\mu_Q \le \nu$, and this measure is nonzero because $\mu_Q(X) \ge \ell > 0$. □

34.5.1 Varying model references

We now let the reference vary. Recall from Definition 31.6.1 the comparison

$$d_\theta(u, v) = 2\,\overline{\mathrm{vol}}_\theta(\max(u, v)) - \overline{\mathrm{vol}}_\theta(u) - \overline{\mathrm{vol}}_\theta(v)$$

and, for $\delta > 0$, the set $\mathcal{M}_\delta(\theta)$ of ordinary models $u = P_\theta[u]$ with $\underline{\mathrm{vol}}_\theta(u) \ge \delta$. For models $\varphi_j \to \varphi$ in this comparison, Theorem 31.7.1 provides increasing lower models whose upper volumes approach that of φ. Projecting finite families of potentials onto these lower models makes their rooftops comparable. To keep the limit in the relative class of φ, domination of the measures is needed only where the potentials lie below one fixed member of that class; domination on all of X gives, in addition, convergence in capacity and convergence of the measures.

Theorem 34.5.1 *Let $\varphi_j, \varphi \in \mathcal{M}_\delta(\theta)$ for some $\delta > 0$, with $d_\theta(\varphi_j, \varphi) \to 0$. Suppose $u_j \in \mathcal{E}_{\varphi_j}$, $u_j \to u \in \mathrm{PSH}(X,\theta)$ in $L^1(X)$, and $\chi \in \mathcal{E}_\varphi$. If one finite non-pluripolar measure ν satisfies*

$$\mathbf{1}_{\{u_j \le \chi\}} \mu_{u_j} \le \nu,$$

then $u \in \mathcal{E}_\varphi$. Moreover, after extraction and relabeling, the potentials

$$\psi_j = P_\theta\left(\inf_{l \ge j} \varphi_l\right), \qquad h_j = P_\theta\left(\inf_{l \ge j} u_l\right)$$

satisfy $\psi_j \in \mathcal{M}_{\delta/2}(\theta)$ and $h_j \in \mathcal{E}_{\psi_j}$. If $\mu_{u_j} \le \nu$ on X, the original sequence converges to u in capacity and $\mu_{u_j} \rightharpoonup \mu_u$.

Proof We first construct the lower models ψ_j, then a common lower bound for rooftops of the u_l in the classes $\mathcal{E}_{\psi_j}$, and finally treat domination on all of X.

By Hartogs' lemma, u_j, u and χ have a common upper bound; subtracting it from all of them preserves the hypothesis, so we may assume that they are nonpositive. The envelope identity then gives $u_j \le \varphi_j$. Since $\varphi_j \to \varphi$ in L^1 by Proposition 31.6.1, it follows that $u \le \varphi$. Put $\Lambda = \nu + \mu_\chi$. Using Lemma 1.23.1, extract a subsequence along which $u_j \to u$ almost everywhere for smooth volume and

$$I_j := \int_X |\mathrm{e}^{u_j} - \mathrm{e}^{u}| \, \mathrm{d}\Lambda \le 2^{-j}.$$

Apply Theorem 31.7.1 to extract further, keeping these estimates, and obtain increasing models $w_j \le \varphi_j, \varphi$ with lower volume at least $\delta/2$ and $d_\theta(w_j, \varphi) \to 0$. For $l \ge j$ we have $w_j \le w_l \le \varphi_l$. Hence ψ_j is not identically $-\infty$, and

$$w_j \le \psi_j \le \varphi.$$

The upper bound follows by passing to the L^1 limit in $\psi_j \le \varphi_l$ as $l \to \infty$. Monotonicity of the ordinary envelope gives $P_\theta[\psi_j] \le \varphi_l$ for all $l \ge j$, so ψ_j is an ordinary model. It increases with j, has lower volume at least $\delta/2$, and

$$\overline{\mathrm{vol}}_\theta(\psi_j) \longrightarrow \overline{\mathrm{vol}}_\theta(\varphi). \tag{34.10}$$

Set $g_j = P_\theta[\psi_j](\chi)$. Proposition 34.5.2, with (φ, η) replaced by (ψ_j, φ), gives $g_j \in \mathcal{E}_{\psi_j}$, $g_j \le \chi$ and $\mu_{g_j} \le \mu_\chi$. Moreover g_j increases with j. For $N \ge j$, consider

$$q_{j,N} = P_\theta(g_j, u_j, \dots, u_N).$$

For each $l \ge j$, the projection $P_\theta[\psi_j](u_l)$ belongs to $\mathcal{E}_{\psi_j}$ by the same proposition, now with upper reference φ_l. The finite rooftop of these projections and g_j lies below $q_{j,N}$. Since $q_{j,N} \le g_j$, the finite-rooftop proposition and upward stability give $q_{j,N} \in \mathcal{E}_{\psi_j}$, and hence $\mu_{q_{j,N}}(X) \ge \delta/2$.

Partition the contact set of $q_{j,N}$ into disjoint pieces, one for each obstacle. On a piece where $q_{j,N} = u_l$, we have $u_l \le g_j \le \chi$, so the hypothesis bounds the measure there by ν. On the piece where $q_{j,N} = g_j$, the measure is bounded by $\mu_{g_j} \le \mu_\chi$.

Consequently $\mu_{q_{j,N}} \le \Lambda$. Put $F_j = \min(e^u, e^{g_j})$. On the u_l piece,

$$|e^{q_{j,N}} - F_j| \le |e^{u_l} - e^u|.$$

On the g_j piece, $g_j = q_{j,N} \le u_N$, so the error is $(e^{g_j} - e^u)_+ \le |e^{u_N} - e^u|$. The u_N piece and the g_j piece are controlled by the two summands ν and μ_χ of Λ, so together they contribute at most I_N. Thus

$$\int_X |e^{q_{j,N}} - F_j|\, \mu_{q_{j,N}} \le \sum_{l=j}^{N} I_l \le 2^{1-j}. \tag{34.11}$$

Since $F_j \ge F_1 > 0$ outside a pluripolar set, we may choose $\epsilon > 0$ with $\Lambda(\{F_1 \le \epsilon\}) < \delta/4$. The lower mass bound and the domination by Λ give

$$\epsilon\delta/4 \le \int_X F_j\, \mu_{q_{j,N}} \le 2^{1-j} + \Lambda(X) e^{\sup_X q_{j,N}}.$$

For every sufficiently large j, this bounds $\sup_X q_{j,N}$ from below, uniformly in N. Hence the decreasing limit q_j as $N \to \infty$ is not identically $-\infty$. Applying Theorem 34.2.1 in the fixed class $\mathcal{E}_{\psi_j}$, with dominating measure Λ, gives $q_j \in \mathcal{E}_{\psi_j}$. Now

$$q_j \le h_j \le \psi_j,$$

so $h_j \in \mathcal{E}_{\psi_j}$ by upward stability. We discard the finitely many earlier indices. Since $h_j \le u \le \varphi$, monotonicity and (34.10) show that u and φ have the same upper volume, and Theorem 31.2.1 gives $u \in \mathcal{E}_\varphi$.

Suppose now that $\mu_{u_j} \le \nu$ on X. The finite rooftops $p_{j,N} = P_\theta(u_j, \dots, u_N)$ are not identically $-\infty$, since they lie above h_j. Partitioning their contact sets gives

$$\mu_{p_{j,N}} \le \nu, \qquad \int_X |e^{p_{j,N}} - e^u|\, \mu_{p_{j,N}} \le 2^{1-j}.$$

For fixed j, they decrease to h_j as $N \to \infty$. These lower envelopes increase to a regularized limit $h \le u \le \varphi$, and their upper volumes tend to that of φ; hence $h \in \mathcal{E}_\varphi$. Both monotone limits hold in capacity by Lemma 1.8.9. Applying Lemma 7.3.1 twice gives

$$\int_X |e^h - e^u|\, \mu_h = 0.$$

Hence $\mu_h(\{h < u\}) = 0$, and zero domination gives $h = u$. The upper envelopes $(\sup_{k \ge j} u_k)^*$ decrease to u by Hartogs' lemma and the almost-everywhere convergence, and squeezing gives convergence in capacity. Finally,

$$\mu_{u_j}(\{u_j \le -T\}) \le \nu(\{h_1 \le -T\}) \longrightarrow 0$$

uniformly in j, and the tail criterion Theorem 8.6.1 gives weak convergence of the measures. Since the extraction can start from any subsequence, both conclusions hold for the original sequence. □

34.6 Weighted measure addition

Given $u, v \in \mathcal{E}_\varphi$ and $\lambda > 0$, we look for $a \in \mathcal{E}_\varphi$ with $\mu_a = e^{\lambda(a-u)}\mu_u + e^{\lambda(a-v)}\mu_v$. This weighted addition is used in Section 34.7 to enlarge the weighted measure of a supersolution. The measure $e^{-\lambda u}\mu_u + e^{-\lambda v}\mu_v$ can have infinite mass, so we approximate the inputs while keeping a common lower bound in $\mathcal{E}_\varphi$ for the approximate solutions. By Lemma 34.3.1, this bound gives convergence in capacity, even when the approximate solutions have different ordinary types; it also localizes the limiting equation away from deep sublevel sets. Exponentially weighted measures are defined off the pole sets and assigned zero mass there.

The next lemma treats exponent one; the general exponent in Theorem 34.6.1 follows by rescaling the background and the potentials.

Lemma 34.6.1 *Let $u, v, \sigma \in \mathrm{PSH}(X, \theta)$ be nonpositive, with $u, v \preceq \varphi$, and let $t_j \uparrow \infty$. Each of the sequences u_j, v_j is either constant, equal to u, respectively v, for every j, or given by the truncations*

$$u_j = \max(u, \sigma - t_j), \qquad v_j = \max(v, \sigma - t_j).$$

Fix $h \in \mathcal{E}_\varphi$ and let $a_j \in \mathrm{PSH}(X, \theta)$ satisfy

$$\mu_{a_j} = e^{a_j - u_j}\mu_{u_j} + e^{a_j - v_j}\mu_{v_j}, \qquad h \le a_j \le \min(u_j, v_j).$$

If $a_j \to a$ in $L^1(X)$, then $a \in \mathcal{E}_\varphi$, the convergence holds in capacity, and

$$a \le \min(u, v), \qquad \mu_a = e^{a-u}\mu_u + e^{a-v}\mu_v.$$

Proof Since u_j and v_j decrease to u and v, the bounds give $h \le a \le \min(u, v) \preceq \varphi$, so $a \in \mathcal{E}_\varphi$. After discarding finitely many indices we may assume $t_j \ge 0$, so all potentials a_j, a are nonpositive. Let E_j be the intersection of the sets $\{u > \sigma - t_j\}$ and $\{v > \sigma - t_j\}$, taken only for those inputs that are truncated; a constant sequence imposes no condition. On E_j, locality gives $\mu_{u_j} = \mu_u$ and $\mu_{v_j} = \mu_v$. Since $a, a_j \le u_j, v_j$, we have

$$e^a \mu_{a_j} \le e^{(a+a_j)/2}(\mu_{u_j} + \mu_{v_j}).$$

If $x \notin E_j$, then one truncated input and its truncation are at most $-t_j$ at x, so $a_j(x), a(x) \le -t_j$. Moreover $|a_j - a|e^{(a+a_j)/4}$ is bounded by an absolute constant, since $a_j, a \le 0$. Thus

$$\int_X |a_j - a| e^a \, \mu_{a_j} \le \int_X |a_j - a| e^{(a+a_j)/2}(\mu_u + \mu_v) + C e^{-t_j/2}. \tag{34.12}$$

The last term uses the uniform bound for the masses of the u_j and v_j given by Proposition 7.5.2. The fixed measure $\mu_u + \mu_v$ is finite by bounded mass. Lemma 1.23.1, applied to e^{a_j} and e^a, shows that $a_j \to a$ in measure for $\mu_u + \mu_v$, since a is finite almost everywhere for this non-pluripolar measure. The integrand in the first term of (34.12) is uniformly bounded and tends to zero in measure, so its integral tends

to zero. Lemma 34.3.1 now gives convergence in capacity. When a subsequence is taken, the corresponding truncation levels t_j are kept.

To identify the equation, fix $T > 0$ and choose a continuous nondecreasing function $g_T\colon [-\infty, 0] \to [0, 1]$ that vanishes on $[-\infty, -2T]$ and equals one on $[-T, 0]$, such that $g_T \uparrow 1$ at finite points as $T \to \infty$. Put $\chi_T = g_T(h)$. Since $h \le a \le u, v$, both u and v exceed $-2T$ on $\{\chi_T > 0\}$. Therefore, once $t_j > 2T$, every truncated input agrees there with its limit, and

$$\chi_T \mu_{a_j} = \chi_T \mathrm{e}^{a_j} (\mathrm{e}^{-u} \mu_u + \mathrm{e}^{-v} \mu_v).$$

Choose a Hermitian form $\Omega \ge \theta$. On the left-hand side, replace a_j by $\max(a_j, -2T)$, which does not change $\chi_T \mu_{a_j}$ because χ_T vanishes where $a_j \le -2T$, and expand the bounded product as

$$(\theta + \mathrm{dd^c} \max(a_j, -2T))^n = \sum_{k=0}^{n} \binom{n}{k} (\Omega + \mathrm{dd^c} \max(a_j, -2T))^k \wedge (\theta - \Omega)^{n-k}.$$

Bounded capacity continuity with the fixed quasi-continuous weight χ_T therefore gives convergence to $\chi_T \mu_a$, after testing against a continuous function. The uncut product may be signed, but its terms have uniformly capacity-controlled total variations.

On the right-hand side, $\chi_T \mathrm{e}^{-u}$ and $\chi_T \mathrm{e}^{-v}$ are bounded by e^{2T}. Fixed-measure exponential convergence therefore gives convergence in total variation to $\chi_T \mathrm{e}^{a} (\mathrm{e}^{-u} \mu_u + \mathrm{e}^{-v} \mu_v)$. Let $T \to \infty$. The limiting right-hand side is at most $\mu_u + \mu_v$, because $a \le u, v$. Both sides ignore the pole set of h, so the cutoff identities prove the global equation. □

34.6.1 Ordinary types and bounded carriers

We first solve the weighted addition equation when one input has the ordinary type of φ, or a measure dominated by the Monge–Ampère measure of a bounded potential. In both cases first moments control the rooftops of the approximating solutions, and these rooftops give the common lower bound required by Lemma 34.6.1. The solution lies below both inputs, so its measure is finite.

Theorem 34.6.1 *Let $u, v \in \mathcal{E}_\varphi$ and $\lambda > 0$. Suppose either $v \sim \varphi$, or there are a Hermitian form ω, a bounded $\psi \in \mathrm{PSH}(X, \omega)$ and a finite constant $A > 0$ such that*

$$\mu_v \le A\,\mathrm{MA}_\omega(\psi).$$

Then there is a unique $a \in \mathcal{E}_\varphi$ satisfying

$$\mu_a = \mathrm{e}^{\lambda(a-u)}\mu_u + \mathrm{e}^{\lambda(a-v)}\mu_v.$$

It satisfies $a \le \min(u, v)$, and hence $\mu_a \le \mu_u + \mu_v$. If both $u, v \sim \varphi$, then also $a \sim \varphi$.

Proof The proof has four steps: two inputs of minimal singularity type, two inputs of the ordinary type of φ, one input of that type, and finally an input with a bounded carrier measure.

It suffices to prove the result for $\lambda = 1$. Indeed, replace (θ, φ, u, v) by $(\lambda\theta, \lambda\varphi, \lambda u, \lambda v)$: envelopes and relative classes scale with the potentials, volumes and Monge–Ampère measures scale by λ^n, and a bounded-carrier domination remains one after changing its constant. Dividing the resulting solution by λ gives the desired equation. Similarly, a common downward translation of u, v and the solution allows us to assume $u, v \le \varphi \le 0$; by the envelope identity, these inequalities hold after a sufficiently large common translation. We restore the translation at the end.

Step 1: inputs of minimal singularity type. First take $\varphi = V$ and $u, v \sim V$. Truncating only the weights, define the finite nonzero measures

$$\tau_j = \mathrm{e}^{-\max(u,-j)}\mu_u + \mathrm{e}^{-\max(v,-j)}\mu_v.$$

The functions $\max(u, -j)$ need not be θ-plurisubharmonic; they occur only as weights. Theorem 34.4.1 gives $a_j \in \mathcal{E}_V$ with $\mu_{a_j} = \mathrm{e}^{a_j}\tau_j$.

Choose $D \ge \frac{1}{2}\|u - v\|_\infty + n\log 2$, where the difference is taken off the common poles, and put $w = (u + v)/2 - D$. Positivity of mixed products gives

$$\mu_w \ge 2^{-n}(\mu_u + \mu_v) \ge \mathrm{e}^w\tau_j.$$

Exponential order in $\mathcal{E}_V$ (Corollary 31.3.1) gives $a_j \ge w$. Since τ_j increases with j, the same order shows that a_j decreases. The envelope identity gives $a_1 \le V + \sup_X a_1$, so $a_1 \sim V$ by the lower bound w. Consequently

$$\mu_{a_j} \le \mathrm{e}^{a_1-u}\mu_u + \mathrm{e}^{a_1-v}\mu_v \le C(\mu_u + \mu_v).$$

The decreasing limit $a \geq w$ is not identically $-\infty$ and has the ordinary type of V. Fixed-dominator continuity (Theorem 34.2.1) gives weak convergence of the left-hand sides, and the right-hand sides converge in total variation by dominated convergence with respect to $\mu_u + \mu_v$. Therefore

$$\mu_a = \mathrm{e}^{a-u}\mu_u + \mathrm{e}^{a-v}\mu_v.$$

Comparing this equation with each input measure, exponential order now gives $a \leq u, v$. This comparison uses the limiting equation; the truncated weights alone would not give it.

Step 2: two inputs of the ordinary type of φ. Suppose $u, v \sim \varphi$, and put $u_j = \max(u, V - j)$ and $v_j = \max(v, V - j)$. Step 1 gives solutions $a_j \sim V$ with

$$\mu_{a_j} = \mathrm{e}^{a_j - u_j}\mu_{u_j} + \mathrm{e}^{a_j - v_j}\mu_{v_j}, \qquad a_j \leq \min(u_j, v_j).$$

The differences $u_j - v_j$ satisfy the same uniform bound as $u - v$. With D as above, $w_j = (u_j + v_j)/2 - D$ satisfies the reverse inequality for this equation, and exponential order gives

$$(u + v)/2 - D \leq w_j \leq a_j \leq \min(u_j, v_j).$$

The fixed lower bound $w = (u + v)/2 - D$ has the ordinary type of φ, and hence lies in $\mathcal{E}_\varphi$. Extract an L^1 limit $a \not\equiv -\infty$; it satisfies $w \leq a \leq \min(u, v)$, so $a \sim \varphi$. Lemma 34.6.1, with $\sigma = V$ and lower bound $h = w$, proves convergence in capacity and the equation. The lemma applies although the approximating solutions have the type of V and the limit has the type of φ.

Step 3: one input of the ordinary type of φ. Now let $u \in \mathcal{E}_\varphi$ and $v \sim \varphi$, and set $u_j = \max(u, \varphi - j)$. Step 2 gives $a_j \sim \varphi$ with

$$\mu_{a_j} = \mathrm{e}^{a_j - u_j}\mu_{u_j} + \mathrm{e}^{a_j - v}\mu_v, \qquad a_j \leq \min(u_j, v).$$

We construct a common lower bound in $\mathcal{E}_\varphi$. Put $\delta = n \log 2 > 0$ and

$$s_j = Q_2(a_j, u_j) \in \mathcal{E}_\varphi, \qquad C_j = \{s_j + u_j = 2a_j\}.$$

By stretching and contact, $s_j \leq a_j$ and

$$2^{-n}\mu_{s_j} + 2^{-n}\mathbf{1}_{C_j}\mu_{u_j} \leq \mathbf{1}_{C_j}\mu_{a_j}. \tag{34.13}$$

Put $D_j = \{s_j \leq u - 2\delta\}$. On $D_j \cap C_j$ we have $a_j = (s_j + u_j)/2 \leq u_j - \delta$, because $u \leq u_j$. Since $\mathrm{e}^{-\delta} = 2^{-n}$, inserting the equation for μ_{a_j} into (34.13) shows that the μ_{u_j} terms cancel there. Also $v \geq \varphi - C \geq u_j - C$ for a fixed constant C. Hence

$$\mathbf{1}_{D_j}\mu_{s_j} \leq 2^n \mathrm{e}^C \mathbf{1}_{D_j \cap C_j} \mathrm{e}^{(s_j - u_j)/2}\mu_v. \tag{34.14}$$

All measures being subtracted here are finite.

For $N \geq 1$ form the finite rooftops

$$R_N = P_\theta(s_1, \dots, s_N), \qquad q_N = P_\theta(u, R_N).$$

They belong to $\mathcal{E}_\varphi$ by the finite-rooftop proposition. On the set $\{R_N \leq u - 2\delta\}$, partition the contact set according to the first index i with $R_N = s_i$. Since $u_i \geq u_N$, (34.14) gives

$$\mathbf{1}_{\{R_N \leq u - 2\delta\}} \mu_{R_N} \leq 2^n \mathrm{e}^C \mathrm{e}^{(R_N - u_N)/2} \mu_v.$$

The contact pieces are disjoint, so there is no factor N. On $\{q_N \leq u - 2\delta\}$ only the contact part $q_N = R_N$ contributes. Since $u \leq u_N$ and $t\mathrm{e}^{-t/2} \leq 2/\mathrm{e}$ for $t \geq 0$, we obtain

$$\int_X (u - q_N)\, \mu_{q_N} \leq 2\delta M + 2^{n+1} \mathrm{e}^{C-1} \mu_v(X).$$

The first term bounds the contribution of the complementary set, including any contact with u. The first-moment criterion Proposition 34.3.1, with fixed reference u, implies that the decreasing sequence q_N has a limit $h \in \mathcal{E}_\varphi$, not identically $-\infty$. For each fixed j and every $N \geq j$, we have $q_N \leq s_j \leq a_j$, so $h \leq a_j$ for every j.

The a_j now have a subsequence converging in L^1 to a limit not identically $-\infty$, bounded above by u and v. Lemma 34.6.1, with reference φ, fixed second input v and common lower bound h, proves the required equation in $\mathcal{E}_\varphi$.

Step 4: a bounded carrier measure for the second input. Finally let $u, v \in \mathcal{E}_\varphi$ and $\mu_v \leq A\nu$, with $\nu = \mathrm{MA}_\omega(\psi)$ and ψ bounded. Again put $u_j = \max(u, \varphi - j)$. By Step 3, applied with the inputs interchanged, there are $a_j \in \mathcal{E}_\varphi$ satisfying the same equations and bounds as in Step 3.

Let $r = P_\theta(u, v) \in \mathcal{E}_\varphi$. On $\{a_j < r\}$ one has $r \leq u_j, v$, so the inequality $t\mathrm{e}^{-t} \leq 1/\mathrm{e}$ gives

$$\int_X (r - a_j)_+ \mu_{a_j} \leq \mathrm{e}^{-1} \big(\mu_{u_j}(X) + \mu_v(X)\big) \leq 2M/\mathrm{e}.$$

Since $a_j \leq \varphi$, the one-sided first-moment criterion gives a uniform lower bound for $\sup_X a_j$. By Lemma 1.23.4,

$$\sup_{\substack{z \in \mathrm{PSH}(X,\theta) \\ \sup_X z = 0}} \int_X (-z)\, \mathrm{d}\nu < \infty.$$

Applying this to $a_j - \sup_X a_j$ and using the lower bound for the suprema, we obtain

$$\sup_j \int_X (u_j - a_j)\, \mu_v \leq \sup_j \int_X (-a_j)\, \mu_v < \infty. \tag{34.15}$$

Keep $s_j = Q_2(a_j, u_j)$, C_j and D_j from Step 3. The same cancellation in (34.13), now using only $a_j \leq v$, gives

$$\mathbf{1}_{D_j} \mu_{s_j} \leq 2^n \mathbf{1}_{D_j \cap C_j} \mathrm{e}^{a_j - v} \mu_v \leq 2^n \mu_v.$$

Take one additional rooftop,

$$q_j = P_\theta(s_j, u - 2\delta) \in \mathcal{E}_\varphi.$$

On its contact set with $u - 2\delta$ its measure is at most μ_u; on its other contact part, where $q_j = s_j \le u - 2\delta$, the preceding inequality applies. Thus

$$\mu_{q_j} \le \mu_u + 2^n \mu_v. \tag{34.16}$$

Moreover the integrand $u - 2\delta - q_j$ vanishes on the first contact part. On the other part, the measure is carried by C_j, where $u_j - s_j = 2(u_j - a_j)$. Therefore

$$\int_X (u - 2\delta - q_j)\, \mu_{q_j} \le 2^{n+1} \int_X (u_j - a_j)\, \mu_v,$$

which is uniformly bounded by (34.15). The first-moment criterion with reference $u - 2\delta$ prevents collapse of the q_j. Extract L^1 limits of the a_j and of the q_j, not identically $-\infty$. The fixed-dominator theorem applies to (34.16) and, after a further extraction, gives

$$h = P_\theta\left(\inf_j q_j\right) \in \mathcal{E}_\varphi.$$

Since $q_j \le s_j \le a_j$, this is a common lower bound for the extracted solutions, and Lemma 34.6.1 gives the equation. Its truncation levels are the original indices of the extracted subsequence.

In every step the solution lies below u and v, so its measure is bounded by $\mu_u + \mu_v$. Two solutions have the same exponentially weighted measure, so Corollary 31.3.1, applied in both directions, gives uniqueness, even when that weighted measure has infinite mass. Restoring the common translation and the exponent finishes the proof.□

34.7 General measures with a model reference

We now prove the existence statements of Theorem 34.1.1. Suppose first that μ is dominated by the Monge–Ampère measure of a bounded potential. The supersolution corollary gives $Q \in \mathcal{E}_\varphi$ with a nonzero weighted measure $\mathrm{e}^{-Q}\mu_Q \le \mu$. Among the potentials $w \in \mathcal{E}_\varphi$ whose weighted measure $\mathrm{e}^{-w}\mu_w$ lies between this measure and μ, we maximize the weighted mass. If a maximizer left part of μ uncovered, weighted addition of a supersolution for the remainder would increase the weighted mass; hence the maximizer solves the equation. Truncating densities then treats every finite non-pluripolar measure.

Theorem 34.7.1 *Let μ be a nonzero finite non-pluripolar measure and let $\lambda > 0$. There is a unique $u \in \mathcal{E}_\varphi$ such that*

$$\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u}\mu.$$

Proof We first take $\lambda = 1$ and suppose that

$$\mu \le A\sigma, \qquad A > 0, \qquad \sigma = \mathrm{MA}_\omega(\psi), \quad \psi \in \mathrm{PSH}(X,\omega) \cap L^\infty(X).$$

Corollary 34.5.1 gives $Q \in \mathcal{E}_\varphi$ such that

$$\nu_0 = \mathrm{e}^{-Q}\mu_Q \le \mu$$

is a finite nonzero measure. Consider

$$\mathcal{T} = \{w \in \mathcal{E}_\varphi : \nu_0 \le \mathrm{e}^{-w}\mu_w \le \mu\}.$$

The family contains Q. For $w \in \mathcal{T}$, exponential order (Corollary 31.3.1) gives $w \le Q$. Also

$$\ell \le \mu_w(X) \le \mathrm{e}^{\sup_X w}\mu(X), \qquad \sup_X w \ge \log\frac{\ell}{\mu(X)}.$$

These bounds prevent the collapse of any sequence in $\mathcal{T}$.

Let

$$\alpha = \sup_{w\in\mathcal{T}} \int_X \mathrm{e}^{-w}\,\mu_w \le \mu(X),$$

and choose $w_j \in \mathcal{T}$ whose weighted masses tend to α. Extract an L^1 limit $w \not\equiv -\infty$. Since

$$\mu_{w_j} \le \mathrm{e}^{\sup_X Q}\mu,$$

the fixed-dominator theorem Theorem 34.2.1 gives $w \in \mathcal{E}_\varphi$, convergence in capacity, and $\mu_{w_j} \rightharpoonup \mu_w$. Put $\nu_j = \mathrm{e}^{-w_j}\mu_{w_j}$. After a further extraction, these finite measures converge weakly to a measure ν, and

$$\nu_0 \le \nu \le \mu, \qquad \nu(X) = \alpha.$$

We claim that $\mu_w = \mathrm{e}^w\nu$. By Lemma 1.23.1,

$$\int_X |\mathrm{e}^{w_j} - \mathrm{e}^w|\,\mathrm{d}\mu \longrightarrow 0.$$

Let f be continuous. Since $\nu_j \le \mu$, replacing e^{w_j} by e^w in $\int_X f\mathrm{e}^{w_j}\,\mathrm{d}\nu_j$ changes the integral by an amount tending to zero. For the remaining integral, approximate the fixed bounded measurable function $f\mathrm{e}^w$ in $L^1(\mu)$ by continuous functions. Since $\nu_j, \nu \le \mu$, the approximation error is uniform in j, while the integrals of each continuous approximation converge by weak convergence. Thus

$$\int_X f\,\mu_{w_j} = \int_X f\mathrm{e}^{w_j}\,\mathrm{d}\nu_j \longrightarrow \int_X f\mathrm{e}^w\,\mathrm{d}\nu.$$

This proves the claim. The measure ν does not charge the poles of w, so $\mathrm{e}^{-w}\mu_w = \nu$. Hence $w \in \mathcal{T}$ attains the maximum α.

Suppose that $\mu - \nu$ were nonzero. Then Corollary 34.5.1, applied to $\mu - \nu$, would give $v \in \mathcal{E}_\varphi$ with

$$0 \neq \nu_v = \mathrm{e}^{-v}\mu_v \leq \mu - \nu.$$

In particular $\mu_v \leq \mathrm{e}^{\sup_X v}\mu \leq A\mathrm{e}^{\sup_X v}\sigma$, so the bounded-carrier hypothesis of Theorem 34.6.1 holds. That theorem, with $\lambda = 1$, gives $a \in \mathcal{E}_\varphi$ with

$$\mathrm{e}^{-a}\mu_a = \mathrm{e}^{-w}\mu_w + \mathrm{e}^{-v}\mu_v = \nu + \nu_v.$$

The right-hand side lies between ν_0 and μ, so $a \in \mathcal{T}$, but its mass exceeds α. This contradicts maximality. Therefore $\nu = \mu$, and w solves the equation.

Now let μ be any nonzero finite non-pluripolar measure. The bounded-carrier decomposition Lemma 15.23.1, applied to ω, writes

$$\mu = f\sigma, \qquad \sigma = \mathrm{MA}_\omega(\psi), \quad \psi \in \mathrm{PSH}(X,\omega) \cap L^\infty(X), \quad 0 \leq f \in L^1(\sigma).$$

Set $\mu_k = \min(f,k)\sigma$. These measures are nonzero, and the first part gives $u_k \in \mathcal{E}_\varphi$ with $\mu_{u_k} = \mathrm{e}^{u_k}\mu_k$. As in the last part of the proof of Theorem 34.4.1, now with $\mathcal{E}_\varphi$ and ℓ in place of $\mathcal{E}_V$ and ℓ_V, the u_k decrease to some $u \in \mathcal{E}_\varphi$ with $\mu_u = \mathrm{e}^u\mu$.

For general $\lambda > 0$, apply the result for exponent one in the background $\lambda\theta$, with model $\lambda\varphi$ and datum $\lambda^n\mu$, and divide the resulting potential by λ. The scaling of the relative class and of the Monge–Ampère measure gives the equation. Uniqueness follows by applying Corollary 31.3.1 to two solutions in both directions. □

Theorem 34.7.2 *For every nonzero finite non-pluripolar measure μ, there are $u \in \mathcal{E}_\varphi$ and $c > 0$ such that*

$$\sup_X u = 0, \qquad \mathrm{MA}_\theta(u) = c\mu.$$

The scalar c is unique and satisfies

$$\frac{\ell}{\mu(X)} \leq c \leq \frac{M}{\mu(X)}.$$

Proof For each integer $j \geq 1$, the preceding theorem gives $u_j \in \mathcal{E}_\varphi$ with $\mu_{u_j} = \mathrm{e}^{u_j/j}\mu$. The zero-exponent part of the proof of Theorem 33.5.1 applies to these u_j, with the comparison potential $s = u_1$, with Corollary 31.3.1 and Theorem 34.2.1 in place of Corollary 33.2.1 and Theorem 33.3.1, and with the lower bound ℓ of (34.1) in place of $v_-(\theta)$; Lemma 1.23.2 holds for every smooth background. It gives $u \in \mathcal{E}_\varphi$ and $c > 0$ with $\sup_X u = 0$ and $\mu_u = c\mu$. Integrating this equation and using (34.1) gives the stated bounds for c, and the scalar comparison in Corollary 31.3.1 gives the uniqueness of c. □

For L^p densities, Section 34.8 proves the stronger conclusion $u \sim \varphi$.

34.7.1 Stability under changes of reference

By Theorem 34.5.1, convergence of the models for d_θ keeps limits in the relative class of the limiting model. For equations, convergence of the densities provides, after extraction, the fixed dominating measure required there. For a positive exponent, uniqueness then identifies the limiting potential; at exponent zero, uniqueness of the scalar identifies the limiting measure.

Theorem 34.7.3 *Let $\varphi_j, \varphi \in \mathcal{M}_\delta(\theta)$ for some $\delta > 0$, with $d_\theta(\varphi_j, \varphi) \to 0$. Let ν be a finite non-pluripolar measure and let f_j, f be nonnegative nonzero functions in $L^1(\nu)$ with $f_j \to f$ in $L^1(\nu)$. Put $\mu_j = f_j \nu$ and $\mu = f\nu$.*

(1) *For a fixed $\lambda > 0$, the solutions*

$$u_j \in \mathcal{E}_{\varphi_j}, \qquad \mathrm{MA}_\theta(u_j) = \mathrm{e}^{\lambda u_j}\mu_j$$

converge in $L^1(X)$ and in capacity to the solution $u \in \mathcal{E}_\varphi$ with datum μ. Their Monge–Ampère measures converge to $\mathrm{MA}_\theta(u)$ in total variation.

(2) *Let $u_j \in \mathcal{E}_{\varphi_j}$ satisfy $\sup_X u_j = 0$ and $\mathrm{MA}_\theta(u_j) = c_j\mu_j$. Then $c_j \to c$, where c is the scalar for the reference φ and the datum μ, and*

$$\mathrm{MA}_\theta(u_j) \longrightarrow c\mu \quad \text{in total variation.}$$

The potentials form a relatively compact family in L^1. Every L^1 cluster limit u lies in $\mathcal{E}_\varphi$, has supremum zero and solves $\mathrm{MA}_\theta(u) = c\mu$. Each subsequence converging in L^1 converges to its limit in capacity.

Proof Write $M_0 = \overline{\mathrm{vol}}_\theta(V_\theta)$. Every solution above has mass between δ and M_0. In the first case let $s_j = \sup_X u_j$. Then

$$s_j \geq \lambda^{-1} \log \frac{\delta}{\mu_j(X)},$$

which is a uniform lower bound since $\mu_j(X) \to \mu(X) > 0$. To bound s_j from above, suppose that $s_j \to +\infty$ along a subsequence. By normalized compactness, after a further extraction, $v_j = u_j - s_j \to v$ in L^1 with $\sup_X v = 0$. The fixed-measure lemma Lemma 1.23.1 and $f_j \to f$ in $L^1(\nu)$ imply

$$\int_X \mathrm{e}^{\lambda v_j} f_j \, \mathrm{d}\nu \longrightarrow \int_X \mathrm{e}^{\lambda v} f \, \mathrm{d}\nu > 0.$$

The limit is positive because $f\nu$ is nonzero and does not charge the poles of v. But the left-hand side equals $\mathrm{e}^{-\lambda s_j}\mu_{u_j}(X) \leq M_0 \mathrm{e}^{-\lambda s_j}$, which tends to zero, a contradiction. Thus s_j is bounded, and the u_j form a relatively compact family in L^1 whose limits are not identically $-\infty$.

Start with any subsequence and extract $u_j \to u$ in L^1. Extract again so that $\sum_j \|f_j - f\|_{L^1(\nu)} < \infty$. The function

$$g = f + \sum_j |f_j - f| \in L^1(\nu)$$

dominates every f_j of this subsequence. If C is a common upper bound for the u_j, then $\mu_{u_j} \le e^{\lambda C} g\nu$. By Theorem 34.5.1, $u \in \mathcal{E}_\varphi$, the potentials converge in capacity, and the measures converge weakly to μ_u. On the other hand,

$$e^{\lambda u_j} f_j \longrightarrow e^{\lambda u} f \quad \text{in } L^1(\nu).$$

Indeed, the error coming from $f_j - f$ is at most $e^{\lambda C}\|f_j - f\|_{L^1(\nu)}$, and Lemma 1.23.1 for the fixed measure $f\nu$ controls the remaining term. Hence $\mu_u = e^{\lambda u}\mu$, with convergence in total variation. By the uniqueness in Theorem 34.7.1, every cluster limit is this solution, and the subsequence criterion gives all three convergences for the original sequence.

In the second case, integrating the equation gives

$$\frac{\delta}{\mu_j(X)} \le c_j \le \frac{M_0}{\mu_j(X)}.$$

Since $\mu_j(X) \to \mu(X) > 0$, these scalars stay in a compact subinterval of $(0, +\infty)$. After extraction, normalized compactness gives $u_j \to u$ in L^1 with $\sup_X u = 0$, and $c_j \to c' > 0$. Extract a summable density error as above. Then $\mu_{u_j} \le Cg\nu$ for a fixed C, so Theorem 34.5.1 gives $u \in \mathcal{E}_\varphi$, convergence in capacity and weak convergence to μ_u. Since $c_j f_j \to c' f$ in $L^1(\nu)$, the limiting equation is $\mu_u = c'\mu$, and the uniqueness of the scalar in Theorem 34.7.2 gives $c' = c$. Hence the whole sequence of scalars converges to c, and $c_j\mu_j \to c\mu$ in total variation. Applying the same argument within any L^1-convergent subsequence proves its convergence in capacity and the equation for its limit. □

34.7.2 Arbitrary weighted sums

With existence for general measures available, weighted addition extends to arbitrary $u, v \in \mathcal{E}_\varphi$. Truncating the exponential weights gives finite data, which Theorem 34.7.1 solves. A first-moment estimate prevents collapse of these solutions, and a rooftop involving their limit localizes the equation before the limit can be compared with the inputs.

Theorem 34.7.4 *For every $u, v \in \mathcal{E}_\varphi$ and $\lambda > 0$, there is a unique $a \in \mathcal{E}_\varphi$ satisfying*

$$\mu_a = \mathrm{e}^{\lambda(a-u)}\mu_u + \mathrm{e}^{\lambda(a-v)}\mu_v.$$

Moreover $a \le \min(u, v)$ and $\mu_a \le \mu_u + \mu_v$.

Proof By the scaling and the common translation used in Theorem 34.6.1, it suffices to take $\lambda = 1$ and $u, v \le 0$. For positive integers j, let

$$\tau_j = \mathrm{e}^{-\max(u,-j)}\mu_u + \mathrm{e}^{-\max(v,-j)}\mu_v.$$

These are nonzero finite non-pluripolar measures, and they increase with j. Only the weights are truncated; the functions $\max(u, -j)$ and $\max(v, -j)$ need not be θ-plurisubharmonic. Theorem 34.7.1 gives $a_j \in \mathcal{E}_\varphi$ with $\mu_{a_j} = \mathrm{e}^{a_j}\tau_j$. Since τ_j increases, exponential order implies that a_j decreases.

Set $r = P_\theta(u, v) \in \mathcal{E}_\varphi$. On $\{a_j < r\}$, one has $r \le u, v$ and

$$(r - a_j)\mathrm{e}^{a_j - \max(u,-j)} \le (r - a_j)\mathrm{e}^{a_j - r} \le \mathrm{e}^{-1},$$

with the same inequality for v. Consequently

$$\int_X (r - a_j)_+ \mu_{a_j} \le \mathrm{e}^{-1}\big(\mu_u(X) + \mu_v(X)\big).$$

Also $a_j \le a_1 \le \varphi + \sup_X a_1$. The one-sided first-moment criterion of Proposition 34.3.1, with fixed reference r, bounds $\sup_X a_j$ from below. Thus the decreasing sequence has an L^1 limit $a \not\equiv -\infty$, and the same criterion puts a in $\mathcal{E}_\varphi$.

We identify the equation before comparing a with the inputs. The rooftop

$$q = P_\theta(a, u, v) \in \mathcal{E}_\varphi$$

lies below a_j, a, u and v.

For $T > 0$, let $\chi_T = g_T(q)$, with g_T as in the proof of Lemma 34.6.1: a continuous nondecreasing cutoff that vanishes where $q \le -2T$, equals one where $q \ge -T$, and increases to one off the pole set of q as $T \to \infty$. On $\{\chi_T > 0\}$ both inputs exceed $-2T$. Hence, for $j > 2T$,

$$\chi_T\mu_{a_j} = \chi_T\mathrm{e}^{a_j}(\mathrm{e}^{-u}\mu_u + \mathrm{e}^{-v}\mu_v). \tag{34.17}$$

The factors $\chi_T\mathrm{e}^{-u}$ and $\chi_T\mathrm{e}^{-v}$ are bounded by e^{2T}. Since $a_j \downarrow a$ and $a_j \le \sup_X a_1$, dominated convergence against the fixed measure $\mu_u + \mu_v$ gives convergence in total variation of the right-hand sides to the corresponding expression with a.

For the left-hand side, $a_j \to a$ in capacity by Lemma 1.8.9. Since $a_j, a \ge q > -2T$ on $\{\chi_T > 0\}$, the bounded potentials $\max(a_j, -2T)$ and $\max(a, -2T)$ agree there with a_j and a. The argument in the proof of Lemma 34.6.1, with these bounded potentials and the fixed quasi-continuous weight χ_T, therefore gives $\chi_T\mu_{a_j} \rightharpoonup \chi_T\mu_a$.

Thus (34.17) holds in the limit. Let $T \to \infty$. Both sides ignore the poles of q, so monotone convergence gives

$$\mu_a = \mathrm{e}^{a-u}\mu_u + \mathrm{e}^{a-v}\mu_v.$$

For $\varepsilon > 0$, the equation implies

$$\mathbf{1}_{\{u<a-\varepsilon\}}\mu_u \le \mathrm{e}^{-\varepsilon}\mathbf{1}_{\{u<a-\varepsilon\}}\mu_a.$$

Strict domination, Theorem 31.3.1, gives $u \ge a - \varepsilon$, and letting $\varepsilon \downarrow 0$ gives $u \ge a$. Similarly $v \ge a$, so $\mu_a \le \mu_u + \mu_v$. Two solutions have the same exponentially weighted measure, so Corollary 31.3.1, applied in both directions, gives uniqueness.□

Corollary 34.7.1 *For fixed $\lambda > 0$, write $u \star_\lambda v$ for the solution a in the theorem. This operation on $\mathcal{E}_\varphi$ is commutative and associative, and, for $c \in \mathbb{R}$,*

$$(u + c) \star_\lambda (v + c) = (u \star_\lambda v) + c, \qquad u \star_\lambda u = u - \lambda^{-1}\log 2.$$

Proof Put $F_\lambda(u) = \mathrm{e}^{-\lambda u}\mu_u$, with zero mass on the poles. The theorem says that $F_\lambda(u \star_\lambda v) = F_\lambda(u) + F_\lambda(v)$. By Corollary 31.3.1, the map F_λ is injective on $\mathcal{E}_\varphi$. Commutativity and associativity therefore follow from those of addition of positive measures, including measures of infinite total mass. Finally, $F_\lambda(u + c) = \mathrm{e}^{-\lambda c}F_\lambda(u)$, which together with injectivity gives both identities. □

34.8 Relative boundedness for L^p data

If the datum has finite moments of sufficiently high order against normalized θ-plurisubharmonic functions, every normalized $u \in \mathcal{E}_\varphi$ whose measure is dominated by the datum differs from φ by a bounded function. We work directly with $u \in \mathcal{E}_\varphi$: truncating u against $\varphi - j$ would introduce a contact measure that need not be controlled by the datum. The concave-obstacle envelopes of Lemma 34.3.2 keep this domination and give a differential inequality for the distribution of $\varphi - u$.

Proposition 34.8.1 *Let μ be a finite non-pluripolar measure. Suppose that, for some real $m > 2n+1$,*

$$A_m := \sup_{\substack{v\in \mathrm{PSH}(X,\theta)\\ \sup_X v=0}} \int_X (-v)^m \,\mathrm{d}\mu < \infty.$$

There is a finite constant C, depending only on $n, \ell, M, m, \mu(X)$ and A_m, such that every $u \in \mathcal{E}_\varphi$ with $\sup_X u = 0$ and $\mu_u \le \mu$ satisfies

$$\varphi - C \le u \le \varphi.$$

Proof As in Proposition 34.3.2, we build a concave weight from the distribution of $\varphi - u$ and apply Lemma 34.3.2. The moment bound on μ now yields a differential inequality for this weight, which bounds the depth of $\varphi - u$ under μ_u.

The envelope identity gives $u \le \varphi \le 0$. Put $x = \varphi - u$, defined off the poles, and $N = 2n+1$. Since $x \le -u$ and $\mu_u \le \mu$, Hölder's inequality gives the uniform first-moment bound

$$B := \int_X x\,\mu_u \le \int_X (-u)\,\mathrm{d}\mu \le \mu(X)^{1-1/m} A_m^{1/m} =: A_1.$$

Let $H(t) = \mu_u(\{x > t\})$ and $T = \sup\{t \ge 0 : H(t) > 0\}$, with $T = 0$ if this set is empty; at this point T may be infinite. If $T < \infty$, zero domination applied to u and $\varphi - t$ for $t > T$ gives $u \ge \varphi - T$. It is therefore enough to bound T. We may assume $T > 1$, and we choose any S with $1 < S < T$.

Define g, F, W and χ as g_s, F_s, W_s and χ_s in the proof of Proposition 34.3.2, with φ, H and S in place of V, $\mu_u(\{x > \cdot\})$ and $T - s$. Since $H(S) > 0$, F is bounded, although its bound may depend on S and u; thus χ satisfies the hypotheses of Lemma 34.3.2. Also $W(t) \ge t$ and $W(t) \le tF(t)^{1/N}$. Let $p = P_\theta(\varphi - W(x)) \in \mathcal{E}_\varphi$ be the envelope of that lemma. The estimate (34.6), now with M in place of M_V and the integral in t over $[0,\infty)$, and the Cauchy–Schwarz estimate in that proof give bounds independent of S and u:

$$\int_X F(x)\,\mu_u \le 3\max(1,M) =: K, \qquad \int_X (\varphi - p)^{1/2}\,\mu_p \le \sqrt{KB}.$$

The identity $P_\theta[p] = \varphi$ implies $p - \sup_X p \le \varphi$. Since $p \le \varphi \le 0$ and $\mu_p(X) \ge \ell$, the last integral is at least $\ell\sqrt{-\sup_X p}$. Consequently

$$0 \le -\sup_X p \le KA_1/\ell^2 =: D.$$

Using $W(x) \le \varphi - p \le -p$, and applying the moment hypothesis to $p - \sup_X p$, we obtain

$$\int_X W(x)^m \,\mathrm{d}\mu \le 2^{m-1}\big(A_m + D^m\mu(X)\big) < C_0, \tag{34.18}$$

where we choose $C_0 = 1 + 2^{m-1}(A_m + D^m\mu(X))$. This constant is independent of S and u.

For $0 < t \le S$, Chebyshev's inequality and $\mu_u \le \mu$ give

$$H(t) \le C_0 W(t)^{-m}, \qquad F'(t) \ge \frac{W(t)^m}{C_0(1+t)^{3/2}} \quad \text{for almost every } t.$$

Since $W' = F^{1/N}$, the derivative of $(W')^{N+1}$ is at least

$$\frac{N+1}{NC_0}\,\frac{W^m W'}{(1+t)^{3/2}}.$$

All functions being differentiated are absolutely continuous on compact intervals. Integrating from 0 to t and using $W(0) = 0$ gives

$$(W'(t))^{N+1} \ge \frac{N+1}{NC_0(m+1)}\,\frac{W(t)^{m+1}}{(1+t)^{3/2}}.$$

In particular

$$W'(t) \ge c_0(1+t)^{-\gamma} W(t)^{\kappa}, \quad \gamma = \frac{3}{2(N+1)} < 1, \quad \kappa = \frac{m+1}{N+1} > 1,$$

with $c_0 > 0$ independent of S and u. Integrating the derivative of $W^{1-\kappa}$ over $[1, S]$ yields

$$(\kappa - 1)c_0 \int_1^S (1+t)^{-\gamma}\,\mathrm{d}t \le W(1)^{1-\kappa} - W(S)^{1-\kappa} \le 1,$$

because $W(1) \ge 1$. Since the integral on the left diverges as $S \to \infty$, every $S < T$ is bounded by a constant independent of S and u, and hence so is T. The zero-domination argument at the beginning of the proof now gives $u \ge \varphi - C$. The case $T \le 1$, which includes $B = 0$, was covered there as well. □

Corollary 34.8.1 *Let $0 \le f \in L^p(X, \mathrm{d}V)$, with $p > 1$ and $\int_X f\,\mathrm{d}V > 0$. For $\lambda > 0$, the solution in Theorem 34.7.1 of*

$$\mathrm{MA}_\theta(u) = \mathrm{e}^{\lambda u} f\,\mathrm{d}V$$

satisfies $u \sim \varphi$. Every normalized solution in $\mathcal{E}_\varphi$ of

$$\mathrm{MA}_\theta(u) = c f\,\mathrm{d}V, \qquad c > 0,$$

also satisfies $u \sim \varphi$. In particular, both existence statements hold in the ordinary singularity type of φ.

Proof Lemma 1.5.1, applied in a Hermitian background dominating θ, gives uniform moments of every finite order against $\mathrm{d}V$ for normalized θ-plurisubharmonic functions. Hölder's inequality therefore implies, for every finite m,

$$\sup_{\substack{v \in \mathrm{PSH}(X,\theta) \\ \sup_X v = 0}} \int_X (-v)^m f\,\mathrm{d}V < \infty.$$

Choose $m > 2n + 1$. For the solution of the exponential equation, set $S = \sup_X u$ and $v = u - S$. Then $v \in \mathcal{E}_\varphi$, $\sup_X v = 0$, and

$$\mu_v = \mathrm{e}^{\lambda u} f\,\mathrm{d}V \le \mathrm{e}^{\lambda S} f\,\mathrm{d}V.$$

The preceding proposition, applied to this fixed dominating measure, shows that v, and hence u, has the ordinary type of φ. For the normalized equation, apply the proposition to the measure $c\,f\,\mathrm{d}V$. □

Part X
Problems and historical notes

Chapter 35
Problems in Hermitian pluripotential theory

Throughout this chapter X is a connected compact complex manifold of dimension $n \geq 1$. The Hermitian forms, volume forms and backgrounds needed for each problem are fixed in the section where the problem is stated.

Many arguments in this book begin with linear control. A uniform bound on Gauduchon traces makes a family of positive currents relatively compact, and Gauduchon duality expresses scalar invariants through optimization over test currents. Most of the questions below arise because such linear control does not determine the nonlinear data of a limit, namely its Monge–Ampère mass and its singularities. Suppose that X fails the bounded mass property of Definition 7.5.1, and let ω be a Hermitian form on X. By Proposition 7.7.1 and Corollary 7.7.1, currents $\omega + \mathrm{dd}^{\mathrm{c}} u$ with integrable coefficients can converge in the coefficient L^1 norm, with Gauduchon trace measures converging in total variation, while their determinant masses are infinite. By Proposition 17.3.3, even the relative envelope class $\mathcal{E}_{\mathrm{env}}(X, \omega, 0)$ of Definition 14.4.1 contains potentials of infinite mass. Passing from linear bounds to an exact equation therefore requires additional control of the nonlinear limit.

Two further requirements recur. Positivity of a scalar invariant does not imply strict positivity: the smooth semipositive background on a projective manifold in Theorem 27.13.1 has positive Gauduchon variational volume but is not big. Likewise, an envelope realizing prescribed divisorial lower bounds may acquire larger orders, and then it does not recover the prescribed singularity type. Thus the mass of a limiting solution, strict positivity of a current, and exactness of its divisorial orders have to be checked in their respective arguments.

35.1 Threshold equations and singular limits

Solving a Monge–Ampère equation with singular data involves two separate steps: one constructs a subsolution at the optimal coefficient, and then passes to a limit of approximate solutions without losing Monge–Ampère mass. Throughout this section, h is a Hermitian form on X and $\mathrm{d}V$ a smooth positive probability volume form. We write

$v_-(h)$ and $v_+(h)$ for the lower and upper masses of h (Definition 7.5.1), $\mathrm{MA}_h^{\mathrm{ext}}(u)$ for the extended Monge–Ampère measure of $u \in \mathrm{PSH}(X,h)$ (Theorem 7.1.1), which may have infinite mass, and $\mathcal{E}_{\mathrm{env}}(X,h,0)$ for the relative envelope class with reference 0. For a nonzero nonnegative density $f \in L^1(\mathrm{d}V)$, the subsolution threshold $b_h(f)$ is the largest $a \geq 0$ for which some $u \in \mathrm{PSH}(X,h)$ satisfies $\mathrm{MA}_h^{\mathrm{ext}}(u) \geq a f \,\mathrm{d}V$ (Theorem 15.27.1); such a u with $a = b_h(f)$ is called a threshold subsolution. Finally, $\mathcal{D}_h$ is the space of density roots $p \in L^n(\mathrm{d}V)$ whose root functional $\rho_h(p) = b_h(|p|^n)^{-1/n}$ is finite (Proposition 15.31.2).

In two regimes the equations are solved for arbitrary finite non-pluripolar measures. For a semipositive big background θ with positive lower mass and a model reference satisfying (14.20), Theorem 33.1.1 solves both the equation with a positive exponent and the normalized equation. For an arbitrary smooth big background θ, Theorems 34.7.1 and 34.7.2 solve both equations under bounded mass and positive lower volume of the model reference. The existence theorem for L^p densities in the minimal ordinary singularity type, in Section 15.15, requires neither mass property; in the second regime, Corollary 34.8.1 also places the solutions for L^p densities in the prescribed ordinary model type.

Under bounded mass the reference may also vary. Suppose the ordinary model references converge for the volume comparison d_θ of Section 31.6 and have a common positive lower bound for their lower volumes, and the densities converge in L^1 of a fixed finite non-pluripolar measure. Then, by Theorems 34.5.1 and 34.7.3, for a fixed positive exponent the solutions converge in capacity and their Monge–Ampère measures converge in total variation. At exponent zero the scalars and the measures converge, and every L^1-convergent subsequence of normalized solutions converges to a solution of the limiting equation; singling out one limit potential requires a further uniqueness or selection argument.

To remove these mass hypotheses, one must replace each place where they enter the compactness arguments. The proof in the semipositive regime uses finite auxiliary contact measures to construct a common lower envelope, and then obtains the limit for a moving measure from convergence in capacity and control of deep sublevel tails. The proof for an arbitrary big background uses finite upper relative volumes to construct a rooftop between classes and to maximize the weighted measure. In either case one must still prove that the measure of the solution is finite; the formal existence of the envelopes does not give this.

35.1.1 Individual density thresholds

One must distinguish solvability for all measures from solvability for one fixed density. By Corollary 33.1.1, the normalized equation $\mathrm{MA}_h^{\mathrm{ext}}(u) = c\mu$ has a solution in $\mathcal{E}_{\mathrm{env}}(X,h,0)$ for every nonzero finite non-pluripolar measure μ exactly when $v_-(h) > 0$. When $v_-(h) = 0$, Theorem 15.26.1 produces a probability density in $L^1(\mathrm{d}V)$, bounded below by a positive constant, that admits no positive scaled subsolution. Such obstructing densities are dense in L^1, although every smooth

strictly positive density is solvable. The compact threefold of Theorem 27.18.1 has $v_-(h) = 0$ for every Hermitian form h, so the obstruction does occur.

For one nonnegative density $f \in L^1(\mathrm{d}V)$, the threshold $b_h(f)$ is the decreasing limit of the normalization constants of the bounded truncations $\min(f, k)$, see (15.80). An argument with upper envelopes and damped continuity shows that a subsolution attains this coefficient, but it proves neither an exact equation nor finiteness of the mass. Smooth comparison tests alone do not detect the threshold, by Theorem 15.27.2. On the other hand, the Gauduchon formula of Theorem 15.28.1 expresses the root functional as a supremum of linear tests, which makes b_h upper semicontinuous in L^1. The densities with positive threshold are those whose roots $f^{1/n}$ lie in $\mathcal{D}_h$, and this space does not depend on the Hermitian form (Proposition 15.28.1). With the norm $\|p\|_{L^n} + \rho_h(p)$ it is a Banach space; it equals L^n when $v_-(h) > 0$, and it is complete for ρ_h alone exactly when $v_+(h) < \infty$, by Propositions 15.31.2 and 15.31.3.

Problem 35.1.1 Suppose that h has zero lower mass, $v_-(h) = 0$, and let $0 \le f \in L^1(\mathrm{d}V)$ be nonzero, with positive subsolution threshold $b_h(f) > 0$. Does there exist $u \in \mathrm{PSH}(X, h)$ such that

$$\mathrm{MA}_h^{\mathrm{ext}}(u) = b_h(f) f \, \mathrm{d}V?$$

Determine the singularity type or envelope class in which such an equation should be posed.

The hypothesis $v_-(h) = 0$ is what makes the problem open. For $\lambda > 0$, write u_λ for the greatest solution of the exponential equation $\mathrm{MA}_h^{\mathrm{ext}}(u) = \mathrm{e}^{\lambda u} f \, \mathrm{d}V$, when it exists (Theorem 15.34.1). When $v_-(h) > 0$, Corollary 33.7.1 solves the threshold equation in $\mathcal{E}_{\mathrm{env}}(X, h, 0)$ for every nonzero nonnegative L^1 density: comparison of finite measures identifies the scalar, and dominated continuity shows that every L^1-convergent subsequence of the normalized greatest exponential solutions $u_\lambda - \sup_X u_\lambda$ converges in capacity. This covers the Hopf example of Chapter 9, which has positive lower mass but not bounded mass. For L^p densities with $p > 1$, the continuous solution of the normalized equation and Proposition 15.27.2 answer the question without any lower mass assumption.

At zero lower mass, Theorem 15.29.1 solves the threshold equation with finite mass for explicit densities on a split product with projective space; these densities are integrable but belong to no L^p with $p > 1$. When the Hermitian factor has zero lower mass, the product background h is nonclosed with $v_-(h) = 0$. The canonical solutions u belong to $\mathcal{E}_{\mathrm{env}}(X, h, 0)$, and for every $0 < \delta < 1$ the envelope $P_{\delta h}(u)$ of Section 1.19 is not identically $-\infty$. More generally, Theorem 15.29.2 solves the exact finite-mass equation for every nonzero integrable density from a closed Kähler factor, with the appropriate change of reference volume. The threshold is the split background's mass divided by the integral of the factor density. On the product of the elliptic threefold with $\mathbb{P}^2$, this gives threshold 240π for every probability density from that factor. An arbitrary density on the product need not have this form, and the general problem remains open.

For product measures from a projective-line factor, Proposition 15.30.1 gives an exact finite-mass equation for every non-polar factor probability. Its supremum-zero

solution is unique among pullbacks from the curve, and it is bounded, or continuous, exactly when the logarithmic potential of the factor probability is. When that potential is finite and continuous, Theorem 15.30.2 identifies the attained coefficient over all subsolutions, constructs the greatest exponential solution for every positive exponent, and shows that the whole family of normalized solutions converges uniformly as $\lambda \downarrow 0$, with error $O(\lambda)$. The trace of Sobolev functions against the measure gives existence, and singular strict domination extends greatestness to unbounded competitors. Weighted circle probabilities in Proposition 15.30.2 give measures singular with respect to smooth volume. The background has zero lower mass when the Hermitian factor does. These product criteria do not classify arbitrary measures on a fixed Hermitian manifold.

A different geometric condition applies directly to singular measures on a fixed Hermitian manifold. The probability ρ of (15.78) is chosen from a coordinate ball before the datum. If ν is finite, nonzero and has bounded density with respect to ρ, Proposition 15.25.1 and Theorem 15.25.1 give a strict Hölder scaled subsolution and a unique bounded exponential solution for every positive exponent. In addition, Corollary 15.25.1 gives a normalized Hölder solution of $\mathrm{MA}_h(w) = c\nu$ with $c > 0$ obtained as part of the solution. This includes bounded densities on one of the spheres and needs neither a product decomposition nor positive lower mass. It is a sufficient condition; it does not classify arbitrary individual measures or identify a homogeneous threshold coefficient.

For the data of Problem 35.1.1, Corollary 15.34.1 shows that the greatest exponential solutions u_λ exist for every $\lambda > 0$, that the scalars $c_\lambda = \mathrm{e}^{\lambda \sup_X u_\lambda}$ converge to $b_h(f)$ as $\lambda \downarrow 0$, and that the measures $\mathrm{e}^{\lambda u_\lambda} f \, \mathrm{d}V$ converge in total variation. What is missing in general is convergence in capacity of the normalized potentials. If $f^{1/n}$ lies in the subspace $\mathcal{D}_h^0$ of density roots with uniformly vanishing tails (Proposition 15.31.4), Proposition 15.34.1 gives a modulus for the errors of the scalars and of the measures; a rate linear in the exponent would require further control.

A power capacity bound (15.95) supplies both root tails and the compactness needed for an exact equation. By Theorem 15.32.1, the threshold is positive and finite, the root belongs to $\mathcal{D}_h^0$, and a subsequence of the normalized solutions for the truncated densities converges uniformly to a continuous solution at $b_h(f)$. The logarithmic density of Theorem 15.32.2 satisfies this condition on every fixed Hermitian background, while belonging to no L^p with $p > 1$. Its reference potential is continuous but not Hölder, and its root tails have the bound (15.98). These conclusions use the capacity assumption on the fixed datum. Whether all positive-threshold roots have vanishing tails, and whether tails alone imply an exact equation at zero lower mass, remain open.

Positivity and integrability of f do not make the threshold positive, and exponential equations can be solvable even when it vanishes. For zero threshold, Theorem 15.34.2 gives a sufficient growth condition in the radial variable, and Theorem 15.34.3 constructs a single strictly positive probability density with $b_h(f) = 0$ whose exponential equation has, for every positive exponent, a greatest solution of finite mass; all its solutions are unbounded. In that construction the projective factors are chosen after the collapsing sequence on the base.

For this density, Proposition 15.34.3 shows that any choice of exact exponential solutions has total mass tending to zero as the exponent decreases to zero; the argument says nothing about convergence of the normalized potentials. In general, the set $\Lambda_h(f)$ of positive exponents for which the exponential equation is solvable is an upper interval, by Proposition 15.34.2. A necessary criterion on the density for solvability at zero threshold, solvability for an arbitrary prescribed concentration family, and an extension to arbitrary non-pluripolar measures are open.

Suppose that $a = \inf \Lambda_h(f)$ is positive. By Proposition 15.34.4, a belongs to $\Lambda_h(f)$ if and only if there are exact solutions at exponents decreasing to a whose suprema are bounded below; if $a \notin \Lambda_h(f)$, the total masses of every such sequence of solutions tend to zero. The common upper barrier of Lemma 15.34.3 also shows that the greatest solutions become monotone in the exponent after a logarithmic correction. Consequently, by Proposition 15.34.5, they are continuous from the right in L^1 and in capacity at every exponent in $\Lambda_h(f)$, and their Monge–Ampère measures converge in total variation. At every interior exponent, Proposition 15.34.6 gives a left limit in L^1 and in capacity that solves the equation at that exponent and lies below the greatest solution; the equation measures converge in total variation. By Proposition 15.34.7, these potentials can differ at only countably many exponents. Whether a positive endpoint can fail to belong to $\Lambda_h(f)$, or whether the left limit can differ from the greatest solution at some interior exponent, remains open.

35.1.2 Dual attainment and exact equations

For the current formulation of the dual problem, suppose $n \geq 2$. By Gauduchon duality, the threshold is determined by a maximization problem over test currents. Let $\mathcal{G}_h$ be the compact convex set of normalized Gauduchon currents, the positive $(n-1, n-1)$-currents T with $\mathrm{dd^c}T = 0$ and $\int_X h \wedge T = 1$ (Section 4.4), and let k_T be the determinant density of T (Theorem 27.7.1). For $0 \leq p \in \mathcal{D}_h$, the root functional $\rho_h(p)$ is the supremum of $D_p(T) = \int_X p\, k_T \,\mathrm{d}V$ over $T \in \mathcal{G}_h$. A maximizing current can turn the threshold inequality into an equality, provided its regularity, or control of its weighted moments, prevents loss of mass in the limit. For roots in the closed order ideal $\mathcal{D}_h^0$, Theorem 27.8.1 and Proposition 27.8.3 show that the supremum is attained and represent all supporting probabilities of the threshold functional by maximizing currents. These probabilities are absolutely continuous, and by Proposition 27.8.4 the trace errors along smooth maximizing sequences tend to zero in total variation. Uniform integrability along a single maximizing sequence already gives attainment and convergence of the determinant-weighted measures of that sequence (Proposition 27.8.2), but it does not give the representation of all supporting probabilities.

Regularity of a maximizer has strong consequences. Let f be a nonzero density. If $D_{f^{1/n}}$ has a smooth strictly positive maximizer, then $b_h(f) > 0$, $f^{1/n}$ is smooth and nonnegative, and every threshold subsolution is smooth and solves the equation exactly, uniquely up to constants (Corollary 27.8.1). For a Kähler background,

Corollary 27.8.3 shows that such a maximizer exists exactly when f is smooth and strictly positive. The strict positivity fails for nonclosed backgrounds: on every compact complex manifold of dimension at least two, Example 27.8.1 gives a smooth strictly positive maximizer for a density vanishing on an open set.

Theorem 27.8.2 weakens the hypothesis to a smooth semipositive maximizer that is strictly positive outside a proper analytic subset $Z \subset X$. It still gives an exact equation on all of X, smoothness of the solutions off Z, and uniqueness of threshold subsolutions up to constants. The proof takes the maximum of two subsolutions and uses a Gauduchon integral to eliminate the divisorial part of their difference; no singularity type is prescribed in advance. The prime divisors along which the threshold solution has positive Lelong number have linearly independent Bott–Chern classes, by Corollary 27.8.7. In this regime write $b = b_h(f)$, with u_λ and c_λ as above. By Proposition 27.8.5, the whole family of normalized solutions converges in L^1 as $\lambda \downarrow 0$, it has a fixed lower barrier, and $b \le c_\lambda \le b\mathrm{e}^{C\lambda}$ for a finite constant C independent of λ. If the envelope $P_{\delta h}(u_0)$ of the normalized threshold solution u_0 is not identically $-\infty$ for some $0 < \delta < 1$, the convergence also holds in capacity (Corollary 27.8.4); for L^p data with $p > 1$ the threshold solution is continuous and the convergence is uniform (Corollary 27.8.5). In Example 27.8.4 the solution is smooth, but the root $f^{1/n}$ is not differentiable on Z. For general L^1 data we do not know whether $P_{\delta h}(u_0) \not\equiv -\infty$ holds automatically, nor how regular the solution is across Z.

The moment criterion of Theorem 27.8.5 requires no regularity of a maximizing current. Suppose that the supremum-normalized h-plurisubharmonic functions have uniformly bounded first moments against the determinant-weighted probabilities of one smooth maximizing sequence. Then the exponential solutions have a fixed lower barrier, and their normalized cluster limits lie below the greatest normalized threshold subsolution Φ_f of Proposition 15.27.1. If, in addition, $P_{\delta h}(\Phi_f) \not\equiv -\infty$ for some $0 < \delta < 1$, the lower-barrier criterion gives convergence in capacity along subsequences and an exact threshold equation, without either mass property. An entropy bound on these test probabilities implies both the bound on first moments and the uniform integrability along one sequence used above. By Proposition 16.6.1, such an entropy bound is equivalent to a uniform bound for H_0, the convex dual of the normalization functional with zero reference (Chapter 16). An entropy bound does not, however, give the condition on $P_{\delta h}(\Phi_f)$.

Attainment by a singular current therefore does not settle the equation by itself. On the products of Theorem 15.29.2, the roots pulled back from the closed Kähler factor form a closed subspace of $\mathcal{D}^0_h$ isomorphic to L^n on that factor, by Proposition 15.31.5; some of these roots belong to no higher L^r space, and their tails decay only as a power of $\log K$. At zero lower mass we do not know in general whether the root-tail condition $f^{1/n} \in \mathcal{D}^0_h$ suffices for an exact threshold equation, or even whether $\mathcal{D}^0_h = \mathcal{D}_h$. By Proposition 27.8.1, the tail condition is equivalent to continuity of the threshold under logarithmic perturbations $\mathrm{e}^{F_j} f$ with F_j uniformly bounded and tending to zero in measure. This is continuity on order intervals, not continuity of the threshold under arbitrary L^1 perturbations.

Convergence in capacity cannot be replaced by L^1 convergence in these arguments. On a curve, the ideal envelopes with finitely many logarithmic poles in Theorem 17.4.1 have zero non-pluripolar mass, yet they converge in L^1 to a smooth potential of positive mass. The existence theorems obtain the stronger convergence from their measure bounds; representing the measure by a carrier, even within a prescribed singularity type, does not provide it.

35.1.3 Regularity of singular solutions

Regularity estimates provide the compactness needed to single out a limit as the exponential parameter tends to zero. In this subsection θ is a smooth big real $(1,1)$-form, and ρ is the strict barrier with analytic singularities fixed in Section 15.12, with pole set Z and regular locus $\Omega = X \setminus Z$. For bounded solutions with L^p density, $p > 1$, Theorem 15.5.1 gives continuity wherever ρ is locally bounded. A locally Hölder continuous solution with minimal singularities is smooth on Ω when the density is smooth and positive there (Theorem 15.17.1), and global L^p control of the density gives the Hölder hypothesis (Theorem 15.16.1 and Corollary 15.17.1).

For a fixed reference η with $\rho \le \eta \le 0$, Corollary 15.14.1 shows that the potentials u with $\eta - C_0 \le u \le \eta$ and uniformly bounded L^p densities form a compact family for uniform convergence of differences on Ω. Since the reference need not have minimal singularities, this compactness does not by itself make the potentials regular. In the minimal type, the convergence of scalars and densities in Proposition 15.16.1 identifies the equation satisfied by every cluster limit of the normalized exponential solutions.

Smoothness of the density on an open set is not enough. On a compact Kähler manifold with a smooth positive volume form, Theorem 29.10.1 proves that the full-mass solution is smooth on the ample locus of every big class, nef or not. Smoothness of the density only on the ample locus, however, does not imply global L^p integrability. On an arbitrary open set where the density is smooth, it does not even imply continuity of a bounded solution with minimal singularities: Theorem 15.20.2 gives a compact Kähler counterexample with constant density on a ball. For locally bounded solutions in any smooth background, Theorem 15.20.1 gives a sufficient condition in terms of smooth boundary traces on small balls. Among prescribed singularity types, the resolved logarithmic types of Theorem 15.21.1 admit Hölder remainders when their residual form is positive, while for exceptional minimal poles Theorems 15.22.1 and 15.22.2 give a Hölder remainder on the target and quantify the loss of integrability caused by the Jacobian. Neither result covers arbitrary prescribed types, and neither derives a global Hölder bound from information on the density confined to the regular locus. At a zero of the density, even smooth data need not give a smooth residual: Proposition 29.11.3 and Example 29.11.3 give a necessary vanishing order and show that it is not sufficient. Beyond these resolved and explicit geometric cases, the regularity questions are open.

For the prescribed radial nonlogarithmic types of Section 15.36, Theorem 15.36.1 controls the residual after the singularity is subtracted by a relative density error $O(r^{\delta})$; the critical C^1 and C^2 thresholds are attained and are sharp by Proposition 15.36.2. Positive cumulative central mass allows smooth crossing of density zeros, and Proposition 15.36.3 gives an open zero annulus with a globally smooth residual. Corollary 15.36.1 transfers this regularity to products with any compact Hermitian factor; the product background is nonclosed with zero lower mass when the factor has zero lower mass. Perturbing the same singular reference as an everywhere radial obstacle instead gives an envelope whose residual is globally $C^{1,1}$ and zero near the pole; its Monge–Ampère measure vanishes on the open regions where it does not touch the obstacle (Theorem 15.37.1 and Corollary 15.37.1). Fixed-support bounded C^2 obstacle families satisfy the derivative stability of Theorem 15.37.2, and Proposition 15.37.1 transfers the full envelope to any fixed positive Hermitian product factor. These explicit radial regimes give neither regularity nor uniqueness for nonsymmetric data, and they do not cover general singular references or ideal weighted envelopes.

35.1.4 Uniqueness and selection

Where the density vanishes, the linearized equation degenerates, and the zero set can separate regions where it is elliptic. The question is whether the difference of two solutions can vary from one such region to another. Throughout this subsection ω is a Hermitian form on X, f is smooth and nonnegative, and potentials are normalized by $\sup_X u = 0$.

On $\mathbb{P}^3$, Theorem 12.6.1 gives a nonclosed Hermitian form and a smooth nonnegative density with two distinct smooth normalized solutions; there the density vanishes on an open set. The construction of Theorem 12.7.1 has uncountably many smooth normalized solutions for a density that is positive off two disjoint real hypersurfaces, and after a modification of the background, Theorem 12.8.1 achieves exactly quadratic vanishing on those hypersurfaces. Thus neither a zero set of zero volume nor vanishing of finite order guarantees uniqueness. All these examples live on $\mathbb{P}^3$, which has the bounded mass property; the degeneracy responsible for nonuniqueness lies in the density and in the solution forms.

These examples restrict any geometric use of the equation. A construction assigning a canonical potential to arbitrary nonnegative data must specify a selection procedure and prove that it is well defined; existence in the minimal ordinary singularity type does not provide one. The scalar, which is unique, and for strictly positive data the corresponding variational volume, remain meaningful without uniqueness of the potential. In low dimension uniqueness holds under additional conditions. On a compact Hermitian surface, Corollary 12.3.1 proves uniqueness for C^2 potentials when the common density is continuous and positive on a dense set, with no differential condition on the metric; in dimension three, the same corollary and Theorem 12.3.1 require in addition a sign condition on $\mathrm{dd}^{\mathrm{c}}\omega$. Both arguments rest on the sign of

an expression quadratic in the difference of the two solution forms. The cofactor identity (12.20) of Proposition 12.2.1 is a different integration mechanism; it gives uniqueness when the cofactor form is dd^{c}-closed.

When $\{f > 0\}$ is connected and dense, Theorem 12.4.1 shows that two bounded normalized solutions with the same positive scalar coincide; the theorem applies to the measure $c f \, \mathrm{d}V$ after the smooth positive ratio $\mathrm{d}V/\omega^n$ is absorbed into the density. In all the counterexamples above whose zero set consists of hypersurfaces, including the real-analytic example of Theorem 12.9.1, the positivity set has three connected components. None of them contradicts Conjecture 35.6.1, which concerns strictly positive densities e^F with F continuous.

For a fixed nonzero density $f \in L^p(\mathrm{d}V)$, $p > 1$, the exponential equation does select a solution. By Theorem 16.12.1, a weakly compact set of norming probabilities determines the limit: as the exponent tends to zero, the suitably shifted exponential solutions converge to the greatest subsolution of the critical equation that satisfies the integral normalizations given by these probabilities. In a Hermitian background the convergence is uniform; in an arbitrary smooth big background it holds in $L^1(X)$ and locally uniformly off the pole set of the strict barrier. The selected limit inherits the symmetries of the data (Proposition 15.19.1), and in the projective example Theorem 15.19.1 identifies it with the explicit symmetric $C^{1,1}$ solution of Proposition 12.6.1. Selection thus coexists with nonuniqueness of the normalized critical equation. For the merely integrable radial densities on zero-lower-mass products, Theorem 15.35.1 determines the whole supremum-normalized limit, even with joint variation of the density parameter and exponent. Scalar centering has a finite limit exactly below the moment boundary of Proposition 15.35.1; at the boundary and above it diverges. For logarithmic product models with positive residual, Theorem 16.13.1 also gives selection under independent variation of the reference coefficients, background, density and exponent; centering follows each varying critical scalar. Selection for arbitrary integrable data and general varying singular reference types remains open.

Dual attainment gives a different criterion. Under the hypothesis of Corollary 27.8.1, a smooth strictly positive maximizing Gauduchon current for $D_{f^{1/n}}$, the normalized threshold solution is unique, and the whole family of normalized solutions with positive exponent converges uniformly to it (Corollary 27.8.2). Since the density of Example 27.8.1 satisfies this hypothesis and vanishes on an open set, the size of the zero set alone does not decide uniqueness; the dual condition also depends on the background.

35.2 Positive Gauduchon volume and bigness

Fix a smooth strictly positive probability form μ on X. For a pseudoeffective smooth real $(1,1)$-form θ, the Gauduchon variational volume $\mathcal{V}_\mu(\theta) = \lambda_\mu(\theta)^n$ of Definition 10.8.1 is an infimum over Gauduchon metrics, whereas bigness asks for one potential u making $\theta + \mathrm{dd}^c u$ strictly positive. Bigness implies $\mathcal{V}_\mu(\theta) > 0$ (Corollary 10.8.1). The converse fails for nef nonclosed backgrounds. For every $n \geq 3$, Theorem 27.13.1 constructs a smooth semipositive background θ on $\mathbb{P}^{n-1} \times \mathbb{P}^1$ that is not big: no current $\theta + \mathrm{dd}^c u$ is bounded below by a positive multiple of a Hermitian form. Its variational volume equals n for the product probability form fixed there, and is therefore positive for every μ. In this example a radial potential compensates the determinant, while a closed test current carried by a degenerate fiber detects the missing strict positivity.

Positivity of $\mathcal{V}_\mu$ does not depend on the choice of μ (Proposition 10.8.2). The infimum over all such μ, however, is a different invariant. For a Hermitian form ω, the infimum of $\mathcal{V}_\mu(\omega)$ is the lower mass $v_-(\omega)$ by Theorem 10.8.2, and it vanishes for every Hermitian form on the threefold of Theorem 27.18.1, although Hermitian forms are big. In dimension one, positive volume and bigness are equivalent, since both mean $\int_X \theta > 0$ (Definition 10.8.1 and Corollary 4.3.1). For a closed nef class on a compact Kähler manifold, $\mathcal{V}_\mu$ is the top self-intersection, and the equivalence is the nef-and-big criterion of Demailly and Păun. On a compact Kähler manifold, Theorem 27.10.1 also gives bigness when a closed nef core of positive volume is supplemented by a semipositive form. Whether positive volume implies bigness for general nef nonclosed backgrounds on surfaces is open. In higher dimension, any such criterion requires geometric hypotheses that exclude the radial obstruction.

35.2.1 Collapse of normalized Gauduchon tests

The root volume and the bigness threshold divide the same pairing by different quantities. In the notation A_θ, A_h of Section 27.12, Proposition 27.12.1 shows that, for $n \geq 2$, a background of positive root volume that is not big forces $D_\mu(\gamma_j) \to 0$ along every sequence of Gauduchon metrics with $A_h(\gamma_j) = 1$ and $A_\theta(\gamma_j) \to 0$, and that a positive lower bound for D_μ along one sequence minimizing the threshold would give bigness. For the radial background of Theorem 27.13.1, the normalized $(n-1)$st powers of a family of balanced test metrics (Definition 4.1.1) converge weakly to the integration current of the degenerate fiber (Proposition 27.13.2), and the volume-minimizing family approaches the bigness threshold only in dimension three, by Proposition 27.13.1.

The quotient D_μ / A_h is always bounded above, by (27.50) with $\theta = h$. What is missing for the converse direction is a lower bound for D_μ along suitable sequences minimizing the threshold. Compactness of normalized potentials does not give compactness of these Gauduchon metrics.

One way to rule out this collapse is to deform the test metrics. Solving a Gauduchon equation replaces the volume density of a test metric while keeping the Aeppli class of γ^{n-1}. When θ is closed, this leaves the numerator A_θ unchanged. For nonclosed θ, integration by parts leaves derivatives of the background, so such a deformation must also control the change of the pairing; this is the additional estimate needed to use the smooth equation against collapse.

35.3 Closed positivity and geometric equations

A closed Kähler current places the manifold in Fujiki's class C, whereas a strictly positive current $\theta + \mathrm{dd}^{\mathrm{c}} u$ in a nonclosed background keeps the torsion $\mathrm{d}\theta$ of that background. This distinction links the passage from positive volume to bigness with the geometry of the manifold, and it limits what a scalar potential equation can prove. Throughout this section μ is a smooth strictly positive probability form on X, and we use balanced and pluriclosed metrics in the sense of Definition 4.1.1. The geometric results from the literature used below are stated, with their precise hypotheses and sources, in Section 36.22.

35.3.1 Nef classes, bigness and the Fujiki class

A compact complex manifold belongs to Fujiki's class C exactly when it carries a closed Kähler current, that is, a closed positive $(1,1)$-current bounded below by a positive multiple of a Hermitian form. The transcendental Grauert–Riemenschneider conjecture asks whether a closed nef real $(1,1)$-class α with

$$\int_X \alpha^n > 0$$

is big. For such a class, bigness is equivalent to $X \in C$; the precise statements and sources are given in Section 36.22.

The obstruction can be expressed through Monge–Ampère mass. Under the same hypotheses on α, one has

$$\alpha \text{ is big} \quad \Longleftrightarrow \quad v_+(h) < \infty$$

for a Hermitian metric h on X, where $v_+(h)$ is the upper mass of Definition 7.5.1. There is also a characterization of C by bounded mass together with positivity of an upper volume of some closed Bott–Chern class; that volume is different from the Gauduchon variational volume $\mathcal{V}_\mu$. Both results are stated in Section 36.22.

The Hopf threefold fails bounded mass, but it is not a counterexample to the conjecture, since a counterexample would also need a closed nef class with positive

top self-intersection. A proof of the conjecture must explain why this closed positivity forces bounded mass.

The elliptic threefold of Theorem 27.18.1 is the complementary test case: it has bounded mass but no big closed class. By Proposition 29.9.1, its nef cone is the nonnegative span of the classes of two smooth semipositive forms of rank one, and every top self-intersection vanishes. This example of zero lower mass therefore does not bear on the conjecture either. The explicit boundary calculation of Proposition 29.9.4 describes its degeneration without identifying a nonclosed metric with a closed class.

The Gauduchon variational volume of Definition 10.8.1 splits the conjecture into two possible steps:

$$\int_X \alpha^n > 0 \quad \overset{?}{\Longrightarrow} \quad \mathcal{V}_\mu(\alpha) > 0 \quad \overset{?}{\Longrightarrow} \quad \alpha \text{ is big.} \tag{35.1}$$

Neither implication is known on arbitrary compact complex manifolds. The second concerns closed nef backgrounds; the projective counterexample of Theorem 27.13.1 has a nonclosed background and does not decide it. On a compact Kähler manifold both implications hold, as recalled in Section 35.2. Without closedness, positivity of $\mathcal{V}_\mu$ cannot characterize C, since every Hermitian metric h on every compact complex manifold satisfies $\mathcal{V}_\mu(h) > 0$.

Problem 35.3.1 Let α be a smooth closed real $(1,1)$-form on X whose class is nef with $\int_X \alpha^n > 0$, and let h be a Hermitian form on X. Prove a positive lower bound for the root volume $\lambda_\mu(\alpha)$ of Definition 10.8.1, and rule out the collapse of Proposition 27.12.1 along a sequence of Gauduchon metrics minimizing the bigness threshold $\kappa_h(\alpha)$.

The test metrics can be normalized by $\int_X h \wedge \gamma^{n-1} = 1$. By Proposition 27.12.1, a positive lower bound for $D_\mu(\gamma)$ along one sequence minimizing the threshold would give the second implication. Since α is closed, the deformation of the test metrics by a Gauduchon equation, described at the end of Section 35.2, keeps the pairing with α; the missing estimate is control of the trace normalization and of the determinant denominator after this change, which prescribing a volume density alone does not give.

35.3.2 Positive lower mass and concentration

Positive lower mass is an integrability property of the Gauduchon tests at an endpoint exponent. It is independent of finite upper mass: the threefold of Theorem 27.18.1 has $v_-(h) = 0$ for every Hermitian metric and carries a balanced metric of finite upper mass, while the normalized Hopf metric has $v_- = 6$ and $v_+ = \infty$.

Let $n \geq 2$, let h be a Hermitian form and $\mathrm{d}V$ a smooth positive volume form. Recall the set $\mathcal{G}_h$ of normalized Gauduchon currents (Section 4.4) and the determinant

measure Γ_0 of (27.23), and write

$$\Gamma_0(T) = k_T \, \mathrm{d}V, \qquad T \in \mathscr{G}_h.$$

By Theorem 27.7.2,

$$v_-(h) > 0 \quad \Longleftrightarrow \quad \sup_{T \in \mathscr{G}_h} \|k_T\|_{L^{n/(n-1)}(\mathrm{d}V)} < \infty, \tag{35.2}$$

and the exact value of the supremum is $v_-(h)^{-1/n}$, including the value $+\infty$. Uniform estimates for every smaller exponent $1 < q < n/(n-1)$ hold without either mass property. In the example of zero lower mass, the endpoint estimate fails already for smooth strictly positive tests. Positivity, the condition $\mathrm{dd}^{\mathrm{c}} T = 0$ and the trace normalization alone therefore do not control concentration at the endpoint exponent.

Problem 35.3.2 Find geometric conditions on X and h, beyond the metric identities in Corollary 27.17.1 and the curvature condition for principal elliptic threefolds in Theorem 27.16.1, that imply the endpoint estimate in (35.2), and explain how they exclude the degeneration of zero lower mass constructed in Section 27.18.

Corollary 27.17.1 proves the endpoint bound when some Hermitian metric g satisfies $\mathrm{dd}^{\mathrm{c}} g = \mathrm{dd}^{\mathrm{c}}(g^2) = 0$, which includes all compact Kähler manifolds and all compact complex surfaces. For a fixed smooth strictly positive density, Theorem 27.5.1 gives a smooth maximizing current with bounded determinant density. However, Proposition 27.17.1 produces maximizing sequences in fixed Kähler classes whose moments diverge at every exponent above the endpoint. Exact maximizers and maximizing sequences therefore require different estimates.

For principal elliptic threefolds over Kähler surfaces, Theorem 27.16.1 gives a geometric sufficient condition: the sum of the squares of the two real curvature forms is nonnegative and not identically zero. A bounded cubic identity then gives positive lower mass, and the determinant bound transfers to every Hermitian background. These manifolds admit no pluriclosed metric; Propositions 27.16.2 and 27.16.3 give explicit families. A primitive pair of curvature forms has the opposite sign, by Proposition 27.16.1, so this criterion leaves the general problem open.

The balanced condition and bounded upper mass do not suffice, even together, as the threefold of Theorem 27.18.1 shows. On this elliptic fibration, the identity (27.84) relates the failure of the lower bound to a vertical Dirichlet integral, and the horizontal curvature forms α, β of Section 27.18 have a positive exact product $\alpha \wedge \beta$, which excludes pluriclosed metrics and closed Kähler currents. These calculations give geometric information about this example, but not a classification of all degenerations of the lower mass.

35.3.3 Lower mass of primitive elliptic bundles

On the primitive elliptic bundles of Section 27.18.5, the variation of the top-degree mass is a vertical Dirichlet integral, and Propositions 27.18.1 and 27.18.2 give the exact upper mass and a positive lower bound for separated potentials. The lower mass over all potentials remains a geometric question, since the collapse construction over a product of curves does not transfer directly to a base without curves.

Problem 35.3.3 Determine the lower Monge–Ampère mass (Definition 7.5.1) of the primitive elliptic bundles other than the threefold of Section 27.18, for instance of their balanced metrics ω from Proposition 27.18.1, in particular for the bundles of Example 27.18.1. Does it vanish whenever $c_1(L) \neq 0$ in real cohomology?

The estimates for separated potentials do not answer this question; the known collapse over $\mathbb{P}^1 \times \mathbb{P}^1$ uses potentials outside that class. On the K3 example every compact curve is vertical, so averaging over such curves cannot produce the horizontal directions used in Lemma 27.18.2. A proof of collapse would therefore need a different construction, while positive lower mass would require a bound for all admissible potentials or for all normalized Gauduchon tests.

35.3.4 Null loci and the singularities of the Chern–Ricci flow

Return to the general manifold X of dimension n. Let ω_0 be a Hermitian metric, taken as the initial metric of the Chern–Ricci flow, and suppose that the flow has a finite maximal time τ. With the Ricci convention of Corollary 27.20.1, the limiting background $\theta_\tau = \omega_0 - \tau \operatorname{Ric}(\omega_0)$ is nef. To identify the region where the flow converges smoothly as the complement of a null locus, one first needs bigness of θ_τ. The dual currents detect the failure of strict positivity, but the degeneration of their determinants does not by itself locate an analytic exceptional set.

For a big smooth background θ, the non-Hermitian locus $E_{\mathrm{nH}}(\theta)$ is the intersection of the singular sets of the strict currents $\theta + \mathrm{dd}^{\mathrm{c}} u$, those bounded below by a positive multiple of a Hermitian form, whose potentials have *almost analytic singularities*; one such current attains the intersection. Here a potential has almost analytic singularities if it is locally a positive multiple of the logarithm of the norm of an ideal up to a bounded remainder, and this remainder becomes smooth after subtracting the divisorial logarithms on a suitable log resolution. This is the class of singularities used in the theorem stated in Section 36.22.

For nef and big θ, define

$$\operatorname{Null}(\theta) = \bigcup_{\substack{V \subset X \text{ irreducible analytic} \\ \dim V > 0,\ \theta|_V \text{ not big}}} V,$$

where bigness on a singular subvariety has its intrinsic analytic meaning. By the theorem recalled in Section 36.22, the equality $E_{\mathrm{nH}}(\theta) = \operatorname{Null}(\theta)$ holds for nef and big smooth backgrounds on arbitrary compact complex manifolds. The numerical criterion recalled there has additional hypotheses: bounded mass on X and positive lower volume on its positive-dimensional submanifolds.

The flow theorem recalled in Section 36.22 identifies the finite-time singularities of the Chern–Ricci flow with the null locus under positive limiting lower volume, for arbitrary initial metrics in dimension two and for Guan–Li initial metrics in higher dimensions, that is, those with $\mathrm{dd}^{\mathrm{c}}\omega_0 = 0$ and $\mathrm{d}\omega_0 \wedge \mathrm{d}^{\mathrm{c}}\omega_0 = 0$. Positivity of $\mathcal{V}_\mu$ cannot replace these hypotheses, as the example below shows.

The calculation with currents in Corollary 27.20.1 applies to arbitrary initial metrics. When $\tau < \infty$, there are normalized Gauduchon currents T annihilating θ_τ:

$$\int_X \theta_\tau \wedge T = 0.$$

If $\mathcal{V}_\mu(\theta_\tau) > 0$, the determinant measures of all such currents vanish, by Proposition 27.20.1. This is a precise statement of degeneracy, but it does not imply that their traces concentrate on analytic sets, since a smooth positive semidefinite matrix can have deficient rank on a whole open set.

Theorem 27.20.3 shows that positivity of the variational volume does not imply bigness at the end of the flow: in every dimension at least three, there is a Chern–Ricci flow with finite maximal time whose limiting background is nef, not big, and has positive $\mathcal{V}_\mu$. Thus being the endpoint of a flow does not exclude the radial obstruction. In this model, Proposition 27.20.2 shows that every normalized annihilating current has its trace supported on the degenerate fiber, with a vertical coefficient matrix of rank one. This support follows from the explicit semipositive representative, not from the vanishing of determinants alone.

If θ_τ is big, whether by assumption or under additional hypotheses, the parabolic theorem recalled in Section 36.22 gives convergence as currents and smooth local convergence off $E_{\mathrm{nH}}(\theta_\tau)$. Combined with nefness and the null-locus theorem, this identifies the analytic exceptional set. The example above does not contradict this conditional theorem, but any further numerical or geometric criterion must impose enough structure to exclude its nonbig endpoint. The local continuity of envelopes and the homogeneous replacement arguments of Theorem 15.7.1 provide an elliptic foundation away from the poles of a strict barrier; the parabolic estimates are a separate input.

Completeness of the limiting metric requires an additional estimate. In Theorem 29.14.1, the metric defined by the unique finite-energy solution of a divisorial exponential equation in a big closed class is incomplete on every nonempty connected

open subset of its ample locus, because its minimal divisorial singularity cancels the prescribed pole. Nefness alone does not help: Theorem 29.14.2 keeps a nonintegrable pole with smooth positive coefficient and a complete cusp, but has a disjoint null curve at finite distance; its solution has zero Lelong numbers and finite energy, but is not of minimal ordinary type. The construction of Theorem 29.14.3 with compatible initial data also gives an incomplete auxiliary metric, with the exact identities for the resolution and the prescribed volume, but its class is not nef. Simultaneous nefness and compatibility remain a separate question, and none of these elliptic examples decides whether the limits of an evolving flow are complete; see Section 36.17.

For integrable poles weakened by logarithmic factors, Corollary 29.15.1 gives a sufficient condition for completeness in terms of the density, the potential and the tangential determinant; the proof controls normal lengths and applies in arbitrary Hermitian backgrounds. The torus models of Examples 29.15.1 and 29.15.2 attain its critical exponent and show that it is sharp, even for a Kähler background and full mass. Their pole coefficients need not extend smoothly and positively across the divisor, so the geometric smoothing question remains separate.

The regularity of the residual also matters. By Theorem 1.18.1, a Hölder residual on a compact manifold that defines a smooth positive metric off its analytic singular set puts that set at finite distance. Continuity alone does not suffice: the prescribed radial densities of Theorem 29.12.2 have bounded continuous minimal remainders and complete regular ends for the range $n < d \leq 2n$ of the exponent d in that theorem. These examples concern the specified densities; they say nothing about compatibility with an evolving Chern–Ricci flow.

35.3.5 Balanced and pluriclosed metrics

The balanced–pluriclosed conjecture asks whether a compact complex manifold carrying both a balanced metric β and a pluriclosed metric ω must be Kähler. The two metrics may differ; a single metric satisfying both conditions is already Kähler by Lemma 4.1.1. The source of the conjecture and its formulation through a flow are given in Section 36.22.

A manifold in Fujiki's class $\mathcal{C}$ that carries a pluriclosed metric is Kähler, by the result recalled there. It would therefore suffice to construct a closed Kähler current from the two metrics. The scalar potential equation alone cannot produce the required closedness.

Indeed, for every distributional potential u,

$$\mathrm{d}(\omega + \mathrm{dd}^{\mathrm{c}} u) = \mathrm{d}\omega.$$

Thus if the pluriclosed metric is not closed, no potential in its background makes it Kähler; the finite-volume equation, including its smooth solution, does not change this differential datum. Similarly, the Gauduchon metric in a conformal class need

not be balanced. Positivity and a scalar normalization cannot replace either missing closedness condition.

The closed form β^{n-1} pairs trivially with exact real two-forms, so it can give a conserved trace along a flow. When $c_1(X) = 0$ in real cohomology, the flow conjecture recalled in Section 36.22 proposes that, in the presence of a balanced metric, the pluriclosed flow exists for all time and converges to a Kähler metric. The invariant examples cited there use the conserved trace. For general metrics, the analytic difficulty is to turn this integral into a pointwise bound.

For a connection with parallel torsion that preserves a Hermitian metric and the complex structure, the proof of Theorem 4.7.1 obtains such a bound. Averaging over the holonomy replaces the two metrics by parallel forms, and along parallel directions of variation the trace constraint holds pointwise; the positive slice of fixed trace is then bounded, and the volume vanishes on its boundary. For a general family, the conserved balanced trace gives no such control.

Problem 35.3.4 Suppose that a compact complex manifold carries a balanced metric and a pluriclosed metric. Beyond the case of a connection with parallel torsion that preserves a Hermitian metric and the complex structure, settled by Theorem 4.7.1, use the two metrics to construct closed positive data, such as a closed Kähler current, or to control a geometric flow that can change the torsion. Determine which estimate, beyond the finite balanced trace, prevents degeneration of the metrics. Can determinant estimates for Gauduchon currents contribute to that estimate?

The complete determinant profile also need not control ellipticity. Under the lower-power dd^c-closed condition, Theorem 27.17.1 computes its exact value on every L^n density root. The nonclosed pluriclosed family of Theorem 27.17.3 keeps this entire profile, the bounded-potential mass extrema and a balanced trace fixed while its least relative eigenvalue tends to zero. Its torsion is fixed and nonzero. The underlying torus is already Kähler; the example shows which scalar data an additional estimate must go beyond, without constructing a pluriclosed-flow trajectory.

Theorem 4.7.2 provides pluriclosed metrics along a deformation of the complex structure with constant Aeppli number $h^{1,1}_A$, while the balanced metric required in Problem 35.3.4 remains a separate geometric condition.

35.3.6 Morse inequalities and the choice of volume

In the closed Kähler theory, subtracting a nef class has a controlled cost in volume. For closed nef classes α, β on a compact Kähler manifold, the transcendental Morse inequality states that

$$\mathrm{Vol}(\alpha - \beta) \geq \int_X \alpha^n - n \int_X \alpha^{n-1} \wedge \beta,$$

where Vol is the usual volume of a class. Its source, and the differentiability of the volume and the duality of cones that follow from it, are recalled in Section 36.22.

To formulate an analogue on an arbitrary compact complex manifold, one must first choose the volume to be estimated.

For nonclosed backgrounds, the intersection numbers on the right-hand side are in general not invariant under a change of representatives. Moreover, the upper volume of a class and $\mathcal{V}_\mu$ have different definitions and can behave differently with respect to finiteness. A Hermitian analogue must therefore specify which invariant it estimates.

The first-variation formula gives a finite quantity to study. Fix smooth real $(1,1)$-forms θ, η and $t > 0$. Suppose that $\theta_s = \theta - s\eta$ has a smooth positive representative for $0 \le s \le t$, and let γ_s be a minimizing Gauduchon metric for its smooth equation. By Proposition 27.1.1 and the fundamental theorem of calculus,

$$\lambda_\mu(\theta - t\eta) = \lambda_\mu(\theta) - \int_0^t \frac{\int_X \eta \wedge \gamma_s^{n-1}}{D_\mu(\gamma_s)}\, \mathrm{d}s. \tag{35.3}$$

The quotient in the integrand does not change when the minimizer is rescaled. This identity holds on the positive segment $0 \le s \le t$ and uses no bound on the upper Monge–Ampère mass. A uniform upper bound for its integrand gives a lower bound for the remaining root volume. On a compact Kähler manifold, Theorem 27.10.1 extends the identity up to a nef endpoint that has no smooth positive representative, and bounds the resulting improper integral, when a closed nef core of positive volume is supplemented by a semipositive form B. Its test metrics keep the prescribed density, and (27.44) isolates the correction involving $\mathrm{dd}^{\mathrm{c}} B$. There the endpoint is big because of the closed core; Example 27.10.1 gives a nonclosed example with an explicit strict current at the endpoint. Positive volume at the endpoint alone cannot provide such a current, by Theorem 27.13.1, so a criterion going beyond the closed core must exclude that obstruction.

For the warped products of Section 27.11, Theorem 27.11.1 computes the cost rate for every choice of minimizing metric and shows that the total subtraction cost, an improper integral, is finite; Proposition 27.11.1 gives the asymptotics of the cost rate from the codimension and vanishing order of the fiber coefficient. These paths can end with positive root volume but no big representative, and Proposition 27.11.2 excludes every closed nef positive-volume core at a zero of that coefficient. Thus exact control of the subtraction cost in this product family does not supply a general bigness criterion.

Problem 35.3.5 In the setting of (35.3), find geometric hypotheses that control the integral on the right-hand side, the cost of the subtraction, up to a nef endpoint, beyond the regime of a closed nef core treated in Theorem 27.10.1. Determine when positive root volume forces bigness in the absence of such a core. For closed backgrounds, compare the resulting estimate with the classical Morse inequalities; for nonclosed backgrounds, the estimate should keep track of the differential data and of the dependence on μ.

The singularity and divisorial constructions of this book become relevant here: they measure how much positivity can be spent on imposed poles while the residual Gauduchon volume stays finite. They do not yet give the moving intersection product

or the birational volume formula of the closed theory. For the nonanalytic ruled type of Example 27.21.1, the improper subtraction cost remains finite as the variational volume vanishes, while the upper ordinary-type volume has a positive limit. This family makes the choice of invariant explicit even for an imposed type with infinite first moment. Recovering a general moving product or birational formula remains a further goal.

35.3.7 Oeljeklaus–Toma manifolds

Oeljeklaus–Toma manifolds are not Kähler, yet they carry a closed semipositive form θ whose potentials are all continuous (Theorem 29.19.1). On a threefold of type $(2,1)$, the natural Gauduchon metric $\omega = \theta + y_1 y_2 \, \mathrm{dd}^{\mathrm{c}} |z|^2$ has an exactly quadratic mass functional, indefinite in sign, and its masses on fiber-invariant potentials are bounded above and below by positive constants (Proposition 29.19.1). The known curvature criteria for finite upper and positive lower mass in dimension three require a Hermitian metric whose dd^{c} has a sign, and these threefolds admit no such metric (Proposition 29.19.2; see Section 36.16 for the criteria). By (29.58), the remaining difficulty lies entirely in the fluctuation of a potential around its fiber average.

Problem 35.3.6 Let X be an Oeljeklaus–Toma threefold of type $(2,1)$, with the Hermitian form ω of Proposition 29.19.1. Does X have the bounded mass property, that is, is $v_+(\omega) < +\infty$? Is the lower mass $v_-(\omega)$ positive? Equivalently, writing $F(u)$ for the fluctuation term in (29.58), is F bounded above on smooth ω-plurisubharmonic functions u, and is $\inf_u \big(\mathrm{M}_\omega(Au) + F(u) \big) > 0$? The same questions can be asked for Oeljeklaus–Toma manifolds of any type (s,t) with $s \geq 2$.

By Proposition 7.5.1, the answer to the first question does not depend on the choice of Hermitian form. A negative answer would require potentials whose fluctuations make $\mathrm{Re}(\partial_{w_1} v \, \overline{\partial_{w_2} v})$ large, while keeping $\omega + \mathrm{dd}^{\mathrm{c}} u \geq 0$. By Lemma 29.19.1, every nonzero Fourier mode on the fiber tori depends on z, so for a single mode $c(y)\cos(\cdots)$ the positivity of the z-coefficient of $\omega + \mathrm{dd}^{\mathrm{c}} u$ bounds $|c|$ by a multiple of $y_1 y_2$ divided by the square of the z-frequency; the frequencies in the x-directions are not controlled pointwise. Whether compactness and the automorphy under the units bound the fluctuation term is the content of the problem. Unlike the Hopf threefold of Chapter 9, the manifold contains no curves along which heat potentials could be concentrated, by the theorem of Ornea and Verbitsky recalled before Corollary 29.19.2.

The rigidity in Theorem 29.19.1 comes from two features of the null foliation of θ: its leaves carry no nonconstant plurisubharmonic functions bounded above, and their closures are real tori on which translations act transitively. For the pullback of a Kähler form under a holomorphic fibration, the null leaves are the compact fibers, and θ-plurisubharmonic functions may have poles.

Problem 35.3.7 Let θ be a smooth closed semipositive real $(1,1)$-form on X whose kernel has constant rank. Suppose that every leaf of the null foliation of θ is Liouville,

in the sense that plurisubharmonic functions bounded above on it are constant, and that the closure of every leaf is a compact real submanifold whose real codimension equals the complex rank of θ. Is every θ-plurisubharmonic function continuous? If not, describe the possible singularity types in $\mathrm{PSH}(X,\theta)$ in terms of the closures of the leaves.

35.4 Closed and twisted classes

A volume attached to a closed Bott–Chern class must not depend on the representative, while a volume equation for a locally conformally Kähler (LCK) form must preserve its Lee class and its twisted differential. The examples below show where ordinary Hermitian perturbations fail to respect these requirements. We use the LCK notions of Definition 30.1.1: for a closed one-form λ, D_λ is the twisted Hessian of (30.6), and $j_X\colon H^1_{\mathrm{dR}}(X,\mathbb{R}) \to H^{1,1}_{\mathrm{BC}}(X,\mathbb{R})$ is the Lee map (30.5); the associated Lee–Gauduchon cone is studied in Section 30.6.

The non-nef and non-Kähler loci of Chapter 29 separate closed currents with small negative errors from strictly positive closed currents. Any numerical restricted volume proposed for closed classes on general compact complex manifolds must be compatible with Proposition 29.5.1. By that proposition, if X fails the bounded mass property, as the Hopf threefold does, then for every Hermitian form ω and every $\varepsilon > 0$ the supremum of $\int_X (T+\varepsilon\omega)^n$ over smooth closed representatives $T \geq -\varepsilon\omega$ of the zero class is already infinite. Since the failure occurs at each fixed error size, letting the error tend to zero afterwards cannot turn this expression into a class invariant.

For $n \geq 2$, the subsolution threshold also constrains divisorial singularities. Let ω be a Hermitian form, $\mathrm{d}V$ a smooth positive probability volume form and $0 \leq f \in L^1(\mathrm{d}V)$ a nonzero density with $b = b_\omega(f) > 0$, and put $p = f^{1/n}$. If a subsolution at the threshold b dominates the current of integration of an effective divisor E, then by Proposition 29.8.1 the weak limits of every smooth maximizing Gauduchon sequence annihilate the class of every component of E. If one such sequence has uniformly integrable determinant-weighted densities, Corollary 29.8.1 excludes divisors whose classes have suitable smooth semipositive representatives, for instance the hyperplane divisor on the Hopf threefold, and Corollary 29.8.2 shows that a component with nef class has zero variational volume for every smooth density. Both conclusions hold under the root-tail condition $p \in \mathcal{D}^0_\omega$. Characterizing the components that can occur would require a realization theorem; note that the weak closure of smooth tests can be smaller than the set of all normalized Gauduchon currents (Example 4.4.1).

For a prescribed logarithmic divisor with a positive Hermitian residual, Theorem 16.9.3 identifies the threshold on the entire one-sided class with that of the residual background. The ordinary class and relative envelope class have the same attained coefficient for smooth strictly positive data. A nonzero divisor then makes it strictly smaller than the global coefficient, and Corollary 16.9.1 excludes divisorial components at the smooth global threshold. The projective product in Example 16.9.1

computes this loss explicitly. These positive-residual results do not give a converse from an annihilating face for general integrable data.

For closed classes on manifolds in Fujiki's class C, Section 29.6 defines a finite restricted invariant on each subvariety, independent of the Kähler resolution used to compute it. It is positive exactly when the class is big on the resolved subvariety, and for a nef class on X the union of the subvarieties where it vanishes is the null locus. For a specified divisorial trace with bounded remainder, Section 29.7 computes the mass of the trace itself and turns positive residual mass into a strict current, more singular than the trace. The nonclosed ruled family of Section 20.10 goes beyond bounded residuals: Theorem 20.10.1 computes both masses over the whole ordinary type, and Proposition 20.10.1 identifies its ambient ideal orders, although the unbounded residual is invisible to those ideals. Positive lower volume gives a strict current in that ordinary type on the surface itself. Under every nonempty finite composition of point blowups, including blowups centered at poles, both volumes are preserved, but Proposition 20.10.3 excludes a strict current in the pulled-back ordinary type, while still allowing a one-sided strict representative. More generally, Theorem 17.7.1 gives an exact ordinary-type strictness criterion for a finite logarithmic divisor plus a residual with arbitrarily small negative curvature. Its ambient ideals and, for divisors not containing the submanifold, all intrinsic traces are given by Proposition 20.2.2, even when the naive residual restriction is identically $-\infty$. Thus a general restricted invariant must specify which singularity relation its strictness criterion uses. Its construction for arbitrary nonclosed backgrounds, and its relation to intrinsic multiplier-ideal trace data outside these families, remain open.

The LCK setting raises separate questions about equations. By Corollary 30.2.1, a compact strictly LCK manifold has no big ordinary Bott–Chern class. Its LCK form belongs instead to a twisted Bott–Chern class, whose representatives are described by the twisted Hessian D_λ. Prescribing the volume form of a Gauduchon LCK metric in such a class leads to the coupled system (30.9), whose second equation is the additional conformal condition imposed by the Gauduchon condition. Problem 30.4.1 asks whether this system has at most one solution when the Lee class and the twisted class are fixed; the nonuniqueness examples for ordinary potentials do not decide it.

On scalar Hopf manifolds the radial case is solved: for every smooth positive radial volume form and every negative Lee period, Theorem 30.4.2 gives exactly one Gauduchon metric in the full positive radial family. By Proposition 30.4.2, this metric has a positive potential in its own Lee gauge, so its twisted class is zero, and the coupled system determines it among radial candidates. For nonconstant radial densities these metrics are not Vaisman (Proposition 30.4.3). Uniqueness among arbitrary LCK candidates is open.

Two geometric conjectures use the Lee form in different ways. Conjecture 30.2.1 asserts that the class $j_X([\lambda])$, for the Lee form λ of a strictly LCK metric, is pseudoeffective, that is, pairs nonnegatively with every Gauduchon metric; the LCK metric itself provides one positive pairing. Conjecture 30.5.1 asks whether a compact complex manifold with vanishing first Bott–Chern class that admits a strictly LCK metric also admits a Vaisman metric. The rigidity theorem Theorem 30.5.1 gives

this when a single strictly LCK metric is both Chern–Ricci flat and Gauduchon; the remaining existence problem is to arrange these two conditions simultaneously.

There is a pointwise sufficient condition for the first conjecture. If the least eigenvalue of the Lee curvature κ_λ relative to the chosen strictly LCK metric is constant, Theorem 30.2.1 shows that it is zero, so that κ_λ is a smooth semipositive representative of the nonzero class $j_X([\lambda])$. In particular this class is nef, and the balanced obstruction applies; compact invariant LCK quotients satisfy the criterion. The proof uses the volume polynomial of Proposition 30.2.3. The two Hopf families show its scope: Example 30.2.2 has constant minimum but varying remaining spectrum, whereas Example 30.2.1 has indefinite Lee curvature but still a smooth semipositive representative of $j_X([\lambda])$. In the latter family the first degenerate form on the ray of Proposition 30.2.3 still has positive total volume. A general LCK metric need not have constant minimum, so the unrestricted pseudoeffectivity conjecture stays open.

The Lee–Gauduchon cone of Section 30.6 makes part of this geometry invariant under bimeromorphic maps. Its dual is the cone of classes in $H^1_{\mathrm{dR}}(X,\mathbb{R})$ whose image under j_X is pseudoeffective, and the strict inequalities defining it exclude $\ker j_X$; the duality and its behavior under modifications are proved in that section. A nonzero pseudoeffective class in the image of j_X obstructs balanced metrics on every manifold bimeromorphic to X. On the Hopf threefold, the classes in $H^1_{\mathrm{dR}}(X,\mathbb{R})$ whose image under j_X is pseudoeffective form a positive half-ray, although the Monge–Ampère masses there are unbounded. This cohomological invariant thus captures a geometric obstruction, while the nonlinear estimates for the scalar equation remain a separate task. The converse of its consequence for balanced metrics fails: the threefold of Example 30.7.1 has zero Lee–Gauduchon cone but no strongly Gauduchon metric.

35.5 Degeneration orders and finite-distance boundaries

Approximating a nef background by positive ones yields two kinds of boundary data: the rate at which the variational volume vanishes, and the distance traveled in the Hessian metric of Section 27.4. These need not detect the same boundary points. Throughout this section μ is a smooth strictly positive probability form on X.

For a nef background θ and a Hermitian form h, the lower and upper degeneration orders $\underline{a}(\theta) \le \overline{a}(\theta)$ of Proposition 27.19.1 are the lower and upper limits of $\log \mathcal{V}_\mu(\theta+\epsilon h)/\log\epsilon$ as $\epsilon \downarrow 0$. They do not depend on μ or on the perturbation direction h, but they need not coincide. On a fixed product of two elliptic curves, Proposition 27.19.5 realizes every pair $(1+\alpha, 1+\beta)$ with $0<\alpha<\beta<1$, as well as irrational common orders, and Proposition 27.19.4 gives zero boundary volume with decay slower than every positive power of ϵ. Equality of the two orders and their integrality therefore require additional geometry, as in the closed Kähler case, where the order is governed by the numerical dimension.

Pulling back by a modification can only increase both orders (Proposition 27.19.6), and a single blowup can increase them strictly, or remove positive limiting volume

while both orders remain zero. Subtracting a positive current gives the same one-sided comparison when the residual stays nef. The perturbative minimal multiplicities of a nef background vanish, whereas in Corollary 27.19.1 the boundary class contains a unique positive current, the current of integration along a divisor; it has minimal singularities, and removing it changes the order. The remaining questions, collected in Problem 27.19.1, concern equality of the orders and geometric restrictions on these behaviors.

The Hessian metric gives boundary information of a different kind. Let V be a finite-dimensional space of smooth real $(1,1)$-forms containing no nonzero form $\mathrm{dd}^{\mathrm{c}} f$ with f smooth, and let $\mathscr{C}$ be the open cone of elements of V that admit a smooth positive representative. By Proposition 27.4.1, the metric $-D^2 \log \lambda_\mu$ on $\mathscr{C}$ splits isometrically as the product of the logarithmic volume direction with the unit-volume hypersurface $\mathscr{S} = \{\lambda_\mu = 1\}$; consequently every path along which the volume tends to zero or to infinity has infinite length. Nevertheless, the blowup example of Example 27.4.1 has boundary points at finite distance, represented by nef classes of positive volume. Completing the metric cone therefore cannot mean simply adjoining all nef classes, nor declaring every nef boundary point infinitely far away.

Problem 35.5.1 Let $\mathscr{C}$ be a cone of backgrounds in a specified finite-dimensional space V as in Section 27.4, with the Hessian metric $-D^2 \log \lambda_\mu$. Beyond the cases where the completion is known, namely invariant cones on compact quotients of complex Lie groups with the invariant probability (Theorem 27.4.1) or, on the Iwasawa manifold, with a fiber-invariant density (Theorem 27.15.2), and the warped product cones of Theorem 27.4.3, describe the metric completion of its unit-volume hypersurface $\mathscr{S} = \{\theta \in \mathscr{C} : \lambda_\mu(\theta) = 1\}$. Which boundary points admit a current representative, and which estimates for the minimizing Gauduchon metrics distinguish boundary points at finite distance from those at infinite distance?

On a compact quotient of a complex Lie group, with μ the normalized invariant probability, Theorem 27.4.1 answers the question for every fixed linear space of invariant backgrounds containing a positive form: the unit-volume hypersurface is complete, and its Cauchy sequences are exactly those whose coefficient matrices converge to a positive matrix. After normalization, nonzero singular limits of the coefficients lie at infinite distance. These assertions depend on the invariant probability and do not classify arbitrary cones of backgrounds or reference densities.

On the Iwasawa manifold, Theorem 27.15.2 extends the same coefficient metric and completion to every smooth reference density that is invariant along the central elliptic fibers; these are exactly the probabilities that maximize the variational volume of each invariant positive background. The minimizing forms need not be invariant, but their coefficient means still satisfy the spectral distance bounds. A reference density that varies along the fibers gives smaller volume, and the completion of its Hessian metric is a separate question.

For the warped product cones of Section 27.4.3, on the product of a compact Kähler manifold with a curve, Theorem 27.4.3 shows that the completion of the unit-volume hypersurface is a closed strip, a closed half-plane or the whole plane. Which ends lie at finite distance is decided by two ratios of real codimension to

vanishing order, attached to the minimum set and the maximum set of the warping coefficient. Every end at finite distance is represented by a smooth semipositive, nef, nonbig form of volume one. The length estimates of Lemma 27.4.2 are bounds for the variance of the traces of the balanced minimizers; they use the product probability and extreme sets consisting of finitely many disjoint strata with power-type vanishing. Other densities and other geometries of the extreme sets remain to be studied.

The cone must be specified in Problem 35.5.1, since allowing additional directions can shorten distances along paths. The finite-dimensional calculations above do not define a distance on the whole infinite-dimensional quotient of backgrounds modulo dd^{c} of smooth functions.

35.6 Weak Gauduchon adjoints

Let θ be a background with a model reference φ and $\mathrm{d}V$ a smooth positive volume form, as in Chapter 16. For a bounded logarithmic density F, the normalized equation $\mathrm{MA}_{\theta}^{\mathrm{ext}}(u) = c_{\varphi}(F)\mathrm{e}^{F}\,\mathrm{d}V$ in the ordinary type of φ determines a unique scalar, and $B_{\varphi}(F) = \log c_{\varphi}(F)$ is a concave functional of F (Section 16.1). For smooth data, differentiating the smooth equation produces a Gauduchon adjoint probability. At a continuous logarithmic density F, the supporting probabilities of B_{φ}, which form the set $\mathcal{A}_{\varphi}(F)$ of (16.3), provide a weak replacement. In the regimes of background and reference considered in Chapter 16, $\mathcal{A}_{\varphi}(F)$ is nonempty, convex and weakly compact, and its members are absolutely continuous (Theorem 16.2.1). If this set consisted of a single probability, the scalar functional would be differentiable at F even where no classical linearized equation is available.

Conjecture 35.6.1 Let ω be a Hermitian form on X and $\mathrm{d}V$ a smooth positive volume form, and take the nonsingular reference $\varphi = 0$. For every real $F \in C(X)$, the set $\mathcal{A}_0(F)$ of supporting probabilities of the normalization functional B_0 at F consists of a single probability measure.

Equivalently, by Proposition 16.2.2, the concave functional B_0 is Gâteaux differentiable at every continuous logarithmic density. The conjecture holds at smooth F, by the linearized equation, and in the closed Kähler setting, where the normalization is explicit. By Proposition 16.2.1, it would imply that for any smooth approximations of F the Gauduchon adjoint probabilities have a unique weak limit.

A further sufficient condition is a bounded solution with continuous, everywhere positive complex Hessian. Then Theorem 16.2.4 identifies a unique stationary supporting probability and the Gâteaux derivative. Proposition 16.2.3 gives its formula on $\mathbb{P}^1$ times any compact Hermitian factor for every positive continuous density of the stated product form; Example 16.2.1 includes a non-Dini density on a non-Kähler threefold. The criterion does not prove that an arbitrary continuous density has a solution with such a Hessian.

For bounded potentials there is a direct result. Suppose that $\omega + \mathrm{dd}^{\mathrm{c}} v$ is Gauduchon for every smooth v with $\omega + \mathrm{dd}^{\mathrm{c}} v > 0$; this holds for every Gauduchon form on a com-

pact surface. Then, by Proposition 4.9.1, the normalized Monge–Ampère measures of bounded ω-plurisubharmonic functions extend the smooth adjoint probabilities and are continuous under convergence in capacity of uniformly bounded potentials. This defines no weak linearized operator, and it does not settle approximation by unbounded potentials in a general background.

A Baire category argument gives a partial result. In every regime of background and reference considered in Chapter 16, Theorem 16.2.2 shows that $\mathcal{A}_\varphi(F)$ is a single probability for F in a dense G_δ subset of $C(X)$, and that at such F all choices of supporting probabilities along uniform approximations of F have the same weak limit. The assertion is generic for the uniform topology and does not exclude exceptional continuous densities.

For $n \geq 2$, the compact formulation with currents gives a more precise target. By Theorem 27.6.1, the supporting probabilities at F are exactly the normalized determinant-weighted measures of the currents $T \in \mathscr{G}_\omega$ maximizing the functional $\mathfrak{D}_F$ of (27.23). The conjecture thus asks whether all maximizers have the same weighted measure. Corollary 27.7.2 bounds the densities of these measures in L^q, for every $1 < q < n/(n-1)$, uniformly as F ranges over a bounded subset of $C(X)$; the endpoint $q = n/(n-1)$ is included when $v_-(\omega) > 0$. Theorem 27.7.3 gives uniform L^∞ bounds for both the adjoint density and the determinant density for smooth logarithmic densities in a small $C^{2,\alpha}$ ball around the logarithmic density of the normalized volume form $\omega^n / \int_X \omega^n$ with respect to $\mathrm{d}V$, even when the lower mass vanishes. By Corollary 27.7.3, this ball contains distinct data with distinct exact maximizing currents. The controlled-Hessian family of Theorem 27.7.4 gives a larger sufficient condition: if the smooth solution forms are uniformly comparable to the background, both densities have uniform two-sided bounds depending only on that comparison and fixed geometry. The divergence of the cofactor uses only the fixed torsion, so derivatives of the varying Hessian do not enter the constants. Example 27.7.1 gives such a smooth family whose densities have unbounded C^1 norms and no common modulus of continuity. Arbitrary bounded families of continuous data remain outside these hypotheses. These estimates alone give compactness but not uniqueness at continuous data. For a fixed smooth density the maximizing current is unique, smooth and strictly positive; nevertheless, Proposition 27.17.1 shows that a sequence approaching the maximum can have unbounded determinant moments above the critical exponent. Another approach to the conjecture would be to show directly that the upper and lower directional derivatives of B_0 agree.

More is known about the functional itself. Relative inequalities between positive residual forms give uniform convergence of the whole scalar functional, by the explicit bound (16.20) of Theorem 16.8.1. Under L^1 convergence of tests with a common L^∞ bound, Corollary 16.8.1 shows that the supporting probabilities have uniformly integrable densities whose weak L^1 cluster limits lie in the limiting supporting set. The adjoint probabilities at smooth tests, together with the value of the functional at zero, determine the whole functional by (16.21). None of these statements decides the conjectured uniqueness at continuous tests.

For a singular reference or a degenerate background, the same uniqueness question makes sense under the hypotheses of Chapter 16. A geometric interpretation of the

supporting probabilities, however, requires a well-defined weak linearization. For a smooth background θ and $u \in \mathrm{PSH}(X,\theta)$, write $\theta_u = \theta + \mathrm{dd}^{\mathrm{c}}u$, and let ρ be a measurable weight. The expression $\mathrm{dd}^{\mathrm{c}}(\rho\theta_u^{n-1})$ is not a definition until multiplication of the product θ_u^{n-1} by ρ has been justified.

An explicit divisorial model shows what can go wrong. For the admissible reference of Lemma 16.2.2, the supporting set is a single probability and the residual linearization is well defined (Proposition 16.2.4). Nevertheless, Theorem 16.2.5 gives decreasing smooth approximations, with positive forms, whose classical adjoint probabilities have different weak limits, each carrying mass one half on the divisor. These potentials converge in capacity and their logarithmic densities converge in every finite L^p, but their exponential densities concentrate and their smooth mass exceeds the relative mass. Within this fixed divisorial type, Corollary 16.2.1 proves recovery, with an explicit estimate, under convergence in capacity and a uniform bound on the residual potentials. On the non-Gauduchon product surface of Section 16.13.4, Theorem 16.13.2 constructs a weak linearization for a logarithmic solution with continuous positive residual and identifies its full supporting set as a singleton. Uniform convergence of the positive residual forms, with the divisor fixed, gives convergence of both the weighted currents and their probabilities in Proposition 16.13.2; pole-filling smoothings still keep unwanted mass on the divisor (Example 16.13.1). For general singular references, adjoints that do not depend on the approximation still require control of the singularity type, of the mass and of the conformal multiplier.

35.7 Topologies on singularity types

A topology on singularity types must keep information that weak convergence of currents can lose. Deficits of upper volume achieve this when bounded mass and positive lower volume provide uniform control, and on curves polar curvature gives a complete description even at zero volume. The problem is to extend such control to degenerations in higher dimension and to backgrounds with unbounded mass. Throughout this section, for a smooth real $(1,1)$-form θ and $u \in \mathrm{PSH}(X,\theta)$, $P_\theta[u]$ is the ordinary envelope of Definition 1.19.1 and $P_\theta[u]_{\mathcal{I}}$ the multiplier-ideal envelope of Definition 17.2.1. An ordinary model is a potential u with $P_\theta[u] = u$, an ideal model is one with $P_\theta[u]_{\mathcal{I}} = u$, and an analytic model is an ordinary model with analytic singularities. The lower and upper singularity volumes $\underline{\mathrm{vol}}_\theta(u) \le \overline{\mathrm{vol}}_\theta(u)$ are the infimum and the supremum of the non-pluripolar masses over the ordinary singularity type of u (Definition 14.3.1).

35.7.1 Weak convergence and singularity data

Weak convergence alone keeps little singularity information. The complete Gauduchon trace metric of Corollary 6.2.1 induces the L^1 topology on normalized potentials, and the ideal metric of Proposition 17.4.1 is the L^1 distance between the canonical ideal representatives $P_\theta[u]_{\mathcal{I}}$. The completion of the ideal metric can acquire points that are not ideal envelopes of their own classes: on a curve, Theorem 17.4.1 approximates every normalized potential by ideal envelopes with finitely many logarithmic poles and zero non-pluripolar mass. In particular, such envelopes can converge to a smooth potential with strictly positive curvature, but not in capacity, since damped continuity would then preserve their zero measures.

A positive lower bound on the mass does not repair this. In Proposition 17.4.2, analytic ideal envelopes of one fixed positive mass converge in the ideal metric to the smooth model 0, whose mass is larger. Nor can the normalization functional recover the lost data: on the closed model domain of Theorem 16.10.1, it records exactly the mass. The examples with positive residual in Examples 16.10.1 and 16.10.2 have identical functionals and supporting sets but different poles, different multiplier ideals and different limits of normalized potentials, also for nonclosed backgrounds.

An order bound gives more control than weak convergence. Let h be a Hermitian form, let $\epsilon_j \geq 0$ tend to zero, and let $a_j \in \mathrm{PSH}(X, \theta + \epsilon_j h)$ satisfy the one-sided bound $u \preceq_{\mathcal{I}} a_j$ in the ideal order of Section 1.12, that is, $\mathcal{I}(ku) \subseteq \mathcal{I}(ka_j)$ for all $k > 0$. Then Proposition 17.5.1 identifies the limit of the ideal envelopes from the convergence of divisorial Lelong numbers, and it shows that along such ordered analytic approximations, including dyadic regularization, the ordinary envelopes converge to $P_\theta[u]$ exactly when $P_\theta[u] = P_\theta[u]_{\mathcal{I}}$. Inclusions of multiplier ideals at fixed weights, as in Corollary 17.5.1, give the required convergence of divisorial numbers. Without the order bound, L^1 convergence together with convergence of every divisorial number is not enough, by Example 17.5.1. These tests make sense even when the upper volume is infinite. In contrast, in a Hermitian background analytic classes are dense for the ideal metric in the whole space (Corollary 17.5.3), so density for that metric cannot characterize equality of the envelopes.

The finite damped measures of Chapter 8 control nonlinear limits throughout the space of potentials, including potentials of infinite mass. Since their weak continuity already follows from convergence in capacity, requiring it in addition imposes no further condition on multiplier ideals.

For a Hermitian form ω, the uniform topology $\tau_{\omega,0}$ of Section 17.6 on ordinary singularity types combines both envelopes with controlled inclusions of multiplier ideals and the total variation of divisorial orders on each fixed compact modification. By Theorem 17.6.1, the closure and the sequential closure of the analytic classes are both the set of ordinary classes with $P_\omega[u] = P_\omega[u]_{\mathcal{I}}$. This characterizes the analytic closure in every Hermitian background, including classes represented by potentials of infinite mass. The topology is not T_0 on ordinary classes. Transport by a smooth gauge change of the background is exact when the obstacle changes with the background, but with the zero obstacle kept fixed it is discontinuous already on curves, even within a cutoff of positive mass (Proposition 17.6.4).

The uniformity of Section 17.6.4, generated by the envelopes $P_{\omega,F}[u]$ and $P_{\omega,F}[u]_I$ below all smooth obstacles F (Definition 1.19.1), is invariant under smooth changes of background, and its analytic closure consists of the classes for which the two envelopes agree below every smooth obstacle (Theorem 17.6.2). On curves this is the same analytic locus, but by Proposition 31.13.4 the topology is incomplete and strictly weaker than polar total variation. In higher dimension, membership of u in the relative envelope class $\mathcal{E}_{\mathrm{env}}(X,\omega,P_\omega[u]_I)$ suffices for equality below every obstacle (Proposition 17.6.6), and under bounded mass and positive lower volume of the ideal envelope, equality at the zero obstacle is already sufficient (Proposition 31.8.1).

On every Hermitian background, Theorem 17.6.3 gives explicit potentials in this sufficient relative envelope class. Their poles may contain a prescribed countable dense set, while their multiplier ideals are trivial at every positive weight. By Proposition 17.6.7, both envelopes equal the unrestricted envelope below every continuous obstacle. Membership in the relative envelope class comes from a fixed potential with stronger poles that the construction provides; it is not deduced from equality below the zero obstacle.

Several geometric cases are known without the volume condition. For every admissible smooth background on a curve, Corollary 31.14.1 shows that equality at the zero obstacle, equality at all smooth obstacles and membership in the relative envelope class of the ideal envelope are equivalent. On products with separated backgrounds, Proposition 32.3.1 computes both envelopes coordinatewise below separated smooth obstacles, also for curve backgrounds of zero area, and Proposition 32.3.6 identifies equality at the zero obstacle with equality at every smooth obstacle, including obstacles that mix the coordinates and factors of zero area. This equivalence persists after adding a unit divisor and an arbitrary compact complex auxiliary factor with a semipositive vertical term, for instance a curve with a nonclosed coefficient, without any mass bound on the ambient manifold (Corollary 32.3.3). For divisorial fibrations over curves, Corollary 32.1.2 gives the same equivalence at zero model volume, even with poles along whole fibers or with disconnected fibers, by reducing the obstacles to the cover formed by the fiber components. In Example 32.1.3 the ambient fourfold fails bounded mass, while the base potential belongs to the relative envelope class required for equality at all obstacles. Whether equality at the zero obstacle implies equality at all obstacles at zero lower volume in general, including backgrounds with unbounded mass, is unresolved.

35.7.2 Volume comparison and analytic approximation

In this subsection X has bounded mass and θ is a smooth big background. For $u,v\in \mathrm{PSH}(X,\theta)$, the comparison $d_\theta(u,v)$ of Definition 31.6.1 is the sum of the deficits of the upper volumes of u and v relative to the upper volume of $\max(u,v)$, and $\mathcal{M}_\delta(\theta)$ denotes the set of ordinary models with lower volume at least $\delta>0$. A common positive lower bound for the volumes converts a small deficit into a stretching

estimate. The resulting quantitative L^1 bound, together with a modulus for three points that replaces the triangle inequality, gives separation and Cauchy completeness on $\mathcal{M}_\delta(\theta)$ (Theorem 31.6.1); the cutoff must be closed, as Example 31.6.1 shows. The positive lower bound is also what prevents the comparison from identifying distinct ordinary models: Example 32.2.4 gives two distinct models at zero lower volume with comparison zero, in positive closed backgrounds in every dimension at least two and in positive nonclosed backgrounds in every dimension at least three. Rooftop continuity gives monotone lower bounds, which yield convergence in capacity (Proposition 31.7.1 and Theorem 31.7.1).

Analytic approximation detects exactly the ideal models. Theorem 31.8.1 corrects analytic regularizations so that they stay in the fixed background; their ordinary models converge for d_θ to the ideal envelope, and their comparison with the original potential tends to the excess of the upper volume of its ideal envelope over its own upper volume. Conversely, at positive lower volume, Theorem 31.8.2 shows that ideal models are closed under this convergence. Although d_θ fails the triangle inequality even on analytic models on a projective manifold, Theorem 31.12.1 constructs a complete metric on the ordinary models of positive lower volume with the same convergent sequences; its Cauchy criterion also controls the reciprocals of the lower volumes, which excludes approach to zero volume. By Corollary 31.12.1, the ideal models form a closed complete subspace, which is the completion of the analytic models of positive volume.

Monotone limits show where positivity enters. For decreasing ordinary models in backgrounds that decrease uniformly to a smooth big form, Theorem 31.5.2 gives continuity of the lower volume even when the limiting volume is zero; if the limiting lower volume is positive, the upper volumes and the Lelong numbers on every fixed modification are continuous as well. For increasing sequences in a fixed background, Theorem 31.5.1 gives continuity of the ordinary envelopes and of both volumes under the same positivity condition. For arbitrary decreasing potentials whose limit has positive lower volume, Proposition 31.5.1 shows that the ordinary envelopes converge exactly when the upper volumes do. Likewise, when the ideal envelope has positive lower volume, Proposition 31.8.1 shows that the ordinary and ideal envelopes of a potential agree exactly when the potential and its ideal envelope have the same upper volume.

There is an explicit divisorial positivity test on $\mathbb{P}^2$. Let κ be the normalized Fubini–Study form, prescribe distinct hyperplanes H_j with summable coefficients $a_j > 0$, and put $A = \sum_j a_j$. If $\mathrm{dd}^{\mathrm{c}}\theta = 0$ and $\theta - A\kappa > 0$, equality with the profile $\sum_j a_j \operatorname{ord}_E(\pi^* H_j)$ on every smooth compact modification specifies one ideal type. By Proposition 31.8.2, its ideal envelope is ordinarily equivalent to the logarithmic series, and every representative of this ordinary type has mass $\int_X (\theta - A\kappa)^2$. The nonclosed example of Example 31.8.1 has infinitely many hyperplanes outside one pencil and positive lower volume. General positivity criteria from restricted ideals or divisorial orders, and the remaining zero-volume implications in Problem 31.8.1, remain open.

The comparison is compatible with changes of background as long as these controls persist. A smooth gauge change, followed by ordinary projection, preserves it

exactly (Proposition 31.10.1). For a fixed potential and backgrounds decreasing to θ, Corollary 31.5.1 gives continuity of the lower volume and, at positive lower volume, of the ordinary envelope. The uniform estimate (31.12) allows the curvature errors of analytic regularization to tend to zero, as in Theorem 31.10.1. For perturbations of the background in both directions, a convex correction produces admissible transported models, with a uniform comparison whose limiting notion of convergence does not depend on the chosen strict reference; this gives the analytic criterion of Theorems 31.11.1 and 31.11.2. None of these arguments covers loss of bigness or the boundary at zero volume in higher dimension.

35.7.3 Polar curvature

Let Y be a compact connected Riemann surface with a smooth positive form θ. For $u \in \mathrm{PSH}(Y,\theta)$, the polar curvature $\mu_u = \mathbf{1}_{\{u=-\infty\}}(\theta + \mathrm{dd}^{\mathrm{c}}u)$ determines the ordinary model of u, and its atomic part determines the ideal model. By Theorem 31.13.1, $d_\theta(u,v)$ is the total variation of $\mu_u - \mu_v$, also at zero non-pluripolar mass. By Corollary 31.13.1, all normalized ordinary models form a complete geodesic space, and the ideal models form the closed subspace obtained by completing the analytic models of positive mass. On ideal models of positive mass this agrees with the topology of the source metric in Theorem 20.5.1 and Corollary 20.5.1, for which this space is not totally bounded. Its completeness therefore has a different content from the compact completion obtained from weak convergence of traces.

Keeping only the atomic part of the polar curvature is a 1-Lipschitz retraction onto the ideal models and selects the unique nearest ideal model; by Proposition 31.13.1, the distance to the analytic closure is exactly the mass of the diffuse part of the polar curvature. The model of zero mass in Example 31.13.2 has trivial multiplier ideals but positive distance from that closure. Convergence in total variation implies convergence in L^1 and in capacity, even at zero mass (Theorem 31.13.2), whereas the moving pole of Example 31.13.1 shows that convergence in capacity does not imply convergence in total variation.

Degeneration of the background is also explicit on curves. Between smooth backgrounds of positive area, Theorem 31.14.1 gives correspondences of the model spaces that rescale by the ratio of the areas and are compatible with ideal projection, also at models of zero mass. Under uniform convergence of the backgrounds, the corresponding normalized representatives vary uniformly in L^1 and in capacity (Corollary 31.14.2). When the limiting area is zero, Theorem 1.8.1 shows that all normalized potentials converge to the unique limiting potential, uniformly over the admissible families, and the unscaled diameters of the model spaces tend to zero.

The joint space of Theorem 31.15.1 treats the smooth background as a further coordinate. It is complete for the combination of smooth convergence of backgrounds and total variation of polar curvature, its analytic closure is exactly the ideal locus, even at zero area, and on each fixed background it restricts to the curve topology

described above. In higher dimension, the realization of compatible singularity data and a replacement for the comparison of finite upper volumes are unresolved.

Finite curve maps give another compatibility within the curve theory. For a finite surjective holomorphic map $r\colon C' \to C$ of degree d between connected compact Riemann surfaces, a smooth base form β of nonnegative area and $\sigma = r^*\beta$, Proposition 31.14.3 commutes both envelopes with pullback below every pulled-back smooth obstacle. The polar metric scales by d, and Proposition 31.14.4 gives a 1-Lipschitz retraction onto the pulled-back models, which replaces the polar measure μ of a model on C' by $d^{-1}r^*r_*\mu$, preserves polar mass and commutes with ideal projection. For a surjective holomorphic map p from a connected compact complex manifold of dimension at least two onto C, with component factorization $p = r \circ q$, Corollary 32.4.2 uses this retraction to describe the closed complete subspace of models for $p^*\beta$ that are constant on the fibers of p. The cubic example Example 31.14.1 shows that this averaging need not select a nearest model. These conclusions do not supply a complete topology for general higher-rank backgrounds.

In dimension at least two, polar curvature on modifications loses information (Sections 32.5.1 to 32.5.3). The polar part $\mathbf{1}_{\{u=-\infty\}}(\omega + \mathrm{dd}^{\mathrm{c}}u)$ of the curvature current vanishes for distinct analytic ideal models with isolated logarithmic poles (Proposition 32.5.1); the blowup at the pole detects the coefficient, but every fixed countable family of modifications misses some such models (Corollaries 32.5.1 and 32.5.2). Normalizing the polar traces on a family of modifications by Gauduchon forms gives the comparison $\delta^{\mathrm{pol}}_{\omega}$ of (32.13). It separates analytic models, recovers the curve topology up to a constant factor and is preserved by smooth gauge transport; its space of trace coordinates is complete, and its Cauchy sequences have compatible weak limits of polar currents (Propositions 32.5.2 to 32.5.4 and Corollary 32.5.3). Whether these limits are realized by ordinary models is open, and independent choices of the Gauduchon forms can fail to induce the positive-volume topology, even for analytic ideal models on $\mathbb{P}^2$ (Proposition 32.5.5).

Even all modifications together do not suffice. If $0 < v_-(\omega) \le v_+(\omega) < +\infty$, for instance for every Kähler form and for every Hermitian form on a surface, Proposition 32.5.7 gives nonideal ordinary models of positive lower volume, with trivial multiplier ideals and first polar currents vanishing on every modification, at positive distance from the zero model; hence no choice of Gauduchon forms makes the polar comparison separate them, and the closure of the analytic models for the polar comparison contains nonideal models. Under the same mass conditions Corollary 32.5.6 gives a complete ordered interval of such models with strictly decreasing upper volume. If $\mathrm{dd}^{\mathrm{c}}(\omega^q) = 0$ for $1 \le q < n$, all representatives have the same mass, and Corollary 32.5.7 computes the interval exactly: volumes $A - Mt^n$ and mutual distances $M|s^n - t^n|$, where $A = \int_X \omega^n$ and $M > 0$ is a constant of the construction. In a general Hermitian background, Proposition 32.5.6 separates the loss Mt^n in the mass of every representative from lower-degree terms coming from the background; on a surface the mass is constant over the bounded-equivalence class exactly when ω is Gauduchon (Corollary 32.5.5).

The location and the shape of the residual mass also matter. Under both mass properties, Proposition 32.5.8 gives two models carrying residual mass r in disjoint

regions, with zero first polar data, at distance at least $2r$; when $\mathrm{dd}^{\mathrm{c}}(\omega^q) = 0$ for $1 \le q < n$, their two singularity volumes also coincide. Under this last condition, Proposition 32.5.9 gives complete families of models of real dimension $n-1$ with the same residual measure, the same two singularity volumes, trivial multiplier ideals and zero first polar currents on every modification, which the comparison distance nevertheless separates. A complete extension must therefore detect singularity information lost even by the full residual measure.

On the curve Y with background θ, the ordinary-type topology $\tau_{\theta,0}$ restricted to ideal models is the topology of total variation of the atomic data, and its canonical metric includes the boundary of zero mass (Theorem 31.13.3). On general ordinary models it is weaker than polar total variation: diffuse polar curvature can concentrate into an atom in the limit of the envelopes, which produces the Cauchy sequence without an ordinary-type limit in Proposition 31.13.2. Characterizing the analytic closure on the whole ordinary space and extending the model topology to a complete one are therefore separate requirements.

Problem 35.7.1 For a smooth big background θ on a manifold with bounded mass, Theorem 31.12.1 gives a complete metrizable topology on the ordinary models of positive lower volume, in which the closure of the analytic models is exactly the set of ideal models. Extend this topology to ordinary models beyond bounded mass and positive lower volume, keeping the closure of the analytic models exactly the set of ideal models. The extension should keep the compatibility with smooth gauge changes and with two-sided perturbations of the background established in Sections 31.10 and 31.11, and it should treat limits of backgrounds that lose bigness or a uniform positive lower bound for the volume. On curves it should recover the complete topology of polar total variation of Corollary 31.13.1 on all models. Determine whether the extension in higher dimension is completely metrizable.

Any such extension must keep the singularity information that weak limits lose. Potentials of infinite mass are already covered by $\tau_{\omega,0}$: for instance, the potential of Proposition 17.3.3 has infinite mass, while its canonical envelopes, the ordinary and the ideal one, both equal zero. On curves, zero volume does not prevent completeness, but in higher dimension the stretching argument still needs a positive lower bound for the volume. There is nevertheless a restricted locus where a complete metric is known at zero volume. On the divisorial product locus of Proposition 32.2.2, polar curvature on the curve factor gives a complete metric whose analytic closure is exactly the set of ideal models; the top-degree volume comparison vanishes identically on this locus and therefore cannot produce this metric. Proposition 32.3.2 gives a complete sum metric, with the same properties of ideal projection and analytic closure, on the locus induced by one or several independent curve models, and Corollary 32.3.1 preserves this locus and metric in a positive nonclosed background on a projective manifold and in its nonbig limit when the auxiliary curve coefficient vanishes. When the curve coefficients vary, Proposition 32.3.3 gives compatible weighted metrics, discarding vanishing coordinates, with uniform L^1 collapse when all curve areas vanish, and Proposition 32.3.4 compares these metrics quantitatively across different sets of active coordinates, with the exact distance for ordered coefficients. Proposition 32.3.5 and Corollary 32.3.2 extend the construction to arbitrary smooth admissible curve backgrounds and identify the limiting singleton, and Proposition 32.3.7 preserves the

restricted complete metric through arbitrary compact complex auxiliary factors with semipositive vertical terms, independently of bounded mass on the ambient manifold.

35.8 Trace realization and recovery

Prescribing divisorial lower bounds and taking an envelope may produce larger orders than those prescribed. Let u be a quasi-plurisubharmonic function on X and $Y \subset X$ a connected compact submanifold of positive dimension along which u has zero generic Lelong number, $\nu(u, Y) = 0$. As in Chapter 20, the trace data $\tau_{Y,u}(E)$ of (20.1) are the asymptotic vanishing orders of the restrictions to Y of the multiplier ideals $\mathcal{I}(tu)$, along prime divisors E on smooth modifications $\pi\colon Z \to Y$. For a smooth real $(1,1)$-form Ω on Y, the trace envelope $\mathcal{T}_\Omega(u; Y)$ of (20.5) is the upper envelope of the nonpositive Ω-plurisubharmonic functions a with $\nu(a \circ \pi, E) \geq \tau_{Y,u}(E)$ for all such E. The realization problem asks when these inequalities are equalities. Theorem 20.6.1 constructs the envelope in the restricted background itself and identifies it as the L^1 limit of the envelopes of the restricted analytic approximants, and Theorem 20.6.2 shows that it pulls back exactly to every compact modification and is there the limit of the envelopes for vanishing Hermitian perturbations. When Y has bounded mass and carries a Hermitian form of positive lower mass, Corollary 31.5.2 proves equality in all divisorial inequalities for sufficiently large backgrounds Ω; when Y is Kähler, equality also follows directly from the closed theory (Theorem 20.6.3). Ambient potentials with analytic singularities have exact ordinary restrictions. In higher dimension, Theorem 20.7.1 realizes all orders, without any mass property, when the data are pulled back from a curve (Section 20.7), with an ordinary type independent of a sufficiently large background. For general data, Theorem 20.6.4 shows that in a larger Hermitian background the envelope has positive finite mass.

Problem 35.8.1 Let u and Y be as above, and suppose that Y fails bounded mass or carries no Hermitian form of positive lower mass. Can the Hermitian background Ω on Y in Theorem 20.6.4 be chosen so that

$$\nu\big(\mathcal{T}_\Omega(u; Y) \circ \pi, E\big) = \tau_{Y,u}(E)$$

for every prime divisor E on every smooth modification π of Y? If so, determine an ordinary or ideal singularity class of this realization that is compatible with changes of background, and prove its addition and approximation properties.

The equality is required on all modifications, which is stronger than realizing the divisorial part of the data on one model. For that part, Corollary 1.17.1 gives a canonical ordinary type by combining a Gauduchon bound for the mass of the divisorial current with a smooth decomposition of currents; it does not determine the trace data of higher codimension, which appear on further modifications. The exceptional divisor of Theorem 24.8.1 describes the discrepancy between the canonical types on two models, together with a cocycle relation and approximation in linear

mass. By Example 24.8.1, a nonzero exceptional term is compatible with an exact trace potential: the term records information absent from the divisorial part on the original model. Enlarging the background provides a competitor with controlled curvature, but it does not automatically exclude additional divisorial orders. In the proof for curve fibrations, every order on a further model reduces to a multiplicity in one variable. Under the two mass properties, Theorem 31.5.2 controls Lelong numbers through the deficit of upper volume and a positive lower volume. The missing step is to extend such control beyond these mass conditions.

Even exact realization does not control the mass of all representatives of the ideal class. In Proposition 20.9.2, on a product with the Hopf threefold, the realizing potential has finite extended mass, while another potential with the same multiplier ideals and the same trace numbers has infinite mass. A proposed characterization by volumes or by continuity must specify which representative, or which class of masses, it uses.

Theorem 20.8.1 answers the realization question for the Hopf pair (H_{n+1}, H_n), $n \geq 3$, when the ambient potential is pulled back from $\mathbb{P}^n$ and has zero generic Lelong number along the projective hyperplane. Although H_n does not have the bounded mass property (Corollary 9.6.1), the same pulled-back potential realizes the trace data on all compact models. Within the invariant backgrounds $a\alpha + b\beta$, Corollary 20.8.1 and Proposition 20.8.2 give an ideal class compatible with changes of background, addition, positive scaling and decreasing analytic approximation. Exactness persists in every smooth background $\Omega \geq a\alpha$, with a as in that theorem. This settles the projective pullback case; general ambient data and background compatibility outside this family remain part of the problem.

35.8.1 Recovery of compatible divisor systems

Prescribing finitely many divisorial poles leaves a smooth residual background. When the residual has a smooth positive representative, Theorem 27.21.1 solves the residual equation for a smooth strictly positive probability measure, and every subsolution at the same scalar whose Lelong numbers along the prescribed divisors are at least the prescribed coefficients differs from this solution by a constant. Consequently a solution in the prescribed ordinary type has the same scalar and a smooth remainder. Passing to compatible data on all modifications requires keeping this positivity and this exact singularity information uniformly.

Passing to all modifications raises two further issues. First, a smooth positive probability measure pulls back to a density that vanishes on the exceptional divisor, and the smooth dual formula cannot be applied to such a density until a degenerate extension has been proved. Second, the residual classes on two models are related explicitly: Corollary 25.2.1 subtracts the Bott–Chern class of the effective exceptional divisor while keeping the prescribed exterior derivative. The approximating coefficients converge in linear trace mass, but this formula gives no transformation rule for the nonlinear volume. Exact pullback preserves singularity volumes by Proposi-

tion 14.3.1; subtracting an exceptional divisorial class changes the background and is a different operation. Assigning an intersection number to a nonclosed form resolves neither the degenerate equation nor the question of volume.

The envelope systems of Chapter 26 are compatible families, over the positive Hermitian perturbations of a background, that record representatives of one multiplier-ideal class (Definition 26.2.1). Their construction gives an order adjunction and an exact criterion for recovery: the limit as the perturbation vanishes must keep the prescribed ideal type. Under bounded mass, for a big background, Theorem 31.5.3 gives a sufficient condition involving only the initial slices: their lower singularity volumes must have a positive limit as the perturbation vanishes. Concavity then gives positive lower volume at every finite parameter, and stretching preserves all local divisorial orders. The ruled-surface example of that chapter has zero residual volume and shows why recovery fails without such a condition.

Problem 35.8.2 Let X fail the bounded mass property, and let θ be a smooth big background on X. Find a condition on a compatible Hermitian envelope system over θ (Definition 26.2.1) that forces recovery of its ideal type at every finite parameter, and that reduces to the criterion of Theorem 31.5.3, a positive limiting lower volume of the initial slices, when bounded mass holds. Specify the invariant and the uniform estimates that replace the finite deficit of upper volume, and describe their behavior under modifications.

On the scalar Hopf threefold, Theorem 31.9.1 gives a sufficient condition despite unbounded ambient upper mass. If the scalar components of the system are invariant under the elliptic fiber action, and the initial member of its transverse limiting curve has positive lower volume on $\mathbb{P}^2$, the system is recovered at every finite parameter and every positive Hermitian perturbation. The symmetry constructs the full transverse system rather than assuming survival of the limiting ideals. Example 31.9.1 gives nonanalytic ideal data with exact positive transverse volumes. The recovery identity and transverse volumes survive modifications in the exact pulled backgrounds, and smooth gauge transport preserves recovery. A criterion for general nonsymmetric systems and arbitrary new Hermitian backgrounds on modifications remains unresolved.

For a fixed finite divisorial support, Proposition 27.21.2 settles the case where the residual stays in the positive cone. If the coefficients converge and the limiting residual background still has a smooth positive representative, then the normalized smooth remainders converge smoothly, the singular solutions converge in capacity, their measures converge in total variation, and their logarithmic coefficients remain exact on every modification. Multiplier ideals can nevertheless jump at an integer coefficient. For divisors of sections moving in fixed finite-dimensional spaces, positive limiting weights and residuals give convergence of the full potentials (Theorem 16.9.2), in each of the two regimes stated there. Recovery beyond the regime of bounded mass and positive volume, degeneration of residual positivity, and estimates uniform over infinitely many models all require further arguments.

35.9 Sections, kernels and the dimension of growth

A Hermitian form is the curvature of a holomorphic line bundle only if it is closed, so the algebraic part of the theory must keep the closed curvature data separate from the Hermitian background. The weighted spaces of sections of Chapter 21, their multiplicative variant and their intrinsic flag bodies do this, and their dimension of growth can be strictly smaller than the complex dimension of the manifold.

For the partial Bergman kernels of Chapter 28, the upper bound by the envelope is proved in general; a matching lower bound is known in the explicit regimes below, and the general problem is open. Let $L \to X$ be a holomorphic line bundle with a smooth Hermitian metric of curvature θ. If X admits a modification from a compact Kähler manifold and $\varphi \in \mathrm{PSH}(X, \theta)$ has positive non-pluripolar mass, Theorem 28.6.1 proves convergence of the kernel measures normalized in full dimension. The proof keeps track of the relative canonical bundle, whose Jacobian factor corrects both the integrability condition of the ideals and the Hilbert norm. This geometric assumption implies the bounded mass property, although the birational transformation identities themselves do not require it. On a holomorphic submersion with descended data, the equilibrium theorem on the base gives a different exact limit on the total space. On the Hopf n-fold, the dimension of the selected space of sections of the mth tensor power grows like m^{n-1}, so normalizing by m^n would erase the limit; adding the vertical Hermitian form recovers a nonzero measure, with the factor computed in (28.15).

Isolated point poles that do not descend are treated in Theorem 28.8.1: their projected jet conditions differ from the multiplier ideals on the base by only finitely many orders. For a single point pole on the Hopf manifold, Theorem 28.9.1 proves the lower asymptotics of the potentials as well as convergence of the kernel measures in total variation. Moving the pole along its fiber leaves every selected space unchanged, so these limits recover the projected equilibrium, while the multiplier-ideal type of the original Hermitian weight can be different.

Theorem 28.11.1 gives a further regime: a logarithmic pole along a section over a hyperplane, with arbitrary compact connected fibers of positive dimension. The growth degree is the dimension of the base. The partial potentials attain the domination envelope of the original singularity on the total space, and the kernel measures converge in total variation for every smooth positive volume used in the norm. The limit on the base is determined by the metric of the line bundle, and its lift keeps exactly the conditional probabilities on the fibers.

For an interior rational Newton polytope with integral affine offset, Theorems 28.16.1 and 28.16.2 prove a lower bound by the envelope and a positive limit of the measures in the relative growth dimension. Here the smooth weight and the sampling probability are invariant under the torus, the logarithmic weight is strictly convex, and the target is the specified envelope of Newton growth. In lower dimension the limit lies on a real moment shell and differs from every mixed measure formed from the curvature of the envelope and a smooth Hermitian form on the ambient manifold. These results do not identify an arbitrary multiplier-ideal envelope.

The multiplier-ideal family of Theorem 28.17.1 has positive growth exponent and potentials converging to its ideal envelope, yet its kernel measures have two different

limits: rounding at two critical divisors makes the leading term of the section count depend on the parity of the degree. Convergence of the normalized dimensions is therefore an additional necessary condition for a finite limit of the measures. This selecting weight has zero top non-pluripolar mass, so it lies outside the positive-mass regime above.

For the two isolated base points of a pencil, Theorems 28.18.1 and 28.18.2 identify the full ideal envelope and prove a global lower bound for the potentials and a positive limit of the measures. The selected spaces consist of three neighboring bands of monomials and have dimension exactly $3m + 1$, although their class of competitors at all weights equals the upper ordinary type class. The specified compact logarithmic box and Gaussian norm give intrinsic mass three. Locally uniform convergence away from the poles, convergence in every finite L^p and uniform convergence after exponentiation coexist with an unbounded additive error at each fixed degree. These spaces are not multiplicative, and this selecting weight also has zero top non-pluripolar mass.

Problem 35.9.1 Let $L \to X$ be a holomorphic line bundle with a smooth Hermitian metric, and consider a series of selected spaces of sections of L^m with weighted L^2 norms, as in Section 28.1, together with its intrinsic growth dimension. Identify hypotheses on the series under which the partial Fubini–Study potentials attain the lower bound given by the envelope and the kernel measures, normalized in the growth dimension, converge. Relate the limiting measure to the Hermitian background while keeping track of the closed curvature and of the growth dimension of the series.

Beyond the regimes treated above (isolated points, horizontal projective poles, descended data, interior Newton polytopes and the pencil example), the problem involves two tasks: producing enough sections with the required ideal constraints, and controlling the resulting measures. A Bergman upper bound gives neither. On a manifold that is not Moishezon, normalization by the ambient degree may already force every kernel measure to vanish, for reasons independent of singularity theory.

Locality of Hölder regularity for extremal functions of compact sets, and regularity of weighted extremal functions under a uniform density condition in capacity, give special cases of the regularity question for envelopes; see Section 36.21. These hypotheses do not cover every singular envelope, and they do not settle the lower-bound problem of Problem 35.9.1.

On a fibration, the comparison of spaces of finite codimension in Lemma 28.8.1 suggests a more specific question, which asks for an analytic description of the constraints on sections after descent.

Problem 35.9.2 Let $\pi\colon X \to B$ be a holomorphic submersion of connected compact complex manifolds with connected fibers, where B is Kähler of dimension $q \geq 1$. Fix a Hermitian form ω on X and $\varphi \in \mathrm{PSH}(X,\omega)$, and a holomorphic line bundle A on B with a smooth Hermitian metric h. Set $\theta = c_1(A,h)$. For positive integers m, identify

$$V_m = H^0\big(X, \pi^* A^m \otimes \mathcal{I}(m\varphi)\big) \subseteq H^0(B, A^m)$$

by holomorphic descent. Find geometric or analytic conditions that produce $\psi \in \mathrm{PSH}(B,\theta)$ of positive non-pluripolar mass such that, for all sufficiently large m,

$$W_m = H^0\big(B, A^m \otimes \mathcal{I}(m\psi)\big) \subseteq V_m,$$
$$\dim V_m - \dim W_m = o(m^q).$$

Determine which features of the singularities of φ along the fibers suffice for this approximation.

For descended continuous weights and probability norms, such an approximation, combined with Lemma 28.8.1 and the equilibrium theorem on the base, would give the limits of the kernel measures and of the determinant functionals, although not by itself convergence of the potentials. For finitely many isolated poles, Theorem 28.8.1 gives the stronger error $O(m^{q-1})$ under its hypothesis on the projected poles. Theorem 28.10.1 gives the same error for a logarithmic singularity along a section over a smooth divisor of the base, under the curvature margin stated there; its base reference is explicitly divisorial, and Example 28.10.1 shows that the order of the error is sharp. The problem does not presuppose that every selecting weight in $\mathrm{PSH}(X,\omega)$ admits such a base reference.

The band criterion of Theorem 28.10.2, which places V_m between two spaces of sections vanishing along divisors with simple normal crossings, requires a global logarithmic lower bound for φ, a semipositive residual with positive top integral on the base after a smooth gauge change, and a single chart over each divisor in which φ has an anisotropic upper bound. It constructs a base reference of positive mass and gives the error $O(m^{q-1})$ for every tensor power. Corollary 28.10.2 computes the selected series exactly, including pole sets of positive dimension inside the fibers, and Example 28.10.2 attains the order of the error. These are sufficient conditions; general selecting weights and necessary conditions remain to be understood.

For several divisors, Example 28.10.3 computes the exact series for simultaneous poles in the fibers and attains the order of the error. The normal-crossing hypothesis provides a separated integrability test on the base; general ideal constraints on the base remain outside these results.

Theorem 28.10.3 allows the divisorial data to live on a specified smooth modification of the base, provided the residual there is semipositive with positive top integral and the pulled-back selecting weight has the required lower bound and witnesses; the transformation rule for adjoint multiplier ideals keeps the Jacobian in the coefficient of the sections. Example 28.10.4 computes the resulting orders of point ideals on projective space and gives a sharp error for a base pole of positive mass in higher codimension. Which general selecting weights admit such modification data is open.

Example 28.10.5 lies outside the regime of strictly positive residuals: contracted exceptional lines obstruct every strictly positive smooth gauge, although the base reference has positive mass. There the exceptional fixed divisor makes the selected and reference spaces identical for every tensor power, which separates the local gap between orders from the global defect of sections; the positive volume of the reference remains part of the sufficient hypotheses.

Proposition 28.10.1 computes this mechanism on the same family of blowups for every positive selecting coefficient, every integral fixed multiplicity of the exceptional divisor and any two anisotropic powers. A coefficient at most the fixed multiplicity leaves the whole space of sections. Above it, the selected spaces impose point jets of linearly growing order. As long as the coefficient stays below the fixed multiplicity plus the projective degree, the explicit reference has positive mass and the defect of sections is $O(m^{q-1})$. At that threshold the reference has zero mass and the sections grow with degree $q-1$; beyond it the series eventually vanishes. These exact model computations leave the general geometric criterion open.

Example 28.10.6 shows that a nonzero sharp defect also occurs outside the regime of strictly positive residuals. On the blowup of projective space at two points, the residual is the sum of the pullbacks of the Fubini–Study forms under the two resolved linear projections; it has degree zero on the line through the two centers but positive top mass 2^q-2. Simultaneous interpolation of point jets gives the defect $2\binom{m+q-2}{q-1}$ for every tensor power. Thus a semipositive reference of positive volume does not force the local gap between orders to be absorbed by fixed divisors. Whether such conditions are necessary in general is open.

35.9.1 Growing vanishing orders

On curves, the localization of Bergman kernels for jet conditions of growing order at a point is settled. Consider a Hermitian holomorphic line bundle L on a compact curve C whose curvature is semipositive and strictly positive off a point x, with local weight $|z|^{2r}+O(|z|^{2r+1})$ at x, and the selected spaces $H^0(C,L^m\otimes\mathfrak{m}_x^{q_m})$ with $q_m\to\infty$ and $q_m=o(m)$. The local model of Section 28.12 has a Gaussian transition at the radius $(q_m/(rm))^{1/(2r)}$, and for $q_m=\lfloor m\lambda\rfloor$ with $\lambda>0$ its limiting kernel measure misses the atom at the pole of the limiting envelope current. On compact curves, Theorem 28.13.3 shows that the uncorrected transition law holds in the optimal universal range $q_m=o(m^{1/(r+1)})$, uniformly in the direction of observation, and gives the exact criterion (28.50) for a fixed weight. In the whole sublinear range, Theorem 28.13.2 centers the transition on the boundary of the curvature quadrature domain, with normal length scale $1/(2\sqrt{m\rho})$, where ρ is the density of the curvature; by Section 28.13.4, no fixed finite Taylor jet of the weight determines this center to within the width of the transition for every sublinear sequence.

In higher dimension, for ordinary point jets of growing total order on the explicit anisotropic product of Section 28.15, Theorem 28.15.1 proves a Gaussian transition throughout the sublinear range of orders. The correct radial center includes the nonzero

remainder terms of the weight, whereas Proposition 28.15.1 gives the sharp, smaller range in which the homogeneous center can be used in this class. Theorem 28.15.2 transfers these laws through a submersion with compact connected fibers, with a pulled-back metric and a matched smooth volume; there the isolated selecting weight varies with the tensor power. Mixed or angular remainders, observation near the coordinate faces, other bases and general transfers through fiber selectors remain open directions.

35.10 Equations on singular spaces

On a resolution of a singular space, a pulled-back Hermitian form degenerates, and the pulled-back density can vanish or become singular. Appendix A develops potentials, envelopes, ideal equivalence and extended Monge–Ampère measures on reduced irreducible compact unibranch spaces, keeping the original nonclosed background. Every potential for the pulled-back background on a resolution descends (Theorem A.2.1), so the equation must be solved in this degenerate background itself, or with estimates that survive the passage to it. On projective cones, Theorems A.6.1 and A.6.2 solve explicit equations, for prescribed angular and radial data, in full ordinary logarithmic classes and in the bounded class of the closed background. The logarithmic construction includes nonclosed deformations that preserve the determinant, and its full solution descends even when the smooth residual is nonconstant on the exceptional fiber. The continuity results discussed in Section 36.21 use additional geometric and integrability hypotheses, notably log terminal singularities.

On the regular locus of a compact reduced irreducible Hermitian space of dimension at least two there is a conformal Gauduchon metric uniformly comparable to the original metric; see Section 36.21. This existence theorem does not by itself give the nonlinear equations asked for below.

In dimension one, Theorem A.7.1 gives a different sufficient condition: the normalization is a torus, the prescribed ordinary type has finitely many logarithmic poles of total weight smaller than the mass of the background, and the measure has a nonnegative L^p density, $p > 1$, with respect to the flat area on the normalization. Convolution with a Green function compensates for the degeneracy of the pulled-back background. The atoms at the poles determine the mass of every potential in the prescribed type and give uniqueness even among solutions that are a priori rough, and the residual descends continuously. The projective cusp of Example A.7.1 has an unbounded density on the curve itself although the data on the normalization are smooth. Every background on a curve is closed. On products of a cuspidal curve with elliptic curves, Theorem A.8.2 solves bounded equations in a nonclosed background all of whose powers are dd^c-closed, with Hölder continuous solutions that are unique in the whole bounded class, for data near the reference density; Corollary A.8.1 gives bounded solutions for all nonzero nonnegative L^p densities on the normalization, $p > 1$.

The family of logarithmic densities in Theorem A.7.2 makes the integrability issue precise. Densities on the normalization with local form $C_0|z|^{-2}(\log(1/|z|))^{-\gamma}$ are integrable for $\gamma > 1$, but they give a solution in the prescribed ordinary type exactly when $\gamma > 2$. For $1 < \gamma \leq 2$, the constructed solution keeps all multiplier ideals and the same non-pluripolar mass, but loses the lower bound defining the type. Multiplier-ideal equivalence alone therefore does not give a bounded residual. A criterion for general integrable data remains to be found.

Problem 35.10.1 Consider reduced irreducible compact unibranch complex spaces with a Hermitian form, as in Appendix A. Beyond the explicit spaces of Appendices A.6 to A.8, determine geometric conditions on such spaces and conditions on the measure that give existence and uniqueness of solutions in specified singularity classes, together with estimates ensuring finite mass and continuity.

In the cone examples, the pulled-back density vanishes on the exceptional divisor, and the logarithmic remainder solves an equation in a strictly positive background; the bounded equation for the closed background uses instead an energy that is coercive off that divisor. More general spaces of higher dimension require a replacement for these two controls. The curve construction uses a Green operator on the normalization; a linear construction by traces on a singular space needs further estimates across the singular locus.

Chapter 36
Historical notes

Local regularization and products provide a common starting point for the results of this book. A smooth local correction makes these methods available for nonclosed backgrounds, including on Kähler manifolds. Global arguments must distinguish a closed class on a non-Kähler manifold from a nonclosed background: the former may lack the Kähler positivity used in classical existence theorems, while the latter loses the cohomological control of Monge–Ampère masses. Gauduchon duality, relative envelopes and the available mass bounds connect the sources discussed below. Their hypotheses also decide whether a solution keeps the ordinary singularity type of its reference or belongs only to its relative envelope class.

These notes identify the sources of the results of the book, with precise locators, and explain how the arguments given in the book relate to the original ones. Published papers are cited in their journal versions, with the statement numbers and pages of the published text; for work available only as a preprint, the locator names the version used (v1, v2, . . .). Where needed, the conversion from the normalization of dd^c or of the Chern–Ricci form used in a source to the conventions fixed at the beginning of the book is indicated next to the citation.

36.1 Plurisubharmonic functions and regularization

Compactness, integrability and regularization are local properties, and they survive the passage to a nonclosed background because a quasi-plurisubharmonic function becomes plurisubharmonic after a smooth local correction. The uniform L^1 bound for normalized potentials in Theorem 1.4.1 is also proved by Dinew [Din16, Proposition 6.2, p. 109]; the compactness follows by propagating local compactness over a connected coordinate cover. Local exponential integrability, as recalled by Zeriahi [Zer01, Section 1, p. 672], becomes uniform over normalized potentials in Lemma 1.5.1, through a contradiction argument with a weighted sum of potentials.

The comparison of convolutions in different coordinate charts in Lemma 1.9.1 and Corollary 1.9.1 is the bounded part of the argument of Błocki and Kołodziej

[BK07, §2, Lemmas 4–5, pp. 2091–2092]. Their full regularization theorem [BK07, Theorem 1, p. 2089] allows unbounded potentials and a positive continuous background. The proof of Theorem 1.11.1 treats only the bounded case, which is all that the construction of products requires; it includes the extraction of a decreasing diagonal sequence and the rescaling that keeps the regularized forms strictly positive.

The smooth regularization of closed real $(1,1)$-currents recalled in Section 1.15.1, which gives smooth representatives $T_j \geq \gamma - C\lambda_j\omega$ with continuous functions λ_j decreasing to the Lelong numbers of T, in the notation there, is due to Demailly [Dem92, Main Theorem 1.1]; see also [Dem94, Theorem 4.1]. The analytic regularization recalled there, with $\nu(S_j, x) \leq \nu(T, x)$, is [Dem11, Corollary 14.13].

Demailly's regularization by the flow of the Chern connection [Dem94, Theorem 4.1] applies to $\mathrm{dd}^{\mathrm{c}}u$ for every $u \in \mathrm{PSH}(X, \theta)$, where θ is a smooth real $(1,1)$-form: the theorem concerns a closed current with an arbitrary smooth lower bound, here $-\theta$, and this lower bound need not be closed. The monotonicity and logarithmic convexity of the regularization enter the estimate (1.42), whose linear error rate follows from the curvature inequality [Dem94, Lemma 4.2, p. 119] and the expansion in [Dem94, Proposition 3.1, equations (3.19), (4.21), and (4.22)]. In the present situation the curvature inequality needs neither a small ϵ term nor an M_ϵ correction, and local trace bounds make the constants uniform in the center and the radius.

The same expansion underlies the truncated Kiselman transform of Lemma 1.21.1; see [Dem94, Theorem 4.1 and Remark 4.4] and Kołodziej–Nguyen [KN19, Lemma 4.1, p. 294]. In Lemma 1.21.3, for bounded potentials, the full base–fiber error is kept as a modulus tending to zero, and, as in [Dem94, Remark 4.4], a term linear in the radius absorbs the mixed and vertical errors. Lemma 1.21.6 applies the same transform, with the linear error of (1.42), to potentials that are only locally bounded off the pole set of a strict analytic potential. This accommodates the unbounded potentials with finite L^1 norm that occur in Theorem 15.7.1: on compact subsets of the complement of the pole set the minimizing radius stays away from zero, so the minimum-principle argument is available away from the singular barrier.

For a continuous potential, the logarithmic derivative of the Chern averages tends to zero; together with Dini's theorem and uniform continuity, this gives the vanishing curvature loss used in Lemma 1.21.4. For a Hölder reference these averages keep a common Hölder bound, and division by $1 + \varepsilon_j$ makes them ω-plurisubharmonic. The uniform approximation of continuous potentials in Lemma 1.10.1 follows from Proposition 1.10.1 and Dini's theorem. Dinh, Ma and Nguyên [DMN17, Theorem 2.1, pp. 551–555] prove quantitative smooth regularization in a Kähler background using its local potentials. The proof here uses Chern averages and the continuous-potential curvature-loss estimate on the Hermitian background. Their quantitative Kähler estimate is not used.

Demailly's regularization in the form of Theorem 1.15.2, whose approximants become divisorial with a smooth remainder after finitely many blowups, is the statement of Demailly and Păun [DP04, Theorem 3.2]. It allows an arbitrary smooth lower bound on a Hermitian manifold, and the smooth remainder after resolution is essential for its applications here. This sequence is in general different from the

multiplier-ideal sequence of Theorem 1.15.1, and we do not assume that a single sequence has the additional properties of both.

The analyticity of the sets where the Lelong number is at least $c > 0$, recalled in Section 1.13.1 and used in Lemma 27.18.1 and Theorem 4.5.1, is Siu's theorem [Siu74, Main Theorem, p. 53]; a smooth local correction makes it applicable to quasi-plurisubharmonic functions. The dimension bound for the Lelong level set and the existence of global irreducible components are stated in Demailly's book [Dem12, Chapter III, Corollary 8.5, p. 174; Chapter II, Theorem 5.3, pp. 102–103]. The non-strict inequality is essential. In Example 1.13.1, a summable series of line currents on the projective plane has the property that the set where the Lelong number exceeds one is a sequence of points accumulating at a point outside that set.

The closedness of polar curvature currents, Proposition 1.25.3, is proved by the complete pluripolar restriction argument of Boucksom, Eyssidieux, Guedj and Zeriahi [BEGZ10, Theorem 1.8, Lemma 1.9 and Remark 1.10, pp. 207–210]. The chain rule for bounded functions allows cutoffs that converge only pointwise, so the complete pluripolar set need not be closed, and a smooth local correction extends the argument to quasi-plurisubharmonic potentials with an arbitrary smooth background. Combined with the recovery of closed currents carried by a Lebesgue-null set from their traces, this removes the hypothesis that the pole set be closed from Corollary 32.5.4 and Proposition 32.5.4. The simultaneous realization of the limiting currents is a separate question.

The Hausdorff estimate of Corollary 1.13.1 follows from the Lelong mass formulas in Demailly's book [Dem12, Chapter III, formulas (5.5)–(5.8), pp. 158–159] and holds in every positive bidimension. When a closed polar current vanishes outside an exceptional locus, the support theorem for currents carried by analytic sets [Dem12, Chapter III, Corollaries 2.11 and 2.14, pp. 141–143] identifies the remaining divisorial part. Full local Monge–Ampère products at an unbounded locus of zero one-dimensional Hausdorff measure rely instead on [Dem12, Chapter III, Theorem 4.5 and Proposition 4.9, pp. 152–156], as stated in Theorem 1.6.3. These products keep their top-degree mass on the pole set, whereas every lower-degree coefficient measure vanishes there. The local maximum argument of Lemma 1.6.6 compares the residual measures and shows that they are invariant under bounded perturbations. Their global consequences for masses and models are discussed in Sections 36.2 and 36.19.

The local reference-integral estimate Lemma 1.8.2 is related to the Hölder estimates for measure functionals of Dinh and Nguyên [DN14, arXiv v1, Lemma 3.5 and Proposition 3.6, pp. 9–10]. Here the tests are bounded plurisubharmonic functions on nested balls. Keeping the cutoff-gradient terms gives a bound uniform in the Hölder norm of the reference; a finite cover yields Corollary 1.8.2 for bounded tests with a common curvature lower bound.

The local moderation theorem of Dinh, Nguyên and Sibony [DNS10, Theorem 1.1 and Corollary 1.2, pp. 466–467; proof pp. 468–471] gives exponential integrability with respect to locally moderate currents. The top-degree estimate Lemma 1.8.3 follows here by applying the reference-integral estimate to consecutive truncations and absorbing their polynomial growth into the exponential tail. Guedj and Zeriahi's extremal-capacity comparison [GZ05, Theorem 5.2, pp. 622–623, and Proposition 7.1,

p. 632] motivates Lemma 1.8.4. Its proof uses conservation of mass only for the closed Fubini–Study form on an auxiliary projective space. Local capacity comparison gives the Hermitian estimate Proposition 1.8.1 with constants uniform for a common Hölder norm.

The convergence against a fixed non-pluripolar measure in Lemma 1.23.1 rests on Cegrell's classical fixed-measure argument, which Kołodziej and Nguyen also state in a form allowing extraction of subsequences [KN25, Lemma 2.1 and Corollary 2.2, p. 116]. Applying that statement to every subsequence gives convergence of the whole bounded sequence.

Let f be a holomorphic function and χ a convex nondecreasing function. The sharp Sobolev criteria for the potentials $\chi(\log|f|)$ in Section 1.13.2 are due to Chen, Shao and Wang [CSW26, v1, Theorems 1.1–1.2, pp. 2–3 and 19–22]; their estimate in terms of the smallest multiplicity and their energy bound on small sublevel sets are [CSW26, v1, Lemma 4.1 and Proposition 3.8, pp. 16–20]. The proofs given here use the regularizations $\chi(\frac{1}{2}\log(|f|^2+\varepsilon^2))$, $\varepsilon>0$, to identify the weak gradient and to keep track of the residual divisor term in the distributional Laplacian, a term that the classical Laplacian omits. The power examples and their endpoint obstruction in Example 1.13.2 are the calculation in normal coordinates and the table of [CSW26, v1, Lemma 2.1 and Section 6.1, pp. 6–7 and 24–25]. These local criteria for derivatives should not be confused with the theorem on products at a small unbounded locus discussed above.

The concentration example of Proposition 1.6.1, the case of one atom and the zero potential in Theorem 1.6.4, is the point-mass case of Qin's [Qin26c, v1, Corollary 3.6, p. 14]. In the one-atom step Proposition 1.6.2, the mass is computed directly from the transverse identity (1.9), which gives a smooth variant of the logarithmic truncation in [Qin26c, v1, Lemma 2.4, pp. 9–10]. The conclusion of Lemma 1.6.8 for sequences with a common local bound agrees with [Qin26c, v1, Proposition 7.2, p. 26]; in the book it follows from the range estimate (1.2).

Theorem 1.6.4, which adds an arbitrary positive Radon measure to the Monge–Ampère measure of a fixed potential in the limit, is Qin's [Qin26c, v1, Theorem B, p. 3] for bounded potentials; the proof given here applies unchanged to locally bounded ones, since the potential enters only through its bounds on compact sets. The addition along finitely many test functions and the passage to a determining family correspond to [Qin26c, v1, Proposition 3.2 and Theorem 3.4, pp. 11–14]. In the proof given here, the smooth ridge gives the limits of the lower-degree currents directly, finite iteration over atoms controls every mixed term, and the one-sided logarithmic bound also gives convergence in local capacity. The smoothness of the approximants is asserted for a smooth background, whereas Qin also treats continuous backgrounds.

The case $b=0$ of Theorem 1.6.5, according to which every vaguely closed set of positive measures is the full cluster set of a sequence of potentials tending to zero, is the case of the zero potential in [Qin26c, v1, Theorem A, pp. 2–3]. The metric for the vague topology and the concatenation of dense sequences are [Qin26c, v1, Lemma 3.3, p. 12 and Lemma 4.1, p. 14], and the comparison example with moving measures is [Qin26c, v1, Example B.1, p. 28]. In the book, finite addition of atoms gives smooth approximants that converge in local capacity directly. For a nonzero background b,

the construction with moving measures in the same theorem corresponds to [Qin26c, v1, Theorem 3.4, pp. 13–14]. The cutoff argument of Lemma 1.6.13 shows that, under convergence in local capacity, every limit measure dominates the Monge–Ampère measure of the background; this gives a complete classification of the cluster sets for locally bounded backgrounds approximated from below. Convergence in local capacity is a stronger requirement than the L^1 convergence in Qin's general classification of reachable sets, whose argument for reaching the zero measure needs additional hypotheses on maximal approximation and on the domain; that argument is not used here.

Let $\phi_1, \dots, \phi_r$ be plurisubharmonic weights with exponents $p_1, \dots, p_r > 1$, as in Theorem 1.16.2. The division estimate with two summands, Lemma 1.16.6, is the case of two weights in Takakura's [Tak26, v1, Theorem 1.3, pp. 2 and 5–6]. Since the relevant kernel is a line, the proof can use the Hessian formula for a scalar function instead of the general curvature theorem for subbundles. The constant in the a priori inequality (1.26) follows from the twisted Bochner identity in Demailly's notes [Dem11, Section 13.A, pp. 123–124] by a weighted Cauchy inequality. Proposition 13.4 there states the weaker constant 2; keeping the full factor $\eta + \lambda$ gives the constant 1 used here. For a single weight, Proposition 1.16.1 is the case $r = 1$ of [Tak26, v1, Theorem 1.2, pp. 1–2 and 6–7]. Applying it successively gives the product constant of Theorem 1.16.2, with no condition on $\sum_i 1/p_i$. The coordinate example in its proof extends [Tak26, v1, Example 3.2, p. 7] and shows that this constant is optimal when $r \le n$. For $2 \le r \le n$ the product is strictly smaller than $1 - r^{-1} \sum_i 1/p_i$, the constant stated in Takakura's theorem. For $r > n$ the product need not improve Takakura's dimension-dependent estimate, and by itself it does not determine the optimal constant.

In dimension one, Theorem 1.16.3 gives the optimal constant for every number r of weights, including the range $\sum_i 1/p_i \ge 1$ in which it vanishes. When $\sum_i 1/p_i < 1$, it equals the dimension-one constant of [Tak26, v1, Theorem 1.2, pp. 1–2], and a common logarithmic pole shows that it is sharp. The local argument in Lemma 1.16.7 uses the logarithmic fundamental solution and the characterization of positive distributions in [Dem12, Chapter I, Sections 4.A–4.C, pp. 30–36], together with Skoda's integrability theorem [Dem11, Lemma 5.6]; in our normalization, $\mathrm{dd^c} \log |z|^2 = \delta_0$. The global lifting in Lemma 1.16.8 is the locally free planar case of vanishing for coherent sheaves. It is proved here with the vector-valued L^2 estimate for complete metrics in [Dem11, Theorem 5.1 and its proof, pp. 33–34], whose Bochner identity is [Dem11, Section 4.C, p. 30]. We do not determine the optimal constant when $n \ge 2$ and $r > n$.

36.2 Mass and energy

On a nonclosed background, the variation of the Monge–Ampère mass and the failure of the usual energy to have a primitive are related manifestations of torsion. Chiose's polarization argument [Chi24, Theorems 1–2, pp. 576–577] gives the characterization

of mass invariance in Theorem 2.2.2. Yi Li constructed energies corrected by torsion terms [Li10, v1, Equation (1.18) and Theorem 1.2, p. 5]. The correction changes the one-form being integrated, in agreement with Theorem 3.1.1, which shows that the uncorrected one-form has no primitive unless ω is closed; Proposition 3.3.1 computes the difference by a first variation.

Cutoffs of diffuse poles separate this variation of the background from a fixed loss of mass at the singularity. Let $n \geq 2$, let H and C be the cutoff and the curvature constant of Lemma 32.5.3, let M be the residual mass of Lemma 32.5.1, fix $0 < t < 1/C$ and put $A = \int_X \omega^n$. By Proposition 32.5.6, the mass of tH is a polynomial in t whose finite torsion coefficients come from Stokes calculations for compactly supported bounded products, and every representative that is ordinarily bounded-equivalent to tH loses the same top residual mass Mt^n, although its lower-degree coefficients can vary. If $\mathrm{dd}^{\mathrm{c}}(\omega^q) = 0$ for $1 \leq q < n$, invariance of the mass under bounded changes and invariance of the residual give the mass $A - Mt^n$ throughout the class (Corollary 32.5.7); this applies to every compact complex surface with a Gauduchon form. On a surface, Corollary 32.5.5 shows that the mass is constant on the class if and only if ω is Gauduchon, since admissible smooth bounded perturbations detect any failure of the Gauduchon condition.

The failure of the bounded mass property on a Hopf threefold is the counterexample of Xia and Zhang [XZ26, v1, Theorem 1.1, p. 2]. There a Gauduchon background converts vertical Dirichlet energy into Monge–Ampère mass, a mechanism developed in Chapter 9. Combined with Theorem 7.6.1, it produces infinite extended mass with trivial multiplier ideals at every positive weight, and the results on integrability and convergence of coefficients in Section 7.7 describe the linear current carried by this construction. Xia and Zhang use the operator $\mathrm{i}\partial\bar\partial$ [XZ26, v1, Section 1, p. 2]; dividing their background forms by 2π and leaving their potentials unchanged preserves positivity and changes the coefficient of the mass identity.

The local matrix, heat potential, Gaussian gradient estimate and completed square used in Section 9.8 are those of Xia and Zhang [XZ26, v1, Lemma 2.2, pp. 4–5; Lemmas 3.1–3.3, pp. 6–9; equation (4.2), p. 10]. The smaller initial time in Theorem 9.8.1 changes the radial length and the integrated defect; Lemmas 9.4.1 and 9.5.1 are proved for all such initial times. The positive fiber flux and the depth estimate give the square-root upper bound for every potential in the class $\mathcal{R}$ of $U(2)$-invariant normalized potentials of that section. The concentration regions of Theorem 9.8.1 then give a single potential with infinite weighted measure, and concentration on a pole divisor at the critical weight; they also give necessary conditions for all normalized potentials, without the symmetry assumption. The optimal sufficient rate for the class of all normalized potentials remains open.

The passage from smooth to bounded and unbounded invariant potentials uses the classical bounded product and capacity theory developed in Sections 1.6 and 1.8. The local product construction and decreasing continuity are Bedford and Taylor's [BT82, Theorem 2.1, pp. 4–7; Theorem 2.4, pp. 7–9; Corollary 2.8, p. 10]; their quasi-continuity and zero-capacity results are [BT82, Theorem 3.5, pp. 13–14; Theorems 6.9–6.10, pp. 25–26]. They use $\mathrm{dd}^{\mathrm{c}} = 2\mathrm{i}\partial\bar\partial$, which is 4π times our operator and changes capacities only by a fixed factor. The local L^1 compactness underlying

Theorem 1.4.1 is also described in [Dem12, Chapter I, Proposition 4.21, pp. 36–37; Proposition 5.9, p. 40]. In the present argument, group convolution and canonical truncation give the original Hermitian capacity convergence, including the boundary mass at a truncation level.

The extension in Theorem 9.6.1 keeps the same threefold heat family. Its proof first bounds the mass on the negative radial half-line by exact integration over the elliptic fiber, then uses Lemma 9.6.2 to compare the angularly integrated densities in different dimensions. The additional angular factor has a fixed positive lower bound on the half-line carrying the divergent mass. For the scalar Hopf manifolds $H_n = (\mathbb{C}^n \setminus \{0\})/\langle z \mapsto \mathrm{e}^{-1} z \rangle$, $n \geq 3$, Corollary 9.6.1 then gives $v_+(h) = +\infty$ for every Hermitian form h, so H_n does not have the bounded mass property.

The failure of positive lower mass is a different phenomenon; it occurs on the elliptic threefold of Section 27.18. Sufficient conditions for a positive lower bound come instead from the sign of the torsion: (27.60) is the three-dimensional case of a formula of Angella, Guedj and Lu [AGL23, proof of Theorem 3.2, p. 4641], who call the condition $\mathrm{dd}^{\mathrm{c}}\omega \leq 0$ plurinegativity.

The completion energy of Section 3.4 is the Dinew–Popovici functional. Hanwen Liu defines it and computes its variations in [Liu26, v1, Section 2.1, pp. 3–5], and proves the identity relating energy and volume along potential variations, together with the indefiniteness of the Hessian, in [Liu26, v1, Theorems 5.1–5.2, pp. 16–19]. Liu assumes that the ambient manifold is Kähler. The proofs in the book use only the finite-dimensional space of closed holomorphic completions and Stokes' theorem, so these statements about potential variations hold on every compact Hermitian symplectic fourfold. Liu's calculation of criticality for products [Liu26, v1, Lemma 2.3 and Theorem 2.4, pp. 7–9] gives the mechanism of Example 3.4.1, where an explicit Fourier form on the square torus replaces the general existence argument on surfaces. Neither the general existence criterion of that paper in terms of the canonical bundle nor its convergence theorem for the global torsion flow is used here. Liu's convention is $\mathrm{d}^{\mathrm{c}} = \mathrm{i}(\bar\partial - \partial)$; rewritten with the book's d^{c} and dd^{c}, the formulas of that paper for potential variations give (3.7) with the coefficients shown there.

Combined with Chiose's criterion, the energy identity gives the equivalence between flatness of the energy and invariance of the mass in Corollary 3.4.1. The calculation with one nonclosed factor in Example 3.4.2 shows that this flatness along potentials implies neither criticality in the full Aeppli class nor the existence of a primitive for the Monge–Ampère one-form.

36.3 Gauduchon metrics and linear mass

Gauduchon metrics replace the cohomological conservation of mass by a linear trace identity. In the form recalled by Tosatti and Weinkove immediately before [TW10, Lemma 2.3], Gauduchon's conformal theorem starts, in dimension $n \geq 2$, from a Hermitian form η and constructs a smooth real function f such that $\gamma = \mathrm{e}^{f}\eta$ is Gauduchon; the multiplier in Theorem 4.1.1 is then $\rho = \mathrm{e}^{(n-1)f}$. The resulting trace

comparison, Corollary 4.2.1, also underlies the first mixed-volume bounds of Guedj and Lu [GL22, Proposition 3.3].

The trace identity has a dual form: positivity can be tested against Gauduchon metrics. For an arbitrary smooth background, the generalized Lamari criterion Theorem 4.3.1 is stated by Boucksom, Guedj and Lu [BGL25, Lemma 1.15], who derive it from the Hahn–Banach theorem; the proof given in the book combines a separation argument with a right inverse of a scalar elliptic operator. In Theorem 4.5.1, Lamari duality combines with normalized compactness and Siu analyticity to turn bounded transverse test pairings into a global hypersurface. Conversely, when an effective Cartier divisor coincides, with multiplicity one, with the local hypersurface of the chosen product chart throughout that chart, its regularized logarithmic potentials in Proposition 4.5.1 bound these pairings uniformly. The same linear control gives the bound (4.3) for the normalization constant, which is used in the weak existence arguments of [BGL25, Section 4].

The support lemma for currents carried by a smooth hypersurface, Lemma 4.6.1, is the local statement used by Dinh and Sibony in [DS04, proof of Theorem 4.1, pp. 965–966], where it is attributed to Bassanelli. The proof given here tests against a quadratic function of the normal coordinate and polarizes horizontal forms; it applies whenever $\mathrm{dd}^{\mathrm{c}} R$ has order zero, R being the positive current supported on the hypersurface. The estimate for pullbacks in Lemma 4.6.2 follows directly from the classical dd^{c}-lemma for currents on compact Kähler manifolds, applied to a fixed closed pushforward; the current that is paired with the distributional primitive is smooth.

The smoothing of positive pluriharmonic forms with continuous coefficients in Lemma 4.6.3 uses the heat equation of the Hodge Laplacian, the Kähler identities and a positive correction tending to zero. For continuous potentials and closed backgrounds on a compact Kähler manifold, Dinh and Sibony construct the product of the curvature current with a positive pluriharmonic current through an energy [DS04, Theorem 5.1, Proposition 5.2 and identity (12), pp. 967–969]. The proofs of Lemma 6.3.1 and Theorem 6.3.1 integrate by parts against the current of bidegree $(n-1, n-1)$ itself, so that no auxiliary power of a Kähler form is needed. They give a positive measure as product, depending jointly continuously on the factors, on any compact Hermitian manifold, for every smooth real background and continuous potential.

The descent of horizontal currents in Lemma 4.6.4 follows directly from the positivity of the coefficient matrix together with the Liouville theorem for harmonic functions on the complex line, or the maximum principle on a compact connected curve. It therefore applies both to bundles with fiber $\mathbb{C}$ and to bundles of compact curves. On the elliptic threefold, the exact vertical trace form gives Lemma 29.9.1 for positive dd^{c}-closed currents, and its closed case gives the calculation of the Bott–Chern cone.

The Kähler criterion under parallel torsion, Theorem 4.7.1, is a theorem of Chen and Zheng [CZ26, v1, Theorem 1.2, pp. 2 and 8]; it settles a case of the conjecture of Fino and Vezzoni recalled in Section 36.22. Their projection by integration of parallel forms and their description of the holonomy are [CZ26, v1, Lemmas 3.1–3.2 and

Propositions 3.3–3.5, pp. 3–6]; Lemma 4.7.1 proves both descriptions again in a form that keeps track of positivity and of commutation with the differential. The slice of fixed trace and the maximum of the volume are [CZ26, v1, Propositions 4.2 and 4.5, Lemmas 4.3–4.4, pp. 7–8]. The inverse of the positive cofactor map proves the step of taking roots directly, and Lemma 4.1.1 proves, by linear algebra of primitive forms, that a Hermitian form which is both balanced and pluriclosed is closed. The additional obstruction of Zhao and Zheng for non-balanced metrics with Bismut-parallel torsion, used in their Corollary 1.3, is a separate ingredient.

The persistence of pluriclosed metrics under deformations, Theorem 4.7.2, which assumes that the Aeppli number $h^{1,1}_A$ is constant, is Cavalcanti's [Cav13, v5, Theorem 8.11, Section 8.3, pp. 41–42]. The fourth-order Aeppli Laplacian and the realization of Aeppli classes by harmonic forms used in Lemma 4.7.2 are due to Schweitzer [Sch07, v1, Section 2.c, pp. 9–10], who constructs the elliptic symbol in Section 2.b, pp. 7–8. The proof given here writes the projection onto $\partial\bar\partial$-closed forms explicitly and obtains its smooth dependence on the parameter from the resolvent and the constancy of the dimension of the kernel. It applies to any finite-dimensional smooth parameter space and keeps the prescribed initial metric. This is also the analytic input stated in [Bar26, v1, Theorem 2.1, p. 7]; Barbaro's construction of deformations of Lie groups there requires a separate argument with transverse cohomology.

For the logarithmic parabolic Monge–Ampère flow on Kähler manifolds, the calculation of L^p norms of the volume ratio is Jiyuan Han's [Han26, v1, Lemma 3.2, pp. 13–14]. The chain rule and Stokes computation in Proposition 4.8.3 keep the additional term coming from the derivative of the Hermitian background. When the forms along the flow remain Gauduchon, convex entropies are nonincreasing (Corollary 4.8.1), and varying the target probability measure gives the exact criterion at the initial time in Proposition 4.8.4. These statements apply on any interval on which a smooth solution exists. Han's uniform estimates for degenerating families of Calabi–Yau manifolds, and the Green-function estimates on which they depend, are not part of this calculation.

The calculation of the total variation in Proposition 4.8.5 is a direct variation of determinants on a flat torus, whose coefficient is given by the two Haar averages of Lemma 4.8.1. Corollary 4.8.2 compares this growth of the distance between two solutions with the initial derivatives of the individual entropies.

36.4 Gauduchon degrees and positivity of vector bundles

The deformation obstruction for convex norms in Theorem 5.1.1 is due to Liu and Wan [LW26, v1, Theorem 3.2 and Sections 3.1–3.3, pp. 8–13], and the normalization of line metrics in Lemma 5.1.1 is their Lemma 3.1, p. 8. The proof combines derivative estimates along the line axes with an average of the horizontal curvature and the formal adjoint on a fixed sphere bundle. The resulting C^4 neighborhood is uniform

over all smooth complex norms, and the positive Gauduchon test forms may differ from one line summand to another.

The regularization of homogeneous functions in Theorem 5.1.2 is a lemma of Kuang-Ru Wu, stated there for vector bundles over compact complex manifolds [Wu23, v1, Section 2.3, Lemma 13, pp. 10–13]. Its gluing step uses the closure under maxima, the stability under smooth perturbations and the semilocal approximation of Greene and Wu [GW79, Definitions 1.1, 1.3–1.4, pp. 48–49; Lemma 4.1 and Theorem 4.1, pp. 80–81], while the local positivity properties are proved directly from a holomorphic shear formula. The replacement in Proposition 5.1.1 assumes that a strictly plurisubharmonic homogeneous function is given, and Corollary 5.1.1 keeps the hypothesis of local strong convexity in the fibers.

The local example Example 5.1.1 adapts the quadratic maximum of Fukuoka and Setti [FS26, v1, Example 3.18, p. 20] to a complex rank-two bundle over a disk. A geometric series of squared holomorphic moduli gives its total-space plurisubharmonicity. Its fiberwise quadratic margin tends to zero at nearby base points, showing why local strong convexity in Theorem 5.1.2 requires a common margin on a total-space neighborhood. The example concerns this distinction between hypotheses; the general real-Finsler smoothing and connection-convergence theorems of that paper are not used.

The existence statement needed here, Kobayashi's characterization of ample bundles in the form stated by Liu and Wan [LW26, v1, Theorem 2.1, p. 7], is proved directly by the tautological construction of Lemma 5.1.6 and Proposition 5.1.2: a metric with positive curvature on the dual of the tautological line bundle gives a homogeneous squared norm that is strictly plurisubharmonic off the zero section. When the dual vector bundle is ample, such a line metric is a root of the Fubini–Study metric pulled back by a very ample embedding. The result is the strictly plurisubharmonic homogeneous function used in their regularization [LW26, v1, Lemma 3.4, pp. 13–14], and it gives Corollary 5.1.2.

The construction of polar norms in Section 5.1.3 gives a proof of Liu and Wan's [LW26, v1, Proposition 3.5, pp. 14–16]. A Schur complement and a holomorphic correction of the second jet give local pluriharmonic supporting functions for the logarithm of the polar square. A uniform upper bound for the real Hessian of the primal squared metric gives strong convexity of the dual through its exact Legendre expression; neither real convexity of the primal metric nor uniqueness of a maximizer is assumed. The tautological construction for an ample dual and the homogeneous regularization then give the smooth replacement and the curvature obstruction of Corollary 5.1.4, under the hypotheses stated there.

On Hirzebruch surfaces, the ample comparison quotient and the two prioritary points of Proposition 5.1.4 and Lemma 5.1.10 are Liu and Wan's [LW26, v1, Proposition 4.2 and Lemma 4.3, pp. 17–20]. The facts about Hirzebruch surfaces that they need are proved directly in Lemma 5.1.9, from sections along the ruling and a very ample embedding. The positive Gauduchon tests for the individual dual line bundles in Proposition 5.1.5 are the divisor calculation in their proof of Theorem 1.1, p. 23. These constructions give the ample point and the test forms used in Theorem 5.1.3 and Corollary 5.1.5.

The stability of extensions in Proposition 5.1.6 is Liu and Wan's [LW26, v1, Proposition 4.4, pp. 20–21], including the equality case on $\mathbb{F}_0$. The proof of Proposition 5.1.7 uses the exact sequences of Hom groups and applies to every extension, while restriction to a fiber of the ruling shows that none of them is ample. The explicit family of Dolbeault operators in Proposition 5.1.8 converges to the split operator in every finite C^m norm. These stable, nonample approximations give the stable point used in the deformation argument of Theorem 5.1.3.

The approximation theorem Theorem 5.1.3 is Liu and Wan's [LW26, v1, Theorem 4.1, p. 17; proof, p. 22]. Its input on the stack of prioritary sheaves is Walter's [Wal98, Proposition 2], which applies to all prioritary torsion-free sheaves of fixed rank at least two with fixed numerical Chern classes, without any semistability assumption. The required openness statements are proved directly by the family arguments in Lemmas 5.1.11 to 5.1.13; on a Hirzebruch surface, slope stability reduces to finitely many classes of line bundles. Parallel transport and compactness, in Lemma 5.1.14, give convergence on the fixed smooth bundle and on its dual in every finite C^m norm. Combining this approximation with the Gauduchon-degree obstruction and the polar construction above gives Corollary 5.1.5, which is Liu and Wan's [LW26, v1, Theorem 1.1, p. 2; proof, p. 23].

The fixed Stein hypersurface of Theorem 5.1.4 and Corollary 5.1.6 corresponds to Liu and Wan's [LW26, v1, Proposition 5.1, pp. 24–27] for explicit algebraic bundles. A positive horizontal direction and the vertical critical hyperplane give the test on the Levi index, and a holomorphic quadratic jet makes the real Hessian nondegenerate without changing the Levi form. Separability of the relevant metric space gives countably many robust local tests, which Lemmas 5.1.15 and 5.1.16 place on a single global graph by Runge and Oka–Weil approximation. The graph is chosen before the metric, whereas the critical point that detects the failure may depend on the metric.

The dual norm of a concave neighborhood, Theorem 5.1.5, is Liu and Wan's [LW26, v1, Lemma 5.2, pp. 27–32]. After a correction by a pure jet, a negative tangent plane of the boundary gives a supporting holomorphic section. Balls of uniform size tangent to the boundary from inside then express the squared dual norm as a supremum of functions with a common margin of strong real convexity, and ampleness together with homogeneous regularization gives a smooth replacement. Let E be an ample bundle of rank r, and write $U_E = \mathbb{P}(\mathcal{O}_X \oplus E) \setminus \mathbb{P}(\mathcal{O}_X)$ for the complement of the zero section in its projective completion. The argument at high levels of an exhaustion in Corollary 5.1.7, which shows that U_E is not r-convex when E^* admits no smooth complex norm with plurisubharmonic square, is their [LW26, v1, Corollary 5.3, p. 32]. Combined with the norm obstruction and the stable ample approximation, it gives Corollary 5.1.8.

The case of positive mean curvature in Theorem 5.1.6 and Corollary 5.1.9 is Liu and Wan's [LW26, v1, Proposition 5.4, pp. 32–33]. The calculation with the reciprocal of the norm gives the more general count, in which s positive horizontal directions of the curvature give $(n + r - s)$-convexity of U_E for a bundle E of rank r. For stable bundles of positive degree, the application uses the existence of Hermitian–Einstein metrics, proved by Uhlenbeck and Yau [UY86] in the Kähler case and extended to Gauduchon metrics by Li and Yau [LY87]. The degree of Definition 5.1.1 pairs the

Bott–Chern first Chern class with the Gauduchon form; with the unscaled curvature F_h of a Hermitian metric h, the integrated trace carries the factor $2\pi n$. The resulting least smooth convexity index three in Corollary 5.1.10 is [LW26, v1, Corollary 5.5, p. 33]; the lower bound uses the dual-norm obstruction above.

Suppose that, for a holomorphic bundle F of rank r over a connected compact base of dimension n, the complement $F^\times$ of the zero section carries a smooth positive plurisubharmonic function that is homogeneous of degree two. Taking maxima over the level spheres, Proposition 5.1.9 then shows that the least smooth convexity index of U_F is $n + r$: a negative radial derivative makes the Levi form restricted to the level set nonpositive on a complex hyperplane. A metric with Griffiths-nonpositive curvature gives such a function in Corollary 5.1.12, and unitary-flat metrics are the case of zero curvature, Corollary 5.1.13. For split bundles on Hopf manifolds, Proposition 5.1.10 combines the weighted curvature with the lower bounds that closed submanifolds, here the restrictions to elliptic fibers, transfer to the ambient manifold in Lemma 5.1.18. It gives the least indices $r + 1$ and $n + r$ in the two pure regimes; in the mixed regime, only separate upper and lower bounds are obtained. For a single line bundle E_a, Example 5.1.2 thus gives the least index two when $0 < |a| < 1$ and $n + 1$ when $|a| \geq 1$, and the sign of the Gauduchon degree distinguishes the three cases $|a| < 1$, $|a| = 1$ and $|a| > 1$, although the de Rham first Chern class vanishes.

For external tensor products, Theorem 5.1.7 applies the same exhaustion with one positive curvature direction from each factor. The positive mean curvatures lifted from the factors control every vector of the tensor product, including the indecomposable ones, and give Corollary 5.1.11. For the tangent bundle of a surface with a nonflat Ricci-flat Kähler metric, Corollary 5.3.1 combines the exhaustion instead with the directional positivity of Theorem 5.3.1: this tangent bundle admits no metric whose mean curvature is positive definite, but its RC-positive metric still gives the required exhaustion.

The mean-curvature criterion for RC-positivity and the conformal construction in Lemma 5.2.1 and Theorem 5.2.1 correspond to Chou and Wu's [CW26, v1, Definition 5 and Lemmas 7–8, pp. 5–6]. The scalar equation in the construction is the Gauduchon compatibility condition of Proposition 4.8.1, and taking the curvature margin over a closed ball makes it uniform in both the base point and the unit fiber vector. Example 5.2.1 shows directly that the order of these two quantifiers matters, and Proposition 5.2.1 gives a global obstruction to interchanging them when the determinant bundle admits a flat Hermitian metric.

The RC-positive tangent metrics on Ricci-flat surfaces in Theorem 5.3.1 are Chou and Wu's [CW26, v1, Theorem 1 and Lemmas 9–15, pp. 7–12]. The pointwise criterion is the invertibility of the anti-self-dual curvature operator. The identities behind (5.5) are Derdziński's [Der83, equations (33)–(35), p. 414], with the nonnegative Laplacian on functions used there replaced by $\operatorname{div}\nabla$ and with the same norm on endomorphisms; summing the three equations for the eigenvalues gives the coefficient 36. The commuting curvature factors in [Der83, Lemma 2, p. 412] also show directly that a nonzero parallel operator with three distinct eigenvalues cannot occur here. This gives the required nonempty open set and a conformal tangent metric whose associated form is not closed.

The symmetrized sectional-curvature average and the integrated trace identity in Lemma 5.4.1 and Proposition 5.4.1 are Xueyuan Wan's [Wan26, v1, Lemma 2.1 and Proposition 3.3 with Lemmas 3.1–3.2, pp. 5–8]. The average is proved here by complex Gaussian moments, and the connection calculation keeps the exact term until Stokes' theorem removes it. The auxiliary metric g is Kähler, while the tangent metric h can be nonclosed. Their identity gives Corollary 5.4.1 directly from the signs of its three terms. This use of the identity does not require the auxiliary Hessian equation or the canonical-bundle positivity argument of that paper. The torus and torus-cover statements of Proposition 5.4.2 and Corollary 5.4.2 follow from this identity and periodic integration by parts: the symmetric connection difference vanishes, forcing the Hermitian coefficients to be constant.

The first-jet characterization of Griffiths positivity on projective manifolds, Theorem 5.5.1, is due to Du and Xie [DX26b, v1, Theorem 1.1 and Proposition 2.1, pp. 2–5]; the identity in normal frames is written in the curvature convention used above. The density with two derivatives that the argument requires is proved directly in Lemmas 5.5.2 and 5.5.3, through normalized projective coordinates and Gaussian convolution. The immersion test without twist, Corollary 5.5.1, is the calculation in [DX26b, v1, Section 3.1, pp. 7–8]. The positive measure in the dual alternative lives on the sphere bundle of the dual bundle, so the condition that it annihilates all metrics is different from Gauduchon duality on the base.

The double cover in Section 5.6, branched over $2q$ points, gives an explicit model with a computable space of sections in which a strictly positive line bundle fails the test without twist. The decomposition of sections into even and odd parts and the projective embedding are proved explicitly. A root of a very ample Fubini–Study metric gives a separately defined weighted family of tests, and Proposition 5.6.1 identifies its first successful level $j = q - 1$, together with obstructions by one or two atoms at the lower levels. The original criterion with a fixed very ample twist m follows the subsequence $j = qm$, so this calculation does not determine its first successful twist. The obstructions at a single level should also be distinguished from an adjoint measure annihilating every smooth metric.

The negative-bundle weight in Proposition 5.7.1 is the construction of Kusakabe and Matsumura [KM26, v1, Lemma 3.2(a,b,d), pp. 20–21]. The complete metric of Corollary 5.7.2 uses the two exhaustion barriers in the first paragraph of their proof of [KM26, v1, Lemma 3.3, p. 21]. Their jet-vanishing condition [KM26, v1, Lemma 3.4, p. 21] follows here from the smooth-center multiplier-ideal formula, which gives the exact ideal and the sharp threshold for every real weight in Corollary 5.7.1. In their argument the logarithmic weight supports weighted $\bar\partial$ estimates and the gluing of local sprays; here its direct Levi calculation relates negative bundle curvature to the singularity along the zero section.

36.5 Bounded products and extended measures

The Bedford–Taylor product of bounded potentials is local even when the background is not closed: expanding against smooth local corrections, as Kołodziej and Nguyen [KN15, Section 1] and Boucksom, Guedj and Lu [BGL25, Section 2.1] do, gives Proposition 1.7.1. Restricting the products of the truncations to the sets where the potential exceeds the truncation level then defines the extended measure of Theorem 7.1.1, by the mechanism of [BGL25, Proposition 2.1]. Finiteness becomes decisive when these restrictions are removed in the category of currents, and [BGL25, Theorem 2.2] uses a uniform bound on the variation for this purpose. An infinite extended positive measure is still given by the local construction, but it is not a global current of order zero.

The independence of the bounded mass property from the metric, and the reduction to smooth potentials, in Proposition 7.5.1 recover the comparison made after [BGL25, Definition 1.16]; decreasing regularization with exact curvature bounds gives the passage from smooth to bounded potentials. The Gauduchon trace bound and the cancellation of the exact top-degree wedge product used in Proposition 8.1.1 also appear in Guedj and Lu's [GL22, Proposition 3.3(i), (vi)], and the estimate in terms of the range of the potential makes the dependence on the oscillation explicit and polynomial.

The form of Moishezon's theorem used in Lemma 7.5.1 is stated in [Moi67, Introduction, p. 1332]; it is used as a classical geometric input. The bounded mass conclusion for a Kähler modification follows Guedj and Lu's argument pulling back upper volumes [GL22, Lemma 3.6].

The local mixed Monge–Ampère inequality, Theorem 1.6.2, is Kołodziej's inequality in the form stated by Dinew [Din09a, Theorem 1.2, p. 2], for bounded plurisubharmonic functions whose Monge–Ampère measures dominate nonnegative integrable densities with respect to Lebesgue measure. The inequality is local, and on relatively compact subdomains the hypotheses of Theorem 1.6.2 reduce to these. Its extension to smooth backgrounds is proved in Section 1.7 and used in Section 15.17.1. Freezing the coefficients of the two smooth backgrounds gives Lemma 1.7.3 and the trace inequalities (15.9) and Lemma 7.4.2, the latter keeping the mass of the linear trace on the pole set. The same argument also recovers the mixed inequality for L^1 densities with respect to ω^n used by Nguyen [Ngu16, Lemma 1.9, p. 255].

36.6 Smooth and continuous equations

On a Hermitian manifold, the smooth Monge–Ampère equation determines a normalization constant together with the potential. For $n \geq 2$, the existence theorem Theorem 10.1.1 is Tosatti and Weinkove's [TW10, Corollary 1]. Their equation, for a smooth function F prescribing the logarithm of the density, is written with $\mathrm{i}\partial\bar{\partial}\varphi$; the substitution $u = 2\pi\varphi$ gives the convention of this book, and a prescribed smooth volume form $\mathrm{d}V$ is absorbed by replacing F with $F + \log(\mathrm{d}V/\omega^n)$. The concavity

argument with Gauduchon metrics for the normalization constant in Theorem 10.7.1 also appears in Collins's [Col21, Theorem 2.13 and its second proof, pp. 1068–1069]. The expression of this constant through Gauduchon quotients in Section 10.8 is discussed with the variational volume in Section 36.15.

The proof of the uniform estimate for the potential in Chapter 10 by weighted integration follows Cherrier and Tosatti–Weinkove [Che87, TW10]: a finite recurrence absorbs the torsion terms before the Sobolev iteration. The calculation of the torsion terms in the second-order estimate of Chapter 10 goes back to Cherrier; its form with completed squares and the maximum-principle argument follow [TW15, Sections 3 and 9, pp. 133–138 and 157–160].

The interpolation in Lemma 10.4.2 is the mechanism of Jiaogen Zhang and Xi Zhang's [ZZ26, v1, Proposition 4.1 and its proof, p. 16]. The fixed scalar operator is uniformly elliptic for a Hermitian metric as well, so the linear implication here does not require closedness. Their nonlinear Hessian application assumes a Kähler metric and a small product of its negative bisectional-curvature bound with the solution's oscillation. Its argument for data close to one in L^1 uses a stability estimate with $q > n/k$ and a uniform L^q bound; the smallness threshold depends on that bound. The interpolation lemma supplies neither this nonlinear trace estimate nor an arbitrary-L^1 stability criterion.

Prescribing the volume form of a Gauduchon metric leads to a different equation, which contains first-order torsion terms. The existence theorem Theorem 10.10.1 is due to Székelyhidi, Tosatti and Weinkove [STW17, Theorem 1.4] when γ is Gauduchon, again with the potential multiplied by 2π. The proof given here does not use that the auxiliary form γ is Gauduchon. The bound for the potential in Section 10.11 follows Székelyhidi's [Szé18, Propositions 10–11 and Remark 12, pp. 346–349], specialized to the determinant of the Gauduchon equation, and the second-order argument in Section 10.14 follows [STW17, Section 3]. The identities (10.21), taken from [STW17, Section 4], explain the structure of the first-order perturbation.

The gradient estimate combines a blow-up argument with a Liouville theorem. In (10.39), where A_j are the rescaling factors of the blow-up, the first-order term is kept before rescaling and tends to zero at the rate A_j^{-1}; the blow-up itself follows [Szé18, Section 6, pp. 366–368], where the equation has no first-order term. The Euclidean argument in Section 10.15 specializes [Szé18, Theorem 20, p. 364] to the Gauduchon cone. Once the equation is uniformly elliptic, Lemma 10.17.1 uses the Evans–Krylov theorem for variable coefficients in the form stated by Tosatti, Wang, Weinkove and Yang [TWWY15, Definition 3.2 and Theorem 3.4, pp. 444–445], who derive it from a theorem of Caffarelli.

For densities in L^p, the existence of continuous solutions of the equation with a normalization constant (Theorem 11.1.1) and of the exponential equation (11.1) is due respectively to Kołodziej and Nguyen [KN15, Theorem 0.1, p. 141] and to Nguyen [Ngu16, Theorem 0.1, p. 241]. Their operator is twice the one used here: for the exponent $\lambda > 0$ in (11.1), the conversion is $u = 2\varphi$, where φ is their solution, and the exponent in their equation is 2λ. These results require neither bounded mass nor positive lower volume. Compactness comes from local capacity estimates:

the arguments of [KN15, Section 5] and [Din16, Section 6, pp. 107–113] lead to Theorem 11.3.2, which is the statement of [KN15, Corollary 5.10, p. 156]. In the book it is deduced from the quantitative stability estimate of Proposition 11.3.1, discussed below. A common bound for the densities is essential here; the L^1 compactness of normalized potentials alone does not suffice.

The modified comparison principle, Theorem 11.2.1, is due to Kołodziej and Nguyen [KN15, Theorem 0.2]. The proof given here follows the finite induction of Guedj and Lu [GL22, Theorem 1.5], with larger dimensional constants.

For measures satisfying a power bound in capacity, Kołodziej and Nguyen [KN18, Proposition 2.4, pp. 2954–2956] give stability measured against the Monge–Ampère measure of the solution. The thin-sublevel estimate and backwards iteration of Lemmas 11.3.1 and 11.3.2 follow the capacity argument in [KN15, v2, Theorem 5.3 and Lemma 5.4, pp. 10–12], using their modified comparison theorem [KN15, v2, Theorem 2.3, pp. 5–6]. The geometric iteration in the book covers every power $1+\alpha$, $\alpha > 0$, including the total-capacity endpoint. Choosing the sublevel height proportional to the cube of the comparison reserve gives the exponent $\alpha/[3n(1+\alpha)+\alpha]$ in Proposition 11.3.1; the square-capacity case has exponent $1/(6n+1)$. The measure can be singular. The preprint [KN15, v2, p. 1] uses $\mathrm{dd}^{\mathrm{c}} = \mathrm{i}\partial\bar\partial/\pi$, twice our operator; a background ω of this book corresponds to their background 2ω with the same potentials, and their top products are 2^n times ours.

The threshold equation in Theorem 15.32.1 combines this iteration with the continuous solutions for bounded densities of Theorem 11.1.1; the root estimate is then the definition $\rho_h(g^{1/n}) = b_h(g)^{-1/n}$. Local L^1 compactness uses Demailly's Propositions 4.21 and 5.9 [Dem12, Chapter I, pp. 36–37 and 40]. The Chern–Levine–Nirenberg inequality on a compact Hermitian manifold is Nguyen's [Ngu16, Proposition 1.1, pp. 245–246]; the passage to the uniform limit uses the continuity of bounded products under convergence in capacity recalled in Sections 1.6 and 1.8. The logarithmic construction in Theorem 15.32.2 uses the exponential volume–capacity estimate (1.14) and gives a datum outside every higher L^p class with an exact continuous threshold solution. The added capacity hypothesis leaves the general root-tail and exact-equation questions open.

For a continuous solution whose measure is dominated by a Hölder reference, Theorem 15.4.2 combines the weighted Chern estimate with the square-capacity stability. The coordinated transform parameter and comparison reserve keep the minimizing radii comparable to the smoothing radius. The resulting exponent is sufficient for the range theorem; no optimal exponent is asserted. The argument assumes continuity of the solution.

The smoothness theorem Theorem 11.5.1 is the strictly positive, globally smooth-data case of Badiane, Lu and Zeriahi's [BLZ26, v1, Theorem 1.1, p. 2; Section 4.2, p. 17]. Their auxiliary second-order estimate is Proposition 4.1, pp. 13–17. The square of the prescribed potential in Lemma 11.5.1 absorbs the derivatives of the frozen density, and bounded exponential comparison identifies the uniform limit with the original solution. The argument then uses the concave uniformly elliptic extension of the logarithmic determinant and the Evans–Krylov theorem for variable coefficients [TWWY15, Definition 3.2, Theorem 3.4, and the paragraph following it,

pp. 444–445]. Only the strictly positive Hermitian background and global smooth positive density are treated here; their paper also treats degenerate backgrounds and data that are smooth only on a regular locus.

The exponential term gives a comparison principle even when the densities have no positive lower bound; Lemma 11.4.2 recovers [Ngu16, Lemma 2.3, p. 256]. Letting the exponent tend to zero then identifies the normalization constant through (11.11), as in [Ngu16, Corollary 0.2, p. 242]. This proves uniqueness of the constant, but not uniqueness of the limiting potential. The weighted mean (11.10), taken from [Ngu16, Proposition 2.5, p. 258], controls the normalization; here its lower bound follows directly from the estimate for sublevel sets.

Equations with prescribed singularities require relative envelopes and separate assumptions on the reference potential and on the ambient metric. The results of Alehyane–Lu–Salouf and of Xuan Li [ALS26, Li26] are therefore treated separately in Chapters 15, 33 and 34; their sources are discussed in Sections 36.10 and 36.20.

36.7 Uniqueness and selection

When the exponent is zero, uniqueness of the normalization constant leaves open the separate question of uniqueness of the potential. Degeneration of the density affects comparison, whereas the family of equations with positive exponent can select a particular solution of the critical equation. The argument by concentration of mass for densities restricted to a set, around (12.38), follows Lu, Phung and Tô [LPT21, proof of Theorem 3.7 and Remark 3.8, pp. 2033–2035]. The exterior-ball argument for (12.19) instead propagates the maximum, following Bardi and Goffi [BG19, Proposition 2.4 and Theorem 2.5, pp. 7–8].

The uniqueness theorem for bounded potentials, Theorem 12.4.1, concerns a continuous nonnegative density whose positivity set is connected and dense, and is proved with energy estimates on level sets and auxiliary test potentials. The mixed inequality and the passage to coefficient densities use the local calculus of Chapter 1, and the continuity of bounded solutions is obtained in the book itself, by a comparison argument, in Corollary 11.4.1. The smooth local Dirichlet problem needed for Proposition 12.4.3 is solved by Guan's theorem [Gua98, Theorem 1.1, pp. 687–688], whose hypothesis is satisfied by a strictly plurisubharmonic quadratic subsolution. The estimate on the real contact set then gives a constant that depends only on the fixed geometry and the L^2 bound, and not on the height of the test density.

For measurable densities that are bounded above and bounded away from zero, uniqueness is derived in Corollary 12.4.1 from the same energy argument on level sets. The earlier counterpart is [KN19, Corollary 3.9, p. 287, and Remark 3.13, p. 289].

36.8 Complex Hessian equations

The cone inequalities and the estimate with a prescribed potential in Chapter 13 follow Fang's [Fan26, v1, Lemma 2.2, Proposition 2.3 and Theorem 3.1, pp. 5–6 and 8–17]. The scalar shift in Lemma 13.1.1 is proved through real roots, including its endpoint. Polarized positivity then gives the mixed-trace and linearized coefficient bounds. The largest-eigenvalue argument keeps the Chern torsion terms and evaluates every derivative of the Hessian root at the original matrix, while a separate positive perturbation supplies its smooth lower support. In the quadratic weight used for the potential, the bound $-3A \le \Psi' \le -A$ determines the choice of the small eigenvalue parameter. The absolute mixed-form estimate below Hessian degree m and the torsion bounds for products are the mechanisms of Kołodziej and Nguyen [KN16, arXiv v2, Lemma 2.1, Lemma 2.3 and Corollary 2.4, pp. 3–6].

The Euclidean Liouville theorem Theorem 13.2.1 is due to Dinew and Kołodziej [DK17, arXiv v2, Theorem 3.2, pp. 8–10]. Their rescaling argument is in Section 2, pp. 6–7, of that preprint; Székelyhidi uses its Hermitian version [Szé18, Section 6, pp. 366–368]. The proof here is that of Theorem 10.15.1 for the cone $C_{n,m}$. It uses constant-cone linear tests and lower viscosity tests to pass to the maximal limit, so it does not require a variable-background Hessian-product convergence theorem, and its averaging argument obtains the uniform directional energy bound with a fixed dependence on the smaller averaging radius. In the rescaling proof of Proposition 13.2.1, interior estimates for the Euclidean Laplacian then bound the real Hessian in L^p for every finite p.

Smooth proper existence and admissible approximation are the mechanisms of Kołodziej and Nguyen [KN16, arXiv v2, Theorem 3.17 and Lemma 3.20, pp. 23–25] and Guedj and Lu [GL25, arXiv v1, Theorem 1.7, pp. 4–5]. Here the smooth trace and gradient estimates first close the proper continuity path. Penalized proper equations then approximate every continuous admissible potential uniformly. This construction precedes the definition of continuous Hessian products and does not use a domination theorem. For a positive exponential coefficient a, rescaling both the metric and the potential reduces the equation to coefficient one. The maximum comparison in Proposition 13.7.1 gives the coefficient a^{-1} for continuous data. The existence assertion in Theorem 13.7.2 then follows from a contraction on the continuous functions with values between the constant barriers, followed by the continuous regularity theorem.

The decreasing approximation theorem Theorem 13.5.1 extends the construction to bounded potentials and recovers the bounded case of Kołodziej and Nguyen's [KN16, arXiv v2, Lemma 3.20, p. 25]. The characterization of bounded admissible potentials by upper tests corresponds to the viscosity description of Guedj and Lu [GL25, arXiv v1, Section 1.1.1, pp. 3–4], which is formulated through smooth strictly elliptic linear operators. Here successive smooth obstacles are chosen below the preceding admissible approximant, and the penalty error is then made smaller than their fixed gap. This gives a decreasing sequence before defining any product of the bounded potential. The same squeeze proves Proposition 13.6.2 after the continuous obstacle envelope has been constructed.

Continuous mixed products and their uniform continuity are [KN16, arXiv v2, Proposition 2.11 and Corollary 2.12, pp. 10–11]. The contact comparison, nonvanishing and domination arguments of Section 13.6 follow Guedj and Lu [GL25, arXiv v1, Lemma 1.10, Theorems 2.1–2.2 and Corollaries 2.3–2.4, pp. 6–7 and 8–9]. The continuous obstacle envelope is constructed by the same penalty solutions, giving continuity and contact support directly. Differentiated background factors are interpreted using the signed lower-degree products. In the mass nonvanishing proof, the telescoping form has degree $n-1$ and keeps each power of the original potential. The normalized linear Green estimate rules out an open set on which the stretched difference envelopes tend to $-\infty$.

The bounded recursion in Section 13.8 follows the construction of Kołodziej and Nguyen [KN26, Lemmas 3.1–3.2, Theorem 3.3 and Proposition 3.7, pp. 125–129]. Their local cone is imposed on $\mathrm{dd}^{\mathrm{c}}u$, whereas the book imposes it on $\omega+\mathrm{dd}^{\mathrm{c}}u$. The proof here therefore expands each raw factor into a shifted admissible factor minus the smooth background before applying the coefficient estimates. The first limit is in the first potential, and the final limit is in the last potential; the top unwedged current remains distributional. The positive shifted wedge has measure coefficients, and Proposition 13.8.2 expresses the derivatives of the shifted currents through derivatives of the background. Permutation invariance does not follow from this ordered construction alone, as Kołodziej and Nguyen also point out in [KN26, Remark 3.6, p. 128]. Their later capacity and weighted-convergence results concern this unshifted local class. The elementary correction in Lemma 13.8.2 places $u+\rho$ in that class while keeping the same Hermitian metric. It preserves the nonclosed background terms and does not itself prove the capacity quasicontinuity needed for higher-degree simultaneous convergence.

The Sobolev capacity in Section 13.9 uses instead the global conformal Gauduchon multiplier of Theorem 4.1.1. The energy identity for u^2 bounds the Sobolev norm, and first-order integration by parts keeps the $\mathrm{d}T$ term of the nonclosed Gauduchon test. Sobolev truncation and mollification prove the capacity bound for the positive linear trace; the second-cone inequality transfers it to every first-Hessian coefficient. Tail convex combinations from Mazur's lemma identify quasicontinuity with the canonical potential pointwise. The cutoff absolute-coordinate potentials in Example 13.9.1 show that a two-Hessian measure can charge a smooth real codimension-two set of zero Sobolev capacity. Thus this capacity estimate cannot supply the higher-degree argument.

The local capacity and decreasing-convergence conclusions are studied in [KN26, Corollaries 3.18–3.19, p. 137, and Lemma 5.1, pp. 143–145]. In Section 13.10, smooth competitors suffice to control the coefficients of the book's ordered currents. A fixed regularized maximum localizes decreasing smooth approximations near the boundary. The smooth energy variation keeps every derivative of the Hermitian background and gives convergence in these local capacities. Outer-open coefficient estimates and bounded continuous extension then prove the weighted convergence needed for Theorem 13.10.1. This proof uses the global approximation already established in the chapter and avoids an identification of gradients in a singular weighted Sobolev completion. It gives decreasing convergence for the book's shifted

class. At degree m the unwedged current converges in distributions, while its positive wedge Q_m converges as a Radon measure.

The coefficient estimate in Lemma 13.1.6 uses Newton's inequality on the list with two coordinates deleted. A related bound appears in [KN26, equation (2-8), p. 111]. The torsion estimate at degree $m-1$ corresponds to [KN26, Lemma 2.4 and Corollary 2.5, pp. 113–114]. Its proof takes absolute values of the diagonal wedge coefficients, which are deleted symmetric functions, rather than of their individual eigenvalue products, and compares them with the nonnegative gradient coefficients. This extends the smooth capacity induction to degree m. Positive shifted currents are then dominated by the corrected local Hessian traces, and Theorem 13.10.2 gives the weighted decreasing convergence studied in [KN26, Lemma 5.1(b), pp. 144–145]. The unwedged current D_m remains only distributional.

The strict locality and maximum inequality in Proposition 13.11.1 and Theorem 13.11.1 correspond to [KN26, Corollary 5.2, p. 145], which concerns local unshifted Hessian products. The arguments here apply to the book's global shifted products. Continuous functions of decreasing potentials supply the strict-set cutoffs; a decreasing constant shift includes the contact set in the measure inequality.

The torsion-weight identity in Lemma 13.11.2 extends the smooth integration-by-parts argument using the local capacity of degree $m-1$ from Section 13.10. Proposition 13.12.2 follows from Theorem 13.12.3, as Corollary 11.2.1 does in the Monge–Ampère case. This is positivity of each individual mass; it does not supply a uniform positive lower bound over the admissible class.

The bounded mass formula in Lemma 13.12.1 extends the integration-by-parts calculation of Theorem 2.2.1 using the recursive currents. The necessity argument in Theorem 13.12.1 is the polarization mechanism of [Chi24, Theorems 1–2, pp. 576–577], which also gives Theorem 2.2.2. At Hessian degree m, only the powers $n-m,\dots,n-1$ occur. Sufficiency follows from the mixed maximum inequality, while Proposition 13.12.1 keeps the exact total-mass difference on arbitrary backgrounds. The example in Example 13.12.1 extends the smooth varying-mass torus calculation of Example 2.3.1.

The mixed torsion comparison and weak modified comparison in Lemma 13.12.2 and Theorem 13.12.2 use the mechanism of Gu and Nguyen [GN18, Lemmas 3.8–3.10, pp. 1197–1200]. Their local construction of bounded products uses a reference metric that is conformal to a Kähler metric on a small ball. Here the global decreasing approximation and weighted torsion currents give the integration-by-parts identity for the chapter's shifted products. A finite mixed-degree induction controls the error by diagonal traces; rescaling the upper potential then absorbs the lower traces on a sufficiently small sublevel set.

The pointwise order in Lemma 13.1.8 uses the canonical linear subharmonic representatives of the admissible class. The positive volume term from Lemma 13.11.1 and weak modified comparison prove Theorem 13.12.3 and Corollary 13.12.1 on arbitrary Hermitian backgrounds. The same strict comparison proves the logarithmic-data estimate in Proposition 13.12.3 for a fixed finite Radon right-hand measure. Together with smooth proper existence, it identifies the bounded solution in Corollary 13.12.2

without a separate continuity argument. The continuous-potential domination results in Propositions 13.6.3 and 13.6.4 use the earlier envelope argument.

The conclusion of Theorem 13.7.1 is the continuous-solution case of Fang's [Fan26, v1, Theorem 4.1, pp. 17–18]. The theorem here assumes continuity of the given admissible weak solution; Fang's statement treats bounded solutions and also invokes a continuity result for them. The uniformly elliptic concave extension uses only a bounded complex Hessian. Its regularity input is the continuous-viscosity version of the Evans–Krylov–Caffarelli estimate in Tosatti, Wang, Weinkove and Yang [TWWY15, Theorem 3.4 and the paragraph following it, pp. 444–445]. It applies directly to the auxiliary limit. The general bounded-solution continuity theory and results for degenerate backgrounds are separate from this argument.

For fully nonlinear concave equations, the uniform estimate under a quantitative subsolution condition is due to Guo and Phong [GP24, v1, Theorem 3.1 and its proof, pp. 8–11]. The strict asymptotic gap of an admissible reference, the matrix bound, the linear stability estimate and the uniform compactness in Section 13.13 follow Guo and Song [GS26, v1, Lemma 2.1, Proposition 2.8 and Lemma 2.9, pp. 4–11]. In the book, the zero-order estimate uses the real contact-set construction of Proposition 10.11.1, and Theorem 11.3.1 bounds both the auxiliary potential and its normalization constant; with these inputs the constants do not depend on a positive lower bound for the right-hand sides. Writing the potential of these sources as $u/(2\pi)$ converts their convention for smooth curvature forms into the normalization of this book.

The gradient estimate in Section 13.14 is Guo and Song's [GS26, v1, Theorem 1.1 and Proposition 3.1 with Lemmas 3.2–3.6, pp. 11–17]; the earlier alternative for convex level sets is Székelyhidi's [Szé18, Propositions 5–6, pp. 342–344]. The tangent-hyperplane argument in Lemma 13.14.1 uses the common strict gap to keep the constants uniform at every positive level. The differentiation is carried out for $U = u/(2\pi)$, where u is the solution, and the final finite net of smooth comparison functions is provided by Corollary 13.13.1.

36.9 Envelopes and comparison

Narzillaev studies weighted extremal functions of a compact set $K \subset \mathbb{C}^n$ with a bounded weight ψ on K and logarithmic growth $\delta \log^+\|z\|$, $\delta > 0$. The concavity in the growth parameter and the alternatives for regular contact in Theorem 14.1.3 are stated in [Nar26, v1, Lemma 3.1, p. 6, and Theorem 1.1, p. 2]; the proof given here also shows directly, by scaling, that the set of parameters at which the unregularized envelope attains the obstacle on K is empty or a ray. Lemma 14.1.1 and Proposition 14.1.5 derive the correspondence with compact envelopes on projective space, the boundedness of the regularized envelope and its agreement with the envelope outside a pluripolar set from the compactness and extension results proved earlier in the book. In the book's convention, the growth $\delta \log^+\|z\|$ corresponds to the projective background $(\delta/2)\omega_{\mathrm{FS}}$ and to the obstacle $\psi - \delta \log(1 + \|z\|^2)/2$; Narzillaev's operator $\mathrm{dd}^{\mathrm{c}} = \mathrm{i}\partial\bar\partial/2$ is π

times the book's. None of the theorems of that preprint on products, plurithinness or weighted counterexamples is used in these proofs.

For the zero obstacle, Corollary 14.1.1 gives the case of compact sets of the classical fact, recalled in [Nar26, v1, Section 2, equation (4), p. 4], that removing a pluripolar set does not change the extremal function. For bounded weights, the corollary is deduced through the correspondence with $\mathbb{P}^n$ from Proposition 1.19.2, which treats compact sets in an arbitrary smooth big background by the correction of pointwise competitors in Lemma 1.19.3.

The estimates for increments in the growth parameter in Lemma 14.1.2 are [Nar26, v1, Lemma 3.1, p. 6], and at a point of regular contact Proposition 14.1.6 and Corollary 14.1.2 recover [Nar26, v1, Proposition 3.2, p. 7]. The proof given here keeps the estimates for the initial and the later slopes at every point of $\mathbb{C}^n$ and every positive growth parameter, including the uniform error for large increments. The representation by probability measures and the attained derivative in Theorems 14.1.4 and 14.1.5 come from [Nar26, v1, Theorems 3.3–3.4, pp. 8–9]; the same argument with the finite envelope extends the representation to every point and every positive parameter, and the attained derivative is the formula for the right derivative in Corollary 14.1.3. The passage to weak limits of these probability measures uses the continuity of the weight. The joint response to changes of the weight and of the growth, and the criteria for differentiability, in Lemma 14.1.3, Proposition 14.1.7, and Corollaries 14.1.3 and 14.1.4, follow directly from the attained representation and the compactness of its set of active pairs. Example 14.1.1 gives an explicit corner in the dependence on the weight, on the closed unit disk.

The relative extremal function of Definition 1.6.1 follows the convention for compact sets in [Nar26, v1, Section 2, equation (8), p. 5]. Lemma 14.1.5, which shows that the relative extremal function equals -1 at a point exactly when the global extremal function vanishes there, is proved by an explicit logarithmic gluing and the logarithmic pole barrier of Lemma 14.1.4. Lemma 14.1.6 and Theorem 14.1.6 give [Nar26, v1, Lemma 4.1 and Theorem 4.2, p. 10], including the local strictness hypothesis $\mathrm{dd}^{\mathrm{c}}\Psi \geq c\omega_0$ near the point, with $\omega_0 = \mathrm{dd}^{\mathrm{c}}\|z\|^2$, a global weight Ψ in the given growth class extending the weight on K, and the closed compact pieces inside the ball. Narzillaev's operator and Euclidean form are both π times those of the book, so the coefficient c and the bound for the defect are unchanged. The pointwise consequence for the Hessian in Corollary 14.1.5 is the observation following that theorem on p. 10.

The planar disks and the quantitative barrier of Lemma 14.1.7 come from [Nar26, v1, Lemma 6.1, pp. 12–13]. Proposition 14.1.8 proves part of the product construction in [Nar26, v1, Lemma 6.2, pp. 13–14], namely the contact defect along the exceptional fiber and the failure of plurithinness at the interior point specified there. The logarithmic series is a decreasing limit of subharmonic functions on disks, and the lower bound for the envelope follows from the invariance under pluripolar exceptions in Corollary 14.1.1. The proof needs neither a general product formula nor a classification at all boundary points.

The formula on normal fibers for the pluricomplex Green function with a divisorial pole, Proposition 1.19.3, is the case of the unit ball of Dao, Nguyen and Tran's

[DNT26, v1, Proposition 4.2, p. 25], with logarithms normalized as $\log|\zeta|^2$ in the normal coordinate ζ; the proof uses the maximum principle in one variable. The bounded minorants of Proposition 1.19.4 converge at every interior point, including the points of the divisor, with an explicit error bound. This is a special model for the approximation by bounded envelopes in [DNT26, v1, Theorem 1.7, p. 5], whose general result assumes an inward deformation in a sector and uses B-regularity for continuity. The calculation of boundary cluster values in Example 1.19.1 concerns the trace of the divisor on the boundary, whereas that theorem asserts continuity inside the domain.

Contact sets connect the order of potentials with the order of their Monge–Ampère measures. For semipositive big backgrounds, Theorem 14.1.2 is due to Alehyane, Lu and Salouf [ALS26, Theorem 2.2, p. 8], who quote it from their earlier work and remark after the statement that the envelope may have infinite mass; the passage from bounded to unbounded potentials in the book removes the bigness assumption. For a general big background, Boucksom, Guedj and Lu [BGL25, Theorem 2.9 and Proposition 2.11] prove that the measure of the envelope is concentrated on the contact set under bounded mass, and local replacement together with a strict singular barrier gives Theorem 15.7.2 without that assumption.

The measure estimate for envelope fixed points, Proposition 14.1.4, is the fixed-point case of [ALS26, Proposition 2.13, p. 14], stated there for semipositive big backgrounds. Let θ be smooth and semipositive, and let $\varphi \in \mathrm{PSH}(X,\theta)$ satisfy $P_\theta[\varphi] = \varphi$ for the ordinary envelope of Definition 1.19.1. The proof given here uses the identity $P_\theta(\varphi+1,0) = \varphi$, where $P_\theta(\varphi+1,0)$ is the largest θ-plurisubharmonic function lying below $\min(\varphi+1,0)$ (Section 1.19); contact with the first obstacle occurs only on the pluripolar set $\{\varphi = -\infty\}$, so only contact with zero contributes to the measure. For bounded potentials, the proof of the domination principle by envelopes, Theorem 14.2.1, is due to Guedj and Lu [GL22, Proposition 2.8].

Difference envelopes give a relative class in which singular potentials can still be compared. The construction appears for semipositive big backgrounds in [ALS26, Proposition 2.4 and Lemma 2.6, pp. 8–9] and for arbitrary big backgrounds in Xuan Li's [Li26, v1, Section 2.1]; the estimate for sublevel sets gives a common upper bound for the competitors, so that the envelope is θ-plurisubharmonic as soon as its family of competitors is nonempty. The relative envelope class of Definition 14.4.1 follows [ALS26, Definition 3.2, p. 16], and the proof of its closure under convex stretching, Lemma 14.4.1, is the argument of [ALS26, Proposition 3.5, p. 19].

The condition (14.20) on the reference and its consequence, positivity of the mass of each individual potential in the class, appear in [ALS26, Proposition 2.11, the discussion after Definition 2.14, and Remark 3.9, pp. 13, 15 and 20]; that paper does not assume bounded mass for its equations with L^p data in a fixed ordinary type. This positivity for individual potentials must be distinguished from a positive infimum of the mass over the whole ordinary singularity type. The mechanism giving the lower bound for the mass in Theorem 14.5.1 is that of [ALS26, Proposition 3.8, p. 20].

For lower semicontinuous obstacles the contact argument requires an approximation. The proof of Proposition 14.1.2 follows Guedj and Lu [GL22, Theorem 2.3] and uses the Bedford–Taylor convergence theorem for increasing bounded sequences

[BT82, Theorem 7.4, p. 27]; smooth local corrections and an expansion against the remaining background extend it to nonclosed backgrounds.

36.10 Equations with prescribed singularities

Relative envelopes allow the solution of an equation to keep prescribed singularities while the measure on the right-hand side changes. Equations with L^p data in a prescribed ordinary type, equations with general measures, and equations under mass bounds on the metric, discussed in Section 36.20, use this framework in different ways. They differ both in the estimates available and in the singularity type of the solution obtained.

36.10.1 Equations of prescribed ordinary type

The equations of prescribed ordinary type in Theorem 15.11.1 are [ALS26, Theorems 5.4 and 5.7, pp. 34 and 37]. Replacing their reference volume ω^n by dV multiplies the density by a smooth positive function and preserves its L^p class, and a change of the normalization of the complex Hessian by a positive constant is absorbed by rescaling both the potential and the exponent.

The two constructions of [ALS26, Lemmas 5.5–5.6, pp. 34–35] are the relative estimate Theorem 15.11.2 and the rooftop supersolution Lemma 15.11.2; together they keep the solution in the ordinary type of the reference. For a big background, the argument with bounded subsolutions of Guedj and Lu [GL23, Lemma 2.1, pp. 270–271, and Theorem 3.4, p. 279] gives Lemma 15.2.1 and Theorem 15.2.1.

The trace calculation with a barrier in Section 15.18 follows Guyett and Mc Gowan [GMG26, v1, Proposition 3.2, pp. 7–12]. Their geometric setting is a small crepant resolution, in Fujiki's class $\mathcal{C}$, of a normal projective Calabi–Yau variety, and their general Theorem 3.1 treats a closed semipositive big background with an upper bound for the mass. In the book, the arguments with bounded subsolutions and exponential comparison of Lemma 15.2.1 and Theorem 15.2.1 give uniform bounds for the potentials and the scalars when the density is a fixed smooth positive function. The trace argument keeps the torsion of the perturbation and also allows a fixed nonzero Chern–Ricci form. Its conclusion is convergence of the potentials along subsequences, in a semipositive background with a strict barrier that is smooth off its closed pole set. The assertions of that paper about metric completions and conifold contractions are not part of this argument.

The bundle and the contraction in Section 15.18.1 form the standard model of a node used in [GMG26, v1, Equation (2.1), p. 4, and Section 4, pp. 15–16]. The logarithmic curvature identity (15.50) gives a strict barrier for this model directly, and a cutoff in a positive collar makes it global. This proves the convergence of the potentials and the assertion about the area of the exceptional curves for these

local resolutions, without using a classification of small resolutions or a theorem on metric completions. Since the canonical bundle is torsion, its local holomorphic trivializations give the Chern–Ricci-flat density.

The equation on the singular space, (15.52), is obtained through the general correspondence of Theorem A.2.1 between potentials on a resolution and potentials on the singular space defined by disk inequalities. For closed backgrounds, the identification in [GMG26, v1, Lemma 3.1 and the proof of Corollary 1.1, pp. 13–14] uses the equation on normal spaces and the estimate for adapted measures of Eyssidieux, Guedj and Zeriahi [EGZ09, Theorem 6.3 and Lemma 6.4, pp. 632–634]. In the standard model of a node, (15.51) gives in Lemma 15.18.4 the exact range $p < 3$ of integrability and the weak L^3 endpoint, for every smooth positive density on the resolution. The bounded energy identity already used in Theorem A.6.2 proves Theorem 15.18.2, using smooth strict positivity only away from the exceptional curves.

The arguments for the metric completion and the distance in Lemma 15.18.6 and Proposition 15.18.1 are those of [GMG26, v1, Lemma 5.2, pp. 23–24, and Proposition 5.1, pp. 24–26]. Compact collars in the regular part separate the points added in the completion, and the first and last visits of a path to the finitely many contracting regions give a distance error controlled by these regions; uniform convergence of the metrics on each compact subset of the regular part suffices. The hypotheses on the diameters of punctured neighborhoods in the limit and on the intrinsic diameters for the approximating metrics are stated explicitly; the conifold estimates and the asymptotics of cones needed to verify them are separate geometric inputs.

For curves, [GMG26, v1, Lemma 4.1, p. 17] bounds the diameter of $\mathbb{P}^1$ by a multiple of $A^{1/3}$, where A is the area, under a uniform upper bound for the metric. Averaging over straight coordinate segments and splitting the kernel given by the reciprocal of the distance give the sharper logarithmic estimate of Lemma 15.18.7 on every fixed compact Riemann surface. The radial example in Example 15.18.1 has the same order, and rescaling it shows that the area alone does not control the diameter. The two radial paths and the connection through the zero section in Proposition 15.18.2 give the mechanism of [GMG26, v1, Proposition 4.2, pp. 20–21], with the uniform bound on the restriction to the curve and the radial bounds stated as explicit hypotheses.

Strict singular barriers extend the comparison argument to potentials with minimal singularities. The development starting in Section 15.12 follows [BGL25, Section 4], where the bounded mass assumption of the earlier sections of that paper is dropped. The equations with minimal singularities, culminating in Theorem 15.15.2, are [BGL25, Theorems 4.6–4.7]. The solution has finite mass, although the masses in its ordinary singularity type are not assumed to be uniformly bounded.

36.10.2 General measures in semipositive backgrounds

The local decomposition of measures used in Lemma 1.6.2 is Cegrell's [Ceg98, Theorem 6.3, p. 209], for compactly supported measures on a bounded hyperconvex domain in $\mathbb{C}^n$, $n \geq 2$, the standing hypothesis of [Ceg98, Section 2, p. 188]; for a non-pluripolar measure its pluripolar part vanishes. The global constructions in Lemma 15.23.1 and Proposition 15.23.1 patch these local representations together.

The bounded-subsolution theorem Theorem 15.9.1, taken from [ALS26, Theorem 5.1, pp. 27–28], solves the exponential equation whose right-hand side has a bounded density with respect to the Monge–Ampère measure of a bounded potential. Truncating the density in such a representation of a general measure brings this theorem into the construction for general measures, and domination then allows passage to the limit. The domination theorem for finite measures, Theorem 33.2.1 in Chapter 33, is the case of finite μ_u in [ALS26, Theorem 3.14, p. 22]: the fixed finite measure prevents collapse, whereas the other members of the envelope class need not have finite mass. The proof of Theorem 33.2.2 also allows the measure being compared to be infinite. The singular strict domination principle, Theorem 15.1.1, follows [ALS26, Theorem 3.11, p. 21].

The homogeneous range result Theorem 15.24.1 is the sufficient direction of Kołodziej and Nguyen's [KN18, Theorem 1.3, p. 2952; proof pp. 2957–2963]. Their weighted regularization estimate is [KN18, Lemma 2.8, p. 2957]. The proof here first approximates a Borel density by continuous functions and then chooses smooth references diagonally. Common Hölder bounds give uniform capacity estimates and a weighted modulus for the varying solution measures; bounds above and below for the normalization constants prevent degeneration before the uniform limit is taken.

The radial bounded product in Lemma 1.6.7 is a direct Bedford–Taylor calculation: the logarithmic derivative of a smooth radial maximum fixes the spherical mass. Summing these local references with geometric weights and controlling a common cutoff gives Proposition 15.25.1. The homogeneous range theorem yields Corollary 15.25.1, with a normalized Hölder solution and an unknown positive coefficient. The bounded-reference theorem yields Theorem 15.25.1 for bounded densities against the fixed spherical probability, including singular data on backgrounds with zero lower mass. This is an individual geometric sufficient condition, distinct from solvability for every non-pluripolar measure and from the exact density-threshold question.

Passing to the limit in the measures requires that the lower envelopes do not collapse. The continuity theorem for measures dominated by a fixed measure, Theorem 33.3.1, is [ALS26, Lemma 4.2 and Theorem 4.1, condition (2), p. 24]; in this case positive lower mass prevents collapse, and no upper bound on the mass is used. The argument with a common envelope preceding Corollary 33.4.1 gives instead the first assertion of [ALS26, Lemma 4.2, p. 24], which concerns approximants of the ordinary singularity type of the reference and assumes no positive lower mass $\nu_-(\theta)$. Under $\nu_-(\theta) > 0$, Proposition 33.4.1 also allows approximants in $\mathcal{E}_\varphi$ of other singularity types.

A common lower barrier also gives Corollary 15.8.1; for approximants in the ordinary singularity type of the reference, this is the corresponding case of [ALS26, Theorem 4.1, p. 24]. In the corollary the approximants need only lie above a translate

of the reference and below a constant, and the reference need not be an envelope fixed point; the condition in a smaller background suffices. Integrals of bounded potentials against varying mixed measures are handled by the torsion calculation in Lemma 1.23.3, which follows Kołodziej and Nguyen [KN25, Lemma 2.3, pp. 116–118].

The resulting equations with general finite non-pluripolar measures, Theorem 33.1.1, are [ALS26, Theorem 5.8 and Corollary 5.9, pp. 38–40]. The constructions with a minimal and with a prescribed reference in Sections 33.5 and 33.6 use the approximations of the density from that proof, and they assume both the condition (14.20) in a smaller background and positive lower mass. The solutions have finite mass, but for arbitrary measures they need not differ boundedly from the reference. For a Hermitian background and the minimal reference, Corollary 33.7.1 identifies the scalar in the equation with the subsolution threshold of Theorem 15.27.1, that is, the largest coefficient for which the scaled density admits a subsolution. The question that remains, Problem 35.1.1 in Chapter 35, concerns zero lower mass. The weighted addition theorem Theorem 15.10.1 is [ALS26, Theorem 5.2, p. 30]; the proof given here localizes the passage to the equation.

36.11 Normalization and adjoint measures

The normalization constant can be studied through its adjoint probability measures even when the potential itself is not unique. The differential method for smooth solutions appears in Collins's work [Col21]; Theorem 16.1.1 replaces differentiation of smooth solutions by mixed products and domination.

Lemma 16.2.1 combines the interior L^p estimates for continuous coefficients and the strong maximum principle for strong solutions [GT01, Theorems 9.11 and 9.6] with the method of continuity [GT01, Theorem 5.2] and compact-perturbation index theory; it does not differentiate the coefficients.

The continuous-Hessian extension in Theorem 16.2.4 uses local bounded Bedford–Taylor continuity and the finite background expansion to linearize the equation along signed potential variations. The supporting-density estimate of Corollary 27.7.2 then allows Sobolev duality with Lemma 16.2.1; the directional derivative is the convex-analytic identity already proved in Proposition 16.2.2. The product formula in Proposition 16.2.3 uses curve Green inversion and the smooth Gauduchon multiplier on the other factor. The non-Dini example uses the same Hopf torsion calculation as (9.1), in complex dimension two.

For the mass quotient and the closed examples in Section 16.10, existence and uniqueness of solutions in a prescribed singularity type are due to Darvas, Di Nezza and Lu [DDNL18, Theorems 1.4(i) and 4.29, pp. 2052 and 2079–2080], and the invariance of the mass under bounded changes of the potential is their [DDNL18, Proposition 2.1, p. 2055]. Their hypotheses are a compact Kähler manifold, a smooth closed big class, a model reference with small unbounded locus, and positive mass. The decomposition $u = \varphi + w$ of an ordinary divisorial type with a positive residual

form, Theorem 16.9.1, concerns the ordinary type itself; it does not assert an identity of model envelopes for every weighted sum of divisors with real coefficients.

The dictionary between toric potentials and convex functions used in Section 16.11 is that of Berman and Berndtsson [BB13, Sections 2.1–2.3, especially Proposition 2.3 and Lemma 2.5, pp. 660–664]. The angular coefficient in (16.24) is computed directly, in the book's conventions for the squared logarithm and the operator. The argument with finite envelopes establishes the identity for ordinary models, and a common translated simplex gives both the witness in a smaller background and the volume estimate in terms of the Hausdorff distance. The two families in Sections 16.11.3 and 16.11.4 satisfy these conditions explicitly; since their references have small unbounded locus, the theorem for prescribed singularity types in closed classes cited above applies. The blowup chart and the normal line of the exceptional divisor used to compute the rank and the mass of the residual are taken from [Dem12, Chapter VII, equation (12.2) and Proposition 12.4, pp. 356–357]. These computations concern fixed modifications and closed backgrounds.

The description of the residual in Proposition 16.2.4 rests on Theorem 16.9.1 and the products of bounded plurisubharmonic functions of Bedford and Taylor [BT82, Section 2, pp. 3–10]. Integration by parts for bounded potentials against a closed positive current is established in [BEGZ10, Theorem 1.14 and Lemma 1.15, pp. 214–217]. In Theorem 16.2.5, the two radial profiles give the coefficients of the divisors before the limit (16.9) of the squares is taken. This distinguishes the relative residual adjoint from the classical adjoints of smooth potentials whose exponential densities concentrate.

For the normalization on split products in Lemma 15.29.1, the bounded equation of [KN15, Theorem 0.1, p. 141] on the closed Kähler factor and Stokes' theorem give the scalar of every bounded factor density, and the original truncations determine the threshold of every integrable factor density for every smooth reference volume. The radial flux in Lemma 15.29.2 and the product formula (15.85) then give Theorem 15.29.1. The elliptic factor is the threefold of Section 27.18, whose balanced metric has volume 24π by Corollary 27.18.3; this gives the fivefold equation with constant 240π. In Theorem 15.29.2, threshold attainment on the closed factor, together with its non-pluripolar mass bound from Stokes' theorem, makes the lower measure an equality for every integrable factor density. The root norm formula of Proposition 15.31.5 then follows from this threshold formula and the definition of the root functional as a power of the threshold; it identifies a closed subspace of the space of density roots with uniformly vanishing tails. The radial family supplies roots outside every higher L^r class, with explicit logarithmic tails. For arbitrary densities on the product that are merely integrable and have positive threshold, the existence of an exact solution remains the problem of Problem 35.1.1.

The same radial flux gives the criterion for exponential growth in Theorem 15.34.2. Disjoint annuli in projective space and a rapidly convergent smooth baseline adapt the concentration in Theorem 15.34.3 to every prescribed sequence of base densities whose scalars collapse. The bounded equations on the products determine the scalars of the components, and for every positive exponent the exponentially weighted pole absorbs their sum. The construction of greatest solutions then gives exact solutions at

zero threshold, which are necessarily unbounded, in particular on the elliptic products of dimensions four and five. The explicit barriers there have masses 96π and 240π, whereas the exact solutions have their own finite positive masses. A general criterion for solvability at zero threshold remains open.

That the set of solvable exponents is an upper interval, Proposition 15.34.2, follows by translating a subsolution downward before increasing its exponent. For the collapse of mass at zero threshold in Proposition 15.34.3, normalized exponential damping gives an L^1 approximation of the fixed density, and upper semicontinuity of the threshold forces the normalization scalars to tend to zero. This controls the masses of the exact solutions without identifying the limits of the normalized potentials.

The criterion for attaining a positive endpoint, Proposition 15.34.4, combines the compactness of normalized potentials with the closure of subsolutions under lower bounds for the density and the existence of greatest solutions. Its common upper barrier comes from the equation of Theorem 11.1.2 with the truncated, bounded density, together with singular strict domination. The corrected order in Lemma 15.34.3 gives a decreasing family as the exponent decreases to a positive solvable exponent. Monotone convergence in capacity and the closedness theorem for positive exponents identify the limit from the right in Proposition 15.34.5, and maximality of the greatest solutions identifies the limiting potential. Approach from below uses the regularized increasing envelope, giving the exact left limit of Proposition 15.34.6. Integrating the corrected family gives a bounded increasing real function on each compact exponent interval. Its jumps detect the difference between the left limit and the greatest solution, which proves Proposition 15.34.7.

For the non-L^p radial family, Lemma 15.35.1 identifies the greatest solutions on the full product through the bounded factor equations and singular strict domination. The proof of Theorem 15.35.1 uses their cumulative fluxes to determine the whole normalized limit, including when the density parameter varies. Local uniform convergence outside the fixed pole set and the Hermitian capacity estimate give capacity convergence on both factors and the product. No general full-mass stability theorem is needed for this radial argument. The finite or divergent first moment determines scalar centering in Proposition 15.35.1. The order characterization of separated critical subsolutions in Corollary 15.35.1 additionally uses Dinew's closed-class full-mass uniqueness theorem [Din09b, arXiv v1, Theorem 1.2, pp. 2 and 5–8]; this input is applied only on the projective factor. It neither proves uniqueness of arbitrary nonclosed critical potentials nor supplies a theorem for varying singular references.

For the logarithmic product references, Proposition 16.13.1 computes the ordinary envelope on the curve and identifies the product envelope by restriction to slices. Divisorial removal transfers the exponential equation and its greatestness to a bounded equation in the positive residual background, as in Lemma 16.13.1. The Green operator on the curve gives uniformly bounded residuals by contraction for small exponents. Its uniformly integrable logarithmic kernel then determines the joint limit and the scalar error in Theorem 16.13.1. The centering uses the critical scalar of each varying datum. The separated order characterization in Corollary 16.13.1 follows from closed curve mass and the distributional equation; maximality of the coefficient in the full

one-sided class is the case of Theorem 16.9.3 with the residual form of the logarithmic reference. This argument uses the earlier bounded existence and domination theorems and does not impose a mass property on the Hermitian factor.

For the product surface in Section 16.13.4, the logarithmic model and Green residual are those of Proposition 16.13.1 and Theorem 16.13.1. The continuous positive residual allows smooth perturbations of the singular solution, and the supporting inequality on all continuous data makes every supporting probability stationary for the weak linearization. Spherical harmonics reduce uniqueness to the signed-measure resolvent lemma Lemma 16.13.2, whose proof uses the round sphere heat operator. This gives Theorem 16.13.2 without any regularity assumption on the stationary probability. The approximation statement Proposition 16.13.2 uses the already proved conformal Gauduchon theorem Theorem 4.1.1 for each smooth residual; uniform tensor convergence transfers the operator and then the weighted products. The explicit pole-filling family in Example 16.13.1 shows why keeping the divisor fixed is part of this approximation statement.

The one-sided divisorial correspondence of Lemma 16.9.1 extends the ordinary correspondence of Theorem 16.9.1 by the nonclosed divisorial-removal lemma and locality of the extended measures. A uniform L^1 bound identifies the ordinary envelope type and its explicit smaller-background competitor. The whole one-sided threshold in Theorem 16.9.3 then follows from Theorem 15.27.1; its smooth attainment uses Theorem 10.1.1, rather than a new prescribed-singularity existence theorem. Strictness follows from the elliptic cofactor identity off the divisor and the strong maximum principle [GT01, Theorem 3.5]. The obstruction in Corollary 16.9.1 is consistent, for $n \geq 2$, with the stronger rigidity of Corollary 27.8.1, while Example 16.9.1 fixes the Fubini–Study factor to compute the precise restricted coefficient. The inverse realization problem for general densities and annihilating faces remains separate.

The curve-measure equations of Section 15.30 begin with the normalized logarithmic kernel on the projective line. Its distributional curvature and the degree-one extended measure give Proposition 15.30.1 by bounded product expansion and truncation. Under continuity of the potential, local Riesz decomposition gives nonpolarity, while rotation smoothing and Dirichlet energy give the canonical Sobolev trace of Lemma 15.30.2. The convex functional in Theorem 15.30.1 yields every positive exponent; the bounded Green density and uniform logarithmic tails give continuity, and fixed barriers give uniform selection. The full-product greatestness and coefficient in Theorem 15.30.2 follow from the singular exponential order of Corollary 15.1.1. The Frostman test includes nonconstant weighted circles. The zero-lower-mass conclusion assumes the factor's vanishing lower mass; the general individual-measure classification is not a consequence of this product construction.

The radial residual criterion in Section 15.36 uses central mass rather than an ordinary-model envelope identity. The local decreasing-limit property gives the plurisubharmonic extension with its pole at the center, while the radial determinant flux identifies its non-pluripolar probability and inverse. The central density error controls the flux and its first derivative without differentiation of that error, giving Theorem 15.36.1. Flat errors, mass-preserving sharp examples and an open zero annulus follow from the same inverse in Propositions 15.36.1 to 15.36.3. The

product statement of Corollary 15.36.1 uses the product formula of Lemma 15.29.1, which rests on bounded multilinearity and pullback by a projection, as proved in Proposition 1.7.1 and Lemma 1.7.2, followed by the non-pluripolar truncation limit. Only its last assertion assumes zero lower mass of the factor, and it makes no assertion about arbitrary nonsymmetric equations.

The singular-obstacle envelopes of Section 15.37 use the same sublogarithmic reference as the preceding radial density criterion. Unitary invariance reduces the full envelope to a convex minorant, and supporting tangents control both ends. On the right tail, the weighted slope and second-derivative estimates in (15.121) give uniform projective Hessian bounds even when contact moves to infinity; interpolation with value contraction gives Theorem 15.37.2. The full product formula in Proposition 15.37.1 follows from exact slicing of every competitor and the product formula of Lemma 15.29.1, obtained by bounded-product expansion before strict-superlevel truncation. It requires no mass property of the Hermitian factor. These results concern this everywhere singular obstacle and the residual in its prescribed ordinary type, while regularity for arbitrary types and compact obstacle sets remains open.

36.12 Perron geometry on Hopf manifolds

The closed-base metric inputs in Section 18.5 are Darvas' Kähler completion and CAT(0) theorems [Dar17, v3, Theorems 6.1 and 9.2, pp. 20–26 and 30–32]. Capacity endpoint attainment and preservation of the squared-energy class are [Dar17, v3, Theorem 5.2 and Corollary 5.4, pp. 18–20]. The full-mass model characterization used in Proposition 18.8.1 is [Dar17, v3, Theorem 4.3, pp. 16–17]. These theorems are applied on the connected compact Kähler base $\mathbb{P}^{n-1}$. The source uses $i\partial\bar\partial$, so potentials are divided by 2π in (18.7); fiber integration contributes the additional factor $\sqrt{2n}$ to distance.

The smooth-endpoint distance formula used in Propositions 18.5.2 and 18.7.1 is [Dar17, v3, equation (15), p. 8, and Theorem 7.2, p. 28]. The bounded Dirichlet description is [Dar17, v3, Section 2.2, equations (16)–(17), p. 9]. Section 7, p. 27, gives a bounded-potential counterexample to using the initial-velocity formula outside its regularity hypotheses. The completed action in the book is therefore defined by metric speed.

The Hopf conclusions use the elliptic fiber action and the descent of basic potentials in Lemma 9.6.1, which rests on bounded products with smooth corrections and finite-level truncation. No bounded-mass property of the Hopf manifold is imposed. Group averaging proves stationarity under all smooth variations, and radial Fourier modes give the negative raw-action directions in Theorem 18.6.1. The explicit diagonal geodesics and singular radial family supply exact distance and energy calculations. Their metric is on basic potentials; its ordinary-model projection is zero throughout the finite-energy domain, so it does not solve the general ordinary-model metric problem.

36.13 Singularity classes and traces

The countable-pole construction of Theorem 17.6.3 uses the local decreasing-limit and canonical-representative theorems in Demailly's book [Dem12, Chapter I, Theorems 5.4 and 5.8, pp. 39–40]. Its exponential integrability follows directly from weighted Jensen's inequality and the sublinear logarithmic profiles. The envelope argument of Proposition 17.6.7, which is the nonseparation argument of Proposition 17.6.5, uses Choquet's lemma and the local upper-envelope theorem [Dem12, Chapter I, Lemma 4.23 and Theorem 5.7, pp. 38–40], through the exact nonclosed-background construction of Section 1.19. The bounded unrestricted envelope has the same trivial multiplier ideals as the constructed potential, so it is a competitor for the ideal-equivalence envelope used here. The fixed stronger potential provides ordinary competitors below every continuous obstacle; the general zero-obstacle implication remains open.

A theory of singularities for nonclosed backgrounds must record torsion as well as divisorial information. The quotient of the space of smooth real $(1,1)$-forms by the forms $\mathrm{dd}^{\mathrm{c}} f$, f smooth, used in Chapter 25, appears in [BGL25, Section 1]; fixing the exterior derivative of the form keeps this datum on each modification.

Resolution makes additivity of traces accessible to approximation by multiplier ideals. The comparison in Proposition 20.1.1 is the argument underlying Cao's additivity theorem [Cao15, proof of Theorem 1.4, pp. 118–119], and it does not require a closed global background. The recovery theorem Theorem 31.5.3 combines the arguments with decreasing models and stretching with the local criterion for multiplier ideals. Uniform exponential integrability bounds each local divisorial order of the auxiliary potentials, so the proof does not require a local modification to extend over the compact manifold. The construction on a ruled surface in Section 26.5, based on Demailly, Peternell and Schneider's [DPS94, Example 1.7], exhibits the boundary behavior relevant to this recovery.

The continuity of model envelopes in closed classes used in Theorem 20.6.3 is due to Darvas and Xia [DX26a]; we use it in the form of [Xia26, Proposition 3.1.12, (3.23)–(3.24)], which covers decreasing model potentials in shrinking backgrounds, the continuity of their masses and, at positive mass, of their divisorial Lelong numbers. The argument for Lelong numbers uses small convex corrections from Darvas, Di Nezza and Lu [DDNL21, Lemma 4.3, p. 154]. It suffices to apply this continuity on the submanifold, because the restricted approximants have a common curvature bound even when the ambient background is not closed.

The projective-base construction of Section 20.8 uses the same positive-mass convergence of decreasing closed models; the statement and proof are in [Xia26, Proposition 3.1.12, pp. 96–97, with Lemma 2.4.4, pp. 67–71]. The ideal-class argument uses its compact Kähler criterion [Xia26, Proposition 3.2.1, pp. 122–123]; the cofinality discussion on p. 497 covers prime divisors on arbitrary compact smooth models by finite smooth blowup models. These are closed-base inputs. The finite extraction proof in Lemma 20.8.1 uses proper mapping for analytic sets and analytic resolution which preserves the whole nonsingular locus, in the form of Włodarczyk's [Wło09, Theorem 2.0.1(1), p. 34]. Its ramification argument uses the local Siu

decomposition and Skoda integrability recalled in the foundations. Elliptic symmetry gives the full Hopf envelope and its mass and approximation formulas; the ideal-class argument remains on the projective base.

The residual condition in (17.9) implies vanishing of point Lelong numbers by scaling the uniform Lelong bound, and vanishing of polar curvature by the Gauduchon trace and bounded-type invariance. The argument persists on every smooth compact modification. The ideal neutrality of Lemma 1.16.4 combines Skoda's integrability theorem [Dem11, Lemma 5.6] with the local ideal-addition calculus; the strong openness used there is stated in [Dem15, arXiv v2, Corollary 2.3, p. 12]. The smoothing step in Lemma 17.7.2 applies Demailly's regularization theorem [Dem92, Main Theorem 1.1], in the form recalled in Section 1.15.1, to the closed current $T = \mathrm{dd}^{\mathrm{c}} q'$ with smooth representative zero and lower bound $\epsilon h - B$. In the source normalization the potential is $q'/2$. The curvature of a fixed tangent metric is bounded by a large multiple of h, and since the Lelong numbers of q' vanish, Dini's theorem makes the continuous functions bounding the loss of positivity converge uniformly to zero. This gives a smooth strictly positive representative of the residual background; the prescribed ordinary type is then recovered with representatives of small negative curvature in Theorem 17.7.1. The ideal bounds in Proposition 20.2.2 use ambient ideals and therefore remain meaningful when the naive residual restriction has poles everywhere on the target.

The full ordinary-type optimization in Theorem 20.10.1 uses the closed comparison theorem of Boucksom, Eyssidieux, Guedj and Zeriahi [BEGZ10, Theorem 1.16, pp. 217–218] in both orders, in degree two. Their Definition 1.2, p. 204, applies because every residual potential is locally bounded outside the divisor. Finiteness and closedness of the auxiliary non-pluripolar products are their [BEGZ10, Proposition 1.6 and Theorem 1.8, pp. 206–209]. Only the dominating Kähler background enters this cohomological comparison; the nonclosed mass is recovered by smooth-background expansion and distributional Stokes. The two ordinary volumes and their divisor and pure-pullback laws are those of [BGL25, Definition 3.1 and Proposition 3.3] in their bounded-mass setting; here the latter identities are proved through the complete competitor correspondence, and the small-curvature sublogarithmic representatives give the extreme sequences. The ambient ideal orders of Proposition 20.10.1 are a case of Proposition 20.2.2, with Lemma 1.16.1 giving the ideal in codimension one. For Proposition 20.10.3, the small-curvature criterion first produces a smooth residual representative; restriction to the last exceptional curve then excludes ordinary-type strictness even over pole centers. The exceptional curvature perturbation constructs a one-sided strict witness inductively. This extends the bounded-center restriction mechanism in Example 29.7.2 to the specified finite blowup family. The separate residual volume and cost in Section 27.21.2 use the all-test dual formula and first variation of this book. The logarithmic degeneration in Example 27.21.1 specializes those formulas to the height distribution on the sphere; pairing against a closed base test excludes every strict current at the endpoint.

36.14 Section growth and partial kernels

The growth of sections links singular potentials to the algebraic dimension through a projective modification. For section rings constructed through the algebraic reduction, see Xiaojun Wu's [Wu24]. The flag construction in Chapter 22 uses a fixed compact flag on an auxiliary modification; it makes no assertion about the centers of arbitrary valuations of the field of meromorphic functions.

The maximum at maximal mapping rank in Proposition 22.6.1 is Qin's [Qin26a, v1, Proposition B.1, pp. 35–36]. The scalar and flag bounds in Corollary 22.6.1 use the section-ratio field and period count already established in this book, with the curvature rank measured on its semipositive locus. The fixed-twist and endpoint deductions, Euler polynomial, surface estimate and sharp product follow [Qin26a, v1, Sections 10.2–10.3 and Example 11.1, pp. 32–34]; they are independent of that paper's all-degree finite-jet estimate, Theorem A.

Principalization and its isomorphism off the base ideal are Włodarczyk's [Wło09, Theorem 2.0.3, p. 35, and Lemma 4.0.3, p. 42]. The image dimension, proper mapping and Chow inputs are given with proofs in [Dem12, Chapter II, Corollary 8.6, Theorem 8.8 and Theorem 8.10, pp. 118–121]. Serre duality uses the compact Hermitian setting of [Dem12, Chapter VI, Theorem 7.3, p. 310]. The analytic Hirzebruch–Riemann–Roch formula used in Proposition 22.6.2 is [Ati64, Theorem (3.4), p. 163]; it applies to compact complex manifolds and requires no Kähler hypothesis.

The formulas for derivatives in the weight and the Gevrey bounds in Proposition 28.5.2, Theorem 28.5.1, and Corollary 28.5.2 follow the mechanism for bounded weights of Marinescu and Xing [MX26, v1, Theorems 1.1–1.2, pp. 2–3, and Section 3, pp. 11–19], applied to the fixed space of selected sections on a compact manifold. The proof uses the inverse of a compressed Gram matrix and its finite jets, while the degree and the multiplier-ideal space stay fixed. The example Example 28.5.1, concerning control at the boundary, is the explicit example on the disk on p. 3 of that preprint. Marinescu and Xing point to the earlier theory of bounded perturbations of Pasternak-Winiarski; its theorems involving the distance to the boundary require additional weighted openness estimates and are not used in these compact arguments.

The estimates for the probability measures in Lemma 28.5.1 and Propositions 28.5.3 and 28.5.4 follow directly from the reproducing identities in finite dimension. Positivity of the symmetric coupling defined by the squared kernel gives the factor one half in the bound for infinitesimal variations. Integrating the logarithm of the odds of the probability of each Borel set then gives a sharper modulus for the global oscillation; the smooth example of rank one, Example 28.5.2, shows that this modulus is optimal as a supremum. Normalizing by the dimension gives a uniform radius of analyticity after rescaling. The calculation on a product in Example 28.5.3 shows that a perturbation of the norm weight of order inverse to the degree already changes the limiting fiber probability; the same survival of a fiber probability reappears in the conditional fiber probabilities of Section 28.11. Its nonzero limit of intrinsic measures and its unchanged limit of logarithmic potentials describe different parts of the kernel data.

In the fiber-integrability criterion of Theorem 28.10.2, Corollary 28.10.2, and Example 28.10.2, a radial calculation on a simplex detects the strict endpoint of

integrability and the weighted gain $p\sum_j b_j^{-1}$, where p and b_j are the exponents of the normal and the fiber variables in the anisotropic model of the selecting potential. A global lower bound and, over each divisor, a single chart with an upper bound then give a bounded divisorial band, whose loss of sections is controlled by the dimension estimate Lemma 28.10.1. This extends the isotropic theorem on sections vanishing along a divisor, without claiming that the condition is necessary for arbitrary selecting potentials.

For finite families of divisors with simple normal crossings, the calculation of the ideal on the base in the proof of Theorem 28.10.2 uses separated normal coordinates, and the same dimension estimate, stated for twists by several line bundles, controls all the mixed twists arising in the restrictions. In the product model of Example 28.10.3, integrability at points where several poles meet is established directly, by Fubini's theorem in distinct pairs of base and fiber coordinates.

The criterion on a modified base, Theorem 28.10.3, combines the same argument with the transformation rule for adjoint ideals, Lemma 1.22.10, and the pushforward identity for non-pluripolar products, Proposition 28.6.1. The projective model with a point pole in Example 28.10.4 makes explicit the Jacobian order $q-1$, where q is the dimension of the base, and the positive residual on the blowup. The criterion assumes that the modification and its divisorial data are given; it does not assert that they exist for an arbitrary selecting potential.

In Theorems 28.10.2 and 28.10.3 the curvature requirements are separated: a smooth semipositive residual gives a positive reference current, and a positive top-degree integral gives it the required mass, while the estimates for integrability and restriction need no strict positivity of the curvature. In Example 28.10.5 the original base is a blowup. Its contracted lines prevent any choice of gauge from making the residual strictly positive, while Hartogs extension shows that, because of the fixed exceptional divisor, the selected space and the reference space coincide.

The calculation of coefficients in Proposition 28.10.1 combines this Hartogs identification with the anisotropic local threshold; the remaining conditions are ordinary jet conditions at points of projective space. Exact counts of homogeneous monomials then identify the range of the references in which the mass is positive, the drop in degree at the boundary of this range, and the tensor power at which a more singular selecting potential removes every section.

The model with two points in Example 28.10.6 keeps positive top-degree volume while the residual has degree zero on the curve joining the two centers. Its nonzero defect, of degree $q-1$, is computed by Lemma 28.10.3, which constructs simultaneous jets from affine linear factors and finite Taylor division and requires the sum of the jet orders to be at most the projective degree plus one. Thus a sharp loss of sections and the obstruction to a smooth strictly positive residual occur in the same explicit family.

The metric on projective space in Section 28.14 is the one in Chiang's [Chi26, v1, Example I.15, pp. 9–10]. The proof of Proposition 28.14.1 computes the global norms of monomials directly, in the convention $|e|_h^2 = \mathrm{e}^{-\phi}$ for a local frame e with weight ϕ, and for a given integration volume; it uses neither the general spectral localization theorem nor the chain of expansions by Newton polyhedra of that preprint. The transition for growing jets on a compact curve, Theorem 28.13.1, is proved in

Section 28.13 by comparison with a model on a shrinking disk, and its full extension to all sublinear orders, at the boundary of the region where the curvature vanishes, is Theorem 28.13.2.

The correction on shrinking scales in Lemma 28.13.2 uses the L^2 estimate with pointwise inverse curvature of [Dem11, Theorem 5.2 and its singular-metric extension, pp. 34–35], whose Hilbert-space estimate comes from the preceding Theorem 5.1 and its proof on pp. 33–34. Applying the estimate to $L^m \otimes K_C^{-1}$, where L is the line bundle on the compact curve C and K_C is the canonical bundle of C, takes the Ricci curvature into account, and a bounded function varying on the scale R of the flat region restores strictly positive curvature inside it. The particular free boundary in Lemma 28.13.1 is constructed by solving an equation for Fourier coefficients in fixed Hölder spaces of finite order and applying the implicit function theorem in Banach spaces. The argument with Gaussian annuli then gives Theorems 28.13.2 and 28.13.3 directly, including slowly growing orders, uniformity in the angle, sharpness on compact curves and the limitation of finite Taylor corrections.

On the explicit compact product in Section 28.15, global orthogonal monomials turn ordinary jet conditions of total order into a selection by a half-space. The transition for one curve factor gives the marginal Gaussian laws used in Theorem 28.15.1, and the calculation of moments at a fixed positive radius, together with the exact trace, gives convergence in total variation. The sharp radial correction in Proposition 28.15.1 follows from the positive first remainder assumed there. Theorem 28.15.2 then uses the integrability calculation at an isolated pole and constancy along compact connected fibers to transfer norms and kernels exactly; the normalization on the base and the matching of volumes are part of its statement.

The selection by an interior Newton polytope in Section 28.16 is set up with an integral affine offset, a polytope in the interior, a strictly convex invariant logarithmic weight and a positive invariant sampling probability on the whole compact manifold. Angular orthogonality gives exact norms, lattice approximation gives the lower bound for the envelope, and uniform concentration gives the shell measure in the relative dimension. The local argument with Hessian minors distinguishes this measure from mixed non-pluripolar curvature in lower dimension. The construction does not identify an arbitrary multiplier-ideal envelope.

The critical product of multiplier ideals in Section 28.17 is computed directly from the integrability formula for smooth centers and the identity for an atom in the plane. Exhausting the curvature mass fiber by fiber identifies its multiplier-ideal envelope. Exact norms given by beta integrals, together with concentration, give two kernel limits depending on parity, whereas dividing by the exact dimension gives a single limit of probability measures. The calculation shows that a fixed growth exponent does not imply convergence of the leading coefficient in the count of sections, even when the growth dimension is positive.

For the selection by multiplier ideals at the base points of a pencil in Section 28.18, the lower bound for Lelong numbers implied by inclusions of multiplier ideals at all weights, together with a radial convex maximum, identifies the full classes of competitors for both the multiplier-ideal and the ordinary envelopes. Restriction to the curves of degree two in the pencil controls the competitors that are not invariant.

Exact norms of truncated Gaussians keep track of the three bands of finite degree, whose count gives the limiting intrinsic mass three. The bounds at the poles also give convergence in every finite L^p and uniform convergence of the exponentials, while excluding a uniform additive error on the whole manifold. The compact sampling box and the obstacle are part of the statement.

36.15 Gauduchon variational volume

The Gauduchon variational volume extracts a homogeneous concave quantity from an elliptic normalization problem. The method goes back to Collins's construction [Col21, Theorem 2.13, pp. 1068–1069]; Section 10.8 develops the dual formula and its dependence on the chosen probability measure, and Chapter 27 studies the volume as a function of the background.

For a Kähler class represented by a closed form θ, this construction meets the Hessian geometry of the Kähler cone: the metric of Section 27.2 is $1/n$ times the Hessian of $-\log \int_X \theta^n$ used by Magnússon [Mag12, v1, Section 1, pp. 2–4]. Completeness and boundary points of positive volume at finite distance are studied in [Mag12, v1, Proposition 4.4 and Lemma 4.5, p. 11]; Section 27.4 gives the argument by homogeneity in the present normalization.

The cones of invariant nonclosed forms in Section 27.4.2 are computed directly. Stokes' theorem on the compact quotient shows that invariant $(n-1,n-1)$-forms are closed, and Theorem 10.8.1, applied to the balanced background itself, then determines the quotient and its normalized minimizer. The resolvent argument of Lemma 27.4.1 and the normalized chord in a prescribed linear subspace give the completeness statement. The Iwasawa example comes with an explicit compact quotient and a proof, by averaging coefficients, that this quotient is not Kähler. The initial construction uses the normalized invariant probability measure.

For the warped cones in Section 27.4.3, a closed equation on the base and an explicit balanced cofactor give the full quotient through Theorem 10.8.1. The pullback of the affine Hessian metric becomes flat, and the variance of the trace at the endpoints gives the three alternatives: a strip, a half-plane or the whole plane. Smooth existence for the equation on the closed base is Theorem 10.1.1, and integration fixes its scalar; the warped background on the product may be nonclosed. That the finite faces are not big is tested against closed top-degree forms on the base, so the conclusion applies to singular current representatives as well. In the computations on projective space and on periodic tori, the density and the cone are given explicitly throughout.

The cones of currents in Proposition 27.4.4 are determined by localized closed test forms and Stokes' theorem for distributions. At the finite faces of the warped cone, Proposition 27.4.5 applies the existence and uniqueness theorem for invariant mass, Theorem 12.5.1, on the closed base, which also fixes the scalar, and expands the pulled-back products. The obstruction to global $C^{1,1}$ solutions is the general bound of currents by determinants in Proposition 1.7.3, derived from the identity for coefficients in Proposition 1.7.2.

The convergence of the whole sequence in Proposition 27.4.6 combines bounds for the reciprocal densities with the compactness for a fixed background in Theorem 11.3.2. Since the base is closed, the uniqueness in Theorem 12.5.1 identifies every cluster point of the potentials. For any bounded competitor at the endpoint pulled back from the base, the density forced away from the extreme set already exhausts the total mass of the closed class, so positivity excludes residual mass on that set before uniqueness is applied.

The alternatives for the trace in Propositions 27.4.7 and 27.4.8 follow from the explicit balanced cofactor and from integrals of reciprocal squares over tubes, while the mass on the closed base controls the remaining cofactor term. When the trace diverges, its normalization concentrates on the extreme components where the ratio is minimal; if there is a single extreme point, the whole sequence converges to a fiber current. For the trace-normalized smooth test metrics, the numerator and the denominator of the volume quotient both tend to zero, while the quotient remains equal to one.

In Section 27.10, bigness of the closed nef core with positive top self-intersection follows from Demailly and Păun's criterion [DP04, Theorem 0.5]; no such criterion for arbitrary nonclosed nef backgrounds is used.

On the warped paths of Section 27.11, the same balanced cofactor as in Section 27.4.3 computes the cost rate for every minimizing test. The finite improper identity for the total subtraction cost follows from the first variation and the limit at the nef endpoint, while the five asymptotic rates of the cost rate in Proposition 27.11.1 come from the first and second inverse moments of the fiber coefficient. The exclusion of every closed nef positive-volume core in Proposition 27.11.2 uses the smooth-test obstruction to bigness and Demailly–Păun's [DP04, Theorem 2.12, pp. 1259–1260], together with the global potential consequence of the Kähler $\partial\bar{\partial}$-lemma. The constant closed case gives exactly the expression in Tosatti's Morse inequality [Tos26, v2, Theorem 1.1, p. 1]; this comparison does not assign a closed cohomology class to a nonconstant warped coefficient.

In the radial model on a projective manifold in Section 27.13, a global bounded potential compensates the coefficient along the fibers and gives the lower bound against every smooth Gauduchon test, and the explicit balanced metrics of the warped construction in Theorem 27.4.2 give matching values after perturbation. In Theorem 27.13.1, the test against closed top-degree forms on the base, through Proposition 27.11.2, excludes every quasi-plurisubharmonic potential whose curvature current is strictly positive. Thus positive volume does not imply bigness for nonclosed nef backgrounds, even on a projective manifold; the construction says nothing about surfaces, or about closed nef classes on general compact complex manifolds.

The limits of test forms in Section 27.13.1 are computed from these explicit balanced metrics by Stokes' theorem for closed forms and radial integration. In dimension three the powers normalized by the trace concentrate on the degenerate fiber at a logarithmic rate, whereas in higher dimension their threshold quotient stays positive. Localized closed forms on the base, with a small Kähler correction, give a different balanced family approaching the fiber in every dimension of the model.

For the invariant backgrounds of Section 27.15.1, Proposition 27.15.2 and Corollary 27.15.1 follow from the formula for the variation of mass proved in the book and from the polynomial continuity of bounded products. On the Iwasawa quotient, Theorem 27.15.2 extends the determinant formula to probability measures invariant along the central fibers, by means of a closed equation on the base for the Schur complement. The exact vertical Dirichlet defect characterizes all maximizing smooth densities, and the completion estimates for this larger family concern the means of the coefficients of the minimizing forms.

The initial metric of the Chern–Ricci flow in Section 27.20.2 is chosen so that, up to a smooth global change of gauge, the radial background is the limiting form at the maximal time, and the maximal-time theorem determines this lifetime. In the normalization $\mathrm{Ric} = -\mathrm{dd}^{\mathrm{c}} \log \omega^n$, time one corresponds to time $1/(2\pi)$ for the Chern–Ricci form $-2\pi \mathrm{dd}^{\mathrm{c}} \log \omega^n$ used in this book. A test against a closed form on the base excludes every positive representative at later times, and the support and the vertical rank of all annihilators at the endpoint follow directly from the semipositive representative and the formula for the trace matrix. The averaged trace is a special case of the trace identity for the flow at the end of Section 27.20.1.

In the criteria of Section 27.8.6, invariance of the mass of bounded potentials and the support of the measure on the contact set prevent the stretched obstacle envelopes from collapsing, and the sublinear condition provides competitors. The uniform bound for Lelong numbers in Lemma 1.17.7 then gives vanishing Lelong numbers by scaling. The argument with smooth roots of rank two extends the scalar first-order equation across the analytic set, and the support theorem for currents carried by analytic sets, recalled in Section 1.17, removes its residual current. The calculations on Hopf manifolds and quadrics give examples in which the dual problem is attained. The general implications involving moments and entropy remain separate questions.

In the example with a point pole, Proposition 27.8.7, radial integration shows directly that the potential has full mass, and the equality case of the arithmetic–geometric mean inequality for matrices, combined with the local strong maximum principle, excludes every smooth semipositive maximizing test form. The argument does not need to identify the potential with the greatest threshold envelope. The thresholds for entropy and weighted integrability in Corollary 27.8.6 follow directly from the computed radial density. They concern the density of this single equation, whereas the question of uniform moment or entropy conditions concerns the determinant-weighted probability measures of a maximizing sequence when the supremum is attained by a smooth semipositive test.

The bound for the determinant at the endpoint, Theorem 27.7.2, is derived within the book from the criterion for mass invariance and the quotient by the lower mass; related criteria for the mass are [GL22, Propositions 3.2 and 3.3(iv)]. The concentrating family in Section 27.17 distinguishes the unique smooth maximizer for one density from maximizing sequences whose higher moments diverge.

The coordinate formula (27.73) identifies the intrinsic determinant measure on singular and degenerate positive currents. The proof of Theorem 27.17.1 applies Tosatti and Weinkove's [TW10, Corollary 1] separately to each smooth positive root. The source uses $\mathrm{i}\partial\bar{\partial}\varphi$; the potential is multiplied by 2π here, and Theorem 2.2.2 fixes

the volume coefficient. The last paragraph of that paper refers the existence argument to Cherrier and Demailly or to the authors' earlier paper. No estimates uniform in the varying right-hand sides are used.

The closed and nonclosed torus families in Theorems 27.17.2 and 27.17.3 preserve the complete determinant data and positive mass extrema while ellipticity degenerates. For the nonclosed family, the lower-power identities hold on the two-coordinate block, which carries the fixed nonzero torsion. In Theorem 27.17.4, the Poisson kernel gives uniformly convergent continuous potentials and exact maximizers for varying densities. The limiting current has rank-deficient absolutely continuous and singular coefficient parts, so its determinant measure vanishes although the bounded potential has positive Monge–Ampère mass. These calculations concern scalar perturbations and current limits; they supply no flow existence or convergence theorem.

The identity (27.67) of Lemma 27.16.1, for bounded potentials in a background g with $\mathrm{dd}^{\mathrm{c}}(g^2) = 0$, uses the calculus of bounded products and of weighted decreasing limits in Proposition 1.7.1 and Lemma 1.6.1, whose classical local foundations are [BT82, Section 2, Theorems 2.1 and 2.4, pp. 4–9]. That the sign $\mathrm{dd}^{\mathrm{c}} g \le 0$ gives positive lower mass is part of the criteria in terms of the sign of the torsion of Angella, Guedj and Lu [AGL23, Theorem 3.2, pp. 4641–4642]. The curvature calculation in Theorem 27.16.1 identifies this sign on principal elliptic threefolds, and the two quotient constructions give explicit mass constants in the present convention for the operator. The conclusion for the determinant of the full current then follows from the endpoint theorem and the change of background in Corollary 27.7.1.

For $q = \mathrm{e}^{2\pi}$, the principal elliptic bundles of Section 27.18.5 are torus bundles of Goldstein and Prokushkin [GP04, Section 2, pp. 69–71], built from the Chern connection of the metric on L chosen there; their two closed 2-forms are a constant multiple of c and 0. Their construction always has the square torus as fiber; other values of q change only the modulus of the fiber. The metric calculation on these bundles gives an upper bound for the mass, but it does not decide whether the lower mass vanishes. The radial calculation of Proposition 27.18.3 bounds the root volume of the specified radial test family at fixed degree; the lower mass over all potentials remains open.

The estimate for local families in Theorem 27.7.3 controls the solution form and its Gauduchon multiplier by two contractions. The elliptic input is the parametrix and Fredholm argument on compact manifolds in Taylor [Tayd, Chapter 7, Sections 4 and 10, especially (10.4)–(10.6)], together with the mapping properties on Hölder spaces in [Tayb, Chapter 13, (8.7), (8.17) and (8.22)]. Invariance of the index is [Tayc, Section 7, Proposition 7.4], and smooth regularity for nonlinear elliptic equations is also formulated in [Taya, Proposition]. The estimates for the equation and for the normalized multiplier give explicit constants for the densities without assuming positive lower mass; instead, the data are restricted to smooth functions in a small $C^{2,\alpha}$ neighborhood.

The bound for the Gauduchon multiplier in Proposition 27.7.1 starts from the conformal theorem of Gauduchon [Gau85, Théorème 2 and its proof, pp. 132–133]. Its general statement has no restriction on the first Betti number; the even-Betti assumption belongs to the surface theorem at the beginning of that article. The

smooth maximum principle is recalled in [Gau85, pp. 131–132] and applied on p. 133. The quantitative estimate here writes the cofactor equation in real divergence form and proves its Harnack bound by power iteration and a dyadic mean-oscillation argument. Since $\mathrm{d}(\omega + \mathrm{dd}^{\mathrm{c}} v) = \mathrm{d}\omega$, the vector coefficient is controlled by the fixed torsion. This gives the bounds for the exact maximizing currents and their supporting probabilities in Theorem 27.7.4 without derivative bounds on the varying solution form. Gauduchon uses $d^c = \mathrm{i}(\bar\partial - \partial)$ and $L = -\operatorname{tr}(dd^c)$; the equation for the positive multiplier is unchanged by this scaling and sign conversion.

The classical analytic input in Section 27.19.2 is Hironaka's elimination of indeterminacies: every modification is dominated by a composition of blowups with smooth centers. The argument by principalization and domination is explained by Abramovich, Karu, Matsuki and Włodarczyk [AKMW02, Sections 1.2.3–1.2.4, pp. 538–539, and Section 5.4, p. 564], where the analytic case is treated explicitly. The curvature of the exceptional divisors, computed in Lemma 1.15.1, then gives the uniform bound for the pairings used here; no factorization into a sequence of blowdowns is needed for this estimate.

For the fixed projective cones in Section 27.19.5, every background is compared with the closed sum of the generators that occur in it, and the intersection polynomial of this sum gives two-sided bounds by powers of the perturbation, also after pullback to any modification whose source is Kähler (Lemma 27.19.2 and Proposition 27.19.10); the ranks of the generators give the exact loss of order when the prescribed effective divisors are removed. Fixed coefficient functions that vanish, cones of residuals without restrictions, and sources that are not Kähler lie outside this construction.

The theorem that the line bundle of a smooth anticanonical curve of self-intersection zero is unitary flat near the curve, Theorem 27.19.3, is due to Chen, Filip, Sun, Tosatti and Zhang [CFS+26, v1, Theorem 1.1, p. 1, with the standing degree-zero hypothesis in Section 1.1]. The proof given here follows their identification of the holomorphic fibers in Proposition 4.1, their elliptic inverse independent of the holonomy in Lemma 5.2, and their weighted horizontal inverse and contraction in Propositions 5.4–5.5. The cutoff in Proposition 27.19.11 is the construction at the end of their proof in Section 6; it applies to every smooth compact divisor whose line bundle is unitary flat near it, in any dimension. The variational bounds in rank one follow from Lemma 27.19.2, and the examples obtained by blowing up nine points on a plane cubic, together with their torsion criterion, follow by restriction to the cubic. Only this local theorem and its direct consequences are used, not the global classification of surfaces in that paper.

The pencil in the case of finite holonomy and the rigidity in the non-torsion case, Proposition 27.19.13 and Theorem 27.19.4, give the case of a smooth reduced divisor of Koike's [Koi26, v2, Lemma 3.5, pp. 15–16]. The Levi-flat level sets of the flat norm have a smooth suspension model, so the density of their leaves and the maximum modulus argument are proved directly. The hypothesis is that the line bundle of the divisor is unitary flat near the smooth compact divisor; the global criteria for semipositivity and the dichotomy between fibrations and the Hartogs phenomenon in Koike's paper are not used. Together with the theorem on anticanonical neighborhoods,

these local results explain the non-Stein and Hartogs properties of the nine-point examples with non-torsion normal bundle.

The question whether a weak Monge–Ampère power admits a scalar conformal factor is raised by Lu [Lu22, Chapter 3, Section 5]. The representation by finite currents in Theorem 27.6.1 gives a supporting probability measure without defining that weak power or its scalar factor.

36.16 Closed classes and singular loci

For closed classes on a Hermitian manifold, almost positive representatives keep the cohomology class separate from the auxiliary metric. The convention in Section 29.1 follows Boucksom's thesis [Bou02, Section 2.1.2]; it does not assign a Bott–Chern class to a nonclosed Hermitian form.

The smooth and analytic regularizations of Section 1.15.1, used for Proposition 29.2.1, have the lower bounds of [Bou04, Theorem 2.1]; see also [Bou02, Theorems 1.2.9–1.2.10]. Smooth regularization loses positivity by a multiple of a continuous approximation of the Lelong numbers, whereas analytic regularization has a curvature error tending uniformly to zero and does not increase Lelong numbers.

The minimal multiplicities of Definition 29.2.1 are defined through these almost positive approximating families, as in [Bou02, Definition 2.1.5]. They must be distinguished from the Lelong numbers of a single current with minimal singularities: in the example of a nef class on a ruled surface in [Bou02, Section 1.2.3], a current with minimal singularities has nonzero Lelong numbers.

The non-Kähler locus of Definition 29.3.1 is defined in [Bou02, Definition 2.1.19], and the proof of Theorem 29.3.1 by finite intersections and regularized maxima is the argument of [Bou02, Theorem 2.1.20].

The equivalence between the existence of a Kähler current and the existence of a Kähler modification, used in Chapter 29, is the classical criterion recalled in [Bou02, Theorem 1.2.11]; it is used as an external geometric input. The equality of the non-Kähler locus and the null locus of a nef and big class is the transcendental Nakamaye theorem of Collins and Tosatti [CT15, Theorem 1.1, p. 1168], which also holds on compact manifolds in Fujiki's class $\mathcal{C}$; Chapter 29 recalls this theorem from the literature, and Proposition 29.11.1 uses it to show that the pullback of a Kähler class by a finite surjective holomorphic map between compact Kähler manifolds is a Kähler class.

Restricted volumes give a numerical test for these loci. The numerical restricted volume (29.1) follows Collins and Tosatti [CT22, Section 2]; the requirement that the analytic singularities have a smooth remainder is part of that definition.

The existence of Kähler models for every subvariety uses Fujiki's theorem that class $\mathcal{C}$ is closed under passing to closed subspaces [Fuj78, Lemma 4.6(2), p. 35], followed by Varouchas's theorem on Kähler modifications in the form stated and explained by Barlet [Bar99, Theorem 10, pp. 31–32]. A common Kähler refinement is obtained by embedded resolution of the graph in a Kähler product, using Bierstone

and Milman [BM97, Theorem 1.6 and Remarks 1.7(2), pp. 215–216]. These are classical geometric inputs. The descent of potentials is proved in Proposition 1.22.4, so the proof of mass invariance does not repeat it.

In Proposition 29.6.2, positive non-pluripolar mass implies bigness by [BEGZ10, Proposition 1.22, p. 222]. The definition of the positive product, its agreement with the cup product for nef classes, and the fact that, for big classes, it is attained by currents with minimal singularities are in [BEGZ10, Definition 1.17, pp. 218–219, and Proposition 1.20, pp. 221–222]. The normalized diagonal current (29.7) uses the mass-concentration theorem of Demailly and Păun [DP04, Proposition 2.6] and makes explicit the coefficient in the construction of [DP04, Theorem 2.12]. These results on Kähler manifolds are used as classical inputs. The identity for the mass with a bounded residual in Lemma 29.7.1, by contrast, follows directly from the book's formula for bounded masses, by comparison of polynomials. The comparison with the numerical restricted volume for nef classes is [CT22, Lemma 2.6], and the final equality in (29.9) uses the theorem [CT15, Theorem 1.1] recalled above.

The rigidity criterion for nef boundary classes on tori, Theorem 29.16.1, develops the flat model of Fu and Ouyang [FO26, v1, Section 3, pp. 5–6]. Testing against Fourier modes proves the vanishing along the kernel directly, and the metric argument uses a finite covering by dense leaves and a character of the circle. The collapse in this theorem is along a fixed positive perturbation $\theta + t\kappa$, where κ is a translation-invariant Kähler form. The calculation with mixed forms in Example 29.16.2 shows that the conclusion of their Lemma 3.1 and the implication (iii)⇒(i) of their Proposition 3.2 fail for arbitrary sequences of translation-invariant Kähler forms whose classes converge to the boundary class; their Theorem 4.4, which concerns positive perturbations, and the distinction made in their Remark 6.4, p. 18, concern a different setting. Their general theorems on collapse, fibrations and lattice dynamics for K3 surfaces are not used in this argument on tori.

The nonrigid class with an irrational ray in Proposition 29.17.1 is the explicit Kummer construction of [FO26, v1, Section 7.1, pp. 18–19, proof of Theorem 1.4(a)]. The resolution of the quotient and its Kähler metric are described in local charts, and the difference of two representatives is given an explicit periodic global dd^{c}-primitive. Periods over translated real two-dimensional subtori detect the irrational ray without extending cohomology classes across the punctures. Fu and Ouyang's conclusions about diameters of Ricci-flat metrics, spectra and limit spaces require additional inputs and are not used in this construction.

The family obtained from four points and its central metric in Section 29.18 are Qin's construction [Qin26b, v1, Theorem A, p. 1, and Section 3, Lemmas 3.1–3.3, pp. 3–4]. Qin's proof of the nonextension theorem on p. 5 uses global L^2 extension for adjoint bundles. Here the conics of degree zero of Lemma 29.18.5 give instead the exact set of restrictions in Theorem 29.18.1: the divisor metric of the negative line multiplied by the pullback of an arbitrary semipositive metric on the base of the pencil. Fubini's theorem, averages over disks and Harnack's inequality force the central weights to be harmonic away from the node, and the converse is constructed from the two homogeneous sections of the pencil. By Corollary 29.18.1, which concerns the least singular metrics among these restrictions, every extendable metric

has logarithmic coefficient at least one along the negative line, whereas the metric constructed here has coefficient one half; this gives Theorem 29.18.2. It shows that the metric has no semipositive extension whose central restriction is at most as singular as the metric itself, without using global adjoint extension and without asserting that this metric has minimal singularities.

Inoue surfaces were introduced by Inoue [Ino74], and their higher-dimensional analogues by Oeljeklaus and Toma [OT05, Section 1 and Proposition 2.5], who proved that the action of $O_K \rtimes U$ used in Section 29.19 is free, properly discontinuous and cocompact, and that the resulting manifolds are not Kähler. Ornea and Verbitsky showed that Oeljeklaus–Toma manifolds with $t = 1$ contain no positive-dimensional complex subvarieties, using a semipositive closed form whose null leaves are Zariski dense [OV11]. The decomposition of Inoue surfaces into three-dimensional tori filled by dense leaves, and a Liouville argument along the leaves for bounded harmonic functions, appear in the work of Fang, Tosatti, Weinkove and Zheng on the Chern–Ricci flow [FTWZ16, Section 2, Lemma 2.1 and its proof, pp. 3166–3168]. The Tricerri metric and the identity $\mathrm{Ric}(\omega_T) = -\alpha$ also appear there; their form $\alpha = \mathrm{i}\,\mathrm{d}w \wedge \mathrm{d}\overline{w}/(4\mathrm{y}^2)$ equals $2\pi\theta$ in the normalization of Section 29.19, and their $\omega_T = 4\alpha+\beta$ is the Tricerri metric used there. The description of all θ-plurisubharmonic functions in Theorem 29.19.1, the mass computations of Proposition 29.19.1 and the absence of plurisigned metrics in Proposition 29.19.2 are proved directly here.

The curvature criteria for finite upper and positive lower mass referred to in Section 29.19 are due to Angella, Guedj and Lu, who proved that, on a compact threefold, a Hermitian metric with $\mathrm{dd}^{\mathrm{c}}\omega \geq 0$ gives finite upper mass and one with $\mathrm{dd}^{\mathrm{c}}\omega \leq 0$ gives positive lower mass; in particular a pluriclosed metric gives both [AGL23, Theorem B, p. 4632]. The general question of finiteness and positivity of these masses is [AGL23, Problem 1.8, p. 4635]. Angella, Dubickas, Otiman and Stelzig showed that an Oeljeklaus–Toma manifold admits a pluriclosed metric only if $s = t$ [ADOS24, Corollary 3, pp. 220 and 228]; Proposition 29.19.2 shows that for type $(s, 1)$ with $s \geq 2$ there is not even a metric whose dd^{c} has a sign.

36.17 Closed classes and metric completeness

The examples Theorems 29.14.1 to 29.14.3 test the completeness question posed as an open problem by Di Nezza, Guedj and Lu [DNGL26, v1, Question 3.5, p. 14], for the solutions whose existence and uniqueness they prove in [DNGL26, v1, Theorem 3.4, p. 14]. The first uses a big class that is not nef and an ordinary solution with minimal singularities. The second has a semipositive nef and big background, a solution of full mass and finite energy with vanishing Lelong numbers, and the prescribed nonintegrable pole with smooth strictly positive coefficient; its cusp is complete, but a disjoint null curve lies at finite normal distance. The third, Theorem 29.14.3, satisfies exactly the identities of [DNGL26, v1, Section 3.1, pp. 12–14] for the initial potential, the resolution, the discrepancy, the class and the prescribed volume. Its class is not nef and has a divisorial reference with minimal singularities, while the

solution adds a cusp with vanishing Lelong numbers along a disjoint curve. These examples answer the completeness question negatively for the elliptic equations, both for a big class that is not nef and for a nef and big class; Theorem 3.4 of that paper asserts existence, uniqueness, the Lelong numbers of the solution and, for nef and big classes, smoothness on a dense Zariski open set, but not completeness. Whether nefness and flow compatibility can hold simultaneously in such an example, and whether the evolving maximal flow itself is complete, remain open. The torus examples for normal lengths in Section 29.15 have instead logarithmically weakened poles whose coefficients tend to zero at the divisor.

The energy argument in Section 29.14 uses the classical finite-energy class of [BEGZ10, Section 2, pp. 223–236], relative to a reference with minimal singularities; finiteness of the mixed masses and of the relative energy comes from [BEGZ10, Proposition 1.6, p. 206, and Section 2.2, pp. 226–230]. Uniqueness follows from the comparison principle for full mass [BEGZ10, Corollary 2.3, p. 224]: the unweighted reference measure is infinite, but the two Monge–Ampère measures being compared are finite.

In Section 29.14.1, the potential with minimal singularities is $V_\theta = 0$, and exact radial identities show that the volume form is smooth and positive across both divisors. The bounded truncations have masses $1 - p(T_k)^2$ for the square and $1 - p(T_k)$ for the mixed product on the transition shell, where p is the slope of the radial profile and T_k the radius at which the truncation at level k begins; their contributions to the energy tend to zero. The products are computed by smooth convex approximation and the continuity along decreasing sequences proved by Bedford and Taylor [BT82, Theorem 2.1, pp. 4–7]. This direct computation of the energy agrees with the criterion by canonical approximation of [BEGZ10, Propositions 2.10–2.11, pp. 228–230]. Here vanishing Lelong numbers do not imply ordinary minimal singularities.

In Section 29.14.2, radial barriers and the inversion of a Volterra operator construct the cusp along the curve for a fixed smooth volume form, and explicit charts on the blowup give a smooth positive residual volume and a Kähler residual at the initial time. The reference with minimal singularities is obtained from the extension across isolated points in Lemma 1.3.8. The canonical truncations at level k have cutoff energy $O(k\mathrm{e}^{-k/2})$, so the continuity of bounded products and the criteria for relative energy cited above apply. The remaining smooth pulled-back metric puts the exceptional curve at finite distance. This is an auxiliary solution of an elliptic equation; it is not identified with the evolving smoothing metric.

Hölder regularity imposes a further restriction on the metric. If a positive residual current has a Hölder continuous potential and is smooth and strictly positive outside a proper analytic set, Theorem 1.18.1 shows that every point of that set lies at finite distance in the metric completion; on a compact manifold, removing a nonempty analytic set therefore gives an incomplete metric. Continuity alone allows complete ends: for the prescribed radial densities of Theorem 29.12.2, boundedness of the minimal remainder and finiteness of the normal length have different logarithmic thresholds. The resulting complete examples with bounded potentials solve equations with prescribed densities; they do not have the compatibility with the flow discussed above.

The mechanism of Dang [Dan26b, v2, Proposition 5.2, pp. 16–17], which bounds distances through a Dini condition on the square root of the modulus of continuity of the potential, is the subject of Theorem 1.18.2. The trace calculation given here keeps the smooth background through the additive term in (1.35) and needs no doubling condition. It gives logarithmic rates and uniform limits of pseudometrics along subsequences in Corollaries 1.18.1 and 1.18.2; a common positive lower bound for the metrics is needed to exclude collapse.

36.18 Locally conformally Kähler geometry

In locally conformally Kähler geometry, the Lee form constrains both the rigidity of metrics and the volume forms that can be prescribed. Ornea and Verbitsky conjecture that the Lee class has positive degree with respect to every Gauduchon metric [OV25a, Conjecture 4.14, p. 7], and relate this to pseudoeffectivity of the Bott–Chern class $\{\kappa_\lambda\}$ [OV25a, Remark 4.16, p. 8]. By Theorem 4.3.1, their conjecture implies the weaker Conjecture 30.2.1, which is the form used here; its consequence for balanced metrics is Proposition 30.2.2. Their theorem on the volume forms of Vaisman metrics [OV24a, Theorem 4.1 and Lemmas 4.2–4.4] requires invariance under both the Lee and the anti-Lee vector fields. The proof of Theorem 30.4.1 uses Theorem 10.1.1: uniqueness gives a basic potential, transverse integration by Stokes' theorem fixes the scalar, and the Lee form of the new metric is parallel because it is closed and its dual vector field is Killing. The uniqueness argument keeps the scale of the candidate metric in (30.13) as an unknown until integration determines it.

The least-eigenvalue criterion Theorem 30.2.1 is proved by applying smooth Stokes to the volume polynomial of Proposition 30.2.3. Nonvanishing of the ordinary Bott–Chern class already follows from the conformal Gauduchon argument and the Lee degree (30.4). The Hopf calculations in Examples 30.2.1 and 30.2.2 distinguish constancy of the least eigenvalue from constancy of the whole spectrum, and distinguish first pointwise degeneration from collapse of total volume. These are explicit compact metrics in every dimension at least two. The criterion gives a smooth semipositive representative under its spectral hypothesis; pseudoeffectivity without that hypothesis remains the conjecture above.

The classification of radial metrics in Section 30.4.2 follows from explicit identities for the Hopf forms and a periodic scalar equation. Its positive periodic solutions are strictly ordered by the constant of integration, and their Lee periods fill the negative half-line. The positive twisted potential is constructed in the gauge given by the Lee form of the metric itself. Nonconstant radial volume forms force the metric to be non-Vaisman, and uniqueness of the metric is asserted only within the radial family considered there.

The proof of Theorem 30.5.1 keeps the divergence terms in the Chern–Ricci-flat specialization of the argument of Barbaro and Otiman [BO25, v2, Section 4, equations (2)–(4), pp. 12–13]. Their Theorem A gives the Gauduchon rigidity conclusion; keeping the scalar-curvature term in the integrated Bianchi identity gives

the divergence identity and the equivalence with holomorphicity of the Lee field. The quotient and the two metrics in Example 30.5.1 are computed directly: the holomorphic volume trivializes the canonical bundle, the Chern–Ricci-flat gauge has nonzero Lee divergence, and compact pluriharmonic rigidity prevents a simultaneous conformal gauge. The conjecture for manifolds, Conjecture 30.5.1, is stated in [BO25, v2, Introduction, p. 2]. The Kodaira example admits a Vaisman metric on its given complex structure, so its conformal obstruction does not refute that conjecture.

The dual description of the Lee–Gauduchon cone and its bimeromorphic invariance in Section 30.6 are due to Ornea and Verbitsky [OV25b, Theorems 3.11 and 4.5, pp. 243–244]. The distinction between vanishing of the Lee–Gauduchon class and the strongly Gauduchon condition in Section 30.7 follows [OV25b, Remark 5.6, Example 5.7 and Remark 5.8, pp. 245–246], and the criterion for the existence of strongly Gauduchon metrics used there is due to Popovici [Pop13, Proposition 1.2, p. 654].

36.19 Volumes under mass bounds

For a nonclosed background, the mass of a single potential does not determine the masses in its singularity type. The lower and upper volumes $\underline{\mathrm{vol}}_\theta(u)$ and $\overline{\mathrm{vol}}_\theta(u)$ of Definition 14.3.1, the infimum and the supremum of the mass over the ordinary singularity type of u, record the information that a single mass loses. Under bounded mass, comparison of these volumes controls envelopes and defines a topology on models of positive lower volume. On curves, polar curvature describes the boundary where the lower volume vanishes; on products and fibrations over curves it gives complete metrics on explicit loci of zero-volume models, while for general models in higher dimension it misses diffuse poles of product type.

36.19.1 Singularity volumes and relative classes

The volumes used in Section 31.1 are those of Boucksom, Guedj and Lu [BGL25, Definition 3.1], expressed in terms of potentials. For closed backgrounds the two volumes coincide on every compact complex manifold with the bounded mass property: by Stokes' theorem the defect term of [BGL25, (2.7)] vanishes for a closed form, so their monotonicity of non-pluripolar masses [BGL25, Theorem 2.8] makes the mass invariant under bounded changes of the potential [BGL25, Example 3.2]. In the big case, the monotonicity theorem Theorem 31.1.1 is [BGL25, Proposition 3.7], and the argument for lower volumes in Proposition 31.1.1 is Li's [Li26, v1, Proposition 2.5].

The order test in Proposition 31.8.2 reduces ideal equivalence to Lelong numbers on local modifications by Theorem 1.16.1 and uses the factorization of a modification of a surface into point blowups; on a projective manifold it is also a case of the valuative

characterization of Darvas and Xia [DX22, Theorem 2.13 and Proposition 2.30], whose Lelong convention differs from the divisor coefficient used here by a factor of two. The countable divisorial domination and envelope arguments are those of Lemma 1.17.5 and Corollary 17.2.2; they identify the whole ordinary type before its mass is computed. The upper bound for the closed top mass uses Witt Nyström's monotonicity theorem [WN19, Theorem 1.2], while the lower bound follows directly from local bounded truncations. Expansion with smooth corrections, as in [BGL25, Section 2.1], then gives the nonclosed mass throughout this type. This computation applies to the prescribed summable hyperplane profile with positive smooth residual; it does not assert mass invariance for arbitrary unbounded types in a pluriclosed background.

The transverse recovery of Theorem 31.9.1 uses the recovery theorem Theorem 31.5.3 on $\mathbb{P}^2$, while the Hopf threefold itself has unbounded upper mass by Theorem 9.0.1. The full ideal envelope is invariant under the elliptic action because its complete competitor family is invariant; the descent of basic potentials in Lemma 9.6.1 then identifies it with the base envelope. The nonanalytic family uses the summable divisorial criterion of Proposition 31.8.2, whose closed mass and Lelong-number inputs are described above. The general exact-pullback identities in Propositions 14.3.1 and 17.2.1 follow from the descent, ideal transform and non-pluripolar product foundations; they do not assert recovery over arbitrary new Hermitian perturbations on a modified manifold.

The relative full-mass class of Definition 31.2.1 is [Li26, v1, Definition 3.2]. The stretching criterion Lemma 31.2.1 is [Li26, v1, Lemma 3.1]; the proof given here keeps track of the deficit of upper volumes and chooses a representative whose mass is close to its upper volume. The case of zero deficit gives Theorem 31.2.1, and a uniform bound for Lelong numbers turns the convex comparisons into the estimate of Proposition 31.2.1.

The domination principle Theorem 31.3.1 is [Li26, v1, Proposition 3.2]; its proof uses that the reference has positive lower volume, but not that it is an envelope fixed point. The estimate for relative obstacles, Lemma 31.3.1, is [Li26, v1, Proposition 3.3]. Together they give the identity for upper volumes in [Li26, v1, Lemma 3.2] and the identity for lower volumes in [Li26, v1, Corollary 3.3], both stated in Theorem 31.3.2. For $u \in \mathrm{PSH}(X,\theta)$, the proof first shows that the increasing limit of the obstacle envelopes has the ordinary singularity type of $P_\theta[u]$, so that domination can be applied before the equality of volumes is established. The resulting characterization of full mass gives the volume criterion in Proposition 31.8.1.

When the pole set is compact of Hausdorff dimension zero, Theorem 31.3.3 shows that ordinary projection preserves the whole residual measure. Bounded mass gives uniform bounds for the traces of all local closed powers of the ordinary rooftop envelopes. Away from the pole set, convergence in capacity of bounded functions identifies the limits, and the Hausdorff estimate removes all mass of the lower-degree coefficients on the pole set; hence the lower-degree global pairings preserve the full total mass. Since the rooftop envelopes all have the same residual measure, positivity together with the local order of residuals forces the model to have this residual as well.

The argument also identifies the weak limit of the full Monge–Ampère measures of these rooftop envelopes. The sources of the local inputs are given in Section 36.1.

The construction of small stretches, Lemma 31.4.1, is [Li26, v1, Proposition 3.1], and its bound for mixed masses gives an explicit uniform scale of stretching. The additive estimate for lower volumes in Proposition 31.4.2 is the case of smooth perturbations of [BGL25, Proposition 3.3(ii)], with the mixed bound of [BGL25, Lemma 2.4] written in terms of potentials. The strict barrier gives the multiplicative comparison at positive lower volume, as in [Li26, v1, Proposition 3.5(ii), Step 1]. Consequently, Corollary 31.5.1 gives continuity of the lower volume of every fixed potential as the background decreases uniformly to θ, while continuity of the upper volume and of the ordinary envelope still requires positive lower volume.

The continuity along increasing sequences, Theorem 31.5.1, is [Li26, v1, Proposition 3.5(i)]: contact identifies the limit of the envelopes and its upper volume, and the deficit estimate then shows that the approximating lower volumes are positive. The continuity along decreasing sequences of models, Theorem 31.5.2, is [Li26, v1, Proposition 3.5(ii)], including its conclusion when the lower volume of the limit vanishes. The uniform small stretch shows that a positive lower bound along the models forces their limit to have positive lower volume, and strict corrections and tail envelopes reduce the variation of the background to the argument for a fixed background. Li states the result for perturbations $\theta + \varepsilon_j\omega$ with $\varepsilon_j \downarrow 0$, whereas the proof here treats arbitrary smooth backgrounds that decrease uniformly in the order of forms. The convergence criterion at positive lower volume, Proposition 31.5.1, is [Li26, v1, Proposition 3.5(iii)]. The deficit bound gives uniform convergence of Lelong numbers on each fixed smooth compact modification, and the envelopes of the restricted analytic approximants give the exact traces of Corollary 31.5.2.

For semipositive backgrounds, Proposition 31.4.1 is the forward direction of [ALS26, Proposition 2.12, pp. 13–14]; the deduction from the strict barrier given here does not use semipositivity. This direction requires neither the identity for ordinary envelopes nor positivity of the lower mass $v_-(\theta)$ of Definition 7.5.1; positivity of the lower mass is used for the converse. Applied to an envelope fixed point, the proposition compares Li's condition (31.5) with the condition (14.20) of Alehyane, Lu and Salouf.

36.19.2 Model topology and analytic approximation

The comparison function d_θ of Definition 31.6.1, defined through the upper volumes of two potentials and of their maximum, is introduced in [Li26, v1, Section 4.2]. That it separates models follows from the characterization by envelopes. The normalized stretching argument of [Li26, v1, Lemma 4.2] gives the quantitative L^1 bound in Proposition 31.6.1, and the three-point estimate (31.7) passes convergence from a subsequence to the whole Cauchy sequence. The construction by maxima of tails in Theorem 31.6.1 follows [Li26, v1, Theorem 4.3], with a fixed $\delta > 0$ and the non-strict lower bound $\underline{\mathrm{vol}}_\theta(u) \geq \delta$. The strict bound $> \delta$ used in version 1 of Li's preprint

is not preserved under limits of Cauchy sequences, as Example 31.6.1 shows. Li explicitly allows d_θ to violate the triangle inequality, and Example 31.12.1 gives an example of such a failure with intersecting divisors.

The criterion for finite rooftop envelopes, Lemma 31.7.1, is [Li26, v1, Lemma 4.1]: after choosing representatives whose masses are close to their upper volumes, a collapse would contradict the lower volume of the common upper bound. Li's argument with softened stretches [Li26, v1, Proposition 4.4] gives Proposition 31.7.1; only the limiting potential needs positive lower volume, and the same comparison gives convergence of the lower volumes of the rooftop envelopes. The two stretching scales in [Li26, v1, Theorem 4.4] produce the monotone bounds of Theorem 31.7.1. Since finite rooftop envelopes are again normalized models, continuity along decreasing models passes their volume bounds to the infinite tails; monotone convergence in capacity then applies, and the subsequence criterion gives convergence of the original sequence.

The results on analytic approximation in Section 31.8 combine this continuity of volumes with the limit of the dyadic multiplier-ideal envelopes. Mixing with a strict analytic potential, with rational coefficients, removes the curvature error in any smooth big background, and two comparisons of backgrounds then give the exact excess of upper volume in (31.10). Conversely, stretching and concavity of the multiplier-ideal envelope show that its fixed points are closed under the upper-volume comparison at positive lower volume. This proves the analytic criterion Theorem 31.8.2 and the Cauchy completeness inherited there. The boundary where the lower volume vanishes in higher dimension, and positivity criteria in terms of ideals, are the separate questions of Problem 31.8.1.

The comparisons of backgrounds in Section 31.10 follow from the same arguments with volumes and envelopes. Transport by a smooth gauge preserves the competitor currents, and ordinary projection corrects the zero obstacle; the explicit scale of small stretches makes the strict barrier uniform on sets of models whose lower volumes are bounded below by a positive constant. This yields (31.12) and extends the analytic criterion to positive background errors that tend to zero. These statements concern singularity volumes and canonical models; they neither identify the mass of the unregularized envelope with a gauge invariant nor assert convergence of arbitrary representatives.

The two-sided transport in Section 31.11 mixes the original model with a strict analytic potential. A common upper background bounds both this transport and the maximum of two transported models, and the resulting volume bounds give uniform comparison, convergence of the transported reference, and equivalence of the limiting comparisons obtained with different strict corrections. The estimate for nearby backgrounds and the three-point modulus then reduce the analytic criterion of that section to Theorem 31.10.1.

The complete metric in Section 31.12 is obtained by the standard construction with finite dyadic chains, carried out directly from the three-point modulus. Stretching first gives continuity of the lower volume, and the reciprocal of the lower volume controls the boundary where it vanishes; the statement about the analytic completion then follows from the approximation criterion above. This metric induces the topology of

the upper-volume comparison; it does not turn the comparison function itself into a metric.

36.19.3 Curves and zero-volume limits

The inverse maximum principle in the plane, Lemma 1.6.4, is the two-dimensional case of the result stated by Brezis and Ponce [BP04, Theorem 3.1], who attribute it to Dupaigne and Ponce; the proof given here uses harmonic barriers on shrinking neighborhoods of a compact polar set. Applied to differences of potentials, it shows that the polar curvature measures reverse the order of the potentials: a potential lying below another, up to a constant, has the larger polar measure. Subtracting a common Green potential gives the formula (1.8) for the maximum. Since these local facts are general, they are proved in Chapter 1.

The classification of models on curves in Section 31.13 combines this order of polar measures with the Green representation of Section 20.3. Every prescribed polar measure gives an ordinary model equivalent to its Green potential, and the characterization of envelopes of positive mass, together with uniqueness of harmonic functions at zero mass, identifies every ordinary envelope with such a model. The formula for the maximum then turns the upper-volume comparison into the total variation of measures, and completeness, geodesics and the nearest ideal model follow from the corresponding operations on measures. The exact defect of analytic approximation is the nonatomic polar mass, which for the Cantor example is the full curvature mass.

For a finite map of curves, the fiber maximum in Lemma 1.22.6 follows from the local branched coordinates and the removable-singularity theorem for subharmonic functions. The weighted trace in Lemma 1.22.7 includes the local degrees at atoms; its agreement with current pullback follows from the Riesz representation on a disk and the Poincaré–Lelong formula. These elementary operations give the envelope identities of Proposition 31.14.3. Applying the curve classification gives the degree-scaled metric embedding and the averaging retraction of Proposition 31.14.4. The component factorization then gives the complete subspace of models descending to the original base in Corollary 32.4.2. The cubic example Example 31.14.1 shows that this averaging retraction is not a nearest-point retraction, even on analytic ideal models.

The continuity statement for curves, Theorem 31.13.2, uses the theorem on convergence in capacity at positive volume in the interior. At zero mass the polar curvature is the whole current, which gives convergence of the currents in total variation. The local estimate Lemma 1.6.3, which follows from the Green representation on a disk, the mean-value estimate for harmonic functions and the weighted Chern–Levine–Nirenberg inequality, gives convergence in capacity at this boundary. Moving a pole distinguishes this topology from that of total variation, while the approximation by finitely many atoms in Theorem 17.4.1 distinguishes weak convergence of the curvature currents from convergence in capacity.

The correspondences between backgrounds in Section 31.14 follow from the same classification by polar measures, which smooth gauges extend to every smooth background of positive area on a curve. Changes of background with equal area preserve total variation, whereas rescaling the area gives homotheties compatible with extraction of atoms and hence with ideal projection. These correspondences are realized by a smooth gauge, obtained from the scalar Poisson equation, followed by ordinary projection. Comparing the new obstacle with its maximum and minimum gives a uniform bound, which yields continuity in L^1 and in capacity up to the boundary of zero mass. Rationality is needed only for the assertion that analytic singularities are preserved under rescaling of the area, because analytic singularities in this book have rational coefficients. At zero area, positivity and Stokes' theorem leave a single normalized potential. The general convergence statement Theorem 1.8.1, proved in Chapter 1, follows from compactness of normalized potentials, uniqueness of harmonic functions and the local estimate relating total variation and capacity. It also gives the collapse of the model representatives as the unscaled metric diameters tend to zero.

36.19.4 Zero-volume models and polar currents

In dimension $n \geq 2$, polar curvature on all modifications still does not detect all ordinary models. The Cantor construction of Lemma 1.6.5 gives the curve potentials of Lemma 32.5.1; their maximum has vanishing first polar currents, but its truncations keep a positive top-degree mass M near the common pole set. A rank bound for smooth maxima gives the construction in every dimension. If $\mathrm{dd}^{\mathrm{c}}(\omega^q) = 0$ for $1 \leq q < n$, for instance for a Kähler form ω, and $A = \int_X \omega^n$, then Corollary 32.5.7 gives a complete ordered interval of diffuse models p_t with volumes $A - Mt^n$ and distances $M|s^n - t^n|$ between p_s and p_t, and every choice of the first polar comparison collapses this interval. Hence the closure of the analytic models for the polar comparison contains models that are not ideal.

For an arbitrary Hermitian background with bounded mass and positive lower mass, the mass bounds for representatives give Proposition 32.5.7. This applies to every Hermitian metric on a compact complex surface, by the existence of Gauduchon metrics, the independence of the bounded mass property from the metric (Proposition 7.5.1) and Corollary 27.18.1. Preservation of residuals and additivity of volumes then give Corollary 32.5.6 for all parameters in the strict range. The upper volume along this interval is strictly decreasing and Lipschitz, although in general the length of the interval is not given by the power formula valid for closed backgrounds, and all first polar coordinates still vanish. In Proposition 32.5.8, two cutoffs with disjoint supports have equal total residual masses and vanishing first polar data, yet their maximum is zero and their distance is positive; in backgrounds with invariant mass, both singularity volumes coincide as well. These scalar data therefore cannot recover the spatial distribution of the residual mass.

A direct calculation with maxima in Lemma 32.5.2 identifies the full residual measure as the tensor product of the measures on the curve factors, multiplied by the product of the positive weights; for a smooth maximum the coefficient reduces to the volume of a simplex. In Proposition 32.5.9, a common cutoff and a uniform curvature bound give a compact family of models indexed by a box of positive weights. Invariance under projection identifies the comparison distance in this family, up to a fixed scale, with the Lebesgue measure of the symmetric difference of coordinate rectangles. The subfamilies on which the product of the weights is constant are complete and have real dimension $n - 1$, although their members have the same full residual measure on X, the same two singularity volumes, the same multiplier ideals and the same first polar currents. The construction takes place in a background with invariant mass and concerns the residual measure on X itself; it does not identify the full residual products on every modification. The problem of describing the complete topology remains open.

36.20 Big backgrounds under mass bounds

Li's theory of equations with general measures [Li26, v1, Theorems 6.1–6.2 and Corollary 6.1] assumes that the manifold has bounded mass and that the Hermitian metric has positive lower mass. It gives Theorem 34.1.1 and the constructions with a model reference in Section 34.7. Under bounded mass, positivity of the lower volume of the model reference implies that the metric has positive lower mass, as shown after Corollary 34.4.1, so expressing the hypothesis through the reference reorganizes this setting without enlarging it.

The relative foundations of this theory use the domination, stretching and obstacle estimates discussed in Section 36.19. In (34.9), the required rooftop envelope is constructed before Lemma 31.3.1 is applied to the relative projection.

Moment and continuity estimates connect these operations on envelopes with existence. The compactness argument with first moments in Proposition 34.3.1, using (34.4), follows [Li26, v1, Proposition 4.2]: one chooses a reference potential in the same singularity type whose mass is sufficiently close to the upper volume, which need not be attained. Continuity for a fixed dominating measure, Theorem 34.2.1, is [Li26, v1, Theorem 4.2]; there the finite rooftop envelopes use only the envelope class, while the two-sided bounds for the mass prevent collapse and control the tails.

The closure theorem for varying references, Theorem 34.5.1, is [Li26, v1, Corollary 4.1]. Projection to ordered lower models gives a fixed relative class for each tail of the sequence, and an exponential contact estimate prevents collapse. The stability for positive exponents in [Li26, v1, Corollary 6.2] assumes uniformly bounded L^p data and the ordinary type of the reference. Theorem 34.7.3 treats instead the relative full-mass class: the models converge for the volume comparison with a common positive lower bound for their lower volumes, and the densities converge in L^1 with respect to a fixed finite non-pluripolar measure. Extracting a summable subsequence produces the dominating measure needed for the closure theorem. For a

positive exponent, uniqueness gives convergence in capacity of the whole sequence of potentials. For exponent zero, the scalars and the measures converge, every L^1 cluster point of the normalized potentials solves the limiting equation, and the corresponding subsequence converges in capacity. In both cases, identifying the right-hand sides in L^1 gives convergence of the measures in total variation.

For the minimal reference, the equations with general measures in Section 34.4.1 are [Li26, v1, Theorems 5.2–5.3]. The estimate for higher moments in Section 34.3.1 follows the concave-weight argument of [Li26, v1, Proposition 5.1 and Lemma 5.1], with the integrability against the fixed singular measure stated explicitly. The smoothing of measures used in the existence proof is collected in Lemmas 1.23.4 and 1.23.5.

The conclusion of Corollary 34.8.1, that for L^p data the solutions have the ordinary type of the reference, is [Li26, v1, Corollary 6.1]. The relative estimate Proposition 34.8.1 starts from a solution in the envelope class and uses a uniform moment of order $m > 2n + 1$. The concave-weight method is related to [Li26, v1, Theorem 5.1], which already assumes the ordinary type of the reference and uses the moment exponent $m > n$; that theorem is not used to deduce the ordinary type of an arbitrary solution in the envelope class.

Addition of measures requires a common lower bound during truncation. The weighted construction in Section 34.6.1 is [Li26, v1, Proposition 6.1]. For unrestricted addition, Theorem 34.7.4 follows [Li26, v1, Theorem 6.3]; an additional rooftop envelope gives the common lower bound needed to pass through the cutoff (34.17).

36.21 Regularity and singular spaces

Regularity in a big class is first a local question on the locus where strict barriers are smooth. For a closed class on a compact Kähler manifold, existence and uniqueness for (29.17) are due to Boucksom, Eyssidieux, Guedj and Zeriahi [BEGZ10, Theorem A, p. 200, equivalently Theorem 3.1, p. 236], and the bounds (29.18) in terms of minimal singularities are [BEGZ10, Theorem B, p. 201, quantitatively Theorem 4.1, p. 245]. The same uniqueness theorem for full mass identifies the solutions in closed classes in Sections 15.21, 15.22, 29.12 and 29.13. Smoothness on the ample locus in the nef case is [BEGZ10, Theorem C, p. 201]; Theorem 29.10.1 gives this smoothness in every closed big class, for a smooth positive volume form.

For a big background that need not be closed, the local Hölder theorem Theorem 15.16.1 is Dang's [Dan26b, v2, Theorem A and Theorem 3.1]. It gives the local input for the argument on the ample locus, while allowing the global potential to be unbounded and the lower curvature bound to be nonclosed. Formulating the equation with a positive scalar and only on the regular locus is equivalent, since the complementary analytic set carries neither smooth volume nor non-pluripolar mass.

The proof of local regularity also needs a quantitative stability estimate. The estimate with a strict barrier in Section 15.14 is [Dan26b, v2, Theorem 3.3 and Remark 3.4] in the case of minimal singularities, with the comparison potential

normalized explicitly; the construction with auxiliary subsolutions in Section 15.3 first treats a semipositive big background.

The interior regularity theorem Theorem 15.17.1 combines this Hölder control with the assumption that the density is smooth and positive on the whole regular locus of one fixed strict analytic barrier. This domain is part of the hypothesis: smoothness of the density on a smaller open set does not give the same conclusion by restriction. The compact Pogorelov example Example 29.11.2 has a smooth positive density on an inner chart, but its density loses regularity along the divisor in the gluing region, so it does not satisfy this hypothesis.

Local smoothing uses the theorem of Guan and Li on the Dirichlet problem [GL10, Theorem 1.1]. In Lemma 15.6.1, the background may be any smooth real $(1,1)$-form and, in local coordinates z, the reference metric is $(\mathrm{i}/2)\partial\bar\partial|z|^2$; for a smooth positive density, a large quadratic correction gives a strict subsolution. The same theorem constructs the approximants in Section 15.17.3 and starts the passage to continuous boundary values and to zero density in Section 15.6; in dimension one it is replaced by the Dirichlet problem for the Poisson equation. Once (15.14) is uniformly elliptic, the Evans–Krylov theorem for variable coefficients [TWWY15, Definition 3.2, Theorem 3.4, and the paragraph following it, pp. 444–445], followed by difference quotients and Schauder estimates, gives smoothness.

The local identification in Section 15.20 uses the local comparison principle of Lemma 15.12.1, whose last step is Bedford and Taylor's uniqueness theorem for the bounded Dirichlet problem [BT82, Corollary 4.4, p. 17]. Degeneration of the density also matters for global stability: Kołodziej and Nguyen [KN21, Theorem 4.1] assume that the density satisfies $f \geq c_0$ for a constant $c_0 > 0$, in addition to their hypotheses on capacity. The densities giving nonuniqueness in Chapter 12 have no such lower bound, even when they are positive almost everywhere.

For compact sets in a Hermitian manifold, Ahn [Ahn26, v3, Theorems 1.1–1.2, p. 3] proves that Hölder continuity of the extremal function is a pointwise local property. The weighted version assumes that both the unweighted extremal function and the weight are Hölder continuous, and that the set is uniformly dense in capacity; the same density condition enters the equivalence with the local and weakly local Hölder properties. These are the regularity settings referred to in Chapter 35.

On singular spaces, regularity on a resolution must be combined with descent. Dang [Dan26b, v2, Theorem B] proves continuity results for bounded solutions on resolutions under additional geometric and integrability hypotheses, notably for log terminal singularities; this result does not cover the arbitrary unibranch singularities in the problem posed in Chapter 35. Pan [Pan25, v1, Theorems A and A'] constructs a Gauduchon metric on the regular locus of a compact reduced irreducible Hermitian space of dimension at least two; it is conformal and uniformly comparable to the given metric, and unique up to scale among such metrics. Control of the conformal factors on a resolution is central to the construction. This result motivates the questions about singular traces in Chapter 35.

The boundary-weighted quadric surface in Appendix A.4.1 is Seo's [Seo26, v1, Example 6.2, pp. 29–30]. The paper uses the weight $\mathrm{e}^{-2\varphi}$, so the potential here is $u = 2\varphi$. The adapted volume comes from a generator of $\omega_X^{[4]}(4D)$ and pulls back to

a constant times $|s|^{-1}$ times Euclidean volume on the finite quotient cover. Splitting the logarithmic quadrant gives the strict monomial test; angular orthogonality and division of convergent series give the full ideal. This proves the explicit example directly, without using the paper's general Newton-body description. The boundary-weighted ideal and the resolution-ideal relation in Proposition A.4.1 use different volume conventions.

The affine toric monomial test in Theorem A.4.1 is the holomorphic-character part of Seo's [Seo26, v1, Theorem 6.1, pp. 28–29], here restricted to full-dimensional cones; the rational boundary need not be effective. Its convex-cone criterion is [Seo26, v1, Lemma 4.4 and Theorem 4.6, pp. 18–19]; the proof here obtains coercivity by a compactness argument and sandwiches the logarithmic neighborhoods between translated cones. The subtraction test of Corollary A.4.1 is the integral calculation behind [Seo26, v1, Proposition 5.10 and Corollary 6.3, pp. 27 and 30]. For this holomorphic-character integrability assertion the exponent can be any positive real number, without assigning a rational boundary to the subtracted weight. The quadric threefold and its whole-germ ideal in Proposition A.4.3 are computed directly from the same criterion and the homogeneous coordinate ring. No general fractional-sheaf generation or ideal-envelope statement is used.

In the construction on unibranch curves in Appendix A.7, the Green function, the exact mass of the pole, uniqueness in the full singularity type, descent through the normalization and the projective cusp are all obtained directly. L^p data on the normalization, $p > 1$, give a continuous residual even when the background coming from a local embedding pulls back to a degenerate form. The example of the cusp also computes the exact threshold for densities on the singular curve. This is a result in dimension one; it neither extends the theorem on log terminal singularities cited above nor gives a general equation on nonclosed singular spaces.

The calculation at the endpoint in Theorem A.7.2 is another direct construction from the Green function of the curve and the radial primitive of Theorem 29.13.1. For one logarithmic family it distinguishes three properties: finite mass of the density, a bounded residual in the ordinary singularity type, and equivalence of multiplier ideals at all weights. It does not characterize arbitrary integrable data.

The equations with nonclosed backgrounds on products with cuspidal curves in Appendix A.8 combine a correction of the reference to a smooth positive form, quantitative Fourier inversion, and a comparison of bounded energies in a background all of whose powers are dd^{c}-closed. For plurisubharmonic functions, the weighted limit Lemma 1.6.1 is quoted from [GZ17, Theorem 3.18]; smooth corrections extend it to quasi-plurisubharmonic functions. The corresponding classical input is Bedford and Taylor's [BT82, Theorem 2.4 and its proof, pp. 7–9]; their operator is 4π times ours, so products of degree r scale by $(4\pi)^{-r}$. The resulting family with small data in all variables has Hölder continuous solutions on the ambient space, unique among all bounded competitors. The existence of bounded solutions for arbitrary nonnegative L^p data on the normalization, $p > 1$, follows separately from Theorem 34.1.1. These results complement the construction with Green functions on curves, without giving a general equation on singular spaces or a sharp criterion for densities on the singular space.

36.22 Geometric problems

A closed strictly positive current has geometric consequences that a positive non-closed background need not have, and this distinction governs the passage from the normalization of equations to the geometric criteria below. The characterization of Fujiki's class by Kähler currents used in Section 35.3 is Demailly and Păun's [DP04, Theorem 0.7]. It shows that a big class forces $X \in C$; conversely, if $X \in C$, the pullback of a closed nef class with positive top self-intersection to a Kähler modification is big by [DP04, Theorem 0.5], and the pushforward of a Kähler current in the pulled-back class is a Kähler current in the original class. In the same paper, the transcendental Grauert–Riemenschneider conjecture is [DP04, Conjecture 0.8].

For nef classes with positive top self-intersection, the equivalence between bigness and bounded mass cited in Section 35.3 is Guedj and Lu's [GL22, Theorem 4.9]. The characterization of Fujiki's class by bounded mass together with positive upper volume of some closed Bott–Chern class is [BGL25, Theorem C]; that upper volume is different from the Gauduchon functional for a fixed probability measure.

The transcendental Morse inequality for differences of nef classes on compact Kähler manifolds, and its consequences for differentiability of the volume and duality of cones, mentioned in Section 35.3, are Tosatti's [Tos26, v2, Theorem 1.1 and Corollary 1.2]; they are quoted from the literature, and their proofs are not reproduced. The qualitative criterion under bounded mass is [GL22, Theorem 4.12].

The non-Hermitian locus in Section 35.3 is that of Dang [Dan26a, Definition 2.12 and Theorem 2.14], where almost analytic singularities have the precise meaning of [Dan26a, Definition 2.8], including a smooth remainder on a suitable log resolution. The qualitative identification of this locus with the null locus, for nef and big backgrounds that may be nonclosed, is [Dan26a, Theorem 1.1], obtained there by observing that the proof of Collins and Tosatti does not use closedness. Theorem A of the same paper gives a numerical criterion under different hypotheses: bounded mass on the ambient manifold and positive lower volume on its submanifolds of positive dimension.

Theorem C of [Dan26a] identifies the singularities of the Chern–Ricci flow at a finite time with the null locus when the limiting lower volume is positive, for arbitrary initial metrics in dimension two and for initial metrics of Guan–Li type in higher dimensions, that is, initial metrics ω_0 with $\mathrm{dd}^{\mathrm{c}}\omega_0 = 0$ and $\mathrm{d}\omega_0 \wedge \mathrm{d}^{\mathrm{c}}\omega_0 = 0$. Once the limiting background is big, [Dan26a, Theorem 4.3] gives convergence as currents and smooth local convergence away from its non-Hermitian locus. This conditional parabolic input should be distinguished from the implication from positive volume to bigness without further hypotheses, which fails by Theorem 27.20.3.

The theorem on the maximal existence time, Theorem 27.20.2, is Tosatti and Weinkove's [TW15, Theorem 1.2 and (1.4), pp. 126–127, and Section 4, pp. 138–142]. The proof given here combines the positive affine path and time-weighted estimates for the potential with Lemma 10.4.1 and classical parabolic regularity. In the conversion to the book's conventions, potentials are multiplied by 2π and time is unchanged. The conditions for geometric smoothing in [DNGL26, v1, Section 3.1, pp. 12–14] include the further compatibility discussed in Section 36.17.

Distance estimates for a twisted flow also depend on its parameters. In [Dan26b, v2, Section 5.2, p. 18], the twist is fixed to be $\mathrm{Ric}(\omega_X)$, for the reference metric ω_X of that paper, and Theorem 5.7 there, p. 21, allows the constants to depend on the endpoint of a finite time interval. The flat flows of Example 27.20.1 show why the formulation for arbitrary twists in Theorem C of the introduction of that paper needs further control: a fixed initial density and a fixed lifetime do not bound the diameter independently of the twist, and a single positive twist can give unbounded diameter over an infinite lifetime.

The conjecture on balanced and pluriclosed metrics referred to in Section 35.3 is stated by Fino and Vezzoni [FV26, Conjecture 1.1]; there the balanced and the pluriclosed metric are allowed to be different. That a compact manifold in Fujiki's class admitting a pluriclosed metric is Kähler is Chiose's [Chi14, Theorem 0.2, p. 3561]. Its case of parallel torsion is discussed in Section 36.3.

The conjectured long-time existence and convergence of the pluriclosed flow on balanced manifolds with $c_1(X) = 0$ is [FV26, Conjecture 1.2]. The discussion in Section 35.3 distinguishes a conserved balanced trace from a pointwise bound for the metric.

36.23 Cotangent torsors

The quotient construction in Proposition B.1.1 follows the mechanism of invariant fibers of Denisi and Höring [DH26, v1, Lemma 2.10, p. 6, Proposition 4.2, p. 12 and pp. 14–18, and Theorem 5.1, p. 19]. Their action has $\kappa = 2\pi i\rho$, where κ is the parameter of Proposition B.1.1 and ρ is theirs, and their function μ_Q equals $1 + 2\sigma$, with σ the function defined before that proposition. The Laurent argument in Lemma B.1.1 proves directly the obstruction for punctured line bundles used in their Example 2.1, while Example B.1.1 computes the degree of an explicit factor of automorphy by its invariant metric. The statement about the quotient includes the two exceptional parameters, which are treated with compact elliptic curves. The computation of classes in Theorem B.2.1 gives their Proposition 4.2 and Lemmas 4.5 and 4.7–4.8, with explicit generators and conventions for projectivization. The flat unitary character in Lemma B.2.2 treats line bundles of every positive degree d directly; the coefficient of the quotient is then $\kappa = 2\pi i d\rho$. This gives their non-Stein conclusion for split elliptic ruled surfaces without the reduction to a finite cover in their Lemma 5.2.

For the split elliptic ruled surface M, the algebra $\bigoplus_{r\geq 0} H^0(M, \mathrm{Sym}^r T_M)$ in Theorem B.3.2 has the bigness and nonfinite generation stated in [DH26, v1, Theorem 1.4, p. 2, with proof on pp. 9–10]. The decomposition by polynomial weights in Lemma B.3.1 gives the exact dimensions and the tautological volume in Theorem B.3.1, and its map to leading coefficients gives a direct proof of nonfinite generation through a quotient by an ideal of square zero.

For canonical extensions, [DH26, v1, Remark 3.8, pp. 10–11] also states bigness and nonfinite generation, leaving the proof to the reader. The homogeneous quadric

in Lemma B.4.1 and the decomposition of coefficients under the unipotent action give the description as a polynomial algebra in Theorem B.4.1. The exact dimensions, the tautological volume and nonfinite generation in Corollary B.4.1 follow for every extension class that is nonzero on the fibers of the ruling, including all Kähler canonical extensions on the split surface. The case of an extension class vanishing on the fibers of the ruling, Proposition B.5.1, follows from the proof of Theorem B.4.1, with the ordinary cotangent quadric of Lemma B.5.1 in place of the homogeneous quadric. Its section algebra is the same as in the case of nonzero restriction, while Proposition B.5.2 identifies the universal cover of the corresponding torsor, which is not Stein, as the product of $\mathbb{C}^2$ with the total space of $\mathcal{O}_{\mathbb{P}^1}(-2)$.

Appendix A
Potentials and equations on unibranch spaces

To solve an equation on a singular complex space, we pass to a resolution and then descend the solution. The pullback background can degenerate along exceptional fibers, and removing a logarithmic pole can leave a residual that does not descend. The examples in this appendix address these two difficulties directly.

Except in the local toric calculations of Appendices A.4.1 and A.4.2, X denotes a reduced irreducible compact complex space of dimension $n \geq 1$ that is *unibranch*: its germ at every point is irreducible, or equivalently, its normalization is a homeomorphism. We write X_{reg} and X_{sing} for the regular and singular loci. Every point of X has a neighborhood that embeds as a closed analytic subset of an open set in some $\mathbb{C}^N$. Smooth functions and forms on X are defined through such local embeddings: they are locally restrictions of smooth functions and forms on the ambient open sets. A Hermitian form on X is a smooth real $(1,1)$-form that is locally the restriction of a smooth strictly positive ambient $(1,1)$-form. Fix a smooth real $(1,1)$-form θ on X.

A *resolution* of X is a proper bimeromorphic holomorphic map $\pi\colon Y \to X$ from a compact complex manifold Y; on Y the pullback $\pi^*\theta$ is an ordinary smooth form, to which the theory of the main text applies. In the last three sections, X is a specific space of this kind.

The basic tool is descent through a resolution. The pullback $\pi^*\theta$ has no component along the fibers of π, so the maximum principle on the connected compact fibers shows that every $\pi^*\theta$-plurisubharmonic function on Y is constant on fibers and descends to X. This correspondence preserves envelopes and convergence. Exceptional sets carry no non-pluripolar mass, so the extended Monge–Ampère measure defined through a resolution does not depend on the resolution. Multiplier ideals computed on resolutions define an ideal envelope on X. Two local toric calculations contrast these ideals with integrability ideals defined by an adapted canonical volume, which may include a boundary weight.

On a projective cone, a logarithmic pole at the vertex becomes a divisor on the incidence resolution, and removing it leaves a strictly positive residual background. This gives uniqueness in the prescribed ordinary class, including for certain nonclosed backgrounds. The bounded equation on the cone uses closedness and an energy

identity. The full logarithmic potential descends even when its smooth residual varies along the contracted divisor.

On a curve whose normalization is a torus, a periodic Green function gives a separate construction. It allows the pulled-back background to vanish at singular points, and it treats every potential in a fixed ordinary type with finitely many logarithmic poles. A projective cusp shows that the residual can be continuous even though the density on the curve is unbounded.

On products of a cuspidal curve with elliptic curves, a smooth potential repairs the degenerate background of the normalization, and two torsion identities fix the masses of all bounded potentials. For data close to the reference density, a positive solution gives uniqueness in the whole bounded class by a dd^{c} energy comparison, even though the background is nonclosed.

We use three facts about unibranch spaces. Resolutions and common smooth refinements of two resolutions exist, and a proper bimeromorphic map to a unibranch space has connected fibers [Xia26, Theorems B.1.1–B.1.3]. The Fornaess–Narasimhan criterion identifies plurisubharmonic functions on X as the upper semicontinuous functions that are subharmonic on holomorphic disks [Xia26, Theorem B.2.1]. The ramified disk lift used below is the argument in [Xia26, proof of Theorem B.2.2].

A.1 Potentials and disk tests

Since X may be singular, θ-plurisubharmonicity is defined by testing along holomorphic disks. Let $\Delta \subset \mathbb{C}$ be the unit disk. For a holomorphic map $f\colon \Delta \to X$, the pullback $f^*\theta$ is a smooth real $(1,1)$-form on Δ and therefore has smooth local potentials. Disk tests thus express the background inequality even when θ has no local potential on X.

Definition A.1.1 An upper semicontinuous function $u\colon X \to [-\infty, +\infty)$, not identically $-\infty$ on any nonempty open subset, is *θ-plurisubharmonic* if it satisfies the following disk condition. For every holomorphic $f\colon \Delta \to X$, the function $u \circ f + a$ is subharmonic or identically $-\infty$ wherever a is a smooth local potential for $f^*\theta$. The set of these functions is denoted by $\mathrm{PSH}(X, \theta)$.

Two choices of a differ by a harmonic function, so the condition is independent of that choice. On every disk where $u \circ f \not\equiv -\infty$, it gives $\mathrm{dd}^{\mathrm{c}}(u \circ f) + f^*\theta \geq 0$ in the sense of distributions.

Lemma A.1.1 *Every member of* $\mathrm{PSH}(X, \theta)$ *is locally the sum of a plurisubharmonic function on X and a smooth function. If X is smooth, this definition agrees with Definition 1.1.1.*

Proof In a local embedding, choose an ambient smooth strictly plurisubharmonic function r with $\mathrm{dd}^{\mathrm{c}} r \geq \theta$ on a smaller neighborhood. For a disk f and a local potential a of $f^*\theta$, the function $r \circ f - a$ is subharmonic. Hence $(u + r) \circ f$ is subharmonic or

identically $-\infty$. The Fornaess–Narasimhan criterion, together with the assumption that u is not identically $-\infty$ on any open set, shows that $u + r$ is plurisubharmonic. The smooth case now follows from Lemma 1.22.1. □

A.2 Descent through a resolution

Let $\pi : Y \to X$ be a resolution. Near a fiber of π, the background $\pi^*\theta$ is dominated by the pullback of a local dd^c-exact form, so a local correction makes a $\pi^*\theta$-plurisubharmonic function plurisubharmonic near the fiber. It is then constant on each compact irreducible component of the fiber, and connectedness of the fiber identifies these constants.

Theorem A.2.1 *Let $\pi : Y \to X$ be a resolution. Pullback $u \mapsto u \circ \pi$ is an order-preserving bijection*

$$\pi^* : \mathrm{PSH}(X, \theta) \longrightarrow \mathrm{PSH}(Y, \pi^*\theta).$$

It preserves suprema of individual functions, boundedness, and ordinary singularity comparisons.

Proof Pullback preserves the disk inequalities. Every nonempty open subset of Y meets the locus where π is biholomorphic, so a pullback is not identically $-\infty$ on any nonempty open set. Injectivity follows from surjectivity of π.

For the converse, take $v \in \mathrm{PSH}(Y, \pi^*\theta)$. Near a fiber, choose the local correction r on X used above. The function $v + r \circ \pi$ is plurisubharmonic on the inverse image of that neighborhood, since

$$\mathrm{dd}^c(v + r \circ \pi) \geq \pi^*(\mathrm{dd}^c r - \theta) \geq 0.$$

Its restriction to each compact irreducible component of the fiber is constant, possibly $-\infty$, by the maximum principle on a resolution of that component. Connectedness of the fiber makes these constants equal: intersecting components have the same value, and the finite intersection graph of the components is connected. Since $r \circ \pi$ is constant on the fiber, so is v.

Thus $v = u \circ \pi$ for a unique function u on X. This function is upper semicontinuous: for each real c, the closed set $\{v \geq c\}$ has closed image under the proper map π, and that image is $\{u \geq c\}$. If u were identically $-\infty$ on a nonempty open set, the same would hold for v on its inverse image. Hence u is not identically $-\infty$ on any nonempty open set.

It remains to verify the disk condition for u with the background θ itself. Let $f : \Delta \to X$ be a holomorphic disk with $u \circ f \not\equiv -\infty$. Near a given point of Δ, take a curve in the fiber product $\Delta \times_X Y$ dominating the disk. Its normalization gives, after shrinking, a finite cover $\rho(t) = t^k$ and a holomorphic lift $f' : \Delta \to Y$ with $\pi \circ f' = f \circ \rho$. Choose a smooth local function a on the original disk with $\mathrm{dd}^c a = f^*\theta$.

The function

$$(u \circ f + a) \circ \rho = v \circ f' + a \circ \rho$$

is subharmonic by Lemma 1.22.1. Subharmonicity descends through ρ by the finite-disk argument following that lemma. Thus $u \circ f + a$ is subharmonic near every point, which proves that pullback is a bijection.

Since π is surjective, pullback preserves suprema, uniform bounds and inequalities up to additive constants; this proves the remaining assertions. □

A.3 Envelopes and normalized compactness

Fix a resolution $\pi\colon Y \to X$. The envelopes of Section 1.19 and Definition 1.19.1 are defined on X by the same formulas: for an obstacle $f\colon X \to [-\infty, +\infty]$, $P_\theta(f)$ is the upper semicontinuous regularization of the supremum of all $u \in \mathrm{PSH}(X, \theta)$ with $u \le f$ pointwise, and $P_\theta[u]$ is defined as in Definition 1.19.1. In the next proposition we assume that the family of these competitors is nonempty and locally uniformly bounded above. The descent bijection identifies the competitors on X and on Y; the remaining issue is whether upper semicontinuous regularization commutes with pullback.

Proposition A.3.1 *For such an obstacle $f\colon X \to [-\infty, +\infty]$, the envelope $P_\theta(f)$ belongs to $\mathrm{PSH}(X, \theta)$ and*

$$P_{\pi^*\theta}(f \circ \pi) = P_\theta(f) \circ \pi.$$

The same identity holds for the ordinary envelope $P_\theta[u]$, for every $u \in \mathrm{PSH}(X, \theta)$.

Proof The preceding theorem identifies the competitor families. Let s be their pointwise supremum on X; its pullback is the pointwise supremum upstairs. The regularization of $s \circ \pi$ on Y is $\pi^*\theta$-plurisubharmonic and not identically $-\infty$, since there is a competitor below it. By the theorem, it descends to a θ-plurisubharmonic function q. Since $q \ge s$ and q is upper semicontinuous, $q \ge s^*$.

For the reverse inequality, fix $x \in X$ and $y \in \pi^{-1}(x)$. For each neighborhood U of x, the unregularized supremum upstairs is bounded by $\sup_U s$ on the open neighborhood $\pi^{-1}(U)$ of y. Therefore $q(x) \le \sup_U s$, and taking the infimum over U gives $q(x) \le s^*(x)$. This proves the identity and shows that $P_\theta(f) = s^*$ is θ-plurisubharmonic. The proof for $P_\theta[u]$ is identical, because pullback preserves the ordinary singularity order together with its constants. □

We give the normalized space $\{u \in \mathrm{PSH}(X, \theta) : \sup_X u = 0\}$ the L^1 topology of the pullbacks to Y. By Theorem 1.4.1 and descent, it is compact. On a fixed resolution, different smooth positive volume forms give equivalent norms. Under a further modification, pullback is continuous by Lemma 1.22.8 and bijective by descent, hence a homeomorphism between the normalized compact spaces. Since any two resolutions are dominated by a common one, the topology does not depend on the resolution; the norms on different models are not compared.

A.4 Ideal envelopes defined on resolutions

Multiplier ideals on smooth models give a singularity relation on X. Call two potentials $u, v \in \mathrm{PSH}(X,\theta)$ *resolution-ideal equivalent* if $u\circ\pi \sim_{\mathcal{I}} v\circ\pi$ on one resolution $\pi\colon Y\to X$ (the ideal type of Section 1.12), that is, if $\mathcal{I}(k\,u\circ\pi) = \mathcal{I}(k\,v\circ\pi)$ for every real $k>0$. By Lemma 1.22.12, applied on a common refinement, this does not depend on the resolution. The ideal projection of Theorem 17.2.1 then defines an envelope in this class.

Proposition A.4.1 *For $u\in\mathrm{PSH}(X,\theta)$, there is a unique potential $P_\theta[u]^{\mathrm{res}}_{\mathcal{I}}$ on X satisfying*

$$P_\theta[u]^{\mathrm{res}}_{\mathcal{I}}\circ\pi = P_{\pi^*\theta}[u\circ\pi]_{\mathcal{I}}$$

on every resolution $\pi\colon Y\to X$, where the right-hand side is the multiplier-ideal envelope of Definition 17.2.1 on Y. It preserves the resolution-ideal class, is idempotent and has supremum zero.

Proof For a fixed resolution, the right-hand side exists by Theorem 17.2.1 and descends by Theorem A.2.1. Under a further smooth modification, the competitor families of the ideal envelopes correspond by the descent theorem and Lemma 1.22.12, and the regularized suprema commute by the argument of Proposition A.3.1. Thus two resolutions give the same pullback on a common refinement, and hence the same function on X. Idempotence, preservation of the class and the normalization follow from the corresponding assertions on Y and surjectivity of π. □

An intrinsic restricted ideal trace on a subspace contained in the singular locus would require a regularization compatible with restriction to that subspace; the resolution-ideal envelope provides neither the regularization nor the trace.

A.4.1 Boundary weights on a quotient surface

The resolution-ideal relation above uses smooth volume forms on smooth models. An integrability ideal on a singular space can instead use a volume form determined by its canonical bundle and a boundary divisor. The following local example shows how this additional weight changes the ideal. In this subsection, replace the compact space fixed above by the affine surface

$$X=\{z_1z_2=z_3^2\}\subset\mathbb{C}^3,\qquad E=\{z_1=z_3=0\},\qquad D=\tfrac12 E.$$

The finite map

$$\pi\colon\mathbb{C}^2\longrightarrow X,\qquad \pi(s,t)=(s^2,t^2,st),$$

identifies X with the quotient by $(s,t)\mapsto(-s,-t)$. It has degree two and is unramified away from the origin. Holomorphic germs at the origin of X correspond to convergent series in s,t containing only monomials s^at^b with $a+b$ even. Indeed, every such

series has the form $F_0(s^2,t^2)+stF_1(s^2,t^2)$ with F_0,F_1 convergent, which is the pullback of $F_0(z_1,z_2)+z_3F_1(z_1,z_2)$.

On X_{reg}, the canonical two-form

$$\Omega=\frac{\mathrm{d}z_1\wedge\mathrm{d}z_3}{z_1}=\frac{\mathrm{d}z_3\wedge\mathrm{d}z_2}{z_2}$$

is defined on the respective charts $z_1\neq0$ and $z_2\neq0$. These charts cover X_{reg}, the formulas agree, and $\pi^*\Omega=2\,\mathrm{d}s\wedge\mathrm{d}t$. The residue form Ω generates the dualizing sheaf of this hypersurface. Since $\operatorname{div}(z_1)=2E$, the meromorphic fourth tensor power

$$\beta=\frac{\Omega^{\otimes4}}{z_1}$$

generates $\omega_X^{[4]}(4D)$, the reflexive fourth power of the dualizing sheaf twisted by $4D$. The fourth root of its squared absolute value defines the *adapted volume* $\mathrm{d}V_D=|\beta|^{2/4}=|z_1|^{-1/2}|\Omega|^2$ on X_{reg}; for zero boundary we use $\mathrm{d}V_0=|\Omega|^2$. Their pullbacks are

$$\pi^*\mathrm{d}V_D=C_D|s|^{-1}\,\mathrm{d}A_s\,\mathrm{d}A_t,\qquad \pi^*\mathrm{d}V_0=C_0\,\mathrm{d}A_s\,\mathrm{d}A_t, \tag{A.1}$$

where $\mathrm{d}A$ denotes Euclidean area and $C_D,C_0>0$ depend only on the convention for absolute values of forms. Replacing a generator by a holomorphic unit multiplies the volume by a positive factor bounded above and below on smaller neighborhoods.

For a boundary $D'\in\{0,D\}$ and a plurisubharmonic function u on this X, define the adapted-volume integrability ideal by

$$\mathcal{I}^{\mathrm{ad}}_{D'}(u)_x=\left\{f\in\mathcal{O}_{X,x}:\int_{U\cap X_{\mathrm{reg}}}|f|^2\mathrm{e}^{-u}\,\mathrm{d}V_{D'}<\infty\right\}, \tag{A.2}$$

where the neighborhood U of x may be chosen for each germ f. This uses the weight convention of Definition 1.12.2. It defines a different object from the resolution-ideal class: the volume in (A.2), including its boundary weight, is part of the definition.

In logarithmic polar coordinates, integrability reduces to the following double integral.

Lemma A.4.1 *Let $A, B, \alpha, \beta, T > 0$. Then*

$$\int_T^\infty \int_T^\infty \exp\bigl(-\alpha p - \beta q + \min(Ap, Bq)\bigr)\, \mathrm{d}q\, \mathrm{d}p < \infty \quad \Longleftrightarrow \quad \frac{\alpha}{A} + \frac{\beta}{B} > 1.$$

Proof Split the quadrant along $q = Ap/B$. On the part $q \geq Ap/B$, set $p_0 = \max\{T, BT/A\}$. After discarding the strip $T < p < p_0$, integration in q gives

$$\frac{1}{\beta} \int_{p_0}^\infty \exp\left(-\left[\alpha - A + \frac{\beta A}{B}\right] p\right) \mathrm{d}p.$$

The discarded strip has finite integral because $\beta > 0$. On the other part, integration in p gives the corresponding tail with coefficient $\beta - B + \alpha B/A$; its discarded strip is finite because $\alpha > 0$. The two coefficients are A and B times $\alpha/A + \beta/B - 1$, and a tail with a nonpositive coefficient diverges. □

Proposition A.4.2 *The plurisubharmonic function*

$$u = \max\{\sqrt{2} \log |z_1|^2,\ 2 \log |z_2|^2\}$$

on X does not have analytic singularities at the origin. Its adapted-volume ideals are

$$\mathcal{I}_D^{\mathrm{ad}}(u) = (z_1, z_3, z_2^2)\mathcal{O}_X, \qquad \mathcal{I}_0^{\mathrm{ad}}(u) = (z_1, z_2, z_3)\mathcal{O}_X.$$

In particular, the same potential requires the generator z_2^2 in place of z_2 when the boundary D is included.

Proof The function is the restriction of a maximum of two plurisubharmonic functions on $\mathbb{C}^3$. On the axes parametrized by $(z_1, z_2, z_3) = (\zeta, 0, 0)$ and $(0, \zeta, 0)$, its logarithmic coefficients are $\sqrt{2}$ and 2. An analytic singularity $c \log \sum_j |f_j|^2 + O(1)$ restricts to a logarithmic coefficient equal to c times an integer on either axis, provided that it is finite off the origin there. The ratio of the two coefficients would therefore be rational, which it is not.

Pullback by the finite degree-two map reduces local integrability to the cover. Neighborhoods of the origin contain and are contained in images of sufficiently small bidisks, so it suffices to integrate over a bidisk in s, t. The pullback weight is

$$u \circ \pi = \max\{2\sqrt{2} \log |s|^2,\ 4 \log |t|^2\}.$$

For $s^a t^b$, $a, b \geq 0$, set $p = -\log |s|^2$ and $q = -\log |t|^2$. By (A.1), angular integration over the bidisk of radius $\mathrm{e}^{-T/2}$ gives, up to a positive constant,

$$\int_T^\infty \int_T^\infty \exp\left(-\left[a + \tfrac{1}{2}\right] p - (b+1)q + \min(2\sqrt{2}\, p, 4q)\right) \mathrm{d}q\, \mathrm{d}p$$

for the boundary D. For zero boundary, replace $a+\frac{1}{2}$ by $a+1$. Thus Lemma A.4.1 gives the respective conditions

$$\frac{a+\frac{1}{2}}{2\sqrt{2}}+\frac{b+1}{4}>1, \qquad \frac{a+1}{2\sqrt{2}}+\frac{b+1}{4}>1. \tag{A.3}$$

These monomial tests determine the entire ideal. Indeed, the measure and the weight on the cover are invariant under separate rotations of s,t. Parseval's identity and Tonelli's theorem show that every nonzero coefficient of a weighted-integrable holomorphic series must satisfy its monomial test. Only exponents with $a+b$ even occur.

In the first test, the minimal allowed exponents, under addition of nonnegative even-total exponents, are

$$(2,0), \quad (1,1), \quad (0,4).$$

To see this, every term with $a\geq 2$ is divisible by s^2, every term with $a=1$ has odd $b\geq 1$ and is divisible by st, and a term with $a=0$ has even b and passes the test exactly when $b\geq 4$. Each of the three displayed exponents passes the test. Splitting a convergent invariant series into these three groups and dividing gives convergent invariant coefficient series on a smaller bidisk. They descend to holomorphic germs on X, bounded on a smaller neighborhood. Since the three generators are weighted-integrable, every such combination is weighted-integrable. This proves the first ideal at the origin. For the second test, every nonconstant even-total monomial is allowed, with minimal exponents $(2,0)$, $(1,1)$, $(0,2)$, and the same argument gives the second ideal.

Away from the origin, u is locally bounded and X is smooth. The only additional volume singularity lies along E: with $z_2\neq 0$, z_3 is a normal coordinate and $z_1=z_3^2/z_2$, so $\mathrm{d}V_D$ is a smooth positive factor times $|z_3|^{-1}$ times smooth volume. This is locally integrable. Both ideals therefore equal $\mathcal{O}_X$ there, as do the ideals generated by the displayed coordinates, which proves the sheaf identities. □

The calculation depends on the prescribed volume as well as on u. It does not change the resolution-ideal equivalence used in Proposition A.4.1. The next calculation expresses the exponent test for a general full-dimensional affine toric cone.

A.4.2 Adapted monomial tests on affine toric spaces

The quotient calculation expresses integrability as a strict inequality on exponents. On an affine toric space the same inequality comes from the recession of a convex function, while the canonical bundle and boundary supply the translation of the exponent. This also applies to cones with more rays than their dimension.

In this subsection let N be a lattice of rank $n \geq 1$, let $M = \operatorname{Hom}(N, \mathbb{Z})$, and let $\sigma \subset N_{\mathbb{R}}$ be a full-dimensional strongly convex rational polyhedral cone. Its dual cone is $\sigma^{\vee} = \{m : \langle m, v\rangle \geq 0 \text{ for all } v \in \sigma\}$. The normal affine toric space

$$X_\sigma = \operatorname{Spec} \mathbb{C}[\sigma^{\vee} \cap M]$$

is understood with its analytic structure. Its dense torus $T = \operatorname{Hom}(M, \mathbb{C}^*)$ has characters χ^m, and its unique closed orbit is a vertex o. Choose nonzero generators $b_1, \dots, b_s$ of the semigroup $\sigma^{\vee} \cap M$; their characters embed X_σ as a closed affine subset of $\mathbb{C}^s$ and all vanish at o.

For each ray ρ of σ, write v_ρ for its primitive lattice generator and D_ρ for the corresponding prime divisor. Fix a boundary $\mathbb{Q}$-divisor $D = \sum_\rho a_\rho D_\rho$ such that $K_{X_\sigma} + D$ is $\mathbb{Q}$-Cartier. With $K_{X_\sigma} = -\sum_\rho D_\rho$, this means that there is $w_D \in M_{\mathbb{Q}}$ with

$$\langle w_D, v_\rho\rangle = 1 - a_\rho \quad \text{for every } \rho. \tag{A.4}$$

Take an integer $q > 0$ with $q w_D \in M$. For torus coordinates $t_1, \dots, t_n$ determined by a basis of M, the meromorphic tensor

$$B_D = \chi^{q w_D} \left(\frac{\mathrm{d}t_1}{t_1} \wedge \cdots \wedge \frac{\mathrm{d}t_n}{t_n} \right)^{\otimes q}$$

generates $\omega_{X_\sigma}^{[q]}(qD)$. Indeed, its order along D_ρ is $-q a_\rho$. The adapted volume $\mathrm{d}V_D = |B_D|^{2/q}$ does not depend on the choice of q, and a change of basis of M changes B_D only by a sign. We use the ideal $\mathcal{I}_D^{\mathrm{ad}}(u)$ defined by the holomorphic-germ integral in (A.2) on this X_σ.

Let u be a plurisubharmonic function invariant under the compact torus of coordinate rotations. The logarithmic map $\operatorname{Log}\colon T \to N_{\mathbb{R}}$ satisfies $\langle m, \operatorname{Log} t\rangle = \log|\chi^m(t)|$. There is a finite convex function g on $N_{\mathbb{R}}$ with

$$u(t) = 2g(\operatorname{Log} t) \qquad (t \in T).$$

To see convexity, pull $u/2$ back by the exponential covering of T. It is independent of the imaginary coordinates, so its distributional complex Hessian is the real Hessian of g up to a positive constant. Since u is finite somewhere on T, convexity along lines shows that g never takes the value $-\infty$. The factor two matches the weight e^{-u}.

For a finite convex function g write

$$(\operatorname{rec} g)(v) = \lim_{t \to \infty} \frac{g(x + tv) - g(x)}{t}.$$

The convex difference quotients increase to this limit in $\mathbb{R} \cup \{+\infty\}$; the limit does not depend on the base point x. The latter fact follows by comparing $g(x + tv)$ with convex combinations of $g(tv/(1-\delta))$ and $g(x/\delta)$, for $0 < \delta < 1$, and then reversing

the translation. Define the Newton set

$$P(u) = \left\{ \ell \in M_{\mathbb{R}} : \sup_{x \in -\sigma} \big(\langle \ell, x \rangle - g(x) \big) < \infty \right\}. \tag{A.5}$$

Only its interior will enter the integrability test; the set itself need not be closed.

Lemma A.4.2 *Let C be a closed full-dimensional convex cone in a Euclidean space, let g be finite and convex, and let ℓ be a linear form. Then*

$$\int_C \exp\big(2\langle \ell, x \rangle - 2g(x)\big)\, \mathrm{d}x < \infty \quad \Longleftrightarrow \quad \langle \ell, v \rangle < (\operatorname{rec} g)(v) \quad (v \in C \setminus \{0\}).$$

This condition is also equivalent to ℓ being an interior point of $\{a : \sup_C(\langle a, x \rangle - g(x)) < \infty\}$. The integral has the same finiteness on every translate of C.

Proof Set $F = g - \ell$. Suppose its recession is positive on $C \setminus \{0\}$. Then there are $\gamma, r > 0$ with $F(x) \geq \gamma |x|$ for all $x \in C$ with $|x| \geq r$. Otherwise there are $x_j \in C$ with $|x_j| \to \infty$ and $F(x_j)/|x_j| \leq 1/j$. After taking a subsequence, $x_j/|x_j| \to v \in C$ with $|v| = 1$. For fixed $t > 0$, convexity gives

$$F\left(t \frac{x_j}{|x_j|} \right) \leq \left(1 - \frac{t}{|x_j|} \right) F(0) + \frac{t}{|x_j|} F(x_j).$$

Continuity gives $F(tv) \leq F(0)$ for every $t > 0$, contradicting positive recession. Since F is bounded below on the compact set $\{x \in C : |x| \leq r\}$, it follows that $F(x) \geq \gamma |x| - A$ on C for a constant A. Exponential decay proves integrability. The same bound persists when ℓ is changed by a sufficiently small linear form, so ℓ lies in the stated interior.

Conversely, if $(\operatorname{rec} F)(v) \leq 0$ for some nonzero $v \in C$, then $F(tv) \leq F(0)$ for $t \geq 0$. Choose a compact ball K in the interior of C. On the half-tube $\{tv + x/2 : t \geq 1,\ x \in K\}$, convexity gives

$$F(tv + x/2) \leq \tfrac{1}{2} F(2tv) + \tfrac{1}{2} F(x).$$

Thus e^{-2F} has a positive lower bound on a set of infinite volume. If instead ℓ is in the stated interior, for each nonzero $v \in C$ one may perturb ℓ within that interior by a linear form positive on v. Dividing its defining upper bound along tv by t gives the strict recession inequality for ℓ. Finally, translating C replaces $g(x)$ by $g(x + b)$ and multiplies the exponential by a constant; recession is unchanged. □

Theorem A.4.1 *For $m \in \sigma^\vee \cap M$, the following are equivalent:*

$$\chi^m \in \mathcal{I}_D^{\mathrm{ad}}(u)_o, \qquad \chi^m \in \Gamma(X_\sigma, \mathcal{I}_D^{\mathrm{ad}}(u)), \qquad m + w_D \in \operatorname{int} P(u).$$

Equivalently, $\langle m + w_D, v \rangle < (\operatorname{rec} g)(v)$ for every nonzero $v \in -\sigma$.

Proof On the torus, polar coordinates turn the weighted monomial integral over a domain invariant under the compact torus into a positive constant times

$$\int_Q \exp\bigl(2\langle m + w_D, x\rangle - 2g(x)\bigr)\,\mathrm{d}x,$$

where Q is the logarithmic image of the domain. The omitted toric divisors have measure zero on the regular locus. A basis of neighborhoods of o is given by $|\chi^{b_j}| < \epsilon$ for every j, with $0 < \epsilon < 1$. Their logarithmic domains satisfy

$$-\sigma - b \;\subset\; Q_\epsilon = \{x : \langle b_j, x\rangle < \log\epsilon \text{ for all } j\} \;\subset\; -\sigma, \tag{A.6}$$

for $b \in \operatorname{int}\sigma$ sufficiently far along an interior ray. Indeed, every $\langle b_j, b\rangle$ is strictly positive, and the b_j generate the dual cone. Lemma A.4.2, applied with $C = -\sigma$ and $\ell = m + w_D$, gives necessity and sufficiency at o.

For a compact subset of X_σ, the embedding coordinates have absolute value at most some $R > 1$. Its logarithmic image is contained in $\{x : \langle b_j, x\rangle \le \log R\}$, which is contained in $b_R - \sigma$ for a sufficiently large $b_R \in \operatorname{int}\sigma$. Under the same strict condition the integral over this translated cone is finite. This proves local integrability everywhere. The converse follows by testing at o, and the interior formulation follows from the lemma. □

Corollary A.4.1 *Let $\mathfrak{a} = (\chi^{a_1}, \dots, \chi^{a_r})$ be a nonzero monomial ideal on X_σ, with $a_j \in \sigma^\vee \cap M$, and let $c > 0$ be real. For the measurable weight*

$$v = u - c\log\left(\sum_j |\chi^{a_j}|^2\right),$$

a holomorphic character χ^m is locally integrable with weight $\mathrm{e}^{-v}\mathrm{d}V_D$ at o, or everywhere on X_σ, exactly when

$$m + w_D + ca_j \in \operatorname{int} P(u) \quad \textit{for every } j.$$

Values on the common zero set do not affect these integrals.

Proof For a finite sum of nonnegative terms and $c > 0$, its c-th power is comparable to the sum of the c-th powers of its terms. Thus the integrand is comparable to

$$\sum_j |\chi^m|^2 |\chi^{a_j}|^{2c} \mathrm{e}^{-u}\,\mathrm{d}V_D.$$

Each term gives the preceding logarithmic integral with exponent $m + w_D + ca_j$. Lemma A.4.2 applies to real exponents, so no rationality of c is needed for this integrability assertion. □

Proposition A.4.3 *Let $X = \{xw = yz\} \subset \mathbb{C}^4$, with vertex o, let $\mathfrak{m}_o = (x, y, z, w)$, and take zero boundary. For $\lambda > 0$ put*

$$u_\lambda = \lambda \log(|x|^2 + |y|^2 + |z|^2 + |w|^2).$$

The adapted-volume ideal at the vertex is

$$\mathcal{I}_0^{\mathrm{ad}}(u_\lambda)_o = \mathfrak{m}_o^{\max\{0,\lfloor\lambda\rfloor-1\}} \mathcal{O}_{X,o}.$$

In particular, the constant germ is integrable exactly when $\lambda < 2$.

Proof On the torus use x, y, z as characters, so $w = yz/x$. The character lattice is $\mathbb{Z}^3$, with semigroup generators

$$(1, 0, 0), \quad (0, 1, 0), \quad (0, 0, 1), \quad (-1, 1, 1).$$

The cone σ has primitive rays $(0, 1, 0)$, $(0, 0, 1)$, $(1, 1, 0)$ and $(1, 0, 1)$. It is full-dimensional with four rays. The residue form $\Omega = \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z/x$ generates the canonical sheaf of this hypersurface; it equals yz times the logarithmic torus form. Hence $w_0 = (0, 1, 1)$, which pairs to one with each primitive ray.

Write $d(m) = m_1 + m_2 + m_3$. Each semigroup generator has degree one. The logarithmic function g differs by a bounded term from λ times the maximum of the four linear forms given by those generators. Their convex hull is the section of $\sigma^\vee$ by $d = 1$, so

$$P(u_\lambda) = \{\ell \in \sigma^\vee : d(\ell) \geq \lambda\}.$$

Since $w_0 \in \operatorname{int}\sigma^\vee$ and $d(w_0) = 2$, the character condition is $d(m) + 2 > \lambda$. The smallest allowed integer degree is $k = \max\{0, \lfloor\lambda\rfloor - 1\}$.

This also gives the ideal for whole holomorphic germs. Lift a germ to a convergent power series in x, y, z, w. On a smaller invariant neighborhood, angular Parseval and Tonelli force every nonzero character coefficient to satisfy the same test. All homogeneous terms of degree less than k therefore vanish in the coordinate ring. They are multiples of the homogeneous relation $xw - yz$, and subtracting those finitely many terms leaves a lift of order at least k. Thus the germ belongs to $\mathfrak{m}_o^k$. Conversely, every degree-k monomial is integrable by the character test, and bounded holomorphic coefficient germs preserve integrability. This proves the identity and its strict endpoint. □

These ideals use the adapted canonical volume, whereas the resolution-ideal equivalence above uses smooth volumes on the models. Return now to the compact unibranch space and background fixed at the beginning of the appendix.

A.5 Extended measures and finite damping

For $u \in \mathrm{PSH}(X,\theta)$ and a resolution $\pi\colon Y \to X$, define

$$\mathrm{MA}^{\mathrm{ext}}_{X,\theta}(u) = \pi_* \mathrm{MA}^{\mathrm{ext}}_{Y,\pi^*\theta}(u\circ\pi), \tag{A.7}$$

where $\mathrm{MA}^{\mathrm{ext}}_{Y,\pi^*\theta}$ is the extended measure of Theorem 7.1.1 on Y.

Theorem A.5.1 *The right-hand side of* (A.7) *is the same for every resolution, including resolutions that modify regular points. It is a positive σ-finite Borel measure, gives zero mass to X_{sing} and to locally pluripolar sets, and may have infinite total mass; for bounded u it is finite. Put $r_n = \max(n-2,0)$. There are finite constants C and, for each $q > r_n$, C_q, depending only on the fixed background data and, for C_q, on q, such that for every nonpositive $u \in \mathrm{PSH}(X,\theta)$ and every $p > 0$,*

$$\int_X \mathrm{e}^{pu}\,\mathrm{MA}^{\mathrm{ext}}_{X,\theta}(u) \le C(1+p^{-1})^{r_n}, \qquad \int_X (1-u)^{-q}\,\mathrm{MA}^{\mathrm{ext}}_{X,\theta}(u) \le C_q.$$

The constants may be obtained on any one fixed resolution.

Proof Given two resolutions, choose a proper analytic subset $S \subset X$ containing X_{sing} and the images of both exceptional sets. Each map is an isomorphism over $X \setminus S$, and there the two measures are given by the same bounded local products, followed by truncation and exhaustion. On each resolution, the inverse image of S is a proper analytic subset and has zero extended mass by Theorem 7.1.1. Hence both pushforwards give S zero mass. This proves independence, including for resolutions whose exceptional sets lie over regular points.

Positivity follows from positivity on Y. On $X \setminus S$ either resolution is a Borel isomorphism, so a countable exhaustion of Y by sets of finite measure pushes forward to one on X; adjoining the null set S proves σ-finiteness. A locally pluripolar subset of X pulls back to a pluripolar set off the exceptional locus, and the exceptional locus itself has zero mass. This gives the asserted vanishing.

If u is bounded, its pullback is bounded on the compact manifold Y, and local bounded-product estimates give finite total mass. For general nonpositive u, apply Theorem 8.6.1 on Y to the smooth background $\pi^*\theta$. Since the weights on Y are the pullbacks of the weights on X, pushforward identifies the weighted integrals exactly, and the constants come from $\pi^*\theta$ and a Hermitian metric on Y. □

We define convergence in capacity on X as convergence in capacity of the pullbacks to a resolution, using a smooth Hermitian metric there. The local capacity comparison of Section 1.8 makes the metric immaterial, and Lemma 1.22.9, applied on a common resolution, makes the choice of resolution immaterial.

By Theorem 8.6.1 and proper pushforward, both damped measures in the theorem depend weakly continuously on nonpositive potentials for this convergence in capacity. For nonpositive potentials $u_j \in \mathrm{PSH}(X,\theta)$ converging in capacity, the undamped measures $\mathrm{MA}^{\mathrm{ext}}_{X,\theta}(u_j)$ converge weakly if, in addition, their total masses are uniformly

bounded and their masses on $\{u_j < -T\}$ tend to zero uniformly in j as $T \to \infty$. Pullback preserves these mass and sublevel conditions.

General equations on singular spaces require estimates for the degeneracy of the pulled-back background and of the density; see Problem 35.10.1. For the cone geometry below, a radial construction gives solutions in a prescribed logarithmic class and in the bounded class of a closed background.

A.6 Equations on projective cones

The residual construction of Section 15.21 has a counterpart on a projective cone, which is a singular space. A logarithmic pole at the vertex becomes a divisor on the incidence resolution, and removing it leaves a positive residual background. The radial equation below determines both the solution and its normalization scalar. Uniqueness among bounded residuals holds because the lifted density is smooth and its positive locus is connected and dense, even though the density vanishes on the exceptional section.

Let $B \subset \mathbb{P}^N$ be a smooth connected, linearly nondegenerate, projectively normal variety of dimension $n-1$, where $N \ge n \ge 2$. Let $X \subset \mathbb{P}^{N+1}$ be its projective cone, with vertex $p = [0 : \cdots : 0 : 1]$, and let $\theta_0 = \omega_{\mathrm{FS}}|_X$ be the restriction of the Fubini–Study form of $\mathbb{P}^{N+1}$. Write the ambient homogeneous coordinates as $[Z : W]$, with $Z = (Z_0, \dots, Z_N)$. The incidence resolution $\pi\colon Y \to X$ is a smooth $\mathbb{P}^1$-bundle $f\colon Y \to B$; its section $E = \pi^{-1}(p)$ is contracted, and π is an isomorphism off E. These cones are reduced, irreducible, compact and normal, hence unibranch, and their vertices are singular.

Indeed, the homogeneous coordinate ring of B is normal by projective normality. Adjoining W preserves normality, and so do the degree-zero projective charts. Algebraic normality passes to the associated analytic space by the algebraic–analytic normality theorem. Smoothness and connectedness of B give irreducibility. In the chart $W = 1$, the ideal of the affine cone has no linear terms, by linear nondegeneracy. Its tangent space at the origin therefore has dimension $N+1 > n$, so p is singular. The incidence construction is the bundle of lines in $\mathcal{O}_B \oplus \mathcal{O}_B(-1)$ and has the stated contraction and isomorphism properties.

Write

$$\omega_B = \omega_{\mathrm{FS}}|_B, \qquad \beta = \pi^*\theta_0, \qquad \sigma = f^*\omega_B,$$
$$d = \int_B \omega_B^{n-1}, \qquad x = \frac{\|Z\|^2}{\|Z\|^2 + |W|^2}.$$

The same symbol x denotes its pullback to Y. Choose $\Psi \in C^\infty(\mathbb{P}^N, \mathbb{R})$ with $\omega_{\mathrm{FS}} + \mathrm{dd^c}\Psi > 0$ and $k \in C^\infty([0,1])$ with $k > 0$. Put $\psi = \Psi|_B$ and define on Y

$$\begin{gathered} s = \frac{\mathrm{e}^\psi x}{1 - x + \mathrm{e}^\psi x}, \qquad \ell = \log(1 - x + \mathrm{e}^\psi x), \qquad \sigma_\psi = f^*(\omega_B + \mathrm{dd^c}\psi), \\ B_\psi = \beta + \mathrm{dd^c}\ell, \qquad \nu = k(s)B_\psi^n, \qquad \mu = \pi_*\nu, \\ I = \int_0^1 k(r) r^{n-1}\,\mathrm{d}r, \qquad M = ndI. \end{gathered} \tag{A.8}$$

These data prescribe the measure μ before the equation is solved. The local form and the fiber integral below describe its degeneracy along E and show that its total mass is M.

Choose a local holomorphic lift $b(z)$ of the embedding of B, and write $S(z) = \sum |b_j(z)|^2$ and $Z = tb(z)$. In these local coordinates (z, t) on Y,

$$\beta = \mathrm{dd^c}\log(1 + |t|^2 S), \qquad \sigma = \mathrm{dd^c}\log S.$$

On $0 < x < 1$ put $\rho = \log(|t|^2 S)$, $\rho_\psi = \rho + \psi$, $h(r) = \log(1 + \mathrm{e}^r)$ and $\zeta_\psi = \mathrm{d}\rho_\psi \wedge \mathrm{d^c}\rho_\psi$. Then $x = h'(\rho)$, $s = h'(\rho_\psi)$ and $\ell = h(\rho_\psi) - h(\rho)$. Differentiation gives

$$B_\psi = s\sigma_\psi + s(1 - s)\zeta_\psi, \qquad \sigma_\psi^n = 0, \qquad \zeta_\psi^2 = 0. \tag{A.9}$$

Near E, a local potential of B_ψ is $\log(1 + \mathrm{e}^\psi |t|^2 S)$; at $t = 0$ its Hessian has rank one, in the transverse direction. At the infinity section, with $v = 1/t$, a local potential is $\log(\mathrm{e}^\psi S + |v|^2)$; its tangential Hessian is σ_ψ and its transverse coefficient is positive. These computations show that B_ψ is smooth and semipositive, and positive off E. Since $n \geq 2$, its top power vanishes on E.

Along a fiber, $\mathrm{d^c}\rho_\psi = \mathrm{d}(\arg t)/(2\pi)$ and $\mathrm{d}\rho_\psi = \mathrm{d}s/[s(1 - s)]$. Base components disappear after wedging with σ_ψ^{n-1}, so (A.9) gives

$$f_*\nu = nI(\omega_B + \mathrm{dd^c}\psi)^{n-1}.$$

Since $\omega_B + \mathrm{dd^c}\psi$ is closed, Stokes' theorem on B gives $\int_B (\omega_B + \mathrm{dd^c}\psi)^{n-1} = d$, hence $\nu(Y) = M$. Smooth measures give E zero mass. Therefore $\mu(X) = M$ and $\mu(\{p\}) = 0$. Taking $\psi = 0$ and $k = 1$ gives $\int_Y \beta^n = d$.

A.6.1 Logarithmic poles at the vertex

The logarithmic pole supplies the tangential positivity that β lacks along E. Fix $0 < a < 1$ and put $\varphi = a \log x$, with value $-\infty$ at p. Set

$$\begin{aligned} \beta_a &= (1-a)\beta + a\sigma, \qquad V = d(1-a^n), \qquad c_a = V/M, \\ P_a(s) &= \left(a^n + c_a n \int_0^s k(r) r^{n-1}\, \mathrm{d}r \right)^{1/n}, \\ Q_a'(s) &= \frac{P_a(s) - a - (1-a)s}{s(1-s)}, \qquad Q_a(1) = 0, \\ q_a &= a\psi + (1-a)\ell + Q_a(s), \qquad A_a = \beta_a + \mathrm{dd^c} q_a. \end{aligned} \tag{A.10}$$

The singularities of Q_a' at $s = 0$ and $s = 1$ are removable. As the proof below shows, the form β_a is strictly positive everywhere, whereas A_a is semipositive and strictly positive off E. Off E, we may therefore view A_a as a positive form on $X \setminus \{p\}$.

Choose a relatively compact coordinate ball $O \subset X \setminus \{p\}$ and a smooth compactly supported endomorphism H of $T^{1,0}O$ that is self-adjoint and trace-free with respect to A_a. For tangent vectors v, w of type $(1,0)$ at the same point of O, define

$$A_H(v, \bar{w}) = A_a(\exp(H)v, \bar{w}), \qquad \eta_H = A_H - A_a, \qquad \theta_H = \theta_0 + \eta_H, \tag{A.11}$$

where η_H is extended by zero. We call H *admissible* if

$$\theta_H > 0, \qquad \beta_{a,H} := \beta_a + \pi^* \eta_H > 0. \tag{A.12}$$

These are explicit open conditions on the prescribed data, and every sufficiently small H satisfies them.

Theorem A.6.1 *Let H be admissible, and let*

$$C_{a,H} = \{ w \in \mathrm{PSH}(X, \theta_H) : \ \varphi - C \le w \le \varphi + C \text{ for some } C < \infty \}$$

be the ordinary class of φ. There is a unique pair (c, u), with $c > 0$ and $u \in C_{a,H}$ normalized by $\sup_X u = 0$, such that

$$\mathrm{MA}^{\mathrm{ext}}_{X,\theta_H}(u) = c\mu.$$

It is given by $c = c_a$ and

$$u \circ \pi = \varphi \circ \pi + q_a - K, \qquad K = \sup_Y (\varphi \circ \pi + q_a).$$

Every member of the class has finite extended mass. The solution has mass V and is the same for every admissible H. For $H = 0$, every member of the class has mass V.

The solution is continuous with values in $[-\infty, +\infty)$, is smooth off p, and tends uniformly to $-\infty$ at p. Its smooth residual $q_a - K$ on Y satisfies

$$(q_a - K)|_E = a\psi + Q_a(0) - K.$$

If ψ is nonconstant, this residual is not the pullback of any function on X. There are arbitrarily small admissible H for which θ_H is not closed.

For every $w \in C_{a,H}$ and every resolution $\tau\colon Z \to X$, including resolutions that modify regular points,

$$\tau_* \left\langle \left(\tau^*\theta_H + \mathrm{dd^c}(w \circ \tau)\right)^n \right\rangle = \mathrm{MA}^{\mathrm{ext}}_{X,\theta_H}(w), \tag{A.13}$$

where $\langle\cdot\rangle$ denotes the non-pluripolar product on Z. The measure on Z is finite. Thus the measure, the normalized descended solution and the scalar are independent of the resolution and of every common refinement.

Proof We first establish the divisor formula

$$\beta + \mathrm{dd^c}(\varphi \circ \pi) = a[E] + \beta_a. \tag{A.14}$$

Indeed, $\log x = \log|t|^2 + \log S - \log(1 + |t|^2 S)$ and $\mathrm{dd^c} \log|t|^2 = [t = 0]$. Along E the kernels of β and σ are complementary, and $\beta > 0$ off E. Thus β_a is a smooth Kähler form. Fiber integration also gives

$$\int_Y \beta_a^n = nd \int_0^1 [a + (1-a)x]^{n-1}(1-a)\,\mathrm{d}x = d(1 - a^n) = V. \tag{A.15}$$

By (A.10), $P_a(0) = a$, $P_a(1) = 1$ and

$$P_a'(s) = \frac{c_a k(s) s^{n-1}}{P_a(s)^{n-1}}.$$

The numerator defining Q_a' vanishes at both ends, so Q_a is smooth. On $0 < x < 1$, a local potential for the full positive current is $F_a(\rho_\psi)$, where

$$F_a(r) = ar + (1-a)h(r) + Q_a(h'(r)).$$

Direct differentiation gives $F_a' = P_a(s)$ and $F_a'' = c_a k(s) s^n (1-s)/P_a(s)^{n-1}$. Consequently

$$A_a = P_a(s)\sigma_\psi + \frac{c_a k(s) s^{n-1}}{P_a(s)^{n-1}}(B_\psi - s\sigma_\psi), \qquad A_a^n = c_a \nu. \tag{A.16}$$

The coefficients are smooth since $P_a \geq a > 0$. On E the restriction is $a\sigma_\psi$; at infinity the horizontal coefficient is one and the transverse coefficient relative to $B_\psi - \sigma_\psi$ is $c_a k(1) > 0$. This proves all the asserted positivity properties, including along both sections.

The functions $\varphi \circ \pi$ and $\varphi \circ \pi + q_a$ are quasi-plurisubharmonic on Y. By (A.14), their curvature currents in the background β are $a[E]+\beta_a$ and $a[E]+A_a$, respectively, so both are positive. The descent theorem Theorem A.2.1 gives $\varphi \in \mathrm{PSH}(X, \theta_0)$ and a potential u_0 with $u_0 \circ \pi = \varphi \circ \pi + q_a$. The bounded residual q_a puts u_0 in the ordinary class of φ and gives the uniform logarithmic asymptotics at p. The finite supremum of $\varphi \circ \pi + q_a$ is the constant K of the theorem.

Let H be admissible. Self-adjointness makes $\exp H$ positive, and $\det \exp H = \exp(\operatorname{tr} H) = 1$. Hence

$$(A_a + \pi^*\eta_H)^n = A_a^n = c_a \nu, \qquad \beta_{a,H} + \mathrm{dd^c}(q_a - K) = A_a + \pi^*\eta_H. \tag{A.17}$$

The function $u = u_0 - K$ does not depend on H. On the regular locus its θ_H-curvature form is the positive form in (A.17); near p the background and the potential are unchanged. Thus u is θ_H-plurisubharmonic. Formula (A.14), together with the deformation, gives $\pi^*\theta_H + \mathrm{dd^c}(u \circ \pi) = a[E] + A_a + \pi^*\eta_H$; removing the divisor proves the equation. The mass of the solution is $c_a M = V$. This calculation uses the preservation of the determinant, rather than a mass identity for the class of a nonclosed form.

Bounded residuals and uniqueness. Let $w \in C_{a,H}$. On $Y \setminus E$, its residual $g = w\circ\pi - \varphi\circ\pi$ is bounded and $\beta_{a,H}$-plurisubharmonic. By Lemma 1.3.7, it extends across E with the same background. Conversely, for any bounded $\beta_{a,H}$-plurisubharmonic g, the function $\varphi \circ \pi + g$ has the positive curvature current $a[E] + \beta_{a,H} + \mathrm{dd^c} g$ in the pulled-back background. It descends by Theorem A.2.1, and its uniform bounds relative to $\varphi \circ \pi$ put the descended potential in $C_{a,H}$. Thus bounded residuals correspond exactly to members of the prescribed ordinary class.

The bounded product is finite on the compact manifold Y and gives E zero mass, by Theorem 1.6.1 and Proposition 1.7.1. Thus

$$\mathrm{MA}^{\mathrm{ext}}_{X,\theta_H}(w) = \pi_*\big(\beta_{a,H} + \mathrm{dd^c} g\big)^n. \tag{A.18}$$

For $H = 0$, closedness of β_a makes every term of positive degree in $\mathrm{dd^c} g$ in the bounded expansion integrate to zero, so (A.15) gives mass V for every member of the class. In the nonclosed case only finiteness is used here.

If w solves the equation with a scalar $c' > 0$, its lifted measure is $c'\nu$, since both sides give E zero mass. Both residuals, g and $q_a - K$, are continuous by Corollary 11.4.1, and the uniqueness of the scalar in Proposition 11.6.1 gives $c' = c_a$. The density $\nu/\beta_{a,H}^n$ is smooth and nonnegative, and its positive locus is exactly $Y \setminus E$. This locus is connected and dense: it is a bundle over the connected base B whose fibers, $\mathbb{P}^1$ minus a point, are connected. Theorem 12.4.1 shows that $g - (q_a - K)$ is constant. The normalization fixes this constant, and both full functions equal $-\infty$ at p. This proves uniqueness in the ordinary class.

The uniform logarithmic asymptotics give the continuity of the full solution with values in $[-\infty, +\infty)$. Its residual restricts to E as $a\psi + Q_a(0) - K$. If ψ is nonconstant, this restriction is nonconstant, so the residual is not a pullback from X, since a pullback is constant on E. This concerns only the residual; the full potential is $-\infty$ on the whole fiber E and does descend.

Nonclosed deformations. Fix $z_0 \in O$ and a smooth A_a-unitary frame on a smaller ball. Let $K_0 = \operatorname{diag}(1, -1, 0, \dots, 0)$ in this frame. Choose a real compactly supported function χ with $\chi(z_0) = 0$ such that $\mathrm{d}\chi(z_0)$ is a nonzero covector in the first two complex directions, and, for $\epsilon \in \mathbb{R}$, put $H = \epsilon\chi K_0$. At z_0,

$$\mathrm{d}\eta_H = \epsilon\, \mathrm{d}\chi \wedge \gamma, \qquad \gamma = A_a(K_0\,\cdot, \cdot).$$

Terms containing derivatives of the frame and of A_a vanish at z_0 because $\chi(z_0) = 0$. The restriction of γ to these four real directions is nondegenerate, so wedge multiplication by γ sends a nonzero one-form to a nonzero three-form. Thus $\mathrm{d}\theta_H(z_0) \neq 0$ for $\epsilon \neq 0$. On the fixed compact support, both strict inequalities (A.12) persist for sufficiently small $|\epsilon|$, and $H \to 0$ smoothly as $\epsilon \to 0$. This proves the existence of arbitrarily small nonclosed admissible choices; admissibility itself does not force nonclosedness.

For any resolution $\tau\colon Z \to X$, Theorem A.5.1 identifies the pushforward of its non-pluripolar product with (A.18), which proves (A.13). Its total mass equals the finite mass on X. Since the ordinary class, the equation and the normalization are intrinsic to X, uniqueness gives the same solution and scalar on every resolution and common refinement. □

A.6.2 The bounded equation

Without a logarithmic pole, the residual background is β, which degenerates along E. The radial construction still gives a bounded solution, and closedness of β gives an energy argument for its uniqueness. Put

$$\begin{gathered} c_b = \frac{d}{M} = \frac{1}{nI}, \qquad R(s) = \left(c_b n \int_0^1 k(sr) r^{n-1}\, \mathrm{d}r \right)^{1/n}, \\ Q'(s) = \frac{R(s) - 1}{1 - s}, \qquad Q(1) = 0. \end{gathered} \tag{A.19}$$

These are the data (A.10) for $a = 0$: then $\beta_0 = \beta$, $V = d$, $c_0 = c_b$, $P_0(s) = sR(s)$, $Q_0 = Q$ and $q_0 = \ell + Q(s)$.

Theorem A.6.2 *For the same cone and prescribed measure, there is a unique pair (c, u), with $c > 0$ and u a bounded θ_0-plurisubharmonic function normalized by $\sup_X u = 0$, such that $\mathrm{MA}^{\mathrm{ext}}_{X,\theta_0}(u) = c\mu$. The scalar is c_b. The smooth function $\ell + Q(s)$ on Y is constant on E and descends to a continuous function on X; its normalization with supremum zero is u.*

Every bounded θ_0-plurisubharmonic function has mass d. Formula (A.13), with θ_0 in place of θ_H, holds on every resolution and common refinement. If $\Psi = 0$ and k is nonconstant, or if ψ is nonconstant, the solution is nonconstant.

Proof Set $P(s) = sR(s)$ with the data in (A.19). Then

$$P(s)^n = c_b n \int_0^s k(r) r^{n-1}\,\mathrm{d}r, \qquad P(0) = 0, \qquad P(1) = 1. \tag{A.20}$$

Smoothness and positivity of R follow from its integral formula. Since $R(1) = 1$, Q is smooth at $s = 1$, and its derivative is also smooth at zero. Put $U = \ell + Q(s)$. It is smooth on Y and equals $Q(0)$ on E. The differentiation that gives (A.16) does not use $a > 0$; for $a = 0$, with $P = P_0$, it gives, for $A = \beta + \mathrm{dd^c} U$,

$$A = P(s)\sigma_\psi + \frac{c_b k(s)}{R(s)^{n-1}}(B_\psi - s\sigma_\psi), \qquad A^n = c_b \nu. \tag{A.21}$$

Its coefficients are smooth at $s = 0$, since $R > 0$. At infinity the horizontal coefficient is one and the transverse coefficient is positive. Thus A is semipositive on Y and strictly positive on $Y \setminus E$. By Theorem A.2.1, $U = u_0 \circ \pi$ for some $u_0 \in \mathrm{PSH}(X, \theta_0)$. The proper surjection π is a quotient map, so continuity of U makes u_0 continuous. Subtracting $K_b = \sup_X u_0$ gives the required solution.

Let w be a bounded θ_0-plurisubharmonic function. Its pullback $g = w \circ \pi$ is bounded and β-plurisubharmonic. Its bounded Monge–Ampère measure gives analytic sets zero mass and pushes forward to $\mathrm{MA}^{\mathrm{ext}}_{X,\theta_0}(w)$. Integrating the bounded expansion and using that β is closed gives $\int_Y (\beta + \mathrm{dd^c} g)^n = \int_Y \beta^n = d$; this forces the scalar of any solution to be $c_b = d/M$. The resolution identity follows from Theorem A.5.1. Now suppose that w solves the equation, and put $q = U - K_b$, $D = \beta + \mathrm{dd^c} g$ and $v = g - q$. The bounded integration-by-parts calculus of Section 1.6 gives

$$0 = \int_Y v(D^n - A^n) = -\sum_{j=0}^{n-1} \int_Y \mathrm{d}v \wedge \mathrm{d^c} v \wedge D^j \wedge A^{n-1-j}.$$

Each integral in the sum is nonnegative, so all of them vanish; in particular, so does the one with the smooth factor A^{n-1}. In the weak-gradient interpretation, strict positivity of A on each relatively compact chart of $Y \setminus E$ forces $\mathrm{d}v = 0$ there. Since $Y \setminus E$ is connected, v is constant almost everywhere on it. The divisor E has volume zero, so Lemma 1.3.3 gives equality everywhere on Y. Surjectivity of π and the normalization give $w = u$ on X.

If $\Psi = 0$ and the solution is constant, then $U = Q(x)$ is constant, so $Q' = 0$ and $R = 1$. Differentiating (A.20) then gives $c_b k = 1$, so k is constant. If ψ is nonconstant,

the restriction of U to the infinity section is ψ, so the solution is nonconstant for every k. □

Example A.6.1 Without the assumption $\Psi = 0$, a nonconstant k does not force a nonconstant solution. Take $\Psi = \log 2$ and $k(s) = 2/(2-s)^{n+1}$. Then $s = 2x/(1+x)$, $\ell = \log(1+x)$ and

$$\frac{B_\psi^n}{\beta^n} = \frac{s^n(1-s)}{x^n(1-x)} = \frac{(2-s)^{n+1}}{2}.$$

Thus $\nu = \beta^n$. Since $\frac{\mathrm{d}}{\mathrm{d}s}(s/(2-s))^n = nk(s)s^{n-1}$, one has $I = 1/n$, $M = d$ and $c_b = 1$. The formulas (A.19) give $R = 1/(2-s)$ and $Q(s) = \log(2-s)$, hence $U = \log 2$. The normalized solution is zero, although k is nonconstant.

A.6.3 The quadric cone

Let $X = \{Z_0 Z_1 = Z_2^2\} \subset \mathbb{P}^3$. Its base conic is the quadratic Veronese image of $\mathbb{P}^1$, so $n = d = 2$. Its homogeneous coordinate ring $\mathbb{C}[\xi^2, \xi\eta, \eta^2]$ is the invariant ring of the sign involution acting on the normal polynomial ring $\mathbb{C}[\xi, \eta]$, and is therefore normal; thus the base is projectively normal. In the chart

$$\pi(z,t) = (t, tz^2, tz), \qquad S = 1 + |z|^2 + |z|^4, \qquad L = 1 + |t|^2 S,$$

one has $\beta = \mathrm{dd}^{\mathrm{c}} \log L$. Differentiation gives

$$\det((\log L)_{j\bar{k}}) = \frac{|t|^2(SS_{z\bar{z}} - |S_z|^2)}{L^3} = \frac{|t|^2(1 + 4|z|^2 + |z|^4)}{L^3}.$$

Writing λ_4 for Euclidean volume, we obtain

$$\frac{\beta^2}{2} = \frac{|t|^2(1 + 4|z|^2 + |z|^4)}{\pi^2(1 + |t|^2(1 + |z|^2 + |z|^4))^3}\, \mathrm{d}\lambda_4. \tag{A.22}$$

The transverse vanishing order on $E = \{t = 0\}$ is exactly two. For $\Psi = 0$ and $k = 1/2$, one has $I = 1/4$ and $M = 1$, so $\mu_0 = \theta_0^2/2$, extended by zero at p, is a probability measure. The logarithmic scalar and the solution mass are $c_a = 2(1 - a^2)$ for every admissible deformation, including the nonclosed ones. This exact normalization follows from the preservation of the determinant.

For $\Psi = 0$ and $k(s) = 1 + s$, one has $I = 5/6$, $M = 10/3$ and $c_b = 3/5$. Put $R(x) = \sqrt{(3+2x)/5}$ and $r_0 = \sqrt{3/5}$. The normalized bounded solution is

$$u(x) = -2(R(x) - r_0) + 2\log\frac{1+R(x)}{1+r_0}. \tag{A.23}$$

Indeed, $u' = -2/[5(1+R)] = (R-1)/(1-x)$, with a removable singularity at $x = 1$. Thus $u = Q(x) - Q(0)$; it is strictly decreasing in x, and $u(p) = \sup_X u = 0$. As a function of x it is smooth on a neighborhood of $[0, 1]$, and x is the restriction of a smooth ambient function, so u is the restriction of a smooth ambient function. Its prescribed measure is $\mu_1 = (1+x)\theta_0^2$, of mass $10/3$, and $\mathrm{MA}^{\mathrm{ext}}_{X,\theta_0}(u) = (3/5)\mu_1$, of mass two. Formula (A.22) also shows how the lift of this measure vanishes along E.

For a real parameter ϵ, the function $\Psi = \epsilon|Z_0|^2/(|Z_0|^2 + |Z_1|^2 + |Z_2|^2)$ is smooth on $\mathbb{P}^2$ and preserves $\omega_{\mathrm{FS}} + \mathrm{dd}^{\mathrm{c}}\Psi > 0$ for sufficiently small $|\epsilon|$. Its restriction to the conic takes the values ϵ and zero at $[1:0:0]$ and $[0:1:0]$. For small nonzero ϵ, the logarithmic residual is therefore nonconstant on E and does not descend. Deformations supported away from p, including nonclosed ones, leave that residual and the full descended solution unchanged.

A.7 Logarithmic equations on normalized curves

On a curve the pulled-back background can be degenerate, but its total mass still determines the equation. A periodic Green function on the normalization compensates for the degeneracy. The normalization also gives direct control of descent, of the atoms at the poles and of continuity.

Let X be a connected compact reduced unibranch curve whose normalization is a torus:

$$\pi : E = \mathbb{C}/\Lambda \longrightarrow X,$$

a finite bijective holomorphic map, where $\Lambda \subset \mathbb{C}$ is a lattice. It is biholomorphic outside a finite set. Let h be a Hermitian form on X in the sense of the beginning of this appendix. In the flat coordinate $z = x + \mathrm{i}y$ on E, set

$$dA = dx\,dy, \qquad V = \int_E dA, \qquad \beta_0 = dA/V, \qquad \beta = \pi^* h = b\beta_0.$$

Here b is smooth and nonnegative, positive over X_{reg}, and may vanish at the other points. Put $H = \int_E \beta > 0$. Every two-form on a curve is closed, and this is essential for the following construction.

Lemma A.7.1 *There is a real periodic distribution G with mean zero, smooth off zero, such that*

$$\mathrm{dd^c} G = \delta_0 - \beta_0, \qquad G(z) = \log|z|^2 + \textit{a smooth function near } 0. \tag{A.24}$$

For $r \in L^p(E, \beta_0)$, $p > 1$, its convolution satisfies

$$\mathrm{dd^c}\big(G * (r\beta_0)\big) = \left(r - \int_E r\beta_0\right)\beta_0.$$

The convolution is bounded and continuous, and it is smooth if r is smooth.

Proof With the normalization of $\mathrm{dd^c}$ fixed in the Conventions, $\mathrm{dd^c} v = (\Delta v)/(4\pi)\, dA$ and $\mathrm{dd^c}\log|z|^2 = \delta_0$, as in Lemma 1.16.7. Cutting off this logarithm in a coordinate disk, with a cutoff equal to one near zero, gives a function L on E with $\mathrm{dd^c} L = \delta_0 + q\beta_0$, where q is smooth and $\int_E q\beta_0 = -1$ by Stokes' theorem.

Let e_ξ be the characters of the torus, with $-\Delta e_\xi = \lambda_\xi e_\xi$. The nonconstant characters have $\lambda_\xi > 0$, growing quadratically on the dual lattice. Define the correction S, of mean zero, by its Fourier coefficients

$$S_\xi = \frac{4\pi}{V}\frac{(q+1)_\xi}{\lambda_\xi} \quad (\xi \neq 0), \qquad S_0 = 0,$$

where $(q+1)_\xi$ is the Fourier coefficient of $q+1$. Integration by parts shows that these coefficients decay faster than any power of λ_ξ. Hence the series and all its derivatives converge uniformly, and $\mathrm{dd^c} S = -(q+1)\beta_0$. Conjugate symmetry of the coefficients makes S real. Subtracting the mean of $L + S$ from $L + S$ proves (A.24). In particular, G belongs to L^r for every finite r.

The convolution uses the group law and the Haar probability β_0. Write $K = G * (r\beta_0)$. Testing against smooth functions and using Fubini's theorem proves its distributional equation. Smooth input gives smooth output by the same Fourier argument. For $p' = p/(p-1)$, Hölder's inequality gives

$$\|K\|_\infty \le \|G\|_{p'}\|r\|_p,$$
$$|K(z+s) - K(z)| \le \|r\|_p \|G(\cdot + s) - G\|_{p'}.$$

Approximating G by smooth functions shows that translation is continuous in $L^{p'}$, and hence that K is continuous. □

Choose finitely many distinct points $q_1, \dots, q_k \in E$ and weights $a_j > 0$, possibly none, with

$$A = \sum_j a_j < H, \qquad M = H - A,$$
$$\Phi = G * ((H-b)\beta_0), \quad \Psi = \Phi + \sum_j a_j G(\cdot - q_j).$$

The function Φ is smooth, and

$$\beta + \mathrm{dd}^{\mathrm{c}}\Psi = M\beta_0 + \sum_j a_j \delta_{q_j}. \tag{A.25}$$

Thus Ψ is a β-plurisubharmonic function, not identically $-\infty$, whose logarithmic poles are exactly the points q_j, with coefficients a_j measured on the normalization. The proper bijection π is a homeomorphism, so we may define $\psi = \Psi \circ \pi^{-1}$; it belongs to $\mathrm{PSH}(X, h)$, as shown in the proof below. The ordinary type of ψ consists of the potentials $w \in \mathrm{PSH}(X, h)$ with

$$w \le \psi + C, \qquad \psi \le w + C$$

for a finite constant C, the inequalities being understood for extended real values.

Theorem A.7.1 *Let $0 \le f \in L^p(E, \beta_0)$ for some finite $p > 1$, with $F = \int_E f\beta_0 > 0$, and set $\nu = \pi_*(f\beta_0)$. Every potential in the ordinary type of ψ has extended non-pluripolar mass M. In this type there is a unique normalized solution of*

$$\mathrm{MA}^{\mathrm{ext}}_{X,h}(u) = c\nu, \qquad \sup_X u = 0, \qquad c = \frac{H - A}{F} > 0.$$

Every member of the type that solves $\mathrm{MA}^{\mathrm{ext}}_{X,h}(w) = c'\nu$ for some constant c' has $c' = c$ and differs from the normalized solution by a constant.

The solution is continuous with real values off its finite pole set, and continuous at the poles as a map into $\mathbb{R} \cup \{-\infty\}$ with the order topology. Its residual relative to ψ extends to a continuous real function on X. If there are no poles, the solution itself is continuous with real values everywhere.

Proof We construct the solution on the normalization with the Green function of Lemma A.7.1 and descend it through the resolution π. The mass and uniqueness assertions then follow from the atoms of the curvature currents at the poles. Set $c = M/F$ and

$$v = G * ((cf - M)\beta_0), \qquad U = \Psi + v.$$

The lemma gives a bounded continuous v and the positive current

$$\beta + \mathrm{dd}^{\mathrm{c}}U = cf\beta_0 + \sum_j a_j \delta_{q_j}. \tag{A.26}$$

In each disk, adding a smooth local primitive of β gives a function with positive distributional Laplacian, and hence a subharmonic representative. Continuity off the poles and the explicit logarithmic values identify this representative with U as defined above. Thus $U \in \mathrm{PSH}(E, \beta)$, and the same argument applies to Ψ. The supremum $S = \sup_E U$ is finite. By Theorem A.2.1, applied to the resolution π, the functions $u = (U - S) \circ \pi^{-1}$ and ψ belong to $\mathrm{PSH}(X, h)$, and $w \circ \pi \in \mathrm{PSH}(E, \beta)$ for every $w \in \mathrm{PSH}(X, h)$.

The homeomorphism π gives continuity off the poles. Near a pole, the smooth logarithmic remainder of Ψ and the bounded continuous function v give continuity with extended values. The residual is the descent of $v - S$, including at the poles, so no difference of two infinite values is involved. The solution belongs to the stated ordinary type.

For the mass and uniqueness assertions, let w be any member of the type and put $W = w \circ \pi \in \mathrm{PSH}(E, \beta)$. The two type inequalities make W locally bounded away from $Q = \{q_j\}$, and at each pole they give

$$W = a_j \log |z - q_j|^2 + O(1),$$

with value $-\infty$ at q_j. The positive Radon measure $T_W = \beta + \mathrm{dd^c} W$ has total mass H by Stokes' theorem. Its atom at q_j has weight exactly a_j. Indeed, choose a smooth local primitive η of β near q_j, and let $m(r)$ be the circular mean of $W + \eta$ around q_j. Since $\mathrm{dd^c}(W + \eta) = T_W$, it satisfies

$$m'(r) = \frac{2T_W(B_r)}{r}$$

for almost every r, by the divergence theorem and regularization. If $\tau = T_W(\{q_j\})$, integration gives $m(r)/\log r^2 \to \tau$, while the bounded logarithmic remainder gives the limit a_j.

Hence the non-pluripolar product of W, defined in dimension one by truncation, is

$$\langle \beta + \mathrm{dd^c} W \rangle = \mathbf{1}_{E \setminus Q} T_W.$$

Indeed, on each relatively compact set avoiding Q, a sufficiently low truncation equals W, and the indicator removes the poles. Its mass is $H - A = M$ for every member of the type. For (A.26), removing the atoms leaves $cf\beta_0$. The definition (A.7), computed on the resolution π, gives the asserted equation on X.

Suppose that w solves the equation with a constant c'. Comparing total masses gives $c'F = M$, so $c' = c$. Since π is a Borel homeomorphism and the atoms are fixed, $T_W = T_U$. Thus $\mathrm{dd^c}(W - U) = 0$, where $W - U$ is an integrable function. Every nonconstant Fourier coefficient of $W - U$ vanishes, so the difference is constant almost everywhere. Subharmonic functions are determined by their averages over shrinking disks, so the values of W and U differ by the same constant everywhere, and the same holds on X. □

The integrability hypothesis on f refers to flat area on E. On X_{reg}, the density of ν with respect to h is $(f/b) \circ \pi^{-1}$; both measures give the finite singular set zero mass. No positive lower bound for b and no bound for the density on X is required.

A.7.1 A projective cusp with a degenerate pullback

The following example has a normalization of genus one and a singularity that is not normal. Its background and its equation can both be computed explicitly.

Example A.7.1 There is a projective unibranch curve with one ordinary cusp, normalized by $E = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$, for which the restricted Fubini–Study background has mass $H = 6$. For the reference with one pole of weight one at the preimage of the cusp, every potential of its ordinary type has mass five. With

$$f(z) = 1 + \tfrac{1}{2}\cos(2\pi \operatorname{Re} z), \qquad \nu = \pi_*(f\beta_0),$$

the normalized equation has $c = 5$. Its residual is continuous on the curve, although the density on the regular locus with respect to the background is unbounded at the cusp. That density belongs to $L^r(X, h)$ exactly for $1 \le r < 2$.

Proof For the square lattice define

$$x(z) = z^{-2} + \sum_{\substack{\lambda \in \mathbb{Z}+\mathrm{i}\mathbb{Z} \\ \lambda \neq 0}} \bigl((z-\lambda)^{-2} - \lambda^{-2}\bigr), \qquad y = x'.$$

The summands are $O(|\lambda|^{-3})$ on compact sets away from the lattice, so the series converges normally. Pairing opposite lattice points shows that x is even. The differentiated series is absolutely and normally convergent, and reindexing shows that y is periodic. Thus $x(z+\omega) - x(z)$ is constant for $\omega = 1, \mathrm{i}$; evaluating at $-\omega/2$ and using evenness shows that this constant is zero. Both functions are meromorphic on E, with their only poles at zero and expansions

$$x = z^{-2} + a_2 z^2 + a_4 z^4 + \cdots, \qquad y = -2z^{-3} + 2a_2 z + 4a_4 z^3 + \cdots.$$

Cancelling the Laurent expansions leaves an elliptic function with no poles and zero constant term, which vanishes by the maximum principle. This gives

$$y^2 = 4x^3 - g_2 x - g_3, \qquad g_2 = 20a_2, \quad g_3 = 28a_4. \tag{A.27}$$

The argument principle on a translated fundamental parallelogram shows that $x - c$ has two zeros, counted with multiplicity, for every finite c. Its fibers are therefore of the form $\{z, -z\}$. If $y(z) = 0$, the local multiplicity is at least two. Distinct points $z, -z$ would then give at least four zeros, so $z = -z$ on E, the multiplicity is exactly two, and $x''(z) \neq 0$. Since y is odd, the map (x, y) is injective and immersive away from zero: use dx where $y \neq 0$, and $dy = x''\,dz$ otherwise.

Define

$$\pi(z) = [1 : x : y : x^2 : x^3], \qquad \pi(0) = [0 : 0 : 0 : 0 : 1].$$

In the last affine chart its holomorphic coordinates are

$$[s^3 : s^2 : t : s : 1], \qquad s = 1/x = z^2 + O(z^6), \qquad t = y/x^3 = -2z^3 + O(z^7).$$

Equation (A.27) becomes

$$t^2 = s^3(4 - g_2 s^2 - g_3 s^3).$$

Dividing t by an analytic square root of the nonvanishing factor gives the ordinary cusp $t'^2 = s^3$. Choose the local parameter ζ so that $s = \zeta^2$ and $t' = \zeta^3$. The local ring of the cusp is $\mathbb{C}\{\zeta^2, \zeta^3\}$, with normalization $\mathbb{C}\{\zeta\}$. To see this directly, note that division by $t'^2 - s^3$ writes every germ as $A(s) + t'B(s)$, and the even and odd parts identify the image ring. Its fraction field contains ζ, which is integral since $\zeta^2 = s$, and the larger ring is a finite module generated by 1 and ζ. An integral meromorphic germ cannot have negative order: in a monic equation, its highest power would have strictly smallest order. Hence $\mathbb{C}\{\zeta\}$ is the full integral closure.

The image is the reduced projective locus

$$\begin{aligned} &Z_1^2 = Z_0 Z_3, \quad Z_1 Z_3 = Z_0 Z_4, \\ &Z_3^2 = Z_1 Z_4, \qquad Z_2^2 = 4Z_0 Z_4 - g_2 Z_0 Z_1 - g_3 Z_0^2. \end{aligned}$$

On $Z_0 \neq 0$ these equations describe the graph of the affine cubic, and since the fibers of x have degree two, both square-root values of y occur. On $Z_0 = 0$ they force $Z_1 = Z_3 = Z_2 = 0$, leaving only $\pi(0)$. In the last chart they give exactly the cusp displayed above. The finite local module at the cusp and the biholomorphic charts elsewhere show that π is finite. Hence π is the bijective normalization of a connected curve that is smooth away from this one cusp.

For the restricted Fubini–Study form, $\beta = b\beta_0$ is positive away from zero and

$$C_1|z|^2 \le b(z) \le C_2|z|^2$$

near zero. Indeed, the derivatives of the coordinates in the last chart vanish at zero, and $ds = 2z\,dz + O(z^5)dz$ gives the leading term. On the complement of zero, the Fubini–Study potential is

$$L = \log(1 + |x|^2 + |y|^2 + |x|^4 + |x|^6) = -6\log|z|^2 + \text{a smooth function near } 0.$$

Globally $\mathrm{dd^c} L = \beta - 6\delta_0$, so integrating gives $H = 6$. Since the normalization has generic degree one, the degree two of x contributes no factor to the measure.

For the pole of weight one, the reference and the continuous correction are

$$\Psi = G * ((6 - b)\beta_0) + G, \qquad v = -\frac{5}{2\pi}\cos(2\pi \operatorname{Re} z).$$

Direct differentiation gives $\beta + \mathrm{dd^c}\Psi = 5\beta_0 + \delta_0$ and $\mathrm{dd^c} v = (5f - 5)\beta_0$. Since $F = 1$, Theorem A.7.1 gives $c = 5$, the mass, and uniqueness in the whole type. The residual is the descent of $v - \sup_E(\Psi + v)$. Finally, f/b is comparable to $|z|^{-2}$ near zero, and

the integral of its rth power against $\beta = b\beta_0$ is comparable to

$$\int_0^{\epsilon} \rho^{3-2r}\, d\rho.$$

This is finite exactly when $r < 2$, with logarithmic divergence at $r = 2$. □

The cusp is not normal, so the example is not covered by hypotheses requiring normal log-terminal singularities. Nor is the curve a projective cone: an irreducible reduced projective cone of dimension one has a single point as base and is a projective line, whereas the normalization here has genus one. Without poles, the same data give $c = 6$ and $v = -(3/\pi)\cos(2\pi \operatorname{Re} z)$, and the solution is continuous with real values even at the cusp. These are sufficient conditions in dimension one; the general problem for singular spaces in Problem 35.10.1 remains open.

A.7.2 A logarithmic endpoint for integrable data

The hypothesis $p > 1$ in Theorem A.7.1 guarantees that the Green correction is bounded and continuous. For merely integrable densities, the solution can leave the ordinary type. The following family of densities locates the boundary exactly, and it also gives continuous residuals for some densities outside every L^p, $p > 1$.

We keep the notation $X, E, h, \beta, \beta_0, H, \Psi, \psi, A, M$ of Appendix A.7, with $A < H$. Fix $q \in E$ and a flat coordinate z centered at q, and put $r = |z|$ and $t = \log(1/r)$, so that $t > 1$ on a small punctured disk around q. For $C_0 > 0$ and $\gamma > 1$, choose a positive density f_γ, smooth away from q, which equals

$$f_\gamma = C_0 r^{-2} t^{-\gamma}$$

on a sufficiently small punctured disk around q, and extend it smoothly and positively to the complement. Set

$$F_\gamma = \int_E f_\gamma \beta_0, \qquad c_\gamma = M/F_\gamma, \qquad \nu_\gamma = \pi_*(f_\gamma \beta_0).$$

By the local computation in the proof of Theorem 29.13.1, the local integral of f_γ is finite exactly for $\gamma > 1$, and f_γ lies outside $L^p(E, \beta_0)$ for every $p > 1$.

Theorem A.7.2 *For this family, the equation* $\mathrm{MA}^{\mathrm{ext}}_{X,h}(u) = c\nu_\gamma$ *with some constant* c *has a solution in the ordinary type of* ψ *if and only if* $\gamma > 2$*. In that case* $c = c_\gamma$*, the normalized solution is unique in the whole type, and its residual is continuous but, on the normalization, not Hölder continuous for any positive exponent.*

For $1 < \gamma \le 2$*, there is a solution* $u_\gamma \in \mathrm{PSH}(X,h)$ *of*

$$\mathrm{MA}^{\mathrm{ext}}_{X,h}(u_\gamma) = c_\gamma \nu_\gamma$$

whose pullback to E *has the same multiplier ideals as* Ψ *at every positive weight, but which violates the lower inequality* $\psi \le u_\gamma + C$ *of the ordinary type for every constant* C*.*

Proof Define on the punctured coordinate disk

$$P_\gamma(t) = \begin{cases} \dfrac{4\pi C_0}{V(\gamma-1)(\gamma-2)} t^{2-\gamma}, & \gamma \neq 2, \\ -\dfrac{4\pi C_0}{V}\log t, & \gamma = 2. \end{cases} \tag{A.28}$$

Since $\frac{\mathrm{i}}{2\pi}\mathrm{d}z \wedge \mathrm{d}\bar z = dA/\pi$ and $\beta_0 = dA/V$, this is the function P_b of (29.31) with $b = \gamma$ and with C_0 replaced by $\pi C_0/V$. The local computation in the proof of Theorem 29.13.1 therefore gives $\mathrm{dd^c} P_\gamma = f_\gamma \beta_0$ off q; the flux of P_γ through small circles tends to zero, so the distributional extension across q adds no atom; and $|P_\gamma(t)|/t \to 0$, so the Lelong number of P_γ at q is zero.

Take a smooth cutoff χ supported in the coordinate disk and equal to one near q, and extend $L = c_\gamma \chi P_\gamma$ by zero. The signed measure

$$\rho\beta_0 = (c_\gamma f_\gamma - M)\beta_0 - \mathrm{dd^c} L$$

has a smooth density ρ on all of E: near q the singular densities cancel, leaving $-M\beta_0$. Its integral is zero by Stokes' theorem and $c_\gamma F_\gamma = M$. Hence Lemma A.7.1 gives a smooth correction $R = G * (\rho\beta_0)$. Set $v_\gamma = L + R$ and $U_\gamma = \Psi + v_\gamma$. Then

$$\beta + \mathrm{dd^c} U_\gamma = c_\gamma f_\gamma \beta_0 + \sum_j a_j \delta_{q_j}. \tag{A.29}$$

As in the proof of Theorem A.7.1, the positive local Laplacians give $U_\gamma \in \mathrm{PSH}(E,\beta)$, and Theorem A.2.1 descends $U_\gamma - \sup_E U_\gamma$ to a potential $u_\gamma \in \mathrm{PSH}(X,h)$. It is smooth away from the original poles and from q. Its non-pluripolar product removes the original atoms, and a possible new value $-\infty$ at q carries no atom. The mass is therefore M, and (A.7) gives the equation on X.

For $\gamma > 2$, $P_\gamma \to 0$, so v_γ extends continuously and boundedly, and the solution belongs to the prescribed ordinary type. Near q, the difference of v_γ from its value at q is $c_\gamma P_\gamma$, a nonzero multiple of $t^{2-\gamma}$, plus a smooth remainder of order $O(r)$. This decays more slowly than every power r^α, $\alpha > 0$, so the residual is not Hölder continuous for any positive exponent. Uniqueness follows from the mass of the whole type, the restoration of the fixed pole atoms and the harmonic-difference argument

used in the proof of Theorem A.7.1; none of these steps requires $p > 1$ once the bounded correction exists.

For $1 < \gamma \le 2$, $P_\gamma \to -\infty$. The correction is bounded above but unbounded below near q, so the inequality $\Psi \le U_\gamma + C$ fails off the pole set. Since its Lelong number is zero, the point atoms of $\beta + \mathrm{dd^c} U_\gamma$ are those of $\beta + \mathrm{dd^c}\Psi$. Applying Lemma 1.16.7 to each positive multiple, after a smooth local correction, shows that all multiplier ideals of U_γ and Ψ on E coincide, also when q is not one of the original poles. On X, this is resolution-ideal equivalence in the sense defined before Proposition A.4.1.

If another solution belonged to the ordinary type, its mass would be M, which forces the constant c_γ. Restoring its prescribed atoms in the extended equation would give exactly (A.29). Its difference from U_γ would then be harmonic on the torus, hence constant, with equality of the values everywhere. This contradicts the unboundedness of the correction near q, so no constant admits a solution in the ordinary type. □

The theorem is an exact criterion for this logarithmic family only; it does not characterize general L^1 data, and it gives no mass or uniqueness statement for the whole resolution-ideal class. On the projective cusp, all these constructions apply with the same, possibly degenerate, pulled-back background. The failure at $\gamma = 2$ occurs already in the bounded ordinary type, that is, when there are no original poles.

A.8 Nonclosed equations on cuspidal products

On a cuspidal curve, the pullback of a Hermitian form to the normalization degenerates at the preimage of the cusp, but adding a smooth potential turns it into a positive form. On the products below we perturb the background by a nonclosed form all of whose powers are $\mathrm{dd^c}$-closed. This fixes the mass of every bounded potential and gives an energy comparison, although the background is not closed. A perturbative solution near a constant-coefficient form then provides a positive solution against which every bounded solution can be compared.

Fix $n \ge 2$. For $j = 1, \dots, n$, let $E_j = \mathbb{C}/(\mathbb{Z} + \mathrm{i}\mathbb{Z})$ with coordinate z_j, and put $\kappa_j = \frac{\mathrm{i}}{2\pi} \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j$. Replace a neighborhood of a point $p \in E_1$ by the cusp with normalization $z \mapsto (z^2, z^3)$; equivalently, the holomorphic functions at the marked point are the functions on E_1 whose first derivative vanishes there. Denote the resulting compact curve by C and its cusp by q. Put

$$X = C \times E_2 \times \cdots \times E_n, \qquad Y = E_1 \times \cdots \times E_n.$$

The normalization $\pi \colon Y \to X$ is finite and bijective, hence a homeomorphism, and has generic degree one. The space X is reduced, irreducible and unibranch, and its singular locus is $\{q\} \times E_2 \times \cdots \times E_n$.

Choose a Hermitian form h_C on C which near q is the restriction of $\mathrm{dd^c}(|x|^2 + |y|^2)$, where $(x, y) = (z^2, z^3)$ are the coordinates of the cusp. Its pullback is $\eta = \rho\kappa_1$, where

ρ is smooth, positive off p, and

$$\rho = 4|z|^2 + 9|z|^4 \quad \text{near } p. \tag{A.30}$$

Write $\lambda > 0$ for the average of ρ on E_1, and set

$$h_0 = h_C + \sum_{j=2}^{n} \kappa_j, \quad H_0 = \pi^* h_0 = \eta + \sum_{j=2}^{n} \kappa_j, \quad K_0 = \lambda\kappa_1 + \sum_{j=2}^{n} \kappa_j, \quad \Omega = K_0^n.$$

Forms on the factors are pulled back to the products without change of notation.

Proposition A.8.1 *There are a smooth real $(1,1)$-form α on X, mixing the factors C and E_2, and a smooth real function v_0 on Y such that, for every sufficiently small nonzero real ε, the Hermitian form $h = h_0 + \varepsilon\alpha$ is not closed and*

$$H = \pi^* h, \qquad k = H + \mathrm{dd^c} v_0 = K_0 + \varepsilon\pi^*\alpha \geq \tfrac{1}{2} K_0,$$
$$\mathrm{dd^c}(H^r) = \mathrm{dd^c}(k^r) = 0 \quad (r \geq 0).$$

Moreover $\int_Y H^n = \int_Y k^n =: M > 0$.

Proof Choose a real smooth function A on C, supported away from q, with a point where $\partial A = 0$ and $\partial\bar\partial A \neq 0$, and a nonconstant real smooth function B on E_2. Let

$$\alpha = \partial A \wedge \bar\partial B + \bar\partial A \wedge \partial B = \partial(A\bar\partial B) + \bar\partial(A\partial B).$$

The form α is real and satisfies $\mathrm{dd^c}\alpha = 0$. At a point of $C \times E_2$ where $\partial A = 0$, $\partial\bar\partial A \neq 0$ and $\partial B \neq 0$,

$$\partial\alpha = -\partial A \wedge \partial\bar\partial B + \partial\bar\partial A \wedge \partial B \neq 0.$$

Thus $\mathrm{d}\alpha \neq 0$. The form α vanishes near the singular locus. Its operator norm relative to H_0 is bounded on its support, and its norm relative to K_0 is bounded everywhere. Choose $|\varepsilon|$ so small that both norms, multiplied by $|\varepsilon|$, are less than $1/2$. This proves positivity of h in the ambient sense, semipositivity of H, and the inequality $K_0 + \varepsilon\pi^*\alpha \geq \frac{1}{2}K_0$.

The equation $\partial_z\partial_{\bar z} v = \lambda - \rho$ on E_1, whose right-hand side has mean zero, has a smooth real periodic solution, obtained by dividing its nonzero Fourier coefficients. Its pullback v_0 satisfies $\mathrm{dd^c} v_0 = (\lambda - \rho)\kappa_1$, which proves the formula for k. The forms H_0 and K_0 are closed. Hence $\mathrm{d}k$ and $\mathrm{d^c}k$ come from the complex surface $E_1 \times E_2$, so that $\mathrm{d}k \wedge \mathrm{d^c}k = 0$ for degree reasons, and $\mathrm{dd^c}k = 0$. The same identities hold for H. The Leibniz rule now gives $\mathrm{dd^c}(k^r) = \mathrm{dd^c}(H^r) = 0$ for all r; this uses both torsion identities. Expanding $(H + \mathrm{dd^c} v_0)^n$ and integrating by parts shows that every term containing $\mathrm{dd^c} v_0$ integrates to zero. Hence the masses agree, and positivity of k makes them positive. □

Fix an integer $s > n$ before making a further choice of ε. Use the Fourier Sobolev norm $\|w\|_{H^t}^2 = \sum_{m\in\mathbb{Z}^{2n}} (1 + |m|^2)^t |\widehat{w}(m)|^2$ on Y, and write H_0^t for the subspace of

real functions in H^t with zero mean with respect to Ω. In the remaining formulas, α denotes its pullback to Y.

Theorem A.8.1 *For sufficiently small nonzero ε, depending on n, s and the fixed forms, there are explicit constants $R, \delta > 0$ such that every $g \in H^s_0$ with $\|g\|_{H^s} \le \delta$ admits a unique $w \in H^{s+2}_0$ with $\|w\|_{H^{s+2}} \le R$ satisfying*

$$(k + \mathrm{dd^c} w)^n = (f_\varepsilon + g)\Omega, \qquad f_\varepsilon = \frac{k^n}{\Omega}, \qquad k + \mathrm{dd^c} w \ge \tfrac{1}{4} K_0.$$

The constants may be chosen as in (A.31). *In particular, w is of class C^2.*

Proof Set $b_1 = \lambda$ and $b_j = 1$ for $j \ge 2$. The operator

$$L_0 w = \frac{n K_0^{n-1} \wedge \mathrm{dd^c} w}{\Omega} = \sum_j b_j^{-1} \partial_{z_j} \partial_{\bar z_j} w$$

has Fourier eigenvalues $\ell(m) = -\pi^2 \sum_j b_j^{-1} (m_{2j-1}^2 + m_{2j}^2)$. It has an inverse $H^s_0 \to H^{s+2}_0$ of norm

$$A_0 = \sup_{m \ne 0} \frac{1 + |m|^2}{|\ell(m)|} \le \frac{2 \max_j b_j}{\pi^2}.$$

Let $L_\varepsilon w = n k^{n-1} \wedge \mathrm{dd^c} w / \Omega$. It preserves zero mean because $\mathrm{dd^c}(k^{n-1}) = 0$. Fix $\varepsilon_0 \le 1$ within the positivity range. The finite expansion of k^{n-1} gives

$$L_\varepsilon - L_0 = \sum_{j=1}^{n-1} \varepsilon^j P_j, \qquad P_j w = n \binom{n-1}{j} \frac{K_0^{n-1-j} \wedge \alpha^j \wedge \mathrm{dd^c} w}{\Omega}.$$

Choose bounds e_j for the norms $\|P_j\|_{H^{s+2} \to H^s}$, which are finite since multiplication by smooth coefficients is bounded on H^s, and set $B = \sum_j \varepsilon_0^{j-1} e_j$. If $|\varepsilon| \le \varepsilon_0$ and $|\varepsilon| A_0 B \le 1/2$, the Neumann series gives $\|L_\varepsilon^{-1}\| \le A := 2A_0$.

The remainder is

$$Q(w) = \sum_{r=2}^{n} \binom{n}{r} \frac{k^{n-r} \wedge (\mathrm{dd^c} w)^r}{\Omega}.$$

For $s > n$, Fourier convolution and the Cauchy–Schwarz inequality, together with $\sum_m (1 + |m|^2)^{-s} < \infty$, make H^s an algebra. Multiplication by smooth coefficients and the finite expansion therefore give a constant C_Q such that, on the unit ball of H^{s+2},

$$\begin{aligned} \|Q(w)\|_{H^s} &\le C_Q \|w\|^2_{H^{s+2}}, \\ \|Q(w) - Q(z)\|_{H^s} &\le C_Q (\|w\|_{H^{s+2}} + \|z\|_{H^{s+2}}) \|w - z\|_{H^{s+2}}. \end{aligned}$$

Each term of $Q(w)$ has zero integral against Ω: for smooth w, its integral is $\int w\, \mathrm{dd^c}(k^{n-r}) \wedge (\mathrm{dd^c} w)^{r-1} = 0$, and Sobolev approximation gives the general case. The same summability gives the Sobolev embedding $H^{s+2} \subset C^2$, and hence a

constant D such that the pointwise operator norm of $\mathrm{dd^c}w$ with respect to K_0 is at most $D\|w\|_{H^{s+2}}$. Choose

$$0 < R \le \min\{1, (4AC_Q)^{-1}, (4D)^{-1}\}, \qquad \delta = R/(2A). \tag{A.31}$$

The equation is equivalent to $L_\varepsilon w + Q(w) = g$. The map $w \mapsto L_\varepsilon^{-1}(g - Q(w))$ takes the closed ball of radius R in H_0^{s+2} into the ball of radius $3R/4$ and has Lipschitz constant at most $1/2$. Its fixed point gives existence and uniqueness in that ball. Since $k \ge K_0/2$ and the operator norm of $\mathrm{dd^c}w$ is at most $1/4$, the form $k + \mathrm{dd^c}w$ is at least $K_0/4$. □

The perturbation g may depend on all torus variables. For example, writing $z_j = x_j + \mathrm{i}y_j$, every sufficiently small nonzero multiple of $\prod_j \cos(2\pi x_j) + \prod_j \cos(2\pi y_j)$ is an admissible perturbation with zero mean. Thus the construction does not force the solution to have separated variables.

Lemma A.8.1 *Every bounded $U \in \mathrm{PSH}(Y, k)$ satisfies $\int_Y (k + \mathrm{dd^c}U)^n = M$.*

Proof The products are the bounded products of Proposition 1.7.1. For $r \ge 1$, the bounded local recursion and Stokes' theorem on the compact manifold Y give

$$\int_Y k^{n-r} \wedge (\mathrm{dd^c}U)^r = \int_Y U\, \mathrm{dd^c}(k^{n-r}) \wedge (\mathrm{dd^c}U)^{r-1} = 0.$$

Expanding the top power proves the assertion. □

Proposition A.8.2 *Let V be a solution w given by Theorem A.8.1. If a bounded $U \in \mathrm{PSH}(Y, k)$ satisfies $(k + \mathrm{dd^c}U)^n = (k + \mathrm{dd^c}V)^n$, then $U - V$ is constant.*

Proof Convolve U, V and k with the same nonnegative radial kernel ϱ_j on the torus, of radius $r_j \downarrow 0$, and write

$$U_j = U * \varrho_j, \quad V_j = V * \varrho_j, \quad k_j = k * \varrho_j, \quad A_j = k_j + \mathrm{dd^c}U_j, \quad B_j = k_j + \mathrm{dd^c}V_j.$$

Translations preserve positivity, so $A_j \ge 0$ and $B_j \ge \beta K_0$ with $\beta = 1/4$. The forms k_j converge smoothly, and V_j converges in C^2. On lifted coordinate balls, choose a common positive quadratic function q with $\mathrm{dd^c}q \ge k$. Convolution gives $q * \varrho_j = q + br_j^2$ for a fixed $b > 0$. The radial averages of the bounded plurisubharmonic functions $U + q$ and $V + q$ decrease as the radii decrease. Hence $U_j + br_j^2 + q$ and $V_j + br_j^2 + q$ decrease to $U + q$ and $V + q$; the identical constants cancel in the difference.

Put $\phi_j = U_j - V_j$ and $T_j = \sum_{r=0}^{n-1} A_j^r \wedge B_j^{n-1-r}$. The identities $\mathrm{dd^c}k_j = 0$ and $\mathrm{d}k_j \wedge \mathrm{d^c}k_j = 0$ persist under convolution, since the torsion is still pulled back from $E_1 \times E_2$. Differentiating the monomials therefore gives $\mathrm{dd^c}T_j = 0$. Also $T_j \ge B_j^{n-1} \ge \beta^{n-1}K_0^{n-1}$. Stokes' theorem applied to $\mathrm{dd^c}(\phi_j^2)$ yields

$$-\int_Y \phi_j (A_j^n - B_j^n) = \int_Y \mathrm{d}\phi_j \wedge \mathrm{d^c}\phi_j \wedge T_j \ge \beta^{n-1} \int_Y \mathrm{d}\phi_j \wedge \mathrm{d^c}\phi_j \wedge K_0^{n-1}. \tag{A.32}$$

We treat the four terms on the left separately. Expand A_j^n with respect to the local plurisubharmonic potential $U_j + br_j^2 + q$ and the smoothly converging coefficient $k_j - \mathrm{dd^c} q$. The weighted limit Lemma 1.6.1 gives $\int U_j A_j^n \to \int U(k + \mathrm{dd^c} U)^n$. Decreasing convergence and the uniform convergence of V_j give the corresponding limit for $\int V_j A_j^n$. Since B_j^n has uniformly converging continuous densities, bounded pointwise convergence and dominated convergence give the limit of $\int U_j B_j^n$. The term $\int V_j B_j^n$ converges by C^2 convergence. A finite partition of unity gives the global statements. Since the two limiting measures are equal, the left-hand side of (A.32) tends to zero.

The flat Fourier Poincaré inequality now shows that the L^2 norm of ϕ_j minus its average tends to zero. Bounded convergence gives $\phi_j \to U - V$ in L^2, so $U - V$ is constant almost everywhere. Local smooth corrections and the canonical plurisubharmonic representatives extend this equality to every point. □

Theorem A.8.2 *Let $f \in H^s(Y)$ be strictly positive and set*

$$\nu = \pi_*(f\Omega), \qquad c = \frac{M}{\int_Y f\Omega}.$$

If $\|cf - f_\varepsilon\|_{H^s} \le \delta$, there is a unique bounded $u \in \mathrm{PSH}(X, h)$ with $\sup_X u = 0$ and

$$\mathrm{MA}^{\mathrm{ext}}_{X,h}(u) = c\nu.$$

The function u is continuous and locally Hölder continuous of exponent $1/3$ in ambient coordinates. On the regular locus the density of ν with respect to h^n is f/J, where $J = H^n/\Omega$; near the cusp this density is $\lambda f/\rho$. Every bounded solution of an equation $\mathrm{MA}^{\mathrm{ext}}_{X,h}(u') = c'\nu$ with a constant c' has $c' = c$.

Proof The function $g = cf - f_\varepsilon$ has zero mean. Let w be the solution given by Theorem A.8.1. The potential $v_0 + w$ is bounded and H-plurisubharmonic. Descent through the resolution π (Theorem A.2.1) gives

$$u = (v_0 + w) \circ \pi^{-1} - \sup_Y (v_0 + w) \in \mathrm{PSH}(X, h).$$

Since the normalization is a homeomorphism, u is continuous. The definition of the extended measure and Theorem A.5.1 give the equation. Any other bounded solution pulls back to a bounded H-plurisubharmonic potential. Since π is a Borel homeomorphism, equality of the pushforward measures means equality on Y. Subtracting v_0 and applying Proposition A.8.2 gives uniqueness. For a bounded solution u', the mass identity of Lemma A.8.1, applied to $u' \circ \pi - v_0 \in \mathrm{PSH}(Y, k)$, forces its scalar to be $M/\nu(X) = c$.

On the regular locus, the change of variables formula gives the density f/J. Near the cusp $\alpha = 0$, so

$$H^n = n!\rho\, \kappa_1 \wedge \cdots \wedge \kappa_n, \qquad \Omega = n!\lambda\, \kappa_1 \wedge \cdots \wedge \kappa_n.$$

The ratio gives the density $\lambda f/\rho$.

For the Hölder assertion, put $d = z - z'$ and $a = z + z'$. Then

$$z^2 - z'^2 = da, \qquad z^3 - z'^3 = d(3a^2 + d^2)/4.$$

If $|a| \geq |d|/4$, the first modulus is at least $|d|^2/4$, which controls $|d|^3$ on a bounded chart. Otherwise the second modulus is at least $13|d|^3/64$. Hence $|z - z'| \leq C(|z^2 - z'^2| + |z^3 - z'^3|)^{1/3}$. The C^1 regularity of $v_0 + w$ in z and in the other product coordinates gives the asserted ambient estimate. □

The smallness condition is needed for the quantitative regularity and for the positive solution used in the uniqueness comparison. Existence alone holds for a larger class of data, since the positive form k places the equation on the normalization in the bounded-mass regime of Chapter 34.

Corollary A.8.1 *For the same X and h, every nonzero nonnegative $f \in L^p(Y, \Omega)$, $p > 1$, admits a bounded $u \in \mathrm{PSH}(X, h)$ with $\sup_X u = 0$ and*

$$\mathrm{MA}^{\mathrm{ext}}_{X,h}(u) = \frac{M}{\int_Y f\Omega}\, \pi_*(f\Omega).$$

Proof The torus Y is Kähler, so it has bounded mass by Lemma 7.5.1. Since $k > 0$, zero is an ordinary model for k: $P_k[0] = 0$. Its lower volume is $M > 0$ by Lemma A.8.1, because the potentials of the same ordinary type as zero are exactly the bounded ones. Hence Theorem 34.1.1, applied on Y with background k and reference zero, gives a bounded k-plurisubharmonic solution for the measure $f\Omega$, and its scalar is $M/\int f\Omega$. Adding v_0 and using Theorems A.2.1 and A.5.1 gives the normalized equation on X. □

The spaces X contain no rational curves. Indeed, a holomorphic map $\mathbb{P}^1 \to X$ lifts locally to Y through the cusp normalization, and the local lifts glue by uniqueness. Every flat holomorphic one-form on Y pulls back to zero on $\mathbb{P}^1$, so the lifted map has zero derivative and is constant. The results of this section depend on this explicit geometry, on the bounded ordinary type and on the dd^{c}-closedness of all powers of the background. Ambient Hölder regularity and uniqueness in the whole bounded class are proved only for data in the small Sobolev neighborhood above. The general problem Problem 35.10.1, including arbitrary pole types and backgrounds whose powers are not dd^{c}-closed, remains open.

Appendix B
Cotangent torsors on split elliptic ruled surfaces

A class in $H^1(M, \Omega^1_M)$ on a complex manifold M determines a holomorphic extension of $\mathcal{O}_M$ by the cotangent bundle and an affine bundle modeled on the cotangent bundle, its cotangent torsor; dually, a Kähler class determines an extension of the tangent bundle by the trivial line bundle, its canonical extension. This appendix computes these objects on the split ruled surface $M = \mathbb{P}_{\text{lines}}(\mathcal{O}_E \oplus A)$ over an elliptic curve E, where A is a line bundle of positive degree. The tautological line bundle of the canonical extension is big, but its section algebra is not finitely generated, and the cotangent torsor is never Stein, although its universal cover is Stein when the class is nonzero on the ruling fibers.

These results give a second instance of the contrast found for the nine-point blowups in Section 27.19.8: a background with a well-defined order of degeneration can have very different holomorphic geometry near its null locus. No other part of the book depends on this appendix.

Appendix B.1 constructs the relevant quotients, and Appendix B.2 identifies them with the cotangent torsors. Appendices B.3 and B.4 compute the symmetric tensors of the tangent bundle and the section algebras of the dual extensions whose class is nonzero on the ruling fibers, and Appendix B.5 treats the classes that vanish there.

B.1 Punctured line bundles and elliptic quotients

In this section we construct the relevant quotients and show that they are not Stein, although their universal covers are. The obstruction comes from fiberwise Laurent expansions on the punctured total space of a line bundle of nonzero degree, rather than from the flat circle levels used in Corollary 27.19.4.

Lemma B.1.1 *Let C be a compact connected Riemann surface, and let $A \to C$ be a holomorphic line bundle of positive degree. Write $A^\times$ for its total space with the zero section removed. Then $A^\times$ is not holomorphically convex. More precisely, put $B = A^*$ and choose a smooth Hermitian norm on B. Every holomorphic function on*

$B^\times$ extends holomorphically to B, and the holomorphic hull in $B^\times$ of

$$K_r = \{v \in B : |v| = r\}, \qquad r > 0,$$

contains $\{v : 0 < |v| \le r\}$ and is not compact.

Proof In a local holomorphic frame of B, a function $f \in \mathcal{O}(B^\times)$ has a fiberwise Laurent expansion

$$f(z,t) = \sum_{k\in\mathbb{Z}} a_k(z)t^k.$$

By the Cauchy integral formula, each coefficient is holomorphic in z, and under a change of frame the coefficients a_k transform as the components of global sections of B^{-k}. For $k < 0$ this line bundle has negative degree, so it has no nonzero section: the zero divisor of a nonzero holomorphic section has nonnegative degree. Thus all negative coefficients vanish. On a smaller base chart, Cauchy estimates on a fixed fiber circle give locally uniform convergence of the remaining power series at $t = 0$. These extensions agree on overlaps and define a holomorphic function on all of B.

The maximum modulus principle on the disk in each fiber gives

$$|f(v)| \le \max_{K_r} |f| \qquad (0 < |v| \le r).$$

The circle bundle K_r is compact because C is compact. A sequence in one fiber whose norms tend to zero lies in the hull and has no convergent subsequence in $B^\times$, so the hull is not compact. Finally, fiberwise inversion identifies $A^\times$ biholomorphically with $B^\times$: a nonzero $v \in A$ corresponds to the unique $\ell \in A^*$ with $\ell(v) = 1$. This proves the assertion for $A^\times$. □

The following quotient isolates this obstruction in an affine-bundle model.

Let $\Lambda = \mathbb{Z} + \mathbb{Z}\tau \subset \mathbb{C}$, with $\operatorname{Im}\tau > 0$, and put $E = \mathbb{C}/\Lambda$. A *factor of automorphy* is a family of nowhere-zero holomorphic functions $b_\lambda(z)$, $\lambda \in \Lambda$, satisfying

$$b_{\lambda+\gamma}(z) = b_\lambda(z+\gamma)b_\gamma(z).$$

This cocycle identity says that $\lambda \cdot (z,t) = (z+\lambda, b_\lambda(z)t)$ is an action of Λ on $\mathbb{C}_z \times \mathbb{C}_t$, and the quotient

$$A = (\mathbb{C}_z \times \mathbb{C}_t)/\Lambda$$

is a holomorphic line bundle over E. Suppose that A has positive degree.

Example B.1.1 The factors

$$b_{m\tau+n}(z) = \exp\bigl(-\pi\mathrm{i}(m^2\tau + 2mz)\bigr)$$

define a line bundle of degree one on E.

Proof For $\gamma = r\tau + s$, the cocycle identity follows from

$$\begin{aligned} b_{m\tau+n}(z+\gamma)b_{r\tau+s}(z) &= b_{(m+r)\tau+n+s}(z)\exp(-2\pi\mathrm{i}ms) \\ &= b_{(m+r)\tau+n+s}(z). \end{aligned}$$

Put $v = \operatorname{Im}\tau$. The squared norm

$$|t|^2 \exp\bigl(-2\pi(\operatorname{Im} z)^2/v\bigr)$$

is invariant under the action, because $\operatorname{Im}(z + m\tau + n) = \operatorname{Im} z + mv$ and

$$|b_{m\tau+n}(z)|^2 = \exp\bigl(2\pi(m^2 v + 2m \operatorname{Im} z)\bigr).$$

Its first Chern form is

$$\frac{\mathrm{i}}{2\pi}\partial\bar\partial\left(\frac{2\pi(\operatorname{Im} z)^2}{v}\right) = \frac{\mathrm{i}}{2v}\,\mathrm{d}z \wedge \mathrm{d}\bar z.$$

This positive form integrates to one over a fundamental parallelogram, which has area v. Thus the degree is one. □

On

$$Q = \{([x_0 : x_1], [w_0 : w_1]) \in \mathbb{P}^1 \times \mathbb{P}^1 : \Delta = x_0 w_1 - x_1 w_0 \neq 0\},$$

the complement of the diagonal, define the holomorphic function $\sigma = x_1 w_0/\Delta$.

Proposition B.1.1 *Fix $\kappa \in \mathbb{C}^*$ and $K \in \mathbb{C}$. For $\lambda = m\tau + n \in \Lambda$, the formula*

$$\begin{aligned}\lambda \cdot (z, u, [x], [w]) = \bigl(&z + \lambda, u + m(K + \kappa\sigma),\\ &[x_0 : b_\lambda(z)x_1], [w_0 : b_\lambda(z)w_1]\bigr)\end{aligned}$$

defines a free and properly discontinuous holomorphic action of Λ on $\mathbb{C}^2 \times Q$. The quotient Z is not Stein, and σ descends to a holomorphic function on Z. If $K \notin \{0, \kappa\}$, the fiber

$$\sigma^{-1}(-K/\kappa) \simeq \mathbb{C} \times A^\times$$

is a closed submanifold of Z that is not holomorphically convex. If $K = 0$ or $K = \kappa$, the quotient contains a compact elliptic curve. In every case the universal cover of Z is the Stein manifold $\mathbb{C}^2 \times Q$.

Proof *The action.* Multiplying x_1 and w_1 simultaneously by $b_\lambda(z)$ multiplies Δ and $x_1 w_0$ by the same factor, so σ is invariant. This invariance and the cocycle identity for b give the group law, including the additivity of the integer m in the u-coordinate. Translation in the z-coordinate makes the action free and properly discontinuous: only finitely many lattice translates of the projection of one compact set meet another fixed compact set. Hence the quotient Z is a complex manifold, and σ descends to a holomorphic function on Z.

The fiber of σ. Set $a = -K/\kappa$. If $a \notin \{0, -1\}$, every point of $\{\sigma = a\} \subset Q$ has $x_0 x_1 \neq 0$, since $\sigma = 0$ where $x_1 = 0$ and $\sigma = -1$ where $x_0 = 0$. In the chart

$$(t, y) \longmapsto ([1 : t], [y : ty + 1]),$$

we have $\Delta = 1$, and the equation $\sigma = a$ becomes $ty = a$. Hence this fiber of σ on Q is $\mathbb{C}^*$, with parameter t, and the group action on its preimage is

$$\lambda \cdot (z, u, t) = (z + \lambda, u, b_\lambda(z)t).$$

This gives the asserted biholomorphism with $\mathbb{C} \times A^\times$. The submanifold $\{u = 0\}$ of this fiber is closed in the fiber, and therefore closed in Z; under fiberwise inversion it is $B^\times$, with $B = A^*$ as in Lemma B.1.1. Let K_r be the compact circle bundle of that lemma. Every holomorphic function on Z restricts to a holomorphic function on $B^\times$, so the hull of K_r in Z contains the same sequence with fiber norms tending to zero. This sequence has no convergent subsequence in the closed submanifold, and hence none in Z. Thus neither the fiber nor Z is holomorphically convex.

The exceptional parameters. For $K = 0$, take $q_+ = ([1 : 0], [0 : 1])$, where $\sigma = 0$. The invariant submanifold $\{u = 0,\ q = q_+\}$ descends to a copy of E. For $K = \kappa$, use $q_- = ([0 : 1], [1 : 0])$, where $\sigma = -1$, with the same conclusion. Every holomorphic function on Z is constant on this compact curve by the maximum modulus principle, so Z is not holomorphically separable, and again not Stein.

The universal cover. The map

$$Q \longrightarrow \{(s, c, d) \in \mathbb{C}^3 : cd = s(s - 1)\}, \qquad ([x], [w]) \longmapsto \left(\frac{x_0 w_1}{\Delta}, \frac{x_0 w_0}{\Delta}, \frac{x_1 w_1}{\Delta}\right)$$

is an isomorphism. Its inverse recovers $[x]$ and $[w]$ as the column space and the row space of the rank-one matrix

$$\begin{pmatrix} c & s \\ s - 1 & d \end{pmatrix}.$$

This matrix never vanishes, and the hypersurface is smooth: simultaneous vanishing of the three partial derivatives of $cd - s(s - 1)$ would give $c = d = 0$ and $s = 1/2$, contradicting its equation. Thus Q is an affine variety and hence Stein. Projection to $[x] \in \mathbb{P}^1$ makes Q a locally trivial bundle of affine lines. The chart above and the chart over $\{x_1 \neq 0\}$, given by $(v, y_\infty) \mapsto ([v : 1], [vy_\infty + 1 : y_\infty])$, are related by

$$v = 1/t, \qquad y_\infty = -t^2 y - t.$$

Since the fibers are contractible, $\pi_1(Q) = \pi_1(\mathbb{P}^1) = 0$. Therefore $\mathbb{C}^2 \times Q$ is simply connected and Stein, and it is the universal cover of Z. □

B.2 Cotangent extensions and their torsors

We now show that the quotients of Proposition B.1.1 are the cotangent torsors of a family of extensions on the ruled surface $M = \mathbb{P}_{\text{lines}}(\mathcal{O}_E \oplus A)$, a family that contains the canonical extensions of all Kähler classes. An extension of the trivial line bundle by the cotangent bundle is determined, up to isomorphism, by a class in $H^1(M, \Omega^1_M)$. This group is two-dimensional: one component comes from the elliptic base and one

from the ruling fibers. In the quotient, the base component appears as the constant K and the ruling component as the coefficient κ of σ.

Keep $E = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$, and let ϑ_E be the holomorphic one-form on E whose pullback to $\mathbb{C}$ is $\mathrm{d}z$. On a complex manifold Y, a holomorphic extension of vector bundles

$$0 \longrightarrow \Omega^1_Y \longrightarrow W \xrightarrow{p_W} \mathcal{O}_Y \longrightarrow 0$$

defines the affine bundle $T_W = p_W^{-1}(1)$, modeled on Ω^1_Y, which we call its cotangent torsor. Choosing local holomorphic sections s_i of W with $p_W(s_i) = 1$, the differences $s_j - s_i$ form a Čech cocycle with values in Ω^1_Y; its class in $H^1(Y, \Omega^1_Y)$ is the class of the extension, and it determines the extension up to isomorphism. Dually, a class $\alpha \in H^1(Y, \Omega^1_Y)$ determines an extension $0 \to \mathcal{O}_Y \to V \to T_Y \to 0$; when α is the class of a Kähler form, V is the *canonical extension* of that Kähler class.

Lemma B.2.1 *Choose local inverses $z_i \colon U_i \to \mathbb{C}$ of the projection $\mathbb{C} \to E$, and on each component of an overlap write*

$$z_i - z_j = m_{ij}\tau + n_{ij}, \qquad m_{ij}, n_{ij} \in \mathbb{Z}.$$

The cocycle $\{m_{ij}\vartheta_E\}$ defines a nonzero class $\eta \in H^1(E, \Omega^1_E)$, which spans this one-dimensional space.

Proof Consider the vector bundle obtained from $\mathbb{C}_z \times \mathbb{C}_a \times \mathbb{C}_b$ by the action

$$(m\tau + n) \cdot (z, a, b) = (z + m\tau + n, a + mb, b).$$

The coordinate b gives a map to $\mathcal{O}_E$, which defines an extension of $\mathcal{O}_E$ by Ω^1_E, the kernel coordinate a corresponding to $a\vartheta_E$. The local lifts $s_i = (z_i, 0, 1)$ satisfy $s_j - s_i = m_{ij}\vartheta_E$, as one sees by expressing s_i in the coordinates over z_j. Thus the extension has class η.

If it split, the pullback of a holomorphic lift of 1 to $\mathbb{C}$ would have the form $(z, g(z), 1)$ with g entire and

$$g(z + 1) = g(z), \qquad g(z + \tau) = g(z) + 1.$$

The derivative g' would be lattice periodic, hence constant by the maximum principle on the compact torus. The first identity then forces this constant to be zero, contradicting the second. Hence $\eta \neq 0$. Serre duality on the compact connected curve gives $H^1(E, \Omega^1_E) \simeq H^0(E, \mathcal{O}_E)^* \simeq \mathbb{C}$, which proves the spanning assertion. □

Lemma B.2.2 *Every holomorphic line bundle A of positive degree d on E is defined by factors of automorphy of the form*

$$b_{m\tau+n}(z) = \chi_{m\tau+n} \exp\bigl(-\pi \mathrm{i} d (m^2\tau + 2mz)\bigr),$$

where $\chi \colon \Lambda \to \mathrm{U}(1)$ is a character. In particular,

$$\frac{b'_{m\tau+n}(z)}{b_{m\tau+n}(z)} = -2\pi \mathrm{i} d m.$$

Proof Let A_d be the dth tensor power of the degree-one bundle of Example B.1.1. Its factors are the stated exponential factors, and its degree is d. The bundle $F = A \otimes A_d^*$ has degree zero. Start with a smooth Hermitian metric on F; its Chern curvature has integral zero. Solving the scalar Poisson equation on the flat compact torus gives a smooth real function whose addition to the weight of the metric makes the curvature vanish. The resulting Chern connection is flat and unitary.

On the simply connected cover $\mathbb{C}$, parallel transport gives a global parallel frame of unit length of the pullback of F. It is holomorphic, because the $(0,1)$-part of the Chern connection is the $\bar\partial$-operator of F. Each lattice element changes this frame by a constant unitary factor, and these factors form a character. Tensoring with the given frame of A_d gives factors of the stated form for A, and differentiation gives the last formula. □

Let $M = \mathbb{P}_{\text{lines}}(\mathcal{O}_E \oplus A)$ be the bundle of lines in $\mathcal{O}_E \oplus A$, and let $f\colon M \to E$ be its projection. Let $C_\infty = \mathbb{P}_{\text{lines}}(A)$ be the section given by the lines in A. Over U_i the line spanned by $(1, t_i e_i)$, where e_i is the local frame of A, defines an affine coordinate t_i away from C_∞, and $v_i = 1/t_i$ is a coordinate away from the other section $\mathbb{P}_{\text{lines}}(\mathcal{O}_E)$. Over $U_i \cap U_j$,

$$t_i = b_{ij} t_j, \quad v_i = b_{ij}^{-1} v_j, \qquad b_{ij} = b_{m_{ij}\tau + n_{ij}}(z_j).$$

Define $\xi \in H^1(M, \Omega^1_M)$ as the logarithmic cocycle of $\mathcal{O}_M(-C_\infty)$. More precisely, use the cover by the charts with indices $(i,0)$ and (i,∞), the local defining functions $q_{i,0} = 1$ and $q_{i,\infty} = v_i$ of C_∞, and the cocycle $\mathrm{d}\log(q_{j,b}/q_{i,a})$. This convention contains no factor $1/(2\pi\mathrm{i})$.

Lemma B.2.3 *There is a direct sum decomposition*

$$H^1(M, \Omega^1_M) = \mathbb{C}\xi \oplus \mathbb{C} f^*\eta.$$

The restriction of a class $c = \rho\xi + K f^*\eta$ to any ruling fiber is nonzero if and only if $\rho \neq 0$.

Proof The relative cotangent sequence is

$$0 \longrightarrow f^*\Omega^1_E \longrightarrow \Omega^1_M \longrightarrow \Omega^1_{M/E} \longrightarrow 0.$$

The two affine charts of a projective line give the direct images

$$f_*\mathcal{O}_M = \mathcal{O}_E, \quad R^1 f_*\mathcal{O}_M = 0, \qquad f_*\Omega^1_{M/E} = 0, \quad R^1 f_*\Omega^1_{M/E} = \mathcal{O}_E.$$

For the last identity, a Laurent cocycle $g(z,t)\,\mathrm{d}t$ can be reduced, by subtracting forms holomorphic on the two charts, to its coefficient of $\mathrm{d}t/t$. Its residue is unchanged by $t_i = b_{ij} t_j$, with differentials taken relative to the base, so the local generators glue to the trivial line bundle. For functions, the two halves of a Laurent series exhaust the cocycle, which gives $R^1 f_*\mathcal{O}_M = 0$. The direct images in degree zero follow because $\mathbb{P}^1$ has no nonconstant holomorphic functions and no nonzero holomorphic one-forms.

The Leray spectral sequences and the fact that the base is a curve now give

$$H^1(M, f^*\Omega^1_E) \simeq H^1(E, \Omega^1_E), \quad H^2(M, f^*\Omega^1_E) = 0, \quad H^1(M, \Omega^1_{M/E}) \simeq \mathbb{C},$$

together with $H^0(M, \Omega^1_{M/E}) = 0$. The cohomology sequence of the relative cotangent sequence is therefore the exact sequence

$$0 \longrightarrow \mathbb{C} f^*\eta \longrightarrow H^1(M, \Omega^1_M) \longrightarrow \mathbb{C} \longrightarrow 0.$$

On a fiber, the class ξ is represented on the two affine charts by $\mathrm{d}\log v = -\mathrm{d}t/t$, whose residue is nonzero, so ξ splits the sequence. The pulled-back class $f^*\eta$ restricts to zero on every fiber. This proves both assertions. □

Theorem B.2.1 *Let W be a holomorphic extension of $\mathcal{O}_M$ by Ω^1_M with class $c = \rho\xi + K f^*\eta$, where $\rho \neq 0$. Then T_W is biholomorphic to the quotient of Proposition B.1.1, with the factors of Lemma B.2.2 and*

$$\kappa = 2\pi \mathrm{i} d\rho.$$

In particular, T_W is not Stein, while its universal cover is Stein. This applies to the cotangent torsor of the canonical extension of every Kähler class on M.

Proof *Step 1: a model torsor.* Over $\mathbb{C}_z \times \mathbb{P}^1$, glue two affine cotangent charts by

$$v = 1/t, \qquad u_\infty = u, \qquad y_\infty = -t^2 y - \rho t.$$

Here (u, y) are the coefficients of the covector $u\,\mathrm{d}z + y\,\mathrm{d}t$, and the linear part of the gluing is the change of these coefficients under $v = 1/t$. If s_0 and s_∞ are the zero sections of the two charts, the affine term says that

$$s_\infty - s_0 = \rho\,\mathrm{d}v/v.$$

For $\lambda = m\tau + n$, give the first chart the lattice action

$$(z, t, u, y) \longmapsto \big(z + \lambda, b_\lambda(z)t, u + Km - (b'_\lambda/b_\lambda)ty, y/b_\lambda(z)\big).$$

Its linear part is the transport of covectors under $(z, t) \mapsto (z + \lambda, b_\lambda(z)t)$, and the cocycle identity for the factors and the invariance of ty give the group law. In the other chart the action becomes

$$(z, v, u_\infty, y_\infty) \longmapsto \big(z + \lambda, b_\lambda^{-1}v, u_\infty + Km + (b'_\lambda/b_\lambda)(vy_\infty + \rho), b_\lambda y_\infty\big),$$

which is holomorphic also at $v = 0$, and the two formulas agree on the overlap. Translation in z makes the action free and properly discontinuous.

Step 2: its class. We compute the class of the quotient torsor using the local lifts z_i. On the two types of overlaps over the base, the zero sections satisfy

$$\begin{aligned} s_{j,0} - s_{i,0} &= Km_{ij} f^*\vartheta_E, \\ s_{j,\infty} - s_{i,\infty} &= Km_{ij} f^*\vartheta_E + \rho\,\mathrm{d}\log(v_j/v_i). \end{aligned}$$

For a mixed overlap, these formulas and $s_{j,\infty} - s_{j,0} = \rho\,\mathrm{d}v_j/v_j$ give

$$s_{j,\infty} - s_{i,0} = Km_{ij}f^*\vartheta_E + \rho\,\mathrm{d}v_j/v_j.$$

These are exactly the cocycles defining $Kf^*\eta + \rho\xi$. Two extensions with the same class have splitting cocycles that differ by a coboundary, and changing their local lifts by that coboundary identifies their inverse images of 1. Thus the quotient torsor is T_W.

Step 3: identification with the quotient. Since $\rho \neq 0$, set $Y = y/\rho$ and $Y_\infty = y_\infty/\rho$. Their transition is $Y_\infty = -t^2Y - t$. The maps

$$(t, Y) \longmapsto ([1 : t], [Y : tY + 1]), \qquad (v, Y_\infty) \longmapsto ([v : 1], [vY_\infty + 1 : Y_\infty])$$

agree on overlaps and identify the glued charts with Q. In the first chart $\sigma = tY = ty/\rho$. Substituting $b'_\lambda/b_\lambda = -2\pi\mathrm{i}dm$ turns the lattice action into the action of Proposition B.1.1 with $\kappa = 2\pi\mathrm{i}d\rho$. That proposition gives the assertions on the Stein property.

Step 4: Kähler classes. The restriction of a Kähler class to a ruling fiber is nonzero, because its integral over the fiber is positive. For the canonical extension $0 \to \mathcal{O}_M \to V \to T_M \to 0$ of a Kähler class, the dual extension of $\mathcal{O}_M$ by Ω^1_M has the negative of that class, which still has nonzero restriction to the fibers; by Lemma B.2.3 its ruling component ρ is nonzero. This proves the final assertion. □

For a holomorphic vector bundle F on a complex manifold Y we use the quotient convention for projectivization: $\mathbb{P}(F)$ parametrizes the one-dimensional quotients of the fibers of F, that is, the lines in F^*, and the projection $\pi\colon \mathbb{P}(F) \to Y$ satisfies $\pi_*\mathcal{O}_{\mathbb{P}(F)}(r) = \operatorname{Sym}^r F$. For the canonical extension V of a Kähler class, the complement $\mathbb{P}(V) \setminus \mathbb{P}(T_M)$ consists of the lines in V^* not contained in Ω^1_M, and it identifies with the cotangent torsor $T_W = p_W^{-1}(1)$ of $W = V^*$ by normalizing a nonzero functional to take the value one on the distinguished copy of $\mathcal{O}_M$.

B.3 Symmetric tensors on split elliptic ruled surfaces

To compute the section algebra of a canonical extension, we first compute that of the tangent bundle of M. A holomorphic section of $\operatorname{Sym}^r T_M$ is the same as a holomorphic function on the total space of the cotangent bundle that is a homogeneous polynomial of degree r on each fiber. The $\mathbb{C}^*$-action on the rulings lifts to the cotangent bundle and decomposes these polynomials into weights. Each weight space is the space of sections of an explicit vector bundle on E, built from powers of A and symmetric powers of a nonsplit bundle U of rank two. Riemann–Roch then gives exact dimensions: the tautological line bundle on $\mathbb{P}(T_M)$ is big, while a square-zero quotient shows that the section algebra is not finitely generated.

Keep E, the line bundle A of degree $d > 0$, the surface M and the normalized factors b_λ of Lemma B.2.2, and put

$$q_\lambda = -b'_\lambda / b_\lambda = 2\pi i d m, \qquad \lambda = m\tau + n.$$

Write $u\,\mathrm{d}z + y\,\mathrm{d}t$ for a covector in the finite ruling chart. Evaluating covectors on the Euler field $t\partial/\partial t$ of the rulings gives the global fiberwise linear function $h = ty$ on the cotangent bundle. The dimensions of the weight spaces measure how many symmetric tensors there are, and their products detect the failure of finite generation.

Let U be the rank-two vector bundle on E whose sections, pulled back to $\mathbb{C}$, are the linear polynomials $F(z, h, u)$ in formal variables h, u with entire coefficients satisfying

$$F(z + \lambda, h, u) = F(z, h, u - q_\lambda h).$$

The invariant polynomial h and the coefficient of u give an exact sequence $0 \to \mathcal{O}_E h \to U \to \mathcal{O}_E \to 0$.

Lemma B.3.1 *For every integer $r \geq 0$, the space $H^0(M, \mathrm{Sym}^r T_M)$ is the direct sum of finitely many weight spaces, with weights $-r \leq \ell \leq r$. The weight-zero space is $H^0(E, \mathrm{Sym}^r U)$. For $1 \leq k \leq r$, the space of weight $-k$ is*

$$H^0(E, A^k \otimes \mathrm{Sym}^{r-k} U),$$

and the space of weight k is $H^0(E, A^{-k} \otimes \mathrm{Sym}^{r-k} U)$. Multiplication adds tensor degrees and weights.

Proof On the cover $\mathbb{C} \times \mathbb{P}^1$, the cotangent coordinates of the two ruling charts are related by

$$v = 1/t, \qquad u_\infty = u, \qquad y_\infty = -t^2 y,$$

and the lattice acts by

$$(z, t, u, y) \longmapsto (z + \lambda, b_\lambda t, u + q_\lambda t y, y/b_\lambda);$$

both formulas follow by preserving the one-form $u\,\mathrm{d}z + y\,\mathrm{d}t$. In the other ruling chart the action is

$$(z, v, u_\infty, y_\infty) \longmapsto (z + \lambda, b_\lambda^{-1} v, u_\infty - q_\lambda v y_\infty, b_\lambda y_\infty),$$

which is holomorphic at $v = 0$ and agrees with the formula in the finite chart. Both formulas preserve $h = ty = -v y_\infty$.

A function that is homogeneous of degree r on the fibers is a sum of terms $g_b(z, t) y^b u^{r-b}$, $0 \leq b \leq r$, with g_b holomorphic on $\mathbb{C}^2$. Since $y = -v^2 y_\infty$, holomorphy at $v = 0$ says that $v^{2b} g_b(z, 1/v)$ is holomorphic there; there is no cancellation between different b, because the monomials in u_∞, y_∞ are linearly independent.

Expanding in t therefore gives

$$g_b(z,t) = \sum_{a=0}^{2b} c_{a,b}(z)t^a,$$

with entire coefficients.

Give t weight 1, y weight -1 and u weight 0, so that $t^a y^b u^{r-b}$ has weight $\ell = a-b$. The chart changes and the lattice action preserve this grading; equivalently, it is the grading by the cotangent lift of the global action $t \mapsto ct$, $c \in \mathbb{C}^*$. The bounds $0 \le a \le 2b$ and $b \le r$ give $-r \le \ell \le r$.

For $\ell = -k < 0$, a homogeneous function of weight ℓ can be written uniquely as $y^k F(z,h,u)$ with F homogeneous of degree $r-k$. Its invariance is equivalent to

$$F(z+\lambda,h,u) = b_\lambda(z)^k F(z,h,u-q_\lambda h),$$

which is the transformation rule for sections of $A^k \otimes \operatorname{Sym}^{r-k} U$. Conversely, each such polynomial defines a function that is holomorphic in both ruling charts, because its exponents satisfy $0 \le a \le 2b$. For $\ell = k > 0$, the condition $a \le 2b$ shows that a function of weight ℓ can be written uniquely as $t^k h^k F(z,h,u)$, with F of the same remaining degree $r-k$. Its invariance involves the factor b_λ^{-k}, which gives $A^{-k} \otimes \operatorname{Sym}^{r-k} U$. Weight zero consists exactly of the homogeneous polynomials $F(z,h,u)$ with the transformation rule of $\operatorname{Sym}^r U$. Products of polynomial functions add degrees and weights, which proves the last assertion. □

Lemma B.3.2 *The extension U does not split. For every $s \ge 0$,*

$$H^0(E, \operatorname{Sym}^s U) = \mathbb{C}h^s.$$

For $k \ge 1$ and $s \ge 0$,

$$\begin{aligned} H^0(E, A^{-k} \otimes \operatorname{Sym}^s U) &= 0, \\ H^1(E, A^k \otimes \operatorname{Sym}^s U) &= 0, \\ h^0(E, A^k \otimes \operatorname{Sym}^s U) &= kd(s+1), \end{aligned}$$

where $h^0 = \dim H^0$. For $s \ge 1$, taking the coefficient of u^s defines a surjection

$$H^0(E, A \otimes \operatorname{Sym}^s U) \longrightarrow H^0(E, A),$$

whose kernel is $hH^0(E, A \otimes \operatorname{Sym}^{s-1} U)$.

Proof Write an invariant polynomial of degree s as

$$F(z,h,u) = \sum_{j=0}^{s} a_j(z) h^{s-j} u^j.$$

If its largest nonzero u-degree is $j > 0$, then a_j is lattice periodic, hence a nonzero constant. Comparing the next coefficients gives

$$a_{j-1}(z+1) = a_{j-1}(z), \qquad a_{j-1}(z+\tau) = a_{j-1}(z) - j(2\pi \mathrm{i} d) a_j,$$

so the derivative a'_{j-1} is lattice periodic, hence constant on the compact torus. The first identity forces this constant to vanish, and then a_{j-1} is constant, contradicting the second identity. Hence $F = a_0 h^s$ with a_0 constant; this also covers $s = 0$. For $s = 1$ this shows that U does not split, since a splitting would give an invariant section with u-coefficient one.

The powers of the invariant line $\mathcal{O}_E h$ give a filtration of $\operatorname{Sym}^s U$ whose $s+1$ successive quotients are trivial line bundles. A line bundle of negative degree on a compact curve has no nonzero section, since the zero divisor of a nonzero section has nonnegative degree. Hence the filtration twisted by A^{-k} has no global sections. Serre duality and $\Omega^1_E \simeq \mathcal{O}_E$ give $H^1(E, A^k) = 0$, and the cohomology sequences of the filtration twisted by A^k then give $H^1(E, A^k \otimes \operatorname{Sym}^s U) = 0$. Riemann–Roch on the curve of genus one, for a bundle of rank $s+1$ and degree $kd(s+1)$, gives the dimension formula.

Finally, the leading coefficient gives the exact sequence of bundles

$$0 \longrightarrow A \otimes \operatorname{Sym}^{s-1} U \xrightarrow{h} A \otimes \operatorname{Sym}^s U \longrightarrow A \longrightarrow 0.$$

Indeed, translating u by a multiple of h does not change the leading coefficient, and a polynomial with zero leading coefficient is divisible by h. The vanishing of $H^1(E, A \otimes \operatorname{Sym}^{s-1} U)$ just proved gives the asserted surjectivity and kernel on global sections. □

For a line bundle $\mathcal{L}$ on a projective manifold Y of dimension N, recall that its volume is $\operatorname{vol}(\mathcal{L}) = \limsup_{r\to\infty} N!\, h^0(Y, \mathcal{L}^r)/r^N$, and that $\mathcal{L}$ is big if and only if $\operatorname{vol}(\mathcal{L}) > 0$.

Theorem B.3.1 *For every integer $r \geq 0$,*

$$h^0(M, \operatorname{Sym}^r T_M) = 1 + \frac{d\, r(r+1)(r+2)}{6}.$$

The tautological line bundle $\mathcal{O}_{\mathbb{P}(T_M)}(1)$ is big and has volume d.

Proof By Lemmas B.3.1 and B.3.2, the positive weights contribute nothing, the weight zero contributes one dimension, and the weight $-k$ contributes $kd(r-k+1)$ dimensions. Thus

$$h^0(M, \operatorname{Sym}^r T_M) = 1 + d \sum_{k=1}^{r} k(r-k+1) = 1 + \frac{d\, r(r+1)(r+2)}{6}.$$

The elliptic curve and its holomorphic line bundle A are algebraic, so M and $\mathbb{P}(T_M)$ are projective. The identity $\pi_* \mathcal{O}_{\mathbb{P}(T_M)}(r) = \operatorname{Sym}^r T_M$ identifies the section spaces.

Since $\mathbb{P}(T_M)$ has dimension three,

$$\operatorname{vol}\bigl(\mathcal{O}_{\mathbb{P}(T_M)}(1)\bigr) = \limsup_{r\to\infty} \frac{6h^0(M, \operatorname{Sym}^r T_M)}{r^3} = d > 0,$$

so the tautological line bundle is big. □

Theorem B.3.2 *The complex algebra*

$$R = \bigoplus_{r\geq 0} H^0(M, \operatorname{Sym}^r T_M)$$

is not finitely generated.

Proof Write $R_{r,k}$ for the component of tensor degree r and weight $-k$. By Lemmas B.3.1 and B.3.2, only $k \geq 0$ occurs, multiplication adds both indices, and the subalgebra of weight zero is $\mathbb{C}[h]$. Let I be the ideal formed by all components with $k > 0$, and put

$$S = R/(hR + I^2).$$

Its component of weight zero is $\mathbb{C}$, and its ideal J of positive weight index satisfies $J^2 = 0$, so $S = \mathbb{C} \oplus J$ as a vector space.

For every $r \geq 1$, products in I^2 have weight index at least two. Consequently the component of S of degree r and weight index one is

$$R_{r,1}/hR_{r-1,1} \simeq H^0(E, A).$$

For $r = 1$ this holds because $R_{0,1} = 0$, and for $r \geq 2$ it is the last assertion of Lemma B.3.2 with $s = r - 1$. Each of these components has dimension $d > 0$, and they lie in distinct degrees, so J is infinite dimensional.

If R were finitely generated, its quotient S would be finitely generated too. Write a finite set of algebra generators of S as $c_i + j_i$, $1 \leq i \leq N$, with $c_i \in \mathbb{C}$ and $j_i \in J$. Because $J^2 = 0$, every polynomial in these generators lies in the finite-dimensional span of $1, j_1, \ldots, j_N$. This contradicts the infinite dimension of J. □

B.4 Section algebras of cotangent extensions

We now pass from the tangent bundle to the dual V of an extension W of $\mathcal{O}_M$ by Ω^1_M with nonzero ruling component, for instance the canonical extension of a Kähler class. Sections of $\operatorname{Sym}^r V$ are fiber-homogeneous polynomial functions on W. Compared with the cotangent bundle there is one more fiber coordinate w, the coordinate of the quotient $\mathcal{O}_M$, and the affine term of the torsor becomes a quadratic relation among the fiber-polynomial functions over the ruling. The weight decomposition of the preceding section still applies and shows that the section algebra of V is a polynomial ring in one variable over the tangent algebra. In particular, $\mathcal{O}_{\mathbb{P}(V)}(1)$ is big and the section algebra of V is not finitely generated.

Keep the notation of the two preceding sections. Let W be an extension of $\mathcal{O}_M$ by Ω^1_M with class $\rho\xi + Kf^*\eta$, where $\rho \neq 0$, and put $V = W^*$. Adding the coordinate w of the quotient $\mathcal{O}_M$ to the charts in the proof of Theorem B.2.1, the fiber coordinates over the elliptic cover change by

$$v = 1/t, \qquad u_\infty = u, \qquad y_\infty = -t^2 y - \rho t w, \qquad w_\infty = w.$$

For $\lambda = m\tau + n$, the lattice acts by

$$(z, t, u, y, w) \longmapsto (z + \lambda, b_\lambda t, u + m(Kw + q_0 t y), y/b_\lambda, w), \qquad q_0 = 2\pi \mathrm{i} d,$$

with the normalized factors b_λ. Fiber-homogeneous polynomial functions on W represent symmetric tensors of V. In the ruling coordinates they satisfy a quadratic relation coming from the affine term of the torsor, while the elliptic monodromy determines which weights descend to M.

Lemma B.4.1 *Over $\mathbb{C} \times \mathbb{P}^1$, the graded algebra of fiber-homogeneous polynomial functions on the pullback of W is*

$$\mathcal{O}(\mathbb{C})[c_-, c_+, h, w, u] / \big(c_- c_+ - h(h + \rho w)\big),$$

where all the variables have degree one and, in the finite ruling chart,

$$c_- = y, \qquad c_+ = t(ty + \rho w), \qquad h = ty.$$

The lattice fixes h and w, multiplies c_- by b_λ^{-1} and c_+ by b_λ, and sends u to $u + m(q_0 h + Kw)$.

Proof Set $s_1 = y$ and $s_2 = ty + \rho w$ in the finite chart. Since $\rho \neq 0$, these are linear coordinates that can replace y, w, with $w = (s_2 - t s_1)/\rho$. In the infinite chart put

$$(s_{1,\infty}, s_{2,\infty}) = (-v y_\infty - \rho w, -y_\infty).$$

The transition is $(s_{1,\infty}, s_{2,\infty}) = t(s_1, s_2)$, so s_1, s_2 are the fiber coordinates of $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$, while u is the coordinate of a trivial line bundle.

By holomorphy in the infinite chart, a polynomial that is homogeneous of degree j in s_1, s_2 has coefficients of degree at most j in t. Distributing the exponent of t among the j fiber factors, every admissible monomial is a product of s_1, s_2, ts_1, ts_2. These four generators satisfy the determinant relation

$$s_1(ts_2) - (ts_1)s_2 = 0,$$

and there are no further relations. Indeed, counting the $j + 1$ monomials in s_1, s_2 and the $j + 1$ admissible powers of t, the space of functions of degree j has dimension $(j + 1)^2$. The polynomial algebra in four variables modulo this nonzero quadratic relation has the same dimension

$$\binom{j+3}{3} - \binom{j+1}{3} = (j + 1)^2$$

in degree j, where the second term is zero for $j < 2$. The natural surjective graded map is therefore an isomorphism in every degree. Including u adjoins a polynomial variable. The argument is uniform in z, and all coefficient functions are entire.

Identify the four generators as $c_- = s_1$, $h = ts_1$, $h + \rho w = s_2$ and $c_+ = ts_2$; the determinant relation becomes the stated quadratic relation. These functions are polynomial also in the infinite chart, where

$$c_- = -v^2 y_\infty - \rho v w, \qquad c_+ = -y_\infty, \qquad h = -v y_\infty - \rho w.$$

The lattice formulas follow by substitution in the finite chart. □

Theorem B.4.1 *There is a degree-preserving isomorphism of complex algebras*

$$\bigoplus_{r \geq 0} H^0(M, \operatorname{Sym}^r V) \simeq \left(\bigoplus_{r \geq 0} H^0(M, \operatorname{Sym}^r T_M) \right) [w],$$

where the variable w on the right has degree one.

Proof *Step 1: weights.* Give c_- weight -1, c_+ weight 1, and h, w, u weight zero. These weights come from the ruling action and are compatible with the quadratic relation and the lattice action. Using the relation to eliminate products $c_- c_+$, a polynomial of weight $-k$, $k \geq 1$, can be written as $c_-^k F(z, h, w, u)$, and a polynomial of weight k as $c_+^k F(z, h, w, u)$, where F has degree $r - k$ in tensor degree r; weight zero consists of the polynomials $F(z, h, w, u)$. These expressions are unique. Indeed, the substitution $c_+ \mapsto h(h + \rho w)/c_-$ embeds the algebra of Lemma B.4.1 into

$$\mathcal{O}(\mathbb{C})[h, w, u, c_-, c_-^{-1}],$$

the reduced expressions involve distinct powers of c_-, and their coefficients vanish only if the original coefficients vanish, because $h(h + \rho w)$ is a nonzero polynomial. This proves that the weight decomposition is a direct sum.

Step 2: coefficient bundles. Let U_3 be the rank-three bundle on E of linear polynomials in h, w, u with the transformation rule

$$F(z + \lambda, h, w, u) = F(z, h, w, u - m(q_0 h + K w)).$$

By the transformation of c_- and c_+ in Lemma B.4.1, the block of weight $-k$ is $H^0(E, A^k \otimes \operatorname{Sym}^{r-k} U_3)$, and the block of weight k is $H^0(E, A^{-k} \otimes \operatorname{Sym}^{r-k} U_3)$; the inverse translation of u appears when invariance is written out. The signs of the twists agree with those in Lemma B.3.1.

Step 3: reduction to the tangent bundle. Since $q_0 \neq 0$, put $H = h + (K/q_0)w$. Then the translation of u is $u \mapsto u - m q_0 H$, and consequently

$$U_3 \simeq U \oplus \mathcal{O}_E w,$$

where the variable h of the bundle U defined above is replaced by H. The symmetric powers of U_3 are the direct sums of the bundles $w^j \operatorname{Sym}^{s-j} U$, $0 \leq j \leq s$. The

vanishing for negative twists in Lemma B.3.2 eliminates every positive weight, and the same lemma identifies the algebra of sections of weight zero with $\mathbb{C}[H, w]$.

In each negative weight, expanding in powers of w identifies the remaining spaces with the corresponding blocks for the tangent bundle in Lemma B.3.1, with a polynomial variable w adjoined. Sending the Euler variable h of those blocks to H and their leading factor y^k to c_-^k gives bijections in every tensor degree and weight. Products of surviving blocks multiply their coefficients and add their powers of c_-, and the eliminated positive weights contribute no products. The bijections therefore preserve multiplication and give the asserted isomorphism of graded algebras. □

Corollary B.4.1 *For every integer $r \geq 0$,*

$$h^0(M, \operatorname{Sym}^r V) = r + 1 + d\binom{r+3}{4}.$$

The tautological line bundle on $\mathbb{P}(V)$ is big, with volume d, and the section algebra of V is not finitely generated. These conclusions hold for the canonical extension of every Kähler class on M.

Proof The isomorphism of Theorem B.4.1 and Theorem B.3.1 give

$$h^0(M, \operatorname{Sym}^r V) = \sum_{j=0}^{r} h^0(M, \operatorname{Sym}^j T_M) = r + 1 + d \sum_{j=0}^{r} \binom{j+2}{3} = r + 1 + d\binom{r+3}{4},$$

where $\binom{2}{3} = 0$. The surface M is projective, and by coherent GAGA the holomorphic bundle V is algebraic. Thus $\mathbb{P}(V)$ is a projective fourfold, and the volume of its tautological line bundle is the limit of $24h^0(M, \operatorname{Sym}^r V)/r^4$, which equals $d > 0$. Hence this line bundle is big. If the section algebra of V were finitely generated, so would be its quotient by w, contradicting Theorem B.3.2.

For the canonical extension $0 \to \mathcal{O}_M \to V \to T_M \to 0$ of a Kähler class, the dual extension of $\mathcal{O}_M$ by Ω^1_M has nonzero ruling component, as shown in the proof of Theorem B.2.1. Its class therefore has $\rho \neq 0$ in the conventions used here, and the same argument applies. □

B.5 Cotangent extensions with zero ruling component

Every Kähler class has a nonzero ruling component, but the extensions with $\rho = 0$, that is, with class $Kf^*\eta$, can be treated in the same way. Their section algebras are the same as for $\rho \neq 0$, but their torsors behave differently: they contain compact rational curves, and their universal covers are not Stein.

Keep the notation of the preceding sections, with the class η and the form ϑ_E of Lemma B.2.1. Let W be an extension of $\mathcal{O}_M$ by Ω^1_M with class $Kf^*\eta$, and put $V = W^*$. The homogeneous charts used above, now with zero ruling component, have

transitions

$$v = 1/t, \qquad u_\infty = u, \qquad y_\infty = -t^2 y, \qquad w_\infty = w.$$

For $\lambda = m\tau + n$, the lattice acts by

$$(z, t, u, y, w) \longmapsto (z + \lambda, b_\lambda t, u + m(q_0 t y + Kw), y/b_\lambda, w), \qquad q_0 = 2\pi \mathrm{i} d.$$

For the differences $z_i - z_j = m_{ij}\tau + n_{ij}$ of local lifts, the zero sections of these charts differ by $Km_{ij} f^* \vartheta_E$ on overlaps over the base and agree on the two ruling charts, so they give the class $Kf^*\eta$. The linear parts are the cotangent transitions, and in the infinite ruling chart the lattice acts holomorphically by $u_\infty \mapsto u_\infty + m(Kw - q_0 v y_\infty)$. Thus the same cocycle computation constructs W also when its ruling component is zero. The fiberwise change of coordinates used for $\rho \neq 0$ is not available, but the polynomial functions on the cotangent bundle give the homogeneous algebra directly.

Lemma B.5.1 *Over $\mathbb{C} \times \mathbb{P}^1$, the graded algebra of fiber-homogeneous polynomial functions on the pullback of W is*

$$O(\mathbb{C})[c_-, c_+, h, u, w]/(c_- c_+ - h^2), \qquad c_- = y, \quad c_+ = t^2 y, \quad h = ty,$$

where all five variables have degree one. The lattice fixes h and w, sends c_- to $b_\lambda^{-1} c_-$ and c_+ to $b_\lambda c_+$, and translates u by $m(q_0 h + Kw)$.

Proof As in the proof of Lemma B.3.1, holomorphy in the infinite ruling chart bounds the degree in t of the coefficient of y^j by $2j$; the variable w does not change between the charts. Distributing the exponent a among j integers in $\{0, 1, 2\}$, every monomial $t^a y^j$ with $0 \leq a \leq 2j$ is a product of y, ty and $t^2 y$. These generators satisfy $y(t^2 y) = (ty)^2$.

In degree j, the space of functions of y and t has dimension $2j + 1$. The polynomial algebra in three variables modulo the nonzero quadratic relation has the same dimension

$$\binom{j+2}{2} - \binom{j}{2} = 2j + 1,$$

where the second term is zero for $j < 2$, so the natural surjective graded map is an isomorphism. Adjoining the two trivial fiber coordinates u, w gives the stated algebra, with entire coefficients in z. In the other chart,

$$c_- = -v^2 y_\infty, \qquad c_+ = -y_\infty, \qquad h = -v y_\infty,$$

so all generators are holomorphic there. Substitution gives their lattice transformations. □

Proposition B.5.1 *There is a degree-preserving isomorphism of algebras*

$$\bigoplus_{r \geq 0} H^0(M, \mathrm{Sym}^r V) \simeq \left(\bigoplus_{r \geq 0} H^0(M, \mathrm{Sym}^r T_M) \right) [w], \qquad \deg w = 1.$$

Consequently the dimension formula, the bigness and volume of the tautological line bundle, and the failure of finite generation in Corollary B.4.1 hold also for this extension.

Proof The proof of Theorem B.4.1 uses Lemma B.4.1 only through its description of the algebra as $O(\mathbb{C})[c_-, c_+, h, w, u]/(c_-c_+ - P)$, with the nonzero polynomial $P = h(h + \rho w)$, and through the lattice action on these generators. Lemma B.5.1 gives the same description with $P = h^2$ and the same lattice action, so that proof applies verbatim and gives the isomorphism. Summing the dimensions for the tangent bundle and taking the quotient by w gives the remaining assertions, exactly as in the proof of Corollary B.4.1. □

Proposition B.5.2 *The cotangent torsor T_W is not Stein. Its universal cover is*

$$\mathbb{C}^2 \times \mathrm{Tot}(O_{\mathbb{P}^1}(-2)),$$

where Tot *denotes the total space of a line bundle; this cover is not Stein either.*

Proof Set $w = 1$. The subset $\{y = 0\}$ of the finite ruling chart agrees with $\{y_\infty = 0\}$ in the other chart, and the lattice acts on it by $u \mapsto u + Km$. It therefore descends to a submanifold of T_W that is a $\mathbb{P}^1$-bundle over

$$B_K = \mathbb{C}^2_{z,u}/\Lambda, \qquad (m\tau + n) \cdot (z, u) = (z + m\tau + n, u + Km);$$

this quotient is a complex manifold because translation in z gives a free and properly discontinuous action. Every fiber of this bundle is a compact connected curve in T_W, on which holomorphic functions are constant. Thus T_W is not holomorphically separable, and hence not Stein.

Before the lattice quotient, the affine charts have no affine term in the ruling direction, and they glue to $\mathbb{C}^2_{z,u} \times \mathrm{Tot}(O_{\mathbb{P}^1}(-2))$, since $O_{\mathbb{P}^1}(-2)$ is the cotangent bundle of $\mathbb{P}^1$. This space is simply connected: contracting the fibers of the line bundle retracts the last factor onto $\mathbb{P}^1$, and $\mathbb{C}^2$ is contractible. The lattice acts freely and properly discontinuously, so this space is the universal cover. Its zero section contains a compact projective line, which again obstructs holomorphic separability and hence the Stein property. □

Thus the classes with zero and with nonzero ruling component have isomorphic graded section algebras, although Theorem B.2.1 gives a Stein universal cover only when the ruling component is nonzero.

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